Quant Interview Notes

Probability, statistics, machine learning, market making, and derivatives for quantitative interviews — with full proofs, worked examples, and a verified problem bank

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Μηδεὶς ἀλόγιστος εἰσίτω

Preface

Who this is for. These notes are written for a strong student or early-career researcher preparing for quantitative-finance and AI-research interviews. Everything needed is developed in place: the document is self-contained given single-variable calculus, elementary linear algebra, and comfort with mathematical notation. Probability is built up from its working tools rather than from measure theory. If you are able to understand and answer all the questions and examples, you will undoubtedly pass just about any assessment/interview in quant or AI research.

The proof policy. Three kinds of statement appear, and the typography distinguishes them honestly:

- **Theorems, Propositions, Lemmas, Corollaries** are proved — either on the spot, or, where a problem in the Problem Bank asks for exactly that proof, by a deferral naming the problem (the full argument then appears in its solution). A proof that would run for more than about three quarters of a page is not printed in its chapter: the chapter gives a sketch of the argument, and the complete proof stands in the appendix at the end of the document, where the sketch points.
- **Facts** are true results whose proofs lie beyond the scope of this document (the Central Limit Theorem, the spectral theorem, and similar); each is stated precisely, attributed inline where a specific source deserves credit, and never silently treated as proven here.
- **Empirical facts** are regularities of markets or computation (the square-root impact law, the variance risk premium); they are labelled as empirical and should be quoted as evidence, not as mathematics.

A fourth numbered environment, **Procedure**, appears occasionally: it is a recipe — steps to execute.

Provenance. Two small marks record where material comes from. A star (★) after a problem, an example, or a statement means that item reproduces, from memory, a question asked in a live 2026 interview (video or on-site); a diamond (⧫) means the same for a timed online assessment. The firms are deliberately not named. Every problem's answer was independently re-derived by two blind solvers before inclusion, and the expository material has been audited by independent recomputation.

Notation

One symbol keeps one meaning throughout. Estimators wear hats; matrices are capital letters.

$P(A)$, $P(A \mid B)$	probability; conditional probability
$\mathbb{E}[X]$, $\mathrm{Var}(X)$, $\mathrm{Cov}(X,Y)$, ρ	expectation, variance, covariance, correlation
$\mathbf{1}_A$, $\mathbf{1}\{A\}$	indicator of the event A
$X \sim \mathcal{N}(\mu, \sigma^2)$	normal with mean μ and *variance* σ^2
$\mathrm{U}(a,b)$, $\mathrm{Exp}(\lambda)$, $\mathrm{Bin}(n,p)$, $\mathrm{Geom}(p)$, $\mathrm{Poisson}(\lambda)$	uniform; exponential; binomial; geometric; Poisson
Φ, φ	standard normal CDF and density
iid	independent and identically distributed
$A^\top$, λ_i, tr, det	transpose, eigenvalues, trace, determinant
$\hat{\beta}$, $\hat{\sigma}^2$	estimators (hats)
$\|x\|$	Euclidean norm
i, N, p, $q = 1-p$, $r = q/p$	random-walk state, absorbing target, step probabilities and their ratio; in the Part *Derivatives*, r is the interest rate and q the dividend yield
log, ln	natural logarithm, written either way ($\log_{10}$ written explicitly)
f', ∂, $\int$	differentiation and integration
★, ⧫	provenance: reproduced from a live 2026 interview (★) or a timed online assessment (⧫); firms not named

Inside the Part *Derivatives* (§§16–23) each symbol below keeps one meaning; where one collides with the table above, the meaning below holds inside the Part, and the section concerned says so in its first Setup.

Part *Derivatives*: prices, rates and claims	
S_t, $S = S_0$; $S_{k,t}$	price of the underlying at time t, and today; price of asset k where two assets appear (first subscript the asset, second the time)
K; t, T, $\tau = T - t$	strike, or delivery price of a forward; current time, maturity and time to maturity, in years
r, $r(T)$, r_i; r_{for}, r_{st}	continuously compounded risk-free rate (not the ratio q/p); T-year zero rate, discounting by $e^{-r(T)T}$; rate of period i (a day, or a tree step), never a compounding frequency; foreign rate; storage cost as a proportional rate
q; b	continuous yield of the underlying (not $1-p$); cost of carry $b = r - q$, equal to $r - q + r_{\mathrm{st}}$ with storage, $r - r_{\mathrm{for}}$ for a currency and 0 for a futures price
F_t, $F = F_0$; F^{fut}; T_F	forward price for delivery at T, which is also the futures price when rates are deterministic; futures price where the two differ; delivery date of a futures contract
basis_t	the basis $S_t - F_t$, written as a word because b is the carry
$V(S,t)$, $V_t = V(S_t,t)$; Π_t	price function of a derivative or position, and its price at time t; value of a trading strategy
$\partial_t V$, $\partial_S V$, $\partial_{SS} V$	partial derivatives, also written $\partial V/\partial S$; a subscript on a price is a time or a label, never a derivative; the order of a mixed derivative is not shown (Lemma 19.54)
C, $\mathcal{P}$; C^{am}, $\mathcal{P}^{\mathrm{am}}$	European call and put (the put is calligraphic because P is probability); American call and put
C^{BSM}, $V^{\mathrm{BSM}}(S,t;\sigma)$	price given by the Black–Scholes–Merton formula; inputs after a semicolon are parameters
$\mathrm{Dig}(K)$, $\mathrm{AoN}(K)$	cash-or-nothing digital paying 1 above K; asset-or-nothing claim paying the underlying above K
f; x^+	payoff function of a claim (in §19 also a function $f(t,x)$ of time and a Brownian value); $\max(x,0)$
d_1, d_2	$[\ln(F/K) \pm \sigma^2\tau/2]/(\sigma\sqrt{\tau})$; the literature's $N(d_1)$ is $\Phi(d_1)$ here

Part *Derivatives*: models and stochastic calculus

σ; μ; $g = \mu - \sigma^2/2$; ν	volatility; real-world expected return; expected log-growth rate; a constant log-drift, $\nu = r - q - \sigma^2/2$ under $\mathbb{Q}$
P, $\mathbb{E}$; $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}$, $\mathbb{Q}(A)$	real-world probability and expectation; risk-neutral probability and expectation (no hats: hats mark estimators)
$S_t^{\mathbb{Q}}$	the risk-neutral price path: S_t with μ replaced by $r-q$
u, d, p, $1-p$	tree up- and down-factors; risk-neutral up- and down-probabilities
Δt, n; J	step length and number of steps (n is a sample size in §§22–23); number of dates in a grid of fixings or simulation steps
W_t, $W_{k,t}$; Z	standard Brownian motion, and motion k of two; a standard normal variable
R_i	log-return $\ln(S_i/S_{i-1})$ of period i, one trading day unless stated
α_t, β_t	drift and diffusion coefficient of an Itô process, $dX_t = \alpha_t\,dt + \beta_t\,dW_t$; always with a subscript or arguments
$\int_{t_0}^{t} \xi_s\,dW_s$; $\mathcal{H}$	Itô integral, the mean-square limit of left-endpoint sums; the class of integrands ξ
$C_E^{1,2}$; $\mathcal{G}_J$, $\mathcal{G} = \mathcal{G}_1$	functions $f(t,x)$ with f, $\partial_t f$, $\partial_x f$, $\partial_{xx} f$ continuous and of exponential growth; payoffs fixed on J dates, of moderate growth (Definition 19.19)
in probability, in mean square, almost surely	the three modes of convergence of a sequence of random variables
$\mathcal{L}V$	the operator $\partial_t V + rS\,\partial_S V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS} V - rV$ of the Black–Scholes–Merton equation
h; $h_{k,t}$	hedge ratio; holding of instrument k at time t in a strategy

Part *Derivatives*: Greeks, volatility and paths

Δ, Γ, Θ, $\mathcal{V}$, $\mathcal{R}$	delta, gamma, theta (per year), vega (per unit of volatility), rho; Δ alone is the Greek, while Δt and ΔS are increments
σ_{imp}, σ_{real}, σ_{loc}	implied, realised and local volatility; $\sigma_{\text{imp}}(K,T)$ is the implied surface
$f_{\mathbb{Q}}(x,T)$	risk-neutral density of S_T at the price x
K_{var}, $K_{\log}$	fair strike of a variance swap on squared returns; fair strike of its log-variance leg
H; t_H, t_ℓ	barrier level; first time the path reaches H, or the level or boundary ℓ
$S_T^{\max}$, $S_T^{\min}$	running maximum and minimum of the path on $[0,T]$
λ	jump intensity, a Poisson rate; in §22 only, the EWMA decay
β	CAPM beta; in §22 a GARCH coefficient, and in §23 the control-variate coefficient β, β^*

Part I
Probability and Games of Chance

1 Expected Value for Trading Games

Expected value is the fair price of a random payoff, and the games in this section — coin patterns, dice markets, re-roll options, shuffled hats — are the standard vehicles by which quantitative interviews test whether a candidate can compute it quickly and without error. Three tools do almost all of the work: linearity of expectation, indicator variables, and conditioning. This section develops those tools formally, records the named distributions used throughout the document, and then works through the classic game families in turn: geometric waiting times, pattern waits, coupon collection, order statistics of dice, the option value of a re-roll, a conditioning trap, and the hat-check problem.

1.1 The Toolkit

This subsection states and proves the three results that dispose of most expected-value games: linearity of expectation, the indicator lemma, and the tower rule. Everything later in the section is an application of these three, so the proofs here repay careful reading.

Setup. Throughout this subsection, all random variables are discrete: each takes values in a countable subset of $\mathbb{R}$. Probabilities are written P, and $P(X = x, Y = y)$ denotes the joint probability mass function of a pair (X, Y) defined on a common probability space. For an event A, the indicator $\mathbf{1}_A$ is the random variable equal to 1 when A occurs and 0 otherwise.

Definition 1.1 (Expectation of a discrete random variable)**.** Let X be a random variable taking values in a countable set $\mathcal{X} \subset \mathbb{R}$. If $\sum_{x\in\mathcal{X}} |x|\, P(X = x) < \infty$, we call X integrable and define its expectation as

$$\mathbb{E}[X] = \sum_{x\in\mathcal{X}} x\, P(X = x).$$

Absolute convergence guarantees that the value does not depend on the order in which the elements of $\mathcal{X}$ are enumerated.

Theorem 1.2 (Linearity of expectation)**.** *Let X and Y be integrable random variables on a common discrete probability space, and let $a, b \in \mathbb{R}$. Then $aX + bY$ is integrable and*

$$\mathbb{E}[aX + bY] = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

No independence assumption is made: the identity holds for arbitrarily dependent X and Y.

Proof. We proceed by direct computation with the joint probability mass function, justifying every rearrangement by absolute convergence. Write $p(x, y) = P(X = x, Y = y)$, with (x, y) ranging over the countable joint range.

Step 1: integrability. By the triangle inequality, and since all terms below are nonnegative (so the sums may be rearranged and grouped freely),

$$\sum_{x,y} |ax + by|\, p(x, y) \;\le\; |a| \sum_{x,y} |x|\, p(x, y) + |b| \sum_{x,y} |y|\, p(x, y) \;=\; |a|\,\mathbb{E}|X| + |b|\,\mathbb{E}|Y| < \infty,$$

where the final equality uses $\sum_y p(x, y) = P(X = x)$ and $\sum_x p(x, y) = P(Y = y)$ (marginalisation). Hence $aX + bY$ is integrable.

Step 2: the identity. Since the double series now converges absolutely, we may split and regroup it:

$$\mathbb{E}[aX+bY] = \sum_{x,y}(ax+by)\,p(x,y) = a\sum_x x\sum_y p(x,y) + b\sum_y y\sum_x p(x,y) = a\,\mathbb{E}[X] + b\,\mathbb{E}[Y].$$

At no point did we use any relationship between X and Y beyond their being defined on the same space: only the marginal sums entered. □

Example 1.3 (Sum of two dice)**.** Let X and Y be the faces of two fair six-sided dice, each with $\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1+2+\cdots+6}{6} = 3.5$. By Theorem 1.2, $\mathbb{E}[X+Y] = 3.5 + 3.5 = 7$, with no need to enumerate the 36 joint outcomes and no need to know whether the dice are independent. The same one-line answer holds even for two dice glued together.

Lemma 1.4 (Indicator lemma)**.** *For any event A, the indicator satisfies $\mathbb{E}[\mathbf{1}_A] = P(A)$.*

Proof. We compute directly from Definition 1.1. The random variable $\mathbf{1}_A$ takes only the values 1 and 0, with probabilities $P(A)$ and $1 - P(A)$ respectively, so

$$\mathbb{E}[\mathbf{1}_A] = 1 \cdot P(A) + 0 \cdot \big(1 - P(A)\big) = P(A).$$ □

Corollary 1.5 (Counting via indicators)**.** *Let $A_1, \dots, A_m$ be events, in any dependence structure whatsoever, and let $X = \sum_{i=1}^m \mathbf{1}_{A_i}$ count how many of them occur. Then*

$$\mathbb{E}[X] = \sum_{i=1}^m P(A_i).$$

Proof. We proceed by induction on m. The case $m = 1$ is Lemma 1.4. For the inductive step, write $X = \big(\sum_{i=1}^{m-1} \mathbf{1}_{A_i}\big) + \mathbf{1}_{A_m}$; both summands are integrable (they are bounded), so Theorem 1.2 applies and gives $\mathbb{E}[X] = \sum_{i=1}^{m-1} P(A_i) + P(A_m)$ by the inductive hypothesis and Lemma 1.4. □

Example 1.6 (Aces without replacement)**.** Draw two cards from a standard 52-card deck without replacement, and let X count the aces drawn. The two draws are dependent, yet by symmetry each draw individually is uniform over the deck, so $P(\text{draw } i \text{ is an ace}) = \frac{4}{52} = \frac{1}{13}$ for $i = 1, 2$. Corollary 1.5 gives $\mathbb{E}[X] = \frac{2}{13} \approx 0.154$. The dependence between the draws, which would complicate any attempt to work with the distribution of X itself, is simply irrelevant to the expectation.

Trading application. Questions of the form "expected number of …" — fills across a set of resting quotes, winning trades in a day, fixed points of a shuffle — are solved by Corollary 1.5: the answer is the sum of the individual probabilities, whatever the dependence between the events. Most game questions in this section fall to writing the payoff as a sum of indicators before doing anything else.

Proposition 1.7 (Tower rule)**.** *Let X be integrable and let Y be a discrete random variable on the same space. For each y with $P(Y = y) > 0$, define the conditional expectation*

$$g(y) = \mathbb{E}[X \mid Y = y] = \sum_x x\,P(X = x \mid Y = y).$$

Then $\mathbb{E}\big[g(Y)\big] = \mathbb{E}[X]$; in the usual shorthand, $\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X]$.

Proof. We expand the outer expectation and swap the order of summation, which is legitimate because the double series converges absolutely ($\sum_{x,y}|x|P(X=x,Y=y) = \mathbb{E}|X| < \infty$):

$$\mathbb{E}\big[g(Y)\big] = \sum_y P(Y=y)\sum_x x\,\frac{P(X=x,Y=y)}{P(Y=y)} = \sum_x x\sum_y P(X=x,Y=y).$$

The inner sum on the right is $P(X = x)$ by marginalisation, so the right-hand side equals $\sum_x x\,P(X=x) = \mathbb{E}[X]$ by Definition 1.1. □

Example 1.8 (A two-stage payout)**.** A fair coin is flipped. On heads, the payout is the face of a fair die; on tails, the payout is a flat 2. Conditioning on the flip Y and applying Proposition 1.7,

$$\mathbb{E}[X] = \tfrac{1}{2} \cdot 3.5 + \tfrac{1}{2} \cdot 2 = 2.75,$$

where 3.5 is the die's mean from Example 1.3. No joint distribution ever needs to be written down.

Intuition. Linearity does the bookkeeping; conditioning does the dynamics. First-step analysis — the workhorse for every waiting-time game below — is nothing but the tower rule of Proposition 1.7 applied at the first move of a sequential game: condition on what the first step does, and let the structure of the game tell you what the conditional expectations are.

1.2 Named Distributions

Seven named distributions cover nearly every random quantity in this document. This subsection records their definitions and collects their means and variances, together with the one-line "hook" worth memorising for each, in Table 1. Two of the table's entries are proved immediately in the flow of this section: the Bernoulli variance below, and the geometric mean as Theorem 1.13 in the next subsection.

Setup. For an integrable random variable X with mean $\mu = \mathbb{E}[X]$ and integrable square, the variance is

$$\operatorname{Var}(X) = \mathbb{E}\big[(X-\mu)^2\big],$$

and the standard deviation is its square root. The reader should verify, by expanding the square and applying Theorem 1.2, the shortcut form

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \mu^2. \tag{1.1}$$

Definition 1.9 (The seven named distributions)**.** Let $p \in [0,1]$ ($p \in (0,1]$ for the geometric), $n \geq 1$ an integer, $\lambda > 0$, $a < b$ reals, $\mu \in \mathbb{R}$, and $\sigma > 0$.

- **Bernoulli(p):** $X \in \{0,1\}$ with $P(X=1) = p$ — a single yes/no trial.
- **Binomial** $\mathrm{Bin}(n,p)$**:** the number of successes in n iid Bernoulli(p) trials; $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$ for $0 \leq k \leq n$.
- **Geometric** $\mathrm{Geom}(p)$**:** the number of iid Bernoulli(p) trials up to and including the first success; $P(X=k) = (1-p)^{k-1}p$ for $k \geq 1$. (This document always counts trials, not failures.)
- **Poisson(λ):** $P(X=k) = e^{-\lambda}\lambda^k/k!$ for integers $k \geq 0$.
- **Uniform** $\mathrm{U}(a,b)$**:** continuous with density $\frac{1}{b-a}$ on (a,b).
- **Exponential** $\mathrm{Exp}(\lambda)$**:** continuous with density $\lambda e^{-\lambda x}$ for $x > 0$; the parameter λ is a rate.
- **Normal** $\mathcal{N}(\mu, \sigma^2)$**:** continuous with density $\frac{1}{\sigma\sqrt{2\pi}} \exp\big(-\frac{(x-\mu)^2}{2\sigma^2}\big)$; the second parameter is the variance.

Proposition 1.10 (Bernoulli mean and variance)**.** *If $X \sim$ Bernoulli(p), then $\mathbb{E}[X] = p$ and $\operatorname{Var}(X) = p(1-p)$.*

Proof. We compute both moments directly. The mean is $\mathbb{E}[X] = 1 \cdot p + 0 \cdot (1-p) = p$, which is Lemma 1.4 in disguise (X is the indicator of its own success event). Because X takes only the values 0 and 1, we have $X^2 = X$ pointwise, hence $\mathbb{E}[X^2] = p$. By the shortcut (1.1),

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1-p). \qquad \square$$

Distribution	$\mathbb{E}$	Var	**Hook**
Bernoulli(p)	p	$p(1-p)$	the atom of everything
Bin(n,p)	np	$np(1-p)$	$\approx \mathcal{N}$ for $np(1-p)$ large
Geom(p)	$1/p$	$(1-p)/p^2$	memoryless
Poisson(λ)	λ	λ	rare-event limit; sums add
U(a,b)	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\mathbb{E}[U^k] = \frac{1}{k+1}$ on $(0,1)$
Exp(λ)	$1/\lambda$	$1/\lambda^2$	memoryless; $\mathbb{E}[X^n] = n!/\lambda^n$
$\mathcal{N}(\mu,\sigma^2)$	μ	σ^2	sums stay normal; $\mathbb{E}[(X-\mu)^4] = 3\sigma^4$

Table 1: The seven named distributions: mean, variance, and the one-line hook worth carrying into an interview.

Example 1.11 (Variance of a directional bet)**.** A strategy wins a fixed unit with probability $p = 0.55$ and loses nothing otherwise, so the per-trade outcome is Bernoulli(0.55). By Proposition 1.10, its variance is $0.55 \times 0.45 = 0.2475$ — barely below the maximum 0.25 attained at $p = 0.5$. Modest edges leave the per-trial variance essentially at its fair-coin worst case, which is why single trials tell you almost nothing and sample sizes matter.

Remark 1.12 (Where the table is proved). The Bernoulli row is Proposition 1.10. The binomial mean np follows in one line from Corollary 1.5, since a Bin(n,p) count is a sum of n success indicators each with probability p. The geometric mean $1/p$ is proved as Theorem 1.13 below. The remaining entries are standard; they are quoted here without proof and re-derived where the document needs them.

1.3 Geometric Waiting Times

How long until the first success? This is the simplest waiting-time game, it introduces first-step analysis — the tower rule of Proposition 1.7 applied at the first move — and every later waiting-time result in this section reduces to it or generalises it.

Setup. Let $X_1, X_2, \ldots$ be iid Bernoulli(p) trials with $0 < p \leq 1$, and let

$$T = \min\{\, n \geq 1 : X_n = 1 \,\}$$

be the number of trials up to and including the first success. Since the first success occurs at trial n exactly when the first $n-1$ trials fail and the n-th succeeds, $P(T = n) = (1-p)^{n-1}p$; that is, $T \sim \text{Geom}(p)$ in the convention of Definition 1.9.

Theorem 1.13 (Expected wait for the first success)**.** *With the setup above,* $\mathbb{E}[T] = 1/p$.

Proof. We proceed by first-step analysis: condition on the outcome of the first trial and apply the tower rule (Proposition 1.7). Figure 1 displays the decomposition.

Step 1: T is integrable. If $p = 1$ then $T \equiv 1$ and there is nothing to prove, so assume $p < 1$. The series $\sum_{n\geq 1} n(1-p)^{n-1}p$ converges by the ratio test (the ratio of consecutive terms tends to $1 - p < 1$), so $\mathbb{E}[T] < \infty$. Finiteness is what makes the algebra of Step 2 legitimate.

Step 2: the recursion. On the event $\{X_1 = 1\}$, which has probability p, we have $T = 1$. On $\{X_1 = 0\}$, one trial has been spent, and the remaining trials $X_2, X_3, \ldots$ are again iid Bernoulli(p), so the residual wait $T - 1$ has the same distribution as T; hence $\mathbb{E}[T \mid X_1 = 0] = 1 + \mathbb{E}[T]$. The tower rule gives

$$\mathbb{E}[T] = p \cdot 1 + (1-p)\big(1 + \mathbb{E}[T]\big) = 1 + (1-p)\,\mathbb{E}[T]. \tag{1.2}$$

Since $\mathbb{E}[T]$ is finite, we may subtract $(1-p)\mathbb{E}[T]$ from both sides of (1.2) to get $p\,\mathbb{E}[T] = 1$, i.e. $\mathbb{E}[T] = 1/p$. □

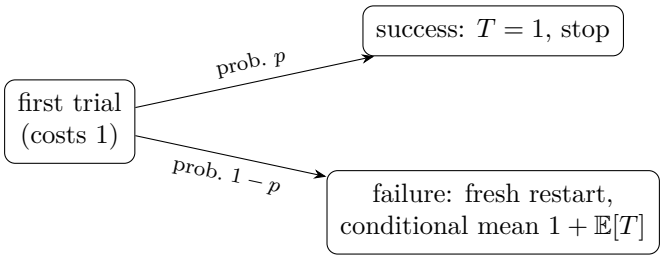


Figure 1: The first-step-analysis tree behind Theorem 1.13. Weighting the two branches by their probabilities yields the recursion (1.2).

Example 1.14 (Coins and dice)**.** For a fair coin, $p = \frac{1}{2}$ and the expected number of flips to the first head is $\mathbb{E}[T] = 2$. For a fair die waiting for the first six, $p = \frac{1}{6}$ and $\mathbb{E}[T] = 6$. As a check on the recursion (1.2) with $p = \frac{1}{2}$: the claimed value 2 satisfies $2 = 1 + \frac{1}{2} \cdot 2$, as it must.

Proposition 1.15 (Memorylessness of the geometric distribution)**.** *With the setup above, assume $p < 1$. Then for all integers $m, n \geq 0$,*

$$P(T > m + n \mid T > m) = P(T > n).$$

Proof. We compute the tail probabilities explicitly. The event $\{T > k\}$ occurs exactly when the first k trials all fail, so $P(T > k) = (1 - p)^k$ by independence. Since $\{T > m + n\} \subset \{T > m\}$,

$$P(T > m + n \mid T > m) = \frac{P(T > m + n)}{P(T > m)} = \frac{(1 - p)^{m+n}}{(1 - p)^m} = (1 - p)^n = P(T > n). \qquad \square$$

Example 1.16 (Failed flips carry no information)**.** For the fair coin, given that no head has appeared in the first two flips, the probability that none appears in the next two is $P(T > 4 \mid T > 2) = (\frac{1}{2})^2 = \frac{1}{4}$, exactly the unconditional $P(T > 2)$. Two wasted flips leave the forecast for the future untouched.

Intuition. The recursion (1.2) works because the process is memoryless: after a failure you are back exactly where you started, one flip poorer. Proposition 1.15 is the precise statement of "back where you started", and $\mathbb{E}[T \mid X_1 = 0] = 1 + \mathbb{E}[T]$ is its expectation-level shadow.

1.4 Pattern Waiting Times: HH versus HT

Any two fixed consecutive flips of a fair coin are HH with probability $\frac{1}{4}$ and HT with probability $\frac{1}{4}$ — yet the expected wait for the first HH differs from the expected wait for the first HT. This subsection resolves the apparent paradox by first-step analysis on a two-state automaton, the same technique as in Theorem 1.13 but with more than one "restart" state.

Setup. A fair coin is flipped repeatedly; flips are iid with $P(\mathrm{H}) = P(\mathrm{T}) = \frac{1}{2}$. Let T_{HH} be the number of flips until the last two flips first read HH, and T_{HT} likewise for HT. For each pattern we track progress with two states: S_0 (no useful progress: either no flips yet, or the last flip was not H) and S_1 (the last flip was H). Write E_0 and E_1 for the expected number of further flips to complete the pattern from S_0 and S_1 respectively; each game starts in S_0.

Theorem 1.17 (Expected pattern waits, HH versus HT)**.** *With the setup above,*

$$\mathbb{E}[T_{\mathrm{HH}}] = 6 \qquad \textit{and} \qquad \mathbb{E}[T_{\mathrm{HT}}] = 4.$$

Proof. We proceed by first-step analysis on the automata of Figure 2, one flip at a time.

Step 1: all expectations are finite. Partition the flips into disjoint blocks of two. Whatever the past, each block independently equals the target pattern with probability $\frac{1}{4}$, so if the pattern has

not appeared by flip $2k$ then in particular none of the first k blocks equalled it: $P(T > 2k) \le (\frac{3}{4})^k$ for either pattern. Geometrically decaying tails give finite expectations, so the linear equations below may be solved by ordinary algebra.

Step 2: the HH system. From S_0, one flip is spent; heads (probability $\frac{1}{2}$) moves to S_1, tails stays in S_0. From S_1, heads completes HH; tails destroys all progress and returns to S_0. Hence

$$E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0, \qquad E_1 = 1 + \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}E_0. \tag{1.3}$$

The first equation of (1.3) rearranges to $E_0 = 2 + E_1$; substituting into the second gives $E_1 = 1 + \frac{1}{2}(2 + E_1)$, so $E_1 = 4$ and $E_0 = 6$. Since the game starts in S_0, $\mathbb{E}[T_{\mathrm{HH}}] = 6$.

Step 3: the HT system. From S_0 the transitions are as before. From S_1, however, tails completes HT, while heads — the failure — leaves the process still holding an H, i.e. still in S_1. This is the crucial structural difference: for HT, failure does not destroy progress. Hence

$$E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0, \qquad E_1 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}\cdot 0. \tag{1.4}$$

The second equation of (1.4) gives $E_1 = 2$; the first then gives $\frac{1}{2}E_0 = 1 + \frac{1}{2}\cdot 2 = 2$, so $E_0 = 4$ and $\mathbb{E}[T_{\mathrm{HT}}] = 4$. □

(Problem P1.10 asks you to reproduce this proof from scratch.)

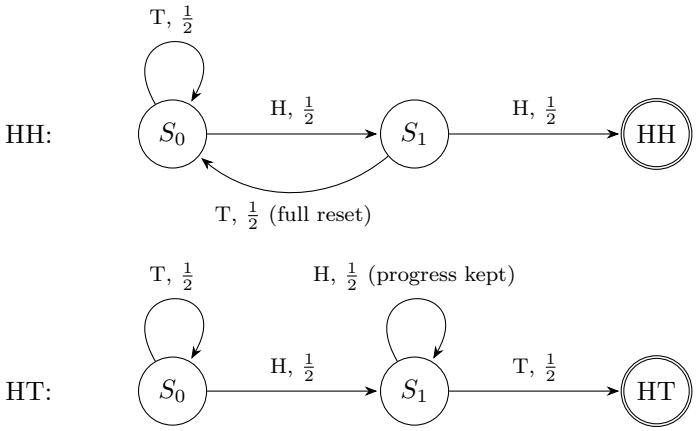


Figure 2: State-transition automata for the waits in Theorem 1.17. The two machines differ only in the failure edge out of S_1: for HH it returns to S_0 (all progress lost), while for HT it loops at S_1 (an extra head still leaves an H on the board).

Example 1.18 (HT as two geometric stages)**.** The HT wait admits an independent numeric check. Because HT progress never resets, the wait splits cleanly: first wait for an H, then — holding an H forever after, since further heads keep the H alive — wait for a T. Each stage is a first-success wait with $p = \frac{1}{2}$, worth 2 flips by Theorem 1.13, so $\mathbb{E}[T_{\mathrm{HT}}] = 2 + 2 = 4$, matching Theorem 1.17. No such decomposition exists for HH, precisely because its failure edge destroys the first stage's work.

Intuition. Self-overlapping patterns such as HH are slower to arrive: a failure at the final step destroys all progress and returns the automaton to S_0, while HT's failure edge preserves progress. The same mathematics governs stop-hunting sequences and losing streaks — "bad" events that reset you entirely are costlier than their probability alone suggests.

1.5 Coupon Collector

How many draws are needed to see every type at least once? The answer combines the two main tools so far — the geometric wait of Theorem 1.13 for each new type, and linearity (Theorem 1.2) to add the stages — and is a fixture of both interviews and applied sampling questions.

Setup. Draws are iid and uniform over n equally likely types. Let T be the number of draws until every type has appeared at least once, and write $H_n = \sum_{k=1}^{n} \frac{1}{k}$ for the n-th harmonic number.

Theorem 1.19 (Coupon collector)**.** *With the setup above,* $\mathbb{E}[T] = nH_n$.

Proof. We decompose the collection into stages and apply linearity. For $j = 0, 1, \ldots, n-1$, let T_j be the number of draws made while exactly j distinct types have been seen, so that $T = \sum_{j=0}^{n-1} T_j$. While exactly j types have been seen, each fresh draw is a new type with probability $\frac{n-j}{n}$, independently of everything drawn so far; the stage therefore ends at a first success of probability $\frac{n-j}{n}$, i.e. $T_j \sim \text{Geom}(\frac{n-j}{n})$, and Theorem 1.13 gives $\mathbb{E}[T_j] = \frac{n}{n-j}$. By Theorem 1.2 — which requires no independence between the stages —

$$\mathbb{E}[T] = \sum_{j=0}^{n-1} \frac{n}{n-j} = n \sum_{k=1}^{n} \frac{1}{k} = nH_n,$$

where the middle step substitutes $k = n - j$. □

(Problem P1.13 asks you to reproduce this proof from scratch.)

Example 1.20 (All six faces of a die)**.** For a fair die, $n = 6$ and

$$\mathbb{E}[T] = 6\Big(1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \tfrac{1}{5} + \tfrac{1}{6}\Big) = 6 \cdot \tfrac{49}{20} = \tfrac{147}{10} = 14.7.$$

The final face alone costs 6 expected rolls — the last stage is the whole first-six wait of Example 1.14 — while the first face is free after one roll.

1.6 Order Statistics of Dice

"Make me a market on the max of two dice" is a staple prompt: it tests whether a candidate reaches for the CDF rather than guessing symmetry. This subsection computes the maximum by CDF differencing and gets the minimum for free by linearity (Theorem 1.2).

Setup. Let X and Y be independent, each uniform on $\{1, \ldots, 6\}$, and set $M = \max(X, Y)$ and $L = \min(X, Y)$.

Proposition 1.21 (Max and min of two dice)**.** *With the setup above, for* $k = 1, \ldots, 6$*,*

$$P(M \le k) = \Big(\frac{k}{6}\Big)^2, \qquad P(M = k) = \frac{2k-1}{36},$$

and consequently

$$\mathbb{E}[M] = \frac{161}{36} \approx 4.47, \qquad \mathbb{E}[L] = \frac{91}{36} \approx 2.53.$$

Proof. We proceed by CDF differencing. The maximum is at most k exactly when both dice are, so independence gives $P(M \le k) = P(X \le k)\,P(Y \le k) = (k/6)^2$. Differencing consecutive CDF values yields the probability mass function:

$$P(M = k) = P(M \le k) - P(M \le k-1) = \frac{k^2 - (k-1)^2}{36} = \frac{2k-1}{36}.$$

The expectation is then

$$\mathbb{E}[M] = \sum_{k=1}^{6} k \cdot \frac{2k-1}{36} = \frac{2\sum_{k=1}^{6} k^2 - \sum_{k=1}^{6} k}{36} = \frac{2 \cdot 91 - 21}{36} = \frac{161}{36}.$$

For the minimum, observe the pointwise identity $L + M = X + Y$: on every outcome, the min and the max are the two rolled values in some order. Theorem 1.2 then gives

$$\mathbb{E}[L] = \mathbb{E}[X] + \mathbb{E}[Y] - \mathbb{E}[M] = 7 - \frac{161}{36} = \frac{91}{36},$$

using $\mathbb{E}[X + Y] = 7$ from Example 1.3. □

(Problem P1.20 runs the same CDF-differencing pipeline on the maximum of four dice.)

Example 1.22 (Sanity checks on the max distribution)**.** The masses $\frac{2k-1}{36}$ for $k = 1, \ldots, 6$ are $\frac{1}{36}, \frac{3}{36}, \frac{5}{36}, \frac{7}{36}, \frac{9}{36}, \frac{11}{36}$; their numerators are the first six odd numbers, summing to $6^2 = 36$, so the pmf sums to 1. The mass has visibly shifted upward — $P(M = 6) = \frac{11}{36} \approx 0.31$ against $P(M = 1) = \frac{1}{36}$ — which is why $\mathbb{E}[M] \approx 4.47$ sits well above the single-die mean 3.5.

Remark 1.23 (Median of five dice)*.* The median (third-smallest) of five iid dice has expectation exactly 3.5, by symmetry: the face swap $k \mapsto 7 - k$ preserves each die's distribution and maps the third-smallest of the five values to the third-largest, which among five values is again the median. Hence the median $\tilde{M}$ and $7 - \tilde{M}$ have the same distribution, so $\mathbb{E}[\tilde{M}] = 7 - \mathbb{E}[\tilde{M}]$, i.e. $\mathbb{E}[\tilde{M}] = 3.5$.

Interview trap. "Make me a market on the max of two dice" — the naive fair value 3.5 ignores that the max is biased up. Before quoting, always ask: is the payoff a *mean*, an *extreme*, or a *median* of the randomness? Extremes shift the fair value (Proposition 1.21); medians resist outliers (Remark 1.23).

1.7 Option Value of a Re-Roll

A die roll with the right to discard and try again is the smallest possible optimal-stopping problem, and it prices an option with nothing but the tower rule. The values 4.25 and $\frac{14}{3}$ below are standard interview answers worth knowing cold.

Setup. A fair die is rolled and the player is paid the face value in dollars. Before the game, the player is granted the right to discard the current roll and re-roll, usable at most k times in total; a discarded roll is gone forever. A fresh roll X is uniform on $\{1, \ldots, 6\}$ with $\mathbb{E}[X] = 3.5$. Let V_k denote the value of the game — the expected payoff under optimal play — with k re-rolls remaining and no roll yet showing, so that $V_0 = 3.5$.

Proposition 1.24 (Optimal re-roll policies and values)**.** *With the setup above:*

(i) With one re-roll, the optimal policy is to keep a first roll in $\{4, 5, 6\}$ *and re-roll* $\{1, 2, 3\}$*, and* $V_1 = \frac{17}{4} = 4.25$*.*

(ii) With two re-rolls, the optimal policy is to keep only $\{5, 6\}$ *on the first roll, then play as in (i); and* $V_2 = \frac{14}{3} \approx 4.67$*.*

Proof. We proceed by backward induction on the number of re-rolls remaining. With $k \geq 1$ re-rolls remaining and a face v showing, the player chooses between stopping (worth v) and discarding, which restarts the game with $k - 1$ re-rolls and is worth V_{k-1}; optimal play takes the larger, so the pre-roll value satisfies the dynamic-programming recursion

$$V_k = \mathbb{E}\big[\max(X, V_{k-1})\big], \qquad V_0 = 3.5, \tag{1.5}$$

by the tower rule (Proposition 1.7) conditioning on the face rolled.

One re-roll. A face v beats the continuation exactly when $v > V_0 = 3.5$, i.e. $v \in \{4, 5, 6\}$, whose average is 5. Evaluating (1.5),

$$V_1 = \tfrac{3}{6} \cdot 5 + \tfrac{3}{6} \cdot 3.5 = 2.5 + 1.75 = 4.25 = \tfrac{17}{4}.$$

Two re-rolls. Now the continuation is worth $V_1 = 4.25$, so only $v \in \{5, 6\}$ (average 5.5) is kept — in particular a 4, kept in the one-re-roll game, is discarded here because $4 < 4.25$. Evaluating (1.5) again,

$$V_2 = \tfrac{2}{6} \cdot 5.5 + \tfrac{4}{6} \cdot 4.25 = \tfrac{11}{6} + \tfrac{17}{6} = \tfrac{28}{6} = \tfrac{14}{3} \approx 4.67. \qquad \square$$

Example 1.25 (What the option itself is worth)**.** Subtracting the no-option value $V_0 = 3.5$ from the results of Proposition 1.24: a single re-roll right is worth $\frac{17}{4} - \frac{7}{2} = \frac{3}{4} = \$0.75$; the two-re-roll right is worth $\frac{14}{3} - \frac{7}{2} = \frac{7}{6} \approx \$1.17$. The second re-roll adds $\frac{14}{3} - \frac{17}{4} = \frac{5}{12} \approx \$0.42$ — total option value grows with each extra chance, though at a diminishing rate.

Intuition. The re-roll is an *option*: worth 0.75 here, and its value grows with the number of chances and with the dispersion of the underlying outcomes. Optionality on a volatile underlying is worth more — the deepest fact in derivatives, visible in a dice game.

1.8 Conditioning Traps

Conditioning on a property of an entire sequence is not the same as altering the mechanism that generates each step. The following die problem is the canonical illustration, and nearly every first instinct about it is wrong.

Setup. A fair die is rolled repeatedly until the first 6 appears. Let T be the number of rolls (including the final 6), and let A be the event that every roll in the sequence — including the final 6 — came up even.

Proposition 1.26 (Conditioned wait: first 6 given all rolls even)**.** *With the setup above,* $P(A) = \frac{1}{4}$ *and*

$$\mathbb{E}[T \mid A] = \frac{3}{2}.$$

Proof. We compute the conditional distribution of T directly from the joint probabilities. The event $\{T = n\} \cap A$ requires the first $n-1$ rolls to land in $\{2, 4\}$ (probability $\frac{2}{6} = \frac{1}{3}$ each) and roll n to be a 6 (probability $\frac{1}{6}$), so by independence

$$P(T = n,\ A) = \Big(\frac{1}{3}\Big)^{n-1} \frac{1}{6}, \qquad n \geq 1. \tag{1.6}$$

Summing the geometric series (1.6) over n gives

$$P(A) = \frac{1/6}{1 - 1/3} = \frac{1}{4}.$$

Dividing (1.6) by $P(A)$ yields the conditional probability mass function

$$P(T = n \mid A) = 4 \cdot \Big(\frac{1}{3}\Big)^{n-1} \frac{1}{6} = \frac{2}{3}\Big(\frac{1}{3}\Big)^{n-1},$$

which is exactly the $\mathrm{Geom}\big(\frac{2}{3}\big)$ pmf of Definition 1.9. By Theorem 1.13, $\mathbb{E}[T \mid A] = \frac{1}{2/3} = \frac{3}{2}$. Alternatively, summing the series directly: $\sum_{n \geq 1} n \cdot \frac{2}{3}(\frac{1}{3})^{n-1} = \frac{2}{3} \cdot \frac{1}{(1-1/3)^2} = \frac{2}{3} \cdot \frac{9}{4} = \frac{3}{2}$. $\square$

Example 1.27 (The first-roll check)**.** From the conditional pmf in the proof of Proposition 1.26, $P(T = 1 \mid A) = \frac{2}{3}$, versus $\frac{1}{3}$ under the naive three-face model discussed below — the conditioning visibly favours stopping immediately, consistent with a mean of $\frac{3}{2} < 3$. As a bound check, the answer must lie between 1 and the naive 3, and $\frac{3}{2}$ does.

Interview trap. The instant answer is 3: "conditioning on all-even reduces the die to the three faces $\{2, 4, 6\}$, a first-success wait with $p = \frac{1}{3}$." That reasoning is wrong — it describes a genuinely three-faced die, a different experiment. Conditioning on "all rolls even" reweights *whole sequences*, not the die's faces: long sequences are penalised because each extra roll must dodge the odd faces, so the conditional law tilts sharply toward short sequences and the mean drops to $\frac{3}{2}$. Trigger to remember: conditioning reweights whole sequences, not just the die's faces.

1.9 Hat-Check Fixed Points

A uniformly random shuffle hands n hats back to their n owners; how many owners get their own hat? The count has mean 1 and variance 1 for every n — a striking constancy that indicator variables (Corollary 1.5) explain in a few lines, and that foreshadows a Poisson limit.

Setup. Let τ be a uniformly random permutation of $\{1, \dots, n\}$, $n \geq 1$, and let

$$X = \#\{\, i : \tau(i) = i \,\}$$

be its number of fixed points — the number of owners who receive their own hat.

Theorem 1.28 (Hat-check moments)**.** *With the setup above,* $\mathbb{E}[X] = 1$ *for every* $n \geq 1$*, and* $\mathrm{Var}(X) = 1$ *exactly, for every* $n \geq 2$*.*

Proof. We use indicators and linearity, computing the second moment through pair indicators. Write $X = \sum_{i=1}^{n} \mathbf{1}_{A_i}$ with $A_i = \{\tau(i) = i\}$.

Mean. The permutations fixing i number $(n-1)!$, so $P(A_i) = \frac{(n-1)!}{n!} = \frac{1}{n}$, and Corollary 1.5 gives $\mathbb{E}[X] = n \cdot \frac{1}{n} = 1$ — for every n, with no independence needed.

Second moment. Expanding the square of the indicator sum, and using that an indicator squared is itself while a product of indicators is the indicator of the intersection,

$$X^2 = \sum_{i=1}^{n} \mathbf{1}_{A_i} + \sum_{i \neq j} \mathbf{1}_{A_i \cap A_j}.$$

For $i \neq j$ (this is where $n \geq 2$ enters), the permutations fixing both i and j number $(n-2)!$, so $P(A_i \cap A_j) = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$. Taking expectations with Theorem 1.2 and Lemma 1.4, the $n(n-1)$ ordered pairs contribute

$$\mathbb{E}[X^2] = n \cdot \frac{1}{n} + n(n-1) \cdot \frac{1}{n(n-1)} = 1 + 1 = 2.$$

By the variance shortcut (1.1), $\mathrm{Var}(X) = 2 - 1^2 = 1$. □

Example 1.29 (All six permutations of three hats)**.** For $n = 3$ the six permutations split as: the identity (3 fixed points), three transpositions (1 fixed point each), and two 3-cycles (0 fixed points). Direct averaging:

$$\mathbb{E}[X] = \frac{3 + 1 + 1 + 1 + 0 + 0}{6} = 1, \qquad \mathbb{E}[X^2] = \frac{9 + 1 + 1 + 1 + 0 + 0}{6} = 2,$$

so $\mathrm{Var}(X) = 2 - 1 = 1$, confirming Theorem 1.28 by brute force.

Fact 1.30 (Poisson limit of fixed points)**.** As $n \to \infty$, the number of fixed points of a uniformly random permutation of n items converges in distribution to Poisson(1) — consistent with Theorem 1.28, since a Poisson(λ) variable has mean and variance both λ (Table 1), here $\lambda = 1$. The underlying matching problem goes back to de Montmort (1713). A proof is beyond our scope.

Intuition. Each indicator $\mathbf{1}_{A_i}$ is a rare event of probability $\frac{1}{n}$, and the pairwise dependence is weak — from the proof of Theorem 1.28, $P(A_i \cap A_j) = \frac{1}{n(n-1)}$ barely exceeds the independent value $\frac{1}{n^2}$. Many weakly dependent rare events with total mean 1 is exactly the regime in which Poisson behaviour emerges, which is why the limit in the fact above should feel inevitable rather than surprising.

Practice: problems P1.1–P1.20 in §27, solutions in §28.

2 Bayesian Updating and Reading Flow

Bayesian updating is the arithmetic of changing one's mind: a prior belief meets evidence and becomes a posterior. This section proves Bayes' theorem from the definition of conditional probability, recasts it in the odds form that turns interview questions into one-line multiplications, and extends it to streams of conditionally independent evidence. The same three results explain how a market maker reads order flow, which is where the section ends.

2.1 Conditional Probability and Bayes' Theorem

Everything in this section rests on a single definition — conditional probability — and on the theorem that follows from it in two lines. We state both carefully, because every later formula in the section is an algebraic rearrangement of them.

Setup. Throughout this section we work on a probability space with probability measure P; every set named below is an event. We write H for a hypothesis under consideration, H^c for its complement, and E for observed evidence. Conditioning on an event requires that event to have positive probability, and each statement below lists exactly the positivity it needs.

Definition 2.1 (Conditional probability)**.** Let A and B be events with $P(B) > 0$. The *conditional probability of A given B* is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Conditioning renormalises: among the outcomes in which B occurs, the conditional probability measures the fraction that also lie in A. Multiplying both sides of the definition by $P(B)$ gives the *multiplication rule*

$$P(A \cap B) = P(A \mid B)\, P(B), \tag{2.1}$$

valid whenever $P(B) > 0$.

Lemma 2.2 (Law of total probability)**.** *Let $B_1, \dots, B_n$ be events that partition the sample space (they are pairwise disjoint and their union is the whole space), with $P(B_i) > 0$ for every i. Then, for any event A,*

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i)\, P(B_i).$$

Proof. We decompose A along the partition and apply additivity. Since the B_i cover the sample space and are pairwise disjoint, the sets $A \cap B_1, \dots, A \cap B_n$ are pairwise disjoint and their union is A. Additivity of P gives $P(A) = \sum_{i=1}^{n} P(A \cap B_i)$, and by the multiplication rule (2.1) each term equals $P(A \mid B_i)\, P(B_i)$. □

Theorem 2.3 (Bayes' theorem)**.** *Let H and E be events with $P(H) > 0$ and $P(E) > 0$. Then*

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E)}. \tag{2.2}$$

If moreover $0 < P(H) < 1$, the denominator expands over the partition $\{H, H^c\}$:

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E \mid H)\, P(H) + P(E \mid H^c)\, P(H^c)}. \tag{2.3}$$

Proof. We compute the probability of the intersection $H \cap E$ in two ways. By the multiplication rule (2.1) with $B = E$ we have $P(H \cap E) = P(H \mid E)\, P(E)$; with $B = H$ we have $P(H \cap E) = P(E \mid H)\, P(H)$. Equating the two expressions and dividing by $P(E) > 0$ yields (2.2). For (2.3),

note that when $0 < P(H) < 1$ the events H and H^c partition the sample space and both have positive probability, so Lemma 2.2 gives $P(E) = P(E \mid H)\,P(H) + P(E \mid H^c)\,P(H^c)$; substituting this into the denominator of (2.2) completes the proof. □

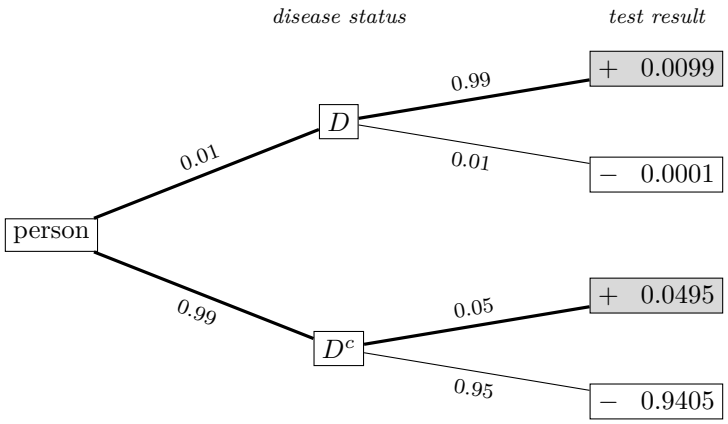
In Bayesian language, $P(H)$ is the *prior*, $P(H \mid E)$ the *posterior*, $P(E \mid H)$ the *likelihood* of the evidence under the hypothesis, and $P(E)$ the *normalising constant*. The theorem inverts a conditional: it converts the easy direction, "probability of the evidence given the hypothesis," into the direction one actually wants, "probability of the hypothesis given the evidence."

Example 2.4 (Rare-disease test)**.** A disease has prevalence 1% in the population. A test has sensitivity 99% (it is positive on 99% of the sick) and specificity 95% (it is negative on 95% of the healthy, so its false-positive rate is 5%). A randomly chosen person tests positive; we compute the probability that the person has the disease.

Let D be the event "has the disease" and let $+$ denote a positive test, so that $P(D) = 0.01$, $P(+ \mid D) = 0.99$, and $P(+ \mid D^c) = 0.05$. By (2.3),

$$\begin{aligned} P(D \mid +) &= \frac{0.99 \times 0.01}{0.99 \times 0.01 + 0.05 \times 0.99} = \frac{0.0099}{0.0099 + 0.0495} \\ &= \frac{0.0099}{0.0594} = \frac{1}{6} \approx 16.7\%. \end{aligned}$$

Despite a 99%-sensitive test, a positive result leaves only a one-in-six chance of disease: the false positives generated by the enormous healthy pool (joint mass 0.0495) outweigh the true positives (joint mass 0.0099) five to one. Figure 3 displays the computation as a probability tree.



$$P(+) = 0.0099 + 0.0495 = 0.0594$$

Figure 3: Probability tree for the rare-disease test of Example 2.4. Edge labels are conditional probabilities; each leaf shows the joint probability of its path, obtained by multiplying along the edges. The two thickened paths (shaded leaves) are the only routes to a positive test, so $P(+) = 0.0594$, and the posterior $P(D \mid +) = 0.0099/0.0594 = 1/6$ is the share of that shaded mass carried by the disease branch.

Intuition. The tree in Figure 3 shows why an accurate test can still leave a small posterior. A positive result can be manufactured in two ways: a sick person (rare — branch mass 0.01) triggering a true positive, or a healthy person (almost everyone — branch mass 0.99) triggering a false alarm. The posterior is a contest between the two joint masses, 0.0099 against 0.0495, and the healthy branch wins five to one — which is exactly the statement that the odds of disease given a positive test are 1:5.

2.2 The Odds Form: the Fast Recipe

Interviews rarely leave time to assemble the denominator of (2.3). The odds form removes it: dividing Bayes' theorem for H by Bayes' theorem for H^c cancels $P(E)$, leaving a recipe — posterior odds equal prior odds times likelihood ratio — that can be executed mentally.

Setup. As before, H is a hypothesis with $0 < P(H) < 1$ and E is evidence with $P(E) > 0$. The next definition introduces the two ratios the recipe runs on.

Definition 2.5 (Odds and likelihood ratio)**.** Let H be an event with $0 < P(H) < 1$. The *odds* of H are

$$\mathrm{O}(H) = \frac{P(H)}{P(H^c)},$$

and, for evidence E with $P(E) > 0$, the *posterior odds* are $\mathrm{O}(H \mid E) = P(H \mid E)/P(H^c \mid E)$ (defined when the denominator is positive). If in addition $P(E \mid H^c) > 0$, the *likelihood ratio* of E for H against H^c is

$$\mathrm{LR}(E) = \frac{P(E \mid H)}{P(E \mid H^c)}.$$

Odds and probabilities carry the same information. Dividing the identity $P(H \mid E)+P(H^c \mid E) = 1$ by $P(H^c \mid E)$ gives $\mathrm{O}(H \mid E) + 1 = 1/P(H^c \mid E)$, and therefore

$$P(H \mid E) = \frac{\mathrm{O}(H \mid E)}{1 + \mathrm{O}(H \mid E)}. \tag{2.4}$$

The same conversion applies to prior odds: odds of 1:5, say, mean a probability of $(1/5)/(1+1/5) = 1/6$.

Corollary 2.6 (Odds form of Bayes' theorem)**.** *Let H be an event with $0 < P(H) < 1$ and let E be evidence with $P(E) > 0$ and $P(E \mid H^c) > 0$. Then*

$$\underbrace{\frac{P(H \mid E)}{P(H^c \mid E)}}_{\text{posterior odds}} = \underbrace{\frac{P(H)}{P(H^c)}}_{\text{prior odds}} \times \underbrace{\frac{P(E \mid H)}{P(E \mid H^c)}}_{\text{likelihood ratio}}, \tag{2.5}$$

that is, $\mathrm{O}(H \mid E) = \mathrm{O}(H) \times \mathrm{LR}(E)$.

Proof. We divide one instance of Bayes' theorem by another. Theorem 2.3 applied to H gives $P(H \mid E) = P(E \mid H)\,P(H)/P(E)$; applied to H^c (which has positive probability since $P(H) < 1$) it gives $P(H^c \mid E) = P(E \mid H^c)\,P(H^c)/P(E)$. The second quantity is positive because $P(E \mid H^c) > 0$ and $P(H^c) > 0$, so we may divide the first equation by the second. The common factor $P(E)$ cancels, leaving exactly (2.5). □

Example 2.7 (Rare-disease test, odds form)**.** We redo Example 2.4 in two lines. The prior odds of disease are $0.01/0.99 = 1/99$, and the likelihood ratio of a positive test is $0.99/0.05 = 19.8$. By (2.5), the posterior odds are

$$\frac{1}{99} \times 19.8 = \frac{19.8}{99} = \frac{1}{5},$$

and by (2.4) the posterior probability is $(1/5)/(6/5) = 1/6$ — the same answer, with no denominator bookkeeping.

Intuition. The odds form is fast because the normalising constant $P(E)$ — the entire denominator of (2.3) — divides out. What remains says that evidence acts on beliefs multiplicatively: the likelihood ratio is the exchange rate at which an observation converts prior odds into posterior odds. A problem that looks like heavy conditioning collapses to one multiplication and one conversion via (2.4).

2.3 Sequential Evidence and Reading Flow

Evidence usually arrives in a stream — repeated coin flips, repeated test results, repeated fills against a standing quote. When the pieces are conditionally independent given each hypothesis, the odds form of Corollary 2.6 iterates: each new piece multiplies the running odds by its own likelihood ratio. We first make the independence assumption explicit, since it is the hypothesis on which the whole multiplication rests.

Setup. As before, H is a hypothesis with $0 < P(H) < 1$. The evidence now consists of n events $E_1, \dots, E_n$; think of E_i as "the i-th observation came out a particular way." The following definition pins down the independence we need.

Definition 2.8 (Conditional independence given an event)**.** Let B be an event with $P(B) > 0$. Events $E_1, \dots, E_n$ are *conditionally independent given* B if for every subset $S \subseteq \{1, \dots, n\}$,

$$P\Big(\bigcap_{i \in S} E_i \;\Big|\; B\Big) = \prod_{i \in S} P(E_i \mid B).$$

Theorem 2.9 (Multiplication of likelihood ratios)**.** *Let H be an event with $0 < P(H) < 1$, and let $E_1, \dots, E_n$ be events that are conditionally independent given H and also conditionally independent given H^c, with $P(E_i \mid H^c) > 0$ for every i. Write $E = E_1 \cap \cdots \cap E_n$ for the compound evidence. Then $P(E) > 0$ and*

$$\mathrm{O}(H \mid E) = \mathrm{O}(H) \times \prod_{i=1}^{n} \mathrm{LR}(E_i), \qquad \textit{where } \mathrm{LR}(E_i) = \frac{P(E_i \mid H)}{P(E_i \mid H^c)}.$$

Proof. We apply the odds form to the compound event E and factor its likelihood ratio using conditional independence. First, E has positive probability: Definition 2.8 applied with $B = H^c$ and $S = \{1, \dots, n\}$ gives $P(E \mid H^c) = \prod_{i=1}^{n} P(E_i \mid H^c) > 0$, and hence $P(E) \geq P(E \cap H^c) = P(E \mid H^c)\, P(H^c) > 0$ by the multiplication rule (2.1). The hypotheses of Corollary 2.6 are therefore satisfied for the pair (H, E), and it gives

$$\mathrm{O}(H \mid E) = \mathrm{O}(H) \times \frac{P(E \mid H)}{P(E \mid H^c)}.$$

Conditional independence given H factors the numerator as $P(E \mid H) = \prod_{i=1}^{n} P(E_i \mid H)$, and conditional independence given H^c factors the denominator as $P(E \mid H^c) = \prod_{i=1}^{n} P(E_i \mid H^c)$. The ratio of the two products is the product of the ratios, namely $\prod_{i=1}^{n} \mathrm{LR}(E_i)$. □

Example 2.10 (The thousand coins)**.** A bag contains 1000 coins: 999 fair and one double-headed. A coin is drawn uniformly at random and flipped 10 times, and all 10 flips come up heads. We compute the probability that the drawn coin is the double-headed one.

Let H be the event that the double-headed coin was drawn, and let E_i be the event that the i-th flip is heads. The prior odds are

$$\mathrm{O}(H) = \frac{1/1000}{999/1000} = \frac{1}{999}.$$

Given H, every flip is heads with probability 1; given H^c, the flips are independent and each is heads with probability $1/2$. In both cases the E_i are conditionally independent given the hypothesis, so Theorem 2.9 applies with

$$\mathrm{LR}(E_i) = \frac{1}{1/2} = 2 \quad \text{for each } i, \qquad \prod_{i=1}^{10} \mathrm{LR}(E_i) = 2^{10} = 1024.$$

The posterior odds are therefore 1024/999, and by (2.4) the posterior probability is

$$P(H \mid E) = \frac{1024/999}{1 + 1024/999} = \frac{1024}{1024 + 999} = \frac{1024}{2023} \approx 50.6\%.$$

Despite ten straight heads, the answer is essentially a coin flip: the prior was roughly 1000:1 against, and the evidence factor $2^{10} = 1024$ only barely overcomes 999.

Remark 2.11 (Correlated evidence). Conditional independence is the load-bearing assumption of Theorem 2.9, and it must hold given *each* hypothesis separately — it neither implies nor follows from unconditional independence of the E_i. When the pieces of evidence share a common mechanism — for instance, several fills that are child slices of a single parent order — the factorisation fails, and multiplying likelihood ratios overstates the evidence.

To read order flow, instantiate the machinery: let H be the event that the counterparty currently trading is informed, and let E_i record the direction of the i-th aggressive fill — for instance, that it lifted the offer rather than hitting the bid. The prior odds come from the population of traders on the venue; each fill's likelihood ratio comes from how much more often informed flow trades in that direction than noise flow.

Trading application. A market maker is a Bayesian machine: every fill is evidence. The observation "the same counterparty lifted the offer three times in a row" carries a likelihood ratio strongly favouring *informed buyer* over *noise*, and by Theorem 2.9 sequential fills multiply likelihood ratios exactly as sequential tests do. The correct response to a posterior that has swung toward "informed" is to move the market up — raise both quotes — not to keep selling at the same price.

Interview trap. Base-rate neglect in reverse: with mostly-noise flow (a high base rate of uninformed traders), one aggressive fill is weak evidence; with thin, professional flow, the same fill is strong evidence. The *prior* on who trades on the venue changes the update — one reason the same quote logic behaves differently across exchanges.

Practice: problems P2.1–P2.6 in §27, solutions in §28.

3 Betting, Bankroll, Ruin and Markov Chains

This section develops the mathematics of repeated betting: the Kelly criterion for sizing a favorable bet, gambler's ruin for survival probabilities and game durations, first-step analysis as the general solving method behind both, stationary distributions as a first contact with the long-run behavior of Markov chains, and the St. Petersburg paradox as the canonical warning about infinite expectations. The same mathematics governs position sizing on a trading desk, the capital buffer of a strategy, and a market maker's inventory, which is why every topic here is a recurring interview theme.

3.1 The Kelly Criterion

A bettor with a genuine edge still faces a sizing decision: staking too little forgoes growth, while staking too much guarantees eventual ruin even though every individual bet has positive expected value. The Kelly criterion resolves the tension by choosing the stake that maximizes the long-run exponential growth rate of the bankroll.

Setup. Fix odds $b > 0$ and a win probability $p \in (0, 1)$, and write $q = 1 - p$. A bet at odds b-to-1 returns b units of profit per unit staked when it wins and loses the unit staked when it loses. A bettor starts with bankroll $W_0 > 0$ and repeatedly stakes the same fraction $f \in [0, 1)$ of the

current bankroll on independent copies of this bet. We write R_k for the gross return (bankroll multiplier) of the k-th bet; the symbol r is reserved for the ratio q/p of the ruin subsections and is not used here.

Definition 3.1 (Fixed-fractional betting and expected log growth)**.** Let $R_1, R_2, \dots$ be iid random variables with

$$R_k = \begin{cases} 1+bf & \text{with probability } p, \\ 1-f & \text{with probability } q, \end{cases}$$

so that the bankroll after n bets is $W_n = W_0 \prod_{k=1}^{n} R_k$. The *expected log growth rate* of the strategy is

$$g(f) = \mathbb{E}\left[\log R_1\right] = p\log(1+bf) + q\log(1-f), \qquad f \in [0,1). \tag{3.1}$$

The next lemma explains why g is the right objective: it is the almost-sure exponential rate at which the bankroll grows or decays.

Lemma 3.2 (Almost-sure growth rate)**.** *Fix $f \in [0,1)$. Then*

$$\frac{1}{n}\log\frac{W_n}{W_0} \longrightarrow g(f) \quad \textit{almost surely as } n \to \infty.$$

Consequently $W_n \to \infty$ almost surely when $g(f) > 0$, and $W_n \to 0$ almost surely when $g(f) < 0$.

Proof. We appeal to the strong law of large numbers. Taking logarithms of the product,

$$\log\frac{W_n}{W_0} = \sum_{k=1}^{n} \log R_k,$$

a sum of iid random variables taking the two finite values $\log(1+bf)$ and $\log(1-f)$, hence with finite mean $g(f)$ by (3.1). The strong law gives $n^{-1}\sum_{k=1}^{n}\log R_k \to g(f)$ almost surely. On the event where the convergence holds, $\log W_n = \log W_0 + n\,(g(f) + o(1))$, which tends to $+\infty$ if $g(f) > 0$ and to $-\infty$ if $g(f) < 0$; exponentiating gives the two consequences. □

Maximizing long-run growth therefore means maximizing g over f. The following theorem describes the shape of g completely and locates its maximizer.

Theorem 3.3 (Kelly criterion)**.** *Let g be as in (3.1) with $b > 0$ and $p \in (0,1)$.*

(i) The function g is strictly concave on $[0,1)$, with $g(0) = 0$, $g'(0) = bp - q$, and $g(f) \to -\infty$ as $f \uparrow 1$.

(ii) If $bp - q > 0$, then g has a unique global maximizer on $[0,1)$, namely

$$f^* = \frac{bp-q}{b},$$

which lies in $(0,1)$, and $g(f^*) > 0$.

(iii) If $bp - q \le 0$, then g is strictly decreasing on $[0,1)$, so the unique maximizer is $f = 0$: the growth-optimal action is not to bet.

Proof. We proceed by elementary calculus: we compute two derivatives, establish strict concavity, and solve the first-order condition.

Step 1 (derivatives and concavity). On $[0,1)$ both logarithms in (3.1) have positive arguments, and

$$g'(f) = \frac{pb}{1+bf} - \frac{q}{1-f}, \qquad g''(f) = -\frac{pb^2}{(1+bf)^2} - \frac{q}{(1-f)^2}.$$

Both terms of g'' are strictly negative for every $f \in [0,1)$, so $g'' < 0$ and g is strictly concave. Evaluating at $f = 0$ gives $g(0) = p\log 1 + q\log 1 = 0$ and $g'(0) = pb - q$. As $f \uparrow 1$ the first term of g tends to the finite limit $p\log(1+b)$ while $q\log(1-f) \to -\infty$, so $g(f) \to -\infty$.

Step 2 (first-order condition). Setting $g'(f) = 0$ and clearing denominators,

$$pb(1-f) = q(1+bf) \iff pb - q = bf(p+q) = bf \iff f = \frac{pb-q}{b}.$$

Step 3 (case $bp - q > 0$). The stationary point $f^* = (bp-q)/b = p - q/b$ is positive by assumption and satisfies $f^* < p < 1$ because $q/b > 0$, so $f^* \in (0,1)$. Strict concavity makes g' strictly decreasing; since $g'(f^*) = 0$, we get $g' > 0$ on $[0, f^*)$ and $g' < 0$ on $(f^*, 1)$, so f^* is the unique global maximizer. Because $g(0) = 0$ and g is strictly increasing on $[0, f^*]$, we conclude $g(f^*) > 0$.

Step 4 (case $bp - q \le 0$). Here $g'(0) = bp - q \le 0$, and g' is strictly decreasing by Step 1, so $g'(f) < 0$ for every $f \in (0,1)$. Hence g is strictly decreasing on $[0,1)$ and its unique maximizer is $f = 0$. □

(Problem P3.1 asks you to reproduce this derivation from scratch.)

The quantity $g'(0) = bp - q$ is the expected profit per unit staked — the *edge* of the bet. The Kelly fraction is therefore "edge over odds": $f^* = (bp-q)/b$.

Corollary 3.4 (Even-money Kelly)**.** *For an even-money bet ($b = 1$) with $p > \frac{1}{2}$, the Kelly fraction is*

$$f^* = p - q = 2p - 1.$$

Proof. Substituting $b = 1$ into Theorem 3.3(ii) gives $f^* = (p-q)/1 = p - q = 2p - 1$. □

Example 3.5 (A 60/40 coin)**.** Consider an even-money bet won with probability $p = 0.6$, so $q = 0.4$ and $b = 1$. By Corollary 3.4, the Kelly fraction is $f^* = 2(0.6) - 1 = 0.2$: stake 20% of the bankroll on each bet. The optimal growth rate is

$$g(0.2) = 0.6\log 1.2 + 0.4\log 0.8 = 0.6(0.18232) + 0.4(-0.22314) \approx 0.02014,$$

so capital compounds by a factor $e^{0.02014} \approx 1.0203$ per bet — about 2.0% — and over 100 bets the typical growth factor is $e^{2.01} \approx 7.5$. Figure 4 plots this growth curve to scale.

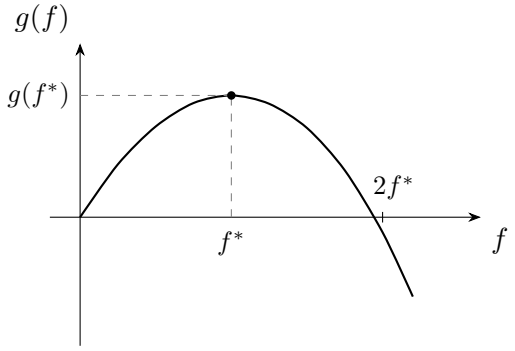


Figure 4: The expected log growth rate $g(f)$ of (3.1) for an even-money bet with $p = 0.6$, drawn to scale. As Theorem 3.3 asserts, the curve is strictly concave, leaves the origin with slope equal to the edge $bp - q = 0.2$, peaks at $f^* = 0.2$, and diverges to $-\infty$ as $f \to 1$. It crosses zero just below $2f^* = 0.4$ (exactly at $2f^*$ in the quadratic approximation of Proposition 3.6); betting beyond the crossing shrinks capital almost surely despite a positive per-bet expectation.

The next proposition makes precise three claims that practitioners quote constantly: half-Kelly keeps about three quarters of the growth, volatility scales linearly in the stake (so half the stake means a quarter of the variance), and growth turns negative at about twice the Kelly fraction.

Proposition 3.6 (Fractional Kelly, small-edge approximation)**.** *Consider an even-money bet ($b = 1$) with edge $\varepsilon = p - q > 0$, so that $f^* = \varepsilon$ by Corollary 3.4. Let $\tilde{g}(f) = \varepsilon f - f^2/2$.*

(i) The polynomial $\tilde{g}$ is the second-order Taylor polynomial of g at $f = 0$.

(ii) The polynomial $\tilde{g}$ is uniquely maximized at $f = \varepsilon = f^*$, with $\tilde{g}(f^*) = \varepsilon^2/2$, and for every $c \geq 0$,

$$\tilde{g}\,(cf^*) = c\,(2 - c)\,\tilde{g}\,(f^*)\,.$$

In particular $\tilde{g}(f^*/2) = \frac{3}{4}\,\tilde{g}(f^*)$ and $\tilde{g}(2f^*) = 0$.

(iii) The variance of the per-bet log return at fraction f is $4pq\,f^2 + O(f^3)$ as $f \downarrow 0$. Hence, to leading order, betting cf^* scales the log-return volatility by c and the variance by c^2.

Consequently, to second order in the edge, half-Kelly earns 75% of the maximal growth at half the volatility (a quarter of the variance), and the growth rate vanishes at twice the Kelly fraction and is negative beyond it.

Proof. We compute Taylor expansions directly.

Part (i). From the proof of Theorem 3.3 with $b = 1$: $g(0) = 0$, $g'(0) = p - q = \varepsilon$, and $g''(0) = -p - q = -1$. The second-order Taylor polynomial at 0 is therefore $\varepsilon f - f^2/2 = \tilde{g}(f)$.

Part (ii). The polynomial $\tilde{g}$ is a downward parabola with $\tilde{g}'(f) = \varepsilon - f$, so its unique maximizer is $f = \varepsilon$ with value $\varepsilon^2 - \varepsilon^2/2 = \varepsilon^2/2$. Substituting $f = c\varepsilon$,

$$\tilde{g}(c\varepsilon) = \varepsilon(c\varepsilon) - \frac{(c\varepsilon)^2}{2} = \varepsilon^2\left(c - \frac{c^2}{2}\right) = c(2-c)\,\frac{\varepsilon^2}{2} = c(2-c)\,\tilde{g}(f^*).$$

Evaluating at $c = \frac{1}{2}$ gives the factor $\frac{1}{2} \cdot \frac{3}{2} = \frac{3}{4}$, and at $c = 2$ the factor $2 \cdot 0 = 0$.

Part (iii). Write L for the per-bet log return, so $L = \log(1 + f)$ with probability p and $L = \log(1 - f)$ with probability q. Expanding, $\log(1 \pm f) = \pm f + O(f^2)$, hence $\mathbb{E}[L] = \varepsilon f + O(f^2)$ and $\mathbb{E}[L^2] = f^2 + O(f^3)$, giving

$$\mathrm{Var}(L) = \mathbb{E}[L^2] - (\mathbb{E}[L])^2 = \left(1 - \varepsilon^2\right) f^2 + O(f^3) = 4pq\,f^2 + O(f^3),$$

where the last equality uses $1 - \varepsilon^2 = (p+q)^2 - (p-q)^2 = 4pq$. Replacing f by cf multiplies the leading term by c^2, which is the claimed scaling of variance; taking square roots gives the scaling of volatility. □

Example 3.7 (Half-Kelly and over-betting on the 60/40 coin)**.** Continue Example 3.5 ($p = 0.6$, $b = 1$, $f^* = 0.2$, $g(f^*) \approx 0.02014$). The exact numbers match Proposition 3.6 closely.

- *Half-Kelly.* At $f = 0.1$: $g(0.1) = 0.6 \log 1.1 + 0.4 \log 0.9 \approx 0.01504$, which is 74.7% of $g(f^*)$ — the predicted three quarters. The exact per-bet log-return standard deviation is 0.0983 at $f = 0.1$ versus 0.1986 at $f = 0.2$ (half), and the variances are 0.00966 versus 0.03946 (a quarter).
- *Twice Kelly.* At $f = 2f^* = 0.4$: $g(0.4) = 0.6 \log 1.4 + 0.4 \log 0.6 \approx -0.00245 < 0$. The exact zero crossing sits at $f \approx 0.389$, just below $2f^*$, exactly as Figure 4 shows. Meanwhile the expected gross return per bet at $f = 0.4$ is $1 + 0.4(p - q) = 1.08$, so the *expected* bankroll grows by 8% per bet even though, by Lemma 3.2, the actual bankroll tends to zero almost surely. The expectation is carried by vanishingly rare lucky paths.

Interview trap. Over-betting is catastrophic while under-betting is merely slow: the growth curve of Figure 4 falls off a cliff to the right of f^* and turns negative beyond roughly $2f^*$. In particular, "the expected value is positive, so bet everything" fails twice over — at f near 1 a single loss is nearly fatal ($g \to -\infty$), and even moderate over-betting produces $g(f) < 0$, which by Lemma 3.2 means almost-sure ruin despite an exponentially growing expected bankroll.

Remark 3.8 (Why practitioners bet less than full Kelly). Three reasons, all standard and all worth stating in an interview.

(1) The win probability p is *estimated*, not known. Plugging an overestimated edge into $f^* = (bp - q)/b$ means betting beyond the true optimum, and Proposition 3.6 shows the penalty is asymmetric: beyond about $2f^*$ the true growth rate is negative.
(2) Log-optimality ignores drawdown pain: a full-Kelly bettor routinely halves the bankroll along the way. Half-Kelly keeps about 75% of the growth at half the volatility — a quarter of the variance — by Proposition 3.6.
(3) Edges correlate: simultaneous "independent" bets often are not independent, so the effective aggregate fraction at risk exceeds what per-bet Kelly arithmetic suggests.

Trading application. A live prediction-market strategy sized positions fixed-fractionally with hard per-market caps — fractional-Kelly logic under parameter uncertainty. That is the shape of a strong sizing answer in an interview: quote Kelly as the theoretical anchor, apply an explicit haircut for estimation error and correlated exposures, and impose hard caps because a model's fitted probability is never the true p.

3.2 First-Step Analysis as a General Method

A single method solves coin-pattern races (HH before TH and their relatives), gambler's ruin, and most "expected time until ..." games: condition on the first move and let the Markov property restart the problem. This subsection names the method once, so that the proofs of §3.3 can invoke it; the worked example shows the companion trick of collapsing symmetric states.

Setup. Consider a time-homogeneous Markov chain on a finite state space, together with a target quantity defined per state: the probability of reaching one absorbing set before another, or the expected number of steps until absorption.

Method (first-step analysis).

(1) *Symmetry first.* Collapse equivalent states before writing any equations: states whose transition structure relative to the target is identical can share a single unknown. This is the biggest practical simplification.
(2) Define one unknown per (collapsed) state — a probability h_i or an expected time m_i.
(3) Write one equation per non-absorbing state by conditioning on the first move: the law of total probability (or total expectation) over the one-step transitions, with the Markov property guaranteeing that after the first step the process is a fresh copy started from the new state. Expected-time equations carry an extra "+1" for the step just taken.
(4) Impose boundary conditions at absorbing states: probabilities 0 or 1, expected times 0.
(5) Solve the resulting linear system.

For expected-time versions the unknowns must be finite for the equations to be valid; finiteness holds whenever an absorbing state is reachable from every state, by the block argument proved as Lemma 3.10 below.

Example 3.9 (Ant on a cube: state collapse)**.** An ant stands on one corner of a cube and walks along edges, choosing among the three incident edges uniformly at random at each step. We compute the expected number of steps to reach the opposite corner.

Collapse. Identify the cube's vertices with binary strings in $\{0,1\}^3$ and the target with the string differing from the start in all three coordinates. Each step flips one uniformly chosen coordinate. If the ant is at Hamming distance d from the target, then d of the three flips decrease the distance and $3 - d$ increase it, so the distance process is itself a Markov chain with

$$P(d \to d-1) = \frac{d}{3}, \qquad P(d \to d+1) = \frac{3-d}{3},$$

and the eight vertices collapse to the four states $d \in \{0,1,2,3\}$. Let m_d denote the expected number of steps to reach $d = 0$ from distance d; the unknowns are finite by the argument of Lemma 3.10.

First-step equations. Conditioning on the first move from each transient state, with boundary condition $m_0 = 0$:

$$m_3 = 1 + m_2, \qquad m_2 = 1 + \tfrac{1}{3}m_3 + \tfrac{2}{3}m_1, \qquad m_1 = 1 + \tfrac{2}{3}m_2 + \tfrac{1}{3}\cdot 0.$$

Solve. Substituting the first equation into the second, $m_2 = 1+\frac{1}{3}(1+m_2)+\frac{2}{3}m_1$, so $\frac{2}{3}m_2 = \frac{4}{3}+\frac{2}{3}m_1$, giving $m_2 = 2+m_1$. The third equation becomes $m_1 = 1+\frac{2}{3}(2+m_1) = \frac{7}{3}+\frac{2}{3}m_1$, so $\frac{1}{3}m_1 = \frac{7}{3}$ and $m_1 = 7$. Back-substituting, $m_2 = 9$ and $m_3 = 10$: the ant needs 10 steps on average. Symmetry reduced an eight-state system to three equations solved in four lines.

Intuition. Hunting for the collapse is step zero of every such problem: the cube's 8 states become 4, a deuce game's score lattice becomes 3 states (lead +1, level, lead −1), and the gambler's-ruin chain is already collapsed. Writing equations before finding the symmetry wastes time and invites algebra errors.

3.3 Gambler's Ruin

Gambler's ruin is the canonical absorption problem and a standing metaphor in trading: the state is a capital buffer or a market maker's inventory, and the absorbing barriers are bankruptcy and a profit target. This subsection proves the three formulas worth knowing cold — the fair-game hitting probability i/N, the fair-game expected duration $i(N-i)$, and the biased-game hitting probability — as applications of the first-step method of §3.2.

Setup. Fix an integer $N \geq 1$ and $p \in (0,1)$, and set $q = 1-p$ and $r = q/p$. Let $(S_n)_{n\geq 0}$ be the random walk on $\{0,1,\ldots,N\}$ started at $S_0 = i$ that, from any interior state, steps up by 1 with probability p and down by 1 with probability q, independently at each step; the states 0 and N are absorbing. Figure 5 shows the chain. Define the absorption time $T = \min\{n \geq 0 : S_n \in \{0,N\}\}$, the hitting probability $h_i = P(S_T = N \mid S_0 = i)$, and the expected duration $D_i = \mathbb{E}[T \mid S_0 = i]$.

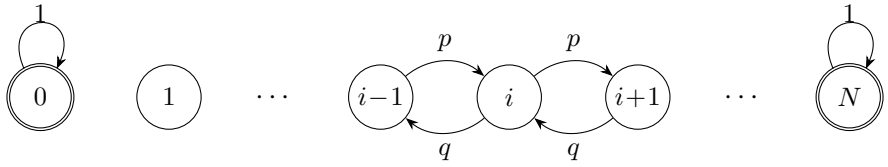


Figure 5: The gambler's-ruin random walk on $\{0,1,\ldots,N\}$. From an interior state i the walk moves up with probability p and down with probability $q = 1-p$ (arrows shown around the generic state i); the boundary states 0 and N are absorbing (double circles, self-loop probability 1).

Before solving for h_i and D_i we verify that the game actually ends, and quickly enough for expected durations to be finite — the hypothesis the first-step method needs.

Lemma 3.10 (Absorption is certain and has finite mean time)**.** *For every starting state $i \in \{0,\ldots,N\}$,*

$$P(T < \infty \mid S_0 = i) = 1 \qquad \textit{and} \qquad \mathbb{E}[T \mid S_0 = i] < \infty.$$

Proof. We use a block argument. Partition time into consecutive blocks of N steps. Suppose the walk is not yet absorbed at the start of a block, so its state lies in $\{1,\ldots,N-1\}$. With probability at least p^N all N steps of the block go up; starting from a state ≥ 1, a run of N consecutive up-moves would carry the walk past level N, which is impossible without first hitting the absorbing

state N. Hence absorption occurs within the block with probability at least $\delta = p^N > 0$, and this bound holds for every block regardless of the past. Iterating,

$$P(T > kN) \le (1-\delta)^k \longrightarrow 0 \quad (k \to \infty),$$

so $T < \infty$ almost surely. For the mean, sum tail probabilities: each n with $kN \le n < (k+1)N$ satisfies $P(T > n) \le P(T > kN) \le (1-\delta)^k$, so

$$\mathbb{E}[T] = \sum_{n \ge 0} P(T > n) \le \sum_{k \ge 0} N(1-\delta)^k = \frac{N}{\delta} < \infty. \qquad \square$$

Theorem 3.11 (Gambler's ruin, fair case)**.** *Suppose $p = \frac{1}{2}$. Then for every $i \in \{0, \dots, N\}$,*

$$h_i = P(\textit{reach } N \textit{ before } 0 \mid S_0 = i) = \frac{i}{N}.$$

Proof. We proceed by first-step analysis (§3.2). Conditioning on the first step from an interior state i, the law of total probability and the Markov property give

$$h_i = \tfrac{1}{2}\, h_{i+1} + \tfrac{1}{2}\, h_{i-1} \qquad (1 \le i \le N-1), \qquad h_0 = 0, \quad h_N = 1. \tag{3.2}$$

Relation (3.2) says that h is *discrete harmonic*: each interior value is the average of its two neighbors. Rearranging,

$$h_{i+1} - h_i = h_i - h_{i-1} \qquad (1 \le i \le N-1),$$

so all successive increments are equal to a common value d, and telescoping from the left boundary gives $h_i = h_0 + i\,d = i\,d$. The right boundary condition $h_N = Nd = 1$ forces $d = 1/N$, hence $h_i = i/N$. Both boundary conditions hold, and the argument derived this form for *every* solution of (3.2), so the solution is unique — the discrete counterpart of the fact that harmonic functions on an interval are affine. $\square$

Theorem 3.12 (Expected duration, fair case)**.** *Suppose $p = \frac{1}{2}$. Then for every $i \in \{0, \dots, N\}$,*

$$D_i = \mathbb{E}[T \mid S_0 = i] = i\,(N-i).$$

Proof. We again proceed by first-step analysis; Lemma 3.10 guarantees the unknowns D_i are finite, so the following equations are valid. Conditioning on the first step (which costs one step and restarts the walk from the new state),

$$D_i = 1 + \tfrac{1}{2}\, D_{i+1} + \tfrac{1}{2}\, D_{i-1} \qquad (1 \le i \le N-1), \qquad D_0 = D_N = 0. \tag{3.3}$$

Multiplying by 2 and rearranging puts (3.3) in the form of a second-order linear recurrence with constant inhomogeneous term:

$$D_{i+1} - 2D_i + D_{i-1} = -2.$$

Homogeneous part. The characteristic equation of $D_{i+1} - 2D_i + D_{i-1} = 0$ is $x^2 - 2x + 1 = (x-1)^2 = 0$, a double root at $x = 1$, so the homogeneous solutions are exactly the affine sequences $\alpha + \beta i$.

Particular solution. Since constants and linear terms already solve the homogeneous equation, we try a quadratic: substituting $D_i = \gamma i^2$ gives

$$\gamma\left[(i+1)^2 - 2i^2 + (i-1)^2\right] = 2\gamma = -2,$$

so $\gamma = -1$ and $-i^2$ is a particular solution.

General solution and boundaries. Every solution of the recurrence has the form $D_i = \alpha + \beta i - i^2$. The boundary condition $D_0 = 0$ forces $\alpha = 0$, and $D_N = \beta N - N^2 = 0$ forces $\beta = N$. Hence $D_i = Ni - i^2 = i(N-i)$. Uniqueness holds because the difference of two solutions of (3.3) solves the homogeneous equation with zero boundary values, i.e. is affine and vanishes at 0 and N, hence is identically zero. $\square$

(Problem P3.3 asks you to reproduce this proof and that of Theorem 3.11 from scratch.)

Example 3.13 (A fair game to 100)**.** Two players flip a fair coin for one unit per flip; one starts with 40 units, the other with 60, and they play until someone is broke, so $i = 40$ and $N = 100$. By Theorem 3.11 the poorer player finishes with everything with probability $h_{40} = 40/100 = 0.40$ — in a fair game, the probability of winning the pot equals the fraction of the total capital contributed. By Theorem 3.12 the game lasts $D_{40} = 40 \times 60 = 2400$ flips on average, far longer than the 100 units at stake might suggest.

Theorem 3.14 (Gambler's ruin, biased case)**.** *Suppose $p \neq \frac{1}{2}$ and set $r = q/p$. Then for every $i \in \{0, \dots, N\}$,*

$$h_i = P(\text{reach } N \text{ before } 0 \mid S_0 = i) = \frac{1 - r^{\,i}}{1 - r^{\,N}}.$$

Proof. We use first-step analysis followed by the standard ansatz for linear recurrences. Conditioning on the first step from an interior state,

$$h_i = p\,h_{i+1} + q\,h_{i-1} \qquad (1 \le i \le N-1), \qquad h_0 = 0, \quad h_N = 1. \tag{3.4}$$

Ansatz. Seek geometric solutions $h_i = x^i$. Substituting into (3.4) and dividing by x^{i-1} gives the characteristic equation

$$px^2 - x + q = 0.$$

The two roots multiply to q/p and sum to $1/p$; the reader should verify that $x = 1$ and $x = q/p = r$ satisfy both, so these are the roots, and they are distinct precisely because $p \neq q$. The general solution of (3.4) is therefore

$$h_i = \alpha \cdot 1^i + \beta\, r^i = \alpha + \beta\, r^i.$$

Boundary conditions. The condition $h_0 = 0$ gives $\alpha + \beta = 0$, so $\alpha = -\beta$, and $h_N = 1$ gives $\beta(r^N - 1) = 1$, so $\beta = 1/(r^N - 1)$. Hence

$$h_i = \frac{r^i - 1}{r^N - 1} = \frac{1 - r^i}{1 - r^N}.$$

Uniqueness follows as in Theorem 3.12: the difference of two solutions of (3.4) is of the form $\alpha + \beta r^i$ with zero values at $i = 0$ and $i = N$, which forces $\alpha = \beta = 0$. □

(Problem P3.4 asks you to reproduce this derivation and push it to the no-upper-target limit of Corollary 3.16.)

Proposition 3.15 (The fair case as a limit)**.** *Fix i and N. As $p \to \frac{1}{2}$ (equivalently $r \to 1$),*

$$\frac{1 - r^{\,i}}{1 - r^{\,N}} \longrightarrow \frac{i}{N},$$

recovering Theorem 3.11.

Proof. We apply l'Hôpital's rule in the variable r at $r = 1$: numerator and denominator both vanish there and both are differentiable, so

$$\lim_{r \to 1} \frac{1 - r^i}{1 - r^N} = \lim_{r \to 1} \frac{-i\, r^{\,i-1}}{-N\, r^{\,N-1}} = \frac{i}{N}.$$

Alternatively, factoring geometric sums, $(1 - r^i)/(1 - r^N) = (1 + r + \cdots + r^{\,i-1})/(1 + r + \cdots + r^{\,N-1})$, whose numerator and denominator tend to i and N respectively. □

Corollary 3.16 (Ruin with no upper target)**.** *Suppose the walk has no upper absorbing barrier (the adversary is infinitely rich, or the strategy simply never stops). Starting from $i \ge 1$:*

(i) if $p > \frac{1}{2}$ (so $r < 1$), the probability of ever hitting 0 is r^i, decaying exponentially in the buffer i;

(ii) if $p \le \frac{1}{2}$, ruin is certain.

Proof. We compute the limit of the two-barrier formulas as $N \to \infty$. Writing T_0 and T_N for the hitting times of 0 and N, the events $\{T_0 < T_N\}$ increase to $\{T_0 < \infty\}$ as $N \to \infty$: if the walk ever hits 0, it visits only finitely many levels beforehand, so $T_0 < T_N$ for all N large enough. By continuity of probability from below, $P(T_0 < \infty) = \lim_{N\to\infty}(1 - h_i)$.

For $p > \frac{1}{2}$ we have $r < 1$, so $r^N \to 0$ and, by Theorem 3.14, $1 - h_i \to 1 - (1 - r^i) = r^i$.

For $p < \frac{1}{2}$ we have $r > 1$; dividing numerator and denominator by r^N,

$$1 - h_i = \frac{r^i - r^N}{1 - r^N} = \frac{r^{\,i-N} - 1}{r^{-N} - 1} \longrightarrow \frac{0 - 1}{0 - 1} = 1.$$

For $p = \frac{1}{2}$, Theorem 3.11 gives $1 - h_i = 1 - i/N \to 1$. □

Example 3.17 (Small edge, large consequence)**.** Two computations show how violently a small bias moves the answer.

- *Two-barrier.* Take $p = 0.48$, $i = 10$, $N = 20$ — a nearly fair game, starting halfway. Then $r = 0.52/0.48 = 13/12$, and $\log(13/12) \approx 0.0800$, so $r^{10} \approx e^{0.800} \approx 2.226$ and $r^{20} \approx 4.957$. By Theorem 3.14,
$$h_{10} = \frac{1 - 2.226}{1 - 4.957} = \frac{1.226}{3.957} \approx 0.310.$$
A two-point deficit in win rate turns a 50% shot (Theorem 3.11 with $i/N = \frac{1}{2}$; also the $p \to \frac{1}{2}$ limit of Proposition 3.15) into a 31% shot.
- *No upper target.* A strategy wins each unit-stake round with probability $p = 0.55$ and never stops. With a buffer of $i = 10$ units, $r = 0.45/0.55 = 9/11$ and $\log(9/11) \approx -0.2007$, so by Corollary 3.16 the lifetime ruin probability is $(9/11)^{10} \approx e^{-2.007} \approx 0.134$. Doubling the buffer to $i = 20$ squares this: $(9/11)^{20} \approx 0.018$, below 2%. Capitalization buys exponential safety.

Intuition. A small negative edge is lethal over many trials: with $p < \frac{1}{2}$ and a distant target, the probability of reaching it decays exponentially and ruin probability tends to 1, by Theorem 3.14 and Corollary 3.16. This is why a market maker obsesses over a fraction of a tick of edge — compounding acts on the *sign* of the edge — and why paying the spread as a taker is a tax that must be beaten before anything else compounds in the right direction.

Trading application. The ruin walk doubles as a capitalization model for a trading operation: the state is the capital buffer in units of per-trade risk, p is the per-trade win probability, and Corollary 3.16 says the lifetime ruin probability of a positive-edge operation is r^i. A desk therefore sizes its buffer so that r^i sits below its ruin tolerance — and treats anything that flips the sign of the edge (fees, spread paid, adverse selection) as an existential threat rather than a cost line.

3.4 Markov Chains and Stationary Distributions

Absorption questions ask where a chain ends; stationary distributions ask how a chain that never ends shares its time among states. This first contact covers the definition and the two-state chain, which is solvable in closed form and already carries the main intuition.

Setup. Let $(X_n)_{n\ge0}$ be a time-homogeneous Markov chain on the finite state space $\{1, \dots, K\}$. Its one-step transition matrix is written $A = (A_{jk})$ with $A_{jk} = P(X_{n+1} = k \mid X_n = j)$; each row of A is nonnegative and sums to 1. We write A rather than the customary P because the symbol P is reserved for probability throughout this document. Distributions over states are row vectors, so that if X_n has distribution μ then X_{n+1} has distribution μA.

Definition 3.18 (Stationary distribution)**.** A row vector $\pi = (\pi_1, \dots, \pi_K)$ is a *stationary distribution* of the chain if

$$\pi_j \geq 0 \ \ (1 \leq j \leq K), \qquad \sum_{j=1}^{K} \pi_j = 1, \qquad \pi A = \pi.$$

If X_0 has distribution π, then by induction X_n has distribution π for every n: the chain is in equilibrium.

Proposition 3.19 (Two-state stationary distribution)**.** *Consider the two-state chain that switches from state* 1 *to state* 2 *with probability* a *and from state* 2 *to state* 1 *with probability* b *(staying put otherwise), i.e.*

$$A = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}, \qquad 0 \leq a, b \leq 1, \quad a + b > 0.$$

Within this subsection a *and* b *denote these switch probabilities; the odds symbol* b *of §3.1 is not used here. The chain has the unique stationary distribution*

$$\pi = \left(\frac{b}{a+b}, \ \frac{a}{a+b} \right).$$

Proof. We solve the defining linear system directly. The first coordinate of $\pi A = \pi$ reads

$$\pi_1(1-a) + \pi_2 \, b = \pi_1 \iff a \, \pi_1 = b \, \pi_2,$$

the *balance equation*: in equilibrium, probability flow from 1 to 2 equals flow from 2 to 1. The second coordinate reads $\pi_1 a + \pi_2(1-b) = \pi_2$, which rearranges to the same equation $a\pi_1 = b\pi_2$ (as it must, since the rows of A sum to one). The system therefore reduces to

$$a \, \pi_1 = b \, \pi_2, \qquad \pi_1 + \pi_2 = 1.$$

Substituting $\pi_2 = 1 - \pi_1$ into the balance equation gives $a\pi_1 = b(1-\pi_1)$, so $(a+b)\pi_1 = b$ and, since $a + b > 0$,

$$\pi_1 = \frac{b}{a+b}, \qquad \pi_2 = \frac{a}{a+b}.$$

Both entries are nonnegative and sum to one, and every step was an equivalence, so this is the unique stationary distribution. □

Example 3.20 (A two-regime order-flow model)**.** Model a venue's order flow as quiet (state 1) or busy (state 2) period by period, switching from quiet to busy with probability $a = 0.05$ and from busy to quiet with probability $b = 0.20$. By Proposition 3.19,

$$\pi = \left(\frac{0.20}{0.25}, \ \frac{0.05}{0.25} \right) = (0.8, \ 0.2) :$$

the venue is quiet 80% of the time in the long run. Verification by direct multiplication: the first entry of πA is $0.8 \times 0.95 + 0.2 \times 0.20 = 0.76 + 0.04 = 0.80$ and the second is $0.8 \times 0.05 + 0.2 \times 0.80 = 0.04 + 0.16 = 0.20$, so $\pi A = \pi$ as required.

Intuition. Time is shared inversely to how easily each state is left. When $a, b > 0$, a visit to state 1 lasts Geom(a) steps (mean $1/a$) and a visit to state 2 lasts Geom(b) steps (mean $1/b$), and the chain alternates visits; the long-run fraction of time in state 1 is therefore

$$\frac{1/a}{1/a + 1/b} = \frac{b}{a+b},$$

matching Proposition 3.19. Sticky states accumulate probability.

3.5 St. Petersburg and Infinite Expected Value

The St. Petersburg game is the standard cautionary tale about treating expected value as a price: its expectation is infinite, yet nobody pays more than about twenty dollars to play. This subsection computes the divergence, proves the bankroll-capped value, and names the classical resolutions.

Setup. A fair coin is flipped until the first head. If the first head occurs on flip n (an event of probability 2^{-n}, $n = 1, 2, \dots$), the player is paid $Y = 2^n$ dollars.

Proposition 3.21 (St. Petersburg divergence)**.** *The expected payoff of the St. Petersburg game is infinite:*

$$\mathbb{E}[Y] = \sum_{n=1}^{\infty} 2^{-n} \cdot 2^n = \sum_{n=1}^{\infty} 1 = \infty.$$

Proof. We compute the defining series of the expectation of a discrete nonnegative random variable. The event $\{Y = 2^n\}$ has probability 2^{-n}, so the n-th term of $\sum_n 2^n P(Y = 2^n)$ equals $2^n \cdot 2^{-n} = 1$. The partial sums are m after m terms and grow without bound, so the series of nonnegative terms diverges and $\mathbb{E}[Y] = \infty$. □

Despite the infinite expectation, essentially nobody will pay more than about \$20 to play once. The sharpest quantitative resolution is that no real counterparty can pay unbounded amounts: capping the payoff at the counterparty's bankroll collapses the value to a small number.

Proposition 3.22 (Capped St. Petersburg)**.** *Fix an integer $M \geq 1$ and suppose the payoff is capped at 2^M dollars, i.e. the player receives $Y_M = \min(Y, 2^M)$. Then*

$$\mathbb{E}[Y_M] = M + 1.$$

Proof. We split the expectation at the cap. For $n \leq M$ the payoff is 2^n, and for $n > M$ it is 2^M:

$$\mathbb{E}[Y_M] = \sum_{n=1}^{M} 2^{-n} \cdot 2^n + 2^M \sum_{n=M+1}^{\infty} 2^{-n} = \underbrace{M}_{\text{one dollar per level}} + 2^M \cdot 2^{-M} = M + 1,$$

where the tail sum is geometric: $\sum_{n=M+1}^{\infty} 2^{-n} = 2^{-M}$. Each uncapped level contributes exactly one dollar of expected value, and the entire capped tail contributes one more. □

(Problem P3.9 asks you to reproduce both computations with a different cap.)

Example 3.23 (A 2^{30} cap)**.** Suppose the counterparty can pay at most 2^{30} dollars — about \$1.07 billion, a very rich casino. Proposition 3.22 with $M = 30$ gives

$$\mathbb{E}[Y_{30}] = 30 + 1 = 31$$

dollars: one dollar of expected value per level up to the cap, plus one dollar for the capped tail. An infinite expectation backed by a billion-dollar bankroll is worth \$31.

Remark 3.24 (Three resolutions). Three classical resolutions of the paradox are worth naming.

(1) *Log utility (Bernoulli, 1738).* A log-utility agent values the uncapped game by $\mathbb{E}[\log_2 Y] = \sum_{n \geq 1} n\, 2^{-n} = 2$ doublings, a certainty equivalent of $2^2 = 4$ dollars — finite and small. This is the same concavity that drives the Kelly criterion of §3.1.

(2) *Finite counterparty bankroll.* No counterparty pays unbounded sums; Proposition 3.22 shows a cap at 2^M collapses the value to $M + 1$ dollars.

(3) *Variance and one-shot reasoning.* The payoff distribution is so heavy tailed that the expectation is carried by astronomically rare outcomes: a single play pays at most \$8 with probability 7/8, and with no finite mean (uncapped) sample averages never stabilize over any feasible number of plays.

Interview trap. Quoting the infinite expectation as a fair price is the trap. *Price* is not *expected value* when distributions are wild — managing the gap between the two is the whole point of risk management. In an interview, the standard follow-up "how much would you pay to play once?" is answered with a small number in the low tens of dollars, justified by the three resolutions of Remark 3.24, never with "infinity".

Practice: problems P3.1–P3.10 in §27, solutions in §28.

Part II
Statistics and Regression

4 Statistics I: Estimators and Inference

Estimation is where probability meets data: a trading signal's mean return, a strategy's volatility, a Sharpe ratio — each is a number computed from a finite sample and therefore a random variable in its own right, with a bias, a variance, and a sampling distribution. This section builds the estimator's error budget (bias, variance, mean squared error), corrects the most famous estimation bias (Bessel's correction), states what the central limit theorem does and does not license, turns hypothesis testing into an explicit procedure with a fully worked coin example, and closes with the standard error of the Sharpe ratio — the single formula that most reliably deflates backtest overconfidence.

4.1 Bias, Variance, and Mean Squared Error

Every estimate carries two kinds of error: a systematic offset (bias) and sampling noise (variance). The mean squared error adds them in exactly one way, and that decomposition is the reason a deliberately biased estimator can beat an unbiased one — the fact that underlies ridge regression and shrinkage throughout statistics and quantitative finance.

Setup. Let $X_1, \dots, X_n$ be data drawn from a distribution that depends on an unknown parameter $\theta \in \mathbb{R}$. All probabilities and expectations in this subsection are taken under the distribution indexed by the true value θ, which is a fixed (non-random) number.

Definition 4.1 (Estimator, bias, and mean squared error)**.** An *estimator* of θ is a function $\hat{\theta} = \hat{\theta}(X_1, \dots, X_n)$ of the data alone (it may not depend on θ), used as a guess for θ. Its *bias* is

$$\operatorname{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta,$$

and the estimator is called *unbiased* when $\operatorname{Bias}(\hat{\theta}) = 0$ for every value of θ. Its *mean squared error* is

$$\operatorname{MSE}(\hat{\theta}) = \mathbb{E}\big[(\hat{\theta} - \theta)^2\big].$$

Theorem 4.2 (Bias–variance decomposition of MSE)**.** *For any estimator $\hat{\theta}$ of θ with $\mathbb{E}[\hat{\theta}^2] < \infty$,*

$$\operatorname{MSE}(\hat{\theta}) = \operatorname{Bias}(\hat{\theta})^2 + \operatorname{Var}(\hat{\theta}). \tag{4.1}$$

Proof. We expand the square after adding and subtracting the mean of the estimator. Write $m = \mathbb{E}[\hat{\theta}]$, so that

$$(\hat{\theta} - \theta)^2 = \big((\hat{\theta} - m) + (m - \theta)\big)^2 = (\hat{\theta} - m)^2 + 2(\hat{\theta} - m)(m - \theta) + (m - \theta)^2.$$

Now take expectations term by term (linearity). The first term gives $\mathbb{E}[(\hat{\theta} - m)^2] = \mathrm{Var}(\hat{\theta})$ by the definition of variance. In the cross term, the factor $m - \theta$ is a deterministic number, so

$$\mathbb{E}\big[2(\hat{\theta} - m)(m - \theta)\big] = 2(m - \theta)\,\mathbb{E}[\hat{\theta} - m] = 2(m - \theta) \cdot 0 = 0.$$

The last term is the constant $(m - \theta)^2 = \mathrm{Bias}(\hat{\theta})^2$. Summing the three terms yields (4.1). □

Example 4.3 (Shrinkage can dominate on MSE)**.** Let $X_1, \ldots, X_{25}$ be iid with mean $\mu = 1$ and variance $\sigma^2 = 4$; here the parameter is $\theta = \mu$. Compare two estimators of μ.

Unbiased estimator. The sample mean $\bar{X} = \frac{1}{25}\sum_{i=1}^{25} X_i$ has $\mathbb{E}[\bar{X}] = 1$ (bias 0) and $\mathrm{Var}(\bar{X}) = \sigma^2/n = 4/25 = 0.16$. By (4.1), $\mathrm{MSE}(\bar{X}) = 0 + 0.16 = 0.16$.

Shrunk estimator. Take $\tilde{\mu} = 0.9\,\bar{X}$. Then $\mathbb{E}[\tilde{\mu}] = 0.9$, so the bias is $0.9 - 1 = -0.1$ and $\mathrm{Bias}^2 = 0.01$; the variance is $0.9^2 \times 0.16 = 0.1296$. By (4.1),

$$\mathrm{MSE}(\tilde{\mu}) = 0.01 + 0.1296 = 0.1396 < 0.16 = \mathrm{MSE}(\bar{X}).$$

The deliberately biased estimator wins: shrinking toward zero costs 0.01 in squared bias but saves 0.0304 in variance. The comparison depends on the true parameter — if instead $\mu = 4$, the squared bias becomes $(0.4)^2 = 0.16$ and $\mathrm{MSE}(\tilde{\mu}) = 0.16 + 0.1296 = 0.2896 > 0.16$, so shrinkage helps when the true parameter is small relative to the sampling noise.

Intuition. Unbiasedness is a constraint, not a virtue in itself. An unbiased estimator is forced to track the data wherever it goes, and pays for that fidelity in variance. Trading a little bias for a large variance reduction lowers total error, and Example 4.3 shows the exchange rate explicitly. Ridge regression and shrinkage estimators institutionalise exactly this trade and can dominate unbiased alternatives on MSE.

4.2 Bessel's Correction

The most common variance estimator divides by $n - 1$ rather than n, and interviewers routinely ask why. The answer is a two-line expectation computation resting on one algebraic identity, which we isolate as a lemma because it recurs across statistics (it is the finite-sample ancestor of the analysis-of-variance decomposition).

Setup. Let $X_1, \ldots, X_n$ be iid random variables with unknown mean μ and unknown variance $\sigma^2 < \infty$, and let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ denote the sample mean. Following universal convention we write s^2 (rather than a hatted symbol) for the Bessel-corrected sample variance defined in Theorem 4.5 below, and $\hat{\sigma}^2_{\mathrm{ML}}$ for the divide-by-n estimator.

Lemma 4.4 (Decomposition around the sample mean)**.** *For any realisation of the data,*

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2 + n(\bar{X} - \mu)^2. \tag{4.2}$$

Proof. We add and subtract the sample mean inside each squared term and show that the cross term vanishes. Writing $X_i - \mu = (X_i - \bar{X}) + (\bar{X} - \mu)$ and expanding the square,

$$\sum_{i=1}^{n}(X_i - \mu)^2 = \sum_{i=1}^{n}(X_i - \bar{X})^2 + 2(\bar{X} - \mu)\sum_{i=1}^{n}(X_i - \bar{X}) + n(\bar{X} - \mu)^2,$$

where the factor $\bar{X} - \mu$ came out of the middle sum because it does not depend on i. The middle sum is

$$\sum_{i=1}^{n}(X_i - \bar{X}) = \sum_{i=1}^{n} X_i - n\bar{X} = n\bar{X} - n\bar{X} = 0$$

by the definition of $\bar{X}$, so the cross term vanishes and (4.2) follows. □

Theorem 4.5 (Bessel's correction)**.** *Under the setup above,*

$$\mathbb{E}\Big[\sum_{i=1}^{n}(X_i-\bar{X})^2\Big]=(n-1)\,\sigma^2,$$

and therefore the sample variance

$$s^2=\frac{1}{n-1}\sum_{i=1}^{n}(X_i-\bar{X})^2$$

is an unbiased estimator of σ^2*.*

Proof. We take expectations on both sides of the identity (4.2) and evaluate each term from the iid assumptions. On the left, each summand has $\mathbb{E}[(X_i-\mu)^2]=\sigma^2$ by the definition of variance, so the left side has expectation $n\sigma^2$. For the final term on the right, the sample mean satisfies $\mathbb{E}[\bar{X}]=\mu$ (linearity) and, by independence,

$$\operatorname{Var}(\bar{X})=\frac{1}{n^2}\sum_{i=1}^{n}\operatorname{Var}(X_i)=\frac{\sigma^2}{n},$$

so $\mathbb{E}[n(\bar{X}-\mu)^2]=n\operatorname{Var}(\bar{X})=\sigma^2$. Substituting both evaluations into the expectation of (4.2) gives

$$n\sigma^2=\mathbb{E}\Big[\sum_{i=1}^{n}(X_i-\bar{X})^2\Big]+\sigma^2,$$

and rearranging yields $\mathbb{E}\big[\sum_i(X_i-\bar{X})^2\big]=(n-1)\sigma^2$. Dividing by $n-1$ shows $\mathbb{E}[s^2]=\sigma^2$. □

(Problem P4.3 asks you to reproduce this proof from scratch.)

Example 4.6 (Bessel's correction on a small sample)**.** Take the sample $\{1,2,6\}$, so $n=3$ and $\bar{X}=3$. The deviations from $\bar{X}$ are $-2,-1,3$, with squared deviations $4,1,9$ summing to 14. The Bessel-corrected estimate is $s^2=14/2=7$, while the divide-by-n estimate is $\hat{\sigma}^2_{\mathrm{ML}}=14/3\approx 4.67$.

The systematic direction of the gap is visible in expectation. Suppose the true variance is $\sigma^2=10$ and $n=5$. Theorem 4.5 gives $\mathbb{E}\big[\sum_i(X_i-\bar{X})^2\big]=4\times 10=40$, so

$$\mathbb{E}[\hat{\sigma}^2_{\mathrm{ML}}]=\frac{40}{5}=8=\frac{n-1}{n}\,\sigma^2,$$

an underestimate of the true variance by 20%, whereas $\mathbb{E}[s^2]=40/4=10$ exactly.

Intuition. The sample mean chases the data: because $\bar{X}$ is computed from the same sample it is being compared against, deviations from $\bar{X}$ are systematically smaller than deviations from the true mean μ — by exactly one degree of freedom. The identity (4.2) makes this precise: the discarded term $n(\bar{X}-\mu)^2$ carries expectation exactly σ^2, one summand's worth of variance.

Remark 4.7 (The MLE divisor and the regression generalisation)*.* The maximum-likelihood estimator of σ^2 for a normal sample is $\hat{\sigma}^2_{\mathrm{ML}}$, which divides by n and is therefore biased low by the factor $(n-1)/n$, as Example 4.6 quantifies. Regression generalises Bessel's correction: with p fitted parameters, the unbiased estimator of the residual variance is $\hat{\sigma}^2=\mathrm{RSS}/(n-p)$, with RSS the residual sum of squares; each fitted parameter absorbs one degree of freedom, just as fitting $\bar{X}$ absorbed one here.

4.3 Sampling Distributions: the Central Limit Theorem and Student's t

Inference needs the distribution of an estimator, not just its first two moments. The central limit theorem supplies a universal normal approximation for averages; Student's t corrects that approximation when the variance itself must be estimated. Both are stated here without proof — what matters for practice is knowing exactly what they assert and, just as importantly, what they do not.

Setup. Let $X_1, X_2, \ldots$ be iid random variables with mean μ and finite variance $\sigma^2 > 0$, and write $\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$ for the sample mean of the first n observations. As throughout, Φ denotes the standard normal CDF, and s^2 is the Bessel-corrected sample variance of Theorem 4.5.

Fact 4.8 (Central Limit Theorem)**.** Under the setup above, the standardised sample mean converges in distribution to a standard normal: for every real x,

$$P\left(\frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \le x\right) \longrightarrow \Phi(x) \qquad \text{as } n \to \infty.$$

A proof is beyond our scope.

Example 4.9 (Normal approximation to a fair-coin count)**.** Let K be the number of heads in 100 flips of a fair coin, so $K \sim \mathrm{Bin}(100, 1/2)$ is the sum of 100 iid Bernoulli variables, each with mean $1/2$ and variance $1/4$. The count has mean $100 \times 1/2 = 50$ and variance $100 \times 1/4 = 25$, so Fact 4.8 licenses the approximation

$$K \approx \mathcal{N}(50,\, 25), \qquad \text{standard deviation } 5.$$

This approximation is the engine of the hypothesis test in Example 4.13 below.

Remark 4.10 (What the CLT does not say)*.* Fact 4.8 is a limit statement about the centre of the distribution, and three misreadings of it are common enough to deserve explicit warnings.

(i) *Nothing about a fixed small n.* The theorem asserts a limit; it attaches no accuracy guarantee to any particular finite sample size.

(ii) *Nothing about the far tails at moderate n.* Fat-tailed summands converge slowly in the tails — exactly where risk lives. A normal approximation can be excellent near the mean and badly wrong at the 4-sigma quantile simultaneously.

(iii) *Nothing for dependent or infinite-variance data.* Both hypotheses (independence, finite variance) are load-bearing, and financial returns violate both in stress: dependence spikes and tails thicken precisely when the approximation is being leaned on hardest.

Fact 4.11 (Student's t and the estimated standard deviation (Student, 1908))**.** Let $X_1, \ldots, X_n$ be iid $\mathcal{N}(\mu, \sigma^2)$. If the true σ were known, the statistic $\sqrt{n}\,(\bar{X}_n - \mu)/\sigma$ would be exactly standard normal. Replacing σ by its estimate s changes the distribution: the statistic

$$T = \frac{\sqrt{n}\,(\bar{X}_n - \mu)}{s}$$

follows Student's t distribution with $n - 1$ degrees of freedom, written t_{n-1}, whose tails are wider than the standard normal's. The widening reflects the extra uncertainty from estimating σ, matters materially for small n, and vanishes in the limit: t_{n-1} converges to $\mathcal{N}(0, 1)$ as $n \to \infty$. A proof is beyond our scope.

4.4 Hypothesis Testing and Confidence Intervals

A hypothesis test is not an inspiration but a checklist: fix the null, fix the statistic, know the statistic's null distribution, and compute how surprising the data would be if the null were true. This subsection states the checklist as a formal procedure, works it end to end on a coin, and proves the duality that makes confidence intervals and tests two views of the same object.

Setup. The data are $X_1, \ldots, X_n$; a *null hypothesis* H_0 is a claim that fully determines the distribution of a chosen statistic; and $\alpha \in (0, 1)$ is a *significance level* fixed before seeing the data (conventionally $\alpha = 0.05$). To avoid a symbol collision we reserve the letter p for p-values in this subsection and write θ for unknown parameters.

Definition 4.12 (Hypothesis-testing procedure)**.** A hypothesis test at significance level α consists of the following steps.

(i) State the null hypothesis H_0.
(ii) Choose a test statistic T, a function of the data.
(iii) Derive the distribution of T under H_0.
(iv) Compute the *p-value*: the probability, under H_0, of a value of T at least as extreme as the one observed. For a two-sided test whose null distribution is symmetric about zero, with observed value t_{obs}, this is $p = P_{H_0}(|T| \geq |t_{\text{obs}}|)$.
(v) Reject H_0 if $p \leq \alpha$; otherwise do not reject.

Equivalently, step (v) rejects exactly when T falls in a *rejection region* chosen to have probability α under H_0 (Figure 6).

Example 4.13 (Sixty heads in one hundred flips)**.** A coin is flipped 100 times and lands heads 60 times. Is it fair? We run Definition 4.12 step by step at $\alpha = 0.05$.

(i) The null hypothesis is that the coin is fair: the heads count satisfies $K \sim \text{Bin}(100, 1/2)$.
(ii) The statistic is the standardised count $z = (K - 50)/5$.
(iii) By Example 4.9, under H_0 the count is approximately $\mathcal{N}(50, 25)$, so z is approximately standard normal.
(iv) The observed value is $z = (60 - 50)/5 = 2$, and the two-sided p-value is

$$p = 2(1 - \Phi(2)) = 2 \times 0.02275 \approx 0.045.$$

(v) Since $0.045 < 0.05$, reject fairness at the 5% level — barely. Equivalently, the observed $z = 2$ lies just beyond the critical value 1.96, inside the shaded rejection region of Figure 6.

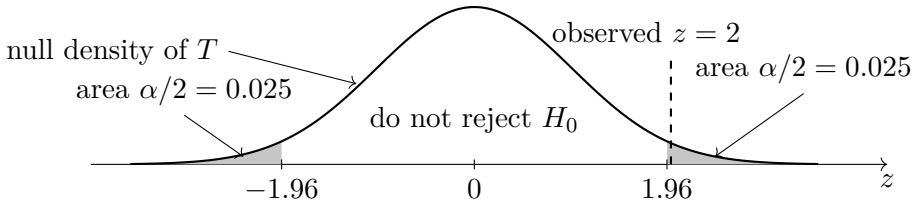


Figure 6: Two-sided rejection region at level $\alpha = 0.05$ under the null density of the test statistic. The shaded tails each carry probability $\alpha/2 = 0.025$ and begin at the critical values ± 1.96. The observed statistic $z = 2$ of Example 4.13 falls just inside the right tail, matching the two-sided p-value of about $0.045 < 0.05$: a rejection, but a marginal one.

Step (iii) of Definition 4.12 is where Fact 4.11 earns its place: when the statistic standardises by an *estimated* rather than known standard deviation, its null distribution is t_{n-1} rather than normal — the z-test becomes a t-test, the critical values widen, and the difference matters for small n.

Proposition 4.14 (Confidence-interval–test duality)**.** *Let $\{P_\theta : \theta \in \Theta\}$ be a family of distributions for the data X, and suppose that for each $\theta_0 \in \Theta$ there is a test of $H_0 : \theta = \theta_0$ with acceptance region $A(\theta_0)$ (the set of data values for which the test does not reject) of level α, meaning $P_{\theta_0}(X \notin A(\theta_0)) \leq \alpha$. Define*

$$C(X) = \{\theta_0 \in \Theta : X \in A(\theta_0)\},$$

the set of null values the data do not reject. Then $C(X)$ is a $(1 - \alpha)$-confidence set:

$$P_\theta(\theta \in C(X)) \geq 1 - \alpha \qquad \text{for every } \theta \in \Theta.$$

Conversely, if $C(X)$ is any $(1 - \alpha)$-confidence set, the test that rejects $H_0 : \theta = \theta_0$ exactly when $\theta_0 \notin C(X)$ has level α.

Proof. The proof is a direct verification from the definitions of level and coverage. For the first claim, fix $\theta \in \Theta$. By the construction of C, the event $\{\theta \in C(X)\}$ is exactly the event $\{X \in A(\theta)\}$, so

$$P_\theta\big(\theta \in C(X)\big) = P_\theta\big(X \in A(\theta)\big) \geq 1 - \alpha,$$

where the inequality is the level condition applied at $\theta_0 = \theta$. For the converse, fix θ_0 and compute the type-I error of the proposed test:

$$P_{\theta_0}(\text{reject}) = P_{\theta_0}\big(\theta_0 \notin C(X)\big) = 1 - P_{\theta_0}\big(\theta_0 \in C(X)\big) \leq 1 - (1-\alpha) = \alpha,$$

where the inequality is the coverage guarantee applied at $\theta = \theta_0$. □

In the standard configuration $\alpha = 0.05$, Proposition 4.14 says: *the 95% confidence interval is exactly the set of null values not rejected at the 5% level.*

Example 4.15 (Confidence interval for the coin's bias)**.** Continue Example 4.13 and write θ for the coin's heads-probability, estimated by $\hat{\theta} = 60/100 = 0.6$. The Wald standard error is

$$\mathrm{SE}(\hat{\theta}) = \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}} = \sqrt{\frac{0.6 \times 0.4}{100}} \approx 0.049,$$

so the 95% Wald interval is

$$0.6 \pm 1.96 \times 0.049 = [0.504,\ 0.696].$$

The fair value $\theta_0 = 0.5$ lies just outside the interval, which by Proposition 4.14 is the same statement as rejection at the 5% level — and indeed the Wald test gives $z = 0.1/0.049 \approx 2.04$ with two-sided $p \approx 0.041$, in agreement with the marginal rejection ($p \approx 0.045$) found in Example 4.13. The small numerical gap arises because Example 4.13 standardised by the null standard error $\sqrt{0.5 \times 0.5/100} = 0.05$ while the Wald interval uses 0.049; the duality is exact when the interval inverts the same family of tests, and here both variants reject, both barely.

4.5 The Sharpe-Ratio Standard Error (the Humility Formula)

A Sharpe ratio computed from data is an estimate, and like every estimate it deserves a standard error. The formula below is short, its consequences are brutal, and quoting it accurately is one of the fastest ways to demonstrate statistical maturity when discussing backtests.

Setup. Let $r_1, \ldots, r_n$ be a strategy's returns, assumed iid at a fixed observation frequency (say daily), with mean μ and standard deviation $\sigma > 0$. The (per-period) Sharpe ratio is $\mathrm{SR} = \mu/\sigma$, estimated by $\widehat{\mathrm{SR}} = \hat{\mu}/\hat{\sigma}$ with the sample mean and sample standard deviation; annualised quantities multiply the daily ones by $\sqrt{252}$.

Fact 4.16 (Standard error of the Sharpe ratio (Lo, 2002))**.** Under the iid setup above, assuming additionally that the returns are normally distributed, the estimated Sharpe ratio has approximate standard error

$$\mathrm{SE}\left(\widehat{\mathrm{SR}}\right) \approx \sqrt{\frac{1 + \mathrm{SR}^2/2}{n}}, \tag{4.3}$$

with SR and n both measured at the observation frequency. The result follows from a delta-method computation; a proof is beyond our scope.

Example 4.17 (One year of daily data)**.** Take one year of daily returns, $n = 252$, from a strategy whose true annual Sharpe ratio is 2; the daily Sharpe ratio is $2/\sqrt{252} \approx 0.126$. By (4.3), the daily-frequency standard error is

$$\mathrm{SE}\left(\widehat{\mathrm{SR}}\right) \approx \sqrt{\frac{1 + 0.126^2/2}{252}} = \sqrt{\frac{1.0079}{252}} \approx 0.0632,$$

and annualising (multiplying by $\sqrt{252} \approx 15.87$) gives an annualised standard error of $0.0632 \times 15.87 \approx 1.0$. *One year of daily data measures an annualised Sharpe ratio to about* ± 1.

Trading application. Reporting a Sharpe ratio of 2.3 versus 2.1 off a year of daily returns is reporting noise: the difference of 0.2 is a fifth of one standard error. Equation (4.3) is the humility check to run before believing any backtest headline, comparing two strategies on short samples, or reallocating between books on the strength of a single year.

Interview trap. "Your signal's t-statistic is 2.5 over 1000 days — do you trade it?" The required first answer is *multiple testing*: if 100 signals were tried, the expected maximum $|t|$ under the null is well above 2.5, so a best-of-many 2.5 is unremarkable. The correction is Bonferroni-style thinking or the Deflated Sharpe Ratio. Only after that come the follow-ups: look-ahead bias and leakage, overlapping observations (which shrink the effective n), heteroskedasticity, and finally costs and capacity.

Practice: problems P4.1–P4.12 in §27, solutions in §28.

5 Regression and Regularisation

Linear regression is the workhorse of quantitative research: it prices hedges, evaluates signals, and underlies almost every factor model. This section develops ordinary least squares and its projection geometry, the Gauss–Markov optimality theorem and what happens when its assumptions fail, the two classical regularisers (ridge and lasso), the inverse-regression identity, weighted least squares, and a dart-throwing problem that ties squared and absolute loss to the mean and the median.

5.1 Ordinary Least Squares

This subsection derives the least-squares estimator from first principles, establishes its interpretation as an orthogonal projection, and specialises it to the one-regressor formulas that every interview expects on demand.

Setup. Throughout this subsection, the data consist of a response vector $y \in \mathbb{R}^n$ and a design matrix $X \in \mathbb{R}^{n \times p}$ of full column rank $p \leq n$, treated as fixed. The working model is

$$y = X\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon] = 0, \qquad \mathrm{Var}(\varepsilon) = \sigma^2 I,$$

with $\beta \in \mathbb{R}^p$ unknown. The least-squares criterion is $L(\beta) = \|y - X\beta\|^2$, and $\mathrm{col}(X) \subset \mathbb{R}^n$ denotes the span of the columns of X.

Theorem 5.1 (Normal equations and the OLS estimator)**.** *The criterion $L(\beta) = \|y - X\beta\|^2$ has a unique minimiser, characterised by the normal equations*

$$X^\top X\beta = X^\top y, \tag{5.1}$$

and given explicitly by

$$\hat{\beta}_{\mathrm{OLS}} = (X^\top X)^{-1} X^\top y. \tag{5.2}$$

Proof. We compute the gradient and Hessian of L and use strict convexity. Expanding the squared norm,

$$L(\beta) = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta,$$

so the gradient is $\nabla_\beta L = -2X^\top y + 2X^\top X\beta$ and the Hessian is $\nabla^2_\beta L = 2X^\top X$. For any $u \neq 0$ we have $u^\top X^\top X u = \|Xu\|^2 > 0$, because full column rank makes $Xu \neq 0$; hence $X^\top X$ is positive definite, so it is invertible and L is strictly convex. Setting the gradient to zero yields (5.1), whose unique solution is (5.2); by strict convexity this stationary point is the unique global minimiser. □

Theorem 5.2 (Projection interpretation and hat-matrix properties)**.** *Define the hat matrix* $H = X(X^\top X)^{-1}X^\top$*, the fitted vector* $\hat{y} = X\hat{\beta}_{\mathrm{OLS}} = Hy$*, and the residual vector* $\hat{\varepsilon} = y - \hat{y}$*. Then:*

(i) $X^\top \hat{\varepsilon} = 0$*: the residual is orthogonal to every column of* X*;*

(ii) $H^\top = H$ *and* $H^2 = H$*: the hat matrix is a symmetric idempotent, i.e. an orthogonal projector;*

(iii) $\mathrm{tr}(H) = p$*;*

(iv) $\hat{y}$ *is the orthogonal projection of* y *onto* $\mathrm{col}(X)$*: for every* $v \in \mathrm{col}(X)$*,*

$$\|y - v\|^2 = \|y - \hat{y}\|^2 + \|\hat{y} - v\|^2 \;\geq\; \|y - \hat{y}\|^2.$$

Proof. We verify each claim directly from the closed form (5.2), using the cyclic property of the trace for (iii).

(i) Substituting the definitions, $X^\top \hat{\varepsilon} = X^\top y - X^\top X(X^\top X)^{-1}X^\top y = X^\top y - X^\top y = 0$.

(ii) Symmetry follows by transposing: $H^\top = X\big((X^\top X)^{-1}\big)^\top X^\top = H$, because $X^\top X$ is symmetric and so is its inverse. Idempotence follows by cancellation:

$$H^2 = X(X^\top X)^{-1}\,(X^\top X)\,(X^\top X)^{-1}X^\top = H.$$

(iii) By the cyclic property of the trace, $\mathrm{tr}(H) = \mathrm{tr}\big[(X^\top X)^{-1}X^\top X\big] = \mathrm{tr}(I_p) = p$.

(iv) The fitted vector satisfies $\hat{y} = X\hat{\beta}_{\mathrm{OLS}} \in \mathrm{col}(X)$. Any $v \in \mathrm{col}(X)$ can be written $v = Xb$, and then $\hat{\varepsilon}^\top(\hat{y} - v) = \hat{\varepsilon}^\top X(\hat{\beta}_{\mathrm{OLS}} - b) = 0$ by (i). Hence the decomposition $y - v = \hat{\varepsilon} + (\hat{y} - v)$ is orthogonal, and the Pythagorean identity gives the display. □

Figure 7 shows the geometry: the fitted vector is the foot of the perpendicular dropped from y onto the plane $\mathrm{col}(X)$.

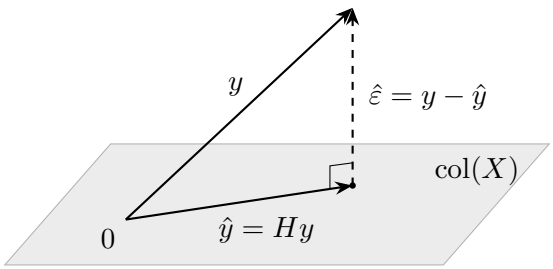


Figure 7: Ordinary least squares as orthogonal projection. The fitted vector $\hat{y} = Hy$ is the point of $\mathrm{col}(X)$ closest to y; the residual $\hat{\varepsilon}$ is perpendicular to every vector in $\mathrm{col}(X)$ (Theorem 5.2).

Intuition. Residuals are orthogonal to every regressor — that one identity generates most regression facts: the Pythagorean decomposition of sums of squares, the vanishing of cross terms in variance calculations, and the proof of Theorem 5.6. The trace identity $\mathrm{tr}(H) = p$ is read as "degrees of freedom used": fitting p coefficients consumes p dimensions of the data, which is why residual variance is estimated with divisor $n - p$.

The one-regressor specialisation below is worth having cold: interviews expect it instantly and build follow-ups on it.

Setup (univariate case). Suppose the regression has an intercept and a single regressor: observations (x_i, y_i) for $i = 1, \ldots, n$, model $y_i = \alpha + \beta x_i + \varepsilon_i$. Write $\bar{x}, \bar{y}$ for the sample means, and

$$S_{xx} = \sum_{i=1}^{n}(x_i - \bar{x})^2, \qquad S_{xy} = \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}), \qquad S_{yy} = \sum_{i=1}^{n}(y_i - \bar{y})^2,$$

assumed positive. The sample standard deviations are $\hat{\sigma}_x = \sqrt{S_{xx}/n}$ and $\hat{\sigma}_y = \sqrt{S_{yy}/n}$, the sample correlation is $\hat{\rho} = S_{xy}/\sqrt{S_{xx}S_{yy}}$, and the coefficient of determination is $R^2 = 1 - \mathrm{RSS}/\mathrm{TSS}$, where $\mathrm{RSS} = \sum_i \hat{\varepsilon}_i^2$ and $\mathrm{TSS} = S_{yy}$.

Corollary 5.3 (Univariate slope and R-squared)**.** *In the univariate regression with intercept,*

$$\hat{\beta} = \frac{S_{xy}}{S_{xx}} = \hat{\rho}\,\frac{\hat{\sigma}_y}{\hat{\sigma}_x}, \qquad \hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}, \qquad R^2 = \hat{\rho}^{\,2}. \tag{5.3}$$

Equivalently, the slope is the ratio of sample covariance to sample variance of the regressor.

Proof. We specialise the normal equations (5.1) to the design whose columns are the all-ones vector and x. The first normal equation reads $\sum_i (y_i - \alpha - \beta x_i) = 0$, which gives $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$. Substituting this into the second normal equation $\sum_i x_i(y_i - \alpha - \beta x_i) = 0$ yields

$$\sum_i x_i\big[(y_i - \bar{y}) - \beta(x_i - \bar{x})\big] = 0.$$

Because $\sum_i (y_i - \bar{y}) = \sum_i (x_i - \bar{x}) = 0$, replacing the factor x_i by $x_i - \bar{x}$ changes nothing, and the equation becomes $S_{xy} = \beta S_{xx}$, i.e. $\hat{\beta} = S_{xy}/S_{xx}$. Dividing numerator and denominator by n and factoring out standard deviations gives $\hat{\beta} = \hat{\rho}\,\hat{\sigma}_y/\hat{\sigma}_x$.

For R^2, the fitted deviations are $\hat{y}_i - \bar{y} = \hat{\beta}(x_i - \bar{x})$, so the explained sum of squares is $\mathrm{ESS} = \hat{\beta}^2 S_{xx} = S_{xy}^2/S_{xx}$. By Theorem 5.2(i) the residual vector is orthogonal to both columns of the design, hence to the fitted deviations, so the cross term in $\mathrm{TSS} = \sum_i \big[\hat{\varepsilon}_i + (\hat{y}_i - \bar{y})\big]^2$ vanishes and $\mathrm{TSS} = \mathrm{RSS} + \mathrm{ESS}$. Therefore

$$R^2 = \frac{\mathrm{ESS}}{\mathrm{TSS}} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \hat{\rho}^{\,2}. \qquad \square$$

(Problem P5.2 asks you to reproduce the derivations of this subsection — estimator, orthogonality, hat-matrix properties, and the univariate reduction — from scratch.)

Example 5.4 (A three-point regression, fully worked)**.** Take $n = 3$ observations $x = (1, 2, 3)$, $y = (1, 2, 2)$, and fit $y_i = \alpha + \beta x_i + \varepsilon_i$. The design and its Gram matrix are

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \qquad X^\top X = \begin{pmatrix} 3 & 6 \\ 6 & 14 \end{pmatrix}, \qquad (X^\top X)^{-1} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}, \qquad X^\top y = \begin{pmatrix} 5 \\ 11 \end{pmatrix},$$

using $\det(X^\top X) = 42 - 36 = 6$. Hence

$$\hat{\beta}_{\mathrm{OLS}} = \frac{1}{6}\begin{pmatrix} 14 & -6 \\ -6 & 3 \end{pmatrix}\begin{pmatrix} 5 \\ 11 \end{pmatrix} = \frac{1}{6}\begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2/3 \\ 1/2 \end{pmatrix}:$$

intercept $2/3$, slope $1/2$. The univariate formula (5.3) confirms this: $\bar{x} = 2$, $\bar{y} = 5/3$, $S_{xy} = 1$, $S_{xx} = 2$, so $\hat{\beta} = 1/2$ and $\hat{\alpha} = 5/3 - 2/2 = 2/3$. The fitted values are $(7/6,\ 5/3,\ 13/6)$ and the residuals $\hat{\varepsilon} = (-1/6,\ 1/3,\ -1/6)$. Orthogonality checks out: $\sum_i \hat{\varepsilon}_i = 0$ and $\sum_i x_i\hat{\varepsilon}_i = -1/6 + 4/6 - 3/6 = 0$. The leverages (diagonal of H) are $h_{11} = 5/6$, $h_{22} = 1/3$, $h_{33} = 5/6$, summing to $2 = p$ as Theorem 5.2(iii) requires. Finally $S_{yy} = 2/3$ and $\mathrm{RSS} = 1/6$, so $R^2 = 1 - (1/6)/(2/3) = 3/4$, matching $\hat{\rho}^{\,2} = S_{xy}^2/(S_{xx}S_{yy}) = 1/(2 \cdot 2/3) = 3/4$.

5.2 The Gauss–Markov Theorem

Having derived the OLS estimator, we now ask in what sense it is optimal. The Gauss–Markov theorem gives the classical answer: under four assumptions — none of which is normality — OLS

has the smallest variance among all linear unbiased estimators. The proof is short and appears in interviews verbatim; the follow-up questions about broken assumptions appear even more often.

Setup. As before, $y = X\beta + \varepsilon$ with $X \in \mathbb{R}^{n\times p}$. The four Gauss–Markov assumptions, conditional on X, are:

(GM1) *Linearity / correct specification:* the model $y = X\beta + \varepsilon$ holds.

(GM2) *Full column rank:* $\operatorname{rank}(X) = p$.

(GM3) *Strict exogeneity:* $\mathbb{E}[\varepsilon \mid X] = 0$.

(GM4) *Spherical errors:* $\operatorname{Var}(\varepsilon \mid X) = \sigma^2 I$ — homoskedastic *and* uncorrelated.

Normality of the errors is *not* assumed. All expectations and variances below are conditional on X; we suppress the conditioning in the notation.

Definition 5.5 (Linear estimator and BLUE)**.** An estimator $\tilde\beta$ of β is *linear* if $\tilde\beta = Cy$ for some matrix $C \in \mathbb{R}^{p\times n}$ that may depend on X but not on y, and *unbiased* if $\mathbb{E}[\tilde\beta] = \beta$ for every $\beta \in \mathbb{R}^p$. A linear unbiased estimator $\hat\beta$ is *BLUE* (best linear unbiased estimator) if, for every competing linear unbiased $\tilde\beta$, the matrix $\operatorname{Var}(\tilde\beta) - \operatorname{Var}(\hat\beta)$ is positive semidefinite — equivalently, $\operatorname{Var}(\ell^\top\tilde\beta) \geq \operatorname{Var}(\ell^\top\hat\beta)$ for every linear combination $\ell \in \mathbb{R}^p$.

Theorem 5.6 (Gauss–Markov)**.** *Under (GM1)–(GM4), the OLS estimator $\hat\beta_{\mathrm{OLS}}$ is linear and unbiased, and it is BLUE: every linear unbiased estimator $\tilde\beta = Cy$ satisfies*

$$\operatorname{Var}(\tilde\beta) - \operatorname{Var}(\hat\beta_{\mathrm{OLS}}) = \sigma^2\Delta\Delta^\top \succeq 0, \qquad \text{where } \Delta = C - (X^\top X)^{-1}X^\top,$$

with equality if and only if $\tilde\beta = \hat\beta_{\mathrm{OLS}}$.

Proof. We decompose an arbitrary competitor as OLS plus a slack matrix that is orthogonal to the design, and show the slack can only add variance.

First, OLS is linear with $C_0 = (X^\top X)^{-1}X^\top$, and it is unbiased because $C_0X = I$ and (GM3) give $\mathbb{E}[\hat\beta_{\mathrm{OLS}}] = C_0X\beta + C_0\,\mathbb{E}[\varepsilon] = \beta$. Its variance is

$$\operatorname{Var}(\hat\beta_{\mathrm{OLS}}) = C_0\,\operatorname{Var}(\varepsilon)\,C_0^\top = \sigma^2 C_0C_0^\top = \sigma^2(X^\top X)^{-1},$$

using (GM4) and $C_0C_0^\top = (X^\top X)^{-1}X^\top X(X^\top X)^{-1} = (X^\top X)^{-1}$.

Now let $\tilde\beta = Cy$ be any linear estimator that is unbiased for every β. Then $\mathbb{E}[\tilde\beta] = CX\beta$ must equal β for all $\beta \in \mathbb{R}^p$, which forces $CX = I$. Write $C = C_0 + \Delta$; subtracting $C_0X = I$ from $CX = I$ gives

$$\Delta X = 0.$$

By (GM4), $\operatorname{Var}(\tilde\beta) = \sigma^2 CC^\top$, and expanding,

$$CC^\top = (C_0+\Delta)(C_0+\Delta)^\top = C_0C_0^\top + C_0\Delta^\top + \Delta C_0^\top + \Delta\Delta^\top.$$

The cross terms vanish because they contain ΔX: indeed $\Delta C_0^\top = \Delta X(X^\top X)^{-1} = 0$, and $C_0\Delta^\top$ is its transpose. Hence

$$\operatorname{Var}(\tilde\beta) = \sigma^2\big[(X^\top X)^{-1} + \Delta\Delta^\top\big] = \operatorname{Var}(\hat\beta_{\mathrm{OLS}}) + \sigma^2\Delta\Delta^\top.$$

The matrix $\Delta\Delta^\top$ is positive semidefinite, since $u^\top\Delta\Delta^\top u = \|\Delta^\top u\|^2 \geq 0$ for every u. Equality of the variance matrices holds if and only if $\Delta\Delta^\top = 0$, i.e. $\Delta = 0$, i.e. $\tilde\beta = \hat\beta_{\mathrm{OLS}}$. The scalar statement follows by sandwiching: $\ell^\top\big[\operatorname{Var}(\tilde\beta) - \operatorname{Var}(\hat\beta_{\mathrm{OLS}})\big]\ell \geq 0$ for every ℓ. □

(Problem P5.5 asks you to reproduce this proof from scratch.)

Intuition. One idea carries the whole proof: *any competitor equals OLS plus a piece orthogonal to the design, and that piece only adds variance.* Unbiasedness pins down the estimator's action on $\operatorname{col}(X)$ exactly; the only remaining freedom is slack in the orthogonal directions, which cannot help and strictly hurts unless it is zero.

Example 5.7 (Estimating a mean: which weights are best?)**.** The simplest instance of Theorem 5.6 takes $X = (1, 1, \ldots, 1)^\top$ and scalar $\beta = \mu$, so the model is $y_i = \mu + \varepsilon_i$ with uncorrelated errors of variance σ^2. Every linear estimator is a weighting $\tilde{\mu} = \sum_i w_i y_i$; unbiasedness for all μ forces $\sum_i w_i = 1$, and then $\mathrm{Var}(\tilde{\mu}) = \sigma^2 \sum_i w_i^2$. The OLS estimator here is the sample mean ($w_i = 1/n$). With $n = 4$ and $\sigma^2 = 1$: equal weights give variance $4 \times (1/4)^2 = 0.25$, while the unequal weighting $(0.4, 0.3, 0.2, 0.1)$ — still unbiased, since the weights sum to 1 — gives $0.16 + 0.09 + 0.04 + 0.01 = 0.30$, strictly worse. The Cauchy–Schwarz inequality shows equal weights are optimal in general: $n \sum_i w_i^2 \geq \left(\sum_i w_i\right)^2 = 1$, so $\mathrm{Var}(\tilde{\mu}) \geq \sigma^2/n$ with equality only at $w_i = 1/n$.

Remark 5.8 (What breaks, what it costs, what fixes it). Interviews rarely stop at the theorem; the follow-up is which assumption a given desk failure violates and what to do. The summary table:

Broken	**Consequence**	**Fix**
exogeneity (omitted variable, leakage)	bias + inconsistency (fatal)	add variable, IV, redesign
heteroskedasticity / autocorrelation	inefficient + wrong SEs	GLS/WLS (then BLUE); robust SEs
near-multicollinearity	variance explodes	ridge, drop/combine features
non-normality	only exact t/F affected	CLT asymptotics, bootstrap

Two gradations matter. A broken (GM3) poisons the estimate itself and no amount of data cures it. Broken (GM4) leaves OLS unbiased and consistent but inefficient, with wrong standard errors — an inference problem, not an estimation catastrophe. Near-multicollinearity violates nothing formally (only exact rank deficiency violates (GM2)) but makes $(X^\top X)^{-1}$ huge, so coefficient variances explode. Non-normality breaks none of (GM1)–(GM4): only exact finite-sample t and F distributions are lost.

The most dangerous violation — exogeneity — has a formula every candidate should be able to derive on demand.

Setup. Suppose the true model involves two centred regressors, $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$ with $\mathbb{E}[\varepsilon \mid x_1, x_2] = 0$, but the analyst regresses y on x_1 alone. Write $\widehat{\mathrm{Cov}}(x_1, x_2) = \frac{1}{n}\sum_i x_{1i}x_{2i}$ and $\widehat{\mathrm{Var}}(x_1) = \frac{1}{n}\sum_i x_{1i}^2 > 0$ for the sample moments of the fixed regressors.

Proposition 5.9 (Omitted-variable bias)**.** *The short-regression slope* $\hat{\beta}_1 = \sum_i x_{1i} y_i / \sum_i x_{1i}^2$ *satisfies*

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 + \beta_2 \frac{\widehat{\mathrm{Cov}}(x_1, x_2)}{\widehat{\mathrm{Var}}(x_1)}. \tag{5.4}$$

The bias is the omitted coefficient times the regression slope of the omitted variable on the included one; it vanishes only if $\beta_2 = 0$ *or the regressors are uncorrelated in sample.*

Proof. We substitute the true model into the estimator and take expectations. Substituting $y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \varepsilon_i$,

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{\sum_i x_{1i}x_{2i}}{\sum_i x_{1i}^2} + \frac{\sum_i x_{1i}\varepsilon_i}{\sum_i x_{1i}^2}.$$

Conditional on the regressors, the last term has expectation zero by exogeneity of the *true* model, and the middle ratio equals $\widehat{\mathrm{Cov}}(x_1, x_2)/\widehat{\mathrm{Var}}(x_1)$ after dividing through by n. Taking expectations gives (5.4). □

Example 5.10 (Phantom alpha, numerically)**.** Suppose the truth is $\beta_1 = 0.2$ (a weak genuine signal) and $\beta_2 = 1.5$ (a strong market effect), and in sample $\widehat{\mathrm{Cov}}(x_1, x_2)/\widehat{\mathrm{Var}}(x_1) = 0.4$. By (5.4), the short regression reports

$$\mathbb{E}[\hat{\beta}_1] = 0.2 + 1.5 \times 0.4 = 0.8$$

— four times the true coefficient. Three quarters of the reported "alpha" is the market loading in disguise, and collecting more data makes the estimate more precise around 0.8, not closer to 0.2.

Trading application. Omit market beta from a return regression while the signal under test correlates with the market, and the backtest rediscovers beta and calls it alpha — the classic way (5.4) bites in practice. The standard cure is residualisation: a systematic equity desk residualises returns against risk-model factors before evaluating signals, precisely so that whatever survives is orthogonal to the omitted-variable directions the risk model spans.

Interview trap. "You just proved OLS is best — so why does ridge regularly beat it out of sample?" Gauss–Markov crowns OLS best *among linear unbiased estimators*. Ridge is biased, so it competes outside that class, and the mean-squared-error decomposition MSE = bias^2 + variance lets it win: a small bias can buy a large variance reduction (Proposition 5.14 makes this exact). Under normality, OLS is best among *all* unbiased estimators — it attains the Cramér–Rao bound and equals the MLE — but it still loses to biased competitors on MSE.

5.3 Ridge Regression

Ridge regression adds an L2 penalty to the least-squares criterion. This buys three things, each proved below: an estimator that exists for *every* design (the mechanical fix for multicollinearity), a transparent shrinkage action, and — the punchline the trap above promised — a strictly smaller mean-squared error than OLS for a well-chosen penalty.

Setup. The data are $y \in \mathbb{R}^n$ and $X \in \mathbb{R}^{n\times p}$; full column rank is *not* assumed in this subsection (that is the point). For a penalty level $\lambda > 0$, the ridge criterion is

$$L_\lambda(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_2^2,$$

and we write $\hat\beta_\lambda$ for its minimiser. Because λ is reserved for the penalty throughout this section, the eigenvalues of the symmetric positive-semidefinite matrix $X^\top X$ are written $\gamma_1 \ge \gamma_2 \ge \cdots \ge \gamma_p \ge 0$ (renaming the document-wide λ_i).

Theorem 5.11 (Ridge estimator and guaranteed invertibility)**.** *For every $\lambda > 0$ and every design X — including rank-deficient designs and $p > n$ — the criterion L_λ is strictly convex with unique minimiser*

$$\hat\beta_\lambda = (X^\top X + \lambda I)^{-1} X^\top y, \tag{5.5}$$

the matrix $X^\top X + \lambda I$ being invertible because its eigenvalues are $\gamma_i + \lambda \ge \lambda > 0$.

Proof. We repeat the gradient–Hessian argument of Theorem 5.1 with the penalised criterion. The gradient is $\nabla_\beta L_\lambda = -2X^\top y + 2(X^\top X + \lambda I)\beta$ and the Hessian is $2(X^\top X + \lambda I)$. By the spectral theorem, the symmetric matrix $X^\top X$ has an orthonormal eigenbasis with eigenvalues $\gamma_i \ge 0$ (nonnegative because $u^\top X^\top X u = \|Xu\|^2 \ge 0$); adding λI shifts every eigenvalue to $\gamma_i + \lambda \ge \lambda > 0$. Hence the Hessian is positive definite regardless of the rank of X: the criterion is strictly convex, the matrix in (5.5) is invertible, and the unique stationary point (5.5) is the unique global minimiser. □

Proposition 5.12 (Orthonormal shrinkage)**.** *If the design is orthonormal, $X^\top X = I$, then*

$$\hat\beta_\lambda = \frac{\hat\beta_{\mathrm{OLS}}}{1+\lambda}:$$

every coefficient is shrunk toward zero by the common factor $1/(1+\lambda)$, and no coefficient is set exactly to zero unless its OLS value is already zero.

Proof. We substitute $X^\top X = I$ into (5.5): the estimator becomes $\hat\beta_\lambda = (I + \lambda I)^{-1} X^\top y = (1+\lambda)^{-1} X^\top y$, and in the orthonormal case (5.2) reduces to $\hat\beta_{\mathrm{OLS}} = X^\top y$. The factor $1/(1+\lambda)$ lies in $(0,1)$ for $\lambda > 0$ and multiplies each coordinate, so a coordinate vanishes exactly when its OLS value does. □

Example 5.13 (Shrinkage, numerically)**.** With an orthonormal design, an OLS coefficient of 0.9 and penalty $\lambda = 0.2$ give the ridge coefficient $0.9/1.2 = 0.75$. Doubling the penalty to $\lambda = 0.4$ gives $0.9/1.4 \approx 0.643$: more penalty, more shrinkage, but never an exact zero.

Proposition 5.14 (Some penalty strictly lowers MSE, orthonormal case)**.** *Assume the orthonormal design $X^\top X = I$ and errors with mean zero and variance $\sigma^2 I$, and let $\mathrm{MSE}(\lambda) = \mathbb{E}\,\|\hat\beta_\lambda - \beta\|^2$. Then*

$$\mathrm{MSE}(\lambda) = \frac{\lambda^2\|\beta\|^2 + p\sigma^2}{(1+\lambda)^2}, \tag{5.6}$$

so $\mathrm{MSE}(0) = p\sigma^2$ recovers OLS and $\mathrm{MSE}'(0) = -2p\sigma^2 < 0$: some $\lambda > 0$ strictly beats OLS. If $\beta \neq 0$, the MSE is uniquely minimised at

$$\lambda^* = \frac{p\sigma^2}{\|\beta\|^2}, \qquad \mathrm{MSE}(\lambda^*) < \mathrm{MSE}(0);$$

if $\beta = 0$, the MSE is strictly decreasing in λ, so any $\lambda > 0$ beats OLS.

Proof. We compute the bias–variance decomposition of $\hat\beta_\lambda$ and optimise over λ by elementary calculus. In the orthonormal case $\hat\beta_{\mathrm{OLS}} = X^\top y = \beta + X^\top\varepsilon$, with $\mathbb{E}[X^\top\varepsilon] = 0$ and $\mathrm{Var}(X^\top\varepsilon) = \sigma^2 X^\top X = \sigma^2 I$. By Proposition 5.12, $\hat\beta_\lambda = \hat\beta_{\mathrm{OLS}}/(1+\lambda)$, so

$$\mathbb{E}[\hat\beta_\lambda] - \beta = \frac{\beta}{1+\lambda} - \beta = -\frac{\lambda}{1+\lambda}\,\beta, \qquad \mathrm{Var}(\hat\beta_\lambda) = \frac{\sigma^2}{(1+\lambda)^2}\,I.$$

Summing squared bias and total variance,

$$\mathrm{MSE}(\lambda) = \left\|\tfrac{\lambda}{1+\lambda}\beta\right\|^2 + \operatorname{tr}\mathrm{Var}(\hat\beta_\lambda) = \frac{\lambda^2\|\beta\|^2 + p\sigma^2}{(1+\lambda)^2},$$

which is (5.6). Differentiating with the quotient rule and simplifying,

$$\mathrm{MSE}'(\lambda) = \frac{2(\lambda\|\beta\|^2 - p\sigma^2)}{(1+\lambda)^3}.$$

At $\lambda = 0$ this equals $-2p\sigma^2 < 0$, so the MSE initially decreases and some positive penalty strictly improves on OLS. If $\beta \neq 0$, the derivative is negative on $(0, \lambda^*)$ and positive on (λ^*, ∞) with $\lambda^* = p\sigma^2/\|\beta\|^2$, so λ^* is the unique minimiser and $\mathrm{MSE}(\lambda^*) < \mathrm{MSE}(0)$ because the MSE strictly decreases on $(0, \lambda^*]$. If $\beta = 0$, the derivative is negative for all $\lambda > 0$. □

Example 5.15 (Halving the MSE)**.** Take $p = 2$, $\beta = (1,1)^\top$, $\sigma^2 = 1$ in the orthonormal setting. Then OLS has $\mathrm{MSE}(0) = p\sigma^2 = 2$, the optimal penalty is $\lambda^* = 2/2 = 1$, and (5.6) gives

$$\mathrm{MSE}(1) = \frac{1\cdot 2 + 2}{(1+1)^2} = 1$$

— half the OLS error, split as squared bias $\frac{1}{4}\|\beta\|^2 = 0.5$ plus variance $\frac{1}{4}\,p\sigma^2 = 0.5$. The accepted small bias bought a variance reduction three times its size.

Fact 5.16 (Ridge improvement for general designs (Hoerl–Kennard, 1970))**.** The orthonormal assumption is a convenience, not the substance: for an arbitrary full-column-rank design and spherical errors, there always exists $\lambda > 0$ such that $\mathbb{E}\|\hat\beta_\lambda - \beta\|^2 < \mathbb{E}\|\hat\beta_{\mathrm{OLS}} - \beta\|^2$; the same one-sided derivative argument goes through eigenvalue by eigenvalue of $X^\top X$. A full proof is beyond our scope.

Remark 5.17 (Bayesian interpretation). Place the prior $\beta \sim \mathcal{N}(0, \tau^2 I)$ on the coefficients and take the Gaussian likelihood $y \mid \beta \sim \mathcal{N}(X\beta, \sigma^2 I)$. Minus twice the log-posterior is, up to constants, $\|y - X\beta\|^2/\sigma^2 + \|\beta\|^2/\tau^2$; multiplying through by σ^2 shows the MAP estimate is exactly ridge with $\lambda = \sigma^2/\tau^2$. A tight prior (small τ) means heavy shrinkage; an uninformative prior ($\tau \to \infty$) recovers OLS.

5.4 The Lasso

Replacing the squared-norm penalty by an absolute-value penalty changes the character of the solution entirely: instead of shrinking every coefficient a little, the lasso sets some coefficients exactly to zero. This subsection proves the soft-thresholding formula that makes the mechanism explicit and gives the geometric picture behind it.

Setup. For $\lambda > 0$ the lasso criterion is

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|_1, \qquad \|\beta\|_1 = \sum_{j=1}^{p} |\beta_j|.$$

The criterion is convex but not differentiable at coordinates equal to zero, and it has no closed-form minimiser in general. In the orthonormal case $X^\top X = I$, however, writing $z = X^\top y = \hat{\beta}_{\mathrm{OLS}}$ and expanding the squared norm,

$$L(\beta) = \|y\|^2 + \sum_{j=1}^{p} \Big[\beta_j^2 - 2\beta_j z_j + \lambda|\beta_j|\Big] = \text{const} + \sum_{j=1}^{p} \Big[(\beta_j - z_j)^2 + \lambda|\beta_j|\Big],$$

so the problem separates into p one-dimensional problems. We write $(a)_+ = \max(a, 0)$ and $\operatorname{sign}(z) \in \{-1, 0, 1\}$ for the sign function.

Theorem 5.18 (Soft-thresholding, orthonormal design)**.** *For any $z \in \mathbb{R}$ and $\lambda > 0$, the function $\psi_z(b) = (b - z)^2 + \lambda|b|$ has the unique minimiser*

$$b^*(z) = \operatorname{sign}(z)\,\big(|z| - \tfrac{\lambda}{2}\big)_+. \tag{5.7}$$

Consequently, in the orthonormal case the lasso solution is coordinatewise soft-thresholding of OLS:

$$\hat{\beta}_j^{\mathrm{lasso}} = \operatorname{sign}(\hat{\beta}_j^{\mathrm{OLS}})\,\big(|\hat{\beta}_j^{\mathrm{OLS}}| - \tfrac{\lambda}{2}\big)_+,$$

and every OLS coefficient with $|\hat{\beta}_j^{\mathrm{OLS}}| \le \lambda/2$ is set exactly to zero — the lasso performs variable selection.

Proof. We argue by case analysis on the sign of the minimiser, using strict convexity to guarantee uniqueness. The function ψ_z is strictly convex, being the sum of the strictly convex quadratic $(b - z)^2$ and the convex $\lambda|b|$; hence it has exactly one minimiser, and any point satisfying a local optimality condition is that minimiser. Three mutually exclusive possibilities cover it.

Case $b^* > 0$. On $(0, \infty)$ the function is differentiable with $\psi_z'(b) = 2(b - z) + \lambda$, so a stationary point requires $b = z - \lambda/2$, which is positive precisely when $z > \lambda/2$.

Case $b^* < 0$. Symmetrically, $\psi_z'(b) = 2(b - z) - \lambda$ on $(-\infty, 0)$, giving $b = z + \lambda/2$, which is negative precisely when $z < -\lambda/2$.

Case $b^* = 0$. The one-sided derivatives at zero are $\psi_z'(0^+) = -2z + \lambda$ and $\psi_z'(0^-) = -2z - \lambda$. A convex function is minimised at 0 if and only if the left derivative is ≤ 0 and the right derivative is ≥ 0, i.e.

$$-2z - \lambda \le 0 \le -2z + \lambda \iff |z| \le \tfrac{\lambda}{2}.$$

The three conditions $z > \lambda/2$, $z < -\lambda/2$, $|z| \le \lambda/2$ partition $\mathbb{R}$, and in each case the minimiser matches (5.7). The coordinatewise claim follows because the separated criterion is minimised by minimising each coordinate's term. □

(Problem P5.8 asks you to reproduce this derivation from scratch.)

Example 5.19 (Soft-thresholding in action)**.** Take $\lambda = 0.8$, so the threshold is $\lambda/2 = 0.4$, and orthonormal-design OLS coefficients $(1.6,\ 0.5,\ -0.2)$. Formula (5.7) gives

$$1.6 \mapsto 1.6 - 0.4 = 1.2, \qquad 0.5 \mapsto 0.5 - 0.4 = 0.1, \qquad -0.2 \mapsto 0 \quad (|-0.2| \le 0.4).$$

The third coefficient is *exactly* zero: the variable is dropped from the model. Ridge at any penalty would have returned three nonzero coefficients (Proposition 5.12).

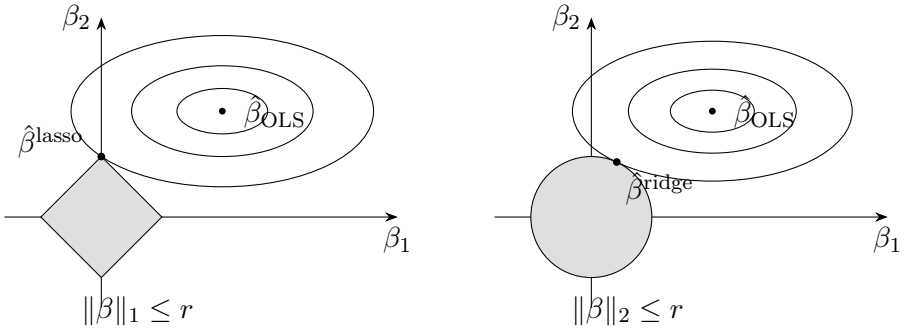


Figure 8: Constrained view of the two penalties for $p = 2$: minimise the quadratic loss (elliptical level sets centred at $\hat{\beta}_{\text{OLS}}$) subject to a norm budget r. Left: the L1 ball is a diamond with corners on the axes; the level sets typically first touch at a corner, where a coordinate is exactly zero. Right: the L2 ball is round, the contact point is generically off-axis, and both coordinates stay nonzero.

Intuition. Figure 8 is the standard geometric picture. The penalised problems are equivalent to minimising $\|y - X\beta\|^2$ subject to $\|\beta\|_1 \le r$ (lasso) or $\|\beta\|_2 \le r$ (ridge) for a budget r matched to λ. The elliptical level sets of the loss expand from $\hat{\beta}_{\text{OLS}}$ until they first touch the constraint set. The L1 diamond presents corners that sit exactly on the coordinate axes, and the expanding ellipse typically lands on a corner — exact zeros. The L2 ball has no corners, so the touching point generically has all coordinates nonzero: ridge shrinks, lasso selects.

Remark 5.20 (Bayesian view)*.* Just as ridge is the MAP estimate under a Gaussian prior (Remark 5.17), the lasso is the MAP estimate under independent Laplace (double-exponential) priors on the coefficients: minus the log of the Laplace density is proportional to $|\beta_j|$, producing the L1 penalty. The Laplace prior's sharp peak at zero is the Bayesian face of variable selection.

5.5 Inverse Regression, Beta and Hedging

Regressing Y on X and regressing X on Y do not give reciprocal slopes, and confusing the two is a classic interview trap with direct hedging consequences. The identity that governs the relationship follows in one line from Corollary 5.3.

Setup. Let the pair (X, Y) have standard deviations $\sigma_X, \sigma_Y > 0$ and correlation ρ (population moments, or sample moments used consistently). By Corollary 5.3, the slope of the regression of Y on X and the slope of the regression of X on Y are

$$B = \rho\,\frac{\sigma_Y}{\sigma_X}, \qquad D = \rho\,\frac{\sigma_X}{\sigma_Y}.$$

Proposition 5.21 (Inverse-regression identity)**.** *The two slopes satisfy*

$$BD = \rho^2 \le 1.$$

Consequently, when $\rho \neq 0$*,*

$$D = \frac{\rho^2}{B} \neq \frac{1}{B} \quad \textit{unless } |\rho| = 1 :$$

the naive reciprocal $1/B$ *overstates the true inverse slope by the factor* $1/\rho^2$*.*

Proof. We multiply the two slope formulas: the standard-deviation ratios cancel, leaving $BD = \rho^2$, and $\rho^2 \le 1$ is the Cauchy–Schwarz inequality for correlations. When $\rho \neq 0$, both slopes are nonzero and division gives $D = \rho^2/B$; this equals $1/B$ precisely when $\rho^2 = 1$, i.e. when the variables are perfectly linearly related, in which case both regression lines coincide. □

Example 5.22 (The factor-of-four error)**.** Take $\rho = 0.5$ and $\sigma_X = \sigma_Y$. Then $B = D = 0.5$, while the naive inversion quotes $1/B = 2$ — overstating the true slope $D = 0.5$ by the factor $1/\rho^2 = 4$. The two correct regression lines are far from being mirror images of each other.

Intuition. Regressing Y on X minimises *vertical* distances to the line; regressing X on Y minimises *horizontal* ones. These are different projections, and each slope is attenuated toward zero by a factor ρ relative to the standard-deviation ratio: noise in the regressor always pulls the fitted slope toward flatness, in whichever direction the regression is run.

Trading application. Pair trading: the hedge ratio depends on which asset is regressed on which. Taking the slope from the convenient regression and inverting it systematically over- or under-hedges the book by the factor $1/\rho^2$; with a typical pair correlation of 0.7, that is a factor of two.

Remark 5.23 (Beta versus correlation). The regression slope $\beta = \rho\,\sigma_Y/\sigma_X$ mixes two ingredients that should be kept mentally separate. Correlation is normalised: it measures direction and tightness of the linear relationship and is dimensionless. Beta additionally carries *relative volatility*: with $\rho \approx 1$ but Y twice as volatile as X, the beta is 2. Conversely a large beta can coexist with a weak correlation if σ_Y/σ_X is large — the relationship is loose, but when Y moves, it moves a lot.

Interview trap. "Which do you hedge with — beta or correlation?" The *beta*: hedging neutralises profit-and-loss magnitude, and magnitude lives in the ratio σ_Y/σ_X. Correlation tells you whether a hedge is worth doing at all (how much residual variance will remain); beta tells you how large to make it. Sizing a hedge by correlation alone ignores the volatility ratio and mis-sizes whenever the two assets have different volatilities.

5.6 Weighted Least Squares and Whitening

When error variances differ across observations, assumption (GM4) fails and OLS, while still unbiased, is no longer efficient. The remedy is to whiten: rescale each observation by its noise level and run OLS on the transformed data. The optimality of the result is an immediate corollary of Gauss–Markov, which is the point of this short subsection.

Setup. Consider $y = X\beta + \varepsilon$ with X of full column rank, $\mathbb{E}[\varepsilon \mid X] = 0$, and *heteroskedastic but uncorrelated* errors: $\mathrm{Var}(\varepsilon \mid X) = \mathrm{diag}(\sigma_1^2, \dots, \sigma_n^2)$ with all $\sigma_i^2 > 0$ known. Let $W = \mathrm{diag}(1/\sigma_1, \dots, 1/\sigma_n)$, and define the whitened data $\tilde{y} = Wy$, $\tilde{X} = WX$: row i of the whole observation is divided by σ_i. The weighted least squares (WLS) estimator is

$$\hat{\beta}_{\mathrm{WLS}} = \arg\min_{\beta} \sum_{i=1}^{n} \frac{\left(y_i - x_i^\top \beta\right)^2}{\sigma_i^2},$$

where $x_i^\top$ is the i-th row of X — OLS on the whitened data, with weights $w_i = 1/\sigma_i^2$.

Theorem 5.24 (WLS is BLUE after whitening)**.** *Under the setup above, the whitened model $\tilde{y} = \tilde{X}\beta + \tilde{\varepsilon}$ with $\tilde{\varepsilon} = W\varepsilon$ satisfies (GM1)–(GM4) with error variance* 1*, and*

$$\hat{\beta}_{\mathrm{WLS}} = (\tilde{X}^\top \tilde{X})^{-1} \tilde{X}^\top \tilde{y}$$

is BLUE among all estimators linear in y and unbiased for β.

Proof. We transform to a homoskedastic model and invoke Theorem 5.6. Multiplying the model by W gives $\tilde{y} = \tilde{X}\beta + \tilde{\varepsilon}$, which is (GM1) for the whitened data. The matrix W is invertible, so

$\tilde{X} = WX$ retains full column rank (GM2). Exogeneity passes through: $\mathbb{E}[\tilde{\varepsilon} \mid X] = W\,\mathbb{E}[\varepsilon \mid X] = 0$ (GM3). Finally

$$\mathrm{Var}(\tilde{\varepsilon} \mid X) = W\,\mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)\,W^\top = I,$$

which is (GM4) with unit variance. The whitened least-squares criterion is

$$\|\tilde{y} - \tilde{X}\beta\|^2 = \sum_{i=1}^{n} \frac{(y_i - x_i^\top \beta)^2}{\sigma_i^2},$$

so its minimiser is exactly $\hat{\beta}_{\mathrm{WLS}}$, with closed form given by Theorem 5.1 applied to $(\tilde{y}, \tilde{X})$. Theorem 5.6 now says this estimator is BLUE among estimators linear in $\tilde{y}$ and unbiased. Because W is invertible, an estimator is linear in $\tilde{y}$ if and only if it is linear in $y = W^{-1}\tilde{y}$, so the two competitor classes coincide, and the BLUE property holds in the original model as claimed. □

Example 5.25 (Precision weighting two measurements)**.** Measure the same unknown μ twice with independent noise: $x_1 = \mu + \varepsilon_1$ with $\mathrm{sd}(\varepsilon_1) = 1$, and $x_2 = \mu + \varepsilon_2$ with $\mathrm{sd}(\varepsilon_2) = 3$. The WLS weights are proportional to $1/\sigma_i^2$, i.e. 1 and 1/9, so

$$\hat{\mu} = \frac{9x_1 + x_2}{10}, \qquad \mathrm{Var}(\hat{\mu}) = \frac{81 \cdot 1 + 1 \cdot 9}{100} = 0.9.$$

Even the noisy measurement helps: variance 0.9 beats the 1.0 of using the good measurement alone. Equal weighting, by contrast, gives $\mathrm{Var}\left((x_1 + x_2)/2\right) = (1 + 9)/4 = 2.5$ — worse than discarding the noisy measurement altogether. Weighting by precision is not a refinement; it is the difference between the noisy observation helping and hurting.

Trading application. Market data are heteroskedastic by regime: residual variance in a returns regression can be several times larger in high-volatility periods. Unweighted OLS lets those crisis observations dominate the fit, because the criterion counts squared errors and crises produce the largest ones. Down-weighting by inverse variance restores efficiency, and heteroskedasticity-robust standard errors repair the inference even when the weights are left alone (Remark 5.8).

Remark 5.26 (From WLS to GLS). Nothing in the proof of Theorem 5.24 needs the covariance to be diagonal. With a general known $\mathrm{Var}(\varepsilon \mid X) = \sigma^2\Omega$, Ω positive definite, whitening with $\Omega^{-1/2}$ in place of W produces a model satisfying (GM1)–(GM4), and the same argument shows the generalised least squares (GLS) estimator $(X^\top\Omega^{-1}X)^{-1}X^\top\Omega^{-1}y$ is BLUE — covering autocorrelated as well as heteroskedastic errors.

Remark 5.27 (Residual weighting is boosting, not GLS). A superficially similar move points in the opposite direction. Fitting a model A, then training a second model B with sample weights $w_i = |y_i - \hat{y}_{A,i}|$, makes B specialise on A's errors — the *boosting* idea in disguise. Safe names for the scheme: boosting-style reweighting, residual fitting, cascading. Avoid calling it "stacking", which properly means training a meta-learner on the models' *outputs*. Distinguish the two weightings sharply: GLS *down*-weights noisy observations to gain efficiency for one model; residual weighting *up*-weights hard observations so a second model specialises. One fights noise, the other chases signal — and the residual-weighting scheme backfires exactly when the "hard" points are hard because they are noise.

5.7 The Dart Problem: Squared Loss, Absolute Loss and Where to Aim

A single physical puzzle packages the deepest fact of this section — that the choice of loss function decides *what* is being estimated. Squared loss points at the mean, absolute loss at the median; a dart game with Gaussian versus Laplace scatter turns the abstraction into an aiming decision, complete with a bifurcation.

Setup. A dart aimed at the point $a \in \mathbb{R}$ lands at $a + Z$, where the scatter Z has density proportional to $\exp(-|z|^p/(2\sigma^p))$: the Gaussian case is $p = 2$ (standard deviation σ) and the

Laplace case is $p = 1$. Two targets sit at $x_1 < x_2$ with separation $d = x_2 - x_1$ and midpoint $m = (x_1 + x_2)/2$. The expected score is proportional to the landing density evaluated at the targets,

$$S(a) = f(x_1 - a) + f(x_2 - a),$$

where f is the scatter density; the question is which aim a maximises S. Before the game itself, we record the two estimation lemmas the interview follow-up is really about.

Lemma 5.28 (Squared loss selects the mean)**.** *Let X be a random variable with $\mathbb{E}[X^2] < \infty$. The function $c \mapsto \mathbb{E}\big[(X - c)^2\big]$ is minimised uniquely at $c = \mathbb{E}[X]$, with minimum value $\mathrm{Var}(X)$.*

Proof. We complete the square around the mean. Adding and subtracting $\mathbb{E}[X]$,

$$\mathbb{E}\big[(X - c)^2\big] = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] + \big(\mathbb{E}[X] - c\big)^2 = \mathrm{Var}(X) + \big(\mathbb{E}[X] - c\big)^2,$$

the cross term vanishing because $\mathbb{E}\big[X - \mathbb{E}[X]\big] = 0$. The second term is nonnegative and zero only at $c = \mathbb{E}[X]$. □

Lemma 5.29 (Absolute loss selects the median)**.** *Let X have $\mathbb{E}|X| < \infty$ and CDF F.*

(i) If F is continuous, then $g(c) = \mathbb{E}|X - c|$ is convex and is minimised exactly at the points c with $F(c) = 1/2$, i.e. at the medians of X.

(ii) For the two-point empirical distribution placing mass $1/2$ at each of $x_1 < x_2$, the function g is constant, equal to $d/2$, on the whole segment $[x_1, x_2]$, and strictly larger outside: every *point of the segment is an optimal aim under absolute loss.*

Proof. We differentiate under the integral sign for (i) and compute directly for (ii).

(i) Convexity holds because $c \mapsto |x - c|$ is convex for each fixed x and expectation preserves convexity. Splitting the expectation at c,

$$g(c) = \int_{-\infty}^{c} (c - x)\, dF(x) + \int_{c}^{\infty} (x - c)\, dF(x),$$

and differentiating each term (the boundary contributions vanish because the integrand is zero at $x = c$),

$$g'(c) = F(c) - \big(1 - F(c)\big) = 2F(c) - 1.$$

A convex function is minimised exactly where its derivative changes sign from ≤ 0 to ≥ 0; since F is nondecreasing and continuous, that set is precisely $\{c : F(c) = 1/2\}$, which is nonempty by the intermediate value theorem.

(ii) Here $g(c) = \frac{1}{2}\big(|x_1 - c| + |x_2 - c|\big)$. For $c \in [x_1, x_2]$ the two distances sum to the fixed separation: $g(c) = \frac{1}{2}\big[(c - x_1) + (x_2 - c)\big] = d/2$. For $c < x_1$, both distances grow by $x_1 - c$ compared to $c = x_1$, giving $g(c) = d/2 + (x_1 - c) > d/2$; the case $c > x_2$ is symmetric. □

With the estimation story in hand — L2 points at the mean, L1 at the median — we can analyse the game, where the answers are close cousins of the lemmas but not identical to them.

Proposition 5.30 (Gaussian dart and the pitchfork at $d = 2\sigma$)**.** *Let $p = 2$, so that*

$$S(a) = e^{-(a - x_1)^2/(2\sigma^2)} + e^{-(a - x_2)^2/(2\sigma^2)},$$

and set $\kappa = d^2/(4\sigma^2)$. Then:

(i) the midpoint m is a critical point of S for every separation d;

(ii) if $d \le 2\sigma$ (i.e. $\kappa \le 1$), the midpoint is the unique global maximiser: aim at the mean of the targets;

(iii) if $d > 2\sigma$ (i.e. $\kappa > 1$), the midpoint is a local minimum*, and S has exactly two global maximisers $m \pm u^*$, where $u^* = (d/2)v^*$ and $v^* \in (0,1)$ is the unique positive solution of*

$$\operatorname{artanh}(v) = \kappa v. \tag{5.8}$$

The optimal aims lie strictly between the midpoint and the targets and approach the targets as $d/\sigma \to \infty$.

Proof. We centre coordinates at the midpoint, characterise all critical points by a transcendental equation, and count its roots; the technique is the standard normal-form analysis of a pitchfork bifurcation.

Step 1 (reduction). Substituting $u = a - m$ places the targets at $\pm d/2$:

$$S(u) = e^{-(u+d/2)^2/(2\sigma^2)} + e^{-(u-d/2)^2/(2\sigma^2)},$$

an even, positive, smooth function of u with $S(u) \to 0$ as $|u| \to \infty$.

Step 2 (critical points lie between the targets). Differentiating,

$$S'(u) = -\frac{1}{\sigma^2}\Big[(u + \tfrac{d}{2})\, e^{-(u+d/2)^2/(2\sigma^2)} + (u - \tfrac{d}{2})\, e^{-(u-d/2)^2/(2\sigma^2)}\Big].$$

At $u = 0$ the two bracketed terms are $(d/2)e^{-d^2/(8\sigma^2)}$ and $-(d/2)e^{-d^2/(8\sigma^2)}$, which cancel: the midpoint is critical for every d, proving (i). For $u \geq d/2$, both factors $u + d/2 > 0$ and $u - d/2 \geq 0$ are nonnegative and the first is strictly positive, so $S'(u) < 0$; by evenness $S'(u) > 0$ for $u \leq -d/2$. Hence every critical point satisfies $|u| < d/2$: candidate aims lie strictly between the targets.

Step 3 (the transcendental characterisation). For $|u| < d/2$, setting $S'(u) = 0$ and moving one term across,

$$(\tfrac{d}{2} + u)\, e^{-(u+d/2)^2/(2\sigma^2)} = (\tfrac{d}{2} - u)\, e^{-(u-d/2)^2/(2\sigma^2)},$$

where both prefactors are positive. Dividing and using $(u + d/2)^2 - (u - d/2)^2 = 2du$,

$$\frac{d/2 + u}{d/2 - u} = \exp\Big(\frac{du}{\sigma^2}\Big).$$

Taking logarithms, halving, and substituting $v = 2u/d \in (-1, 1)$ with the identity $\operatorname{artanh}(v) = \frac{1}{2}\log\frac{1+v}{1-v}$ turns the condition into exactly (5.8), $\operatorname{artanh}(v) = \kappa v$.

Step 4 (counting roots). Let $h(v) = \operatorname{artanh}(v) - \kappa v$ on $(-1, 1)$; h is odd with $h(0) = 0$, and $h'(v) = \frac{1}{1-v^2} - \kappa$, where $1/(1 - v^2)$ increases strictly from 1 at $v = 0$ to $+\infty$ as $v \to 1^-$. If $\kappa \leq 1$, then $h'(v) \geq 1 - \kappa \geq 0$ with equality at most at $v = 0$, so h is strictly increasing and $v = 0$ is its only zero: the midpoint is the only critical point. If $\kappa > 1$, then $h' < 0$ near 0 and $h' > 0$ near 1, with h' strictly increasing; hence h strictly decreases to a negative minimum and then strictly increases to $+\infty$, so it has exactly one zero $v^* \in (0, 1)$, plus the symmetric $-v^*$: three critical points $0, \pm u^*$ with $u^* = (d/2)v^*$.

Step 5 (classification). Because S is continuous, positive, and vanishes at infinity, its supremum is attained, necessarily at a critical point. For $\kappa \leq 1$ the only critical point is $u = 0$, proving (ii). For $\kappa > 1$, differentiating once more and evaluating at the midpoint gives

$$S''(0) = \frac{2}{\sigma^2}\,(\kappa - 1)\, e^{-d^2/(8\sigma^2)} > 0,$$

so the midpoint is a strict local minimum, and the maximum is attained at the remaining critical points $\pm u^*$, which carry equal values by evenness: exactly two global maximisers, strictly between midpoint and targets by Step 2. Finally, as $\kappa \to \infty$ the solution of (5.8) satisfies $v^* \to 1$ (for fixed $v < 1$, κv eventually exceeds $\operatorname{artanh}(v)$), so the optimal aims approach the targets. □

(Problem P5.16 asks you to reproduce the core of this analysis.)

Example 5.31 (Aiming numbers for a supercritical separation)**.** Take $\sigma = 1$ and targets at 0 and 2.5, so $d = 2.5 > 2\sigma$ and $\kappa = 2.5^2/4 = 1.5625$. Solving (5.8) numerically: at $v = 0.88$, $\operatorname{artanh}(0.88) = 1.3758$ against $\kappa v = 1.3750$, so $v^* \approx 0.880$ and $u^* = 1.25 \times 0.880 = 1.10$. The optimal aims are $m \pm u^* = 1.25 \pm 1.10$, i.e. 0.15 and 2.35 — about 0.15 *inside* each target. The scores confirm the ranking:

$$S(0.15) = e^{-0.15^2/2} + e^{-2.35^2/2} \approx 0.9888 + 0.0632 = 1.0520,$$

against $S(0) = 1 + e^{-3.125} \approx 1.0439$ at a target and $S(1.25) = 2e^{-0.78125} \approx 0.9157$ at the midpoint. Aim slightly inside a target: better than the target itself, and much better than the middle. For a subcritical separation, say $d = 1.5$ ($\kappa = 0.5625 \le 1$), part (ii) applies and the unique optimum is the midpoint 0.75.

Proposition 5.32 (Laplace dart: aim at a target)**.** *Let $p = 1$ and measure distances in units of the Laplace scale, so a target at distance r contributes e^{-r} and*

$$S(a) = e^{-|a-x_1|} + e^{-|a-x_2|}.$$

Then S is maximised exactly at the two targets $a \in \{x_1, x_2\}$, and the endpoint beats the midpoint by

$$S(x_1) - S(m) = \left(1 - e^{-d/2}\right)^2 > 0 \quad \textit{for every } d > 0. \tag{5.9}$$

Proof. We show monotonicity outside the segment $[x_1, x_2]$ and strict convexity inside it; the maximum of a convex function on an interval sits at an endpoint. Without loss of generality $x_1 = 0$ and $x_2 = d$.

For $a \le 0$, both distances shrink as a increases: $S(a) = e^a + e^{a-d} = e^a(1 + e^{-d})$, strictly increasing, so $S(a) < S(0)$ for $a < 0$; symmetrically $S(a) < S(d)$ for $a > d$.

For $a \in [0, d]$, the score is $S(a) = e^{-a} + e^{a-d}$, with $S''(a) = e^{-a} + e^{a-d} > 0$: strictly convex. A strictly convex function on $[0, d]$ satisfies $S(a) < \max\{S(0), S(d)\}$ for every interior a, and here the endpoint values are equal, $S(0) = S(d) = 1 + e^{-d}$. Hence the global maximisers are exactly the two targets. The interior stationary point, from $S'(a) = -e^{-a} + e^{a-d} = 0$, is the midpoint $a = d/2$, where $S(d/2) = 2e^{-d/2}$ — the *worst* aim on the segment. The gap is

$$S(0) - S(d/2) = 1 + e^{-d} - 2e^{-d/2} = \left(1 - e^{-d/2}\right)^2,$$

a perfect square, strictly positive for all $d > 0$. □

(Problem P5.17 asks you to reproduce both the median characterisation of Lemma 5.29 and this endpoint computation.)

Example 5.33 (Laplace aiming numbers)**.** With separation $d = 2$ scale units, aiming at a target scores $S(x_1) = 1 + e^{-2} \approx 1.1353$, while the midpoint scores $S(m) = 2e^{-1} \approx 0.7358$. The gap is $(1 - e^{-1})^2 \approx 0.632^2 \approx 0.400$, matching (5.9): with Laplace tails, committing to one target beats hedging between them by a wide margin.

Intuition. The estimation lemmas and the game agree in spirit but not in letter, and the contrast is the insight. Under absolute loss, *estimation* is indifferent across the whole segment between the targets (Lemma 5.29(ii)); the Laplace *game* flips to the extreme points, because maximising a density sum rewards concentration — the Laplace peak is sharp, so being exactly on one target buys more than being mediocre for both. The Gaussian's flat top around its centre rewards compromise, which is why the midpoint (the mean, per Lemma 5.28) wins — but only until the targets separate past $d = 2\sigma$, where the compromise collapses in a pitchfork and the optimum breaks symmetry toward one target.

Interview trap. "Does this remind you of anything in regression?" The expected mapping: **L2 loss** $\leftrightarrow$ conditional *mean*, outlier-sensitive (the square in the Gaussian exponent is the squared loss of OLS); **L1 loss** $\leftrightarrow$ conditional *median*, robust (quantile regression at the median). Then keep *losses* distinct from *penalties*: losses (L2 versus L1 on the residuals) decide what functional you predict — mean versus median; penalties (ridge versus lasso on the coefficients) decide how estimates behave — shrinkage versus selection. Conflating the two families because both carry the labels "L1/L2" is a known trap.

Practice: problems P5.1–P5.17 in §27, solutions in §28.

Part III
Linear Algebra and Calculus

6 Linear Algebra, PCA and Covariance

6.1 Symmetric Matrices, Eigenvalues and Positive Semidefiniteness

This subsection assembles the spectral toolkit on which the rest of the section runs: the diagonalisation of symmetric matrices, the notion of positive semidefiniteness, and the trace and determinant identities that let a hand-computed eigensystem be checked in seconds. Interviewers use exactly these facts as rapid-fire warm-ups before the PCA derivation.

Setup. Throughout this subsection, vectors $x, y \in \mathbb{R}^p$ are columns, $A \in \mathbb{R}^{p\times p}$ is a symmetric matrix ($A = A^\top$), and $\|x\| = \sqrt{x^\top x}$ denotes the Euclidean norm. The standard basis vectors are $e_1, \dots, e_p$.

Definition 6.1 (Quadratic form; positive semidefiniteness)**.** The *quadratic form* of a symmetric matrix A is the map $x \mapsto x^\top Ax = \sum_{i,j} A_{ij}x_i x_j$. The matrix A is *positive semidefinite* (PSD) if $x^\top Ax \geq 0$ for every $x \in \mathbb{R}^p$, and *positive definite* (PD) if $x^\top Ax > 0$ for every $x \neq 0$.

Fact 6.2 (Spectral theorem)**.** Every symmetric matrix $A \in \mathbb{R}^{p\times p}$ has p real eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$ (listed with multiplicity) and an orthonormal basis of corresponding eigenvectors $q_1, \dots, q_p$. Equivalently,

$$A = Q\Lambda Q^\top, \qquad Q = (q_1 \mid \cdots \mid q_p), \quad Q^\top Q = I, \quad \Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_p).$$

This is the structural fact the entire section leans on. A proof is beyond our scope.

Proposition 6.3 (Trace and determinant read off the eigenvalues)**.** *For a symmetric matrix A with eigenvalues $\lambda_1, \dots, \lambda_p$,*

$$\operatorname{tr} A = \sum_{i=1}^{p} \lambda_i, \qquad \det A = \prod_{i=1}^{p} \lambda_i.$$

Proof. We exploit the invariance of trace and determinant under the factorisation of Fact 6.2. First, for any matrices B, C of compatible sizes, $\operatorname{tr}(BC) = \sum_i \sum_j B_{ij}C_{ji} = \sum_j \sum_i C_{ji}B_{ij} = \operatorname{tr}(CB)$. Applying this with $B = Q\Lambda$ and $C = Q^\top$ gives

$$\operatorname{tr} A = \operatorname{tr}(Q\Lambda Q^\top) = \operatorname{tr}(\Lambda Q^\top Q) = \operatorname{tr}\Lambda = \sum_i \lambda_i.$$

For the determinant, multiplicativity gives $\det A = \det Q \cdot \det\Lambda \cdot \det Q^\top = (\det Q)^2 \det\Lambda$, and $Q^\top Q = I$ forces $(\det Q)^2 = 1$, so $\det A = \det\Lambda = \prod_i \lambda_i$. □

Intuition. These two identities are the fastest sanity checks available: after any hand computation of an eigensystem, confirm that the eigenvalues sum to the trace and multiply to the determinant. A mismatch localises the arithmetic error immediately.

Theorem 6.4 (PSD equivalences)**.** *Let A be symmetric with eigenvalues $\lambda_1, \dots, \lambda_p$. Then A is positive semidefinite if and only if $\lambda_i \geq 0$ for all i, and positive definite if and only if $\lambda_i > 0$ for all i.*

Proof. We expand the quadratic form in the eigenbasis furnished by Fact 6.2. Write $x = \sum_i c_i q_i$ with $c_i = q_i^\top x$; orthonormality gives $\|x\|^2 = \sum_i c_i^2$ and

$$x^\top A x = \sum_{i=1}^{p} \lambda_i c_i^2. \tag{6.1}$$

If every $\lambda_i \geq 0$, then (6.1) is a sum of nonnegative terms, so $x^\top A x \geq 0$ for all x. Conversely, if A is PSD, take $x = q_j$: then $c_i = \mathbf{1}_{\{i=j\}}$ and $q_j^\top A q_j = \lambda_j \geq 0$. For the definite case: if every $\lambda_i > 0$ and $x \neq 0$, some $c_j \neq 0$, so the sum in (6.1) is strictly positive; conversely $x = q_j$ gives $\lambda_j = q_j^\top A q_j > 0$. □

Intuition. In the eigenbasis the quadratic form is a weighted sum of squared coordinates, each weighted by an eigenvalue. For a PD matrix the level set $\{x : x^\top A x = 1\}$ is an ellipsoid whose axes point along the eigenvectors, with semi-axis length $1/\sqrt{\lambda_i}$ along q_i: large eigenvalues squeeze the ellipsoid, small ones stretch it.

Example 6.5 (The two-by-two staple)**.** Consider

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

The characteristic polynomial is $(2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 3)(\lambda - 1)$, so $\lambda_1 = 3$ and $\lambda_2 = 1$, with eigenvectors $q_1 = (1,1)^\top/\sqrt{2}$ and $q_2 = (1,-1)^\top/\sqrt{2}$: indeed $A(1,1)^\top = (3,3)^\top$ and $A(1,-1)^\top = (1,-1)^\top$. Both eigenvalues are positive, so A is positive definite by Theorem 6.4. The checks of Proposition 6.3 pass: $\operatorname{tr} A = 4 = 3 + 1$ and $\det A = 4 - 1 = 3 = 3 \cdot 1$. A shortcut worth memorising: a symmetric 2×2 matrix with equal diagonal entries d and off-diagonal c has eigenvalues $d \pm c$ with eigenvectors $(1, \pm 1)^\top$.

6.2 Covariance Matrices Are Positive Semidefinite

The single most quoted one-liner of quant linear algebra: every covariance matrix is PSD, because a quadratic form in a covariance matrix is a variance. This subsection proves it, then explains why matrices *estimated* entrywise in practice can nevertheless fail the property, and what to do about it.

Setup. Let $Z = (Z_1, \dots, Z_p)^\top$ be a random vector with finite second moments, mean vector $\mu = \mathbb{E}[Z]$, and covariance matrix

$$\Sigma = \mathbb{E}\big[(Z - \mu)(Z - \mu)^\top\big], \qquad \Sigma_{ij} = \operatorname{Cov}(Z_i, Z_j).$$

Theorem 6.6 (Covariance matrices are PSD)**.** *Every covariance matrix Σ is symmetric positive semidefinite.*

Proof. We evaluate the quadratic form and recognise it as a variance. Symmetry is immediate from $\operatorname{Cov}(Z_i, Z_j) = \operatorname{Cov}(Z_j, Z_i)$. For any fixed $x \in \mathbb{R}^p$, linearity of expectation lets the (deterministic) vector x pass inside:

$$x^\top \Sigma x = \mathbb{E}\big[x^\top (Z - \mu)(Z - \mu)^\top x\big] = \mathbb{E}\big[\big(x^\top (Z - \mu)\big)^2\big] = \operatorname{Var}(x^\top Z) \geq 0.$$

By Theorem 6.4, all eigenvalues of Σ are nonnegative. □

Interview trap. "Prove that a covariance matrix is positive semidefinite" is a standard interview opener, and the expected answer is exactly the one-line identity $x^\top \Sigma x = \mathrm{Var}(x^\top Z) \geq 0$: the quadratic form is the variance of the portfolio with weights x, and variances cannot be negative. Produce it without hesitation; attempts to argue entrywise (for example from the signs or sizes of individual correlations) stall.

Example 6.7 (Portfolio-variance check)**.** Let $Z = (Z_1, Z_2)^\top$ with $\mathrm{Var}(Z_1) = \mathrm{Var}(Z_2) = 2$ and $\mathrm{Cov}(Z_1, Z_2) = 1$, so that Σ is the matrix of Example 6.5. The spread portfolio $x = (1, -1)^\top$ has

$$\mathrm{Var}(Z_1 - Z_2) = x^\top \Sigma x = 2 - 2(1) + 2 = 2 \geq 0.$$

The eigen route agrees: x is proportional to the eigenvector q_2, so $x^\top \Sigma x = \lambda_2 \|x\|^2 = 1 \cdot 2 = 2$.

Remark 6.8 (Why a sample covariance can fail PSD in practice). Theorem 6.6 covers the population covariance, and also any sample covariance computed from a *single* complete panel: with a centred data matrix X (rows are observations), the estimate $\hat{\Sigma} = \frac{1}{n} X^\top X$ satisfies $x^\top \hat{\Sigma} x = \|Xx\|^2/n \geq 0$. The guarantee breaks when the entries are estimated *pairwise on different samples* — asynchronous data across assets, or pairwise-complete observations — because then no single dataset generates all the entries at once: the single-quadratic-form structure underlying the proof is gone, and the assembled matrix can have negative eigenvalues. Three standard repairs:

1. *Shrinkage / regularisation:* blend the estimate toward a PSD target, in the simplest form by adding a small positive multiple of the identity;
2. *Eigenvalue clipping:* floor the negative eigenvalues at zero and renormalise (for a correlation matrix, rescale the diagonal back to one);
3. *Factor structure:* impose a low-rank-plus-diagonal model, which is PSD by construction.

Note the rhyme with ridge regression: adding a positive multiple of I is the same medicine, there applied to a near-singular $X^\top X$.

Interview trap. Feasibility itself can be the question: with $\rho_{AB} = 0.9$ and $\rho_{AC} = -0.9$, no joint distribution allows $\rho_{BC} = 0.8$ — the PSD constraint forces $\rho_{BC} \in [-1, -0.62]$. A risk report can still display such a matrix if its entries were estimated pairwise, which is precisely the failure mode of Remark 6.8. Problem P6.4 works this through in full.

6.3 Weighted Inner Products and A-Orthogonality

Ordinary orthogonality is orthogonality with respect to the identity; timed online assessments ⧫ probe the general weighted version, in which a symmetric positive definite matrix reweights the whole geometry. This subsection defines the weighted inner product, derives the matrix criterion for maps that preserve it, and classifies the standard candidates for a concrete diagonal weight matrix.

Setup. Let $A \in \mathbb{R}^{p \times p}$ be symmetric positive definite (Definition 6.1), and let $x, y \in \mathbb{R}^p$.

Definition 6.9 (A-inner product; A-orthogonal matrix)**.** The *A-inner product* of x and y is

$$\langle x, y \rangle_A = x^\top A y.$$

A matrix $Q \in \mathbb{R}^{p \times p}$ is *A-orthogonal* if it preserves this inner product: $\langle Qx, Qy \rangle_A = \langle x, y \rangle_A$ for all $x, y \in \mathbb{R}^p$. The reader should verify, using the symmetry and positive definiteness of A, that $\langle \cdot, \cdot \rangle_A$ is a genuine inner product; the choice $A = I$ recovers the Euclidean case.

Proposition 6.10 (Matrix criterion for A-orthogonality)**.** *A matrix Q is A-orthogonal if and only if*

$$Q^\top A Q = A.$$

In particular, for $A = I$ the criterion reads $Q^\top Q = I$: ordinary orthogonality.

Proof. We reduce equality of bilinear forms to equality of matrices by testing on standard basis vectors. For any x, y,

$$\langle Qx, Qy\rangle_A = (Qx)^\top A(Qy) = x^\top (Q^\top AQ)y.$$

If $Q^\top AQ = A$, the right side equals $x^\top Ay = \langle x, y\rangle_A$ for all x, y. Conversely, suppose $x^\top (Q^\top AQ)y = x^\top Ay$ for all x, y; choosing $x = e_i$ and $y = e_j$ reads off the (i, j) entry, $(Q^\top AQ)_{ij} = A_{ij}$, for every i, j, hence $Q^\top AQ = A$. □

Example 6.11 (A-orthogonality for weights $2, 3, 5$ ♦)**.** Let $A = \mathrm{diag}(2, 3, 5)$, whose diagonal weights are pairwise distinct. We test four natural candidates against the criterion of Proposition 6.10.

1. *Permutation swapping $e_1 \leftrightarrow e_3$:* here $Q^\top AQ = \mathrm{diag}(5, 3, 2) \neq A$. Not A-orthogonal — a permutation *reorders* the weights, and can only succeed when the swapped weights are equal.
2. *Sign flip $Q = \mathrm{diag}(1, -1, 1)$:* here $Q^\top AQ = \mathrm{diag}(1^2 \cdot 2,\ (-1)^2 \cdot 3,\ 1^2 \cdot 5) = A$. This one is A-orthogonal: signs square away, so axis reflections are A-orthogonal for *any* diagonal A.
3. *Permutation swapping $e_1 \leftrightarrow e_2$:* here $Q^\top AQ = \mathrm{diag}(3, 2, 5) \neq A$. Not A-orthogonal.
4. *Rotation by angle t in the x–y plane:* the upper 2×2 block of $Q^\top AQ$ is $R_t^\top \mathrm{diag}(2, 3) R_t$, whose off-diagonal entry is $(3 - 2)\sin t \cos t = \sin t \cos t \neq 0$ for generic t; even at $t = \pi/2$ the block becomes $\mathrm{diag}(3, 2) \neq \mathrm{diag}(2, 3)$. Not A-orthogonal — a rotation *mixes* unequal weights, and preserves $\langle \cdot, \cdot \rangle_A$ only on blocks where A is a multiple of the identity.

Conclusion: with distinct diagonal weights, axis sign flips qualify, while axis permutations and in-plane rotations do not. *(Problem P6.10 asks you to reproduce this classification from scratch.)*

Intuition. An A-orthogonal map is a rigid motion of the geometry in which direction i carries weight A_{ii}. Equal weights everywhere (A proportional to I) leave the full rotation group available; unequal weights break the rotational symmetry, and the only surviving axis symmetries are the reflections that keep each axis in place.

6.4 Principal Component Analysis

Principal component analysis answers one question: along which direction does centred data vary the most? The answer is a variational problem solved completely by the spectral decomposition of Fact 6.2. Interviews ask for the derivation, not the recipe — this subsection gives the rigorous argument and the explained-variance bookkeeping that follows from it.

Setup. Let $X \in \mathbb{R}^{n \times p}$ be a data matrix whose n rows are observations of p variables, with each column *centred* (summing to zero). The sample covariance matrix is

$$\hat{\Sigma} = \frac{1}{n} X^\top X$$

(the $1/(n-1)$ convention rescales every entry by the same factor and changes no direction; we return to it in Proposition 6.18). For a unit vector $w \in \mathbb{R}^p$, the projected data $Xw \in \mathbb{R}^n$ has sample variance $\frac{1}{n}\|Xw\|^2 = w^\top \hat{\Sigma} w$, so "the direction of maximal variance" means the maximiser of the quadratic form over the unit sphere. The theorem is stated for an arbitrary symmetric matrix Σ and applied with $\Sigma = \hat{\Sigma}$.

Theorem 6.12 (Variational characterisation of the first principal direction)**.** *Let $\Sigma \in \mathbb{R}^{p \times p}$ be symmetric with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p$ and orthonormal eigenvectors $q_1, \ldots, q_p$ as in Fact 6.2. Then*

$$\max_{\|w\|=1} w^\top \Sigma w = \lambda_1, \quad \text{attained at } w = q_1, \qquad \min_{\|w\|=1} w^\top \Sigma w = \lambda_p, \quad \text{attained at } w = q_p.$$

If $\lambda_1 > \lambda_2$, the maximiser is unique up to sign.

Proof. We expand in the eigenbasis, as in the proof of Theorem 6.4. Write $w = \sum_i c_i q_i$; the constraint $\|w\| = 1$ reads $\sum_i c_i^2 = 1$, and by (6.1), $w^\top \Sigma w = \sum_i \lambda_i c_i^2$. The right side is a weighted average of the eigenvalues with nonnegative weights c_i^2 summing to one, hence

$$w^\top \Sigma w = \sum_i \lambda_i c_i^2 \le \lambda_1 \sum_i c_i^2 = \lambda_1,$$

with equality if and only if all the weight sits on indices i with $\lambda_i = \lambda_1$. The choice $w = q_1$ attains the bound, proving the maximum; the minimum statement is identical with the inequality reversed and $w = q_p$. If $\lambda_1 > \lambda_2$, equality forces $c_1^2 = 1$ and $c_i = 0$ for $i \ge 2$, that is $w = \pm q_1$. □

Remark 6.13 (The Lagrangian route, made honest). The derivation usually quoted in interviews maximises $w^\top \Sigma w$ subject to $w^\top w = 1$ via the Lagrangian $L(w, \lambda) = w^\top \Sigma w - \lambda(w^\top w - 1)$: setting $\nabla_w L = 2\Sigma w - 2\lambda w = 0$ yields $\Sigma w = \lambda w$, so the stationary points are exactly the eigenvectors, the multiplier is the eigenvalue, and the objective at an eigenvector equals that eigenvalue. This is worth reproducing fluently — but stationarity alone does not say *which* eigenvector is the maximiser. The eigenbasis expansion in the proof of Theorem 6.12 is what certifies that the top eigenvector wins.

Definition 6.14 (Principal directions and components)**.** Let $\hat{\Sigma}$ be the sample covariance of a centred data matrix X, with eigenvalues sorted $\lambda_1 \ge \cdots \ge \lambda_p$ and orthonormal eigenvectors $q_1, \dots, q_p$. The vectors q_i are the *principal directions* of the data; the projected series $Xq_i \in \mathbb{R}^n$ is the *i-th principal component* (PC i).

Corollary 6.15 (Explained variance)**.** *The sample variance of the i-th principal component equals λ_i, the total sample variance of the data equals $\operatorname{tr} \hat{\Sigma} = \sum_{j=1}^p \lambda_j$, and the fraction of total variance explained by PC i is*

$$\frac{\lambda_i}{\sum_{j=1}^p \lambda_j}.$$

Proof. The sample variance of Xq_i is $q_i^\top \hat{\Sigma} q_i = \lambda_i q_i^\top q_i = \lambda_i$, using the eigenvector property and $\|q_i\| = 1$. The total sample variance of the data is the sum of the column variances of X, which is exactly the sum of the diagonal entries $\operatorname{tr} \hat{\Sigma}$; by Proposition 6.3 this equals $\sum_j \lambda_j$. The ratio of the two gives the displayed fraction. □

Example 6.16 (PCA of the two-by-two staple)**.** Suppose two centred variables have sample covariance matrix $\hat{\Sigma} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, the matrix of Example 6.5. The first principal direction is $q_1 = (1, 1)^\top / \sqrt{2}$ with variance $\lambda_1 = 3$; the second is $q_2 = (1, -1)^\top / \sqrt{2}$ with variance $\lambda_2 = 1$. By Corollary 6.15, PC 1 explains $3/(3 + 1) = 75\%$ of total variance and PC 2 the remaining 25%; the check $\operatorname{tr} \hat{\Sigma} = 4 = 3 + 1$ passes. Figure 9 shows a data cloud with this covariance structure and its principal axes. *(Problem P6.5 asks you to reproduce this derivation from scratch; Problem P6.6 applies it to an equicorrelated covariance matrix.)*

Interview trap. Principal component analysis is defined on the *covariance*, so the data must be centred first. Without centring, $X^\top X$ equals the centred scatter plus the rank-one term $n\bar{x}\bar{x}^\top$ built from the mean vector $\bar{x}$; when the mean sits far from the origin this term dominates, and the leading "principal direction" chases the direction of the mean rather than the direction of maximal variance, with an inflated explained-variance share.

6.5 Computation via the SVD

In practice principal directions are not computed by eigendecomposing a covariance matrix: they are read off the singular value decomposition of the data matrix itself. This subsection proves the link and records the numerical reason for preferring it.

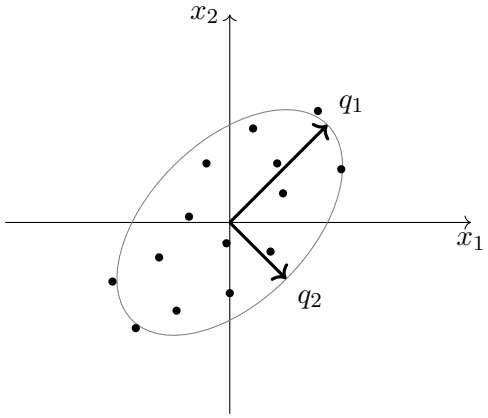


Figure 9: A centred data cloud with covariance as in Example 6.16. The principal axes point along the eigenvectors $q_1 = (1,1)^\top/\sqrt{2}$ and $q_2 = (1,-1)^\top/\sqrt{2}$ of the sample covariance (each eigenvector is defined only up to sign), with arrow lengths proportional to $\sqrt{\lambda_1} = \sqrt{3}$ and $\sqrt{\lambda_2} = 1$; the gray curve is the one-standard-deviation ellipse.

Setup. Let $X \in \mathbb{R}^{n\times p}$ be a centred data matrix with $n \geq p$.

Fact 6.17 (Singular value decomposition)**.** Every matrix $X \in \mathbb{R}^{n\times p}$ with $n \geq p$ admits a thin factorisation

$$X = USV^\top, \qquad U \in \mathbb{R}^{n\times p},\ U^\top U = I_p, \qquad V \in \mathbb{R}^{p\times p} \text{ orthogonal}, \qquad S = \operatorname{diag}(s_1, \ldots, s_p)$$

with singular values $s_1 \geq s_2 \geq \cdots \geq s_p \geq 0$. A proof is beyond our scope.

Proposition 6.18 (The SVD–PCA link)**.** *Let $X = USV^\top$ be as in Fact 6.17. Then*

$$X^\top X = VS^2V^\top,$$

so the columns $v_1, \ldots, v_p$ of V are the eigenvectors of $X^\top X$ with eigenvalues $s_1^2, \ldots, s_p^2$: the principal directions of the data are the columns of V, and the sample variance along v_i is s_i^2/n under the $1/n$ covariance convention and $s_i^2/(n-1)$ under the $1/(n-1)$ convention. The explained-variance fraction of PC i is $s_i^2/\sum_j s_j^2$ under either convention.

Proof. We substitute the factorisation and cancel:

$$X^\top X = (USV^\top)^\top(USV^\top) = VS\,(U^\top U)\,SV^\top = VS^2V^\top,$$

using $U^\top U = I_p$ and the symmetry of the diagonal factor S. Since V is orthogonal and S^2 is diagonal with nonnegative entries, this *is* a spectral decomposition of $X^\top X$ in the sense of Fact 6.2: the columns of V are orthonormal eigenvectors and the eigenvalues are s_i^2. Dividing by n gives $\hat{\Sigma} = V(S^2/n)V^\top$ — the same eigenvectors, with eigenvalues s_i^2/n, which by Corollary 6.15 are the sample variances along the principal directions; the $1/(n-1)$ variant differs only by the scalar $n/(n-1)$. In the explained-variance ratio of Corollary 6.15 the normalisation cancels, leaving $s_i^2/\sum_j s_j^2$. □

Remark 6.19 (Pick one convention). The two normalisations $1/n$ and $1/(n-1)$ produce identical directions and identical explained-variance shares; only the per-direction variance figures differ, by the factor $n/(n-1)$. Either is defensible — what is not defensible is mixing them within one computation. Pick one and stay consistent.

Example 6.20 (Variances from singular values)**.** Suppose $n = 5$ observations of $p = 2$ centred variables give singular values $s_1 = 4$ and $s_2 = 2$. Under the $1/(n-1)$ convention the sample

variances along the principal directions are $16/4 = 4$ and $4/4 = 1$, and PC 1 explains $4/(4+1) = 80\%$ of total variance — equivalently $s_1^2/(s_1^2+s_2^2) = 16/20 = 80\%$, where the normalisation has cancelled as promised by Proposition 6.18. *(Problem P6.7 asks you to reproduce this proof and computation from scratch.)*

Remark 6.21 (Conditioning: decompose X, not $X^\top X$). Given a numerical library offering both routines, compute the SVD of X rather than the eigendecomposition of $X^\top X$. Forming the product explicitly squares the condition number, $\kappa(X^\top X) = \kappa(X)^2$, so the directions belonging to small singular values drown in floating-point error, and rounding can even return small negative eigenvalues for a matrix that is PSD in exact arithmetic (compare Remark 6.8). The SVD works on X directly, is backward stable, and never materialises the $p \times p$ product.

6.6 Orthogonalising Correlated Signals (Gram–Schmidt)

A recurring desk problem: a candidate signal overlaps an existing one, and only the part of the candidate *not* already captured should be kept. The answer is a single Gram–Schmidt step — project and keep the residual — and it is simultaneously an ordinary least squares regression.

Setup. Two signals recorded over n periods are vectors $a, b \in \mathbb{R}^n$ (entry a_t is the first signal's value in period t), with $a \neq 0$. The Euclidean inner product is $\langle a, b\rangle = \sum_{t=1}^n a_t b_t$, and $\mathbf{1} \in \mathbb{R}^n$ denotes the all-ones vector. A signal is *demeaned* if its entries sum to zero, that is $\langle a, \mathbf{1}\rangle = 0$; for demeaned signals the sample covariance is $\langle a, b\rangle/n$ (or the $1/(n-1)$ variant, Remark 6.19), so zero inner product is the same statement as zero sample correlation.

Proposition 6.22 (Gram–Schmidt residual)**.** *Define the projection coefficient and residual*

$$\hat\beta = \frac{\langle a, b\rangle}{\langle a, a\rangle}, \qquad b_\perp = b - \hat\beta\, a. \tag{6.2}$$

Then:

- *(i) the residual is orthogonal to a: $\langle a, b_\perp\rangle = 0$;*
- *(ii) norms split (Pythagoras): $\|b\|^2 = \hat\beta^2\|a\|^2 + \|b_\perp\|^2$;*
- *(iii) if a and b are both demeaned, then $b_\perp$ is demeaned and (when $b_\perp \neq 0$) the sample correlation between a and $b_\perp$ is zero;*
- *(iv) the coefficient $\hat\beta$ is the OLS slope of the regression of b on a without intercept, and $b_\perp$ is its residual vector.*

Proof. We argue by direct computation. (i) By bilinearity and (6.2),

$$\langle a, b_\perp\rangle = \langle a, b\rangle - \hat\beta\langle a, a\rangle = \langle a, b\rangle - \langle a, b\rangle = 0.$$

(ii) Expanding $b = \hat\beta a + b_\perp$ and using (i) to kill the cross term,

$$\|b\|^2 = \hat\beta^2\|a\|^2 + 2\hat\beta\langle a, b_\perp\rangle + \|b_\perp\|^2 = \hat\beta^2\|a\|^2 + \|b_\perp\|^2.$$

(iii) If $\langle a, \mathbf{1}\rangle = \langle b, \mathbf{1}\rangle = 0$, then $\langle b_\perp, \mathbf{1}\rangle = \langle b, \mathbf{1}\rangle - \hat\beta\langle a, \mathbf{1}\rangle = 0$, so $b_\perp$ is demeaned; the sample covariance of a and $b_\perp$ is then proportional to $\langle a, b_\perp\rangle = 0$, and dividing by the (nonzero) sample standard deviations gives sample correlation zero. (iv) The least squares problem $\min_\beta \|b - \beta a\|^2$ has derivative $-2\langle a, b - \beta a\rangle$, which vanishes at $\beta = \langle a, b\rangle/\langle a, a\rangle = \hat\beta$; the second derivative $2\langle a, a\rangle > 0$ confirms a minimum. The residual vector $b - \hat\beta a$ is exactly $b_\perp$, and (i) is the familiar OLS orthogonality of residuals to regressors. □

Figure 10 shows the geometry: the projection $\hat\beta a$ and the residual $b_\perp$ form the two legs of a right triangle with hypotenuse b, which is precisely statement (ii).

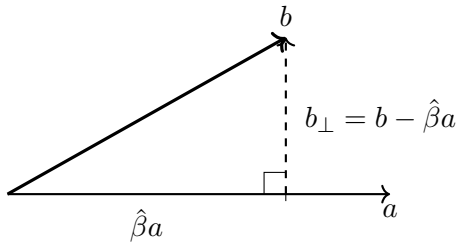


Figure 10: The Gram–Schmidt step of Proposition 6.22: the candidate signal b decomposes into its projection $\hat{\beta}a$ onto the incumbent signal a and the orthogonal residual $b_\perp$. Pythagoras splits $\|b\|^2 = \hat{\beta}^2\|a\|^2 + \|b_\perp\|^2$.

Remark 6.23 (Demeaning caveat). Orthogonality equals zero correlation *only for demeaned signals*. In general the sample covariance is $\frac{1}{n}\langle a, b\rangle - \bar{a}\bar{b}$ with $\bar{a}, \bar{b}$ the sample means, so with nonzero means two orthogonal vectors can be correlated and two uncorrelated signals can fail to be orthogonal. Demean first — or, equivalently, include an intercept in the regression of (iv), after which the residuals are orthogonal to the constant column as well.

Example 6.24 (Orthogonalising a candidate signal)**.** Let a and b be demeaned with unit sample variance and sample correlation 0.8, so that $\hat{\beta} = 0.8$ and $b_\perp = b - 0.8\,a$ by (6.2). The residual's sample variance is

$$1 - 2(0.8)(0.8) + (0.8)^2(1) = 1 - 1.28 + 0.64 = 0.36 = 1 - 0.8^2,$$

the general pattern $1 - \rho^2$ for unit-variance signals with correlation ρ. In words: 64% of the candidate's variance was already contained in the incumbent, and only 36% of it is new information. *(Problem P6.8 asks you to reproduce this computation from scratch.)*

Trading application. Suppose a desk runs a production model whose output cannot be modified — an untouchable black box — and a candidate signal overlaps it. Orthogonalise the demeaned candidate against the demeaned incumbent output via (6.2): what survives is the candidate's *incremental* information. The combined book then aggregates cleanly — by Proposition 6.22(iii) the two components are uncorrelated, so variances add with no covariance term — and combined risk stops double-counting the shared exposure.

6.7 Beta Neutralisation (Market-Neutral Signals)

The Gram–Schmidt step of Proposition 6.22 applied in the *cross-section* strips market exposure out of a trading signal. The recipe rests on one empirical fact about equity return covariances and one line of algebra.

Setup. Consider a universe of p assets. A cross-sectional signal is now a vector $b \in \mathbb{R}^p$ whose i-th entry scores asset i (note the change of arena: entries index assets, not time periods). Let $m \in \mathbb{R}^p$ denote the first principal direction of the asset-return covariance matrix, estimated as in Definition 6.14 (in practice via Proposition 6.18).

Fact 6.25 (PC1 as the market mode, empirical)**.** For broad equity universes, the first principal direction of the return covariance matrix empirically has same-sign weights on every asset: it is the *market mode*, the direction in which the whole market moves together. This is an empirical regularity, not a theorem, although positively correlated toy models reproduce it exactly — for an equicorrelated covariance with correlation $\rho > 0$, PC1 is exactly the equal-weight vector (Problem P6.6 derives this).

Corollary 6.26 (Beta-neutral projection)**.** *The projected signal*

$$b' = b - \frac{\langle b, m\rangle}{\langle m, m\rangle}\, m$$

satisfies $\langle b', m\rangle = 0$*: the neutralised signal carries no exposure along the market mode. The neutrality holds by construction, not by a regression estimate.*

Proof. This is Proposition 6.22(i) with the incumbent vector taken to be m and the candidate taken to be b. □

Example 6.27 (Neutralising a three-asset signal)**.** Take the market mode $m = (1, 1, 1)^\top$ and the raw signal $b = (2, -1, 1)^\top$. The two inner products are $\langle b, m\rangle = 2 - 1 + 1 = 2$ and $\langle m, m\rangle = 3$, so

$$b' = b - \tfrac{2}{3}\,(1, 1, 1)^\top = \left(\tfrac{4}{3},\ -\tfrac{5}{3},\ \tfrac{1}{3}\right)^\top.$$

The check passes: $\langle b', m\rangle = \frac{4}{3} - \frac{5}{3} + \frac{1}{3} = 0$, that is, the neutralised weights sum to zero. The position is long–short balanced against the equal-weight market portfolio: a broad move lifting all three assets equally produces zero profit or loss to first order. *(Problem P6.9 asks you to reproduce and extend this computation.)*

Trading application. The full recipe for a market-neutral signal: (1) run PCA on the asset returns (Proposition 6.18); (2) inspect the first principal direction's weights — same-sign loadings identify it as the market mode per Fact 6.25; (3) project the signal off it by Corollary 6.26. The result is beta-neutral by construction. When m has exactly equal weights the projection reduces to cross-sectional demeaning of the signal; an estimated PC1 instead removes a loading-weighted average, giving each name credit for its actual market sensitivity.

Practice: problems P6.1–P6.10 in §27, solutions in §28.

7 Calculus for Quant Interviews

Assessments test calculus at two speeds. Reflex questions ask for a derivative or a standard antiderivative and are graded on speed; reasoning questions ask whether an improper integral converges, what a moment equals, or why a root-finder stalls, and are graded on the one-sentence justification that accompanies the verdict. This section builds the toolkit in that order: derivative fluency, integration technique, the convergence decision, the bridge from integrals to moments, Taylor expansions, and Newton's method. Material sourced from real assessments carries its provenance glyph where it appears.

7.1 The Derivative Toolkit

Differentiation questions reward fluency: the rules and the core table below should come without visible thought, because they are the substrate for every optimisation, sensitivity, and Taylor question later in the section. The one identity worth proving is the logistic derivative, which turns every "sensitivity of a price to news" question into a two-second multiplication.

Throughout this subsection, u, v, and g denote real functions differentiable wherever the rules below are applied, and a denotes a positive constant.

Fact 7.1 (Differentiation rules and the core table)**.** Let u and v be differentiable at x, with $v(x) \neq 0$ where v appears in a denominator, and let f be differentiable at $g(x)$. Then

$$(uv)' = u'v + uv', \qquad \left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}, \qquad \frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x).$$

The core table, for n real and $a > 0$:

$$(x^n)' = nx^{n-1}, \qquad (e^{ax})' = ae^{ax}, \qquad (\ln x)' = \frac{1}{x}, \qquad (a^x)' = a^x \ln a,$$

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x, \qquad (\sqrt{x}\,)' = \frac{1}{2\sqrt{x}}.$$

These are standard first-course results; we use them without proof.

In an interview, narrate the chain rule from the outside in: "the derivative of the outside, evaluated at the inside, times the derivative of the inside." The narration prevents the classic slip of forgetting the inner factor.

The logistic (sigmoid) function maps log-odds to a probability, and it is the standard model of a prediction-market contract price responding to news. In this subsection only, the symbol p denotes this price function (elsewhere in the document p is a probability parameter).

Proposition 7.2 (Logistic derivative identity)**.** *Let $p(x) = (1 + e^{-x})^{-1}$ for $x \in \mathbb{R}$. Then*

$$p'(x) = p(x)(1 - p(x)) \qquad \text{for all } x,$$

and the derivative attains its maximum value $\frac{1}{4}$ at $x = 0$, where $p = \frac{1}{2}$.

Proof. We differentiate directly with the chain rule and then recognise the factors. Writing $p(x) = (1 + e^{-x})^{-1}$,

$$p'(x) = -(1 + e^{-x})^{-2} \cdot (-e^{-x}) = \frac{e^{-x}}{(1 + e^{-x})^2} = \frac{1}{1 + e^{-x}} \cdot \frac{e^{-x}}{1 + e^{-x}}.$$

The first factor is $p(x)$. For the second, note

$$1 - p(x) = 1 - \frac{1}{1 + e^{-x}} = \frac{e^{-x}}{1 + e^{-x}},$$

which proves the identity. For the maximum, set $s = p(x)$; as x ranges over $\mathbb{R}$, the value s ranges over the open interval $(0, 1)$, and $p'(x) = s(1 - s)$. The map $s \mapsto s(1 - s)$ is a downward parabola with vertex at $s = \frac{1}{2}$, where it equals $\frac{1}{4}$. Since p is continuous and strictly increasing with $p(0) = \frac{1}{2}$, the maximum sensitivity $\frac{1}{4}$ is attained exactly at $x = 0$. □

(Problem P7.3 asks you to reproduce this proof from scratch.)

Example 7.3 (Sensitivity at eighty cents)**.** A contract trades at $p = 0.80$. By Proposition 7.2, the price moves at rate $p(1 - p) = 0.80 \times 0.20 = 0.16$ price units per unit of log-odds. At $p = 0.50$ the rate would be $0.50 \times 0.50 = 0.25$, the maximum possible: a 50-cent contract moves most per unit of news.

Trading application. Any "sensitivity of price to log-odds" question reduces to: compute p, multiply $p(1 - p)$. The qualitative fact that sensitivity peaks at $p = \frac{1}{2}$ is itself a common follow-up: contracts near even odds are where news moves prices fastest.

The optimisation recipe. Interview optimisation questions are graded in three parts. (1) Set $f' = 0$ and solve. (2) *Justify* that the critical point is a maximum: by the sign of the second derivative, by a sign change of f', or by boundary behaviour (if $f \to 0$ at both ends of the domain and there is a single interior critical point, that point is the maximum). (3) Report both the maximiser x^* and the optimal value $f(x^*)$.

Example 7.4 (The recipe on a decaying objective)**.** Maximise $f(x) = xe^{-x}$ over $x > 0$. Step 1: the product rule gives $f'(x) = (1-x)e^{-x}$, which vanishes only at $x^* = 1$. Step 2: the factor e^{-x} is positive, so f' changes sign from positive to negative at $x = 1$; alternatively, $f(x) \to 0$ as $x \downarrow 0$ and as $x \to \infty$, and $x = 1$ is the only interior critical point, so it is the maximum. Step 3: the maximiser is $x^* = 1$ with value $f(1) = e^{-1} \approx 0.368$.

Interview trap. The justification sentence in step (2) is graded. A stated critical point without the one-line argument for *why* it is a maximum, or an answer that reports x^* but not $f(x^*)$, leaves points on the table even when the calculus is flawless.

7.2 Integration: Substitution, Parts, Symmetry

Two named integrals anchor a large share of quant calculus: the Gaussian integral and the Gamma integral. This subsection proves both and then organises the three techniques — substitution, parts, symmetry — that evaluate almost everything an assessment asks.

Setup. Throughout this subsection x, u and t are real variables and n is a nonnegative integer. Integrals over unbounded ranges are improper Riemann integrals, evaluated as limits of integrals over bounded ranges (the convergence theory is developed in §7.3).

Theorem 7.5 (Gaussian integral)**.**

$$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}.$$

Proof. We use the polar-coordinates squaring trick. First, the integral converges: for $|x| \geq 1$ we have $x^2 \geq |x|$, hence $e^{-x^2} \leq e^{-|x|}$, and $\int_{-\infty}^{\infty} e^{-|x|}dx = 2$ is finite. Let I denote the integral; the integrand is positive, so $I > 0$. Since the name of the integration variable is immaterial,

$$I^2 = \Big(\int_{-\infty}^{\infty} e^{-x^2}dx\Big)\Big(\int_{-\infty}^{\infty} e^{-y^2}dy\Big) = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy,$$

where writing the product as a double integral over the plane is justified by Tonelli's theorem for nonnegative integrands. Now change to polar coordinates with radius s, $x = s\cos\theta$, $y = s\sin\theta$ with area element $s\,ds\,d\theta$:

$$I^2 = \int_0^{2\pi}\int_0^{\infty} e^{-s^2}\,s\,ds\,d\theta.$$

The factor s is exactly the derivative factor a substitution wants: with $u = s^2$, $du = 2s\,ds$, the inner integral is $\frac{1}{2}\int_0^{\infty} e^{-u}du = \frac{1}{2}$. Hence $I^2 = 2\pi\cdot\frac{1}{2} = \pi$, and since $I > 0$ we conclude $I = \sqrt{\pi}$. □

Example 7.6 (Normalising the standard normal density)**.** The substitution $x = \sqrt{2}\,u$, $dx = \sqrt{2}\,du$, converts Theorem 7.5 into the normalisation of the standard normal density:

$$\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2}\int_{-\infty}^{\infty} e^{-u^2}\,du = \sqrt{2\pi} \approx 2.5066,$$

so $\varphi(x) = (2\pi)^{-1/2}e^{-x^2/2}$ integrates to exactly 1. Every "find the constant that makes this a density" question with a Gaussian kernel is this computation with different constants.

Theorem 7.7 (Gamma integral at the integers)**.** *For every integer $n \geq 0$,*

$$\int_0^{\infty} x^n e^{-x}\,dx = n!$$

(in Gamma-function notation, $\Gamma(n+1) = n!$).

Proof. We proceed by induction on n, with integration by parts supplying the inductive step.

Base case. For $n = 0$, $\int_0^\infty e^{-x}dx = \lim_{t\to\infty}(1 - e^{-t}) = 1 = 0!$.

Inductive step. Assume $\int_0^\infty x^n e^{-x}dx = n!$. Integrating by parts on $[0, t]$ with $u = x^{n+1}$ and $dv = e^{-x}dx$ (so $du = (n+1)x^n dx$, $v = -e^{-x}$),

$$\int_0^t x^{n+1}e^{-x}\,dx = \big[-x^{n+1}e^{-x}\big]_0^t + (n+1)\int_0^t x^n e^{-x}\,dx.$$

The boundary term vanishes in the limit: from the exponential series, $e^t \geq t^{n+2}/(n+2)!$ for $t > 0$, so $t^{n+1}e^{-t} \leq (n+2)!/t \to 0$ as $t \to \infty$. Letting $t \to \infty$ and applying the inductive hypothesis,

$$\int_0^\infty x^{n+1}e^{-x}\,dx = (n+1)\cdot n! = (n+1)!,$$

which completes the induction. □

Example 7.8 (Gamma integral in action)**.** Theorem 7.7 with $n = 3$ gives

$$\int_0^\infty x^3 e^{-x}\,dx = 3! = 6$$

in one line — no triple integration by parts required. Check by a single parts step: $\int_0^\infty x^3e^{-x}dx = 3\int_0^\infty x^2e^{-x}dx = 3\cdot 2! = 6$. A rescaled variant follows from the substitution $u = 2x$:

$$\int_0^\infty x^2 e^{-2x}\,dx = \frac{1}{2^3}\int_0^\infty u^2 e^{-u}\,du = \frac{2!}{2^3} = 0.25,$$

the pattern developed systematically in Proposition 7.23.

Substitution: spot the g' factor. If the integrand contains $g(x)$ *and* a constant multiple of $g'(x)$, substitute $u = g(x)$. The screaming tell is a factor $x\,dx$ next to anything built from x^2. Two habits keep definite integrals honest: transform the limits immediately, and never return to the x variable.

Example 7.9 (The substitution reflex)**.** Consider $\int_0^\infty xe^{-x^2}\,dx$. The factor $x\,dx$ beside e^{-x^2} calls for $u = x^2$, $du = 2x\,dx$; the limits $x = 0$ and $x = \infty$ become $u = 0$ and $u = \infty$:

$$\int_0^\infty xe^{-x^2}\,dx = \frac{1}{2}\int_0^\infty e^{-u}\,du = \frac{1}{2},$$

where the final integral is the $n = 0$ case of Theorem 7.7.

Parts: u is what dies when differentiated. In $\int u\,dv$, choose as u the factor that simplifies under differentiation. The LIATE mnemonic orders the candidates: log, inverse trig, algebraic, trig, exponential — $\ln x$ is always u, and a power of x against e^x or $\cos x$ is u. The prototype is the antiderivative of the logarithm: with $u = \ln x$ and $dv = dx$,

$$\int \ln x\,dx = x\ln x - \int x\cdot\frac{1}{x}\,dx = x\ln x - x + C.$$

(Problem P7.5 asks you to reproduce this computation.) The reflex classics, each verifiable in ten seconds by differentiating the right-hand side:

$$\int x\cos x\,dx = x\sin x + \cos x + C, \qquad \int xe^x\,dx = (x-1)e^x + C,$$

$$\int x^2e^x\,dx = e^x(x^2 - 2x + 2) + C, \qquad \int x\ln x\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C.$$

For $\int x^n e^x dx$ and its relatives, tabular integration (repeated parts laid out in a table) takes n passes for degree n.

Proposition 7.10 (Vanishing of odd integrals)**.** *Let $f : \mathbb{R} \to \mathbb{R}$ be odd, meaning $f(-x) = -f(x)$ for all x, and absolutely integrable, meaning $\int_{-\infty}^{\infty} |f(x)|\,dx < \infty$. Then*

$$\int_{-\infty}^{\infty} f(x)\,dx = 0.$$

The same conclusion holds for an odd integrable f on a bounded symmetric interval $[-a, a]$.

Proof. We substitute $x \mapsto -x$ on the negative half-line. With $u = -x$,

$$\int_{-\infty}^{0} f(x)\,dx = \int_{\infty}^{0} f(-u)\,(-du) = \int_{0}^{\infty} f(-u)\,du = -\int_{0}^{\infty} f(u)\,du,$$

using oddness in the last step. Absolute integrability guarantees that both half-line integrals are finite numbers, so their sum is legitimately

$$\int_{-\infty}^{\infty} f(x)\,dx = -\int_{0}^{\infty} f(u)\,du + \int_{0}^{\infty} f(x)\,dx = 0.$$

On a bounded interval the same substitution applies and finiteness is automatic. □

Example 7.11 (Symmetry with the caveat)**.** Consider $\int_{-\infty}^{\infty} \sin(2x)\,e^{-x^2}\,dx$. The integrand is odd times even, hence odd, and it is dominated in absolute value by e^{-x^2}, whose integral is $\sqrt{\pi}$ by Theorem 7.5; Proposition 7.10 therefore gives the value 0. The absolute-integrability hypothesis is not decoration: "odd, so zero" alone would equally "prove" $\int_{-\infty}^{\infty} x\,dx = 0$, yet that integral diverges (each half-line piece is infinite; only the symmetric principal value is zero). In an interview, state the caveat explicitly — it is the difference between a right answer and a right argument.

Interview trap. After *any* antiderivative, differentiate the result back — a ten-second check that catches most sign errors. And a positive integrand can never produce a negative or zero answer: if it does, something upstream is wrong. Both checks return in the improper integrals below.

7.3 Improper Integrals: the Convergence Decision

Converge-or-diverge questions are a reported assessment topic ♦, and the expected deliverable is a verdict plus a one-line justification — that is *full* credit. This subsection defines convergence, proves the tests that decide it (the two p-tests, direct comparison, limit comparison, and the logarithmic refinements), and ends with the decision procedure in speaking order.

Definition 7.12 (Improper integral and convergence)**.** Let f be continuous on $[a, \infty)$. The improper integral $\int_a^{\infty} f(x)\,dx$ *converges* to L if $\lim_{t\to\infty} \int_a^t f(x)\,dx = L$ exists finitely, and *diverges* otherwise. Similarly, if f is continuous on $(a, b]$ but unbounded near a, then $\int_a^b f(x)\,dx = \lim_{t\downarrow a} \int_t^b f(x)\,dx$ when the limit exists finitely. An integral whose domain contains several *problem points* — infinite endpoints, finite endpoints where the integrand is unbounded, and interior singularities — is split into pieces containing exactly one problem point each; the integral converges if and only if *every* piece converges.

Rule one is built into the definition: before any test, list every problem point — $\pm\infty$, endpoint blow-ups, *and* interior singularities — and split there. Missing an interior singularity is the classic fatal error; it reappears in the trap at the end of this subsection.

Proposition 7.13 (p-test at infinity)**.** *For real p, the integral $\int_1^{\infty} x^{-p}\,dx$ converges if and only if $p > 1$, in which case its value is $\frac{1}{p-1}$.*

Proof. We evaluate the antiderivative directly. For $p \neq 1$, an antiderivative of x^{-p} is $x^{1-p}/(1-p)$, so

$$\int_1^t x^{-p}\,dx = \frac{t^{1-p}-1}{1-p}.$$

If $p > 1$ then $1-p < 0$, so $t^{1-p} \to 0$ as $t \to \infty$ and the limit is $\frac{-1}{1-p} = \frac{1}{p-1}$, finite. If $p < 1$ then $t^{1-p} \to \infty$ and the integral diverges. In the boundary case $p = 1$, $\int_1^t x^{-1}dx = \ln t \to \infty$: logarithmic divergence. □

Proposition 7.14 (p-test at zero)**.** *For real p, the integral $\int_0^1 x^{-p}\,dx$ converges if and only if $p < 1$, in which case its value is $\frac{1}{1-p}$.*

Proof. We evaluate the same antiderivative at the other problem point. For $p \neq 1$,

$$\int_t^1 x^{-p}\,dx = \frac{1-t^{1-p}}{1-p}.$$

As $t \downarrow 0$, the term t^{1-p} tends to 0 precisely when $1-p > 0$, giving the value $\frac{1}{1-p}$; when $p > 1$ it blows up. At $p = 1$, $\int_t^1 x^{-1}dx = -\ln t \to \infty$: again logarithmic divergence. □

Intuition. At infinity the integrand must decay *fast*; at zero a *mild* blow-up is affordable. The function $1/x$ fails both tests — it is log-divergent at each end — and it is the exact boundary case of each: whenever $p = 1$ appears, the verdict is logarithmic divergence.

Example 7.15 (The p-tests with values)**.** The tail integral $\int_1^\infty x^{-3/2}dx$ has $p = \frac{3}{2} > 1$: it converges, with value $\frac{1}{3/2-1} = 2$ by Proposition 7.13. The endpoint integral $\int_0^1 x^{-1/2}dx$ has $p = \frac{1}{2} < 1$: it converges, with value $\frac{1}{1-1/2} = 2$ by Proposition 7.14.

Proposition 7.16 (Direct comparison test)**.** *Let f and g be continuous on $[a,\infty)$ with $0 \le f(x) \le g(x)$ for all $x \ge a$.*

(i) If $\int_a^\infty g$ converges, then $\int_a^\infty f$ converges, and $\int_a^\infty f \le \int_a^\infty g$.
(ii) If $\int_a^\infty f$ diverges, then $\int_a^\infty g$ diverges.

The same statements hold on an interval whose problem point is a finite endpoint.

Proof. We use monotone boundedness. Define $F(t) = \int_a^t f(x)\,dx$. Since $f \ge 0$, the function F is nondecreasing in t. For every t,

$$F(t) \le \int_a^t g(x)\,dx \le \int_a^\infty g(x)\,dx,$$

the second inequality because $g \ge 0$ makes $t \mapsto \int_a^t g$ nondecreasing with the improper integral as its limit. A nondecreasing function that is bounded above converges, as $t \to \infty$, to its supremum, which is finite; hence $\int_a^\infty f = \lim_{t\to\infty} F(t)$ exists and is at most $\int_a^\infty g$. Statement (ii) is the contrapositive of (i). At a finite problem point the argument is identical with t tending to that point. □

Figure 11 shows the geometry: the region under f sits inside the region under g, so a finite total area for g forces a finite total area for f.

Example 7.17 (Comparison from below)**.** Consider $\int_1^\infty \frac{2+\sin x}{x}\,dx$. Since $\sin x \ge -1$, the integrand satisfies $\frac{2+\sin x}{x} \ge \frac{1}{x} \ge 0$ on $[1,\infty)$, and $\int_1^\infty \frac{dx}{x}$ diverges by Proposition 7.13 with $p = 1$. Part (ii) of Proposition 7.16 gives divergence.

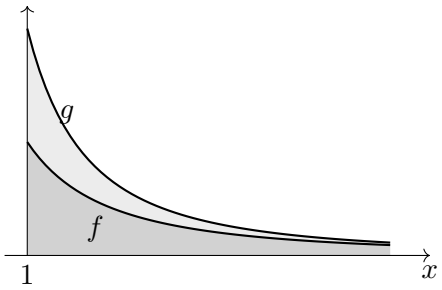


Figure 11: The direct comparison test (Proposition 7.16). Here $f(x) = \frac{1}{x(x+1)}$ (dark region) is dominated by $g(x) = x^{-2}$; the light band is the gap between the two graphs. The area under f is contained in the area under g, so a finite area under g forces a finite area under f.

Corollary 7.18 (Exponentials beat polynomials)**.** *For every real $k \geq 0$, every rate $\lambda > 0$, and every $a > 0$, both*

$$\int_a^\infty x^k e^{-\lambda x}\,dx \qquad \text{and} \qquad \int_a^\infty x^k e^{-\lambda x^2}\,dx$$

converge.

Proof. We compare against a pure exponential. Choose an integer $j > k$. For $x > 0$ the exponential series gives $e^{\lambda x/2} \geq (\lambda x/2)^j / j!$, hence

$$x^k e^{-\lambda x/2} \leq j!\,(2/\lambda)^j\, x^{k-j} \longrightarrow 0 \quad \text{as } x \to \infty,$$

so there is a constant B with $x^k e^{-\lambda x/2} \leq B$ for all $x \geq a$. Therefore

$$x^k e^{-\lambda x} = \left(x^k e^{-\lambda x/2}\right) e^{-\lambda x/2} \leq B\, e^{-\lambda x/2}, \qquad \int_a^\infty e^{-\lambda x/2}\,dx = \frac{2}{\lambda} e^{-\lambda a/2} < \infty,$$

and Proposition 7.16(i) gives convergence of the first integral. For the second, on $x \geq \max(a, 1)$ we have $\lambda x^2 \geq \lambda x$, so $x^k e^{-\lambda x^2} \leq x^k e^{-\lambda x}$ and the first case applies; the remaining piece over $[a, \max(a, 1)]$ is a proper integral of a continuous function. The displayed bound also shows the pointwise fact that a polynomial times $e^{-\lambda x}$ or $e^{-\lambda x^2}$ tends to zero at infinity; we reuse this when discarding boundary terms in integration by parts. □

Proposition 7.19 (Limit comparison test)**.** *Let f and g be continuous on $[a, \infty)$ with $f \geq 0$ and $g > 0$, and suppose*

$$\lim_{x \to \infty} \frac{f(x)}{g(x)} = c \in (0, \infty).$$

Then $\int_a^\infty f$ and $\int_a^\infty g$ either both converge or both diverge.

Proof. We run the epsilon argument for the limit and hand the result to the direct comparison test. Apply the definition of the limit with $\varepsilon = c/2 > 0$: there exists $M \geq a$ such that $|f(x)/g(x) - c| < c/2$ for all $x \geq M$. Multiplying through by $g(x) > 0$,

$$\frac{c}{2}\, g(x) \;<\; f(x) \;<\; \frac{3c}{2}\, g(x) \qquad \text{for all } x \geq M. \tag{7.1}$$

Each of the two improper integrals splits at M into a proper integral over $[a, M]$ (a continuous function on a compact interval, hence finite) plus a tail over $[M, \infty)$, so each converges if and only if its tail converges.

Suppose $\int_M^\infty g$ converges. Then so does $\int_M^\infty \frac{3c}{2} g$ (a constant multiple), and the right-hand inequality in (7.1) together with Proposition 7.16(i) gives convergence of $\int_M^\infty f$.

Suppose instead $\int_M^\infty g$ diverges. Then $\int_M^\infty \frac{c}{2} g$ diverges (a positive constant multiple of a divergent integral of a nonnegative function), and the left-hand inequality in (7.1) together with Proposition 7.16(ii) gives divergence of $\int_M^\infty f$.

The two cases together prove the claim. The same argument, run as x tends to a finite problem point, gives the analogous statement there. □

In practice the limit comparison test is applied by replacing the integrand with its *asymptotic shape* — the simplest function with the same leading behaviour at the problem point — and it is usually faster than hunting for a pointwise domination.

Example 7.20 (Asymptotic shape in action)**.** (a) For $\int_1^\infty \frac{dx}{x(x+1)}$: at infinity the integrand behaves like x^{-2}, and indeed $\frac{1/(x(x+1))}{1/x^2} = \frac{x}{x+1} \to 1 \in (0,\infty)$. Since $\int_1^\infty x^{-2}dx$ converges ($p=2$), Proposition 7.19 gives convergence. Here the value is also available: partial fractions give $\frac{1}{x(x+1)} = \frac{1}{x} - \frac{1}{x+1}$, so

$$\int_1^T \frac{dx}{x(x+1)} = \ln T - \ln(T+1) + \ln 2 \longrightarrow \ln 2 \approx 0.693.$$

(b) For $\int_1^\infty \frac{x}{1+x^2}\,dx$: the asymptotic shape is $1/x$, and $\frac{x/(1+x^2)}{1/x} = \frac{x^2}{1+x^2} \to 1$. Since $\int_1^\infty \frac{dx}{x}$ diverges ($p=1$), the integral diverges.

Proposition 7.21 (Logarithmic refinements)**.** *The integral $\int_2^\infty \frac{dx}{x \ln x}$ diverges, while $\int_2^\infty \frac{dx}{x(\ln x)^2}$ converges, with value $\frac{1}{\ln 2}$.*

Proof. We substitute $u = \ln x$, $du = dx/x$, which transforms both integrals into p-integrals; the limits $x = 2$ and $x = t$ become $u = \ln 2$ and $u = \ln t$. For the first integral,

$$\int_2^t \frac{dx}{x \ln x} = \int_{\ln 2}^{\ln t} \frac{du}{u} = \ln\ln t - \ln\ln 2 \longrightarrow \infty,$$

so it diverges (doubly logarithmically slowly). For the second,

$$\int_2^t \frac{dx}{x(\ln x)^2} = \int_{\ln 2}^{\ln t} \frac{du}{u^2} = \frac{1}{\ln 2} - \frac{1}{\ln t} \longrightarrow \frac{1}{\ln 2} \approx 1.4427.$$

□

Intuition. The substitution $u = \ln x$ is the general device: it turns every $\int^\infty \frac{dx}{x(\ln x)^q}$ into the p-test with exponent q. The borderline $p = 1$ integrand $1/x$ thus acquires a whole ladder of refinements, and the first rung $1/(x \ln x)$ still diverges.

Interview trap. Three standing errors. (1) "The integrand tends to 0" proves *nothing*: $1/x$ tends to zero and diverges — the question is always *how fast*. (2) Never apply the fundamental theorem of calculus across an *interior* singularity. The canonical wrong answer is $\int_{-1}^1 x^{-2}dx = [-x^{-1}]_{-1}^1 = -2$: the integrand blows up at the interior point 0, so one must split there, and each half diverges (Proposition 7.14 with $p = 2$ after shifting the endpoint); moreover a positive integrand can never produce -2 — the positive-integrand check of §7.2. (3) Whenever the exponent $p = 1$ appears, the answer is logarithmic divergence.

The decision procedure, in speaking order. List the problem points and split. For a tail at infinity: compare to x^{-p} (need $p > 1$) or spot exponential decay (Corollary 7.18). For a finite blow-up at a: compare to $(x-a)^{-p}$ (need $p < 1$). For a messy integrand: limit-compare against its asymptotic shape (Proposition 7.19). If the integral converges and a value is asked: substitution, parts, partial fractions, symmetry, or a named result (Theorems 7.5 and 7.7). For a converge-or-diverge question, the verdict plus a one-line justification is full credit.

7.4 Moments as Integrals: the Probability Bridge

Moment computations are improper integrals in probability clothing, and the named integrals of §7.2 do all the heavy lifting. This subsection states the transfer principle (LOTUS) and derives the moment formulas — exponential, normal, uniform — that assessments expect on demand.

Throughout, X denotes a continuous random variable with density f_X (nonnegative, integrating to 1), and k and n denote nonnegative integers.

Fact 7.22 (Law of the unconscious statistician (LOTUS))**.** Let g be a function for which the integral below converges absolutely, that is, $\int |g(x)|\, f_X(x)\, dx < \infty$. Then

$$\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x)\, f_X(x)\, dx.$$

There is no need to find the density of $g(X)$ first. A full proof requires measure-theoretic machinery and is beyond our scope.

Proposition 7.23 (Exponential moments)**.** *Let $X \sim \mathrm{Exp}(\lambda)$ with rate $\lambda > 0$, that is, with density $f_X(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Then for every integer $n \geq 0$,*

$$\mathbb{E}[X^n] = \frac{n!}{\lambda^n}.$$

In particular $\mathbb{E}[X] = 1/\lambda$, $\mathbb{E}[X^2] = 2/\lambda^2$ (so $\mathrm{Var}(X) = 1/\lambda^2$), and $\mathbb{E}[X^3] = 6/\lambda^3$.

Proof. We substitute the integral into the Gamma form. By Fact 7.22 (the integrand is nonnegative, and finiteness follows from Corollary 7.18),

$$\mathbb{E}[X^n] = \int_0^{\infty} x^n \lambda e^{-\lambda x}\, dx.$$

Substituting $u = \lambda x$ (so $x = u/\lambda$, $dx = du/\lambda$, limits unchanged in form),

$$\mathbb{E}[X^n] = \int_0^{\infty} \frac{u^n}{\lambda^n}\, \lambda\, e^{-u}\, \frac{du}{\lambda} = \frac{1}{\lambda^n} \int_0^{\infty} u^n e^{-u}\, du = \frac{n!}{\lambda^n},$$

the last equality by Theorem 7.7. The variance follows: $\mathrm{Var}(X) = 2/\lambda^2 - (1/\lambda)^2 = 1/\lambda^2$. □

Example 7.24 (Exponential moments at rate two)**.** Let $X \sim \mathrm{Exp}(2)$. Then $\mathbb{E}[X] = 1/2 = 0.5$, $\mathbb{E}[X^2] = 2/4 = 0.5$, $\mathrm{Var}(X) = 0.5 - 0.25 = 0.25$, and $\mathbb{E}[X^3] = 6/8 = 0.75$.

Proposition 7.25 (Normal moments up to order four)**.** *Let $X \sim \mathcal{N}(0, \sigma^2)$ with density $\varphi_\sigma(x) = (2\pi\sigma^2)^{-1/2} e^{-x^2/(2\sigma^2)}$.*

(i) Every odd moment vanishes: $\mathbb{E}[X^k] = 0$ for odd k.
(ii) $\mathbb{E}[X^2] = \sigma^2$.
(iii) $\mathbb{E}[X^4] = 3\sigma^4$; equivalently, the kurtosis $\mathbb{E}[X^4]/\sigma^4$ equals 3.

Proof. Part (i) is symmetry: for odd k the integrand $x^k \varphi_\sigma(x)$ is an odd function (odd power times even density), and it is absolutely integrable by Corollary 7.18, so Proposition 7.10 gives 0.

For the even moments we use an integration-by-parts reduction. The density satisfies $\varphi_\sigma'(x) = -(x/\sigma^2)\, \varphi_\sigma(x)$, so $-\sigma^2 \varphi_\sigma$ is an antiderivative of $x\, \varphi_\sigma(x)$. For $k \geq 2$, write $x^k \varphi_\sigma = x^{k-1} \cdot (x\, \varphi_\sigma)$ and integrate by parts on $[-t, t]$ with $u = x^{k-1}$, $dv = x\, \varphi_\sigma(x)\, dx$:

$$\int_{-t}^{t} x^k \varphi_\sigma(x)\, dx = \Big[-\sigma^2 x^{k-1} \varphi_\sigma(x) \Big]_{-t}^{t} + (k-1)\, \sigma^2 \int_{-t}^{t} x^{k-2} \varphi_\sigma(x)\, dx.$$

The boundary term is a polynomial times a Gaussian and tends to zero as $t \to \infty$ (the pointwise bound in the proof of Corollary 7.18). Letting $t \to \infty$,

$$\mathbb{E}[X^k] = (k-1)\,\sigma^2\,\mathbb{E}[X^{k-2}] \qquad (k \geq 2). \tag{7.2}$$

The base of the reduction is $\mathbb{E}[X^0] = \int \varphi_\sigma = 1$: substituting $x = \sigma u$ reduces this to Example 7.6. Applying (7.2) with $k = 2$ and then $k = 4$:

$$\mathbb{E}[X^2] = 1 \cdot \sigma^2 \cdot 1 = \sigma^2, \qquad \mathbb{E}[X^4] = 3\sigma^2 \cdot \mathbb{E}[X^2] = 3\sigma^4.$$

The kurtosis is $3\sigma^4/\sigma^4 = 3$. □

(Problem P7.11 asks you to reproduce the fourth-moment derivation.)

Example 7.26 (Fourth moment of a return model)**.** Model a daily return as $X \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 0.02$. Then $\mathbb{E}[X^4] = 3\,(0.02)^4 = 4.8 \times 10^{-7}$, and the ratio $\mathbb{E}[X^4]/\sigma^4 = 3$ regardless of σ: under normality, kurtosis is always exactly 3.

Proposition 7.27 (Uniform moments)**.** *Let $U \sim \mathrm{U}(0,1)$. Then for every integer $k \geq 0$,*

$$\mathbb{E}[U^k] = \frac{1}{k+1}.$$

Proof. We integrate directly against the density, which is 1 on $[0,1]$: $\mathbb{E}[U^k] = \int_0^1 u^k\,du = \frac{1}{k+1}$. □

Example 7.28 (Variance of a squared uniform)**.** By Proposition 7.27, $\mathbb{E}[U^2] = \frac{1}{3}$ and $\mathbb{E}[U^4] = \frac{1}{5}$, so

$$\mathrm{Var}(U^2) = \mathbb{E}[U^4] - \big(\mathbb{E}[U^2]\big)^2 = \frac{1}{5} - \frac{1}{9} = \frac{4}{45} \approx 0.089.$$

Remark 7.29 (Normalisation questions). Questions of the form "find c making $c\,h(x)$ a density" are improper-integral evaluations in probability clothing: first confirm $\int h$ converges (the tests of §7.3), then set $c = 1/\int h$. The evaluation is typically Theorem 7.7 or Theorem 7.5 after a substitution.

7.5 Taylor Expansions and Limits

Limit questions are usually expansion questions in disguise, and so are most of the mental approximations a desk relies on: compounding, log returns against simple returns, the first- and second-order sensitivity of a price. All of them rest on one theorem, which replaces a smooth function near a point by a polynomial and bounds the error. This subsection proves that theorem with the integration by parts of §7.2, derives the standard expansions from it, and turns them into a procedure for limits.

Setup. Throughout this subsection x and t are real variables, x_0 is a fixed real number (the expansion point), $n \geq 0$ is an integer (the order of the expansion), m is a positive integer (a number of compounding periods), j and k are nonnegative integers, and a, α are fixed real constants. The function f is defined on an open interval I containing x_0 and has $n+1$ continuous derivatives there; $f^{(k)}$ denotes its k-th derivative, with $f^{(0)} = f$. As $x \to x_0$, the symbol $O(|x - x_0|^k)$ denotes a quantity bounded by a constant multiple of $|x - x_0|^k$ in a neighbourhood of x_0; when $x_0 = 0$ it is written $O(x^k)$. Such terms combine as powers do, $O(x^j) + O(x^k) = O(x^{\min(j,k)})$ and $x^j \cdot O(x^k) = O(x^{j+k})$. As $m \to \infty$, $O(m^{-k})$ denotes a quantity bounded by a constant multiple of m^{-k} for all large m.

Definition 7.30 (Taylor polynomial and remainder)**.** The *Taylor polynomial of order* n of f at x_0 is

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}\,(x - x_0)^k,$$

and the *remainder* is $R_n(x) = f(x) - T_n(x)$. Differentiating term by term shows that T_n and its first n derivatives take the same values at x_0 as f and its first n derivatives.

Theorem 7.31 (Taylor's theorem with integral remainder)**.** *Let f have $n+1$ continuous derivatives on an open interval I containing x_0. For every $x \in I$,*

$$f(x) = T_n(x) + R_n(x), \qquad R_n(x) = \frac{1}{n!}\int_{x_0}^{x} (x - t)^n f^{(n+1)}(t)\,dt. \tag{7.3}$$

If moreover $|f^{(n+1)}(t)| \le M$ for every t between x_0 and x, then

$$|R_n(x)| \le \frac{M\,|x - x_0|^{n+1}}{(n+1)!}. \tag{7.4}$$

Proof. We proceed by induction on n, integrating by parts at each step. For $n = 0$ the claim reads $f(x) = f(x_0) + \int_{x_0}^{x} f'(t)\,dt$, which is the fundamental theorem of calculus. Suppose that, for some $n \ge 1$, (7.3) holds at order $n-1$ for every function with n continuous derivatives on I; the function f is one of them, because it has $n+1$. Integrate the remainder R_{n-1} by parts with $u = f^{(n)}(t)$ and $dv = (x-t)^{n-1}\,dt$, so that $v = -(x-t)^n/n$:

$$\int_{x_0}^{x} (x - t)^{n-1} f^{(n)}(t)\,dt = \Big[-\frac{(x-t)^n}{n}\, f^{(n)}(t) \Big]_{t=x_0}^{t=x} + \frac{1}{n}\int_{x_0}^{x} (x-t)^n f^{(n+1)}(t)\,dt.$$

The bracket equals $(x - x_0)^n f^{(n)}(x_0)/n$, because the upper limit contributes zero. Dividing by $(n-1)!$,

$$R_{n-1}(x) = \frac{f^{(n)}(x_0)}{n!}\,(x - x_0)^n + \frac{1}{n!}\int_{x_0}^{x} (x-t)^n f^{(n+1)}(t)\,dt.$$

The first term on the right is $T_n(x) - T_{n-1}(x)$, so $R_n(x) = R_{n-1}(x) - \big(T_n(x) - T_{n-1}(x)\big)$ is the remaining integral, which is (7.3) at order n. For the bound, replace the integrand by its absolute value and $|f^{(n+1)}|$ by M: $|R_n(x)| \le \frac{M}{n!}\,\big|\int_{x_0}^{x} |x-t|^n\,dt\big| = \frac{M}{n!}\cdot\frac{|x-x_0|^{n+1}}{n+1}$, which is (7.4). □

Corollary 7.32 (Lagrange form and order of the remainder)**.** *For every $x \in I$ there is a point ξ between x_0 and x with*

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}\,(x - x_0)^{n+1},$$

and $R_n(x) = O(|x - x_0|^{n+1})$ as $x \to x_0$.

Proof. We apply the intermediate value theorem to the integral form. For $x = x_0$ both sides vanish and $\xi = x_0$ serves. Let $x \ne x_0$, and let c and C be the minimum and the maximum of the continuous function $f^{(n+1)}$ on the closed interval between x_0 and x. The weight $(x-t)^n/n!$ keeps one sign on that interval, and its integral from x_0 to x is $w = (x - x_0)^{n+1}/(n+1)! \ne 0$. By (7.3) the ratio $R_n(x)/w$ therefore lies between c and C, and the intermediate value theorem gives a point ξ of the interval with $f^{(n+1)}(\xi) = R_n(x)/w$. For the order, take M to be the maximum of $|f^{(n+1)}|$ on a closed interval around x_0 inside I and apply (7.4). □

Example 7.33 (Approximating $e^{0.1}$ with an error bar)**.** Take $f(x) = e^x$, $x_0 = 0$ and $n = 2$. Every derivative of f is e^x, so $T_2(x) = 1 + x + x^2/2$ and $T_2(0.1) = 1.105$. On $[0, 0.1]$ the third derivative satisfies $e^t \le e < 3$, so by (7.4) the error is at most $3 \times 0.1^3/3! = 0.0005$. The true value is $e^{0.1} = 1.105171$, an error of 0.000171, inside the bound.

Intuition. Order one is the tangent line, $f(x) \approx f(x_0) + f'(x_0)(x - x_0)$; order two adds the curvature term $\frac{1}{2}f''(x_0)(x - x_0)^2$. In trading language the first derivative is a delta and the second a gamma or a convexity. The error of stopping at order n shrinks at least like $|x - x_0|^{n+1}$: halving the distance divides a first-order error by about four and a second-order error by about eight.

Proposition 7.34 (Standard expansions near zero)**.** *As $x \to 0$,*

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + O(x^4), \qquad \ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} + O(x^4),$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}\,x^2 + O(x^3), \qquad \frac{1}{1-x} = 1 + x + x^2 + O(x^3),$$

$$\sin x = x - \frac{x^3}{6} + O(x^5), \qquad \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + O(x^6).$$

Proof. We apply Theorem 7.31 at $x_0 = 0$ to each function, computing its derivatives at 0. Every derivative of e^x equals 1 at 0. The first three derivatives of $\ln(1+x)$ are $(1+x)^{-1}$, $-(1+x)^{-2}$ and $2(1+x)^{-3}$, with values 1, -1, 2 at 0, which give the coefficients 1, $-\frac{1}{2}$ and $\frac{2}{6} = \frac{1}{3}$. The first two derivatives of $(1+x)^\alpha$ are $\alpha(1+x)^{\alpha-1}$ and $\alpha(\alpha-1)(1+x)^{\alpha-2}$, with values α and $\alpha(\alpha-1)$ at 0; the expansion of $1/(1-x)$ is the case $\alpha = -1$ with x replaced by $-x$. The derivatives of $\sin x$ at 0 cycle through $0, 1, 0, -1$ and those of $\cos x$ through $1, 0, -1, 0$; because the coefficient of x^4 in the sine expansion and that of x^5 in the cosine expansion vanish, $T_3 = T_4$ for the sine and $T_4 = T_5$ for the cosine, so the remainders are of orders five and six. In every case the next derivative is continuous near 0 (for $\ln(1+x)$ and $(1+x)^\alpha$, on $x > -1$), so the remainder has the stated order by Corollary 7.32. □

Procedure 7.35 (Limit by expansion)**.** To evaluate the limit of a quotient that presents as $\frac{0}{0}$:

1. Name the small quantity. For $x \to 0$ it is x; for $x \to x_0$ substitute $y = x - x_0$; for $x \to \infty$ substitute $y = 1/x$.
2. Find the order of the denominator: expand it with Proposition 7.34 as far as its first term that does not cancel. The power of that term is the order.
3. Expand every term of the numerator through that order, carrying the O-term of the next order.
4. Divide. If the numerator's first surviving term has the same order as the denominator, the limit is the ratio of the two coefficients, and the next term of the quotient is the rate at which the limit is approached. If every term of the numerator through that order cancels, the limit is 0. If a term of lower order survives, the quotient is unbounded.

Example 7.36 (A second-order limit)**.** Compute $\lim_{x\to 0} \frac{1-\cos x}{x^2}$. The denominator has order two. By Proposition 7.34 the numerator is $\frac{x^2}{2} + O(x^4)$, so the ratio is $\frac{1}{2} + O(x^2) \to \frac{1}{2}$. Numerical sanity check at $x = 0.1$: $(1 - \cos 0.1)/0.01 = 0.49958$, already within 0.1 percent of the limit.

Example 7.37 (A limit in which the leading terms cancel)**.** Compute $\lim_{x\to 0} \frac{\sin x - x\cos x}{x^3}$. The denominator has order three, so expand the numerator to order three: $\sin x = x - \frac{x^3}{6} + O(x^5)$ and $x\cos x = x - \frac{x^3}{2} + O(x^5)$. The terms of order one cancel and the numerator is $\frac{x^3}{3} + O(x^5)$, so the limit is $\frac{1}{3}$. Carrying one more term on each side ($\frac{x^5}{120}$ for the sine, $\frac{x^5}{24}$ for $x\cos x$) gives the quotient $\frac{1}{3} - \frac{x^2}{30} + O(x^4)$. At $x = 0.1$ this predicts 0.333000, and direct evaluation gives $(0.099833417 - 0.099500417)/0.001 = 0.333000$.

Fact 7.38 (L'Hôpital's rule (L'Hôpital, 1696))**.** Let g and h be functions differentiable on an open interval around x_0 with x_0 removed, with $h'(x) \neq 0$ there, and suppose that $g(x)$ and $h(x)$ both tend to 0, or both tend to $\pm\infty$, as $x \to x_0$. If $g'(x)/h'(x) \to L$, where L is a real number or $\pm\infty$, then $g(x)/h(x) \to L$. A proof is beyond our scope.

Interview trap. L'Hôpital's rule applies only to the indeterminate forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$ — check the form *before* differentiating, at every round, because applying the rule to a non-indeterminate form silently produces nonsense. And reach for the expansion first: it is usually faster and shows more structure (for instance the next-order term, which L'Hôpital hides).

Proposition 7.39 (Compounding limit and its first correction)**.** *For fixed real a, as $m \to \infty$,*

$$\Big(1 + \frac{a}{m}\Big)^m = e^a\Big(1 - \frac{a^2}{2m} + O(m^{-2})\Big),$$

and in particular $\big(1 + \frac{a}{m}\big)^m \to e^a$.

Proof. We take logarithms and expand. For $m > |a|$ the quantity $x = a/m$ satisfies $|x| < 1$ and tends to 0, so Proposition 7.34 gives

$$m \ln\Big(1 + \frac{a}{m}\Big) = m\Big(\frac{a}{m} - \frac{a^2}{2m^2} + O(m^{-3})\Big) = a - \frac{a^2}{2m} + O(m^{-2}).$$

Exponentiating, $\big(1 + \frac{a}{m}\big)^m = e^a e^w$ with $w = -\frac{a^2}{2m} + O(m^{-2})$, and $e^w = 1 + w + O(w^2) = 1 - \frac{a^2}{2m} + O(m^{-2})$ by the same proposition. □

Example 7.40 (Fifty periods at one percent)**.** Estimate $(1.01)^{50}$ without a calculator. Here $a = 0.5$ and $m = 50$. The limit alone gives $e^{0.5} \approx 1.6487$. The correction factor of Proposition 7.39 is $1 - 0.25/100 = 0.9975$, so $(1.01)^{50} \approx 1.6487 \times 0.9975 = 1.6446$. The exact value is 1.64463: the limit alone is 0.25 percent too high, and the corrected estimate is right to four decimal places.

(Problem P7.13 works this computation at different numbers.)

Trading application. The expansion $\ln(1 + x) = x - \frac{x^2}{2} + O(x^3)$ is the bridge between simple and log returns. A simple return of 1% is a log return of $0.01 - 0.00005 = 0.995\%$; at 10% the log return is $\ln 1.1 = 9.53\%$, about half a percentage point lower. Taking expectations of the second-order expansion for a per-period return R with mean μ and standard deviation σ, both small, gives $\mathbb{E}[\ln(1 + R)] \approx \mu - \frac{1}{2}(\sigma^2 + \mu^2) \approx \mu - \frac{1}{2}\sigma^2$: to second order, the growth rate of compounded wealth is the mean return minus half the variance. With $\mu = 0.1\%$ and $\sigma = 2\%$ per day the growth rate is about $0.001 - 0.0002 = 0.08\%$ per day, a fifth below the mean. Maximising this second-order growth rate over the bet size is the small-edge form of the Kelly criterion (§3.1).

Interview trap. Keeping too few terms returns the indeterminate form the expansion was meant to resolve. In Example 7.37 the first-order approximations $\sin x \approx x$ and $\cos x \approx 1$ turn the numerator into $x - x$. With the O-terms carried this is $O(x^3)$, which says nothing about the limit; with them dropped it reads 0 and suggests, wrongly, that the limit is 0. The order to which the numerator must be expanded is set by the denominator.

Interview trap. An expansion is a *local* statement. The approximation $\ln(1 + x) \approx x$ is excellent at $x = 0.01$ and poor at $x = 0.5$, where it returns 0.5 against the true 0.405; and no number of extra terms rescues it once $x > 1$, where the Taylor polynomials of $\ln(1 + x)$ move further from the function as the order grows. Before expanding, check that the quantity called small is small.

7.6 Newton's Method: Convergence Rates

Root-finding convergence rates are a reported assessment topic ♦, and the graded distinction is between a *simple* root, where Newton's method is quadratic, and a *multiple* root, where it silently degrades to linear. This subsection proves both rates, using the Taylor expansions of §7.5 as the engine.

Throughout, f is a real function, r denotes a root ($f(r) = 0$), and $e_k = x_k - r$ denotes the error of the k-th iterate.

Definition 7.41 (Newton iteration)**.** Given a differentiable function f and a starting point x_0, the Newton iteration is

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}, \qquad k = 0, 1, 2, \ldots,$$

defined as long as $f'(x_k) \neq 0$. Geometrically, x_{k+1} is the point where the tangent line to the graph at $(x_k, f(x_k))$ crosses the horizontal axis; Figure 12 shows two steps.

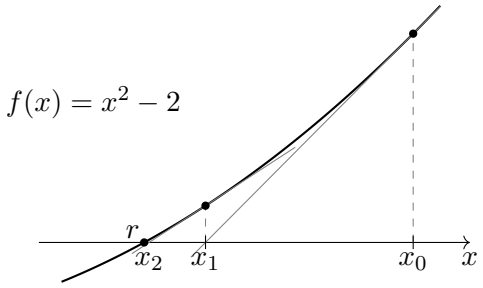


Figure 12: Two Newton steps on $f(x) = x^2 - 2$ from $x_0 = 2.6$. Each gray tangent line is followed to its axis crossing: $x_1 \approx 1.685$, then $x_2 \approx 1.436$. The filled dot on the axis marks the root $r = \sqrt{2} \approx 1.414$; after two steps the iterate almost coincides with it.

Theorem 7.42 (Quadratic convergence at a simple root)**.** *Let f be twice continuously differentiable on an open interval containing r, with $f(r) = 0$ and $f'(r) \neq 0$ (a* simple *root). Then there exist $\delta > 0$ and $C \geq 0$ such that for every starting point x_0 with $|x_0 - r| \leq \delta$, the Newton iterates are defined, remain within δ of r, converge to r, and satisfy*

$$|x_{k+1} - r| \leq C\,|x_k - r|^2 \qquad \text{for all } k.$$

Consequently, once the error is small, the number of correct digits roughly doubles at every step.

Proof. We apply Taylor's theorem in the Lagrange form (Corollary 7.32) and then trap the iterates.

Step 1: a safe interval. Since f' is continuous and $f'(r) \neq 0$, choose $\delta_0 > 0$ such that $I = [r - \delta_0,\, r + \delta_0]$ lies in the domain and $|f'(x)| \geq |f'(r)|/2 > 0$ for all $x \in I$. Set $K = \max_{x \in I} |f''(x)|$ and $C = K/|f'(r)|$.

Step 2: the one-step error identity. Suppose $x_k \in I$. Expanding f about x_k and evaluating at r, Corollary 7.32 gives, for some ξ_k between r and x_k,

$$0 = f(r) = f(x_k) - f'(x_k)\,e_k + \tfrac{1}{2} f''(\xi_k)\,e_k^2.$$

Dividing by $f'(x_k) \neq 0$ and rearranging, the Newton update satisfies

$$x_{k+1} - r \;=\; e_k - \frac{f(x_k)}{f'(x_k)} \;=\; \frac{f''(\xi_k)}{2f'(x_k)}\,e_k^2, \qquad \text{hence} \qquad |e_{k+1}| \leq \frac{K}{2 \cdot |f'(r)|/2}\,e_k^2 = C\,e_k^2. \tag{7.5}$$

Step 3: trapping and convergence. If $K = 0$ then f'' vanishes on I, so by (7.5) the very first step lands exactly on r. Otherwise set $\delta = \min(\delta_0,\, 1/(2C))$. If $|e_k| \leq \delta$, then by (7.5)

$$|e_{k+1}| \leq C\,|e_k| \cdot |e_k| \leq C\delta\,|e_k| \leq \tfrac{1}{2}|e_k| \leq \delta,$$

so the next iterate stays in the safe interval and the error at least halves. By induction $|e_k| \leq 2^{-k}|e_0| \to 0$, and the quadratic bound in (7.5) holds at every step. For the digit count, take base-10 logarithms in (7.5): $-\log_{10}|e_{k+1}| \geq 2(-\log_{10}|e_k|) - \log_{10} C$, so the number of correct digits roughly doubles per step once it dominates $\log_{10} C$. □

Example 7.43 (Square root of two: digits doubling)**.** Take $f(x) = x^2 - 2$, whose positive root is $r = \sqrt{2} = 1.4142136$. The iteration is $x_{k+1} = x_k - (x_k^2 - 2)/(2x_k) = (x_k + 2/x_k)/2$. From $x_0 = 1.5$:

$$x_1 = 1.4166667, \qquad x_2 = 1.4142157,$$

with errors $|e_0| = 8.58 \times 10^{-2}$, $|e_1| = 2.45 \times 10^{-3}$, $|e_2| = 2.1 \times 10^{-6}$: roughly 1, then 3, then 6 correct digits — doubling per step. The ratios $|e_1|/e_0^2 = 0.33$ and $|e_2|/e_1^2 = 0.35$ approach $|f''(r)/(2f'(r))| = 1/(2\sqrt{2}) \approx 0.354$, exactly as the error identity (7.5) predicts.

Proposition 7.44 (Linear convergence at a multiple root)**.** *Fix an integer $m \geq 2$ and consider the model function $f(x) = (x - r)^m$. For every $x_k \neq r$, the Newton iterate satisfies*

$$x_{k+1} - r = \Big(1 - \frac{1}{m}\Big)(x_k - r)\,:$$

convergence is only linear*, with the error multiplied by $\frac{m-1}{m}$ at every step (at a double root, the error merely halves). The modified iteration $x_{k+1} = x_k - m\, f(x_k)/f'(x_k)$ reaches r in a single step for this model.*

Proof. We compute the update directly. With $f(x) = (x - r)^m$ we have $f'(x) = m(x - r)^{m-1}$, so for $x_k \neq r$,

$$\frac{f(x_k)}{f'(x_k)} = \frac{(x_k - r)^m}{m(x_k - r)^{m-1}} = \frac{x_k - r}{m}, \qquad x_{k+1} - r = (x_k - r) - \frac{x_k - r}{m} = \Big(1 - \frac{1}{m}\Big)(x_k - r).$$

For the modified step, the subtracted quantity is $m \cdot (x_k - r)/m = x_k - r$, so $x_{k+1} = x_k - (x_k - r) = r$ exactly. □

Remark 7.45 (General multiplicity). A root of multiplicity m means $f(x) = (x - r)^m h(x)$ with h continuously differentiable and $h(r) \neq 0$. The model of Proposition 7.44 captures the leading behaviour: near such a root the error ratio tends to $\frac{m-1}{m}$, and when m is known, the modified step $x - mf/f'$ restores quadratic convergence. The practical diagnostic runs in reverse: unexpectedly slow, linear-looking Newton convergence is itself evidence of sitting on a multiple root.

Example 7.46 (Double root: errors halve)**.** Take $f(x) = (x - 1)^2$ from $x_0 = 2$. The iterates are 2, 1.5, 1.25, 1.125, … with errors 1, 0.5, 0.25, 0.125: the ratio is exactly $\frac{1}{2} = \frac{m-1}{m}$ for $m = 2$, as Proposition 7.44 predicts. The modified step with $m = 2$ jumps home immediately: $x_1 = 2 - 2\, f(2)/f'(2) = 2 - 2 \cdot \frac{1}{2} = 1$.

Interview trap. "Newton's method always converges quadratically" is the textbook overstatement: quadratic convergence requires a *simple* root and a sufficiently good start. When a function has several roots, the slowest is the one of *highest multiplicity*. Assessment instance ♦: for $f(x) = \tanh(x^4 - x^2)$, the roots are where $x^4 - x^2 = x^2(x - 1)(x + 1) = 0$, namely $x = 0$ and $x = \pm 1$. Near zero, $\tanh z = z + O(z^3)$ gives $f(x) = -x^2 + O(x^4)$: a double root, where Newton merely halves the error (Proposition 7.44) while it converges quadratically at the simple roots ± 1 (Theorem 7.42). The slowest root is therefore the one of smallest absolute value.

Practice: problems P7.1–P7.17 in §27, solutions in §28.

Part IV
Machine Learning and Deep Learning

8 ML Theory: Trees, Ensembles, Bias–Variance

This section develops the three results that anchor machine-learning theory questions in quant interviews: the decomposition of prediction error into bias, variance, and irreducible noise; the

variance of an average of correlated predictors; and the reading of gradient boosting as gradient descent in function space. Decision trees are the running example throughout, because the two great ensemble designs — random forests and gradient boosting — are engineering responses to the two reducible terms of the first decomposition.

8.1 The Bias–Variance Decomposition

Almost every qualitative model-selection question — why a deep tree overfits, why averaging helps, why a deliberately biased estimator can beat an unbiased one — reduces to a single identity for the expected squared prediction error. This subsection states and proves that identity, with the cross-term cancellations argued in full, since those cancellations are precisely what an interviewer probes.

Setup. Fix a test point x_0. The data-generating model is

$$y = f(x_0) + \varepsilon,$$

where f is an unknown deterministic function, and the test noise ε satisfies $\mathbb{E}[\varepsilon] = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$. A training set D is drawn at random (from whatever process generates training data), independently of ε, and a learning algorithm maps D to a fitted predictor $\hat{f}_D$. We abbreviate $\hat{f} := \hat{f}_D(x_0)$, a random variable whose randomness comes entirely from D. All expectations below are over the joint randomness of D and ε.

Definition 8.1 (Bias, variance, and irreducible error at a test point)**.** In the setup above, the *bias* of the predictor at x_0 is

$$\mathrm{Bias}(\hat{f}) = \mathbb{E}[\hat{f}] - f(x_0),$$

the systematic offset of the average fitted prediction from the truth; the *variance* is

$$\mathrm{Var}(\hat{f}) = \mathbb{E}\big[(\hat{f} - \mathbb{E}[\hat{f}])^2\big],$$

the fluctuation of the fitted prediction across training sets; and the noise variance σ^2 is called the *irreducible error*, since no predictor — however trained — can remove it.

Theorem 8.2 (Bias–variance decomposition)**.** *In the setup above, the expected squared prediction error at x_0 satisfies*

$$\mathbb{E}\big[(y - \hat{f})^2\big] = \mathrm{Bias}(\hat{f})^2 + \mathrm{Var}(\hat{f}) + \sigma^2. \tag{8.1}$$

Proof. We proceed by centering: we add and subtract $\mathbb{E}[\hat{f}]$ inside the square and expand the resulting three-term sum. Write

$$y - \hat{f} = \underbrace{\varepsilon}_{A} + \underbrace{\big(f(x_0) - \mathbb{E}[\hat{f}]\big)}_{B} + \underbrace{\big(\mathbb{E}[\hat{f}] - \hat{f}\big)}_{C},$$

where A is the test noise, B is a deterministic constant (it involves no randomness at all), and C is a mean-zero function of the training set: indeed $\mathbb{E}[C] = \mathbb{E}[\hat{f}] - \mathbb{E}[\hat{f}] = 0$. Expanding the square,

$$(y - \hat{f})^2 = A^2 + B^2 + C^2 + 2AB + 2AC + 2BC.$$

We take expectations term by term. The three squared terms give exactly the three terms of (8.1):

$$\mathbb{E}[A^2] = \mathrm{Var}(\varepsilon) = \sigma^2, \qquad B^2 = \big(f(x_0) - \mathbb{E}[\hat{f}]\big)^2 = \mathrm{Bias}(\hat{f})^2, \qquad \mathbb{E}[C^2] = \mathrm{Var}(\hat{f}),$$

where the middle equality holds because squaring removes the sign difference between B and $\mathrm{Bias}(\hat{f}) = -B$. It remains to show that each cross term vanishes, and the three cancellations have two distinct causes.

- $\mathbb{E}[AB] = B\,\mathbb{E}[\varepsilon] = 0$: the constant B factors out, and the noise has mean zero.
- $\mathbb{E}[AC] = \mathbb{E}[A]\,\mathbb{E}[C] = 0$: here $A = \varepsilon$ depends only on the test noise while C depends only on the training set D, and the two are independent by assumption, so the expectation factorizes; each factor is zero.
- $\mathbb{E}[BC] = B\,\mathbb{E}[C] = 0$: the constant factors out, and C has mean zero by construction — this is exactly what centering at $\mathbb{E}[\hat{f}]$ buys.

Summing the six expectations yields (8.1). □

(Problem P8.1 asks you to reproduce this proof from scratch.)

Example 8.3 (A shrinkage estimator beats the unbiased one)**.** Take a location model: the true value is $f(x_0) = 1$, the noise variance is $\sigma^2 = 1$, and the training set consists of $n = 4$ independent noisy observations of $f(x_0)$, with sample mean $\bar{y}$. Consider the family of shrinkage predictors $\hat{f}_\alpha = \alpha\bar{y}$ for $\alpha \in [0, 1]$. Then $\mathbb{E}[\hat{f}_\alpha] = \alpha$, so

$$\mathrm{Bias}(\hat{f}_\alpha)^2 = (1-\alpha)^2, \qquad \mathrm{Var}(\hat{f}_\alpha) = \alpha^2 \cdot \frac{\sigma^2}{n} = \frac{\alpha^2}{4},$$

and by Theorem 8.2 the risk is $R(\alpha) = (1-\alpha)^2 + \alpha^2/4 + 1$. Three instructive values:

$$R(1) = 0 + 0.25 + 1 = 1.25, \qquad R(0.8) = 0.04 + 0.16 + 1 = 1.20, \qquad R(0) = 1 + 0 + 1 = 2.$$

Setting $R'(\alpha) = -2(1-\alpha) + \alpha/2 = 0$ gives $\alpha^* = 0.8$: the optimal predictor deliberately accepts squared bias 0.04 to save 0.09 of variance, and beats the unbiased choice $\alpha = 1$ outright. The constant predictor $\alpha = 0$ shows the opposite failure: zero variance, but crippling bias. The irreducible $\sigma^2 = 1$ is untouched by every choice.

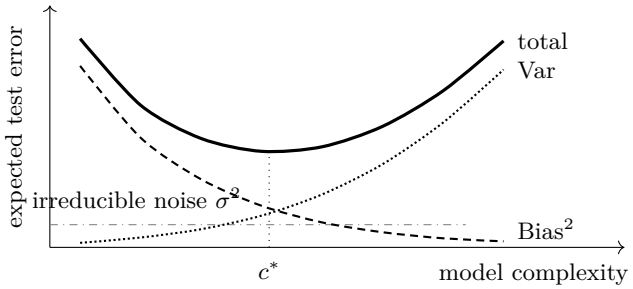


Figure 13: The bias–variance trade-off of Theorem 8.2 as model complexity grows: squared bias falls (dashed), variance rises (dotted), the irreducible noise σ^2 is a constant floor (dash-dotted), and their sum — the expected test error (solid) — is U-shaped, minimized at an interior complexity c^*.

Intuition. Figure 13 is the canonical picture behind (8.1). A more flexible model class tracks f more closely on average (bias falls) but its fit swings more from one training set to the next (variance rises); the noise floor never moves. Every regularisation device in this section — tree depth limits, averaging, shrinkage, early stopping — is a way of walking along this curve toward c^*.

Interview trap. When an interviewer asks for this derivation, the graded step is the cross-term argument, not the expansion. Asserting "the cross terms vanish" without a reason loses the points; the two reasons are distinct and must both be named: the terms containing ε die because the test noise is *independent of the training randomness* and has mean zero, while the remaining term dies because $\hat{f} - \mathbb{E}[\hat{f}]$ is *mean zero by centering* against a deterministic factor.

8.2 Decision Trees and Split Criteria

A decision tree grows by greedily choosing, at each node, the split that most reduces an impurity measure of the node's label distribution. Interviews test the two standard impurity measures on small numeric cases, and the bias–variance behaviour of trees of different depths — the property that motivates both ensemble designs of the next two subsections.

Setup. Consider a classification node holding n training samples distributed over K classes, with p_k the proportion of the node's samples in class k, so $\sum_{k=1}^{K} p_k = 1$. A candidate split partitions the node's samples into a left child L and a right child R holding n_L and n_R samples respectively, with $n_L + n_R = n$.

Definition 8.4 (Gini impurity, entropy, and impurity decrease)**.** The *Gini impurity* of the node is

$$G = 1 - \sum_{k=1}^{K} p_k^2,$$

and its *entropy*, measured in bits, is

$$H = -\sum_{k=1}^{K} p_k \log_2 p_k,$$

with the convention $0 \log_2 0 = 0$. (Entropy in natural-log units differs only by the constant factor $\log 2$ and ranks all splits identically.) For an impurity measure $I \in \{G, H\}$, the *impurity decrease* of the candidate split is the parent impurity minus the sample-weighted child impurity,

$$\Delta I = I_{\text{parent}} - \frac{n_L}{n} I_L - \frac{n_R}{n} I_R,$$

and the tree greedily selects the split maximizing ΔI; when $I = H$ the quantity ΔH is called the *information gain*. For regression trees the impurity of a node is the mean squared deviation of the node's targets from the node mean, and splits maximize the same weighted decrease (MSE reduction).

Example 8.5 (A ten-sample split)**.** A node holds 10 samples, 5 positive and 5 negative, and a candidate split sends 5 samples left (4 positive, 1 negative) and 5 right (1 positive, 4 negative); Figure 14 shows the split. The parent has $p = (0.5, 0.5)$, so

$$G_{\text{parent}} = 1 - 0.5^2 - 0.5^2 = 0.5, \qquad H_{\text{parent}} = -2 \cdot 0.5 \log_2 0.5 = 1 \text{ bit}.$$

Each child has proportions $(0.8, 0.2)$, hence

$$G_{\text{child}} = 1 - 0.8^2 - 0.2^2 = 1 - 0.64 - 0.04 = 0.32,$$

$$H_{\text{child}} = -0.8 \log_2 0.8 - 0.2 \log_2 0.2 \approx 0.8 \times 0.322 + 0.2 \times 2.322 \approx 0.722 \text{ bits}.$$

Both children hold half the samples, so the weighted child impurities equal the single-child values, and

$$\Delta G = 0.5 - 0.32 = 0.18, \qquad \Delta H = 1 - 0.722 = 0.278 \text{ bits}.$$

Both criteria would accept this split over any candidate with a smaller decrease.

Intuition. Depth is the tree's complexity dial in the sense of Figure 13. A deep tree carves feature space finely and fits the training data almost exactly: low bias, *high variance* — a slightly different training set produces very different splits. A shallow tree is the reverse: stable but crude, high bias and low variance. Trees are therefore the canonical high-variance base learner, and this single property dictates both ensemble designs that follow: averaging many deep trees to cut variance (§8.3), or adding many shallow trees sequentially to cut bias (§8.4).

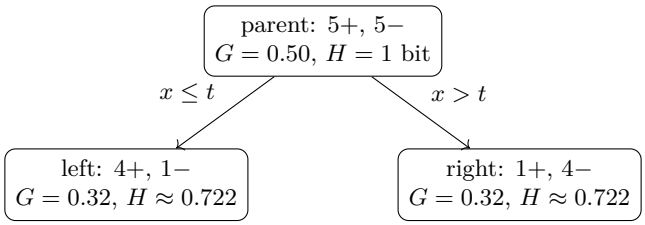


Figure 14: The split of Example 8.5: a threshold t on one feature partitions a mixed node into two purer children, for an impurity decrease of $\Delta G = 0.18$ (Gini) or $\Delta H \approx 0.278$ bits (information gain).

8.3 Bagging and Random Forests

Bagging fits many deep trees on bootstrap resamples of the training data and averages their predictions. Whether — and how much — averaging helps is settled by one covariance computation, whose limit explains the defining design choice of random forests.

Setup. At a fixed test point, let $X_1, \dots, X_N$ be the predictions of N models (for instance N trees fit on N bootstrap resamples), and let $\bar{X} = \frac{1}{N}\sum_{i=1}^{N} X_i$ be the ensemble prediction. Assume the predictions are exchangeable with common variance $\mathrm{Var}(X_i) = v$ and common pairwise correlation ρ, so that $\mathrm{Cov}(X_i, X_j) = \rho v$ for all $i \neq j$. We write v for the variance of an individual model's prediction — a different quantity from the irreducible noise σ^2 of §8.1, so it gets its own symbol.

Proposition 8.6 (Ensemble variance and the correlation floor)**.** *In the setup above,*

$$\mathrm{Var}(\bar{X}) = \rho v + \frac{1-\rho}{N}\, v. \tag{8.2}$$

For $\rho < 1$ this is strictly decreasing in N, and for any fixed $\rho \in [0, 1)$,

$$\lim_{N\to\infty} \mathrm{Var}(\bar{X}) = \rho v.$$

Proof. We compute the variance of the sum by bilinearity of covariance and then rescale. Expanding,

$$\mathrm{Var}\Big(\sum_{i=1}^{N} X_i\Big) = \sum_{i=1}^{N} \mathrm{Var}(X_i) + \sum_{i\neq j} \mathrm{Cov}(X_i, X_j) = Nv + N(N-1)\,\rho v,$$

since the diagonal contributes N variance terms and there are exactly $N(N-1)$ ordered pairs (i, j) with $i \neq j$, each contributing ρv. Dividing by N^2,

$$\mathrm{Var}(\bar{X}) = \frac{v}{N} + \frac{N-1}{N}\,\rho v = \frac{v}{N} + \rho v - \frac{\rho v}{N} = \rho v + \frac{1-\rho}{N}\, v,$$

which is (8.2). Monotonicity is immediate: the first term does not depend on N, and for $\rho < 1$ the coefficient $(1-\rho)v > 0$ multiplies the strictly decreasing sequence $1/N$. The same observation gives the limit: the second term tends to zero, and the first term ρv survives. □

(Problem P8.4 asks you to reproduce this computation from scratch.)

Example 8.7 (The floor in numbers)**.** Take $v = 1$ and $\rho = 0.2$. By (8.2),

$$\mathrm{Var}(\bar{X}) = \begin{cases} 1.00 & N = 1, \\ 0.2 + 0.8/10 = 0.28 & N = 10, \\ 0.2 + 0.8/100 = 0.208 & N = 100, \end{cases}$$

with limit 0.2: the first ten trees do almost all the work, and no number of additional trees crosses the floor $\rho v = 0.2$. Now compare two stylised design choices. Plain bagging with highly correlated trees, say $\rho = 0.5$ and $v = 1$, has floor 0.5. Decorrelating the trees at the price of making each one individually worse — say $\rho = 0.2$ but $v = 1.2$ — gives floor $\rho v = 0.24$, and already at $N = 100$ the decorrelated ensemble achieves $0.24 + 0.96/100 \approx 0.25$, half the correlated ensemble's 0.505. Attacking ρ beats polishing the $1/N$ term.

Intuition. Correlation between learners is the floor on ensemble variance: as $N \to \infty$ the $1/N$ term in (8.2) dies and ρv survives, so averaging can never remove the shared component of the trees' errors. Bagging alone leaves trees highly correlated — fit on overlapping resamples, they keep chasing the same strong features near the root. This is why random forests additionally sample a random subset of features at *every split*: the restriction forces different trees to use different features, attacking ρ itself — which averaging cannot — at the acceptable price of slightly worse individual trees, as Example 8.7 quantifies.

8.4 Boosting and Gradient Boosting

The second ensemble design runs in the opposite direction: instead of averaging deep trees fit in parallel, boosting adds *shallow* trees *sequentially*, each fit to what the current ensemble still gets wrong. Gradient boosting (Friedman, 2001) makes this precise: the procedure is gradient descent in function space, and for squared loss the descent direction is literally the residual vector.

Setup. Fix a training sample $(x_1, y_1), \dots, (x_n, y_n)$ and a per-sample loss $\ell(y, u)$, differentiable in its second argument (the prediction u). A candidate model F enters the training objective only through its values at the training points, so we identify F with the vector $\big(F(x_1), \dots, F(x_n)\big) \in \mathbb{R}^n$ and write the empirical risk as

$$R(F) = \sum_{i=1}^{n} \ell\big(y_i, F(x_i)\big).$$

Base learners h are drawn from a class $\mathcal{H}$ of weak models, typically shallow trees, and $\eta \in (0, 1]$ is a learning rate.

Definition 8.8 (Gradient boosting)**.** Initialize F_0 (for instance the constant minimizing R). At each round $t = 1, 2, \dots$, compute the *pseudo-residuals*

$$g_i^{(t)} = - \left. \frac{\partial\, \ell(y_i, u)}{\partial u} \right|_{u = F_{t-1}(x_i)}, \qquad i = 1, \dots, n,$$

fit a base learner $h_t \in \mathcal{H}$ to the targets $g^{(t)} = (g_1^{(t)}, \dots, g_n^{(t)})$ by least squares (a projection of the direction $g^{(t)}$ onto the base class), and update

$$F_t = F_{t-1} + \eta\, h_t. \tag{8.3}$$

Proposition 8.9 (Residuals are the negative gradient under squared loss)**.** *Take the squared loss $\ell(y, u) = \frac{1}{2}(y - u)^2$. Then the pseudo-residuals of Definition 8.8 are the ordinary residuals,*

$$g_i^{(t)} = y_i - F_{t-1}(x_i), \qquad i = 1, \dots, n,$$

and the vector $g^{(t)}$ equals $-\nabla R$ evaluated at the current value vector $\big(F_{t-1}(x_1), \dots, F_{t-1}(x_n)\big)$. Consequently, "fit the next tree to the residuals" is steepest descent on R over $\mathbb{R}^n$, with each step projected onto the base class $\mathcal{H}$ and scaled by η.

Proof. We differentiate directly. For the squared loss,

$$\frac{\partial\, \ell(y, u)}{\partial u} = \frac{\partial}{\partial u}\, \tfrac{1}{2}(y - u)^2 = -(y - u),$$

so the pseudo-residual at sample i is

$$g_i^{(t)} = -\big[-(y_i - u)\big]_{u=F_{t-1}(x_i)} = y_i - F_{t-1}(x_i),$$

the residual. Because R is a sum of per-sample terms, each depending on a single coordinate $F(x_i)$, the gradient of R with respect to the value vector is the vector of per-sample derivatives, $\big(\partial \ell(y_i, u)/\partial u\big|_{u=F(x_i)}\big)_{i=1}^n$; its negative at F_{t-1} is exactly $g^{(t)}$. The update (8.3) changes the value vector by $\eta\,(h_t(x_i))_{i=1}^n$, where h_t is the least-squares approximation to $g^{(t)}$ within $\mathcal{H}$ — a gradient step of length η taken in the closest direction the base class can represent. □

(Problem P8.6 asks you to reproduce this derivation from scratch.)

Example 8.10 (Two rounds of boosting on one observation)**.** Take a single training observation with $y = 8$, squared loss, initial model $F_0 = 0$, learning rate $\eta = 0.25$, and base learners rich enough to fit any target exactly (with $n = 1$, a single constant leaf suffices). By Proposition 8.9, each round fits the current residual $r_t = y - F_t$:

$$r_0 = 8, \quad F_1 = 0 + 0.25 \times 8 = 2; \qquad r_1 = 6, \quad F_2 = 2 + 0.25 \times 6 = 3.5; \qquad r_2 = 4.5.$$

In general each update (8.3) replaces the residual by $(1 - \eta)$ times itself, so $r_t = 8 \times 0.75^t$ and $F_t = 8\,(1 - 0.75^t)$: geometric convergence to the target, which is exactly gradient descent with step size η on the quadratic $R(F) = \frac{1}{2}(8 - F)^2$. With $\eta = 1$ one round would fit exactly — the point of shrinking η is that real base learners must also generalize away from the training points, and small steps plus early stopping keep the ensemble from fitting noise.

Intuition. Boosting attacks *bias*: each round fits precisely what the current ensemble still misses, so weak, shallow, high-bias trees suffice as base learners. Its variance is controlled by the shrinkage η, early stopping on a validation set, tree depth, and L1/L2 penalties on the leaf weights — the regularisation toolkit implemented in XGBoost and LightGBM.

Interview trap. "Would you rather have one very deep tree or many small ones?" The graded answer runs through the decomposition (8.1). One deep tree overfits: low bias, unchecked variance. *Many deep* trees averaged — a random forest — keep the low bias and cut the variance toward the floor of Proposition 8.6. *Many shallow* trees added sequentially — gradient boosting — start from high bias and remove it round by round (Proposition 8.9). The choice between the ensembles is which error component must be attacked, plus the tuning budget: forests are nearly tuning-free, boosting buys lower bias at the cost of the knobs listed above.

Trading application. Feature guidance for LightGBM and tree models generally, as posed in a live quantitative interview ★. Good tree features are monotone-informative and interaction-revealing raw quantities with low-cardinality encodings. Because a tree splits on thresholds, its splits are invariant to monotone transforms of a feature: scaling and normalisation are wasted effort. What trees do *not* get for free are relations *across* columns — engineer those explicitly as differences and ratios (spreads, imbalances, a price relative to its moving average). Two standing dangers: high-cardinality categorical features invite the tree to memorise identities rather than learn structure, and every feature must be constructed *point-in-time* — time-safe, using only information available at prediction time — or the backtest silently leaks the future.

Interview trap. "Give five examples of regularisation" expects breadth across both model worlds, not five variants of one idea. For neural networks: L1/L2 weight penalties, dropout, early stopping, data augmentation. For trees: limits on depth and leaf count, minimum child samples, row and feature subsampling, L1/L2 penalties on leaf weights, and early stopping on boosting rounds. An answer drawn entirely from one column signals a narrow toolkit.

Practice: problems P8.1–P8.7 in §27, solutions in §28.

9 Deep Learning

Deep-learning questions in quant interviews concentrate on a small set of mechanisms: how gradients are actually computed and why they vanish with depth, why attention scores carry a $1/\sqrt{d_k}$ factor, where the transformer's cost profile breaks down on high-frequency data, and a layer of engineering judgment about the numerical computing stack. The first two admit genuine mathematics and are proved below; the rest are engineering facts and are presented honestly as such, in structured remarks and applications.

9.1 Backpropagation and Vanishing Gradients

Backpropagation is the chain rule organised efficiently: gradients are computed layer by layer, back to front, and intermediate sensitivities are reused so that the entire gradient costs a small constant times one forward pass. This subsection works the mechanics explicitly on a two-layer scalar network, then isolates what goes wrong with depth — vanishing gradients — and why each standard architectural fix restores a usable gradient path.

Setup. Throughout this section the symbol σ denotes the logistic (sigmoid) activation function

$$\sigma(z) = \frac{1}{1+e^{-z}},$$

not a standard deviation; where a standard deviation is needed in this section it is written $\mathrm{Var}(\cdot)^{1/2}$. A *scalar chain network* of depth L maps an input $a_0 \in \mathbb{R}$ through

$$z_\ell = w_\ell\, a_{\ell-1}, \qquad a_\ell = \sigma(z_\ell), \qquad \ell = 1, \dots, L,$$

where $w_1, \dots, w_L$ are scalar weights, z_ℓ is the *pre-activation* of layer ℓ, and a_ℓ its *activation*. For the worked example we take depth $L = 2$ and attach the squared loss $\mathcal{L} = \frac{1}{2}(a_2 - y)^2$ with a fixed target y.

Lemma 9.1 (Sigmoid derivative bound)**.** *The sigmoid satisfies $\sigma'(z) = \sigma(z)\,(1-\sigma(z))$ for every $z \in \mathbb{R}$, and consequently*

$$0 < \sigma'(z) \le \tfrac{1}{4},$$

with equality exactly at $z = 0$.

Proof. We differentiate directly and then complete the square. Writing $\sigma(z) = (1+e^{-z})^{-1}$ and applying the chain rule,

$$\sigma'(z) = \frac{e^{-z}}{(1+e^{-z})^2} = \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1+e^{-z}} = \sigma(z)\,(1-\sigma(z)).$$

Now set $u = \sigma(z) \in (0,1)$. Completing the square gives

$$u(1-u) = \tfrac{1}{4} - \left(u - \tfrac{1}{2}\right)^2 \le \tfrac{1}{4},$$

with equality if and only if $u = \frac{1}{2}$, that is, at $z = 0$; positivity holds because $u \in (0,1)$. □

The next example is the whole of backpropagation in miniature: every partial derivative is written out, and Figure 15 displays the computational graph with the gradient flowing backward along it.

Example 9.2 (Backpropagation through a two-layer network)**.** Take input $a_0 = 1$, target $y = 0$, and weights $w_1 = 0$, $w_2 = 1$ in the depth-two network of the Setup.

Forward pass (left to right in Figure 15): the network evaluates

$$z_1 = w_1 a_0 = 0, \quad a_1 = \sigma(0) = 0.5, \quad z_2 = w_2 a_1 = 0.5, \quad a_2 = \sigma(0.5) = 0.6225,$$

and the loss is $\mathcal{L} = \frac{1}{2}(0.6225 - 0)^2 = 0.1937$. Every intermediate value is stored.

Backward pass (right to left): the chain rule is applied from the loss backward, one local factor at a time. Each partial is written out in full:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial a_2} &= a_2 - y = 0.6225, \\
\frac{\partial a_2}{\partial z_2} &= \sigma'(z_2) = a_2(1 - a_2) = 0.6225 \times 0.3775 = 0.2350, \\
\delta_2 := \frac{\partial \mathcal{L}}{\partial z_2} &= \frac{\partial \mathcal{L}}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} = 0.6225 \times 0.2350 = 0.1463, \\
\frac{\partial \mathcal{L}}{\partial w_2} &= \delta_2 \cdot \frac{\partial z_2}{\partial w_2} = \delta_2 \, a_1 = 0.1463 \times 0.5 = 0.0731, \\
\frac{\partial \mathcal{L}}{\partial a_1} &= \delta_2 \cdot \frac{\partial z_2}{\partial a_1} = \delta_2 \, w_2 = 0.1463 \times 1 = 0.1463, \\
\frac{\partial a_1}{\partial z_1} &= \sigma'(z_1) = a_1(1 - a_1) = 0.5 \times 0.5 = 0.25, \\
\delta_1 := \frac{\partial \mathcal{L}}{\partial z_1} &= \frac{\partial \mathcal{L}}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} = 0.1463 \times 0.25 = 0.0366, \\
\frac{\partial \mathcal{L}}{\partial w_1} &= \delta_1 \cdot \frac{\partial z_1}{\partial w_1} = \delta_1 \, a_0 = 0.0366 \times 1 = 0.0366.
\end{aligned}$$

Two observations carry the whole subject. First, the sensitivity δ_2 was computed *once* and used *twice* — for the weight gradient $\partial \mathcal{L}/\partial w_2$ and to continue the backward sweep to a_1; this reuse of intermediates is what makes backpropagation cheap. Second, propagating one layer further back multiplied the sensitivity by $w_2 \, \sigma'(z_1) = 0.25$, so $\delta_1 = 0.25 \, \delta_2$: a single backward step through a sigmoid shrank the signal by a factor of four. Proposition 9.3 shows how this attenuation compounds with depth.

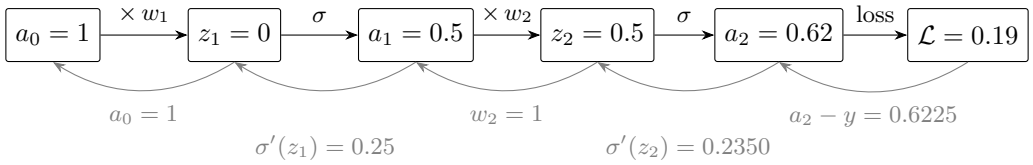


Figure 15: The computational graph of Example 9.2 (node values rounded to two decimals; the final edge applies the squared loss $\frac{1}{2}(a_2 - y)^2$). The forward pass (black, left to right) stores every intermediate value; the backward pass (gray, right to left) multiplies the running sensitivity by one local factor per edge. Weight gradients branch off the running product: $\partial \mathcal{L}/\partial w_2 = \delta_2 \, a_1$ and $\partial \mathcal{L}/\partial w_1 = \delta_1 \, a_0$.

Intuition. Backpropagation is not a new differentiation rule; it is bookkeeping. The chain rule is organised back to front so that each layer's sensitivity δ_ℓ is computed once and reused by every parameter feeding that layer, which is why the full gradient costs a small constant times one forward pass. Differentiating naively — one perturbed forward pass per parameter, as finite differences would — pays the forward-pass cost once per parameter instead of once in total.

Proposition 9.3 (Vanishing gradients in a deep sigmoid chain)**.** *Consider the scalar chain network of the Setup with depth $L \geq 1$ and $|a_0| \leq 1$. Then*

$$\frac{\partial a_L}{\partial w_1} = \Big[\prod_{\ell=2}^{L} w_\ell \, \sigma'(z_\ell) \Big] \, \sigma'(z_1) \, a_0, \tag{9.1}$$

a product containing one derivative factor $\sigma'(z_\ell)$ for every layer. If moreover $|w_\ell| \le 1$ for all ℓ, then

$$\Big|\frac{\partial a_L}{\partial w_1}\Big| \le 4^{-L} :$$

the gradient reaching the first layer decays exponentially in the depth.

Proof. We prove (9.1) by induction on the depth L. For the base case $L = 1$ we have $a_1 = \sigma(w_1 a_0)$, so the chain rule gives $\partial a_1/\partial w_1 = \sigma'(z_1)\, a_0$, which is (9.1) with the empty product equal to 1. For the inductive step, suppose (9.1) holds at depth $L-1$. Since $a_L = \sigma(w_L\, a_{L-1})$ and the weight w_L does not depend on w_1, the chain rule gives

$$\frac{\partial a_L}{\partial w_1} = \sigma'(z_L)\, w_L\, \frac{\partial a_{L-1}}{\partial w_1},$$

and substituting the inductive hypothesis extends the product by the factor $w_L\, \sigma'(z_L)$, which is exactly (9.1) at depth L.

For the bound, Lemma 9.1 gives $|\sigma'(z_\ell)| \le \frac{1}{4}$ for each of the L sigmoid factors in (9.1), while $|w_\ell| \le 1$ for the $L-1$ weight factors and $|a_0| \le 1$. Taking absolute values and multiplying the bounds yields $|\partial a_L/\partial w_1| \le (\frac{1}{4})^L = 4^{-L}$. □

(Problem P9.1 asks you to reproduce this derivation and bound from scratch.)

Example 9.4 (A depth-twenty chain)**.** For depth $L = 20$ the bound of Proposition 9.3 gives

$$\Big|\frac{\partial a_{20}}{\partial w_1}\Big| \le 4^{-20} = 2^{-40} \approx 9.1 \times 10^{-13}.$$

For mental arithmetic: $2^{40} = (2^{10})^4 = 1024^4 \approx (10^3)^4 = 10^{12}$, so the bound is roughly 10^{-12}. At that scale the first layer receives essentially no learning signal at any learning rate for which the last layers remain stable: the early layers are frozen while the late layers train.

Intuition. Nothing in the mechanism is specific to the sigmoid. Any deep composition whose local derivative factors are uniformly below 1 in magnitude — say bounded by $c < 1$ — has gradients decaying like c^L, and factors uniformly above 1 produce the mirror image, exploding gradients. Saturating activations manufacture small factors by construction: their derivative is near zero on most of the input range, so deep products of such factors collapse exponentially.

Remark 9.5 (Architectural fixes and the gradient path each one preserves). Every standard fix attacks a specific factor of the product (9.1); naming *which* factor is what an interviewer listens for.

- *ReLU.* With $\mathrm{ReLU}(z) = \max(z, 0)$ the derivative is exactly 1 wherever the unit is active ($z > 0$), so the per-layer shrink factor of at most $\frac{1}{4}$ becomes 1: an active path passes the gradient through undiminished.
- *Residual connections.* A residual block computes $h + f(h)$, whose derivative $1 + f'(h)$ contains an identity term. Composed over depth, there is a gradient path built purely from identity terms — additive rather than multiplicative — so signal reaches early layers without traversing the full product of derivatives.
- *LSTM/GRU gating.* The cell state is updated *additively* through gates; when the forget gate sits near 1, gradient propagation through time is a near-identity, sum-like highway rather than a product of Jacobians at every step.
- *Batch normalisation.* Normalising pre-activations keeps each z_ℓ in the non-saturated region of the activation, so the factors $\sigma'(z_\ell)$ are not driven toward 0 in the first place.

9.2 Self-Attention

Scaled dot-product attention is the most frequently requested formula in modern deep-learning interviews, and the follow-up is nearly always the same: why divide by $\sqrt{d_k}$? The complete answer is a one-line variance computation (Proposition 9.7) combined with an understanding of softmax saturation — the same pathology as the saturated sigmoid factors behind Proposition 9.3.

Setup. Fix a sequence length T and head dimensions d_k (queries and keys) and d_v (values). The inputs are matrices $Q, K \in \mathbb{R}^{T\times d_k}$ and $V \in \mathbb{R}^{T\times d_v}$, whose rows q_i, k_j, v_j are the query, key, and value vectors of individual tokens. For a vector $z \in \mathbb{R}^T$ the softmax is defined componentwise by

$$\operatorname{softmax}(z)_i = \frac{e^{z_i}}{\sum_{j=1}^{T} e^{z_j}},$$

and applied to a matrix it acts row by row. The resulting attention weights are written α_{ij}.

Definition 9.6 (Scaled dot-product attention)**.** Given Q, K, V as in the Setup, *scaled dot-product attention* is the map

$$\operatorname{Attention}(Q, K, V) = \operatorname{softmax}\Big(\frac{QK^\top}{\sqrt{d_k}}\Big) V, \tag{9.2}$$

with the softmax applied row-wise. The (i, j) entry of $QK^\top$ is the *score* $q_i \cdot k_j$, measuring the match between query i and key j; row i of the output is the convex combination $\sum_{j=1}^{T} \alpha_{ij}\, v_j$ of the value vectors, with weights α_{ij} given by the softmax of the scaled scores.

Proposition 9.7 (Variance of an unscaled attention score)**.** *Let $q, k \in \mathbb{R}^{d_k}$, and suppose the $2d_k$ components $q_1, \dots, q_{d_k}, k_1, \dots, k_{d_k}$ are jointly independent, each with mean 0 and variance 1. Then the unscaled score $s = q \cdot k = \sum_{i=1}^{d_k} q_i k_i$ satisfies*

$$\mathbb{E}[s] = 0, \qquad \operatorname{Var}(s) = d_k.$$

Proof. We compute directly, using independence to factor every mixed moment. Each term has mean $\mathbb{E}[q_i k_i] = \mathbb{E}[q_i]\,\mathbb{E}[k_i] = 0$, so $\mathbb{E}[s] = 0$ by linearity of expectation. Each term's variance is

$$\operatorname{Var}(q_i k_i) = \mathbb{E}[q_i^2 k_i^2] - \big(\mathbb{E}[q_i k_i]\big)^2 = \mathbb{E}[q_i^2]\,\mathbb{E}[k_i^2] - 0 = 1 \cdot 1 = 1,$$

where the second equality uses independence of q_i and k_i and the fact that mean 0 and variance 1 force $\mathbb{E}[q_i^2] = \mathbb{E}[k_i^2] = 1$. For $i \neq j$ the terms are uncorrelated:

$$\operatorname{Cov}(q_i k_i,\ q_j k_j) = \mathbb{E}[q_i k_i q_j k_j] - 0 = \mathbb{E}[q_i]\,\mathbb{E}[k_i]\,\mathbb{E}[q_j]\,\mathbb{E}[k_j] = 0,$$

again by joint independence. The variance of a sum of uncorrelated terms is the sum of their variances, which is d_k. □

(Problem P9.2 asks you to reproduce this computation.)

Corollary 9.8 (Unit variance under $\sqrt{d_k}$ scaling)**.** *In the setting of Proposition 9.7, the scaled score $s/\sqrt{d_k}$ has mean 0 and variance 1, for every head dimension d_k.*

Proof. The claim follows from the scaling rule for variance. With $c = 1/\sqrt{d_k}$, linearity gives $\mathbb{E}[cs] = 0$, and $\operatorname{Var}(cs) = c^2 \operatorname{Var}(s) = d_k/d_k = 1$ by Proposition 9.7. □

Unscaled scores therefore grow like $\sqrt{d_k}$ in standard deviation, while the scaled scores stay at unit scale regardless of head dimension. The next example shows what the unscaled growth does to the softmax.

Example 9.9 (Saturation at $d_k = 64$)**.** Take $d_k = 64$, a standard head dimension. By Proposition 9.7, unscaled scores have standard deviation $\sqrt{64} = 8$, so the *gap* between two competing logits — the difference of two independent scores — has standard deviation $8\sqrt{2} \approx 11.3$. For two logits the softmax reduces to a sigmoid of the gap: with $\Delta = z_1 - z_2$,

$$\text{softmax}(z_1, z_2)_1 = \frac{e^{z_1}}{e^{z_1} + e^{z_2}} = \frac{1}{1 + e^{-\Delta}} = \sigma(\Delta),$$

the sigmoid of the Setup of §9.1. At a typical unscaled gap $\Delta \approx 11$,

$$\sigma(11) = \frac{1}{1 + e^{-11}} \approx 1 - 1.7 \times 10^{-5},$$

essentially one-hot: at initialisation, attention freezes onto single random tokens. After dividing by $\sqrt{d_k} = 8$, the gap has standard deviation $\sqrt{2} \approx 1.4$, and $\sigma(1.4) \approx 0.80$: weights like (0.80, 0.20), a diffuse distribution that training can still reshape. Figure 16 shows both regimes on the softmax curve.

Remark 9.10 (Softmax saturation kills the gradient). The softmax Jacobian is $\partial\alpha_i/\partial z_j = \alpha_i(\mathbf{1}\{i = j\} - \alpha_j)$. At a near-one-hot output every entry is close to 0: for each pair (i, j), either $\alpha_i \approx 0$ or $\mathbf{1}\{i = j\} - \alpha_j \approx 0$ (diagonal entries $\alpha_i(1 - \alpha_i)$ die because α_i is near 0 or 1; off-diagonal entries $-\alpha_i\alpha_j$ die because at most one weight is non-negligible). Almost no gradient flows back to the scores, so the attention pattern cannot learn its way out of its random initial assignment — the same mechanism as the vanished sigmoid factors of Proposition 9.3, where a saturated squashing function contributes a near-zero derivative factor. Dividing by $\sqrt{d_k}$ keeps the score variance at 1 for every head dimension (Corollary 9.8), which keeps the softmax in its sensitive regime at initialisation.

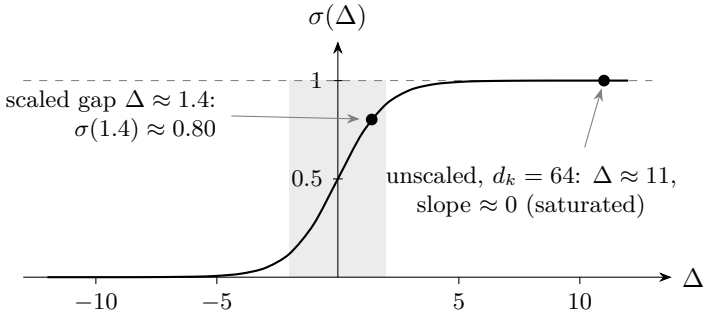


Figure 16: Softmax saturation for two competing logits: the larger logit's weight is $\sigma(\Delta)$, the sigmoid of the gap Δ (Example 9.9). The shaded band $|\Delta| \leq 2$ is the sensitive regime where the curve has usable slope. Scaled scores (unit variance, Corollary 9.8) place typical gaps inside the band; unscaled scores at $d_k = 64$ place them near $\Delta \approx 11$, deep in the flat tail where the gradient through the softmax is essentially zero.

9.3 Transformer Limitations (Tick-Data Question)

A recurring interview prompt asks why one should not run a transformer directly on raw tick data. A precise answer separates attention's memory cost from its compute cost, and training cost from streaming-inference cost. These are engineering facts about implementations rather than theorems, and they are stated as such. Throughout this subsection T denotes the attention sequence length, as in §9.2 (the symbol L already denotes network depth in this section).

Remark 9.11 (What exactly is quadratic). Naive attention materialises the full $T \times T$ score matrix $QK^\top$ of (9.2): both memory and compute are $O(T^2)$ per head per layer. FlashAttention removes the memory problem, not the compute problem: it streams tiles of the score matrix through

fast on-chip memory without ever materialising the whole matrix, reducing attention memory to $O(T)$ — but every query–key pair is still computed, so *compute remains* $O(T^2)$. The scale is easy to underestimate: a liquid instrument prints on the order of 5×10^5 ticks per day, so a full-day window means $T^2 = 2.5 \times 10^{11}$ score entries per head per layer — roughly 500 GB at two bytes per entry if materialised, far beyond any single accelerator's memory.

Interview trap. Claiming that "FlashAttention makes attention linear" is a half-truth that interviewers probe. FlashAttention linearises *memory* only; compute stays quadratic in the sequence length. Be precise about which resource each statement refers to, and state both halves.

Trading application. For tick data the quadratic profile is the wrong shape twice over. Training must attend over years of history at $O(T^2)$ compute per sequence; and streaming inference re-attends over a growing window each time a new tick arrives, so the per-tick cost grows with the session. The standard alternatives, in the order worth naming:

- *State-space models.* S4 trains in $O(T \log T)$ and Mamba in $O(T)$; both run recurrently at inference — a fixed-size state updated in $O(1)$ per step — which is the right computational shape for live trading.
- *Linear and sparse attention* variants, which cut the quadratic cost by approximating or restricting the score matrix.
- *The pragmatic first answer:* aggregate ticks into bars or engineered features before any sequence model sees the data, shrinking T by orders of magnitude while injecting domain knowledge.

9.4 "The Model Worked 2005–2018, Then Died"

Another open-ended standard: a systematic strategy performed well for over a decade, then stopped working. The question tests whether a candidate organises hypotheses systematically rather than guessing a single cause; the expected structure is the checklist below.

Remark 9.12 (Five failure modes of a dead strategy). Five distinct hypotheses explain a strategy that worked and then died, and a complete answer names all five:

1. *Non-stationarity / regime change* — the statistical relationship the model exploits weakened as the market changed.
2. *Overfitting all along* — the historical performance was in-sample luck; no real edge ever existed.
3. *Alpha decay* — the edge was real but has been arbitraged away as other participants found it.
4. *Look-ahead bias* — the backtest used information not available at the time, so the historical performance was never achievable live.
5. *Structural change* — regulation, new participants, or market-structure shifts broke the mechanism the model relied on.

Trading application. In an interview, a strong opening response commits to a diagnostic plan rather than a verdict: suspect regime change or overfitting first (Remark 9.12, items 1 and 2), check feature stationarity over time, re-run walk-forward validation, and test whether the model over-relies on patterns with no causal story.

9.5 numpy, PyTorch and GPUs

Numerical-stack questions ★ test whether a candidate understands why the standard tools are fast, one level below the API. The three standard questions — why numpy beats lists, why PyTorch exists at all, and what a GPU actually buys — are collected here as engineering facts.

Remark 9.13 (Why numpy arrays beat Python lists). A Python list of floats is an array of *pointers* to individually heap-allocated Python objects: every element is boxed with a type tag and a reference count, the elements live wherever the allocator placed them, and every operation passes through the interpreter's dynamic dispatch — the interpreter loop pays that dispatch cost on every single element. A numpy array is one *contiguous, typed* block of memory, and a vectorised operation on it runs as a compiled C loop with SIMD instructions over cache-friendly memory: no per-element boxing, no pointer chasing, no per-element reference counting, and no per-element interpreter dispatch. The speedup is a memory-layout and compiled-code story, not a parallelism story.

Remark 9.14 (Why PyTorch does not simply use numpy). Tensors carry three things numpy arrays lack. First, *autograd*: every operation records a node in the computational graph — exactly the structure drawn in Figure 15 — so the backward pass of Example 9.2 runs automatically over arbitrary programs. Second, *device placement*: the same tensor API executes on GPU through CUDA, with asynchronous kernel launches. Third, kernels *fused and optimised* for deep-learning workloads and shapes. The numpy library, by contrast, is CPU-only, records no graph, and executes synchronously.

Fact 9.15 (GPU versus CPU speedup, empirical)**.** For large parallel workloads — big matrix multiplications above all — a GPU delivers on the order of 10–100× over a CPU. The advantage shrinks or disappears for small, branchy, or memory-bound workloads, and host–device transfers over PCIe can consume the entire gain. These are measured engineering regularities, not derivable constants.

Interview trap. Quoting "10–100×" without qualification. The expected answer volunteers the caveats unprompted: the range holds for large parallel workloads such as big matrix multiplications, and it shrinks or dies when the workload is small, branchy, or memory-bound — or when PCIe transfer time dominates the computation it feeds.

Practice: problems P9.1–P9.11 in §27, solutions in §28.

Part V

Trading, Portfolios, Market Making, and Market Structure

10 Sharpe, Loss Probabilities and Model Combination

This section connects the single most quoted performance statistic in systematic trading — the Sharpe ratio — to the quantity a risk manager actually cares about: the probability of losing money over a given horizon. We first derive the exact loss probability under a Gaussian model and establish how the Sharpe ratio scales with the observation horizon. We then work through a complete assessment-style loss calculation and audit every assumption behind it, quantify the model risk with Cantelli's distribution-free bound, and finish with the algebra of combining strategies: how variances add, why uncorrelated legs earn a $\sqrt{2}$ Sharpe improvement, and how to decorrelate a new model from an existing black box.

10.1 The Sharpe Ratio and the Probability of Loss

This subsection derives the identity $P(R < 0) = \Phi(-\mathrm{SR})$ and the square-root-of-time scaling law for the Sharpe ratio. Together they answer the two most common warm-up questions on strategy

risk: "how often does a Sharpe-s strategy lose money?" and "why does the same strategy look so much worse on a monthly statement than on an annual one?"

Setup. Let the annual P&L of a strategy be a random variable R (measured in currency units; the scale plays no role) with

$$R \sim \mathcal{N}(\mu, \sigma^2), \qquad \mu \in \mathbb{R}, \quad \sigma > 0,$$

where μ is the annual mean and σ the annual volatility. Recall that Φ denotes the standard normal CDF and that a standardised normal variable $Z = (R - \mu)/\sigma$ satisfies $Z \sim \mathcal{N}(0, 1)$.

Definition 10.1 (Sharpe ratio)**.** The (annualised) *Sharpe ratio* of the strategy is

$$\mathrm{SR} = \frac{\mu}{\sigma}.$$

Throughout this document the ratio is taken directly as mean over volatility, with no risk-free adjustment; interview problems intend this convention unless they state a risk-free rate explicitly.

Theorem 10.2 (Loss probability under normality)**.** *Let $R \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma > 0$ and $\mathrm{SR} = \mu/\sigma$. Then*

$$P(R < 0) = \Phi(-\mathrm{SR}).$$

Proof. We standardise. Set $Z = (R-\mu)/\sigma$, so that $Z \sim \mathcal{N}(0, 1)$. Dividing by the positive constant σ preserves the inequality, so the events coincide:

$$\{R < 0\} = \Big\{\frac{R - \mu}{\sigma} < \frac{0 - \mu}{\sigma}\Big\} = \{Z < -\mathrm{SR}\}.$$

Taking probabilities and using the definition of the standard normal CDF gives $P(R < 0) = P(Z < -\mathrm{SR}) = \Phi(-\mathrm{SR})$. □

Setup for time scaling. Now let the strategy's P&L over each of T consecutive unit periods be $X_1, \ldots, X_T$, assumed iid with per-period mean μ_1 and per-period standard deviation $\sigma_1 > 0$, and write $S_T = X_1 + \cdots + X_T$ for the cumulative P&L. The per-period Sharpe ratio is $\mathrm{SR}_1 = \mu_1/\sigma_1$.

Proposition 10.3 (Sharpe time-scaling under iid returns)**.** *Under the setup above,*

$$\mathbb{E}[S_T] = T\mu_1 \qquad \textit{and} \qquad \sqrt{\mathrm{Var}(S_T)} = \sigma_1\sqrt{T},$$

and hence the horizon-T Sharpe ratio is

$$\mathrm{SR}_T \;=\; \frac{\mathbb{E}[S_T]}{\sqrt{\mathrm{Var}(S_T)}} \;=\; \mathrm{SR}_1\sqrt{T}.$$

In particular, taking the month as the unit period and the year as $T = 12$ months, $\mathrm{SR}_{\text{annual}} = \sqrt{12}\,\mathrm{SR}_{\text{monthly}}$, so a strategy with annual Sharpe ratio SR has monthly Sharpe ratio $\mathrm{SR}_M = \mathrm{SR}/\sqrt{12}$; and over a horizon of T years, $\mathrm{SR}_T = \mathrm{SR}_{\text{annual}}\sqrt{T}$.

Proof. We compute the mean by linearity of expectation and the variance by expanding the square. Linearity gives

$$\mathbb{E}[S_T] = \sum_{t=1}^{T} \mathbb{E}[X_t] = T\mu_1,$$

which holds with no independence assumption. For the variance, write $\bar{X}_t = X_t - \mu_1$ for the centred increments, so that $S_T - T\mu_1 = \sum_t \bar{X}_t$, and expand:

$$\begin{aligned}\operatorname{Var}(S_T) = \mathbb{E}\Big[\Big(\sum_{t=1}^{T} \bar{X}_t\Big)^2\Big] &= \sum_{t=1}^{T} \mathbb{E}[\bar{X}_t^2] + \sum_{s \neq t} \mathbb{E}[\bar{X}_s \bar{X}_t] \\ &= T\sigma_1^2 + \sum_{s \neq t} \operatorname{Cov}(X_s, X_t).\end{aligned}$$

Independence makes every cross term vanish, leaving $\operatorname{Var}(S_T) = T\sigma_1^2$ and $\sqrt{\operatorname{Var}(S_T)} = \sigma_1\sqrt{T}$. Dividing,

$$\mathrm{SR}_T = \frac{T\mu_1}{\sigma_1\sqrt{T}} = \sqrt{T}\,\frac{\mu_1}{\sigma_1} = \mathrm{SR}_1\sqrt{T}. \qquad \square$$

Example 10.4 (Loss frequencies at Sharpe one)**.** Consider a strategy with annual Sharpe ratio 1 and normally distributed, iid returns. By Theorem 10.2, the probability of a losing *year* is

$$\Phi(-1) \approx 15.9\%.$$

By Proposition 10.3, the monthly Sharpe ratio is $1/\sqrt{12} \approx 0.289$, so the probability of a losing *month* is

$$\Phi(-0.289) \approx 38.6\%.$$

The same strategy loses money in roughly one year in six but in nearly two months in five — nothing about the strategy has changed, only the clock on which it is read.

Intuition. Shrinking the horizon shrinks the drift linearly but the noise only as the square root, so the signal-to-noise ratio of the P&L degrades like $\sqrt{T}$. At short horizons the drift is invisible inside the noise and the loss frequency drifts toward $1/2$; as $T \to \infty$ it tends to 0. Watching a good strategy at too fine a time scale makes it feel like a coin flip.

Interview arithmetic in this area runs through a handful of standard normal lower-tail values. Table 2 collects the four worth memorising.

k	1	2	3	4
$\Phi(-k)$	15.9%	2.28%	0.135%	3.2×10^{-5}

Table 2: Normal lower-tail anchors (memorise).

10.2 A Worked Loss Probability and Its Assumption Audit

The identity of Theorem 10.2 turns a Sharpe ratio into a loss probability in one line, and assessment questions exploit exactly that. The example below is a real assessment question; the audit that follows it is the part of the answer that separates a complete response from a mere calculation.

Example 10.5 (Loss probability at Sharpe two ★)**.** A strategy has an annual Sharpe ratio of 2 and annual P&L volatility of \$50M. What is the probability of losing at least \$100M in a year?

Solution. Work in units of \$M and model the annual P&L as $X \sim \mathcal{N}(\mu, \sigma^2)$ with $\sigma = 50$. Definition 10.1 recovers the mean from the two given numbers:

$$\mu = \mathrm{SR} \times \sigma = 2 \times 50 = 100.$$

A loss of at least \$100M is the event $\{X \leq -100\}$. Standardising as in the proof of Theorem 10.2, the threshold sits at

$$z = \frac{-100 - 100}{50} = -4$$

standard deviations, so by Table 2

$$P(X \le -100) = \Phi(-4) \approx 3.2 \times 10^{-5}.$$

Figure 17 locates this threshold in the standard normal density.

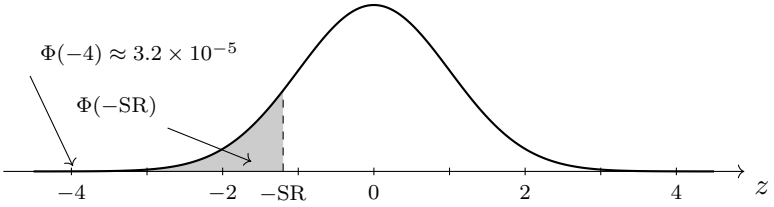


Figure 17: The standard normal density with the lower tail beyond $-$SR shaded (the threshold is drawn at $z = -1.2$ so the region is visible; for a Sharpe-1 strategy it would sit at -1). By Theorem 10.2 the shaded mass $\Phi(-\mathrm{SR})$ is the probability of a losing period. The mark at $z = -4$ locates the \$100M loss threshold of Example 10.5: essentially no Gaussian mass lies beyond it, which is exactly why the assumption audit of Remark 10.7 — and the distribution-free bound of Theorem 10.8 — matter.

The number $\Phi(-4)$ is only as good as the model behind it, and an interviewer will press on exactly that. One of the assumptions concerns what "losing \$100M" means: terminal P&L at year end, or the running P&L ever touching $-$\$100M during the year. The following classical result quantifies the difference.

Fact 10.6 (Reflection bound for path losses)**.** Let $(W_t)_{t\ge 0}$ be a driftless Brownian motion, $\sigma > 0$, and $a > 0$. By the reflection principle (classical, going back to Bachelier, 1900), the driftless P&L path σW_t satisfies

$$P\Big(\min_{0\le t\le T} \sigma W_t \le -a\Big) \;=\; 2\,P(\sigma W_T \le -a) \;=\; 2\,\Phi\Big(\frac{-a}{\sigma\sqrt{T}}\Big)\,:$$

the path probability is *exactly* twice the terminal one. Adding a positive drift $\mu > 0$ pushes the path probability strictly *above* twice the terminal probability: conditional on the running P&L touching the loss level $-a$, the upward drift makes finishing below it less likely than $1/2$, so the path-to-terminal ratio exceeds two and can be arbitrarily large — the path-versus-terminal correction is even larger than the reflection factor of two. The exact formula for the drifted process $X_t = \mu t + \sigma W_t$ is

$$P\Big(\min_{0\le t\le T} X_t \le -a\Big) \;=\; \Phi\Big(\frac{-a-\mu T}{\sigma\sqrt{T}}\Big) \;+\; e^{-2\mu a/\sigma^2}\,\Phi\Big(\frac{-a+\mu T}{\sigma\sqrt{T}}\Big),$$

whose first term is the terminal probability $P(X_T \le -a)$. As a consistency check, at $\mu = 0$ the exponential factor equals 1 and both normal terms reduce to $\Phi(-a/(\sigma\sqrt{T}))$, so the formula collapses to exactly the driftless identity above. Every claim of this Fact is proved in Theorem 19.63 of §19, once Brownian motion and the Itô integral have been built.

Remark 10.7 (Demolishing the assumptions)*.* A complete answer to Example 10.5 names its assumptions and states how each one fails in reality:

1. *Normality.* Real P&L distributions are fat-tailed; the Gaussian badly underprices four-sigma events, so the true tail probability is higher than $\Phi(-4)$.
2. *Independence across time.* Returns exhibit autocorrelation in their scale — volatility clustering — so the iid aggregation implicit in the annual figure understates the tails.
3. *Constant volatility.* Volatility moves in regimes. Even if the P&L is conditionally normal within each regime, a scale mixture of normals is unconditionally fat-tailed.

4. *Stationarity of the Sharpe ratio itself.* The quoted SR = 2 is an estimate, typically a selected one (the best of many strategies), and edges decay; the true mean is likely lower, moving the loss threshold closer than four sigmas.
5. *Path versus terminal.* If "losing \$100M" means the drawdown ever touching −\$100M during the year, the probability is strictly larger — for a driftless process exactly twice the terminal probability by Fact 10.6, and *more* than twice it under the positive drift of the example.

Every correction pushes in the same direction: the real loss probability is *higher* than $\Phi(-4)$.

10.3 Cantelli's Inequality: the Distribution-Free Bound

If the Gaussian answer of Example 10.5 is the optimistic end of the model spectrum, the pessimistic end is the best bound that uses *no* distributional assumption at all — only the mean and the variance. That bound is Cantelli's inequality, the one-sided sharpening of Chebyshev, and the gap between the two ends is the model-risk conversation the interviewer is fishing for.

Setup. Let X be any random variable with mean μ and finite variance $\sigma^2 \in (0, \infty)$; no other assumption is made on its distribution. Fix $k > 0$.

Theorem 10.8 (Cantelli's inequality, one-sided Chebyshev)**.** *Under the setup above,*

$$P(X - \mu \le -k\sigma) \;\le\; \frac{1}{1+k^2}.$$

Proof. We apply Markov's inequality to a shifted square and then optimise the shift. Let $Y = \mu - X$, so that $\mathbb{E}[Y] = 0$, $\mathrm{Var}(Y) = \sigma^2$, and the event of interest is $\{Y \ge k\sigma\}$.

Fix any $u \ge 0$. On the event $\{Y \ge k\sigma\}$ we have $Y + u \ge k\sigma + u > 0$, and squaring both sides of this inequality between positive quantities gives $(Y+u)^2 \ge (k\sigma + u)^2$. The variable $(Y+u)^2$ is nonnegative, so Markov's inequality applies:

$$P(Y \ge k\sigma) \;\le\; P\big((Y+u)^2 \ge (k\sigma+u)^2\big) \;\le\; \frac{\mathbb{E}[(Y+u)^2]}{(k\sigma+u)^2} \;=\; \frac{\sigma^2 + u^2}{(k\sigma+u)^2}, \tag{10.1}$$

where the last equality expands $\mathbb{E}[(Y+u)^2] = \mathbb{E}[Y^2] + 2u\,\mathbb{E}[Y] + u^2 = \sigma^2 + u^2$, using $\mathbb{E}[Y] = 0$.

The bound (10.1) holds for every $u \ge 0$, so we may minimise the right-hand side over u. Differentiating with respect to u,

$$\frac{d}{du}\,\frac{\sigma^2+u^2}{(k\sigma+u)^2} = \frac{2u(k\sigma+u)^2 - 2(k\sigma+u)(\sigma^2+u^2)}{(k\sigma+u)^4} = \frac{2\,(uk\sigma - \sigma^2)}{(k\sigma+u)^3},$$

which is negative for $u < \sigma/k$ and positive for $u > \sigma/k$; the minimiser is $u^* = \sigma/k$. Substituting,

$$\frac{\sigma^2 + \sigma^2/k^2}{(k\sigma + \sigma/k)^2} = \frac{\sigma^2\,(k^2+1)/k^2}{\sigma^2\,(k^2+1)^2/k^2} = \frac{1}{1+k^2}. \qquad \square$$

(Problem P10.3 asks you to reproduce this proof from scratch; it also shows the bound is tight, by exhibiting a two-point distribution that attains it, so no better distribution-free bound exists.)

Example 10.9 (The distribution-free ceiling at four sigma)**.** Return to the numbers of Example 10.5: annual mean P&L \$100M, volatility \$50M, distribution now *unknown*. Losing at least \$100M means $X \le -100 = \mu - 4\sigma$, a $k = 4$ shortfall, so Theorem 10.8 gives

$$P(X - \mu \le -4\sigma) \;\le\; \frac{1}{1+16} = \frac{1}{17} \approx 5.9\%.$$

The Gaussian model gave $\Phi(-4) \approx 3.2 \times 10^{-5}$; the distribution-free ceiling is larger by over three orders of magnitude (a factor of roughly two thousand).

Trading application. The gap between 3.2×10^{-5} and 5.9% *is* the model-risk conversation. Neither number is "the answer": the Gaussian figure is the best case of one specific thin-tailed model, the Cantelli figure is the worst case over all distributions with that mean and volatility, and every number in between corresponds to a choice of tail assumption. In an interview, naming that gap — and saying which way reality leans, per Remark 10.7 — is the point of the question.

10.4 Combining Models: $\mathrm{Var}(A+B)$ and the Root-Two Rule

A desk rarely runs one strategy. This subsection establishes the variance algebra of running two strategies side by side and proves the classic diversification result: two uncorrelated legs of equal Sharpe and equal risk combine to a Sharpe $\sqrt{2}$ times as large.

Setup. In this section, A and B denote the P&Ls of two strategies over a common period — random variables with finite, strictly positive variances, *not* matrices (the document-wide matrix notation is suspended here; no matrix appears in this section). Write $\sigma_A^2 = \mathrm{Var}(A)$, $\sigma_B^2 = \mathrm{Var}(B)$, and $\rho = \mathrm{Corr}(A,B) = \mathrm{Cov}(A,B)/(\sigma_A\sigma_B)$.

Proposition 10.10 (Variance of a sum)**.** *For any random variables A and B with finite variances,*

$$\mathrm{Var}(A+B) \;=\; \mathrm{Var}(A) + \mathrm{Var}(B) + 2\,\mathrm{Cov}(A,B).$$

Proof. We expand the square and apply linearity of expectation. Write $\bar{A} = A - \mathbb{E}[A]$ and $\bar{B} = B - \mathbb{E}[B]$ for the centred variables, and note $A + B - \mathbb{E}[A+B] = \bar{A} + \bar{B}$ by linearity. Then

$$\begin{aligned}\mathrm{Var}(A+B) = \mathbb{E}\big[(\bar{A}+\bar{B})^2\big] &= \mathbb{E}[\bar{A}^2] + 2\,\mathbb{E}[\bar{A}\bar{B}] + \mathbb{E}[\bar{B}^2] \\ &= \mathrm{Var}(A) + 2\,\mathrm{Cov}(A,B) + \mathrm{Var}(B),\end{aligned}$$

where the middle equality expands the square inside the expectation and the last one applies the definitions of variance and covariance term by term. □

Intuition. The covariance term is the whole game. Positive correlation makes risk superadditive: the two legs are partly the same bet, and running both buys redundancy. Negative correlation is diversification in its strongest form — hedging — and the combined risk can fall below that of either leg alone. Zero correlation is the neutral case in which risks add in quadrature, and it is where the cleanest Sharpe arithmetic lives.

Theorem 10.11 (Root-two Sharpe combination)**.** *Let A and B be uncorrelated ($\rho = 0$) with a common mean $\mu > 0$ and a common standard deviation $\sigma > 0$ — equal Sharpe ratios* $\mathrm{SR}_1 = \mu/\sigma$ *and equal volatilities. Then the combined book $A + B$ has Sharpe ratio*

$$\mathrm{SR}_{A+B} \;=\; \sqrt{2}\,\mathrm{SR}_1.$$

Proof. We compute the mean and the variance of the sum directly. Linearity of expectation gives $\mathbb{E}[A+B] = 2\mu$. Proposition 10.10 with $\mathrm{Cov}(A,B) = 0$ gives $\mathrm{Var}(A+B) = \sigma^2 + \sigma^2 = 2\sigma^2$, hence a combined volatility of $\sigma\sqrt{2}$. Dividing,

$$\mathrm{SR}_{A+B} = \frac{2\mu}{\sigma\sqrt{2}} = \sqrt{2}\,\frac{\mu}{\sigma} = \sqrt{2}\,\mathrm{SR}_1.$$ □

Corollary 10.12 (Quadrature rule for equal-risk uncorrelated legs)**.** *Let $A_1, \dots, A_N$ be pairwise uncorrelated, each with mean $\mu > 0$ and standard deviation $\sigma > 0$. Then the book $A_1 + \cdots + A_N$ has Sharpe ratio*

$$\mathrm{SR}_N \;=\; \sqrt{N}\,\mathrm{SR}_1 \;=\; \sqrt{\textstyle\sum_{i=1}^{N} \mathrm{SR}_i^2},$$

the last equality holding because all the SR_i *equal* SR_1*.*

Proof. We repeat the argument of Theorem 10.11 with N summands. The mean is $N\mu$ by linearity. Expanding the square of the centred sum as in the proof of Proposition 10.3, all $N(N-1)$ cross terms are covariances of distinct legs and vanish by pairwise uncorrelatedness, so the variance is $N\sigma^2$ and the volatility $\sigma\sqrt{N}$. Hence $\mathrm{SR}_N = N\mu/(\sigma\sqrt{N}) = \sqrt{N}\,\mu/\sigma$, and $\sqrt{N}\,\mathrm{SR}_1 = \sqrt{N\,\mathrm{SR}_1^2} = \sqrt{\sum_i \mathrm{SR}_i^2}$. □

Example 10.13 (Two legs at three correlations)**.** Strategies A and B each have annual mean P&L \$20M and annual volatility \$20M (Sharpe 1 each). Working in \$M, Proposition 10.10 gives $\mathrm{Var}(A+B) = 400 + 400 + 2\rho \cdot 400 = 800(1+\rho)$, while the combined mean is always 40.

- At $\rho = 0.6$: variance 1280, volatility $\sqrt{1280} \approx$ \$35.8M, Sharpe $40/35.8 \approx 1.12$ — redundancy, a modest gain over a single leg.
- At $\rho = 0$: variance 800, volatility $\approx$ \$28.3M, Sharpe $40/28.3 \approx 1.41 = \sqrt{2}$, confirming Theorem 10.11.
- At $\rho = -0.6$: variance 320, volatility $\approx$ \$17.9M — *below either leg's* \$20M — and Sharpe $40/17.9 \approx 2.24$: hedging with positive carry.

Remark 10.14 (Risk-balanced weights, not a raw sum). For uncorrelated legs with *equal* Sharpe ratios but *unequal* volatilities, the quadrature formula $\mathrm{SR} = \sqrt{\sum_i \mathrm{SR}_i^2} = \sqrt{N}\,\mathrm{SR}_1$ requires risk-balanced weights, not a raw sum. A raw dollar sum weights each leg by its volatility: a leg with tiny volatility contributes almost nothing to the book's P&L, while a high-volatility leg dominates the book's risk. Scaling each leg to a common volatility (weights proportional to $1/\sigma_i$) restores the symmetric setting of Corollary 10.12, and only then does the quadrature formula hold. The equal-Sharpe hypothesis is essential: with unequal Sharpe ratios, the equal-risk book earns $(\sum_i \mathrm{SR}_i)/\sqrt{N}$, which by the Cauchy–Schwarz inequality is strictly below $\sqrt{\sum_i \mathrm{SR}_i^2}$ unless all the SR_i are equal; attaining the quadrature value in general requires allocating risk in proportion to each leg's Sharpe ratio — the mean–variance weights $w_i \propto \mu_i/\sigma_i^2$ — which coincide with $1/\sigma_i$ weights only in the equal-Sharpe case.

10.5 Decorrelating a New Model from a Black Box

Corollary 10.12 makes low correlation between models directly valuable, which raises a practical question: a desk already runs a profitable production model whose internals cannot be inspected, and a newly built model correlates substantially with it — how should the new model be decorrelated? This subsection gives the exact one-line answer, two alternatives, and the interview trap lurking next to them.

Setup. Let A denote the output (or P&L) of the production black-box model — observable, but with hidden internals — and let B denote the output of the new model. Both are random variables with finite variances, $\mathrm{Var}(A) > 0$, and $\rho = \mathrm{Corr}(A, B)$. The new model is trained on features $X_1, \ldots, X_p$, and its training loss is denoted $\mathcal{L}$.

Proposition 10.15 (Exact decorrelation by regression residual)**.** *Set*

$$\beta^* = \frac{\mathrm{Cov}(A,B)}{\mathrm{Var}(A)} \qquad \textit{and} \qquad B_\perp = B - \beta^* A.$$

Then $\mathrm{Cov}(B_\perp, A) = 0$*, and* β^* *is the unique coefficient with this property. Moreover*

$$\mathrm{Var}(B_\perp) \;=\; \mathrm{Var}(B) - \frac{\mathrm{Cov}(A,B)^2}{\mathrm{Var}(A)} \;=\; (1-\rho^2)\,\mathrm{Var}(B).$$

Proof. We use bilinearity of covariance. For any coefficient β,

$$\mathrm{Cov}(B - \beta A,\, A) = \mathrm{Cov}(B, A) - \beta\,\mathrm{Var}(A),$$

which vanishes if and only if $\beta = \mathrm{Cov}(A,B)/\mathrm{Var}(A) = \beta^*$; this proves both the decorrelation and the uniqueness. For the variance, apply Proposition 10.10 to the sum $B + (-\beta^* A)$:

$$\mathrm{Var}(B_\perp) = \mathrm{Var}(B) + (\beta^*)^2 \mathrm{Var}(A) - 2\beta^* \mathrm{Cov}(A,B) = \mathrm{Var}(B) - \frac{\mathrm{Cov}(A,B)^2}{\mathrm{Var}(A)},$$

substituting β^* and simplifying. Writing $\mathrm{Cov}(A,B)^2 = \rho^2 \mathrm{Var}(A)\mathrm{Var}(B)$ gives the final form. □

Example 10.16 (Orthogonalising at correlation 0.7)**.** Normalise both model outputs to unit variance and suppose $\rho = 0.7$. Then $\beta^* = \mathrm{Cov}(A,B) = 0.7$, the residual model is $B_\perp = B - 0.7A$, and by Proposition 10.15

$$\mathrm{Var}(B_\perp) = 1 - 0.7^2 = 0.51,$$

a standard deviation of about 0.71. Nearly half of the new model's output variance was redundant exposure to the black box; what remains is exactly its incremental information, and by Theorem 10.11 it is the uncorrelated part that improves the book's Sharpe.

Three levers are available, in decreasing order of exactness:

1. *Orthogonalise the outputs.* Regress B on A and trade the residual $B_\perp$ of Proposition 10.15 — exact, and one line of algebra.
2. *Feature filtering.* Drop features X_j with high $\mathrm{Corr}(X_j, A)$ before training — crude, for the reasons the trap below makes precise.
3. *A correlation penalty in the loss.* Train B with $\mathcal{L} = \mathrm{MSE} + \lambda\,\mathrm{Corr}(A,B)^2$ — a continuous trade-off between standalone accuracy and orthogonality, tuned through λ.

Interview trap. "I dropped every feature correlated with A, then retrained. Thoughts?" Two independent failures: (i) the procedure discards genuinely predictive features — a feature can correlate with A's output and still carry incremental information A does not use; (ii) pairwise feature correlations do *not* bound the correlation of the trained model's combined output with A, so the procedure does not even achieve its stated goal. The cleaner answers are the exact one — orthogonalise the output — or the tunable one — constrain the loss.

Practice: problems P10.1–P10.6 in §27, solutions in §28.

11 Portfolio Theory: Mean–Variance Optimisation, Diversification, and the Capital Market Line

This section extends the two-strategy variance algebra of §10.4 — Proposition 10.10 and Theorem 10.11 — to n assets held in weights, following Markowitz (1952). It derives the minimum-variance and mean–variance-efficient portfolios by Lagrangian, shows that the frontier is a hyperbola spanned by two funds, adds a risk-free asset to obtain the tangency (maximum-Sharpe) portfolio and the capital market line, reads the capital asset pricing model off the tangency algebra, decomposes portfolio risk into Euler contributions (risk parity), and closes with the reason the optimiser fails when fed estimated inputs. The $r_f = 0$ convention of Definition 10.1 is kept until §11.4, where a risk-free rate enters. Two running examples recur: a two-asset pair with volatilities 10% and 20% and means 5% and 10%, and a hand-invertible three-asset universe with equal volatilities, means $(4\%, 7\%, 8\%)$ and $r_f = 2\%$, introduced in §11.2.

11.1 Portfolios, Weights and the Two-Asset Diversification Formula

The warm-up block at every quant fund is "write the variance of a two-asset portfolio", "why does diversification work", and the sharper follow-up "asset B has the same mean and twice the

volatility of asset A — should you hold any of it?". This subsection fixes the notation for the whole section, proves the portfolio mean and variance formulas (the quadratic form of Theorem 6.6; the n-fold form of Proposition 10.10), and settles the two-asset case in closed form, including the degenerate lines at $\rho = \pm 1$.

Setup. There are $n \geq 2$ assets with one-period return vector $R = (R_1, \ldots, R_n)^\top$, each R_i having finite second moments. The mean vector is $\mu = \mathbb{E}[R] \in \mathbb{R}^n$ — a *vector* throughout this section, unlike the scalar μ of §10. The covariance matrix is $\Sigma \in \mathbb{R}^{n \times n}$ with entries

$$\Sigma_{ij} = \mathrm{Cov}(R_i, R_j) = \rho_{ij}\, \sigma_i \sigma_j, \qquad \sigma_i = \sqrt{\Sigma_{ii}} > 0,$$

where ρ_{ij} is the correlation of R_i and R_j (so $\rho_{ii} = 1$). The all-ones vector is $\mathbf{1} \in \mathbb{R}^n$ and e_i is the i-th standard basis vector. The asset count is n, not N, because N is the random-walk target of the notation table and the leg count of §10.4. By Theorem 6.6, Σ is symmetric positive semidefinite. *Standing assumption:* wherever Σ^{-1} appears, Σ is positive definite (PD), which by Theorem 6.4 means that every eigenvalue is strictly positive, equivalently that no nonzero w has $w^\top \Sigma w = 0$; since $w^\top \Sigma w = \mathrm{Var}(w^\top R)$ by the identity displayed in the proof of Theorem 6.6, the assumption says that no riskless combination of the assets exists. The pairs with $\rho_{12} = \pm 1$ of Proposition 11.4(iv) below are the canonical failures; Remark 6.8 explains why an *estimated* Σ can fail even semidefiniteness.

Symbol notice for this section. The scalars A, B, C, D are the frontier constants of §11.3, so a generic positive-definite matrix is written G. Lagrange multipliers are θ_1, θ_2 because λ is an eigenvalue; β is the population CAPM beta of §11.5, whose sample estimate is the $\hat{\beta}$ of §5. Every other local symbol is declared in the Setup of the subsection that uses it.

Definition 11.1 (Portfolio, weights and portfolio return)**.** A *portfolio* is a weight vector $w \in \mathbb{R}^n$, entry w_i being the fraction of capital placed in asset i. It is *fully invested* if $\mathbf{1}^\top w = 1$, and *long-only* if additionally $w_i \geq 0$ for every i; short positions $w_i < 0$ are allowed unless "long-only" is stated. Its return is the random variable

$$R_w = w^\top R = \sum_{i=1}^{n} w_i R_i,$$

itself a return. The *Sharpe ratio* of a portfolio with $w^\top \Sigma w > 0$ is

$$\mathrm{SR}(w) = \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}},$$

the $r_f = 0$ convention of Definition 10.1; Definition 11.22 in §11.4 generalises it to a risk-free rate.

Proposition 11.2 (Mean and variance of a portfolio)**.** *For every $w \in \mathbb{R}^n$,*

$$\mathbb{E}[R_w] = w^\top \mu \qquad \textit{and} \qquad \mathrm{Var}(R_w) = w^\top \Sigma w,$$

and, written out,

$$\mathrm{Var}(R_w) = \sum_{i=1}^{n} w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j\, \rho_{ij} \sigma_i \sigma_j. \tag{11.1}$$

In particular $\mathrm{Var}(R_w) \geq 0$ *for every* w*.*

Proof. We use linearity of expectation for the mean and cite the covariance identity of Theorem 6.6 for the variance. Linearity gives $\mathbb{E}[R_w] = \sum_i w_i\, \mathbb{E}[R_i] = w^\top \mu$. The identity $x^\top \Sigma x = \mathrm{Var}(x^\top Z)$ displayed in the proof of Theorem 6.6, read with $Z = R$ and $x = w$, gives $\mathrm{Var}(R_w) = w^\top \Sigma w \geq 0$. Writing the quadratic form as the double sum $\sum_{i,j} w_i w_j \Sigma_{ij}$ and splitting it into diagonal terms ($\Sigma_{ii} = \sigma_i^2$) and off-diagonal terms ($\Sigma_{ij} = \rho_{ij} \sigma_i \sigma_j$) gives (11.1). □

The case $n = 2$ is Proposition 10.10 applied to the two legs w_1R_1 and w_2R_2. The formula does not need the budget constraint; only the Sharpe ratio of Definition 11.1 is meant for fully invested portfolios.

Example 11.3 (The 50/50 mix)**.** Take $\sigma_1 = 10\%$, $\sigma_2 = 20\%$, $\mu = (5\%, 10\%)^\top$, $\rho_{12} = 0.3$ and $w = (0.5, 0.5)^\top$. By Proposition 11.2 the mean is $0.5(0.05) + 0.5(0.10) = 7.5\%$, the plain average of the two means, and the variance is

$$0.25(0.01) + 0.25(0.04) + 2(0.25)(0.3)(0.10)(0.20) = 0.0025 + 0.0100 + 0.0030 = 0.0155,$$

a volatility of $\sqrt{0.0155} = 12.45\%$, against the weight-average of the volatilities, $0.5(10\%) + 0.5(20\%) = 15\%$. The mean is exactly the weighted average; the volatility is strictly below it, and the 2.55-point gap is diversification. Example 6.7 evaluates the same quadratic form on the spread weights $(1, -1)^\top$, which do not sum to one: the formula is the same, only the interpretation as a return on capital is lost.

Setup (two assets). Let $n = 2$ with volatilities $\sigma_1, \sigma_2 > 0$ and correlation $\rho = \rho_{12} \in [-1, 1]$. A fully invested portfolio is $(w, 1 - w)$ for a single real number w, and we write $\sigma_p^2(w)$ for its variance and $\mu_p(w) = w\mu_1 + (1 - w)\mu_2$ for its mean.

Proposition 11.4 (Two-asset variance, the minimum-variance mix, and the degenerate lines)**.** *In the two-asset setup:*

(i) The variance is

$$\sigma_p^2(w) = w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\,\rho\,\sigma_1\sigma_2, \tag{11.2}$$

a quadratic in w whose leading coefficient $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ equals $\mathrm{Var}(R_1 - R_2)$ and is strictly positive when $|\rho| < 1$. For fixed $w \in (0, 1)$, the variance is strictly increasing in ρ.

(ii) If $|\rho| < 1$, the variance has the unique minimiser

$$w^* = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}, \qquad \sigma_p^2(w^*) = \frac{\sigma_1^2\sigma_2^2(1 - \rho^2)}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}.$$

(iii) Assume $|\rho| < 1$, so that w^* is defined, and label the assets so that $\sigma_1 \le \sigma_2$. The higher-volatility asset receives positive weight ($w^* < 1$) if and only if $\rho < \sigma_1/\sigma_2$, and then $w^* \in (0, 1)$: no shorting is involved. For $\rho \ge \sigma_1/\sigma_2$ the long-only minimum is asset 1 alone, while the unconstrained minimiser holds a nonpositive position in asset 2.

(iv) At $\rho = +1$ the quadratic is the perfect square $(w\sigma_1 + (1-w)\sigma_2)^2$, so $\sigma_p(w) = |w\sigma_1 + (1-w)\sigma_2|$, which vanishes at the short position $w = \sigma_2/(\sigma_2 - \sigma_1)$ when $\sigma_1 \ne \sigma_2$. At $\rho = -1$ it is $(w\sigma_1 - (1 - w)\sigma_2)^2$, so $\sigma_p(w) = |w\sigma_1 - (1 - w)\sigma_2|$, which vanishes at the long-only weight $w = \sigma_2/(\sigma_1 + \sigma_2)$. Suppose $\mu_1 \ne \mu_2$ and, at $\rho = +1$, $\sigma_1 \ne \sigma_2$. In the (σ, mean) plane the attainable set with shorts allowed is then a V with vertex on the vertical axis: at $\rho = -1$ the vertex lies between the asset points and the long-only mixes trace parts of both arms; at $\rho = +1$ the vertex is the short position, outside $[0, 1]$, and the long-only mixes trace the straight segment joining the two asset points. In the degenerate cases the V collapses: for $\sigma_1 = \sigma_2$ at $\rho = +1$ the volatility is constant and the attainable set is a vertical line; for $\mu_1 = \mu_2$ (with $\sigma_1 \ne \sigma_2$ at $\rho = +1$) the mean is constant and it is a horizontal ray from $(0, \mu_1)$, traversed twice. Both correlations make Σ singular, and a riskless combination whose mean differs from any available risk-free rate would be an arbitrage.

Sketch of the proof. The proof is the calculus of one quadratic. Formula (11.1) at the weights $(w, 1 - w)$ is (11.2); collecting powers of w gives the quadratic (A.1),

$$\sigma_p^2(w) = (\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)\,w^2 - 2(\sigma_2^2 - \rho\sigma_1\sigma_2)\,w + \sigma_2^2.$$

Its leading coefficient is $(\sigma_1 - \sigma_2)^2 + 2(1-\rho)\sigma_1\sigma_2 > 0$ when $|\rho| < 1$, so the derivative vanishes only at w^* (the second bracket over the first), a strict global minimum since the second derivative is positive. In (iii), $w^* < 1$ reduces to $\rho\sigma_1\sigma_2 < \sigma_1^2$, that is $\rho < \sigma_1/\sigma_2$, and then $\sigma_1 \le \sigma_2$ gives $w^* > 0$; otherwise $w^* \ge 1$ and the convex quadratic decreases on $[0, 1]$. In (iv), completing the square makes $\sigma_p(w)$ the absolute value of an affine function of w; with $\mu_p(w)$ affine, the image is a V when neither slope is zero, as (iv) assumes. The complete proof is in Appendix A.1. □

Example 11.5 (Ten and twenty percent volatility at six correlations)**.** Take the pair of Example 11.3: $\sigma_1 = 10\%$, $\sigma_2 = 20\%$, $\mu = (5\%, 10\%)^\top$. The variance is the quadratic (A.1), $\sigma_p^2(w) = Vw^2 - 2cw + \sigma_2^2$ with $V = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ and $c = \sigma_2^2 - \rho\sigma_1\sigma_2$, so that $w^* = c/V$; here $V = 0.05 - 0.04\rho$ and $c = 0.04 - 0.02\rho$.

- At $\rho = 0.3$: $w^* = (0.04 - 0.006)/(0.05 - 0.012) = 0.034/0.038 = 0.8947$; minimum variance $0.0004 \times 0.91/0.038 = 0.009579$, volatility 9.79% — below asset 1's 10% — with mean $0.8947(5\%) + 0.1053(10\%) = 5.53\%$.
- At $\rho = 0$: $w^* = 0.04/0.05 = 0.8$; variance $0.0004/0.05 = 0.008$, volatility 8.94%, mean 6.0%. The shortcut $w^* = \sigma_2^2/(\sigma_1^2 + \sigma_2^2)$ is this case only.
- At $\rho = 0.5 = \sigma_1/\sigma_2$ exactly: $w^* = 0.03/0.03 = 1$, the threshold of part (iii); asset 1 alone.
- At $\rho = 0.6$: $w^* = 0.028/0.026 = 1.077 > 1$; the unconstrained minimiser shorts the volatile asset (7.7% short, volatility 9.92%), and the long-only minimum is asset 1 alone at 10%.
- At $\rho = -1$: part (iv) gives $w = 0.20/0.30 = 2/3$, volatility 0, mean $\frac{2}{3}(5\%) + \frac{1}{3}(10\%) = 6.67\%$; absence of arbitrage would force that number to equal the risk-free rate.
- At $\rho = +1$: $w = 0.20/0.10 = 2$, i.e. long 200% of asset 1 and short 100% of asset 2, volatility 0, mean $2(5\%) - 10\% = 0$.

Two readings. The minimum-variance weight leans toward the lower-volatility asset and moves further toward it as ρ rises (0.80, 0.89, 1 at $\rho = 0$, 0.3, 0.5). Reducing variance moved the mean — 5.53% at $\rho = 0.3$ against 7.5% for the 50/50 mix — because minimum variance and maximum Sharpe are different objectives, a point made precise in §11.4. Example 10.13 is the equal-volatility, equal-weight cousin of this computation.

(Problem P11.1 works this computation at different numbers.)

Figure 18 draws the long-only mixes of Example 11.5 in the (σ, mean) plane at $\rho = 1$, $\rho = 0.3$, $\rho = 0$ and $\rho = -1$.

Intuition. The cross term is the only place correlation enters. The diagonal terms of (11.1) are what the assets contribute alone; the off-diagonal terms are what they contribute *together*, and they can be negative. Diversification is the observation that the off-diagonal terms are smaller than the diagonal ones whenever $\rho < 1$, so the whole is less volatile than the weighted sum of its parts — while the mean, being linear, is exactly the weighted sum of its parts.

Interview trap. Diversification does not lower expected return. Since $\mathbb{E}[R_w] = w^\top\mu$ is linear in w, a long-only mix has mean exactly the weighted average of the asset means (7.5% for the 50/50 mix, 5.53% for the minimum-variance mix at $\rho = 0.3$, both between 5% and 10%); only the variance is nonlinear in w, and only its cross term carries ρ. What moves the mean is the choice of weights, not the act of combining.

Interview trap. "Never hold an asset with the same mean and twice the volatility" is wrong whenever $\rho < \sigma_1/\sigma_2$: at $\rho = 0.3$ a 10.5% allocation to the 20%-volatility asset cuts the volatility from 10.0% to 9.79%. The test compares ρ with the volatility *ratio*, not one volatility with the other; "diversification requires negative correlation" fails for the same reason, and the shortcut $w^* = \sigma_2^2/(\sigma_1^2 + \sigma_2^2)$ is the $\rho = 0$ case only (0.80 against 0.8947 at $\rho = 0.3$).

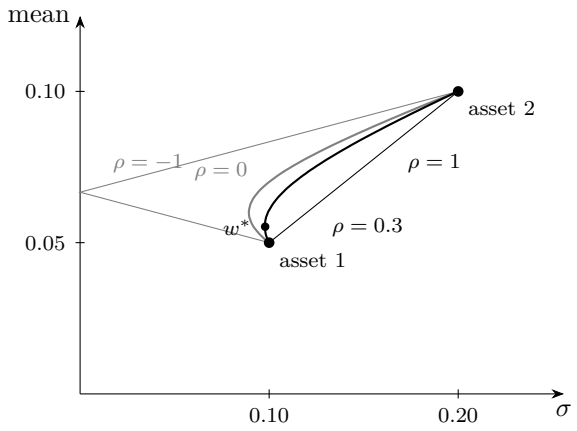


Figure 18: Long-only mixes of the two assets of Example 11.5 (volatilities 10% and 20%, means 5% and 10%) in the (σ, mean) plane. At $\rho = 1$ the mixes lie on the straight segment joining the assets; at $\rho = 0.3$ (black) and $\rho = 0$ (gray) they trace curves bulging leftward, and the curves nest leftward as ρ falls; at $\rho = -1$ they form a V touching the vertical axis at mean 6.67%. The leftmost point of each curve is the minimum-variance mix of Proposition 11.4(ii) — marked w^* at $(9.79\%, 5.53\%)$ for $\rho = 0.3$ — and the $\rho = \pm 1$ curves are the straight lines of part (iv). With two assets every fully invested portfolio lies on its curve, so each curve is also the $n = 2$ version of the frontier of Figure 19.

11.2 The Global Minimum-Variance Portfolio and the Diversification Floor

"Derive the minimum-variance portfolio" is the standard Lagrangian derivation at quant funds, and "how does risk fall with the number of assets, and what stops it?" is the follow-up; AI-lab interviews ask the identical computation as ensemble variance. This subsection first proves one lemma with three tools that every later theorem reuses, then proves $w_g = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top\Sigma^{-1}\mathbf{1})$ by Lagrangian with the convexity certificate that Remark 6.13 warns is missing from the naive Lagrangian, and shows that equal-weight diversification bottoms out at the average covariance — the formula of Proposition 8.6.

Setup. The covariance matrix Σ is positive definite (standing assumption of §11.1). The feasible set is $\{w \in \mathbb{R}^n : \mathbf{1}^\top w = 1\}$, an affine subspace. We write

$$A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1},$$

a scalar shown positive in Lemma 11.6, in which G denotes a generic symmetric positive-definite matrix. The running three-asset universe, used through §11.5, has

$$\Sigma = 0.01\begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}, \qquad \Sigma^{-1} = 25\begin{pmatrix} 3 & -2 & 1 \\ -2 & 4 & -2 \\ 1 & -2 & 3 \end{pmatrix},$$

with $\det\Sigma = 0.01^3\,(8-2-2) = 4\times10^{-6}$: every asset has volatility $\sqrt{0.02} = 14.14\%$, $\rho_{12} = \rho_{23} = 0.5$ and $\rho_{13} = 0$. The reader should verify the inverse by multiplying: each diagonal entry of $\Sigma\Sigma^{-1}$ is $0.25 \times 4 = 1$ and every off-diagonal entry cancels. The mean vector $\mu = (4\%, 7\%, 8\%)^\top$ is not needed until §11.3.

Lemma 11.6 (Three tools: the inverse of a positive-definite matrix, weighted Cauchy–Schwarz, and the gradient of a quadratic form)**.** *Let $G \in \mathbb{R}^{n\times n}$ be symmetric positive definite. Then:*

(i) The matrix G is invertible and G^{-1} is symmetric positive definite. Consequently $A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1} > 0$, and $\mu^\top\Sigma^{-1}\mu > 0$ whenever $\mu \neq 0$.

(ii) For all $x, y \in \mathbb{R}^n$,

$$(x^\top G y)^2 \;\le\; (x^\top G x)(y^\top G y),$$

with equality if and only if x and y are linearly dependent — the Cauchy–Schwarz inequality for the G-inner product $\langle x, y\rangle_G = x^\top G y$ of Definition 6.9.

(iii) For any symmetric G and any $a \in \mathbb{R}^n$,

$$\nabla_w (w^\top G w) = 2\,G w, \qquad \nabla_w (a^\top w) = a. \tag{11.3}$$

Proof. We expand in the eigenbasis for part (i), use the discriminant of a nonnegative quadratic for part (ii), and differentiate the double sum termwise for part (iii).

Part (i). By Fact 6.2, $G = Q\Lambda Q^\top$ with $Q^\top Q = I$ and $\Lambda = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$, and every $\lambda_i > 0$ by Theorem 6.4. The matrix $Q\Lambda^{-1}Q^\top$ is then well defined and satisfies $G\,(Q\Lambda^{-1}Q^\top) = Q\Lambda Q^\top Q\Lambda^{-1}Q^\top = QQ^\top = I$, so G is invertible with $G^{-1} = Q\Lambda^{-1}Q^\top$, which is symmetric and has eigenvalues $1/\lambda_i > 0$ with the same eigenvectors; by Theorem 6.4 again it is positive definite. Applying this with $G = \Sigma$, the quadratic form of Σ^{-1} is positive at every nonzero vector, in particular at $\mathbf{1}$ and at $\mu \neq 0$.

Part (ii). If $y = 0$ both sides vanish and x, y are dependent, so assume $y \neq 0$. For real t define

$$q(t) = (x - ty)^\top G (x - ty) = x^\top G x - 2t\, x^\top G y + t^2\, y^\top G y,$$

using the symmetry of G to merge the two cross terms. Positive definiteness gives $q(t) \ge 0$ for every t, and the leading coefficient $y^\top G y$ is strictly positive, so q is a quadratic that never goes below zero: its discriminant is nonpositive,

$$4(x^\top G y)^2 - 4(x^\top G x)(y^\top G y) \le 0,$$

which is the inequality. If equality holds, the discriminant is zero and q has a real double root t_0 with $q(t_0) = 0$, i.e. $(x - t_0 y)^\top G (x - t_0 y) = 0$, so $x = t_0 y$ by positive definiteness. Conversely, if $x = ty$ then both sides equal $t^2 (y^\top G y)^2$.

Part (iii). Write $w^\top G w = \sum_{j,k} G_{jk} w_j w_k$. Differentiating with respect to w_i, the terms containing w_i are $G_{ii} w_i^2$ and, for each $k \neq i$, the pair $G_{ik} w_i w_k + G_{ki} w_k w_i = 2 G_{ik} w_i w_k$ by symmetry; their derivatives sum to $2G_{ii} w_i + 2\sum_{k \neq i} G_{ik} w_k = 2(Gw)_i$. The second formula is the derivative of $\sum_j a_j w_j$. □

Part (ii) is used with $G = \Sigma^{-1}$ in Lemma 11.15 and with $G = \Sigma$ in Theorem 11.23; part (iii) drives every Lagrangian in this subsection and the next, Proposition 11.33 and Theorem 11.45.

Theorem 11.7 (Global minimum-variance portfolio)**.** *Under the setup above, the problem of minimising the variance $w^\top \Sigma w$ over the fully invested portfolios, $\mathbf{1}^\top w = 1$, has the unique solution*

$$w_g = \frac{\Sigma^{-1}\mathbf{1}}{A}, \qquad A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1} > 0, \tag{11.4}$$

with minimum variance $\sigma_g^2 = w_g^\top \Sigma w_g = 1/A$. At the optimum $\Sigma w_g = (1/A)\,\mathbf{1}$: every asset has the same covariance $1/A$ with the portfolio.

Proof. We proceed by the Lagrangian to locate the candidate, then certify optimality by completing the square — the honesty point of Remark 6.13, which observes that stationarity alone does not identify a minimum. Positivity of A is Lemma 11.6(i).

Candidate. The Lagrangian is $L(w, \theta) = \tfrac{1}{2} w^\top \Sigma w - \theta(\mathbf{1}^\top w - 1)$. By (11.3), stationarity in w reads $\Sigma w = \theta \mathbf{1}$, so $w = \theta \Sigma^{-1}\mathbf{1}$; the budget constraint gives $1 = \theta\, \mathbf{1}^\top \Sigma^{-1} \mathbf{1} = \theta A$, hence $\theta = 1/A$ and $w = w_g$.

Certificate. Let v be any feasible portfolio and write $v = w_g + d$; feasibility of both v and w_g gives $\mathbf{1}^\top d = 0$. Expanding the quadratic form,

$$v^\top \Sigma v = w_g^\top \Sigma w_g + 2\, d^\top \Sigma w_g + d^\top \Sigma d.$$

By the stationarity equation, $\Sigma w_g = (1/A)\mathbf{1}$, so $d^\top \Sigma w_g = (1/A)\,\mathbf{1}^\top d = 0$, and

$$v^\top \Sigma v = \frac{1}{A} + d^\top \Sigma d \;\geq\; \frac{1}{A},$$

with equality if and only if $d = 0$ by positive definiteness. Hence w_g is the unique global minimiser, and no second-order argument is needed: the objective is strictly convex on the affine feasible set, which is why stationarity suffices here, unlike on the sphere constraint of Theorem 6.12. The value is

$$w_g^\top \Sigma w_g = \frac{1}{A^2}\,\mathbf{1}^\top \Sigma^{-1} \Sigma \Sigma^{-1} \mathbf{1} = \frac{1}{A^2}\,\mathbf{1}^\top \Sigma^{-1} \mathbf{1} = \frac{1}{A}.$$

Finally, $\Sigma w_g = (1/A)\mathbf{1}$ is the stationarity equation at $\theta = 1/A$, and the covariance of asset i with R_{w_g} is $\mathrm{Cov}(R_i, w_g^\top R) = e_i^\top \Sigma w_g = 1/A$ by bilinearity. □

(Problem P11.3 asks you to reproduce this proof from scratch.)

Example 11.8 (The minimum-variance portfolio of the running universe)**.** Summing the rows of Σ^{-1} in the setup, $\Sigma^{-1}\mathbf{1} = 25\,(2, 0, 2)^\top = (50, 0, 50)^\top$, so $A = 100$ and by (11.4)

$$w_g = (0.5,\ 0,\ 0.5)^\top, \qquad \sigma_g^2 = 1/100, \qquad \sigma_g = 10\%,$$

against 14.14% for each asset alone. The equal-covariance property checks: $\Sigma w_g = 0.01\,(1, 1, 1)^\top = (1/A)\mathbf{1}$. Asset 2 receives zero weight because its covariance with the equal mix of assets 1 and 3 is $0.01(0.5+0.5) = 0.01$, exactly the variance of that mix: at the margin it adds nothing. Consistency with Proposition 11.4 on assets 1 and 3 alone ($\rho = 0$, equal volatilities): $w^* = 1/2$ and variance $0.02/2 = 0.01$.

(Problem P11.3 works this computation at different numbers.)

In the language of §11.5, where the ratio of an asset's covariance with a portfolio to the portfolio's variance is called its beta, the equal-covariance property says that every asset has beta $(1/A)/(1/A) = 1$ to the minimum-variance portfolio.

Corollary 11.9 (Inverse-variance weights for uncorrelated assets)**.** *If the assets are uncorrelated, so that Σ is the diagonal matrix $\mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$, then the weights are the normalised precisions,*

$$w_{g,i} = \frac{\sigma_i^{-2}}{\sum_{j=1}^n \sigma_j^{-2}}, \qquad \sigma_g^2 = \frac{1}{\sum_{j=1}^n \sigma_j^{-2}},$$

and σ_g^2 lies strictly below the smallest σ_j^2 — the parallel-resistor combination of the variances.

Proof. We substitute into (11.4). With Σ diagonal, $\Sigma^{-1} = \mathrm{diag}(\sigma_1^{-2}, \ldots, \sigma_n^{-2})$, so $\Sigma^{-1}\mathbf{1}$ has entries σ_i^{-2} and $A = \sum_j \sigma_j^{-2}$; the weights and $\sigma_g^2 = 1/A$ follow. Since every summand of A is positive and $n \geq 2$, $A > \sigma_j^{-2}$ for each j, i.e. $1/A < \sigma_j^2$. □

Example 11.10 (Three uncorrelated assets)**.** Three uncorrelated assets have variances 0.01, 0.02, 0.04 (volatilities 10%, 14.14%, 20%). Their precisions are 100, 50, 25, so $A = 175$ and by Corollary 11.9

$$w_g = \left(\tfrac{4}{7}, \tfrac{2}{7}, \tfrac{1}{7}\right)^\top = (0.571, 0.286, 0.143)^\top, \qquad \sigma_g^2 = \tfrac{1}{175} = 0.005714, \qquad \sigma_g = 7.56\%,$$

below the best single asset's 10%. The equal-covariance property holds in every coordinate: $\Sigma w_g = (0.01 \cdot \tfrac{4}{7},\ 0.02 \cdot \tfrac{2}{7},\ 0.04 \cdot \tfrac{1}{7})^\top = \tfrac{1}{175}\mathbf{1}$.

Remark 11.11 (Precision weights are BLUE). Corollary 11.9 is also the best linear unbiased combination of n independent unbiased estimates of one quantity: unbiasedness of $w^\top R$ forces $\mathbf{1}^\top w = 1$, independence makes Σ diagonal, and Theorem 11.7 minimises the variance under exactly that constraint. Example 5.7 is the equal-variance case, $w_i = 1/n$.

Interview trap. Minimum variance is not maximum Sharpe: $w_g = \Sigma^{-1}\mathbf{1}/A$ never looks at μ, and it coincides with the maximum-Sharpe portfolio only when every asset has the same expected excess return (Corollary 11.25, proved in §11.4). Answering a "best portfolio" question with the minimum-variance portfolio answers a different question. Nor is the result "equal weights": it equalises every asset's *covariance with the portfolio*, and asset 2 of Example 11.8 receives no weight at all.

Setup (equal weights). The *equal-weight* portfolio is $w = \frac{1}{n}\mathbf{1}$. Write $\bar{v} = \frac{1}{n}\sum_i \Sigma_{ii}$ for the average variance and $\bar{c} = \frac{1}{n(n-1)}\sum_{i\neq j}\Sigma_{ij}$ for the average covariance of the n assets present; both depend on n. The *equicorrelated model* is the covariance structure with a common variance $\Sigma_{ii} = v > 0$ and a common correlation $\rho_{ij} = \rho$ for all $i \neq j$, i.e. $\Sigma = v\big[(1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top\big]$.

Proposition 11.12 (Equal-weight diversification and the systematic floor)**.** *Under the setup above:*

(i) For any covariance matrix,

$$\operatorname{Var}\Big(\frac{1}{n}\mathbf{1}^\top R\Big) = \frac{\bar{v}}{n} + \Big(1 - \frac{1}{n}\Big)\bar{c},$$

so that $\operatorname{Var}(\frac{1}{n}\mathbf{1}^\top R) - \bar{c} \to 0$ *as* $n \to \infty$ *when* $\bar{v}$ *stays bounded: the equal-weight variance converges to the average covariance.*

(ii) In the equicorrelated model, Σ *is positive definite if and only if* $-1/(n-1) < \rho < 1$*, and then*

$$\operatorname{Var}\Big(\frac{1}{n}\mathbf{1}^\top R\Big) = \rho v + \frac{1-\rho}{n}\,v,$$

which is (8.2) of Proposition 8.6 with $X_i = R_i$*: an equal-weight portfolio of* n *correlated assets is an average of* n *correlated random variables, so the mathematics is identical. For* $\rho \geq 0$ *the floor* ρv *(volatility* $\sqrt{\rho}\sqrt{v}$*) is the* systematic *(undiversifiable) variance and* $(1-\rho)v/n$ *the* idiosyncratic *part — vocabulary that §11.5 formalises through beta.*

(iii) In the equicorrelated model the equal-weight portfolio is *the global minimum-variance portfolio of Theorem 11.7:* $w_g = \frac{1}{n}\mathbf{1}$ *and* $1/A = \rho v + (1-\rho)v/n$*.*

Proof. We expand the quadratic form at $w = \frac{1}{n}\mathbf{1}$ for part (i), identify the result with the cited formula for part (ii), and compute $\Sigma\mathbf{1}$ by a row sum for part (iii).

Part (i). Formula (11.1) at $w_i = 1/n$ gives

$$\frac{1}{n^2}\Big(\sum_i \Sigma_{ii} + \sum_{i\neq j}\Sigma_{ij}\Big) = \frac{1}{n^2}\big(n\bar{v} + n(n-1)\bar{c}\big) = \frac{\bar{v}}{n} + \frac{n-1}{n}\,\bar{c}.$$

The difference between this variance and $\bar{c}$ is $(\bar{v}-\bar{c})/n$, and $|\bar{c}| \leq \bar{v}$ because $|\Sigma_{ij}| \leq \sigma_i\sigma_j \leq \frac{1}{2}(\sigma_i^2+\sigma_j^2)$; so the difference tends to 0 when $\bar{v}$ stays bounded.

Part (ii). For d with $\mathbf{1}^\top d = 0$, $\Sigma d = v(1-\rho)d + v\rho\,\mathbf{1}(\mathbf{1}^\top d) = v(1-\rho)d$; and summing a row, $\Sigma\mathbf{1} = v(1-\rho)\mathbf{1} + v\rho\,n\,\mathbf{1} = v(1+(n-1)\rho)\mathbf{1}$. Since $\mathbb{R}^n$ is spanned by $\mathbf{1}$ and the $(n-1)$-dimensional space $\{d : \mathbf{1}^\top d = 0\}$, the eigenvalues of Σ are $v(1-\rho)$ with multiplicity $n-1$ and $v(1+(n-1)\rho)$ once; by Theorem 6.4, Σ is positive definite if and only if both are positive, i.e. $\rho < 1$ and $\rho > -1/(n-1)$. Substituting $\bar{v} = v$ and $\bar{c} = \rho v$ into part (i) gives $v/n + (1-1/n)\rho v = \rho v + (1-\rho)v/n$, and comparing with (8.2) — cited, not re-derived — shows the two formulas coincide term by term.

Part (iii). From $\Sigma\mathbf{1} = v(1+(n-1)\rho)\mathbf{1}$ and invertibility, $\Sigma^{-1}\mathbf{1} = \mathbf{1}/\big(v(1+(n-1)\rho)\big)$, a multiple of $\mathbf{1}$. Hence $A = n/\big(v(1+(n-1)\rho)\big)$ and (11.4) gives $w_g = \frac{1}{n}\mathbf{1}$, with $1/A = v(1+(n-1)\rho)/n = \rho v + (1-\rho)v/n$. The row-sum identity is the same eigenvector computation behind the equicorrelated case of Fact 6.25: for $\rho > 0$, $\mathbf{1}$ is the eigenvector of the largest eigenvalue $v(1+(n-1)\rho)$. □

Example 11.13 (The diversification floor in numbers)**.** Take $\sigma = 20\%$ for every asset and $\rho = 0.3$, so $v = 0.04$ and $\rho v = 0.012$. By Proposition 11.12(ii) the equal-weight volatility is $\sqrt{0.012 + 0.028/n}$:

n	1	2	10	30	100
volatility	20.0%	16.1%	12.2%	11.4%	11.1%

with limit $\sqrt{0.3} \times 20\% = 10.95\%$. Ten names capture $(20.0 - 12.2)/(20.0 - 10.95) \approx 87\%$ of the achievable reduction; going from 30 to 100 names removes only 0.3 points of volatility; no n breaks below 10.95%. Example 8.7 ($v = 1$, $\rho = 0.2$) is the same table read as ensemble variance in §8.3.

(Problem P11.4 works this computation at different numbers.)

Intuition. Averaging kills the diagonal at rate $1/n$ and leaves the average off-diagonal entry standing. Whatever the assets share — the market, a sector, a common factor — survives in every one of them and therefore in their average; only what is specific to each asset cancels. The floor is the shared part, which is why ρv is called systematic or market risk and the $1/n$ term idiosyncratic risk.

Interview trap. "With enough assets all risk diversifies away" is false for $\rho > 0$: the limit is the average covariance ρv, and 100 names are already within 1.5% of that floor at $\rho = 0.3$ (Example 11.13). The right one-liner: averaging removes idiosyncratic variance at rate $1/n$ and leaves the systematic part intact — identical to the random-forest floor of Proposition 8.6.

11.3 The Mean–Variance Frontier and Two-Fund Separation

"Sketch the efficient frontier and explain its shape" is a whiteboard staple, written tests ask for the variance minimisation with two constraints, and "why does every investor hold the same two funds?" is the one-sentence theory question. Fixing a target mean m and minimising variance subject to two linear constraints produces the frontier; the solution is affine in m, which is the whole content of two-fund separation (Tobin, 1958; in the risky-only form used here, Merton, 1972); and expressed through four constants the frontier is a hyperbola in the (σ, mean) plane whose vertex is the minimum-variance portfolio of Theorem 11.7. Figure 19 is built here and completed in §11.4 with the capital market line.

Setup. The covariance matrix Σ is positive definite, and the *frontier constants* are the scalars

$$A = \mathbf{1}^\top\Sigma^{-1}\mathbf{1}, \qquad B = \mathbf{1}^\top\Sigma^{-1}\mu, \qquad C = \mu^\top\Sigma^{-1}\mu, \qquad D = AC - B^2, \tag{11.5}$$

local to this section as announced in §11.1. *Standing assumption for this and the later subsections:* μ is not a multiple of $\mathbf{1}$. Otherwise every fully invested portfolio has the same mean B/A and the frontier is the single point w_g. A target mean is a real number m — a scalar here, whereas §6 used m for the market-mode vector. The mixing coefficient between two frontier portfolios is τ.

Definition 11.14 (Frontier portfolio and efficient frontier)**.** For a target mean m, the *frontier portfolio* $w(m)$ is the minimiser of $w^\top\Sigma w$ over $\{w : \mathbf{1}^\top w = 1,\ \mu^\top w = m\}$ (unique by Theorem 11.16 below). The *mean–variance frontier* is the curve $m \mapsto (\sigma(m), m)$ it traces, where $\sigma(m)^2 = w(m)^\top\Sigma w(m)$. The *efficient frontier* is its part with $m \geq B/A$; the part with $m < B/A$ is called *inefficient*.

Lemma 11.15 (Signs of the frontier constants)**.** *The constant A is positive; C is positive whenever $\mu \neq 0$; and $D = AC - B^2 > 0$ exactly when μ and $\mathbf{1}$ are linearly independent — in particular under the standing assumption.*

Proof. We read all three claims off Lemma 11.6 with $G = \Sigma^{-1}$, which is symmetric positive definite by part (i) of that lemma. The constants A and C are the quadratic form of Σ^{-1} at $\mathbf{1} \neq 0$ and at μ, hence positive when the vector is nonzero. The constant D is the Gram determinant of $\mathbf{1}$ and μ in the Σ^{-1}-inner product, $D = \langle \mathbf{1}, \mathbf{1} \rangle_{\Sigma^{-1}} \langle \mu, \mu \rangle_{\Sigma^{-1}} - \langle \mathbf{1}, \mu \rangle^2_{\Sigma^{-1}}$, which is nonnegative by Lemma 11.6(ii) with $x = \mathbf{1}$, $y = \mu$, and zero if and only if the two vectors are linearly dependent; under the standing assumption they are independent, so $D > 0$. □

Theorem 11.16 (Minimum variance at a target mean)**.** *Under the setup above:*

(i) For each $m \in \mathbb{R}$, the problem $\min_w w^\top \Sigma w$ subject to $\mathbf{1}^\top w = 1$ and $\mu^\top w = m$ has the unique solution

$$w(m) = \theta_1 \, \Sigma^{-1}\mathbf{1} + \theta_2 \, \Sigma^{-1}\mu, \qquad \theta_1 = \frac{C - Bm}{D}, \quad \theta_2 = \frac{Am - B}{D},$$

equivalently

$$w(m) = g + h\, m, \qquad g = \frac{C\,\Sigma^{-1}\mathbf{1} - B\,\Sigma^{-1}\mu}{D}, \quad h = \frac{A\,\Sigma^{-1}\mu - B\,\Sigma^{-1}\mathbf{1}}{D}, \tag{11.6}$$

where the fixed vectors g, h satisfy $\mathbf{1}^\top g = 1$, $\mathbf{1}^\top h = 0$, $\mu^\top g = 0$, $\mu^\top h = 1$.

(ii) The minimum variance is

$$\sigma^2(m) = w(m)^\top \Sigma w(m) = \frac{Am^2 - 2Bm + C}{D}; \tag{11.7}$$

at $m = B/A$ the solution is w_g and the variance is $1/A$.

(iii) (Frontier membership from the first-order condition.) Any fully invested w with $\Sigma w \in \operatorname{span}(\mathbf{1}, \mu)$ is the frontier portfolio at its own mean: $w = w(\mu^\top w)$.

Proof. We proceed by the Lagrangian with two multipliers and certify optimality by completing the square, as in Theorem 11.7.

Candidate. With $L(w, \theta_1, \theta_2) = \frac{1}{2} w^\top \Sigma w - \theta_1(\mathbf{1}^\top w - 1) - \theta_2(\mu^\top w - m)$, the gradient formula (11.3) makes stationarity read

$$\Sigma w = \theta_1 \mathbf{1} + \theta_2 \mu, \qquad \text{i.e.} \qquad w = \theta_1 \Sigma^{-1}\mathbf{1} + \theta_2 \Sigma^{-1}\mu. \tag{11.8}$$

Imposing the two constraints on this w and using (11.5),

$$\mathbf{1}^\top w = A\theta_1 + B\theta_2 = 1, \qquad \mu^\top w = B\theta_1 + C\theta_2 = m,$$

a 2×2 linear system in (θ_1, θ_2) with determinant $AC - B^2 = D > 0$ by Lemma 11.15. Cramer's rule gives $\theta_1 = (C \cdot 1 - B \cdot m)/D$ and $\theta_2 = (A \cdot m - B \cdot 1)/D$. Collecting the terms in m yields (11.6); the four identities for g and h are direct substitutions: $\mathbf{1}^\top g = (CA - B^2)/D = 1$, $\mathbf{1}^\top h = (AB - BA)/D = 0$, $\mu^\top g = (CB - BC)/D = 0$, $\mu^\top h = (AC - B^2)/D = 1$.

Certificate. Let w be any portfolio satisfying both constraints and write $w = w(m) + d$. Then $\mathbf{1}^\top d = 0$ and $\mu^\top d = 0$, so by (11.8)

$$d^\top \Sigma w(m) = \theta_1 \, \mathbf{1}^\top d + \theta_2 \, \mu^\top d = 0,$$

hence

$$w^\top \Sigma w = w(m)^\top \Sigma w(m) + d^\top \Sigma d \;\geq\; w(m)^\top \Sigma w(m),$$

with equality if and only if $d = 0$ by positive definiteness. So $w(m)$ is the unique minimiser, proving (i).

Part (ii). Using (11.8) once more,

$$w(m)^\top \Sigma w(m) = w(m)^\top(\theta_1 \mathbf{1} + \theta_2 \mu) = \theta_1 + \theta_2 m = \frac{C - Bm + Am^2 - Bm}{D},$$

which is (11.7). At $m = B/A$: $\theta_2 = 0$ and $\theta_1 = (C - B^2/A)/D = (AC - B^2)/(AD) = 1/A$, so $w(B/A) = \Sigma^{-1}\mathbf{1}/A = w_g$ and $\sigma^2(B/A) = 1/A$.

Part (iii). Such a w satisfies (11.8) for some scalars θ_1, θ_2, and it is feasible for the target $m = \mu^\top w$. The certificate above used only (11.8) and feasibility, so it applies verbatim: every w' feasible for that m has $w'^\top \Sigma w' \geq w^\top \Sigma w$ with equality only at $w' = w$. By the uniqueness just proved, $w = w(m)$. □

Example 11.17 (The frontier of the running universe)**.** With Σ^{-1} from the setup of §11.2 and $\mu = (4\%, 7\%, 8\%)^\top$,

$$\Sigma^{-1}\mu = 25\,(0.06,\ 0.04,\ 0.14)^\top = (1.5,\ 1,\ 3.5)^\top,$$

so $A = 100$, $B = 1.5 + 1 + 3.5 = 6$, $C = 0.04(1.5) + 0.07(1) + 0.08(3.5) = 0.41$ and $D = 41 - 36 = 5$. The minimum-variance mean is $B/A = 6\%$. From (11.6),

$$g = \tfrac{1}{5}\big(0.41\,(50, 0, 50)^\top - 6\,(1.5, 1, 3.5)^\top\big) = (2.3,\ -1.2,\ -0.1)^\top,$$
$$h = \tfrac{1}{5}\big(100\,(1.5, 1, 3.5)^\top - 6\,(50, 0, 50)^\top\big) = (-30,\ 20,\ 10)^\top,$$

with the checks $\mathbf{1}^\top g = 1$, $\mathbf{1}^\top h = 0$, $\mu^\top g = 0.092 - 0.084 - 0.008 = 0$, $\mu^\top h = -1.2 + 1.4 + 0.8 = 1$; so $w(m) = (2.3 - 30m,\ -1.2 + 20m,\ -0.1 + 10m)^\top$ and $\sigma^2(m) = (100m^2 - 12m + 0.41)/5$.

- Target $m = 6\%$: $w = (0.5, 0, 0.5)^\top$, variance $(0.36 - 0.72 + 0.41)/5 = 0.01$, volatility 10% — the minimum-variance portfolio.
- Target $m = 7\%$: $w = (0.2, 0.2, 0.6)^\top$, variance $(0.49 - 0.84 + 0.41)/5 = 0.012$, volatility 10.95%. Direct check by (11.1): $0.01\,(0.08 + 0.08 + 0.72 + 0.08 + 0.24) = 0.012$.
- Target $m = 8\%$: $w = (-0.1, 0.4, 0.7)^\top$, variance $(0.64 - 0.96 + 0.41)/5 = 0.018$, volatility 13.42% — a short position appears.
- Target $m = 4\%$: $w = (1.1, -0.4, 0.3)^\top$, variance $(0.16 - 0.48 + 0.41)/5 = 0.018$, the same volatility as $m = 8\%$. Asset 1 alone (mean 4%, variance 0.02) is beaten even by this portfolio.

(Problem P11.5 works this computation at different numbers.)

Corollary 11.18 (The frontier is a hyperbola with vertex at the minimum-variance portfolio)**.** *The frontier variance satisfies*

$$\sigma^2(m) = \frac{1}{A} + \frac{A}{D}\Big(m - \frac{B}{A}\Big)^2, \tag{11.9}$$

so in the (σ, mean) plane the frontier is the right-hand branch of the hyperbola

$$A\sigma^2 - \frac{A^2}{D}\Big(m - \frac{B}{A}\Big)^2 = 1,$$

with vertex $(1/\sqrt{A},\ B/A)$ — the minimum-variance portfolio, $w(B/A) = w_g$ — and asymptotes $m = B/A \pm \sqrt{D/A}\,\sigma$. Against variance rather than volatility it is a parabola. The portfolios $w(B/A + t)$ and $w(B/A - t)$ share the same variance, so the lower branch $m < B/A$ is dominated.

Proof. We complete the square in (11.7) and factor the resulting difference of squares for the asymptotes. Using $C - B^2/A = D/A$: $Am^2 - 2Bm + C = A(m - B/A)^2 + D/A$, and dividing by D gives (11.9). Multiplying through by A and rearranging gives the hyperbola equation; the vertex

is where the squared term vanishes, $m = B/A$ and $\sigma^2 = 1/A$, which is w_g by Theorem 11.16(ii). For the asymptotes write $t = m - B/A$ and read (11.9) as $\sigma(m)^2 - (A/D)t^2 = 1/A$, i.e.

$$\Big(\sigma(m) - \sqrt{A/D}\,|t|\Big)\Big(\sigma(m) + \sqrt{A/D}\,|t|\Big) = \frac{1}{A}.$$

As $|t| \to \infty$ the second factor tends to infinity (it is at least $\sqrt{A/D}\,|t|$), so the first factor tends to zero: the horizontal distance from the curve to the line $\sigma = \sqrt{A/D}\,|m - B/A|$, equivalently $m = B/A \pm \sqrt{D/A}\,\sigma$, tends to zero, which is what makes these lines the asymptotes. Finally, reading (11.9) at $m = B/A \pm t$ gives the same variance $1/A + (A/D)t^2$ for both, with means differing by $2t$; for $t > 0$ the upper one has strictly larger mean at equal volatility, so the lower one is dominated. □

The dominated branch is the inefficient part of Definition 11.14: every portfolio $w(m)$ with $m < B/A$ has the portfolio $w(2B/A - m)$, of the same volatility and strictly higher mean, directly above it.

Example 11.19 (The hyperbola of the running universe)**.** With $A = 100$, $B = 6$ and $D = 5$ from Example 11.17, (11.9) reads $\sigma^2 = 0.01 + 20(m - 0.06)^2$: vertex $(10\%, 6\%)$ and asymptote slope $\sqrt{D/A} = \sqrt{0.05} = 0.2236$. The targets $m = 4\%$ and $m = 8\%$ sit at equal distance 0.02 from the vertex and share the variance $0.01 + 20(0.0004) = 0.018$, volatility 13.42%; the 8% portfolio dominates the 4% one. Far from the vertex the curve hugs its asymptote: at $m = 30\%$ it gives $\sigma = \sqrt{0.01 + 1.152} = 107.8\%$ against $0.24/0.2236 = 107.3\%$ on the asymptote, a gap of half a point.

Theorem 11.20 (Two-fund separation)**.** *Let $w(m)$ be the frontier map of (11.6). Then:*

(i) The frontier map is affine in the target mean: $w(m) = w_g + (m - B/A)\,h$, with the fixed direction h of (11.6), which satisfies $\mathbf{1}^\top h = 0$.

(ii) For any two distinct targets $m_1 \neq m_2$ and every m,

$$w(m) = (1 - \tau)\,w(m_1) + \tau\,w(m_2), \qquad \tau = \frac{m - m_1}{m_2 - m_1},$$

and conversely every combination $(1 - \tau)\,w(m_1) + \tau\,w(m_2)$, $\tau \in \mathbb{R}$, is the frontier portfolio at mean $(1 - \tau)m_1 + \tau m_2$: any two frontier portfolios span the frontier — for instance the minimum-variance portfolio and, when $B \neq 0$, the "μ-fund" $\Sigma^{-1}\mu/B$, which has mean C/B.

Proof. We show that the frontier map is affine in m and use the fact that an affine map sends a line to a line.

Part (i). We verify $g = w_g - (B/A)h$ from the definitions in (11.6):

$$\begin{aligned} w_g - \frac{B}{A}h &= \frac{\Sigma^{-1}\mathbf{1}}{A} - \frac{B}{A}\cdot\frac{A\,\Sigma^{-1}\mu - B\,\Sigma^{-1}\mathbf{1}}{D} = \Big(\frac{1}{A} + \frac{B^2}{AD}\Big)\Sigma^{-1}\mathbf{1} - \frac{B}{D}\,\Sigma^{-1}\mu \\ &= \frac{D + B^2}{AD}\,\Sigma^{-1}\mathbf{1} - \frac{B}{D}\,\Sigma^{-1}\mu = \frac{C}{D}\,\Sigma^{-1}\mathbf{1} - \frac{B}{D}\,\Sigma^{-1}\mu = g, \end{aligned}$$

since $D + B^2 = AC$. Then $w(m) = g + hm = w_g + (m - B/A)h$, and $\mathbf{1}^\top h = 0$ was shown in Theorem 11.16(i).

Part (ii). Substituting part (i),

$$\begin{aligned} (1 - \tau)\,w(m_1) + \tau\,w(m_2) &= w_g + \big[(1 - \tau)(m_1 - B/A) + \tau(m_2 - B/A)\big]h \\ &= w_g + \big((1 - \tau)m_1 + \tau m_2 - B/A\big)h, \end{aligned}$$

and the choice $\tau = (m - m_1)/(m_2 - m_1)$ makes $(1 - \tau)m_1 + \tau m_2 = m$, so the right-hand side is $w(m)$. The combination is fully invested because both inputs are and the coefficients sum

to one. The converse reads the same identity right to left: for any τ the combination equals $w_g + (m' - B/A)h = w(m')$ with $m' = (1-\tau)m_1 + \tau m_2$. For the named second fund, $\Sigma^{-1}\mu/B$ is fully invested when $B \neq 0$ and satisfies $\Sigma(\Sigma^{-1}\mu/B) = \mu/B \in \operatorname{span}(\mathbf{1}, \mu)$, so by Theorem 11.16(iii) it is the frontier portfolio at its own mean $\mu^\top\Sigma^{-1}\mu/B = C/B$; it differs from w_g because $\Sigma^{-1}\mu$ and $\Sigma^{-1}\mathbf{1}$ are independent (multiply by Σ and use the standing assumption). □

The two multipliers of Theorem 11.16 are the two funds: a desk needs only two products — the minimum-variance portfolio and one other frontier portfolio — to span every efficient portfolio, and investors differ only in how they mix them (Merton, 1972; the version with a risk-free asset, Tobin, 1958, is Proposition 11.33 below).

Example 11.21 (Two-fund checks in the running universe)**.** From the table of Example 11.17, $w(7\%) = \frac{1}{2}w(6\%) + \frac{1}{2}w(8\%) = (0.2, 0.2, 0.6)^\top$, as Theorem 11.20(ii) predicts for $\tau = \frac{1}{2}$. With $m_1 = 6\%$, $m_2 = 7\%$ and $m = 8\%$, $\tau = (0.08 - 0.06)/(0.07 - 0.06) = 2$ gives $w(8\%) = -w(6\%) + 2\,w(7\%) = (-0.5, 0, -0.5)^\top + (0.4, 0.4, 1.2)^\top = (-0.1, 0.4, 0.7)^\top$: a short position in the minimum-variance fund. The direction is $h = (-30, 20, 10)^\top$, so raising the target by one point moves 0.3 of the capital out of asset 1 into assets 2 and 3.

(Problem P11.6 works this computation at different numbers.)

Figure 19 draws this frontier, its asymptotes and the three asset points; the risk-free rate, the tangency portfolio and the capital market line on the same figure are explained in §11.4.

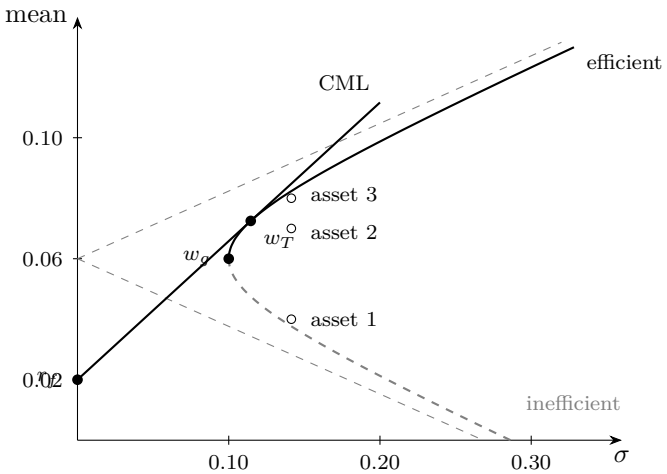


Figure 19: The mean–variance frontier of the running universe (Example 11.17) in the (σ, mean) plane. The vertex $w_g = (0.5, 0, 0.5)^\top$ at $(10\%, 6\%)$ is Theorem 11.7; the curve $\sigma^2 = 0.01 + 20(m - 0.06)^2$ is Theorem 11.16 and Corollary 11.18, with its asymptotes of slope ± 0.2236 dashed in gray; the upper branch is efficient, the lower (dashed) branch is dominated. The three assets stack vertically at volatility 14.14% because all three volatilities are equal, and lie strictly inside the hyperbola. The straight line from $r_f = 2\%$ on the vertical axis is the capital market line (11.11) of Corollary 11.27 in §11.4; it lies weakly above the hyperbola everywhere and touches it only at the tangency portfolio w_T at $(11.46\%, 7.25\%)$.

Intuition. Two constraints, two multipliers, two funds. The first-order condition $\Sigma w = \theta_1\mathbf{1} + \theta_2\mu$ says that on the frontier every asset's covariance with the portfolio is an affine function of its own mean — assets with higher expected return are allowed to co-move more with the book, in exact proportion. Everything else (the hyperbola, the vertex, the spanning by two funds) is bookkeeping on that one equation.

Interview trap. Sketching the frontier as a parabola in the (σ, mean) plane, or with its vertex on the vertical axis, is wrong: it is a hyperbola with straight-line asymptotes whose vertex is the

minimum-variance portfolio at $\sigma = 1/\sqrt{A} > 0$, and it is a parabola only when plotted against *variance*. Calling the whole hyperbola "efficient" is the companion error: only the branch $m \geq B/A$ is, and in Example 11.19 the $m = 4\%$ portfolio is dominated by the $m = 8\%$ one at the same 13.42% volatility.

Interview trap. Two-fund separation is a statement about the weights being affine in the target mean, not about any two portfolios: the spanning funds must both lie on the frontier, and the mix may short one of them ($w(8\%)$ is -100% of the minimum-variance fund plus 200% of $w(7\%)$ in Example 11.21). Adding an asset never moves the frontier inward; it can leave the frontier unchanged at a single point but never on an interval (unless the new asset's weight is identically zero), since a frontier weight such as asset 2's $-1.2 + 20m$ is affine in m and vanishes only at $m = 6\%$.

11.4 The Risk-Free Asset, the Tangency Portfolio and the Capital Market Line

The single most-asked question in this area — "you run uncorrelated strategies with Sharpe ratios 0.5, 1 and 1; what is the best combined Sharpe ratio and how do you weight them?" — and its theory cousin "what does a risk-free asset do to the frontier?" are the same computation. Mixing cash with any risky portfolio traces a ray from $(0, r_f)$, and the best ray is tangent to the frontier: its risky portfolio maximises the Sharpe ratio, with weights proportional to $\Sigma^{-1}(\mu - r_f\mathbf{1})$ and squared Sharpe ratio $(\mu - r_f\mathbf{1})^\top\Sigma^{-1}(\mu - r_f\mathbf{1})$, which for uncorrelated strategies is the interview classic $\mathrm{SR}^2 = \sum_i \mathrm{SR}_i^2$ of which Theorem 10.11 and Corollary 10.12 are special cases. The proof is one application of Lemma 11.6(ii). The subsection ends with the mean–variance utility problem, whose solution is the same direction scaled by risk aversion — the premise of the CAPM Fact in §11.5 and the bridge to the Kelly remark of §11.7.

Setup. Fix a real number r_f, the *risk-free rate*: the deterministic return of an additional asset (cash) with zero variance. A portfolio is now a risky weight vector $w \in \mathbb{R}^n$ that need not sum to one; the remainder $1 - \mathbf{1}^\top w$ earns r_f, negative values meaning borrowing at r_f. Its return is

$$R_p = (1 - \mathbf{1}^\top w)\, r_f + w^\top R = r_f + w^\top(R - r_f\mathbf{1}).$$

The *excess-mean vector* is $\mu_e = \mu - r_f\mathbf{1}$, assumed nonzero. By Proposition 11.2, with the deterministic component contributing zero variance and zero covariance, $\mathbb{E}[R_p] = r_f + w^\top\mu_e$ and $\mathrm{Var}(R_p) = w^\top\Sigma w$. With the frontier constants (11.5), put

$$k = \mathbf{1}^\top\Sigma^{-1}\mu_e = B - r_f A.$$

Standing assumption: $r_f < B/A$, the risk-free rate lies below the minimum-variance mean; since $A > 0$ this is equivalent to $k > 0$. The case $r_f \geq B/A$ receives one sentence after Theorem 11.23. Risk aversion is $\gamma > 0$.

Definition 11.22 (Excess Sharpe ratio, tangency portfolio and capital market line)**.** The *(excess) Sharpe ratio* of a risky position w with $w^\top\Sigma w > 0$ is

$$\mathrm{SR}(w) = \frac{w^\top\mu_e}{\sqrt{w^\top\Sigma w}} = \frac{\mathbb{E}[R_p] - r_f}{\sqrt{\mathrm{Var}(R_p)}}.$$

It reduces to Definition 10.1 at $r_f = 0$ — §10 took $\mathrm{SR} = \mu/\sigma$ with no risk-free adjustment, and that convention is the $r_f = 0$ case of this one — and is invariant under $w \mapsto tw$ for $t > 0$, both numerator and denominator scaling by t. The *tangency portfolio* is a fully invested ($\mathbf{1}^\top w = 1$) maximiser of SR. The *capital market line* (CML) is the set of (σ, mean) pairs obtained by mixing the risk-free asset with the tangency portfolio.

Theorem 11.23 (Maximum Sharpe ratio and the tangency portfolio)**.** *Under the setup above, with Σ positive definite and $\mu_e \neq 0$:*

(i) For every w with $w^\top \Sigma w > 0$,

$$w^\top \mu_e \;\le\; \sqrt{w^\top \Sigma w}\,\sqrt{\mu_e^\top \Sigma^{-1} \mu_e},$$

with equality if and only if w is a positive multiple of $\Sigma^{-1}\mu_e$. Hence

$$\sup_w \mathrm{SR}(w) = \mathrm{SR}^* := \sqrt{\mu_e^\top \Sigma^{-1} \mu_e} = \sqrt{C - 2r_f B + r_f^2 A},$$

attained exactly on the ray $\{t\,\Sigma^{-1}\mu_e : t > 0\}$.

(ii) Under $r_f < B/A$ the unique tangency portfolio is

$$w_T = \frac{\Sigma^{-1}\mu_e}{k} = \frac{\Sigma^{-1}(\mu - r_f \mathbf{1})}{B - r_f A}, \tag{11.10}$$

with mean $\mu_T = (C - r_f B)/k$ and variance $\sigma_T^2 = (C - 2r_f B + r_f^2 A)/k^2 = (\mathrm{SR}^*)^2/k^2$.

(iii) The portfolio w_T is the frontier portfolio at $m = \mu_T$ and lies on the efficient branch: $\mu_T - B/A = D/(Ak) > 0$.

Proof. We apply the Cauchy–Schwarz inequality of Lemma 11.6(ii) in the Σ-inner product, with the second vector chosen so that its Σ-product with w is the excess mean.

Part (i). Put $y = \Sigma^{-1}\mu_e$, so that $\Sigma y = \mu_e$ and $w^\top \mu_e = w^\top \Sigma y$. Lemma 11.6(ii) with $G = \Sigma$, $x = w$ gives

$$\begin{aligned}(w^\top \mu_e)^2 = (w^\top \Sigma y)^2 &\le (w^\top \Sigma w)(y^\top \Sigma y) \\ &= (w^\top \Sigma w)\,(\mu_e^\top \Sigma^{-1} \Sigma \Sigma^{-1} \mu_e) = (w^\top \Sigma w)\,(\mu_e^\top \Sigma^{-1} \mu_e),\end{aligned}$$

with equality if and only if w and y are linearly dependent. Both $w^\top \Sigma w$ and $\mu_e^\top \Sigma^{-1}\mu_e$ are strictly positive (positive definiteness of Σ and, by Lemma 11.6(i), of Σ^{-1}, with $w \neq 0$ and $\mu_e \neq 0$), so taking square roots gives $|w^\top \mu_e| \le \sqrt{w^\top \Sigma w}\,\mathrm{SR}^*$, and dividing by $\sqrt{w^\top \Sigma w}$ gives $\mathrm{SR}(w) \le \mathrm{SR}^*$. Equality in the inequality for $w^\top \mu_e$ itself (not its absolute value) requires $w = ty$ with $w^\top \mu_e = t\, y^\top \Sigma y > 0$, i.e. $t > 0$; negative multiples have Sharpe ratio $-\mathrm{SR}^*$. Every positive multiple attains SR^* by scale invariance. Finally, substituting $\mu_e = \mu - r_f \mathbf{1}$, i.e. $\Sigma^{-1}\mu_e = \Sigma^{-1}\mu - r_f \Sigma^{-1}\mathbf{1}$, and expanding with (11.5),

$$\begin{aligned}\mu_e^\top \Sigma^{-1} \mu_e &= \mu^\top \Sigma^{-1}\mu - 2r_f\, \mathbf{1}^\top \Sigma^{-1}\mu + r_f^2\, \mathbf{1}^\top \Sigma^{-1}\mathbf{1} \\ &= C - 2r_f B + r_f^2 A.\end{aligned}$$

Part (ii). The ray $\{ty : t > 0\}$ meets the hyperplane $\mathbf{1}^\top w = 1$ exactly when $t\,\mathbf{1}^\top y = tk = 1$, which has the unique solution $t = 1/k$ because $k > 0$ under the standing assumption; so $w_T = y/k$ is the unique fully invested point of the ray, hence the unique tangency portfolio, and $k = \mathbf{1}^\top \Sigma^{-1}\mu - r_f\, \mathbf{1}^\top \Sigma^{-1}\mathbf{1} = B - r_f A$. The mean is $w_T^\top \mu = \mu^\top \Sigma^{-1}(\mu - r_f \mathbf{1})/k = (C - r_f B)/k$ and the variance is $w_T^\top \Sigma w_T = y^\top \Sigma y/k^2 = \mu_e^\top \Sigma^{-1}\mu_e/k^2$.

Part (iii). From (11.10), $\Sigma w_T = \mu_e/k = (1/k)\mu - (r_f/k)\mathbf{1} \in \mathrm{span}(\mathbf{1}, \mu)$, and w_T is fully invested, so Theorem 11.16(iii) makes it the frontier portfolio at its own mean μ_T. For efficiency,

$$\mu_T - \frac{B}{A} = \frac{C - r_f B}{k} - \frac{B}{A} = \frac{A(C - r_f B) - B(B - r_f A)}{Ak} = \frac{AC - B^2}{Ak} = \frac{D}{Ak} > 0$$

by Lemma 11.15, so w_T sits on the upper branch of Corollary 11.18. □

(Problem P11.8 asks you to reproduce this proof from scratch.)

When $r_f \geq B/A$ the maximum-Sharpe ray of part (i) still exists, but its fully invested normalisation is undefined ($k = 0$) or lies on the inefficient branch with Sharpe ratio $-\mathrm{SR}^*$ ($k < 0$), so no tangency point sits on the efficient frontier; the case receives no further treatment.

Example 11.24 (Tangency in the running universe at $r_f = 2\%$)**.** With $\mu = (4\%, 7\%, 8\%)^\top$ and $r_f = 2\%$, $\mu_e = (2\%, 5\%, 6\%)^\top$; with Σ^{-1} from the setup of §11.2, $\Sigma^{-1}\mu_e = 25\,(0.02,\ 0.04,\ 0.10)^\top = (0.5,\ 1,\ 2.5)^\top$ (the three row sums being $0.06-0.10+0.06$, $-0.04+0.20-0.12$ and $0.02-0.10+0.18$), so $k = 4 = B - r_f A = 6 - 2$, and by (11.10) $w_T = (0.125, 0.25, 0.625)^\top$. The squared maximal Sharpe ratio is

$$\begin{aligned}(\mathrm{SR}^*)^2 = \mu_e^\top \Sigma^{-1}\mu_e &= 0.02(0.5) + 0.05(1) + 0.06(2.5) = 0.21 \\ &= C - 2r_f B + r_f^2 A = 0.41 - 0.24 + 0.04,\end{aligned}$$

so $\mathrm{SR}^* = 0.458$. The tangency mean is $(C - r_f B)/k = (0.41 - 0.12)/4 = 7.25\%$ and the variance $0.21/16 = 0.013125$ (volatility 11.46%); direct check, $(7.25\% - 2\%)/11.46\% = 0.458$. It lies on the frontier: $\sigma^2(0.0725) = 0.01 + 20(0.0125)^2 = 0.013125$ by (11.9), and $\mu_T - B/A = 1.25\% = D/(Ak) = 5/400$.

(Problem P11.9 works this computation at different numbers.)

Corollary 11.25 (Minimum variance equals maximum Sharpe only under equal excess means)**.** *Under the setup above, with the standing assumption of §11.3 suspended for this statement (so that $\mu_e \propto \mathbf{1}$, i.e. $D = 0$, is allowed), $w_T = w_g$ if and only if μ_e is a multiple of $\mathbf{1}$, i.e. if and only if every asset has the same expected excess return. Under that standing assumption the two portfolios therefore differ, and the tangency portfolio sits strictly above the minimum-variance portfolio on the efficient branch.*

Proof. We compare (11.4) with (11.10) after multiplying by the invertible matrix Σ; both formulas remain valid when $D = 0$, because the proofs of Theorem 11.7 and of Theorem 11.23(ii) use only positive definiteness and $k > 0$. The equality $\Sigma^{-1}\mu_e/k = \Sigma^{-1}\mathbf{1}/A$ holds if and only if $\mu_e = (k/A)\mathbf{1}$, a multiple of $\mathbf{1}$. Conversely, if $\mu_e = \kappa\,\mathbf{1}$ for a scalar κ, then $k = \kappa A$, so $\kappa > 0$ and $\Sigma^{-1}\mu_e/k = \kappa\,\Sigma^{-1}\mathbf{1}/(\kappa A) = w_g$. Since $\mu_e = \mu - r_f\mathbf{1}$ is a multiple of $\mathbf{1}$ exactly when μ is, the standing assumption rules this out, and then $\mu_T - B/A = D/(Ak) > 0$ by Theorem 11.23(iii). □

Example 11.26 (Minimum variance against maximum Sharpe)**.** In the running universe the minimum-variance portfolio $w_g = (0.5, 0, 0.5)^\top$ has Sharpe ratio $(6\% - 2\%)/10\% = 0.40$, against $\mathrm{SR}^* = 0.458$ at the tangency portfolio of Example 11.24; the two differ because the excess-mean vector $(2\%, 5\%, 6\%)^\top$ is not a multiple of $\mathbf{1}$. Replacing μ by $(6\%, 6\%, 6\%)^\top$ instead gives $\mu_e = 0.04\,\mathbf{1}$, so that $\Sigma^{-1}\mu_e = 0.04\,(50, 0, 50)^\top = (2, 0, 2)^\top$ and $k = 4$, hence $w_T = (0.5, 0, 0.5)^\top = w_g$, as Corollary 11.25 predicts.

Corollary 11.27 (Capital market line)**.** *Under the setup above:*

(i) Holding a fraction $a \geq 0$ of wealth in w_T and $1 - a$ in the risk-free asset gives mean $r_f + a(\mu_T - r_f)$ and volatility $a\sigma_T$, so the attainable pairs form the line

$$\text{mean} = r_f + \mathrm{SR}^*\,\sigma, \qquad \sigma \geq 0. \tag{11.11}$$

(ii) Every portfolio (risky part w, remainder at r_f) satisfies $\mathbb{E}[R_p] - r_f \leq \mathrm{SR}^*\sqrt{\mathrm{Var}(R_p)}$, with equality if and only if w is a nonnegative multiple of $\Sigma^{-1}\mu_e$. In particular every risky frontier portfolio satisfies $m - r_f \leq \mathrm{SR}^*\,\sigma(m)$ with equality only at $m = \mu_T$: the CML lies weakly above the hyperbola of Corollary 11.18 and touches it at exactly one point (Figure 19).

Proof. We compute the moments of the mix by Proposition 11.2 and then rewrite Theorem 11.23(i).

Part (i). The risky weight vector of the mix is $a\,w_T$ and the risk-free asset contributes zero variance and zero covariance, so the mean is $r_f + a\,w_T^\top \mu_e = r_f + a(\mu_T - r_f)$ and the variance is $a^2 w_T^\top \Sigma w_T = a^2\sigma_T^2$, giving volatility $a\sigma_T$ for $a \geq 0$. Eliminating $a = \sigma/\sigma_T$ yields mean $= r_f + ((\mu_T - r_f)/\sigma_T)\sigma$, and $(\mu_T - r_f)/\sigma_T = \mathrm{SR}(w_T) = \mathrm{SR}^*$ by Theorem 11.23.

Part (ii). For $w = 0$ both sides vanish. For $w \neq 0$, $\mathbb{E}[R_p] - r_f = w^\top \mu_e$ and $\sqrt{\mathrm{Var}(R_p)} = \sqrt{w^\top \Sigma w}$, so the inequality is Theorem 11.23(i), with equality exactly on the ray $t\,\Sigma^{-1}\mu_e$, $t > 0$. Applied to $w = w(m)$, a fully invested portfolio, the equality case identifies $w(m)$ as a positive multiple of $\Sigma^{-1}\mu_e$ that sums to one, hence as w_T by Theorem 11.23(ii), i.e. $m = \mu_T$. □

Example 11.28 (The capital market line of the running universe)**.** With $r_f = 2\%$ and $\mathrm{SR}^* = 0.458$ from Example 11.24, the line (11.11) reads mean $= 0.02 + 0.458\,\sigma$. At the volatility 13.42% of the frontier portfolio $w(8\%)$ it delivers $8.15\% > 8\%$, at the 10.95% of $w(7\%)$ it delivers $7.02\% > 7\%$, and at $\sigma_T = 11.46\%$ it returns 7.25%, the tangency point itself: the line beats the hyperbola everywhere except where it touches. The 8.15% point is held as $a = 0.1342/0.1146 = 1.17$ times the tangency portfolio, financed by borrowing 17% of wealth at r_f.

Figure 19 shows the line (11.11): from $r_f = 2\%$ on the vertical axis, through the tangency point at $(11.46\%, 7.25\%)$, with slope $\mathrm{SR}^* = 0.458$; the hyperbola lies under it everywhere except at w_T.

Corollary 11.29 (Sharpe ratios of uncorrelated strategies add in quadrature)**.** *Under the setup above:*

(i) If $\Sigma = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_n^2)$ *and* $\mathrm{SR}_i = (\mu_i - r_f)/\sigma_i$ *denotes the Sharpe ratio of asset* i *alone, then*

$$(\mathrm{SR}^*)^2 = \sum_{i=1}^{n} \mathrm{SR}_i^2,$$

attained at weights $w_i \propto (\mu_i - r_f)/\sigma_i^2 = \mathrm{SR}_i/\sigma_i$*, i.e. at* risk budgets $\sigma_i w_i \propto \mathrm{SR}_i$*.*

(ii) For every Σ*,* $\mathrm{SR}^* \geq \max_i \mathrm{SR}_i$*.*

Special cases: with $n = 2$*, equal Sharpe ratios and* $r_f = 0$*,* $\mathrm{SR}^* = \sqrt{2}\,\mathrm{SR}_1$*, which is Theorem 10.11: the equal-volatility hypothesis of that theorem is not needed for the value* $\sqrt{2}\,\mathrm{SR}_1$ *itself, only for the raw sum of the two legs to attain it, since equal volatilities put the raw sum on the tangency ray; with* n *legs of equal Sharpe ratio,* $\mathrm{SR}^* = \sqrt{n}\,\mathrm{SR}_1$*, which is Corollary 10.12; and the weights* $w_i \propto \mu_i/\sigma_i^2$ *are exactly the mean–variance weights promised in Remark 10.14.*

Proof. We evaluate Theorem 11.23(i) for a diagonal matrix and at the basis vectors.

Part (i). With $\Sigma^{-1} = \mathrm{diag}(\sigma_1^{-2}, \ldots, \sigma_n^{-2})$, $\mu_e^\top \Sigma^{-1} \mu_e = \sum_i (\mu_i - r_f)^2/\sigma_i^2 = \sum_i \mathrm{SR}_i^2$, and $\Sigma^{-1}\mu_e$ has entries $(\mu_i - r_f)/\sigma_i^2 = \mathrm{SR}_i/\sigma_i$; multiplying entry i by σ_i gives the risk budget $\sigma_i w_i \propto \mathrm{SR}_i$. For the special cases set every SR_i equal to SR_1 and count the terms: $\sqrt{2\,\mathrm{SR}_1^2}$ and $\sqrt{n\,\mathrm{SR}_1^2}$. In the setting of Theorem 10.11 the two legs have equal volatility, so SR_i/σ_i is the same for both and the raw sum of the two legs in that theorem is a positive multiple of the tangency direction.

Part (ii). Taking $w = e_i$ in the supremum of Theorem 11.23(i) gives $\mathrm{SR}(e_i) = (\mu_i - r_f)/\sigma_i = \mathrm{SR}_i \leq \mathrm{SR}^*$. □

Example 11.30 (The interview classic: three uncorrelated strategies)**.** Three uncorrelated strategies have Sharpe ratios 0.5, 1.0, 1.0 and volatilities 10%, 20%, 5%; their returns are self-financing P&L streams, so $r_f = 0$ and the means are 5%, 20%, 5%. By Corollary 11.29, $\mathrm{SR}^* = \sqrt{0.25 + 1 + 1} = 1.5$, at weights proportional to $\mu_i/\sigma_i^2 = (5, 5, 20)$, i.e. $(1/6, 1/6, 2/3)$. The risk allocations $\sigma_i w_i = (1.67\%, 3.33\%, 3.33\%)$ are proportional to $(0.5, 1, 1) = \mathrm{SR}_i$. The combined mean is $5/6 + 20/6 + 10/3 = 7.5\%$ and the volatility $\sqrt{0.0167^2 + 0.0333^2 + 0.0333^2} = 5.0\%$, Sharpe ratio 1.5. Equal capital weights instead give mean 10%, volatility $\sqrt{0.0333^2 + 0.0667^2 + 0.0167^2} = 7.64\%$

and Sharpe ratio 1.31. The "average Sharpe" 0.83 and the "sum of Sharpes" 2.5 are both wrong. The two-strategy version — Sharpe 1 with Sharpe 0.5, uncorrelated — gives $\sqrt{1.25} = 1.118$: not 0.75, not 1.5.

(Problem P11.7 works this computation at different numbers.)

Corollary 11.31 (Two correlated strategies and the correlation hurdle)**.** *Let $n = 2$ with Sharpe ratios $\mathrm{SR}_1 = \mu_{e,1}/\sigma_1$, $\mathrm{SR}_2 = \mu_{e,2}/\sigma_2$ and correlation $\rho \in (-1, 1)$. Then*

$$(\mathrm{SR}^*)^2 = \frac{\mathrm{SR}_1^2 - 2\rho\,\mathrm{SR}_1\mathrm{SR}_2 + \mathrm{SR}_2^2}{1-\rho^2},$$

attained at risk budgets $\sigma_1 w_1 \propto \mathrm{SR}_1 - \rho\,\mathrm{SR}_2$ and $\sigma_2 w_2 \propto \mathrm{SR}_2 - \rho\,\mathrm{SR}_1$. In particular the maximum-Sharpe position in strategy 2 is long if and only if $\mathrm{SR}_2 > \rho\,\mathrm{SR}_1$ (the correlation hurdle*), and $\mathrm{SR}^* > \mathrm{SR}_1$ whenever $\mathrm{SR}_2 \neq \rho\,\mathrm{SR}_1$.*

Proof. We evaluate Theorem 11.23(i) with the explicit two-by-two inverse. Write $\Sigma = \Delta K \Delta$ with $\Delta = \mathrm{diag}(\sigma_1, \sigma_2)$ and $K = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, so that $K^{-1} = \frac{1}{1-\rho^2}\left(\begin{smallmatrix} 1 & -\rho \\ -\rho & 1 \end{smallmatrix}\right)$, and put $s = (\mathrm{SR}_1, \mathrm{SR}_2)^\top$, so that $\mu_e = \Delta s$. Then $\Sigma^{-1} = \Delta^{-1}K^{-1}\Delta^{-1}$ and $\mu_e^\top \Sigma^{-1} \mu_e = s^\top K^{-1} s$, which is the displayed formula. Next, $\Sigma^{-1}\mu_e = \Delta^{-1}K^{-1}s$, whose i-th entry multiplied by σ_i is $(K^{-1}s)_i = (\mathrm{SR}_i - \rho\,\mathrm{SR}_j)/(1-\rho^2)$ with $j \neq i$; by Theorem 11.23(i) the Sharpe ratio is maximised exactly on the ray of positive multiples of this vector, so the sign claim reads off its second entry; no fully invested normalisation is used, so the standing assumption $r_f < B/A$ plays no role here. Finally

$$(\mathrm{SR}^*)^2 - \mathrm{SR}_1^2 = \frac{\mathrm{SR}_1^2 - 2\rho\,\mathrm{SR}_1\mathrm{SR}_2 + \mathrm{SR}_2^2 - (1-\rho^2)\mathrm{SR}_1^2}{1-\rho^2} = \frac{(\mathrm{SR}_2 - \rho\,\mathrm{SR}_1)^2}{1-\rho^2} \;\geq\; 0,$$

with equality only at $\mathrm{SR}_2 = \rho\,\mathrm{SR}_1$. □

Example 11.32 (Adding a signal to a book)**.** A book has Sharpe ratio 1.5; a candidate signal has Sharpe ratio 0.8. At $\rho = 0.6$ the hurdle is $0.6 \times 1.5 = 0.9 > 0.8$: the optimiser shorts the signal, and even then $(\mathrm{SR}^*)^2 = (2.25 - 1.44 + 0.64)/0.64 = 2.266$, $\mathrm{SR}^* = 1.505$ — no material gain, so a long-only desk does not add it. At $\rho = 0.3$ the hurdle is $0.45 < 0.8$: add it long, with $(\mathrm{SR}^*)^2 = (2.25 - 0.72 + 0.64)/0.91 = 2.385$, $\mathrm{SR}^* = 1.544$. At $\rho = 0$ the quadrature rule gives $\sqrt{2.89} = 1.70$.

Proposition 11.33 (Mean–variance utility and the tangency direction)**.** *Fix a* risk aversion *$\gamma > 0$ and let $w \in \mathbb{R}^n$ be unconstrained (the risk-free asset absorbs $1 - \mathbf{1}^\top w$, borrowing allowed). The* mean–variance utility

$$U(w) = \mathbb{E}[R_p] - \tfrac{\gamma}{2}\,\mathrm{Var}(R_p) = r_f + w^\top \mu_e - \tfrac{\gamma}{2}\, w^\top \Sigma w$$

has the unique maximiser

$$w^\star = \frac{1}{\gamma}\,\Sigma^{-1}\mu_e,$$

a multiple of the tangency direction — hence of w_T — with $\mathbf{1}^\top w^\star = k/\gamma$ in risky assets, and maximal value $U(w^\star) = r_f + (\mathrm{SR}^*)^2/(2\gamma)$.

Proof. We complete the square. Expanding the quadratic form in $w - w^\star$ and using $\Sigma w^\star = \mu_e/\gamma$,

$$\begin{aligned} \frac{\gamma}{2}(w - w^\star)^\top \Sigma (w - w^\star) &= \frac{\gamma}{2}\, w^\top \Sigma w - \gamma\, w^\top \Sigma w^\star + \frac{\gamma}{2}\,(w^\star)^\top \Sigma w^\star \\ &= \frac{\gamma}{2}\, w^\top \Sigma w - w^\top \mu_e + \frac{1}{2\gamma}\,\mu_e^\top \Sigma^{-1} \mu_e, \end{aligned}$$

so that

$$U(w) = r_f + \frac{1}{2\gamma}\,\mu_e^\top \Sigma^{-1}\mu_e - \frac{\gamma}{2}(w - w^\star)^\top \Sigma (w - w^\star).$$

As a check on the algebra, (11.3) gives $\nabla_w U = \mu_e - \gamma\Sigma w$, which vanishes exactly at $w^\star$. Since Σ is positive definite the subtracted term is nonnegative and vanishes if and only if $w = w^\star$, so $w^\star$ is the unique maximiser and the maximal value is the remaining constant, $r_f + (\mathrm{SR}^*)^2/(2\gamma)$. Finally $\mathbf{1}^\top w^\star = \mathbf{1}^\top \Sigma^{-1}\mu_e/\gamma = k/\gamma$. □

This is two-fund separation with a risk-free asset: every mean–variance investor holds the same risky portfolio w_T and differs only in the scale $1/\gamma$ — the premise of Fact 11.40 in §11.5. The case $\gamma = 1$ is revisited as the Kelly connection in Remark 11.59.

Example 11.34 (Utility scaling in the running universe)**.** With $\Sigma^{-1}\mu_e = (0.5, 1, 2.5)^\top$ and $k = 4$ from Example 11.24, Proposition 11.33 gives:

- With $\gamma = 1$: $w^\star = (0.5, 1, 2.5)^\top$, a 400% gross risky exposure financed by borrowing 300% at r_f; excess mean $w^{\star\top}\mu_e = 0.01 + 0.05 + 0.15 = 21\%$ (total mean 23%), variance $\mu_e^\top \Sigma^{-1}\mu_e = 0.21$, volatility 45.8%, value $r_f + 0.21/2 = r_f + 0.105$.
- With $\gamma = 4$: $w^\star = (0.125, 0.25, 0.625)^\top$, exactly the fully invested tangency portfolio (100% risky).
- With $\gamma = 10$: $w^\star = (0.05, 0.10, 0.25)^\top$, 40% in risky assets and 60% in cash; excess mean 2.1%, volatility $\sqrt{0.21}/10 = 4.58\%$, value $0.02 + 0.21/20 = 3.05\%$.

The risky mix is $(0.125, 0.25, 0.625)^\top$ in every case: the mean–variance weights fix a *direction*, and the leverage $1/\gamma$ is a separate decision. "Full Kelly on the tangency portfolio" ($\gamma = 1$) means 400% gross exposure at 45.8% volatility; Remark 3.8 and §11.7 give the reasons practice sets γ well above 1.

Intuition. Uncorrelated strategies are orthogonal directions, and Sharpe ratios are the lengths of the edge along each: the best combination is the diagonal of the box, and its length is the root of the sum of squares. Correlation tilts the axes; Σ^{-1} is exactly the change of basis that makes them orthogonal again, which is why the general answer is $\sqrt{\mu_e^\top \Sigma^{-1}\mu_e}$.

Interview trap. Sharpe ratios do not average and do not add. Uncorrelated strategies combine in quadrature, and only with risk budgets proportional to SR_i: equal capital weights give 1.31 instead of 1.5 in Example 11.30, and two uncorrelated Sharpe-1 strategies give 1.41, not 1 and not 2. Quoting a combined Sharpe ratio without saying "uncorrelated" is the companion error: correlation changes the answer in both directions, and with unequal Sharpe ratios a large ρ can even exceed quadrature because Σ^{-1} turns the weaker leg into a hedge (Corollary 11.31 at Sharpe ratios 1 and 0.5, $\rho = 0.9$: $\mathrm{SR}^* = 1.357$ against 1.118, the weaker leg shorted).

Interview trap. Minimum variance is not maximum Sharpe. The portfolios $w_g = \Sigma^{-1}\mathbf{1}/A$ and $w_T \propto \Sigma^{-1}\mu_e$ coincide only when every asset has the same expected excess return (Corollary 11.25); in the running universe they are $(0.5, 0, 0.5)^\top$ and $(0.125, 0.25, 0.625)^\top$ with Sharpe ratios 0.40 and 0.458, and the tangency portfolio always sits strictly higher on the efficient branch.

Interview trap. The tangency direction is $\Sigma^{-1}(\mu - r_f\mathbf{1})$, not $\Sigma^{-1}\mu$: "the optimal portfolio" without naming r_f is incomplete. Raising r_f raises the tangency portfolio's mean, $\mu_T - B/A = D/(Ak)$ growing as $k = B - r_f A$ shrinks, and once r_f reaches B/A no tangency point remains on the efficient branch (Theorem 11.23(iii) and the sentence after its proof).

11.5 Beta, the CAPM and the Security Market Line

The rapid-fire block: "what is beta?", "do you hedge with beta or with correlation?", "state the CAPM in one line", "what is alpha?", "what is the difference between the CML and the SML?".

The tangency algebra of §11.4 contains, as an identity, the statement that every asset's expected excess return is its beta to the tangency portfolio times the tangency portfolio's expected excess return; the CAPM (Sharpe, 1964; Lintner, 1965) is the economic claim that in equilibrium the market portfolio *is* the tangency portfolio, which turns the identity into a pricing model. This subsection defines beta as the population form of the regression slope of Corollary 5.3, proves the systematic/idiosyncratic decomposition and the security-market-line identity, and states the equilibrium as a Fact.

Setup. A *reference portfolio* (a benchmark, or the market) has return R_M with mean μ_M and variance $\sigma_M^2 > 0$. An asset or portfolio has return R_i with mean μ_i, volatility $\sigma_i > 0$ and correlation $\rho_{iM} = \mathrm{Corr}(R_i, R_M)$ with the reference. The risk-free rate is r_f; excess returns are $R_i - r_f$ and $R_M - r_f$. A portfolio w over the n assets is fully invested, $\mathbf{1}^\top w = 1$. Beta here is a population quantity; its sample estimate is the $\hat{\beta}$ of §5, whose subsection on inverse regression, beta and hedging (Proposition 5.21, Remark 5.23) already contrasts beta with correlation. The residual return of asset i is ε_i, with variance $\sigma_{\varepsilon,i}^2$; Jensen's alpha is α_i.

Definition 11.35 (Beta with respect to a reference portfolio)**.** The *beta* of asset i to the reference portfolio is

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_M)}{\mathrm{Var}(R_M)} = \rho_{iM}\,\frac{\sigma_i}{\sigma_M},$$

the population slope of the regression of R_i on R_M: it is the population form of (5.3) in Corollary 5.3 with $x = R_M$ and $y = R_i$, and the coefficient β^* of Proposition 10.15 with the black-box output taken to be R_M and the new model taken to be R_i. The beta of a portfolio is a weighted average of betas, $\beta_w = w^\top \beta$: by bilinearity of covariance, $\mathrm{Cov}(w^\top R, R_M)/\sigma_M^2 = \sum_i w_i \,\mathrm{Cov}(R_i, R_M)/\sigma_M^2 = \sum_i w_i \beta_i$.

Proposition 11.36 (Systematic and idiosyncratic variance)**.** *Define $\alpha_i = \mu_i - r_f - \beta_i(\mu_M - r_f)$ and the residual $\varepsilon_i = (R_i - r_f) - \alpha_i - \beta_i(R_M - r_f)$. Then:*

(i) The residual satisfies $\mathbb{E}[\varepsilon_i] = 0$ and $\mathrm{Cov}(\varepsilon_i, R_M) = 0$, and β_i is the unique coefficient with the latter property.

(ii) The variance splits as

$$\mathrm{Var}(R_i) = \beta_i^2 \sigma_M^2 + \sigma_{\varepsilon,i}^2, \tag{11.12}$$

with $\sigma_{\varepsilon,i}^2 = (1 - \rho_{iM}^2)\,\sigma_i^2$, so the systematic *share of variance, $\beta_i^2\sigma_M^2/\sigma_i^2$, equals ρ_{iM}^2, the R^2 of Corollary 5.3.*

(iii) The portfolio beta is $\beta_w = w^\top \beta$, and if the residuals $\varepsilon_1, \ldots, \varepsilon_n$ are pairwise uncorrelated, then

$$\mathrm{Var}(R_w) = \beta_w^2 \sigma_M^2 + \sum_{i=1}^{n} w_i^2\, \sigma_{\varepsilon,i}^2.$$

With equal weights and a common residual variance σ_ε^2, the idiosyncratic part is $\sigma_\varepsilon^2/n \to 0$ while the systematic part $\bar{\beta}^2\sigma_M^2$ persists, $\bar{\beta}$ being the average beta: this is the floor of Proposition 11.12 identified with market risk. In particular, equal betas β and equal volatilities put Σ in the equicorrelated model with $\rho v = \beta^2\sigma_M^2$.

Proof. We reduce parts (i) and (ii) to Proposition 10.15 and prove part (iii) by bilinearity of covariance.

Part (i). By bilinearity, $\mathrm{Cov}(\varepsilon_i, R_M) = \mathrm{Cov}(R_i, R_M) - \beta_i \,\mathrm{Var}(R_M) = 0$ by the definition of β_i, the constants r_f, α_i, $\beta_i r_f$ contributing nothing. Uniqueness is exactly as in Proposition 10.15 with the black box taken to be R_M and the new model R_i: $\mathrm{Cov}(R_i - bR_M, R_M) = \mathrm{Cov}(R_i, R_M) - b\sigma_M^2$ vanishes only at $b = \beta_i$. Up to the constant mean shift, ε_i is the $B_\perp$ of that proposition. Taking expectations, $\mathbb{E}[\varepsilon_i] = (\mu_i - r_f) - \alpha_i - \beta_i(\mu_M - r_f) = 0$ by the definition of α_i.

Part (ii). Rearranging, $R_i = r_f + \alpha_i + \beta_i(R_M - r_f) + \varepsilon_i$ has random part $\beta_i R_M + \varepsilon_i$, so Proposition 10.10 gives $\mathrm{Var}(R_i) = \beta_i^2\sigma_M^2 + \sigma_{\varepsilon,i}^2 + 2\beta_i \mathrm{Cov}(R_M, \varepsilon_i)$, whose cross term vanishes by part (i); this is (11.12). Since constants change no variance, $\sigma_{\varepsilon,i}^2 = \mathrm{Var}(B_\perp) = (1 - \rho_{iM}^2)\sigma_i^2$ by Proposition 10.15; equivalently, substituting $\beta_i = \rho_{iM}\sigma_i/\sigma_M$ into (11.12) gives $\beta_i^2\sigma_M^2 = \rho_{iM}^2\sigma_i^2$, and the systematic share is ρ_{iM}^2.

Part (iii). Bilinearity gives $\beta_w = w^\top\beta$ as in Definition 11.35. Summing the decompositions of part (ii) with weights w_i and using $\sum_i w_i = 1$, $R_w = r_f + \sum_i w_i\alpha_i + \beta_w(R_M - r_f) + \sum_i w_i\varepsilon_i$. The residual sum is uncorrelated with R_M by part (i), and its variance has no cross terms by the pairwise-uncorrelatedness hypothesis, so $\mathrm{Var}(R_w) = \beta_w^2\sigma_M^2 + \sum_i w_i^2\sigma_{\varepsilon,i}^2$. At $w = \frac{1}{n}\mathbf{1}$ with $\sigma_{\varepsilon,i}^2 = \sigma_\varepsilon^2$ the second term is $n \cdot \sigma_\varepsilon^2/n^2 = \sigma_\varepsilon^2/n$ and $\beta_w = \bar\beta$. For the last claim, with a common beta β and $i \neq j$, bilinearity and part (i) give $\mathrm{Cov}(R_i, R_j) = \mathrm{Cov}(\beta R_M + \varepsilon_i, \beta R_M + \varepsilon_j) = \beta^2\sigma_M^2$, while $\mathrm{Var}(R_i) = \beta^2\sigma_M^2 + \sigma_\varepsilon^2 = v$ is common; so Σ is the equicorrelated model of §11.2 with $\rho v = \beta^2\sigma_M^2$, and the floor ρv of Proposition 11.12(ii) is the shared factor exposure. □

Example 11.37 (One stock decomposed)**.** A stock has $\rho_{iM} = 0.6$, $\sigma_i = 30\%$, and the market has $\sigma_M = 15\%$. By Definition 11.35, $\beta_i = 0.6 \times 0.30/0.15 = 1.2$. By (11.12) the systematic variance is $1.44 \times 0.0225 = 0.0324$ (volatility 18%), the idiosyncratic variance is $0.09 - 0.0324 = 0.0576$ (volatility 24%), and the systematic share is $0.0324/0.09 = 0.36 = \rho_{iM}^2$. Beta is not correlation: a second stock with $\rho_{iM} = 0.9$ and $\sigma_i = 10\%$ has $\beta_i = 0.9 \times 0.10/0.15 = 0.6$ — tighter co-movement, half the beta — and its hedge ratio is 0.6, not 0.9 (Remark 5.23; the direction of the regression matters, Proposition 5.21). Its residual share is $1 - 0.81 = 19\%$ against the first stock's 64%: the better-hedged stock is the one with the smaller hedge.

(Problem P11.10 works this computation at different numbers.)

Setup (tangency reference). As in §11.4: Σ is positive definite, $r_f < B/A$, and the tangency portfolio w_T of (11.10) has return R_T, mean μ_T and variance σ_T^2. It is now taken as the reference portfolio, so that $\beta_{w,T} = \mathrm{Cov}(R_w, R_T)/\mathrm{Var}(R_T)$.

Proposition 11.38 (The tangency portfolio prices every asset (the SML identity))**.** *If the reference portfolio is the tangency portfolio ($R_M = R_T$), then for every fully invested portfolio w — in particular every single asset, $w = e_i$ —*

$$\mu_w - r_f = \beta_{w,T}\,(\mu_T - r_f), \qquad \mu_w = w^\top\mu :$$

every alpha measured against the tangency portfolio is zero. No equilibrium assumption is used.

Proof. We compute both covariances from (11.10), which gives $\Sigma w_T = \mu_e/k$. By bilinearity and $\mathbf{1}^\top w = 1$,

$$\mathrm{Cov}(R_w, R_T) = w^\top\Sigma w_T = \frac{w^\top\mu_e}{k} = \frac{w^\top\mu - r_f\,\mathbf{1}^\top w}{k} = \frac{\mu_w - r_f}{k}$$

and

$$\mathrm{Var}(R_T) = w_T^\top\Sigma w_T = \frac{w_T^\top\mu_e}{k} = \frac{\mu_T - r_f}{k}.$$

The second quantity is positive ($\mu_T - r_f = \mathrm{SR}^*\sigma_T > 0$), so dividing, $\beta_{w,T} = (\mu_w - r_f)/(\mu_T - r_f)$, which rearranges to the claim. □

Example 11.39 (The SML identity in the running universe)**.** Against the tangency portfolio of Example 11.24 ($r_f = 2\%$, $w_T = (0.125, 0.25, 0.625)^\top$) the covariances are $\mathrm{Cov}(R_i, R_T) = (\Sigma w_T)_i$ with $\Sigma w_T = 0.01\,(0.5,\ 1.25,\ 1.5)^\top = (0.005, 0.0125, 0.015)^\top = \mu_e/4$ (the row sums being $0.25+0.25$, $0.125+0.5+0.625$ and $0.25+1.25$), and $\mathrm{Var}(R_T) = 0.013125$, so the betas are $(0.381, 0.952, 1.143)^\top$. Then $\beta_i \times (7.25\% - 2\%) = (2\%, 5\%, 6\%)^\top = \mu_e$ exactly: every alpha is zero, as Proposition 11.38 requires. Against the minimum-variance portfolio instead, every beta equals 1 (the equal-covariance property of Theorem 11.7) while the excess means differ, so w_g does not price the assets.

Dividing the identity by the volatility σ_w and substituting $\beta_{w,T} = \rho_{w,T}\,\sigma_w/\sigma_T$ gives $\mathrm{SR}(w) = \rho_{w,T}\,\mathrm{SR}^*$: zero alpha against the tangency portfolio is the correlation hurdle of Corollary 11.31 met with equality, the "book" being the tangency portfolio itself.

Fact 11.40 (Capital asset pricing model (Sharpe, 1964; Lintner, 1965))**.** Suppose that all investors choose portfolios by mean–variance criteria over one common period from the same μ and Σ (homogeneous beliefs), can lend and borrow without limit at r_f, and trade in frictionless markets that clear. Then, since by Proposition 11.33 every investor holds the risky assets in the proportions w_T, the aggregate value-weighted market portfolio M equals the tangency portfolio, and Proposition 11.38 becomes the pricing relation

$$\mathbb{E}[R_i] - r_f = \beta_i\,(\mathbb{E}[R_M] - r_f) \qquad \text{for every asset } i,$$

with β_i measured against the market. The equilibrium step — market clearing, existence, and the aggregation of individual demands into the market portfolio — is what the statement asserts without argument. A proof is beyond our scope. The CAPM's mathematical content is one first-order condition, proved above as Proposition 11.38; its economic content — what empirical tests of the CAPM actually test — is the claim that the market portfolio is mean–variance efficient.

Definition 11.41 (Security market line and alpha)**.** The *security market line* (SML) is the map $\beta \mapsto r_f + \beta(\mu_M - r_f)$ in the $(\beta, \text{expected return})$ plane, the line through $(0, r_f)$ and $(1, \mu_M)$. The *alpha* of asset i (Jensen's alpha) is

$$\alpha_i = \mu_i - r_f - \beta_i(\mu_M - r_f),$$

its vertical distance from the line.

Example 11.42 (An alpha of 4.8 percent)**.** With $r_f = 2\%$ and $\mu_M = 8\%$, the first stock of Example 11.37 ($\beta = 1.2$) has CAPM value $2\% + 1.2 \times 6\% = 9.2\%$; a forecast mean of 14% is an alpha of 4.8%. The second stock ($\beta = 0.6$) has CAPM value $2\% + 0.6 \times 6\% = 5.6\%$, and an expected return of 5.6% puts it on the line with zero alpha. Figure 20 draws both.

Remark 11.43 (Reading an alpha)*.* Under Fact 11.40 every alpha is zero, so a nonzero measured alpha is mispricing, an omitted risk factor, or estimation noise. The pair (α_i, β_i) is the population intercept and slope of the regression of *excess* returns on excess market returns, the population form of (5.3); with raw returns instead, the intercept is $\alpha_i + r_f(1 - \beta_i)$, which equals α_i only when $\beta_i = 1$.

Interview trap. Beta is not correlation: $\beta_i = \rho_{iM}\sigma_i/\sigma_M$ carries the volatility ratio, so a $\rho = 0.6$ stock can have $\beta = 1.2$ while a $\rho = 0.9$ stock has $\beta = 0.6$. Hedge with beta (it sets the magnitude); judge the hedge's quality with correlation (the residual share is $1 - \rho^2$) — Remark 5.23 restated.

Interview trap. A high beta or a high volatility does not buy a higher Sharpe ratio. Dividing the pricing relation of Fact 11.40 by σ_i and substituting $\beta_i = \rho_{iM}\sigma_i/\sigma_M$ gives $\mathrm{SR}_i = \rho_{iM}\,\mathrm{SR}_M \le \mathrm{SR}_M$: a higher beta buys more expected return only with proportionally more systematic risk, and the idiosyncratic part carries no premium because Proposition 11.36(iii) diversifies it away at rate $1/n$.

Interview trap. CML versus SML: different horizontal axes (σ against β) and different membership (only efficient portfolios against every asset). An individual stock lies *below* the CML — its idiosyncratic variance is unrewarded — but *on* the SML under the CAPM; drawing the SML in the (σ, mean) plane is the most common whiteboard error in this block. A related confusion of arenas: the beta-neutral projection of Corollary 6.26 acts on a signal vector *across* assets, whereas the beta of this subsection is one asset's exposure *through time* — the same projection algebra of Proposition 10.15, a different index.

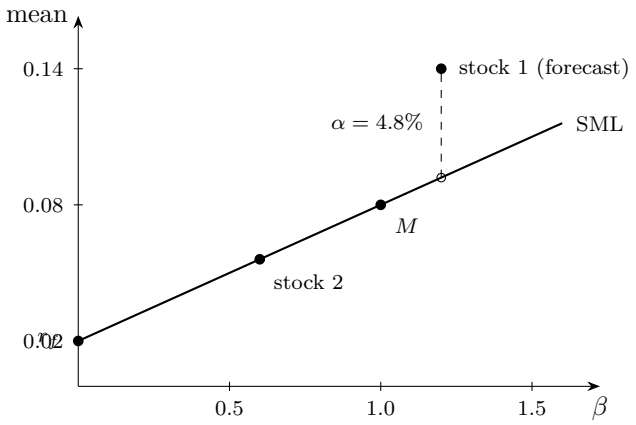


Figure 20: The security market line of Example 11.42: expected return against *beta*, not against σ (that is the CML of Figure 19). The line passes through $(0, r_f) = (0, 2\%)$ and $(1, \mu_M) = (1, 8\%)$. Stock 1 ($\beta = 1.2$) is forecast at 14% against a CAPM value of 9.2% (open circle); the dashed gap is its alpha of 4.8%, the vertical distance from the line. Stock 2 ($\beta = 0.6$) is drawn at its CAPM value $2\% + 0.6 \times 6\% = 5.6\%$, i.e. on the line. Under Fact 11.40 every asset sits on it.

11.6 Risk Contributions and Risk Parity

The desk questions: "decompose the book's volatility by position", "what are the risk-parity weights for a bond/equity portfolio, and do they depend on the correlation?", and the classic "60/40 is balanced, isn't it?". Capital weights say where the money is, not where the risk comes from. Because portfolio volatility is homogeneous of degree one in the weights, Euler's theorem splits it exactly into per-asset contributions, and equalising those contributions defines risk parity. The key surprise is that the two-asset risk-parity weights $w_i \propto 1/\sigma_i$ hold for every correlation.

Setup. The covariance matrix Σ is positive definite. A position vector $w \in \mathbb{R}^n$, $w \neq 0$, need not sum to one (a book's exposures) until risk parity is normalised; the portfolio volatility is $\sigma_p(w) = \sqrt{w^\top \Sigma w} > 0$, so its partial derivatives exist. The beta of asset i to the portfolio is $\beta_{i,p} = \operatorname{Cov}(R_i, R_w)/\operatorname{Var}(R_w)$, Definition 11.35 with the portfolio itself as reference. In the equicorrelated case we write $\Delta = \operatorname{diag}(\sigma_1, \dots, \sigma_n)$ and $K = (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top$ for the equicorrelation matrix (the equicorrelated model of §11.2 with $v = 1$), so that $\Sigma = \Delta K \Delta$; a common Sharpe ratio is SR_0.

Definition 11.44 (Marginal and total risk contributions)**.** The *marginal risk contribution* of asset i is $\mathrm{MRC}_i = \partial\sigma_p/\partial w_i$, the *(total) risk contribution* is $\mathrm{RC}_i = w_i\,\mathrm{MRC}_i$, and the *risk share* of asset i is RC_i/σ_p.

Theorem 11.45 (Euler decomposition of portfolio volatility)**.** *For every $w \neq 0$:*

(i) The marginal contributions are

$$\mathrm{MRC}_i = \frac{(\Sigma w)_i}{\sigma_p} = \beta_{i,p}\,\sigma_p,$$

so that $\mathrm{RC}_i = w_i(\Sigma w)_i/\sigma_p$ and the risk share of asset i equals $w_i\beta_{i,p}$.

(ii) The volatility is the sum of the total contributions,

$$\sigma_p(w) = \sum_{i=1}^{n} w_i\,\frac{\partial\sigma_p}{\partial w_i} = \sum_{i=1}^{n} \mathrm{RC}_i; \tag{11.13}$$

equivalently the risk shares sum to one, $\sum_i w_i\beta_{i,p} = 1$.

Proof. We differentiate the square root by the chain rule for part (i) and proceed by homogeneity — Euler's theorem for functions homogeneous of degree one — for part (ii).

Part (i). By the chain rule and the gradient formula (11.3), $\nabla_w \sqrt{w^\top \Sigma w} = 2\Sigma w/(2\sqrt{w^\top \Sigma w}) = \Sigma w/\sigma_p$, whose i-th entry is the first equality. Since $(\Sigma w)_i = \sum_j \Sigma_{ij} w_j = \mathrm{Cov}(R_i, R_w)$ by bilinearity and $\sigma_p^2 = \mathrm{Var}(R_w)$, we have $(\Sigma w)_i/\sigma_p = \beta_{i,p}\sigma_p^2/\sigma_p = \beta_{i,p}\sigma_p$. Multiplying by w_i gives RC_i, and dividing by σ_p gives the risk share.

Part (ii). For $t > 0$, $\sigma_p(tw) = \sqrt{t^2\, w^\top \Sigma w} = t\, \sigma_p(w)$. Differentiating both sides with respect to t, the chain rule on the left gives $\sum_i w_i\, (\partial \sigma_p/\partial w_i)(tw)$ and the right gives $\sigma_p(w)$; setting $t = 1$ yields (11.13). Direct check with part (i): $\sum_i w_i (\Sigma w)_i/\sigma_p = w^\top \Sigma w/\sigma_p = \sigma_p$. Dividing (11.13) by σ_p and using the risk-share identity of part (i) gives $\sum_i w_i \beta_{i,p} = 1$. The argument used only degree-one homogeneity, so it applies verbatim to any risk measure that is positively homogeneous of degree one in the positions; nothing further is claimed here about other measures. □

(Problem P11.11 asks you to reproduce this proof from scratch.)

Example 11.46 (Where the risk of 60/40 comes from)**.** Take bonds with $\sigma_B = 5\%$, equities with $\sigma_E = 15\%$ and $\rho = 0.2$, so

$$\Sigma = \begin{pmatrix} 0.0025 & 0.0015 \\ 0.0015 & 0.0225 \end{pmatrix},$$

and the classic 60/40 portfolio $w = (0.4, 0.6)^\top$ (40% bonds, 60% equities). Then $\Sigma w = (0.0019, 0.0141)^\top$ and $\sigma_p = \sqrt{0.4(0.0019) + 0.6(0.0141)} = \sqrt{0.00922} = 9.60\%$. By Theorem 11.45(i) the marginal contributions are $\mathrm{MRC} = (0.0019, 0.0141)/0.0960 = (0.0198, 0.147)$: one extra point of equity weight adds 0.147 points of volatility, 7.4 times the bond marginal. The total contributions are $\mathrm{RC} = (0.4 \times 0.0198,\ 0.6 \times 0.147) = (0.79\%, 8.81\%)$, summing to 9.60% — the Euler check of part (ii) — so equities carry 91.8% of the risk (99.5% at $\rho = -0.2$). The betas to the portfolio are $(\Sigma w)_i/\sigma_p^2 = (0.206, 1.529)$, with $0.4(0.206) + 0.6(1.529) = 1.000$ as part (ii) requires.

(Problem P11.11 works this computation at different numbers.)

Definition 11.47 (Risk parity)**.** A long-only, fully invested portfolio $w > 0$, $\mathbf{1}^\top w = 1$, is a *risk-parity* (equal-risk-contribution) portfolio if $\mathrm{RC}_1 = \cdots = \mathrm{RC}_n$, equivalently if $w_i(\Sigma w)_i$ takes the same value for every i (σ_p is a common factor). The condition is scale-free — the defining equations are invariant under $w \mapsto tw$ — so the normalisation is imposed last.

Proposition 11.48 (Risk-parity weights: two assets for every correlation, and equicorrelated assets)**.** *Let Σ be positive definite. Then:*

(i) For $n = 2$ and any $\rho \in (-1, 1)$, $\mathrm{RC}_1 = \mathrm{RC}_2$ holds for positive weights if and only if $w_1\sigma_1 = w_2\sigma_2$, so the unique risk-parity portfolio is

$$w = \Big(\frac{\sigma_2}{\sigma_1 + \sigma_2},\ \frac{\sigma_1}{\sigma_1 + \sigma_2}\Big)^\top, \qquad \sigma_p = \frac{\sigma_1\sigma_2\sqrt{2(1+\rho)}}{\sigma_1 + \sigma_2},$$

independent of ρ in its weights.

(ii) For $n \geq 2$ with a common pairwise correlation ρ, $-1/(n-1) < \rho < 1$, and volatilities σ_i, the weights $w_i \propto 1/\sigma_i$ are risk parity.

(iii) If in addition every excess Sharpe ratio $(\mu_i - r_f)/\sigma_i$ equals a common $\mathrm{SR}_0 > 0$, the same portfolio is the tangency portfolio of Theorem 11.23, with $\mathrm{SR}^* = \mathrm{SR}_0\sqrt{n/(1 + (n-1)\rho)}$. At $\rho = 0$ this is $\mathrm{SR}_0\sqrt{n}$ — the proof behind Remark 10.14 and Corollary 10.12: equal-Sharpe uncorrelated legs sized to equal volatility are simultaneously risk parity and maximum Sharpe.

Proof. We write out the equal-contribution equations and watch the correlation term cancel; for the equicorrelated case we factor Σ through its correlation matrix.

Part (i). With $(\Sigma w)_1 = w_1\sigma_1^2 + w_2\rho\sigma_1\sigma_2$ and $(\Sigma w)_2 = w_2\sigma_2^2 + w_1\rho\sigma_1\sigma_2$,

$$w_1(\Sigma w)_1 = w_1^2\sigma_1^2 + w_1w_2\rho\sigma_1\sigma_2, \qquad w_2(\Sigma w)_2 = w_2^2\sigma_2^2 + w_1w_2\rho\sigma_1\sigma_2.$$

The common term $w_1w_2\rho\sigma_1\sigma_2$ cancels identically, leaving $w_1^2\sigma_1^2 = w_2^2\sigma_2^2$, and for positive weights $w_1\sigma_1 = w_2\sigma_2$: equal volatility budgets. Normalising $w_1 + w_2 = 1$ gives the stated weights. Substituting $w = \sigma_2/(\sigma_1 + \sigma_2)$ into (11.2),

$$\sigma_p^2 = \frac{\sigma_2^2\sigma_1^2 + \sigma_1^2\sigma_2^2 + 2\rho\,\sigma_1\sigma_2\,\sigma_1\sigma_2}{(\sigma_1 + \sigma_2)^2} = \frac{\sigma_1^2\sigma_2^2\,(2 + 2\rho)}{(\sigma_1 + \sigma_2)^2},$$

and taking the square root gives the volatility.

Part (ii). Since the defining equations are scale-invariant, take $w_i = t/\sigma_i$ with $t > 0$, so that $\Delta w = t\mathbf{1}$. Summing a row of K, $K\mathbf{1} = (1+(n-1)\rho)\mathbf{1}$, hence $(\Sigma w)_i = \sigma_i(K\Delta w)_i = t\,\sigma_i(1+(n-1)\rho)$ and $w_i(\Sigma w)_i = t^2(1 + (n-1)\rho)$, the same for every i. (The computation shows more: inverse-volatility weights are risk parity whenever the correlation matrix has equal row sums, equicorrelation being the standard case.)

Part (iii). The eigenvalues of K are $1 + (n-1)\rho$ once (eigenvector $\mathbf{1}$) and $1 - \rho$ with multiplicity $n - 1$ (eigenvectors orthogonal to $\mathbf{1}$), by the computation in the proof of Proposition 11.12(ii) with $v = 1$; on the stated range both are positive, so K is invertible and $K^{-1}\mathbf{1} = \mathbf{1}/(1 + (n-1)\rho)$. With $\Sigma = \Delta K\Delta$, $\Sigma^{-1} = \Delta^{-1}K^{-1}\Delta^{-1}$. The hypothesis says $\mu_e = \mathrm{SR}_0\,\Delta\mathbf{1}$, so

$$\Sigma^{-1}\mu_e = \mathrm{SR}_0\,\Delta^{-1}K^{-1}\Delta^{-1}\Delta\mathbf{1} = \mathrm{SR}_0\,\Delta^{-1}K^{-1}\mathbf{1} = \frac{\mathrm{SR}_0}{1 + (n-1)\rho}\,\Delta^{-1}\mathbf{1},$$

whose entries are proportional to $1/\sigma_i$: by Theorem 11.23 the tangency portfolio is the inverse-volatility portfolio of part (ii). Its squared Sharpe ratio is $\mu_e^\top\Sigma^{-1}\mu_e = \mathrm{SR}_0^2\,\mathbf{1}^\top\Delta\,\Delta^{-1}K^{-1}\Delta^{-1}\Delta\,\mathbf{1} = \mathrm{SR}_0^2\,\mathbf{1}^\top K^{-1}\mathbf{1} = \mathrm{SR}_0^2\,n/(1 + (n-1)\rho)$. □

Example 11.49 (Bonds and equities at risk parity)**.** For the bond/equity pair of Example 11.46 ($\sigma_B = 5\%$, $\sigma_E = 15\%$, $\rho = 0.2$), Proposition 11.48(i) gives $w = (0.15, 0.05)/0.20 = (0.75, 0.25)^\top$. Check: $\Sigma w = (0.001875 + 0.000375,\ 0.001125 + 0.005625)^\top = (0.00225, 0.00675)^\top$, so $w_i(\Sigma w)_i = 0.0016875$ for both assets; $\sigma_p = \sqrt{0.003375} = 5.81\%$ (closed form $0.05(0.15)\sqrt{2.4}/0.20$), each side contributing 2.905%. The same weights are risk parity at $\rho = -0.2$ ($\sigma_p = 4.74\%$) and at $\rho = 0.6$ (6.71%): the split never moves, only the total. Levering the mix up to the 9.60% volatility of 60/40 requires $9.60/5.81 = 1.65$ times the capital.

For general Σ with $n \geq 3$ the equations $w_i(\Sigma w)_i = w_j(\Sigma w)_j$ contain the terms $w_iw_k\Sigma_{ik}$ and $w_jw_k\Sigma_{jk}$ for every third asset k; their sums cancel under the symmetry of Proposition 11.48(ii) (equal row sums of the correlation matrix; termwise under equicorrelation) but not in general; the system is nonlinear and solved numerically, and that it has exactly one positive solution is the next fact.

Fact 11.50 (Existence and uniqueness of the risk-parity portfolio (Maillard, Roncalli and Teiletche, 2010; Spinu, 2013))**.** For any positive-definite Σ there is exactly one portfolio with positive weights and equal risk contributions; it is the unique minimiser of $\frac{1}{2}w^\top\Sigma w - \frac{1}{n}\sum_i \log w_i$ over $w > 0$, rescaled to sum to one — a strictly convex problem solved numerically for $n \geq 3$. A proof is beyond our scope.

Risk parity is thus distinct from the *inverse-volatility* portfolio $w_i \propto 1/\sigma_i$ (the two coincide in the cases of Proposition 11.48), from the inverse-variance portfolio of Corollary 11.9, and from the minimum-variance portfolio, which equalises the *marginal* contributions rather than the totals, as the next remark records.

Remark 11.51 (Minimum variance, risk parity and maximum Sharpe are three different portfolios). At the minimum-variance portfolio all *marginal* contributions are equal: from $\Sigma w_g = (1/A)\mathbf{1}$ in Theorem 11.7 and $\sigma_g = 1/\sqrt{A}$, $\mathrm{MRC}_i = (1/A)/(1/\sqrt{A}) = \sigma_g$ for every i, so $\mathrm{RC}_i = w_{g,i}\sigma_g$ and the risk shares equal the capital weights. At risk parity all *total* contributions $w_i\mathrm{MRC}_i$ are equal. At the tangency portfolio $\Sigma w_T = \mu_e/k$ makes the ratio of expected excess return to marginal risk, $\mu_{e,i}/\mathrm{MRC}_i = k\sigma_T$, equal across assets. Risk parity and maximum Sharpe coincide under the symmetry of Proposition 11.48(iii); minimum variance joins them only when the volatilities are equal as well, in which case each of the three is the equal-weight portfolio by Proposition 11.12(iii). With unequal volatilities $w_g \propto \Delta^{-1}K^{-1}\Delta^{-1}\mathbf{1}$ is not inverse-volatility: for $n = 2$, $\rho = 0.5$ and $(\sigma_1, \sigma_2) = (1, 2)$, $w_g = (1, 0)^\top$ (the threshold $\rho = \sigma_1/\sigma_2$ of Proposition 11.4(iii)) against $(2/3, 1/3)^\top$ for the other two. For two assets with $|\rho| < 1$, the risk-parity weight $\sigma_2/(\sigma_1 + \sigma_2)$ equals the minimum-variance weight $w^* = c/V$ of Proposition 11.4(ii), where $V = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$ and $c = \sigma_2^2 - \rho\sigma_1\sigma_2$ are the coefficients of (A.1), exactly when $\sigma_1 = \sigma_2$: the reader should verify that $\sigma_2 V - c(\sigma_1 + \sigma_2) = \sigma_1\sigma_2(1 + \rho)(\sigma_1 - \sigma_2)$, which also vanishes in the limit $\rho \to -1$, where w^* tends to the riskless weight of part (iv).

Example 11.52 (Three portfolios for one pair)**.** For two uncorrelated assets with volatilities 5% and 15%: minimum variance, which here is inverse variance, $(\sigma_2^2, \sigma_1^2)/(\sigma_1^2 + \sigma_2^2) = (0.90, 0.10)$; risk parity $(\sigma_2, \sigma_1)/(\sigma_1 + \sigma_2) = (0.75, 0.25)$; equal weight $(0.50, 0.50)$. Their volatilities are 4.74%, 5.30% and 7.91%. In the running universe the minimum-variance marginal contributions are $(0.01, 0.01, 0.01)/0.10 = (0.10, 0.10, 0.10)$, all equal to σ_g, while its risk shares are the capital weights $(0.5, 0, 0.5)$: asset 2's marginal risk matches the others', and its total contribution is zero because its position is.

Intuition. Volatility is a length, and Euler's theorem says a length is the sum of its shadows on the positions. Each shadow is the position size times how fast the length grows when that position is nudged. Minimum variance makes every nudge equally costly at the margin; risk parity makes every position's shadow equally long; the tangency portfolio makes every position's shadow proportional to its expected excess return. Three different symmetries, three different portfolios.

Interview trap. Capital weights are not risk weights: "60/40 is balanced" is 40/60 in capital and 8/92 in risk at $\rho = 0.2$ (1/99 at $\rho = -0.2$). The standalone volatilities $w_i\sigma_i$ ignore correlation and do not sum to σ_p (for positive weights they exceed it whenever $\rho < 1$), so their shares are not risk contributions either; only the Euler contributions $w_i(\Sigma w)_i/\sigma_p$ sum to σ_p, and the sum is a one-line arithmetic check.

Interview trap. "Risk parity means inverse-volatility weights" is true for two assets at any correlation and for an equicorrelated universe (Proposition 11.48(i)–(ii)); with three or more assets and unequal correlations the exact solution is numerical (Fact 11.50) and inverse volatility is only the naive approximation. Like minimum variance, risk parity uses no expected returns: it is a robustness choice (§11.7), not an optimality claim, and the leverage needed to reach a target volatility is a separate decision.

11.7 Estimation Error: Why Naive Markowitz Fails in Practice

The open-ended follow-up at every fund, and the covariance-estimation question at research labs: "so why does nobody run raw Markowitz — what breaks, and what do you do instead?". Every formula in this section takes μ and Σ as known; in practice both are estimated, and the optimiser multiplies the estimation error by Σ^{-1}. The provable core is that Σ^{-1} amplifies expected-return error by up to $1/\lambda_{\min}$ along the smallest-eigenvalue direction, with a closed-form two-asset sensitivity showing one percentage point of expected return moving a 50/50 portfolio to 91/9. The literature results — error maximisation, shrinkage, constraints as shrinkage, and the $1/N$ horse race — are Facts with attribution; a Remark ties mean–variance to the Kelly criterion of §3.1; a Procedure states the practical answer.

Setup. The true μ_e and Σ are unknown. The *plug-in* tangency portfolio is $\hat{w} \propto \hat{\Sigma}^{-1}\hat{\mu}_e$, with $\hat{\mu}_e$ and $\hat{\Sigma}$ estimated from T periods of returns. A *perturbation* $\delta \in \mathbb{R}^n$ is a change in the expected-return vector. By Fact 6.2, $\Sigma = Q\Lambda Q^\top$ with orthonormal eigenvectors $q_1, \dots, q_n$ and eigenvalues $\lambda_1 \geq \cdots \geq \lambda_n$; positive definiteness gives $\lambda_{\min} = \lambda_n > 0$ by Theorem 6.4. Here λ denotes an eigenvalue, as in the notation table; $z_1, \dots, z_n$ denote the coordinates of a vector in the eigenbasis; ν denotes a vector of Karush–Kuhn–Tucker multipliers and ζ a shrinkage intensity. The unnormalised tangency direction is written $y(\mu_e) = \Sigma^{-1}\mu_e$ — the vector y of the proof of Theorem 11.23 — so that $w(\cdot)$ keeps its meaning as the frontier map of (11.6).

Proposition 11.53 (Σ^{-1} amplifies expected-return error)**.** *Under the setup above:*

(i) For the unnormalised tangency direction $y(\mu_e) = \Sigma^{-1}\mu_e$, a perturbation δ of the expected-return vector changes the weights by $\Sigma^{-1}\delta$, and

$$\|\Sigma^{-1}\delta\| \;\leq\; \frac{\|\delta\|}{\lambda_{\min}},$$

with equality if and only if δ lies in the eigenspace of $\lambda_{\min}$ (the direction q_n when λ_n is simple).

(ii) Take two assets with a common volatility σ, correlation $\rho \in (-1, 1)$, and excess means μ_1, μ_2 with $\mu_1 + \mu_2 > 0$. The eigenvalues of Σ are $\sigma^2(1 \pm \rho)$ with eigenvectors $(1, \pm 1)^\top/\sqrt{2}$, and the tangency weights normalised to sum to one satisfy

$$w_1 - w_2 = \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2} \cdot \frac{1 + \rho}{1 - \rho}\,: \tag{11.14}$$

the relative gap between the two mean inputs is amplified by $(1 + \rho)/(1 - \rho)$, which for $\rho \geq 0$ is the condition number $\lambda_{\max}/\lambda_{\min}$ of Σ and is unbounded as $\rho \to 1$ (for $\rho < 0$ the factor is below one and the gap is damped).

Proof. We expand the perturbation in the eigenbasis for part (i) and invert the 2×2 matrix explicitly for part (ii).

Part (i). By linearity, $y(\mu_e + \delta) - y(\mu_e) = \Sigma^{-1}\delta$. Write $\delta = \sum_i z_i q_i$ in the orthonormal eigenbasis of Fact 6.2, so that $\|\delta\|^2 = \sum_i z_i^2$. From $\Sigma q_i = \lambda_i q_i$ and $\lambda_i > 0$, multiplying by Σ^{-1} and dividing by λ_i gives $\Sigma^{-1} q_i = q_i/\lambda_i$, hence $\Sigma^{-1}\delta = \sum_i (z_i/\lambda_i) q_i$ and, by orthonormality,

$$\|\Sigma^{-1}\delta\|^2 = \sum_i \frac{z_i^2}{\lambda_i^2} \;\leq\; \frac{1}{\lambda_{\min}^2} \sum_i z_i^2 = \frac{\|\delta\|^2}{\lambda_{\min}^2},$$

since each $1/\lambda_i^2 \leq 1/\lambda_{\min}^2$. Equality holds if and only if $z_i = 0$ for every i with $\lambda_i > \lambda_{\min}$, i.e. all the weight sits on the minimal eigenvalue — the argument of Theorem 6.12 applied to Σ^{-1}, whose largest eigenvalue is $1/\lambda_{\min}$.

Part (ii). With $\Sigma = \sigma^2 \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$, direct inversion gives

$$\Sigma^{-1} = \frac{1}{\sigma^2(1 - \rho^2)} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}, \qquad \Sigma^{-1}\mu_e \propto (\mu_1 - \rho\mu_2,\; \mu_2 - \rho\mu_1)^\top.$$

The sum of the two entries is $(1 - \rho)(\mu_1 + \mu_2) > 0$ and their difference is $(1 + \rho)(\mu_1 - \mu_2)$. The normalised weights are the entries divided by their sum, so $w_1 - w_2$ is the difference divided by the sum, which is (11.14). The eigenvalues are the $d \pm c$ shortcut of Example 6.5 with $d = \sigma^2$, $c = \rho\sigma^2$: $\sigma^2(1 + \rho)$ on $(1, 1)^\top$ and $\sigma^2(1 - \rho)$ on $(1, -1)^\top$. For $\rho \geq 0$ the larger is $\sigma^2(1 + \rho)$ and the ratio $\lambda_{\max}/\lambda_{\min}$ is $(1 + \rho)/(1 - \rho)$; for $\rho < 0$ the roles swap and the same expression is $\lambda_{\min}/\lambda_{\max} < 1$. □

The directions belonging to small eigenvalues are where inversion amplifies error, and they are also where the inputs are worst: Fact 11.56 below records that the smallest sample eigenvalues are biased downward, so the sample covariance is least reliable exactly there, and Remark 6.21 adds that floating-point inversion is least accurate there too. In a portfolio those directions are the near-arbitrage spreads between highly correlated assets.

Example 11.54 (One percentage point flips the portfolio)**.** Two assets, $\sigma = 20\%$ each, $\rho = 0.8$, $r_f = 0$, so

$$\Sigma = \begin{pmatrix} 0.04 & 0.032 \\ 0.032 & 0.04 \end{pmatrix}, \qquad \Sigma^{-1} = \begin{pmatrix} 69.4 & -55.6 \\ -55.6 & 69.4 \end{pmatrix},$$

with $\det \Sigma = 0.000576$.

- For $\mu_e = (5\%, 5\%)^\top$: $\Sigma^{-1}\mu_e = (0.694, 0.694)^\top$, weights 50/50.
- For $\mu_e = (6\%, 5\%)^\top$: $\Sigma^{-1}\mu_e = (1.389, 0.139)^\top$, weights $(90.9\%, 9.1\%)$ — one point of expected return moves 41 points of weight. Check against (11.14): $(0.01/0.11)(1.8/0.2) = 0.818 = 0.909 - 0.091$; the amplification factor 9 is the condition number $0.072/0.008$.
- For $\mu_e = (6\%, 4\%)^\top$: $\Sigma^{-1}\mu_e = (1.944, -0.556)^\top$, weights $(140\%, -40\%)$ — a short position appears. Read as positions, $(140\%, -40\%)$ is a 100% directional holding plus a 40% long–short spread between two assets that are 80% correlated: the optimiser has found a near-arbitrage in the sample means, along exactly the low-eigenvalue direction $(1, -1)^\top/\sqrt{2}$ of part (i). That spread is the position to challenge in a review, and it is the first thing a long-only constraint removes (Fact 11.57).

At $\rho = 0$ the same $(6\%, 5\%)$ gives weights proportional to $(0.06, 0.05)/0.04$, i.e. $(54.5\%, 45.5\%)$, a 4.5-point move, nine times smaller: the amplification is entirely the factor $(1 + \rho)/(1 - \rho)$. The eigenvalues are 0.072 and 0.008, so $1/\lambda_{\min} = 125$ against $1/\lambda_{\max} = 13.9$, and the perturbation $(1, 0)^\top = \frac{1}{2}\big((1, 1)^\top + (1, -1)^\top\big)$ half-loads the spread direction. The statistical scale of the shock: with $\sigma = 20\%$ and ten annual observations, the average of T iid annual returns has standard deviation $\sigma/\sqrt{T}$ (the variance computation in the proof of Proposition 10.3), i.e. $0.20/\sqrt{10} = 6.3$ points — six times the one-point change that already flipped the weights. Expected returns are the least identifiable input, so fixing Σ alone is not enough.

(Problem P11.12 works this computation at different numbers.)

Interview trap. The inverse Σ^{-1} amplifies estimation error — precisely along the small-eigenvalue directions, the near-arbitrage spreads between highly correlated assets where the sample covariance is least reliable (its smallest eigenvalues are biased downward, Fact 11.56) and where the optimiser bets hardest: a factor of 9 at $\rho = 0.8$, unbounded as $\rho \to 1$. The three-part answer to "why is Markowitz not used naively": the inputs are estimated, the inverse amplifies the error along low-variance directions, and the remedies are shrinkage, constraints and the $1/N$ benchmark.

Interview trap. The mean vector, not the covariance, is the dangerous input: the standard error of a mean shrinks only with calendar time, $\sigma/\sqrt{T}$ — 6.3 points against the 1-point shock that flipped the weights in Example 11.54 — so a shrunk Σ with a raw $\hat{\mu}$ still produces error-maximised weights. Minimum-variance and risk-parity portfolios avoid μ entirely, which is a large part of their practical appeal.

Fact 11.55 (Error maximisation (Michaud, 1989), empirical)**.** A mean–variance optimiser fed sample estimates systematically overweights assets whose returns are overestimated and whose variances and covariances are underestimated, and underweights the reverse, because those are exactly the inputs that make an asset look attractive. The in-sample frontier therefore lies above the true frontier, and the realised out-of-sample performance of the "optimal" portfolio is typically worse than that of naive alternatives — Michaud's "estimation-error maximiser". The mechanism is selection: the optimiser picks the maximum of noisy estimates, which is biased upward — the

selection effect flagged in item 4 of Remark 10.7 — amplified along every low-variance direction by Proposition 11.53. A proof is beyond our scope.

Fact 11.56 (Shrinkage covariance estimation (Ledoit and Wolf, 2004), with empirical evidence)**.** Replace the sample covariance $\hat{\Sigma}$ by $\hat{\Sigma}_{\text{shrunk}} = \zeta F + (1-\zeta)\hat{\Sigma}$, where F is a structured target (a scaled identity, or the constant-correlation matrix) and $\zeta \in [0,1]$ is the shrinkage intensity; the intensity ζ^* minimising expected Frobenius loss has an explicitly estimable closed form. The estimator is positive definite even when n exceeds the sample size — repair (1) of Remark 6.8 — and pulls the sample eigenvalues, which are biased outward (the largest too large, the smallest too small), toward their mean; empirically it lowers the out-of-sample variance of the minimum-variance portfolio built from it. It is the same medicine as ridge regression (Theorem 5.11, Remark 5.17) and the same bias–variance trade as Theorem 8.2 and Example 8.3. A proof is beyond our scope.

Fact 11.57 (Weight constraints as shrinkage (Jagannathan and Ma, 2003), with empirical evidence)**.** The no-short-sale minimum-variance portfolio — minimise $w^\top \hat{\Sigma} w$ subject to $\mathbf{1}^\top w = 1$ and $w \geq 0$ — equals the unconstrained minimum-variance portfolio of the modified matrix $\hat{\Sigma} - (\nu\mathbf{1}^\top + \mathbf{1}\nu^\top)$, where $\nu \geq 0$ is the vector of Lagrange multipliers of the sign constraints. Imposing the constraint is therefore equivalent to shrinking the covariances of the assets that would otherwise have been shorted, and empirically the constrained sample minimum-variance portfolio performs as well out of sample as unconstrained portfolios built from shrunk or factor-based covariance matrices. The argument runs through the Karush–Kuhn–Tucker conditions. A proof is beyond our scope. The interview one-liner: constraints are implicit shrinkage.

Fact 11.58 ($1/N$ versus optimised portfolios (DeMiguel, Garlappi and Uppal, 2009), empirical)**.** Across fourteen optimising models (sample-based mean–variance, Bayesian and shrinkage variants, minimum variance, constrained versions) and seven equity datasets, none consistently beat the equal-weight $1/N$ rule out of sample on Sharpe ratio, certainty-equivalent return or turnover. For 25 assets the sample-based mean–variance rule needs an estimation window of about 3,000 months (250 years) to outperform $1/N$, and about 6,000 months for 50 assets. This is evidence, not mathematics: at realistic sample sizes the gains from optimisation are smaller than the losses from estimation error. Proposition 11.12 explains the strength of the benchmark: $1/N$ captures the bulk of diversification with zero estimation error, and it is itself the minimum-variance and the risk-parity portfolio under equal volatilities and equal correlations. A proof is beyond our scope.

Remark 11.59 (Mean–variance as second-order Kelly).

1. *Expansion.* For a portfolio return R_w small enough that $\ln(1+x) = x - x^2/2 + O(x^3)$ applies (Proposition 7.34), the expected log growth of Definition 3.1 generalises to

$$\mathbb{E}[\log(1+R_w)] = w^\top \mu - \tfrac{1}{2}(w^\top \Sigma w + (w^\top \mu)^2) + \dots,$$

using $\mathbb{E}[R_w^2] = \operatorname{Var}(R_w) + (\mathbb{E}[R_w])^2$. Dropping the third-order terms and the $(w^\top\mu)^2$ term (second order in the small means) leaves $w^\top\mu - \frac{1}{2}w^\top\Sigma w$: the objective of Proposition 11.33 with $\gamma = 1$ and $r_f = 0$. With a risk-free asset the same expansion of $\log(1+r_f+w^\top(R-r_f\mathbf{1}))$, treating r_f as small of the same order as the excess means, gives $w^\top\mu_e - \frac{1}{2}w^\top\Sigma w$ up to a constant and terms of that order.

2. *Maximiser.* Its maximiser $\Sigma^{-1}\mu_e$ is the multi-asset, continuous-return Kelly allocation: full mean–variance with $\gamma = 1$ is full Kelly.
3. *One asset.* For one asset it reads $f = \mu_e/\sigma^2$, "edge over variance", the analogue of the edge-over-odds rule $f^* = (bp-q)/b$ of Theorem 3.3. Proposition 3.6 is exactly the one-asset case of this quadratic approximation: in that proposition's notation, with $\varepsilon = p - q$ the edge of an even-money bet, $\tilde{g}(f) = \varepsilon f - f^2/2$, the bet having return of size f with mean εf and variance f^2 to leading order.
4. *Fractional Kelly.* Fractional Kelly is $\gamma > 1$, and log utility itself is item (1) of Remark 3.24. The estimation-error discussion of this subsection is Remark 3.8(1) in matrix form: an

overestimated edge in any direction of Σ^{-1} means over-betting in that direction, with the asymmetric penalty that remark describes — which is why practice sets γ well above 1.

5. *Running universe.* Full Kelly is $w = \Sigma^{-1}\mu_e = (0.5, 1, 2.5)^\top$ of Example 11.34, at 400% gross exposure and 45.8% volatility. The value of the quadratic objective, $(\mathrm{SR}^*)^2/2 = 0.105$, is the second-order estimate of the growth in excess of r_f; the dropped term $(\mathbb{E}[R_p])^2/2 = 0.23^2/2 = 0.026$ is of the same order at this leverage, so the figure is indicative only, not a growth rate.

Interview trap. Full mean–variance with $\gamma = 1$ is full Kelly, and Remark 3.8's reasons for betting less carry over verbatim: a desk running the unconstrained $\Sigma^{-1}\mu_e$ is over-betting whenever any edge is overestimated, and by Proposition 11.53 the overestimate is largest exactly along the spread directions where the position is largest.

Procedure 11.60 (A robust allocation pipeline (the interview answer))**.** Given T periods of returns on n assets:

1. Estimate Σ with shrinkage (Fact 11.56) or a factor structure (repair (3) of Remark 6.8).
2. Avoid or shrink μ: default to the minimum-variance, risk-parity or $1/N$ portfolio when views are weak, and shrink strategy Sharpe ratios toward a common prior, remembering their standard error from Fact 4.16.
3. Impose long-only, position and turnover constraints (Fact 11.57).
4. Size total leverage by fractional Kelly (Remark 11.59, Remark 3.8), i.e. a risk aversion γ well above 1.
5. Benchmark out of sample against $1/N$ (Fact 11.58) and report the gap between in-sample and out-of-sample Sharpe ratios.

Trading application. Procedure 11.60 is the answer to "what do you do instead of raw Markowitz": each step closes one channel through which Σ^{-1} turns estimation noise into positions, and the last step is the only honest measure of whether the optimiser earned its complexity.

Practice: problems P11.1–P11.12 in §27, solutions in §28.

12 Market Making I: the Economics of the Spread

A market maker earns the right to trade by quoting two prices at once: a bid at which it stands ready to buy and an ask at which it stands ready to sell. The gap between the two — the spread — is not arbitrary; it is the priced sum of identifiable costs and risks, plus a margin. This section names the components of the spread (§12.1), prices the subtlest component — adverse selection — exactly in a two-value Glosten–Milgrom miniature that is a standard interview derivation (§12.2), converts the theory into a measurement a trading desk can run on its own fills (§12.3), and closes with the decision rules for adjusting quotes when conditions change (§12.4).

12.1 Where the Spread Comes From

The most basic market-making interview question is also the most important: where does the bid–ask spread come from? The answer is a four-part decomposition, worth stating precisely because every quoting decision in this section and the next addresses one of its parts.

Setup. Consider a market maker quoting a single asset. All prices are in the asset's currency units.

Definition 12.1 (Bid, ask, and quoted spread)**.** A market maker posts a *bid* b, the price at which it stands ready to buy, and an *ask* $a \geq b$, the price at which it stands ready to sell (in any live

market the inequality is strict). The *quoted spread* is $s = a - b$, and the *mid-price* (or *mid*) is $m = (a + b)/2$. A counterparty who sells to the maker at b is said to *hit the bid*; a counterparty who buys from the maker at a is said to *lift the ask*.

Definition 12.2 (Spread decomposition)**.** The economically sustainable spread decomposes into four components, each the compensation for a distinct cost borne by the maker:

$$s = \underbrace{\text{adverse selection}}_{\text{informed pickoffs}} + \underbrace{\text{inventory risk}}_{\text{price moves while held}} + \underbrace{\text{order processing and fees}}_{\text{infrastructure, exchange fees}} + \underbrace{\text{margin}}_{\text{profit}}.$$

Adverse selection is the expected loss to better-informed counterparties who choose to trade against the quote; inventory risk is the exposure to price moves while the maker holds the position a fill hands it; order processing covers infrastructure, exchange and clearing fees; the margin is the maker's profit.

Volatility widens the first two components: when prices move more, an informed counterparty's private information is worth more per fill (the pickoff profit per trade grows), and any inventory a fill hands the maker is riskier to carry. Competition between makers compresses the margin, and squeezes operational slack out of the processing component, but it cannot push the spread below the actuarial cost of the first two components: a maker quoting inside that cost loses money in expectation and exits. In §12.2 we compute the adverse-selection part of that floor exactly, in the smallest model that produces it (Theorem 12.4).

12.2 Adverse Selection: a Glosten–Milgrom Miniature

This subsection isolates the adverse-selection component of Definition 12.2 in a two-value miniature of the Glosten–Milgrom sequential-trade model. The derivation uses nothing beyond conditional probability, Bayes' rule, and conditional expectation, and it delivers the cleanest available answer to the question of why a spread must exist at all.

Setup. The following assumptions are in force for the rest of this subsection.

- An asset has an unknown value $V \in \{99, 101\}$ with $P(V = 101) = P(V = 99) = \frac{1}{2}$, so the prior mean is $\mathbb{E}[V] = 100$.
- A market maker posts a bid b and an ask a as in Definition 12.1, and a single trader arrives to trade one unit against these quotes.
- With probability $\pi \in [0, 1)$ the trader is *informed*: it knows V, buys (lifts the ask) when $V = 101$, and sells (hits the bid) when $V = 99$ — it trades only in the profitable direction.
- With probability $1 - \pi$ the trader is *noise*: it buys or sells with probability $\frac{1}{2}$ each, independently of V. The trader's type is independent of V.
- The maker is risk neutral and pays no fees and no inventory costs. The value V is revealed after the trade, so the maker's realized profit is $a - V$ on a lifted ask and $V - b$ on a hit bid.
- We write "buy" for the event that the arriving trader buys at the ask and "sell" for the event that it sells at the bid, and we restrict attention to quotes with $99 < b \le a < 101$, so that an informed trader strictly profits by trading on its information. The reader should verify that the quotes found in Theorem 12.4 satisfy this restriction for every $\pi \in [0, 1)$.

Definition 12.3 (Zero-profit quotes)**.** Quotes (b, a) are *zero-profit* if each side earns zero expected profit conditional on being traded against:

$$\mathbb{E}[\, a - V \mid \text{buy}\,] = 0 \qquad \text{and} \qquad \mathbb{E}[\, V - b \mid \text{sell}\,] = 0.$$

Since $\mathbb{E}[\, a - V \mid \text{buy}\,] = a - \mathbb{E}[V \mid \text{buy}]$ by linearity, the conditions say exactly that $a = \mathbb{E}[V \mid \text{buy}]$ and $b = \mathbb{E}[V \mid \text{sell}]$: each quote must equal the value of the asset conditional on the side that

trades at it. Competition between makers drives quotes to this point — any maker quoting a strictly profitable side is undercut.

Theorem 12.4 (Glosten–Milgrom breakeven quotes)**.** *Under the setup above — values* 99 *and* 101 *equiprobable, informed fraction* $\pi \in [0,1)$*, noise flow buying and selling with probability* $\frac{1}{2}$ *each — the unique zero-profit quotes are*

$$a = 100 + \pi, \qquad b = 100 - \pi,$$

and the breakeven spread is therefore $s = a - b = 2\pi$*.*

Proof. We proceed by conditioning: first on the trader's type to obtain the probability of a buy in each state of V, then by Bayes' rule to obtain the posterior on V given a buy, and finally by computing the conditional expectation that the zero-profit condition of Definition 12.3 equates to the ask.

Step 1 (buy probabilities in each state). Condition on $V = 101$: with probability π the trader is informed and buys with certainty; with probability $1 - \pi$ it is a noise trader and buys with probability $\frac{1}{2}$, independently of V. Conditioning on $V = 99$, an informed trader never buys. By the law of total probability,

$$P(\text{buy} \mid V = 101) = \pi + \frac{1-\pi}{2} = \frac{1+\pi}{2}, \qquad P(\text{buy} \mid V = 99) = \frac{1-\pi}{2}. \tag{12.1}$$

Averaging over the equiprobable states gives the unconditional buy probability

$$P(\text{buy}) = \frac{1}{2} \cdot \frac{1+\pi}{2} + \frac{1}{2} \cdot \frac{1-\pi}{2} = \frac{1}{2}.$$

Step 2 (Bayes' rule). Using (12.1),

$$P(V = 101 \mid \text{buy}) = \frac{P(\text{buy} \mid V = 101)\, P(V = 101)}{P(\text{buy})} = \frac{\frac{1+\pi}{2} \cdot \frac{1}{2}}{\frac{1}{2}} = \frac{1+\pi}{2}.$$

Step 3 (conditional expectation). Consequently

$$\mathbb{E}[V \mid \text{buy}] = 101 \cdot \frac{1+\pi}{2} + 99 \cdot \frac{1-\pi}{2} = \frac{101+99}{2} + \frac{101-99}{2}\,\pi = 100 + \pi.$$

Step 4 (zero profit pins the ask). For any quotes in the restricted class $99 < b \le a < 101$, the behavior of both trader types — hence every conditional probability computed above — does not depend on the exact quote levels, so $\mathbb{E}[V \mid \text{buy}] = 100 + \pi$ regardless of a. The maker's expected profit conditional on a lift, $\mathbb{E}[\, a - V \mid \text{buy}\,] = a - (100 + \pi)$, is strictly increasing in a and vanishes if and only if $a = 100 + \pi$: the zero-profit ask exists and is unique.

Step 5 (the bid, by the mirrored computation). The model is invariant under simultaneously relabeling the states $99 \leftrightarrow 101$ and the sides buy $\leftrightarrow$ sell; carrying out the same three steps on the sell side gives

$$P(\text{sell} \mid V = 99) = \frac{1+\pi}{2}, \qquad P(\text{sell} \mid V = 101) = \frac{1-\pi}{2}, \qquad P(\text{sell}) = \frac{1}{2},$$

so that $P(V = 99 \mid \text{sell}) = \frac{1+\pi}{2}$ and

$$\mathbb{E}[V \mid \text{sell}] = 99 \cdot \frac{1+\pi}{2} + 101 \cdot \frac{1-\pi}{2} = 100 - \pi,$$

and the zero-profit bid is uniquely $b = 100 - \pi$. The breakeven spread is $s = a - b = (100 + \pi) - (100 - \pi) = 2\pi$. □

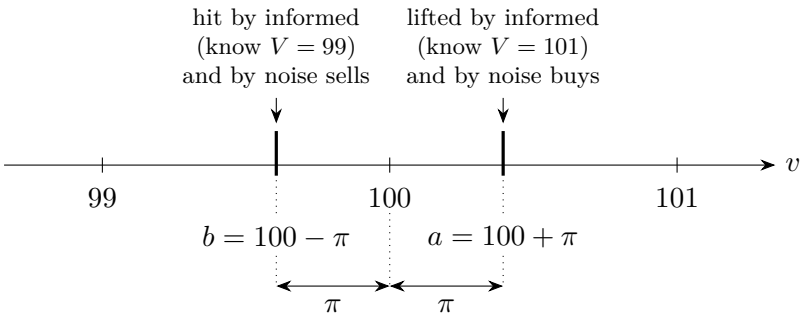


Figure 21: The Glosten–Milgrom miniature of Theorem 12.4 on the value axis, drawn for $\pi = 0.4$. The asset is worth 99 or 101 with equal probability, so the prior mean is 100. The zero-profit quotes sit symmetrically at $100 \pm \pi$: each quote equals the expected value of the asset conditional on the side that trades at it, so every fill breaks even in expectation and the quoted spread is $a - b = 2\pi$.

(Problem P12.2 asks you to reproduce this proof from scratch, at $\pi = 0.3$.)

Figure 21 shows the geometry of the result on the value axis: the two possible values, the prior mean, the two quotes sitting symmetrically a distance π from it, and the flow that trades at each quote.

Example 12.5 (Breakeven quotes with 30 percent informed flow)**.** Take $\pi = 0.3$. By (12.1), $P(\text{buy} \mid V = 101) = 0.65$ and $P(\text{buy} \mid V = 99) = 0.35$, so $P(\text{buy}) = 0.5$ and Bayes' rule gives

$$P(V = 101 \mid \text{buy}) = \frac{0.65 \times 0.5}{0.5} = 0.65.$$

The zero-profit ask is therefore

$$a = \mathbb{E}[V \mid \text{buy}] = 101(0.65) + 99(0.35) = 65.65 + 34.65 = 100.30,$$

and by Theorem 12.4 the bid is $b = 99.70$, for a breakeven spread of $s = 0.60 = 2\pi$.

Direct profit check. Conditional on a lift, the asset the maker delivers is worth 100.30 on average, so an ask of 100.30 earns exactly zero per lifted offer; a maker who instead quoted the prior mean 100 would lose 0.30 in expectation on every lifted offer, with zero fees, zero inventory cost, and no profit margin in the model at all. As a small-case check, setting $\pi = 0$ collapses the quotes to $a = b = 100$ and the spread to zero.

Intuition. Even a maker with zero directional opinion must quote a spread: the flow that chooses to trade against a fixed quote is conditionally biased against it. The spread is an insurance premium, and Theorem 12.4 prices it by conditional expectation — each quote sits at the value of the asset given the bad news implicit in being traded with.

Interview trap. "Incoming buys and sells are equally likely, so a maker with no directional opinion can quote an arbitrarily tight spread around fair value and still break even." The premise is true — in the model, $P(\text{buy}) = \frac{1}{2}$ unconditionally — but the conclusion confuses unconditional flow with the flow that actually trades: a resting ask is lifted disproportionately in the state $V = 101$. Quotes must be set at the expected value conditional on being traded with, not at the unconditional mean, and any spread below 2π loses money in expectation even with zero fees, zero inventory cost, and no margin. This is also why competition cannot drive spreads to zero: it can compress only the margin component of Definition 12.2, never the actuarial floor.

Remark 12.6 (Limiting cases of the breakeven spread). The breakeven spread 2π is increasing in the informed fraction, and its two limits are instructive. As $\pi \to 0$ the spread tends to zero: facing pure noise flow, a maker can quote at the mid and break even — the claim in the trap above is

exactly this special case. As $\pi \to 1$ the quotes tend to the two possible values themselves, $b \to 99$ and $a \to 101$, and the spread tends to 2, the full width of the uncertainty: every fill breaks even exactly, and with no noise flow left there is no counterparty from whom a margin could be earned — the market is at the edge of shutting down. For every $\pi \in [0, 1)$ the quotes remain strictly inside $(99, 101)$, consistent with the restriction in the Setup.

12.3 Measuring Adverse Selection: Mark-Outs

Theorem 12.4 prices adverse selection in a model where the informed fraction π is known. A desk must instead estimate the toxicity of its actual flow from its own fills. The standard estimator — and the concrete, operational answer to the interview question "what is adverse selection?" — is the mark-out.

Setup. Fix a horizon $\tau > 0$, measured in seconds, and write m_u for the mid-price of Definition 12.1 at time u. Consider a fill executed at time t against the maker's quotes: either the maker's ask a is lifted (the maker sells at a) or its bid b is hit (the maker buys at b).

Definition 12.7 (Mark-out)**.** The *mark-out* of the fill at horizon τ is the mid-price change τ seconds after the fill, signed so that a negative value means the market moved against the maker's trade:

$$M_\tau = \begin{cases} a - m_{t+\tau} & \text{if the ask was lifted (the maker sold at } a\text{),} \\ m_{t+\tau} - b & \text{if the bid was hit (the maker bought at } b\text{).} \end{cases}$$

Desk practice averages M_τ over fills within buckets — by counterparty, by venue, and by size — and monitors the averages across several horizons τ.

The average mark-out is realized adverse selection, and it estimates the informed-flow premium of the model. To see the link, return to the miniature of §12.2 and suppose that after each trade the market impounds the trade's information, so that the post-trade mid equals the conditional expectation of V given the observed side. A maker who ignores Theorem 12.4 and quotes both sides at the prior mean 100 records $M_\tau = 100 - (100 + \pi) = -\pi$ on every lifted ask and $M_\tau = (100 - \pi) - 100 = -\pi$ on every hit bid: the average mark-out recovers exactly the premium π that the maker failed to charge. At the zero-profit quotes the same computation gives $M_\tau = 0$ on both sides. A persistently negative mark-out bucket is therefore under-priced toxicity in that bucket.

Example 12.8 (Mark-out computation)**.** A maker's ask is lifted at $a = 100.30$; five seconds later the mid stands at $m_{t+5} = 100.34$. The mark-out is

$$M_5 = 100.30 - 100.34 = -0.04,$$

an adverse move of four cents on a price near 100 — about 4 basis points against the maker. Separately, the maker's bid is hit at $b = 99.70$ and the mid five seconds later is 99.71, giving $M_5 = 99.71 - 99.70 = +0.01$: a mildly profitable fill. If fills of the first kind concentrate in a single counterparty bucket — say one counterparty's fills mark 4 basis points against the maker at five seconds while all other flow marks flat — that counterparty's flow is toxic: its trades systematically precede adverse moves.

Trading application. A market-maker dashboard's mark-outs-by-counterparty panel is exactly the estimator of Definition 12.7: average post-fill mid drift, bucketed by counterparty, venue, and size. In an interview, this panel is the concrete answer to "what is adverse selection and how would you measure it?" — realized adverse selection is the average mark-out on the desk's own fills, and the bucketing identifies which flow is informed.

12.4 Widen, Skew, or Pull

A maker that concludes its current quotes are wrong has exactly three controls: *widen* (increase the spread s around an unchanged mid), *skew* (shift both quotes in the same direction, moving the effective mid), and *pull* (withdraw quotes entirely). Which control answers which diagnosis follows directly from the spread decomposition of Definition 12.2 and the mark-out evidence of §12.3.

Remark 12.9 (Quote-adjustment decision rules). The standard mapping from diagnosis to control is the following.

- *Volatility spikes: widen.* Both volatility-sensitive components of Definition 12.2 — adverse selection and inventory risk — grow with volatility. The flow itself has not become more informed, so the maker keeps quoting, at a spread that re-prices the risk.
- *Mark-outs deteriorate in an identifiable bucket: widen or pull for that bucket.* Flow that consistently marks against the maker is toxic; where the venue permits differentiation, widen against that bucket, cap the size shown to it, or stop quoting to it.
- *Toxic flow that cannot be distinguished: wider by default, for everyone.* Unidentifiable toxic flow raises the average informed fraction — the effective π of Theorem 12.4 — of all flow, and the breakeven spread rises with it.
- *News or a jump in uncertainty imminent: pull first.* Re-quote wide once the news is out, and tighten as the picture clears.
- *Inventory builds up: skew.* Skewing is driven by inventory risk rather than adverse selection; it is the subject of the next section.

Intuition. Pulling is not an admission that quoting has failed; it is the limiting zero-profit quote. Around an unscheduled announcement, the informed — or merely faster — fraction of flow spikes, so the breakeven spread 2π of Theorem 12.4 can momentarily exceed any spread that would still be filled: quoting nothing is then the only quote that does not lose money in expectation, and being on the wrong side of a single gap through a stale quote costs many spreads.

Practice: problems P12.1–P12.4 in §27, solutions in §28.

13 Market Making II: Inventory and Skewing

The preceding section priced the spread; this section decides where to place it. A maker who has been filled on one side is no longer indifferent to direction, and the center of the quote pair must move away from the mid to reflect the position being carried. We develop the reservation price of Avellaneda and Stoikov, write the make-me-a-market interview ritual as an explicit procedure, work through a calibration game exactly as played in a live assessment, and close with how a desk sheds inventory when skewing alone is too slow.

13.1 Inventory Risk and the Reservation Price

A market maker's quotes are supposed to earn the spread, but every fill also changes the book: inventory accumulates, and inventory converts price risk into profit-and-loss risk for as long as it is held. The central result of this subsection says that a maker carrying inventory should quote around a *shifted* center — the reservation price — rather than around the mid, and it makes the size of the shift explicit.

Setup. Fix a current time t and a risk horizon $T > t$. The mid price of the traded asset is m, and over the remaining horizon the mid is modeled as driftless with variance rate σ^2: writing m_T

for the terminal mid, $\mathbb{E}[m_T] = m$ and $\mathrm{Var}(m_T) = \sigma^2(T-t)$, where σ^2 has units of price squared per unit time. The maker's signed inventory is q (in units of the asset; positive means long, negative means short) and the maker's cash is c. The risk-aversion parameter is $\gamma > 0$, with units $1/(\text{price} \times \text{units})$ so that $q\gamma\sigma^2(T-t)$ is a price. Quotes are a bid and an ask placed at half-width $\delta > 0$ around a chosen center. Within this section the symbols q and r denote inventory and the reservation price; they are unrelated to the Markov-chain notation $q = 1-p$ and $r = q/p$ used elsewhere in the document.

Fact 13.1 (Reservation price (Avellaneda–Stoikov, 2008))**.** Consider a market maker with exponential utility of terminal wealth (risk aversion γ) who quotes a bid and an ask around a driftless diffusion mid with variance rate σ^2, whose quotes are filled by market orders arriving with an intensity that decays with the quote's distance from the mid. The maker's indifference price for inventory — the *reservation price* — is

$$r = m - q\,\gamma\,\sigma^2\,(T-t), \tag{13.1}$$

and the optimal bid and ask straddle r, not the mid m. For $q > 0$ the correction is negative and both quotes shift *down*: the ask moves toward the mid (the long is offered out more eagerly) and the bid backs away (further buying is discouraged). For $q < 0$ the shift is upward, symmetrically. The full derivation, via the Hamilton–Jacobi–Bellman equation of the underlying stochastic control problem, is beyond our scope.

Although the control-theoretic proof is out of scope, every factor in (13.1) can be recovered from a one-line variance-penalty argument. The following proposition is that argument. It is a *heuristic* relative to the dynamic model — it freezes the book and ignores future quoting entirely — but it reproduces the formula exactly and explains why each of q, γ, σ^2, and $(T-t)$ belongs in the correction.

Proposition 13.2 (Inventory-penalty heuristic)**.** *Adopt the setup above and freeze the book: the maker holds cash c and inventory q from t to T with no further trading, and values random terminal wealth W by the mean–variance certainty equivalent*

$$\mathrm{CE}[W] = \mathbb{E}[W] - \frac{\gamma}{2}\,\mathrm{Var}(W).$$

Then the marginal indifference price for inventory — the price per unit at which selling an infinitesimal quantity leaves the certainty equivalent unchanged — equals

$$r(q) = m - q\,\gamma\,\sigma^2\,(T-t),$$

in agreement with (13.1).

Proof. We compute the certainty equivalent of the frozen book as a function of inventory and differentiate. Terminal wealth is $W(q) = c + q\,m_T$, so by the setup $\mathbb{E}[W(q)] = c + qm$ and $\mathrm{Var}(W(q)) = q^2\sigma^2(T-t)$, giving

$$\mathrm{CE}(q) = c + q\,m - \frac{\gamma}{2}\,q^2\sigma^2(T-t). \tag{13.2}$$

The mean grows linearly in q, but the variance penalty grows quadratically: each additional unit is worth less to the book than the one before. Now sell an infinitesimal quantity $\mathrm{d}q$ at a unit price r: inventory falls to $q - \mathrm{d}q$ and cash rises by $r\,\mathrm{d}q$. To first order the certainty equivalent changes by $(r - \partial\,\mathrm{CE}\,/\partial q)\,\mathrm{d}q$, so indifference requires the sale price to equal the marginal certainty-equivalent value of inventory:

$$r(q) = \frac{\partial\,\mathrm{CE}}{\partial q} = m - q\,\gamma\,\sigma^2(T-t).$$

This is (13.1). The reader should verify from (13.2) that the indifference price for selling one *whole* unit is $\mathrm{CE}(q) - \mathrm{CE}(q-1) = m - \gamma\sigma^2(T-t)\,(q - \frac{1}{2})$, which agrees with the marginal version up to the half-unit convention. □

(Problem P13.2 asks you to reproduce this argument from scratch and to stress-test each factor of the correction.)

Example 13.3 (Reservation-price quoting, numeric)**.** A maker quotes a coin whose mid is $m = 50.00$. The book is long $q = +40$ units, the risk aversion is $\gamma = 0.002$ per (price unit $\times$ unit), the variance rate is $\sigma^2 = 1.5$ squared price units per day, and $T - t = 0.5$ days remain in the risk horizon. The inventory shift in (13.1), computed stepwise, is

$$q\,\gamma\,\sigma^2(T-t) = 40 \times 0.002 \times 1.5 \times 0.5 = 0.08 \times 1.5 \times 0.5 = 0.12 \times 0.5 = 0.06,$$

so the reservation price is $r = 50.00 - 0.06 = 49.94$. With a fixed half-width $\delta = 0.10$ the quotes are 49.84 bid at 50.04 ask. Compared with mid-centered quotes (49.90 bid at 50.10 ask), *both* quotes sit 0.06 lower: the ask now undercuts, so the offer is more likely to be lifted and the long is sold down; the bid has backed off, so the book is less likely to be hit into an even larger long. The maker leans toward shedding inventory while remaining two-sided. If instead the book is short $q = -40$ units, the shift flips sign by linearity: $r = 50.06$, and the quotes become 49.96 bid at 50.16 ask, now leaning to buy the short back. Figure 22 sketches the geometry.

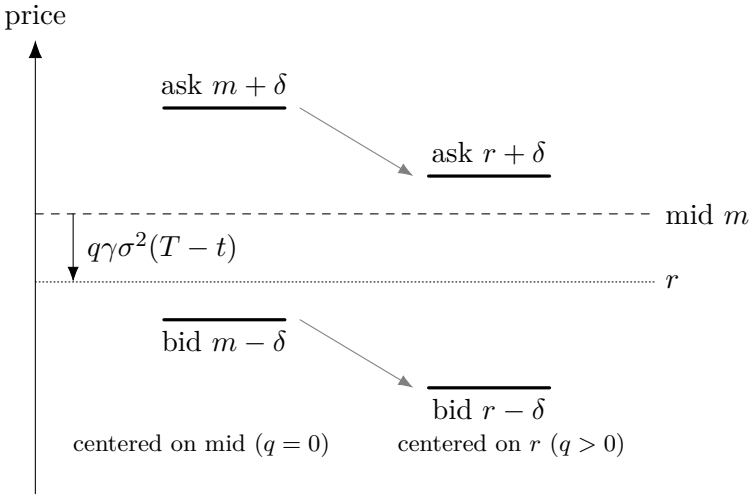


Figure 22: Mid price versus reservation price for positive inventory. A maker long $q > 0$ units centers the quote pair on the reservation price $r = m - q\gamma\sigma^2(T-t)$ of (13.1) rather than on the mid m: both quotes shift down by the same amount, so the ask moves toward the mid (the long is offered out more eagerly) while the bid backs away (further accumulation is discouraged). The half-width δ is unchanged — skewing relocates the quote pair; it does not widen it.

Intuition. The reservation price is where the maker is genuinely indifferent, and quotes should always straddle the point of indifference. To a long book the marginal unit is worth *less* than mid, because it adds variance the maker must carry to the horizon; the quoting center therefore drops below mid by exactly the marginal risk cost $q\gamma\sigma^2(T-t)$. Skewing is not a directional bet — it is the book repricing its own risk: sell more eagerly, buy less eagerly, and never go one-sided.

13.2 The Make-Me-a-Market Ritual

"Make me a market on X" is one of the most common live tests at trading firms, and underneath it sits exactly the machinery of §13.1: a fair-value estimate, a defensible width, Bayesian updating on the flow, and an inventory skew. The graded behavior can be written down as a procedure.

Procedure 13.4 (Make-me-a-market ritual)**.** Given a request to make a market on a quantity X:

1. **Compute a fair value silently.** Form the estimate before speaking; the market is quoted around it, not reasoned toward out loud.

2. **Quote two-sided, at a width that can be defended.** The width prices the uncertainty in the estimate (the spread economics of the preceding section); a width chosen at random is indefensible in the follow-up.
3. **After each fill, update the level and skew.** A fill is information: move the level in the direction of the flow (a Bayesian update on what the counterparty's willingness to trade reveals) *and* shift the center away from the accumulated inventory, as in (13.1).
4. **Track and narrate inventory aloud.** A maker long two units might say: "long two now, so 6.6 at 7.4." The narration shows the count is being kept and ties the skew to the position.
5. **Never freeze.** A committed, updatable price beats a perfect silent one.

Interview trap. The repeated-lift trap: the interviewer announces "I buy. I buy again. I buy again — same price?" Keeping the same offer through repeated lifts ignores both the information in the flow (an offer that keeps getting lifted is evidently cheap) and the maker's growing short position. Move the market — step 3 of Procedure 13.4 applies after *every* fill, not once.

13.3 The Calibration Game: Making a Market on an Estimate

A harder variant of the ritual fuses Fermi estimation with quoting: the player must first manufacture the fair value from scratch, state an honest confidence interval around it, and then trade on it. The game grades calibration and bookkeeping, not trivia. Throughout this subsection, k denotes thousands and M denotes millions, and cash is measured in the same price units as the quotes.

Definition 13.5 (Calibration game)**.** The interviewer names a quantity whose value the quoting player does not know precisely. The player must, within roughly 15 seconds, state a point estimate and a 95% confidence interval for the quantity, and then quote a two-sided market on it. The interviewer trades against the quotes at will, one unit per quote — a counterparty *buy* lifts the offer, meaning the player has *sold* one unit at the ask; a counterparty *sell* hits the bid, meaning the player has *bought* one unit at the bid — and may stop the game at any time. The game ends with the closing question: "are you long or short, and how much?" Grading rewards process — the four behaviors collected in Remark 13.7 below — not proximity of the estimate to the true value.

Example 13.6 (A calibration game as played: the complete ledger)**.** The quantity: the number of Bitcoin blocks mined since 2012, given that a block arrives every 10 minutes on average. Prices are quoted in blocks. The quoting player's rounds, with the counterparty trading one unit per quote, were as follows.

Round	Quote (bid–ask)	Counterparty	Player's fill	Cash flow	Position	Cash
1	100k–200k	buys	sold 1 at 200k	+0.2M	−1	+0.2M
2	400k–600k	buys	sold 1 at 600k	+0.6M	−2	+0.8M
3	900k–1M	buys	sold 1 at 1M	+1.0M	−3	+1.8M
4	4M–6M	sells	bought 1 at 4M	−4.0M	−2	−2.2M
5	2M–3M	no trade	—	0	−2	−2.2M

Closing accounting. Three units were sold and one was bought, so the position is **short 2 units**. Cash is $+0.2 + 0.6 + 1.0 - 4.0 = -2.2$M. Marked at the final mid $(2\text{M} + 3\text{M})/2 = 2.5$M, the mark-to-market equity is

$$\text{cash} + q \times \text{mid} = -2.2\text{M} + (-2)(2.5\text{M}) = -7.2\text{M} :$$

the book is down badly — three sales printed far below value before the single buy printed above it.

The flow read. Three consecutive buys said the offer was cheap, and the player correctly jumped the market in multiples rather than ticks (100k–200k to 400k–600k to 900k–1M to 4M–6M) until

the flow flipped. The flip is the information payoff: the counterparty bought at the 1M offer and later sold at the 4M bid, so they were acting as if the value lay in the bracket $(1\text{M}, 4\text{M})$ — and the final quote 2M–3M correctly sits inside that bracket.

The underlying Fermi chain. At 10 minutes per block: 6 blocks per hour $\times 24 = 144$ per day; $\times 365 \approx 52.6$k per year; $\times 14$ years ≈ 735k blocks. The game grades process, not proximity to this number.

Remark 13.7 (Graded behaviors in the calibration game). Four behaviors are graded, in order of importance.

1. **Width equals confidence.** A 95% interval on a number the player does not actually know must be *embarrassingly wide* — order-of-magnitude wide. A $\pm 10\%$ band is a memorized-fact interval; on a true unknown it will simply not contain the answer. The one losing move is refusing to give an interval at all (for instance, "the volatility is not known, so no interval can be given"): the interval *is* the statement of the player's uncertainty.
2. **Move with the flow.** Repeated buys mean the offer is cheap: jump the market up aggressively, in multiples rather than ticks, until the flow flips. The flip brackets the counterparty's view: a buy at offer a_1 followed later by a sell at bid b_2 means the counterparty acts as if the value lies in (a_1, b_2), and the next quote should sit inside that bracket.
3. **Track inventory out loud.** Every lift of the offer is a sale by the player; every hit of the bid is a purchase. Keep the running count: the closing question is the whole point of the game.
4. **Know the rough profit and loss.** Cash collected minus cash paid, with the open position marked at the final mid — exactly the closing accounting of Example 13.6.

Interview trap. The 15-second freeze: the components of the estimate are usually available — 10 minutes per block, 14 years — so the failure mode is not missing knowledge but an unpracticed chain. The drill is the rate-times-time chain said fast: per hour, then per day, then per year, then total. Training regimen: pick any quantity, produce a point estimate plus an honest 95% interval in 15 seconds, and score interval *coverage* over many repetitions — roughly 19 out of 20 intervals should contain the truth. Missing half the time means the intervals are far too narrow; never missing means they are too wide.

13.4 Hedging Inventory

Skewing sheds inventory only through fill probabilities, which is gradual; a maker can instead remove the risk directly by hedging the inventory with a closely related instrument, often on another venue. The crypto version of this is the cleanest illustration.

Spot inventory hedges with perpetual futures on the same or another venue: a maker absorbing spot buy flow on venue A shorts the perp on venue B, leaving the combined book approximately flat in the underlying. The position then carries the funding rate instead of directional risk (perpetual futures and their funding mechanics are treated in the section on crypto market structure). The hedge is not free of risk: the residual exposures are basis moves — the spot–perp spread can shift against the pair — and venue and counterparty risk on the leg now held at B.

Trading application. On a desk of any size, inventory management becomes a *routing and financing* problem, not just a quoting problem: which venue to hedge on, what funding carry to pay while the hedge is on, and where collateral sits. Quoting decides how inventory arrives; routing and financing decide where it goes and what it costs to hold in the meantime.

Practice: problems P13.1–P13.6 in §27, solutions in §28.

14 Market Microstructure

Market microstructure studies how trading itself moves prices and how the visible limit-order book informs the next price move. This section develops three tools that recur constantly in market-making and execution interviews: the empirical square-root law of market impact, Kyle's price-impact coefficient, and the pair of order-book signals — imbalance and the volume-adjusted mid — that summarize a book snapshot in one number each. All three connect a forecast or a quote screen to realizable profit, which is why the section closes each thread with the practical consequence for trading.

14.1 The Square-Root Law of Market Impact

Executing a large order moves the price against the trader, and the size of that move is the single most important number in execution. This subsection states the empirical law governing it, derives the law's exponents by a dimensional-analysis heuristic, and draws the practical consequence: at sufficient size, impact consumes the alpha that motivated the trade.

The setup is as follows. A *metaorder* is a large parent order of Q shares, executed over some horizon as a sequence of smaller child orders. The market in that asset trades an average of V shares per day, and the asset's daily volatility is σ. The *impact* I of the metaorder is the expected displacement of the price, in the direction of the trade, between the start and the end of the execution. Impact and volatility must be measured in the same price convention — both in currency units, or both in relative (return) units; the numerical examples below use relative units and quote results in basis points (bp), where $1\,\text{bp} = 0.01\%$. The constant Y appearing below is dimensionless.

Fact 14.1 (Square-root impact law, empirical)**.** Across asset classes, market epochs, and execution styles, the impact of a metaorder of Q shares satisfies

$$I \;=\; Y\,\sigma\sqrt{\frac{Q}{V}}, \qquad Y \approx 1, \tag{14.1}$$

where σ is daily volatility and V is average daily volume. The scaling and the near-universality of $Y \approx 1$ are documented in large metaorder studies (Almgren et al., 2005; Toth et al., 2011). The law is an empirical regularity: no derivation from first principles is settled, and a proof is beyond our scope.

Although the law itself is empirical, its exponents are not arbitrary. The following proposition — a *heuristic*, in the sense that its hypothesis is a modelling assumption rather than an established premise — shows that once one accepts a pure power law in the three variables Q, V, σ, the square root is forced by unit consistency alone.

Proposition 14.2 (Square-root exponents by dimensional analysis; heuristic)**.** *Suppose the impact of a metaorder depends on the trade size Q, the average daily volume V, and the daily volatility σ only, through a pure power law*

$$I \;=\; Y\,Q^{\alpha}\,V^{\beta}\,\sigma^{\gamma}$$

with Y a dimensionless constant. Then necessarily $\gamma = 1$, $\beta = -\frac{1}{2}$, and $\alpha = \frac{1}{2}$; that is, $I = Y\sigma\sqrt{Q/V}$ is the only dimensionally consistent power law in these three variables, recovering the form of (14.1).

Proof. We proceed by dimensional analysis, matching units on both sides of the ansatz. The units of the four quantities are

$$[\,I\,] = \text{price}, \qquad [\,Q\,] = \text{shares}, \qquad [\,V\,] = \frac{\text{shares}}{\text{day}}, \qquad [\,\sigma\,] = \frac{\text{price}}{\text{day}^{1/2}},$$

and Y carries no units. Substituting these into $I = YQ^{\alpha}V^{\beta}\sigma^{\gamma}$ gives the units equation

$$\text{price} \;=\; \text{shares}^{\alpha}\left(\frac{\text{shares}}{\text{day}}\right)^{\beta}\left(\frac{\text{price}}{\text{day}^{1/2}}\right)^{\gamma} \;=\; \text{shares}^{\,\alpha+\beta}\;\text{day}^{\,-\beta-\gamma/2}\;\text{price}^{\,\gamma}.$$

The three units are independent, so the exponents must match one unit at a time. Matching price exponents gives $\gamma = 1$. Matching shares exponents gives $\alpha + \beta = 0$. Matching day exponents gives $-\beta - \gamma/2 = 0$, hence $\beta = -\gamma/2 = -\frac{1}{2}$, and then $\alpha = -\beta = \frac{1}{2}$. Substituting back yields $I = Y\sigma\sqrt{Q/V}$. As a check, the units of $\sigma\sqrt{Q/V}$ are

$$\frac{\text{price}}{\text{day}^{1/2}} \times \left(\frac{\text{shares}}{\text{shares/day}}\right)^{1/2} = \frac{\text{price}}{\text{day}^{1/2}} \times \text{day}^{1/2} = \text{price},$$

as required. □

(Problem P14.1 asks you to reproduce this proof from scratch and to name the hidden assumption.)

Remark 14.3 (What the ansatz assumes). The hypothesis of Proposition 14.2 does the heavy lifting. It declares that Q, V, and σ are the *only* relevant variables — excluding tick size, quoted spread, execution duration, and participation rate as independent quantities — and that the dependence is a pure power law with a universal dimensionless Y. Dimensional analysis then shows the square root is the only consistent power law in those three variables, but it cannot justify the variable selection itself. What justifies it is the empirical record: $Y \approx 1$ holds across markets and epochs, which is why the law is stated as Fact 14.1 and the derivation only as a heuristic.

Example 14.4 (Impact at two sizes)**.** Consider a stock with daily volatility $\sigma = 2\%$ and average daily volume $V = 16$ million shares, and take $Y = 1$ in (14.1).

- A metaorder of $Q = 160{,}000$ shares has $Q/V = 0.01$, so $\sqrt{Q/V} = 0.10$ and

$$I \approx 1 \times 2\% \times 0.10 = 0.20\% = 20 \text{ bp}.$$

- A metaorder of $Q = 1.6$ million shares has $Q/V = 0.1$, so $\sqrt{Q/V} \approx 0.316$ and $I \approx 2\% \times 0.316 \approx 63$ bp.

Ten times the size produces only about 3.16 times the impact — impact is concave in Q. The total dollar cost, however, scales like $Q \times I \propto Q^{3/2}$ (up to a size-independent factor reflecting that the average fill is displaced by a fraction of the final impact): the tenfold-larger order costs roughly 31.6 times as many dollars, and the *per-share* cost still grows like $\sqrt{Q}$.

Remark 14.5 (Slicing and the Almgren–Chriss trade-off). Because per-share cost grows like $\sqrt{Q}$, large parent orders are never sent at once: they are sliced over time on a schedule (TWAP spreads the order evenly in time; VWAP spreads it in proportion to expected volume). Scheduling opens a genuine trade-off, formalized by Almgren and Chriss (2000): trade too big and impact eats the alpha; trade too small and the position misses the forecast move; trade too slowly and the un-executed remainder bears the risk of adverse price moves while the forecast decays. Optimal execution balances impact cost against this timing risk.

Intuition. Two structural features of (14.1) are worth internalizing. First, doubling both Q and V leaves impact unchanged: what matters is the trade's size *relative to the market's turnover*, not its absolute size. Second, the square root means each additional share moves the price less than the one before — large orders are worked where and when liquidity is available, so the market absorbs them more gently than a linear extrapolation from small trades would predict.

Interview trap. A standard interview question runs: "The signal says the stock rises 1% over the next two hours, and the desired position is large — can the alpha be captured?" The tempting

answer is yes, since the forecast is genuine. The correct answer is *no* at sufficient size: with impact growing as $\sqrt{Q}$ by (14.1), the trader's own impact — paid once entering and again exiting — can exceed the forecast move, so the net capture (alpha minus round-trip impact and costs) turns negative. Any claimed expected profit from a forecast must be quoted *after* subtracting the impact of getting in and out at the intended size.

14.2 Kyle's Lambda

The square-root law (Fact 14.1) describes the total impact of a worked metaorder. At the granularity of individual trades, impact is instead summarized by a single regression coefficient, universally called Kyle's lambda. Interviews probe three things: the definition, how to estimate it, and how a linear coefficient can coexist with the concave square-root law.

The setup is as follows. Trading is observed over short intervals indexed by $t = 1, 2, \ldots$ (one-minute bars, say). Let S_t denote the price at the end of interval t, and let x_t denote the *signed volume* in interval t: the sum of trade sizes with buyer-initiated (aggressor-buy) trades counted positive and seller-initiated trades counted negative. Let ε_t be a noise term uncorrelated with x_t. In this section S denotes a price (the document-wide P is reserved for probability) and λ denotes Kyle's lambda, not an eigenvalue; no eigenvalues appear in this section, so the canonical symbol is retained.

Definition 14.6 (Kyle's lambda, via the price-impact regression)**.** Kyle's lambda is the coefficient λ in the linear price-impact model

$$S_t \;=\; S_{t-1} + \lambda\, x_t + \varepsilon_t, \tag{14.2}$$

that is, the expected price move per unit of signed order flow. It is estimated in practice as the OLS slope of a regression of price changes $S_t - S_{t-1}$ on signed volume x_t. A large λ means small flow moves the price a lot: λ is a direct measure of *illiquidity*, the inverse of market depth.

Example 14.7 (Estimating and using lambda)**.** A researcher regresses one-minute price changes on one-minute signed volume for two stocks and finds that 10,000 shares of net signed flow move stock A by 1.5 bp on average and stock B by 0.3 bp. Then

$$\lambda_A = 0.15 \text{ bp per 1,000 shares}, \qquad \lambda_B = 0.03 \text{ bp per 1,000 shares},$$

so stock A is five times as illiquid as stock B — equivalently, B is five times as deep. Under the linear model (14.2), a 20,000-share buy in stock A is expected to move its price by $\lambda_A \times 20 = 3.0$ bp.

Trading application. Three standard uses of an estimated λ: comparing liquidity across assets, venues, and time; feeding pre-trade cost models and position sizing (an impact budget: how much flow a given price concession buys); and monitoring for liquidity droughts — a spike in λ means the same flow suddenly moves the price much further, an early warning that depth has evaporated.

Fact 14.8 (Equilibrium origin of lambda (Kyle, 1985))**.** The regression coefficient of Definition 14.6 has a theoretical pedigree. In Kyle's single-period auction model — one informed trader who knows the terminal value $v \sim \mathcal{N}(\mu, \sigma_v^2)$, noise traders submitting exogenous flow $u \sim \mathcal{N}(0, \sigma_u^2)$, and a competitive risk-neutral market maker who observes only the aggregate flow — there is a unique linear equilibrium in which the market maker sets the price as a linear function of the flow with slope

$$\lambda \;=\; \frac{\sigma_v}{2\,\sigma_u}.$$

Impact is steep when there is much private information per unit of noise trading, and markets are deep when noise trading is plentiful: λ is the price of adverse selection, compensating the market maker for trading against informed flow. A proof is beyond our scope.

Remark 14.9 (Reconciling a linear lambda with concave impact). The model (14.2) is linear, yet Fact 14.1 says total impact is concave. Empirically the two are reconciled by letting the fitted slope depend on trade size: estimated within size buckets,

$$\lambda(Q) \;\propto\; Q^{-1/2}, \qquad \text{so total impact} \qquad \lambda(Q)\,Q \;\propto\; \sqrt{Q},$$

which is exactly the square-root law (14.1). A single OLS regression estimates an *average* slope over whatever size distribution was sampled; extrapolating that slope linearly to sizes far beyond the sample overstates the impact of large orders. For sizing a large trade, the concave prediction is the one to trust.

14.3 Order-Book Imbalance and the Volume-Adjusted Mid

The displayed limit-order book carries information about the next price move, and two one-number summaries of the top of the book are the workhorses of short-horizon prediction. This subsection defines both, computes them on a deliberately lopsided book, and closes with a naming distinction that matters in interviews at high-frequency firms.

The setup is as follows. At the top of the book, the best bid quotes a price S_b with displayed size q_b (shares resting at the bid), and the best ask quotes a price $S_a > S_b$ with displayed size q_a. The *mid price* is $m = (S_b + S_a)/2$. Here q_b and q_a denote displayed queue sizes — not the Markov-chain probability q of the document-wide notation — and S again denotes a price.

Definition 14.10 (Order-book imbalance)**.** The *order-book imbalance* of the top of the book is

$$\text{OBI} \;=\; \frac{q_b - q_a}{q_b + q_a} \;\in\; [-1, 1].$$

An imbalance of $+1$ means all displayed size is on the bid, -1 means all on the ask, and 0 means a balanced book.

Trading application. A positive imbalance signals buying pressure: heavy resting demand below and thin supply above make the next mid-price move more likely up than down. Order-book imbalance is the workhorse short-horizon (tick-scale) predictor on desks that trade against the book.

Definition 14.11 (Volume-adjusted mid price (VAMP))**.** The *volume-adjusted mid price* is

$$S_{\text{VAMP}} \;=\; \frac{S_a\, q_b + S_b\, q_a}{q_b + q_a},$$

the convex combination of the two quoted prices in which each price is weighted by the *opposite* side's displayed size.

The reader should verify from Definition 14.11 that the two weights sum to one, so S_{VAMP} always lies between S_b and S_a; that a balanced book ($q_b = q_a$) gives $S_{\text{VAMP}} = m$; and that $S_{\text{VAMP}} \to S_a$ as $q_b/q_a \to \infty$. The next example shows why the opposite-side weighting is the correct one.

Example 14.12 (A one-sided book)**.** Consider the book with best bid $S_b = 99$ carrying $q_b = 1{,}000$ shares and best ask $S_a = 100$ carrying $q_a = 1$ share. Then

$$\text{OBI} \;=\; \frac{1{,}000 - 1}{1{,}000 + 1} \;=\; \frac{999}{1{,}001} \;\approx\; +0.998,$$

$$S_{\text{VAMP}} \;=\; \frac{100 \times 1{,}000 + 99 \times 1}{1{,}001} \;=\; \frac{100{,}099}{1{,}001} \;\approx\; 99.999,$$

while the plain mid is $m = 99.50$. The interpretation: to push the price down one tick, sellers must consume 1,000 resting shares, while a single share of buying lifts the entire ask — the

next move is overwhelmingly likely to be up, so the "true" price sits essentially at the ask. Weighting the ask price by the *bid* size encodes exactly this: a heavy bid pins value toward the ask. The plain mid (99.50) ignores the pressure entirely, and the naive *same*-side weighting $(99\times1{,}000+100\times1)/1{,}001 \approx 99.001$ would drag the estimate toward the crowded side — backwards. Figure 23 shows this book together with two deeper levels on each side.

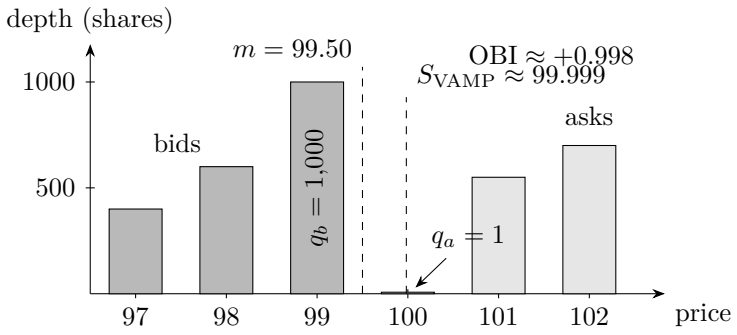


Figure 23: The limit-order book of Example 14.12: bid depth (dark bars, prices 97–99) against ask depth (light bars, prices 100–102), with a 1,000-share best bid facing a single displayed share at the best ask. The imbalance is extreme (OBI $\approx +0.998$), and the volume-adjusted mid sits essentially at the ask, far from the plain mid $m = 99.50$. As defined here, OBI and S_{VAMP} use top-of-book sizes only; the deeper levels are shown for context.

Remark 14.13 (Weighted mid versus the formal micro-price). Naming precision matters here. The quantity S_{VAMP} of Definition 14.11 is the volume-weighted (imbalance-weighted) *mid*. The formal *micro-price* of Stoikov (2018) is defined differently, as the limit of expected future mid prices conditional on the current state of the book; the weighted mid is a first-order proxy for it, not the object itself. In an interview at a high-frequency or market-making firm, stating the distinction — "this is the weighted mid, a first-order approximation to Stoikov's micro-price" — signals fluency with the literature rather than a memorized formula.

Practice: problems P14.1–P14.5 in §27, solutions in §28.

15 Crypto Market Structure

Crypto markets run continuously, fragment across hundreds of venues, and host two instrument classes with no direct equity-market analogue: perpetual futures on centralized exchanges and automated market makers on-chain. This section develops the small amount of arithmetic the subject genuinely requires — funding-rate annualization, the constant-product trade-cost formula, and the impermanent-loss formula — and organizes the surrounding market-structure facts (the venue landscape, liquidation cascades, maximal extractable value) that interviewers use to test whether a candidate has actually watched these markets trade.

15.1 The Centralized-Exchange Landscape

Centralized crypto exchanges (CeFi) differ from equity venues in ways that reshape what liquidity provision means in practice. The observations in this subsection are qualitative, but each carries a direct operational consequence for a trading desk.

Crypto trades 24 hours a day, 7 days a week, on hundreds of venues, and the same asset is listed on essentially all of them. There is no primary listing exchange, no opening or closing auction, and no market close at which risk can be handed off.

Remark 15.1 (Fair value is a cross-venue estimate). Because the same asset trades on many venues simultaneously, no single venue's last price is authoritative. Fair value is an *estimate* aggregated across venues, typically a liquidity-weighted combination of prices wherever the asset trades, and every quote a market maker posts on one venue leans on that cross-venue estimate. The spot *index* against which derivatives venues settle and compute funding (Definition 15.3) is exactly such a composite.

Trading application. Liquidity provision on CeFi venues is inventory *logistics* as much as quoting. To quote on a venue, a desk must hold inventory on that venue; withdrawal and transfer times (blockchain confirmations plus exchange processing) are frictions that any cross-venue arbitrage must cross, so a price discrepancy that closes faster than assets can move is capturable only with inventory pre-positioned on both venues. Fee tiers shift where the edge lies: maker rebates and taker fees differ across venues and volume tiers, and the difference can flip a marginal quote from unprofitable to profitable.

Remark 15.2 (Custody is a first-class risk). Since the failure of the FTX exchange (2022), *where inventory sits overnight* is a first-class risk decision, not an operational afterthought: assets held on a venue are exposed to the venue's own credit. Desks cap per-venue balances and treat the allocation of inventory across venues as part of the risk book.

15.2 Perpetual Futures and Funding

The perpetual future is the dominant crypto derivative. It has no expiry, so nothing contractual forces convergence to spot at a terminal date; a periodic cash flow between longs and shorts does the anchoring instead. Interviewers routinely ask a candidate to annualize a printed funding rate on the spot, which is the content of Proposition 15.4.

Setup. Let S denote the venue's spot index for the underlying (a cross-venue composite, Remark 15.1) and let F denote the perpetual's traded price. Write ϕ for the funding rate per interval; a position's *notional* is its size times the mark price. We assume three funding intervals per day and a 365-day year, hence $N = 3 \times 365 = 1095$ intervals per year.

Definition 15.3 (Perpetual future and funding rate)**.** A *perpetual future* (perp) is a linear futures contract with no expiry date. Once per *funding interval* — commonly 8 hours, hence three intervals per day — every open position pays or receives a *funding payment* equal to ϕ times its notional, where the venue sets the rate ϕ as a clamped (capped and floored) function of the perp's average premium $(F - S)/S$ over the interval. By convention, a positive rate means longs pay shorts, and a negative rate means shorts pay longs.

Intuition. The funding flow is what ties the perp to spot in the absence of an expiry. If the perp trades above the index, the premium is positive, so funding is positive: holding a long now *costs* ϕ per interval, which pushes longs to close and pays shorts to lean against the premium — pressure on the perp price is *downward*, back toward the index. Below the index the mechanism mirrors. The anchor is economic rather than contractual.

Proposition 15.4 (Funding annualization)**.** *Suppose the funding rate is a constant $\phi > -1$ per 8-hour interval. Then, as a fraction of a fixed notional:*

(i) the funding accrued per day is 3ϕ*, and the* simple *(non-compounded) annualized rate is* $N\phi = 1095\,\phi$*;*

(ii) if each payment is reinvested into the position, the compounded *annualized rate is* $(1+\phi)^N - 1$*;*

(iii) the compounded rate is never below the simple rate: $(1+\phi)^N - 1 \geq N\phi$*, with equality if and only if* $\phi = 0$*.*

Proof. Parts (i) and (ii) are direct computation; part (iii) is Bernoulli's inequality, which we prove by induction.

(i) Over one day the position pays or receives three payments of ϕ times notional; over a year, $N = 1095$ of them. Held against a fixed notional, the payments sum to $1095\,\phi$.

(ii) If each payment is reinvested, the account is multiplied by $(1+\phi)$ each interval, so after N intervals it holds $(1+\phi)^N$ times its starting value, a gain of $(1+\phi)^N - 1$.

(iii) We claim $(1+\phi)^n \geq 1 + n\phi$ for every integer $n \geq 1$ and every $\phi > -1$. The base case $n = 1$ holds with equality. For the inductive step, assume $(1+\phi)^n \geq 1+n\phi$. Since $1+\phi > 0$, multiplying both sides by $(1+\phi)$ preserves the inequality:

$$(1+\phi)^{n+1} \;\geq\; (1+n\phi)(1+\phi) \;=\; 1 + (n+1)\phi + n\phi^2 \;\geq\; 1 + (n+1)\phi.$$

The final inequality discards $n\phi^2 \geq 0$, which is strict unless $\phi = 0$; hence equality throughout forces $\phi = 0$, and conversely $\phi = 0$ gives equality. Applying the claim with $n = N$ and subtracting 1 proves (iii). □

(Problem P15.4 asks you to reproduce this arithmetic from scratch at a different rate.)

Example 15.5 (The 0.01% anchor)**.** Take $\phi = 0.01\% = 10^{-4}$ per 8-hour interval, the standard reference rate on major venues. Per day: $3 \times 0.01\% = 0.03\%$. Simple annualized, by Proposition 15.4(i):

$$1095 \times 0.0001 = 0.1095 \approx 11\% \text{ per year (simple).}$$

Compounded, by part (ii): $(1.0001)^{1095} = e^{1095\ln(1.0001)} \approx e^{0.1095} \approx 1.1157$, i.e. about 11.6% per year. As a self-check on the gap, the second-order term of $e^x \approx 1 + x + x^2/2$ at $x = 0.1095$ predicts an excess of $0.1095^2/2 \approx 0.6$ percentage points over the simple figure — which is exactly the difference $11.57\% - 10.95\%$. The desk anchor worth memorizing: *0.01% per 8 hours is about 11% a year.*

Remark 15.6 (The compounding caveat). Two caveats attach to the 11% anchor. First, it is the *simple* figure; reinvestment lifts it to about 11.6%, and Proposition 15.4(iii) says the lift is always upward. Second, and more importantly, the rate is not constant: it resets every interval as a function of the premium, so annualizing one printed rate extrapolates a snapshot. A funding rate is a state variable, not a locked-in yield.

Trading application. *Cash-and-carry.* Buy spot, short an equal notional of the perp, and collect the funding while it is positive: a delta-neutral yield. The yield is capped by *basis risk* (no expiry ever forces convergence, so an unwind at an adverse basis can surrender accumulated carry) and by *venue risk* (both legs and the collateral sit on an exchange, Remark 15.2). The funding *sign* is also a positioning gauge: persistently positive funding means leveraged longs are paying a steady premium to stay long — crowded positioning — and at extremes it reads as a contrarian signal.

15.3 Liquidations and Cascades

Perpetual positions are levered: the trader posts margin far smaller than notional, and the venue force-closes (*liquidates*) the position when its equity falls to a maintenance threshold. For leverage L, an adverse move of roughly $1/L$ exhausts the initial margin, so every levered position carries a *liquidation price* that is known in advance — and clusters of similar positions carry clusters of nearby liquidation prices.

Remark 15.7 (Liquidation cascades). A liquidation is a *forced market order in the direction of the move*: a levered long liquidated in a falling market is closed with a forced sell. When price trades through a dense cluster of liquidation prices, the first tranche of forced sells pushes price lower, which triggers the next tranche, and so on — a mechanical positive-feedback loop called a *cascade*, producing relentless one-way flow and gapping prices.

Trading application. For a market maker a cascade is both danger and opportunity. Danger: the flow is toxic in the mark-out sense even though it is forced rather than informed — it is all same-side and it marks against every fill while the cascade runs, and prices gap through quote levels. Opportunity: liquidity premia spike, because the forced flow carries no information about fundamentals and must transact regardless of price; whoever still has risk capacity is paid unusually well for providing liquidity into it. In an interview, describing cascade dynamics accurately is a credible signal of having actually watched crypto trade.

15.4 Constant-Product Automated Market Makers

On-chain venues (DeFi) replace the limit order book with a deterministic pricing curve: a passive pool of two assets quotes every trade size algorithmically. The constant-product design is the canonical one, and its trade-cost formula is a one-line derivation that interviewers expect a candidate to produce on demand.

Setup. In the remainder of this section, p denotes a *price* and r a *gross price ratio*; these local meanings supersede the document-wide Markov-chain conventions for p and r. A pool holds reserves $x > 0$ of asset X (say ETH) and $y > 0$ of asset Y (say a dollar stablecoin), with invariant constant $k = xy$. We ignore trading fees throughout. Prices of X are quoted in units of Y.

Definition 15.8 (Constant-product AMM)**.** A *constant-product automated market maker* is a pool with reserves (x, y) that accepts any trade leaving the product of its reserves unchanged: a fee-free trade moving the reserves to (x', y') is admissible if and only if $x'y' = xy = k$. The pool's *marginal price* of X is $p = y/x$, the cost per unit of an infinitesimal purchase (justified in Proposition 15.9).

Proposition 15.9 (Cost of a buy from a constant-product pool)**.** *Consider a constant-product pool with reserves (x, y) and invariant $k = xy$. A trader who buys Δx units of X, where $0 < \Delta x < x$, must pay*

$$\Delta y \;=\; \frac{k}{x - \Delta x} - y \;=\; \frac{k\,\Delta x}{x\,(x - \Delta x)} \;>\; 0. \tag{15.1}$$

Moreover the cost per unit tends to y/x as $\Delta x \to 0$, which justifies calling $p = y/x$ the marginal price.

Proof. We compute directly from the invariant. After the trade the pool holds $x - \Delta x$ units of X and $y + \Delta y$ units of Y, and admissibility requires

$$(x - \Delta x)(y + \Delta y) = k.$$

Solving for the payment: $y + \Delta y = k/(x - \Delta x)$, hence $\Delta y = k/(x - \Delta x) - y$. For the second form, substitute $y = k/x$ and combine over a common denominator:

$$\Delta y = \frac{k}{x - \Delta x} - \frac{k}{x} = \frac{k\,x - k\,(x - \Delta x)}{x\,(x - \Delta x)} = \frac{k\,\Delta x}{x\,(x - \Delta x)},$$

which is positive since $0 < \Delta x < x$. Dividing by Δx gives the per-unit cost $k/(x(x - \Delta x))$, which tends to $k/x^2 = y/x$ as $\Delta x \to 0$. □

(Problem P15.7 asks you to reproduce this derivation and run the numbers on a live pool; Problem P15.8 applies it three times in a row.)

Corollary 15.10 (Price impact is built into the curve)**.** *In the setting of Proposition 15.9, write $f = \Delta x/x \in (0, 1)$ for the fraction of the reserve purchased. Then the average price paid and the post-trade marginal price are*

$$\frac{\Delta y}{\Delta x} = \frac{p}{1 - f}, \qquad \frac{y + \Delta y}{x - \Delta x} = \frac{p}{(1 - f)^2},$$

both strictly increasing and convex in f, and both diverging as $f \uparrow 1$. In particular the total cost $\Delta y = p\,x\,f/(1-f)$ grows super-linearly in trade size: a constant-product pool has price impact built into its geometry, with no order book required.

Proof. We substitute $k = px^2$ (from $p = y/x$ and $k = xy$) into the formulas of Proposition 15.9. The average price is

$$\frac{\Delta y}{\Delta x} = \frac{k}{x(x-\Delta x)} = \frac{p\,x^2}{x^2(1-f)} = \frac{p}{1-f},$$

and the post-trade reserves are $x' = x(1-f)$ and $y' = k/x'$, so the new marginal price is $y'/x' = k/x'^2 = p/(1-f)^2$. Each expression is p times a positive integer power of $(1-f)^{-1}$, hence strictly increasing and convex on $(0,1)$ and unbounded as $f \uparrow 1$. Finally $\Delta y = \Delta x \cdot p/(1-f) = p\,x\,f/(1-f)$, whose second derivative in f is $2px/(1-f)^3 > 0$: the cost is convex, i.e. super-linear. □

Figure 24 shows the geometry: a buy slides the state point along the hyperbola $xy = k$ into its steepening region.

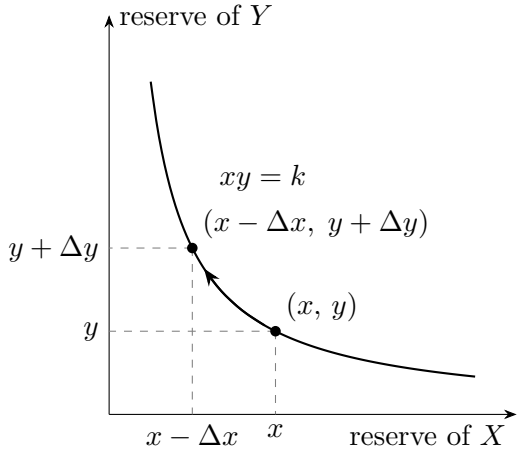


Figure 24: A buy of Δx units of X from a constant-product pool moves the state along the hyperbola $xy = k$ from (x, y) to $(x - \Delta x,\ y + \Delta y)$; the buyer pays $\Delta y = k/(x - \Delta x) - y$ by (15.1). The curve steepens as x shrinks, so successive units cost more: price impact is the geometry of the curve itself (Corollary 15.10).

Example 15.11 (Buying 10 ETH from a 100-ETH pool)**.** A fee-free ETH/USDC pool holds $x = 100$ ETH and $y = 200{,}000$ USDC, so $k = 2 \times 10^7$ and the marginal price is $p = 200{,}000/100 =$ \$2,000 per ETH. A trader buys $\Delta x = 10$ ETH, a fraction $f = 0.1$ of the reserve. By (15.1),

$$\Delta y = \frac{2 \times 10^7}{90} - 200{,}000 = 222{,}222.22 - 200{,}000 = 22{,}222.22 \text{ USDC},$$

an average price of $22{,}222.22/10 =$ \$2,222.22 per ETH, matching Corollary 15.10: $p/(1-f) = 2{,}000/0.9 = 2{,}222.22$. The pool's new marginal price is $p/(1-f)^2 = 2{,}000/0.81 =$ \$2,469.14. Two self-checks: the invariant survives, $90 \times 222{,}222.22 = 2.0 \times 10^7$; and the reader should verify, from the two formulas of Corollary 15.10, that the average price is always the geometric mean of the pre- and post-trade marginal prices — here $\sqrt{2{,}000 \times 2{,}469.14} = 2{,}222.2$.

Remark 15.12 (Arbitrage is the AMM's price discovery). The pool has no view, no oracle, and no order book: after any trade its marginal price deviates from the price on centralized exchanges, and arbitrageurs trade against the pool until the two agree again, pocketing the difference. That external arbitrage is the entire price-discovery mechanism of an AMM, and it is what keeps AMM prices honest. Its cost to the passive side is quantified in Theorem 15.14 as impermanent loss.

15.5 Impermanent Loss

A liquidity provider (LP) deposits both assets into a constant-product pool and passively takes the other side of every trade, including the arbitrage flow of Remark 15.12. The classic interview derivation of this subsection prices what that passivity costs: relative to simply holding the deposited coins, the LP loses a deterministic function of the price move, negative in *both* directions.

Setup. Fix a fee-free constant-product pool with invariant k (Definition 15.8), and assume arbitrageurs keep the pool's marginal price equal to the external market price at all times. An LP deposits x_0 units of X and y_0 units of Y when the external price is p_0; the price then moves to $p_1 = r\,p_0$ for some gross ratio $r > 0$. Write V_{LP} for the value of the pool position at the new price, V_{HODL} for the value of simply holding the deposited coins (x_0 of X and y_0 of Y) to the same date, and define the *impermanent loss* $\mathrm{IL}(r) = V_{\mathrm{LP}}/V_{\mathrm{HODL}} - 1$. All values are in units of Y.

Lemma 15.13 (Arbitrage-aligned reserves)**.** *A constant-product pool with invariant k and marginal price p has uniquely determined reserves*

$$x = \sqrt{k/p}, \qquad y = \sqrt{k\,p},$$

the value of the reserves at price p is $p\,x + y = 2\sqrt{k\,p}$, and the pool always holds equal values of the two assets ($p\,x = y$).

Proof. We solve the two defining equations simultaneously. The constraints are $xy = k$ and $y/x = p$ with $x, y > 0$. Substituting $y = p\,x$ into the invariant gives $p\,x^2 = k$, whose unique positive root is $x = \sqrt{k/p}$; then $y = p\,x = \sqrt{k\,p}$. The value at price p is $p\,x + y = \sqrt{k\,p} + \sqrt{k\,p} = 2\sqrt{k\,p}$, and the equal-split identity $p\,x = \sqrt{k\,p} = y$ is the middle step of that computation. □

Theorem 15.14 (Impermanent loss)**.** *In the setup above, for every $r > 0$:*

(i) the pool position is worth $V_{\mathrm{LP}} = V_0\sqrt{r}$, where $V_0 = p_0x_0 + y_0 = 2y_0$ is the deposit value;

(ii) the hold benchmark is worth $V_{\mathrm{HODL}} = V_0\,\dfrac{1+r}{2}$;

(iii) consequently

$$\mathrm{IL}(r) \;=\; \frac{2\sqrt{r}}{1+r} - 1 \;\le\; 0, \tag{15.2}$$

with equality if and only if $r = 1$.

Proof. We compute both terminal values in closed form via Lemma 15.13 and compare them with the AM–GM inequality.

Step 1: the deposit. At price p_0 the aligned pool holds $x_0 = \sqrt{k/p_0}$ and $y_0 = \sqrt{k\,p_0}$, and by the equal-split identity $p_0x_0 = y_0$, so the deposit is automatically a 50/50 split with value $V_0 = 2y_0 = 2\sqrt{k\,p_0}$.

Step 2: the pool after the move. Arbitrageurs re-align the pool to the new price $p_1 = r\,p_0$, so by Lemma 15.13 the reserves rebalance to

$$x_1 = \sqrt{k/p_1} = \frac{x_0}{\sqrt{r}}, \qquad y_1 = \sqrt{k\,p_1} = y_0\sqrt{r}.$$

For $r > 1$ the pool now holds *fewer* units of X: it sold X into the rally. The position's value is

$$V_{\mathrm{LP}} = p_1x_1 + y_1 = 2\sqrt{k\,p_1} = 2\sqrt{k\,p_0}\,\sqrt{r} = V_0\sqrt{r},$$

proving (i): the LP's value grows like $\sqrt{r}$, not like r.

Step 3: the benchmark. Holding the deposited coins instead,

$$V_{\mathrm{HODL}} = p_1x_0 + y_0 = r\,p_0x_0 + y_0 = (r+1)\,y_0 = V_0\,\frac{1+r}{2},$$

proving (ii).

Step 4: the comparison. Dividing,

$$\frac{V_{\mathrm{LP}}}{V_{\mathrm{HODL}}} = \frac{V_0\sqrt{r}}{V_0(1+r)/2} = \frac{2\sqrt{r}}{1+r},$$

which gives the formula in (15.2). By the AM–GM inequality applied to the pair $\{1, r\}$, we have $\sqrt{r} = \sqrt{1 \cdot r} \le (1+r)/2$, with equality if and only if $1 = r$; hence the ratio is at most 1 and $\mathrm{IL}(r) \le 0$, vanishing exactly at $r = 1$. □

(Problem P15.9 asks you to reproduce this proof from scratch and extend it to the symmetry $\mathrm{IL}(r) = \mathrm{IL}(1/r)$ *and the small-move law; Problem P15.10 applies it to a benchmark trap.)*

Example 15.15 (Impermanent loss at $r = 2$ and $r = 4$)**.** First the pure ratios from (15.2). A doubling, $r = 2$: $\mathrm{IL}(2) = 2\sqrt{2}/3 - 1 = 0.9428 - 1 = -5.72\% \approx -5.7\%$. A quadrupling, $r = 4$: $\mathrm{IL}(4) = 2 \cdot 2/5 - 1 = -20\%$.

Now in dollars. Deposit \$5,000 of ETH and \$5,000 of USDC when ETH trades at \$2,000: $x_0 = 2.5$ ETH, $y_0 = 5{,}000$ USDC, $V_0 = \$10{,}000$, $k = 12{,}500$. ETH doubles to \$4,000 ($r = 2$): the pool rebalances to $x_1 = 2.5/\sqrt{2} = 1.7678$ ETH and $y_1 = 5{,}000\sqrt{2} = 7{,}071.07$ USDC, so

$$V_{\mathrm{LP}} = 1.7678 \times 4{,}000 + 7{,}071.07 = \$14{,}142.14 = V_0\sqrt{2}, \qquad V_{\mathrm{HODL}} = 2.5 \times 4{,}000 + 5{,}000 = \$15{,}000,$$

a shortfall of \$857.86 ≈ \$858, i.e. 858/15,000 = 5.72% — matching IL(2). If instead ETH quadruples to \$8,000 ($r = 4$): $V_{\mathrm{LP}} = 10{,}000\sqrt{4} = \$20{,}000$ against $V_{\mathrm{HODL}} = 2.5 \times 8{,}000 + 5{,}000 = \$25{,}000$, a \$5,000 shortfall: exactly 20%. Figure 25 plots the curve through these points.

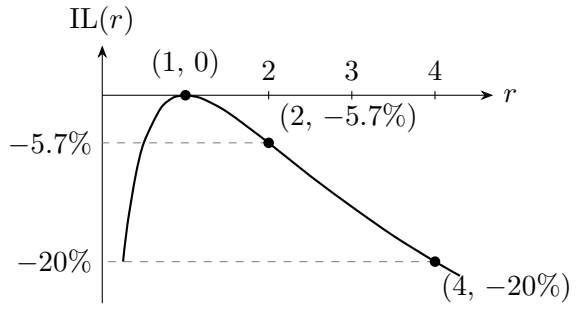


Figure 25: Impermanent loss $\mathrm{IL}(r) = 2\sqrt{r}/(1+r) - 1$ of (15.2) against the gross price ratio r. The curve touches zero only at $r = 1$ and is negative in both directions — the left branch shows that a sell-off hurts the LP too. Marked points: $(1, 0)$, $(2, -5.7\%)$, $(4, -20\%)$.

Intuition. The mechanism behind Theorem 15.14 is visible in Step 2 of the proof: as price rises the pool mechanically *sells* the appreciating asset (from x_0 down to $x_0/\sqrt{r}$), and as price falls it mechanically buys the depreciating one. A position that sheds winners and accumulates losers on every move, and whose value ($V_0\sqrt{r}$, concave in r) lags a linear benchmark in both directions, is *short gamma*: the LP has sold an option to the market, and the trading fees a real pool charges are the premium collected for it. Hence the one-line summary: *an LP profits precisely when fees exceed the option value given away.*

Interview trap. Two standard traps. First, the benchmark: an LP position can be up substantially in absolute terms and still have lost — in Example 15.15 the $r = 2$ position is up 41% on the deposit yet lags holding the same coins by \$858; quoting profit against the deposit hides the loss, and the honest benchmark is always *holding the deposited assets.* Second, the name: "impermanent" suggests the loss heals if one waits, but it is a property of the current price ratio — withdraw at any $r \neq 1$ and the loss is permanent.

15.6 Maximal Extractable Value and Decentralized Venues

Public-blockchain execution creates an adversarial ordering game with no CeFi analogue, and interviewers probe it with one-liners. Two facts suffice at this level.

Remark 15.16 (MEV and the sandwich attack). *Maximal extractable value* (MEV) is profit extracted by controlling the order of transactions within a block. The canonical instance is the *sandwich*: an attacker brackets a victim's pending AMM trade, front-running it with a buy (pushing the pool up the curve of Figure 24), letting the victim execute at the worsened price, then back-running with a sell that captures the victim's extra slippage. The attack is possible because pending transactions are visible in the *public mempool* before inclusion, and block producers — or searchers bidding to them — choose the ordering. Sending an order to a public mempool broadcasts intent to adversaries who can trade around it.

Trading application. Major market-making firms quote on decentralized venues alongside centralized ones; the two market structures share one trading business. In crypto-facing interviews, DeFi literacy — AMM mechanics, funding, liquidations, MEV — is table stakes, not a specialty.

Practice: problems P15.1–P15.10 in §27, solutions in §28.

Part VI
Derivatives

16 Forwards, Futures and No-Arbitrage Pricing

This chapter prices forward and futures contracts and then hedges with them. A single argument carries every price: two strategies with the same payoff must cost the same (Lemma 16.8). Along the way five standard interview items become answerable: pricing a forward on a dividend-paying stock; spotting the arbitrage in a quoted forward and writing down its cash flows; deciding whether the futures price is an unbiased forecast of the spot price at delivery; deciding which variable is regressed on which to obtain a hedge ratio; and annualising a crypto basis. Interest-rate conventions come first, because every price below is a discounted payoff.

16.1 Interest-Rate Conventions: Compounding, Zero Rates and Discount Factors

"Six percent compounded quarterly: what is that continuously?" is a mental-arithmetic warm-up, and every formula of this Part discounts at a continuously compounded rate. This subsection defines compounding frequencies, the zero rate and the discount factor, and proves the conversion between frequencies. The continuous limit is cited from Proposition 7.39 and not proved again; the desk shortcut for the conversion receives an exact error bracket from the integral remainder (7.3) of Theorem 7.31, with the derivatives computed in Proposition 7.34 (§7.5). Only conventions are treated here.

Setup. Amounts are in one currency, and time t is measured in years; $T > 0$ is a horizon. An amount invested gets no letter, because in this Part A is an event, as in the Notation table. The number of compounding periods per year is $m > 0$, the m of Proposition 7.39: a positive integer in practice, although the statements below hold for every positive real m. A rate compounded m times a year is written r_m, always with $r_m > -m$, so that $1 + r_m/m > 0$; a continuously compounded rate is written r. The T-year zero rate is $r(T)$ and the discount factor is $e^{-r(T)T}$, which gets no letter of its own (D_i are dividends). *Standing simplification of the chapter:* the

curve is flat, $r(T) = r$ for every T, unless a statement says otherwise. The natural logarithm is written ln. As $x \to 0$, the symbol $O(x^3)$ denotes a quantity bounded by a constant multiple of $|x|^3$ for all x in a neighbourhood of 0, as in §7.5.

Symbol notices for the Part. This is the first Setup of the Part *Derivatives*. Five conventions hold from here to the end of §23.

(1) The letter r is a continuously compounded interest rate in this Part, not the random-walk ratio q/p of the Notation table; q is a yield, not $1 - p$, and a down-probability is always written $1 - p$.
(2) The letter b is the cost of carry, not the Kelly odds of §3.1; the basis $S_t - F_t$ is written as the word basis_t.
(3) The letter Π is the value of a strategy (Π^A and Π^B in Lemma 16.8); p is never a price.
(4) The symbols ΔS, ΔF and Δt are increments; Δ alone is the Greek of §21.
(5) The symbol ln is the natural logarithm of the Notation table; partial derivatives are written with the partial sign, never as subscripts, because a subscript t is a time.

Notices for this chapter. The letter V is the value of a contract and V_t its value at t: always a price, never a derivative. The storage-cost rate is r_{st}, not s, because from §19 on s is a time, and s is the sample standard deviation of §4. The option prices C and $\mathcal{P}$ are not used in this chapter.

Definition 16.1 (Compounding frequency, continuous compounding, zero rate and discount factor)**.**

(i) An amount invested at a rate r_m *compounded m times a year* ($r_m > -m$) grows by the factor $(1 + r_m/m)^{mT}$ in T years: interest of r_m/m per unit invested is credited at the end of each period of length $1/m$ and earns interest afterwards. When mT is not an integer, the same formula is the convention.

(ii) An amount invested at a *continuously compounded* rate r grows by the factor e^{rT} in T years. This is the limit of (i) as $m \to \infty$ through the positive integers, the setting of Proposition 7.39, with $r_m = r$ held fixed: that proposition with $a = r$ gives $(1+r/m)^m \to e^r$, and the map $x \mapsto x^T$ is continuous at $e^r > 0$, so

$$\Big(1 + \frac{r}{m}\Big)^{mT} = \Big(\Big(1 + \frac{r}{m}\Big)^m\Big)^T \longrightarrow e^{rT} \qquad (m \to \infty).$$

The proof of Proposition 7.39, which takes logarithms, uses no integrality of m, so the same limit holds as $m \to \infty$ through the positive reals.

(iii) The *T-year zero rate* $r(T)$ is the continuously compounded rate earned on money invested today and repaid, with all of its interest, in a single payment at T. The *discount factor* $e^{-r(T)T}$ is the amount that must be invested today to hold 1 at T.

Derivatives are priced with a risk-free rate; it is taken as given in these notes.

Proposition 16.2 (Converting between compounding frequencies)**.** *Let $m > 0$.*

(i) Let $r_m > -m$ and let r be real. The two quotes give the same growth over every horizon, $(1 + r_m/m)^{mT} = e^{rT}$ for all $T > 0$, if and only if

$$r = m \ln\Big(1 + \frac{r_m}{m}\Big), \qquad \textit{equivalently} \qquad r_m = m(e^{r/m} - 1). \tag{16.1}$$

Equality at a single horizon $T > 0$ already implies equality at every horizon.

(ii) Let $r \neq 0$ be real and, for each $m > 0$, let $r_m = m(e^{r/m} - 1)$ be its m-times-compounded equivalent. Then $r_m > r$ for every $m > 0$, and the map $m \mapsto r_m$ is strictly decreasing on $(0, \infty)$ with $r_m \to r$ as $m \to \infty$. Every finite-frequency quote therefore exceeds the continuous rate, which is the smallest quote of a given growth. The statement covers negative rates. (For $r = 0$ every quote is 0.)

(iii) (Desk shortcuts with exact error brackets; m is fixed and r, r_m are related by (16.1).) For $r > 0$ the shortcut $r + r^2/(2m)$ understates r_m by an amount strictly between two explicit ends:

$$\frac{r^3}{6m^2} \;<\; r_m - \Big(r + \frac{r^2}{2m}\Big) \;<\; e^{r/m}\,\frac{r^3}{6m^2}.$$

For $r_m > 0$ the shortcut $r_m - r_m^2/(2m)$ understates r in the same way:

$$\frac{r_m^3}{3m^2(1+r_m/m)^3} \;<\; r - \Big(r_m - \frac{r_m^2}{2m}\Big) \;<\; \frac{r_m^3}{3m^2}.$$

For rates of either sign,

$$\Big|\, r_m - r - \frac{r^2}{2m}\Big| \le e^{|r|/m}\,\frac{|r|^3}{6m^2} \quad \textit{for every real } r,$$
$$\Big|\, r - r_m + \frac{r_m^2}{2m}\Big| \le \frac{8\,|r_m|^3}{3m^2} \quad \textit{whenever } |r_m| \le \frac{m}{2}.$$

Hence $r_m - r = r^2/(2m) + O(r^3)$ as $r \to 0$, and $r - r_m + r_m^2/(2m) = O(r_m^3)$ as $r_m \to 0$.

Sketch of the proof. We take logarithms for (i), compare integrands for (ii), and apply Taylor's theorem with the integral remainder for (iii). In (i), both sides of $(1 + r_m/m)^{mT} = e^{rT}$ are positive, because $r_m > -m$, so taking logarithms and dividing by $T > 0$ turns the equality into $r = m\ln(1 + r_m/m)$, which does not involve T. In (ii), the fundamental theorem of calculus gives $r_m = m(e^{r/m} - 1) = r\int_0^1 e^{(r/m)s}\,ds$. The integral tends to its value 1 at $r/m = 0$, as a difference quotient of the exponential function, and it is strictly increasing in r/m, because its integrand is at every $s > 0$ and a continuous nonnegative integrand that is positive somewhere has a positive integral. Since $r \neq 0$, either $r > 0$ and the integral falls to 1 from above as m grows, or $r < 0$ and it rises to 1 while the negative factor r reverses the order; either way r_m decreases strictly to r, hence exceeds it. In (iii), Theorem 7.31 for the exponential function at r/m, multiplied by m, writes $r_m - r - r^2/(2m)$ as $m/2$ times the integral of $(r/m - s)^2 e^s$ from 0 to r/m; for $r > 0$ and $0 < s < r/m$ the factor e^s lies strictly between 1 and $e^{r/m}$, and $(r/m - s)^2$ integrates to $(r/m)^3/3$, which gives the first bracket. The expansion of $\ln(1 + x)$ at $x = r_m/m$, allowed because $r_m > -m$, gives the second bracket in the same way. For either sign the bound (7.4) applies, with third derivatives at most $e^{|r|/m}$ for the exponential function and, when $|r_m| \le m/2$, at most $2 \times 2^3 = 16$ for the logarithm. The complete proof is in Appendix B.1. □

Two comments on the proof. The leading correction $r^2/(2m)$ is the first correction $a^2/(2m)$ of Proposition 7.39 seen from the other side: there the quoted rate a is held fixed and the growth $(1 + a/m)^m$ falls short of e^a by the factor $1 - a^2/(2m)$, up to terms of order m^{-2}; here the growth is held fixed and the quote compounded m times a year must exceed the continuous rate by $r^2/(2m)$, up to the cubic terms bracketed in part (iii). The integral form of the remainder is used, rather than the Lagrange form of Corollary 7.32, because the point which that corollary supplies may be an endpoint of the interval, and an endpoint would give only non-strict brackets.

Example 16.3 (Five conversions)**.** All rates are per year.

- *Quarterly to continuous.* A rate of 6% compounded quarterly ($m = 4$, $r_m = 0.06$) is $r = 4\ln(1.015) = 5.955445\%$ continuous, by (16.1). The shortcut $0.06 - 0.06^2/8 = 5.955000\%$ understates it by 4.45×10^{-6}, strictly inside the bracket of Proposition 16.2(iii), whose ends are $0.06^3/(48 \times 1.015^3) = 4.30 \times 10^{-6}$ and $0.06^3/48 = 4.50 \times 10^{-6}$.
- *Continuous to annual.* A rate of 5% continuous is $e^{0.05} - 1 = 5.12711\%$ with annual compounding ($m = 1$). The shortcut $0.05 + 0.05^2/2 = 5.12500\%$ misses by 2.110×10^{-5}, which part (iii) brackets strictly between $0.05^3/6 = 2.083 \times 10^{-5}$ and $e^{0.05} \times 0.05^3/6 = 2.190 \times 10^{-5}$. Adding the cubic term $r^3/6$ to the shortcut gives 5.12708%.

- *A discount factor.* A two-year zero rate of 4.5% gives the discount factor $e^{-0.09} = 0.91393$, so 1,000 due in two years is worth 913.93 today.
- *A funding rate.* The perpetual-funding anchor of Example 15.5, 0.01% per 8 hours, is a rate of 10.95% compounded $m = 1095$ times a year. Its continuous equivalent is $1095 \ln(1.0001) = 10.949\%$ and its annual-compounding equivalent is $1.0001^{1095} - 1 = 11.57\%$ (Proposition 15.4(ii)). The ordering continuous $<$ 1095 times $<$ annual is Proposition 16.2(ii), applied with $m = 1095$ and $m = 1$.
- *A negative rate.* A rate of -3% continuous is $e^{-0.03} - 1 = -2.9554\%$ with annual compounding and $12(e^{-0.03/12} - 1) = -2.9963\%$ compounded monthly: both quotes lie above -3% and fall towards it as m grows, as Proposition 16.2(ii) asserts.

(Problem P16.1 works this computation at different numbers.)

Remark 16.4 (Scope: what the same discount factors also price). Bond prices, forward rates, forward-rate agreements, interest-rate futures, duration and swaps are all priced from the discount factors $e^{-r(T)T}$. None of them is treated in these notes.

Intuition. The gap $r_m - r$ of about $r^2/(2m)$ is interest on interest within the year (compare Example 7.40). At 5% a continuously compounded account credits interest all year long, and the interest credited early itself earns interest for the rest of the year, about $0.05 \times 0.05/2 = 0.125\%$ of the principal in total; an account that compounds once a year must quote that much more, 5.125%, to keep up. With m periods a year the same effect operates inside each period only, and it is m times smaller. Part (iii) of Proposition 16.2 turns "about" into a bracket with explicit ends.

Interview trap. Quotes with different compounding frequencies cannot be compared by their headlines. A rate of 6.00% compounded monthly is $1.005^{12} - 1 = 6.168\%$ with annual compounding, so a loan at 6.00% monthly costs more than a loan at 6.15% annual despite the lower headline. Convert both quotes to one convention before comparing: continuously compounded, the two loans cost $12 \ln(1.005) = 5.985\%$ and $\ln(1.0615) = 5.968\%$.

Interview trap. Conventions must not be mixed inside one formula: the factor e^{rT} needs the continuous rate. Using the annually compounded 5.127% in e^{rT} in place of the continuous 5% overstates a one-year forward price on an asset worth 100 by about $r^2T/2$ of the price, here 0.13, which is more than many quoted arbitrage margins. A nonzero funding rate annualised by simple multiplication is likewise neither the continuous nor the annually compounded rate. The simple figure is the rate compounded 1095 times a year, so it exceeds the continuous rate, by Proposition 16.2(ii) with $m = 1095$, and it falls short of the annually compounded rate, by Proposition 15.4(iii). Annualising one printed rate, moreover, extrapolates a snapshot (Remark 15.6).

16.2 The No-Arbitrage Toolkit and Forward Contracts

The first derivatives question at a trading firm is usually "what is a forward, and how do you price it?", and every price in this Part comes from one principle: no strategy may cost nothing, never lose, and gain with positive probability. This subsection defines the forward contract, states the principle precisely and proves the dominance lemma that every later price, bound and parity relation of this chapter and of §17 cites and never proves again.

Setup. *Dates and states.* The dates are 0 (today) and $T > 0$ (the maturity), with intermediate dates $t \in [0, T]$. The state of the world at T is random, with real-world probability P. The probability P matters only through which events have probability zero and which have positive probability: the proofs use only the phrases "with probability one" and "with positive probability". An event or a quantity is *known at* t if it is determined by what has been observed up to date t. The underlying of a contract is an asset whose price at time t is S_t.

Market. The market is *frictionless*: there are no transaction costs, taxes, bid–offer spreads or costs of margin, and positions are divisible. There is a *cash account* in which 1 deposited at time 0 is worth e^{rt} at time t, open for lending and for borrowing at the same continuously compounded rate r (Definition 16.1), so that an amount x deposited at t is worth $xe^{r(T-t)}$ at T; a negative deposit is a loan. *Short selling* is allowed for every asset: the short seller borrows a unit, sells it, later buys a unit back and returns it, and pays the lender any income the unit produces in the meantime.

Strategies. A *strategy* is set up at time 0 and holds positions in assets, contracts and the cash account. Its holdings change at most J times before T, for a number J fixed in advance. Each change happens at a date that is either fixed in advance (a settlement date of §16.6, a rebalancing date of a later chapter) or a random time that is known when it occurs (the exercise time of an American option in §17), and the new holdings depend only on what is known at that date. The reinvested holding of Proposition 16.15(i) in §16.3, whose unit count changes continuously but is a function of time fixed in advance, is held as a single position and uses none of the J changes. A limit of this kind is needed. Without it the classical doubling strategy would be allowed. It trades at the dates $T(1-2^{-j})$, $j=0,1,2,\dots$, and after each losing period it scales up its next position so that one favourable enough period recovers every earlier loss and leaves the gain that the first position aimed at, at the price of losses without bound in between. In the Black–Scholes–Merton price model of §20 such a strategy gains with probability one from nothing. This is a classical example that nothing below uses, and §20 excludes it through its own definition of a strategy (Definition 20.1). A strategy may also hold nothing until a date t and start at t, on an event known at t, provided that the positions it opens at t cost nothing in total (every purchase is financed from the cash account), so that its value at t is zero; this is a *late start*. A strategy is *self-financing* if no cash is added or withdrawn after time 0: every intermediate cash flow (income, storage costs, the settlement of a futures contract) is paid into the cash account or financed from it.

Values. The *value* Π_t of a self-financing strategy at t is what its positions fetch at t; Π_0 is its *price* and Π_T its *payoff*, its value at T in each state. In this frictionless market every position, an existing forward contract included, can be bought and sold at t at its value, and values at t are known at t. A position opened at t and managed in the same way from then on, called a *position held from* t, has a value Π_t at t that need not be zero. It is not a strategy by itself; it enters Lemma 16.8(v) only inside combinations whose value at t is zero, which are strategies with a late start. The symbols Π_t^A and Π_t^B are the values of the two strategies or positions compared in Lemma 16.8, called strategy A and strategy B in words. The letters are names and appear in symbols only as these superscripts, while an italic A standing alone is an event, as in reading (c) of Definition 16.7.

Bookkeeping. Every proof below uses three closure rules, which are assumed here. The sum of two self-financing strategies (both held at once), the negative of one (every position reversed, which is how a strategy is sold short), and a strategy together with a deposit in the cash account or a loan from it are again self-financing strategies. Values add, so that prices add and payoffs add. (The bounds on the numbers of changes of holdings add as well, so a sum of strategies is again a strategy in the sense above.) The same rules hold for positions held from a date t.

The standing assumption that there is no arbitrage is stated immediately after Definition 16.7, once the term is defined. Lemma 16.8 is stated for self-financing strategies, and not only for portfolios that are bought and held, because the proofs of Proposition 16.15 and Theorem 16.28 trade at intermediate dates; it covers positions held from any date t because Proposition 16.17 values a contract in the middle of its life.

Definition 16.5 (Forward contract, delivery price and forward price)**.** A *forward contract* is an agreement made at time 0 to trade one unit of the underlying at the *maturity* T for the *delivery price* K. The *long* side buys the unit and so receives S_T-K at T; the *short* side sells it and receives $K-S_T$. The two payoffs sum to zero in every state. The *forward price* $F=F_0$ for

delivery at T is the delivery price that makes a newly written contract cost nothing (there is only one such price, as shown after Lemma 16.8): a new contract struck at $K = F$ is a position of value zero for both sides, and nothing is paid at time 0 to enter it. The forward price F_t at a later date $t < T$ is defined in the same way for contracts written at t for delivery at T. Once a contract is struck its K stays fixed, while the forward price for the same delivery date keeps moving. The *value* V of an existing contract is its price as a position; it is computed in §16.4.

Example 16.6 (A gold forward at two outcomes)**.** Consider a one-year forward on 100 ounces of gold at the delivery price $K = 2{,}500$ per ounce. If gold trades at $S_T = 2{,}620$ at maturity, the long gains $100 \times (2{,}620 - 2{,}500) = 100 \times 120 = 12{,}000$ and the short loses 12,000. If it trades at $S_T = 2{,}410$, the long loses $100 \times 90 = 9{,}000$ and the short gains 9,000. The two payoffs sum to zero in every state. Figure 26 draws both payoffs per ounce against S_T and marks the two outcomes.

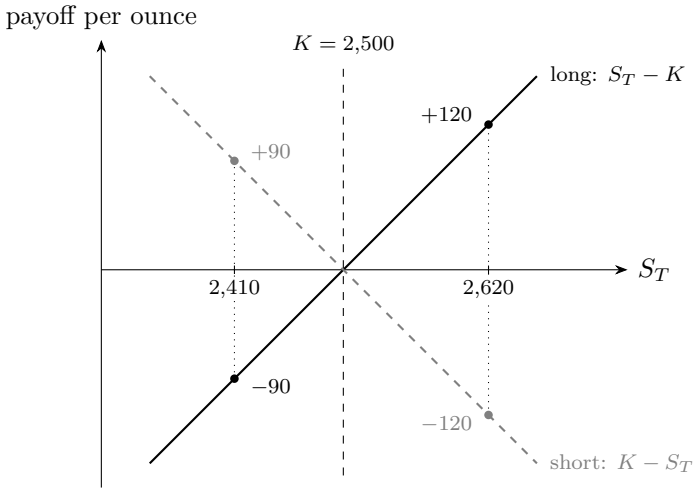


Figure 26: Payoff per ounce at maturity of the long forward (solid black, slope 1) and of the short forward (dashed gray, slope -1) of Example 16.6, against the price S_T of the underlying. Both lines cross zero at the delivery price K, and the dots mark the outcomes $S_T = 2{,}410$ and $S_T = 2{,}620$. The lines are mirror images in the horizontal axis, because the contract is zero-sum, and no premium shifts them, unlike the option payoffs of the next chapter.

Interview trap. The forward *price* and the forward *value* are different objects. The forward price is a delivery price, the K that makes a new contract cost nothing; the value of an existing contract is a profit or loss. At inception the value is 0 and the price is F; afterwards both move, and they differ (§16.4). The sentence "the forward is worth F" mixes the two.

Interview trap. A forward is zero-sum state by state: every gain of the long is the loss of the short. "Both sides hedge, so both win" is a statement about the usefulness of the contract to each side, not about payoffs.

Interview trap. A forward is not an option. Its payoff is linear and can be negative for either side, and no premium is paid, because neither side holds a right: both have an obligation.

Definition 16.7 (Arbitrage)**.** An *arbitrage* is a self-financing strategy whose price is at most zero and whose payoff is nonnegative with probability one and positive with positive probability:

$$\Pi_0 \le 0, \qquad P(\Pi_T \ge 0) = 1, \qquad P(\Pi_T > 0) > 0.$$

With finitely many states, each of positive probability, this says that the payoff is nonnegative in every state and positive in at least one. Three readings are used repeatedly below; each is one line from the definition.

(a) A strategy with a strictly negative price, $\Pi_0 < 0$, and a payoff that is nonnegative with probability one becomes an arbitrage once its proceeds $-\Pi_0$ are deposited in the cash account: the price becomes zero and the payoff gains $-\Pi_0 e^{rT} > 0$, so it is positive with probability one.

(b) A strategy with price zero whose payoff is a positive constant is an arbitrage.

(c) Let a self-financing strategy hold nothing before a date $t \le T$ and start at t on an event A known at t, with value zero at t (the late start of the Setup). If, outside an event of probability zero, its payoff is nonnegative and is positive on A, and if $P(A) > 0$, then the strategy is an arbitrage: its price at time 0 is zero, because it holds nothing then, and its payoff is positive with probability at least $P(A)$. The date $t = T$ is allowed; the positions opened at T are then financed in full from the cash account, and the payoff is what the trades at T leave once that loan is repaid.

Definition 20.1(v) in §20 extends the definition to strategies that trade in continuous time.

Standing assumption. From this point to the end of the Part, the market admits no arbitrage.

Intuition. An arbitrage is a lottery ticket handed out free: it costs nothing, it cannot lose, and it sometimes wins. If one existed, every trader would take it at the largest possible size, and the prices that created it would move until it was gone; the standing assumption says that this has already happened. These notes have met the notion once before: in Proposition 11.4(iv), a riskless combination of two assets with correlation $+1$ or -1 whose mean differs from the risk-free rate is an arbitrage.

Interview trap. An arbitrage is not a trade with positive expected value. A trade with an expected gain and a possible loss is a bet, however good the odds. The definition allows no losing state, and it needs no view of where the market is going: it never mentions an expectation, only events of probability zero and of positive probability.

Lemma 16.8 (Dominance and the law of one price)**.** *Let strategy A and strategy B be self-financing strategies in the market of the Setup, with values Π_t^A and Π_t^B, hence with prices Π_0^A, Π_0^B and payoffs Π_T^A, Π_T^B. (The letters are names, used in symbols only as these superscripts.)*

(i) If $\Pi_T^A \ge \Pi_T^B$ with probability one, then $\Pi_0^A \ge \Pi_0^B$.

(ii) If moreover $\Pi_T^A > \Pi_T^B$ with positive probability, then $\Pi_0^A > \Pi_0^B$.

(iii) If $\Pi_T^A = \Pi_T^B$ with probability one, then $\Pi_0^A = \Pi_0^B$ (replication: the law of one price).

(iv) A sure amount c paid at T, that is, a strategy whose payoff equals the constant c with probability one, has price ce^{-rT}.

(v) (From a date $t \in [0, T)$.) Let A and B now be positions held from t and managed from t on with no cash added or withdrawn, each either a strategy of the Setup or a position opened at t, with values Π_t^A, Π_t^B at t and payoffs Π_T^A, Π_T^B. If $\Pi_T^A \ge \Pi_T^B$ with probability one, then $\Pi_t^A \ge \Pi_t^B$ with probability one. If $\Pi_T^A = \Pi_T^B$ with probability one, then $\Pi_t^A = \Pi_t^B$ with probability one. In particular, a position held from t whose payoff is an amount c known at t (a number, or a quantity fixed by what is known at t, such as $F_t - K$) is worth $ce^{-r(T-t)}$ at t, with probability one.

Sketch of the proof. We argue by contradiction: each violation is turned into an arbitrage, which the standing assumption excludes. For (i), if $\Pi_0^A < \Pi_0^B$, hold strategy A, sell strategy B short and deposit the difference: by the bookkeeping rules this is a self-financing strategy of price zero, with payoff

$$\Pi_T^A - \Pi_T^B + (\Pi_0^B - \Pi_0^A)e^{rT} \ge (\Pi_0^B - \Pi_0^A)e^{rT} > 0 \qquad \text{with probability one.}$$

In (ii) the deposit may be zero; the payoff is then only nonnegative, but it is at least $\Pi_T^A - \Pi_T^B$, hence positive with positive probability. Part (iii) is (i) in both directions, and (iv) is (iii) against a deposit of ce^{-rT}. Part (v) makes the same trade at t, only on the event $\{\Pi_t^A < \Pi_t^B\}$, known at

t: a late start of value zero, so the event has probability zero, by reading (c) of Definition 16.7; in its last clause B is a deposit of $ce^{-r(T-t)}$ made at t. The complete proof is in Appendix B.2. □

Lemma 16.8 is the engine of this chapter and of §17; later chapters cite it and never prove it again. The replication of a variance swap in §22 cites part (iii) for a strategy rebalanced at finitely many dates, which the Setup allows. Proposition 16.17, and note (4) after Theorem 16.11, use the last clause of part (v).

That clause also shows that the forward price of Definition 16.5 is unique. Suppose that new contracts for delivery at T struck at K and at K' both cost nothing at a date $t \in [0, T)$. The long contract at K together with the short contract at K' is a position held from t of value 0 at t, and it pays $(S_T - K) + (K' - S_T) = K' - K$ at T, an amount known at t. By the last clause of part (v), $0 = (K' - K)e^{-r(T-t)}$, so $K' = K$.

Intuition. Part (iii) is the working tool. To price a claim, build from traded instruments a strategy with the same payoff; the claim then costs what the strategy costs, whatever anyone believes about the future. Parts (i) and (ii) do the same job for bounds: a payoff that is never smaller cannot be cheaper.

Example 16.9 (The lemma at work: a mispriced sure payment, and a first bound)**.** *Part (iv).* Let $r = 5\%$. A sure 100 paid in two years must cost $100e^{-0.10} = 90.48$. Suppose that it is quoted at 89.00. Buy it and borrow 89.00 from the cash account. At $T = 2$ the claim pays 100 and the loan is repaid with $89e^{0.10} = 98.36$, which leaves a sure 1.64 at zero cost: an arbitrage, by reading (b) of Definition 16.7. The trade is written as a table of cash flows at time 0 and at time T, a format used throughout this chapter:

position	cash flow at 0	cash flow at $T = 2$
buy the claim at 89.00	−89.00	+100.00
borrow 89.00	+89.00	−98.36
net	0.00	+1.64

Part (i). Let the underlying pay no income and cost nothing to store, so that holding one unit is a self-financing strategy with price S_0 and payoff S_T, and consider a claim that pays $\max(S_T, 100)$ at T. Its payoff is at least S_T, the payoff of one unit of the asset, and at least 100, the payoff of a sure 100. By part (i) the claim costs at least as much as each of the two, and by part (iv) the sure 100 costs $100e^{-rT}$. With $r = 5\%$ and $T = 1$ the claim therefore costs at least $\max(S_0, 100e^{-0.05}) = \max(S_0, 95.12)$: at least 95.12 when $S_0 = 90$, and at least 98 when $S_0 = 98$. No model of the price S_T was used. This is a preview of the model-free option bounds of the next chapter.

16.3 The Forward Price of an Investment Asset

"A stock trades at 100, rates are 5 percent, and the one-year forward is quoted at 108: what do you do?" and "price a forward on a stock paying two dividends" are the two most frequent derivatives questions at junior level. Both are answered by the cash-and-carry argument, which is proved here from Lemma 16.8: first without income, then with known cash income, then with a known yield.

Setup. The market is that of §16.2: frictionless, with a cash account at the flat continuously compounded rate r, with short sales as defined there, and without arbitrage. The underlying has price S_t at time t, and $S = S_0$ is its price today. A forward contract has maturity T and delivery price K, and $F = F_0$ is the forward price for delivery at T (Definition 16.5). Two kinds of income are considered.

Known cash income. The asset pays finitely many known amounts D_i per unit at known dates $t_i \in (0, T)$. The present value of this income is

$$I = \sum_i D_i e^{-rt_i}.$$

Known yield. Let n_t be the number of units held at time t. The asset pays a *known continuous yield* q if a holding of n_t units over a short interval $[t, t+dt]$ pays the income $qS_t n_t\,dt$. A *reinvested holding* spends this income at once on the asset, at the price S_t. The income then buys $qn_t\,dt$ units, so the unit count n_t of a reinvested holding is a differentiable function of t that obeys

$$\frac{dn_t}{dt} = q\,n_t. \tag{16.2}$$

The yield q may be any real number; a negative q is a proportional holding cost, paid by selling units.

There are no storage costs in this subsection; they enter in §16.5. The distinction between investment and consumption assets is defined first, because it decides whether the reverse leg of the argument can be executed.

Definition 16.10 (Investment and consumption assets)**.** An *investment asset* is one whose holders, in large enough number, own it only for its return: equities and equity indices, government bonds, currencies, and gold and silver are of this kind. Its holders sell it, or lend it to a short seller, whenever a position that delivers the same exposure is cheaper. A *consumption asset* is owned chiefly as an input to production or for use: energy, base metals and grains. For an investment asset the reverse cash-and-carry (sell the asset now, deposit the proceeds, buy the asset back forward) can always be executed, by a short sale or by a holder selling. For a consumption asset it cannot, and the convenience yield of §16.5 prices the difference.

In the market of §16.2 short selling is allowed for every asset. Definition 16.10 says for which assets that assumption describes a real market, and the three results below carry the hypothesis "investment asset" because each of their proofs sells short the asset or a strategy that holds it.

Theorem 16.11 (Forward price of an investment asset without income)**.** *Let the underlying be an investment asset (Definition 16.10) that pays no income and costs nothing to store, traded at S at time 0 in the market of §16.2, and let F be the forward price for delivery at T. Then*

$$F = Se^{rT}. \tag{16.3}$$

Proof. We argue by contradiction, twice: each strict inequality between F and Se^{rT} is turned into an arbitrage, written as a table of cash flows at time 0 and of values at time T.

Step 1 ($F > Se^{rT}$ gives an arbitrage: cash-and-carry). At time 0 borrow S from the cash account, buy one unit of the asset, and go short one forward at F, which costs nothing (Definition 16.5).

position	cash flow at 0	value at T
borrow S	$+S$	$-Se^{rT}$
buy one unit	$-S$	$+S_T$
short one forward at F	0	$F - S_T$
net	0	$F - Se^{rT}$

In practice the unit is delivered under the forward for F, and the loan, which has grown to Se^{rT}, is repaid out of F. Between 0 and T the positions produce no cash flow, because the asset pays no income and costs nothing to store, so the strategy is self-financing. Its price is 0 and its payoff is

the constant $F - Se^{rT} > 0$, whatever S_T is. This is an arbitrage, by reading (b) of Definition 16.7. Hence $F \le Se^{rT}$.

Step 2 ($F < Se^{rT}$ gives an arbitrage: reverse cash-and-carry). At time 0 sell one unit short, deposit the proceeds S in the cash account, and go long one forward at F.

position	cash flow at 0	value at T
sell one unit short	$+S$	$-S_T$
deposit S	$-S$	$+Se^{rT}$
long one forward at F	0	$S_T - F$
net	0	$Se^{rT} - F$

In practice the deposit returns Se^{rT}, of which F is paid under the forward; the unit taken in delivery is returned to the lender, and no income is owed to the lender, because the asset pays none. The strategy is self-financing, its price is 0, and its payoff is the constant $Se^{rT} - F > 0$: an arbitrage again. Hence $F \ge Se^{rT}$, and the two steps together give (16.3). □

(Problem P16.2 asks you to reproduce this proof from scratch.)

Four notes on the theorem and its proof.

(1) *A one-line check by replication.* One unit of the asset together with a short forward at F costs $S + 0$ and pays $S_T + (F - S_T) = F$ with probability one, so $S = Fe^{-rT}$ by Lemma 16.8(iv).

(2) *No short sale is needed for an investment asset.* In Step 2 a holder of the asset can take the place of the short seller. The holder sells the unit, deposits S and goes long the forward, and at T owns the unit again together with $Se^{rT} - F > 0$ in cash. Holding the asset outright would then be dominated, and Definition 16.10 says that holders make the switch.

(3) *The rate is the zero rate of the date T.* Only a loan or a deposit from 0 to T was used, so r may be read as the zero rate $r(T)$ when the curve is not flat.

(4) *From a date t.* Keep the hypotheses of the theorem, let $t \in [0, T)$, and let F_t be the forward price at t for delivery at T. One unit of the asset together with a short forward struck at t at F_t is a position held from t. It is worth $S_t + 0$ at t, it produces no cash flow before T, and its payoff $S_T + (F_t - S_T) = F_t$ is known at t. By the last clause of Lemma 16.8(v), $S_t = F_t e^{-r(T-t)}$, that is,

$$F_t = S_t e^{r(T-t)}. \tag{16.4}$$

This form is used in §16.4 and in §16.8.

Example 16.12 (A forward quoted at 108 against a fair price of 105.13)**.** Let a stock that pays no dividend trade at $S = 100$, and let $r = 5\%$ and $T = 1$. A stock is an investment asset and costs nothing to store, so by (16.3) the forward price is $100e^{0.05} = 105.13$, so a quote of 108 is rich: sell the forward and carry the asset.

position	cash flow at 0	cash flow at $T = 1$
borrow 100 and repay the loan at T	+100.00	−105.13
buy one unit	−100.00	
short one forward at 108 and deliver the unit at T	0.00	+108.00
net	0.00	+2.87

The profit of 2.87 is the same in every state. At a quote of 103 the forward is cheap, and the reverse trade (sell the asset short, deposit 100, buy the forward) locks in $105.13 - 103 = 2.13$.

(Problem P16.2 works this computation at different numbers.)

Interview trap. The difference $F - S$ is not a profit. Buying the asset at 100 and selling it forward at the fair 105.13 "earns" 5.13 only if the funding leg is ignored: the 100 has to be borrowed, and the interest on it is $100(e^{0.05} - 1) = 5.13$. The gap $F - S$ is the carry of the position, interest paid minus income received; a book that marks it as P&L counts its own funding bill as a gain.

Interview trap. A quote away from Se^{rT} is an arbitrage only in the frictionless market of §16.2. Let cash be borrowed at the continuously compounded rate r_{bor} and lent at the rate $r_{\text{len}} < r_{\text{bor}}$. Step 1 of the proof of Theorem 16.11 borrows and Step 2 lends, so the two steps prove only

$$Se^{r_{\text{len}}T} \le F \le Se^{r_{\text{bor}}T}.$$

With $S = 100$, $T = 1$, $r_{\text{len}} = 4.5\%$ and $r_{\text{bor}} = 5.5\%$, every quote between 104.60 and 105.65 survives. A bid–offer spread on the asset widens the band further, because Step 1 buys at the offer and Step 2 sells at the bid.

Proposition 16.13 (Forward price with known cash income)**.** *Let the underlying be an investment asset that pays the known amounts D_i at the known dates $t_i \in (0, T)$ of the Setup, with present value $I = \sum_i D_i e^{-rt_i}$, and that has no other income and no storage cost. Then*

$$F = (S - I)e^{rT}.$$

Proof. We replicate a sure payment and apply Lemma 16.8(iv). At time 0 buy one unit of the asset, at the cost S; for each i borrow $D_i e^{-rt_i}$ from the cash account, a loan that has grown to exactly $D_i e^{-rt_i} e^{rt_i} = D_i$ at the date t_i; and go short one forward at F, at no cost. The package costs $S - \sum_i D_i e^{-rt_i} = S - I$. At each date t_i the dividend D_i received on the unit is paid into the cash account, where it extinguishes the i-th loan. No cash is added or withdrawn, so the package is a self-financing strategy. At T it holds the unit and no debt, and the short forward pays $F - S_T$, so its payoff is $S_T + (F - S_T) = F$ with probability one. By Lemma 16.8(iv) its price is Fe^{-rT}. Hence $S - I = Fe^{-rT}$, which is the claim. □

The same result follows from the two arbitrage tables of Theorem 16.11 with the dividends added. If $F > (S - I)e^{rT}$: borrow S, buy the unit, use each dividend to repay part of the loan, and deliver the unit for F. At T the loan stands at $Se^{rT} - \sum_i D_i e^{r(T-t_i)} = (S - I)e^{rT}$, which leaves $F - (S - I)e^{rT} > 0$. If $F < (S - I)e^{rT}$: sell the unit short, deposit S, pay each D_i to the lender of the unit out of the deposit, take delivery under a long forward at F, and return the unit. At T the deposit stands at $(S - I)e^{rT}$, which leaves $(S - I)e^{rT} - F > 0$. The obligation of the short seller to pay the dividends is what makes I appear on both legs.

Example 16.14 (A stock paying two dividends)**.** Let a stock trade at $S = 75$, and let $r = 4.5\%$ and $T = 1$, with dividends of 0.80 in 4 months and in 10 months. The present values of the dividends are $0.80e^{-0.045\times 4/12} = 0.78809$ and $0.80e^{-0.045\times 10/12} = 0.77056$, so $I = 1.55865$ and

$$F = (75 - 1.55865)e^{0.045} = 76.8217,$$

against $75e^{0.045} = 78.45$ if the dividends were ignored. The desk shortcut "spot plus interest minus dividends", $75 + 75(0.045) - 1.60 = 76.775$, misses by $76.8217 - 76.775 = 0.0467$. The miss has two parts: interest on interest on the spot leg, $75(e^{0.045} - 1.045) = +0.0771$, less the interest that the dividends would earn from their payment dates to T, $0.80(e^{0.045\times 8/12} - 1) + 0.80(e^{0.045\times 2/12} - 1) = 0.0304$. The second part is of first order in r, about $0.80 \times 0.045 \times (8/12 + 2/12) = 0.0300$, so the shortcut errs at first order in r whenever dividends are paid before T.

(Problem P16.3 works this computation at different numbers.)

Interview trap. The dividends must be discounted. Subtracting the raw sum 1.60 instead of $I = 1.55865$ misprices the forward of Example 16.14 by $(1.60 - 1.55865)e^{0.045} = 0.0433$, about 0.04. In the reverse arbitrage, moreover, the short seller owes every dividend to the lender of the stock: a short position pays the income that the asset produces.

Proposition 16.15 (Forward price with a known yield, and the reinvested holding)**.** *Let the underlying pay a known continuous yield q, any real number, in the sense of the Setup: the unit count n_t of a reinvested holding is differentiable and obeys (16.2).*

(i) (The reinvested holding, or prepaid forward.) The strategy that buys e^{-qT} units at time 0 and reinvests all income holds $n_t = e^{-q(T-t)}$ units at every $t \in [0,T]$, whatever path the price takes. It is self-financing, its value at t is $S_t e^{-q(T-t)}$, and its payoff is S_T.

(ii) If the asset is an investment asset, then $F = Se^{(r-q)T}$.

Proof. We solve the unit-count equation for (i) and replicate a sure payment for (ii).

Part (i). Let n_t be differentiable on $[0,T]$ (with one-sided derivatives at the two ends) with $dn_t/dt = qn_t$. A differentiable function is continuous, so n_t is continuous in t, and so is its derivative qn_t. By the product rule,

$$\frac{d}{dt}(n_t e^{-qt}) = \Big(\frac{dn_t}{dt} - q\,n_t\Big)e^{-qt} = 0 \qquad (0 \le t \le T),$$

and the zero function is continuous. Let $0 < s < t < T$. The case of order zero of Theorem 7.31, which is the fundamental theorem of calculus, applied to $n_t e^{-qt}$ on the open interval $(0,T)$, gives $n_t e^{-qt} - n_s e^{-qs}$ as the integral from s to t of this derivative, which is 0. So $n_t e^{-qt}$ takes a single value on $(0,T)$. It is continuous on $[0,T]$, as a product of continuous functions, so letting t decrease to 0 and increase to T shows that it takes the same value at 0 and at T, and that value is $n_0 e^0 = n_0$. Hence $n_t e^{-qt} = n_0$ for every $t \in [0,T]$, that is, $n_t = n_0 e^{qt}$. With $n_0 = e^{-qT}$ this is $n_t = e^{-q(T-t)}$, and $n_T = 1$. The price S_t cancelled when the income $qS_t n_t\,dt$ was converted into $qn_t\,dt$ units at the price S_t; this is why no price appears in (16.2) and why the count does not depend on the path. All income is spent on units and no cash enters or leaves, so the strategy is self-financing. Its value at t is the value of the units it holds, $n_t S_t = S_t e^{-q(T-t)}$; in particular its price is Se^{-qT} and its payoff is $n_T S_T = S_T$.

Part (ii). Hold the strategy of part (i) and go short one forward at F. The package is self-financing, it costs $Se^{-qT} + 0$, and it pays $S_T + (F - S_T) = F$ with probability one. By Lemma 16.8(iv), $Se^{-qT} = Fe^{-rT}$, that is, $F = Se^{(r-q)T}$. □

The same formula serves stock indices here, currencies in §16.5 (with q the foreign interest rate r_{for}, Proposition 16.19) and proportional storage costs (with $q = -r_{\text{st}}$, Proposition 16.21). Later chapters cite part (i) for the instrument "one unit with income reinvested": §20 writes its price $\tilde{S}_t = S_t e^{-q(T-t)}$, and §23 uses it for each of two assets.

Example 16.16 (An index forward, and the reinvested holding in numbers)**.** An index stands at 4,000 and pays a dividend yield of $q = 1.5\%$; let $r = 4\%$ and $T = 0.5$. By Proposition 16.15(ii),

$$F = 4{,}000\,e^{(0.04-0.015)\times 0.5} = 4{,}000\,e^{0.0125} = 4{,}050.31.$$

Part (i) in numbers: buying $e^{-0.015\times 0.5} = e^{-0.0075} = 0.992528$ units of the index costs 3,970.11, and reinvesting the dividends brings the holding to exactly one unit at T. With a short forward the package pays F for certain, and $3{,}970.11\,e^{0.04\times 0.5} = 3{,}970.11\,e^{0.02} = 4{,}050.31$ recovers the same forward price. A quote of 4,070 is rich by 19.69 index points: buy the basket of index stocks and sell the forward, the trade known as index arbitrage.

(Problem P16.4 works this computation at different numbers.)

Intuition. Carry is financing minus income: $F/S = e^{(r-q)T}$. The forward price rises with r and falls with q and with I; for the yield formula, $\partial_r F = TF$ and $\partial_q F = -TF$. It depends on neither the volatility nor the expected return of the asset.

The continuous yield is the limit of discrete reinvestment, just as continuous compounding is the limit of discrete compounding. Suppose that income of $(q/m)S_{t_j}$ per unit is paid at each of m

equally spaced dates t_j a year and is reinvested at the price S_{t_j} of that date. Each payment buys q/m units per unit held, whatever the price, so the unit count grows by the factor $(1+q/m)^{mT}$ over $[0,T]$, which tends to e^{qT} as $m \to \infty$ by Definition 16.1(ii) (Proposition 7.39): the yield compounds like a rate. With $q = 1.5\%$ and $T = 0.5$ the factor is 1.0075000 for $m = 2$, 1.0075141 for $m = 4$ and 1.0075280 for $m = 252$, against $e^{0.0075} = 1.0075282$.

Interview trap. The forward price is not a forecast of S_T. The real-world drift of the asset appears nowhere in $F = Se^{(r-q)T}$, so two traders who disagree completely about $\mathbb{E}[S_T]$ agree on F. Proposition 16.31 in §16.6 measures the gap between the two.

Interview trap. "The forward price is always above the spot price" fails whenever $q > r$. An index at 4,000 with $q = 3\%$ and $r = 2\%$ has the one-year forward price $4{,}000\, e^{-0.01} = 3{,}960.20 < 4{,}000$, and a currency whose interest rate exceeds the domestic rate trades at a forward discount (§16.5).

Interview trap. A forward below Se^{rT} on a stock that pays nothing is often the price of borrowing the stock, not an arbitrage. A holder who lends the stock out earns a fee at the rate r_{fee} a year on its value. When the fee accrues continuously and is reinvested in the stock, it is a yield in the sense of the Setup, so by Proposition 16.15(ii) the forward price is $Se^{(r-r_{\text{fee}})T}$: 103.05 with $S = 100$, $r = 5\%$, $r_{\text{fee}} = 2\%$ and $T = 1$, against 105.13 without the fee. A short seller pays the same fee to the lender. The reverse cash-and-carry therefore shorts $e^{-r_{\text{fee}}T} = 0.9802$ units, deposits the proceeds 98.02 and pays the fee by selling further units short, so that it owes exactly one unit at T. Its deposit then stands at $98.02\, e^{0.05} = 103.05$, and at that quote it earns nothing.

16.4 Valuing a Forward Contract During Its Life

A forward is worth zero only on the day it is written; afterwards it is marked to market every day, and "you went long a forward six months ago and spot has moved: what is the contract worth now?" tests whether the forward price and the forward value are kept apart. One comparison with a newly struck contract, through the last clause of Lemma 16.8(v), gives the answer. The value turns out to be linear in the spot price, with a slope equal to the discount factor for the yield, which is the delta of a forward that the option chapters use.

Setup. The market is that of §16.2, with the flat continuously compounded rate r. The current time is $t \in [0,T)$, the maturity is T, and $\tau = T - t$ is the time to maturity. The contract to be valued is a long forward for delivery at T with delivery price K, agreed at or before t; at $t = 0$ an off-market K, different from the forward price, is allowed. The current forward price for delivery at T is F_t (Definition 16.5), and S_t is the current price of the underlying. For an asset with the known cash income of §16.3,

$$I_t = \sum_{i\,:\,t<t_i<T} D_i e^{-r(t_i-t)}$$

is the present value at t of the income still to be paid before T. At a dividend date $t = t_i$ the dividend D_i is not in this sum, and S_{t_i} is the price just after D_i has been paid (the ex-dividend price), so that a buyer at t_i does not receive D_i. For an asset with a known yield, q is that yield. Storage costs enter in §16.5; part (i) below covers them, and part (ii) covers only a proportional storage cost, counted in the yield. The value at t of the contract is V_t. In this Part V_t is always a price at t, never a partial derivative. Values at a date $t > 0$ depend on what has happened up to t, and the equalities between them below hold with probability one, as in Lemma 16.8(v).

Proposition 16.17 (Value of a forward contract)**.**

(i) For any underlying on which a new forward for delivery at T can be struck at t, whether it is an investment asset or a consumption asset, the long forward of the Setup is worth

$$V_t = (F_t - K)e^{-r\tau}$$

at t, and the short forward is worth $-(F_t - K)e^{-r\tau}$.

(ii) Let the underlying be an investment asset with no storage cost, except, in the case of a known yield, a proportional storage cost counted in that yield. Then the forward prices of Theorem 16.11, Proposition 16.13 and Proposition 16.15 hold at the date t with S_t, I_t and τ in the places of S, I and T, and substituting them in (i) gives

$$V_t = S_t - Ke^{-r\tau}, \qquad V_t = S_t - I_t - Ke^{-r\tau}, \qquad V_t = S_t e^{-q\tau} - Ke^{-r\tau}$$

for an asset with no income, with known cash income and with a known yield, respectively. The yield q may be negative, as in the Setup of §16.3: for an investment asset whose only storage cost is proportional, at the rate r_{st} of §16.5, q is its income yield (zero if it pays none) minus r_{st}.

Proof. We replicate a payment known at t and apply the last clause of Lemma 16.8(v).

Part (i). At t hold the existing long forward and write a new short forward for delivery at T at the current forward price F_t, which costs nothing (Definition 16.5). The package is a position held from t, and it is worth $V_t + 0$ at t. It produces no cash flow before T, and at T the two contracts pay

$$(S_T - K) + (F_t - S_T) = F_t - K$$

in every state, an amount known at t. The existing contract can be bought and sold at t at its value V_t (Setup of §16.2). By the last clause of Lemma 16.8(v), $V_t = (F_t - K)e^{-r\tau}$. The package and its reversal consist of forward contracts only, and the comparison position of that clause is a deposit in the cash account, so the argument never sells a unit of the underlying short. This is why (i) holds for any underlying, consumption commodities included: strategies made of forwards and cash can always be reversed (Setup of §16.5). The short forward is the negative of the long one, and values add, so it is worth $-V_t$.

Part (ii). It remains to compute F_t in the three cases. Each computation repeats at the date t the argument given at the date 0, with the last clause of Lemma 16.8(v) in the place of part (iv).

No income. By (16.4), $F_t = S_t e^{r\tau}$, so $V_t = (S_t e^{r\tau} - K)e^{-r\tau} = S_t - Ke^{-r\tau}$.

Known cash income. At t buy one unit, borrow $D_i e^{-r(t_i - t)}$ for each date $t_i \in (t, T)$, and go short one forward at F_t. This position held from t is worth $S_t - I_t$ at t. Each dividend extinguishes its loan, as in the proof of Proposition 16.13, so no cash is added or withdrawn, and the payoff is $S_T + (F_t - S_T) = F_t$, an amount known at t. Hence $S_t - I_t = F_t e^{-r\tau}$, that is, $F_t = (S_t - I_t)e^{r\tau}$, and $V_t = S_t - I_t - Ke^{-r\tau}$.

Known yield. At t buy $e^{-q\tau}$ units, reinvest all income from then on, and go short one forward at F_t. The unit count solves (16.2) on $[t, T]$ and equals $e^{-q\tau}$ at t. By the computation in the proof of Proposition 16.15(i), the product of the count at time s and e^{-qs} is constant on $[t, T]$, so the count is $e^{-q\tau}e^{q(s-t)} = e^{-q(T-s)}$ at each $s \in [t, T]$, and it is 1 at T. The position is therefore worth $S_t e^{-q\tau}$ at t and pays $S_T + (F_t - S_T) = F_t$. Hence $S_t e^{-q\tau} = F_t e^{-r\tau}$, that is, $F_t = S_t e^{(r-q)\tau}$, and $V_t = S_t e^{-q\tau} - Ke^{-r\tau}$. The computation uses the yield only through (16.2), so it holds for q of either sign. An investment asset whose only storage cost is proportional is such an asset: a holding of n_s units receives its income at the yield $q + r_{\mathrm{st}}$ and pays the storage $r_{\mathrm{st}} S_s n_s \, ds$ by selling $r_{\mathrm{st}} n_s \, ds$ units (Setup of §16.5), so the count of a reinvested holding grows at the net rate q and obeys (16.2); a short position in that holding is the reversal of rule (2) of the same Setup, which is available because the asset is an investment asset. □

At $t = 0$ with an off-market delivery price K, part (i) reads $V_0 = (F - K)e^{-rT}$. This is the form that §17 uses for put–call parity.

Example 16.18 (Marking a forward nine months on)**.** A two-year long forward on a stock that pays no dividend is written at $K = 110.25$ when $S = 100$ and the flat rate is 5% a year with annual compounding, that is, $r = \ln 1.05 = 4.879\%$ continuous (Proposition 16.2(i)). The contract is at market, because $100e^{2r} = 100 \times 1.05^2 = 110.25$ by (16.3), and it costs nothing.

Nine months later the stock trades at $S_t = 107$ and the flat rate is unchanged, so $\tau = 1.25$ and $e^{r\tau} = 1.05^{1.25} = 1.06289$. The forward price is $F_t = 107 \times 1.05^{1.25} = 113.73$, and by Proposition 16.17(i)

$$V_t = \frac{113.73 - 110.25}{1.05^{1.25}} = 3.27,$$

which equals $107 - 110.25/1.05^{1.25} = 107 - 103.73$, the form of part (ii) for an investment asset with no income. Had the stock fallen to 96 instead, the contract would be worth $96 - 103.73 = -7.73$.

(Problem P16.3 revalues a short forward on a dividend-paying stock.)

Interview trap. "It cost nothing, so it cannot lose" is false. A contract that cost nothing has a negative value as soon as $F_t < K$: the long forward of Example 16.18 is worth -7.73 when the stock trades at 96.

Interview trap. The difference $F_t - K$ is paid at T, so it must be discounted: the mark in Example 16.18 is 3.27, not $113.73 - 110.25 = 3.48$. The same move would already have been credited in full, as variation margin, to a futures position (§16.6): 3.48, against a forward mark of $3.48\, e^{-r\tau} = 3.27$. The two marks of one exposure differ by the factor $e^{-r\tau}$.

Trading application. The value $V_t = S_t e^{-q\tau} - K e^{-r\tau}$ is linear in S_t with slope $e^{-q\tau}$. A long forward is therefore replicated by the reinvested holding of Proposition 16.15(i), whose unit count, $e^{-q(T-s)}$ at each time $s \in [t, T]$, is fixed in advance and does not depend on the path of S, financed in part by a loan of $K e^{-r\tau}$: at T the holding is one unit and the loan has grown to K. The opposite position hedges the forward. It never trades in response to a move of S, unlike the hedge of an option, and for $q = 0$ it is a single purchase held to T.

16.5 Currencies, Commodities and the Cost of Carry

FX desks ask covered interest parity on the spot ("EURUSD is 1.10, dollar rates are 4 percent and euro rates 2.5 percent: what is the six-month forward?"), and commodity desks ask why the gold forward is pinned by carry while the wheat forward is not. Both answers follow from Proposition 16.15 and Proposition 16.13 once a currency is read as an asset that pays a yield and storage as negative income. For a consumption commodity the reverse arbitrage breaks, which leaves an upper bound; the gap below that bound is the convenience yield.

Setup. The market is that of §16.2, with the flat continuously compounded rate r, except for the short-selling assumption, which this subsection does not make for commodities (last paragraph of this Setup). The forward price for delivery at T is F (Definition 16.5).

Currencies. Prices are quoted in the domestic currency, and S is the domestic price of one unit of the foreign currency (domestic per foreign); a quote given the other way round is inverted first. The domestic and foreign continuously compounded rates are r and r_{for}. Besides the domestic cash account there is a foreign cash account, open for lending and for borrowing, in which one unit of foreign currency deposited at time 0 is worth $e^{r_{\text{for}} t}$ units of foreign currency at time t. The foreign rate is written r_{for}, and not r_f as in much of the literature, because r_f is the risk-free rate of §11.4.

Storage. A commodity is *storable* if a unit bought today can be kept until T at a known cost. Two descriptions of that cost are used. With *discrete storage*, keeping one unit from 0 to T costs known amounts $U_j \geq 0$, due at known dates $t_j \in [0, T]$ (a payment at $t_j = 0$ is made in advance, a payment at $t_j = T$ in arrears), with present value

$$U = \sum_j U_j e^{-r t_j}.$$

With *proportional storage* at the rate $r_{\text{st}} \geq 0$, a holding of n_t units over a short interval $[t, t+dt]$ costs $r_{\text{st}} S_t n_t \, dt$, with n_t the unit count of §16.3. This is the known yield of §16.3 with $q = -r_{\text{st}}$: a proportional holding cost, paid by selling units. The storage rate is the literature's u, renamed because u is the up-factor of the trees of §18; it is not s, for the reason given in the first Setup of the chapter. The commodity pays no income.

Carry. The cost of carry is b and the convenience yield is y; both are defined in Definition 16.23. The cost of carry is the literature's c, renamed because c is a generic constant in this Part and would sit beside the call price C; it is unrelated to the Kelly odds b of §3.1.

Short sales, and how the lemma is used. For a commodity the short-selling assumption of §16.2 is *not* made: Definition 16.10 decides whether the reverse leg can be executed. Three rules replace it; Definition 16.7 is unchanged.

(1) Every strategy of this subsection can be held.

(2) A strategy that holds a commodity can be *reversed* (sold short) only if the commodity is an investment commodity. The reversal is then the short sale of §16.2 with storage costs as negative income: the lender of the unit, who no longer stores it, pays the short seller each storage cost as it falls due, and a holder who sells a unit and buys it back forward plays both roles. Strategies made only of forwards and of positions in the two cash accounts can always be reversed.

(3) The proof of Lemma 16.8(i) holds strategy A and reverses strategy B, so its conclusion $\Pi_0^A \geq \Pi_0^B$ needs only that A can be held and that B can be reversed. Parts (iii) and (iv), which apply part (i) in both directions, need both strategies to be reversible.

Proposition 16.19 (Covered interest parity)**.** *Let S be the spot exchange rate (domestic per foreign), and let riskless deposits earn the continuously compounded rate r at home and r_{for} abroad, as in the Setup. Then the forward exchange rate for delivery at T is*

$$F = Se^{(r - r_{\text{for}})T}.$$

Proof. We give two routes: the foreign currency is first read as an asset with a known yield, and then a sure payment is replicated directly.

First route (a known yield). A holding of foreign currency is kept in the foreign cash account. By the Setup, n_0 units deposited at time 0 have grown to $n_t = n_0 e^{r_{\text{for}} t}$ units at time t, whatever the exchange rate does in between. This unit count is differentiable and obeys $dn_t/dt = r_{\text{for}} n_t$, which is (16.2) with $q = r_{\text{for}}$. The foreign currency is therefore an asset with the known yield $q = r_{\text{for}}$ in the sense of §16.3, whose income, the interest, is paid in units of the asset itself. It is an investment asset (Definition 16.10), and a foreign deposit is reversed by borrowing abroad. Proposition 16.15(ii) gives $F = Se^{(r - r_{\text{for}})T}$.

Second route (a sure payment). At time 0 convert one unit of domestic currency into $1/S$ units of foreign currency, deposit them abroad, and sell forward at F the $e^{r_{\text{for}}T}/S$ units of foreign currency that the deposit will hold at T; the forward sale costs nothing (Definition 16.5). No cash is added or withdrawn before T, so the strategy is self-financing, and its price is 1. At T the deposit holds $e^{r_{\text{for}}T}/S$ units, worth $S_T e^{r_{\text{for}}T}/S$ in domestic currency, and the short forwards pay $(e^{r_{\text{for}}T}/S)(F - S_T)$. The payoff in domestic currency is therefore

$$\frac{e^{r_{\text{for}}T}}{S} S_T + \frac{e^{r_{\text{for}}T}}{S} (F - S_T) = \frac{F e^{r_{\text{for}}T}}{S}$$

with probability one, a sure amount. The strategy consists of a foreign deposit and of forwards, so it can be held and reversed, and Lemma 16.8(iv) gives its price as $Fe^{r_{\text{for}}T} e^{-rT}/S$. Setting this equal to 1 gives $F = Se^{(r - r_{\text{for}})T}$. In words: a unit that is converted, deposited abroad and sold forward pays exactly what the same unit pays when it is deposited at home, e^{rT}. □

Example 16.20 (A euro forward quoted at 1.12)**.** Let the spot rate be $S = 1.10$ dollars per euro, with $r = 4\%$ in dollars, $r_{\text{for}} = 2.5\%$ in euros and $T = 0.5$. By Proposition 16.19 the forward rate is

$$F = 1.10\, e^{(0.04-0.025)\times 0.5} = 1.10\, e^{0.0075} = 1.10828,$$

so a quote of 1.12 is rich: sell euros forward and carry them. Borrow 1,000,000 dollars, convert them into euros, deposit the euros, and sell forward at 1.12 the euros that the deposit will hold at T.

	cash flow at 0		cash flow at $T = 0.5$	
position	USD	EUR	USD	EUR
borrow USD at 4%	+1,000,000.00		−1,020,201.34	
convert at 1.10	−1,000,000.00	+909,090.91		
deposit EUR at 2.5%		−909,090.91		+920,525.87
sell EUR forward at 1.12			+1,030,988.97	−920,525.87
net	0.00	0.00	+10,787.63	0.00

The loan grows to $1{,}000{,}000\, e^{0.02} = 1{,}020{,}201.34$ dollars and the deposit to $909{,}090.91\, e^{0.0125} = 920{,}525.87$ euros, which the forward turns into $920{,}525.87 \times 1.12 = 1{,}030{,}988.97$ dollars. The riskless profit is 10,787.63 dollars at T, that is, $1.12 - 1.10828 = 0.01172$ dollars per euro delivered. A mental check of the forward points: $F - S$ is about $S(r - r_{\text{for}})T = 1.10 \times 0.015 \times 0.5 = 0.00825$, against the exact 0.00828. Read the other way, the quote 1.12 implies the euro rate

$$r_{\text{for}} = r - \frac{1}{T}\ln\frac{F}{S} = 0.04 - \frac{\ln(1.12/1.10)}{0.5} = 0.396\%,$$

the rate at which the quote would be fair, against the 2.5% available on deposit: the gap between the two rates is the arbitrage, seen as a rate.

(Problem P16.4 works a covered-interest-parity arbitrage and an implied foreign rate at different numbers.)

Interview trap. The formula is written for a quote in domestic currency per unit of foreign currency. For a quote the other way round (yen per dollar, for a dollar-based desk) the two rates swap places, and mixing them up reverses the direction of the arbitrage. In Example 16.20, quoted in euros per dollar, the spot rate is $1/1.10 = 0.90909$ and the forward rate is $0.90909\, e^{(0.025-0.04)\times 0.5} = 0.90230 = 1/1.10828$; with the rates in the wrong order the formula returns 0.91593. A rule that survives every quoting convention: the currency with the higher interest rate trades at a forward discount. Here the dollar, at 4% against 2.5%, buys fewer euros forward (0.90230) than spot (0.90909).

Proposition 16.21 (Commodity forwards: equality for investment commodities, a bound for consumption commodities)**.** *Let the underlying be a storable commodity with spot price S and no income, whose storage costs are either paid at known dates, with present value U, or proportional at the rate r_{st}, as in the Setup.*

(i) If it is an investment commodity (Definition 16.10: holders sell spot and buy forward whenever that is cheaper), then

$$F = (S+U)e^{rT} \quad \textit{with discrete storage}, \qquad F = Se^{(r+r_{\text{st}})T} \quad \textit{with proportional storage}.$$

(ii) For every storable commodity, consumption commodities included,

$$F \le (S+U)e^{rT} \quad \textit{with discrete storage}, \qquad F \le Se^{(r+r_{\text{st}})T} \quad \textit{with proportional storage}.$$

Sketch of the proof. The proof is a cash-and-carry argument: (ii) by contradiction, then (i) by replicating a sure payment. With discrete storage, suppose that $F > (S+U)e^{rT}$. Borrow $S+U$, buy one unit and store it, deposit $U_j e^{-rt_j}$ for each storage date t_j, which grows to the cost U_j due then, and go short one forward at F. The price is zero and the payoff is

$$S_T + (F - S_T) - (S+U)e^{rT} = F - (S+U)e^{rT} > 0,$$

a sure gain, hence an arbitrage. No unit is sold short, so (ii) covers consumption commodities. Without its loan the package costs $S+U$ and pays F; an investment commodity lets it be reversed too, so Lemma 16.8(iv) gives $S+U = Fe^{-rT}$, which is (i). With proportional storage the stored unit becomes the reinvested holding of Proposition 16.15(i) with $q = -r_{\text{st}}$: $e^{r_{\text{st}}T}$ units at time 0, storage paid by selling units, one unit at T. The complete proof is in Appendix B.3. □

For a consumption commodity the direction of part (i) that part (ii) does not cover fails. It needs someone to sell or lend physical inventory now and to buy it back forward. A holder whose plant consumes the good declines to part with it at the carry price, so no trade enforces a lower bound on F.

Example 16.22 (A gold forward with storage paid in arrears)**.** Gold trades at 2,400 per ounce. Storage costs 3 per ounce per half-year, paid at the end of each half-year ($t = 0.5$ and $t = 1$), and $r = 4.5\%$ with $T = 1$. The present value of the storage is

$$U = 3e^{-0.0225} + 3e^{-0.045} = 5.801,$$

and by Proposition 16.21(i)

$$F = (2{,}400 + 5.801)\, e^{0.045} = 2{,}516.54,$$

which is also $2{,}400\, e^{0.045} + 3e^{0.0225} + 3 = 2{,}510.47 + 3.07 + 3$: the financed purchase plus each storage bill carried to delivery. This is an equality, not a bound, because gold is an investment asset. A quote of 2,530 would be sold against a stored ounce for a sure 13.46 per ounce. A quote of 2,500 would lead holders to sell the ounce, deposit what they would have spent on storage, and buy the ounce back forward, for a sure 16.54.

Definition 16.23 (Convenience yield, cost of carry, contango and backwardation)**.** Let the underlying have the spot price $S > 0$ and the forward price $F > 0$ for delivery at T.

(i) (Cost of carry.) For an underlying whose income and storage are quoted as rates, a yield q and storage at the proportional rate r_{st}, the *cost of carry* is

$$b = r - q + r_{\text{st}},$$

the cost, as a rate, of holding one unit: financing r, plus storage r_{st}, minus income q. A commodity stored at a proportional rate has $b = r + r_{\text{st}}$ ($q = 0$), a currency $b = r - r_{\text{for}}$ ($q = r_{\text{for}}$, Proposition 16.19), an index with dividend yield q has $b = r - q$, and an asset with neither income nor storage $b = r$.

(ii) (Convenience yield.) The *convenience yield* y of such an underlying is the number defined by

$$F = Se^{(b-y)T}, \tag{16.5}$$

that is, $y = b - \ln(F/S)/T$. For a commodity ($q = 0$) this reads $F = Se^{(r+r_{\text{st}}-y)T}$. For a commodity with no income whose storage is paid at known dates, with present value U, the convenience yield is defined instead by $Fe^{yT} = (S+U)e^{rT}$. Like an implied volatility, y is not observed directly: it is read off the quote F.

(iii) (Two values, proved after the definition.) For every storable commodity, $y \geq 0$; for every investment asset for which y is defined in (ii), $y = 0$.

(iv) (Contango and backwardation.) The market is in *contango* when $F > S$ and in *backwardation* when $F < S$. For an underlying quoted in rates these are, by (16.5), the cases $b > y$ and $y > b$. A whole curve of forward prices is in contango or in backwardation according as F increases or decreases with the delivery date T.

We prove the two values of Definition 16.23(iii) by substituting the forward-price results into (16.5) and into its discrete-storage form. *Storable commodities.* With proportional storage, $b = r + r_{\text{st}}$ and Proposition 16.21(ii) reads $Se^{(b-y)T} \le Se^{bT}$; dividing by $Se^{bT} > 0$ gives $e^{-yT} \le 1$, hence $y \ge 0$. With discrete storage, the definition of y and the same proposition give $Fe^{yT} = (S+U)e^{rT} \ge F$, hence $e^{yT} \ge 1$ and $y \ge 0$, because $F > 0$. *Investment assets.* An investment asset with the yield q and proportional storage at the rate r_{st} has the net yield $q - r_{\text{st}}$, a proportional holding cost being a negative yield as in Proposition 16.21. Proposition 16.15(ii), applied to this net yield, gives $F = Se^{(r-q+r_{\text{st}})T} = Se^{bT}$, and comparison with (16.5) gives $e^{-yT} = 1$, that is, $y = 0$. A currency is the case $q = r_{\text{for}}$, $r_{\text{st}} = 0$ (Proposition 16.19). For an investment commodity with discrete storage, Proposition 16.21(i) gives $Fe^{yT} = (S+U)e^{rT} = F$, hence $y = 0$. An investment asset with discrete cash income, such as a stock that pays known dividends, has no convenience yield in the sense of (ii), and (iii) says nothing about it.

Sections 18 and 20 also treat a futures price as an underlying with $b = 0$; they prove there that a futures position, which costs nothing to hold, behaves like an asset whose yield is $q = r$. Nothing in this chapter uses that convention.

Example 16.24 (Wheat below full carry)**.** Wheat trades at 6.00 per bushel. Storage costs 0.30 per bushel per year, paid in two instalments of 0.15, at $t = 0$ and at $t = 0.5$; let $r = 4\%$ and $T = 1$. The present value of the storage is $U = 0.15 + 0.15e^{-0.02} = 0.2970$, and the upper bound of Proposition 16.21(ii) is

$$(6.00 + 0.2970)\, e^{0.04} = 6.2970\, e^{0.04} = 6.554.$$

A quoted one-year forward price of 6.30 lies below the bound and implies, by Definition 16.23(ii), the convenience yield

$$y = \frac{1}{T} \ln \frac{(S+U)e^{rT}}{F} = \ln \frac{6.554}{6.30} = 3.95\% \text{ a year.}$$

Since $6.30 > 6.00$ the market is still in contango, although it trades below full carry: a positive convenience yield and contango coexist. No arbitrage is available at 6.30. The cash-and-carry loses $6.554 - 6.30$ per bushel, and the reverse trade needs a holder willing to part with physical wheat.

(Problem P16.5 works storage bounds and convenience yields at different numbers.)

Intuition. The cost-of-carry formula *prices* gold and only *bounds* wheat, for one reason. Gold has a vast stock held purely as an investment, and its holders sell spot and buy forward as soon as the forward is cheap, so both arbitrages operate and the forward price is pinned (Example 16.22). Wheat is held to be milled, so only the cash-and-carry operates, and the forward price can sit anywhere below $(S+U)e^{rT}$; the convenience yield measures the gap, 3.95% a year in Example 16.24. "Storage costs are the difference" is the wrong answer to the interview question: both goods are costly to store, and storage enters both formulas in the same way.

Intuition. The convenience yield is the implied dividend of holding inventory: a physical stock can be drawn down on the day it is needed, and a forward position cannot. No one receives it as a cash flow. It is read off F, as an implied volatility is read off an option price, and it is largest when stocks are thin, because that is when a unit on hand is worth most.

Interview trap. "Backwardation means that the market expects the price to fall" confuses two comparisons. By Definition 16.23, $F < S$ says that $y > b$: inventory is scarce, or the income of the asset exceeds its financing cost. It does not say that $\mathbb{E}[S_T] < S$. Expectations enter only through the risk premium of Proposition 16.31 in §16.6.

16.6 Futures: Daily Settlement, Convergence and the Forward-Futures Relation

"What is the difference between a forward and a future?" and "is the futures price the market's forecast of the spot price?" are the two interview questions of this subsection. Daily settlement is the only contractual difference between the two contracts. The subsection runs a margin account, proves that the futures price converges to the spot price at delivery, and proves that daily settlement leaves the price unchanged when interest rates are deterministic. It then states what changes when rates are random, with a proof promised for §18, ties the futures price to the expected spot price through the CAPM of §11.5, and closes with the crypto application of annualising and locking in the basis of a dated future, which is placed here because the lock-in uses convergence.

Setup. The market is that of §16.2, with one change: the interest rate may differ from day to day.

Days and rates. The interval $[0, T]$ is cut into n days of equal length $\Delta t = T/n$; day i runs from $(i-1)\Delta t$ to $i\Delta t$, and the count n is local to this subsection (the unit count n_t of a reinvested holding, §16.3, is a different object). The continuously compounded rate that applies to day i is r_i, and the rates $r_1, \dots, r_n$ are *deterministic*: known at time 0. *Notice:* with i a day, r_i is the rate of day i, not a rate compounded i times a year; the notation r_m of §16.1 is not used here.

Cash account. The cash account is

$$B_i = \exp\Big(\sum_{k=1}^{i} r_k \, \Delta t\Big) \quad (i = 1, \dots, n), \qquad B_0 = 1,$$

the value at the end of day i of 1 deposited at time 0. Hence 1 deposited at the end of day i is worth B_n/B_i at T, and $B_n = e^{r(T)T}$ (Definition 16.1(iii)). The letter r without a subscript is this zero rate, $e^{rT} = B_n$; when all the daily rates are equal it is their common value, and then $B_i = e^{ri\Delta t}$. The letter B is money in the bank throughout this Part; it is unrelated to the frontier constant B of §11.3.

The toolkit with this cash account. Definition 16.7, Lemma 16.8(i)–(iii) and the two comparison clauses of its part (v) hold verbatim with this cash account, for trades made at the settlement dates, because their proofs use only that a positive deposit grows to a positive amount: here by the factor $B_n/B_i > 0$ from the end of day i to T, the factor e^{rT} of reading (a) becoming B_n. In part (iv) of the lemma, e^{-rT} becomes $1/B_n$; in the last clause of part (v), at the end of day i, $e^{-r(T-t)}$ becomes B_i/B_n.

Futures. The futures price at time t for delivery at T is F_t^{fut} (Definition 16.25). It is written F_t once Theorem 16.28 has shown that it equals the forward price at every settlement date when rates are deterministic. The settlement price at the end of day i is F_i, so that $F_0 = F_0^{\text{fut}}$ and $F_n = F_T^{\text{fut}}$: in this subsection an integer subscript on F is a day, and F_0 is the futures price, which equals the forward price F only once Theorem 16.28 is proved. Positions are divisible, and the margin account earns the rate of the cash account, so margin plays no role in values (Definition 16.25).

Expected return and beta. The real-world expected rate of return of the asset is μ. It is stated as a continuously compounded rate and defined, for $S > 0$ and $0 < \mathbb{E}[S_T] < \infty$, by

$$\mathbb{E}[S_T] = S e^{\mu T}.$$

This is the μ of the Notation table, the drift of §§18–20; it is the growth rate of the expected price, not the expected log-growth rate, which for the lognormal model of §19, with volatility σ, is $g = \mu - \sigma^2/2$. The beta of the asset, β, is that of Definition 11.35, measured against a market portfolio whose simple return over $[0, T]$ is R_M, and $r_f = e^{rT} - 1$ is the riskless simple return over $[0, T]$. The letters R_M and r_f are those of §11.5; this chapter uses them only where results of that section are quoted, here and in §16.8.

Definition 16.25 (Futures contract, daily settlement, margin and basis)**.** A *futures contract* is an exchange-traded, standardised contract to trade one unit of the underlying at the delivery date T, whose *futures price* F_t^{fut} is *settled daily*. At the end of each day the change in the settlement price, times the size of the position, is paid into or out of the holder's margin account: a long position of one contract receives $F_i - F_{i-1}$ at the end of day i, and a short position pays it. The payment is the *variation margin*, and the procedure is called *marking to market*. After each settlement the open position is worth zero and runs on with the new settlement price as its reference, so a position can be opened, enlarged or closed at any settlement at no cost.

A position still open at the delivery date T is settled by delivery: the long receives one unit and pays the final settlement price F_T^{fut}, and the short delivers one unit and receives that price. A single delivery date is assumed; a delivery window is discussed after Proposition 16.27.

Opening a position requires an *initial margin*. When the balance of the margin account falls below the *maintenance margin*, a *margin call* obliges the holder to bring the account back up to the initial margin, not merely to the maintenance level; only the balance above the initial margin is free to be taken out. Margin is collateral that earns interest, not a price: entering a futures position costs nothing.

The *basis* at time t is

$$\text{basis}_t = S_t - F_t^{\text{fut}},$$

written as a word because b is the cost of carry. It is defined here because Figure 27 marks it; §16.7 extends it to the asset that a hedger actually holds.

Of the differences from a forward contract (exchange trading through a clearing house that stands between the two sides, standard terms, and the habit of closing positions before delivery), only daily settlement changes the stream of payments; Theorem 16.28 shows when it leaves the price unchanged.

Example 16.26 (A margin account over five days)**.** A trader goes short 2 bitcoin futures contracts of 5 bitcoin each at 62,000 per bitcoin. The position is 10 bitcoin, so each move of 1 in the futures price is worth 10, and the short side loses when the price rises. The initial margin is $2 \times 100{,}000 = 200{,}000$ and the maintenance margin is $2 \times 80{,}000 = 160{,}000$. The account can therefore lose 40,000 before a call: until the first margin call, and while nothing is deposited or withdrawn, a call is triggered by a settlement price above $62{,}000 + 40{,}000/10 = 66{,}000$. The settlement prices of the next five days are 61,100, 63,900, 66,600, 65,200 and 63,500.

day	settlement price	gain of the day	balance before call	margin call	balance
0	62,000				200,000
1	61,100	+9,000	209,000		209,000
2	63,900	−28,000	181,000		181,000
3	66,600	−27,000	154,000	46,000	200,000
4	65,200	+14,000	214,000		214,000
5	63,500	+17,000	231,000		231,000

On day 3 the balance of 154,000 is below the maintenance margin of 160,000, and the call of 46,000 restores it to the initial 200,000, not merely to the maintenance level. The call moves the trigger: the balance is again 200,000, now at the price 66,600, so the next call needs a settlement above $66{,}600 + 40{,}000/10 = 70{,}600$. In total the trader has deposited $200{,}000 + 46{,}000 = 246{,}000$ against a final balance of 231,000, a net of $-15{,}000 = (62{,}000 - 63{,}500) \times 10$: the price change times the position, as it must be, since margin is collateral and not a cost. Of the final balance, $231{,}000 - 200{,}000 = 31{,}000$ exceeds the initial margin and can be withdrawn. Interest on the balance is left out of the table.

(Problem P16.6 works this computation at different numbers.)

Interview trap. Margin is not a down payment. The 200,000 of Example 16.26 is collateral: it stays the trader's money and earns interest. Entering a futures position costs nothing, which is why a futures price will turn out to have zero drift under the risk-neutral probability of the option chapters (§18 and §20).

Interview trap. Daily settlement can turn a perfectly hedged position into a liquidity problem. A short futures hedge on physical inventory pays variation margin today when the price rallies, while the matching gain on the inventory arrives only when the inventory is sold. The P&L nets to zero; the cash flows arrive on different dates, and the cash for the margin calls has to be found in between.

Proposition 16.27 (Convergence of the futures price to spot at delivery)**.** *For a futures contract with a single delivery date T, in the market of the Setup of this subsection, and equally when the daily rates are random,*

$$F_T^{\text{fut}} = S_T \qquad \textit{with probability one,}$$

so that $\text{basis}_T = 0$.

Proof. We argue by contradiction, with an arbitrage at the delivery date: reading (c) of Definition 16.7 with $t = T$, which holds with the cash account of this subsection (Setup) and with any cash account, because a loan made at T and repaid at T carries no interest. A futures position opened at T costs nothing to open (Definition 16.25) and, by the delivery term of that definition, is an immediate exchange of one unit for the final settlement price F_T^{fut}.

The event $\{F_T^{\text{fut}} > S_T\}$. This event is known at T. On it, borrow S_T from the cash account, buy one unit at the spot price, open a short futures position, deliver the unit for F_T^{fut}, and repay the loan. What is left is $F_T^{\text{fut}} - S_T > 0$.

The event $\{F_T^{\text{fut}} < S_T\}$. This event is also known at T. On it, open a long futures position, borrow F_T^{fut} from the cash account to take delivery, sell the unit at the spot price S_T, and repay the loan. What is left is $S_T - F_T^{\text{fut}} > 0$.

Each of the two is a strategy that holds nothing before T and that opens, on an event known at T, positions financed in full from the cash account, so that its value when it starts is zero. Its payoff is positive on the event and zero off it. If the event had positive probability, the strategy would be an arbitrage, by reading (c) of Definition 16.7 with $t = T$. Both events therefore have probability zero, which is the claim. Neither strategy sells the underlying short, so the result holds for consumption assets as well as for investment assets. □

The proof trades only at the date T, and its loan is repaid at that same date, so it uses nothing about the daily rates, which is why the statement covers random rates, as §18 uses it.

When the short side may choose the delivery date within a window, a rule of thumb applies. It is an argument from carry that is not proved here and is used nowhere below: the short delivers early when $b > y$ and late when $y > b$, so that the futures price tracks the forward price for that date.

Most index and crypto futures are *cash-settled*: no unit changes hands, and the contract defines the final settlement price as the level of the spot index at T. For them $\text{basis}_T = 0$ holds by the terms of the contract rather than by the arbitrage of the proof; what remains is the gap between the settlement index and the price at which a hedger actually trades spot at the fixing.

Figure 27 draws the convergence for a futures price that starts above the spot price and for one that starts below it.

Theorem 16.28 (Futures and forward prices coincide under deterministic rates)**.** *Let the daily rates $r_1, \dots, r_n$ be deterministic, that is, known at time 0. Let a futures contract, settled at the end of each of the n days as in Definition 16.25, and forward contracts on the same underlying, both*

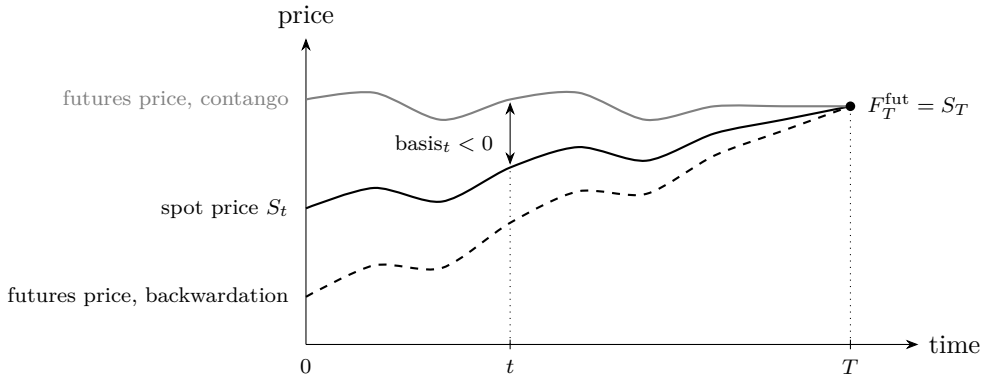


Figure 27: Convergence of the futures price to the spot price at the delivery date T (Proposition 16.27). The solid black curve is one path of the spot price S_t. The solid gray curve is a futures price that trades above spot (contango) and the dashed curve a futures price that trades below spot (backwardation); both meet the spot path at T. The double arrow at the interior time t marks $\text{basis}_t = S_t - F_t^{\text{fut}}$ in the contango case, where it is negative. The paths are illustrative.

for delivery at T, trade in the market of the Setup of this subsection (daily deterministic rates, cash account B_i), and let F be the forward price at time 0. Then

$$F_0^{\text{fut}} = F,$$

and at the end of every day $k < n$ the futures price F_k equals, with probability one, the forward price for delivery at T written at that date.

Sketch of the proof. We compare two self-financing strategies and apply the law of one price, Lemma 16.8(iii). The first, the rolling position, holds B_i futures contracts during day i, a number known in advance because the rates are deterministic, and carries each settlement to T in the cash account, where the gain of day i has become

$$B_i\,(F_i - F_{i-1})\,\frac{B_n}{B_i} = (F_i - F_{i-1})\,B_n.$$

The gains telescope to $(F_n - F_0)B_n$. Futures cost nothing to enter or to resize, so with a deposit of F_0 at time 0 this strategy costs F_0^{fut} and pays $F_nB_n = S_TB_n$, by Proposition 16.27. The second is long B_n forwards struck at F, with B_n known at time 0 for the same reason, and deposits F: it costs F and pays $B_n(S_T - F) + FB_n = S_TB_n$. Equal payoffs give $F_0^{\text{fut}} = F$. The appendix repeats this from the end of day k, with B_i/B_k contracts, B_n/B_k forwards and part (v) of the lemma. The complete proof is in Appendix B.4. □

(Problem P16.7 asks you to reproduce the complete proof from scratch.)

With deterministic rates the futures price and the forward price therefore agree at every settlement date, and F_t stands for both from here on.

Where deterministic rates are used. For a general position of x_i contracts on day i, the gain of day i carried to T is $x_i(F_i - F_{i-1})B_n/B_i$, and the sum telescopes only if the factor x_iB_n/B_i is the same for every i, that is, only if x_i is proportional to B_i/B_n. With random rates the ratio B_i/B_n depends on the rates of the days after day i, which are not known at the end of day $i-1$, when x_i must be chosen; and the number B_n of forwards in the second strategy (Step 5 of the complete proof, Appendix B.4) is not known at time 0. Remark 16.30 says what happens then.

Example 16.29 (The rolling position over three days)**.** Let delivery be three days away ($n = 3$) and let the rate be the same on each day, with $r\,\Delta t = 0.0002$ per day, so that $B_1 = e^{0.0002}$, $B_2 = e^{0.0004}$ and $B_3 = e^{0.0006}$. The futures price starts at $F_0 = 2{,}400$ and settles at 2,410, 2,385

and 2,395. The rolling position of Theorem 16.28 holds $e^{0.0002}$, $e^{0.0004}$ and $e^{0.0006}$ contracts on the three days.

day i	$F_i - F_{i-1}$	contracts B_i	settlement $B_i(F_i - F_{i-1})$	carried to T
1	$+10$	1.00020	$+10.0020$	$+10.0060$
2	-25	1.00040	-25.0100	-25.0150
3	$+10$	1.00060	$+10.0060$	$+10.0060$
sum	-5			-5.0030

Each carried gain is $(F_i - F_{i-1})e^{0.0006}$, and the reinvested gains sum to

$$10e^{0.0006} - 25e^{0.0006} + 10e^{0.0006} = -5e^{0.0006} = -5.0030,$$

exactly $(F_3 - F_0)e^{3r\Delta t}$. This is the payoff of $e^{0.0006}$ forward contracts struck at 2,400 when the asset ends at $S_T = F_3 = 2{,}395$. A constant position of one contract would instead carry $10e^{0.0004} - 25e^{0.0002} + 10 = -5.0010$ to T, a number that depends on the order in which the gains and losses arrive and not only on $F_3 - F_0$.

(Problem P16.7 works this computation at different numbers.)

Remark 16.30 (Futures and forwards when rates are random). When interest rates are random the futures price and the forward price can differ. We prove in §18, Proposition 18.29(iv), for an asset that pays no income on a binomial tree whose one-period rate changes from node to node (the no-arbitrage condition of that tree holding at every node), that

$$F^{\text{fut}} - F = -\frac{\text{Cov}^{\mathbb{Q}}(1/B_n,\, S_T)}{\mathbb{E}^{\mathbb{Q}}[1/B_n]}.$$

Here $F^{\text{fut}} = F_0^{\text{fut}}$ and F are the futures and forward prices today, written as §18 writes them. Also, $1/B_n$ is the discount factor accumulated along the path by the bank account B_n, random on that tree (it is deterministic in the Setup of this subsection), and $\mathbb{Q}$ is the risk-neutral probability of the tree; both are defined in §18. In words, the gap is minus the risk-neutral covariance of the path discount factor with the terminal price, divided by the expected discount factor. The two prices therefore coincide exactly when this covariance is zero, in particular when rates are deterministic, and the futures price is the higher one exactly when the covariance is negative, that is, when high values of S_T come with a low path discount factor, which means high rates along the path (Cox, Ingersoll and Ross, 1981). Example 18.30 there, a two-step tree over one year with $S = 60$, $u = 1.12$, $d = 0.92$, a rate of 4% at the root and of 6% after an up-move or 2% after a down-move, has $F^{\text{fut}} = 62.5142$ against $F = 62.4468$, a gap of 0.0675, about 0.1%, which is why F stands for both prices in this Part.

Intuition. Read the formula of Remark 16.30 from the right. On paths where rates are high the path discount factor $1/B_n$ is low. If the asset tends to end high on those same paths, the covariance of $1/B_n$ with S_T is negative and $F^{\text{fut}} > F$. In cash terms, daily settlement pays the long holder on the paths where cash compounds fastest, which the once-settled forward cannot match; a contract that is free to enter cannot carry such an edge for nothing, so the futures price sits higher, by the amount that §18 computes.

Interview trap. "Futures and forward prices are always equal" is guaranteed only with deterministic rates (Theorem 16.28). With random rates they can differ (Remark 16.30, proved in §18 for an asset that pays no income): the futures price is the higher of the two exactly when the risk-neutral covariance of the path discount factor with S_T is negative, that is, when the asset tends to be high when rates are high. In the one-year example of the remark the gap is 0.0675 on 62.45, about 0.1%.

Proposition 16.31 (Futures price versus expected future spot price)**.** *Let the underlying be an investment asset that pays no income and costs nothing to store, with $S > 0$ and $0 < \mathbb{E}[S_T] < \infty$, and let μ be its real-world expected rate of return, $\mathbb{E}[S_T] = Se^{\mu T}$ (Setup). Then the forward price is*

$$F = \mathbb{E}[S_T]\, e^{-(\mu - r)T},$$

and with deterministic rates the futures price is the same number (Theorem 16.28). Hence $F < \mathbb{E}[S_T]$ exactly when $\mu > r$, $F = \mathbb{E}[S_T]$ exactly when $\mu = r$, and $F > \mathbb{E}[S_T]$ exactly when $\mu < r$.

Suppose moreover that the simple returns of the asset and of the market over $[0, T]$ have finite variances, with $\mathrm{Var}(R_M) > 0$, and that the CAPM holds over the single period $[0, T]$ (Fact 11.40, which is cited and not proved) with a positive market premium, $\mathbb{E}[R_M] - r_f > 0$. Then $\mu > r$ exactly when the beta of the asset (Definition 11.35) is positive, and $\mu = r$ exactly when $\beta = 0$.

Proof. We substitute the forward price of Theorem 16.11 and compare exponents.

The forward price. By Theorem 16.11, with r read as the zero rate of the date T (note (3) after that theorem), $F = Se^{rT}$. Since $S = \mathbb{E}[S_T]e^{-\mu T}$ by the definition of μ,

$$F = Se^{rT} = \mathbb{E}[S_T]\, e^{-\mu T} e^{rT} = \mathbb{E}[S_T]\, e^{-(\mu - r)T}.$$

The map $x \mapsto e^{-xT}$ is strictly decreasing, with the value 1 at $x = 0$, and $\mathbb{E}[S_T] > 0$. Hence the factor $e^{-(\mu - r)T}$ is below 1, equal to 1 or above 1 according as $\mu - r$ is positive, zero or negative, which gives the three cases.

The CAPM clause. Read the CAPM over the period $[0, T]$. The simple return of the asset over the period is $S_T/S - 1$, with expectation $\mathbb{E}[S_T]/S - 1 = e^{\mu T} - 1$, and the riskless simple return is that of the cash account, $r_f = e^{rT} - 1$. The pricing relation of Fact 11.40 reads

$$e^{\mu T} - e^{rT} = (e^{\mu T} - 1) - r_f = \beta\,(\mathbb{E}[R_M] - r_f).$$

The map $x \mapsto e^{xT}$ is strictly increasing, so $\mu > r$ if and only if the left side is positive, and, the premium $\mathbb{E}[R_M] - r_f$ being positive, the right side is positive if and only if $\beta > 0$. In the same way, $\mu = r$ if and only if the left side is zero, if and only if $\beta = 0$. □

Under the CAPM clause, an asset with positive systematic risk therefore has a futures price below its expected spot price. The settlement payments of one long futures contract held to delivery add up to $F_n - F_0 = S_T - F$, by telescoping and by Proposition 16.27, and their expectation is $\mathbb{E}[S_T] - F = S(e^{\mu T} - e^{rT})$, which is positive when $\mu > r$. The long side pays nothing to enter and earns the risk premium of the asset; the short side pays it.

Example 16.32 (One forward price, two expected spot prices)**.** Let $S = 100$, $r = 4\%$ and $T = 1$, so that $F = 100e^{0.04} = 104.08$ whatever μ is. A positive-beta asset with $\mu = 9\%$ has $\mathbb{E}[S_T] = 100e^{0.09} = 109.42$, above F by the factor $e^{0.05} = 1.0513$; a long position expects to gain $109.42 - 104.08 = 5.34$. A negative-beta asset with $\mu = 2\%$, one that hedges the market, has $\mathbb{E}[S_T] = 100e^{0.02} = 102.02$, below F; a long position expects to lose $104.08 - 102.02 = 2.06$, which is the price of the insurance that the asset provides.

Interview trap. "The futures price is the market's unbiased forecast of the spot price" holds only when $\mu = r$, which under the CAPM clause of Proposition 16.31 means zero beta. For a positive-beta asset without income, $F < \mathbb{E}[S_T]$, as in the first case of Example 16.32. A stock index pays the yield q, so the proposition as proved does not cover it, but the same comparison holds with the yield in both places. By the unit count of Proposition 16.15(i), one unit bought at 0 with its dividends reinvested is e^{qT} units at T, so the total return of the index over $[0, T]$ is $(S_T/S)e^{qT} - 1$, with expectation $e^{(\mu+q)T} - 1$. The CAPM then reads $e^{(\mu+q)T} - e^{rT} = \beta(\mathbb{E}[R_M] - r_f)$, so a positive beta, with the positive market premium of the proposition, gives $\mu + q > r$, that is, $F = Se^{(r-q)T} < Se^{\mu T} = \mathbb{E}[S_T]$.

Remark 16.33 (Normal backwardation and contango (Keynes, 1930; Hicks, 1939)). The older account of the gap between F and $\mathbb{E}[S_T]$ is a story about hedging pressure, and it comes without a proof. If hedgers are net short, the speculators who take the long side demand a premium, so $F < \mathbb{E}[S_T]$: this is Keynes's *normal backwardation*. If hedgers are net long, $F > \mathbb{E}[S_T]$: *normal contango*. It is an equilibrium story, not a theorem; Proposition 16.31 is its modern version, in which the premium is paid for systematic risk. In this remark the two words compare F with $\mathbb{E}[S_T]$; in Definition 16.23 they compare F with S.

Interview trap. Backwardation and contango have two meanings: F against S (Definition 16.23, the usage of the market) and F against $\mathbb{E}[S_T]$ (Remark 16.33, the usage of Keynes). The first can be read off a screen and the second cannot. Interviewers use both, so the answer should begin by saying which one is meant.

Example 16.34 (Annualising and locking in a crypto basis)**.** Bitcoin trades at 60,000 spot, and a dated future that expires in three months ($T = 0.25$) trades at 61,200. The future stands 1,200, or 2%, above spot; in the sign convention of Definition 16.25, $\text{basis}_0 = -1{,}200$. Annualised, the premium is

$$\frac{1}{T}\ln\frac{F}{S} = \frac{\ln 1.02}{0.25} = 7.92\% \text{ continuously compounded,}$$

or $0.02/0.25 = 8.00\%$ as a simple rate. With deterministic rates the futures price equals the forward price (Theorem 16.28), so by (16.5) this is $b - y$, the financing rate that the market implies. Buying one bitcoin spot and shorting the future locks in 1,200 per bitcoin at expiry whatever the price then, by convergence at expiry (Proposition 16.27, or the cash-settlement terms of the contract): at $S_T = 50{,}000$ the spot leg loses 10,000 and the short future gains 11,200; at $S_T = 75{,}000$ the spot leg gains 15,000 and the short future loses 13,800. Interest on the variation margin of the short leg is left aside. At a funding rate of 4% the quote lies above the cash-and-carry bound $60{,}000\, e^{0.01} = 60{,}603$ of Proposition 16.21(ii) with $U = 0$, since bitcoin pays no income and costs nothing to store. The implied convenience yield $y = 4\% - 7.92\% = -3.92\%$ is negative, which Definition 16.23(iii) excludes in the frictionless market of this chapter. A desk that funds the purchase at 4% pays $60{,}000(e^{0.01} - 1) = 603$ of interest and keeps 597. In the model this would be an arbitrage; in practice it pays for what the model leaves out: the capital tied up as margin, funding above the risk-free rate, and the venue risk of the Trading application below.

(Problem P16.10 works this computation at different numbers.)

Trading application. The basis of a dated future is locked in by convergence. The perpetual future of Definition 15.3 has no expiry, so its carry floats: a funding rate of 0.01% per 8 hours is 10.95% a year as a simple rate (Example 15.5, Proposition 15.4) only while the rate stays there (Remark 15.6); it can turn negative, and nothing forces the perpetual to converge to spot. Both trades carry venue risk on the leg held at the exchange (Remark 15.2), and the variation margin of the short future is the liquidity trap of this subsection: in the rally to 75,000 of Example 16.34 the short leg must find 13,800 in cash per bitcoin before the spot leg is sold.

16.7 Hedging with Futures: Basis Risk and the Minimum-Variance Hedge Ratio

"You hedge a regional crude grade with benchmark crude futures: how many contracts?" and its follow-up "which do you regress on which?" are standard trap questions at trading firms. This subsection shows that a hedge locks in the futures price plus the final basis, and it proves the minimum-variance hedge ratio. That ratio is the regression slope of Corollary 5.3 in §5, so the residual risk is given by Proposition 10.15 and the cost of regressing the wrong way round by Proposition 5.21; both are cited, not derived again.

Setup. A hedger holds, or will buy or sell, $n_A > 0$ units of an asset with spot price S_t, and hedges with futures contracts of $n_F > 0$ units each, with futures price F_t and delivery date T. Rates are deterministic, with the flat continuously compounded rate r where a number is needed, so F_t is also the forward price at every settlement date (Theorem 16.28). The basis is $\text{basis}_t = S_t - F_t$, as in Definition 16.25, with S_t now the spot price of the asset actually hedged.

Vocabulary. A *short hedge* shorts futures against a holding of the asset or against a planned sale. A *long hedge* buys futures against a planned purchase. A *cross hedge* uses futures on a different but correlated asset, whose own spot price is written S'_t; a prime marks another object of the same kind.

Dates. The hedge is opened at t_1 and closed at t_2, with $t_1 < t_2 \le T$, and S_i, F_i and basis_i are the values at t_i for $i = 1, 2$; in this subsection an integer subscript on S or on F is one of these two dates. The date t_1 is the present: S_1 and F_1 are known numbers, and all variances and covariances below are those of quantities still unknown at t_1.

Hedge ratio and moments. The *hedge ratio* h is the size of the futures position, in units of the futures' underlying, per unit of exposure. The price changes over the life of the hedge are $\Delta S = S_2 - S_1$ and $\Delta F = F_2 - F_1$. They have finite variances, standard deviations $\sigma_S > 0$ and $\sigma_F > 0$, and correlation ρ. They are written with Δ, and not with a differential d, because d is the down-factor of the trees of §18. Here σ_S and σ_F are standard deviations of price changes over the life of the hedge, in price units; they are not volatilities.

Standing simplification. Interest on the variation margin over the life of the hedge is ignored: a long futures position of h units gains $h\,\Delta F$, counted as received at t_2. Remark 16.39 discusses the correction.

Optimal quantities and values. The minimum-variance hedge ratio is h^* and the corresponding number of contracts is N_F^*, a signed number. When $\rho > 0$ it is the number of contracts sold in a short hedge, or bought in a long hedge; when $\rho < 0$ it is negative, and the hedge takes the other side of the futures, $|N_F^*|$ contracts. The values of the position and of one contract are $V_A = S_1 n_A$ and $V_F = F_1 n_F$; the subscripts are labels.

Proposition 16.35 (A hedge locks in the futures price plus the final basis)**.** *Hedge one unit of the asset with one unit of futures, opened at t_1 at the price F_1 and closed at t_2 at the price F_2, under the standing simplification of the Setup. A short hedger, who sells the unit at t_2, realises for it*

$$S_2 + (F_1 - F_2) = F_1 + \text{basis}_2.$$

A long hedger, who buys the unit at t_2, pays for it

$$S_2 - (F_2 - F_1) = F_1 + \text{basis}_2.$$

The only quantity unknown at t_1 is therefore basis_2*, the* basis risk *of the hedge. It is zero when the hedge is closed at the delivery date T of a futures contract on the asset itself, because* $\text{basis}_T = 0$ *(Proposition 16.27).*

Proof. The proof is one line of arithmetic for each hedger, regrouped. The short hedger sells the unit for S_2 and gains $F_1 - F_2$ on the short futures position, so the total received is

$$S_2 + F_1 - F_2 = F_1 + (S_2 - F_2) = F_1 + \text{basis}_2.$$

The long hedger pays S_2 for the unit and gains $F_2 - F_1$ on the long futures position, so the net cost is

$$S_2 - (F_2 - F_1) = F_1 + (S_2 - F_2) = F_1 + \text{basis}_2.$$

In both expressions F_1 is known at t_1 and basis_2 is the only term that is not. If $t_2 = T$ and the futures contract is written on the asset hedged, then $\text{basis}_2 = S_T - F_T = 0$ with probability one, by Proposition 16.27, and both hedgers lock in F_1 exactly.

For a cross hedge, let S_2' be the spot price at t_2 of the futures' own underlying. Then

$$\text{basis}_2 = (S_2 - S_2') + (S_2' - F_2) :$$

the spread between the two assets is added to the ordinary basis of the futures contract, and only the second term vanishes at delivery. □

Example 16.36 (A smelter's locked-in price)**.** An aluminium smelter will sell 2,000 tonnes in September and shorts November futures at 2,450 per tonne. In September the spot price is 2,380 and the November future trades at 2,405, a basis of $2{,}380 - 2{,}405 = -25$. By Proposition 16.35 the smelter realises

$$2{,}380 + (2{,}450 - 2{,}405) = 2{,}380 + 45 = 2{,}425 = 2{,}450 - 25$$

per tonne, 4,850,000 in total, against 4,760,000 without the hedge. Had the final basis been -10, the realised price would have been 2,440. The hedge is closed two months before the November delivery date, so the basis has not yet converged to zero, and its value in September is the one number that the smelter could not know when the hedge was opened.

Interview trap. A hedge locks in F_1 plus the final basis, not today's spot price S_1. The smelter of Example 16.36 secured 2,450 plus a basis still to be revealed, whatever the spot price was on the day the futures were sold.

Theorem 16.37 (Minimum-variance hedge ratio)**.** *Under the Setup, let ΔS and ΔF have finite variances, standard deviations $\sigma_S > 0$ and $\sigma_F > 0$, and correlation ρ, and let the hedged change per unit of exposure be $\Delta S - h\,\Delta F$.*

(i) Over all real h, the variance $\operatorname{Var}(\Delta S - h\,\Delta F)$ has the unique minimiser

$$h^* = \frac{\operatorname{Cov}(\Delta S, \Delta F)}{\operatorname{Var}(\Delta F)} = \rho\,\frac{\sigma_S}{\sigma_F}. \tag{16.6}$$

(ii) The minimum is $(1-\rho^2)\sigma_S^2$, so the hedge effectiveness*, the variance removed by the hedge divided by the unhedged variance, is ρ^2.*

(iii) Hedging n_A units with contracts of n_F units each takes $N_F^* = h^* n_A / n_F$ contracts.

Proof. We complete the square. By Proposition 10.10, applied to the two variables ΔS and $-h\,\Delta F$,

$$\operatorname{Var}(\Delta S - h\,\Delta F) = \operatorname{Var}(\Delta S) + \operatorname{Var}(-h\,\Delta F) + 2\operatorname{Cov}(\Delta S, -h\,\Delta F).$$

By bilinearity of covariance, $\operatorname{Var}(-h\,\Delta F) = h^2 \operatorname{Var}(\Delta F) = h^2\sigma_F^2$ and $\operatorname{Cov}(\Delta S, -h\,\Delta F) = -h\operatorname{Cov}(\Delta S, \Delta F) = -h\rho\,\sigma_S\sigma_F$, the last step being the definition of the correlation. Hence

$$\operatorname{Var}(\Delta S - h\,\Delta F) = \sigma_S^2 - 2h\rho\,\sigma_S\sigma_F + h^2\sigma_F^2 = \sigma_F^2\Big(h - \rho\,\frac{\sigma_S}{\sigma_F}\Big)^2 + (1-\rho^2)\,\sigma_S^2. \tag{16.7}$$

The second equality is checked by expanding the square: $\sigma_F^2(h^2 - 2h\rho\,\sigma_S/\sigma_F + \rho^2\sigma_S^2/\sigma_F^2) = h^2\sigma_F^2 - 2h\rho\,\sigma_S\sigma_F + \rho^2\sigma_S^2$, and adding $(1-\rho^2)\sigma_S^2$ restores the term σ_S^2.

Parts (i) and (ii). In (16.7) the first term is nonnegative, and since $\sigma_F > 0$ it vanishes only at $h = \rho\,\sigma_S/\sigma_F$, which equals $\operatorname{Cov}(\Delta S, \Delta F)/\operatorname{Var}(\Delta F)$ because $\operatorname{Cov}(\Delta S, \Delta F) = \rho\,\sigma_S\sigma_F$ and $\operatorname{Var}(\Delta F) = \sigma_F^2$. This h^* is therefore the unique minimiser, and the minimum is the second term, $(1-\rho^2)\sigma_S^2$. The unhedged variance is the value at $h = 0$, namely σ_S^2, so the effectiveness is

$$\frac{\sigma_S^2 - (1-\rho^2)\,\sigma_S^2}{\sigma_S^2} = \rho^2.$$

(Equivalently, the derivative of the quadratic in h, $2h\sigma_F^2 - 2\rho\,\sigma_S\sigma_F$, vanishes at h^*, and its second derivative $2\sigma_F^2$ is positive; the completed square gives the global minimum and its value in one line.)

Part (iii). A hedge of $h\,n_A/n_F$ contracts is a futures position of $h\,n_A$ units against n_A units of exposure, so its hedge ratio is h and the change in the value of the hedged position is $n_A\,\Delta S - h\,n_A\,\Delta F = n_A(\Delta S - h\,\Delta F)$, with variance $n_A^2\,\mathrm{Var}(\Delta S - h\,\Delta F)$. By part (i) this is smallest exactly when $h = h^*$, that is, with $h^* n_A/n_F$ contracts. □

(Problem P16.8 asks you to reproduce this proof from scratch.)

Three earlier results say the same thing in the language of regression; they are cited, not derived again.

(1) The ratio h^* is the population counterpart of the slope (5.3) of Corollary 5.3, with ΔF as the regressor and ΔS as the response. The ordinary least-squares regression of observed changes ΔS on observed changes ΔF, with an intercept, therefore estimates h^*, and its R^2, the squared sample correlation, estimates the hedge effectiveness.

(2) Part (ii) is Proposition 10.15 with the black-box output taken to be ΔF and the new model taken to be ΔS: the decorrelating coefficient $\mathrm{Cov}(\Delta F, \Delta S)/\,\mathrm{Var}(\Delta F)$ of that proposition is h^*, its residual is the hedged change $\Delta S - h^*\Delta F$, which is uncorrelated with ΔF, and the residual variance is $(1-\rho^2)\,\mathrm{Var}(\Delta S)$.

(3) Regressing ΔF on ΔS and inverting the slope overstates h^* in absolute value by the factor $1/\rho^2$, for $\rho \neq 0$ (Proposition 5.21, with ΔS in the role of X and ΔF in the role of Y: its B is then the slope of the regression of ΔF on ΔS, its D is h^*, and $1/B = h^*/\rho^2$).

Example 16.38 (A cross hedge of 432 contracts)**.** A trading desk holds 600,000 barrels of a regional crude grade that has no futures contract of its own, to be sold in two weeks, and hedges with futures on a benchmark crude, in contracts of 1,000 barrels. Over two weeks the standard deviations of the *price changes* are $\sigma_S = 3.2$ dollars per barrel for the regional grade and $\sigma_F = 4.0$ dollars per barrel for the futures, with correlation $\rho = 0.9$. By (16.6) in Theorem 16.37,

$$h^* = 0.9 \times \frac{3.2}{4.0} = 0.72,$$

the effectiveness is $0.9^2 = 0.81$, and the residual standard deviation is $\sqrt{0.19} \times 3.2 = 1.39$ dollars per barrel against 3.2 unhedged. In units, the holding equals $600{,}000/1{,}000 = 600$ contracts, so the hedge shorts $0.72 \times 600 = 432$ contracts. Regressing ΔF on ΔS and inverting the slope would give $1/(0.9 \times 4.0/3.2) = 1/1.125 = 0.889$, too large by the factor $1/0.81 = 1.235$ (Proposition 5.21, Example 5.22). Figure 28 draws both lines through a sample of twelve two-week periods.

(Problem P16.8 works this computation at different numbers.)

Intuition. The hedger minimises vertical distances in Figure 28: what is left after the hedge is the risk in S that the futures do not explain. This is why the regression runs ΔS on ΔF. The inverted regression minimises horizontal distances, the risk in F that the asset does not explain, which is the intuition of Proposition 5.21; that is a different problem, and its answer is too steep for this one.

Remark 16.39 (Tailing the hedge, and units versus values). Two identities complete the sizing of a hedge. Each is shown here, and nothing later cites this remark for a claim.

(a) Units versus returns. Let $R_S = \Delta S/S_1$ and $R_F = \Delta F/F_1$ be the percentage changes over the life of the hedge. (The subscripts are letters, as in R_A and R_M of §16.8; the indexed R_i of this Part is a log-return.) Since S_1 and F_1 are known at t_1, bilinearity of covariance gives $\mathrm{Cov}(\Delta S, \Delta F) = S_1 F_1\,\mathrm{Cov}(R_S, R_F)$ and $\mathrm{Var}(\Delta F) = F_1^2\,\mathrm{Var}(R_F)$. Hence

$$h^* = \frac{S_1}{F_1}\,h_R^* \qquad \text{with} \qquad h_R^* = \frac{\mathrm{Cov}(R_S, R_F)}{\mathrm{Var}(R_F)},$$

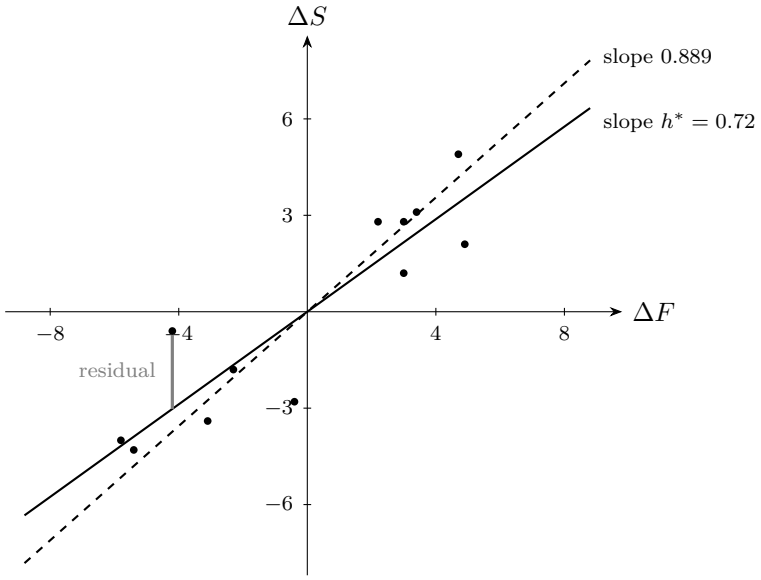


Figure 28: Twelve simulated pairs of two-week price changes $(\Delta F, \Delta S)$, in dollars per barrel. Their sample means are zero, and their sample standard deviations (computed with the divisor 11, one less than the number of points, as for the sample standard deviation of §4) and correlation are 4.0, 3.2 and 0.90, the values of Example 16.38. The solid line has slope $h^* = 0.72$, the slope of the regression of ΔS on ΔF for these points (0.720), which minimises the vertical distances to the line; the vertical residual of one point is marked in gray. The dashed line has slope 0.889: it is the regression of ΔF on ΔS, whose slope is 1.125 for these points, inverted and drawn in the same axes; it minimises the horizontal distances. Both lines pass through the point of means. Sizing the hedge with the dashed line overstates it by the factor $1/\rho^2$.

the hedge ratio computed from returns, and therefore

$$N_F^* = h^* \frac{n_A}{n_F} = h_R^* \frac{S_1 n_A}{F_1 n_F} = h_R^* \frac{V_A}{V_F}$$

exactly. A ratio estimated from price changes multiplies a ratio of units; a ratio estimated from returns multiplies a ratio of values. In Example 16.38, at $S_1 = 78.00$ and $F_1 = 80.00$ dollars per barrel, the two standard deviations are $3.2/78 = 4.103\%$ and $4.0/80 = 5.000\%$ of price, so $h_R^* = 0.9 \times 4.103/5.000 = 0.7385$ and

$$N_F^* = 0.7385 \times \frac{600{,}000 \times 78}{1{,}000 \times 80} = 0.7385 \times 585 = 432.0,$$

the same answer. Mixing the two routes mis-sizes the hedge: $0.7385 \times 600 = 443.1$ is too large by the factor $F_1/S_1 = 1.026$, and $0.72 \times 585 = 421.2$ is too small by the factor $S_1/F_1 = 0.975$.

(b) Tailing. When a futures hedge is held for many days, each day's variation margin earns interest until the horizon, so the hedge ratio for one day is computed afresh each day; this is *tailing the hedge*. The rate r, the carry b and the convenience yield y are those of §16.5, and the cash account B_i, the number n of daily settlements to T and the day length Δt are those of §16.6. Take a hedge of the futures' own underlying (no cross hedge, so that $S'_t = S_t$) and the cost-of-carry model with $y = 0$, in which $F_t = S_t e^{b(T-t)}$ at every date t. This is (16.5) at each date: the forward-price results hold from any date (note (4) after Theorem 16.11, Proposition 16.17(ii)), and the futures price equals the forward price at every settlement date, by Theorem 16.28. Over one day of length Δt, with $\Delta S = S_{t+\Delta t} - S_t$ and $\Delta F = F_{t+\Delta t} - F_t$,

$$\Delta F = e^{b(T-t-\Delta t)}\,\Delta S - F_t\bigl(1 - e^{-b\Delta t}\bigr)$$

exactly. Indeed, since $F_t = e^{b(T-t)}S_t$, the right side is $e^{b(T-t-\Delta t)}S_{t+\Delta t} - e^{b(T-t-\Delta t)}S_t - e^{b(T-t)}S_t + e^{b(T-t-\Delta t)}S_t = e^{b(T-t-\Delta t)}S_{t+\Delta t} - e^{b(T-t)}S_t$, which is $F_{t+\Delta t} - F_t$. The last term of the identity, $F_t(1-e^{-b\Delta t})$, is known at t. Over one day, therefore, $e^{-b(T-t-\Delta t)} = e^{b\Delta t}S_t/F_t$ units of futures per unit of exposure hedge the change exactly: the hedged change $\Delta S - e^{-b(T-t-\Delta t)}\Delta F = S_t(e^{b\Delta t}-1)$ is known at t. Since $b\,\Delta t = 1.6 \times 10^{-4}$ for $b = 4\%$ and $\Delta t = 1/252$, the one-day hedge is about $S_t/F_t = e^{-b(T-t)}$. When $b = r$ (no income and no storage), the rate is constant, and $t = (i-1)\Delta t$ is the end of day $i-1$, the one-day hedge is $e^{-r(T-i\Delta t)} = B_i/B_n$: exactly the position, per forward contract replicated, that the proof of Theorem 16.28 holds during day i.

Interview trap. "Which do you regress on which?" The dependent variable is the thing being hedged: regress ΔS on ΔF. Inverting the regression of ΔF on ΔS overstates the hedge by the factor $1/\rho^2$ (Proposition 5.21): 0.889 instead of 0.72 in Example 16.38.

Interview trap. The slope h^* and the effectiveness ρ^2 are different numbers. A unit slope can leave a large residual: $\rho = 0.8$, $\sigma_S = 5$ and $\sigma_F = 4$ give $h^* = 0.8 \times 5/4 = 1$ and a hedged standard deviation of $5\sqrt{1-0.64} = 3$. A zero residual can come with a slope far from one: $\rho = 1$ and $\sigma_F = 1.25\,\sigma_S$ give a perfect hedge at $h^* = 1/1.25 = 0.8$. The two are both equal to one exactly when the final basis is known in advance. Then $\Delta S - \Delta F = \text{basis}_2 - \text{basis}_1$ is a constant, so $0 = \text{Var}(\Delta S - \Delta F) = (\sigma_S - \sigma_F)^2 + 2(1-\rho)\sigma_S\sigma_F$, which forces $\sigma_S = \sigma_F$, $\rho = 1$ and $h^* = 1$; conversely, $h^* = 1$ and $\rho^2 = 1$ give $\rho = 1$, because $h^* = \rho\,\sigma_S/\sigma_F$ is positive, and then $\sigma_S = \sigma_F$; these make that variance zero, so $\Delta S - \Delta F$ equals its mean with probability one and $\text{basis}_2 = \text{basis}_1 + \mathbb{E}[\Delta S - \Delta F]$ is known at t_1.

Interview trap. Hedging fixes the outcome; it does not improve it. Had the spot price ended at 2,600 with the same final basis of -25, so that the November future stood at 2,625, the smelter of Example 16.36 would still have realised $2{,}600 + (2{,}450 - 2{,}625) = 2{,}425$ per tonne, 175 below the unhedged sale. A hedge is chosen to lower the variance of the outcome, not to raise its expected value. Whether a firm should hedge at all is a question of corporate finance, outside these notes.

Interview trap. Hedge with beta, not with correlation (Remark 5.23). Sizing the hedge of Example 16.38 by $\rho = 0.9$ alone would short $0.9 \times 600 = 540$ contracts instead of 432: the correlation says how good the hedge is, and the slope $\rho\,\sigma_S/\sigma_F$ says how large it must be.

16.8 Stock-Index Futures: Hedging an Equity Portfolio and Changing Its Beta

"Hedge a 27-million portfolio with beta 1.15 using index futures" and "now cut its beta to 0.4" are quick-numeric staples, and the follow-up "what does the hedged book earn?" separates candidates. The answer is Theorem 16.37 in return form, with the CAPM beta of Definition 11.35 as the hedge ratio, and the hedged book earns the risk-free rate plus alpha.

Setup. The rate r is flat and constant, and the hedge runs over a horizon of length Δt, from a settlement date t to the settlement date $t + \Delta t$.

Portfolio and index. A portfolio has the value $V_A > 0$ at t and the simple return R_A over the horizon. The index level is S_t; the index is the underlying of the futures and pays the dividend yield q. (The letter I is not used for the index, because I is the present value of income.) The total return of the index over the horizon, price change plus dividends reinvested, is R_M.

Futures. Index futures with delivery date $T \geq t + \Delta t$ have the price F_t, the percentage change $R_F = (F_{t+\Delta t} - F_t)/F_t$ over the horizon, and the contract value $V_F = F_t \times \text{multiplier} > 0$. The signed number of futures contracts held is N_F: a long position if $N_F > 0$, a short of $-N_F$ contracts if $N_F < 0$. It is written N_F and not N, which the Notation table keeps for the absorbing target. The futures position gains $N_F V_F R_F$ over the horizon; interest on the variation margin is ignored, as in §16.7. How R_F is tied to R_M is the hypothesis (H) of Proposition 16.40, and the exact relation in the market of this chapter is derived after its proof. The minimum-variance number of

contracts is N_F^*, found in Proposition 16.40(i); it is the N_F^* of Theorem 16.37(iii) in return form, a signed number. The hedge sells N_F^* contracts when $\beta > 0$; when $\beta < 0$, N_F^* is negative and the hedge buys $|N_F^*|$ contracts. A target beta chosen by the hedger is β': a prime marks another object of the same kind, and a star would mark an optimum, which a target is not.

The beta decomposition. The returns R_A and R_M have finite variances, and $\operatorname{Var}(R_M) > 0$. The decomposition of Proposition 11.36 is used with the constant $r\,\Delta t$ in the place of the r_f of §11.5:

$$R_A - r\,\Delta t = \alpha + \beta\,(R_M - r\,\Delta t) + \varepsilon, \tag{16.8}$$

in which $\beta = \operatorname{Cov}(R_A, R_M)/\operatorname{Var}(R_M)$ is the beta of Definition 11.35, α is the constant $\mathbb{E}[R_A] - r\,\Delta t - \beta(\mathbb{E}[R_M] - r\,\Delta t)$, and the residual ε, defined by (16.8), satisfies $\mathbb{E}[\varepsilon] = 0$ and $\operatorname{Cov}(\varepsilon, R_M) = 0$ (Proposition 11.36(i)). The proof of that proposition uses only that r_f is a constant, so it holds with any constant in its place; (16.8) is an identity, not an assumption. The letters α, β and ε are those of §11.5, with the index as the reference portfolio: β is the beta of Definition 11.35 against the index, ε a regression residual, and α the alpha of Definition 11.41 computed with $r\,\Delta t$ in the place of r_f. Jensen's alpha proper, computed with the exact riskless return, is written α' and appears after the proof. In this Part the coefficients of an Itô process always carry a subscript or arguments, so no confusion arises. The constant $r\,\Delta t$ is the riskless return over the horizon to first order. The symbol r_f keeps the meaning it has in §11.5, the exact riskless simple return, here $r_f = e^{r\Delta t} - 1$; it appears only after the proof.

Proposition 16.40 (Hedging and re-gearing an equity portfolio with index futures)**.** *Let the portfolio, the index, the futures position and the decomposition (16.8) be those of the Setup, and model the futures return over the horizon by the first-order relation that it equals the excess return of the index,*

$$R_F = R_M - r\,\Delta t. \tag{H}$$

The relation (H) takes the place of the exact carry relation, which the note after the proof derives together with the error of (H). Then:

(i) the variance of the P&L of the hedged position, $V_A R_A + N_F V_F R_F$, is minimised over all real N_F only at $N_F = -\beta V_A/V_F$, that is, by selling

$$N_F^* = \beta\,\frac{V_A}{V_F}$$

contracts (buying $|N_F^*|$ when $\beta < 0$), and the minimum is $V_A^2 \operatorname{Var}(\varepsilon)$;

(ii) holding N_F contracts gives the position, per unit of the capital V_A, the beta $\beta + N_F V_F/V_A$, so moving the beta to a target β' takes

$$N_F = (\beta' - \beta)\,\frac{V_A}{V_F}$$

contracts, a short when this number is negative;

(iii) the fully hedged position, with $N_F = -N_F^*$, returns $r\,\Delta t + \alpha + \varepsilon$ per unit of the capital V_A.

Proof. We substitute the decomposition (16.8) and complete the square. With N_F contracts the P&L over the horizon is $V_A R_A + N_F V_F R_F$, the futures costing nothing to enter. By (16.8), $R_A = r\,\Delta t + \alpha + \beta(R_M - r\,\Delta t) + \varepsilon$, and by (H), $R_F = R_M - r\,\Delta t$. Hence

$$\begin{aligned} V_A R_A + N_F V_F R_F &= V_A(r\,\Delta t + \alpha + \varepsilon) + \beta V_A\,(R_M - r\,\Delta t) + N_F V_F\,(R_M - r\,\Delta t) \\ &= V_A(r\,\Delta t + \alpha + \varepsilon) + (\beta V_A + N_F V_F)(R_M - r\,\Delta t). \end{aligned} \tag{16.9}$$

Part (ii). Divide (16.9) by V_A to obtain the return of the position per unit of capital. By bilinearity of covariance, its covariance with R_M is

$$\operatorname{Cov}(\varepsilon, R_M) + \Big(\beta + \frac{N_F V_F}{V_A}\Big)\operatorname{Var}(R_M) = \Big(\beta + \frac{N_F V_F}{V_A}\Big)\operatorname{Var}(R_M),$$

because $\mathrm{Cov}(\varepsilon, R_M) = 0$ and the remaining terms are constants. Dividing by $\mathrm{Var}(R_M) > 0$, the beta of the position (Definition 11.35) is $\beta + N_F V_F / V_A$. It equals β' if and only if $N_F V_F / V_A = \beta' - \beta$, that is, $N_F = (\beta' - \beta) V_A / V_F$.

Part (i). Constants do not change a variance, so by (16.9) the variance of the P&L is that of $V_A \varepsilon + (\beta V_A + N_F V_F) R_M$. By Proposition 10.10 and bilinearity,

$$\mathrm{Var}\left(V_A R_A + N_F V_F R_F\right) = V_A^2 \,\mathrm{Var}(\varepsilon) + \left(\beta V_A + N_F V_F\right)^2 \mathrm{Var}(R_M) + 2 V_A \left(\beta V_A + N_F V_F\right) \mathrm{Cov}(\varepsilon, R_M),$$

and the last term is zero. The middle term is a square times $\mathrm{Var}(R_M) > 0$: it is nonnegative and vanishes only when $\beta V_A + N_F V_F = 0$, that is, at $N_F = -\beta V_A / V_F$, a sale of $N_F^* = \beta V_A / V_F$ contracts. The minimum is the remaining term, $V_A^2 \,\mathrm{Var}(\varepsilon)$.

Part (iii). At $N_F = -\beta V_A / V_F$ the second term of (16.9) vanishes, and the P&L is $V_A (r\,\Delta t + \alpha + \varepsilon)$, a return of $r\,\Delta t + \alpha + \varepsilon$ on V_A. □

The proposition is Theorem 16.37 in return form. Under (H) the return ratio of Remark 16.39(a) is $\mathrm{Cov}(R_A, R_F) / \mathrm{Var}(R_F) = \mathrm{Cov}(R_A, R_M) / \mathrm{Var}(R_M) = \beta$, and it is converted into contracts by the ratio of values V_A / V_F. The direct proof is given so that part (iii) comes out of the same line.

What the assumption (H) approximates. Let the index pay the continuous yield q, in the market of this chapter. Then $F_t = S_t e^{(r-q)(T-t)}$ at both ends of the horizon: the forward price at a date t is $S_t e^{(r-q)(T-t)}$ (Proposition 16.17(ii), whose proof runs the argument of Proposition 16.15(ii) from the date t with the last clause of Lemma 16.8(v), as in note (4) after Theorem 16.11), and the futures price equals the forward price at every settlement date (Theorem 16.28; the rate is constant, hence deterministic); when the horizon ends at the delivery date, $t + \Delta t = T$, as in Example 16.41, the formula at that end reads $F_T = S_T$, which is Proposition 16.27, the two results just cited covering only dates before T. Dividing the two prices,

$$\frac{F_{t+\Delta t}}{F_t} = \frac{S_{t+\Delta t}}{S_t}\, e^{-(r-q)\Delta t}.$$

One unit of the index bought at t, with dividends reinvested, has grown to $e^{q\Delta t}$ units at $t + \Delta t$, by the unit count of Proposition 16.15(i), so the total return of the index is given by $1 + R_M = (S_{t+\Delta t} / S_t) e^{q\Delta t}$. Combining the two relations,

$$1 + R_F = \frac{F_{t+\Delta t}}{F_t} = (1 + R_M)\, e^{-q\Delta t} e^{-(r-q)\Delta t} = (1 + R_M)\, e^{-r\Delta t}$$

exactly, that is,

$$R_F = R_M - r\,\Delta t + \Big[\left(e^{-r\Delta t} - 1 + r\,\Delta t\right) + R_M \left(e^{-r\Delta t} - 1\right)\Big].$$

The bracket is of second order. By the bound (7.4) for the exponential function at the orders 1 and 0, whose derivatives are at most $e^{|r|\Delta t}$ between 0 and $-r\,\Delta t$, it is at most $e^{|r|\Delta t}((r\,\Delta t)^2/2 + |R_M|\,|r|\,\Delta t)$ in absolute value. The assumption (H) is the first-order version of the exact identity, and in this market it is only an approximation. Subtracting (H) from the exact identity gives $R_M(1 - e^{-r\Delta t}) = e^{-r\Delta t} - 1 + r\,\Delta t$. When $r \neq 0$ the factor $1 - e^{-r\Delta t}$ is not zero, so both relations together would make R_M a constant, against $\mathrm{Var}(R_M) > 0$: here (H) holds exactly only when $r = 0$, which is why Proposition 16.40 takes it as a model rather than deriving it.

The exact hedge. With $r_f = e^{r\Delta t} - 1$ the exact identity reads

$$R_F = (1 + R_M) e^{-r\Delta t} - 1 = e^{-r\Delta t}(1 + R_M - e^{r\Delta t}) = e^{-r\Delta t}(R_M - r_f),$$

an affine function of R_M. Write the decomposition with r_f in the place of $r\,\Delta t$: $R_A = r_f + \alpha' + \beta(R_M - r_f) + \varepsilon$, where $\alpha' = \mathbb{E}[R_A] - r_f - \beta(\mathbb{E}[R_M] - r_f)$ is Jensen's alpha of Definition 11.41, with the index as the reference portfolio. The beta is unchanged, and so is the residual, which in both

decompositions equals $R_A - \mathbb{E}[R_A] - \beta(R_M - \mathbb{E}[R_M])$, as subtracting from each decomposition its own expectation shows. Substituting both relations, the P&L is

$$V_A R_A + N_F V_F R_F = V_A(r_f + \alpha' + \varepsilon) + (\beta V_A + N_F V_F e^{-r\Delta t})(R_M - r_f).$$

By Proposition 10.10 and $\mathrm{Cov}(\varepsilon, R_M) = 0$, exactly as in part (i) of the proof, its variance is

$$V_A^2 \,\mathrm{Var}(\varepsilon) + \left(\beta V_A + N_F V_F e^{-r\Delta t}\right)^2 \mathrm{Var}(R_M),$$

which is minimised only at $N_F = -e^{r\Delta t}\beta V_A / V_F$, a sale of $e^{r\Delta t} N_F^*$ contracts. At that N_F the second term of the P&L vanishes and the hedged return is $r_f + \alpha' + \varepsilon$. In the same way the beta of the position is $\beta + e^{-r\Delta t} N_F V_F / V_A$, and a target β' takes $N_F = e^{r\Delta t}(\beta' - \beta) V_A / V_F$ contracts. These counts are exact under the standing simplification of the Setup, which ignores interest on the variation margin and is exact for a single settlement at the end of the horizon. With daily settlement over the horizon, the margin interest that the simplification ignores is of the same order $r\,\Delta t$ as the correction, so the refinement from N_F^* to $e^{r\Delta t} N_F^*$ is meaningful only together with tailing (Remark 16.39). The two alphas differ by a term of second order,

$$\alpha - \alpha' = (r_f - r\,\Delta t) - \beta\,(r_f - r\,\Delta t) = (1 - \beta)(e^{r\Delta t} - 1 - r\,\Delta t).$$

Over two months at $r = 4.5\%$ the factor is $e^{0.0075} = 1.00753$, so Example 16.41 would sell 208.6 contracts instead of 207. The count $\beta V_A / V_F$ is the first-order answer, and it is the one that is quoted.

Example 16.41 (Hedging and re-gearing a 27-million portfolio)**.** A portfolio is worth $V_A = 27{,}000{,}000$ and has $\beta = 1.15$. Two-month index futures trade at 6,000 with a multiplier of 25, so $V_F = 150{,}000$ and $V_A / V_F = 180$. The index itself stands at about 5,973: with $r = 4.5\%$ and $q = 1.8\%$, Proposition 16.15(ii) gives $5{,}973\, e^{(0.045 - 0.018)\times 2/12} = 5{,}973\, e^{0.0045} = 5{,}999.9$.

- *To hedge*, sell $N_F^* = 1.15 \times 180 = 207$ contracts.
- *To cut the beta to* 0.4, the target $\beta' = 0.4$ gives $N_F = (0.4 - 1.15) \times 180 = -135$, a short of 135 contracts.
- *To raise the beta to* 1.6, $N_F = (1.6 - 1.15) \times 180 = +81$, a purchase of 81 contracts.

Outcome check for the full hedge. Take $\alpha = 0$ and $\varepsilon = 0$, with $r = 4.5\%$ and a two-month horizon, so that $r\,\Delta t = 0.75\%$, and work under the first-order model (H). Suppose that the futures fall by 6%, from 6,000 to 5,640, so that $R_M = -6\% + 0.75\% = -5.25\%$. The portfolio returns $0.75\% + 1.15 \times (-5.25\% - 0.75\%) = -6.15\%$, a loss of 1,660,500, while the 207 short contracts gain $207 \times 360 \times 25 = 1{,}863{,}000$. The net is +202,500, which is 0.75% of 27,000,000: the hedged book earns $r\,\Delta t$, as part (iii) of Proposition 16.40 says. In the market of this chapter the exact carry identity of the note after the proposition holds instead of (H): the exact hedge sells $e^{0.0075} \times 207 = 208.6$ contracts and, with $\alpha' = 0$ and $\varepsilon = 0$, earns $r_f = e^{0.0075} - 1 = 0.753\%$ whatever the futures do, that is, 203,261.

(Problem P16.9 works this computation at different numbers.)

Interview trap. Forgetting beta. Shorting $V_A / V_F = 180$ contracts hedges a portfolio of beta 1. At $\beta = 1.15$ it leaves, by part (ii) of Proposition 16.40, a beta of $1.15 - 180 \times 150{,}000 / 27{,}000{,}000 = 0.15$: fifteen hundredths of a market exposure remain in place.

Interview trap. A hedged equity book does not earn zero. To first order it earns $r\,\Delta t + \alpha$ plus noise; exactly, in the market of this chapter, it earns $e^{r\Delta t} - 1 + \alpha' + \varepsilon$ (Proposition 16.40 and the note after it). The hedge strips out the market, not the alpha (Definition 11.41). Apply Fact 11.40 over the horizon, with the exact riskless return $r_f = e^{r\Delta t} - 1$, taking the index as the market portfolio M of the Fact and the portfolio as a combination of the assets that the Fact prices. Jensen's alpha α' is then zero, so the exactly hedged book earns r_f in expectation, and the first-order alpha of the Setup is $(1 - \beta)(e^{r\Delta t} - 1 - r\,\Delta t)$, which is zero up to terms of second order: -4.2×10^{-6} for $\beta = 1.15$ over two months at 4.5%.

Interview trap. The contract value V_F is the futures price times the multiplier, not the index level times the multiplier. With the index at 5,973 the wrong value $5{,}973 \times 25 = 149{,}325$ gives $1.15 \times 27{,}000{,}000/149{,}325 = 207.9$, that is, 208 contracts instead of 207.

Trading application. *Inventory hedging.* A market maker who absorbs spot flow on one venue and hedges with a correlated future or perpetual on another (§13.4) is solving exactly the problem of Theorem 16.37. The hedge is sized by h^*, not by 1, and the residual variance $(1-\rho^2)\sigma_S^2$ is the basis risk that remains on the book.

Trading application. *Stack and roll.* An exposure that runs beyond the listed maturities is hedged with short-dated futures, which are rolled into the next contract before each expiry. Each roll realises a basis, by Proposition 16.35, and the daily settlement of a large stack of contracts can create funding needs that force the hedge to be cut at the worst moment: the liquidity trap of §16.6. For a long position the same arithmetic is the *roll yield*. With an unchanged curve in contango each contract converges down to spot, so every roll costs $F - S$ per unit: $6.30 - 6.00 = 0.30$ per bushel a year at the wheat curve of Example 16.24 if spot stays at 6.00. The roll yield is negative in contango and positive in backwardation.

Practice: problems P16.1–P16.11 in §27, solutions in §28.

17 Options: Payoffs, Bounds, Put–Call Parity and Trading Strategies

The first option in these notes was the re-roll of Proposition 1.24 (§1.7), a right without an obligation, worth 0.75 (Example 1.25) because its holder throws away bad outcomes. This section derives everything about European and American calls and puts that follows from no arbitrage alone, before any model of the underlying: payoffs, model-free bounds, put–call parity, early exercise, the shape of the price as a function of the strike, and the standard spreads and combinations. The tools are Lemma 16.8 and the forward prices of §16. The few signs that need a model are flagged: the European ones are proved in §20 and §21, and the American ones are model statements, computable on a tree by Procedure 18.16 in §18, where the early-exercise boundary is computed, and not proved in this Part.

17.1 Calls, Puts and Payoffs

This subsection defines the four basic positions (long or short, a call or a put) and computes their payoff and profit at expiry. Every later strategy is a sum of them, and "draw the P and L of a short put; where is the breakeven?" opens most options interviews.

Setup. The underlying is a stock, an index, a currency or a futures contract, with price $S_t \geq 0$ at time t; today is $t = 0$ and $S = S_0$. The expiry is $T > 0$, in years, and the strike is $K > 0$. The continuously compounded interest rate is r, and $x^+ = \max(x, 0)$ is the positive part of a real number x. The prices today of a European call and a European put are C and $\mathcal{P}$, written $C(K)$ and $\mathcal{P}(K)$ when the strike varies and $C(K,T)$ and $\mathcal{P}(K,T)$ when the expiry varies too; the prices of the American call and put are C^{am} and $\mathcal{P}^{\mathrm{am}}$, with $C^{\mathrm{am}}(K,T)$ and $\mathcal{P}^{\mathrm{am}}(K,T)$ likewise. The premium of a generic option, its price at time 0, is V_0, and V_t is its price at a date t, as in the Notation table. *Carry convention:* a premium paid at time 0 is carried to T at the factor e^{rT} when a profit is computed; profit diagrams drop the carry and say so.

Symbol notices for this section.

- Here r is the interest rate, not the ratio q/p of the Notation table.

- The put is the calligraphic $\mathcal{P}$, because P is probability and p is the tree probability of §18; put–call parity is read "C minus $\mathcal{P}$".
- The literature, Hull's textbook included, writes c, p for European and C, P for American prices.
- A quoted volatility in parentheses, such as "(about 25% implied)", is a forward pointer to the implied volatility of Definition 20.37 in §20, the volatility at which that section's formula reproduces the quote. Option prices in the examples and problems of this section are data, and no argument of the section uses the parenthesis.

Definition 17.1 (European and American calls and puts; holder, writer, premium, payoff and profit)**.** A *European call* with strike K and expiry T gives its holder the right, and not the obligation, to buy one unit of the underlying for K at time T. A *European put* gives the right, and not the obligation, to sell one unit for K at time T. An *American* call or put gives the same right at any one time in $[0, T]$ that its holder chooses; using the right is called *exercising* the option.

The *holder* (the long side) pays the *premium* V_0 at time 0 to the *writer* (the short side), who carries the obligation: to deliver the unit for K when a call is exercised, and to take it for K when a put is exercised. A holder exercises at T only when exercise brings in money, so the *payoffs* at expiry of the long side are

$$(S_T - K)^+ \quad \text{for a call,} \qquad (K - S_T)^+ \quad \text{for a put,}$$

and the payoffs of the short side are their negatives. The *profit* at T of the long side is its payoff minus the premium carried to T, $V_0 e^{rT}$; the profit of the short side is its payoff plus $V_0 e^{rT}$, that is $V_0 e^{rT}$ minus the payoff of the long side, so the two profits sum to zero. A profit diagram conventionally drops the carry, plotting the long payoff minus V_0 and the short payoff plus V_0; a diagram drawn that way must say so.

Definition 17.2 (Moneyness, intrinsic value and time value)**.** A call is *in the money* when $S > K$, *at the money* when $S = K$ and *out of the money* when $S < K$; for a put the two inequalities are exchanged. *At the money spot* means $K = S$ and *at the money forward* means $K = F$, where F is the forward price for delivery at T (for an investment asset without income or storage cost, $F = Se^{rT}$ by Theorem 16.11); the two differ by the carry. The *intrinsic value* of an option is the value of exercising it at once when that is worth doing, $(S - K)^+$ for a call and $(K - S)^+$ for a put, and its *time value* is its price minus its intrinsic value. At a later date t the same words are used with S_t and the forward price F_t of that date.

The time value of an American option is never negative (Proposition 17.6(i), proved in §17.2), while that of a European option can be negative (Proposition 17.18 in §17.4).

Remark 17.3 (Listed and over-the-counter options)*.* Listed equity options cover a multiplier of units (typically 100 shares per contract) and are typically American; index options are settled in cash on a multiplier and are typically European; currency options trade mostly over the counter; and an option on a futures contract delivers, when it is exercised at a time t, a futures position together with a cash amount, $F_t^{\text{fut}} - K$ for a call and $K - F_t^{\text{fut}}$ for a put, where F_t^{fut} is the price at t of the futures contract, whose delivery date may follow T (with deterministic rates it equals the forward price for that delivery date, and §20.8, where such options are treated, writes it F_t). Writers post margin because they carry the obligation, while buyers risk only the premium. Over-the-counter options are bilateral contracts, tailored to the two parties, and carry counterparty risk. Every other exchange mechanic is outside the scope of these notes.

Example 17.4 (Payoff, profit and breakeven of the four positions)**.** A stock that pays no income trades at $S = 99.70$, the rate is $r = 4\%$, and two three-month options ($T = 0.25$) are struck at $K = 100$: a call bought for 4.20 and a put bought for 3.50.

Moneyness. The forward price is $F = 99.70\, e^{0.01} = 100.70$. The call is therefore out of the money spot ($99.70 < 100$) and in the money forward ($100.70 > 100$), and the put is the reverse. The

intrinsic values are 0 for the call and $100 - 99.70 = 0.30$ for the put, so the time values are 4.20 and $3.50 - 0.30 = 3.20$.

Profit. The carry factor is $e^{0.01} = 1.01005$, so the premiums carried to T are $4.20 \times 1.01005 = 4.24$ and $3.50 \times 1.01005 = 3.54$. The profit of each position at three values of S_T is the payoff of Definition 17.1 less the carried premium for the long side, and the opposite for the short side:

Position	$S_T = 90$	$S_T = 100$	$S_T = 110$
long call	-4.24	-4.24	$+5.76$
long put	$+6.46$	-3.54	-3.54
short call	$+4.24$	$+4.24$	-5.76
short put	-6.46	$+3.54$	$+3.54$

For instance the long put at $S_T = 90$ pays $(100 - 90)^+ = 10$ and has cost 3.54 at T, a profit of 6.46.

Breakevens. The long call breaks even where $S_T - 100 = 4.24$, at $S_T = 104.24$, and the long put where $100 - S_T = 3.54$, at $S_T = 96.46$. With the carry ignored, the diagram convention, the breakevens are 104.20 and 96.50.

Extremes. The long call loses at most its carried premium, 4.24, and the loss of the short call has no upper limit, because S_T has none. The short put earns at most 3.54, and it loses most at $S_T = 0$, where it pays 100 and keeps 3.54: a loss of 96.46. Both options trade near 19.4% implied (§20), and the two premiums are consistent with put–call parity, which is checked at the end of Example 17.10.

(Problem P17.1 works this computation at different numbers.)

Figure 29 draws the payoff and the profit of the four positions of Example 17.4, arranged by direction: the top row holds the two positions that gain when the underlying rises. Each short panel is the mirror image, in the horizontal axis, of the long panel in the same column, because the two sides of one contract receive opposite amounts in every state.

Interview trap. Writing a put is not buying a call. Both positions gain when the underlying rises, but in Example 17.4 the short put collects at most its premium (3.54) and can lose up to K minus that premium (96.46), while the long call risks its premium (4.24) for a gain without limit; the two share the top row of Figure 29. The payoff of the long call exceeds that of the short put by $(S_T - K)^+ + (K - S_T)^+ = |S_T - K|$, the payoff of a long straddle (Definition 17.32 in §17.7).

Interview trap. The loss on a short call is unbounded; the loss on a short put is bounded by K minus the premium carried to T. "Short options have unlimited risk" is true of the call only.

Interview trap. "At the money" is ambiguous: it may mean spot ($K = S$) or forward ($K = F$). In Example 17.4 the 100 call is out of the money spot and in the money forward. Put–call parity (§17.3) cares only about the forward.

Interview trap. Profit is not payoff. A long call breaks even at K plus the premium carried to T, not at K, and not at K plus the premium paid. Ignoring the carry is harmless over weeks (104.20 against 104.24 over three months above) and wrong over years.

17.2 Model-Free Bounds

No arbitrage alone confines every option price to a band. The band answers the interview staple "what are the maximum and minimum values of a call, and of a put?", and its lower edge drives the early-exercise results of §17.4. Every European bound is one dominance argument (Lemma 16.8), written once in forward form so that the income, yield and currency cases are substitutions of the forward price and not new proofs. American options need one more technique, exercise matching,

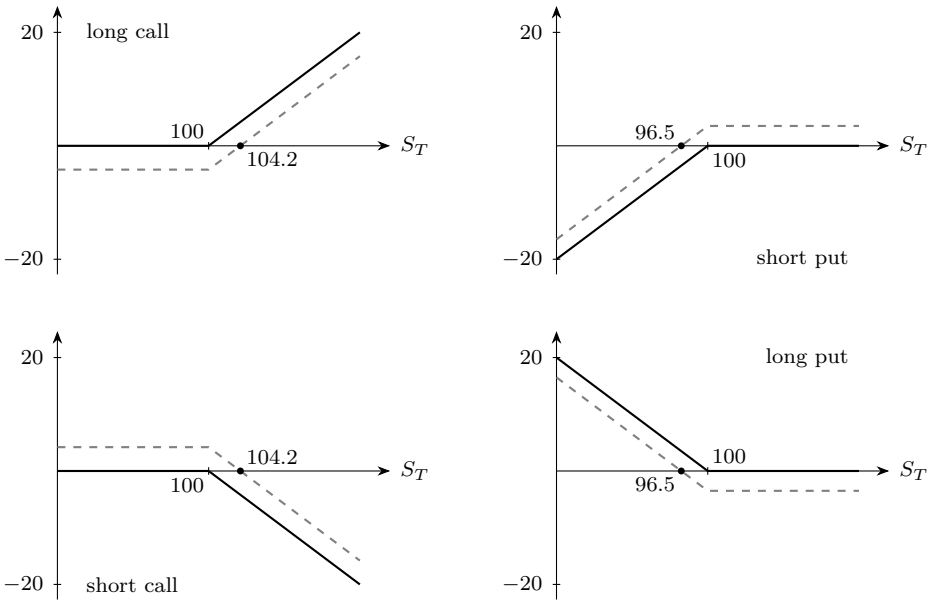


Figure 29: The four basic positions at expiry, for the options of Example 17.4 ($K = 100$; the horizontal axis runs from $S_T = 80$ to $S_T = 120$). Top row: the long call and the short put, the positions that gain when the underlying rises; bottom row: the short call and the long put. Solid black: the payoff, with its kink at the strike. Dashed gray: the profit with the carry ignored, that is the payoff shifted by the premium (4.20 for the call, 3.50 for the put), down for the long side and up for the short side. The dots mark the breakevens 104.2 and 96.5 of that convention; carrying the premiums to expiry moves them to 104.24 and 96.46.

which Lemma 17.5 states and proves before the first bound, together with the fact that every bound of this section also holds at later dates.

Setup. *Market.* The standing assumptions are those of §16.2: a frictionless market; one continuously compounded rate r for borrowing and lending, assumed nonnegative ($r \geq 0$) throughout this section; short sales allowed, any income of the borrowed unit being paid to its lender; self-financing strategies, which may hold nothing until a date t and start then, on an event known at t; and no arbitrage in the sense of Definition 16.7. Limited liability holds: $S_t \geq 0$ at every date. A zero-coupon bond paying K at T costs Ke^{-rT} (Lemma 16.8(iv)). Most European statements of this section do not use the assumption $r \geq 0$. The European bounds of Proposition 17.6(i)–(iii), Theorem 17.9, Corollaries 17.11 and 17.12, Proposition 17.23(i), the band (17.4), Propositions 17.24, 17.29 and 17.35, and Lemma 17.5(i) applied to European claims are proved from Definition 16.7, Lemma 16.8, the forward prices of §16 and one another, none of which places a sign on r, so these statements hold for every real r. The monotonicity of the European call in the expiry, Theorem 17.16(iii), does use $r \geq 0$, and every statement that involves an American option keeps the standing assumption.

Income. Three cases are declared once for the section.

(N) *No income*: the underlying pays nothing before T.

(I) *Known income*: the underlying pays known cash dividends $D_i > 0$ at ex-dividend dates $0 < t_i < T$, with present value $I = \sum_i D_i e^{-rt_i}$. The instant immediately before t_i is written t_i^-: the share trades cum dividend at $S_{t_i^-}$ and ex dividend at S_{t_i}, and an option that expires at t_i^- settles on $S_{t_i^-}$.

(Y) *Known yield*: the underlying pays a continuous yield q in the sense of §16.3; for a currency, $q = r_{\text{for}}$, the foreign rate. The letter q is always the yield, never the down-probability $1 - p$.

In the three cases the underlying is an investment asset (Definition 16.10) that costs nothing to store, so that the forward price for delivery at T is

$$F = Se^{rT} \text{ in case (N)}, \qquad F = (S - I)e^{rT} \text{ in case (I)}, \qquad F = Se^{(r-q)T} \text{ in case (Y)},$$

by Theorem 16.11, Proposition 16.13 and Proposition 16.15(ii) (Proposition 16.19 for a currency). A futures contract belongs to none of the three cases, because entering one costs nothing, so that no unit of it is bought for a price S; options on futures are treated in §20.8, and in this section a futures price enters only through the forward form (17.3) of put–call parity, as the Setup of §17.3 explains. A consumption asset, which need not be available for short sale (Definition 16.10), is likewise covered in this section only by that forward form. The *prepaid forward* is the position that pays S_T at T: one share in case (N); one share together with the loans $D_i e^{-rt_i}$, each repaid by its dividend, in case (I); and e^{-qT} shares with the yield reinvested in case (Y) (Proposition 16.15(i)). It costs S, $S - I$ or Se^{-qT}, which is $e^{-rT}F$ in each case by the formula for F. Cash received before T is reinvested at the rate r until T, so that positions are compared at T.

Later dates. At a fixed date $t \in [0, T)$ the remaining life is $\tau = T - t$. Prices at t carry the subscript t: C_t, $\mathcal{P}_t$, C_t^{am}, $\mathcal{P}_t^{\text{am}}$, so that $C = C_0$. The forward price at t for delivery at T is F_t, and I_t is the present value at t of the dividends that go ex in (t, T).

American options. An *exercise policy* is a rule that decides, from what is known at each date of $[0, T]$, whether to exercise then. The holder chooses the policy and the writer must honour whichever policy is chosen. The holder may exercise out of the money, and may let the option lapse. A written American option that has not been exercised before T therefore costs its writer at most the European payoff at T, $(S_T - K)^+$ for a call and $(K - S_T)^+$ for a put: exercise at T in the money costs exactly that amount, a lapse costs nothing, and exercise at T out of the money hands the writer a gain. The exercise time that a policy produces is written t_e; the letter τ keeps its meaning $T - t$.

Proof techniques. Two techniques are named here and reused in the whole section. *Dominance at T* (Lemma 16.8) serves European claims: two payoffs are compared at T and the comparison passes to the prices. *Exercise matching* serves American claims: hold the dominating position, sell the dominated one, and at the exercise time t_e chosen by the counterparty act so that the cash flows at t_e are covered; then hold everything that remains, cash included, to T (to the later expiry when two expiries are compared). Exercise matching is the arbitrage form of dominance. It uses payoffs only, never an option price at the random time t_e, and Lemma 17.5(ii) shows that it proves a price inequality.

For instance, Step 1 of the proof of Proposition 17.6 (Appendix C.1) proves $C \geq S - Ke^{-rT}$ in case (N): if the call were cheaper, buying it and the bond and selling the share short would bring in money today and leave a nonnegative amount at T. Part (i) of the next lemma says that three months later $C_t \geq S_t - Ke^{-r\tau}$, by the same trade opened then, and Example 17.7 runs it in numbers. Part (ii) says that a position short American options proves a price relation once it is safe under every exercise policy.

Lemma 17.5 (Bounds at later dates; exercise matching)**.** *Assume the standing assumptions of the Setup. The* horizon *of a relation or of a position is a date at or after every expiry it involves: T for the options of strike K and expiry T, and T_2 when two expiries $T_1 < T_2$ are compared.*

(i) Later dates. *Let a relation between prices (of options, the underlying, forwards and bonds) have the following property. Whatever the date at which the prices are quoted, and whatever the values then of the spot price, of the forward price, of the income still to come (its amounts and dates) and of the remaining lives, if the quoted prices violate the relation there is a position opened at that date such that:*

(a) the position is specified by these inputs, the strikes, the rate r and the quoted prices alone;

(b) opening it brings in a positive amount;

(c) on every path along which the price of the underlying stays nonnegative, it is worth a nonnegative amount at the horizon, under every exercise policy of the holders if it is short American options.

For European claims such a position is a dominance position of Lemma 16.8; for American claims it is a position as in (ii). Then, at each fixed date $t \geq 0$ earlier than every expiry of the relation, the relation holds with probability one with S_t, F_t, I_t and the prices quoted at t in place of S, F, I and the prices at time 0, and with each expiry replaced by its remaining life at t (so that T becomes $\tau = T - t$).

(ii) Exercise matching. *Let a position be set up at time 0, possibly short American options, whose later trades depend only on what is known at each date, the exercise times chosen by the holders of those options included, and whose holdings change at most a number of times fixed in advance (at the exercise times of the finitely many options it holds or has written, and at finitely many other dates), as for a strategy of §16.2. Suppose that setting it up brings in a positive amount and that, for every exercise policy of those holders, the position is worth a nonnegative amount at its horizon with probability one. Then the position, with its inflow deposited in the cash account until the horizon, is an arbitrage whatever policy the holders follow (reading (a) of Definition 16.7). Consequently the price relation whose failure produced the position holds.*

Proof. We argue by contradiction with the absence of arbitrage, using readings (a) and (c) of Definition 16.7, with the horizon in the role of the maturity T of that definition. Part (ii) is proved first, because part (i) uses it.

Part (ii). Fix an exercise policy for each holder. With the policies fixed, the position is a self-financing strategy in the sense of §16.2. Its holdings change only at dates at which the change depends on what is known then: the exercise times are such dates, because a policy decides from what is known at each date. Each option is exercised at most once, so by hypothesis the holdings change at most a number of times fixed in advance, the bound that the Setup of §16.2 imposes on a strategy; a reinvested holding counts as one position, as it does there. No cash is added or withdrawn after time 0, every cash flow at an exercise being paid from or into the cash account. The price of the strategy is strictly negative, because setting it up brings in a positive amount, and its payoff at the horizon is nonnegative with probability one. By reading (a) of Definition 16.7, once the proceeds are deposited in the cash account until the horizon, the strategy has price zero and a payoff that is positive with probability one: it is an arbitrage. This holds for every choice of policies, hence for the policies that the holders actually follow, which the arbitrageur does not choose and need not know in advance. The standing assumption excludes arbitrage, so the relation cannot fail.

Part (i). Fix a date $t \geq 0$ earlier than every expiry of the relation, and let A be the event that the relation fails at t. The event A is known at t, because it is read off the inputs S_t, F_t, I_t of that date and the prices quoted then. On A the hypothesis, applied at the date t with these inputs and the remaining lives at t, supplies a position opened at t. By item (a) the position is determined by what is known at t; by item (b) it brings in a positive amount at t; and by item (c) and limited liability it is worth a nonnegative amount at the horizon with probability one, under every exercise policy from t on if it is short American options.

Consider the strategy that holds nothing before t and, at t and on A only, opens this position and deposits its positive inflow in the cash account until the horizon. The positions opened at t, the deposit included, cost nothing in total, so the value of the strategy at t is zero: this is a late start in the sense of §16.2. Its payoff is zero off A; on A, outside an event of probability zero, it is at least the deposited inflow grown to the horizon, which is positive. If the position is short American options, fix the holders' policies as in part (ii), so that the strategy is self-financing whatever they are. By reading (c) of Definition 16.7 the strategy is an arbitrage unless $P(A) = 0$. Hence $P(A) = 0$, which is the claim. For European claims with horizon T the conclusion is also

Lemma 16.8(v), applied to the two positions of the dominance argument held from t. □

Every bound, parity relation and comparison below that is stated at time 0 is used at later dates through part (i); the few statements of §17.4 that are made at a date t are either derived from such a bound through part (i) or proved there by the late-start trade of the proof of part (i). Each proof of a statement made at time 0 is written for arbitrary values of the spot price, the forward price, the income and the remaining lives, specifies its position by these inputs and the quoted prices, changes the holdings of that position at most a number of times fixed in advance (at the exercise times of the options involved and at finitely many other dates), and bounds the final value using only the standing assumptions, none of which depends on the date; so each has the property that part (i) requires. In part (ii) the matching must succeed under *every* policy, because the writer does not choose the exercise time: a position that is safe only under some policies proves nothing. Example 17.7 carries part (i) to a later date in numbers, and Example 17.15 runs part (ii) on a quoted put.

Proposition 17.6 (Model-free bounds and comparisons)**.** *Assume the standing assumptions of the Setup, in particular $r \geq 0$, and consider European and American calls and puts with strike K and expiry T on one underlying.*

(i) Comparisons. *European prices are nonnegative: $C \geq 0$ and $\mathcal{P} \geq 0$. An American option is worth at least the European option of the same type with the same strike and any expiry $T' \leq T$ at which one trades, an instant t_i^- before an ex-date included; in particular $C^{\mathrm{am}} \geq C$ and $\mathcal{P}^{\mathrm{am}} \geq \mathcal{P}$. An American option is worth at least its intrinsic value,*

$$C^{\mathrm{am}} \geq (S-K)^+, \qquad \mathcal{P}^{\mathrm{am}} \geq (K-S)^+,$$

so the time value of an American option is nonnegative. American prices are nondecreasing in the expiry: for $T_1 < T_2$,

$$C^{\mathrm{am}}(K,T_1) \leq C^{\mathrm{am}}(K,T_2), \qquad \mathcal{P}^{\mathrm{am}}(K,T_1) \leq \mathcal{P}^{\mathrm{am}}(K,T_2).$$

(ii) Upper bounds. *The European call satisfies $C \leq e^{-rT}F$, that is $C \leq S$ in case (N), $C \leq S - I$ in case (I) and $C \leq Se^{-qT}$ in case (Y). The American call satisfies $C^{\mathrm{am}} \leq S$ in every income case, provided $q \geq 0$ in case (Y). The puts satisfy $\mathcal{P} \leq Ke^{-rT}$ and $\mathcal{P}^{\mathrm{am}} \leq K$.*

(iii) European lower bounds. *In every income case,*

$$C \geq e^{-rT}(F-K)^+, \qquad \mathcal{P} \geq e^{-rT}(K-F)^+. \tag{17.1}$$

Written out, the two bounds are $\max(S - Ke^{-rT}, 0)$ and $\max(Ke^{-rT} - S, 0)$ in case (N), $\max(S - I - Ke^{-rT}, 0)$ and $\max(Ke^{-rT} + I - S, 0)$ in case (I), and $\max(Se^{-qT} - Ke^{-rT}, 0)$ and $\max(Ke^{-rT} - Se^{-qT}, 0)$ in case (Y).

Each bound also holds with probability one at every date t earlier than the expiries it involves, with S_t, F_t, I_t, the remaining lives (τ for the expiry T) and the prices at t in place of S, F, I, the expiries and the prices at time 0.

Sketch of the proof. The European bounds come from dominance at T (Lemma 16.8(i)), the American ones from exercise matching (Lemma 17.5(ii)). In case (N), by limited liability and $K > 0$ the share pays at least the call and the bond paying K at least the put; the call with the bond pays $(S_T - K)^+ + K = \max(S_T, K)$, as does the put with the share, and this is at least S_T and at least K; and no option pays a negative amount. So $C \leq S$, $\mathcal{P} \leq Ke^{-rT}$, $C \geq S - Ke^{-rT}$, $\mathcal{P} \geq Ke^{-rT} - S$, $C \geq 0$ and $\mathcal{P} \geq 0$. With income, the comparisons without a share stand, the prepaid forward replaces the share in $C \leq e^{-rT}F$, and by the identity $x^+ - x = (-x)^+$ a call with a short forward at delivery price K pays $(K - S_T)^+ \geq 0$ and costs $C - e^{-rT}(F-K)$ (Proposition 16.17(i)); a put with a long forward is symmetric, which gives (17.1). A failed

American bound yields a trade that brings in money and ends nonnegative under every exercise policy: buy the American option, write the European one and exercise at T' if the European one finishes in the money; buy the option and exercise it at once; write the call and hold the share, income only adding because $D_i > 0$ and $q \geq 0$; write the put and deposit K, enough at any exercise time because $r \geq 0$; buy the T_2 option, write the T_1 option and exercise the T_2 option when the T_1 option is exercised. Each trade is written for arbitrary inputs, so Lemma 17.5(i) gives the bounds at later dates. The complete proof is in Appendix C.1. □

For an asset without income (case (N)) the proposition answers the interview question in three lines:

$$\begin{aligned} \max(S - Ke^{-rT}, 0) \leq C \leq C^{\text{am}} \leq S, \\ \max(Ke^{-rT} - S, 0) \leq \mathcal{P} \leq Ke^{-rT}, \\ \max(K - S, 0) \leq \mathcal{P}^{\text{am}} \leq K, \end{aligned}$$

together with $\mathcal{P} \leq \mathcal{P}^{\text{am}}$. In every income case the two call bounds squeeze together as the strike falls: by (ii) and (iii), $e^{-rT}(F - K) \leq C(K) \leq e^{-rT}F$ for $K < F$, so $C(K) \to e^{-rT}F$ as $K \to 0$, and a call struck near zero is worth the prepaid forward, the share itself in case (N). The two examples below put numbers on the lower edge of the call and on the effect of a dividend.

Example 17.7 (A call quoted below its lower bound)**.** A stock that pays no dividends trades at $S = 72$, and a call on it has $K = 68$ and $T = 0.75$; the rate is $r = 5\%$. Then $Ke^{-rT} = 68\,e^{-0.0375} = 65.50$, and by (17.1) in case (N) the call is worth at least $72 - 65.50 = 6.50$. A call quoted at 6.00 is therefore an arbitrage. The trade is the dominance position of Step 1 of the proof of Proposition 17.6 (Appendix C.1), the call and the bond held long and the share sold short, with its inflow deposited: buy the call, sell the share short, and lend the net proceeds $72.00 - 6.00 = 66.00$, which are the price 65.50 of the bond plus the inflow 0.50. At T the loan returns $66\,e^{0.0375} = 68.52$.

Trade	$t = 0$	T, if $S_T \leq 68$	T, if $S_T > 68$
Buy the call	-6.00	0	$S_T - 68$
Short the share	$+72.00$	$-S_T$	$-S_T$
Lend 66.00	-66.00	$+68.52$	$+68.52$
Net	0	$68.52 - S_T \geq 0.52$	0.52

If $S_T > 68$, the call is exercised: the 68 paid buys the share that closes the short, and 0.52 is kept. If $S_T \leq 68$, the call lapses and the share is bought in the market for S_T, which leaves $68.52 - S_T \geq 0.52$. The trade costs nothing and pays at least 0.52 in every state. The present value of 0.52 is $0.52\,e^{-0.0375} = 0.50 = 6.50 - 6.00$, the size of the violation.

A later date. Three months on, suppose the stock trades at $S_t = 75$. The remaining life is $\tau = 0.5$, and by Lemma 17.5(i) the floor is $75 - 68\,e^{-0.025} = 75 - 66.3211 = 8.6789$. A quote below 8.6789 on that day, 8.67 or less in cents, is the same trade, started then.

(Problem P17.2 works this computation at different numbers.)

Example 17.8 (Bounds with a dividend)**.** A stock trades at $S = 66$ and pays one dividend of 1.40 in three months; the rate is $r = 5\%$ and the options expire at $T = 0.75$. The present value of the dividend is $I = 1.40\,e^{-0.0125} = 1.3826$, or 1.383 to three decimals, so the prepaid forward costs $S - I = 64.62$.

A call struck at 60. Here $Ke^{-rT} = 60\,e^{-0.0375} = 57.7917$. By Proposition 17.6(iii) in case (I) the European call is worth at least $66 - 1.3826 - 57.7917 = 6.8257$, against $66 - 57.7917 = 8.2083$ if

the dividend were ignored. By part (ii) it is worth at most $S - I = 64.62$, against 66 without the dividend; the American call keeps the upper bound 66.

A put struck at 72. Here $Ke^{-rT} = 72\,e^{-0.0375} = 69.3500$. The European put is worth at least $69.3500 + 1.3826 - 66 = 4.73$, against $69.35 - 66 = 3.35$ without the dividend. Its upper bound is 69.35, and that of the American put is 72.

Here, where every bound is positive, the dividend lowers both bounds of the European call by I and raises the lower bound of the put by I; in general the upper bound of the call falls by I, while each lower bound, a positive part, moves by I only where both the old and the new bound are positive. This is the model-free half of the dividend row of Table 3 in §17.5.

Figure 40 in §20 draws a model price of the call between the two bounds of case (N).

Interview trap. The lower bound uses the discounted strike: $S - Ke^{-rT}$, not $S - K$. In Example 17.7, $S - K = 4$ would make a quote of 6.00 look comfortably fair; the true floor is 6.50.

Interview trap. "An option is always worth at least its intrinsic value" holds for American options only. A European put can trade below its intrinsic value (Proposition 17.18 in §17.4), and a deep in-the-money European call on an asset with a yield $q > 0$ must: its upper bound Se^{-qT} (Proposition 17.6(ii)) lies below $S - K$ as soon as $K < S(1 - e^{-qT})$. With $S = 100$, $q = 5\%$ and $T = 1$ that is every strike below $100(1 - e^{-0.05}) = 4.877$, and at those strikes the American call is strictly dearer than the European one (Proposition 17.20(iv), proved in §17.4, of which this is the numeric case). Nor does the floor of the European call protect its intrinsic value: $Se^{-qT} - Ke^{-rT}$ lies below $S - K$ as soon as $S(1 - e^{-qT}) > K(1 - e^{-rT})$, which holds for every $S > K$ when $q \geq r$ and $q > 0$. At $q = r = 0$ the floor and the intrinsic value coincide.

Interview trap. The European put is bounded by Ke^{-rT} and the American put by the undiscounted K, which it attains once the underlying has fallen to 0 (parts (i) and (ii) of Proposition 17.6 at that date). Quoting Ke^{-rT} as the maximum of an American put is wrong.

17.3 Put–Call Parity and Synthetic Positions

Put–call parity is the most-asked identity in options interviews, usually in the form "here are a call, a put and the spot; is there an arbitrage?". It prices a put from a call, defines the synthetic forward, contains no volatility, and holds for any underlying once it is written in forward form; for American options only a band survives. The proof is the law-of-one-price half of Lemma 16.8, and the forward form uses Proposition 16.17.

Setup. The standing assumptions are those of §17.2. A European call and a European put have the same strike K and the same expiry T and are written on the same underlying, which is in one of the income cases (N), (I), (Y) of §17.2; a currency is case (Y) with $q = r_{\text{for}}$, its forward price being given by Proposition 16.19. Part (iii) of Corollary 17.11 is not restricted to the income cases: it needs only a forward for delivery at T trading at F, on any underlying, investment or consumption asset. The forward price for delivery at T is F; when interest rates are deterministic, the futures price of a contract that expires with the options may replace it (Theorem 16.28). For the American band the underlying is in case (N), the rate satisfies $r \geq 0$, the prices of the American call and put with strike K and expiry T are C^{am} and $\mathcal{P}^{\text{am}}$, and t_e is the exercise time produced by an exercise policy.

Theorem 17.9 (Put–call parity)**.** *Assume the standing assumptions of §17.2, and let the underlying pay no income (case (N)). A European call and a European put with the same strike K and the same expiry T satisfy*

$$C - \mathcal{P} = S - Ke^{-rT}. \tag{17.2}$$

Proof. We use the law of one price: two portfolios with the same payoff in every state cost the same (Lemma 16.8(iii)). The *fiduciary call* holds one call and a zero-coupon bond paying K at T; the bond costs Ke^{-rT} by Lemma 16.8(iv), so the portfolio costs $C + Ke^{-rT}$. The *protective put* holds one put and one share and costs $\mathcal{P} + S$. Both are held unchanged from 0 to T and are self-financing, because in case (N) nothing they hold pays or costs anything before T. Their payoffs at T are $(S_T - K)^+ + K$ and $(K - S_T)^+ + S_T$, and in the two cases of the final price they read:

	fiduciary call, $(S_T - K)^+ + K$	protective put, $(K - S_T)^+ + S_T$
if $S_T \geq K$	$(S_T - K) + K = S_T$	$0 + S_T = S_T$
if $S_T < K$	$0 + K = K$	$(K - S_T) + S_T = K$

This is the check made for the third and fourth rows of the table in Step 1 of the proof of Proposition 17.6: both portfolios pay $\max(S_T, K)$ in every state. By Lemma 16.8(iii) they cost the same, $C + Ke^{-rT} = \mathcal{P} + S$, which is (17.2). □

(Problem P17.3 asks you to reproduce this proof from scratch.)

Read from left to right, (17.2) says that a call minus a put is a forward contract with delivery price K: by Proposition 16.17(ii), a long forward with delivery price K on an asset without income is worth $S - Ke^{-rT}$. The identity also holds at every date $t < T$, as $C_t - \mathcal{P}_t = S_t - Ke^{-r\tau}$ with probability one, by Lemma 17.5(i); here that is Lemma 16.8(v) applied to the two portfolios held from t.

Example 17.10 (A parity arbitrage, with its cash flows)**.** A stock that pays no dividends trades at $S = 44$; a call and a put on it have $K = 45$ and $T = 0.5$, and the rate is $r = 5\%$. Then $Ke^{-rT} = 45\,e^{-0.025} = 43.889$, and parity requires $C - \mathcal{P} = 44 - 43.889 = 0.111$. The quotes are $C = 3.15$ and $\mathcal{P} = 2.80$, so $C - \mathcal{P} = 0.35$. The put is cheap relative to the call by $0.35 - 0.111 = 0.239$: parity prices it at $3.15 - 0.111 = 3.039$.

The trade. Buy the cheap portfolio and sell the dear one. The protective put (the share and the put) costs $44 + 2.80 = 46.80$; writing the call brings in 3.15; the remaining $46.80 - 3.15 = 43.65$ is borrowed, so the trade costs nothing. At T the loan has grown to $43.65\,e^{0.025} = 44.755$.

Trade	$t = 0$	T, if $S_T \leq 45$	T, if $S_T > 45$
Buy the share	-44.00	S_T	S_T
Buy the put	-2.80	$45 - S_T$	0
Write the call	$+3.15$	0	$-(S_T - 45)$
Borrow 43.65	$+43.65$	-44.755	-44.755
Net	0	0.245	0.245

The share and the two options are worth exactly 45 in both columns: below 45 the put is exercised and the share is delivered for 45, and above 45 the call is exercised against the position and the share is delivered for 45. Repaying 44.755 leaves 0.245 in every state, which is the mispricing carried to T: $0.239\,e^{0.025} = 0.245$. Desks call this position, long the share and the put and short the call, a *conversion*. The opposite position, short the share and the put and long the call, captures a cheap call and is a *reversal*; it needs the short sale that the application after Corollary 17.11 prices.

Two volatilities. The violation shows up as two implied volatilities for one strike and one expiry: 25.0% for the call and 23.1% for the put. Quotes that satisfy parity carry a single implied volatility (Proposition 22.3 in §22).

A consistent pair. The two premiums of Example 17.4 satisfy parity: $4.20 - 3.50 = 0.70$, and $99.70 - 100\,e^{-0.01} = 99.70 - 99.00498 = 0.69502$, which is 0.70 to the cent.

(Problem P17.3 works this computation at different numbers.)

Interview trap. Parity contains no volatility, no probabilities and no model: it holds for any distribution of S_T. A call and a put with the same strike and expiry that are quoted at two different implied volatilities are a parity violation, not a smile.

Corollary 17.11 (Parity with income, with a yield, and in forward form)**.** *Assume the standing assumptions of §17.2, and let a European call and a European put have the same strike K and the same expiry T.*

(i) With known income (case (I)), $C - \mathcal{P} = S - I - Ke^{-rT}$.

(ii) With a yield q (case (Y)), $C - \mathcal{P} = Se^{-qT} - Ke^{-rT}$; for a currency, $C - \mathcal{P} = Se^{-r_{\text{for}}T} - Ke^{-rT}$.

(iii) For any underlying, investment or consumption asset, on which a forward contract for delivery at T trades at the forward price F,

$$C - \mathcal{P} = e^{-rT}(F - K). \tag{17.3}$$

In particular $C = \mathcal{P}$ if and only if $K = F$.

Proof. We argue by replication, proving (iii) first. The identity $x^+ - x = (-x)^+$ of Step 2 of the proof of Proposition 17.6, rearranged as $x^+ - (-x)^+ = x$, shows at $x = S_T - K$ that one call held long and one put held short pay

$$(S_T - K)^+ - (K - S_T)^+ = S_T - K$$

in every state, which is the payoff of a long forward contract with delivery price K. That forward is worth $e^{-rT}(F - K)$ at time 0 by Proposition 16.17(i), which requires only that a forward for delivery at T can be struck at the price F; consumption assets are therefore included. Neither of the two strategies compared, the option pair and the forward, holds the asset: the option pair can be bought or written, and a forward entered on either side, so both strategies can be held and reversed, which is what Lemma 16.8(iii) needs for a consumption asset (Setup of §16.5). By Lemma 16.8(iii) the pair of options costs the same as the forward, which is (17.3). The right side vanishes exactly when $K = F$, which is the last claim of (iii).

For (i) and (ii), substitute the forward price of the Setup of §17.2: $e^{-rT}F = S - I$ in case (I) by Proposition 16.13, $e^{-rT}F = Se^{-qT}$ in case (Y) by Proposition 16.15(ii), and $e^{-rT}F = Se^{-r_{\text{for}}T}$ for a currency by Proposition 16.19. In case (N), $e^{-rT}F = S$ by Theorem 16.11, and (17.3) is (17.2) again. □

The forward form (17.3) is the one that §20 uses for indices, currencies and futures (Corollary 20.41, Corollary 20.43, Theorem 20.52). Like (17.2), each form holds at every date $t < T$ with S_t, I_t, F_t and τ (Lemma 17.5(i)).

Trading application. Read backwards, parity turns a call and a put into an *implied forward*, $F = K + e^{rT}(C - \mathcal{P})$, and hence into an implied dividend or an implied cost of borrowing the stock. In Example 17.10 the quotes imply a forward of $45 + 0.35\,e^{0.025} = 45.359$ against the cash-and-carry forward $44\,e^{0.025} = 45.114$; the gap, 0.245, is the gain locked in there, and it is a genuine arbitrage because the trade buys the share and needs no short sale. An implied forward *below* the cash-and-carry forward is different: the arbitrage would sell the share short, and a violation of that sign that persists on a hard-to-borrow stock is the price of the short sale. The borrow fee acts as a yield in (17.3).

Corollary 17.12 (Synthetic positions)**.** *Assume the standing assumptions of §17.2, and let a European call and a European put have the same strike K and the same expiry T. "The share" means the prepaid forward of the Setup of §17.2, which pays S_T and costs $e^{-rT}F$; in case (N) it is an actual share. Two positions are said to be equal when they have the same payoff in every state, and equal positions have the same price.*

(i) A long call together with a short put is a long forward contract with delivery price K: the pair pays $S_T - K$ and costs $e^{-rT}(F-K)$, which is zero exactly when $K = F$.

(ii) A covered call*, long the share and short a call, pays* $\min(S_T, K)$ *and equals a short put together with a bond paying K. Built with an actual share in case (I) it also collects the dividends, and it then equals the short put, the bond and the dividend stream, which is worth I.*

(iii) A protective put*, long the share and long a put, pays* $\max(S_T, K)$ *and equals a long call together with a bond paying K (and the dividend stream, when it is built with an actual share in case (I)).*

(iv) A synthetic share*, a long call, a short put and a bond paying K, pays S_T and equals the share.*

The reverse positions are the negatives: for instance a short call together with a long put is a short forward with delivery price K.

Proof. We check payoff identities at $x = S_T$ on the two cases $x \geq K$ and $x < K$, and then apply the law of one price (Lemma 16.8(iii)).

Payoff	if $x \geq K$	if $x < K$	equals
$(x-K)^+ - (K-x)^+$	$x - K$	$-(K-x)$	$x - K$
$x - (x-K)^+$	$x - (x-K)$	x	$\min(x, K)$
$K - (K-x)^+$	K	$K - (K-x)$	$\min(x, K)$
$x + (K-x)^+$	x	$x + (K-x)$	$\max(x, K)$
$K + (x-K)^+$	$K + (x-K)$	K	$\max(x, K)$
$(x-K)^+ - (K-x)^+ + K$	x	x	x

The first row is the payoff of the pair in (i), which is that of a long forward with delivery price K; its price is $e^{-rT}(F-K)$ by Proposition 16.17(i), and this is zero exactly when $K = F$. The second and third rows show that the covered call (the share less the call) and the short put with the bond both pay $\min(S_T, K)$. The fourth and fifth rows show that the protective put (the share plus the put) and the long call with the bond both pay $\max(S_T, K)$. The last row shows that the synthetic share pays S_T, as the share does. In each case the two positions have the same payoff in every state, so Lemma 16.8(iii) equates their prices:

$$e^{-rT}F - C = Ke^{-rT} - \mathcal{P}, \qquad e^{-rT}F + \mathcal{P} = C + Ke^{-rT}, \qquad C - \mathcal{P} + Ke^{-rT} = e^{-rT}F,$$

each of which is (17.3) rearranged, and (17.2) in case (N).

In case (I) an actual share is the prepaid forward together with the dividend stream: the holder of the share receives each D_i at t_i, while in the prepaid forward each D_i repays a loan. A sure amount D_i at t_i, deposited until T, is the sure amount $D_i e^{r(T-t_i)}$ at T, whose price is $D_i e^{-rt_i}$ by Lemma 16.8(iv); the stream is therefore worth I. Adding it to both sides of (ii) and (iii) gives the statements for an actual share. Reversing every holding of two equal positions reverses both payoffs, so the negatives are equal as well. □

Example 17.13 (An index in forward form, and the synthetic forward)**.** An index stands at $S = 5000$ and pays a dividend yield $q = 1\%$; the rate is $r = 4.5\%$ and $T = 0.5$. The forward price is $F = 5000\, e^{0.0175} = 5088.27$, and the discount factor is $e^{-0.0225} = 0.97775$.

- At $K = 5000$, (17.3) gives $C - \mathcal{P} = 0.97775 \times 88.27 = 86.31$. The call that is at the money spot is dearer than the put, because $F > S$.
- At $K = 5100$, $C - \mathcal{P} = 0.97775 \times (5088.27 - 5100) = -11.47$: the put is dearer.
- At $K = F = 5088.27$ the call and the put cost the same.

By Corollary 17.12(i), buying the call and writing the put at $K = 5000$ is a long forward with delivery price 5000, and it costs 86.31; the same pair struck at $K = F$ is a forward at the forward price and costs nothing. The form of Corollary 17.11(ii) gives the same number: $5000\,e^{-0.005} - 5000\,e^{-0.0225} = 4975.062 - 4888.756 = 86.31$.

(Problem P17.3 works this computation at different numbers.)

Interview trap. At $K = F$ the call and the put are worth the same. At $K = S$, at the money spot, the call is dearer whenever $F > S$: with a yield, whenever $r > q$; on a stock without dividends, whenever $r > 0$, the difference being $S - Se^{-rT}$. "An at-the-money call and put cost the same" is the $K = F$ statement.

Interview trap. With dividends, use $S - I$ or Se^{-qT} in parity, never S. Forgetting the dividend manufactures a fake arbitrage of size I, or $S(1 - e^{-qT})$ with a yield.

Proposition 17.14 (The American parity band)**.** *Assume the standing assumptions of §17.2 with $r \geq 0$, and let the underlying pay no income (case (N)). An American call and an American put with the same strike K and the same expiry T satisfy*

$$S - K \;\leq\; C^{\mathrm{am}} - \mathcal{P}^{\mathrm{am}} \;\leq\; S - Ke^{-rT}.$$

Proof. We use exercise matching (Lemma 17.5(ii)), once for each edge. The proof does not use the early-exercise theorem of §17.4 (Theorem 17.16), which is why the band can be stated here. Both edges use the identity $x^+ - (-x)^+ = x$ of the proof of Corollary 17.11.

Lower edge. Suppose that $\mathcal{P}^{\mathrm{am}} + S > C^{\mathrm{am}} + K$. Write the American put, sell one share short, buy the American call and deposit K. The inflow is $\mathcal{P}^{\mathrm{am}} + S - C^{\mathrm{am}} - K > 0$. The call is held to T and exercised there only if it is in the money, so it pays $(S_T - K)^+$.

- If the put is exercised against the position at some time $t_e < T$, the deposit is worth $Ke^{rt_e} \geq K$, because $r \geq 0$. Pay K out of it, receive the share and return it to its lender, which closes the short sale; no dividend is owed in case (N). The position then holds the call and the cash $K(e^{rt_e} - 1) \geq 0$, and both are worth a nonnegative amount at T.
- If the put is not exercised before T, its writer owes at most $(K - S_T)^+$ at T (Setup of §17.2). The deposit is worth Ke^{rT}, the call pays $(S_T - K)^+$ and closing the short sale costs S_T, so the position is worth at least

  $$Ke^{rT} + (S_T - K)^+ - (K - S_T)^+ - S_T = Ke^{rT} + (S_T - K) - S_T = K(e^{rT} - 1) \geq 0,$$

  by the identity at $x = S_T - K$.

Every exercise policy leaves a nonnegative value at T after a positive inflow, which Lemma 17.5(ii) excludes. Hence $\mathcal{P}^{\mathrm{am}} + S \leq C^{\mathrm{am}} + K$, that is $S - K \leq C^{\mathrm{am}} - \mathcal{P}^{\mathrm{am}}$. The American call was held as a European one, so the band requires no European option to trade.

Upper edge. Suppose that $C^{\mathrm{am}} + Ke^{-rT} > \mathcal{P}^{\mathrm{am}} + S$. Write the American call, borrow Ke^{-rT}, buy the American put and buy one share. The inflow is $C^{\mathrm{am}} + Ke^{-rT} - \mathcal{P}^{\mathrm{am}} - S > 0$. The loan is worth $Ke^{-r\tau}$ at a date t, where $\tau = T - t$, and K at T. The put is held to T and exercised there only if it is in the money, so it pays $(K - S_T)^+$.

- If the call is exercised against the position at some time $t_e < T$, deliver the share for K and repay the loan, which is then worth $Ke^{-r(T-t_e)} \leq K$, because $r \geq 0$. The position keeps the cash $K - Ke^{-r(T-t_e)} \geq 0$ and the put, and both are worth a nonnegative amount at T.
- If the call is not exercised before T, its writer owes at most $(S_T - K)^+$ at T. The put pays $(K - S_T)^+$, the share is worth S_T and the loan costs K, so the position is worth at least

  $$(K - S_T)^+ + S_T - K - (S_T - K)^+ = (K - S_T) + S_T - K = 0,$$

  by the identity at $x = K - S_T$.

Every exercise policy leaves a nonnegative value at T after a positive inflow, so $C^{\text{am}} + Ke^{-rT} \le \mathcal{P}^{\text{am}} + S$, that is $C^{\text{am}} - \mathcal{P}^{\text{am}} \le S - Ke^{-rT}$. □

(Problem P17.5 extends the band to known income: $S - I - K \le C^{\text{am}} - \mathcal{P}^{\text{am}} \le S - Ke^{-rT}$.)

By Lemma 17.5(i) the band holds at every date $t < T$, with S_t and τ in place of S and T. Its upper edge is the European value of $C - \mathcal{P}$ and its lower edge is that value with the strike left undiscounted, so its width $K(1 - e^{-rT})$ is the interest on the strike over the life of the options.

Example 17.15 (The band for an American put, and a quote outside it)**.** A stock that pays no dividends trades at $S = 126$; American options on it have $K = 120$ and $T = 0.5$, the rate is $r = 4\%$, and the American call is quoted at 11.40 (about 18.5% implied). Solving the two inequalities of Proposition 17.14 for the put,

$$C^{\text{am}} - S + Ke^{-rT} \;\le\; \mathcal{P}^{\text{am}} \;\le\; C^{\text{am}} - S + K,$$

and with $Ke^{-rT} = 120\,e^{-0.02} = 117.6238$ the American put lies in $[11.40 - 126 + 117.6238,\ 11.40 - 126 + 120] = [3.0238,\ 5.40]$. The band is $K(1 - e^{-rT}) = 120 - 117.6238 = 2.3762$ wide.

Exercise matching in numbers. A put quoted at 5.50 lies above the band. Following the proof of the lower edge, write the put (+5.50), sell the share short (+126.00), buy the call (−11.40) and deposit 120 (−120.00): the trade collects 0.10 today. If the put is exercised at any time $t_e < T$, the deposit pays the 120 and the share received closes the short sale, which leaves the call and the interest earned on the deposit. If the put is not exercised before T, the position is worth at least $120(e^{0.02} - 1) = 2.42$ at T. No exercise policy of the put's holder produces a loss, and the 0.10 collected today is kept in every case.

The example is taken up again in Example 17.17 in §17.4, once the identity $C^{\text{am}} = C$ has identified the lower end of the band as the European put.

(Problem P17.5 works this computation at different numbers.)

Interview trap. "Parity holds for American options" is false. Only the band of Proposition 17.14 holds, and the short American leg of a parity trade can be assigned early, at a time its writer does not choose.

17.4 Early Exercise

Exercising an American option is an optimal-stopping problem, the re-roll again: Proposition 1.24 stops when the value of stopping beats the value of continuing, by the recursion (1.5). Model-free arguments decide when early exercise can never be optimal; when it can be, the tree of §18 (Procedure 18.16) locates the boundary. This subsection answers the whiteboard question "why is an American call on a non-dividend stock never exercised early, while an American put can be?", and reopens it for calls on assets with income: just before an ex-dividend date, and at any time on an asset with a yield. It builds on Proposition 17.6 and Proposition 17.14.

Setup. The standing assumptions are those of §17.2. The options are American calls and puts with strike K and expiry T, together with the European options of the same strike and expiry; the underlying is in case (N) unless a statement says otherwise, and $r \ge 0$; each statement says where it needs $r > 0$. At a date t the remaining life is $\tau = T - t$, and t_e is the exercise time that an exercise policy produces (Setup of §17.2).

Vocabulary. The *exercise value* of an option at a date t is what exercise delivers then: $S_t - K$ for a call and $K - S_t$ for a put, a negative number when the option is out of the money. Exercise at a date $t < T$ is called *strictly suboptimal* when the price of the option at t exceeds its exercise value. The holder then receives more by selling the option than by exercising it, and a holder who wants

the share buys it in the market and keeps the option. When the price is only known to be at least the exercise value, exercise gains nothing over selling. The *early-exercise premium* is the excess of the American price over the European one, $C^{\text{am}} - C$ or $\mathcal{P}^{\text{am}} - \mathcal{P}$.

Discrete dividends, for parts (i) to (iii) of Proposition 17.20 (case (I)). The underlying pays n dividends $D_1, \dots, D_n > 0$ at the ex-dates $0 < t_1 < \cdots < t_n < T$; here n counts dividends, and $t_0 = 0$ and $t_{n+1} = T$ are written for the two ends. The price is assumed to drop by exactly the dividend on each ex-date,

$$S_{t_i} = S_{t_i^-} - D_i,$$

the exact-drop assumption of a frictionless market without taxes, where t_i^- is the instant immediately before t_i (Setup of §17.2); limited liability, $S_{t_i} \geq 0$, then gives $S_{t_i^-} \geq D_i$. No interest accrues between t_i^- and t_i. Exercise at t_i^- delivers the share cum dividend, so that the exerciser receives D_i at t_i, and an option that expires at t_i^- settles on $S_{t_i^-}$.

A yield, for part (iv) of the same proposition (case (Y)). The underlying pays a continuous yield $q > 0$; for a currency, $q = r_{\text{for}}$.

Theorem 17.16 (No early exercise of an American call without dividends)**.** *Assume the standing assumptions of §17.2 with $r \geq 0$, and let the underlying pay no income (case (N)).*

(i) An American call and a European call with the same strike K and the same expiry T have the same price: $C^{\text{am}} = C$.

(ii) If $r > 0$ and $K > 0$, exercise at any date $t < T$ is strictly suboptimal: $C_t^{\text{am}} > S_t - K$ with probability one. Moreover a holder who wants to keep the exposure to the share does strictly better, in every state, by holding the call to T than by exercising it at t and financing the strike with a loan.

(iii) The European call price is nondecreasing in the expiry: for two expiries $T_1 < T_2$, the same strike and the same underlying,

$$C(K, T_1) \leq C(K, T_2).$$

Sketch of the proof. Part (i) is exercise matching (Lemma 17.5(ii)). If $C^{\text{am}} > C$, write the American call and buy the European one. On exercise at $t_e < T$, deliver a borrowed share, on which nothing is owed in case (N), and deposit the K received: at T the position is worth

$$(S_T - K)^+ - S_T + Ke^{r(T-t_e)} \;\geq\; (S_T - K)^+ - (S_T - K) = (K - S_T)^+ \;\geq\; 0,$$

because $r \geq 0$; otherwise the European call pays what is owed. Hence $C^{\text{am}} \leq C$, the reverse of Proposition 17.6(i). When $r > 0$ and $K > 0$, part (i) and (17.1) at a date t give $C_t^{\text{am}} = C_t \geq S_t - Ke^{-r\tau} > S_t - K$, and exercise with a loan leaves $S_T - Ke^{r\tau} < (S_T - K)^+$ at T. Part (iii) is the same trade, long the T_2 call and short the T_1 call, then dominance at T_2 (Lemma 16.8(i)). The complete proof is in Appendix C.2. □

(Problem P17.6 asks you to reproduce the complete proof from scratch.)

Part (iii) is the proof of part (i) run at the near expiry. Example 17.26 shows that the monotonicity fails for puts and for calls on assets with income, and Proposition 22.15(i) in §22 is its $r = 0$ case. With part (i), the upper edge of Proposition 17.14 also follows in one line:

$$C^{\text{am}} - \mathcal{P}^{\text{am}} = C - \mathcal{P}^{\text{am}} \;\leq\; C - \mathcal{P} = S - Ke^{-rT},$$

by $\mathcal{P}^{\text{am}} \geq \mathcal{P}$ (Proposition 17.6(i)) and (17.2). On a tree the theorem can be checked node by node (§18.5).

Example 17.17 (Sell, do not exercise)**.** Take the numbers of Example 17.15: a stock without dividends at $S = 126$, American options with $K = 120$ and $T = 0.5$, a rate $r = 4\%$, and the American call quoted at 11.40.

Exercise against sale. Exercising the call pays $126 - 120 = 6.00$. Selling it pays 11.40. Even the model-free floor $S - Ke^{-rT} = 126 - 117.6238$ of Proposition 17.6(iii), which bounds the American call because $C^{\text{am}} = C$ (Theorem 17.16(i)), exceeds the exercise value 6.00, by the interest $K(1 - e^{-rT}) = 120 - 117.6238 = 2.3762$ on the strike.

Holding against exercising with a loan. A holder who exercises and borrows 120 owes $120\, e^{0.02} = 122.42$ at T and so ends with $S_T - 122.42$. A holder who keeps the call ends with $(S_T - 120)^+ \geq S_T - 120$, at least 2.42 more in every state, and more still when $S_T < 120$.

The band of Example 17.15, read again. Since $C^{\text{am}} = C = 11.40$, parity (17.2) prices the European put at $\mathcal{P} = 11.40 - 126 + 117.6238 = 3.0238$, which is the lower end of the band $[3.0238, 5.40]$ found there for the American put. The width of the band therefore caps the early-exercise premium of the put: $\mathcal{P}^{\text{am}} - \mathcal{P} \leq 5.40 - \mathcal{P} = K(1 - e^{-rT})$, which is 2.3762 to four decimals.

(Problem P17.6 works this computation at different numbers.)

Intuition. The final display of the proof of Theorem 17.16(i) prices what exercise gives up. Against a holder who exercises at t_e and borrows the strike, the holder of the call is ahead at T by

$$(S_T - K)^+ - \left(S_T - Ke^{r(T-t_e)}\right) = \left(Ke^{r(T-t_e)} - K\right) + (K - S_T)^+,$$

the interest on the strike plus a put struck at K that the call carries inside it. The second term does not involve r, which is why part (i) of the theorem holds at $r = 0$ as well. The practical form is "sell, do not exercise": a holder who wants out sells the call, and a holder who wants the stock buys it and keeps the call.

Interview trap. An in-the-money American call on an asset without income is worth more alive than dead when $r > 0$ (Theorem 17.16(ii)), and never less at $r = 0$ (Proposition 17.6(i)). The answer to "you hold a deep in-the-money call and want the stock, or have turned bearish" is to sell the call, or to keep it and trade the stock; it is not to exercise and throw away the time value.

Proposition 17.18 (The American put carries an early-exercise premium)**.** *Assume the standing assumptions of §17.2, and let the underlying pay no income (case (N)). Consider puts with strike K and expiry T.*

(i) If $r > 0$ and $S < K(1 - e^{-rT})$, the European put trades strictly below its intrinsic value and the American put does not:

$$\mathcal{P} \;\leq\; Ke^{-rT} \;<\; K - S \;\leq\; \mathcal{P}^{\text{am}}.$$

So $\mathcal{P}^{\text{am}} > \mathcal{P}$, a strictly positive early-exercise premium. Moreover immediate exercise beats every exercise policy that never exercises before T: such a policy pays at most the European payoff $(K - S_T)^+$ at T, and exercising now and buying the European put out of the proceeds leaves $K - S - \mathcal{P} > 0$ in cash on top of that payoff. The same holds, with probability one, at each date $t < T$ with S_t and τ in place of S and T.

(ii) Let $r > 0$ and let $t < T$ be a date. On the event $\{S_t = 0\}$, exercising at t is worth at least as much at T as any other exercise policy, and strictly more in every state of that event in which the other policy does not exercise at t; and, with probability one on that event, $\mathcal{P}_t^{\text{am}} = K$.

(iii) If $r = 0$, then $\mathcal{P}^{\text{am}} = \mathcal{P}$.

Sketch of the proof. Parts (i) and (ii) use the bounds of Proposition 17.6 and compare values at T; part (iii) is exercise matching (Lemma 17.5(ii)). The chain of (i) is $\mathcal{P} \leq Ke^{-rT}$, the hypothesis

rearranged, and $\mathcal{P}^{\text{am}} \geq (K-S)^+$; deposited, the $K-S$ from exercise grows to more than K, the most a put pays at T. In (ii), where $S_t = 0$, exercise pays K at once, a later exercise at most K, and $r > 0$ makes the earlier K worth more at T; the price lies between the intrinsic value K and the cap K. In (iii), if $\mathcal{P}^{\text{am}} > \mathcal{P}$, write the American put and buy the European one; on exercise, borrow the K to pay and keep the share, which with the put pays $S_T + (K-S_T)^+ = \max(S_T, K) \geq K$ at T and repays the loan because $r = 0$. The complete proof is in Appendix C.3. □

No arbitrage also makes zero an absorbing price; the argument is not needed above. For two dates $t < t' \leq T$, the event $\{S_t = 0,\ S_{t'} > 0\}$ has probability zero. Consider the strategy that holds nothing before t and, on the event $\{S_t = 0\}$, which is known at t, takes one share at its price of zero, sells it at t' and deposits the proceeds $S_{t'}$ until T. Its value at t is zero, so it is a strategy with a late start, and its price at time 0 is zero, as in reading (c) of Definition 16.7. Its payoff is $S_{t'}e^{r(T-t')} \geq 0$ on the event and 0 off it: nonnegative in every state, and positive exactly on $\{S_t = 0,\ S_{t'} > 0\}$. By Definition 16.7 the strategy is an arbitrage unless that event has probability zero.

Example 17.19 (A deep in-the-money put)**.** A stock without dividends has fallen to $S = 1$; puts on it have $K = 100$ and $T = 1$, and the rate is $r = 5\%$.

The European put. By Proposition 17.6(ii) it is worth at most $100\,e^{-0.05} = 95.123$, which is below its intrinsic value $100 - 1 = 99$: its time value is at most $95.123 - 99 = -3.877$. The condition of Proposition 17.18(i) is $S < 100(1 - e^{-0.05}) = 4.877$, and it holds.

The American put. It is worth at least 99, so the early-exercise premium $\mathcal{P}^{\text{am}} - \mathcal{P}$ is at least $99 - 95.123 = 3.877$. A holder who exercises now and buys the European put keeps at least 3.877 in cash and still receives $(100 - S_T)^+$ at T. The premium is also at most $K(1 - e^{-rT})$, because

$$\mathcal{P}^{\text{am}} \ \leq\ C^{\text{am}} - S + K \ =\ C - S + K \ =\ \mathcal{P} + K(1 - e^{-rT}),$$

by the lower edge of Proposition 17.14, Theorem 17.16(i) and (17.2). The premium of this put therefore lies in $[99 - 100\,e^{-0.05},\ 100(1 - e^{-0.05})]$, which is $[3.877, 4.877]$ to three decimals.

At zero. If the stock falls to $S = 0$, the put is exercised at once and is worth exactly 100 (Proposition 17.18(ii)). In numbers, with one year left: the 100 received and deposited is worth $100\,e^{0.05} = 105.13$ at T; exercise at a later time t_e is worth at most $100\,e^{0.05(1-t_e)} < 105.13$ at T, for instance 102.53 at $t_e = 0.5$; and exercise at T is worth at most 100.

(Problem P17.6 works this computation at different numbers.)

Intuition. Proposition 17.18 reads as a balance. Part (iii) shows that without interest exercise never gains, and part (ii) that once the put pays its cap K only interest is left to earn. In between, exercise earns interest on the strike it brings in and gives up the call struck at K that the unexercised put carries inside it. By the identity $x^+ - x = (-x)^+$ at $x = K - S_T$, the put pays

$$(K - S_T)^+ = (K - S_T) + (S_T - K)^+,$$

the strike less the share, which is what exercise delivers, together with a call: its holder loses nothing more if the share recovers above K, and exercise surrenders that protection. This is the mirror image of the intuition after Theorem 17.16, and it is the insurance of the trap below. Hence a higher rate, a lower volatility and a lower spot all favour exercise. The first raises the interest that exercise earns on K. The second cheapens the embedded call that exercise gives up. The third makes that call less valuable still, because the share would have further to recover. Where the interest and the embedded call balance is the early-exercise boundary, which a model locates (Procedure 18.16).

Interview trap. With $r = 0$ and no dividends, neither the American call nor the American put carries an early-exercise premium: $C^{\text{am}} = C$ and $\mathcal{P}^{\text{am}} = \mathcal{P}$ (Theorem 17.16(i), Proposition 17.18(iii)). Early exercise then never gains, though it can tie: at $r = 0$ a put at $S_t = 0$ is

worth K whether it is exercised or held. The early-exercise premium of a put exists only because of interest on K: it vanishes at $r = 0$, and in case (N) it never exceeds $K(1 - e^{-rT})$, by the chain $\mathcal{P}^{\mathrm{am}} \le C - S + K = \mathcal{P} + K(1 - e^{-rT})$ of Example 17.19. An answer that cites "insurance" as the reason *to* exercise a put has the trade-off backwards: insurance is the reason to wait.

Proposition 17.20 (American calls on assets with income)**.** *Assume the standing assumptions of §17.2, and consider American and European calls with strike $K > 0$ and expiry T. Parts (i) to (iii) concern an underlying that pays the discrete dividends of the Setup (case (I)), with $r > 0$ in parts (i) and (ii). Part (iv) concerns an underlying with a continuous yield $q > 0$ (case (Y)). Parts (iii) and (iv) need only $r \ge 0$. Every inequality between prices holds with probability one.*

(i) Between ex-dates. *Let $t \in [t_i, t_{i+1})$ for one $i \in \{0, \dots, n\}$, and let $t' = t_{i+1}$, so that $t' = T$ when $i = n$. Then*

$$C_t^{\mathrm{am}} \;\ge\; S_t - Ke^{-r(t'-t)} \;>\; S_t - K,$$

so exercise at t is strictly suboptimal.

(ii) Just before an ex-date. *For each $i \in \{1, \dots, n\}$,*

$$C_{t_i^-}^{\mathrm{am}} \;\ge\; S_{t_i^-} - D_i - Ke^{-r(t_{i+1}-t_i)}.$$

Consequently, if $D_i < K\big(1 - e^{-r(t_{i+1}-t_i)}\big)$, exercise at t_i^- is strictly suboptimal; and if D_i equals that threshold, the price of the call is at least its exercise value, so exercise gains nothing over selling.

(iii) A dividend above the strike. *If $D_i > K$, then $S_{t_i^-} > K$ and*

$$C_{t_i}^{\mathrm{am}} \;\le\; S_{t_i} = S_{t_i^-} - D_i \;<\; S_{t_i^-} - K :$$

an instant after the ex-date, the unexercised call is worth strictly less than exercise paid an instant before it. Moreover $C_{t_i^-}^{\mathrm{am}} > C_{t_i^-}$, a strictly positive early-exercise premium just before the ex-date. Pathwise, exercising at t_i^- and keeping the share leaves, once D_i has arrived and K is paid, the share and $D_i - K > 0$ in cash, which is strictly more than any policy that continues past t_i^- receives, at whatever time it exercises.

(iv) A yield. *At any date $t < T$ at which $K < S_t(1 - e^{-q\tau})$,*

$$C_t \;\le\; S_t e^{-q\tau} \;<\; S_t - K \;\le\; C_t^{\mathrm{am}},$$

so $C_t^{\mathrm{am}} > C_t$, a strictly positive early-exercise premium, and immediate exercise beats every policy that never exercises before T. In particular this holds for a call on a currency ($q = r_{\mathrm{for}} > 0$), whatever the domestic rate $r \ge 0$ is.

Sketch of the proof. Parts (i) and (ii) use exercise matching with the share, in the late-start trade of the proof of Lemma 17.5(i); parts (iii) and (iv) chain the bounds of Proposition 17.6 at a later date. In (i), on the event that the bound fails at t, buy the call, sell the share short and deposit the proceeds, more than $Ke^{-r(t'-t)}$; just before t' that part has grown to K, which exercises the call or, if cheaper, buys the share back before any dividend goes ex, and the excess is kept: an arbitrage unless the event has probability zero. The second inequality uses $r > 0$. In (ii) the trade opens at t_i^- with D_i more on deposit for the lender, and the bound minus the exercise value $S_{t_i^-} - K$ is $K\big(1 - e^{-r(t_{i+1}-t_i)}\big) - D_i$. In (iii) the exact drop and $D_i > K$ give the display, and $C_{t_i^-} \le S_{t_i^-} - I_{t_i^-} \le S_{t_i^-} - D_i < S_{t_i^-} - K \le C_{t_i^-}^{\mathrm{am}}$ gives the premium; in (iv) the chain is the hypothesis rearranged, and the policies compare as for the put in Proposition 17.18(i). The complete proof is in Appendix C.4. □

Three comments complete the picture. First, at the last ex-date t_n, for a dividend D_n strictly between the threshold $K\big(1 - e^{-r(T-t_n)}\big)$ of part (ii) and the strike, the results of this section do

not decide whether exercise at t_n^- beats continuing, and two markets give the two answers: a market without volatility, proved free of arbitrage below, and the Black–Scholes–Merton market, free of arbitrage by its assumption (A6) (Remark 20.14). In a market without volatility the price grows at the rate r between ex-dates and drops by the dividend on each ex-date. Every position in the share, the cash account and calls priced as the present values of their sure payoffs then grows at r, dividends included, so there is no arbitrage. From t_n on there is no income, so the American call at t_n is the European one (Theorem 17.16(i) at that date), and its payoff is a sure amount: it is worth $\bigl(S_{t_n} - Ke^{-r(T-t_n)}\bigr)^+$. When this is positive, exercise at t_n^-, which pays $S_{t_n} + D_n - K$, beats it exactly when D_n exceeds the threshold $K\bigl(1 - e^{-r(T-t_n)}\bigr)$. In the Black–Scholes–Merton market (Theorem 20.22 in §20) the call at t_n, $e^{-r(T-t_n)}\bigl[F_{t_n}\Phi(d_1) - K\Phi(d_2)\bigr]$ with Φ the standard normal distribution function and d_1 and d_2 the two arguments of (20.10) taken at the remaining life $T - t_n$, tends to $e^{-r(T-t_n)}F_{t_n} = S_{t_n}$ as the volatility grows (no income is paid after t_n), because $d_1 \to \infty$ and $d_2 \to -\infty$; at a large enough volatility it is worth more than $S_{t_n} + D_n - K$ whenever $D_n < K$. Deciding a given case needs a model, such as the escrowed-dividend model of §20.7, in which Remark 20.48 discusses Black's approximation and Example 20.49 computes the American call on the stock of Example 17.21 below. At an earlier ex-date t_i, $i < n$, the threshold of part (ii) is not the only model-free test. Proposition 17.6(i) with $T' = T$ and Proposition 17.6(iii) in case (I), both at the instant t_i^- (Lemma 17.5(i)), give

$$C^{\mathrm{am}}_{t_i^-} \;\ge\; C_{t_i^-} \;\ge\; S_{t_i^-} - I_{t_i^-} - Ke^{-r(T-t_i)},$$

which exceeds the exercise value $S_{t_i^-} - K$ whenever $I_{t_i^-} < K\bigl(1 - e^{-r(T-t_i)}\bigr)$, even when D_i lies above the threshold of part (ii); exercise at t_i^- is then strictly suboptimal. With $K = 60$, $r = 4\%$, dividends 1.19 at $t_1 = 0.25$ and 0.01 at $t_2 = 0.75$, and $T = 1$, the first dividend exceeds $60(1 - e^{-0.02}) = 1.188$, yet $I_{t_1^-} = 1.19 + 0.01\,e^{-0.02} = 1.1998 < 60(1 - e^{-0.03}) = 1.773$, so exercise at t_1^- is strictly suboptimal. At the last ex-date the two tests coincide, because $I_{t_n^-} = D_n$ and $t_{n+1} = T$. Second, with a yield $q > 0$ the argument of part (i) fails at every time, because the share sold short would owe the yield to its lender, and part (iv) shows that early exercise then beats holding to T in deep in-the-money states; Proposition 20.54 in §20.8 proves the analogue for futures (where $q = r$), and Example 18.28 in §18.7 shows it on a tree. Third, the two European calls of Black's approximation (Remark 20.48 in §20), expiring at t_n^- and at T, are lower bounds on the American call when both are prices in one market, by Proposition 17.6(i) with $T' = t_n^-$ and $T' = T$. Black's own convention prices the first on the dividends of its own life, and that number need not be a lower bound, as the remark shows. Part (i) is why only the instants t_i^- and the expiry T are candidate exercise dates.

Example 17.21 (Which ex-dates allow early exercise)**.** An American call has $K = 60$ and $T = 1$; the rate is $r = 4\%$, and the stock pays a dividend of 1.20 at $t_1 = 0.25$ and again at $t_2 = 0.75$.

The thresholds of part (ii). Before t_1 the next date is t_2, half a year later, and the threshold is $60(1 - e^{-0.04\times0.5}) = 1.188$. Before t_2 the next date is the expiry, a quarter later, and the threshold is $60(1 - e^{-0.04\times0.25}) = 0.597$. The dividend 1.20 exceeds both, so part (ii) rules exercise out at neither date, though only narrowly at the first. At the first date the other test of the first comment after Proposition 17.20 fails as well: $I_{t_1^-} = 1.20 + 1.20\,e^{-0.02} = 2.376$ exceeds $60(1 - e^{-0.03}) = 1.773$. Early exercise therefore cannot be ruled out at either date. Had the first dividend been $1.10 < 1.188$, exercise before it would be ruled out while the second date stayed open. Exercise at any other time is ruled out by part (i). "Cannot be ruled out" does not mean that exercise beats continuing: deciding needs a model, such as the escrowed-dividend model of §20.7 (Example 20.49 works this stock in it, at $S = 60$ and a volatility of 25%).

The two ends of the undecided range. Had the second dividend been exactly the threshold $60(1 - e^{-0.01})$, which is 0.597 to three decimals, the call would be worth at least its exercise value just before t_2, and exercise would gain nothing. At the other extreme, part (iii): suppose a special dividend of 65 on a stock priced at 100 just before its ex-date, with the call struck at

$60 < 65$. Exercise is worth $100 - 60 = 40$, while the call an instant later is worth at most the ex-dividend share, $100 - 65 = 35$. For a dividend at the second date, the last ex-date, strictly between $60(1 - e^{-0.01})$ and 60, the results of this section do not decide, and the two markets described after Proposition 17.20 show that either answer can occur.

Part (iv) in numbers. An index stands at $S = 100$ and pays a yield $q = 5\%$; the rate is $r = 4\%$, one year is left, and a call is struck at $K = 3$, below $100(1 - e^{-0.05}) = 4.877$. The European call is worth at most $100\, e^{-0.05} = 95.123$ and the American call at least $100 - 3 = 97$, so the early-exercise premium is at least $97 - 95.123 = 1.877$. Exercising, selling the index and buying the European call leaves at least 1.877 in cash today, which deposited at 4% is at least $1.877\, e^{0.04} = 1.95$ at T, on top of the European payoff.

(Problem P17.7 works this computation at different numbers.)

Remark 17.22 (Equity as a call on the firm). The equity of a firm whose only debt is a zero-coupon bond of face value K maturing at T pays, at T, the value of the firm's assets minus K when the assets cover the debt and nothing otherwise (limited liability: the debt holders take the assets), that is the positive part of the asset value at T minus K, so equity is a call on the firm's assets struck at K (Merton, 1974); credit risk is outside the scope of these notes, and no later result uses this remark.

Interview trap. When $r > 0$, early exercise of a call on a stock with discrete dividends can gain only at the instant before an ex-date (Proposition 17.20(i)). The threshold condition of part (ii) only rules exercise *out*, and it is not the only test that does: when it fails, exercise may still lose to continuing, and before the last ex-date the European call expiring at T may still rule it out (the first comment after Proposition 17.20). Among the results of this section, only a dividend at or above the strike shows exercise to be at least as good as continuing. Above the strike, part (iii) makes exercise strictly better; at $D_i = K$ the bound $C^{\mathrm{am}}_{t_i} \le S_{t_i} = S_{t_i^-} - K$ of its proof makes continuing gain nothing over exercise.

Interview trap. "Never exercise early" is a theorem for calls on assets without income only. American puts when $r > 0$ (Proposition 17.18(i)), calls immediately before an ex-dividend date (Proposition 17.20(iii), when the dividend exceeds the strike), calls on an asset with a yield, currencies included (Proposition 17.20(iv)), and calls on futures (Proposition 20.54 in §20.8) can all carry a strictly positive early-exercise premium.

17.5 What Moves an Option Price: the Strike Curve and the Sign Table

An option price depends on the strike, the expiry, the dividends, the spot and the rate, and, once a model is chosen, on the volatility. "What happens to a call when each of them goes up?" and "is the call price convex in the strike, and what does a negative butterfly mean?" are first-round questions. This subsection proves the strike structure in full (European prices are monotone, Lipschitz and convex in the strike, American prices monotone and Lipschitz) and tabulates the signs, marking each one as proved model-free in this section or proved under a model in a later one. The two strike results are the model-free ancestors of the Breeden–Litzenberger theorem (Theorem 22.7 in §22): the digital is minus the slope of the call curve, and the risk-neutral density is e^{rT} times its curvature. The subsection builds on Lemma 16.8, Proposition 17.6 and Theorem 17.16.

Setup. The standing assumptions are those of §17.2, with $r \ge 0$; the underlying may be in any income case unless a statement restricts it. The underlying and the expiry T are fixed, and the prices of the European call and put are read as functions of the strike, $C(K)$ and $\mathcal{P}(K)$ for $K > 0$; the American prices are $C^{\mathrm{am}}(K)$ and $\mathcal{P}^{\mathrm{am}}(K)$. The claim $\mathrm{Dig}(K)$ is the cash-or-nothing digital of the Notation table, which pays 1 at T if $S_T > K$ and nothing otherwise; the same symbol denotes

its price. Three strikes $K_1 < K_2 < K_3$ define the weight

$$w = \frac{K_3 - K_2}{K_3 - K_1} \in (0,1), \qquad \text{so that} \qquad K_2 = wK_1 + (1-w)K_3,$$

since $wK_1 + (1-w)K_3 = K_3 - w(K_3 - K_1) = K_3 - (K_3 - K_2)$. The literature often writes this weight λ; in this Part λ is the jump intensity of §23. The derivative of the call curve in the strike is written $\partial_K C(K)$. In this subsection the volatility σ of §20 enters only the rows of Table 3 that are flagged as model statements, the model prices of Example 17.26 and the model prices of the trap on rates; no proof of this subsection uses it.

Proposition 17.23 (Call-spread and put-spread bounds: the strike curve is monotone and Lipschitz)**.** *Assume the standing assumptions of §17.2, in particular $r \geq 0$, and consider options with a common expiry T and two strikes $K_1 < K_2$ on one underlying.*

(i) European.

$$0 \;\leq\; C(K_1) - C(K_2) \;\leq\; e^{-rT}(K_2 - K_1), \qquad 0 \;\leq\; \mathcal{P}(K_2) - \mathcal{P}(K_1) \;\leq\; e^{-rT}(K_2 - K_1).$$

So C is nonincreasing and $\mathcal{P}$ nondecreasing in the strike, and both are Lipschitz with constant e^{-rT}. At a strike where C is differentiable, $-e^{-rT} \leq \partial_K C(K) \leq 0$; at a strike where $\mathcal{P}$ is differentiable, $0 \leq \partial_K \mathcal{P}(K) \leq e^{-rT}$.

(ii) American.

$$0 \;\leq\; C^{\mathrm{am}}(K_1) - C^{\mathrm{am}}(K_2) \;\leq\; K_2 - K_1, \qquad 0 \;\leq\; \mathcal{P}^{\mathrm{am}}(K_2) - \mathcal{P}^{\mathrm{am}}(K_1) \;\leq\; K_2 - K_1,$$

with the width undiscounted.

Sketch of the proof. The European bounds are dominance at T (Lemma 16.8(i)), the American ones exercise matching (Lemma 17.5(ii)). In every state the call spread, long the K_1 call and short the K_2 call, pays

$$0 \;\leq\; (S_T - K_1)^+ - (S_T - K_2)^+ \;\leq\; K_2 - K_1,$$

so its price lies between 0 and that of a bond paying $K_2 - K_1$ at T. The put spread is symmetric, and a difference quotient gives the slopes. In (ii), when an inequality fails, buy the cheaper option, write the dearer one and, for the upper bounds, deposit $K_2 - K_1$, with money left over; then exercise the bought option whenever the written one is exercised. The shares cancel, so income plays no part, and the strikes leave a gain of $K_2 - K_1$ or a shortfall of that size, which the deposit covers because $r \geq 0$. The complete proof is in Appendix C.5. □

The European bound is the discounted width and the American bound the undiscounted one, for the reason that separates Ke^{-rT} from K in the bounds of the two puts (Proposition 17.6(ii)): an American spread can be assigned today, and then its width is paid today.

Proposition 17.23 comes with a band for the digital, in the form in which later sections cite it. The claim $\mathrm{Dig}(K)$ pays 0 or 1, so Lemma 16.8(i), applied against the empty portfolio and against a bond paying 1 (Lemma 16.8(iv)), gives

$$0 \;\leq\; \mathrm{Dig}(K) \;\leq\; e^{-rT}. \tag{17.4}$$

This is the slope band of part (i) again, and no coincidence: Proposition 23.2 in §23 brackets the digital between two scaled call spreads and shows that, where the call curve is differentiable, $\mathrm{Dig}(K) = -\partial_K C(K)$, so that part (i) becomes (17.4). A tight call spread is the model-free hedge of a digital, and so of a binary contract on a prediction market. Example 17.25, after Proposition 17.24, tests quoted prices against the band of part (i) and against the convexity proved there.

Proposition 17.24 (Convexity in the strike: every butterfly has a nonnegative price)**.** *Assume the standing assumptions of §17.2, and consider European options with a common expiry T on one underlying. For strikes $K_1 < K_2 < K_3$ and the weight $w = (K_3 - K_2)/(K_3 - K_1)$,*

$$C(K_2) \;\le\; w\,C(K_1) + (1-w)\,C(K_3), \qquad \mathcal{P}(K_2) \;\le\; w\,\mathcal{P}(K_1) + (1-w)\,\mathcal{P}(K_3).$$

Equivalently, C and $\mathcal{P}$ are convex functions of the strike on $(0,\infty)$, and every butterfly *(long w options at K_1, short one at K_2, long $1-w$ at K_3) has a nonnegative price. With equally spaced strikes, $w = \frac{1}{2}$ and*

$$C(K_1) - 2\,C(K_2) + C(K_3) \;\ge\; 0.$$

Proof. We use dominance at T (Lemma 16.8(i)): the butterfly has a nonnegative payoff. For $x \ge 0$ define the local function

$$\chi(x) = w\,(x - K_1)^+ + (1-w)\,(x - K_3)^+,$$

the payoff of the two bought calls when $S_T = x$. Both terms are nonnegative, so $\chi(x) \ge 0$. A positive part is at least its argument and the weights are positive, so

$$\chi(x) \;\ge\; w\,(x - K_1) + (1-w)\,(x - K_3) = x - \big(wK_1 + (1-w)K_3\big) = x - K_2,$$

by the identity $K_2 = wK_1 + (1-w)K_3$ of the Setup. The two bounds together give $\chi(x) \ge \max(x - K_2, 0) = (x - K_2)^+$ for every $x \ge 0$.

The call butterfly is held from 0 to T and pays $\chi(S_T) - (S_T - K_2)^+ \ge 0$ in every state. Its price is $w\,C(K_1) - C(K_2) + (1-w)\,C(K_3)$, and Lemma 16.8(i), applied against the empty portfolio, shows that this price is nonnegative, which is the inequality for calls. The payoff is a *tent*. It is zero for $S_T \le K_1$. On $[K_1, K_2]$ it equals $w(S_T - K_1)$ and rises with slope w to $w(K_2 - K_1)$ at K_2. On $[K_2, K_3]$ it equals

$$w(S_T - K_1) - (S_T - K_2) = K_2 - wK_1 - (1-w)S_T = (1-w)(K_3 - S_T),$$

using $K_2 - wK_1 = (1-w)K_3$, and falls with slope $-(1-w)$ to zero at K_3. Beyond K_3 it equals $(S_T - K_2) - (S_T - K_2) = 0$.

For puts the same argument runs with $(K - x)^+$ in place of $(x - K)^+$: the function $w\,(K_1 - x)^+ + (1-w)\,(K_3 - x)^+$ is nonnegative and at least $w(K_1 - x) + (1-w)(K_3 - x) = K_2 - x$, hence at least $(K_2 - x)^+$. The put butterfly therefore has a nonnegative payoff, and its price $w\,\mathcal{P}(K_1) - \mathcal{P}(K_2) + (1-w)\,\mathcal{P}(K_3)$ is nonnegative.

For the equivalent forms, a function is convex on $(0,\infty)$ when its value at every point $wK_1 + (1-w)K_3$ with $K_1 < K_3$ and $0 < w < 1$ is at most w times its value at K_1 plus $1-w$ times its value at K_3. Every such point is a strike K_2 between K_1 and K_3, and its weight is $w = (K_3 - K_2)/(K_3 - K_1)$ by the Setup, so the two displayed inequalities are the convexity of C and of $\mathcal{P}$. At $w = \frac{1}{2}$ the inequality for calls, multiplied by 2, is the last display. □

(Problem P17.8 asks you to reproduce this proof from scratch.)

Proposition 17.23 and Proposition 17.24 are the model-free ancestors of Theorem 22.7 in §22: there the digital is minus the slope of the call curve (22.2), and the risk-neutral density is e^{rT} times its curvature (22.3). A slope in $[-e^{-rT}, 0]$ and a nonnegative curvature are what a probability and a density require. Example 22.8 cites both propositions for its butterfly and digital-band checks.

Example 17.25 (Testing a strike curve: the slope and the curvature)**.** The rate is $r = 4\%$, and three-month calls ($T = 0.25$) on one underlying are quoted at three strikes:

Strike	90	100	110
Call price	12.0	6.5	2.0

Slope. The two call spreads cost $12.0 - 6.5 = 5.5$ and $6.5 - 2.0 = 4.5$. Both lie in $[0,\ 10\,e^{-0.01}] = [0, 9.90]$, as Proposition 17.23(i) requires.

Curvature. The butterfly costs $12.0 - 2(6.5) + 2.0 = 1.0 \geq 0$, consistent with Proposition 17.24.

Unequal spacing. Add a 120 call quoted at 0.9. The triple 90, 100, 120 has $w = (120-100)/(120-90) = \frac{2}{3}$, and the bound is $\frac{2}{3}(12.0) + \frac{1}{3}(0.9) = 8.3 \geq 6.5$: still consistent.

A violation that the slope test misses. Suppose the quotes are 11.0, 6.8 and 2.0 instead. The spreads, 4.2 and 4.8, still pass the band test, but the butterfly costs $11.0 - 13.6 + 2.0 = -0.6$. Buy one 90 call and one 110 call for 13.0 and sell two 100 calls for 13.6: the trade brings in 0.60 today, and it pays the tent $(S_T - 90)^+ - 2(S_T - 100)^+ + (S_T - 110)^+$, which is never negative:

S_T	85	95	100	105	115
Payoff of the butterfly	0	5	10	5	0

The slope test alone misses a violation of the curvature; a strike curve must pass both.

(Problem P17.8 works this computation at different numbers.)

Interview trap. "A call spread can cost anything up to its width." A European call spread costs at most its *discounted* width $e^{-rT}(K_2 - K_1)$; a quote above that is an arbitrage. Every digital is capped likewise, at e^{-rT}, by (17.4).

Interview trap. A negative butterfly price is free money or a stale quote. With unequal strikes the butterfly must be weighted by w and $1 - w$: the plain $1 : -2 : 1$ test applied to unequally spaced strikes gives wrong conclusions. In Example 17.25 the plain test on 90, 100, 120 would compute $12.0 - 13.0 + 0.9 = -0.1$ and report a violation where there is none.

Table 3 collects the signs. Each row moves one input alone. A row is *model-free* when its signs are proved in this section from no arbitrage alone, and *under the model* when they are properties of the Black–Scholes–Merton price of §20, proved there or in §21.

Input that rises	C	C^{am}	$\mathcal{P}$	$\mathcal{P}^{\mathrm{am}}$	Status
strike K	$-$	$-$	$+$	$+$	model-free (Proposition 17.23)
expiry T	$+$ or ?	$+$	?	$+$	model-free (Theorem 17.16(iii), Proposition 17.6(i)); the European call is $+$ in case (N) and ? with income
dividends before T	$-$	$-$	$+$	$+$	$C - \mathcal{P}$ model-free (Corollary 17.11); each sign under the model; †
spot S	$+$	$+$	$-$	$-$	under the model (Proposition 21.2); †
rate r, at fixed S	$+$	$+$	$-$	$-$	under the model (Proposition 21.9); †
volatility σ	$+$	$+$	$+$	$+$	under the model (Proposition 20.35); †

Table 3: Comparative statics of the four prices: the sign of the change when the input of the row rises alone. Every sign is weak: $+$ means nondecreasing and $-$ nonincreasing. The model-free entries are weak inequalities, and strictness needs a model. A question mark means that no arbitrage argument of this section fixes a sign: Example 17.26 gives both directions, the second in the Black–Scholes–Merton market (arbitrage-free by its assumption (A6), Remark 20.14). In a row marked † the entry for $\mathcal{P}^{\mathrm{am}}$, and the entry for C^{am} on an asset with income, are model statements, computable on a tree by Procedure 18.16 (with a yield on the trees of §18.7; with cash dividends on the escrowed-dividend tree that Remark 18.31 describes and Example 20.49 uses), and not proved in this Part; the entry for C^{am} on an asset without income follows from $C^{\mathrm{am}} = C$ (Theorem 17.16(i)), except in the dividend row, where every American entry is a model statement.

Strike. A higher strike makes a call worth less and a put worth more, for European and for American options, by Proposition 17.23. This row is model-free in all four columns, and Proposition 17.24

adds the convexity of the two European curves.

Expiry. American options are worth at least as much with a later expiry (Proposition 17.6(i)), and so is the European call on an asset without income (Theorem 17.16(iii)). The European put, and the European call on an asset with income, have no sign that an arbitrage argument of this section fixes: Example 17.26 moves each of them in both directions, downwards by the bounds of Proposition 17.6 and upwards in the Black–Scholes–Merton market.

Dividends. What is model-free is the difference: by Corollary 17.11, $C - \mathcal{P}$ falls by exactly the increase in the present value I of the dividends. The bounds move too: the upper bound of the European call falls by I, and the lower bounds of the European call and put move by I, down and up, wherever both the old and the new bound are positive (Example 17.8). That the call falls and the put rises, each separately, is a model statement: dividends lower the forward price, and the European prices move with the forward (Corollary 20.41 and Theorem 20.52, with Proposition 21.2). The American entries stand as the caption of Table 3 says.

Spot. Under the model a call gains and a put loses when the spot rises with everything else fixed: the delta of a European call is positive and that of a European put negative (Proposition 21.2 in §21). No arbitrage alone says nothing here, because two different spot prices describe two different markets. The American entries stand as the caption of Table 3 says.

Rate. With the spot held fixed, a higher rate raises the European call and lowers the European put under the model (Proposition 21.9 in §21). The American entries stand as the caption of Table 3 says.

Volatility. Under the model both European prices rise with the volatility: their common vega is positive (Proposition 20.35 in §20, restated in Proposition 21.9). The American entries stand as the caption of Table 3 says.

Example 17.26 (Expiry: five cases behind the expiry row)**.** Each case backs one entry of the expiry row of Table 3.

(1) A European put that falls with the expiry. Take the deep put of Example 17.19: $S = 1$, $K = 100$, $r = 5\%$, no dividends. By Proposition 17.6 its price lies in $[Ke^{-rT} - S,\ Ke^{-rT}]$, which is $[94.12, 95.12]$ at $T = 1$ and $[89.48, 90.48]$ at $T = 2$. The two-year European put is worth at most $100\,e^{-0.1}$ and the one-year put at least $100\,e^{-0.05} - 1$, and $100\,e^{-0.1} = 90.48 < 94.12 = 100\,e^{-0.05} - 1$: the longer put is worth strictly less.

(2) A European call that falls with the expiry, across a dividend. A stock at 100 announces a special dividend of 65 that goes ex at $t_1 = 0.5$; the rate is $r = 4\%$ and $K = 40$. The call that expires at $T = 0.25$, before the dividend, is worth at least $100 - 40\,e^{-0.01} = 60.398$ (Proposition 17.6(iii), case (N)). The call that expires at $T = 1$ is worth at most $S - I = 100 - 65\,e^{-0.02} = 36.29$ (Proposition 17.6(ii), case (I)). The longer European call is worth less. American options cannot do this (Proposition 17.6(i)).

(3) The + of the European call without income, as a trade. A stock without income trades at $S = 100$, the rate is $r = 4\%$, and two calls struck at $K = 100$ are quoted: 4.50 for three months and 4.30 for one year. The quotes violate Theorem 17.16(iii). Buy the one-year call and write the three-month call, which brings in 0.20 today. If the three-month call ends in the money at $T_1 = 0.25$, sell one share short at S_{T_1}, pay its holder $S_{T_1} - 100$ out of the proceeds and deposit the remaining 100, which is worth $100\,e^{0.03} = 103.05$ at $T_2 = 1$. The position is then worth

$$(S_{T_2} - 100)^+ - S_{T_2} + 103.05 \;=\; (100 - S_{T_2})^+ + 3.05 \;>\; 0,$$

by the identity $x^+ - x = (-x)^+$ at $x = S_{T_2} - 100$. If the three-month call ends out of the money, the one-year call is simply kept. This is the proof of Theorem 17.16(iii) in numbers.

(4) A European put that rises with the expiry. Let $r = q = 0$ and $K = S$. Parity (17.2) gives $\mathcal{P} = C$ at every expiry. In a Black–Scholes–Merton market with $\sigma = 20\%$ and $S = K = 100$ (Theorem 20.22 in §20; arbitrage-free by its assumption (A6), Remark 20.14), the at-the-money

put is worth 3.99 at three months and 5.64 at six: the put rises with the expiry. With case (1), the European put has no model-free sign in T.

(5) A yield, in both directions. Let $S = 100$, $q = 5\%$, $r = 4\%$ and $K = 10$. The one-year call is worth at least $100\,e^{-0.05} - 10\,e^{-0.04} > 85.51$ (Proposition 17.6(iii), case (Y)), and the ten-year call at most $100\,e^{-0.5} < 60.66$ (Proposition 17.6(ii)): the longer call is worth less. In the other direction, the Black–Scholes–Merton market with $q = 1\%$, $r = 4\%$, $\sigma = 20\%$ and $S = K = 100$ (Corollary 20.41(i); arbitrage-free by its assumption (A6), Remark 20.14) prices the at-the-money call at 4.35 at three months and 6.34 at six. With case (2), the European call on an asset with income has no model-free sign in T.

Intuition. Option payoffs are convex in S_T: they keep the good outcomes and cut off the bad ones. Dispersion therefore raises their value, as the re-roll already showed (§1.7), and this is why volatility helps the call and the put alike, and why more time helps whenever nothing else works against it. A higher rate lowers the present value of the strike, which the call pays and the put receives, and raises the forward price; dividends do the opposite to the forward.

Interview trap. "Longer-dated options are always worth more" holds for American options and for European calls on assets without income. It fails for a deep in-the-money European put, for a European call across a large dividend, and for a deep in-the-money European call on a high-yield asset (cases (1), (2) and (5) of Example 17.26).

Interview trap. "Higher rates raise every option price" is false: under the model puts fall (Proposition 21.9). The rate row of Table 3 holds the spot fixed. With $S = K = 100$, $\sigma = 20\%$, $T = 1$ and no dividends, the Black–Scholes–Merton call (Theorem 20.22 in §20) is worth 9.41 at $r = 3\%$ and 9.93 at $r = 4\%$, but 8.16 at $r = 4\%$ if the spot has meanwhile fallen to 97: the table does not sign a move of two inputs.

Interview trap. Under the model, higher volatility raises both the call and the put (Proposition 20.35); it is not bullish for one and bearish for the other. And convexity in the *strike* is model-free (Proposition 17.24), while convexity in the spot is a property of a model and is not claimed in this section.

17.6 Spreads: Bull, Bear, Box, Butterfly and Calendar

A spread combines options of one type on one underlying. The model-free strike results of §17.5 become trades here: a call spread is a bull spread, and convexity is a butterfly with a nonnegative price. Put–call parity (Theorem 17.9, in the forward form of Corollary 17.11) turns the box into a synthetic zero-coupon bond, which makes it a rate trade and not an option trade. Interviewers ask "what is a box spread?" and "what does this butterfly cost, and what does it pay?".

Setup. The standing assumptions are those of §17.2. The options are written on one underlying and have a common expiry T, unless a statement says otherwise; calendar spreads use two expiries $T_1 < T_2$. The strikes are $K_1 < K_2 < K_3$, and prices are those of time 0. The profit of a position is its payoff minus its cost, with the carry convention of §17.1; the examples and the figure of this subsection ignore the carry and say so. A position that costs a positive amount is a *debit*, and one that brings money in is a *credit*. The price of the box spread on $K_1 < K_2$ (Definition 17.27) is written $\mathrm{Box}(K_1, K_2)$, an upright claim name like $\mathrm{Dig}(K)$, because B is the bank holding of §18; the continuously compounded rate that a box price implies is written r_{box}.

Definition 17.27 (Spreads: bull, bear, box, butterfly, calendar, diagonal)**.** All options below are European unless stated, of one type on one underlying.

- A *bull spread* is long the K_1 option and short the K_2 option. Built with calls it pays

$$(S_T - K_1)^+ - (S_T - K_2)^+ = \min\left((S_T - K_1)^+,\ K_2 - K_1\right),$$

and its price $C(K_1) - C(K_2)$ is nonnegative (Proposition 17.23(i)): it is a debit as soon as $S_T > K_1$ has positive probability, because its payoff is then positive with positive probability (Lemma 16.8(ii)). Built with puts it pays $(K_1 - S_T)^+ - (K_2 - S_T)^+$, which lies in $[-(K_2 - K_1), 0]$ and is negative where $S_T < K_2$; its price $\mathcal{P}(K_1) - \mathcal{P}(K_2)$ is at most 0, and it is a credit as soon as $S_T < K_2$ has positive probability. Either way the profit rises with S_T between the strikes and is flat outside them.

- A *bear spread* is the reverse: short the K_1 option and long the K_2 option. Built with puts it pays $\min\big((K_2 - S_T)^+,\ K_2 - K_1\big)$ and is a debit; built with calls it is a credit, under the same provisos.
- Market usage calls a debit spread *aggressive* when both strikes are out of the money and *conservative* when both are in the money. By Proposition 17.23(i) its price lies in $[0, e^{-rT}(K_2 - K_1)]$. The names describe the payoff: with both strikes out of the money it reaches the full width only if the underlying moves through both strikes, while with both in the money it would pay the full width at today's price; where the price sits in the band is a model question.
- A *box* is a bull call spread together with a bear put spread on the same strikes $K_1 < K_2$. It pays $K_2 - K_1$ in every state, as Proposition 17.29 proves below.
- A *butterfly* is long one option at K_1 and one at K_3 and short two at the midpoint $K_2 = \frac{1}{2}(K_1 + K_3)$, with calls or with puts. With unequal spacing it is long w at K_1, short one at K_2 and long $1 - w$ at K_3, with the weight $w = (K_3 - K_2)/(K_3 - K_1)$ of Proposition 17.24.
- A *calendar spread* has one strike K and two expiries: it is short the option that expires at T_1 and long the option that expires at $T_2 > T_1$. The *reverse calendar* is the opposite position.
- A *diagonal spread* is long one option and short another of the same type whose strike and expiry both differ.

Example 17.28 (A butterfly and a bull spread from three quotes)**.** Three-month calls at the strikes 95, 100 and 105 are quoted at 7.71, 4.75 and 2.68. These are the prices that a model with 22% volatility gives at $S = 100$ and $r = 3\%$, but the example uses only the quotes; the carry is ignored throughout.

The butterfly. Buying one 95 call and one 105 call and writing two 100 calls costs $7.71 - 2(4.75) + 2.68 = 0.89$. Its payoff is

Range of S_T	$S_T \le 95$	$95 \le S_T \le 100$	$100 \le S_T \le 105$	$S_T \ge 105$
Payoff	0	$S_T - 95$	$105 - S_T$	0

the tent of Proposition 17.24 with $w = \frac{1}{2}$, scaled by two. The profit is largest at $S_T = 100$, where it is $5 - 0.89 = 4.11$. The loss is largest outside $[95, 105]$, where it is the premium 0.89. The breakevens are $95 + 0.89 = 95.89$ and $105 - 0.89 = 104.11$. Divided by its peak payoff, the price $0.89/5 = 0.178$ is, up to discounting, the weight that the quotes put on finishing near 100 (0.179 once the discount at 3% over three months is undone); Theorem 22.7 in §22 makes this reading exact. By Proposition 17.29(iii), proved below, the put butterfly on the same strikes costs the same 0.89.

The bull spread. From the same quotes, the 95/105 bull call spread costs $7.71 - 2.68 = 5.03$. It pays nothing below 95, $S_T - 95$ between the strikes and 10 above 105. It breaks even at $95 + 5.03 = 100.03$, makes at most $10 - 5.03 = 4.97$ and loses at most 5.03.

Figure 30 draws the payoff and the profit of both positions.

Interview trap. The maximum loss of a long equally spaced butterfly $(1 : -2 : 1)$ is its premium, and its maximum gain is $K_2 - K_1$ minus the premium (0.89 and 4.11 in Example 17.28); with the weights w, -1, $1 - w$ the peak payoff is $w(K_2 - K_1)$ (Proposition 17.24). An answer of "the butterfly loses if the stock moves" must quantify both, and name the breakevens.

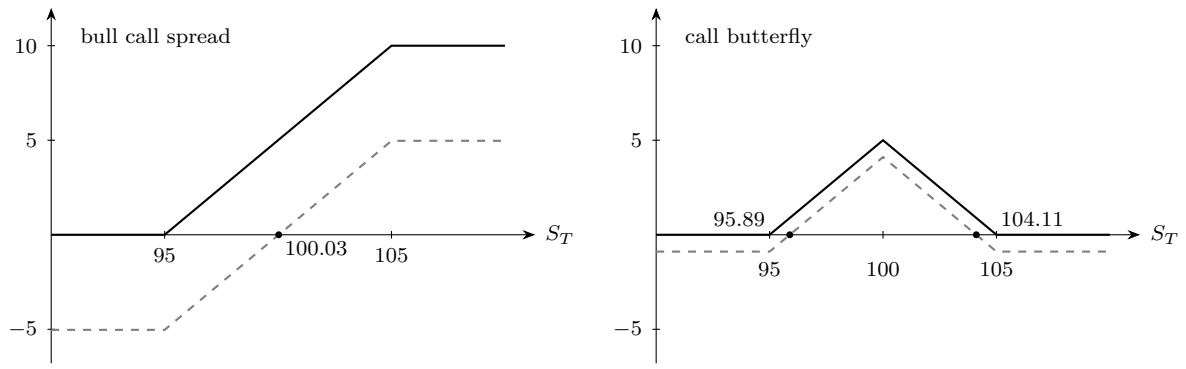


Figure 30: The two spreads of Example 17.28, on the same scales (the horizontal axis runs from $S_T = 90$ to $S_T = 110$). Left: the 95/105 bull call spread; its payoff (solid black) is zero up to 95, rises one for one and is flat at 10 from 105 on, and its profit (dashed gray) runs from -5.03 to $+4.97$. Right: the 95/100/105 call butterfly; its payoff (solid black) is a tent of height 5, and its profit (dashed gray) is the tent lowered by the premium 0.89. The dots mark the breakevens: 100.03 on the left, 95.89 and 104.11 on the right. The carry is ignored.

Proposition 17.29 (The box is a zero-coupon bond; spread parity)**.** *Assume the standing assumptions of §17.2, and consider European options with a common expiry T on one underlying, in any of the income cases (N), (I), (Y).*

(i) For strikes $K_1 < K_2$, the box (a bull call spread together with a bear put spread on K_1 and K_2) pays $K_2 - K_1$ in every state. Its price is therefore

$$\mathrm{Box}(K_1, K_2) = e^{-rT}(K_2 - K_1),$$

and a quoted box price $\mathrm{Box}(K_1, K_2) > 0$ implies the continuously compounded rate

$$r_{\mathrm{box}} = \frac{1}{T} \ln \frac{K_2 - K_1}{\mathrm{Box}(K_1, K_2)}.$$

(ii) Spread parity. *The bull call spread costs exactly $e^{-rT}(K_2 - K_1)$ more than the bull put spread on the same strikes, whose price is at most 0, and negative, a credit, as soon as $S_T < K_2$ has positive probability:*

$$[C(K_1) - C(K_2)] - [\mathcal{P}(K_1) - \mathcal{P}(K_2)] = e^{-rT}(K_2 - K_1).$$

(iii) For strikes $K_1 < K_2 < K_3$ and the weight $w = (K_3 - K_2)/(K_3 - K_1)$, the call butterfly and the put butterfly with the weights w, -1, $1 - w$ cost the same.

Proof. We compute the payoff of the box strike by strike, and we price it as a sure amount by Lemma 16.8(iv). Parts (ii) and (iii) then follow from parity in its forward form (17.3).

Part (i). The bull call spread is long the K_1 call and short the K_2 call, and the bear put spread is long the K_2 put and short the K_1 put. Held from 0 to T and grouped by strike, the four options pay

$$[(S_T - K_1)^+ - (K_1 - S_T)^+] - [(S_T - K_2)^+ - (K_2 - S_T)^+] = (S_T - K_1) - (S_T - K_2) = K_2 - K_1,$$

by the identity $x^+ - (-x)^+ = x$ of the proof of Corollary 17.11, at $x = S_T - K_1$ and at $x = S_T - K_2$. The box pays the sure amount $K_2 - K_1$, so it costs $e^{-rT}(K_2 - K_1)$ by Lemma 16.8(iv). The options pay nothing before T, so the income of the underlying plays no part. Solving $\mathrm{Box}(K_1, K_2) = e^{-r_{\mathrm{box}}T}(K_2 - K_1)$ for the rate gives the formula for r_{box}.

The same price follows from parity. Regrouped by strike, the box is a long call and a short put at K_1 together with a short call and a long put at K_2: two opposite synthetic forwards (Corollary 17.12(i)). By (17.3) at each strike,

$$\begin{aligned}\mathrm{Box}(K_1,K_2) &= \big[C(K_1)-\mathcal{P}(K_1)\big]-\big[C(K_2)-\mathcal{P}(K_2)\big]\\ &= e^{-rT}(F-K_1)-e^{-rT}(F-K_2)=e^{-rT}(K_2-K_1).\end{aligned}$$

The forward price F cancels, and with it the spot and the income.

Part (ii). Regroup the first line of the last display by option type: it is $[C(K_1)-C(K_2)]-[\mathcal{P}(K_1)-\mathcal{P}(K_2)]$, the bull call spread minus the bull put spread, and the display says that it equals $e^{-rT}(K_2-K_1)$. The bull put spread pays $(K_1-S_T)^+-(K_2-S_T)^+\le 0$, which is negative where $S_T<K_2$. Its price $\mathcal{P}(K_1)-\mathcal{P}(K_2)$ is therefore at most 0 by Proposition 17.23(i), and negative as soon as $S_T<K_2$ has positive probability, by Lemma 16.8(ii) applied to the empty portfolio and the bull put spread.

Part (iii). By (17.3), $C(K)-\mathcal{P}(K)=e^{-rT}(F-K)$ at every strike K. The call butterfly minus the put butterfly therefore costs

$$\begin{aligned}&w\big[C(K_1)-\mathcal{P}(K_1)\big]-\big[C(K_2)-\mathcal{P}(K_2)\big]+(1-w)\big[C(K_3)-\mathcal{P}(K_3)\big]\\ &\quad= e^{-rT}\Big(\big[w-1+(1-w)\big]F-\big[wK_1-K_2+(1-w)K_3\big]\Big)=0,\end{aligned}$$

because $w-1+(1-w)=0$ and $wK_1+(1-w)K_3=K_2$. The parity term is an affine function of the strike, and the weights of a butterfly annihilate affine functions. □

Example 17.30 (The rate implied by a box)**.** A stock that pays no dividends trades at $S=100$, the rate is $r=4\%$, and four six-month European options ($T=0.5$) are quoted:

$$C(95)=10.79,\qquad C(105)=5.78,\qquad \mathcal{P}(95)=3.91,\qquad \mathcal{P}(105)=8.70$$

(about 25% implied). Parity holds strike by strike to the cent: $C-\mathcal{P}$ is 6.88 at 95, against $100-95\,e^{-0.02}=6.88$, and -2.92 at 105, against $100-105\,e^{-0.02}=-2.92$.

The box. The bull call spread costs $10.79-5.78=5.01$ and the bear put spread costs $8.70-3.91=4.79$, so $\mathrm{Box}(95,105)=9.80$. It pays 10 at T, and Proposition 17.29(i) gives

$$r_{\mathrm{box}}=\frac{1}{0.5}\,\ln\frac{10}{9.80}=4.04\%,$$

the 4% behind the four quotes, up to the rounding of the quotes to the cent. A change of one cent in the box price, to 9.79 or to 9.81, moves the implied rate by about 20 basis points at this expiry and this width, which is why box rates are read off tight markets and wide strikes.

Spread parity. The bull put spread, long the 95 put and short the 105 put, costs $3.91-8.70=-4.79$: a credit. By part (ii), $5.01-(-4.79)=9.80=e^{-rT}\times 10$, to the cent.

Who trades it. A desk that can lend only at 3.5% buys the box, and lends at 4.04%; a desk that borrows at 4.5% sells it, and borrows at 4.04%.

(Problem P17.9 works this computation at different numbers.)

Trading application. The box rate is a financing rate: buying a box lends $\mathrm{Box}(K_1,K_2)$ until T at the rate r_{box}, and selling one borrows at it. On a crypto options venue the box rate can be compared with the two other financing rates the market quotes, the annualised basis of a dated future (§16.6, Example 16.34) and the funding of a perpetual future (Definition 15.3); a persistent gap between them reflects the different margin, collateral and counterparty terms of the three instruments, not an arbitrage.

Interview trap. A box is riskless only with European legs. A seller of a box is short the K_2 put. With American options on a single stock that put, once deep in the money, can be assigned early (Proposition 17.18): the seller then pays K_2 for the share and, delivering it through the long K_1 put, has paid $K_2 - K_1$ before T, and the fixed payment at T on which the pricing rests no longer holds. Selling a listed American box above $e^{-rT}(K_2 - K_1)$ is therefore not an arbitrage.

Interview trap. A bull spread does not always cost money. Built with puts it is a credit; its maximum loss is then $K_2 - K_1$ minus the credit, and the call version and the put version differ in cost by exactly $e^{-rT}(K_2 - K_1)$ (Proposition 17.29(ii)).

Remark 17.31 (A calendar spread has no model-free value). At T_1 the near leg of a calendar spread pays its intrinsic value, but the far leg is still a live option whose value depends on the volatility to come. The value of the calendar at T_1, and hence its value today, is therefore not fixed by no arbitrage. Two markets with the same S, the same strike $K = S$, the same rates $r = q = 0$ and the same expiries $T_1 < T_2$ show it: the first is proved free of arbitrage below; the second is the Black–Scholes–Merton market of §20, free of arbitrage by its assumption (A6), which is consistent with (A1)–(A5) by the theorem named in Remark 20.14 and not proved in this Part.

A market in which the price never moves. If $S_t = S$ at every date and $r = 0$, every strategy in the share and the cash account has a constant value. Both calls pay $(S - K)^+ = 0$; priced at 0 at every date, they keep every strategy in the share, the cash account and the two calls at a constant value, so the extended market has no arbitrage, and the calendar is worth 0.

The Black–Scholes–Merton market. With a volatility $\sigma > 0$, the at-the-money call is worth $S\big(2\Phi(\sigma\sqrt{T}/2) - 1\big)$, where Φ is the standard normal distribution function (Theorem 20.22 in §20, at $K = S$ and $r = q = 0$; Proposition 20.29 states this form). This is strictly increasing in T because Φ is strictly increasing. The calendar costs $C(K, T_2) - C(K, T_1) > 0$. With $S = K = 100$, $\sigma = 20\%$, $T_1 = 0.25$ and $T_2 = 0.5$ the two calls are worth 3.99 and 5.64: a calendar worth 1.65, against 0 in the first market.

The one model-free statement is a sign: on an asset without income and with $r \geq 0$, the call calendar costs at least 0, by Theorem 17.16(iii). Outside that case the sign is lost as well: by cases (1), (2) and (5) of Example 17.26, a put calendar, or a call calendar across a large dividend or on a high-yield asset, can have a negative price. A long calendar is a bet on the term structure of implied volatility: it is short the near-dated gamma and long the far-dated vega. It is taken up with Proposition 22.15 and the forward variance of §22.3.

Interview trap. A calendar spread cannot be priced from the spot, the strike and the rate alone. An interview answer that produces a number from those inputs has assumed a volatility, and should say which.

17.7 Combinations, Packages and Replicating Any Payoff

Combinations mix calls and puts (straddles, strangles, strips, straps, risk reversals, collars); packages add bonds or the underlying (covered calls, protective puts, principal-protected notes). Straddles and strangles are the volatility trades of interview questions, the principal-protected note is the classic "price this structured product" question, and "build me this payoff from calls" tests the replication result proved here. That result is the discrete ancestor of the Carr–Madan replication (Lemma 22.34 in §22) and of the static replication of exotic options (§23.1). The subsection builds on Corollary 17.12, Proposition 17.6 and Lemma 16.8.

Setup. The standing assumptions are those of §17.2. Calls and puts are European, written on one underlying, with a common expiry T; the strikes of a combination are K, or $K_1 \leq K_2$. The prepaid forward pays S_T and costs $e^{-rT}F$, as in the Setup of §17.2; in case (N) it is one share. Profits ignore the carry in the examples and the figure of this subsection.

For the replication result there are m strikes $K_1 < \cdots < K_m$ (here m counts strikes), with $K_0 = 0$, and a payoff function $f : [0, \infty) \to \mathbb{R}$ that is continuous and *piecewise linear with kinks at the strikes*: it is affine on each of the intervals $[0, K_1]$, $[K_1, K_2]$, $\ldots$, $[K_{m-1}, K_m]$ and $[K_m, \infty)$. Its slope is a_0 on $[0, K_1]$ and a_j on the interval that starts at K_j, for $j = 1, \ldots, m$. The slopes are written a_j because s is a time from §19 on and the sample standard deviation of §4, and the payoff is written f because g is the expected log-growth rate in the notation of this Part.

A principal-protected note has a stated principal; its *participation rate*, the fraction of the growth of the underlying that the note pays on top of the principal, is written in words and has no symbol.

Definition 17.32 (Combinations and packages)**.** All options have the same expiry T and the same underlying.

- A *straddle* is long a call and a put at the same strike K; it pays $|S_T - K|$. Selling both is a *short straddle*.
- A *strangle* is long a put at K_1 and a call at $K_2 > K_1$; it pays $(K_1 - S_T)^+ + (S_T - K_2)^+$.
- A *strip* is long one call and two puts at K, and a *strap* is long two calls and one put at K.
- A *risk reversal* is long the K_2 call and short the K_1 put, or the reverse, with $K_1 \le K_2$.
- A *collar* is long the share, long a put at K_1 and short a call at $K_2 > K_1$; it is *zero-cost* when the two premiums offset.
- The *covered call* (long the share, short a call) and the *protective put* (long the share, long a put) are packages. By Corollary 17.12, in which the share is the prepaid forward of the Setup of §17.2 (an actual share in case (N)), the covered call is a short put together with a bond paying K, and the protective put is a long call together with a bond paying K (with the dividend stream as well when an actual share pays known income, Corollary 17.12(ii)–(iii)).
- A *principal-protected note* is a zero-coupon bond that pays the principal at T together with a participation in an at-the-money call on the underlying, or in a bull spread.

Example 17.33 (A straddle against a strangle)**.** Take $r = 0$ for simplicity, a stock at 100, and three-month options quoted near 20% implied.

The straddle. An at-the-money straddle at $K = 100$ is bought for 8.0. Its profit is $|S_T - 100| - 8$. The breakevens are 92 and 108, and the loss is largest at $S_T = 100$, where the whole 8 is lost: the underlying must move 8% either way before the position makes money.

The strangle. A 95 put costs 1.89 and a 105 call costs 2.06, so the 95/105 strangle is bought for 3.95. Its profit is $(95 - S_T)^+ + (S_T - 105)^+ - 3.95$. The breakevens are $95 - 3.95 = 91.05$ and $105 + 3.95 = 108.95$, and the loss is capped at 3.95 on the whole interval $[95, 105]$. The strangle costs less than half as much as the straddle (3.95 against 8.0) and needs a bigger move.

Where the 8.0 *comes from.* The price of the at-the-money straddle is about $0.8\,\sigma\sqrt{T}\,S$, with $\sigma = 0.20$ the implied volatility of the quotes and $T = 0.25$: $0.8 \times 0.20 \times 0.5 \times 100 = 8.0$, an approximation made precise by Proposition 20.29 in §20.

Figure 31 draws the two profits on the same axes.

Interview trap. A long straddle is a bet that realised volatility will exceed the implied volatility priced into it, not merely a bet on "a big move": a move that the market expects is already in the premium. Unhedged, the straddle needs $|S_T - K|$ to exceed the premium. Delta-hedged over one rebalancing interval, its P and L is, up to a third-order remainder, gamma times half the realised-minus-implied variance (proved under the model, with $r = q = 0$, in Proposition 21.17 in §21).

Interview trap. A strangle is cheaper than a straddle but needs a larger move: compare the breakevens (91.05 and 108.95 against 92 and 108 in Example 17.33), not the premiums. Short straddles and short strangles have an unbounded loss on the upside.

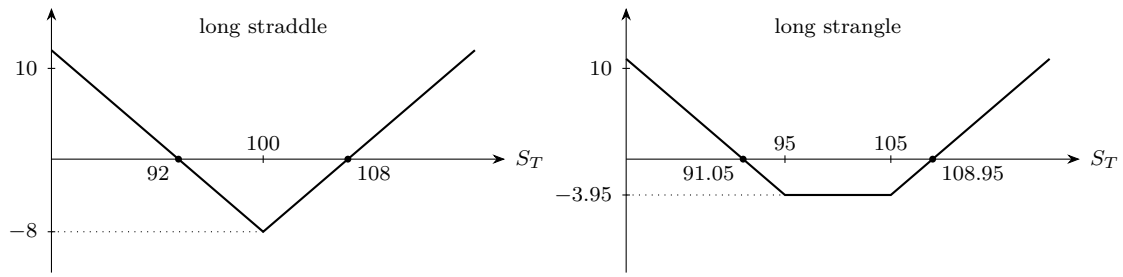


Figure 31: Profit at expiry of the two positions of Example 17.33. The two panels share one scale; the horizontal range, S_T from 80 to 120, is that of Figure 29; the carry is ignored. Left: the long straddle struck at 100 and bought for 8.0, a V with its lowest point -8 at $S_T = 100$ and breakevens 92 and 108. Right: the long 95/105 strangle bought for 3.95, flat at -3.95 between the strikes, with breakevens 91.05 and 108.95. The strangle loses less in the middle and needs a bigger move to make money.

Interview trap. A risk reversal is a forward only when its two strikes coincide (Corollary 17.12(i)). A zero-cost collar is not a free hedge: it pays for the put by selling the upside above K_2.

Trading application. Writing covered calls month after month is a short-volatility programme. By Corollary 17.12(ii) the covered call is a short put together with a bond, so the writer collects option premium and is short the insurance; on average that premium has exceeded what the insurance later cost, which is the variance risk premium of Fact 21.29 in §21.

Example 17.34 (A principal-protected note)**.** A five-year note has a principal of 1,000 and is linked to an index that pays a dividend yield of 2%; the five-year zero rate is 4%.

The budget. The zero-coupon bond that pays 1,000 at $T = 5$ costs $1{,}000\, e^{-0.20} = 818.73$, which leaves a budget of $1{,}000 - 818.73 = 181.27$ for options.

The participation. A five-year at-the-money call on 1,000 of the index costs 184.66 (a stated price; about 18% implied, and every option price below is stated at that same implied volatility). The budget buys $181.27/184.66 = 98.2\%$ of one call, so the note can pay the principal plus 98.2% of the rise of the index. An issuer margin of 15 leaves 166.27, and the participation falls to $166.27/184.66 = 90.0\%$.

Two structures, priced at the same implied volatility. (a) A cap at 1,300. The 1,300 call costs 89.68, so the 1,000/1,300 bull call spread costs $184.66 - 89.68 = 94.98$. The budget buys $181.27/94.98 = 190.9\%$ of the rise of the index up to a rise of 30%; the largest extra payment, reached at a rise of 300 or more, is $300 \times 181.27/94.98 = 572.55$. After the margin of 15 the participation is $166.27/94.98 = 175.1\%$. (b) A strike of 1,100. The 1,100 call costs 146.10, and the budget buys $181.27/146.10 = 124.1\%$ of the rise above 1,100, or $166.27/146.10 = 113.8\%$ after the margin. Each structure buys participation by giving up part of the payoff, (a) the growth beyond 1,300 and (b) the first 10%; and a lower rate shrinks the budget, which is $1{,}000(1 - e^{-5r})$. A higher volatility works the same way from the other side: it raises the price of the call that the budget must buy, and at 21% instead of 18% the five-year participation falls from 98.2% to 87.8%. A longer life restores participation, because the budget grows faster than the call: a seven-year note, its call priced at the same implied volatility as above, has a budget of $1{,}000(1 - e^{-0.28}) = 244.22$ against a call of 216.14, a participation of 113.0%.

Without a yield. With no yield, parity (17.2) at $K = S = 1{,}000$ gives

$$C = \left(S - Ke^{-rT}\right) + \mathcal{P} = 181.27 + \mathcal{P},$$

the budget plus the put. At the same implied volatility the call then costs $252.91 = 181.27 + 71.64$, where 71.64 is the at-the-money put. The put is worth a positive amount as soon as $S_T < K$ has positive probability (Lemma 16.8(ii), applied to the put and the empty portfolio), as in any model

with positive volatility; Proposition 17.6(i) and (iii) give only the weak form $\mathcal{P} \geq 0$. Whenever $S_T < K$ has positive probability, the call therefore costs strictly more than the budget, and full, uncapped participation is impossible on an asset without yield. Only in a market where $S_T \geq K$ with probability one, such as a price growing deterministically at the rate r, is the put worth 0 and the call worth exactly the budget.

(Problem P17.12 works this computation at different numbers.)

Interview trap. A principal-protected note is not free protection. In Example 17.34 the protection is paid for: the 181.27 that buys the option is the present value of five years of interest on the 1,000; the dividends of the index, worth $1{,}000(1 - e^{-0.10}) = 95.16$ today, are not passed on; and the 1,000 at maturity is a promise of the issuer, whose credit risk the holder bears.

Proposition 17.35 (Replicating a piecewise-linear payoff)**.** *Assume the standing assumptions of §17.2, and let $f : [0, \infty) \to \mathbb{R}$ be continuous and affine on each of $[0, K_1]$, $[K_1, K_2]$, ..., $[K_m, \infty)$, with slope a_0 on $[0, K_1]$ and slope a_j on the interval that starts at K_j, as in the Setup. Then, for every $x \geq 0$,*

$$f(x) = f(0) + a_0\, x + \sum_{j=1}^{m} (a_j - a_{j-1})\,(x - K_j)^+. \tag{17.5}$$

Consequently the claim that pays $f(S_T)$ at T is replicated by a bond paying $f(0)$, a_0 units of the prepaid forward (a_0 shares in case (N)) and $a_j - a_{j-1}$ calls struck at each kink K_j, a negative number being a short position. Its price is

$$f(0)\, e^{-rT} + a_0\, e^{-rT} F + \sum_{j=1}^{m} (a_j - a_{j-1})\, C(K_j).$$

Proof. We proceed by induction over the intervals, from left to right, and then apply the law of one price (Lemma 16.8(iii)). Write $\chi(x)$ for the right-hand side of (17.5), a local function, and write $[K_m, K_{m+1}]$ for the last interval $[K_m, \infty)$.

Step 1 (χ is affine on each interval, with the slope of f). Fix $j \in \{0, \dots, m\}$ and let x lie in the interval that starts at K_j. For $i \leq j$ the point x is at least $K_j \geq K_i$, so $(x - K_i)^+ = x - K_i$. For $i > j$ it is at most $K_{j+1} \leq K_i$, so $(x - K_i)^+ = 0$. On this interval, therefore,

$$\chi(x) = f(0) + a_0\, x + \sum_{i=1}^{j} (a_i - a_{i-1})\,(x - K_i),$$

an affine function of x whose slope is the telescoping sum

$$a_0 + \sum_{i=1}^{j} (a_i - a_{i-1}) = a_j.$$

By hypothesis f is affine on the same interval with the same slope a_j.

Step 2 (induction). On $[0, K_1]$ the sum in Step 1 is empty and $\chi(x) = f(0) + a_0\, x$. The function f is affine there with slope a_0 and value $f(0)$ at 0, so $f(x) = f(0) + a_0\, x = \chi(x)$ on $[0, K_1]$. Suppose that $\chi = f$ on $[0, K_j]$ for some $j \in \{1, \dots, m\}$. Then $\chi(K_j) = f(K_j)$, and on the interval that starts at K_j both functions are affine with slope a_j (Step 1), so that for x in that interval

$$\chi(x) = \chi(K_j) + a_j(x - K_j) = f(K_j) + a_j(x - K_j) = f(x).$$

Hence $\chi = f$ on $[0, K_{j+1}]$. After m steps, $\chi = f$ on $[0, \infty)$, which is (17.5).

Step 3 (price). Hold from 0 to T a bond paying $f(0)$, a_0 units of the prepaid forward and $a_j - a_{j-1}$ calls struck at K_j for each j. The payoff is

$$f(0) + a_0\, S_T + \sum_{j=1}^{m} (a_j - a_{j-1})\,(S_T - K_j)^+ = f(S_T),$$

by (17.5) at $x = S_T \geq 0$. The claim and the replica pay the same in every state, so Lemma 16.8(iii) equates their prices. The bond costs $f(0)\,e^{-rT}$ by Lemma 16.8(iv). The prepaid forward costs $e^{-rT}F$ (Setup of §17.2, by Theorem 16.11, Proposition 16.13 or Proposition 16.15(i)–(ii), according to the income case). The calls cost $C(K_j)$ each. Adding the three gives the stated price. □

(Problem P17.11 works this construction at other numbers.)

The weights $a_j - a_{j-1}$ are the jumps of the slope of f at the kinks: a discrete second derivative. Puts or forwards can replace calls through parity in forward form (17.3), which is (17.2) in case (N). A payoff with a jump, such as a digital, is not of this form and is only bracketed by call spreads (Proposition 23.2 in §23). Letting the kinks fill the whole line turns the sum into an integral. Expanded about the forward price F instead of 0, with puts below F and calls above it, this is Lemma 22.34 in §22, where the weight at the strike K is the second derivative of the payoff there, times dK. A row of narrow butterflies, each the tent of Proposition 17.24 centred at one strike, is a special case of (17.5).

Example 17.36 (A condor built from four calls)**.** The payoff f is zero below 90, rises one for one to 10 at 100, is flat to 110, falls one for one to zero at 120, and is zero above.

The weights. The kinks are $K_1 = 90$, $K_2 = 100$, $K_3 = 110$ and $K_4 = 120$; the slopes are $a_0 = 0$, $a_1 = 1$, $a_2 = 0$, $a_3 = -1$ and $a_4 = 0$; and $f(0) = 0$. The jumps of the slope are $+1$, -1, -1, $+1$, so by (17.5)

$$f(S_T) = (S_T - 90)^+ - (S_T - 100)^+ - (S_T - 110)^+ + (S_T - 120)^+,$$

with no bond and no share. This is a long call *condor*: a 90/100 bull spread together with a 110/120 bear spread.

A check. At $S_T = 115$ the four calls pay $25 - 15 - 5 + 0 = 5$, and the payoff described is $10 - (115 - 110) = 5$.

The price. With the six-month quotes $C(90) = 14.11$, $C(100) = 8.01$, $C(110) = 4.06$ and $C(120) = 1.86$ ($S = 100$, $r = 4\%$; about 25% implied) the condor costs

$$14.11 - 8.01 - 4.06 + 1.86 = 3.90.$$

Its payoff lies in $[0, 10]$, so by Lemma 16.8(i) and (iv) its price must lie in $[0,\ 10\,e^{-0.02}] = [0, 9.80]$, and it does.

A second instance. The straddle payoff is $|S_T - K| = K - S_T + 2(S_T - K)^+$: here $f(0) = K$, $a_0 = -1$ and $a_1 = +1$, so the weight at the kink is 2. The straddle is a bond paying K, one share sold short and two calls. In case (N) its price is $Ke^{-rT} - S + 2C$, which by (17.2) is $C + \mathcal{P}$: a call and a put.

Interview trap. Replication needs the value at zero (bonds) and the slope at zero (a_0 units of the underlying), and not only calls. A payoff that falls from $S_T = 0$, such as the straddle of Example 17.36, needs a short position in the underlying, or puts; forgetting either term misprices the claim.

Practice: problems P17.1–P17.12 in §27, solutions in §28.

18 Binomial Trees and Risk-Neutral Valuation

The binomial tree is the one model in which every pricing idea of this Part — replication, the risk-neutral measure, delta hedging, early exercise, the passage to Black–Scholes–Merton — can be proved with nothing beyond a two-by-two linear system, sums over the paths of the tree and Taylor expansions; the passage to the limit adds only the normal moment generating function and

one classical Fact (Fubini–Tonelli). This section builds the model in that order. Two trees run throughout: a one-step tree on which a stock at 40 moves to 50 or to 36, and a two-step tree that starts at 80 with $u = 1.15$ and $d = 0.9$. The section ends by proving that tree prices converge to the expectations that §20 evaluates in closed form (Theorem 18.37 in §18.8).

18.1 One-Step Replication and the No-Arbitrage Condition

The first question of almost every options interview is a one-step tree. A stock that can take only two values one period ahead is the smallest market in which an option has a forced price: two instruments, the stock and a riskless bond, span two states, so every payoff is a stock-and-bond portfolio, and the cost of that portfolio is the only price that leaves no arbitrage. This subsection proves the no-arbitrage condition on the two factors of the tree, solves the replication system and finds the risk-neutral weights inside the solution, resting on the law of one price, Lemma 16.8.

Setup. The price of a stock today is $S > 0$. After one step of length $\Delta t > 0$ (in years) the price is Su or Sd, where the *up-factor* u and the *down-factor* d satisfy $0 < d < u$ and are always written as bare letters. The first outcome is the *up state* (an up-move), the second the *down state* (a down-move). The stock pays no dividend. The continuously compounded interest rate is r, so that a riskless bond (a money-market account) worth 1 today is worth $e^{r\Delta t}$ at time Δt. The market is frictionless: short sales and fractional units are allowed, borrowing and lending take place at the same rate r, and there are no costs. The real-world probability of the up state is p_{real}, with $0 < p_{\text{real}} < 1$: both moves are possible.

A *portfolio* (Δ, B) holds Δ shares and B currency units in the bond; either number may be negative (a short sale, a loan). Its value today is $\Pi_0 = \Delta S + B$, and its payoff at time Δt is

$$\Delta Su + Be^{r\Delta t} \text{ in the up state,} \qquad \Delta Sd + Be^{r\Delta t} \text{ in the down state.}$$

With a single step there is no intermediate trading date, so the self-financing strategies of §16.2 that trade the stock and the bond are exactly these portfolios. Arbitrage is as in Definition 16.7. With two states, each of positive probability, the definition reads: a portfolio whose value today is at most zero and whose payoff is nonnegative in both states and positive in at least one. A portfolio with negative value and nonnegative payoffs becomes one once the proceeds are put in the bond, because its value is then zero and both payoffs are positive.

A *derivative* pays V_u in the up state and V_d in the down state at time Δt. Typically $V_u = f(Su)$ and $V_d = f(Sd)$ for a *payoff function* f, for instance $f(x) = (x - K)^+$ for a call struck at K and $f(x) = (K - x)^+$ for a put. Its price today is V. The prices today of a European call and a European put struck at K are written C and $\mathcal{P}$. The *risk-neutral up-probability* p is found inside the solution (Theorem 18.3), and $1 - p$ is the down-probability.

Symbol notices for this section.

(1) The letter r is the interest rate, not the ratio q/p of the Notation table.
(2) The value of the derivative is V (much of the literature writes f for it), and f is its payoff function.
(3) A bare d is the down-factor, never the d_1 or d_2 of the Black–Scholes–Merton formula; the down-probability is always written $1 - p$, because q is the dividend yield in this Part.
(4) The symbol Δ is a number of shares here, and §21 calls the same quantity the option's delta; the step Δt is a single symbol, and in this section ΔS is the value of the shares, Δ times S, never an increment of the price as in §16 and §21.
(5) The letter B is money in the bond, as B_t is the bank holding of a strategy in §20.
(6) Binomial coefficients are written $\binom{n}{j}$, never $C(n, j)$, because C is the call.
(7) The put is $\mathcal{P}$, because P is the real-world probability.

(8) The natural logarithm is written ln, and the literature's $N(d_1)$ is written $\Phi(d_1)$ (§18.8).

Proposition 18.1 (No arbitrage on a one-step tree)**.** *In the one-step market of the Setup, with* $0 < p_{\text{real}} < 1$*, the market of the stock and the bond admits no arbitrage if and only if*

$$d < e^{r\Delta t} < u. \tag{18.1}$$

Proof. We prove both directions by explicit construction: an arbitrage portfolio when the condition fails, and a sign argument on an arbitrary portfolio when it holds.

Only if. Suppose that (18.1) fails. Since $d < u$, either $e^{r\Delta t} \le d$ or $e^{r\Delta t} \ge u$. In the first case borrow S and buy one share: the portfolio $(\Delta, B) = (1, -S)$ has value $S - S = 0$ today, pays $Su - Se^{r\Delta t} = S(u - e^{r\Delta t}) > 0$ in the up state, because $u > d \ge e^{r\Delta t}$, and pays $S(d - e^{r\Delta t}) \ge 0$ in the down state. The up state has probability $p_{\text{real}} > 0$, so this portfolio is an arbitrage. In the second case short one share and lend S: the portfolio $(-1, S)$ has value 0 today, pays $S(e^{r\Delta t} - u) \ge 0$ in the up state and $S(e^{r\Delta t} - d) > 0$ in the down state, because $d < u \le e^{r\Delta t}$. The down state has probability $1 - p_{\text{real}} > 0$, so this portfolio is an arbitrage as well.

If. Assume (18.1), and let (Δ, B) be any portfolio with value $\Pi_0 = \Delta S + B \le 0$ and nonnegative payoffs in both states. Substituting $B = \Pi_0 - \Delta S$, the two payoffs are

$$\Delta S\,(u - e^{r\Delta t}) + \Pi_0 e^{r\Delta t} \quad \text{(up)}, \qquad \Delta S\,(d - e^{r\Delta t}) + \Pi_0 e^{r\Delta t} \quad \text{(down)}.$$

Nonnegativity of the first gives $\Delta S\,(u - e^{r\Delta t}) \ge -\Pi_0 e^{r\Delta t} \ge 0$; since $S > 0$ and $u - e^{r\Delta t} > 0$, this forces $\Delta \ge 0$. Nonnegativity of the second gives $\Delta S\,(d - e^{r\Delta t}) \ge -\Pi_0 e^{r\Delta t} \ge 0$; since $d - e^{r\Delta t} < 0$, this forces $\Delta \le 0$. Hence $\Delta = 0$, both payoffs equal $\Pi_0 e^{r\Delta t}$, and their nonnegativity together with $\Pi_0 \le 0$ gives $\Pi_0 = 0$. Both payoffs therefore vanish, neither is positive, and the portfolio is not an arbitrage. □

The hypothesis $0 < p_{\text{real}} < 1$ is what makes the "only if" direction true. With $p_{\text{real}} = 1$ and $e^{r\Delta t} = u$ the condition fails, yet there is no arbitrage: the up-state payoff of every portfolio is $\Delta S\,(u - e^{r\Delta t}) + \Pi_0 e^{r\Delta t} = \Pi_0 e^{r\Delta t}$, so a portfolio of value $\Pi_0 \le 0$ whose payoff is nonnegative with probability one pays exactly 0 in the up state, which is then the only possible one.

Example 18.2 (A tree that contains an arbitrage)**.** A stock at $S = 70$ moves to 72.10 or to 67.90 over six months, so $u = 1.03$ and $d = 0.97$, while $r = 9\%$ and $\Delta t = 0.5$. Then $e^{r\Delta t} = e^{0.045} = 1.046028 > u$: the bond beats the stock in both states and (18.1) fails. The second construction of the proof applies. Short one share and lend the 70 received. At time Δt the loan returns $70e^{0.045} = 73.222$, and buying the share back costs 72.10 or 67.90: a riskless profit of 1.122 in the up state and 5.322 in the down state, from zero outlay.

Figure 32 shows the one-step tree with the labels of the theorem that follows.

Theorem 18.3 (One-step replication and the binomial pricing formula)**.** *In the one-step market of the Setup, assume* $d < e^{r\Delta t} < u$*, and let a derivative pay* V_u *in the up state and* V_d *in the down state.*

(i) The portfolio of

$$\Delta = \frac{V_u - V_d}{S(u - d)} \quad \textit{shares} \qquad \textit{and} \qquad B = e^{-r\Delta t}\,\frac{uV_d - dV_u}{u - d} \quad \textit{in the bond} \tag{18.2}$$

pays V_u *in the up state and* V_d *in the down state, and no other portfolio of stock and bond does. Its cost is*

$$\Delta S + B = e^{-r\Delta t}\big[pV_u + (1 - p)V_d\big], \qquad p = \frac{e^{r\Delta t} - d}{u - d} \in (0, 1). \tag{18.3}$$

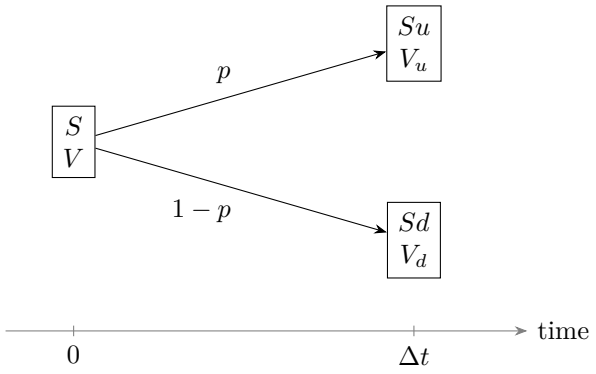


Figure 32: The one-step tree of Theorem 18.3. Each box shows the stock price above the value of the derivative; the branches carry the risk-neutral weights p and $1-p$, and the price at the root is $V = e^{-r\Delta t}[pV_u + (1-p)V_d]$. In Example 18.4 the stock moves from 40 to 50 or 36, a call struck at 45 is worth 5 or 0 at time Δt, and its price at the root is 1.809.

(ii) If the market of the stock, the bond and the derivative admits no arbitrage, the price of the derivative is

$$V = \Delta S + B = e^{-r\Delta t}\big[pV_u + (1-p)V_d\big]. \tag{18.4}$$

Proof. We solve a two-by-two linear system, regroup the cost of its solution, and then invoke the law of one price.

Step 1 (replication). A portfolio (Δ, B) pays V_u in the up state and V_d in the down state exactly when

$$\Delta Su + Be^{r\Delta t} = V_u, \qquad \Delta Sd + Be^{r\Delta t} = V_d. \tag{18.5}$$

This is a linear system in the unknowns Δ and B. Its determinant is $Su \cdot e^{r\Delta t} - Sd \cdot e^{r\Delta t} = S(u-d)e^{r\Delta t} \neq 0$, so it has exactly one solution. Subtracting the second equation from the first gives $\Delta S(u-d) = V_u - V_d$, which is the Δ of (18.2). Substituting it into the first equation,

$$Be^{r\Delta t} = V_u - \Delta Su = V_u - \frac{u(V_u - V_d)}{u-d} = \frac{(u-d)V_u - uV_u + uV_d}{u-d} = \frac{uV_d - dV_u}{u-d},$$

which is the B of (18.2).

Step 2 (the cost). With these values, and writing $1 = e^{-r\Delta t}e^{r\Delta t}$ in front of the first fraction,

$$\Delta S + B = \frac{V_u - V_d}{u-d} + e^{-r\Delta t}\,\frac{uV_d - dV_u}{u-d} = e^{-r\Delta t}\,\frac{V_u\,(e^{r\Delta t} - d) + V_d\,(u - e^{r\Delta t})}{u-d}.$$

The coefficient of V_u is $(e^{r\Delta t} - d)/(u-d) = p$, and that of V_d is $(u - e^{r\Delta t})/(u-d) = ((u-d) - (e^{r\Delta t} - d))/(u-d) = 1-p$, which is (18.3). The assumption $d < e^{r\Delta t} < u$ says that $0 < e^{r\Delta t} - d < u - d$, that is, $0 < p < 1$: this is condition (18.1) of Proposition 18.1 rearranged, and the condition holds exactly when $p \in (0,1)$. Steps 1 and 2 prove part (i); they use no price of the derivative.

Step 3 (the price). Buying the derivative today and holding it to time Δt is a strategy with price V and payoff V_u or V_d. The portfolio of Step 1 is a strategy with price $\Delta S + B$ and the same payoff in both states. Lemma 16.8(iii) therefore gives $V = \Delta S + B$, and the arbitrage behind it is the following. If $V > \Delta S + B$, sell the derivative and buy the portfolio: the position has value $(\Delta S + B) - V < 0$ today and payoff $V_u - V_u = 0$ or $V_d - V_d = 0$, so it becomes an arbitrage once the proceeds $V - (\Delta S + B)$ are put in the bond. If $V < \Delta S + B$, buy the derivative and sell the portfolio, that is, hold $(-\Delta, -B)$: the position has value $V - (\Delta S + B) < 0$ today and payoff zero in both states, again an arbitrage. Neither is possible, so $V = \Delta S + B$, and (18.3) turns this into (18.4). □

(Problem P18.3 asks you to reproduce this proof from scratch.)

The riskless-portfolio route found in many textbooks is the same equation. A position that is long Δ shares and short one derivative is worth $\Delta Su - V_u$ in the up state and $\Delta Sd - V_d$ in the down state, and the two are equal exactly when $\Delta S(u-d) = V_u - V_d$, the Δ of (18.2). The position is then riskless, so by Lemma 16.8(iv) it is worth $e^{-r\Delta t}(\Delta Su - V_u)$ today: $\Delta S - V = e^{-r\Delta t}(\Delta Su - V_u) = -B$, by the first equation of (18.5).

Example 18.4 (A call, a mispriced quote and a put on the tree from 40 to 50 or 36)**.** Take $S = 40$, $u = 1.25$, $d = 0.9$ (the price moves to 50 or to 36), $r = 6\%$ and $\Delta t = 0.5$. Then $e^{r\Delta t} = e^{0.03} = 1.0304545$ lies between d and u, so the tree contains no arbitrage. A call struck at $K = 45$ pays $V_u = 5$ and $V_d = 0$.

Replication. By (18.2), $\Delta = 5/(40 \times 0.35) = 5/14 = 0.3571$ shares and

$$B = e^{-0.03}\,\frac{1.25 \times 0 - 0.9 \times 5}{0.35} = -\frac{90}{7}\,e^{-0.03} = -12.477,$$

a loan. The shares cost $\frac{5}{14} \times 40 = 14.286$, so the call is worth $V = 14.286 - 12.477 = 1.809$.

Risk-neutral weights. Here $p = (1.0304545 - 0.9)/0.35 = 0.1304545/0.35 = 0.372727$, and (18.4) gives $V = e^{-0.03} \times 0.372727 \times 5 = 0.970446 \times 1.863636 = 1.809$, the same number.

Hedger's check. A position that is long 5/14 share and short one call is worth $17.857 - 5 = 12.857$ in the up state and $12.857 - 0 = 12.857$ in the down state. It is riskless, so its value today is the present value $12.857\,e^{-0.03} = 12.477$; hence $14.286 - V = 12.477$, the same price again.

A quote of 2.00. Sell the call at 2.00, buy 5/14 share for 14.286 and borrow 12.477: the trade pays $2.00 - 14.286 + 12.477 = 0.191$ today. At time Δt the loan has grown to 12.857, and the position is worth $17.857 - 5 - 12.857 = 0$ in the up state and $12.857 - 0 - 12.857 = 0$ in the down state. The 0.191 is kept whatever happens.

The put. On the same tree a put struck at 45 pays $V_u = 0$ and $V_d = 9$. Then $\Delta = -9/14 = -0.6429$, a short position worth 25.714, and $B = e^{-0.03} \times 1.25 \times 9/0.35 = +31.193$, a deposit; the put costs $31.193 - 25.714 = 5.479$, and the weights give the same number, $e^{-0.03} \times 0.627273 \times 9 = 5.479$. Put–call parity (Theorem 17.9) holds: $C - \mathcal{P} = 1.809 - 5.479 = -3.670$, and $S - Ke^{-r\Delta t} = 40 - 45e^{-0.03} = 40 - 43.670 = -3.670$.

(Problem P18.1 works this computation at different numbers.)

Example 18.5 (The 120/80 classic)**.** A stock at $S = 100$ moves to 120 or to 80, and $r = 0$; price a call struck at 100. The call pays $V_u = 20$ and $V_d = 0$, so $\Delta = 20/40 = 1/2$ and $B = (1.2 \times 0 - 0.8 \times 20)/0.4 = -40$: buy half a share for 50 and borrow 40, a cost of 10. Equivalently $p = (1 - 0.8)/0.4 = 1/2$ and $V = \frac{1}{2} \times 20 = 10$. The put struck at 100 is also worth $\frac{1}{2} \times 20 = 10$, as parity requires at $r = 0$, where $C - \mathcal{P} = S - K = 0$. A digital that pays 1 after an up-move is worth $p = 1/2$. None of these prices changes if the "true" up-probability is 0.75, because p_{real} appears nowhere in the computation.

(Problem P18.2 runs the full mental-arithmetic round on this tree.)

Interview trap. The real-world up-probability never enters. In Example 18.4, weighting the payoff 5 by a believed 70% up-probability and discounting at r gives $0.7 \times 5 \times e^{-0.03} = 3.40$, a price that the hedge of the example arbitrages at once: sell at 3.40, replicate for 1.81, keep 1.59. The interview question "why does the expected return of the stock not matter?" has a short answer. In each of the two states the option equals a fixed stock-and-bond portfolio, so its price is that portfolio's cost. Two economies with the same S, u, d and r but different expected returns give the same option price: the replicating portfolio (18.2) is built from those four numbers and the two payoffs, and nothing else.

Interview trap. Delta is a share count, the slope $(V_u - V_d)/(Su - Sd)$ of the payoff across the two states; it is not a probability. In Example 18.4, $\Delta = 0.357$ while $p = 0.373$. The two coincide only in special cases such as the symmetric 120/80 tree of Example 18.5, where both equal $\frac{1}{2}$.

Interview trap. Signs. Replicating a call struck at $K > 0$ borrows ($B \leq 0$) and replicating a put lends ($B \geq 0$), strictly once the option finishes in the money in at least one state; the Δ of a put is never positive, and it is negative once the put can finish in the money. A call or a put that finishes out of the money in both states is worth nothing, and its replica is empty. The riskless portfolio (long Δ shares, short one option) and the replica (long Δ shares plus the bond) are the same equation with the sign of the option flipped; mixing them up flips the bond position.

Interview trap. Check $d < e^{r\Delta t} < u$ before pricing anything. When the condition fails, the formula returns a weight p outside $(0, 1)$ and a "price" in a market that already contains an arbitrage: on the tree of Example 18.2, $p = (1.046028 - 0.97)/0.06 = 1.267$.

Interview trap. "What if the stock can take three values?" Then the stock and the bond no longer span the payoffs, and an option has no forced price. Let $S = 100$ move to 120, 100 or 80, each with positive probability, with $r = 0$, and price the call struck at 100, which pays 20, 0 or 0. Half a share bought with 40 borrowed costs 10 and pays 20, 10 or 0: at least the call in every state, and more in the middle one. So a call price $x \geq 10$ admits an arbitrage (sell the call and buy that portfolio: value $x - 10 \geq 0$ received today, payoffs 0, 10, 0), and so does $x \leq 0$ (buy the call: it costs nothing or less and pays 20 in the up state). Every price strictly between 0 and 10 is free of arbitrage. Given such a price x, put $w = x/20$, so that $0 < w < \frac{1}{2}$. The weights w, $1 - 2w$, w of the three states are positive, sum to 1, which is the price of the bond at $r = 0$, price the stock $(120w + 100(1 - 2w) + 80w = 100)$, and give the call the price $20w = x$. Prices and payoffs add over a portfolio, so under such weights the price of every portfolio of the three instruments is the weighted sum of its payoffs, and a portfolio whose payoff is nonnegative, and positive in some state, has a positive price: none is an arbitrage. The upper end, 10, is the price on the two-state tree of Example 18.5. Two states for two instruments is what makes Theorem 18.3 a pricing theorem rather than a bound.

18.2 Risk-Neutral Valuation

The weights p and $1 - p$ of Theorem 18.3 are positive and sum to one, so they define a probability, and under it every traded asset grows at the riskless rate. This subsection names that probability, proves the "everything earns r" property and its uniqueness, recovers the forward price of Theorem 16.11 on the tree, and then shows with real-world numbers why a real-world expectation discounted at r is the wrong price. Interviewers test it with "is p the market's forecast?" and "price a claim on $S_{\Delta t}^2$".

Setup. The market is the one-step market of §18.1, with $d < e^{r\Delta t} < u$. The random prices at time Δt are $S_{\Delta t}$ for the stock (Su or Sd) and $V_{\Delta t}$ for a derivative (V_u or V_d). The real-world probability is P, with expectation $\mathbb{E}$ and up-probability $p_{\text{real}} \in (0, 1)$. The *real-world expected return* μ of the stock, a continuously compounded rate, is defined by

$$\mathbb{E}[S_{\Delta t}] = Se^{\mu\Delta t}.$$

For a derivative with $V > 0$ and $\mathbb{E}[V_{\Delta t}] > 0$, the real-world expected return μ_V is defined by

$$V = e^{-\mu_V \Delta t}\, \mathbb{E}[V_{\Delta t}] :$$

it is the rate at which the real-world expected payoff must be discounted to give the price. Both conditions hold, for example, for any payoff $f \geq 0$ that is positive in at least one state, because p and p_{real} lie in $(0, 1)$. The letter is the μ of the stock with the asset as a subscript, and μ is the real-world drift of Proposition 16.31. The risk-neutral probability, expectation and variance are

written $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}$ and $\mathrm{Var}^{\mathbb{Q}}$, without hats, because hats denote estimators. The extension to several steps is declared in the Setup of §18.3.

Definition 18.6 (Risk-neutral probability on a one-step tree)**.** Assume $d < e^{r\Delta t} < u$. The *risk-neutral up-probability* is

$$p = \frac{e^{r\Delta t} - d}{u - d}.$$

The *risk-neutral probability* $\mathbb{Q}$ is the probability on the two states with $\mathbb{Q}(\text{up}) = p$ and $\mathbb{Q}(\text{down}) = 1 - p$, and $\mathbb{E}^{\mathbb{Q}}$ is its expectation: for a random quantity X equal to X_u in the up state and to X_d in the down state, $\mathbb{E}^{\mathbb{Q}}[X] = pX_u + (1-p)X_d$. Because $0 < e^{r\Delta t} - d < u - d$, that is, $p \in (0,1)$ (as recorded in (18.3); by Proposition 18.1 this is exactly the absence of arbitrage), $\mathbb{Q}$ gives positive probability to both states, so it agrees with P on which events are impossible. The real-world probability p_{real} plays no role in the definition.

On the tree of Example 18.4, $\mathbb{Q}(\text{up}) = 0.372727$ and $\mathbb{Q}(\text{down}) = 0.627273$, whatever p_{real} is. The continuous-time counterpart is Definition 20.6 in §20, where $\mathbb{E}^{\mathbb{Q}}$ is the expectation of the price path with its drift replaced by r, with no change of measure; risk-neutral valuation is proved there as Theorem 20.7, and Theorem 18.37 in §18.8 connects the tree's $\mathbb{Q}$ to that law.

Proposition 18.7 (Risk-neutral growth and the pricing rule)**.** *In the one-step market of the Setup, with $d < e^{r\Delta t} < u$:*

(i) The stock satisfies $\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = Se^{r\Delta t}$, that is, $pu + (1-p)d = e^{r\Delta t}$, and p is the only up-probability with this property.

(ii) If the market of the stock, the bond and a derivative admits no arbitrage, the price of the derivative is $V = e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}[V_{\Delta t}]$. Hence the $\mathbb{Q}$-expected value at time Δt of the stock, of the bond and of every derivative is its price today multiplied by $e^{r\Delta t}$: under $\mathbb{Q}$ every asset whose price is not zero has expected gross return $e^{r\Delta t}$.

(iii) Under the hypothesis of (ii), a forward contract with delivery price K and delivery at Δt is worth $S - Ke^{-r\Delta t}$, whatever u and d are, and the forward price is $F = \mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = Se^{r\Delta t}$, in agreement with Theorem 16.11.

Proof. We compute directly from the definition of p for part (i) and rearrange Theorem 18.3 for parts (ii) and (iii).

Part (i). By the definition of p,

$$\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = pSu + (1-p)Sd = S\big(d + p\,(u-d)\big) = S\big(d + e^{r\Delta t} - d\big) = Se^{r\Delta t}.$$

For uniqueness, let the up state have probability x. The expected price is then $xSu + (1-x)Sd = Sd + xS(u-d)$, an affine function of x with slope $S(u-d) \neq 0$, so it takes the value $Se^{r\Delta t}$ at exactly one x, which is p.

Part (ii). By (18.4), $V = e^{-r\Delta t}[pV_u + (1-p)V_d] = e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}[V_{\Delta t}]$, so $\mathbb{E}^{\mathbb{Q}}[V_{\Delta t}] = Ve^{r\Delta t}$. The stock satisfies the same identity by part (i). The bond is worth 1 today and $e^{r\Delta t}$ in both states, so its $\mathbb{Q}$-expected value is $e^{r\Delta t}$. Dividing each identity by a nonzero price gives the expected gross return $e^{r\Delta t}$.

Part (iii). The long side of the forward receives $S_{\Delta t} - K$ at time Δt (Definition 16.5), so the contract is a derivative with $V_u = Su - K$ and $V_d = Sd - K$. By part (ii), Theorem 1.2 and part (i), its value is

$$e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}[S_{\Delta t} - K] = e^{-r\Delta t}\big(\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] - K\big) = S - Ke^{-r\Delta t},$$

in which u and d no longer appear. The forward price is the delivery price at which a new contract costs nothing (Definition 16.5): $S - Ke^{-r\Delta t} = 0$ exactly at $K = Se^{r\Delta t} = \mathbb{E}^{\mathbb{Q}}[S_{\Delta t}]$, which is the forward price of Theorem 16.11 with $T = \Delta t$. □

On the 120/80 tree of Example 18.5, where $r = 0$ and $p = \frac{1}{2}$, the three parts read as follows. The risk-neutral mean of the stock is $\frac{1}{2} \times 120 + \frac{1}{2} \times 80 = 100 = S$, which is the forward price at a zero rate; the call is worth $\frac{1}{2} \times 20 + \frac{1}{2} \times 0 = 10$; and a forward with delivery price 90 is worth $\frac{1}{2} \times 30 + \frac{1}{2} \times (-10) = 10 = 100 - 90$. Part (ii) also prices the digital of that example at sight: a claim that pays 1 in the up state is worth $e^{-r\Delta t}\,\mathbb{Q}(\text{up}) = e^{-r\Delta t}p$, the discounted risk-neutral probability of being paid. Theorem 20.17 in §20 proves the continuous-time form of this statement.

Remark 18.8 (The real-world discount rate of an option). Let a derivative be priced by Theorem 18.3, with replicating portfolio (Δ, B). In both states $V_{\Delta t} = \Delta \cdot S_{\Delta t} + Be^{r\Delta t}$, by (18.5). Both sides take two values, so Theorem 1.2 applies to their real-world expectations: $\mathbb{E}[V_{\Delta t}] = \Delta\,\mathbb{E}[S_{\Delta t}] + Be^{r\Delta t}$. Subtracting $Ve^{r\Delta t} = (\Delta S + B)e^{r\Delta t}$ gives the one-line identity

$$\mathbb{E}[V_{\Delta t}] - Ve^{r\Delta t} = \Delta\big(\mathbb{E}[S_{\Delta t}] - Se^{r\Delta t}\big). \tag{18.6}$$

For $V > 0$, dividing (18.6) by V,

$$\frac{\mathbb{E}[V_{\Delta t}]}{V} - e^{r\Delta t} = \frac{\Delta S}{V}\left(\frac{\mathbb{E}[S_{\Delta t}]}{S} - e^{r\Delta t}\right): \tag{18.7}$$

the option's expected excess gross return is its *elasticity* $\Delta S/V$ times that of the stock. Equivalently, when μ_V is defined, moving $e^{r\Delta t}$ to the right and using $V = \Delta S + B$,

$$e^{\mu_V \Delta t} = \frac{\mathbb{E}[V_{\Delta t}]}{V} = \frac{\Delta S}{V}\,e^{\mu\Delta t} + \frac{B}{V}\,e^{r\Delta t}, \qquad \frac{\Delta S}{V} + \frac{B}{V} = 1, \tag{18.8}$$

a weighted average of the two gross returns, with weights that sum to one but need not lie between 0 and 1.

Suppose that the stock carries a positive risk premium, $\mu > r$. A call with strike $K > 0$ and $Su > K$ pays $V_u = Su - K > 0$ and $V_d = (Sd - K)^+ \geq 0$, so $V > 0$ and $\mathbb{E}[V_{\Delta t}] > 0$; and $uV_d - dV_u$ equals $-K(u - d)$ when $Sd \geq K$ and $-d\,(Su - K)$ when $Sd < K$, so $B < 0$ by (18.2). Its elasticity is therefore $\Delta S/V = 1 - B/V > 1$, and subtracting $e^{\mu\Delta t}$ from both sides of (18.8) gives $e^{\mu_V\Delta t} - e^{\mu\Delta t} = (\Delta S/V - 1)(e^{\mu\Delta t} - e^{r\Delta t}) > 0$: the call is a levered long position in the stock, and $\mu_V > \mu$. A put with $Sd < K$ pays $V_d = K - Sd > 0$ and $V_u = (K - Su)^+ < V_d$, so $V > 0$, $\mathbb{E}[V_{\Delta t}] > 0$ and $\Delta < 0$. Its elasticity is negative, and (18.7), whose last bracket is $e^{\mu\Delta t} - e^{r\Delta t} > 0$, gives $e^{\mu_V\Delta t} < e^{r\Delta t}$: the put is a short position in the stock, and $\mu_V < r$, possibly negative. A call with $K \geq Su$ pays nothing, is worth nothing and has no μ_V.

The same slope governs beta. Let R_M be any reference return on the same probability space with $\mathrm{Var}(R_M) > 0$, as in Definition 11.35. The option's return over the step,

$$\frac{V_{\Delta t}}{V} - 1 = \frac{\Delta S}{V}\left(\frac{S_{\Delta t}}{S} - 1\right) + \frac{B}{V}(e^{r\Delta t} - 1),$$

is an affine function of the stock's return with slope $\Delta S/V$. By bilinearity of the covariance, the constant term contributing nothing,

$$\mathrm{Cov}\left(\frac{V_{\Delta t}}{V} - 1,\ R_M\right) = \frac{\Delta S}{V}\ \mathrm{Cov}\left(\frac{S_{\Delta t}}{S} - 1,\ R_M\right),$$

and dividing by $\mathrm{Var}(R_M)$ shows that the option's beta is $\Delta S/V$ times that of the stock.

The rate μ_V depends on the payoff, and it is an output of the price, never an input. The risk-neutral probability is the computational device that makes the single rate r correct for every asset at once; it is not a forecast, and it is not a statement that investors are risk-neutral.

Example 18.9 (Real-world expected returns on the tree from 40 to 50 or 36)**.** Take the tree of Example 18.4, on which $p = 0.372727$ and $e^{r\Delta t} = 1.030455$, and let $p_{\text{real}} = 0.7$. To six decimals the call struck at 45 is worth 1.808557 and the put struck at 45 is worth 5.478606.

The stock. Under $\mathbb{Q}$ the mean of $S_{\Delta t}$ is $0.372727 \times 50 + 0.627273 \times 36 = 18.636 + 22.582 = 41.218 = 40e^{0.03}$, the six-month forward price (Proposition 18.7). Under P it is $0.7 \times 50 + 0.3 \times 36 = 45.8$, so $e^{\mu \Delta t} = 45.8/40 = 1.145$ and $\mu = 2 \ln 1.145 = 27.08\%$ a year. The stock's expected excess gross return is $1.145 - 1.030455 = 0.114545$.

The call. The call has real-world expected payoff $0.7 \times 5 = 3.5$, against its price 1.809. Discounting 3.5 at r gives $3.5e^{-0.03} = 3.397$, and discounting it at μ gives $3.5/1.145 = 3.057$. Both are wrong. The rate that reproduces the price is the call's own expected return, $\mu_V = 2\ln(3.5/1.808557) = 2 \ln 1.935244 = 132.0\%$ a year. The identity (18.7) checks it: the call's elasticity is $\Delta S/V = 14.28571/1.808557 = 7.8990$, and its expected excess gross return, $1.935244 - 1.030455 = 0.90479$, is $7.8990 \times 0.114545 = 0.90479$.

The put, from the identity. The put's elasticity is $\Delta S/V = -25.71429/5.478606 = -4.6936$. By (18.7) its expected excess gross return is $-4.6936 \times 0.114545 = -0.53763$, so $e^{\mu_V \Delta t} = 1.030455 - 0.53763 = 0.4928$ and $\mu_V = 2 \ln 0.4928 = -141.5\%$ a year: the holder of the put expects to lose money. The direct computation agrees: the put's expected payoff is $0.3 \times 9 = 2.7$, and $2.7/5.478606 = 0.4928$.

(Problem P18.4 works this computation at different numbers.)

Example 18.10 (A claim on the square of the price)**.** On the tree of Example 18.4, a claim pays $S_{\Delta t}^2$ at time Δt: 2500 in the up state and 1296 in the down state. By Proposition 18.7(ii),

$$\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}^2] = 0.372727 \times 2500 + 0.627273 \times 1296 = 1744.76, \qquad V = e^{-0.03} \times 1744.76 = 1693.20.$$

Replication check. By (18.2), $\Delta = (2500 - 1296)/14 = 86$ shares and $B = e^{-0.03}(1.25 \times 1296 - 0.9 \times 2500)/0.35 = -1800\, e^{-0.03} = -1746.80$, and $86 \times 40 - 1746.80 = 1693.20$.

Closed form. Expanding $(u+d)(pu+(1-p)d) = pu^2 + (1-p)d^2 + ud$ and using $pu + (1-p)d = e^{r\Delta t}$ (Proposition 18.7(i)) gives $pu^2 + (1-p)d^2 = (u+d)e^{r\Delta t} - ud$, so

$$V = e^{-r\Delta t} S^2 [pu^2 + (1-p)d^2] = S^2 [(u+d) - ud\, e^{-r\Delta t}] = 1600 \times 1.058249 = 1693.20.$$

What the naive answers miss. The guesses $S^2 = 1600$ and $S^2 e^{r\Delta t} = 1648.73$ (the square of the forward price $Se^{r\Delta t}$, discounted) are both too low. By the shortcut (1.1) under $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}^2] = (Se^{r\Delta t})^2 + \mathrm{Var}^{\mathbb{Q}}(S_{\Delta t})$, so $V = S^2 e^{r\Delta t} + e^{-r\Delta t}\, \mathrm{Var}^{\mathbb{Q}}(S_{\Delta t})$. The price $S_{\Delta t}$ equals Sd plus $S(u-d)$ times the indicator of the up state; adding a constant does not change a variance and a constant factor multiplies it by the square of the factor, both directly from the definition of the variance, so Proposition 1.10 gives $\mathrm{Var}^{\mathbb{Q}}(S_{\Delta t}) = p(1-p)(Su - Sd)^2 = 0.372727 \times 0.627273 \times 14^2 = 45.825$, and the missing term is $0.970446 \times 45.825 = 44.47 = 1693.20 - 1648.73$. It is the convexity of the payoff, priced.

(Problem P18.6 extends this example to n steps and to the limit.)

Intuition. Holding S, u, d and r fixed while changing p_{real} changes the expected return of the stock but not its price. That is coherent only if the market's risk premium adjusts: a higher p_{real} at an unchanged price means that investors demand a higher expected return for bearing the same two-state risk. The option, which is a stock-and-bond portfolio state by state, inherits whatever premium the stock carries, scaled by its elasticity.

Interview trap. "Risk-neutral" does not mean that investors are risk-neutral, or that p is anyone's forecast: in Example 18.9, $p = 0.373$ while the real-world up-probability is 0.7. In an interview, say instead: p and $1-p$ are the unique weights under which every traded asset earns r, so that one discount rate prices everything.

Interview trap. Discounting a real-world expected payoff at r (3.397) or at the stock's μ (3.057) overprices the call of Example 18.9. Only the call's own $\mu_V = 132\%$ a year works, and it can be computed only after the price is known, which is why pricing is done under $\mathbb{Q}$. The identity

(18.7) gives the sign of every option's excess return at once: it is the sign of the elasticity times that of the stock's. A put has a negative elasticity, so on a stock with a positive risk premium its expected return lies below r and can be negative (-142% a year in the example): the put is a hedge, and its holders accept an expected return below r, in the example an expected loss.

Interview trap. The risk-neutral mean $\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}]$ is the forward price for delivery at Δt, not today's price: under $\mathbb{Q}$ the stock drifts at r. In Example 18.9 it is 41.218, against $S = 40$.

Interview trap. A claim on $S_{\Delta t}^2$ is not worth the square of the forward price, discounted. Convexity adds the discounted $\mathbb{Q}$-variance: 44.47 on top of 1648.73 in Example 18.10.

18.3 Multi-Step Trees and the Binomial Pricing Formula

Chaining one-step trees gives a recombining lattice on which any European payoff is priced by stepping backward, one application of Theorem 18.3 per node. The backward recursion then collapses into a single binomial sum, the discounted $\mathbb{Q}$-expectation of the payoff with the number of up-moves binomially distributed (Definition 1.9): the discrete ancestor of the Black–Scholes–Merton formula. Two-step puts with a parity check are standard written-test items.

Setup. The maturity $T > 0$ is split into $n \geq 1$ steps of length $\Delta t = T/n$. At every step the stock price is multiplied by u or by d, with the same u, d and r at each step and $d < e^{r\Delta t} < u$; the stock pays no dividend, and at every step the market is that of §18.1. A *path* is a sequence of n moves, each up or down, so there are 2^n paths. Because the order of the factors does not change a product, the tree *recombines*: the price after i steps depends only on the number j of up-moves among them. *Node* (i, j), with $0 \leq j \leq i \leq n$, is the state at time $i\Delta t$ after j up-moves. It carries the price

$$S_{i,j} = Su^j d^{i-j},$$

and it is reached by $\binom{i}{j}$ of the 2^i sequences of i moves, the number of ways to choose which j of them are up. An up-move leads from node (i, j) to node $(i + 1, j + 1)$, where the price is $S_{i,j}u$, and a down-move to node $(i + 1, j)$, where it is $S_{i,j}d$; these are the two *children* of (i, j). The nodes (n, j) are *terminal*. The random price at time $i\Delta t$ is $S_{i\Delta t}$, so that $S_T = S_{n\Delta t}$.

For a European payoff function f, paid as $f(S_T)$ at time T, the *backward-induction values* are the numbers defined by

$$V_{n,j} = f(S_{n,j}), \qquad V_{i,j} = e^{-r\Delta t}\big[pV_{i+1,j+1} + (1-p)V_{i+1,j}\big] \quad (0 \leq j \leq i < n), \tag{18.9}$$

with the p of Definition 18.6. They are numbers computed from the tree, with no trading assumption. That $V_{i,j}$ is the price of the derivative at node (i, j) when the derivative trades at every node is Proposition 18.11(i). When the derivative trades only today, Proposition 18.13 proves that $V_{0,0}$ is its price and that each $V_{i,j}$ is the value at node (i, j) of a self-financing portfolio of stock and bond that replicates it.

The real-world probability P gives every path positive probability: whatever the path so far, the next move is up with positive probability and down with positive probability. This is all that the arbitrage arguments at interior nodes use. Under the risk-neutral probability $\mathbb{Q}$ the moves are independent, each up with probability p. Precisely, $\mathbb{Q}$ is defined on the n-step tree by giving each path the product of its one-step probabilities, so that a path with j up-moves has probability $p^j(1-p)^{n-j}$, and $\mathbb{E}^{\mathbb{Q}}$ is the corresponding weighted sum over the 2^n paths. These probabilities sum to one, because expanding the product $\prod_{i=1}^{n}\big(p + (1-p)\big) = 1$ gives exactly one term per path. The number of up-moves among the n steps is the random variable Y_n.

Symbol notice. The letter n counts tree steps throughout this section, and binomial coefficients follow notice (6) of §18.1. The up-move count is called Y_n because J is the number of time steps of a simulated path in §23. Two subscripts on a price name a node, $S_{i,j}$ (step i, j up-moves), not

an asset and a time as in the Notation table's $S_{k,t}$; a single subscript, as in $S_{i\Delta t}$, is a time. The same reading holds for $V_{i,j}$, and later for $\Delta_{i,j}$ and $B_{i,j}$.

Proposition 18.11 (European price on an n-step tree)**.** *On the n-step tree of the Setup, let $V_{i,j}$ be the backward-induction values (18.9) of a European payoff f.*

(i) If the derivative trades at every node and there is no arbitrage, then on every path that reaches node (i,j) its price at time $i\Delta t$ is $V_{i,j}$, whichever moves led there.

(ii) At every node (i,j), with $m = n - i$ steps remaining,

$$V_{i,j} = e^{-rm\Delta t} \sum_{k=0}^{m} \binom{m}{k} p^k (1-p)^{m-k} f(S_{i,j} u^k d^{m-k}). \tag{18.10}$$

In particular, at the root,

$$V_{0,0} = e^{-rn\Delta t} \sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j} f(S u^j d^{n-j}) = e^{-rT}\, \mathbb{E}^{\mathbb{Q}}[f(S_T)], \tag{18.11}$$

where $S_T = S u^{Y_n} d^{n-Y_n}$ and $Y_n \sim \mathrm{Bin}(n,p)$ under $\mathbb{Q}$.

Sketch of the proof. Part (i) is a backward induction on i from the payoff at maturity; part (ii) is an induction on the number m of remaining steps. If (i) holds at level $i+1$, the step out of node (i,j) is a one-step tree with payoffs $V_{i+1,j+1}$ and $V_{i+1,j}$. By Theorem 18.3(i) its replica costs $V_{i,j}$, and the law of one price (Lemma 16.8(v)) makes this the price, because the derivative trades at the node and every path has positive probability. For (ii), put the formula at the two children into (18.9) and add the coefficients by Pascal's rule,

$$\binom{m-1}{k-1} + \binom{m-1}{k} = \binom{m}{k};$$

no trading assumption is used. Grouping the 2^n paths by their number of up-moves gives the expectation. The complete proof is in Appendix D.1. □

The induction step of part (ii) is the tower rule (Proposition 1.7) in disguise: it conditions the $\mathbb{Q}$-expectation of the discounted payoff on the first move out of the node. The mean np and the variance $np(1-p)$ of the count Y_n are recorded in Table 1. Part (i) assumes that the derivative trades at every node. §18.4 removes that assumption: by Proposition 18.13, the value $V_{0,0}$ is forced even when the derivative trades only today.

Put–call parity holds exactly on every European tree, for two reasons. First, tree values are linear in the payoff: by (18.11) and Theorem 1.2, applied to $\mathbb{E}^{\mathbb{Q}}$, and because $(x-K)^+ - (K-x)^+ = x - K$ for every real x, the root value of the call minus that of the put is $e^{-rT}(\mathbb{E}^{\mathbb{Q}}[S_T] - K)$. Second, the tree prices the stock correctly: for the payoff $f(x) = x$ the backward-induction values are the node prices themselves, by backward induction on i, because $V_{n,j} = S_{n,j}$ and, if the children of (i,j) carry $V_{i+1,j+1} = S_{i,j}u$ and $V_{i+1,j} = S_{i,j}d$, then $V_{i,j} = e^{-r\Delta t} S_{i,j}(pu + (1-p)d) = S_{i,j}$ by Proposition 18.7(i); hence $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[S_T] = S$ by (18.11). With C and $\mathcal{P}$ the root values of the call and the put struck at K,

$$C - \mathcal{P} = S - K e^{-rT}, \tag{18.12}$$

the relation of Theorem 17.9, here an identity between numbers computed on the tree that needs no arbitrage argument.

Example 18.12 (A two-step put and call with a parity check)**.** Take $S = 80$, $u = 1.15$, $d = 0.9$, $r = 3\%$ and $\Delta t = 0.25$, so that $n = 2$ and $T = 0.5$, and the strike $K = 80$. The stock nodes are 80; then 92 and 72; then 105.8, 82.8 and 64.8. Since $e^{r\Delta t} = e^{0.0075} = 1.0075282$ lies between 0.9 and 1.15, the weights are $p = (1.0075282 - 0.9)/0.25 = 0.1075282/0.25 = 0.430113$ and $1 - p = 0.569887$,

and the one-step discount factor is $e^{-r\Delta t} = 0.992528$. Figure 33 shows the tree with the values computed below. Each amount below is rounded separately, so the last digit of a sum or difference of rounded amounts can differ by one from the rounded exact value.

European put. The terminal values are 0, 0 and $80 - 64.8 = 15.2$. By (18.9) the up node is worth 0, the down node $0.992528 \times 0.569887 \times 15.2 = 8.59756$, and the root $e^{-r\Delta t}(1-p) \times 8.59756 = 4.8630$. The closed form (18.11) has a single nonzero term, $j = 0$: $e^{-0.015}(1-p)^2 \times 15.2 = 0.985112 \times 0.324771 \times 15.2 = 4.8630$.

European call. The terminal values are 25.8, 2.8 and 0. The up node is worth $0.992528\,(0.430113 \times 25.8 + 0.569887 \times 2.8) = 12.5978$, the down node $0.992528 \times 0.430113 \times 2.8 = 1.1953$, and the root $0.992528\,(0.430113 \times 12.5978 + 0.569887 \times 1.1953) = 6.0541$. The closed form gives $e^{-0.015}[p^2 \times 25.8 + 2p(1-p) \times 2.8] = 0.985112\,(4.77292 + 1.37265) = 6.0541$.

Parity. The difference is $C - \mathcal{P} = 6.0541 - 4.8630 = 1.191$, and $S - Ke^{-rT} = 80 - 80e^{-0.015} = 80 - 78.809 = 1.191$, as the tree identity (18.12) guarantees and Theorem 17.9 requires.

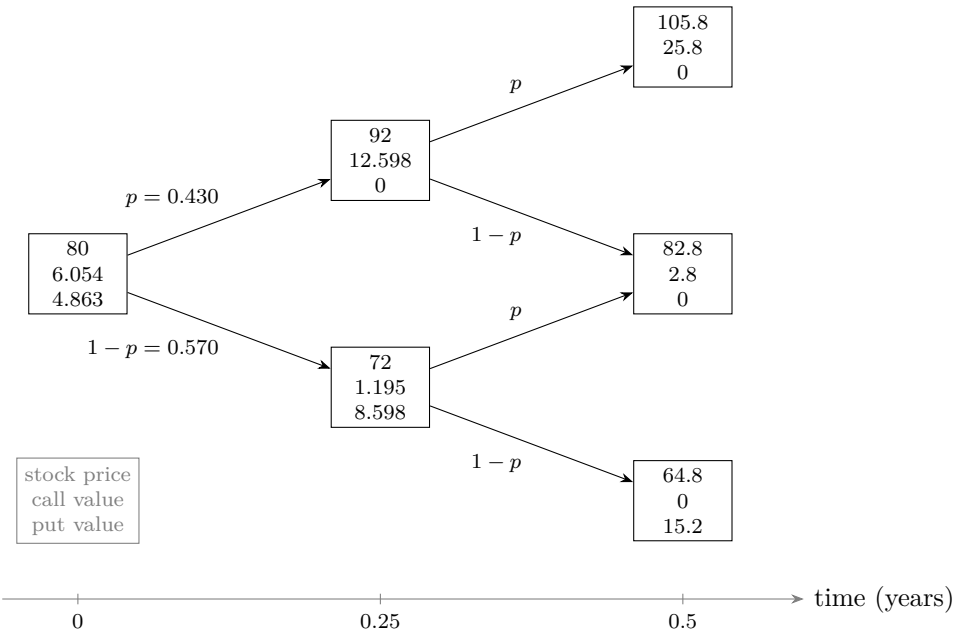


Figure 33: The recombining two-step tree of Example 18.12: $S = 80$, $u = 1.15$, $d = 0.9$, $r = 3\%$, $\Delta t = 0.25$. Each box shows, from top to bottom, the stock price, the value of the European call struck at 80 and the value of the European put struck at 80, as the gray box indicates. Up-branches carry the risk-neutral weight $p = 0.430$ and down-branches $1 - p = 0.570$. The middle terminal node is reached by two paths.

Interview trap. A tree has 2^n paths but only $n + 1$ terminal nodes, and the binomial coefficient counts the paths into each node. Dropping the factor 2 on the middle node of the two-step tree of Example 18.12 gives a call price of 5.378 instead of 6.054; it is the most common arithmetic slip on a two-step tree.

Interview trap. The backward induction (18.9) prices payoffs of S_T alone, because a node forgets how it was reached. A payoff that depends on the path is priced over the 2^n paths instead. On the tree of Example 18.12, the lookback call paying $\max(S, S_{\Delta t}, S_T) - 80$ pays 25.8, 12, 2.8 and 0 on the paths up-up, up-down, down-up and down-down; the two paths into the middle node pay different amounts. With $p^2 = 0.184997$ and $p(1-p) = 0.245116$, it is worth $e^{-0.015}\,[p^2 \times 25.8 + p(1-p)(12 + 2.8)] = 8.2756$, against 6.0541 for the European call. Proposition 18.29(i) in §18.7 proves that such path sums price every claim that is a function of the path; its rates may in particular be constant, as here.

Interview trap. Discount once per backward step, by $e^{-r\Delta t}$ at every node, or once by e^{-rT} on the terminal sum, never both.

Interview trap. The tree recombines because the same u and d multiply the price at every node, so the order of the moves does not matter: $Sud = Sdu$. A cash dividend breaks this, as Remark 18.31 in §18.7 shows. Recombination keeps the number of terminal nodes at $n+1$, and the number of nodes in all at $(n+1)(n+2)/2$, about $n^2/2$. A daily tree over one year, $n = 252$, has 253 terminal nodes and 32,131 nodes in all, so backward induction performs 31,878 one-step valuations; following the paths instead would mean 2^{252}, about 7.2×10^{75} of them.

Interview trap. Put–call parity holds exactly on a European tree, by (18.12). A mismatch beyond rounding is an arithmetic error, never an arbitrage; parity is the debugging check to run after pricing a call and a put by hand.

18.4 Delta and Dynamic Replication

On a tree with more than one step a fixed position in the stock and the bond replicates only a payoff that is affine on the terminal prices; for an option whose payoff is not, such as the two-step call of Example 18.12, the hedge has to be rebalanced as delta changes from node to node. This subsection proves that the rebalanced hedge is self-financing and replicates every European payoff on every path. That removes the "traded at every node" assumption of Proposition 18.11 and makes the backward-induction value a price rather than a convention; it is the discrete form of the delta hedge studied in §21, and it comes before American options so that their price can be proved, in §18.5, with the same hedge plus withdrawals.

Setup. The tree, the probabilities P and $\mathbb{Q}$ and the backward-induction values $V_{i,j}$ are those of §18.3; the $V_{i,j}$ are numbers, defined by (18.9) with no trading assumption. A *trading strategy* in the stock and the bond starts from a capital at time 0 and assigns to each non-terminal node (i,j), $i < n$, a holding of $\Delta_{i,j}$ shares and $B_{i,j}$ currency units in the bond, chosen with knowledge of the path so far and kept until the next move. (A strategy may in general use the whole path so far; the strategies of this subsection use it only through the node that the path has reached.) One step later the holding chosen at (i,j) is worth

$$\Delta_{i,j}S_{i,j}u + B_{i,j}e^{r\Delta t} \text{ after an up-move}, \qquad \Delta_{i,j}S_{i,j}d + B_{i,j}e^{r\Delta t} \text{ after a down-move};$$

this is the value of the strategy *entering* the next node, and the value entering the root $(0,0)$ is the initial capital. The strategy is *self-financing* if no money is added or withdrawn after time 0: at every non-terminal node the holding carried into the node, valued at that node's prices, pays exactly for the holding chosen there,

$$\text{value entering } (i,j) \;=\; \Delta_{i,j}S_{i,j} + B_{i,j},$$

whichever path led to the node. The *value* of a self-financing strategy at a node is this common number, and its *payoff* is its value entering the terminal node that the path reaches. This is a self-financing strategy in the sense of §16.2, rebalanced at the dates $\Delta t, 2\Delta t, \dots, (n-1)\Delta t$.

Proposition 18.13 (Dynamic replication on an n-step tree)**.** *Let $V_{i,j}$ be the backward-induction values (18.9) of a European payoff f on the n-step tree of §18.3. For every non-terminal node (i,j) set*

$$\Delta_{i,j} = \frac{V_{i+1,j+1} - V_{i+1,j}}{S_{i,j}(u-d)}, \qquad B_{i,j} = V_{i,j} - \Delta_{i,j}S_{i,j}. \tag{18.13}$$

(i) The strategy that starts from the capital $V_{0,0}$ and holds $(\Delta_{i,j}, B_{i,j})$ at node (i,j) is self-financing; it is worth $V_{i,j}$ at every node (i,j), and $f(S_T)$ at maturity on every path.

(ii) Consequently, if there is no arbitrage, the price of the derivative at time 0 *is* $V_{0,0}$*: any other price is an arbitrage, even if the derivative trades only at time* 0*.*

(iii) The holdings $\Delta_{i,j}$ *take one and the same value* c' *at every non-terminal node if and only if the payoff is affine on the terminal prices, that is,* $f(S_{n,j}) = c + c'S_{n,j}$ *for* $j = 0, \dots, n$ *and some constants* c *and* c'*. In that case* $V_{i,j} = ce^{-r(n-i)\Delta t} + c'S_{i,j}$ *at every node, and the position held at time* 0 *is never traded again.*

Sketch of the proof. We proceed by induction along a path. By Theorem 18.3(i), applied to the step out of node (i, j), the holding (18.13) costs $V_{i,j}$, by (18.9), and is worth the value of the child reached,

$$\Delta_{i,j}S_{i,j}u + B_{i,j}e^{r\Delta t} = V_{i+1,j+1}, \qquad \Delta_{i,j}S_{i,j}d + B_{i,j}e^{r\Delta t} = V_{i+1,j},$$

which is (D.1) of the complete proof. A strategy that enters node (i, j) with $V_{i,j}$ therefore buys this holding with no money added or withdrawn and enters the next node with that node's value: this is (i). The law of one price (Lemma 16.8(iii)) gives (ii), by an arbitrage that trades the derivative at time 0 only. Part (iii) is an induction on the remaining steps, with $pu + (1-p)d = e^{r\Delta t}$, and a telescoping sum for the converse. The complete proof is in Appendix D.2. □

Part (iii) covers a forward, $f(x) = x - K$, and a call whose strike lies at or below the lowest terminal price $S_{n,0}$, for which $(x-K)^+ = x - K$ at every terminal price: their replicas are bought once and held. For other payoffs the holding changes from node to node. In Example 18.14, which follows, it is 0.5701 shares at the root and then 1 or 0.1556, so the hedge is rebalanced after the first move.

Example 18.14 (Hedging the two-step call: deltas and cash)**.** Take the call struck at 80 of Example 18.12, with the node values of Figure 33 and $e^{r\Delta t} = 1.007528$. Amounts are rounded separately, as in that example.

Root. The holding is $\Delta_{0,0} = (12.59776 - 1.19532)/(92 - 72) = 0.57012$ shares, which cost $0.57012 \times 80 = 45.610$, and $B_{0,0} = 6.0541 - 45.610 = -39.556$: a loan.

Up node ($S_{1,1} = 92$). Here $\Delta_{1,1} = (25.8 - 2.8)/(105.8 - 82.8) = 1$ exactly, and $B_{1,1} = 12.598 - 92 = -79.402 = -80e^{-0.0075}$. Both children are in the money, so the call has become a forward: one share, financed by a loan of the present value of the strike.

Down node ($S_{1,0} = 72$). Here $\Delta_{1,0} = (2.8 - 0)/(82.8 - 64.8) = 2.8/18 = 0.1556$ shares, which cost 11.200, and $B_{1,0} = 1.1953 - 11.200 = -10.005$.

Self-financing check. After one step the root loan of 39.55568 has grown to $39.55568 \times 1.007528 = 39.85346$, and the 0.570122 shares are worth $0.570122 \times 92 = 52.45122$ at the up node and $0.570122 \times 72 = 41.04878$ at the down node. The position entering the next node is therefore worth $52.45122 - 39.85346 = 12.59776$ after an up-move and $41.04878 - 39.85346 = 1.19532$ after a down-move: the call values 12.5978 and 1.1953 of Example 18.12, as (D.1) says.

Rebalancing. After an up-move the hedger buys $1 - 0.570122 = 0.42988$ share at 92 for $0.42988 \times 92 = 39.549$, financed by borrowing: the debt rises from 39.853 to $39.853 + 39.549 = 79.402$. After a down-move the hedger sells $0.570122 - 0.155556 = 0.41457$ share at 72 for $0.41457 \times 72 = 29.849$ and repays part of the loan: the debt falls from 39.85346 to $39.85346 - 29.84878 = 10.00468$.

Terminal check. From the up node the debt grows to $79.402 \times 1.007528 = 80.000$, the strike, and the single share is worth 105.8 or 82.8: the position ends at $105.8 - 80 = 25.8$ or $82.8 - 80 = 2.8$. From the down node the debt grows to $10.005 \times 1.007528 = 10.080$, and the $2.8/18 = 7/45$ share is worth $\frac{7}{45} \times 82.8 = 12.880$ or $\frac{7}{45} \times 64.8 = 10.080$: the position ends at $12.880 - 10.080 = 2.8$ or $10.080 - 10.080 = 0$. On each of the four paths the strategy ends with exactly the payoff of the call.

Remark 18.15 (Greeks read off a tree). A tree that has been built for a price also yields estimates of the sensitivities of that price. They are stated here as estimates, without proof. In words: *delta*

is the share holding of Proposition 18.13; *gamma* is the rate of change of delta with the stock price; *theta* is the rate of change of the value with calendar time; *vega* and *rho* are the sensitivities of the value to the volatility σ, which enters a tree through u and d (§18.6), and to the rate r. Their formal definition is Definition 21.1. Their closed forms under the Black–Scholes–Merton model are proved in §21.1 and §21.2 (delta, gamma, theta) and in §21.3 (rho). Vega is Proposition 20.35 of §20, which §21.3 recalls. On a tree with $n \geq 2$ steps:

- Delta at time 0 is the first-step slope $\Delta_{0,0}$.
- Gamma is estimated by
$$\Gamma \approx \frac{\Delta_{1,1} - \Delta_{1,0}}{\frac{1}{2}\left(S_{2,2} - S_{2,0}\right)},$$
the second derivative of the parabola through the three values at time $2\Delta t$. Since $S_{2,2} - S_{2,1} = S_{1,1}(u-d)$ and $S_{2,1} - S_{2,0} = S_{1,0}(u-d)$, the holdings (18.13) are the slopes $\Delta_{1,1} = (V_{2,2} - V_{2,1})/(S_{2,2} - S_{2,1})$ and $\Delta_{1,0} = (V_{2,1} - V_{2,0})/(S_{2,1} - S_{2,0})$, and the parabola
$$x \mapsto V_{2,0} + \Delta_{1,0}\left(x - S_{2,0}\right) + \frac{\Delta_{1,1} - \Delta_{1,0}}{S_{2,2} - S_{2,0}}\left(x - S_{2,0}\right)\left(x - S_{2,1}\right)$$
takes the values $V_{2,0}$, $V_{2,1}$ and $V_{2,2}$ at the three prices (at $S_{2,2}$ the reader should verify this using $V_{2,2} - V_{2,0} = \Delta_{1,1}(S_{2,2} - S_{2,1}) + \Delta_{1,0}(S_{2,1} - S_{2,0})$). Its second derivative, twice the coefficient of x^2, is the estimate.
- Theta is estimated by $\Theta \approx (V_{2,1} - V_{0,0})/(2\Delta t)$, the change in value over two steps at an unchanged stock price. The estimate is valid only when $ud = 1$, so that $S_{2,1} = S$, as on the Cox–Ross–Rubinstein tree of §18.6.
- Vega and rho are estimated by rebuilding the tree with σ or r changed by a small amount, keeping n fixed, and dividing the change in $V_{0,0}$ by that amount. The number of steps must not change, because the tree price zigzags with n (Fact 18.40, stated in §18.8). For the at-the-money call of Example 18.39 in that subsection ($S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$, on the Cox–Ross–Rubinstein tree), moving from 100 to 101 steps changes the price by 0.037, from 10.431 to 10.468, as much as raising σ by 0.1 percentage point. A bump of σ from 20% to 21% raises the 100-step price by 0.375, but by 0.414 if the bumped tree has 101 steps: a vega about 10% too high.

For an American option the same formulas are read off the American node values $V^{\mathrm{am}}_{i,j}$ of §18.5, with $\Delta^{\mathrm{am}}_{i,j}$ in place of $\Delta_{i,j}$. On the call of Example 18.14, gamma is about $(1 - 0.1556)/(\frac{1}{2}(105.8 - 64.8)) = 0.8444/20.5 = 0.041$ per unit of stock price; theta is not read off this tree, because $ud = 1.035 \neq 1$. This is how a trading desk obtains Greeks for American puts, and for American calls on underlyings that pay dividends, which have no closed form.

Interview trap. On a two-step tree, can the hedge chosen at time 0 be left alone until maturity? For the call of Example 18.14, no: the root holding of 0.5701 shares replicates the call over the first step only, and the replica of the second step needs 1 share at the up node and 0.1556 share at the down node, so the hedge must be traded after the first move. By Proposition 18.13(iii) the answer is yes exactly for a payoff that is affine on the terminal prices, such as a forward.

Trading application. Whoever delta-hedges a short call buys shares after up-moves and sells them after down-moves: the hedger buys high and sells low. In Example 18.14 that is 0.4299 share bought at 92, or 0.4146 share sold at 72. The premium received for the call pays for that slippage; this is the gamma story that Proposition 21.17 in §21 makes precise.

Interview trap. Self-financing means that no cash is injected or withdrawn after time 0. It does not mean that the cash account is zero: the hedge of Example 18.14 carries a loan throughout, 39.556 at the root and then 79.402 or 10.005.

Interview trap. Delta is the slope between the two child values, $(V_u - V_d)/(Su - Sd)$, never V/S: at the root of Example 18.14 it is 0.5701, against $6.0541/80 = 0.0757$. The delta of a put is

never positive, and it is negative at every node from which the put can finish in the money (each term of (18.10) is at least as large at the down child as at the up child, and the term of the lowest reachable price is strictly larger), so its replica is short shares and long the bond (Problem P18.7).

18.5 American Options and Early Exercise

An American option adds a decision at every node, to exercise now for the exercise value or to continue and hold a claim worth the discounted risk-neutral expectation of its two values one step later, and the backward pass takes the larger of the two: this is the dynamic programme of the re-roll game of §1.7 (Proposition 1.24, recursion (1.5)), the first option of this document, with discounting and $\mathbb{Q}$-expectations. This subsection states the recipe and proves that its output is the arbitrage-free price, by the seller's hedge of §18.4 with withdrawals added. It then checks on the tree the two model-free results of §17.4 that interviewers ask about more often than the recipe: an American call on a stock that pays no dividend is never exercised early (Theorem 17.16), while an American put may be (Proposition 17.18).

Setup. The tree is that of §18.3: n steps of length $\Delta t = T/n$, nodes (i,j) with prices $S_{i,j} = Su^j d^{i-j}$, no dividend, $d < e^{r\Delta t} < u$, the risk-neutral up-probability p of Definition 18.6, and the probabilities P and $\mathbb{Q}$ on the 2^n paths. Trading strategies are those of §18.4; a strategy may also use, at each node, whether the holder of an American claim has exercised so far, because the exercise time is a random time known when it occurs (Setup of §16.2). An *American claim* with payoff function f can be exercised by its holder at any node. Exercise at node (i,j) pays the *exercise value* $f(S_{i,j})$ at once and ends the claim, and a holder who has not exercised before maturity receives $f(S_{n,j})$ at time T. For a put $f(x) = (K-x)^+$ and for a call $f(x) = (x-K)^+$; the exercise value is then the intrinsic value of the option.

The *American node values* $V^{\mathrm{am}}_{i,j}$ are the numbers produced by Procedure 18.16 below; the qualifier is a superscript, as in C^{am}. The symbol $V_{i,j}$ without a superscript remains the European backward-induction value (18.9) of the same payoff function f. For $i < n$ the *continuation value* at node (i,j) is

$$V^{\mathrm{cont}}_{i,j} = e^{-r\Delta t}\bigl[pV^{\mathrm{am}}_{i+1,j+1} + (1-p)V^{\mathrm{am}}_{i+1,j}\bigr], \tag{18.14}$$

computed from the American values of the two children. (The continuation value is not written W, which is Brownian motion from §19 on, nor c, which is a generic constant.)

A *path prefix* of length i is a sequence of i moves, the first i moves of some path; it ends at the node (i,j), where j is its number of up-moves. An *exercise rule* attaches to every path prefix of length $i < n$ one of two decisions, "exercise" or "continue". On a given path the rule exercises at the step i^*, the smallest $i < n$ at which the prefix of length i of the path carries the decision "exercise", and $i^* = n$ if there is no such i. The decision at step i therefore uses the first i moves and nothing later; two paths into the same node may be treated differently; and $i^* \le n$ on every path. A holder who follows the rule receives $f(S_{i^*\Delta t})$ at time $i^*\Delta t$, where $S_{i^*\Delta t}$ is the price at that time on the path. The *risk-neutral value* of the rule is

$$\mathbb{E}^{\mathbb{Q}}\bigl[e^{-ri^*\Delta t} f(S_{i^*\Delta t})\bigr],$$

the sum over the 2^n paths of the $\mathbb{Q}$-probability of the path times its discounted exercise value. The star marks the step that the rule selects. The prices at time 0 of an American call and of an American put are C^{am} and $\mathcal{P}^{\mathrm{am}}$.

Procedure 18.16 (Backward induction for an American option)**.** To value an American claim with payoff function f on an n-step tree:

1. Build the tree: compute p, the one-step discount factor $e^{-r\Delta t}$ and the price $S_{i,j} = Su^j d^{i-j}$ at every node.

2. At maturity set $V^{\mathrm{am}}_{n,j} = f(S_{n,j})$ for $0 \le j \le n$.
3. For $i = n-1, n-2, \dots, 0$ and each j with $0 \le j \le i$, compute the continuation value $V^{\mathrm{cont}}_{i,j}$ of (18.14) from the *American* values $V^{\mathrm{am}}_{i+1,j+1}$ and $V^{\mathrm{am}}_{i+1,j}$ of the two children, and set

$$V^{\mathrm{am}}_{i,j} = \max\left(f(S_{i,j}),\ V^{\mathrm{cont}}_{i,j}\right). \tag{18.15}$$

   Mark (i,j) as an *exercise node* when $f(S_{i,j}) > V^{\mathrm{cont}}_{i,j}$.
4. Report the value $V^{\mathrm{am}}_{0,0}$, the exercise nodes and the *early-exercise premium* $V^{\mathrm{am}}_{0,0} - V_{0,0}$, where $V_{0,0}$ is the European value of Proposition 18.11.

Step 3 is the recursion (1.5) of Proposition 1.24: the value is the larger of stopping now and of the expected value of continuing optimally. In the re-roll game the expectation ran over the face of a die and the conditioning was the tower rule (Proposition 1.7); here it runs over the next move of the tree, under $\mathbb{Q}$, and is discounted over one step. The same recursion is reused on a finite-difference grid in §23.

Two properties follow from the recursion alone. First, $V^{\mathrm{am}}_{i,j} \ge f(S_{i,j})$ at every node, with equality at maturity. Second, $V^{\mathrm{am}}_{i,j} \ge V_{i,j}$ at every node, by backward induction on i: the two agree at $i = n$, and if $V^{\mathrm{am}}_{i+1,j+1} \ge V_{i+1,j+1}$ and $V^{\mathrm{am}}_{i+1,j} \ge V_{i+1,j}$ then, because p and $1-p$ are positive,

$$V^{\mathrm{am}}_{i,j} \ \ge\ V^{\mathrm{cont}}_{i,j} \ \ge\ e^{-r\Delta t}\left[pV_{i+1,j+1} + (1-p)V_{i+1,j}\right] = V_{i,j}.$$

The early-exercise premium is therefore never negative. The Procedure produces numbers; that $V^{\mathrm{am}}_{0,0}$ is a price is the content of the next proposition.

Proposition 18.17 (Backward induction gives the American price)**.** *Let $V^{\mathrm{am}}_{i,j}$ and $V^{\mathrm{cont}}_{i,j}$ be the numbers produced by Procedure 18.16 for an American claim with payoff function f on the n-step tree of the Setup, and for $i < n$ set*

$$\Delta^{\mathrm{am}}_{i,j} = \frac{V^{\mathrm{am}}_{i+1,j+1} - V^{\mathrm{am}}_{i+1,j}}{S_{i,j}(u-d)}. \tag{18.16}$$

(i) Seller's hedge. *A seller starts a hedging portfolio with the capital $V^{\mathrm{am}}_{0,0}$. At every node (i,j) with $i < n$ that is reached without the holder having exercised there or earlier, the seller moves $V^{\mathrm{am}}_{i,j} - V^{\mathrm{cont}}_{i,j} \ge 0$ from the portfolio to a separate deposit and holds $\Delta^{\mathrm{am}}_{i,j}$ shares and $V^{\mathrm{cont}}_{i,j} - \Delta^{\mathrm{am}}_{i,j} S_{i,j}$ in the bond. No money is added or withdrawn, so the portfolio and the deposit together form a self-financing position. On entering every node (i,j) with $i \le n$ that is reached without an earlier exercise, the portfolio is worth exactly $V^{\mathrm{am}}_{i,j} \ge f(S_{i,j})$, before any transfer. Any exercise is therefore covered.*

(ii) Holder's rule. *Let the holder exercise at the first step $i < n$ at which $f(S_{i,j}) \ge V^{\mathrm{cont}}_{i,j}$, and at $i = n$ if there is none. Before that step the hedge of (i) transfers nothing, and at that step its portfolio is worth exactly the exercise value. Hence the hedge of (i), started from $V^{\mathrm{am}}_{0,0}$, is a self-financing strategy in the stock and the bond that replicates the payoff $f(S_{i^*\Delta t})$ of this rule on every path.*

(iii) The price. *Consequently, if there is no arbitrage, the price of the American claim at time 0 is $V^{\mathrm{am}}_{0,0}$: any other price is an arbitrage, even if the claim trades only at time 0. Moreover*

$$V^{\mathrm{am}}_{0,0} = \max_{\text{exercise rules}} \mathbb{E}^{\mathbb{Q}}\left[e^{-ri^*\Delta t} f(S_{i^*\Delta t})\right], \tag{18.17}$$

and the rule of (ii) attains the maximum.

Sketch of the proof. We proceed by induction on i, as for Proposition 18.13, with transfers to the deposit as the one new feature. By Theorem 18.3(i), which needs $d < e^{r\Delta t} < u$, the holding of (i) costs $V^{\mathrm{cont}}_{i,j}$ and is worth, one step later, the American value of the child reached, so the

portfolio enters each node with $V^{\text{am}}_{i,j}$; and (18.15) gives $V^{\text{am}}_{i,j} \geq V^{\text{cont}}_{i,j}$, a nonnegative transfer, and $V^{\text{am}}_{i,j} \geq f(S_{i,j})$, which covers every exercise. The rule of (ii) continues only where the first is an equality and exercises only where the second is, so the hedge replicates its payoff with no transfer. Any other price at time 0 is an arbitrage: a seller above $V^{\text{am}}_{0,0}$ runs the hedge, a buyer below it exercises by that rule and runs the hedge reversed. Finally, by backward induction over path prefixes, the risk-neutral value of any exercise rule, seen from a prefix that ends at (i, j) before it exercises, is $f(S_{i,j})$ or a discounted average, with the positive weights p and $1-p$, of its values one move later, so it is at most $V^{\text{am}}_{i,j}$, with equality for the rule of (ii): this is (18.17). The complete proof is in Appendix D.3. □

This proposition is what turns the maximum of step 3 from a recipe into a theorem. The rule of (ii) exercises at a tie, $f(S_{i,j}) = V^{\text{cont}}_{i,j}$, where exercising and continuing are worth the same, whereas the Procedure marks a node only when exercise is strictly better. The rule that exercises at the first marked node on the path, and at maturity if there is none, attains the maximum as well, by the equality induction at the end of Step 5 of the proof of Proposition 18.17 (Appendix D.3), run for this rule. At a prefix where it has not yet exercised and now exercises, $f(S_{i,j}) > V^{\text{cont}}_{i,j}$, and its value $f(S_{i,j})$ equals $V^{\text{am}}_{i,j}$ by (18.15). Where it continues, $f(S_{i,j}) \leq V^{\text{cont}}_{i,j}$, and its value is $e^{-r\Delta t}[p\, y_u + (1-p)\, y_d]$, with y_u and y_d its values from the prefix extended by an up-move and by a down-move (as in that Step), which equals $e^{-r\Delta t}[pV^{\text{am}}_{i+1,j+1} + (1-p)V^{\text{am}}_{i+1,j}] = V^{\text{cont}}_{i,j} = V^{\text{am}}_{i,j}$ once the equality holds at level $i+1$, again by (18.15).

Remark 18.18 (The tree confirms the no-early-exercise theorem node by node). Theorem 17.16 is model-free, so it holds on every arbitrage-free tree, and nothing new is claimed here: the tree only shows where the continuation value beats exercise at each node. Let $r \geq 0$ and $K > 0$. Let $C_{i,j}$ be the European call value at node (i, j), that is, the $V_{i,j}$ of (18.9) for $f(x) = (x-K)^+$; let $V^{\text{am}}_{i,j}$ be the node value of the American call from Procedure 18.16; and let $\tau_i = (n-i)\Delta t$ be the time to maturity at step i. Then at every node with $i < n$ the continuation value of the American call is at least its exercise value, so the Procedure never needs to exercise: $V^{\text{am}}_{i,j} = C_{i,j}$ at every node, and $C^{\text{am}} = C$. With $r > 0$, exercise is strictly worse than continuing at every node with $i < n$ and $S_{i,j} > K$.

The argument is a backward induction on i that rests on one bound. By the node-wise formula (18.10), with $m = n - i$ steps to go,

$$
\begin{aligned}
C_{i,j} &= e^{-r\tau_i} \sum_{k=0}^{m} \binom{m}{k} p^k (1-p)^{m-k} \left(S_{i,j} u^k d^{m-k} - K\right)^+ \\
&\geq e^{-r\tau_i} \Big[S_{i,j} \sum_{k=0}^{m} \binom{m}{k} (pu)^k \left((1-p)d\right)^{m-k} - K \Big] = e^{-r\tau_i} \Big[S_{i,j} \left(pu + (1-p)d\right)^m - K \Big] \\
&= S_{i,j} - K e^{-r\tau_i} \;\geq\; S_{i,j} - K.
\end{aligned}
$$

The first inequality uses $(z-K)^+ \geq z - K$ and the fact that the weights $\binom{m}{k} p^k (1-p)^{m-k}$ are positive and sum to one. That sum and the equality that follows are both the binomial theorem, $\sum_{k=0}^{m} \binom{m}{k} y_1^k y_2^{m-k} = (y_1 + y_2)^m$: expanding the product of m factors $(y_1 + y_2)$ gives one term for each choice of y_1 or y_2 in every factor, and $\binom{m}{k}$ of these terms contain y_1 exactly k times. With $y_1 = p$ and $y_2 = 1-p$ the weights sum to 1; with $y_1 = pu$ and $y_2 = (1-p)d$ the sum is $\left(pu + (1-p)d\right)^m = e^{rm\Delta t} = e^{r\tau_i}$, by Proposition 18.7(i). The last inequality uses $r \geq 0$, and it is strict when $r > 0$ and $i < n$. Since also $C_{i,j} \geq 0$, the bound gives $C_{i,j} \geq (S_{i,j} - K)^+ = f(S_{i,j})$, with strict inequality when $r > 0$, $i < n$ and $S_{i,j} > K$.

Now the induction. At $i = n$, $V^{\text{am}}_{n,j} = f(S_{n,j}) = C_{n,j}$. If $V^{\text{am}}_{i+1,j+1} = C_{i+1,j+1}$ and $V^{\text{am}}_{i+1,j} = C_{i+1,j}$, the continuation value at (i, j) is $e^{-r\Delta t}[pC_{i+1,j+1} + (1-p)C_{i+1,j}] = C_{i,j}$, by the defining recursion (18.9), and it is at least the exercise value, by the bound. Hence $V^{\text{am}}_{i,j} = \max\left(f(S_{i,j}), C_{i,j}\right) = C_{i,j}$. No conditional expectation was needed.

Remark 18.19 (Bermudan options). A Bermudan option may be exercised only on a stated set of dates. On the tree, the maximum of step 3 of Procedure 18.16 is taken only on the time slices of those dates, and at every other node the value is the continuation value. Its price lies between the European and the American price: the recursion is increasing in the values of the children, so backward induction on i places the Bermudan node values between $V_{i,j}$ and $V^{\mathrm{am}}_{i,j}$, and the proof of Proposition 18.17, with exercise allowed only on those slices, makes them prices. Bermudan options are not pursued further.

Example 18.20 (An American put on the two-step tree)**.** Take the stock tree of Example 18.12 (Figure 33): $S = 80$, $u = 1.15$, $d = 0.9$, $r = 3\%$ and $\Delta t = 0.25$, with $p = 0.430113$, $1-p = 0.569887$ and $e^{-r\Delta t} = 0.992528$. Amounts are rounded separately, as in that example.

Strike 80*: in the money is not enough.* The European put values are those of Figure 33. At the down node ($S_{1,0} = 72$) the exercise value $80 - 72 = 8$ is below the continuation value 8.598. At the up node both are 0, and at the root the exercise value is 0. Nothing is exercised anywhere, the American values equal the European ones, and $\mathcal{P}^{\mathrm{am}} = \mathcal{P} = 4.863$.

The strike at which the down node is exercised. For $72 < K \le 82.8$ the put at the down node pays 0 after an up-move and $K - 64.8$ after a down-move, so its continuation value is $e^{-r\Delta t}(1-p)(K-64.8) = 0.565629\,(K-64.8)$. Exercise is strictly better exactly when $K - 72 > 0.565629\,(K-64.8)$, that is, when

$$K > \frac{72 - 0.565629 \times 64.8}{1 - 0.565629} = \frac{35.347}{0.434371} = 81.38.$$

For $K > 82.8$ both children are in the money, the continuation value is $Ke^{-r\Delta t} - 72 < K - 72$, and exercise is better as well.

Strike 85*.* The terminal put values are 0, 2.2 and 20.2. European values: the up node is worth $0.992528 \times 0.569887 \times 2.2 = 1.2444$, the down node $0.992528\,(0.430113 \times 2.2 + 0.569887 \times 20.2) = 12.3649$, and the root $0.992528\,(0.430113 \times 1.2444 + 0.569887 \times 12.3649) = 7.5252$. American values: at the up node the exercise value is 0, so $V^{\mathrm{am}}_{1,1} = 1.2444$. At the down node both children (82.8 and 64.8) are in the money, so the continuation value is $85e^{-0.0075} - 72 = 12.365$, below the exercise value $85 - 72 = 13$: exercise, and $V^{\mathrm{am}}_{1,0} = 13$. At the root the continuation value $0.992528\,(0.430113 \times 1.2444 + 0.569887 \times 13) = 7.8844$ exceeds the exercise value 5: hold. Hence $\mathcal{P}^{\mathrm{am}} = 7.8844$ against the European 7.5252, an early-exercise premium of 0.3592, all of it created at the down node.

The seller's hedge at strike 85*.* By (18.16) the seller starts with $\Delta^{\mathrm{am}}_{0,0} = (1.2444 - 13)/(92 - 72) = -0.58778$ shares, a short position worth 47.022, and $7.8844 + 47.022 = 54.907$ in the bond. After a down-move the bond has grown to $54.907 \times 1.007528 = 55.320$ and the short shares cost $0.58778 \times 72 = 42.320$ to buy back, so the portfolio is worth 13.000, the exercise value. If the holder fails to exercise there, the seller moves $13 - 12.365 = 0.635$ to the separate deposit; this is $85\,(1 - e^{-0.0075})$, the interest on the strike over one step, which the holder has given away. The remaining 12.365 buys the holding $\Delta^{\mathrm{am}}_{1,0} = (2.2 - 20.2)/(82.8 - 64.8) = -1$ share and $12.365 + 72 = 84.365$ in the bond, which is worth $85 - 82.8 = 2.2$ or $85 - 64.8 = 20.2$ at maturity.

Call check for Remark 18.18. For the call struck at 85 the terminal values are 20.8, 0 and 0. At the up node the continuation value $0.992528 \times 0.430113 \times 20.8 = 8.8795$ exceeds the exercise value $92 - 85 = 7$, nothing is exercised, and $C^{\mathrm{am}} = C = 0.992528 \times 0.430113 \times 8.8795 = 3.791$. For the call struck at 80 the up node is worth $12.598 = 92 - 80e^{-0.0075}$, more than the exercise value 12.

(Problem P18.5 works this computation at different numbers.)

Intuition. Why can a put on a tree be exercised early, but not a call on a stock without dividends? At the down node of Example 18.20 the put struck at 85 finishes in the money in both states. Holding it is therefore holding a short forward, worth $Ke^{-r\Delta t} - S = 12.365$, which is less than $K - S = 13$ by the interest on K over the step; the insurance that the put would otherwise keep is

worthless there. The mirror image is a call with both children in the money: it is a long forward worth $S - Ke^{-r\Delta t} > S - K$, so waiting earns the interest instead of giving it up (Theorem 17.16, Proposition 17.18). At strike 80 the up child of the down node is out of the money, and the insurance left in the put, $8.598 - (80e^{-0.0075} - 72) = 1.195$, is worth more than the interest on K, $80\,(1 - e^{-0.0075}) = 0.598$: the put is held.

Interview trap. "Exercise whenever the option is in the money" is wrong. In Example 18.20 the put struck at 80 is 8 in the money at the down node and is still held, and the put struck at 85 is 5 in the money at the root but worth 7.88 alive.

Interview trap. Check every node, not only the root, and feed the *American* values of the children, not their European values, into the continuation value. Comparing the European value at the root with the exercise value there misses the premium created at interior nodes: for the put struck at 85 of Example 18.20 it returns $\max(5, 7.5252) = 7.5252$ instead of 7.8844.

Interview trap. An American option is always worth at least its European twin, because $V^{\text{am}}_{i,j} \geq V_{i,j}$ at every node. The premium is zero for calls on stocks without dividends when $r \geq 0$ (Remark 18.18). It is not zero for puts in general, nor for calls once the underlying has a yield or is a futures price (§18.7).

Trading application. American puts, and American calls on underlyings with a yield or on futures prices, have no closed-form price. Procedure 18.16, run on the Cox–Ross–Rubinstein tree of §18.6 with the Greeks of Remark 18.15 read off the American node values, is the standard desk pricer for them.

18.6 Matching Volatility: the Cox–Ross–Rubinstein Tree

So far u and d were given; a tree becomes a model of a real underlying once the two factors are tied to its volatility, that is, chosen so that the log-return over one step has variance $\sigma^2\Delta t$. The choice of Cox, Ross and Rubinstein, $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$, achieves this to the order that matters. The proof is a Taylor expansion (Theorem 7.31 and Proposition 7.34 in §7.5), and the side condition under which the probabilities are positive is a favourite interview trap.

Setup. The *volatility* $\sigma > 0$ is the annualised standard deviation of the log price: the tree is built so that the log-return over one step has a variance close to $\sigma^2\Delta t$, and over n steps, a period of $n\Delta t$ years, close to $\sigma^2 n\Delta t$. The step is $\Delta t > 0$, and the tree of §18.3 is built with

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = \frac{1}{u} = e^{-\sigma\sqrt{\Delta t}}, \qquad p = \frac{e^{r\Delta t} - d}{u - d}; \tag{18.18}$$

this is the *Cox–Ross–Rubinstein tree*, or CRR tree. The *log-return of step i* is

$$R_i = \ln \frac{S_{i\Delta t}}{S_{(i-1)\Delta t}},$$

the R_i of the Notation table taken over one tree step instead of one trading day. It equals $+\sigma\sqrt{\Delta t}$ after an up-move and $-\sigma\sqrt{\Delta t}$ after a down-move; in particular $R_1 = \ln(S_{\Delta t}/S)$. Under $\mathbb{Q}$ every R_i equals $\sigma\sqrt{\Delta t}$ with probability p and $-\sigma\sqrt{\Delta t}$ with probability $1 - p$, because each move is up with probability p whatever the earlier moves (Setup of §18.3). The local abbreviation

$$x = \sigma\sqrt{\Delta t}$$

is used in this subsection and declared again in §18.8; with it $u = e^x$, $d = e^{-x}$ and $\Delta t = x^2/\sigma^2$.

Orders of magnitude are those of §7.5. As $\Delta t \to 0$, the symbol $O(\Delta t^k)$ denotes a quantity bounded by a constant multiple of Δt^k for all sufficiently small $\Delta t > 0$; the constant depends only on r, σ

and, in part (iv) below, μ, and the exponent k may be a half-integer. Because $x = \sigma\sqrt{\Delta t}$, a term $O(x^k)$ is a term $O(\Delta t^{k/2})$, and a term $O(\Delta t^2)$ is also $O(\Delta t^{3/2})$. For part (iv) of the proposition below, the stock has a real-world drift μ, and each move is up with the real-world probability

$$p_{\text{real}} = \frac{e^{\mu\Delta t} - d}{u - d},$$

whatever the earlier moves. Then $\mathbb{E}[S_{\Delta t}] = S\big(d + p_{\text{real}}(u-d)\big) = Se^{\mu\Delta t}$, so this μ is the real-world expected return of §18.2. Both p and p_{real} lie in $(0,1)$ once Δt is small enough; Corollary 18.23 below gives the exact condition for p, and its proof, run with μ in place of r, gives $\sigma > |\mu|\sqrt{\Delta t}$ for p_{real}.

The number $r - \sigma^2/2$ that appears below is the *risk-neutral log-drift*; it is named ν in §18.8. Proposition 18.21(ii) shows that the expected one-step log-return under $\mathbb{Q}$ is $(r - \sigma^2/2)\Delta t$ to leading order, so $r - \sigma^2/2$ is the yearly expected log-growth rate of the price under $\mathbb{Q}$. It is the analogue of the expected log growth rate of Definition 3.1, the mean logarithm of the bankroll multiplier of one bet (that definition's betting fraction f and bankroll multipliers R_k are unrelated to the payoff f and the log-returns R_i of this section); with the real-world drift μ in place of r it is the $g = \mu - \sigma^2/2$ of the Notation table.

Proposition 18.21 (Moments of the CRR step; Cox, Ross and Rubinstein, 1979)**.** *On the CRR tree (18.18), as $\Delta t \to 0$:*

(i) the up-probability is $p = \dfrac{1}{2} + \dfrac{(r - \sigma^2/2)\sqrt{\Delta t}}{2\sigma} + O(\Delta t)$*;*

(ii) the mean of the one-step log-return is $\mathbb{E}^{\mathbb{Q}}[R_1] = (r - \sigma^2/2)\Delta t + O(\Delta t^{3/2})$*;*

(iii) its variance is $\operatorname{Var}^{\mathbb{Q}}(R_1) = \sigma^2\Delta t + O(\Delta t^2)$*;*

(iv) the same three statements hold with r replaced by any constant rate, in particular by μ, with p_{real} in place of p and the real-world mean and variance in place of $\mathbb{E}^{\mathbb{Q}}$ and $\operatorname{Var}^{\mathbb{Q}}$. The variance of the one-step log-return is therefore $\sigma^2\Delta t + O(\Delta t^2)$ under both probabilities, and only the drift changes.

Every R_i has the law of R_1, so (ii) and (iii) hold for each step.

Sketch of the proof. We compute the two moments exactly and then expand p in powers of $x = \sigma\sqrt{\Delta t}$. Since R_1 is x or $-x$, its mean is $(2p-1)\,x$ and its variance is $x^2 - (2p-1)^2x^2 = 4p(1-p)\,x^2$; these are (D.3) and (D.4) of the complete proof. With $r\Delta t = (r/\sigma^2)\,x^2$, Proposition 7.34 gives

$$p = \frac{e^{r\Delta t} - e^{-x}}{e^{x} - e^{-x}} = \frac{x\big[1 + (r/\sigma^2 - 1/2)\,x + O(x^2)\big]}{2x\big[1 + O(x^2)\big]} = \frac{1}{2} + \frac{1}{2}\Big(\frac{r}{\sigma^2} - \frac{1}{2}\Big)x + O(x^2),$$

which is (i). Multiplying $2p - 1$ by x gives (ii); the squared mean is then $O(\Delta t^2)$, which gives (iii). The rate enters only through $e^{r\Delta t} = 1 + r\Delta t + O(\Delta t^2)$, so any constant, such as μ, may replace it: this is (iv). The complete proof is in Appendix D.4. □

(Problem P18.8 asks you to reproduce the complete proof from scratch.)

Carrying the cubic terms of the exponentials sharpens (i) and (ii). Put $a = r/\sigma^2 - 1/2$. Proposition 7.34 gives $e^{r\Delta t} = 1 + (r/\sigma^2)\,x^2 + O(x^4)$ and $e^{\pm x} = 1 \pm x + x^2/2 \pm x^3/6 + O(x^4)$, so the numerator and the denominator of p (Steps 2 and 3 of the proof of Proposition 18.21, Appendix D.4) become

$$e^{r\Delta t} - e^{-x} = x\Big[1 + ax + \frac{x^2}{6} + O(x^3)\Big], \qquad e^{x} - e^{-x} = 2x\Big[1 + \frac{x^2}{6} + O(x^3)\Big].$$

The expansion $1/(1-y) = 1 + y + O(y^2)$, applied with $y = -x^2/6 + O(x^3)$, turns the reciprocal of the second bracket into $1 - x^2/6 + O(x^3)$, and

$$p = \frac{1}{2}\Big[1 + ax + \frac{x^2}{6} + O(x^3)\Big]\Big[1 - \frac{x^2}{6} + O(x^3)\Big] = \frac{1}{2} + \frac{1}{2}\,ax + O(x^3),$$

because the two terms in $x^2/6$ cancel and every other product is $O(x^3)$. Hence the error of (i) is $O(\Delta t^{3/2})$: the term in x^2 of p vanishes. By (D.3), $\mathbb{E}^{\mathbb{Q}}[R_1] = (2p-1)\,x = ax^2 + O(x^4) = (r - \sigma^2/2)\Delta t + O(\Delta t^2)$, so the error of (ii) is $O(\Delta t^2)$.

Example 18.22 (CRR parameters for a monthly step)**.** Take $\sigma = 20\%$, $r = 5\%$ and $\Delta t = 1/12$. Then $x = 0.2/\sqrt{12} = 0.057735$, $u = e^x = 1.05943$, $d = e^{-x} = 0.94390$ and $e^{r\Delta t} = 1.004175$, so

$$p = \frac{1.004175 - 0.943900}{1.059434 - 0.943900} = \frac{0.060275}{0.115534} = 0.52171.$$

The expansion (i) gives $0.5 + 0.03 \times 0.288675/0.4 = 0.52165$, where $0.03 = r - \sigma^2/2$ and $0.288675 = \sqrt{\Delta t}$. The gap is 6×10^{-5}. The term in x^2 of p vanishes (the sharper expansion after the proof of Proposition 18.21), so the gap is of order $x^3 = 1.9 \times 10^{-4}$: here it is 0.31 times x^3.

The one-step log-return under $\mathbb{Q}$. By (D.3) the mean is $(2p-1)\,x = 0.043420 \times 0.057735 = 0.0025068$, against $(r - \sigma^2/2)\Delta t = 0.0025$. By (D.4) the variance is $4p(1-p)\,x^2 = 0.0033270$, against $\sigma^2 \Delta t = 0.0033333$.

Part (iv). With a real-world drift $\mu = 12\%$, $p_{\text{real}} = (e^{0.01} - 0.943900)/0.115534 = 0.57256$, and the variance of the one-step log-return is $4 \times 0.57256 \times 0.42744 \times x^2 = 0.0032631$: still $\sigma^2 \Delta t$ to first order, while the mean has moved from 0.0025068 to $(2p_{\text{real}} - 1)\,x = 0.145118 \times 0.057735 = 0.0083784$, against $(\mu - \sigma^2/2)\Delta t = 0.0083333$.

Positivity. Here $|r|\sqrt{\Delta t} = 0.05 \times 0.288675 = 0.0144$, far below $\sigma = 0.20$, so the condition of Corollary 18.23 below holds.

Intuition. Part (iv) in numbers: in Example 18.22, replacing $p = 0.52171$ by $p_{\text{real}} = 0.57256$ multiplies the one-step mean of the log-return by 3.3, from 0.0025068 to 0.0083784, while the variance moves only from 0.0033270 to 0.0032631, 0.2% and 2.1% below $\sigma^2 \Delta t = 0.0033333$. The volatility sits in u and d, which neither probability enters, so one calibration of the tree serves under both. This is the discrete shadow of Definition 20.6 in §20, where the risk-neutral price path is the real one with μ replaced by r and the same σ. No proof in this Part changes the probability measure; Remark 20.14 in §20 relates this description to the change of measure of the literature, which is named there and not used.

Corollary 18.23 (When the CRR probabilities are valid)**.** *On the CRR tree (18.18),* $p \in (0,1)$ *if and only if*

$$\sigma > |r|\sqrt{\Delta t}.$$

For an underlying with a continuous yield q*, whose risk-neutral up-probability is* $p = (e^{(r-q)\Delta t} - d)/(u-d)$*, the condition is* $\sigma > |r-q|\sqrt{\Delta t}$*; for a futures price, where* $p = (1-d)/(u-d)$*, it always holds. (We prove in §18.7, Proposition 18.25, that these are the risk-neutral probabilities of the two cases.) The condition is exact, not asymptotic.*

Proof. We rewrite the condition on p as a pair of inequalities between exponentials and take logarithms. Let b be a real number and $p = (e^{b\Delta t} - d)/(u-d)$; the three cases of the statement are $b = r$, $b = r - q$ and $b = 0$, and b is the cost of carry of §18.7. Since $u - d > 0$, the condition $0 < p < 1$ is $0 < e^{b\Delta t} - d < u - d$, that is, $d < e^{b\Delta t} < u$, which on the CRR tree reads

$$e^{-\sigma\sqrt{\Delta t}} < e^{b\Delta t} < e^{\sigma\sqrt{\Delta t}}.$$

The logarithm is increasing, so this is $-\sigma\sqrt{\Delta t} < b\Delta t < \sigma\sqrt{\Delta t}$, that is, $|b|\Delta t < \sigma\sqrt{\Delta t}$; dividing by $\sqrt{\Delta t} > 0$ gives $|b|\sqrt{\Delta t} < \sigma$. For $b = 0$ the condition reads $0 < \sigma$, which always holds. No expansion was used, so the condition is exact. □

For $b = r$ the inequalities $d < e^{r\Delta t} < u$ are the no-arbitrage condition (18.1) of Proposition 18.1. When the condition of the corollary fails, p lies outside $(0,1)$: it equals 0 or 1 when $\sigma = |b|\sqrt{\Delta t}$,

and is otherwise negative or above one. This signals an arbitrage inside the model. For $b = r$, by Proposition 18.1, the bond of the tree dominates the stock ($e^{r\Delta t} \geq u$) or is dominated by it ($e^{r\Delta t} \leq d$); for a yield the arbitrage is between the bond and the reinvested holding of Proposition 18.25(i), whose factors are $ue^{q\Delta t}$ and $de^{q\Delta t}$; for a futures price the condition never fails.

Example 18.24 (A tree with a negative probability)**.** Take $\sigma = 2\%$, $r = 5\%$ and $\Delta t = 1$. Then $u = e^{0.02} = 1.0202$ is below $e^{r\Delta t} = e^{0.05} = 1.0513$, and

$$p = \frac{1.0512711 - 0.9801987}{1.0202013 - 0.9801987} = \frac{0.0710724}{0.0400027} = 1.777, \qquad 1 - p = -0.777,$$

because $\sigma = 2\%$ is below $|r|\sqrt{\Delta t} = 5\%$. The bond beats the stock in both states, so shorting the stock and lending is an arbitrage on this tree (Proposition 18.1): with $S = 100$, the loan returns 105.127 after a year and the share is bought back for 102.020 or 98.020, a profit of 3.107 or 7.107 from no outlay. The remedy is a shorter step. The condition $\sigma > |r|\sqrt{\Delta t}$ reads $\Delta t < (\sigma/|r|)^2 = 0.16$ years. With monthly steps it reads $\sigma > 5\%/\sqrt{12} = 1.44\%$, which $\sigma = 2\%$ satisfies; then $p = 0.860$.

Interview trap. Forgetting the square root of Δt. On a monthly step, $u = e^{0.20}$ builds a tree whose annual log-volatility is about $0.20\sqrt{12} = 69\%$, and $u = e^{0.20/12}$ builds one with about $0.20/\sqrt{12} = 5.8\%$. Only $u = e^{0.20\sqrt{1/12}}$ gives 20%: the variance of the log-return over n steps is n times that of one step (we prove this in §18.8, Proposition 18.32), so the standard deviation of one step must scale with $\sqrt{\Delta t}$.

Interview trap. The factors u and d carry the volatility, and p carries the drift: σ fixes the node prices through u and d, and the rate r (or the carry $r - q$) enters the risk-neutral dynamics only through p.

Interview trap. A negative probability is not a numerical curiosity: the bond of the tree dominates its stock, or is dominated by it, as in Example 18.24. The remedy is a smaller Δt, never clipping p to $[0, 1]$.

Interview trap. The mean of the log-return is $r - \sigma^2/2$ per year, not r. Under $\mathbb{Q}$ the mean of the price grows at the rate r (Proposition 18.7(i)), while the mean of its logarithm grows at $r - \sigma^2/2$, which is also the growth rate of the median of the price in the lognormal limit of §18.8.

18.7 Trees for Indices, Currencies and Futures

An index that pays a dividend yield, a foreign currency that earns its own interest rate and a futures price that costs nothing to hold differ from a stock without dividends only through what holding them earns, and on the tree that changes only p. This subsection proves the general formula by replication, written with the cost of carry b so that one line covers all four underlyings, and specialises it with Proposition 16.15 and Proposition 16.19. It then shows the consequence that interviewers like, that American calls on futures, and on currencies with a positive foreign rate, can be exercised early, unlike calls on stocks without dividends, and finally lets the interest rate itself move on the tree and proves the exact gap between futures and forward prices that Remark 16.30 announced: futures prices exceed forward prices when the asset and the rates rise together.

Setup. The underlying has the price S today and Su or Sd after a step of length Δt, as before. It pays a continuous yield q that is reinvested in the underlying: one unit held at time 0 has grown to $e^{q\Delta t}$ units at time Δt (Proposition 16.15(i)). The *cost of carry* is $b = r - q$. Thus $b = r$ for a stock without dividends; $b = r - r_{\text{for}}$ for a currency, whose yield is the foreign interest rate r_{for} and whose price S is the domestic price of one unit of foreign currency; and $b = 0$ for

a futures price. Storage costs do not appear in this section. A *futures price* $F > 0$ moves to Fu or to Fd over a step. A futures position costs nothing to enter, and the gain $F_{\Delta t} - F$ of a long contract is settled at the end of the step, after which the position is worth zero again (Definition 16.25, with one settlement per step). With deterministic rates the futures price equals the forward price (Theorem 16.28), so no superscript distinguishes them until the rates become random. The number of futures contracts in a hedge is Δ_F, a symbol local to this subsection. In Examples 18.27 and 18.28 the factors are the CRR factors $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$ of §18.6, where σ is the volatility of the underlying or of the futures price; Example 18.30 states its own factors. As before, $\mathbb{Q}$ is the probability under which each move is up with probability p, for the p of the case at hand.

Setup for Proposition 18.29. This Setup serves that proposition only. An asset that pays no income has the price $S > 0$ today, and at each of n steps of length Δt, with $T = n\Delta t$, its price is multiplied by u or by d, where $0 < d < u$. The one-period interest rate may change from node to node. In this setting a *node* at time $k\Delta t$ is the path so far, that is, the sequence of the first k moves: the tree is drawn without recombination, with 2^k nodes at time $k\Delta t$. The price $S_{k\Delta t}$ at time $k\Delta t$ depends only on the number of up-moves so far, and $S_T = S_{n\Delta t}$; the rates, a claim and its values may depend on the whole path so far. The rate r_i is the continuously compounded rate of step i, from time $(i-1)\Delta t$ to time $i\Delta t$: one unit deposited at a node at time $(i-1)\Delta t$ is worth $e^{r_i\Delta t}$ at time $i\Delta t$, after either move. It is set at the node at time $(i-1)\Delta t$, so it depends on the first $i-1$ moves only, and r_1 is today's rate, a number. These are the rates $r_1, \dots, r_n$ of the rolling argument of Theorem 16.28, now random. At every node $d < e^{r_i\Delta t} < u$ for the rate r_i set there, and

$$p_i = \frac{e^{r_i\Delta t} - d}{u - d} \in (0,1)$$

is the $\mathbb{Q}$-probability of an up-move on step i; like r_i, it depends on the first $i-1$ moves. The $\mathbb{Q}$-probability of a path is the product, over its n steps, of p_i for an up-move on step i and $1 - p_i$ for a down-move, and $\mathbb{E}^{\mathbb{Q}}$, $\mathrm{Var}^{\mathbb{Q}}$ and $\mathrm{Cov}^{\mathbb{Q}}$ are the corresponding weighted sums over the 2^n paths; in particular $\mathrm{Cov}^{\mathbb{Q}}(X,Y) = \mathbb{E}^{\mathbb{Q}}[XY] - \mathbb{E}^{\mathbb{Q}}[X]\,\mathbb{E}^{\mathbb{Q}}[Y]$ for two functions X and Y of the path. For $0 \le k \le n$,

$$B_k = \exp\big(\Delta t\,(r_1 + \cdots + r_k)\big), \qquad B_0 = 1,$$

is the value at time $k\Delta t$ of one unit deposited at time 0 and rolled over at the one-period rates along the path. It is the bank account B_i of §16.6, now random, and $1/B_n$ is the *path discount factor*. The futures price for delivery at T, settled at the end of every step, is F^{fut} today; at a later date it is a number attached to each node of that date. The forward price for delivery at T is F. Every path has positive real-world probability. Lemma 16.8 (parts (i) to (iii), and part (v) without its last clause) and Definition 16.7 are used on this tree with its own bank account in place of e^{rt}: their proofs use only that a deposit grows to a positive amount.

Symbol notice. The bank account B_k, a function of the path, is neither the bond holding B of a one-step replica nor the node holding $B_{i,j}$ of §18.4; all three are money in the bond, and the letter B is money in the bank throughout this Part. An integer subscript on r names a step, like the day-i rate of §16.6, never a compounding frequency.

Proposition 18.25 (Risk-neutral probability with a yield, for a currency and for a futures price)**.** *Consider a one-step tree as in §18.1, for the underlyings of the Setup.*

(i) Yield. *Let the underlying pay a continuous yield q, and let $d < e^{(r-q)\Delta t} < u$. If the market of the underlying, the bond and a derivative that pays V_u or V_d at time Δt admits no arbitrage, the derivative has the price*

$$V = e^{-r\Delta t}\big[pV_u + (1-p)V_d\big], \qquad p = \frac{e^{(r-q)\Delta t} - d}{u - d} = \frac{e^{b\Delta t} - d}{u - d} \in (0,1). \tag{18.19}$$

The replica holds $e^{-q\Delta t}(V_u - V_d)/(S(u-d))$ units of the underlying at the start of the step, which the reinvested income brings to the $\Delta = (V_u - V_d)/(S(u-d))$ of Theorem 18.3 at its end, and $e^{-r\Delta t}(uV_d - dV_u)/(u-d)$ in the bond. Moreover $\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = Se^{(r-q)\Delta t}$, and this is the forward price for delivery at Δt on the tree, in agreement with Proposition 16.15(ii).

(ii) Currency. *For a foreign currency, (i) holds with $q = r_{\text{for}}$: the up-probability is $p = (e^{(r-r_{\text{for}})\Delta t} - d)/(u-d)$, and $\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = Se^{(r-r_{\text{for}})\Delta t}$ is the forward exchange rate of Proposition 16.19.*

(iii) Futures. *Let a futures price $F > 0$ move to Fu or to Fd. The market of the futures contract and the bond admits no arbitrage if and only if $d < 1 < u$. In that case, if the market of the futures contract, the bond and a derivative that pays V_u or V_d at time Δt admits no arbitrage, the derivative has the price*

$$V = e^{-r\Delta t}[pV_u + (1-p)V_d], \qquad p = \frac{1-d}{u-d} \in (0,1). \tag{18.20}$$

Its replica holds $\Delta_F = (V_u - V_d)/(F(u-d))$ futures contracts and the whole of V in the bond, and $\mathbb{E}^{\mathbb{Q}}[F_{\Delta t}] = F$: a futures price has no risk-neutral drift. This is the formula of (i) at $q = r$, that is, at $b = 0$.

Sketch of the proof. We reduce (i) and (ii) to Theorem 18.3 and solve the replication system again for (iii). Held with its income reinvested, the underlying is an asset without income that moves from S to $Sue^{q\Delta t}$ or $Sde^{q\Delta t}$, and the hypothesis of (i) is (18.1) for these factors, so the theorem gives the replica and the price, with

$$p = \frac{e^{r\Delta t} - de^{q\Delta t}}{ue^{q\Delta t} - de^{q\Delta t}} = \frac{e^{(r-q)\Delta t} - d}{u-d};$$

a currency is the case $q = r_{\text{for}}$. In (iii) contracts are free, so a position costs its bond holding B and pays $Be^{r\Delta t}$ plus $\Delta_F F(u-1)$ or $\Delta_F F(d-1)$: the argument of Proposition 18.1 gives $d < 1 < u$, and equating the payoffs to V_u and V_d gives the replica, whose cost $B = e^{-r\Delta t}[pV_u + (1-p)V_d]$, with $p = (1-d)/(u-d)$, is the price. The complete proof is in Appendix D.5. □

Part (iii) is the constant-rate case of Proposition 18.29(ii), proved later in this subsection, which shows that a futures price has no drift under $\mathbb{Q}$ even when the rates move on the tree. Its continuous-time form is Proposition 20.51 in §20.

The backward induction (18.9) and Procedure 18.16 were set up, and Proposition 18.11, parts (i) and (ii) of Proposition 18.13 and Proposition 18.17 were proved, for a stock without dividends; the next corollary states them for the underlyings of the Setup.

Corollary 18.26 (Pricing and hedging on an n-step tree with a yield or on a futures price)**.** *Let the n-step tree of §18.3, with n steps of length $\Delta t = T/n$ and the same u, d and r at every step, carry, in place of the stock without dividends, one of the following two underlyings of the Setup of this subsection.*

(a) A yield. *The underlying pays a continuous yield q that is reinvested in it, so that the units held at a node have grown by the factor $e^{q\Delta t}$ one step later; the cost of carry is $b = r - q$, and the node prices are $S_{i,j} = Su^j d^{i-j}$, with $S > 0$. A foreign currency is the case $q = r_{\text{for}}$ (Proposition 18.25(ii)).*

(b) A futures price. *The cost of carry is $b = 0$, and the node prices are $F_{i,j} = Fu^j d^{i-j}$, with $F > 0$. A futures position costs nothing to enter; its gain over a step is settled at the end of the step and paid into the bond, after which the position is worth zero.*

Let the condition $d < e^{r\Delta t} < u$ of the Setup of §18.3 be replaced by $d < e^{b\Delta t} < u$ (for a futures price, $d < 1 < u$), let every path keep a positive real-world probability, and set

$$p = \frac{e^{b\Delta t} - d}{u-d},$$

the p of (18.19) or (18.20); under $\mathbb{Q}$ a path with j up-moves has the probability $p^j(1-p)^{n-j}$. For a futures price, F, $F_{i,j}$ and the futures prices $F_{i\Delta t}$ and F_T along a path stand in place of S, $S_{i,j}$, $S_{i\Delta t}$ and S_T in the rest of this statement and in every result cited there. For a European payoff function f, let $V_{i,j}$ be the backward-induction values (18.9) computed with this p, and at every non-terminal node (i,j) let $\Delta_{i,j}$ be given by (18.13). Then the following hold.

(i) Proposition 18.11: its part (i) when the derivative trades at every node and there is no arbitrage, and its part (ii), (18.11) included, with $Y_n \sim \mathrm{Bin}(n,p)$ under $\mathbb{Q}$.

(ii) Parts (i) and (ii) of Proposition 18.13, when the holding at each non-terminal node (i,j) is the following, in place of $\Delta_{i,j}$ shares and $B_{i,j}$ in the bond.

- *With a yield: $e^{-q\Delta t}\Delta_{i,j}$ units of the underlying, bought at the node and held over the step with their income reinvested, and*

$$V_{i,j} - e^{-q\Delta t}\Delta_{i,j}S_{i,j} = e^{-r\Delta t}\,\frac{uV_{i+1,j} - dV_{i+1,j+1}}{u-d}$$

  in the bond.
- *With a futures price: $\Delta_{i,j}$ futures contracts, which cost nothing, and the whole node value $V_{i,j}$ in the bond.*

(iii) For an American claim with payoff function f, whose exercise at node (i,j) pays $f(S_{i,j})$: Procedure 18.16, run with this p, produces values with $V^{\mathrm{am}}_{i,j} \geq f(S_{i,j})$ and $V^{\mathrm{am}}_{i,j} \geq V_{i,j}$ at every node; and Proposition 18.17 holds when the seller's hedge holds at a node the holding of (ii) with the American values $V^{\mathrm{am}}_{i+1,j+1}$ and $V^{\mathrm{am}}_{i+1,j}$ of the children in place of $V_{i+1,j+1}$ and $V_{i+1,j}$, the $\Delta^{\mathrm{am}}_{i,j}$ of (18.16) in place of $\Delta_{i,j}$ and $V^{\mathrm{cont}}_{i,j}$ in place of $V_{i,j}$.

Sketch of the proof. We rerun the carried-over proofs after proving, for the new underlyings, the one fact about the market that they take from Theorem 18.3(i): the holding chosen at a node is worth, one step later, the value V_u or V_d of the child reached (backward-induction or American), with no money added or withdrawn, and costs $e^{-r\Delta t}[pV_u + (1-p)V_d]$, which is $V_{i,j}$ or $V^{\mathrm{cont}}_{i,j}$. Put $\Delta = (V_u - V_d)/(S_{i,j}(u-d))$, with $F_{i,j}$ for a futures price. With a yield, the $e^{-q\Delta t}\Delta$ units grow to Δ units, so with $e^{-r\Delta t}(uV_d - dV_u)/(u-d)$ in the bond the holding pays V_u or V_d, and since $e^{-q\Delta t} = e^{-r\Delta t}e^{b\Delta t}$ it costs $e^{-r\Delta t}[(e^{b\Delta t}-d)V_u + (u-e^{b\Delta t})V_d]/(u-d) = e^{-r\Delta t}[pV_u + (1-p)V_d]$. With a futures price the contracts are free, the bond holding $e^{-r\Delta t}[pV_u + (1-p)V_d]$ is the whole cost, and the settlement $\Delta F_{i,j}(u-1)$ or $\Delta F_{i,j}(d-1)$ added to the grown bond is V_u or V_d, because $p = (1-d)/(u-d)$. The rest of each proof is unchanged: it uses p only through $0 < p < 1$, which is the hypothesis $d < e^{b\Delta t} < u$. The complete proof is in Appendix D.6. □

Two statements do not carry over, because each uses $pu + (1-p)d = e^{r\Delta t}$, which a yield replaces by $e^{(r-q)\Delta t}$ and a futures price by 1: the bound of Remark 18.18, and part (iii) of Proposition 18.13, through Step 4 of its proof. For the payoff $c + c'x$ the induction of that Step, run with the new identity, gives with a yield $V_{i,j} = ce^{-r(n-i)\Delta t} + c'e^{-q(n-i)\Delta t}S_{i,j}$ and $\Delta_{i,j} = c'e^{-q(n-i-1)\Delta t}$, which changes from step to step when $c'q \neq 0$. The $e^{-q\Delta t}\Delta_{i,j} = c'e^{-q(n-i)\Delta t}$ units bought at a node are, however, exactly the units carried in from the previous node with their income reinvested, and the bond holding $ce^{-r(n-i)\Delta t}$ is the previous one grown by $e^{r\Delta t}$, so with a yield the position is still never traded. For a futures price the induction gives $V_{i,j} = e^{-r(n-i)\Delta t}(c + c'F_{i,j})$ and $\Delta_{i,j} = c'e^{-r(n-i-1)\Delta t}$ contracts, a hedge that is traded at every step: on the tree of Example 18.28 below, with $f(x) = x - 40$, it holds $e^{-0.015} = 0.98511$ contracts at the root and 1 at time Δt, and $V_{0,0} = e^{-0.03}(45-40) = 4.8522$.

Example 18.27 (An index with a dividend yield)**.** An index has volatility $\sigma = 18\%$ and dividend yield $q = 2\%$, with $r = 4\%$ and monthly CRR steps, $\Delta t = 1/12$. Then $u = e^{0.051962} = 1.05334$, $d = 0.94937$ and $e^{(r-q)\Delta t} = 1.001668$, so by (18.19)

$$p = \frac{1.001668 - 0.949365}{1.053335 - 0.949365} = \frac{0.052303}{0.103970} = 0.50306.$$

With $q = 0$ the same tree would have $p = (1.003339 - 0.949365)/0.103970 = 0.51913$: the yield lowers p and changes nothing else. The probability is valid by Corollary 18.23, since $|r - q|\sqrt{\Delta t} = 0.02 \times 0.288675 = 0.0058$ is far below $\sigma = 0.18$.

Example 18.28 (A call on a futures price that is exercised early)**.** A futures price stands at $F = 45$, with $\sigma = 25\%$, $r = 3\%$ and $T = 1$. Calls and puts struck at $K = 40$ are priced on a two-step CRR tree, $\Delta t = 0.5$, with the backward induction (18.9) and Procedure 18.16 carried over by Corollary 18.26. Amounts are rounded separately, as in Example 18.12. Here $u = e^{0.176777} = 1.19336$ and $d = 0.83797$, and by (18.20)

$$p = \frac{1 - 0.83797}{1.19336 - 0.83797} = \frac{0.16203}{0.35539} = 0.45592, \qquad 1 - p = 0.54408,$$

and the one-step discount factor is $e^{-r\Delta t} = e^{-0.015} = 0.985112$. The futures nodes are 45; then 53.701 and 37.709; then 64.085, 45.000 and 31.598.

European call. The terminal values are 24.085, 5 and 0. The up node is worth $0.985112\,(0.45592 \times 24.085 + 0.54408 \times 5) = 13.497$, the down node $0.985112 \times 0.45592 \times 5 = 2.246$, and the root $0.985112\,(0.45592 \times 13.497 + 0.54408 \times 2.246) = 7.266$.

American call. At the up node the exercise value $53.701 - 40 = 13.701$ beats the continuation value 13.497, and the call is exercised. Both children are in the money there, so continuing is worth exactly $e^{-r\Delta t}(53.701 - 40)$, the exercise value discounted over one step, because $pu + (1-p)d = 1$. At the down node the exercise value is 0 and the call continues. At the root the continuation value $0.985112\,(0.45592 \times 13.701 + 0.54408 \times 2.246) = 7.357$ exceeds the exercise value 5: hold. Hence $C^{\text{am}} = 7.357$ against $C = 7.266$; to four decimals the early-exercise premium is $7.3574 - 7.2658 = 0.0916$. On two steps the only interior decision nodes are those at time Δt, so the early exercise sits at the up node.

Puts and parity. The put struck at 40 pays 0, 0 and 8.402. Its down node is worth $0.985112 \times 0.54408 \times 8.402 = 4.503$, more than the exercise value $40 - 37.709 = 2.291$, and its root $0.985112 \times 0.54408 \times 4.503 = 2.414$: the put is never exercised early and is worth 2.414, European and American alike. The European prices satisfy $C - \mathcal{P} = 7.266 - 2.414 = 4.852 = e^{-0.03}(45 - 40)$. The identity is exact on the tree. By (18.11), which holds on this tree by Corollary 18.26(i), and Theorem 1.2, applied to $\mathbb{E}^{\mathbb{Q}}$, and because $(x - K)^+ - (K - x)^+ = x - K$ for every real x, the difference is $C - \mathcal{P} = e^{-rT}\big(\mathbb{E}^{\mathbb{Q}}[F_T] - K\big)$, where F_T is the futures price at T. By the binomial theorem, as in Remark 18.18, $\mathbb{E}^{\mathbb{Q}}[F_T] = F\big(pu + (1-p)d\big)^2 = F$, because $pu + (1-p)d = 1$ by Proposition 18.25(iii). This is the futures form of Corollary 17.11.

Contrast with a stock. A call struck at 40 on a stock without dividends at $S = 45$, on the same tree with $p = (e^{0.015} - d)/(u - d) = 0.49844$, is worth 8.233 and is never exercised early (Remark 18.18). With both children in the money, a call on such a stock is worth $S - Ke^{-r\Delta t} > S - K$ if it is continued; a call on a futures price is worth only $e^{-r\Delta t}(F - K) < F - K$.

(Problem P18.9 works this computation at different numbers.)

Interview trap. A futures tree uses the growth factor 1, not $e^{r\Delta t}$: $p = (1 - d)/(u - d)$. Using $(e^{r\Delta t} - d)/(u - d)$ treats the futures contract as if it cost F to hold. A futures position is free, so its expected gain under $\mathbb{Q}$ is zero. In Example 18.28 the wrong weight is 0.49844 instead of 0.45592.

Interview trap. For a currency the yield is the foreign rate: the growth factor is $e^{(r - r_{\text{for}})\Delta t}$, not $e^{(r_{\text{for}} - r)\Delta t}$. Swapping the two rates reverses the drift: $e^{(r_{\text{for}} - r)\Delta t}$ is the risk-neutral growth factor of the reciprocal rate $1/S$, the foreign price of one unit of domestic currency, seen from the foreign side (Proposition 18.25(ii) with the roles of the two currencies exchanged). Applied to S, it misprices the option.

Interview trap. American calls on futures can be exercised early, and so can American calls on any underlying with a positive yield q. One step before maturity, at a node where both children

are in the money, continuing is worth $e^{-r\Delta t}[p\,(Su-K)+(1-p)(Sd-K)] = Se^{-q\Delta t} - Ke^{-r\Delta t}$, by (18.19) and $pu+(1-p)d = e^{(r-q)\Delta t}$, so exercise wins once $S(1-e^{-q\Delta t}) > K(1-e^{-r\Delta t})$, which holds for every large enough S when $q>0$. It happens most readily for a currency with $r_{\text{for}} > r$ or a high-yield index. Theorem 17.16 needs the absence of a yield, and a futures price behaves like an asset with the yield $q = r$.

Interview trap. A futures option is hedged with futures contracts, which cost nothing, so the whole premium sits in the bond. With a yield, the number of units bought at the start of a step is $e^{-q\Delta t}\Delta$, not Δ: the reinvested income supplies the rest.

Proposition 18.29 (Futures and forward prices on a tree with random rates; Cox, Ingersoll and Ross, 1981)**.** *Let an asset that pays no income, with price $S>0$ today, move on an n-step tree with step Δt and $T = n\Delta t$, its price being multiplied by u or by d at every step, where $0<d<u$. A node at time $k\Delta t$ is the path so far, so that the tree, drawn without recombination, has 2^k nodes at time $k\Delta t$; the rates and the claims below may depend on the whole path so far. The rate r_i of step i, from $(i-1)\Delta t$ to $i\Delta t$, is set at the node at time $(i-1)\Delta t$, with $d < e^{r_i\Delta t} < u$ there, and $p_i = (e^{r_i\Delta t}-d)/(u-d)$. The $\mathbb{Q}$-probability of a path is the product of its factors p_i (up-move on step i) and $1-p_i$ (down-move); $B_n = \exp(\Delta t\,(r_1+\cdots+r_n))$, and $1/B_n$ is the path discount factor. Every path has positive real-world probability, and there is no arbitrage.*

(i) Pricing by path sums. *A claim that pays X at time T, where X is any function of the path, has the price $\mathbb{E}^{\mathbb{Q}}[X/B_n]$ at time 0: any other price is an arbitrage, even if the claim trades only at time 0. In particular $\mathbb{E}^{\mathbb{Q}}[S_T/B_n] = S$, and $\mathbb{E}^{\mathbb{Q}}[1/B_n]$ is the price of a zero-coupon bond that pays 1 at time T.*

(ii) No drift for futures. *At every node at time $k\Delta t$ with $k<n$ the futures price for delivery at T, settled at the end of every step, is the weighted average of the futures prices at the two children of the node, with the weights p_i and $1-p_i$ of the step that leaves the node. Hence $F^{\text{fut}} = \mathbb{E}^{\mathbb{Q}}[S_T]$, whatever the rates do.*

(iii) Forward price. *The forward price for delivery at T is $F = S/\mathbb{E}^{\mathbb{Q}}[1/B_n]$.*

(iv) The gap.

$$F^{\text{fut}} - F = -\,\frac{\operatorname{Cov}^{\mathbb{Q}}(1/B_n,\ S_T)}{\mathbb{E}^{\mathbb{Q}}[1/B_n]}. \tag{18.21}$$

The futures price therefore exceeds the forward price exactly when S_T and the discount factor $1/B_n$ have a negative covariance under $\mathbb{Q}$ (as when high asset prices arrive with high rates), lies below it exactly when the covariance is positive, and coincides with it when the rates are deterministic, which recovers Theorem 16.28 on the tree.

Sketch of the proof. We argue by backward induction over the nodes, with sums over paths in place of conditional expectations. The value of X at a node, a sum over the completions of the path of X times their factors p_i or $1-p_i$ and $e^{-r_i\Delta t}$, is X at the end and $\mathbb{E}^{\mathbb{Q}}[X/B_n]$ at the root, and splitting the sum by the first move gives $V = e^{-r_{k+1}\Delta t}[p_{k+1}V_u + (1-p_{k+1})V_d]$ at a node at time $k\Delta t$ with value V and children of values V_u and V_d. By Theorem 18.3(i) at the rate of the node, which is where $d < e^{r_{k+1}\Delta t} < u$ enters, V buys a holding worth V_u or V_d one step later; chained, these holdings replicate X, and Lemma 16.8(iii) gives (i), the node values of S_T being the prices since $p_{k+1}u + (1-p_{k+1})d = e^{r_{k+1}\Delta t}$. A futures contract held over a step costs nothing, so its replica does too (Lemma 16.8(v); every node has positive probability), and setting that cost to zero makes the futures price at a node the undiscounted p_{k+1}-weighted average of those at its children; from S_T at delivery (Proposition 16.27) this gives (ii). By (i) a forward with delivery price K is worth $S - K\,\mathbb{E}^{\mathbb{Q}}[1/B_n]$, which gives (iii), and (iv) is (ii) minus (iii), with $S = \mathbb{E}^{\mathbb{Q}}[S_T/B_n]$ and the definition of the covariance. The complete proof is in Appendix D.7. □

Two observations complete the picture. First, nothing in the complete proof (Appendix D.7) uses that u and d are the same at every node: with factors that depend on the node, each

node at time $k\Delta t$ satisfying $d < e^{r_{k+1}\Delta t} < u$ for its own factors and the rate r_{k+1} set there, the statement and the proof are unchanged (the price is then S times a product of positive factors along the path, which may depend on the whole path so far; the proof uses only that it is positive). Second, $\mathbb{E}^{\mathbb{Q}}[1/B_n]$ is the zero-coupon price $e^{-r(T)T}$, where $r(T)$ is the T-year zero rate (Definition 16.1), so (iii) is Theorem 16.11 with r read as the zero rate $r(T)$, and the no-drift statement of Proposition 18.25(iii) is the constant-rate case of (ii). Part (iv) is the result that Remark 16.30 announced, and Proposition 20.51 in §20 cites part (ii) as the model-free discrete statement behind the zero drift of a futures price.

Example 18.30 (Futures above forwards when rates rise with the asset)**.** Take $S = 60$, with $u = 1.12$ and $d = 0.92$ at every node, $\Delta t = 0.5$ and two steps, so that $T = 1$. The rate of the first step is $r_1 = 4\%$; the rate r_2 of the second step is 6% after an up-move and 2% after a down-move, so the rates rise with the asset. The growth factors $e^{r_i\Delta t}$ at the three nodes are 1.0202, 1.0305 and 1.0101, all between d and u. Amounts are rounded separately, so the last digit of a sum of rounded amounts can differ by one.

Probabilities. At the root $p_1 = (e^{0.02} - 0.92)/0.2 = 0.501007$. For the second step $p_2 = (e^{0.03} - 0.92)/0.2 = 0.552273$ after an up-move and $p_2 = (e^{0.01} - 0.92)/0.2 = 0.450251$ after a down-move.

Futures prices. The asset is at 67.2 or 55.2 after one step, and at 75.264, 61.824 or 50.784 at T. One step before delivery the futures price is $0.552273 \times 75.264 + 0.447727 \times 61.824 = 69.2465 = 67.2e^{0.03}$ after an up-move, and $0.450251 \times 61.824 + 0.549749 \times 50.784 = 55.7548 = 55.2e^{0.01}$ after a down-move. At the root, $F^{\text{fut}} = 0.501007 \times 69.2465 + 0.498993 \times 55.7548 = 62.5142 = \mathbb{E}^{\mathbb{Q}}[S_T]$.

Path sums. The four paths have the following probabilities, discount factors and terminal prices; $1/B_2$ is $e^{-0.05}$ after a first up-move and $e^{-0.03}$ after a first down-move.

path	up, up	up, down	down, up	down, down
$\mathbb{Q}$-probability	0.27669	0.22431	0.22467	0.27432
$1/B_2$	0.95123	0.95123	0.97045	0.97045
S_T	75.264	61.824	61.824	50.784

Then $\mathbb{E}^{\mathbb{Q}}[1/B_2] = 0.96082$, the price of the one-year zero-coupon bond, which corresponds to a zero rate of 3.997%; $\mathbb{E}^{\mathbb{Q}}[S_T/B_2] = 60.000$, the check of part (i); and $F = 60/0.96082 = 62.447$.

The gap. To five decimals $F^{\text{fut}} = 62.51424$ and $F = 62.44678$, so $F^{\text{fut}} - F = 0.0675$. On the other side of (18.21), with $\mathbb{E}^{\mathbb{Q}}[1/B_2] = 0.9608181$ to seven digits, $\text{Cov}^{\mathbb{Q}}(1/B_2, S_T) = 60.00000 - 0.9608181 \times 62.51424 = -0.06481$, and $0.06481/0.96082 = 0.0675$.

Other rate patterns. Swapping the two rates (2% after an up-move, 6% after a down-move) gives $F^{\text{fut}} = 62.389$, below $F = 62.444$. A flat 4% gives $62.449 = 60e^{0.04}$ for both. A swing of four percentage points in the rate over a year moves the futures price about 0.1% away from the forward price (0.108% here), which is why F stands for both prices in the rest of this Part.

Intuition. Why is the futures price the higher one when the rates rise with the asset? A long futures position is paid its gains at once and finances its losses at once. When high prices come with high rates, the gains are reinvested at high rates and the losses are financed at low ones, an advantage that a forward, settled once at T, does not have; since both contracts cost nothing to enter, the futures price must be the higher one to offset it. The formulas say the same. Part (ii) says that the futures price is the plain risk-neutral mean of S_T: each change of the futures price is paid at the end of its step and averages to zero there, so no discount factor enters. A forward pays all of $S_T - K$ at T, and part (i) discounts that amount along the path that produced it. In Example 18.30 the highest terminal price, 75.264, arrives on a path discounted by 0.95123, and the lowest, 50.784, on a path discounted by 0.97045: the high prices are discounted more. The discounted mean $\mathbb{E}^{\mathbb{Q}}[S_T/B_2] = 60$ is therefore below $\mathbb{E}^{\mathbb{Q}}[S_T]\,\mathbb{E}^{\mathbb{Q}}[1/B_2] = 62.51424 \times 0.9608181 = 60.0648$, and the forward price $S/\mathbb{E}^{\mathbb{Q}}[1/B_2]$ lies below $\mathbb{E}^{\mathbb{Q}}[S_T]$, the futures price.

Interview trap. The futures price is the undiscounted $\mathbb{Q}$-mean of S_T; the forward price is S divided by the zero-coupon price $\mathbb{E}^{\mathbb{Q}}[1/B_n]$. They agree only when $1/B_n$ and S_T are uncorrelated

under $\mathbb{Q}$, in particular when the rates are deterministic. The sign is about the discount factor: rates that rise with the asset make $1/B_n$ fall with it, a negative covariance, and put the futures price above the forward price. The size, 0.1% for the one-year contract of Example 18.30, is why the rest of this Part writes F for both.

Remark 18.31 (Discrete dividends). Unlike a dividend equal to a known fraction of the price, which scales every later node by the same number and so keeps the tree recombining, a known cash dividend D_1 paid at a date t_1 breaks recombination. The price $Su - D_1$ followed by a down-move differs from the price $Sd - D_1$ followed by an up-move: $(Su - D_1)d - (Sd - D_1)u = D_1(u - d) \neq 0$. On the tree of Example 18.12, a dividend $D_1 = 2$ paid just after the first step gives $(92 - 2) \times 0.9 = 81.0$ after up-then-down and $(72 - 2) \times 1.15 = 80.5$ after down-then-up, a gap of $0.5 = D_1(u - d)$. One remedy splits the price before t_1 into the discounted dividend and the rest,

$$S_t = \left(S_t - D_1 e^{-r(t_1 - t)}\right) + D_1 e^{-r(t_1 - t)} \qquad (t < t_1),$$

and lets the tree carry the rest alone, which pays no dividend and so recombines; its value today is $S - I$, where $I = D_1 e^{-rt_1}$ is the present value of the dividend (§16). The volatility σ is then that of the rest, and the price at a node before t_1 is recovered by adding the discounted dividend back. With discrete dividends an American call may be exercised just before an ex-dividend date (§17.4). This remark is descriptive, and trees with discrete dividends are not developed further.

18.8 The Limit: From the Tree to Black–Scholes–Merton

As the number of steps grows, the log-price of the Cox–Ross–Rubinstein tree becomes a sum of many small independent moves, its law approaches a normal law, and tree prices approach the Black–Scholes–Merton prices; Theorem 18.37 proves this, and Theorem 20.22 in §20 evaluates the limit expectations in closed form. Example 18.39, Fact 18.40, Figure 34 and the traps at the end of the subsection carry what an interview asks about the limit: the size and the zigzag of the error, the averaging of two consecutive trees, and the lognormal law of the price. The subsection builds on the step moments of Proposition 18.21. Fact 18.34 and Lemma 18.35 are the tools that §§19–23 reuse, and §19 and §20 cite the conclusions of Theorem 18.37.

Setup. The maturity $T > 0$ is split into n steps of length $\Delta t = T/n$, and the tree is the CRR tree (18.18) of §18.6. With the local abbreviation $x = \sigma\sqrt{\Delta t}$, declared again here,

$$u = e^{x}, \qquad d = e^{-x}, \qquad p = \frac{e^{r\Delta t} - d}{u - d}.$$

The volatility is $\sigma > 0$, the rate r is any real number, and n is large enough that $\sigma > |r|\sqrt{\Delta t}$, so that $p \in (0, 1)$ and $\mathbb{Q}$ is a probability (Corollary 18.23). The probability $\mathbb{Q}$ is defined on the 2^n paths as in §18.3: the probability of a path is the product of one factor per step, p for an up-move and $1 - p$ for a down-move. Under $\mathbb{Q}$, the variables $R_1, \ldots, R_n$ are the one-step log-returns of §18.6, each equal to x after an up-move and to $-x$ after a down-move, and $Y_n \sim \mathrm{Bin}(n, p)$ is the number of up-moves (Proposition 18.11). Part (ii) of Proposition 18.32 below verifies that

$$\ln \frac{S_T}{S} = R_1 + \cdots + R_n = (2Y_n - n)\, x.$$

A *random variable on the tree* is a function of the path. It is *computed from the moves of a set of steps* if it takes the same value on any two paths that agree on those steps: R_i is computed from the move of step i, and $S_{k\Delta t}$ from the moves of steps $1, \ldots, k$. The n-step prices of the European call and put struck at $K > 0$ (Proposition 18.11) are C_n and $\mathcal{P}_n$; the subscript is the number of steps, a label, not a date.

The *risk-neutral log-drift* is

$$\nu = r - \frac{\sigma^2}{2}.$$

By Proposition 18.21(ii) the expected one-step log-return under $\mathbb{Q}$ is $\nu\Delta t$ to leading order, so ν is the yearly expected log-growth rate of the price under $\mathbb{Q}$. It is the analogue of the expected log growth rate of Definition 3.1, and with the real-world drift μ in place of r it is the $g = \mu - \sigma^2/2$ of the Notation table.

The variables Z and Z' are independent standard normal variables, with density φ and distribution function Φ; the literature writes $N(d_1)$ for what is $\Phi(d_1)$ here. The symbol $\mathbb{E}$ applied to a function of Z is the Gaussian integral of Fact 7.22,

$$\mathbb{E}\,\psi(Z) = \int_{-\infty}^{\infty} \psi(z)\,\varphi(z)\,dz$$

whenever the integral converges absolutely, and likewise for Z'; where both variables appear, as in (18.29) of Lemma 18.35 below, the expectation is that of a function of the two independent variables, and a probability P of an event of them is the expectation of its indicator; Fact 18.34(ii) below computes such an expectation as an iterated integral. The symbol $\mathbb{E}^{\mathbb{Q}}$ always refers to the tree. The *moment generating function* of a random variable X is the function $\theta \mapsto \mathbb{E}\,e^{\theta X}$ of the real exponent θ; no letter is given to it. A *test function* ψ is a real function of one real variable, or of two variables (or of two vectors of variables) where stated; ψ_1, ψ_2, ψ_+, ψ_- and ψ_ε are test functions of one variable. Its derivatives are ψ', ψ'' and ψ''', and $\sup|\psi'|$ is the supremum of $|\psi'|$ over the real line. The letters c, c' are generic constants, declared again where they are used. As $n \to \infty$, the symbol $O(n^{-k})$ denotes a quantity bounded by a constant multiple of n^{-k} for all large n, as in §7.5.

The complete proof of Theorem 18.37 (Appendix D.10) declares its own symbols: Ψ_k, the smoothing of ψ over the last $n-k$ steps, and $\Psi_k[\chi]$, the same smoothing applied to another function χ; the hybrid expectations $\mathcal{M}_k$; a Taylor error $\mathcal{E}_k$; a smoothing kernel X with density f_X and distribution function F_X (the subscript names the variable, and F_X is not a forward price); and the smoothed positive part $(z)^+_\varepsilon$.

Proposition 18.32 (Independent steps, and the mean and variance of the CRR log-price)**.** *Under $\mathbb{Q}$ on the n-step CRR tree of the Setup, with $x = \sigma\sqrt{\Delta t}$:*

(i) Product rule for separate steps. *If X is computed from the moves of one set of steps and X' from the moves of a disjoint set of steps, then*

$$\mathbb{E}^{\mathbb{Q}}[XX'] = \mathbb{E}^{\mathbb{Q}}[X]\;\mathbb{E}^{\mathbb{Q}}[X'], \tag{18.22}$$

and the same holds for finitely many factors computed from pairwise disjoint sets of steps. In particular, for functions ψ_1 and ψ_2 of one variable and steps $i \neq k$,

$$\mathbb{E}^{\mathbb{Q}}\big[\psi_1(R_i)\,\psi_2(R_k)\big] = \mathbb{E}^{\mathbb{Q}}\psi_1(R_1)\;\mathbb{E}^{\mathbb{Q}}\psi_2(R_1); \tag{18.23}$$

for every real θ,

$$\mathbb{E}^{\mathbb{Q}}e^{\theta(R_1+\cdots+R_n)} = \big(\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}\big)^n; \tag{18.24}$$

and, if X is computed from the moves of steps $1,\dots,k-1$, then for every function ψ of two variables

$$\mathbb{E}^{\mathbb{Q}}\psi(X,R_k) = \mathbb{E}^{\mathbb{Q}}\big[\,p\,\psi(X,x) + (1-p)\,\psi(X,-x)\,\big]: \tag{18.25}$$

step k can be averaged out first.

(ii) Exact moments. *On every path $\ln(S_T/S) = R_1 + \cdots + R_n = (2Y_n - n)\,x$, and*

$$\mathbb{E}^{\mathbb{Q}}\Big[\ln\frac{S_T}{S}\Big] = n(2p-1)\,x, \qquad \mathrm{Var}^{\mathbb{Q}}\Big(\ln\frac{S_T}{S}\Big) = 4np(1-p)\,x^2, \tag{18.26}$$

while $\mathbb{E}^{\mathbb{Q}}Y_n = np$ and $\mathrm{Var}^{\mathbb{Q}}\,Y_n = np(1-p)$.

(iii) Limits. *As $n \to \infty$ with $\Delta t = T/n$,*

$$\mathbb{E}^{\mathbb{Q}}\Big[\ln \frac{S_T}{S}\Big] = \nu T + O(n^{-1/2}), \qquad \mathrm{Var}^{\mathbb{Q}}\Big(\ln \frac{S_T}{S}\Big) = \sigma^2 T + O(n^{-1}).$$

Sketch of the proof. For (i) we group the sum over the paths that defines $\mathbb{E}^{\mathbb{Q}}[XX']$. The $\mathbb{Q}$-probability of a path is a product of one factor per step; the two sets of steps being disjoint, it splits into the factors of the steps of X, those of X' and those of the remaining steps, which sum to 1 over the moves of those steps, and what is left is $\mathbb{E}^{\mathbb{Q}}[X]\,\mathbb{E}^{\mathbb{Q}}[X']$. The particular cases follow by induction on the number of factors and by splitting on the two values of R_k. For (ii), $R_1 + \cdots + R_n$ and $\ln(S_T/S)$ both equal $(2Y_n - n)\,x$; linearity gives the mean, and in the square of the sum of the n centred steps each cross term vanishes by (18.23) and each square contributes the $4p(1-p)\,x^2$ of (D.4); the moments of Y_n, an affine function of the log-price, follow. Part (iii) is Proposition 18.21 multiplied by $n = T/\Delta t$. The complete proof is in Appendix D.8. □

Example 18.33 (The moments of a monthly tree over one year)**.** Carry the monthly tree of Example 18.22 over one year: $\sigma = 20\%$, $r = 5\%$, $T = 1$ and $n = 12$, so that $p = 0.52171$ and one step has the mean 0.0025068 and the variance 0.0033270 computed there. By (18.26) the log-price $\ln(S_T/S)$ has the exact mean $12 \times 0.0025068 = 0.03008$ and the exact variance $12 \times 0.0033270 = 0.03992$, against $\nu T = 0.03$ and $\sigma^2 T = 0.04$. The number of up-moves has the mean $12 \times 0.52171 = 6.2605$ and the variance $12 \times 0.52171 \times 0.47829 = 2.9943$. A sum over the $2^{12} = 4096$ paths, each weighted by its $\mathbb{Q}$-probability, returns the same four numbers. Example 18.39 follows the two moments of the log-price as n grows.

Part (ii) proves, on the tree, the binomial row of Table 1, whose variance Remark 1.12 quoted without proof: the weights of Proposition 18.11 are the $\mathrm{Bin}(n,p)$ masses of Definition 1.9, and the argument used nothing about p beyond $0 < p < 1$. The rate $n^{-1/2}$ for the mean in (iii) is what the orders stated in Proposition 18.21 give. The sharper expansion after its proof, $\mathbb{E}^{\mathbb{Q}}[R_1] = \nu\Delta t + O(\Delta t^2)$, makes the error of the mean $O(n^{-1})$ too, as the last line of Example 18.39 shows.

Intuition. Measured in units of x, the log-price $\ln(S_T/S)/x = 2Y_n - n$ is the position after n steps of the biased walk of Theorem 3.14, which moves up one unit with probability p and down one unit otherwise. Refining the tree shrinks the step of the walk like $\sqrt{\Delta t}$ and its bias $2p-1$ like $\sqrt{\Delta t}$ as well, while the number of steps grows like $1/\Delta t$. The variance, n steps of size $x^2 = \sigma^2\Delta t$, stays near $\sigma^2 T$, and the mean, n steps of bias times size, stays near νT. Only the product rule (i) is needed to add the variances; nothing in it says yet that the sum is close to normal. That is the work of the rest of the subsection.

The next two statements are tools for the whole Part. The first is the Fubini–Tonelli theorem, which the Part states without proof (as it quotes Fact 7.22 of §7.4), and which §§19–23 use whenever two expectations, or an expectation and an integral, are exchanged.

Fact 18.34 (Fubini–Tonelli (Fubini, 1907; Tonelli, 1909))**.** The order of two integrations, or of an expectation and an integration, may be exchanged in the following three situations.

(i) *Iterated integrals.* Let $\psi(y, y') \geq 0$ be a Borel measurable function (the class of functions that Lebesgue's theory integrates) of two variables y and y', each ranging over a Euclidean space of any finite dimension; continuity outside finitely many hyperplanes suffices in this Part. Its two iterated integrals, first over y' and then over y or the other way round, are equal, possibly both infinite, and they equal its integral over the product space:

$$\int \Big[\int \psi(y,y')\,dy'\Big]dy = \int \Big[\int \psi(y,y')\,dy\Big]dy'.$$

If the iterated integral of $|\psi|$ is finite, a Borel measurable function ψ of either sign has finite iterated integrals, equal to each other and to its integral over the product space; the inner

integral is then finite outside a set of volume zero. Sums replace integrals in any discrete coordinate.

(ii) *Independent variables.* Let X and Y be independent random vectors and let ψ be Borel measurable, as in (i), with $\psi \ge 0$ or $\mathbb{E}|\psi(X,Y)| < \infty$. Then

$$\mathbb{E}\,\psi(X,Y) = \mathbb{E}\,\bar{\psi}(X), \qquad \text{where} \qquad \bar{\psi}(y) = \mathbb{E}\,\psi(y,Y);$$

in the case $\mathbb{E}|\psi(X,Y)| < \infty$, the number $\bar{\psi}(y)$ is finite for all y outside a set of probability zero under the law of X. Here X and Y are generic letters; Y is not the up-move count Y_n.

(iii) *Expectation of a time integral.* If X_s, $t_0 \le s \le t$, are random variables whose paths $s \mapsto X_s$ are piecewise continuous, and either $X_s \ge 0$ for every s or $\int_{t_0}^{t} \mathbb{E}|X_s|\,ds < \infty$, then

$$\mathbb{E}\int_{t_0}^{t} X_s\,ds = \int_{t_0}^{t} \mathbb{E}X_s\,ds.$$

Here X_s is a generic process, a family of random variables indexed by the time s.

A proof is beyond our scope.

A proof needs Lebesgue integration on product spaces, because the Part integrates over unbounded domains and against joint laws that are neither discrete nor given by a density (a Brownian value together with the indicator of a hitting event, for instance). Two special cases are already in this document. The proof of Theorem 7.5 used (i) for a nonnegative integrand. On a tree, (ii) with Y the move of a single later step is (18.25), proved by grouping a finite sum. For X and Y computed from any two disjoint sets of steps it follows in the same way from (18.22) and Theorem 1.2: write $\psi(X,Y) = \sum_{y'} \psi(X,y')\,\mathbf{1}\{Y = y'\}$ over the finitely many values y' of Y, so that $\mathbb{E}^{\mathbb{Q}}\psi(X,Y) = \sum_{y'} \mathbb{E}^{\mathbb{Q}}\psi(X,y')\,\mathbb{Q}(Y = y') = \mathbb{E}^{\mathbb{Q}}\bar{\psi}(X)$. The Fact is stated here, at its first use (Lemma 18.35(ii) below), once for the whole Part.

The second tool is the Gaussian computation behind every lognormal formula of the Part. In it, μ_X and σ_X are the mean and the standard deviation of a normal variable X: the subscript names the variable, as in μ_V, and they are not the drift μ and the volatility σ of the underlying. The real numbers a_1 and a_2 are two smoothing scales.

Lemma 18.35 (The normal moment generating function, and composing Gaussian smoothings)**.** *Let $Z \sim \mathcal{N}(0,1)$, with density φ.*

(i) For every real θ,

$$\mathbb{E}\,e^{\theta Z} = e^{\theta^2/2}. \tag{18.27}$$

Consequently:

- *if $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$, that is, $X = \mu_X + \sigma_X Z$ with $\sigma_X \ge 0$, then $\mathbb{E}\,e^{\theta X} = e^{\theta\mu_X + \theta^2\sigma_X^2/2}$, the moment generating function of the normal law;*
- *the absolute exponential moment satisfies $\mathbb{E}\,e^{c|Z|} \le 2e^{c^2/2}$ for every constant $c \ge 0$;*
- *every function ψ that is continuous except at finitely many points and satisfies $|\psi(y)| \le c\,e^{c|y|}$ for all real y, with one generic constant c, has $\mathbb{E}|\psi(X)|^k < \infty$ for every normal X and every $k \ge 1$.*

(ii) Gaussian smoothing composes. *Let ψ be a function on the real line, continuous except at finitely many points, with $|\psi(y)| \le c\,e^{c|y|}$ for all real y, and let a_1, a_2 be real numbers. Then*

$$\int \Big[\int \psi(a_1 z + a_2 z')\,\varphi(z')\,dz' \Big] \varphi(z)\,dz = \int \psi\big((a_1^2 + a_2^2)^{1/2} z\big)\,\varphi(z)\,dz, \tag{18.28}$$

all integrals being over the real line and absolutely convergent. In words: smoothing by $\mathcal{N}(0, a_2^2)$ and then by $\mathcal{N}(0, a_1^2)$ is smoothing by $\mathcal{N}(0, a_1^2 + a_2^2)$. With Fact 18.34(ii), for independent standard normal variables Z and Z',

$$\mathbb{E}\,\psi(a_1 Z + a_2 Z') = \mathbb{E}\,\psi\big((a_1^2 + a_2^2)^{1/2} Z\big), \tag{18.29}$$

and $a_1Z + a_2Z'$ has the law of $(a_1^2 + a_2^2)^{1/2}Z$, which is $\mathcal{N}(0, a_1^2 + a_2^2)$. Replacing ψ by a shifted function $y \mapsto \psi(y_0 + y)$, which belongs to the same class, adds the constant mean y_0 on both sides of (18.28) and of (18.29).

Sketch of the proof. For (i) we complete the square: since $\theta z - z^2/2 = \theta^2/2 - (z-\theta)^2/2$,

$$\int_{-\infty}^{\infty} e^{\theta z}\varphi(z)\,dz = e^{\theta^2/2}\int_{-\infty}^{\infty}\varphi(z-\theta)\,dz = e^{\theta^2/2},$$

by a translation and $\int\varphi = 1$ (Theorem 7.5). The consequences follow with $\theta\sigma_X$ in place of θ, with $e^{c|z|} \le e^{cz} + e^{-cz}$, and with the growth bound on ψ. For (ii) we rotate the plane. By that bound, (i) and Fact 18.34(i), the left side of (18.28) is an absolutely convergent integral over the plane. The rotation that turns $a_1z + a_2z'$ into $(a_1^2+a_2^2)^{1/2}$ times the first new coordinate preserves $z^2 + z'^2$, hence $\varphi(z)\varphi(z')$, and integrating out the second new coordinate leaves the right side; that a rotation leaves the integral unchanged is proved by writing it as three shears, each a translation of one coordinate with the other fixed. Fact 18.34(ii) then gives (18.29), and the indicator of a half-line gives the law. The complete proof is in Appendix D.9. □

Example 18.36 (Exponential moments of a normal variable, and a sum of two normals)**.** *Part (i).* With $\theta = 2$, $\mathbb{E}\,e^{2Z} = e^2 = 7.3891$. With $c = 1$ the bound of the lemma reads $\mathbb{E}\,e^{|Z|} \le 2e^{1/2} = 3.2974$. The exact value follows by completing the square on the half-line $z > 0$ and doubling, because $e^{|z|}\varphi(z)$ is an even function:

$$\mathbb{E}\,e^{|Z|} = 2\int_0^{\infty} e^z\varphi(z)\,dz = 2e^{1/2}\int_0^{\infty}\varphi(z-1)\,dz = 2e^{1/2}\int_{-1}^{\infty}\varphi(z)\,dz = 2e^{1/2}\,\Phi(1) = 2.7743.$$

The bound overstates it by 19%, which is harmless: the lemma is used to show that expectations are finite. Now take $r = 5\%$, $\sigma = 20\%$ and $T = 1$, so that $\nu = 0.05 - 0.02 = 0.03$. The variable $\nu T + \sigma\sqrt{T}\,Z \sim \mathcal{N}(0.03, 0.04)$ has

$$\mathbb{E}\,e^{\nu T + \sigma\sqrt{T}\,Z} = e^{0.03 + 0.04/2} = e^{0.05} = 1.05127 = e^{rT}:$$

the limit law of Theorem 18.37 below is still risk-neutral, as every tree is.

Part (ii). For independent standard normal variables Z and Z', the sum $Z + Z'$ has the law of $\sqrt{2}\,Z$ (take $a_1 = a_2 = 1$), so

$$P(Z + Z' > 1) = P(\sqrt{2}\,Z > 1) = 1 - \Phi(1/\sqrt{2}) = 0.2398,$$

against $1 - \Phi(1) = 0.1587$ for a single standard normal variable.

Theorem 18.37 (Convergence of the CRR tree: the law of the log-price and European prices; Cox, Ross and Rubinstein, 1979)**.** *Fix $S > 0$, $K > 0$, $T > 0$, $\sigma > 0$ and a real r, and put $\nu = r - \sigma^2/2$. For every n with $\sigma > |r|\sqrt{T/n}$, so that $p \in (0,1)$ and $\mathbb{Q}$ is a probability (Corollary 18.23), let C_n and $\mathcal{P}_n$ be the n-step CRR prices of the European call and put struck at K (Proposition 18.11 with $\Delta t = T/n$, $u = e^x$, $d = e^{-x}$ and $x = \sigma\sqrt{\Delta t}$), so that $\ln(S_T/S) = R_1 + \cdots + R_n$ under $\mathbb{Q}$, with steps that are independent in the sense of Proposition 18.32(i). Let $Z \sim \mathcal{N}(0,1)$.*

(i) Smooth test functions, with a rate. *There are constants c_0 and n_0, depending only on r, σ and T, such that for every function ψ on the real line with bounded continuous derivatives ψ', ψ'', ψ''' and every $n \ge n_0$,*

$$\Big|\,\mathbb{E}^{\mathbb{Q}}\psi\Big(\ln\frac{S_T}{S}\Big) - \mathbb{E}\,\psi(\nu T + \sigma\sqrt{T}\,Z)\Big| \le c_0\,\big(\sup|\psi'| + \sup|\psi'''|\big)\,n^{-1/2}. \tag{18.30}$$

Such a ψ grows at most linearly, $|\psi(y)| \le |\psi(0)| + \sup|\psi'|\,|y|$, so both expectations are finite.

(ii) Prices. *As $n \to \infty$,*

$$\mathcal{P}_n \to e^{-rT}\,\mathbb{E}\big(K - Se^{\nu T + \sigma\sqrt{T}\,Z}\big)^+ \qquad \text{and} \qquad C_n \to e^{-rT}\,\mathbb{E}\big(Se^{\nu T + \sigma\sqrt{T}\,Z} - K\big)^+. \tag{18.31}$$

(iii) Parity on every tree. *For every n,*

$$C_n - \mathcal{P}_n = S - Ke^{-rT}. \tag{18.32}$$

(iv) Moment generating functions. *For every real θ, as $n \to \infty$,*

$$\mathbb{E}^{\mathbb{Q}} e^{\theta \ln(S_T/S)} \to \mathbb{E}\, e^{\theta(\nu T + \sigma\sqrt{T}\, Z)} = e^{\theta\nu T + \theta^2\sigma^2 T/2},$$

with equality for every n at $\theta = 1$, where both sides equal e^{rT}: $\mathbb{E}^{\mathbb{Q}} S_T = Se^{rT}$.

(v) The law. *For every real y, as $n \to \infty$,*

$$\mathbb{Q}\Big(\ln \frac{S_T}{S} \le y\Big) \to \Phi\Big(\frac{y - \nu T}{\sigma\sqrt{T}}\Big) :$$

the law of $\ln(S_T/S)$ converges to $\mathcal{N}(\nu T, \sigma^2 T)$.

The two limits in (ii) are the Black–Scholes–Merton prices, and §20 evaluates them in closed form (Theorem 20.22). The central limit theorem in the form of Fact 4.8 does not apply as stated, because the law of a step changes with n: the steps form a triangular array, unlike the coin flips with a fixed probability of Example 4.9. The proof is Lindeberg's replacement argument, a self-contained proof of the central limit theorem in the form needed here. It swaps the tree steps one at a time for normal steps with the same mean and variance, which gives the $n^{-1/2}$ bound (i) for smooth test functions, with an explicit constant for the replacement part; the put follows by smoothing its kink, and the call by parity. The rate $n^{-1/2}$ of (i) is what this proof gives. The true error of the call is of order $1/n$ and, at the money, zigzags with the parity of n; that finer statement and the American case are recorded as Fact 18.40, stated after Example 18.39, and the zigzag is why practitioners average two consecutive trees.

Sketch of the proof. The proof is Lindeberg's replacement argument. Let $\mathcal{M}_k$ be the expectation of ψ of the log-price when the first k steps are those of the tree and the last $n-k$ are normal steps with the same mean and variance, together one normal variable (Lemma 18.35(ii)); $\mathcal{M}_n$ is the tree expectation. As the first two moments match, each replacement changes only a third-order Taylor term, and $|\mathcal{M}_n - \mathcal{M}_0| \le \frac{1}{6}(5 + 16\varphi(0))\, \sigma^3 T^{3/2} \sup|\psi'''|\, n^{-1/2}$, where the constant $\frac{1}{6}(5 + 16\varphi(0))$ is 1.897. The normal law in $\mathcal{M}_0$ has the mean and standard deviation of the tree, within $O(n^{-1/2})$ of νT and $\sigma\sqrt{T}$ for $n \ge n_0$ (Proposition 18.21), which costs $O(n^{-1/2}) \sup|\psi'|$ by the mean value theorem: this is (i). The put follows by applying (i) to its payoff smoothed within ε, for every $\varepsilon > 0$, and the call by the parity (iii), which is linearity with $\mathbb{E}^{\mathbb{Q}} S_T = Se^{rT}$ and holds in the limit by Lemma 18.35(i). For (iv), by (18.24) the left side is the n-th power of $\mathbb{E}^{\mathbb{Q}} e^{\theta R_1} = 1 + (\theta\nu + \theta^2\sigma^2/2)\Delta t + O(\Delta t^{3/2})$, which compounds to $e^{\theta\nu T + \theta^2\sigma^2 T/2}$ (Proposition 7.39, extended to a convergent sequence). For (v), (i) is applied to two smooth functions that enclose the indicator of a half-line and differ only on an interval of length 4ε. The complete proof is in Appendix D.10. □

(Problem P18.6 asks you to reproduce the proof of part (iv) from scratch.)

Example 18.38 (The five parts of the theorem at 10, 100 and 1000 steps)**.** Take $S = K = 100$, $r = 5\%$, $\sigma = 20\%$ and $T = 1$, so that $\nu = 0.03$ and $\sigma\sqrt{T} = 0.2$. The condition $\sigma > |r|\sqrt{T/n}$ holds for every $n \ge 1$.

Part (i). Take $\psi = \sin$, for which $\sup|\psi'| = \sup|\psi'''| = 1$. The limit is $\mathbb{E}\sin(0.03 + 0.2Z) = 0.029402$, by numerical integration. The tree values $\mathcal{M}_n = \mathbb{E}^{\mathbb{Q}} \sin(\ln(S_T/S))$, their distance to the limit and the two pieces into which the proof splits that distance are as follows, where the hybrid $\mathcal{M}_0 = \mathbb{E}\sin\big(n(2p-1)\,x + 2x(np(1-p))^{1/2} Z\big)$ is the expectation under the normal law with the

exact mean and variance of the tree, also computed by numerical integration.

n	$\mathcal{M}_n$	$\mathcal{M}_n - \text{limit}$	$\mathcal{M}_0$	$\lvert\mathcal{M}_n - \mathcal{M}_0\rvert$	replacement bound	$\lvert\mathcal{M}_0 - \text{limit}\rvert$
10	0.029538	1.4×10^{-4}	0.029499	3.9×10^{-5}	4.8×10^{-3}	9.8×10^{-5}
100	0.029415	1.4×10^{-5}	0.029411	3.9×10^{-6}	1.5×10^{-3}	9.8×10^{-6}
1000	0.029403	1.4×10^{-6}	0.029403	3.9×10^{-7}	4.8×10^{-4}	9.8×10^{-7}

The replacement part $|\mathcal{M}_n - \mathcal{M}_0|$ is far inside its bound $1.897\,\sigma^3 T^{3/2} n^{-1/2}$, which holds at every n here (Steps 3 and 4 of the complete proof, Appendix D.10). The rest is the mismatch of mean and variance (Step 5 there), whose constant is not explicit. Both parts fall like $1/n$ here, faster than the $n^{-1/2}$ that the proof guarantees.

Part (iv). At $\theta = 1$, $\mathbb{E}^{\mathbb{Q}} e^{\ln(S_T/S)} = e^{0.05} = 1.051271$ at every n: the tree is risk-neutral at every step. At $\theta = 2$ the limit is $e^{2\nu T + 2\sigma^2 T} = e^{(2r+\sigma^2)T} = e^{0.14} = 1.150274$, and

n	10	100	1000	limit
$\mathbb{E}^{\mathbb{Q}} e^{2\ln(S_T/S)}$	1.149684	1.150214	1.150268	1.150274
$n(\mathbb{E}^{\mathbb{Q}} e^{2R_1} - 1)$	0.140464	0.140046	0.140005	0.14

Parts (ii) and (iii). The limit of the put prices is $e^{-rT}\,\mathbb{E}(K - Se^{\nu T + \sigma\sqrt{T}\,Z})^+ = 5.5735$, by numerical integration here and in closed form by Theorem 20.22 in §20. The tree gives

n	10	100	1000	limit
$\mathcal{P}_n$	5.3764	5.5536	5.5715	5.5735
$\mathcal{P}_n - \text{limit}$	−0.197	−0.020	−0.002	
C_n	10.2534	10.4306	10.4486	10.4506

The error falls like $1/n$, again faster than the $n^{-1/2}$ of the proof (Fact 18.40 below). At every n, $C_n - \mathcal{P}_n = 4.8771 = 100 - 100e^{-0.05}$, so the calls approach 10.4506 with the same errors as the puts.

Part (v). At $y = 0$ the limit is $\Phi(-\nu\sqrt{T}/\sigma) = \Phi(-0.15) = 0.4404$. The tree gives $\mathbb{Q}(S_T \le S) = 0.4799$ at $n = 100$ and 0.4402 at $n = 101$. For even n the middle terminal node sits exactly at S and carries the probability 0.079 at $n = 100$, so at this lattice point the error, 0.0395, is about half that atom. The distribution function of the tree jumps by the atom at the node while the limit Φ is continuous, so the error at the node or just below it is at least half the atom; ten times as many steps divide the atom only by about $\sqrt{10} = 3.16$ (at $n = 1000$ it is 0.025, and the error at S is 0.012). Near a lattice point a distribution function therefore cannot converge faster than $n^{-1/2}$, whereas smooth test functions do better (part (i), and the $1/n$ errors above). The theorem states (v) without a rate.

Intuition. Lindeberg's replacement swaps the coin-flip steps of the tree, one at a time, for normal steps with the same mean and variance. Each swap changes the expectation of a smooth function of the total only through third-order terms, of size $\Delta t^{3/2}$, because the first two moments agree. The n swaps cost $n\Delta t^{3/2} = T^{3/2}/\sqrt{n}$, and after the last swap the total is exactly normal (Lemma 18.35(ii)). The kink of the put is smoothed at the scale ε before the argument, and the smoothing is removed after it. The payoff of the call is unbounded, so the call comes through parity, which holds exactly on every tree: (18.12) is Theorem 17.9 computed on the tree, and part (iii) of the theorem reads it on the CRR tree.

Example 18.39 (Convergence of an at-the-money call)**.** Take $S = K = 100$, $r = 5\%$, $\sigma = 20\%$ and $T = 1$, as in Example 18.38. The n-step CRR prices of the call, from Proposition 18.11, are

n	1	2	3	5	10	50	100	limit
C_n	12.162	9.541	11.044	10.806	10.253	10.411	10.431	10.4506
$n \times (C_n - \text{limit})$	+1.71	−1.82	+1.78	+1.78	−1.97	−1.99	−2.00	

The limit 10.4506 is the call limit of Theorem 18.37(ii); its closed form is computed in §20.

The price at $n = 1$ by hand. Here $u = e^{0.2} = 1.221403$, $d = 0.818731$ and $e^{r\Delta t} = e^{0.05} = 1.051271$, so $p = (1.051271 - 0.818731)/0.402672 = 0.57749$. The call pays $122.1403 - 100 = 22.1403$ after an up-move and nothing after a down-move, and $C_1 = e^{-0.05} \times 0.57749 \times 22.1403 = 12.162$.

The zigzag. Odd n lie above the limit and even n below it, and n times the error settles near $+1.75$ for odd n (it is $+1.754$ at $n = 101$) and near -2.00 for even n: the error is of order $1/n$, with a sign that follows the parity of n. For even n the middle terminal node sits exactly on the strike, $Su^{n/2}d^{n/2} = S = K$, where the kink of the payoff loses value; for odd n the terminal nodes straddle the strike. A tree with 101 steps gives 10.468, and the average of the trees with 100 and 101 steps, 10.449, is within 0.002 of the limit.

The moments. By (18.26), the mean of $\ln(S_T/S)$ under $\mathbb{Q}$ is 0.03100, 0.03010 and 0.03001 at $n = 1$, 10 and 100, against $\nu T = 0.03$, and its variance is 0.03904, 0.03991 and 0.03999, against $\sigma^2 T = 0.04$. Both errors are divided by ten when n is multiplied by ten.

(Problem P18.11 works this computation at different numbers.)

Fact 18.40 (Rate of convergence of the CRR tree (Leisen and Reimer, 1996; Diener and Diener, 2004; Amin and Khanna, 1994))**.** The error of the n-step CRR call price, C_n minus its limit in Theorem 18.37(ii), is of order $1/n$ and oscillates: with the parity of n when the strike is at the money, irregularly otherwise (Leisen and Reimer, 1996; Diener and Diener, 2004). American tree prices also converge as $n \to \infty$; the limit is the American price of the Black–Scholes–Merton model, the value of an optimal-stopping problem for the risk-neutral lognormal price path, which this Part does not define (Amin and Khanna, 1994). A proof is beyond our scope.

A proof of the rate needs an Edgeworth expansion of the lattice binomial law, with its lattice correction: the replacement argument of Theorem 18.37 gives only $n^{-1/2}$, and only for smooth payoffs. The American case needs the convergence of optimal-stopping problems. No proof in this document relies on this Fact. Example 18.39 illustrates it: n times the error of the call is -1.995 at $n = 50$ and -1.997 at $n = 100$. Figure 34 shows the first thirty prices.

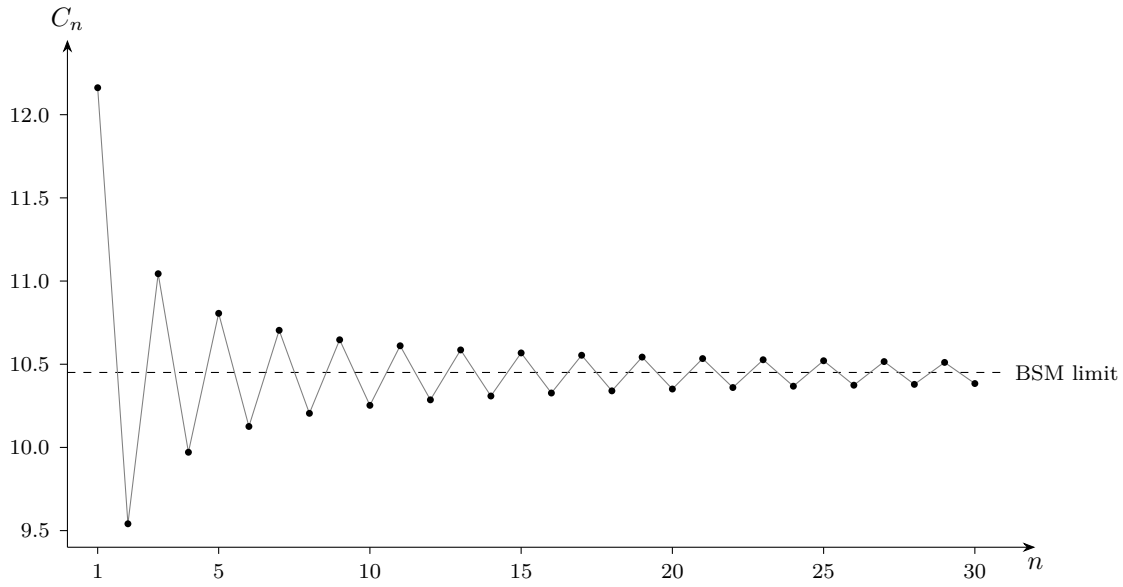


Figure 34: The n-step CRR price C_n of the call of Example 18.39 ($S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$) for $n = 1, \ldots, 30$. Odd n lie above the limit and even n below it, with an error of about $2/n$ in absolute value, between $1.7/n$ and $2/n$ (Fact 18.40). The dashed line is the call limit 10.4506 of Theorem 18.37(ii), the Black–Scholes–Merton price.

Interview trap. More steps is not monotonically closer. In Figure 34 the tree with $n = 10$ steps ($C_{10} = 10.253$) is slightly further from the limit (error 0.1972) than the tree with $n = 9$ steps ($C_9 = 10.647$, error 0.1966), and the sign alternates. Quoting a 50-step tree to the cent is off by 4 cents. Compare odd n with odd n and even with even, or average two consecutive values of n.

Interview trap. The error of the tree is a deterministic $O(1/n)$ (Fact 18.40). It is not the statistical error of a Monte Carlo estimate, which shrinks like one over the square root of the number of simulated paths (§23).

Interview trap. In the limit S_T is lognormal, and $\ln S_T$ is normal with the mean $\ln S+(r-\sigma^2/2)T$, not $\ln S + rT$ (Theorem 18.37(v); Theorem 19.40 in §19, with μ replaced by r). With the numbers of Example 18.39, the limit law of S_T has the median $100e^{0.03} = 103.05$ and the mean $100e^{0.05} = 105.13$. The real-world drift is absent from the limit for the same reason as on one step: it is absent from every tree.

Interview trap. The tree converges because the first two moments of a step match those of its normal surrogate, not because each step looks normal: every step is a two-point variable. Step 3 of the proof of Theorem 18.37 cancels exactly the first-order and second-order Taylor terms and leaves third-order terms of total size $n\Delta t^{3/2} = T^{3/2}/\sqrt{n}$. For the American put the limit has no closed form, and the tree of §18.5 is the standard method for it.

Interview trap. A distribution function converges slowly at a lattice point, at the rate $n^{-1/2}$: the error there is about half the atom of the node. In Example 18.38, $\mathbb{Q}(S_T \le S) = 0.480$ on a 100-step tree against the limit $\Phi(-0.15) = 0.440$, because the middle node, of probability 0.079, sits exactly at S; at 101 steps it is 0.440. Smooth payoffs converge faster in practice: the errors of part (i) in Example 18.38 fall like $1/n$. The proof of Theorem 18.37 works with smooth test functions because its Step 3 expands them to second order with a remainder in the third derivative. It reaches the put (Step 6) by smoothing the kink first, and the distribution function (Step 9) by smoothing the jump of the indicator first.

Practice: problems P18.1–P18.11 in §27, solutions in §28.

19 Brownian Motion, Itô's Lemma and the Lognormal Model

This section builds the continuous-time model behind the Black–Scholes–Merton formula. It explains why variance, and not volatility, adds over time (§19.1); constructs Brownian motion (§19.2) and the Itô integral (§19.3); defines the price processes of the Part (§19.4); proves Itô's lemma, which is the chain rule plus a convexity term (§19.5); shows that a price following a geometric Brownian motion is lognormal (§19.6); separates the expected return from the growth rate, which is where volatility drag, leveraged products and the Kelly criterion live (§19.7); treats the correlated motions that two-asset payoffs need (§19.8); and ends with hitting times and the reflection principle (§19.9).

Sections 20 to 23 rely above all on the following results of this section: Itô's lemma (Theorem 19.30 and Corollary 19.31, with their parameter and stopped clauses); the Itô integral (Theorem 19.23) and its class of integrands (Definition 19.22); the growth classes (Definition 19.19, Lemma 19.20); the freezing lemma and the limit tools (Lemmas 19.17 and 19.12); the lognormal law and the Gaussian shift (Theorem 19.40, Lemma 19.38); the Gaussian smoothing lemma and the reflection theorem (Lemma 19.61, Theorem 19.63); the equality of mixed partial derivatives (Lemma 19.54); and the two-asset results (Proposition 19.52, Theorem 19.55, Proposition 19.57).

Reading the section. For interview preparation the essential parts are §19.1; in §19.2, the definition, Propositions 19.8 and 19.10 with their examples and traps, and the statement of Theorem 19.14; Example 19.24 and the traps after it; §19.4; the statements of Theorem 19.30 and Corollary 19.31 and everything after them up to the end of §19.7; the statements, examples and traps of §19.8;

and the statement of Theorem 19.63 with its examples. The complete proofs of Lemma 19.12, Theorems 19.14, 19.23, 19.30 and 19.55, Lemma 19.61 and Theorem 19.63 are the foundation of Sections 20 to 23; the section gives their sketches, and the complete proofs, in Appendix E, can wait for a second reading.

Symbol notice for this section. Seven conventions hold throughout the section.

1. The letter r is the continuously compounded risk-free rate, not the ratio q/p of the general notation table, and q is a continuous yield, not $1-p$. The literature's $N(d)$ is written $\Phi(d)$ here, and ln is the natural logarithm.
2. A subscript t is a time (S_t, W_t, F_t; and $V_t = V(S_t, t)$ is a value at time t). Partial derivatives are written with the partial sign, $\partial_t f$, $\partial_x f$, $\partial_{xx} f$, never as subscripts.
3. The letter ξ is the Itô integrand, so the intermediate point of Corollary 7.32 (written ξ in §7) is written with a star here: t^*, x^*, u^*.
4. The symbols α_t and β_t, always with a subscript or with arguments, are the drift and diffusion coefficients of an Itô process. They are not the GARCH coefficients of §22 or a CAPM beta.
5. The letter ν is a constant log-drift, not the number of degrees of freedom of Student's t distribution (§4), for which the statistics literature uses the same letter.
6. In §19.2, W_1 and W_4 are values of one motion at times 1 and 4. From §19.8 on, a first subscript names the motion or the asset ($W_{k,t}$, $S_{k,t}$), except in §19.9, which has a single motion and reads a single subscript as a time again.
7. The probability P is countably additive in this Part, an axiom stated once in the Setup of §19.2; Sections 1 to 15 used countable additivity only implicitly, in sums over the values of a discrete variable.

19.1 How Volatility Scales with Horizon

Volatility quoted for one horizon must constantly be converted to another: annual to daily on a trading desk, daily to monthly in a risk report. The conversion follows a square-root law, and interviews test both the rule ("16 vol is 1 percent a day") and the reason ("why not multiply by 252?"). This subsection proves the law under independent returns and records the conversion constants worth memorising. The fact proved here, that variance and not volatility adds over time, is the property that the Wiener process of §19.2 builds into continuous time.

Setup. Trading periods are indexed $i = 1, \dots, n$, and $S_i > 0$ is the price at the end of period i, with S_0 the price today. The *log-return* of period i is $R_i = \ln(S_i/S_{i-1})$, and the volatility of one period is $\sigma_{\text{per}} = \sqrt{\operatorname{Var}(R_i)}$, assumed finite and the same for every i. Log-returns are the right object here because they add across periods: the n-period log-return is exactly

$$\ln(S_n/S_0) = R_1 + \cdots + R_n,$$

whereas simple returns compound rather than add. The n-period volatility means $\sqrt{\operatorname{Var}(\ln(S_n/S_0))}$.

Two random variables X and Y are *independent* if, for every pair of sets of real numbers, the probability that X lies in the first and Y in the second is the product of the two probabilities; the Setup of §19.2 extends this to random vectors and to several of them. The one consequence used in this subsection is the *product rule for means*: if X and Y are independent and $\mathbb{E}|X|$ and $\mathbb{E}|Y|$ are finite, then

$$\mathbb{E}|XY| = \mathbb{E}|X|\,\mathbb{E}|Y| \qquad \text{and} \qquad \mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]. \tag{19.1}$$

Indeed, Fact 18.34(ii) applied to the nonnegative function $|x|\,|y|$, whose mean over the second variable at a fixed x is $|x|\,\mathbb{E}|Y|$, gives $\mathbb{E}|XY| = \mathbb{E}\big[|X|\,\mathbb{E}|Y|\big] = \mathbb{E}|X|\,\mathbb{E}|Y|$, which is finite. Applied then to xy, whose mean over the second variable at a fixed x is $x\,\mathbb{E}[Y]$, it gives $\mathbb{E}[XY] = \mathbb{E}\big[X\,\mathbb{E}[Y]\big] = \mathbb{E}[X]\,\mathbb{E}[Y]$. In both steps the constant comes out of the expectation by linearity

(Theorem 1.2, which this Part uses for general random variables, as the Setup of §19.2 declares). In particular independent variables with finite variances are uncorrelated, because a finite second moment gives a finite mean ($|x| \le (1+x^2)/2$, and a variable dominated by one with a finite mean has a finite mean, the domination property declared with linearity in the same Setup). Lemma 19.17 extends the rule to functions $\psi_1(X)$ and $\psi_2(Y)$ of independent random vectors.

Proposition 19.1 (Square-root-of-time volatility scaling)**.** *Let $R_1, \dots, R_n$ be iid log-returns with finite variance σ_{per}^2. Then*

$$\operatorname{Var}\big(\ln(S_n/S_0)\big) = n\,\sigma_{\text{per}}^2, \qquad \sqrt{\operatorname{Var}\big(\ln(S_n/S_0)\big)} = \sigma_{\text{per}}\sqrt{n}\,:$$

variance grows linearly in the horizon, and the n-period volatility is $\sigma_{\text{per}}\sqrt{n}$.

Proof. We apply the variance computation of Proposition 10.3 to the log-returns. That proposition treats iid P&L increments $X_1, \dots, X_T$ with standard deviation σ_1. Read with R_i in place of X_t, with n in place of T and with σ_{per} in place of σ_1, the expansion displayed in its proof is

$$\begin{aligned}\operatorname{Var}\big(\ln(S_n/S_0)\big) = \operatorname{Var}\Big(\sum_{i=1}^{n} R_i\Big) &= \sum_{i=1}^{n}\operatorname{Var}(R_i) + \sum_{i\neq j}\operatorname{Cov}(R_i,R_j)\\ &= n\,\sigma_{\text{per}}^2 + \sum_{i\neq j}\operatorname{Cov}(R_i,R_j).\end{aligned} \tag{19.2}$$

The expansion squares the centred sum and uses only linearity of expectation, so it holds for any common finite variance, $\sigma_{\text{per}} = 0$ included; the hypothesis $\sigma_1 > 0$ of that proposition serves only its last step, the division that forms a Sharpe ratio. For $i \neq j$ the variables R_i and R_j are independent with finite variances, hence uncorrelated by the product rule (19.1) of the Setup: $\mathbb{E}[R_iR_j] = \mathbb{E}[R_i]\,\mathbb{E}[R_j]$, so every cross term vanishes, as in that proof. What remains is $\operatorname{Var}(\ln(S_n/S_0)) = n\,\sigma_{\text{per}}^2$, and taking the square root, which is monotone on $[0,\infty)$, gives the volatility statement.

The computation needs only that the cross terms of (19.2) vanish, that is, uncorrelated returns with a common finite variance. Independence and identical distributions are sufficient, not necessary. □

Proposition 10.3 applies the same computation to the mean, which grows like n, and hence to the Sharpe ratio, which grows like $\sqrt{n}$.

(Problem P19.1 asks you to reproduce this scaling argument, and its failure modes, from scratch.)

Corollary 19.2 (Annualization convention)**.** *Let the daily log-returns be iid with volatility σ_{daily} (the σ_{per} of a one-day period), and count 252 trading days in a year. The annual volatility σ_{annual}, which is the volatility of the 252-day log-return, is then*

$$\sigma_{\text{annual}} = \sqrt{252}\,\sigma_{\text{daily}} \approx 15.9\,\sigma_{\text{daily}}.$$

In particular a 16% annual volatility corresponds to a typical daily move of about 1%: the desk anchor "16 vol is 1 percent a day".

Proof. We specialise Proposition 19.1: take one period to be a trading day and $n = 252$. Numerically $\sqrt{252} = 15.87$, and $16\%/15.87 = 1.01\%$. □

The corollary returns in §21, as the breakeven daily move of a long option position, and in §22, where a historical estimate of daily volatility is annualised.

Remark 19.3 (What the scaling rule assumes). Proposition 19.1 needs uncorrelated increments with a common variance. The departure from the rule is read off the cross terms of (19.2); for two periods that expansion is Proposition 10.10, $\operatorname{Var}(R_1 + R_2) = 2\sigma_{\text{per}}^2 + 2\operatorname{Cov}(R_1, R_2)$. Markets break the first assumption in measurable ways. Momentum (positive autocorrelation) makes the cross terms positive, so long-horizon volatility exceeds the square-root extrapolation; mean reversion makes them negative, so it falls short. The *variance ratio* $\operatorname{Var}(\ln(S_n/S_0))/(n\,\sigma_{\text{per}}^2)$ measures exactly this departure from one. Volatility clustering acts differently. In the standard clustering model, the GARCH model of §22, returns stay uncorrelated and, when the model is stationary, their unconditional variance is constant, so the rule still holds for the unconditional variance. What moves is the conditional variance, the quantity that a point-in-time daily estimate measures, which is why such an estimate extrapolates poorly to long horizons. Intraday and day-of-week seasonality, in which the variance itself differs from one period to the next, breaks the common-variance assumption at short horizons. The Wiener process of §19.2 builds both assumptions in exactly.

Example 19.4 (Scaling a 16 percent annual volatility)**.** An asset carries $\sigma_{\text{annual}} = 16\%$. By Corollary 19.2 its daily volatility is $16\%/\sqrt{252} = 16\%/15.87 = 1.01\%$, and by Proposition 19.1 with twelve monthly periods its monthly volatility is $16\%/\sqrt{12} = 4.6\%$. The inverse check closes the loop: $1.01\% \times 15.87 = 16.0\%$. Both conversions divide by the square root of the number of periods per year, never by the number of periods itself.

Interview trap. The classic annualisation error is to scale volatility by the horizon instead of its square root: multiplying the daily volatility of Example 19.4, $16\%/\sqrt{252}$ (1.01% after rounding), by 252 "annualises" it to $16\% \times \sqrt{252} = 254\%$. Variance is the additive quantity in Proposition 19.1, not volatility: the factor 252 belongs to the variance, so applying it to the standard deviation double-counts time. By (19.2) the variance reaches $n^2\sigma_{\text{per}}^2$ only if each of the $n(n-1)$ cross covariances takes its largest possible value σ_{per}^2 (the Cauchy–Schwarz inequality, Lemma 19.12(ii) of the next subsection, applied to the centred returns), that is, only if every pair of daily returns has correlation one: a 254% figure assumes that the days never offset each other at all. The correct multiplier for volatility is $\sqrt{252} = 15.9$, which gives back 16%.

Variance per unit of time. Read as a rate, the rule says that variance accrues in proportion to elapsed time. An annual variance of $0.16^2 = 0.0256$ is $0.0256/252 = 0.000102 = (1.01\%)^2$ per trading day and $5 \times 0.0256/252 = 0.000508$ per five-day week, a weekly volatility of $\sqrt{0.000508} = 2.25\%$, which is the daily volatility times $\sqrt{5}$. Time is counted in trading days here; the empirical reason why a weekend adds little variance is recorded in §22.4. A process whose increments are normal with variance proportional to elapsed time, and independent across disjoint periods, is exactly the object defined next.

19.2 The Wiener Process

Interviewers ask about Brownian motion directly ("what is $\mathbb{E}[W_s W_t]$?", "why is $(dW)^2 = dt$?", "what is the probability that the path touches 1 before time 1?") and indirectly, through every continuous-time pricing argument. The object is motivated by two ideas, the Markov property and a limit of finer and finer random walks.

A process is *Markov* if its future, given its present value, does not depend on the route by which the present was reached. The Cox–Ross–Rubinstein tree of §18 is the discrete example: each move out of a node depends only on the node. For prices the property is motivated by an equilibrium argument, the weak form of market efficiency: if a signal read off the past path, such as a run of three rising days, predicted returns, traders would trade on it until it stopped predicting them. The two are different statements, efficiency being a claim about forecasts and the Markov property a claim about conditional laws. The property concerns the path, not its statistics: past returns still estimate the size of future moves (§22.4), and a Markov price may drift.

The continuous-time limit comes from refining the iid grid of §19.1 while the variance per unit of time stays fixed. Let T be a horizon in years and σ^2 a variance per year: n periods of length T/n, each with variance $\sigma^2 T/n$, have total variance $\sigma^2 T$ whatever n is (Proposition 19.1). The Cox–Ross–Rubinstein tree of §18, with n steps of length $\Delta t = T/n$, is calibrated in this way: each of its steps has variance $\sigma^2 \Delta t$ to leading order (Proposition 18.21), and its log-price converges in law to a normal variable (Theorem 18.37). The central limit theorem (Fact 4.8) is the reason the limiting increments are normal. The limiting object is the Wiener process.

This subsection defines the Wiener process, proves its covariance, scaling and quadratic variation (the precise content of $(dW)^2 = dt$), collects the limit tools that the rest of the Part uses (Lemma 19.12), and proves that its paths have unbounded variation and are nowhere differentiable (Theorem 19.14), which is the reason the integral of §19.3 has to be built by hand. The touching probability of the third question is computed in §19.9, by Theorem 19.63.

Setup. Time $t \geq 0$ is measured in years. A *stochastic process* is a family $(X_t)_{t \geq 0}$ of random variables on one probability space, and its *path* is the map $t \mapsto X_t$. The probability P is *countably additive*:

$$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots \qquad \text{for disjoint events } A_1, A_2, \ldots$$

(Kolmogorov, 1933). This is an axiom of the probability model, not a theorem. Sections 1 to 15 used countable additivity only implicitly, in sums of probabilities over the countably many values of a discrete variable (the marginal sums in the proof of Theorem 1.2, the geometric law of §1.3); the Part states the axiom once and collects its consequences in Lemma 19.12(0). Two working properties of expectation are declared beside it. Expectation is *linear* (Theorem 1.2, proved in §1 for discrete variables and used in this Part for general random variables). Expectation is also *monotone*: if $X \leq Y$ pointwise and both means exist, then $\mathbb{E}[X] \leq \mathbb{E}[Y]$; in particular a variable X with $|X| \leq Y$ for a variable Y with $\mathbb{E}[Y] < \infty$ has $\mathbb{E}|X| \leq \mathbb{E}[Y] < \infty$. When both means are finite, monotonicity follows from linearity, because $\mathbb{E}[Y] - \mathbb{E}[X] = \mathbb{E}[Y - X]$ and the mean of a nonnegative variable is nonnegative. The domination clause, which the Part uses to show that a mean is finite, is declared with linearity; for discrete variables it is the comparison test for series of nonnegative terms.

The letters A, A_n and $A_{n,M}$ denote events, and A^{c} is the complement of A. An event is *determined by* a random vector X if it has the form $\{X \text{ lies in a given set}\}$. Random vectors $X_1, \ldots, X_k$ are *mutually independent* if

$$P(A_1 \cap \cdots \cap A_k) = P(A_1) \cdots P(A_k)$$

for all events $A_1, \ldots, A_k$ such that each A_j is determined by X_j; this identity is the *product rule* for the events, and for $k = 2$ the two vectors are called *independent*. Leaving out some of the vectors keeps the others mutually independent, because an event determined by any vector may be the sure event. An infinite family of random vectors is mutually independent if every finite subfamily is. Several Brownian motions are mutually independent if, for every finite set of times, the vectors of their values at those times are mutually independent; they are independent of a family of random variables if these vectors and any finitely many members of the family are mutually independent. Two further properties of independence are used without proof.

(a) (Grouping) If $X_1, \ldots, X_k$ are mutually independent random vectors and $\{1, \ldots, k\}$ is split into disjoint groups, the vectors formed by the groups are mutually independent. In particular, functions of disjoint groups are mutually independent, because an event determined by a function of a vector is determined by the vector itself.

(b) (Intervals suffice) Random vectors $X_1, \ldots, X_k$ are mutually independent as soon as the product rule holds for all events of the form $A_j = \{X_j \text{ lies in a product of intervals}\}$.

For events that are intersections of events determined by single vectors X_j, property (a) is the definition. For a general event determined by a group, and for (b), a proof needs the uniqueness

of a probability fixed by its values on products of sets (Dynkin's π-λ theorem), which is measure theory, part of the product-measure machinery behind Fact 19.7. The product rule for means (19.1) of §19.1 holds for independent random variables in this sense.

Throughout, $Z \sim \mathcal{N}(0,1)$, with density φ and distribution function Φ. As in Lemma 18.35(i), $X \sim \mathcal{N}(\mu_X, \sigma_X^2)$ with $\sigma_X \geq 0$ means that X has the law of $\mu_X + \sigma_X Z$; here μ_X and σ_X are the mean and the standard deviation of X, not the drift and the volatility of an asset. For constants c and c',

$$cX + c' \sim \mathcal{N}(c\mu_X + c',\ c^2\sigma_X^2). \tag{19.3}$$

Indeed, applying one function to two variables with the same law gives two variables with the same law, so $cX + c'$ has the law of $(c\mu_X + c') + c\sigma_X Z$. Moreover $-Z$ has the law of Z, because $P(-Z \leq x) = 1 - \Phi(-x) = \Phi(x)$ by the symmetry of φ, so $c\sigma_X Z$ has the law of $|c|\sigma_X Z$ (this is also Lemma 18.35(ii) with $a_2 = 0$). Hence $cX + c'$ has the law of $(c\mu_X + c') + |c|\sigma_X Z$, which is (19.3) by the definition just recalled. For $c = 0$ or $\sigma_X = 0$ the law in (19.3) is that of the constant $c\mu_X + c'$.

For a partition $0 = t_0 < t_1 < \cdots < t_n = t$ of $[0,t]$ write

$$\Delta t_i = t_i - t_{i-1}, \qquad \Delta W_i = W_{t_i} - W_{t_{i-1}}, \qquad \Delta t_{\max} = \max_{1 \leq i \leq n} \Delta t_i,$$

where n counts the partition steps and $\Delta t_{\max}$ is the *mesh* of the partition (the mesh is never written with a lower-case delta, which §23 keeps for a jump parameter). The sums

$$\mathrm{QV}_n = \sum_{i=1}^{n} (\Delta W_i)^2 \qquad \text{and} \qquad \mathrm{FV}_n = \sum_{i=1}^{n} |\Delta W_i|$$

are the *quadratic-variation sum* and the *first-variation sum* of the partition. Their names are upright because an italic Q would sit beside the risk-neutral measure $\mathbb{Q}$ of §18 and §20. Until §19.8, and again in §19.9, W_1, W_4 and the like are values of one motion at the times 1 and 4.

Definition 19.5 (Standard Brownian motion (Wiener process))**.** A *standard Brownian motion*, or *Wiener process*, is a stochastic process $(W_t)_{t \geq 0}$ with four properties.

(1) The process starts at zero: $W_0 = 0$.
(2) For $0 \leq s < t$ the increment $W_t - W_s$ has the law $\mathcal{N}(0, t-s)$.
(3) For every finite grid of times $0 \leq t_0 < t_1 < \cdots < t_k$ the increments $W_{t_1} - W_{t_0}, \ldots, W_{t_k} - W_{t_{k-1}}$ are mutually independent.
(4) The path $t \mapsto W_t$ is continuous.

The symbol dW_t stands for the increment of W over an infinitesimal interval dt. It is shorthand only: every statement written with it is made precise by an integral identity (Theorem 19.23, Definition 19.25) or, for the multiplication rules of Table 4, by a statement about sums over partitions (Propositions 19.10 and 19.52(ii)).

Example 19.6 (Two tail probabilities)**.** Properties (1) and (2) give $W_t = W_t - W_0 \sim \mathcal{N}(0,t)$ for $t > 0$, so W_t has standard deviation $\sqrt{t}$ and $W_t/\sqrt{t}$ is standard normal. Hence

$$P(W_1 > 1.5) = P(Z > 1.5) = \Phi(-1.5) = 0.0668,$$

between the one- and two-sigma anchors 15.9% and 2.28% of Table 2. At $t = 4$ the standard deviation is $\sqrt{4} = 2$, so a move of two units is a one-sigma event:

$$P(|W_4| > 2) = P(|Z| > 1) = 2\,\Phi(-1) = 31.7\%.$$

Figure 35 shows the band ± 2 at $t = 4$.

Intuition. Independent increments make W Markov: the future value $W_t = W_s + (W_t - W_s)$ is the present value plus an increment that is independent of the path up to time s (Proposition 19.8(iv) makes this precise). Sampled on a grid of step Δt, W is the Gaussian random walk of §19.1 with variance Δt per step: variance accrues at rate one, which is Proposition 19.1 in continuous time. By the central limit theorem (Fact 4.8), any iid walk with that variance per step looks the same at long horizons, which is why W is the universal model; Theorem 18.37 proves this for the binomial tree, with a rate. Figure 35 shows one path together with the envelopes $\pm\sqrt{t}$ and $\pm 2\sqrt{t}$.

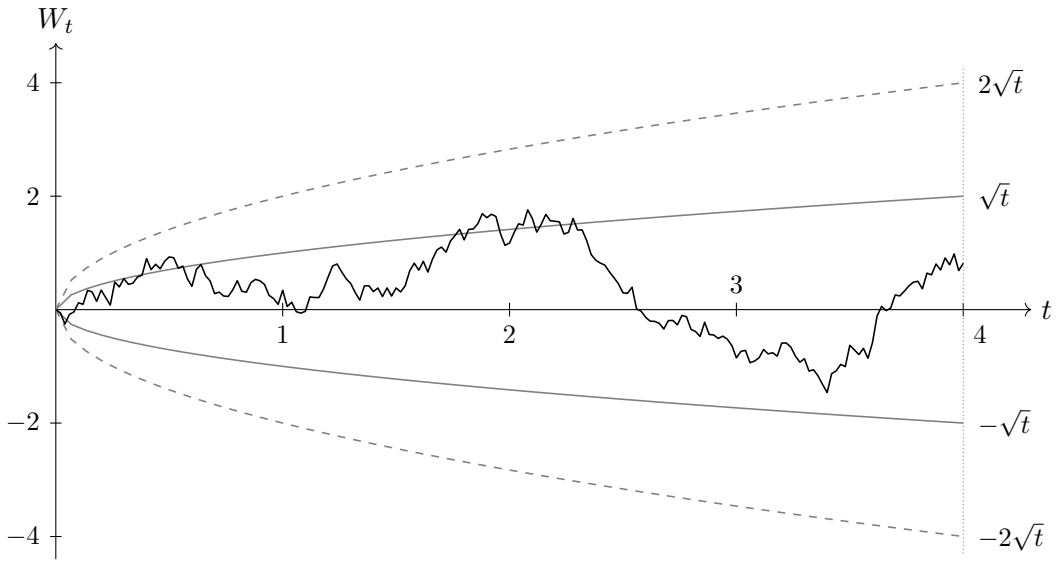


Figure 35: One Brownian path on $[0, 4]$, sampled in 200 steps of 0.02, with the envelopes $\pm\sqrt{t}$ (solid gray) and $\pm 2\sqrt{t}$ (dashed). At each *fixed* t about 68% and 95% of paths lie inside the two bands; the statement is pointwise, not for the whole path, whose probabilities are those of Theorem 19.63. At $t = 4$ the inner band is ± 2 (Example 19.6). The path is equally rough at every scale (Proposition 19.8(ii)).

Interview trap. "Markov means unpredictable, so a Markov price has zero expected return." The Markov property says that the path so far adds nothing once the price today is known. A drift is allowed, and weak-form efficiency says nothing about the size of the expected return μ.

Fact 19.7 (Existence of Brownian motion (Wiener, 1923))**.** On some probability space, with P countably additive as declared in the Setup, there is a process with the four properties of Definition 19.5. Moreover, on one probability space there exist finitely many standard Brownian motions together with any countable family of random variables with prescribed laws (normal variables, Poisson counts and the like), the motions and the variables being mutually independent in the sense of the Setup (the product construction of Kolmogorov, 1933). A proof is beyond our scope.

A proof needs a measure-theoretic construction on the space of paths (Lévy's series of Schauder functions, with a Borel–Cantelli argument for its uniform convergence, or Kolmogorov's extension and continuity theorems) and product measures for the independent family. The second assertion of the Fact supplies the second motion of Proposition 19.52 and Theorem 19.55, the two-asset clause of Definition 20.6, and the normal and Poisson variables of Proposition 23.60.

One consistency check is immediate. For $s < t < u$ the increments over $[s, t]$ and $[t, u]$ are independent with laws $\mathcal{N}(0, t - s)$ and $\mathcal{N}(0, u - t)$, and their sum $W_u - W_s$ must have the law $\mathcal{N}(0, u - s)$. That a sum of independent normal variables is normal, with the variances added, is Lemma 18.35(ii), so properties (2) and (3) agree on adjacent intervals; that all four properties hold together is the content of Fact 19.7.

Proposition 19.8 (Covariance, scaling, symmetry and restart of Brownian motion)**.** *Let W be a standard Brownian motion.*

(i) (Covariance) For all $s, t \geq 0$: $\mathbb{E}[W_t] = 0$, $\mathbb{E}[W_t^2] = t$, and

$$\mathrm{Cov}(W_s, W_t) = \mathbb{E}[W_s W_t] = \min(s, t);$$

hence $\mathrm{Corr}(W_s, W_t) = \sqrt{s/t}$ for $0 < s \leq t$.

(ii) (Scaling) For every constant $c > 0$ the process $t \mapsto c^{-1/2} W_{ct}$ is a standard Brownian motion.

(iii) (Symmetry) The process $-W$ is a standard Brownian motion.

(iv) (Restart) For every $u \geq 0$ the process $s \mapsto W_{u+s} - W_u$ is a standard Brownian motion, and every finite family of its increments is independent of every finite family of the values $W_{s'}$, $0 \leq s' \leq u$.

Sketch of the proof. We expand a product for (i) and check Definition 19.5 for (ii) to (iv). For $0 < s < t$, write $W_t = W_s + (W_t - W_s)$: the two terms are independent increments with mean zero (properties (1) to (3)), so by the product rule (19.1)

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}[W_s]\,\mathbb{E}[W_t - W_s] = s.$$

In (ii) to (iv) an increment of the new process is a constant times an increment of W, hence normal by (19.3), with variance $t - s$ (in (ii), $c^{-1}\,c\,(t-s)$); independence over a grid and continuity are inherited from W. The last clause of (iv) merges all the times into one grid: the values up to u and the later increments are functions of its increments before and after u, two independent groups (grouping property (a)). The complete proof is in Appendix E.1. □

Part (ii) is the continuous-time form of Proposition 19.1, and the reason a path looks the same at every zoom. Part (iii) is used in Example 19.33, Corollary 19.49 and Theorem 19.63; part (iv) in Theorems 19.14, 19.23 and 19.30. Part (ii) is used again only in the caption of Figure 35.

Example 19.9 (Covariances of Brownian values)**.** By Proposition 19.8(i), $\mathbb{E}[W_2 W_5] = \min(2, 5) = 2$ and $\mathrm{Corr}(W_2, W_5) = 2/\sqrt{2 \times 5} = 2/\sqrt{10} = 0.632$. By Proposition 10.10,

$$\mathrm{Var}(W_2 + W_5) = 2 + 5 + 2 \times 2 = 11,$$

not 7: the two values share the increment over $[0, 2]$. In the same way $\mathrm{Corr}(W_1, W_4) = \sqrt{1/4} = 0.5$.

Interview trap. Two slips are standard. "The standard deviation of W_t is t": it is $\sqrt{t}$, that is 2 and not 4 at $t = 4$. "The values W_s and W_t are independent": they share the increment over $[0, s]$. The covariance is $\min(s, t)$, and $\mathrm{Var}(W_s + W_t) = s + t + 2s = t + 3s$ for $s \leq t$, which is 11 and not 7 for $W_2 + W_5$. Only increments over disjoint intervals are independent.

Proposition 19.10 (Quadratic variation of Brownian motion)**.** *Let W be a standard Brownian motion and let $0 = t_0 < t_1 < \cdots < t_n = t$ be a partition of $[0, t]$ with mesh $\Delta t_{\max}$.*

(i) The quadratic-variation sum $\mathrm{QV}_n = \sum_{i=1}^n (\Delta W_i)^2$ has

$$\mathbb{E}[\mathrm{QV}_n] = t \qquad \text{and} \qquad \mathrm{Var}(\mathrm{QV}_n) = 2\sum_{i=1}^n (\Delta t_i)^2 \leq 2t\,\Delta t_{\max},$$

so $\mathbb{E}\big[(\mathrm{QV}_n - t)^2\big] \to 0$ as $\Delta t_{\max} \to 0$: the sum converges to t in mean square (the modes of convergence are collected in Lemma 19.12(iii)).

(ii) The sum $\sum_{i=1}^n \Delta t_i\,\Delta W_i$ has mean 0 and second moment $\sum_{i=1}^n (\Delta t_i)^3 \leq t\,\Delta t_{\max}^2$, which tends to 0 as $\Delta t_{\max} \to 0$.

(iii) The deterministic sum satisfies $\sum_{i=1}^n (\Delta t_i)^2 \leq t\,\Delta t_{\max}$, which tends to 0 as $\Delta t_{\max} \to 0$.

Proof. We compute the mean and the variance of each sum from the normal moments of the increments. By Definition 19.5 the increments $\Delta W_1, \dots, \Delta W_n$ are independent and $\Delta W_i \sim \mathcal{N}(0, \Delta t_i)$; equivalently $\Delta W_i = \sqrt{\Delta t_i}\, Z_i$, where $Z_i = \Delta W_i/\sqrt{\Delta t_i}$, $i = 1, \dots, n$, are iid standard normal variables.

Part (iii). Each Δt_i is at most $\Delta t_{\max}$, and the cells add up to $[0, t]$, so

$$\sum_{i=1}^{n} (\Delta t_i)^2 \le \Delta t_{\max} \sum_{i=1}^{n} \Delta t_i = t\, \Delta t_{\max}.$$

Part (i). By Proposition 7.25(ii) and (iii), $\mathbb{E}[(\Delta W_i)^2] = \Delta t_i$ and $\mathbb{E}[(\Delta W_i)^4] = 3(\Delta t_i)^2$. Hence

$$\mathbb{E}[\mathrm{QV}_n] = \sum_{i=1}^{n} \Delta t_i = t, \qquad \mathrm{Var}\left((\Delta W_i)^2\right) = 3(\Delta t_i)^2 - (\Delta t_i)^2 = 2(\Delta t_i)^2.$$

The squares $(\Delta W_i)^2$ are functions of mutually independent variables taken one at a time, hence mutually independent by the grouping property (a) of the Setup; they have finite variances, so any two of them are uncorrelated by the product rule (19.1). The variance of their sum expands as in (19.2), which is Proposition 10.10 extended to n terms as in the proof of Proposition 10.3, now with unequal variances: it is the sum of the variances plus the sum of the cross covariances, and every cross covariance vanishes. Therefore

$$\mathrm{Var}(\mathrm{QV}_n) = \sum_{i=1}^{n} \mathrm{Var}\left((\Delta W_i)^2\right) = 2 \sum_{i=1}^{n} (\Delta t_i)^2 \le 2t\, \Delta t_{\max}$$

by part (iii). Since $\mathbb{E}[\mathrm{QV}_n] = t$, the mean-square error $\mathbb{E}[(\mathrm{QV}_n - t)^2]$ is this variance, and it tends to 0 with the mesh.

Part (ii). Each term $\Delta t_i\, \Delta W_i$ has mean zero and variance $(\Delta t_i)^2\, \Delta t_i = (\Delta t_i)^3$. The terms are constant multiples of the mutually independent increments, hence uncorrelated by the grouping property (a) and the product rule (19.1), so by the same expansion the sum has mean zero and its second moment is its variance,

$$\sum_{i=1}^{n} (\Delta t_i)^3 \le \Delta t_{\max}^2 \sum_{i=1}^{n} \Delta t_i = t\, \Delta t_{\max}^2. \qquad \square$$

(Problem P19.2 asks you to reproduce this proof from scratch.)

Parts (i) to (iii) are the precise content of the bookkeeping rules $(dW)^2 = dt$, $dt\, dW = 0$ and $(dt)^2 = 0$: summed over a fine partition, the products on the left behave, in mean square, like the quantities on the right. Table 4 collects the rules; its last row is filled in by Proposition 19.52(ii).

Product	Rule	Made precise by
$dt \cdot dt$	0	Proposition 19.10(iii)
$dt \cdot dW_t$	0	Proposition 19.10(ii)
$dW_t \cdot dW_t$	dt	Proposition 19.10(i)
$dW_{1,t} \cdot dW_{2,t}$	$\rho\, dt$	Proposition 19.52(ii)

Table 4: The multiplication rules of stochastic calculus. Each is a statement about sums over a partition whose mesh tends to zero. In the last row $W_{1,t}$ and $W_{2,t}$ are the values at time t of two Brownian motions with correlation ρ, which §19.8 names W_1 and W_2 (from there on a first subscript names the motion, except in the single-motion §19.9).

The weighted version of the third row, in which each $(\Delta W_i)^2$ is multiplied by a weight fixed at the start of its cell, is Step 4 of the proof of Theorem 19.30.

Example 19.11 (Quadratic and first variation on a grid)**.** On $[0,1]$ with n equal steps, $\Delta t_i = 1/n$, so Proposition 19.10(i) gives $\mathbb{E}[\mathrm{QV}_n] = 1$ and $\mathrm{Var}(\mathrm{QV}_n) = 2n(1/n)^2 = 2/n$. With $n = 4$ the variance is 0.5 (standard deviation 0.71); with $n = 400$ it is 0.005 (standard deviation 0.071). For comparison, the smooth path $s \mapsto s$ has sum of squared increments $n(1/n)^2 = 1/n$, which tends to 0.

Over the same grids the first variation $\mathrm{FV}_n = \sum_i |\Delta W_i|$ has mean

$$\mathbb{E}[\mathrm{FV}_n] = n\sqrt{1/n}\,\mathbb{E}|Z| = \sqrt{2n/\pi}, \qquad \mathbb{E}|Z| = 2\int_0^\infty z\,\varphi(z)\,dz = 2\,\varphi(0) = \sqrt{2/\pi},$$

where the first equality for $\mathbb{E}|Z|$ is Fact 7.22 with the symmetry of φ, and the integral is evaluated with the antiderivative $-\varphi$ of $z\,\varphi(z)$. The mean of FV_n is 1.60 at $n = 4$ and 15.96 at $n = 400$. The sum of squares settles at the elapsed time while the path length explodes; Theorem 19.14(i) makes the explosion a theorem.

Interview trap. "The rule $(dW)^2 = dt$ means that each squared increment equals dt." Each $(\Delta W)^2 = \Delta t\, Z^2$ is random, with mean Δt and standard deviation $\sqrt{2}\,\Delta t$, larger than its mean. Only the sum over a fine partition is deterministic in the limit, because its variance $2\sum_i (\Delta t_i)^2$ vanishes.

Lemma 19.12 (Continuity of probability, Markov, Cauchy–Schwarz and the modes of convergence)**.** *Let X, Y, X_n and Y_n $(n = 1, 2, \dots)$ be random variables on one probability space.*

(0) (Continuity of probability) For events $A_1 \subset A_2 \subset \cdots$,

$$P\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \lim_{n\to\infty} P(A_n);$$

for events $A_1 \supset A_2 \supset \cdots$, $P(\bigcap_n A_n) = \lim_n P(A_n)$; and for any events $A_1, A_2, \dots$, $P(\bigcup_n A_n) \le \sum_n P(A_n)$ (countable subadditivity).

(i) (Markov, Chebyshev) If $X \ge 0$ and $a > 0$, then $P(X \ge a) \le \mathbb{E}[X]/a$. Hence $P(|X - Y| \ge \varepsilon) \le \mathbb{E}[(X-Y)^2]/\varepsilon^2$ for every $\varepsilon > 0$.

(ii) (Cauchy–Schwarz) If $\mathbb{E}[X^2]$ and $\mathbb{E}[Y^2]$ are finite, then

$$\big(\mathbb{E}|XY|\big)^2 \le \mathbb{E}[X^2]\,\mathbb{E}[Y^2].$$

In particular $\big(\mathbb{E}[(X+Y)^2]\big)^{1/2} \le \big(\mathbb{E}[X^2]\big)^{1/2} + \big(\mathbb{E}[Y^2]\big)^{1/2}$, so the mean-square distance $\big(\mathbb{E}[(X-Y)^2]\big)^{1/2}$ obeys the triangle inequality.

(iii) (Modes of convergence) The sequence X_n converges to X in probability *if $P(|X_n - X| > \varepsilon) \to 0$ for every $\varepsilon > 0$;* in mean square *if $\mathbb{E}[(X_n - X)^2] \to 0$; and* almost surely *if $P(X_n \to X) = 1$, that is, if $X_n \to X$ outside an event of probability zero. Mean-square convergence and almost-sure convergence each imply convergence in probability. Two limits in probability of the same sequence agree with probability one. If $X_n \to X$ and $Y_n \to Y$ in probability, then $X_n + Y_n \to X + Y$ and $X_n Y_n \to XY$ in probability.*

(iv) (Fourth-moment lemma) If $X_n \to X$ in probability, $\sup_n \mathbb{E}[X_n^4] < \infty$ and $\mathbb{E}[X^4] < \infty$, then $\mathbb{E}[(X_n - X)^2] \to 0$. In particular $\mathbb{E}[X_n] \to \mathbb{E}[X]$ and $\mathbb{E}[X_n^2] \to \mathbb{E}[X^2]$.

Sketch of the proof. We cut a union into disjoint pieces for (0), bound an indicator for (i), minimise a square for (ii), and split on a small event for (iii) and (iv). An increasing union is the disjoint union of A_1 and of the parts of each A_n outside A_{n-1}, to which countable additivity, the axiom of the Setup, applies; complements and the same cutting give the other two claims of (0). Taking expectations in $a\,\mathbf{1}\{X \ge a\} \le X$, valid because $X \ge 0$, gives (i), and Chebyshev's inequality is the case of $(X - Y)^2$. The quadratic $c \mapsto \mathbb{E}[(|X| - c\,|Y|)^2]$ in a real number c is nonnegative, and its minimum is the inequality of (ii). In (iii), Chebyshev's inequality and (0) give the two

implications, and unions of events of small probability give the rest. In (iv), (ii) bounds the part of $\mathbb{E}[(X_n - X)^2]$ on the event $\{|X_n - X| > \varepsilon\}$ by $(\mathbb{E}[(X_n - X)^4])^{1/2}$, bounded by the hypothesis, times $P(|X_n - X| > \varepsilon)^{1/2}$, which tends to zero; so $\mathbb{E}[(X_n - X)^2] \le 2\varepsilon^2$ for all large n, and two more uses of (ii) give the means and the second moments. The complete proof is in Appendix E.2. □

Example 19.13 (Chebyshev's bound for the quadratic variation)**.** Take the quadratic variation of Example 19.11 with $n = 400$ equal steps of $[0, 1]$, for which $\mathbb{E}[\mathrm{QV}_{400}] = 1$ and $\mathrm{Var}(\mathrm{QV}_{400}) = 2/400$. Chebyshev's inequality (Lemma 19.12(i), with the constant 1 in the role of Y and $\varepsilon = 0.1$) gives

$$P(|\mathrm{QV}_{400} - 1| \ge 0.1) \le \frac{\mathrm{Var}(\mathrm{QV}_{400})}{0.1^2} = \frac{2/400}{0.01} = 0.5,$$

and with $n = 4{,}000$ steps the bound is $(2/4000)/0.01 = 0.05$. The bound falls like $1/n$, which is the quantitative content of the statement that $\mathrm{QV}_n \to t$ in probability (Lemma 19.12(iii)).

Part (iv) of Lemma 19.12 is the Part's substitute for the dominated convergence theorem: it is cited wherever a limit passes through an expectation. The fourth-moment bounds that it asks for come from exponential growth (Lemma 19.20(iii)) and from normal moments.

Theorem 19.14 (Brownian paths have unbounded variation and are nowhere differentiable (Paley, Wiener and Zygmund, 1933))**.** *Let W be a standard Brownian motion. For $0 \le s < s'$, the* total variation *of the path on $[s, s']$ is the supremum of the sums $\sum_{i=1}^{n} |W_{t_i} - W_{t_{i-1}}|$ over all partitions $s = t_0 < t_1 < \cdots < t_n = s'$; write TV_t for the total variation on $[0, t]$, the supremum of FV_n over all partitions of $[0, t]$. There is an event of probability zero outside which the following hold.*

- *(i) For every $t > 0$, $\mathrm{TV}_t = \infty$. More generally the path has infinite total variation on every interval $[s, s']$ with $0 \le s < s'$; in particular it is monotone on no interval.*
- *(ii) The path is differentiable at no point of $[0, \infty)$; at 0, it has no right derivative.*

Sketch of the proof. Part (i) is forced by the quadratic variation and part (ii) by a counting bound. For (i), take the dyadic partitions of $[0, t]$ into n cells and a bound $M > 0$. Since $\mathrm{QV}_n \le \max_i |\Delta W_i| \, \mathrm{FV}_n$, a first variation of at most M forces a small quadratic variation or a large increment, and Chebyshev's inequality (with Proposition 19.10(i)) and Markov's inequality for fourth powers give

$$P(\mathrm{FV}_n \le M) \le P(\mathrm{QV}_n \le t/2) + P\Big(\max_{1 \le i \le n} |\Delta W_i| > \frac{t}{2M}\Big) \le \frac{8}{n} + \frac{48 M^4}{n\, t^2},$$

the first inequality being (E.1). As $\mathrm{TV}_t \ge \mathrm{FV}_n$, the paths with $\mathrm{TV}_t \le M$ lie in an event of probability zero, and countable unions with the restart property (Proposition 19.8(iv)) cover every M and every interval. For (ii), a path with a derivative at a point of $[0, 1)$ (from the right at 0) has, for an integer M and all large n, three consecutive increments of size at most $7M/n$ on the grid of step $1/n$. These are independent $\mathcal{N}(0, 1/n)$ variables, so the event $A_{n,M}$ that this happens at one of the n positions has probability at most $n\,(14M/\sqrt{2\pi n})^3 = 174.2\, M^3/\sqrt{n}$, which tends to zero; countable unions over M and over the unit intervals finish the proof. The complete proof is in Appendix E.3. □

Example 19.15 (The two bounds of the proof in numbers)**.** Take $[0, 1]$ with $n = 2^{10} = 1{,}024$ dyadic cells and $M = 2$ in Step 1 of the proof of Theorem 19.14. Each increment has standard deviation $\sqrt{1/1024} = 1/32$, so a cell move above $t/(2M) = 1/4$ is an 8-sigma event, of probability $2\,\Phi(-8) = 1.24 \times 10^{-15}$; finite subadditivity over the 1,024 cells bounds the probability that the largest move exceeds $1/4$. (The fourth-moment bound of that step gives only $48M^4/(n\,t^2) = 48 \times 16/1024 = 0.75$ for this probability: it serves the limit, not the number.) By (E.1),

$$P(\mathrm{FV}_n \le 2) \le \frac{8}{n} + n \cdot 2\,\Phi(-8) = \frac{8}{1024} + 1024 \times 2\,\Phi(-8) = 0.0078 + 1.27 \times 10^{-12},$$

about 0.0078. By the formula of Example 19.11, $\mathbb{E}[\mathrm{FV}_n] = \sqrt{2 \times 1024/\pi} = 25.5$, and the standard deviation of FV_n is $\sqrt{1 - 2/\pi} = 0.60$, because each of the n independent terms $|\Delta W_i|$ has variance $(1/n)(1 - 2/\pi)$. A first variation below 2 is very unlikely, and its typical size is about 25.5. The bound $174.2\, M^3/\sqrt{n}$ of Step 4 of the same proof is 1.74 at $M = 1$ and $n = 10^4$, which says nothing, and 0.0174 at $n = 10^8$. The differentiability argument is qualitative, and its force is that it holds for every M at once.

Intuition. Part (i) is the reason the integral of §19.3 is built as a mean-square limit of left-endpoint sums and not as a Riemann–Stieltjes integral, which cannot be defined against every continuous integrand when the integrator has infinite variation (a classical theorem of analysis, recalled in the opener of §19.3). Smoothness and quadratic variation exclude each other. A path with $|W_s - W_{s'}| \le c\,|s - s'|$ on $[0, t]$, for a constant c, would have $\mathrm{QV}_n \le c^2 \sum_i (\Delta t_i)^2 \le c^2\, t\, \Delta t_{\max} \to 0$ along every sequence of partitions whose mesh tends to zero, while $\mathrm{QV}_n \to t$ in probability (Proposition 19.10(i), Lemma 19.12(iii)). So paths with this property, for one constant c, cannot fill an event of positive probability (Problem P19.2(e)). This is a weaker form of part (i), because such a path has $\mathrm{FV}_n \le c\,t$ for every partition, so its total variation on $[0, t]$ is at most $c\,t$. By the mean value theorem it covers in particular every path that is differentiable on $[0, t]$ with $|W'| \le c$ throughout, a much smaller class than the paths that part (ii) excludes.

19.3 The Itô Integral

"Compute $\int_0^t W\, dW$" is a standard interview question, and its answer, $(W_t^2 - t)/2$, surprises anyone who applies the ordinary chain rule. The integral against dW is also the profit of a trading strategy: holding ξ_s units of an asset whose price moves by dW_s earns $\int \xi_s\, dW_s$. A classical theorem of analysis, not proved here, says that the Riemann–Stieltjes integral exists against every continuous integrand only for an integrator of finite variation, so Theorem 19.14(i) rules out the integral of calculus; and for $\int W\, dW$ the point at which the integrand is sampled changes the limit. The left end point of each step gives the gain of a strategy that decides before the move, and the right end point does not (Example 19.24). This subsection builds the integral for exactly the integrands the Part uses, as the mean-square limit of left-endpoint sums, and proves that it has mean zero and obeys the Itô isometry. It is the only definition of the stochastic integral in the document. It builds on Proposition 19.10, on Lemma 19.12 and on two tools of §18, the Fubini–Tonelli Fact 18.34 and the normal moment generating function of Lemma 18.35. It also states once, for the whole Part, the growth conditions that every later hypothesis cites (Definition 19.19).

Setup. The number $T > 0$ is a horizon, W is a standard Brownian motion, and $0 \le t_0 < t \le T$. A partition $t_0 < t_1 < \cdots < t_n = t$ of $[t_0, t]$ has steps $\Delta t_i = t_i - t_{i-1}$, increments $\Delta W_i = W_{t_i} - W_{t_{i-1}}$ and mesh $\Delta t_{\max} = \max_i \Delta t_i$, as in §19.2, where t_0 was 0; n counts its steps. For real numbers a and b, $a \wedge b = \min(a, b)$. The vector $Y = (W_{s_1}, \ldots, W_{s_m})$ collects values of W at fixed earlier times $s_j \le t_0$; m counts them, and $m = 0$ means that Y is absent. A *box* is a product of intervals, each of which may be a single point or unbounded, and $\mathcal{D}$ denotes one. The function $\ell(s)$ is a continuous boundary on $[0, T]$, and t_ℓ is the first time at which W reaches it (Definition 19.22(c)). The letters ξ and ζ denote integrands. The process and the function that generates it share the letter, $\xi_s = \xi(s, W_s, Y)\, \mathbf{1}\{s < t_\ell\}$, as in $V_t = V(S_t, t)$ for a price. The class of integrands is $\mathcal{H}$, and $\mathcal{I}_n(\xi)$ is a left-endpoint sum (Definition 19.22). The letters c, c', c'' are generic constants, re-declared where they are used. In Definition 19.19 and Lemma 19.20, d counts the real or positive variables of a function; it is not the down-factor of a tree (§18), and no tree factor appears in this section.

Independence is that of the Setup of §19.2, recalled here for two vectors. An event is *determined by* a random vector X if it has the form $\{X \text{ lies in a given set}\}$. Two random vectors X and Y are *independent* if $P(A \cap A') = P(A)\, P(A')$ for every event A determined by X and every event A' determined by Y; this identity is called the product rule for the two events. In Lemma 19.17, X

and Y are generic independent random vectors, and $\bar\psi(x) = \mathbb{E}\,\psi(x,Y)$ is the average of a function ψ over the second one. Where a second standard Brownian motion $W_\perp$ is present, it is *independent of* W: every finite vector of values of W is independent of every finite vector of values of $W_\perp$ (Fact 19.7 supplies such a pair). An integrand may then also depend on the values of $W_\perp$, and $W_{\perp,s}$ is its value at time s. A second motion is first needed in §19.8, where W is called W_1.

Fact 19.16 (Completeness of mean square (Riesz, 1907; Fischer, 1907))**.** Let $X_1, X_2, \dots$ be random variables on one probability space with $\mathbb{E}[X_n^2] < \infty$ for every n and $\mathbb{E}[(X_n - X_m)^2] \to 0$ as $m, n \to \infty$. Then there is a random variable X with $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[(X_n - X)^2] \to 0$, and any two such X agree with probability one. A proof is beyond our scope.

Existence needs Lebesgue integration: one extracts a subsequence that converges almost surely and identifies its limit with Fatou's lemma. Uniqueness alone is elementary, because two mean-square limits of one sequence are limits in probability, and those agree with probability one (Lemma 19.12(iii)). The Fact is used once in this section, for the existence of the Itô integral in Theorem 19.23, and again in the proofs of Proposition 21.22 and Corollary 21.24 (Appendix G), for hedge integrals that run up to expiry.

Lemma 19.17 (Freezing an independent variable)**.** *Let X and Y be independent random vectors, and let ψ be a Borel measurable function on the product of their ranges (as in Fact 18.34(i); continuity outside finitely many hyperplanes suffices in this Part), with $\psi \ge 0$, or with $\mathbb{E}|\psi(X,Y)| < \infty$. Put*

$$\bar\psi(x) = \mathbb{E}\,\psi(x,Y),$$

the mean of ψ with its first argument frozen at x. Then

$$\mathbb{E}\,\psi(X,Y) = \mathbb{E}\,\bar\psi(X). \tag{19.4}$$

In the integrable case, $\bar\psi(x)$ is finite outside a set of values x that X avoids with probability one. In particular, if ψ_1 and ψ_2 are Borel measurable and $\mathbb{E}|\psi_1(X)|$ and $\mathbb{E}|\psi_2(Y)|$ are finite, then $\mathbb{E}|\psi_1(X)\,\psi_2(Y)| = \mathbb{E}|\psi_1(X)|\,\mathbb{E}|\psi_2(Y)|$ is finite and

$$\mathbb{E}\big[\psi_1(X)\,\psi_2(Y)\big] = \mathbb{E}\,\psi_1(X)\,\mathbb{E}\,\psi_2(Y). \tag{19.5}$$

For discrete X and Y and $\mathbb{E}|\psi(X,Y)| < \infty$, (19.4) is the tower rule (Proposition 1.7) with conditioning on X, because $\mathbb{E}[\psi(X,Y) \mid X = x] = \bar\psi(x)$ when Y is independent of X.

Proof. We reduce the lemma to the Fubini–Tonelli Fact of §18. Identity (19.4) is Fact 18.34(ii). Its content is that the joint law of independent vectors is the product of their laws. When X and Y have densities f_X and f_Y, the joint density is $f_X(x)\,f_Y(y)$, and (19.4) is the exchange of iterated integrals of Fact 18.34(i):

$$\mathbb{E}\,\psi(X,Y) = \int \Big[\int \psi(x,y)\,f_Y(y)\,dy\Big] f_X(x)\,dx = \int \bar\psi(x)\,f_X(x)\,dx = \mathbb{E}\,\bar\psi(X).$$

When one of the two vectors is discrete, the corresponding integral is a sum.

The finiteness clause. Let $\mathbb{E}|\psi(X,Y)| < \infty$, and write $\overline{|\psi|}(x) = \mathbb{E}|\psi(x,Y)|$, a number in $[0,\infty]$. Identity (19.4), applied to the nonnegative function $|\psi|$, reads $\mathbb{E}\,\overline{|\psi|}(X) = \mathbb{E}|\psi(X,Y)| < \infty$. A nonnegative variable with a finite mean is finite with probability one. To see it, fix an integer $M \ge 1$ and apply Markov's inequality (Lemma 19.12(i)) to the real nonnegative variable $\min\big(\overline{|\psi|}(X), M\big)$, which is at least M exactly when $\overline{|\psi|}(X)$ is:

$$\begin{aligned} P\big(\overline{|\psi|}(X) = \infty\big) \le P\big(\overline{|\psi|}(X) \ge M\big) &= P\Big(\min\big(\overline{|\psi|}(X), M\big) \ge M\Big) \\ &\le \frac{\mathbb{E}\big[\min\big(\overline{|\psi|}(X), M\big)\big]}{M} \le \frac{\mathbb{E}|\psi(X,Y)|}{M}, \end{aligned}$$

the last step by monotonicity of expectation and (19.4) for $|\psi|$. The right side tends to 0 as $M \to \infty$, so the left side is zero. Wherever $\overline{|\psi|}(x)$ is finite, $\bar{\psi}(x)$ is finite, with $|\bar{\psi}(x)| \le \overline{|\psi|}(x)$ by monotonicity of expectation.

The product rule. Apply (19.4) first to the nonnegative function $|\psi_1(x)|\,|\psi_2(y)|$. With x frozen, the factor $|\psi_1(x)|$ is a constant, so by linearity of expectation (Theorem 1.2) the frozen mean is $|\psi_1(x)|\,\mathbb{E}|\psi_2(Y)|$, and (19.4) reads $\mathbb{E}|\psi_1(X)\,\psi_2(Y)| = \mathbb{E}|\psi_1(X)|\,\mathbb{E}|\psi_2(Y)|$, which is finite. The function $\psi_1(x)\,\psi_2(y)$ therefore falls under the integrable case. Its frozen mean is $\psi_1(x)\,\mathbb{E}\,\psi_2(Y)$, and (19.4) becomes (19.5).

Discrete variables. Let X and Y be discrete with $\mathbb{E}|\psi(X,Y)| < \infty$. For every x with $P(X = x) > 0$, independence gives $P(Y = y \mid X = x) = P(Y = y)$. Grouping the outcomes by the value of Y,

$$\mathbb{E}[\psi(X,Y) \mid X = x] = \sum_y \psi(x,y)\,P(Y = y \mid X = x) = \sum_y \psi(x,y)\,P(Y = y) = \bar{\psi}(x),$$

and Proposition 1.7, with $\psi(X,Y)$ as its integrable variable and X as its conditioning variable, reads $\mathbb{E}\,\bar{\psi}(X) = \mathbb{E}\,\psi(X,Y)$. In this case the lemma needs no Fact. □

Identity (19.5) extends the product rule for means (19.1) of §19.1, stated there for two random variables, to functions of two independent random vectors. The lemma replaces every use of a tower rule with a continuous conditioning variable in Sections 19 to 23: the recipe is to integrate out the independent variable first. The expectation form of Proposition 22.15(i) and Proposition 22.27 in §22, and Proposition 23.60 in §23, are examples.

Example 19.18 (Freezing one Brownian value)**.** Let W_1 and W_2 be the values of W at times 1 and 2. The increment $W_2 - W_1$ is independent of $W_1 = W_1 - W_0$ (Definition 19.5(3)), and $\{W_2 > W_1\} = \{W_2 - W_1 > 0\}$. Lemma 19.17 with $X = W_1$, $Y = W_2 - W_1$ and $\psi(x,y) = e^x\,\mathbf{1}\{y > 0\}$ gives the frozen mean $\bar{\psi}(x) = e^x\,P(W_2 - W_1 > 0) = e^x/2$, because the increment has the law $\mathcal{N}(0,1)$, which is symmetric about zero. Hence

$$\mathbb{E}\big[e^{W_1}\,\mathbf{1}\{W_2 > W_1\}\big] = \mathbb{E}\big[e^{W_1}\big] \times \tfrac{1}{2} = e^{1/2} \times \tfrac{1}{2} = 0.824,$$

where $\mathbb{E}[e^{W_1}] = e^{1/2} = 1.649$ is Lemma 18.35(i) with $\theta = 1$. The dependent pair (W_1, W_2) was first rewritten as a function of the independent pair $(W_1, W_2 - W_1)$. Every application of the lemma to Brownian values starts with that rewriting.

Definition 19.19 (Growth classes)**.** The four classes below are stated once for the whole Part. In each bound, $c \ge 0$ is one generic constant, which serves as multiplier and as exponent.

(i) (Exponential growth) A function of a time s in an interval and of real variables $x_1, \dots, x_d$ (Brownian values or log prices) has *exponential growth* if, for some constant $c \ge 0$, its absolute value is at most

$$c\,e^{c(|x_1| + \cdots + |x_d|)}$$

for all real $x_1, \dots, x_d$ and every time s of the interval: the bound is uniform in the time.

(ii) (Moderate growth) A function of a time and of positive variables $x_1, \dots, x_d$ (prices) has *moderate growth* if, for some constant $c \ge 0$, its absolute value is at most

$$c \prod_{j=1}^{d} (x_j^c + x_j^{-c})$$

for all $x_1, \dots, x_d > 0$, uniformly in the time.

(iii) (The class $C_E^{1,2}$) For $0 \le t_0 < T$, $C_E^{1,2}([t_0, T] \times \mathbb{R})$ is the class of continuous functions $f(t,x)$ on $[t_0, T] \times \mathbb{R}$ whose partial derivatives $\partial_t f$, $\partial_x f$ and $\partial_{xx} f$ exist and are continuous on $[t_0, T] \times \mathbb{R}$ (one-sided in t at t_0 and at T), with f and all three derivatives of exponential growth.

(iv) (Payoff classes) For an integer $J \geq 1$, $\mathcal{G}_J$ is the class of payoffs f on $(0,\infty)^J$ that are continuous outside finitely many coordinate hyperplanes $\{x_j = \ell\}$, each at a fixed level $\ell > 0$, and have moderate growth. The class $\mathcal{G} = \mathcal{G}_1$ consists of the payoffs $f(x)$ that are continuous except at finitely many prices and satisfy $|f(x)| \leq c\,(x^c + x^{-c})$ for all $x > 0$.

A function without a time argument falls under (i) and (ii) as a function that is constant in time.

Exponential growth is what Gaussian expectations tolerate: $\mathbb{E}\, e^{c|Z|} \leq 2e^{c^2/2}$ is finite for every c (Lemma 18.35(i)), and Lemma 19.20(iii) turns this into moment bounds. Moderate growth in prices is the same condition read through the exponential map from log prices to prices (Lemma 19.20(i)). One constant is enough, because a larger c absorbs both a multiplier and an exponent (first paragraph of the complete proof of Lemma 19.20, in Appendix E.4); so neither the call price C nor an index k appears in a growth bound. The level of a discontinuity is written ℓ, as in the Notation table, never with a growth constant. Price variables are written x, never s, which is a time.

Lemma 19.20 (Moderate growth in prices is exponential growth in Brownian coordinates)**.** *The growth conditions of Definition 19.19 are related as follows.*

(i) Let $S_0 > 0$, g real, $\sigma > 0$ and $T > 0$, and put $S(t,y) = S_0\, e^{gt+\sigma y}$. A function $V(x,t)$ of a price $x > 0$ and a time $t \in [0,T]$ has moderate growth if and only if $(t,y) \mapsto V(S(t,y),t)$ has exponential growth on $[0,T] \times \mathbb{R}$. The same holds coordinate by coordinate for a function of several prices, the j-th price being replaced by $S(t_j, y_j)$ with its own time $t_j \in [0,T]$ (and, where needed, its own constants S_0, g and σ): moderate growth in the prices is exponential growth in $(y_1,\dots,y_d)$, uniformly in all the times. Sums and products of functions of moderate growth have moderate growth.

(ii) A function f on $(0,\infty)^J$ is in $\mathcal{G}_J$ if and only if $(y_1,\dots,y_J) \mapsto f(e^{y_1},\dots,e^{y_J})$ is continuous outside finitely many coordinate hyperplanes and has exponential growth. The hyperplanes are the sets $\{y_j = \ln \ell\}$ that the exponential map sends to the hyperplanes $\{x_j = \ell\}$ of f.

(iii) Sums and products of functions of exponential growth have exponential growth. A function of exponential growth in $(x_1,\dots,x_d)$, evaluated at values $W_{s_1},\dots,W_{s_d}$ of standard Brownian motions (one motion, or two independent ones) with every $s_j \in [0,T]$, has finite moments of every order, bounded uniformly in the times s_j and in the time argument of the function.

Sketch of the proof. We bound explicitly, one coordinate at a time. For (i), at $x = S(t,y)$ with $0 \leq t \leq T$,

$$x^c + x^{-c} \leq (S_0^c + S_0^{-c})\, e^{c|g|T}\, e^{c\sigma|y|},$$

so moderate growth in x gives exponential growth in y, uniformly in t; the converse inverts the map, $y = (\ln(x/S_0) - gt)/\sigma$, which uses $\sigma > 0$. Part (ii) is the case $S_0 = 1$, $g = 0$, $\sigma = 1$. In (iii), a power of a bound $c\, e^{c(|x_1|+\cdots+|x_d|)}$ is at most a constant times a sum of terms $e^{c'|x_j|}$ (c' a constant), each of mean at most $2\, e^{c'^2T/2}$ at $x_j = W_{s_j}$ by Lemma 18.35(i), because $s_j \leq T$; only the law of each W_{s_j} enters. Sums and products of exponential bounds are exponential bounds, and moderate growth reduces to that case through (i) with $S_0 = 1$, $g = 0$, $\sigma = 1$; one constant suffices because $x^c + x^{-c}$ and $e^{c|y|}$ do not decrease as $c \geq 0$ grows. The complete proof is in Appendix E.4. □

Example 19.21 (Which payoffs are in the classes)**.** Let $K > 0$ be a strike and $H > 0$ a barrier level. Since $x^c + x^{-c} \geq 2$ for every $x > 0$ and $c \geq 0$, a bounded payoff has moderate growth. So the put $(K - x)^+ \leq K$ and the digital $\mathbf{1}\{x > K\}$ are in $\mathcal{G}$; the digital has one discontinuity, at K. The call satisfies $(x - K)^+ \leq x \leq x + 1/x$, which is moderate growth with $c = 1$. A power x^a with a real exponent a is at most $x^{|a|} + x^{-|a|}$. The log contract $-\ln(x/F)$, with $F > 0$ a constant, is covered by $|\ln x| \leq x + 1/x$. Two-fixing payoffs such as $\mathbf{1}\{x_1 > H\}\,(x_2 - K)^+$ and $(x_2/x_1 - 1)^+$ are in $\mathcal{G}_2$: the first is at most x_2 and jumps only across the hyperplane $\{x_1 = H\}$, the second is continuous and at most $x_2\, x_1^{-1}$, and both bounds are at most $(x_1 + x_1^{-1})(x_2 + x_2^{-1})$. By contrast

e^{x^2} has neither kind of growth: a bound $e^{x^2} \le c\,e^{c|x|}$, or $e^{x^2} \le c\,(x^c + x^{-c}) \le 2c\,e^{cx}$ for $x \ge 1$, would force $x^2 - cx$ to stay bounded as $x \to \infty$.

In numbers, take the call along $S(t,y) = S_0 e^{gt+\sigma y}$ with $S_0 = 100$, $g = 3\%$, $\sigma = 20\%$ and $t \le 1$:

$$(S(t,y) - K)^+ \le S(t,y) \le 100\,e^{0.03}\,e^{0.2|y|} = 103.05\,e^{0.2|y|},$$

exponential growth, as Lemma 19.20(i) says. The moment bound of Lemma 19.20(iii) at the exponent 0.4 and the time 1 is

$$\mathbb{E}\,e^{0.4|W_1|} \le 2\,e^{0.4^2/2} = 2\,e^{0.08} = 2.167,$$

against the exact value $2\,e^{0.08}\,\Phi(0.4) = 1.420$, which follows by completing the square as in Lemma 18.35(i) and is confirmed by numerical quadrature. The bound is crude, but it is finite, which is all the proofs need.

Definition 19.22 (Brownian integrands and left-endpoint sums)**.** Fix $0 \le t_0 < t \le T$ and a standard Brownian motion W. A *Brownian integrand* on $[t_0, t]$ is a process

$$\xi_s = \xi(s, W_s, Y)\,\mathbf{1}\{s < t_\ell\}, \qquad t_0 \le s \le t,$$

in which the process and the function that generates it share the letter ξ, with the following three ingredients.

(a) The function $\xi(s,x,y)$ has exponential growth in (x,y) (Definition 19.19(i)) and is *piecewise continuous in* y: the range of y is covered by finitely many disjoint boxes, and for each box $\mathcal{D}$ the function is continuous on $[t_0,t] \times \mathbb{R} \times \mathcal{D}$.

(b) The vector $Y = (W_{s_1}, \dots, W_{s_m})$ consists of values of W at fixed times $s_j \le t_0$. It may be absent, and then $\xi_s = \xi(s, W_s)\,\mathbf{1}\{s < t_\ell\}$ with $\xi(s,x)$ continuous on $[t_0,t] \times \mathbb{R}$.

(c) The time t_ℓ is either $+\infty$ (no stopping: the indicator is 1) or the first time at which W reaches a continuous boundary ℓ on $[0,T]$. For a boundary approached from below,

$$t_\ell = \inf\{s \in [0,T] : W_s \ge \ell(s)\} \qquad \text{with } \ell(0) > 0,$$

and for a boundary approached from above,

$$t_\ell = \inf\{s \in [0,T] : W_s \le \ell(s)\} \qquad \text{with } \ell(0) < 0,$$

the infimum of the empty set being $+\infty$. The time t_ℓ is computed from time 0, even when $t_0 > 0$.

Two motions. When a standard Brownian motion $W_\perp$ independent of W is present, ξ may take $W_{\perp,s}$ as a further argument, $\xi_s = \xi(s, W_s, W_{\perp,s}, Y)\,\mathbf{1}\{s < t_\ell\}$, the continuity and the growth of (a) then holding in that argument too, and Y may include values of $W_\perp$ at times $s_j \le t_0$.

The class of Brownian integrands on $[t_0,t]$ is written $\mathcal{H}$. Finite sums of its elements, possibly with different vectors Y and different times t_ℓ, form its *span*. For a partition $t_0 < t_1 < \cdots < t_n = t$ with mesh $\Delta t_{\max} = \max_i \Delta t_i$, the *left-endpoint sum* of ξ is

$$\mathcal{I}_n(\xi) = \sum_{i=1}^{n} \xi_{t_{i-1}}\,\Delta W_i.$$

With two motions, the left-endpoint sum of ξ against $W_\perp$ is $\mathcal{I}_n^\perp(\xi) = \sum_{i=1}^n \xi_{t_{i-1}}\,\Delta W_{\perp,i}$, with $\Delta W_{\perp,i} = W_{\perp,t_i} - W_{\perp,t_{i-1}}$.

The integrand is decided at the left end of each step, before the increment is seen. This is what makes $\mathcal{I}_n(\xi)$ the gain of a trading strategy, which holds $\xi_{t_{i-1}}$ units over the i-th step. The piecewise continuity in y admits payoffs that jump on hyperplanes of earlier fixings, such as a discretely monitored barrier. The proofs use continuity in (s,x) for each fixed y only, and the boxes make each indicator $\mathbf{1}\{Y \in \mathcal{D}\}$ a function of earlier values. A constant multiple of an element of $\mathcal{H}$ is in $\mathcal{H}$, and so is its product with a continuous function of (s, W_s) of exponential growth (Lemma 19.20(iii)).

Properties of the hitting time. Let t_ℓ be as in (c), for a boundary approached from below; for a boundary approached from above the arguments are the same with the inequalities reversed. The hitting time has the three properties

$$t_\ell > 0; \qquad W_{t_\ell} = \ell(t_\ell) \ \text{ whenever } t_\ell \le T; \qquad P(t_\ell = s) = 0 \ \text{ for every } s \in [0,T], \tag{19.6}$$

and, for every $s \in [0,T]$, the description

$$\{t_\ell > s\} = \{W_{s'} < \ell(s') \text{ for every } s' \in [0,s]\}, \tag{19.7}$$

which Theorems 19.23, 19.30 and 19.63 use. We argue path by path from the continuity of W and of ℓ for the first two properties and for (19.7), and use the continuous distribution function of W_s for the third property. The function $s \mapsto \ell(s) - W_s$ is continuous on $[0,T]$ and equals $\ell(0) > 0$ at $s = 0$, so it is positive on an interval $[0,\varepsilon)$ with $\varepsilon > 0$. No time of that interval belongs to the set $\{s \in [0,T] : W_s \ge \ell(s)\}$ whose infimum is t_ℓ, and therefore $t_\ell \ge \varepsilon > 0$. Suppose next that $t_\ell \le T$, that is, that the set is not empty. By the definition of an infimum there are times u_k of the set with $u_k \to t_\ell$, and $W_{u_k} \ge \ell(u_k)$ gives $W_{t_\ell} \ge \ell(t_\ell)$ in the limit, by continuity. Every time $s < t_\ell$ lies outside the set, so $W_s < \ell(s)$; letting s increase to t_ℓ, which is possible because $t_\ell > 0$, gives $W_{t_\ell} \le \ell(t_\ell)$. Hence $W_{t_\ell} = \ell(t_\ell)$: at the hitting time the path sits on the boundary. For the third property, $t_\ell = 0$ never happens, and for $s > 0$ the event $\{t_\ell = s\}$ is contained in $\{W_s = \ell(s)\}$. That event has probability zero: $W_s \sim \mathcal{N}(0,s)$ has the continuous distribution function $y \mapsto \Phi(y/\sqrt{s})$, so $P(W_s = y) \le \Phi(y/\sqrt{s}) - \Phi((y-\varepsilon)/\sqrt{s})$ for every $\varepsilon > 0$, and the right side tends to 0 with ε. For (19.7), if $t_\ell > s$, then every $s' \le s$ is smaller than t_ℓ and lies outside the set. Conversely, if no time of $[0,s]$ belongs to the set, then $t_\ell \ge s$, and $t_\ell \ne s$ because t_ℓ, when it is at most T, belongs to the set by the second property. This proves (19.6) and (19.7).

Examples in and out of the class. On $[t_0,t]$ the processes W_s, $2\sigma e^{\sigma W_s}$ (for a constant $\sigma > 0$), $W_{s_1} e^{W_s}$ and $\mathbf{1}\{W_{s_1} > 0\}\, W_s$, with $s_1 \le t_0$, are in $\mathcal{H}$. Their generating functions are x, $2\sigma e^{\sigma x}$, $y\, e^x$ and $\mathbf{1}\{y > 0\}\, x$. Each has exponential growth, by $|x| \le e^{|x|}$; the first three are continuous, and the fourth is continuous on each of the two boxes $\{y > 0\}$ and $\{y \le 0\}$. A barrier gives a stopped example. Let $S_s = S_0 e^{gs + \sigma W_s}$ be a price path as in Lemma 19.20(i) and let $H \ne S_0$ be a barrier level. The price reaches H at the first time at which W reaches the continuous boundary $\ell(s) = (\ln(H/S_0) - gs)/\sigma$, for which $\ell(0) \ne 0$. If $V(S,t)$ is a price function whose delta $\partial_S V$ is continuous and of moderate growth on $(0,\infty) \times [0,T']$, with $T' < T$, then

$$\xi_s = \mathbf{1}\{s < t_\ell\}\, \partial_S V(S_s, s)\, \sigma S_s, \qquad 0 \le s \le T',$$

is in $\mathcal{H}$ on $[0,T']$: the function $x\, \partial_S V(x,s)$ has moderate growth as a product, and Lemma 19.20(i) turns moderate growth in the price into exponential growth in the Brownian coordinate. The price functions that Lemma 19.61 produces, and with them those of §20, have such a delta before expiry: part (i) of that lemma bounds their derivatives in the Brownian coordinate, and Lemma 19.20(i) reads the bound in the price. The restriction $T' < T$ is needed, because on $[0,T]$ itself the delta of a digital, or of a knock-out whose payoff does not vanish at the barrier, is unbounded near expiry, at the strike or at the barrier. By contrast, the running maximum $W_s^{\max} = \max_{0 \le s' \le s} W_{s'}$ is not in $\mathcal{H}$: it depends on the whole past of the path, not on finitely many values at fixed times. The Part never needs it as an integrand.

Theorem 19.23 (The Itô integral and the Itô isometry (Itô, 1944))**.** *Let ξ and ζ be Brownian integrands on $[t_0,t]$, that is, elements of the class $\mathcal{H}$ of Definition 19.22.*

(i) (Existence) As $\Delta t_{\max} \to 0$, the left-endpoint sums $\mathcal{I}_n(\xi)$ converge in mean square. The limit is written

$$\int_{t_0}^{t} \xi_s \, dW_s$$

and called the Itô integral *of ξ. It has a finite second moment, and it does not depend on the sequence of partitions, up to an event of probability zero.*

(ii) (Mean zero) The integral has mean zero: $\mathbb{E}\int_{t_0}^{t}\xi_s\,dW_s = 0$.

(iii) (Itô isometry)

$$\mathbb{E}\Big[\Big(\int_{t_0}^{t}\xi_s\,dW_s\Big)^2\Big] = \int_{t_0}^{t}\mathbb{E}[\xi_s^2]\,ds, \tag{19.8}$$

and more generally

$$\mathbb{E}\Big[\int_{t_0}^{t}\xi_s\,dW_s\int_{t_0}^{t}\zeta_s\,dW_s\Big] = \int_{t_0}^{t}\mathbb{E}[\xi_s\zeta_s]\,ds.$$

(iv) (Linearity, additivity, stopping) The integral is linear in ξ. *It extends by linearity to the span of* $\mathcal{H}$, *where the left-endpoint sums of a finite sum are the sums of the left-endpoint sums, and (i) to (iii) hold unchanged on the span. It is additive over adjacent intervals: for* $t_0 < t' < t$, *with probability one,*

$$\int_{t_0}^{t}\xi_s\,dW_s = \int_{t_0}^{t'}\xi_s\,dW_s + \int_{t'}^{t}\xi_s\,dW_s.$$

And let $t_0 \le t' < t$, *let* t_ℓ *be a hitting time as in Definition 19.22(c), and let* ξ *be an element of the span of* $\mathcal{H}$ *with* $\xi_s = 0$ *whenever* $s \ge t_\ell$ *(for instance, each of its summands carries the factor* $\mathbf{1}\{s < t_\ell\}$*): on the event* $\{t_\ell \le t'\}$ *the integral of* ξ *over* $[t', t]$ *is zero with probability one.*

(v) (Mean-square continuity) The map $s \mapsto \xi_s$ *is continuous in mean square on* $[t_0, t]$: $\mathbb{E}[(\xi_u - \xi_s)^2] \to 0$ *as* $u \to s$. *Consequently* $s \mapsto \mathbb{E}[\xi_s^2]$ *and* $s \mapsto \mathbb{E}[\xi_s\zeta_s]$ *are bounded and continuous on* $[t_0, t]$, *and every* ds*-integral in (iii) is an ordinary Riemann integral.*

(vi) (Two independent motions) When a standard Brownian motion $W_\perp$ *independent of* W *is present, the left-endpoint sums* $\mathcal{I}_n^{\perp}(\xi)$ *against* $W_\perp$ *converge in mean square in the same way to an integral* $\int_{t_0}^{t}\xi_s\,dW_{\perp,s}$, *for which (i) to (v) hold as well, and the cross expectation vanishes:*

$$\mathbb{E}\Big[\int_{t_0}^{t}\xi_s\,dW_s\int_{t_0}^{t}\zeta_s\,dW_{\perp,s}\Big] = 0,$$

whichever of the two motions the integrands ξ *and* ζ *depend on.*

Sketch of the proof. We compute exactly for left-endpoint sums, show that they form a Cauchy sequence in mean square, and take the limit that Fact 19.16 provides. The increment ΔW_i is independent of the weight $\xi_{t_{i-1}}$ and of everything earlier, hitting-time indicators included, so in the square of the sum each diagonal term factorises and each cross term has mean zero:

$$\mathbb{E}[\mathcal{I}_n(\xi)] = 0, \qquad \mathbb{E}\big[\mathcal{I}_n(\xi)^2\big] = \sum_{i=1}^{n}\mathbb{E}[\xi_{t_{i-1}}^2]\,\Delta t_i,$$

the moments being bounded by exponential growth (Lemma 19.20(iii)). The continuity of ξ and of the path, with Lemma 19.12(iv), gives (v), uniformly in time; the indicator of $\{s < t_\ell\}$ fails to converge only on $\{t_\ell = s\}$, of probability zero by (19.6). The difference of the sums of two partitions is a sum of the same kind over their common refinement, so by the identity and (v) the sums are Cauchy, with a limit that does not depend on the partitions. Mean zero and the isometry pass to the limit; the product form is the isometry for $\xi \pm \zeta$; the rest of (iv) holds for the sums and passes to the limit; and (vi) uses the two independent increments of each cell. The complete proof is in Appendix E.5. □

The stochastic integral of a Brownian integrand is the only one the Part needs. Its extension to general integrands determined by the past is clause (ii) of Fact 19.29.

Example 19.24 (The integral of W against dW)**.** The integrand $\xi_s = W_s$ is in $\mathcal{H}$ on $[0,1]$: its generating function $\xi(s,x) = x$ is continuous, and $|x| \le e^{|x|}$. Take n equal steps. Since $W_{t_i}^2 - W_{t_{i-1}}^2 = 2W_{t_{i-1}}\,\Delta W_i + (\Delta W_i)^2$, each term of the left-endpoint sum is

$$W_{t_{i-1}}\,\Delta W_i = \tfrac{1}{2}(W_{t_i}^2 - W_{t_{i-1}}^2 - (\Delta W_i)^2),$$

and the sum telescopes to

$$\mathcal{I}_n(W) = \tfrac{1}{2}(W_1^2 - \mathrm{QV}_n).$$

By Proposition 19.10(i), $\mathbb{E}[(\mathrm{QV}_n - 1)^2] = \mathrm{Var}(\mathrm{QV}_n) = 2/n$, so

$$\mathbb{E}\Big[\Big(\mathcal{I}_n(W) - \tfrac{1}{2}(W_1^2 - 1)\Big)^2\Big] = \tfrac{1}{4}\,\mathbb{E}\big[(1 - \mathrm{QV}_n)^2\big] = \frac{1}{2n} \to 0,$$

and by Theorem 19.23(i) the mean-square limit is the Itô integral:

$$\int_0^1 W_s\,dW_s = \tfrac{1}{2}(W_1^2 - 1).$$

Its mean is $(1-1)/2 = 0$, as (ii) requires, and by the isometry (19.8) its variance is $\int_0^1 \mathbb{E}[W_s^2]\,ds = \int_0^1 s\,ds = 1/2$. The direct check uses Proposition 7.25: $(\mathbb{E}[W_1^4] - 2\,\mathbb{E}[W_1^2] + 1)/4 = (3-2+1)/4 = 1/2$. The n-step sum has mean-square error $1/(2n)$, which is 0.125 at $n = 4$ and 0.001 at $n = 500$; the corresponding mean-square distances $(\mathbb{E}[(X-Y)^2])^{1/2}$ of Lemma 19.12(ii) are 0.354 and 0.0316. A simulation of 200,000 paths at $n = 500$ returns the error 0.00100.

Right-endpoint sums behave differently. Since $W_{t_i}\,\Delta W_i = \tfrac{1}{2}(W_{t_i}^2 - W_{t_{i-1}}^2 + (\Delta W_i)^2)$, they telescope to $\tfrac{1}{2}(W_1^2 + \mathrm{QV}_n)$ and converge in mean square to $\tfrac{1}{2}(W_1^2 + 1)$, whose mean is 1, not 0. The end point matters, which is the first sign that dW is not an ordinary differential. On $[0,t]$ the same computation gives $\int_0^t W_s\,dW_s = (W_t^2 - t)/2$, and Theorem 19.30, applied to $f(t,x) = x^2$, returns the same value (Example 19.32).

Interview trap. "By the chain rule, $\int_0^t W\,dW = W_t^2/2$." The Itô integral is $(W_t^2 - t)/2$ (Example 19.24). The left-endpoint sums carry the quadratic variation $\mathrm{QV}_n \to t$ with a minus sign, because $(dW)^2 = dt$ is of first order and does not vanish in the limit. Theorem 19.30 is the corrected chain rule.

Interview trap. "The end point does not matter in the limit, as for a Riemann integral." Left-endpoint sums of W against dW have mean 0. Right-endpoint sums have mean

$$\mathbb{E}\Big[\sum_{i=1}^n W_{t_i}\,\Delta W_i\Big] = \sum_{i=1}^n \Big(\mathbb{E}[W_{t_{i-1}}\,\Delta W_i] + \mathbb{E}[(\Delta W_i)^2]\Big) = \sum_{i=1}^n \Delta t_i = t,$$

and converge to $(W_t^2 + t)/2$. A strategy sampled at the right end point decides after seeing the move. In trading terms it is look-ahead bias, the integral analogue of a backtest that peeks.

Intuition. The isometry (19.8) says that the variance of a trading gain is the integral of the mean squared position times the variance rate of the asset, which is one for W. Holding one unit of W for a year gives the variance $\int_0^1 1\,ds = 1$. Holding W_s units at time s gives $\int_0^1 s\,ds = 1/2$, the variance of $(W_1^2 - 1)/2$.

Trading application. The integral $\int \xi_s\,dW_s$ is the P&L of holding ξ_s units of an asset whose price is W, with each holding decided before the move it earns. Section 20 builds continuous-time trading on exactly this class (Definition 20.1), so the question "what can a strategy hold?" has the answer "on each cell between deterministic switch dates, a finite sum of Brownian integrands that lies in the span of $\mathcal{H}$ up to any time before the cell ends".

19.4 Generalised Wiener Processes, Itô Processes and Geometric Brownian Motion

A price is not a driftless process with unit variance rate, and choosing its model is itself an interview question. Why proportional rather than additive shocks for a stock? What does a one-week move look like? When is a normal (Bachelier) model the right one? This subsection adds a drift and a scale to Brownian motion, lets both depend on the path (the Itô process, defined through the integral of §19.3 and motivated by its Euler scheme), and defines geometric Brownian motion, the model used for every price in the rest of this Part, by its explicit formula.

Setup. The number $T > 0$ is a horizon and W is a standard Brownian motion. Time is measured in years unless an example states another unit. The symbols α_t and β_t are the drift and the diffusion coefficient of an Itô process; $\alpha(x,t)$ and $\beta(x,t)$ are the same coefficients as functions of a real state x and of the time t; and in the constant case of Definition 19.25(i) they are constants α and β. *Symbol notice:* α_t and β_t, always with a subscript or with arguments and bare only for the constants of (i), are not the GARCH coefficients of §22 or a CAPM beta. The letters a and b that the literature often uses for the two coefficients are avoided, because b is the Part's cost of carry. A process X starts at the value X_0, a number. A step has length $\Delta t > 0$, and $\Delta W = W_{t+\Delta t} - W_t$ is the increment of W over it. A geometric Brownian motion has three parameters: a starting price $S_0 > 0$, a real number μ (the expected rate of return) and a volatility $\sigma > 0$. The letter r is re-declared: it is the continuously compounded risk-free rate, not the random-walk ratio q/p of the general notation table.

Definition 19.25 (Generalised Wiener process and Itô process)**.** The equation $dX = \alpha_t\,dt + \beta_t\,dW$ is shorthand for an integral identity, in three degrees of generality.

(i) (Generalised Wiener process, or arithmetic Brownian motion) For constants α and β, the equation $dX = \alpha\,dt + \beta\,dW$ *means*

$$X_t = X_0 + \alpha t + \beta W_t, \qquad t \geq 0.$$

Every increment $X_t - X_s = \alpha(t-s) + \beta(W_t - W_s)$, $s < t$, is an affine image of the normal increment $W_t - W_s \sim \mathcal{N}(0, t-s)$ of Definition 19.5(2). By (19.3) of the Setup of §19.2, with $c = \beta$ and $c' = \alpha(t-s)$,

$$X_t - X_s \sim \mathcal{N}\big(\alpha(t-s),\ \beta^2(t-s)\big)$$

(the constant $\alpha(t-s)$ when $\beta = 0$). Increments of X over disjoint intervals are independent, because they are functions of independent increments of W (Definition 19.5(3) and the grouping property (a) of the Setup of §19.2). In particular $X_T \sim \mathcal{N}(X_0 + \alpha T, \beta^2 T)$. The drift grows like T and the noise like $\sqrt{T}$, so the drift dominates at long horizons, by the arithmetic of Proposition 10.3.

(ii) (Itô process) For processes α_s and β_s, $0 \leq s \leq T$, the equation $dX = \alpha_t\,dt + \beta_t\,dW$ is shorthand for

$$X_t = X_0 + \int_0^t \alpha_s\,ds + \int_0^t \beta_s\,dW_s \qquad \text{for every } t \in [0,T], \tag{19.9}$$

each identity holding with probability one. Here α_s is determined by the past of W at each time s (a function of the path of W up to time s, and of that of $W_\perp$ when a second motion is present), and $s \mapsto \alpha_s$ is piecewise continuous on every path, so that the first integral is a Riemann integral computed path by path; and β, restricted to $[0,t]$, lies in the class $\mathcal{H}$ of Definition 19.22 or in its span, so that the second integral is the Itô integral of Theorem 19.23. The process α_t is the *drift* and β_t the *diffusion coefficient* of X.

Clause (iii) is motivated by the *Euler scheme*. For functions $\alpha(x,t)$ and $\beta(x,t)$, step forward by

$$X_{t+\Delta t} = X_t + \alpha(X_t,t)\,\Delta t + \beta(X_t,t)\,\Delta W,$$

with $\Delta W \sim \mathcal{N}(0,\Delta t)$ independent of the past. Summed over the steps of a partition, the last terms form the left-endpoint sum $\sum_i \beta(X_{t_{i-1}},t_{i-1})\,\Delta W_i$, which is exactly the kind of sum that defines the integral in (ii), here evaluated along the Euler iterates. The scheme motivates clause (iii); the Part does not prove that it converges to a solution.

(iii) (SDE form) For functions $\alpha(x,t)$ and $\beta(x,t)$, continuous in (x,t), the equation $dX = \alpha(X_t,t)\,dt + \beta(X_t,t)\,dW$ means a process X with continuous paths, determined at each time by the path of W up to that time, with $\int_0^T \mathbb{E}[\beta(X_s,s)^2]\,ds < \infty$ when $s \mapsto \beta(X_s,s)$ is not in the span of $\mathcal{H}$, such that, for every $t \in [0,T]$, with probability one,

$$X_t = X_0 + \int_0^t \alpha(X_s,s)\,ds + \int_0^t \beta(X_s,s)\,dW_s.$$

The stochastic integral is that of Theorem 19.23 when $s \mapsto \beta(X_s,s)$ is in the span of $\mathcal{H}$, and that of Fact 19.29(ii) otherwise. Existence and uniqueness of such a process for Lipschitz coefficients are Fact 19.29(i), which no proof in this Part uses.

In clause (iii) the coefficients read only the current state and the clock, so the Euler step taken at time t uses the path only through X_t: clause (iii) is the continuous-time form of a Markov model. For the Lipschitz coefficients of Fact 19.29(i) the solution itself is Markov; this is not proved here, and no later result uses it. With the state frozen at $X_t = x$, one Euler step is $\alpha(x,t)\,\Delta t + \beta(x,t)\,\Delta W$, which has the law $\mathcal{N}\big(\alpha(x,t)\,\Delta t,\ \beta(x,t)^2\,\Delta t\big)$ by (19.3). Over many steps the coefficients move with the path, and the law of X_t is in general not normal: the geometric Brownian motion of Definition 19.27, which solves the equation with $\alpha(x,t) = \mu x$ and $\beta(x,t) = \sigma x$ (Corollary 19.34(ii)), is lognormal at every horizon (§19.6). Clause (i) is the constant-coefficient case of (ii): the constant integrand β is in $\mathcal{H}$, every left-endpoint sum of it telescopes to $\beta\,(W_t - W_0) = \beta W_t$, and therefore $\int_0^t \beta\,dW_s = \beta W_t$; the Riemann integral of the constant α is αt.

Example 19.26 (An excess-margin balance with drift)**.** The excess margin of a futures account, the balance above the maintenance level in thousands of dollars, follows a generalised Wiener process with time in weeks: it starts at $X_0 = 10$, with drift $\alpha = 0.5$ per week and variance rate $\beta^2 = 8$ per week. A negative value means a margin call. By Definition 19.25(i), after 8 weeks $X_8 \sim \mathcal{N}(10 + 0.5 \times 8,\ 8 \times 8) = \mathcal{N}(14, 64)$, with standard deviation 8, so

$$P(X_8 < 0) = \Phi\Big(\frac{-14}{8}\Big) = \Phi(-1.75) = 4.01\%,$$

about 4.0%. After 30 weeks $X_{30} \sim \mathcal{N}(25, 240)$, with standard deviation 15.49, so $P(X_{30} < 0) = \Phi(-1.614) = 5.33\%$. The mean grows like t and the standard deviation like $\sqrt{t}$, so the probability of a negative balance first rises and then falls. It is largest where the ratio $(X_0 + \alpha t)/(\beta\sqrt{t}) = X_0\,t^{-1/2}/\beta + \alpha\,t^{1/2}/\beta$ is smallest. The derivative of the ratio in t, $(-X_0\,t^{-3/2} + \alpha\,t^{-1/2})/(2\beta)$, changes sign from negative to positive at $t^* = X_0/\alpha = 20$ weeks, between the two horizons, where the ratio is $2\sqrt{X_0\alpha}/\beta = 2\sqrt{5}/\sqrt{8} = 1.581$ and the probability is $\Phi(-1.581) = 5.69\%$.

(Problem P19.3 works this computation at different numbers.)

Why proportional shocks. The return of a stock over a short step should not depend on the price level: a move of two percent means the same to a holder at a price of 25 as at a price of 250. The natural discretisation is therefore

$$\frac{\Delta S}{S} = \mu\,\Delta t + \sigma\sqrt{\Delta t}\;Z, \qquad Z \sim \mathcal{N}(0,1),$$

a return over the step Δt with the law $\mathcal{N}(\mu\,\Delta t, \sigma^2\Delta t)$. It is the Euler scheme with $\alpha(x,t)=\mu x$ and $\beta(x,t)=\sigma x$. After n steps of length T/n the Euler price is the product

$$S_0 \prod_{i=1}^{n} \Big(1 + \frac{\mu T}{n} + \sigma\sqrt{\frac{T}{n}}\, Z_i\Big),$$

with $Z_1,\dots,Z_n$ iid standard normal. Its factors are functions of the mutually independent variables Z_i, each with mean $1+\mu T/n$ and a finite absolute mean. For $k=1,\dots,n-1$ the product of the first k factors is a function of $(Z_1,\dots,Z_k)$, which is independent of Z_{k+1} by the grouping property (a) of the Setup of §19.2. So the product rule (19.5) of Lemma 19.17, applied for $k=1,\dots,n-1$, shows that each partial product has a finite absolute mean and gives the mean $S_0(1+\mu T/n)^n$. By Proposition 7.39 this tends to $S_0 e^{\mu T}$ as $n\to\infty$, with a relative error of about $-(\mu T)^2/(2n)$; Example 7.40 shows the size of such a correction. The limit anticipates Proposition 19.42(i).

Definition 19.27 (Geometric Brownian motion)**.** A *geometric Brownian motion* with drift μ and volatility $\sigma>0$, started at $S_0>0$, is the process

$$S_t = S_0 \exp\Big(\big(\mu - \tfrac{1}{2}\sigma^2\big)\,t + \sigma W_t\Big), \qquad t\ge 0. \tag{19.10}$$

The process is positive by construction. Corollary 19.34 proves, in §19.5, that S satisfies $dS = \mu S\,dt + \sigma S\,dW$ in the sense of Definition 19.25(ii), that is, (19.9) with $\alpha_s=\mu S_s$ and $\beta_s=\sigma S_s$. This is why μ is called the instantaneous expected rate of return and σ the volatility. The square σ^2 is the variance rate: by (19.10) the log-return over a step is $\ln(S_{t+\Delta t}/S_t) = (\mu-\sigma^2/2)\,\Delta t + \sigma\,\Delta W$, with standard deviation $\sigma\sqrt{\Delta t}$, consistent with Corollary 19.2. The term $-\sigma^2/2$ in the exponent is explained by Corollary 19.34, and it makes $\mathbb{E}[S_t]=S_0 e^{\mu t}$ (Proposition 19.42(i)). Fact 19.29(i) adds that every continuous solution of $dS=\mu S\,dt+\sigma S\,dW$ started at S_0 that is determined at each time by the path of W up to that time agrees with (19.10) for all $t\in[0,T]$ outside an event of probability zero, for every horizon T (the coefficients μx and σx are Lipschitz and of linear growth); nothing in the Part uses this, because the Part starts from (19.10). In the risk-neutral world of §18 (Definition 18.6) the drift μ is replaced by r, which §20 makes precise (Definition 20.6).

Example 19.28 (A one-week move)**.** Take $S=80$, $\mu=9\%$, $\sigma=35\%$ and one week of five trading days, $\Delta t = 5/252$, the trading-day count of §19.1. By the discretisation above, the change ΔS over the week is approximately normal with mean $S\mu\,\Delta t = 80\times 0.09\times 5/252 = 0.14$ and standard deviation $S\sigma\sqrt{\Delta t} = 80\times 0.35\times\sqrt{5/252} = 3.94$. A 95% band is the mean plus or minus 1.96 standard deviations, 0.14 ± 7.73; computed before rounding, it is $[-7.59,\ 7.87]$. The weekly drift is about one twenty-eighth of the weekly noise, the ratio of the standard deviation to the mean being $\sigma\sqrt{252/5}/\mu = 27.6$. This is the signal-to-noise point of §10.1, in price form.

(Problem P19.4 works this computation at different numbers.)

Fact 19.29 (Stochastic differential equations and general Itô processes (Itô, 1944; Itô, 1951))**.** The three statements below go beyond the integrands of Theorem 19.23.

(i) (Existence and uniqueness) If $\alpha(x,t)$ and $\beta(x,t)$ are continuous in (x,t), Lipschitz in x uniformly in t, and of linear growth in x, the equation $dX=\alpha(X_t,t)\,dt+\beta(X_t,t)\,dW$ has, for each starting value X_0, a continuous solution determined at each time by the path of W up to that time, and two such solutions with the same starting value agree for all $t\in[0,T]$ outside an event of probability zero.

(ii) (General integrands) The Itô integral of Theorem 19.23 extends to every integrand ξ that is determined by the past of W (at each time s, ξ_s is a function of the path of W up to s) and has $\int_0^t \mathbb{E}[\xi_s^2]\,ds<\infty$. The extended integral has mean zero and obeys the isometry (19.8).

(iii) (Chain rule for a general Itô process) Let

$$X_t = X_0 + \int_0^t \alpha_s \, ds + \int_0^t \beta_s \, dW_s,$$

where the coefficients α_s and β_s are determined by the past of W, $s \mapsto \alpha_s$ is piecewise continuous on every path, β satisfies the condition of (ii), the stochastic integral is that of (ii), and X is taken with continuous paths. This is Definition 19.25(ii) with a larger class of coefficients; the solution of (i) is the case $\alpha_s = \alpha(X_s, s)$, $\beta_s = \beta(X_s, s)$. Let $\mathcal{O}$ be an open interval that contains every value of X, and let $V(x,t)$, defined on $\mathcal{O} \times [0,T]$, have continuous partial derivatives $\partial_t V$, $\partial_x V$ and $\partial_{xx} V$ there; let the integrand $\beta_s \, \partial_x V(X_s, s)$ satisfy the condition of (ii). Then, for every $t \in [0,T]$, with probability one,

$$V(X_t, t) = V(X_0, 0) + \int_0^t \left(\partial_t V + \alpha_s \, \partial_x V + \tfrac{1}{2} \, \beta_s^2 \, \partial_{xx} V \right) ds + \int_0^t \beta_s \, \partial_x V \, dW_s,$$

the derivatives of V being evaluated at (X_s, s). For two such processes X_1 and X_2, with coefficients $\alpha_{k,s}$ and $\beta_{k,s}$, driven by Brownian motions W_1 and W_2 with correlation ρ (Definition 19.51; here the first subscript names the process or the motion), a function $V(x_1, x_2, t)$ on $\mathcal{O}_1 \times \mathcal{O}_2 \times [0,T]$, with $\mathcal{O}_k$ an open interval containing every value of X_k, with continuous $\partial_t V$ and continuous first and second partial derivatives in (x_1, x_2) obeys, under the same integrability condition, with (ii) extended to integrands determined by the past of (W_1, W_2), the two-dimensional analogue. Its ds-integrand is

$$\partial_t V + \alpha_{1,s} \, \partial_1 V + \alpha_{2,s} \, \partial_2 V + \tfrac{1}{2} \, \beta_{1,s}^2 \, \partial_{11} V + \rho \, \beta_{1,s} \beta_{2,s} \, \partial_{12} V + \tfrac{1}{2} \, \beta_{2,s}^2 \, \partial_{22} V,$$

and its stochastic integrals are $\int_0^t \beta_{1,s} \, \partial_1 V \, dW_{1,s} + \int_0^t \beta_{2,s} \, \partial_2 V \, dW_{2,s}$.

A proof is beyond our scope.

Part (ii) needs the approximation of a general integrand determined by the past with step integrands, which rests on measurability with respect to the past. Part (i) needs Picard iteration with Doob's maximal inequality. Part (iii) needs the proof of Theorem 19.30, with the integral of (ii) in place of that of Theorem 19.23, for the process stopped when it leaves a bounded interval, where the moment bounds hold, followed by a limit that removes the stopping (localisation). The ds-integral of (iii) is in general a Lebesgue integral computed path by path, because its integrand contains β_s^2, which need not be piecewise continuous in s. Corollary 19.31 and Theorem 19.55(iii) prove the formulas of (iii), with no Fact, when each process is an explicit function of time and of a Brownian value.

No proof in this Part relies on Fact 19.29. Every process the Part differentiates is an explicit function of time and of Brownian values, which Theorem 19.30 covers. The Fact is cited only to name what general stochastic volatility (§21, §23) and local volatility (§22, §23) would need.

Interview trap. Modelling a stock by arithmetic Brownian motion. A stock at 25 with a dollar volatility of 9 a year (36% of the price at the start) and no drift has $S_2 \sim \mathcal{N}(25,\ 2 \times 9^2) = \mathcal{N}(25, 162)$ under Definition 19.25(i), with standard deviation $9\sqrt{2} = 12.73$, and so

$$P(S_2 < 0) = \Phi(-25/12.73) = \Phi(-1.964) = 2.5\% :$$

the model gives a negative price a visible probability. The same 9 dollars on a stock at 250 would be a 3.6% volatility, so the additive model says that the dollar risk does not depend on the price level, while investors think in percentage returns. Prices are scale-free, which is why returns, not changes, are modelled. Bachelier's (1900) normal model is still the right one for spreads, for rates near or below zero, and for a P&L, all of which can be negative.

Interview trap. "Every Itô process is normal at every horizon." Constant coefficients guarantee it (Definition 19.25(i)), but normality is a special feature, not a rule. Once the coefficients move with the path, the law need not be normal: a geometric Brownian motion is an Itô process with $\beta_t = \sigma S_t$ (Corollary 19.34(ii)), and its level is lognormal (§19.6) and never negative. Nor does a moving coefficient rule normality out: Itô's lemma, proved in §19.5 (Theorem 19.30 with $f(t,x) = tx$, worked in Example 19.32), shows that the process tW_t has the drift W_t and the diffusion coefficient t, yet $tW_t \sim \mathcal{N}(0, t^3)$ at every horizon, by (19.3).

Intuition. The drift μ sets the real-world distribution of the price, but derivative prices will turn out not to depend on it (§18, §20); σ is the parameter that prices options. Nor is μ the rate at which a typical path grows: that rate is $\mu - \sigma^2/2$ (§19.7).

19.5 Itô's Lemma

The most common stochastic-calculus requests in an interview are "apply Itô's lemma to $\ln S$", "what is $d(W_t^2)$?" and "where does the $-\sigma^2/2$ come from?". Itô's lemma is also the engine of the Black–Scholes–Merton equation of §20 (Theorem 20.3), because the value of a derivative is a function of the underlying price and of time. This subsection proves the lemma for functions of time and of a Brownian value (Theorem 19.30), with the two clauses that later sections need: a parameter clause, in which earlier fixings are frozen, and a stopped clause, which runs up to a hitting time. The proof is a second-order Taylor expansion in which $(dW)^2 = dt$ survives; it is where the Taylor subsection §7.5 and Proposition 19.10 pay off. The subsection then derives the classical form for a function of an Itô process (Corollary 19.31) and proves the dynamics of the log price and of a forward price.

Setup. The number $T > 0$ is a horizon and W is a standard Brownian motion. The function $f(t,x)$ is a function of time and of a Brownian value, time first, usually in the class $C_E^{1,2}([0,T] \times \mathbb{R})$ of Definition 19.19(iii); its derivative in the time argument is written $\partial_t f$ whatever letter the time carries. The process X is an Itô process, $dX = \alpha_t\, dt + \beta_t\, dW$ (Definition 19.25), with values in an open interval $\mathcal{O}$, and the *outer function* $V(x,t)$ is defined on $\mathcal{O} \times [0,T]$, state first and time second, as in $V(S,t)$; along the path, $V_t = V(X_t, t)$. Partial derivatives are written $\partial_t V$, $\partial_x V$ and $\partial_{xx} V$, never as subscripts, because subscripts index time in this section (S_t, W_t, F_t). Where the outer function takes the state x (Corollary 19.31), the Brownian argument of f is written y, as in $f(t,y)$, and the composite

$$\tilde{V}(y,t) = V\big(f(t,y),\ t\big)$$

is V written in the Brownian variable, state first and time second like V; a tilde marks the same function in a transformed variable.

Partitions, steps Δt_i, increments ΔW_i, the mesh $\Delta t_{\max}$, the sum QV_n and the left-endpoint sums $\mathcal{I}_n$ are those of §19.2 and §19.3, and $a \wedge b = \min(a,b)$. In the proof of Theorem 19.30, $\mathcal{T}_1, \dots, \mathcal{T}_4$ are its four sums (time, gradient, quadratic and remainder); a_i are the weights of its Step 4, and

$$\mathrm{QV}_n^a = \tfrac{1}{2} \sum_{i=1}^{n} a_i \left((\Delta W_i)^2 - \Delta t_i \right)$$

is their weighted quadratic-variation sum; M_W is a bound on one path and osc_n an oscillation, both declared where they are used; and t^+ is the first partition point at or after $t \wedge t_\ell$, where t_ℓ is a hitting time as in Definition 19.22(c). The starred points are those supplied by the fundamental theorem of calculus with the intermediate value theorem ($t_i^* \in [t_{i-1}, t_i]$, a closed cell, because the last one may end at T) or by Corollary 7.32 (x_i^* between $W_{t_{i-1}}$ and W_{t_i}, with $u^* \in [0,1]$). That corollary calls its point ξ, which is the integrand here. The letter χ denotes a local function of one real variable, declared in place. The letter q is the continuous yield of an asset, not $1 - p$; r is the risk-free rate; and F_t is the forward price at time t for delivery at T.

Theorem 19.30 (Itô's lemma (Itô, 1951))**.** *Let $T > 0$, let W be a standard Brownian motion, and let $f \in C_E^{1,2}([0,T] \times \mathbb{R})$ (Definition 19.19(iii)). An integral over an interval of length zero is zero.*

(i) For every $t \in [0,T]$, with probability one,

$$f(t, W_t) = f(0,0) + \int_0^t \left(\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\right)(s, W_s)\, ds + \int_0^t \partial_x f(s, W_s)\, dW_s. \tag{19.11}$$

The first integral is a Riemann integral computed path by path. The second is the Itô integral of Theorem 19.23; its integrand $\partial_x f(s, W_s)$ is in the class $\mathcal{H}$. In shorthand,

$$df(t, W_t) = \left(\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\right) dt + \partial_x f\, dW.$$

(ii) (Parameter clause) Let $0 \le t_0 \le t \le T$, and let $Y = (W_{s_1}, \dots, W_{s_m})$ collect values of W at fixed times $s_j \le t_0$. In this clause $f(s, x, y)$ is a function of a time $s \in [t_0, t]$, of a real number x and of a value y of Y, with the following hypotheses in place of the one above. It is piecewise continuous in y in the sense of Definition 19.22(a): the range of y is covered by finitely many disjoint boxes, and for each box $\mathcal{D}$ the function f and its partial derivatives $\partial_t f$, $\partial_x f$ and $\partial_{xx} f$ in (s, x) exist and are continuous in (s, x, y) on $[t_0, t] \times \mathbb{R} \times \mathcal{D}$ (one-sided in s at t_0 and at t). Let these four functions have exponential growth in (x, y). Then, with probability one,

$$f(t, W_t, Y) = f(t_0, W_{t_0}, Y) + \int_{t_0}^t \left(\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\right)(s, W_s, Y)\, ds + \int_{t_0}^t \partial_x f(s, W_s, Y)\, dW_s. \tag{19.12}$$

(iii) (Stopped clause) Let ℓ be a continuous function on $[0,T]$, and let t_ℓ be the first time at which W reaches it, as in Definition 19.22(c): either $\ell(0) < 0$ and $t_\ell = \inf\{s \in [0,T] : W_s \le \ell(s)\}$, or $\ell(0) > 0$ and $t_\ell = \inf\{s \in [0,T] : W_s \ge \ell(s)\}$, the infimum of the empty set being $+\infty$. Then, for every $t \in [0,T]$, with probability one,

$$\begin{aligned} f(t \wedge t_\ell,\ W_{t \wedge t_\ell}) = f(0,0) &+ \int_0^t \mathbf{1}\{s < t_\ell\} \left(\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\right)(s, W_s)\, ds \\ &+ \int_0^t \mathbf{1}\{s < t_\ell\}\, \partial_x f(s, W_s)\, dW_s. \end{aligned} \tag{19.13}$$

The first integral is the Riemann integral over $[0, t \wedge t_\ell]$, and the second is the Itô integral of a stopped integrand, which is in $\mathcal{H}$ by Definition 19.22(c).

Sketch of the proof. The proof is a Taylor expansion summed over a partition of $[0,t]$, in which the squared increments survive. We telescope over the cells, and Taylor's theorem, to first order in time and to second order in space, writes the change of $f(t, W_t)$ over the i-th cell as

$$\partial_t f(t_i^*, W_{t_i})\, \Delta t_i + \partial_x f(t_{i-1}, W_{t_{i-1}})\, \Delta W_i + \tfrac{1}{2}\,\partial_{xx} f(t_{i-1}, x_i^*)\, (\Delta W_i)^2,$$

with the starred points of the Setup. On every path the first terms sum to $\int_0^t \partial_t f(s, W_s)\, ds$ in the limit, by uniform continuity, and the second form the left-endpoint sum of $\partial_x f(s, W_s)$, which is in $\mathcal{H}$ by exponential growth and converges in mean square to its Itô integral (Theorem 19.23(i)). In the third, $(dW)^2 = dt$ becomes exact: with $a_i = \partial_{xx} f(t_{i-1}, W_{t_{i-1}})$, the centred squares $(\Delta W_i)^2 - \Delta t_i$ are independent of the past, with mean zero and variance $2(\Delta t_i)^2$, so $\mathbb{E}[(\mathrm{QV}_n^a)^2]$ is $\frac{1}{2}\sum_i \mathbb{E}[a_i^2]\,(\Delta t_i)^2$, which tends to zero by Lemma 19.20(iii), and $\frac{1}{2}\sum_i a_i\,(\Delta W_i)^2$ has the limit of $\frac{1}{2}\sum_i a_i\,\Delta t_i$. Replacing x_i^* by $W_{t_{i-1}}$ costs at most $\frac{1}{2}\,\mathrm{QV}_n$ times an oscillation of $\partial_{xx} f$ that tends to zero by continuity. The limits in probability give (19.11); clause (ii) freezes Y, and clause (iii) multiplies the i-th term by $\mathbf{1}\{t_{i-1} < t_\ell\}$, which is known at the start of the cell. The complete proof is in Appendix E.6. □

(Problem P19.6 asks you to reproduce Steps 1 to 6 of the proof of Theorem 19.30 (Appendix E.6) for W_t^2 and W_t^3, where every sum can be computed by hand.)

The classical form of the lemma, for a function $V(X_t, t)$ of an Itô process, is Corollary 19.31. The general statement, for coefficients determined by the past and for solutions of stochastic differential equations, is Fact 19.29(iii), which no proof in this Part uses. That the stochastic integrals of the theorem have mean zero is Theorem 19.23(ii).

Intuition. For a general Itô process X with $dX = \alpha_t\,dt + \beta_t\,dW$ the same computation is a second-order Taylor expansion of $V(X_t, t)$ along the segment that joins (X_t, t) to $(X_t + \Delta X, t + \Delta t)$. For a V with continuous second derivatives in (x, t), Corollary 7.32 at order one, applied to $\chi(u) = V(X + u\,\Delta X,\ t + u\,\Delta t)$, $0 \le u \le 1$, gives, with the mixed derivative taken in either order (Lemma 19.54, proved in §19.8),

$$\Delta V \approx \partial_t V\,\Delta t + \partial_x V\,\Delta X + \tfrac{1}{2}\,\partial_{xx} V\,(\Delta X)^2 + \partial_{xt} V\,\Delta X\,\Delta t + \tfrac{1}{2}\,\partial_{tt} V\,(\Delta t)^2,$$
$$(\Delta X)^2 \approx \beta_t^2\,(\Delta W)^2 + 2\alpha_t\beta_t\,\Delta t\,\Delta W + \alpha_t^2\,(\Delta t)^2.$$

Table 4 keeps the first term of the square as $\beta_t^2\,\Delta t$ and kills the other two, and with them the last two terms of ΔV, of sizes $(\Delta t)^{3/2}$ and $(\Delta t)^2$. (The proof of Theorem 19.30 needs no second time derivative, because its Step 1 moves time and space one at a time.) The error of the second-order expansion is smaller than $(\Delta X)^2$ by a factor that vanishes with the step (Step 5 of that proof); for a V with a third derivative in the state it has size $|\Delta X|^3 \sim (\Delta t)^{3/2}$, and the sum of $1/\Delta t$ such terms vanishes with Δt. Step 4 of that proof is $(dW)^2 = dt$ made honest. The surviving term $\frac{1}{2}\,\beta_t^2\,\partial_{xx} V\,dt$ is of second order in ΔX but of *first* order in Δt, because ΔX has size $\sqrt{\Delta t}$. It is the convexity term that a first-order chain rule misses, and it is positive for a convex V: convexity earns from motion. In §21 it becomes the gamma gain of a delta-hedged option: $\frac{1}{2}\,\Gamma\,(\Delta S)^2$ over one rebalancing interval (Proposition 21.17), and $\frac{1}{2}\,\Gamma_t\,S_t^2\,\sigma_{\text{real}}^2\,dt$ in continuous time, where σ_{real} is the volatility that the underlier realises (Proposition 21.22). The theta that the option pays at its implied volatility is set against it.

Corollary 19.31 (Itô's lemma for a function of an Itô process)**.** *Let $f(t, y)$ be in $C_E^{1,2}([0, T] \times \mathbb{R})$ (Definition 19.19(iii); the Brownian argument of f is written y because x is the state of V), with all its values in an open interval $\mathcal{O}$, and let $X_t = f(t, W_t)$. By Theorem 19.30(i), X is an Itô process, $dX = \alpha_t\,dt + \beta_t\,dW$, with*

$$\alpha_t = \Big(\partial_t f + \tfrac{1}{2}\,\partial_{yy} f\Big)(t, W_t), \qquad \beta_t = \partial_y f(t, W_t).$$

Let the outer function $V(x, t)$ be defined on $\mathcal{O} \times [0, T]$ and have continuous partial derivatives $\partial_t V$, $\partial_x V$ and $\partial_{xx} V$ there (one-sided in t at 0 and at T). Suppose that the composite $\tilde{V}(y, t) = V(f(t, y), t)$ and its partial derivatives $\partial_t \tilde{V}$, $\partial_y \tilde{V}$ and $\partial_{yy} \tilde{V}$, which exist and are continuous by the proof of the corollary (Appendix E.7), have exponential growth; that is, $(t, y) \mapsto \tilde{V}(y, t)$ is in $C_E^{1,2}([0, T] \times \mathbb{R})$.

(i) (Chain rule) For every $t \in [0, T]$, with probability one,

$$V(X_t, t) = V(X_0, 0) + \int_0^t \Big(\partial_t V + \alpha_s\,\partial_x V + \tfrac{1}{2}\,\beta_s^2\,\partial_{xx} V\Big)(X_s, s)\,ds + \int_0^t \beta_s\,\partial_x V(X_s, s)\,dW_s. \tag{19.14}$$

In shorthand,

$$dV(X_t, t) = \Big(\partial_t V + \alpha_t\,\partial_x V + \tfrac{1}{2}\,\beta_t^2\,\partial_{xx} V\Big)\,dt + \beta_t\,\partial_x V\,dW,$$

every derivative of V being evaluated at (X_t, t): the process $V(X_t, t)$ is again an Itô process in the sense of Definition 19.25(ii).

(ii) (Expectation identity) For every $t \in [0,T]$,

$$\mathbb{E}\, V(X_t,t) = V(X_0,0) + \int_0^t \mathbb{E}\Big[\Big(\partial_t V + \alpha_s\, \partial_x V + \tfrac{1}{2}\,\beta_s^2\, \partial_{xx} V\Big)(X_s,s)\Big]\, ds : \tag{19.15}$$

the drift is the expected rate of change.

(iii) (Geometric Brownian motion) Let $X = S$ be the geometric Brownian motion of Definition 19.27, so that $f(t,y) = S_0\, e^{gt+\sigma y}$ with $g = \mu - \sigma^2/2$, $\mathcal{O} = (0,\infty)$, $\alpha_t = \mu S_t$ and $\beta_t = \sigma S_t$. The hypothesis on the composite then holds whenever V, $\partial_t V$, $\partial_x V$ and $\partial_{xx} V$ are continuous and of moderate growth on $(0,\infty)\times[0,T]$ (Definition 19.19(ii)), and, with the state variable of V written S as in $V(S,t)$, the chain rule (i) reads

$$dV(S_t,t) = \Big(\partial_t V + \mu S_t\, \partial_S V + \tfrac{1}{2}\,\sigma^2 S_t^2\, \partial_{SS} V\Big)\, dt + \sigma S_t\, \partial_S V\, dW,$$

with the derivatives of $V(S,t)$ evaluated at (S_t,t).

(iv) (Parameter and stopped clauses) The two further clauses of Theorem 19.30 carry over. Let $0 \le t_0 \le t \le T$, and let f and V depend in addition on the value y of a vector Y of Brownian values at times at most t_0; in this clause the Brownian argument of f is written y'. Suppose that, for each such y, the functions $(s,y') \mapsto f(s,y',y)$ on $[t_0,t]\times\mathbb{R}$ and $(x,s) \mapsto V(x,s,y)$ on $\mathcal{O}\times[t_0,t]$ satisfy the hypotheses above, with $[t_0,t]$ in place of $[0,T]$, and that the composite $(s,y',y) \mapsto V(f(s,y',y),s,y)$ satisfies the hypotheses that Theorem 19.30(ii) puts on its function, with y' in the place of its x. Then (19.14) holds on $[t_0,t]$ with Y frozen, α_s and β_s now depending on Y: the integrals start at t_0 and the starting value is $V(X_{t_0},t_0)$. For a hitting time t_ℓ as in Theorem 19.30(iii), (19.14) and (19.15) hold with $V(X_{t\wedge t_\ell}, t\wedge t_\ell)$ on the left and with every integrand multiplied by $\mathbf{1}\{s < t_\ell\}$.

Sketch of the proof. The chain rule of calculus reduces the corollary to Theorem 19.30, applied to the composite $\tilde{V}(y,t) = V(f(t,y),t)$, which is in $C_E^{1,2}$ by hypothesis. With the derivatives of V taken at $(f(t,y),t)$, the chain rule gives $\partial_y \tilde{V} = \partial_x V\, \partial_y f$ and

$$\partial_t \tilde{V} + \tfrac{1}{2}\,\partial_{yy}\tilde{V} = \partial_t V + \partial_x V\,\Big(\partial_t f + \tfrac{1}{2}\,\partial_{yy} f\Big) + \tfrac{1}{2}\,\partial_{xx} V\,(\partial_y f)^2.$$

At $y = W_s$ and $t = s$ the right side is the drift $\partial_t V + \alpha_s\,\partial_x V + \frac{1}{2}\,\beta_s^2\,\partial_{xx} V$ and $\partial_y \tilde{V}$ is $\beta_s\,\partial_x V$, so the theorem for the composite is (i). The appendix proves the chain rule in t, where both arguments of V move, with the mean value theorem on the open interval $\mathcal{O}$. For (ii), take expectations: the stochastic integral has mean zero (Theorem 19.23(ii)), and the expectation passes inside the ds-integral by Fact 18.34(iii), the drift having a mean bounded in time by exponential growth. For (iii), $f(t,y) = S_0\, e^{gt+\sigma y}$ has $\partial_t f = g\, f$, $\partial_y f = \sigma f$ and $\partial_{yy} f = \sigma^2 f$, so $\alpha_t = \mu S_t$ and $\beta_t = \sigma S_t$, and Lemma 19.20(i) turns moderate growth in the price into exponential growth in y. Part (iv) applies clause (ii) or (iii) of the theorem to the composite. The complete proof is in Appendix E.7. □

Example 19.32 (Functions of Brownian motion)**.** Take $X = W$, that is, $f(t,y) = y$, so that $\alpha_t = 0$ and $\beta_t = 1$; the corollary is then Theorem 19.30 itself. The functions x^2, x^3, $x^3 - 3tx$, e^x and tx used below are in $C_E^{1,2}([0,T]\times\mathbb{R})$ for every T: each of them, and each of its derivatives, is a polynomial in (t,x) or equals e^x, and $|x|^k \le k!\, e^{|x|}$.

The square. For $f(t,x) = x^2$: $\partial_t f = 0$, $\partial_x f = 2x$ and $\partial_{xx} f = 2$, so (19.11) reads

$$W_t^2 = t + \int_0^t 2W_s\, dW_s, \qquad \text{in shorthand} \qquad d(W_t^2) = dt + 2W_t\, dW_t.$$

Taking expectations, $\mathbb{E}[W_t^2] = t$ by Theorem 19.23(ii), in agreement with Proposition 19.8(i). Solving for the integral, $\int_0^t W_s\, dW_s = (W_t^2 - t)/2$, the value that Example 19.24 found from the left-endpoint sums.

The cube. For $f(t,x) = x^3$: $\partial_x f = 3x^2$ and $\partial_{xx} f = 6x$, so $d(W_t^3) = 3W_t\,dt + 3W_t^2\,dW_t$, with drift $3W_t$. For $f(t,x) = x^3 - 3tx$: $\partial_t f = -3x$, $\partial_x f = 3x^2 - 3t$ and $\partial_{xx} f = 6x$, so the drift is $-3x + \frac{1}{2}\cdot 6x = 0$ and

$$d(W_t^3 - 3tW_t) = 3(W_t^2 - t)\,dW_t.$$

The exponential. For $f(t,x) = e^x$ every derivative in x is e^x, so $d(e^{W_t}) = \frac{1}{2}\,e^{W_t}\,dt + e^{W_t}\,dW_t$. Let $\chi(t) = \mathbb{E}\,e^{W_t}$. The expectation identity (19.15) gives

$$\chi(t) = 1 + \tfrac{1}{2}\int_0^t \chi(s)\,ds.$$

The function χ is continuous: as $s' \to s$, $e^{W_{s'}} \to e^{W_s}$ on every path, and the fourth moments $\mathbb{E}\,e^{4W_{s'}}$ are bounded for $s' \le T$ (Lemma 19.20(iii)), so $\mathbb{E}\,e^{W_{s'}} \to \mathbb{E}\,e^{W_s}$ by the fourth-moment lemma (Lemma 19.12(iv)), as in Step 3 of the proof of Theorem 19.23. By the fundamental theorem of calculus, χ is then differentiable, with $\chi' = \chi/2$ and $\chi(0) = 1$. Hence $\big(\chi(t)\,e^{-t/2}\big)' = \big(\chi'(t) - \chi(t)/2\big)\,e^{-t/2} = 0$, and

$$\mathbb{E}\,e^{W_t} = \chi(t) = e^{t/2}.$$

At $t = 1$ this is $e^{0.5} = 1.6487$, the number of Example 7.40. It agrees with Lemma 18.35(i), used in Example 19.18, and Lemma 19.38(i) confirms it once more in §19.6 (Example 19.39).

The time integral. For $f(t,x) = tx$: $\partial_t f = x$, $\partial_x f = t$ and $\partial_{xx} f = 0$, so (19.11) reads

$$tW_t = \int_0^t W_s\,ds + \int_0^t s\,dW_s.$$

The integral of the constant t over $[0,t]$ is $\int_0^t t\,dW_s = tW_t$ (the paragraph after Definition 19.25), so linearity (Theorem 19.23(iv)) gives

$$\int_0^t W_s\,ds = \int_0^t (t-s)\,dW_s,$$

the stochastic integral of a deterministic continuous integrand, which is in $\mathcal{H}$. By Theorem 19.23(ii) and (iii), the time integral of the path has mean 0, variance $\int_0^t (t-s)^2\,ds = t^3/3$, and covariance $\int_0^t (t-s)\cdot 1\,ds = t^2/2$ with $W_t = \int_0^t 1\,dW_s$. At $t = 1$ the variance is $1/3 = 0.3333$, the standard deviation 0.5774, and the correlation with W_1 is $(1/2)/\sqrt{1/3} = \sqrt{3}/2 = 0.8660$, the same at every t. Its law is $\mathcal{N}(0, t^3/3)$, the answer an interviewer expects. Let $t > 0$ (at $t = 0$ the integral is 0), take a partition $0 = t_0 < t_1 < \cdots < t_n = t$ with $n \ge 2$, and write $Y_n = \sum_{i=1}^n W_{t_{i-1}}\,\Delta t_i$ for its left-endpoint Riemann sum and $Y = \int_0^t W_s\,ds$ (names local to this paragraph). Since $W_{t_{i-1}} = \sum_{j=1}^{i-1} \Delta W_j$, exchanging the order of the two finite sums gives

$$Y_n = \sum_{j=1}^{n-1} \Delta W_j \sum_{i=j+1}^{n} \Delta t_i = \sum_{j=1}^{n-1} (t - t_j)\,\Delta W_j,$$

a linear combination, with positive coefficients, of the mutually independent increments of Definition 19.5(3). Each term is normal with mean 0 and variance $(t-t_j)^2\,\Delta t_j$, by (19.3). The sum of the first k terms is a function of $\Delta W_1, \ldots, \Delta W_k$, hence independent of ΔW_{k+1} by the grouping property (a) of the Setup of §19.2. So Lemma 19.38(ii) in the next subsection, applied one term at a time, shows that Y_n is normal with mean 0 and variance

$$\operatorname{Var}(Y_n) = \sum_{j=1}^{n} (t - t_j)^2\,\Delta t_j,$$

the term $j = n$ being zero. This is a Riemann sum, with right end points as tags, of the continuous function $s \mapsto (t-s)^2$, so it tends to $\int_0^t (t-s)^2\,ds = t^3/3$ as $\Delta t_{\max} \to 0$. On every path $Y_n \to Y$ as $\Delta t_{\max} \to 0$, because the path is continuous; hence $Y_n \to Y$ in probability (Lemma 19.12(iii)). For

every real x and every $\varepsilon > 0$, the event $\{Y \le x\}$ is contained in $\{Y_n \le x+\varepsilon\} \cup \{|Y_n - Y| > \varepsilon\}$, and the event $\{Y_n \le x-\varepsilon\}$ in $\{Y \le x\} \cup \{|Y_n - Y| > \varepsilon\}$, so

$$P(Y \le x) \le P(Y_n \le x+\varepsilon) + P(|Y_n - Y| > \varepsilon), \qquad P(Y_n \le x-\varepsilon) \le P(Y \le x) + P(|Y_n - Y| > \varepsilon).$$

Along a sequence of partitions whose meshes tend to zero, $P(Y_n \le x \pm \varepsilon) = \Phi\big((x \pm \varepsilon)/\sqrt{\mathrm{Var}(Y_n)}\big)$ tends to $\Phi\big((x \pm \varepsilon)/\sqrt{t^3/3}\big)$, by the continuity of Φ, and $P(|Y_n - Y| > \varepsilon) \to 0$. Hence

$$\Phi\Big(\frac{x-\varepsilon}{\sqrt{t^3/3}}\Big) \le P(Y \le x) \le \Phi\Big(\frac{x+\varepsilon}{\sqrt{t^3/3}}\Big),$$

and letting $\varepsilon \to 0$ gives $P(Y \le x) = \Phi\big(x/\sqrt{t^3/3}\big)$, by the continuity of Φ once more. The time integral has the distribution function of $\sqrt{t^3/3}\,Z$, which is the law $\mathcal{N}(0, t^3/3)$. The time average $\frac{1}{t}\int_0^t W_s\,ds$ has variance $t/3$, a third of that of W_t. This is the factor behind the log-variance $\sigma^2 T/3$ that the geometric average of Proposition 23.28 approaches as its fixings become dense.

Interview trap. "Which of these has zero drift?" By Example 19.32 and Theorem 19.30: $W_t^2 - t$, yes; W_t^3, no (drift $3W_t$); $W_t^3 - 3tW_t$, yes; $e^{W_t - t/2}$, yes, because $f(t,x) = e^{x-t/2}$ has $\partial_t f + \frac{1}{2}\,\partial_{xx} f = -\frac{1}{2}f + \frac{1}{2}f = 0$; tW_t, no, because $f(t,x) = tx$ has $\partial_t f + \frac{1}{2}\,\partial_{xx} f = x$, a drift W_t. The error behind the second answer is "W_t^3 has zero drift because W does". Zero drift, with an integrand in $\mathcal{H}$, means a constant expectation, by the expectation identity (19.15), which rests on Theorem 19.23(ii). Interviewers usually ask this as "which of these is a martingale?". In the general theory a driftless Itô process with a square-integrable integrand is a *martingale*: its expected future value, given the path so far, is its present value. This Part uses no conditional expectation given a path and proves only the consequence it needs, a constant expectation; the answers above are the answers to the martingale question.

Example 19.33 (Powers of a geometric Brownian motion)**.** Let S be the geometric Brownian motion of Definition 19.27, so that $\alpha_t = \mu S_t$ and $\beta_t = \sigma S_t$ by Corollary 19.31(iii). For a real exponent a, the function $V(x,t) = x^a$ on $\mathcal{O} = (0,\infty)$ has $\partial_t V = 0$, $\partial_x V = a\,x^{a-1}$ and $\partial_{xx} V = a(a-1)\,x^{a-2}$. Each is a constant multiple of a power of x, and a power $x^{a'}$ with any real exponent a' has moderate growth because $x^{a'} \le x^{|a'|} + x^{-|a'|}$. The corollary gives the drift $\mu S \cdot a\,S^{a-1} + \frac{1}{2}\,\sigma^2 S^2 \cdot a(a-1)\,S^{a-2}$ and the diffusion coefficient $\sigma S \cdot a\,S^{a-1}$, that is,

$$\frac{d(S^a)}{S^a} = \Big(a\mu + \tfrac{1}{2}\,a(a-1)\,\sigma^2\Big)\,dt + a\sigma\,dW.$$

The explicit formula agrees: by (19.10), $S_t^a = S_0^a \exp(agt + a\sigma W_t)$ with $g = \mu - \sigma^2/2$. For a real exponent $a \neq 0$ this is a geometric Brownian motion with volatility $|a|\sigma$ and drift $ag + \frac{1}{2}a^2\sigma^2 = a\mu + \frac{1}{2}a(a-1)\sigma^2$, driven by W when $a > 0$ and by $-W$ when $a < 0$ (Proposition 19.8(iii)). For $a = 0$ the power is the constant 1, which has volatility 0 and is not a geometric Brownian motion.

With $\mu = 10\%$ and $\sigma = 20\%$, so that $\sigma^2 = 4\%$:

- the square S^2 drifts at $2 \times 10\% + 4\% = 24\%$, with volatility 40%;
- the reciprocal $1/S$ ($a = -1$) drifts at $-10\% + 4\% = -6\%$, which is $\sigma^2 - \mu$, and its coefficient of dW is -20%: a volatility of 20% with the shock reversed;
- the square root $\sqrt{S}$ ($a = \frac{1}{2}$) drifts at $5\% - 0.5\% = 4.5\%$, with volatility 10%.

(Problem P19.5 works this computation at different numbers.)

Interview trap. Siegel's paradox: by Example 19.33 the reciprocal $1/S$ drifts at $\sigma^2 - \mu$, not at $-\mu$. For an exchange rate with a volatility of 10%, $\sigma^2 = 1\%$: if the rate quoted one way has an expected appreciation of $\mu = 0.4\%$ a year, the rate quoted the other way has an expected appreciation of $1\% - 0.4\% = 0.6\%$ a year. Both are positive. Quoting the reciprocal rate does not simply flip the sign of the drift.

Interview trap. Writing an independent noise term for dV on the whiteboard. By Corollary 19.31(iii), the form in which §20 applies the corollary to the value of a derivative, the diffusion coefficient of $V(S_t,t)$ is $\beta_t\,\partial_S V(S_t,t)=\sigma S_t\,\partial_S V(S_t,t)$: the derivative is driven by the *same* W as the underlying. That shared shock is what lets a position of $\partial_S V(S_t,t)$ shares cancel the risk of a derivative in §20 (Theorem 20.3), and it is the answer to "why does Itô matter for pricing?".

Corollary 19.34 (Positivity, the log price and the GBM equation)**.** *Let S be the geometric Brownian motion of Definition 19.27, with drift μ, volatility $\sigma>0$ and $S_0>0$, and let $g=\mu-\sigma^2/2$.*

(i) For every $t\ge 0$, $S_t>0$ and $\ln S_t=\ln S_0+gt+\sigma W_t$: the log price is a generalised Wiener process with drift g and coefficient σ (Definition 19.25(i)).

(ii) For every $t\ge 0$, with probability one,

$$S_t=S_0+\int_0^t \mu S_s\,ds+\int_0^t \sigma S_s\,dW_s;$$

that is, $dS=\mu S\,dt+\sigma S\,dW$ in the sense of Definition 19.25(ii).

(iii) Corollary 19.31 with the outer function $V(x,t)=\ln x$ returns $d\ln S=(\mu-\sigma^2/2)\,dt+\sigma\,dW$, in agreement with (i).

Proof. We read (i) off the definition and apply Itô's lemma for (ii) and (iii).

Part (i). By (19.10), S_t is $S_0>0$ times an exponential, which is positive, and taking logarithms gives $\ln S_t=\ln S_0+(\mu-\sigma^2/2)\,t+\sigma W_t$. This is Definition 19.25(i) with $X_0=\ln S_0$ and the constants g and σ.

Part (ii). Fix $t>0$ and a horizon $T\ge t$, and apply Theorem 19.30(i) to $f(t,x)=S_0\,e^{gt+\sigma x}$, for which $f(t,W_t)=S_t$ and $f(0,0)=S_0$. The proof of Corollary 19.31 showed that $f\in C_E^{1,2}([0,T]\times\mathbb{R})$, with $\partial_t f=g\,f$, $\partial_x f=\sigma f$ and $\partial_{xx} f=\sigma^2 f$. So the drift is $(g+\sigma^2/2)\,f=\mu f$ and the diffusion coefficient is σf; evaluated along the path they are μS_s and σS_s, and (19.11) is the identity of (ii). The integrand σS_s is in $\mathcal{H}$, and $s\mapsto\mu S_s$ is continuous on every path, as Definition 19.25(ii) requires.

Part (iii). The function $V(x,t)=\ln x$ on $\mathcal{O}=(0,\infty)$ has $\partial_t V=0$, $\partial_x V=1/x$ and $\partial_{xx}V=-1/x^2$. All four functions are continuous and of moderate growth: $|\ln x|\le x+1/x$ (Example 19.21), $1/x\le x+1/x$ and $1/x^2\le x^2+1/x^2$. So Corollary 19.31 applies, with $\alpha_t=\mu S_t$ and $\beta_t=\sigma S_t$: the drift of $\ln S$ is

$$\mu S\cdot\frac{1}{S}+\tfrac12\,\sigma^2S^2\cdot\Big(-\frac{1}{S^2}\Big)=\mu-\tfrac12\,\sigma^2,$$

and its diffusion coefficient is $\sigma S\cdot(1/S)=\sigma$. Hence $\ln S_t=\ln S_0+\int_0^t(\mu-\sigma^2/2)\,ds+\int_0^t\sigma\,dW_s=\ln S_0+gt+\sigma W_t$, the integral of the constant σ being σW_t (the paragraph after Definition 19.25). This is (i) again. □

Example 19.35 (The drift of a log price)**.** Let $\mu=10\%$ and $\sigma=30\%$. Then $\sigma^2/2=0.09/2=4.5\%$, and by Corollary 19.34 the log price drifts at $10\%-4.5\%=5.5\%$ a year. The median path therefore grows at 5.5% a year while the mean grows at 10%; both rates are proved in §19.6 (Proposition 19.42(i) and (ii)).

Part (iii) is the computation an interviewer expects. No uniqueness theorem for the equation $dS=\mu S\,dt+\sigma S\,dW$ is used: the Part defines S by its formula, and (ii) shows that the formula solves the equation.

Intuition. The whiteboard version is a Taylor expansion of the logarithm. By Proposition 7.34,

$$\ln\Big(1+\frac{\Delta S}{S}\Big)=\frac{\Delta S}{S}-\frac12\Big(\frac{\Delta S}{S}\Big)^2+O\Big(\Big|\frac{\Delta S}{S}\Big|^3\Big).$$

Over a short step, $\Delta S/S\approx\mu\,\Delta t+\sigma\,\Delta W$ has mean about $\mu\,\Delta t$, and its square is about $\sigma^2(\Delta W)^2$, which Table 4 replaces by $\sigma^2\,\Delta t$. So the log-return has mean about $(\mu-\sigma^2/2)\,\Delta t$: the simple

return averages $\mu\,\Delta t$, and the concave logarithm takes off half the variance. The one-step log mean $(r-\sigma^2/2)\,\Delta t$ of the Cox–Ross–Rubinstein tree (Proposition 18.21(ii)) is the discrete shadow of the same drift.

Interview trap. "By the chain rule $d\ln S = dS/S$, so $\ln S$ drifts at μ." This misses the convexity term $\frac{1}{2}\cdot(-1/S^2)\cdot(\sigma S)^2 = -\sigma^2/2$ of Corollary 19.34(iii). The logarithm is concave, so the term is negative: in Example 19.35 the drift is 5.5%, not 10%. This is why the median grows more slowly than the mean.

Corollary 19.36 (Dynamics of a forward price)**.** *Let S be the geometric Brownian motion of Definition 19.27, let r and q be constants (the risk-free rate and the continuous yield of the asset; $q=0$ for a stock that pays no dividend), let $T>0$, and let*

$$F_t = S_t\,e^{(r-q)(T-t)}, \qquad 0\le t\le T,$$

be the forward price at time t for delivery at T of an investment asset (Theorem 16.11 for $q=0$ and Proposition 16.15(ii) in general, each applied from the date t with $T-t$ in the place of T). Then, in the sense of Definition 19.25(ii),

$$dF = (\mu - r + q)\,F\,dt + \sigma F\,dW,$$

and for a stock that pays no dividend $dF = (\mu-r)\,F\,dt+\sigma F\,dW$. The forward price is a geometric Brownian motion with the same volatility as S, started at $F_0 = S_0\,e^{(r-q)T}$; its drift $\mu+q-r$ is the expected total return of the asset, $\mu+q$, in excess of the risk-free rate.

Proof. We apply Corollary 19.31 to S and the outer function $V(x,t) = x\,e^{(r-q)(T-t)}$ on $(0,\infty)\times[0,T]$, for which $F_t = V(S_t,t)$. Its derivatives are $\partial_x V = e^{(r-q)(T-t)}$, $\partial_{xx}V = 0$ and $\partial_t V = -(r-q)\,V$. The factor $e^{(r-q)(T-t)}$ is at most $e^{|r-q|T}$ on $[0,T]$, so V and its three derivatives are continuous and at most a constant times $x+1/x$ in absolute value, which is moderate growth. With $\alpha_t = \mu S_t$ and $\beta_t = \sigma S_t$, the drift of F is

$$\partial_t V + \mu S\,\partial_x V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{xx}V = -(r-q)\,F + \mu S\,e^{(r-q)(T-t)} = (\mu - r + q)\,F,$$

and its diffusion coefficient is $\sigma S\,e^{(r-q)(T-t)} = \sigma F$. Equivalently, by (19.10),

$$F_t = S_0\,e^{(r-q)T}\exp\Big(\big(\mu - r + q - \tfrac{1}{2}\sigma^2\big)\,t + \sigma W_t\Big),$$

which is the process of Definition 19.27 with drift $\mu - r + q$, volatility σ and starting value $S_0\,e^{(r-q)T}$; Corollary 19.34(ii), applied to it, gives the same equation. □

When the drift of the asset is the risk-neutral one, $\mu = r - q$, the forward price has no drift. That a futures price has zero drift under $\mathbb{Q}$ is proved in §20 (Proposition 20.51), and it is the model behind Black's formula for options on futures. By Theorem 16.28, under deterministic rates the futures price equals the forward price F_t at every settlement date, so on that grid of dates it has the law of the geometric Brownian motion above.

Example 19.37 (A six-month forward)**.** A stock that pays no dividend trades at $S = 100$, with $r = 5\%$ and $T - t = 0.5$. The forward price is $F = 100\,e^{0.05\times 0.5} = 100\,e^{0.025} = 102.53$. With $\mu = 12\%$ and $\sigma = 25\%$, Corollary 19.36 gives

$$dF = 0.07\,F\,dt + 0.25\,F\,dW\ :$$

a long forward position carries the full 25% volatility of the stock but only its 7% excess drift. The forward costs nothing to enter, so no capital is tied up on which the rate r would be earned.

19.6 The Lognormal Law

"What is the median of S_T?", "what is the probability that the stock ends above 110 in a year?", "give a 95% range for the price" and "compute $\mathbb{E}[e^{\sigma W_T}]$" all reduce to one fact: under geometric Brownian motion, $\ln S_T$ is normal. This subsection first proves a Gaussian lemma that does every exponential-moment computation of this Part. Section 20 cites it for the asset-or-nothing price and for $\Phi(d_1)$, §22 for the calendar condition, and §23 for Margrabe's formula. The subsection then derives the lognormal law and computes its mean, variance, median, mode, central interval and real-world exceedance probabilities. It builds on Definition 19.27, on Lemma 19.17 and on the two tools of §18, Fact 18.34 and Lemma 18.35.

Setup. Throughout, $Z \sim \mathcal{N}(0,1)$, with density φ and distribution function Φ. The real number θ is a shift. A *test function* ψ is a function on the real line, continuous except at finitely many points, of exponential growth: for some constant $c \geq 0$,

$$|\psi(z)| \leq c\, e^{c|z|} \qquad \text{for all real } z,$$

the class of Lemma 18.35(i). Bounded functions qualify, such as indicators of intervals, and so do nonnegative functions of exponential growth, such as call-type payoffs $(e^z - 1)^+$, and functions of either sign such as $\psi(z) = z$, because $|z| \leq e^{|z|}$. In Step 2 of the proof of Lemma 19.38 a test function is a function on the plane, continuous except on finitely many lines, with $|\psi(z_1', z_2')| \leq c\, e^{c(|z_1'|+|z_2'|)}$. For such functions every integral below converges absolutely, so Fact 7.22 applies exactly as stated. Integrals without limits run over the whole real line.

A pair (X, Y) of random variables is *jointly normal* if X and Y are affine functions of one pair (Z_1, Z_2) of independent standard normal variables,

$$X = \mathbb{E}[X] + a_{X,1}Z_1 + a_{X,2}Z_2, \qquad Y = \mathbb{E}[Y] + a_{Y,1}Z_1 + a_{Y,2}Z_2,$$

with real coefficients; degenerate cases, in which a coefficient vector vanishes or the two vectors are proportional, are allowed. The coefficient vectors $a_X = (a_{X,1}, a_{X,2})$ and $a_Y = (a_{Y,1}, a_{Y,2})$ are used only in proofs. In the proof of the lemma, $\sigma_Y = \sqrt{\operatorname{Var}(Y)}$, $c_{XY} = \operatorname{Cov}(X,Y)/\operatorname{Var}(Y)$ and $\sigma_\perp$ are constants local to the proof, ψ_1 is an inner average, and Z_1', Z_2' are rotated variables, with z_1', z_2' as their integration variables (a prime marks another object of the same kind, never a derivative here).

A geometric Brownian motion has the parameters μ, $\sigma > 0$ and $S_0 > 0$ of Definition 19.27; $T > 0$ is a horizon and $g = \mu - \sigma^2/2$. The mean $\ln S_0 + gT$ and the standard deviation $\sigma\sqrt{T}$ of $\ln S_T$ are written out, with no letters of their own. The number $K > 0$ is a price level, the strike of a later option. *Symbol notice:* apart from the number d of variables in Definition 19.19 and Lemma 19.20, the letter d without a subscript is reserved to Proposition 19.42(iii) and its applications, where it is the real-world counterpart of the d_2 of §20. No tree, and so no tree factor d, appears in this section. In the examples, W_1 and W_2 are the values of one motion at the times 1 and 2.

Lemma 19.38 (Gaussian shift, rotation and the jointly normal pair)**.** *Let Z, Z_1 and Z_2 be standard normal variables, Z_1 and Z_2 independent, let θ be a real number and let ψ be a test function as in the Setup.*

(i) (Shift) The exponential moment is $\mathbb{E}\, e^{\theta Z} = e^{\theta^2/2}$, and

$$\mathbb{E}\big[e^{\theta Z}\, \psi(Z)\big] = e^{\theta^2/2}\, \mathbb{E}\big[\psi(Z+\theta)\big],$$

both means being finite.

(ii) (Rotation) If (c_1, c_2) is a unit vector, $c_1^2 + c_2^2 = 1$, then $c_1Z_1 + c_2Z_2$ and $-c_2Z_1 + c_1Z_2$ are independent standard normal variables. Hence, for all real c_1 and c_2,

$$c_1Z_1 + c_2Z_2 \sim \mathcal{N}\big(0,\ c_1^2 + c_2^2\big)$$

(the constant 0 *when* $c_1 = c_2 = 0$*); every linear combination* $cX + c'Y$ *of a jointly normal pair* (X, Y)*, with real constants* c *and* c'*, is normal; and the sum of two independent normal variables is normal, with the means added and the variances added.*

(iii) (Pair) For a jointly normal pair (X, Y) *with* $\mathrm{Var}(Y) > 0$*,*

$$\mathbb{E}\, e^Y = e^{\mathbb{E}[Y]+\mathrm{Var}(Y)/2} \qquad \textit{and} \qquad \mathbb{E}\big[e^Y\, \psi(X)\big] = \mathbb{E}\big[e^Y\big]\, \mathbb{E}\big[\psi(X + \mathrm{Cov}(X, Y))\big].$$

Sketch of the proof. We complete the square for (i), rotate the plane for (ii), and combine the two for (iii). Since $\theta z - z^2/2 = \theta^2/2 - (z-\theta)^2/2$, the weighted density $e^{\theta z}\varphi(z)$ is $e^{\theta^2/2}\varphi(z-\theta)$, and the substitution $z \mapsto z + \theta$ gives (i), the growth of ψ making the integrals converge. For a unit vector (c_1, c_2), the rotation $(Z_1', Z_2') = (c_1 Z_1 + c_2 Z_2,\ -c_2 Z_1 + c_1 Z_2)$ preserves the sum of squares, hence the joint density of (Z_1, Z_2), and it preserves double integrals (identity (D.9) in the proof of Lemma 18.35(ii)); so (Z_1', Z_2') has the joint density of (Z_1, Z_2), which is (E.14) and the first assertion of (ii), and the others follow by scaling. For (iii), the rotation along the coefficient vector of Y, which needs $\mathrm{Var}(Y) > 0$, gives (E.15),

$$Y = \mathbb{E}[Y] + \sigma_Y\, Z_1', \qquad X = \mathbb{E}[X] + c_{XY}\, \sigma_Y\, Z_1' + \sigma_\perp\, Z_2',$$

with a constant $\sigma_\perp$. Freezing Z_2' and applying (i) with $\theta = \sigma_Y$ turns e^Y into the constant $e^{\mathbb{E}[Y]+\sigma_Y^2/2}$, which is $\mathbb{E}\, e^Y$, and shifts X by $c_{XY}\, \sigma_Y^2 = \mathrm{Cov}(X, Y)$. The complete proof is in Appendix E.8. □

In words: weighting by the exponential of a normal variable is the same as shifting the variable under the expectation, by θ in (i) and by the covariance in (iii), at the price of the factor $\mathbb{E}\, e^{\theta Z}$ or $\mathbb{E}\, e^Y$. The lemma is cited in §20 for the asset-or-nothing price (Proposition 20.19), whose proof also gives the reading of $\Phi(d_1)$ as a reweighted exercise probability (Proposition 20.26), in §22 for the calendar condition (Proposition 22.15), and in §23 for Margrabe's formula (Theorem 23.32).

(Problem P19.7 asks you to reproduce Steps 1 and 3 of the proof of Lemma 19.38 (Appendix E.8) from scratch, with the rotation of Step 2 given.)

Example 19.39 (The shift in numbers)**.** With $\theta = 0.4$, Lemma 19.38(i) gives $\mathbb{E}\, e^{0.4Z} = e^{0.08} = 1.0833$ and, for the indicator $\psi(z) = \mathbf{1}\{z > 0\}$,

$$\mathbb{E}\big[e^{0.4Z}\, \mathbf{1}\{Z > 0\}\big] = e^{0.08}\, P(Z + 0.4 > 0) = 1.0833 \times \Phi(0.4) = 1.0833 \times 0.6554 = 0.7100.$$

The tilt moves mass to the right: reweighted by $e^{0.4Z}/\mathbb{E}\, e^{0.4Z}$, the event $\{Z > 0\}$ has probability 0.655 instead of 0.5.

For the pair, take $X = W_1$ and $Y = W_2$. They are jointly normal, because $W_1 = Z_1$ and $W_2 = Z_1 + Z_2$ with $Z_1 = W_1$ and $Z_2 = W_2 - W_1$ independent standard normal variables (Definition 19.5(2) and (3)). Here $\mathbb{E}[Y] = 0$, $\mathrm{Var}(Y) = 2$ and $\mathrm{Cov}(X, Y) = \min(1, 2) = 1$ (Proposition 19.8(i)). Part (iii) gives $\mathbb{E}\, e^{W_2} = e^{2/2} = e$ and

$$\mathbb{E}\big[e^{W_2}\, \mathbf{1}\{W_1 > 0\}\big] = e \times P(W_1 + 1 > 0) = e \times \Phi(1) = 2.7183 \times 0.8413 = 2.287.$$

With $\theta = 1$, part (i) gives $\mathbb{E}\, e^{W_1} = e^{1/2} = 1.6487$, the value that Example 19.32 obtained from Itô's lemma. The test function $\psi(z) = z$, for which $|z| \le e^{|z|}$, gives $\mathbb{E}\big[Z e^{\theta Z}\big] = e^{\theta^2/2}\, \mathbb{E}[Z + \theta] = \theta\, e^{\theta^2/2}$, so $\mathbb{E}\big[W_1 e^{W_1}\big] = e^{1/2} = 1.6487$ as well.

An orthant and a bridge. With $W_1 = Z_1$ and $W_2 = Z_1 + Z_2$ as above, the event $\{W_1 > 0,\ W_2 > 0\}$ is $\{Z_1 > 0,\ Z_1 + Z_2 > 0\}$, the event that the point (Z_1, Z_2) lies in a wedge of the plane with its vertex at the origin and an opening angle of $3\pi/4$. Applied with $c_1 = c_2 = 1/\sqrt{2}$ to the indicator of a wedge, a bounded test function on the plane, (E.14) says that turning the wedge through the angle $\pi/4$ about the origin does not change its probability. The axes and the two diagonals cut the plane into eight wedges of opening $\pi/4$, each obtained from the next by such a turn, so they have equal probabilities. The four lines between them have probability zero, because a nonzero

combination of Z_1 and Z_2 is normal with a positive variance by part (ii), and so takes the value 0 with probability zero. Each wedge therefore has probability 1/8, and the event, made of three of them, has

$$P(W_1 > 0,\ W_2 > 0) = \tfrac{3}{8} = 0.375,$$

against $\frac{1}{4}$ for two independent values. In (E.15) with $X = W_1$ and $Y = W_2$,

$$c_{XY} = \frac{\operatorname{Cov}(W_1, W_2)}{\operatorname{Var}(W_2)} = \frac{1}{2}, \qquad \sigma_\perp^2 = \operatorname{Var}(W_1) - c_{XY}^2 \operatorname{Var}(W_2) = 1 - \tfrac{1}{4} \times 2 = \tfrac{1}{2},$$

the second because (E.15) gives $\operatorname{Var}(X) = c_{XY}^2 \sigma_Y^2 + \sigma_\perp^2$. So

$$W_1 = \tfrac{1}{2}\, W_2 + \sigma_\perp Z_2',$$

with the residual $\sigma_\perp Z_2'$ independent of $W_2 = \sqrt{2}\, Z_1'$. This is how the bridge question "given the endpoint W_2, what is the law of the midpoint W_1?" is answered: half the endpoint plus an independent $\mathcal{N}(0, \frac{1}{2})$ error, whose variance $\frac{1}{2} = 1 \times (2-1)/2$ is the Brownian-bridge variance $s(t-s)/t$ at $s = 1$, $t = 2$.

Theorem 19.40 (The lognormal law under geometric Brownian motion)**.** *Let S be the geometric Brownian motion of Definition 19.27, with drift μ, volatility $\sigma > 0$ and $S_0 > 0$, and let $T > 0$. Then*

$$\ln S_T \sim \mathcal{N}\Big(\ln S_0 + (\mu - \tfrac{1}{2}\sigma^2)\, T,\ \sigma^2 T \Big);$$

equivalently,

$$S_T = S_0 \exp\Big((\mu - \tfrac{1}{2}\sigma^2)\, T + \sigma \sqrt{T}\, Z \Big) \qquad \text{with } Z = W_T/\sqrt{T} \sim \mathcal{N}(0,1).$$

The law of S_T is called lognormal.

Proof. We read the law off the defining formula. By (19.10) and Corollary 19.34(i), $\ln S_T = \ln S_0 + gT + \sigma W_T$ with $g = \mu - \sigma^2/2$: the log price is a generalised Wiener process, and Definition 19.25(i) gives the law of its value at T. In detail, by Definition 19.5(1) and (2), $W_T \sim \mathcal{N}(0, T)$, so $Z = W_T/\sqrt{T}$ is standard normal and $\sigma W_T = \sigma\sqrt{T}\, Z$. Thus $\ln S_T = (\ln S_0 + gT) + \sigma\sqrt{T}\, Z$ is an affine image of a standard normal variable, and by (19.3) in the Setup of §19.2 it is normal with mean $\ln S_0 + gT$ and variance $\sigma^2 T$. Exponentiating gives the formula for S_T. □

Example 19.41 (The law of a six-month price)**.** Let $S_0 = 60$, $\mu = 10\%$, $\sigma = 32\%$ and $T = 0.5$. Then $\sigma^2 = 0.1024$, so $g = 0.10 - 0.0512 = 4.88\%$ and $gT = 0.0244$; the variance of the log price is $\sigma^2 T = 0.0512$ and its standard deviation is $\sigma\sqrt{T} = 0.2263$. With $\ln 60 = 4.0943$, Theorem 19.40 gives

$$\ln S_T \sim \mathcal{N}\big(4.0943 + 0.0244,\ 0.0512\big) = \mathcal{N}\big(4.1187,\ 0.2263^2\big).$$

The proof is short because Definition 19.27 gives S_T explicitly. Sections 20 to 23 cite the statement as the lognormal law.

Intuition. With μ replaced by r, this is the limit law of the Cox–Ross–Rubinstein tree (Theorem 18.37). Lognormality is the central limit theorem (Fact 4.8) at work: $\ln(S_T/S_0)$ is the sum of many small independent log-returns, as in §19.1, so it is close to normal whatever the law of a single return, and the price is the exponential of a normal variable.

Proposition 19.42 (Moments, quantiles and exceedance probabilities of the lognormal law)**.** *Let S_T be as in Theorem 19.40, and let $g = \mu - \sigma^2/2$.*

(i) The mean is $\mathbb{E}[S_T] = S_0\, e^{\mu T}$ and the variance is $\operatorname{Var}(S_T) = S_0^2\, e^{2\mu T}(e^{\sigma^2 T} - 1)$.

(ii) The median of S_T is $S_0\, e^{gT}$, and the mode of its density is $S_0\, e^{(\mu - 3\sigma^2/2)T}$. Hence mode $<$ median $<$ mean.

(iii) For every $K > 0$,

$$P(S_T > K) = \Phi(d) \qquad \text{and} \qquad P(S_T < K) = \Phi(-d), \qquad d = \frac{\ln(S_0/K) + \left(\mu - \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}.$$

These are the real-world *probabilities that a call, and a put, struck at K are exercised.*

(iv) The central 95% interval of S_T, between its 2.5% and 97.5% quantiles, is

$$\Big[S_0\, e^{gT - \Phi^{-1}(0.975)\,\sigma\sqrt{T}},\ S_0\, e^{gT + \Phi^{-1}(0.975)\,\sigma\sqrt{T}}\Big],$$

where $\Phi^{-1}(0.975) = 1.959964\ldots$ is the 97.5% quantile of the standard normal law, written 1.96 in examples.

Proof. We use the Gaussian shift for the moments and the monotonicity of the exponential for the quantiles. Throughout, $S_T = S_0\, e^{gT}\, e^{\sigma\sqrt{T}\,Z}$ with Z standard normal (Theorem 19.40).

Part (i). Lemma 19.38(i) with $\theta = \sigma\sqrt{T}$ gives $\mathbb{E}\, e^{\sigma\sqrt{T}\,Z} = e^{\sigma^2 T/2}$, so

$$\mathbb{E}[S_T] = S_0\, e^{gT + \sigma^2 T/2} = S_0\, e^{\mu T},$$

because $g + \sigma^2/2 = \mu$. With $\theta = 2\sigma\sqrt{T}$ it gives $\mathbb{E}\, e^{2\sigma\sqrt{T}\,Z} = e^{2\sigma^2 T}$, so

$$\mathbb{E}[S_T^2] = S_0^2\, e^{2gT + 2\sigma^2 T} = S_0^2\, e^{2\mu T + \sigma^2 T},$$

because $2g + 2\sigma^2 = 2\mu + \sigma^2$. Then $\operatorname{Var}(S_T) = \mathbb{E}[S_T^2] - (\mathbb{E}[S_T])^2 = S_0^2\, e^{2\mu T}\big(e^{\sigma^2 T} - 1\big)$.

Parts (ii) and (iv), quantiles. For a price $x > 0$, since the logarithm is increasing,

$$P(S_T \le x) = P\big(\ln S_0 + gT + \sigma\sqrt{T}\,Z \le \ln x\big) = \Phi\Big(\frac{\ln x - \ln S_0 - gT}{\sigma\sqrt{T}}\Big).$$

The function Φ is continuous and strictly increasing, with limits 0 and 1 at $-\infty$ and $+\infty$, so it takes every value in $(0,1)$ exactly once; write $\Phi^{-1}(0.975)$ for the point at which it takes the value 0.975. Then $\Phi(0) = \frac{1}{2}$ by symmetry, $\Phi\big(\Phi^{-1}(0.975)\big) = 0.975$ and $\Phi\big(-\Phi^{-1}(0.975)\big) = 1 - 0.975 = 0.025$, exactly. So $P(S_T \le x) = \frac{1}{2}$ exactly when $\ln x = \ln S_0 + gT$: the median is $S_0\, e^{gT}$. In the same way $P(S_T \le x)$ equals 2.5% and 97.5% at $\ln x = \ln S_0 + gT \mp \Phi^{-1}(0.975)\,\sigma\sqrt{T}$, which gives (iv). Numerically $\Phi^{-1}(0.975) = 1.959964$, and $\Phi(1.96) = 0.9750021$, so the rounded value 1.96 widens the interval only by the factors $e^{\mp 3.6\times 10^{-5}\,\sigma\sqrt{T}}$ at its lower and upper end. Quantiles pass through the increasing exponential: the quantiles of S_T are the exponentials of those of $\ln S_T$.

Part (ii), mode. Differentiating the distribution function in x gives the density of S_T,

$$f_{S_T}(x) = \frac{1}{x\,\sigma\sqrt{T}}\,\varphi\Big(\frac{\ln x - \ln S_0 - gT}{\sigma\sqrt{T}}\Big), \qquad x > 0.$$

Its logarithm is $-\ln x - (\ln x - \ln S_0 - gT)^2/(2\sigma^2 T)$ plus a constant, with derivative

$$-\frac{1}{x} - \frac{\ln x - \ln S_0 - gT}{\sigma^2 T\, x} = -\frac{1}{x}\Big(1 + \frac{\ln x - \ln S_0 - gT}{\sigma^2 T}\Big),$$

which is positive for $\ln x < \ln S_0 + gT - \sigma^2 T$ and negative for $\ln x > \ln S_0 + gT - \sigma^2 T$. The density therefore increases up to $x = S_0\, e^{gT - \sigma^2 T} = S_0\, e^{(\mu - 3\sigma^2/2)T}$ and decreases afterwards: this point is its unique maximum, the mode. The three exponents satisfy $(\mu - 3\sigma^2/2)\,T < (\mu - \sigma^2/2)\,T < \mu T$, because $\sigma^2 T > 0$, which orders the mode, the median and the mean.

Part (iii). We standardise, as in the proof of Theorem 10.2. Since the logarithm is increasing and $\sigma\sqrt{T} > 0$,

$$\{S_T > K\} = \{\ln S_0 + gT + \sigma\sqrt{T}\,Z > \ln K\} = \{Z > -d\},$$

and $P(Z > -d) = 1 - \Phi(-d) = \Phi(d)$ by the symmetry of the normal law. Likewise $\{S_T < K\} = \{Z < -d\}$, of probability $\Phi(-d)$. The two probabilities add up to one, because $P(S_T = K) = P(Z = -d) = 0$. □

The risk-neutral probability $\Phi(d_2)$ of §20 is (iii) with μ replaced by $r-q$ (by r for an asset without income). An option price depends on that probability, not on the real-world $\Phi(d)$. Taking $K = \mathbb{E}[S_T] = S_0\, e^{\mu T}$ in (iii) gives $d = (-\mu T + \mu T - \sigma^2 T/2)/(\sigma\sqrt{T}) = -\sigma\sqrt{T}/2$, so

$$P\big(S_T > \mathbb{E}[S_T]\big) = \Phi\big(-\tfrac{1}{2}\,\sigma\sqrt{T}\big) < \tfrac{1}{2}\,:$$

a lognormal price ends below its mean more often than above it.

Example 19.43 (Mean, median, mode and tail of a six-month price)**.** Continue Example 19.41: $S_0 = 60$, $\mu = 10\%$, $\sigma = 32\%$, $T = 0.5$, with $gT = 0.0244$, $\sigma^2 T = 0.0512$ and $\sigma\sqrt{T} = 0.2263$. By Proposition 19.42:

- the mean is $60\, e^{0.05} = 63.08$ and the median is $60\, e^{0.0244} = 61.48$;
- the mode is $60\, e^{0.0244-0.0512} = 60\, e^{-0.0268} = 58.41$;
- the standard deviation is $63.08 \times \sqrt{e^{0.0512}-1} = 63.08 \times 0.2292 = 14.46$;
- with the quantile rounded to 1.96, $1.96 \times 0.2263 = 0.4435$, and the central 95% interval is $\big[60\, e^{0.0244-0.4435},\ 60\, e^{0.0244+0.4435}\big] = [39.46,\ 95.80]$. It is asymmetric about the median: 22.02 below it and 34.32 above it;
- for $K = 66$,

$$d = \frac{\ln(60/66) + 0.0244}{0.22627} = \frac{-0.09531 + 0.0244}{0.22627} = \frac{-0.07091}{0.22627} = -0.3134,$$

  so $P(S_T > 66) = \Phi(-0.3134) = 37.7\%$;
- the probability of ending above the mean is $P\big(S_T > \mathbb{E}[S_T]\big) = \Phi(-0.2263/2) = \Phi(-0.113) = 45.5\%$.

(Problem P19.8 works this computation at different numbers.)

Figure 36 shows a lognormal density at a longer horizon, where the three marks are far enough apart to be seen: for $\sigma\sqrt{T} = 0.6$ the density is visibly skewed to the right.

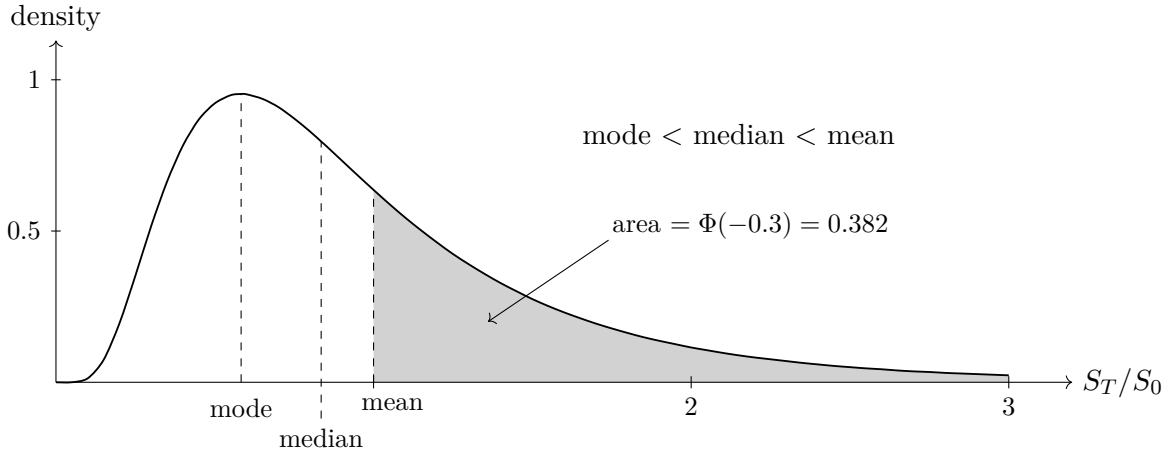


Figure 36: The lognormal density of S_T/S_0 for $\mu = 0$, $\sigma = 30\%$ and $T = 4$, so that $\ln(S_T/S_0) \sim \mathcal{N}(-0.18,\ 0.6^2)$. The dashed lines mark the mode $e^{-0.54} = 0.583$, where the density peaks at 0.953, the median $e^{-0.18} = 0.835$ and the mean 1 (Proposition 19.42(ii)). The shaded area to the right of the mean, continued beyond the plot, is $P(S_T > \mathbb{E}[S_T]) = \Phi(-0.3) = 0.382$, less than one half. In Example 19.43 the three values are $58.41 < 61.48 < 63.08$.

Interview trap. "The expected price is $S_0\, e^{\mu T}$, so half the paths end above it." Half the paths end above the *median* $S_0\, e^{gT}$, which is lower. The probability of ending above the mean is $\Phi(-\sigma\sqrt{T}/2) < \frac{1}{2}$: 45.5% in Example 19.43, 38.2% in Figure 36, and 21.2% for $\sigma = 80\%$ over four years, where $\sigma\sqrt{T}/2 = 0.8$. The mean is pulled up by the right tail.

Interview trap. "A 95% range is the mean plus or minus 1.96 standard deviations." In Example 19.43 that gives $63.076 \pm 1.96 \times 14.457 = [34.74,\ 91.41]$ against the true $[39.46,\ 95.80]$. At long horizons the symmetric range goes negative: with the same parameters over five years the mean is 98.92 and the standard deviation 80.89, so its lower end is -59.6. Build the interval in log space and map it through the exponential.

Interview trap. Quoting $\Phi(d_2)$ for "the probability that the call finishes in the money" without naming the measure. Under the real-world measure the probability is the $\Phi(d)$ of Proposition 19.42(iii), which depends on μ. The number $\Phi(d_2)$ is the risk-neutral probability of §20, and only it enters the price.

19.7 Expected Return versus Growth Rate

"A fund reports an 11 percent average annual return with 18 percent volatility: what did its investors earn?", "why do leveraged ETFs decay?" and "what leverage maximises growth?" are one question. The answer is the gap between the expected rate of return μ and the growth rate $g = \mu - \sigma^2/2$ at which a typical path compounds. This subsection identifies g with the expected log growth of the Kelly criterion (§3.1, Definition 3.1). It then proves the volatility drag, for sample returns and for the model, and verifies the growth of a continuously rebalanced leveraged position. It builds on the lognormal law (Theorem 19.40, Proposition 19.42), on the Gaussian shift (Lemma 19.38) and on Corollary 19.31.

Setup. The price S is the geometric Brownian motion of Definition 19.27, with drift μ, volatility $\sigma > 0$ and $S_0 > 0$; $T > 0$ is a horizon in years, and $g = \mu - \sigma^2/2$. The *realised growth rate* over $[0, T]$ is

$$\hat{g}_T = \frac{1}{T} \ln \frac{S_T}{S_0},$$

the continuously compounded return per year that the path delivered. It wears a hat because it is the natural estimator of g. For statements about samples, $R_1, \dots, R_n$ are the log-returns of n periods, as in §19.1; here they may be any real numbers. The simple return of period i is $e^{R_i} - 1$, the average log-return is $\bar{R} = \frac{1}{n}\sum_{i=1}^{n} R_i$ (a bar marks an average), and

$$\mathrm{AM} = \frac{1}{n}\sum_{i=1}^{n} (e^{R_i} - 1), \qquad \mathrm{GM} = e^{\bar{R}} - 1$$

are the *arithmetic mean return* and the *geometric mean return*. The second is the compound rate per period, because $(1 + \mathrm{GM})^n = e^{R_1 + \cdots + R_n} = S_n/S_0$. A period has length Δt. The constant L is a leverage factor (the letter of §15); h_s is a holding, the number of units of S held at time s; and $V_t = V(S_t, t)$ is the value of a leveraged position, with an explicit function V as in §19.5. The value is not called W, which is the Brownian motion here, and the bankroll W_n of Definition 3.1 does not appear. The letter r is the risk-free rate.

Proposition 19.44 (Distribution of the realised growth rate)**.** *Let S be the geometric Brownian motion of Definition 19.27 and let $T > 0$.*

(i) The realised growth rate is normal: $\hat{g}_T \sim \mathcal{N}(g,\ \sigma^2/T)$.

(ii) Its mean is $\mathbb{E}[\hat{g}_T] = g = \mu - \sigma^2/2$, whereas $\frac{1}{T}\ln \mathbb{E}[S_T/S_0] = \mu$. The gap is $\mu - g = \sigma^2/2 > 0$.

(iii) Its mean-square distance from g is $\mathbb{E}\big[(\hat{g}_T - g)^2\big] = \sigma^2/T$, which tends to 0 as $T \to \infty$: the realised growth rate converges to g in mean square.

Proof. We compute directly from the explicit formula. By (19.10), $\ln(S_T/S_0) = gT + \sigma W_T$, so

$$\hat{g}_T = g + \frac{\sigma}{T} W_T.$$

By Definition 19.5(1) and (2), $W_T \sim \mathcal{N}(0,T)$. By (19.3) in the Setup of §19.2, the constant multiple $(\sigma/T)\,W_T$ is normal, with the variance multiplied by $(\sigma/T)^2$, and adding the constant g shifts the mean. So $\hat{g}_T$ is normal with mean g and variance $(\sigma/T)^2\,T = \sigma^2/T$. This is (i), and (iii) is the statement that the variance of $\hat{g}_T$, which is its mean-square distance from its mean g, equals σ^2/T. For (ii), $\mathbb{E}[\hat{g}_T] = g$ by (i), and taking logarithms in Proposition 19.42(i) gives $\frac{1}{T}\ln \mathbb{E}[S_T/S_0] = \frac{1}{T}\ln e^{\mu T} = \mu$. The difference is $\mu - (\mu - \sigma^2/2) = \sigma^2/2$, which is positive. □

The number g is the expected log growth of Definition 3.1 per unit of time: $\mathbb{E}\ln(S_T/S_0) = gT$. So "expected return" means μ for the price and g for its logarithm. The gap $\sigma^2/2$ between the logarithm of a mean and the mean of a logarithm is often called the Jensen gap; here it is computed exactly, and no general inequality is needed. Part (iii) is the analogue, for a geometric Brownian motion, of Lemma 3.2: over a long horizon the path compounds at g, not at μ.

Example 19.45 (Four years of realised growth)**.** Let $\mu = 15\%$, $\sigma = 20\%$ and $T = 4$. Then $g = 0.15 - 0.02 = 13\%$ and $\sigma/\sqrt{T} = 0.20/2 = 10\%$, so by Proposition 19.44 $\hat{g}_4 \sim \mathcal{N}\big(13\%,\ (10\%)^2\big)$. A 95% band for the realised growth rate is $13\% \pm 1.96 \times 10\% = 13\% \pm 19.6\% = [-6.6\%,\ 32.6\%]$, and

$$P(\hat{g}_4 < 0) = \Phi(-0.13/0.10) = \Phi(-1.3) = 9.7\%,$$

the loss probability of Theorem 10.2, read on log-returns. To pin g down to $\pm 2\%$ at the 95% level, the horizon must satisfy $1.96\,\sigma/\sqrt{T} = 0.02$, that is, $T = (1.96 \times 0.20/0.02)^2 = 384$ years of data. This is the humility about estimated drifts of Fact 4.16.

Interview trap. "A positive expected return makes a gain likely over a long horizon." With $\mu = 10\%$ and $\sigma = 50\%$, $g = 0.10 - 0.125 = -2.5\%$. By Proposition 19.42(iii) with $K = S_0$ and $T = 30$, $d = gT/(\sigma\sqrt{T}) = -0.274$ and $P(S_{30} < S_0) = \Phi(0.274) = 60.8\%$, although $\mathbb{E}[S_{30}] = S_0\,e^{3} = 20.1\,S_0$; the median is $S_0\,e^{-0.75} = 0.47\,S_0$. As $T \to \infty$ with $g < 0$, $d \to -\infty$ for every level $K > 0$, so $S_T \to 0$ in probability while $\mathbb{E}[S_T] \to \infty$. The typical (median) investor compounds at g, not at μ.

Proposition 19.46 (Volatility drag)**.** *Let* AM *and* GM *be the arithmetic and the geometric mean return of the Setup.*

(i) (Sample) For any real log-returns $R_1, \dots, R_n$, GM $\le$ AM*, with equality if and only if all the R_i are equal.*

(ii) (Model) Let S be the geometric Brownian motion of Definition 19.27, and let a period run from t to $t + \Delta t$. The population arithmetic mean return of the period is

$$\mathbb{E}\Big[\frac{S_{t+\Delta t}}{S_t}\Big] - 1 = e^{\mu\,\Delta t} - 1,$$

and the population geometric mean return is

$$\exp\Big(\mathbb{E}\,\ln\frac{S_{t+\Delta t}}{S_t}\Big) - 1 = e^{g\,\Delta t} - 1.$$

The logarithm of the ratio of the two gross returns is exactly

$$\ln\frac{e^{\mu\,\Delta t}}{e^{g\,\Delta t}} = \tfrac{1}{2}\,\sigma^2\,\Delta t,$$

that is, $\sigma^2/2$ per unit of time, and the difference of the two returns is

$$(e^{\mu\,\Delta t} - 1) - (e^{g\,\Delta t} - 1) = e^{\mu\,\Delta t}(1 - e^{-\sigma^2\Delta t/2}) = \tfrac{1}{2}\,\sigma^2\,\Delta t + O((\Delta t)^2) \qquad \text{as } \Delta t \to 0.$$

Proof. We use the tangent-line inequality of the exponential for (i) and a direct computation for (ii).

Part (i). For every real x, $e^x \geq 1 + x$, with equality only at $x = 0$: the function $e^x - 1 - x$ has the derivative $e^x - 1$, which is negative for $x < 0$ and positive for $x > 0$, so its minimum, 0, is attained only at $x = 0$. Apply the inequality to $x = R_i - \bar{R}$, multiply by $e^{\bar{R}} > 0$ and average over i:

$$\frac{1}{n}\sum_{i=1}^{n} e^{R_i} = e^{\bar{R}}\,\frac{1}{n}\sum_{i=1}^{n} e^{R_i - \bar{R}} \;\geq\; e^{\bar{R}}\Big(1 + \frac{1}{n}\sum_{i=1}^{n}(R_i - \bar{R})\Big) = e^{\bar{R}},$$

because the deviations $R_i - \bar{R}$ add up to zero. The left side is $1 + \text{AM}$ and the right side is $1 + \text{GM}$. Equality holds if and only if it holds in each of the n inequalities, that is, if and only if $R_i - \bar{R} = 0$ for every i.

Part (ii). By (19.10),

$$\frac{S_{t+\Delta t}}{S_t} = \exp\left(g\,\Delta t + \sigma\,(W_{t+\Delta t} - W_t)\right),$$

and $W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t)$ has the law of $\sqrt{\Delta t}\,Z$ (Definition 19.5(2)). Lemma 19.38(i) with $\theta = \sigma\sqrt{\Delta t}$ gives

$$\mathbb{E}\Big[\frac{S_{t+\Delta t}}{S_t}\Big] = e^{g\,\Delta t}\,e^{\sigma^2 \Delta t/2} = e^{\mu\,\Delta t},$$

while $\mathbb{E}\,\ln(S_{t+\Delta t}/S_t) = g\,\Delta t$, because the increment has mean zero. So the population arithmetic and geometric mean returns are $e^{\mu\,\Delta t} - 1$ and $e^{g\,\Delta t} - 1$; the logarithm of the ratio of the gross returns is $(\mu - g)\,\Delta t = \sigma^2\,\Delta t/2$, and the difference of the returns is

$$e^{\mu\,\Delta t} - e^{g\,\Delta t} = e^{\mu\,\Delta t}\big(1 - e^{-\sigma^2 \Delta t/2}\big).$$

By Proposition 7.34, $1 - e^{-\sigma^2 \Delta t/2} = \sigma^2\,\Delta t/2 + O((\Delta t)^2)$ and $e^{\mu\,\Delta t} = 1 + O(\Delta t)$, and the product is $\sigma^2\,\Delta t/2 + O((\Delta t)^2)$. □

A more general second-order rule is quoted here and not proved. For any per-period return with a small mean and a small standard deviation s, the compound rate is about the mean minus $s^2/2$. No numbered result below uses it; Examples 19.47 and 19.48 and the trap after them use it only as an approximation, and the two examples compare it with an exact value. It is the variance-drag computation of the application after Example 7.40 in §7.5, and item (1) of Remark 11.59. Lemma 3.2 shows, for a repeated bet, that the mean log-return is the almost-sure long-run compound rate.

Example 19.47 (A coin that returns plus or minus 50 percent)**.** A fair coin returns +50% or −50% each round, so wealth is multiplied by 1.5 or by 0.5. The arithmetic mean return is $\frac{1}{2}(0.5) + \frac{1}{2}(-0.5) = 0$. Over two rounds with one gain and one loss wealth is multiplied by $1.5 \times 0.5 = 0.75$, a geometric mean return of $\sqrt{0.75} - 1 = -13.4\%$ per round.

Figure 37 shows four rounds. The mean wealth is still 1, but 11 of the 16 paths, 68.75%, end below 1, and the median is 0.5625. After 100 rounds the median path has 50 gains and 50 losses and is worth $0.75^{50} = 5.7 \times 10^{-7}$, while the mean is still 1.

The expected log-return per round is $\frac{1}{2}\ln 1.5 + \frac{1}{2}\ln 0.5 = \frac{1}{2}\ln 0.75 = -14.4\%$, and the compound rate per round, the geometric mean return, is $e^{-0.1438} - 1 = -13.4\%$. The second-order rule, the mean minus half the variance, gives $0 - 0.5^2/2 = -12.5\%$, a smaller drag than either exact figure. The expansion is local, as the last trap of §7.5 warns, and a 50% move is not small.

Example 19.48 (What the fund's investors earned)**.** A fund's annual returns average 11% with a volatility of 18%. By the second-order rule quoted after Proposition 19.46, its investors compound at about

$$11\% - \tfrac{1}{2} \times 0.18^2 = 11\% - 1.62\% = 9.38\%,$$

about 9.4% a year. The exact figure depends on the law of the returns. If the annual gross return is lognormal, model it as S_1/S_0 for a geometric Brownian motion. Proposition 19.42(i) gives its mean

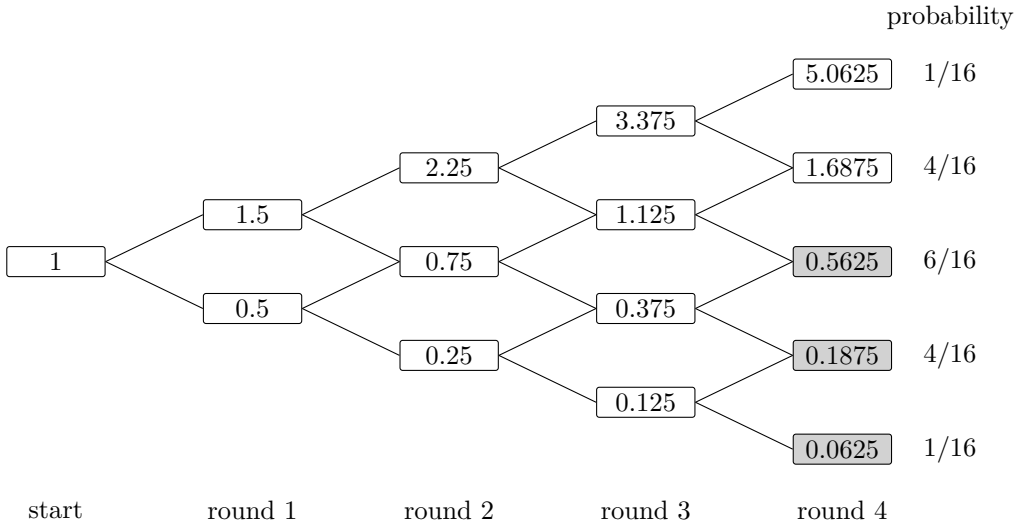


Figure 37: Four rounds of the coin of Example 19.47: at each round wealth is multiplied by 1.5 or by 0.5, each with probability $\frac{1}{2}$. The mean stays 1 at every round, while the median ends at 0.5625 and 11/16 of the paths (the three shaded nodes) end below the start.

$e^{\mu} = 1.11$ and its squared ratio of standard deviation to mean, $e^{\sigma^2} - 1 = (0.18/1.11)^2 = 0.02630$, so $\sigma^2 = \ln 1.02630 = 0.02596$. The compound rate is then

$$e^{g} - 1 = e^{\mu - \sigma^2/2} - 1 = 1.11\, e^{-0.01298} - 1 = 9.57\%,$$

about 9.6%. This is a simple annual rate, the geometric mean return of Proposition 19.46. As a continuously compounded rate it is $g = \ln 1.11 - 0.01298 = 0.10436 - 0.01298 = 9.14\%$, the expected log growth that the expansion of §7.5 approximates. Either way, by the rule or by the lognormal model in simple annual terms, the investors earned about a point and a half less than the reported average.

Interview trap. Quoting the arithmetic average of annual returns as "the return". A fund that makes +100% and then −50% averages +25% a year and has earned nothing, since $2 \times 0.5 = 1$. Investors compound at the geometric mean, which lies below the arithmetic mean (Proposition 19.46(i)), by about half the variance of the returns when these are small.

Corollary 19.49 (Growth of a continuously rebalanced leveraged position)**.** *Let S be the geometric Brownian motion of Definition 19.27, let $V_0 > 0$ and $T > 0$, and let $L \neq 0$ be a constant. Put*

$$V(x,t) = V_0 \Big(\frac{x}{S_0}\Big)^{L} e^{-L(L-1)\,\sigma^2 t/2}, \qquad V_t = V(S_t, t), \quad 0 \le t \le T.$$

(i) (Explicit path) The value is

$$V_t = V_0 \exp\Big(\big(L\mu - \tfrac{1}{2}\,L^2\sigma^2\big)\,t + L\sigma\, W_t\Big), \qquad \text{so that} \qquad \frac{V_T}{V_0} = \Big(\frac{S_T}{S_0}\Big)^{L} e^{-L(L-1)\,\sigma^2 T/2}:$$

the process V is a geometric Brownian motion with drift $L\mu$ and volatility $|L|\sigma$, driven by W when $L > 0$ and by $-W$ when $L < 0$ (Proposition 19.8(iii)).

(ii) (Trading gains) For every $t \in [0,T]$, with probability one,

$$V_t = V_0 + \int_0^t h_s\, \mu S_s\, ds + \int_0^t h_s\, \sigma S_s\, dW_s, \qquad h_s = \frac{L\,V_s}{S_s};$$

in shorthand, $dV = h\, dS$. These are the gains of holding $h = LV/S$ units of S, that is, a fraction L of the value in the asset, with the remainder $(1-L)\,V$ in cash at a zero rate and with no costs.

(iii) (Growth rate) The growth rate of V, $\frac{1}{T}\,\mathbb{E}\ln(V_T/V_0)$, is

$$g_L = L\mu - \tfrac{1}{2}\,L^2\sigma^2 = L\,(\mu - \tfrac{1}{2}\,L\sigma^2).$$

As a function of a real number L, g_L vanishes at $L=0$ and at $L=2\mu/\sigma^2$, and it is negative exactly when L lies outside the closed interval between these two numbers: for $\mu>0$, when $L>2\mu/\sigma^2$ or $L<0$; for $\mu<0$, when $L>0$ or $L<2\mu/\sigma^2$; for $\mu=0$, for every $L\neq 0$. It is largest at $L^*=\mu/\sigma^2$, where $g_{L^*}=\mu^2/(2\sigma^2)$; for $\mu=0$ this is $L=0$, the all-cash position $V_t=V_0$, with growth 0.

(iv) (Financing) If the cash $(1-L)\,V$ earns, and a borrowing ($L>1$) costs, the constant rate r, let instead

$$V(x,t) = V_0\left(\frac{x}{S_0}\right)^{L} e^{((1-L)\,r - L(L-1)\,\sigma^2/2)\,t}, \qquad V_t = V(S_t,t).$$

Then, with the same holding $h_s = L\,V_s/S_s$,

$$V_t = V_0 + \int_0^t \Big(h_s\,\mu S_s + r\,(V_s - h_s S_s)\Big)\,ds + \int_0^t h_s\,\sigma S_s\,dW_s,$$

in shorthand $dV = h\,dS + r\,(V-hS)\,dt$. The process V is a geometric Brownian motion with drift $r+L(\mu-r)$ and volatility $|L|\sigma$; its growth rate, again written g_L, is $g_L = r+L(\mu-r)-\frac{1}{2}\,L^2\sigma^2$, which is largest at $L^*=(\mu-r)/\sigma^2$, where it equals $r+(\mu-r)^2/(2\sigma^2)$.

Sketch of the proof. We verify each claim with Itô's lemma (Corollary 19.31(iii)), whose growth hypothesis holds because V is a power of x times a bounded function of t. Part (i) raises (19.10) to the power L. For (ii), $\partial_t V = -\frac{1}{2}L(L-1)\sigma^2\,V$, $x\,\partial_x V = LV$ and $x^2\,\partial_{xx}V = L(L-1)\,V$, so the time derivative cancels the convexity term in the drift,

$$\partial_t V + \mu x\,\partial_x V + \tfrac{1}{2}\,\sigma^2 x^2\,\partial_{xx}V = L\mu\,V,$$

and the diffusion coefficient is $\sigma x\,\partial_x V = L\sigma\,V$; these are $h\,\mu S$ and $h\,\sigma S$ for $h = LV/S$. For (iii), the exponent of (i) has mean $g_L\,T$, and g_L is a quadratic in L with negative leading coefficient, zeros 0 and $2\mu/\sigma^2$ and derivative $\mu - L\sigma^2$. Part (iv) adds $(1-L)\,r\,V = r\,(V-hS)$ to $\partial_t V$. The complete proof is in Appendix E.9. □

The corollary is a verification: the process V is given by a formula and Itô's lemma shows that it earns the stated gains, so no uniqueness theorem for the equation $dV/V = L\,dS/S$ is used. The $L^*=(\mu-r)/\sigma^2$ of the financing clause (iv) is the "edge over variance" of Remark 11.59(3), and the continuous-time form of the Kelly criterion (Theorem 3.3).

Example 19.50 (A three-times leveraged product)**.** An index has $\mu = 8\%$ and $\sigma = 25\%$, so $\sigma^2 = 6.25\%$, and it grows at $g = 8\% - 3.125\% = 4.875\%$, about 4.9% a year. By Corollary 19.49:

- a 3× product grows at $g_3 = 3\times 8\% - \frac{1}{2}\times 9\times 6.25\% = 24\% - 28.125\% = -4.125\%$, about -4.1% a year: it shrinks while the index grows;
- its expected one-year value is up $e^{0.24} - 1 = 27.1\%$ (Proposition 19.42(i) for the drift 24%), while its median is down $1 - e^{-0.04125} = 4.0\%$;
- if the index ends the year flat, $S_T = S_0$, the 3× product is down $1 - e^{-3\times 2\times 0.0625/2} = 1 - e^{-0.1875} = 17.1\%$, and the −1× product, for which $L(L-1)=2$, is down $1-e^{-0.0625} = 6.1\%$;
- the growth-optimal leverage is $L^* = 0.08/0.0625 = 1.28$, with $g_{L^*} = 0.08^2/(2\times 0.0625) = 5.12\%$, and the break-even leverage is $2\mu/\sigma^2 = 2.56$.

With financing at $r = 3\%$ (Corollary 19.49(iv)), $L^* = (0.08-0.03)/0.0625 = 0.8$, with growth $0.03 + 0.05^2/(2\times 0.0625) = 5.0\%$, and the 3× product grows at $0.03 + 3\times 0.05 - 0.28125 = -10.1\%$: the $2r = 6\%$ a year paid on the two borrowed units adds to the decay.

(Problem P19.10 works this computation at different numbers.)

Interview trap. "A 3× product returns three times the index over a year." Only over one rebalancing interval. Under geometric Brownian motion, $V_T/V_0 = (S_T/S_0)^L\, e^{-L(L-1)\sigma^2 T/2}$ (Corollary 19.49(i)), and the shortfall is the factor $e^{-L(L-1)\sigma^2 T/2}$: a 17.1% loss on a flat year in Example 19.50. The numbers L and $1-L$ have the same product $L(L-1)$, so the 2× and the −1× products decay alike.

Trading application. Leveraged-ETF decay, rebalancing and Kelly sizing are one formula, $g_L = L\mu - \frac{1}{2}L^2\sigma^2$. Beyond $L^* = \mu/\sigma^2$, extra leverage raises the expected return $L\mu$ and lowers the long-run growth rate; beyond $2\mu/\sigma^2$, for $\mu > 0$, the position has a positive expected return and a negative growth rate.

19.8 Correlated Brownian Motions

Every two-asset question runs through a pair of correlated Brownian motions: "generate two correlated normals", "what is the volatility of EURGBP, given those of EURUSD and GBPUSD and their correlation?", "is a basket of two lognormal stocks lognormal?". This subsection defines such a pair and builds it from two independent motions, by the two-dimensional Cholesky factor that §23 uses to simulate correlated paths (Procedure 23.36). It proves that $dW_1\, dW_2 = \rho\, dt$ in the mean-square sense; that a mixed partial derivative does not depend on the order of differentiation (Lemma 19.54); the two-dimensional Itô lemma with its product rule (Theorem 19.55); and the volatility of a ratio and of a product of two geometric Brownian motions (Proposition 19.57). Margrabe's formula for an exchange option (Theorem 23.32) uses the theorem for its replication and the proposition for its volatility, and the two-asset market of §20 (Definitions 20.1 and 20.6) is driven by the pair of motions built here. The subsection builds on the Itô integral and on Itô's lemma (Theorems 19.23 and 19.30), on the Gaussian lemma (Lemma 19.38) and on the mean value theorem of §7.5.

Setup. From here on a first subscript $k = 1, 2$ names a motion or an asset, and the time is the second subscript: $W_{k,t}$ is the value of the motion W_k at time t, and $S_{k,t}$ is the price of asset k. The W_1 and W_4 of §19.2 were values of one motion at the times 1 and 4; that usage stops here, until §19.9, which has a single motion and declares it again. Differentials are written dW_1 and dW_2, and $dW_{k,s}$ inside an integral. The numbers μ_k, $\sigma_k > 0$ and $g_k = \mu_k - \sigma_k^2/2$ are the drift, the volatility and the growth rate of asset k, which is why §19.1 wrote σ_{per} for the volatility of one period. The process $W_\perp$ is a standard Brownian motion independent of W_1 in the sense of the Setup of §19.3: every finite vector of values of one motion is independent of every finite vector of values of the other. Fact 19.7 supplies such a pair. The value of $W_\perp$ at time t is $W_{\perp,t}$, and $\tilde{W}$ is an auxiliary standard Brownian motion. The number $\rho \in [-1, 1]$ is a correlation, and $T > 0$ is a horizon. For a grid $0 \le t_0 < t_1 < \cdots < t_n$, and in particular for a partition $0 = t_0 < t_1 < \cdots < t_n = t$ of $[0, t]$ with steps $\Delta t_i = t_i - t_{i-1}$ and mesh $\Delta t_{\max}$,

$$\Delta W_{k,i} = W_{k,t_i} - W_{k,t_{i-1}}, \qquad \Delta W_{\perp,i} = W_{\perp,t_i} - W_{\perp,t_{i-1}},$$

where n counts the cells. A pair of random variables is *jointly normal* in the sense of the Setup of §19.6: both are affine functions of one pair of independent standard normal variables. The Cholesky factor of the correlation matrix of two variables is described in words, as the lower-triangular matrix with rows $(1, 0)$ and $(\rho, \sqrt{1-\rho^2})$, because the letter L is the leverage factor of §19.7.

The function $f(t, x_1, x_2)$ is a function of time and of two Brownian values, with partial derivatives $\partial_t f$, $\partial_1 f$, $\partial_2 f$, $\partial_{11} f$, $\partial_{12} f$ and $\partial_{22} f$. Here ∂_1 and ∂_2 differentiate in x_1 and in x_2, and the order of the mixed derivative is not shown, by Lemma 19.54. In that lemma $\psi(x_1, x_2)$ is a function of two real variables (the Part's letter for a test function), $\partial_2\partial_1\psi$ means: differentiate in x_1, then in x_2, and a starred letter (x_1^*, x_1^{**}, u^*) is a point supplied by the mean value theorem, which is Corollary 7.32 at order zero. Increments of the two variables are written Δx_1 and Δx_2, and χ is a local function of one real variable u, as in §19.5.

In the process form of the Itô lemma, $X_{k,t} = X_k(t, W_{k,t})$, $k = 1, 2$, are Itô processes generated by explicit functions $X_k(t, y)$; process and function share the letter, as in $V_t = V(S_t, t)$. Their coefficients are $\alpha_{k,t}$ and $\beta_{k,t}$, and each X_k maps $[0, T] \times \mathbb{R}$ into an open interval $\mathcal{O}_k$. The outer function is $V(x_1, x_2, t)$ on $\mathcal{O}_1 \times \mathcal{O}_2 \times [0, T]$, and

$$\tilde{V}(y_1, y_2, t) = V\big(X_1(t, y_1),\ X_2(t, y_2),\ t\big)$$

is its composite: the function V written in the Brownian variables, state first as for V. Its partial derivatives are written $\partial_t \tilde{V}$, $\partial_{y_1} \tilde{V}$, $\partial_{y_1 y_2} \tilde{V}$ and so on. In one trap, w_1 and w_2 are the weights of two assets in a basket.

Definition 19.51 (Correlated Brownian motions)**.** Two standard Brownian motions W_1 and W_2 on one probability space *have correlation* ρ, where $-1 \le \rho \le 1$, if the following two conditions hold.

(1) For all $0 \le s < t$ the pair of increments $(W_{1,t} - W_{1,s},\ W_{2,t} - W_{2,s})$ is jointly normal, with covariance $\rho\,(t - s)$.
(2) For every grid $0 \le t_0 < t_1 < \cdots < t_n$ the n pairs of increments $(\Delta W_{1,i}, \Delta W_{2,i})$, $i = 1, \dots, n$, are mutually independent random vectors.

The definition is abbreviated $dW_1\, dW_2 = \rho\, dt$, the last row of Table 4. Proposition 19.52(ii) makes the shorthand precise.

Each increment in (1) has variance $t - s$, so the correlation coefficient of the two increments is $\rho\,(t - s)/(t - s) = \rho$ over every interval, whatever its length. Condition (2) is the two-dimensional form of the independent increments of Definition 19.5(3).

Proposition 19.52 (Cholesky construction and cross-variation of correlated Brownian motions)**.** *Let W_1 and $W_\perp$ be independent standard Brownian motions (Fact 19.7), let $-1 \le \rho \le 1$, and put*

$$W_{2,t} = \rho\, W_{1,t} + \sqrt{1 - \rho^2}\, W_{\perp,t}, \qquad t \ge 0.$$

(i) (Construction) The process W_2 is a standard Brownian motion, the pair (W_1, W_2) has correlation ρ in the sense of Definition 19.51, and $\operatorname{Cov}(W_{1,t}, W_{2,t}) = \rho\, t$ for every $t \ge 0$.

(ii) (Cross-variation) Let W_1 and W_2 be any two standard Brownian motions with correlation ρ in the sense of Definition 19.51, $-1 \le \rho \le 1$, for instance the pair of (i). For a partition $0 = t_0 < t_1 < \cdots < t_n = t$ of $[0, t]$ with mesh $\Delta t_{\max}$,

$$\mathbb{E}\Big[\Big(\sum_{i=1}^{n} \Delta W_{1,i}\, \Delta W_{2,i} - \rho\, t\Big)^2\Big] = (1 + \rho^2) \sum_{i=1}^{n} (\Delta t_i)^2 \le (1 + \rho^2)\, t\, \Delta t_{\max},$$

so $\sum_i \Delta W_{1,i}\, \Delta W_{2,i} \to \rho\, t$ in mean square as $\Delta t_{\max} \to 0$.

(iii) (Converse) Let $|\rho| < 1$, and let W_1 and W_2 be any two standard Brownian motions with correlation ρ in the sense of Definition 19.51. Then

$$W_\perp = \frac{W_2 - \rho\, W_1}{\sqrt{1 - \rho^2}}$$

is a standard Brownian motion independent of W_1, and $W_2 = \rho\, W_1 + \sqrt{1 - \rho^2}\, W_\perp$: every correlated pair with $|\rho| < 1$ arises from the construction.

Sketch of the proof. We check Definition 19.5 for (i), rotate a jointly normal pair for (iii), and expand a square for (ii). In (i), the increment of W_2 over $[s, t]$ is ρ times that of W_1 plus $\sqrt{1 - \rho^2}$ times the independent increment of $W_\perp$, with the weights of a unit vector, hence $\mathcal{N}(0, t - s)$ by Lemma 19.38(ii), with covariance $\rho\,(t - s)$; disjoint cells use disjoint groups of increments. In

(iii), which needs $|\rho| < 1$, Definition 19.51(1) makes the increments of W_1 and $W_\perp$ over a cell a rotation of two independent standard normal variables, up to a sign and the scale $\sqrt{\Delta t_i}$, so they are independent with the law $\mathcal{N}(0, \Delta t_i)$, and condition (2) makes the cells independent. For (ii), (iii) reduces every pair with $|\rho| < 1$ to the construction, where

$$\sum_{i=1}^{n} \Delta W_{1,i}\, \Delta W_{2,i} - \rho\, t = \rho \sum_{i=1}^{n} \left((\Delta W_{1,i})^2 - \Delta t_i \right) + \sqrt{1-\rho^2} \sum_{i=1}^{n} \Delta W_{1,i}\, \Delta W_{\perp,i};$$

the two sums are uncorrelated, with second moments $2\sum_i (\Delta t_i)^2$ (Proposition 19.10(i)) and $\sum_i (\Delta t_i)^2$, so the total is $(1+\rho^2)\sum_i (\Delta t_i)^2$. For $\rho = \pm 1$, $\Delta W_{2,i} = \rho\, \Delta W_{1,i}$ with probability one, and Proposition 19.10(i) gives the claim. The complete proof is in Appendix E.10. □

The numerator $W_{2,t} - \rho\, W_{1,t}$ of (iii) is the decorrelation residual of Proposition 10.15, with $W_{1,t}$ in the role of the black-box output; the regression coefficient $\operatorname{Cov}(W_{1,t}, W_{2,t}) / \operatorname{Var}(W_{1,t})$ of that proposition is $\rho\, t / t = \rho$. In matrix form, the map $(Z_1, Z_2) \mapsto \left(Z_1,\ \rho Z_1 + \sqrt{1-\rho^2}\, Z_2 \right)$ multiplies a pair of independent standard normal variables by the lower-triangular matrix with rows $(1, 0)$ and $(\rho, \sqrt{1-\rho^2})$. This matrix times its transpose has the rows $(1 \cdot 1 + 0 \cdot 0,\ 1 \cdot \rho + 0 \cdot \sqrt{1-\rho^2}) = (1, \rho)$ and $(\rho \cdot 1 + \sqrt{1-\rho^2} \cdot 0,\ \rho^2 + (1-\rho^2)) = (\rho, 1)$, which are the rows of the correlation matrix. So the matrix is the two-dimensional Cholesky factor of the correlation matrix, and Procedure 23.36 simulates correlated paths with it. For $\rho = 0$, part (iii) says that $W_2 = W_\perp$ is independent of W_1: two Brownian motions with correlation zero in the sense of Definition 19.51 are independent.

Example 19.53 (One correlated draw)**.** Take $\rho = 0.6$, so that $\sqrt{1-\rho^2} = \sqrt{1-0.36} = 0.8$. Independent standard normal draws $Z_1 = 0.8$ and $Z_\perp = -1.2$ give

$$Z_2 = \rho\, Z_1 + \sqrt{1-\rho^2}\, Z_\perp = 0.6 \times 0.8 + 0.8 \times (-1.2) = 0.48 - 0.96 = -0.48,$$

so $(0.8, -0.48)$ is one draw of a pair of standard normal variables with correlation 0.6. As a check, $\operatorname{Var}(Z_2) = 0.36 + 0.64 = 1$ and $\operatorname{Cov}(Z_1, Z_2) = 0.6$. For the motions, Propositions 10.10 and 19.52(i) give

$$\operatorname{Var}(W_{1,t} + W_{2,t}) = t + t + 2\rho\, t = (2 + 2\rho)\, t = 3.2\, t,$$

between the $2t$ of two independent motions and the $4t$ of one motion counted twice.

(Problem P19.11 works this computation at different numbers.)

Interview trap. Generating correlated normals as $Z_2 = \rho\, Z_1 + (1-\rho)\, Z_\perp$. The variance of this Z_2 is $\rho^2 + (1-\rho)^2$, which is not 1: at $\rho = 0.6$ it is 0.52, a standard deviation of 0.72, and the correlation with Z_1 is $0.6/0.72 = 0.83$ instead of 0.6. Both the scale and the correlation come out wrong. The second weight is $\sqrt{1-\rho^2}$, which makes the squares of the two weights add up to one.

The two-dimensional Itô lemma expands a function of two Brownian values to second order, and its cross term is written with one mixed derivative. That is legitimate because of the following classical theorem of calculus, which Sections 1 to 15 do not state.

Lemma 19.54 (Equality of mixed partial derivatives (Schwarz, 1873))**.** *Let $\psi(x_1, x_2)$ be a function on an open set of the plane such that ψ, $\partial_1\psi$, $\partial_2\psi$, $\partial_2\partial_1\psi$ and $\partial_1\partial_2\psi$ exist and are continuous there. Then*

$$\partial_2\partial_1\psi = \partial_1\partial_2\psi$$

at every point of the set. For a function of three or more variables the same holds for each pair of variables, the others being held fixed, because a slice of an open set is an open set of the plane. In particular a function $f(t, x_1, x_2)$ whose partial derivatives of order at most two in (x_1, x_2) exist and are continuous on $[0, T] \times \mathbb{R}^2$ has $\partial_2\partial_1 f = \partial_1\partial_2 f$, written $\partial_{12} f$, for each fixed t, the slice being the whole plane. A price function $V(S, t, \sigma)$ whose partial derivatives up to order two exist and are continuous on an open set of points (S, t, σ) has $\partial_\sigma\partial_S V = \partial_S\partial_\sigma V$, written $\partial_{S\sigma} V$, and likewise

$\partial_{St}V$ and $\partial_{t\sigma}V$ (§21). If such a function is defined only at the points of an open set with $t \geq 0$, its derivatives at $t = 0$ being one-sided, and if the two mixed derivatives of a pair that contains t are continuous up to $t = 0$, then they also agree at $t = 0$.

Sketch of the proof. We apply the mean value theorem twice to a double difference, once in each order. For nonzero increments Δx_1 and Δx_2 small enough to stay in the open set, the double difference

$$\psi(x_1 + \Delta x_1,\ x_2 + \Delta x_2) - \psi(x_1 + \Delta x_1,\ x_2) - \psi(x_1,\ x_2 + \Delta x_2) + \psi(x_1, x_2)$$

equals $\Delta x_1\, \Delta x_2\, \partial_2\partial_1\psi(x_1^*, x_2^*)$, by that theorem in x_1 and then, for $\partial_1\psi$, in x_2; in the other order it equals $\Delta x_1\, \Delta x_2\, \partial_1\partial_2\psi(x_1^{**}, x_2^{**})$, the starred points lying in the rectangle spanned by the increments. Dividing by $\Delta x_1\, \Delta x_2$ and letting the increments tend to zero gives the equality at (x_1, x_2) by the continuity of the two mixed derivatives. The other clauses fix the remaining variables and, at $t = 0$, take a limit from $t > 0$. The complete proof is in Appendix E.11. □

The continuity of the mixed derivatives is used in the final limit (Step 4 of the complete proof, in Appendix E.11) and cannot be dropped. The function $\psi(x_1, x_2) = x_1 x_2\,(x_1^2 - x_2^2)/(x_1^2 + x_2^2)$, with $\psi(0, 0) = 0$, vanishes on both axes, and for $x_2 \neq 0$ and $x_1 \neq 0$ respectively

$$\partial_1\psi(0, x_2) = \lim_{x_1 \to 0} \frac{\psi(x_1, x_2)}{x_1} = \frac{x_2\,(-x_2^2)}{x_2^2} = -x_2, \qquad \partial_2\psi(x_1, 0) = \lim_{x_2 \to 0} \frac{\psi(x_1, x_2)}{x_2} = \frac{x_1\, x_1^2}{x_1^2} = x_1,$$

while both derivatives vanish at the origin. So $\partial_2\partial_1\psi(0, 0) = -1$ and $\partial_1\partial_2\psi(0, 0) = 1$: the two orders disagree at a point where the mixed derivatives are not continuous. The functions of two or more variables that this Part differentiates twice have continuous second partial derivatives in the variables in which they are differentiated, by hypothesis or by proof: the function f of Theorem 19.55 by hypothesis, and the Black–Scholes–Merton price functions before expiry, which are infinitely differentiable in (S, t) (Theorem 20.7(i)). Where a price is also differentiated in σ (§21), the lemma needs its second partial derivatives to be continuous in (S, t, σ) jointly. For the call and the put this holds by their explicit formula (Theorem 20.22): for $S > 0$, $t < T$ and $\sigma > 0$ the numbers d_1 and d_2 are infinitely differentiable functions of (S, t, σ), and so are $\Phi(d_1)$, $\Phi(d_2)$ and the prices built from them by sums and products; for any other price function differentiated in σ, this continuity is a hypothesis of the result that uses it, as in Proposition 21.14. The order of differentiation is therefore never shown: $\partial_{12}f$ and $\partial_{S\sigma}V$ denote either order. A numerical check of the lemma, the vanna of an at-the-money call computed in both orders, is Example 21.13; §21 cites the lemma in Proposition 21.14, and the proof of Proposition 21.27, in Appendix G.8, uses it again.

Theorem 19.55 (Two-dimensional Itô lemma (Itô, 1951))**.** *Let $T > 0$, let W_1 and $W_\perp$ be independent standard Brownian motions, and let $f(t, x_1, x_2)$ be continuous on $[0, T] \times \mathbb{R}^2$, with partial derivatives $\partial_t f$, $\partial_1 f$, $\partial_2 f$, $\partial_{11} f$, $\partial_2\partial_1 f$, $\partial_1\partial_2 f$ and $\partial_{22} f$ that exist and are continuous on $[0, T] \times \mathbb{R}^2$ (one-sided in t at 0 and at T). Let f and each of these derivatives have exponential growth (Definition 19.19(i)). By Lemma 19.54 the two mixed derivatives agree, and they are written $\partial_{12} f$. The Itô integrals below are those of Theorem 19.23 with $W = W_1$ and with the second motion $W_\perp$; an integral over an interval of length zero is zero.*

(i) (Independent motions) For every $t \in [0, T]$, with probability one,

$$\begin{aligned} f(t, W_{1,t}, W_{\perp,t}) = f(0,0,0) &+ \int_0^t \left(\partial_t f + \tfrac{1}{2}\,\partial_{11} f + \tfrac{1}{2}\,\partial_{22} f\right) ds \\ &+ \int_0^t \partial_1 f\, dW_{1,s} + \int_0^t \partial_2 f\, dW_{\perp,s}, \end{aligned} \tag{19.16}$$

every integrand being evaluated at $(s, W_{1,s}, W_{\perp,s})$. The first integral is a Riemann integral computed path by path, and the integrands of the other two are in the class $\mathcal{H}$ (Definition 19.22, two motions).

(ii) (Correlated motions) Let $-1 \le \rho \le 1$ *and* $W_2 = \rho\, W_1 + \sqrt{1-\rho^2}\, W_\perp$ *(Proposition 19.52). For an integrand* ξ *of the class* $\mathcal{H}$ *with two motions, or of its span, put*

$$\int_0^t \xi_s\, dW_{2,s} = \rho \int_0^t \xi_s\, dW_{1,s} + \sqrt{1-\rho^2} \int_0^t \xi_s\, dW_{\perp,s}.$$

This integral is the mean-square limit of the left-endpoint sums $\sum_i \xi_{t_{i-1}}\, \Delta W_{2,i}$*, and it has mean zero. If* $\xi_s = \xi(s, W_{2,s})$ *for a continuous function* $\xi(s,x)$ *of exponential growth, it agrees with probability one with the integral that Theorem 19.23 builds on the Brownian motion* W_2 *alone. With this integral, for every* $t \in [0,T]$*, with probability one,*

$$\begin{aligned} f(t, W_{1,t}, W_{2,t}) = f(0,0,0) &+ \int_0^t \Big(\partial_t f + \tfrac{1}{2}\, \partial_{11} f + \rho\, \partial_{12} f + \tfrac{1}{2}\, \partial_{22} f \Big)\, ds \\ &+ \int_0^t \partial_1 f\, dW_{1,s} + \int_0^t \partial_2 f\, dW_{2,s}, \end{aligned} \tag{19.17}$$

every integrand being evaluated at $(s, W_{1,s}, W_{2,s})$*.*

(iii) (Process form) Let W_2 *be as in (ii). For* $k = 1, 2$ *let* $X_k(t,y)$ *be a function of the class* $C_E^{1,2}([0,T] \times \mathbb{R})$ *with all its values in an open interval* $\mathcal{O}_k$*, and put* $X_{k,t} = X_k(t, W_{k,t})$*. By Theorem 19.30(i), applied to the Brownian motion* W_k *(for* $k = 2$*, a standard Brownian motion by Proposition 19.52(i)),* $dX_k = \alpha_{k,t}\, dt + \beta_{k,t}\, dW_k$ *with*

$$\alpha_{k,t} = \Big(\partial_t X_k + \tfrac{1}{2}\, \partial_{yy} X_k \Big)(t, W_{k,t}), \qquad \beta_{k,t} = \partial_y X_k(t, W_{k,t}).$$

Let $V(x_1, x_2, t)$ *be continuous on* $\mathcal{O}_1 \times \mathcal{O}_2 \times [0,T]$*, with partial derivatives* $\partial_t V$*,* $\partial_1 V$*,* $\partial_2 V$*,* $\partial_{11} V$*,* $\partial_2 \partial_1 V$*,* $\partial_1 \partial_2 V$ *and* $\partial_{22} V$ *that exist and are continuous there (one-sided in* t *at* 0 *and at* T*); the mixed derivatives agree by Lemma 19.54 and are written* $\partial_{12} V$*. Suppose that the composite* $\tilde{V}(y_1, y_2, t) = V(X_1(t, y_1), X_2(t, y_2), t)$ *and its partial derivatives* $\partial_t \tilde{V}$*,* $\partial_{y_1} \tilde{V}$*,* $\partial_{y_2} \tilde{V}$*,* $\partial_{y_1 y_1} \tilde{V}$*,* $\partial_{y_1 y_2} \tilde{V}$ *and* $\partial_{y_2 y_2} \tilde{V}$*, which exist and are continuous by the proof of the theorem (Appendix E.12), have exponential growth. Then, for every* $t \in [0,T]$*, with probability one,*

$$\begin{aligned} V(X_{1,t}, X_{2,t}, t) = V(X_{1,0}, X_{2,0}, 0) &+ \int_0^t \Big(\partial_t V + \alpha_{1,s}\, \partial_1 V + \alpha_{2,s}\, \partial_2 V + \tfrac{1}{2}\, \beta_{1,s}^2\, \partial_{11} V \\ &\qquad + \rho\, \beta_{1,s} \beta_{2,s}\, \partial_{12} V + \tfrac{1}{2}\, \beta_{2,s}^2\, \partial_{22} V \Big)\, ds \\ &+ \int_0^t \beta_{1,s}\, \partial_1 V\, dW_{1,s} + \int_0^t \beta_{2,s}\, \partial_2 V\, dW_{2,s}, \end{aligned} \tag{19.18}$$

every derivative of V *being evaluated at* $(X_{1,s}, X_{2,s}, s)$*. In shorthand,*

$$\begin{aligned} dV = \Big(\partial_t V + \alpha_{1,t}\, \partial_1 V + \alpha_{2,t}\, \partial_2 V + \tfrac{1}{2}\, \beta_{1,t}^2\, \partial_{11} V + \rho\, \beta_{1,t} \beta_{2,t}\, \partial_{12} V + \tfrac{1}{2}\, \beta_{2,t}^2\, \partial_{22} V \Big)\, dt \\ + \beta_{1,t}\, \partial_1 V\, dW_1 + \beta_{2,t}\, \partial_2 V\, dW_2. \end{aligned}$$

Two cases are used most.

(a) (Two geometric Brownian motions) If $X_{k,t} = S_{k,t} = S_{k,0} \exp(g_k t + \sigma_k W_{k,t})$*,* $k = 1, 2$*, the hypothesis on the composite holds whenever* V *and the six derivatives* $\partial_t V$*,* $\partial_1 V$*,* $\partial_2 V$*,* $\partial_{11} V$*,* $\partial_{12} V$ *and* $\partial_{22} V$ *are continuous and of moderate growth on* $(0, \infty)^2 \times [0,T]$ *(Definition 19.19(ii)); then* $\alpha_{k,t} = \mu_k S_{k,t}$ *and* $\beta_{k,t} = \sigma_k S_{k,t}$*.*

(b) (Product rule) For $V(x_1, x_2, t) = x_1 x_2$ *the hypothesis on the composite always holds, and (19.18) reads*

$$\begin{aligned} X_{1,t} X_{2,t} = X_{1,0} X_{2,0} &+ \int_0^t \Big(X_{1,s}\, \alpha_{2,s} + X_{2,s}\, \alpha_{1,s} + \rho\, \beta_{1,s} \beta_{2,s} \Big)\, ds \\ &+ \int_0^t X_{2,s}\, \beta_{1,s}\, dW_{1,s} + \int_0^t X_{1,s}\, \beta_{2,s}\, dW_{2,s}, \end{aligned}$$

in shorthand $d(X_1 X_2) = X_1\, dX_2 + X_2\, dX_1 + \rho\, \beta_{1,t} \beta_{2,t}\, dt$*.*

(iv) (Every correlated pair) Let W_1 and W_2 be any two standard Brownian motions with correlation ρ in the sense of Definition 19.51. If $|\rho| < 1$, then $W_\perp = (W_2 - \rho\, W_1)/\sqrt{1-\rho^2}$ is a standard Brownian motion independent of W_1 with $W_2 = \rho\, W_1 + \sqrt{1-\rho^2}\, W_\perp$, and (ii) and (iii) apply to the pair. If $\rho = \pm 1$, then with probability one $W_{2,s} = \rho\, W_{1,s}$ for every $s \geq 0$; for $\xi_s = \xi(s, W_{1,s}, W_{2,s})$ with ξ continuous and of exponential growth, the left-endpoint sums $\sum_i \xi_{t_{i-1}}\, \Delta W_{2,i}$ converge in mean square to $\int_0^t \xi_s\, dW_{2,s} = \rho \int_0^t \xi_s\, dW_{1,s}$; and with this integral (19.17) and (iii) hold.

Sketch of the proof. We repeat the proof of Theorem 19.30 with a Taylor expansion in two space variables, whose only new term in a cell is the cross term $\partial_{12} f\, \Delta W_{1,i}\, \Delta W_{\perp,i}$, with one mixed derivative by Lemma 19.54 and $\partial_{12} f$ taken at the start of the cell. Its sum tends to zero in mean square, which is $dW_1\, dW_\perp = 0$: the two increments of a cell have mean zero and are independent of each other and of the past, so the second moment of the sum is $\sum_i \mathbb{E}[(\partial_{12} f)^2]\, (\Delta t_i)^2$. The squared terms and the gradient terms are treated as in one dimension, the latter with Theorem 19.23(vi). For (ii), apply (i) to $f(t, y_1, \rho\, y_1 + \sqrt{1-\rho^2}\, y_2)$, with y_1 the value of W_1 and y_2 that of $W_\perp$; half the sum of its second derivatives in y_1 and y_2 is

$$\tfrac{1}{2}(\partial_{11} f + 2\rho\, \partial_{12} f + \rho^2\, \partial_{22} f) + \tfrac{1}{2}\,(1-\rho^2)\, \partial_{22} f = \tfrac{1}{2}\, \partial_{11} f + \rho\, \partial_{12} f + \tfrac{1}{2}\, \partial_{22} f,$$

and the stochastic integrals recombine into the integral against W_2. Part (iii) is (ii) for the composite $\tilde{V}$, as in Corollary 19.31, the product rule being the case $V = x_1 x_2$; and (iv) takes $W_\perp$ from Proposition 19.52(iii) when $|\rho| < 1$, and uses $W_2 = \rho\, W_1$, with probability one, when $\rho = \pm 1$. The complete proof is in Appendix E.12. □

In the product rule the term $\rho\, \beta_{1,t} \beta_{2,t}\, dt$ is the two-dimensional form of the convexity term of Theorem 19.30: it is $dX_1\, dX_2$ computed with the last row of Table 4. The ordinary product rule of calculus is the case in which one of the two factors has no dW term. The same formulas for processes driven by general correlated noise, with coefficients determined by the past, are part of Fact 19.29(iii).

Example 19.56 (The product rule at work)**.** *(a) Two Brownian motions.* Take $X_k(t, y) = y$, so that $X_{k,t} = W_{k,t}$, $\mathcal{O}_k = \mathbb{R}$, $\alpha_{k,t} = 0$ and $\beta_{k,t} = 1$. The product rule of Theorem 19.55(iii)(b) reads

$$W_{1,t}\, W_{2,t} = \rho\, t + \int_0^t W_{2,s}\, dW_{1,s} + \int_0^t W_{1,s}\, dW_{2,s},$$

in shorthand $d(W_1 W_2) = W_1\, dW_2 + W_2\, dW_1 + \rho\, dt$. Both integrals have mean zero: the first by Theorem 19.23(ii), its integrand $W_{2,s} = \rho\, W_{1,s} + \sqrt{1-\rho^2}\, W_{\perp,s}$ being in the class $\mathcal{H}$ with two motions (Definition 19.22); the second by Theorem 19.55(ii). So $\mathbb{E}[W_{1,t}\, W_{2,t}] = \rho\, t$, which recovers Proposition 19.52(i): the covariance is $0.6 \times 2 = 1.2$ at $\rho = 0.6$ and $t = 2$.

(b) Two geometric Brownian motions. Let S_1 and S_2 be geometric Brownian motions driven by two standard Brownian motions W_1 and W_2 with correlation $\rho = 0.5$, a pair to which the theorem applies by its part (iv), with $\mu_1 = 10\%$, $\mu_2 = 5\%$, $\sigma_1 = 30\%$, $\sigma_2 = 20\%$ and $S_{1,0} = S_{2,0} = 100$. With $\alpha_{k,t} = \mu_k S_{k,t}$ and $\beta_{k,t} = \sigma_k S_{k,t}$ (Theorem 19.55(iii)(a)) the product rule (iii)(b) gives

$$d(S_1 S_2) = S_1 S_2 \Big((\mu_1 + \mu_2 + \rho\, \sigma_1 \sigma_2)\, dt + \sigma_1\, dW_1 + \sigma_2\, dW_2 \Big),$$

with the drift $\mu_1 + \mu_2 + \rho\, \sigma_1 \sigma_2 = 0.10 + 0.05 + 0.03 = 0.18$. Let $\chi(t) = \mathbb{E}[S_{1,t}\, S_{2,t}]$. Take expectations in the integral form, as in the proof of Corollary 19.31: the two stochastic integrals have mean zero, and Fact 18.34(iii) exchanges the expectation with the ds-integral, because the path $s \mapsto S_{1,s}\, S_{2,s}$ is continuous and its mean is bounded on $[0, t]$ (Lemma 19.20(iii)). Hence

$$\chi(t) = \chi(0) + 0.18 \int_0^t \chi(s)\, ds.$$

The function χ is continuous, by the fourth-moment argument of Example 19.32, so it is differentiable with $\chi' = 0.18\,\chi$, and as in that example $\chi(t) = \chi(0)\,e^{0.18\,t}$. At $t = 1$,

$$\begin{aligned}\mathbb{E}[S_{1,1}\,S_{2,1}] &= 10^4\,e^{0.18} = 11{,}972,\\ \operatorname{Cov}(S_{1,1}, S_{2,1}) &= 10^4\,e^{0.18} - 10^4\,e^{0.10}\,e^{0.05} = 10^4\,e^{0.15}(e^{0.03} - 1) = 353.8,\end{aligned}$$

the means $100\,e^{\mu_k}$ being those of Proposition 19.42(i). The drift 18% of the product and its volatility $\sqrt{0.09 + 0.04 + 0.06} = \sqrt{0.19} = 43.6\%$ are those of Proposition 19.57(ii) below. The same mean follows from the Gaussian lemma. The variable $\sigma_1 W_{1,1} + \sigma_2 W_{2,1}$ is normal with mean zero and variance $0.09 + 0.04 + 0.06 = 0.19$ (Lemma 19.38(ii) and Proposition 10.10), so by Lemma 19.38(i), $\mathbb{E}\,e^{\sigma_1 W_{1,1} + \sigma_2 W_{2,1}} = e^{0.19/2} = e^{0.095}$ and

$$\mathbb{E}[S_{1,1}\,S_{2,1}] = 10^4\,e^{g_1 + g_2}\,e^{0.095} = 10^4\,e^{0.055 + 0.03 + 0.095} = 10^4\,e^{0.18}.$$

Proposition 19.57 (Volatility of a ratio and of a product of geometric Brownian motions)**.** *Let W_1 and W_2 be standard Brownian motions with correlation ρ (Definition 19.51), and for $k = 1, 2$ let*

$$S_{k,t} = S_{k,0} \exp\left(g_k t + \sigma_k W_{k,t}\right), \qquad g_k = \mu_k - \tfrac{1}{2}\,\sigma_k^2,$$

be the geometric Brownian motion with drift μ_k, volatility $\sigma_k > 0$ and starting price $S_{k,0} > 0$ driven by W_k (Definition 19.27). Put

$$\sigma = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2}.$$

(i) (Ratio) If $\sigma > 0$, that is, unless $\rho = 1$ and $\sigma_1 = \sigma_2$, the ratio S_1/S_2 is a geometric Brownian motion with volatility σ and drift $\mu_1 - \mu_2 + \sigma_2^2 - \rho\,\sigma_1\sigma_2$, started at $S_{1,0}/S_{2,0}$. If $\sigma = 0$, then for each $t \geq 0$, with probability one,

$$\frac{S_{1,t}}{S_{2,t}} = \frac{S_{1,0}}{S_{2,0}}\,e^{(\mu_1 - \mu_2)\,t}.$$

(ii) (Product) If $\sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\,\sigma_1\sigma_2} > 0$, that is, unless $\rho = -1$ and $\sigma_1 = \sigma_2$, the product $S_1 S_2$ is a geometric Brownian motion with that volatility and with drift $\mu_1 + \mu_2 + \rho\,\sigma_1\sigma_2$, started at $S_{1,0}S_{2,0}$. Otherwise, for each $t \geq 0$, with probability one,

$$S_{1,t}\,S_{2,t} = S_{1,0}\,S_{2,0}\,e^{(\mu_1 + \mu_2 - \sigma_1^2)\,t}.$$

Sketch of the proof. We identify the process that drives the logarithm of the ratio. By (19.10),

$$\ln\frac{S_{1,t}}{S_{2,t}} = \ln\frac{S_{1,0}}{S_{2,0}} + (g_1 - g_2)\,t + \sigma_1 W_{1,t} - \sigma_2 W_{2,t}.$$

The continuous process $\sigma_1 W_1 - \sigma_2 W_2$ starts at zero, and by Definition 19.51 and Lemma 19.38(ii) its increments are independent over a grid and normal, with mean zero and variance $(\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2)(t - s) = \sigma^2(t - s)$ over $[s, t]$. If $\sigma > 0$, dividing it by σ gives a standard Brownian motion $\tilde{W}$, and the ratio is (19.10) driven by $\tilde{W}$, with volatility σ and drift $g_1 - g_2 + \frac{1}{2}\sigma^2$, the stated one. If $\sigma = 0$, the identity $\sigma^2 = (\sigma_1 - \sigma_2)^2 + 2\,(1 - \rho)\,\sigma_1\sigma_2$ forces $\sigma_1 = \sigma_2$ and $\rho = 1$, so $g_1 - g_2 = \mu_1 - \mu_2$, and the driving term, of variance zero, vanishes with probability one. The product is the same with a plus sign and $1 + \rho$ in place of $1 - \rho$. The complete proof is in Appendix E.13. □

The product clause is cross-checked by the product rule of Theorem 19.55(iii)(b) in Example 19.56(b). The direct route to the ratio, through Theorem 19.55(iii) with $V(x_1, x_2, t) = x_1/x_2$, is Problem P19.11. Theorem 23.32 cites the proposition for the volatility of an exchange option.

Example 19.58 (The volatility of a ratio at four correlations)**.** Take $\sigma_1 = 30\%$ and $\sigma_2 = 20\%$. At $\rho = 0.5$, Proposition 19.57(i) gives

$$\sigma^2 = 0.09 + 0.04 - 2 \times 0.5 \times 0.30 \times 0.20 = 0.13 - 0.06 = 0.07, \qquad \sigma = \sqrt{0.07} = 26.5\%.$$

At $\rho = 0.9$, $\sigma^2 = 0.13 - 0.108 = 0.022$ and $\sigma = 14.8\%$; at $\rho = 0$, $\sigma = \sqrt{0.13} = 36.1\%$; at $\rho = -0.5$, $\sigma = \sqrt{0.19} = 43.6\%$. At $\rho = 0.5$ the drift of the ratio exceeds $\mu_1 - \mu_2$ by $\sigma_2^2 - \rho\,\sigma_1\sigma_2 = 0.04 - 0.03 = 1\%$ a year. This σ is the volatility that enters Margrabe's formula for the option to exchange asset 2 for asset 1 (Theorem 23.32).

(Problem P19.11 works this computation at different numbers.)

Example 19.59 (A foreign stock in domestic currency)**.** A foreign stock has a volatility of 30% in its own currency, the exchange rate (domestic units per foreign unit) has a volatility of 10%, and the correlation between the two is -0.3. The value of the stock in domestic currency is the product of the two prices, so by Proposition 19.57(ii) its volatility is

$$\sqrt{0.09 + 0.01 + 2 \times (-0.3) \times 0.30 \times 0.10} = \sqrt{0.09 + 0.01 - 0.018} = \sqrt{0.082} = 28.6\%,$$

below the 30% of the stock alone: with a negative correlation the currency is a partial hedge.

Example 19.60 (A basket of two geometric Brownian motions that is not one)**.** Is the basket $S_1 + S_2$ of two geometric Brownian motions a geometric Brownian motion? A geometric Brownian motion is lognormal at each $t > 0$ (Theorem 19.40), and every lognormal variable X satisfies

$$\left(\mathbb{E}[X^2]\right)^3 = \left(\mathbb{E}[X]\right)^3 \mathbb{E}[X^3].$$

Indeed, write $\ln X = \mathbb{E}[\ln X] + \sqrt{\operatorname{Var}(\ln X)}\, Z$ with Z standard normal. Lemma 19.38(i) gives

$$\mathbb{E}[X^k] = \exp\left(k\,\mathbb{E}[\ln X] + \tfrac{1}{2}\,k^2 \operatorname{Var}(\ln X)\right) \qquad \text{for } k = 1, 2, 3.$$

On the left of the identity the exponent is $3\left(2\,\mathbb{E}[\ln X] + 2\operatorname{Var}(\ln X)\right)$, and on the right it is $3\left(\mathbb{E}[\ln X] + \tfrac{1}{2}\operatorname{Var}(\ln X)\right) + 3\,\mathbb{E}[\ln X] + \tfrac{9}{2}\operatorname{Var}(\ln X)$. Both equal $6\,\mathbb{E}[\ln X] + 6\operatorname{Var}(\ln X)$.

Let S_1 and S_2 be driven by W_1 and W_2 with correlation ρ. For integers $a, b \geq 0$ the variable $a\,\sigma_1 W_{1,t} + b\,\sigma_2 W_{2,t}$ is normal with mean zero and variance $(a^2\sigma_1^2 + 2ab\,\rho\,\sigma_1\sigma_2 + b^2\sigma_2^2)\,t$ (Definition 19.51, Lemma 19.38(ii) and Proposition 10.10), so by Lemma 19.38(i)

$$\mathbb{E}\big[S_{1,t}^a\, S_{2,t}^b\big] = S_{1,0}^a\, S_{2,0}^b \exp\Big((a\,g_1 + b\,g_2)\,t + \tfrac{1}{2}(a^2\sigma_1^2 + 2ab\,\rho\,\sigma_1\sigma_2 + b^2\sigma_2^2)\,t\Big),$$

and the binomial expansion of $(S_{1,t} + S_{2,t})^k$ gives the moments of the basket. Take the parameters of Example 19.56(b) at $t = 1$: $S_{1,0} = S_{2,0} = 100$, $\mu_1 = 10\%$, $\mu_2 = 5\%$, $\sigma_1 = 30\%$, $\sigma_2 = 20\%$ and $\rho = 0.5$. The mixed moments $\mathbb{E}\big[S_{1,1}^a\, S_{2,1}^b\big]$ are 110.52 and 105.13 for $(a, b) = (1, 0)$ and $(0, 1)$; 13,364.27, 11,972.17 and 11,502.74 for $(2, 0)$, $(1, 1)$ and $(0, 2)$, the middle one being the 11,972 of Example 19.56(b); and 1,768,267, 1,491,825, 1,349,859 and 1,309,964 for $(3, 0)$, $(2, 1)$, $(1, 2)$ and $(0, 3)$. With the binomial coefficients $1, 2, 1$ and $1, 3, 3, 1$, the basket $X = S_{1,1} + S_{2,1}$ has

$$\mathbb{E}[X] = 215.644, \qquad \mathbb{E}[X^2] = 48{,}811.4, \qquad \mathbb{E}[X^3] = 1.16033 \times 10^7,$$

so that

$$\frac{\left(\mathbb{E}[X^2]\right)^3}{\left(\mathbb{E}[X]\right)^3 \mathbb{E}[X^3]} = 0.99947,$$

not 1; arithmetic with 40 digits returns the same ratio, $0.9994671\ldots$, so the gap is not a rounding error. The variable X is not lognormal, and therefore the basket is not a geometric Brownian motion, whatever motion might drive it.

Interview trap. "The volatility of the ratio is $|\sigma_1 - \sigma_2|$", which is true only at $\rho = 1$, or "it is $\sqrt{\sigma_1^2 + \sigma_2^2}$", which is true only at $\rho = 0$. The correlation term carries the answer: in Example 19.58 the two shortcuts give 10% and 36.1%, against 26.5% at $\rho = 0.5$.

Interview trap. "A basket $S_1 + S_2$ of two geometric Brownian motions is a geometric Brownian motion." Theorem 19.55(iii) with $V = x_1 + x_2$, which applies to any pair of correlated motions by part (iv), gives $d(S_1 + S_2) = (\mu_1 S_1 + \mu_2 S_2)\,dt + \sigma_1 S_1\,dW_1 + \sigma_2 S_2\,dW_2$, and by Proposition 19.52(i) the variance rate of the two noise terms, divided by the square of the basket, is

$$w_1^2\,\sigma_1^2 + w_2^2\,\sigma_2^2 + 2\rho\,w_1 w_2\,\sigma_1\sigma_2, \qquad w_k = \frac{S_k}{S_1 + S_2}.$$

This percentage variance rate moves as the weights move (unless $\rho = 1$ and $\sigma_1 = \sigma_2$, when it is the constant $\sigma_1^2\,(w_1 + w_2)^2 = \sigma_1^2$), which explains why a basket does not keep a constant volatility. The *proof* that a basket is in general not a geometric Brownian motion is the moment test of Example 19.60: a geometric Brownian motion is lognormal at each t, every lognormal X has $(\mathbb{E}[X^2])^3 = (\mathbb{E}[X])^3\,\mathbb{E}[X^3]$, and the basket of that example gives 0.99947 in place of 1. Products and ratios of geometric Brownian motions are geometric Brownian motions (Proposition 19.57, when the combined volatility is positive); sums in general are not (Example 19.60; in the degenerate case $\rho = 1$, $\sigma_1 = \sigma_2$, $\mu_1 = \mu_2$, where $g_1 = g_2$, Theorem 19.55(iv) gives $W_2 = W_1$ with probability one and the sum is $(S_{1,0} + S_{2,0})\,e^{g_1 t + \sigma_1 W_{1,t}}$, a geometric Brownian motion), so the lognormal formulas of §20 do not apply to basket options and arithmetic Asian options, which §23 prices numerically.

Interview trap. Correlating price levels instead of returns. The rule $dW_1\,dW_2 = \rho\,dt$ is a statement about increments. Two independent geometric Brownian motions with the same positive drift often show a large sample correlation between their price levels over a long window, because both trend upward: the spurious correlation of trending series. The correlation ρ is estimated from returns, never from levels.

Trading application. Currency crosses. The rate EURJPY is the product of EURUSD and USDJPY, and EURGBP is the ratio of EURUSD to GBPUSD. By Proposition 19.57, the volatility σ of a cross and the volatilities σ_1 and σ_2 of its two legs determine the correlation of the legs: for a ratio, $\rho = (\sigma_1^2 + \sigma_2^2 - \sigma^2)/(2\sigma_1\sigma_2)$. Desks read the implied correlation of two currency pairs off the three quoted implied volatilities of the triangle in this way; Problem P19.11 works one case.

19.9 Hitting Times and the Reflection Principle

"What is the probability that a Brownian motion touches 1 before time 1?", "what is the expected maximum of W over $[0, T]$?" and "how much more likely is a stop-loss to be hit during the year than to be breached at the end of it?" are staple interview questions, and the same computation prices the one-touch and barrier options of §23.2. The reflection principle answers them: a path that reaches a level ℓ and ends at x is matched with the path reflected after the hitting time, which ends at $2\ell - x$ (Figure 38). This subsection proves the principle, with drift, by the method of images, an analytic form of the same symmetry. A Gaussian smoothing solves a backward heat equation (Lemma 19.61); its mirror image about the level solves the same equation and cancels it on the level; and the stopped clause of Itô's lemma (Theorem 19.30(iii)) carries the difference up to the hitting time. No strong Markov property and no change of measure is needed. As a by-product the subsection proves every claim of the reflection Fact of §10.2 (Fact 10.6), which was stated there without proof. The subsection builds on Theorem 19.30(iii), Theorem 19.23(ii), Lemmas 19.12 and 19.20, Proposition 19.8(iii) and Lemma 18.35. Lemma 19.61 is also the smoothness half of risk-neutral valuation in §20 (Theorem 20.7).

Setup. The motion W is a standard Brownian motion and $T > 0$ is a horizon. The number ν is a real constant, the drift of the motion below; it is the Part's letter for a constant log-drift, not

the number of degrees of freedom of Student's law. The number $\sigma > 0$ is a volatility, and $\ell > 0$ is a level; the letter a, which §10.2 uses for a loss level, appears only in part (iii) of Theorem 19.63 and in its check in Example 19.65. In part (iii), in its proof and in that check, as in the Fact of §10.2, μ and σ are the drift and the volatility of an arithmetic P&L $\mu t + \sigma W_t$, in money units per year and per square root of a year (the roles of μ_{N} and σ_{N} in the arithmetic model of §23); they are not the expected return and the log-price volatility of an asset. The process $X_t = \nu t + W_t$ is a Brownian motion with drift ν, and

$$X_T^{\max} = \max_{0 \le t \le T} X_t$$

is its running maximum on $[0, T]$, which exists because the path is continuous on a closed interval; the notation follows the pattern of $S_T^{\max}$, and $W_T^{\max}$ is the case $\nu = 0$. Only one motion appears in this subsection, so, as in §19.2, a single subscript on W or X is again a time: W_1 and X_1 are the values of W and X at the time 1, and $W_1^{\max}$ and $X_1^{\max}$ are their running maxima on $[0, 1]$. The *hitting time* of the level is

$$t_\ell = \inf\{t \in [0, T] : X_t \ge \ell\},$$

equal to $+\infty$ if the level is not reached by time T. A hitting time is never written τ, which stays the time to maturity. The function $\Psi(t, x)$ is the Gaussian smoothing of a test function ψ in calendar time (Lemma 19.61); the literature often writes $u(t, x)$ for a solution of the heat equation, a letter that this Part keeps for the up-factor of a tree. Its image about the level is Ψ^{im}, and $\Psi^{\mathrm{out}} = \Psi - \Psi^{\mathrm{im}}$, on the pattern of the knock-out price V^{out} of §23. Inside proofs, $\tau = T - t$ abbreviates the time to the horizon. The density of $\mathcal{N}(0, s^2)$ is φ_s, the notation of Proposition 7.25, and $\varphi = \varphi_1$. *Symbol notice:* in the proof of Lemma 19.61, and only there, $\varphi^{(i,j)}(\tau, w)$ denotes the Gaussian kernel $\varphi_{\sigma\sqrt{\tau}}(w - \nu\tau)$ differentiated i times in τ and j times in w. This is a declared local exception to the partial sign of the Part, made because some fifty kernel derivatives, at many different points, appear in that proof; the superscript counts derivatives, as in the $f^{(k)}$ of §7.5. In Theorem 19.63(iii), $\tilde{\mu} = \mu\sqrt{T}/\sigma$ and $\tilde{\ell} = a/(\sigma\sqrt{T})$ are a standardised drift and a standardised level; a tilde marks a standardised companion. The letters c, c', c'' are generic constants and ε, ε' small positive numbers. The commentary after the examples declares, where it uses them, a hitting time on the whole half-line $[0, \infty)$, a second level $\ell' > 0$ below the starting point and the step Δx of a random walk.

Lemma 19.61 (Gaussian smoothing solves the backward equation)**.** *Let ψ be a function on the real line, continuous except at finitely many points, with $|\psi(y)| \le c\, e^{c|y|}$ for all real y and one constant $c \ge 0$; equivalently, $\psi(y) = f(e^y)$ for a payoff f of the class $\mathcal{G}$ (Lemma 19.20(ii)). Let ν be real, $\sigma > 0$ and $T > 0$, let $Z \sim \mathcal{N}(0, 1)$, and put*

$$\Psi(t, x) = \mathbb{E}\,\psi\big(x + \nu(T - t) + \sigma\sqrt{T - t}\,Z\big) \quad \text{for } t < T, \qquad \Psi(T, x) = \psi(x).$$

(i) (Smoothness) The function Ψ is infinitely differentiable on $[0, T) \times \mathbb{R}$: its partial derivatives of every order exist and are continuous there. The same formula defines $\Psi(t, x)$ for $t < 0$, and Ψ is infinitely differentiable on the whole open set of the points (t, x) with $t < T$, so its derivatives at $t = 0$ are two-sided. For every $T' < T$, each partial derivative of Ψ is bounded in absolute value by $c'\, e^{c|x|}$ on $[0, T'] \times \mathbb{R}$, with the constant c of ψ and a constant c' that depends on T' and on the derivative.

(ii) (Equation) On $[0, T) \times \mathbb{R}$,

$$\partial_t \Psi + \nu\, \partial_x \Psi + \tfrac{1}{2}\sigma^2\, \partial_{xx} \Psi = 0. \tag{19.19}$$

(iii) (Boundary values) There is a constant c'' with $|\Psi(t, x)| \le c''\, e^{c|x|}$ on $[0, T] \times \mathbb{R}$, and $\Psi(t, x) \to \psi(x^*)$ as $(t, x) \to (T, x^*)$ in $[0, T] \times \mathbb{R}$, at every point x^* at which ψ is continuous.

Sketch of the proof. We write Ψ as an integral against a Gaussian kernel, check the equation on the kernel, and differentiate under the integral sign. With $\tau = T - t$, the substitution

$y = x + \nu\tau + \sigma\sqrt{\tau}\,z$ gives $\Psi(t,x) = \int \psi(y)\,\varphi_{\sigma\sqrt{\tau}}(y - x - \nu\tau)\,dy$, so (t,x) enters only through the kernel at $w = y - x$, and differentiating its logarithm gives, in the notation of the Setup,

$$\varphi^{(1,0)} = -\nu\,\varphi^{(0,1)} + \tfrac{1}{2}\,\sigma^2\,\varphi^{(0,2)}.$$

A derivative in t is minus a derivative in τ, and one in x minus one in w, so this identity integrated against ψ is (19.19); Taylor's theorem and a Gaussian envelope of the kernel derivatives, integrable against ψ by its growth bound, justify the differentiation. Undoing the substitution writes each derivative of Ψ as a mean of $\psi(x + \nu\tau + \sigma\sqrt{\tau}\,Z)$ times a polynomial in Z, such as $\partial_x\Psi = \mathbb{E}[\psi\,Z]/(\sigma\sqrt{\tau})$ and $\partial_{xx}\Psi = \mathbb{E}[\psi\,(Z^2 - 1)]/(\sigma^2\tau)$, which gives the bounds of (i) for $\tau \geq T - T'$ and, without the polynomial, that of (iii). Chebyshev's and the Cauchy–Schwarz inequalities, with the continuity of ψ at x^*, give the limit at the horizon. The complete proof is in Appendix E.14. □

In words, $\Psi(t,x)$ is the mean of ψ at the horizon for a motion with drift ν and volatility σ that stands at x at time t. The lemma is stated in calendar time t; the time to the horizon $T - t$ is abbreviated τ only inside the proof. Theorem 19.63 uses the lemma with $\sigma = 1$, and Theorem 20.7 with $\nu = r - \sigma^2/2$ and x a log price.

Example 19.62 (Smoothing a square)**.** Take $\psi(y) = y^2$, which is continuous and satisfies $y^2 \leq 2e^{|y|} \leq 2e^{2|y|}$, the growth bound of the lemma with $c = 2$. The variable $x + \nu(T-t) + \sigma\sqrt{T-t}\,Z$ is normal with mean $x + \nu(T-t)$ and variance $\sigma^2(T-t)$, and the mean of a square is the square of the mean plus the variance, so

$$\Psi(t,x) = \big(x + \nu(T-t)\big)^2 + \sigma^2\,(T-t).$$

Its derivatives are $\partial_t\Psi = -2\nu\,\big(x + \nu(T-t)\big) - \sigma^2$, $\partial_x\Psi = 2\,\big(x + \nu(T-t)\big)$ and $\partial_{xx}\Psi = 2$, and

$$\partial_t\Psi + \nu\,\partial_x\Psi + \tfrac{1}{2}\,\sigma^2\,\partial_{xx}\Psi = \Big(-2\nu\,\big(x + \nu(T-t)\big) - \sigma^2\Big) + 2\nu\,\big(x + \nu(T-t)\big) + \sigma^2 = 0,$$

as Lemma 19.61(ii) says; and $\Psi(t,x) \to x^2$ as $t \to T$, as (iii) says. With $\nu = 0.1$, $\sigma = 0.3$, $T - t = 0.5$ and $x = 0.2$,

$$\Psi = (0.2 + 0.05)^2 + 0.09 \times 0.5 = 0.0625 + 0.045 = 0.1075,$$

and a numerical quadrature of the defining expectation returns 0.10750. The formulas of Step 5 of the proof of Lemma 19.61 can be checked on the same numbers: $\mathbb{E}[\psi\,Z]/(\sigma\sqrt{T-t})$ and $\mathbb{E}[\psi\,(Z^2-1)]/(\sigma^2(T-t))$, computed by quadrature, return 0.5000 and 2.0000, the values of $\partial_x\Psi = 2 \times 0.25$ and $\partial_{xx}\Psi = 2$. With $\psi(y) = (e^y - K)^+$ for a strike $K > 0$ and $\nu = r - \sigma^2/2$, the function $e^{-r(T-t)}\,\Psi(t, \ln S)$ is the Black–Scholes–Merton price of a call (Theorem 20.7).

Theorem 19.63 (The reflection principle with drift (Bachelier, 1900, for zero drift))**.** *Let W be a standard Brownian motion, let ν be real, $T > 0$ and $\ell > 0$, and let $X_t = \nu t + W_t$, with running maximum $X_T^{\max} = \max_{0 \leq t \leq T} X_t$.*

(a) For every bounded function ψ on the real line that is continuous except at finitely many points and vanishes on $[\ell, \infty)$,

$$\mathbb{E}\big[\psi(X_T)\,\mathbf{1}\{X_T^{\max} < \ell\}\big] = \mathbb{E}\,\psi(X_T) - e^{2\nu\ell}\,\mathbb{E}\,\psi(2\ell + \nu T + W_T). \tag{19.20}$$

(b) Consequently, for every $x \leq \ell$,

$$P\big(X_T^{\max} \geq \ell,\ X_T \leq x\big) = e^{2\nu\ell}\,\Phi\Big(\frac{x - 2\ell - \nu T}{\sqrt{T}}\Big). \tag{19.21}$$

Three consequences follow.

(i) (Zero drift) For $\nu = 0$ and $x \le \ell$,

$$P(W_T^{\max} \ge \ell,\ W_T \le x) = P(W_T \ge 2\ell - x),$$
$$P(W_T^{\max} \ge \ell) = 2\,P(W_T \ge \ell) = 2\,\Phi\Big(-\frac{\ell}{\sqrt{T}}\Big).$$

The maximum $W_T^{\max}$ has the law of $|W_T|$, with density $2\,\varphi(y/\sqrt{T})/\sqrt{T}$ on $(0,\infty)$, and $\mathbb{E}\,W_T^{\max} = \sqrt{2T/\pi}$.

(ii) (The maximum with drift) The probability that the path reaches the level by time T is

$$P(X_T^{\max} \ge \ell) = \Phi\Big(\frac{\nu T - \ell}{\sqrt{T}}\Big) + e^{2\nu\ell}\,\Phi\Big(\frac{-\ell - \nu T}{\sqrt{T}}\Big).$$

(iii) (The Fact of §10.2) Let μ be real, $\sigma > 0$, and let $a > 0$ be a loss level. Then

$$P\Big(\min_{0\le t\le T}(\mu t + \sigma W_t) \le -a\Big) = \Phi\Big(\frac{-a - \mu T}{\sigma\sqrt{T}}\Big) + e^{-2\mu a/\sigma^2}\,\Phi\Big(\frac{-a + \mu T}{\sigma\sqrt{T}}\Big),$$

which is $2\,\Phi(-a/(\sigma\sqrt{T}))$ at $\mu = 0$. With $\tilde\mu = \mu\sqrt{T}/\sigma$ and $\tilde\ell = a/(\sigma\sqrt{T})$, the ratio of this path probability to the terminal probability $P(\mu T + \sigma W_T \le -a) = \Phi(-\tilde\mu - \tilde\ell)$ is

$$1 + e^{-2\tilde\mu\tilde\ell}\,\frac{\Phi(\tilde\mu - \tilde\ell)}{\Phi(-\tilde\mu - \tilde\ell)}.$$

For a fixed $\tilde\ell$ this ratio is strictly increasing in $\tilde\mu$. It equals 2 at $\mu = 0$, exceeds 2 exactly when $\mu > 0$, and is below 2 when $\mu < 0$. For $\tilde\mu + \tilde\ell > 0$ it is at least

$$1 + \Phi(\tilde\mu - \tilde\ell)\,(\tilde\mu + \tilde\ell)\,\sqrt{2\pi}\,e^{(\tilde\mu - \tilde\ell)^2/2},$$

so it grows without bound as $\tilde\mu \to \infty$. Given that the path touches the loss level, the conditional probability that it ends at or below it is the reciprocal of the ratio, which is below $1/2$ exactly when $\mu > 0$.

Sketch of the proof. We use the method of images, with the stopped clause of Theorem 19.30. With Ψ the smoothing of ψ of Lemma 19.61 for $\sigma = 1$, so that $\Psi(0,x) = \mathbb{E}\,\psi(x + \nu T + W_T)$, put

$$\Psi^{\mathrm{im}}(t,x) = e^{-2\nu(x-\ell)}\,\Psi(t, 2\ell - x), \qquad \Psi^{\mathrm{out}} = \Psi - \Psi^{\mathrm{im}}.$$

The mirror reverses the sign of $\partial_x\Psi$, and the factor $e^{-2\nu(x-\ell)}$ adds $2\nu\,\partial_x\Psi$, which restores the drift term, and two terms $\pm 2\nu^2\,\Psi$ that cancel; so Ψ^{out} solves (19.19), vanishes on the level and, ψ vanishing at the mirror points, tends at the horizon to ψ at its points of continuity below the level. Then $\Psi^{\mathrm{out}}(t, X_t)$ has no drift, and Theorem 19.30(iii) gives $\mathbb{E}\,\Psi^{\mathrm{out}}(T' \wedge t_\ell, X_{T'\wedge t_\ell}) = \Psi^{\mathrm{out}}(0,0)$ for $T' < T$, the stopped values being bounded at or below the level. As $T' \to T$ they tend, with probability one, to 0 on the paths that reach the level and to $\psi(X_T)$ on the others, and Lemma 19.12(iv) gives (a), since $\Psi^{\mathrm{out}}(0,0) = \mathbb{E}\,\psi(X_T) - e^{2\nu\ell}\,\mathbb{E}\,\psi(2\ell + \nu T + W_T)$; for $\nu = 0$ the image term is, by the symmetry of W_T, the mean of ψ at the reflected end point $2\ell - W_T$. Part (b) is (a) for $\psi(y) = \mathbf{1}\{y < x\}$, subtracted from $P(X_T < x)$; (ii) adds $P(X_T > \ell)$ to (b) at $x = \ell$, which gives $2\,P(W_T \ge \ell) = P(|W_T| \ge \ell)$ when $\nu = 0$, so that the maximum has the law of $|W_T|$, as (i) states; and (iii) is (ii) for $-(\mu t + \sigma W_t)/\sigma$, with the drift $-\mu/\sigma$ and the level a/σ, the claims on its ratio resting on the inequality $\varphi(y)/\Phi(y) > -y$. The complete proof is in Appendix E.15. □

Part (iii) contains both displays of Fact 10.6, for μ of either sign, its claim that a positive drift pushes the path probability above twice the terminal one and without bound, and its conditional reading: every claim of the Fact is proved here, and §10.2 keeps its text.

Figure 38 shows the classical picture behind part (i). For zero drift, reflecting a path about the level after its hitting time produces a path that is just as likely, and the proof of Theorem 19.63 replaces this path-by-path argument, which needs the strong Markov property, with a computation on the image function.

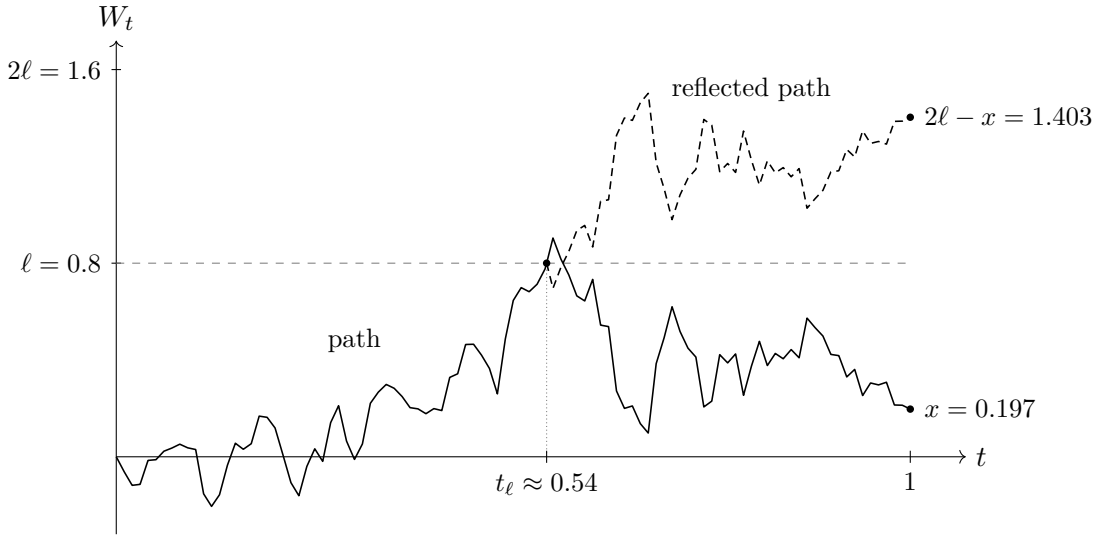


Figure 38: The reflection principle. A Brownian path on $[0,1]$ (solid, 100 steps of 0.01) first reaches the level $\ell = 0.8$ (gray dashed line) at $t_\ell \approx 0.54$ and ends at $x = 0.197$; reflected about the level after t_ℓ (black dashed), it ends at $2\ell - x = 1.403$. Reflecting after t_ℓ maps the paths that touch ℓ and end below x onto the paths that end above $2\ell - x$, which is why $P(W_1^{\max} \geq \ell,\ W_1 \leq x) = P(W_1 \geq 2\ell - x)$ for a driftless motion (Theorem 19.63(i)). The proof of the theorem replaces this path-by-path picture with the image function Ψ^{im}, which also handles a drift.

Example 19.64 (The running maximum over one year)**.** By Theorem 19.63(i) with $T = 1$ and $\ell = 1$,

$$P(W_1^{\max} \geq 1) = 2\,\Phi(-1) = 31.7\%, \qquad \text{against} \qquad P(W_1 \geq 1) = \Phi(-1) = 15.9\%,$$

and $\mathbb{E}\,W_1^{\max} = \sqrt{2/\pi} = 0.80$. A driftless P&L with an annual volatility of 10 (millions) is the process $10\,W$, whose running maximum is 10 times that of W: its expected running peak within the year is $10 \times 0.798 = 8.0$ million, and it touches $+10$ million at some time in the year with probability 31.7%.

Example 19.65 (A maximum with drift, and the check of the Fact)**.** Let $\nu = 0.5$, $\ell = 1$ and $T = 1$, so that $e^{2\nu\ell} = e$. By Theorem 19.63(ii),

$$\begin{aligned} P\big(X_1^{\max} \geq 1\big) &= \Phi(-0.5) + e\,\Phi(-1.5) \\ &= 0.30854 + 2.71828 \times 0.066807 = 0.30854 + 0.18160 = 0.49014, \end{aligned}$$

and a simulation of 200,000 paths with 100 steps each, corrected for crossings between grid dates, returns 0.491 (standard error 0.001). By (19.21) with $x = 0$,

$$P\big(X_1^{\max} \geq 1,\ X_1 \leq 0\big) = e\,\Phi(-2.5) = 2.71828 \times 0.00621 = 0.01688.$$

The terminal probability is $P(X_1 \geq 1) = \Phi(-0.5) = 0.309$. The path probability is 1.59 times as large, since $0.49014/0.30854 = 1.59$: with a drift *toward* the level the factor is below 2.

For the Fact of §10.2, take $\mu = 0.1$, $\sigma = 1$, $a = 1$ and $T = 1$ in part (iii):

$$\begin{aligned} P\Big(\min_{0 \leq t \leq 1} (0.1\,t + W_t) \leq -1\Big) &= \Phi(-1.1) + e^{-0.2}\,\Phi(-0.9) \\ &= 0.1357 + 0.8187 \times 0.1841 = 0.1357 + 0.1507 = 0.2864, \end{aligned}$$

against the terminal probability $\Phi(-1.1) = 0.1357$. The ratio, computed before rounding, is 2.111, which is the value of $1 + e^{-2\tilde{\mu}\tilde{\ell}}\,\Phi(\tilde{\mu} - \tilde{\ell})/\Phi(-\tilde{\mu} - \tilde{\ell})$ at $\tilde{\mu} = 0.1$ and $\tilde{\ell} = 1$: with a drift *away from*

the level the factor exceeds 2. Given that the path touches -1, it ends at or below -1 with conditional probability $1/2.111 = 0.474$, less than $1/2$. At $\tilde{\mu} = 2$ and the same $\tilde{\ell}$ the ratio is 12.4, above the lower bound 11.4 of part (iii).

Interview trap. "The probability of touching a level equals the probability of ending beyond it." For a driftless path the touching probability is exactly twice as large (Theorem 19.63(i)): 31.7% against 15.9% for the level 1 on $[0, 1]$ in Example 19.64. Half of the paths that touch the level end below it.

A second slip: "Brownian motion reaches 1 for sure, so it does so in a finite expected time." Let $\ell = 1$ and let $t_\ell = \inf\{t \geq 0 : W_t \geq \ell\}$ be the hitting time on the whole half-line. By the continuity of the path, $\{t_\ell \leq T\}$ is the event $\{W_T^{\max} \geq 1\}$, so by (i) $P(t_\ell \leq T) = 2\,\Phi(-1/\sqrt{T})$, which increases to 1 as $T \to \infty$: it is 0.920 at $T = 100$ and 0.992 at $T = 10^4$. But $t_\ell \geq T\,\mathbf{1}\{t_\ell > T\}$ on every path, so by the monotonicity of expectation, and because $2\,\Phi(y) - 1 = 2\int_0^y \varphi(z)\,dz \geq 2y\,\varphi(y)$ for $y > 0$, φ being decreasing on $[0, \infty)$,

$$\mathbb{E}\,t_\ell \geq T\,P(t_\ell > T) = T\left(2\,\Phi\Big(\frac{1}{\sqrt{T}}\Big) - 1\right) \geq 2\sqrt{T}\;\varphi\Big(\frac{1}{\sqrt{T}}\Big) \geq 2\sqrt{T}\;\varphi(1) \qquad \text{for } T \geq 1.$$

The middle expression is 7.97 at $T = 100$ and 79.8 at $T = 10^4$, and the right side grows without bound: no finite number bounds $\mathbb{E}\,t_\ell$, and the expected hitting time is infinite.

Interview trap. "The touching probability is always twice the terminal one." Only without drift. A drift toward the level makes the factor smaller than 2 (1.59 in Example 19.65), and a drift away from it makes the factor larger than 2, without bound (2.11 in the same example; Theorem 19.63(iii)). A stop-loss below a strategy with a positive expected P&L is hit during the year more than twice as often as it is breached at the end of the year.

Intuition. Two-sided exits are the continuous form of the gambler's ruin of §3.3. Let $\ell > 0$ be an upper and $-\ell'$, with $\ell' > 0$, a lower level, and let the step $\Delta x > 0$ divide both ℓ and ℓ'. A fair walk with steps $\pm\Delta x$, one step every $(\Delta x)^2$ units of time, has variance 1 per unit of time, the scaling of §19.2. Counted in steps from the lower level, it starts at $\ell'/\Delta x$, and the upper level is $(\ell + \ell')/\Delta x$ steps away. Theorem 3.11 gives the probability $(\ell'/\Delta x)/((\ell + \ell')/\Delta x) = \ell'/(\ell + \ell')$ of reaching ℓ before $-\ell'$, whatever Δx is, and Theorem 3.12 an expected duration of $(\ell'/\Delta x)(\ell/\Delta x)$ steps, that is, $\ell\ell'$ units of time. With the up-probability $p = (1 + \nu\,\Delta x)/2$ the mean step is $\nu\,(\Delta x)^2$, a drift ν per unit of time, and the formula of Theorem 3.14, whose ratio of the down- to the up-probability is $(1 - \nu\,\Delta x)/(1 + \nu\,\Delta x)$, tends as $\Delta x \to 0$ to

$$\frac{1 - e^{-2\nu\ell'}}{1 - e^{-2\nu(\ell + \ell')}},$$

because $\big((1 - \nu\,\Delta x)/(1 + \nu\,\Delta x)\big)^{1/\Delta x} \to e^{-2\nu}$. At $\ell = \ell' = 1$ and $\nu = 0.5$ the limit is 0.731, and the walk with $\Delta x = 0.1$ already gives 0.7312. The Part does not prove these limits for Brownian motion itself. A proof would apply Itô's lemma, stopped at the first exit from the corridor, to $f(t, x) = x$ and $x^2 - t$ for the fair case and to $e^{-2\nu x}$ with drift, which needs a stopped clause like that of Theorem 19.30(iii) for two boundaries at once.

Trading application. A one-touch that pays 1 if the price reaches a barrier, a knock-out option, and a drawdown or stop-loss limit on a P&L are all computed with Theorem 19.63. The principle returns in §23.2, for the one-touch (Proposition 23.19) and for every single-barrier option (Theorem 23.16); their complete proofs (Appendix I) reuse Steps 1 to 4 of the proof of Theorem 19.63, with the price written as an explicit function of W.

Practice: problems P19.1–P19.11 in §27, solutions in §28.

20 The Black–Scholes–Merton Model

20.1 Assumptions and the Black–Scholes–Merton Equation

Deriving the Black–Scholes–Merton equation at a whiteboard is the most requested derivation in options interviews, and the follow-up "where did the expected return go?" checks that it was understood. Every price in this section comes from one argument: a derivative and its underlying are driven by the same Brownian motion, so some mix of the two is riskless over an instant and must earn the riskless rate. This subsection states the assumptions, defines continuous-time trading strategies precisely (the bookkeeping that makes "riskless over an instant" exact), proves the equation with Corollary 19.31 as the engine (the argument is the continuous-time limit of the one-step replication of Theorem 18.3), and shows how to test whether a given function of the price and of time can be the price of a traded derivative. The derivation asked for at the whiteboard is Step 1 of the proof of Theorem 20.3; Definition 20.1 and Steps 2 and 3 make it exact.

Setup (for the whole section). The declarations of this paragraph hold throughout §20; later subsections re-declare what they use. The price of the underlying at time t is S_t, and $S = S_0$ is today's price. The strike of an option is K. The current time is t, the maturity is $T > 0$, in years, and $\tau = T - t$ is the time to maturity; for a price computed at time 0 the letter T stands for τ. The continuously compounded interest rate is r, the real-world expected rate of return of the underlying is μ, and its volatility is $\sigma > 0$. A standard Brownian motion (Definition 19.5) is written W_t, and $Z \sim \mathcal{N}(0,1)$ is a standard normal variable. The price function of a derivative is $V(S,t)$, its price at time t is $V_t = V(S_t, t)$, and its partial derivatives are $\partial_t V$, $\partial_S V$ and $\partial_{SS} V$. The prices of a European call and of a European put are C and $\mathcal{P}$, and Φ and φ are the standard normal distribution function and density.

Symbol notices for this section.

- Here r is the interest rate, not the ratio q/p of the Notation table. Wherever q appears it is the continuous dividend yield (first among the instruments listed before Definition 20.1, then in Definition 20.6; developed in §20.7), and a down-probability is written $1 - p$, never q.
- The literature writes $N(d_1)$ and $N(d_2)$; this Part writes $\Phi(d_1)$ and $\Phi(d_2)$, and ln is the natural logarithm, the log of the Notation table.
- The put is the calligraphic $\mathcal{P}$, because P is probability, so put–call parity reads $C - \mathcal{P} = S - Ke^{-rT}$.
- The symbol V_t is the price at time t; the time derivative of the price function is $\partial_t V$. Subscripts are times, never derivatives.
- The calligraphic $\mathcal{L}$ is the Black–Scholes–Merton operator, defined in Theorem 20.3; the italic L is the leverage factor of §19.
- The bank holding of a strategy is B_t, as B is the bond holding of §18.

Standing assumptions. The model is the following list, cited from here on as (A1)–(A6).

(A1) The underlying is the geometric Brownian motion of Definition 19.27,

$$S_t = S_0 \exp\left((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right),$$

with constant μ and constant $\sigma > 0$; it satisfies $dS = \mu S\,dt + \sigma S\,dW$ (Corollary 19.34).

(A2) The underlying pays no dividends on $[0, T]$ (relaxed in §20.7).

(A3) Money is borrowed and lent at one constant rate r: one unit of currency deposited at time s is worth $e^{r(t-s)}$ at a later time t.

(A4) Trading takes place at every instant and costs nothing: no commissions, spreads or taxes are paid, and any real number of units may be held.

(A5) Any instrument may be sold short; the seller receives the whole sale price at once and may invest it like any other cash.

(A6) There is no arbitrage: no strategy of Definition 20.1, stated next, is an arbitrage in the sense of its part (v), which is the continuous-time form of Definition 16.7.

Definition 20.1 declares the price $A_{k,t} = a_k(t, W_t)$ of a traded instrument, its holding $h_{k,t}$, the bank holding B_t and the value Π_t of a strategy; Theorem 20.3 declares the operator $\mathcal{L}$. Smoothness of the price function V is a hypothesis of that theorem, not an assumption about the market.

Setup (instruments and their discounted prices). The market consists of a bank account and finitely many traded instruments, indexed by k; their number gets no letter. By (A3), one unit of currency deposited at time s is worth $e^{r(t-s)}$ at time t. Instrument k has the price

$$A_{k,t} = a_k(t, W_t),$$

an explicit function of time and of the Brownian value, with $a_k \in C_E^{1,2}([0,T'] \times \mathbb{R})$ (Definition 19.19(iii)) for every $T' < T$. Where two independent Brownian motions W_1 and $W_\perp$ drive the market, as in §23.4, the price is $A_{k,t} = a_k(t, W_{1,t}, W_{\perp,t})$, and the function $a_k(t, x_1, x_2)$ is continuous on $[0,T'] \times \mathbb{R}^2$ for every $T' < T$, with continuous first partial derivatives in t, x_1 and x_2 and continuous second partial derivatives in x_1 and x_2, all of exponential growth (the hypothesis of Theorem 19.55(i)). The instruments met in this Part are the following.

- The stock of (A1), with $a_k(t,x) = S_0 e^{(\mu - \sigma^2/2)t + \sigma x}$. It is in the class, and in $C_E^{1,2}([0,T] \times \mathbb{R})$ itself, because each of its partial derivatives is a constant multiple of a_k and $|a_k(t,x)| \le S_0 e^{|\mu - \sigma^2/2|T} e^{\sigma|x|}$ on $[0,T] \times \mathbb{R}$ (Lemma 19.20(i)).
- The reinvested holding $\tilde{S}_t = S_t e^{-q(T-t)}$ of an asset with yield q (Proposition 16.15; §20.7).
- A forward contract for delivery at a date $T_F \ge T$, struck at K. Its price $(F_t - K)e^{-r(T_F - t)}$ (Proposition 16.17(i), which holds for consumption assets too) is explicit once the forward price F_t is a geometric Brownian motion. The prepaid forward $\tilde{F}_t = e^{-r(T_F - t)} F_t$ of §20.8 is one such forward plus $K e^{-r(T_F - t)}$ in the bank.
- A second asset, or a derivative quoted at $V_t = V(S_t, t)$, as in Theorem 20.3.

The function $(t,x) \mapsto e^{-rt} a_k(t,x)$ is again in $C_E^{1,2}([0,T'] \times \mathbb{R})$, because e^{-rt} is bounded on $[0,T']$; its partial derivatives in t, in x and twice in x are $e^{-rt}(\partial_t a_k - r a_k)$, $e^{-rt}\,\partial_x a_k$ and $e^{-rt}\,\partial_{xx} a_k$. Theorem 19.30(i) applied to it gives, for every $t < T$, with probability one,

$$e^{-rt} A_{k,t} = A_{k,0} + \int_0^t e^{-rs}(\alpha_{k,s} - r A_{k,s})\, ds + \int_0^t e^{-rs} \beta_{k,s}\, dW_s, \tag{20.1}$$

where

$$\alpha_{k,s} = (\partial_t a_k + \tfrac{1}{2}\,\partial_{xx} a_k)(s, W_s) \qquad \text{and} \qquad \beta_{k,s} = \partial_x a_k(s, W_s)$$

are the drift and the diffusion coefficient of instrument k in the sense of Definition 19.25(ii). With two motions, Theorem 19.55(i) gives the same identity with $W = W_1$, with $\alpha_{k,s} = (\partial_t a_k + \frac{1}{2}\,\partial_{11} a_k + \frac{1}{2}\,\partial_{22} a_k)(s, W_{1,s}, W_{\perp,s})$ and $\beta_{k,s} = \partial_1 a_k(s, W_{1,s}, W_{\perp,s})$, and with one more term on the right, the integral of $e^{-rs}\,\partial_2 a_k(s, W_{1,s}, W_{\perp,s})$ against $dW_{\perp,s}$.

For a holding process h and times $t' \le t$, write

$$\int_{t'}^t h_s\, d(e^{-rs} A_{k,s}) := \int_{t'}^t h_s\, e^{-rs}(\alpha_{k,s} - r A_{k,s})\, ds + \int_{t'}^t h_s\, e^{-rs} \beta_{k,s}\, dW_s, \tag{20.2}$$

a pathwise Riemann integral plus an Itô integral (Theorem 19.23); with two motions the matching integral against $dW_\perp$ is added. Both integrals are defined when h is as in part (i) of the definition below; this is Check 1, proved in Appendix F.1.

Definition 20.1 (Trading strategies, self-financing and arbitrage in continuous time)**.** Let the instruments, their prices $A_{k,t}$ and the integral (20.2) be as in the Setup.

(i) (Strategy.) A *trading strategy* on $[t_0, T)$, usually with $t_0 = 0$, consists of three things. First, deterministic *switch dates* $t_0 < t_1 < \cdots < t_J = T$. Second, an *initial capital* Π_{t_0}: a number when $t_0 = 0$, and when $t_0 > 0$ a function of exponential growth (Definition 19.19(i)) of a vector of Brownian values at fixed times $\le t_0$, the kind of vector that Definition 19.22(b) calls Y. Third, for each instrument k, a *holding* $h_{k,s}$, $t_0 \le s < T$, which on each cell $[t_{j-1}, t_j)$ is a finite sum of processes of the form allowed in Definition 19.22,

$$\xi(s, W_s, Y)\, \mathbf{1}\{s < t_\ell\}, \qquad Y \text{ a vector of Brownian values at times} \le t_{j-1},$$

and is *admissible*: for every $T' < t_j$, its restriction to $[t_{j-1}, T']$ lies in the span of the class $\mathcal{H}$ on $[t_{j-1}, T']$. Holdings may therefore depend on earlier fixings, stop at a hitting time, and grow without bound as a switch date approaches.

(ii) (Value.) The *value* Π_t of the strategy is defined cell by cell, in discounted form: for $t \in [t_{j-1}, t_j)$,

$$e^{-rt}\,\Pi_t = e^{-rt_{j-1}}\,\Pi_{t_{j-1}} + \sum_k \int_{t_{j-1}}^{t} h_{k,s}\, d\big(e^{-rs} A_{k,s}\big), \tag{20.3}$$

and Π_{t_j} is the mean-square limit of Π_t as t increases to t_j, which is required to exist. (When the holdings stay admissible up to t_j and, if $t_j = T$, the instruments held have price functions in $C_E^{1,2}([0,T] \times \mathbb{R})$, the limit exists and is given by the same identity at $t = t_j$; this is Check 2, proved in Appendix F.1.) The *bank holding* of the strategy is $B_t = \Pi_t - \sum_k h_{k,t} A_{k,t}$, and its *terminal value* is Π_T. Identity (20.3) is the *self-financing* condition. It says that money is never added or withdrawn: measured in units of the bank account, the value changes only by trading gains.

(iii) (Static positions.) A fixed number of units of a claim, bought or sold at t_0 at its quoted price and held to T, may be added to a strategy. The cost enters the initial capital and the payoff enters the terminal value.

(iv) (Closure.) Three operations produce strategies; this is Check 3, proved in Appendix F.1. First, sums and scalar multiples of strategies on $[t_0, T)$ are strategies: holdings, initial capitals and values add, and the switch dates of a sum are those of all its terms. Second, a strategy on $[t_0, t')$, which is this definition with t' in the place of T, followed by a strategy on $[t', T)$ whose initial capital at t' equals the value $\Pi_{t'}$ of the first, is a strategy on $[t_0, T)$ with t' among its switch dates. Third, let t_ℓ be a hitting time as in Definition 19.22(c), and suppose that no term of a holding already carries a stopping indicator, or that every indicator $\mathbf{1}\{s < t_{\ell'}\}$ carried by a term has its boundary approached from the same side as ℓ. Then multiplying every holding by $\mathbf{1}\{s < t_\ell\}$ gives holdings of the kind allowed in (i); as soon as the limits required in (ii) exist for them, they define a strategy, which holds no instrument from t_ℓ on and whose whole value then sits in the bank with constant discounted value.

(v) (Arbitrage.) An *arbitrage* is a strategy on $[0, T)$, possibly with static positions, whose initial capital is at most zero and whose value at some deterministic time $T' \le T$ is nonnegative with probability one and positive with positive probability. When the strategy has static positions, $T' = T$, because (iii) defines their value only at the start and at T. This is the continuous-time form of Definition 16.7.

Assumption (A6) says that no arbitrage exists among the strategies built from the instruments traded in the market at hand. The definition itself uses (A3)–(A5) only. It is bookkeeping, and it applies to any instruments with explicit prices, including an option marked at a model value (Proposition 21.22 in §21 uses it so). Boundaries approached from opposite sides are left out of the stopping clause of (iv) because they would give a two-sided exit time, which the class of

Definition 19.22 does not contain and no result of this Part needs: every barrier result uses a single barrier.

Example 20.2 (The simplest strategy: a forward replicated by one share and a loan)**.** Take $S_0 = K = 100$, $r = 5\%$ and $T = 1$. Consider the strategy on $[0, T)$ with no switch date before T that holds one share throughout, $h_s = 1$, from the initial capital

$$\Pi_0 = S_0 - Ke^{-rT} = 100 - 95.12 = 4.88.$$

The constant holding is admissible on the closed interval $[0, T]$. By (20.2) with $h = 1$ and by (20.1) for the stock, $\int_0^t 1\, d(e^{-rs}S_s) = e^{-rt}S_t - S_0$ with probability one, for each t, so (20.3) reads

$$e^{-rt}\,\Pi_t = (S_0 - Ke^{-rT}) + (e^{-rt}S_t - S_0) = e^{-rt}S_t - Ke^{-rT}.$$

Hence $\Pi_t = S_t - Ke^{-r(T-t)}$ and $B_t = \Pi_t - S_t = -Ke^{-r(T-t)}$: the value of a long forward struck at K (Proposition 16.17), replicated by one share and a loan that grows at the rate r. At $t = 0.5$ with $S_t = 110$ the value is $110 - 100e^{-0.025} = 110 - 97.53 = 12.47$ and the bank holding is -97.53. At T the value is $S_T - 100$, the forward payoff, with probability one: the holding is admissible up to T and the stock's price function is in $C_E^{1,2}([0, T] \times \mathbb{R})$, so by Definition 20.1(ii) and its Check 2 the value at T is the identity at $t = T$. The price function $V(S, t) = S - Ke^{-r(T-t)}$ reappears in Example 20.5(1) as a solution of equation (20.4).

Definition 20.1 relies on three claims, used only inside proofs. *Check 1*: the integrals in (ii) exist; the Itô integrand stays in the span of $\mathcal{H}$, and the Riemann integrand is continuous on each path except at finitely many hitting times. *Check 2*: under the two conditions of (ii), the limit Π_{t_j} exists and is given by (20.3) at $t = t_j$, by second-moment bounds on both integrals. *Check 3*: the closure properties of (iv) hold, by the additivity and linearity of the integrals; for stopping, two boundaries approached from the same side reduce to one. Their proofs are in Appendix F.1.

Reading the definition. No claim made here is relied on later. In the shorthand of Definition 19.25, identity (20.3) reads $d(e^{-rt}\,\Pi_t) = \sum_k h_{k,t}\, d(e^{-rt}A_{k,t})$. Before discounting, it is the whiteboard form

$$d\Pi = \sum_k h_k\, dA_k + rB\, dt,$$

gains plus interest: since $d(e^{-rt}\,\Pi) = e^{-rt}(d\Pi - r\Pi\, dt)$ and $d(e^{-rt}A_k) = e^{-rt}(dA_k - rA_k\, dt)$, the two forms differ by $r(\Pi - \sum_k h_k A_k)\, dt = rB\, dt$. Step 1 of the proof of Theorem 20.3 argues in the whiteboard form. Proofs use the discounted form because every term in it is an integral of a known integrand: the undiscounted form would need Π_s inside a pathwise integral, and the Itô integral is constructed for each fixed t only. Over a partition, a strategy in one instrument k, with price $A_{k,t}$, that trades only at the partition times and finances each trade from the bank holds $h_{k,t_{i-1}}$ units and $B_{t_{i-1}}$ in the bank over $[t_{i-1}, t_i]$, so its discounted value changes by

$$\begin{aligned} &e^{-rt_i}(h_{k,t_{i-1}}A_{k,t_i} + B_{t_{i-1}}e^{r(t_i - t_{i-1})}) - e^{-rt_{i-1}}(h_{k,t_{i-1}}A_{k,t_{i-1}} + B_{t_{i-1}}) \\ &\qquad = h_{k,t_{i-1}}(e^{-rt_i}A_{k,t_i} - e^{-rt_{i-1}}A_{k,t_{i-1}}). \end{aligned}$$

The left-endpoint sums of these terms are the discounted gains under the discrete self-financing rule of §16.2 and of the trees of §18 (Proposition 18.13), and "holding $\partial_S V$ fixed over dt" is the left endpoint. The leveraged position of Corollary 19.49 is the strategy holding LV_t/S_t shares, with the cash at zero rate in its first clause and at the rate r in its financing clause. Unlike the static replications of §16 and §17, replication in this section is dynamic, and exact only with continuous rebalancing; the error of discrete rebalancing is quantified in Proposition 21.27 (§21).

Theorem 20.3 (The Black–Scholes–Merton equation)**.** *Assume (A1)–(A6). Suppose that a derivative paying $f(S_T)$ at T can be traded at every time $t < T$ at the price $V_t = V(S_t, t)$, where the function V is continuous on $(0, \infty) \times [0, T)$ and has continuous partial derivatives $\partial_t V$, $\partial_S V$*

and $\partial_{SS}V$ there, and where V, $\partial_t V$, $\partial_S V$ and $\partial_{SS}V$ have moderate growth (Definition 19.19(ii)) on $(0,\infty)\times[0,T']$ for every $T'<T$. Define the Black–Scholes–Merton operator

$$\mathcal{L}V = \partial_t V + rS\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS}V - rV.$$

Then $\mathcal{L}V = 0$ on $(0,\infty)\times[0,T)$, that is,

$$\partial_t V + rS\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS}V = rV. \tag{20.4}$$

The contract supplies the terminal condition $V(S,T)=f(S)$, which the theorem does not use. The operator $\mathcal{L}$ is named once here; later sections write the equation out in full. With a yield q its term rS becomes $(r-q)S$ (Corollary 20.45 in §20.7). The italic L stays the leverage factor.

Sketch of the proof. The argument is a hedging argument, which the appendix makes exact by turning any violation of the equation into an arbitrage. At a whiteboard (Step 1 of the complete proof), Itô's lemma in the form of Corollary 19.31, whose hypothesis is the moderate growth assumed of V and its derivatives, gives $dV = (\partial_t V + \mu S\,\partial_S V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS}V)\,dt + \sigma S\,\partial_S V\,dW$. The portfolio $\Pi = -V + (\partial_S V)\,S$, short one derivative and long $\partial_S V$ shares, with that holding kept fixed over dt, changes by

$$d\Pi = -dV + \partial_S V\,dS = -(\partial_t V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS}V)\,dt,$$

because $dS = \mu S\,dt + \sigma S\,dW$ by (A1): the dW terms cancel, and so do the μ terms. A riskless portfolio must earn the rate r, so $d\Pi = r\Pi\,dt = r(-V + S\,\partial_S V)\,dt$, and equating the two expressions gives $\mathcal{L}V = 0$. Steps 2 and 3 replace the words "kept fixed over dt" by a strategy of Definition 20.1 that, from zero capital, holds $\mathcal{L}V(S_t,t)$ units of the derivative and $-\mathcal{L}V\,\partial_S V$ shares up to a date $T'<T$. In (20.3) its two dW integrands cancel and its two ds integrands add up to $e^{-rs}(\mathcal{L}V)^2$, so $e^{-rt}\,\Pi_t = \int_0^t e^{-rs}\big(\mathcal{L}V(S_s,s)\big)^2\,ds \geq 0$ for $t\leq T'$. If $\mathcal{L}V$ were nonzero at one point, the continuity of the partial derivatives would keep it away from zero on a small rectangle of prices and times. With positive probability the lognormal price of (A1) is inside that rectangle at its first time and, its path being continuous, stays inside for a while, so the value at T', the last time of the rectangle, would be positive with positive probability: an arbitrage, which (A6) excludes. The complete proof is in Appendix F.2. □

(Problem P20.1 asks you to reproduce the complete proof from scratch.)

Intuition. The hedge ratio $\partial_S V$ is the continuous version of the tree hedge ratio $(V_u - V_d)/(S(u-d))$ of Theorem 18.3: the change of the derivative per unit change of the stock. For the running call of Example 20.24 ($S=K=100$, $r=5\%$, $\sigma=20\%$, $T=1$) the hedge ratio is $\partial_S V = \Phi(0.35) = 0.6368$ (Proposition 21.2 in §21), so a short call is hedged with 0.6368 shares. After a move of the stock to 105, with no time elapsed, the ratio is 0.7237 and another 0.0869 shares must be bought; after a move to 95 it is 0.5373, and 0.0996 shares must be sold. The difference quotient across that move, $(13.8579 - 7.5109)/10 = 0.6347$, formed from the call prices at $S=105$ and at $S=95$ as the tree ratio is formed from V_u and V_d, is within 0.0021 of $\partial_S V$. The hedge ratio moves with the stock, which is why the riskless mix lasts only an instant.

Intuition. Step 2 of the proof of Theorem 20.3 is a machine that converts any violation of the equation into riskless profit. Hold $\mathcal{L}V$ units of the derivative, hedge them with $-\mathcal{L}V\,\partial_S V$ shares, and the hedged position earns the square of the violation, $(\mathcal{L}V)^2$, per unit of time in discounted money. The equation says exactly that this machine earns nothing. And because the hedge cancels the dW term identically, the proof needs no estimate of a stochastic integral.

Remark 20.4 (What the equation contains, and what selects the solution). The drift μ has disappeared from (20.4): two investors who disagree about μ but agree on σ agree on every derivative price. The equation is linear and runs backward in time. Every derivative whose price is a function $V(S_t,t)$ with the regularity of Theorem 20.3 solves the same equation, and the contract

enters only through its payoff f, which the equation does not contain. Which solution is the price is settled by replication, not by a uniqueness theorem: Theorem 20.7 in §20.2 exhibits the solution that tends to f at maturity and that a self-financing strategy delivers, and the law of one price fixes the time-0 price. Uniqueness among solutions of exponential growth in $\ln S$, for continuous payoffs, is true and classical (Tychonoff, 1935), but it is not used anywhere in this Part; for the discontinuous payoffs of the class $\mathcal{G}$ of Definition 19.19(iv), such as digitals, it would need a further argument. In the log variable $x = \ln S$, and for the undiscounted value $\Psi(t,x) = e^{r(T-t)}V(e^x,t)$, the equation becomes the heat equation with a constant drift, $\partial_t\Psi + \nu\,\partial_x\Psi + \frac{1}{2}\sigma^2\,\partial_{xx}\Psi = 0$ with $\nu = r - \sigma^2/2$, by the change of variables (F.2) of Steps 1 and 2 of the proof of Theorem 20.7; this is why the equation is called parabolic. The log variable alone does not suffice: $V(e^x,t)$ satisfies the same equation with the extra term $-rV(e^x,t)$, and the discount factor removes it. No later result uses this remark.

Example 20.5 (Which functions are prices)**.** Take $r = 5\%$ and $\sigma = 20\%$. Theorem 20.3 gives a test: a function of S and t with the continuity and the moderate growth required there can be the price of a traded derivative only if $\mathcal{L}V = 0$.

(1) The forward value. For $V = S - Ke^{-r\tau}$: $\partial_t V = -rKe^{-r\tau}$, $\partial_S V = 1$ and $\partial_{SS}V = 0$, so

$$\mathcal{L}V = -rKe^{-r\tau} + rS - r\big(S - Ke^{-r\tau}\big) = 0.$$

The value of Proposition 16.17 solves the equation, as it must: Example 20.2 replicates it.

(2) The power family. For $V = S^a$ with a real, a price that does not depend on t, substitution gives $rS\,\partial_S V = ra\,S^a$ and $\frac{1}{2}\sigma^2S^2\,\partial_{SS}V = \frac{1}{2}\sigma^2a(a-1)\,S^a$, so

$$\mathcal{L}S^a = \big(ra + \tfrac{1}{2}\sigma^2a(a-1) - r\big)S^a = (a-1)\big(r + \tfrac{1}{2}\sigma^2a\big)\,S^a.$$

This vanishes only for $a = 1$, the stock itself, and for $a = -2r/\sigma^2 = -0.10/0.04 = -2.5$. Term by term for $S^{-2.5}$: $rS\,\partial_S V = -2.5\,rV = -0.125\,V$ and $\frac{1}{2}\sigma^2S^2\,\partial_{SS}V = 0.02\times(-2.5)(-3.5)\,V = 0.175\,V$, which sum to $0.05\,V = rV$. So $S^{-2.5}$ is a possible price, constant in time and decreasing in S, and no power of S other than S itself and $S^{-2.5}$ is.

(3) A non-example and its repair. The function $V = S^2$ leaves the residual $\mathcal{L}V = (r+\sigma^2)S^2 = 0.09\,S^2$. The factor $e^{(r+\sigma^2)\tau}$ repairs it: for $V = S^2e^{(r+\sigma^2)\tau}$ the new term $\partial_t V = -(r+\sigma^2)V$ cancels the residual. The repaired price is worth $10{,}000\,e^{0.09} = 10{,}941.74$ at $S = 100$ and $\tau = 1$, the number that §20.2 recovers as a risk-neutral expectation (Example 20.10).

(4) Step 2 of the proof of Theorem 20.3 in numbers. A market that quoted S_t^2 itself as the price of a traded claim paying S_T^2 at T would violate the equation, and the strategy of that step, which from zero capital holds $\mathcal{L}V$ units of the claim and $-\mathcal{L}V\,\partial_S V$ shares, turns the violation into money:

$$e^{-rt}\,\Pi_t = \int_0^t e^{-rs}\,\big(0.09\,S_s^2\big)^2\,ds = \int_0^t e^{-rs}\,0.0081\,S_s^4\,ds.$$

At $S_0 = 100$ the residual is $\mathcal{L}V = 0.09\times100^2 = 900$. Scale the strategy by $1/900$, so that it holds one claim at time 0. It then starts with one claim worth 10,000, with $-\mathcal{L}V\,\partial_S V/900 = -2S_0 = -200$ shares worth $-20{,}000$, and with $+10{,}000$ in the bank, a total of zero. It earns at the rate $(\mathcal{L}V)^2/900 = 900$ a year at $S = 100$, about $900/252 = 3.57$ a trading day, with no risk.

Interview trap. "The hedged portfolio is riskless, so put it on and hold it to expiry." It is riskless only over dt: $\partial_S V$ changes with S and t, and a hedge that is not rebalanced keeps the exposure to the curvature $\partial_{SS}V$, the gamma of §21. The profit and loss of discrete rebalancing is computed in §21, at $r = q = 0$: over one interval in Proposition 21.17, and over the life of the option, with its exact mean and variance, in Proposition 21.27.

Interview trap. "A stock with a higher expected return has a dearer call." At fixed S and σ, the drift μ does not enter the price. A higher μ raises the real-world chance of exercise, but today's S already reflects it, and the hedge removes the exposure to it.

Interview trap. Dropping the second-order Itô term, that is, treating $(dS)^2$ as zero, gives $\partial_t V + rS\,\partial_S V = rV$, an equation with no volatility in it. With the call payoff it is solved, away from the kink, by $e^{-r\tau}(Se^{r\tau} - K)^+ = (S - Ke^{-r\tau})^+$, the zero-volatility value of Remark 20.28(1). Everything the call is worth above that value, 5.57 of the 10.45 of Example 20.24, comes from the term $\frac{1}{2}\sigma^2 S^2\,\partial_{SS} V$, the second-order term of Taylor's theorem (Theorem 7.31) that a first-order chain rule misses.

Interview trap. Sign bookkeeping. The portfolio of the whiteboard argument (Step 1 of the proof of Theorem 20.3) is *short* the derivative and *long* $\partial_S V$ shares. The opposite convention $\Pi = V - (\partial_S V)\,S$ gives the same equation, but switching conventions in the middle of the derivation produces a sign error in the term $rS\,\partial_S V$.

Trading application. Which assumptions fail in practice, and where the Part repairs them. Assumption (A1) is the weakest. Options on one underlying with different strikes and expiries trade at different implied volatilities (§22.1, Fact 22.5), which a single constant σ cannot produce. A volatility that varies independently of the price shocks enters a European price only through its path average σ_{avg}: under the pricing convention of Proposition 23.58, the price is the Black–Scholes–Merton price averaged over the law of σ_{avg}, not the price at an average volatility: $C = \mathbb{E}[C^{\text{BSM}}(\sigma_{\text{avg}})]$, where $C^{\text{BSM}}(\sigma)$ is the call formula of Theorem 20.22 at the volatility σ (§20.6); jumps and stochastic volatility, which produce smiles, are treated in Proposition 23.60, Fact 23.63 and Remark 23.64 in §23. Assumption (A4) fails because hedges are rebalanced at discrete dates and every trade has a cost; the error of discrete rebalancing is computed, at $r = q = 0$, in Proposition 21.27 in §21. In place of the one constant rate of (A3), practice uses the zero rate $r(T)$ of the option's maturity. The formula survives these failures as a quoting convention: the market quotes the volatility at which the formula reproduces each price (§20.6).

20.2 Risk-Neutral Valuation

Because μ is absent from Theorem 20.3, the price must be the same in a fictitious world in which every asset earns r, and in that world a price is a discounted expectation: the risk-neutral valuation of Definition 18.6 carried to continuous time. This subsection makes that sentence precise and proves it. It defines the risk-neutral expectation as an ordinary expectation of the price path rebuilt with μ replaced by r (Definition 20.6: no probability measure is changed). It proves risk-neutral valuation of European payoffs by exhibiting the strategy that replicates them (Theorem 20.7: Gaussian smoothing, Itô's lemma and the law of one price). It derives the one distribution that the rest of the section needs: under $\mathbb{Q}$ the terminal price is lognormal with mean equal to the forward price. It extends the rule to payoffs observed on finitely many dates (Corollary 20.11), and it records the continuously monitored case as the one Fact of the section. Along the way it answers two standard interview requests: price a claim paying S_T^2 or $\ln S_T$, and say where the risk-neutral median sits.

Setup. The risk-neutral probability, the risk-neutral expectation and the risk-neutral probability of an event A are $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}$ and $\mathbb{Q}(A)$, all defined in Definition 20.6, and $S_t^{\mathbb{Q}}$ is the risk-neutral price path. The symbol $W^{\mathbb{Q}}$ stands for the Brownian motion W itself and is written only in the shorthand "under $\mathbb{Q}$, $dS = rS\,dt + \sigma S\,dW^{\mathbb{Q}}$"; no hats are used, because hats denote estimators. The companion process $\tilde{W}_t = W_t + (\mu - r)t/\sigma$, the Brownian motion of the literature's change of measure, appears only in Remark 20.14 and in the trap after it. The risk-neutral log-drift is $\nu = r - \sigma^2/2$, and $r - q - \sigma^2/2$ with a yield q: it is the expected log-growth rate $g = \mu - \sigma^2/2$ of §19, the continuous-time form of the growth rate of Definition 3.1, with μ replaced by r. A payoff $f(S_T)$ paid at T has f in the class $\mathcal{G}$ of Definition 19.19(iv): f is continuous on $(0, \infty)$ except at finitely many prices, and $|f(S)| \le c\,(S^c + S^{-c})$ for $S > 0$, with one constant c. The generic constants are c, c', c'', because C is the call. A discontinuity hyperplane $\{x_j = \ell\}$ of a payoff on several dates sits at a fixed level ℓ, a letter for levels, never at a growth constant.

The Gaussian smoothing of Lemma 19.61 is $\Psi(t,x)$, in calendar time t and in the log coordinate $x = \ln S$. In Theorem 20.7 the letter x is a log price only and prices are written S; the price variables $x_1, \dots, x_J$ of Corollary 20.11 are declared there and follow Definition 19.19. As before $\tau = T - t$, and $F = Se^{r\tau}$ is the forward price of Theorem 16.11. The payoff function is f, as in Proposition 18.11, and h is a holding (Definition 20.1).

Definition 20.6 (The risk-neutral law of the price path)**.** The objects below are built from the Brownian motion that drives the price; the probability P is not changed.

(i) (One asset.) Assume (A1), for an underlying with a continuous yield q; under (A2), $q = 0$, and the yield enters in §20.7. The *risk-neutral price path* is

$$S_t^{\mathbb{Q}} = S_0 \exp\left((r - q - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right), \qquad 0 \le t \le T,$$

the price path with μ replaced by $r - q$, built from the same Brownian motion W. Let X be a quantity computed from a path on $[0,T]$ by a fixed rule (from its values at finitely many times, its running extremes, a hitting time); write $X(S)$ for its value on the price path, which is the random variable X, and $X(S^{\mathbb{Q}})$ for its value on the risk-neutral path. The *risk-neutral expectation* of X is

$$\mathbb{E}^{\mathbb{Q}} X = \mathbb{E}\, X(S^{\mathbb{Q}}),$$

defined whenever $X(S^{\mathbb{Q}}) \ge 0$ or $\mathbb{E}\,|X(S^{\mathbb{Q}})| < \infty$. Likewise, if an event A is described by the price path as the event that the path lies in a given set of paths, its *risk-neutral probability* $\mathbb{Q}(A)$ is the probability that $S^{\mathbb{Q}}$ lies in the same set; for instance $\mathbb{Q}(S_T > K) = P(S_T^{\mathbb{Q}} > K)$. Both are attached to the rule, not to the random variable alone; every quantity of this Part to which they are applied is given by such a rule.

(ii) (Two assets.) For two assets with yields q_1, q_2, volatilities σ_1, σ_2 and correlation ρ, driven by W_1 and $W_2 = \rho W_1 + \sqrt{1-\rho^2}\, W_\perp$ (Proposition 19.52, Fact 19.7), the risk-neutral price paths are

$$S_{k,t}^{\mathbb{Q}} = S_{k,0} \exp\left((r - q_k - \tfrac{1}{2}\sigma_k^2)t + \sigma_k W_{k,t}\right), \qquad k = 1, 2,$$

and $\mathbb{E}^{\mathbb{Q}}$ and $\mathbb{Q}$ are defined in the same way for quantities computed from both paths.

The phrase "under $\mathbb{Q}$, $dS = (r-q)S\,dt + \sigma S\,dW^{\mathbb{Q}}$" is shorthand: the path $S^{\mathbb{Q}}$ satisfies this equation by Corollary 19.34 with μ replaced by $r - q$, and $W^{\mathbb{Q}} = W$.

No probability measure is changed in this definition: $\mathbb{E}^{\mathbb{Q}}$ is an ordinary expectation of a different function of the same Brownian motion, or motions. For instance, with $S_0 = 100$, $\mu = 10\%$, $r = 5\%$, $q = 0$ and $\sigma = 20\%$, on a path on which the Brownian motion takes the value 0.5 at time 1, the price after one year is $S_1 = 100\, e^{0.08+0.10} = 119.72$, while $S_1^{\mathbb{Q}} = 100\, e^{0.03+0.10} = 113.88$: the same Brownian value, read through a different drift. On a tree, Definition 18.6 defines a genuine probability, and Theorem 18.37 shows that the expectations of smooth test functions of $\ln(S_T/S)$ on the tree converge to those of $\ln(S_T^{\mathbb{Q}}/S)$ (its part (i)), and that the tree prices of the call and the put converge to e^{-rT} times the risk-neutral expectations of their payoffs (its part (ii)).

Theorem 20.7 (Risk-neutral valuation of European payoffs, by replication (Black and Scholes, 1973; Merton, 1973))**.** *Let r be real, $\sigma > 0$ and $T > 0$, and let $f \in \mathcal{G}$ (Definition 19.19(iv)): f is continuous on $(0,\infty)$ except at finitely many points, and $|f(S)| \le c\,(S^c + S^{-c})$ for $S > 0$. Put $\nu = r - \sigma^2/2$ and define, for $0 \le t < T$ and $S > 0$,*

$$V(S,t) = e^{-r(T-t)}\, \mathbb{E}\, f\left(S \exp\left(\nu(T-t) + \sigma\sqrt{T-t}\, Z\right)\right), \qquad Z \sim \mathcal{N}(0,1), \tag{20.5}$$

and $V(S,T) = f(S)$. Then:

(i) the function V is infinitely differentiable on $(0,\infty) \times [0,T)$, every partial derivative of V has moderate growth on $(0,\infty) \times [0,T']$ for every $T' < T$, and $\mathcal{L}V = 0$ there, that is, V solves equation (20.4);

(ii) there is a constant c' with $|V(S,t)| \le c'\,(S^{c'} + S^{-c'})$ on $(0,\infty)\times[0,T]$, and $V(S,t) \to f(S^*)$ as $(S,t) \to (S^*,T)$ at every continuity point S^* of f.

Parts (i) and (ii) concern the function V alone. Their proof (Steps 1 and 2 of the complete proof in Appendix F.3) uses Lemma 19.61 and none of (A1)–(A6), so they hold for every real r and every $\sigma > 0$, for instance at an implied volatility that differs from the volatility of the stock, which is how §21 applies them. Assume now (A1)–(A6), with r and σ the rate and the volatility of the market. Then:

(iii) the strategy of Definition 20.1 with the initial capital $V(S_0,0)$ that holds $\partial_S V(S_t,t)$ shares on $[0,T)$, so that its bank holding is $V(S_t,t) - S_t\,\partial_S V(S_t,t)$, has the value $V(S_t,t)$ at every $t < T$, and its terminal value, the mean-square limit as $t \to T$, is $f(S_T)$;

(iv) consequently the arbitrage-free time-0 price of the claim paying $f(S_T)$ at T is

$$V(S_0,0) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[f(S_T)\big]$$

(Definition 20.6): at any other time-0 price there is an arbitrage;

(v) (parameter clause) let $0 \le t_0 < T$, and let $Y = (S_{s_1},\dots,S_{s_m})$ be the prices observed at $m \ge 1$ times $0 < s_1 < \cdots < s_m \le t_0$; a price fixed at time 0 is a known number and is absorbed into f. For a payoff $f(S_T;Y)$ with $(S,y) \mapsto f(S;y)$ in $\mathcal{G}_{m+1}$ (Definition 19.19(iv)), let $V(S,t;y)$ be defined by (20.5) with $f(\cdot\,;y)$ in place of f when y lies off the finitely many hyperplanes $\{y_j = \ell\}$ of f, each at a fixed level ℓ, and put $V(S,t;y) = 0$ on them; with probability one Y avoids them, because each S_{s_j} has a density. Then $V(\cdot,\cdot\,;y)$ satisfies (i) and (ii) for every y off the hyperplanes, and the strategy that starts at t_0 with the capital $V(S_{t_0},t_0;Y)$ and holds $\partial_S V(S_t,t;Y)$ shares on $[t_0,T)$ has the value $V(S_t,t;Y)$ for $t \in [t_0,T)$ and the terminal value $f(S_T;Y)$. So the time-t_0 replication cost of the claim is $V(S_{t_0},t_0;Y)$. Nothing is claimed about a market price at t_0: (A6) speaks about strategies, and every price statement of this Part is made at time 0.

The yield case follows by Corollary 20.41 (reduction to the reinvested asset); payoffs on finitely many dates are Corollary 20.11; single barriers are Theorem 23.16 (§23); other continuously monitored payoffs are the residual Fact 20.13.

Sketch of the proof. The proof is a verification: Gaussian smoothing produces a smooth solution of the equation, Itô's lemma shows that holding $\partial_S V$ shares delivers the payoff, and the law of one price fixes the price. For (i) and (ii), Steps 1 and 2 of the complete proof write $V(S,t) = e^{-r(T-t)}\,\Psi(t,\ln S)$, the identity (F.2), with Ψ the Gaussian smoothing of $x \mapsto f(e^x)$. Since f is in $\mathcal{G}$, Lemma 19.61 makes Ψ an infinitely differentiable solution of $\partial_t\Psi + \nu\,\partial_x\Psi + \frac{1}{2}\sigma^2\,\partial_{xx}\Psi = 0$ before T, with derivatives of exponential growth, that tends to the payoff at its continuity points; the chain rule turns these facts into (i) and (ii). For (iii), Itô's lemma (Corollary 19.31) for $e^{-rt}V(S_t,t)$ under (A1), with $\mathcal{L}V = 0$, gives the integrands $e^{-rs}(\mu - r)S\,\partial_S V$ in ds and $e^{-rs}\sigma S\,\partial_S V$ in dW, which are $\partial_S V$ times those of the discounted stock in (20.1): this is the self-financing identity (20.3) of the strategy. As the lognormal S_T is a continuity point of f with probability one, (ii) and a fourth-moment bound give the terminal value $f(S_T)$. For (iv), Step 5 of the complete proof sells a claim quoted above $V(S_0,0)$ and runs the strategy, or buys one quoted below it and runs the opposite strategy, banking the difference: an arbitrage, which (A6) excludes. And (20.5) at $t = 0$ is $e^{-rT}\,\mathbb{E}^{\mathbb{Q}} f(S_T)$ by Definition 20.6. Part (v) runs the same argument with the earlier prices frozen. The complete proof is in Appendix F.3. □

Example 20.10 works the theorem on three payoffs, once Proposition 20.8 has supplied the law of S_T under $\mathbb{Q}$; the commentary on the theorem follows that example.

Proposition 20.8 (The risk-neutral law of the terminal price)**.** *Assume (A1) and (A2), and let $\mathbb{Q}$ be as in Definition 20.6, with $F = Se^{rT}$ and $\nu = r - \sigma^2/2$. (With a yield q, replace r by $r - q$ in the drift; Corollary 20.41.)*

(i) Under $\mathbb{Q}$, $S_T = F\exp\big(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z\big)$ with $Z \sim \mathcal{N}(0,1)$; that is, $\ln S_T \sim \mathcal{N}\big(\ln S + (r - \frac{1}{2}\sigma^2)T,\ \sigma^2 T\big)$ under $\mathbb{Q}$. At a time $t < T$, $S_T^{\mathbb{Q}} = S_t^{\mathbb{Q}} \exp\big(\nu\tau + \sigma(W_T - W_t)\big)$, and the second factor is independent of $S_t^{\mathbb{Q}}$.

(ii) The risk-neutral mean of the terminal price is the forward price, $\mathbb{E}^{\mathbb{Q}}[S_T] = F$. Hence $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[S_T - K] = S - Ke^{-rT}$; under (A1)–(A6), Theorem 20.7(iv) identifies this number with the price of the forward contract struck at K, the forward value of Proposition 16.17.

(iii) The $\mathbb{Q}$-median of S_T is $Fe^{-\sigma^2 T/2} < F$, and $\mathbb{Q}(S_T > F) = \Phi\big(-\frac{1}{2}\sigma\sqrt{T}\big) < \frac{1}{2}$.

(iv) Let V be the price function (20.5) of Theorem 20.7. Along the risk-neutral path,

$$dV(S_t^{\mathbb{Q}}, t) = rV\,dt + \sigma S_t^{\mathbb{Q}}\,\partial_S V\,dW,$$

with V and $\partial_S V$ evaluated at $(S_t^{\mathbb{Q}}, t)$, and $\mathbb{E}\,V(S_t^{\mathbb{Q}}, t) = e^{rt}\,V(S_0, 0)$ for every $t < T$. Under $\mathbb{Q}$ every derivative grows at the rate r in expectation, as every traded asset does on the tree (Proposition 18.7).

Proof. We read each part off the explicit formula for $S^{\mathbb{Q}}$ and, for (iv), apply Itô's lemma along the risk-neutral path.

Part (i). By Definition 20.6 with $q = 0$, $S_T^{\mathbb{Q}} = S\exp\big((r - \frac{1}{2}\sigma^2)T + \sigma W_T\big) = F\exp\big(-\frac{1}{2}\sigma^2 T + \sigma W_T\big)$, and W_T has the law of $\sqrt{T}\,Z$ (Definition 19.5(2)). This is Theorem 19.40 with μ replaced by r. For the form at $t < T$, divide the formula at T by the formula at t: $S_T^{\mathbb{Q}}/S_t^{\mathbb{Q}} = \exp\big(\nu\tau + \sigma(W_T - W_t)\big)$, and the increment $W_T - W_t$ is independent of W_t, hence of $S_t^{\mathbb{Q}}$ (Definition 19.5(3)).

Part (ii). By Lemma 19.38(i) with $\theta = \sigma\sqrt{T}$, $\mathbb{E}e^{\theta Z} = e^{\theta^2/2}$, so $\mathbb{E}^{\mathbb{Q}}S_T = Fe^{-\sigma^2 T/2}e^{\sigma^2 T/2} = F$; this is also Proposition 19.42(i) with $\mu = r$. The payoff $f(S) = S - K$ is in $\mathcal{G}$ (it is continuous, and $|S - K| \le (1 + K)(S + S^{-1})$ because $S + S^{-1} \ge 1$), so under (A1)–(A6) Theorem 20.7(iv) prices the claim paying $S_T - K$ at $e^{-rT}(\mathbb{E}^{\mathbb{Q}}S_T - K) = e^{-rT}(F - K) = S - Ke^{-rT}$, expectation being linear.

Part (iii). The median of Z is 0 and the exponential is increasing, so the median of $F\exp(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z)$ is $Fe^{-\sigma^2 T/2}$. By (i), $\{S_T > F\} = \{-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z > 0\} = \{Z > \frac{1}{2}\sigma\sqrt{T}\}$ under $\mathbb{Q}$, an event of probability $1 - \Phi(\frac{1}{2}\sigma\sqrt{T}) = \Phi(-\frac{1}{2}\sigma\sqrt{T})$, which is below $\Phi(0) = \frac{1}{2}$ because $\sigma > 0$.

Part (iv). The path $S^{\mathbb{Q}}$ is the geometric Brownian motion of Definition 19.27 with drift r, so its drift and diffusion coefficients are $rS_t^{\mathbb{Q}}$ and $\sigma S_t^{\mathbb{Q}}$. By Theorem 20.7(i), V and its partial derivatives have moderate growth on $(0, \infty) \times [0, T']$ for every $T' < T$. Corollary 19.31 with the outer function $(S, t) \mapsto V(S, t)$ gives

$$dV = \big(\partial_t V + rS\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS}V\big)\,dt + \sigma S\,\partial_S V\,dW$$

along $S = S_t^{\mathbb{Q}}$, and the dt coefficient is rV by (20.4). The same corollary with the outer function $(S, t) \mapsto e^{-rt}V(S, t)$ has the drift $e^{-rt}(\partial_t V + rS\,\partial_S V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS}V - rV) = e^{-rt}\,\mathcal{L}V = 0$. Its expectation identity, which rests on Theorem 19.23(ii) for the dW term and on Fact 18.34(iii) for the exchange of the expectation and the ds integral on $[0, T']$, therefore gives $\mathbb{E}\,e^{-rt}V(S_t^{\mathbb{Q}}, t) = V(S_0, 0)$ for every $t \le T'$, and $T' < T$ was arbitrary. □

Example 20.9 (The risk-neutral law of the running stock in numbers)**.** Take the running stock of this section: $S = 100$, $r = 5\%$, $\sigma = 20\%$ and $T = 1$. The forward price is $F = 100\,e^{0.05} = 105.13$, which is also $\mathbb{E}^{\mathbb{Q}}S_T$ by Proposition 20.8(ii). The $\mathbb{Q}$-median is $105.13\,e^{-0.02} = 103.05$, and $\mathbb{Q}(S_T > F) = \Phi(-0.1) = 0.460$ by part (iii): the risk-neutral price is more likely than not to finish below its own mean. By part (i), $\ln S_T \sim \mathcal{N}(\ln 100 + 0.03,\ 0.2^2)$ under $\mathbb{Q}$, so the price whose logarithm lies two standard deviations below the mean of $\ln S_T$, $100\,e^{0.03 - 0.40} = 69.07$, has $\mathbb{Q}(S_T < 69.07) = \Phi(-2) = 2.28\%$ (Table 2). The forward contract struck at K is worth $e^{-rT}(\mathbb{E}^{\mathbb{Q}}S_T - K) = S - Ke^{-rT}$, in agreement with Example 20.5(1) and with the replicating strategy of Example 20.2; at $K = 100$ this is $100 - 95.12 = 4.88$.

Example 20.10 (A power claim and two log claims)**.** Keep $S = 100$, $r = 5\%$, $\sigma = 20\%$ and $T = 1$, so that $\nu = 0.03$ and $F = 105.13$.

(1) The claim paying S_T^2. The payoff $f(S) = S^2$ is in $\mathcal{G}$. By (20.5) and Lemma 19.38(i) with $\theta = 2\sigma\sqrt{\tau}$,

$$V(S,t) = e^{-r\tau}\,\mathbb{E}\big[S^2 e^{2\nu\tau + 2\sigma\sqrt{\tau}\,Z}\big] = S^2 e^{-r\tau + 2\nu\tau + 2\sigma^2\tau} = S^2 e^{(r+\sigma^2)\tau},$$

because $2\nu + 2\sigma^2 - r = r + \sigma^2$. Today the claim is worth $10{,}000\,e^{0.09} = 10{,}941.74$. This is the solution of the equation found in Example 20.5(3), as Theorem 20.7(i) says it must be. Against $e^{-rT}F^2 = 10{,}512.71$, the discounted square of the forward price, the gap 429.03 is a convexity premium: $\mathbb{E}^{\mathbb{Q}}S_T^2 > (\mathbb{E}^{\mathbb{Q}}S_T)^2$. Replication, by Theorem 20.7(iii): hold $\partial_S V = 2Se^{(r+\sigma^2)\tau}$ shares, 218.83 of them at time 0, with $V - S\,\partial_S V = -S^2 e^{(r+\sigma^2)\tau}$ in the bank, $-10{,}941.74$ at time 0. The bank holding is always minus the value of the claim: the claim is replicated by shares worth twice its value, half of them bought with borrowed money.

(2) The claim paying $\ln(S_T/S)$. The payoff $f(x) = \ln(x/S)$ is in $\mathcal{G}$. For $x > 0$, $|\ln x| \le x + 1/x$ (for $x \ge 1$, $\ln x \le x - 1$; for $x < 1$, $|\ln x| = \ln(1/x) \le 1/x$) and $x + 1/x \ge 2$, so $|\ln(x/S)| \le |\ln x| + \ln S \le (1 + \ln S)(x + 1/x)$. With $S = 100$ the factor is $1 + \ln 100 = 5.61 < 6$, and $x + 1/x \le x^6 + x^{-6}$, because $x^a + x^{-a}$ increases with $a \ge 0$ (its derivative in a is $\ln x\,(x^a - x^{-a}) \ge 0$); so $|f(x)| \le 6\,(x^6 + x^{-6})$, the bound of $\mathcal{G}$ with $c = 6$. By Proposition 20.8(i), $\mathbb{E}^{\mathbb{Q}}\ln(S_T/S) = \nu T$, so the claim is worth $e^{-rT}(r - \frac{1}{2}\sigma^2)T = 0.951 \times 0.03 = 0.0285$.

(3) The log contract paying $\ln(S_T/F)$. By the same part, $\mathbb{E}^{\mathbb{Q}}\ln(S_T/F) = -\frac{1}{2}\sigma^2 T$, so the contract is worth $e^{-rT}(-\frac{1}{2}\sigma^2 T) = 0.951 \times (-0.02) = -0.0190$. Equivalently $(2/T)\,\mathbb{E}^{\mathbb{Q}}\big[-\ln(S_T/F)\big] = \sigma^2 = 0.04$: twice the undiscounted value of the short log contract, per year, is the variance. This is the seed of the variance swap of Theorem 22.36 in §22.

Interview trap. "Risk-neutral valuation assumes that investors are risk-neutral." It assumes nothing about preferences. The price is the cost of replication (Theorem 20.7(iii)–(iv)), and $\mathbb{E}^{\mathbb{Q}}$ is an expectation that part (iv) of the theorem proves equal to that cost; compare Remark 18.8, which makes the point on a one-step tree.

Intuition. The drift moves the expected payoff of a claim and the rate at which that payoff must be discounted together, and the price stays where it is. For the running call of Example 20.27 in §20.4, the risk-neutral expected payoff is $e^{rT}C = 10.99$. With $\mu = 10\%$ the real-world expected payoff is 14.67, and the only continuously compounded rate that discounts it to the price 10.45 is 33.9% a year; with $\mu = 15\%$ these become 19.00 and 59.8%. The call's own discount rate depends on its leverage over the stock (Remark 18.8) and not on μ alone, and replication fixes the price before either number is computed.

Interview trap. "Risk-neutral valuation needs a uniqueness theorem for the equation." It does not. The price is the initial capital of a replicating strategy (Theorem 20.7(iii)–(iv)), and any other price is an arbitrage. Uniqueness of the solution (for continuous payoffs, Remark 20.4) is true but is used nowhere.

Intuition. Parts (i), (ii) and (iv) of Theorem 20.7 together are the Black–Scholes case of the Feynman–Kac representation (Kac, 1949): the discounted expectation (20.5) of the payoff along the path whose drift is r solves the backward equation (20.4) and tends to the payoff at maturity. An interview request to solve the equation is answered by this representation, or by the change of variables $V(S,t) = e^{-r(T-t)}\Psi(t, \ln S)$ of Steps 1 and 2 of the proof (F.2), which turns (20.4) into a heat equation with a constant drift that Gaussian smoothing solves.

Interview trap. Mixing the measures. "Under $\mathbb{Q}$ the expected price is $Se^{\mu T}$": it is the forward price $F = Se^{rT}$, and $Se^{\mu T}$ is the real-world mean of Proposition 19.42. "Discount the real-world expected payoff at r", or "at μ": both give wrong prices. For a one-year at-the-money call on the running stock with $\mu = 10\%$, Example 20.27 in §20.4 computes them, 13.95 and 13.27 against the correct 10.45.

Interview trap. "Half the risk-neutral paths end above the forward." Fewer than half do: the mean of a lognormal variable exceeds its median, so $\mathbb{Q}(S_T > F) = \Phi(-\frac{1}{2}\sigma\sqrt{T}) < \frac{1}{2}$, which is 0.460 in Example 20.9.

Setup (several dates). The fixing dates are $0 = t_0 < t_1 < \cdots < t_J = T$, so that J counts the dates, and $\Delta t_j = t_j - t_{j-1}$. The price variables are $x_1, \ldots, x_J$, where x_j stands for S_{t_j}. The value at t_j of the claim, as a function of the prices fixed so far, is $V^{(j)}(x_1, \ldots, x_j)$: a capital letter with the stage as a parenthesised superscript, because a lower-case italic letter would be mistaken for ν.

Corollary 20.11 (Payoffs observed on finitely many dates)**.** *Assume (A1)–(A6), and let $f \in \mathcal{G}_J$ (Definition 19.19(iv)): f is continuous outside finitely many hyperplanes $\{x_j = \ell\}$, each at a fixed level ℓ, and has moderate growth. Put $\nu = r - \sigma^2/2$ and $V^{(J)} = f$, and for $j = J, \ldots, 2$ let*

$$V^{(j-1)}(x_1, \ldots, x_{j-1}) = e^{-r\Delta t_j}\, \mathbb{E}\, V^{(j)}\Big(x_1, \ldots, x_{j-1},\ x_{j-1} \exp\big(\nu\Delta t_j + \sigma\sqrt{\Delta t_j}\, Z\big)\Big),$$

and finally $V^{(0)} = e^{-rt_1}\, \mathbb{E}\, V^{(1)}\big(S_0 \exp(\nu t_1 + \sigma\sqrt{t_1}\, Z)\big)$, with $Z \sim \mathcal{N}(0,1)$. Then the claim paying $f(S_{t_1}, \ldots, S_{t_J})$ at T has the arbitrage-free time-0 price

$$V^{(0)} = e^{-rT}\, \mathbb{E}^{\mathbb{Q}} f(S_{t_1}, \ldots, S_{t_J})$$

(Definition 20.6). It is replicated in stages. On each interval $[t_{j-1}, t_j)$ with $j \geq 2$, run the strategy of Theorem 20.7(v) with the maturity t_j and the earlier prices $Y = (S_{t_1}, \ldots, S_{t_{j-1}})$, for the European claim that pays

$$V^{(j)}(S_{t_1}, \ldots, S_{t_j}) \quad \text{at } t_j;$$

on the first interval, where Y is empty, run the strategy of Theorem 20.7(iii) for the claim that pays $V^{(1)}(S_{t_1})$ at t_1. This covers discretely monitored barriers, lookbacks and Asians, forward starts and cliquets. With a yield q the same holds with $\nu = r - q - \sigma^2/2$ and the reinvested holding as the hedge instrument, as Corollary 20.41(i) proves in §20.7.

Sketch of the proof. The proof is a backward induction on the fixing dates that chains the one-date strategies of Theorem 20.7. Each stage value $V^{(j)}$ inherits from $f \in \mathcal{G}_J$ the continuity outside finitely many hyperplanes and the moderate growth that part (v) of the theorem asks of a payoff (the appendix supplies the moment bounds). On $[t_{j-1}, t_j)$ part (v), or part (iii) when $j = 1$, replicates the claim paying $V^{(j)}$ at t_j from a capital that is, with probability one, the expectation that defines $V^{(j-1)}$: each stage ends with the capital the next one needs. Integrating out one independent Brownian increment at a time (Lemma 19.17; part (d) of the complete proof) identifies $V^{(0)}$ with $e^{-rT}\, \mathbb{E}^{\mathbb{Q}} f(S_{t_1}, \ldots, S_{t_J})$, and any other time-0 price is an arbitrage against the chained strategy, which (A6) excludes. The complete proof is in Appendix F.4. □

Example 20.12 (One leg of a cliquet: a forward-start at-the-money call)**.** Take $r = 5\%$ and $\sigma = 20\%$, and the claim paying $(S_{t_2}/S_{t_1} - 1)^+$ at $t_2 = 1.5$, with $t_1 = 0.5$: a forward-start at-the-money call on one unit of notional, the building block of a cliquet. Here $J = 2$, $\Delta t_2 = 1$ and $\nu = 0.03$, and the payoff $f(x_1, x_2) = (x_2/x_1 - 1)^+$ is in $\mathcal{G}_2$: it is continuous, and $0 \leq f \leq x_2/x_1 \leq (x_1 + x_1^{-1})(x_2 + x_2^{-1})$.

Stage 2. The value at t_1 is

$$V^{(1)}(x_1) = e^{-r}\, \mathbb{E}\Big(\frac{x_1 e^{\nu+\sigma Z}}{x_1} - 1\Big)^+ = e^{-0.05}\, \mathbb{E}\big(e^{0.03+0.2Z} - 1\big)^+ = 0.104506,$$

the price of a one-year at-the-money call on an asset worth one unit (Theorem 20.22 in §20.4 gives the closed form), the same for every x_1.

Stage 1. Since $V^{(1)}$ is constant, $V^{(0)} = e^{-0.025} \times 0.104506 = 0.101926$ per unit of notional: today's one-year at-the-money call on one unit, discounted over the half-year wait.

Replication. On $[0, 0.5)$ the strategy holds only the bank, because $V^{(1)}$ does not depend on x_1 and the hedge ratio of the first stage is zero; the capital 0.101926 grows to 0.104506. At t_1 it switches to the hedge of $1/S_{t_1}$ calls struck at S_{t_1}, which costs exactly 0.104506. Directly, the risk-neutral expectation of the payoff is $e^{0.05} \times 0.104506$, and the price is $e^{-0.075} \times e^{0.05} \times 0.104506 = 0.101926$, as part (d) of the proof in Appendix F.4 says.

Fact 20.13 (Risk-neutral valuation of continuously monitored payoffs (Harrison and Pliska, 1981))**.** Assume (A1)–(A6), and let X be a payoff paid at T that depends on the path of S on $[0, T]$ through continuous monitoring (a lookback on the running maximum $S_T^{\max}$, a continuously averaged Asian, a double or curved barrier), with $\mathbb{E}^{\mathbb{Q}} X^2 < \infty$. Then X is the terminal value of a self-financing strategy whose holdings are general past-determined processes, outside the class of Definition 20.1, with the initial capital $e^{-rT}\,\mathbb{E}^{\mathbb{Q}} X$, where $\mathbb{E}^{\mathbb{Q}}$ is as in Definition 20.6. If (A6) is extended to such strategies whose discounted value is square-integrable under the measure named in Remark 20.14, an admissibility condition that excludes doubling strategies, then the arbitrage-free price of X is $e^{-rT}\,\mathbb{E}^{\mathbb{Q}} X$.

A proof is beyond our scope: replicating a general path functional needs the martingale representation theorem (every square-integrable payoff is the terminal value of a stochastic integral with a general past-determined integrand), and such integrands lie outside the class $\mathcal{H}$ of Definition 19.22; the price statement also needs Girsanov's theorem, to show that the extended (A6) is consistent.

This Part proves the rule for payoffs on finitely many dates (Corollary 20.11) and for single barriers (Theorem 23.16 in §23). No proof relies on Fact 20.13.

Remark 20.14 (Why replacing μ by r works, and what is not proved)*.* This is a scope note; it carries no claim that is used later. *(1)* The literature constructs $\mathbb{Q}$ as a probability measure (Girsanov, 1960) under which the process $\tilde{W}_t = W_t + (\mu - r)t/\sigma$ is a standard Brownian motion; the literature writes this process $W_t^{\mathbb{Q}}$, while in this Part $W^{\mathbb{Q}}$ is W itself. Since $S_t = S_0 \exp\left((r - \frac{1}{2}\sigma^2)t + \sigma\tilde{W}_t\right)$, the price path has the same law under that measure as $S^{\mathbb{Q}}$ has under P, so every $\mathbb{E}^{\mathbb{Q}}$ of this Part is an expectation under that measure. Measure changes are not developed here. Lemma 19.38(i) is the one-dimensional version of the computation: $\mathbb{E}\left[e^{\theta Z - \theta^2/2}\,\psi(Z)\right] = \mathbb{E}\,\psi(Z + \theta)$. *(2)* Strategies whose value has no mean-square limit, such as continuous-time doubling, are excluded by Definition 20.1, which takes terminal values as mean-square limits and holdings of exponential growth. That no arbitrage exists within the class, so that (A6) is consistent with (A1)–(A5), for the market of the stock and the bank and for claims quoted at $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}$ of their payoffs, follows from Girsanov's theorem: under that measure the discounted value of every strategy, mean-square limits included, has constant expectation, the step from mean square under P to expectations under the new measure being the Cauchy–Schwarz inequality with the density, which has moments of every order. The theorem is named here and not used. *(3)* Two converse statements are named and not used: that pricing every claim by $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}$ leaves the market free of arbitrage (the first fundamental theorem of asset pricing; Harrison and Pliska, 1981, which needs a notion of admissible strategies), and that every claim can be replicated (completeness; martingale representation, Fact 20.13).

Interview trap. "Under $\mathbb{Q}$ the volatility changes", or "under $\mathbb{Q}$ the stock really grows at r". Only the drift used for pricing changes. In this Part $\mathbb{E}^{\mathbb{Q}}$ is the ordinary expectation of the price path rebuilt from the *same* Brownian motion with μ replaced by r (Definition 20.6); σ and the real-world paths are untouched, and the price is fixed by the strategy of Theorem 20.7(iii), however $\mathbb{Q}$ is described. In the language of the literature (Remark 20.14) the measure changes, and the process $\tilde{W}_t = W_t + (\mu - r)t/\sigma$ is a Brownian motion under the new measure. The volatility σ is the same under both measures, and both descriptions give the same numbers.

20.3 Digital Options under Black-76: the Building Blocks

A cash-or-nothing digital pays one unit of currency if the underlying settles above a strike, and nothing otherwise. It is the simplest nontrivial derivative, and it has exactly the payoff of a

prediction-market binary contract ("pays 1 if the price settles above K"), which is why pricing it from the quotes of an options venue becomes a live relative-value trade; §22.2 in §22 takes that trade up. With its companion, the asset-or-nothing claim, the digital is also the building block from which §20.4 assembles the call and the put. This subsection reads the Black-76 lognormal model as a model of any forward price, proves that the digital is worth "the risk-neutral probability, discounted", the single most quoted formula in options interviews, and prices the asset-or-nothing claim with the Gaussian shift of Lemma 19.38.

Setup. Today's forward or futures price for delivery at a date $T_F \geq T$ is $F > 0$, and F_T is its value at the expiry $T > 0$ of the option. When $T_F = T$ the contract delivers at the expiry and $F_T = S_T$, the price of the underlying itself. The delivery date is written T_F, not T', which in this Part is a date before maturity inside a proof. The strike is $K > 0$, the volatility is $\sigma > 0$, and r is the continuously compounded interest rate, as in the section-wide Setup. The numbers d_2 and d_1 are defined in Theorem 20.17 and in Proposition 20.19. The time-0 prices of the cash-or-nothing digital and of the asset-or-nothing claim struck at K are $\mathrm{Dig}(K)$ and $\mathrm{AoN}(K)$. A standard normal variable is Z, with distribution function Φ and density $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$; the literature writes $N(d_2)$ for the $\Phi(d_2)$ of this subsection. In proofs $\theta = \sigma\sqrt{T}$ is the exponent of the Gaussian shift.

Prices in this subsection are discounted risk-neutral expectations, $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\,\cdot\,]$. This pricing rule is written into Definition 20.16, and it is a theorem in each case in which the model is used: Theorem 20.7(iv) for a non-dividend stock, and its two reductions, Corollary 20.41 for an asset with a yield (proved in §20.7) and Theorem 20.52 for a futures price (proved in §20.8).

Definition 20.15 (Cash-or-nothing digital)**.** The *cash-or-nothing digital* (or *binary*) with strike K and expiry T is the contract paying $\mathbf{1}\{F_T > K\}$ at time T: one unit of currency if the value at expiry of the forward or futures price exceeds the strike, and zero otherwise. For an option on the underlying itself, $T_F = T$ and the payoff is $\mathbf{1}\{S_T > K\}$. Its price at time 0 is written $\mathrm{Dig}(K)$. The payoff function $x \mapsto \mathbf{1}\{x > K\}$ is in the class $\mathcal{G}$ of Definition 19.19(iv): it has one discontinuity, at $x = K$, and it is bounded by $1 \leq x + x^{-1}$. So Theorem 20.7 and its reductions price it.

Definition 20.16 (Black-76 lognormal model)**.** The *Black-76 model* (Black, 1976, who stated it for futures prices) for the forward or futures price of the Setup consists of a law and a pricing rule.

(i) (Law.) Under $\mathbb{Q}$, the value at expiry is lognormal with mean F:

$$F_T = F\exp\big(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z\big), \qquad Z \sim \mathcal{N}(0,1) \text{ under } \mathbb{Q}. \tag{20.6}$$

That is, $\mathbb{E}^{\mathbb{Q}} f(F_T) = \mathbb{E}\, f\big(F\exp(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z)\big)$ for every function f for which the right side is defined, and the risk-neutral probability $\mathbb{Q}(A)$ of an event A described by F_T is the probability of the same event described by the right side of (20.6).

(ii) (Pricing rule.) The time-0 price of a claim paying $f(F_T)$ at T, with $f \in \mathcal{G}$ (Definition 19.19(iv)), is $e^{-rT}\,\mathbb{E}^{\mathbb{Q}} f(F_T)$.

The drift-free check: the model has mean F, and its prices are finite. By Lemma 19.38(i) with $\theta = \sigma\sqrt{T}$, $\mathbb{E}\, e^{\theta Z} = e^{\theta^2/2}$, so

$$\mathbb{E}^{\mathbb{Q}} F_T = F e^{-\sigma^2 T/2}\,\mathbb{E}\, e^{\sigma\sqrt{T}\,Z} = F e^{-\sigma^2 T/2}\, e^{\sigma^2 T/2} = F.$$

The correction $-\frac{1}{2}\sigma^2 T$ in the exponent is exactly what makes the mean equal to F, as befits a forward or futures price, which costs nothing to enter and therefore earns no risk-neutral drift; Proposition 20.51 in §20.8 proves that last statement. The same lemma, with $\theta = \pm c\,\sigma\sqrt{T}$, shows that the expectation in (ii) is finite for every $f \in \mathcal{G}$: if $|f(x)| \leq c\,(x^c + x^{-c})$, then $|f(F_T)| \leq c\,(F_T^c + F_T^{-c})$ and $\mathbb{E}^{\mathbb{Q}} F_T^{\pm c} = F^{\pm c}\exp\big(\mp\tfrac{1}{2}c\,\sigma^2 T + \tfrac{1}{2}c^2\sigma^2 T\big)$.

Where the model holds. Each case is proved in this section. For a non-dividend stock under (A1)–(A6), take $T_F = T$, so that $F_T = S_T$, and $F = Se^{rT}$ (Theorem 16.11): the law (i) is

Proposition 20.8(i), and the pricing rule (ii) is Theorem 20.7(iv). For an asset with a yield q, $F = Se^{(r-q)T}$ (Proposition 16.15(ii)), and both parts are Corollary 20.41, proved in §20.7. For a futures price that itself follows (A1), the model of the Setup of §20.8, they are Proposition 20.51 and Theorem 20.52, proved in §20.8. This includes an underlying with no investable spot position, such as a consumption commodity, whose spot trades but cannot be carried as a reinvested holding. Until those two subsections, the results of §20.3, §20.4 and §20.5 are statements about the model of Definition 20.16; for a non-dividend stock they are already consequences of (A1)–(A6).

Theorem 20.17 (Digital price under Black-76)**.** *Under the Black-76 model of Definition 20.16, the cash-or-nothing digital of Definition 20.15 is worth*

$$\mathrm{Dig}(K) = e^{-rT}\,\Phi(d_2), \qquad \textit{where} \qquad d_2 = \frac{\ln(F/K) - \sigma^2 T/2}{\sigma\sqrt{T}}. \tag{20.7}$$

In words: the price of the digital is the risk-neutral probability of settling above the strike, discounted.

Proof. We compute the expectation of an indicator in three steps: the event $\{F_T > K\}$ is reduced to an event about the standard normal variable Z, its probability is evaluated, and the result is discounted.

Step 1: the event. Take logarithms in (20.6), the logarithm being strictly increasing, and divide by $\sigma\sqrt{T} > 0$:

$$F_T > K \iff -\frac{\sigma^2 T}{2} + \sigma\sqrt{T}\,Z > \ln\frac{K}{F} \iff Z > \frac{\ln(K/F) + \sigma^2 T/2}{\sigma\sqrt{T}} = -d_2,$$

where the last equality is the definition of d_2 in (20.7) with both signs changed.

Step 2: the integral. Since Z has the density φ,

$$\mathbb{Q}(F_T > K) = P(Z > -d_2) = \int_{-d_2}^{\infty} \varphi(z)\,dz = \int_{-\infty}^{d_2} \varphi(z)\,dz = \Phi(d_2).$$

The third equality substitutes $-z$ for z: the limits $-d_2$ and ∞ become d_2 and $-\infty$, the sign of dz restores their order, and $\varphi(-z) = \varphi(z)$. The computation used nothing about d_2, so $P(Z > -y) = \Phi(y)$ for every real y; consequently $\Phi(-y) = P(Z \le -y) = 1 - \Phi(y)$. Both identities are reused below.

Step 3: the discount. The payoff is the indicator of the event $\{F_T > K\}$, and the expectation of an indicator is the probability of its event. The payoff function is in $\mathcal{G}$ (Definition 20.15), so the pricing rule of Definition 20.16(ii) gives

$$\mathrm{Dig}(K) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[\mathbf{1}\{F_T > K\}\big] = e^{-rT}\,\mathbb{Q}(F_T > K) = e^{-rT}\,\Phi(d_2). \qquad \square$$

Figure 39 shows the two objects of the proof side by side: the lognormal density of F_T, with the probability mass above the strike, the area $\Phi(d_2)$, shaded, and the step payoff of the digital drawn above it.

Example 20.18 (Pricing a digital on a futures contract)**.** A futures contract trades at $F = 100$. Price the digital that pays 1 if the futures price settles above $K = 120$ in $T = 0.25$ years (three months), with volatility $\sigma = 40\%$ and rate $r = 2\%$. The pieces of (20.7) are $\sigma\sqrt{T} = 0.40 \times 0.5 = 0.20$ (a 20% three-month volatility), $\ln(F/K) = -\ln 1.2 = -0.18232$ and $\sigma^2 T/2 = 0.16 \times 0.25/2 = 0.02$. Then

$$d_2 = \frac{-0.18232 - 0.02}{0.20} = -1.0116 \approx -1.01, \qquad \Phi(-1.01) = 1 - \Phi(1.01) = 1 - 0.844 = 0.156,$$

and the price is $e^{-0.02\times 0.25} \times 0.156 = 0.995 \times 0.156 = 0.155$: about 15.5 cents per unit of notional. (With d_2 unrounded, $\Phi(-1.0116) = 0.1559$ and the price is 0.1551.)

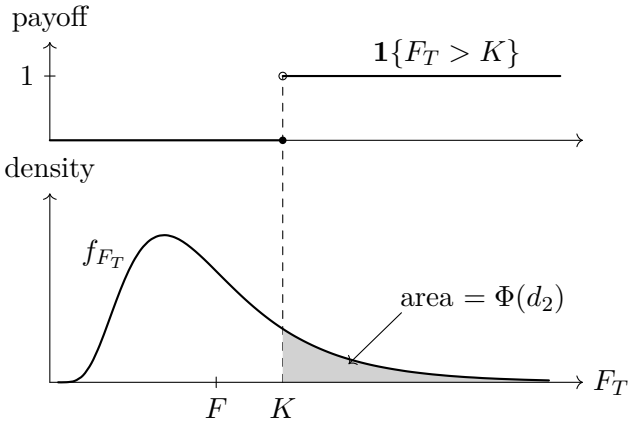


Figure 39: The cash-or-nothing digital under Black-76 (Theorem 20.17), drawn for $F = 1$, $\sigma\sqrt{T} = 0.5$ and $K = 1.4$. Bottom: the lognormal density f_{F_T} of the value at expiry of the forward price, centred so that $\mathbb{E}^{\mathbb{Q}}[F_T] = F$; the shaded area above the strike is $\mathbb{Q}(F_T > K) = \Phi(d_2) = 0.178$, about 0.18. Top: the step payoff $\mathbf{1}\{F_T > K\}$ of the digital; the contract is worth the discounted shaded area, $e^{-rT}\,\Phi(d_2)$.

Two sanity checks. First, the required log-move 0.182 against a three-month volatility of 0.20 is about 0.91 standard deviations; the $-\sigma^2 T/2$ shift of the risk-neutral median adds another 0.10; and a 1.01-standard-deviation upper tail should be in the mid-teens, a little under the anchor $\Phi(-1) = 15.9\%$ of Table 2. Second, the price respects its structural bounds: it lies in $(0, e^{-rT})$, inside the model-free band $0 \le \mathrm{Dig}(K) \le e^{-rT}$ of (17.4), and well below the 0.458 of an at-the-money-forward digital, $e^{-rT}\,\Phi(-0.10)$ at $K = F$, which is a little under half the discount factor.

Intuition. The quantity d_2 is the log-distance from the strike to the risk-neutral *median* of F_T, measured in units of the total volatility $\sigma\sqrt{T}$. By (20.6) the median of F_T is $Fe^{-\sigma^2 T/2}$, because the median of Z is 0 and the exponential is increasing, and

$$d_2 = \frac{\ln\left(Fe^{-\sigma^2 T/2}/K\right)}{\sigma\sqrt{T}}.$$

A digital struck at the median is worth half the discount factor, and every volatility unit of distance moves the price along the normal distribution function.

Interview trap. The precise statement is "the *risk-neutral* probability, discounted". Calling $\Phi(d_2)$ "the probability that the option finishes in the money" without the qualifier invites the follow-up "under which measure?". Under $\mathbb{Q}$ the forward price has no drift, while its real-world law generally has one: for a stock under (A1), the real-world probability of finishing above K is that of Proposition 19.42(iii), Φ evaluated at d_2 with μ in the place of r, which contains μ. The distinction is not pedantry. When the digital is compared with a prediction-market quote (§22.2), part of any gap can be a legitimate risk premium between the real-world law and the risk-neutral law, and not a mispricing. And with a volatility skew, that is, when the volatility at which the market prices options varies with the strike (§22), the digital is in general not $e^{-rT}\,\Phi(d_2)$, even at the implied volatility of its own strike: on an open interval of strikes on which the implied volatility is differentiable, Corollary 22.10 adds the vega-times-skew term $-\mathcal{V}\,\sigma'_{\mathrm{imp}}(K)$, in which $\mathcal{V} > 0$ is the vega of §20.6 and $\sigma'_{\mathrm{imp}}(K)$ is the slope of the implied volatility in the strike, so the correction vanishes only at a strike where that slope is zero.

Setup (the asset-or-nothing claim). The *asset-or-nothing claim* with strike K and expiry T pays $F_T\,\mathbf{1}\{F_T > K\}$ at T: the value at expiry of the forward price if it ends above the strike,

and nothing otherwise. For an option on the underlying itself ($T_F = T$) it pays $S_T\,\mathbf{1}\{S_T > K\}$, one unit of the asset if the asset ends above the strike. Its price at time 0 is $\mathrm{AoN}(K)$. The payoff function $x \mapsto x\,\mathbf{1}\{x > K\}$ is in $\mathcal{G}$: it has one discontinuity, at $x = K$, and it is bounded by $x \le x + x^{-1}$.

Proposition 20.19 (Asset-or-nothing price)**.** *Under the Black-76 model of Definition 20.16,*

$$\mathrm{AoN}(K) = e^{-rT} F\,\Phi(d_1), \qquad \text{where} \qquad d_1 = d_2 + \sigma\sqrt{T} = \frac{\ln(F/K) + \sigma^2 T/2}{\sigma\sqrt{T}}.$$

For a claim on an asset with a yield q, where $T_F = T$ and $F = Se^{(r-q)T}$, the factor $e^{-rT}F$ equals Se^{-qT}, so $\mathrm{AoN}(K) = Se^{-qT}\,\Phi(d_1)$; for a non-dividend stock, $\mathrm{AoN}(K) = S\,\Phi(d_1)$.

Proof. We use the Gaussian shift. Put $\theta = \sigma\sqrt{T}$. By (20.6), $F_T = Fe^{\theta Z - \theta^2/2}$ under $\mathbb{Q}$, and by Step 1 of the proof of Theorem 20.17, $\{F_T > K\} = \{Z > -d_2\}$. The payoff function is in $\mathcal{G}$ (Setup), so the pricing rule and the law of Definition 20.16 give

$$\mathrm{AoN}(K) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T > K\}\big] = e^{-rT} F\,e^{-\theta^2/2}\,\mathbb{E}\big[e^{\theta Z}\,\mathbf{1}\{Z > -d_2\}\big].$$

Lemma 19.38(i), applied to the bounded nonnegative function $\psi(z) = \mathbf{1}\{z > -d_2\}$, says that $\mathbb{E}[e^{\theta Z}\psi(Z)] = e^{\theta^2/2}\,\mathbb{E}\,\psi(Z+\theta)$. Hence

$$e^{-\theta^2/2}\,\mathbb{E}\big[e^{\theta Z}\,\mathbf{1}\{Z > -d_2\}\big] = P(Z + \theta > -d_2) = P\big(Z > -(d_2+\theta)\big) = \Phi(d_2+\theta) = \Phi(d_1),$$

where the third equality is Step 2 of the same proof with $y = d_2 + \theta$. Multiplying by $e^{-rT}F$ gives the formula. The yield case is a substitution: $e^{-rT}\,Se^{(r-q)T} = Se^{-qT}$.

An equivalent route completes the square in the lognormal integral. Since $\theta z - \frac{1}{2}\theta^2 - \frac{1}{2}z^2 = -\frac{1}{2}(z-\theta)^2$, the integrand $e^{\theta z - \theta^2/2}\varphi(z)$ equals $\varphi(z-\theta)$, and the expectation is $\int_{-d_2}^{\infty}\varphi(z-\theta)\,dz = \int_{-d_1}^{\infty}\varphi(z)\,dz = \Phi(d_1)$, the constant $1/\sqrt{2\pi}$ of φ being the one fixed by Theorem 7.5. Problem P20.4, which rebuilds Theorem 20.22 from the digitals, asks for this route. □

Corollary 20.20 (Digital parity)**.** *Let the* digital put *with strike K and expiry T pay $\mathbf{1}\{F_T \le K\}$ at T, with time-0 price* $\mathrm{Dig}^{\mathrm{put}}(K)$*, and let the* asset-or-nothing put *pay $F_T\,\mathbf{1}\{F_T \le K\}$ at T, with time-0 price* $\mathrm{AoN}^{\mathrm{put}}(K)$*.*

(i) (Model-free.) If the digital and the digital put both trade at time 0, then for every $K > 0$

$$\mathrm{Dig}(K) + \mathrm{Dig}^{\mathrm{put}}(K) = e^{-rT}.$$

If the asset-or-nothing claim and the asset-or-nothing put both trade at time 0, and forward contracts for delivery at T_F can be entered at time 0 at the forward price F, then for every $K > 0$

$$\mathrm{AoN}(K) + \mathrm{AoN}^{\mathrm{put}}(K) = e^{-rT}F :$$

the asset-or-nothing call and put together are worth the discounted forward price.

(ii) Under the Black-76 model of Definition 20.16,

$$\mathrm{Dig}^{\mathrm{put}}(K) = e^{-rT}\,\Phi(-d_2), \qquad \mathrm{AoN}^{\mathrm{put}}(K) = e^{-rT}F\,\Phi(-d_1).$$

Proof. We use the law of one price for (i) and the pricing rule of the model for (ii).

Part (i), the digitals. One digital and one digital put, bought at time 0 and held to T, pay $\mathbf{1}\{F_T > K\} + \mathbf{1}\{F_T \le K\} = 1$ in every state. By Lemma 16.8(iv), a position whose payoff is the constant 1 has the price e^{-rT}. So the two prices sum to e^{-rT}, whatever the law of F_T is: no model is needed.

Part (i), the asset-or-nothing claims. The pair pays $F_T\,\mathbf{1}\{F_T > K\} + F_T\,\mathbf{1}\{F_T \le K\} = F_T$ at T. We replicate that payment. At time 0 enter $e^{r(T_F-T)}$ long forwards for delivery at T_F with the delivery price F, which cost nothing (Definition 16.5), and deposit $e^{-rT}F$ in the bank. When $T_F = T$ there is one forward, and at T it pays $F_T - F$, since $F_T = S_T$. When $T_F > T$, the forward price at T for delivery at T_F is F_T, so at T each forward is worth $(F_T - F)e^{-r(T_F-T)}$ by Proposition 16.17(i), which holds for investment and consumption assets alike, and the forwards are closed there for $F_T - F$ in total. In both cases the deposit has grown to F, so the package is worth $F_T - F + F = F_T$ at T, and it cost $e^{-rT}F$. By Lemma 16.8(iii) the pair of claims, which pays the same amount, has the same price $e^{-rT}F$. No model of F_T was used. When F is a futures price, the constant rate of (A3) makes it the forward price for the same delivery date as well (Theorem 16.28), so the hypothesis on forwards holds.

Part (ii). The payoff functions $x \mapsto \mathbf{1}\{x \le K\}$ and $x \mapsto x\,\mathbf{1}\{x \le K\}$ are in $\mathcal{G}$: each has one discontinuity, and they are bounded by $1 \le x + x^{-1}$ and by $K \le K(x + x^{-1})$. So the pricing rule of Definition 20.16(ii) applies. By Steps 1 and 2 of the proof of Theorem 20.17,

$$\mathbb{Q}(F_T \le K) = P(Z \le -d_2) = \Phi(-d_2),$$

so $\mathrm{Dig}^{\mathrm{put}}(K) = e^{-rT}\,\Phi(-d_2)$. In the model, digital parity is the identity $\Phi(d_2) + \Phi(-d_2) = 1$. Next, $F_T\,\mathbf{1}\{F_T \le K\} = F_T - F_T\,\mathbf{1}\{F_T > K\}$, and $\mathbb{E}^{\mathbb{Q}}F_T = F$ by the drift-free check after Definition 20.16. Since expectation is linear, the proof of Proposition 20.19 gives $\mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T \le K\}\big] = F - F\,\Phi(d_1) = F\,\Phi(-d_1)$, so $\mathrm{AoN}^{\mathrm{put}}(K) = e^{-rT}F\,\Phi(-d_1)$, in agreement with the model-free sum of (i). Finally, under (20.6) the boundary event $\{F_T = K\} = \{Z = -d_2\}$ has probability zero, so in the model a digital put that pays on $\{F_T < K\}$ has the same price. □

Example 20.21 (The asset-or-nothing claim and the digital put on the same futures)**.** Keep the inputs of Example 20.18: $F = 100$, $K = 120$, $\sigma\sqrt{T} = 0.20$, $d_2 = -1.0116$ and $e^{-rT} = 0.9950$. Then $d_1 = -1.0116 + 0.20 = -0.8116$ and $\Phi(-0.8116) = 0.2085$, so by Proposition 20.19

$$\mathrm{AoN}(120) = 0.995 \times 100 \times 0.2085 = 20.75,$$

against $F \times \mathrm{Dig}(120) = 100 \times 0.1551 = 15.51$. The asset-or-nothing claim is worth more than "F times the digital", because it pays most in the highest states. Dividing the two prices gives the risk-neutral mean of F_T over the states in which the claims pay, the expected price given exercise:

$$\frac{\mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T > K\}\big]}{\mathbb{Q}(F_T > K)} = \frac{\mathrm{AoN}(120)}{\mathrm{Dig}(120)} = \frac{F\,\Phi(d_1)}{\Phi(d_2)} = \frac{100 \times 0.20851}{0.15586} = 133.78.$$

By Corollary 20.20 the digital put is worth $0.995 - 0.155 = 0.840$ (to four digits, $0.1551 + 0.8399 = 0.9950 = e^{-0.005}$), and the asset-or-nothing put is worth $99.50 - 20.75 = 78.75$. The difference

$$\mathrm{AoN}(120) - 120\,\mathrm{Dig}(120) = 20.75 - 18.61 = 2.14$$

is the price of the call struck at 120, as Theorem 20.22 in §20.4 proves (2.1366 before rounding).

(Problem P20.4 works this computation at different numbers.)

Interview trap. "The digital put is $1 - \mathrm{Dig}(K)$." It is $e^{-rT} - \mathrm{Dig}(K)$: digital parity holds with the discount factor, not with 1 (Corollary 20.20). The pair pays one unit at T, and one unit at T is worth e^{-rT} today.

Interview trap. "The asset-or-nothing claim is F times the digital." It is $e^{-rT}F\,\Phi(d_1)$, not $e^{-rT}F\,\Phi(d_2)$: the payment F_T is largest exactly in the states in which the claim pays (20.75 against 15.51 in Example 20.21).

Trading application. A digital is hedged with calls. With no model, for $0 < \varepsilon < K$, tight call spreads bracket it,

$$\frac{C(K-\varepsilon) - C(K)}{\varepsilon} \;\ge\; \mathrm{Dig}(K) \;\ge\; \frac{C(K) - C(K+\varepsilon)}{\varepsilon},$$

where $C(K)$ is the call price as a function of the strike, so that $\mathrm{Dig}(K) = -\partial_K C(K)$ wherever the call curve is differentiable in the strike: Proposition 23.2 in §23 proves both, and the bracket sits inside the band (17.4). The upper spread is the hedge of a sold digital, and so of a sold prediction-market binary. Under Black-76, Theorem 20.17 turns the identity into $\partial_K C(K) = -e^{-rT}\,\Phi(d_2)$: for the call of Example 20.21, minus the strike slope at $K = 120$ is the digital price 0.1551.

20.4 The Black–Scholes–Merton Formula

A call pays the asset above the strike and costs the strike above the strike:

$$(F_T - K)^+ = F_T\,\mathbf{1}\{F_T > K\} - K\,\mathbf{1}\{F_T > K\}.$$

Pricing the two pieces with §20.3 gives the Black–Scholes–Merton formula in the forward form that desks use, valid for every underlying whose forward price is lognormal under $\mathbb{Q}$. The stock formula is its first corollary, and the decomposition answers the most common conceptual question about the formula: what $\Phi(d_1)$ and $\Phi(d_2)$ are, and why they differ.

Setup. The model is the Black-76 model of Definition 20.16, with F, F_T, T_F, K, σ, r and Z as in the Setup of §20.3. The European call and put with strike K and expiry T pay $(F_T - K)^+$ and $(K - F_T)^+$ at T, where $x^+ = \max(x, 0)$. Their time-0 prices C and $\mathcal{P}$ are functions of F, K, σ, T and r. Both payoff functions are in the class $\mathcal{G}$ of Definition 19.19(iv): they are continuous, $(x-K)^+ \le x \le x + x^{-1}$ and $(K-x)^+ \le K \le K(x + x^{-1})$. The numbers d_1 and d_2 are those of §20.3. At a general time $t < T$ the formulas are read with $\tau = T - t$ in the place of T and with the forward price at t in the place of F. For a non-dividend stock, the formula at (S_t, t) is then the value at t of the replicating strategy, that is, the time-t replication cost (Theorem 20.7(iii), as the paragraph after Corollary 20.23 shows); no claim is made about a market price at t. For an asset with a yield and for a futures price the price functions at t are (20.12) of Corollary 20.41(i) and (20.15) of Theorem 20.52. In practice r is the zero rate $r(T)$ for the maturity of the option. The identity $\Phi(-y) = 1 - \Phi(y)$ (Step 2 of the proof of Theorem 20.17) is used throughout; Table 2 lists the tail values of Φ worth memorising, and the examples quote other values of Φ to four digits. The real-world drift μ and the real-world probability P of the section-wide Setup appear only in Example 20.27 and in the traps that follow it.

Theorem 20.22 (Black–Scholes–Merton formula, forward form)**.** *Under the Black-76 model of Definition 20.16, the European call and the European put with strike K and expiry T are worth*

$$C = e^{-rT}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big] \tag{20.8}$$

and

$$\mathcal{P} = e^{-rT}\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big], \tag{20.9}$$

where

$$d_1 = \frac{\ln(F/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = \frac{\ln(F/K) - \sigma^2 T/2}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}. \tag{20.10}$$

The two prices satisfy $C - \mathcal{P} = e^{-rT}(F - K)$ for every $\sigma > 0$. The same identity holds with no model of F_T, for every real r and every delivery date $T_F \ge T$, between the time-0 prices of a traded call and a traded put, whenever forward contracts for delivery at T_F can be entered at time 0 at the price F.

Proof. We decompose each payoff into an asset-or-nothing piece and a cash-or-nothing piece, and price the pieces with §20.3.

The call. For every $x > 0$,

$$(x - K)^+ = x\,\mathbf{1}\{x > K\} - K\,\mathbf{1}\{x > K\},$$

because both sides equal $x-K$ when $x>K$ and 0 otherwise. Put $x=F_T$. The three payoff functions are in $\mathcal{G}$, so the pricing rule of Definition 20.16(ii) applies to each of the three claims. Expectation is linear (Theorem 1.2, used in this Part for general random variables, as the Setup of §19.2 declares; here each expectation is an integral against the density φ, and the integral is linear). Hence

$$\begin{aligned} C = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(F_T-K)^+\big] &= e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T>K\}\big] - K\,e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[\mathbf{1}\{F_T>K\}\big] \\ &= \mathrm{AoN}(K) - K\,\mathrm{Dig}(K). \end{aligned}$$

Proposition 20.19 and Theorem 20.17 give $C = e^{-rT}F\,\Phi(d_1) - K\,e^{-rT}\,\Phi(d_2)$, which is (20.8).

The put. For every $x>0$,

$$(K-x)^+ = K\,\mathbf{1}\{x\le K\} - x\,\mathbf{1}\{x\le K\},$$

because both sides equal $K-x$ when $x\le K$ and 0 otherwise. The two pieces are the payoffs of K digital puts and of one asset-or-nothing put (Corollary 20.20). Directly: $\mathbf{1}\{x\le K\} = 1-\mathbf{1}\{x>K\}$, so by linearity, by the two results just used and by $\mathbb{E}^{\mathbb{Q}}F_T = F$,

$$\mathbb{E}^{\mathbb{Q}}\big[\mathbf{1}\{F_T\le K\}\big] = 1-\Phi(d_2) = \Phi(-d_2), \qquad \mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T\le K\}\big] = F - F\,\Phi(d_1) = F\,\Phi(-d_1).$$

The pricing rule then gives $\mathcal{P} = e^{-rT}\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big]$, which is (20.9).

The parity clause. Subtract (20.9) from (20.8) and use $\Phi(y)+\Phi(-y)=1$ twice:

$$C-\mathcal{P} = e^{-rT}\Big[F\big(\Phi(d_1)+\Phi(-d_1)\big) - K\big(\Phi(d_2)+\Phi(-d_2)\big)\Big] = e^{-rT}(F-K).$$

This identity also holds with no model. One call held long and one put held short pay $(F_T-K)^+ - (K-F_T)^+ = F_T-K$ at T, which is the payoff of the package of forwards and deposit in the proof of Corollary 20.20(i), worth $e^{-rT}F$, less a deposit paying K, worth Ke^{-rT}; the package is built from forwards for delivery at T_F entered at the price F, the hypothesis of the clause, and neither its construction nor Lemma 16.8 uses a sign of r. By Lemma 16.8(iii) and (iv), $C-\mathcal{P} = e^{-rT}(F-K)$ whatever the law of F_T. For $T_F=T$ and $r\ge 0$, the standing assumption of §17.2, this is the parity of Corollary 17.11(iii). So the two formulas satisfy model-free put–call parity identically, for every σ. □

(Problem P20.4 asks you to reproduce this proof from scratch.)

Corollary 20.23 (Non-dividend-paying stock)**.** *Assume (A1)–(A6). The European call and the European put with strike K and expiry T on the stock are worth*

$$C = S\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2), \qquad \mathcal{P} = Ke^{-rT}\,\Phi(-d_2) - S\,\Phi(-d_1),$$

with

$$d_1 = \frac{\ln(S/K) + (r+\sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T} = \frac{\ln(S/K) + (r-\sigma^2/2)T}{\sigma\sqrt{T}}.$$

Proof. We substitute the forward price of the stock into Theorem 20.22. Under (A1)–(A6) the Black-76 model holds for the stock, with $T_F=T$, $F_T=S_T$ and $F=Se^{rT}$, the forward price of Theorem 16.11: its law (i) is Proposition 20.8(i), and its pricing rule (ii) is Theorem 20.7(iv). Then $\ln(F/K) = \ln(S/K) + rT$ and $e^{-rT}F = S$. Substituting these in (20.8), (20.9) and (20.10) gives the statement. □

The formula as a price function. At a time $t<T$ write $\tau = T-t$. For the call payoff $f(x) = (x-K)^+$, the price function (20.5) of Theorem 20.7 is $V(S,t) = e^{-r\tau}\,\mathbb{E}\big(S\exp(\nu\tau+\sigma\sqrt{\tau}\,Z) - K\big)^+$, and

$$S\exp\big(\nu\tau+\sigma\sqrt{\tau}\,Z\big) = Se^{r\tau}\exp\big(-\tfrac{1}{2}\sigma^2\tau+\sigma\sqrt{\tau}\,Z\big)$$

is the right side of (20.6) with $Se^{r\tau}$ in the place of F and τ in the place of T. The computation that proved Theorem 20.22 used only that expression, so

$$V(S,t) = S\,\Phi(d_1) - Ke^{-r\tau}\,\Phi(d_2), \qquad d_1 = \frac{\ln(S/K) + (r+\sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau}.$$

The call formula, read as a function of (S,t), is therefore the function V of Theorem 20.7. By part (i) of that theorem it solves equation (20.4), and by part (iii) the strategy that holds $\partial_S V$ shares replicates the call: the formula at (S_t,t) is the value at t of that strategy, the time-t replication cost, and no claim is made about a market price at t. Example 20.25 checks both in numbers. When $r \geq 0$ the American call on a non-dividend stock has the same time-0 price, $C^{\text{am}} = C$ (Theorem 17.16(i)), and for $r > 0$ its early exercise is strictly suboptimal (part (ii) of that theorem); no formula of this kind is available for the American put, whose price Procedure 18.16 computes on a tree.

Example 20.24 (The running example: a one-year at-the-money call)**.** Take the running stock of this section, $S = 100$, $r = 5\%$, $\sigma = 20\%$, with no dividends, and a call and a put with $K = 100$ and $T = 1$. The forward price is $F = 100\,e^{0.05} = 105.13$, and

$$d_1 = \frac{0.05 + 0.02}{0.20} = 0.35, \qquad d_2 = 0.35 - 0.20 = 0.15,$$

with $\Phi(0.35) = 0.6368$, $\Phi(0.15) = 0.5596$ and $Ke^{-rT} = 95.12$. By Corollary 20.23,

$$C = 100 \times 0.6368 - 95.12 \times 0.5596 = 63.68 - 53.23 = 10.45 :$$

the asset-or-nothing claim minus K digitals. The forward form (20.8) gives the same number:

$$e^{-0.05}\,(105.13 \times 0.6368 - 100 \times 0.5596) = 0.9512 \times 10.99 = 10.45.$$

For the put, $\Phi(-0.15) = 0.4404$ and $\Phi(-0.35) = 0.3632$, so

$$\mathcal{P} = 95.12 \times 0.4404 - 100 \times 0.3632 = 41.89 - 36.32 = 5.57.$$

Parity holds: $10.45 - 5.57 = 4.88 = 100 - 95.12$ (Theorem 17.9), the forward value of Example 20.2. The risk-neutral probability of exercise is $\Phi(d_2) = 0.560$, against $\Phi(d_1) = 0.637$; Proposition 20.26 below says what the second number is. As in Example 20.21, their ratio gives the risk-neutral mean of S_T over the exercise states, $F\,\Phi(d_1)/\Phi(d_2) = 105.13 \times 0.6368/0.5596 = 119.63$.

This closes the promise made in Theorem 18.37, whose part (ii) says that the n-step CRR call prices converge to $e^{-rT}\,\mathbb{E}\big(Se^{\nu T + \sigma\sqrt{T}\,Z} - K\big)^+$, the number just computed; here n is the number of tree steps, as in §18. For this option, the CRR prices C_n of Example 18.39 at $n = 1, 2, 3, 5, 10, 50, 100$ are 12.16, 9.54, 11.04, 10.81, 10.25, 10.41 and 10.43, and at $n = 500$, a depth outside that table, the tree gives 10.45: they oscillate into 10.45. At $n = 100$ the error is -0.020 (-0.01997 before rounding), of the order $1/n$ recorded in Fact 18.40. Theorem 18.37 itself proves the rate $n^{-1/2}$ for smooth test functions only (its part (i)); for the call it proves convergence without a rate (its part (ii)).

(Problem P20.3 works this computation at different numbers.)

Interview trap. Writing the stock formula as $S\,\Phi(d_1) - K\,\Phi(d_2)$, with the discount on the strike forgotten. The forward form (20.8) puts e^{-rT} on *both* legs, which is the cleanest way to remember it: compute the bracket in forward terms, then discount the whole of it, as the second display of Example 20.24 does.

Example 20.25 (The equation and the replicating strategy, checked on the running call)**.** Take the call of Example 20.24 at time 0: $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ and $C = 10.4506$.

This example checks Theorem 20.7(i) and the strategy of its part (iii) on the price function $V(S,t) = S\,\Phi(d_1) - Ke^{-r\tau}\,\Phi(d_2)$. Central differences of that function, with the steps 0.01 in S and 10^{-5} in t, give

$$\partial_S C = 0.6368, \qquad \partial_{SS} C = 0.018762, \qquad \partial_t C = -6.4140 \text{ a year},$$

in agreement to five digits with the closed forms that §21 proves (Propositions 21.2 and 21.6). The left side of equation (20.4) is then

$$\begin{aligned} \partial_t C + rS\,\partial_S C + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} C &= -6.41403 + 5 \times 0.63683 + 200 \times 0.018762 \\ &= -6.41403 + 3.18415 + 3.75240 = 0.52252, \end{aligned}$$

and the right side is $rC = 0.05 \times 10.45058 = 0.52253$. The last digits differ only through the rounding of the three terms: before rounding, the central differences sum to 0.5225291, against $rC = 0.5225292$. Equation (20.4) holds at this point, as Theorem 20.7(i) says it must.

The same check can be done by hand from the closed forms of Proposition 21.6, with $\varphi(d_1) = \varphi(0.35) = 0.37524$:

$$\partial_{SS} C = \frac{\varphi(d_1)}{S\sigma\sqrt{T}} = \frac{0.37524}{20} = 0.018762,$$

$$\partial_t C = -\frac{S\sigma\,\varphi(d_1)}{2\sqrt{T}} - rKe^{-rT}\,\Phi(d_2) = -3.75240 - 2.66162 = -6.41403.$$

The term $\frac{1}{2}\sigma^2 S^2\,\partial_{SS} C = 3.75240$ cancels the first part of $\partial_t C$, and with $\partial_S C = \Phi(d_1)$ (Proposition 21.2) what is left is $rS\,\Phi(d_1) - rKe^{-rT}\,\Phi(d_2) = rC$: the equation balances because the gamma term pays exactly for the part of the time decay that volatility causes.

The replicating strategy of Theorem 20.7(iii) holds $\partial_S C = 0.6368$ shares, worth 63.6831, and $C - S\,\partial_S C = 10.4506 - 63.6831 = -53.2325$ in the bank at time 0. That is the reading "hold $\Phi(d_1)$ shares and borrow $Ke^{-rT}\,\Phi(d_2)$" of the Intuition after Proposition 20.26: $\Phi(d_1) = 0.6368$ and $Ke^{-rT}\,\Phi(d_2) = 95.1229 \times 0.559618 = 53.2325$.

Figure 40 plots the call price of Corollary 20.23 against the stock price for the running inputs, at two maturities, between the model-free bounds of Proposition 17.6.

Proposition 20.26 ($\Phi(d_1)$ as a reweighted exercise probability)**.** *Under the law (20.6) of the Black-76 model, the two normal probabilities of Theorem 20.22 have the following meanings.*

(i) The number $\Phi(d_2)$ is the risk-neutral probability that the call is exercised: $\Phi(d_2) = \mathbb{Q}(F_T > K)$.

(ii) The number $\Phi(d_1)$ is the probability of exercise when each outcome is reweighted by F_T/F, the terminal value of the underlying relative to its risk-neutral mean:

$$\Phi(d_1) = \mathbb{E}^{\mathbb{Q}}\Big[\frac{F_T}{F}\,\mathbf{1}\{F_T > K\}\Big].$$

The weights F_T/F are positive and have $\mathbb{Q}$-mean 1.

(iii) The reweighted probability is the larger one: $\Phi(d_1) > \Phi(d_2)$.

The literature calls the reweighted probability of (ii) the probability of exercise under the share measure*; no measure theory is needed for the statement.*

Proof. We read each part off a proof of §20.3. Part (i) is Steps 1 and 2 of the proof of Theorem 20.17. For part (ii), the proof of Proposition 20.19 showed that $\mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T > K\}\big] = F\,\Phi(d_1)$; divide by $F > 0$. The weights are positive because $F_T > 0$, and $\mathbb{E}^{\mathbb{Q}}[F_T/F] = 1$ because $\mathbb{E}^{\mathbb{Q}} F_T = F$ (the drift-free check after Definition 20.16). For part (iii), $d_1 - d_2 = \sigma\sqrt{T} > 0$, and Φ is strictly increasing because its derivative φ is positive. □

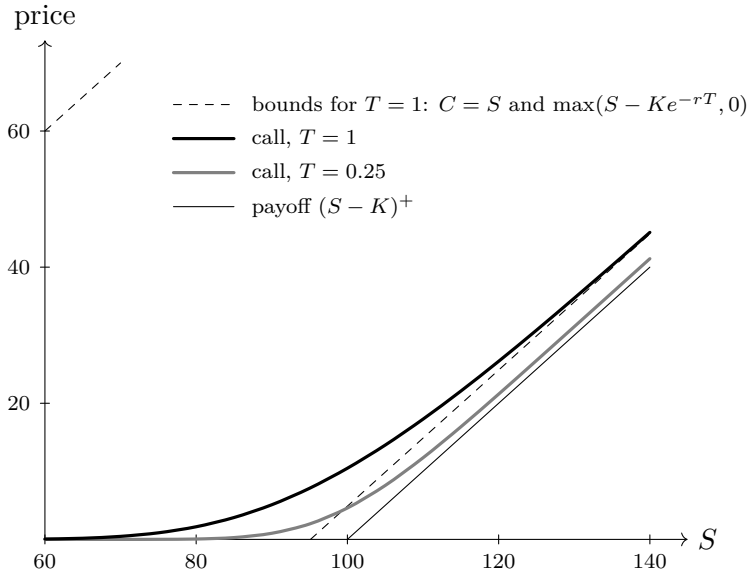


Figure 40: The call price of Corollary 20.23 against S, for $K = 100$, $r = 5\%$ and $\sigma = 20\%$. Thick black: $T = 1$ (prices 0.05, 0.44, 1.86, 5.09, 10.45, 17.66, 26.17, 35.44, 45.11 at $S = 60, 70, \dots, 140$). Thick gray: $T = 0.25$ (prices 0.00, 0.00, 0.06, 0.90, 4.61, 11.99, 21.35, 31.25, 41.24). Dashed: the bounds of Proposition 17.6 for $T = 1$, the upper bound $C = S$ (inside the frame only for $S \le 70$) and the lower bound $\max(S - Ke^{-rT}, 0)$, which leaves the axis at $Ke^{-rT} = 95.12$. Thin black: the payoff $(S - K)^+$, the limit $T \to 0$. In the figure each curve is convex and lies strictly between its own bounds (those of the $T = 0.25$ curve are not drawn), and the price rises with T, as Theorem 17.16(iii) requires. The excess over the payoff is largest at the strike (10.45 at $S = 100$ for $T = 1$), and deep in the money the $T = 1$ curve hugs its lower bound (45.11 against 44.88 at $S = 140$).

Intuition. The strike is paid exactly when the option is exercised, so the weight of the K-leg is the exercise probability $\Phi(d_2)$. The asset is received in the states with a high F_T, which the reweighting by F_T/F favours, so the weight $\Phi(d_1)$ of the asset leg is larger. Read as a replicating portfolio for a non-dividend stock, the formula says: hold $\Phi(d_1)$ shares and borrow $Ke^{-rT}\,\Phi(d_2)$. That $\Phi(d_1)$ is the number of shares, the delta, is proved in §21 (Proposition 21.2); Example 20.25 checked it numerically.

Example 20.27 (Three probabilities of exercise)**.** Take the stock and the call of Example 20.24, and let the real-world drift be $\mu = 10\%$. Three numbers answer to the name "the probability that the call is exercised", and they differ.

- The real-world probability. By Proposition 19.42(iii),

$$P(S_T > K) = \Phi\Big(\frac{\ln(S/K) + (\mu - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\Big) = \Phi\Big(\frac{0.10 - 0.02}{0.20}\Big) = \Phi(0.40) = 0.655;$$

  the argument 0.40 is d_2 with μ in the place of r.
- The risk-neutral probability, $\mathbb{Q}(S_T > K) = \Phi(d_2) = \Phi(0.15) = 0.560$: the same expression with μ replaced by r.
- The reweighted probability of Proposition 20.26(ii), $\Phi(d_1) = \Phi(0.35) = 0.637$.

Only the last two enter the price. The real-world expected payoff is computed as the call price was. Under P, $S_T = Se^{\mu T}\exp\big(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z\big)$ (Theorem 19.40), which is the right side of (20.6) with $Se^{\mu T}$ in the place of F. The computation that proved Theorem 20.22 therefore gives,

with 0.40 in the place of d_2 and $0.40 + \sigma\sqrt{T} = 0.60$ in the place of d_1,

$$\begin{aligned}\mathbb{E}\big[(S_T - K)^+\big] &= Se^{\mu T}\,\Phi(0.40 + \sigma\sqrt{T}) - K\,\Phi(0.40)\\ &= 110.517 \times \Phi(0.60) - 100 \times \Phi(0.40) = 80.208 - 65.542 = 14.67,\end{aligned}$$

with $\Phi(0.60) = 0.72575$ and $\Phi(0.40) = 0.65542$. Discounted at r this is $14.67\,e^{-0.05} = 13.95$, and discounted at μ it is $14.67\,e^{-0.10} = 13.27$. Both differ from the price 10.45. The continuously compounded rate that discounts 14.67 to 10.45 is $\ln(14.67/10.45) = 33.9\%$ a year. That is the real-world expected return of the call, more than three times that of the stock: the call is a position in 63.68 of stock financed by a loan of 53.23. This is the check promised by the trap in §20.2; Remark 18.8, which follows the tree's risk-neutral probability of Definition 18.6, makes the same point on a one-step tree.

(Problem P20.5 works this computation at different numbers.)

Interview trap. "The probability that the call finishes in the money is $N(d_1)$." That probability is $\Phi(d_2)$, the literature's $N(d_2)$, and only under $\mathbb{Q}$. The number $\Phi(d_1)$, the literature's $N(d_1)$, is the reweighted probability of Proposition 20.26 and, as §21 proves (Proposition 21.2), the delta of the call. The three candidate numbers of Example 20.27, 0.655, 0.560 and 0.637, all differ.

Interview trap. "The probability that the stock ends above K is $\Phi(d_2)$." Only under $\mathbb{Q}$. The real-world probability has μ in the place of r (Proposition 19.42(iii)) and is larger for a stock with $\mu > r$: 0.655 against 0.560 in Example 20.27.

20.5 Sanity Checks and the At-the-Money Approximation

Pricing a one-year at-the-money call on a stock with 20% volatility without paper is the most common mental-arithmetic question about options; the answer, about $0.4\,S\sigma\sqrt{T}$ plus a carry correction, is the first-order Taylor expansion of the formula at the forward price. This subsection first checks the formula against its limits and against the model-free bounds of Proposition 17.6, then proves the at-the-money rule with a two-sided error bound from Theorem 7.31, and reads the at-the-money straddle as the discounted expected absolute move.

Setup. The model is the Black-76 model of Definition 20.16, and C, $\mathcal{P}$, d_1 and d_2 are as in Theorem 20.22. The total volatility $\sigma\sqrt{T}$ and the log-moneyness $\ln(F/K)$ are written out and get no letters of their own: an italic letter for the first would sit beside the log-drift ν, and m counts the earlier fixings in Theorem 20.7(v). In these terms (20.10) reads

$$d_1 = \frac{\ln(F/K)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2}, \qquad d_2 = \frac{\ln(F/K)}{\sigma\sqrt{T}} - \frac{\sigma\sqrt{T}}{2}.$$

An option is *at the money forward* when $K = F$, and *at the money spot* when $K = S$. The *straddle* with strike K is one call and one put with that strike and the same expiry, worth $C + \mathcal{P}$. The function $\chi(y) = \Phi(y) - \frac{1}{2}$ is a local auxiliary function of the proof of Proposition 20.29. The linear rule of Proposition 20.29 is written $C^{\mathrm{lin}} = e^{-rT} F\,\sigma\sqrt{T}/\sqrt{2\pi}$, a price with a qualifier, written as a superscript as in C^{am}. A normal (price-unit) volatility is written σ_{N}; it is used only in Remark 20.31 and in Example 20.32(4). As in §7.5, $O((\sigma\sqrt{T})^3)$ denotes a quantity bounded by a constant multiple of $(\sigma\sqrt{T})^3$.

Remark 20.28 (Limits of the formula and the no-arbitrage bounds). Every limit below is read off (20.8) from the signs of d_1 and d_2 in the form of the Setup, using $\Phi(y) \to 1$ as $y \to \infty$, $\Phi(y) \to 0$ as $y \to -\infty$ and $\Phi(0) = \frac{1}{2}$. No proposition rests on this remark: Proposition 20.38 proves again the two limits in σ that it needs.

(1) As $\sigma \to 0$, with F, K and T fixed. If $F > K$, then $\ln(F/K)/(\sigma\sqrt{T}) \to +\infty$ while $\sigma\sqrt{T}/2 \to 0$, so d_1 and d_2 tend to $+\infty$ and $C \to e^{-rT}(F - K)$. If $F < K$, both tend to $-\infty$ and $C \to 0$. If

$F = K$, then $d_1 = \sigma\sqrt{T}/2$ and $d_2 = -\sigma\sqrt{T}/2$ tend to 0, both terms of the bracket in (20.8) tend to $F/2$, and $C \to 0$. In all three cases

$$C \to e^{-rT}(F-K)^+,$$

the discounted payoff at the forward price: without volatility the value at expiry is known today.

(2) As $T \to 0$, with S, K and σ fixed, for an asset with a yield q ($q = 0$ for a non-dividend stock), so that $F = Se^{(r-q)T} \to S$ and $e^{-rT} \to 1$. If $S > K$, then $\ln(F/K) \to \ln(S/K) > 0$ while $\sigma\sqrt{T} \to 0$, so d_1 and d_2 tend to $+\infty$ and $C \to S - K$. If $S < K$, both tend to $-\infty$ and $C \to 0$. If $S = K$, then $\ln(F/K) = (r-q)T$, so d_1 and d_2 equal $(r-q)\sqrt{T}/\sigma \pm \sigma\sqrt{T}/2$ and tend to 0, and $C \to \frac{1}{2}S - \frac{1}{2}K = 0$. So $C \to (S-K)^+$: at expiry the formula returns the payoff.

(3) As $\sigma \to \infty$, with F, K and T fixed: $\ln(F/K)/(\sigma\sqrt{T}) \to 0$ and $\sigma\sqrt{T}/2 \to \infty$, so $d_1 \to +\infty$ and $d_2 \to -\infty$, and

$$C \to e^{-rT}F,$$

which is Se^{-qT} for an asset with a yield and S for a non-dividend stock: for $r \geq 0$, the standing assumption of Proposition 17.6, this is the upper bound of its part (ii).

(4) As $S \to \infty$ and as $S \to 0$, for the non-dividend stock of Corollary 20.23, with K, σ, T and r fixed. By the parity clause of Theorem 20.22, $C - (S - Ke^{-rT}) = \mathcal{P}$, and

$$0 \leq \mathcal{P} \leq Ke^{-rT}\,\Phi(-d_2)\,:$$

the put is the discounted expectation of a nonnegative payoff, and the term $S\,\Phi(-d_1)$ subtracted in its formula is nonnegative. As $S \to \infty$, $d_2 \to +\infty$, so $\mathcal{P} \to 0$ and $C - (S - Ke^{-rT}) \to 0$: deep in the money the call approaches its lower bound. As $S \to 0$, $0 \leq C \leq S\,\Phi(d_1) \leq S \to 0$.

(5) Strict bounds. For every value of $\sigma\sqrt{T}$ in $(0, \infty)$,

$$e^{-rT}(F-K)^+ < C < e^{-rT}F.$$

For the upper bound, (20.8) and $1 - \Phi(d_1) = \Phi(-d_1)$ give $e^{-rT}F - C = e^{-rT}\big[F\,\Phi(-d_1) + K\,\Phi(d_2)\big] > 0$; in payoff terms, $(F_T - K)^+ < F_T$. For the lower bound, the call payoff is at least $K\,\mathbf{1}\{F_T > 2K\}$, because $F_T - K > K$ on that event, and the put payoff is at least $\frac{1}{2}K\,\mathbf{1}\{F_T \leq K/2\}$, because $K - F_T \geq K/2$ on that event. Expectation is monotone (Setup of §19.2). The event $\{F_T > 2K\}$ has the $\mathbb{Q}$-probability $\Phi(d_2)$ computed at the strike $2K$ (Steps 1 and 2 of the proof of Theorem 20.17), and the event $\{F_T \leq K/2\}$ has the $\mathbb{Q}$-probability $\Phi(-d_2)$ computed at the strike $K/2$ (the proof of Corollary 20.20(ii)); both are positive, because Φ is positive everywhere. So $C > 0$ and $\mathcal{P} > 0$, and the parity clause of Theorem 20.22 gives $C - e^{-rT}(F-K) = \mathcal{P} > 0$. The two outer terms are model-free bounds: for $r \geq 0$ they are those of Proposition 17.6(ii)–(iii) for an option on the underlying ($T_F = T$), and, for every real r, those of Proposition 20.54(i) for an option on a futures price. So the formula stays strictly inside them and never attains them. The gap to the upper bound closes in limit (3), and the gap to the lower bound in limits (1), (2) and (4); as $S \to 0$ in (4) both bounds tend to 0, so both gaps close there.

Interview trap. "As $\sigma \to \infty$ the call becomes worthless", or "infinitely valuable". It tends to $e^{-rT}F$, which is S for a non-dividend stock, the upper bound of Proposition 17.6(ii) when $r \geq 0$. The asset-or-nothing leg tends to the whole asset while $\mathbb{Q}(F_T > K) = \Phi(d_2) \to 0$: the holder ends up with the value of the asset and almost never pays the strike (Remark 20.28(3)).

Proposition 20.29 (At-the-money-forward approximation)**.** *Under the Black-76 model of Definition 20.16, let $K = F$. Then*

$$C = \mathcal{P} = e^{-rT}F\left[2\,\Phi\Big(\frac{\sigma\sqrt{T}}{2}\Big) - 1\right],$$

and

$$e^{-rT}F\,\frac{\sigma\sqrt{T}}{\sqrt{2\pi}}\left(1 - \frac{\sigma^2 T}{24}\right) \;\leq\; C \;\leq\; e^{-rT}F\,\frac{\sigma\sqrt{T}}{\sqrt{2\pi}}. \tag{20.11}$$

Hence, with $C^{\text{lin}} = e^{-rT} F\,\sigma\sqrt{T}/\sqrt{2\pi}$ *the linear rule,*

$$C = C^{\text{lin}} + O((\sigma\sqrt{T})^3), \qquad 0 \;\le\; C^{\text{lin}} - C \;\le\; e^{-rT} F\,\frac{(\sigma\sqrt{T})^3}{24\sqrt{2\pi}} = C^{\text{lin}}\,\frac{\sigma^2 T}{24}\,:$$

the linear rule exceeds the price by at most the fraction $\sigma^2 T/24$ *of the rule,* $(C^{\text{lin}} - C)/C^{\text{lin}} \le \sigma^2 T/24$. *Measured against the price, when* $\sigma^2 T < 24$,

$$\frac{C^{\text{lin}} - C}{C} \;\le\; \frac{\sigma^2 T/24}{1 - \sigma^2 T/24}.$$

Since $1/\sqrt{2\pi} = 0.3989$, *the call and the put are each worth about* $0.4\,e^{-rT} F\,\sigma\sqrt{T}$, *and the straddle about* $0.8\,e^{-rT} F\,\sigma\sqrt{T}$.

Sketch of the proof. The proof is a Taylor expansion of the normal distribution function at 0 (Theorem 7.31). At $K = F$, $d_1 = \sigma\sqrt{T}/2 = -d_2$, so (20.8) and $\Phi(-y) = 1 - \Phi(y)$ give exactly $C = 2e^{-rT} F\,\chi(\sigma\sqrt{T}/2)$, and the put equals the call by parity. For $y \ge 0$, $\chi(y) = \int_0^y \varphi(z)\,dz \le \varphi(0)\,y$, as $\varphi \le \varphi(0) = 1/\sqrt{2\pi}$. For the lower bound, $\chi''(0) = 0$ and $\chi''' \ge -\varphi(0)$ (the appendix locates the extremes of χ'''), so the integral remainder (7.3) gives

$$\chi(y) = \varphi(0)\,y + \frac{1}{2}\int_0^y (y - z)^2\,\chi'''(z)\,dz \;\ge\; \varphi(0)\,y\left(1 - \frac{y^2}{6}\right).$$

At $y = \sigma\sqrt{T}/2$, multiplied by $2e^{-rT} F$, the two bounds are (20.11), since $y^2/6 = \sigma^2 T/24$; the error statements follow by subtraction and division. The complete proof is in Appendix F.5. □

(Problem P20.6 asks you to reproduce the complete proof from scratch.)

Example 20.32 works the rule and its error bar in numbers, once the next two statements have added the straddle and the normal model.

Corollary 20.30 (The straddle prices the expected absolute move)**.** *Under the Black-76 model of Definition 20.16:*

(i) for every strike $K > 0$ *the straddle is worth exactly* $C + \mathcal{P} = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\,|F_T - K|$;

(ii) at $K = F$,

$$\sqrt{\frac{2}{\pi}}\,e^{-rT} F\,\sigma\sqrt{T}\left(1 - \frac{\sigma^2 T}{24}\right) \;\le\; C + \mathcal{P} \;\le\; \sqrt{\frac{2}{\pi}}\,e^{-rT} F\,\sigma\sqrt{T},$$

and $\sqrt{2/\pi}\;F\sigma\sqrt{T} = \mathbb{E}\,|F\sigma\sqrt{T}\,Z|$ *is the mean absolute deviation of a normal move with standard deviation* $F\sigma\sqrt{T}$, *with* $\sqrt{2/\pi} = 0.798$.

By (i) the at-the-money-forward straddle is exactly the discounted expected absolute move of the forward price, $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}|F_T - F|$. *By (ii), to the order of Proposition 20.29, that move is* $\sqrt{2/\pi}\;F\sigma\sqrt{T}$, *the mean absolute deviation of a normal move with standard deviation* $F\sigma\sqrt{T}$.

Proof. We add the two payoffs for (i), and for (ii) we double Proposition 20.29 and compute $\mathbb{E}|Z|$.

Part (i). For every real x, $|x| = x^+ + (-x)^+$: for $x \ge 0$ the right side is $x + 0$, and for $x < 0$ it is $0 + (-x)$. At $x = F_T - K$ this reads $|F_T - K| = (F_T - K)^+ + (K - F_T)^+$. The payoff function $x \mapsto |x - K|$ is in $\mathcal{G}$: it is continuous, and $|x - K| \le x + K \le (1 + K)(x + x^{-1})$. By the pricing rule of Definition 20.16(ii) and the linearity of expectation, $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}|F_T - K| = C + \mathcal{P}$.

Part (ii). At $K = F$ the put equals the call (Proposition 20.29), so $C + \mathcal{P} = 2C$, and doubling (20.11) gives the two bounds, because $2/\sqrt{2\pi} = \sqrt{2/\pi}$. For the last claim, by Fact 7.22 and the symmetry of φ,

$$\mathbb{E}|Z| = \int_{-\infty}^{\infty} |z|\,\varphi(z)\,dz = 2\int_0^{\infty} z\,\varphi(z)\,dz = 2\Big[-\varphi(z)\Big]_{z=0}^{z\to\infty} = 2\,\varphi(0) = \sqrt{\frac{2}{\pi}} = 0.798,$$

because the derivative of $-\varphi(z)$ is $z\,\varphi(z)$ (Step 2 of the proof of Proposition 20.29) and $\varphi(z) \to 0$ as $z \to \infty$; the constant $1/\sqrt{2\pi}$ in $\varphi(0)$ is the one fixed by Theorem 7.5. Hence $\mathbb{E}\,|F\sigma\sqrt{T}\,Z| = F\sigma\sqrt{T}\;\mathbb{E}|Z| = \sqrt{2/\pi}\;F\sigma\sqrt{T}$. The same integral gives $\mathbb{E}|Z| = \sqrt{2/\pi}$ in Example 19.11 of §19, and Theorem 19.63(i) evaluates it once more for the expected running maximum of a Brownian motion, $\mathbb{E}\,W_T^{\max} = \sqrt{2T/\pi}$. □

Remark 20.31 (The normal model; Bachelier, 1900). Suppose instead, in this remark only, that under $\mathbb{Q}$ the value at expiry is normal,

$$F_T = F + \sigma_{\mathrm{N}}\sqrt{T}\,Z,$$

with a volatility σ_{N} measured in price units, and that prices are again discounted $\mathbb{Q}$-expectations. In this model prices can become negative. The at-the-money-forward call is then worth exactly

$$e^{-rT}\,\mathbb{E}\big[(\sigma_{\mathrm{N}}\sqrt{T}\,Z)^+\big] = e^{-rT}\,\sigma_{\mathrm{N}}\sqrt{T}\;\mathbb{E}[Z^+] = e^{-rT}\,\sigma_{\mathrm{N}}\sqrt{\frac{T}{2\pi}},$$

since $\mathbb{E}[Z^+] = \int_0^\infty z\,\varphi(z)\,dz = \varphi(0) = 1/\sqrt{2\pi}$, the integral computed in the proof of Corollary 20.30. Matching $\sigma_{\mathrm{N}} = F\sigma$ reproduces the leading term of Proposition 20.29. Near the money, normal and lognormal quotes are therefore related by $\sigma_{\mathrm{N}} \approx F\sigma$: this is the bridge between a volatility quoted in price units (in basis points, for rates) and a lognormal volatility. The normal model survives for spreads and for rates, which can be negative, as §19 notes.

Example 20.32 (The at-the-money rule in the head)**.** *(1) The rule and its error bar.* Take $F = 100$, $\sigma = 20\%$, $T = 1$, $r = 0$ and $K = F$. The 0.4 rule gives $0.4 \times 100 \times 0.20 = 8.00$. The rule with $\varphi(0) = 1/\sqrt{2\pi} = 0.39894$ gives $C^{\mathrm{lin}} = 0.39894 \times 100 \times 0.20 = 7.979$. The exact value is $100\,(2\,\Phi(0.1) - 1) = 100 \times 0.07966 = 7.966$. The error of the $\varphi(0)$ rule, $7.97885 - 7.96557 = 0.01328$, sits just inside the bound $100 \times 0.2^3/(24\sqrt{2\pi}) = 0.01330$ of Proposition 20.29; said differently, the lower bound of (20.11), 7.9655, is within 2×10^{-5} of the exact value. The straddle is worth 15.93, against the normal mean absolute deviation $100 \times 0.2 \times 0.798 = 15.96$ of Corollary 20.30.

(2) One day. A one-day at-the-money straddle on 100 at 16 vol, with $r = 0$: $\sigma\sqrt{T} = 0.16/\sqrt{252} = 0.01008$, the "16 vol is one percent a day" anchor of Corollary 19.2. The straddle is worth about $0.8 \times 100 \times 0.01008 = 0.806$ (exactly 0.804): four-fifths of the daily standard deviation, the expected absolute daily move.

(3) Reading a straddle backwards. A one-month at-the-money-forward straddle quoted at 6.40 on a forward price of 100, with $r = 0$, implies

$$\sigma \approx \frac{6.40}{0.798 \times 100 \times \sqrt{1/12}} = \frac{6.40}{23.04} = 27.8\%,$$

and the exact inversion of $C + \mathcal{P} = 200\,(2\,\Phi(\sigma\sqrt{T}/2) - 1)$ gives 27.79%.

(4) Bachelier. With $\sigma_{\mathrm{N}} = 20$ a year, the value $F\sigma$ for $F = 100$ and $\sigma = 20\%$, and with $T = 1$ and $r = 0$, Remark 20.31 gives $20/\sqrt{2\pi} = 7.98$, against the lognormal 7.97 of part (1).

(Problem P20.6 works this computation at different numbers.)

Figure 41 shows the exact at-the-money-forward price per unit of discounted forward price, $2\,\Phi(\sigma\sqrt{T}/2) - 1$, against the linear rule, which is its tangent at 0.

Interview trap. The rule is a small-volatility statement, and doubling the volatility only approximately doubles an at-the-money price. Measured against the exact price, the linear rule overprices by 1% at $\sigma\sqrt{T} = 0.5$, by 4% at 1 and by 17% at 2 (Figure 41), within the bound $(\sigma^2 T/24)/(1 - \sigma^2 T/24)$ of Proposition 20.29, which is 20% at $\sigma\sqrt{T} = 2$. The error matters for long-dated or very high-volatility options: 80 vol over one year is already $\sigma\sqrt{T} = 0.8$, where the rule is 2.7% too high.

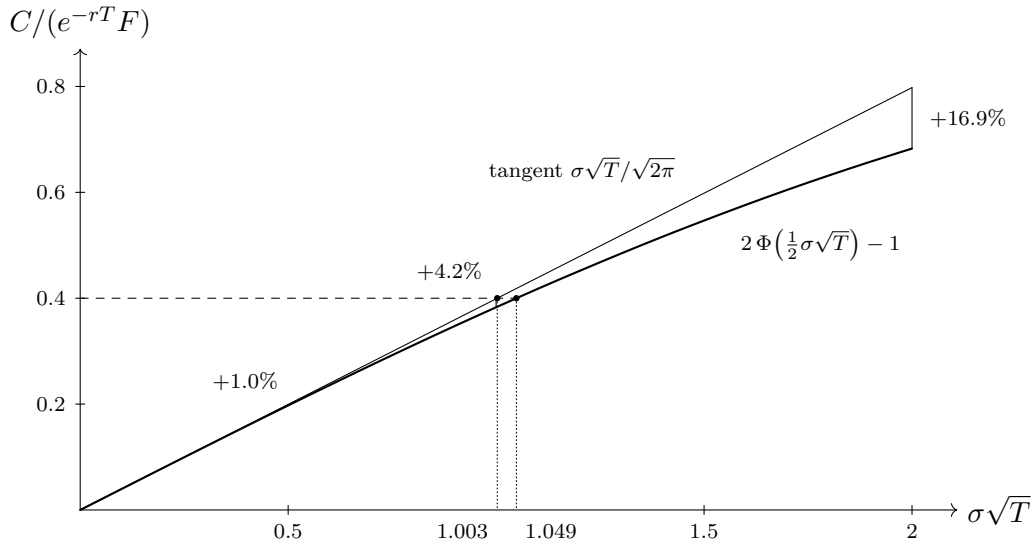


Figure 41: The at-the-money-forward price per unit of discounted forward price, $2\,\Phi(\sigma\sqrt{T}/2) - 1$ (thick), against $\sigma\sqrt{T}$, with its tangent at 0, the linear rule $\sigma\sqrt{T}/\sqrt{2\pi}$ (thin). The tangent lies above the curve (Proposition 20.29): measured against the price, the linear rule overstates it by 1.0% at $\sigma\sqrt{T} = 0.5$, by 4.2% at 1 and by 16.9% at 2, within the bounds $(\sigma^2T/24)/(1 - \sigma^2T/24) = 1.1\%$, 4.3% and 20.0% of that proposition. The dashed level 0.40 meets the tangent at $\sigma\sqrt{T} = 1.003$ and the curve at $\sigma\sqrt{T} = 1.049$: inverting a quoted price with the linear rule gives a volatility to the left of the true one. That first point is the Brenner–Subrahmanyam start of Newton's iteration in Proposition 20.38(iii).

Example 20.33 (The interview question: at the money spot)**.** The interview question asks for an at-the-money *spot* call, $K = S$, not $K = F$. The identity

$$C = \tfrac{1}{2}\,(C + \mathcal{P}) + \tfrac{1}{2}\,(C - \mathcal{P})$$

is exact. Its second half needs no model: $\frac{1}{2}(C - \mathcal{P}) = \frac{1}{2}(S - Ke^{-rT})$ on a non-dividend stock, by parity (Theorem 17.9). Its first half is half a straddle, which changes little when the strike moves from F to S, so its at-the-money-forward value $0.4\,e^{-rT}F\,\sigma\sqrt{T} = 0.4\,S\sigma\sqrt{T}$ can replace it. For the one-year call and put of Example 20.24, with $S = K = 100$, $r = 5\%$ and $\sigma = 20\%$:

$$C \approx 0.4 \times 100 \times 0.20 + \tfrac{1}{2} \times 4.88 = 8.00 + 2.44 = 10.44, \qquad \mathcal{P} \approx 8.00 - 2.44 = 5.56,$$

against the exact 10.45 and 5.57. The exact half-straddle at $K = S$ is $(10.45 + 5.57)/2 = 8.01$, against 7.97 at $K = F = 105.13$: moving the strike from the forward price to the spot price changes it by less than 0.05. The reason is that the straddle is flattest in the strike at the risk-neutral median $Fe^{-\sigma^2T/2} = 103.05$ of S_T, where the risk-neutral probabilities of finishing above and below the strike are equal. The strike slope of the straddle is $e^{-rT}\big(1 - 2\,\Phi(d_2)\big)$, which vanishes there because $d_2 = 0$ at that strike: the strike slope of the call is minus the digital, $-e^{-rT}\,\Phi(d_2)$ (Proposition 23.2 in §23 with Theorem 20.17), and by parity that of the put is larger by e^{-rT}. The slope is -0.113 at $K = 100$ and $+0.076$ at $K = F$, so the straddle barely moves between the two strikes. The half-carry term $(S - Ke^{-rT})/2 = 2.44$ is about $SrT/2 = 2.50$, by the expansion $e^{-rT} = 1 - rT + O\big((rT)^2\big)$ of Proposition 7.34. In words: an at-the-money spot call on a non-dividend stock is worth about $0.4\,S\sigma\sqrt{T}$ plus half the carry SrT, and the put about $0.4\,S\sigma\sqrt{T}$ minus half the carry.

(Problem P20.6 works this computation at different numbers.)

Interview trap. At the money *forward* is not at the money *spot*. The 0.4 rule is for $K = F$. At $K = S$ on a non-dividend stock the call exceeds the put by $S - Ke^{-rT}$, which is 4.88 at

$r = 5\%$ and $T = 1$ (Theorem 17.9), so quoting 8.00 for both is wrong by about 2.4 each way. The half-carry correction of Example 20.33 repairs it.

20.6 Vega and Implied Volatility

Every input of Theorem 20.22 can be observed except the volatility: the forward price, the strike, the expiry and the rate are read off the market or the contract, and σ is not. Option markets therefore quote an option by the one number that reproduces its price, the implied volatility. This subsection proves that the number exists and is unique, because the price is continuous and strictly increasing in σ between the no-arbitrage bounds $e^{-rT}(F-K)^+$ and $e^{-rT}F$. It computes the slope, the vega, through one identity that §21 reuses for every Greek. And it inverts the formula by Newton's method (Definition 7.41), started from the at-the-money rule of §20.5.

Setup. The model is the Black-76 model of Definition 20.16, with the forward or futures price $F > 0$ for delivery at $T_F \geq T$ (Setup of §20.3), the strike $K > 0$, the expiry $T > 0$ and a real rate r; the numbers d_1 and d_2 are those of (20.10). A quoted (market) call price is C^{mkt}, and a quoted put price is $\mathcal{P}^{\mathrm{mkt}}$. The price (20.8) of Theorem 20.22, read as a function of the volatility at fixed F, K, T and r, is $C^{\mathrm{BSM}}(\sigma)$, in full $C^{\mathrm{BSM}}(K,T;\sigma)$; likewise $\mathcal{P}^{\mathrm{BSM}}(\sigma)$ is the price (20.9). The *vega* is the derivative $\mathcal{V} = \partial_\sigma C^{\mathrm{BSM}}$. It is written with a calligraphic letter because ν is the log-drift of this Part, and like every Greek it takes no argument: its value at a volatility $\sigma^{(k)}$ is written $\partial_\sigma C^{\mathrm{BSM}}(\sigma^{(k)})$, and the second derivative in σ is $\partial_{\sigma\sigma} C^{\mathrm{BSM}}$. The implied volatility is σ_{imp}. The Newton iterates are $\sigma^{(k)}$, $k = 0, 1, 2, \ldots$, with the start $\sigma^{(0)}$ and the limit $\sigma^{(\infty)}$: a parenthesised superscript counts iterations, as it counts stages in $V^{(j)}$, because σ_1 and σ_2 are the volatilities of the two assets of Definition 20.6(ii). The index k is local to the iteration, as it is in Definition 7.41. The inverse function of Φ is Φ^{-1}; it exists on $(0,1)$ because Φ is continuous and strictly increasing. One notice: §7.6 calls the root of its function r. Here the root is σ_{imp}, and r stays the interest rate.

Lemma 20.34 (The key identity)**.** *For all $F > 0$, $K > 0$, $\sigma > 0$ and $T > 0$, with d_1 and d_2 as in (20.10),*

$$F\,\varphi(d_1) = K\,\varphi(d_2).$$

Equivalently, for an asset with a yield q, where $F = Se^{(r-q)T}$,

$$Se^{-qT}\,\varphi(d_1) = Ke^{-rT}\,\varphi(d_2).$$

Proof. We compare the exponents of the two densities through a difference of squares. By (20.10), $d_1 - d_2 = \sigma\sqrt{T}$ and $d_1 + d_2 = 2\ln(F/K)/(\sigma\sqrt{T})$, so

$$d_1^2 - d_2^2 = (d_1 - d_2)(d_1 + d_2) = \sigma\sqrt{T} \cdot \frac{2\ln(F/K)}{\sigma\sqrt{T}} = 2\ln\frac{F}{K}.$$

Since $\varphi(y) = e^{-y^2/2}/\sqrt{2\pi}$,

$$\frac{\varphi(d_2)}{\varphi(d_1)} = \exp\Big(\frac{d_1^2 - d_2^2}{2}\Big) = \exp\Big(\ln\frac{F}{K}\Big) = \frac{F}{K},$$

and multiplying by $K\,\varphi(d_1)$ gives the first identity. For the second, multiply both sides of the first by e^{-rT} and use $e^{-rT}F = Se^{-qT}$. □

The identity is what keeps every first derivative of the formula short. When (20.8) is differentiated in any of its inputs, the chain rule produces the two terms $F\,\varphi(d_1)\,\partial d_1$ and $K\,\varphi(d_2)\,\partial d_2$, and the identity merges them into one. The vega below is the first instance; §21 uses the lemma for the delta, the gamma, the theta and the rho. Example 20.36 checks the identity in numbers.

Proposition 20.35 (Sensitivity to volatility: vega)**.** *Under the Black-76 model of Definition 20.16, at fixed F, K, T and r, the call price $C^{\mathrm{BSM}}(\sigma)$ of (20.8) and the put price $\mathcal{P}^{\mathrm{BSM}}(\sigma)$ of (20.9) are differentiable in σ on $(0,\infty)$, with the same derivative, the vega:*

$$\mathcal{V} = \partial_\sigma C^{\mathrm{BSM}} = \partial_\sigma \mathcal{P}^{\mathrm{BSM}} = e^{-rT} F\,\varphi(d_1)\,\sqrt{T} = K e^{-rT}\,\varphi(d_2)\,\sqrt{T} \;>\; 0.$$

For an asset with a yield q, where $e^{-rT}F = Se^{-qT}$, the vega is $Se^{-qT}\,\varphi(d_1)\,\sqrt{T}$. At the money forward, $K = F$, it equals $e^{-rT}F\,\varphi(\sigma\sqrt{T}/2)\,\sqrt{T}$, and

$$\varphi(0)\,e^{-rT}F\sqrt{T}\,\Big(1-\frac{\sigma^2 T}{8}\Big) \;\le\; e^{-rT}F\,\varphi\Big(\frac{\sigma\sqrt{T}}{2}\Big)\sqrt{T} \;\le\; \varphi(0)\,e^{-rT}F\sqrt{T},$$

with $\varphi(0) = 0.3989$: about $0.4\,e^{-rT}F\sqrt{T}$ per unit of volatility.

Proof. We differentiate (20.8) by the chain rule and merge the two terms by Lemma 20.34. In the form of the Setup of §20.5,

$$d_1 = \frac{\ln(F/K)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2}, \qquad d_2 = \frac{\ln(F/K)}{\sigma\sqrt{T}} - \frac{\sigma\sqrt{T}}{2},$$

so both are differentiable in $\sigma > 0$, with

$$\partial_\sigma d_1 = -\frac{\ln(F/K)}{\sigma^2\sqrt{T}} + \frac{\sqrt{T}}{2}, \qquad \partial_\sigma d_2 = -\frac{\ln(F/K)}{\sigma^2\sqrt{T}} - \frac{\sqrt{T}}{2}, \qquad \partial_\sigma d_1 - \partial_\sigma d_2 = \sqrt{T}.$$

The derivative of Φ is φ (Step 2 of the proof of Proposition 20.29). The chain rule applied to (20.8) therefore gives

$$\partial_\sigma C^{\mathrm{BSM}} = e^{-rT}\big[F\,\varphi(d_1)\,\partial_\sigma d_1 - K\,\varphi(d_2)\,\partial_\sigma d_2\big].$$

By Lemma 20.34, $K\,\varphi(d_2) = F\,\varphi(d_1)$, so the bracket is $F\,\varphi(d_1)\,(\partial_\sigma d_1 - \partial_\sigma d_2) = F\,\varphi(d_1)\,\sqrt{T}$, and

$$\partial_\sigma C^{\mathrm{BSM}} = e^{-rT} F\,\varphi(d_1)\,\sqrt{T}.$$

The same lemma turns this into $Ke^{-rT}\,\varphi(d_2)\,\sqrt{T}$. Both forms are positive, because $\varphi > 0$. For the put, the parity clause of Theorem 20.22 shows that $\mathcal{P}^{\mathrm{BSM}}(\sigma) = C^{\mathrm{BSM}}(\sigma) - e^{-rT}(F-K)$ for every σ, and $e^{-rT}(F-K)$ does not depend on σ; so the put is differentiable in σ with the same derivative. The form for an asset with a yield is the substitution $e^{-rT}F = Se^{-qT}$.

At $K = F$, $\ln(F/K) = 0$ and $d_1 = \sigma\sqrt{T}/2$, which gives the at-the-money value. For the two bounds, $\varphi(y) = \varphi(0)\,e^{-y^2/2}$, and $1 - x \le e^{-x} \le 1$ for every $x \ge 0$: the function $e^{-x} - 1 + x$ vanishes at 0 and has the nonnegative derivative $1 - e^{-x}$ on $[0,\infty)$. Put $x = y^2/2$ with $y = \sigma\sqrt{T}/2$, so that $x = \sigma^2 T/8$, and multiply by $\varphi(0)\,e^{-rT}F\sqrt{T}$. □

The at-the-money vega is the derivative in σ of the linear rule $e^{-rT}F\,\sigma\sqrt{T}/\sqrt{2\pi}$ of Proposition 20.29, up to the factor between $1 - \sigma^2 T/8$ and 1. Proposition 21.9 in §21 reads the vega per volatility point and across maturities, and relates it to the gamma.

Example 20.36 (The key identity and the vega of the running call)**.** Take the one-year at-the-money-spot call of Example 20.24: $S = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$, so that $F = 105.127$, $d_1 = 0.35$, $d_2 = 0.15$ and $C = 10.45$. The densities are $\varphi(0.35) = 0.37524$ and $\varphi(0.15) = 0.39448$.

The key identity. The two sides of Lemma 20.34 are

$$F\,\varphi(d_1) = 105.127 \times 0.37524 = 39.448 \qquad \text{and} \qquad K\,\varphi(d_2) = 100 \times 0.39448 = 39.448.$$

The vega. By Proposition 20.35, with $e^{-rT}F = S$,

$$\mathcal{V} = S\,\varphi(d_1)\,\sqrt{T} = 100 \times 0.37524 \times 1 = 37.52$$

per unit of volatility; the second form gives the same number, $Ke^{-rT}\,\varphi(d_2)\,\sqrt{T} = 95.123\times 0.39448 = 37.52$. A unit of volatility is 100 volatility points, so the vega is 0.375 per volatility point: one point of volatility moves the 10.45 call by about 0.38. Repricing confirms it: at $\sigma = 21\%$ the formula gives 10.826, against 10.451 at 20%, a change of 0.376. The vega is close to the $0.4\,S\sqrt{T} = 40$ per unit of the at-the-money rule, the option being at the money spot and not forward.

Definition 20.37 (Implied volatility)**.** Consider a European call with strike K, expiry T and market price C^{mkt}, on an underlying whose market forward or futures price is F, for delivery at $T_F \geq T$ as in Definition 20.16, with the rate r. An *implied volatility* of the quote is a number $\sigma_{\text{imp}} > 0$ at which the formula of Theorem 20.22 reproduces the market price:

$$C^{\text{BSM}}(\sigma_{\text{imp}}) = C^{\text{mkt}}.$$

For a European put with market price $\mathcal{P}^{\text{mkt}}$ the defining equation is $\mathcal{P}^{\text{BSM}}(\sigma_{\text{imp}}) = \mathcal{P}^{\text{mkt}}$. Proposition 20.38(ii) below proves that a call quote has exactly one implied volatility when $e^{-rT}(F-K)^+ < C^{\text{mkt}} < e^{-rT}F$ and none otherwise, so that for such quotes σ_{imp} is *the* implied volatility. Viewed across strikes and expiries, $\sigma_{\text{imp}}(K,T)$ is the implied-volatility surface of Definition 22.1 in §22.

The definition does not assume that the market obeys the model. It is a change of units, from a price to a volatility, made with the formula. For the running call, a quote of 10.45 has $\sigma_{\text{imp}} = 20.0\%$, and by Example 20.36 a quote of 10.83 has $\sigma_{\text{imp}} = 21.0\%$. Example 20.40 computes an implied volatility that is not known in advance.

Proposition 20.38 (Existence and uniqueness of implied volatility, and the at-the-money Newton guarantee)**.** *Fix $F > 0$, $K > 0$, $T > 0$ and a real r, and let $C^{\text{BSM}}(\sigma)$ and $\mathcal{P}^{\text{BSM}}(\sigma)$ be the prices (20.8) and (20.9) as functions of $\sigma \in (0,\infty)$.*

(i) The function $\sigma \mapsto C^{\text{BSM}}(\sigma)$ is continuous and strictly increasing on $(0,\infty)$, with

$$\lim_{\sigma\to 0} C^{\text{BSM}}(\sigma) = e^{-rT}(F-K)^+ \qquad \text{and} \qquad \lim_{\sigma\to\infty} C^{\text{BSM}}(\sigma) = e^{-rT}F.$$

(ii) Every call price strictly inside the no-arbitrage bounds

$$e^{-rT}(F-K)^+ \;<\; C^{\text{mkt}} \;<\; e^{-rT}F$$

has exactly one implied volatility. A price on or outside these bounds has none. When the forward or futures price is nonnegative at T, a price outside them is an arbitrage, whatever the sign of r; for $r \geq 0$ and an option on the underlying itself ($T_F = T$) these are the bounds of Proposition 17.6(ii)–(iii). Likewise a put price has exactly one implied volatility when $e^{-rT}(K-F)^+ < \mathcal{P}^{\text{mkt}} < e^{-rT}K$ and none otherwise; it is the implied volatility of the call price $\mathcal{P}^{\text{mkt}} + e^{-rT}(F-K)$, which then lies strictly inside the call bounds.

(iii) Let $K = F$ and $0 < C^{\text{mkt}} < e^{-rT}F$. Then C^{BSM} is concave in σ: its derivative $\partial_\sigma C^{\text{BSM}}$ is strictly decreasing on $(0,\infty)$. The Brenner–Subrahmanyam start

$$\sigma^{(0)} = \frac{\sqrt{2\pi}\,C^{\text{mkt}}}{e^{-rT}F\sqrt{T}}$$

satisfies $0 < \sigma^{(0)} \leq \sigma_{\text{imp}}$. And Newton's iteration for the equation $C^{\text{BSM}}(\sigma) = C^{\text{mkt}}$,

$$\sigma^{(k+1)} = \sigma^{(k)} + \frac{C^{\text{mkt}} - C^{\text{BSM}}(\sigma^{(k)})}{\partial_\sigma C^{\text{BSM}}(\sigma^{(k)})}, \qquad k = 0,1,2,\ldots,$$

started at $\sigma^{(0)}$, increases monotonically to σ_{imp}: $\sigma^{(k)} \leq \sigma^{(k+1)} \leq \sigma_{\text{imp}}$ for every k, and $\sigma^{(k)} \to \sigma_{\text{imp}}$ as $k \to \infty$.

Sketch of the proof. Parts (i) and (ii) combine the positive vega of Proposition 20.35 with the intermediate value theorem; part (iii) uses the concavity of the at-the-money price to trap the Newton iterates below the root. The formula is continuous in σ and, its derivative being positive, strictly increasing; its limits, read off those of d_1 and d_2, are those of Remark 20.28(1) and (3), derived again in the complete proof. This is (i). So the formula takes each value strictly between the bounds exactly once and no other value, which is (ii) for the call; the put reduces to the call by the parity clause of Theorem 20.22. A quote outside the bounds is an arbitrage by Lemma 16.8: forwards and a deposit costing $e^{-rT}F$ pay the forward price at T, which dominates the call payoff when it is nonnegative, and the call payoff dominates both 0 and that price less K. For (iii), one more differentiation shows that the at-the-money vega $e^{-rT}F\sqrt{T}\,\varphi(\sigma\sqrt{T}/2)$ decreases in σ. The upper bound of (20.11) at $\sigma^{(0)}$ equals C^{mkt}, so $C^{\mathrm{BSM}}(\sigma^{(0)}) \le C^{\mathrm{mkt}}$ and $\sigma^{(0)} \le \sigma_{\mathrm{imp}}$ by (i). If $\sigma^{(k)} \le \sigma_{\mathrm{imp}}$, the mean value theorem with the decreasing vega gives $C^{\mathrm{mkt}} - C^{\mathrm{BSM}}(\sigma^{(k)}) \le \partial_\sigma C^{\mathrm{BSM}}(\sigma^{(k)})\,(\sigma_{\mathrm{imp}} - \sigma^{(k)})$, the tangent lying above the concave graph, so the Newton step, nonnegative by (i), lands at or below σ_{imp}. The iterates are nondecreasing and bounded, so they converge; by continuity the limit solves the equation, and by (ii) it is σ_{imp}. The complete proof is in Appendix F.6. □

Figure 41 is the picture of part (iii). The at-the-money price is concave in $\sigma\sqrt{T}$; the Brenner–Subrahmanyam start is the point where the tangent at 0 meets the level of the quote, to the left of the root; and every later Newton tangent lies above the curve, so it meets the level of the quote to the left of the root again. That arbitrage-free quotes of a call and a put with one strike and one expiry, which satisfy parity, have the same implied volatility is proved in §22 (Proposition 22.3).

Procedure 20.39 (Implied volatility by Newton's method)**.** The inputs are F, K, T, r and a quoted call price C^{mkt}.

1. Check that C^{mkt} lies strictly inside the interval $\big(e^{-rT}(F-K)^+,\ e^{-rT}F\big)$. Otherwise stop: there is no implied volatility, and a quote outside the interval is an arbitrage (Proposition 20.38(ii)).
2. Start at $\sigma^{(0)} = \sqrt{2\pi}\,C^{\mathrm{mkt}}/(e^{-rT}F\sqrt{T})$ (Brenner and Subrahmanyam, 1988), the inversion of the linear term of Proposition 20.29.
3. Iterate
$$\sigma^{(k+1)} = \sigma^{(k)} - \frac{C^{\mathrm{BSM}}(\sigma^{(k)}) - C^{\mathrm{mkt}}}{\partial_\sigma C^{\mathrm{BSM}}(\sigma^{(k)})},$$
with the vega $\partial_\sigma C^{\mathrm{BSM}} = e^{-rT}F\,\varphi(d_1)\,\sqrt{T}$ of Proposition 20.35, evaluated at $\sigma^{(k)}$. This is the iteration of Definition 7.41 for the function $\sigma \mapsto C^{\mathrm{BSM}}(\sigma) - C^{\mathrm{mkt}}$.
4. Safeguard and stop. Keep a bracket $[\sigma', \sigma'']$ with $C^{\mathrm{BSM}}(\sigma') < C^{\mathrm{mkt}} < C^{\mathrm{BSM}}(\sigma'')$, chosen before the first step so that it contains $\sigma^{(0)}$. After each evaluation of the formula at an iterate $\sigma^{(k)}$, replace σ' by $\sigma^{(k)}$ if $C^{\mathrm{BSM}}(\sigma^{(k)}) < C^{\mathrm{mkt}}$, and σ'' by $\sigma^{(k)}$ if $C^{\mathrm{BSM}}(\sigma^{(k)}) > C^{\mathrm{mkt}}$. If the Newton step of step 3 falls outside the current bracket, or if the vega $\partial_\sigma C^{\mathrm{BSM}}(\sigma^{(k)})$ is tiny (deep in or out of the money), reject it and take the midpoint of the bracket as the next iterate (a bisection step). Stop when $|C^{\mathrm{BSM}}(\sigma^{(k)}) - C^{\mathrm{mkt}}|$ is below the tolerance.

For a quoted put, apply the procedure to the call price $\mathcal{P}^{\mathrm{mkt}} + e^{-rT}(F-K)$ (Proposition 20.38(ii)).

Why the procedure works. The function $\sigma \mapsto C^{\mathrm{BSM}}(\sigma) - C^{\mathrm{mkt}}$ is twice continuously differentiable on $(0,\infty)$, because its derivative $e^{-rT}F\,\sqrt{T}\,\varphi(d_1)$ is a continuously differentiable function of σ, and its derivative at the root is positive (Proposition 20.35). The root σ_{imp} is therefore simple, and Theorem 7.42 applies: from a start close enough to σ_{imp} the iterates converge, and each error is at most a constant times the square of the previous one. At the money forward the convergence is also global and monotone from the start of step 2, by Proposition 20.38(iii). Away from the money neither guarantee covers the first steps, and the first step can overshoot the root (Problem P20.7 works a case), which is why the safeguard of step 4 matters. A first

bracket exists by the proof of Proposition 20.38(ii): the inequalities $C^{\mathrm{BSM}}(\sigma') < C^{\mathrm{mkt}}$ and $C^{\mathrm{BSM}}(\sigma'') > C^{\mathrm{mkt}}$ hold for every σ' small enough and every σ'' large enough, so the bracket can be taken to contain $\sigma^{(0)}$. The root stays bracketed at every step: since C^{BSM} is strictly increasing, $C^{\mathrm{BSM}}(\sigma') < C^{\mathrm{mkt}} = C^{\mathrm{BSM}}(\sigma_{\mathrm{imp}}) < C^{\mathrm{BSM}}(\sigma'')$ gives $\sigma' < \sigma_{\mathrm{imp}} < \sigma''$, and each update of step 4 replaces an end of the bracket by an evaluated point at which $C^{\mathrm{BSM}} - C^{\mathrm{mkt}}$ has the same sign as at that end, so the two strict inequalities persist. Every iterate lies in the current bracket, Newton steps that would leave it being rejected, and a bisection step halves the bracket.

Example 20.40 (One Newton step at the money)**.** Take $S = K = 100$, $T = 1$ and $r = 0$, so that $K = F$, and a quoted call price $C^{\mathrm{mkt}} = 12.00$.

Step 1. The quote lies strictly inside $(0, 100)$, so an implied volatility exists.

Step 2. Since $\sqrt{2\pi} = 2.506628$, the start is $\sigma^{(0)} = 2.506628 \times 12.00/100 = 0.300795$, that is, 0.30080 to five digits.

Step 3. At the money forward, $d_1 = \sigma\sqrt{T}/2$, so

$$C^{\mathrm{BSM}}(\sigma^{(0)}) = 100\,\big(2\,\Phi(0.150398) - 1\big) = 11.9549,$$
$$\partial_\sigma C^{\mathrm{BSM}}(\sigma^{(0)}) = 100\,\varphi(0.150398) = 39.4456,$$

that is, 11.955 and 39.45, and

$$\sigma^{(1)} = 0.300795 + \frac{12.0000 - 11.9549}{39.4456} = 0.300795 + \frac{0.0451}{39.4456} = 0.300795 + 0.001143 = 0.301938.$$

The exact value. At the money forward the formula can be inverted in closed form. By Proposition 20.29, $C^{\mathrm{mkt}} = e^{-rT}F\,[2\,\Phi(\sigma_{\mathrm{imp}}\sqrt{T}/2) - 1]$, so

$$\sigma_{\mathrm{imp}} = \frac{2}{\sqrt{T}}\,\Phi^{-1}\Big(\frac{1 + e^{rT}C^{\mathrm{mkt}}/F}{2}\Big) = 2\,\Phi^{-1}(0.56) = 2 \times 0.150969 = 0.301938.$$

One step gives six correct digits: the error falls from $\sigma_{\mathrm{imp}} - \sigma^{(0)} = 1.1 \times 10^{-3}$ to $\sigma_{\mathrm{imp}} - \sigma^{(1)} = 4.9 \times 10^{-8}$, and a second step leaves an error below 10^{-14}, the squaring of the error that Theorem 7.42 describes. The iterates satisfy $\sigma^{(0)} < \sigma^{(1)} \le \sigma_{\mathrm{imp}}$, as Proposition 20.38(iii) guarantees. The implied volatility is 30.19%; the linear rule alone, the start, gives 30.08%, to the left of it, as in Figure 41.

(Problem P20.7 works this computation at different numbers.)

Interview trap. "Implied volatility is the market's forecast of realised volatility." It is a quoting convention for a price: the number that makes one model reproduce one price. On average, implied volatility exceeds the volatility realised afterwards (Fact 21.29 in §21), and options with different strikes on one underlying and one expiry give different numbers (the surface of Definition 22.1 in §22), which no single forecast could do.

Interview trap. "A call quoted above the stock price just has a very high implied volatility." No volatility reproduces it. For every σ the formula stays strictly below $e^{-rT}F$, which is S for a non-dividend stock, so a price on or outside the bounds has no implied volatility, and one outside them is an arbitrage (Proposition 20.38(ii)).

Interview trap. Running Newton's method from a fixed guess on a deep out-of-the-money option. The vega is tiny there, so a one-cent error in the price is a large error in the volatility, and the first step can shoot to a negative or absurd σ. Bracket the root and fall back on bisection (step 4 of Procedure 20.39). Desks read the volatility of each strike off its out-of-the-money option, the call above the forward price and the put below it, and convert through parity (Proposition 22.3).

20.7 Dividends, Indices and Currencies

Income on the underlying lowers its forward price, and in forward form that is the only change: Theorem 20.22 prices options on dividend-paying stocks, on stock indices and on currencies once

the right forward price is inserted. This subsection derives Merton's formula for a continuous yield and the Garman–Kohlhagen formula for currencies as corollaries, each by reducing to a traded asset that pays no income, the reinvested holding, and applying Theorem 20.7 to it. It treats known cash dividends by the escrowed-dividend convention. It extends the pricing equation to a yield and to a futures price by a change of variables, the form that §21 needs. It shows how a desk reads the forward price and the implied dividend yield off one call–put pair. And it records Black's approximation for American calls.

Setup. The continuous yield of the underlying is q, any real number, in the sense of Proposition 16.15: a holding whose income is reinvested in the asset grows at the rate q in units. Symbol notice: here q is the yield, not the $1-p$ of the Notation table. The foreign interest rate is r_{for}; the symbol r_f belongs to §11. Known cash dividends are $D_1, \dots, D_n$, paid at the ex-dividend dates $0 < t_1 < \cdots < t_n < T$; in this subsection n counts the dividends. Their present value is $I = \sum_{i=1}^{n} D_i e^{-rt_i}$, and

$$I_t = \sum_{i\,:\,t_i > t} D_i\, e^{-r(t_i - t)}$$

is the present value at t of the dividends still to be paid before T, so that $I_0 = I$ and $I_T = 0$.

The *reinvested holding* is $\tilde{S}_t = S_t e^{-q(T-t)}$: the price of $e^{-q(T-t)}$ units of the asset with all income reinvested, a position that is worth S_T at T (Proposition 16.15(i)). The price function of a derivative written in that variable is $\tilde{V}$, so that $V(S,t) = \tilde{V}(Se^{-q(T-t)}, t)$; both functions are defined in Corollary 20.41(i). The price variable of $\tilde{V}$ is written $\tilde{S}$, and its partial derivatives are $\partial_t \tilde{V}$, $\partial_{\tilde{S}} \tilde{V}$ and $\partial_{\tilde{S}\tilde{S}} \tilde{V}$. A tilde marks the same object written in a transformed variable; the rescaled payoff of the proofs is $\tilde{f}$ for the same reason. In Corollary 20.45(ii) the same pattern serves a futures price: $T_F \geq T$ is the delivery date, $\tilde{F} = Fe^{-r(T_F - t)}$ is the prepaid forward of §20.8, and $V^{\text{fut}}(F,t)$ is a price written as a function of the futures price.

Two models are used. In the *yield model*, (A1) holds for the asset, so that $S_t = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right)$; in place of (A2) the asset pays the yield q, and the traded instrument is its reinvested holding; and (A3)–(A6) hold. Then

$$\tilde{S}_t = S_0 e^{-qT} \exp\left((\mu + q - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right)$$

is a geometric Brownian motion with volatility σ that pays no income. In the *escrowed-dividend model*, for a stock with the known dividends above and with $S_0 > I$, the stock price net of the dividends still to be paid is a geometric Brownian motion with volatility σ,

$$S_t - I_t = (S_0 - I) \exp\left((\mu - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right), \qquad 0 \leq t \leq T;$$

the traded instrument is the position of one share together with the loans that the dividends still to come repay (the position in the proof of Proposition 16.13), worth $S_t - I_t$ at t, as the proof of Corollary 20.41(ii) checks; and (A3)–(A6) hold. In both models F is the forward price for delivery at T and $S = S_0$, and with a yield the risk-neutral log-drift is $\nu = r - q - \sigma^2/2$. Put–call parity in every setting is Corollary 17.11 when $r \geq 0$, the standing assumption of §17.2, and the parity clause of Theorem 20.22 proves it without a model for every real r: the difference $C - \mathcal{P}$ equals $Se^{-qT} - Ke^{-rT}$ with a yield, $S - I - Ke^{-rT}$ with known dividends and $Se^{-r_{\text{for}}T} - Ke^{-rT}$ for a currency, and all three are $e^{-rT}(F - K)$.

Corollary 20.41 (Options on an asset with a yield or known dividends; Merton, 1973)**.** *The two parts use the two models of the Setup.*

(i) (Continuous yield.) Assume the yield model. The European call and the European put with strike K and expiry T are worth

$$C = Se^{-qT}\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2), \qquad \mathcal{P} = Ke^{-rT}\,\Phi(-d_2) - Se^{-qT}\,\Phi(-d_1),$$

with

$$d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = \frac{\ln(S/K) + (r - q - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}\,:$$

the formulas of Theorem 20.22 with $F = Se^{(r-q)T}$*. A stock index is the case in which* q *is its dividend yield. More generally, let* $f \in \mathcal{G}$*, put* $\nu = r - q - \sigma^2/2$ *and define, for* $0 \le t < T$ *and* $S > 0$*,*

$$V(S,t) = e^{-r(T-t)}\,\mathbb{E}\,f\big(S\exp\big(\nu(T-t) + \sigma\sqrt{T-t}\,Z\big)\big), \qquad Z \sim \mathcal{N}(0,1), \tag{20.12}$$

and $V(S,T) = f(S)$*. Then:*

(a) the identity $V(S,t) = \tilde{V}(Se^{-q(T-t)}, t)$ *holds on* $(0,\infty)\times[0,T]$*, where* $\tilde{V}$ *is the function (20.5) of Theorem 20.7 for the payoff* f*, with its price variable written* $\tilde{S}$*; and for* $t < T$ *the call payoff* $f(x) = (x-K)^+$ *and the put payoff* $f(x) = (K-x)^+$ *have the price functions*

$$V(S,t) = Se^{-q(T-t)}\,\Phi(d_1) - Ke^{-r(T-t)}\,\Phi(d_2) \qquad \textit{(call)},$$
$$V(S,t) = Ke^{-r(T-t)}\,\Phi(-d_2) - Se^{-q(T-t)}\,\Phi(-d_1) \qquad \textit{(put)},$$

where d_1 *and* d_2 *are the numbers above with* $T-t$ *in the place of* T*: the call and put formulas, read with* $T-t$ *for* T*, are the price functions of the two options;*

(b) the function V *is infinitely differentiable on* $(0,\infty)\times[0,T)$*; every partial derivative of* V *has moderate growth on* $(0,\infty)\times[0,T']$ *for every* $T' < T$*; there is a constant* c' *with* $|V(S,t)| \le c'\,(S^{c'} + S^{-c'})$ *on* $(0,\infty)\times[0,T]$*; and* $V(S,t) \to f(S^*)$ *as* $(S,t) \to (S^*,T)$ *at every continuity point* S^* *of* f*. These are the properties (i) and (ii) of Theorem 20.7 other than its equation:* V *solves equation (20.13) in the place of (20.4), which Corollary 20.45(i) proves below from (a) by the chain rule;*

(c) the strategy of Definition 20.1 with the initial capital $V(S_0,0)$ *that holds*

$$\partial_{\tilde{S}}\tilde{V}(\tilde{S}_t,t) = e^{q(T-t)}\,\partial_S V(S_t,t)$$

units of the reinvested holding $\tilde{S}$ *on* $[0,T)$*, that is,* $\partial_S V(S_t,t)$ *units of the asset with the income reinvested, has the value* $V(S_t,t)$ *at every* $t < T$ *and the terminal value* $f(S_T)$*;*

(d) the claim paying $f(S_T)$ *at* T *has the arbitrage-free time-*0 *price*

$$V(S_0,0) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}} f(S_T),$$

where $\mathbb{E}^{\mathbb{Q}}$ *is that of Definition 20.6(i) with the yield* q*;*

(e) Theorem 20.7(v) and Corollary 20.11 hold in the yield model with four changes: $\mathbb{E}^{\mathbb{Q}}$ *is that of Definition 20.6(i) with the yield* q*; the log-drift is* $\nu = r - q - \sigma^2/2$ *in their price functions and recursions; for each* y *off the hyperplanes, the price function* $V(\cdot,\cdot\,;y)$ *of Theorem 20.7(v) has the properties of (b) in the place of (i) and (ii), its equation being (20.13) in the place of (20.4); and the hedge instrument is the reinvested holding* $\tilde{S}$*, of which each strategy holds* $e^{q(T-t)}$ *times the derivative in* S *of its price function.*

(ii) (Known cash dividends.) Assume the escrowed-dividend model. The claim paying $f(S_T)$ *at* T*, with* $f \in \mathcal{G}$*, has the arbitrage-free time-*0 *price*

$$e^{-rT}\,\mathbb{E}\,f\big((S-I)\exp\big((r - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z\big)\big).$$

In particular the European call and put are given by the formulas of Corollary 20.23 with S *replaced by* $S - I$*,*

$$C = (S-I)\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2), \qquad \mathcal{P} = Ke^{-rT}\,\Phi(-d_2) - (S-I)\,\Phi(-d_1),$$

with

$$d_1 = \frac{\ln\left((S-I)/K\right) + (r+\sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}\,:$$

the formulas of Theorem 20.22 with $F = (S-I)e^{rT}$.

Sketch of the proof. Both parts are reductions to a traded position that pays no income, to which Theorem 20.7 applies. For (i), the reinvested holding $\tilde{S}_t = S_t e^{-q(T-t)}$ pays no income (Proposition 16.15(i)), is worth S_T at T and, in the yield model, is a geometric Brownian motion with the volatility σ. The theorem, whose conclusions do not involve the drift $\mu + q$, therefore prices and replicates $f(\tilde{S}_T) = f(S_T)$ with a price function $\tilde{V}$ and a holding of $\partial_{\tilde{S}}\tilde{V}$ units of $\tilde{S}$. The one computation, Step 3 of the complete proof, is the identity $Se^{-q(T-t)}\, e^{(r-\sigma^2/2)(T-t)} = Se^{\nu(T-t)}$, which turns $\tilde{V}(Se^{-q(T-t)}, t)$ into (20.12): the identity of (a). The chain rule then gives the holding $e^{q(T-t)}\,\partial_S V$ of (c) and the properties (b), and $t = 0$ gives the price (d). With (d), the law (20.6) of S_T under $\mathbb{Q}$, where $F = Se^{(r-q)T}$, makes the Black-76 model hold, and Theorem 20.22 gives the call and the put, and at a time $t < T$ the same formulas with $T - t$ in the place of T. For (e), each earlier price is a fixed multiple of $\tilde{S}$ at its date, and the same substitution applies. Part (ii) is the same reduction with one share and the loans that its dividends repay, a position worth $S_t - I_t$ that pays no net income and, in the escrowed-dividend model, is a geometric Brownian motion with the volatility σ: $S - I$ takes the place of S, and $F = (S-I)e^{rT}$. The complete proof is in Appendix F.7. □

Reading the hedge. No claim made here is relied on later. A holding of h units of $\tilde{S}$ is $he^{-q(T-t)}$ units of the asset whose income is reinvested in the asset. It can equally be read as $he^{-q(T-t)}$ units of the asset whose income is paid into the bank. Corollary 19.31, applied to the function $(x,t) \mapsto xe^{-q(T-t)}$ of the process $e^{-rt}S_t$, gives in the shorthand of Definition 19.25

$$d\big(e^{-rt}\tilde{S}_t\big) = e^{-q(T-t)}\left[d\big(e^{-rt}S_t\big) + q\,e^{-rt}S_t\,dt\right],$$

because the function has the derivatives $qxe^{-q(T-t)}$ in t, $e^{-q(T-t)}$ in x and 0 twice in x. Multiplied by h, the discounted gain on h units of $\tilde{S}$ is the discounted price gain on $he^{-q(T-t)}$ units of the asset plus the discounted income qS_t per unit on those units. With $h = e^{q(T-t)}\,\partial_S V$, as in (c), the strategy is $\partial_S V$ units of the asset with the dividends paid into the bank. In (ii), the volatility σ is that of the stock net of the escrowed dividends, $S - I$, and not that of S.

Example 20.42 works part (i) on a stock index. Example 20.49 works part (ii) on a stock with two dividends; it comes after Remark 20.48, which it also illustrates.

Example 20.42 (An index option with a dividend yield)**.** A stock index stands at $S = 3{,}000$ and pays a dividend yield $q = 2\%$; the rate is $r = 4\%$ and the volatility $\sigma = 18\%$. Price the call and the put with $K = 3{,}000$ and $T = 0.5$.

The prices. The forward price is $F = 3{,}000\,e^{0.01} = 3{,}030.15$, and $\sigma\sqrt{T} = 0.1273$, so

$$d_1 = \frac{\ln(F/K)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2} = \frac{0.0100}{0.1273} + 0.0636 = 0.1422, \qquad d_2 = 0.1422 - 0.1273 = 0.0149,$$

with $\Phi(d_1) = 0.5565$ and $\Phi(d_2) = 0.5060$ (0.556542 and 0.505955 before rounding). The two discounted quantities are $Se^{-qT} = 2{,}970.15$ and $Ke^{-rT} = 2{,}940.60$. By Corollary 20.41(i),

$$C = 2{,}970.15 \times 0.556542 - 2{,}940.60 \times 0.505955 = 1{,}653.01 - 1{,}487.81 = 165.20,$$
$$\mathcal{P} = 2{,}940.60 \times 0.494045 - 2{,}970.15 \times 0.443458 = 1{,}452.79 - 1{,}317.14 = 135.65.$$

Parity holds: $C - \mathcal{P} = 29.55 = 2{,}970.15 - 2{,}940.60 = Se^{-qT} - Ke^{-rT}$, which is also $e^{-0.02}\,(F-K) = 0.9802 \times 30.15$.

The dividend trap in numbers. Without the yield the forward price would be $3{,}000\,e^{0.02} = 3{,}060.60$, and the same formulas would give $C = 182.28$ and $\mathcal{P} = 122.87$: ignoring a 2% yield overprices the call by about 17.1 and underprices the put by about 12.8.

A mental check. The at-the-money-forward value $0.4\,e^{-rT}F\,\sigma\sqrt{T} = 0.4 \times 2{,}970.15 \times 0.1273 = 151.2$ is close to the half-straddle $(C + \mathcal{P})/2 = 150.4$, and by the identity of Example 20.33 the call is the half-straddle plus half the forward value: $150.4 + 29.55/2 = 165.2$.

The replicating strategy. By (a), the price function of the call is $V(S,t) = Se^{-q(T-t)}\,\Phi(d_1) - Ke^{-r(T-t)}\,\Phi(d_2)$, with d_1 and d_2 written with $T - t$. By (a) and Proposition 21.2 in §21, which proves it with Lemma 20.34, $\partial_S V = e^{-q(T-t)}\,\Phi(d_1)$; a central difference of the formula, with the step 1 in S, confirms it at time 0, giving 0.5510. By (c), at time 0 the strategy holds $e^{qT}\,\partial_S C = \Phi(d_1) = 0.5565$ units of the reinvested holding, each worth $\tilde{S}_0 = 3{,}000\,e^{-0.01} = 2{,}970.15$. That is $\partial_S C = e^{-qT}\,\Phi(d_1) = 0.5510$ units of the index, worth 1,653.01, with $165.20 - 1{,}653.01 = -1{,}487.81$ in the bank, the amount $-Ke^{-rT}\,\Phi(d_2)$. The pricing equation of this call is checked in numbers after Corollary 20.45.

(Problem P20.8 works this computation at different numbers.)

Corollary 20.43 (Currency options; Garman and Kohlhagen, 1983)**.** *Let S_t be the domestic price of one unit of a foreign currency, let r be the domestic rate and r_{for} the foreign rate, and assume the yield model of the Setup with $q = r_{\text{for}}$: (A1) holds for the exchange rate, the traded instrument is a foreign deposit, and (A3)–(A6) hold. Then the European call and the European put on one unit of the foreign currency, with strike K and expiry T, are worth*

$$C = Se^{-r_{\text{for}}T}\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2), \qquad \mathcal{P} = Ke^{-rT}\,\Phi(-d_2) - Se^{-r_{\text{for}}T}\,\Phi(-d_1),$$

with

$$d_1 = \frac{\ln(S/K) + (r - r_{\text{for}} + \sigma^2/2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

In forward form these are the formulas of Theorem 20.22 with $F = Se^{(r-r_{\text{for}})T}$.

Proof. We reduce to Corollary 20.41(i) by showing that a foreign deposit is the reinvested holding of an asset with the yield r_{for}. A deposit of $e^{-r_{\text{for}}(T-t)}$ units of the foreign currency at time t grows at the foreign rate to one unit at T, the interest being paid in the currency itself. In domestic currency it is worth $\tilde{S}_t = S_t e^{-r_{\text{for}}(T-t)}$ at t and $\tilde{S}_T = S_T$ at T. It is a domestic asset that pays no income, and under (A1) for the exchange rate it is a geometric Brownian motion with volatility σ. This is the reinvested holding of Proposition 16.15(i) for the foreign currency, an asset whose unit count grows at the rate $q = r_{\text{for}}$ (the first route in the proof of Proposition 16.19). The forward exchange rate is $F = Se^{(r-r_{\text{for}})T}$, by covered interest parity (Proposition 16.19). So the proof of Corollary 20.41(i) applies word for word with $q = r_{\text{for}}$, and its formulas with $q = r_{\text{for}}$ are the stated ones. □

Example 20.44 (A put on the euro, and a range forward)**.** The euro trades at $S = 1.0850$ dollars. The dollar is the domestic currency, with $r = 4.5\%$; the euro rate is $r_{\text{for}} = 2.5\%$, and the volatility of the exchange rate is $\sigma = 7.5\%$. Price the put on one euro with $K = 1.0700$ and $T = 0.5$.

The forward rate is $F = 1.0850\,e^{0.01} = 1.0959$ (Proposition 16.19); it lies above the spot rate because the domestic rate is the higher one. With $\sigma\sqrt{T} = 0.05303$ and $\ln(F/K) = 0.02392$,

$$d_1 = \frac{0.02392}{0.05303} + 0.026517 = 0.47758, \qquad d_2 = 0.47758 - 0.05303 = 0.42455,$$

that is, $d_1 = 0.4776$ and $d_2 = 0.4245$ to four digits, and $\Phi(-d_1) = 0.3165$, $\Phi(-d_2) = 0.3356$ (0.31647 and 0.33558 before rounding). By Corollary 20.43 in forward form,

$$\mathcal{P} = e^{-0.0225}\,[1.0700 \times 0.33558 - 1.0959 \times 0.31647] = 0.97775 \times 0.01225 = 0.01198$$

dollars per euro, which is 11,976 dollars on a notional of one million euros. The call with the same strike is worth 0.03730, and parity holds: $C - \mathcal{P} = 0.02533 = e^{-rT}(F - K) = 0.97775 \times 0.025904$, which is also $Se^{-r_{\text{for}}T} - Ke^{-rT} = 1.07152 - 1.04619$.

(Problem P20.8 works this computation at different numbers.)

Trading application. A *range forward*. An importer whose home currency is the dollar must buy euros in six months. In the market of Example 20.44 the importer sells the put struck at 1.0700 and buys the call struck at 1.1234. The two premiums offset: the call is worth 0.01197 against 0.01198 for the put, and they are equal at the strike 1.12338. Per euro bought at the spot rate S_T, the importer pays $(1.0700 - S_T)^+$ on the short put and receives $(S_T - 1.1234)^+$ on the long call, so the all-in purchase rate is

$$S_T + (1.0700 - S_T)^+ - (S_T - 1.1234)^+,$$

the middle one of the three numbers S_T, 1.0700 and 1.1234. At $S_T = 1.04$ the put costs 0.03 per euro and the rate is 1.07; at $S_T = 1.10$ neither option pays and the rate is 1.10; at $S_T = 1.15$ the call pays 0.0266 and the rate is 1.1234. The rate never leaves $[1.0700, 1.1234]$, and the package costs nothing at inception. This is the buyer's mirror image of the zero-cost collar of §17.7, now priced: the cap is paid for by giving up any rate below the floor, and the forward rate 1.0959 lies between the two strikes.

Interview trap. Currencies: r_{for} enters exactly where q entered in Corollary 20.41, and the domestic rate r alone does the discounting. Swapping the two rates, or quoting the exchange rate the wrong way up (foreign per domestic), prices a different option. A put on one currency is a call on the other: a put on one unit of foreign currency struck at K is K calls on the domestic currency struck at $1/K$, because $(K - S_T)^+ = K\,S_T\,(1/S_T - 1/K)^+$, and a call on one unit of domestic currency struck at $1/K$ pays $(1/S_T - 1/K)^+$ units of foreign currency, each worth S_T (Problem P20.8 works the prices).

Corollary 20.45 (The pricing equation with a yield, and for a futures price)**.** *Both parts are statements about functions; no model assumption is needed.*

(i) (Yield.) Let q be real. For every payoff $f \in \mathcal{G}$, the price function (20.12) of Corollary 20.41(i) satisfies

$$\partial_t V + (r - q)S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V = rV \qquad \text{on } (0,\infty)\times[0,T). \tag{20.13}$$

More generally, (20.13) holds for $V(S,t) = \tilde{V}(Se^{-q(T-t)}, t)$ whenever $\tilde{V}(\tilde{S}, t)$ is a function on $(0,\infty)\times[0,T)$ with continuous partial derivatives $\partial_t\tilde{V}$, $\partial_{\tilde{S}}\tilde{V}$ and $\partial_{\tilde{S}\tilde{S}}\tilde{V}$ that solves equation (20.4) in the variable $\tilde{S}$,

$$\partial_t \tilde{V} + r\tilde{S}\,\partial_{\tilde{S}}\tilde{V} + \tfrac{1}{2}\sigma^2\tilde{S}^2\,\partial_{\tilde{S}\tilde{S}}\tilde{V} = r\tilde{V}.$$

(ii) (Futures price.) Consider a derivative on a futures price F_t for delivery at $T_F \geq T$, with deterministic rates, and write $\tilde{F} = Fe^{-r(T_F - t)}$ for the prepaid forward of §20.8. Let $\tilde{V}(\tilde{F}, t)$ be a function on $(0,\infty)\times[0,T)$ with continuous partial derivatives $\partial_t\tilde{V}$, $\partial_{\tilde{F}}\tilde{V}$ and $\partial_{\tilde{F}\tilde{F}}\tilde{V}$ that solves equation (20.4) in the variable $\tilde{F}$; Theorem 20.52 in §20.8 shows that the price of a futures option, written in the variable $\tilde{F}$, is such a function. Then the price written as a function of the futures price,

$$V^{\text{fut}}(F,t) = \tilde{V}\big(Fe^{-r(T_F - t)}, t\big),$$

satisfies

$$\partial_t V^{\text{fut}} + \tfrac{1}{2}\sigma^2 F^2\,\partial_{FF} V^{\text{fut}} = rV^{\text{fut}} \qquad \text{on } (0,\infty)\times[0,T): \tag{20.14}$$

the yield equation with $q = r$, that is, with the cost of carry $b = 0$. For a futures on an asset with a yield q, where $F_t = S_t e^{(r-q)(T_F - t)}$ (Proposition 16.17(ii), the forward price

of Proposition 16.15(ii) at the date t with the delivery date T_F, together with the equality of futures and forward prices in the Setup of §20.8), the same price in the spot variable, $V^{\text{fut}}(Se^{(r-q)(T_F-t)},t) = \tilde{V}(Se^{-q(T_F-t)},t)$, solves (20.13), by (i) with T_F in the place of T.

Proof. We change variables and apply the chain rule; no new hedging argument is needed.

Part (i). The substitution in Step 3 of the proof of Corollary 20.41 is an identity between two Gaussian expectations and uses no model assumption. It shows, for every real r and q and every $\sigma > 0$, that the price function (20.12) is $V(S,t) = \tilde{V}(Se^{-q(T-t)},t)$, where $\tilde{V}$ is the function (20.5); by Theorem 20.7(i), which is a property of the function alone, $\tilde{V}$ is infinitely differentiable and solves (20.4) in the variable $\tilde{S}$. So the first claim is a case of the second. For the second, the partial derivatives of $\tilde{V}$ are continuous, so the chain rule applies to the composition with $(S,t) \mapsto (Se^{-q(T-t)},t)$, which sends $(0,\infty)\times[0,T)$ into itself. Write $\tilde{S} = Se^{-q(T-t)}$, the derivatives of $\tilde{V}$ being evaluated at $(\tilde{S},t)$. Differentiating in S,

$$\partial_S V = e^{-q(T-t)}\,\partial_{\tilde{S}}\tilde{V}, \qquad \partial_{SS} V = e^{-2q(T-t)}\,\partial_{\tilde{S}\tilde{S}}\tilde{V},$$

so that

$$S\,\partial_S V = \tilde{S}\,\partial_{\tilde{S}}\tilde{V} \qquad \text{and} \qquad S^2\,\partial_{SS} V = \tilde{S}^2\,\partial_{\tilde{S}\tilde{S}}\tilde{V}.$$

Differentiating in t at a fixed S, the variable $\tilde{S}$ increases at the rate $q\tilde{S}$, because the derivative of $e^{-q(T-t)}$ in t is $q\,e^{-q(T-t)}$. Hence

$$\partial_t V = \partial_t\tilde{V} + q\tilde{S}\,\partial_{\tilde{S}}\tilde{V} = \partial_t\tilde{V} + qS\,\partial_S V, \qquad \text{that is,} \qquad \partial_t\tilde{V} = \partial_t V - qS\,\partial_S V.$$

Substituting the three relations into the equation for $\tilde{V}$ gives

$$(\partial_t V - qS\,\partial_S V) + rS\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V = rV,$$

which is (20.13). The computation used only the derivative of $e^{-q(T-t)}$, so it holds with any fixed date in the place of T.

Part (ii). This is part (i) with $\tilde{F}$ in the place of $\tilde{S}$, F in the place of S, $q = r$ and T_F in the place of T. Explicitly, with $\tilde{F} = Fe^{-r(T_F-t)}$,

$$F\,\partial_F V^{\text{fut}} = \tilde{F}\,\partial_{\tilde{F}}\tilde{V}, \qquad F^2\,\partial_{FF} V^{\text{fut}} = \tilde{F}^2\,\partial_{\tilde{F}\tilde{F}}\tilde{V},$$

and at a fixed F the variable $\tilde{F}$ increases at the rate $r\tilde{F}$, so $\partial_t V^{\text{fut}} = \partial_t\tilde{V} + rF\,\partial_F V^{\text{fut}}$. Substituting $\partial_t\tilde{V} = \partial_t V^{\text{fut}} - rF\,\partial_F V^{\text{fut}}$ into equation (20.4) written in the variable $\tilde{F}$,

$$(\partial_t V^{\text{fut}} - rF\,\partial_F V^{\text{fut}}) + rF\,\partial_F V^{\text{fut}} + \tfrac{1}{2}\sigma^2 F^2\,\partial_{FF} V^{\text{fut}} = rV^{\text{fut}}:$$

the two terms $rF\,\partial_F V^{\text{fut}}$ cancel, which leaves (20.14). For the last sentence, when $F = Se^{(r-q)(T_F-t)}$ the prepaid variable is $\tilde{F} = Se^{(r-q)(T_F-t)}e^{-r(T_F-t)} = Se^{-q(T_F-t)}$, so the price in the spot variable is $\tilde{V}(Se^{-q(T_F-t)},t)$, and part (i) with T_F in the place of T gives (20.13). □

(Problem P20.2 asks you to reproduce this proof from scratch.)

Part (i) in numbers. Take the index call of Example 20.42, with $S = K = 3{,}000$, $q = 2\%$, $r = 4\%$, $\sigma = 18\%$, $T = 0.5$ and $C = 165.20$. At time 0, central differences of its price function, with the steps 0.01 in S and 10^{-5} in t, give $\partial_S C = 0.551004$, $\partial_{SS} C = 0.00102399$ and $\partial_t C = -175.750$ a year. The closed forms of Propositions 21.2 and 21.6 in §21 give the same numbers, $\partial_S C = e^{-qT}\,\Phi(d_1) = 0.551004$, $\partial_{SS} C = e^{-qT}\varphi(d_1)/(S\sigma\sqrt{T}) = 0.00102399$ and

$$\partial_t C = -\frac{Se^{-qT}\sigma\,\varphi(d_1)}{2\sqrt{T}} + qSe^{-qT}\,\Phi(d_1) - rKe^{-rT}\,\Phi(d_2) = -149.298 + 33.060 - 59.512 = -175.750.$$

Then

$$\begin{aligned}(r-q)S\,\partial_S C &= 0.02\times 3{,}000\times 0.551004 = 33.060,\\ \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS}C &= 0.5\times 0.0324\times 3{,}000^2\times 0.00102399 = 149.298,\\ \partial_t C+(r-q)S\,\partial_S C+\tfrac{1}{2}\sigma^2 S^2\,\partial_{SS}C &= -175.750+33.060+149.298 = 6.608,\end{aligned}$$

and $rC = 0.04\times 165.20 = 6.608$, which is (20.13). The closed forms show why it balances: the term $\frac{1}{2}\sigma^2 S^2\,\partial_{SS}C = 149.298$ cancels the first part of $\partial_t C$, and what is left, $qSe^{-qT}\Phi(d_1)+(r-q)Se^{-qT}\Phi(d_1)-rKe^{-rT}\Phi(d_2) = r\big[Se^{-qT}\Phi(d_1)-Ke^{-rT}\Phi(d_2)\big]$, is rC. Example 20.53 checks part (ii) on a futures call.

No hedge with futures contracts was needed for (ii): the hedge instrument is the prepaid forward, as Theorem 20.52 shows. Equation (20.13) is the yield form of (20.4), satisfied by every price function of Corollary 20.41(i); it is the form on which Corollary 21.8 in §21 rests, and Theorem 23.32 in §23 uses (ii) at $r = 0$.

Remark 20.46 (Implied forwards and implied dividend yields). Parity in forward form, $C-\mathcal{P} = e^{-rT}(F-K)$ (Corollary 17.11(iii) for $r \geq 0$, and the parity clause of Theorem 20.22 for every real r), holds in every setting of this subsection and contains no σ: it is model-free. Read backwards, a liquid call–put pair with one strike and one expiry gives the forward price,

$$F = K + e^{rT}\,(C-\mathcal{P}).$$

For an index, $F = Se^{(r-q)T}$ (Proposition 16.15(ii)), and solving for the yield gives the *implied dividend yield*

$$q = r - \frac{\ln(F/S)}{T}.$$

In Example 20.47 one call–put pair fixes $F = 5{,}030.09$ and $q = 1.60\%$, and both quotes then carry the same 16.0% volatility: Theorem 20.22 never asks for a dividend forecast.

Example 20.47 (A forward and a dividend yield read off two quotes)**.** An index stands at 5,000. The three-month call and put struck at 5,000 trade at 173.76 and 143.97, and $r = 4\%$. By Remark 20.46, with $T = 0.25$ and $C-\mathcal{P} = 29.79$,

$$F = 5{,}000 + e^{0.01}\times 29.79 = 5{,}000 + 30.09 = 5{,}030.09$$

and

$$q = 4\% - \frac{\ln(5{,}030.09/5{,}000)}{0.25} = 4\% - 2.40\% = 1.60\%,$$

with no dividend forecast. With this forward price, Procedure 20.39 returns the same implied volatility, 16.0%, for both quotes: at $F = 5{,}030.09$ and $\sigma = 16\%$ the formulas of Theorem 20.22 give 173.76 and 143.97.

(Problem P20.8 works this computation at different numbers.)

Remark 20.48 (American calls on dividend payers: Black's approximation; Black, 1975). This remark records a named heuristic; no further formula is given. Let $r > 0$ and $K > 0$. Early exercise of an American call on a stock with discrete dividends can then be optimal only just before an ex-dividend date, and there it gains nothing when $D_i \leq K\big(1-e^{-r(t_{i+1}-t_i)}\big)$, with $t_{n+1} = T$ (Proposition 17.20 in §17.4). Black (1975) approximates the American price by the larger of two European calls priced by Corollary 20.41(ii): one expiring at T, and one expiring just before the last ex-dividend date, at t_n^-.

Within one model, the escrowed-dividend model with horizon T, each of the two European calls is a lower bound on the American price. The holder of the American call may follow the policy "exercise just before the last ex-dividend date if in the money, otherwise let the call lapse" or the policy "hold to T", and each pays at least what the corresponding European call pays (Proposition 17.6(i)

with $T' = t_n^-$ and with $T' = T$, as the sentences after the proof of Proposition 17.20 record). So the larger of the two single-model prices is a lower bound, and that version of the approximation errs low: it discards the holder's choice at t_n^-, which is made once $S_{t_n^-}$ is known.

Black's convention instead prices each call on the dividends of its own life: the call expiring at t_n^- is priced with the escrowed-dividend model of $[0, t_n)$, in which D_n is not escrowed. That number is not the price of the same call in the model with horizon T (in Example 20.49 below the two are 5.34 and 5.24), so it is not a proved lower bound, and the approximation can then exceed the American price. Take $S = K = 100$, $r = 5\%$, $\sigma = 30\%$, one dividend of 20 at $t_1 = 0.5$ and $T = 0.51$. Black's convention prices the call expiring at t_1^- on the whole stock, at 9.63. In the escrowed model with horizon T the same call is a call on $S - I = 80.49$ struck at $K - D_1 = 80$, worth 8.00, and the American call, computed on a binomial tree in that model, is worth about 8.00 as well. With a continuous yield, exercising a call early gives up the interest on K and gains the income qS that the asset pays from then on, so a high yield favours the early exercise of calls; a put makes the reverse exchange, so a low yield favours the early exercise of puts. Proposition 18.25 and Procedure 18.16 in §18 price both on trees.

Example 20.49 (A stock with two dividends, and Black's approximation)**.** Take the stock of Example 17.21: $K = 60$, $r = 4\%$, $T = 1$ and dividends of 1.20 at $t_1 = 0.25$ and $t_2 = 0.75$; let $S = 60$ and $\sigma = 25\%$ in the escrowed-dividend model.

European prices. The present value of the dividends is $I = 1.20\,e^{-0.01} + 1.20\,e^{-0.03} = 1.1881 + 1.1645 = 2.3526$, and the forward price is

$$F = (60 - 2.3526)\,e^{0.04} = 57.6474 \times 1.04081 = 60.000,$$

more precisely 60.00004. The option struck at the spot price is therefore at the money *forward* to four decimals: by parity $C - \mathcal{P} = e^{-rT}(F - K) = 0.00004$, and Corollary 20.41(ii) gives $C = 5.7346$ and $\mathcal{P} = 5.7345$. Both round to the at-the-money-forward value of Proposition 20.29,

$$e^{-rT} F\,[2\,\Phi(\sigma\sqrt{T}/2) - 1] = 57.647 \times (2\,\Phi(0.125) - 1) = 57.647 \times 0.09948 = 5.73,$$

so $C = \mathcal{P} = 5.73$ to the precision shown; the 0.4 rule gives $0.4 \times 57.647 \times 0.25 = 5.76$.

The effect of the dividends. Without dividends, Corollary 20.23 gives $d_1 = 0.285$, $d_2 = 0.035$ and

$$C = 60 \times 0.61218 - 57.647 \times 0.51396 = 36.73 - 29.63 = 7.10, \qquad \mathcal{P} = 4.75.$$

The dividends lower the call by 1.37 and raise the put by 0.98, and $C - \mathcal{P}$ moves from $S - Ke^{-rT} = 2.35$ to $S - I - Ke^{-rT} = 0.00$.

Black's approximation. The European call expiring just before the second ex-dividend date sees the first dividend only. In the escrowed-dividend model of its own life only the first dividend is escrowed, with present value $1.20\,e^{-0.01} = 1.1881$, and the maturity is 0.75, so it is the call of Corollary 20.41(ii) on $60 - 1.1881 = 58.81$ struck at 60, worth 5.34. Since $5.34 < 5.73$, Black's approximation of the American call is 5.73. In the single escrowed model with horizon T, the stock just before t_2 is the escrowed part plus the dividend D_2 about to be paid, so the same call pays $\big((S_{t_2} - I_{t_2}) + D_2 - K\big)^+$: it is a call on $S - I = 57.65$ struck at $K - D_2 = 58.80$ with maturity 0.75, worth 5.24, and the answer 5.73 is unchanged. In that single model both European prices are lower bounds (Remark 20.48), so 5.73 is a lower bound on the American price. The thresholds of Example 17.21, $60\,(1 - e^{-0.02}) = 1.188$ before t_1 and $60\,(1 - e^{-0.01}) = 0.597$ before t_2, are both below the dividend 1.20, so early exercise cannot be ruled out at either date. A binomial tree for the escrowed part $S_t - I_t$, with the stock rebuilt at each node as the escrowed part plus the present value of the dividends not yet paid and early exercise tested at every node, gives 5.83 for the American call with 1,000 to 8,000 steps: here Black's approximation errs low by about 0.10.

(Problem P20.9 works this computation at different numbers.)

Interview trap. Explaining the effect of a dividend by a payment to, or withheld from, the option holder. A dividend lowers the call and raises the put because it lowers the forward price; nothing is paid to the option holder in either case. With S unchanged, a known dividend lowers F by its value carried forward to T (Proposition 16.13); and in the model F_T is F times a factor that does not depend on F, so the call payoff rises with F and the put payoff falls. In Example 20.49 the call loses 1.37 and the put gains 0.98.

Interview trap. Double counting: applying a yield q *and* subtracting the present value of the same dividends. One or the other describes the income, never both. And with the escrowed convention σ is the volatility of $S - I$, slightly above that of S: in Example 20.49, a volatility of 25% for $S - I = 57.65$ is a volatility of $25\% \times 57.65/60 = 24.0\%$ for the stock at time 0.

Interview trap. "Black's approximation is the American price." It is a heuristic: the larger of two European prices, which leaves out the holder's decision at t_n^-. Priced in one model it is the larger of two lower bounds and errs low (5.73 against 5.83 in Example 20.49); under Black's two-model convention neither direction is guaranteed (Remark 20.48).

20.8 Futures Options and Black's Model

Options on futures dominate commodity option markets, and Black's (1976) observation is that under the risk-neutral law a futures price has no drift, because a futures position costs nothing to enter. A futures price therefore fits the forward-lognormal model of Definition 20.16 with no adjustment, and Theorem 20.22 prices European futures options directly. The hedge is built from forward contracts and bonds, the prepaid forward of the Setup, so no unit of the underlying is ever held, and the results cover a consumption commodity, whose convenience yield cannot be collected by holding a reinvested position. This subsection closes the promise made after Definition 20.16, and it shows the one place where futures options behave differently from stock options: an American futures call can be optimally exercised early.

Setup. The futures price at time t for delivery at $T_F \geq T$ is F_t, settled daily (Definition 16.25), and $F = F_0$; the delivery date is written T_F, not T'. The symbol F^{fut} is needed only where a futures price must be told apart from a forward price, which is not the case here: rates are deterministic by (A3), so at every settlement date the futures price equals the forward price for delivery at T_F (Theorem 16.28, applied from that date). The model below treats settlement as continuous, so this equality holds at every $t \leq T$, and F_t denotes both prices. The European call and put on the futures, with strike $K > 0$ and expiry T, have the prices C and $\mathcal{P}$; the American call and put have the prices C^{am} and $\mathcal{P}^{\text{am}}$.

The model. Assumptions (A3)–(A6) hold, and (A1) is assumed for the futures price itself:

$$F_t = F \exp\left((\mu_F - \tfrac{1}{2}\sigma^2)t + \sigma W_t\right), \qquad 0 \leq t \leq T,$$

with a constant real-world drift μ_F and a volatility $\sigma > 0$. This covers a futures on an underlying that has no investable spot position, such as a consumption commodity: its spot trades, so that the futures price converges to it at delivery (Proposition 16.27) and the futures-forward equality above applies, but it cannot be carried as a reinvested holding. A futures on an asset with a yield q that satisfies (A1) is a case of the model, as the second check below shows.

The hedge instrument is the *prepaid forward*

$$\tilde{F}_t = e^{-r(T_F - t)}\, F_t,$$

the value of one long forward for delivery at T_F together with a deposit, as the first check below shows. A position of h units of $\tilde{F}$ is h long forwards plus bonds; when the forwards are struck at the current futures price they cost nothing, and the whole value of the position sits in the bonds.

A tilde marks the prepaid companion of a price, and the payoff written in the prepaid variable is $\tilde{f}$.

The risk-neutral path of the futures price is $F_t^{\mathbb{Q}}$, defined in Proposition 20.51. The price of a derivative written as a function of the futures price is $V^{\text{fut}}(F,t)$ (Corollary 20.45(ii), Theorem 20.52), with the partial derivatives $\partial_t V^{\text{fut}}$, $\partial_F V^{\text{fut}}$ and $\partial_{FF} V^{\text{fut}}$; the same price in the prepaid variable is $\tilde{V}(\tilde{F},t)$.

Check: the prepaid forward is a traded instrument. By Proposition 16.17(i), which holds for consumption assets too, with the futures-forward equality of this Setup, one long forward for delivery at T_F struck at any K is worth $(F_t-K)e^{-r(T_F-t)}$ at t. Together with a deposit worth $Ke^{-r(T_F-t)}$ at t, it is worth

$$(F_t-K)\,e^{-r(T_F-t)}+Ke^{-r(T_F-t)}=\tilde{F}_t$$

at every t. As a process,

$$\tilde{F}_t=\tilde{F}_0\exp\left((\mu_F+r-\tfrac{1}{2}\sigma^2)t+\sigma W_t\right),\qquad \tilde{F}_0=e^{-rT_F}F,$$

a geometric Brownian motion with volatility σ that pays no income. Its price function is in $C_E^{1,2}$ for the reason given for the stock in the Setup of §20.1 (Lemma 19.20(i)), so $\tilde{F}$ is an instrument of Definition 20.1.

Check: a futures on an asset with a yield. When the futures is written on an asset with a yield q that satisfies (A1), Proposition 16.17(ii) (the forward price of Proposition 16.15(ii) at the date t, with the delivery date T_F), with the futures-forward equality of this Setup, gives $F_t=S_te^{(r-q)(T_F-t)}$, the cost of carry being $b=r-q$, and F follows the model with $\mu_F=\mu-r+q$ (Corollary 19.36 with T_F in the place of T). Then $\tilde{F}_t=S_te^{-q(T_F-t)}$ is the reinvested holding of the asset over $[t,T_F]$ (Proposition 16.15(i) with the horizon T_F), which is the $\tilde{S}_t$ of §20.7 when $T_F=T$.

Definition 20.50 (Futures option)**.** A *call on a futures contract* with strike K and expiry $T\le T_F$ gives its holder, on exercise at a time t, a long futures position at the latest settlement price together with the cash amount F_t-K. A *put on a futures contract* gives a short futures position together with the cash amount $K-F_t$. Since a futures position is worth zero after each settlement (Definition 16.25), exercise is worth $(F_t-K)^+$ for the call and $(K-F_t)^+$ for the put. A European futures option can be exercised only at T, so it pays $(F_T-K)^+$ or $(K-F_T)^+$ at T; an American futures option can be exercised at any time up to T. Listed futures options are usually American. Options on interest-rate futures are out of scope (Remark 20.57).

For instance, the holder of a call struck at 78 who exercises when the futures settles at 80 receives a long futures position at 80 and 2 in cash. Closing the futures position at once costs nothing, and keeping it is a separate decision: the option has paid 2.

Proposition 20.51 (A futures price has zero drift under $\mathbb{Q}$)**.** *Assume the model of the Setup.*

(i) Let

$$\tilde{F}_t^{\mathbb{Q}}=\tilde{F}_0\exp\left((r-\tfrac{1}{2}\sigma^2)t+\sigma W_t\right)$$

be the risk-neutral path of the prepaid forward: Definition 20.6(i) applied to $\tilde{F}$, which pays no income, so that its drift μ_F+r is replaced by r. Define the risk-neutral path of the futures price *by*

$$F_t^{\mathbb{Q}}=e^{r(T_F-t)}\,\tilde{F}_t^{\mathbb{Q}},\qquad 0\le t\le T,$$

and define $\mathbb{E}^{\mathbb{Q}}$ and $\mathbb{Q}$ for quantities computed from the futures path as in Definition 20.6, with $F^{\mathbb{Q}}$ in the place of $S^{\mathbb{Q}}$. Then

$$F_t^{\mathbb{Q}}=F\exp\left(-\tfrac{1}{2}\sigma^2t+\sigma W_t\right).$$

When the futures is written on an asset with a yield q, this is the path rebuilt from the path $S^{\mathbb{Q}}$ of Definition 20.6(i), $F_t^{\mathbb{Q}} = S_t^{\mathbb{Q}} e^{(r-q)(T_F-t)}$, so the two descriptions of $\mathbb{E}^{\mathbb{Q}}$ agree. Consequently, for $0 \le t < t+\Delta t \le T$,

$$F_{t+\Delta t}^{\mathbb{Q}} = F_t^{\mathbb{Q}}\, Y \qquad \text{with} \qquad Y = \exp\big(-\tfrac{1}{2}\sigma^2\Delta t + \sigma(W_{t+\Delta t} - W_t)\big),$$

where the random variable Y is independent of $F_t^{\mathbb{Q}}$ and $\mathbb{E}\,Y = 1$. In particular $\mathbb{E}^{\mathbb{Q}} F_T = F$: the futures price has zero drift.

(ii) Under $\mathbb{Q}$, $dF = \sigma F\, dW^{\mathbb{Q}}$, and $F_T = F\exp\big(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z\big)$ with $Z \sim \mathcal{N}(0,1)$: exactly the law (20.6).

Proof. We substitute the explicit formulas and read each claim off the resulting geometric Brownian motion.

Part (i). Since $\tilde{F}_0 = e^{-rT_F} F$,

$$F_t^{\mathbb{Q}} = e^{r(T_F-t)}\, e^{-rT_F} F\, e^{rt} \exp\big(-\tfrac{1}{2}\sigma^2 t + \sigma W_t\big) = F\exp\big(-\tfrac{1}{2}\sigma^2 t + \sigma W_t\big),$$

because $r(T_F - t) - rT_F + rt = 0$. For a futures on an asset with a yield q, substitute $S_t^{\mathbb{Q}} = S\exp\big((r-q-\tfrac{1}{2}\sigma^2)t + \sigma W_t\big)$:

$$S_t^{\mathbb{Q}}\, e^{(r-q)(T_F-t)} = S e^{(r-q)T_F} \exp\big(-\tfrac{1}{2}\sigma^2 t + \sigma W_t\big),$$

the factors $e^{(r-q)t}$ and $e^{-(r-q)t}$ cancelling, and $Se^{(r-q)T_F} = F$ by Proposition 16.15(ii) with Theorem 16.28. So $S_t^{\mathbb{Q}} e^{(r-q)(T_F-t)}$ is the same path. The futures price is the function $S_t e^{(r-q)(T_F-t)}$ of the price of the asset, and $F^{\mathbb{Q}}$ is the same function of $S^{\mathbb{Q}}$; so a quantity computed from the futures path has the same risk-neutral expectation whether it is rebuilt from $F^{\mathbb{Q}}$ or, as Definition 20.6(i) prescribes for the asset, from $S^{\mathbb{Q}}$.

For the increment, divide the formula at $t+\Delta t$ by the formula at t: $F_{t+\Delta t}^{\mathbb{Q}} / F_t^{\mathbb{Q}} = \exp\big(-\tfrac{1}{2}\sigma^2\Delta t + \sigma(W_{t+\Delta t} - W_t)\big) = Y$. The variable Y is a function of an increment of W after t, so it is independent of W_t, and hence of $F_t^{\mathbb{Q}}$ (Definition 19.5(3)). The increment $W_{t+\Delta t} - W_t$ has the law of $\sqrt{\Delta t}\,Z$ (Definition 19.5(2)), so by Lemma 19.38(i) with $\theta = \sigma\sqrt{\Delta t}$,

$$\mathbb{E}\,Y = e^{-\sigma^2\Delta t/2}\,\mathbb{E}\,e^{\sigma\sqrt{\Delta t}\,Z} = e^{-\sigma^2\Delta t/2}\, e^{\sigma^2\Delta t/2} = 1.$$

With $t = 0$ and $\Delta t = T$, $\mathbb{E}^{\mathbb{Q}} F_T = \mathbb{E}\,F_T^{\mathbb{Q}} = F\,\mathbb{E}\,Y = F$.

Part (ii). By (i), $F^{\mathbb{Q}}$ is the geometric Brownian motion of Definition 19.27 with the drift 0 and the volatility σ, started at F. Corollary 19.34(ii) with $\mu = 0$ gives $dF^{\mathbb{Q}} = \sigma F^{\mathbb{Q}}\, dW$, which is what the shorthand "under $\mathbb{Q}$, $dF = \sigma F\, dW^{\mathbb{Q}}$" means. For a futures on an asset with a yield q the same equation is Corollary 19.36, applied to the path $S^{\mathbb{Q}}$, whose drift is $r - q$, with T_F in the place of T: the drift $(r-q) - r + q$ of the forward price vanishes. Finally W_T has the law of $\sqrt{T}\,Z$ (Definition 19.5(2)), so $F_T^{\mathbb{Q}} = F\exp(-\tfrac{1}{2}\sigma^2 T + \sigma W_T)$ has the law of the right side of (20.6). □

Equivalently, a futures price behaves like an asset with the yield $q = r$, that is, with the cost of carry $b = 0$: this is the case recorded in Corollary 20.45(ii). The discrete counterpart of Proposition 20.51 is Proposition 18.29(ii) in §18. On a binomial tree for an asset that pays no income, with rates that may depend on the path, the futures price at every node is the $\mathbb{Q}$-weighted average of the futures prices at the two children of the node. Example 20.53 gives the numbers, once the next theorem has turned the law into prices.

Theorem 20.52 (Black's model for futures options; Black, 1976)**.** *Assume the model of the Setup, whether or not the underlying of the futures can be held as an investment. Let $f \in \mathcal{G}$, and define, for $0 \le t < T$ and $F > 0$,*

$$V^{\mathrm{fut}}(F,t) = e^{-r(T-t)}\,\mathbb{E}\,f\big(F\exp\big(-\tfrac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}\,Z\big)\big), \qquad Z \sim \mathcal{N}(0,1), \tag{20.15}$$

and $V^{\mathrm{fut}}(F,T) = f(F)$. Then:

(i) the function V^{fut} has the properties (i) and (ii) of Theorem 20.7 in the variable F: it is infinitely differentiable on $(0,\infty)\times[0,T)$; every partial derivative of it has moderate growth on $(0,\infty)\times[0,T']$ for every $T'<T$; there is a constant c' with $|V^{\text{fut}}(F,t)|\le c'\,(F^{c'}+F^{-c'})$ on $(0,\infty)\times[0,T]$; and $V^{\text{fut}}(F,t)\to f(F^*)$ as $(F,t)\to(F^*,T)$ at every continuity point F^* of f. It solves the futures equation (20.14) of Corollary 20.45(ii);

(ii) the strategy of Definition 20.1 with the initial capital $V^{\text{fut}}(F,0)$ that holds

$$e^{r(T_F-t)}\,\partial_F V^{\text{fut}}(F_t,t)$$

units of the prepaid forward $\tilde F$ on $[0,T)$ has the value $V^{\text{fut}}(F_t,t)$ at every $t<T$ and the terminal value $f(F_T)$. Consequently the arbitrage-free time-0 price of the claim paying $f(F_T)$ at T is

$$V^{\text{fut}}(F,0)=e^{-rT}\,\mathbb{E}^{\mathbb{Q}}f(F_T)\,:$$

the Black-76 model of Definition 20.16 holds for the futures price;

(iii) in particular the European call and put on the futures are priced by Theorem 20.22, with F the current futures price and σ the volatility of the futures price (equal to that of the underlying when F_t is a deterministic multiple of S_t), and $C-\mathcal{P}=e^{-rT}(F-K)$; at a time $t<T$ their price functions (20.15) are

$$V^{\text{fut}}(F,t)=e^{-r(T-t)}\big[F\,\Phi(d_1)-K\,\Phi(d_2)\big] \qquad \textit{(call)},$$
$$V^{\text{fut}}(F,t)=e^{-r(T-t)}\big[K\,\Phi(-d_2)-F\,\Phi(-d_1)\big] \qquad \textit{(put)},$$

with d_1 and d_2 of (20.10) at $T-t$ in the place of T.

Sketch of the proof. The proof is a reduction to Theorem 20.7, with the prepaid forward $\tilde F$ in the role of the stock. By the check of the Setup, $\tilde F$, a forward contract and a deposit, is a traded instrument that pays no income and, under the model, a geometric Brownian motion with the volatility σ; and the claim pays $f\big(e^{r(T_F-T)}\,\tilde F_T\big)$, a payoff that is again in $\mathcal{G}$. The theorem, in whose conclusions the drift does not appear, gives a price function $\tilde V(\tilde F,t)$ with its properties (i) and (ii), a replicating strategy that holds $\partial_{\tilde F}\tilde V$ units of $\tilde F$, and the arbitrage-free price $\tilde V(\tilde F_0,0)$. The one computation is the return to the futures price: with $\tilde F=Fe^{-r(T_F-t)}$, the argument of f in the formula for $\tilde V$ is

$$e^{r(T_F-T)}\,Fe^{-r(T_F-t)}\exp\big((r-\tfrac12\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z\big)=F\exp\big(-\tfrac12\sigma^2(T-t)+\sigma\sqrt{T-t}\,Z\big),$$

because $r(T_F-T)-r(T_F-t)+r(T-t)=0$; so $\tilde V(Fe^{-r(T_F-t)},t)=V^{\text{fut}}(F,t)$. The chain rule turns the holding into $e^{r(T_F-t)}\,\partial_F V^{\text{fut}}$ units of $\tilde F$, carries the properties of $\tilde V$ to V^{fut} and, through Corollary 20.45(ii), gives the futures equation. At $t=0$ the price is $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}f(F_T)$ by Proposition 20.51(ii), so both clauses of Definition 20.16 hold for the futures price. That definition is the only hypothesis of Theorem 20.22, which gives the call, the put and their parity, and at a time $t<T$ the same formulas with $T-t$ in the place of T. The complete proof is in Appendix F.8. □

Three remarks on the proof. First, the hedge is held in $\tilde F$, that is, in forward contracts for delivery at T_F and in bonds (Setup). No unit of the underlying is held, which is why the theorem covers futures on consumption commodities. Second, for a futures on an asset with a yield q, $\tilde F_t=S_te^{-q(T_F-t)}$ is the reinvested holding of the asset over $[t,T_F]$, so the same hedge can be held in the asset with its income reinvested, and the price agrees with Corollary 20.41(i) applied to the payoff $f\big(S_Te^{(r-q)(T_F-T)}\big)$: there the argument of f is $Se^{(r-q)T_F}\exp(-\frac12\sigma^2T+\sigma\sqrt{T}\,Z)$, and $Se^{(r-q)T_F}=F$. Third, the parity $C-\mathcal{P}=e^{-rT}(F-K)$ is model-free; Proposition 20.54(i) proves it for every delivery date $T_F\ge T$ and every real r.

Example 20.53 (A commodity futures option, and its hedge without the commodity)**.** A commodity futures trades at $F=80$. Price the call and the put on it with $K=80$ and $T=0.5$, with $\sigma=35\%$ and $r=4\%$.

The prices. Here $\sigma\sqrt{T} = 0.2475$ and $K = F$, so $d_1 = 0.1237 = -d_2$ and $\Phi(0.1237) = 0.5492$ (0.54924 before rounding). By Theorem 20.52(iii) and Proposition 20.29,

$$C = \mathcal{P} = e^{-0.02} \times 80 \times (2 \times 0.54924 - 1) = 0.9802 \times 80 \times 0.09848 = 7.72.$$

The call and the put are equal by parity, because $K = F$, whatever the carry of the commodity is. The 0.4 rule gives $0.4 \times 0.9802 \times 80 \times 0.2475 = 7.76$, and the rule with $\varphi(0)$ gives 7.74.

The law. By Proposition 20.51, the risk-neutral mean of F_T is 80, while its $\mathbb{Q}$-median is $80\, e^{-\sigma^2 T/2} = 80\, e^{-0.030625} = 77.59$.

The hedge. Let the futures deliver at $T_F = 0.75$, three months after the option expires. By Proposition 21.2 in §21, read for an asset with the yield $q = r$ and with F in the place of S, the derivative of the call formula (20.8) in F is $\partial_F C = e^{-rT}\,\Phi(d_1) = 0.9802 \times 0.54924 = 0.5384$. The prepaid forward is worth $\tilde{F}_0 = e^{-0.03} \times 80 = 77.64$, and by Theorem 20.52(ii) the strategy holds

$$e^{rT_F}\,\partial_F C = e^{r(T_F - T)}\,\Phi(d_1) = e^{0.01} \times 0.54924 = 0.5548$$

units of it: the futures delta 0.5384 divided by the sensitivity $e^{-rT_F} = 0.9704$ of the prepaid forward to the futures price. The units are worth $0.5548 \times 77.64 = 43.07$, and the bank holds $7.72 - 43.07 = -35.35$, which is $-e^{-rT} K\,\Phi(d_2)$. Written with forwards struck at today's futures price 80, which cost nothing, the same hedge is 0.5548 long forwards for delivery at T_F, with the whole premium 7.72 in the bank. No unit of the commodity is held at any time.

The futures equation. Corollary 20.45(ii) and Theorem 20.52(i) say that $\partial_t C + \frac{1}{2}\sigma^2 F^2\,\partial_{FF} C = rC$. At time 0, central differences of the call formula of Theorem 20.52(iii), with the steps 0.01 in F and 10^{-5} in t, give $\partial_{FF} C = 0.0196000$ and $\partial_t C = -7.37430$ a year. The closed forms of Proposition 21.6 with the yield $q = r$ give the same numbers: $\partial_{FF} C = e^{-rT}\varphi(d_1)/(F\sigma\sqrt{T}) = 0.98020 \times 0.395900/19.799 = 0.0196000$ and $\partial_t C = -e^{-rT} F\sigma\,\varphi(d_1)/(2\sqrt{T}) + rC = -7.68320 + 0.30890 = -7.37430$. So

$$\partial_t C + \tfrac{1}{2}\sigma^2 F^2\,\partial_{FF} C = -7.37430 + 0.5 \times 0.1225 \times 6{,}400 \times 0.0196000 = -7.37430 + 7.68320 = 0.30890,$$

which is $rC = 0.04 \times 7.7225 = 0.30890$: the term $\frac{1}{2}\sigma^2 F^2\,\partial_{FF} C$ cancels the first part of $\partial_t C$ exactly, and no term in $\partial_F C$ appears.

(Problem P20.10 works this computation at different numbers.)

Interview trap. "Under $\mathbb{Q}$ the futures price drifts at r, like a stock." It has zero drift. A stock costs its price to hold, and under $\mathbb{Q}$ that outlay must earn r; entering a futures position costs nothing, so there is nothing to earn a return on. The futures price behaves like an asset with the yield $q = r$ (Proposition 20.51).

Interview trap. "An at-the-money futures call is worth more than the put because the commodity is in contango." At $K = F$ the call and the put are equal by parity, whatever the carry: the carry is already inside F (Example 20.53).

Proposition 20.54 (American futures calls can be exercised early)**.** *Let rates be deterministic, with a constant real rate r, and let C, $\mathcal{P}$ and C^{am} be the time-0 prices of the European call, the European put and the American call on a futures price, with strike $K > 0$ and expiry $T \le T_F$; the futures price is positive.*

(i) (Model-free, for every real r.) The European call satisfies

$$e^{-rT}(F - K)^+ \;\le\; C \;<\; e^{-rT} F \qquad \text{and} \qquad C = e^{-rT}(F - K) + \mathcal{P},$$

while $C^{\text{am}} \ge (F - K)^+$. So the European call is worth less than the exercise value $F - K$ exactly when

$$\mathcal{P} < (1 - e^{-rT})(F - K).$$

(ii) Let $r > 0$. The condition of (i) holds whenever $K \le F\,(1 - e^{-rT})$, and, under Black's model (Theorem 20.52), for every σ, whenever F is large enough. In any state in which it holds, $C^{\text{am}} > C$: the early-exercise premium is strictly positive, and exercising at once is worth strictly more than any policy that never exercises before T. No policy that never exercises before T is optimal: immediate exercise beats every such policy.

For $r > 0$, contrast Theorem 17.16: the lower bound $S - Ke^{-rT}$ of a call on a non-dividend stock exceeds $S - K$, while, for $F > K$, the lower bound $e^{-rT}(F - K)$ of the futures call falls short of $F - K$.

Sketch of the proof. The proof replicates statically and then compares payoffs (Lemma 16.8). Under deterministic rates F is also the forward price, so $e^{r(T_F - T)}$ forwards struck at F cost nothing and, with $e^{-rT}F$ on deposit, are worth F_T at T: a claim paying F_T costs $e^{-rT}F$. The call pays less, by $\min(F_T, K) > 0$: the upper bound. A call less a put pays $F_T - K$: the parity and, as $C \ge 0$ and $\mathcal{P} \ge 0$, the lower bound. Immediate exercise is worth $F - K$, so $C^{\text{am}} \ge (F - K)^+$. By the parity,

$$C - (F - K) = \mathcal{P} - \big(1 - e^{-rT}\big)(F - K),$$

which is negative exactly under the condition of (i). For $r > 0$ the upper bound gives it when $K \le F(1 - e^{-rT})$, and under Black's model it holds for large F, as the put tends to 0. Where it holds, exercising and buying the European call for C leaves the payoff of waiting until T plus $F - K - C > 0$. The complete proof is in Appendix F.9. □

Example 20.55 (A deep in-the-money futures call)**.** Take $F = 150$, $K = 100$, $r = 5\%$, $T = 1$ and $\sigma = 20\%$. Then

$$d_1 = \frac{\ln 1.5 + 0.02}{0.20} = 2.13, \qquad d_2 = 2.13 - 0.20 = 1.93,$$

and the put of Theorem 20.52(iii) is worth $e^{-0.05}\left[100\,\Phi(-d_2) - 150\,\Phi(-d_1)\right] = 2.565 - 2.382 = 0.183$, about 0.18 (computed with the unrounded $d_1 = 2.1273$ and $d_2 = 1.9273$); its $\Phi(-1.93) = 2.7\%$ sits next to the 2.28% anchor of Table 2. By parity the European call is worth

$$C = e^{-0.05} \times 50 + 0.18 = 47.56 + 0.18 = 47.74 \;<\; 50 = F - K.$$

Exercising now and investing the 50 beats holding the call to expiry, and the American call is worth at least 50.

The crude sufficient condition of Proposition 20.54(ii) does not apply here: it asks for $K \le F(1 - e^{-rT}) = 150 \times 0.04877 = 7.32$. The parity condition is what decides: $\mathcal{P} = 0.18 < (1 - e^{-0.05}) \times 50 = 2.44$. The conclusion depends on the volatility and on the moneyness. At $\sigma = 40\%$ the European call is worth $51.30 > 50$, and the comparison no longer favours immediate exercise. At $\sigma = 20\%$ the European call falls below the exercise value for every F above about 126.3. For contrast, a call on a non-dividend stock at 150 with the same strike is worth at least $150 - 100\,e^{-0.05} = 54.88 > 50$ (Proposition 17.6(iii)), which is why it is never exercised early.

(Problem P20.11 works this computation at different numbers.)

Interview trap. "A futures pays no dividends, so an American futures call is never exercised early." False whenever $r > 0$. Deep in the money the European call is worth about $e^{-rT}(F - K)$, less than the $F - K$ obtained by exercising now: exercise brings the cash forward, and the futures position that comes with it keeps the exposure at no cost (Example 20.55).

Remark 20.56 (Spot versus futures options). *(i) European options.* When the futures contract delivers at the expiry of the option, $T_F = T$, the futures price at T is the spot price, $F_T = S_T$ (Proposition 16.27). The European futures option then has the payoff of the European spot option, so the two have the same price (Lemma 16.8(iii)): the European futures option *is* the European spot option. Since $F_T = S_T$, Theorem 20.22 prices the spot option from the quoted F alone; whatever the asset pays or costs to hold reaches the price only through that quote.

(ii) American options. For American options the equality fails, and exercise matching compares them through their exercise values. Consider two American options with expiry T whose exercise values are cash amounts (a futures position received on exercise is worth zero and is closed at no cost; a share received or delivered is valued at its price), the first option's exercise value being at least the second's at every date $t \le T$. Then the first is worth at least the second. Indeed, if it were cheaper, buy the first, write the second and deposit the difference. If the second is exercised against the position at a time $t_e \le T$, exercise the first at the same time: the net cash flow at t_e is the first exercise value less the second, which is nonnegative. If the second is never exercised, let the first lapse. A positive inflow followed by a nonnegative value at T under every exercise policy is an arbitrage (Lemma 17.5(ii), stated there for $r \ge 0$; its proof, reading (a) of Definition 16.7, uses no sign of r, because the inflow, deposited until T, stays positive at every rate).

This comparison applies four times. The spot call is worth $S_t - K$ on exercise at t (pay K, receive the asset), the spot put $K - S_t$, the futures call $F_t - K$ and the futures put $K - F_t$ (Definition 20.50). Suppose first that $F_t \ge S_t$ at every date $t \le T$; for an investment asset with a cost of carry $b \ge 0$ this holds, because $F_t = S_t e^{b(T_F - t)}$. Then $F_t - K \ge S_t - K$, so the American futures call is worth at least the American spot call; and $K - S_t \ge K - F_t$, so the American futures put is worth at most the American spot put. If instead $F_t \le S_t$ at every date, the spot call's exercise value dominates the futures call's and the futures put's dominates the spot put's, so both inequalities reverse. Trees for futures options are in §18 (Proposition 18.25).

Interview trap. "An option on the futures is the same as an option on the spot." Only for European options whose expiry is the delivery date of the futures, $T_F = T$. American futures options and American spot options differ (Remark 20.56), and when the futures delivers after the option expires, F_T is not S_T.

Remark 20.57 (Interest-rate options). Black's model is also the market model for caps, floors and swaptions, with a forward rate or a forward swap rate in the role of F; interest-rate options are out of the scope of this Part.

Practice: problems P20.1–P20.12 in §27, solutions in §28.

21 The Greeks and Hedging

21.1 Delta and Delta Hedging

The writer of an option has two static choices, both of them bets on where the price ends, taken from opposite sides: doing nothing loses without bound if the option ends deep in the money, and buying the full underlying against it loses on the shares if the option ends out of the money, because by Corollary 17.12(ii), for a share that pays no dividend, that covered position is a bond paying K plus a written put, so the call risk has only become a put risk. The working answer is to hold, at every instant, minus the option's sensitivity to the underlying: this is the continuous-time reading of the node-by-node replicating holding of Theorem 18.3, and the options desk's version of the inventory hedge of §13.4. This subsection defines the five Greeks of a book, proves the delta of European calls and puts and of the underlying, forwards and futures, and sizes the hedge of a written block, which answers the opening questions of an options interview: "what is the delta of an at-the-money call?" and "how many shares hedge 1,200 of them, and what is traded after a one-point move?".

Setup. Time is measured in years. The underlying has price $S > 0$ at the current time t. An option on it has strike $K > 0$ and maturity $T > t$, and $\tau = T - t$ is its time to maturity; pricing at time 0 writes T for τ. The continuously compounded risk-free rate is r, and q is the continuous yield of the underlying: $q = 0$ for a stock that pays no dividend, q is the dividend yield for an

index, and $q = r_{\text{for}}$, the foreign rate, for a currency; for an option on a futures price, replace S by the futures price and q by r. The volatility is $\sigma > 0$, the forward price for delivery at T is $F = Se^{(r-q)\tau}$ (Proposition 16.15), and

$$d_1 = \frac{\ln(F/K) + \sigma^2\tau/2}{\sigma\sqrt{\tau}} = \frac{\ln(S/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau}. \tag{21.1}$$

The functions Φ and φ are the standard normal CDF and density, so $\Phi' = \varphi$ and $\varphi(-x) = \varphi(x)$; the literature writes them $N(d_1)$ and $N'(d_1)$. The European call and put prices C and $\mathcal{P}$ are the price functions of Corollary 20.41(i)(a), and the corollary obtains its closed form through Theorem 20.22: for $0 \le t < T$ they are the closed forms of that corollary with $\tau = T - t$ in the place of T,

$$C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad \mathcal{P} = Ke^{-r\tau}\Phi(-d_2) - Se^{-q\tau}\Phi(-d_1). \tag{21.2}$$

Two properties of these formulas are used throughout the section. Since $\Phi(x) + \Phi(-x) = 1$, subtracting the second formula from the first gives put–call parity in its yield form,

$$C - \mathcal{P} = Se^{-q\tau} - Ke^{-r\tau}, \tag{21.3}$$

which is Corollary 17.11(ii), the yield case of Theorem 17.9. And Lemma 20.34, with τ in place of T, is the key identity

$$Se^{-q\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2). \tag{21.4}$$

The value of a derivative, or of a book, is written $V(S, t, \sigma, r)$, a function of the price, the time, the volatility and the rate, and Π is the value of a hedged portfolio. A *book* is a finite collection of positions on one underlying: w_j units (negative for a short position) of an instrument of value V_j, for a total value $\sum_j w_j V_j$. The volatility is annualised with 252 trading days (Corollary 19.2), so one trading day is $\Delta t = 1/252$; systems that quote theta per calendar day divide by 365 instead.

Symbol notices for this section.

- The put is the calligraphic $\mathcal{P}$, because P is probability and p the tree probability. Vega is $\mathcal{V}$, because ν is the constant log-drift of this Part (and the customary letter for the degrees of freedom of Student's distribution). Rho is $\mathcal{R}$, because ρ is correlation.
- The symbol Δ alone is the Greek; Δt and ΔS are increments. The numbers d_1 and d_2 are not the down-factor d of a tree.
- A Greek takes no argument: Γ_t is its value at time t along a price path, $\partial_{SS}V(S, t)$ is the function, and a subscript label names a book or an instrument, as in Δ_Π and Γ_1.
- The letter r is the interest rate, not the ratio q/p of the Notation table, which never appears in this Part; q is the yield, not $1 - p$. The logarithm ln is the natural logarithm.
- Partial derivatives are written $\partial_S V$, $\partial_{SS}V$ and $\partial_t V$, or $\partial V/\partial S$ in a display, and never as subscripts, because a subscript is a time: $V_t = V(S_t, t)$ is the value at time t.

Through the yield q, every formula of the section covers index, currency and futures options as well as stock options. The single exception is the rho of a futures option, which Problem P21.3 derives.

Definition 21.1 (The Greeks of a position; delta-neutral books and delta hedging)**.** Let a position have value $V(S, t, \sigma, r)$, twice continuously differentiable in S and differentiable in t, in σ and in r, each with the other arguments held fixed.

(i) The *Greeks* of the position are delta, gamma, theta, vega and rho:

$$\Delta = \partial_S V, \qquad \Gamma = \partial_{SS}V, \qquad \Theta = \partial_t V, \qquad \mathcal{V} = \partial_\sigma V, \qquad \mathcal{R} = \partial_r V.$$

In a display the first two may also be written $\partial V/\partial S$ and $\partial^2 V/\partial S^2$. Theta is per year of calendar time; per trading day it is $\Theta/252$. Vega is per unit of volatility; per volatility

point (a change of 0.01 in σ) it is $\mathcal{V}/100$. Rho is per unit of rate; per percentage point it is $\mathcal{R}/100$. Along a price path the Greeks carry the time as a subscript, $\Delta_t = \partial_S V(S_t, t)$ and $\Gamma_t = \partial_{SS} V(S_t, t)$, in the same way as $V_t = V(S_t, t)$ is the value at time t.

(ii) For a book of value $\Pi = \sum_j w_j V_j$, every Greek is the w-weighted sum of the Greeks of the instruments, because differentiation is linear: $\Delta_\Pi = \sum_j w_j \, \partial_S V_j$, and likewise for the other four.

(iii) A book is *delta-neutral* when its delta is 0. *Delta hedging* a position of delta Δ_Π means holding $-\Delta_\Pi$ units of the underlying: $-\Delta$ units for each option held, and $+\Delta$ units for each option written. Neutrality holds only at the current (S, t), because Δ moves with both. The hedge is therefore rebalanced as the price and the time move (*dynamic hedging*); it is not set once and left.

Proposition 21.2 (Delta of European calls and puts)**.** *For the call and put prices (21.2) of Corollary 20.41, at every $S > 0$ and $\tau > 0$,*

$$\Delta_C = e^{-q\tau}\Phi(d_1) \in (0, e^{-q\tau}), \qquad \Delta_{\mathcal{P}} = e^{-q\tau}(\Phi(d_1) - 1) = -e^{-q\tau}\Phi(-d_1) \in (-e^{-q\tau}, 0).$$

Proof. We differentiate the price formulas directly and cancel the density terms with the key identity. By (21.1), $d_1 = [\ln S - \ln K + (r - q + \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$, and $d_1 - d_2 = \sigma\sqrt{\tau}$ does not depend on S, so

$$\partial_S d_1 = \partial_S d_2 = \frac{1}{S\sigma\sqrt{\tau}}.$$

The product rule and the chain rule, with $\Phi' = \varphi$, applied to the call formula in (21.2) give

$$\begin{aligned} \partial_S C &= e^{-q\tau}\Phi(d_1) + Se^{-q\tau}\varphi(d_1)\,\partial_S d_1 - Ke^{-r\tau}\varphi(d_2)\,\partial_S d_2 \\ &= e^{-q\tau}\Phi(d_1) + \frac{Se^{-q\tau}\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)}{S\sigma\sqrt{\tau}}. \end{aligned}$$

The numerator of the last fraction is zero by (21.4), so $\Delta_C = e^{-q\tau}\Phi(d_1)$. Since $0 < \Phi(d_1) < 1$, it lies strictly between 0 and $e^{-q\tau}$.

For the put, differentiate the parity identity (21.3) in S: its right side has derivative $e^{-q\tau}$, so $\Delta_C - \Delta_{\mathcal{P}} = e^{-q\tau}$ and $\Delta_{\mathcal{P}} = e^{-q\tau}(\Phi(d_1) - 1) = -e^{-q\tau}\Phi(-d_1)$, which lies strictly between $-e^{-q\tau}$ and 0. Differentiating the put formula directly gives the same result by the same cancellation: since $\varphi(-x) = \varphi(x)$,

$$\partial_S \mathcal{P} = -Ke^{-r\tau}\varphi(d_2)\,\partial_S d_2 - e^{-q\tau}\Phi(-d_1) + Se^{-q\tau}\varphi(d_1)\,\partial_S d_1 = -e^{-q\tau}\Phi(-d_1). \qquad \square$$

(Problem P21.2 asks you to reproduce this proof from scratch.)

Figure 42 plots the call delta against the price at three maturities. The delta rises from 0 far below the strike to $e^{-q\tau}$ far above it, and as expiry approaches the curve steepens toward the step $\mathbf{1}\{S > K\}$, the slope of the payoff $(S - K)^+$.

Example 21.3 (Hedging 1,200 written calls)**.** A dealer writes 1,200 nine-month at-the-money calls on a stock that pays no dividend: $S = K = 60$, $\tau = 0.75$, $\sigma = 30\%$, $r = 3\%$, $q = 0$. Then $\sigma\sqrt{\tau} = 0.30\sqrt{0.75} = 0.25981$ and, since $\ln(S/K) = 0$,

$$d_1 = \frac{(0.03 + 0.045)(0.75)}{0.25981} = 0.2165, \qquad d_2 = 0.2165 - 0.2598 = -0.0433.$$

By Proposition 21.2, $\Delta_C = \Phi(0.2165) = 0.5857$, and by (21.2) the call is worth $60(0.58570) - 60e^{-0.0225}(0.48273) = 35.142 - 28.319 = 6.823$.

The hedge. The block brings in 8,187 and the written book has delta -702.8, so the dealer buys 703 shares.

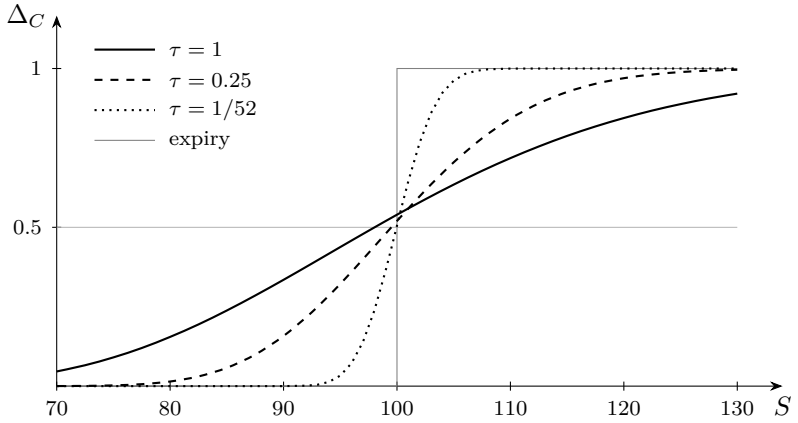


Figure 42: Call delta $\Phi(d_1)$ against the price S for $K = 100$, $\sigma = 20\%$ and $r = q = 0$: one year (solid), three months (dashed) and one week (dotted) before expiry, with the expiry step $\mathbf{1}\{S > K\}$ in thin gray. At the strike the three curves pass through $\Phi(\sigma\sqrt{\tau}/2) = 0.540$, 0.520 and 0.506, slightly above one half, while the curve steepens toward the step. The put delta is the same curve shifted down by $e^{-q\tau}$, which is 1 here.

After a move. If the price jumps to 61, then $d_1 = 0.2801$ and $\Delta_C = 0.6103$: the book delta is -732.4 and 29 more shares are bought. The gamma of §21.2 predicts the size of this trade: the gamma of one call is 0.0250 (Proposition 21.6(i)), so a one-point move changes the delta of the block by about $1{,}200 \times 0.0250 \times 1 = 30$. If the price falls to 59 instead, $\Delta_C = 0.5603$ and 31 shares are sold.

The cost of the move. The call is worth 6.82278 at 60, 7.42085 at 61 and 6.24970 at 59. With the exact hedge of 702.84 shares, the jump to 61 changes the value of the hedged book by $-1{,}200(7.42085 - 6.82278) + 702.84 = -14.8$, and the fall to 59 changes it by $-1{,}200(6.24970 - 6.82278) - 702.84 = -15.1$. Both are close to $-\frac{1}{2}(1{,}200)(0.0250)(1)^2 = -15.0$; with the 703 shares actually held, the two numbers are -14.7 and -15.3. The writer's hedge buys after rises and sells after falls, and the hedged written option loses on a move in either direction; §21.4 quantifies the loss. For the first trap below: $\Phi(d_2) = 0.483$, against $\Delta_C = 0.586$.

(Problem P21.1 works this computation at different numbers.)

Interview trap. "Delta is the probability of finishing in the money." The risk-neutral probability of exercise is $\Phi(d_2)$, the price of the digital before discounting (Theorem 20.17). The delta is $e^{-q\tau}\Phi(d_1)$, and $\Phi(d_1) > \Phi(d_2)$ is the exercise probability with each outcome reweighted by the terminal price (Proposition 20.26). In Example 21.3 the two numbers are 0.586 and 0.483: the delta is above one half while the probability of exercise is below it. For a one-year option struck at the forward at 20% volatility, with $q = 0$, they are 0.540 and 0.460.

Interview trap. "The delta of an at-the-money call is one half." At the money forward ($K = F$, so $d_1 = \sigma\sqrt{\tau}/2$) the delta is $e^{-q\tau}\Phi(\sigma\sqrt{\tau}/2)$, slightly above one half when $q = 0$: it is 0.540 at $\sigma\sqrt{\tau} = 0.2$, which the mental rule $\Phi(x) \approx 0.5 + 0.4x$ (the tangent of Φ at 0, of slope $\varphi(0) = 0.399$) gives as $0.5 + 0.4 \times 0.1$. At the money spot ($K = S$, with $q = 0$) d_1 gains the term $r\sqrt{\tau}/\sigma$, here 0.0866, which lifts the delta to 0.586 in Example 21.3. A large yield can pull the delta below one half. At the money forward with $q = 0$, the delta tends to one half only as $\sigma\sqrt{\tau} \to 0$, as the three values at the strike in Figure 42 show.

Interview trap. "Call delta minus put delta is one." By parity it is $e^{-q\tau}$ (Proposition 21.2), which is 0.990 for $q = 1\%$ over one year and equals one only without a yield. The same slip writes the put delta as $\Phi(d_1) - 1$, without the factor $e^{-q\tau}$.

Setup (linear contracts). A forward or a futures contract on the underlying has delivery date $T_F > t$, and $\tau_F = T_F - t$. The underlying is an investment asset (a stock, an index or a currency), as Propositions 16.15(ii) and 16.17(ii) require. Rates are deterministic, so that at every daily settlement the futures price equals the forward price for the same delivery date (Theorem 16.28, applied at time 0 and, to the days that remain, at the end of every day); between two settlements the futures price is taken to be that forward price as well. A futures position is re-marked to zero value at each daily settlement, which credits to the long side the change in the futures price since the previous settlement. Between two settlements the position is therefore owed the change of the futures price since the last one, and its value is defined as this accrued, unsettled credit: the futures price minus F', where F' is the futures price fixed at the last settlement, a constant until the next one. With this value, Definition 21.1 applies to futures positions as to any other instrument.

Proposition 21.4 (Delta of the underlying, forwards and futures)**.** *In the setup above:*

(i) One unit of the underlying has $\Delta = 1$.

(ii) A long forward contract on one unit with delivery price K *is worth* $Se^{-q\tau_F} - Ke^{-r\tau_F}$ *(Proposition 16.17(ii), the yield form, with Proposition 16.15), so its delta is* $\Delta = e^{-q\tau_F}$.

(iii) The futures price is $Se^{(r-q)\tau_F}$*, so between settlements a long futures position on one unit is worth* $Se^{(r-q)\tau_F} - F'$*. Its delta, which is the settlement credit per unit move in* S*, is* $\Delta = e^{(r-q)\tau_F}$.

(iv) All three positions have $\Gamma = 0$ *and* $\mathcal{V} = 0$.

Consequently a book of delta Δ_Π *is made delta-neutral by* $-\Delta_\Pi$ *units of the underlying, or by* $-\Delta_\Pi e^{q\tau_F}$ *forward contracts, or by* $-\Delta_\Pi e^{-(r-q)\tau_F}$ *futures contracts, each contract being on one unit.*

Proof. We differentiate the stated values in S at fixed t. (i) One unit is worth S, and $\partial_S S = 1$. (ii) The derivative of $Se^{-q\tau_F} - Ke^{-r\tau_F}$ in S is $e^{-q\tau_F}$. (iii) By Proposition 16.15(ii), which holds at the date t with S and τ_F in the places of the time-0 price and maturity (Proposition 16.17(ii)), the forward price for delivery at T_F is $Se^{(r-q)\tau_F}$; the futures price equals it at every settlement by Theorem 16.28, because rates are deterministic, and between settlements by the Setup. The value of the Setup is therefore $Se^{(r-q)\tau_F} - F'$, and since F' does not move between settlements, its derivative in S is $e^{(r-q)\tau_F}$: a move of the price from S to $S + \Delta S$ moves the futures price, and with it the credit due at the next settlement, by $e^{(r-q)\tau_F}\Delta S$. (iv) The three values S, $Se^{-q\tau_F} - Ke^{-r\tau_F}$ and $Se^{(r-q)\tau_F} - F'$ are linear in S, so their second derivatives in S vanish, and none of them contains σ, so their derivatives in σ vanish.

For the consequence, add w units of an instrument of delta $\Delta_1 \neq 0$ to the book. By Definition 21.1(ii), which applies to a futures position through its value in the Setup, the new delta is $\Delta_\Pi + w\Delta_1$, which is zero exactly when $w = -\Delta_\Pi/\Delta_1$. The three instruments have $\Delta_1 = 1$, $e^{-q\tau_F}$ and $e^{(r-q)\tau_F}$, which gives the three hedges. □

The forward and futures deltas differ by the factor $e^{r\tau_F}$, although the forward price and the futures price coincide, because the gain on the forward is paid at T_F and the gain on the futures is paid today.

Example 21.5 (One delta, three hedges)**.** A book of index options has delta +2,000 index units. The index stands at 5,000, with $r = 4\%$ and dividend yield $q = 1\%$. By Proposition 21.4 the book is made delta-neutral in any of three ways.

- *Spot:* sell 2,000 units of the index.
- *Six-month forward* ($\tau_F = 0.5$): sell $2{,}000\, e^{0.01\times 0.5} = 2{,}010.0$ units.
- *Six-month futures:* sell $2{,}000\, e^{-0.03\times 0.5} = 1{,}970.2$ units. With a contract on 50 times the index this is 39.4 contracts, rounded to 39.

The forward and the futures have the same price, $5{,}000\,e^{0.03\times0.5} = 5{,}075.6$, and the two hedges still differ by about 40 units (39.8 to one decimal).

(Problem P21.3 works this computation at different numbers.)

Interview trap. "The forward price and the futures price are equal, so hedging with either takes the same number of units." The deltas are $e^{-q\tau_F}$ and $e^{(r-q)\tau_F}$ (Proposition 21.4): daily settlement pays the gain on the futures now and the gain on the forward at maturity. In Example 21.5 the hedge is 2,010 units through forwards against 1,970 through futures.

21.2 Gamma, Theta and the Theta–Gamma Identity

A delta hedge is exact only for an instant: gamma measures how fast it goes stale, and theta measures what the position pays or earns as time passes. This subsection proves the gamma and theta formulas, shows that the Black–Scholes–Merton equation (Theorem 20.3) ties delta, gamma and theta together for any book of European options, so that "theta is the rent paid for gamma", the most useful single sentence of an options-trading interview, and explains why gamma and theta both explode at the strike as expiry approaches. On the way it answers two standard questions: "why does theta accelerate into expiry?" and "what is the gamma of a digital the day before expiry?".

Setup. The notation is that of §21.1: the price $S > 0$ at the time t, the strike K, the time to maturity $\tau = T - t > 0$, the rate r, the yield q, the volatility $\sigma > 0$, the forward price $F = Se^{(r-q)\tau}$, the numbers d_1 and d_2 of (21.1) and the prices C and $\mathcal{P}$ of (21.2). Gamma and theta are those of Definition 21.1: $\Gamma = \partial_{SS}V$, and $\Theta = \partial_t V$ per year, $\Theta/252$ per trading day. An option is *at the money forward* when $K = F$, so that $d_1 = \sigma\sqrt{\tau}/2$ and $d_2 = -\sigma\sqrt{\tau}/2$. In Corollary 21.8 a book is a finite collection of European calls, puts and forward contracts on one underlying, an investment asset with yield q as in the Setup (linear contracts) of §21.1, plus cash, each option being priced by Corollary 20.41 at one and the same volatility σ. The symbol $O(\sigma^2\tau)$ denotes a quantity bounded in absolute value by a constant multiple of $\sigma^2\tau$, and c is a positive constant declared where it is used. A starred letter, such as y^*, is a point supplied by Corollary 7.32; that corollary writes the point ξ, a letter which in this Part is the Itô integrand of §19.

Proposition 21.6 (Gamma and theta of European options)**.** *For the call and put prices (21.2), at every $S > 0$ and $\tau > 0$:*

(i) The call and the put have the same gamma,

$$\Gamma = \frac{e^{-q\tau}\varphi(d_1)}{S\sigma\sqrt{\tau}}.$$

(ii) Their thetas are

$$\Theta_C = -\frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}} + qSe^{-q\tau}\Phi(d_1) - rKe^{-r\tau}\Phi(d_2),$$
$$\Theta_{\mathcal{P}} = -\frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}} - qSe^{-q\tau}\Phi(-d_1) + rKe^{-r\tau}\Phi(-d_2).$$

(iii) (Near expiry.) At the money forward,

$$\Gamma = \frac{e^{-q\tau}}{S\sigma\sqrt{2\pi\tau}}\,e^{-\sigma^2\tau/8}, \qquad 1 - \frac{\sigma^2\tau}{8} \le e^{-\sigma^2\tau/8} \le 1,$$

so that $\Gamma = \frac{e^{-q\tau}}{S\sigma\sqrt{2\pi\tau}}(1 + O(\sigma^2\tau))$, about $0.4/(S\sigma\sqrt{\tau})$. At the money forward with $r = q = 0$, the call and the put have the theta

$$\Theta = -\frac{S\sigma}{2\sqrt{2\pi\tau}}\,e^{-\sigma^2\tau/8} = -\frac{S\sigma}{2\sqrt{2\pi\tau}}(1 + O(\sigma^2\tau)),$$

about $-0.2\,S\sigma/\sqrt{\tau}$. *Both grow like* $\tau^{-1/2}$ *as* $\tau \to 0$. *At any fixed* $S \neq K$, *in contrast,* $\Gamma \to 0$ *as* $\tau \to 0$, *and so does the density term* $-Se^{-q\tau}\varphi(d_1)\sigma/(2\sqrt{\tau})$ *of theta. Hence* $\Theta \to 0$ *when* $r = q = 0$, *while in general*

$$\Theta_C \to (qS - rK)\,\mathbf{1}\{S > K\}, \qquad \Theta_{\mathcal{P}} \to (rK - qS)\,\mathbf{1}\{S < K\} \qquad (\tau \to 0),$$

and the theta of a long forward contract worth $Se^{-q\tau} - Ke^{-r\tau}$ *tends to* $qS - rK$.

Sketch of the proof. We differentiate the closed forms and cancel the density terms with the key identity (21.4). Part (i) is $\partial_S \Delta_C$; the put delta $\Delta_C - e^{-q\tau}$ has the same derivative. For (ii), $\Theta = -\partial_\tau V$, and in $\partial_\tau C$ the identity merges the two density terms into

$$Se^{-q\tau}\varphi(d_1)\,\partial_\tau(d_1 - d_2) = \frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}};$$

the put follows from parity (21.3). For (iii), at the money forward $\varphi(d_1) = \varphi(0)\,e^{-\sigma^2\tau/8}$, and the exponential function, increasing and above its tangent at 0, puts $e^{-\sigma^2\tau/8}$ in $[1 - \sigma^2\tau/8, 1]$; at a fixed $S \neq K$, $|d_1|$ and $|d_2|$ are at least a constant times $1/\sqrt{\tau}$ for small τ, so the tangent bound sends $\varphi(d_1)/\sqrt{\tau}$ to 0, and $\Phi(d_1)$, $\Phi(d_2)$ tend to $\mathbf{1}\{S > K\}$. The complete proof is in Appendix G.1. □

The limits away from the strike are the carry of the position that the option becomes at expiry: an in-the-money call becomes a long forward, whose theta tends to $qS - rK$ by part (iii); an in-the-money put becomes a short forward, whose theta tends to $rK - qS$; and an out-of-the-money option becomes nothing. The limits can be watched in numbers. At $S = 110$, $K = 100$, $r = 5\%$, $q = 0$ and $\sigma = 20\%$, the call theta is -4.9979, -4.99975 and -4.999975 a year at $\tau = 10^{-2}$, 10^{-3} and 10^{-4}, tending to $-rK = -5$. The theta of the in-the-money put at $S = 90$ tends to $+5$ (it is $+4.99975$ at $\tau = 10^{-3}$). With $q = 2\%$ the call theta at $S = 110$ tends to $qS - rK = -2.8$ (it is -2.79998 at $\tau = 10^{-4}$). At $r = q = 0$ all of these thetas tend to 0.

Example 21.7 (Gamma and theta of a three-month at-the-money option, and the run into expiry)**.** Take $S = K = 100$, $\sigma = 20\%$, $\tau = 0.25$ and $r = q = 0$, so that the option is at the money forward: $\sigma\sqrt{\tau} = 0.1$, $d_1 = 0.05$ and $\varphi(0.05) = 0.3984$.

Three months out. By Proposition 21.6(i)–(ii),

$$\Gamma = \frac{0.3984}{100 \times 0.2 \times 0.5} = 0.03984, \qquad \Theta = -\frac{100 \times 0.3984 \times 0.2}{2 \times 0.5} = -7.97 \text{ a year.}$$

The approximation of part (iii) gives $\Gamma \approx 0.3989/10 = 0.0399$. Per trading day the theta is $-7.97/252 = -0.0316$, and per calendar day $-7.97/365 = -0.0218$. The call is worth $100\,(2\Phi(0.05) - 1) = 3.99$, so one trading day costs about 0.8% of its value.

A first look at the rent. Here $\frac{1}{2}\sigma^2 S^2\Gamma = \frac{1}{2}(0.04)(10{,}000)(0.03984) = 7.97 = -\Theta$. The equality is not a coincidence of the numbers: Corollary 21.8 below proves it for every book when $r = q = 0$.

The run into expiry. For the same strike one week before expiry ($\tau = 1/52$), $\Gamma = 0.144$ and the theta is -0.114 a trading day. One trading day before expiry ($\tau = 1/252$) the call is worth 0.503, its gamma is 0.317, about $\sqrt{63} = 7.9$ times the three-month value, and $\Theta = -63.3$ a year, which is -0.251 a trading day: the linearised decay of the last day is half of the remaining value.

Acceleration. The at-the-money value scales like $\sqrt{\tau}$ (Proposition 20.29), so a 20-trading-day option is worth 2.25 and a 5-trading-day option 1.12: the last week costs half of the four-week premium.

Figure 43 plots gamma against the price. The location of the peak takes one line of calculus. By part (i), $\ln\Gamma = -q\tau - \frac{1}{2}d_1^2 - \ln S - \ln(\sigma\sqrt{2\pi\tau})$, so

$$\partial_S \ln\Gamma = -d_1\,\partial_S d_1 - \frac{1}{S} = -\frac{1}{S}\Big(\frac{d_1}{\sigma\sqrt{\tau}} + 1\Big),$$

which is positive where $d_1 < -\sigma\sqrt{\tau}$ and negative where $d_1 > -\sigma\sqrt{\tau}$. Since d_1 increases with S, gamma peaks where $d_1 = -\sigma\sqrt{\tau}$, that is, where $\ln(F/K) = -\frac{3}{2}\sigma^2\tau$, or $S = Ke^{-(r-q+3\sigma^2/2)\tau}$: slightly below the strike when $r - q + 3\sigma^2/2 > 0$, as at $r = q = 0$. Substituting this d_1 and this S in (i), with $\varphi(\sigma\sqrt{\tau}) = e^{-\sigma^2\tau/2}/\sqrt{2\pi}$, gives the height of the peak,

$$\frac{e^{(r-2q+\sigma^2)\tau}}{K\sigma\sqrt{2\pi\tau}},$$

which at $r = q = 0$ is proportional to $1/\sqrt{\tau}$ up to the factor $e^{\sigma^2\tau}$. The size of theta peaks elsewhere. At $r = q = 0$ part (ii) gives $|\Theta| = S\sigma\varphi(d_1)/(2\sqrt{\tau})$ for the call and for the put, so

$$\partial_S \ln|\Theta| = \frac{1}{S} - d_1\,\partial_S d_1 = \frac{1}{S}\Big(1 - \frac{d_1}{\sigma\sqrt{\tau}}\Big),$$

which is positive where $d_1 < \sigma\sqrt{\tau}$ and negative where $d_1 > \sigma\sqrt{\tau}$. So $|\Theta|$ peaks where $d_1 = \sigma\sqrt{\tau}$, that is, where $\ln(S/K) = \sigma^2\tau/2$, or $S = Ke^{\sigma^2\tau/2}$: just above the strike, at 100.50 three months and at 102.02 one year before expiry for $K = 100$ and $\sigma = 20\%$.

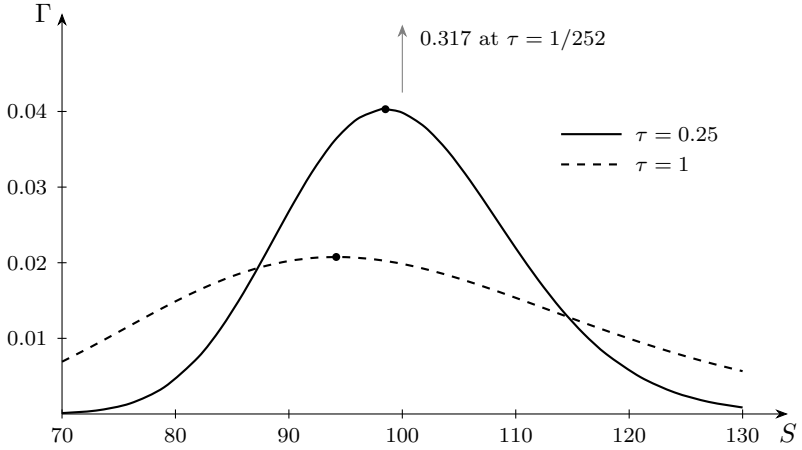


Figure 43: Gamma against the price S for $K = 100$, $\sigma = 20\%$ and $r = q = 0$: three months (solid, peak 0.040 at $S = 98.5$) and one year (dashed, peak 0.021 at $S = 94.2$) before expiry. The dots mark the peaks, at $S = Ke^{-(r-q+3\sigma^2/2)\tau}$, slightly below the strike. The short-dated curve is about twice as tall and half as wide: to leading order the peak is proportional to $1/\sqrt{\tau}$ and the width to $\sqrt{\tau}$, and the exact peaks are in the ratio $2e^{-0.03} = 1.94$. The arrow at $S = 100$ points off the scale, to the value 0.317 one trading day before expiry. The put has the same curve, and at $r = q = 0$ the size of theta, $|\Theta| = \frac{1}{2}\sigma^2S^2\Gamma$, is also a bell-shaped curve (Corollary 21.8), peaked just above the strike, at $S = Ke^{\sigma^2\tau/2}$.

Corollary 21.8 (The theta–gamma identity for a book)**.** *Consider a book of European calls, puts and forward contracts on one underlying, an investment asset with yield q as in the Setup (linear contracts) of §21.1, plus cash, each option being priced by Corollary 20.41 at one volatility σ. Let V be the value of the book and Δ, Γ, Θ its Greeks. For every $S > 0$ and every t before the earliest expiry and the earliest delivery date in the book,*

$$\Theta + (r - q)S\Delta + \tfrac{1}{2}\sigma^2S^2\Gamma = rV. \tag{21.5}$$

In particular:

- *(a) At $r = q = 0$, $\Theta = -\frac{1}{2}\sigma^2S^2\Gamma$ for every such book, whatever its delta: theta is the rent paid for gamma.*
- *(b) When $q = 0$, shares also satisfy (21.5): one share has $\Theta = 0$, $\Delta = 1$, $\Gamma = 0$ and value S. A book that is delta-hedged with the underlying, of total value Π, therefore satisfies $\Theta + \frac{1}{2}\sigma^2S^2\Gamma = r\Pi$.*

Sketch of the proof. We check the identity on each instrument and add up, by linearity in the positions. For an option it is the pricing equation with a yield (Corollary 20.45(i)),

$$\partial_t V + (r-q)S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V = rV,$$

read with $\Theta = \partial_t V$, $\Delta = \partial_S V$ and $\Gamma = \partial_{SS} V$ (Step 1 of the complete proof, valid at any volatility). A forward contract on an investment asset and cash satisfy the same equation, by differentiating their values. Since every option is priced at the one σ and t precedes every expiry and delivery date, these identities, weighted by the positions w_j, add up to that of the book. Part (a) is the case $r = q = 0$. For (b), at $q = 0$ a share has the left side rS, which is r times its value, and a delta-hedged book has $\Delta = 0$. The complete proof is in Appendix G.2. □

(Problem P21.5 asks you to verify the identity by substitution from scratch.)

Example 21.7 met part (a) on one option: $-\Theta = 7.97 = \frac{1}{2}\sigma^2 S^2 \Gamma$. With $r = 5\%$, Example 20.25 checks the identity with numbers on a one-year call, and Example 21.12 below checks it on the put with the same strike. Two limits of the identity should be kept in mind. A share that pays dividends violates it by its dividend stream: with $\Theta = 0$, $\Delta = 1$ and $\Gamma = 0$, the left side of (21.5) is $(r-q)S$, not rS, which is why part (b) assumes $q = 0$. And under a volatility smile (§22) each option is marked at its own implied volatility; by Step 1 of the proof of Corollary 21.8 (Appendix G.2) it then satisfies the identity at that volatility, so the book pays its rent option by option, each at its own volatility.

Interview trap. Pin risk. An at-the-money option one trading day before expiry ($S = K = 100$, 20% volatility) has a gamma of 0.317 per point, eight times its three-month value, and a theta of -0.25 a day (Example 21.7). Its gamma predicts that a one-point move swings the delta by about 0.32 (the exact change is 0.284 up and 0.288 down, since the gamma falls off within the point), against 0.04 three months out, and as $\tau \to 0$ the delta tends to the step of Figure 42. A writer whose options sit at the strike into expiry cannot re-hedge smoothly, and does not know until the close whether the stock will be delivered. Figure 43 shows the peak growing.

Interview trap. The gamma of a digital near expiry. Differentiating $\mathrm{Dig}(K) = e^{-r\tau}\Phi(d_2)$ (Theorem 20.17) in S, with $\partial_S d_2 = 1/(S\sigma\sqrt{\tau})$, gives the delta $e^{-r\tau}\varphi(d_2)/(S\sigma\sqrt{\tau})$, which has the shape of a vanilla gamma. Differentiating once more, with $\varphi'(x) = -x\varphi(x)$ and $d_2 + \sigma\sqrt{\tau} = d_1$,

$$\partial_{SS}\,\mathrm{Dig}(K) = \frac{e^{-r\tau}}{\sigma\sqrt{\tau}}\,\partial_S\Big(\frac{\varphi(d_2)}{S}\Big) = -\frac{e^{-r\tau}\varphi(d_2)}{S^2\sigma\sqrt{\tau}}\Big(\frac{d_2}{\sigma\sqrt{\tau}} + 1\Big) = -\frac{e^{-r\tau}\varphi(d_2)\,d_1}{S^2\sigma^2\tau}.$$

This gamma changes sign where $d_1 = 0$, that is, at $S = Ke^{-(r-q+\sigma^2/2)\tau}$, which is just below the strike when $r - q + \sigma^2/2 > 0$ (in particular at $r = q = 0$); and at a fixed number of standard deviations from the strike it grows like $1/\tau$, against $1/\sqrt{\tau}$ for a vanilla option. One trading day before expiry, on $K = 100$ at 20% volatility and $r = q = 0$, the delta is 0.317 per unit of payout at $S = 100$, and the gamma is $+0.147$ at $S = 99$ and -0.144 at $S = 101$. Dealers therefore hedge a digital as a tight call spread; §23.1 proves the bracket that justifies it (Proposition 23.2). Problem P21.2(d) asks for the same computation at other numbers.

Interview trap. "Why does theta accelerate into expiry?" At the money, with $r = q = 0$, theta is about $-S\sigma/(2\sqrt{2\pi\tau})$ (Proposition 21.6(iii)), which grows like $\tau^{-1/2}$, because the at-the-money time value is about $0.4\,S\sigma\sqrt{\tau}$, a concave function of τ. It is the rent for a gamma that grows at the same rate (Corollary 21.8). Away from the strike, at $r = q = 0$, both decay to zero: a deep out-of-the-money option one day before expiry has almost no theta. With rates, an in-the-money option keeps the carry theta $qS - rK$ or $rK - qS$ of Proposition 21.6(iii). Equivalently, since the at-the-money value is about $0.4\,S\sigma\sqrt{\tau}$ (Proposition 20.29) and the theta about $-0.2\,S\sigma/\sqrt{\tau}$, $\Theta \approx -C/(2\tau)$: one trading day costs the value divided by twice the number of trading days left, $3.99/126 = 0.032$ for the option of Example 21.7, against the exact 0.0316.

Interview trap. "Theta is always negative." A deep in-the-money European put earns the interest on its strike. With $q = 0$, Proposition 21.6(ii) gives $\Theta_{\mathcal{P}} = -S\varphi(d_1)\sigma/(2\sqrt{\tau}) + rKe^{-r\tau}\Phi(-d_2)$, and as $S \to 0$ the first term tends to 0 and $\Phi(-d_2) \to 1$, so $\Theta_{\mathcal{P}} \to rKe^{-r\tau} > 0$ when $r > 0$. For $S = 60$, $K = 100$, $r = 5\%$, $\sigma = 20\%$ and $\tau = 1$, $\Theta_{\mathcal{P}} = +4.5$ a year. A deep in-the-money call on an underlying with a high yield behaves in the same way: for large S its density term is small and $\Phi(d_1)$ and $\Phi(d_2)$ are close to 1, so its theta is close to $qSe^{-q\tau} - rKe^{-r\tau}$, the theta of the forward that it has become, which is positive when the yield term is the larger of the two.

Interview trap. "A delta-neutral book can be long gamma and collect theta." Not on one underlying at one flat volatility with $r = q = 0$: by Corollary 21.8(a), positive theta means negative gamma, whatever the delta. Each hypothesis is needed. With $r > 0$, the deep in-the-money put of the previous trap, delta-hedged with shares (which have no theta and no gamma), is long gamma with a positive theta. Under a smile each option pays rent at its own implied volatility (the paragraph after Corollary 21.8), so at $r = q = 0$ a book long one option of gamma Γ_1 marked at the implied volatility σ, short one option of a smaller gamma Γ_2 marked at another implied volatility σ', and delta-hedged with shares, which change neither its gamma nor its theta, is long gamma and has $\Theta = -\frac{1}{2}S^2(\sigma^2\Gamma_1 - \sigma'^2\Gamma_2)$, which is positive when $\sigma'^2\Gamma_2 > \sigma^2\Gamma_1$.

21.3 Vega, Rho and the Greeks through Parity

The Black–Scholes–Merton model assumes a constant volatility, yet desks mark options at an implied volatility that moves (Definition 20.37), so the sensitivity to σ is the second risk after the price; rho prices the least volatile input of the model. This subsection proves the vega and rho formulas and the relation $\mathcal{V} = \sigma\tau S^2\Gamma$, which makes long-dated options the vega trade, short-dated options the gamma trade, and a calendar spread the way to separate the two. It also proves the model-free result that a call and a put at one strike share their gamma and their vega, a standard interview question.

Setup. The notation is that of §21.1. Vega, $\mathcal{V} = \partial_\sigma V$, is the derivative with respect to the volatility at which the option is marked, with S, t, r and q held fixed. Rho, $\mathcal{R} = \partial_r V$, is the derivative in r with S, t, q and σ held fixed, so that the forward price $Se^{(r-q)\tau}$ moves with r. The vega of part (i), with its value and its two-sided bound at the money forward, is Proposition 20.35 of §20, recalled here in spot form; it is cited, not re-proved, and part (ii) uses the cancellation pattern of Proposition 21.2.

Proposition 21.9 (Vega and rho of European options; vega as gamma times maturity)**.** *For the call and put prices (21.2), at every $S > 0$ and $\tau > 0$:*

(i) The call and the put have the same vega,

$$\mathcal{V} = Se^{-q\tau}\varphi(d_1)\sqrt{\tau} > 0,$$

which is Proposition 20.35 in spot form, since $e^{-r\tau}F = Se^{-q\tau}$. At the money forward, $F = K$, the same proposition gives

$$\mathcal{V} = Se^{-q\tau}\sqrt{\tau}\,\varphi(\sigma\sqrt{\tau}/2) = Se^{-q\tau}\sqrt{\frac{\tau}{2\pi}}\,e^{-\sigma^2\tau/8},$$

with the factor $e^{-\sigma^2\tau/8}$ between $1 - \sigma^2\tau/8$ and 1: about $0.4\,Se^{-q\tau}\sqrt{\tau}$ per unit of volatility. Up to that factor, this is the derivative in σ of the approximation of Proposition 20.29.

(ii) The rhos are

$$\mathcal{R}_C = K\tau e^{-r\tau}\Phi(d_2) > 0, \qquad \mathcal{R}_{\mathcal{P}} = -K\tau e^{-r\tau}\Phi(-d_2) < 0.$$

(iii) For every European call and put, $\mathcal{V} = \sigma\tau S^2\Gamma$, and at fixed S and σ the ratio $\mathcal{V}/\Gamma = \sigma\tau S^2$ depends on the expiry alone. At the money forward,

$$\Gamma = \frac{e^{-(q+\sigma^2/8)\tau}}{S\sigma\sqrt{2\pi\tau}}, \qquad \mathcal{V} = S\sqrt{\frac{\tau}{2\pi}}\, e^{-(q+\sigma^2/8)\tau}, \qquad 1 - \Big(|q| + \frac{\sigma^2}{8}\Big)\tau \le e^{-(q+\sigma^2/8)\tau} \le e^{|q|\tau} :$$

up to the common factor $e^{-(q+\sigma^2/8)\tau}$, which tends to 1 as $\tau \to 0$, gamma is proportional to $\tau^{-1/2}$ and vega to $\tau^{1/2}$.

Sketch of the proof. We cite part (i), differentiate the closed form for (ii) and divide two formulas for (iii). At fixed S, r and q the forward price F does not contain σ, so the vega at fixed S is the vega at fixed F of Proposition 20.35, bound included. For (ii), $\partial_r d_1 = \partial_r d_2$, so in $\partial_r C$ the density terms cancel by the key identity (21.4) and only the derivative of the factor $e^{-r\tau}$ survives; the put follows from parity (21.3). For (iii), the density cancels in the ratio,

$$\frac{\mathcal{V}}{\Gamma} = Se^{-q\tau}\varphi(d_1)\sqrt{\tau}\cdot\frac{S\sigma\sqrt{\tau}}{e^{-q\tau}\varphi(d_1)} = \sigma\tau S^2,$$

and at the money forward the common factor is bounded because the exponential function is increasing and lies above its tangent at 0. The complete proof is in Appendix G.3. □

Example 21.10 (Vega, rho and a calendar spread)**.** Take a one-year at-the-money option on $S = K = 100$ at 20% volatility, with $r = q = 0$: $d_1 = 0.10$, $d_2 = -0.10$ and $\varphi(0.10) = 0.39695$.

Vega. By Proposition 21.9(i), $\mathcal{V} = 100 \times 0.39695 \times 1 = 39.7$ per unit of volatility, which is 0.397 per volatility point; the 0.4 rule gives 40. A rise of the volatility from 20% to 21% therefore adds about 0.40 to the price of the call or of the put (exact repricing gives 0.397).

Check of (iii). The gamma is $0.39695/(100 \times 0.2 \times 1) = 0.01985$, and $\sigma\tau S^2\Gamma = 0.2 \times 1 \times 10{,}000 \times 0.01985 = 39.7$.

Rho. By part (ii), $\mathcal{R}_C = 100 \times 1 \times \Phi(-0.10) = 46.0$ per unit of rate, which is 0.46 per percentage point, and $\mathcal{R}_{\mathcal{P}} = -100\,\Phi(0.10) = -54.0$.

A calendar spread. Buy this one-year option and sell the three-month option of Example 21.7, which has vega $100 \times 0.39844 \times 0.5 = 19.9$, gamma 0.0398 and theta -7.97 a year; the one-year option has theta $-100 \times 0.39695 \times 0.2/2 = -3.97$ a year. The spread has

$$\mathcal{V} = 39.7 - 19.9 = +19.8, \quad \Gamma = 0.0198 - 0.0398 = -0.0200, \quad \Theta = -3.97 + 7.97 = +4.0 \text{ a year.}$$

It is long vega, short gamma, and collects theta.

Interview trap. "Long vega and long gamma are the same bet." They share a sign for a single long option, but $\mathcal{V}/\Gamma = \sigma\tau S^2$ grows with the maturity (Proposition 21.9(iii)), so the calendar spread of Example 21.10 is long vega (+19.8) and short gamma (−0.020): long implied volatility and short realised movement. The same relation answers "are long-dated options the gamma trade?". It is the reverse: gamma is short-dated and vega is long-dated.

Interview trap. Units. A vega of 39.7 is per unit of volatility, and 0.397 per volatility point; quoting one for the other is an error by a factor of 100. The same holds for rho per unit of rate against rho per percentage point, and theta per year against theta per day differs by a factor of 252 (or 365). A rule of thumb keeps the scale honest: at the money the price is about proportional to σ (Proposition 20.29), so $\mathcal{V} \approx C/\sigma$, and one volatility point is worth about $1/(100\sigma)$ of the price. For the one-year option of Example 21.10, worth 7.97, one point is worth 0.397, which is 5% of the price at 20% volatility.

Proposition 21.11 (Call and put Greeks through parity)**.** *Consider any pricing model in which the prices C and $\mathcal{P}$ of a European call and a European put with the same strike K and the same*

expiry T, on an underlying with yield q, are functions of S, t, r and of finitely many parameters of the model, the yield q, the strike K and the expiry T being held fixed and not counted among the parameters. Let these functions be twice differentiable in S and differentiable in t, in r and in each parameter, and let parity hold identically in these variables on an open set of their values (Corollary 17.11(ii)):

$$C - \mathcal{P} = Se^{-q\tau} - Ke^{-r\tau}, \qquad \tau = T - t.$$

Then

$$\Delta_C - \Delta_{\mathcal{P}} = e^{-q\tau}, \qquad \Gamma_C = \Gamma_{\mathcal{P}}, \qquad \Theta_C - \Theta_{\mathcal{P}} = qSe^{-q\tau} - rKe^{-r\tau}, \qquad \mathcal{R}_C - \mathcal{R}_{\mathcal{P}} = K\tau e^{-r\tau},$$

and the call and the put have equal derivatives in every parameter of the model; in particular $\mathcal{V}_C = \mathcal{V}_{\mathcal{P}}$ when the volatility is a parameter.

Proof. We differentiate the parity identity. Since it holds at every point of an open set, each partial derivative of its left side equals the same partial derivative of its right side, the function $Se^{-q(T-t)} - Ke^{-r(T-t)}$ of S, t and r. That function is linear in S with slope $e^{-q\tau}$, which gives the delta relation, and its second derivative in S is zero, which gives the equality of the gammas. Its derivative in t is $qSe^{-q\tau} - rKe^{-r\tau}$, and its derivative in r is $K\tau e^{-r\tau}$. Finally it contains none of the parameters of the model, because q, K and T are not among them, so its derivative in any parameter is zero, and the derivatives of C and of $\mathcal{P}$ in that parameter are equal. □

The exclusion of the yield is needed: in q itself the derivatives of the call and the put differ by $\partial_q(Se^{-q\tau}) = -\tau Se^{-q\tau}$. For options marked with the Black–Scholes–Merton formula, "the same model" means the same σ. That is what parity forces on implied volatilities: §22 proves that a call and a put with the same strike and expiry have the same implied volatility (Proposition 22.3).

Example 21.12 (A put's Greeks from the call's)**.** Take the one-year option of Example 20.24, for which $C = 10.45$ and $\mathcal{P} = 5.57$: $S = K = 100$, $\sigma = 20\%$, $\tau = 1$, $r = 5\%$ and $q = 0$, so that $d_1 = 0.35$, $d_2 = 0.15$, $\varphi(0.35) = 0.37524$, $\Phi(0.15) = 0.55962$ and $Ke^{-r\tau} = 95.123$. The call's Greeks come from Propositions 21.2, 21.6 and 21.9; its delta, gamma and theta are the values that Example 20.25 found by central differences. The put's Greeks come from Proposition 21.11.

- *Delta:* the call has $\Delta_C = \Phi(0.35) = 0.637$, so the put has $\Delta_{\mathcal{P}} = 0.637 - 1 = -0.363$.
- *Gamma and vega,* the same for both: the gamma is $0.37524/(100 \times 0.2) = 0.0188$ and the vega is $100 \times 0.37524 = 37.5$, the value of Example 20.36.
- *Theta:* the call has $\Theta_C = -3.752 - 0.05 \times 95.123 \times 0.55962 = -3.752 - 2.662 = -6.414$, that is -6.41 a year, the first term being $-100 \times 0.37524 \times 0.2/2$. The difference $\Theta_C - \Theta_{\mathcal{P}}$ is $-rKe^{-r\tau} = -4.756$, so the put has $\Theta_{\mathcal{P}} = -6.414 + 4.756 = -1.658$, that is -1.66 a year.
- *Rho:* the call has $\mathcal{R}_C = 95.123 \times 0.55962 = 53.23$, that is 53.2. The difference $\mathcal{R}_C - \mathcal{R}_{\mathcal{P}}$ is $K\tau e^{-r\tau} = 95.12$, so the put has $\mathcal{R}_{\mathcal{P}} = 53.23 - 95.12 = -41.89$, that is -41.9.

Direct evaluation of the put formulas of Propositions 21.6(ii) and 21.9(ii) gives the same -1.66 and -41.9. At $r = 5\%$ the theta and rho differences are visible; at $r = q = 0$, as in Example 21.10, the two thetas coincide.

The theta–gamma identity at $r = 5\%$. For the call, (21.5) with $q = 0$ is the equation that Example 20.25 checks. For the put, with the unrounded gamma 0.018762 and delta -0.363169,

$$\Theta_{\mathcal{P}} + rS\Delta_{\mathcal{P}} + \tfrac{1}{2}\sigma^2 S^2 \Gamma = -1.6579 - 1.8158 + 3.7524 = 0.2787 = 0.05 \times 5.5735 = r\mathcal{P},$$

as Corollary 21.8 requires.

Interview trap. "A call and a put at the same strike have different gammas, since one gains on the way up and the other on the way down." Their difference is a forward position, linear in S and free of σ, so the gamma and the vega coincide in every model (Proposition 21.11). Only the delta (by $e^{-q\tau}$), the theta and the rho differ.

21.4 The Profit-and-Loss Explain

Every desk splits the profit and loss of the day into delta, gamma, theta and vega pieces, and interviewers ask for the split with numbers: "the book is delta-neutral, long gamma and short theta; the spot moves two points and the volatility rises a point; what was the P and L?". The split is a second-order Taylor expansion along a segment, and proving it with Theorem 7.31 and its integral remainder (7.3) states exactly what is left out: the cross Greeks and a remainder of third order. The cross Greeks come out with the right factors because a mixed partial derivative does not depend on the order of differentiation, which is Schwarz's theorem, Lemma 19.54, proved from the mean value theorem in §19, where the two-dimensional Itô lemma first needs it; the subsection opens by checking it on the vanna of an at-the-money call.

Setup. The value of a position or of a book is a function $V(S,t,\sigma)$ of the price, the time and the implied volatility σ at which the book is marked; the rate is held fixed and is not shown. The function V has continuous partial derivatives up to order three on an open convex set $\mathcal{O}$ of points (S,t,σ), which contains the start (S,t,σ) and the end $(S+\Delta S, t+\Delta t, \sigma+\Delta\sigma)$ of the move. For the call and put prices (21.2) this holds with $\mathcal{O}$ the set of points with $S>0$, $t<T$ and $\sigma>0$, which is open and convex: the formulas are built from Φ, exponentials, logarithms, square roots and quotients, all infinitely differentiable there. For a general payoff, Corollary 20.41(i)(b) gives the smoothness in (S,t). That corollary assumes the yield model, but its clause (b) is a statement about the function (20.12) alone: Step 4 of its proof derives it from identity (a), a substitution made in Step 3, and from clauses (i) and (ii) of Theorem 20.7, which hold for every real r and every $\sigma>0$, so it holds at the volatility at which the book is marked. At $q=0$ its smoothness clause is that of Theorem 20.7(i). A book of options and forward contracts inherits the property before its earliest expiry, so Lemma 19.54 always applies to such books.

The change in value is $\Delta V = V(S+\Delta S, t+\Delta t, \sigma+\Delta\sigma) - V(S,t,\sigma)$, and the Greeks Δ, Γ, Θ and $\mathcal{V}$ are evaluated at the start. Increments are written ΔS, Δt and $\Delta\sigma$, as everywhere in this Part, so that $\Delta\,\Delta S$ is the Greek times the move. The start is also written as the point $x=(x_1,x_2,x_3)=(S,t,\sigma)$, and the three increments form the vector $\Delta x=(\Delta x_1,\Delta x_2,\Delta x_3)=(\Delta S,\Delta t,\Delta\sigma)$, so that the end is $x+\Delta x$ and the segment between them is the set of points $x+u\Delta x$ with $0\le u\le 1$. The partial derivative in the j-th variable is $\partial_j V$, with $\partial_1=\partial_S$, $\partial_2=\partial_t$ and $\partial_3=\partial_\sigma$, and sums run over indices $j,k,l\in\{1,2,3\}$, which are unrelated to the positions w_j of §21.1. Second partial derivatives are written $\partial_{S\sigma}V$, $\partial_{\sigma\sigma}V$, $\partial_{St}V$, $\partial_{t\sigma}V$ and $\partial_{tt}V$; the order of differentiation is not shown because, by Lemma 19.54, it does not matter. The constants M_2 and M_3 bound the absolute values of all second and of all third partial derivatives of V on the segment from the start to the end. The unexplained part of the change is $\mathcal{E}$, and $\mathcal{E}_3$ is its third-order Taylor part; both are calligraphic because an italic letter would read as an expectation, and R_i are log-returns. The proof uses an auxiliary function χ of one real variable u. No tree appears in this section, so u is never the up-factor: it is the parameter of a Taylor segment, here and in Step 5 of the proof of Proposition 21.27. Finally ε and ε' are small positive numbers, and $O(\cdot)$ denotes a quantity bounded in absolute value by a constant multiple of its argument for all sufficiently small moves from a fixed start.

Example 21.13 (Vanna in both orders)**.** The mixed derivative $\partial_{S\sigma}V$ is called *vanna*. Take the three-month at-the-money call of Example 21.7: $S=K=100$, $\sigma=20\%$, $\tau=0.25$ and $r=q=0$, so that $d_1=0.05$, $d_2=-0.05$ and $\varphi(0.05)=0.39844$.

First S, then σ. The delta is $\Phi(d_1)$ (Proposition 21.2). Write $d_1=\ln(F/K)/(\sigma\sqrt{\tau})+\sigma\sqrt{\tau}/2$ and differentiate in σ with F fixed:

$$\partial_\sigma d_1 = -\frac{\ln(F/K)}{\sigma^2\sqrt{\tau}} + \frac{\sqrt{\tau}}{2} = -\frac{1}{\sigma}\Big(\frac{\ln(F/K)}{\sigma\sqrt{\tau}} - \frac{\sigma\sqrt{\tau}}{2}\Big) = -\frac{d_2}{\sigma}.$$

Hence $\partial_\sigma\partial_S C = \varphi(d_1)\,\partial_\sigma d_1 = -\varphi(d_1)\,d_2/\sigma = 0.39844\times 0.05/0.2 = 0.0996$.

First σ, then S. The vega is $S\varphi(d_1)\sqrt{\tau}$ (Proposition 21.9(i)). Since $\varphi'(x) = -x\varphi(x)$ and $S\,\partial_S d_1 = 1/(\sigma\sqrt{\tau})$,

$$\partial_S\partial_\sigma C = \sqrt{\tau}\,(\varphi(d_1) - S\,d_1\varphi(d_1)\,\partial_S d_1) = \sqrt{\tau}\,\varphi(d_1)\Big(1 - \frac{d_1}{\sigma\sqrt{\tau}}\Big) = \frac{\varphi(d_1)\,(\sigma\sqrt{\tau} - d_1)}{\sigma} = -\frac{\varphi(d_1)\,d_2}{\sigma},$$

the same 0.0996, as Lemma 19.54 says it must be. With a yield, both orders give $-e^{-q\tau}\varphi(d_1)\,d_2/\sigma$ by the same two computations. The put has the same vanna: its delta differs from the call's by $e^{-q\tau}$, which does not contain σ, and its vega is the call's.

Desk check by bumping. The delta is 0.519939 at 20% volatility and 0.520935 at 21%: a change of 0.000996 per volatility point, which is 0.0996 per unit of volatility. The vega per unit of volatility is 19.7225 at $S = 99$ and 19.9227 at $S = 101$: a central difference of $(19.9227 - 19.7225)/2 = 0.1001$ per point.

On a book. For the 5,000 such calls explained in Example 21.15 below, a move of two points together with a rise of one volatility point contributes the vanna term $5{,}000 \times 0.0996 \times 2 \times 0.01 = +9.96$.

Proposition 21.14 (Profit-and-loss explain)**.** *In the setup above,*

$$\Delta V = \Delta\,\Delta S + \tfrac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t + \mathcal{V}\,\Delta\sigma + \mathcal{E}, \tag{21.6}$$

where

$$\mathcal{E} = \partial_{S\sigma}V\,\Delta S\,\Delta\sigma + \tfrac{1}{2}\partial_{\sigma\sigma}V\,(\Delta\sigma)^2 + \partial_{St}V\,\Delta S\,\Delta t + \partial_{t\sigma}V\,\Delta t\,\Delta\sigma + \tfrac{1}{2}\partial_{tt}V\,(\Delta t)^2 + \mathcal{E}_3,$$

all derivatives being evaluated at the start, and

$$|\mathcal{E}_3| \le \frac{M_3}{6}\,(|\Delta S| + |\Delta t| + |\Delta\sigma|)^3.$$

Consequently:

(i) (Diffusive scaling.) If $|\Delta S| \le \varepsilon$, $|\Delta t| \le \varepsilon^2$ and $|\Delta\sigma| \le \varepsilon^2$ for some $0 < \varepsilon \le 1$, then

$$|\mathcal{E}| \le \tfrac{9}{2}\,(M_2 + M_3)\,\varepsilon^3.$$

This is the case of price moves of the order of $\sqrt{\Delta t}$, the scaling of Proposition 19.10, with volatility moves of second order: the remainder is then of third order in the price move, and the cross term in $\Delta S\,\Delta t$ is inside it.

(ii) (No volatility move.) If $\Delta\sigma = 0$ and $0 \le \Delta t \le 1$, then

$$|\mathcal{E}| \le M_2(|\Delta S|\,\Delta t + (\Delta t)^2) + \tfrac{2}{3}M_3(|\Delta S|^3 + (\Delta t)^3),$$

so that $\mathcal{E} = O(|\Delta S|^3 + |\Delta S|\,\Delta t + (\Delta t)^2)$ as the moves tend to zero with the start fixed.

Sketch of the proof. The proof is Taylor's theorem in one variable, applied to the value along the segment from the start to the end. Put $\chi(u) = V(x + u\,\Delta x)$, so that $\Delta V = \chi(1) - \chi(0)$. The segment lies in $\mathcal{O}$ because $\mathcal{O}$ is convex, and χ is defined on an open interval that contains $[0, 1]$ because $\mathcal{O}$ is open. Since V has continuous partial derivatives up to order three, the chain rule makes χ three times continuously differentiable, with $\chi'(u) = \sum_j \partial_j V\,\Delta x_j$, $\chi''(u) = \sum_{j,k}\partial_k\partial_j V\,\Delta x_j\Delta x_k$ and a third derivative of 27 analogous terms, the derivatives of V being taken at $x + u\,\Delta x$. Theorem 7.31 at order two, with its integral remainder (7.3), gives

$$\Delta V = \chi'(0) + \tfrac{1}{2}\chi''(0) + \mathcal{E}_3, \qquad \mathcal{E}_3 = \frac{1}{2}\int_0^1 (1-u)^2\,\chi'''(u)\,du.$$

The first term is $\Delta\,\Delta S + \Theta\,\Delta t + \mathcal{V}\,\Delta\sigma$. In $\chi''(0)$ each mixed derivative appears twice, once in each order, and with one value by Lemma 19.54; hence in $\frac{1}{2}\chi''(0)$ the three cross terms carry the

factor 1 and the three squares the factor $\frac{1}{2}$. The square in ΔS is the gamma term, and the other five terms are those of $\mathcal{E}$. On the segment $|\chi'''(u)| \le M_3\,(|\Delta S| + |\Delta t| + |\Delta\sigma|)^3$, so the bound (7.4) gives the bound on $\mathcal{E}_3$, since $3! = 6$. Parts (i) and (ii) bound the second-order terms of $\mathcal{E}$ one by one with M_2 and add this bound. The appendix does the arithmetic of the constants and shows that one pair of them serves every small move from a fixed start, as the $O(\cdot)$ form requires. The complete proof is in Appendix G.4. □

The function χ is the device of Step 1 of the proof of Theorem 19.30 and of the Intuition after it, where the segment also moves in time, and of Step 1 of the proof of Theorem 19.55.

Example 21.15 (Explaining a day on a long-gamma book)**.** A book is long 5,000 three-month at-the-money calls on $S = 100$ at $\sigma = 20\%$, with $r = q = 0$, and is delta-hedged: the delta of one call is 0.51994 (Example 21.13), so the book is short $5{,}000 \times 0.51994 = 2{,}599.7$ shares, about 2,600.

The Greeks of the book. One call has $\Gamma = 0.039844$ and $\Theta = -7.9689$ a year (the unrounded values of Example 21.7) and vega $100 \times 0.39844 \times 0.5 = 19.922$ (Proposition 21.9(i)); the shares add delta only (Proposition 21.4(i) and (iv); a share has $\Theta = 0$, Corollary 21.8(b)). The book therefore has

$$\Delta = 0, \qquad \Gamma = 199.22 \text{ per point}, \qquad \Theta = -39{,}844 \text{ a year}, \qquad \mathcal{V} = 99{,}611 \text{ per unit of volatility},$$

that is, a theta of -158.1 a trading day and a vega of 996.1 per volatility point. Check of Corollary 21.8(b), which at $r = 0$ reads $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = 0$ for a book delta-hedged with shares: $\frac{1}{2}(0.04)(10{,}000)(199.22) = 39{,}844 = -\Theta$.

The day. The spot moves to 102, the implied volatility to 21%, and one trading day passes: $\Delta S = 2$, $\Delta\sigma = 0.01$ and $\Delta t = 1/252$.

The explain. By (21.6):

$$\underbrace{0}_{\text{delta}} + \underbrace{\tfrac{1}{2}(199.22)(2)^2}_{\text{gamma: } +398.4} + \underbrace{(-39{,}844)/252}_{\text{theta: } -158.1} + \underbrace{99{,}611 \times 0.01}_{\text{vega: } +996.1} = +1{,}236.4.$$

Full revaluation. Each call goes from 3.9878 to 5.2707 (at $S = 102$, $\tau = 0.25 - 1/252$, $\sigma = 21\%$), a gain of $5{,}000 \times 1.282926 = 6{,}414.6$, and the 2,599.7 short shares lose 5,199.4: the hedged book makes +1,215.2.

The residual. The unexplained -21.2, which is 1.7% of the explained total, is the omitted part $\mathcal{E}$. Its two largest terms are the vanna term $\partial_{S\sigma}V\,\Delta S\,\Delta\sigma = +9.96$ (Example 21.13) and, in $\mathcal{E}_3$, the fall of gamma as the volatility rises, $\frac{1}{2}\,\partial_{SS\sigma}V\,(\Delta S)^2\,\Delta\sigma = -19.97$ (a Greek takes no argument, so the derivative is written on V; the number is a central difference of the book gamma in σ). The other terms of $\mathcal{E}$ add up to -11.2. The largest of them, $\partial_{t\sigma}V\,\Delta t\,\Delta\sigma = -7.89$, is of second order: the vega of the book falls as the day passes. Together with $\partial_{St}V\,\Delta S\,\Delta t = -1.58$ and $\frac{1}{2}\partial_{tt}V\,(\Delta t)^2 = -0.63$, it almost cancels the vanna term, so that the five second-order terms of $\mathcal{E}$ add up to only -0.20 and the residual is in fact of third order, $\mathcal{E}_3 = -21.0$. These numbers are the leading pieces, not a full decomposition.

Without the volatility move. If the volatility stays at 20%, the explain is $398.4 - 158.1 = +240.3$ and the full revaluation is +236.0.

Breakeven. With no change in volatility, the gamma gain pays the theta of the day when $\frac{1}{2}\Gamma\,(\Delta S)^2 = |\Theta|\,\Delta t$, that is, when $|\Delta S| = \sqrt{2 \times 158.1/199.22} = 1.26$ points. This is $20\%/\sqrt{252}$ of the spot: the daily move at which the realised volatility equals the implied one, which is the link to §21.5.

(Problem P21.6 works this computation at different numbers.)

Remark 21.16 (Beyond the four Greeks). When the volatility move is of the same order as the price move, the terms $\partial_{S\sigma}V\,\Delta S\,\Delta\sigma$ and $\frac{1}{2}\partial_{\sigma\sigma}V\,(\Delta\sigma)^2$ of $\mathcal{E}$ join the second order; desks call these two

Greeks vanna and volga, and charm is $\partial_{St}V$, the coefficient of the term in $\Delta S\,\Delta t$. The truncation $\Delta\,\Delta S + \frac{1}{2}\Gamma\,(\Delta S)^2$ is the delta–gamma approximation behind the quadratic value-at-risk model, which is outside the scope of these notes.

Interview trap. "A delta-neutral book has no exposure to the underlying." It still carries $\frac{1}{2}\Gamma\,(\Delta S)^2$, which is symmetric in the move. A book with a large short gamma and no delta loses on a large move in either direction: the hedged written calls of Example 21.3 lost about 15 on a one-point move each way. A long-gamma book, for its part, loses money on every day quieter than the breakeven move (1.26 points in Example 21.15), though less than its theta unless the price does not move at all, so "theta is the P and L of a quiet day" is right only for a day with no move.

Interview trap. Units in the explain. A vega per unit of volatility multiplies $\Delta\sigma$ in units (0.01 for one volatility point), and a theta per year multiplies Δt in years (1/252 for a trading day). Multiplying a vega per volatility point by a move of 1 point is right; multiplying it by 0.01 is an error by a factor of 100.

Interview trap. "The explain is exact." It is local. The residual collects the second-order volatility terms, vanna when the spot and the volatility move together and volga when the volatility moves by a large amount, which is why books with a large skew exposure report them. And the third derivatives in the remainder blow up near expiry, as gamma itself does (Proposition 21.6(iii)), so for short-dated books and for moves of several standard deviations desks reprice in full instead of explaining.

21.5 The Variance Risk Premium and Delta-Hedged Profit and Loss

What does it mean to "trade volatility"? Mechanically, it means holding a delta-hedged option position, whose profit and loss is gamma-weighted realised-minus-implied variance; empirically, implied volatility systematically sits above realised volatility, so the side that is short options usually collects. This subsection proves the one-interval identity from Proposition 21.14 and Corollary 21.8, and then, with the stochastic calculus of §19 and the price function of Theorem 20.7, three exact statements about a whole hedging programme in the Black–Scholes–Merton model when the realised volatility differs from the implied one: on the mean, for every grid and every hedge (Proposition 21.19), on continuous rebalancing (Proposition 21.22, extended to a volatility re-set along the path by Corollary 21.24) and on discrete rebalancing (Proposition 21.27), leaving as Facts only the stochastic-volatility form (Fact 21.26) and the empirical premium (Fact 21.29).

Setup. The interview questions answered in this subsection are the breakeven daily move of a long 25-vol straddle, "you are long gamma and the stock rallies, what do you do?", "does hedging more often reduce the expected loss on an option bought too dear?" and the straddle bought at 25 implied that realises 20. An option on an underlier with price S is bought at the implied volatility σ_{imp} (Definition 20.37) and marked at σ_{imp} throughout, so its value is $V(S,t) = V^{\mathrm{BSM}}(S,t;\sigma_{\mathrm{imp}})$. Here $V^{\mathrm{BSM}}(S,t;\sigma)$ denotes the function (20.5) of Theorem 20.7 at the volatility σ. Its payoff, a function f of the class $\mathcal{G}$ of Definition 19.19(iv), is named in the sentence and never written as a subscript.

Scope of the citations of Theorem 20.7. Clauses (i) and (ii) of that theorem give the smoothness of the price function, the moderate growth of every derivative before expiry, the pricing equation (20.4), the growth bound and the limit at expiry. They are statements about the function (20.5) at any volatility $\sigma > 0$, and Steps 1 and 2 of their proof use Lemma 19.61, the growth facts of Lemma 19.20(i) and (ii) and the chain rule, and none of the assumptions (A1)–(A6) of §20. They therefore hold for $V^{\mathrm{BSM}}(\cdot,\cdot\,;\sigma_{\mathrm{imp}})$ whatever volatility the underlier realises. Every citation of them in this subsection is of this kind, and the first citation in each statement repeats the reminder in a parenthesis.

The Greeks Γ and Θ are those of V, at the volatility σ_{imp}, with Θ per year. Along a price path,

$\Delta_t = \partial_S V(S_t, t)$ and $\Gamma_t = \partial_{SS} V(S_t, t)$. The hedge is rebalanced at the dates $0 = t_0 < t_1 < \cdots < t_n$, where n, a count local to this subsection, is the number of rebalancing dates: the hedge is set at $t_0, \ldots, t_{n-1}$ and closed at t_n. The dates are equally spaced unless a statement says otherwise, with $\Delta t = t_{i+1} - t_i$; one trading day is $\Delta t = 1/252$. Over $(t_i, t_{i+1}]$ the hedge is short a number h_i of units of the underlier, fixed at t_i, with $h_i = \Delta_{t_i}$ for the delta hedge, and the move of the interval is $\Delta S_i = S_{t_{i+1}} - S_{t_i}$. No symbol is used for a simple return: the identities are written with $(\Delta S)^2$, and R_i keeps its one meaning, the log-return of §19.1.

In the model statements the underlier is the geometric Brownian motion $S_t = S_0 \exp\left((\mu - \sigma_{\text{real}}^2/2)\, t + \sigma_{\text{real}} W_t\right)$ of Definition 19.27, where μ is the real-world expected rate of return of the underlier (its drift) and W is a standard Brownian motion (Definition 19.5). The volatility actually realised, σ_{real}, is therefore a constant of the model in the propositions of this subsection; Corollary 21.24 lets it depend on the path at finitely many dates, and in §22 it is random. Rates are $r = q = 0$ unless a statement says otherwise. The symbol $\mathcal{E}$ is the omitted part of Proposition 21.14, $\mathcal{E}_n$ is the hedging error after n rebalancing dates, and ε_i is its contribution from one interval. The letters c, c' and c'' are generic constants, re-declared where they are used; X and Y are generic random vectors, ψ and ψ_1 are test functions, and $\bar{\psi}(x) = \mathbb{E}\,\psi(x, Y)$, as in Lemma 19.17; Z and Z' are independent standard normal variables; and a starred letter, such as y^*, is the point supplied by Corollary 7.32.

One interval. The identity behind the phrase "trading volatility" concerns a single rebalancing interval.

Proposition 21.17 (Delta-hedged profit and loss over one interval)**.** *Let $r = q = 0$. Let V be the price function, at one constant volatility $\sigma_{\text{imp}} > 0$, of a European call, of a European put, or of a book of the kind of Corollary 21.8 on one underlier: calls, puts, forward contracts and cash. Let T be the earliest expiry or delivery date in the book, and let one rebalancing interval run from t to $t + \Delta t < T$, with $\Delta t \geq 0$, while the underlier moves from $S > 0$ to $S + \Delta S > 0$. The position is long the book and holds $-\Delta$ units of the underlier, where Δ is the delta of the book at the start, kept over the interval. Its P and L over the interval is*

$$\tfrac{1}{2}\Gamma\left[(\Delta S)^2 - \sigma_{\text{imp}}^2 S^2\, \Delta t\right] + \mathcal{E}, \tag{21.7}$$

where Γ is the gamma of the book at the start and $\mathcal{E}$ is the remainder of Proposition 21.14 with $\Delta\sigma = 0$. In particular $\mathcal{E} = O\big(|\Delta S|^3 + |\Delta S|\,\Delta t + (\Delta t)^2\big)$ as the moves tend to zero with the start fixed.

Proof. We combine the P and L explain with the theta–gamma identity, in three lines.

Line 1 (the explain). In the variables (S, t, σ) the value V is infinitely differentiable on the open convex set $\mathcal{O} = (0, \infty) \times (-\infty, T) \times (0, \infty)$. Indeed, each option formula (21.2) of Corollary 20.41 is defined and infinitely differentiable for every time to its own expiry $\tau > 0$ and every $\sigma > 0$, and $t < T$ makes every such τ positive; at $r = q = 0$ a forward contract with delivery price K is worth $S - K$, and cash is constant. The set $\mathcal{O}$ contains the start $(S, t, \sigma_{\text{imp}})$ and the end $(S + \Delta S, t + \Delta t, \sigma_{\text{imp}})$ of the move. Proposition 21.14 with $\Delta\sigma = 0$ therefore gives

$$\Delta V = \Delta\, \Delta S + \tfrac{1}{2}\Gamma\, (\Delta S)^2 + \Theta\, \Delta t + \mathcal{E},$$

with $\mathcal{E}$ of the stated order by its part (ii).

Line 2 (the hedge). The $-\Delta$ units of the underlier, held over the interval, gain $-\Delta\, \Delta S$. At $r = q = 0$ no interest accrues on the cash that finances the position and the underlier pays no income, so the P and L of the interval is $\Delta V - \Delta\, \Delta S = \frac{1}{2}\Gamma\, (\Delta S)^2 + \Theta\, \Delta t + \mathcal{E}$: the delta term is cancelled.

Line 3 (the rent). The book is priced at the one volatility σ_{imp} and t is before its earliest expiry and delivery date, so Corollary 21.8(a), applied to V itself, gives $\Theta = -\frac{1}{2}\sigma_{\text{imp}}^2 S^2 \Gamma$. Hence $\Theta\, \Delta t = -\frac{1}{2}\Gamma\, \sigma_{\text{imp}}^2 S^2\, \Delta t$, and substituting in Line 2 gives (21.7). □

(Problem P21.8 asks you to reproduce this proof from scratch.)

In words: over one interval, the P and L of a delta-hedged position is gamma-weighted realised-minus-implied variance. The three lines use only that V, as a function of (S, t, σ), has continuous partial derivatives up to order three on an open convex set containing the move, and that, at $r = q = 0$, $\partial_t V + \frac{1}{2}\sigma_{\text{imp}}^2 S^2 \, \partial_{SS} V = 0$ at the start; they apply verbatim to any such price function.

The bracket of (21.7) vanishes at the *breakeven move* $|\Delta S| = \sigma_{\text{imp}} S\sqrt{\Delta t}$, the move that the implied volatility allots to the interval: for one trading day it is $\sigma_{\text{imp}}/\sqrt{252}$ of the price, about 1% at 16 vol (Corollary 19.2). Figure 44 plots the leading term against the move of the day. A long option loses its theta on a day without a move, breaks even at the implied move, and gains quadratically beyond it; the short option has the mirror image. The constant in $\mathcal{E}$ is built from the second and third derivatives of V, and gamma itself grows without bound at the strike as expiry approaches (Proposition 21.6(iii)). So (21.7) is a statement about one interval; the next three propositions say what happens over a whole hedging programme, to expiry included.

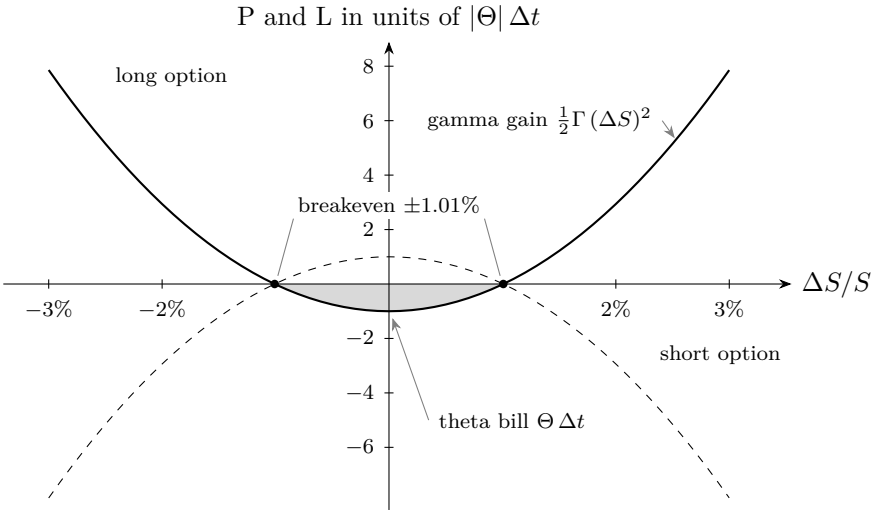


Figure 44: The one-day P and L of a delta-hedged long option, $\frac{1}{2}\Gamma\,[(\Delta S)^2 - \sigma_{\text{imp}}^2 S^2 \Delta t]$, against the relative move $\Delta S/S$ of the day, for $\sigma_{\text{imp}} = 16\%$. The vertical axis is in units of the theta bill of the day, $|\Theta|\,\Delta t$, so the solid curve is $\big((\Delta S/S)/(\sigma_{\text{imp}}\sqrt{\Delta t})\big)^2 - 1$. Its minimum, at no move, is the whole bill. It is negative on the gray band between the breakeven moves $\pm 0.16/\sqrt{252} = \pm 1.01\%$ (dots), and beyond them the gamma gain exceeds the bill: a 3% day earns 7.9 times the bill. The short option is the mirror image (dashed).

Example 21.18 (Breakeven move of a long 25-vol straddle, and a gamma scalp)**.** Take $S = 100$, $\sigma_{\text{imp}} = 25\%$ and $r = q = 0$.

The breakeven move. By (21.7), a hedged long option breaks even over one trading day when $|\Delta S| = \sigma_{\text{imp}} S\sqrt{\Delta t} = 0.25 \times 100/\sqrt{252} = 1.575$ points, which is 1.575% of the price. At 16 vol the same computation gives 1.01%, the rule "16 vol is 1% a day" of Corollary 19.2.

A one-month straddle. A one-month at-the-money straddle has $K = 100$ and $\tau = 1/12$, so $\sqrt{\tau} = 0.2887$, $d_1 = \sigma_{\text{imp}}\sqrt{\tau}/2 = 0.0361$ and $\varphi(d_1) = 0.3987$. By Proposition 21.6(i)–(ii) each of its two options has the gamma $0.3987/(100 \times 0.25 \times 0.2887) = 0.0552$ and the theta $-100 \times 0.3987 \times 0.25/(2 \times 0.2887) = -17.26$ a year. The straddle therefore has $\Gamma = 0.1105$ and a theta of -34.53 a year, which is -0.137 a trading day. At the breakeven move the gamma gain is $\frac{1}{2}(0.1105)(1.575)^2 = 0.137$: it pays the theta of the day exactly, as the proposition says.

Two days. The implied variance of one day is $\sigma_{\text{imp}}^2 S^2\,\Delta t = 2.48$ in squared points. By the leading term, a two-point day earns $\frac{1}{2}(0.1105)(4 - 2.48) = +0.084$, and a one-point day loses $\frac{1}{2}(0.1105)(1 - 2.48) = -0.082$, in either direction. Full revaluation of the hedged straddle one day later gives $+0.083$ after a two-point rise and -0.083 after a one-point rise, and $+0.090$ and -0.081

after falls of the same sizes; the differences are the remainder $\mathcal{E}$, which, unlike the leading term, depends on the direction of the move.

A gamma scalp. After a move of +2 the delta of the straddle has risen by about $\Gamma \times 2 = 0.221$, so the hedger sells 0.221 units at 102. When the price returns to 100 the delta is back where it started, and the 0.221 units are bought back. The two trades lock in $0.221 \times 2 = 0.442 = \Gamma\,(\Delta S)^2$, that is, $\frac{1}{2}\Gamma\,(\Delta S)^2$ for each of the two moves, against two days of theta, 0.274. A long-gamma hedger sells rallies and buys dips.

(Problem P21.7 works this computation at different numbers.)

Interview trap. "You are long gamma and the stock rallies: what do you do?" Sell. The rally raised the delta of the position by about $\Gamma\,\Delta S$, so the hedger sells into strength and buys back on dips, locking in $\frac{1}{2}\Gamma\,(\Delta S)^2$ for each move (Example 21.18). A short-gamma hedger is forced to buy high and sell low, which is where the premium that it collected goes.

Interview trap. "A delta-hedged long option makes money whenever the stock moves." Only on days with $|\Delta S| > \sigma_{\text{imp}} S\sqrt{\Delta t}$: 1.57% of the price at 25 vol, and 1% at 16 vol. The breakeven is a daily move, so it is annualised by multiplying by $\sqrt{252}$, never by 252.

A whole programme: the mean. Over the whole life of the option the mean of the P and L is computed exactly, for any grid and any hedge.

Proposition 21.19 (The expected P and L of a delta-hedged option, for every rebalancing grid)**.** *Let $r = q = 0$, and let the underlier follow the driftless geometric Brownian motion $S_t = S_0 \exp\big(-\sigma_{\text{real}}^2 t/2 + \sigma_{\text{real}} W_t\big)$ with $\sigma_{\text{real}} > 0$. A trader buys the European claim that pays $f(S_T)$ at T, with $f \in \mathcal{G}$ (Definition 19.19(iv)), at its Black–Scholes–Merton price $V^{\text{BSM}}(S_0, 0; \sigma_{\text{imp}})$ for some $\sigma_{\text{imp}} > 0$. The trader hedges it at any dates $0 = t_0 < t_1 < \cdots < t_n = T$ by holding $-h_i$ units of the underlier over $(t_i, t_{i+1}]$, where h_i is any Borel measurable function of $(S_{t_0}, \dots, S_{t_i})$ with $\mathbb{E}\,|h_i\,(S_{t_{i+1}} - S_{t_i})| < \infty$. Then the expected P and L at T is exactly*

$$V^{\text{BSM}}(S_0, 0; \sigma_{\text{real}}) - V^{\text{BSM}}(S_0, 0; \sigma_{\text{imp}}),$$

the difference of the Black–Scholes–Merton prices at the two volatilities, for every grid and every such hedge. The delta hedge $h_i = \Delta_{t_i} = \partial_S V^{\text{BSM}}(S_{t_i}, t_i; \sigma_{\text{imp}})$ qualifies, and so does $h_i = 0$, no hedge at all. Assumption (A6) is not used: the claim may be bought at any volatility, and the proposition is an identity between expectations.

Sketch of the proof. We take the expectation of the P and L term by term, integrating out each future move with Lemma 19.17. At $r = 0$ the P and L is the payoff, minus the premium, plus the gains of the hedge:

$$f(S_T) - V^{\text{BSM}}(S_0, 0; \sigma_{\text{imp}}) - \sum_{i=0}^{n-1} h_i\,(S_{t_{i+1}} - S_{t_i}).$$

Without drift, $V^{\text{BSM}}(S_0, 0; \sigma_{\text{real}})$ at $r = 0$ is, by its definition (20.5), the mean of f over the law of S_T (Step 2 of the complete proof). A hedge term (Step 3) is $h_i S_{t_i}\,(S_{t_{i+1}}/S_{t_i} - 1)$: the first factor is a function of the prices up to t_i, and the ratio is independent of them with mean 1, again because there is no drift, so the lemma gives the term, integrable by hypothesis, the mean 0. The appendix checks the integrability for the delta hedge. The complete proof is in Appendix G.5. □

The hedge changes the spread of the P and L, not its mean. With a real-world drift $\mu \neq 0$ the ratio $S_{t_{i+1}}/S_{t_i}$ has the mean $e^{\mu(t_{i+1} - t_i)} \neq 1$, so a hedge term need not have mean zero, and $\mathbb{E}\,f(S_T)$ is no longer the price at σ_{real}: the identity is about the driftless case. When the realised and the implied volatility agree, Proposition 21.27 measures the spread.

Example 21.20 (Same mean, different spread)**.** A one-year at-the-money call on $S_0 = K = 100$, with $r = \mu = 0$, is bought at $\sigma_{\text{imp}} = 25\%$ for 9.948 (the formula of Theorem 20.22) while the realised volatility is 20%, at which the call is worth 7.966. Proposition 21.19 gives the expected P and L $7.966 - 9.948 = -1.982$, whatever the hedge. Without a hedge the P and L is $(S_T - 100)^+ - 9.948$ with $S_T = 100\, e^{-0.02+0.2Z}$, so its standard deviation, 13.15, and its probability of a profit, $P(S_T > 109.948) = 28.3\%$, are exact lognormal integrals. A simulation with 1,000,000 paths for each hedged strategy, with exact lognormal steps, shows what the hedge does change:

hedge at the 25-vol delta	mean	standard error	standard deviation	profit probability
none (exact)	-1.982	–	13.15	28.3%
weekly ($n = 52$)	-1.981	0.001	1.16	2.0%
daily ($n = 252$)	-1.983	0.001	0.80	under 0.01%

The two simulated means are within two standard errors of -1.982. The hedge buys certainty, not expected value: the trader who paid 25 vol for 20-vol movement loses 1.98 on average however often the hedge is rebalanced.

Remark 21.21 (Lifetime profit and loss and its path dependence). Summing (21.7) over the rebalancing intervals suggests

$$\text{lifetime P and L} \;\approx\; \sum_i \tfrac{1}{2}\Gamma_{t_i}\left[(\Delta S_i)^2 - \sigma_{\text{imp}}^2 S_{t_i}^2\, \Delta t\right],$$

a gamma-weighted sum of realised-minus-implied variance: the one-line explanation of "trading vol". The weights $\Gamma_{t_i} S_{t_i}^2$ make the sum path-dependent: variance realised while the price is near the strike, where gamma is largest, counts most. The sign $\approx$ is not claimed as a theorem here, because the remainders of (21.7) have constants that blow up at expiry. Three exact statements stand in for it, all proved in this subsection: the mean, for every grid, is Proposition 21.19; continuous rebalancing, the formal limit in which $(\Delta S_i)^2$ becomes $\sigma_{\text{real}}^2 S_t^2\, dt$, is the pathwise identity of Proposition 21.22 below; and the error of discrete rebalancing at the right volatility is Proposition 21.27 below.

One approximation is kept, and labelled as an approximation. Freeze the gamma weight of a straddle at its at-the-money value, as if the price stayed pinned at the strike: $\Gamma_t S_t^2 \approx 2\varphi(0)\, S/(\sigma_{\text{imp}}\sqrt{T-t})$, by Proposition 21.6(iii) for each of the two options. Since $\int_0^T (T-t)^{-1/2}\, dt = 2\sqrt{T}$, integrating $\frac{1}{2}\Gamma_t S_t^2\, (\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2)$ over the life gives the loss

$$S\,(\sigma_{\text{imp}}^2 - \sigma_{\text{real}}^2)\,\frac{2\sqrt{T}}{\sigma_{\text{imp}}\sqrt{2\pi}}.$$

For a two-month at-the-money straddle on $S = 100$ bought at 30 implied with 21 realised, this is 4.98, against the exact mean loss 2.93 of Proposition 21.19; the straddle is that of Example 21.30, after Fact 21.29. The ratio is $(\sigma_{\text{real}} + \sigma_{\text{imp}})/\sigma_{\text{imp}} = 1.7$ to first order. This is the "more if the price sat near the strike" of the trap that follows Fact 21.29, quantified. A variance swap removes the gamma weights (§22.7).

Continuous rebalancing. When the hedge is adjusted continuously, the P and L becomes a pathwise integral.

Proposition 21.22 (Continuous-time hedged P and L at constant realised volatility)**.** *Let the underlier follow $S_t = S_0 \exp\left((\mu - \sigma_{\text{real}}^2/2)\, t + \sigma_{\text{real}} W_t\right)$ with constants μ and $\sigma_{\text{real}} > 0$, let $q = 0$, and let the bank pay and charge a constant rate r. Let $V(S,t) = V^{\text{BSM}}(S,t;\sigma_{\text{imp}})$ be the price function, at the rate r and at a volatility $\sigma_{\text{imp}} > 0$, of a European call or put with strike K, expiry T and payoff f: the function (20.5) of Theorem 20.7 at the volatility σ_{imp}. (Clauses (i) and (ii) of that theorem, cited below for the smoothness, the growth, the pricing equation and the limit at*

T, are statements about this function at any volatility $\sigma > 0$, and Steps 1 and 2 of their proof use none of (A1)–(A6), which this proposition does not assume.) Write $\Gamma_t = \partial_{SS} V(S_t, t)$.

A trader buys the option at time 0 for $V(S_0, 0)$ with borrowed money, holds $-\partial_S V(S_t, t)$ units of the underlier at every time $t < T$, keeps the balance in the bank, and marks the option at $V(S_t, t)$. Precisely, let Π_t be the value of the strategy of Definition 20.1 with two instruments, the underlier and the option quoted at its mark $V(S_t, t)$, on the single cell $[0, T)$, with the initial capital 0 and the holdings 1 (option) and $-\partial_S V(S_t, t)$ (underlier). Assumption (A6) is not made, and the mark is not claimed to be a price. Then:

(i) for every $t < T$, with probability one,

$$e^{-rt}\,\Pi_t = \int_0^t e^{-rs}\,\tfrac{1}{2}\Gamma_s S_s^2\,(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)\,ds, \tag{21.8}$$

a pathwise Riemann integral;

(ii) as $t \to T$, the right side of (21.8) converges on every path, monotonically, and Π_t converges in mean square to the terminal value Π_T of Definition 20.1(ii), in which the option is settled at its payoff $f(S_T)$; with probability one,

$$\Pi_T = e^{rT} \int_0^T e^{-rs}\,\tfrac{1}{2}\Gamma_s S_s^2\,(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)\,ds,$$

and this integral is positive on every path if $\sigma_{\text{real}} > \sigma_{\text{imp}}$ and negative on every path if $\sigma_{\text{real}} < \sigma_{\text{imp}}$, because $\Gamma_s > 0$ for a call or a put;

(iii) if $\mu = r = 0$, then $\mathbb{E}\,\Pi_T = V^{\text{BSM}}(S_0, 0; \sigma_{\text{real}}) - V^{\text{BSM}}(S_0, 0; \sigma_{\text{imp}})$.

Sketch of the proof. We apply Itô's lemma to the discounted mark and use the pricing equation at the marking volatility. With the initial capital 0, Definition 20.1(ii) makes $e^{-rt}\,\Pi_t$ the discounted mark minus the premium, $e^{-rt}\,V(S_t, t) - V(S_0, 0)$, minus the discounted gains of the hedge, $\int_0^t \partial_S V(S_s, s)\,d(e^{-rs} S_s)$: this is the bookkeeping (G.3) of the complete proof. Before expiry V is smooth, with derivatives of moderate growth, so Corollary 19.31, along the underlier of drift μS and diffusion coefficient $\sigma_{\text{real}} S$, gives the discounted mark the Itô integrand $e^{-rs}\sigma_{\text{real}} S\,\partial_S V$, which the hedge gains share, and the drift e^{-rs} times the left side of

$$\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma_{\text{real}}^2 S^2\,\partial_{SS} V - rV = (\mu - r) S\,\partial_S V + \tfrac{1}{2}(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2) S^2\,\partial_{SS} V,$$

the equality being the pricing equation (20.4) at σ_{imp}, which V solves whatever the path does. The hedge gains have the drift $e^{-rs}(\mu - r) S\,\partial_S V$, so subtracting leaves (i). For (ii), a call or a put has $0 < \Gamma_s S_s^2 \le S_s / (\sigma_{\text{imp}} \sqrt{2\pi (T - s)})$, so the right side converges monotonically on every path; on the left, $0 \le V \le S + K e^{|r| T}$, $|\partial_S V| \le 1$, the moments of the price and the Itô isometry give a limit in mean square, and the two limits agree with probability one. For (iii), at $\mu = r = 0$ the hedge gains have mean zero, also in the limit, and $\mathbb{E}\,f(S_T)$ is the price at σ_{real} (Step 2 of the proof of Proposition 21.19). The complete proof is in Appendix G.6. □

The same terminal value is reached by buying the option as a static position (Definition 20.1(iii)) and running only the part of the strategy that holds the underlier and the bank: the static position contributes the payoff $f(S_T)$ at T, which is what the mark converges to. So if the option could actually be bought at a volatility $\sigma_{\text{imp}} < \sigma_{\text{real}}$, the package would start from zero capital and end with $\Pi_T > 0$ with probability one: an arbitrage (Definition 20.1(v)). If it could be sold at $\sigma_{\text{imp}} > \sigma_{\text{real}}$, the opposite position would be one. In a market that satisfies (A6) and whose realised volatility is the constant σ_{real}, the option cannot trade at any $\sigma_{\text{imp}} \neq \sigma_{\text{real}}$, in agreement with Theorem 20.7. A volatility that is re-set along the path at finitely many dates is the subject of Corollary 21.24 below, proved by running the proof of Proposition 21.22 cell by cell (Appendix G.7); the general statement, with a stochastic volatility in continuous time, is Fact 21.26.

Example 21.23 (Every path loses)**.** Take the option of Example 21.20: a one-year at-the-money call on $S_0 = K = 100$ bought at $\sigma_{\text{imp}} = 25\%$, with the realised volatility 20% and $\mu = r = 0$. It is now hedged continuously at the 25-vol delta. By Proposition 21.22(ii) the terminal P and L is

$$\Pi_T = \tfrac{1}{2}\,(0.04 - 0.0625) \int_0^1 \Gamma_t S_t^2\, dt = -0.01125 \int_0^1 \Gamma_t S_t^2\, dt$$

with probability one; the integral is positive on every path, so Π_T is negative on every path.

The mean. By part (iii), $\mathbb{E}\,\Pi_T = 7.966 - 9.948 = -1.982$. The same number comes out of the gamma weights. With the gamma at 25 vol, $\Gamma_t S_t^2 = S_t\,\varphi(d_1)/(0.25\sqrt{1-t})$, and S_t is lognormal at 20 vol. Weighting the law of $\ln S_t$ by S_t raises its mean by $0.04\,t$, and completing the square in the Gaussian integral then gives

$$\mathbb{E}\big[\Gamma_t S_t^2\big] = \frac{S_0\,\varphi(\bar\sigma_t/2)}{\bar\sigma_t}, \qquad \bar\sigma_t^2 = 0.0625\,(1-t) + 0.04\,t,$$

where $\bar\sigma_t^2$ is the total variance that the hedged option sees: the implied variance still to run plus the realised variance already run. This rises from 158.3 at $t = 0$ to 197.9 at $t = 0.99$, and its integral over $[0, 1]$ is 176.18. The expectation enters the time integral by Fact 18.34(iii), because the integrand is nonnegative and, after it is set to zero on the event $S_1 = K$ of probability zero, every path of it, extended by zero at $t = 1$, is continuous on $[0, 1]$. So $\mathbb{E}\,\Pi_T = -0.01125 \times 176.18 = -1.982$. The formula also explains the answer. Since $d(\bar\sigma_t^2)/dt = 0.04 - 0.0625$, and the at-the-money call $100\,\big(2\Phi(\bar\sigma/2) - 1\big)$ has the derivative $100\,\varphi(\bar\sigma/2)/(2\bar\sigma)$ in the total variance $\bar\sigma^2$, the integrand $-0.01125\,\mathbb{E}[\Gamma_t S_t^2]$ is the rate of change of that call as $t \mapsto \bar\sigma_t^2$ runs from 0.0625 to 0.04. The mean P and L is therefore the call at the realised volatility minus the call at the implied one: the gamma rent walks the total variance from the implied to the realised value.

The spread. A simulation of 1,000,000 paths, with the time integral computed on a grid of 1,000 points refined toward expiry, gives a standard deviation of 0.67. Of the paths, 90% end between -3.05 and -0.90, and only one in a thousand loses less than 0.46. The daily hedge of Example 21.20 had the same mean and the larger standard deviation 0.80: rebalancing once a day adds noise, which Proposition 21.27 measures when the realised and the implied volatility agree; for unequal volatilities the comparison here is numerical only.

Interview trap. "Hedging more often reduces the expected loss on an option bought too dear." When the underlier has no real-world drift, it does not move the mean at all. By Proposition 21.19 (driftless, $r = q = 0$) the expected P and L is $V^{\text{BSM}}(S_0, 0; \sigma_{\text{real}}) - V^{\text{BSM}}(S_0, 0; \sigma_{\text{imp}})$ for every grid and every hedge: -1.98 in Example 21.20 with no hedge, with a weekly hedge and with a daily hedge. Rebalancing only narrows the spread, from 13.15 unhedged to 1.16 weekly and 0.80 daily; the continuous hedge of Proposition 21.22 has the standard deviation 0.67 (Example 21.23). With a drift, the hedge also strips out the directional return, which is no part of the volatility bet: $\mathbb{E}[h_i(S_{t_{i+1}} - S_{t_i})] = \big(e^{\mu(t_{i+1}-t_i)} - 1\big)\,\mathbb{E}[h_i S_{t_i}]$ by Lemma 19.17, as in Step 3 of the proof of Proposition 21.19.

Setup (a volatility re-set along the path). Fix deterministic re-set dates $0 = t_0 < t_1 < \cdots < t_J = T$, where J is the number of cells. On the cell $[t_{j-1}, t_j)$ the volatility is $\sigma_j = \sigma_j(Y_j)$, a function of the earlier Brownian values $Y_j = (W_{t_1}, \ldots, W_{t_{j-1}})$; the first one, σ_1, is a constant. Each function σ_j is piecewise continuous in the sense of Definition 19.22(a), that is, its domain is covered by finitely many disjoint boxes on each of which it is continuous, and $0 < \sigma_j \le c$ for one constant c. The value and the function share the letter σ_j, as $\xi_s = \xi(s, W_s, Y)$ does in Definition 19.22; write $\sigma(s) = \sigma_j$ for $s \in [t_{j-1}, t_j)$. The underlier is defined cell by cell: for $j = 1, \ldots, J$,

$$S_t = S_{t_{j-1}} \exp\big((\mu - \sigma_j^2/2)(t - t_{j-1}) + \sigma_j\,(W_t - W_{t_{j-1}})\big), \qquad t \in [t_{j-1}, t_j],$$

with constants μ and r and with $q = 0$. Let $V = V^{\text{BSM}}(\cdot, \cdot\,; \sigma_{\text{imp}})$, Γ_s, f and the trader be as in Proposition 21.22: a call or a put marked at the volatility σ_{imp}, bought for $V(S_0, 0)$ with borrowed money and hedged by a short position of $\partial_S V(S_s, s)$ units of the underlier.

On each cell the price of this underlier is a function of (t, W_t, Y_j), not of (t, W_t) alone, so Definition 20.1 does not admit it as an instrument. The P and L is therefore defined directly, by the bookkeeping (G.3): for t in the cell $[t_{j-1}, t_j)$,

$$e^{-rt}\,\Pi_t = e^{-rt}\,V(S_t,t) - V(S_0,0) \\ - \sum_{i=1}^{j}\Big[\int_{t_{i-1}}^{t_i\wedge t}\partial_S V(S_s,s)\,e^{-rs}(\mu-r)S_s\,ds + \int_{t_{i-1}}^{t_i\wedge t}\partial_S V(S_s,s)\,e^{-rs}\sigma_i S_s\,dW_s\Big], \tag{21.9}$$

where $t_i \wedge t = \min(t_i, t)$. Each bracket is a pathwise Riemann integral plus an Itô integral (Theorem 19.23), whose integrand is a function of (s, W_s, Y_i) of the kind that Definition 19.22(a)–(b) allows; Step 1 of the proof of Corollary 21.24, in Appendix G.7, verifies the admissibility. The sum is $\int_0^t \partial_S V(S_s,s)\,d(e^{-rs}S_s)$ written out cell by cell.

Corollary 21.24 (Continuous-time hedged P and L when the volatility is re-set along the path)**.** *In the Setup above, with the P and L Π_t defined by (21.9):*

(a) for every $t < T$, with probability one,

$$e^{-rt}\,\Pi_t = \int_0^t e^{-rs}\,\tfrac{1}{2}\Gamma_s S_s^2\,\big(\sigma(s)^2 - \sigma_{\text{imp}}^2\big)\,ds,$$

which is (21.8) with $\sigma(s)^2$ in place of σ_{real}^2;

(b) as $t \to T$ the right side of (a) converges on every path, and Π_t converges in mean square to a limit Π_T, the value with the option settled at its payoff $f(S_T)$; with probability one,

$$\Pi_T = e^{rT}\int_0^T e^{-rs}\,\tfrac{1}{2}\Gamma_s S_s^2\,\big(\sigma(s)^2 - \sigma_{\text{imp}}^2\big)\,ds.$$

In particular this integral is positive on every path along which $\sigma(s) \geq \sigma_{\text{imp}}$ for all s, with strict inequality on at least one cell, and negative on every path with the reverse inequalities, because $\Gamma_s > 0$ for a call or a put;

(c) if $\mu = r = 0$, then $\mathbb{E}\,\Pi_T = \mathbb{E}\,f(S_T) - V(S_0,0)$.

Sketch of the proof. We run the proof of Proposition 21.22 on each cell with the earlier Brownian values Y_j frozen, and add the cells. By induction on the cells, on the cell j the logarithm $\ln S_t$ is a function of (t, W_t, Y_j), continuous on each box and, since every volatility is at most c, at most a constant multiple of $1 + |W_t| + |W_{t_1}| + \cdots + |W_{t_{j-1}}|$ in absolute value. Moderate growth in the price therefore becomes exponential growth in the Brownian values: this makes the integrands of (21.9) admissible and is the hypothesis of the parameter clause of Itô's lemma (Theorem 19.30(ii)). With Y_j frozen, σ_j is a constant, and that clause with the pricing equation at σ_{imp} gives, as in the proposition with σ_j for σ_{real}, for $t_{j-1} < t \leq t_j$ with $t < T$, with probability one,

$$e^{-rt}\,V(S_t,t) - e^{-rt_{j-1}}\,V(S_{t_{j-1}},t_{j-1}) - [j\text{-th bracket of (21.9)}] = \int_{t_{j-1}}^{t} e^{-rs}\,\tfrac{1}{2}\Gamma_s S_s^2\,\big(\sigma_j^2 - \sigma_{\text{imp}}^2\big)\,ds.$$

Adding these identities over the cells, the discounted marks at the inner dates telescope, which gives (a). For (b), the limit on the last cell is taken as in the proposition, with c in the place of σ_{real}; integrating out each increment of W first (Lemma 19.17) gives $\mathbb{E}\,S_t^4 \leq S_0^4\,e^{(4|\mu|+6c^2)t}$, and $\Gamma_s S_s^2 > 0$ gives the sign cell by cell. For (c), at $\mu = r = 0$ the brackets are Itô integrals of mean zero, and the mean passes to the limit. The complete proof is in Appendix G.7. □

This is the exact form of "the replication cost is the gamma-weighted realised variance along the path" for a market whose volatility depends on what it has done, and the synthetic-option trap of §21.7 rests on it. The general continuous-time statement, with σ_t any bounded process determined by the past, is Fact 21.26 below; the gap between the two is the class of integrands (Fact 19.29(ii)).

Example 21.25 (Calm or wild: a volatility chosen by the path)**.** A one-year at-the-money call on $S_0 = K = 100$, with $\mu = r = q = 0$, is bought at $\sigma_{\text{imp}} = 20\%$ for 7.966 and hedged continuously at the 20-vol delta. The market realises 20% over the first half-year. At $t_1 = 0.5$ it turns wild, 30%, if the stock has fallen, and calm, 10%, if it has risen: a crude picture of an equity market, which is more volatile after a fall than after a rise. In the notation of the Setup before Corollary 21.24, $J = 2$ and $\sigma_1 = 0.2$. Since $S_{0.5} = 100\, e^{-0.01 + 0.2\, W_{0.5}}$ is below 100 exactly when $W_{0.5} < 0.05$, the second volatility is $\sigma_2 = 0.3$ on the box $W_{0.5} < 0.05$ and $\sigma_2 = 0.1$ on the box $W_{0.5} \geq 0.05$. The wild branch has the probability $\Phi(0.05/\sqrt{0.5}) = 0.5282$.

Path by path. By Corollary 21.24(b) the first half-year contributes exactly zero on every path, and, with probability one, the terminal P and L is

$$\Pi_T = \tfrac{1}{2}\,(0.09 - 0.04) \int_{0.5}^{1} \Gamma_s S_s^2\, ds \qquad \text{if the stock fell over the first half-year,}$$

$$\Pi_T = \tfrac{1}{2}\,(0.01 - 0.04) \int_{0.5}^{1} \Gamma_s S_s^2\, ds \qquad \text{if it rose.}$$

The integral is positive on every path, so the sign of the P and L is fixed at mid-year, path by path.

Averages. For a frozen value of $W_{0.5}$, the expectation of $\Gamma_s S_s^2$ over the later increment is a Gaussian integral, computed in closed form by completing the square; Lemma 19.17 justifies integrating the later increment out first. Quadrature in $W_{0.5}$ and in s then gives the average P and L $+2.0618$ on the wild branch and -1.7637 on the calm one. Hence

$$\mathbb{E}\,\Pi_T = 0.5282 \times 2.0618 - 0.4718 \times 1.7637 = +0.257.$$

Check by part (c). The mean payoff $\mathbb{E}\,(S_T - 100)^+$ is computed branch by branch from the Black–Scholes–Merton formula over the second half-year, at 30 vol and at 10 vol, and integrated over $W_{0.5}$: it is 8.22243. Minus the premium 7.96557, this is 0.25686, which is again $+0.257$.

Read from the other side. A dealer who creates this call synthetically at 20 vol, holding the delta and financing it from the premium, falls short of the payoff by 2.06 on average where the market turned wild, and has 1.76 to spare where it turned calm.

Fact 21.26 (Continuous-time hedged P and L under stochastic volatility (El Karoui, Jeanblanc-Picqué and Shreve, 1998))**.** Suppose that the underlier follows $dS_t = \mu_t S_t\, dt + \sigma_t S_t\, dW_t$, where the drift μ_t and the instantaneous volatility σ_t are bounded processes determined by the past of W: a general Itô process, the form that Fact 19.29 names. Let r be constant and $q = 0$. An agent who buys a European option whose payoff f is convex and lies in $\mathcal{G}$ (Definition 19.19(iv)) at its Black–Scholes–Merton price at the volatility σ_{imp}, and hedges it continuously with the delta at σ_{imp}, financing at r, ends with a P and L whose value discounted to time 0 is, with probability one,

$$\int_0^T e^{-rt}\, \tfrac{1}{2} \Gamma_t S_t^2\, (\sigma_t^2 - \sigma_{\text{imp}}^2)\, dt,$$

where $\Gamma_t \geq 0$ is the gamma at σ_{imp}: the identity (21.8) with σ_t in place of σ_{real}. Consequently a convex payoff of the class $\mathcal{G}$ sold at a volatility above every instantaneous realised volatility does not lose, with probability one, and gains whenever its gamma is positive on a set of times of positive length; for a call or a put the gamma is positive at every time before expiry. A proof is beyond our scope.

Its proof would need what this document does not build: the Itô integral and Itô's lemma for a general Itô process (Fact 19.29(ii)–(iii)), whose integrands fall outside the class of Definition 19.22, a drift μ_t that is piecewise continuous on every path, the P and L defined directly by the bookkeeping (G.3) as in Corollary 21.24, a moment bound for a general Itô process with bounded volatility (by localisation or an exponential martingale) and, for a payoff other than a call or a put,

the integrability of $\Gamma_t S_t^2$ near expiry; with these, Steps 1 to 4 of the proof of Proposition 21.22 carry over, the case of a volatility re-set at finitely many dates from the past of the path is Corollary 21.24, proved in Appendix G.7, and nothing argued in this section rests on the Fact.

Intuition. Fact 21.26 is the formal continuous-time limit of the lifetime sum of Remark 21.21: each squared move $(\Delta S_i)^2$ becomes $\sigma_t^2 S_t^2\,dt$, by the rule $(dW)^2 = dt$ of Proposition 19.10, and the sum becomes an integral with no remainder. Its reading is one of robustness. A writer who sold convex payoffs at a volatility above every realised instantaneous volatility does not lose, with probability one, and gains whenever gamma is positive on a set of times of positive length. The writer's model need not be right about the volatility; it only has to be on the right side of it.

Discrete rebalancing at the right volatility. When the realised volatility equals the implied one, what is left of the P and L is the error of rebalancing at discrete dates.

Proposition 21.27 (The discrete-hedging error: exact mean, exact variance and the $n^{-1/2}$ law, to expiry included (Boyle and Emanuel, 1980; Bertsimas, Kogan and Lo, 2000))**.** *Let $r = q = 0$, and let the underlier follow $S_t = S_0 \exp\big(-\sigma^2 t/2 + \sigma W_t\big)$: the real-world drift is zero, and the realised volatility equals the implied volatility $\sigma > 0$. Let $V(S,t) = V^{\mathrm{BSM}}(S,t;\sigma)$ for a payoff $f \in \mathcal{G}$ (Definition 19.19(iv)), and write $\Delta_s = \partial_S V(S_s, s)$ and $\Gamma_s = \partial_{SS} V(S_s, s)$. Fix a hedge horizon $T' \in (0, T]$, where $T' = T$ is allowed for a call or a put only, and an integer $n \geq 1$; put $t_i = iT'/n$ and $\Delta t = T'/n$. The discretely hedged position is long the claim, bought at $V(S_0, 0)$, short Δ_{t_i} units of the underlier over $(t_i, t_{i+1}]$, and closed at T' at the model value $V(S_{T'}, T')$, which is the payoff $f(S_T)$ when $T' = T$. Its* hedging error *is*

$$\mathcal{E}_n = V(S_{T'}, T') - V(S_0, 0) - \sum_{i=0}^{n-1} \Delta_{t_i}\,(S_{t_{i+1}} - S_{t_i}).$$

Then:

(i) the hedging error has mean zero, $\mathbb{E}\,\mathcal{E}_n = 0$, for every n;

(ii) the error is the sum $\mathcal{E}_n = \sum_{i=0}^{n-1} \varepsilon_i$ of the one-interval errors $\varepsilon_i = V(S_{t_{i+1}}, t_{i+1}) - V(S_{t_i}, t_i) - \Delta_{t_i}\,(S_{t_{i+1}} - S_{t_i})$, where $\mathbb{E}\,\varepsilon_i = 0$ and $\mathbb{E}[\varepsilon_i \varepsilon_j] = 0$ for $i \neq j$, and exactly

$$\operatorname{Var} \mathcal{E}_n = \sigma^2 \int_0^{T'} \mathbb{E}\big[(\Delta_s - \Delta_{t(s)})^2\, S_s^2\big]\,ds, \tag{21.10}$$

where $t(s)$ is the last grid date at or before s;

(iii) as $n \to \infty$,

$$n \operatorname{Var} \mathcal{E}_n \longrightarrow \tfrac{1}{2}\sigma^4\, T' \int_0^{T'} \mathbb{E}\big[\Gamma_t^2 S_t^4\big]\,dt,$$

and the integral is finite, also for $T' = T$, although Γ blows up at the strike. To leading order the standard deviation of the hedging error is therefore a constant times $n^{-1/2}$. For $T' < T$ the two sides differ by $O(n^{-1/2})$.

Sketch of the proof. We integrate out one move at a time (Lemma 19.17), then use the Itô isometry and a Taylor expansion of the delta on each interval. The underlier is driftless with the volatility σ at which V is computed, so at $r = 0$ the price function is its own expectation one step ahead: $\mathbb{E}\,V(xY, t_{i+1}) = V(x, t_i)$ for $x > 0$ when Y has the law of $S_{t_{i+1}}/S_{t_i}$, and $\mathbb{E}\,Y = 1$. That ratio is independent of the prices up to t_i, so $\mathbb{E}[\psi_1\,\varepsilon_i] = 0$ for every Borel measurable function ψ_1 of those prices with an integrable product: $\psi_1 = 1$ and $\psi_1 = \varepsilon_j$, $j < i$, give the mean and the orthogonality, and $\operatorname{Var} \mathcal{E}_n = \sum_i \mathbb{E}\,\varepsilon_i^2$ because the ε_i telescope to $\mathcal{E}_n$. For the same reason $V(S_t, t)$ has no drift (the pricing equation and Itô's lemma), so on a cell that ends before T both differences in ε_i are Itô integrals and the isometry gives

$$\varepsilon_i = \int_{t_i}^{t_{i+1}} (\Delta_s - \Delta_{t_i})\,\sigma S_s\,dW_s, \qquad \mathbb{E}\,\varepsilon_i^2 = \sigma^2 \int_{t_i}^{t_{i+1}} \mathbb{E}\big[(\Delta_s - \Delta_{t_i})^2\, S_s^2\big]\,ds;$$

on the last cell when $T' = T$ the second formula is a limit as the right end tends to T. Summing gives (21.10). For (iii) with $T' < T$, Taylor's theorem, $\Delta_s - \Delta_{t_i} = \Gamma_{t_i}(S_s - S_{t_i})$ plus higher order, and the moments of the ratio S_s/S_{t_i}, independent of S_{t_i}, make the integrand $\sigma^2(s-t_i)\,\mathbb{E}[\Gamma_{t_i}^2 S_{t_i}^4] + O((s-t_i)^{3/2})$; so $n\operatorname{Var}\mathcal{E}_n$ is $\frac{1}{2}\sigma^4 T'$ times a left Riemann sum of the Lipschitz function $t \mapsto \mathbb{E}[\Gamma_t^2 S_t^4]$, hence of its integral, up to $O(n^{-1/2})$. For $T' = T$ and a call or a put, $\mathbb{E}[\Gamma_t^2 S_t^4]$ is at most a constant times $(T-t)^{-1/2}$ near expiry, and the cells within a short time of expiry add to $n\operatorname{Var}\mathcal{E}_n$ at most a constant times its square root, while the other cells behave as for $T' < T$; letting that time shrink gives the limit, with no rate. The complete proof is in Appendix G.8. □

Intuition. A back-of-envelope reading of (iii) freezes $\Gamma_t S_t^2$ at its initial value ΓS^2. Then $\frac{1}{2}\sigma^4 T \int_0^T \Gamma^2 S^4\,dt = \frac{1}{2}(\Gamma S^2\sigma^2 T)^2$, so the standard deviation of the hedging error is about

$$\frac{\Gamma S^2\sigma^2 T}{\sqrt{2n}} = \frac{\mathcal{V}\,\sigma}{\sqrt{2n}},$$

by Proposition 21.9(iii). The scaling is right, and the constant runs low because gamma moves along the path: by about the factor $\sqrt{\pi/2}$ at the money (Example 21.28). Rebalancing four times as often halves the error, against transaction costs that grow with the number of rebalancings. Contrast this with the stop-loss rule: §21.7 shows that its expected shortfall is the time value of the option at every monitoring frequency (Proposition 21.34), with a spread that barely falls (Example 21.35).

Example 21.28 (The $n^{-1/2}$ law with numbers)**.** Take a one-year call with $S_0 = K = 100$, $\sigma = 20\%$ and $r = q = \mu = 0$.

Hedged over $T' = 0.9$. The call is hedged at n equally spaced dates and marked at the model value at T'. For each t the expectation $\mathbb{E}[\Gamma_t^2 S_t^4]$ is a Gaussian integral, in closed form by completing the square, and quadrature in t gives $\frac{1}{2}\sigma^4 T' \int_0^{0.9} \mathbb{E}[\Gamma_t^2 S_t^4]\,dt = 31.86$. By Proposition 21.27(iii) the standard deviation of the hedging error is about $\sqrt{31.86/n}$: 0.78 at $n = 52$ and 0.36 at $n = 252$. (Over 0.9 years these steps are not weekly and daily.) A simulation of 1,000,000 paths gives 0.78 and 0.355, with means within 0.001 of zero, as part (i) requires.

Hedged to expiry. With $T' = T$, and so with weekly and daily rebalancing, the limit constant is 49.68, which gives 0.98 at $n = 52$ and 0.44 at $n = 252$. The simulation gives 0.95 and 0.44: the approach to the limit is slower when the last cells sit at the strike.

The constant in closed form. At $S_0 = K$ the Gaussian integral can be done exactly. Write z_t for the value of d_1 at the price S_t and the time to expiry $T - t$. At $r = q = 0$, $\Gamma_t S_t = \varphi(z_t)/(\sigma\sqrt{T-t})$ (Proposition 21.6(i)), so $\mathbb{E}[\Gamma_t^2 S_t^4] = \mathbb{E}\big[S_t^2 e^{-z_t^2}\big]/(2\pi\sigma^2(T-t))$. By the completion of the square in Step 6(a) of the proof of Proposition 21.27 (Appendix G.8), the density of $\ln S_t$ at x, times e^{2x}, is $S_0^2 e^{\sigma^2 t}$ times the normal density with mean $\ln S_0 + 3\sigma^2 t/2$ and variance $\sigma^2 t$. Under that normal law z_t is normal with mean $\sigma(T+2t)/(2\sqrt{T-t})$ and variance $t/(T-t)$, and completing the square once more, in $\mathbb{E}\,e^{-z_t^2}$ (a normal variable $X = \mu_X + \sigma_X Z$ has $\mathbb{E}\,e^{-X^2} = e^{-\mu_X^2/(1+2\sigma_X^2)}/\sqrt{1+2\sigma_X^2}$), gives

$$\mathbb{E}\big[\Gamma_t^2 S_t^4\big] = \frac{S_0^2\, e^{-\sigma^2 T^2/(4(T+t))}}{2\pi\sigma^2\sqrt{T^2 - t^2}} \qquad (0 \le t < T).$$

The exponential lies between $e^{-\sigma^2 T/4}$ and 1, and $\int_0^T (T^2 - t^2)^{-1/2}\,dt = \pi/2$, so at $T' = T$ the limit constant $\frac{1}{2}\sigma^4 T \int_0^T \mathbb{E}[\Gamma_t^2 S_t^4]\,dt$ lies between $e^{-\sigma^2 T/4}\sigma^2 S_0^2 T/8$ and $\sigma^2 S_0^2 T/8$: here $49.50 \le 49.68 \le 50$, and $12.47 \le 12.48 \le 12.5$ for the three-month at-the-money call at 20 vol, whose constant reappears in Example 21.35. Since the vega is about $S_0\sqrt{T/(2\pi)}$ (Proposition 21.9(i)), the standard deviation of the error is about $\sigma S_0\sqrt{T/(8n)} = \sqrt{\pi/4}\,\mathcal{V}\sigma/\sqrt{n}$: the frozen-gamma rule of the Intuition times $\sqrt{\pi/2} = 1.25$.

The frozen-gamma rule. The rule $\mathcal{V}\sigma/\sqrt{2n}$ of the Intuition, at $T' = T$ and with the vega 39.695 of Example 21.10, has the constant $\frac{1}{2}(\mathcal{V}\sigma)^2 = \frac{1}{2}(39.695 \times 0.2)^2 = 31.5$ against 49.68. It gives 0.78

and 0.35 against the limits 0.98 and 0.44: it runs low by the factor $\sqrt{49.68/31.5} = 1.26$, close to the $\sqrt{\pi/2} = 1.25$ just found. Rebalancing four times as often halves the error; from $n = 52$ to $n = 252$ the factor is $\sqrt{252/52} = 2.2$.

Interview trap. "Hedged means riskless." The hedge removes the direction, not the volatility bet. The position is long realised variance and pays implied variance as rent. Its mark moves with the implied volatility, the vega term of Proposition 21.14. And with discrete rebalancing the P and L carries noise of order $n^{-1/2}$ (Proposition 21.27).

The premium. The identities above say how a delta-hedged option earns or loses; the data say on which side the money usually ends.

Fact 21.29 (Variance risk premium, empirical)**.** Across major option markets, equity indices most prominently, implied volatility systematically exceeds the volatility subsequently realised over the option's life, by a few volatility points on average on equity indices. By (21.7), through the lifetime sum of Remark 21.21, and, in the constant-volatility driftless model, exactly in the mean by Proposition 21.19, a systematically long, delta-hedged option book therefore loses money on average, and the short side earns a steady carry punctuated by occasional violent drawdowns. This is an empirical regularity, documented in delta-hedged-gains and variance-swap studies (Bakshi–Kapadia, 2003; Carr–Wu, 2009), not a theorem; there is nothing to prove, only data.

Intuition. The identity and the regularity fit together as an insurance market. A long option position is long realised variance and pays implied variance as rent: by (21.7), gamma gains accrue when the underlier moves and theta is the bill, priced at σ_{imp}. Options function as insurance against large moves, and the seller's losses arrive precisely in high-volatility crisis states. So buyers pay a premium above actuarial value, implied volatility sits above realised volatility on average, and the gap persists because bearing it is genuinely risky rather than free money.

Interview trap. "You buy a one-month straddle at 25 implied and delta-hedge daily; realised volatility comes in at 20. Do you make or lose money?" The tempting answer is that a hedged position is flat; the correct answer is that the position *loses*. By (21.7) the daily $(\Delta S_i)^2$ falls short of the theta rent $\sigma_{\text{imp}}^2 S_{t_i}^2 \, \Delta t$ on average, and to first order the loss is about $(1 - 20/25) = 20\%$ of the premium paid; exactly, in expectation and in the constant-volatility driftless model, it is the difference of the two Black–Scholes–Merton prices (Proposition 21.19). The loss is larger if the shortfall in realised volatility happened while the price sat near the strike, where the gamma weights concentrate. Problem P21.8 works the arithmetic in full.

Example 21.30 (The same arithmetic at other numbers)**.** Take $S = 100$, a two-month expiry, so that $T = 1/6$ and $\sqrt{T} = 0.4082$, and $r = q = \mu = 0$, the underlier following the driftless geometric Brownian motion of Proposition 21.19. An at-the-money straddle is bought at $\sigma_{\text{imp}} = 30\%$ and delta-hedged daily, and the realised volatility comes in at $\sigma_{\text{real}} = 21\%$.

The premium. By the rule of Proposition 20.29 for a straddle (Corollary 20.30), it costs about $0.8 \times 100 \times 0.30 \times 0.4082 = 9.80$. The exact price, from the first formula of that proposition, is $2 \times 100 \, (2\Phi(0.06124) - 1) = 9.77$, since $\sigma_{\text{imp}}\sqrt{T}/2 = 0.06124$.

The first-order loss. It is $(1 - 21/30) \times 9.77 = 2.93$, which is 30% of the premium.

A vega check. The vega of the straddle is $2 \times 100 \times \varphi(0.0612) \times 0.4082 = 32.5$ per unit of volatility (Proposition 21.9(i)). Times the shortfall of 0.09, this is 2.93.

The exact check. The payoff $|S_T - K|$ of the straddle is in the class $\mathcal{G}$, so Proposition 21.19 applies. The expected P and L of the hedged straddle, for daily or any other rebalancing, is the straddle at 21 vol minus the straddle at 30 vol: $6.84 - 9.77 = -2.93$, about 2.9% of the spot price and 30% of the premium.

Interview trap. "Do you hedge at the implied or at the realised volatility?" Both can be done, and they give different P and L. Hedging with the delta at the realised volatility, a known

constant of the model here, makes the total P and L a sure amount. Proposition 21.22(ii) with the marking volatility set to σ_{real} has a zero integrand, so the option marked at σ_{real}, minus its hedge, tracks exactly; the trader, who paid the price at the implied volatility, ends with $e^{rT}\left[V^{\text{BSM}}(S_0,0;\sigma_{\text{real}}) - V^{\text{BSM}}(S_0,0;\sigma_{\text{imp}})\right]$ with probability one. But the day-to-day P and L against the mark of the market at σ_{imp} swings, because the hedge is not the delta of the mark. Hedging at the implied volatility gives a P and L that accrues smoothly: its discounted value $e^{-rt}\Pi_t$ grows at the rate $e^{-rt}\,\frac{1}{2}\Gamma_t S_t^2\,(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2)$ per unit of time, which has one sign on every path (Proposition 21.22(i)–(ii)); at $r = 0$ this is the rate of the P and L itself. But its total depends on where the path spent its time, and only its mean equals the sure amount (Proposition 21.22(iii), with $\mu = r = 0$). For the straddle of Example 21.30, with 30 implied, 21 realised and $r = 0$ (the proposition is applied to the call and to the put separately and the results are added, since strategies add by Definition 20.1(iv)), the hedge at the realised volatility locks in the loss $6.84 - 9.77 = -2.93$ with probability one, and the hedge at the implied volatility loses 2.93 on average. Desks hedge at the implied volatility in practice because the realised volatility is not known in advance.

21.6 Gamma- and Vega-Neutral Books

The underlying hedges delta and nothing else, because shares, forwards and futures have zero gamma and zero vega (Proposition 21.4(iv)), so curvature and volatility risk are reduced only by trading other options, and with two traded options a book can be made delta-, gamma- and vega-neutral at once. Interviewers pose the question as linear algebra, "solve for the two option positions, then the stock". The two lessons behind it are why two options of the same expiry cannot do the job, and why the delta is fixed last.

Setup. A book on one underlying, whose price is $S > 0$, has the Greeks Δ_Π, Γ_Π and $\mathcal{V}_\Pi$, computed by additivity (Definition 21.1(ii)). Two options on the same underlying can be traded. Option i, for $i = 1, 2$, has the Greeks Δ_i, Γ_i and $\mathcal{V}_i$ per option and the time to maturity τ_i, with $\Gamma_i \neq 0$ and $\mathcal{V}_i \neq 0$; for a European call or put both are positive (Propositions 21.6(i) and 21.9(i)). In this subsection the index i names a traded option; it is not the date index of §21.5. The hedger adds w_1 options of the first kind, w_2 of the second and w_S units of the underlying, a negative number being a sale. One unit of the underlying has delta 1, gamma 0 and vega 0 (Proposition 21.4(i) and (iv)), so by Definition 21.1(ii) the hedged book has

$$\begin{aligned} \text{gamma} \quad & \Gamma_\Pi + w_1\Gamma_1 + w_2\Gamma_2, \\ \text{vega} \quad & \mathcal{V}_\Pi + w_1\mathcal{V}_1 + w_2\mathcal{V}_2, \\ \text{delta} \quad & \Delta_\Pi + w_1\Delta_1 + w_2\Delta_2 + w_S. \end{aligned} \tag{21.11}$$

The vega of the book and the vegas of the two options are quoted in one unit, either per unit of volatility or per volatility point. Where the relation between vega and gamma is used, every option, in the book and traded, is a European call or put priced by (21.2) at one flat volatility $\sigma > 0$.

Procedure 21.31 (Neutralising gamma and vega, then delta)**.** To make a book gamma-neutral and vega-neutral with traded options, and then delta-neutral with the underlying:

1. Add up the Greeks Δ_Π, Γ_Π and $\mathcal{V}_\Pi$ of the book.
2. Adjust gamma and vega first, and with options: the underlying and the linear contracts on it have zero gamma and zero vega.
3. *One target.* To remove the gamma, trade $w_1 = -\Gamma_\Pi/\Gamma_1$ options. To remove the vega instead, trade $w_1 = -\mathcal{V}_\Pi/\mathcal{V}_1$ options. A single option fixes one of the two.
   Two targets. Solve the system

$$w_1\Gamma_1 + w_2\Gamma_2 = -\Gamma_\Pi, \qquad w_1\mathcal{V}_1 + w_2\mathcal{V}_2 = -\mathcal{V}_\Pi \tag{21.12}$$

by Cramer's rule:

$$w_1 = \frac{\Gamma_2\mathcal{V}_\Pi - \mathcal{V}_2\Gamma_\Pi}{\Gamma_1\mathcal{V}_2 - \Gamma_2\mathcal{V}_1}, \qquad w_2 = \frac{\mathcal{V}_1\Gamma_\Pi - \Gamma_1\mathcal{V}_\Pi}{\Gamma_1\mathcal{V}_2 - \Gamma_2\mathcal{V}_1}. \tag{21.13}$$

A unique solution exists exactly when the determinant $\Gamma_1\mathcal{V}_2 - \Gamma_2\mathcal{V}_1$ is not zero.

4. Trade $w_S = -(\Delta_\Pi + w_1\Delta_1 + w_2\Delta_2)$ units of the underlying.
5. The result is neutral only at the current state. Run the procedure again as S, t and σ move.

Why the procedure works. Each step is read off (21.11).

The order of the steps. The position w_S appears in the delta alone. Whatever w_S is, the gamma and the vega of the hedged book are set by w_1 and w_2; and once these are chosen, the delta is zero exactly when $w_S = -(\Delta_\Pi + w_1\Delta_1 + w_2\Delta_2)$. These are steps 2 and 4.

One target. With $w_2 = 0$, the gamma $\Gamma_\Pi + w_1\Gamma_1$ is zero exactly when $w_1 = -\Gamma_\Pi/\Gamma_1$, and the vega $\mathcal{V}_\Pi + w_1\mathcal{V}_1$ is zero exactly when $w_1 = -\mathcal{V}_\Pi/\mathcal{V}_1$. The two numbers agree exactly when $\Gamma_\Pi\mathcal{V}_1 = \mathcal{V}_\Pi\Gamma_1$. So a single option makes a book gamma-neutral and vega-neutral at once exactly when $\Gamma_\Pi\mathcal{V}_1 = \mathcal{V}_\Pi\Gamma_1$: when $\Gamma_\Pi = 0$ this says $\mathcal{V}_\Pi = 0$, and when $\Gamma_\Pi \neq 0$ it says that the vega-to-gamma ratio of the option equals that of the book, $\mathcal{V}_1/\Gamma_1 = \mathcal{V}_\Pi/\Gamma_\Pi$.

Two targets. Multiply the first equation of (21.12) by $\mathcal{V}_2$ and the second by Γ_2, and subtract; then multiply the second by Γ_1 and the first by $\mathcal{V}_1$, and subtract. Every solution therefore satisfies

$$(\Gamma_1\mathcal{V}_2 - \Gamma_2\mathcal{V}_1)\, w_1 = \Gamma_2\mathcal{V}_\Pi - \mathcal{V}_2\Gamma_\Pi, \qquad (\Gamma_1\mathcal{V}_2 - \Gamma_2\mathcal{V}_1)\, w_2 = \mathcal{V}_1\Gamma_\Pi - \Gamma_1\mathcal{V}_\Pi.$$

When the determinant is not zero, these equations are (21.13), and substituting that pair into the two equations of (21.12) confirms that it solves them: this is Cramer's rule for a 2×2 system, as in the proof of Theorem 11.16. When the determinant is zero, $\mathcal{V}_1/\Gamma_1 = \mathcal{V}_2/\Gamma_2$, so the left side of the second equation is $\mathcal{V}_1/\Gamma_1$ times the left side of the first, for every w_1 and w_2. The system then has no solution when $\mathcal{V}_\Pi \neq (\mathcal{V}_1/\Gamma_1)\,\Gamma_\Pi$, and when $\mathcal{V}_\Pi = (\mathcal{V}_1/\Gamma_1)\,\Gamma_\Pi$ every one of the infinitely many pairs that solve the first equation solves the second. In neither case is the solution unique. This is step 3.

The weights through the ratios. Let the determinant be nonzero and $\Gamma_\Pi \neq 0$. The numerators of (21.13) are $-\Gamma_2\Gamma_\Pi\,(\mathcal{V}_2/\Gamma_2 - \mathcal{V}_\Pi/\Gamma_\Pi)$ and $-\Gamma_1\Gamma_\Pi\,(\mathcal{V}_\Pi/\Gamma_\Pi - \mathcal{V}_1/\Gamma_1)$, and the determinant is $\Gamma_1\Gamma_2\,(\mathcal{V}_2/\Gamma_2 - \mathcal{V}_1/\Gamma_1)$. Hence

$$w_1 = -\frac{\Gamma_\Pi}{\Gamma_1} \cdot \frac{\mathcal{V}_2/\Gamma_2 - \mathcal{V}_\Pi/\Gamma_\Pi}{\mathcal{V}_2/\Gamma_2 - \mathcal{V}_1/\Gamma_1}, \qquad w_2 = -\frac{\Gamma_\Pi}{\Gamma_2} \cdot \frac{\mathcal{V}_\Pi/\Gamma_\Pi - \mathcal{V}_1/\Gamma_1}{\mathcal{V}_2/\Gamma_2 - \mathcal{V}_1/\Gamma_1}. \tag{21.14}$$

The first factors are the two gamma-only hedges of the one-target case, and the second factors add up to 1. The gamma of the book is thus split between the two options in the proportions in which the vega-to-gamma ratio of the book divides the interval between theirs. Both proportions are positive exactly when the ratio of the book lies strictly between the ratios of the two options. When Γ_1 and Γ_2 are positive, as for calls and puts, both hedges then have the sign of $-\Gamma_\Pi$: a book that is short gamma buys both options.

Two options of one expiry cannot separate gamma from vega. Under one flat volatility, Proposition 21.9(iii) gives $\mathcal{V}_i = \sigma\tau_i S^2\Gamma_i$ per unit of volatility, so the determinant is

$$\Gamma_1\mathcal{V}_2 - \Gamma_2\mathcal{V}_1 = \Gamma_1\,\sigma\tau_2 S^2\Gamma_2 - \Gamma_2\,\sigma\tau_1 S^2\Gamma_1 = \sigma S^2\,\Gamma_1\Gamma_2\,(\tau_2 - \tau_1).$$

Since σ, S, Γ_1 and Γ_2 are positive, it vanishes exactly when $\tau_1 = \tau_2$, whatever the two strikes and whether the options are calls or puts. Conversely, the one-target paragraph shows when a single option suffices. A book of European options that all expire on one date, with time to maturity τ, has $\mathcal{V}_\Pi = \sigma\tau S^2\,\Gamma_\Pi$: each of its options satisfies this relation (Proposition 21.9(iii)), the underlying and the linear contracts in it add zero to both sides, and both Greeks are additive. A traded option of the same maturity has $\mathcal{V}_1 = \sigma\tau S^2\,\Gamma_1$, hence $\Gamma_\Pi\mathcal{V}_1 = \mathcal{V}_\Pi\Gamma_1$, and it neutralises the gamma and the vega of that book together.

Example 21.32 (Two listed options, then the stock)**.** Take $S = 100$, $\sigma = 20\%$ and $r = q = 0$, and quote every vega per volatility point. A book has

$$\Delta_\Pi = +400, \qquad \Gamma_\Pi = -600 \text{ per point}, \qquad \mathcal{V}_\Pi = -6{,}000 \text{ per volatility point}.$$

These are, to the nearest round numbers, the Greeks of 21,300 written six-month at-the-money calls held against 11,650 shares: one such call has delta 0.528, gamma 0.0281 and vega 0.281 per volatility point. The vega-to-gamma ratio of the book is $6{,}000/600 = 10$, which is $\sigma\tau S^2/100$ at $\tau = 0.5$. Two at-the-money options are listed:

	delta	gamma	vega per volatility point	vega over gamma
option 1, three months	0.52	0.040	0.20	5
option 2, one year	0.54	0.020	0.40	20

The entries are rounded from the values of Examples 21.7, 21.13 and 21.10 (0.5199, 0.0398 and 0.199 for the three-month option; $0.5398 = \Phi(0.10)$, 0.0198 and 0.397 for the one-year option), and the two ratios are $\sigma\tau_i S^2/100$ at $\tau_1 = 0.25$ and $\tau_2 = 1$.

Gamma alone, with option 1. Buy $w_1 = 600/0.040 = 15{,}000$ options. The vega becomes $-6{,}000 + 15{,}000 \times 0.20 = -3{,}000$ and the delta $400 + 15{,}000 \times 0.52 = 8{,}200$: sell 8,200 shares.

Vega alone, with option 2. Buy $w_2 = 6{,}000/0.40 = 15{,}000$ options. The gamma becomes $-600 + 15{,}000 \times 0.020 = -300$ and the delta $400 + 15{,}000 \times 0.54 = 8{,}500$: sell 8,500 shares.

Both. The system (21.12) is

$$0.040\,w_1 + 0.020\,w_2 = 600, \qquad 0.20\,w_1 + 0.40\,w_2 = 6{,}000,$$

with the determinant $0.040 \times 0.40 - 0.020 \times 0.20 = 0.016 - 0.004 = 0.012$. In the form $\sigma S^2\,\Gamma_1\Gamma_2\,(\tau_2 - \tau_1)/100$ it is $0.2 \times 10{,}000 \times 0.040 \times 0.020 \times 0.75/100 = 0.012$. By (21.13),

$$w_1 = \frac{600 \times 0.40 - 6{,}000 \times 0.020}{0.012} = \frac{240 - 120}{0.012} = 10{,}000,$$
$$w_2 = \frac{0.040 \times 6{,}000 - 0.20 \times 600}{0.012} = \frac{240 - 120}{0.012} = 10{,}000 :$$

buy 10,000 of each. The delta becomes $400 + 5{,}200 + 5{,}400 = 11{,}000$: sell 11,000 shares.

Checks. The gamma is $10{,}000 \times 0.040 + 10{,}000 \times 0.020 - 600 = 400 + 200 - 600 = 0$, and the vega is $2{,}000 + 4{,}000 - 6{,}000 = 0$.

Reading. The ratio of the book, 10, lies between 5 and 20, which is why both positions are purchases. By (21.14), w_1 is the gamma-only hedge 15,000 times $(20 - 10)/(20 - 5) = 2/3$, and w_2 is the gamma-only hedge with option 2, $600/0.020 = 30{,}000$, times $(10 - 5)/(20 - 5) = 1/3$. A six-month at-the-money option, with gamma 0.02814 and vega 0.2814 per volatility point, has the ratio 10 of the book, so it alone neutralises both: $600/0.02814 = 21{,}322$ of them bring $21{,}322 \times 0.2814 = 6{,}000$ of vega. That trade buys back the written calls, of which the book held about 21,300.

(Problem P21.9 works this computation at different numbers.)

Interview trap. "Hedge the gamma with more stock." Shares have zero gamma and zero vega (Proposition 21.4(iv)), so trading them changes the delta only. Gamma and vega move only through options.

Interview trap. Neutralise the delta last. Every option bought for its gamma or its vega brings its own delta: in Example 21.32 the stock trade grows from 400 shares to 11,000. A delta hedge set before the options are traded has to be set again.

Interview trap. "Gamma-neutral means vega-neutral." Removing the gamma with option 1 alone left −3,000 of vega per volatility point (Example 21.32). One option removes both only when its vega-to-gamma ratio is that of the book.

Interview trap. "Any two options will do." Under one flat volatility, two options of the same expiry give a singular system: both have the vega-to-gamma ratio $\sigma\tau S^2$, and the determinant $\sigma S^2\,\Gamma_1\Gamma_2\,(\tau_2-\tau_1)$ is zero. The two hedges need different maturities.

Remark 21.33 (Vega neutrality across expiries is approximate). Summing vegas across maturities takes one scenario as the risk: every implied volatility moves by the same amount. Implied volatilities do not move in parallel: the front of the curve is the more mobile end, and the curve also tilts; §22.3 takes up the term structure of implied volatility. A book that is vega-neutral in aggregate can therefore still lose when the curve changes shape. Risk reports keep a separate vega for each expiry, or weight the vega of each expiry by the typical size of its volatility moves before adding them. A calendar spread built from the two options of Example 21.10, but with two three-month options sold for each one-year option bought, is exactly such a position: its net vega is $39.7 - 2 \times 19.9 = -0.1$, and it still gains or loses whenever the three-month and the one-year implied volatilities move by different amounts.

Trading application. The three neutralities are maintained at different speeds, set by the cost of the instrument that moves each one. The underlying is the cheapest and deepest market on the book, so the delta is re-centred inside its limit every day. Gamma and vega move only through options, whose spreads are wide and whose depth is small. They are therefore held inside limits (a vega limit is an amount of money per volatility point) and adjusted when an option can be traded near its model value, not on a schedule. Because the options sold to clients outnumber those bought from them, the residual that a dealer manages is usually short gamma and short vega. Size helps: opposite deltas cancel inside a large book, so a single trade in the underlying hedges many options, and the cost of re-hedging per option falls. Finally, the Greeks are derivatives at one point, and the explain of Proposition 21.14 is a Taylor expansion around it (Remark 21.16), so risk reports add a grid of full revaluations over joint moves of the spot and of the volatility. The largest loss on such a grid need not lie at the extreme spot moves. Revalue the delta-hedged long-gamma book of Example 21.15 one trading day later, at spot moves of −6 to +6 points in steps of 2 and at volatilities of 18%, 20% and 22%. It loses most, 2,135, with the spot unchanged and the volatility at 18%, while the two 18% cells with a six-point move show gains of 1,913 (down) and 1,569 (up).

21.7 Stop-Loss Rules, Synthetic Options and Portfolio Insurance

Hedging an option and creating it synthetically are the same trade with opposite signs: the delta strategy that neutralises a written put also manufactures a bought one. This subsection disposes of the tempting stop-loss rule, sizes index-put insurance for a portfolio whose beta is not one, replicates those puts with bonds or with index futures, and records why dynamic insurance fails exactly when it is needed, answering on the way "why does the stop-loss hedge not work?", "does a choppy path or a smooth one, ending at the same price, cost the replicator more?" and "how many index puts, at what strike, insure a beta-1.4 portfolio?". It builds on the beta of Definition 11.35, the CAPM of Fact 11.40, the index-futures hedge of Proposition 16.40, the cascades of Remark 15.7 and Proposition 21.19.

Setup. The subsection has three settings.

The stop-loss rule. A dealer has written a European call with strike K and expiry T, and $r = q = 0$. The underlier follows the risk-neutral price path of Definition 20.6, which at $r = q = 0$ is $S_t = S_0 \exp\big(-\sigma^2 t/2 + \sigma W_t\big)$, with W a standard Brownian motion (Definition 19.5) and $\sigma > 0$. The price is observed at the monitoring dates $0 = t_0 < t_1 < \cdots < t_n = T$; as in §21.5, the count n

is the number of dates $t_0, \dots, t_{n-1}$ at which the holding is set, and the last observation, at $t_n = T$, settles the position. The stop-loss holding over $(t_i, t_{i+1}]$ is $h_i = \mathbf{1}\{S_{t_i} > K\}$. The price of the call at time 0 is C, given by (21.2) with $r = q = 0$ and $\tau = T$. Its *intrinsic value* is $(S_0 - K)^+$, and its *time value* is $C - (S_0 - K)^+$. In Remark 21.36, $\varepsilon > 0$ is the typical overshoot of a crossing of the strike, the distance by which the price has passed K when a monitoring date reveals the crossing; the trades of the rule are then in effect executed in the band $K \pm \varepsilon$, and ε is its half-width. As in §21.5, Y_i is a price ratio and Z is a standard normal variable.

Portfolio insurance. An equity portfolio is worth $V_A = V_{A,0}$ today and $V_{A,T}$ at the horizon. The first subscript names the position and the second the time, because V_T alone would read as the price of a derivative. Its CAPM beta to an index is $\beta > 0$ (Definition 11.35), and its dividend yield is q_A. The index stands at S today and at S_T at the horizon, and one *index unit* is a position worth the index level; the dividend yield of the index is q and its volatility is σ. The horizon is τ years away, the floor is a value $V_{\text{floor}} < V_A$, and the insurance consists of European index puts on n_{put} index units, with strike K, expiring at the horizon. In the one-period CAPM statement, rates and yields accrue simply over the horizon: the riskless return is $r\tau$, the index pays the dividends $q\tau$ per unit of its value, and the portfolio pays $q_A\tau$. The total return of the index over the horizon is R_M, as in §11.5 and §16.8, and a is a constant defined in Proposition 21.37.

The synthetic put. The model is the yield model of §20.7, the model of Corollary 20.41(i): (A1) of §20 holds for an asset of price S with the continuous yield $q \geq 0$; in place of (A2) the asset pays the yield q, and the traded instrument is its reinvested holding $\tilde{S}$, defined below; and (A3)–(A6) hold, with a constant rate r and a constant volatility $\sigma > 0$. The put price $\mathcal{P}$ and its delta $\Delta_{\mathcal{P}}$ are those of (21.2) and of Proposition 21.2, with $\tau = T - t$. The *reinvested holding* is $\tilde{S}_t = S_t e^{-q(T-t)}$, the price of $e^{-q(T-t)}$ units of the asset whose dividends are reinvested in the asset. It pays no income, it is worth S_T at T, and it is the instrument that the strategies of Definition 20.1 trade when the asset has a yield (§20.7). Futures contracts mature at T_F, and $\tau_F = T_F - t$.

Symbol notice. The letter β is the CAPM beta in this subsection, as in §16.8. The coefficients of an Itô process always carry a subscript or arguments (β_t, $\beta(x,t)$ in §19), and none appears in this chapter, so the bare letter is free. The strike of the written call and the strike of the index put are both written K; the two settings never meet.

Proposition 21.34 (The stop-loss rule leaves the time value as expected shortfall)**.** *Let $r = q = 0$, let the underlier follow the risk-neutral path of Definition 20.6, $S_t = S_0 \exp\big(-\sigma^2 t/2 + \sigma W_t\big)$ with $\sigma > 0$, and fix monitoring dates $0 = t_0 < t_1 < \cdots < t_n = T$. The* stop-loss rule *for the writer of a European call with strike K and expiry T starts from the capital $(S_0 - K)^+$ and holds*

$$h_i = \mathbf{1}\{S_{t_i} > K\}$$

units of the underlier over $(t_i, t_{i+1}]$, financed at the zero rate: the writer is covered while the last observed price is above the strike, and naked otherwise. The terminal value of the rule is $(S_0 - K)^+ + \sum_{i=0}^{n-1} h_i \,(S_{t_{i+1}} - S_{t_i})$, and the writer owes the payoff $(S_T - K)^+$. Then the expected shortfall of the rule is the time value of the call,

$$\mathbb{E}^{\mathbb{Q}}\Big[(S_T - K)^+ - (S_0 - K)^+ - \sum_{i=0}^{n-1} h_i \,(S_{t_{i+1}} - S_{t_i})\Big] = C - (S_0 - K)^+, \tag{21.15}$$

for every monitoring grid.

Sketch of the proof. We take the expectation of the shortfall term by term, as in the proof of Proposition 21.19. At $r = q = 0$ the risk-neutral path is the path of the underlier itself, and S_T is lognormal with the forward price S_0, so Theorem 20.22 gives $\mathbb{E}^{\mathbb{Q}}(S_T - K)^+ = C$. The holding h_i is a function of S_{t_i} and is bounded, so the hedge term is integrable; the ratio $S_{t_{i+1}}/S_{t_i}$ is independent of the prices up to t_i with mean 1, so Lemma 19.17 gives

$$\mathbb{E}^{\mathbb{Q}}\big[h_i \,(S_{t_{i+1}} - S_{t_i})\big] = 0 \qquad \text{for every } i \text{ and every grid.}$$

By linearity the expected shortfall is $C - (S_0 - K)^+ - 0$: the grid enters through the hedge terms alone. The complete proof is in Appendix G.9. □

(Problem P21.10 asks you to reproduce the complete proof from scratch.)

Nothing in the proof uses the form of h_i beyond two properties: h_i is a Borel measurable function of the prices observed up to t_i, and the hedge terms are integrable. The first is essential: a holding that peeks at the next price, such as $\mathbf{1}\{S_{t_{i+1}} > S_{t_i}\}$, has $\mathbb{E}[h_i(S_{t_{i+1}} - S_{t_i})] = \mathbb{E}\,(S_{t_{i+1}} - S_{t_i})^+ > 0$. By Proposition 21.19 with $\sigma_{\text{real}} = \sigma_{\text{imp}} = \sigma$, the conclusion (21.15) holds for every strategy that starts from the intrinsic value and whose holdings h_i are Borel measurable functions of the prices up to t_i with $\mathbb{E}\,|h_i\,(S_{t_{i+1}} - S_{t_i})| < \infty$: on average, each of them falls short of the payoff by the time value. What the stop-loss rule adds is the appearance that every trade is done at K. Remark 21.36, after the next example, takes that appearance apart.

Example 21.35 (A stop-loss hedge that never converges)**.** Take the three-month at-the-money call of Example 21.7: $S_0 = K = 100$, $\sigma = 20\%$, $T = 0.25$ and $r = q = 0$. It has no intrinsic value, so its price is all time value, $C = 100\,(2\Phi(0.05) - 1) = 3.988$. By Proposition 21.34 the expected shortfall of the stop-loss rule is 3.988 for every monitoring grid.

The simulation. The rule is simulated with exact lognormal steps on 10,000,000 paths for each of six grids of n equally spaced dates; $n = 63$ is daily monitoring and $n = 504$ is eight checks a trading day. On the same paths the call is also delta-hedged at the same dates from the capital C, which gives the hedging error $\mathcal{E}_n$ of Proposition 21.27.

number of dates n	16	32	63	126	252	504
stop-loss rule: mean shortfall	3.99	3.99	3.99	3.99	3.99	3.99
stop-loss rule: standard deviation	3.28	3.15	3.08	3.05	3.03	3.02
stop-loss rule: mean number of trades	2.3	3.4	4.8	6.9	9.9	14.1
delta hedge: standard deviation of $\mathcal{E}_n$	0.844	0.605	0.435	0.310	0.220	0.156
the line $\sqrt{12.4801/n}$	0.883	0.625	0.445	0.315	0.223	0.157

The standard errors are about 0.001 for each mean shortfall and for each standard deviation of the stop-loss rule, and at most 0.0003 for each standard deviation of the delta hedge.

The stop-loss rule. Monitored daily, the rule has the mean shortfall 3.99 and the standard deviation 3.08. Monitored eight times a day, it has the mean 3.99 and the standard deviation 3.02. The mean does not move, as the proposition says, and the spread barely does: it falls from 3.28 at $n = 16$ to 3.02 at $n = 504$ (Figure 45).

The trades. A trade is a change of the holding h_i from one monitoring date to the next. The rule trades 4.8 times per path on average at $n = 63$ and 14.1 times at $n = 504$. Refining the monitoring eight times multiplies the trades by 2.9, close to $\sqrt{8}$, while the typical move of the price between two dates, $\sigma S_0\sqrt{T/n}$, shrinks from 1.26 to 0.45.

The delta hedge. The mean error is 0.000 at every n, with a standard error of 0.0001 at $n = 63$, as Proposition 21.27(i) requires. The standard deviation is 0.435 at $n = 63$ and 0.156 at $n = 504$, a ratio of 2.8, close to $\sqrt{8} = 2.83$. The limit constant of Proposition 21.27(iii) for this option, computed by quadrature from the closed form of Example 21.28, which brackets it between 12.47 and 12.5, is $\frac{1}{2}\sigma^4 T \int_0^T \mathbb{E}[\Gamma_t^2 S_t^4]\,dt = 12.4801$, which gives $\sqrt{12.4801/n} = 0.445$ and 0.157 at these two grids.

Remark 21.36 (The stop-loss rule is not free)*.* The rule looks costless because each trade seems to be done at the strike. Proposition 21.34 rules this out: the expected shortfall is the time value for every grid, so a finer grid cannot lower the cost and can only change how it is paid. The payment is an overshoot, and this part is exact. The shortfall telescopes into the sum over i of $(S_{t_{i+1}} - K)^+ - (S_{t_i} - K)^+ - h_i\,(S_{t_{i+1}} - S_{t_i})$. If $S_{t_i} > K$, then $h_i = 1$ and the term is $(S_{t_{i+1}} - K)^+ - (S_{t_i} - K) - (S_{t_{i+1}} - S_{t_i}) = (S_{t_{i+1}} - K)^+ - (S_{t_{i+1}} - K) = (K - S_{t_{i+1}})^+$, since

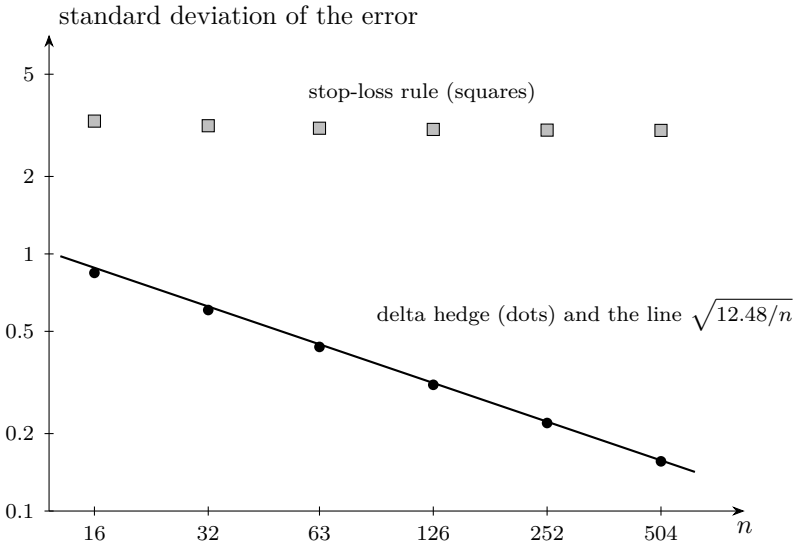


Figure 45: The standard deviation of the hedging error against the number n of rebalancing or monitoring dates, on logarithmic axes, for the three-month at-the-money call of Example 21.35 ($S_0 = K = 100$, $\sigma = 20\%$, $r = \mu = 0$, realised volatility equal to implied; 10,000,000 simulated paths for each point). The dots are the delta hedge started from C. They approach the line $\sqrt{12.48/n}$, of slope $-1/2$, which is the limit of Proposition 21.27(iii). The squares are the shortfall of the stop-loss rule. Rebalancing four times as often halves the delta-hedging error. Refining the stop-loss monitoring leaves its mean shortfall at the time value 3.99 (Proposition 21.34) and its spread near 3: in the simulation the spread falls only from 3.28 to 3.02 and shows no sign of shrinking toward zero.

$x^+ - x = (-x)^+$; if $S_{t_i} \le K$, then $h_i = 0$ and $(S_{t_i} - K)^+ = 0$, so the term is $(S_{t_{i+1}} - K)^+$. So, on every path,

$$\begin{aligned}(S_T - K)^+ - (S_0 - K)^+ - \sum_{i=0}^{n-1} h_i \,(S_{t_{i+1}} - S_{t_i}) \\ = \sum_{i=0}^{n-1} \Big(\mathbf{1}\{S_{t_i} > K\}\,(K - S_{t_{i+1}})^+ + \mathbf{1}\{S_{t_i} \le K\}\,(S_{t_{i+1}} - K)^+ \Big).\end{aligned}$$

A term is positive exactly when the monitored price passes from $\{S \le K\}$ to $\{S > K\}$, or from $\{S > K\}$ to $\{S < K\}$, between two dates, and it is then the distance by which the price has passed the strike when the crossing is seen: the rule buys above K and sells below it, and a crossing seen at T is settled at S_T. The shortfall is therefore nonnegative on every path, and it is zero exactly on the paths with no such crossing.

How many crossings the time value pays for is an approximation, not a theorem. Write ε for the typical overshoot, as in the Setup. If each round trip, a crossing up followed by a crossing down, gives up about 2ε, then the expected number of round trips is about

$$\frac{C - (S_0 - K)^+}{2\varepsilon},$$

which grows without bound as $\varepsilon \to 0$. For the option of Example 21.35 this is about $3.988/0.1 = 40$ round trips, or 80 trades, for a band of ± 0.05, and about $3.988/0.02 = 200$ round trips for a band of ± 0.01. The simulation of Example 21.35 agrees with the count. With daily monitoring the 4.8 trades are about 2.4 round trips, which is $3.988/(2\varepsilon)$ for $\varepsilon = 0.83$; with eight checks a day the 14.1 trades are about 7.0 round trips, which gives $\varepsilon = 0.28$. In both cases ε is about two thirds of the typical move between two dates, 1.26 and 0.45. By the identity above, the mean shortfall

is also the mean number of crossings, those seen at T included, times the average overshoot per crossing; the mean numbers of crossings, computed by quadrature, are 4.87 and 14.10, so the average overshoots are 0.82 and 0.28. The mean shortfall cannot shrink as the monitoring tightens (Proposition 21.34), and in the simulation of Example 21.35 its standard deviation stays near 3, while the delta-hedging error of Proposition 21.27 falls like $n^{-1/2}$ (Figure 45).

Interview trap. "The stop-loss hedge costs only the intrinsic value." Its expected shortfall under the pricing measure is the whole time value of the option, at every monitoring frequency (Proposition 21.34): 3.99 on a three-month at-the-money option at 20 vol. In the simulation its standard deviation is near 3, and finer monitoring hardly reduces it: 3.08 monitored daily and 3.02 at eight checks a day (Example 21.35, Figure 45). The slippage at $K \pm \varepsilon$ does not vanish as the monitoring tightens: by the count of Remark 21.36, an approximation, the crossings multiply like $1/\varepsilon$.

Proposition 21.37 (Index puts for a portfolio with beta)**.** *Suppose that over the horizon the portfolio lies on the CAPM line with no residual: its total return, dividends included, is*

$$r\tau + \beta\,(R_M - r\tau), \qquad \text{where} \qquad R_M = \frac{S_T}{S} - 1 + q\tau$$

is the total return of the index, and it pays out the dividends $q_A V_A \tau$. (This is the decomposition of Proposition 11.36, with the riskless return $r\tau$ in the place of r_f, in which the alpha of Definition 11.41 is zero, as Fact 11.40 asserts, and the residual is set to zero.) Let $\beta > 0$ and put

$$a = 1 + r\tau - q_A\tau - \beta\,(1 + r\tau - q\tau).$$

Then:

(i) the portfolio is worth $V_{A,T} = V_A\,(a + \beta\,S_T/S)$ at the horizon, an affine and increasing function of S_T;

(ii) if $V_{\text{floor}} > aV_A$, then puts on

$$n_{\text{put}} = \frac{\beta V_A}{S} \text{ index units, struck at } K = \frac{S}{\beta}\Big(\frac{V_{\text{floor}}}{V_A} - a\Big) > 0,$$

give

$$V_{A,T} + n_{\text{put}}\,(K - S_T)^+ = \max\,(V_{A,T},\ V_{\text{floor}}) \qquad \text{for every } S_T > 0;$$

(iii) if $V_{\text{floor}} \le aV_A$, then $V_{A,T} > V_{\text{floor}}$ for every $S_T > 0$, and no put is needed.

Proof. We substitute the hypothesis and solve one linear equation.

Part (i). The total return of the portfolio is its change in value plus the dividends it pays, per unit of its initial value: $(V_{A,T} - V_A + q_A V_A \tau)/V_A$. Setting this equal to $r\tau + \beta(R_M - r\tau)$ and solving for $V_{A,T}$,

$$\begin{aligned} V_{A,T} &= V_A\Big[1 + r\tau + \beta\Big(\frac{S_T}{S} - 1 + q\tau - r\tau\Big) - q_A\tau\Big] \\ &= V_A\Big[1 + r\tau - q_A\tau - \beta\,(1 + r\tau - q\tau) + \beta\,\frac{S_T}{S}\Big] = V_A\Big(a + \beta\,\frac{S_T}{S}\Big). \end{aligned}$$

This function of S_T is affine, with the slope $\beta V_A/S > 0$, so it is increasing.

Part (ii). By (i), $V_{A,T} = V_{\text{floor}}$ exactly when $a + \beta S_T/S = V_{\text{floor}}/V_A$, that is, when $S_T = K$; and $K > 0$ because $V_{\text{floor}} > aV_A$. If $S_T \ge K$, the puts pay nothing and $V_{A,T} \ge V_{\text{floor}}$ because $V_{A,T}$ increases with S_T: both sides of the claim equal $V_{A,T}$. If $S_T < K$, then $V_{A,T} < V_{\text{floor}}$ for the same reason, and with $n_{\text{put}} = \beta V_A/S$,

$$\begin{aligned} V_{A,T} + n_{\text{put}}\,(K - S_T) &= V_A a + \frac{\beta V_A}{S}\,S_T + \frac{\beta V_A}{S}\,(K - S_T) = V_A a + \frac{\beta V_A}{S}\,K \\ &= V_A a + V_A\Big(\frac{V_{\text{floor}}}{V_A} - a\Big) = V_{\text{floor}}\ : \end{aligned}$$

both sides equal V_{floor}. The number of puts is the slope of $V_{A,T}$ in S_T, so that below the strike the puts gain exactly what the portfolio loses.

Part (iii). By (i) and $S_T > 0$, $V_{A,T} = V_A a + \beta V_A S_T/S > V_A a \geq V_{\text{floor}}$. □

Beta enters the insurance twice. The number $n_{\text{put}} = \beta V_A/S$ is proportional to β. For the strike, substitute a:

$$\frac{K}{S} = \frac{1}{\beta}\Big(\frac{V_{\text{floor}}}{V_A} - 1 - r\tau + q_A\tau\Big) + 1 + r\tau - q\tau = 1 + (r-q)\tau - \frac{1}{\beta}\Big(1 + (r-q_A)\tau - \frac{V_{\text{floor}}}{V_A}\Big).$$

Whenever the floor lies below $V_A\,(1 + (r - q_A)\tau)$, the value that the riskless return would give the portfolio, the last bracket is positive, and K increases with β toward $S\,(1 + (r-q)\tau)$. That is the index level at which the index earns exactly the riskless return, $R_M = r\tau$, and at which every portfolio on the CAPM line earns $r\tau$ whatever its beta. Under that condition, and with $r \geq 0$, the standing assumption of Proposition 17.23, both effects push the premium the same way: a put is worth at least as much at a higher strike (Proposition 17.23(i)), so the premium $n_{\text{put}}\,\mathcal{P}$ grows at least in proportion to β, and in Example 21.38 it grows faster. When the floor lies above $V_A\,(1 + (r - q_A)\tau)$, the bracket is negative and the strike falls as β rises. For $\beta = 1$ and $q_A = q$ the constant a is zero and $K = S\,V_{\text{floor}}/V_A$: the puts are struck as far below the index, in proportion, as the floor is below the portfolio. Finally, a real portfolio carries the residual of Proposition 11.36, which the hypothesis sets to zero. The floor therefore holds only on the CAPM line, and the residual is not insured.

Example 21.38 (Insuring a beta-1.4 portfolio with index puts)**.** A portfolio worth $V_A = 40$ million has $\beta = 1.4$ to an index that stands at $S = 5{,}000$. The rate is $r = 4\%$, both dividend yields are $q = q_A = 1\%$, the horizon is $\tau = 0.5$, and the floor is 36 million, which is 90% of the value.

The mapping. Here $r\tau = 0.02$ and $q\tau = q_A\tau = 0.005$, so $a = 1.015 - 1.4 \times 1.015 = -0.406$, and by Proposition 21.37(i) the portfolio is worth $40\,(-0.406 + 1.4\,S_T/5{,}000)$ million at the horizon.

The puts. By part (ii), $n_{\text{put}} = 1.4 \times 40{,}000{,}000/5{,}000 = 11{,}200$ index units, which is 112 contracts on 100 times the index, struck at

$$K = \frac{5{,}000\,(0.9 + 0.406)}{1.4} = 4{,}664.286.$$

Checks. If the index ends at 4,400, the portfolio is worth $40\,(-0.406 + 1.4 \times 0.88) = 33.04$ million and the puts pay $11{,}200 \times 264.29 = 2.96$ million: 36.00 million in total. At 4,000 the two numbers are 28.56 and $11{,}200 \times 664.29 = 7.44$ million, again 36.00 million. At 5,200 the portfolio is worth 42.0 million and the puts expire worthless.

The cost. At an index volatility of 20%, (21.1) gives

$$d_1 = \frac{\ln(5{,}000/4{,}664.286) + (0.03 + 0.02)(0.5)}{0.2\sqrt{0.5}} = \frac{0.069503 + 0.025000}{0.141421} = 0.668239,$$

and $d_2 = d_1 - 0.2\sqrt{0.5} = 0.526817$. Then (21.2) prices one put at $4{,}664.286\,e^{-0.02}\,\Phi(-0.526817) - 5{,}000\,e^{-0.005}\,\Phi(-0.668239) = 1{,}367.74 - 1{,}253.67 = 114.07$ per index unit. (The put is a difference of two large terms, so d_1 and d_2 are kept to six decimals.) The insurance costs $11{,}200 \times 114.07 = 1.28$ million, which is 3.2% of the portfolio.

Against a beta of one. A portfolio with $\beta = 1$ and the same floor has $a = 0$. It needs puts on 8,000 index units struck at $5{,}000 \times 0.9 = 4{,}500$, each costing 72.60: 0.58 million in all. The cost ratio, 2.2, exceeds the beta ratio, 1.4: beta scales the number of puts, and through the CAPM mapping it also lifts each strike toward the index, so each put is dearer as well. Here the floor, $0.9\,V_A$, lies below $V_A\,(1 + (r - q_A)\tau) = 1.015\,V_A$, the condition under which the strike rises with β.

(Problem P21.11 works this computation at different numbers.)

Interview trap. "Insure a beta-1.4 portfolio with as many index puts as its value buys, struck 10% below the index." It needs β times as many, 11,200 index units and not 8,000, and a higher strike, the index level at which the portfolio reaches its floor under the CAPM mapping: 4,664, not 4,500 (Example 21.38). Even then the floor holds only on the CAPM line: the idiosyncratic residual of Proposition 11.36 is not insured.

Proposition 21.39 (Synthetic put)**.** *Assume the yield model of §20.7, the model of Corollary 20.41(i): (A1) holds for an asset of price S with the continuous yield $q \geq 0$; in place of (A2) the asset pays the yield q and the traded instrument is its reinvested holding $\tilde{S}$; and (A3)–(A6) hold, with a constant rate r and a constant volatility $\sigma > 0$. Let $\mathcal{P}(S,t)$ be the price function (21.2) of the European put with strike K and expiry T, and let $\Delta_{\mathcal{P}} = e^{-q\tau}\big(\Phi(d_1) - 1\big)$ be its delta, both evaluated at (S_t, t) along the path, with $\tau = T - t$. Consider the strategy of Definition 20.1 that starts from the capital $\mathcal{P}(S_0, 0)$ and holds, at every time $t < T$,*

$$\Delta_{\mathcal{P}}\, e^{q\tau} \text{ units of the reinvested holding } \tilde{S}_t = S_t e^{-q(T-t)}.$$

One unit of $\tilde{S}$ is $e^{-q\tau}$ units of the asset, so the strategy holds $\Delta_{\mathcal{P}}$ units of the asset, a short position, and its bank holding is $\mathcal{P} - \Delta_{\mathcal{P}} S = K e^{-r\tau}\Phi(-d_2) > 0$. This strategy replicates the put: its value is $\mathcal{P}(S_t, t)$ at every $t < T$, and its terminal value is $(K - S_T)^+$. Consequently:

(a) (Selling the asset.) Add the strategy to the position of an owner of one unit of the asset, that is, to the strategy that holds one unit of the asset at every time. The sum is a strategy. It holds $1 + \Delta_{\mathcal{P}} = 1 - e^{-q\tau}\big(1 - \Phi(d_1)\big)$ units of the asset, a fraction of the unit between 0 and 1, and it holds $\mathcal{P} - \Delta_{\mathcal{P}} S$ in bonds on top of the owner's bank balance. Its value is the value of the owner's position plus $\mathcal{P}(S_t, t)$ at every $t < T$, and plus $(K - S_T)^+$ at T.

(b) (Selling futures.) The delta $1 + \Delta_{\mathcal{P}}$ of the asset and the put together is also obtained by keeping the unit of the asset and selling

$$e^{-q\tau}\big(1 - \Phi(d_1)\big)\, e^{-(r-q)\tau_F}$$

futures contracts, each on one unit. For the insured portfolio of Proposition 21.37 the put is on the index: the n_{put} puts have the delta of a short position of $n_{\text{put}}\,|\Delta_{\mathcal{P}}|$ index units, and in index futures that delta is the fraction $|\Delta_{\mathcal{P}}|$ of the full hedge of $N_F^* = \beta V_A / V_F$ contracts of Proposition 16.40(i), where V_F is the contract value of §16.8, the futures price times the multiplier of the contract.

Sketch of the proof. We cite the replication theorem in its yield form and add strategies and deltas. In the yield model, the strategy that starts from the price of a claim and holds $e^{q\tau}$ times its delta in units of $\tilde{S}$ is worth the claim before T and its payoff at T (Corollary 20.41(i)(c)); for the put it is the strategy of the statement, whose bank holding, its value minus that of its holding, is

$$\mathcal{P} - \Delta_{\mathcal{P}} S = \mathcal{P} + S e^{-q\tau}\Phi(-d_1) = K e^{-r\tau}\Phi(-d_2).$$

For (a), holdings and values of a sum of strategies add (Definition 20.1(iv)), and $q \geq 0$ keeps $1 + \Delta_{\mathcal{P}}$ between 0 and 1. For (b), a short futures contract has the delta $-e^{(r-q)\tau_F}$, hence the number sold; for the $n_{\text{put}} = \beta V_A / S$ puts of Proposition 21.37, futures on $n_{\text{put}}\,|\Delta_{\mathcal{P}}|\, e^{-(r-q)\tau_F}$ index units are $|\Delta_{\mathcal{P}}|\, \beta V_A / V_F = |\Delta_{\mathcal{P}}|\, N_F^*$ contracts. The complete proof is in Appendix G.10. □

In words, part (a) says that the owner holds the asset and the put synthetically by keeping the fraction $1 + \Delta_{\mathcal{P}}$ of the asset and holding the rest of the combined value $S + \mathcal{P}$ in bonds; the capital $\mathcal{P}(S_0, 0)$ beyond the asset is the premium of the put. Two comments on part (b). It is stated for the delta only: that the futures version replicates the put exactly in continuous time would need a self-financing rule for daily-settled futures, which Definition 20.1 does not include. And it has a reading in the language of §16.8. By Proposition 16.40(ii), under its assumption (H), a short position of $|\Delta_{\mathcal{P}}|\, N_F^*$ contracts moves the beta of the position from β to $\beta - |\Delta_{\mathcal{P}}|\,\beta = \beta\,(1 + \Delta_{\mathcal{P}})$. Synthetic insurance is a beta that is lowered as the index falls and restored as it rises.

Intuition. Positive gamma (Proposition 21.6(i)) makes the delta of the put fall toward $-e^{-q\tau}$ as the price drops and climb toward 0 as it recovers. The short position of the replication is therefore enlarged at low prices and reduced at high ones, and each swing realises a small loss: the gamma scalp of Example 21.18 with the sign reversed. On a path that ends above the strike the put pays nothing, and these realised losses have used up the whole starting capital $\mathcal{P}(S_0, 0)$. The premium of synthetic insurance is paid in trading losses rather than up front.

Example 21.40 (The same insurance, created synthetically)**.** Take the 11,200 puts of Example 21.38, for which $d_1 = 0.668239$ and $q\tau = 0.005$. By Proposition 21.39 each has

$$\Delta_{\mathcal{P}} = e^{-0.005}\left(\Phi(0.668239) - 1\right) = 0.99501 \times (-0.25199) = -0.2507.$$

Selling the index. The replication of the 11,200 puts is short $11{,}200 \times 0.2507 = 2{,}808$ index units, which is $2{,}808 \times 5{,}000 = 14.04$ million of index exposure. It starts from the price of the puts, $11{,}200 \times 114.07 = 1.28$ million, and the sale proceeds and this capital go into bills: 15.32 million, the bank holding $11{,}200 \times Ke^{-r\tau}\Phi(-d_2)$ of Proposition 21.39. Held beside the portfolio, the replication has the delta of the 11,200 bought puts. Equivalently, in delta and on the CAPM line of Proposition 21.37, it is the sale of 10.03 million of the portfolio itself into bills: the portfolio carries $\beta = 1.4$ units of index exposure per unit of value, so 14.04 million of index exposure is the exposure of $14.04/1.4 = 10.03$ million of it, which is 25.1%, the fraction $|\Delta_{\mathcal{P}}|$.

Selling futures. Alternatively the portfolio is kept and futures that mature at the horizon are sold, so that $\tau_F = \tau$. By part (b) the position is short $2{,}808\, e^{-0.03\times 0.5} = 2{,}766$ index units of futures, which is 55.3 contracts on 50 times the index. In the terms of Proposition 16.40: the futures price is $5{,}000\, e^{0.015} = 5{,}075.6$, one contract has the contract value $V_F = 253{,}778$, the full hedge is $N_F^* = 1.4 \times 40{,}000{,}000/253{,}778 = 220.7$ contracts, and $0.2507 \times 220.7 = 55.3$. The beta of the position falls from 1.4 to $1.4 \times 0.7493 = 1.05$.

A round trip. Suppose the index falls 3% to 4,850 on the next trading day and returns to 5,000 the day after. On the first day $\Delta_{\mathcal{P}} = -0.3236$, the hedge grows to 3,624 index units, and 816 more are sold at 4,850. On the second day $\Delta_{\mathcal{P}} = -0.2499$, the hedge is back at 2,799 units, and 825 are bought back at 5,000. The short position gains $2{,}808 \times 150 = 0.42$ million on the fall and loses $3{,}624 \times 150 = 0.54$ million on the recovery, a net loss of 0.12 million. Counting the interest on the bills and the dividends owed on the short units, the replication is worth 1.16 million after the two days. The 11,200 bought puts, with the index back where it started, are worth 1.26 million: they have lost only their two days of time decay, 0.016 million. The swing cost the synthetic insurance about 0.10 million that bought puts did not pay, because the replication sold at the low and bought back at the high. In the model, synthetic insurance pays the premium of bought insurance through such trading losses; a path that swings more than the volatility of 20%, as this one does with two 3% days, makes it pay more.

(Problem P21.11 works this computation at different numbers.)

Interview trap. "Two price paths end at the same level above the strike: one climbs smoothly past it, the other whipsaws across it many times. On which is a call dearer to manufacture by delta hedging?" The whipsaw. The shortfall of a replication run at σ_{imp} is the gamma-weighted realised-minus-implied variance along the path (Remark 21.21), and in continuous time this is exact. For a calm path at a low constant realised volatility against a wild one at a high constant volatility, it is Proposition 21.22(ii): a replication run at the volatility σ_{imp} falls short of the payoff with probability one when the realised volatility is higher, and has money to spare with probability one when it is lower: added to the hedged book of that proposition, the replication gives the option itself, so its shortfall is the terminal value Π_T there (Definition 20.1(iv)). For a volatility that the path itself switches, it is Corollary 21.24: in Example 21.25 the synthetic call falls short of its payoff by 2.06 on average where the market turned wild, and has 1.76 to spare where it turned calm. The replicator buys high and sells low on every swing.

Remark 21.41 (Dynamic insurance fails when it is needed). The synthetic put of Proposition 21.39 requires selling the asset, or index futures, as prices fall, because its delta becomes more negative (Example 21.40). When many insurers follow the same rule, their forced sales into a falling market push prices lower and trigger further sales: the positive-feedback mechanism of Remark 15.7, with a hedging rule in the place of a margin rule. The replication, for its part, assumed continuous prices and unlimited liquidity. The equity crash of October 1987 is the textbook instance, in which both assumptions failed at once: prices moved in gaps rather than continuously, and the selling that the rules called for could not be done near the prices of the model, so portfolios that relied on the rule ended below their floors. The replication argument assumes that the hedger's own trades do not move the price; the episode shows what happens when that assumption fails for hedgers whose combined trades are large against the volume that the market trades.

Interview trap. "Synthetic insurance is as good as bought puts." It is, in calm and liquid markets. In a gap or in a crowded sell-off the trades that it requires cannot be executed (Remarks 21.41 and 15.7), whereas a bought put pays regardless.

Practice: problems P21.1–P21.11 in §27, solutions in §28.

22 Volatility: Smiles, Surfaces, Estimation and Variance Swaps

Volatility is the one input of the Black–Scholes–Merton formula that no screen displays, and it reaches a desk in two forms: the *implied* volatility that option prices quote, and the *realised* volatility that a price path delivers. This section develops both and ends with the contract that exchanges one for the other. The implied side comes first: the vocabulary of smiles and skews and the reason calls and puts share one implied volatility (§22.1), the risk-neutral density and the digital prices that a strike curve encodes (§22.2), and the conditions that tie the surface together across expiries (§22.3). The realised side follows in rising difficulty: the historical estimator and its standard error (§22.4), the EWMA and GARCH(1,1) updating rules (§22.5), and the GARCH forecast with its term structure and its estimation by maximum likelihood (§22.6). The variance swap closes the section (§22.7), because its strike is an integral over the implied side and its payoff a statistic of the realised side.

22.1 Implied Volatility Smiles and Skews

Implied volatility (Definition 20.37) is the unit in which options are quoted, and the first thing a desk sees is that it is not one number: every strike and every expiry carries its own. This subsection fixes the vocabulary of that surface, proves the one model-free statement that interviewers always probe (a European call and a European put with the same strike and expiry share one implied volatility, by put–call parity alone: Theorem 17.9, in the forward form of Corollary 17.11), and records the empirical shapes by asset class with their three standard explanations. It ends with a binary event whose two-point law produces a frown.

Setup. This Setup serves the whole section and carries its symbol notices. An underlying trades today at the price $S > 0$. The continuously compounded risk-free rate is r and the continuous dividend yield of the underlying is q. For each expiry $T > 0$, measured in years, the forward price for delivery at T is

$$F = F(T) = Se^{(r-q)T}$$

(Theorem 16.11 when $q = 0$, Proposition 16.15 in general). The market prices of the European call and the European put struck at $K > 0$ and expiring at T are $C(K, T)$ and $\mathcal{P}(K, T)$, written C and $\mathcal{P}$ when the strike and the expiry are fixed; put–call parity is read "C minus $\mathcal{P}$". Market prices

are arbitrage-free under the standing assumptions of §17.2. The prices given by the Black–Scholes–Merton formula in its forward (Black-76) form, Theorem 20.22 under the model of Definition 20.16, are

$$C^{\mathrm{BSM}}(K,T;\sigma) = e^{-rT}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big], \qquad \mathcal{P}^{\mathrm{BSM}}(K,T;\sigma) = e^{-rT}\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big],$$

$$d_1 = \frac{\ln(F/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = \frac{\ln(F/K) - \sigma^2 T/2}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T},$$

where the volatility $\sigma > 0$ after the semicolon is a parameter, and Φ and φ are the standard normal distribution function and density. The vega $\mathcal{V} = \partial_\sigma C^{\mathrm{BSM}} = \partial_\sigma \mathcal{P}^{\mathrm{BSM}} = e^{-rT} F\,\varphi(d_1)\sqrt{T}$ is the same for the call and the put, and positive (Proposition 20.35; Proposition 21.9 lists it among the Greeks). The implied surface is written $\sigma_{\mathrm{imp}}(K,T)$ and is defined below. Symbols that serve one subsection only are declared in that subsection's Setup.

Four notices on symbols hold throughout the section.

1. The letter r is the interest rate (not the random-walk ratio q/p of the general notation table) and q is the dividend yield, so a down-probability, where one appears, is written $1-p$. The options literature writes $N(d_1)$ for the quantity written $\Phi(d_1)$ here. The put is the calligraphic $\mathcal{P}$, because P is the real-world probability, and ln is the natural logarithm, which the general notation table also writes log.
2. In this section only, λ is the decay factor of the exponentially weighted moving average (the jump intensity of §23 is unrelated), and ω, α, β are the GARCH coefficients. This β is not the CAPM beta of Definition 11.35, and this α is neither the Itô drift α_t of Definition 19.25 nor Jensen's alpha of Definition 11.41.
3. The letter n is a sample size, never a number of tree steps; the GARCH day index is i, as in the log-return R_i.
4. Total implied variance is written $\sigma_{\mathrm{imp}}(K,T)^2 T$ in full; it has no letter. Remark 22.9, in §22.2 below, keeps the letters a, b, ϱ, m_0, ς of the parametrisation it describes, and its b is not the cost of carry.

Definition 22.1 (Implied-volatility surface, smile, skew, term structure, moneyness and the FX quoting triple)**.** The following terms describe the implied volatilities of the European options on one underlying, across strikes and expiries.

(i) *Surface.* The *implied volatility* $\sigma_{\mathrm{imp}}(K,T)$ is that of Definition 20.37 for the European option struck at K and expiring at T: the $\sigma > 0$ with

$$C^{\mathrm{BSM}}(K,T;\sigma) = C(K,T).$$

It exists when the price lies strictly inside the bounds of Proposition 17.6, $e^{-rT}(F-K)^+ < C(K,T) < e^{-rT}F$, and it is then unique (Proposition 20.38, which rests on the strict monotonicity in σ supplied by Proposition 20.35). By Proposition 22.3, proved next, the put gives the same number. The map $(K,T) \mapsto \sigma_{\mathrm{imp}}(K,T)$ is the *implied-volatility surface*.

(ii) *Shapes.* At a fixed expiry T, the strike profile $K \mapsto \sigma_{\mathrm{imp}}(K,T)$ is called a *smile* when it is U-shaped with its minimum near the money, a *skew* when it is essentially monotone (for equities, decreasing), and a *frown* when it is inverted, with its maximum near the money. At a fixed strike, or at a fixed moneyness, the maturity profile $T \mapsto \sigma_{\mathrm{imp}}$ is the *term structure* of implied volatility.

(iii) *Moneyness coordinates.* Three coordinates locate a strike relative to the forward. The first is the ratio K/F. The second is the *standardised moneyness* $\ln(K/F)/(\sigma_{\mathrm{ATM}}\sqrt{T})$, the log-distance from the forward in units of the at-the-money *total volatility* $\sigma_{\mathrm{ATM}}\sqrt{T}$, where σ_{ATM} is the at-the-money implied volatility of (iv) (a volatility times the square root of the time to expiry is called a total volatility). The third is delta: a "25-delta call" is a strike

whose call has forward delta $\Phi(d_1) = 0.25$, and a "25-delta put" a strike whose put has forward delta $\Phi(d_1) - 1 = -0.25$, with d_1 computed at that strike's own implied volatility. For a flat smile, at one volatility σ, the quantity $d_1 = [\ln(F/K) + \sigma^2 T/2]/(\sigma\sqrt{T})$ strictly decreases from $+\infty$ to $-\infty$ as K runs over $(0, \infty)$, so exactly one strike has each delta. These are the deltas of Proposition 21.2 without their factor e^{-qT}.

(iv) *At the money.* The phrase must come with its convention: $K = F$ (at the money forward), $K = S$ (at the money spot), or the 50-delta strike. The symbol σ_{ATM} denotes the at-the-money implied volatility under the stated convention.

(v) *The FX quoting triple.* Write $\sigma_{25\mathrm{c}}$ and $\sigma_{25\mathrm{p}}$ for the implied volatilities at the 25-delta call strike and the 25-delta put strike. For each expiry the currency market quotes three numbers: the at-the-money volatility σ_{ATM}; the 25-delta *risk reversal*

$$\mathrm{RR}_{25} = \sigma_{25\mathrm{c}} - \sigma_{25\mathrm{p}},$$

the tilt of the smile, whose sign says which wing is dear; and the 25-delta *butterfly*

$$\mathrm{BF}_{25} = \tfrac{1}{2}(\sigma_{25\mathrm{c}} + \sigma_{25\mathrm{p}}) - \sigma_{\mathrm{ATM}},$$

its curvature. These two numbers are quotes in volatility, named after the positions that trade them: RR_{25} is quoted for the 25-delta risk reversal of Definition 17.32 (long the 25-delta call, short the 25-delta put), and BF_{25} for a 25-delta strangle against the at-the-money straddle. Everywhere else in this section a butterfly is the three-strike position of Proposition 17.24. Adding the definition of BF_{25} to half of that of RR_{25}, and subtracting half of the definition of RR_{25} from that of BF_{25}, inverts the two definitions:

$$\sigma_{25\mathrm{c}} = \sigma_{\mathrm{ATM}} + \mathrm{BF}_{25} + \tfrac{1}{2}\mathrm{RR}_{25}, \qquad \sigma_{25\mathrm{p}} = \sigma_{\mathrm{ATM}} + \mathrm{BF}_{25} - \tfrac{1}{2}\mathrm{RR}_{25}.$$

Markets differ on whether the delta is a spot or a forward delta and on whether it is adjusted for the premium; those details are not needed here.

Example 22.2 (One strike in three coordinates; decoding an FX quote)**.** *(i) One strike, three coordinates, two expiries.* Take $F = 100$, $K = 110$, an implied volatility of 18% at this strike and at the money, and $r = q = 0$. The ratio $K/F = 1.10$ is the same at every expiry, and $\ln(K/F) = 0.09531$. At $T = 0.25$ the total volatility is $0.18 \times 0.5 = 0.09$, so the standardised moneyness is $0.09531/0.09 = 1.06$, and

$$d_1 = \frac{-0.09531 + 0.00405}{0.09} = -1.014, \qquad \Phi(-1.014) = 0.155 :$$

the strike is a 16-delta call. At $T = 1$ the total volatility is 0.18, the standardised moneyness is $0.09531/0.18 = 0.53$, and $d_1 = (-0.09531 + 0.0162)/0.18 = -0.440$ with $\Phi(-0.440) = 0.330$: a 33-delta call. The same strike is twice as far out of the money at three months as at one year, which is why smiles of different expiries are compared in standardised moneyness or in delta, never in K.

(ii) Decoding an FX quote. A three-month quote reads $\sigma_{\mathrm{ATM}} = 10.0\%$, $\mathrm{RR}_{25} = -1.0\%$, $\mathrm{BF}_{25} = +0.4\%$. By Definition 22.1(v),

$$\sigma_{25\mathrm{c}} = 10.0 + 0.4 - 0.5 = 9.9\%, \qquad \sigma_{25\mathrm{p}} = 10.0 + 0.4 + 0.5 = 10.9\%,$$

and the check is $\mathrm{RR}_{25} = 9.9 - 10.9 = -1.0$ and $\mathrm{BF}_{25} = 10.4 - 10.0 = 0.4$. The negative risk reversal says that the put wing is the dear one. The strikes follow from the forward deltas, each wing at its own volatility. For the call, $\Phi(d_1) = 0.25$ gives $d_1 = -0.6745$, and $\sigma\sqrt{T} = 0.099 \times 0.5 = 0.0495$, so $\ln(F/K) = d_1\,\sigma\sqrt{T} - \sigma^2 T/2 = -0.03339 - 0.00123 = -0.0346$ and $K/F = e^{0.0346} = 1.035$. For the put, $\Phi(d_1) = 0.75$ gives $d_1 = 0.6745$, and $\sigma\sqrt{T} = 0.109 \times 0.5 = 0.0545$, so $\ln(F/K) = 0.03676 - 0.00149 = 0.0353$ and $K/F = e^{-0.0353} = 0.965$.

Proposition 22.3 (A European call and put with the same strike and expiry have the same implied volatility)**.** *Let C and $\mathcal{P}$ be arbitrage-free market prices (the standing assumptions of §17.2) of a European call and a European put on the same underlying, with the same strike $K > 0$ and the same expiry $T > 0$, and let F be the forward price of the underlying for delivery at T, at which C^{BSM} and $\mathcal{P}^{\mathrm{BSM}}$ are evaluated (with known income, case (I) of §17.2, $F = (S - I)e^{rT}$ by Proposition 16.13). Then for every trial volatility $\sigma > 0$ the two pricing errors of the Black–Scholes–Merton formula are equal:*

$$C^{\mathrm{BSM}}(K, T; \sigma) - C \;=\; \mathcal{P}^{\mathrm{BSM}}(K, T; \sigma) - \mathcal{P}.$$

Consequently, if the call has an implied volatility σ_{imp}, then σ_{imp} is also an implied volatility of the put, and the put has no other.

Proof. We subtract two parity identities, one satisfied by the market prices and one by the formula.

Step 1: market parity. The prices are arbitrage-free, so Corollary 17.11(iii) applies (for a stock that pays no dividend it is Theorem 17.9, because $e^{-rT}F = S$ there):

$$C - \mathcal{P} = e^{-rT}(F - K).$$

Step 2: formula parity. Fix $\sigma > 0$. By Theorem 20.22, with d_1 and d_2 computed at this σ,

$$\begin{aligned} C^{\mathrm{BSM}}(K, T; \sigma) - \mathcal{P}^{\mathrm{BSM}}(K, T; \sigma) &= e^{-rT}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big] - e^{-rT}\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big] \\ &= e^{-rT}\Big[F\,\big(\Phi(d_1) + \Phi(-d_1)\big) - K\,\big(\Phi(d_2) + \Phi(-d_2)\big)\Big] \\ &= e^{-rT}(F - K), \end{aligned}$$

because $\Phi(-x) = 1 - \Phi(x)$ for every real x.

Step 3: the errors agree. Subtracting the identity of Step 1 from that of Step 2 gives

$$\big(C^{\mathrm{BSM}}(K, T; \sigma) - C\big) - \big(\mathcal{P}^{\mathrm{BSM}}(K, T; \sigma) - \mathcal{P}\big) = 0 \qquad \text{for every } \sigma > 0,$$

which is the displayed equality of the statement.

Step 4: the implied volatilities agree. At $\sigma = \sigma_{\mathrm{imp}}$ the call error vanishes, by the definition of the call's implied volatility; by Step 3 the put error vanishes too, so σ_{imp} reprices the put. It is the put's only implied volatility. Indeed, by Step 2 the function $\sigma \mapsto \mathcal{P}^{\mathrm{BSM}}(K, T; \sigma)$ differs from $\sigma \mapsto C^{\mathrm{BSM}}(K, T; \sigma)$ by the amount $e^{-rT}(F - K)$, which does not depend on σ; its derivative in σ is therefore the call's vega, which is positive by Proposition 20.35. A function with a positive derivative on $(0, \infty)$ is strictly increasing there, by the mean value theorem, and a strictly increasing function takes the value $\mathcal{P}$ at most once. (Proposition 20.38(ii) reaches the same uniqueness for the put, also through parity.) □

Step 3 holds at every trial volatility and uses no law of S_T, and Step 4 runs the same way from the put to the call, so one surface $\sigma_{\mathrm{imp}}(K, T)$ serves both option types (Definition 22.1(i)). The statement is about European options only. For American options parity is a band, not an equality (Proposition 17.14; early exercise is the subject of §17.4), so small gaps between the implied volatilities of an American call and an American put can exist.

Example 22.4 (One implied volatility for the call and the put)**.** A stock that pays no dividend trades at $S = 100$; take $K = 95$, $T = 0.5$ and $r = 3\%$, so that $Ke^{-rT} = 95\,e^{-0.015} = 93.5856$ and $F = 100\,e^{0.015} = 101.51$. The market quotes the call at $C = 9.80$ and the put at $\mathcal{P} = 3.3856$. These quotes satisfy parity, since $C - \mathcal{P} = 6.4144 = S - Ke^{-rT}$. Compare both with the formula at two trial volatilities:

trial volatility	C^{BSM}	$C^{\mathrm{BSM}} - C$	$\mathcal{P}^{\mathrm{BSM}}$	$\mathcal{P}^{\mathrm{BSM}} - \mathcal{P}$
20%	9.2511	−0.5489	2.8367	−0.5489
25%	10.4969	+0.6969	4.0825	+0.6969

At each trial volatility the two pricing errors are equal, as Proposition 22.3 requires, and they change sign together between 20% and 25%. At 20%, $d_1 = 0.5395$ and the vega is $\mathcal{V} = e^{-rT} F \varphi(d_1)\sqrt{T} = 100 \times 0.3449 \times 0.7071 = 24.39$ per unit of volatility, so one Newton step from 20% gives $20\% + 0.5489/24.39 = 22.25\%$ for both options. Exact inversion of the formula gives 22.23% for the call and 22.23% for the put.

Interview trap. "The put is dear because its implied volatility is above the call's at the same strike and expiry." For European options this cannot happen, by Proposition 22.3. The skew is a function of the strike, not of the option type: a low strike is dear for the out-of-the-money put and for the in-the-money call alike. A gap between the two implied volatilities, both computed at the market forward F, is a parity arbitrage (sell the leg with the higher volatility, buy the leg with the lower one, and hedge with the forward; Problem P22.1 builds the trade), not a view on volatility. A gap computed at an assumed dividend yield or borrow cost says instead that the assumed forward is wrong: the pair itself gives $F = K + e^{rT}(C - \mathcal{P})$ (Remark 20.46), and at that forward the two implied volatilities coincide. For American options on single stocks parity is a band (Proposition 17.14), so small gaps can exist there.

Fact 22.5 (Smile shapes by asset class, empirical (equity skew: Rubinstein, 1994))**.** At a fixed expiry, the strike profile of equity-index implied volatility, and of most single-stock implied volatility, has sloped down since the crash of October 1987, with low strikes well above the at-the-money level and high strikes below it; before the crash it was close to flat. Major currency pairs show a smile that is more nearly symmetric about the at-the-money forward, and whose tilt (the risk reversal of Definition 22.1(v)) moves with the market. For the major asset classes the strike profile is flatter at longer expiries. Realised returns are fatter-tailed than the lognormal model allows (Remark 10.7).

Fact 22.5 records empirical regularities, not a theorem: no mathematical proof is possible, only evidence from market data, and the text quotes it as evidence.

Intuition. Three explanations are standard. (1) *Symmetric fat tails.* Jumps of either sign, and a volatility that moves independently of the price, put more probability in both tails than a lognormal law with one volatility does, and so lift both wings: this is the currency smile. (2) *The leverage effect.* Take a firm whose debt is fixed and whose assets move with a volatility of 20%. A change in the value of the assets passes one for one to the equity, so the volatility of the equity is the asset volatility times the ratio of the assets to the equity. With assets of 100 and debt of 50 it is $20\% \times 100/50 = 40\%$; after the assets fall to 80 it is $20\% \times 80/30 = 53.3\%$. Equity volatility therefore rises as the share price falls, a negative correlation between spot and volatility that fattens the left tail and thins the right. (3) *The crash-insurance premium*, called crashophobia by Rubinstein. Persistent demand for downside protection concentrates the premium of Fact 21.29 in out-of-the-money index puts. Explanations (2) and (3) tilt the curve; (1) in this symmetric form cannot, while jumps with a negative mean tilt it as (3) does. The binary-event frown of Example 22.6 below is the opposite case: a law with no tails at all.

Figure 46 draws the two shapes of Fact 22.5 against the ratio K/F. The equity curve is not a sketch: its points are the one-year implied volatilities of the market used again in Figure 47, so that the two figures describe one set of prices.

Example 22.6 (A binary event produces a frown)**.** A stock trades at $S = 120$. An announcement tomorrow will send the price to 135 or to 108, and nothing else moves it until the options expire at the end of the week, $T = 5/252$; the rate is $r = 0$, so $F = S = 120$. The market is the one-step tree of Definition 18.6, with one step to the expiry, down-factor $d = 108/120 = 0.9$ and up-factor $u = 135/120 = 1.125$, so that $d < e^{r\Delta t} = 1 < u$. The risk-neutral up-probability solves $120 = 135p + 108(1 - p)$:

$$p = \frac{120 - 108}{135 - 108} = \frac{12}{27} = \frac{4}{9}.$$

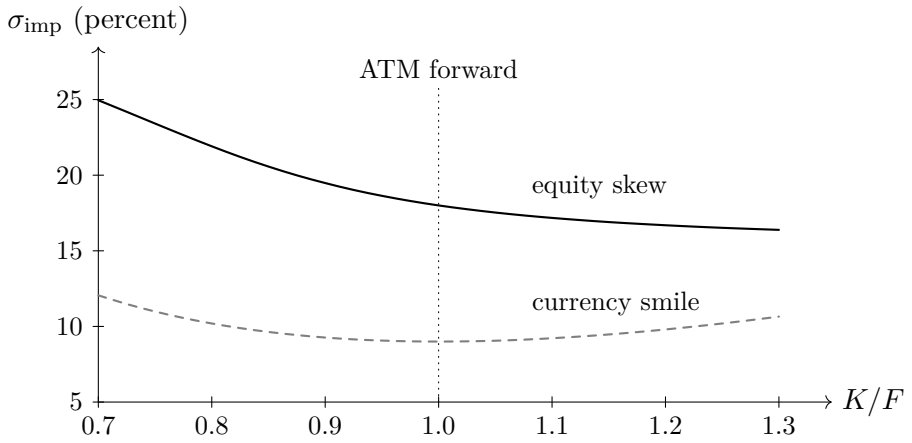


Figure 46: Implied volatility against the ratio K/F at one expiry. Solid: an equity-index skew, the one-year implied volatilities of the two-lognormal mixture of Figure 47: 25.0, 21.9, 19.5, 18.0, 17.2, 16.7, 16.4 percent at $K/F = 0.7, 0.8, \ldots, 1.3$. Dashed: a currency smile symmetric in $\ln(K/F)$, here $9 + 24\,(\ln(K/F))^2$ percent, which is 9.0 at the money and about 12 at $K/F = 0.7$. The dotted line marks the at-the-money forward strike $K = F$.

By Proposition 18.7(ii), at $r = 0$ each option is worth its $\mathbb{Q}$-expected payoff. The calls struck at 114, 120 and 126 pay only in the up state, so they are worth $21p = 9.33$, $15p = 6.67$ and $9p = 4.00$. By parity at $r = 0$, $\mathcal{P} = C - (120 - K)$, the puts are worth 3.33, 6.67 and 10.00; the direct computation of the first, $6(1 - p) = 3.33$, agrees.

Inverting Theorem 20.22 at each strike gives the total volatilities

$$\sigma_{\text{imp}}\sqrt{T} = 0.125, \quad 0.139, \quad 0.134 \qquad \text{at } K = 114,\ 120,\ 126,$$

a profile with its peak at the money. These three numbers do not depend on the value given to T: at $r = 0$ the formula contains σ and T only through the product $\sigma\sqrt{T}$, whose square, the total implied variance, is defined in §22.3 (Definition 22.13). Annualised over one week, $\sigma_{\text{imp}} = \sigma_{\text{imp}}\sqrt{T}/\sqrt{5/252}$ is 89.1%, 98.9% and 95.1%. As a cross-check, the at-the-money approximation of Proposition 20.29, $15p = 6.667 \approx 0.4 \times 120 \times \sigma_{\text{imp}}\sqrt{T}$, gives $\sigma_{\text{imp}}\sqrt{T} \approx 0.139$.

At the edges of the range $[108, 135]$ the time value vanishes. For $108 \le K \le 120$ the call is worth $4(135 - K)/9$ against an intrinsic value of $120 - K$, a time value of

$$\frac{4(135 - K)}{9} - (120 - K) = \frac{5(K - 108)}{9},$$

and for $120 \le K \le 135$ its whole value $4(135 - K)/9$ is time value. Strictly between the edges the price therefore lies strictly inside the bounds $(F - K)^+ < C < F$, and each strike has its implied volatility (Proposition 20.38(ii)). The time value tends to 0 as K decreases to 108 or increases to 135, and the implied volatility tends to 0 with it. To see this, fix $\sigma' > 0$. The formula's time value $K \mapsto C^{\text{BSM}}(K, T; \sigma') - (F - K)^+$ is continuous in K, because d_1 and d_2 are continuous in K on $(0, \infty)$ and Φ is continuous, and it is positive at $K = 108$, because by Proposition 20.38(i) the formula's price strictly increases in the volatility from its limit $e^{-rT}(F - K)^+$ as $\sigma \to 0$; so it is at least some $\varepsilon > 0$ on a neighbourhood of 108. At every strike $K > 108$ of that neighbourhood at which the market's time value $5(K - 108)/9$ is below ε, the market price is below $C^{\text{BSM}}(K, T; \sigma')$, and the strict monotonicity in σ of Proposition 20.38(i) gives $\sigma_{\text{imp}}(K, T) < \sigma'$. The volatility σ' was arbitrary, so $\sigma_{\text{imp}}(K, T) \to 0$ as K decreases to 108; the same argument works at 135. The fall is steep only at the very edge: $\sigma_{\text{imp}}\sqrt{T}$ is still 0.082 at $K = 109$ and 0.083 at $K = 134$. At and beyond the edges the call sits on its lower bound $(F - K)^+$, being worth $120 - K$ for $K \le 108$ and 0 for $K \ge 135$, so no implied volatility exists there (Proposition 20.38(ii)); desks quote it as 0.

The profile is a frown, because a two-point law has thinner tails than any lognormal law with the same spread. One volatility for every strike would put mass in tails that this law does not have: at the at-the-money total volatility 0.1394 the formula charges 9.93 for the 114 call, which the tree prices at 9.33, and 4.25 for the 126 call, which it prices at 4.00.

(Problem P22.2 works this computation at different numbers.)

Interview trap. "Symmetric fat tails explain the equity skew." Symmetric fat tails lift both wings and produce a smile. A skew needs asymmetry: the leverage effect or the crash premium of the Intuition above.

Interview trap. Skew is read in implied volatility at matched moneyness, ideally at matched delta, never in dollars. Equidistant out-of-the-money puts and calls have different dollar prices even under a flat smile: with $F = 100$, $T = 1$, $r = 0$ and a volatility of 20% at every strike, the 90 put costs 3.59 and the 110 call 4.29. The gap says nothing about skew.

Interview trap. "The smile proves the Black–Scholes–Merton model wrong, so the formula is useless." The formula survives as the quoting map from a volatility to a price: by Proposition 20.38(ii) every call price strictly inside the no-arbitrage bounds has exactly one implied volatility, so a quote in volatility carries the same information as a quote in money (the Trading application of §22.3, below, uses the formula this way). The smile is the market's correction to the lognormal assumption of the model, and parity and the no-arbitrage bounds still constrain the prices.

22.2 The Breeden–Litzenberger Density and the Prediction-Market Link

Theorem 20.17 prices a digital under one lognormal law, and the smile of §22.1 says that the market's law is not lognormal. This subsection proves that the whole risk-neutral density, and with it a digital at every strike, can be read off the curve of listed call prices with no parametric model of the distribution, only a continuous density with a finite mean (the continuous versions of the call-spread and butterfly bounds of Propositions 17.23 and 17.24), and it turns the slope identity into an exact vega-times-skew correction to the flat-volatility digital. The theorem is the engine of the prediction-market application below and a standard derivation request in options and crypto market-making interviews ("price a digital off the smile").

Setup. Fix an expiry $T > 0$ and let $S_T > 0$ denote the settlement price of the underlying at T. Prices are computed under the risk-neutral probability $\mathbb{Q}$, with expectation $\mathbb{E}^{\mathbb{Q}}$. Under $\mathbb{Q}$ the law of S_T has a density $f_{\mathbb{Q}}$ that is continuous on $(0, \infty)$, so that

$$\mathbb{Q}(K_1 < S_T \le K_2) = \int_{K_1}^{K_2} f_{\mathbb{Q}}(x)\,dx \quad (0 \le K_1 < K_2), \qquad \mathbb{Q}(S_T > K) = \int_K^{\infty} f_{\mathbb{Q}}(x)\,dx \quad (K \ge 0),$$

and the mean $\mathbb{E}^{\mathbb{Q}}[S_T] = \int_0^{\infty} x\, f_{\mathbb{Q}}(x)\,dx$ is finite. The integration variable is x, not s, which is a time from §19 on. The *pricing rule* is assumed for every European claim, not for calls only, because the digital is priced by it: a claim that pays a function of S_T at T, with finite $\mathbb{Q}$-expected absolute payoff, trades today at its discounted $\mathbb{Q}$-expected payoff. For the call struck at $K > 0$ the rule reads

$$C(K) \;=\; e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_T - K)^+\big] \;=\; e^{-rT}\int_K^{\infty} (x - K)\, f_{\mathbb{Q}}(x)\,dx, \tag{22.1}$$

where $C(K)$ abbreviates $C(K, T)$, the expiry being fixed. The derivatives of the call curve in the strike are written $\partial_K C(K)$ and $\partial_{KK} C(K)$, as in §17.5: they are partial derivatives of $C(K, T)$ at the fixed expiry. The second equality is Fact 7.22 applied to the payoff $x \mapsto (x - K)^+$, whose integral against the density converges absolutely because $0 \le (x - K)^+ \le x$ for $x > 0$. The

cash-or-nothing digital of Definition 20.15, with S_T in place of F_T, pays $\mathbf{1}\{S_T > K\}$ at T. The expectation of an indicator is the probability of its event, so the rule gives

$$\mathrm{Dig}(K) = e^{-rT}\,\mathbb{Q}(S_T > K).$$

A forward contract entered at the forward price F of §22.1 pays $S_T - F$ at T and costs nothing, so the rule also gives $\mathbb{E}^{\mathbb{Q}}[S_T] = F$: the density has its mean at the forward. Finally, $\Delta K > 0$ denotes a strike increment or the spacing of listed strikes (not h, which is the hedge ratio of §16). Remark 22.9, below, uses the log-moneyness $y = \ln(K/F)$ and letters of its own, declared there.

Theorem 22.7 (Breeden–Litzenberger)**.** *Under the Setup of this subsection, the call-price curve $K \mapsto C(K)$ is twice differentiable on $(0,\infty)$, and*

$$\partial_K C(K) \;=\; -\,e^{-rT}\,\mathbb{Q}(S_T > K), \tag{22.2}$$

$$\partial_{KK} C(K) \;=\; e^{-rT} f_{\mathbb{Q}}(K), \qquad \textit{equivalently} \qquad f_{\mathbb{Q}}(K) \;=\; e^{rT}\,\partial_{KK} C(K). \tag{22.3}$$

In particular the cash-or-nothing digital struck at K is worth the negative slope of the call curve, $\mathrm{Dig}(K) = -\,\partial_K C(K)$. *The result is due to Breeden and Litzenberger (1978).*

Sketch of the proof. We compute the difference quotient of the call curve exactly and squeeze it between two probabilities. Splitting the integrals of (22.1) at $K + \Delta K$, which the finite mean permits, and using $0 \le x - K \le \Delta K$ between the strikes gives

$$-\,e^{-rT}\,\mathbb{Q}(S_T > K) \;\le\; \frac{C(K+\Delta K) - C(K)}{\Delta K} \;\le\; -\,e^{-rT}\,\mathbb{Q}(S_T > K + \Delta K),$$

the right-hand sandwich (H.2) of Step 2 of the complete proof; at $K - \Delta K$ it is the left-hand sandwich (H.3), and the two give the bracket (H.4). The density is continuous, hence bounded near K, so the two ends of each sandwich differ by at most a constant times ΔK, which gives (22.2); differentiating the tail probability by the fundamental theorem of calculus, again by continuity, gives (22.3). The complete proof is in Appendix H.1. □

(Problem P22.3 asks you to reproduce the complete proof from scratch and apply it to listed quotes.)

The sandwiches of Step 2 rest on the pricing density of the Setup. The bracket (H.4) that they give for the digital, $(C(K) - C(K+\Delta K))/\Delta K \le \mathrm{Dig}(K) \le (C(K-\Delta K) - C(K))/\Delta K$ for $0 < \Delta K < K$, holds far more widely: in any arbitrage-free market, with no density at all, by dominance alone. That version is proved in §23 as Proposition 23.2, where the upper call spread is also the hedge of a sold digital; it sharpens the call-spread bounds of Proposition 17.23. No result of the present section uses it.

Example 22.8 (Digital and density from three listed calls)**.** With $r = 0$, calls on some underlying trade at $C(90) = 11.20$, $C(100) = 5.00$ and $C(110) = 1.80$. Estimate the digital struck at 100 and the risk-neutral density there, replacing the derivatives in (22.2) and (22.3) by centred finite differences with spacing $\Delta K = 10$:

$$\mathrm{Dig}(100) \approx -\,\frac{C(110) - C(90)}{2\,\Delta K} = -\,\frac{1.80 - 11.20}{20} = 0.47,$$

$$f_{\mathbb{Q}}(100) \approx \frac{C(90) - 2\,C(100) + C(110)}{(\Delta K)^2} = \frac{11.20 - 10.00 + 1.80}{100} = \frac{3.00}{100} = 0.030$$

per unit of price. Three checks confirm that the numbers are sensible.

First, the two one-sided slopes bracket the digital itself, not only the centred estimate. By (H.4) at $K = 100$ and $r = 0$, the left-hand sandwich gives $\mathbb{Q}(S_T > 100) \le (11.20 - 5.00)/10 = 0.62$ and the right-hand sandwich gives $\mathbb{Q}(S_T > 100) \ge (5.00 - 1.80)/10 = 0.32$, so $\mathrm{Dig}(100)$ lies in $[0.32, 0.62]$. This is derived under the pricing density of the Setup; Proposition 23.2, in §23, shows

that the bracket needs no density. The interval sits inside $[0,1]$, the model-free band of a digital at $r=0$: each of the two call spreads, divided by its width ΔK, lies in $[0,e^{-rT}]=[0,1]$ by Proposition 17.23(i).

Second, the numerator 3.00 of the second difference is the price of the 90/100/110 butterfly, which is nonnegative by Proposition 17.24. This convexity of C in K is what keeps the density estimate from coming out negative.

Third, a density of 0.030 over a 10-wide bucket puts about 30% of the mass near 100, consistent with a digital near one half. The sandwiches give a rigorous but coarser statement. At $K=100$, (H.3) gives $\mathbb{Q}(S_T>90)\geq(11.20-5.00)/10=0.62$ and (H.2) gives $\mathbb{Q}(S_T>110)\leq(5.00-1.80)/10=0.32$, so at least 30% of the mass lies in the 20-wide interval $(90,110]$ on which the butterfly pays. No 10-wide bucket is bounded this way.

(Problem P22.3 works this computation at different numbers.)

Remark 22.9 (Fitting a smooth smile before differentiating: SVI (Gatheral, 2004)). A second difference divides the noise of the quotes by $(\Delta K)^2$. Three calls 5 apart, each off by up to 0.05, move a density estimate by up to $(0.05+2\times0.05+0.05)/25=0.008$, against a density of 0.04 in Problem P22.3. Desks therefore fit the total implied variance $\sigma_{\text{imp}}(K,T)^2T$ of each expiry as a smooth function of the log-moneyness $y=\ln(K/F)$ before applying (22.2) and (22.3). The standard form is the five-parameter SVI ("stochastic volatility inspired") slice

$$\sigma_{\text{imp}}(K,T)^2\,T \;=\; a+b\Big(\varrho\,(y-m_0)+\sqrt{(y-m_0)^2+\varsigma^2}\,\Big),$$

with a the level, $b\geq0$ the wing slope, ϱ the tilt ($|\varrho|<1$), m_0 the shift and $\varsigma>0$ the curvature. These are the letters of that parametrisation and are used in this remark only: b is not the cost of carry, and ϱ is not a correlation of this document. For $b>0$ the derivative of the slice in y, $b(\varrho+(y-m_0)/\sqrt{(y-m_0)^2+\varsigma^2}\,)$, increases in y and vanishes where $(y-m_0)/\sqrt{(y-m_0)^2+\varsigma^2}=-\varrho$, that is at $y=m_0-\varrho\varsigma/\sqrt{1-\varrho^2}$; there the square root equals $\varsigma/\sqrt{1-\varrho^2}$, and the slice attains its minimum $a+b\varsigma\sqrt{1-\varrho^2}$. A total variance cannot be negative, so this minimum must be nonnegative, although the level a itself may be negative. The slice is asymptotically linear in y in both wings, with slope $b(\varrho+1)$ as $y\to+\infty$ and $b(\varrho-1)$ as $y\to-\infty$, because

$$\sqrt{(y-m_0)^2+\varsigma^2}-|y-m_0| \;=\; \frac{\varsigma^2}{\sqrt{(y-m_0)^2+\varsigma^2}+|y-m_0|}\;\longrightarrow\;0 \qquad \text{as } |y|\to\infty,$$

so the slice differs from $a+b(\varrho\,(y-m_0)+|y-m_0|)$ by an amount that vanishes in both wings. A fitted slice is checked for butterfly arbitrage, that is, for the convexity in K of the call prices it implies (Proposition 17.24), and the slices of different expiries are checked against one another for calendar arbitrage (Proposition 22.15, proved in §22.3). Parametric smile models beyond this remark are outside the scope of the section.

Setup for the digital off the smile. Keep the Setup of this subsection, with F, T and r fixed, and write $C^{\text{BSM}}(K;\sigma)$ for the Black-76 call price $C^{\text{BSM}}(K,T;\sigma)$ of §22.1 (the model of Definition 20.16), as a function of the strike and of the volatility. On an open interval of strikes the market curve is generated by a smile: $C(K)=C^{\text{BSM}}(K;\sigma_{\text{imp}}(K))$, where $\sigma_{\text{imp}}(K)$ abbreviates the implied volatility $\sigma_{\text{imp}}(K,T)$ of Definition 22.1 and is differentiable in K on the interval. Its derivative $\sigma'_{\text{imp}}(K)$ is the *skew* at K, in units of volatility per unit of strike.

Corollary 22.10 (The digital off the smile: flat-volatility price plus a vega-times-skew correction)**.** *Under the Setup of this subsection, suppose that on an open interval of strikes $C(K)=C^{\text{BSM}}(K;\sigma_{\text{imp}}(K))$, with σ_{imp} differentiable there. Then for every K in the interval*

$$\text{Dig}(K) \;=\; e^{-rT}\,\Phi(d_2) \;-\; \mathcal{V}\,\sigma'_{\text{imp}}(K),$$

where d_1 and d_2 are evaluated at $\sigma=\sigma_{\text{imp}}(K)$, and $\mathcal{V}=e^{-rT}F\,\varphi(d_1)\sqrt{T}$ is the vega (Proposition 20.35) of the call struck at K, taken at its own implied volatility.

Sketch of the proof. We differentiate the identity $C(K) = C^{\mathrm{BSM}}(K; \sigma_{\mathrm{imp}}(K))$ by the chain rule, because $\mathrm{Dig}(K) = -\,\partial_K C(K)$ by (22.2):

$$\partial_K C(K) \;=\; \partial_K C^{\mathrm{BSM}} \;+\; \partial_\sigma C^{\mathrm{BSM}}\, \sigma'_{\mathrm{imp}}(K),$$

with the partial derivatives of the formula taken at $(K, \sigma_{\mathrm{imp}}(K))$. The first is $-\,e^{-rT}\,\Phi(d_2)$, the terms from d_1 and d_2 cancelling by $F\,\varphi(d_1) = K\,\varphi(d_2)$ (Lemma 20.34), and the second is the vega (Proposition 20.35). The appendix proves this chain rule by the mean value theorem, using the differentiability of σ_{imp} and the continuity of the two partial derivatives. The complete proof is in Appendix H.2. □

A downward skew, $\sigma'_{\mathrm{imp}}(K) < 0$, makes every cash-or-nothing call dearer than flat-volatility pricing says, by the vega of the call times the size of the skew. By digital parity (Corollary 20.20: the digital that pays above K and the digital that pays at or below K are together worth e^{-rT}), it makes every "below K" digital cheaper by the same amount. The "skew correction" of desk language is therefore an exact term, not a rule of thumb.

Example 22.11 (An at-the-money digital under a downward skew)**.** Take $F = K = 100$, $T = 1$, $r = 0$, an implied volatility $\sigma_{\mathrm{imp}}(100) = 20\%$ and a skew $\sigma'_{\mathrm{imp}}(100) = -0.002$ per unit of strike: one strike point up costs 0.2 volatility points. Then $d_1 = 0.1$ and $d_2 = -0.1$. The flat-volatility digital is $\Phi(-0.1) = 0.4602$. The vega is $\mathcal{V} = 100 \times \varphi(0.1) \times 1 = 100 \times 0.3970 = 39.7$, so the correction of Corollary 22.10 is $-39.7 \times (-0.002) = +0.0794$ and

$$\mathrm{Dig}(100) = 0.4602 + 0.0794 = 0.5396,$$

that is, 0.540 against a flat-volatility 0.460. As a check, price a tight call spread with each call at its own volatility, 20.2% at $K = 99$ and 19.8% at $K = 101$: $(C(99) - C(101))/2 = (8.5146 - 7.4356)/2 = 0.5395$. The same spread at a flat 20% is worth $(8.4357 - 7.5153)/2 = 0.4602$. By digital parity at $r = 0$ the "below 100" digital is worth $1 - 0.540 = 0.460$, not the 0.540 of flat-volatility pricing. Flat-volatility pricing misses 8 cents per unit of notional, about 17% of the flat price.

(Problem P22.4 works this computation at different numbers.)

Figure 47 shows the density behind the equity skew of Figure 46. Both curves are built from Black-76 laws (Definition 20.16) at $T = 1$: a lognormal law with a given mean and a given volatility is the law of that definition with the mean as its forward and the volatility as its σ, and under it a call has the expected payoff $F\,\Phi(d_1) - K\,\Phi(d_2)$ of Theorem 20.22, computed with those two inputs. The solid curve is the density of S_T/F under a mixture of two such laws: with probability 0.9 the law with mean 31/30 and volatility 0.15, and with probability 0.1 the law with mean 0.70 and volatility 0.25. The mixture has mean $0.9 \times 31/30 + 0.1 \times 0.70 = 1$, so its forward is F. Its call prices are the sums, with the weights 0.9 and 0.1, of two Black-76 prices, and inverting the formula at each strike gives the equity curve of Figure 46, with the at-the-money implied volatility 18.0%. The dashed curve is the *flat law*: the lognormal law with mean 1 whose volatility is that at-the-money value, under which every strike would carry the at-the-money implied volatility. The shaded bands are the tails beyond $0.7F$ and $1.3F$, the two ends of the range of Figure 46.

Intuition. Tails can be read off the smile through two integrals. For a positive settlement price, $(K - S_T)^+ = \int_0^K \mathbf{1}\{S_T \le x\}\,dx$ and $(S_T - K)^+ = \int_K^\infty \mathbf{1}\{S_T > x\}\,dx$; taking expectations and exchanging the order of integration (Fact 18.34(i), the integrand being nonnegative) gives

$$\mathcal{P}(K) = e^{-rT} \int_0^K \mathbb{Q}(S_T \le x)\,dx, \qquad C(K) = e^{-rT} \int_K^\infty \mathbb{Q}(S_T > x)\,dx.$$

A put is the area under the distribution function to the left of its strike, and a call the area under the tail probability to the right of it. In Figure 47 the mixture's distribution function lies above

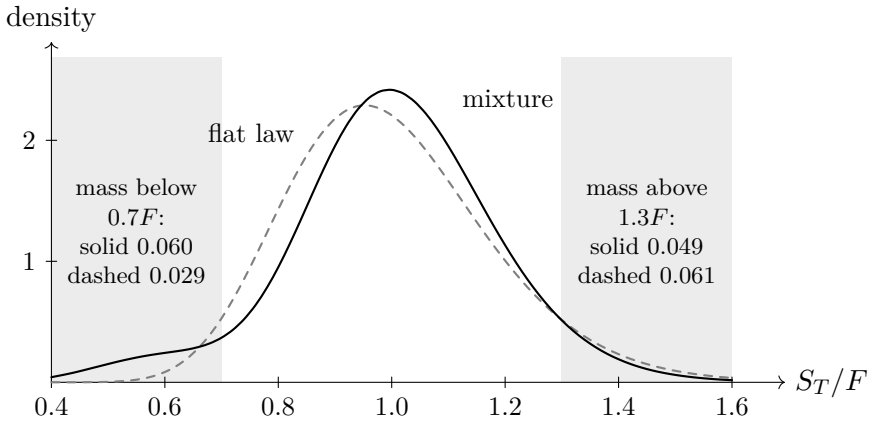


Figure 47: Two risk-neutral densities of S_T/F at $T = 1$, both with mean 1. Solid: the mixture $0.9\times$ lognormal (mean 31/30, volatility 0.15) + $0.1\times$ lognormal (mean 0.70, volatility 0.25), whose implied volatilities are the equity curve of Figure 46: 25.0% at $K/F = 0.7$, 18.0% at the money, 16.4% at $K/F = 1.3$. Dashed: the flat law, the lognormal density with the at-the-money volatility 18.0%. The shaded bands are the tails beyond $0.7F$ and $1.3F$: the mixture puts 0.060 of its mass below $0.7F$, against 0.029 for the flat law, and 0.049 above $1.3F$, against 0.061. At $0.6F$ the two densities are 0.243 and 0.085; at $1.4F$ they are 0.190 and 0.232. On the plotted range the curves cross near 0.66, 0.95 and 1.30.

that of the flat law at every price up to $0.7F$, so the mixture's $0.7F$ put is dearer, $0.0071F$ against $0.0013F$ before discounting, and it carries the higher implied volatility, 25.0% against 18.0%, because the formula's put price increases with the volatility (Proposition 20.35). Beyond $1.3F$ the mixture has less mass, and its $1.3F$ call is cheaper, $0.0043F$ against $0.0066F$, at 16.4%. An equity skew is a fat left tail with a thin right tail; a currency smile is two fat tails; the binary-event frown of Example 22.6 is no tails at all. For the equity-skew mixture, whose right tail is thin, the heavy left tail pushes the median above F, to $1.003F$, whereas the median $Fe^{-\sigma^2 T/2}$ of a lognormal law is below F (Proposition 20.8(iii)). The at-the-money upside digital is correspondingly dearer than its flat-volatility price, as Corollary 22.10 predicts from the downward skew at F.

Trading application. A cash-or-nothing digital has exactly the payoff of a prediction-market binary contract: "pays 1 if the price settles above K." Theorem 22.7 therefore prices such contracts from any liquid strike curve, with no parametric model of the distribution: interpolate the call quotes of a listed crypto options venue, differentiate at the contract's strike, and compare $-\partial_K C(K)$ with the prediction-market quote for the same event and settlement, an options-venue-to-prediction-market comparison. A persistent gap between the two is a relative-value signal, tradeable by dealing the rich leg against a tight call spread on the options venue. Using the slope of the *actual* curve rather than $e^{-rT}\Phi(d_2)$ at a single implied volatility matters: the smile makes implied volatility vary with the strike, and the curve's slope carries exactly the vega-times-skew term of Corollary 22.10, which the flat-volatility formula of Theorem 20.17 misses. The comparison measures a gap, not yet a mispricing: the gap can include a legitimate risk premium between the risk-neutral and the real-world probability (the trap on the phrase "the risk-neutral probability, discounted" in §20.3), as well as differences in settlement index, fixing time and fees.

The digitals of this application are short-dated, and for them the meaning of T needs one more caveat.

Remark 22.12 (The vol clock: intraday seasonality). Empirically, crypto underliers exhibit strong intraday and day-of-week volatility seasonality: some hours are reliably far busier than others. For short-dated options — hours to a few days — measuring time to expiry in calendar units then misprices, because the constant-variance-rate assumption behind calendar scaling (Proposition 19.1, with its assumptions in Remark 19.3) fails at that horizon. Practice replaces calendar time with a

deformed vol clock: time is measured in expected accumulated variance, so that hours expected to be quiet contribute little clock and busy hours contribute much, and T in formulas such as (20.7) is read off that clock rather than the calendar.

Interview trap. "The digital is $e^{-rT}\Phi(d_2)$ at the strike's own implied volatility." This misses the term $-\mathcal{V}\,\sigma'_{\text{imp}}(K)$ of Corollary 22.10, which is 8 cents on a 46-cent digital in Example 22.11. The digital is the slope of the actual strike curve: differentiate that curve, or price the tight call spread.

Interview trap. "A fatter left tail makes the upside digital cheaper." Not for the equity skew. When a fat left tail comes with a thin right tail, as in the mixture of Figure 47, the mean condition $\mathbb{E}^{\mathbb{Q}}[S_T] = F$ is met by extra mass around and just above F, where the solid curve lies above the dashed one, and $\mathbb{Q}(S_T > F)$ rises: it is 0.507 for the mixture, against $\Phi(-0.09) = 0.464$ for the flat law at the at-the-money volatility of 18.0% (Proposition 20.8(iii)). What decides the sign in general is the skew at F. By Corollary 22.10, whose vega is positive, the at-the-money digital is above its flat-volatility price $e^{-rT}\Phi(d_2)$ exactly when $\sigma'_{\text{imp}}(F) < 0$. A fat left tail paired with a heavier right tail can tilt the smile upward at F, and the digital is then cheaper.

Interview trap. Second differences of raw quotes amplify noise. A negative butterfly price signals an arbitrage or a stale quote, not a negative probability. Fit a smooth slice, checked for arbitrage, before differentiating (Remark 22.9).

22.3 Term Structure, Surfaces and the Calendar Condition

Interviewers move from the strike to the expiry with two questions: "the six-month trades at 20 vol and the one-year at 13: is there an arbitrage?" and "how does the skew change your delta?". This subsection answers both: it measures the term structure in total implied variance, states the model-free *calendar condition*, whose price form at a fixed strike is Theorem 17.16(iii) (the one model-free statement about a calendar spread in §17.6, which leaves the rest of the spread as a bet on the term structure), carries it to total implied variance, proves its carry case at a fixed forward moneyness by a dominance argument in the style of §17, extracts the forward volatility between two expiries, and defines the two textbook rules for how the smile moves with the spot, with the hedge ratio that each implies. Remark 21.33 on vega neutrality across expiries and the local-volatility trap of §23 point here.

Setup. The notation is that of §22.1. Throughout, $T_1 < T_2$ are two expiries and K, K_1, K_2 are positive strikes. Total implied variance is written $\sigma_{\text{imp}}(K,T)^2T$ in full. It receives no letter, because w is a weight in §17 and §21, and a letter for its square root would sit beside the log-drift ν. The forward volatility $\sigma_{\text{fwd}}(T_1,T_2)$ is defined in Definition 22.13. Two market settings are used.

- *Zero carry*: $r = q = 0$, so that $F(T) = S$ for every expiry T, and $C(K,T)$ is the price of the call struck at K and expiring at T. This is the setting of parts (i) and (ii) of Proposition 22.15 and of the zero-carry case of its part (iii).
- *Carry*: r and q are constants and $F(T) = Se^{(r-q)T}$. This is the setting of part (iv) and of the carry case of part (iii).

With zero carry, $r = q = 0$, so that $F(T) = S$ at every expiry, the Black-76 price of §22.1 depends on the pair (σ,T) only through the total volatility $\sigma\sqrt{T}$. Indeed

$$d_1 = \frac{\ln(F/K) + (\sigma\sqrt{T})^2/2}{\sigma\sqrt{T}}, \qquad d_2 = \frac{\ln(F/K) - (\sigma\sqrt{T})^2/2}{\sigma\sqrt{T}}$$

contain σ and T only in the product $\sigma\sqrt{T}$, and at $r = 0$ the price is $F\,\Phi(d_1) - K\,\Phi(d_2)$ with no discount factor, the same F at every expiry. A volatility σ over T years and a volatility $\sigma\sqrt{T}$ over

one year therefore give the same price:

$$C^{\mathrm{BSM}}(K,T;\sigma) = C^{\mathrm{BSM}}(K,1;\sigma\sqrt{T}) \qquad (r=q=0). \tag{22.4}$$

With carry the forward $F(T)$ itself moves with the expiry, and the two sides would be computed with different forwards; the carry case is therefore handled at a fixed ratio $K/F(T)$, in part (iv) of Proposition 22.15 (Step 3 of its proof, Appendix H.3). In the expectation form of part (i), after that proposition, X and Y are independent random variables and $\bar{\psi}(x) = \mathbb{E}\,\psi(x,Y)$ is the frozen expectation of Lemma 19.17. The symbol $\mathcal{S}_T$ denotes the smile as a function of the moneyness K/S under the sticky-delta rule of Definition 22.17, whose own Setup precedes it.

Definition 22.13 (Total implied variance, forward variance and forward volatility)**.** Let $\sigma_{\mathrm{imp}}(K,T)$ be the implied volatility of Definition 22.1.

(i) The *total implied variance* at (K,T) is $\sigma_{\mathrm{imp}}(K,T)^2\,T$. It is the variance of the logarithm of the settlement price under the Black-76 law of Definition 20.16 that reproduces the option's price: under that law $\ln F_T = \ln F - \sigma^2 T/2 + \sigma\sqrt{T}\,Z$ with $Z \sim \mathcal{N}(0,1)$ and $\sigma = \sigma_{\mathrm{imp}}(K,T)$, and the variance of the right-hand side is $\sigma^2 T$.

(ii) Let $T_1 < T_2$, and let K_1 and K_2 be strikes at the same forward moneyness, $K_1/F(T_1) = K_2/F(T_2)$, whose implied volatilities exist. With zero carry $F(T_1) = F(T_2) = S$, so $K_1 = K_2 = K$ is one fixed strike. The *forward variance* between T_1 and T_2 is

$$\frac{\sigma_{\mathrm{imp}}(K_2,T_2)^2\,T_2 - \sigma_{\mathrm{imp}}(K_1,T_1)^2\,T_1}{T_2 - T_1}.$$

(iii) When the forward variance is nonnegative, its square root is the *forward volatility*

$$\sigma_{\mathrm{fwd}}(T_1,T_2) = \sqrt{\frac{\sigma_{\mathrm{imp}}(K_2,T_2)^2\,T_2 - \sigma_{\mathrm{imp}}(K_1,T_1)^2\,T_1}{T_2 - T_1}}.$$

It is the constant volatility over $[T_1,T_2]$ that, appended to $\sigma_{\mathrm{imp}}(K_1,T_1)$ on $[0,T_1]$, reproduces the total variance at T_2:

$$\sigma_{\mathrm{imp}}(K_1,T_1)^2\,T_1 + \sigma_{\mathrm{fwd}}(T_1,T_2)^2\,(T_2 - T_1) = \sigma_{\mathrm{imp}}(K_2,T_2)^2\,T_2.$$

The forward volatility depends on the common forward moneyness as well as on the two expiries; the moneyness is fixed in context and is not shown.

Variance, not volatility, adds across periods (Proposition 19.1), which is why the term structure is measured, differenced and interpolated in total variance.

Example 22.14 (Forward volatility, and interpolation in total variance)**.** The at-the-money-forward implied volatilities are 20% at six months and 18% at one year, with $r = q = 0$. The total implied variances are

$$0.20^2 \times 0.5 = 0.0200, \qquad 0.18^2 \times 1 = 0.0324.$$

The forward variance is $(0.0324 - 0.0200)/0.5 = 0.0248$, and the forward volatility is $\sigma_{\mathrm{fwd}}(0.5,1) = \sqrt{0.0248} = 15.7\%$. The one-year quote of 18% is an average, in variance, of 20% for the first half-year and 15.7% for the second: $0.5 \times 0.0400 + 0.5 \times 0.0248 = 0.0324$. The market prices the second half-year below both quoted volatilities.

For a ten-month quote, $T = 5/6$, interpolate the total variance linearly in T: the expiry lies two thirds of the way from the first node to the second, so the total variance is $0.0200 + \frac{2}{3} \times (0.0324 - 0.0200) = 0.02827$, and $\sigma_{\mathrm{imp}} = \sqrt{0.02827/0.8333} = 18.42\%$. Linear interpolation of the volatility itself gives $20\% + \frac{2}{3} \times (18\% - 20\%) = 18.67\%$. Interpolation in total variance is the one that preserves the calendar condition of Proposition 22.15, proved next, between the nodes: a linear interpolation between nondecreasing node values is nondecreasing. Problem P22.6 shows interpolation that is linear in volatility breaking the condition.

(Problem P22.6 works this computation at different numbers.)

Proposition 22.15 (The calendar condition: call prices and total implied variance are nondecreasing in the expiry; forward volatility is real)**.** *Assume that prices are arbitrage-free in the sense of Lemma 16.8 (the market of §16.2, in which a self-financing strategy may trade at a later date on what is known then), that the underlying can be bought or sold short at any date, the short seller paying any income of the borrowed units to their lender, and that European calls of every expiry trade at time* 0*. Let* $T_1 < T_2$*.*

(i) Prices, zero carry. *If* $r = q = 0$*, then* $C(K, T_1) \le C(K, T_2)$ *for every* $K > 0$*. No model is assumed.*

(ii) Total variance, zero carry. *If* $r = q = 0$ *and both implied volatilities exist, that is, both prices lie strictly inside the bounds of Proposition 17.6 (Proposition 20.38), then*

$$\sigma_{\text{imp}}(K, T_1)^2\, T_1 \;\le\; \sigma_{\text{imp}}(K, T_2)^2\, T_2.$$

(iii) Forward volatility. *Every forward variance of Definition 22.13 whose two implied volatilities exist is nonnegative: at a fixed strike with zero carry, and at a fixed forward moneyness with carry. The forward volatility* $\sigma_{\text{fwd}}(T_1, T_2)$ *is therefore a real number.*

(iv) Carry. *Let* r *and* q *be constants,* $F(T) = Se^{(r-q)T}$*, and let* $K_1, K_2 > 0$ *be strikes at the same forward moneyness,* $K_1/F(T_1) = K_2/F(T_2)$*. Then*

$$C(K_2, T_2) \;\ge\; e^{-q(T_2-T_1)}\, C(K_1, T_1), \qquad \textit{equivalently} \qquad \frac{e^{rT_1}\, C(K_1, T_1)}{F(T_1)} \;\le\; \frac{e^{rT_2}\, C(K_2, T_2)}{F(T_2)},$$

and, when the two implied volatilities exist, $\sigma_{\text{imp}}(K_1, T_1)^2\, T_1 \le \sigma_{\text{imp}}(K_2, T_2)^2\, T_2$*.*

Sketch of the proof. We order the prices by a dominance argument and carry the order to the implied volatilities by monotonicity. For (i) (Step 1 of the complete proof), write the T_1 call, buy the T_2 call and, if $S_{T_1} > K$, sell one share short at T_1 to pay the holder, keeping K in cash: at T_2 the position is worth

$$(S_{T_2} - K)^+ - S_{T_2} + K = (K - S_{T_2})^+ \;\ge\; 0$$

in that case and $(S_{T_2} - K)^+$ otherwise, so by Lemma 16.8(i) its cost $C(K, T_2) - C(K, T_1)$ is nonnegative. For (ii) (Step 2), by (22.4) the two prices are values of one function of the total volatility, strictly increasing because the vega is positive. Part (iv) (Step 3) repeats both steps with carry, at a fixed forward moneyness, and (iii) follows from (ii) and (iv). The complete proof is in Appendix H.3. □

(Problem P22.6 asks you to prove part (i), with the strategy of Step 1, and to deduce part (ii); to apply parts (i) to (iii) at zero carry to a forward volatility, a calendar arbitrage and two interpolations of the term structure; and to apply part (iv) at a positive rate, without proving it, by naming the two strikes it compares.)

The expectation form. A model that prices by expectation produces part (i) in a second way, which is the form in which the inequality is usually quoted. Let $r = q = 0$, and suppose that there is a probability under which S_{T_1} and $Y > 0$ are independent random variables with $\mathbb{E}[Y] = 1$, $S_{T_2} = S_{T_1}Y$, and $C(K, T) = \mathbb{E}[(S_T - K)^+]$ at $T = T_1, T_2$. Fix $x > 0$. Pointwise, $(xY - K)^+ \ge xY - K$ and $(xY - K)^+ \ge 0$. Both sides have finite means, because $0 \le (xY - K)^+ \le xY$. Expectation is linear (Theorem 1.2) and monotone (Setup of §19.2), and $\mathbb{E}[Y] = 1$, so

$$\bar{\psi}(x) := \mathbb{E}\big[(xY - K)^+\big] \;\ge\; \max\big(x\,\mathbb{E}[Y] - K,\; 0\big) = (x - K)^+.$$

Lemma 19.17, applied to the independent variables $X = S_{T_1}$ and Y with the nonnegative function $\psi(x, y) = (xy - K)^+$, gives

$$C(K, T_2) = \mathbb{E}\big[(S_{T_1}Y - K)^+\big] = \mathbb{E}\big[\bar{\psi}(S_{T_1})\big] \;\ge\; \mathbb{E}\big[(S_{T_1} - K)^+\big] = C(K, T_1).$$

The Black–Scholes–Merton model with $r = q = 0$ is an instance. By Definition 20.6 and Theorem 20.7, $C(K,T) = \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = \mathbb{E}[(S_T^{\mathbb{Q}} - K)^+]$ with $S_t^{\mathbb{Q}} = S_0 \exp(-\sigma^2 t/2 + \sigma W_t)$, so

$$S_{T_1}^{\mathbb{Q}} = S_0\, e^{-\sigma^2 T_1/2 + \sigma W_{T_1}}, \qquad S_{T_2}^{\mathbb{Q}} = S_{T_1}^{\mathbb{Q}}\, e^{-\sigma^2 (T_2 - T_1)/2 + \sigma (W_{T_2} - W_{T_1})}.$$

The increments W_{T_1} and $W_{T_2} - W_{T_1}$ are independent by Definition 19.5(3). To avoid any step of the form "a function of independent variables is independent", apply Lemma 19.17 to these two increments themselves, $X = W_{T_1}$ and $Y = W_{T_2} - W_{T_1}$, with

$$\psi(x,y) = \Big(S_0\, e^{-\sigma^2 T_1/2 + \sigma x}\; e^{-\sigma^2 (T_2 - T_1)/2 + \sigma y} - K \Big)^+ \;\geq\; 0.$$

The second exponential factor has mean 1: the increment $W_{T_2} - W_{T_1}$ has the law of $\sqrt{T_2 - T_1}\, Z$ (Definition 19.5(2)), and $\mathbb{E}\, e^{\theta Z} = e^{\theta^2/2}$ with $\theta = \sigma\sqrt{T_2 - T_1}$ (Lemma 19.38(i)). The two pointwise inequalities above, with $S_0 e^{-\sigma^2 T_1/2 + \sigma x}$ in the role of x and the positive factor $e^{-\sigma^2 (T_2 - T_1)/2 + \sigma y}$, of mean 1 when y is replaced by $W_{T_2} - W_{T_1}$, in the role of the factor Y of the general argument, give $\bar\psi(x) \geq (S_0 e^{-\sigma^2 T_1/2 + \sigma x} - K)^+$, and the lemma gives

$$C(K, T_2) = \mathbb{E}\, \psi(W_{T_1}, W_{T_2} - W_{T_1}) = \mathbb{E}\, \bar\psi(W_{T_1}) \;\geq\; \mathbb{E}[(S_{T_1}^{\mathbb{Q}} - K)^+] = C(K, T_1).$$

Part (i) is cited by Proposition 23.27(ii) in §23, which bounds an Asian call by the vanilla call of the last fixing date. Part (iv) at $r = q = 0$ is part (i): then $K_2 = K_1$ and the scaling factor is 1.

Example 22.16 (A calendar arbitrage)**.** Keep the six-month quote of Example 22.14, 20% at $K = F = 100$ with $r = q = 0$, but let the one-year option trade at 13%. The total implied variances are 0.0200 at six months and $0.13^2 \times 1 = 0.0169$ at one year: the total variance falls with the expiry, and the forward variance $(0.0169 - 0.0200)/0.5 = -0.0062$ is negative, so no forward volatility exists. Part (ii) of Proposition 22.15 follows from its part (i), by Step 2 of the proof, so the prices must violate part (i), and they do. At the money, $d_1 = \sigma\sqrt{T}/2 = -d_2$, and at $r = 0$ the formula of Theorem 20.22 reads $C^{\mathrm{BSM}} = F\,(2\,\Phi(\sigma\sqrt{T}/2) - 1)$, so the six-month call costs $100\,(2\,\Phi(0.070711) - 1) = 5.6372$ and the one-year call $100\,(2\,\Phi(0.065) - 1) = 5.1826$.

Sell the six-month call, buy the one-year call and collect $5.6372 - 5.1826 = 0.4546$, about 0.45. Then run Step 1 of the proof. At six months, if $S_{0.5} > 100$, pay the holder $S_{0.5} - 100$ by selling one share short, which leaves 100 in cash; if $S_{0.5} \leq 100$, do nothing. At one year the position is worth $(100 - S_1)^+$ in the first case and $(S_1 - 100)^+$ in the second. Three paths show the mechanics: $S_{0.5} = 112$ and $S_1 = 95$ leave $0 - 95 + 100 = 5$; $S_{0.5} = 112$ and $S_1 = 130$ leave $30 - 130 + 100 = 0$; $S_{0.5} = 93$ and $S_1 = 108$ leave 8. The position is never negative, so the 0.45 collected at the start is a riskless profit.

The one-year volatility may fall to $20\% \times \sqrt{0.5} = 14.14\%$ and no further: at that volatility the one-year total variance is 0.0200, the two total volatilities agree, and both calls cost 5.6372.

(Problem P22.6 works this computation at different numbers.)

Interview trap. "A longer-dated implied volatility can be anything." It can be lower than a shorter-dated one: inverted term structures are common in stressed markets. What cannot fall is the total variance. At a fixed strike with zero carry, $\sigma_{\mathrm{imp}}(K, T_2) \geq \sigma_{\mathrm{imp}}(K, T_1)\sqrt{T_1/T_2}$, which is the 14.14% of Example 22.16; with carry, $\sigma_{\mathrm{imp}}(K_2, T_2) \geq \sigma_{\mathrm{imp}}(K_1, T_1)\sqrt{T_1/T_2}$ at a fixed forward moneyness $K_1/F(T_1) = K_2/F(T_2)$. Below that bound the calendar spread that is short the near call and long the far call is an arbitrage, whatever the model: Proposition 22.15(i) and (ii) with zero carry, and (iv) with carry, where the spread is one far call bought against $e^{-q(T_2 - T_1)}$ near calls written.

Interview trap. "Interpolate the surface linearly in volatility." Interpolate the total variance in T at a fixed moneyness: 18.42% against 18.67% at ten months in Example 22.14. Interpolation that is linear in volatility can create a calendar arbitrage between two nodes that are themselves consistent.

Interview trap. With carry, comparing two expiries at the same strike K compares two different moneynesses, because the forward moves with the expiry. The calendar condition holds at a fixed ratio $K/F(T)$ (Proposition 22.15(iv)), not at a fixed K.

Trading application. The surface is an interpolation device. Traders quote volatilities, not prices. A desk fits each expiry's slice with a smooth curve (Remark 22.9), interpolates in total variance along the maturity at a fixed moneyness, and converts the resulting volatility into a price with the Black–Scholes–Merton formula, used as a change of units: any price strictly inside the no-arbitrage bounds can be written as a volatility (Proposition 20.38), and the constraints on the quotes are the model-free ones, the butterfly condition across strikes (Proposition 17.24) and the calendar condition across expiries (Proposition 22.15). How much a model can add to the surface is settled by Theorem 22.7. Under the pricing rule of its Setup, the call curve of an expiry fixes the risk-neutral density at that expiry, so two such models that reproduce the same call curves give every European claim at a quoted expiry the same price: the discounted expectation of its payoff under that one density. They can differ only on claims whose payoff depends on the path, or on the joint law of the prices at several expiries, the products of §23.

The calendar condition constrains the surface at one instant. The second interview question, how the skew changes a delta, is about how the surface moves when the spot does.

Setup for the smile dynamics. Let $r = q = 0$, fix an expiry T and a strike K, and let the spot price $S > 0$ vary. With zero carry $F = S$, and the Black–Scholes–Merton call price is written as a function of the spot,

$$C^{\mathrm{BSM}}(S, K; \sigma) = S\,\Phi(d_1) - K\,\Phi(d_2), \qquad d_1, d_2 = \frac{\ln(S/K) \pm \sigma^2 T/2}{\sigma\sqrt{T}}.$$

The implied volatility at the strike K when the spot is S is written $\sigma_{\mathrm{imp}}(K; S)$, so that the market call is

$$C(S, K) = C^{\mathrm{BSM}}\big(S, K; \sigma_{\mathrm{imp}}(K; S)\big).$$

In this Setup the arguments of C and of C^{BSM} are the spot and the strike, the expiry being fixed. The surface of today gives $\sigma_{\mathrm{imp}}(K; S)$ at today's spot only. A rule for how it depends on S completes the description; the rule is assumed differentiable in S, and in K where a derivative in the strike is taken. At today's spot, $\partial_K \sigma_{\mathrm{imp}}$ is the skew $\sigma'_{\mathrm{imp}}(K)$ of the Setup of Corollary 22.10; the partial sign is used because $\sigma_{\mathrm{imp}}(K; S)$ also depends on the spot.

Definition 22.17 (Sticky strike and sticky delta (Derman, 1999))**.** Under the Setup above:

(i) *Sticky strike.* The implied volatility of every fixed strike is unchanged when the spot moves: $\sigma_{\mathrm{imp}}(K; S) = \sigma_{\mathrm{imp}}(K)$ for every S.

(ii) *Sticky delta*, or *sticky moneyness.* There is a fixed differentiable function $\mathcal{S}_T : (0, \infty) \to (0, \infty)$, the smile in moneyness, with

$$\sigma_{\mathrm{imp}}(K; S) = \mathcal{S}_T(K/S) \qquad \text{for all } K, S > 0,$$

so that the smile moves with the spot. With zero carry, $d_1 = [-\ln(K/S) + \sigma^2 T/2]/(\sigma\sqrt{T})$ depends on K and S only through the moneyness K/S and through σ. Under this rule the Black–Scholes–Merton delta $\Phi(d_1)$ of an option is therefore a function of its moneyness alone: a fixed moneyness is a fixed delta, which gives the rule its name.

(iii) The *smile-adjusted delta* of the call is the total derivative dC/dS of the market price $S \mapsto C(S, K)$ along the rule.

Which rule a market follows is an empirical matter and depends on the regime (Derman, 1999). The definition claims nothing about it.

The smile-adjusted delta follows from the chain rule. For any rule $\sigma_{\text{imp}}(K;S)$ that is differentiable in S,

$$\frac{dC}{dS} \;=\; \Delta^{\text{BSM}} \;+\; \mathcal{V}\,\frac{d\sigma_{\text{imp}}}{dS}, \tag{22.5}$$

where $\Delta^{\text{BSM}} = \partial_S C^{\text{BSM}} = \Phi(d_1)$ (Proposition 21.2 at $q = 0$) and $\mathcal{V} = \partial_\sigma C^{\text{BSM}} = S\,\varphi(d_1)\sqrt{T}$ (Proposition 21.9(i)) are both evaluated at the point $(S, \sigma_{\text{imp}}(K;S))$, and $d\sigma_{\text{imp}}/dS$ is the derivative of $S \mapsto \sigma_{\text{imp}}(K;S)$ at the fixed strike. The qualifier BSM on the delta is a superscript, as on a price. The justification of (22.5) is the argument of Step 2 in the proof of Corollary 22.10, with the spot in the role of the strike: split the increment of the market price into a move of the spot at the new volatility and a move of the volatility at the old spot, and apply the mean value theorem to each. That argument uses only two facts, and both hold here. The partial derivatives $\partial_S C^{\text{BSM}} = \Phi(d_1)$ and $\partial_\sigma C^{\text{BSM}} = S\,\varphi(d_1)\sqrt{T}$ exist and are continuous on $(0,\infty)^2$, because d_1 is continuous there and Φ and φ are continuous; and the rule $S \mapsto \sigma_{\text{imp}}(K;S)$ is differentiable, hence continuous.

Under sticky strike, $d\sigma_{\text{imp}}/dS = 0$ and the smile-adjusted delta is Δ^{BSM}. Under sticky delta, differentiate $\mathcal{S}_T(K/S)$ in S and in K:

$$\frac{d\sigma_{\text{imp}}}{dS} = -\frac{K}{S^2}\,\mathcal{S}_T'(K/S), \qquad \partial_K \sigma_{\text{imp}} = \frac{1}{S}\,\mathcal{S}_T'(K/S), \qquad \text{so} \qquad \frac{d\sigma_{\text{imp}}}{dS} = -\frac{K}{S}\,\partial_K \sigma_{\text{imp}}.$$

For a downward skew, $\partial_K \sigma_{\text{imp}} < 0$, this derivative is positive, and by (22.5) the smile-adjusted delta of a call exceeds its Black–Scholes–Merton delta by $\mathcal{V}\,(K/S)\,|\partial_K \sigma_{\text{imp}}|$. The same holds for a put. The market put is $\mathcal{P}^{\text{BSM}}$ at the same implied volatility (Proposition 22.3), the put has the same vega as the call (Proposition 20.35), and its Black–Scholes–Merton delta is $\Phi(d_1) - 1$, so its smile-adjusted delta is $\Phi(d_1) - 1 + \mathcal{V}\,d\sigma_{\text{imp}}/dS$. The delta of calls and puts alike therefore rises by $\mathcal{V}\,(K/S)\,|\partial_K \sigma_{\text{imp}}|$. The put's Black–Scholes–Merton delta is $\Phi(d_1) - 1 = -\Phi(-d_1) < 0$, and the increase shrinks the size of this negative delta but never turns it positive when the put prices are free of arbitrage.

Why the put's delta stays nonpositive. The argument combines a homogeneity identity with the convexity of the put in the strike. With zero carry, d_1 and d_2 depend on S and K only through S/K, so the formula's put satisfies $\mathcal{P}^{\text{BSM}}(cS, cK; \sigma) = c\,\mathcal{P}^{\text{BSM}}(S, K; \sigma)$ for $c > 0$; under sticky delta the volatility $\mathcal{S}_T(K/S)$ does not change when S and K are scaled together, so the market put $\mathcal{P}(S,K)$ satisfies $\mathcal{P}(cS, cK) = c\,\mathcal{P}(S,K)$ as well. For such a function the smile-adjusted delta is given by Euler's relation

$$\frac{d\mathcal{P}}{dS} \;=\; \frac{\mathcal{P} - K\,\partial_K \mathcal{P}}{S},$$

which the formulas above confirm directly. At $r = q = 0$ the market put is $\mathcal{P}^{\text{BSM}} = K\,\Phi(-d_2) - S\,\Phi(-d_1)$ at the implied volatility $\sigma_{\text{imp}}(K;S)$, and by the formula parity of Step 2 in the proof of Proposition 22.3 it equals $C - (S - K)$ at every strike, so $\partial_K \mathcal{P} = \partial_K C + 1$. At the fixed spot, the chain rule (H.5) and the strike derivative of Step 3 in the proof of Corollary 22.10, which use only the formula and the differentiability of the smile, give $\partial_K C = -\Phi(d_2) + \mathcal{V}\,\partial_K \sigma_{\text{imp}}$, so $\partial_K \mathcal{P} = \Phi(-d_2) + \mathcal{V}\,\partial_K \sigma_{\text{imp}}$. Hence

$$\frac{\mathcal{P} - K\,\partial_K \mathcal{P}}{S} = -\Phi(-d_1) - \frac{K}{S}\,\mathcal{V}\,\partial_K \sigma_{\text{imp}} = \Phi(d_1) - 1 + \mathcal{V}\,\frac{d\sigma_{\text{imp}}}{dS},$$

the smile-adjusted put delta found above. The same computation for the call gives $dC/dS = (C - K\,\partial_K C)/S$. Moreover $\mathcal{P}(K) \le K\,\partial_K \mathcal{P}(K)$ at every strike, where $\mathcal{P}(K)$ abbreviates $\mathcal{P}(S,K)$ at the fixed spot. The market put is convex in the strike (Proposition 17.24). For $0 < x < K < K + \Delta K$, multiplying the convexity inequality $\mathcal{P}(K) \le w\,\mathcal{P}(x) + (1-w)\,\mathcal{P}(K + \Delta K)$, with $w = \Delta K/(K + \Delta K - x)$, by $K + \Delta K - x$ and rearranging gives the chord inequality

$$\frac{\mathcal{P}(K) - \mathcal{P}(x)}{K - x} \;\le\; \frac{\mathcal{P}(K + \Delta K) - \mathcal{P}(K)}{\Delta K}.$$

Letting $\Delta K \to 0$ gives $(\mathcal{P}(K) - \mathcal{P}(x))/(K - x) \le \partial_K \mathcal{P}(K)$, and letting then $x \to 0$, with $0 \le \mathcal{P}(x) \le x$ (Proposition 17.6(i) and (ii) at $r = 0$), gives $\mathcal{P}(K)/K \le \partial_K \mathcal{P}(K)$. The smile-adjusted put delta $(\mathcal{P} - K\,\partial_K \mathcal{P})/S$ is therefore at most 0. A skew steep enough to push it above zero is an arbitrage. At $S = 100$, $K = 70$, $T = 1$, with the smile $\sigma_{\text{imp}}(K'; 100) = 0.30 - 0.0085\,(K' - 70)$ near that strike, the formula gives the Black–Scholes–Merton delta -0.0903 and the increase $+0.0969$, a smile-adjusted delta of $+0.0066$. But then the put costs $\mathcal{P} = 1.430$, more than $K\,\partial_K \mathcal{P} = 70 \times 0.0110 = 0.773$, and on that smile the second derivative of the put in the strike is -0.0011 at $K = 70$, so a tight butterfly there has a negative price.

Example 22.18 (One call, two hedge ratios)**.** Reuse the smile of Example 22.11: $S = K = F = 100$, $T = 1$, $r = q = 0$, an at-the-money implied volatility of 20% and a skew $\partial_K \sigma_{\text{imp}} = \sigma'_{\text{imp}}(100) = -0.002$ per unit of strike. Then $d_1 = 0.1$, $\Delta^{\text{BSM}} = \Phi(0.1) = 0.540$ and $\mathcal{V} = 100 \times \varphi(0.1) = 39.7$.

Under sticky strike the smile-adjusted delta is 0.540. Under sticky delta, $d\sigma_{\text{imp}}/dS = -(K/S) \times (-0.002) = +0.002$ per unit of spot, and by (22.5)

$$\frac{dC}{dS} = 0.540 + 39.7 \times 0.002 = 0.540 + 0.079 = 0.619.$$

As a check, take near the money the smile linear in the strike, $\mathcal{S}_T(K/S) = 0.20 - 0.2\,(K/S - 1)$ for $0.5 \le K/S \le 1.5$ (extended outside that range to a positive differentiable function on $(0, \infty)$, which the check does not use), so that $\sigma_{\text{imp}}(K; 100) = 0.20 - 0.002\,(K - 100)$ there, and move the spot by one point each way under sticky delta. At $S = 101$ the 100 strike has moneyness 0.9901, volatility 20.198% and price 8.594; at $S = 99$ it has moneyness 1.0101, volatility 19.798% and price 7.356. The central difference is $(8.594 - 7.356)/2 = 0.619$. Under sticky strike the volatility stays at 20%, the two prices are 8.515 and 7.436, and the central difference is 0.540. A second check uses the relation $dC/dS = (C - K\,\partial_K C)/S$ proved before the example: with $C = 7.9656$ and $-\partial_K C = \text{Dig}(100) = 0.5396$ from Example 22.11, it gives $(7.9656 + 100 \times 0.5396)/100 = 0.619$.

A dealer who is short 10,000 such calls buys 5,398 shares under one rule and 6,192 under the other. The assumed dynamics of the smile move the hedge by 794 shares, about 15%.

(Problem P22.7 works this computation at different numbers.)

Intuition. Under sticky delta a rally drags the whole skew to the right. A fixed strike then sits relatively lower on a downward-sloping smile and gains volatility, an extra gain that the Black–Scholes–Merton delta ignores. Under sticky strike nothing moves at a fixed strike. The term structure has dynamics of its own. In the GARCH model, a short-dated volatility above its long-run level goes with a falling term structure, and one below it with a rising one (Corollary 22.28(i), proved in §22.6).

Interview trap. "Delta is a model-free number once the price is known", or "the skew lowers your delta". Under a skew the hedge ratio depends on the assumed dynamics of the smile: 0.540 against 0.619 in Example 22.18, and under sticky delta a downward skew raises the delta. In an interview, state the rule first and the delta second.

Interview trap. "A vega-neutral book across expiries is hedged." Vega neutrality is neutrality to a parallel shift of the surface. In the GARCH model a shock to today's volatility moves the short-dated model volatilities more than the long-dated ones (Corollary 22.28(ii), proved in §22.6). To the extent that implied volatilities respond in the same way, the hedge holds only after the vegas have been weighted by expiry, with the weights computed in the Trading application after Example 22.29. Remark 21.33 points to this subsection for that reason.

22.4 Historical Volatility and Its Standard Error

Implied volatility is a price; realised volatility is a statistic, and before any model of its dynamics come the plain estimator and its error bar. The interview questions are mechanical: "estimate

the volatility from these closes", "annualise it", "how precise is it?". This subsection defines the close-to-close estimator with the annualisation of Corollary 19.2, proves its standard error, bias and mean-square error from the normal moments (Proposition 7.25 and the reduction (7.2)) and Taylor's theorem (Theorem 7.31), and records the empirical reason why time is counted in trading days.

Setup. Closing prices $S_0, S_1, \dots, S_n$ are observed at equally spaced dates, one period of Δt years apart: $\Delta t = 1/252$ for daily data and $\Delta t = 1/52$ for weekly data. The log-return of period i is $R_i = \ln(S_i/S_{i-1})$, as in §19.1. Across an ex-dividend date it is $R_i = \ln\big((S_i + D_i)/S_{i-1}\big)$, where D_i is the dividend paid in period i, so that the fall of the price on the ex-date is not counted as a return. The average return is $\bar{R} = \frac{1}{n}\sum_{i=1}^{n} R_i$; a bar marks an average. The letter n is the sample size, the number of returns, never a number of tree steps. The volatility of one period is $\sigma_{\text{per}} = \sqrt{\text{Var}(R_i)}$, the σ_{per} of §19.1 (it does not carry a subscript Δ, because Δ alone is the Greek). The annual volatility is defined as $\sigma = \sigma_{\text{per}}/\sqrt{\Delta t}$. When the returns are uncorrelated with a common variance, Proposition 19.1 with $1/\Delta t$ periods in a year (its proof uses only these two properties) identifies σ as the volatility of the annual log-return. Estimators wear hats.

Definition 22.19 (Historical (close-to-close) volatility)**.** Let $R_1, \dots, R_n$ be the log-returns of n consecutive periods of length Δt.

(i) The *zero-mean estimator* of the one-period variance is

$$\hat{\sigma}^2_{\text{per}} = \frac{1}{n}\sum_{i=1}^{n} R_i^2,$$

and $\hat{\sigma}_{\text{per}} = \sqrt{\hat{\sigma}^2_{\text{per}}}$ estimates the one-period volatility. This is the convention for daily and weekly data.

(ii) The *demeaned* alternative ($n \geq 2$) is the sample variance

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(R_i - \bar{R})^2,$$

which is unbiased under iid sampling, by Theorem 4.5; s is the sample standard deviation.

(iii) The *annualised estimate*, the historical volatility, is $\hat{\sigma} = \hat{\sigma}_{\text{per}}/\sqrt{\Delta t}$: it is $\hat{\sigma}_{\text{per}}\sqrt{252}$ for daily data (Corollary 19.2) and $\hat{\sigma}_{\text{per}}\sqrt{52}$ for weekly data. Time is counted in trading days (Fact 22.22, later in this subsection).

One computation justifies the zero-mean convention. A drift of 10% a year is $0.10/252 = 0.04\%$ a day, and its square, 1.6×10^{-7}, is 0.16% of the variance 10^{-4} of an asset that moves 1% a day: the squared mean that the zero-mean estimator leaves in is negligible. Estimating the mean, on the other hand, adds noise of the same order as the mean itself. The standard error of $\bar{R}$ is $\sigma_{\text{per}}/\sqrt{n}$, which for a year of daily data is $1\%/\sqrt{252} = 0.063\%$ a day, larger than the mean of 0.04% a day that it is meant to remove.

Proposition 22.20 (Unbiasedness, standard error and bias of the historical estimator)**.** *Let $R_1, \dots, R_n$ be iid $\mathcal{N}(0, \sigma^2_{\text{per}})$ with $\sigma_{\text{per}} > 0$, and let $\hat{\sigma}^2_{\text{per}} = \frac{1}{n}\sum_{i=1}^{n} R_i^2$ and $\hat{\sigma}_{\text{per}} = \sqrt{\hat{\sigma}^2_{\text{per}}} \geq 0$ be the estimators of Definition 22.19(i).*

(i) The variance estimator is unbiased, with

$$\mathbb{E}[\hat{\sigma}^2_{\text{per}}] = \sigma^2_{\text{per}}, \qquad \text{Var}\,(\hat{\sigma}^2_{\text{per}}) = \frac{2\,\sigma^4_{\text{per}}}{n}.$$

(ii) Exactly,

$$\hat{\sigma}_{\text{per}} - \sigma_{\text{per}} \;=\; \frac{\hat{\sigma}^2_{\text{per}} - \sigma^2_{\text{per}}}{2\,\sigma_{\text{per}}} \;-\; \frac{(\hat{\sigma}_{\text{per}} - \sigma_{\text{per}})^2}{2\,\sigma_{\text{per}}}. \tag{22.6}$$

The first term on the right, the linear term*, has mean* 0 *and standard deviation exactly* $\sigma_{\mathrm{per}}/\sqrt{2n}$*. The second, the* remainder*, is nonpositive and has a mean between* $-\sigma_{\mathrm{per}}/n$ *and* 0*. Consequently*

$$-\frac{\sigma_{\mathrm{per}}}{n} \;\le\; \mathbb{E}[\hat{\sigma}_{\mathrm{per}}] - \sigma_{\mathrm{per}} \;\le\; 0\,:$$

the volatility estimator is biased low, by at most σ_{per}/n*.*

(iii) Exactly, $\mathbb{E}\big[(\hat{\sigma}_{\mathrm{per}} - \sigma_{\mathrm{per}})^2\big] \le 2\,\sigma_{\mathrm{per}}^2/n$*.*

(iv) The standard error of $\hat{\sigma}_{\mathrm{per}}$ *is* $\sigma_{\mathrm{per}}/\sqrt{2n}$ *up to a term of order* $1/n$*:*

$$\Big|\sqrt{\mathrm{Var}\,(\hat{\sigma}_{\mathrm{per}})} - \frac{\sigma_{\mathrm{per}}}{\sqrt{2n}}\Big| \;\le\; \sigma_{\mathrm{per}}\sqrt{\frac{3}{n^2} + \frac{12}{n^3}}.$$

The relative standard error $\sqrt{\mathrm{Var}(\hat{\sigma}_{\mathrm{per}})}/\sigma_{\mathrm{per}}$ *is therefore* $1/\sqrt{2n} + O(1/n)$*. Annualisation does not change it, because* $\hat{\sigma}$ *and* σ *are the per-period quantities divided by the same* $\sqrt{\Delta t}$*.*

Sketch of the proof. We compute moments of the squared returns and transfer them to the square root by the exact identity (22.6). For (i), the mean follows by linearity, the normal moments (Proposition 7.25) give $\mathrm{Var}(R_i^2) = 3\sigma_{\mathrm{per}}^4 - \sigma_{\mathrm{per}}^4 = 2\sigma_{\mathrm{per}}^4$, and the squares of independent returns are uncorrelated, so their average has the variance $2\sigma_{\mathrm{per}}^4/n$. Part (iii) is next: since $\hat{\sigma}_{\mathrm{per}} \ge 0$ and $\sigma_{\mathrm{per}} > 0$, the difference of squares gives

$$\big|\hat{\sigma}_{\mathrm{per}} - \sigma_{\mathrm{per}}\big| = \frac{\big|\hat{\sigma}_{\mathrm{per}}^2 - \sigma_{\mathrm{per}}^2\big|}{\hat{\sigma}_{\mathrm{per}} + \sigma_{\mathrm{per}}} \le \frac{\big|\hat{\sigma}_{\mathrm{per}}^2 - \sigma_{\mathrm{per}}^2\big|}{\sigma_{\mathrm{per}}},$$

and squaring, taking expectations and using (i) bounds the mean square of the left side by $\mathrm{Var}(\hat{\sigma}_{\mathrm{per}}^2)/\sigma_{\mathrm{per}}^2 = 2\sigma_{\mathrm{per}}^2/n$. For (ii), the identity is the same difference of squares rearranged: the Taylor expansion of order one of the square root at σ_{per}^2, with its remainder in closed form. By (i) the linear term has mean 0 and variance $\mathrm{Var}(\hat{\sigma}_{\mathrm{per}}^2)/(4\sigma_{\mathrm{per}}^2) = \sigma_{\mathrm{per}}^2/(2n)$; the remainder is nonpositive and, by (iii), its mean is at least $-\sigma_{\mathrm{per}}/n$.

For (iv), the triangle inequality for the mean-square distance (Lemma 19.12(ii)) shows that the standard deviations of $\hat{\sigma}_{\mathrm{per}}$ and of the linear term differ by at most the root-mean-square of the remainder, whose mean square is at most $\mathbb{E}[(\hat{\sigma}_{\mathrm{per}}^2 - \sigma_{\mathrm{per}}^2)^4]/(4\sigma_{\mathrm{per}}^6)$, by the fourth power of the display. The appendix computes this fourth moment, $\sigma_{\mathrm{per}}^8(12/n^2 + 48/n^3)$, by expanding the fourth power of the sum of the centred squares over ordered quadruples of indices: by independence a term in which some index occurs exactly once has mean 0, and the others need the normal moments up to order eight, from (7.2). The complete proof is in Appendix H.4. □

The bound of (iv) is of order $1/n$ against a leading term of order $1/\sqrt{n}$, but it is loose for small samples. At $n = 26$ it allows a gap of $0.072\,\sigma_{\mathrm{per}}$, whereas the exact standard deviation of $\hat{\sigma}_{\mathrm{per}}$, computed numerically from the law of $\hat{\sigma}_{\mathrm{per}}$, is $0.1380\,\sigma_{\mathrm{per}}$ against the first-order value $0.1387\,\sigma_{\mathrm{per}}$. Two consequences of the proposition are used below, and each is proved here.

The demeaned estimator carries the same first-order error. Let $n \ge 2$, so that s^2 is defined. Expanding the square, $\sum_i (R_i - \bar{R})^2 = \sum_i R_i^2 - 2\bar{R}\sum_i R_i + n\bar{R}^2 = n\,\hat{\sigma}_{\mathrm{per}}^2 - n\bar{R}^2$, so

$$s^2 - \hat{\sigma}_{\mathrm{per}}^2 = \frac{n\,\hat{\sigma}_{\mathrm{per}}^2 - n\bar{R}^2}{n-1} - \hat{\sigma}_{\mathrm{per}}^2 = \frac{\hat{\sigma}_{\mathrm{per}}^2 - n\bar{R}^2}{n-1}.$$

By the inequality $(x-y)^2 \le 2x^2 + 2y^2$ for real x and y,

$$\mathbb{E}\big[(\hat{\sigma}_{\mathrm{per}}^2 - n\bar{R}^2)^2\big] \;\le\; 2\,\mathbb{E}[\hat{\sigma}_{\mathrm{per}}^4] + 2n^2\,\mathbb{E}[\bar{R}^4].$$

The first mean is $\mathrm{Var}(\hat{\sigma}_{\mathrm{per}}^2) + \big(\mathbb{E}[\hat{\sigma}_{\mathrm{per}}^2]\big)^2 = \sigma_{\mathrm{per}}^4\,(1 + 2/n)$, by (i) and (1.1). For the second, expand $\big(\sum_i R_i\big)^4$ as the sum of the products $R_a R_b R_c R_d$ over the n^4 ordered quadruples of indices a, b, c,

d, as in Step 1 of part (iv) of the proof of Proposition 22.20 (Appendix H.4). A term in which some index occurs exactly once, say a, has mean 0, by the product rule of Lemma 19.17 and $\mathbb{E}[R_a] = 0$. In every other quadruple one index occurs four times or two distinct indices occur twice each. The n terms R_a^4 have mean $3\sigma_{\text{per}}^4$, and the $3n(n-1)$ terms $R_a^2 R_b^2$ with $a \neq b$ ($n(n-1)/2$ pairs $\{a, b\}$, and $\binom{4}{2} = 6$ places for the two copies of a) have mean σ_{per}^4 (Proposition 7.25), so

$$\mathbb{E}[\bar{R}^4] = \frac{3n + 3n(n-1)}{n^4}\,\sigma_{\text{per}}^4 = \frac{3\,\sigma_{\text{per}}^4}{n^2}.$$

Hence $\mathbb{E}[(\hat{\sigma}_{\text{per}}^2 - n\bar{R}^2)^2] \le 2\sigma_{\text{per}}^4(1 + 2/n) + 6\sigma_{\text{per}}^4 = \sigma_{\text{per}}^4(8 + 4/n)$, and the root-mean-square of $s^2 - \hat{\sigma}_{\text{per}}^2$ is at most

$$\frac{\sigma_{\text{per}}^2\sqrt{8 + 4/n}}{n-1},$$

which is of order $1/n$, against the sampling error $\sigma_{\text{per}}^2\sqrt{2/n}$ of (i), of order $1/\sqrt{n}$. For a year of daily data, $n = 252$, the two numbers are $0.011\,\sigma_{\text{per}}^2$ and $0.089\,\sigma_{\text{per}}^2$. Remark 4.7 discusses the divisor.

An approximate confidence interval. The squares R_i^2 are iid with mean σ_{per}^2 and variance $2\sigma_{\text{per}}^4$, so by Fact 4.8 their average $\hat{\sigma}_{\text{per}}^2$ is approximately normal for large n, with standard deviation $\sigma_{\text{per}}^2\sqrt{2/n}$: with probability about 0.95, the ratio $\hat{\sigma}_{\text{per}}^2/\sigma_{\text{per}}^2$ lies between $1 - 1.96\sqrt{2/n}$ and $1 + 1.96\sqrt{2/n}$. Taking square roots of the endpoints with $\sqrt{1+x} = 1 + x/2 + O(x^2)$ (Proposition 7.34), and noting that $\frac{1}{2}\sqrt{2/n} = 1/\sqrt{2n}$, the ratio $\hat{\sigma}_{\text{per}}/\sigma_{\text{per}}$ lies between $1 - 1.96/\sqrt{2n}$ and $1 + 1.96/\sqrt{2n}$ to first order. Solving for the unknown volatility, with $1/(1 \pm x) = 1 \mp x + O(x^2)$ from the same proposition, and annualising, the approximate 95% interval for σ is

$$\hat{\sigma}\left(1 \pm \frac{1.96}{\sqrt{2n}}\right).$$

Example 22.21 (Error bars on a historical volatility)**.** Half a year of weekly closes, $n = 26$ log-returns, gives the zero-mean estimate $\hat{\sigma}_{\text{per}} = 2.0\%$ a week. The annualised estimate is $\hat{\sigma} = 2.0\% \times \sqrt{52} = 14.4\%$. By Proposition 22.20(ii) the first-order relative standard error is $1/\sqrt{2 \times 26} = 1/\sqrt{52} = 0.139$, which is $14.4 \times 0.139 = 2.0$ volatility points. At $n = 26$ the bound of part (iv), $\sqrt{3/676 + 12/17{,}576} = 0.072$ in relative terms, is loose; the exact relative standard error under the normal model, computed numerically, is 0.1380, against the first-order 0.1387. The approximate 95% interval is $14.4\,(1 \pm 1.96 \times 0.139)$, about 10.5% to 18.3%. The bias of part (ii) is at most $14.4/26 = 0.55$ volatility points, below the standard error.

One year of daily data, $n = 252$, on an asset with a volatility of 16% pins the estimate to $16/\sqrt{504} = 0.71$ volatility points to first order, and part (iv) puts the exact standard error within $16 \times 0.0069 = 0.11$ points of that value. To know the volatility of a 25%-volatility asset to within one point (one standard error) requires $25/\sqrt{2n} \le 1$, that is $n \ge 25^2/2 = 312.5$: 313 trading days.

The contrast is with the drift. Over the same year the average log-return per year is $\ln(S_n/S_0)$, whose standard deviation is one annual volatility: the drift is known only to ± 16 percentage points, whatever the sampling frequency.

Intuition. The mean log-return over a span telescopes: $\sum_i R_i = \ln(S_n/S_0)$, so the average return per year is $\ln(S_n/S_0)/(n\,\Delta t)$, a function of the first and the last price only. Its standard deviation is $\sigma/\sqrt{n\,\Delta t}$, which depends on the calendar span $n\,\Delta t$ and not on the number of observations inside it. Sampling more often sharpens the volatility estimate, never the drift estimate (Merton, 1980). This is the same reason the Sharpe ratio is hard to pin down (Fact 4.16).

Interview trap. "Sample as often as possible." Proposition 22.20 assumes that the observed log-returns are the returns of the price itself. At intraday frequencies the observed prices bounce between the bid and the ask. Suppose that each observed log price is the true one plus an error with mean 0 and variance σ_{noise}^2, the errors of different times being independent of each other and

of the price. An observed return is then the true return plus the difference of two independent errors, and its mean square is that of the true return plus $2\sigma^2_{\text{noise}}$, whatever the length of the interval: the cross terms have mean zero by the product rule of Lemma 19.17. Over a day cut into m intervals, the expected sum of the m observed squared returns is therefore the expected sum of the true ones, which is the daily variance when the true returns have mean zero and are uncorrelated (Proposition 10.10, applied one return at a time, then has no covariance terms), plus $2m\,\sigma^2_{\text{noise}}$: it grows with the sampling frequency instead of settling down. With an error of one basis point, $\sigma_{\text{noise}} = 0.0001$, and $m = 1{,}000$ returns a day, the excess is $2 \times 1{,}000 \times 10^{-8} = 2 \times 10^{-5}$, a fifth of the variance 10^{-4} of an asset that moves 1% a day; with one return a second over a session of six and a half hours, $m = 23{,}400$, it is 4.68×10^{-4}, nearly five times that variance. This is why realised variances are computed from returns over intervals of minutes, not from every tick.

Interview trap. Annualise by the square root of the number of periods per year of the data: $\sqrt{52}$ for weekly returns and $\sqrt{252}$ for trading-day returns. Applying $\sqrt{365}$ to trading-day returns overstates the volatility by the factor $\sqrt{365/252} = 1.20$, so that a 20%-volatility asset is reported at 24%. And never multiply by 252 itself: that is the trap of §19.1.

Interview trap. The standard error of the variance is not that of the volatility. The relative standard error of $\hat\sigma^2_{\text{per}}$ is $\sqrt{2/n}$, by Proposition 22.20(i); that of $\hat\sigma_{\text{per}}$ is $1/\sqrt{2n}$ to first order, half as large (part (iv)). At $n = 26$ the two are 0.277 and 0.139.

Interview trap. Fat tails widen the error bar. For returns with mean zero, so that $\mathbb{E}[R_i^2] = \sigma^2_{\text{per}}$, when the kurtosis $\mathbb{E}[R_i^4]/\sigma^4_{\text{per}}$ exceeds 3, the variance $\text{Var}(R_i^2) = \mathbb{E}[R_i^4] - \sigma^4_{\text{per}}$ exceeds $2\sigma^4_{\text{per}}$, and the normal formula understates the standard error. For iid returns with mean zero and a kurtosis of 6, $\text{Var}(R_i^2) = 5\sigma^4_{\text{per}}$, and the standard error of $\hat\sigma^2_{\text{per}}$ is $\sqrt{5/2} = 1.58$ times its normal value.

Interview trap. Demeaning daily returns before squaring is not "more correct". The mean is estimated with a standard error $\sigma_{\text{per}}/\sqrt{n}$ that is comparable to the mean itself. The bias that demeaning removes is the squared mean $(\mathbb{E}[R_i])^2$ in $\mathbb{E}[R_i^2] = \text{Var}(R_i) + (\mathbb{E}[R_i])^2$, by (1.1), and it is negligible for daily data: 1.6×10^{-7}, which is 0.16% of the variance of an asset that moves 1% a day (the computation after Definition 22.19). And when the mean is zero, demeaning changes the estimate only by a term of order $1/n$ (the first consequence proved after Proposition 22.20). Demeaning matters for monthly data, where the squared mean is no longer negligible beside the variance.

Fact 22.22 (Volatility is generated by trading: the weekend effect, empirical (French and Roll, 1986))**.** Measured from close to close, a span that contains a weekend carries little more return variance than one ordinary trading day, far from the three days' worth that calendar time would give: return variance accrues while the exchange trades. Time in volatility estimates and in the lives of options is therefore counted in trading days, 252 to a year, the convention of Corollary 19.2.

Fact 22.22 is a regularity of market data, quoted as evidence: it is not a theorem and no mathematical proof is possible. Remark 22.12 is the intraday version of the same effect.

Trading application. *Window choice.* A window of n daily returns estimates the volatility with a relative standard error of about $1/\sqrt{2n}$ (Proposition 22.20(iv)): 8.9% at $n = 63$, a quarter's worth of trading days, 6.3% at $n = 126$ and 4.5% at $n = 252$. That gain in precision is real only if the variance was the same throughout the window. Volatility clusters (Fact 22.25 in §22.5): the conditional variance, the quantity a point-in-time estimate measures, moves from one period to the next (Remark 19.3), so a long window averages over periods whose variance differed from today's. The weighted estimators of §22.5 trade the two effects continuously: they down-weight old returns instead of cutting them off, and their effective sample size is computed there.

Range estimators. Write S_i^{hi} and S_i^{lo} for the high and the low of day i; these local symbols avoid

H, the barrier level, and L, the leverage factor, of this Part. The Parkinson estimator

$$\hat{\sigma}^2_{\text{Park}} = \frac{1}{4\ln 2} \cdot \frac{1}{n} \sum_{i=1}^{n} \Big[\ln \frac{S_i^{\text{hi}}}{S_i^{\text{lo}}} \Big]^2$$

estimates the daily variance when the log price is a driftless, continuously monitored Brownian motion, that is, a geometric Brownian motion whose log-drift, not its price drift, is zero. Its variance is roughly one-fifth of that of the close-to-close estimator, so it is about five times as efficient (Parkinson, 1980). Discrete trading and overnight gaps bias it down, and a drift biases it up, negligibly for daily data. The estimator is stated with its attribution and is not proved: a proof needs the law of the range of Brownian motion, and nothing later uses the result.

22.5 EWMA and GARCH(1,1): Updating a Volatility Estimate

The estimator of Definition 22.19 weights a return from six months ago like yesterday's, yet volatility clusters. Risk and options desks therefore run two recursions instead, the exponentially weighted moving average (EWMA) and GARCH(1,1), and "EWMA versus GARCH" is a standard interview question: both update today's variance from yesterday's variance and yesterday's squared return, but only GARCH pulls the estimate back towards a long-run level. This subsection defines both, records the clustering that justifies them, and works one update with each on the same day; the forecasts that GARCH produces are the subject of §22.6.

Setup. This Setup is shared with §22.6. Days are indexed by i, the period index of the log-return R_i; the letter n remains the sample size of §22.4. The quantity σ_i^2 is the variance of the return of day i, computed from the data through day $i-1$, so it is known at the end of day $i-1$; its square root σ_i is a daily volatility. The model is

$$R_i = \sigma_i Z_i,$$

where the shocks Z_i are iid $\mathcal{N}(0,1)$, and Z_i is independent of everything observed before day i, that is, of σ_i^2 and of $Z_{i-1}, Z_{i-2}, \ldots$. With the past held fixed, R_i therefore has mean 0 and variance σ_i^2. In the unrolled formulas below, $m \geq 1$ is a number of days and j is a lag in days.

The symbol notice of §22.1 is repeated here, where the letters are used. In this section only, λ is the decay factor of the EWMA; it is not an eigenvalue, and the jump intensity of §23 is unrelated. The letters ω, α, β are the GARCH parameters. This β is a persistence parameter, not the CAPM beta of Definition 11.35, and this α is neither the Itô drift α_t of Definition 19.25 nor Jensen's alpha of Definition 11.41. The long-run variance is $\bar{\sigma}^2$; a bar marks an average.

Definition 22.23 (Exponentially weighted moving average (RiskMetrics, 1996))**.** Let $0 < \lambda < 1$. The *exponentially weighted moving average* (EWMA) of the squared returns is the sequence of variances defined, from a given starting value, by the recursion

$$\sigma_i^2 = \lambda\, \sigma_{i-1}^2 + (1-\lambda)\, R_{i-1}^2. \tag{22.7}$$

The number λ is the *decay factor*. The convention for daily data is $\lambda = 0.94$.

Unrolling the recursion shows which returns the estimate contains. For every number of days $m \geq 1$ such that day $i-m$ is not earlier than the starting day,

$$\sigma_i^2 = (1-\lambda) \sum_{j=1}^{m} \lambda^{j-1} R_{i-j}^2 + \lambda^m\, \sigma_{i-m}^2. \tag{22.8}$$

The proof is by induction on m. For $m = 1$, (22.8) is (22.7). If it holds for some m and day $i-m$ is later than the starting day, so that the recursion applies at day $i-m$, substitute $\sigma_{i-m}^2 = \lambda\, \sigma_{i-m-1}^2 + (1-\lambda)\, R_{i-m-1}^2$ into its last term:

$$\lambda^m\, \sigma_{i-m}^2 = (1-\lambda)\, \lambda^m\, R_{i-(m+1)}^2 + \lambda^{m+1}\, \sigma_{i-(m+1)}^2.$$

The first term on the right is the term $j = m + 1$ of the sum, and the second is the new last term: this is (22.8) at $m + 1$.

Formula (22.8) is read as follows. The weights decline by the factor λ per day. They sum to $(1-\lambda)\sum_{j=1}^{m} \lambda^{j-1} = 1 - \lambda^m$, by the geometric series, so together with the weight λ^m of the starting value they sum to one. The newest return carries the weight $1 - \lambda$, which is 6% at $\lambda = 0.94$. The weight λ^m of the starting value vanishes as m grows: it is $0.94^{100} = 0.002$ after one hundred days. Read backwards, (22.7) carries this whole weighted history in one number, updated each day from the latest return alone.

The decay factor sets the length of the memory, in two senses. First, the weights halve every $\ln(1/2)/\ln\lambda$ days: 6.6, 11.2 and 22.8 days at $\lambda = 0.90$, 0.94 and 0.97. Second, it fixes the sampling error. Suppose that $R_{i-1}, \ldots, R_{i-m}$ are iid $\mathcal{N}(0, \sigma^2_{\text{per}})$, as in Proposition 22.20. Their squares are then uncorrelated with variance $2\sigma^4_{\text{per}}$ (the proof of part (i) of that proposition), so the variance of a weighted sum of them is the sum of the variances of its terms, by Proposition 10.10 and induction on the number of terms, and multiplying a variable by a constant multiplies its variance by the square of the constant, by linearity of expectation. The weighted sum in (22.8) therefore has the variance

$$2\,\sigma^4_{\text{per}}\,(1-\lambda)^2 \sum_{j=1}^{m} \lambda^{2(j-1)} = 2\,\sigma^4_{\text{per}}\,\frac{(1-\lambda)(1-\lambda^{2m})}{1+\lambda},$$

by the geometric series with ratio λ^2 and $1 - \lambda^2 = (1-\lambda)(1+\lambda)$. The equal-weight estimator of n returns has the variance $2\sigma^4_{\text{per}}/n$, by Proposition 22.20(i). As the history lengthens, $\lambda^{2m} \to 0$ and the weights of the returns sum to one, and the EWMA has the sampling variance of an equal-weight window of $(1+\lambda)/(1-\lambda)$ returns, its *effective sample size*: 19.0, 32.3 and 65.7 days at $\lambda = 0.90$, 0.94 and 0.97. At the convention $\lambda = 0.94$ the EWMA is as precise as an equal-weight window of about a month and a half of trading days, not of a year: it pays for its responsiveness with sampling error.

Definition 22.24 (GARCH(1,1) (Bollerslev, 1986))**.** Let $\omega > 0$, $\alpha \geq 0$ and $\beta \geq 0$ with $\alpha + \beta < 1$. The *GARCH(1,1)* variances are defined, from a given starting value, by the recursion

$$\sigma_i^2 = \omega + \alpha\,R_{i-1}^2 + \beta\,\sigma_{i-1}^2. \tag{22.9}$$

The *long-run variance* is

$$\bar{\sigma}^2 = \frac{\omega}{1-\alpha-\beta},$$

so that $\omega = (1-\alpha-\beta)\,\bar{\sigma}^2$ and the update is a weighted average,

$$\sigma_i^2 = (1-\alpha-\beta)\,\bar{\sigma}^2 + \alpha\,R_{i-1}^2 + \beta\,\sigma_{i-1}^2,$$

with the weight $1 - \alpha - \beta$ on the long-run variance, α on yesterday's squared return and β on yesterday's variance. The three weights are nonnegative and sum to one. The sum $\alpha + \beta$ is the *persistence*.

The name of $\bar{\sigma}^2$ is justified in two stages. First, $\bar{\sigma}^2$ is the level that the update leaves in place: if yesterday's variance and yesterday's squared return are both equal to $\bar{\sigma}^2$ (a squared return equal to its expected value σ_{i-1}^2), the weighted-average form gives $\sigma_i^2 = \bar{\sigma}^2$. Second, Proposition 22.27 in §22.6 proves that the expected future variance converges to $\bar{\sigma}^2$ as the horizon grows.

The recursion unrolls as for the EWMA. Since $\alpha \geq 0$ and $\alpha + \beta < 1$, the parameter β is below 1, and for every $m \geq 1$ such that day $i - m$ is not earlier than the starting day

$$\sigma_i^2 = \omega\,\frac{1-\beta^m}{1-\beta} + \alpha \sum_{j=1}^{m} \beta^{j-1}\,R_{i-j}^2 + \beta^m\,\sigma_{i-m}^2.$$

The proof is again by induction on m. For $m = 1$ the formula is (22.9). If it holds for some m and day $i - m$ is later than the starting day, substituting $\sigma^2_{i-m} = \omega + \alpha\, R^2_{i-m-1} + \beta\, \sigma^2_{i-m-1}$ into the last term turns $\beta^m\, \sigma^2_{i-m}$ into $\omega\, \beta^m + \alpha\, \beta^m\, R^2_{i-(m+1)} + \beta^{m+1}\, \sigma^2_{i-(m+1)}$, and

$$\omega\, \frac{1 - \beta^m}{1 - \beta} + \omega\, \beta^m = \omega\, \frac{1 - \beta^{m+1}}{1 - \beta},$$

which is the formula at $m + 1$. The squared return of j days ago carries the weight $\alpha\, \beta^{j-1}$: GARCH(1,1) is an EWMA with a pull towards $\bar{\sigma}^2$. The EWMA itself is the boundary case $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$, which has $\alpha + \beta = 1$ and no anchor, and which Definition 22.24 excludes. The ARCH model (Engle, 1982) is the finite-lag predecessor of GARCH; with one lag it is the case $\beta = 0$.

Fact 22.25 (Volatility clusters, empirical (Mandelbrot, 1963; Engle, 1982))**.** Large price changes tend to be followed by large changes, of either sign, and small changes by small ones. Daily returns are nearly uncorrelated, while their squares are positively autocorrelated over many lags. Fitted to daily data, the GARCH(1,1) persistence $\alpha + \beta$ is close to one, in the high 0.9s, across equity indices, currencies and commodities, and the standardised squares R^2_i/σ^2_i of a good fit show little remaining autocorrelation.

Fact 22.25 is an empirical regularity, not a theorem: no mathematical proof is possible, only evidence. What clusters is the conditional variance, the quantity that a point-in-time estimate measures; Remark 19.3 records why this makes such an estimate extrapolate poorly, and clustering is the second item of Remark 10.7.

Example 22.26 (One day, two updates)**.** Yesterday's estimate was $\sigma_{i-1} = 1.2\%$ a day, a variance of 0.000144. Yesterday's return was $R_{i-1} = -2.5\%$, with square 0.000625: a move of $2.5/1.2 = 2.1$ standard deviations.

EWMA with $\lambda = 0.94$. By (22.7),

$$\begin{aligned} \sigma^2_i &= 0.94 \times 0.000144 + 0.06 \times 0.000625 \\ &= 0.00013536 + 0.00003750 = 0.00017286, \qquad \sigma_i = 1.315\%. \end{aligned}$$

GARCH(1,1) with $\omega = 0.00000484$, $\alpha = 0.07$, $\beta = 0.89$. The persistence is 0.96 and the long-run variance is $0.00000484/(1 - 0.96) = 0.000121$, a long-run volatility of 1.1% a day. By (22.9),

$$\begin{aligned} \sigma^2_i &= 0.00000484 + 0.07 \times 0.000625 + 0.89 \times 0.000144 \\ &= 0.00000484 + 0.00004375 + 0.00012816 = 0.00017675, \qquad \sigma_i = 1.329\%. \end{aligned}$$

A move of 2.1 standard deviations lifts a volatility of 1.2% only to 1.315% under the EWMA and to 1.329% under GARCH. Under the EWMA the variance moves by $(1 - \lambda)(R^2_{i-1} - \sigma^2_{i-1}) = 0.06 \times 0.000481 = 0.0000289$, a small fraction of the surprise. A return of +2.5% would have produced the same two numbers, because only the square enters.

(Problem P22.9 works this computation at different numbers.)

Trading application. The EWMA has a single parameter, fixed by convention at $\lambda = 0.94$ for daily data, and no long-run level to estimate, which suits a risk system that must cover many assets with one rule. GARCH fits its long-run level, and that level is what gives it a term structure of forecasts (§22.6), the reason it is preferred where options of several expiries are priced. The same recursions update covariances: for the log-returns $R_{1,i}$ and $R_{2,i}$ of two assets on day i (first subscript the asset, second the day, the rule of this Part), the EWMA covariance is

$$\sigma_{12,i} = \lambda\, \sigma_{12,i-1} + (1 - \lambda)\, R_{1,i-1}\, R_{2,i-1}.$$

A covariance matrix assembled pair by pair with different schemes or different decay factors can fail to be positive semidefinite, for the reason of Remark 6.8: no single weighted quadratic form generates all the entries. With one λ for every entry the guarantee returns. The unrolling (22.8) uses only the linearity of the recursion, so it applies entry by entry and writes the matrix as $1-\lambda$ times the sum, with the weights λ^{j-1}, of the outer products of the vectors of daily log-returns of the days $i-j$ with themselves, plus λ^m times the starting matrix. For any vector of weights, the quadratic form of each outer product is the square of the weighted sum of that day's log-returns, which is nonnegative, so the matrix is positive semidefinite whenever the starting matrix is. One λ is therefore used for every entry. Correlation estimation and multivariate GARCH are outside the scope of this section.

Interview trap. The GARCH β is a persistence parameter, the weight on yesterday's variance, not the CAPM beta of Definition 11.35. And it is the sum $\alpha+\beta$, not β alone, that governs the speed of mean reversion (Proposition 22.27 in §22.6).

Interview trap. In the EWMA the newest squared return carries the weight $1-\lambda$, which is 6% at $\lambda=0.94$, not the weight λ. Moving λ trades memory against sampling error, by the computation after Definition 22.23: at $\lambda=0.90$ the weights halve every 6.6 days and the estimate is as precise as 19.0 equally weighted days, while at $\lambda=0.97$ they halve every 22.8 days and it is as precise as 65.7.

Interview trap. "A big move raises the volatility by the size of the move." Under the EWMA the variance moves by $(1-\lambda)(R_{i-1}^2-\sigma_{i-1}^2)$, so the 2.1-standard-deviation day of Example 22.26 moves a volatility of 1.2% by about 0.11 points, to 1.315%.

Interview trap. The update uses R_{i-1}^2 and is blind to the sign of the return. GARCH(1,1) therefore cannot produce the leverage effect behind the equity skew (the Intuition after Fact 22.5): a fall and a rise of the same size raise the variance equally. Asymmetric variants of the model exist; they are not developed here.

22.6 GARCH Forecasts, the Model Term Structure and Maximum Likelihood

The reason to prefer GARCH(1,1) to the EWMA is that it forecasts: its expected variance decays geometrically towards the long-run level, which gives a half-life, a model term structure of volatility for options of every life, and the rule that a shock to today's volatility moves short-dated volatilities more than long-dated ones. This subsection proves the forecast formula by induction with the product rule for independent variables (Lemma 19.17), derives the term structure and the half-life, which closes the promise made in §22.3, and states the maximum-likelihood recipe that fits the parameters to data.

Setup. The Setup and the symbol notice of §22.5 stay in force. Days are indexed by i, and σ_i^2 is the variance of the return of day i, known at the end of day $i-1$. The letter λ is the EWMA decay factor, ω, α, β are the GARCH parameters of Definition 22.24, and $\bar\sigma^2=\omega/(1-\alpha-\beta)$ is the long-run variance. The persistence is always written $\alpha+\beta$; it receives no letter.

Today is the end of day $i-1$. Expectations are taken with the information through day $i-1$ held fixed, so that $\sigma_i^2>0$ is a known number and the only randomness is in the future shocks $Z_i, Z_{i+1}, \dots$. The integers $k\ge 0$ and $j\ge 0$ count trading days ahead: the variance k days ahead is σ_{i+k}^2, and $k=0$ gives today's known variance. An option with k trading days to expiry has a life of $T=k/252$ years. The symbol $\sigma_{\mathrm{GARCH}}(k)$ is the annualised volatility that the model assigns to such an option; it is defined in Corollary 22.28. In the proofs, X and Y are independent random vectors, $\bar\psi(x)=\mathbb{E}\,\psi(x,Y)$ is the frozen mean of Lemma 19.17, and χ is a local auxiliary function of one variable.

Proposition 22.27 (GARCH(1,1) variance forecasts revert geometrically to the long-run level)**.**

Let $\omega \geq 0$, $\alpha \geq 0$ and $\beta \geq 0$ be numbers. Let $\sigma_i^2 > 0$ be a known number, and let the shocks $Z_i, Z_{i+1}, \ldots$ be iid with mean 0 and variance 1; normality is not needed. For $j \geq 0$ let the returns and the variances of the days from i on be generated by

$$R_{i+j} = \sigma_{i+j}\, Z_{i+j}, \qquad \sigma_{i+j+1}^2 = \omega + \alpha\, R_{i+j}^2 + \beta\, \sigma_{i+j}^2,$$

the second equation being the recursion (22.9) with these parameters.

(i) The recursion in expectation, with no condition on $\alpha + \beta$. *For every $k \geq 0$ the expected variance $\mathbb{E}[\sigma_{i+k}^2]$ is finite, and for every $k \geq 1$*

$$\mathbb{E}[\sigma_{i+k}^2] = \omega + (\alpha + \beta)\, \mathbb{E}[\sigma_{i+k-1}^2]. \tag{22.10}$$

(ii) Reversion. *If moreover $\omega > 0$ and $\alpha + \beta < 1$, as in Definition 22.24, with $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$, then for every $k \geq 0$, with the convention $(\alpha+\beta)^0 = 1$,*

$$\mathbb{E}[\sigma_{i+k}^2] \;=\; \bar{\sigma}^2 + (\alpha + \beta)^k\, (\sigma_i^2 - \bar{\sigma}^2). \tag{22.11}$$

The forecasts converge to $\bar{\sigma}^2$ as $k \to \infty$. The long-run variance $\bar{\sigma}^2$ is the unique fixed point of the map $x \mapsto \omega + (\alpha + \beta)\, x$, and the fixed point is attracting.

Sketch of the proof. We proceed by induction on k, with the product rule for independent variables (Lemma 19.17). By the recursion, the variance of a day is a function of the earlier shocks, nonnegative because $\omega, \alpha, \beta \geq 0$, and those shocks are independent of the day's shock, whose mean square is 1. The lemma therefore gives

$$\mathbb{E}[Z_{i+k-1}^2\, \sigma_{i+k-1}^2] = \mathbb{E}[\sigma_{i+k-1}^2]$$

(Step (a) of the complete proof), and the expectation of the recursion is (22.10), whatever $\alpha + \beta$. For (ii), subtracting $\bar{\sigma}^2 = \omega + (\alpha + \beta)\, \bar{\sigma}^2$ shows that each day multiplies the deviation from $\bar{\sigma}^2$ by $\alpha + \beta$; only here does $\alpha + \beta < 1$ enter, to define $\bar{\sigma}^2$ and to send $(\alpha + \beta)^k$ to 0. The complete proof is in Appendix H.5. □

(Problem P22.9 asks you to reproduce the complete proof from scratch.)

No expectation in the proof was conditioned on a continuous variable: the tower rule of Proposition 1.7 is proved for a discrete conditioning variable only, and Lemma 19.17 takes its place. In words, (22.11) says that each day the forecast keeps the fraction $\alpha + \beta$ of the previous day's deviation from the long-run level. Part (i) covers the EWMA as well, which has $\omega = 0$ and $\alpha + \beta = 1$; it is treated after the corollary. Example 22.29, after the corollary, evaluates the formula.

Why the average of the daily variances. An option with k trading days to expiry is exposed to the log-return over those k days, the sum $R_i + \cdots + R_{i+k-1}$ (§19.1), and under the model of Proposition 22.27, with any parameters $\omega, \alpha, \beta \geq 0$, its variance is the sum of the expected daily variances:

$$\operatorname{Var}\Big(\sum_{j=0}^{k-1} R_{i+j}\Big) = \sum_{j=0}^{k-1} \mathbb{E}[\sigma_{i+j}^2].$$

The proof applies the product rule of Lemma 19.17 three times, each time to the vector X of the shocks before a day and to the shock Y of that day, which are independent because the joint law of the shocks is the product of their marginal laws (Step (a) of the proof of Proposition 22.27, Appendix H.5). By the recursion, the variance of day $i + j$ is a nonnegative function $\sigma_{i+j}^2(Z_i, \ldots, Z_{i+j-1})$ of the shocks before that day, a constant for $j = 0$ (the Preliminary of that proof); write $\sigma_{i+j}(x)$ for the square root of this function at a point x. Every $\mathbb{E}[\sigma_{i+j}^2]$ is finite, by part (i).

(1) *Each return has mean zero.* For $j = 0$, $R_i = \sigma_i Z_i$ with σ_i a constant. For $j \geq 1$, $R_{i+j} = \sigma_{i+j}(X)\, Y$ with $X = (Z_i, \ldots, Z_{i+j-1})$ and $Y = Z_{i+j}$. The two factors have finite absolute means, since $\sigma_{i+j} \leq 1 + \sigma_{i+j}^2$ and $|Y| \leq 1 + Y^2$, so $\mathbb{E}[R_{i+j}] = \mathbb{E}[\sigma_{i+j}]\, \mathbb{E}[Z_{i+j}] = 0$.

(2) *Each return has the second moment of its variance:* $\mathbb{E}[R_{i+j}^2] = \mathbb{E}[\sigma_{i+j}^2\, Z_{i+j}^2] = \mathbb{E}[\sigma_{i+j}^2]$, by Step (a).

(3) *Two different returns are uncorrelated.* Let $j < j'$. The product $R_{i+j}\, \sigma_{i+j'}$ is a function $\psi_1(X)$ of the shocks $X = (Z_i, \ldots, Z_{i+j'-1})$ before the day $i + j'$, and $R_{i+j} R_{i+j'} = \psi_1(X)\, Y$ with $Y = Z_{i+j'}$. The inequality $|ab| \leq \frac{1}{2}(a^2 + b^2)$ for real numbers gives $\mathbb{E}|\psi_1(X)| \leq \frac{1}{2}(\mathbb{E}[R_{i+j}^2] + \mathbb{E}[\sigma_{i+j'}^2]) < \infty$, and $|Y| \leq 1 + Y^2$ as in (1), so $\mathbb{E}[R_{i+j} R_{i+j'}] = \mathbb{E}[\psi_1(X)]\, \mathbb{E}[Z_{i+j'}] = 0$.

Expand the square of the sum and use linearity of expectation: the cross terms vanish by (3) and the squares contribute the right-hand side by (2). By (1) the sum has mean zero, so its second moment is its variance. The variance of the k-day log-return, per day and annualised with 252 trading days, is therefore the squared volatility of an option with k days to expiry under the model; the next corollary computes it.

Corollary 22.28 (Average variance over k days, the GARCH term structure, and the half-life)**.** *Under the hypotheses of Proposition 22.27(ii), let* $0 < \alpha + \beta < 1$ *and let* $k \geq 1$.

(i) Term structure. *The expected average daily variance over the* k *days* $i, \ldots, i + k - 1$ *is*

$$\frac{1}{k} \sum_{j=0}^{k-1} \mathbb{E}[\sigma_{i+j}^2] \;=\; \bar{\sigma}^2 + (\sigma_i^2 - \bar{\sigma}^2)\, \frac{1 - (\alpha + \beta)^k}{k\,(1 - \alpha - \beta)}. \tag{22.12}$$

The model volatility of a k-day option *is its annualised square root,*

$$\sigma_{\mathrm{GARCH}}(k) = \sqrt{252 \times \frac{1}{k} \sum_{j=0}^{k-1} \mathbb{E}[\sigma_{i+j}^2]}\,,$$

with 252 *trading days in a year (Corollary 19.2): the volatility that a desk using the model assigns to an option with* k *trading days to expiry, by the paragraph before this corollary. The map* $k \mapsto \sigma_{\mathrm{GARCH}}(k)$ *is the* model term structure. *With* $\bar{\sigma} = \sqrt{\bar{\sigma}^2}$ *the long-run daily volatility, it is strictly decreasing in* k *when* $\sigma_i > \bar{\sigma}$*, strictly increasing when* $\sigma_i < \bar{\sigma}$*, and constant when* $\sigma_i = \bar{\sigma}$.

(ii) Short-dated volatilities react more. *The* averaging factor $\big(1 - (\alpha + \beta)^k\big)/\big(k\,(1 - \alpha - \beta)\big)$ *lies in* $(0, 1]$ *and is strictly decreasing in* k*. A change in today's variance* σ_i^2 *moves the expected average variance over* k *days by the averaging factor times the change. It moves the volatility as well, by less at longer horizons: if today's variance is raised from* $x_0 > 0$ *to* $x_1 > x_0$*, all else fixed, then* $\sigma_{\mathrm{GARCH}}(k)$ *rises by a positive amount that is strictly decreasing in* k.

(iii) Half-life. *The deviation of* $\mathbb{E}[\sigma_{i+k}^2]$ *from* $\bar{\sigma}^2$ *is halved after* $\ln(1/2)/\ln(\alpha + \beta)$ *days.*

Sketch of the proof. We average the forecast (22.11) with a geometric series, which gives (22.12), and compare horizons by monotonicity. By the same series the averaging factor is

$$\frac{1 - (\alpha + \beta)^k}{k\,(1 - \alpha - \beta)} = \frac{1}{k} \sum_{j=0}^{k-1} (\alpha + \beta)^j,$$

the average of the first k terms of a sequence that starts at 1 and, as $0 < \alpha + \beta < 1$, is positive and strictly decreasing: it lies in $(0, 1]$ and falls when a term is appended. This gives the variance claim of (ii) and, through the increasing square root, the direction of the term structure. For the volatility claim, the derivative of $\sigma_{\mathrm{GARCH}}(k)$ in today's variance is $\sqrt{252}/2$ times the averaging factor over the square root of the expected average daily variance, a positive ratio that increases strictly with the averaging factor, hence decreases strictly in k; it is integrated from x_0 to x_1. Part (iii) solves $(\alpha + \beta)^k = 1/2$. The complete proof is in Appendix H.6. □

The EWMA forecasts flat. The EWMA of Definition 22.23 is the recursion (22.9) with $\omega = 0$, $\alpha = 1 - \lambda$ and $\beta = \lambda$, so that $\alpha + \beta = 1$: it lies outside part (ii) of the proposition and outside the corollary, and it has no long-run variance. Part (i) of Proposition 22.27 applies, and (22.10) reads $\mathbb{E}[\sigma^2_{i+k}] = \mathbb{E}[\sigma^2_{i+k-1}]$. By induction $\mathbb{E}[\sigma^2_{i+k}] = \sigma_i^2$ for every $k \geq 0$: the forecast, the expected average variance and the term structure are all flat at today's level.

The model's total variance. By the paragraph before the corollary, the sum of the expected daily variances over k days is the variance of the k-day log-return; dividing by k gives the average of (22.12), and multiplying by 252 expresses it per year. With $T = k/252$,

$$\sigma_{\mathrm{GARCH}}(k)^2\, T = \sum_{j=0}^{k-1} \mathbb{E}[\sigma^2_{i+j}],$$

the model's counterpart of the total implied variance of Definition 22.13. Every term of the sum is positive, because $\mathbb{E}[\sigma^2_{i+j}] = (1-(\alpha+\beta)^j)\bar{\sigma}^2 + (\alpha+\beta)^j\sigma_i^2$. The model's total variance therefore increases with the horizon, as the calendar condition of Proposition 22.15 requires of total implied variance, and the model's forward variance between two horizons is 252 times the average expected daily variance of the days between them.

Example 22.29 (A GARCH forecast, a half-life and a term structure)**.** Take the parameters of Example 22.26, $\omega = 0.00000484$, $\alpha = 0.07$ and $\beta = 0.89$. The persistence is $\alpha + \beta = 0.96$ and the long-run variance is $\bar{\sigma}^2 = 0.00000484/0.04 = 0.000121$: a long-run volatility of 1.1% a day, or $1.1\% \times \sqrt{252} = 17.46\%$ a year. Today's volatility is $\sigma_i = 0.8\%$ a day, which is 12.70% annualised, so $\sigma_i^2 = 0.000064$ and the deviation from the long-run variance is -0.000057.

Forecast. By (22.11), with $0.96^{15} = 0.5421$,

$$\mathbb{E}[\sigma^2_{i+15}] = 0.000121 - 0.5421 \times 0.000057 = 0.0000901,$$

a volatility of $\sqrt{0.0000901} = 0.949\%$ a day fifteen days ahead. By Corollary 22.28(iii) the half-life is $\ln(1/2)/\ln 0.96 = 17.0$ trading days.

Term structure. For a 21-day option the averaging factor is $(1 - 0.96^{21})/(21 \times 0.04) = 0.6853$, the expected average daily variance is $0.000121 - 0.6853 \times 0.000057 = 0.00008194$ by (22.12), and

$$\sigma_{\mathrm{GARCH}}(21) = \sqrt{252 \times 0.00008194} = 14.37\%.$$

The same computation gives 15.88% for 63 days and 17.05% for 252 days. The term structure slopes up from the one-day 12.70% towards the long-run 17.46%, because today's level is below the long-run one (Corollary 22.28(i)).

A shock. Let today's volatility be 0.9% a day in place of 0.8%: the variance rises by $0.000081 - 0.000064 = 0.000017$. The 21-day average variance rises by $0.6853 \times 0.000017 = 0.0000117$ and the 252-day average by $0.0992 \times 0.000017 = 0.0000017$.

horizon k (trading days)	1	21	63	252
averaging factor	1.000	0.685	0.367	0.099
$\sigma_{\mathrm{GARCH}}(k)$ from $\sigma_i = 0.8\%$	12.70%	14.37%	15.88%	17.05%
$\sigma_{\mathrm{GARCH}}(k)$ from $\sigma_i = 0.9\%$	14.29%	15.36%	16.37%	17.17%
change (volatility points)	+1.59	+0.99	+0.49	+0.12

The response shrinks with the life of the option, as Corollary 22.28(ii) proves. Under the EWMA the forecast is flat (Proposition 22.27(i), as shown after the corollary), and the two volatility rows would read 12.70% and 14.29% at every horizon, a change of 1.59 points whatever the life of the option.

(Problem P22.9 works this computation at different numbers.)

Trading application. Corollary 22.28(ii) says that, in the model, a move in today's volatility moves the short-dated model volatilities more than the long-dated ones: in Example 22.29 a move of 1.59 points in the one-day volatility moves the 21-day volatility by 0.99 points and the 252-day volatility by 0.12. A vega hedge built for a parallel shift of every expiry is therefore not neutral to the move that the model predicts. Scaling the vega of each expiry by its own response, relative to that of the one-day volatility, gives the weights $0.99/1.59 = 0.62$ at 21 days and $0.12/1.59 = 0.08$ at 252 days, and a book whose weighted vegas net to zero is neutral, to first order, to a move of that shape. This is why Remark 21.33 on vega neutrality across expiries points to §22.3. The model volatilities are real-world forecasts, so the weights hedge the implied surface only to the extent that implied volatilities respond to a shock as the model volatilities do.

Interview trap. EWMA forecasts are flat. With $\omega = 0$ and $\alpha+\beta = 1$, part (i) of Proposition 22.27 applies and (22.10) gives $\mathbb{E}[\sigma^2_{i+k}] = \sigma^2_i$ for every k. The EWMA has no long-run level and cannot produce a term structure; after a spike it overstates the variance that matters for a long-dated option. GARCH reverts.

Interview trap. "The GARCH term structure is the market's." It is a real-world forecast of realised variance. Implied volatilities are prices, and they carry the variance risk premium of Fact 21.29 on top of any forecast, so the model volatility $\sigma_{\mathrm{GARCH}}(k)$ of a k-day option is not its implied volatility.

Interview trap. "With $\alpha + \beta = 0.99$ the half-life is $1/(1 - 0.99) = 100$ days." The half-life is $\ln 2/(-\ln 0.99) = 69$ days. The number $1/(1 - \alpha - \beta) = \sum_{k\geq 0}(\alpha + \beta)^k$ is the mean lifetime of a deviation: the sum, over all future days, of the fraction of today's deviation that is still present. The half-life is the mean lifetime times a factor that tends to $\ln 2 = 0.69$ as the persistence tends to one.

Interview trap. Indexing. The variance σ^2_i is known at the end of day $i - 1$, so "k days ahead" counts from it: $k = 0$ is today's variance, not a forecast, and a 21-day option averages the days $j = 0, \ldots, 20$. The horizon k is in trading days, while the life of the option in the pricing formula stays in years, $T = k/252$.

Procedure 22.30 (Fitting EWMA and GARCH(1,1) by maximum likelihood)**.** The data are n daily log-returns $R_1, \ldots, R_n$, where n is the sample size. The forecast of (22.11) starts at the end of the sample, at day $n + 1$.

1. *Write the likelihood.* Model R_i, with the past held fixed, as $\mathcal{N}(0, \sigma^2_i)$. The likelihood of the sample is the product over the days of the normal densities $(2\pi\sigma^2_i)^{-1/2} \exp\big(-R^2_i/(2\sigma^2_i)\big)$ at the observed returns, read as a function of the parameters, which enter through the variances σ^2_i. Its logarithm is
$$\sum_{i=1}^{n} \Big[-\tfrac{1}{2}\ln(2\pi) - \tfrac{1}{2}\ln\sigma^2_i - \frac{R^2_i}{2\,\sigma^2_i} \Big].$$
Dropping the constant and multiplying by 2, the parameters that maximise it are those that maximise
$$\sum_{i=1}^{n} \Big(-\ln\sigma^2_i - \frac{R^2_i}{\sigma^2_i} \Big).$$
2. *Give the recursion a starting value.* The first day needs a variance σ^2_1: the zero-mean estimate $\frac{1}{n}\sum_{i=1}^{n} R^2_i$ of Definition 22.19(i) over the whole sample, or the first squared return R^2_1. The choice fades out of the recursion, with the weight β^m after m days for GARCH(1,1) and λ^m for the EWMA, by the unrolled forms of §22.5.
3. *Run the recursion.* For candidate parameters, compute $\sigma^2_2, \ldots, \sigma^2_n$ by (22.7) for the EWMA or by (22.9) for GARCH(1,1), and evaluate the objective of step 1.
4. *Search over the parameters* with a numerical optimiser, keeping $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$ and $\alpha + \beta < 1$. Nothing makes the objective concave, so a search started from one point may

stop at a local maximum; restarting from a spread of starting points guards against it. A moment condition cuts the search to two dimensions. By Proposition 22.27(ii) the forecasts revert to $\bar{\sigma}^2$, so *variance targeting* sets $\bar{\sigma}^2$ equal to the sample second moment $\frac{1}{n}\sum_{i=1}^{n} R_i^2$ and $\omega = \bar{\sigma}^2\,(1-\alpha-\beta)$, and leaves α and β to the optimiser. When the best fit presses against the boundary of the constraints, with ω near zero and $\alpha+\beta$ near one together (under variance targeting $\omega = \bar{\sigma}^2(1-\alpha-\beta)$ goes to zero with $1-\alpha-\beta$, so $\alpha+\beta$ near one suffices), it is approaching the EWMA, the boundary case of §22.5; the EWMA, with its single parameter λ, is then fitted instead.

5. *Check the fit.* Under the model the standardised returns R_i/σ_i are the iid shocks Z_i. Their sample kurtosis should therefore be close to the normal value 3 assumed in step 1; a value well above 3 is the case of the fat-tail trap of §22.4. Their squares should also have lost the autocorrelation that the raw squares R_i^2 carry (Fact 22.25), which a portmanteau statistic such as that of Ljung and Box (1978) measures over many lags at once. Out of sample, the one-day forecasts σ_i^2 should track the squared returns R_i^2 that follow them.

One further step of the recursion after the last day gives σ_{n+1}^2, the starting value of the forecast.

Procedure 22.30 is a recipe, not a claim: nothing is asserted about the existence or the uniqueness of the maximiser. In the one case where the maximisation can be done by hand, it returns an estimator that is already familiar.

Example 22.31 (Maximum likelihood with a constant variance)**.** Let the variance be the same on every day, $\sigma_i^2 = \sigma_{\text{per}}^2$, and let at least one return be different from zero. The objective of step 1 becomes

$$-\,n\ln\sigma_{\text{per}}^2 - \frac{1}{\sigma_{\text{per}}^2}\sum_{i=1}^{n} R_i^2,$$

a function of the single positive variable σ_{per}^2; the variance itself, not its square root, is the variable. Its derivative with respect to σ_{per}^2 is

$$-\frac{n}{\sigma_{\text{per}}^2} + \frac{1}{\sigma_{\text{per}}^4}\sum_{i=1}^{n} R_i^2 = \frac{n}{\sigma_{\text{per}}^4}\,(\hat{\sigma}_{\text{per}}^2 - \sigma_{\text{per}}^2), \qquad \hat{\sigma}_{\text{per}}^2 = \frac{1}{n}\sum_{i=1}^{n} R_i^2.$$

The derivative is positive for $\sigma_{\text{per}}^2 < \hat{\sigma}_{\text{per}}^2$ and negative for $\sigma_{\text{per}}^2 > \hat{\sigma}_{\text{per}}^2$, so the objective increases up to $\hat{\sigma}_{\text{per}}^2$ and decreases after it: $\hat{\sigma}_{\text{per}}^2$ is the unique maximiser. As a check, the second derivative $n/\sigma_{\text{per}}^4 - 2n\,\hat{\sigma}_{\text{per}}^2/\sigma_{\text{per}}^6$ equals $-n/\hat{\sigma}_{\text{per}}^4 < 0$ at that point. The maximum-likelihood estimator is the zero-mean estimator of Definition 22.19(i). It is the known-mean counterpart of the divide-by-n estimator $\hat{\sigma}_{\text{ML}}^2$ of Remark 4.7, which divides the squared deviations from the sample mean by n and is biased low; with the mean known to be zero, dividing by n carries no bias (Proposition 22.20(i)).

With the four returns $+1.0\%$, -2.0%, $+1.5\%$ and -0.5%,

$$\hat{\sigma}_{\text{per}}^2 = \frac{0.000100 + 0.000400 + 0.000225 + 0.000025}{4} = \frac{0.000750}{4} = 0.0001875,$$

and $\hat{\sigma}_{\text{per}} = 1.37\%$ a day. At a variance of 0.0001 the derivative is $4\times(0.0001875-0.0001)/0.0001^2 = +35{,}000$, and at 0.0003 it is $4\times(0.0001875-0.0003)/0.0003^2 = -5{,}000$: the objective rises before the estimate and falls after it. These four returns happen to average zero, so the demeaned estimator of Definition 22.19(ii) differs only through its divisor: $s^2 = 0.000750/3 = 0.00025$ and $s = 1.58\%$ a day.

Interview trap. Stationarity. The long-run variance $\bar{\sigma}^2 = \omega/(1-\alpha-\beta)$ exists only for $\alpha+\beta<1$. Part (i) of Proposition 22.27 has no condition on $\alpha+\beta$, and by induction its recursion (22.10) gives, at $\alpha+\beta=1$ with $\omega>0$, $\mathbb{E}[\sigma_{i+k}^2] = \sigma_i^2 + k\,\omega$, and for $\alpha+\beta>1$

$$\mathbb{E}[\sigma_{i+k}^2] = (\alpha+\beta)^k\Big(\sigma_i^2 + \frac{\omega}{\alpha+\beta-1}\Big) - \frac{\omega}{\alpha+\beta-1},$$

which grows geometrically: there is no finite long-run variance and the forecast does not revert. A fitted persistence at or above one is a signal to use the EWMA (step 4 of Procedure 22.30), not a forecast to be trusted at long horizons.

22.7 Realised Variance, Variance Swaps and Volatility Swaps

A delta-hedged option is a dirty bet on volatility, because its P and L depends on where along the path the variance was realised (Proposition 21.17 over one interval, Remark 21.21 over the life, and exactly in continuous time Proposition 21.22), and a variance swap is the clean one. The interview questions are "what is the fair variance strike?", "why are the options weighted by $1/K^2$?" and "is a volatility swap struck at the square root of the variance strike?". This subsection defines the contracts and answers the three questions with Taylor's theorem with integral remainder (7.3), which expands a smooth payoff into options as the continuous version of Proposition 17.35, with an exact pathwise identity that needs no stochastic calculus, and with the Black–Scholes–Merton law of Definition 20.6.

Setup. Unless a statement says otherwise, $r = q = 0$ in this subsection, so that the forward price is $F = S_0$ and option prices are undiscounted. The contracts are monitored at the dates

$$0 = t_0 < t_1 < \cdots < t_n = T,$$

where n, the number of returns, is the sample size of this subsection; for daily monitoring $\Delta t = 1/252$ and $n = 252\,T$. The price S_{t_i} at every monitoring date is positive, and the log-return of period i is $R_i = \ln(S_{t_i}/S_{t_{i-1}})$.

The market. The market is that of §16.2, with no arbitrage. The underlier can be bought and sold at every monitoring date, it pays no income, and cash is borrowed and lent at the zero rate. European claims are priced as in the Setup of §22.2: under $\mathbb{Q}$ the law of S_T has a density $f_{\mathbb{Q}}$, continuous on $(0, \infty)$, the density that Theorem 22.7 reads off the call curve, and every European payoff $f(S_T)$ with $\mathbb{E}^{\mathbb{Q}}|f(S_T)| < \infty$ trades today at $\mathbb{E}^{\mathbb{Q}}[f(S_T)]$. Here f is a payoff function, a function of the price $x > 0$. In particular the call and the put struck at K are worth

$$C(K) = \mathbb{E}^{\mathbb{Q}}\big[(S_T - K)^+\big], \qquad \mathcal{P}(K) = \mathbb{E}^{\mathbb{Q}}\big[(K - S_T)^+\big].$$

The asset itself is a claim that pays S_T, and its mean $\mathbb{E}^{\mathbb{Q}}[S_T]$ is finite, as in that Setup. One unit bought today and held to T costs S_0 and pays S_T, so by the law of one price (Lemma 16.8(iii)) and the pricing rule, $\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 = F$. One further assumption is made, that the log claim has a price:

$$\mathbb{E}^{\mathbb{Q}}|\ln S_T| = \int_0^\infty |\ln x|\, f_{\mathbb{Q}}(x)\, dx < \infty.$$

Path-dependent legs. Where a floating leg depends on the whole monitored path, a *pricing law* $\mathbb{Q}$ *of the monitored path* is a joint law of $(S_{t_1}, \ldots, S_{t_n})$ under which the fair strike of the leg is its expectation. In the Black–Scholes–Merton model, with the law of Definition 20.6, this is a theorem (Corollary 20.11); in general it is an assumption, and each statement says where it is used. Parts (i) and (ii) of Theorem 22.36 do not use it.

Symbols. Definition 22.32 declares the realised volatility σ_{real}, which is a random variable in this section (it is a constant of the model in §21), the fair strikes K_{var}, $K_{\log}$ and K_{vol}, and the notionals N_{var} and N_{vega}: an N with an upright subscript is a notional, never a count. The integration variable over strikes is K, and x is a price, the argument of a payoff, except in the reading after Theorem 22.36 and in the trap on jumps, where $1 + x$ is the factor by which a jump multiplies the price. A starred letter such as R_i^* is a point supplied by Taylor's theorem. The Taylor theorems of §7.5 call the order of an expansion n; here n is the number of returns, and the order is named in words. The letters χ, χ_1 and χ_2 denote local auxiliary functions of one variable. The local

volatility $\sigma_{\mathrm{loc}}(S_t, t)$ and the average variance σ^2_{avg} along a path appear only in the Intuition after Corollary 22.38, where they are defined; the average variance is not written V_T, which would read as a price at T. With interest rates, market practice replaces each option price by its forward value, e^{rT} times the price, and S_0 by F; this is stated without proof and is used nowhere below.

Definition 22.32 (Realised variance and log-variance, variance swap, volatility swap, fair strikes and notionals)**.** Let the monitoring dates and the log-returns $R_1, \dots, R_n$ be those of the Setup.

(a) *Realised variance.* The *realised variance* over $[0, T]$ is

$$\sigma^2_{\mathrm{real}} = \frac{1}{T} \sum_{i=1}^{n} R_i^2,$$

with the zero-mean convention of Definition 22.19; its square root σ_{real} is the *realised volatility*. For equally spaced dates, $\Delta t = T/n$, it is $\hat{\sigma}^2_{\mathrm{per}}/\Delta t$, the square of the annualised zero-mean estimate $\hat{\sigma}$ of Definition 22.19(iii). For daily dates, $n = 252\,T$, it is $(252/n)\sum_{i=1}^{n} R_i^2$, the average squared daily return annualised with 252 trading days (Corollary 19.2). The factor $252/n$ belongs to the daily case only; the definition, for any monitoring dates, divides by T.

(b) *Realised log-variance.* The *realised log-variance* is

$$\frac{1}{T} \sum_{i=1}^{n} 2\left(e^{R_i} - 1 - R_i\right).$$

Each term is nonnegative, because $e^R - 1 - R = \frac{1}{2} e^{R^*} R^2$ for a point R^* between 0 and R (Corollary 7.32 at order one). By Proposition 7.34, $2\left(e^R - 1 - R\right) = R^2 + R^3/3 + O(R^4)$ as $R \to 0$, so the realised log-variance agrees with σ^2_{real} up to terms of cubic order in the returns.

(c) *Contracts.* Each of the three swaps is settled in cash at T and costs nothing at inception. The *variance swap* pays the long side $N_{\mathrm{var}}\left(\sigma^2_{\mathrm{real}} - \text{strike}\right)$, with the strike in variance units (a strike of 0.04 is "20 vol") and N_{var} the *variance notional*. The *log-variance swap* pays N_{var} times the realised log-variance minus its strike. The *volatility swap* pays $N_{\mathrm{vega}}\left(\sigma_{\mathrm{real}} - \text{strike}\right)$, with N_{vega} the *vega notional*.

(d) *Fair strikes, at $r = 0$.* The fair strike of a swap is the strike at which the contract is worth nothing, that is, the price today of its floating leg, a sure amount at T being worth itself at $r = 0$ (Lemma 16.8(iv)). Under a pricing law $\mathbb{Q}$ of the monitored path,

$$K_{\mathrm{var}} = \mathbb{E}^{\mathbb{Q}}\big[\sigma^2_{\mathrm{real}}\big], \qquad K_{\mathrm{vol}} = \mathbb{E}^{\mathbb{Q}}\big[\sigma_{\mathrm{real}}\big].$$

(In the Black–Scholes–Merton model these are the arbitrage-free strikes: Theorem 22.36(iii), proved below, shows it.) The third strike needs the law of S_T only:

$$K_{\log} = \frac{2}{T}\, \mathbb{E}^{\mathbb{Q}}\Big[- \ln \frac{S_T}{F} \Big].$$

Theorem 22.36(ii), proved below, identifies $K_{\log}$, in every model that satisfies the Setup, as the fair strike of the log-variance swap and as a strip of option prices weighted by $1/K^2$. "The strip" always means $K_{\log}$, never K_{var}.

(e) *Market conventions.* A variance-swap strike is quoted in volatility points, as $100\sqrt{\text{strike}}$: the strike 0.04 is quoted "20". The trade is sized by its vega notional, the sensitivity of the payoff to σ_{real} at the strike. The derivative of $N_{\mathrm{var}}\,\sigma^2_{\mathrm{real}}$ with respect to σ_{real}, at $\sigma_{\mathrm{real}} = \sqrt{K_{\mathrm{var}}}$, is $2\sqrt{K_{\mathrm{var}}}\,N_{\mathrm{var}}$, so, when $K_{\mathrm{var}} > 0$, the vega notional N_{vega} is matched by the variance notional

$$N_{\mathrm{var}} = \frac{N_{\mathrm{vega}}}{2\sqrt{K_{\mathrm{var}}}}.$$

The conversion holds in decimal units or in points, provided both sides use the same units: a volatility point is 0.01 and a variance point is 0.0001. The quoted number $100\sqrt{K_{\mathrm{var}}}$ is not 100 times the volatility-swap strike K_{vol}, which lies below $\sqrt{K_{\mathrm{var}}}$ unless σ_{real} is constant under $\mathbb{Q}$ (Proposition 22.40, proved below).

Example 22.33 (Variance swap against volatility swap: notionals and P and L)**.** Both swaps are struck at 20 vol: the variance swap at $K_{\text{var}} = 0.04$ and the volatility swap at 0.20. The vega notional is 10,000 per volatility point, that is, $N_{\text{vega}} = 1{,}000{,}000$ per unit of volatility. By Definition 22.32(e),

$$N_{\text{var}} = \frac{1{,}000{,}000}{2 \times 0.2} = 2{,}500{,}000 \text{ per unit of variance},$$

which is 250 per variance point. A dealer is short the variance swap. If the realised volatility is 15%, the realised variance is 0.0225 and the dealer receives $2{,}500{,}000 \times (0.04 - 0.0225) = +43{,}750$. If it is 35%, the realised variance is 0.1225 and the dealer receives $2{,}500{,}000 \times (0.04 - 0.1225) = -206{,}250$. The short volatility swap with the matched notional pays $1{,}000{,}000 \times (0.20 - 0.15) = +50{,}000$ and $1{,}000{,}000 \times (0.20 - 0.35) = -150{,}000$.

The seller of variance gains less in the calm outcome and loses more in the turbulent one, by 6,250 and by 56,250. These two numbers are exactly $N_{\text{var}}\,(\sigma_{\text{real}} - 0.2)^2$: $2{,}500{,}000 \times 0.05^2 = 6{,}250$ and $2{,}500{,}000 \times 0.15^2 = 56{,}250$. The reason is one line of algebra. With $N_{\text{vega}} = 2\sqrt{K_{\text{var}}}\,N_{\text{var}}$, the two long payoffs differ by

$$N_{\text{var}}(\sigma_{\text{real}}^2 - K_{\text{var}}) - 2\sqrt{K_{\text{var}}}\,N_{\text{var}}(\sigma_{\text{real}} - \sqrt{K_{\text{var}}}) = N_{\text{var}}(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2,$$

by completing the square. The identity is recorded as part of Proposition 22.40 below, and Figure 48 draws the two payoffs.

(Problem P22.11 works this computation at different numbers.)

Interview trap. Variance notional is not vega notional: $N_{\text{var}} = N_{\text{vega}}/(2\sqrt{K_{\text{var}}})$. A swap of 10,000 per volatility point struck at 20 vol is 250 per variance point. Settling a quote "on 1,000,000 of vega" as 1,000,000 of variance notional mis-sizes the trade by the factor $2\sqrt{K_{\text{var}}}$, which is 0.4 at 20 vol.

Additivity in time: seasoned and forward-starting swaps. For two monitoring dates $s < s'$ write

$$\sigma_{\text{real}}^2[s, s'] = \frac{1}{s' - s} \sum_{s < t_i \le s'} R_i^2$$

for the realised variance of the periods in $(s, s']$, so that $\sigma_{\text{real}}^2 = \sigma_{\text{real}}^2[0, T]$, and write $K_{\text{var}}(s, s')$ for the strike at which a variance swap on the periods of $(s, s']$, entered at s, costs nothing at s; in particular $K_{\text{var}} = K_{\text{var}}(0, T)$. Let t be a monitoring date with $0 < t < T$. Splitting the sum of Definition 22.32(a) at t gives, on every path,

$$\sigma_{\text{real}}^2 = \frac{t}{T}\,\sigma_{\text{real}}^2[0, t] + \frac{T - t}{T}\,\sigma_{\text{real}}^2[t, T],$$

and at t the first term is known. Hold a seasoned long variance swap, entered at time 0 with the strike K and the notional N_{var}, and at t sell $(T - t)/T$ times N_{var} of a fresh swap on $(t, T]$ struck at $K_{\text{var}}(t, T)$. The floating legs on $(t, T]$ cancel, and the two positions together pay at T

$$N_{\text{var}}\Big[\frac{t}{T}\,\sigma_{\text{real}}^2[0, t] + \frac{T - t}{T}\,K_{\text{var}}(t, T) - K\Big],$$

an amount known at t. The fresh swap costs nothing at t, so by Lemma 16.8(v) at $r = 0$ this amount is the value of the seasoned swap at t, with no model of the path. The same split at time 0 prices a forward-starting swap, which pays the variance realised over $(T_1, T_2]$ for two monitoring dates $0 < T_1 < T_2$. A long position in $T_2/(T_2 - T_1)$ units of the swap on $(0, T_2]$ and a short position in $T_1/(T_2 - T_1)$ units of the swap on $(0, T_1]$, each at its fair strike, cost nothing. The

swap on $(0, T_1]$ settles at T_1; carrying its cash flow to T_2 in the cash account at the zero rate (the self-financing bookkeeping of §16.2), the position pays at T_2

$$\sigma^2_{\text{real}}[T_1, T_2] - \frac{K_{\text{var}}(0, T_2)\, T_2 - K_{\text{var}}(0, T_1)\, T_1}{T_2 - T_1},$$

because the squared returns of $(0, T_2]$, less those of $(0, T_1]$, are those of $(T_1, T_2]$:

$$T_2\, \sigma^2_{\text{real}}[0, T_2] - T_1\, \sigma^2_{\text{real}}[0, T_1] = (T_2 - T_1)\, \sigma^2_{\text{real}}[T_1, T_2].$$

By the law of one price (Lemma 16.8(iii)) the fraction is the fair strike of the forward-starting swap: the swap counterpart of the forward variance of Definition 22.13. In numbers, give the short swap of Example 22.33 a life of one year, with daily dates, the strike 0.04 and the notional 2,500,000. After six months of realised volatility at 25%, with a fresh six-month swap quoted at 22 vol, the floating leg is locked at $0.5 \times 0.0625 + 0.5 \times 0.0484 = 0.05545$, which is 23.55 vol, and the short is marked at $2{,}500{,}000 \times (0.04 - 0.05545) = -38{,}625$. With a six-month swap at 20 vol and a one-year swap at 21 vol, the forward-starting swap on the second half-year is struck at $(0.0441 \times 1 - 0.04 \times 0.5)/0.5 = 0.0482$, that is, 21.95 vol.

The strike of the log-variance leg will be read off option prices. The tool is an expansion of an arbitrary smooth payoff into options.

Lemma 22.34 (Carr–Madan replication of a smooth payoff)**.** *Let f be twice continuously differentiable on $(0, \infty)$ and let $F > 0$. Then for every $x > 0$,*

$$f(x) = f(F) + f'(F)\,(x - F) + \int_0^F f''(K)\,(K - x)^+\, dK + \int_F^\infty f''(K)\,(x - K)^+\, dK. \qquad (22.13)$$

For a fixed x, the integrand of the first integral vanishes for $K \le x$ and that of the second for $K \ge x$. Each integral is therefore the integral of a continuous function over a closed bounded interval, $[x, F]$ or $[F, x]$, or it is zero: both are proper integrals, although f'' may be unbounded near 0 and the second range is unbounded.

Proof. We apply Taylor's theorem with integral remainder and rewrite the remainder according to the side of F on which x lies. The function f has two continuous derivatives on the open interval $(0, \infty)$, which contains F. Theorem 7.31 at the expansion point F and order one, that is, (7.3) with its integration variable written K because t is a time in this Part, gives for every $x > 0$

$$f(x) = f(F) + f'(F)\,(x - F) + \int_F^x (x - K)\, f''(K)\, dK.$$

Case $x > F$. For $F \le K \le x$ the weight $x - K$ equals $(x - K)^+$, and for $K > x$ the positive part $(x - K)^+$ is zero, so the remainder equals $\int_F^\infty (x - K)^+ f''(K)\, dK$, the call integral. The put integral vanishes, because $(K - x)^+ = 0$ for $K \le F < x$.

Case $x < F$. Reversing the limits of integration,

$$\int_F^x (x - K)\, f''(K)\, dK = \int_x^F (K - x)\, f''(K)\, dK = \int_0^F (K - x)^+ f''(K)\, dK,$$

because $(K - x)^+$ equals $K - x$ on $[x, F]$ and is zero on $(0, x]$: the remainder is the put integral. The call integral vanishes, because $(x - K)^+ = 0$ for $K \ge F > x$.

Case $x = F$. The remainder is an integral over an interval of length zero, and both integrands of (22.13) vanish identically: $(K - F)^+ = 0$ for $K \le F$ and $(F - K)^+ = 0$ for $K \ge F$. □

Read at $x = S_T$, the lemma says that any smooth payoff is a bond that pays $f(F)$, a position of $f'(F)$ forward contracts with the delivery price F, and a strip of out-of-the-money options, puts

below F and calls above it, with $f''(K)\,dK$ options at the strike K. It is the continuous version of Proposition 17.35. For a piecewise-linear payoff the second derivative is concentrated at the kinks, with weights equal to the changes of slope, and the integrals become a finite sum of puts at the kinks below F and calls at the kinks above F: the replication of that proposition, once put–call parity has turned its calls below F into puts and a forward position.

Example 22.35 (A squared payoff as a strip)**.** Take $f(x) = x^2$ and $F = 100$, so that $f(F) = 10{,}000$, $f'(F) = 200$ and $f'' = 2$ at every strike. Lemma 22.34 reads

$$x^2 = 10{,}000 + 200\,(x - 100) + 2\int_0^{100} (K - x)^+\,dK + 2\int_{100}^{\infty} (x - K)^+\,dK.$$

At $x = 110$ the put integral is zero and the call integral is $2\int_{100}^{110}(110 - K)\,dK = 2 \times 50 = 100$, so the right side is $10{,}000 + 2{,}000 + 0 + 100 = 12{,}100 = 110^2$. At $x = 90$ the call integral is zero and the put integral is $2\int_{90}^{100}(K - 90)\,dK = 100$, so the right side is $10{,}000 - 2{,}000 + 100 + 0 = 8{,}100 = 90^2$. A squared payoff is a bond, a forward position and a strip of options with the same weight at every strike. The log payoff of the next theorem has the weights $1/K^2$.

Theorem 22.36 (The fair variance strike (Neuberger, 1994; Carr and Madan, 1998; Demeterfi, Derman, Kamal and Zou, 1999))**.** *Assume the Setup of this subsection: $r = q = 0$; monitoring dates $0 = t_0 < \cdots < t_n = T$ at which the underlier, whose price is positive, can be traded; the pricing density $f_{\mathbb{Q}}$ and the pricing rule for European claims; $F = S_0 = \mathbb{E}^{\mathbb{Q}}[S_T]$; and $\mathbb{E}^{\mathbb{Q}}|\ln S_T| < \infty$.*

(i) A pathwise identity. *For every positive price path,*

$$\sum_{i=1}^{n} 2\,(e^{R_i} - 1 - R_i) \;=\; 2\sum_{i=1}^{n} \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} \;-\; 2\ln\frac{S_T}{S_0}. \tag{22.14}$$

(ii) The log-variance strike is the strip. *The log-variance swap of Definition 22.32 costs nothing at inception exactly when its strike is $K_{\log}$, and*

$$K_{\log} \;=\; \frac{2}{T}\,\mathbb{E}^{\mathbb{Q}}\Big[-\ln\frac{S_T}{F}\Big] \;=\; \frac{2}{T}\Big[\int_0^F \frac{\mathcal{P}(K)}{K^2}\,dK + \int_F^{\infty} \frac{C(K)}{K^2}\,dK\Big]. \tag{22.15}$$

This holds in every model that satisfies the Setup (positive prices at the monitoring dates, trading at those dates, a continuous risk-neutral density of S_T with mean F and a finite mean of $|\ln S_T|$), continuous or with jumps: only the law of S_T enters. More generally, for any probability law of S_T on $(0,\infty)$ with a continuous density, mean F and a finite mean of $|\ln S_T|$, the second equality of (22.15) holds with $\mathbb{E}^{\mathbb{Q}}$ the expectation under that law and $C(K)$ and $\mathcal{P}(K)$ the expectations of the call and the put payoffs under it.

(iii) The squared-return strike. *In the Black–Scholes–Merton model at the volatility $\sigma > 0$, that is, under (A1)–(A6) of §20.1 and the law of Definition 20.6 with $r = q = 0$, the strikes K_{var} and K_{vol} of Definition 22.32(d) are the arbitrage-free strikes of the variance and the volatility swaps. With equally spaced dates, $\Delta t = T/n$,*

$$K_{\log} = \sigma^2, \qquad K_{\text{var}} = \sigma^2 + \frac{\sigma^4\,\Delta t}{4},$$

so the fair strike of the squared-return swap exceeds the strip by $\sigma^4\Delta t/4$. Moreover

$$\mathbb{E}^{\mathbb{Q}}\Big[\big(\sigma^2_{\text{real}} - \sigma^2\big)^2\Big] = \frac{2\,\sigma^4\,\Delta t + \sigma^6\,(\Delta t)^2}{T} + \frac{\sigma^8\,(\Delta t)^2}{16},$$

which tends to 0 as $\Delta t \to 0$ with T fixed: the realised variance converges to σ^2 in mean square. In general, whenever one pricing law $\mathbb{Q}$ of the monitored path prices both floating legs by their expectations and both strikes are finite,

$$K_{\text{var}} - K_{\log} \;=\; -\,\frac{1}{3\,T}\,\mathbb{E}^{\mathbb{Q}}\Big[\sum_{i=1}^{n} e^{R_i^*}\,R_i^3\Big], \tag{22.16}$$

where R_i^* is a point between 0 and R_i. The summand is a function of R_i alone, because $\frac{1}{3}e^{R_i^*}R_i^3 = 2\,(e^{R_i} - 1 - R_i) - R_i^2$, so the expectation is that of a function of the returns.

Sketch of the proof. The argument is a replication: the floating leg of the log-variance swap is a dynamic leg, earned by trading at no cost, plus $2/T$ log claims, replicated by a strip of options. Part (i) is algebra: $e^{R_i} - 1 = (S_{t_i} - S_{t_{i-1}})/S_{t_{i-1}}$, and the log-returns telescope to $\ln(S_T/S_0)$. For (ii), borrowing $2/T$ at each date t_{i-1}, buying $2/(T\,S_{t_{i-1}})$ units and selling them at t_i costs nothing and pays $1/T$ times the first term on the right of (22.14); the rest is $2/T$ log claims, each paying $-\ln(S_T/F)$. Lemma 22.34 with $f(x) = -\ln(x/F)$, whose second derivative $1/K^2$ supplies the weights, gives (H.9) of the complete proof,

$$-\ln\frac{x}{F} = -\frac{x-F}{F} + \int_0^F \frac{(K-x)^+}{K^2}\,dK + \int_F^\infty \frac{(x-K)^+}{K^2}\,dK.$$

At $x = S_T$ the first term has mean zero, as $\mathbb{E}^{\mathbb{Q}}[S_T] = F$, and the means of the integrals are the two integrals of (22.15), by an exchange of integrations that uses the continuous density and the finite mean of $|\ln S_T|$: only the law of S_T enters. By the pricing rule the replication costs $K_{\log}$, and the law of one price forces that strike. For (iii), the mean $-\frac{1}{2}\sigma^2 T$ of $\ln(S_T/F)$ gives $K_{\log} = \sigma^2$; Corollary 20.11, under (A1)–(A6), prices both legs by their expectations; and returns that are independent and normal with mean $-\frac{1}{2}\sigma^2\Delta t$ and variance $\sigma^2\Delta t$ give $\mathbb{E}^{\mathbb{Q}}[R_i^2] = \sigma^2\Delta t + \frac{1}{4}\sigma^4(\Delta t)^2$, hence K_{var}, and with the normal fourth moment the mean square about σ^2. The gap (22.16) sums the identity that ends the statement over the returns and averages it under a pricing law of the path, under which the mean of each leg is its fair strike, $K_{\log}$ for the log-variance leg by (ii). The complete proof is in Appendix H.7. □

(Problem P22.10 asks you to prove part (i) and, with Lemma 22.34 given, part (ii) without its last sentence; to compute the strip from five quoted strikes and split it into the at-the-money variance, the effect of the skew and the discretisation error; to explain the weights $1/K^2$ and the lift from the skew; and to evaluate the gap $\sigma^4\Delta t/4$ of part (iii) at 21% vol with daily dates, without proving the rest of part (iii).)

Reading the gap. Only the Black–Scholes–Merton value of the gap (22.16) is proved: there it is $\sigma^4\Delta t/4$, the first statement of (iii), which is 1.6×10^{-6} for daily dates at 20 vol. A jump is different. A period in which the price is multiplied by $1 + x$, with $x > -1$, has the log-return $R_i = \ln(1 + x)$, and its contribution to $T\,(K_{\text{var}} - K_{\log})$ in (22.16), before the expectation is taken, is

$$-\tfrac{1}{3}\,e^{R_i^*}R_i^3 = R_i^2 - 2\,(e^{R_i} - 1 - R_i) = \big(\ln(1+x)\big)^2 - 2\,\big(x - \ln(1+x)\big) = -\frac{x^3}{3} + O(x^4),$$

by $\ln(1 + x) = x - x^2/2 + x^3/3 + O(x^4)$ (Proposition 7.34): the square of the logarithm is $x^2 - x^3 + O(x^4)$ and twice $x - \ln(1 + x)$ is $x^2 - \frac{2}{3}x^3 + O(x^4)$. To leading order in the jump size, and neglecting the diffusive part of the same period, the jump therefore enters $K_{\text{var}} - K_{\log}$ as $-x^3/(3T)$ weighted by its probability, and this is not small for a large jump (the trap on jumps below). Under a stochastic-volatility diffusion the gap is not $\sigma^4\Delta t/4$, and it is not computed here.

Scope. Parts (i) and (ii) are exact in discrete monitoring and hold in every model that satisfies the Setup, jumps included; no stochastic calculus was used. The literature states the result for continuous monitoring under a diffusion with a volatility that may itself be random. That statement is mentioned in the Intuition after Corollary 22.38 as the limit of the present one; its proof would need Itô's lemma for a general Itô process (Fact 19.29), and no result of this Part relies on it.

Example 22.37 (The identity on a three-day path, and the Black–Scholes–Merton gap)**.** *A path.* Over three trading days the price goes 100, 101.5, 99.8, 100.6, so $T = 3/252$. The log-returns are

$$R_1 = \ln\frac{101.5}{100} = 0.014889, \qquad R_2 = \ln\frac{99.8}{101.5} = -0.016891, \qquad R_3 = \ln\frac{100.6}{99.8} = 0.007984.$$

The left side of (22.14) is

$$\sum_{i=1}^{3} 2\left(e^{R_i} - 1 - R_i\right) = 0.0002227750 + 0.0002836934 + 0.0000639154 = 0.0005703838.$$

On the right side the dynamic leg is

$$2\sum_{i=1}^{3} \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} = 2\left(0.0150000000 - 0.0167487685 + 0.0080160321\right) = 0.0125345272,$$

and the log leg is $-2\ln(100.6/100) = -0.0119641434$. Their sum is 0.0005703838: the two sides agree to every digit shown, as part (i) says they must on every path. Beside them, the squared returns sum to $\sum_i R_i^2 = 0.0005707091$. The squared-return sum exceeds the log-variance sum, the left side of (22.14), by 3.3×10^{-7}, close to the cubic term $-\sum_i R_i^3/3 = 3.4 \times 10^{-7}$ of part (iii). Annualised by the factor $1/T = 84$, the realised log-variance is 0.047912, which is 21.89 vol, and the realised variance is 0.047940, which is 21.90 vol.

In terms of the strategy of the proof, $2/T = 168$: the dynamic leg holds $168/S_{t_{i-1}}$ units, that is 1.680, 1.655 and 1.683 units on the three days, and earns $168 \times 0.0062673 = 1.0529$; the 168 log claims pay $168 \times (-0.0059821) = -1.0050$; and the total is the realised log-variance, 0.0479.

The model. In the Black–Scholes–Merton model at 20 vol, with daily dates over one year ($\Delta t = 1/252$, $T = 1$), part (iii) gives $K_{\log} = 0.04$ and

$$K_{\text{var}} = 0.04 + \frac{0.2^4}{4 \times 252} = 0.04 + 0.0000015873 = 0.04000159,$$

that is, 20.0000 against 20.0004 vol. The root-mean-square sampling error of one year of daily realised variance is

$$\sqrt{\frac{2\,\sigma^4\,\Delta t + \sigma^6(\Delta t)^2}{T} + \frac{\sigma^8(\Delta t)^2}{16}} = \sqrt{0.0000126994} = 0.00356,$$

a realised volatility of about 20 ± 0.9 volatility points, since $0.00356/(2 \times 0.2) = 0.0089$ to first order. It matches the first-order standard error $20/\sqrt{504} = 0.89$ of Proposition 22.20(iv) for a year of daily data. The gap between the two strikes is more than two thousand times smaller than the noise in one year of realised variance.

Interview trap. Jumps. A jump that multiplies the price by $1 + x$ is a period with the log-return $R = \ln(1 + x)$. It adds $R^2 = (\ln(1 + x))^2$ to the sum behind the realised variance, but $2\left(e^R - 1 - R\right) = 2\left(x - \ln(1 + x)\right)$ to the sum behind the log leg, which is what the strip replicates. The gap for the period is $-\frac{1}{3}e^{R^*}R^3 = -x^3/3 + O(x^4)$ (Theorem 22.36(iii) and the reading after it), and it enters both annualised legs divided by T; to leading order in the jump size, and neglecting the diffusive part of the same period, it enters the strikes weighted by the probability of the jump. For a fall of 20%, $x = -0.2$: the realised variance gains 0.0498 and the log leg 0.0463, a gap of 0.0035, of which the leading term $-x^3/3$ accounts for 0.0027. On a one-year swap the realised variance then exceeds the realised log-variance by 0.0035, against the dealer who sold variance and hedged with the strip. This is the reason variance swaps on single names are capped.

Interview trap. "The strip is the variance-swap strike." The strip is the log-variance strike $K_{\log}$. The strike K_{var} of the squared-return swap differs from it by a term of cubic order, (22.16): $\sigma^4\Delta t/4$ in the Black–Scholes–Merton model, which is negligible, but, to leading order in the jump size and neglecting the diffusive part of the same period, $-x^3/(3T)$ for each period with a jump of $1 + x$, weighted by its probability, which is not.

Corollary 22.38 (With a flat smile the log-variance strike is σ^2)**.** *Assume the Setup of this subsection, with $r = q = 0$. If every European call and put expiring at T is priced by the Black-76 model of Definition 20.16 at one volatility $\sigma > 0$, then*

$$K_{\log} = \sigma^2.$$

If moreover (A1)–(A6) of §20.1 hold at the same volatility σ, with the law of Definition 20.6 at that volatility as the law of the monitored path, with n equally spaced monitoring dates, then $K_{\text{var}} = \sigma^2 + \sigma^4 \Delta t/4$.

Proof. We evaluate the strip under the lognormal law that generates the Black-76 prices. By (22.15), $K_{\log}$ is $2/T$ times a strip that involves option prices only. By hypothesis and Definition 20.16 at $r = 0$, these prices are the expectations of the call and the put payoffs under the law of

$$F \exp\big(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\, Z\big), \qquad Z \sim \mathcal{N}(0,1).$$

This is the law of the terminal price under Definition 20.6 at $r = q = 0$ (Proposition 20.8(i)). Its density is the lognormal density of the proof of Proposition 19.42(ii) with $\mu = 0$, continuous on $(0, \infty)$; its mean is F, by Proposition 20.8(ii); and the mean of the absolute value of its logarithm is at most $|\ln F| + \frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,\mathbb{E}|Z| < \infty$. The last sentence of Theorem 22.36(ii), applied to this law, says that the strip of its option prices equals the mean of minus the logarithm of the variable over F:

$$K_{\log} = \frac{2}{T}\,\mathbb{E}\Big[\frac{\sigma^2 T}{2} - \sigma\sqrt{T}\, Z\Big] = \frac{2}{T} \cdot \frac{\sigma^2 T}{2} = \sigma^2,$$

because $\mathbb{E}[Z] = 0$; this is the computation of Example 20.10(3), and Corollary 19.34 performs it for the path form of the model. The clause on K_{var} is Theorem 22.36(iii) at this σ. □

The strip is a model-free average of the smile, with the weights $1/K^2$. Under a flat smile it returns the one volatility there is. Under a skew it generally differs from the at-the-money variance σ_{ATM}^2, and the direction depends on the shape of the smile, not only on its tilt. Three smiles with $F = 100$, $T = 1$ and $\sigma_{\text{ATM}} = 20\%$ show it; every strip below is (22.15) integrated numerically, each linear smile is kept at 1% where the line would fall below it (which changes no digit shown), and the call curve of each smile is decreasing and convex in the strike.

(1) *Linear in log-moneyness*, $\sigma_{\text{imp}}(K) = \sigma_{\text{ATM}} - \gamma \ln(K/F)$, where the tilt γ is a number local to this paragraph. To first order in γ, each option price moves by its vega $F\varphi(d_1)\sqrt{T}$ (the vega of §22.1 at $r = 0$) times $-\ln(K/F)$. In the variable $y = \ln(K/F)$, the weight $F\varphi(d_1)\sqrt{T}\,dK/K^2$ is $\sqrt{T}\,\varphi\big((y + \sigma_{\text{ATM}}^2 T/2)/(\sigma_{\text{ATM}}\sqrt{T})\big)\,dy$, which is $\sigma_{\text{ATM}} T$ times the $\mathcal{N}(-\sigma_{\text{ATM}}^2 T/2, \sigma_{\text{ATM}}^2 T)$ density of y. The slope of the strip at $\gamma = 0$ is therefore $(2/T)\,\sigma_{\text{ATM}} T$ times the mean $\sigma_{\text{ATM}}^2 T/2$ of $-y$ under that law, that is, $\sigma_{\text{ATM}}^3 T = 0.008$: a downward tilt, $\gamma > 0$, lifts the strip at first order. At $\gamma = 0.05$ the strip is 0.040719, against 0.0404 at first order.

(2) *Linear in the strike*, $\sigma_{\text{imp}}(K) = \sigma_{\text{ATM}} - \gamma\,(K - F)/F$. The same computation weighs $1 - e^y$ in place of $-y$, and the mean of e^y under $\mathcal{N}(-\sigma_{\text{ATM}}^2 T/2, \sigma_{\text{ATM}}^2 T)$ is 1, so the first-order effect cancels. At $\gamma = 0.05$ the strip is 0.040294: the lift is of second order in the tilt.

(3) *A downward skew below the at-the-money variance.* With $y = \ln(K/F)$, let the smile before its floor be

$$\sigma_{\text{raw}}(y) = 0.2 - 0.01\, y - 0.29\,\big[0.05 \ln\big(1 + e^{y/0.05}\big) - 0.05 \ln 2\big],$$

and let $\sigma_{\text{imp}}(K) = 0.03 + 0.01 \ln\big(1 + e^{(\sigma_{\text{raw}}(y) - 0.03)/0.01}\big)$. The slope is -0.01 per unit of log-moneyness well below F and -0.3 above it, the two joined smoothly over about 0.1 in y. The volatility levels off at 3% far above F, and $\sigma_{\text{imp}}(F) = 20.00\%$. The smile is decreasing in K, and its strip is 0.0375, which is 19.36 vol, below 0.04: the cheap calls above F outweigh the slightly rich puts below it.

A downward tilt lifts the strip at first order in case (1) and at second order in case (2). The equity skew of Problem P22.10 falls from 26% at the strike 70 to 20% at the forward and 16% at 130, more steeply below the forward than above it, and its strip is 0.0443, which is 21.0 vol, about one volatility point above the at-the-money variance. The tilt, not the asymmetry, does most of this: the smile linear in the strike at the slope of the put side, which coincides with the skew below

the forward, falls to 14% at 130 and is flat beyond the quotes like the skew, has the strip 0.0434, which is 20.8 vol, so the shallower fall above the forward adds only 0.0009, about a fifth of the lift and 0.2 of a volatility point. None of this is a theorem: no inequality between $K_{\log}$ and σ^2_{ATM} holds for every skew. At $\sigma = 20\%$ with daily dates the two strikes of the corollary are $K_{\log} = 0.04$ and $K_{\text{var}} = 0.04000159$, the 20.0000 and 20.0004 vol of Example 22.37.

Example 22.39 (A strip from five listed strikes)**.** Let $F = 100$, $T = 1$, $r = 0$, and let the smile be flat at 20%. Out-of-the-money options are listed at the strikes 80, 90, 100, 110 and 120, with the spacing $\Delta K = 10$. Their Black-76 prices are $\mathcal{P}(80) = 1.186$, $\mathcal{P}(90) = 3.589$, $C(100) = \mathcal{P}(100) = 7.966$ (the 0.4 rule of Proposition 20.29 gives 8.0), $C(110) = 4.292$ and $C(120) = 2.147$. Replace the two integrals of (22.15) by one sum over the listed strikes, each strike contributing $\Delta K \times \text{price}/K^2$:

strike K	80	90	100	110	120	sum
price	1.186	3.589	7.966	4.292	2.147	
$\Delta K \times \text{price}/K^2$	0.001853	0.004431	0.007966	0.003547	0.001491	0.019288

The discretised strip is $K_{\log} \approx (2/T) \times 0.019288 = 0.0386$, which is 19.6 vol, against the exact $K_{\log} = 0.04$, 20.0 vol, of Corollary 22.38.

Two errors of opposite sign are at work. The five buckets cover the strikes from 75 to 125, and the wings beyond them are missing: by numerical integration their exact contributions to $K_{\log}$ are 0.001275 below 75 and 0.001992 above 125, together 0.003267. Inside $[75, 125]$ the coarse sum overstates the exact integral, 0.038576 against 0.036733, an excess of 0.001843, because the out-of-the-money price is peaked at $K = F$ and a rectangle of width 10 at the peak covers more than the area under it. The net shortfall is $0.003267 - 0.001843 = 0.001424$, the distance from 0.038576 to 0.04. A strike every 1 from 20 to 400 gives 0.04002.

The balance between the two errors depends on the expiry. At $T = 0.25$ the same grid of five strikes overshoots, to 0.0466: the spacing is then about one standard deviation of S_T, $100 \times 0.2 \times \sqrt{0.25} = 10$ to first order (exactly 10.03), and the at-the-money bucket supplies more than two thirds of the sum. A spacing of 5 from 60 to 140 gives 0.0417, and a spacing of 1 from 20 to 400 gives 0.0401.

(Problem P22.10 works this computation at different numbers.)

Intuition. *Why* $1/K^2$. In the Black–Scholes–Merton model at $r = 0$ and the volatility σ_{imp}, the claim that pays $-(2/T)\ln(S_T/F)$ has the price function (Theorem 20.7)

$$V(S,t) = -\frac{2}{T}\Big[\ln\frac{S}{F} - \frac{\sigma^2_{\text{imp}}\,(T-t)}{2}\Big], \qquad \text{so} \qquad \partial_S V = -\frac{2}{T\,S}, \qquad \tfrac{1}{2}\,S^2\,\partial_{SS}V = \frac{1}{T}.$$

Its dollar gamma is the same at every spot and at every time. The dynamic leg of Theorem 22.36, which holds $2/(T\,S_{t_{i-1}})$ units, is exactly the delta hedge of a long position in this claim. The claim also satisfies the theta–gamma relation behind (21.7), since $\partial_t V = -\sigma^2_{\text{imp}}/T = -\frac{1}{2}\sigma^2_{\text{imp}} S^2\,\partial_{SS}V$. Over the monitoring interval from t_{i-1} to t_i, write $\Delta S_i = S_{t_i} - S_{t_{i-1}}$. Directly from the price function, $V(S_{t_i}, t_i) - V(S_{t_{i-1}}, t_{i-1}) = -(2/T)\,R_i - \sigma^2_{\text{imp}}(t_i - t_{i-1})/T$ and $\partial_S V(S_{t_{i-1}}, t_{i-1})\,\Delta S_i = -(2/T)(e^{R_i} - 1)$, so the delta-hedged P and L of the claim over the interval is exactly

$$V(S_{t_i}, t_i) - V(S_{t_{i-1}}, t_{i-1}) - \partial_S V(S_{t_{i-1}}, t_{i-1})\,\Delta S_i = \frac{1}{T}\Big[2\,(e^{R_i} - 1 - R_i) - \sigma^2_{\text{imp}}\,(t_i - t_{i-1})\Big],$$

the per-period term of (22.14) less the implied variance of the period. Its leading term,

$$\frac{1}{T}\Big[\Big(\frac{\Delta S_i}{S_{t_{i-1}}}\Big)^2 - \sigma^2_{\text{imp}}\,(t_i - t_{i-1})\Big],$$

is the gamma-weighted form $\frac{1}{2}\Gamma\,[(\Delta S)^2 - \sigma^2_{\text{imp}} S^2\,\Delta t]$ of (21.7) with $\Gamma = \partial_{SS}V = 2/(T\,S^2)$, whose gamma weight $\frac{1}{2}\Gamma S^2 = 1/T$ is the same at every spot: the variance counts wherever along the

path it was realised. Summed over the intervals, the claim bought at $V(S_0,0)=\sigma_{\text{imp}}^2$ and hedged at the monitoring dates earns exactly the realised log-variance minus σ_{imp}^2, on every path.

An option struck at K has its dollar gamma concentrated near K. For a call or a put at $r=q=0$ and the volatility σ_{imp}, $\Gamma=\varphi(d_1)/(S\sigma_{\text{imp}}\sqrt{T-t}\,)$ at the time t (Proposition 21.6(i), whose τ is $T-t$), and d_1 depends on S and K only through the ratio K/S. So $S\,\Gamma=\chi(K/S)$ for a positive local function χ of one variable, and the dollar gamma of the option is $\frac{1}{2}S^2\,\Gamma=\frac{1}{2}S\,\chi(K/S)$. A strip with the power weights K^{-c}, for a real constant c, has the dollar gamma

$$\int_0^\infty \frac{S}{2}\,\chi\Big(\frac{K}{S}\Big)\,K^{-c}\,dK=\frac{S^{2-c}}{2}\int_0^\infty \chi(z)\,z^{-c}\,dz,$$

by the substitution $K=zS$. The integral on the right is finite and positive, because $\chi(z)=\varphi(d_1)/(\sigma_{\text{imp}}\sqrt{T-t})$ with d_1 affine in $\ln z$ decays faster than every power of z at 0 and at ∞, and it does not depend on S, so the dollar gamma is the same at every spot exactly when $c=2$. With the weights $2/(T\,K^2)$ of (22.15) the constant is $1/T$, as it must be: by (H.9) the strip differs from the log claim by a forward position and a constant, which have no gamma. Among power weights, $1/K^2$ is the one that turns peaked option gammas into a constant dollar gamma.

The continuous limit. The following is mentioned, not proved. Under a local-volatility model $dS_t=\sigma_{\text{loc}}(S_t,t)\,S_t\,dW_t^{\mathbb{Q}}$, the strip $K_{\log}$, which does not depend on the monitoring grid, equals $\mathbb{E}^{\mathbb{Q}}[\sigma_{\text{avg}}^2]$, and K_{var} tends to it as the monitoring grid is refined, where

$$\sigma_{\text{avg}}^2=\frac{1}{T}\int_0^T \sigma_{\text{loc}}(S_t,t)^2\,dt$$

is the average variance along the path. This is the continuously monitored statement of the literature. Its proof needs Itô's lemma for a general Itô process (Fact 19.29), and no result of this Part relies on it.

Trading application. A VIX-style volatility index is 100 times the square root of the discretised strip of Theorem 22.36 at a horizon of 30 days, obtained from the two listed expiries that bracket 30 days by interpolation and annualisation. It approximates $100\sqrt{K_{\log}}$, the log-variance strike $K_{\log}$ in volatility points, up to the discretisation errors of Example 22.39. The strike $K_{\log}$ in turn equals the variance-swap strike K_{var} up to the cubic-order gap of Theorem 22.36(iii), which is $\sigma^4\Delta t/4$ in the Black–Scholes–Merton model and larger when the price can jump. Derivatives on the index are outside the scope of this section. A short variance swap is the way to harvest the premium of Fact 21.29 without gamma weights, whereas the delta-hedged option of Proposition 21.17 earns a gamma-weighted variance and so depends on where the variance was realised.

Interview trap. "A delta-hedged straddle is a variance swap." Its P and L is weighted by gamma (Proposition 21.17), and the dollar gamma of a straddle is concentrated near its strike: the variance realised far from the strike earns little. Among strips with power weights K^{-c}, only $c=2$ gives a dollar gamma that does not depend on the spot (the Intuition above).

Interview trap. A coarse strip misstates $K_{\log}$: it overshoots when the at-the-money bucket is wide and undershoots when the wings are truncated (0.0466 and 0.0386 against 0.04 in Example 22.39). Desks therefore integrate a fitted smile, extrapolated into the wings (Remark 22.9), not the raw listed strikes. And "the fair variance strike is the at-the-money variance" is false under a skew. For the equity skew of Problem P22.10 the strip, and with it K_{var} up to the cubic-order gap of (22.16), sits about one volatility point above the at-the-money variance; the direction is not guaranteed for every skew (the paragraph after Corollary 22.38).

Proposition 22.40 (The fair volatility strike is at most the square root of the variance strike)**.** *Let $\mathbb{Q}$ be a pricing law of the monitored path under which σ_{real} has a finite second moment, and let $K_{\text{var}}=\mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}^2]$ and $K_{\text{vol}}=\mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}]$ be the fair strikes of Definition 22.32(d). Here K_{var} is the strike of the squared-return swap, not the strip. Then*

$$K_{\text{vol}}\;=\;\sqrt{K_{\text{var}}-\operatorname{Var}^{\mathbb{Q}}(\sigma_{\text{real}})}\;\le\;\sqrt{K_{\text{var}}},$$

with equality if and only if $\mathrm{Var}^{\mathbb{Q}}(\sigma_{\mathrm{real}}) = 0$. *Moreover, let a variance swap be struck at* K_{var} *and a volatility swap at* $\sqrt{K_{\mathrm{var}}}$*, with the vega-matched notionals* $N_{\mathrm{vega}} = 2\sqrt{K_{\mathrm{var}}}\,N_{\mathrm{var}}$. *Then on every path the long variance-swap payoff minus the long volatility-swap payoff is*

$$N_{\mathrm{var}}\left(\sigma_{\mathrm{real}} - \sqrt{K_{\mathrm{var}}}\right)^2 \;\geq\; 0.$$

In the Black–Scholes–Merton model K_{var} *exceeds* $K_{\log}$ *only by* $\sigma^4\Delta t/4$ *(Theorem 22.36(iii)), so there the strip is quoted for* K_{var}.

Proof. We apply the variance shortcut to the realised volatility and complete a square. By (1.1) with $X = \sigma_{\mathrm{real}}$ under $\mathbb{Q}$,

$$0 \;\leq\; \mathrm{Var}^{\mathbb{Q}}(\sigma_{\mathrm{real}}) = \mathbb{E}^{\mathbb{Q}}[\sigma_{\mathrm{real}}^2] - \left(\mathbb{E}^{\mathbb{Q}}[\sigma_{\mathrm{real}}]\right)^2 = K_{\mathrm{var}} - K_{\mathrm{vol}}^2,$$

so $K_{\mathrm{vol}}^2 = K_{\mathrm{var}} - \mathrm{Var}^{\mathbb{Q}}(\sigma_{\mathrm{real}})$. The realised volatility is nonnegative, hence so is its mean K_{vol}, and taking the nonnegative square root gives the formula. The variance is nonnegative and the square root is increasing, which gives the inequality; equality holds exactly when the variance is zero. For the payoffs, on every path

$$\begin{aligned} N_{\mathrm{var}}(\sigma_{\mathrm{real}}^2 - K_{\mathrm{var}}) - 2\sqrt{K_{\mathrm{var}}}\,N_{\mathrm{var}}(\sigma_{\mathrm{real}} - \sqrt{K_{\mathrm{var}}}) &= N_{\mathrm{var}}(\sigma_{\mathrm{real}}^2 - 2\,\sigma_{\mathrm{real}}\sqrt{K_{\mathrm{var}}} + K_{\mathrm{var}}) \\ &= N_{\mathrm{var}}(\sigma_{\mathrm{real}} - \sqrt{K_{\mathrm{var}}})^2, \end{aligned}$$

and a square is nonnegative. The last clause of the statement is Theorem 22.36(iii). □

A long variance swap is therefore a long volatility swap plus a position that gains whenever the realised volatility ends away from the strike, in either direction: a long position in the volatility of volatility. The volatility swap struck at $\sqrt{K_{\mathrm{var}}}$ is not fair to its buyer whenever $\mathrm{Var}^{\mathbb{Q}}(\sigma_{\mathrm{real}}) > 0$: taking expectations in the payoff identity, the variance swap being fair, the buyer's expected payoff is $-N_{\mathrm{var}}\,\mathbb{E}^{\mathbb{Q}}[(\sigma_{\mathrm{real}} - \sqrt{K_{\mathrm{var}}})^2]$, and $\mathbb{E}^{\mathbb{Q}}[(\sigma_{\mathrm{real}} - \sqrt{K_{\mathrm{var}}})^2] = \mathrm{Var}^{\mathbb{Q}}(\sigma_{\mathrm{real}}) + \left(K_{\mathrm{vol}} - \sqrt{K_{\mathrm{var}}}\right)^2 > 0$, so for a positive notional that expected payoff is negative.

Example 22.41 (The convexity gap in numbers)**.** Let $K_{\mathrm{var}} = 0.04$, and let the risk-neutral standard deviation of σ_{real} be 4 volatility points, so that $\mathrm{Var}^{\mathbb{Q}}(\sigma_{\mathrm{real}}) = 0.04^2 = 0.0016$. By Proposition 22.40,

$$K_{\mathrm{vol}} = \sqrt{0.04 - 0.0016} = \sqrt{0.0384} = 19.60\%,$$

which is 0.40 volatility points below $\sqrt{K_{\mathrm{var}}} = 20\%$. A variance swap quoted "20" and a fair volatility swap on the same underlier and dates are struck 0.4 points apart.

Figure 48 shows the payoff identity of Proposition 22.40. The parabola of the variance swap touches the straight line of the vega-matched volatility swap at the strike and lies above it everywhere else.

Intuition. To second order in the spread of the realised variance about its mean, the gap has a closed form. The expansion of the square root at K_{var} (Theorem 7.31; it is the expansion of $(1+\cdot)^{1/2}$ in Proposition 7.34, applied to the relative deviation $(\sigma_{\mathrm{real}}^2 - K_{\mathrm{var}})/K_{\mathrm{var}}$) reads

$$\sigma_{\mathrm{real}} \approx \sqrt{K_{\mathrm{var}}} + \frac{\sigma_{\mathrm{real}}^2 - K_{\mathrm{var}}}{2\sqrt{K_{\mathrm{var}}}} - \frac{(\sigma_{\mathrm{real}}^2 - K_{\mathrm{var}})^2}{8\,K_{\mathrm{var}}^{3/2}}.$$

Take $\mathbb{Q}$-expectations: the linear term has mean zero, because $\mathbb{E}^{\mathbb{Q}}[\sigma_{\mathrm{real}}^2] = K_{\mathrm{var}}$, and

$$K_{\mathrm{vol}} \approx \sqrt{K_{\mathrm{var}}} - \frac{\mathrm{Var}^{\mathbb{Q}}\left(\sigma_{\mathrm{real}}^2\right)}{8\,K_{\mathrm{var}}^{3/2}},$$

the market's *convexity adjustment* (Problem P22.12). In Example 22.41, a standard deviation of 0.04 for σ_{real} is, to first order, a standard deviation of $2 \times 0.2 \times 0.04 = 0.016$ for σ_{real}^2, and the adjustment is $0.016^2/(8 \times 0.04^{3/2}) = 0.000256/0.064 = 0.004$: the 0.4 volatility points of the example.

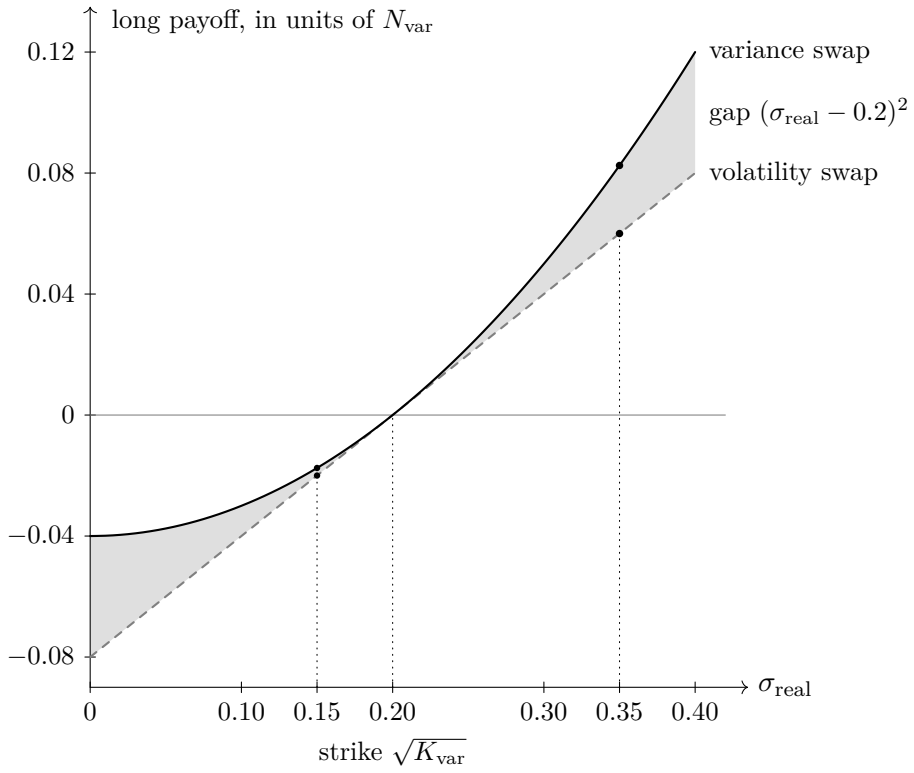


Figure 48: Long payoffs against the realised volatility σ_{real}, in units of the variance notional N_{var}, for swaps struck at 20 vol ($K_{\text{var}} = 0.04$). Solid: the variance swap, $\sigma_{\text{real}}^2 - 0.04$. Dashed: the vega-matched volatility swap, $0.4\,(\sigma_{\text{real}} - 0.2)$, tangent to the parabola at the strike $\sigma_{\text{real}} = \sqrt{K_{\text{var}}} = 0.2$. The shaded gap between them is $(\sigma_{\text{real}} - 0.2)^2$, the payoff identity of Proposition 22.40. The dots mark the outcomes $\sigma_{\text{real}} = 0.15$ and 0.35 of Example 22.33: the long payoffs are -0.0175 and -0.0200 at 0.15, and $+0.0825$ and $+0.0600$ at 0.35, with gaps 0.0025 and 0.0225.

Interview trap. "The fair volatility strike is the square root of the variance strike." It is lower whenever the realised volatility is uncertain under the pricing law, $\text{Var}^{\mathbb{Q}}(\sigma_{\text{real}}) > 0$ (Proposition 22.40), and the gap is the price of the volatility of volatility. The market's habit of quoting a variance strike "in vol", as $100\sqrt{K_{\text{var}}}$, does not make that number a volatility-swap strike.

Practice: problems P22.1–P22.12 in §27, solutions in §28.

23 Exotic Options and Numerical Methods

This section prices the standard exotic options, then builds the three numerical engines that are used when no closed form exists (simulation, variance reduction and finite differences), and ends with the repairs to the Black–Scholes–Merton model that exotic options force. Every price in it is proved, except where a Fact is named or, in §23.8, a pricing convention is stated for risks that cannot be traded. European payoffs and payoffs fixed on finitely many dates are priced by the replication results of §20 (Theorem 20.7 and Corollary 20.11); single barriers are priced by Theorem 23.16, proved in §23.2; and every identity that can be model-free is proved from the dominance lemma (Lemma 16.8) and the model-free results of §§16–22 that rest on it. A Fact remains only where a proof needs machinery outside this document (the discrete-monitoring correction, the forward equation of local volatility, Heston's formula), and each one says what its proof would need.

Symbol notices for this section.

1. The literature's $N(d_1)$ is $\Phi(d_1)$ here, and the standard normal density is φ.
2. The letter λ is the jump intensity, a Poisson rate, not the EWMA decay of §22; κ and δ are Merton's jump parameters only.
3. The letter n is the number of simulated paths (a sample size) and J the number of time steps or fixings; the reinvested holding of an asset with yield q is $\tilde{S}_t = S_t e^{-q(T-t)}$.
4. The letter H is the barrier and $\mathcal{H}$ the class of integrands of §19; two assets carry two subscripts, $S_{k,t}$ (asset first, time second).
5. In §23.7, i is the imaginary unit (i indexes time steps) and ω a frequency, not the GARCH constant of §22.

Partial derivatives are written $\partial_t V$, $\partial_S V$, $\partial_{SS} V$, never as subscripts, because V_t is the price at time t.

23.1 Exotics from Vanilla Building Blocks

Many exotic options are vanilla options in disguise: a digital is squeezed between two call spreads, which is also how a desk hedges it; a gap option is an asset-or-nothing claim minus a scaled digital; a forward-start call is a scaled at-the-money call, by homogeneity; a chooser is a call plus a put, by put–call parity at the choice date. Each decomposition is at once the price and the hedge, so the interview requests "hedge a digital" and "price a chooser" are answered by exactly the proofs below. The prices of the binary claims themselves are in §20.3 and are not repeated.

Setup. This is the first Setup of the section; its symbols keep their meaning in every later subsection, except where a later Setup or statement redeclares one locally and says so (for example the ratio volatility σ of §23.4, with its own d_1 and d_2, and the vector of normals Z of §23.6).

(a) The underlying has price S_t at time t, with $S = S_0$ today, and pays a continuous yield q (the yield, never the down-probability $1-p$). The continuously compounded risk-free rate r (not the ratio q/p of the Notation table) and the volatility $\sigma > 0$ are constants, and $b = r - q$ is the cost of carry. Times are in years: T is a maturity and τ a time to maturity.

(b) The risk-neutral probability and expectation of Definition 20.6 are written $\mathbb{Q}$ and $\mathbb{E}^{\mathbb{Q}}$. Under $\mathbb{Q}$, $dS = bS\,dt + \sigma S\,dW^{\mathbb{Q}}$, so that

$$S_T = S \exp\left((b - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z\right), \qquad Z \sim \mathcal{N}(0,1),$$

by Theorem 19.40 with $\mu = b$, and the forward price for delivery at T is $F = Se^{bT}$.

(c) The pricing rule: a claim that pays the amount X at T is worth $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[X]$ today. The rule is proved for European payoffs $X = f(S_T)$ (Theorem 20.7; with a yield, Corollary 20.41) and for payoffs fixed on finitely many dates (Corollary 20.11); for single barriers it is proved in §23.2 below (Theorem 23.16(v)). Fact 20.13 would be needed only to price a continuously monitored payoff other than a single barrier, and no proof of this section uses it.

(d) Today's prices of the European call and put with strike K and expiry T are $C(K,T)$ and $\mathcal{P}(K,T)$ (the put is calligraphic because P is probability). They are written $C(K)$ and $\mathcal{P}(K)$ when the expiry is fixed, and $C(S,K;\tau)$ and $\mathcal{P}(S,K;\tau)$ when the spot and the time to maturity matter; C_{t_1} and $\mathcal{P}_{t_1}$ are their values at a date t_1. The price of the cash-or-nothing digital, which pays $\mathbf{1}\{S_T > K\}$ at T (Definition 20.15), is $\mathrm{Dig}(K)$, and that of the asset-or-nothing claim, which pays $S_T\,\mathbf{1}\{S_T > K\}$, is $\mathrm{AoN}(K)$. As usual $x^+ = \max(x,0)$, ln is the natural logarithm, and d_1, d_2 are the quantities $[\ln(F/K) \pm \sigma^2 T/2]/(\sigma\sqrt{T})$ of Theorem 20.22.

(e) The date t_1 is an intermediate date, $0 < t_1 < T$; the forward-start ratio is $\eta > 0$, and $c_\eta = C(1,\eta;T-t_1)$.

A statement called *model-free* assumes only the standing assumptions of §16.2 (a frictionless market, one rate r for borrowing and lending, short sales, no arbitrage) and that the claims it

names trade; its proof uses Lemma 16.8 and the model-free results proved from it in §§16–22 (put–call parity, the reinvested holding, the calendar condition), and no model. A statement made *under BSM* assumes (A1) and (A3)–(A6) of §20.1, with (A2) replaced by the continuous yield q as in §20.7; then $C(S,K;\tau)$ and $\mathcal{P}(S,K;\tau)$ are the formulas of Corollary 20.41(i).

Definition 23.1 (Gap, forward-start, chooser and compound options)**.** Let $0 < t_1 < T$.

(i) A *gap call* with trigger K_1 and payment strike K_2 pays $(S_T - K_2)\,\mathbf{1}\{S_T > K_1\}$ at T; its price is written $\mathrm{Gap}(K_1, K_2)$, in the style of Dig and AoN. For $K_1 = K_2$ it is the vanilla call. The *gap put* pays $(K_2 - S_T)\,\mathbf{1}\{S_T < K_1\}$.

(ii) A *forward-start call* with ratio $\eta > 0$ gives its holder, at t_1, a European call that expires at T and is struck at ηS_{t_1}; it therefore pays $(S_T - \eta S_{t_1})^+$ at T. For $\eta = 1$ the call is struck at the money on the date t_1. The *forward-start put* pays $(\eta S_{t_1} - S_T)^+$.

(iii) A *simple chooser* with strike K, choice date t_1 and expiry T lets its holder declare, at t_1, that the contract is a European call or a European put, in either case with strike K and expiry T. The holder declares the put exactly when $\mathcal{P}_{t_1} > C_{t_1}$, so the chooser is worth $\max(C_{t_1}, \mathcal{P}_{t_1})$ at t_1.

(iv) A *compound option* is an option that expires at t_1, with strike K', on a vanilla option with strike K and expiry T. The *call-on-call* is the right to buy the call at t_1 for K', the *put-on-call* the right to sell it for K'; the *call-on-put* and the *put-on-put* are the same rights on the put. (The compound strike is primed so that K_1 and K_2 keep one meaning.)

Proposition 23.2 (Call-spread bracket of a digital)**.** *Fix the expiry T and a strike K, and let $0 < \varepsilon < K$. The following is model-free: if the calls struck at $K - \varepsilon$, K and $K + \varepsilon$ and the digital struck at K trade, then*

$$\frac{C(K-\varepsilon) - C(K)}{\varepsilon} \;\geq\; \mathrm{Dig}(K) \;\geq\; \frac{C(K) - C(K+\varepsilon)}{\varepsilon}. \tag{23.1}$$

If calls trade at every strike near K and the call curve $K \mapsto C(K)$ is differentiable at K, both bounds tend to $-\partial_K C(K)$ as $\varepsilon \to 0$, so that $\mathrm{Dig}(K) = -\partial_K C(K)$.

Proof. We compare payoffs state by state and apply the dominance lemma. Call the claim that pays $1/\varepsilon$ times the payoff of the call spread from $K - \varepsilon$ to K the *upper spread*, and the same for the spread from K to $K + \varepsilon$ the *lower spread*. As functions of $x = S_T$ their payoffs are piecewise linear (Proposition 17.35):

$$\text{upper spread:}\quad \frac{(x-K+\varepsilon)^+ - (x-K)^+}{\varepsilon} = \begin{cases} 0, & x \leq K-\varepsilon, \\ (x-K+\varepsilon)/\varepsilon, & K-\varepsilon < x < K, \\ 1, & x \geq K, \end{cases}$$

$$\text{lower spread:}\quad \frac{(x-K)^+ - (x-K-\varepsilon)^+}{\varepsilon} = \begin{cases} 0, & x \leq K, \\ (x-K)/\varepsilon, & K < x < K+\varepsilon, \\ 1, & x \geq K+\varepsilon. \end{cases}$$

The digital pays $\mathbf{1}\{x > K\}$. Compare the three payoffs on three ranges. For $x \leq K$ the digital and the lower spread pay 0 and the upper spread pays a nonnegative amount. For $K < x < K + \varepsilon$ the digital and the upper spread pay 1 and the lower spread pays $(x - K)/\varepsilon < 1$. For $x \geq K + \varepsilon$ all three pay 1. Hence, in every state,

$$\text{upper spread} \;\geq\; \mathbf{1}\{S_T > K\} \;\geq\; \text{lower spread}.$$

Each of the three claims is a static position in traded claims, so Lemma 16.8(i) turns the two payoff inequalities into the two price inequalities of (23.1); the price of the upper spread is $(C(K-\varepsilon) - C(K))/\varepsilon$ and that of the lower spread $(C(K) - C(K+\varepsilon))/\varepsilon$, because prices add.

For the limit, suppose that C is differentiable at K. Then

$$\frac{C(K-\varepsilon)-C(K)}{\varepsilon} = -\,\frac{C(K-\varepsilon)-C(K)}{-\varepsilon} \;\to\; -\partial_K C(K),$$
$$\frac{C(K)-C(K+\varepsilon)}{\varepsilon} = -\,\frac{C(K+\varepsilon)-C(K)}{\varepsilon} \;\to\; -\partial_K C(K)$$

as $\varepsilon \to 0$, since both are difference quotients of C at K with the sign changed. The number $\mathrm{Dig}(K)$ does not depend on ε and lies between them for every small ε, so it equals their common limit. □

The identity $\mathrm{Dig}(K) = -\partial_K C(K)$ is the model-free content of (22.2) in Theorem 22.7, which obtains it from a pricing density. The digital itself lies in the band (17.4) of §17, $0 \le \mathrm{Dig}(K) \le e^{-rT}$. The dominance argument given there does not use the sign of r, so the band holds for every r. The payoffs of the two scaled spreads lie between 0 and 1 as well, so the same argument puts the whole bracket inside that band. Figure 49 shows the three payoffs.

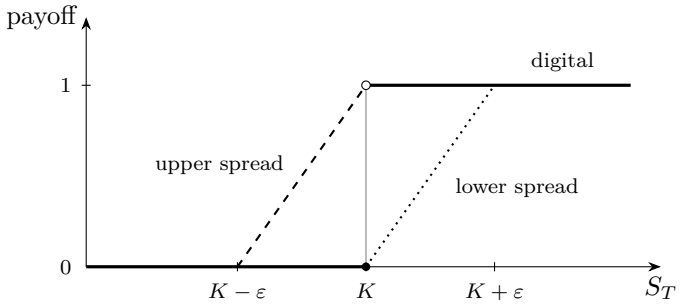


Figure 49: Payoffs near the strike. The digital is the unit step at K (thick; it pays 0 at $S_T = K$). The upper spread rises linearly from 0 at $K-\varepsilon$ to 1 at K (dashed) and the lower spread from 0 at K to 1 at $K+\varepsilon$ (dotted); away from its ramp each spread coincides with the step. The two ramps sandwich the step state by state, which is Proposition 23.2.

Example 23.3 (Bracketing a digital)**.** Let $S = K = 100$, $r = 2\%$, $q = 0$, $\sigma = 20\%$ and $T = 0.5$, under BSM. Here $r = \sigma^2/2$, so the risk-neutral log-drift vanishes, $d_2 = 0$ at $K = 100$, and by Theorem 20.17

$$\mathrm{Dig}(100) = e^{-0.01}\,\Phi(0) = \tfrac{1}{2}e^{-0.01} = 0.4950.$$

The formula of Theorem 20.22 gives $C(95) = 8.9470$, $C(99) = 6.6297$, $C(100) = 6.1207$, $C(101) = 5.6395$ and $C(105) = 3.9856$. The brackets (23.1) are therefore

$$\varepsilon = 5:\ [0.4270,\ 0.5653], \qquad \varepsilon = 1:\ [0.4811,\ 0.5090], \qquad \varepsilon = 0.1:\ [0.4936,\ 0.4964],$$

each containing 0.4950 and closing on it as ε shrinks. A desk that is short a digital paying 1,000,000 buys 1,000,000 of the 99/100 call spreads at 0.5090 each, a cost of 509,030 against the fair value 495,025 of the digital: 1.4 cents per unit of payout is the price of never being short at expiry.

(Problem P23.1 works this computation at different numbers.)

Trading application. The upper spread pays at least as much as the digital in every state: it *super-replicates*, and it is the hedge of a *sold* digital, chosen so that the seller is never short at expiry. The lower spread *sub-replicates* and hedges a bought digital. Their two prices are a natural offer and bid for the digital, and the width of the bracket, which shrinks with ε, is the cost of a hedge that needs no trading before expiry.

Interview trap. "A digital is worth $e^{-rT}\Phi(d_2)$ at the implied volatility of its strike." Only where the skew vanishes at that strike. The model-free price is $-\partial_K C(K)$ (Proposition 23.2); the term the formula misses is the vega-times-skew correction of Corollary 22.10, which is 8 cents on a 46-cent digital in Example 22.11.

Interview trap. Hedging a sold digital with a spread centred on the strike, or with the lower spread, leaves the desk short if S_T lands just above K: the digital then pays 1 and the hedge pays less. The seller holds the spread from $K-\varepsilon$ to K.

Remark 23.4 (Pin risk and why digitals are hedged statically). A digital with time τ to expiry has the delta

$$\frac{e^{-r\tau}\,\varphi(d_2)}{S\sigma\sqrt{\tau}}$$

and a gamma that changes sign near the strike; both are derived in §21.2. Their size on a book is what makes a dynamic hedge impractical. Take a digital paying 1,000,000 with $S=K=100$, $\sigma=20\%$ and $r=0$. Its delta hedge is about 316,600 units of the underlying (31.7 million of stock) one trading day before expiry, against 69,100 units one month before and 39,800 units three months before; and on that last day the digital itself, per unit of payout, is worth 0.211 at $S=99$ and 0.783 at $S=101$. A one-point move in either direction on that day cuts the hedge from 316,600 to about 231,000 units, a trade of more than eight million of stock, and a move back reverses it: a position of this kind cannot be rebalanced smoothly. Holding the upper spread of Proposition 23.2 instead leaves a residual payoff that is never negative, so nothing has to be traded into expiry; the width of the bracket is the price of that safety.

Proposition 23.5 (Price of a gap option)**.** *Model-free: if the gap call, the call struck at K_1 and the digital struck at K_1 trade, then*

$$\mathrm{Gap}(K_1,K_2)=C(K_1)-(K_2-K_1)\,\mathrm{Dig}(K_1), \tag{23.2}$$

and, when $\mathrm{Dig}(K_1)>0$ *and the asset-or-nothing claim trades, the gap call has a negative price exactly when $K_2>K_1+C(K_1)/\mathrm{Dig}(K_1)=\mathrm{AoN}(K_1)/\mathrm{Dig}(K_1)$. Under BSM,*

$$\mathrm{Gap}(K_1,K_2)=Se^{-qT}\,\Phi(d_1)-K_2e^{-rT}\,\Phi(d_2),$$

with d_1 and d_2 evaluated at the trigger *K_1, not at the payment strike.*

Proof. We decompose the payoff. For every $x\ge 0$,

$$(x-K_2)\,\mathbf{1}\{x>K_1\}=(x-K_1)^+-(K_2-K_1)\,\mathbf{1}\{x>K_1\}:$$

both sides vanish for $x\le K_1$, and for $x>K_1$ the right side is $(x-K_1)-(K_2-K_1)=x-K_2$. So the gap call and the portfolio of one call struck at K_1 and $-(K_2-K_1)$ digitals struck at K_1 pay the same amount in every state, and Lemma 16.8(iii) gives (23.2). In the same way $(x-K_1)^+=x\,\mathbf{1}\{x>K_1\}-K_1\mathbf{1}\{x>K_1\}$ gives $C(K_1)=\mathrm{AoN}(K_1)-K_1\mathrm{Dig}(K_1)$, so that

$$\mathrm{Gap}(K_1,K_2)=\mathrm{AoN}(K_1)-K_2\,\mathrm{Dig}(K_1).$$

When $\mathrm{Dig}(K_1)>0$, this is negative exactly when K_2 exceeds the ratio $\mathrm{AoN}(K_1)/\mathrm{Dig}(K_1)$, and that ratio equals $K_1+C(K_1)/\mathrm{Dig}(K_1)$ by the identity for $C(K_1)$. Under BSM the forward-lognormal model of Definition 20.16 holds with $F=Se^{(r-q)T}$ (Corollary 20.41), so Proposition 20.19 gives $\mathrm{AoN}(K_1)=e^{-rT}F\,\Phi(d_1)=Se^{-qT}\Phi(d_1)$ and Theorem 20.17 gives $\mathrm{Dig}(K_1)=e^{-rT}\Phi(d_2)>0$, both with strike K_1. □

Example 23.6 (Two gap calls)**.** In the market of Example 23.3 ($S=100$, $r=2\%$, $q=0$, $\sigma=20\%$, $T=0.5$), take the trigger $K_1=100$ and the payment strike $K_2=105$. At the trigger, $d_2=0$ and $d_1=\sigma\sqrt{T}=0.1414$, so $\mathrm{AoN}(100)=100\,\Phi(0.1414)=55.6231$ and $\mathrm{Dig}(100)=0.495025$, and

$$\mathrm{Gap}(100,105)=55.6231-105\times 0.495025=55.6231-51.9776=3.6455,$$

which is also $C(100)-5\,\mathrm{Dig}(100)=6.12065-2.47512=3.64553$, as (23.2) requires. This is below the vanilla $C(105)=3.986$, because the gap call pays a negative amount when S_T ends in $(100,105)$. Its price turns negative once $K_2>55.6231/0.495025=112.36$. In the other direction, with trigger 110 and payment strike 100 the holder receives $S_T-100\ge 10$ whenever $S_T>110$, and (23.2) gives $C(110)+10\,\mathrm{Dig}(110)=2.473+2.477=4.950$.

(Problem P23.2 works this computation at different numbers.)

Interview trap. "An option cannot have a negative price." A vanilla cannot, because its payoff is never negative. A gap call with K_2 well above K_1 obliges its holder to pay $K_2 - S_T$ when $K_1 < S_T < K_2$: it is an obligation as well as a right, and beyond $K_2 = \mathrm{AoN}(K_1)/\mathrm{Dig}(K_1)$ the obligation outweighs the right.

Lemma 23.7 (Homogeneity of the BSM price)**.** *Let r and q be real numbers and let $\sigma > 0$ and $\tau > 0$, all four fixed, and let $C(S,K;\tau)$ and $\mathcal{P}(S,K;\tau)$ be the BSM prices of Corollary 20.41(i). For every constant $c > 0$ and all $S, K > 0$,*

$$C(cS, cK; \tau) = c\,C(S,K;\tau), \qquad \mathcal{P}(cS, cK; \tau) = c\,\mathcal{P}(S,K;\tau).$$

Proof. We read the property off the formula. In Corollary 20.41(i),

$$C(S,K;\tau) = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_{1,2} = \frac{\ln(S/K) + (r - q \pm \sigma^2/2)\tau}{\sigma\sqrt{\tau}},$$

and d_1, d_2 depend on S and K only through $\ln(S/K)$, which is unchanged when both are multiplied by c. So replacing (S,K) by (cS,cK) leaves $\Phi(d_1)$ and $\Phi(d_2)$ unchanged and multiplies each of the two terms $Se^{-q\tau}\Phi(d_1)$ and $Ke^{-r\tau}\Phi(d_2)$ by c. The put formula $\mathcal{P}(S,K;\tau) = Ke^{-r\tau}\Phi(-d_2) - Se^{-q\tau}\Phi(-d_1)$ is treated in the same way. □

Homogeneity says that the option does not care about the currency unit. The proof read it off the formula, but the formula is not needed: in any model in which the call with life τ is priced as $e^{-r\tau}\,\mathbb{E}^{\mathbb{Q}}(S_\tau - K)^+$ and $S_\tau = SY$ for a random variable Y whose risk-neutral law does not depend on S,

$$C(cS, cK; \tau) = e^{-r\tau}\,\mathbb{E}^{\mathbb{Q}}(cSY - cK)^+ = c\,e^{-r\tau}\,\mathbb{E}^{\mathbb{Q}}(SY - K)^+ = c\,C(S,K;\tau).$$

Intuition. Under local volatility (§23.8) the volatility depends on the price level, so the law of the ratio S_τ/S changes with S and the scaling argument above breaks down. This is why forward-start options, which are priced by homogeneity, are sensitive to the model.

Proposition 23.8 (Forward-start option (Rubinstein, 1991))**.** *Under BSM, the forward-start call of Definition 23.1(ii) has the arbitrage-free price*

$$e^{-qt_1}\,C(S, \eta S; T - t_1) = e^{-qt_1} c_\eta S$$

today: e^{-qt_1} times today's call struck at ηS with the residual life $T - t_1$. With $q = 0$ it is exactly that call, whatever t_1. It is replicated by buying $e^{-qt_1}c_\eta$ units of the asset today and reinvesting every dividend in the asset, which gives $e^{-q(t_1 - t)}c_\eta$ units at each time $t \le t_1$, and from t_1 on by the delta strategy of the call struck at ηS_{t_1}. The forward-start put is worth $e^{-qt_1}\,\mathcal{P}(S, \eta S; T - t_1)$, by the same argument with $\mathcal{P}$ in place of C.

Sketch of the proof. We replicate in two stages joined at t_1: a static holding, then a delta hedge. At t_1 the holder owns a call with spot S_{t_1} and strike ηS_{t_1}, and by homogeneity (Lemma 23.7) the capital that replicates it is linear in the price:

$$C(S_{t_1}, \eta S_{t_1}; T - t_1) = c_\eta S_{t_1}.$$

A static holding supplies it (Step 2 of the complete proof): $e^{-qt_1}c_\eta$ units of the asset bought today, with the dividends reinvested, are worth $e^{-q(t_1 - t)}c_\eta S_t$ at each $t \le t_1$. From t_1 on, the delta strategy of Theorem 20.7(v), run in the reinvested asset, turns that capital into the payoff; BSM enters here and in the homogeneity. The two stages form one self-financing strategy, and (A6) makes its cost $e^{-qt_1}c_\eta S$ the price. The appendix checks the growth condition of that theorem. The complete proof is in Appendix I.1. □

The same price follows from Corollary 20.11 with the two fixing dates t_1 and T, applied to the reinvested asset (the reduction is written out in Step 1 of the proof of Proposition 23.28, in Appendix I.5). The static stage of the replication (Step 2 of the proof of Proposition 23.8) also gives the Greeks before the strike is set: at a time $t < t_1$ the replicating strategy is worth $e^{-q(t_1-t)}c_\eta S_t$, so the delta is $e^{-q(t_1-t)}c_\eta$ (today $e^{-qt_1}c_\eta$) and the gamma is zero. The risk of the option before t_1 is not to the price but to the volatility that will rule between t_1 and T, through c_η.

Example 23.9 (A forward-start call and put)**.** Let $S = 100$, $r = 5\%$, $q = 2\%$, $\sigma = 20\%$, $t_1 = 0.5$ and $T = 1.5$, so that the option lives for one year from the date on which its strike is set. The one-year at-the-money call is $C(100, 100; 1) = 9.227$ (with $d_1 = 0.25$ and $d_2 = 0.05$), so $c_\eta = 0.09227$ at $\eta = 1$, and by Proposition 23.8 the forward-start call is worth

$$9.227\, e^{-0.02\times 0.5} = 9.227 \times 0.99005 = 9.135.$$

The replicating holding today is $e^{-0.01}c_\eta = 0.0914$ units of the asset, growing with the reinvested dividends to $c_\eta = 0.0923$ units at t_1; at each date before t_1 that holding is the delta, and the gamma is zero. Homogeneity can be checked on the formula: $C(200, 200; 1) = 18.454 = 2\times 9.227$. A forward-start put struck at 110% of S_{t_1} ($\eta = 1.1$) is worth $\mathcal{P}(100, 110; 1)\, e^{-0.01} = 11.80395 \times 0.99005 = 11.6865$. With $q = 0$ the call would be worth today's one-year at-the-money call, 10.451, for every t_1 (100 times the 0.104506 of Example 20.12).

(Problem P23.3 works this computation at different numbers.)

Interview trap. A forward-start option "is cheaper because it starts later", or "has no Greeks before its strike is set". With $q = 0$ it is worth today's call with the same moneyness and the same residual life, whatever t_1. Before t_1 its value is linear in S, with delta $e^{-q(t_1-t)}c_\eta$ at time t and gamma zero; but its vega is to the *forward* volatility between t_1 and T, which makes it a forward-smile product, one that a local-volatility model misprices (§23.8). These answers concern the contract of Definition 23.1(ii), whose size is one unit of the asset. The cliquet leg of Example 20.12 pays $(S_T/S_{t_1} - 1)^+$ on a fixed cash notional. Its value at t_1, $C(1, 1; T - t_1)$, does not depend on S_{t_1}, so with $q = 0$ it is worth $e^{-rt_1}C(1, 1; T - t_1)$ per unit of notional (0.101926 against 0.104506 in that example). That contract is cheaper by the discount over the wait, and before t_1 its replication holds only the bank, with delta and gamma zero. The two contracts differ by the factor S_{t_1}, which the static holding of Proposition 23.8 pays for.

Proposition 23.10 (Simple chooser by put–call parity (Rubinstein, 1991))**.** *Let $\tau = T - t_1$, and let $\tilde{K} = Ke^{-(r-q)\tau}$ be the* shifted strike*, a companion of K (it is not the compound strike K' of Definition 23.1(iv)). The following is model-free, for an asset with a continuous yield q: if the chooser of Definition 23.1(iii), the call and the put with strike K and expiry T, and the call and the put with strike $\tilde{K}$ and expiry t_1 trade, then the chooser is worth*

$$C(K, T) + e^{-q\tau}\,\mathcal{P}(\tilde{K}, t_1) \tag{23.3}$$

today: one call to T plus $e^{-q\tau}$ puts that expire at t_1 with the shifted strike $\tilde{K}$. Moreover

$$\max\big(C(K,T),\, \mathcal{P}(K,T)\big) \;\le\; \textit{chooser} \;\le\; C(K,T) + \mathcal{P}(K,T).$$

A model is needed only to price the pieces.

Sketch of the proof. We apply put–call parity at the choice date and make it static. At t_1 parity is $\mathcal{P}_{t_1} - C_{t_1} = Ke^{-r\tau} - S_{t_1}e^{-q\tau} = e^{-q\tau}(\tilde{K} - S_{t_1})$, so the put is declared exactly when $S_{t_1} < \tilde{K}$ and pays the call payoff plus $K - S_T$. Buy the call to T and $e^{-q\tau}$ puts struck at $\tilde{K}$ expiring at t_1. When $S_{t_1} < \tilde{K}$ the puts pay exactly the cost of depositing $Ke^{-r\tau}$ and shorting $e^{-q\tau}$ units of the asset, which with the yield q reinvested owe one unit at T: together $K - S_T$. Equal payoffs give

equal prices (Lemma 16.8(iii)). For the bounds: the chooser pays at most the sum of both payoffs, option prices are nonnegative, and parity today at both expiries gives

$$C(K,T) + e^{-q\tau}\,\mathcal{P}(\tilde{K},t_1) = \mathcal{P}(K,T) + e^{-q\tau}\,C(\tilde{K},t_1) \geq \mathcal{P}(K,T).$$

The complete proof is in Appendix I.2. □

The last display is a second decomposition of the chooser, into a put to T plus $e^{-q\tau}$ calls expiring at t_1. *(Problem P23.2 derives this alternative decomposition by replication.)*

Example 23.11 (A six-month choice on a one-year option)**.** Let $S = K = 80$, $r = 3\%$, $q = 1\%$, $\sigma = 25\%$, $T = 1$ and $t_1 = 0.5$, with the pieces priced under BSM. The call to T is $C(80,1) = 8.610$. The shifted strike is $\tilde{K} = 80\,e^{-0.02\times 0.5} = 79.204$, the put with that strike and six months to run is worth 4.798, and its weight is $e^{-0.01\times 0.5} = 0.99501$, a contribution of 4.774. By (23.3) the chooser is worth

$$8.610 + 4.774 = 13.384,$$

between $\max(C,\mathcal{P}) = 8.610$ and the straddle $C + \mathcal{P} = 8.6099 + 7.0416 = 15.6515$. Moving the choice date moves the price between these bounds: $t_1 = 0.25$ gives 11.784 and $t_1 = 0.75$ gives 14.615. In the limit $t_1 \to 0$ the short-dated put tends to its intrinsic value and the chooser to $\max(C,\mathcal{P})$; as $t_1 \to T$ it tends to the straddle.

(Problem P23.2 works this computation at different numbers.)

Interview trap. "The chooser is a straddle." The holder receives the *better* of the two options at t_1, not both: $\max(C_{t_1},\mathcal{P}_{t_1})$, not $C_{t_1} + \mathcal{P}_{t_1}$. In Example 23.11 the chooser costs 13.38 against 15.65 for the straddle, and the straddle is reached only as $t_1 \to T$, when the choice is made knowing the outcome.

Remark 23.12 (Compound options)*.* The four compound options of Definition 23.1(iv) obey a parity of their own, and it is model-free. We compare the two sides through their values at t_1, so that nothing depends on what is done when exercise is a matter of indifference. Hold a call-on-call and be short a put-on-call, with the same K' and t_1, and assume only that each compound option is exercised when it is strictly in the money (the call-on-call when $C_{t_1} > K'$, the put-on-call when $C_{t_1} < K'$) and not when it is strictly out of the money; at $C_{t_1} = K'$ either choice is allowed to each holder. When $C_{t_1} > K'$ the position exercises its call-on-call, and when $C_{t_1} < K'$ the owner of the put-on-call exercises and the position, as its writer, takes the call; either way it ends t_1 with one call with strike K and expiry T, bought for K'. When $C_{t_1} = K'$ it ends with k such calls, $k \in \{0,1,2\}$, bought for K' each. Valuing the calls at their market price C_{t_1}, the position is therefore worth, in every state,

$$k\,(C_{t_1} - K') = C_{t_1} - K' \qquad \text{at } t_1,$$

with $k = 1$ off the event $\{C_{t_1} = K'\}$ and both sides zero on it. Sell the calls at t_1 and keep the proceeds in the cash account: the position then pays $(C_{t_1} - K')\,e^{r(T-t_1)}$ at T. The portfolio that is long one call with strike K and expiry T and has borrowed $K'e^{-rt_1}$ today, and that at t_1 sells the call and repays K', pays the same amount at T in every state. By Lemma 16.8(iii),

$$\text{call-on-call} - \text{put-on-call} = C(K,T) - K'e^{-rt_1},$$

and likewise call-on-put − put-on-put $= \mathcal{P}(K,T) - K'e^{-rt_1}$. Under BSM each compound option has a closed form in the bivariate normal distribution function (Geske, 1979), which this document does not derive.

Remark 23.13 (Packages and the deferred premium)*.* Call a claim a *package* when its payoff is that of a position in the underlying, zero-coupon bonds, forwards and European options that is set up today and never traded again. Every continuous piecewise-linear European payoff is a

package (Proposition 17.35), every twice continuously differentiable one is a continuum of them (Lemma 22.34), and the digital is the degenerate limit of Proposition 23.2. The date on which the premium changes hands has no bearing on value: by Lemma 16.8(iv) (a sure amount c paid at T is worth ce^{-rT}), an amount V due today and the amount Ve^{rT} due at T are the same obligation, so a dealer can always quote a contract with nothing payable upfront. The contracts that are exotic in substance are those whose payoff depends on the path of the underlying or on a choice, and they occupy the rest of the section, beginning with the barrier.

23.2 Barrier Options and the Reflection Principle

Barrier options are the most traded exotics and the richest source of structural interview questions: what a knock-in and a knock-out add up to, which barriers make an option worthless, why a one-touch costs about two digitals, and which way discrete monitoring moves the price. The model-free identities come from splitting the paths into those that reach the barrier and those that do not; the prices come from one idea, the method of images (Theorem 23.16), of which the closed forms and the one-touch identities are corollaries. The one-touch on an arithmetic underlying is the reflection principle itself (Theorem 19.63, which also proves every claim of Fact 10.6).

Setup. The symbols of §23.1 are kept. In addition:

(a) The price path $t \mapsto S_t$ is continuous on $[0,T]$. In Theorem 23.16 the barrier is written on a price S_t that is either an asset with yield q, with cost of carry $b = r - q$, or a forward or futures price for delivery at a date $T_F \geq T$, with $b = 0$. In both cases the real-world path is $S_t = S_0 \exp(gt + \sigma W_t)$, with a constant g and a standard Brownian motion W (Definition 19.27), and the risk-neutral path is that of Definition 20.6 (for a futures price, Proposition 20.51(i)), in which g is replaced by $b - \sigma^2/2$.

(b) The running extremes of the path are $S_T^{\min} = \min_{0\leq t\leq T} S_t$ and $S_T^{\max} = \max_{0\leq t\leq T} S_t$; the barrier is $H > 0$ and the strike K. Monitoring is continuous, the date T included, unless stated otherwise; *discrete monitoring* observes the path only on the dates $k\,\Delta t$, $k = 1, 2, \ldots, T/\Delta t$, where the number $T/\Delta t$ of observation dates is an integer, so that T is itself an observation date.

(c) The first time at which the path reaches the barrier is t_H: $t_H = \inf\{t \in [0,T] : S_t \leq H\}$ for a down barrier and $t_H = \inf\{t \in [0,T] : S_t \geq H\}$ for an up barrier, with $t_H = +\infty$ when the barrier is not reached by T. It is written t_H because τ is the time to maturity.

(d) The European price function of the claim that pays $f(S_T)$ at T is $V(S,t)$ (Theorem 20.7 with Corollary 20.41, or Theorem 20.52), and $V^{\text{out}}(S,t)$, $V^{\text{in}}(S,t)$ are the price functions of the knock-out and knock-in versions of that claim. The exponent $1 - 2b/\sigma^2$ that appears in them is always written out; it receives no letter, because λ is the jump intensity of §23.8 and α_t the drift of an Itô process.

(e) The hedge instrument is $\tilde{S}_t = S_t e^{-(r-b)(T-t)}$. For an asset with yield q it is the reinvested holding $S_t e^{-q(T-t)}$ of the notices. For a futures price F_t it is $e^{-r(T-t)}F_t$, that is, $e^{r(T_F-T)}$ units of the prepaid forward $\tilde{F}_t = e^{-r(T_F-t)}F_t$ of §20.8, a position in forward contracts and bonds; when an asset with yield q stands behind the futures, this is $e^{(r-q)(T_F-T)}$ units of the reinvested holding of that asset. In every case $\tilde{S}$ is an instrument of Definition 20.1 that pays no income.

(f) The price of the one-touch of Definition 23.14 is written $\text{OT}(H)$, in the style of Dig and AoN.

(g) The letter ℓ denotes the boundary that a Brownian path must reach, as in Definition 19.22(c): the straight line $\ell(t) = (\ln(H/S_0) - gt)/\sigma$ in Theorem 23.16, and the level $\ell = (H - S_0)/\sigma_{\text{N}}$, as in Theorem 19.63, in Proposition 23.19(i).

(h) Proposition 23.19(i) concerns an *arithmetic* underlying with absolute (normal) volatility $\sigma_{\text{N}} > 0$, as in the normal model of Remark 20.31: its real-world path is $S_t = S_0 + \mu_{\text{N}} t + \sigma_{\text{N}} W_t$,

with a constant real-world drift μ_{N}, and at a zero interest rate its risk-neutral path is $S_0 + \sigma_{\mathrm{N}} W_t$. In Proposition 23.19(ii) the forward or futures price is written F_t, with $F = F_0$.

Definition 23.14 (Barrier options, rebates, one-touch and monitoring)**.** A *down* barrier is a level $H < S$ and an *up* barrier a level $H > S$. A *knock-out* option pays the vanilla payoff at T only on the paths that stay clear of the barrier until T; a *knock-in* option pays it only on the paths that reach the barrier by T. Down or up, out or in, call or put: there are eight types, each paid at T, for example

$$\begin{aligned} \text{down-and-out call:} &\quad (S_T - K)^+ \mathbf{1}\{S_T^{\min} > H\}, \\ \text{down-and-in call:} &\quad (S_T - K)^+ \mathbf{1}\{S_T^{\min} \le H\}, \\ \text{up-and-out call:} &\quad (S_T - K)^+ \mathbf{1}\{S_T^{\max} < H\}, \\ \text{up-and-in put:} &\quad (K - S_T)^+ \mathbf{1}\{S_T^{\max} \ge H\}. \end{aligned}$$

A *rebate* is a fixed amount paid to the holder when a knock-out option is knocked out, or at T when a knock-in option was never activated. The *one-touch* pays 1 at T if the barrier is reached by T, that is $\mathbf{1}\{S_T^{\max} \ge H\}$ for an up barrier and $\mathbf{1}\{S_T^{\min} \le H\}$ for a down barrier; its price is written $\mathrm{OT}(H)$. The *no-touch* pays 1 at T on the complementary event. Under discrete monitoring, $S_T^{\min}$ and $S_T^{\max}$ are replaced by the minimum and the maximum of the path over the observation dates.

Proposition 23.15 (In–out parity and degenerate barriers)**.** *The following statements are model-free.*

(i) For European barrier options without rebates that share their type (call or put), strike K, barrier H, expiry T and monitoring rule, the knock-in and the knock-out are together worth the vanilla option, whenever the three claims trade.

(ii) For a down barrier with $K \le H < S$, the down-and-out put is worthless and the down-and-in put is worth the vanilla put.

(iii) For an up barrier with $S < H \le K$, the up-and-out call is worthless and the up-and-in call is worth the vanilla call.

Parts (ii) and (iii) also hold under discrete monitoring, because T is an observation date (Setup (b)).

Proof. We split the paths by whether they reach the barrier and apply the law of one price. (i) On every path exactly one of the events $\{S_T^{\min} \le H\}$ and $\{S_T^{\min} > H\}$ occurs, so $\mathbf{1}\{S_T^{\min} \le H\} + \mathbf{1}\{S_T^{\min} > H\} = 1$, and likewise $\mathbf{1}\{S_T^{\max} \ge H\} + \mathbf{1}\{S_T^{\max} < H\} = 1$; the same holds for the extremes over a set of observation dates. Multiplying by the vanilla payoff shows that the payoffs of the knock-in and the knock-out add up to the vanilla payoff in every state, and Lemma 16.8(iii) gives equal prices.

(ii) The down-and-out put pays $(K - S_T)^+ \mathbf{1}\{S_T^{\min} > H\}$. The minimum over $[0, T]$ is at most the value at T, so on the event $\{S_T^{\min} > H\}$ we have $S_T \ge S_T^{\min} > H \ge K$ and the put pays nothing; off that event the indicator is zero. The claim pays zero in every state, so it is worth zero (Lemma 16.8(iii), against the empty portfolio), and (i) then gives the knock-in. In terms of paths: a path that ends below $K \le H$ started above H, so by the intermediate value theorem it crossed H on the way and was knocked out. Under discrete monitoring the observed minimum is still at most S_T, because T is an observation date, and the argument is unchanged.

(iii) is the mirror image: the up-and-out call pays $(S_T - K)^+ \mathbf{1}\{S_T^{\max} < H\}$, and on $\{S_T^{\max} < H\}$ we have $S_T \le S_T^{\max} < H \le K$, so the call pays nothing. □

Example 23.18 below puts numbers on part (i).

Interview trap. "A down-and-in call with the barrier above the strike is just the vanilla call." False: a path that rises from S without ever coming down to H pays the vanilla but not the

knock-in. The degenerate pairs are those of Proposition 23.15: the down-and-out and down-and-in *puts* when $K \le H$, and the up-and-out and up-and-in *calls* when $H \le K$. In each of them the option can finish in the money only by crossing the barrier.

Interview trap. "In–out parity holds for American barrier options too." It does not: the holder of the knock-in can exercise only after the barrier has been touched and the holder of the knock-out only before, so the two exercise policies are tied to the barrier and the two American values need not add up to the American vanilla. Rebates break even the European identity, unless they are added back.

The prices rest on the *method of images*: the pricing equation keeps its form when the spot is reflected across the barrier in log space and the result is multiplied by a power of S/H, so the vanilla price minus its image vanishes on the barrier, and the delta hedge of that difference, stopped at the first touch, replicates the knock-out. The proof uses the stopped clause of Itô's lemma (Theorem 19.30(iii)); in this subsection only the correction for discrete monitoring (Fact 23.21) stays a Fact.

Theorem 23.16 (Barrier options by the method of images (Merton, 1973; Reiner and Rubinstein, 1991))**.** *Let the barrier be written on a price S_t of one of two kinds:*

- *an asset with a continuous yield q, under (A1) and (A3)–(A6) of §20.1, with (A2) replaced by that yield as in §20.7; its cost of carry is $b = r - q$;*
- *a futures price for delivery at $T_F \ge T$, in the model of the Setup of §20.8: (A1) holds for the futures price itself, (A3)–(A6) hold, and the prepaid forward is traded. Here $b = 0$, and no traded spot asset is needed.*

Let $H > 0$ and $S_0 > H$, and let f belong to the class $\mathcal{G}$ of Definition 19.19(iv) (continuous except at finitely many points, with $|f(x)| \le c(x^c + x^{-c})$) and satisfy $f(x) = 0$ for $x \le H$. Let $V(S,t)$ be the European price function of the claim paying $f(S_T)$ at T (Corollary 20.41(i) for an asset, Theorem 20.52 for a futures price):

$$V(S,t) = e^{-r(T-t)}\,\mathbb{E}\, f\big(S\exp\big((b-\tfrac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t}\,Z\big)\big) \quad \text{for } t < T,$$

and $V(S,T) = f(S)$. Define, for $S > 0$,

$$V^{\mathrm{out}}(S,t) = V(S,t) - \Big(\frac{S}{H}\Big)^{1-2b/\sigma^2} V\Big(\frac{H^2}{S},\, t\Big) \quad \text{for } t < T, \qquad V^{\mathrm{out}}(S,T) = f(S). \tag{23.4}$$

Then:

(i) The function V^{out} solves the pricing equation

$$\partial_t V^{\mathrm{out}} + bS\,\partial_S V^{\mathrm{out}} + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V^{\mathrm{out}} = rV^{\mathrm{out}} \tag{23.5}$$

on $(0,\infty)\times[0,T)$; it vanishes on the barrier, $V^{\mathrm{out}}(H,t) = 0$ for every $t < T$; and $V^{\mathrm{out}}(S,t) \to f(S^*)$ as $(S,t) \to (S^*,T)$, at every continuity point $S^* > H$ of f.

(ii) The down-and-out claim, which pays $f(S_T)\,\mathbf{1}\{S_T^{\min} > H\}$ at T, is replicated by the strategy with initial capital $V^{\mathrm{out}}(S_0,0)$ that holds

$$\partial_S V^{\mathrm{out}}(S_t,t)\, e^{(r-b)(T-t)}$$

units of the hedge instrument $\tilde{S}_t$ until the hitting time t_H, and nothing afterwards. Its arbitrage-free price today is $V^{\mathrm{out}}(S_0,0)$, which is nonnegative when $f \ge 0$.

(iii) The down-and-in claim, which pays $f(S_T)\,\mathbf{1}\{S_T^{\min} \le H\}$ at T, has the arbitrage-free price

$$V^{\mathrm{in}}(S_0,0) = V(S_0,0) - V^{\mathrm{out}}(S_0,0) = \Big(\frac{S_0}{H}\Big)^{1-2b/\sigma^2} V\Big(\frac{H^2}{S_0},\, 0\Big).$$

(iv) The same holds for an up barrier. If $S_0 < H$ and $f(x) = 0$ for $x \geq H$, then, with the same formula (23.4) and with $t_H = \inf\{t \in [0,T] : S_t \geq H\}$, parts (i)–(iii) hold for the up-and-out claim paying $f(S_T)\,\mathbf{1}\{S_T^{\max} < H\}$ and the up-and-in claim paying $f(S_T)\,\mathbf{1}\{S_T^{\max} \geq H\}$, the limit in (i) being taken at continuity points $S^* < H$.

(v) The two prices are risk-neutral expectations. With $\mathbb{E}^{\mathbb{Q}}$ as in Definition 20.6 for an asset, and as in Proposition 20.51(i) for a futures price,

$$V^{\mathrm{out}}(S_0,0) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[f(S_T)\,\mathbf{1}\{S_T^{\min} > H\}\big],$$
$$V^{\mathrm{in}}(S_0,0) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[f(S_T)\,\mathbf{1}\{S_T^{\min} \leq H\}\big],$$

and in the case (iv) the same holds with $\mathbf{1}\{S_T^{\max} < H\}$ and $\mathbf{1}\{S_T^{\max} \geq H\}$ in place of the two indicators.

Sketch of the proof. We use the method of images to build a solution of the pricing equation that vanishes on the barrier, and then replicate the claim by the delta hedge of that solution, stopped at the barrier. The image term of (23.4) solves (23.5): by the chain rule, $S\,\partial_S$ acts on a function of the reflected spot H^2/S as minus the same operator in that variable, and once the equation for V is used at the reflected spot, all that is left is the first derivative of V and V itself, with the coefficients

$$-b - b + \tfrac{1}{2}\sigma^2 \cdot \frac{4b}{\sigma^2} = 0 \qquad \text{and} \qquad \Big(1 - \frac{2b}{\sigma^2}\Big)(b - b) = 0,$$

the first of which vanishes for no other exponent; by linearity V^{out} solves the equation as well. On the barrier the reflected spot is H and the power is 1, so $V^{\mathrm{out}}(H,t) = 0$; near T the reflected spot of a point above H lies below H, where f vanishes by hypothesis, so the image term tends to zero. This is (i). For (ii), Itô's lemma with its stopped clause (Theorem 19.30(iii)), applied to $e^{-rt}\,V^{\mathrm{out}}$ along the hedge instrument, leaves by (i) only the gains of the stated holding: the strategy is worth $V^{\mathrm{out}}(S_t,t)$ before t_H and $V^{\mathrm{out}}(H,t_H) = 0$ from t_H on, its value tends to $f(S_T)\,\mathbf{1}\{S_T^{\min} > H\}$ as $t \to T$, and (A6) makes its initial capital the price. Part (iii) subtracts this strategy from the one that replicates the vanilla claim, (iv) is the mirror image, and (v) takes expectations in the same stopped identity on the risk-neutral path, where only a stochastic integral of mean zero is left. The appendix supplies the growth bounds, resting on $f \in \mathcal{G}$, that make the holding admissible and the limit at T a limit in mean square. The complete proof is in Appendix I.3. □

Intuition. The image term $(S/H)^{1-2b/\sigma^2}\,V(H^2/S,t)$ is itself a solution of the pricing equation. It is the price function of a claim whose payoff $(S_T/H)^{1-2b/\sigma^2} f(H^2/S_T)$ lives entirely on the dead side of the barrier, and the reflection makes it equal to the vanilla on the barrier itself. (The reader should verify, using Lemma 19.38(i) with $\theta = (1 - 2b/\sigma^2)\,\sigma\sqrt{T-t}$ and then the symmetry of Z, which replaces Z by $-Z$, that $e^{-r(T-t)}$ times the expectation of that payoff, for $S_T = S\exp((b - \frac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t}\,Z)$, is the image term.) Subtracting it leaves the payoff on the live side untouched and kills the value at the barrier, which is all a knock-out is. The power $(S/H)^{1-2b/\sigma^2}$ corrects for the log-drift $\nu = b - \sigma^2/2$ of the price under $\mathbb{Q}$: the exponent is $-2\nu/\sigma^2$, and it is zero exactly when $\nu = 0$, the case in which the log price is a driftless Brownian motion and plain reflection suffices. The proof uses no reflection principle for S, no strong Markov property and no pricing rule for path-dependent payoffs: the stopped delta hedge does all the work.

Corollary 23.17 (Down-and-in and down-and-out calls in closed form (Merton, 1973; Reiner and Rubinstein, 1991))**.** *Assume (A1) and (A3)–(A6) of §20.1 with (A2) replaced by a continuous yield q (§20.7), continuous monitoring, $H \leq K$ and $H < S$. Then the down-and-in call is worth*

$$\Big(\frac{H}{S}\Big)^{2(r-q)/\sigma^2-1} C\Big(\frac{H^2}{S}, K; T\Big) = Se^{-qT}\Big(\frac{H}{S}\Big)^{2(r-q)/\sigma^2+1}\Phi(y)$$
$$- Ke^{-rT}\Big(\frac{H}{S}\Big)^{2(r-q)/\sigma^2-1}\Phi(y - \sigma\sqrt{T}),$$

with

$$y = \frac{\ln\left(H^2/(SK)\right) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}},$$

and the down-and-out call is worth $C(S,K;T)$ minus that amount. The other single-barrier cases follow from Theorem 23.16 in the same way and are not listed.

Proof. We substitute in Theorem 23.16. The payoff $f(x) = (x-K)^+$ is continuous with $0 \le f(x) \le x$, so $f \in \mathcal{G}$, and it vanishes on $(0,H]$ because $H \le K$. Part (iii) of the theorem, with $b = r - q$ and $V(S,0) = C(S,K;T)$, gives the price

$$\Big(\frac{S}{H}\Big)^{1-2(r-q)/\sigma^2} C\Big(\frac{H^2}{S}, K; T\Big) = \Big(\frac{H}{S}\Big)^{2(r-q)/\sigma^2-1} C\Big(\frac{H^2}{S}, K; T\Big),$$

which is the first form. For the second, evaluate the formula of Corollary 20.41(i) at the spot H^2/S: its d_1 is

$$\frac{\ln\left((H^2/S)/K\right) + (r - q + \sigma^2/2)T}{\sigma\sqrt{T}} = y,$$

its d_2 is $y - \sigma\sqrt{T}$, and $C(H^2/S, K; T) = (H^2/S)e^{-qT}\Phi(y) - Ke^{-rT}\Phi(y - \sigma\sqrt{T})$. Multiplying by the power factor, and using $(H/S)^{2(r-q)/\sigma^2-1}\, H^2/S = S\,(H/S)^{2(r-q)/\sigma^2+1}$, gives the second form. The down-and-out call follows from in–out parity (Proposition 23.15(i)), or from part (ii) of the theorem. □

Example 23.18 (A down barrier below and above the strike)**.** Let $S = K = 100$, $H = 95$, $r = 3\%$, $q = 0$, $\sigma = 20\%$ and $T = 0.5$, under BSM with continuous monitoring. The vanilla call is worth $C(100, 100; 0.5) = 6.371028$. The exponent is $1 - 2r/\sigma^2 = 1 - 0.06/0.04 = -0.5$ and the reflected spot is $H^2/S = 9025/100 = 90.25$, at which the call struck at 100 is worth $C(90.25, 100; 0.5) = 2.176364$. By Theorem 23.16(iii) the down-and-in call is worth

$$V^{\text{in}} = \Big(\frac{100}{95}\Big)^{-0.5} \times 2.176364 = 0.974679 \times 2.176364 = 2.121257,$$

and the $\Phi(y)$ form of Corollary 23.17, with $y = -0.5486$, returns the same value. In–out parity (Proposition 23.15(i)) gives the down-and-out call:

$$V^{\text{out}} = 6.371028 - 2.121257 = 4.249771,$$

that is, 6.3710, 2.1213 and 4.2498 to four decimals: the barrier removes a third of the value of the call.

A barrier above the strike. Keep $H = 95$ and lower the strike to $K = 90$. The down-and-out call pays $(S_T - 90)^+ \mathbf{1}\{S_T^{\min} > 95\}$. A path that survives ends above 95, so this is $f(S_T)\,\mathbf{1}\{S_T^{\min} > 95\}$ with

$$f(x) = (x - 90)^+ \mathbf{1}\{x > 95\} = (x - 95)^+ + 5 \cdot \mathbf{1}\{x > 95\},$$

a payoff in $\mathcal{G}$ that vanishes for $x \le 95$. Its European price function is a call struck at 95 plus five digitals struck at 95, and Theorem 23.16 applies with this f. At the spot, $V(100,0) = 9.2511 + 3.2248 = 12.4759$; at the reflected spot, $V(90.25,0) = 3.68317 + 1.83085 = 5.51402$, so the image term is $0.97468 \times 5.51402 = 5.3744$ and, by part (ii) of the theorem, the down-and-out call is worth $12.4759 - 5.3744 = 7.1015$. The vanilla call struck at 90 is worth 12.7993, so by in–out parity the down-and-in call is worth $12.7993 - 7.1015 = 5.6978$. This is *not* the image term: by part (iii) the image term 5.3744 prices $f(S_T)\,\mathbf{1}\{S_T^{\min} \le 95\}$, which pays nothing when S_T ends in $(90, 95]$. The difference $5.6978 - 5.3744 = 0.3234 = 12.7993 - 12.4759$ is the value of the claim paying $(S_T - 90)^+ \mathbf{1}\{S_T \le 95\}$, every path of which has touched the barrier.

A check by simulation. By Theorem 23.16(v) each knock-out price is a risk-neutral expectation, so it can be estimated by averaging simulated risk-neutral payoffs (§23.5 studies the method). A

path simulated on 10 time steps of length $\Delta t = 0.05$ is seen only at the grid dates, and a crossing between two of them is accounted for by the *Brownian-bridge correction*: a step from $S_{(k-1)\Delta t}$ to $S_{k\Delta t}$, both above H, is counted as having crossed the barrier in between with probability

$$\exp\Big(- \frac{2\ln(S_{(k-1)\Delta t}/H)\,\ln(S_{k\Delta t}/H)}{\sigma^2 \Delta t}\Big),$$

whatever the drift, and the payoff of each path is multiplied by the product, over its steps, of one minus these numbers. For a single step this number is the image density of Theorem 19.63(b), read for a lower level by the symmetry $W \mapsto -W$, divided by the density of the endpoint; the product over several steps is used here and not proved. A run of 10,000,000 paths gives 4.2505 for the first down-and-out call and 7.1022 for the second, with standard errors 0.0027 and 0.0037: each is within one standard error of its closed form.

(Problem P23.4 works this computation at different numbers.)

Interview trap. "Reflect the spot across the barrier and subtract the vanilla there." The image term is $(S/H)^{1-2b/\sigma^2}\,V(H^2/S,t)$, and without the power it does not solve the pricing equation (Step 1 of the proof of Theorem 23.16). The power equals 1 only when the log-drift $\nu = b - \sigma^2/2$ is zero. In Example 23.18 it is $(100/95)^{-0.5} = 0.975$, and dropping it overstates the knock-in by 2.6% (2.176 against 2.121).

Interview trap. "Vega is always positive." Not for a knock-out near its barrier. With $K = 100$, $H = 120$, $r = 3\%$, $q = 0$ and $T = 0.5$, Theorem 23.16(iv) with $f(x) = (x-K)^+\,\mathbf{1}\{x < H\}$ prices the up-and-out call at 2.97, 2.14 and 1.46 for $\sigma = 15\%$, 20% and 25% when $S = 100$: there more volatility means more knock-outs. At $S = 80$, far below the barrier, the same option is worth 0.085, 0.300 and 0.483: there volatility is needed to reach the money. The vega changes sign as the spot moves toward the barrier. Near the barrier and close to expiry its delta is negative as well, and large: the value falls from about $H - K$ just below the barrier to zero on it (a *reverse* knock-out), so delta and gamma jump there, and such books are hedged statically rather than by delta.

Proposition 23.19 (A one-touch is worth about two digitals)**.** *(i) (Exact, driftless arithmetic case.) Consider a market with $r = 0$ in which (A3)–(A6) of §20.1 hold and in which an arithmetic underlying with real-world path $S_t = S_0 + \mu_{\mathrm{N}} t + \sigma_{\mathrm{N}} W_t$ (Setup (h)) trades and pays no income. Let $H > S_0$. For the risk-neutral path $S_0 + \sigma_{\mathrm{N}} W_t$,*

$$\mathbb{Q}(S_T^{\max} \geq H) = 2\,\mathbb{Q}(S_T \geq H) = 2\,\Phi\Big(\frac{S_0 - H}{\sigma_{\mathrm{N}}\sqrt{T}}\Big),$$

and these probabilities are the prices: the digital paying $\mathbf{1}\{S_T > H\}$ at T is worth $\mathrm{Dig}(H) = \Phi((S_0 - H)/(\sigma_{\mathrm{N}}\sqrt{T}))$, and $\mathrm{OT}(H) = 2\,\mathrm{Dig}(H)$.

(ii) (Exact, lognormal zero-carry case.) Let F_t be a futures price in the model of Theorem 23.16 (so $b = 0$, with any r), let the one-touch pay at T, and let $\mathrm{Dig}(H)$, $C(H)$ and $\mathcal{P}(H)$ be the prices of the digital, the call and the put on F_T with strike H and expiry T, with $d_{1,2} = [\ln(F/H) \pm \sigma^2 T/2]/(\sigma\sqrt{T})$. For an up barrier $H > F$,

$$\mathrm{OT}(H) = e^{-rT}\Big[\Phi(d_2) + \frac{F}{H}\,\Phi(d_1)\Big] = 2\,\mathrm{Dig}(H) + \frac{C(H)}{H},$$

and for a down barrier $H < F$,

$$\mathrm{OT}(H) = e^{-rT}\Big[\Phi(-d_2) + \frac{F}{H}\,\Phi(-d_1)\Big] = 2(e^{-rT} - \mathrm{Dig}(H)) - \frac{\mathcal{P}(H)}{H},$$

where $e^{-rT} - \mathrm{Dig}(H)$ is the price of the digital put (Corollary 20.20).

Sketch of the proof. For (i) we read the probabilities off the reflection principle and then show, by a replication stopped at the barrier, that they are prices; (ii) is Theorem 23.16 applied to the no-touch. In (i) the risk-neutral path reaches H by T exactly when W reaches the level ℓ, positive because $H > S_0$, and Theorem 19.63(i) gives

$$\mathbb{Q}(S_T^{\max} \geq H) = 2\,P(W_T \geq \ell) = 2\,\mathbb{Q}(S_T \geq H) = 2\,\Phi\Big(\frac{S_0 - H}{\sigma_{\mathrm{N}}\sqrt{T}}\Big),$$

the factor 2 being the pairing of paths of Figure 50. These are prices because the function $1 - 2\,\Phi((S-H)/(\sigma_{\mathrm{N}}\sqrt{T-t}))$, a normal distribution function minus its mirror image about the barrier, vanishes on the barrier, tends to 1 below it as $t \to T$, and has a derivative in t that cancels $\frac{1}{2}\sigma_{\mathrm{N}}^2$ times its second derivative in S. By the stopped clause of Itô's lemma (Theorem 19.30(iii)), holding until t_H as many units of the underlying as the derivative of this function in S therefore replicates the no-touch from the capital $1 - 2\,\Phi(-\ell/\sqrt{T})$, whatever the drift μ_{N}. The one-touch is one unit in the bank (which pays no interest, as $r = 0$) minus the no-touch, the digital is replicated in the same way without stopping, and (A6) makes each cost a price. In (ii) the up no-touch is the up-and-out version of the digital put, whose payoff vanishes from H upward, so Theorem 23.16(iv) applies with $b = 0$ and the exponent 1: the digital put is worth $e^{-rT}\,\Phi(-d_2)$, at the reflected price H^2/F its d_2 becomes $-d_1$, and the no-touch is worth $e^{-rT}[\Phi(-d_2) - (F/H)\,\Phi(d_1)]$. Subtracting from e^{-rT}, the price of one unit paid at T, gives the first form, the formulas for $C(H)$ and $\mathrm{Dig}(H)$ give the second, and the down barrier is the mirror image. The complete proof is in Appendix I.4. □

(Problem P23.5 asks you to reproduce the complete proof from scratch.)

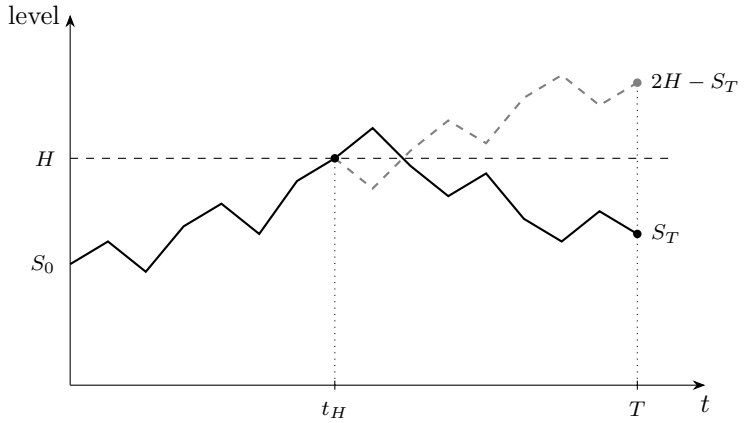


Figure 50: The reflection principle for a driftless arithmetic underlying. A path (solid) starts at S_0 below the barrier H, first reaches it at t_H and ends at $S_T < H$; its mirror image about H after t_H (gray, dashed) ends at $2H - S_T > H$. Every path that touches H and ends below it is paired in this way with one that ends above it, so that $\mathbb{Q}(\text{touch}) = 2\,\mathbb{Q}(S_T \geq H)$ (Proposition 23.19(i), from Theorem 19.63(i)).

Example 23.20 (A one-touch on a futures price)**.** A bitcoin futures price stands at $F = 60{,}000$; the barrier is $H = 70{,}000$, the horizon $T = 30/365$, the volatility $\sigma = 60\%$ and $r = 0$. Then $\sigma\sqrt{T} = 0.1720$, $d_2 = -0.9822$ and $d_1 = -0.8101$, and the digital that pays if the futures settles above 70,000 is worth $\mathrm{Dig}(H) = \Phi(d_2) = 0.1630$ (these are the numbers of Problem P22.5). The rule of thumb gives $2 \times 0.1630 = 0.326$ for the one-touch. The exact value, by Proposition 23.19(ii), is

$$\mathrm{OT}(H) = \Phi(d_2) + \frac{F}{H}\,\Phi(d_1) = 0.1630 + \tfrac{6}{7} \times 0.2089 = 0.1630 + 0.1791 = 0.3421,$$

that is, $0.3260 + C(H)/H$ with $C(H) = 1{,}125$ and $C(H)/H = 0.0161$. In the arithmetic model with $\sigma_{\mathrm{N}} = 0.6 \times 60{,}000 = 36{,}000$, part (i) gives $\mathrm{Dig}(H) = \Phi(-0.9689) = 0.1663$ and

$\mathrm{OT}(H) = 0.3326$, exactly twice. The mirror case: the down one-touch at $H = 50{,}000$ is worth $2 \times 0.165050 - 0.013955 = 0.3161$, a little less than twice the digital put, 0.3301. The shortfall is larger for a barrier nearer the money in standard deviations: with $F = 100$, $H = 90$, $\sigma = 20\%$, $T = 1$ and $r = 0$, the no-touch is worth $\Phi(d_2) - (F/H)\,\Phi(-d_1) = 0.66524 - 0.29488 = 0.3704$ and the one-touch 0.6296, against 0.6695 for two digital puts.

(Problem P23.5 works this computation at different numbers.)

Intuition. For a lognormal price with zero carry the log price drifts at $-\sigma^2/2$ under $\mathbb{Q}$: away from an upper barrier and toward a lower one. A path that has just touched an upper barrier is therefore a little less likely than one half to finish beyond it, so touching is a little *more* than twice as likely as finishing beyond, and the up one-touch exceeds two digitals by $C(H)/H$; for a lower barrier the drift helps the path stay beyond, and the down one-touch falls short of two digital puts by $\mathcal{P}(H)/H$. Near the money both corrections are of relative size of order $\sigma\sqrt{T}$, so "a one-touch is about twice the digital" is exact to leading order.

Trading application. On a prediction market, the contract "touches 70,000 within 30 days" is fair near 34 cents when the contract "settles above 70,000 in 30 days" is fair near 16 cents, in the market of Example 23.20. Quoting the first at twice the second is right to within two cents; quoting the two at the same price gives away half the value.

Interview trap. "A one-touch is a digital", or "exactly two digitals". The event of touching the barrier contains the event of finishing beyond it and is about twice as likely (0.342 against 0.163 in Example 23.20). It is exactly twice as likely when the risk-neutral drift of the price (arithmetic model) or of the log price (lognormal model, cost of carry $b = \sigma^2/2$) is zero: in the second case the exponent $1 - 2b/\sigma^2$ is 0, and Theorem 23.16(iv) with $f(x) = \mathbf{1}\{x < H\}$ gives $\mathrm{OT}(H) = 2e^{-rT}\,\Phi(d_2) = 2\,\mathrm{Dig}(H)$. With zero carry in the lognormal model the log price drifts at $-\sigma^2/2$, and the factor is slightly more than two for an upper barrier and slightly less for a lower one.

Fact 23.21 (Discrete-monitoring correction (Broadie, Glasserman and Kou, 1997))**.** Under BSM, let a single-barrier down call or up put, knock-in or knock-out, be monitored on the dates $k\,\Delta t$, $k = 1, \ldots, T/\Delta t$, an integer number of them (Setup (b)). Its price is approximated, to good accuracy, by the price of the same option with continuous monitoring (Theorem 23.16; Corollary 23.17 for the down-and-in and down-and-out calls) with the barrier moved away from the spot: H is replaced by

$$He^{-0.5826\,\sigma\sqrt{\Delta t}} \text{ for a down barrier,} \qquad He^{+0.5826\,\sigma\sqrt{\Delta t}} \text{ for an up barrier,}$$

where $0.5826 = -\zeta(1/2)/\sqrt{2\pi}$ and ζ is the Riemann zeta function, with $\zeta(1/2) = -1.4604$ (a function written with its argument, not an integrand of §19). The error of the approximation is small compared with $\sqrt{\Delta t}$ as $\Delta t \to 0$.

A proof is beyond our scope. It needs the expected overshoot of a Gaussian random walk over a level, which is computed through Spitzer's identity and involves the Riemann zeta function, together with a corrected diffusion approximation for the hitting probability. The statement is asymptotic as $\Delta t \to 0$, and no proof in this Part relies on it. The same shift is used in practice for the other single-barrier payoffs (up calls, down puts, and the one-touch of Example 23.20); the Fact as stated does not cover them, and there the shifted price is an approximation with no error bound given here.

Example 23.22 (Monitoring frequency)**.** Take the first market of Example 23.18 ($S = K = 100$, $H = 95$, $r = 3\%$, $\sigma = 20\%$, $T = 0.5$), in which the continuously monitored down-and-out call is worth 4.250 and the vanilla 6.371. With daily monitoring, $\Delta t = 1/252$, Fact 23.21 moves the barrier to $95\,e^{-0.5826\times 0.2\times\sqrt{1/252}} = 94.305$, and the formulas of Corollary 23.17 at that barrier give

1.763 for the down-and-in call and 4.608 for the down-and-out call. A simulation of 2,000,000 paths observed on the 126 daily dates gives 4.606 for the latter, with standard error 0.006. Weekly monitoring ($\Delta t = 1/52$) moves the barrier to 93.48 and the knock-out to 4.973; monthly monitoring ($\Delta t = 1/12$) to 91.86 and 5.516, an approximate figure at so coarse a spacing. The rarer the observation, the dearer the knock-out and the cheaper the knock-in. This ordering holds path by path whenever the coarser dates are among the finer ones, as the monthly dates here are among the daily ones, 21 trading days apart; Remark 23.26, in §23.3 below, makes the comparison path by path.

The same shift, used outside the Fact as a practical approximation, can be applied to the one-touch of Example 23.20 when it is observed once a day, 365 days a year (that market never closes): the barrier moves up to $70{,}000\, e^{0.5826\times 0.6\times\sqrt{1/365}} = 71{,}293$, and Proposition 23.19(ii) at that barrier gives 0.289, against 0.342 with continuous monitoring. A simulation of the 30 daily observations over 60,000,000 paths gives 0.2913, with standard error 0.00006.

(Problem P23.4 works this computation at different numbers.)

Interview trap. Which way does the monitoring frequency move the price? Adding observation dates to a grid gives the path more chances to be seen below the barrier: it can only lower the observed minimum. So, path by path, finer monitoring makes a down-and-out call *cheaper* and a down-and-in call *dearer* (Lemma 16.8(i); Remark 23.26 below). Continuous monitoring contains every date, so a daily-monitored knock-out is dearer than the continuously monitored one (4.61 against 4.25 in Example 23.22), and pricing it with the continuous formula underprices it. The shift of the barrier away from the spot in Fact 23.21 encodes exactly this.

Remark 23.23 (Static replication of a knock-in). With zero cost of carry a down-and-in call can be hedged without any trading before the barrier is touched. Let F_t be a futures price in the model of Theorem 23.16 ($b = 0$), and write $C(F, K; \tau)$ and $\mathcal{P}(F, K; \tau)$ for the prices of Theorem 20.52(iii) with futures price F, strike K and life τ.

Put–call symmetry. For every $F, K > 0$ and $\tau > 0$,

$$C(F, K; \tau) = \frac{K}{F}\, \mathcal{P}\Big(F, \frac{F^2}{K}; \tau\Big).$$

Indeed, with d_1, d_2 those of the call, the put struck at F^2/K has $[\ln(K/F) + \sigma^2\tau/2]/(\sigma\sqrt{\tau}) = -d_2$ in the place of d_1 and $-d_2 - \sigma\sqrt{\tau} = -d_1$ in the place of d_2, so by Theorem 20.22 it is worth $e^{-r\tau}[(F^2/K)\,\Phi(d_1) - F\,\Phi(d_2)] = (F/K)\, C(F, K; \tau)$.

The knock-in is a put position. Let $H \le K$ and $F > H$. By Theorem 23.16(iii) with $b = 0$ (exponent 1), then symmetry at the futures price H^2/F, then homogeneity with the factor F^2/H^2 (Lemma 23.7 with $q = r$, the case in which the formulas of Corollary 20.41(i) are those of Theorem 20.22 with F in the place of S), the down-and-in call has the time-0 price

$$V^{\text{in}}(F, 0) = \frac{F}{H}\, C\Big(\frac{H^2}{F}, K; T\Big) = \frac{F}{H} \cdot \frac{KF}{H^2}\, \mathcal{P}\Big(\frac{H^2}{F}, \frac{H^4}{F^2 K}; T\Big) = \frac{K}{H}\, \mathcal{P}\Big(F, \frac{H^2}{K}; T\Big).$$

The down-and-in call is worth K/H puts struck at H^2/K (Bowie and Carr, 1994). The same algebra, with T replaced by any $\tau > 0$, gives the identity of functions

$$\frac{F}{H}\, C\Big(\frac{H^2}{F}, K; \tau\Big) = \frac{K}{H}\, \mathcal{P}\Big(F, \frac{H^2}{K}; \tau\Big), \qquad F > 0,\ \tau > 0;$$

it is a statement about the two formulas, not a price statement at a later date, and what makes the puts a hedge at every date is the strategy below.

The static hedge. Buy the K/H puts and hold them. If the barrier is never touched they expire worthless, because $H^2/K \le H < F_T$, and so does the knock-in. At the first touch, $F = H$, they

are worth exactly the vanilla call, $(K/H)\,\mathcal{P}(H, H^2/K;\tau) = C(H, K;\tau)$ by symmetry at $F = H$, and they are exchanged for it at no cost; the call then pays what the activated knock-in pays.

Why the exchange is a strategy. In the model, hold the replicating strategy of the K/H puts before t_H and that of the call from t_H on: with $h^{\mathcal{P}}_s$ and h^{C}_s the two delta holdings of the hedge instrument, the holding is $h^{\mathcal{P}}_s\,\mathbf{1}\{s < t_H\} + h^{C}_s - h^{C}_s\,\mathbf{1}\{s < t_H\}$, a finite sum of admissible holdings (Definition 20.1(i)), started from the price of the puts. Write $\mathcal{P}_t$ and C_t for the model values at time t of the K/H puts and of the call. By Theorem 19.30, clauses (i) and (iii), through Corollary 19.31, the three integrals of the holding against the discounted hedge instrument over $[0, t]$ are

$$e^{-r(t\wedge t_H)}\,\mathcal{P}_{t\wedge t_H} - \mathcal{P}_0, \qquad e^{-rt}\,C_t - C_0, \qquad -\big(e^{-r(t\wedge t_H)}\,C_{t\wedge t_H} - C_0\big).$$

Adding them to the initial capital $\mathcal{P}_0$, the discounted value of the strategy is $e^{-rt}\,\mathcal{P}_t$ on $\{t < t_H\}$ and

$$e^{-rt}\,C_t + e^{-rt_H}\big(\mathcal{P}_{t_H} - C_{t_H}\big) = e^{-rt}\,C_t \qquad \text{on } \{t \ge t_H\},$$

the bracket vanishing by put–call symmetry at $F = H$. The value is continuous at the switch, and its terminal value, the limit as $t \to T$ taken as in Step 4 of the proof of Theorem 23.16, is $(F_T - K)^+\,\mathbf{1}\{t_H \le T\}$, the payoff of the down-and-in call.

Away from zero carry no single vanilla option does the job. The image claim, which pays $(S_T/H)^{1-2b/\sigma^2}(H^2/S_T - K)^+$, still hedges statically but is not a vanilla; static hedges built from vanillas instead match the value of the barrier option along the barrier with a strip of vanilla options of several maturities, unwound when the barrier is hit (Derman, Ergener and Kani, 1995). The call-spread hedge of Proposition 23.2 is the simplest static hedge of all.

Example 23.24 (Hedging a knock-in with puts)**.** Let a futures price stand at $F = 100$, with $K = 100$, $H = 90$, $\sigma = 25\%$, $T = 1$ and $r = 0$. The hedge of Remark 23.23 is $K/H = 1.111$ puts struck at $H^2/K = 81$, which cost 2.7716. The down-and-in call is worth, by Theorem 23.16(iii) with zero carry, $(100/90)\,C(81, 100; 1) = 2.7716$, the same number, out of a vanilla value of 9.948. At the touch, $F = 90$, the call struck at 100 and the 1.111 puts struck at 81 agree at every residual life: both are worth 5.2721 for $\tau = 1$, 2.8412 for $\tau = 0.5$ and 0.3182 for $\tau = 0.1$.

With the barrier at the strike, $H = K$, the hedge is one put struck at K: at zero carry the down-and-in call is worth the vanilla put, and the down-and-out call is worth $C - \mathcal{P} = e^{-rT}(F - K)$, the value of a forward contract (Theorem 20.52(iii)). In this market with $K = H = 90$ the down-and-in call is worth $\mathcal{P}(100, 90; 1) = 5.2721$ and the down-and-out call $15.2721 - 5.2721 = 10.0000$.

23.3 Lookback and Asian Options

Lookback options pay on the extreme of the path and are the dearest standard options a client can buy; Asian options pay on an average of the path and are among the cheapest. This subsection proves the dominance bounds behind both statements, and then the effective-volatility formula that answers the question "why is an Asian option cheaper?", by pricing the geometric Asian call on J fixings in closed form. That closed form returns in §23.6 as a control variate for the arithmetic Asian, which has no closed form.

Setup. The symbols of §23.1 are kept. In addition:

(a) The *fixing dates* are $0 < t_1 < t_2 < \cdots < t_J = T$, and $t_0 = 0$. They are *equally spaced* when $t_j = jT/J$, which is assumed only where stated. The letter J counts the fixings, because n is the number of simulated paths in this section; when an Asian option is simulated, the time steps of a path are its fixing dates.

(b) The *arithmetic average* and the *geometric average* of the fixings are

$$\bar{S}_T = \frac{1}{J}\sum_{j=1}^{J} S_{t_j}, \qquad G_T = \Big(\prod_{j=1}^{J} S_{t_j}\Big)^{1/J},$$

so that $\ln G_T = \frac{1}{J}\sum_j \ln S_{t_j}$. The running extremes $S_T^{\max}$ and $S_T^{\min}$ are those of §23.2.

(c) Every Asian option of this subsection pays at T. Its payoff is a function of the J prices $S_{t_1}, \dots, S_{t_J}$, so under BSM it is priced by Corollary 20.11 as soon as that function belongs to the class $\mathcal{G}_J$ of Definition 19.19(iv). A continuously monitored lookback would need Fact 20.13; lookbacks therefore appear only in dominance statements, which need no model.

(d) Under $\mathbb{Q}$ (Definition 20.6(i)) the price at a fixing date is $S_{t_j} = S\exp\big((b-\frac{1}{2}\sigma^2)t_j + \sigma W_{t_j}\big)$, where W_{t_j} is the value of the Brownian motion at t_j. The increments between consecutive fixings are $\Delta W_k = W_{t_k} - W_{t_{k-1}}$, $k = 1, \dots, J$.

Definition 23.25 (Lookback and Asian options)**.** All the following claims are paid at T.

(i) The *floating lookback call* pays $S_T - S_T^{\min}$ and the *floating lookback put* pays $S_T^{\max} - S_T$: the holder buys at the lowest price of the period, or sells at the highest. The *fixed-strike lookback call* pays $(S_T^{\max} - K)^+$ and the *fixed-strike lookback put* pays $(K - S_T^{\min})^+$.

(ii) The *arithmetic average-price call* pays $(\bar{S}_T - K)^+$ and the *arithmetic average-price put* pays $(K - \bar{S}_T)^+$. These are the contracts meant by "an Asian option" when nothing else is said. The *geometric average-price call* pays $(G_T - K)^+$.

(iii) The *average-strike call* pays $(S_T - \bar{S}_T)^+$ and the *average-strike put* pays $(\bar{S}_T - S_T)^+$. They are only named here; Problem P23.6 relates them to the average-price options.

Under discrete monitoring the extremes in (i) are taken over the observation dates, as in Definition 23.14.

Remark 23.26 (Lookback dominance and monitoring). Three comparisons hold path by path. Lemma 16.8(i) turns each of them into a comparison of prices, in any arbitrage-free market in which the claims concerned trade; no pricing rule for a lookback is needed.

Against the vanilla. Since $S_T^{\max} \geq S_T$, the fixed-strike lookback call pays $(S_T^{\max} - K)^+ \geq (S_T - K)^+$ in every state, so it is worth at least the vanilla call. In the same way $S_T^{\min} \leq S_T$ makes the fixed-strike lookback put worth at least the vanilla put. The floating call and the floating put together pay $(S_T - S_T^{\min}) + (S_T^{\max} - S_T) = S_T^{\max} - S_T^{\min}$, the range of the path.

Discrete against continuous monitoring. A maximum over a set of observation dates is at most the maximum over all of $[0, T]$, and a minimum over the dates is at least the minimum over $[0, T]$. Each of the four payoffs of Definition 23.25(i) is nondecreasing in the observed maximum or nonincreasing in the observed minimum. Discrete monitoring therefore can only lower a lookback payoff, path by path, and hence (Lemma 16.8(i)) it can only lower a lookback price.

Finer against coarser monitoring. The same argument compares two grids of observation dates when one contains the other, as the daily dates ($\Delta t = 1/252$) contain the monthly ones, 21 trading days apart. Adding observation dates can only raise the observed maximum and lower the observed minimum. It can therefore only raise a lookback price. It can only lower the price of a knock-out option with a nonnegative payoff, because the indicators $\mathbf{1}\{\text{observed minimum} > H\}$ and $\mathbf{1}\{\text{observed maximum} < H\}$ can only fall; and by in–out parity (Proposition 23.15(i)) it can only raise the price of the knock-in. The two families move in opposite directions. For two grids neither of which contains the other (every third trading day against every fifth), neither observed extreme dominates the other, and no path-by-path comparison holds.

A number. For a quick estimate, take the driftless arithmetic underlying of Proposition 23.19(i), whose risk-neutral path is $S_0 + \sigma_{\mathrm{N}} W_t$ with $\sigma_{\mathrm{N}} > 0$. Its running maximum is $S_T^{\max} = S_0 + \sigma_{\mathrm{N}} W_T^{\max}$, and Theorem 19.63(i) gives $\mathbb{E}\, W_T^{\max} = \sqrt{2T/\pi}$, while $\mathbb{E}\, W_T = 0$. Hence

$$\mathbb{E}^{\mathbb{Q}}[S_T^{\max} - S_T] = \sigma_{\mathrm{N}}\sqrt{2T/\pi},$$

exactly twice the at-the-money call of that model, $\sigma_{\mathrm{N}}\sqrt{T/(2\pi)}$ (Remark 20.31 at $r = 0$). With $\sigma_{\mathrm{N}} = 20$ and $T = 1$ the two numbers are 15.96 and 7.98. That the first number is the price of the floating lookback put is not claimed: it would need a pricing rule for continuously monitored

payoffs in the arithmetic model, and the Part states such a rule only for the lognormal model, as Fact 20.13, which no proof uses.

Under BSM the continuously monitored lookbacks have closed forms (Goldman, Sosin and Gatto, 1979), which this document does not state.

Interview trap. "A lookback is a cheap way to buy at the low." It is the dearest standard payoff: it dominates the vanilla state by state (Remark 23.26), and the seller charges for the hindsight. The monitoring frequency acts on it the other way round from a knock-out: removing observation dates, so that the coarser grid sits inside the finer one, makes a lookback cheaper and a knock-out dearer.

Proposition 23.27 (Bounds for Asian calls)**.** *Let the fixing dates be $0 < t_1 < \cdots < t_J = T$, equally spaced or not. The following statements are model-free.*

(i) If the arithmetic average-price call and the European calls with strike K that expire on the fixing dates trade, then

$$\textit{arithmetic average-price call} \;\le\; \frac{1}{J}\sum_{j=1}^{J} e^{-r(T-t_j)}\,C(K,t_j). \tag{23.6}$$

The right side is the average of the vanilla calls that expire on the fixing dates, the payoff of each being deposited in the bank until T.

(ii) If $r = q = 0$, European calls of every expiry trade and the underlying can be bought or sold short at any date (the hypotheses of Proposition 22.15), then the arithmetic average-price call is worth at most the vanilla call $C(K,T)$.

(iii) If the geometric and the arithmetic average-price calls on the same fixing dates trade, the geometric one is worth at most the arithmetic one.

Proof. We compare payoffs path by path and apply the dominance lemma. The two payoff inequalities are finite forms of Jensen's inequality, and both are proved here directly.

(i). The positive part is subadditive and positively homogeneous: for real numbers a and a' and a constant $c > 0$,

$$(a+a')^+ \le a^+ + a'^+, \qquad (ca)^+ = c\,a^+.$$

For the first, $a + a' \le a^+ + a'^+$ because $a \le a^+$ and $a' \le a'^+$, and $0 \le a^+ + a'^+$; so the larger of $a + a'$ and 0 is at most $a^+ + a'^+$. The second is $\max(ca, 0) = c\max(a, 0)$. Applying subadditivity $J - 1$ times, and homogeneity with $c = 1/J$,

$$(\bar{S}_T - K)^+ = \Big(\frac{1}{J}\sum_{j=1}^{J}(S_{t_j} - K)\Big)^+ \le \frac{1}{J}\sum_{j=1}^{J}(S_{t_j} - K)^+ \tag{23.7}$$

on every path. The right side is the payoff of a strategy. For each j, buy today $e^{-r(T-t_j)}/J$ calls with strike K and expiry t_j, and at t_j deposit in the bank, until T, whatever they pay. The calls of expiry t_j pay $e^{-r(T-t_j)}(S_{t_j} - K)^+/J$ at t_j, and the deposit grows to $(S_{t_j} - K)^+/J$ at T; for $j = J$ the factor is 1 and nothing is deposited. The strategy trades at finitely many dates, adds or withdraws no money, costs the right side of (23.6), and by (23.7) pays at least as much as the Asian call in every state. Lemma 16.8(i) gives (23.6).

(ii). With $r = q = 0$ every discount factor in (23.6) equals 1, and $C(K,t_j) \le C(K,T)$ for every j by Proposition 22.15(i), which is model-free. An average of J numbers, each at most $C(K,T)$, is at most $C(K,T)$.

(iii). We prove the inequality between the geometric and the arithmetic mean from the tangent line of the logarithm. The function $\chi(y) = y - 1 - \ln y$ on $(0,\infty)$ has the derivative $1 - 1/y$, which

is negative on $(0,1)$ and positive on $(1,\infty)$. So χ has its minimum at $y=1$, where $\chi(1)=0$, and $\ln y \le y-1$ for every $y>0$. Taking $y=x/m$ with $x,m>0$,

$$\ln x \le \ln m + \frac{x-m}{m},$$

which says that the tangent to the logarithm at m lies above its graph. Take $m=\bar S_T$ and $x=S_{t_j}$, and average over j:

$$\ln G_T = \frac{1}{J}\sum_{j=1}^{J} \ln S_{t_j} \le \ln \bar S_T + \frac{1}{\bar S_T}\Big(\frac{1}{J}\sum_{j=1}^{J} S_{t_j} - \bar S_T\Big) = \ln \bar S_T,$$

because the bracket is zero by the definition of $\bar S_T$. Hence $G_T \le \bar S_T$ on every path. The function $x \mapsto (x-K)^+$ is nondecreasing, so $(G_T-K)^+ \le (\bar S_T - K)^+$ on every path, and Lemma 16.8(i) gives the inequality between the prices. □

Part (i) is the precise form of the saying that an option on an average is worth no more than the average of the options, and with (ii) it is the model-free reason why an Asian call costs no more than the vanilla call when $r=q=0$. Example 23.30 below puts numbers on all three parts, once the geometric Asian has a price.

Proposition 23.28 (Geometric Asian in closed form (Kemna and Vorst, 1990))**.** *Assume BSM, for an asset with a continuous yield q and cost of carry $b=r-q$, and let the J fixing dates be equally spaced, $t_j = jT/J$. Put*

$$\sigma_G^2 = \sigma^2\,\frac{(J+1)(2J+1)}{6J^2}, \qquad F_G = S\exp\Big((b-\tfrac12\sigma^2)T\,\frac{J+1}{2J} + \tfrac12\sigma_G^2 T\Big). \tag{23.8}$$

(i) Under $\mathbb{Q}$, the variable $\ln G_T$ is normal with mean $\ln S + (b-\frac12\sigma^2)T(J+1)/(2J)$ and variance $\sigma_G^2 T$, and $\mathbb{E}^{\mathbb{Q}}\,G_T = F_G$.

(ii) The geometric average-price call has the arbitrage-free price

$$e^{-rT}\big[F_G\,\Phi(d_1) - K\,\Phi(d_2)\big], \qquad d_{1,2} = \frac{\ln(F_G/K) \pm \sigma_G^2 T/2}{\sigma_G\sqrt{T}}, \tag{23.9}$$

where, in this proposition and its proof, d_1 and d_2 are formed with F_G and σ_G in the places of F and σ. This is the vanilla formula of Theorem 20.22 with the forward F_G and the volatility σ_G.

(iii) As $J\to\infty$, $\sigma_G \to \sigma/\sqrt{3}$ and $F_G \to S\exp\big((\frac12 b - \frac{1}{12}\sigma^2)T\big)$, and the price (23.9) converges to the same formula evaluated at these limits: the vanilla formula with the volatility $\sigma/\sqrt{3}$ and an adjusted forward.

That the limit in (iii) is the price of the continuously averaged contract would need Fact 20.13, and is not claimed.

Sketch of the proof. We find the law of $\ln G_T$ from a sum of independent normal increments. Under BSM the price is $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(G_T-K)^+]$ (Corollary 20.11). The random part of $\ln G_T$ is $(\sigma/J)\sum_j W_{t_j}$, where ΔW_k occurs for each $j \ge k$:

$$\operatorname{Var}\Big(\sum_j W_{t_j}\Big) = \operatorname{Var}\Big(\sum_{k=1}^{J}(J-k+1)\,\Delta W_k\Big) = \frac{T}{J}\sum_{k=1}^{J}(J-k+1)^2 = \frac{T(J+1)(2J+1)}{6},$$

the increments being independent and, for equally spaced dates, $\mathcal{N}(0,T/J)$; their weighted sum is normal. Times $(\sigma/J)^2$ this is $\sigma_G^2 T$, and the dates sum to $T(J+1)/2$, which gives the mean; the exponential of that mean plus half of $\sigma_G^2 T$ is F_G, the mean of the lognormal G_T: hence (i). The expectation is then the vanilla formula at a zero rate, (I.7): this is (ii), and (iii) is its continuity in (F_G,σ_G). The complete proof is in Appendix I.5. □

Example 23.29 (Effective volatility of a geometric average)**.** Let $\sigma = 30\%$. By (23.8), the factor $(J+1)(2J+1)/(6J^2)$ and the effective volatility σ_G are as follows.

fixings J	1	4	12	252	$J \to \infty$
factor	1	0.46875	0.37616	0.33532	1/3
σ_G	30.00%	20.54%	18.40%	17.37%	17.32%

One fixing is the vanilla option. With daily fixings the effective volatility is already within about 0.05 of a percentage point of its limit $30\%/\sqrt{3} = 17.32\%$.

Now let $S = K = 100$, $r = 5\%$, $q = 0$ and $T = 1$. The limit formula of Proposition 23.28(iii) has the forward $F_G = 100\, e^{0.025-0.0075} = 101.77$ and the log-variance $\sigma^2 T/3 = 0.03$, so $d_1 = 0.1876$ and $d_2 = 0.0144$, and its value is

$$e^{-0.05}\big[101.77\,\Phi(0.1876) - 100\,\Phi(0.0144)\big] = 7.50,$$

against 14.23 for the vanilla call: little more than half. With monthly fixings ($J = 12$) the forward is $F_G = 101.98$ and the log-variance $\sigma_G^2 T = 0.0339$, and by (23.9) the geometric average-price call is worth 8.02.

(Problem P23.6 works this computation at different numbers.)

Intuition. The weight of the k-th increment in $\ln G_T$ is $(J-k+1)/J$: a shock that arrives early moves every later fixing, and a shock that arrives late moves only the last ones. The weights fall linearly from 1 to $1/J$, and the mean square of a weight that falls linearly from 1 to 0 is $\int_0^1 y^2\,dy = \frac{1}{3}$. This is the whole content of "an Asian option has volatility $\sigma/\sqrt{3}$": averaging over the life of the option keeps one third of the variance.

Example 23.30 (The Asian sandwich)**.** Let $S = K = 100$, $\sigma = 30\%$, $T = 1$ and $r = q = 0$, with quarterly fixings ($J = 4$), so that every part of Proposition 23.27 applies.

The geometric Asian. By (23.8), the log-variance is $\sigma_G^2 T = 0.09 \times 0.46875 = 0.0422$, the mean term is $-0.045 \times \frac{5}{8} = -0.0281$, and

$$F_G = 100\, e^{-0.0281+0.0211} = 99.30, \qquad d_1 = 0.0685, \qquad d_2 = -0.1369.$$

So (23.9) gives $99.30\,\Phi(0.0685) - 100\,\Phi(-0.1369) = 7.81$.

The vanilla calls. The at-the-money calls expiring after 3, 6, 9 and 12 months are worth 5.98, 8.45, 10.34 and 11.92, and their average is 9.17.

The arithmetic Asian. A simulation of 200,000 paths (Procedure 23.36 in §23.5) gives 8.19, with standard error 0.031; on the same paths the control variate of Example 23.46, in §23.6, sharpens this to 8.175, with standard error 0.0013.

The four numbers are ordered as Proposition 23.27 requires:

$$\underbrace{7.81}_{\text{geometric}} \;\le\; \underbrace{8.175}_{\text{arithmetic}} \;\le\; \underbrace{9.17}_{\text{average of vanillas}} \;\le\; \underbrace{11.92}_{\text{vanilla}}.$$

The geometric lower bound is tight, within five percent of the arithmetic price; the two upper bounds are loose.

(Problem P23.6 works this computation at different numbers.)

Interview trap. "Asian options are cheaper because averaging lowers the expected price." The average of the forward prices of the fixing dates is of the same order as the forward price for T, and equal to it when $r = q$: in Example 23.30 the expected average is 100, exactly the forward, yet the Asian call costs 8.175 against 11.92. What averaging lowers is the variance: $\sigma_G^2 T$, which tends

to $\sigma^2 T/3$, against $\sigma^2 T$. Averaging also changes the hedge. Once j of the J fixings are known, only the fraction $(J-j)/J$ of the average still moves with the price, so the delta decays toward zero over the last fixings. An Asian option is therefore easier to hedge near expiry than a vanilla option, and harder to manipulate at settlement.

Interview trap. "The geometric Asian prices the arithmetic one." It is a lower bound, by the inequality between the two means (7.81 against 8.175 in Example 23.30), not an estimate. That it is a close and exactly priced lower bound, almost perfectly correlated with the arithmetic payoff, is precisely what makes it an excellent control variate (Proposition 23.45 in §23.6).

Remark 23.31 (The arithmetic Asian). The arithmetic average $\bar{S}_T$ is a sum of lognormal variables, and such a sum is not lognormal; this is the reason why no closed form is known for the arithmetic average-price call. Three routes are used in practice. The first fits a lognormal law to the first two moments of $\bar{S}_T$ and applies the formula of Theorem 20.22 with the fitted forward and volatility (Turnbull and Wakeman, 1991); the moment formulas are not reproduced here. The second simulates, with the geometric Asian of Proposition 23.28 as a control variate; this is done in §23.6 (Proposition 23.45). The third brackets the price between the bounds of Proposition 23.27.

Once the first $j < J$ fixings are known, they move the effective strike:

$$\bar{S}_T - K = \frac{J-j}{J}\Big(\frac{1}{J-j}\sum_{i=j+1}^{J} S_{t_i} - K_{\text{eff}}\Big), \qquad K_{\text{eff}} = \frac{JK - \sum_{i=1}^{j} S_{t_i}}{J-j}.$$

So the claim is $(J-j)/J$ average-price calls on the remaining fixings struck at K_{eff}. Since $K_{\text{eff}} - K = j(K - \frac{1}{j}\sum_{i=1}^{j} S_{t_i})/(J-j)$, the effective strike lies below K exactly when the observed fixings average more than K. If $K_{\text{eff}} \le 0$, the positive part never binds, because the mean of the remaining fixings is positive, and the claim pays $(J-j)/J$ times the bracket of the display: a linear payoff, with no optionality left. Nothing later in this section relies on this remark.

23.4 Exchange Options and Margrabe's Formula

The option to exchange one risky asset for another is the one two-asset exotic with a closed form. It sits inside stock-for-stock takeover offers, outperformance options and best-of payoffs, and its derivation is a standard whiteboard exercise: count the payoff in units of the asset given up, and the interest rate drops out. That derivation computes a risk-neutral expectation. A verification step then shows that holding the two prepaid forwards in the right ratios, with no bond at all, replicates the option, and this is what makes the formula a price.

Setup. The symbols of §23.1 are kept, with the following additions.

(a) There are two assets, with prices $S_{1,t}$ and $S_{2,t}$ at time t (first subscript the asset, second the time; today's prices are $S_{1,0}$ and $S_{2,0}$), continuous yields q_1 and q_2, and volatilities $\sigma_1 > 0$ and $\sigma_2 > 0$. They are driven by the standard Brownian motions W_1 and $W_2 = \rho W_1 + \sqrt{1-\rho^2}\, W_\perp$, where W_1 and $W_\perp$ are independent and $-1 \le \rho \le 1$ is the correlation (Proposition 19.52, Fact 19.7); $W_{k,t}$ is the value of the motion k at time t.

(b) The real-world paths are $S_{k,t} = S_{k,0}\exp(g_k t + \sigma_k W_{k,t})$, $k = 1, 2$, with constants g_1 and g_2, the expected log-growth rates of the two assets. The risk-neutral paths are those of Definition 20.6(ii), in which g_k is replaced by $r - q_k - \frac{1}{2}\sigma_k^2$; so under $\mathbb{Q}$, by Theorem 19.40 for each asset,

$$S_{k,T} = S_{k,0}\exp\big((r - q_k - \tfrac{1}{2}\sigma_k^2)T + \sigma_k W_{k,T}\big), \qquad k = 1, 2.$$

(c) The *two-asset BSM market* is the market in which both assets trade and pay their yields, and (A3)–(A6) of §20.1 hold. Its instruments are the reinvested holdings, or prepaid forwards, $\tilde{S}_{k,t} = S_{k,t}e^{-q_k(T-t)}$. They pay no income (Proposition 16.15(i)), they are instruments of Definition 20.1, being explicit functions of $(t, W_{1,t}, W_{\perp,t})$, and $\tilde{S}_{k,T} = S_{k,T}$.

(d) In this subsection σ without a subscript is the volatility of the ratio $S_{1,t}/S_{2,t}$ (Proposition 19.57):

$$\sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2.$$

It is assumed that $\sigma > 0$, which excludes only the case $\rho = 1$, $\sigma_1 = \sigma_2$.

(e) The *exchange option* pays $(S_{1,T} - S_{2,T})^+$ at T: it is the right to receive asset 1 in exchange for asset 2.

(f) For $x > 0$ and $0 \le t < T$,

$$\tilde V(x,t) = x\,\Phi\big(d_1(x,t)\big) - \Phi\big(d_2(x,t)\big), \qquad d_{1,2}(x,t) = \frac{\ln x \pm \sigma^2(T-t)/2}{\sigma\sqrt{T-t}}, \tag{23.10}$$

and $\tilde V(x,T) = (x-1)^+$. This is the price of Theorem 20.22, at a zero rate, of a call with the forward x, the strike 1 and the volatility σ. It will turn out to be the value of the exchange option counted in units of asset 2, as a function of the ratio of the two prepaid forwards. (This use of $\tilde V$ is local to the subsection. A tilde marks the same price in transformed variables; here the transformation changes the unit of account as well as the argument.)

(g) The candidate price function of the two prepaid-forward prices x_1 and x_2 is

$$V(x_1,x_2,t) = x_2\,\tilde V\Big(\frac{x_1}{x_2},\,t\Big), \qquad x_1, x_2 > 0, \quad 0 \le t \le T, \tag{23.11}$$

and ∂_1, ∂_2 denote differentiation in x_1, x_2. (The letter G is not used for this function, because G_T is the geometric average of §23.3.)

Theorem 23.32 (Margrabe's formula (Margrabe, 1978))**.** *In the two-asset BSM market of the Setup, with $\sigma > 0$, the exchange option has the arbitrage-free price*

$$S_{1,0}e^{-q_1T}\,\Phi(d_1) - S_{2,0}e^{-q_2T}\,\Phi(d_2) = V(\tilde S_{1,0}, \tilde S_{2,0}, 0), \tag{23.12}$$

with

$$d_{1,2} = \frac{\ln\big(S_{1,0}e^{-q_1T}/(S_{2,0}e^{-q_2T})\big) \pm \sigma^2T/2}{\sigma\sqrt{T}}.$$

This price equals $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(S_{1,T} - S_{2,T})^+]$, and no interest rate appears in it. The option is replicated by the strategy that holds, at each time $t < T$,

$$\Phi(d_1) \text{ units of } \tilde S_{1,t} \quad \text{and} \quad -\Phi(d_2) \text{ units of } \tilde S_{2,t}, \qquad \text{with } d_{1,2} = d_{1,2}(\tilde S_{1,t}/\tilde S_{2,t},\,t),$$

that is, $\Phi(d_1)$ units of the prepaid forward of asset 1 and $-\Phi(d_2)$ units of the prepaid forward of asset 2, and nothing in the bank at any time.

Sketch of the proof. We compute the risk-neutral expectation by factoring out the asset given up and applying a Gaussian shift, and then verify, by a replication that holds no bond, that this expectation is the price. Under $\mathbb{Q}$ the payoff is $S_{2,T}\,(S_{1,T}/S_{2,T} - 1)^+$, and the logarithms of the ratio and of $S_{2,T}$ form a jointly normal pair, the first with the variance σ^2T and free of the rate. Lemma 19.38(iii) takes the factor $S_{2,T}$ out of the expectation at the price of shifting the log ratio by its covariance with $\ln S_{2,T}$, which is $(\rho\sigma_1\sigma_2 - \sigma_2^2)T$:

$$\mathbb{E}^{\mathbb{Q}}\big[(S_{1,T} - S_{2,T})^+\big] = S_{2,0}e^{(r-q_2)T}\,\mathbb{E}\Big[\Big(\frac{\tilde S_{1,0}}{\tilde S_{2,0}}\,e^{-\sigma^2T/2+\sigma\sqrt{T}\,Z} - 1\Big)^+\Big],$$

with Z standard normal. By the lognormal expectation (I.7), the call formula of Theorem 20.22 read at a zero rate, which holds for every positive forward, strike and volatility and so needs $\sigma > 0$, the last mean is $\tilde V(\tilde S_{1,0}/\tilde S_{2,0}, 0)$, and the discount factor e^{-rT} removes the rate and leaves (23.12). For the replication, V of (23.11) is homogeneous of degree one in (x_1, x_2), with $\partial_1 V = \Phi(d_1)$ and

$\partial_2 V = -\Phi(d_2)$, so by Euler's identity (I.10), $x_1\,\partial_1 V + x_2\,\partial_2 V = V$, the stated holdings are worth V at every date, with nothing in the bank. The two-dimensional Itô lemma (Theorem 19.55) shows that the discounted value changes only by the gains of these holdings, the remaining terms adding up to $x_2[\partial_t \tilde{V} + \frac{1}{2}\sigma^2 x^2\,\partial_{xx}\tilde{V}]$ at $x = x_1/x_2$, which is zero by (I.8), the pricing equation of $\tilde{V}$ at a zero rate. The value tends to the payoff at T, and (A6) makes the initial capital the price. The appendix supplies the growth bounds and the limit in mean square. The complete proof is in Appendix I.6. □

(Problem P23.7 asks you to reproduce Steps 1 to 3 of the complete proof in Appendix I.6, with Lemma 19.38 and the lognormal expectation (I.7) given as tools, and, from its Step 4, to derive the deltas $\partial_1 V = \Phi(d_1)$ and $\partial_2 V = -\Phi(d_2)$ (I.12), Euler's identity (I.10) and the vanishing drift term of part (d) directly from the closed form $V = x_1\,\Phi(d_1) - x_2\,\Phi(d_2)$; the rest of Step 4, namely the growth conditions, the self-financing identity (I.14), the terminal value and the arbitrage argument, is left to that proof.)

Remark 23.33 (Pricing in units of an asset). The proof priced the payoff in units of asset 2 rather than in cash. In those units the exchange option is a call on the ratio $S_{1,T}/S_{2,T}$ struck at 1, and its replication holds the prepaid forward on asset 2 where the replication of a vanilla call holds the bond. This is the change of numéraire in its simplest form; the theory of martingales and numéraires is outside the scope of this document.

Remark 23.34 (Other multi-asset payoffs). For any two numbers, $\max(x_1, x_2) = x_2 + (x_1 - x_2)^+$ and $\min(x_1, x_2) = x_1 - (x_1 - x_2)^+$. Hence the *best-of* claim, which pays $\max(S_{1,T}, S_{2,T})$ at T, has the payoff of one prepaid forward on asset 2 plus one exchange option, and the *worst-of* claim, which pays $\min(S_{1,T}, S_{2,T})$, that of one prepaid forward on asset 1 minus one exchange option. By the law of one price (Lemma 16.8(iii)) and Proposition 16.15(i), in any arbitrage-free market in which these claims trade,

$$\text{best-of} = S_{2,0}e^{-q_2 T} + \text{exchange option}, \qquad \text{worst-of} = S_{1,0}e^{-q_1 T} - \text{exchange option}.$$

Baskets, spread options with a nonzero strike, rainbow options on more than two assets, quantos, cliquets (strips of forward-start options) and shout options are outside the scope of this document beyond this sentence; baskets are valued by matching moments or by simulation (§23.5).

Example 23.35 (Exchanging two assets at five correlations)**.** Let $S_{1,0} = S_{2,0} = 100$, $\sigma_1 = 30\%$, $\sigma_2 = 20\%$, $\rho = 0.5$ and $T = 1$, with no yields. Then, as in Example 19.58, $\sigma^2 = 0.07$ and $\sigma = 26.46\%$, while $x_0 = 1$ and $d_1 = -d_2 = \sigma\sqrt{T}/2 = 0.1323$. By Theorem 23.32 the exchange option is worth

$$100\,\Phi(0.1323) - 100\,\Phi(-0.1323) = 100 \times (0.5526 - 0.4474) = 10.52.$$

The at-the-money rule of Proposition 20.29, applied in units of asset 2 with the ratio volatility, gives $0.4 \times 100 \times 0.2646 = 10.58$. The replicating portfolio at time 0 holds $\Phi(0.1323) = 0.5526$ units of asset 1 and $-\Phi(-0.1323) = -0.4474$ units of asset 2, worth $55.26 - 44.74 = 10.52$, with no cash: this is Euler's identity $x_1\,\partial_1 V + x_2\,\partial_2 V = V$ (I.10) in numbers. So is the pricing equation of $\tilde{V}$ at a zero rate, $\partial_t \tilde{V} + \frac{1}{2}\sigma^2 x^2\,\partial_{xx}\tilde{V} = 0$ (I.8): at $x = 1$ and $t = 0$, $\partial_t \tilde{V} = -0.0523$ and $\frac{1}{2}\sigma^2 x^2\,\partial_{xx}\tilde{V} = 0.035 \times 1.4947 = +0.0523$, so the time decay of the option, counted in units of asset 2, is exactly paid for by the gamma of the ratio.

Correlation. With the other inputs unchanged:

ρ	0.9	0.8	0.5	0	−0.5
ratio volatility σ	14.83%	18.44%	26.46%	36.06%	43.59%
exchange option	5.91	7.35	10.52	14.31	17.25

The ratio volatilities at $\rho = 0.9$, 0.5, 0 and -0.5 are those of Example 19.58, given here to two decimals; the one at $\rho = 0.8$ is $\sqrt{0.13 - 0.096} = 18.44\%$. The more the two assets move together, the less the right to swap them is worth.

The rate and the yields. A simulation of 4,000,000 pairs of risk-neutral terminal prices at $r = 5\%$ returns 10.527, with standard error 0.009, against 10.524 from the formula, in which r does not appear. With the normals held fixed, the discounted payoff does not depend on r at all, path by path: both terminal prices carry the factor e^{rT}, and the discount removes it. Yields do matter: with $S_{1,0} = 105$ and $q_1 = 2\%$ the ratio is $x_0 = 105\, e^{-0.02}/100 = 1.0292$, so $d_1 = 0.2411$ and $d_2 = -0.0235$, and the option is worth 12.20.

Best-of. By Remark 23.34, the claim that pays the better of the two assets at T is worth $100 + 10.52 = 110.52$ today, and the claim that pays the worse of the two $100 - 10.52 = 89.48$.

(Problem P23.7 works this computation at different numbers.)

Interview trap. Using $\sigma_1 - \sigma_2$ or $\sigma_1 + \sigma_2$ as the volatility of the exchange. The variance of the ratio is $\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$; it equals $(\sigma_1 - \sigma_2)^2$ only at $\rho = 1$ and $(\sigma_1 + \sigma_2)^2$ only at $\rho = -1$. The option gets *cheaper* as the correlation rises: 14.31 at $\rho = 0$ and 5.91 at $\rho = 0.9$ in Example 23.35.

Interview trap. "There is no r in the formula, so carry does not matter." The rate is absent because no position in the hedge earns or pays it: the strategy of Theorem 23.32 holds only $\tilde{S}_{1,t}$ and $\tilde{S}_{2,t}$. The yields q_1 and q_2 stay, through those prepaid forwards.

Interview trap. "It is a call on asset 1 struck at the forward price of asset 2, priced with σ_1." The strike of that call would be random, and correlated with the underlying. The correct reduction counts the payoff in units of asset 2: there it is $(x - 1)^+$ in the ratio $x = \tilde{S}_{1,T}/\tilde{S}_{2,T}$ of the two prepaid forwards at T, a ratio whose volatility is σ and whose value today is $x_0 = \tilde{S}_{1,0}/\tilde{S}_{2,0}$. The yields enter only through x_0, and no bond appears in the hedge.

23.5 Monte Carlo Valuation

When no closed form exists, risk-neutral valuation becomes an average: simulate the underlying under $\mathbb{Q}$ and average the discounted payoffs. The error of that average is the error of a sample mean, so everything about it follows from §4 and §10: the estimate is unbiased, its standard error is the sample standard deviation of the payoffs divided by $\sqrt{n}$, and its rate of convergence does not depend on the dimension of the problem. Interviewers ask for the error bar ("how many paths for a cent?") and for Greeks that are not drowned in noise.

Setup. The symbols of §23.1 are kept. In addition:

(a) The quantity to be estimated is $V = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\text{payoff}]$, the discounted risk-neutral expectation of a payoff made at T. By the pricing rule of §23.1 it is the price today of the claim, for every payoff for which that rule is proved. In this subsection V without arguments is that number, and $V(S, 0)$ is the same quantity as a function of the spot.

(b) A simulation draws n independent paths (n is a sample size, as in §4). Each path has J time steps of length $\Delta t = T/J$, with the grid dates $k\,\Delta t$, $k = 1, \ldots, J$.

(c) The discounted payoff of the path i is X_i. The variables $X_1, \ldots, X_n$ are independent and identically distributed, with mean V and variance $\sigma_X^2 < \infty$; here σ_X is the standard deviation of one discounted payoff, not a volatility.

(d) The *Monte Carlo estimate* is the sample mean $\hat{V}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$, and s is the Bessel-corrected sample standard deviation of Theorem 4.5: $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \hat{V}_n)^2$.

(e) A *bump* of the spot is written ΔS, an increment; it is not called h, which is the hedge ratio of §16. In Proposition 23.39 and Example 23.40, Y is the discounted payoff of one simulated path at the spot S, and X that of one simulated path at the spot $S + \Delta S$, the two being computed either from independent draws or from the same normal variables, as stated there.

(f) The triangular factor of a correlation matrix receives no letter of its own, because L is the leverage factor of this Part. Its entries are written γ_{ik}, those of the correlation matrix ρ_{ik}, and those of a general symmetric matrix a_{ik}.

Procedure 23.36 (Risk-neutral Monte Carlo)**.** To estimate $V = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[\text{payoff}]$ under BSM:

1. *Draw.* Generate independent standard normal variables: one for each path if the payoff depends on S_T alone, and otherwise one for each time step of each path, and of each asset.
2. *Build the paths.* For a payoff that depends on S_T alone, set

$$S_T = S\exp\left((b - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z\right)$$

in one step, which is exact (Theorem 19.40). For a path-dependent payoff, step the *logarithm* of the price along the grid, with a new normal Z_k for the step k:

$$\ln S_{k\Delta t} = \ln S_{(k-1)\Delta t} + (b - \tfrac{1}{2}\sigma^2)\,\Delta t + \sigma\sqrt{\Delta t}\,Z_k, \qquad k = 1, \dots, J.$$

For constant parameters this is exact in law at the grid dates, unlike an Euler step on the price itself. For two assets with correlation ρ, draw two independent normals Z and Z' for each step, and drive the first asset with Z and the second with $\rho Z + \sqrt{1-\rho^2}\,Z'$ (Proposition 19.52). For more than two assets, multiply the vector of independent normals by a lower-triangular matrix whose product with its own transpose is the correlation matrix. That factor is built column by column. Its first column is the first column of the matrix divided by the square root of the *pivot*, the $(1,1)$ entry. The outer product of that column with itself is then subtracted from the matrix, which empties its first row and column, and the construction is repeated on the block that remains.
3. *Evaluate.* Compute the payoff on each path and discount it: this is X_i.
4. *Report.* Report the estimate $\hat{V}_n$ together with its standard error $s/\sqrt{n}$.

Three claims stand behind the procedure. The simulated path has, step by step, the law of the risk-neutral path at the grid dates, so the procedure estimates V; for a payoff of $\mathcal{G}_J$ of one asset on those dates, V is the arbitrage-free price (Corollary 20.11). By the bilinearity of the covariance, the triangular factor turns independent standard normals into standard normals with the prescribed correlations. By induction on the dimension, the factor exists exactly when the matrix is positive semidefinite, a zero pivot then giving a zero column. Their proofs are in Appendix I.7.

Interview trap. Stepping the price itself, $S_{t+\Delta t} = S_t(1 + b\,\Delta t + \sigma\sqrt{\Delta t}\,Z)$ (an Euler step), instead of its logarithm. The Euler step replaces a lognormal factor by a normal one. It changes the law of the price at every finite Δt, which is a discretisation bias, and it produces a negative price whenever $Z < -(1 + b\,\Delta t)/(\sigma\sqrt{\Delta t})$. With $\sigma = 60\%$, $b = 0$ and a single annual step this has probability $\Phi(-1.667) = 4.8\%$. The log step of Procedure 23.36 has neither defect when the parameters are constant.

Proposition 23.37 (Monte Carlo error)**.** *Let $X_1, \dots, X_n$, with $n \geq 2$, be independent and identically distributed with mean V and variance $\sigma_X^2 < \infty$, and let $\hat{V}_n$ and s be as in the Setup.*

(i) The estimate is unbiased: $\mathbb{E}\,\hat{V}_n = V$.

(ii) Its variance is $\operatorname{Var}\hat{V}_n = \sigma_X^2/n$, so its standard deviation is $\sigma_X/\sqrt{n}$. Moreover $\mathbb{E}\,s^2 = \sigma_X^2$, so $s/\sqrt{n}$ estimates that standard deviation.

(iii) If $\sigma_X > 0$, then $P(|\hat{V}_n - V| \leq 1.96\,\sigma_X/\sqrt{n}) \to 2\,\Phi(1.96) - 1$ as $n \to \infty$, a limit equal to 0.9500 to four decimals. If moreover $\mathbb{E}\,X_1^4 < \infty$, the same holds with the estimate s in the place of the unknown σ_X:

$$P(|\hat{V}_n - V| \leq 1.96\,s/\sqrt{n}) \to 2\,\Phi(1.96) - 1,$$

so $\hat{V}_n \pm 1.96\,s/\sqrt{n}$ is an approximate 95% confidence interval for V.

The rate $n^{-1/2}$ does not mention the dimension of the simulated path, that is, the number of time steps or of assets: only the constant σ_X depends on the problem.

Sketch of the proof. We compute the mean and the variance of a sample mean, apply the central limit theorem, and pass from σ_X to s by a sandwich. Linearity of expectation gives (i). Independent payoffs have zero covariances, so their sum has the variance $n\sigma_X^2$ and

$$\operatorname{Var} \hat{V}_n = \frac{n\sigma_X^2}{n^2} = \frac{\sigma_X^2}{n};$$

with $\mathbb{E}\, s^2 = \sigma_X^2$ (Theorem 4.5) this is (ii). For (iii), Fact 4.8, applied to the error standardised by $\sigma_X > 0$, gives the limit $2\,\Phi(1.96) - 1$, and likewise with any positive number for 1.96. By Chebyshev's inequality, $\hat{V}_n \to V$ and, the fourth moment being finite, the mean of the squares X_i^2 tends to $\mathbb{E}\, X_1^2$, both in probability; since s^2 is $n/(n-1)$ times the difference between that mean and $\hat{V}_n^2$, it follows that $s/\sigma_X \to 1$. Outside an event of vanishing probability, s lies between two multiples of σ_X as close to it as we wish, and the continuity of Φ closes the sandwich. The complete proof is in Appendix I.8. □

The hypothesis $\mathbb{E}\, X_1^4 < \infty$ of (iii) holds for every discounted payoff $X_1 = e^{-rT} f(S_{t_1}, \dots, S_{t_J})$ with $f \in \mathcal{G}_J$ under BSM. Each S_{t_j} is $S \exp((b - \frac{1}{2}\sigma^2)t_j + \sigma y_j)$ at $y_j = W_{t_j}$; a function of moderate growth in the prices is a function of exponential growth in $(y_1, \dots, y_J)$ (Lemma 19.20(i)), and such a function, evaluated at Brownian values, has finite moments of every order (Lemma 19.20(iii)). The interval with s is the plug-in interval of §4.4, built in the same way as the interval of Example 4.15; part (iii) is what justifies its nominal level when n is large.

Example 23.38 (How many paths for a cent)**.** A simulation of $n = 10{,}000$ paths returns the sample standard deviation $s = 14.7$ for the discounted payoff. The standard error of the estimate is $14.7/\sqrt{10{,}000} = 0.147$, and the approximate 95% interval is $\hat{V}_n \pm 1.96 \times 0.147 = \hat{V}_n \pm 0.288$. A standard error of one cent, 0.01, needs

$$n = \Big(\frac{14.7}{0.01}\Big)^2 = 2{,}160{,}900$$

paths: about 2.16 million, or 216 times as many. Every extra decimal digit of accuracy costs 100 times the paths.

(Problem P23.8 works this computation at different numbers.)

Interview trap. "To halve the error, double the paths." Quadruple them: the error falls like $n^{-1/2}$ (Proposition 23.37(ii)).

Interview trap. "More time steps reduce the Monte Carlo error." More time steps reduce the discretisation bias, not the sampling error, which is $\sigma_X/\sqrt{n}$ whatever J is. Under constant parameters the log step is exact at the grid dates, so for a payoff fixed on those dates there is no bias to reduce. A barrier or a lookback simulated on J dates, however, is monitored discretely, and the difference from the continuously monitored contract is a bias, not noise. For a single-barrier down call or up put, knock-in or knock-out, its size is the one that Fact 23.21 describes (shift the barrier with $\Delta t = T/J$); for a lookback, Remark 23.26 gives its sign (the discretely monitored lookback is never dearer). In Example 23.22, paths with 126 daily steps price the knock-out at 4.61 however many of them are drawn, against 4.25 for continuous monitoring.

Proposition 23.39 (Common random numbers for bump-and-revalue Greeks)**.** *Let $\Delta S > 0$. Let Y be the discounted payoff of a claim on one simulated path at the spot S, and X that of one simulated path at the spot $S + \Delta S$, the two paths being computed either from independent draws, as in (i), or from the same normal variables, as in (ii); both payoffs have finite variance. The* bump-and-revalue *estimator of the delta from this pair of runs is $(X - Y)/\Delta S$, and*

$$\operatorname{Var}\Big(\frac{X - Y}{\Delta S}\Big) = \frac{\operatorname{Var} X + \operatorname{Var} Y - 2\operatorname{Cov}(X, Y)}{(\Delta S)^2}. \tag{23.13}$$

(i) If the two runs use independent draws, the covariance vanishes and the variance equals $(\operatorname{Var} X + \operatorname{Var} Y)/(\Delta S)^2 \geq \operatorname{Var} Y/(\Delta S)^2$*. Hence, if* $\operatorname{Var} Y > 0$ *(a number that does not depend on* ΔS*), the variance is at least* $\operatorname{Var} Y\,(\Delta S)^{-2}$ *and tends to infinity as* $\Delta S \to 0$*.*

*(ii) If the two runs use the same normal variables (*common random numbers*) and the claim is a European call under BSM, the variance is at most* $e^{-2rT}\,\mathbb{E}^{\mathbb{Q}}[(S_T/S)^2] = e^{(\sigma^2-2q)T}$*, for every* ΔS*.*

Proof. We apply the formula for the variance of a sum and, for the bound, compare the two payoffs path by path. Proposition 10.10 applied to the sum of X and $-Y$, with $\operatorname{Var}(-Y) = \operatorname{Var} Y$ and $\operatorname{Cov}(X,-Y) = -\operatorname{Cov}(X,Y)$, gives $\operatorname{Var}(X-Y) = \operatorname{Var} X + \operatorname{Var} Y - 2\operatorname{Cov}(X,Y)$, and dividing by $(\Delta S)^2$ gives (23.13).

(i). Independent variables with finite variances have zero covariance (the product rule of Lemma 19.17, as in the proof of Proposition 23.37(ii)), so (23.13) becomes $(\operatorname{Var} X + \operatorname{Var} Y)/(\Delta S)^2$, which is at least $\operatorname{Var} Y/(\Delta S)^2$ because $\operatorname{Var} X \geq 0$. The variance of Y is that of the payoff at the spot S, whatever the bump; if it is positive, $\operatorname{Var} Y\,(\Delta S)^{-2}$ tends to infinity as $\Delta S \to 0$.

(ii). With the same normal Z in both runs, the two terminal prices are

$$(S+\Delta S)\exp\left((b-\tfrac12\sigma^2)T+\sigma\sqrt{T}\,Z\right) = S_T\,\frac{S+\Delta S}{S} \qquad \text{and} \qquad S_T = S\exp\left((b-\tfrac12\sigma^2)T+\sigma\sqrt{T}\,Z\right),$$

which differ by $S_T\,\Delta S/S$. The positive part does not increase distances: $|a^+ - a'^+| \leq |a-a'|$ for real a, a'. Indeed, if $a \geq a'$ then $0 \leq a^+ - a'^+ \leq a - a'$, as the three cases $a' \geq 0$, $a \geq 0 > a'$ and $0 > a$ show, and the other order is symmetric. So, on every path,

$$|X-Y| = e^{-rT}\left|\Big(S_T\,\frac{S+\Delta S}{S} - K\Big)^+ - (S_T-K)^+\right| \leq e^{-rT}\,\Delta S\,\frac{S_T}{S}.$$

A variance is at most a second moment, so

$$\operatorname{Var}\Big(\frac{X-Y}{\Delta S}\Big) \leq \frac{\mathbb{E}[(X-Y)^2]}{(\Delta S)^2} \leq e^{-2rT}\,\mathbb{E}^{\mathbb{Q}}[(S_T/S)^2],$$

and $\mathbb{E}^{\mathbb{Q}}[(S_T/S)^2] = e^{(2b-\sigma^2)T}\,\mathbb{E}\,e^{2\sigma\sqrt{T}Z} = e^{(2b+\sigma^2)T}$ by Lemma 18.35(i). With $b = r-q$ the bound is $e^{(\sigma^2-2q)T}$. □

Example 23.40 (A bumped delta with and without common numbers)**.** Let $S = K = 100$, $\sigma = 20\%$, $r = q = 0$, $T = 1$ and $\Delta S = 1$, for a European call. The discounted payoff has the standard deviation 13.15 at the spot 100 and 13.61 at the spot 101, and under common random numbers the correlation of the two payoffs is 0.9996. By (23.13), the variance of $X - Y$ for one pair of runs is $13.61^2 + 13.15^2 = 358$ with independent draws and 0.35 with common ones, a thousand times less; the bound of Proposition 23.39(ii) is $e^{0.04} = 1.04$. With $n = 10{,}000$ paths the standard error of the estimated delta is $\sqrt{358/10{,}000} = 0.19$ in the first case, which is useless for a delta of 0.54, and $\sqrt{0.35/10{,}000} = 0.006$ in the second. The estimator has the expectation $(8.5153 - 7.9656)/1 = 0.5497$, about 0.550: the forward difference of the two prices. It exceeds the BSM delta $\Phi(0.1) = 0.5398$ by 0.0099, which is about $\Delta S \times \Gamma/2 = 0.0099$, the gamma being $\varphi(0.1)/20 = 0.0198$: this is a bias of the bump, not of the sampling.

A central difference, $\big(V(S+\Delta S,0) - V(S-\Delta S,0)\big)/(2\Delta S)$ with common numbers for both runs, removes that first-order bias: its expectation is $(8.51527 - 7.43571)/2 = 0.53978$, against the delta 0.53983, and its variance for one pair of runs is 0.345, as low as before. For the gamma, the pathwise route of Remark 23.41 below fails. Along a path, $S_T/S = \exp((b-\frac12\sigma^2)T + \sigma\sqrt{T}\,Z)$ does not depend on the spot, so the pathwise delta of the call, $e^{-rT}\,\mathbf{1}\{S_T > K\}\,S_T/S$, has the derivative zero in S wherever it is differentiable: the obstacle that the remark meets for the delta of a digital. The gamma is taken instead from the central second difference with common numbers, or from a likelihood-ratio weight (named here, not derived).

(Problem P23.8 works this computation at different numbers.)

Interview trap. Bumping with independent random numbers. The variance of the estimated delta then grows like $(\Delta S)^{-2}$, and the Greek is drowned in noise (a standard error of 0.19 on a delta of 0.54 in Example 23.40). The base run and the bumped run must reuse the same normal variables.

Remark 23.41 (Pathwise and likelihood-ratio Greeks). Two estimators of the delta need no bump. Let $f \in \mathcal{G}$ be a European payoff under BSM, let $S_T = S\exp((b-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,Z)$ be the simulated terminal price, and let $V(S,0) = e^{-rT}\,\mathbb{E}\,f(S_T)$ be the price as a function of the spot, which is differentiable (Corollary 20.41(i)).

The pathwise estimator differentiates the payoff along the path. It is $e^{-rT}f'(S_T)\,S_T/S$; for a call, $e^{-rT}\,\mathbf{1}\{S_T > K\}\,S_T/S$. It is unbiased for the delta when f is Lipschitz, $|f(x)-f(x')| \le c\,|x-x'|$, and continuously differentiable except at finitely many points, as calls, puts and spreads are. Indeed, with common normals the bumped terminal price is $S_T(S+\Delta S)/S$, so for a bump $\Delta S \neq 0$

$$\frac{V(S+\Delta S,0)-V(S,0)}{\Delta S} = e^{-rT}\,\mathbb{E}\Big[\frac{f\big(S_T(S+\Delta S)/S\big)-f(S_T)}{\Delta S}\Big].$$

The variable S_T has a density, so with probability one it avoids the finitely many points at which f is not differentiable, and then the quotient inside the expectation tends to $f'(S_T)\,S_T/S$ as $\Delta S \to 0$. The quotients are bounded in absolute value by $c\,S_T/S$, whose fourth moment is finite (Lemma 18.35(i)), and so is the limit: $|f'(S_T)\,S_T/S| \le c\,S_T/S$, because $|f'| \le c$ wherever f is differentiable, by the Lipschitz bound. Convergence with probability one implies convergence in probability (Lemma 19.12(iii)). Along any sequence of bumps that tends to zero, Lemma 19.12(iv) therefore passes the limit through the expectation, and the left side tends to the delta: $\partial_S V(S,0) = e^{-rT}\,\mathbb{E}[f'(S_T)\,S_T/S]$. (A general Lipschitz function is differentiable only almost everywhere, by Rademacher's theorem, which is not used here.) The pathwise estimator is useless for a digital: the derivative of the payoff is zero wherever it exists, so the estimator is identically zero, while the delta is not.

The likelihood-ratio estimator differentiates the density instead of the payoff. It is

$$e^{-rT}\,f(S_T)\,\frac{Z}{S\sigma\sqrt{T}},$$

and it is unbiased for every $f \in \mathcal{G}$, discontinuous payoffs included. Indeed, replacing S by $Se^{\theta\sigma\sqrt{T}}$ has the same effect on S_T as replacing Z by $Z+\theta$. Put $\psi(z) = f\big(S\exp((b-\frac{1}{2}\sigma^2)T+\sigma\sqrt{T}\,z)\big)$, which is continuous except at finitely many points and has exponential growth, $|\psi(z)| \le ce^{c|z|}$ (Lemma 19.20(ii)). So ψ is a test function in the sense of the Setup of §19.6, a class that admits functions of either sign, and the Gaussian shift of Lemma 19.38(i) applies to it, both means being finite. Read from right to left, it gives

$$V\big(Se^{\theta\sigma\sqrt{T}},0\big) = e^{-rT}\,\mathbb{E}\,\psi(Z+\theta) = e^{-rT}\,\mathbb{E}\big[e^{\theta Z-\theta^2/2}\,\psi(Z)\big],$$

so that the difference quotient of the left side in θ, at $\theta = 0$, is $e^{-rT}\,\mathbb{E}\big[\psi(Z)\,(e^{\theta Z-\theta^2/2}-1)/\theta\big]$. As $\theta \to 0$ the quotient $(e^{\theta Z-\theta^2/2}-1)/\theta$ tends to Z. By the mean value theorem it equals $(Z-\theta^*)\,e^{\theta^* Z-(\theta^*)^2/2}$ for some θ^* between 0 and θ, so for $|\theta| \le 1$ it is at most $(|Z|+1)\,e^{|Z|} \le e^{2|Z|}$ in absolute value. The fourth moments of $\psi(Z)$ times the quotient are therefore at most $c^4\,\mathbb{E}\,e^{(4c+8)|Z|}$, which is finite (Lemma 18.35(i)); the limit satisfies $|Z\,\psi(Z)| \le c\,e^{(c+1)|Z|}$, whose fourth moment is finite for the same reason. Along any sequence of shifts that tends to zero, Lemma 19.12(iv) gives the derivative at $\theta = 0$: it is $e^{-rT}\,\mathbb{E}[Z\,\psi(Z)] = e^{-rT}\,\mathbb{E}[Z\,f(S_T)]$. By the chain rule the same derivative is $\partial_S V(S,0)\cdot S\sigma\sqrt{T}$. Hence $\partial_S V(S,0) = e^{-rT}\,\mathbb{E}[f(S_T)\,Z]/(S\sigma\sqrt{T})$. The factor $Z/(S\sigma\sqrt{T})$ is the derivative in S of the logarithm of the density of $\ln S_T$, the *score*.

The price of this generality is variance. For the call of Example 23.40, both estimators have the mean 0.540, and the standard deviation on one path is 0.59 for the pathwise estimator and 1.34 for the likelihood-ratio estimator.

Remark 23.42 (American options by simulation). A forward simulation does not know the exercise rule of an American option, because that rule is decided backward in time: whether to exercise at a date depends on the value of continuing, which depends on the later decisions. Regressing the realised continuation values on functions of the state at each exercise date, working backward from maturity, gives a workable exercise rule and an estimate of the price (Longstaff and Schwartz, 2001). The method is named here and not derived. In one or two dimensions, trees (Procedure 18.16) and finite differences (§23.7) remain the default.

Interview trap. "Monte Carlo cannot price American options." A plain forward simulation cannot, because it does not know when to exercise; simulation with a regression estimate of the continuation value can (Remark 23.42).

23.6 Variance Reduction

Because the error of a simulation falls only like $n^{-1/2}$ (Proposition 23.37), the cheap route to accuracy is not more paths but less variance per path. Antithetic pairs and control variates are the two techniques that candidates are asked to derive, together with the cases in which each of them fails. Both reduce to the covariance algebra of §10: the variance of a sum (Proposition 10.10) and the regression residual (Proposition 10.15).

Setup. The symbols of §23.1 and §23.5 are kept. In addition:

(a) The variable Z is a standard normal variable or, for a path, a vector $Z = (Z_1, \dots, Z_m)$ of m independent standard normal variables: the normals that drive one simulated path in Procedure 23.36. The vector $-Z$ has every coordinate of Z with its sign changed, and Z' is an independent copy of Z.

(b) In this subsection f is the discounted payoff of the claim *written as a function of the normals of the path*, so that $f(Z)$ is the variable X_i of §23.5 for one path. It satisfies $\mathbb{E}\, f(Z)^2 < \infty$. Its mirror image is $\tilde{f}(z) = f(-z)$, and the *antithetic correlation* is $\rho_A = \operatorname{Corr}(f(Z), f(-Z))$.

(c) A function on $\mathbb{R}^m$ is *nondecreasing in each coordinate* if raising one coordinate, the others being fixed, never lowers its value, and *nonincreasing in each coordinate* if that never raises its value. It has *exponential growth* (Definition 19.19(i)) if its absolute value at z is at most $c\, e^{c(|z_1|+\cdots+|z_m|)}$, for a constant $c \geq 0$. In the proof of Proposition 23.43, ψ_1 and ψ_2 are two such functions, and $\bar{\psi}_1$, $\bar{\psi}_2$ are their means over the last normal (the bar of Lemma 19.17).

(d) For control variates, X is the discounted payoff whose mean is wanted (the *target*) and Y is a second discounted payoff computed on the same path (the *control*), whose mean $\mathbb{E}\, Y$ is *known* in closed form. Both variances are finite and positive, $0 < \operatorname{Var} X < \infty$ and $0 < \operatorname{Var} Y < \infty$. In this subsection $\rho = \operatorname{Corr}(X, Y)$, not the correlation of two assets.

(e) The *control-variate coefficient* is a real number β, with optimal value β^*. *Notice:* this β is a regression coefficient, the β^* of Proposition 10.15. It is not the CAPM beta, which this section does not use, and not the diffusion coefficient β_t of an Itô process.

(f) For importance sampling, θ is a real shift, the exponent of Lemma 19.38.

An *antithetic pair* is the pair of payoffs $f(Z)$ and $f(-Z)$ computed from one draw of the normals. A simulation with n pairs reports the mean of the n *pair averages* $\frac{1}{2}(f(Z) + f(-Z))$, and costs $2n$ payoff evaluations.

Proposition 23.43 (Antithetic variates)**.** *Let* $\mathbb{E}\, f(Z)^2 < \infty$ *and* $\operatorname{Var} f(Z) > 0$ *(a payoff that is constant with probability one needs no simulation), and let* $\rho_A = \operatorname{Corr}(f(Z), f(-Z))$.

(i) The pair average is unbiased for $\mathbb{E}\, f(Z)$*, and*

$$\operatorname{Var}\Big(\frac{f(Z)+f(-Z)}{2}\Big) = \tfrac{1}{2}\Big(\operatorname{Var} f(Z) + \operatorname{Cov}\big(f(Z), f(-Z)\big)\Big) = \frac{\operatorname{Var} f(Z)}{2}\,(1+\rho_A).$$

(ii) The average of two independent draws, $\frac{1}{2}(f(Z)+f(Z'))$*, has the variance* $\operatorname{Var} f(Z)/2$*. Hence, for the same two payoff evaluations, the antithetic pair has the smaller variance if and only if* $\rho_A < 0$*.*

(iii) If Z *is a scalar and* f *is monotone (nondecreasing, or nonincreasing), then* $\rho_A \le 0$*. If* $Z = (Z_1, \dots, Z_m)$ *is a vector of independent standard normal variables and* f *has exponential growth and is nondecreasing in each coordinate, or nonincreasing in each coordinate, then again* $\rho_A \le 0$*. An Asian call and a discretely monitored fixed-strike lookback call, simulated under BSM by Procedure 23.36, are such functions of the normals of their path.*

Sketch of the proof. We compute the variance of a sum, and obtain the sign of the covariance by comparing two independent draws, first for one normal and then, by induction on the dimension, for a vector. Since $-Z$ has the law of Z, the payoff $f(-Z)$ has the mean and the variance of $f(Z)$: the pair average is unbiased, and by Proposition 10.10 the sum $f(Z)+f(-Z)$ has the variance $2\operatorname{Var} f(Z) + 2\operatorname{Cov}(f(Z), f(-Z))$, which gives (i); for two independent draws the covariance is zero, which gives (ii). For (iii) with one normal and f nondecreasing, $\tilde f$ is nonincreasing, so between any two points the increments of f and of $\tilde f$ have a product that is never positive. Taking its mean at Z and the independent copy Z', where the mixed products factor,

$$0 \ge \mathbb{E}\Big[\big(f(Z)-f(Z')\big)\big(\tilde f(Z)-\tilde f(Z')\big)\Big] = 2\,\mathbb{E}\big[f(Z)\tilde f(Z)\big] - 2\,\mathbb{E}\,f(Z)\;\mathbb{E}\,\tilde f(Z) = 2\operatorname{Cov}\big(f(Z), f(-Z)\big),$$

so $\rho_A \le 0$; the four products are integrable because $\mathbb{E}\,f(Z)^2 < \infty$. With two nondecreasing functions the same comparison gives $\operatorname{Cov}(\psi_1(Z), \psi_2(Z)) \ge 0$ for one normal, and induction on m extends this association inequality, (I.17), to functions of exponential growth that are nondecreasing in each coordinate: freeze the first $m-1$ normals (Lemma 19.17), apply the one-dimensional case to the last one, and then the induction hypothesis to $\bar\psi_1$ and $\bar\psi_2$, which are again nondecreasing in each coordinate; exponential growth keeps every mean finite. With $\psi_1 = f$ and $\psi_2 = -\tilde f$ the inequality reads $\operatorname{Cov}(f(Z), f(-Z)) \le 0$. The two path payoffs qualify because every grid price of Procedure 23.36, hence the average of the prices, their maximum and a call payoff on either, is nondecreasing in each normal and of exponential growth. The complete proof is in Appendix I.9. □

Example 23.44 (Antithetic correlations of six payoffs and of an Asian call)**.** Let $S = 100$, $\sigma = 20\%$, $r = q = 0$ and $T = 1$, so that one normal Z gives $S_T = 100\exp(-0.02 + 0.2\,Z)$. The antithetic correlations of six payoffs, computed by numerical integration against the normal density, and the factor $1+\rho_A$ by which the variance of a pair differs from that of two independent draws (Proposition 23.43(i)–(ii)), are as follows.

payoff	ρ_A	factor $1+\rho_A$
call, $K = 100$	-0.367	0.633
put, $K = 100$	-0.584	0.416
digital, $K = 100$	-0.852	0.148
call, $K = 70$ (deep in the money)	-0.928	0.072
call, $K = 130$ (far out of the money)	-0.044	0.956
straddle, $K = 100$	$+0.899$	1.899

The first five payoffs are monotone in Z, so their correlations are nonpositive by the proposition; here all five are negative. The gain is large when the payoff is close to linear in Z (the call struck at 70 is almost a forward) and negligible when the payoff vanishes on most paths (the call struck at 130 pays only when $Z > 1.41$, on 7.9% of them, and its mirror image, which pays only when $Z < -1.41$, never pays on the same draw). The digital can be checked by hand. It pays when $Z > 0.1$, with probability $\Phi(-0.1) = 0.4602$, and its mirror image pays when $Z < -0.1$; the two never pay together, so the covariance is $-\Phi(-0.1)^2$, the variance is $\Phi(-0.1)\,\Phi(0.1)$, and

$\rho_A = -\Phi(-0.1)/\Phi(0.1) = -0.852$. The straddle, $|S_T - 100|$, is not monotone: it is large for large Z and for very negative Z alike, so $f(Z)$ and $f(-Z)$ are large together, $\rho_A = +0.899$, and antithetic sampling nearly doubles the variance (a factor of 1.90).

A path payoff. For the quarterly arithmetic Asian call of Example 23.30 ($S = K = 100$, $\sigma = 30\%$, $r = q = 0$, $T = 1$, four normals per path) the payoff is nondecreasing in each normal. A simulation of 4,000,000 antithetic pairs gives $\rho_A = -0.355$, a variance factor of 0.645 per pair: nonpositive, as part (iii) of the proposition guarantees, and here negative.

(Problem P23.9 works this computation at different numbers.)

The inequality (I.17), which says that $\operatorname{Cov}(\psi_1(Z), \psi_2(Z)) \geq 0$ for two functions ψ_1 and ψ_2 of exponential growth that are nondecreasing in each coordinate, is the association inequality of Harris (1960) and of Esary, Proschan and Walkup (1967), here for independent normal variables: two quantities that both rise with every source of noise are positively correlated. Antithetic sampling turns it round. The payoff of a monotone claim rises with every normal, its mirror image falls with every normal, and the two are negatively correlated.

Interview trap. "Antithetic sampling always helps." It helps for payoffs that are monotone in the normal, or in each normal of a path (Proposition 23.43(iii)). For a straddle $\rho_A = +0.90$ and the variance per pair nearly doubles (Example 23.44); for a far out-of-the-money option the gain is a few percent.

Interview trap. Quoting a standard error computed from the $2n$ payoffs when n antithetic pairs were drawn. The two payoffs of a pair are not independent; the independent units are the n pair averages, and the standard error is their sample standard deviation divided by $\sqrt{n}$. Treating the $2n$ payoffs as independent reports $\operatorname{Var} f(Z)/(2n)$ where the truth is $\operatorname{Var} f(Z)\,(1 + \rho_A)/(2n)$. For the call struck at 100 in Example 23.44 the reported standard error is too large by the factor $1/\sqrt{0.633} = 1.26$, which wastes the gain; for the straddle it is too small by the factor $\sqrt{1.899} = 1.38$, which hides the loss.

Proposition 23.45 (Control variates)**.** *Let $0 < \operatorname{Var} X < \infty$ and $0 < \operatorname{Var} Y < \infty$, and let $\rho = \operatorname{Corr}(X, Y)$. For every real β, the variable*

$$X - \beta\,(Y - \mathbb{E}\,Y)$$

is unbiased for $\mathbb{E}\,X$, and

$$\operatorname{Var}\left(X - \beta\,(Y - \mathbb{E}\,Y)\right) = \operatorname{Var} X - 2\beta \operatorname{Cov}(X, Y) + \beta^2 \operatorname{Var} Y. \tag{23.14}$$

This variance is smallest at

$$\beta^* = \frac{\operatorname{Cov}(X, Y)}{\operatorname{Var} Y},$$

and only there, and its smallest value is $(1 - \rho^2) \operatorname{Var} X$.

Proof. We compute the mean and the variance of a linear combination, locate the vertex of a quadratic in β, and read the smallest value off the regression residual of Proposition 10.15. By linearity of expectation (Theorem 1.2),

$$\mathbb{E}\big[X - \beta\,(Y - \mathbb{E}\,Y)\big] = \mathbb{E}\,X - \beta\,(\mathbb{E}\,Y - \mathbb{E}\,Y) = \mathbb{E}\,X.$$

Adding the constant $\beta\,\mathbb{E}\,Y$ does not change a variance, so the variance in (23.14) is that of the sum of X and $-\beta Y$. Proposition 10.10, with $\operatorname{Var}(-\beta Y) = \beta^2 \operatorname{Var} Y$ and $\operatorname{Cov}(X, -\beta Y) = -\beta \operatorname{Cov}(X, Y)$ (the covariance is bilinear), gives (23.14). Its right side is a quadratic in β with the positive leading coefficient $\operatorname{Var} Y$. Since $\operatorname{Var} Y\,(\beta - \beta^*)^2 = \beta^2 \operatorname{Var} Y - 2\beta \operatorname{Cov}(X, Y) + \operatorname{Cov}(X, Y)^2/\operatorname{Var} Y$, it equals $\operatorname{Var} Y\,(\beta - \beta^*)^2$ plus a term that does not depend on β, so it is smallest at $\beta = \beta^*$ and only there, the first term being positive for every other β. At β^* the variable is $X - \beta^* Y$ plus

the constant $\beta^* \mathbb{E}\, Y$, so its variance is that of $X - \beta^* Y$. Proposition 10.15, read with $A = Y$ and $B = X$, has the same coefficient $\beta^* = \operatorname{Cov}(Y, X)/\operatorname{Var} Y$ and the residual $B_\perp = X - \beta^* Y$, whose variance it gives as $(1 - \rho^2) \operatorname{Var} X$. The residual is the part of the target that is uncorrelated with the control. □

(Problem P23.9 asks you to reproduce this proof from scratch.)

With n paths the control-variate estimate is the sample mean of the variables $X_i - \beta\,(Y_i - \mathbb{E}\, Y)$. By Proposition 23.37 it is unbiased, with the variance (23.14) divided by n; at β^* that is $(1 - \rho^2) \operatorname{Var} X/n$. In practice β^* is not known. It is the population form of the regression slope of Corollary 5.3, a covariance divided by the variance of the regressor, and it is estimated by regressing the X_i on the Y_i. An estimate from an independent pilot run keeps the estimator exactly unbiased, whenever that estimate has a finite mean: it is then independent of the main paths, and the mean of its product with $Y_i - \mathbb{E}\, Y$ is the product of the two means, which is zero (Lemma 19.17). Estimating the slope on the same paths introduces a bias of order $1/n$ (Glasserman, 2004), which is negligible beside the standard error of order $n^{-1/2}$; this is a known result, named here and not used.

Example 23.46 (Two control variates)**.** *The terminal price as a control.* Take the call of Example 23.44 ($S = K = 100$, $\sigma = 20\%$, $r = q = 0$, $T = 1$) as the target, $X = (S_T - 100)^+$, and the terminal price as the control, $Y = S_T$, whose mean is the forward price $Se^{bT} = 100$. Numerical integration gives the standard deviations 13.15 for X and 20.20 for Y and the covariance 236.45. Hence

$$\rho = \frac{236.45}{13.15 \times 20.20} = 0.890, \qquad 1 - \rho^2 = 0.208, \qquad \beta^* = \frac{236.45}{20.20^2} = 0.579.$$

The control removes 79% of the variance: about five times fewer paths for the same accuracy.

The geometric Asian as a control. Take the quarterly arithmetic Asian call of Example 23.30 as the target and, on the same path, the geometric Asian call as the control; its mean is exact, 7.806 by Proposition 23.28. Over 200,000 paths the plain estimate is 8.19, with standard error 0.031. The sample correlation of the two payoffs is 0.99914, so $1 - \rho^2 = 0.0017$: the variance falls about 580-fold. The slope estimated on the same paths is 1.04, and the control-variate estimate is

$$8.175, \qquad \text{with standard error } 0.0013,$$

which is the standard error that the plain estimate would reach with $580 \times 200{,}000 = 116$ million paths. A run of 10,000,000 paths with the control, its slope taken from an independent pilot run, gives 8.1752, with standard error 0.0002.

A rule of thumb. The variance that remains is the fraction $1 - \rho^2$. A correlation of 0.99 leaves 0.0199, about a fifty-fold reduction; 0.9 leaves 0.19; 0.7 leaves 0.51, so it removes only half of the variance; and 0.5 leaves 0.75, so it removes only a quarter. A control is worth its cost only when its correlation with the target is very high.

(Problem P23.9 works this computation at different numbers.)

Interview trap. Using $\beta = 1$, the textbook estimator $X - Y + \mathbb{E}\, Y$, instead of β^*. By (23.14) the choice $\beta = 1$ is optimal only when $\operatorname{Cov}(X, Y) = \operatorname{Var} Y$, and for a weakly correlated or badly scaled control it can *increase* the variance. With the terminal price as the control for the call struck at 100 in Example 23.46, $\beta = 1$ leaves 63% of the variance where $\beta^* = 0.58$ leaves 21%. For the call struck at 130, $\rho = 0.56$ and $\beta^* = 0.13$: the optimal coefficient removes 31% of the variance, and $\beta = 1$ multiplies it by 14. For the Asian pair, whose two payoffs have nearly the same scale ($\beta^* = 1.04$), $\beta = 1$ costs little: a standard error of 0.0017 in place of 0.0013.

Interview trap. Using a control whose mean is itself estimated. The estimator subtracts $\beta\,(Y - \mathbb{E}\, Y)$, and an error in $\mathbb{E}\, Y$ goes straight into the answer, as a bias if the value is wrong and as extra noise if it is simulated. The gain needs a mean in closed form: the forward price for S_T, a BSM price for a vanilla option, the formula of Proposition 23.28 for the geometric Asian.

Remark 23.47 (Importance sampling and quasi-random numbers). *Importance sampling* moves the draws to where the payoff is. Instead of Z it uses the shifted draw $Z+\theta$, which has the law $\mathcal{N}(\theta,1)$ and is concentrated on the paying region when θ is chosen well (for a deep out-of-the-money digital, a shift that puts the strike near the centre of the draws), and it multiplies the payoff of each draw by the weight $\exp\bigl(-\theta(Z+\theta)+\frac{1}{2}\theta^2\bigr)$. The estimator stays unbiased. For a scalar payoff f that is continuous except at finitely many points and has exponential growth, $|f(z)|\le c\,e^{c|z|}$ (as every payoff of moderate growth in the price has, in its normal, by Lemma 19.20(i)), the function $\psi(z)=f(z)\,e^{-\theta z}$ is again continuous except at finitely many points and has exponential growth (with the constant $c+|\theta|$), so Lemma 19.38(i), read from right to left with this test function, gives

$$\mathbb{E}\Bigl[f(Z+\theta)\,e^{-\theta(Z+\theta)+\theta^2/2}\Bigr]=e^{\theta^2/2}\,\mathbb{E}\bigl[\psi(Z+\theta)\bigr]=\mathbb{E}\bigl[e^{\theta Z}\,\psi(Z)\bigr]=\mathbb{E}\,f(Z).$$

For the digital that pays when $Z>3$, an event of probability 0.00135, the shift $\theta=3$ divides the variance per draw by about 220. *Quasi-random* (low-discrepancy) sequences replace the random draws by points that fill the unit cube evenly. In low dimension their error approaches the order n^{-1}, up to logarithmic factors (the Koksma–Hlawka inequality: Koksma, 1942; Hlawka, 1961; for an integrand of bounded variation and a standard low-discrepancy sequence the error is at most of order $(\ln n)^m/n$ in dimension m), a known result that is named here and not used; the price is that the simple standard error $s/\sqrt{n}$ is no longer available. *Stratified sampling* draws a fixed share of the paths from each of several slices of the range of a normal, which removes the variance between the slices. *Moment matching* shifts and rescales the drawn normals, subtracting their sample mean and dividing by their sample standard deviation, so that the sample mean and the sample variance are exactly 0 and 1.

23.7 Finite Differences in Brief

Trees discretise the law of the underlying; finite differences discretise the pricing equation (Theorem 20.3; with a yield, Corollary 20.45). In the log price the explicit scheme is literally a trinomial tree; a sufficient condition for its stability in the maximum norm is that its branch weights be nonnegative; and American exercise is the same node-by-node maximum as in Procedure 18.16. The implicit and Crank–Nicolson schemes are stable for every time step, which an energy estimate (a summation by parts and the Cauchy–Schwarz inequality) proves in a few lines, and the Lax argument shows that all three converge for smooth solutions, the explicit one under its positivity condition. The interview questions concern stability ("why does an explicit grid produce negative prices?", "why does Crank–Nicolson oscillate?") and which engine suits which product.

Setup. The symbols of §23.1 are kept, and $r\ge 0$ throughout this subsection. In addition:

(a) *The log variable.* Put $x=\ln S$ and $\tilde V(x,t)=V(e^x,t)$: the same price function, written in the log price (a tilde marks the same object in a transformed variable). Since $V(S,t)=\tilde V(\ln S,t)$, the chain rule gives $\partial_S V=\partial_x\tilde V/S$ and $\partial_{SS}V=(\partial_{xx}\tilde V-\partial_x\tilde V)/S^2$, that is,

$$S\,\partial_S V=\partial_x\tilde V,\qquad S^2\,\partial_{SS}V=\partial_{xx}\tilde V-\partial_x\tilde V.$$

Substituting in the pricing equation (20.13), $\partial_t V+bS\,\partial_S V+\frac{1}{2}\sigma^2S^2\,\partial_{SS}V=rV$ with $b=r-q$, turns it into an equation with constant coefficients:

$$\partial_t\tilde V+\nu\,\partial_x\tilde V+\tfrac{1}{2}\sigma^2\,\partial_{xx}\tilde V-r\tilde V=0,\qquad \nu=b-\tfrac{1}{2}\sigma^2. \tag{23.15}$$

(b) *The grid.* Fix a space step $\Delta x>0$ and a number J of time steps of length $\Delta t=T/J$ (J counts time steps, as for a simulated path). The nodes are $x_j=\ln S+j\,\Delta x$ for the integers $|j|\le j_{\max}$, so that the spot is the node $j=0$, and $t_i=i\,\Delta t$ for $i=0,\dots,J$. The index i is a time and j a position. The nodes with $|j|<j_{\max}$ are *interior*; $j=\pm j_{\max}$ are the two *boundary* nodes. A *grid vector* is a list $w=(w_j)$, $|j|\le j_{\max}$, of real numbers; in this

subsection w and z denote grid vectors. The grid value V_j^i approximates $\tilde V(x_j,t_i)$, and V^i is the grid vector of these values at t_i; the superscript is a time index, not a power. The *maximum norm* of a grid vector is $\max_j |w_j|$.

(c) *Terminal and boundary rows.* The terminal row is the payoff, $V_j^J = f(e^{x_j})$. The two boundary rows $V^i_{\pm j_{\max}}$ are prescribed from the behaviour of the claim far from the money. For a put with strike K they are 0 at the top and, at the bottom, the discounted intrinsic value $Ke^{-r(T-t_i)} - e^{x_{-j\max}}e^{-q(T-t_i)}$ (the value of the short forward position that a deep in-the-money European put resembles), or the intrinsic value $K - e^{x_{-j\max}}$ for an American put. In practice the boundary rows are placed four to five standard deviations $\sigma\sqrt{T}$ of the log price from the spot, $j_{\max}\,\Delta x \approx 4\sigma\sqrt{T}$ to $5\sigma\sqrt{T}$, where the risk-neutral probability of ending beyond them is about $2\,\Phi(-4) = 6.3\times10^{-5}$ to $2\,\Phi(-5) = 5.7\times10^{-7}$; the error of this truncation is not analysed here.

(d) *Difference operators.* For a grid vector w and an interior node j,

$$(A_0 w)_j = \tfrac12\sigma^2\,\frac{w_{j+1}-2w_j+w_{j-1}}{\Delta x^2} + \nu\,\frac{w_{j+1}-w_{j-1}}{2\,\Delta x}, \qquad (Aw)_j = (A_0w)_j - r\,w_j,$$

the central differences that replace $\frac12\sigma^2\,\partial_{xx} + \nu\,\partial_x$ and $\frac12\sigma^2\,\partial_{xx} + \nu\,\partial_x - r$. The results A_0w and Aw are lists of interior values. On grid vectors that vanish at the two boundary nodes, A_0 and $A = A_0 - rI$ are square matrices of size $2j_{\max}-1$ acting on the interior values (capital letters; in this subsection A is a matrix, not an event, and I is the identity matrix, not the present value of income of §16), and $A^\top$ is the transpose of A.

(e) *Three schemes.* Each scheme replaces $\partial_t\tilde V$ by $(V_j^{i+1}-V_j^i)/\Delta t$ and steps backward in time from the terminal row, for $i = J-1,\dots,0$; they differ in the time level at which the space differences are taken. The *explicit* scheme takes them at the known level $i+1$ and the discount term at level i: $(V_j^{i+1}-V_j^i)/\Delta t + (A_0V^{i+1})_j - rV_j^i = 0$, that is,

$$(1+r\,\Delta t)\,V_j^i = p_u\,V_{j+1}^{i+1} + p_m\,V_j^{i+1} + p_d\,V_{j-1}^{i+1}, \tag{23.16}$$

with the weights

$$p_u = \frac12\Big(\frac{\sigma^2\,\Delta t}{\Delta x^2} + \frac{\nu\,\Delta t}{\Delta x}\Big), \qquad p_m = 1 - \frac{\sigma^2\,\Delta t}{\Delta x^2}, \qquad p_d = \frac12\Big(\frac{\sigma^2\,\Delta t}{\Delta x^2} - \frac{\nu\,\Delta t}{\Delta x}\Big)$$

(read "p up, p middle, p down"). In vector form, on the interior nodes, $(1+r\,\Delta t)V^i = (I + \Delta t\,A_0)V^{i+1}$. The *implicit* scheme takes the space differences at the unknown level i, and the *Crank–Nicolson* scheme averages the two levels:

$$(I - \Delta t\,A)\,V^i = V^{i+1}, \tag{23.17}$$

$$(I - \tfrac12\Delta t\,A)\,V^i = (I + \tfrac12\Delta t\,A)\,V^{i+1}, \tag{23.18}$$

both on the interior nodes. In these two schemes the unknown interior values of V^i are coupled, and each step solves a tridiagonal linear system. The boundary values of V^i are known; they occur in the equations of the two interior nodes next to the boundary and are moved to the right-hand side, so each step is an affine map of V^{i+1}.

(f) *Reading the grid.* For an American option, V_j^i is replaced by $\max\big(V_j^i, f(e^{x_j})\big)$ at every node after each step, as in Procedure 18.16. The price is V_0^0. The delta and the gamma are read off the neighbouring nodes through (a):

$$\Delta \approx \frac{V_1^0 - V_{-1}^0}{2\,\Delta x\,S}, \qquad \Gamma \approx \frac{1}{S^2}\Big(\frac{V_1^0 - 2V_0^0 + V_{-1}^0}{\Delta x^2} - \frac{V_1^0 - V_{-1}^0}{2\,\Delta x}\Big).$$

Figure 51 shows the grid and the three nodes that the explicit scheme (23.16) uses for one value.

Proposition 23.48 (The explicit scheme is a trinomial tree; positivity and stability)**.** *Let p_u, p_m and p_d be the weights of the explicit scheme (23.16).*

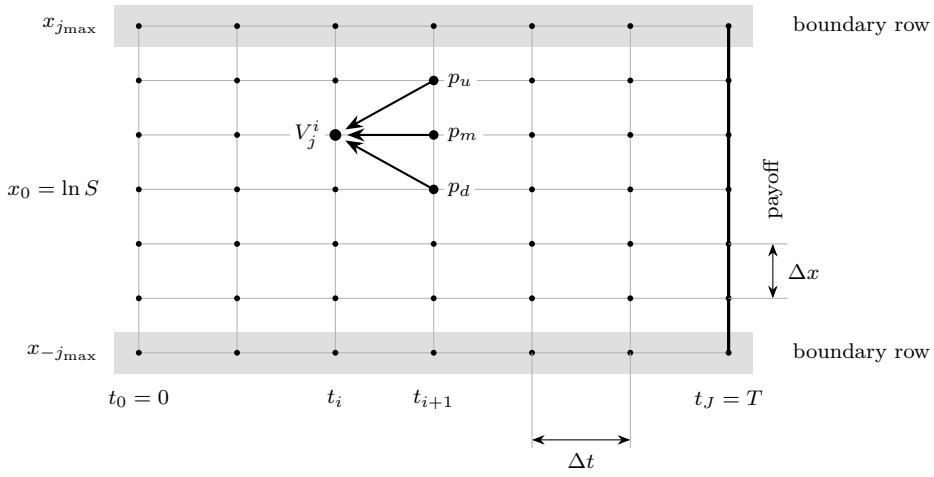


Figure 51: The finite-difference grid in the variables (t, x), $x = \ln S$. Columns are the time levels t_i, from 0 to $T = J\,\Delta t$, and rows are the nodes x_j, spaced Δx apart. The values in the last column are the payoff and those in the two shaded rows are the prescribed boundary values. The explicit scheme (23.16) computes the value V_j^i from the three values V_{j+1}^{i+1}, V_j^{i+1} and V_{j-1}^{i+1} of the next time level, with the weights p_u, p_m and p_d: one step of a trinomial tree, run backward.

(i) The weights satisfy

$$p_u + p_m + p_d = 1, \qquad \Delta x\,(p_u - p_d) = \nu\,\Delta t, \qquad \Delta x^2\,(p_u + p_d) = \sigma^2\,\Delta t.$$

So, when they are nonnegative, they are the probabilities of a three-point step of the log price (up by Δx, unchanged, down by Δx) whose mean is $\nu\,\Delta t$ and whose second moment is $\sigma^2\,\Delta t$. The mean is exactly that of the increment of the risk-neutral log price over Δt, which has the law $\mathcal{N}(\nu\,\Delta t, \sigma^2\,\Delta t)$; the second moment of that increment is $\sigma^2\,\Delta t + \nu^2\,\Delta t^2$, and it is matched up to the term $\nu^2\,\Delta t^2$, which is of second order in Δt and so lies below the first-order accuracy of the scheme (Proposition 23.51(iv), proved below; the variance of the three-point step is $\sigma^2\,\Delta t - \nu^2\,\Delta t^2$). Each step of (23.16) is then a discounted expectation over one trinomial step.

(ii) The three weights are nonnegative if and only if

$$\sigma^2\,\Delta t \le \Delta x^2 \qquad \text{and} \qquad |\nu|\,\Delta x \le \sigma^2.$$

(iii) When they are, the scheme is stable in the maximum norm. If V^i and V'^i are two solutions of (23.16) with the same boundary rows (a prime marks the second solution) and $w^i = V^i - V'^i$, then for every $i < J$

$$\max_j |w_j^i| \le \frac{\max_j |w_j^{i+1}|}{1 + r\,\Delta t}.$$

(iv) (A growing sawtooth.) On the infinite grid (all integers j), one step of (23.16) multiplies the sawtooth $w_j^{i+1} = \varepsilon\,(-1)^j$, with $\varepsilon > 0$, by

$$\frac{1 - 2\sigma^2\,\Delta t/\Delta x^2}{1 + r\,\Delta t}.$$

This factor is below -1 exactly when $\sigma^2\,\Delta t/\Delta x^2 > 1 + \frac{1}{2} r\,\Delta t$, and then the sawtooth grows geometrically. In particular, with $r = 0$ the factor is below -1 whenever $\sigma^2\,\Delta t > \Delta x^2$; and with $r > 0$, for a fixed ratio $\sigma^2\,\Delta t/\Delta x^2 > 1$, it is below -1 as soon as $\Delta t < 2(\sigma^2\,\Delta t/\Delta x^2 - 1)/r$.

Proof. We verify the identities by direct computation and then use the triangle inequality. Write $a = \sigma^2\,\Delta t/\Delta x^2$ for the ratio that occurs in all three weights, so that $p_m = 1 - a$ and $p_u + p_d = a$.

(i). The sum is $p_u + p_d + p_m = a + (1-a) = 1$. The difference $p_u - p_d$ is $\nu\,\Delta t/\Delta x$, so $\Delta x\,(p_u - p_d) = \nu\,\Delta t$; and $\Delta x^2\,(p_u + p_d) = \Delta x^2 a = \sigma^2\,\Delta t$. If the weights are nonnegative they are the probabilities of a step that takes the values Δx, 0 and $-\Delta x$. Its mean is $\Delta x\,p_u - \Delta x\,p_d = \nu\,\Delta t$, its second moment is $\Delta x^2 p_u + \Delta x^2 p_d = \sigma^2\,\Delta t$, and its variance is the second moment minus the squared mean, $\sigma^2\,\Delta t - \nu^2\,\Delta t^2$. Under $\mathbb{Q}$ the log price moves over a time Δt by $\nu\,\Delta t + \sigma\sqrt{\Delta t}\,Z$ (Procedure 23.36), a normal variable with mean $\nu\,\Delta t$, variance $\sigma^2\,\Delta t$ and second moment $\sigma^2\,\Delta t + \nu^2\,\Delta t^2$. Finally, dividing (23.16) by $1 + r\,\Delta t$ shows that V_j^i is the mean of the values of V^{i+1} at the three nodes that the step reaches from x_j, discounted by the factor $1/(1 + r\,\Delta t)$.

(ii). The weight $p_m = 1 - a$ is nonnegative if and only if $a \le 1$, that is, $\sigma^2\,\Delta t \le \Delta x^2$. The weights p_u and p_d are $\frac{1}{2}(a \pm \nu\,\Delta t/\Delta x)$, and both are nonnegative if and only if $a \ge |\nu|\,\Delta t/\Delta x$; multiplying by $\Delta x^2/\Delta t > 0$, this reads $\sigma^2 \ge |\nu|\,\Delta x$.

(iii). The scheme is linear and the two solutions have the same boundary rows, so w satisfies the same recursion with boundary rows equal to zero:

$$(1 + r\,\Delta t)\,w_j^i = p_u\,w_{j+1}^{i+1} + p_m\,w_j^{i+1} + p_d\,w_{j-1}^{i+1}$$

at every interior node, and $w_j^i = 0$ at the two boundary nodes. By the triangle inequality, with nonnegative weights that sum to one,

$$(1 + r\,\Delta t)\,|w_j^i| \le (p_u + p_m + p_d)\,\max_k |w_k^{i+1}| = \max_k |w_k^{i+1}|,$$

the maximum being taken over all the nodes. Since $1 + r\,\Delta t \ge 1 > 0$, dividing by it and taking the maximum over j gives the claim.

(iv). Substitute the sawtooth in the right side of (23.16):

$$p_u\,(-1)^{j+1} + p_m\,(-1)^j + p_d\,(-1)^{j-1} = (p_m - p_u - p_d)\,(-1)^j = (1 - 2a)\,(-1)^j,$$

the drift terms $\pm\nu\,\Delta t/(2\,\Delta x)$ cancelling in $p_u + p_d$. Dividing by $1 + r\,\Delta t$ gives $w_j^i = \varepsilon\,(-1)^j\,(1 - 2a)/(1 + r\,\Delta t)$: the same sawtooth, multiplied by the factor $(1 - 2a)/(1 + r\,\Delta t)$ of the statement. Since $1 + r\,\Delta t > 0$, this factor is below -1 exactly when $1 - 2a < -(1 + r\,\Delta t)$, that is, when $a > 1 + \frac{1}{2}r\,\Delta t$. Then, after k steps, the amplitude is ε times the k-th power of the absolute value of the factor, which exceeds one, so it grows geometrically. With $r = 0$ the condition reads $a > 1$, that is, $\sigma^2\,\Delta t > \Delta x^2$. With $r > 0$ and $a > 1$ fixed, it reads $r\,\Delta t < 2(a - 1)$, that is, $\Delta t < 2(a - 1)/r$. □

Parts (ii) to (iv) settle the explicit scheme. Nonnegative weights give stability in the maximum norm. A negative middle weight gives a growing sawtooth when $r = 0$, and, when $r > 0$, once $\sigma^2\,\Delta t/\Delta x^2 > 1 + \frac{1}{2}r\,\Delta t$. (In the narrow band $1 < \sigma^2\,\Delta t/\Delta x^2 \le 1 + \frac{1}{2}r\,\Delta t$, with p_u and p_d nonnegative, one step multiplies the maximum norm of a difference by at most $(p_u + |p_m| + p_d)/(1 + r\,\Delta t) = (2\sigma^2\,\Delta t/\Delta x^2 - 1)/(1 + r\,\Delta t) \le 1$, by the triangle inequality as in (iii); so the condition of (ii) is sufficient for stability, not necessary.) The American step does not disturb part (iii). For real numbers a, a' and y, $\max(a, y) = y + (a - y)^+$, so

$$|\max(a, y) - \max(a', y)| = |(a - y)^+ - (a' - y)^+| \le |a - a'|,$$

by the inequality $|a^+ - a'^+| \le |a - a'|$ proved in the proof of Proposition 23.39(ii), applied to $a - y$ and $a' - y$. Replacing both solutions by their maximum with the same intrinsic value therefore does not increase their distance at any node, and the maximum over the nodes in (iii) is not increased by the exercise step.

Example 23.49 (One explicit step at three nodes)**.** Let $\sigma = 20\%$, $r = 4\%$, $q = 0$, $\Delta t = 1/12$ and $\Delta x = 0.1$, which is $\sigma\sqrt{3\,\Delta t}$. Then $\sigma^2\,\Delta t/\Delta x^2 = 1/3$, $\nu = b - \frac{1}{2}\sigma^2 = 0.04 - 0.02 = 0.02$ and $\nu\,\Delta t/\Delta x = 0.01667$, so

$$p_u = \tfrac{1}{2}(0.33333 + 0.01667) = 0.17500, \quad p_m = 0.66667, \quad p_d = \tfrac{1}{2}(0.33333 - 0.01667) = 0.15833.$$

They sum to 1; $\Delta x\,(p_u - p_d) = 0.1 \times 0.01667 = 0.02/12 = \nu\,\Delta t$; and $\Delta x^2\,(p_u + p_d) = 0.01/3 = 0.04/12 = \sigma^2\,\Delta t$, while the exact second moment of the log increment adds $(0.02/12)^2 = 2.8 \times 10^{-6}$. The discount factor is $1/(1 + 0.04/12) = 0.99668$.

Consider a put with $K = 100$ one step before expiry, on the nodes $S = 100\,e^{j\,\Delta x}$, $j = -2, \ldots, 2$, that is 81.87, 90.48, 100, 110.52 and 122.14. The payoffs are 18.1269, 9.5163, 0, 0 and 0. One step of (23.16) gives the following values.

- At $S = 90.48$: $0.99668 \times (0.66667 \times 9.5163 + 0.15833 \times 18.1269) = 0.99668 \times 9.2143 = 9.184$. The exact one-month BSM put is worth 9.290. The value 9.184 is below the intrinsic value 9.516, so an American put is exercised at this node and takes the value 9.516.
- At $S = 100$: $0.99668 \times 0.15833 \times 9.5163 = 1.502$, against the BSM value 2.137: one coarse step beside the kink of the payoff is a poor approximation.
- At $S = 110.52$: 0, against the BSM value 0.089.

Grids therefore use many small steps.

(Problem P23.10 works this computation at different numbers.)

Example 23.50 (Violating the stability condition)**.** Keep $\sigma = 20\%$, $r = 4\%$ and $\Delta x = 0.1$. By Proposition 23.48(ii) the middle weight is nonnegative only if $\Delta t \le \Delta x^2/\sigma^2 = 0.25$ years. At $\Delta t = 0.3$, $\sigma^2\,\Delta t/\Delta x^2 = 1.2$ and the weights are $p_u = 0.63$, $p_m = -0.2$ and $p_d = 0.57$. By Proposition 23.48(iv) the sawtooth is multiplied at each step by $(1 - 2 \times 1.2)/(1 + 0.04 \times 0.3) = -1.4/1.012 = -1.383$; the ratio 1.2 exceeds $1 + \frac{1}{2}r\,\Delta t = 1.006$, so the factor is below -1 (it is also the factor of Proposition 23.51(iii) below at the frequency π). An error of 10^{-6} in the sawtooth mode, the size of a rounding error, becomes 6.6×10^{-4} after 20 steps and about 11 after 50. At $\Delta t = 0.5$ the middle weight is $p_m = -1$. Halving Δx to 0.05 forces $\Delta t \le 0.0625$: twice the nodes and four times the steps, eight times the work.

(Problem P23.10 works this computation at different numbers.)

Interview trap. Refining the space grid without refining the time grid. In the explicit scheme, halving Δx requires dividing Δt by four, or p_m turns negative and, once $\sigma^2\,\Delta t/\Delta x^2$ exceeds $1 + \frac{1}{2}r\,\Delta t$, errors grow geometrically (Proposition 23.48(iv)). Negative or oscillating option values in an explicit grid are the signature of a violated stability condition, not of a wrong payoff.

Interview trap. "The explicit scheme is just a worse tree." When its three weights are nonnegative (Proposition 23.48(ii)) it *is* a trinomial tree (Proposition 23.48(i)), with the same accuracy of order Δt. The reason to prefer the implicit and Crank–Nicolson schemes is not accuracy per step but the freedom to take large time steps on a fine space grid.

Setup for Proposition 23.51. The symbols above are kept. In addition:

(g) *Norms.* For grid vectors, $\langle w, z\rangle = \Delta x \sum_{|j|<j_{\max}} w_j z_j$ (a sum over the interior nodes) and $\|w\| = \langle w, w\rangle^{1/2}$: this is Δx times the Euclidean inner product of the interior values, so that $\|w\|^2$ approximates the integral of the square of a function on the grid.

(h) *Complex numbers.* In part (iii), i is the imaginary unit, $\mathrm{i}^2 = -1$ (upright, because i indexes time steps), and grid vectors may have complex entries. For real a and a', the real part of $a + \mathrm{i}a'$ is $\mathrm{Re}(a + \mathrm{i}a') = a$ and its modulus is $|a + \mathrm{i}a'| = (a^2 + a'^2)^{1/2} \ge |a|$; the modulus of a product or of a quotient is the product or the quotient of the moduli. For a real ω, $e^{\mathrm{i}\omega} = \cos\omega + \mathrm{i}\sin\omega$ (Euler's formula, taken here as the definition of the left side), and the addition formulas for the sine and the cosine give $e^{\mathrm{i}(\omega+\omega')} = e^{\mathrm{i}\omega}e^{\mathrm{i}\omega'}$. The number $\omega \in [-\pi, \pi]$ is a *frequency* (not the GARCH constant of §22), and $\Lambda_0(\omega)$, $\Lambda(\omega)$ are eigenvalues (capital, because λ is the jump intensity). A vector (w_j) on the infinite grid is *bounded* if $\sup_j |w_j| < \infty$.

(i) *Errors.* When an exact solution $\tilde V$ of (23.15) is given, $\tilde V^i$ is the grid vector of its values $\tilde V(x_j,t_i)$. The *local truncation error* ε^i, with components ε^i_j at the interior nodes, is what the exact solution leaves over when it is inserted in a scheme, divided by Δt: on the interior nodes, $\Delta t\,\varepsilon^i$ is the left side minus the right side of the vector form of the scheme in Setup (e), with $\tilde V^i$ and $\tilde V^{i+1}$ in the places of V^i and V^{i+1}; for example $\Delta t\,\varepsilon^i = (I-\Delta t\,A)\,\tilde V^i - \tilde V^{i+1}$ for the implicit scheme. (The three formulas are written out in Step 4 of the proof of Proposition 23.51, Appendix I.10.) The *global error* is $\mathcal{E}^i = \tilde V^i - V^i$ (calligraphic, an approximation error).

Proposition 23.51 (Stability and accuracy of the implicit and Crank–Nicolson schemes (Crank and Nicolson, 1947))**.** *In the notation of the Setup, with $r \ge 0$:*

(i) (Energy identity.) For every grid vector w that vanishes at the two boundary nodes,

$$\langle Aw, w\rangle = \Delta x\Big[-\frac{\sigma^2}{2\,\Delta x^2}\sum_{j=-j_{\max}}^{j_{\max}-1}(w_{j+1}-w_j)^2 - r\sum_{|j|<j_{\max}} w_j^2\Big] \le 0, \tag{23.19}$$

the first sum running over all the pairs of adjacent nodes, boundary nodes included.

(ii) (Stability for every time step.) For every $\Delta t>0$ the implicit step (23.17) and the Crank–Nicolson step (23.18) are well defined: given V^{i+1} and the boundary values, there is exactly one V^i. If V^i and V'^i are two solutions of the same scheme with the same boundary values and $w^i = V^i - V'^i$, then

$$\|w^i\| \le \|w^{i+1}\| \qquad \text{for every } i<J.$$

If moreover $|\nu|\,\Delta x \le \sigma^2$, the implicit scheme also satisfies $\max_j|w^i_j| \le \max_j|w^{i+1}_j|/(1+r\,\Delta t)$.

(iii) (Amplification factors.) On the infinite grid (all integers j), for every real ω the mode $w_j = e^{\mathrm{i}j\omega}$ satisfies $A_0 w = \Lambda_0(\omega)\,w$ and $Aw = \Lambda(\omega)\,w$, with

$$\Lambda_0(\omega) = -\frac{\sigma^2}{\Delta x^2}(1-\cos\omega) + \mathrm{i}\,\frac{\nu}{\Delta x}\sin\omega, \qquad \Lambda(\omega) = \Lambda_0(\omega) - r, \qquad \operatorname{Re}\Lambda(\omega) \le -r \le 0.$$

For each scheme, the vector equal to the following amplification factor *times w solves one backward step from $V^{i+1}=w$; for the implicit and Crank–Nicolson schemes it is the only bounded solution:*

$$\frac{1+\Delta t\,\Lambda_0}{1+r\,\Delta t}\ \ \textit{(explicit)}, \qquad \frac{1}{1-\Delta t\,\Lambda}\ \ \textit{(implicit)}, \qquad \frac{1+\frac12\Delta t\,\Lambda}{1-\frac12\Delta t\,\Lambda}\ \ \textit{(Crank–Nicolson)}.$$

The last two have modulus at most one, for every $\Delta t>0$ and every ω. The explicit factor at $\omega=\pi$ is the real number $(1-2\sigma^2\,\Delta t/\Delta x^2)/(1+r\,\Delta t)$, which is below -1 exactly when $\sigma^2\,\Delta t/\Delta x^2 > 1+\frac12 r\,\Delta t$: the sawtooth of Proposition 23.48(iv).

(iv) (Accuracy.) Let $\tilde V$ be a function defined on an open set that contains the rectangle $[x_{-j_{\max}}, x_{j_{\max}}]\times[0,T]$, with continuous partial derivatives $\partial_t^k\partial_x^m\tilde V$ (k derivatives in t and m in x) for all integers $k,m\ge 0$ with $2k+m\le 6$, each bounded on the rectangle. Suppose that $\tilde V$ solves (23.15) on the rectangle and takes the values of the scheme at the terminal nodes and at the boundary nodes. (The error then vanishes on those rows. The error made by cutting the domain of the true price function down to the rectangle is a separate approximation, which is not analysed.) Then there is a constant c, which depends only on σ, ν, r and on bounds for these derivatives, such that at every interior node and every step

$$|\varepsilon^i_j| \le c\,(\Delta t+\Delta x^2)\ \ \textit{(explicit, implicit)}, \qquad |\varepsilon^i_j| \le c\,(\Delta t^2+\Delta x^2)\ \ \textit{(Crank–Nicolson)},$$

and the global error at time 0 is of the same order:

- *for the implicit and Crank–Nicolson schemes and every* $\Delta t > 0$*,*

$$\|\mathcal{E}^0\| \le T\,(2j_{\max}\,\Delta x)^{1/2}\,\max_{i,j}|\varepsilon^i_j|,$$

where $2j_{\max}\,\Delta x$ *is the width of the rectangle; hence also at each node, with the bound divided by* $\Delta x^{1/2}$*, because* $|\mathcal{E}^0_j| \le \|\mathcal{E}^0\|/\Delta x^{1/2}$*;*

- *for the explicit scheme under the positivity condition of Proposition 23.48(ii), and for the implicit scheme when* $|\nu|\,\Delta x \le \sigma^2$*,*

$$\max_j|\mathcal{E}^0_j| \le T\,\max_{i,j}|\varepsilon^i_j|.$$

Sketch of the proof. The argument is an energy estimate for stability and the Lax argument for accuracy. For (i), summation by parts turns the sum of $(w_{j+1}-2w_j+w_{j-1})\,w_j$ into minus the sum of the squared differences, the centred first differences contribute nothing, and $r \ge 0$ gives the sign. For (ii), (i) and the Cauchy–Schwarz inequality give, for w vanishing at the boundary,

$$\|w\|^2 \le \langle (I-\Delta t\,A)w, w\rangle \le \|(I-\Delta t\,A)w\|\,\|w\|,$$

so, with the same bound for the transpose, the matrix of the implicit step is invertible and its inverse does not increase the norm, whatever Δt. For Crank–Nicolson the difference of two solutions satisfies $w^i - w^{i+1} = \frac{1}{2}\Delta t\,A(w^i + w^{i+1})$, and the inner product with $w^i + w^{i+1}$ gives $\|w^i\| \le \|w^{i+1}\|$ by (i); when $|\nu|\,\Delta x \le \sigma^2$, the triangle inequality at a node where $|w^i_j|$ is largest gives the maximum-norm bound. For (iii), the mode turns the two differences into $(2\cos\omega - 2)\,w_j$ and $2\,\mathrm{i}\sin\omega\;w_j$, and $\operatorname{Re}\Lambda \le 0$ bounds the moduli. For (iv), Taylor's theorem bounds ε^i through the bounded derivatives of the hypothesis, the leading terms cancelling by (23.15), and an expansion about the midpoint of the step gives Crank–Nicolson its second order in Δt. The global error vanishes on the terminal and boundary rows and solves the scheme with the source $\Delta t\,\varepsilon^i$, so stability gives $\|\mathcal{E}^i\| \le \|\mathcal{E}^{i+1}\| + \Delta t\,\|\varepsilon^i\|$, and adding over the J steps gives the bounds, those in the maximum norm from the stability in that norm. The complete proof is in Appendix I.10. □

A payoff with a kink violates the smoothness hypothesis of part (iv) near expiry, so the rates of that part are not reached at once for a call or a put. The Crank–Nicolson factor, which is close to -1 for high frequencies on a fine grid (Example 23.52), explains the oscillations that this scheme shows there; the standard cure, named only, is to take a few implicit steps first.

Example 23.52 (Amplification factors of the three schemes)**.** On the grid of Example 23.49 ($\sigma = 20\%$, $r = 4\%$, $\nu = 0.02$, $\Delta x = 0.1$), the sawtooth mode $\omega = \pi$ has $\Lambda_0(\pi) = -2 \times 0.04/0.01 = -8$ and $\Lambda(\pi) = -8.04$. The amplification factors of Proposition 23.51(iii) at $\omega = \pi$ are as follows.

	explicit	implicit	Crank–Nicolson
$\Delta t = 1/12$	0.332	0.599	0.498
$\Delta t = 0.3$	−1.383	0.293	−0.093

For instance, at $\Delta t = 1/12$ the explicit factor is $(1 - 8/12)/(1 + 0.04/12)$ and the implicit one is $1/(1 + 8.04/12)$. At $\Delta t = 0.3$, beyond the explicit limit 0.25 of Example 23.50, the explicit sawtooth grows and the other two decay. Over all frequencies $\omega \in [-\pi, \pi]$ the moduli of the implicit and Crank–Nicolson factors stay below 0.997 at $\Delta t = 1/12$ and below 0.99 at $\Delta t = 0.3$; the largest values are attained at $\omega = 0$, where both factors are close to the discount factor of one step.

On a fine grid, $\Delta x = 0.01$ with the monthly step $\Delta t = 1/12$, $\Lambda_0(\pi) = -800$, and the factors at $\omega = \pi$ are -0.942 for Crank–Nicolson, 0.015 for the implicit scheme and -65.4 for the explicit scheme. Crank–Nicolson is stable, but the high-frequency error left by a kink dies slowly and alternates in sign from one step to the next; the implicit scheme removes it in a single step.

(Problem P23.10 works this computation at different numbers.)

Interview trap. "Crank–Nicolson is unconditionally stable, so it is always the best choice." Stability bounds the error; it does not damp it. On a fine grid the Crank–Nicolson factor of the sawtooth mode is about -0.94 per step (Example 23.52), so the kink of a payoff leaves an oscillation in the price, and especially in the gamma, that dies slowly. The implicit scheme damps it at once, which is why a few implicit steps are taken first.

Remark 23.53 (Trees, grids and dimension). With the spacing $\Delta x = \sigma\sqrt{3\,\Delta t}$ the weights of (23.16) are $p_m = 2/3$ and $p_u, p_d = 1/6 \pm \nu\,\Delta t/(2\,\Delta x)$: the standard trinomial tree. (With this spacing the fourth moment of the step, $\Delta x^4\,(p_u + p_d) = 9\sigma^4\Delta t^2/3 = 3\sigma^4\Delta t^2$, is that of a normal variable with variance $\sigma^2\,\Delta t$ and mean zero.) The explicit scheme is the backward induction of a trinomial tree: the same computation as the binomial method of §18 (Procedure 18.16), with three branches and with the discount factor $1/(1 + r\,\Delta t)$ in the place of $e^{-r\Delta t}$. Grids handle early exercise, but not path dependence: the state must be the current price alone, so a lookback or an Asian option needs an extra state variable. And they suffer beyond two or three state variables, because the number of nodes is the product of the numbers of nodes in each variable; there the error of order $n^{-1/2}$ of Proposition 23.37, which does not depend on the dimension, wins.

Interview trap. "Finite differences price an Asian or a lookback option directly." Not on the variables (S, t) alone. The value of a path-dependent claim depends on the running average or the running extreme as well as on the current price, so the grid needs an extra state variable. Simulation is the natural engine for path dependence, and grids and trees for early exercise in one or two state variables.

Remark 23.54 (Bermudan and other nonstandard American features). In a backward induction the whole exercise rule sits in one line, the maximum with the intrinsic value, and editing that line changes the contract. Taking the maximum only at a chosen set of time levels prices a *Bermudan* option; a set of levels that begins at a later date prices an option with a *lock-out* period; and taking it against a strike that depends on the time level prices a contract whose strike is reset on a schedule. The same edit applies on a tree (Procedure 18.16; Remark 18.19). The value of a Bermudan option lies between the European and the American value, by exercise matching (Lemma 17.5(ii)). If the Bermudan option traded above the American one, sell the Bermudan, buy the American, and exercise it whenever the counterparty exercises the Bermudan, which is possible because every Bermudan exercise date is an American exercise date. If it traded below the European one, sell the European, buy the Bermudan, and exercise the Bermudan only at expiry, a permitted date (expiry being among the exercise dates of the contract), where it pays the European payoff. In each case, with the two options on the same underlying with the same strike and expiry, the matched exercises pay and receive the same intrinsic value, so the position brings in a positive amount today and is worth a nonnegative amount at T whatever policy the holder of the claim sold follows: an arbitrage, by that Lemma.

23.8 Beyond Black–Scholes–Merton

Vanilla prices pin down only the risk-neutral distribution of the price at each expiry (Theorem 22.7). Barriers, forward starts and cliquets also depend on how the smile moves, so for them the flat-volatility model is replaced by one of three repairs: *local volatility*, which fits every vanilla price exactly, and *stochastic volatility* and *jumps*, which generate smiles from the dynamics of the price. The examples compute what each repair implies, and the interview question behind them is "which model explains which smile?".

Setup. The symbols of §23.1 are kept. In addition:

(a) *Surfaces and densities.* In the statements about local volatility, $r = q = 0$. The *call surface* is the function $C(K, T)$, today's price of the European call with strike K and expiry T, and

$\sigma_{\text{imp}}(K,T)$ is the implied surface of Definition 22.1. The risk-neutral density of the price at a date, at the price level $x > 0$, is written $f_{\mathbb{Q}}(x,T)$, with the date shown (Theorem 22.7); the date is written t in Fact 23.55, where it is the running time of a model, and T in Proposition 23.56, where it is the expiry of a call. Partial derivatives are written $\partial_T C$, $\partial_{KK} C$, $\partial_t f_{\mathbb{Q}}$, $\partial_T f_{\mathbb{Q}}$, ∂_x and ∂_{xx}, with the fraction form in some displays.

(b) *Local volatility.* A *local volatility function* is a function $\sigma_{\text{loc}}(x,t) > 0$ of the price level and of time; σ_{loc}^2 is the *local variance*. The *local-volatility model* is the equation $dS = \sigma_{\text{loc}}(S_t,t)\, S_t\, dW^{\mathbb{Q}}$ for the risk-neutral price at zero rates: the volatility is a known function of the price and the date.

(c) *Average volatility and BSM prices.* The *average volatility* along a path is $\sigma_{\text{avg}} \geq 0$, with $\sigma_{\text{avg}}^2 = \frac{1}{T}\int_0^T \sigma_t^2\, dt$, where σ_t is the volatility at time t (the subscript is a word, because $\bar{\sigma}^2$ is the long-run variance of the GARCH model in §22). The BSM price today, at the volatility σ, of the claim named in the sentence is $V^{\text{BSM}}(S,0;\sigma)$, the other inputs being fixed; it is written $V^{\text{BSM}}(S,0;\sigma,r)$ when the rate also varies, and $C^{\text{BSM}}(S,K;\sigma)$ or $C^{\text{BSM}}(S,K;\sigma,r)$ for the call (here the spot and the strike are shown and the expiry is fixed; Definition 22.1 of §22 writes $C^{\text{BSM}}(K,T;\sigma)$ at a fixed spot, so $C^{\text{BSM}}(100,100;\sigma)$ below has $S = K = 100$).

(d) *Jumps.* The *jump intensity* is $\lambda \geq 0$, a Poisson rate (not the EWMA decay of §22). In Merton's model each jump multiplies the price by a factor whose logarithm has the law $\mathcal{N}\big(\ln(1+\kappa) - \frac{1}{2}\delta^2,\ \delta^2\big)$, so that $\kappa > -1$ is the *mean proportional jump* (the mean of the factor is $1+\kappa$, by Lemma 18.35(i)) and $\delta \geq 0$ is the standard deviation of the log jump. These two are often written k and s in the literature; here k is an index and s the sample standard deviation of §23.5. The number of jumps by T is $N_T \sim \text{Poisson}(\lambda T)$, a *jump count* is j (in this subsection j is this count, not a grid position), the variables $Z_0, Z_1, Z_2, \ldots$ are independent standard normal variables, and $\sigma_{(j)}$, $r_{(j)}$ are the volatility and the rate given j jumps (the subscript is in parentheses, because σ_1 and σ_2 are the volatilities of two assets in §23.4).

(e) *The pricing convention.* When the volatility is random, or the price jumps, the extra risk is not traded, so prices are not fixed by replication. This subsection adopts the convention that prices are discounted expectations under a chosen risk-neutral law $\mathbb{Q}$: the assumption of Hull and White for volatility risk and of Merton for jump risk. It is a modelling convention, not a theorem. It is stated once here and used by Propositions 23.58 and 23.60.

What the tools of this Part can prove is proved below: Dupire's formula from the forward equation, by two integrations by parts and Breeden–Litzenberger; the mixing formula for an independent average variance; and Merton's jump formula, the last two by freezing an independent variable (Lemma 19.17). The forward equation itself and Heston's correlated model stay Facts.

Fact 23.55 (The local-volatility model and its forward equation (Kolmogorov, 1931; Dupire, 1994))**.** Let $r = q = 0$, and let the local volatility $\sigma_{\text{loc}}(x,t)$ be bounded, bounded away from zero, and smooth as a function of $\ln x$ and t, with bounded derivatives of every order. Then the equation $dS = \sigma_{\text{loc}}(S_t,t)\, S_t\, dW^{\mathbb{Q}}$ has a unique continuous positive solution from each $S_0 > 0$, and for every $t > 0$ the law of S_t has a smooth density $f_{\mathbb{Q}}(x,t)$ that solves the *forward equation* (the Fokker–Planck equation)

$$\frac{\partial f_{\mathbb{Q}}}{\partial t} = \frac{1}{2}\,\frac{\partial^2}{\partial x^2}\Big(\sigma_{\text{loc}}(x,t)^2\, x^2\, f_{\mathbb{Q}}\Big), \tag{23.20}$$

with the decay and integrability of hypotheses (b)–(d) of Proposition 23.56, stated next. Conversely, let a smooth arbitrage-free call surface, defined for $T > 0$, with $\partial_{KK} C > 0$, $\partial_T C > 0$ and $C(K,T) \to (S-K)^+$ as $T \to 0$ for every $K > 0$, be given, and suppose that its *Dupire local variance* $2\,\partial_T C/(K^2\,\partial_{KK} C)$, as a function of (K,T), satisfies the conditions just imposed on σ_{loc}^2. Then that local variance defines such a model, and the model reproduces every vanilla price of the surface. Implied trees (Derman and Kani, 1994; Rubinstein, 1994) are the discrete form of this model.

A proof is beyond our scope. It needs the existence theory of Fact 19.29 for an equation whose coefficient depends on the state; the regularity theory of parabolic equations, for the existence, the smoothness and the decay of the density; and, for the converse, the uniqueness of the solution of the forward equation $\partial_T C = \frac{1}{2}\sigma_{\text{loc}}^2 K^2\,\partial_{KK} C$ with the initial condition $C(K,0) = (S-K)^+$, which the call prices of the model and the given surface both solve. The computation that turns the forward equation of the density into Dupire's formula is elementary, and it is proved next.

Proposition 23.56 (Dupire's formula from the forward equation (Dupire, 1994))**.** *Let $r = q = 0$. Suppose that, for every T in an open interval of expiries, the risk-neutral law of S_T has a density $f_{\mathbb{Q}}(x,T)$, $x>0$, continuous in (x,T), and that $\sigma_{\text{loc}}(x,T) > 0$ is a continuous function such that:*

(a) for each T the function $x \mapsto \sigma_{\text{loc}}(x,T)^2\,x^2\,f_{\mathbb{Q}}(x,T)$ is twice continuously differentiable, $f_{\mathbb{Q}}$ is continuously differentiable in T, and the forward equation (23.20) holds, with the expiry T in the place of the time t:

$$\frac{\partial f_{\mathbb{Q}}}{\partial T} = \frac{1}{2}\,\frac{\partial^2}{\partial x^2}\Big(\sigma_{\text{loc}}^2\,x^2\,f_{\mathbb{Q}}\Big);$$

(b) for each T, as $x \to \infty$, both $\sigma_{\text{loc}}^2\,x^2\,f_{\mathbb{Q}}$ and $x\,\partial_x(\sigma_{\text{loc}}^2\,x^2\,f_{\mathbb{Q}})$ tend to 0;

(c) the density $f_{\mathbb{Q}}$ is twice continuously differentiable in T, and for each T of the interval there are a number $\varepsilon > 0$ and a continuous function $\psi \geq 0$ on $(0,\infty)$ with $\int_0^\infty (1+x)\,\psi(x)\,dx < \infty$ such that $[T-\varepsilon, T+\varepsilon]$ lies in the interval and $|\partial_{TT} f_{\mathbb{Q}}(x,s)| \leq \psi(x)$ for all $x>0$ and all $s \in [T-\varepsilon, T+\varepsilon]$;

(d) the integral $\int_0^\infty x\,f_{\mathbb{Q}}(x,T)\,dx$ is finite for every T of the interval, that is, $\mathbb{E}^{\mathbb{Q}}\,S_T < \infty$: call prices are finite.

Then the call price $C(K,T) = \int_K^\infty (x-K)\,f_{\mathbb{Q}}(x,T)\,dx$ satisfies, for every $K>0$ and every T of the interval,

$$\partial_T C(K,T) = \tfrac{1}{2}\,\sigma_{\text{loc}}(K,T)^2\,K^2\,\partial_{KK} C(K,T).$$

Hence, wherever $\partial_{KK} C > 0$,

$$\sigma_{\text{loc}}(K,T)^2 = \frac{2\,\partial_T C(K,T)}{K^2\,\partial_{KK} C(K,T)}. \tag{23.21}$$

In particular, on the set where $\partial_{KK} C > 0$, at most one local variance is compatible with a given call surface.

Sketch of the proof. We differentiate the call price under the integral sign and integrate by parts twice. The bound ψ of (c) controls the difference quotient of $f_{\mathbb{Q}}$ in T, and (d) keeps call prices finite, so $\partial_T C$ is the integral of $(x-K)\,\partial_T f_{\mathbb{Q}}$; by the forward equation (a),

$$\partial_T C(K,T) = \frac{1}{2}\int_K^\infty (x-K)\,\partial_{xx}(\sigma_{\text{loc}}^2\,x^2\,f_{\mathbb{Q}})\,dx = \tfrac{1}{2}\,\sigma_{\text{loc}}(K,T)^2\,K^2\,f_{\mathbb{Q}}(K,T).$$

The second equality is the two integrations: $x-K$ vanishes at K, the boundary terms at infinity vanish by (b), and only the bracket at K is left. Last, $f_{\mathbb{Q}} = \partial_{KK} C$ by (22.3), whose proof needs only a continuous density with a finite mean. Dividing where $\partial_{KK} C > 0$ gives (23.21), whose right side the call surface fixes. The complete proof is in Appendix I.11. □

That a local-volatility model with such a density exists, and that the output of the formula reproduces the surface, is the residual Fact 23.55.

A consistency check. On the flat surface of the BSM model with $r = q = 0$ and volatility σ, $C = S\,\Phi(d_1) - K\,\Phi(d_2)$ with $d_{1,2} = \ln(S/K)/(\sigma\sqrt{T}) \pm \frac{1}{2}\sigma\sqrt{T}$. The strike derivative is $\partial_K C = -\text{Dig}(K) = -\Phi(d_2)$ (Proposition 23.2 and Theorem 20.17), and $\partial_K d_2 = -1/(K\sigma\sqrt{T})$,

so $\partial_{KK}C = \varphi(d_2)/(K\sigma\sqrt{T})$. The price depends on σ and T only through the product $\sigma\sqrt{T}$, so $\partial_T C = \frac{\sigma}{2T}\,\partial_\sigma C = \frac{\sigma}{2T}\,K\,\varphi(d_2)\sqrt{T}$, by the vega of Proposition 20.35. Hence

$$\frac{2\,\partial_T C}{K^2\,\partial_{KK}C} = \frac{K\,\varphi(d_2)\,\sigma/\sqrt{T}}{K\,\varphi(d_2)/(\sigma\sqrt{T})} = \sigma^2 :$$

on a flat surface the formula returns the constant volatility, as it must. The proposition does apply to this model. The computation shows that the BSM call surface satisfies $\partial_T C = \frac{1}{2}\sigma^2 K^2\,\partial_{KK}C$ at every $K > 0$ and $T > 0$. Differentiating this identity twice in K, exchanging the order of the derivatives in K and T (Lemma 19.54, the call price having continuous partial derivatives of every order for $T > 0$) and using $f_{\mathbb{Q}} = \partial_{KK}C$ (Theorem 22.7) gives $\partial_T f_{\mathbb{Q}} = \frac{1}{2}\sigma^2\,\partial_{xx}(x^2 f_{\mathbb{Q}})$: the lognormal density satisfies the forward equation (23.20) with $\sigma_{\text{loc}} = \sigma$. Hypothesis (d) holds because $\mathbb{E}^{\mathbb{Q}}\,S_T = S$. Hypotheses (b) and (c) hold because every partial derivative of the lognormal density in x or in T is the density times a polynomial in $\ln(x/S)$ and $1/x$ whose coefficients are continuous in $T > 0$, and the factor $\exp\big(-(\ln(x/S))^2/(2\sigma^2 T)\big)$ of the density makes x^2 times such a product tend to zero as $x \to \infty$ and makes it integrable against $(1 + x)\,dx$, uniformly for T in a closed interval of $(0, \infty)$. So the formula had to return σ^2.

Example 23.57 (Local volatility from five quotes)**.** Let $r = q = 0$ and $S = 100$, and let the quotes be generated from the implied surface $\sigma_{\text{imp}}(K) = 20\% - 10\%\,\ln(K/100)$, the same at every expiry: a skew that falls by about one volatility point for each 10% of strike. Five call prices are needed at the money.

strike and expiry (K, T)	$(100, 0.9)$	$(100, 1)$	$(100, 1.1)$	$(95, 1)$	$(105, 1)$
call price $C(K, T)$	7.5581	7.9656	8.3530	10.7117	5.7131

Central differences with the steps $\Delta T = 0.1$ and $\Delta K = 5$ give

$$\partial_T C \approx \frac{8.3530 - 7.5581}{0.2} = 3.9745, \qquad \partial_{KK}C \approx \frac{10.71171 - 2 \times 7.96557 + 5.71308}{25} = 0.019746,$$

the second with the quotes carried to five decimals, because a second difference loses digits. By (23.21),

$$\sigma_{\text{loc}}^2 = \frac{2 \times 3.9745}{10^4 \times 0.019746} = 0.04026, \qquad \sigma_{\text{loc}} = 20.06\%.$$

In the limit of very fine differences the value is 20.001%: local and implied volatility are about equal at the money, not equal. The remaining 0.06% is the error of the steps: a flat 20% surface returns 20.06% with the same steps and exactly 20% in the limit, which is the consistency check above.

With the same steps, the formula gives a local volatility of 22.2% at $K = 90$ (in the limit 22.16%, against an implied volatility of 21.05%) and 18.2% at $K = 110$ (in the limit 18.14%, against 19.05%). From $K = 90$ to $K = 110$ the local volatility falls by 4.0 points and the implied volatility by 2.0: the local skew is about twice the implied skew.

In the wings the density in the denominator is tiny, and noise in the quotes blows the estimate up. At $K = 200$ the three calls needed are worth less than 10^{-6} each and their second difference is 2.4×10^{-8}, while rounding three quotes to four decimals can move a second difference by as much as $4 \times 0.00005/25 = 8 \times 10^{-6}$.

Interview trap. "Dupire's formula is a model of future volatility." It is a calibration identity: the one function $\sigma_{\text{loc}}(K, T)$ that makes a local-volatility model reproduce today's call surface (Proposition 23.56). The local-volatility surface is not a forecast of realised volatility.

Interview trap. "Local volatility equals implied volatility at the money." About equal, not equal: 20.001% against 20% in Example 23.57. Away from the money the local skew is about twice the implied skew, because the implied volatility at a strike is roughly an average of the local volatilities along the paths that end near that strike.

Interview trap. "A model that fits every vanilla price prices every exotic." Vanilla quotes fix one marginal law per expiry (Theorem 22.7) and are silent on how the price at one expiry is tied to the price at another. Any model fitted to the surface therefore prices single-date payoffs alike, and a local-volatility model supplies one particular tie between the dates, chosen for convenience and not observed. A forward start (Proposition 23.8), a cliquet or the hedge of a barrier option inherits that tie: in a local-volatility model the forward smiles flatten and the smile moves against the spot, unlike the sticky rules of §22.3, which fix the motion of the smile by assumption (which rule a market follows is an empirical matter, Definition 22.17).

Proposition 23.58 (Mixing formula for an independent average variance (Hull and White, 1987))**.** *Adopt the pricing convention of the Setup. Let r and q be constants, let $F = Se^{(r-q)T}$, and suppose that under $\mathbb{Q}$*

$$S_T = F \exp\big(-\tfrac{1}{2}\sigma_{\text{avg}}^2 T + \sigma_{\text{avg}}\sqrt{T}\, Z\big),$$

where Z is a standard normal variable and the average volatility $\sigma_{\text{avg}} \geq 0$ is a random variable independent of Z. Let f be a payoff of the class $\mathcal{G}$, with $|f(x)| \leq c\,(x^c + x^{-c})$ (Definition 19.19(iv)), and suppose that

$$\mathbb{E}\, \exp\big(\tfrac{1}{2} c\,(c+1)\, \sigma_{\text{avg}}^2 T\big) < \infty,$$

which holds for every c when σ_{avg} is bounded. Then

$$e^{-rT}\, \mathbb{E}^{\mathbb{Q}}\, f(S_T) = \mathbb{E}\big[V^{\text{BSM}}(S, 0; \sigma_{\text{avg}})\big],$$

where $V^{\text{BSM}}(S, 0; \sigma)$ is the BSM price of the claim paying $f(S_T)$ at T, at the volatility $\sigma > 0$ and with S, r, q and T fixed, and where $V^{\text{BSM}}(S, 0; 0) = e^{-rT} f(F)$, the discounted payoff at the forward. In particular, for the call and for every $\sigma_{\text{avg}} \geq 0$ independent of Z,

$$C = \mathbb{E}\big[C^{\text{BSM}}(S, K; \sigma_{\text{avg}})\big].$$

As an identity between expectations the statement needs no convention; it is a statement about prices under the convention.

Proof. We freeze the independent volatility. Put

$$\psi(\sigma, z) = f\big(F \exp(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\, z)\big), \qquad \bar{\psi}(\sigma) = \mathbb{E}\, \psi(\sigma, Z), \qquad \sigma \geq 0,$$

so that $f(S_T) = \psi(\sigma_{\text{avg}}, Z)$.

Step 1 (integrability). Fix $\sigma \geq 0$ and put $S_T^{(\sigma)} = F \exp(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\, Z)$. By Lemma 18.35(i), for the exponent $c \geq 0$ of the growth bound,

$$\mathbb{E}\big[(S_T^{(\sigma)})^c\big] = F^c\, e^{-c\sigma^2 T/2 + c^2\sigma^2 T/2} = F^c\, e^{c(c-1)\sigma^2 T/2}, \qquad \mathbb{E}\big[(S_T^{(\sigma)})^{-c}\big] = F^{-c}\, e^{c(c+1)\sigma^2 T/2}.$$

Since $c\,(c-1) \leq c\,(c+1)$, both are at most $(F^c + F^{-c})\, e^{c(c+1)\sigma^2 T/2}$, and the growth bound of f gives

$$\mathbb{E}\, |\psi(\sigma, Z)| \leq 2c\,(F^c + F^{-c})\, e^{c(c+1)\sigma^2 T/2}.$$

Apply Lemma 19.17 to the nonnegative function $|\psi|$, with the independent variables $X = \sigma_{\text{avg}}$ and $Y = Z$: the mean of $|f(S_T)| = |\psi(\sigma_{\text{avg}}, Z)|$ is the mean of the function $\sigma \mapsto \mathbb{E}\, |\psi(\sigma, Z)|$ evaluated at σ_{avg}, and by the last display it is at most $2c\,(F^c + F^{-c})\, \mathbb{E}\, e^{c(c+1)\sigma_{\text{avg}}^2 T/2} < \infty$, by the hypothesis. For the call no hypothesis is needed: $0 \leq (x-K)^+ \leq x$, so $0 \leq \psi(\sigma, Z) \leq S_T^{(\sigma)}$, whose mean is F for every σ, and $\mathbb{E}\, |f(S_T)| \leq F$.

Step 2 (freezing). Since $\psi(\sigma_{\text{avg}}, Z)$ is integrable, Lemma 19.17 gives

$$\mathbb{E}^{\mathbb{Q}}\, f(S_T) = \mathbb{E}\, \psi(\sigma_{\text{avg}}, Z) = \mathbb{E}\, \bar{\psi}(\sigma_{\text{avg}}).$$

For a fixed $\sigma > 0$, $F \exp(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\, Z) = S \exp((r - q - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\, Z)$, so $e^{-rT}\, \bar{\psi}(\sigma)$ is the price function of Theorem 20.7, in the yield form of Corollary 20.41(i), at the spot S and the time 0: it is $V^{\text{BSM}}(S, 0; \sigma)$. For $\sigma = 0$, $\bar{\psi}(0) = f(F)$. Multiplying by e^{-rT} gives the statement, and the call is the case $f(x) = (x-K)^+$. □

Two cases show what the proposition does and does not cover. *A deterministic volatility.* Let the volatility σ_t be deterministic and constant on each interval $[t_{k-1}, t_k)$, for dates $0 = t_0 < t_1 < \cdots < t_m = T$, and suppose that under $\mathbb{Q}$ the log price moves over that interval by $(r - q - \frac{1}{2}\sigma_{t_{k-1}}^2)(t_k - t_{k-1}) + \sigma_{t_{k-1}}\sqrt{t_k - t_{k-1}}\, Z_k$, with independent standard normal variables $Z_1, \ldots, Z_m$. Then $\ln S_T$ is $\ln S$ plus a sum of independent normal variables, hence normal (Lemma 19.38(ii)), with the mean $\ln F - \frac{1}{2}\sigma_{\text{avg}}^2 T$ and the variance $\sigma_{\text{avg}}^2 T$, where $\sigma_{\text{avg}}^2 = \frac{1}{T}\int_0^T \sigma_t^2\, dt = \frac{1}{T}\sum_k \sigma_{t_{k-1}}^2 (t_k - t_{k-1})$ is the constant of Setup (c). So S_T has the law of the proposition with a constant σ_{avg}, and the proposition gives the BSM price at the root-mean-square volatility (Problem P23.11(a) uses this case). That this price is also forced by replication, without any convention, is not claimed here. The claim would need instruments whose price on each interval is the price at the start of the interval times an explicit geometric Brownian motion in the Brownian increment since then, that is, a function of an earlier Brownian value as well as of (t, W_t); Definition 20.1 admits only prices that are functions of (t, W_t). *A volatility process.* In the model of Hull and White, in which the volatility is a process independent of the Brownian motion that drives the price, the law of S_T given the whole path of the volatility has the form assumed above. Conditioning on a path is beyond this Part, which is why the proposition is stated for the average variance.

Example 23.59 (A smile from two volatility scenarios)**.** Let $S = 100$, $r = q = 0$ and $T = 1$, and let σ_{avg}^2 be 0.01 or 0.07 with probability one half each: a volatility of 10.00% or 26.46%, with a mean variance of 0.04, that is, 20%. By Proposition 23.58 the at-the-money call is worth

$$\tfrac{1}{2}\, C^{\text{BSM}}(100, 100; \sqrt{0.01}) + \tfrac{1}{2}\, C^{\text{BSM}}(100, 100; \sqrt{0.07}) = \tfrac{1}{2} \times 3.9878 + \tfrac{1}{2} \times 10.5243 = 7.2560,$$

against 7.9656 at a flat 20%: an implied volatility of 18.21%. The same average at other strikes, inverted for the implied volatility (Definition 20.37), gives the following.

strike K	70	80	90	100	110	120	130
implied volatility	22.5%	20.7%	18.9%	18.2%	18.8%	20.1%	21.3%

(The mixture price of a put is the same average of BSM puts, so put–call parity holds for the mixture prices and the call and the put at a strike have the same implied volatility, by Proposition 22.3.) The result is a smile with its minimum near the money, from stochastic volatility that is uncorrelated with the price. At the money the reason is visible in the formula: with $r = q = 0$ and $K = S$ the BSM call is $S\,(2\,\Phi(\frac{1}{2}\sigma\sqrt{T}) - 1)$, a concave function of the variance σ^2 (the function Φ is concave and increasing on $[0, \infty)$, and the square root is concave), so the average of two such prices lies below the price at the average variance.

(Problem P23.11 works this computation at different numbers.)

Interview trap. "Stochastic volatility explains the equity skew." Only with a negative correlation between the price and its volatility (Fact 23.63 below). Stochastic volatility that is uncorrelated with the price gives a nearly symmetric smile, with its minimum near the money (Example 23.59).

Proposition 23.60 (Merton's jump-diffusion price (Merton, 1976))**.** *Adopt the pricing convention of the Setup: jump risk cannot be hedged, and Merton prices it as diversifiable. Let $\sigma > 0$, $\lambda \geq 0$, $\kappa > -1$ and $\delta \geq 0$, and suppose that under $\mathbb{Q}$*

$$S_T = S \exp\Big((r - q - \lambda\kappa - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\, Z_0 + \sum_{i=1}^{N_T} \big(\ln(1+\kappa) - \tfrac{1}{2}\delta^2 + \delta Z_i \big) \Big), \tag{23.22}$$

where $N_T \sim \text{Poisson}(\lambda T)$ is the number of jumps by T (when $\lambda = 0$, $N_T = 0$ and only the term $j = 0$ of the series below remains, with the convention $0^0 = 1$) and $Z_0, Z_1, Z_2, \ldots$ are standard

normal variables, the whole family $N_T, Z_0, Z_1, \dots$ being independent (Fact 19.7 supplies such a family). The model is specified at the single date T; no jump process in time is constructed. Put

$$\lambda' = \lambda\,(1+\kappa), \qquad \sigma_{(j)}^2 = \sigma^2 + \frac{j\,\delta^2}{T}, \qquad r_{(j)} = r - \lambda\kappa + \frac{j\,\ln(1+\kappa)}{T}, \qquad j = 0, 1, 2, \dots$$

(i) The forward price is that of the model without jumps: $\mathbb{E}^{\mathbb{Q}}\, S_T = S e^{(r-q)T}$.

(ii) For every payoff f of the class $\mathcal{G}$,

$$e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\, f(S_T) = \sum_{j=0}^{\infty} e^{-\lambda' T}\,\frac{(\lambda' T)^j}{j!}\; V^{\mathrm{BSM}}\big(S, 0; \sigma_{(j)}, r_{(j)}\big),$$

where $V^{\mathrm{BSM}}(S, 0; \sigma, r)$ is the BSM price of the claim paying $f(S_T)$ with that volatility and with that rate used throughout, in the drift and in the discounting, the yield q being unchanged. For the call,

$$C = \sum_{j=0}^{\infty} e^{-\lambda' T}\,\frac{(\lambda' T)^j}{j!}\; C^{\mathrm{BSM}}\big(S, K; \sigma_{(j)}, r_{(j)}\big).$$

Sketch of the proof. We condition on the number of jumps. On the event $\{N_T = j\}$ the exponent of (23.22) is a constant plus a sum of $j+1$ independent normal variables, hence normal, with the variance $\sigma^2 T + j\,\delta^2 = \sigma_{(j)}^2 T$ and the mean $(r_{(j)} - q - \frac{1}{2}\sigma_{(j)}^2)T$: given j jumps, S_T has the lognormal law of the BSM model with the rate $r_{(j)}$, the volatility $\sigma_{(j)} \geq \sigma > 0$ and the yield q. Since N_T is independent of the normals, Lemma 19.17 and the Poisson probabilities show that $\mathbb{E}^{\mathbb{Q}}[f(S_T)\,\mathbf{1}\{N_T = j\}]$ is $e^{-\lambda T}(\lambda T)^j/j!$ times $e^{r_{(j)}T}\,V^{\mathrm{BSM}}(S, 0; \sigma_{(j)}, r_{(j)})$, the factor $e^{r_{(j)}T}$ undoing the discounting inside the BSM price. Discounting at r, with $e^{r_{(j)}T} = e^{rT}e^{-\lambda\kappa T}(1+\kappa)^j$,

$$e^{-rT}\cdot e^{-\lambda T}\,\frac{(\lambda T)^j}{j!}\cdot e^{r_{(j)}T} = e^{-\lambda(1+\kappa)T}\,\frac{\big(\lambda T\,(1+\kappa)\big)^j}{j!} = e^{-\lambda' T}\,\frac{(\lambda' T)^j}{j!},$$

which is why the weights are Poisson probabilities with the intensity λ'. Summing over j gives (ii). The appendix justifies exchanging the infinite sum with the expectation, through the density of S_T and a bound on the mean of $|f(S_T)|$ given j jumps that grows at most geometrically in j; this is where the growth condition of the class $\mathcal{G}$ enters. Part (i) is (ii) for the payoff $f(x) = x$: the BSM price of the claim paying S_T is Se^{-qT} whatever the rate and the volatility, and the weights sum to one. The complete proof is in Appendix I.12. □

(Problem P23.11 asks you to rederive the series of Proposition 23.60(ii) for the call only, by the same conditioning on the number of jumps, and to price a put from a call by put–call parity, which rests on part (i); the series for a general payoff of the class $\mathcal{G}$, with the growth bound of Step 4, is left to the complete proof in Appendix I.12.)

Each term of the series is the BSM price in the scenario with j jumps: the variance of j jumps is added to the diffusion variance, and the rate $r_{(j)}$ is r less the compensation $\lambda\kappa$ of the drift plus the mean growth that j jumps bring. The weights are Poisson probabilities with the intensity λ' in the place of λ, because the price is discounted at r whereas the formula of the scenario discounts at $r_{(j)}$ (the identity displayed in the sketch of the proof of Proposition 23.60). Put–call parity, $C - \mathcal{P} = Se^{-qT} - Ke^{-rT}$, follows from (i) and the linearity of the expectation, since $(x-K)^+ - (K-x)^+ = x - K$.

Example 23.61 (Checking the series)**.** Let $\lambda = 0.5$, $\kappa = -10\%$, $\delta = 15\%$ and $\sigma = 12\%$: a jump every two years on average, which lowers the price by 10% on average. Let $S = K = 100$, $r = q = 0$ and $T = 0.25$. Then $\lambda' = 0.45$ and $\lambda' T = 0.1125$, and the first terms of the series of Proposition 23.60 are as follows.

jumps j	weight	$\sigma_{(j)}$	$r_{(j)}$	$C^{\mathrm{BSM}}(S,K;\sigma_{(j)},r_{(j)})$	term
0	0.8936	12.00%	+5.00%	3.0510	2.7263
1	0.1005	32.31%	−37.14%	2.9681	0.2984
2	0.0057	44.09%	−79.29%	2.4394	0.0138
3	0.0002	53.33%	−121.43%	1.9608	0.0004

The rate of the scenario without jumps is $r - \lambda\kappa = +5\%$: with no jump the price must drift up to pay for the jumps that did not come. The series sums to $C = 3.0389$, an implied volatility of 15.24%. A simulation of 4,000,000 draws of the law (23.22) gives 3.0360, with standard error 0.0022, and a mean terminal price of 99.998, with standard error 0.004, against the forward 100 of part (i).

With $r = 4\%$ and $T = 1$ the series gives $C = 8.8169$; the simulation gives 8.8167, with standard error 0.0055, and a mean terminal price of 104.073, with standard error 0.009, against $Se^{rT} = 104.081$.

(Problem P23.11 works this computation at different numbers.)

Example 23.62 (The smile of a jump model)**.** Keep the model of Example 23.61 ($\lambda = 0.5$, $\kappa = -10\%$, $\delta = 15\%$, $\sigma = 12\%$, $S = 100$, $r = q = 0$). The prices of Proposition 23.60, inverted for the implied volatility, give the following smiles.

strike K	70	85	100	115	130
three months	31.1%	23.5%	15.2%	15.1%	19.2%
two years	19.5%	18.1%	17.0%	16.2%	15.8%

The premium of the 85 strike over the money falls from 8.3 volatility points at three months to 1.1 at two years. Jumps make the short-dated skew, and it fades with maturity, as the jump risk averages out over the longer horizon.

Interview trap. "Jumps explain the long-dated smile." The smile of a jump model decays with maturity (Example 23.62: 8.3 points of skew at three months, 1.1 at two years). Curvature that persists at long maturities comes from stochastic volatility.

Fact 23.63 (Correlated stochastic volatility (Heston, 1993))**.** In the model in which the variance of the price follows a mean-reverting square-root process, driven by a Brownian motion that is correlated with the one that drives the price, European option prices have a semi-closed form: an integral of an explicit characteristic function. A negative correlation between the price and its variance produces a downward skew.

A proof is beyond our scope. The square-root equation has a coefficient that is not Lipschitz, so even the existence of the variance process lies outside Fact 19.29; and the formula needs the characteristic function of an affine process, obtained from Riccati equations, and Fourier inversion. No proof of this Part relies on it.

Remark 23.64 (Which feature makes which smile). The following are observations, illustrated by Example 23.59 (computed from Proposition 23.58) and Example 23.62 (computed from Proposition 23.60); they are not theorems, and nothing later uses them. Stochastic volatility that is uncorrelated with the price makes a roughly symmetric smile, with its minimum near the money, that persists at long maturities (the general statement, symmetry in the log-moneyness, is due to Renault and Touzi, 1996). A negative correlation between the price and its volatility tilts that smile into the equity skew of Fact 22.5 (Fact 23.63). Jumps make steep short-dated smiles that flatten with maturity, as the jump risk averages out. The constant-elasticity-of-variance model (Cox, 1975), whose local volatility is a negative power of the price, makes a skew by raising the volatility as the price falls. The variance-gamma model (Madan, Carr and Chang, 1998) is named only; no formula of either of these two models is given here.

Remark 23.65 (Out of scope). Convertible bonds and real options are not treated in this document.

Practice: problems P23.1–P23.11 in §27, solutions in §28.

Part VII
Interview Craft: Cost Models and Open-Ended Questions

24 Algorithmic Cost Models

This section develops the two cost-model calculations that live quant and AI assessments ask most often: when a low-rank factorisation of a matrix beats the dense product, and when pre-sorting or hashing a dataset beats scanning it. Both questions reduce to a single move — write down the two competing cost expressions and solve for the crossover — which we isolate as a named pattern in Remark 24.9.

24.1 Matrix Multiplication and the Decomposition Crossover

Every linear layer, factor exposure, and covariance update is ultimately a matrix product, so the first quantity to control is what a product costs. This subsection counts that cost exactly, then answers the question interviewers build on top of it: if a dense matrix is replaced by a low-rank factorisation, at what rank does the factored product become cheaper — and why does one of the two ways of associating the factored product *never* win?

Setup. Let N, P, R, and K denote positive integers. Here N, P, R, K are integer matrix dimensions; in particular P is not the probability operator of the notation table and R is not a return — no probabilities or returns appear in this section. All matrices in this subsection are real and dense: A is $N \times P$ and B is $P \times R$, so the product AB is $N \times R$. The cost model counts *scalar multiplications*: the cost of an algorithm is the number of scalar multiplications it performs, with scalar additions tracked alongside for completeness. This is the standard unit in numerical linear algebra, and the two counts agree to leading order in any case, as the next proposition shows.

Proposition 24.1 (Matrix multiplication count)**.** *Computing the product AB of an $N \times P$ matrix A and a $P \times R$ matrix B directly from the definition costs exactly NPR scalar multiplications and $NR(P-1)$ scalar additions. In total this is $2NPR - NR$ floating-point operations, which is $2NPR$ flops to leading order.*

Proof. We proceed by a counting argument. The product AB has NR entries, and entry (i, j) is the dot product of row i of A with column j of B:

$$(AB)_{ij} \;=\; \sum_{k=1}^{P} a_{ik}\, b_{kj}.$$

Evaluating one such length-P dot product takes P multiplications (one per term) and $P-1$ additions (to sum P terms). Since the NR output entries are computed independently, the totals are $NR \cdot P = NPR$ multiplications and $NR(P-1)$ additions. Summing the two counts gives $NPR + NR(P-1) = 2NPR - NR$ operations; the deficit NR is lower order in P, so the flop count is $2NPR$ to leading order. □

(Problem P24.1 asks you to reproduce this proof from scratch.)

Example 24.2 (Counting a small product by hand)**.** Take $N = 2$, $P = 3$, $R = 2$, so A is 2×3 and B is 3×2. Proposition 24.1 predicts $NPR = 2 \cdot 3 \cdot 2 = 12$ multiplications and $NR(P-1) = 2 \cdot 2 \cdot 2 = 8$ additions. Direct verification: the product has $NR = 4$ entries, each a length-3 dot product costing 3 multiplications and 2 additions, giving $4 \cdot 3 = 12$ multiplications and $4 \cdot 2 = 8$ additions, as predicted. The exact operation total is $12 + 8 = 20$, against the leading-order figure $2NPR = 24$; the gap is exactly $NR = 4$.

Intuition. The formula NPR is best remembered as a picture, not a symbol string: the output has $N \times R$ cells, and filling each cell means running a length-P dot product across the inner dimension. Output area times inner length. The flop convention $2NPR$ simply counts the multiply and the add of each multiply-accumulate step separately.

Setup (decomposition). Suppose now that a dense $N \times P$ matrix M has been replaced by a rank-K factorisation

$$M = UV, \qquad U \in \mathbb{R}^{N \times K}, \qquad V \in \mathbb{R}^{K \times P},$$

and that the object we actually need is the product MB with a data block $B \in \mathbb{R}^{P \times R}$ (for a single input vector, $R = 1$). Matrix multiplication is associative, so the factored product can be evaluated in two orders, $(UV)B$ or $U(VB)$, which give the same matrix at very different prices. The next theorem prices both orders and locates the rank below which the factorisation pays.

Theorem 24.3 (Decomposition crossover)**.** *Under the multiplication-count cost model of Proposition 24.1, with $M = UV$ as above:*

(i) the order $(UV)B$ costs $NKP + NPR$ multiplications, which exceeds the dense cost NPR of MB for every $K \geq 1$ — this association order never beats the dense product;

(ii) the order $U(VB)$ costs $KPR + NKR = KR(N+P)$ multiplications;

(iii) with the correct association $U(VB)$, the factored product beats the dense product if and only if

$$K \;<\; \frac{NP}{N+P}, \tag{24.1}$$

a threshold that does not depend on R.

Proof. We apply Proposition 24.1 to each subproduct and compare the resulting cost expressions.

Part (i). Forming UV multiplies an $N \times K$ matrix by a $K \times P$ matrix, costing NKP by Proposition 24.1, and the result is the full dense $N \times P$ matrix M. Multiplying it by B then costs NPR, for a total of

$$\text{cost}\,[(UV)B] \;=\; NKP + NPR. \tag{24.2}$$

The second term alone already equals the dense cost of MB, so the total exceeds it by $NKP > 0$ whenever $K \geq 1$. The mechanism deserves stating in words: reassembling the factors recreates the dense matrix, hence the entire dense bill NPR, and the assembly cost NKP is paid on top. No choice of K rescues this order.

Part (ii). Forming VB multiplies a $K \times P$ matrix by a $P \times R$ matrix, costing KPR; the result is only $K \times R$. Multiplying U $(N \times K)$ by this small block costs NKR. The total is

$$\text{cost}\,[U(VB)] \;=\; KPR + NKR \;=\; KR(N+P). \tag{24.3}$$

Part (iii). We compare (24.3) with the dense cost NPR and solve for K:

$$KR(N+P) \;<\; NPR \iff K(N+P) \;<\; NP \iff K \;<\; \frac{NP}{N+P},$$

where the first equivalence divides both sides by $R > 0$. The variable R cancels, which proves the threshold is independent of the width of the data block. □

(Problem P24.3 asks you to reproduce this proof from scratch.)

Figure 52 shows all three cost expressions as functions of K at fixed N, P, R: the dense cost is flat, the correctly associated factored cost is a line through the origin that crosses it at $K^* = NP/(N+P)$, and the misassociated cost (24.2) starts *at* the dense cost when $K = 0$ and only climbs from there.

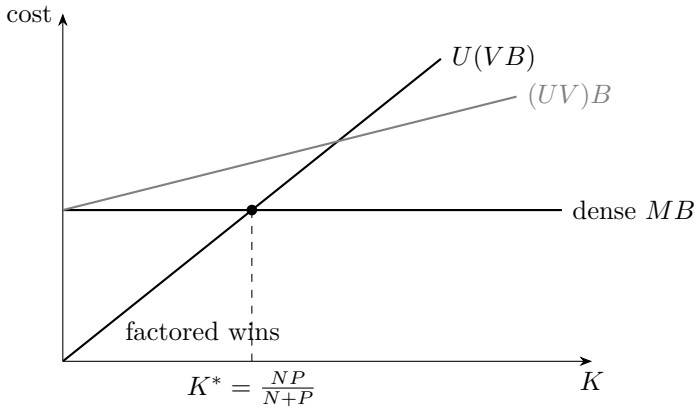


Figure 52: Multiplication cost as a function of the factor rank K, at fixed N, P, R. The dense product MB costs NPR regardless of K. The correctly associated factored product $U(VB)$ costs $KR(N+P)$, a straight line through the origin that crosses the dense cost at $K^* = NP/(N+P)$; for $K < K^*$ the factored path wins. The misassociated product $(UV)B$ costs $NKP + NPR$: it starts at the dense cost and rises, so it loses to the dense product for every $K \geq 1$.

Remark 24.4 (Harmonic-mean form of the threshold). The harmonic mean of N and P is $H = 2NP/(N+P)$, so the threshold (24.1) reads $K < H/2$: the break-even rank is half the harmonic mean of the two dense dimensions. For a square matrix, $N = P$ gives $K < N/2$ — any factorisation with rank below half the side length is a win. Because R cancels, the comparison is effectively per column of data: pushing more data through the layer changes both bills proportionally and moves the crossover not at all.

Example 24.5 (Low-rank arithmetic at LoRA scale)**.** A dense 4096×4096 layer is replaced by a rank-64 factorisation, so $N = P = 4096$ and $K = 64$; take a single input vector, $R = 1$.

- *Dense cost:* $NPR = 4096 \cdot 4096 = 16{,}777{,}216$ multiplications.
- *Factored cost, correct order $U(Vx)$:* $KR(N+P) = 64 \cdot 8192 = 524{,}288$ multiplications.
- *Speedup:* $16{,}777{,}216/524{,}288 = 32$, an exact 32-fold reduction.
- *Break-even rank:* $K^* = NP/(N+P) = 4096^2/8192 = 2048$, which is $N/2$ as Remark 24.4 predicts for a square layer; the speedup can be read off as $K^*/K = 2048/64 = 32$, consistent with the direct division.
- *Misassociated order $(UV)x$:* $NKP + NPR = 4096 \cdot 64 \cdot 4096 + 16{,}777{,}216 = 1{,}073{,}741{,}824 + 16{,}777{,}216 = 1{,}090{,}519{,}040$ multiplications — 65 times the dense cost, instead of 32 times cheaper.

Trading application. This crossover is the economics behind LoRA-style adapters: a rank-K update with $K \ll \min(N, P)$ sits far below the threshold (24.1), so the factored path is cheap — rank 64 against a break-even of 2048 in Example 24.5. The operational rule that comes with it: keep the factors separate and always multiply the data into V first. Re-multiplying the factors back together before touching the data forfeits the entire saving and adds an assembly bill on top, by Theorem 24.3(i).

Interview trap. Association order is the hidden half of the question. An interviewer who asks for the cost of the factored product expects both orders priced: quoting only $(UV)B$ at $NKP + NPR$,

or asserting that "the factorisation makes it cheaper" without naming the order, misses that one order *always* loses — for every rank, by Theorem 24.3(i), and by a factor of 65 at the scale of Example 24.5.

24.2 Membership Queries: Scan vs Sort vs Hash

The second house calculation prices a data-structure decision instead of a linear-algebra one: given a batch of membership queries against a static dataset, when does the up-front cost of sorting (or hashing) pay for itself? The move is the same as in Theorem 24.3(iii): write both total costs and solve for the crossover.

Setup. A static dataset holds N elements, with $N \geq 2$, and must answer q membership queries ("is x in the dataset?"). In this subsection q denotes the number of queries — not the complement probability $1 - p$ of the notation table; the symbol is re-purposed locally. Logarithms in cost expressions are written $\log_2$ because the counts arise from repeated halving; inside $O(\cdot)$ the base only shifts a constant factor. Three designs compete, with the following idealised comparison counts:

$$\begin{aligned} &\text{scan: } qN, \\ &\text{sort + binary search: } N\log_2 N + q\log_2 N = (N+q)\log_2 N, \\ &\text{hash set: } O(N) \text{ expected build} + O(q) \text{ expected probes.} \end{aligned} \tag{24.4}$$

Here a linear scan costs N comparisons per query with no preprocessing; a comparison sort costs $N\log_2 N$ comparisons once, after which each binary search costs $\log_2 N$ comparisons; and a hash set costs $O(N)$ expected time to build, then $O(1)$ expected time per probe, for $O(N+q)$ expected in total.

Proposition 24.6 (Scan-versus-sort crossover)**.** *Under the cost model (24.4), sorting once and binary-searching each query beats scanning per query if and only if*

$$q \;>\; \frac{N\log_2 N}{N - \log_2 N}, \tag{24.5}$$

and the threshold satisfies $N\log_2 N/(N-\log_2 N) \to \log_2 N$ *in ratio as* $N \to \infty$*: for large* N*, sorting pays for itself after roughly* $\log_2 N$ *queries.*

Proof. We compare the two total costs in (24.4) directly and solve the resulting inequality for q. Sorting wins exactly when

$$(N+q)\log_2 N \;<\; qN \quad\Longleftrightarrow\quad N\log_2 N \;<\; q\,(N - \log_2 N).$$

Since $2^N > N$ for every $N \geq 1$, we have $N > \log_2 N$, so the factor $N - \log_2 N$ is positive and dividing by it preserves the inequality, giving (24.5). For the asymptotic claim, divide the threshold by $\log_2 N$:

$$\frac{1}{\log_2 N}\cdot\frac{N\log_2 N}{N-\log_2 N} \;=\; \frac{N}{N-\log_2 N} \;=\; \Bigl(1 - \tfrac{\log_2 N}{N}\Bigr)^{-1} \;\longrightarrow\; 1 \qquad (N\to\infty),$$

because $\log_2 N/N \to 0$. The threshold is therefore $\log_2 N$ up to a factor tending to one. □

(Problem P24.5 asks you to reproduce this derivation from scratch.)

Example 24.7 (Crossover at $N = 1024$)**.** Take $N = 1024$, so $\log_2 N = 10$. The exact threshold (24.5) is

$$q \;>\; \frac{1024\cdot 10}{1024 - 10} \;=\; \frac{10{,}240}{1014} \;\approx\; 10.1,$$

against the approximation $\log_2 N = 10$ — the approximation is tight already at this size. Checking the two integers either side of the threshold: at $q = 10$, scanning costs $10 \cdot 1024 = 10{,}240$ while sort-plus-search costs $(1024 + 10) \cdot 10 = 10{,}340$, so scanning still wins; at $q = 11$, scanning costs $11 \cdot 1024 = 11{,}264$ while sort-plus-search costs $(1024 + 11) \cdot 10 = 10{,}350$, and sorting has already paid for itself. Ten queries against a thousand elements — a strikingly small number.

Fact 24.8 (Expected cost of hashing)**.** Under the standard uniform-hashing assumptions, building a hash set over N elements takes $O(N)$ expected time (amortised $O(1)$ per insertion with table doubling) and each membership probe takes $O(1)$ expected time, so answering q queries costs $O(N + q)$ expected in total — asymptotically better than both alternatives in (24.4). The guarantees are expected-case, not worst-case: adversarial or heavily colliding keys degrade probes toward linear time. The analysis is classical (Knuth, 1998); a proof is beyond our scope.

Trading application. The hash set beats both scanning and sorting asymptotically, but the win is bought with three concessions: extra memory for the table, the loss of all ordering information, and worst-case degradation under collisions. In an interview, frame it as "expected $O(1)$ lookups, amortised build" — the qualifiers are the point, because they show the caveats are understood rather than papered over.

Interview trap. Quoting the hash table as flatly "$O(1)$" drops both qualifiers that make the claim true. The lookup bound is *expected*, not guaranteed — collisions, and in the worst case adversarial keys, push probes toward linear time — and the build is *amortised* over insertions. An answer that prices hashing without saying "expected" and "amortised" invites exactly the follow-up it fails to anticipate.

Remark 24.9 (The house pattern: equate the two costs)*.* Every crossover question in this section is the same calculation wearing different clothes: write down the total cost of each competing design as an expression in the problem's parameters, set the two expressions equal, and solve for the variable the question asks about. Theorem 24.3(iii) is this pattern solved for the rank K; Proposition 24.6 is the same pattern solved for the query count q. Live assessments have asked exactly this shape ★, and the pattern generalises to any "at what point does design A beat design B" question: two cost lines, one intersection, solve.

Practice: problems P24.1–P24.6 in §27, solutions in §28.

25 Forecasting Returns: the Open-Ended Question

"How would you predict the returns of an index?" is a standard closing question in quantitative interviews, and it is graded differently from every puzzle that precedes it: the expected answer is a *process*, not a model name. This section fixes that process as a four-step numbered procedure (Procedure 25.1), supplies the rationale behind each step, examines the one genuine architectural fork inside the pipeline — modelling the index directly versus modelling its constituents and aggregating (§25.2) — and makes the validation protocol and its metrics precise (§25.3). Figure 53 shows the whole pipeline at a glance.

25.1 The Pipeline as a Procedure

This subsection states the pipeline formally and gives the rationale for its first two steps; the modelling fork (Step 3) and the validation protocol (Step 4) carry enough content to receive their own subsections below.

The setup is as follows. Fix an equity index and a forecast horizon, and let time run in discrete periods of that horizon. The quantity to be forecast is the index return R_{t+1} over the period from t to $t + 1$. The forecaster observes, at time t, a feature vector $X_t \in \mathbb{R}^d$ and issues a forecast

$\hat{R}_{t+1}$ of R_{t+1}. The standing assumption on all features is *measurability at forecast time*: every component of X_t must be computable from information actually available when the forecast is issued. Violations of this assumption are the look-ahead bias that the interview-trap block of §25.3 names first.

Procedure 25.1 (Index-return forecasting pipeline)**.** Given the task of forecasting the return of an equity index, proceed in four steps.

1. **Define the target.** Fix the return convention — log or simple — and the horizon (one minute, one day, one week), and state *why*: every downstream choice depends on the target.
2. **Explore the data, then build features in layers.** Exploratory analysis first: stationarity, outliers, and regime breaks; the correlation matrix; PCA for the factor structure. Then assemble X_t in four layers: (i) *endogenous* — past returns, realized volatility, volume, spread, plus order-book features when the horizon is intraday; (ii) *exogenous lead–lag* — markets that move earlier (the US close leads the European open; overnight index futures; rates, FX, commodities); (iii) *composition-specific* — features implied by what the index actually holds (the CAC 40's heavy luxury-sector weight makes China sentiment a feature; TotalEnergies makes Brent one); (iv) *sentiment* — news NLP and positioning data.
3. **Choose the modelling route: direct or factorial** (Definition 25.4). In either route, fit the simplest credible model first — ridge regression on standardized features — and escalate to gradient-boosted trees (LightGBM) or sequence models only if the added capacity is justified out of sample.
4. **Validate against a baseline, walking forward.** Establish a trivial baseline and ask first whether the information coefficient (Definition 25.6) is positive at all; split the data chronologically (`TimeSeriesSplit` / walk-forward), never shuffled; score the information coefficient, the hit rate, and the Sharpe ratio of the simulated strategy; and name the deployment risks collected in the interview-trap block of §25.3 without being prompted.

(Problem P25.1 asks for exactly this procedure, narrated end to end under interview conditions.)

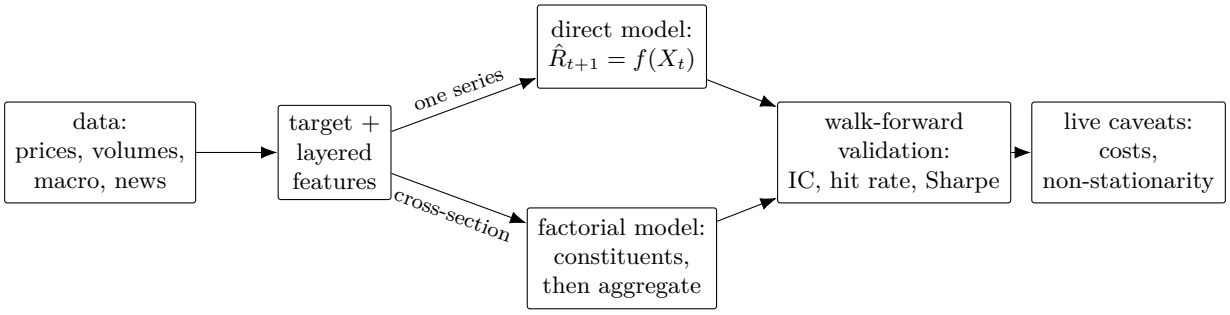


Figure 53: The forecasting pipeline of Procedure 25.1. Data feed layered features; the modelling stage forks into the direct route (one model fitted to the index series) and the factorial route (constituent-level models aggregated by index weights, Definition 25.4); both routes are scored by the same walk-forward validation, and whatever survives still faces the live caveats collected in the interview-trap block of §25.3.

Interview trap. The model-name answer. Replying "an LSTM" or "gradient boosting" to this question fails regardless of which model is named, because the question grades process, not taste in architectures. The strong answer walks Procedure 25.1 in order — target first, model third — and volunteers the risks of §25.3 unprompted.

Remark 25.2 (Rationale for Step 1: the target determines everything downstream). Two choices are made in Step 1, and neither is cosmetic. First, the return convention: log returns add across time, which suits multi-period and volatility work, while simple returns add across assets, which is what a weighted aggregation over index constituents requires — so the convention chosen here

interacts directly with the fork of Step 3 (see (25.1)). Second, the horizon: it sets the noise level of the target, determines which features exist at all (order-book features are meaningful intraday and not at a weekly horizon), governs how quickly any signal decays, and fixes how often the resulting strategy must trade — hence its exposure to transaction costs. Stating the "why" aloud is part of what is graded: a forecaster who cannot say why the horizon is one day has not yet defined the problem.

Remark 25.3 (Rationale for Step 2: explore before modelling, layer the features). The exploratory pass earns its place ahead of any model. Stationarity checks decide whether tools that assume a stable distribution are usable at all; outliers decide the loss function and winsorization policy; detected regime breaks warn that validation must span regimes rather than sit inside one; and the correlation matrix with a PCA on top exposes the factor structure that Step 3's factorial route will reuse. The layers themselves are an audit device: each layer draws on a different source of information — the index's own history, other markets that trade earlier (an empirical lead–lag regularity, as with the US close leading the European open), the macro exposures implied by the index's actual holdings, and text or positioning data. Walking the layers in order shows the coverage is systematic rather than anecdotal, and the composition-specific layer is what distinguishes forecasting *this* index from forecasting a generic one.

25.2 The Modelling Fork: Direct versus Factorial

Step 3 of Procedure 25.1 contains the pipeline's one genuine architectural decision: fit one model to the index series itself, or model the constituents and aggregate. This subsection defines both routes precisely and gives the rationale for preferring the second.

The setup is as follows. The index has n constituents with weights $w_1, \dots, w_n$, positive and summing to one, held fixed over the forecast horizon for simplicity. Constituent i has simple return $R_{i,t+1}$ over the period from t to $t+1$, and the index simple return is the weighted aggregate

$$R_{t+1} \;=\; \sum_{i=1}^{n} w_i \, R_{i,t+1}, \tag{25.1}$$

which is an identity for simple returns (and only approximate for log returns — the convention choice of Step 1 at work). The symbol $R_{m,t+1}$ denotes the market return over the same period, and β_i the market beta of constituent i.

Definition 25.4 (Direct and factorial forecasts)**.** Fix the target R_{t+1} and the feature vector X_t of §25.1.

1. The *direct route* fits a single model f to the index series itself and forecasts $\hat{R}_{t+1} = f(X_t)$.
2. The *factorial route* decomposes each constituent by the market model

$$R_{i,t+1} \;=\; \beta_i \, R_{m,t+1} + \alpha_{i,t+1}, \tag{25.2}$$

   where $\alpha_{i,t+1}$ is the beta-neutralized residual return of constituent i; builds forecasts at the constituent level — for the market component and for the residuals — and aggregates them into an index forecast through (25.1): $\hat{R}_{t+1} = \sum_{i=1}^{n} w_i \, \hat{R}_{i,t+1}$.

Remark 25.5 (Rationale for Step 3: why the factorial route is often better). Three observations favor the factorial route.

1. *PCA reveals the structure the route exploits.* An eigendecomposition of the constituent correlation matrix typically shows the first principal component approximating the market mode and the next few components (roughly PC2 through PC5) approximating sector modes. This decomposition supplies the market factor used for the residualization in (25.2) and compresses the n correlated constituent series into a small number of drivers.

2. *The beta-neutralized residual is the cleaner target.* At short horizons the raw constituent series is dominated by the common market mode, which is the noisiest and hardest component to predict; stripping it out by (25.2) and forecasting the residual $\alpha_{i,t+1}$ gives a prediction target with cleaner structure than the raw series.
3. *The cross-section multiplies the data.* Modelling n related series instead of one provides many more observations of the shared structure per unit of calendar time than the single index series does.

The aggregation step itself relies on (25.1) holding exactly, which is one more reason the return convention is fixed in Step 1 before any model is chosen.

(Problem P25.3 asks for this case to be argued in full, including the ways the weighted aggregation can fail.)

Intuition. The direct route asks one model to predict the hardest object in the market — the market mode itself — from a single noisy history. The factorial route replaces that one hard problem with many easier, related ones plus a known aggregation rule: model what has structure (the residuals, the factor loadings), and let the arithmetic of (25.1) do the rest.

25.3 Validation: Protocol, Metrics, and the Risks to Name

Step 4 is where forecasting pipelines are separated from curve-fitting exercises. This subsection defines the three evaluation metrics precisely, gives the rationale for the walk-forward protocol and for rejecting mean squared error as the headline metric, works a fully numeric example, and closes with the structured list of risks that a complete answer names unprompted.

The setup is as follows. The model is evaluated on a window of T periods, producing pairs $(\hat{R}_t, R_t)$ for $t = 1, \ldots, T$, where $\hat{R}_t$ denotes the forecast of R_t issued one period earlier; every forecast obeys the measurability assumption of §25.1. Sample means over the window are written $\bar{\hat{R}}$ and $\bar{R}$. The sign function takes values in $\{-1, 0, +1\}$. The position held in period t is written w_t, with the convention $w_0 = 0$ — a traded position, unrelated to the constituent weights w_i of §25.2 — and $c \geq 0$ denotes the per-unit transaction cost charged on position changes.

Definition 25.6 (Forecast evaluation metrics: IC, hit rate, strategy Sharpe)**.** Over the evaluation window:

1. The *information coefficient* (IC) is the Pearson sample correlation between forecasts and realizations,
$$\mathrm{IC} \;=\; \frac{\sum_{t=1}^{T}(\hat{R}_t - \bar{\hat{R}})(R_t - \bar{R})}{\sqrt{\sum_{t=1}^{T}(\hat{R}_t - \bar{\hat{R}})^2}\,\sqrt{\sum_{t=1}^{T}(R_t - \bar{R})^2}}.$$
2. The *hit rate* is the fraction of periods on which the forecast calls the direction correctly, $H = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\mathrm{sign}(\hat{R}_t) = \mathrm{sign}(R_t)\}$.
3. The *strategy Sharpe* is computed by trading the forecast: take the position $w_t = \mathrm{sign}(\hat{R}_t)$ each period, record the per-period profit and loss net of transaction costs, $\Pi_t = w_t R_t - c\,|w_t - w_{t-1}|$ (the per-unit cost c charged on position changes), and report the Sharpe ratio of the series $\Pi_1, \ldots, \Pi_T$ — its sample mean over its sample standard deviation, annualized by the square root of the number of periods per year (the Sharpe conventions of §10).

Remark 25.7 (Rationale for Step 4, part one: why these metrics and not MSE)*.* Raw mean squared error is direction-blind: it measures distance between forecast and realization, while trading profit depends on the *sign* of the forecast through the position w_t. A squared-error criterion can therefore strictly prefer a money-losing forecast to a money-making one — Example 25.9 exhibits the failure numerically. The three metrics of Definition 25.6 are chosen to respect the use case: the IC measures monotone co-movement, the hit rate scores direction itself, and the strategy Sharpe is the economic bottom line after costs.

Remark 25.8 (Rationale for Step 4, part two: baseline first, walk forward, never shuffle). The baseline comes first because it defines the null: trivial forecasts (identically zero, or the historical mean) cost nothing and encode no skill, so the first question for any candidate model is whether it achieves an out-of-sample IC greater than zero at all. The evaluation must be chronological: train on a window of past data, test on strictly later data, then roll the window forward — the scheme implemented by `TimeSeriesSplit` and generically called walk-forward validation. Shuffled cross-validation is never acceptable for this problem: shuffling places observations from the test period's future inside the training set, which is look-ahead bias built directly into the protocol, and it produces validation scores that cannot be earned in live trading, where the future is genuinely unavailable.

Example 25.9 (Scoring a five-day toy forecast)**.** A daily index forecast is evaluated over $T = 5$ days. Returns are in percent.

Day t	1	2	3	4	5
Realized R_t	$+1.0$	-0.5	$+0.2$	-0.8	$+0.6$
Forecast $\hat{R}_t$	$+0.4$	-0.1	-0.3	-0.2	$+0.5$
Direction hit	yes	yes	no	yes	yes
P&L $\Pi_t = w_t R_t$ (at $c = 0$)	$+1.0$	$+0.5$	-0.2	$+0.8$	$+0.6$

Hit rate. Four of the five signs agree, so $H = 4/5 = 80\%$.

Information coefficient. The sample means are $\bar{\hat{R}} = 0.3/5 = 0.06$ and $\bar{R} = 0.5/5 = 0.10$. The centered cross-products sum to

$$(0.34)(0.90) + (-0.16)(-0.60) + (-0.36)(0.10) + (-0.26)(-0.90) + (0.44)(0.50) = 0.820,$$

while the centered sums of squares are 0.5320 for the forecasts and 2.24 for the realizations, so

$$\mathrm{IC} = \frac{0.820}{\sqrt{0.5320 \times 2.24}} = \frac{0.820}{1.0916} \approx 0.75.$$

Strategy Sharpe (before costs). The P&L series has sample mean $2.7/5 = 0.54\%$ per day and sample standard deviation $\sqrt{0.8320/4} \approx 0.456\%$, giving a per-day Sharpe of $0.54/0.456 \approx 1.18$ and an annualized Sharpe of $1.18 \times \sqrt{252} \approx 18.8$.

Why MSE is the wrong headline metric. Consider day 1, on which the realized return is $+1.0$. A forecast of -0.2 has squared error $(-1.2)^2 = 1.44$ and puts the strategy short, losing 1.0; a forecast of $+2.4$ has the *larger* squared error $(1.4)^2 = 1.96$ yet puts the strategy long, making $+1.0$. Squared error prefers the money-losing forecast, which is the direction-blindness of Remark 25.7 in one line.

Calibration of the numbers. The toy window is five days, so an IC of 0.75 and an annualized Sharpe of 18.8 carry no evidential weight — and in a real pipeline a number like 18.8 is precisely the "too-beautiful" Sharpe that the trap block below says to treat as a bug. Good daily-horizon index pipelines sit only a few points of hit rate above 50%; a backtest printing far above that level is evidence of leakage, not of edge.

Interview trap. The risks to name unprompted. Volunteering the deployment risks — rather than waiting to be asked — is what separates a process answer from a model-name answer. Four must be named.

1. *Look-ahead bias.* Any feature, label, or preprocessing step that touches information from after the forecast time violates the measurability assumption of §25.1; the walk-forward protocol of Remark 25.8 exists to prevent the protocol-level version of it.
2. *Overfitting.* A too-beautiful Sharpe is a bug: an implausibly good backtest is evidence of leakage or overfitting, not of edge, and the correct response is to hunt for the error (Example 25.9 calibrates what "implausible" means at a daily horizon).

3. *Transaction costs.* A gross edge at high turnover can be a net loss; the strategy Sharpe of Definition 25.6 is scored after costs for exactly this reason.
4. *Non-stationarity.* The relationships the model learns drift, and regimes break; a model validated inside one regime may fail in the next, which is why regime breaks are hunted in Step 2's exploratory pass and why walk-forward performance is examined for stability across sub-periods rather than only in aggregate.

(Problem P25.2 applies exactly this checklist to a strategy that "backtests very well.")

Practice: problems P25.1–P25.3 in §27, solutions in §28.

Part VIII
Computer Science for Research Engineering

26 Computer Science for Research Engineers: Python Data Structures, Parallelism, and Time

Research-lab on-site rounds for research-engineering roles contain one distinctive interview: a single long-form "algorithmic reasoning" question in escalating parts, answered in pseudocode on paper, with no live coding and a grade for the quality of the reasoning rather than for syntax. Two topics decide its later parts more than any other — the difference between wall-clock time and CPU time, and race conditions — and this section is built around them, developing in order Python's containers as CPython implements them, a few algorithmic patterns, the two clocks and their ratio, the speedup laws, associativity as the licence to parallelise, Python's threads and processes, race conditions and deadlock, and distributed aggregation (§26.1 to §26.8). It builds on the cost models of Section 24: the crossover pattern of Remark 24.9 reappears solved for the number of workers, and the collision caveat of Fact 24.8 receives its defence. Very little code appears; the objects of study are costs, clocks and interleavings, which is what a paper-and-pencil format examines. Implementation figures quoted below (container capacities, resize points, bytecode listings, floating-point identities, small traces and race measurements) were recomputed on CPython 3.11.9 and, where the behaviour is version-dependent, checked against the 3.12 sources; every other number is arithmetic the reader can reproduce.

26.1 Cost Qualifiers and Python's Containers as CPython Implements Them

Part (a) of every long-form algorithmic-reasoning question is "pick a container and price it", and the grade is on the qualifiers — worst-case, amortised, expected — that a candidate attaches unprompted. This subsection defines the three qualifiers, states how CPython 3.12 lays out `list`, `tuple`, `dict`, `set`, `deque` and `heapq`, proves the two results behind the cost table (any geometric growth factor gives amortised $O(1)$ append; `heapify` is linear) and the quadratic cost of repeated string concatenation, and ends in the one cost table to reproduce on paper. It builds on the membership cost model of §24.2: Fact 24.8 priced hashing with a collision caveat, and the hash-randomisation Fact below is the defence to that caveat.

Setup. Costs are counted in the RAM model: one unit per pointer read or write, per comparison, or per hash evaluation, with memory access at unit cost. The symbol n denotes the number of elements currently stored in a container, m the length of a sequence of operations, ℓ the length of a slice, and k the number of largest items requested in a top-k query; none of these is a probability or a random-walk quantity of the notation table. The symbol Ψ denotes an amortisation potential; Φ

keeps its document-wide meaning and does not appear here. A CPython container stores *pointers* to boxed objects, never the values themselves (Remark 9.13); on a 64-bit build a pointer occupies 8 bytes. Every implementation claim refers to CPython 3.12 unless a version is named, and is an implementation choice rather than a language guarantee unless it is stated as one; measurements quoted from CPython 3.11 are marked as such. The symbol $g > 1$ denotes a growth factor and $c_0 \geq 1$ an initial capacity.

Definition 26.1 (Worst-case, amortised and expected cost)**.** Let an operation on a container be applied repeatedly.

(i) Its *worst-case cost* is a bound that holds for every single application, whatever the state of the container.

(ii) Its *amortised cost* is a bound a such that any sequence of m applications, starting from an empty container, costs at most am in total (the *aggregate method*); one expensive application is paid for by many cheap ones, and the bound says nothing about any single call.

(iii) Its *expected cost* is the average over the algorithm's own randomness — for a hash table, the hash function — on a *fixed* input; it says nothing about the cost on an adversarial input.

The three are independent: `append` is amortised $O(1)$ yet worst-case $O(n)$ (a resize), and a `dict` probe is expected $O(1)$ yet worst-case $O(n)$ (collisions). Fact 24.8 already attached "expected" and "amortised" to hashing; every row of Table 5 below carries exactly one of the three.

Fact 26.2 (CPython container layouts (CPython 3.12: Objects/listobject.c and its sibling object files; Lib/heapq.py))**.** Five layouts, read from the CPython source files.

(i) *list and tuple.* A `list` is a header holding a pointer to one contiguous array of object pointers, the current length, and an allocated capacity. When an append finds the array full, `list_resize` allocates a new capacity of newsize $+ \lfloor \text{newsize}/8 \rfloor + 6$ rounded down to a multiple of 4 and copies every pointer across; the formula is unchanged from 3.9 to 3.12. The capacities of a list grown by appends are $4, 8, 16, 24, 32, 40, 52, 64, \ldots$, a growth ratio tending to 9/8 (read through `sys.getsizeof` on CPython 3.11). Hence indexing and item assignment cost $O(1)$, `pop()` at the end amortised $O(1)$ (`list_resize` shrinks the allocation whenever the new size falls below half the capacity, so an occasional realloc copy occurs: popping from a 100-element list drops the capacity from 100 to 60 at length 49, then to 36 at 29, read through `sys.getsizeof` on CPython 3.11), `append` amortised $O(1)$ (Theorem 26.3 below), `insert(0, x)` and `pop(0)` $\Theta(n)$, the membership test `x in L` $O(n)$, and a slice of length ℓ $O(\ell)$. A `tuple` stores its pointer array inline in the object (the `ob_item[]` member of `PyTupleObject`, declared in `Include/cpython/tupleobject.h`) with no spare capacity; it is immutable, and hashable exactly when every element is.

(ii) *dict.* A `dict` is an open-addressing hash table in the *compact* layout adopted in 3.6 (Hettinger's 2012 proposal): a sparse *index* array of 1-, 2-, 4- or 8-byte slots points into a dense *entries* array of (hash, key, value) triples stored in insertion order. A lookup hashes the key, probes the index along a hash-perturbed sequence, and compares the stored hash and then the key. The usable slots are 2/3 of the index size, and growth is geometric: resizes occur after 5, 10, 21, 42, 85 and 170 insertions, two thirds of successive powers of two (read through `sys.getsizeof` on CPython 3.11). Insertion order was an implementation detail in 3.6 and is a *language guarantee* since 3.7. Keys must be *hashable* (the `__hash__`/`__eq__` contract: a hash constant over the object's lifetime and equal on objects that compare equal): `int`, `str`, `bytes`, tuples of hashables and `frozenset` qualify; `list`, `dict` and `set` do not. Get, set and delete cost expected $O(1)$ and worst-case $O(n)$ under collisions (Fact 24.8); iteration costs $O(n)$ in insertion order.

(iii) *set.* A `set` is its *own* open-addressing table of (hash, key) slots, with no values and no dense ordered array: add, membership and discard are expected $O(1)$, union is expected $O(|A| + |B|)$, intersection is expected $O(\min(|A|, |B|))$ because the smaller set is iterated and

the larger one probed, and there is *no* ordering promise at all; `frozenset` is the hashable version.

(iv) *deque.* A `collections.deque` is a doubly linked list of fixed-size blocks of 64 object pointers: `append`, `appendleft`, `pop` and `popleft` cost $O(1)$ worst case, indexing near the middle costs $O(n)$ (documented), and `maxlen` turns it into a ring buffer that discards from the far end. A `queue.Queue` is a deque plus a lock and condition variables.

(v) *heapq.* The `heapq` module keeps a binary min-heap in a plain list: the children of index i sit at $2i+1$ and $2i+2$, the parent at $\lfloor (i-1)/2 \rfloor$, and the invariant is $h[i] \le h[2i+1]$ and $h[i] \le h[2i+2]$ wherever those indices exist. The list is a complete binary tree of height $\lfloor \log_2 n \rfloor$, so `heappush` (append as the last leaf, then sift up) and `heappop` (move the last leaf to the root, then sift down) each walk one root-to-leaf path in $O(\log n)$, and the minimum is $h[0]$ in $O(1)$. The invariant is local, so there is no search, no sorted iteration and no decrease-key. The library's documented lazy-deletion idiom replaces decrease-key: push a fresh (priority, item) entry and skip the stale one when it surfaces; the heap then holds one entry per *update* rather than per item, which is what the Dijkstra bound of §26.2 charges for.

These are properties of the cited implementation, checked against its source; a proof is beyond our scope.

Theorem 26.3 (Dynamic-array costs)**.** *Consider a dynamic array of initial capacity $c_0 \ge 1$ that, whenever an append finds it full, allocates a larger array and copies its contents across, and let n appends be performed on an initially empty array.*

(i) If each resize multiplies the capacity by a fixed factor $g > 1$ (rounded up to an integer), the total number of pointer copies is less than $g\,n/(g-1)$, so the amortised cost of an append — copies plus the one direct write — is at most $1 + g/(g-1)$, a constant independent of n.

(ii) If instead each resize adds a fixed constant $c \ge 1$ to the capacity (take $c_0 = c$), the total number of copies is $n^2/(2c) + O(n)$: quadratic in n.

(iii) Inserting at index i of a contiguous array of length n moves the $n - i$ pointers behind it, and deleting index i moves $n - i - 1$; hence `insert(0, x)` and `pop(0)` cost $\Theta(n)$, and draining n items from the front with `pop(0)` costs $n(n-1)/2$ pointer moves.

Proof. We use the aggregate method of Definition 26.1(ii) for (i) and (ii), give a second proof of (i) by the potential method for $g = 2$, and argue (iii) from contiguity.

Part (i), aggregate method. Let $s_0 < s_1 < \cdots < s_M$ be the sizes at which resizes occur. A resize at size s_j copies s_j pointers, and afterwards the capacity is at least $g s_j$, so the next resize occurs at a size $s_{j+1} \ge g s_j$; going backwards from the last resize, $s_j \le s_M / g^{M-j}$. The last resize is triggered by an append into a full array of size s_M, so $s_M \le n - 1 < n$. The copies therefore total

$$\sum_{j=0}^{M} s_j \;\le\; s_M \sum_{i=0}^{M} g^{-i} \;<\; s_M \,\frac{g}{g-1} \;<\; \frac{g\,n}{g-1},$$

the middle step summing the geometric series. Adding the n direct writes and dividing by n gives an amortised cost below $1 + g/(g-1)$.

Part (i), potential method for $g = 2$. Define the potential $\Psi = 2 \cdot \text{size} - \text{capacity}$. An append that finds room costs 1 and raises Ψ by 2: amortised cost 3. An append that finds the array full at size s — capacity s, so $\Psi = s$ — copies s pointers, doubles the capacity to $2s$ and writes the new element; it costs $s + 1$ and leaves $\Psi = 2(s+1) - 2s = 2$, an amortised cost of $(s+1) + (2 - s) = 3$ as well. The potential starts at $-c_0$ (size 0, capacity c_0) and never drops below that value: before the first resize the capacity is still c_0 and the size is non-negative, and after any resize $\Psi \ge 2$. The total actual cost of n appends equals the total amortised cost minus the net rise of Ψ, and the net

rise is non-negative, so the total is at most $3n$. Both methods give a constant per append; the aggregate method shows that the constant is $1 + g/(g-1)$ for every $g > 1$.

Part (ii). With $c_0 = c$ and additive growth the array is full at sizes $c, 2c, \ldots, Mc$, where $M = \lfloor (n-1)/c \rfloor$, and the j-th resize copies jc pointers. By the arithmetic series the copies total $c\,M(M+1)/2$; since $(n-1)/c - 1 < M \le (n-1)/c$, this total lies between $(n-1)(n-1-c)/(2c)$ and $(n-1)(n-1+c)/(2c)$, and both bounds equal $n^2/(2c) + O(n)$.

Part (iii). The list invariant is contiguity: element j lives at address base $+ 8j$. Inserting at index i requires the elements $i, \ldots, n-1$ to occupy slots $i+1, \ldots, n$ afterwards, so each of the $n - i$ of them moves once (a single `memmove` in C, but $n - i$ pointer copies in the cost model); deleting index i moves the $n - i - 1$ elements behind it down by one slot. With $i = 0$ both counts are $\Theta(n)$, and draining from length n by `pop(0)` moves $(n-1) + (n-2) + \cdots + 0 = n(n-1)/2$ pointers, by the arithmetic series once more. □

(Problem P26.2 asks you to reproduce this proof from scratch.)

Example 26.4 (Appending a million pointers)**.** Take $n = 10^6$ appends from $c_0 = 4$ with doubling, $g = 2$. Resizes occur at sizes $4, 8, 16, \ldots, 2^{19} = 524{,}288$ (the array would next be full at $2^{20} > 10^6$, a size never reached), and the copies sum to $4 + 8 + \cdots + 2^{19} = 2^{20} - 4 = 1{,}048{,}572$. With the 10^6 direct writes the total is 2,048,572 writes, about 2.05 per append, inside the bound $1 + g/(g-1) = 3$ of Theorem 26.3(i). CPython's ratio of 9/8 gives the bound $1 + (9/8)/(1/8) = 10$ writes per append: the same order with a larger constant, bought with one eighth of slack memory after a resize instead of up to one half. Additive growth by $c = 100$ slots costs about $n^2/(2c) = 10^{12}/200 = 5 \times 10^9$ copies by part (ii), three and a half orders of magnitude worse for the same million appends. For part (iii), draining a queue of 10^6 items with `pop(0)` moves $n(n-1)/2 \approx 5 \times 10^{11}$ pointers; `deque.popleft` (Fact 26.2(iv)) moves none.

Fact 26.5 (Hash randomisation for str and bytes (PEP 456, 2013; Python/pyhash.c))**.** The hash of a `str` or `bytes` object is a keyed function of the SipHash family with a per-process random key. Randomisation has been on by default since 3.3, as the response to the 2011 hash-collision denial-of-service attack; SipHash-2-4 became the default function in 3.4 (PEP 456) and the faster SipHash-1-3 in 3.11 (`sys.hash_info` reports `siphash13` on CPython 3.11); the environment variable `PYTHONHASHSEED` fixes the key. Integers are *not* randomised: `hash(n)` equals n for small non-negative n, and numeric hashes are reduced modulo $2^{61} - 1$ on 64-bit builds. This is the defence to the adversarial caveat of Fact 24.8: an attacker who cannot predict the key cannot precompute strings that share a probe chain and so cannot drive `dict` operations to $O(n)$. The flood it prevents: with $n = 10^5$ engineered keys sharing one chain, each lookup costs about n probes, so 10^5 lookups cost about 10^{10} probes instead of the expected 10^5. Two consequences are carried forward: a set of strings iterates in a run-dependent order (Example 26.6, the smallest form of the determinism issue of §26.7), and `hash(word) % M` takes different values in different worker processes unless the seed is fixed or a stable hash from `hashlib` is used — the shuffle trap of §26.8. These are properties of the cited implementation; a proof is beyond our scope.

Example 26.6 (Run-dependent set order)**.** The one-line program

```
print(list({'apple', 'banana', 'cherry'}))
```

run four times on CPython 3.11 with `PYTHONHASHSEED` unset, printed `banana, apple, cherry`; then `apple, banana, cherry`; then `banana, apple, cherry` twice. With `PYTHONHASHSEED=0` it printed `banana, cherry, apple` on every run, and with `PYTHONHASHSEED=1` `cherry, apple, banana` on every run. A list or a dict built from the same three strings returns `apple, banana, cherry` on every run: both keep insertion order, and neither consults the hash for its iteration order.

Theorem 26.7 (heapify runs in linear time)**.** *Building a heap on n items in place by sifting down the internal nodes from index $\lfloor n/2 \rfloor - 1$ down to 0 performs at most $2n$ swaps. By contrast, n successive* `heappush` *calls perform $\Theta(n \log n)$ swaps in the worst case.*

Proof. We count swaps by the height of the node they start from. Define the *height* of a node as the number of edges on the longest downward path from it to a leaf, so that leaves have height 0 and the root has height $H = \lfloor \log_2 n \rfloor$. A sift-down from a node of height h moves it down one level per swap, so it makes at most h swaps.

Step 1: at most $\lceil n/2^{h+1} \rceil$ nodes have height h. By induction on h. The internal nodes are the indices $0, \ldots, \lfloor n/2 \rfloor - 1$, so there are $\lceil n/2 \rceil$ leaves, which is the case $h = 0$. For $h \geq 1$, delete all the leaves: what remains is the array prefix of length $n' = \lfloor n/2 \rfloor$, itself a complete binary tree, in which every surviving node has height exactly one less than before (its longest downward path lost its final edge, and every new leaf was an old internal node). The nodes of height h in the original tree are the nodes of height $h - 1$ in the new one, so by the induction hypothesis there are at most $\lceil n'/2^h \rceil \leq \lceil n/2^{h+1} \rceil$ of them.

Step 2: summation. For $1 \leq h \leq H$ we have $n/2^{h+1} \geq 1/2$ because $n \geq 2^H \geq 2^h$, and $\lceil x \rceil \leq 2x$ whenever $x \geq 1/2$; hence $\lceil n/2^{h+1} \rceil \leq n/2^h$. Nodes of height 0 contribute no swaps. The total is therefore at most

$$\sum_{h=1}^{H} h \cdot \frac{n}{2^h} \;\leq\; n \sum_{h \geq 1} \frac{h}{2^h} \;=\; 2n,$$

where the series is evaluated by differentiating the geometric series: $\sum_{h \geq 0} x^h = 1/(1-x)$ for $|x| < 1$ gives $\sum_{h \geq 1} h x^{h-1} = 1/(1-x)^2$, so $\sum_{h \geq 1} h x^h = x/(1-x)^2$, which equals $(1/2)/(1/4) = 2$ at $x = 1/2$.

The contrast. Pushing a strictly decreasing sequence makes every new element climb to the root: the i-th push sifts up through $\lfloor \log_2 i \rfloor$ levels, and the last $n/2$ pushes each climb at least $\lfloor \log_2 n \rfloor - 1$ levels, for $\Theta(n \log n)$ swaps in total. □

Example 26.8 (heapify against repeated pushes)**.** For $n = 15$, a full tree of height 3, the heights $0, 1, 2, 3$ hold $8, 4, 2, 1$ nodes. Sifting down the internal nodes costs at most $4 \times 1 + 2 \times 2 + 1 \times 3 = 11$ swaps, and $11 \leq 2n = 30$. Running the procedure on the reversed sequence $15, 14, \ldots, 1$ makes exactly 11 swaps, while 15 successive pushes of that sequence make 34. For $n = 2^{20}$ the bound is $2n \approx 2.1 \times 10^6$ swaps, against about $n \log_2 n \approx 2.1 \times 10^7$ for pushes.

Example 26.9 (Top-k with a size-k min-heap)**.** To find the k largest of n items, keep a min-heap of size k: for each item, compare it with the root; if it is larger, replace the root and sift down (a `heapreplace`). The invariant "the heap holds the k largest items of the prefix seen so far" holds after the first k items and is preserved by each step, because an item that fails to beat the heap's minimum cannot be among the k largest, and one that beats it displaces exactly the item that is no longer among them. The cost is $O(n \log k)$ time and $O(k)$ memory, against $O(n \log n)$ and $O(n)$ for sorting. For $k = 100$ of $n = 10^7$ items: $n \log_2 k \approx 10^7 \times 6.64 = 6.6 \times 10^7$ heap steps and 100 slots, against $n \log_2 n \approx 2.3 \times 10^8$ and 10^7 slots. The same heap merges M sorted shard lists in $O(Mk \log M)$, which the top-k result of §26.8 uses.

Fact 26.10 (Timsort and bisect (Peters, 2002; Objects/listsort.txt))**.** `list.sort` and `sorted` run Timsort, a merge sort over detected monotone runs: it is stable, uses $O(n \log n)$ comparisons in the worst case and $O(n)$ on input that is already sorted or reverse-sorted, needs $O(n)$ auxiliary memory in the worst case, and evaluates `key=` once per element. Version 3.11 changed the run-merging policy to powersort (Munro and Wild, 2018) without changing these bounds. `bisect_left` and `bisect_right` give $O(\log n)$ search on a sorted list (the loop analysed in §26.2), but `insort` pays the $O(n)$ shift of Theorem 26.3(iii). The standard library has no balanced binary search tree, so the classical "ordered map" row — insert, delete, search, predecessor, successor and range queries all in $O(\log n)$, the last three being what a dict cannot answer — has no stdlib implementation:

a sorted list with `bisect` is the read-mostly substitute, and the third-party `sortedcontainers` package the general one. A proof of the Timsort bounds is beyond our scope.

Proposition 26.11 (Repeated string concatenation is quadratic; join is linear)**.** *Strings are immutable, so* `s = s + t` *allocates a fresh object and copies both operands. Building a string from n pieces of length L by repeated concatenation copies $L\,n(n+1)/2 = \Theta(n^2 L)$ characters, while* `"".join(pieces)` *copies $\Theta(nL)$ characters.*

Proof. We count characters copied and sum an arithmetic series. After $i-1$ concatenations the accumulated string has length $(i-1)L$, and the i-th concatenation copies it together with the new piece, iL characters in all. Summing over $i = 1, \ldots, n$ gives $L(1+2+\cdots+n) = L\,n(n+1)/2$. The `join` method first sums the lengths of the pieces (n reads), allocates once, and copies each of the nL characters exactly once. □

Fact 26.12 (In-place string concatenation (Python/specialize.c, CPython 3.11–3.12; Python/ceval.c since 2.4))**.** CPython has extended the left operand of `s += t` in place, instead of copying it, since 2.4 (`unicode_concatenate`) when the string has a single owner; in 3.11 and 3.12 the specialisation (`BINARY_OP_INPLACE_ADD_UNICODE`) fires only when the target is a local variable (the next instruction is `STORE_FAST` to the same name), so the quadratic cost of Proposition 26.11 is fully visible for globals, attributes and container items. The optimisation hides that cost in a micro-benchmark of a local accumulator; it is an implementation choice, not a language guarantee, and it disappears as soon as another reference to the string exists. These are properties of the cited implementation; a proof is beyond our scope.

Example 26.13 (Joining a hundred thousand pieces)**.** With $n = 10^5$ pieces of $L = 10$ characters, repeated concatenation copies $10 \times 10^5 \times (10^5+1)/2 \approx 5.0 \times 10^{10}$ characters by Proposition 26.11, while `join` copies 10^6. The ratio is 50,000.

Table 5 collects the costs proved or recorded above; it is the table to reproduce, qualifiers included, when a question opens with "which container, and what does it cost?".

The three-way membership decision of Example 24.7 now acquires its qualifiers: with $N = 1024$ elements and $q = 100$ queries, scanning costs $100 \times 1024 = 102{,}400$ comparisons and sort-plus-search costs $(1024 + 100) \times 10 = 11{,}240$ by (24.4), while a `set` costs 1024 expected insertions plus 100 expected probes, about 1,124 operations — a further factor of ten, bought with the loss of order, the hashability requirement, and an expected rather than a worst-case bound. A candidate who says "dict lookup is expected $O(1)$, list append is amortised $O(1)$, deque popleft is $O(1)$ worst case" has answered the qualifier question before it is asked.

Trading application. A tick-data pipeline makes the same decision several times over. Per-symbol state (last price, running volume) lives in a dict keyed by symbol: expected $O(1)$ per update. A rolling window of the last w trades is a `deque` with `maxlen = w`: $O(1)$ per tick, the oldest trade falling off the far end without a shift. A "top ten movers" panel is a size-10 min-heap over the symbol universe (Example 26.9), not a sort. A limit order book, which needs the best bid and best ask *and* the next levels in price order, is the one structure on the list for which the standard library has no direct answer: a sorted list with `bisect` works while the book is small, and a balanced tree or a price-indexed array is the production answer (Fact 26.10).

Interview trap. Four wrong sentences about containers. "A hash map is $O(1)$": the trap after Fact 24.8; its defence against the adversarial case is Fact 26.5, and the missing qualifier "keys must be hashable" is new here. "A list is a fine queue": `pop(0)` is $\Theta(n)$ by Theorem 26.3(iii), so a breadth-first search written with a list as its queue costs $O(V^2)$ instead of $O(V+E)$; the queue is a deque, a list is a stack. "Sets keep insertion order like dicts": a set is a separate table with no ordering promise, a set of strings iterates in a run-dependent order (Example 26.6), and a dict's insertion order is not sorted order. "`heapq` gives sorted iteration": only the minimum is $O(1)$;

the maximum costs $O(n)$ (it lies among the $\lceil n/2 \rceil$ leaves), the k-th smallest $O(k \log k)$, and the underlying list is not sorted.

26.2 A Little Algorithms: Invariants, Costs, and the Recursion Limit

Long-form reasoning questions rarely name an algorithm, but a strong answer recognises one of half a dozen patterns and states its cost unprompted, and the parallel material that follows reuses them as bricks: breadth-first layers become schedule levels (§26.4), prefix sums become scans (§26.5), hash sets replace sorts. This short subsection proves the invariant behind each pattern in the containers of §26.1 and records Python's recursion limit as the practical reason to write searches iteratively.

Setup. For binary search, $a[0], \ldots, a[n-1]$ with $n \geq 1$ is a list sorted in non-decreasing order and t is a target value; intervals are half-open, $[lo, hi)$ denoting the indices $lo \leq i < hi$. A graph $G = (V, E)$ is stored as adjacency lists (a dict mapping each vertex to a list of its out-neighbours); the letters V and E also denote the numbers of vertices and edges, and $\deg v$ is the out-degree of v. For Dijkstra's algorithm every edge (u, v) carries a weight $w(u, v) \geq 0$, $\delta(v)$ denotes the true shortest-path distance from a fixed source s (with every weight equal to 1 in the unweighted case), and heap entries (d, v) are compared as tuples, so that the heap's minimum is well defined even among equal distances. In the sliding-window example $s_{\max}$ is a threshold on window sums. Logarithms in operation counts are base 2. The cost model is that of §26.1.

Proposition 26.14 (Binary search)**.** *Initialise $lo = 0$ and $hi = n$, and while $lo < hi$ set $mid = \lfloor (lo + hi)/2 \rfloor$ and move lo to $mid + 1$ if $a[mid] < t$, or hi to mid otherwise. Then:*

(i) the invariant "every $a[i]$ with $i < lo$ is $< t$, and every $a[i]$ with $i \geq hi$ is $\geq t$" holds before every iteration;

(ii) the loop terminates after at most $\lfloor \log_2 n \rfloor + 1$ iterations with $lo = hi$ equal to the number of elements strictly below t (the `bisect_left` contract);

(iii) the target t is present if and only if $lo < n$ and $a[lo] = t$.

Proof. We proceed by induction on the number of iterations for (i) and by bounding the interval length for (ii). The invariant holds initially because no index satisfies $i < 0$ or $i \geq n$. Suppose it holds before an iteration. If $a[mid] < t$ then, by sortedness, every $a[i]$ with $i \leq mid$ is at most $a[mid] < t$, so setting $lo = mid + 1$ preserves the first half of the invariant and leaves the second untouched. Otherwise $a[mid] \geq t$, so every $a[i]$ with $i \geq mid$ is $\geq t$ and setting $hi = mid$ preserves the second half. This proves (i).

For (ii), write $\ell = hi - lo$ for the interval length. Since $lo \leq mid < hi$ whenever $\ell \geq 1$, the new length is either $hi - mid - 1 = \lceil \ell/2 \rceil - 1 \leq \lfloor \ell/2 \rfloor$ or $mid - lo = \lfloor \ell/2 \rfloor$; in both cases the length at least halves, so after j iterations $\ell \leq \lfloor n/2^j \rfloor$, which is 0 once $2^j > n$, that is, once $j = \lfloor \log_2 n \rfloor + 1$. On exit $lo = hi$, and the invariant then says that the elements below t are exactly $a[0], \ldots, a[lo-1]$, so lo counts them.

For (iii), if t is present, sortedness places all its copies immediately after the elements below it, so $a[lo] = t$; conversely $a[lo] = t$ exhibits it. □

Python's arbitrary-precision integers remove the C overflow hazard of $lo + hi$; what the proof settles is the loop test and the ± 1 decisions, which follow from the half-open invariant rather than from memory.

Example 26.15 (A binary-search trace)**.** Take $a = [2, 3, 5, 7, 11, 13, 17, 19]$ and $t = 13$. Iteration 1: $(lo, hi) = (0, 8)$, $mid = 4$, $a[4] = 11 < 13$, so $lo = 5$. Iteration 2: $(5, 8)$, $mid = 6$, $a[6] = 17 \geq 13$, so $hi = 6$. Iteration 3: $(5, 6)$, $mid = 5$, $a[5] = 13 \geq 13$, so $hi = 5$. The loop exits with $lo = 5$

and $a[5] = 13$: found in 3 iterations, within the bound $\lfloor \log_2 8 \rfloor + 1 = 4$. For $t = 4$ the trace is $(0, 8) \to (0, 4) \to (0, 2) \to (2, 2)$, exiting with $lo = 2$, the insertion point between 3 and 5; `bisect_left` returns 5 and 2 respectively. At $n = 10^6$ each query costs at most $\lfloor \log_2 10^6 \rfloor + 1 = 20$ iterations, so 10^6 queries cost about 2×10^7 comparisons against 10^{12} for repeated scanning.

Proposition 26.16 (BFS, DFS and Kahn's order)**.** *Let a search mark each vertex when it is first* discovered *— on insertion into its container, not on removal — and scan each vertex's adjacency list once, when the vertex is removed from the container.*

(i) Breadth-first search with a deque as a FIFO queue, depth-first search with a list as a stack, and Kahn's topological ordering with in-degree counters and a container of in-degree-zero vertices all run in $O(V + E)$; the only difference among them is the container that decides what to expand next. Kahn's waves*, obtained with a FIFO queue so that wave t is scanned in full before any vertex of wave $t + 1$ — the vertices whose in-degree reaches zero while wave t is being scanned form wave $t + 1$, the sources forming wave 1 — are the schedule levels of Definition 26.40 in §26.4: wave t holds exactly the vertices at which the longest directed path ending there has t vertices.*

(ii) In breadth-first search from s, every vertex $v \neq s$ reachable from s is discovered while a vertex at distance $\delta(v) - 1$ is being scanned; hence the layer in which v is discovered equals $\delta(v)$, and the discovery pointers form a shortest-path tree.

Proof. We count container operations for (i) and use induction on the distance for (ii).

Part (i). Marking on discovery ensures that each vertex enters its container at most once, hence is removed at most once, hence has its adjacency list scanned at most once. The total work is $\sum_v (1 + \deg v) = V + E$ container operations and edge inspections, each $O(1)$ for a deque, for a list used as a stack, and for a decrement of an in-degree counter (Table 5); Kahn's algorithm additionally initialises the in-degrees in one $O(E)$ pass over the edges. For the waves, we argue by strong induction on t that a vertex enters wave t exactly when t is the number of vertices on a longest directed path ending at it. With a FIFO queue every vertex of wave t is dequeued before any vertex appended during the scan of wave t, so the waves are scanned in order: wave 1 in full, then wave 2 in full, and so on. Wave 1 consists of the sources, for which that longest path is the vertex alone. A vertex v with predecessors has its in-degree reach zero when the last of its predecessors is scanned; by the induction hypothesis each predecessor u is scanned in the wave numbered by its own longest ending path, and by the FIFO order the last one scanned lies in the highest-numbered of those waves, of number ℓ say, which is the longest ending path among the predecessors; so v enters wave $\ell + 1$, which is the length of the longest path ending at v.

Part (ii). Call a vertex *pending* if it has been discovered but not yet dequeued. The claim, proved by induction on d, is: at the moment the first vertex at distance d is dequeued, every vertex at distance $\leq d$ has been discovered in the layer equal to its distance, every pending vertex has distance d or $d + 1$, and the pending vertices sit in the queue in non-decreasing order of distance. For $d = 0$ the queue holds only s. Assume the claim for d and follow the phase during which the distance-d vertices are dequeued. Each of them scans its out-neighbours: a neighbour already marked is ignored; an unmarked neighbour has distance exactly $d + 1$ (it cannot be closer, since everything closer is already marked, and it cannot be farther, since it is adjacent to a vertex at distance d), and it is discovered now and appended behind the pending distance-$(d + 1)$ vertices. Every vertex at distance $d + 1$ is adjacent to some vertex at distance d, which is dequeued during this phase, so it is discovered by the end of the phase. When the last distance-d vertex has been dequeued, the pending vertices are therefore exactly the distance-$(d + 1)$ vertices, all discovered in layer $d + 1$, which is the claim for $d + 1$. Since a reachable $v \neq s$ is discovered while a vertex at distance $\delta(v) - 1$ is being scanned, its discovery pointer leads to such a vertex, and following the pointers back to s traces a path of length $\delta(v)$. □

The three searches of part (i) are one loop with three containers: FIFO order gives layers, LIFO

order gives depth-first branches, and "in-degree just reached zero" gives a topological order. The computational graph of Figure 15 is such a DAG, and its topological order is the order of the forward pass.

Example 26.17 (Kahn's waves on an eight-task DAG)**.** Consider the eight tasks A to H with the nine edges A→B, A→C, A→D, B→E, C→E, D→F, E→G, F→G, G→H, so $V = 8$ and $E = 9$; the DAG is drawn in Figure 55 of §26.4, where it is scheduled. The in-degrees are A 0, B 1, C 1, D 1, E 2, F 1, G 2, H 1. Kahn's waves are $\{A\}$, then $\{B, C, D\}$, then $\{E, F\}$ (E's in-degree reaches zero only after both B and C are scanned), then $\{G\}$ (after both E and F), then $\{H\}$: 8 unit tasks in 5 waves. Each vertex is enqueued once (8 enqueues) and each edge decrements one counter (9 decrements), confirming $V + E = 17$ container operations. In the language of §26.4 the work is 8, the span 5 and the parallelism 1.6.

Fact 26.18 (Dijkstra's algorithm with a binary heap (Dijkstra, 1959; lazy deletion per the heapq documentation))**.** Dijkstra's algorithm maintains $dist[v]$, initially ∞ except $dist[s] = 0$, and a min-heap of pairs (d, v) initially holding $(0, s)$; it repeatedly pops the minimum (d, u), skips it as *stale* if $d > dist[u]$, and otherwise *settles* u: for each edge (u, v) with $dist[u] + w(u, v) < dist[v]$ it sets $dist[v]$ to that value and pushes $(dist[v], v)$. With non-negative weights every vertex is settled at most once, with $dist[u] = \delta(u)$ at that moment; each edge is relaxed at most once, so the heap receives at most $E + 1$ pushes and the total cost is $O((V + E) \log V)$. The stale-entry skip is the lazy deletion of Fact 26.2(v) standing in for decrease-key. The correctness argument uses $w \geq 0$ at exactly one inequality — that the first unsettled vertex on a shortest path to u is no farther from the source than u — which is why negative edges break it. A proof is beyond our scope.

Proposition 26.19 (Linear passes)**.** *Let a scan maintain two indices lo and hi in $\{0, \ldots, n\}$, each moving monotonically — lo never decreases, and hi either never decreases (a window) or never increases (inward-moving pointers) — such that each iteration moves at least one index by at least one and does $O(1)$ work. Then the scan performs at most $2n$ iterations and costs $O(n)$.*

Proof. We use a potential: $lo + hi$ when hi never decreases, and $lo + (n - hi)$ when it never increases. Either is an integer in $[0, 2n]$ that rises by at least one per iteration; hence there are at most $2n$ iterations, each $O(1)$. □

Three instances: the longest window with sum at most $s_{\max}$ in an array of non-negative numbers (advance hi, then advance lo while the window sum exceeds $s_{\max}$); pair-sum on a sorted array (pointers at both ends, moving inward); and merging two sorted lists of lengths n and m in $O(n + m)$, the brick behind the k-way shard merge of §26.8. The companion pattern "sort first, then one linear pass" (duplicates, interval merging, grouping) costs $O(n \log n)$ and competes with hash-set membership at expected $O(n)$; Proposition 24.6 gives the crossover shape and §24.2 the cases in which order is worth paying for.

Example 26.20 (A sliding-window trace)**.** Take $[2, 1, 5, 1, 3, 2]$ and $s_{\max} = 7$. Advancing hi from 0: the window sums are 2, 3, 8 (exceeds 7: drop the 2, $lo = 1$, sum 6), 7, 10 (drop the 1 and then the 5, $lo = 3$, sum 4), 6. The right pointer visits 6 positions and the left pointer moves 3 times: 9 iterations, within $2n = 12$. The longest windows have length 3: $[1, 5, 1]$ with sum 7 and $[1, 3, 2]$ with sum 6.

Fact 26.21 (CPython's recursion limit (sys.getrecursionlimit; library reference))**.** CPython raises `RecursionError` when the depth of Python frames exceeds the recursion limit, whose default is 1000 (CPython 3.11 and 3.12); the limit is adjustable but fragile, and 3.12 bounds C-level recursion by a separate mechanism. A recursive depth-first search on a chain of 5,000 vertices therefore raises at depth 1000, while the explicit-stack version pushes and pops 5,000 entries and scans 4,999 edges, $V + E = 9{,}999$ units of work, and finishes. The fix is the iterative version with an explicit list as the stack — the call stack made visible — pushing children in reverse order to

reproduce the recursive visiting order, and keeping (node, state) pairs when a post-order action is needed; the same rewrite applies to tree traversals and to divide-and-conquer on long inputs. These are properties of the cited implementation; a proof is beyond our scope.

Fact 26.22 (Master theorem (Bentley, Haken and Saxe, 1980; CLRS form))**.** Let $T(n) = a\,T(n/b)+\Theta(n^d)$ with $a \geq 1$, $b > 1$ and $d \geq 0$. Then $T(n) = \Theta(n^d)$ if $d > \log_b a$; $T(n) = \Theta(n^d \log n)$ if $d = \log_b a$; and $T(n) = \Theta(n^{\log_b a})$ if $d < \log_b a$. Instances: binary search, $(a, b, d) = (1, 2, 0)$, gives $\Theta(\log n)$, consistent with Proposition 26.14; merge sort, $(2, 2, 1)$, gives $\Theta(n \log n)$, consistent with Fact 26.10. The hypotheses — equal-size subproblems and a polynomial driving term — are part of any quotation of the theorem. A proof is beyond our scope.

Interview trap. Four wrong moves. An off-by-one in binary search: write the invariant first (Proposition 26.14(i)); it decides the loop test and the ± 1, and nothing is remembered. "Dijkstra needs decrease-key": lazy deletion with stale-entry skipping suffices; and "Dijkstra works with negative edges" fails at the one inequality that uses $w \geq 0$ (Fact 26.18). "Recursion is fine in Python": the default depth is 1000 (Fact 26.21); say "iterative with an explicit stack" unprompted. Quoting the master theorem for unequal splits or a non-polynomial driving term: the hypotheses are part of the answer.

26.3 Wall-Clock Time versus CPU Time

"This job took 100 seconds" is ambiguous until the clock is named, and the ratio of the two clocks is the fastest single diagnostic of where a program spends its life; on-site rounds have asked exactly this ★ ("100 seconds of wall time, 40 seconds of CPU: what happened?"). This subsection defines the clocks, proves the accounting inequality that relates CPU time to wall time and turns it into a diagnosis table, separates latency from throughput, and ends with how to measure without fooling oneself. Every later subsection reads its ratio r.

Setup. A process runs on a machine with $C \geq 1$ cores over a wall-clock interval of length $W > 0$ seconds; the operating system schedules the process's threads onto cores. Let m be the number of threads the process has (its maximum over the interval). For each instant t of the interval, $k(t) \in \{0, 1, \ldots, C\}$ denotes the number of cores executing a thread of this process at t; the function k is piecewise constant. Let U and S be the user and system CPU time the process accumulates over the interval (defined below), $r = (U + S)/W$ the *CPU-to-wall ratio*, and $\rho = (U + S)/(CW)$ the *utilisation*. In this subsection r is this ratio and not q/p, ρ is a utilisation and not a correlation, S is system time and not a P&L, and κ denotes a value of r above 1. A *pipeline* is a chain of J stages with steady-state rates $\lambda_1, \ldots, \lambda_J$ items per second, every item passing through every stage in order. Time is in seconds.

Definition 26.23 (Wall-clock time, CPU time, latency and throughput)**.** *Wall-clock time* (elapsed time, real time) is the length W of the interval between two readings of a clock that advances with physical time whatever the process does. *CPU time* is the integral

$$U + S \;=\; \int_0^W k(t)\,dt,$$

the number of core-seconds consumed by all the process's threads; it splits into *user time* U, spent executing the program's own code, and *system time* S, spent in the kernel on the program's behalf (system calls, page faults, I/O setup). Per-thread CPU time is the same integral restricted to one thread. *Waiting* — blocked on I/O, on a lock, on a network reply, on a GPU kernel, or in `time.sleep` — accrues no CPU time at all; a thread that *spins* while it waits, polling instead of blocking, is not waiting in this sense and does accrue CPU time (the CUDA runtime's default wait, Fact 26.35 below). Nor does time spent *runnable but not scheduled*: the process was ready and no core was given to it (other tenants, more runnable threads than cores); from the CPU-time

accounting this is indistinguishable from waiting, and the two must be told apart by other means. *Latency* is the wall time from the submission of one item of work to its completion; *throughput* is the number of items completed per unit of wall time in steady state.

Remark 26.24 (Latency and throughput under batching). A batch of B items processed with a fixed overhead c and a per-item cost τ has

$$\text{latency} \;=\; c + B\tau, \qquad \text{throughput} \;=\; \frac{B}{c + B\tau}, \tag{26.1}$$

and the throughput tends to $1/\tau$ as B grows while the latency grows linearly: batching buys throughput with latency.

Fact 26.25 (Python's clocks and the shell time command (PEP 418, 2012)). `time.time()` returns seconds since the epoch from the system clock, which the operating system may step forwards or backwards (NTP corrections, manual changes), so its differences are not reliable intervals. `time.perf_counter()` is monotonic with the highest available resolution and is the clock for measuring intervals (the `timeit` module uses it); `time.monotonic()` is monotonic and, on some platforms and versions, of coarser resolution (on Windows before 3.13 it reads `GetTickCount64`, resolution 15.6 ms, while `perf_counter` reads `QueryPerformanceCounter`; on Linux both read `CLOCK_MONOTONIC`), and `time.get_clock_info` reports which. `time.process_time()` returns the user plus system CPU time of the current process, summed over *all* of its threads, and `time.thread_time()` (added in 3.7) the calling thread's alone. `process_time` covers only the calling process: CPU consumed by `multiprocessing` workers, or by the BLAS threads of another process, is invisible to it, so a parent that times a process pool reads r near 0. The POSIX `time` command prints `real` (wall), `user` and `sys`, the last two summed over the process and the children it waited for, and, on Unix, `resource.getrusage(resource.RUSAGE_CHILDREN)`, read after the children have been waited for, gives the same figure from Python. These are properties of the cited implementation; a proof is beyond our scope.

Theorem 26.26 (The accounting inequality and the three regimes). *Under the setup above:*

(i) The CPU time satisfies $0 \le U + S \le \min(C, m)\, W$*; hence* $\rho \in [0, 1]$ *and* $W \ge (U + S)/C$.

(ii) If $r < 1$*, the process was on no core at all for at least a fraction* $1 - r$ *of the interval: it* waited*. If the process has a single thread, then* $r \le 1$*, and* $r = 1$ *means that thread occupied a core throughout. If* $r = \kappa > 1$*, then an average of* κ *cores were busy on this process over the interval, and necessarily* $\kappa \le \min(C, m)$.

Proof. We integrate the busy-core count. At every instant a core executes at most one thread and a thread runs on at most one core, so $0 \le k(t) \le \min(C, m)$ pointwise; integrating over $[0, W]$ gives (i), and dividing by CW and by C gives the two consequences.

For (ii), let $I = \{t : k(t) \ge 1\}$ be the set of instants at which the process is on some core. Since $k \ge 1$ on I and $k \ge 0$ elsewhere, the measure of I is at most $\int_0^W k(t)\,dt = rW$, so the process is on no core for a set of instants of measure at least $W - rW = (1 - r)W$. For a single thread, $k(t) \le 1$ pointwise gives $U + S \le W$, and $U + S = W$ forces $k = 1$ almost everywhere. The last statement restates $r = (1/W) \int_0^W k(t)\,dt$ as the time average of k, bounded by (i). □

This is the inequality candidates get backwards: CPU time exceeds wall time exactly when more than one core is busy, and the bound is $U + S \le C\,W$, not $U + S \le W$.

Example 26.27 (Two threads on two cores: $r = 1.1$). Thread 1 is busy on $[0, 3]$ and $[6, 8]$, thread 2 on $[1, 7]$, on a machine with $C = 2$ cores, over $W = 10$ s: both cores are busy on $[1, 3]$ and $[6, 7]$ (3 s, $k = 2$), one core on $[0, 1]$, $[3, 6]$ and $[7, 8]$ (5 s, $k = 1$), and none on $[8, 10]$ ($k = 0$). The CPU time is $2 \times 3 + 5 = 11$ s, the sum of the threads' busy lengths $5 + 6$; it exceeds the wall time, $r = 1.1$, $\rho = 11/20 = 0.55$, and the inequality of Theorem 26.26(i) holds as $11 \le \min(C, m) \times 10 = 20$.

Reading r is the diagnosis, and Table 6 is the one to reproduce.

Remark 26.28 (Locating the off-core time). The ratio r says how much of the interval was off-core; four cheap readings say why. A large system share (`sys`) points at system calls or page faults (`strace -c`; `/usr/bin/time -v` reports major and minor faults). Many involuntary context switches, or a load average above the core count, point at contention with other processes: the program was runnable and descheduled. Many voluntary context switches with low system time point at blocking on locks, sleeps or I/O completion; a sampling profiler (`py-spy dump`) shows the frames that are blocked. A device-bound program shows the device utilisation near 100% (`nvidia-smi`, or the framework profiler) whatever r reads, because the host may be spinning on the wait (Fact 26.35), and a network-bound one shows latency times request count equal to the wall time. Wall time is what the user waits for; CPU-seconds is what a shared cluster bills and what energy is proportional to.

Proposition 26.29 (Pipeline throughput is the slowest stage)**.** *Let a pipeline of J stages have steady-state rates $\lambda_1, \dots, \lambda_J$ items per second, every item passing through every stage in order, each stage processing whenever it has input, and the first stage never starved of input. Then the steady-state throughput of the pipeline equals* $\min_i \lambda_i$.

Proof. We bound the throughput above by counting completions and attain the bound by a queue argument. Every item passes through stage i, so over an interval of length t the number of items completing the pipeline is at most the number completing stage i, which is at most $\lambda_i t$; dividing by t bounds the throughput by λ_i for every i, hence by $\min_i \lambda_i$. For attainment, let i^* attain the minimum. By induction along the chain, each stage delivers at the smaller of its own rate and its input rate — the first stage, never starved, at λ_1 — so every stage before i^* delivers at a rate at least λ_{i^*}, and in steady state the input queue of stage i^* is never empty and that stage runs continuously at its rate λ_{i^*}; every stage after it consumes at least as fast as it receives, so its output passes through at the same rate. The pipeline therefore completes λ_{i^*} items per second. □

Remark 26.30 (Backpressure). A *bounded* queue in front of the slowest stage fills, and the faster stages block when they find it full — *backpressure* — so they run at the slow stage's rate and memory stays bounded; an unbounded queue grows without limit. This is the shape of a CPU data loader feeding a GPU and of the `queue.Queue` and `asyncio.Queue` designs of §26.6.

Example 26.31 (Diagnosing 100 seconds of wall time and 40 of CPU ★)**.** With $W = 100$ s and $U + S = 40$ s, $r = 0.4$, so by Theorem 26.26(ii) the process was on no core for at least 60 s: it waited. It cannot be a single-threaded CPU-bound program (that gives r near 1), and more CPU cores cannot help the 60 s. Parallelising the 40 s of compute perfectly over 4 cores leaves $10 + 60 = 70$ s, a speedup of $100/70 = 1.43$, which §26.4 names as Amdahl's law with $p = 0.4$ and $P = 4$, ceiling $1/0.6 = 1.67$. Three candidates explain the 60 s, and Remark 26.28 separates them: I/O or a network wait (voluntary context switches, low system time, blocked frames in a sampling profiler), a GPU (device utilisation near 100%), or contention with other tenants (involuntary context switches, a load average above the core count). The fixes differ: if the wait is I/O, overlap it (threads, `asyncio`, prefetching) and the 60 s shrinks toward the compute; if the wait is a GPU, the CPU is not the bottleneck and the number to measure is device time (Fact 26.35 below); if it is contention, the machine, not the program, is the problem.

Example 26.32 (Utilisation of eight workers)**.** Eight worker processes report a total of $U+S = 700$ core-seconds over $W = 100$ s: $r = 7$, $\rho = 700/800 = 87.5\%$, and 100 core-seconds were idle; the accounting inequality holds as $700 \le 8 \times 100$. Two shapes explain the idle 12.5%: a straggler (seven workers finish at $600/7 \approx 85.7$ s and the eighth at 100 s, so $7 \times 85.7 + 100 = 700$ and the idle time is $7 \times 14.3 = 100$ core-seconds) or every worker idling 12.5% of the time at a per-step barrier (eight workers busy 87.5 s each). The fixes differ — rebalance the tasks in the first case, shorten the synchronised section in the second — and §26.4 re-reads both as a span term.

Example 26.33 (CPU time exceeding wall time)**.** A numpy product of two 8000×8000 double-precision matrices on 8 BLAS threads reports `real` 1.9 s, `user` 14.6 s, `sys` 0.2 s: $r = 14.8/1.9 = 7.8$ cores on average. By Proposition 24.1 the work is $2 \times 8000^3 = 1.02 \times 10^{12}$ flops, so the aggregate rate is $1.02 \times 10^{12}/1.9 = 539$ GFLOP/s, about 69 GFLOP/s per busy core — plausible for vectorised double precision. No thread was created by the Python program: the BLAS library underneath numpy started the eight. CPU time above wall time is the signature of parallelism, not an accounting error.

Trading application. A daily research job that reloads a year of tick data, computes features and fits a model reads `real` 100 s, `user` 38 s, `sys` 2 s. By the reasoning of Example 26.31 the job is waiting for most of its life — almost certainly on the data reads — and the lever is to overlap or cache the loading (columnar files, a prefetching reader, a process that keeps the frame in memory), not to optimise the model fit. The opposite reading, `user` far above `real`, means that the BLAS underneath the fit is already using every core, and the next lever is algorithmic (a smaller cross-validation grid, a cheaper solver), not more threads.

Example 26.34 (Latency against throughput under batching, and a starved GPU)**.** A batched model call has launch overhead $c = 5$ ms and per-item cost $\tau = 100$ ns. By (26.1), a batch of $B = 10^5$ items has latency $5 + 10 = 15$ ms and throughput $10^5/0.015 = 6.7 \times 10^6$ items per second; a batch of $B = 10^3$ has latency 5.1 ms and throughput 2.0×10^5 per second. Batching buys 34× the throughput for 2.9× the latency. Overhead equals compute at $B^* = c/\tau = 50{,}000$ items, the house crossover of Remark 24.9 solved for the batch size. The pipeline reading, by Proposition 26.29: a GPU step of 20 ms per batch ($\lambda_2 = 50$ batches per second) fed by 4 loader workers at 8 batches per second each ($\lambda_1 = 32$ per second in total) runs at 32 batches per second with the GPU busy 64% of the time, while the training process shows a low r; $\lceil 50/8 \rceil = 7$ workers are needed before the GPU becomes the bottleneck, and a bounded prefetch queue of depth 2 to 4 (Remark 26.30) keeps memory flat once it does.

Fact 26.35 (Asynchronous GPU execution requires synchronisation before timing (CUDA C Programming Guide; `cudaSetDeviceFlags`, CUDA Runtime API))**.** A kernel launch returns control to the host before the kernel has executed — the asynchrony named in Remark 9.14 — so a `perf_counter` difference taken around an unsynchronised launch measures the launch (microseconds), and the device time surfaces later, at the next synchronising operation: `.item()`, a copy to host, `torch.cuda.synchronize()`, or CUDA events recorded on the stream. Unsynchronised timings therefore underestimate GPU work by orders of magnitude. Whether the wait itself shows up as CPU time depends on the runtime's wait policy, not on the device: with the default flag (`cudaDeviceScheduleAuto`) the CUDA runtime *spins* on the host thread while waiting whenever the machine has more logical processors than active CUDA contexts, so a synchronising wait is charged to user time and r stays near 1 although the CPU does no useful work; only a blocking wait (`cudaDeviceScheduleBlockingSync`, or the yield the runtime falls back to when the host is oversubscribed) leaves r well below 1 in the sense of Theorem 26.26(ii). A low r therefore points at a GPU-bound program only under a blocking wait; the reliable signature is device utilisation near 100%, read from the device (Remark 26.28), and the 10–100× of Fact 9.15 is measurable only with the synchronisation in place. The same kernel timed two ways: `perf_counter` around the bare launch, 0.05 ms; `perf_counter` after `torch.cuda.synchronize()`, 12.3 ms. The first number is the trap and the second is the honest wall-clock figure. These are properties of the cited platform; a proof is beyond our scope.

Remark 26.36 (Measuring time correctly). Six steps, in order.

1. Read `time.perf_counter()` for wall time and `time.process_time()` for CPU time immediately around the region of interest, never `time.time()`; record both and their ratio r (Table 6). For a multi-process job, take the CPU figure from the shell `time` command or from `getrusage`, never from the parent's `process_time` (Fact 26.25).

2. Warm up: discard the first call, whose cost includes imports, allocator growth, cache misses, kernel autotuning and CUDA context creation.
3. Repeat, and report the minimum or the median together with the spread, never a single run.
4. Synchronise the accelerator before reading the clock (Fact 26.35).
5. Pin the environment and state it: the BLAS thread counts (the `OMP_NUM_THREADS` and `MKL_NUM_THREADS` variables), other tenants on the machine, CPU frequency scaling, and the core count, because $\kappa > 1$ changes the reading.
6. Measure the quantity the question asks for — latency per item or throughput — and say which; convert a measured time into a rate with a flop count (Proposition 24.1) as a plausibility check.

Interview trap. Five wrong sentences about time. "CPU time cannot exceed wall time": it does whenever more than one core is busy; the inequality is $U + S \le C\,W$, and $U + S > W$ is the signature of parallelism (Theorem 26.26, Example 26.33). "Wall time measures my algorithm": it includes waiting on I/O, locks, other tenants and unsynchronised device work; the ratio r says how much and Remark 26.28 says on what (Example 26.31). "`time.time()` is fine for benchmarks": the epoch clock can step; `perf_counter` is the interval clock (Fact 26.25). "The GPU op took 50 microseconds" is an unsynchronised launch timing (Fact 26.35), and "the GPU is at 30%, so the model is small" misreads a starved pipeline whose loader is the minimum-rate stage (Proposition 26.29, Example 26.34). "The parent shows $r = 0.1$, so my pool is waiting": the workers' CPU time is not in the parent's `process_time` (Fact 26.25).

26.4 Speedup Laws: Amdahl's Law and the Work–Span Model

"How much faster with P machines?" has two textbook answers, both proved here: Amdahl's law for a program with a serial fraction, and the greedy-scheduler bound of the work–span model for a computation with a dependency graph. The diagnostics of §26.3 supply their inputs — the serial fraction is the part of the run where $r = 1$, and the span is what a straggler or a barrier exposes — and interviewers use both to kill "just add machines" and to read a scaling measurement as a diagnosis.

Setup. Let $T_1 > 0$ be the running time of a program on one processor. A fraction $p \in [0, 1)$ of T_1 parallelises perfectly (it divides exactly by the number of processors) and the remaining fraction $1 - p$ does not parallelise at all; $P \ge 1$ is the number of processors, T_P the running time on P of them, and the speedup is $S(P) = T_1/T_P$. In this subsection p is the parallel fraction and P the processor count, not the step probability and the probability operator of the notation table, and S is a speedup, not system time. For the work–span model, a computation is a directed acyclic graph (DAG) of unit-time *tasks* (a task of duration d is a chain of d unit tasks), with an edge $u \to v$ meaning that v may start only after u has finished; schedules on such a DAG are defined in Definition 26.40 below. In Corollary 26.43(iii), J counts independent tasks; it is not the stage count of §26.3.

Theorem 26.37 (Amdahl's law (Amdahl, 1967) and its inversions)**.** *Under the setup above:*

(i) The parallel running time is $T_P = (1 - p)\,T_1 + p\,T_1/P$, so that

$$S(P) \;=\; \frac{1}{(1 - p) + p/P}. \tag{26.2}$$

(ii) For $0 < p < 1$ the speedup is strictly increasing in P, satisfies $S(P) < 1/(1 - p)$ for every P, and tends to $1/(1 - p)$ as $P \to \infty$; for $p = 0$ it equals 1 for every P.

(iii) Inversions: a target speedup s with $1 \le s < 1/(1 - p)$ requires $P = p/(1/s - (1 - p))$ processors; for $p \ge 1/2$ the speedup equals half its ceiling exactly at $P = p/(1 - p) \ge 1$ (for

$p < 1/2$ *the ceiling is below* 2 *and* $S(1) = 1$ *already exceeds half of it); and a measured speedup* S *on* $P > 1$ *processors implies the serial fraction*

$$1 - p \;=\; \frac{P/S - 1}{P - 1} \qquad \textit{(Karp and Flatt, 1990).}$$

Proof. We compute directly from the hypotheses. Split the serial run into its serial part $(1-p)T_1$ and its parallel part pT_1; by hypothesis the parallel part takes pT_1/P on P processors and the serial part is unchanged, which gives (i), and $S(P) = T_1/T_P$ is (26.2). For (ii), with $p > 0$ the term p/P is positive, strictly decreases in P, and tends to 0, so the denominator of (26.2) strictly decreases toward $1-p$ from above; hence the speedup strictly increases, stays below $1/(1-p)$, and converges to it; with $p = 0$ the denominator is 1. For (iii), solve $1/s = (1-p) + p/P$ for P, which is possible exactly when $1/s > 1 - p$; set $S(P) = 1/(2(1-p))$, that is $(1-p) + p/P = 2(1-p)$, which gives $P = p/(1-p)$, an admissible processor count $P \geq 1$ exactly when $p \geq 1/2$; and rewrite $1/S = (1-p) + p/P = (1-p)(1-1/P) + 1/P$, so that $1 - p = (1/S - 1/P)/(1-1/P) = (P/S - 1)/(P-1)$. □

The Karp–Flatt identity of (iii) turns a scaling measurement into a diagnosis: the serial fraction it returns from several values of P should be constant, and one that grows with P signals overhead — communication, contention, load imbalance — rather than a fixed serial part.

Example 26.38 (Amdahl arithmetic)**.** With $p = 0.95$: $S(8) = 1/(0.05 + 0.11875) = 5.93$; $S(32) = 1/(0.05 + 0.0297) = 1/0.0797 = 12.5$; $S(64) = 15.4$; $S(128) = 17.4$; the ceiling is 20. Reaching $15\times$ needs $P = 0.95/(1/15 - 0.05) = 0.95/0.01667 = 57$ processors, and half the ceiling ($10\times$) is reached at $P = 0.95/0.05 = 19$. With $p = 0.9$ and $P = 100$, $S = 1/(0.1 + 0.009) = 9.17$ against a ceiling of 10. The inversion: a measured $12.5\times$ at $P = 32$ implies $1 - p = (32/12.5 - 1)/31 = 1.56/31 = 0.050$, and then $S(64) = 1/(0.050 + 0.950/64) = 15.4$, not 25. A backtest sweep that scaled $3\times$ on 8 processes has, by the same inversion, $1 - p = (8/3 - 1)/7 = 0.238$ and a ceiling of $1/0.238 = 4.2\times$ however many cores are added; the lever is the serial part (loading the panel, writing the results), not more processes. The $p = 0.4$ of Example 26.31 has ceiling $1/0.6 = 1.67$.

Figure 54 plots (26.2) for three values of p: the curves bend away from the ideal line almost immediately and flatten under their ceilings.

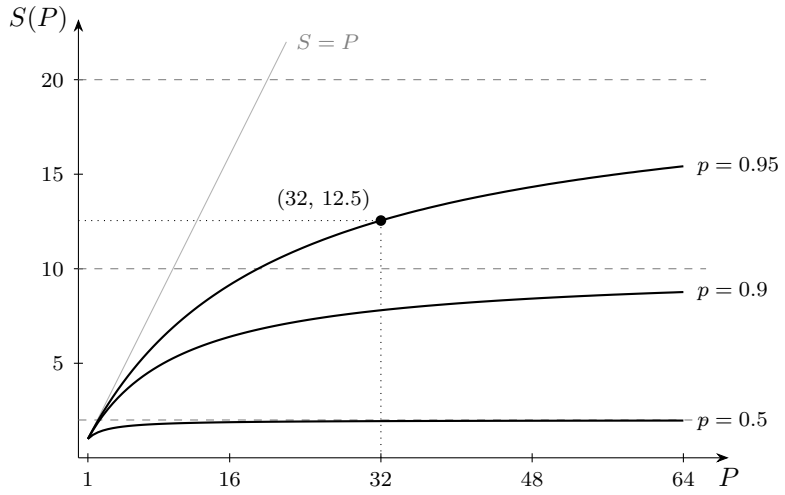


Figure 54: Amdahl speedup (26.2) against the processor count for $p = 0.5$, 0.9 and 0.95, with the ceilings $1/(1-p) = 2$, 10 and 20 as dashed lines and the ideal $S = P$ in light gray. The marked point is $S(32) = 12.5$ at $p = 0.95$ (Example 26.38); doubling to 64 processors reaches only 15.4.

Remark 26.39 (Gustafson's scaled speedup (Gustafson, 1988)). Hold the *parallel* run at unit time with serial part s' and parallel part $1 - s'$. The same work on one processor would take $s' + (1 - s')P$, so the *scaled speedup* is $s' + (1 - s')P$, linear in P. Amdahl's law is fixed-size scaling and Gustafson's is fixed-time scaling: the same identity read from different baselines. The practical content is that the serial fraction of a training run — data loading, logging, a single-device optimiser step — is an engineering choice, not physics, and growing the problem with the machine keeps it small.

Definition 26.40 (Work, span and parallelism)**.** For a computation DAG of unit-time tasks, the *work* T_1 is the number of tasks, the *span* T_∞ is the number of tasks on a longest directed path (a *critical path*; there may be several), and the *parallelism* is T_1/T_∞. The *level* of a task is the number of tasks on a longest directed path ending at it, equivalently the earliest step at which any schedule can execute it. A *schedule* on P processors assigns each task to a step, respecting the edges, with at most P tasks per step; its length T_P is its number of steps. A task is *ready* at a step if all its predecessors have executed in earlier steps. A *greedy* schedule executes P ready tasks whenever at least P are ready, and all ready tasks otherwise; a step is *complete* if P tasks execute in it and *incomplete* otherwise.

The parallelism T_1/T_∞ is the largest number of processors that can be kept fully busy on average. Figure 55 shows the DAG of Example 26.17, with $T_1 = 8$ and $T_\infty = 5$; its levels are Kahn's waves (Proposition 26.16(i)), and the computational graph of Figure 15 is a DAG of this kind.

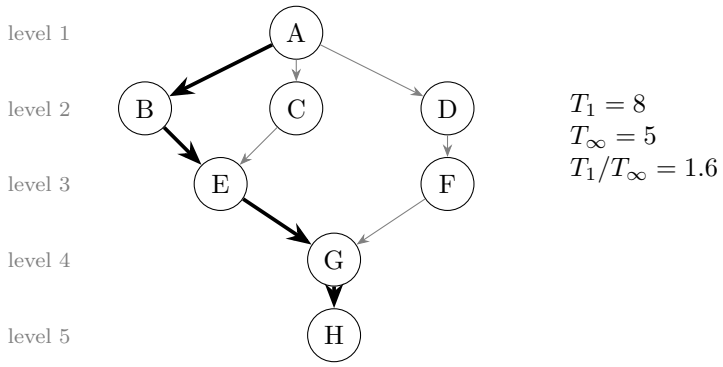


Figure 55: The eight-task DAG of Example 26.17, scheduled in Example 26.42: a critical path, A–B–E–G–H (thick), has length $T_\infty = 5$ — one of three of that length, the other two running through C and through D–F — and the remaining edges are gray. The levels are the earliest steps at which tasks can run, which are Kahn's waves. With three processors the greedy schedule finishes in 5 steps, so the span law of Theorem 26.41(i) is tight and a fourth processor buys nothing.

Theorem 26.41 (Work and span laws and the greedy-scheduler bound (Graham, 1969; Brent, 1974))**.** *For a computation DAG with work T_1 and span T_∞ scheduled on P processors:*

(i) every schedule satisfies $T_P \geq T_1/P$ (the work law) and $T_P \geq T_\infty$ (the span law);
(ii) every greedy schedule satisfies $T_P \leq \lfloor T_1/P \rfloor + T_\infty$.

Proof. We prove (i) by counting, and (ii) by classifying the steps of a greedy schedule as complete or incomplete.

Part (i). Each step executes at most P tasks, so $T_1 \leq P\,T_P$. The tasks of a longest path must execute in distinct steps in path order, because each is a predecessor of the next, so $T_P \geq T_\infty$.

Part (ii). A complete step executes P tasks, so there are at most $\lfloor T_1/P \rfloor$ complete steps. Consider an incomplete step and the DAG R of tasks not yet executed before it. Fewer than P tasks are ready, so the greedy rule executes *every* ready task. Take any longest path in R; its first task

has no unexecuted predecessor (otherwise prepending that predecessor would give a longer path in R), so it is ready and executes in this step. A step executes only tasks with no unexecuted predecessor, and any path in R contains at most one such task, namely its first; so every path in R shrinks by at most one during the step, every longest path shrinks by exactly one, and the span of the remaining DAG decreases by exactly one at every incomplete step. The span starts at T_∞ and cannot go below 0, so there are at most T_∞ incomplete steps. Adding the two counts gives $T_P \le \lfloor T_1/P \rfloor + T_\infty$. □

(Problem P26.6 asks you to reproduce this proof from scratch.)

Intuition. Amdahl's law is a span statement. A program with serial fraction $1-p$ is the DAG "a chain of $(1-p)T_1$ tasks followed by pT_1 independent tasks": its span is $(1-p)T_1 + 1$, so the span law of Theorem 26.41(i) reproduces the ceiling $1/(1-p)$, and the greedy schedule on this DAG — the chain, then the independent tasks P at a time — takes $(1-p)T_1 + \lceil pT_1/P \rceil$ steps, which is Amdahl's T_P up to rounding. The span is the general notion and the serial fraction its simplest instance; a per-step barrier — an all-reduce, a simulation time step — contributes to the span in the same way.

Example 26.42 (Scheduling the DAG of Figure 55)**.** The tasks A to H have the edges of Example 26.17: work $T_1 = 8$, span $T_\infty = 5$ along the critical path A–B–E–G–H (one of three of that length), parallelism 1.6. With $P = 2$, the lower bounds of Theorem 26.41(i) are $\max(8/2, 5) = 5$, the greedy bound of (ii) is $\lfloor 8/2 \rfloor + 5 = 9$, and the greedy schedule $\{\mathrm{A}\}, \{\mathrm{B}, \mathrm{C}\}, \{\mathrm{D}, \mathrm{E}\}, \{\mathrm{F}\}, \{\mathrm{G}\}, \{\mathrm{H}\}$ takes 6 steps: the steps $\{\mathrm{B}, \mathrm{C}\}$ and $\{\mathrm{D}, \mathrm{E}\}$ are complete, the other four are incomplete, and each incomplete step reduces the remaining span by exactly one ($5 \to 4$ at $\{\mathrm{A}\}$, then $3 \to 2 \to 1 \to 0$ at the last three). With $P = 3$, the bound is $\lfloor 8/3 \rfloor + 5 = 7$ and the greedy schedule $\{\mathrm{A}\}, \{\mathrm{B}, \mathrm{C}, \mathrm{D}\}, \{\mathrm{E}, \mathrm{F}\}, \{\mathrm{G}\}, \{\mathrm{H}\}$ takes $5 = T_\infty$ steps, so the span law is tight and a fourth processor buys nothing.

Corollary 26.43 (Consequences of the greedy bound)**.** *In the setting of Theorem 26.41:*

(i) every greedy schedule satisfies $T_P \le 2\max(T_1/P, T_\infty) \le 2\,T_P^{\mathrm{opt}}$, where T_P^{opt} is the length of an optimal schedule;

(ii) if $P \le \varepsilon\, T_1/T_\infty$ for some $\varepsilon > 0$, then every greedy schedule has speedup $T_1/T_P \ge P/(1+\varepsilon)$: near-linear speedup whenever P is well below the parallelism;

(iii) for independent tasks of integer durations $d_1, \ldots, d_J$ with $t_{\max} = \max_i d_i$, handed to whichever worker is free, the finish time satisfies $T_P \le T_1/P + t_{\max}$ where $T_1 = \sum_i d_i$.

Proof. We substitute into Theorem 26.41 for (i) and (ii), and give two arguments for (iii).

Parts (i) and (ii). Each of the two terms of the greedy bound is at most $\max(T_1/P, T_\infty)$, and the lower bounds of Theorem 26.41(i) show that $\max(T_1/P, T_\infty) \le T_P^{\mathrm{opt}}$; together these give (i). If $P \le \varepsilon T_1/T_\infty$, then $T_\infty \le \varepsilon T_1/P$, so the greedy bound gives $T_P \le (1+\varepsilon)\, T_1/P$ and hence $T_1/T_P \ge P/(1+\varepsilon)$.

Part (iii), in the DAG model. Each task of duration d_i is a chain of d_i unit tasks, and the chains are independent, so $T_1 = \sum_i d_i$ and $T_\infty = t_{\max}$; Theorem 26.41(ii) gives $T_P \le \lfloor T_1/P \rfloor + t_{\max}$.

Part (iii), non-preemptively (Graham's argument). In the DAG model a chain may be paused between its unit tasks; a worker that takes a task and keeps it until it finishes is the non-preemptive case, and the same bound holds. Let task j be the task that finishes last, started at time s_j. Before s_j no worker was ever idle, since an idle worker would have taken task j (or an earlier task) at once; so by time s_j the P workers had performed Ps_j units of work, none of it on task j, whence $Ps_j \le T_1 - d_j$. Therefore $T_P = s_j + d_j \le (T_1 - d_j)/P + d_j \le T_1/P + t_{\max}$. □

Fact 26.44 (Longest-processing-time-first (Graham, 1969))**.** Handing out independent tasks in decreasing order of duration to whichever worker is free finishes within a factor $4/3 - 1/(3P)$ of the

optimal finish time, against the factor $2 - 1/P$ guaranteed for an arbitrary order; the improvement comes from the last-finishing task being small. A proof is beyond our scope.

The finish time is the *maximum* over workers, not the mean. That is why a straggler set the wall time of Example 26.32: read through Theorem 26.41(ii) and Corollary 26.43(iii), its $T_1 = 700$ s gives $T_1/P = 87.5$ s against an observed 100 s, so a span-like term of at least 12.5 s is present — a task with $t_{\max} \geq 12.5$ s, or a per-step barrier — and adding workers shrinks only the 87.5 s term. It is also why many more tasks than workers are drawn from a queue in §26.8.

Example 26.45 (Reduction trees and the chunked reduction)**.** Summing $n = 8$ values as a balanced binary tree has work 7 and span $3 = \log_2 8$. For $n = 2^{20}$ the work is about 10^6 and the span 20, a parallelism of about 5×10^4; on $P = 64$ cores, Corollary 26.43(ii) with $\varepsilon = 64 \times 20/(2^{20} - 1) = 0.0012$ guarantees a greedy speedup of at least $64/1.0012 = 63.9$. The P-worker *chunked* reduction — each worker folds a contiguous block of $\lceil n/P \rceil$ values, then the P partial results merge in a tree — has span at most $\lceil n/P \rceil + \lceil \log_2 P \rceil$, which is $16{,}384 + 6 = 16{,}390$ at $n = 2^{20}$ and $P = 64$; this is the shape reused by the block scan of §26.5 and the all-reduce of §26.6.

Proposition 26.46 (The overhead crossover: how many workers)**.** *Suppose a total amount of work $W_0 > 0$ divides perfectly among P workers, but each worker adds a fixed cost $c > 0$ (spawning, importing, loading a model, gathering its result) that is paid serially, so that $T(P) = cP + W_0/P$. Then T is minimised at $P^* = \sqrt{W_0/c}$, where $T^* = 2\sqrt{W_0 c}$ and the two terms are equal; past P^* every added worker costs more in coordination than it removes in compute.*

Proof. We apply the arithmetic–geometric mean inequality: for positive a and b, $a + b \geq 2\sqrt{ab}$, with equality if and only if $a = b$. With $a = cP$ and $b = W_0/P$, $T(P) \geq 2\sqrt{cP \cdot W_0/P} = 2\sqrt{W_0 c}$, with equality if and only if $cP = W_0/P$, that is $P = \sqrt{W_0/c}$. Equivalently, $T'(P) = c - W_0/P^2$ vanishes at P^* and $T''(P) = 2W_0/P^3 > 0$ there. □

This is the "equate the two costs" pattern of Remark 24.9 solved for the worker count, the crossover shape that cost-model assessments favour.

Example 26.47 (Six workers, not sixteen)**.** With $W_0 = 150$ s of work and $c = 4$ s per worker, $P^* = \sqrt{37.5} = 6.1$, so six workers, with $T^* = 2\sqrt{600} = 49$ s. Checking: $T(1) = 154$ s, $T(4) = 16 + 37.5 = 53.5$ s, $T(6) = 24 + 25 = 49$ s, $T(8) = 32 + 18.75 = 50.75$ s, $T(16) = 64 + 9.4 = 73.4$ s — sixteen workers are slower than six. A second instance: $W_0 = 10$ s and $c = 0.01$ s give $P^* = 31.6$ and $T^* = 0.63$ s.

Interview trap. Four wrong sentences about speedup. "Just add machines": the ceiling $1/(1-p)$, the span law and the overhead crossover P^* all cap it, and past P^* the job gets slower (Example 26.47). "The work divides, so the speedup is P": the span law forbids beating T_∞, and a chain of dependencies is unaffected by P. "The mean load is the finish time": the finish time is the maximum, and the $t_{\max}$ term of Corollary 26.43(iii) is the price of a large task handed out late. "95% parallel is basically all parallel, and 12.5× on 32 cores means 25× on 64": at most 20× ever, and 15.4× on 64 by the inversion of Example 26.38.

26.5 Associativity: the Licence to Parallelise, Floating Point, Merged Statistics, and Scans

Every parallel aggregation — private partial sums merged at the end, a gradient all-reduce, a MapReduce combiner, a prefix scan — rests on one algebraic property, and interviewers probe it with "can you parallelise this loop, and why is the answer the same?", "why is my parallel sum different every run?" and "how do you combine the means and variances from ten shards?". This subsection proves that an associative operation may be evaluated by any chunking or tree and

a commutative one in any order, shows that floating-point addition lacks the property, proves that the mean and the variance become associative merges once the state is lifted, and turns a dependent loop into a scan. The fix ladder of §26.7 and the all-reduce of §26.6 cite it backward.

Setup. Let $\mathcal{S}$ be a set with a binary operation written $\oplus$ and, where needed, an identity element e with $e \oplus x = x \oplus e = x$. For a sequence $x_1, \dots, x_n$ in $\mathcal{S}$ the *left fold* is

$$\operatorname{fold}(x_1, \dots, x_n) \;=\; (\cdots((x_1 \oplus x_2) \oplus x_3) \oplus \cdots) \oplus x_n,$$

with $\operatorname{fold}(x_1) = x_1$. A *bracketing* of $x_1, \dots, x_n$ is any full parenthesisation of $x_1 \oplus x_2 \oplus \cdots \oplus x_n$ with the operands in their given order; a *chunking* partitions the sequence into contiguous blocks, folds each block separately, and combines the block results by some bracketing; a permutation σ reorders the operands (σ is a permutation here, not a volatility). Where workers appear, P counts them as in §26.4. The *inclusive scan* of $x_1, \dots, x_n$ is the sequence $y_i = x_1 \oplus \cdots \oplus x_i$, and the *exclusive scan* is $y_i = e \oplus x_1 \oplus \cdots \oplus x_{i-1}$ (so $y_1 = e$); the sequential loop $y_i = y_{i-1} \oplus x_i$ has work $\Theta(n)$ and, as written, span $\Theta(n)$, because each output depends on the previous one. IEEE 754 binary64 has unit roundoff $u = 2^{-53} \approx 1.1 \times 10^{-16}$. For a finite multiset D of real numbers with $n \geq 1$ elements, the *summary state* is $\operatorname{st}(D) = (n, m, M_2)$, where m is the mean of D and $M_2 = \sum_{x \in D}(x - m)^2$ is the sum of squared deviations from the mean; the population variance is M_2/n and the sample variance $M_2/(n-1)$, the divisor of Theorem 4.5. In this subsection n is a count of elements and m a mean, not a thread count.

Definition 26.48 (Associative and commutative operations; the classification list)**.** The operation $\oplus$ is *associative* if $(a \oplus b) \oplus c = a \oplus (b \oplus c)$ for all $a, b, c \in \mathcal{S}$, and *commutative* if $a \oplus b = b \oplus a$ for all a, b. A *monoid* is a set with an associative operation and an identity e; the identity is what lets empty chunks merge harmlessly. The classification to quote aloud: integer addition and multiplication, max, min, logical and, logical or, set union and gcd are associative and commutative; string concatenation, "last non-null", "first value" and the matrix product are associative but not commutative (the associativity of the matrix product is what Theorem 24.3 relies on); subtraction and division are neither; the mean and the variance as written are neither; floating-point addition is commutative but not associative; and the median is neither and cannot be merged without shipping data or approximating.

Two entries of the list are developed below: Proposition 26.55 repairs the mean and the variance by lifting the state, and Fact 26.52 is the floating-point entry.

Theorem 26.49 (Fold invariance)**.** *Let $\oplus$ be a binary operation on $\mathcal{S}$ and $x_1, \dots, x_n$ a sequence in $\mathcal{S}$.*

(i) If $\oplus$ is associative, every bracketing of $x_1 \oplus \cdots \oplus x_n$ equals $\operatorname{fold}(x_1, \dots, x_n)$; hence every contiguous chunking and every reduction tree computes the same value.

(ii) If $\oplus$ is associative and commutative, every bracketing of every permutation $x_{\sigma(1)} \oplus \cdots \oplus x_{\sigma(n)}$ equals $\operatorname{fold}(x_1, \dots, x_n)$; hence partial results may be merged in any arrival order.

Proof. We proceed in three named steps: a concatenation identity for folds, a strong induction on n for bracketings, and a decomposition of permutations into adjacent transpositions.

Step 1 (folds concatenate). Abbreviate $F_a^b = \operatorname{fold}(x_a, \dots, x_b)$ for $a \leq b$. For associative $\oplus$ and $1 \leq k < n$,

$$F_1^k \oplus F_{k+1}^n \;=\; F_1^n. \tag{26.3}$$

By induction on $n - k$. If $n - k = 1$ the right-hand fold is $F_{k+1}^{k+1} = x_{k+1}$ and (26.3) is the definition of the left fold. For $n - k \geq 2$, write $F_{k+1}^n = F_{k+1}^{n-1} \oplus x_n$ by definition; then

$$\begin{aligned} F_1^k \oplus F_{k+1}^n \;=\; F_1^k \oplus (F_{k+1}^{n-1} \oplus x_n) &= (F_1^k \oplus F_{k+1}^{n-1}) \oplus x_n \\ &= F_1^{n-1} \oplus x_n \;=\; F_1^n, \end{aligned}$$

regrouping once by associativity, then applying the induction hypothesis to the shorter gap $(n-1)-k$, and finally using the definition of the left fold.

Step 2 (part (i)). By strong induction on n. For $n = 1$ there is nothing to prove. For $n \geq 2$, any bracketing has an outermost $\oplus$ that splits the operands as

$$(\text{a bracketing of } x_1, \ldots, x_k) \;\oplus\; (\text{a bracketing of } x_{k+1}, \ldots, x_n)$$

for some $1 \leq k < n$; by the induction hypothesis the two sides equal $\mathrm{fold}(x_1, \ldots, x_k)$ and $\mathrm{fold}(x_{k+1}, \ldots, x_n)$, and (26.3) joins them into $\mathrm{fold}(x_1, \ldots, x_n)$. A chunking followed by a tree of merges is itself a bracketing, so the consequence follows.

Step 3 (part (ii)). Every permutation is a product of adjacent transpositions (bubble sort exhibits the factorisation), so it suffices to show that swapping x_j and x_{j+1} does not change the fold. By part (i), $\mathrm{fold}(x_1, \ldots, x_n)$ equals the bracketing in which $x_j \oplus x_{j+1}$ is formed first as a pair; commutativity replaces that pair by $x_{j+1} \oplus x_j$ without changing its value; the result is a bracketing of the transposed sequence, which equals the fold of the transposed sequence by part (i). Induction on the number of transpositions finishes the argument. □

Rung 3 of the fix ladder of §26.7 — private per-worker state plus a merge at the end — is therefore correct for any associative merge and any assignment of contiguous chunks to workers, and needs commutativity as well whenever workers finish in an unscheduled order, which they do. Associative but non-commutative operations (concatenation, last non-null, the matrix product) admit any tree but not an unordered merge, so their partial results must carry their chunk position.

Example 26.50 (Three ways to fold eight integers)**.** Take $x = [3, 1, 7, 0, 4, 1, 6, 3]$ under integer addition. The left fold gives 25; the balanced tree $((3+1)+(7+0))+((4+1)+(6+3)) = 11+14 = 25$; two chunks of four folded separately and then merged give $11+14 = 25$, as Theorem 26.49(i) promises. Under subtraction the left fold gives -19 and the balanced tree $((3-1)-(7-0))-((4-1)-(6-3)) = -5-0 = -5$: subtraction is not associative, and no parallel version of the loop exists as written.

Example 26.51 (Last non-null: a tree is fine, an unordered merge is not)**.** Chunks $[\mathrm{None}, 5]$, $[\mathrm{None}, \mathrm{None}]$ and $[7, \mathrm{None}]$ under the merge "last non-null", which is associative but not commutative, fold to 5, None and 7. Any tree that respects chunk order gives 7; merging in the arrival order chunk 3, chunk 1, chunk 2 gives 5, the wrong answer. Commutativity is exactly what an unordered merge needs (Theorem 26.49(ii)).

Fact 26.52 (Floating-point addition is not associative (IEEE 754; Kahan, 1965; Higham, 2002))**.** Rounding breaks associativity, and the damage is bounded as follows.

(i) Each floating-point addition rounds its exact result to the nearest representable number, so the sequence of roundings, and hence the computed sum, depends on the bracketing. Three binary64 identities:

$$(0.1+0.2)+0.3 = 0.6000000000000001 \quad \text{but} \quad 0.1+(0.2+0.3) = 0.6,$$
$$(10^{16}+1.0)+1.0 = 10^{16} \quad \text{but} \quad 10^{16}+(1.0+1.0) = 10^{16}+2,$$
$$(10^{20}+(-10^{20}))+1.0 = 1.0 \quad \text{but} \quad 10^{20}+((-10^{20})+1.0) = 0.0,$$

the middle pair because the spacing of doubles near 10^{16} is 2, so that $10^{16}+1.0$ rounds back to 10^{16} while $10^{16}+2$ is representable (the value printed as $1.0000000000000002 \times 10^{16}$).

(ii) To first order in u, the forward error of the recursive (left-to-right) sum of $x_1, \ldots, x_n$ is at most $(n-1)\,u \sum_i |x_i|$; pairwise (balanced-tree) summation has error at most about $\lceil \log_2 n \rceil\, u \sum_i |x_i|$, and a blocked variant of it is what `numpy.sum` uses on contiguous floating-point arrays; Kahan's compensated summation, which carries the rounding error of each addition in a correction term, has error at most $(2u + O(nu^2)) \sum_i |x_i|$, independent of n to first order; and `math.fsum` returns the correctly rounded sum (Shewchuk's algorithm).

Consequently Theorem 26.49 does *not* apply, and a parallel sum whose bracketing depends on the thread count, the chunking or the arrival order is run-to-run nondeterministic in its low bits, which chaotic downstream computations — an optimiser trajectory, a simulation — amplify. Proofs of the bounds are beyond our scope.

Remark 26.53 (Fixes for a nondeterministic floating-point sum). In decreasing order of strength. A *fixed reduction tree*: a bracketing that does not depend on the worker count or the arrival order (or a framework's deterministic-algorithms flag), at a cost in speed. *Compensated or pairwise summation*, which shrinks the discrepancy without removing it. *Integer or fixed-point accumulation* when exactness is required, since integer addition is associative. *Tolerance-based tests*: $|a-b| \le \mathrm{atol} + \mathrm{rtol}\,|b|$ (`numpy.allclose`) or the symmetric $|a-b| \le \max(\mathrm{rel_tol}\cdot\max(|a|,|b|),\ \mathrm{abs_tol})$ (`math.isclose`, whose defaults are $\mathrm{rel_tol} = 10^{-9}$ and $\mathrm{abs_tol} = 0$), never equality. Remark 6.21 is the sibling numerical caution for linear algebra.

Example 26.54 (Bracketings in binary64 and binary32)**.** The first identity of Fact 26.52(i) comes from the intermediate roundings

$$0.1 + 0.2 = 0.30000000000000004, \qquad 0.2 + 0.3 = 0.5 \text{ exactly},$$

after which adding 0.3 to the first gives 0.6000000000000001 and adding 0.1 to the second gives 0.6. In binary32, the value 0.1 is stored as 0.100000001490116, so the exact sum of one million copies is 100000.0015; five ways of adding them give

Method	Result
sequential left-to-right loop, binary32	100958.34 (relative error 1%)
`numpy.sum`, blocked pairwise	100000.09
bottom-up pairwise tree (adjacent pairs, level by level)	100000.0 (the correctly rounded binary32 sum)
top-down halving of the same array	99999.99 (two balanced trees, two answers)
`math.fsum` of the stored binary32 inputs	100000.0015 (exact, in binary64)

and `math.fsum` of one million binary64 copies of 0.1 returns 100000.0 exactly. At a billion terms of magnitude about 1 the bounds of Fact 26.52(ii) read: recursive $(n-1)\,u\sum|x_i| \approx 10^9 \times 1.1 \times 10^{-16} \times 10^9 = 1.1 \times 10^2$ absolute on a sum of 10^9, relative 10^{-7}; pairwise about $30\,u = 3.3 \times 10^{-15}$ relative; Kahan about $2u = 2.2 \times 10^{-16}$. These are worst cases, but the *difference* between two bracketings is what a parallel run exposes: a test asserting equality between an 8-thread and a 16-thread sum fails on such inputs, while a relative tolerance of 10^{-6} passes.

Proposition 26.55 (Mean and variance as associative merges (Chan, Golub and LeVeque, 1979))**.** *For the mean, the state (n, s) with s the sum merges as $(n_A, s_A) \oplus (n_B, s_B) = (n_A + n_B, s_A + s_B)$, and the mean of the merged multiset is s/n. For the variance, write $\delta = m_B - m_A$ and $n = n_A + n_B$, and define on summary states*

$$(n_A, m_A, M_2^A) \oplus (n_B, m_B, M_2^B) \;=\; \Big(n,\ m_A + \delta\,\frac{n_B}{n},\ M_2^A + M_2^B + \delta^2\,\frac{n_A n_B}{n}\Big).$$

Then $\mathrm{st}(A) \oplus \mathrm{st}(B) = \mathrm{st}(A \uplus B)$ for all non-empty multisets A and B, where $\uplus$ is multiset union. Consequently $\oplus$ is associative and commutative on the summary states realised by data, and the variance is read off the final state as M_2/n or $M_2/(n-1)$.

Proof. We verify the merge identity by expanding a square, then obtain associativity and commutativity from the homomorphism property.

Mean. The mean of $A \uplus B$ is $(n_A m_A + n_B m_B)/n = m_A + n_B(m_B - m_A)/n = m_A + \delta\, n_B/n$, as the merge states; for the sum-based state the claim is the additivity of sums.

Squared deviations. Write m for the merged mean. Split $M_2(A \uplus B) = \sum_{x\in A}(x-m)^2 + \sum_{x\in B}(x-m)^2$ and expand the first sum around m_A:

$$\begin{aligned}\sum_{x\in A}(x-m)^2 &= \sum_{x\in A}\left[(x-m_A)+(m_A-m)\right]^2 \\ &= M_2^A + 2(m_A-m)\sum_{x\in A}(x-m_A) + n_A(m_A-m)^2 \\ &= M_2^A + n_A(m_A-m)^2,\end{aligned}$$

because the deviations from m_A sum to zero — the cancellation in the proof of Lemma 4.4. Likewise $\sum_{x\in B}(x-m)^2 = M_2^B + n_B(m_B-m)^2$. Now $m_A - m = -\delta\, n_B/n$ and $m_B - m = \delta - \delta\, n_B/n = \delta\, n_A/n$, so the two constant terms add to

$$n_A\,\frac{\delta^2 n_B^2}{n^2} + n_B\,\frac{\delta^2 n_A^2}{n^2} = \delta^2\,\frac{n_A n_B(n_B+n_A)}{n^2} = \delta^2\,\frac{n_A n_B}{n},$$

which is the merge's third component.

Associativity and commutativity. Every summary state that a fold produces is $\mathrm{st}(D)$ for the non-empty multiset D of data merged so far, and the merge identity says $\mathrm{st}(A)\oplus\mathrm{st}(B) = \mathrm{st}(A\uplus B)$. Hence for any three multisets

$$(\mathrm{st}(A)\oplus\mathrm{st}(B))\oplus\mathrm{st}(C) = \mathrm{st}(A\uplus B\uplus C) = \mathrm{st}(A)\oplus(\mathrm{st}(B)\oplus\mathrm{st}(C)),$$

and $\mathrm{st}(A)\oplus\mathrm{st}(B) = \mathrm{st}(A\uplus B) = \mathrm{st}(B\uplus A) = \mathrm{st}(B)\oplus\mathrm{st}(A)$, because multiset union is associative and commutative. No separate algebraic check of the formula is needed. □

Example 26.56 (Merging two shards' statistics)**.** Shard $A = \{1,2,3\}$ has state $(3,2,2)$ and shard $B = \{10,14\}$ has $(2,12,8)$; $\delta = 10$ and $n = 5$. The merged mean is $2 + 10\times 2/5 = 6$ and $M_2 = 2 + 8 + 100\times 3\times 2/5 = 130$. Direct check on $\{1,2,3,10,14\}$: mean 6, squared deviations $25+16+9+16+64 = 130$. Merging in the other order, $\delta = -10$: mean $12 - 10\times 3/5 = 6$ and $M_2 = 8 + 2 + 100\times 6/5 = 130$, confirming commutativity. For unequal shards $\{10\}$ and $\{2,2,2\}$, the size-weighted mean is $16/4 = 4$; the unweighted mean of the shard means, $(10+2)/2 = 6$, is wrong.

Interview trap. The naive state $(n, \sum x, \sum x^2)$ is also a homomorphism in exact arithmetic — all three components are plain sums — but it computes the variance as $\mathbb{E}[x^2] - m^2$, a difference of two nearly equal large numbers. For $x = 10^8+1,\ 10^8+2,\ 10^8+3$ it returns 0.0 in binary64 against the true population variance $2/3$, which the merged-deviation state of Proposition 26.55 returns correctly. The merged-deviation form, which never forms $\sum x^2$, is the house answer.

Proposition 26.57 (Parallel prefix sum by blocks)**.** *Let $(\mathcal{S}, \oplus, e)$ be a monoid — associative with identity e, so that the exclusive scan of phase (2) and the offset $o_1 = e$ of the proof are defined — and let P workers hold contiguous blocks of at most $\lceil n/P\rceil$ elements each. The three-phase scheme — (1) each worker folds its block; (2) an exclusive scan of the P block totals is computed, sequentially in $O(P)$ or by a tree in $O(\log P)$ span; (3) each worker rescans its block, seeded with its block's offset from phase (2) — computes all n inclusive prefixes with work $2n + O(P)$ and span $2\lceil n/P\rceil + O(\log P)$.*

Proof. We use the concatenation identity (26.3) from the proof of Theorem 26.49. Let position i lie in block j, whose first element is x_a, and let o_j be the block's offset, the exclusive scan value $t_1 \oplus \cdots \oplus t_{j-1}$ of the block totals $t_1, \ldots, t_{j-1}$ (with $o_1 = e$). Repeated application of (26.3) shows that $o_j = \mathrm{fold}(x_1, \ldots, x_{a-1})$, the fold of everything before block j, for $j \geq 2$; for $j = 1$, with $a = 1$, the offset is e and the identity property gives $e \oplus \mathrm{fold}(x_1, \ldots, x_i) = y_i$ directly, so take $j \geq 2$

below. The seeded rescan computes $\mathrm{fold}(o_j, x_a, \ldots, x_i)$, which by (26.3) applied to the sequence $(o_j, x_a, \ldots, x_i)$ with $k = 1$ equals

$$\begin{aligned} o_j \oplus \mathrm{fold}(x_a, \ldots, x_i) &= \mathrm{fold}(x_1, \ldots, x_{a-1}) \oplus \mathrm{fold}(x_a, \ldots, x_i) \\ &= \mathrm{fold}(x_1, \ldots, x_i) \;=\; y_i, \end{aligned}$$

using (26.3) once more for the last step. For the costs, phase (1) performs at most n operations across the workers with span $\lceil n/P \rceil$, phase (2) performs $O(P)$ operations, and phase (3) performs n operations with span $\lceil n/P \rceil$; adding gives the claim. The dependency of the sequential loop in the Setup was an artifact of the bracketing. □

Fact 26.58 (Work-efficient scan (Blelloch, 1990; Hillis and Steele, 1986))**.** Blelloch's *up-sweep* (a reduction tree that stores each subtree's fold) followed by a *down-sweep* on the same tree (the root is set to e; each node passes its own value to its left child and its value $\oplus$ the left child's stored fold to its right child) computes the exclusive scan for any monoid with work $O(n)$ and span $O(\log n)$. Hillis and Steele's scheme — in round j, every position i absorbs position $i - 2^j$ — achieves span $O(\log n)$ with work $O(n \log n)$: span-optimal but not work-efficient, and slower than the sequential loop when $P \ll n$. A proof is beyond our scope.

Example 26.59 (A block-scan trace and a Blelloch trace)**.** Scan $[3, 1, 4, 1, 5, 9, 2, 6]$ with $P = 2$: the block totals are 9 and 22; the exclusive offsets are 0 and 9; the seeded rescans give $[3, 4, 8, 9]$ and $[14, 23, 25, 31]$, the correct inclusive prefixes. Work: $3 + 3$ additions in phase (1), 1 in phase (2), $4 + 4$ in phase (3), 15 in all against 7 sequentially; span $3 + 1 + 4 = 8$ against 7 here, but about $2n/P + \log_2 P$ in general — for $n = 2^{20}$ and $P = 64$ about 32,800 against 10^6. The Blelloch scheme of Fact 26.58 on $[3, 1, 7, 0, 4, 1, 6, 3]$: the up-sweep forms the pair sums $[4, 7, 5, 9]$, then $[11, 14]$, then the root 25; the down-sweep, with the root reset to 0, yields the exclusive scan $[0, 3, 4, 11, 11, 15, 16, 22]$; 14 additions at depth 6 against 7 additions at depth 7 sequentially.

Remark 26.60 (What scans are for; nonlinear recurrences)*.* Scans compute running totals, output offsets in count-then-fill layouts, stream compaction, and linear recurrences $s_{t+1} = A_t s_t + b_t$, whose (A, b) pairs compose associatively as affine maps (the state-space models of sequence modelling). A *nonlinear* recurrence such as a physics time step has no associative composition, keeps span $\Theta(T)$ over T steps, and the honest answer says so; the parallelism then lives within each step.

Interview trap. Four wrong sentences about associativity. "A parallel floating-point sum is deterministic": only with a fixed bracketing; thread count, chunking and arrival order all change the rounding sequence (Fact 26.52), the fix should be named in the same breath (Remark 26.53), and `assert a == b` after a parallel run is the same error in test form. "Mean is associative, average the averages": the merge needs the counts, either (sum, count) or the summary state of Proposition 26.55; the unweighted average is wrong for unequal shards (Example 26.56). "Variance from the sum and the sum of squares": exact in algebra, catastrophic cancellation in floating point (0.0 against 2/3 in the trap after Example 26.56). "Prefix sums are inherently sequential": the dependency is the bracketing, and associativity gives a scan with span $O(\log n)$ (Proposition 26.57, Fact 26.58); the honest answer for a nonlinear recurrence is span $\Theta(n)$ (Remark 26.60).

26.6 Parallelism in Python: Threads, Processes, the GIL, and Vectorisation

Python offers three ways to do two things at once — threads, processes and coroutines — and the interpreter's global lock decides which of them buys wall-clock time; "does multithreading speed up my numpy code?" and "why did my thread pool not help?" are standard screens. This subsection states the global interpreter lock and its removal as Facts with version qualifiers, proves the one consequence that follows from the lock and from the accounting inequality of Theorem 26.26 (CPU-bound pure-Python threads cannot speed up), prices multiprocessing, places vectorisation as the first form of parallelism, and ends with data-parallel training as the application of Theorem 26.49, using the diagnostics of §26.3 to classify a workload before any tool is chosen.

Setup. A workload is *CPU-bound* if a single-threaded run shows r near 1 in the sense of Theorem 26.26, and *I/O-bound* if r is well below 1. *Concurrency* means that several tasks are in progress during overlapping intervals; *parallelism* means $k(t) > 1$, more than one core busy at once. A *thread* (the `threading` module) shares its process's interpreter and heap; a *process* (`multiprocessing`, or the `ProcessPoolExecutor` of `concurrent.futures`) has its own interpreter and memory; a *coroutine* (`asyncio`) is cooperatively scheduled on one thread and yields control only at an `await`. As in §26.4, P counts workers and p is a parallel fraction, and m counts threads as in §26.3. For the training application, θ denotes the model parameters, $\ell_i(\theta)$ the loss on example i, and ∇ the gradient with respect to θ.

Fact 26.61 (The global interpreter lock and the free-threaded build (Python/ceval_gil.c; PEP 703, 2023; PEP 779, 2025))**.** The lock, and the build without it.

(i) In the default CPython build a thread must hold the process-wide global interpreter lock (GIL) to execute Python bytecode, so at most one thread executes bytecode at any instant, however many cores exist. A thread holding the lock is asked to release it after the *switch interval* (`sys.getswitchinterval()`, default 0.005 s on CPython 3.11 and 3.12) whenever another thread is waiting for it. The lock is released around blocking system calls — the `Py_BEGIN_ALLOW_THREADS` macros around file and socket I/O, `sleep` and lock waits — and by extension modules that opt out during C-level work: numpy for many array and BLAS operations (its `NPY_BEGIN_THREADS` macros), and the `hashlib`, `zlib` and `socket` modules among others. The GIL protects the interpreter's own data structures, not the program's invariants.

(ii) CPython 3.13 ships an experimental build without the GIL (the `python3.13t` executable; per-object locks and biased reference counting, PEP 703) alongside the default build, and extensions must be rebuilt for it. PEP 779 (2025) sets the criteria for moving that build from experimental to supported status, and the 3.14 release notes describe the build as supported; it remains a separate build rather than the default. No figure for its single-threaded overhead is claimed here.

Consequence: threads overlap *waits* and GIL-releasing C work; they do not parallelise pure-Python CPU work; and no correctness argument may depend on the lock in either direction (§26.7). These are properties of the cited implementation; a proof is beyond our scope.

Proposition 26.62 (No CPU-bound speedup from threads under the GIL)**.** *Under Fact 26.61(i), if m threads execute only Python bytecode, never block, and their bytecode work totals T_1 seconds of core time, then the wall time of the multi-threaded run satisfies $W \geq T_1$ whatever the number of cores; the run has $r \leq 1$, and Amdahl's law (26.2) applies with $p = 0$.*

Proof. We integrate the busy-core count as in Theorem 26.26. By Fact 26.61(i) the intervals during which different threads execute bytecode are pairwise disjoint, so the number of cores executing bytecode for this process is at most 1 at every instant. Integrating over the run gives $U + S \leq W$, that is, $r \leq 1$. All T_1 seconds of bytecode work must be performed during the run, so $T_1 \leq U + S \leq W$. □

A multi-threaded pure-Python program showing r near 1 on an eight-core machine is GIL-bound, and in practice the lock hand-offs make it somewhat slower than the single-threaded run (Fact 26.65 below); the fix is processes or C-level work, not more threads.

Example 26.63 (Threads against processes on two workloads)**.** A CPU-bound pure-Python job of 40 s: four threads take about 40 s of wall time (r near 1, often slightly more wall from hand-offs, by Proposition 26.62); four processes take about 10 s plus roughly 0.3 s of start-up and pickling. An I/O-bound job of 100 network requests, each with 0.5 s of latency and 5 ms of CPU work: serially 50 s; with 20 threads, five rounds of 0.5 s, about 2.5 s of wall time with 0.5 s of CPU time

in total, $r = 0.2$. The threads win despite the GIL, because the waits overlap and the lock is released during each of them.

Fact 26.64 (multiprocessing: start methods, pickling, memory (library reference))**.** Worker processes are separate interpreters, each with its own GIL. Three start methods exist. `fork` (Unix only) duplicates the parent through copy-on-write, so large read-only data is inherited free; it is unsafe when the parent has running threads, and 3.12 warns when it is used in that situation. `spawn` (always on Windows; the macOS default since 3.8) starts a fresh interpreter that re-imports the main module, so everything sent to the worker must be picklable. `forkserver` forks workers from a clean helper process. The Linux default was `fork` through 3.13, and the 3.14 changelog documents a change of the default to `forkserver` on platforms other than macOS and Windows. Arguments and results cross process boundaries by pickling through pipes, at a cost proportional to their size; lambdas and open handles cannot cross; and no memory is shared by default, the `shared_memory` submodule (3.8) being the escape hatch. These are properties of the cited implementation; a proof is beyond our scope.

Fact 26.65 (Costs of processes, locks and GIL hand-offs, empirical)**.** Three magnitudes recur in the arithmetic of this section. Starting a worker process costs tens to hundreds of milliseconds, plus the module import time under `spawn`, and a round trip of a small task through a process pool costs of the order of 50–100 microseconds. An uncontended `threading.Lock` acquisition and release costs tens of nanoseconds, and contention adds the wait. A multi-threaded pure-Python program on several cores runs no faster, and usually somewhat slower, than the single-threaded run, the difference being the cost of the GIL hand-offs of Fact 26.61(i). These are typical figures on current hardware, quoted as evidence and not as bounds; a proof is beyond our scope.

Example 26.66 (The price of pickling and of tiny tasks)**.** Shipping an 800 MB numpy array to 8 workers by pickling copies 6.4 GB, about 3.2 s at 2 GB/s before any work is done; a fork-inherited read-only array or a `shared_memory` block costs nothing per worker. Splitting a 10^6-item job into 8 chunks pays 8 round trips; splitting it into 10^6 tasks pays 10^6 round trips at 50–100 microseconds each (Fact 26.65), 50–100 s of pure overhead. This is the cP term of Proposition 26.46 with the per-task cost in place of the per-worker one.

Remark 26.67 (Coroutines and vectorisation). (a) `asyncio` is concurrency, not parallelism: one thread, cooperative scheduling, a coroutine yielding only at `await`, so thousands of pending I/O operations coexist cheaply, but one CPU-bound coroutine stalls the whole event loop (`run_in_executor` hands such work to a thread or process pool); a bounded `asyncio.Queue` between producer and consumer is the backpressure pattern of Remark 26.30. (b) Vectorisation is the first parallelism. Remark 9.13 explains the compiled-loop win as a memory-layout and dispatch story; on top of it, SIMD instructions process several lanes per instruction (4 doubles per AVX2 instruction, 8 per AVX-512), and BLAS libraries (OpenBLAS, MKL) run large products on several threads by default, controlled by `OMP_NUM_THREADS` and `MKL_NUM_THREADS`. This is why a "single-threaded" numpy script shows $r = 8$ (Example 26.33), and why *oversubscription* — eight processes each starting eight BLAS threads — collapses throughput while r stays high. A GPU kernel is the same idea with thousands of lanes (Fact 9.15, with its caveats). The order of escalation for a slow Python loop is: vectorise, then BLAS or GPU, then processes.

Proposition 26.68 (Data-parallel gradients)**.** *Let the batch loss be $L(\theta) = \frac{1}{B}\sum_{i=1}^{B} \ell_i(\theta)$ over B examples split into P shards $I_1, \dots, I_P$ of sizes $B_1, \dots, B_P$ with $\sum_j B_j = B$, and let worker j compute its shard-mean gradient $g_j = \frac{1}{B_j}\sum_{i \in I_j} \nabla \ell_i(\theta)$. Then*

$$\sum_{j=1}^{P} \frac{B_j}{B}\, g_j \;=\; \nabla L(\theta);$$

with equal shards this is the plain mean of the g_j, that is, an all-reduce sum divided by P.

Proof. We use linearity of the gradient and regroup the sum over the batch as a sum over shards:

$$\nabla L(\theta) = \frac{1}{B}\sum_{i=1}^{B}\nabla \ell_i(\theta) = \frac{1}{B}\sum_{j=1}^{P}\sum_{i\in I_j}\nabla \ell_i(\theta)$$
$$= \sum_{j=1}^{P}\frac{B_j}{B}\cdot\frac{1}{B_j}\sum_{i\in I_j}\nabla \ell_i(\theta) = \sum_{j=1}^{P}\frac{B_j}{B}\,g_j.$$

With $B_j = B/P$ for all j the weights are $1/P$. □

Remark 26.69 (Data-parallel training as an all-reduce (Patarasuk and Yuan, 2009)). Data-parallel training instantiates Proposition 26.68 once per step: each of P workers computes the gradient on its shard, an all-reduce forms the sum, and every replica applies the identical update. The ring all-reduce (a reduce-scatter phase followed by an all-gather phase) has each worker send and receive $2\,\frac{P-1}{P}\,D$ bytes for a gradient of D bytes, in $2(P-1)$ steps; the per-worker traffic is independent of P to leading order and the scheme is bandwidth-optimal, and it requires an associative and commutative reduction (Theorem 26.49(ii)). The per-step wall time is $T_1/P + T_{\text{sync}}$, a greedy-bound-shaped budget whose synchronisation term is the span (Theorem 26.41). One process per GPU is the rule because of Fact 26.61; in floating point the reduction tree changes the low bits (Fact 26.52); and the training loop being parallelised is that of Section 9.

Example 26.70 (One data-parallel step)**.** Take $P = 8$ GPUs, 120 ms of compute per step on one shard, and a model of 350 million float32 parameters, 1.4 GB of gradient. The ring all-reduce moves $2 \times \frac{7}{8} \times 1.4 = 2.45$ GB per GPU, about 12 ms at 200 GB/s, so a step takes 132 ms and the efficiency is $120/132 = 91\%$. Over a 10 GB/s network link the same all-reduce takes 245 ms and the efficiency falls to $120/365 = 33\%$: the synchronisation term dominates. For the merge itself, shards of sizes 3 and 1 with shard-mean gradients 1 and 5 (one coordinate shown) merge to $(3 \times 1 + 1 \times 5)/4 = 2$, the full-batch gradient by Proposition 26.68; the unweighted mean 3 is wrong — the same error as the unweighted shard means of Example 26.56.

Remark 26.71 (Choosing the concurrency tool). Six steps, the first of which decides the rest.

1. Measure r on a single-threaded run and read it off Table 6.
2. If waiting dominates, use threads (a few tasks) or `asyncio` (thousands of waits) so that the waits overlap; the wall time becomes the maximum of the waits rather than their sum.
3. If CPU-bound and the hot loop is numpy, BLAS or another GIL-releasing extension, check whether r already exceeds 1 before adding anything (Example 26.33).
4. If CPU-bound pure Python, first vectorise (Remark 26.67); then use multiprocessing with chunked tasks — many more tasks than workers, each chunk large enough to amortise its pickling (Fact 26.65) — priced with Proposition 26.46.
5. Before promising a speedup, compare the result with the Amdahl ceiling of the measured serial fraction (Theorem 26.37(iii)).
6. For training, run one process per GPU (Proposition 26.68, Remark 26.69).

Trading application. A parameter sweep over 10^4 backtest configurations is pure Python around a numpy core. Remark 26.71 reads: measure r first (a core already driving r to 8 means more processes will oversubscribe); then hand the configurations to a process pool in chunks of about 50 — many more tasks than workers, each chunk large enough to amortise its pickling, with the shared price panel fork-inherited or placed in shared memory rather than pickled per task; and expect at most the Amdahl ceiling of the serial part — loading the panel, writing the results table — however many cores the machine has.

Interview trap. Four wrong sentences about Python parallelism. "The GIL makes Python thread-safe": it serialises bytecodes, not statements; the trap of §26.7 works the consequence. "More threads means faster" and its mirror "threads never help in Python": for CPU-bound

pure Python, no (Proposition 26.62); for I/O waits and for GIL-releasing C code, yes, up to the waiting fraction, and numpy-heavy threads scale. "`multiprocessing` shares my big array for free": every task pickles its arguments and `spawn` re-imports the module (Fact 26.64); price it with Proposition 26.46. "`asyncio` is parallel": it is single-threaded concurrency; one CPU-bound coroutine blocks everything (Remark 26.67).

26.7 Race Conditions, the Lost Update, and Deadlock

A race condition is the failure no test reliably reproduces and no interviewer lets pass: two workers touch the same state without an agreed order, and the result depends on the scheduler. On-site rounds have asked the canonical form ★: "two threads increment a shared counter — what can go wrong, how wrong can it get, does the GIL save you, what does a lock cost, and is it safe to retry?". This subsection defines data races, shows on Python bytecode that the lock of Fact 26.61 does not make an increment atomic, works the lost update on paper (Figure 56), proves how low the counter can go, builds the fix ladder on Theorem 26.37 and Theorem 26.49, and proves that lock ordering prevents deadlock.

Setup (the interleaving model). Threads $t_1, \ldots, t_K$ share memory. Each thread executes a fixed sequence of *atomic steps* — one memory read, one memory write, or one interpreter bytecode — and an *execution* is any interleaving of the threads' sequences that preserves each thread's own order (sequential consistency; weaker hardware memory models are out of scope and are said to be). A shared integer c starts at 0. Each of the $K \geq 2$ threads — named A and B when $K = 2$ — performs $N \geq 1$ increments; one increment is the three steps *load* (read c into a private temporary), *add* (add one to the temporary, a private step), and *store* (write the temporary back to c). The final value of c is the value written by the last store of the execution. In this subsection N is a number of increments and K a number of threads, not the random-walk target of the notation table.

Definition 26.72 (Data race, race condition and atomicity)**.** Two accesses to the same memory location *conflict* if they are made by different threads and at least one is a write. A *data race* is a pair of conflicting accesses that no synchronisation orders, so that some execution places them adjacent in either order. A program has a *race condition* if its outcome depends on the interleaving; a data race is the usual cause but not the only one — a check-then-act on a file name, or a size check followed by an append on a structure that is locked during each operation but not across the pair, is a race condition without a data race. A *critical section* is a region of code that must appear atomic to other threads; an operation is *atomic* if no execution observes an intermediate state of it; *mutual exclusion* is the property that at most one thread is inside a given critical section at any instant.

Fact 26.73 (An augmented assignment is not atomic (the dis module; Python/ceval.c, CPython 3.11–3.12))**.** Inside a function that declares `global counter`, the statement `counter += 1` compiles, after the function's `RESUME`, to

```
LOAD_GLOBAL counter,  LOAD_CONST 1,  BINARY_OP +=,  STORE_GLOBAL counter
```

(`dis` output on CPython 3.11, identical on 3.12) — logically load, add, store — and the language guarantees no atomicity for the statement. An attribute target uses `LOAD_ATTR` and `STORE_ATTR` in the same shape, and a subscript target such as `counts[w] += 1` uses `BINARY_SUBSCR` and `STORE_SUBSCR`, so a word count updated from several threads is the lost update of Example 26.74 below with a dict in place of the integer. CPython transfers the GIL only at its *eval-breaker* check points, which in 3.11 and 3.12 sit at function entry and resumption, at backward jumps, and at calls; none of these lies between the load and the store of the plain statement above, whereas in 3.9 and earlier the check was polled before most bytecodes, and the placement may change

again. Whether a particular straight-line statement in a particular version can be split between its load and its store is therefore an implementation accident; any statement with a call or a GIL-releasing operation between its load and its store — `counter += f()`, `counter += int(1)` — can be split in every version; and the Python FAQ's list of operations that happen to be atomic in CPython because their mutation is a single bytecode or one C-level call made while the GIL is held (`list.append`, dict item assignment) is an implementation property, not a guarantee. On the free-threaded build of Fact 26.61(ii) each built-in container operation still runs under a per-object lock (PEP 703), so a single `append` or item assignment cannot corrupt the container; what is lost there is any accident that made a compound statement such as `+=` appear atomic. The precise statement is the weak one: the interleaving model permits the split, a correct program never depends on its absence, and a test that fails to reproduce the lost update is not evidence of atomicity. These are properties of the cited implementation; a proof is beyond our scope.

Example 26.74 (The lost update ★)**.** Start from $c = 0$ with one increment per thread. Thread A loads 0 and adds, holding 1 in its temporary; the scheduler switches; thread B loads 0, adds, and stores 1; the scheduler switches back; thread A stores its stale 1. Two increments were executed and $c = 1$: B's update is lost. Figure 56 draws this interleaving; the fully alternating one (load A, load B, add A, add B, store A, store B) also ends at 1. Proposition 26.75 below scales this up: $N = 1000$ increments per thread can end at $c = 2$ instead of 2000.

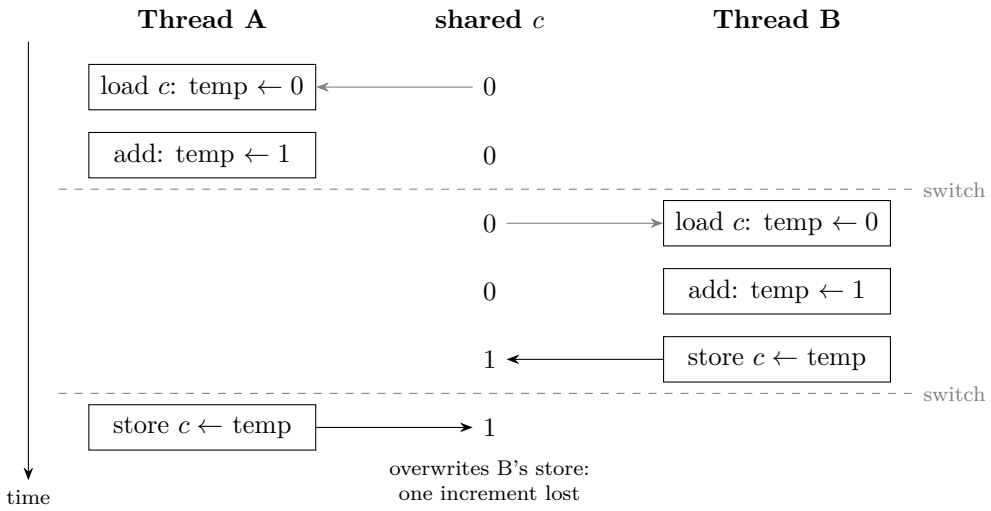


Figure 56: The lost update of Example 26.74. Time runs downward; the middle column shows the shared value after each step. Thread A loads 0 and computes 1, the scheduler switches (dashed line) at the first hand-off, thread B loads the same 0, computes 1 and stores it, and after the second hand-off thread A stores its stale 1 over B's. Two increments, final value 1. Nothing in the picture depends on the GIL: the two switches are the hand-offs Fact 26.73 permits.

Proposition 26.75 (How low a racy counter can go)**.** *Let $K \geq 2$ threads each perform $N \geq 2$ load–add–store increments of c from 0 in the interleaving model. Then every reachable final value v satisfies $2 \leq v \leq KN$, and both bounds are attained; in particular two threads performing N increments each can finish with $c = 2$. For $N = 1$ the reachable set is $\{1, \ldots, K\}$, with minimum 1.*

Proof. We prove the upper bound by an invariant, the lower bound by examining the last store, and attainment by exhibiting schedules.

Upper bound. The invariant is: at every moment, c is at most the number of stores completed so far. It holds initially ($0 \leq 0$). A store writes temp + 1, where temp was loaded at an earlier moment, when c was at most the number of stores completed by then, which is at most the number

completed just before this store; so the value written is at most that number plus one, which is the number of stores completed after this store. The run has KN stores, so $v \le KN$.

Lower bound. Every store writes temp $+ 1 \ge 1$, since every loaded value is at least 0; hence once any store has completed, $c \ge 1$ forever. The last store of the run is some thread's N-th store, and in that thread's own order its N-th load comes after its $(N-1)$-th store, which exists because $N \ge 2$. At the moment of that load, therefore, at least one store had completed and $c \ge 1$; the thread loaded at least 1 and its final store wrote at least 2. So $v \ge 2$.

Attaining 2. Thread A loads 0. All other threads then complete all their increments except that thread B withholds its last one. Thread A stores 1. Thread B loads 1. Thread A completes its remaining $N-1$ increments in isolation, bringing c to N. Thread B adds and stores 2, the last store of the run. Every step respects each thread's own order, so this is an execution, and $v = 2$.

Attaining KN. Run the threads one after another with no interleaving; each increment then reads the previous store, and $v = KN$.

The case $N = 1$. Each thread's single store writes 1 plus what it loaded; running the threads one after another gives K, and letting j threads load 0 before any store, then interleaving the rest sequentially, gives every value from 1 to K; no store can write more than the number of stores completed, by the invariant, and none can write less than 1. □

(Problem P26.10 asks you to reproduce this proof from scratch.)

Example 26.76 (Reachable counter values for small K and N)**.** Brute-force enumeration of all interleavings gives the reachable sets $\{1, 2\}$ for $K = 2$, $N = 1$; $\{2, 3, 4\}$ for $K = 2$, $N = 2$; $\{2, \ldots, 6\}$ for $K = 2$, $N = 3$ and for $K = 3$, $N = 2$; and $\{2, \ldots, 8\}$ for $K = 2$, $N = 4$. The value 2 for $K = 2$, $N = 2$ is the schedule of the proof, step by step: A loads 0; B loads 0, adds and stores 1; B loads 1 (the start of its second increment) and stops; A adds and stores 1; A loads 1, adds and stores 2; B adds its stale 1 and stores 2. Four increments, final value 2. The damage in the worst case is near-total, not a small fraction.

Fact 26.77 (Lost updates measured on CPython 3.11, empirical)**.** Two threads each performing 10^6 increments of a global integer, three runs per setting on CPython 3.11.9: the plain statement `counter += 1` finished at exactly 2,000,000 both at the default switch interval and under `sys.setswitchinterval(1e-6)`, because no check point of Fact 26.73 lies inside its load–store window on that version; the statement `counter += int(1)`, which puts a call inside the window, lost 0%, 14% and 35% of its increments in the three runs at the default interval — one run lost nothing — and 37%, 38% and 43% at the shortened one. On 3.9 and earlier the plain statement also loses updates. These are measurements on one machine, quoted as evidence and not as bounds; a proof is beyond our scope.

Intuition. Why the loss is a fraction, not a handful. A hand-off that lands inside a load–store window does not lose one increment: the stalled thread's later store overwrites every increment the other thread completed during its whole quantum — thousands at the 5 ms interval — so each such landing discards a quantum, and the expected loss is a constant fraction of the run, of the order of the fraction of check points that sit inside the window. Shortening the switch interval makes the landings more frequent and the loss larger, as Fact 26.77 shows. A test that observes 2,000,000 has observed one schedule on one version, while Proposition 26.75 says that another schedule ends at 2: rarity is a property of the scheduler, the proposition a property of the program.

Procedure 26.78 (The fix ladder, best last)**.** Four rungs, in increasing order of preference.

1. *A lock around the critical section* (`threading.Lock` as a context manager). Correct, but the section runs serially: if it is a fraction s of the work, Theorem 26.37 with $1 - p = s$ caps the speedup at $1/s$ regardless of the worker count; each acquisition costs tens of nanoseconds

plus contention (Fact 26.65). Hold locks for the shortest possible span. *Striping* — one lock per shard or key range, chosen by hash — shrinks the serialised fraction s in proportion to the number of stripes but not to zero, costs memory per stripe, and reintroduces the lock-ordering discipline of Proposition 26.83 below whenever an operation touches two stripes.

2. *An atomic read-modify-write*: a fetch-and-add, or a compare-and-swap loop (read old, compute new, `CAS(old, new)`, retry on failure), available in C and in CUDA's `atomicAdd`, not for Python integers in the standard library. No blocking, but every update still funnels through one contended memory location (one cache line), and floating-point atomics reduce in a nondeterministic order (Fact 26.52).
3. *Private per-thread state plus an associative merge at the end.* Each worker accumulates into its own variable, and a single merge follows the join; no synchronisation during the work. This is correct for any chunking by Theorem 26.49(i) and for any arrival order, given commutativity, by (ii); it is the pattern behind combiners (§26.8) and the all-reduce (Remark 26.69), and it is the answer interviewers want.
4. *Immutability and message passing.* Transfer ownership through a `queue.Queue` — a deque plus a lock and condition variables (Fact 26.2(iv)), bounded for backpressure — instead of sharing; a value that is never mutated cannot race.

Example 26.79 (A lock against private counters)**.** Eight workers — processes, or threads on a free-threaded build — each perform 10^7 increments. With one shared lock, 8×10^7 serialised acquisitions at no less than the 25 ns uncontended cost each — eight-way contention adds cache-line transfers and waits on top — cost at least 2 s however many workers there are; a single thread performing the same 8×10^7 increments unlocked at about 20 ns each takes 1.6 s, so the locked eight-worker version is slower than doing nothing in parallel, a speedup below 1. If the locked section is instead 10% of each task, Theorem 26.37 with $1 - p = 0.1$ caps the speedup at $10\times$ regardless of P. With private counters, each worker performs 10^7 local increments in parallel and the merge is one eight-term sum, for a speedup close to 8.

Example 26.80 (A race without a counter: a time-stepped simulation)**.** Let P workers advance N particles by one time step, each worker owning a slice of the particles: it computes the forces on its particles from the positions of *all* particles, then writes its particles' new positions. If worker A writes $x[0]$ at time $t + 1$ while worker B is still computing the pair $(5, 0)$ from what it believes is time t, then B mixes two time levels and the result depends on scheduling: a race condition on state in the sense of Definition 26.72, with no counter in sight. The fix is double-buffering — read x_t, write x_{t+1}, swap — plus one barrier per step, and the barrier cannot be removed because it *is* the time dependency (the nonlinear-recurrence span of Remark 26.60). A second race hides in the force accumulation itself: with Newton's third law, computing the pair (i, j) once and writing both $F[i] \mathrel{+}= f$ and $F[j] \mathrel{-}= f$ makes two workers write $F[j]$ — the lost update in disguise. Rung 3 of Procedure 26.78 is the answer: per-worker arrays F_w merged by addition after the barrier, an associative and commutative merge; the alternative of computing every pair twice does $2\times$ the work with no conflicts, and which wins is a crossover in P. With k neighbours per particle, pair cost t_p and per-vector accumulation cost t_a, the third-law version saves $Nkt_p/(2P)$ per worker and pays a merge of PNt_a if one thread sums the P copies, so it wins while $P^2 < kt_p/(2t_a)$: for $k = 50$, $t_p = 20$ ns and $t_a = 3$ ns, up to $P = 12$; if instead every worker merges its own slice of the P copies, the merge costs Nt_a per worker and the third-law version wins while $P < kt_p/(2t_a) \approx 167$.

Definition 26.81 (Deadlock and the wait-for graph)**.** A set of threads is *deadlocked* if each of them is waiting for a resource held by another member of the set, so that none can ever proceed. The *wait-for graph* has an edge from each waiting thread to the holder of the lock it awaits; a deadlock is a cycle in it.

Fact 26.82 (Coffman conditions (Coffman, Elphick and Shoshani, 1971))**.** Four conditions are jointly necessary for deadlock: *mutual exclusion* (a resource is held by at most one thread),

hold-and-wait (a thread holds resources while waiting for more), *no preemption* (a held resource is released only by its holder), and *circular wait* (a cycle in the wait-for graph). Breaking any one of them prevents deadlock: try-locks with release on failure break hold-and-wait, timeouts break no-preemption, and a global lock order breaks circular wait — the standard practical choice, proved next. A proof of necessity is beyond our scope.

Proposition 26.83 (A global lock order prevents deadlock)**.** *Fix a total ranking of all locks. If every thread acquires locks in strictly increasing rank — it never waits for a lock of lower rank than one it already holds — then no deadlock can occur.*

Proof. We argue by contradiction through the wait-for cycle. Suppose threads $t_1, \dots, t_k$ are deadlocked, with t_i holding lock l_i and waiting for l_{i+1} held by t_{i+1}, indices taken modulo k. The discipline gives $\operatorname{rank}(l_i) < \operatorname{rank}(l_{i+1})$ for every i, since t_i waits for l_{i+1} while holding l_i. Composing these strict inequalities around the cycle yields $\operatorname{rank}(l_1) < \operatorname{rank}(l_1)$, which is impossible. □

Example 26.84 (Two locks suffice)**.** Thread A acquires L_1 then L_2; thread B acquires L_2 then L_1. The interleaving "A takes L_1; B takes L_2; A waits for L_2; B waits for L_1" satisfies all four Coffman conditions — each lock has one holder, each thread holds one lock while waiting, neither lock is taken away, and the wait-for graph is the cycle $A \to B \to A$ — and both threads wait forever. Impose the order $L_1 < L_2$: B must now take L_1 first, so it blocks on L_1 while holding nothing, A completes and releases both locks, and B proceeds. This is Proposition 26.83 in action.

Definition 26.85 (Determinism, idempotence, and safe retry)**.** A task is *deterministic* if its output is a function of its input alone — no dependence on scheduling, on the clock, or on unseeded randomness. A task is *idempotent* if executing it twice leaves the same state as executing it once.

Fault tolerance by re-execution — the task restarts of MapReduce (§26.8), a data-loader worker respawn, a job-array retry — is correct only for idempotent tasks. Writing `results[task_id] = y`, or writing to a per-task temporary file and then renaming it atomically into place, is idempotent; `results.append(y)`, `counter += 1` and "append a line to a shared log" are not. At-least-once delivery combined with an idempotent handler behaves as exactly-once. Random numbers keyed by (seed, task id) make a re-execution reproduce the original and the task deterministic in the sense of Definition 26.85.

Example 26.86 (A duplicated shard)**.** A shard task that appends its 100 partial word counts to a shared list is re-executed after a timeout, and the original also completes; the list now holds 200 records, 100 of them duplicates, and the word "the" is counted twice. With `results[shard_id] = partial` — or write-to-temporary-then-rename — the re-execution overwrites its own entry with an identical value and the total is correct.

Trading application. A backtest sweep that writes each configuration's Sharpe ratio into one shared results dict from several threads is the counter of Example 26.74 in disguise if any entry is ever read-modified-written (a running best, a count of configurations tried). Rung 3 of Procedure 26.78 is the design: each worker keeps its own results dict keyed by configuration id, the merge after the join is a union of dicts with disjoint keys, a retried configuration overwrites its own key (idempotent by Definition 26.85), and the rolling best is computed once, at the end, from the merged dict.

Interview trap. Four wrong sentences about races. "The GIL makes Python thread-safe": it serialises bytecodes, not statements; `+=` is four bytecodes, and the GIL changes hands between them whenever a check point — a call — sits in the window, or on 3.9 and earlier (Fact 26.73, Fact 26.77). "I ran it a thousand times and the count was right, so it is atomic": absence of the bug is not absence of the race; check-point placement is an implementation accident, the free-threaded build removes the lock, and Proposition 26.75 bounds what another schedule can do. "Just add a lock": correct, but it serialises the critical section (Amdahl), and two locks already

suffice for deadlock (Example 26.84); the house answer is private state plus an associative merge (Procedure 26.78). "Retry the failed task": only if the task is idempotent; appends and increments are not (Definition 26.85).

26.8 Distributed Aggregation: MapReduce, Combiners, and Top-k Across Shards

The long-form on-site question escalates from one machine to many ("count word frequencies across 1000 machines; now top-k; now it is skewed"), and its last part is where candidates most often fail: they select each shard's top-k and merge. This subsection assembles §26.1 (a dict per machine), §26.5 (associative merges), §26.4 (the finish time is the maximum) and §26.7 (idempotent retries) into the MapReduce pattern, identifies the shuffle as the bottleneck, and proves that top-k must aggregate before it selects.

Setup. There are n records whose keys are drawn from d distinct keys, spread over M machines with about n/M records each; h is a hash partition from keys to $\{1, \dots, M\}$; each machine holds a local dict; communication is measured in records crossing the network; k is the number of top items requested; d_j denotes the number of distinct keys present on machine j, so that $d_j \le \min(n/M, d)$; and keys are ranked by a strict total order (count first, then key identity) so that ties are broken identically everywhere. A hot key may be *salted* into J sub-keys. In this subsection M is a machine count, k a rank and J a salt count, not a heap height or a stage count.

Procedure 26.87 (MapReduce word count with combiners)**.** Four phases; the third is the one to name as the bottleneck.

1. *Map.* Each machine emits one (word, 1) pair per occurrence in its own data: embarrassingly parallel, $O(n/M)$ expected dict updates per machine (Fact 26.2(ii), Fact 24.8).
2. *Combine.* Each machine pre-reduces its own emissions into a local dict of (word, local count). This is legal exactly because integer addition is associative and commutative (Theorem 26.49(ii)): the partial counts will later be merged in an arbitrary order.
3. *Shuffle.* Each (word, partial count) record travels to machine h(word): an all-to-all network exchange, after which every occurrence of a word has its partial counts on exactly one machine.
4. *Reduce.* Each owner folds the partials of its keys and emits (word, total).

Communication. Without the combiner, n records cross the network. With it, machine j sends at most $d_j \le \min(n/M, d)$ records — one per distinct key it holds — so the total is at most Md: the combiner collapses shuffle traffic from the number of records to the number of distinct keys per machine. The shuffle's all-to-all phase is the bottleneck and must be named as such.

Example 26.88 (Word-count costs at scale)**.** Take $n = 10^{10}$ words over $M = 1000$ machines with $d = 10^6$ distinct words: 10^7 words are mapped per machine, about 1 s at 100 ns per expected dict update. Without a combiner, 10^{10} records cross the network; with one, at most 10^6 per machine and at most 10^9 in total. At 8 bytes per record the cluster-wide shuffle volume is 80 GB against at most 8 GB, and each machine's share is 80 MB against at most 8 MB — 0.8 s against 0.08 s at a 100 MB/s share of the network, so the combiner turns a shuffle that costs as much as the map into one that costs a tenth of it.

Fact 26.89 (MapReduce fault tolerance and combiners (Dean and Ghemawat, 2004))**.** The framework re-executes the tasks of failed machines and speculatively re-executes straggler tasks near the end of a job, which requires tasks to be deterministic and idempotent (Definition 26.85); combiner functions are the optional local reducer run on each mapper's output before the shuffle. These are properties of the published system; a proof is beyond our scope.

Remark 26.90 (Skew, hot keys and stragglers). Hash partitioning balances *keys*, not *occurrences*: a hot key floods its owner. The combiner caps that key's partials at M, one per mapper; for aggregates that stay hot, *salting* splits the key into J sub-keys aggregated in two rounds. The finish time is the maximum over machines (Corollary 26.43(iii)), so the standard levers are many more tasks than workers drawn from a work queue, speculative re-execution of stragglers, and power-of-two-choices assignment (send each task to the less loaded of two random workers).

Example 26.91 (A hot key)**.** The word "the" occurs 10^9 times in the corpus of Example 26.88. Without a combiner, 10^9 records (8 GB) converge on one reducer, which at 100 MB/s receives for 80 s while every other machine finishes in under a second: a straggler by construction. With the combiner, at most 1000 partials arrive. With salt $J = 10$, the ten sub-keys land on ten machines and a second aggregation of ten partials finishes the job.

Example 26.92 (The select-then-aggregate counterexample)**.** Let $k = 1$ with three shards. The word a appears $2, 2, 2$ times on the three shards (total 6); the words b, c and d appear 3 times each, each on one shard only. The local top-1 lists are b, c and d, and the true global top-1, a, is never reported, whatever the tie-breaking rule: select-then-aggregate is wrong.

Proposition 26.93 (Aggregate, then select)**.** *Let the M shards hold raw occurrence counts under the setup above.*

(i) Selecting each shard's local top-k of raw occurrences *and merging the lists can miss the global winner; Example 26.92 is a witness, and (i) needs no proof beyond it.*

(ii) After hash-partitioning by key, so that every key's complete *count lives on exactly one machine, the union of the machines' local top-k lists contains the global top-k. Hence "aggregate, select locally, merge M lists of k" is exact, with communication $O(\sum_j d_j) \le O(\min(n, Md))$ records for the shuffle plus $O(Mk)$ for the merge, and a merge cost of $O(Mk \log M)$ by a heap of M list heads (Example 26.9; the two-list case is the merge of Proposition 26.19).*

Proof. We prove (ii) by a ranking argument under the strict total order of the Setup. Let x be any key in the global top-k, so that at most $k - 1$ keys precede x in the order. The owner of x ranks its keys by the same order on the same complete counts, since after partitioning it holds every occurrence of each of its keys; so at most $k - 1$ keys precede x locally, and x is in the owner's local top-k list. Thus every global top-k key appears in the union of the local lists, with its exact count, and the coordinator that merges the M lists selects the global top-k correctly. Merging M sorted lists of length k with a heap of M heads costs $O(Mk \log M)$, each of the Mk outputs being one pop and one push on a heap of size at most M. □

The invariant to state aloud: *after key-partitioning, local top-k lists are globally exhaustive candidates.* Three refinements and one trap. Threshold algorithms (Fagin, Lotem and Naor, 2001) fetch the local lists incrementally and stop as soon as no unseen key can enter the top-k. For streaming data, keep a *mergeable* heavy-hitter summary per shard (Misra–Gries or Space-Saving with s counters; Agarwal, Cormode, Huang, Phillips, Wei and Yi, 2013): any merge tree, in any order, preserves an additive error of at most n/s in every estimated count, so every key whose global count exceeds n/s survives the merge and the candidate set is exhaustive above that threshold. A count-min sketch merges cell-wise exactly — cell-wise addition is associative and commutative (Theorem 26.49) — but a per-shard heap of *local* leaders is select-then-aggregate again: the key a of Example 26.92 enters no local heap and is never queried against the merged sketch. The coordinator's $O(Mk)$ load is never the bottleneck; the shuffle is. And a home-made shuffle that partitions by `hash(word) % M` in separate Python processes is wrong unless `PYTHONHASHSEED` is fixed or a stable hash from `hashlib` is used, because string hashes are randomised per process (Fact 26.5): the same word would be sent to different owners by different mappers, and no machine would hold its complete count.

Example 26.94 (The same shards aggregated first)**.** Repartition the shards of Example 26.92 by key: a's total 6 lives on one machine, which reports it as its local top-1; the coordinator merges three lists of one and returns a. The shuffle moves $\sum_j d_j = 6$ records (a from three shards, b, c and d from one each), more than the $d = 4$ distinct keys, and the merge handles $Mk = 3$, as Proposition 26.93(ii) states.

Trading application. "Which ten symbols traded the most notional today, across a feed sharded by exchange?" is Proposition 26.93 verbatim: a symbol that trades moderately on every venue can out-total one that is large on a single venue, so per-shard top tens are not exhaustive candidates. Aggregate per symbol first — a shuffle by symbol with local combiners summing notional, an associative and commutative merge — then select. The same holds for "top signals by information coefficient across date shards" when the statistic is a size-weighted mean (Proposition 26.55).

Interview trap. Four wrong sentences about distributed aggregation. "Top-k of each shard, then merge": wrong (Example 26.92); aggregate first (Example 26.94). "Hash partitioning balances the load": it balances keys, not occurrences; skew lives in hot keys (Remark 26.90). "Combiners always apply": only for associative and commutative reducers; a median cannot be combined without shipping values or approximating (Definition 26.48). "`hash(word) % M` partitions consistently across my worker processes": not for string keys under per-process hash randomisation (Fact 26.5).

Practice: problems P26.1–P26.12 in §27, solutions in §28.

Part IX
Problems and Solutions

27 Problem Bank

Two-plus problems per subsection of the body, mixed difficulty, no coding. A star (★) after a problem means it reproduces, from memory, a question asked in a live 2026 interview; a diamond (♦) means the same for a timed online assessment; the firms are deliberately not named. Work them closed-book, under a rough 5-minute clock each; full worked solutions are in §28. Every answer was re-derived by two independent blind solvers before inclusion.

27.1 Problems — §1 Expected Value for Trading Games

P1.1 *Independence and the product rule* ♦

The probability I order a burger from the office Deliveroo on a given day is 0.4. The probability my stomach hurts on a given day is 0.1. Assuming the two events are independent, what is the probability that I order a burger *and* my stomach hurts that day?

P1.2 *Binomial counting* ♦

A fair coin is flipped 5 times. What is the probability of exactly 3 heads?

P1.3 *Two aces (hypergeometric draw)* ♦

Cards are dealt from a well-shuffled standard 52-card deck, and a specific player receives two of them. What is the probability that this player is dealt two aces?

P1.4 *Dice outcome counting* ♦

Two fair eight-sided dice (faces numbered 1–8) are thrown. What is the probability that the sum of the two faces is greater than or equal to 14?

P1.5 *Binomial at-least* ♦

In foosball I have a 25% chance of scoring each time I shoot, independently of all other shots. I shoot 3 times. What is the probability that I score 2 or more goals?

P1.6 *Distinct faces in ten rolls*

A fair die is rolled 10 times. What is the expected number of *distinct* faces that appear? Give an exact expression and a numeric value — and explain why your calculation is legitimate even though the quantities you sum are not independent.

P1.7 *Gold and silver balls* ♦

A bag contains 100,000 gold balls and 100,000 silver balls. Employees take turns drawing balls from the bag at random, one at a time, without replacement; each employee stops their turn as soon as they draw a gold ball, and then the next employee begins. This continues until all the gold balls have been drawn. What is the expected proportion of gold balls among all the balls pulled out of the bag? Give the closest simple value and justify it.

P1.8 *Geometric waiting time* ♦

What is the expected number of times I need to roll a fair die in order to get a 6?

P1.9 *Geometric waiting time: memorylessness*

You are working a resting limit order. Each second, independently of everything that has happened so far, the order is filled with probability $p = 0.05$.

(a) What is your expected wait, in seconds, until the fill?
(b) You have already waited 30 seconds without a fill. What is your expected *additional* wait from this moment?
(c) A different game: a fair coin is flipped repeatedly. What is the expected number of flips until you have seen *both* a head and a tail at least once?

P1.10 *Pattern waiting times: HH versus HT*

A fair coin is flipped until the pattern HH first appears; in a separate game, until HT first appears. Compute both expected waiting times by first-step analysis. Then explain, in one sentence, why the answers differ even though HH and HT are equally likely to occupy any two fixed flips.

P1.11 *Pattern waiting times: the mental closed-form arsenal* ♦

A timed online numerical test — no pen and paper allowed — hits you with a rapid block of expected-number-of-rolls questions: expected rolls or flips to reach a given outcome or pattern. Under those conditions the hard part is not the concepts but tracking states and transitions purely in your head. State *from memory*, each with a one-line justification you could reconstruct under pressure:

(a) expected trials until a first success of probability p;
(b) expected flips of a fair coin until a run of k consecutive heads;
(c) expected rolls of a fair die until a run of k consecutive sixes;
(d) expected flips until the first appearance of HH, of HT, and of HTH;
(e) expected rolls of a die until every face has been seen.

Briefly: why does memorising these dominate deriving them live in this test format?

P1.12 *Pattern races: which comes first*

A single fair coin is flipped repeatedly.

(a) Your colleague notes that $\mathbb{E}[\text{flips to HT}] = 4$ while $\mathbb{E}[\text{flips to HH}] = 6$, and concludes that in

a head-to-head race on the same coin, HT beats HH more often than not. What is the actual probability that HT appears before HH, and what exactly is wrong with the reasoning?

(b) Now race TH against HH on one coin. A counterparty offers you an even-money bet on HH appearing first. Do you take it, and what would the fair odds be?

P1.13 *Coupon collector*

You roll a fair die repeatedly. How many rolls do you expect to need until every face 1–6 has appeared at least once? Give the exact value, then the rough growth rate of the answer for an n-faced die.

P1.14 *Coupon collector variant: subset of coupons amid noise*

Same fair die, but now you only care about the faces 1, 2 and 3. How many rolls do you expect to need until each of 1, 2 and 3 has appeared at least once? (Faces 4–6 still come up as usual — they are simply useless to you.) Before computing: should the answer be the 3-face coupon-collector number $3(1 + \frac{1}{2} + \frac{1}{3}) = 5.5$?

P1.15 *Order statistics of dice: make-a-market ladder*

An interviewer rolls two fair dice X and Y behind a screen and asks you to make a market — state your fair value and how you would quote around it — on each of:

(a) $\max(X, Y)$;
(b) $\min(X, Y)$;
(c) $|X - Y|$.

P1.16 *Option value of a re-roll*

You roll a fair die and are paid its face value in dollars.

(a) Before rolling, you are granted the right to discard the first roll and re-roll once. What is the optimal policy and the value of the game?
(b) With the right to re-roll up to two times?
(c) What is the single re-roll option itself worth? And if using the re-roll costs a fee of \$0.50 payable at the moment you use it, what is the game worth — is it simply the answer to (a) minus \$0.50?

P1.17 *First 6 given all rolls even*

A fair die is rolled repeatedly until the first 6 appears. You are told that every roll in the sequence — including the final 6 — came up even. Given this, what is the expected number of rolls? Most candidates instantly answer 3, reasoning that the conditioning "reduces the die to the three even faces $\{2, 4, 6\}$", making the wait geometric with $p = 1/3$. Give the correct answer and explain precisely why the instant answer is wrong.

P1.18 *Hat-check fixed points*

At the end of a conference dinner, the coat-check tickets of n traders are shuffled and handed back uniformly at random, one ticket each. Let X be the number of traders who receive their own ticket.

(a) Compute $\mathbb{E}[X]$.
(b) Compute $\mathrm{Var}(X)$ for $n \geq 2$.
(c) For large n, a colleague offers you an even-money bet that *nobody* gets their own ticket back. Which side do you take, and what is your edge?

P1.19 *Random chord longer than the radius* ♦

Two points are chosen independently and uniformly at random on a circle. What is the probability

that the length of the chord joining them is larger than the circle's radius?

(A) $1/\pi$
(B) $2/3$
(C) $1/2$
(D) $3/4$
(E) $\pi/4$

P1.20 *Max of four dice, exactly* ♦

You roll 4 fair six-sided dice and keep the highest value. What is the probability that the highest value is exactly 3?

(A) 80/1296
(B) 1/81
(C) 65/1296
(D) 1/16

27.2 Problems — §2 Bayesian Updating and Reading Flow

P2.1 *Bayes odds form, base rates* ♦

In a large population, 0.1% of people are infected with a disease. A test is positive for 100% of infected people and for 10% of uninfected people. A randomly chosen person tests positive. What is the probability that they are infected? Give the exact fraction, then an approximation you could say out loud.

P2.2 *Conditioning on the decisive event* ♦

The probability I beat my arch-nemesis at chess is 0.4, the probability they beat me is 0.2, and the probability we draw is 0.4. If we draw, we replay, and keep replaying until somebody wins. What is the probability that I win overall?

P2.3 *Odds form, prior vs evidence* ♦

A gag coin with heads on both sides is dropped into a bag containing 99 fair coins. One coin is chosen at random from the hundred and flipped 5 times, coming up heads all 5 times. What is the probability that the gag coin was chosen?

P2.4 *Sequential likelihood ratios*

A bag holds two dice. One is fair; the other is loaded so that a six appears with probability 1/2 (the other five faces equally likely). You pick one die at random and start rolling it.

(a) The first roll is a six. What is the probability you hold the loaded die?
(b) The first three rolls are all sixes. Now what is it?
(c) How many consecutive sixes would you need to be at least 99% sure?
(d) After three sixes your fourth roll is a two. Does your probability stay where it was, and if not, where does it go?

P2.5 *Correlated evidence*

You make markets in a mid-cap altcoin perp. Historically 10% of the accounts that trade aggressively against you are informed. Each aggression from an informed account lifts your offer (rather than hitting your bid) with probability 0.8; from a noise account, with probability 0.2. Your junior sees the offer lifted three times within one second and computes: prior odds $1:9$, likelihood ratio $(0.8/0.2)^3 = 64$, posterior odds $64:9$, hence $P(\text{informed}) \approx 88\%$ — and pulls

every quote. The execution log then shows that all three fills were child slices of a single iceberg order worked by one account. What is wrong with the 88%, and what is a defensible update?

P2.6 *Flow reading, sequential update to a threshold*

You quote both sides of a crypto perp. Base rate: 5% of aggressive counterparties are informed buyers; the rest are noise. Each aggression from an informed buyer lifts your offer (rather than hitting your bid) with probability 0.9; a noise account lifts or hits with probability 0.5 each. Aggressions are conditionally independent given the type.

(a) One account lifts your offer twice in a row. What is the probability you are facing an informed buyer?
(b) After how many consecutive lifts does the posterior first exceed 90%?
(c) As the posterior climbs, what is the correct quoting response — and what exactly is wrong with "keep filling them at the same offer"?

27.3 Problems — §3 Betting, Bankroll, Ruin and Markov Chains

P3.1 *Kelly criterion*

You can bet any fraction of your bankroll, repeatedly, on independent even-money flips of a coin that lands your way with probability $p = 0.55$.

(a) For the general bet paying b-to-1 with win probability p, derive the fraction f^* that maximises the long-run growth rate, and check it is a maximum.
(b) Evaluate f^* for the coin above, and for a bet paying 2-to-1 that wins 40% of the time.
(c) For the coin, compute the growth rate at f^* and at $2f^*$. What is the moral?

P3.2 *Kelly under estimation error*

A junior trader backtests a signal over 25 independent trades and wins 15 of them ($\hat{p} = 0.60$, even-money payoffs). He computes full Kelly $f^* = 2\hat{p} - 1 = 20\%$ and starts betting a fifth of the book on every trade.

(a) What is wrong, before any further calculation?
(b) Suppose the true win probability is 0.55. Compute the growth rate of his 20% sizing and compare it with correct Kelly sizing and with betting half of his *estimated* Kelly.
(c) Quantify the noise in $\hat{p}$ itself. What sizing discipline does this justify?

P3.3 *Gambler's ruin, fair case*

You and a rival flip a fair coin for \$1 a flip; you start with \$30, they start with \$70, and you play until one of you is broke.

(a) What is the probability you end up with all \$100?
(b) What is the expected number of flips?
(c) You switch to \$5 per flip. What changes?

P3.4 *Gambler's ruin, biased*

(a) At a casino you win each \$1 round with probability 0.49 and lose it with probability 0.51. You start with \$50 and commit to playing until you either reach \$100 or go broke. Estimate your probability of reaching \$100, cleanly enough to do at a whiteboard.
(b) A market-making strategy earns $+1$ tick with probability 0.51 and -1 tick with probability 0.49 on each trade, and trades forever — there is no upper target. Starting with a 20-tick

capital buffer, what is the probability it ever goes bust? What buffer keeps lifetime ruin below 1%?

P3.5 *First-step analysis / ruin reduction*

You win each point against an opponent independently with probability 0.6. Starting level, the match continues until one player leads by two points; that player wins (deuce rules). What is your probability of winning the match? Give two independent derivations, and comment on why the answer is not 0.6.

P3.6 *First-step analysis, pattern race*

A fair coin is flipped repeatedly. What is the probability that the pattern HH appears before the pattern TH? Answer with a structural argument, verify by first-step analysis, and explain why comparing the two patterns' expected waiting times (6 flips for HH, 4 for TH) does not answer the question.

P3.7 *Stationary distribution, two-state chain*

A market is either trending or choppy day to day. From trending it switches to choppy with probability 0.1; from choppy it switches to trending with probability 0.3; otherwise it stays where it is.

(a) In the long run, what fraction of days are trending?
(b) Today is trending. What is the expected number of days until the *next* trending day?
(c) Verify your stationary vector is actually stationary.

P3.8 *Stationary distribution on a graph, return time*

A knight sits on a corner square of an otherwise empty chessboard and makes a uniformly random legal knight move at each step, forever. What is the expected number of moves until it first returns to its starting corner? (No 64-state linear system allowed — find the structure.)

P3.9 *St. Petersburg, capped EV*

A casino flips a fair coin until the first head; if the first head arrives on flip n, you are paid 2^n dollars.

(a) Show the expected payoff is infinite.
(b) The casino's bankroll actually caps the payout at 2^{20} dollars (about a million). What is the fair price now?
(c) The interviewer pushes: "so how much would you personally pay to play once?" Name the considerations that justify an answer far below even the capped expectation.

P3.10 *Utility shape, risk aversion* ♦

What shape should the utility function of a rational, risk-averse investor take? State the two defining properties, justify each, and give one consequence of each: why does such an investor decline a fair \$100 coin flip, and what does this shape have to do with Kelly betting and the St. Petersburg paradox?

27.4 Problems — §4 Statistics I: Estimators and Inference

P4.1 *Second moment of the standard normal* ♦

Let X be a standard normal random variable. What is $E[X^2]$?

P4.2 *Bias-variance-MSE trade-off in variance estimation*

Let $X_1, \ldots, X_n$ be i.i.d. $\mathcal{N}(\mu, \sigma^2)$ with both parameters unknown, and let $S = \sum_{i=1}^{n}(X_i - \bar{X})^2$.

Consider the family of variance estimators $\hat{\sigma}_c^2 = S/c$. Two standard choices are $c = n-1$ (unbiased) and $c = n$ (the maximum-likelihood estimator). Which value of c minimises the mean squared error $E[(\hat{\sigma}_c^2 - \sigma^2)^2]$? Rank the three candidate divisors by MSE, and state the moral about unbiasedness.

P4.3 *Bessel's correction and the bias of s*

You estimate the variance of a strategy's daily returns from an i.i.d. sample $X_1, \ldots, X_n$ with unknown mean μ and variance σ^2.

(a) Show that $E\big[\sum_{i=1}^{n}(X_i - \bar{X})^2\big] = (n-1)\sigma^2$, so the unbiased variance estimator divides by $n-1$.

(b) If instead the true mean μ were known and you centred the deviations on μ, what divisor makes the estimator unbiased?

(c) Your colleague concludes that $s = \sqrt{s^2}$, with s^2 the Bessel-corrected estimator from (a), is therefore an unbiased estimator of the volatility σ. Is that right?

P4.4 *Sum of i.i.d. normals* ♦

For independent random variables $X_1, \ldots, X_k$, each distributed $\mathcal{N}(m, s^2)$, what is the distribution of the sum $X_1 + \cdots + X_k$?

P4.5 *Sum of normals without independence* ♦

One random variable is distributed $\mathcal{N}(5, 4)$ and another is distributed $\mathcal{N}(3, 9)$ (means 5 and 3, variances 4 and 9). What is the distribution of their sum?

P4.6 *Distribution recognition (binomial)* ♦

Every IOC (immediate-or-cancel) buy order placed at the best offer on the Coinbase SOL/EUR order book fills with probability 0.7, independently of all other orders. If 50 such IOC orders are sent, what is the distribution of the number of orders filled?

P4.7 *Normal approximation to the binomial* ♦

What Normal distribution should you use to approximate a Binomial($n = 100,\ p = 0.2$) distribution?

P4.8 *CLT - what it does not say*

A junior quant on your desk argues: "Our daily P&L is fat-tailed, but annual P&L is the sum of 252 daily P&Ls, so by the central limit theorem it is normal, and I have computed our 4-sigma annual loss probability from the normal table." Give at least three distinct reasons this argument fails, and say in which direction the normal-table answer errs.

P4.9 *Confidence interval: variance vs sd* ♦

A variable is normally distributed with mean 1000 and variance 100. What is the approximate 95% confidence interval?

P4.10 *Hypothesis testing recipe*

A strategy was up on 140 of the last 252 trading days, and your PM declares the hit rate "obviously better than a coin flip." Run the test properly: state the null hypothesis, the test statistic and its distribution under the null, compute the two-sided p-value, and give your conclusion at the 5% level. What changes with a one-sided test — and what is the danger of switching to one after seeing the data?

P4.11 *Sharpe ratio standard error (the humility formula)*

A backtest over 3 years of daily data shows an annualised Sharpe ratio of 1.2. Using the i.i.d. standard-error formula $\mathrm{SE}(\widehat{\mathrm{SR}}) \approx \sqrt{(1 + \mathrm{SR}^2/2)/n}$ (with SR and n at the observation frequency),

estimate the standard error of the *annualised* Sharpe estimate. Is the Sharpe statistically distinguishable from 0? From 1? Roughly how many years of daily data would you need to pin the annualised Sharpe down to ± 0.25 (one standard error)?

P4.12 *Multiple testing / deflated-Sharpe logic*

Your team backtested 100 signal variants this quarter. The best one shows a t-statistic of 2.5 on 1000 days of returns, and a colleague notes "$p < 1.3\%$, let's allocate." Suppose, for simplicity, that all 100 backtests are independent and the signals are pure noise. (a) What is the probability that the *best* of the 100 shows $|t| \geq 2.5$? (b) Roughly what t-statistic should the best-of-100 need to clear for 5% family-wise significance? (c) What is the practical recipe when someone shows you the best result of many trials?

27.5 Problems — §5 Regression and Regularisation

P5.1 *OLS univariate slope and R-squared*

Desk quickie, no calculator. You regress daily returns of stock Y on daily returns of stock X (with an intercept). Sample statistics: correlation $\rho = 0.4$, $\sigma_X = 1\%$ daily, $\sigma_Y = 3\%$ daily. (a) What slope does the regression give? (b) What is the R^2? (c) X is up 2% today; ignoring the (tiny) intercept, what move does the regression predict for Y? (d) One sentence: how can the slope exceed 1 while the fit explains so little of Y's variance?

P5.2 *OLS derivation and projection*

From first principles, for $y = X\beta + \varepsilon$ with $X \in \mathbb{R}^{n \times p}$ of full column rank: (a) derive the OLS estimator and confirm you have found a minimum; (b) show the residual vector is orthogonal to every column of X; (c) show the hat matrix $H = X(X^\top X)^{-1}X^\top$ satisfies $H^\top = H$, $H^2 = H$ and $\operatorname{tr}(H) = p$, and interpret the trace; (d) check that your matrix formula reduces to $\widehat{\operatorname{Cov}}(x, y)/\widehat{\operatorname{Var}}(x)$ in the univariate centred case.

P5.3 *Gauss–Markov assumptions and violations*

(a) State the four Gauss–Markov assumptions and the theorem's exact conclusion. Is normality of the errors assumed? (b) For each desk failure below, name the assumption broken (if any), the consequence for OLS, and the standard fix: (i) the "alpha" signal you regress on was built with data that leaks future information; (ii) residual variance triples in high-volatility regimes; (iii) two of your factors have sample correlation 0.98; (iv) residuals are fat-tailed but independent and homoskedastic.

P5.4 *Omitted-variable bias*

The true daily return process of a stock is $y = \beta_1 x_1 + \beta_2 x_2 + \varepsilon$, where x_1 is your new signal and x_2 is the market return, with $\mathbb{E}[\varepsilon \mid x_1, x_2] = 0$. In truth $\beta_1 = 0$ (the signal has no alpha) and $\beta_2 = 1$. Your backtest regresses y on x_1 alone, and in the sample $\widehat{\operatorname{Cov}}(x_1, x_2)/\widehat{\operatorname{Var}}(x_1) = 0.3$. (a) Derive the expectation of the estimated coefficient on x_1 in general. (b) What number does your backtest report as "alpha"? (c) Name the finance mistake and the standard fix.

P5.5 *Gauss–Markov BLUE proof + ridge follow-up*

(a) Prove the Gauss–Markov theorem: under linearity, full rank, strict exogeneity and spherical errors, any estimator $\tilde{\beta} = Cy$ that is linear in y and unbiased for every β satisfies $\operatorname{Var}(\tilde{\beta}) - \operatorname{Var}(\hat{\beta}_{\mathrm{OLS}}) \succeq 0$. (b) The interviewer follows up: "Ridge regularly beats OLS out of sample. Doesn't that contradict the theorem you just proved?" (c) Verify the theorem's content in the simplest case: estimating a mean μ from n i.i.d. observations with variance σ^2, which linear unbiased weighting $\sum_i w_i y_i$ is best, and why?

P5.6 *Ridge derivation, invertibility, shrinkage*

(a) Derive the ridge estimator minimising $\|y - X\beta\|^2 + \lambda\|\beta\|_2^2$ for $\lambda > 0$. (b) Explain why $(X^\top X + \lambda I)$ is always invertible, even when $X^\top X$ is singular (duplicated features, or $p > n$). (c) In the orthonormal-design case $X^\top X = I$, show ridge is pure shrinkage of OLS; if an OLS coefficient is 1.2 and $\lambda = 0.5$, what is the ridge coefficient? (d) What is the Bayesian interpretation of ridge, and how does λ relate to the prior?

P5.7 *Regularisation menu* ★

"Give five examples of regularization in machine learning." Give the five with one line each on the mechanism — and have a second batch ready in case the interviewer pushes with "give me more," including tree-model regularisers and at least one finance-specific example.

P5.8 *Lasso soft-thresholding*

In the orthonormal-design case the lasso objective separates coordinate by coordinate into $\min_b (b - \hat{\beta}_j)^2 + \lambda|b|$, where $\hat{\beta}_j$ is the OLS coefficient. (a) Solve this one-dimensional problem exactly, handling the kink at zero carefully. (b) Apply it with $\lambda = 1$ to OLS coefficients $(2.0,\ 0.6,\ -0.3)$. (c) Explain why no positive λ ever produces an exact zero in ridge, yet lasso does.

P5.9 *Lasso vs ridge in alpha research*

You have 500 candidate alpha signals, many of them highly correlated with one another, and about three years of daily observations ($n \approx 750$) for a cross-sectional return model. (a) Give the geometric picture for why an L1 penalty produces exact zeros while L2 does not. (b) Which penalty do you reach for here, and what is the known failure mode of lasso with groups of correlated features? (c) What does elastic net do, and why is it the pragmatic default in exactly this setting?

P5.10 *Inverse regression*

A colleague regresses Y on X and gets slope 0.5. For a different report he needs the slope of X regressed on Y, and — short on time — quotes $1/0.5 = 2$. The sample correlation between X and Y is 0.5. (a) What exactly is wrong with the shortcut? (b) Given $\sigma_X = \sigma_Y$, compute the actual X-on-Y slope. (c) Prove the general identity linking the two slopes, and state the one case where naive inversion is legitimate.

P5.11 *Hedge-ratio inversion, market application*

Stock S has daily volatility 2%, ETF A has daily volatility 1%, and their daily-return correlation is 0.6. You are long \$1M of S and hedge by shorting A. (a) What notional of A minimises the variance of the hedged position? (b) An intern instead regresses A's returns on S's returns, takes that slope, and inverts it to get the hedge ratio. What notional does he short, and by what factor is he over-hedged? (c) Compute the daily P&L volatility in dollars of the unhedged position, the correctly hedged position, and the intern's position. What is the punchline?

P5.12 *Beta vs correlation*

(a) True or false: "a stock with beta 2 on the index must be highly correlated with the index." If false, construct an explicit counterexample. (b) In hedging, which quantity sizes the hedge and which tells you whether the hedge is worth doing? (c) Quick numbers: a stock has daily vol 10%, the index 1%, correlation 0.2. Compute the stock's beta and the fraction of its variance the index explains — and reconcile the two numbers in one sentence.

P5.13 *Choosing a hedge instrument*

You must hedge a \$1M long position in stock Y (daily vol 4%) with exactly one of two futures: F_1 has correlation 0.9 with Y and daily vol 8%; F_2 has correlation 0.6 with Y and daily vol 1%. (a) For each instrument, compute the minimum-variance hedge ratio (in \$ notional) and the residual daily vol of the hedged position. (b) Which instrument do you choose, and what single quantity

decides it? (c) A desk rule of thumb says "hedge with the higher-beta instrument." Critique it using your numbers.

P5.14 *WLS precision weighting*

You observe the same unknown quantity μ twice with independent noise: $x_1 = \mu + \varepsilon_1$ with $\mathrm{sd}(\varepsilon_1) = 1$, and $x_2 = \mu + \varepsilon_2$ with $\mathrm{sd}(\varepsilon_2) = 2$. (a) Find the best (minimum-variance) linear unbiased combination and its variance — is it better than using the good measurement alone? (b) Show this is exactly weighted least squares, and state the general weighting rule. (c) In a returns regression whose residual variance is four times higher in crisis months, what does unweighted OLS do wrong, and what are the fixes?

P5.15 *Residual weighting vs GLS*

Teammate 1 fits model A to daily returns. Teammate 2 then trains a second model B on the same data using sample weights $w_i = |y_i - \hat{y}_{A,i}|$ and averages the two models' predictions. Teammate 3 objects: "that is just WLS, and WLS weights are supposed to be *inverse*-variance — your weights are upside-down." Adjudicate: what is B actually doing, which classical technique does the scheme resemble, what is the right way to combine the two models, and in what market regime does the scheme backfire?

P5.16 *Dart problem, Gaussian, pitchfork*

You throw a single dart. Aiming at point a on the real line, the dart lands at $a + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$. Two targets sit at x_1 and $x_2 = x_1 + d$, and your expected score is proportional to the landing density summed over the targets:

$$S(a) = e^{-(a-x_1)^2/(2\sigma^2)} + e^{-(a-x_2)^2/(2\sigma^2)}.$$

(a) Show the midpoint $m = (x_1 + x_2)/2$ is a critical point for every d. (b) For which d is the midpoint optimal, and what happens to the optimal aim as d grows past the threshold? (c) Follow-up: "what does the square in the exponent have to do with regression?"

P5.17 *Dart problem, L1, median vs endpoint*

(a) For a random variable X with continuous CDF F, show that c minimises $g(c) = \mathbb{E}|X - c|$ exactly when $F(c) = 1/2$, i.e. c is a median. With an empirical distribution on just two points $x_1 < x_2$, which aims are optimal under absolute loss? (b) Now a dart game with Laplace tails: aiming at a, the expected score contributed by a target at distance r is e^{-r} (distances in units of the Laplace scale). With targets at x_1 and $x_2 = x_1 + d$, so $S(a) = e^{-|a-x_1|} + e^{-|a-x_2|}$, prove the optimal aim is at one of the targets, and compute how much an endpoint beats the midpoint by. (c) Follow-up: "does this remind you of anything in regression?" Give the mapping and the classic conflation to avoid.

27.6 Problems — §6 Linear Algebra, PCA and Covariance

P6.1 *Vector norms / reverse triangle inequality* ♦

Vector A has length 5 and vector B has length 7. What is the lowest possible value of $|A - B|$? Justify your answer and describe the configuration that attains it.

P6.2 *Matrix transformations rapid-fire* ♦

Rapid-fire, multiple-choice pace — one line of justification each.

- **(a)** You want to apply a rotation R to a column vector x, then a shear S to the result. Which expression represents the combined transformation: SR or RS?

- **(b)** The matrix $\begin{pmatrix} \cos t & 0 & \sin t \\ 0 & 1 & 0 \\ -\sin t & 0 & \cos t \end{pmatrix}$ represents a rotation about which axis of $\mathbb{R}^3$?
- **(c)** If the columns of a square matrix A are linearly dependent, what must be true of $\det(A)$?
- **(d)** If A and B are invertible matrices, what is $(AB)^{-1}$?
- **(e)** A 2×2 matrix has determinant -2. Describe accurately the geometric effect of the transformation on the 2D plane.

P6.3 *Eigenvectors of a 2D rotation* ♦

Consider the rotation by 90 degrees counter-clockwise in the standard real plane $\mathbb{R}^2$. What is true of its eigenvectors? (One tempting multiple-choice option read: "exactly one eigenvector, along the axis of rotation.")

P6.4 *Correlation-matrix feasibility*

A risk report shows the following correlations for three strategies A, B, C, each entry estimated pairwise on a *different* (pairwise-complete) sample of dates:

$$\rho_{AB} = 0.9, \qquad \rho_{AC} = -0.9, \qquad \rho_{BC} = 0.8.$$

- **(a)** Show that no joint distribution of (A, B, C) can have this correlation matrix.
- **(b)** Explain how the pairwise estimation procedure can produce such a matrix anyway.
- **(c)** Derive the feasible range of ρ_{BC} given the other two entries, and name two standard repairs used in practice.

P6.5 *PCA derivation and explained variance*

(a) A dataset is centred and has sample covariance matrix Σ. Derive the direction w with $\|w\| = 1$ that maximises the variance of the projected data, and identify the maximised value. **(b)** For $\Sigma = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, find both principal directions and the fraction of total variance explained by the first principal component. **(c)** If you skip the centring step and the data mean is far from the origin, what does the leading "principal component" of $X^\top X$ actually capture?

P6.6 *Equicorrelated covariance / PC1 as market mode*

A universe of $n = 100$ stocks has, for simplicity, identical variance σ^2 and identical pairwise correlation $\rho = 0.3$, so the covariance matrix is

$$\Sigma = \sigma^2[(1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top],$$

where $\mathbf{1}$ is the all-ones vector. **(a)** Find all eigenvalues of Σ with their multiplicities (hint: $\mathbf{1}\mathbf{1}^\top$ has rank one). **(b)** Identify PC1 and compute the fraction of total variance it explains for the given numbers. **(c)** Use the result to justify the claim "PC1 is the market mode," and state the consequence for a signal whose weights do not sum to zero.

P6.7 *PCA via the SVD / conditioning*

Let X be an $n \times p$ centred data matrix with thin SVD $X = USV^\top$. **(a)** Show that the principal directions of the data are the columns of V, and express the sample variance along the i-th direction in terms of the singular value s_i (use the $1/(n-1)$ convention). **(b)** With $n = 5$ observations and singular values $s_1 = 4$, $s_2 = 2$, give the variance along each principal direction and the share of variance explained by PC1. **(c)** Your library offers both an SVD of X and an eigendecomposition of $X^\top X$. Which do you call for PCA, and why?

P6.8 *Gram-Schmidt / orthogonalising correlated signals*

Your desk runs a production signal A that you cannot modify. Your candidate signal B has correlation $\rho = 0.8$ with A; both are demeaned and scaled to unit variance. **(a)** Write the

orthogonalised signal $B_\perp$ via the projection formula and verify $\mathrm{Corr}(A, B_\perp) = 0$. **(b)** Compute $\mathrm{Var}(B_\perp)$. What fraction of B's variance was already contained in A? **(c)** Why must the signals be demeaned for "orthogonal" to mean "uncorrelated"? **(d)** What does $B_\perp$ represent economically, and when is trading $B_\perp$ rather than raw B the right call?

P6.9 *Beta neutralisation via PC1*

You run PCA on the returns of three assets and find the first principal component is proportional to $M = (1, 1, 1)^\top$ — the market mode. Your raw signal is $S = (2, -1, 1)^\top$. **(a)** Beta-neutralise S by projecting off M: compute $S' = S - \frac{\langle S, M\rangle}{\langle M, M\rangle} M$. **(b)** Verify neutrality and interpret the property the new weights satisfy. **(c)** Show that when the market mode has exactly equal weights this projection coincides with a familiar simpler operation, and explain why the two differ for a real, estimated PC1.

P6.10 *A-orthogonal matrices* ♦

Define $\langle x, y\rangle_A = x^\top A y$, where A is symmetric positive definite. A matrix Q is *A-orthogonal* if $\langle Qx, Qy\rangle_A = \langle x, y\rangle_A$ for all x, y. Take $A = \mathrm{diag}(2, 3, 5)$. Which of the following is A-orthogonal?

(A) the permutation swapping $e_1 \leftrightarrow e_3$
(B) $Q = \mathrm{diag}(1, -1, 1)$
(C) the permutation swapping $e_1 \leftrightarrow e_2$
(D) the rotation by angle t in the x–y plane

27.7 Problems — §7 Calculus for Quant Interviews

P7.1 *Derivatives toolkit: power rule* ♦

What is the derivative of $\sqrt{x}$ with respect to x?

P7.2 *Derivatives toolkit: chain-rule cycle* ♦

What is the fourth derivative of $\sin(4x)$?

P7.3 *Derivatives toolkit: logistic identity*

A prediction-market contract's price is modelled as $p(x) = \dfrac{1}{1 + e^{-x}}$, where x is the log-odds implied by the news flow. (a) Show that $\dfrac{dp}{dx} = p(1 - p)$. (b) The contract currently trades at 80 cents; by how much does the price move per small unit of log-odds? (c) At what price is the contract most sensitive to news, and why does a trader care?

P7.4 *Derivatives toolkit: optimisation recipe*

You are quoting a two-sided market and must choose your half-spread $s > 0$ (in ticks). Wider quotes earn more per fill but are filled less often: model the fill probability as $e^{-s/2}$, so the expected profit per quote is $f(s) = s\, e^{-s/2}$. Find the half-spread that maximises expected profit, justify that it is a maximum, and report the maximum value.

P7.5 *Integration by parts* ♦

Compute the indefinite integral $\displaystyle\int \ln x \, dx$.

P7.6 *Integration: substitution to Gamma + symmetry caveat*

(a) Evaluate $\displaystyle\int_0^\infty x^3 e^{-x^2} dx$. (b) Evaluate $\displaystyle\int_{-\infty}^\infty x^3 e^{-x^2} dx$, and explain why the symmetry argument you use is legitimate here but would *not* justify writing $\int_{-\infty}^{\infty} x\, dx = 0$.

P7.7 *Improper integrals: p-tests, comparison, log refinement*

For each integral, state *converge* or *diverge* with a one-line justification (verdict plus one line is full credit):

- (a) $\displaystyle\int_1^\infty \frac{dx}{x^{3/2}}$
- (b) $\displaystyle\int_0^1 \frac{dx}{\sqrt{1-x}}$
- (c) $\displaystyle\int_2^\infty \frac{dx}{x\ln x}$
- (d) $\displaystyle\int_1^\infty \frac{2+\sin x}{x}\,dx$
- (e) $\displaystyle\int_0^\infty x^7 e^{-x/10}\,dx$

P7.8 *Improper integrals: interior singularity*

A candidate writes

$$\int_{-1}^{1} \frac{dx}{x^2} = \Big[-\frac{1}{x}\Big]_{-1}^{1} = (-1) - (1) = -2.$$

The interviewer raises an eyebrow. What is wrong, and what is the correct conclusion?

P7.9 *Improper integrals: combined p-tests via limit comparison*

Find all real values of p for which

$$I(p) = \int_0^\infty \frac{dx}{x^p(1+x)}$$

converges.

P7.10 *Moments as integrals: normalisation via Gamma*

A nonnegative random variable X has density $f(x) = c\,x e^{-x/2}$ for $x \geq 0$ (and 0 otherwise). Find c, $\mathbb{E}[X]$, and $\mathrm{Var}(X)$.

P7.11 *Moments as integrals: Gaussian fourth moment*

Daily returns on a book are modelled as $X \sim \mathcal{N}(0, \sigma^2)$. (a) Derive $\mathbb{E}[X^4]$ from the integral definition. (b) On real daily returns your desk measures $\widehat{\mathbb{E}[X^4]}\,/\,\hat{\sigma}^4 \approx 9$. What does the mismatch tell you, and give one practical consequence for risk management.

P7.12 *Taylor micro-kit: limit*

Compute $\displaystyle\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$.

P7.13 *Taylor micro-kit: compounding estimate*

Without a calculator, estimate $(1.02)^{30}$ — e.g. the growth factor of 30 periods of a steady 2% return. State the approximation you use and how you would sanity-check the result.

P7.14 *Change of variables in an ODE* ♦

Suppose $dx/dt = x$. Define $\tau = 1/t$ (for $t > 0$). Which equation does $x(\tau)$ satisfy?

(A) $dx/d\tau = -x$
(B) $dx/d\tau = -x/\tau$
(C) $dx/d\tau = -x/\tau^2$
(D) $dx/d\tau = \tau x$

P7.15 *Differentiate a parameter-dependent integral* ♦

Let $F(x) = \int_x^{2x} s\,x\,ds$ (the integration variable is s; x appears both in the limits and in the integrand). Compute $F'(x)$.

(A) $2x\int_0^x s\,ds$
(B) $\frac{9}{2}x^2$
(C) x^2
(D) $x^2/2$

P7.16 *Probability by integration* ♦

Let $X, Y \sim \text{Uniform}(0,1)$ be i.i.d. What is $P(X^2 > Y)$?

(A) $1/2$
(B) $1/\pi$
(C) $1/3$
(D) $1/4$
(E) $\pi/4$

P7.17 *Newton's method: the slowest root* ♦

We use Newton iteration to find the roots of $f(x) = \tanh(x^4 - x^2)$. Assuming correct initialization near each root, which root has the slowest convergence?

(A) The largest one.
(B) Newton's method always converges quadratically.
(C) The smallest one.
(D) The one with the smallest absolute value.

27.8 Problems — §8 ML Theory: Trees, Ensembles, Bias–Variance

P8.1 *Bias-variance decomposition*

Let $y = f(x) + \varepsilon$ with $\mathbb{E}[\varepsilon] = 0$, $\text{Var}(\varepsilon) = \sigma^2$, and ε independent of the training data. A learning algorithm trained on a random training set D produces a predictor $\hat{f}_D$. At a fixed test point x_0, derive the decomposition of the expected squared prediction error $\mathbb{E}_{D,\varepsilon}[(y - \hat{f}_D(x_0))^2]$ into bias squared, variance, and irreducible noise, justifying precisely why each cross term vanishes. Then: if you collect more training data (same model class), which of the three terms shrink?

P8.2 *Decision trees: Gini and information gain*

A classification node holds 16 samples: 8 positive, 8 negative. A candidate split sends 8 samples left (6 positive, 2 negative) and 8 right (2 positive, 6 negative). (a) Compute the Gini impurity of the parent, the weighted Gini after the split, and the impurity decrease. (b) Compute the information gain of the split in bits. (c) Which of these numbers change if you first apply the monotone transform $x \mapsto \log x$ to the split feature?

P8.3 *Trees cannot extrapolate*

You train a gradient-boosted tree model on 2015–2019 data to predict next-day realised volatility of an equity index. One feature is the VIX level, which ranged between 10 and 30 in the training window. In March 2020 the VIX prints 80. Roughly what will the model output, and why? What would a linear regression trained on the same data do differently? Name two feature-engineering changes that would have made the tree model behave better out of range.

P8.4 *Ensemble variance formula and the correlation floor*

N models produce predictions $X_1, \ldots, X_N$, each with variance σ^2 and common pairwise correlation ρ. (a) Derive $\mathrm{Var}\left(\frac{1}{N}\sum_i X_i\right)$. (b) Take $N \to \infty$ and explain what the limit means for random forests — in particular, why do they subsample *features at every split* rather than only bootstrapping rows? (c) How negative can ρ be for the covariance structure to remain valid, and what is the ensemble variance at that boundary? (d) With $\sigma^2 = 1$ and $\rho = 0.2$, how many models are needed for the ensemble variance to be within 0.05 of its large-N floor?

P8.5 *Bootstrap and out-of-bag*

A bootstrap sample draws n times with replacement from n training points. (a) What is the probability that a given training point never appears in the sample, and what is its limit for large n? (b) On average, what fraction of *distinct* training points does each tree in a bagged ensemble see? (c) What practical free lunch does this give random forests?

P8.6 *Boosting as functional gradient descent*

Gradient boosting builds $F_t = F_{t-1} + \eta\, h_t$. (a) For squared loss $L(y, F) = \frac{1}{2}(y - F)^2$, show that the negative functional gradient evaluated at the current model is exactly the residual — so "fit the next tree to the residuals" is literally gradient descent in function space. (b) What does the fitting target become under absolute loss $L = |y - F|$? (c) Boosting and bagging both combine many trees: contrast which error component each attacks, and list the knobs that control overfitting in boosting. Does adding more trees eventually overfit in each case?

P8.7 *Good features for tree models* ★

What makes a good feature for LightGBM — or for tree-based models in general?

27.9 Problems — §9 Deep Learning

P9.1 *Backprop mechanics and vanishing gradients*

Consider a deep scalar chain $a_\ell = \sigma(w_\ell\, a_{\ell-1})$ for $\ell = 1, \ldots, L$ with input a_0, $|a_0| \le 1$, and sigmoid activation σ. (a) Backpropagation and finite differences both produce gradients: what exactly does backprop reuse that makes it cheap, and roughly what would finite differences cost per gradient step for a network with 10^7 parameters and a 10 ms forward pass? (b) Write $\partial a_L / \partial w_1$ as a product and show why it decays exponentially in depth for sigmoid activations; give a numeric bound for $L = 20$ when every $|w_\ell| \le 1$. (c) For each of ReLU, residual connections, LSTM gating, and batch normalisation, state in one line which factor of that product it repairs.

P9.2 *Self-attention and the sqrt(dk) scaling*

Scaled dot-product attention computes $\mathrm{softmax}(QK^\top/\sqrt{d_k})V$. (a) Assuming query and key components are independent with zero mean and unit variance, compute $\mathrm{Var}(q \cdot k)$. (b) For $d_k = 64$, what is the standard deviation of an *unscaled* score, and what does softmax do to logits at that scale — give a concrete two-logit example? (c) Why does that kill training — what happens to the gradient through the softmax? (d) What property does dividing by $\sqrt{d_k}$ restore?

P9.3 *Transformers on tick data*

A desk proposes running a transformer directly on raw ticks: a liquid instrument prints about 500,000 ticks per day and they want a full-day context window. (a) How many entries does the attention score matrix have per head per layer, and how much memory is that at 2 bytes per entry? (b) Someone suggests FlashAttention as the fix — what does it actually fix, and what does it not fix? (c) In live trading a new tick arrives and you want an updated prediction: compare the per-tick inference cost shape of the transformer (with a KV cache) against a state-space model such as S4/Mamba. (d) What would a pragmatic first system do instead?

P9.4 *The model stopped working*

A systematic strategy performed well from 2005 to 2018, then decayed to zero and stayed there. Give five distinct hypotheses for what happened, and for each one a concrete diagnostic you would run. Which two hypotheses do you check first, and why?

P9.5 *Numpy vs lists, and why PyTorch exists* ★

Why are numpy arrays faster than Python lists? And given that, why doesn't PyTorch just use numpy arrays by default?

P9.6 *GPU vs CPU speedup* ★

What order of magnitude of speedup do you get from GPU versus CPU parallelisation? When does that answer change?

P9.7 *Python class variables* ♦

In Python, what is true of a class (static) variable?

P9.8 *Training bottleneck diagnosis* ♦

You are training a model and observe low GPU and CPU utilisation while the hard drive runs at 100% capacity reading data files. What standard term describes this system state?

P9.9 *Fail fast vs fail silent* ♦

A data pipeline takes 3 days to run. If it detects an unexpected null/NaN in a critical feature column during the first hour, why is it generally better to crash immediately ("fail loudly") rather than skip the bad data and continue ("fail silently")?

P9.10 *Immutability and concurrency* ♦

Functional programming encourages immutability — data cannot be changed once created. Why is this property highly valuable for multi-threaded systems?

P9.11 *Shared-node OOM forensics* ♦

A job on a shared cluster node runs without issue, but neighbouring jobs on the same node crash with out-of-memory (OOM) errors. What is the most likely cause?

27.10 Problems — §10 Sharpe, Loss Probabilities and Model Combination

P10.1 *P(loss) = Phi(-SR) and time scaling*

A strategy's annual returns are normally distributed with annualised Sharpe ratio 1.5 (take the mean/vol ratio directly, no risk-free adjustment). (a) What is the probability of a losing year? (b) Assuming returns are i.i.d. across time, what is the probability of a losing month? (c) Why does the monthly number look so much worse when the strategy has not changed?

P10.2 *Sharpe to loss probability with assumption demolition* ★

A strategy has an annual Sharpe ratio of 2 and annual P&L volatility of \$50M. What is the probability of losing \$100M? State your assumptions and justify them. Is the true probability higher or lower in reality?

P10.3 *Cantelli's inequality: proof, application, tightness*

(a) Prove Cantelli's inequality (the one-sided Chebyshev bound): for any random variable X with mean μ and variance σ^2, and any $k > 0$,

$$P(X - \mu \leq -k\sigma) \;\leq\; \frac{1}{1+k^2}.$$

(b) A strategy has annual mean P&L \$100M and volatility \$50M, distribution otherwise unknown. Give the best distribution-free bound on the probability of losing at least \$100M in a year, and compare it with the Gaussian answer. (c) Show the bound is tight by exhibiting a two-point distribution that achieves it.

P10.4 *Sharpe vs EV at maximum capacity* ♦

Assuming all strategies are at maximum capacity (none can be scaled up), which strategy would a market-making firm most prefer?

(A) Expected value \$5,000/day with standard deviation \$1/day.
(B) Expected value \$20,000/day with standard deviation \$30,000/day.
(C) Expected value \$50,000/day with standard deviation \$100,000/day.
(D) Expected value \$100,000/day with standard deviation \$400,000/day.

P10.5 *Var(A+B) and the Sharpe of combined strategies*

Strategies A and B each have annualised expected P&L \$10M and annualised volatility \$10M (Sharpe 1 each). You run both at full size, so the book's P&L is $A + B$.

(a) If $\mathrm{Corr}(A, B) = 0.5$, what are the book's volatility and Sharpe?
(b) If $\mathrm{Corr}(A, B) = 0$?
(c) If $\mathrm{Corr}(A, B) = -0.3$?
(d) Derive the rule for N uncorrelated strategies with equal Sharpe and equal vol, and explain why the general claim $\mathrm{SR}_{\text{total}} = \sqrt{\sum_i \mathrm{SR}_i^2}$ requires risk-balanced weights rather than a raw sum.

P10.6 *Decorrelating a new model from a black box*

Your desk already runs a profitable black-box model A — you can observe its outputs and P&L but not its internals. You have built a new model B whose output correlates 0.7 with A's. (a) A colleague proposes: "drop every feature from B's training set that is correlated with A's output, then retrain." Give two reasons this is a poor primary approach. (b) What would you do instead? Give the one-line exact method and one training-time alternative, and state the trade-off in each case.

27.11 Problems — §11 Portfolio Theory: Mean–Variance Optimisation, Diversification, and the Capital Market Line

P11.1 *Two-asset diversification: the same-mean, triple-volatility asset*

Asset 1 has expected return 4% and volatility 10%; asset 2 has expected return 10% and volatility 30%; their correlation is $\rho = 0.2$. All portfolios are fully invested (weights sum to one). **(a)** Compute the expected return and the volatility of the 50/50 mix, and compare the volatility with the weighted average of the two volatilities. **(b)** Find the weight w in asset 1 that minimises the portfolio variance, the minimum volatility, and the expected return of that portfolio. **(c)** For which values of ρ does the minimum-variance portfolio hold a strictly positive amount of asset 2? At $\rho = 0.5$, what is the long-only minimum-variance portfolio? **(d)** A colleague objects that asset 2 has three times the volatility of asset 1, so no risk-minimising investor should touch it. Is the objection right at $\rho = 0.2$? Does the minimum-variance mix earn less than asset 1 alone?

P11.2 *Eight portfolio claims: true or false, with a counterexample*

Rapid-fire. Decide whether each claim is true or false; for every false claim give a one-line numeric counterexample or the corrected statement.

(i) Diversifying across assets lowers the portfolio's expected return along with its variance.

(ii) Diversification only works when the assets are negatively correlated.

(iii) With enough assets, the variance of an equal-weight portfolio can be driven as close to zero as desired.

(iv) Two uncorrelated strategies with Sharpe ratio 1 each can be combined into a strategy with Sharpe ratio 2.

(v) The minimum-variance portfolio is the maximum-Sharpe portfolio.

(vi) Beta is the correlation of an asset with the market.

(vii) A 60/40 equity/bond portfolio is balanced because its capital is roughly evenly split.

(viii) Once the covariance matrix has been shrunk, plugging sample means into the mean–variance optimiser is safe.

P11.3 *The global minimum-variance portfolio from scratch*

Let Σ be a symmetric positive-definite $n \times n$ covariance matrix and $\mathbf{1}$ the all-ones vector. **(a)** Show that the problem $\min_w w^\top \Sigma w$ subject to $\mathbf{1}^\top w = 1$ has the unique solution $w_g = \Sigma^{-1}\mathbf{1}/A$ with $A = \mathbf{1}^\top \Sigma^{-1}\mathbf{1}$, that the minimum variance is $1/A$, and that every asset has the same covariance $1/A$ with the optimal portfolio. Locate the candidate with a Lagrangian, then prove that it is the global minimum (stationarity alone is not a proof). **(b)** Apply the result to three assets with

$$\Sigma = \begin{pmatrix} 0.01 & 0 & 0 \\ 0 & 0.04 & 0.02 \\ 0 & 0.02 & 0.02 \end{pmatrix}$$

(volatilities 10%, 20%, 14.1%; asset 1 uncorrelated with the others): give the weights, the minimum volatility, and verify the equal-covariance property. **(c)** Explain in one sentence, using the covariance structure, why one asset receives zero weight.

P11.4 *The diversification floor*

A universe of n stocks has, for simplicity, a common volatility of 25% and a common pairwise correlation $\rho = 0.2$. **(a)** Derive the variance of the equal-weight portfolio as a function of n and identify its limit as $n \to \infty$. **(b)** Give the equal-weight volatility for $n = 1, 5, 20$ and in the limit. **(c)** How many stocks are needed before the diversifiable (idiosyncratic) part of the variance is one tenth of the undiversifiable part? **(d)** An ensemble of n regression models has prediction errors with common variance v and common pairwise correlation ρ. What is the variance of the averaged prediction, and why is this the same computation?

P11.5 *Frontier constants and the hyperbola of a three-asset universe*

Three uncorrelated assets have variances 0.04, 0.04, 0.02 (volatilities 20%, 20%, 14.1%) and expected returns 5%, 9%, 11%. Write Σ for the (diagonal) covariance matrix, μ for the mean vector, $\mathbf{1}$ for the all-ones vector, and

$$A = \mathbf{1}^\top \Sigma^{-1}\mathbf{1}, \quad B = \mathbf{1}^\top \Sigma^{-1}\mu, \quad C = \mu^\top \Sigma^{-1}\mu, \quad D = AC - B^2.$$

(a) Compute A, B, C, D. **(b)** Give the global minimum-variance portfolio, its expected return and its volatility. **(c)** The frontier portfolio at target mean m (minimum variance subject to $\mathbf{1}^\top w = 1$ and $\mu^\top w = m$) is $w(m) = \theta_1 \Sigma^{-1}\mathbf{1} + \theta_2 \Sigma^{-1}\mu$ with $\theta_1 = (C - Bm)/D$ and $\theta_2 = (Am - B)/D$, and its variance is $(Am^2 - 2Bm + C)/D$. Compute $w(6\%)$ and $w(12\%)$ and their volatilities, and say which of the two is efficient. **(d)** Write the frontier as a curve in the (σ, m) plane: give its vertex and the slope of its asymptotes.

P11.6 *Two-fund separation in action*

Three uncorrelated assets have variances 0.04, 0.04, 0.02 and expected returns 5%, 9%, 11%. Two frontier portfolios (minimum variance at their target means) are known:

$$w(6\%) = (0.75,\ 0.25,\ 0)^\top, \qquad w(12\%) = (-0.25,\ 0.25,\ 1.00)^\top,$$

and both have variance 0.025. **(a)** Without setting up any optimisation, find the frontier portfolios with target means 10% and 3%. **(b)** Deduce the expected return of the global minimum-variance portfolio, then the portfolio itself and its variance. **(c)** Asset 2 has weight 0.25 in both given portfolios. Show that it has weight 0.25 in every frontier portfolio and explain why. **(d)** A colleague proposes to span the frontier with two "funds": the 50/50 mix of assets 1 and 2, and asset 3 alone. Show that this fails, and identify the single frontier portfolio the pair does reach.

P11.7 *Combining uncorrelated strategies, and the correlation hurdle*

A desk runs three uncorrelated, self-financing strategies (take $r_f = 0$) with annual Sharpe ratios 0.6, 0.8, 1.2 and annual volatilities 8%, 16%, 4%. **(a)** What is the best achievable Sharpe ratio of a combination, and what capital weights (summing to one) achieve it? Express the answer also as risk budgets $w_i\sigma_i$. **(b)** What Sharpe ratio do equal capital weights achieve? **(c)** The combined book of (a), with Sharpe ratio SR_1, is now offered a fourth strategy with Sharpe ratio 0.6 and correlation ρ with the book. For $\rho = 0.5$ and for $\rho = 0.2$, should the desk add it long, and what is the best Sharpe ratio of book plus strategy?

P11.8 *Maximum Sharpe ratio by Cauchy–Schwarz*

Let Σ be a symmetric positive-definite $n \times n$ covariance matrix and $\mu_e \neq 0$ the vector of expected excess returns. For a position w with $w^\top \Sigma w > 0$ define $\mathrm{SR}(w) = w^\top \mu_e / \sqrt{w^\top \Sigma w}$. **(a)** Prove that $\mathrm{SR}(w) \le \sqrt{\mu_e^\top \Sigma^{-1} \mu_e}$ for every such w, with equality if and only if w is a positive multiple of $\Sigma^{-1}\mu_e$. (You may use that $\langle x, y \rangle_\Sigma = x^\top \Sigma y$ is an inner product.) **(b)** Deduce: if Σ is diagonal with entries σ_i^2 and $\mathrm{SR}_i = \mu_{e,i}/\sigma_i$, then the maximal squared Sharpe ratio is $\sum_i \mathrm{SR}_i^2$, attained at risk budgets $\sigma_i w_i \propto \mathrm{SR}_i$; and two uncorrelated legs with equal Sharpe ratio combine to $\sqrt{2}$ times it. **(c)** Two strategies have Sharpe ratios 1.0 and 0.5, volatilities 10% and 20%, correlation 0.9, and $r_f = 0$. Find the maximal Sharpe ratio and the sign of the optimal position in the weaker strategy. Explain why the answer exceeds the uncorrelated value $\sqrt{1.25} = 1.118$.

P11.9 *Tangency, the capital market line and zero alphas*

Three uncorrelated assets have variances 0.04, 0.04, 0.02 and expected returns 5%, 9%, 11%; the risk-free rate is $r_f = 1\%$, and borrowing at r_f is allowed. (For this universe $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1} = 100$, $B = \mathbf{1}^\top \Sigma^{-1} \mu = 9$ and $C = \mu^\top \Sigma^{-1} \mu = 0.87$.) **(a)** Find the tangency (maximum-Sharpe) portfolio, its expected return, volatility and Sharpe ratio. **(b)** Write the equation of the capital market line. What expected return does it deliver at 10% volatility, and how is that point held? Compare with the minimum-variance portfolio $(0.25, 0.25, 0.5)$, which has volatility 10% and mean 9%. **(c)** Compute the beta of each asset with respect to the tangency portfolio and show that every asset's expected excess return equals its beta times the tangency portfolio's expected excess return. **(d)** Why is the Sharpe ratio of the minimum-variance portfolio lower than that of the tangency portfolio, and for which μ would the two portfolios coincide? **(e)** Suppose the risk-free rate rose to 9% with μ and Σ unchanged. What happens to the tangency portfolio?

P11.10 *Beta, correlation and the hedge ratio*

The market has volatility $\sigma_M = 20\%$ and expected return 9%; the risk-free rate is 3%. Stock A has volatility 40% and correlation 0.5 with the market; stock B has volatility 15% and correlation 0.8 with the market. **(a)** Compute the beta of each stock. Which stock co-moves more tightly with the market, and which has the larger beta? **(b)** Split each stock's variance into a systematic and an idiosyncratic part, and give the systematic share. **(c)** What expected return does the CAPM assign to each stock? An analyst forecasts 12% for A and 6.6% for B: what are the alphas? **(d)** You hold one unit of each stock and hedge with the market. How many units of the market do you short against each, and what is the volatility of each hedged position? Which hedge is better, and in what sense?

P11.11 *Where a book's risk comes from: Euler contributions and risk parity*

Let Σ be positive definite and $\sigma_p(w) = \sqrt{w^\top \Sigma w}$ for $w \neq 0$. **(a)** Show that $\partial \sigma_p / \partial w_i = (\Sigma w)_i / \sigma_p$, and prove that $\sigma_p = \sum_i w_i \, \partial \sigma_p / \partial w_i$ using only the fact that $\sigma_p(tw) = t\,\sigma_p(w)$ for $t > 0$. **(b)** Bonds have volatility 6%, equities 18%, and their correlation is 0.3. For the 50/50 portfolio compute the portfolio volatility, each asset's marginal contribution, each asset's total risk contribution, and the risk shares; check that the contributions add up. **(c)** Find the long-only weights at which the two risk contributions are equal (risk parity), the resulting volatility, and the leverage needed to bring that portfolio to the volatility of the 50/50 mix. **(d)** Show that the risk-parity weights in (c) do not depend on the correlation, and say what does.

P11.12 *One percentage point of expected return: error amplification by the inverse covariance*

Two assets have volatility 15% each and correlation $\rho = 0.6$; take $r_f = 0$, so the tangency weights are proportional to $\Sigma^{-1}\mu_e$, normalised to sum to one. **(a)** Compute Σ^{-1} and the tangency weights when the expected returns are $(6\%, 6\%)$, when they are $(7\%, 6\%)$, and when they are $(8\%, 4\%)$. **(b)** Show that for two assets with a common volatility the normalised tangency weights satisfy

$$w_1 - w_2 = \frac{\mu_1 - \mu_2}{\mu_1 + \mu_2} \cdot \frac{1+\rho}{1-\rho},$$

and identify the amplification factor with the eigenvalues of Σ. **(c)** Prove that for any positive-definite Σ and any perturbation δ of the expected-return vector, $\|\Sigma^{-1}\delta\| \le \|\delta\| / \lambda_{\min}$, with equality exactly when δ lies in the eigenspace of the smallest eigenvalue. Evaluate both sides for $\delta = (0.01, 0)$ and for $\delta = (0.01, -0.01)/\sqrt{2}$. **(d)** With twenty years of annual data, what is the standard error of an estimated annual mean here, and what does it say about the $(7\%, 6\%)$ versus $(6\%, 6\%)$ comparison? Name three remedies used in practice.

27.12 Problems — §12 Market Making I: the Economics of the Spread

P12.1 *Spread decomposition*

An interviewer asks: "Where does the bid-ask spread come from? Name its components, say which of them widen when volatility rises, and which get compressed by competition between market makers."

P12.2 *Glosten-Milgrom zero-profit quotes*

A friend claims: "A market maker with no directional opinion can quote an arbitrarily tight spread around fair value and still break even, because incoming buys and sells are equally likely." Test the claim in a miniature model: the asset's value is $V \in \{99, 101\}$, equally likely. A fraction $\pi = 0.3$ of order flow is informed (knows V and trades in the profitable direction); the rest is noise, buying or selling with probability $\frac{1}{2}$ each. (a) Where must a zero-profit maker set the ask and the bid? (b) What is the breakeven spread, and what exactly is wrong with the friend's claim?

P12.3 *Glosten-Milgrom with asymmetric prior*

Take the standard Glosten–Milgrom miniature — $V \in \{99, 101\}$, informed fraction $\pi = 0.2$, noise traders buy/sell with probability $\frac{1}{2}$ — but make the prior asymmetric: $P(V = 101) = p = 0.6$. (a) Compute the zero-profit ask and bid. (b) Is the spread still 2π? Is the quote midpoint still the prior mean? Which quote updates further from the prior mean, and why? (c) Sanity-check your quotes in the limits $p \to 1$ and $\pi \to 0$.

P12.4 *Mark-outs*

You run a market-making book. (a) An interviewer says: "Adverse selection is a nice theory word — how would you actually *measure* it in your fills?" (b) Your measurement then shows that fills against counterparty X mark 4 bp against you 5 seconds after trading, while all other flow marks

flat. (c) Separately, volatility doubles on a macro print. (d) Separately, the exchange announces it will publish unscheduled news in 30 seconds. What do you do to your quotes in each of (b), (c), (d), and why?

27.13 Problems — §13 Market Making II: Inventory and Skewing

P13.1 *Reservation price computation*

You are quoting a coin whose mid is $s = 100$. Your inventory is $q = +60$ units (long), your risk-aversion parameter is $\gamma = 0.001$, the variance rate is $\sigma^2 = 2.5$ per day (in squared price units), and $T - t = 0.4$ days remain in your risk horizon. (a) Compute the Avellaneda–Stoikov reservation price $r = s - q\gamma\sigma^2(T - t)$. (b) If you keep a fixed half-spread of 0.10 around r, what are your quotes, and what are they doing in trading terms? (c) What are the quotes if instead you are short 60 units?

P13.2 *Reservation-price anatomy, skew vs widen*

An interviewer pushes on the reservation-price formula $r = s - q\gamma\sigma^2(T - t)$. (a) Justify in one line why each factor q, γ, σ^2, $(T - t)$ belongs in the correction. (b) What happens to the skew as $T - t \to 0$, and does that make trading sense? (c) Why does the maker shift *both* quotes rather than only improving the ask when long? (d) Skewing and widening are both defensive moves — what different risk does each address?

P13.3 *Fermi estimation + interval calibration* ★

You have 15 seconds. Bitcoin blocks arrive on average every 10 minutes. Estimate how many blocks have been mined since 2012, and give a 95% confidence interval on your estimate. (After you answer, the interviewer will make you trade a market on this number — both the point estimate and, especially, the interval are graded.)

P13.4 *Make-a-market game: position and P&L accounting* ★

You have just estimated the number of Bitcoin blocks mined since 2012 (a quantity you do not know precisely). The interviewer says: "Now make a market on your estimate." One unit trades per quote, prices in blocks. The sequence, as played:

- You quote 100k–200k — interviewer **BUYS**.
- You quote 400k–600k — **BUYS**.
- You quote 900k–1M; asked "how confident are you?" you say 10% — **BUYS**.
- You quote 4M–6M — **SELLS**.
- You quote 2M–3M — no trade; game stops.

Closing question (the one the game builds to): are you long or short, and how much? Also give your cash, your mark-to-market P&L at your final mid, and what the buy-buy-buy-then-sell pattern told you about the interviewer's view while it was happening.

P13.5 *Pre-trade risk controls / position limits* ♦

What is the best way to stop trading algorithms from accidentally entering very large positions which may cause extreme losses during price movements?

(A) Have traders monitor the algorithms' positions on a dashboard and intervene manually.
(B) Send an automated alert to the team whenever a position becomes large.
(C) Enforce hard position limits that automatically block the algorithm from trading further on the side that would increase the breach.
(D) Backtest the algorithm thoroughly before deploying it.

P13.6 *Inventory hedging with perps*

You make markets in spot BTC on venue A, and persistent one-way retail flow has left you long a large spot inventory. Skewing your quotes has not shed it fast enough, and dumping the inventory through your own market would move the price against you and against your own resting quotes. A colleague says: "just pull your quotes until it clears." Propose a better solution using instruments or venues beyond your own book, and name the residual risks you now carry.

27.14 Problems — §14 Market Microstructure

P14.1 *Square-root impact law by dimensional analysis*

Empirically, executing a metaorder of Q shares in a market with daily volume V and daily volatility σ moves the price by $\Delta P \approx Y\sigma\sqrt{Q/V}$ with $Y \approx 1$. Derive the exponents by dimensional analysis: posit $I = Y\,Q^{\alpha}V^{\beta}\sigma^{\gamma}$ with units $[I]$ = price, $[Q]$ = shares, $[V]$ = shares/day, $[\sigma]$ = price/$\sqrt{\text{day}}$, and show that α, β, γ are forced. Then say what extra modelling assumption the ansatz itself sneaked in.

P14.2 *Impact vs alpha capture*

A stock has daily volatility 2% and average daily volume 10 million shares. Your signal says the stock will rise 25 bp over the next two hours, and your desired position is 250,000 shares. (a) Using the square-root law with $Y = 1$, estimate your market impact. (b) Can you capture the alpha? (c) A colleague suggests slicing the order over five days to shrink the impact — what is the catch?

P14.3 *Kyle's lambda and reconciliation with concave impact*

You regress one-minute price changes on signed volume, $\Delta P_t = \lambda Q_t + \varepsilon_t$, and find that trades of 10,000 shares move the price 2 bp on average. (a) What does λ measure, and what would you use it for? (b) Predict the average move for a 40,000-share trade under (i) the linear Kyle model and (ii) the empirical square-root law. (c) The two predictions disagree — how does the empirical literature reconcile a linear λ with concave total impact, and which prediction should you trust?

P14.4 *OBI and VAMP computation*

Top of book: bid 99 with 1,000 shares, ask 100 with 100 shares. (a) Compute the order-book imbalance $\text{OBI} = \frac{q_b - q_a}{q_b + q_a}$ and the volume-adjusted mid $P_{\text{VAMP}} = \frac{P_a q_b + P_b q_a}{q_b + q_a}$. (b) The VAMP formula weights each price by the *opposite* side's volume — explain the intuition using this book. (c) An interviewer at an HFT-adjacent shop asks: "so is that the micro-price?" Give the precise answer.

P14.5 *OBI signal monetisation*

A junior researcher backtests the rule "when OBI > 0.8, cross the spread and buy; exit after 10 seconds" on top-of-book data for a liquid crypto pair, ignoring costs, and reports a 58% hit rate on next-tick direction. He proposes going live. Give at least three distinct reasons this strategy loses money live *even if the 58% is real*, and describe how a market maker would actually monetise the same signal.

27.15 Problems — §15 Crypto Market Structure

P15.1 *Bitcoin mechanics* ★

Walk me through the basics of how Bitcoin works: how are new blocks mined, how are blocks linked into a chain, how does a user get a transaction included, and how do miners get paid?

P15.2 *Cross-exchange arbitrage, leg risk* ♦

You run a strategy that fires simultaneous orders — buy on centralized exchange A, sell on

centralized exchange B — whenever the same coin shows a small, fleeting price discrepancy between the two venues. What is the biggest risk in executing this arbitrage for very fast, small opportunities?

- (a) Trading fees eat the margin.
- (b) Network latency makes you miss many of the opportunities.
- (c) Only one of the two legs fills, leaving an unhedged position.
- (d) Withdrawal and transfer times lock up your capital between venues.

P15.3 *The market-maker mandate* ♦

A token project engages a market-making firm as a designated liquidity provider for its token. Which of the following does the market maker promise the project: higher prices, or higher volumes?

P15.4 *Perpetual funding arithmetic*

A perpetual future on ETH trades above its spot index, and the venue prints a funding rate of 0.015% per 8-hour interval. (a) Who pays whom, and which way does the mechanism push the perp price? (b) Annualise the rate, simple and (approximately) compounded. (c) You are long $2M of the perp; roughly what does funding cost you per week if the rate persists?

P15.5 *Cash-and-carry*

Funding on a major venue's BTC perpetual has run at an annualised +30% for two weeks. A colleague proposes: buy $10M of spot BTC, short $10M of the perp, collect 30% a year delta-neutral — "free money." Present the strongest version of the trade, then explain what is wrong with the free-money claim (give at least four distinct risks), and say what the persistently positive funding is telling you about market positioning.

P15.6 *Liquidation price and cascades*

A trader opens a 10x leveraged long on a BTC perpetual at $50,000, posting initial margin equal to 10% of entry notional. The venue liquidates the account when equity (initial margin plus mark-to-market P&L) falls to the maintenance requirement of 0.5% of position notional marked at the current price. (a) At approximately what price is the position liquidated? Give the general rule of thumb for leverage L. (b) Venue data shows a dense cluster of similar longs with liquidation prices just below $45,500. Describe what happens if price trades down into the cluster, and how you would react as a market maker quoting this book.

P15.7 *Constant-product AMM pricing and impact*

An ETH/USDC constant-product pool ($xy = k$, no fees) holds $x = 100$ ETH and $y = 200{,}000$ USDC. (a) What price does the pool currently quote? (b) A trader buys 10 ETH from the pool: how much USDC do they pay, what average price do they get, and what is the pool's new marginal price? (c) Suppose ETH trades at $2,000 on a centralized exchange throughout. What happens next, and why is that mechanism essential to how an AMM tracks fair value?

P15.8 *MEV sandwich attack*

Same pool as the previous problem: $x = 100$ ETH, $y = 200{,}000$ USDC, $xy = k$, no fees. A victim's order to buy 10 ETH sits in the public mempool with a loose slippage tolerance. An attacker sandwiches it: front-runs with their own 10 ETH buy, lets the victim's buy execute, then back-runs by selling 10 ETH. (a) Compute the attacker's profit, ignoring gas and pool fees. (b) Show the attacker ends flat in ETH and identify exactly whose money the profit is. (c) What limits this attack in practice, and what does its existence imply about sending orders to a public mempool?

P15.9 *Impermanent loss*

You deposit equal dollar values of ETH and USDC into a constant-product AMM ($xy = k$, no fees); arbitrageurs always keep the pool price equal to the external market price. The ETH price then moves from p_0 to $r\,p_0$. (a) Derive the pool's holdings and the value of the LP position as a function of r. (b) Show that $V_{\text{LP}}/V_{\text{HODL}} = 2\sqrt{r}/(1+r)$, conclude $\text{IL}(r) = 2\sqrt{r}/(1+r) - 1 \le 0$, and evaluate at $r = 2$ and $r = 4$. (c) Show $\text{IL}(r) = \text{IL}(1/r)$ and that for a small move $r = 1+\varepsilon$, $\text{IL} \approx -\varepsilon^2/8$. What option-like position does this make the LP, and when is providing liquidity rational?

P15.10 *Impermanent-loss benchmark*

You deposited \$5,000 of ETH and \$5,000 of USDC into a constant-product pool when ETH traded at \$2,000. ETH now trades at \$4,000 and your LP position is worth \$14,142 — up 41% — so you conclude that providing liquidity beat holding. What is wrong with that conclusion, exactly how much did LPing cost you, and what would have to be true for the LP position to have genuinely won?

27.16 Problems — §16 Forwards, Futures and No-Arbitrage Pricing

P16.1 *Quoted rates, compounding frequencies and the cheaper loan*

All rates are per year. **(a)** A deposit pays 8% compounded semiannually. Find the equivalent continuously compounded rate and the equivalent rate with annual compounding. **(b)** A rate of 4.5% is continuously compounded. Find the equivalent rate compounded monthly, and compare it with the shortcut $r + r^2/(2m)$, where r is the continuous rate and m the number of compounding periods a year. **(c)** One loan is quoted at 7.90% with annual compounding, another at 7.70% compounded monthly. Which loan is cheaper for the borrower? Convert both quotes to one convention, and give the difference in interest on 100,000 borrowed for one year and repaid, with all its interest, in a single payment at the end of the year. **(d)** A perpetual future charges funding of 0.03% every 8 hours, that is, 1095 times a year, which is quoted as $1095 \times 0.03\% = 32.85\%$ a year. Find the continuously compounded rate that gives the same growth over each 8-hour period. Which of the two figures is larger, and why?

P16.2 *Both cash-and-carry arbitrages, and the no-arbitrage band with frictions*

A market is frictionless: there are no transaction costs, cash is lent and borrowed at one continuously compounded rate r, and short selling is allowed. An investment asset trades at S today, pays no income and costs nothing to store. A forward contract for delivery at T costs nothing to enter at the forward price F. **(a)** Assuming that the market admits no arbitrage, prove that $F = Se^{rT}$ by exhibiting an arbitrage, written as a table of cash flows at time 0 and at time T, in each of the cases $F > Se^{rT}$ and $F < Se^{rT}$. **(b)** Let $S = 45$, $r = 6\%$ and $T = 0.75$ (nine months). Compute the forward price. The forward is quoted at 46.50: state the trade and the sure profit per unit at T. **(c)** Now add frictions. Cash is lent at $r_{\text{len}} = 5.5\%$ and borrowed at $r_{\text{bor}} = 6.5\%$ (both continuously compounded, and the proceeds of a short sale are lent at r_{len}); the asset is sold at the bid 44.95 and bought at the offer 45.05; the forward trades at a single price. Find the interval of forward quotes that admits neither arbitrage. Is the quote 46.50 still an arbitrage, and if so, for how much per unit at T?

P16.3 *A short forward on a dividend-paying stock, priced and revalued*

A stock trades at $S = 80$ and pays a dividend of 1.20 per share in 3 months and again in 9 months. The continuously compounded rate is $r = 4\%$ for all maturities. A trader goes *short* a one-year forward on one share, struck at the forward price rounded to the cent. **(a)** Find the delivery price K and the value of the trader's position on the day it is written, to the cent. **(b)** Five months later the first dividend has been paid, the stock trades at 76 and the rate is unchanged. Find the

forward price for the original delivery date and the value of the trader's short position.

P16.4 *Three mispriced quotes: an index future, an implied foreign rate and a currency forward*

Rates are continuously compounded, markets are frictionless, short selling is allowed, and a futures price is treated as a forward price (interest on variation margin is ignored). **(a)** A stock index stands at 4,800 and pays a continuous dividend yield of $q = 1.8\%$; the rate is $r = 4.5\%$. Compute the four-month futures price. The future is quoted at 4,830: describe the arbitrage, including the number of units of the index basket traded today, and give the sure profit per unit of index at delivery. **(b)** Sterling trades at 1.2700 dollars spot and at 1.2620 dollars for delivery in six months. The dollar rate is 4.3%. Which sterling interest rate do the two quotes imply? **(c)** The yen trades at 0.6700 dollars per 100 yen spot; the yen rate is 0.5% and the dollar rate 4.3%. Compute the three-month forward rate. The forward is quoted at 0.6720 dollars per 100 yen: build the arbitrage on a yen loan sized to repay exactly 500 million yen at delivery, list the cash flows in both currencies today and at delivery, and give the profit in dollars at delivery.

P16.5 *Storage, an upper bound and the convenience yield: silver versus crude*

Rates are continuously compounded and known, so futures and forward prices coincide; interest on futures variation margin is ignored, so a futures position pays what a forward pays; and storage costs are known in advance. Silver is held in large quantities purely as an investment; crude oil is held chiefly to be refined. **(a)** Silver trades at 30.00 per ounce. Storing an ounce costs 0.36 a year, paid in advance in instalments of 0.12 today, in 4 months and in 8 months. With $r = 5\%$, find the futures price for delivery in 10 months (all three instalments fall due before delivery). **(b)** Crude oil trades at 75 per barrel. Storing a barrel costs 1.20 per half-year, paid at the end of each half-year. With $r = 4.5\%$, what does the absence of arbitrage imply for the one-year futures price? **(c)** The one-year crude future is quoted at 73.80. Is there an arbitrage? Compute the convenience yield that the quote implies, and say whether the market is in contango or in backwardation. **(d)** Suppose instead that the one-year crude future is quoted at 82. State the trade and its profit per barrel at delivery. **(e)** Why does the same carry argument pin the silver price and only bound the crude price? Is the cost of storage the reason?

P16.6 *A short futures position through four settlement days*

A trader goes short 3 crude oil futures contracts of 1,000 barrels each at 72.40 per barrel. The initial margin is 4,000 per contract and the maintenance margin is 3,100 per contract. Interest on the margin account is ignored. **(a)** Before any margin call or withdrawal, above which settlement price does the trader receive a margin call, and at or below which settlement price can 3,000 be withdrawn? **(b)** The next four settlement prices are 72.85, 73.55, 72.90 and 71.60, and nothing is withdrawn from the account. Run the margin account day by day: the gain of the day, the margin call if there is one, and the closing balance. What has the trader gained in total? **(c)** Day 4 is the delivery date. Compare the result with that of a short forward contract on 3,000 barrels at the delivery price 72.40. What differs?

P16.7 *Futures prices equal forward prices when rates are known*

The interval $[0, T]$ is cut into n days of length Δt. The continuously compounded rate of day i is r_i, and all the rates are known at time 0. The cash account is $B_0 = 1$ and $B_i = \exp\big(\sum_{k=1}^{i} r_k \, \Delta t\big)$, so that 1 deposited at the end of day i is worth B_n / B_i at T; it is open for deposits and for loans at the same rates, and every settlement is deposited in it or financed from it until T. A futures contract for delivery at T is settled daily: one long contract receives $F_i - F_{i-1}$ at the end of day i, where F_i is the settlement price of day i, $F_0 = F_0^{\text{fut}}$ is today's futures price and $F_n = S_T$. Entering or resizing a futures position costs nothing. A forward contract for delivery at T costs nothing to enter at the forward price F and pays $S_T - F$ at T; forward contracts for delivery at T can also be written at the end of any day k, at no cost, at the forward price of that date. Positions in both contracts may be long or short and of any real size. There is no arbitrage. **(a)** Prove that $F_0^{\text{fut}} = F$, and that at the end of every day $k < n$ the settlement price F_k equals the

forward price for delivery at T written at that date. Say where the proof uses that the rates are known in advance. **(b)** Check the central step on numbers: $n = 3$, $r_i\,\Delta t = 0.0002$ on each day, and settlement prices $F_0 = 100$, $F_1 = 101$, $F_2 = 99.5$, $F_3 = 100.8$. For the rolling position that holds B_i contracts long during day i, give each day's position, settlement and value at T, and the total. Compare with a constant position of one contract. **(c)** One futures contract and one forward contract are both entered long at 100, with $n = 3$ and $r_i\,\Delta t = 0.0002$. Along the path $100, 104, 104, 100$ of settlement prices, and then along $100, 96, 96, 100$, which contract ends ahead at T, and by how much? What does this say about whether daily settlement helps or hurts a hedger? **(d)** Suppose instead that rates are random and tend to be high when the asset price is high. Which is higher, the futures price or the forward price? Argue in two sentences.

P16.8 *The minimum-variance hedge ratio: from statistics, from data, and the wrong regression*

A hedger is exposed to the change ΔS of a spot price over the life of a hedge and can trade futures whose price changes by ΔF over the same period. The two changes have finite variances, standard deviations $\sigma_S > 0$ and $\sigma_F > 0$, and correlation ρ. With a hedge ratio h (units of futures sold per unit of exposure) the hedged change per unit of exposure is $\Delta S - h\,\Delta F$. **(a)** Derive the hedge ratio that minimises $\mathrm{Var}(\Delta S - h\,\Delta F)$, prove that it is the unique global minimiser, give the minimum variance and the fraction of the unhedged variance that the hedge removes, and show that, with h^* the minimising ratio, an exposure of n_A units hedged with futures contracts of n_F units each takes $h^* n_A / n_F$ contracts. **(b)** Let $\sigma_S = 0.45$ and $\sigma_F = 0.60$ per unit, and $\rho = 0.85$. Find the hedge ratio, the fraction of variance removed and the residual standard deviation, and the number of futures contracts of 20,000 units each that hedges an exposure of 900,000 units. **(c)** Six past periods gave the pairs $(\Delta F, \Delta S) = (2, 1)$, $(-1, -1)$, $(3, 2)$, $(0, 1)$, $(-2, -2)$, $(4, 2)$. Estimate the hedge ratio and the correlation from them. **(d)** A colleague regresses ΔF on ΔS in the same data and inverts the slope. Which ratio does that give, and by what factor does it misstate the hedge?

P16.9 *Index futures: hedge, re-gear, and judge the manager*

An equity portfolio worth 50 million has a beta of 0.9 against a stock index. Three-month futures on the index trade at 4,000 with a multiplier of 50. Model the futures return over the hedging horizon as the total return of the index minus $r\,\Delta t$, where r is the continuously compounded rate and Δt the horizon, and ignore interest on variation margin. **(a)** How many contracts hedge the market exposure of the portfolio, and on which side? **(b)** Starting from the unhedged portfolio, which futures trade raises its beta to 1.4? **(c)** With the hedge of (a) in place, the futures price falls by 6% over the three months. Take $r = 4.8\%$, and suppose that the alpha and the residual return of the portfolio over the period are zero. Compute the profit or loss of the portfolio and of the futures, and the return of the hedged book. What does a hedged book earn? **(d)** Another manager runs a portfolio with a beta of 0.3 against the index. It returned -6% in a year in which the index returned -25% and the riskless return was 4%. Did the manager beat the CAPM benchmark? Compute the realised alpha.

P16.10 *Crypto carry: a dated future against the perpetual*

Bitcoin trades at 64,000 spot, and a dated future that expires in three months trades at 65,600. Dollars are borrowed at a continuously compounded rate of 5%. **(a)** Annualise the premium of the future over spot, as a continuously compounded rate and as a simple rate. Estimate first without a calculator, using $\ln(1 + x) \approx x - x^2/2$. By how much does the continuous figure exceed the dollar rate? **(b)** Describe the trade that locks the premium in, and verify at expiry prices of 50,000 and of 80,000 that it earns the same amount per bitcoin. What is left after financing the spot purchase at 5%? **(c)** A perpetual future on the same venue currently charges funding of 0.012% per 8 hours, paid by longs to shorts. Annualise it, with three payments a day and a 365-day year, as a simple rate and as a compounded annual rate with every payment reinvested. The perpetual carry trade buys one bitcoin spot and shorts one bitcoin of the perpetual, collecting the funding. Why is its carry not comparable like for like with the answer to (a)? **(d)** The day

after the trade of (b) is opened, the dated future rallies by 25%. How much cash per bitcoin must the short futures leg pay in variation margin, and when is it recovered? Name one risk carried by both the dated and the perpetual trade that none of the rates above measures.

P16.11 *Ten claims about futures, forecasts and hedges: true or false*

Decide whether each claim is true or false, and justify the verdict in one or two lines, with the numbers in brackets where they are given. Rates are continuously compounded unless stated.

(i) The futures price of an asset is the market's unbiased forecast of its spot price at delivery. [An investment asset without income or storage cost trades at 50; rates are deterministic, the one-year riskless simple return is 3% and the expected excess return of the market is 6%. Take the CAPM as given, and compare the one-year futures price with the expected spot price for a beta of 1.5 and for a beta of -0.5.]

(ii) A minimum-variance hedge ratio of 1 means that the hedge is perfect. [The changes of the spot price and of the futures price over the life of the hedge have correlation $\rho = 0.6$ and standard deviations $\sigma_S = 5$ and $\sigma_F = 3$.]

(iii) If the basis on the day the hedge is closed is known in advance, the minimum-variance hedge ratio is 1.

(iv) A perfectly hedged position always ends up better off than an imperfectly hedged one. [A producer sells one unit at the delivery date. A short hedge with futures on the asset itself is opened at a futures price of 100 and held to delivery; a second producer hedges half the unit; the spot price at delivery is 120.]

(v) A short hedge with futures on the asset itself, held to the delivery date of the futures contract, locks in today's spot price. [Spot 97, futures 100.]

(vi) The cost-of-carry formula prices copper futures exactly as it prices gold futures. [For both metals take $r = 4\%$, proportional storage at 2% a year, a spot price of 9,000 and a one-year futures quote of 8,820.]

(vii) A futures price always exceeds the spot price. [An index at 5,000 with a dividend yield of 4%; $r = 2.5\%$; delivery in six months.]

(viii) A forward contract that cost nothing to enter is worth nothing until it matures. [A long forward with delivery price 60; six months before delivery the forward price for that date is 57, and $r = 4\%$.]

(ix) Futures and forward prices for the same delivery date are always equal.

(x) A fully hedged equity portfolio earns nothing. [A portfolio of 20 million with a beta of 1.2, an alpha of 0.5% over the quarter, measured on excess returns, and no residual return; index futures at 5,000 with a multiplier of 50; $r = 5\%$; the futures fall by 8% over the quarter. Model the futures return as the total return of the index minus $r\,\Delta t$, and ignore interest on variation margin.]

Closing part. A commodity trades at 80 spot, its six-month and one-year futures trade at 82 and 84, and its expected spot price in one year is 88. Is this market in backwardation? Answer in the market's sense of the word and in Keynes's sense.

27.17 Problems — §17 Options: Payoffs, Bounds, Put–Call Parity and Trading Strategies

P17.1 *Writing a put versus buying a call*

A stock that pays no dividends trades at $S = 46$, and the continuously compounded rate is $r = 4\%$. Three-month European options ($T = 0.25$) with strike $K = 45$ are quoted at 3.05 for the call and 1.60 for the put. One trader writes the put and another buys the call. Profit is measured at expiry, with each premium carried to T at the factor e^{rT}. **(a)** Tabulate the profit of the short

put and of the long call at $S_T = 38$, 42, 45, 48 and 52. **(b)** For each position give the breakeven, the values of S_T for which the option is exercised, the maximum gain and the maximum loss. **(c)** Show that in every state the profit of the long call minus the profit of the short put equals the profit of a long straddle struck at 45, bought at the quoted prices with its premium carried to T in the same way. For which values of S_T does the short put do better than the long call? **(d)** A third trader holds both positions, the long call and the short put. Give the profit as a function of S_T and identify the position.

P17.2 *Lower bounds and the discounted strike*

A stock that pays no dividends trades at $S = 33$, the continuously compounded rate is $r = 4\%$, and every option below expires in nine months ($T = 0.75$). The market is frictionless, money is borrowed and lent at the rate r, and short sales are allowed. The four quotes are separate situations: in each part the option quoted is the only option that trades. For each quote decide whether it is an arbitrage. If it is, give the trade, a table of its cash flows today and at expiry (or the cash flow at exercise), and the amount it locks in; if it is not, say why not. **(a)** A European put with strike 38 quoted at 3.60. **(b)** A European call with strike 30 quoted at 3.50. **(c)** An American put with strike 38 quoted at 4.90. **(d)** A European put with strike 38 quoted at 4.90.

P17.3 *Put–call parity from scratch, in forward form, and an arbitrage on the spot*

Throughout, the market is frictionless, money is borrowed and lent at one continuously compounded rate r, short sales are allowed and there is no arbitrage. A European call and a European put on the same underlying have the same strike K and the same expiry T; their prices are C and $\mathcal{P}$. **(a)** The underlying trades at S and pays no income before T. Prove that $C - \mathcal{P} = S - Ke^{-rT}$ by comparing two portfolios that pay the same amount in every state. **(b)** Let the underlying be any asset on which a forward contract for delivery at T can be entered, at no cost, at the forward price F. Show that $C - \mathcal{P} = e^{-rT}(F - K)$, and that the call and the put cost the same exactly when $K = F$. Then take an index at $S = 80$ with dividend yield $q = 2\%$, $r = 3\%$ and $T = 1$, so that $F = Se^{(r-q)T}$: compute $C - \mathcal{P}$ at $K = 80$, say which option is dearer there, and give the strike at which the two cost the same. **(c)** A stock without dividends trades at $S = 58$, the rate is $r = 3\%$, and six-month options ($T = 0.5$) struck at $K = 60$ are quoted at $C = 2.95$ and $\mathcal{P} = 4.80$. Which option is cheap relative to the other, and by how much? Give the arbitrage with a table of its cash flows today and at T, and the amount it locks in. **(d)** Drop the assumption that short sales cost nothing: suppose that quotes like those of (c) persist on a stock that is hard to borrow, and that the lender of the share charges the short seller a fee at a constant yearly rate, continuously on the value of the borrowed share. Why is the trade of (c) not free money? What is the smallest such fee at which the trade of (c), adapted to pay the fee, no longer makes money?

P17.4 *Parity with a discrete dividend*

A stock trades at $S = 52$ and pays one dividend of 0.90 in three months; the continuously compounded rate is $r = 4\%$. Six-month European options ($T = 0.5$) with strike $K = 50$ are quoted at $C = 4.60$ and $\mathcal{P} = 2.80$. The market is frictionless, and a short seller pays the dividend to the lender of the share. **(a)** Which value of $C - \mathcal{P}$ does put–call parity require? Which option is cheap relative to the other, and by how much? **(b)** Give the arbitrage with a dated table of its cash flows (today, the dividend date and expiry) and the profit it locks in. **(c)** Before anyone trades, the company announces that the dividend will be 1.50 instead of 0.90; the share price stays at 52 and the rate at 4%. By how much does the value of $C - \mathcal{P}$ required by parity change? What does the absence of arbitrage say about the call and the put separately? **(d)** Suppose instead that the two options are American and quoted at the same prices, with the original dividend. American prices on a stock whose dividends have present value I satisfy only the band $S - I - K \le C^{\text{am}} - \mathcal{P}^{\text{am}} \le S - Ke^{-rT}$. Do the quotes violate it? Say what can go wrong with the trade of (b).

P17.5 *Bounds on an American put from an American call, with and without dividends*

American options on a stock have strike $K = 52$ and nine months to expiry ($T = 0.75$). The stock trades at $S = 55$, the continuously compounded rate is $r = 4\%$, and the American call is quoted at 5.80. The market is frictionless, short sales are allowed (a short seller pays any dividend to the lender of the share), there is no arbitrage, and the writer of an option does not choose when it is exercised. **(a)** The stock pays no dividends. Take as known the band $S - K \le C^{\text{am}} - \mathcal{P}^{\text{am}} \le S - Ke^{-rT}$ and find the interval in which the American put must trade. **(b)** The stock pays no dividends, as in (a), and the put is quoted at 3.00. Give a trade that collects money today, and show that it is worth a nonnegative amount at T whatever the holder of the put does, exercise at an arbitrary time $t_e < T$ included. **(c)** Now let the stock pay known dividends D_i at dates $t_i < T$, with present value $I = \sum_i D_i e^{-rt_i}$, and let $r \ge 0$. Prove that

$$S - I - K \;\le\; C^{\text{am}} - \mathcal{P}^{\text{am}} \;\le\; S - Ke^{-rT}.$$

(For the lower edge: against a written put and a share sold short, hold the call to expiry and deposit $K + I$.) **(d)** The stock pays one dividend of 0.70 in four months. With the same call quote, find the interval for the American put that the band of (c) gives.

P17.6 *Calls are never exercised early; puts can be*

Throughout, the market is frictionless, money is borrowed and lent at one continuously compounded rate $r \ge 0$, short sales are allowed, any income of a borrowed share being paid to its lender, there is no arbitrage, and the writer of an American option does not choose when it is exercised. The American and European options below share the strike $K > 0$ and the expiry T, and $\tau = T - t$ is the remaining life at a date t. Take as known: an American option is worth at least the European option of the same type and at least its intrinsic value; on an asset without income, $C_t \ge S_t - Ke^{-r\tau}$ and $\mathcal{P}_t \le Ke^{-r\tau}$; on an asset with a continuous yield q, $C_t \le S_t e^{-q\tau}$. **(a)** The stock pays no dividends. Prove, by writing the American call and buying the European call, that $C^{\text{am}} = C$, and that for $r > 0$ exercising the call at a date $t < T$ pays strictly less than selling it. Run the same kind of trade between two expiries to show that $C(K, T_1) \le C(K, T_2)$, where $C(K, T_1)$ and $C(K, T_2)$ are the prices of European calls with strike K that expire at $T_1 < T_2$. Which two advantages does a holder give up by exercising early, and which of them survives at $r = 0$? **(b)** A stock without dividends trades at 140, an American call on it has $K = 100$ and one month to expiry, and $r = 4\%$. Its holder has turned bearish. Compare (1) exercising and selling the share, (2) selling the call, and (3) keeping the call and selling the share short. **(c)** Which step of that exercise-matching proof of $C^{\text{am}} = C$ fails for a put? Show that for $S = 4$, $K = 60$, $r = 6\%$, $T = 2$ and no dividends the American put is worth strictly more than the European put, and find the spot prices for which the same argument works. Which step of it fails for a call on an asset with a continuous yield $q > 0$? Show that for $S = 150$, $K = 10$, $q = 6\%$ and $T = 2$ the American call is worth strictly more than the European call. **(d)** Show that $\mathcal{P}^{\text{am}} = \mathcal{P}$ when $r = 0$ and the stock pays no dividends. Can early exercise of the put nevertheless tie with selling it?

P17.7 *Ex-dividend dates and the American call*

An American call on a stock has strike $K = 80$ and one year to expiry ($T = 1$), and the continuously compounded rate is $r = 5\%$. The stock pays known dividends of 0.60 at $t_1 = 0.25$ and 1.50 at $t_2 = 0.75$, and on each ex-dividend date its price drops by exactly the dividend. The market is frictionless, a short seller pays the dividend to the lender of the share, and there is no arbitrage. Take as known the following bound: at a date t such that no dividend goes ex between t and t', where t' is the next ex-dividend date or the expiry, the call is worth at least $S_t - Ke^{-r(t'-t)}$. **(a)** Derive a lower bound for the call the instant before an ex-dividend date, when the share still carries the dividend D. Deduce the largest dividend for which the call is certain to be worth at least what exercise pays at that instant, evaluate this threshold at each of the two dates, and conclude. **(b)** How large can the second dividend be if exercise just before t_2 is to be ruled out by (a)? **(c)** Why does exercise at any time before expiry, other than the instants just before the two

ex-dividend dates, pay strictly less than selling the call? **(d)** The stock trades at 120 the instant before the second ex-dividend date. Compare what exercise pays with the bound of (a). What decides between exercising and waiting?

P17.8 *Convexity in the strike: proof and arbitrage*

European options on one stock with a common expiry T trade in a frictionless market without arbitrage, with continuously compounded rate r. **(a)** With $r = 3\%$ and $T = 0.5$, calls are quoted at $C(40) = 6.10$, $C(45) = 4.40$ and $C(55) = 0.90$. Check that the 40/45 and 45/55 call spreads satisfy $0 \le C(K_1) - C(K_2) \le e^{-rT}(K_2 - K_1)$ for $K_1 < K_2$. **(b)** Find the weight w with $45 = 40w + 55(1 - w)$ and show that the quotes violate $C(45) \le w\,C(40) + (1 - w)\,C(55)$. Give an arbitrage that uses whole numbers of options: its cash flow today and its payoff at $S_T = 35$, 40, 45, 50, 55 and 60. **(c)** Prove that for any strikes $K_1 < K_2 < K_3$ and $w = (K_3 - K_2)/(K_3 - K_1)$,

$$C(K_2) \le w\,C(K_1) + (1 - w)\,C(K_3) \qquad \text{and} \qquad \mathcal{P}(K_2) \le w\,\mathcal{P}(K_1) + (1 - w)\,\mathcal{P}(K_3).$$

(d) The stock of (a) pays no dividends and trades at $S = 45$. Price the three puts by put–call parity and show that they violate convexity by the same amount as the calls. Why must the two amounts agree? **(e)** On another stock, with $r = 4\%$ and six months to expiry, the European 100/105 call spread (long the 100 call, short the 105 call) is quoted at 5.10. Give the arbitrage. What do the bounds of (a) and the inequality of (c) say about the price of a digital option paying 1 if $S_T > K$, and about the risk-neutral density of S_T?

P17.9 *The box spread as a loan*

An index stands at 4100. European options on it with nine months to expiry ($T = 0.75$) are quoted at

$$C(4000) = 345.60, \qquad C(4200) = 244.60, \qquad \mathcal{P}(4000) = 167.40, \qquad \mathcal{P}(4200) = 260.20.$$

The 4000/4200 box is the bull call spread (long the 4000 call, short the 4200 call) together with the bear put spread (long the 4200 put, short the 4000 put). **(a)** What does the box cost, and what does it pay at expiry? **(b)** Which continuously compounded rate does it imply? **(c)** A desk can otherwise lend at 3.8% and borrow at 4.6% (continuously compounded). Which side of the box does it trade to lend, and which to borrow? What does it gain at expiry, per box, in each case? **(d)** Show that the box is a long synthetic forward struck at 4000 plus a short synthetic forward struck at 4200. By how much, exactly, does the cost of the bull call spread exceed the cost of the bull put spread (long the 4000 put, short the 4200 put)? **(e)** Taking the rate implied in (b) as the market rate r, find the forward price of the index that put–call parity implies at each strike, and the continuous dividend yield q of the index. **(f)** Why is the same box, built from American options on a single stock, not riskless for its seller?

P17.10 *Name the position*

All options below are European, on one underlying, with a common expiry T; profits ignore the carry of the premiums. For each item, name the position, write its payoff at expiry and answer the question attached. **(a)** Long a 95 put and a 105 call, short a 100 put and a 100 call, at the quotes $\mathcal{P}(95) = 1.74$, $\mathcal{P}(100) = 3.73$, $C(100) = 4.23$ and $C(105) = 2.22$. Which two combinations make it up? Show that its payoff is that of a 95/100/105 call butterfly (long one 95 call and one 105 call, short two 100 calls) minus 5. Is it opened for a debit or for a credit, and what are its maximum gain, its maximum loss and its breakevens? **(b)** Long a 110 call and short a 90 put. What does the position become when both strikes are moved to one strike K? **(c)** A position in options struck at 100, where the call is quoted at 5 and the put at 4, costs 14. It loses 14 if $S_T = 100$, gains two points for each point of S_T above 100 and one point for each point below. Which position is it, and where does it break even? **(d)** Long one share that pays no dividends before T, long a 90 put and short a 115 call. Write the position as a bond plus a spread, and give the range of its payoff. **(e)** A payoff is 0 for $S_T \le 80$, rises one for one to 20 at $S_T = 100$

and stays at 20 above. Build it from calls; build it again from puts and a bond. With $r = 4\%$ (continuously compounded), $T = 0.5$ and the call version costing 9.78, how much does the put spread of the second version bring in?

P17.11 *Build it from calls*

A stock that pays no dividends trades at $S = 100$, the continuously compounded rate is $r = 3\%$, and six-month European calls ($T = 0.5$) are quoted at

$$C(90) = 13.79, \qquad C(100) = 7.76, \qquad C(110) = 3.90, \qquad C(130) = 0.73.$$

(a) A claim pays $f(S_T)$ at T, where f equals 10 for $S_T \le 90$, falls one for one to 0 at $S_T = 100$, is 0 on $[100, 110]$, rises two for one to 40 at $S_T = 130$, and equals 40 above. Replicate the claim with a zero-coupon bond and the four calls, price it, and check the replica at $S_T = 95$, 120 and 150. **(b)** Rewrite the bond and the two calls with the lowest strikes as a spread of puts, and price that spread. **(c)** Replicate the payoff $|S_T - 100|$ with a bond, the share and the 100 call, price it, and confirm the price with put–call parity. **(d)** Why can a continuous payoff be matched as closely as desired on a range of prices once enough strikes trade, and what does the replication become when calls trade at every strike?

P17.12 *Participation rate of a principal-protected note*

An issuer sells a four-year note with a principal of 10,000, linked to an index that pays a dividend yield of 2%; the four-year continuously compounded rate is $r = 3.5\%$. At maturity the note repays the principal and adds the participation rate times the gain, if any, on 10,000 invested in the index at today's level (dividends excluded). The issuer keeps 1% of the principal as a fee at the outset and spends the remainder on a zero-coupon bond that repays the principal and on European options on 10,000 of the index that expire with the note, whose prices are stated below as percentages of that notional. **(a)** An at-the-money call costs 14.0%. Which participation rate can the issuer offer? **(b)** A 100/140 bull call spread (long the at-the-money call, short the call struck at 140% of today's level) costs 10.6%. Which participation rate can a note offer whose index gain is capped at 40%, and what is the largest amount that it can pay? **(c)** If instead $r = 1.5\%$, the at-the-money call costs 10.65%. Which participation rate does the uncapped note then offer? **(d)** Back at $r = 3.5\%$, which is also the six-year rate: a six-year uncapped note faces an at-the-money call at 17.0%, and a four-year uncapped note in a more volatile market faces one at 17.1%. Compute the two participation rates, and say what a longer life, a higher volatility and a cap each do to participation. **(e)** Now suppose that the index pays no dividends and that the issuer takes no fee. Show with put–call parity that the uncapped at-the-money participation rate is below 100% whatever the volatility, as soon as the index has a positive probability of finishing below today's level. Check the argument on four-year at-the-money prices of 19.25% for the call and 6.19% for the put, at $r = 3.5\%$.

27.18 Problems — §18 Binomial Trees and Risk-Neutral Valuation

P18.1 *One-step call and put: the hedge, the risk-neutral check and a mispriced quote*

A stock trades at $S = 64$. Three months from now ($\Delta t = 0.25$) its price is either 73.60 or 56.00, each with positive probability. The stock pays no dividend, the continuously compounded interest rate is $r = 4\%$, and the market is frictionless (short sales and fractional holdings are allowed, and money is borrowed and lent at the rate r). A European call and a European put on the stock are both struck at $K = 66$ and expire at the end of the step; their prices are C and $\mathcal{P}$. **(a)** Check that the tree admits no arbitrage. Find the portfolio of shares and bond that replicates the call, and the price C. **(b)** Compute the risk-neutral up-probability p and price the call again with it. Price the put both ways, and check put–call parity. **(c)** A dealer stands ready to buy or sell the call at 4.20. Give the arbitrage: the trades made today, the amount they bring in, and the value of the position in each state at the end of the step.

P18.2 *Up 120, down 80: the mental-arithmetic round*

A stock trades at $S = 100$ and will be worth either 120 or 80 at the end of one period. It pays no dividend, the market is frictionless and, except in part (f), the interest rate is zero. All options are European and expire at the end of the period. Work without a calculator. **(a)** Price the call struck at 100 twice: with the risk-neutral up-probability, and by building its replicating portfolio of shares and bond. **(b)** Price the put struck at 100 and the straddle struck at 100 (the call plus the put). What does put–call parity say here? **(c)** Price the calls struck at 90 and at 110. **(d)** Price the digital that pays 1 if the stock goes up and nothing otherwise. **(e)** A counterparty is convinced that the stock goes up with probability 0.8 and therefore bids 16 for the call struck at 100. What is the call worth to someone who can trade the stock, and how is the bid turned into a riskless profit? **(f)** Now let 1 placed in the bond grow to 1.05 over the period. Reprice the call and the put struck at 100 and check parity.

P18.3 *Re-derive one-step replication and the no-arbitrage band*

A stock is worth $S > 0$ today and Su or Sd after one step of length Δt, where $0 < d < u$; the two outcomes have the real-world probabilities p_{real} and $1 - p_{\text{real}}$, both positive. The stock pays no dividend. One unit placed in a riskless bond grows to $e^{r\Delta t}$. The market is frictionless: short sales and fractional holdings are allowed, and money is borrowed and lent at the rate r. A portfolio (Δ, B) holds Δ shares and B in the bond. An arbitrage is a portfolio whose value today is at most zero and whose payoff is nonnegative in both states and positive in at least one. In (b) and (c) the derivative is traded too, and a portfolio may also hold it; the definition of an arbitrage is unchanged. **(a)** Prove that the market of the stock and the bond admits no arbitrage if and only if $d < e^{r\Delta t} < u$, and write down an arbitrage portfolio in each case where the condition fails. **(b)** Assume $d < e^{r\Delta t} < u$. A derivative pays V_u in the up state and V_d in the down state. Find the unique portfolio (Δ, B) that replicates it, show that its cost is $e^{-r\Delta t}[pV_u + (1-p)V_d]$ with $p = (e^{r\Delta t} - d)/(u - d)$, and explain why this cost is the only price V of the derivative that leaves no arbitrage. **(c)** Derive the same price by the riskless-portfolio route: choose Δ so that the position long Δ shares and short one derivative is worth the same in both states. **(d)** Show that $0 < p < 1$ and that $pSu + (1-p)Sd = Se^{r\Delta t}$, and say where p_{real} enters the price.

P18.4 *The discount rate the real world would need*

A stock trades at $S = 100$ and moves to 125 or to 85 over six months ($\Delta t = 0.5$). It pays no dividend, the continuously compounded interest rate is $r = 5\%$, the market is frictionless (short sales and fractional holdings are allowed, and money is borrowed and lent at the rate r), and the real-world probability of the up-move is $p_{\text{real}} = 0.6$. A European call and a European put are struck at $K = 100$ and expire at the end of the step. For an asset with price $V > 0$ today and random value $V_{\Delta t}$ at the end of the step, the real-world expected return μ_V is the continuously compounded rate defined by $V = e^{-\mu_V \Delta t}\,\mathbb{E}[V_{\Delta t}]$, where $\mathbb{E}$ is the real-world expectation; for the stock it is written μ. **(a)** Price the call and the put by risk-neutral valuation and check put–call parity. **(b)** Compute μ for the stock and μ_V for the call and for the put, in percent per year. **(c)** Let a derivative with the payoffs V_u and V_d, priced without arbitrage, be replicated by Δ shares and B in the bond. Prove that $\mathbb{E}[V_{\Delta t}] - Ve^{r\Delta t} = \Delta\,(\mathbb{E}[S_{\Delta t}] - Se^{r\Delta t})$, and check the identity on the call and on the put, giving the elasticity $\Delta S/V$ of each. **(d)** Why is the rate of the call far above that of the stock, and the rate of the put far below r? What does discounting the real-world expected payoff of the call at r, or at μ, give?

P18.5 *Two-step puts: European, American, parity and the strike for immediate exercise*

A stock trades at $S = 60$ and pays no dividend. Over each of two six-month steps ($\Delta t = 0.5$, so $T = 1$) its price is multiplied by $u = 1.12$ or by $d = 0.88$. The continuously compounded interest rate is $r = 7\%$. The options below expire at T and, except in part (d), are struck at $K = 62$. **(a)** Compute the risk-neutral up-probability and the price of the European put, giving its value at both nodes at time Δt. **(b)** Price the European call and check put–call parity. **(c)** Price

the American put by backward induction. At which nodes is it exercised early, and what is the early-exercise premium? **(d)** Keep the tree and let the strike vary over $60 < K \le 67.2$. Write the continuation value of the American put at time 0 (the value of not exercising at time 0 and exercising optimally afterwards) as a function of K, and find the strikes for which exercising at once is at least as good as continuing.

P18.6 *A claim on the square of the price: one step, n steps and the limit*

A stock that pays no dividend trades at S, and the continuously compounded interest rate is r. A claim pays the square of the stock price at its maturity. **(a)** *One step.* Let $S = 60$ move to 66 or to 57 ($u = 1.1$, $d = 0.95$) over $\Delta t = 0.25$, with $r = 4\%$. Price the claim that pays $S_{\Delta t}^2$ by the risk-neutral expectation and by replication, and show that both equal $S^2[(u+d) - ud\,e^{-r\Delta t}]$. **(b)** *n steps.* On an n-step tree with the same u, d, r and Δt at every step, show that the claim that pays S_T^2 at $T = n\Delta t$ is worth $V_n = S^2[(u+d) - ud\,e^{-r\Delta t}]^n$. Evaluate V_2 with the data of (a) and compare it with $S^2 e^{rT}$. **(c)** *The limit.* Now fix T and a volatility $\sigma > 0$, and build the tree with $\Delta t = T/n$, $u = e^x$, $d = e^{-x}$, $x = \sigma\sqrt{\Delta t}$ and the risk-neutral up-probability $p = (e^{r\Delta t} - d)/(u - d)$, for every n with $\sigma > |r|\sqrt{T/n}$, so that $0 < p < 1$. Let R_1 be the log-return of the first step, equal to x with risk-neutral probability p and to $-x$ otherwise, and put $\nu = r - \sigma^2/2$. Prove that, for every real θ,

$$\left(\mathbb{E}^{\mathbb{Q}} e^{\theta R_1}\right)^n \to e^{\theta\nu T + \theta^2\sigma^2 T/2} \qquad \text{as } n \to \infty,$$

where $\mathbb{E}^{\mathbb{Q}}$ is the risk-neutral expectation. You may use the expansion $e^y = 1 + y + y^2/2 + y^3/6 + O(y^4)$ as $y \to 0$, Taylor's theorem with the Lagrange form of the remainder, and the expansion $(2p-1)\,x = \nu\Delta t + O(\Delta t^{3/2})$ of the mean log-return of a step. Deduce the limit of the price V_n of the claim on this tree, and evaluate V_1, V_{12} and the limit for $S = 60$, $\sigma = 20\%$, $r = 4\%$ and $T = 1$. **(d)** In (b) and in (c) the claim is worth more than $S^2 e^{rT}$, the discounted square of the forward price Se^{rT}. Identify the difference exactly, and evaluate it in the limit of (c).

P18.7 *Delta, the cash account and the self-financing check on a two-step put*

A stock trades at $S = 120$ and pays no dividend. Over each of two six-month steps ($\Delta t = 0.5$) its price is multiplied by $u = 1.1$ or by $d = 0.92$. The continuously compounded interest rate is $r = 2.5\%$. A European put with strike $K = 125$ expires after the two steps. **(a)** Compute the risk-neutral up-probability and the value of the put at the three nodes that precede maturity. **(b)** A trader who has sold the put covers it by running its replicating strategy in shares and the bond. Give the number of shares and the amount in the bond at time 0 and at each of the two nodes at time Δt. Why is the holding at the down node (the node reached by a first down-move) exactly -1 share? **(c)** Check that the strategy is self-financing: the position set up at time 0 must be worth, at each node at time Δt, exactly the cost of the position held from that node. Which trade is made at each node? Why can no position in shares and the bond that is set up at time 0 and never traded again replicate the put? **(d)** Estimate the gamma of the put from the tree, as the share holding at the up node minus the share holding at the down node, divided by half the distance between the highest and the lowest terminal price. **(e)** If the put were American, at which node would it be exercised early, and what would it be worth today?

P18.8 *Re-derive the CRR moments, then sharpen them*

On a Cox–Ross–Rubinstein tree with volatility $\sigma > 0$, continuously compounded interest rate r and step Δt, the factors are $u = e^x$ and $d = e^{-x}$ with $x = \sigma\sqrt{\Delta t}$, and the risk-neutral up-probability is $p = (e^{r\Delta t} - d)/(u - d)$. The log-return R_1 of one step equals x with probability p and $-x$ with probability $1 - p$; its risk-neutral mean and variance are $\mathbb{E}^{\mathbb{Q}}[R_1]$ and $\mathrm{Var}^{\mathbb{Q}}(R_1)$. You may use $e^y = 1 + y + y^2/2 + y^3/6 + O(y^4)$ and $1/(1-y) = 1 + y + O(y^2)$ as $y \to 0$. **(a)** Show that

$\mathbb{E}^{\mathbb{Q}}[R_1] = (2p-1)\,x$ and $\mathrm{Var}^{\mathbb{Q}}(R_1) = 4p(1-p)\,x^2$ exactly, and prove that, as $\Delta t \to 0$,

$$p = \frac{1}{2} + \frac{(r - \sigma^2/2)\sqrt{\Delta t}}{2\sigma} + O(\Delta t),$$

$$\mathbb{E}^{\mathbb{Q}}[R_1] = \Big(r - \frac{\sigma^2}{2}\Big)\Delta t + O(\Delta t^{3/2}), \qquad \mathrm{Var}^{\mathbb{Q}}(R_1) = \sigma^2 \Delta t + O(\Delta t^2).$$

(b) Carry the cubic terms of the exponentials and show that the error of the expansion of p is in fact $O(\Delta t^{3/2})$, and that of the expansion of $\mathbb{E}^{\mathbb{Q}}[R_1]$ is $O(\Delta t^2)$. **(c)** Under the real-world probability each move is up with probability $p_{\text{real}} = (e^{\mu \Delta t} - d)/(u - d)$, where μ is the expected return of the stock. What becomes of the three expansions, and what does the answer say about the volatility under the two probabilities? **(d)** Check (a) to (c) numerically for weekly steps, $\Delta t = 1/52$, with $\sigma = 30\%$, $r = 2\%$ and $\mu = 12\%$.

P18.9 *Futures options: a zero-cost hedge, and a call exercised early*

Throughout, a futures position costs nothing to enter, and the gain or loss of a contract over a step of the tree is settled at the end of the step. The continuously compounded interest rate is r, and the options pay on the futures price itself. **(a)** A futures price stands at $F = 60$ and will be 66 or 54 in three months ($\Delta t = 0.25$); $r = 5\%$. A European call struck at $K = 58$ pays $(F_{\Delta t} - 58)^+$ in three months. Find the replicating position in futures contracts and the bond, the risk-neutral up-probability and the price of the call. How many contracts hedge one short call? **(b)** A futures price stands at $F = 120$ with volatility $\sigma = 20\%$; $r = 4\%$. On a two-step Cox–Ross–Rubinstein tree over $T = 0.5$ (so $\Delta t = 0.25$, $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$), price the European call, the American call, the European put and the American put, all struck at $K = 115$. Say where early exercise occurs, give the early-exercise premium, and check the parity relation $C - \mathcal{P} = e^{-rT}(F - K)$ for the European pair. **(c)** An American call on a stock that pays no dividend is never exercised early when $r > 0$. Why is the call of (b) exercised early all the same?

P18.10 *A currency tree: CRR parameters, admissibility and an American call exercised early*

An exchange rate S is the domestic price of one unit of a foreign currency. The domestic and the foreign continuously compounded interest rates are r and r_{for}; one unit of foreign currency on deposit grows to $e^{r_{\text{for}}\Delta t}$ units over a step of length Δt. All trees are Cox–Ross–Rubinstein trees, with $u = e^{\sigma\sqrt{\Delta t}}$ and $d = 1/u$ for the volatility σ of the exchange rate. **(a)** Take $S = 1.25$, $\sigma = 10\%$, $r = 2\%$, $r_{\text{for}} = 6\%$ and two steps over $T = 0.5$. Compute u, d and the risk-neutral up-probability. Price the European and the American call struck at $K = 1.20$, and the European and the American put with the same strike. Say where early exercise occurs, and check put–call parity for the European pair, $C - \mathcal{P} = Se^{-r_{\text{for}}T} - Ke^{-rT}$. **(b)** With the strike lowered to $K = 1.18$, what should the holder of the American call do at time 0? **(c)** A managed currency has $\sigma = 2\%$, with $r = 1\%$ and $r_{\text{for}} = 5\%$. For which step lengths Δt does the risk-neutral up-probability lie strictly between 0 and 1? Compute it for $\Delta t = 1/12$, 0.25 and 0.5, and for $\Delta t = 0.5$ exhibit the arbitrage that the tree contains.

P18.11 *Counting steps: oscillation and averaging on the CRR tree*

A stock trades at $S = 60$ and pays no dividend; the continuously compounded interest rate is $r = 3\%$ and the volatility is $\sigma = 25\%$. A European call struck at $K = 60$ expires in nine months, $T = 0.75$. Its price on a Cox–Ross–Rubinstein tree with n steps ($\Delta t = T/n$, $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, risk-neutral up-probability $p = (e^{r\Delta t} - d)/(u - d)$) is written C_n. As $n \to \infty$ these prices converge to 5.8097, the Black–Scholes–Merton price. A computer returns $C_3 = 6.2353$, $C_4 = 5.5011$, $C_{50} = 5.7841$ and $C_{51} = 5.8342$. **(a)** Compute C_1 and C_2 by hand, and confirm the given C_3 and C_4 with the same formula. **(b)** For the six trees, tabulate the error $C_n - 5.8097$ and n times the error. What is the pattern? **(c)** Explain the sign of the error for even and for odd n. Estimate the limit from C_{50} and C_{51} alone, and estimate how many steps a single tree needs for an error below one cent. **(d)** Is the risk-neutral up-probability above or below $\frac{1}{2}$ on these trees, although $r > 0$? Give the exact criterion and its form to leading order in Δt.

27.19 Problems — §19 Brownian Motion, Itô's Lemma and the Lognormal Model

P19.1 *Sqrt-t vol scaling*

An index has an annualised volatility of 20%. (a) What is its typical daily move? Its typical weekly move? (b) A colleague annualises a 1.26% daily vol by multiplying by 252 and reports 318% — what exactly went wrong? (c) State the assumption behind the square-root-of-time rule and give one realistic way markets violate it.

P19.2 *Brownian moments and quadratic variation from scratch*

Let W be a standard Brownian motion: $W_0 = 0$; for $0 \le s < t$ the increment $W_t - W_s$ has the law $\mathcal{N}(0, t-s)$; for every grid $0 \le t_0 < t_1 < \cdots < t_k$ the increments over its cells are mutually independent; and the paths are continuous. For a partition $0 = t_0 < t_1 < \cdots < t_n = t$ of $[0,t]$ write $\Delta t_i = t_i - t_{i-1}$, $\Delta W_i = W_{t_i} - W_{t_{i-1}}$, $\Delta t_{\max} = \max_i \Delta t_i$,

$$\mathrm{QV}_n = \sum_{i=1}^{n} (\Delta W_i)^2 \qquad \text{and} \qquad \mathrm{FV}_n = \sum_{i=1}^{n} |\Delta W_i|.$$

For $Z \sim \mathcal{N}(0,1)$ you may use $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$, $\mathbb{E}|Z| = \sqrt{2/\pi}$ and $\mathbb{E}|Z|^3 = 2\sqrt{2/\pi} = 1.596$. **(a)** Prove that $\mathbb{E}[W_s W_t] = \min(s,t)$ for all $s, t \ge 0$. Deduce $\mathrm{Corr}(W_1, W_4)$ and $\mathrm{Var}(W_1 + W_2 + W_3)$, where $W_1, \dots, W_4$ are the values of the motion at the times $1, \dots, 4$. **(b)** Prove that $\mathbb{E}[\mathrm{QV}_n] = t$ and $\mathrm{Var}(\mathrm{QV}_n) = 2\sum_i (\Delta t_i)^2 \le 2t\,\Delta t_{\max}$, so that $\mathrm{QV}_n \to t$ in mean square as $\Delta t_{\max} \to 0$. Prove also that $\sum_i \Delta t_i\,\Delta W_i$ has mean 0 and second moment $\sum_i (\Delta t_i)^3 \le t\,\Delta t_{\max}^2$, and that $\sum_i (\Delta t_i)^2 \le t\,\Delta t_{\max}$. Which three rules of stochastic calculus do these statements make precise? **(c)** For 100 equal steps on $[0,2]$, give the mean, the variance and the standard deviation of QV_n. **(d)** Show that for n equal steps $\mathbb{E}[\mathrm{FV}_n] = \sqrt{2nt/\pi}$, and evaluate it at $t = 1$, $n = 400$. Show that for every partition $\mathbb{E}\big[\sum_i |\Delta W_i|^3\big] \le 1.596\, t\, \sqrt{\Delta t_{\max}}$. What do the two results, read together with (b), say about a Brownian path and about a Taylor expansion along it? **(e)** Let $c > 0$. Show that a path with $|W_s - W_{s'}| \le c\,|s - s'|$ for all $s, s' \in [0,t]$ has $\mathrm{QV}_n \le c^2\, t\, \Delta t_{\max}$ for every partition, and deduce that the paths with this property lie in an event of probability zero. What does this say about the differentiability of Brownian paths?

P19.3 *A cash balance as a generalised Wiener process*

A proprietary trading firm models its free cash X_t, in millions of dollars with t in months, as a generalised Wiener process $dX = \alpha\,dt + \beta\,dW$, that is, $X_t = X_0 + \alpha t + \beta W_t$ with W a standard Brownian motion. Today $X_0 = 4$; the drift is $\alpha = 0.25$ per month and the variance rate is $\beta^2 = 1$ per month. A negative balance forces the firm to draw on a credit line. **(a)** What is the law of X_3? **(b)** What is the probability that the balance is negative at the end of month 12? **(c)** What starting balance would make that probability 1%? **(d)** With $X_0 = 4$, at which horizon $t > 0$, within the first year or beyond it, is $P(X_t < 0)$ largest, and how large is it there? **(e)** What is the probability that the balance touches zero at some time during the first 12 months? You may use the formula, valid for a level $a > 0$, a drift μ, a volatility $\sigma > 0$ and a horizon T,

$$P\Big(\min_{0 \le t \le T} (\mu t + \sigma W_t) \le -a\Big) = \Phi\Big(\frac{-a - \mu T}{\sigma\sqrt{T}}\Big) + e^{-2\mu a/\sigma^2}\, \Phi\Big(\frac{-a + \mu T}{\sigma\sqrt{T}}\Big).$$

Compare with (b). **(f)** Suppose instead that during the second half of the year, for $6 < t \le 12$, the drift is -0.1 per month and the variance rate is 2 per month, that is, $X_t = X_6 - 0.1\,(t-6) + \sqrt{2}\,(W_t - W_6)$ for $6 < t \le 12$, with $X_t = 4 + 0.25\,t + W_t$ for $t \le 6$. What is the law of X_{12}, and what is $P(X_{12} < 0)$? You may use $\Phi(-2.02) = 0.0217$, $\Phi^{-1}(0.99) = 2.326$, $\Phi(-2) = 0.0228$, $\Phi(-0.289) = 0.3863$, $\Phi(-1.155) = 0.1240$ and $e^{-2} = 0.1353$.

P19.4 *Which process for a price?*

Four discretised models are proposed for the change ΔS of a stock price S over a short step Δt (in years), with $Z \sim \mathcal{N}(0,1)$:

$$(1)\quad \Delta S = \alpha\,\Delta t + \beta\sqrt{\Delta t}\;Z, \qquad (2)\quad \Delta S = \mu S\,\Delta t + \beta\sqrt{\Delta t}\;Z,$$
$$(3)\quad \Delta S = \alpha\,\Delta t + \sigma S\sqrt{\Delta t}\;Z, \qquad (4)\quad \Delta S = \mu S\,\Delta t + \sigma S\sqrt{\Delta t}\;Z.$$

The constants are $\alpha = 4$ dollars a year, $\beta = 16$ dollars per square root of a year, $\mu = 10\%$ and $\sigma = 40\%$ a year, so that the four models agree for a stock at $S = 40$. **(a)** For each model say which quantities are constant: the dollar drift or the percentage drift, the dollar noise or the percentage noise. Then give the mean and the standard deviation of next week's move ($\Delta t = 1/52$) for a stock at 16 and for a stock at 160, in dollars and as a percentage of the price. Which model should describe a stock, and why? **(b)** In continuous time model (1) reads $S_t = S_0 + \alpha t + \beta W_t$, with W a standard Brownian motion. Starting from $S_0 = 40$, find the law of S_2 and the probability $P(S_2 < 0)$ that the price at time $t = 2$ is negative (the value at that date, not a dip below zero at some earlier time); you may use $\Phi(-2.12) = 0.0170$. What is $P(S_2 < 0)$ under the continuous-time form of model (4), the geometric Brownian motion $S_t = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\right)$? **(c)** Name three financial quantities for which the arithmetic model (1) is the right choice, and say what they have in common.

P19.5 *Itô's lemma on functions of a stock price*

A stock price follows the geometric Brownian motion $S_t = S_0 \exp\left((\mu - \frac{1}{2}\sigma^2)\,t + \sigma W_t\right)$, that is, $dS = \mu S\,dt + \sigma S\,dW$, where W is a standard Brownian motion, $\mu = 9\%$ and $\sigma = 30\%$. **(a)** Apply Itô's lemma to S^a for a real exponent $a \neq 0$, and write the result in the form $d(S^a)/S^a = \dots$. What kind of process is S^a? Give its drift and its volatility for $a = 3$, $a = -1$ and $a = \frac{1}{2}$. **(b)** An exchange rate, in dollars per euro, is a geometric Brownian motion with a drift of 0.5% a year and a volatility of 10%. Find the drift of the reciprocal rate, in euros per dollar, and comment. **(c)** Let $r = 4\%$ and $q = 2\%$ be constants and T a fixed date. Find the processes followed by the discounted price $e^{-rt}S_t$ and by the forward price $F_t = S_t\,e^{(r-q)(T-t)}$. **(d)** Find the process followed by $\ln S$, and say what is wrong with the argument "by the chain rule $d\ln S = dS/S$, so $\ln S$ drifts at 9%".

P19.6 *Itô's lemma on Brownian motion from the sums, and which processes have zero drift*

Let W be a standard Brownian motion and fix $t > 0$. Divide $[0,t]$ into n equal steps of length $\Delta t = t/n$, with the points $t_i = i\,\Delta t$, and write $\Delta W_i = W_{t_i} - W_{t_{i-1}}$ and $\mathrm{QV}_n = \sum_{i=1}^n (\Delta W_i)^2$. You may use the following. For $Z \sim \mathcal{N}(0,1)$, $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$ and $\mathbb{E}[Z^6] = 15$. For a polynomial ξ, the left-endpoint sums $\sum_i \xi(W_{t_{i-1}})\,\Delta W_i$ converge in mean square, as $n \to \infty$, to the Itô integral $\int_0^t \xi(W_s)\,dW_s$, which has mean zero. Convergence in mean square, or on every path, implies convergence in probability; limits in probability add, and two limits in probability of one sequence agree with probability one. In (d) and (e) you may take expectations in Itô's lemma: for the functions f used there, $\mathbb{E}f(t,W_t) = f(0,0) + \int_0^t \mathbb{E}\big[(\partial_t f + \frac{1}{2}\,\partial_{xx} f)(s,W_s)\big]\,ds$. **(a)** Prove the exact identities

$$W_t^2 = 2\sum_{i=1}^n W_{t_{i-1}}\,\Delta W_i + \mathrm{QV}_n, \qquad W_t^3 = 3\sum_{i=1}^n W_{t_{i-1}}^2\,\Delta W_i + 3\sum_{i=1}^n W_{t_{i-1}}\,(\Delta W_i)^2 + \sum_{i=1}^n (\Delta W_i)^3.$$

Then show that

$$\mathbb{E}\big[(\mathrm{QV}_n - t)^2\big] = \frac{2t^2}{n}, \qquad \mathbb{E}\Big[\Big(3\sum_{i=1}^n W_{t_{i-1}}\big((\Delta W_i)^2 - \Delta t\big)\Big)^2\Big] = \frac{9t^3(n-1)}{n^2},$$
$$\mathbb{E}\Big[\Big(\sum_{i=1}^n (\Delta W_i)^3\Big)^2\Big] = \frac{15t^3}{n^2},$$

evaluate the three at $t = 1$ and $n = 100$, and let $n \to \infty$ to conclude that $d(W^2) = dt + 2W\,dW$ and $d(W^3) = 3W\,dt + 3W^2\,dW$. **(b)** With Itô's lemma, $df(t, W_t) = (\partial_t f + \frac{1}{2}\,\partial_{xx} f)\,dt + \partial_x f\,dW$, find $d(e^W)$, $d(tW)$ and $d(\sin W)$. **(c)** Which of the following processes have zero drift: $W_t^2 - t$, W_t^3, $W_t^3 - 3tW_t$, $e^{\sigma W_t - \sigma^2 t/2}$ (with σ a constant), tW_t, $e^{t/2}\cos W_t$? **(d)** Compute $d(W^4)$ and deduce $\mathbb{E}[W_t^4]$. Check the result against the law of W_t. **(e)** Let $\chi(t) = \mathbb{E}\,e^{2W_t}$. Use $d(e^{2W})$ to obtain a differential equation for χ, solve it, and check the result against $\mathbb{E}\,e^{\theta Z} = e^{\theta^2/2}$.

P19.7 *The Gaussian tilt from scratch*

Let $Z \sim \mathcal{N}(0,1)$, with density $\varphi(z) = e^{-z^2/2}/\sqrt{2\pi}$ and distribution function Φ, and let θ and c be real numbers. In (a) and (b), ψ is a function on the real line, continuous except at finitely many points, of exponential growth: $|\psi(z)| \le c'\,e^{c'|z|}$ for all real z, for some constant $c' \ge 0$, a growth constant unrelated to the threshold c of (a) (bounded functions qualify). **(a)** By completing the square, prove that

$$\mathbb{E}\big[e^{\theta Z}\,\psi(Z)\big] = e^{\theta^2/2}\,\mathbb{E}\big[\psi(Z+\theta)\big],$$

and deduce $\mathbb{E}\,e^{\theta Z} = e^{\theta^2/2}$ and $\mathbb{E}\big[e^{\theta Z}\,\mathbf{1}\{Z > c\}\big] = e^{\theta^2/2}\,\Phi(\theta - c)$. **(b)** Let Z_1 and Z_2 be independent standard normal variables, and let

$$X = \mathbb{E}[X] + a_{X,1}Z_1 + a_{X,2}Z_2, \qquad Y = \mathbb{E}[Y] + a_{Y,1}Z_1 + a_{Y,2}Z_2,$$

with real coefficients and $(a_{Y,1}, a_{Y,2}) \neq (0,0)$: a jointly normal pair, written with its means. Prove that

$$\mathbb{E}\,e^{Y} = e^{\mathbb{E}[Y] + \mathrm{Var}(Y)/2} \qquad \text{and} \qquad \mathbb{E}\big[e^{Y}\,\psi(X)\big] = \mathbb{E}\big[e^{Y}\big]\,\mathbb{E}\big[\psi(X + \mathrm{Cov}(X,Y))\big].$$

You may use two facts. A rotation of the pair, $Z_1' = c_1 Z_1 + c_2 Z_2$ and $Z_2' = -c_2 Z_1 + c_1 Z_2$ with $c_1^2 + c_2^2 = 1$, is again a pair of independent standard normal variables. And the mean of a function of two independent variables may be computed by first averaging over one of them with the other frozen, provided that the function is nonnegative or the mean of its absolute value is finite. **(c)** A price follows the geometric Brownian motion $S_t = S_0 \exp\big((\mu - \frac{1}{2}\sigma^2)\,t + \sigma W_t\big)$, $t \ge 0$, with $S_0 > 0$, a real drift μ, a volatility $\sigma > 0$ and W a standard Brownian motion; for a horizon $T > 0$, $S_T = S_0 \exp\big((\mu - \frac{1}{2}\sigma^2)\,T + \sigma\sqrt{T}\,Z\big)$ with $Z = W_T/\sqrt{T}$. Show that $\mathbb{E}[S_T^a] = S_0^a \exp\big(a\mu T + \frac{1}{2}\,a(a-1)\,\sigma^2 T\big)$ for every real a, and reconcile this with the drift $a\mu + \frac{1}{2}\,a(a-1)\,\sigma^2$ that Itô's lemma gives for the process S^a. **(d)** Show that, for $K > 0$,

$$P(S_T > K) = \Phi(d), \qquad \mathbb{E}\big[S_T\,\mathbf{1}\{S_T > K\}\big] = S_0\,e^{\mu T}\,\Phi\big(d + \sigma\sqrt{T}\big),$$
$$d = \frac{\ln(S_0/K) + (\mu - \frac{1}{2}\sigma^2)\,T}{\sigma\sqrt{T}}.$$

Evaluate both, and $\mathbb{E}\big[(S_T - K)^+\big]$, for $S_0 = 100$, $\mu = 8\%$, $\sigma = 25\%$, $T = 1$ and $K = 110$, using $\Phi(-0.18624) = 0.42613$ and $\Phi(0.06376) = 0.52542$. Is the last number, discounted, the price of the call?

P19.8 *Mean, median, mode and range of a future price*

A stock trades at $S_0 = 80$. Its price follows a geometric Brownian motion with $\mu = 10\%$ and $\sigma = 40\%$, so that $S_T = S_0 \exp\big((\mu - \frac{1}{2}\sigma^2)\,T + \sigma\sqrt{T}\,Z\big)$ with $Z \sim \mathcal{N}(0,1)$. You may use $\ln 80 = 4.3820$ and the standard normal quantile $\Phi^{-1}(0.975) = 1.96$. **(a)** For $T = 0.25$, give the law of $\ln S_T$; then the mean, the median, the mode and the standard deviation of S_T; and its central 95% interval, between the 2.5% and the 97.5% quantiles. **(b)** Give the mean, the standard deviation and the central 95% interval of the price one trading day ahead, $T = 1/252$. **(c)** Prove that, with $g = \mu - \frac{1}{2}\sigma^2$, the central 95% interval of S_T is $\big[S_0\,e^{gT - 1.96\,\sigma\sqrt{T}},\ S_0\,e^{gT + 1.96\,\sigma\sqrt{T}}\big]$, and explain why it is not symmetric about the mean. How wrong is the range "mean plus or minus 1.96 standard deviations" in (a), and in (b)?

P19.9 *Real-world exercise probabilities*

A stock that pays no dividend trades at $S_0 = 80$ and follows a geometric Brownian motion with expected return $\mu = 14\%$ and volatility $\sigma = 25\%$, so that $\ln S_T \sim \mathcal{N}\big(\ln S_0 + (\mu - \frac{1}{2}\sigma^2)\,T,\ \sigma^2 T\big)$.

(a) A European call and a European put have the strike $K = 90$ and the maturity $T = 0.75$. Find the real-world probability that each is exercised. **(b)** Find the probability that the price has at least doubled after three years. **(c)** Find the probability that the stock shows a loss after one year, that is, ends the year below S_0. **(d)** The risk-free rate is $r = 4\%$. The risk-neutral probability that the call of (a) is exercised is obtained by replacing μ with r. Compute it. Why does it differ from the answer to (a)? If the expected return is re-estimated at $\mu = 20\%$, what happens to each of the two probabilities and to the price of the call? Which of the two can be read off market prices? You may use $\Phi(-0.1673) = 0.4336$, $\Phi(-0.8473) = 0.1984$, $\Phi(-0.4350) = 0.3318$, $\Phi(-0.5137) = 0.3037$ and $\Phi(0.0406) = 0.5162$.

P19.10 *Average versus compounded returns, and the leveraged product*

(a) A fund returns $+18\%$, -12%, $+27\%$, -6% and $+8\%$ in five successive years. Compute the arithmetic mean return, the geometric mean return (the compound rate per year), and the estimate "arithmetic mean minus half the variance", using the variance of the five returns about their mean with divisor 5. **(b)** A manager reports an average annual return of 18% with a volatility of 30%. At about what rate did the investors compound, and why is the reported figure misleading? **(c)** An index follows a geometric Brownian motion with $\mu = 7\%$ and $\sigma = 18\%$. Give the law of its realised growth rate over five years, $\hat{g}_5 = \frac{1}{5}\ln(S_5/S_0)$, a 95% interval for it, and $P(\hat{g}_5 < 0)$, using $\Phi(-0.668) = 0.252$. How many years of data would pin the growth rate down to $\pm 1\%$ at the 95% level? **(d)** A product with leverage L is rebalanced continuously so that the fraction L of its value is always invested in the index of (c), the rest being held or borrowed at a zero rate: $dV/V = L\,dS/S$. Find its growth rate $g_L = \frac{1}{T}\,\mathbb{E}\ln(V_T/V_0)$ for $L = 1, 2, 3, -1$ and -2, the leverage that maximises g_L with the growth it attains, and the nonzero leverage at which the growth is zero. **(e)** The index rises 10% on one day and falls 10% on the next. What is the value of one dollar in the index and in the $2\times$, $-1\times$ and $3\times$ products, each rebalanced at the close of every day? **(f)** Over a year in which the index ends where it started, how much do the $2\times$, $-1\times$, $3\times$ and $-2\times$ products of (d) lose?

P19.11 *Correlated Brownian motions: construction, ratios and the FX triangle*

Throughout, $W_{k,t}$ is the value at time t of the Brownian motion W_k, and two standard Brownian motions have correlation ρ if their increments over each interval $[s, t]$ are jointly normal with covariance $\rho\,(t - s)$ and, for every grid $0 \le t_0 < t_1 < \cdots < t_n$, their pairs of increments over the n cells are mutually independent. **(a)** Let W_1 and $W_\perp$ be independent standard Brownian motions and let $\rho = -0.4$. Find the weight $w > 0$ for which $W_2 = \rho\,W_1 + w\,W_\perp$ is a standard Brownian motion, prove that it is one, and prove that $\mathrm{Cov}(W_{1,t}, W_{2,t}) = \rho\,t$. Turn the independent standard normal draws $Z_1 = 1.2$ and $Z_\perp = -0.7$ into one draw of a pair of standard normal variables with correlation -0.4. **(b)** Let $X_{k,t} = X_{k,0} + \alpha_k t + \beta_k W_{k,t}$, $k = 1, 2$, be two generalised Wiener processes with constant coefficients, driven by Brownian motions with an arbitrary correlation $\rho \in [-1, 1]$. Show that $X_1 + X_2$ is again a process of this kind, and find its drift and its variance rate. **(c)** Two stocks follow $S_{k,t} = S_{k,0}\exp\left((\mu_k - \frac{1}{2}\sigma_k^2)\,t + \sigma_k W_{k,t}\right)$ with $S_{1,0} = S_{2,0} = 50$, $\mu_1 = 6\%$, $\mu_2 = 10\%$, $\sigma_1 = 25\%$, $\sigma_2 = 35\%$, and motions with correlation 0.5. Is the value of a portfolio of one share of each a geometric Brownian motion? Proceed as follows: prove that every lognormal variable X satisfies $(\mathbb{E}[X^2])^3 = (\mathbb{E}[X])^3\,\mathbb{E}[X^3]$, and test this identity on $X = S_{1,1} + S_{2,1}$, the value after one year. Then use the two-dimensional Itô lemma to explain what goes wrong. **(d)** Apply the two-dimensional Itô lemma to $V(x_1, x_2) = x_1/x_2$ to find the drift and the volatility of the ratio S_1/S_2 of two such stocks, and state the volatility of the product $S_1 S_2$. Evaluate the two volatilities for $\sigma_1 = 25\%$, $\sigma_2 = 35\%$ and $\rho = 0.4$. **(e)** With $\sigma_1 = 25\%$ and $\sigma_2 = 35\%$, for which correlations is the ratio less volatile than the less volatile of the two stocks? **(f)** The implied volatilities of the exchange rates EURUSD, GBPUSD and EURGBP are 8%, 9% and 6%, and EURGBP is the ratio of EURUSD to GBPUSD. Which correlation between EURUSD and GBPUSD do the three quotes imply?

27.20 Problems — §20 The Black–Scholes–Merton Model

P20.1 *Deriving the Black–Scholes–Merton equation*

A stock pays no dividends and follows $dS = \mu S\,dt + \sigma S\,dW$ with constants μ and $\sigma > 0$. Money is borrowed and lent at one constant rate r, trading is continuous and free, short sales are allowed, and there is no arbitrage. A derivative that pays $f(S_T)$ at T trades at every time $t < T$ at the price $V_t = V(S_t, t)$, where V is continuous on $(0,\infty)\times[0,T)$ with continuous partial derivatives $\partial_t V$, $\partial_S V$ and $\partial_{SS} V$ there, and V and these three derivatives have moderate growth on $(0,\infty)\times[0,T']$ for every $T' < T$, so that Itô's lemma applies on every interval $[0,T']$. Write

$$\mathcal{L}V = \partial_t V + rS\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V - rV.$$

(a) (The whiteboard argument.) Form the portfolio that is short one derivative and long $\partial_S V$ shares, apply Itô's lemma, and derive $\mathcal{L}V = 0$. Say where the absence of arbitrage enters, why the portfolio must be rebalanced, and why μ drops out. **(b)** (The exact argument.) A self-financing strategy that holds $h_{k,t}$ units of each instrument k, with price $A_{k,t}$, and keeps the rest of its value Π_t in the bank satisfies $d(e^{-rt}\,\Pi_t) = \sum_k h_{k,t}\,d(e^{-rt}A_{k,t})$. Consider the strategy with zero initial capital that holds $\mathcal{L}V(S_t,t)$ units of the derivative and $-\mathcal{L}V(S_t,t)\,\partial_S V(S_t,t)$ shares. Show that

$$e^{-rt}\,\Pi_t = \int_0^t e^{-rs}\,\big(\mathcal{L}V(S_s,s)\big)^2\,ds,$$

with no stochastic integral left, and conclude that $\mathcal{L}V = 0$ at every point of $(0,\infty)\times[0,T)$. **(c)** Take $r = 5\%$ and $\sigma = 20\%$, and write $\tau = T - t$. Compute $\mathcal{L}V$ for each of the following functions and decide which of them can be the price of a traded derivative: (1) $e^{-r\tau}$; (2) S^2; (3) $S^{-2.5}$; (4) $\ln S$; (5) $e^{-r\tau}\big[\ln S + (r - \tfrac{1}{2}\sigma^2)\tau\big]$. For each function that passes, name the payoff at T of the claim that it prices.

P20.2 *The yield equation by a change of variables; power and log claims on a stock with a yield*

Let $\tilde V(\tilde S, t)$ be a function on $(0,\infty)\times[0,T)$ with continuous partial derivatives $\partial_t \tilde V$, $\partial_{\tilde S}\tilde V$ and $\partial_{\tilde S\tilde S}\tilde V$ that solves the Black–Scholes–Merton equation of an asset without income,

$$\partial_t \tilde V + r\tilde S\,\partial_{\tilde S}\tilde V + \tfrac{1}{2}\sigma^2\tilde S^2\,\partial_{\tilde S\tilde S}\tilde V = r\tilde V.$$

(a) An asset pays a continuous yield q. The holding of $e^{-q(T-t)}$ units of it, with the income reinvested, is worth $\tilde S = Se^{-q(T-t)}$ and pays no income. Show by the chain rule that $V(S,t) = \tilde V(Se^{-q(T-t)}, t)$ satisfies

$$\partial_t V + (r-q)S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V = rV.$$

(b) A futures price F for delivery at $T_F \ge T$ has the prepaid forward $\tilde F = Fe^{-r(T_F - t)}$. Show that $V^{\text{fut}}(F,t) = \tilde V(Fe^{-r(T_F-t)}, t)$ satisfies $\partial_t V^{\text{fut}} + \tfrac{1}{2}\sigma^2 F^2\,\partial_{FF}V^{\text{fut}} = rV^{\text{fut}}$. Which yield does a futures price behave as if it had? **(c)** A stock trades at $S = 50$ and pays the yield $q = 1\%$; its volatility is $\sigma = 30\%$ and the rate is $r = 3\%$. Under the risk-neutral law, $S_T = S\exp\big((r - q - \tfrac{1}{2}\sigma^2)\tau + \sigma\sqrt{\tau}\,Z\big)$ with $Z \sim \mathcal{N}(0,1)$ and $\tau = T - t$. Price, as discounted risk-neutral expectations, the claim that pays S_T^3 and the claim that pays $\ln S_T$ at T, for $\tau = 2$ (use $\mathbb{E}\,e^{\theta Z} = e^{\theta^2/2}$), and confirm that each price function satisfies the equation of (a). **(d)** With the same r, q and σ, find the exponents a for which the time-independent function S^a satisfies the equation of (a). Compare with the case $q = 0$, and explain why S itself is no longer a solution once $q > 0$.

P20.3 *A call and a put from the formula*

A stock that pays no dividends trades at $S = 80$ with volatility $\sigma = 25\%$; the continuously compounded rate is $r = 4\%$. European options on it have strike $K = 84$ and expiry $T = 0.5$, and are priced with the Black–Scholes–Merton model. Use $\Phi(0.0745) = 0.5297$ and $\Phi(0.2513) = 0.5992$.

(a) Compute d_1, d_2, the call price C and the put price $\mathcal{P}$. **(b)** Verify put–call parity on the two prices. **(c)** What is the price of the American call with the same strike and expiry? **(d)** A buyer finances the call premium by borrowing at the rate r until expiry. Above which terminal stock price does the purchase make money?

P20.4 *The formula assembled from digitals*

Under the Black-76 model, with today's forward price $F > 0$, volatility $\sigma > 0$, expiry $T > 0$ and continuously compounded rate r, the value at expiry T of the forward price is $F_T = F \exp\big(-\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z\big)$, with $Z \sim \mathcal{N}(0,1)$ under the risk-neutral law $\mathbb{Q}$, and a claim that pays $f(F_T)$ at T is worth $e^{-rT}\,\mathbb{E}^{\mathbb{Q}} f(F_T)$ today. Write Φ and φ for the standard normal distribution function and density. Let $K > 0$ and

$$d_1 = \frac{\ln(F/K) + \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}.$$

(a) Show that $\mathbb{Q}(F_T > K) = \Phi(d_2)$, and price the cash-or-nothing digital that pays $\mathbf{1}\{F_T > K\}$. **(b)** Show, by completing the square in the integral against the normal density φ, that the asset-or-nothing claim that pays $F_T\,\mathbf{1}\{F_T > K\}$ is worth $e^{-rT} F\,\Phi(d_1)$. **(c)** Deduce the call price $C = e^{-rT}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big]$, derive the put price $\mathcal{P}$ directly from the put payoff, and show that the two formulas satisfy $C - \mathcal{P} = e^{-rT}(F - K)$ for every $\sigma > 0$. **(d)** Show that $\Phi(d_1) > \Phi(d_2)$, and say what each of the two numbers is the probability of. **(e)** Take $F = 50$, $K = 45$, $\sigma = 30\%$, $T = 0.5$ and $r = 3\%$, with $\Phi(0.60274) = 0.72666$ and $\Phi(0.39061) = 0.65196$. Compute the price of the cash-or-nothing digital, the price of the asset-or-nothing claim, F times the price of the digital, the call and the put.

P20.5 *Three exercise probabilities and a cash digital*

A stock that pays no dividends trades at $S = 50$. Its real-world expected return is $\mu = 12\%$ and its volatility $\sigma = 30\%$; the rate is $r = 4\%$. A European call and a European put on it have strike $K = 55$ and expiry $T = 1$, in the Black–Scholes–Merton model. Use $\Phi(0.3344) = 0.63096$, $\Phi(0.0677) = 0.52699$ and $\Phi(0.0344) = 0.51372$. **(a)** Compute the probability that the call is exercised under the real-world law and under the risk-neutral law, and compute $\Phi(d_1)$. Give the two probabilities that the put is exercised. **(b)** Which of the three call numbers of (a) (the two probabilities that the call is exercised, and $\Phi(d_1)$) enter the price of the call, which one is the number of shares held by the strategy that replicates the call, and which one plays no part in pricing or hedging? **(c)** Price the digital that pays 100 at T if $S_T > 55$. **(d)** A colleague says: "the call has a 49% chance of finishing in the money, because $N(d_1) = 0.49$." Correct the sentence.

P20.6 *The at-the-money rule, with its error bar*

Under the Black-76 model a European call and a European put with strike K and expiry T on a forward price F are worth $C = e^{-rT}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big]$ and $\mathcal{P} = C - e^{-rT}(F - K)$, with $d_1 = \big[\ln(F/K) + \sigma^2 T/2\big]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$. **(a)** Let $K = F$. Show that $C = \mathcal{P} = e^{-rT} F\,\big[2\,\Phi(\sigma\sqrt{T}/2) - 1\big]$, and prove, with Taylor's theorem for Φ at 0, that

$$C^{\mathrm{lin}}\Big(1 - \frac{\sigma^2 T}{24}\Big) \;\le\; C \;\le\; C^{\mathrm{lin}}, \qquad \text{where} \qquad C^{\mathrm{lin}} = e^{-rT} F\,\frac{\sigma\sqrt{T}}{\sqrt{2\pi}},$$

so that C is about $0.4\,e^{-rT} F\,\sigma\sqrt{T}$. Deduce that $0 \le C^{\mathrm{lin}} - C \le e^{-rT} F (\sigma\sqrt{T})^3/(24\sqrt{2\pi})$ and, when $\sigma^2 T < 24$, that $(C^{\mathrm{lin}} - C)/C \le (\sigma^2 T/24)/(1 - \sigma^2 T/24)$. **(b)** An index has the three-month forward price $F = 4{,}000$ and the volatility $\sigma = 15\%$; $r = 0$. For the at-the-money-forward call with $T = 0.25$, give the 0.4 rule, the rule with $1/\sqrt{2\pi} = 0.39894228$, and the exact value, using $\Phi(0.0375) = 0.5149568299$. Is the error of the second rule inside the bound of (a)? **(c)** A forward price is 200, the volatility is 32% and $r = 0$. Estimate the one-week at-the-money-forward straddle ($T = 1/52$) with the rule, and compare with the exact value, using $\Phi(0.022188) = 0.50885101$. **(d)**

A stock that pays no dividends trades at $S = 50$, with $r = 6\%$, $\sigma = 30\%$ and $T = 1$. Estimate the at-the-money *spot* call and put ($K = S$) from the identity $C = \frac{1}{2}(C + \mathcal{P}) + \frac{1}{2}(C - \mathcal{P})$, replacing the half-straddle by its at-the-money-forward estimate $0.4\, S\sigma\sqrt{T}$. Compare with the exact prices, using $\Phi(0.35) = 0.63683$, $\Phi(0.05) = 0.51994$ and $\Phi(0.15) = 0.55962$, and split the error into its three sources.

P20.7 *Implied volatility by Newton's method*

The implied volatility of a quoted call price C^{mkt} is computed by Newton's method,

$$\sigma^{(k+1)} = \sigma^{(k)} + \frac{C^{\text{mkt}} - C^{\text{BSM}}(\sigma^{(k)})}{\partial_\sigma C^{\text{BSM}}(\sigma^{(k)})}, \qquad \sigma^{(0)} = \frac{\sqrt{2\pi}\, C^{\text{mkt}}}{e^{-rT} F \sqrt{T}},$$

where $C^{\text{BSM}}(\sigma) = e^{-rT}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big]$ is the Black–Scholes–Merton price, with the forward price F, $d_1 = [\ln(F/K) + \sigma^2 T/2]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$, and where the vega is $\partial_\sigma C^{\text{BSM}} = e^{-rT} F\, \varphi(d_1)\sqrt{T}$. **(a)** (At the money forward.) Let $F = K = 100$, $T = 1$, $r = 0$ and $C^{\text{mkt}} = 30$. Compute $\sigma^{(0)}$, $\sigma^{(1)}$ and $\sigma^{(2)}$, rounding $\sigma^{(0)}$ and $\sigma^{(1)}$ to four decimals before the next step, and compare with the exact implied volatility. Use $\sqrt{2\pi} = 2.506628$, $\Phi(0.3760) = 0.6465416$, $\varphi(0.3760) = 0.371715$, $\Phi(0.3853) = 0.6499924$, $\varphi(0.3853) = 0.370402$ and $\Phi^{-1}(0.65) = 0.3853205$. In which direction do the iterates move, and why? **(b)** (Out of the money.) A stock that pays no dividends trades at $S = 100$; let $K = 110$, $T = 0.5$, $r = 3\%$ and $C^{\text{mkt}} = 3.50$, so that $e^{-rT}F = 100$. The formula and its vega take the following values.

σ	0.0500	0.1241	0.2593	0.2350
$C^{\text{BSM}}(\sigma)$	0.0146	0.8916	4.1448	3.5053
$\partial_\sigma C^{\text{BSM}}(\sigma)$	2.225	19.30	26.57	26.04

Run three Newton steps from $\sigma^{(0)}$, rounding each iterate to four decimals. What happens at the first step? Then take one Newton step from the fixed guess $\sigma = 0.05$ instead, and say what a safeguard must do with it. **(c)** In the market of (a) the call is quoted at 101. What is its implied volatility? Which trade does the quote offer?

P20.8 *Index and currency options, and the forward implied by parity*

(a) A stock index stands at $S = 5{,}200$ and pays a dividend yield $q = 1.5\%$; the rate is $r = 4.5\%$ and the volatility $\sigma = 16\%$. In the Black–Scholes–Merton model with a yield, price the European call and put with strike $K = 5{,}100$ and expiry $T = 0.25$, using $\Phi(0.376476) = 0.6467185$ and $\Phi(0.296476) = 0.6165667$. Give the model-free lower bound of the call. **(b)** A trader sees only the index level, the rate and the two option prices of (a). Recover the forward price and the dividend yield from them. **(c)** One unit of a foreign currency costs $S = 0.68$ units of the domestic currency. The domestic rate is $r = 4.5\%$, the foreign rate $r_{\text{for}} = 2\%$ and the volatility of the exchange rate $\sigma = 11\%$. Price the European call and put on one unit of the foreign currency with strike $K = 0.70$ and expiry $T = 0.75$, using $\Phi(0.059835) = 0.523856$ and $\Phi(0.155098) = 0.561628$. **(d)** Show that the put of (c) has the payoff of K calls on one unit of the *domestic* currency struck at $1/K$ units of foreign currency, and recover its price by applying the formula from the side of a foreign investor.

P20.9 *Discrete dividends and the American call*

A stock trades at $S = 45$ and pays the dividends $D_1 = D_2 = 0.80$ in two months and in eight months ($t_1 = 2/12$, $t_2 = 8/12$); the rate is $r = 5\%$. Options on it have strike $K = 45$ and expiry $T = 10/12$. In the escrowed-dividend model, the stock price net of the present value of the dividends still to be paid before the horizon is a geometric Brownian motion with volatility $\sigma = 30\%$, so a European option is priced by the Black–Scholes–Merton formula with S replaced by $S - I$, where I is the present value of the dividends paid during its life. **(a)** Compute I, the forward

price F, and the European call and put, using $\Phi(0.15965) = 0.563422$ and $\Phi(0.11422) = 0.545468$. Without dividends the same formula gives $C = 5.77$ and $\mathcal{P} = 3.94$: by how much do the dividends move each price, and why in that direction? **(b)** Exercising an American call just before the ex-dividend date t_i gains nothing when $D_i \le K(1 - e^{-r(t_{i+1}-t_i)})$, with $t_3 = T$. Apply the test at both dates. **(c)** Black's approximation takes the larger of the European call expiring at T and the European call expiring just before t_2, the second being priced in the escrowed-dividend model of its own life, in which only D_1 is escrowed. Compute it, using $\Phi(0.18594) = 0.573754$ and $\Phi(0.05901) = 0.523528$. **(d)** In the single escrowed-dividend model with horizon T, which European call has the payoff of "exercise just before t_2"? Price it, using $\Phi(0.18708) = 0.574201$ and $\Phi(0.05787) = 0.523074$, and say whether the answer of (c) changes.

P20.10 *Black's formula and a futures-option parity arbitrage*

A commodity futures contract for delivery in one year trades at $F = 60$; the rate is $r = 5\%$. European options on the futures expire on the delivery date, $T = 1$. **(a)** The call and the put struck at $K = 58$ are quoted at $C = 5.10$ and $\mathcal{P} = 3.00$. Which value of $C - \mathcal{P}$ does put–call parity require? Construct the arbitrage, with its cash flows today and at T, and give the profit that it locks in today. **(b)** Black's formula returns 5.10 for this call at $\sigma = 18.24\%$ and 3.00 for this put at $\sigma = 17.34\%$. What do the two implied volatilities say about the quotes? **(c)** Price the call and the put struck at $K = 60$ with Black's formula at $\sigma = 25\%$, using $\Phi(0.125) = 0.549738$. A colleague argues that the call must be worth more than the put because the commodity market is in contango. Is that right?

P20.11 *When to exercise an American futures call*

A futures price is $F = 120$ and the rate is $r = 6\%$. A European call, a European put and their American counterparts on the futures have strike $K = 100$ and expiry $T = 0.5$. Exercising a futures call at a time t brings $F_t - K$ in cash and a long futures position worth zero; exercising a futures put brings $K - F_t$ in cash and a short futures position worth zero; a claim that pays F_T at T is worth $e^{-r(T-t)}F_t$ at time t. **(a)** Give the model-free lower bounds of the European and of the American futures call, and compare them. Then state, in terms of the European put, when the European call is worth less than the exercise value $F - K$. **(b)** Price the European call with Black's formula at $\sigma = 25\%$, using $\Phi(1.11975) = 0.868590$ and $\Phi(0.94298) = 0.827154$, and at $\sigma = 10\%$, using $\Phi(2.61377) = 0.995523$ and $\Phi(2.54306) = 0.994506$. In each case, does the comparison with $F - K$ rule out immediate exercise of the American call? **(c)** Under Black's model the European put is worth 0.794 when $F = 126.87$ and $\sigma = 25\%$, and 0.296 when $F = 110.00$ and $\sigma = 10\%$. Show that these are the futures prices at which the European call equals its exercise value, and prove that for every σ there are states in which holding the American call to expiry is not optimal. **(d)** Prove that, for $r \ge 0$,

$$Fe^{-rT} - K \;\le\; C^{\text{am}} - \mathcal{P}^{\text{am}} \;\le\; F - Ke^{-rT},$$

and evaluate the band here. **(e)** Suppose that the futures is written on an investment asset with a nonnegative cost of carry, so that $F_t \ge S_t$ at every date. Compare the American futures call and put with the American call and put on the asset itself, with the same strike and expiry; the spot call and put are worth $S_t - K$ and $K - S_t$ on exercise at t.

P20.12 *Nine claims about the Black–Scholes–Merton model: true or false*

Rapid-fire. Decide whether each claim about the Black–Scholes–Merton model is true or false, and give the result or the number that settles it. Unless an item gives its own inputs, use a stock without dividends at $S = 60$ with $\sigma = 30\%$ and $r = 4\%$, and European options on it with $K = 60$ and $T = 0.5$, for which $d_1 = 0.2003$, $d_2 = -0.0118$, $\Phi(d_1) = 0.5794$, $\Phi(d_2) = 0.4953$, $C = 5.63$ and $\mathcal{P} = 4.45$.

(i) $\Phi(d_1)$, the literature's $N(d_1)$, is the probability that the call finishes in the money.

(ii) If the expected return of the stock is higher, the call is dearer.

(iii) As $\sigma \to \infty$ the call becomes worthless.

(iv) Doubling the volatility exactly doubles the price of an at-the-money-forward option. (For numbers, the at-the-money-forward call on the stock above, at $\sigma = 30\%$ and at $\sigma = 60\%$.)

(v) A known dividend paid before expiry raises the price of the put. (For numbers, a dividend of 1.50 paid at $t = 0.25$ on the stock above, with $\sigma = 30\%$ the volatility of the stock net of the present value of the dividend.)

(vi) An at-the-money-forward call and put on a futures price are worth the same, whatever the carry of the underlying.

(vii) A futures contract pays no dividends, so an American futures call is never exercised early. (For numbers, a futures price $F = 90$, with $K = 60$, $r = 4\%$, $T = 0.5$ and $\sigma = 30\%$.)

(viii) A call quoted above the stock price simply has a very high implied volatility.

(ix) The digital put, which pays 1 if $S_T \le K$, is worth $1 - \mathrm{Dig}(K)$.

27.21 Problems — §21 The Greeks and Hedging

P21.1 *Hedging a written option book through a move*

A dealer's book holds four option positions on a stock that trades at 80 and pays no dividend. Each option is on one share, a negative quantity is a written position, and the risk system reports these Greeks per option, rounded:

Position	Quantity	Delta	Gamma
Three-month calls, strike 85	−18,000	0.37	0.031
Three-month puts, strike 80	−9,000	−0.47	0.033
Six-month calls, strike 90	+7,000	0.33	0.021
Six-month puts, strike 70	+6,000	−0.23	0.018

The passage of time and changes of volatility are ignored throughout. **(a)** The three-month call has a delta of 0.37. Say what this number means, and give the position in the stock that delta-hedges the 18,000 written calls on their own. **(b)** Compute the delta and the gamma of the whole book, and the trade in the stock that makes the book delta-neutral. **(c)** With that hedge in place the stock jumps from 80 to 82. Estimate the delta of the hedged book after the jump, and give the trade that restores neutrality. **(d)** Estimate the profit or loss of the hedged book over the jump. What changes if the stock falls to 78 instead?

P21.2 *Deltas from the key identity, the at-the-money delta, and a digital near expiry*

A European call and a European put with strike K and time to maturity τ, on an asset with price S and continuous yield q, have the Black–Scholes–Merton prices

$$C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad \mathcal{P} = Ke^{-r\tau}\Phi(-d_2) - Se^{-q\tau}\Phi(-d_1),$$

$$d_1 = \frac{\ln(S/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau},$$

where r is the continuously compounded rate, σ the volatility, S, K, σ and τ are positive, and Φ and φ are the standard normal distribution function and density. **(a)** Show that $Se^{-q\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$, and use this identity to prove that $\partial_S C = e^{-q\tau}\Phi(d_1)$, which lies in $(0, e^{-q\tau})$. Then derive the put delta $\partial_S \mathcal{P} = -e^{-q\tau}\Phi(-d_1)$, which lies in $(-e^{-q\tau}, 0)$, in two ways: by differentiating the put formula, and from put–call parity. **(b)** A stock that pays no dividend trades at 80; $r = 5\%$, $\sigma = 30\%$ and $\tau = 0.75$. For the options struck at the money spot, $K = 80$, compute d_1, the delta of the call, the delta of the put, and the risk-neutral probability $\Phi(d_2)$ that the call is exercised. Compute also the delta of the call struck at the forward price, $K = 80e^{0.0375}$. **(c)** Check the

mental rule $\Phi(x) \approx 0.5 + 0.4x$ on the three values $\Phi(d_1)$, $\Phi(d_2)$ and the forward-struck delta of (b), and say where the rule comes from. **(d)** A cash-or-nothing digital call pays 1 at expiry if $S_T > K$ and is worth $\mathrm{Dig}(K) = e^{-r\tau}\Phi(d_2)$. Derive its delta and its gamma, and find the price at which the gamma changes sign. Evaluate them for $K = 50$, $r = q = 0$ and $\sigma = 31.75\%$ one trading day before expiry ($\tau = 1/252$, so that $\sigma\sqrt{\tau} = 0.02$): the delta at $S = 50$, and the gamma at $S = 49.5$ and at $S = 50.5$. Compare with the gamma of the ordinary call with the same strike at $S = 50$.

P21.3 *Hedging with spot, forwards and futures; the Greeks of a futures option*

A dealer has written 2,000 European calls on a gold futures price. Each call is on one ounce, has strike $K = 2{,}450$ and expires in four months ($\tau = 1/3$); the underlying futures contract is for delivery in six months, and its price is $F = 2{,}400$ an ounce. The continuously compounded rate is $r = 4\%$ for every maturity and the volatility of the futures price is $\sigma = 16\%$. Gold is treated as an investment asset with no storage cost and no lease rate, so that the forward and futures prices for delivery in τ_F years both equal $Se^{r\tau_F}$, where S is the spot price. The futures call is worth

$$C = e^{-r\tau}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big], \qquad d_1 = \frac{\ln(F/K) + \sigma^2\tau/2}{\sigma\sqrt{\tau}}, \qquad d_2 = d_1 - \sigma\sqrt{\tau},$$

which is the yield formula $xe^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, with $d_1 = [\ln(x/K) + (r - q + \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$ and $d_2 = d_1 - \sigma\sqrt{\tau}$, of a price x with a continuous yield q, evaluated at $x = F$ and $q = r$. **(a)** Compute the price of one call and the delta of the written position with respect to the six-month futures price. **(b)** How many ounces must the dealer buy to make the position delta-neutral with (i) the six-month futures, (ii) spot gold, (iii) a one-year futures contract, (iv) a six-month forward contract? Explain why (i) and (iv) differ although the six-month futures price and the six-month forward price are equal. **(c)** Derive the vega of the futures call, and compute the vega of the position per volatility point. **(d)** Derive the rho of the futures call, that is, its derivative in r with the futures price F held fixed, and compute it for the position per percentage point. The rho of a call on an asset with yield q, at fixed price and fixed yield, is $K\tau e^{-r\tau}\Phi(d_2)$; explain why setting $q = r$ in that expression does not give the rho of the futures call.

P21.4 *The Greeks of a written currency-option block, interpreted*

A bank has written five-month European calls on 40 million units of a foreign currency. The spot exchange rate is $S = 1.15$ units of domestic currency per unit of foreign currency, the strike is $K = 1.17$, the domestic rate is $r = 4\%$ and the foreign rate is $r_{\text{for}} = 2.5\%$, both continuously compounded, and the volatility of the exchange rate is $\sigma = 8\%$. A currency option is priced by the Black–Scholes–Merton formula for an asset with the continuous yield $q = r_{\text{for}}$. **(a)** Compute, per unit of foreign currency, the price of the call to six decimals (carry the normal distribution function to seven digits) and its delta, gamma, vega, theta (per year) and rho, the sensitivity to the domestic rate. **(b)** Derive the sensitivity of the call to the foreign rate, $\partial_{r_{\text{for}}} C$, and evaluate it. **(c)** Restate the six sensitivities for the bank's written position in desk units: units of foreign currency for the delta, units of foreign currency per move of 0.01 in the exchange rate for the gamma, domestic currency per volatility point for the vega, per trading day (252 a year) for the theta, and per percentage point for the two rate sensitivities. Which spot trade makes the position delta-neutral? **(d)** Revalue the call by hand, to six decimals, with the spot at 1.16 and all other inputs unchanged; the same revaluation with the spot at 1.14 gives 0.013925. Use the two bumped prices and the price of (a), rounded to six decimals, to check the delta and the gamma.

P21.5 *Theta is rent for gamma: the identity, its variants, parity Greeks and the run into expiry*

A European call with strike K and time to maturity $\tau = T - t$, on an asset with price S and continuous yield q, with continuously compounded risk-free rate r and volatility σ, is worth $C = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, where $d_1 = [\ln(S/K) + (r - q + \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$ and $d_2 = d_1 - \sigma\sqrt{\tau}$. Its delta, gamma and theta (the derivative in t, per year) are

$$\Delta_C = e^{-q\tau}\Phi(d_1), \qquad \Gamma = \frac{e^{-q\tau}\varphi(d_1)}{S\sigma\sqrt{\tau}}, \qquad \Theta_C = -\frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}} + qSe^{-q\tau}\Phi(d_1) - rKe^{-r\tau}\Phi(d_2).$$

(a) Verify by substitution that the call satisfies $\Theta + (r-q)S\Delta + \frac{1}{2}\sigma^2 S^2\Gamma = rV$, where V is its value. Show that a long forward contract with delivery price K and delivery date T, worth $Se^{-q\tau} - Ke^{-r\tau}$, satisfies the same identity, and deduce that it holds for the put with the same strike and for every book of European calls, puts and forward contracts on one asset, all valued at the same volatility, plus cash. **(b)** State the form that the identity takes for a book of European currency options, where the foreign rate is r_{for}, and for a book of European options on a futures price F, whose Greeks are taken with respect to F (theta at fixed F). Justify each. **(c)** Use put–call parity to express the delta, gamma, vega and theta of a put through those of the call with the same strike and expiry. **(d)** A delta-neutral book of European options on a stock at 50 that pays no dividend, with $r = 0$ and every option valued at $\sigma = 30\%$, has a theta of +9,000 a year. What is its gamma? What does the book earn on a trading day (1/252 of a year) on which neither the stock nor the implied volatility moves, which one-day move of the stock, with the volatility unchanged, wipes that out, and what does the book lose on a two-point day? Is it long or short options, on balance? **(e)** For an at-the-money call on the same stock ($S = K = 50$, $\sigma = 30\%$, $r = q = 0$), compute the theta per trading day one month ($\tau = 1/12$) and one trading day ($\tau = 1/252$) before expiry, and explain why theta accelerates as expiry approaches.

P21.6 *Explaining a day's profit and loss*

A dealer has written 20,000 three-month European puts struck at 45 on a stock that trades at 50 and pays no dividend, and has delta-hedged them by selling short the 4,369.2 shares of their delta. The puts are marked at an implied volatility of 30%, and $r = 0$. The hedged book has a zero delta and

$$\Gamma = -786.40 \text{ per point}, \qquad \Theta = +88{,}470 \text{ a year}, \qquad \mathcal{V} = -147{,}450 \text{ per unit of volatility},$$

that is, a vega of −1,474.5 per volatility point. Its cross sensitivity to the price and the volatility is $\partial_{S\sigma}V = +12{,}335$, and the sensitivity of its gamma to the volatility is $\partial_{SS\sigma}V = +1{,}343$. One trading day is 1/252 of a year. **(a)** Check that the stated theta and gamma are consistent with each other. **(b)** Over one trading day the stock falls to 48.5. In scenario A the implied volatility rises to 32% on the same day; in scenario B it stays at 30%. Predict the change in the value of the book in each scenario from its Greeks, term by term. **(c)** Full revaluation gives −3,821.4 in scenario A and −547.3 in scenario B. Compute the two residuals. How much of the residual of scenario A is the cross term $\partial_{S\sigma}V\,\Delta S\,\Delta\sigma$, and how much is the term $\frac{1}{2}\,\partial_{SS\sigma}V\,(\Delta S)^2\,\Delta\sigma$? **(d)** Which term of (b) is the largest in scenario A? A six-month at-the-money call on the same stock, marked at the same 30% implied volatility, has a vega of 0.1403 per volatility point, a gamma of 0.0374 and a delta of 0.542. Which trade in this call, made at the start of the day, would have removed that term, and what would it have done to the gamma of the book and to the share hedge? Would it have removed the cross term of (c)? **(e)** Estimate the loss of the hedged book if the stock gaps 5 points overnight, in either direction, before the hedge can be adjusted, and express it in days of theta. What does this say about a book with a zero delta and a large negative gamma?

P21.7 *Breakeven moves and gamma scalping*

Throughout, $r = q = 0$ and a year has 252 trading days. A delta-hedged long option position marked at the implied volatility σ_{imp} earns, to leading order, $\frac{1}{2}\Gamma\,[(\Delta S)^2 - \sigma_{\text{imp}}^2 S^2\,\Delta t]$ over an interval of length Δt in which the price moves by ΔS. **(a)** Give the breakeven move of a delta-hedged long at-the-money straddle, as a percentage of the price, over one trading day and over one week of five trading days, at implied volatilities of 16%, 25% and 40%. **(b)** A trader is long 1,000 two-month at-the-money straddles on a stock at 80 (strike 80, implied volatility 40%), delta-hedged. One straddle has a gamma of 0.0609 per point. Suppose the stock rallies 2%. Which trade re-hedges the position? **(c)** In fact the trader re-hedges only at each close. The closing prices of the next four days, and the value, delta and gamma of one straddle at each close, marked at 40%, are:

Day	Close	Value	Delta	Gamma
0	80.0	10.412	0.065	0.0609
1	83.0	10.751	0.243	0.0568
2	81.0	10.256	0.125	0.0610
3	81.5	10.198	0.156	0.0610
4	78.0	9.910	−0.068	0.0656

For each of the four days give the profit and loss of the hedged position of 1,000 straddles and the trade made at the close. Compare the four-day total with $\sum_i \frac{1}{2}\Gamma_{t_i}[(\Delta S_i)^2 - \sigma_{\text{imp}}^2 S_{t_i}^2\,\Delta t]$, where t_i is the close of day i, $\Delta S_i = S_{t_{i+1}} - S_{t_i}$, $\Delta t = 1/252$ and Γ_{t_i} is the gamma of the 1,000 straddles at the close of day i. **(d)** When the realised volatility equals the implied volatility σ, the standard deviation of the hedging error of an option hedged at n equally spaced dates until its expiry T is roughly $\Gamma S^2\sigma^2 T/\sqrt{2n}$, with the gamma frozen at its initial value. Evaluate this for the position of (b) at day 0, for daily rebalancing ($n = 42$) and for rebalancing seven times a day ($n = 294$).

P21.8 *Variance risk premium, delta-hedged P&L*

You buy a 1-month at-the-money straddle on an index at 25% implied volatility and delta-hedge daily; realised volatility over the month comes in at 20%. (a) Do you make or lose money? Derive the standard expression for the daily P&L of a delta-hedged option that justifies your answer. (b) Roughly how much do you lose as a fraction of the premium paid? (You may use the approximation that an ATM straddle costs about $0.8\,S\,\sigma\sqrt{T}$.) (c) This outcome is the norm rather than the exception — name the phenomenon, explain why it persists, and describe the risk profile of the trader on the other side of your trade.

P21.9 *Delta-, gamma- and vega-neutral with two traded options*

An index stands at $S = 2{,}000$; $r = q = 0$, and every option is valued at the flat volatility $\sigma = 20\%$. A dealer's over-the-counter book consists of three positions in nine-month European options on the index, and two European calls on the index are listed, a three-month call struck at 2,000 and an eighteen-month call struck at 2,160. The Greeks are per option, each option is on one index unit, vegas are per volatility point, and a negative quantity is a written position:

	Quantity	Delta	Gamma	Vega
Book: nine-month calls, strike 2,000	−4,000	0.53	0.00115	6.9
Book: nine-month puts, strike 1,750	+2,000	−0.20	0.00080	4.8
Book: nine-month calls, strike 2,225	−1,000	0.30	0.00100	6.0
Listed option 1: three-month call, strike 2,000		0.52	0.0020	4.0
Listed option 2: eighteen-month call, strike 2,160		0.42	0.0008	9.6

The index itself can be traded in index units, each with a delta of 1. **(a)** Compute the delta, the gamma and the vega of the book. **(b)** Make the book gamma-neutral with option 1 alone and then delta-neutral with the index. Which vega is left? **(c)** Starting again from the book, make it vega-neutral with option 2 alone and then delta-neutral. Which gamma is left? **(d)** Starting again from the book, make it gamma-, vega- and delta-neutral with the two listed options and the index. **(e)** Under which condition does a single further option on the index remove the gamma and the vega of this book together? Name such an option and the quantity needed.

P21.10 *The stop-loss rule is not free; synthetic options along two paths*

A dealer has written a European call with strike K and expiry T on a stock that pays no dividend, and $r = 0$. The price follows $S_t = S_0 \exp(-\sigma^2 t/2 + \sigma W_t)$, where W is a standard Brownian motion and $\sigma > 0$, so that $\mathbb{E}\,(S_T - K)^+$ is the Black–Scholes–Merton price C of the call. The *stop-loss rule* observes the price at the dates $0 = t_0 < t_1 < \cdots < t_n = T$, starts from the capital $(S_0 - K)^+$

and holds $h_i = \mathbf{1}\{S_{t_i} > K\}$ shares over $(t_i, t_{i+1}]$: the writer owns the share while the last observed price is above the strike, and owns nothing otherwise. **(a)** Describe the trades that the rule makes for a call written out of the money. Why does the rule look free, and why is it a poor hedge? **(b)** Prove that the expected shortfall of the rule,

$$\mathbb{E}\Big[(S_T - K)^+ - (S_0 - K)^+ - \sum_{i=0}^{n-1} h_i \left(S_{t_{i+1}} - S_{t_i}\right)\Big],$$

equals the time value $C - (S_0 - K)^+$ of the call, whatever the monitoring dates. Two facts may be used: an increment $W_t - W_s$, $s < t$, is independent of the path up to s and has the law $\mathcal{N}(0, t-s)$; and $\mathbb{E}\, e^{\theta Z} = e^{\theta^2/2}$ for a standard normal variable Z. **(c)** Take $S_0 = 95$, $K = 100$, $\sigma = 20\%$ and $T = 0.25$. Compute the time value. If the purchases are in effect made at 100.05 and the sales at 99.95, estimate the expected number of round trips (a purchase followed by a sale) and of trades. **(d)** A stock is at 40, and a trader creates a call struck at 45 synthetically, by holding its delta in shares. Along one path the stock rises steadily to 60; along another it swings wildly and also ends at 60. On which path does the synthetic call cost more, and why? **(e)** Explain why creating an option synthetically is the reverse of hedging it.

P21.11 *Insuring a low-beta portfolio: index puts, then synthetically*

An equity portfolio worth $V_A = 25$ million has a CAPM beta of $\beta = 0.8$ to an index and a dividend yield of $q_A = 2\%$. The index stands at $S = 5{,}000$, with a dividend yield of $q = 1.5\%$ and a volatility of $\sigma = 18\%$, and the rate is $r = 4\%$. The manager wants the portfolio to be worth at least 23 million in nine months ($\tau = 0.75$). Assume that over the horizon the total return of the portfolio, dividends included, is $r\tau + \beta\,(R_M - r\tau)$ with no residual, where $R_M = S_T/S - 1 + q\tau$ is the total return of the index and rates and yields accrue simply over the horizon. Price index options with the Black–Scholes–Merton formula for an asset with a continuous yield, reading r and q as continuously compounded, here and in the futures price $Se^{(r-q)\tau_F}$ of a contract for delivery in τ_F years. **(a)** Express the value of the portfolio at the horizon as a function of the index level S_T. On how many index units must European puts be bought, and at which strike? What does the insurance cost? **(b)** Repeat (a) for a portfolio with $\beta = 1$ and the same dividend yield of 2%, and compare the ratio of the two costs with the ratio of the betas. **(c)** For the portfolio with $\beta = 0.8$ the puts are to be created synthetically. Give the delta of the put, the initial short position in the index in index units, the bills that the replicating strategy holds (its bank holding), and the equivalent sale of part of the portfolio into bills. Alternatively, how many one-year index futures contracts, each on 10 times the index, are sold at the start? **(d)** Why is the floor only approximate when the portfolio is not the index itself, and why did dynamic insurance of this kind fail in a market crash?

27.22 Problems — §22 Volatility: Smiles, Surfaces, Estimation and Variance Swaps

P22.1 *Call and put at different implied volatilities*

A stock that pays no dividend trades at $S = 80$, and the continuously compounded interest rate is $r = 3\%$. European options on it with strike $K = 80$ and expiry $T = 0.5$ are quoted in volatility: the call at an implied volatility of 24%, which is a price of $C = 5.986$, and the put at 27%, which is a price of $\mathcal{P} = 5.462$. Both implied volatilities are computed at the forward price $F = Se^{rT}$. The market is frictionless, short sales are allowed, and money is borrowed and lent at the rate r. Use $80\,e^{-0.015} = 78.809$. **(a)** Which option is dear relative to the other, and by how much? Give the trade that captures the difference, with a table of its cash flows today and at T in the two cases $S_T \le 80$ and $S_T > 80$, and the amount it locks in. **(b)** Check the amount of (a) against the vega of the two options multiplied by the gap between the two implied volatilities. Use $\varphi(0.173) = 0.393$. **(c)** Does the argument of (a) assume that the stock price is lognormal?

(d) Suppose that 5.986 and 5.462 are instead the prices of an American call and an American put with the same strike and expiry. Are the two quotes an arbitrage?

P22.2 *Earnings tomorrow: the binary frown*

A stock trades at $S = 40$ and pays no dividend. An announcement tomorrow will send the price to 46 or to 36, and nothing else moves it before the options considered here expire; the interest rate is $r = 0$. **(a)** Find the risk-neutral probability of the move to 46. **(b)** Price the European calls struck at 38, 40, 42 and 44, and the puts with the same strikes by put–call parity. Check two of the puts directly. **(c)** At $r = 0$ the Black–Scholes–Merton formula contains the volatility and the time to expiry only through the total volatility $\sigma\sqrt{T}$. Inverting it at the four call prices gives the implied total volatilities $\sigma_{\text{imp}}\sqrt{T} = 13.17\%$, 15.05%, 15.13% and 13.50%. Confirm the at-the-money value with the rule $C \approx 0.4\,F\,\sigma\sqrt{T}$. Name the shape of this strike profile and explain it. What happens to the implied volatility as the strike decreases to 36 or increases to 46, and what at strikes $K \le 36$ or $K \ge 46$? **(d)** Name the shape (smile, frown, downward skew, upward skew) of the implied-volatility profile produced by each of the following risk-neutral laws of the price at expiry, compared with a lognormal law with the same mean: (i) both tails thinner; (ii) a heavier right tail and a thinner left tail, as for a commodity prone to price spikes; (iii) a currency that its central bank holds inside a narrow band; (iv) a volatility that rises when the price falls.

P22.3 *Breeden-Litzenberger*

(a) Starting from $C(K) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_T - K)^+\big]$, derive the price of a cash-or-nothing digital (paying 1 if $S_T > K$) and the Breeden–Litzenberger identity for the risk-neutral density $f_{\mathbb{Q}}(K)$. (b) With $r = 0$, listed calls trade at $C(95) = 7.00$, $C(100) = 4.00$, $C(105) = 2.00$. Estimate the price of the digital struck at 100 and the risk-neutral density at 100. (c) Which no-arbitrage condition guarantees your density estimate is non-negative, and what is this construction used for in practice?

P22.4 *A digital off the skew*

The forward price of an index for delivery in six months is $F = 200$; the expiry is $T = 0.5$ and the interest rate is $r = 0$. The call struck at $K = 210$ has an implied volatility of 18%, and around that strike the smile slopes down by 0.15 volatility points per unit of strike: $\sigma'_{\text{imp}}(210) = -0.0015$. Assume that the risk-neutral law of the settlement has a continuous density with a finite mean, and that the smile is differentiable around 210. A cash-or-nothing digital pays 1 at T if the index settles above 210. Use $\ln(200/210) = -0.04879$, $\Phi(0.447) = 0.6726$ and $\varphi(0.320) = 0.3790$. **(a)** Price the digital with the Black-76 formula at the single volatility 18%. **(b)** Compute the vega of the 210 call, and the price of the digital on this smile. **(c)** On the smile, the 205 call (implied volatility 18.75%) costs 8.388 and the 215 call (17.25%) costs 4.311; at a flat 18% they would cost 7.968 and 4.681. Check (a) and (b) with the 205/215 call spread. **(d)** Price the digital that pays 1 if the index settles at or below 210, at the flat volatility 18% and on the smile. In which direction does the skew move it? What does the answer say about where the risk-neutral law puts its mass above 210, compared with the lognormal law at 18%?

P22.5 *Black-76 digital, prediction-market link*

BTC futures for delivery in 30 days trade at $F = 60{,}000$; the 70,000-strike implied volatility on the options venue is 60% and rates are approximately zero. (a) Under Black-76, price the cash-or-nothing digital paying \$1 if the future settles above 70,000 (take $T = 30/365$). You may use $\ln(7/6) \approx 0.154$ and $\Phi(0.98) \approx 0.837$. (b) A prediction market quotes "BTC above 70k in 30 days" at 25 cents. Interpret the gap and name the main caveats you would check before trading it.

P22.6 *The calendar condition: prove it, then trade it*

Except in (e), $r = q = 0$, so that the forward price equals the spot price S at every expiry. Markets are frictionless, the underlying can be bought or sold short at any date, and there is no arbitrage. $C(K,T)$ is the price today of the European call struck at K and expiring

at T, and $\sigma_{\text{imp}}(K,T)$ is its implied volatility. **(a)** Let $T_1 < T_2$. Prove, with no model, that $C(K,T_1) \le C(K,T_2)$ for every strike $K > 0$: exhibit a strategy that buys the later call, writes the earlier one, trades once more at T_1, and is worth a nonnegative amount at T_2 in every state. Deduce that $\sigma_{\text{imp}}(K,T_1)^2\,T_1 \le \sigma_{\text{imp}}(K,T_2)^2\,T_2$ whenever both implied volatilities exist. In (b) to (d) the strike is at the money, $K = S = 100$, and the implied volatilities are 30% at three months and 22% at six months. **(b)** Compute the forward volatility between three and six months. **(c)** The six-month quote drops to 20%. Show that the forward variance is negative, price both calls, state the arbitrage and the amount it collects, and give the lowest six-month volatility that the three-month quote allows. At the money and at $r = 0$ the formula reads $C = S\,[2\,\Phi(\sigma\sqrt{T}/2) - 1]$; use $\Phi(0.0750) = 0.5299$ and $\Phi(0.0707) = 0.5282$. **(d)** Back at 22% for six months: quote the 4.5-month volatility by interpolating linearly in total variance, and then by interpolating linearly in volatility. Show that the second quote creates an arbitrage against the six-month option. Which interpolation should a desk use, and why? **(e)** Now let $r = 5\%$, with $q = 0$ and $S = 100$. Which three-month strike and which six-month strike does the calendar condition compare when the three-month option is struck at the money forward?

P22.7 *Sticky strike or sticky delta: hedging a skewed put*

A stock trades at $S = 100$, with $r = q = 0$. Around the money the three-month smile ($T = 0.25$) is linear in the strike, $\sigma_{\text{imp}}(K) = 0.20 - 0.006\,(K - 100)$, so that the European put struck at $K = 95$ has an implied volatility of 23%; it costs $\mathcal{P} = 2.408$. Use $\ln(100/95) = 0.05129$, $\Phi(0.5035) = 0.6927$ and $\varphi(0.5035) = 0.3514$. **(a)** Compute the Black–Scholes–Merton delta and the vega of the put. **(b)** Two rules describe how the smile moves with the spot. Under *sticky strike* every strike keeps its implied volatility. Under *sticky delta* the implied volatility is a fixed function of the moneyness K/S, here $0.20 - 0.6\,(K/S - 1)$ near the money (for $0.8 \le K/S \le 1.2$, extended outside that range to a positive differentiable function, which the problem does not use), which is the smile above when $S = 100$. Compute the smile-adjusted delta $d\mathcal{P}/dS$ of the put under each rule. **(c)** The spot falls at once to 98, the expiry still three months away. Give the implied volatility of the 95 strike under each rule, and pick the new price of the put from these Black–Scholes–Merton values at $S = 98$: 2.870 at a volatility of 21.837%, 3.085 at 23.000% and 3.301 at 24.163%. **(d)** A book is long one put and holds the Black–Scholes–Merton hedge of (a) in shares. Compute its P and L on the fall to 98 under each rule. Why must the rule be stated before a hedge ratio is quoted?

P22.8 *Volatility from ten closes*

A stock that pays no dividend in the period closes on eleven consecutive trading days at

$$50.0,\quad 50.8,\quad 49.9,\quad 50.4,\quad 51.2,\quad 50.6,\quad 50.9,\quad 51.8,\quad 51.1,\quad 50.3,\quad 51.0.$$

The ten daily log-returns $R_i = \ln(S_i/S_{i-1})$ are, in percent,

$$1.587,\quad -1.788,\quad 0.997,\quad 1.575,\quad -1.179,\quad 0.591,\quad 1.753,\quad -1.361,\quad -1.578,\quad 1.382.$$

Use $\sqrt{252} = 15.87$. **(a)** Compute the zero-mean estimate of the daily volatility and annualise it. Then compute the demeaned estimate, the sample standard deviation with divisor $n - 1$, and annualise it. **(b)** Under iid normal returns with mean zero, give the first-order standard error of the annualised zero-mean estimate, in volatility points. **(c)** How many daily returns are needed to estimate the volatility of an asset whose volatility is about 25% to within one volatility point, taken as one standard error? **(d)** No exchange holiday falls in the period, so two of the ten returns run from a Friday close to a Monday close. Does each of them count as one period or as three, and why?

P22.9 *EWMA and GARCH: update, prove the forecast, use it*

Daily log-returns follow $R_i = \sigma_i Z_i$, where the shocks Z_i are iid with mean 0 and variance 1, each independent of everything observed before its day, and where σ_i^2, the variance of the return of day i, is known at the end of day $i - 1$. Yesterday's estimate of the volatility was 1.4% a day; the

stock then closed at 78.40, after 80.00 the day before. Use $\ln 0.98 = -0.02020$, $0.98^{10} = 0.8171$, $0.98^{20} = 0.6676$, $0.98^{50} = 0.3642$, $0.98^{100} = 0.1326$, $\ln 2 = 0.6931$ and $\sqrt{252} = 15.87$. **(a)** Update the variance with the exponentially weighted moving average $\sigma_i^2 = \lambda\,\sigma_{i-1}^2 + (1-\lambda)\,R_{i-1}^2$ at the decay factor $\lambda = 0.94$, and with the GARCH(1,1) rule $\sigma_i^2 = \omega + \alpha\,R_{i-1}^2 + \beta\,\sigma_{i-1}^2$ at $\omega = 0.0000025$, $\alpha = 0.06$ and $\beta = 0.92$. Give the two new daily volatilities, and the long-run daily and annual volatilities of the GARCH model. **(b)** Consider the GARCH(1,1) rule with $\omega > 0$, $\alpha \geq 0$, $\beta \geq 0$ and $\alpha + \beta < 1$, and write $\bar\sigma^2 = \omega/(1-\alpha-\beta)$. Today's variance σ_i^2 is a known number. Prove that for every $k \geq 0$, with the convention $(\alpha+\beta)^0 = 1$,

$$\mathbb{E}\big[\sigma_{i+k}^2\big] = \bar\sigma^2 + (\alpha+\beta)^k\,(\sigma_i^2 - \bar\sigma^2).$$

Show also that $\bar\sigma^2$ is the unique fixed point of the map $x \mapsto \omega + (\alpha+\beta)\,x$, and that it is attracting. **(c)** Start from the GARCH variance of (a). Compute the square root of the expected variance 10 and 50 trading days ahead, and the half-life of a deviation from the long-run variance. The model assigns to an option with k trading days to expiry the annualised volatility $\sqrt{252}$ times the square root of the average of the expected daily variances of those k days, today's included: compute it for $k = 20$ and $k = 100$. What volatility does the moving average assign to the same two options? **(d)** On the next day the price does not move. Update both estimates once more, and explain why GARCH now falls further than the moving average.

P22.10 *The log-variance strike from five quotes*

Throughout, $r = q = 0$, so that the forward price is $F = S_0$ and option prices are undiscounted; the market is frictionless and admits no arbitrage. A price is observed at the dates $0 = t_0 < t_1 < \cdots < t_n = T$; it is positive at each of them and the underlier, which pays no income, can be traded at each of them. Write $R_i = \ln(S_{t_i}/S_{t_{i-1}})$. Under the risk-neutral probability $\mathbb{Q}$ the law of S_T has a continuous density, the mean F and $\mathbb{E}^{\mathbb{Q}}|\ln S_T| < \infty$, and every European claim on S_T is worth its $\mathbb{Q}$-expected payoff; $C(K)$ and $\mathcal{P}(K)$ are the prices of the call and the put struck at K. A *log-variance swap* costs nothing at inception and pays at T the realised log-variance $\frac{1}{T}\sum_{i=1}^{n} 2\,(e^{R_i} - 1 - R_i)$ minus a strike. **(a)** Prove that on every path

$$\sum_{i=1}^{n} 2\,(e^{R_i} - 1 - R_i) = 2\sum_{i=1}^{n} \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} - 2\ln\frac{S_T}{S_0},$$

and deduce that the swap costs nothing exactly when its strike is

$$K_{\log} = \frac{2}{T}\,\mathbb{E}^{\mathbb{Q}}\Big[-\ln\frac{S_T}{F}\Big] = \frac{2}{T}\Big[\int_0^F \frac{\mathcal{P}(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\Big].$$

The following expansion may be used without proof: if f is twice continuously differentiable on $(0,\infty)$, then for every $x > 0$

$$f(x) = f(F) + f'(F)\,(x-F) + \int_0^F f''(K)\,(K-x)^+\,dK + \int_F^\infty f''(K)\,(x-K)^+\,dK.$$

(b) Let $F = 100$ and $T = 1$. Five options are quoted, out of the money except at the forward: the puts struck at 70 and 85 at 0.843 and 2.985 (implied volatilities 26% and 23%), the call struck at 100 (equal to the put there, by put–call parity at $K = F$) at 7.966 (20%), and the calls struck at 115 and 130 at 2.256 and 0.385 (17.5% and 16%). Approximate the two integrals by one sum over the five strikes, each strike contributing $\Delta K \times \text{price}/K^2$ with $\Delta K = 15$, and quote the resulting strike in volatility points. **(c)** With all five options priced at a flat 20% the same discretisation gives the strike 0.0431. A fine numerical integration of the strip, with the smile interpolated linearly in the strike between the five quotes and held flat beyond them, gives 0.0443. Using the fine integration to measure the effect of the skew, split the result of (b) into the at-the-money variance, the effect of the skew and the discretisation error. **(d)** Explain why the options are weighted by $1/K^2$. Explain why this skew puts $K_{\log}$ above the at-the-money variance 0.04, and

say whether every downward-sloping smile does. Finally, in the Black–Scholes–Merton model with daily dates, $\Delta t = 1/252$, and a volatility of 21%, by how much does the fair strike K_{var} of the swap that pays $\frac{1}{T}\sum_i R_i^2$ exceed $K_{\log}$?

P22.11 *Selling variance: notionals and P and L*

A dealer sells a three-month variance swap struck at 25 volatility points, sized at a vega notional of 50,000 per volatility point. At expiry the swap pays its buyer the variance notional times the difference between the realised variance and the strike, where the realised variance is the annualised average of the squared daily log-returns. The interest rate is zero. One volatility point is 0.01 and one variance point is 0.0001, so a volatility of 25 points is a variance of 625 points. **(a)** Convert the vega notional into the variance notional, per variance point. **(b)** Compute the dealer's P and L when the realised volatility is 20%, 25% and 40%, and compare each with the estimate "vega notional times the difference in volatility points". **(c)** The swap covers 62 daily returns. After 31 of them the realised volatility has been 30% so far, and a new variance swap on the remaining 31 returns is struck at 28 volatility points. Mark the dealer's position to market. **(d)** Had the dealer sold instead a volatility swap struck at 25 with the same vega notional, what would the loss at a realised volatility of 40% be? Which of the two losses is larger, and why?

P22.12 *The volatility swap is worth less than the square root*

Under a pricing law $\mathbb{Q}$ of the monitored path, the realised volatility $\sigma_{\text{real}} \geq 0$ of an underlier over a fixed period has a finite second moment. The fair strike of the variance swap on that period is $K_{\text{var}} = \mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}^2]$, that of the volatility swap is $K_{\text{vol}} = \mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}]$, and the interest rate is zero. **(a)** Prove that $K_{\text{vol}} \leq \sqrt{K_{\text{var}}}$ in two ways: from the fact that a variance is nonnegative, and from the inequality $\sigma_{\text{real}}^2 - K_{\text{var}} \geq 2\sqrt{K_{\text{var}}}\,(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})$, which is to be proved first. When does equality hold? **(b)** Let $K_{\text{var}} = 0.09$, that is 30 volatility points, and let the standard deviation of the realised variance σ_{real}^2 under $\mathbb{Q}$ be 0.03. Estimate K_{vol} by expanding the square root to second order about K_{var}. **(c)** A trader is long a variance swap struck at K_{var} with the variance notional N_{var}, and short a volatility swap struck at $\sqrt{K_{\text{var}}}$ with the vega notional $2\sqrt{K_{\text{var}}}\,N_{\text{var}}$. Show that the combined payoff is nonnegative on every path. To what is the trader exposed, and why is the position not an arbitrage? **(d)** When does the adjustment of (b) vanish?

27.23 Problems — §23 Exotic Options and Numerical Methods

P23.1 *Hedging a short digital with call spreads*

A stock trades at 100 and the interest rate is zero. Three European calls on it with the same expiry T are listed, at the prices $C(98) = 4.10$, $C(100) = 3.00$ and $C(102) = 2.10$, where $C(K)$ is the price of the call struck at K. A *digital* struck at 100 pays 1 at T if $S_T > 100$ and nothing otherwise; its price is Dig(100). No model is assumed in parts (a) to (d). **(a)** By comparing payoffs state by state, show that

$$\frac{C(98) - C(100)}{2} \;\geq\; \text{Dig}(100) \;\geq\; \frac{C(100) - C(102)}{2},$$

and give the two numbers. **(b)** A desk sells digitals with a total payout of 1,000,000 (one million digitals, each paying 1) and hedges them with one of the two call spreads of (a), from 98 to 100 or from 100 to 102, so that it is never short at expiry. Which spread does it hold, in what quantities and at what cost? How much does this hedge cost above the value of the digitals taken at the midpoint of the bracket of (a)? **(c)** Another dealer bids 0.60 for one digital. Construct an arbitrage and give its cash flows today and at T. **(d)** Suppose that calls were listed at every strike and that the call curve $K \mapsto C(K)$ were differentiable at 100. To what do the two bounds of (a), written with a strike spacing ε in the place of 2, tend as $\varepsilon \to 0$, and why? **(e)** Now take another stock and assume for it the Black–Scholes–Merton model with $\sigma = 30\%$ and a zero rate, in

which a digital struck at K with time τ to expiry is worth $\Phi(d_2)$, $d_2 = [\ln(S/K) - \sigma^2\tau/2]/(\sigma\sqrt{\tau})$, where S is the price of the stock. A desk has sold digitals struck at 50 on it, with a total payout of 2,000,000, and, unlike the desk of (b), holds no call spread. One trading day before expiry ($\tau = 1/252$), with $S = K = 50$, how many shares would delta-hedge the sold digitals? Compare the value of that stock position with the payout. Use $\sqrt{252} = 15.8745$ and $\varphi(0) = 0.39894$.

P23.2 *Gap and chooser options by parity*

A stock that pays no dividends trades at $S = 40$; the continuously compounded rate is $r = 5\%$, the volatility is $\sigma = 30\%$, and the pieces below are priced in the Black–Scholes–Merton model. Write $C(K,T)$ and $\mathcal{P}(K,T)$ for today's prices of the European call and put with strike K and expiry T, written $C(K)$ and $\mathcal{P}(K)$ when the expiry is $T = 0.75$, Dig(K) for the price of the claim that pays $\mathbf{1}\{S_T > K\}$ at $T = 0.75$, and AoN(K) for that of the claim that pays $S_T\,\mathbf{1}\{S_T > K\}$. Data for the expiry $T = 0.75$ and the strike 40: $\Phi(d_1) = 0.60805$, $\Phi(d_2) = 0.50576$, $e^{-rT} = 0.96319$, $C(40, 0.75) = 4.8363$ and $\mathcal{P}(40, 0.75) = 3.3641$. Data for the expiry 0.25 and the strike 39.0124: the call is worth 3.1563 and the put 1.6841. **(a)** A *gap call* pays $(S_T - 45)\,\mathbf{1}\{S_T > 40\}$ at $T = 0.75$: the trigger is 40 and the payment strike 45. Without a model, express its price through $C(40)$ and Dig(40), and through AoN(40) and Dig(40). Then price it. Above which payment strike, the trigger staying at 40, is the price negative? **(b)** A *simple chooser* lets its holder declare, at $t_1 = 0.25$, that the contract is a European call or a European put, in either case with strike 40 and expiry $T = 0.75$. Using put–call parity at t_1, show that the chooser is worth $C(40, 0.75) + \mathcal{P}(\tilde{K}, 0.25)$ with $\tilde{K} = 40\,e^{-r(T-t_1)} = 39.0124$, and price it. Compare with the straddle. **(c)** Derive the second decomposition, into the put to T plus a call that expires at t_1 with strike $\tilde{K}$, by building a self-financing strategy that pays what the chooser pays. Check it on the data. **(d)** A variant lets the holder declare at any time up to t_1. Is it worth more than the simple chooser?

P23.3 *Forward-start options with a percentage strike*

A stock trades at $S = 40$ and pays a continuous dividend yield $q = 1.5\%$; the continuously compounded rate is $r = 3\%$ and the volatility is $\sigma = 35\%$, in the Black–Scholes–Merton model. A *forward-start call* gives its holder, in one year ($t_1 = 1$), a European call that expires one year later ($T = 2$) and is struck at the stock price S_{t_1} of that date. Write $C(S, K; \tau)$ and $\mathcal{P}(S, K; \tau)$ for the model prices of the call and the put with spot S, strike K and life τ. Data: $C(40, 40; 1) = 5.7316$, $\mathcal{P}(40, 36; 1) = 3.2283$ and $e^{-0.015} = 0.98511$. **(a)** Show that $C(cS, cK; \tau) = c\,C(S, K; \tau)$ for every $c > 0$, and deduce that at t_1 the option will be worth a fixed multiple of S_{t_1}. Give the multiple. **(b)** Deduce today's price of the forward-start call, and describe the position, held from today to t_1, that replicates it over that period. **(c)** Give the delta and the gamma of the forward-start call today. To which market variable is it exposed before t_1? **(d)** Price the forward-start put with the same dates that is struck at 90% of S_{t_1}. **(e)** How does the price of (b) change if the strike is set in two years instead of one, the life after that date remaining one year? What if the stock paid no dividends?

P23.4 *Barrier rapid-fire: parity, degenerate barriers and monitoring*

(a) In the Black–Scholes–Merton model let $S = K = 100$, $r = 5\%$, $q = 0$, $\sigma = 25\%$ and $T = 1$, and let a down barrier stand at $H = 90$, monitored continuously. The vanilla call is worth 12.336, and the same call with the spot 81 in the place of 100 is worth 3.4352. Compute the down-and-in call by the method of images and the down-and-out call by in–out parity; use $(10/9)^{-0.6} = 0.93874$. Now let the barrier be observed only at the close of each of the 252 trading days of the year. With the discrete-monitoring correction, whose constant is 0.5826, to what level does the barrier move, and in which direction does each of the two prices move? Use $\sqrt{252} = 15.8745$ and $e^{-0.009175} = 0.99087$; the continuous formula at the moved barrier gives 9.527 for the down-and-out call. **(b)** True or false? Give a one-line reason for each. The options are European and monitored continuously unless stated otherwise; S is the spot, K the strike, H the barrier, and there are no rebates.

(i) A down-and-out put with $K \le H < S$ is worthless.

(ii) A down-and-in call with $K \le H < S$ is worth the vanilla call.

(iii) An up-and-out call with $S < H \le K$ is worthless.

(iv) For American barrier options, the knock-in and the knock-out are together worth the American vanilla option.

(v) Observing the barrier on each of the 252 trading days, rather than on every 21st trading day only, makes a down-and-out call dearer.

(vi) The same change makes a floating lookback call, which pays S_T minus the lowest observed price, cheaper.

(vii) Over the same year, observing the barrier on every third trading day rather than on every seventh never raises the payoff of a down-and-out call, on any path.

(viii) When $r > q$, a down-and-out call whose barrier is watched on the futures price for delivery at T is worth the same as the one whose barrier is watched on the spot price, because the two prices coincide at T.

(ix) An up-and-out call has a positive vega at every spot.

P23.5 *A one-touch is worth about two digitals*

A *one-touch* with barrier H pays 1 at T if the underlying price reaches H at some time of $[0,T]$ (continuous monitoring), and the *no-touch* pays 1 at T otherwise; a *digital* pays 1 at T if the price ends above H, and a *digital put* if it ends below. Their prices are written $\mathrm{OT}(H)$ and $\mathrm{Dig}(H)$. Two results may be used without proof.

(R) *Reflection.* For a standard Brownian motion W and a level $\ell > 0$, $P\big(\max_{0\le t\le T} W_t \ge \ell\big) = 2\,P(W_T \ge \ell)$.

(M) *Images at zero carry.* Let F_t be a futures price with constant volatility σ, and let $V(F,0)$ be today's price, as a function of today's futures price F, of the European claim that pays $f(F_T)$ at T. If f vanishes on and beyond the barrier H, the claim that pays $f(F_T)$ only when the barrier is not reached by T is worth $V(F,0) - (F/H)\,V(H^2/F,0)$.

(a) Let $r = 0$, and let the underlying be a traded asset that pays no income, in a market without arbitrage, with the real-world price $S_t = S_0 + \mu_{\mathrm{N}} t + \sigma_{\mathrm{N}} W_t$ for a constant drift μ_{N} and an absolute volatility $\sigma_{\mathrm{N}} > 0$ (an arithmetic price), W being a standard Brownian motion; its risk-neutral path is $S_0 + \sigma_{\mathrm{N}} W_t$. Let $H > S_0$. For the risk-neutral path and its running maximum $S_T^{\max} = \max_{0\le t\le T} S_t$, show that $\mathbb{Q}(S_T^{\max} \ge H) = 2\,\mathbb{Q}(S_T \ge H) = 2\,\Phi\big((S_0 - H)/(\sigma_{\mathrm{N}}\sqrt{T})\big)$. **(b)** Show that this probability is a price. Let

$$V^{\mathrm{out}}(S,t) = 1 - 2\,\Phi\Big(\frac{S-H}{\sigma_{\mathrm{N}}\sqrt{T-t}}\Big), \qquad t < T.$$

Verify that $\partial_t V^{\mathrm{out}} + \frac{1}{2}\sigma_{\mathrm{N}}^2\,\partial_{SS} V^{\mathrm{out}} = 0$, that $V^{\mathrm{out}}(H,t) = 0$, and that $V^{\mathrm{out}}(S,t) \to 1$ as $t \to T$ for every $S < H$. Explain why the strategy that starts with the capital $V^{\mathrm{out}}(S_0,0)$ and holds $\partial_S V^{\mathrm{out}}(S_t,t)$ units of the underlying until the barrier is first reached, and nothing afterwards, replicates the no-touch whatever the value of μ_{N}, and conclude, by absence of arbitrage, that $\mathrm{OT}(H) = 2\,\mathrm{Dig}(H)$. **(c)** Now let F_t be a futures price with volatility σ, at any rate r, and let $d_{1,2} = [\ln(F/H) \pm \sigma^2 T/2]/(\sigma\sqrt{T})$, so that $\mathrm{Dig}(H) = e^{-rT}\Phi(d_2)$, the digital put is worth $e^{-rT}\Phi(-d_2)$, and the call and the put struck at H are worth $C(H) = e^{-rT}[F\,\Phi(d_1) - H\,\Phi(d_2)]$ and $\mathcal{P}(H) = e^{-rT}[H\,\Phi(-d_2) - F\,\Phi(-d_1)]$. Use (M) to show that, for an up barrier $H > F$,

$$\mathrm{OT}(H) = e^{-rT}\Big[\Phi(d_2) + \frac{F}{H}\,\Phi(d_1)\Big] = 2\,\mathrm{Dig}(H) + \frac{C(H)}{H},$$

and find the corresponding formula for a down barrier $H < F$. **(d)** Gold futures trade at $F = 2{,}000$, with $\sigma = 18\%$, $T = 0.25$ and $r = 0$. Price the one-touch with the up barrier 2,200 and the one-touch with the down barrier 1,800, and compare each with twice the corresponding

digital. Use $\ln 1.1 = 0.09531$, $\ln(10/9) = 0.10536$, $\Phi(-1.1040) = 0.13480$, $\Phi(-1.0140) = 0.15529$, $\Phi(-1.1257) = 0.13015$ and $\Phi(-1.2157) = 0.11205$. **(e)** Explain the signs of the two corrections. A desk has sold the up one-touch of (c) and holds two digitals struck at H: what is its position worth at the moment the barrier is touched, with a time τ left to expiry?

P23.6 *Asian options: effective volatility, bounds and average-price parity*

A stock trades at $S = 50$ and pays a continuous dividend yield $q = 1\%$; the rate is $r = 3\%$ and the volatility $\sigma = 35\%$, in the Black–Scholes–Merton model; the cost of carry is $b = r - q = 2\%$. Asian options with expiry $T = 1$ and strike $K = 50$ are written on the J fixings S_{t_j}, $t_j = jT/J$, with arithmetic average $\bar{S}_T = \frac{1}{J}\sum_j S_{t_j}$ and geometric average $G_T = \big(\prod_j S_{t_j}\big)^{1/J}$. **(a)** For $J = 2$, compute by hand the variance of $\ln G_T$ under $\mathbb{Q}$ as a multiple of $\sigma^2 T$, by writing $W_{T/2}$ and W_T through two independent increments. Check the result against the general factor $(J+1)(2J+1)/(6J^2)$. **(b)** For $J = 4$, compute the effective volatility σ_G, the forward $F_G = \mathbb{E}^{\mathbb{Q}} G_T$ and the price of the geometric average-price call, which pays $(G_T - K)^+$. Use $e^{0.00293} = 1.00293$, $\Phi(0.13204) = 0.55252$, $\Phi(-0.10759) = 0.45716$ and $e^{-0.03} = 0.97045$. **(c)** The vanilla calls struck at 50 that expire after 0.25, 0.5, 0.75 and 1 year are worth 3.5947, 5.1266, 6.3102 and 7.3099. Give a model-free upper bound for the arithmetic average-price call with $J = 4$, which pays $(\bar{S}_T - K)^+$, and bracket its price. Use $e^{-0.0225} = 0.97775$, $e^{-0.015} = 0.98511$ and $e^{-0.0075} = 0.99253$. **(d)** To what value does σ_G tend as $J \to \infty$? The formula of (b) then gives 3.900. Why is the quarterly price higher? **(e)** Write C^{avg} and $\mathcal{P}^{\text{avg}}$ for the prices of the arithmetic average-price call and put (payoffs $(\bar{S}_T - K)^+$ and $(K - \bar{S}_T)^+$), C^{as} and $\mathcal{P}^{\text{as}}$ for the average-strike call and put (payoffs $(S_T - \bar{S}_T)^+$ and $(\bar{S}_T - S_T)^+$), and C, $\mathcal{P}$ for the vanilla call and put with strike K and expiry T. Prove, without a model, that $C^{\text{avg}} + C^{\text{as}} - C = \mathcal{P}^{\text{avg}} + \mathcal{P}^{\text{as}} - \mathcal{P}$. **(f)** Why is an Asian option easier to delta hedge near expiry than a vanilla option?

P23.7 *Exchanging one asset for another*

Two assets pay continuous yields q_1 and q_2 and have, under the risk-neutral measure, the terminal prices

$$S_{k,T} = S_{k,0} \exp\big((r - q_k - \tfrac{1}{2}\sigma_k^2)T + \sigma_k \sqrt{T}\, Z_k\big), \qquad k = 1, 2,$$

where Z_1 and $Z_2 = \rho Z_1 + \sqrt{1-\rho^2}\, Z_\perp$ are standard normal, Z_1 and $Z_\perp$ being independent. Here $T > 0$, $\sigma_1, \sigma_2 > 0$ and $-1 \le \rho \le 1$, and the *ratio volatility* σ is given by $\sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$; it is assumed that $\sigma > 0$, which excludes only $\rho = 1$ with $\sigma_1 = \sigma_2$. The *exchange option* pays $(S_{1,T} - S_{2,T})^+$ at T. Two tools may be used without proof.

(G) If X and Y are linear combinations, plus constants, of the same independent normal variables, with $\operatorname{Var} Y > 0$, then X is normal, $\mathbb{E}\, e^Y = e^{\mathbb{E}[Y] + \operatorname{Var}(Y)/2}$, and, for every function ψ of at most exponential growth, $\mathbb{E}[e^Y \psi(X)] = \mathbb{E}[e^Y]\, \mathbb{E}\big[\psi(X + \operatorname{Cov}(X, Y))\big]$.

(L) For $x > 0$, $\theta > 0$ and a standard normal Z, $\mathbb{E}\big[(x\, e^{-\theta^2/2 + \theta Z} - 1)^+\big] = x\, \Phi(d_1) - \Phi(d_2)$ with $d_{1,2} = (\ln x \pm \theta^2/2)/\theta$.

(a) Show that

$$e^{-rT}\, \mathbb{E}^{\mathbb{Q}}\big[(S_{1,T} - S_{2,T})^+\big] = S_{1,0} e^{-q_1 T}\, \Phi(d_1) - S_{2,0} e^{-q_2 T}\, \Phi(d_2),$$

where

$$d_{1,2} = \frac{\ln\big(S_{1,0} e^{-q_1 T} / (S_{2,0} e^{-q_2 T})\big) \pm \sigma^2 T/2}{\sigma\sqrt{T}}, \qquad \sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2,$$

and say where the rate r disappears. **(b)** At a time $t < T$ let $V(x_1, x_2, t) = x_1\, \Phi(d_1) - x_2\, \Phi(d_2)$, with $d_{1,2} = [\ln(x_1/x_2) \pm \sigma^2(T-t)/2]/(\sigma\sqrt{T-t})$, be the candidate value of the option, as a function of the prices $x_k = S_{k,t} e^{-q_k(T-t)}$ of the two reinvested holdings (the reinvested holding of asset k is $e^{-q_k(T-t)}$ units of the asset at t, its income reinvested in the asset, so that it pays nothing and is exactly one unit at T). Show that $x_1\, \varphi(d_1) = x_2\, \varphi(d_2)$, that $\partial_1 V = \Phi(d_1)$ and $\partial_2 V = -\Phi(d_2)$ (derivatives in x_1 and x_2), and that $V = x_1\, \partial_1 V + x_2\, \partial_2 V$ (Euler's identity). Show also that

$\partial_t V + \frac{1}{2}(\sigma_1^2 x_1^2\,\partial_{11}V + 2\rho\sigma_1\sigma_2 x_1 x_2\,\partial_{12}V + \sigma_2^2 x_2^2\,\partial_{22}V) = 0$, the drift term that the two-dimensional Itô lemma attaches to V. What does Euler's identity say about the cash in the delta hedge, and why is r absent from the price while the yields are not? **(c)** A contract gives the right to receive one share of A in exchange for two shares of B in six months. Share A trades at 60, with volatility 25% and no dividends; share B trades at 30, with volatility 35% and a dividend yield of 2%; the correlation is 0.6. Price the contract and describe the replicating portfolio today. Use $\Phi(0.15) = 0.55962$, $\Phi(0.05) = 0.51994$ and $e^{-0.01} = 0.99005$. **(d)** Price the claim that pays, in six months, the more valuable of one share of A and two shares of B.

P23.8 *The Monte Carlo error budget*

A simulation prices a path-dependent option from $n = 40{,}000$ independent paths. The sample standard deviation of the discounted payoffs is $s = 22.4$. **(a)** Give the standard error of the estimated price and the half-width of the approximate 95% confidence interval. **(b)** How many paths are needed for a standard error of 0.02? By what factor must the number of paths grow to halve a standard error? **(c)** For a second claim, the delta is estimated by bump and revalue with the bump $\Delta S = 0.5$: for each pair of paths, X is the discounted payoff at the spot $S + \Delta S$ and Y the discounted payoff at the spot S, with $\operatorname{Var} X = 180$ and $\operatorname{Var} Y = 170$. Compute the variance of $(X - Y)/\Delta S$ for one pair, and the standard error of the delta estimated from 10,000 pairs, (i) when the two runs use independent random numbers, and (ii) when they use the same random numbers and $\operatorname{Corr}(X, Y) = 0.999$. **(d)** Three assets have the correlations $\rho_{12} = 0.6$, $\rho_{13} = 0.3$ and $\rho_{23} = 0.5$. Find the lower-triangular array γ_{ik} with positive diagonal entries for which the combinations $\sum_k \gamma_{ik} Z_k$ of three independent standard normal variables Z_1, Z_2, Z_3 are standard normal with these correlations, and verify it.

P23.9 *Antithetic pairs and control variates*

A simulation estimates $\mathbb{E}\,X$, where X is the discounted payoff of an option on one simulated path. On the same path a second discounted payoff Y, the *control*, is computed, whose mean $\mathbb{E}\,Y$ is known in closed form. Both variances are finite and positive, and $\rho = \operatorname{Corr}(X, Y)$. **(a)** Show that, for every real β, the variable $X - \beta\,(Y - \mathbb{E}\,Y)$ is unbiased for $\mathbb{E}\,X$, and compute its variance. Find the coefficient β^* that minimises this variance, prove that it is the only minimiser, and show that the minimum is $(1 - \rho^2)\operatorname{Var} X$. **(b)** A pilot run gives $\operatorname{Var} X = 64$, $\operatorname{Var} Y = 49$ and $\operatorname{Cov}(X, Y) = 54.6$. Compute ρ, β^*, the variance per path at β^* and at $\beta = 1$, and the factor by which the control at β^* cuts the number of paths needed for a given accuracy. **(c)** Antithetic sampling. Let $f(Z)$ be the discounted payoff written as a function of the standard normal variable Z that drives the path, with $0 < \operatorname{Var} f(Z) < \infty$, and let $\rho_A = \operatorname{Corr}(f(Z), f(-Z))$. Show that the pair average $\frac{1}{2}(f(Z) + f(-Z))$ is unbiased with variance $\frac{1}{2}\operatorname{Var} f(Z)\,(1 + \rho_A)$, and compare it with the average of two independent draws. When does the pair help? **(d)** Compute ρ_A by hand for (i) the digital $f(z) = \mathbf{1}\{z > 0.5\}$, using $\Phi(0.5) = 0.69146$, and (ii) the payoff $f(z) = |z|$, an idealised straddle that is symmetric in the normal. What does antithetic sampling do in each case? **(e)** A trial combines the two techniques: it draws an antithetic pair and applies the control to each member. How many payoff evaluations does one trial need, and from which numbers is the standard error computed?

P23.10 *One explicit finite-difference step and the stability limit*

In the log price $x = \ln S$ the Black–Scholes–Merton equation has constant coefficients, and on a grid with space step Δx and time step Δt the explicit scheme reads

$$\frac{V_j^{i+1} - V_j^i}{\Delta t} + \tfrac{1}{2}\sigma^2\,\frac{V_{j+1}^{i+1} - 2V_j^{i+1} + V_{j-1}^{i+1}}{\Delta x^2} + \nu\,\frac{V_{j+1}^{i+1} - V_{j-1}^{i+1}}{2\,\Delta x} - r\,V_j^i = 0, \qquad \nu = r - q - \tfrac{1}{2}\sigma^2,$$

where V_j^i is the value at the time $i\,\Delta t$ and the node $x_j = x_0 + j\,\Delta x$ (the superscript is a time index). Take $\sigma = 30\%$, $r = 5\%$, $q = 0$ and $\Delta x = 0.06$. **(a)** Write the scheme as $(1 + r\,\Delta t)\,V_j^i = p_u\,V_{j+1}^{i+1} + p_m\,V_j^{i+1} + p_d\,V_{j-1}^{i+1}$ and compute the three weights for $\Delta t = 0.02$.

Check their sum, and the mean and the second moment of the step of the log price that they describe. **(b)** What is the largest Δt for which no weight is negative? Compute the weights for $\Delta t = 0.05$, and the factor by which one step then multiplies a sawtooth error $\varepsilon\,(-1)^j$ on the infinite grid (all integers j), before and after the division by $1 + r\,\Delta t$. How large has an error of 10^{-6} become after 40 steps? **(c)** For the same sawtooth and $\Delta t = 0.05$, consider the implicit scheme $(I - \Delta t\,A)V^i = V^{i+1}$ and the Crank–Nicolson scheme $(I - \frac{1}{2}\Delta t\,A)V^i = (I + \frac{1}{2}\Delta t\,A)V^{i+1}$, where $(Aw)_j = \frac{1}{2}\sigma^2\,(w_{j+1} - 2w_j + w_{j-1})/\Delta x^2 + \nu\,(w_{j+1} - w_{j-1})/(2\,\Delta x) - r\,w_j$ for a grid vector w and I is the identity. Show that, on the infinite grid, A multiplies the sawtooth by the number $\Lambda(\pi) = -2\sigma^2/\Delta x^2 - r$ (the sawtooth is the mode of frequency π), and compute the factor by which each scheme multiplies it in one step. **(d)** An American put with strike 50 is one step ($\Delta t = 0.02$) from expiry, on the nodes $S_j = 50\,e^{j\,\Delta x}$, which for $j = -2, \dots, 2$ are 44.3460, 47.0882, 50, 53.0918 and 56.3748. Compute its value at the nodes 47.0882 and 50 after one step, exercise test included. The exact European put at $S = 47.0882$ with 0.02 years to run is worth 2.939: what does the comparison show? State also the two boundary rows that a full grid for this put carries. **(e)** The space step is halved to $\Delta x = 0.03$. What happens to the largest time step of (b), and to the amount of work?

P23.11 *Which model makes which smile, and Merton's formula*

Throughout, prices are discounted risk-neutral expectations, $S = 100$ and $r = q = 0$; $C^{\mathrm{BSM}}(S, K; \sigma)$ is the Black–Scholes–Merton call, which at the money ($K = S$, $r = q = 0$) equals $S\,[2\,\Phi(\frac{1}{2}\sigma\sqrt{T}) - 1]$. **(a)** The volatility is 18% for the first three months and 26% for the following nine, and over each period the log price moves by an independent normal variable whose variance is the squared volatility times the length of the period and whose mean is minus half that variance. Which single volatility prices a one-year European option through the BSM formula, and why is it not the time-weighted average of the two volatilities? Give the at-the-money call both ways, using $\Phi(0.12124) = 0.54825$ and $\Phi(0.12) = 0.54776$. **(b)** Match each smile with the model feature that produces it. Smiles: (1) flat in the strike at each maturity, at a level that depends on the maturity; (2) symmetric about the money, with its minimum near the money, at every maturity; (3) a steep downward skew at short maturities that fades as the maturity grows; (4) a downward skew that persists at long maturities; (5) an exact fit of today's whole surface, with forward smiles that flatten. Features: (A) local volatility; (B) a deterministic volatility that depends on time; (C) jumps with a negative mean; (D) stochastic volatility uncorrelated with the price; (E) stochastic volatility negatively correlated with the price. Each feature is used exactly once. **(c)** Let the average volatility σ_{avg} over the year be 10% or 30% with probability one half each, independently of the standard normal variable Z that drives the price, so that $S_T = 100 \exp\big(-\frac{1}{2}\sigma_{\mathrm{avg}}^2 + \sigma_{\mathrm{avg}} Z\big)$. Compute the one-year at-the-money call and its implied volatility, and compare the latter with the root-mean-square volatility. The same model prices the call struck at 125 at an implied volatility of 22.91%. Using put–call symmetry at zero carry, $C^{\mathrm{BSM}}(S, K; \sigma) = (K/S)\,\mathcal{P}^{\mathrm{BSM}}(S, S^2/K; \sigma)$ for every σ, find the implied volatility at the strike 80. Use $\Phi(0.05) = 0.51994$, $\Phi(0.15) = 0.55962$ and $\Phi(0.09988) = 0.53978$. **(d)** In Merton's jump model the risk-neutral price at T is

$$S_T = S \exp\Big((r - q - \lambda\kappa - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z_0 + \sum_{i=1}^{N_T} \big(\ln(1+\kappa) - \tfrac{1}{2}\delta^2 + \delta Z_i \big) \Big),$$

where $N_T \sim \mathrm{Poisson}(\lambda T)$ counts the jumps and $N_T, Z_0, Z_1, \dots$ are independent, the Z_i being standard normal, with $\sigma > 0$, $\lambda \geq 0$, $\kappa > -1$ and $\delta \geq 0$. By conditioning on the number of jumps, show that the call is worth

$$\sum_{j=0}^{\infty} e^{-\lambda' T}\,\frac{(\lambda' T)^j}{j!}\; C^{\mathrm{BSM}}(S, K; \sigma_{(j)}, r_{(j)}),$$

with

$$\lambda' = \lambda(1+\kappa), \qquad \sigma_{(j)}^2 = \sigma^2 + \frac{j\,\delta^2}{T}, \qquad r_{(j)} = r - \lambda\kappa + \frac{j\ln(1+\kappa)}{T},$$

where $C^{\mathrm{BSM}}(S, K; \sigma, r)$ is the BSM call with that volatility and with that rate used both in the drift and in the discounting, the yield q being unchanged. Then let $\lambda = 1$, $\kappa = -5\%$, $\delta = 10\%$, $\sigma = 20\%$ and $T = 0.5$. Compute $\lambda' T$, the weights of $j = 0, 1, 2, 3$, $\sigma_{(1)}$ and $r_{(1)}$, and the at-the-money call, given that the scenario prices $C^{\mathrm{BSM}}(100, 100; \sigma_{(j)}, r_{(j)})$ for $j = 0, \ldots, 5$ are 6.8887, 5.7413, 4.8665, 4.1656, 3.5889 and 3.1066; use $e^{-0.475} = 0.62189$ and $\ln 0.95 = -0.05129$. In the same model the call struck at 90 is worth 12.428: what is the put struck at 90 worth, and which property of the model gives it without a second series? **(e)** Why does a local-volatility model, fitted exactly to today's vanilla prices, misprice a forward-start option?

27.24 Problems — §24 Algorithmic Cost Models

P24.1 *Matmul operation count* ★

What is the complexity — the number of operations — of matrix multiplication? Cover the count for a general $(m \times n)(n \times p)$ product and for the square $n \times n$ case, say which operation is conventionally counted and why, and mention what is known below the naive exponent.

P24.2 *Why count multiplications* ♦

When estimating the speed of matrix operations in AI, why is it common practice to count the number of multiplications rather than the number of additions?

P24.3 *Low-rank decomposition crossover* ★

Multi-step, pen and paper. **(a)** What is the cost, in multiplications, of multiplying an $N \times P$ matrix by a $P \times R$ matrix? **(b)** A dense $N \times P$ matrix M is replaced by a low-rank factorisation $M = UV$, with U of size $N \times K$ and V of size $K \times P$. To apply M to a $P \times R$ block of data B you may compute $(UV)B$ or $U(VB)$: cost both association orders, state which one you must use, and derive the condition on K under which the factored form beats the dense product MB.

P24.4 *Low-rank speedup, numeric*

An inference engineer replaces a dense 4096×4096 layer with a rank-64 factorisation $(4096 \times 64)(64 \times 4096)$ and claims each forward pass through the layer becomes about 32 times cheaper. **(a)** Verify or refute the claim with a multiplication count for a single input vector. **(b)** What is the break-even rank? **(c)** Name the first-order cost of this trade that the multiplication count does not capture.

P24.5 *Membership: scan vs sort vs hash* ★

Multi-step. You must answer q membership queries ("is x in the dataset?") against a static dataset of N items. Compare three designs: (i) a linear scan per query; (ii) sort once, then binary-search each query; (iii) build a hash set once, then probe each query. Give the total cost of each, derive roughly how many queries make sorting pay for itself relative to scanning, and state when each design is the right choice.

P24.6 *CS fundamentals rapid-fire* ♦

Rapid-fire, multiple-choice pace — one line of justification each.

- **(a)** A slow function always returns the same output for the same input; you store results in a dictionary so repeat calls return instantly. What is this optimisation technique called? (One tempting option read: "dynamic programming.")
- **(b)** A brute-force nearest-neighbour search compares the query against every record. If the dataset size doubles, roughly what happens to the search time?
- **(c)** A stack (LIFO) undergoes, in order: Push(10), Push(20), val = Pop(), Push(40), Push(30), Push(val), Pop(), Pop(). What is the value of the last item popped?

- **(d)** A recursive function calls itself to depth N without tail-call optimisation. What is the impact on space complexity?
- **(e)** Simplify the Boolean expression $[(A \vee B) \wedge (A \vee \neg B)] \vee [(C \wedge D) \wedge (C \wedge \neg D)]$ to its most minimal form.

27.25 Problems — §25 Forecasting Returns: the Open-Ended Question

P25.1 *Open-ended process question*

An interviewer closes your first round with: "Suppose you wanted to predict tomorrow's return of a major equity index — say the Euro Stoxx 50. Walk me through, end to end, how you would approach it." Give the full process you would narrate, including the design choices you would flag explicitly and the risks you would name unprompted.

P25.2 *Backtest critique* ♦

A strategy of buying Ethereum at 9am every day and closing the position at 7pm backtests very well over the past seven years. What would be the main issue in executing this strategy?

P25.3 *Open-ended design choice: direct vs factorial*

You are building a weekly return forecast for the CAC 40. A colleague fits models directly to the index return series. You propose instead modelling the 40 constituents and aggregating the forecasts with index weights. Make the case for the constituent (factorial) route, explain where PCA enters, and name at least two ways the aggregation can fail.

27.26 Problems — §26 Computer Science for Research Engineers: Python Data Structures, Parallelism, and Time

P26.1 *Pricing Python's containers, qualifiers included*

Costs are counted in the RAM model: one unit per pointer read or write, per comparison, or per hash evaluation. **(a)** For each of the following six CPython operations give its cost and the one qualifier that belongs to it (worst case, amortised, or expected), with a one-clause reason: `L.append(x)`; `D[k]` on a dict; `dq.popleft()` on a deque; `L.insert(0, x)`; `heapq.heapify(L)` on n items; `x in S` on a set. **(b)** A queue is implemented as a list, appending at the back and removing from the front with `pop(0)`, which shifts every remaining pointer down one slot. The queue is filled with $n = 200{,}000$ items and then drained with no interleaved appends. How many pointer moves does the draining cost? How many for a `deque`? A breadth-first search on a star graph with $V = 100{,}000$ vertices and $E = V - 1$ edges uses that list as its queue: the centre is expanded first and enqueues the $V - 1$ leaves, which are then removed one by one. Give the pointer moves of the list version against the $V + E$ container operations of the deque version. **(c)** A report is built from $n = 50{,}000$ pieces of $L = 20$ characters by `s = s + piece`, where `s` is a module-level global. How many characters are copied in total? How many by `"".join(pieces)`? Would the answer change if `s` were a local variable?

P26.2 *Dynamic arrays from scratch: geometric against additive growth*

A dynamic array starts with capacity $c_0 \geq 1$ and, whenever an append finds it full, allocates a larger array and copies every stored pointer across; n appends are performed on an initially empty array, and the cost of an append is the pointers copied plus the one direct write. **(a)** Suppose each resize multiplies the capacity by a fixed factor $g > 1$ (rounded up to an integer). Prove that the total number of pointer copies over the n appends is less than $g\,n/(g-1)$, so that the amortised cost of an append is at most $1 + g/(g-1)$. **(b)** Suppose instead each resize adds a fixed constant $c \geq 1$ to the capacity, with $c_0 = c$. Show that the copies total $n^2/(2c) + O(n)$. **(c)** Take

$c_0 = 4$, doubling ($g = 2$) and $n = 3{,}000{,}000$. At which sizes do resizes occur, how many are there, how many pointers are copied in total, and how many writes does an append cost on average? Compare with the bound of (a) for $g = 2$ and for $g = 9/8$ (CPython's asymptotic growth ratio). How many copies would additive growth by $c = 1000$ cost for the same n?

P26.3 *Resizes, heaps and binary search by the numbers*

(a) Model a hash table as follows: the index array starts at 8 slots; when inserting a new key would make the number of stored keys exceed $\lfloor 2s/3 \rfloor$, where s is the current index size, the table first doubles s and re-inserts every stored key. Insert 10^6 distinct keys into an empty table. How many resizes occur, how many re-insertions do they perform in total, what is the amortised number of re-insertions per insertion, and what is the final index size? **(b)** A binary min-heap is stored in a list, the children of index i at $2i + 1$ and $2i + 2$. For $n = 31$ items (a full tree of height 4), bound the number of swaps made by `heapify`, which sifts down the internal nodes from index $\lfloor n/2 \rfloor - 1$ to 0. How many swaps do 31 successive pushes of the strictly decreasing sequence $31, 30, \ldots, 1$ make? (A push appends the item as the last leaf and swaps it upward while it is smaller than its parent.) **(c)** The $k = 50$ largest of $n = 2 \times 10^7$ items are wanted. Give the heap-step count and memory of the size-k min-heap method against the comparison count and memory of sorting, using $\log_2 50 = 5.64$ and $\log_2(2 \times 10^7) = 24.25$. **(d)** Binary search with half-open interval $[lo, hi)$: start at $(0, n)$; while $lo < hi$, set $mid = \lfloor (lo + hi)/2 \rfloor$ and move lo to $mid + 1$ if $a[mid] < t$, else hi to mid. How many iterations at most for $n = 10^6$? Trace it on $a = [1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144]$ with $t = 50$: list the (lo, hi, mid) triples, the exit value of lo, and what that value means. **(e)** A recursive depth-first search is run on a path graph of 2500 vertices from one end, with CPython's default recursion limit of 1000. What happens, and how many units of work (V pushes and pops plus E edge scans) does the iterative version with an explicit stack perform instead?

P26.4 *Waves, work and span on a ten-task DAG*

Ten unit-time tasks A to J have the thirteen dependency edges A→B, A→C, A→D, B→E, C→E, C→F, D→F, D→G, E→H, F→H, G→I, H→J, I→J (an edge $u \to v$ means v may start only after u has finished). **(a)** Give every task's in-degree and run Kahn's algorithm with a FIFO queue: list the waves (wave 1 is the set of sources; wave $t + 1$ is the set of tasks whose in-degree reaches zero while wave t is scanned). How many container operations does the run perform? **(b)** Give the work T_1, the span T_∞, the parallelism, and the number of critical paths. **(c)** A schedule on P processors assigns tasks to steps, respecting the edges, at most P per step; a greedy schedule runs P ready tasks whenever at least P are ready and all ready tasks otherwise. Give a greedy schedule for $P = 2$ and one for $P = 3$, the number of complete steps (exactly P tasks) and incomplete steps in each, and check each against the bound $\lfloor T_1/P \rfloor + T_\infty$. Show that no 2-processor schedule beats your $P = 2$ length, and say what a fourth processor buys.

P26.5 *Reading the two clocks, then Amdahl*

A machine has C cores; a process's CPU time is the integral over the interval of the number of cores executing its threads, r is CPU time divided by wall time, and utilisation is r/C. Answer each report. **(a)** A single-process job on $C = 4$: `real` 240 s, `user` 60 s, `sys` 12 s. Compute r and the utilisation. For how long, at least, was the process on no core, and what does that say about it? If the compute were spread perfectly over the four cores with the off-core time unchanged, what would the wall time and the speedup be, and what is the ceiling over any number of cores? **(b)** A numpy script on $C = 8$: `real` 2.5 s, `user` 18.0 s, `sys` 0.5 s. Is the report consistent with the accounting inequality? What is the smallest number of threads the process can have had, and who created them? Would launching a second copy of the script at the same time be expected to finish both in about 2.5 s? **(c)** Two threads on $C = 2$ cores over a $W = 12$ s interval: thread 1 is busy on $[0, 5]$ and $[9, 12]$, thread 2 on $[2, 8]$ and $[10, 12]$. Compute the CPU time, r, the utilisation, and the lengths of time with two, one and zero cores busy. **(d)** A related job measured a speedup of 6.0 on 12 processes. Compute the serial fraction implied by the Karp–Flatt identity

$1 - p = (P/S - 1)/(P - 1)$ and the speedup it predicts on 24 processes. The job is then run on 24 and measures 7.0: what serial fraction does that read as, and what does the change between the two readings indicate?

P26.6 *The greedy-scheduler bound from scratch*

A computation is a directed acyclic graph of unit-time tasks; its work T_1 is the number of tasks and its span T_∞ the number of tasks on a longest directed path. A schedule on P processors assigns each task to a step so that every task runs in a later step than each of its predecessors, with at most P tasks per step; its length T_P is its number of steps. A task is ready at a step if all its predecessors ran in earlier steps; a greedy schedule runs P ready tasks whenever at least P are ready and all ready tasks otherwise. **(a)** Prove that every schedule satisfies $T_P \geq T_1/P$ and $T_P \geq T_\infty$. **(b)** Prove that every greedy schedule satisfies $T_P \leq \lfloor T_1/P \rfloor + T_\infty$. **(c)** Deduce that a greedy schedule is within a factor 2 of optimal, and that if $P \leq \varepsilon\, T_1/T_\infty$ then its speedup T_1/T_P is at least $P/(1+\varepsilon)$. **(d)** Eight independent tasks of durations $9, 7, 6, 5, 4, 4, 3, 2$ (total 40) are handed to $P = 3$ workers, each task going to the worker that becomes free first, either in the order listed (longest first) or in the reverse order (shortest first). Give both finish times, the bound $T_1/P + t_{\max}$ from (b), the lower bound from (a), and say which order is optimal and why the other loses.

P26.7 *Which reductions parallelise, and why the answer is the same*

A reduction may be evaluated by any bracketing of the operands in their given order if its operation is associative, and by any bracketing of any reordering if it is also commutative. **(a)** Classify each of the following eleven reductions as *associative and commutative*, *associative only*, or *neither*, with a small counterexample for each property that fails; for each *neither*, say whether a lifted state repairs it and give the state: integer sum; max; string concatenation; "last non-null" ($a \oplus b = b$ if b is non-null, else a); subtraction; the pairwise mean $(x, y) \mapsto (x + y)/2$; the median; floating-point sum; the matrix product; "argmax, ties to the smaller index" on (index, value) pairs; the number of distinct elements. **(b)** Fold $x = [8, 3, 5, 2, 7, 1]$ under subtraction as a left fold and as the tree $((8 - 3) - (5 - 2)) - (7 - 1)$. Then fold the same list with the lifted state (first element, sum of the rest) on two chunks of three and read off the result. **(c)** In IEEE binary64, doubles near 10^{16} are spaced 2 apart and a tie rounds to the neighbour with even significand; 10^{16} has an even significand. Evaluate $(10^{16} + 1) + 1$ and $10^{16} + (1 + 1)$. Then sum the list $[10^{16}, 1, 1, 1, 1]$ three ways: sequentially left to right; with $P = 2$ workers holding contiguous chunks of $\lceil 5/2 \rceil = 3$ and 2 elements, chunk totals merged in chunk order; and with $P = 3$ workers holding chunks of $2, 2, 1$, merged in chunk order and then in the order (chunk 1 + chunk 3) + chunk 2. Give the exact sum for comparison and say what a test asserting equality between the 2-worker and 3-worker results would do.

P26.8 *Merging shard statistics and shard gradients*

(a) Three shards report (count, mean, M_2), where M_2 is the sum of squared deviations from the shard's own mean: shard 1 reports $(4, 10, 20)$, shard 2 reports $(6, 15, 54)$ and shard 3 reports $(2, 30, 8)$. Merge them into the count, mean and M_2 of the union of the data, and give the population variance and the sample variance. Show that the unweighted average of the three means and the unweighted average of the three population variances are both wrong. **(b)** A colleague proposes the state (count, $\sum x$, $\sum x^2$) instead, computing the variance as $\sum x^2/n - m^2$, and tests it in binary64 on $x = 10^8 + 1,\ 10^8 + 2,\ 10^8 + 3$. Doubles near 3×10^{16} are spaced 4 apart. Give the true value of $\sum x^2 - nm^2$ and explain why the floating-point evaluation cannot be trusted to return it. **(c)** In data-parallel training a batch is split into two shards of 5 and 3 examples whose shard-mean gradients (one coordinate shown) are 2.0 and -2.0. Give the full-batch gradient and the wrong unweighted answer. A model of 10^9 float32 parameters (4 GB of gradient) trains on $P = 4$ GPUs with 200 ms of compute per step; a ring all-reduce moves $2\,(P-1)/P$ times the gradient size per GPU. Give the bytes moved per GPU, the step time and the efficiency at 300 GB/s per GPU, and again at 25 GB/s.

P26.9 *Threads, processes, pools and a data loader, priced*

On the default CPython build at most one thread executes Python bytecode at any instant; the lock is released during blocking I/O and inside C extensions that opt out. **(a)** A pure-Python CPU-bound job takes 60 s single-threaded, of which 8 s (loading and writing) is inherently serial and 52 s divides perfectly. Give the wall time and the CPU-to-wall ratio r with 6 threads on an 8-core machine; the wall time and speedup with 6 processes (start-up ignored); and the speedup ceiling over any number of processes. **(b)** An I/O-bound job issues 200 network requests, each waiting 0.4 s for its reply and costing 5 ms of CPU. Give the serial wall time, the wall time with 25 threads (requests issued in rounds of 25), the total CPU time, and r for the threaded run. **(c)** A read-only 2 GB numpy array is needed by 16 worker processes. How many bytes are copied if it is pickled to each worker (at 2 GB/s, how long), and if it is fork-inherited or placed in shared memory? A job of 10^6 small items is split into 10^6 pool tasks at 75 μs of round-trip overhead each: what is the overhead, and what is it with 100 chunks? **(d)** A pool does $W_0 = 400$ s of perfectly divisible work, but each worker adds a fixed $c = 1$ s (spawn, import, result gathering) paid serially, so $T(P) = cP + W_0/P$. Find the optimal worker count and time, and evaluate $T(10)$, $T(40)$ and $T(80)$. **(e)** A training step takes 40 ms on the GPU per batch; 3 loader workers each produce 6 batches per second. What are the pipeline throughput, the GPU's busy fraction, and the number of workers at which the GPU becomes the bottleneck? Separately, a batched model call has launch overhead $c = 2$ ms and per-item cost $\tau = 50$ ns: give latency and throughput for $B = 10^4$ and $B = 10^5$, and the batch size at which overhead equals compute.

P26.10 *How low can a racy counter go?*

Threads share an integer c that starts at 0. Each thread executes a fixed sequence of atomic steps in its own order, and an execution is any interleaving of the threads' sequences. One increment is three steps: *load* (copy c into a private temporary), *add* (add one to the temporary), *store* (write the temporary to c). $K \geq 2$ threads each perform $N \geq 2$ increments; the final value v of c is the value written by the last store. **(a)** Prove that $v \leq KN$. **(b)** Prove that $v \geq 2$. **(c)** For $K = 2$ and $N = 3$, write out step by step an interleaving that ends with $c = 2$, and one that ends with $c = 6$. **(d)** For $N = 1$ and $K = 3$, list every reachable final value with an interleaving for each, and explain why 2 is not the minimum in that case. **(e)** Eight workers (processes, or threads on a build without the interpreter lock) each perform 10^7 increments of a shared counter. A lock costs at least 25 ns per uncontended acquisition and an unlocked increment 20 ns. Give the least wall time of the locked version, the wall time of a single thread doing all 8×10^7 increments unlocked, and the wall time with private per-worker counters merged at the end. If instead a lock guards 4% of each worker's time, what is the speedup ceiling?

P26.11 *Deadlock with three locks, and what is safe to retry*

(a) Thread A acquires lock L_1 then L_2; thread B acquires L_2 then L_3; thread C acquires L_3 then L_1; each releases both of its locks when its work is done. Exhibit an interleaving in which all three threads wait for ever, give the wait-for graph, and check the four Coffman conditions (mutual exclusion, hold-and-wait, no preemption, circular wait) on it. Can any two of the three threads deadlock while the third is absent? **(b)** Impose the global order $L_1 < L_2 < L_3$ and rewrite the one thread that violates it. Starting from the same first two steps (A takes L_1, B takes L_2), show that the run now completes, giving the order in which the threads finish. **(c)** A MapReduce-style framework re-executes a shard task after a timeout, and the original also completes, so the task's output action runs twice on the same input. For each action say whether the final result is still correct, that is, whether the action is idempotent: (i) `results[shard_id] = partial`; (ii) `results.append(partial)`; (iii) `total += partial`; (iv) write `partial` to a temporary file, then rename it atomically onto `results/shard_id`; (v) `best = max(best, partial_best)`; (vi) append one line per shard to a shared log; (vii) `credit(account, 10)`. With 40 shards of 250 partial counts each, how many records does the list of (ii) hold after one duplicated shard, and by how much is the merged total of a word with 7 occurrences in that shard wrong?

P26.12 *Top-k across shards and the shuffle*

Words are counted across shards; a shard's local top-k is by raw occurrences on that shard; ties are broken by a fixed total order on words used everywhere. **(a)** Construct three shards of word counts such that, for $k = 2$, the union of the three local top-2 lists contains neither of the two globally most frequent words, whatever the tie-break; verify the claim. **(b)** Repartition your shards by word, so that every word's complete count lives on exactly one machine, and give the local top-2 lists then. State in one sentence the invariant that makes the coordinator's merge of the local lists exact, and prove it. How many records does the repartition move for your example, against the number of distinct words and the number of raw occurrences? **(c)** At scale: $n = 4 \times 10^9$ words over $M = 200$ machines, $d = 5 \times 10^5$ distinct words, 16 bytes per record, 200 MB/s of network per machine, 100 ns per dict update. Give the map time per machine; the records crossing the network without a combiner and the upper bound with one; and each machine's shuffle volume and time in both cases. **(d)** One word occurs 8×10^8 times. Without a combiner, how many records converge on its owner and how long does that owner spend receiving them, against the others' shuffle time? With the combiner? With a salt of $J = 8$ sub-keys, what does the second round merge? **(e)** The coordinator merges $M = 200$ local top-10 lists with a heap of list heads: how many heap steps, using $\log_2 200 = 7.64$? Mappers running as separate Python processes partition by `hash(word) % M` with `PYTHONHASHSEED` unset: what goes wrong, and what are two fixes?

27.27 Problems — Puzzle Classics

PZ.1 *Monty Hall*

A game show has three doors: one hides a car, two hide goats. You pick a door. The host — who knows where the car is and always opens a goat door from the two you did not pick — opens one, revealing a goat, and offers you the chance to switch to the remaining closed door. Should you switch, and what is your probability of winning the car if you do? Would your answer change if the host had opened one of the other doors uniformly at random and it merely *happened* to show a goat?

PZ.2 *Two ropes, 45 minutes*

You have two ropes and a lighter. Each rope takes exactly 60 minutes to burn from one end to the other, but they burn *non-uniformly*: half the rope may take 5 minutes, the rest 55. How do you measure exactly 45 minutes?

PZ.3 *Clock-hands angle* ♦

Mental only, 30 seconds. What is the angle between the hands of a clock at 09:30?

PZ.4 *100 lockers*

100 lockers are all closed. Person 1 toggles every locker; person 2 toggles lockers $2, 4, 6, \ldots$; person k toggles every k-th locker; and so on through person 100. After all 100 passes, how many lockers are open, and which ones?

PZ.5 *25 horses, 5 lanes*

You have 25 horses and a five-lane track with no stopwatch: each race ranks its five runners, nothing more. Each horse always runs at its own constant speed. What is the minimum number of races needed to find the fastest three horses, in order?

PZ.6 *Sixteen gamers, four per game* ♦

In a gaming competition, 4 players play each game and each game produces exactly one winner. Assuming the best player always wins any game they play, how many games are needed to find the best gamer in a group of 16 employees?

PZ.7 *Prisoners' hats in a line*

n prisoners stand in a line, each wearing a black or white hat, each seeing only the hats *in front* of them. Starting from the back (the prisoner who sees all $n-1$ others), each must call out "black" or "white"; a prisoner is freed iff they call their own hat's color. Everyone hears every call. The prisoners may agree on a strategy beforehand. How many can be guaranteed freedom, and by what strategy?

PZ.8 *Hats: guess or pass*

Three players are each given a hat, independently red or blue with probability 1/2. Each player sees the other two hats but not their own. Simultaneously, and with no communication once the hats are on, each player must either name a colour or pass. The team wins if at least one player correctly names their own hat colour and nobody names wrongly; passing is always safe. The team agrees on a strategy in advance. A single designated guesser wins with probability 1/2 — find a strategy that does strictly better, compute its winning probability, and prove no strategy can beat it.

PZ.9 *Broken stick triangle*

A stick is broken at two points chosen independently and uniformly at random along its length. What is the probability that the three pieces can form a triangle?

PZ.10 *Four points on a quarter arc*

Four points are chosen independently and uniformly at random on a circle. What is the probability that all four lie on some arc spanning a quarter of the circumference? Give also the general formula for n points and an arc that is a fraction $f \le 1/2$ of the circumference.

PZ.11 *Random walk on a solid*

An ant stands on one vertex of a regular octahedron and walks along edges, choosing uniformly at random among the available edges at every vertex. What is the expected number of steps for it to reach the opposite (antipodal) vertex? (The octahedron has 6 vertices, each of degree 4; each vertex is adjacent to every vertex except its antipode.)

PZ.12 *Bold vs timid play*

You walk into a casino with \$100 and must walk out with \$200 by midnight — anything less is worthless to you. You may stake any amounts on even-money bets that win with probability $p = 18/38$ (roulette red). Compare two plans: (a) one bold bet of \$100; (b) timid play, betting \$1 at a time until you reach \$200 or go bust. Compute both success probabilities, state the general lesson, and give the trading analogue.

PZ.13 *Fair coin to probability 1/3* ★

Using only a fair coin, construct an event that occurs with probability exactly 1/3. Follow-ups to be ready for: what is the expected number of flips your construction uses, and can exact probability 1/3 be achieved with a fixed, predetermined number of flips?

28 Solutions

28.1 Solutions — §1 Expected Value for Trading Games

P1.1 *Independence and the product rule* ♦

Independence is given, so the joint probability is the product:

$$P(\text{burger} \cap \text{stomach}) = P(\text{burger})\,P(\text{stomach}) = 0.4 \times 0.1 = 0.04.$$

The entire content of the question is knowing that independence licenses multiplication. Without the independence assumption, all that could be said is $0 \le P(A \cap B) \le \min(0.4,\, 0.1) = 0.1$ — the word "assuming" is doing real work.

Self-check. Consistency bounds: $P(A \cap B) = 0.04 \le \min(0.4, 0.1)$, and inclusion-exclusion gives $P(A \cup B) = 0.4 + 0.1 - 0.04 = 0.46 \le 1$. Both hold, so 0.04 is a legal joint probability for these marginals.

Answer: 0.04.

P1.2 *Binomial counting* ♦

Each of the $2^5 = 32$ sequences is equally likely; the favourable ones are the $\binom{5}{3} = 10$ sequences with heads in exactly 3 of the 5 positions:

$$P(\text{exactly 3 H}) = \frac{\binom{5}{3}}{2^5} = \frac{10}{32} = \frac{5}{16} = 0.3125.$$

Self-check. The full binomial row is $1, 5, 10, 10, 5, 1$, which sums to 32 as it must; and by the symmetry $H \leftrightarrow T$, $P(3 \text{ heads}) = P(2 \text{ heads}) = 10/32$, consistent with the row.

Answer: $5/16 = 0.3125$.

P1.3 *Two aces (hypergeometric draw)* ♦

The player's two cards are a uniformly random 2-subset of the deck (which other players also get cards does not matter — a specific player's hand is exchangeable with any other 2-subset). Favourable hands: $\binom{4}{2} = 6$ pairs of aces out of $\binom{52}{2} = 1326$ possible hands:

$$P = \frac{\binom{4}{2}}{\binom{52}{2}} = \frac{6}{1326} = \frac{1}{221} \approx 0.45\%.$$

Self-check. Sequentially: first card an ace with probability $4/52 = 1/13$, then second ace $3/51 = 1/17$, product $\frac{1}{13} \cdot \frac{1}{17} = \frac{1}{221}$ — same number by a different route.

Answer: $\binom{4}{2}/\binom{52}{2} = 1/221 \approx 0.45\%$.

P1.4 *Dice outcome counting* ♦

There are $8 \times 8 = 64$ equally likely ordered outcomes. Count the favourable sums:

- sum = 14: $(6,8), (7,7), (8,6)$ — 3 outcomes;
- sum = 15: $(7,8), (8,7)$ — 2 outcomes;
- sum = 16: $(8,8)$ — 1 outcome.

Total 6 of 64:

$$P(\text{sum} \ge 14) = \frac{6}{64} = \frac{3}{32} \approx 9.4\%.$$

Self-check. Reflect each die by $x \mapsto 9 - x$: this bijection maps $\{\text{sum} \ge 14\}$ onto $\{\text{sum} \le 4\}$, which has $1 + 2 + 3 = 6$ outcomes (sums $2, 3, 4$) — the same count from the other tail.

Answer: $3/32 \approx 9.4\%$.

P1.5 *Binomial at-least* ♦

"2 or more" out of 3 means *both* the exactly-2 case and the all-3 case — under time pressure the second term is exactly what gets dropped:

$$P(\ge 2) = \binom{3}{2}\left(\tfrac{1}{4}\right)^2\left(\tfrac{3}{4}\right) + \left(\tfrac{1}{4}\right)^3 = \frac{9}{64} + \frac{1}{64} = \frac{10}{64} = \frac{5}{32} \approx 15.6\%.$$

On the live test this question was the one confirmed miss: any answer built from a single term (or from mangled terms) is wrong because the event is a union of two disjoint outcomes.

Self-check. Complement: $P(0) = (3/4)^3 = 27/64$ and $P(1) = 3 \cdot \frac{1}{4} \cdot (3/4)^2 = 27/64$, so $P(\geq 2) = 1 - \frac{27+27}{64} = \frac{10}{64}$ — the same value computed from the other side of the distribution.

Answer: $5/32 = 10/64 \approx 15.6\%$.

P1.6 *Distinct faces in ten rolls*

Let I_k be the indicator that face k appears at least once in the 10 rolls. The number of distinct faces is $D = \sum_{k=1}^{6} I_k$, and

$$\mathbb{E}[I_k] = P(\text{face } k \text{ appears}) = 1 - \left(\tfrac{5}{6}\right)^{10},$$

so by linearity

$$\mathbb{E}[D] = 6\Big(1 - \big(\tfrac{5}{6}\big)^{10}\Big) \approx 6\,(1 - 0.1615) \approx 5.03.$$

Legitimacy: linearity of expectation holds for *any* random variables, dependent or not. The I_k here are negatively dependent (one face appearing crowds out others), but that only matters for the variance of D, never for its mean. This is the master move of the whole section: write the count as a sum of indicators and sum the probabilities.

Self-check. Small case $n = 2$ rolls: the formula gives $6(1 - (5/6)^2) = 6 \cdot \frac{11}{36} = \frac{11}{6}$; directly, $\mathbb{E}[D] = 2 - P(\text{second roll repeats the first}) = 2 - \frac{1}{6} = \frac{11}{6}$. Match. Bounds: $5.03 < 6$ (cannot exceed the number of faces) and it is close to 6 minus about one face — consistent with the coupon-collector fact that 10 rolls usually leaves roughly one face unseen.

Answer: $6(1 - (5/6)^{10}) \approx 5.03$ distinct faces. □

P1.7 *Gold and silver balls* ♦

The turn structure is a red herring: who is drawing changes nothing about *which* balls leave the bag. Drawing without replacement just reveals a uniformly random permutation of all 200,000 balls one position at a time, and the process stops exactly when the 100,000th — i.e. the *last* — gold ball is revealed. So the balls pulled out are precisely every ball up to and including the last gold in a random permutation.

All 100,000 golds are pulled by construction. The only randomness is how many silvers are left behind, namely those lying *after* the last gold. For a fixed silver ball, consider just its position relative to the 100,000 golds: all 100,001 relative orderings are equally likely, so it lies after every gold with probability $\frac{1}{100{,}001}$. By linearity over the silvers,

$$\mathbb{E}[\#\text{silver left in bag}] = \frac{100{,}000}{100{,}001} \approx 1.$$

So on average $\approx 199{,}999$ balls are pulled, of which 100,000 are gold: the proportion is

$$\approx \frac{100{,}000}{199{,}999} \approx 0.5000025,$$

a hair *above* one half. (Strictly, the expected ratio is not the ratio of expectations, but with 2×10^5 balls the pulled count concentrates completely relative to any plausible answer spacing.) The closest simple value is 0.5.

Self-check. Tiny case, 1 gold + 1 silver: the two orders are GS (pull only the gold; proportion 1) and SG (pull both; proportion $\frac{1}{2}$), so the expected proportion is $\frac{3}{4}$ — above $\frac{1}{2}$, as our analysis predicts, and the excess shrinks like $1/N$: at $N = 10^5$ it is invisible. Also, the expected-silvers-left formula gives $\frac{1}{2}$ for this tiny case ($S/(G+1) = 1/2$), matching the direct average of $\{1, 0\}$ silvers left.

Answer: 0.5 (exactly: a shade above — about 0.5000025, since on average only ≈ 1 silver ball is never pulled).

P1.8 *Geometric waiting time* ♦

The number of rolls is geometric with success probability $p = \frac{1}{6}$. First-step analysis: one roll is spent; with probability $\frac{5}{6}$ nothing has changed (memorylessness) and the expected remaining count is again E:

$$E = 1 + \tfrac{5}{6}E \quad \Longrightarrow \quad \tfrac{1}{6}E = 1 \quad \Longrightarrow \quad E = 6.$$

In general $E = 1/p$: the single most-used closed form in this family.

Self-check. Direct series: $E = \sum_{k\geq 1} k(\frac{5}{6})^{k-1}\frac{1}{6} = \frac{1}{6}\cdot\frac{1}{(1-5/6)^2} = \frac{1}{6}\cdot 36 = 6$. Same value without the recursion.

Answer: 6 rolls.

P1.9 *Geometric waiting time: memorylessness*

(a) Geometric waiting time: $\mathbb{E}[T] = 1/p = 1/0.05 = 20$ seconds.

(b) Still 20 seconds. The geometric distribution is memoryless: $P(T > s + t \mid T > s) = P(T > t)$, because the per-second fill events are independent with constant probability — the 30 fruitless seconds carry no information about the future. If waiting *did* change your forecast, the per-second fill probability would have to depend on the past, contradicting the model. (The trading caveat worth saying out loud: real fill processes are *not* memoryless — a long wait often does carry adverse information, e.g. your quote has gone stale or you are deep in the queue. The clean answer holds only under the stated independence.)

(c) The first flip shows one of the two faces for free. From then on you wait for the *opposite* face, a geometric wait with $p = \frac{1}{2}$, so

$$\mathbb{E} = 1 + \frac{1}{1/2} = 3.$$

Self-check. For (c), compute from the distribution: both faces first complete at flip $k \geq 2$ with probability $P(T = k) = 2\cdot(1/2)^k = (1/2)^{k-1}$ (the first $k-1$ flips all equal, times 2 choices of which face repeats). Then $\mathbb{E}[T] = \sum_{k\geq 2} k(1/2)^{k-1} = \big(\sum_{k\geq 1} k(1/2)^{k-1}\big) - 1 = 4 - 1 = 3$. For (b), a one-line sanity: the process restarts identically each second, so conditioning on any fill-free past leaves the same law — consistent with (a).

Answer: (a) 20 s; (b) 20 s (memorylessness); (c) 3 flips.

P1.10 *Pattern waiting times: HH versus HT*

HT. States: S_0 (no H held yet), S_1 (currently hold an H). From S_1, a T finishes; an H *stays in* S_1 — failure does not destroy progress:

$$E_1 = 1 + \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}E_1 \;\Rightarrow\; E_1 = 2, \qquad E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0 \;\Rightarrow\; E_0 = 2 + E_1 = 4.$$

HH. Same states, but now from S_1 a T sends you all the way back to S_0:

$$E_1 = 1 + \tfrac{1}{2}\cdot 0 + \tfrac{1}{2}E_0, \qquad E_0 = 1 + \tfrac{1}{2}E_1 + \tfrac{1}{2}E_0.$$

The second equation gives $E_0 = 2 + E_1$; substituting into the first, $E_1 = 1 + \frac{1}{2}(2 + E_1)$, so $E_1 = 4$ and $E_0 = 6$.

The sentence. The per-window probabilities are identical; the waiting times differ because of what a *failure* does to your progress: after H, a bad flip for HT (another H) leaves you still holding an H, while a bad flip for HH (a T) resets you to zero — self-overlapping patterns are slower to arrive. (Same mathematics as losing streaks that reset a strategy entirely: full-reset events cost more than their probability suggests.)

Self-check. Conway's overlap rule: $\mathbb{E}[\text{wait}] = \sum 2^k$ over every length k at which the pattern's prefix equals its suffix. HH overlaps with itself at $k = 2$ and $k = 1$: $4 + 2 = 6$. HT only at $k = 2$: 4. Both match the state solutions.

Answer: $\mathbb{E}[\text{flips to HH}] = 6$; $\mathbb{E}[\text{flips to HT}] = 4$. □

P1.11 *Pattern waiting times: the mental closed-form arsenal* ♦

The arsenal, with one-line reconstructions:

(a) $E = 1/p$. Justification: $E = 1+(1-p)E$ (one trial spent; failure restarts by memorylessness).

(b) $E_k = 2^{k+1} - 2$. Justification: to extend a run of $k-1$ heads, flip once more; tails restarts from scratch: $E_k = E_{k-1}+1+\frac{1}{2}E_k$, i.e. $E_k = 2E_{k-1}+2$ with $E_1 = 2$. Values: $2, 6, 14, 30, \ldots$

(c) $E_k = \dfrac{6^{k+1} - 6}{5}$. Same recursion with the die: $E_k = E_{k-1} + 1 + \frac{5}{6}E_k$, i.e. $E_k = 6E_{k-1} + 6$ with $E_1 = 6$. Values: $6, 42, 258, \ldots$

(d) HH = 6, HT = 4, HTH = 10. Justification: state equations (see the HH/HT problem), or Conway's overlap rule $\sum 2^k$ over self-overlaps — HTH overlaps at $k = 3$ and $k = 1$: $8 + 2 = 10$.

(e) Coupon collector: $6(1 + \frac{1}{2} + \cdots + \frac{1}{6}) = 14.7$. Justification: with j faces unseen the wait is geometric with $p = j/6$; sum $6/j$.

Two neighbours from the same first-step-analysis family are worth keeping equally warm for such tests (they live with the Markov-chain material): fair-game ruin duration $i(N - i)$ and the ant-on-a-cube answer 10.

Why memorise. Each of these is a 60–120 second, multi-equation derivation if done honestly — and doing it *mentally* means holding two or three unknowns and their coefficients in working memory while a clock runs. A stored closed form turns the question into arithmetic on the given parameters, and the one-line recursions above are the fallback if the test throws a variant (biased coin, different alphabet). Under a no-paper format, the binding constraint is working memory, not understanding; the closed forms are how you buy it back.

Self-check. Each run formula must reduce to (a) at $k = 1$: $2^2 - 2 = 2 = 1/(1/2)$ and $(6^2 - 6)/5 = 6 = 1/(1/6)$ — both pass. And HTH by direct states (no progress / H / HT): $e_2 = 1 + \frac{1}{2}\cdot 0 + \frac{1}{2}e_0$, $e_1 = 1+\frac{1}{2}e_1+\frac{1}{2}e_2$, $e_0 = 1+\frac{1}{2}e_1+\frac{1}{2}e_0$ gives $e_1 = 2+e_2$, $e_0 = 2+e_1 = 4+e_2$, then $e_2 = 1+\frac{1}{2}(4+e_2)$ so $e_2 = 6$ and $e_0 = 10$, agreeing with the overlap rule.

Answer: (a) $1/p$; (b) $2^{k+1} - 2$; (c) $(6^{k+1} - 6)/5$; (d) HH 6, HT 4, HTH 10; (e) 14.7.

P1.12 *Pattern races: which comes first*

(a) Neither pattern can complete before the first H appears (each needs an H strictly before its final flip). At the first H, nothing has completed, and the *very next flip decides the race*: H completes HH, T completes HT. So

$$P(\text{HT before HH}) = \tfrac{1}{2},$$

despite the waiting times being 4 versus 6. The flaw: expected waiting times are *marginal* statistics of two separate games, while a race depends on the joint structure on one coin. The gap between 6 and 4 comes entirely from what happens *after a failure* (HH resets fully, HT keeps its H) — and in a race, the moment one pattern completes the game is over, so the post-failure behaviour that inflates $\mathbb{E}[\text{HH}]$ never gets a chance to matter.

(b) Claim: HH beats TH only when the game's first two flips are HH. If they are, HH has won at flip 2 (TH cannot have appeared inside one flip). Otherwise a T occurs before any HH has completed; consider the *first H after that T*. All flips between that T and this H are tails, so this H is immediately preceded by a T — it completes TH. And no HH can have completed earlier,

since there were no heads between the T and this H. Hence

$$P(\text{HH first}) = P(\text{first two flips are HH}) = \tfrac{1}{4},$$

so TH wins with probability $\frac{3}{4}$. Decline the even-money bet on HH emphatically — fair odds are 3:1 against HH. (This asymmetry — every pattern is beaten by some other pattern chosen in response — is the Penney-game structure; the practical lesson is that "my signal fires eventually" and "my signal fires *first*" are different claims.)

Self-check. (a): enumerate from the first H — there is no path on which both patterns stay alive past the next flip, so $\frac{1}{2}$ is airtight. (b): condition on the first two flips: HH ($\frac{1}{4}$) gives HH the win; HT, TH, TT ($\frac{3}{4}$ total) each contain a T before HH can have completed, and the argument above then hands the win to TH; in the TH case it has already won at flip 2. The two computations agree, and (b) is consistent with the known Penney result that TH beats HH exactly 3:1.

Answer: (a) 1/2 — expected waiting times do not determine race odds; (b) do not take it: $P(\text{HH first}) = 1/4$, fair odds 3:1 against HH.

P1.13 *Coupon collector*

Decompose the collection into stages by how many faces remain unseen. While j faces are unseen, each roll reveals a new face with probability $j/6$, so that stage lasts a geometric time with mean $6/j$. By linearity, summing over $j = 6, 5, \ldots, 1$ (equivalently $j = 1, \ldots, 6$):

$$\mathbb{E}[T] = \sum_{j=1}^{6} \frac{6}{j} = 6\Big(1 + \tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4} + \tfrac{1}{5} + \tfrac{1}{6}\Big) = 6 \cdot \frac{49}{20} = \frac{147}{10} = 14.7 \text{ exactly.}$$

The stages are independent geometrics, but linearity would not care if they weren't. For an n-faced die, $\mathbb{E}[T] = nH_n \approx n \ln n + \gamma n + \frac{1}{2}$ with $\gamma \approx 0.577$: slightly super-linear, and dominated by the long hunt for the last few faces ($6/1 + 6/2 = 9$ of the 14.7 rolls are spent on the last two).

Self-check. Two checks. (i) $n = 2$ "die" (a coin, collect both faces): formula gives $2H_2 = 3$, matching the direct answer $1 + 2 = 3$ from the geometric subsection. (ii) Asymptotic consistency at $n = 6$: $6(\ln 6 + 0.577) + 0.5 \approx 6 \times 2.369 + 0.5 \approx 14.7$ — the approximation already reproduces the exact value to one decimal.

Answer: $6H_6 = 147/10 = 14.7$ rolls; for n faces, $nH_n \approx n(\ln n + 0.577)$. □

P1.14 *Coupon collector variant: subset of coupons amid noise*

No — the noise faces do not help you, but they do consume rolls. While j of the three target faces remain unseen, each roll hits a new target with probability $j/6$ (not $j/3$), so the stage means are $6/3, 6/2, 6/1$:

$$\mathbb{E}[T] = \frac{6}{3} + \frac{6}{2} + \frac{6}{1} = 2 + 3 + 6 = 11.$$

Note the clean structure: $\sum_j 6/j = 2\sum_j 3/j$, i.e. exactly double the 3-face collector 5.5. That is thinning: only half of all rolls land in $\{1, 2, 3\}$ at all, and conditional on landing there the process looks like the 3-face collector, so every stage — and hence the total — is stretched by the factor 2. Recognising "coupon collector run at acceptance rate q" as "divide by q" is the fast route in an interview.

Self-check. Inclusion-exclusion on the maximum. $T = \max(T_1, T_2, T_3)$ where T_i is the first appearance of face i, each geometric with mean 6. Using $\max = \sum T_i - \sum_{i<j} \min(T_i, T_j) + \min(T_1, T_2, T_3)$ and the fact that the min over a set of faces is the wait for *any* of them (geometric: mean 3 for two faces, 2 for three faces):

$$\mathbb{E}[T] = 3(6) - 3(3) + 2 = 18 - 9 + 2 = 11.$$

Same answer by an entirely different decomposition. Bounds also pass: 11 sits between the 3-face collector 5.5 (too optimistic) and the full 6-face collector 14.7 (which demands more than we need).

Answer: 11 rolls. □

P1.15 *Order statistics of dice: make-a-market ladder*

(a) CDF first: $P(\max \le k) = (k/6)^2$, so $P(\max = k) = \frac{k^2-(k-1)^2}{36} = \frac{2k-1}{36}$ and

$$\mathbb{E}[\max] = \sum_{k=1}^{6} k\,\frac{2k-1}{36} = \frac{161}{36} \approx 4.47.$$

The trap to dodge out loud: quoting 3.5 "because dice average 3.5" ignores that an extreme is biased away from the mean — always ask whether the payoff is a mean, an extreme, or a median of the randomness.

(b) No new sum needed: $\min + \max = X + Y$ pointwise, so by linearity

$$\mathbb{E}[\min] = 7 - \frac{161}{36} = \frac{91}{36} \approx 2.53.$$

(c) Also no new sum: $|X - Y| = \max - \min$ pointwise, so

$$\mathbb{E}\,|X - Y| = \frac{161}{36} - \frac{91}{36} = \frac{70}{36} = \frac{35}{18} \approx 1.94.$$

Quoting: centre a two-sided market on the fair value with a spread you can defend — e.g. 4.3 bid at 4.6 offer on the max — wide enough to cover the payoff's dispersion and the risk that the counterparty (the interviewer) knows something; tighten only under pressure. The structural lesson is that (b) and (c) cost nothing once (a) is done: pointwise identities plus linearity replace two fresh distributions.

Self-check. (c) by brute count: over the 36 ordered pairs, $|i-j| = 0, 1, 2, 3, 4, 5$ occurs $6, 10, 8, 6, 4, 2$ times, so $\sum |i - j| = 0 + 10 + 16 + 18 + 16 + 10 = 70$ and $\mathbb{E} = 70/36 = 35/18$ — matches. Consistency: $\mathbb{E}[\max] + \mathbb{E}[\min] = 252/36 = 7 = \mathbb{E}[X + Y]$, as the identity demands.

Answer: (a) $161/36 \approx 4.47$; (b) $91/36 \approx 2.53$; (c) $35/18 \approx 1.94$.

P1.16 *Option value of a re-roll*

(a) A fresh roll is worth 3.5, so keep any face beating that: keep $\{4, 5, 6\}$ (conditional mean 5), re-roll $\{1, 2, 3\}$:

$$V_1 = \tfrac{1}{2} \cdot 5 + \tfrac{1}{2} \cdot 3.5 = 4.25.$$

(b) Now the continuation after discarding the first roll is the one-re-roll game, worth 4.25 — so at the first stage keep only $\{5, 6\}$ (mean 5.5):

$$V_2 = \tfrac{1}{3} \cdot 5.5 + \tfrac{2}{3} \cdot 4.25 = \tfrac{11}{6} + \tfrac{17}{6} = \tfrac{28}{6} = \tfrac{14}{3} \approx 4.67.$$

The keep-threshold rises with remaining optionality: more chances make you pickier.

(c) The option value is $V_1 - 3.5 = 0.75$. With a \$0.50 usage fee, re-rolling nets $3.5 - 0.5 = 3.0$, so keep any face ≥ 3: keep $\{3, 4, 5, 6\}$ (mean 4.5) with probability $\frac{2}{3}$, re-roll $\{1, 2\}$:

$$V_1^{\text{fee}} = \tfrac{2}{3} \cdot 4.5 + \tfrac{1}{3} \cdot 3.0 = 3 + 1 = 4.00.$$

This is *not* $4.25 - 0.50 = 3.75$: the fee is contingent — you only pay it when you exercise, and the fee itself changes the exercise policy (you now keep a 3), so the option still adds $4.00 - 3.5 = 0.50$ of value, not 0.25. Never net a contingent cost against an unconditional value. Market framing: the re-roll is a call-like claim on a volatile underlying; its value grows with dispersion and with the number of exercise dates, and shrinks — but less than one-for-one — with exercise costs.

Self-check. Ladder monotonicity and diminishing returns: $3.5 < 4.25 < 4.67 < 6$ (must be bounded by the max face) with increments 0.75, then $\frac{5}{12} \approx 0.42$ — optionality has diminishing marginal

value, as it should. Fee case tie-check: at face 3 you are indifferent (keep 3.0 vs re-roll net 3.0); computing the value with the other tie-break, re-rolling on 3 as well, gives $\frac{1}{2} \cdot 5 + \frac{1}{2} \cdot 3.0 = 4.00$ — identical, confirming the indifference and the value 4.00.

Answer: (a) keep $\{4, 5, 6\}$: value 4.25; (b) keep $\{5, 6\}$ first stage: value $14/3 \approx 4.67$; (c) option worth 0.75; with the \$0.50 usage fee the game is worth 4.00 (keep $\{3, 4, 5, 6\}$) — the option still adds 0.50, not 0.25. □

P1.17 *First 6 given all rolls even*

Work from the joint law. The sequence has $N = n$ with all rolls even iff the first $n - 1$ rolls land in $\{2, 4\}$ and roll n is a 6:

$$P(N = n,\ \text{all even}) = \left(\tfrac{2}{6}\right)^{n-1} \tfrac{1}{6} = \left(\tfrac{1}{3}\right)^{n-1} \tfrac{1}{6}, \qquad P(\text{all even}) = \frac{1/6}{1 - 1/3} = \frac{1}{4}.$$

Hence the conditional distribution is

$$P(N = n \mid \text{all even}) = 4 \cdot \left(\tfrac{1}{3}\right)^{n-1} \tfrac{1}{6} = \tfrac{2}{3}\left(\tfrac{1}{3}\right)^{n-1},$$

which is geometric with success probability $\frac{2}{3}$, so

$$\mathbb{E}[N \mid \text{all even}] = \frac{1}{2/3} = \frac{3}{2}.$$

Why 3 is wrong. "Roll a 3-face die $\{2, 4, 6\}$ until a 6" is a *different experiment*: there, each roll is guaranteed even by construction. Here, the conditioning is a statement about the *whole realised sequence*: a length-n sequence must dodge the odd faces n times, so long sequences are exponentially down-weighted — $P(\text{all even} \mid N = n) = (2/5)^{n-1}$ decays in n (the $n - 1$ non-6 rolls are uniform on five faces). Conditioning reweights sequences, not the die's faces, and it shifts mass towards short sequences: the conditional mean drops below even the unconditional geometric structure the naive answer assumes. The general trigger: whenever you condition on an event whose probability depends on the length (or path) of the realisation, you may not just "restrict the alphabet".

Self-check. The conditional distribution sums to 1: $\sum_{n \geq 1} \frac{2}{3}(1/3)^{n-1} = \frac{2/3}{2/3} = 1$. And $P(N = 1 \mid \text{all even}) = \frac{2}{3}$, versus $\frac{1}{3}$ under the naive 3-face model — the conditioning visibly favours stopping immediately, consistent with a mean of $1.5 < 3$. As a bound, the answer must be ≥ 1 and ≤ 3 (the naive model maximally favours long survival); 3/2 sits inside.

Answer: 3/2.

P1.18 *Hat-check fixed points*

(a) Let I_i indicate that trader i gets their own ticket; $P(I_i = 1) = \frac{1}{n}$ (their ticket is a uniform draw). By linearity,

$$\mathbb{E}[X] = n \cdot \tfrac{1}{n} = 1 \quad \text{for every } n,$$

independence nowhere required.

(b) Second moment via pairs: $X^2 = \sum_i I_i + \sum_{i \neq j} I_i I_j$, and for $i \neq j$,

$$P(I_i = I_j = 1) = \frac{1}{n} \cdot \frac{1}{n-1} = \frac{(n-2)!}{n!},$$

(the permutation must fix both positions). So

$$\mathbb{E}[X^2] = 1 + n(n-1) \cdot \frac{1}{n(n-1)} = 2 \qquad \Longrightarrow \qquad \mathrm{Var}(X) = 2 - 1^2 = 1 \text{ exactly, for all } n \geq 2.$$

The positive correlation between fixed points exactly offsets the small-p Bernoulli variance shortfall — mean 1, variance 1, for every $n \geq 2$.

(c) Mean 1 and variance 1, with dependence vanishing as n grows: X converges to Poisson(1), so

$$P(X = 0) \to e^{-1} \approx 0.368, \qquad P(X \geq 1) \to 1 - e^{-1} \approx 0.632.$$

Take the side "at least one match": at even money your edge is $0.632 - 0.368 \approx 26$ cents per dollar staked. (The general lesson for making markets on "no coincidence" events: rare-per-trial matches with many trials are Poisson, and $P(\text{none}) = e^{-\lambda}$ with λ the expected count — here $\lambda = 1$ regardless of n.)

Self-check. Exact tiny case $n = 2$: the two permutations give $X = 2$ (identity) or $X = 0$ (swap), each with probability $\frac{1}{2}$: $\mathbb{E}[X] = 1$, $\mathbb{E}[X^2] = \frac{1}{2}(4 + 0) = 2$, Var $= 1$ — all three formulas verified exactly. And the derangement probability converges fast: at $n = 4$ it is $9/24 = 0.375$, already within 0.007 of e^{-1}, so the "large n" claim in (c) is safe for any realistic dinner.

Answer: (a) 1; (b) 1; (c) bet on at least one match: it wins with probability $\approx 1 - 1/e \approx 0.632$, an edge of about 26 cents per dollar at even money. □

P1.19 *Random chord longer than the radius* ♦

By rotational symmetry, fix the first point and let the second sit at central angle $\theta \sim \mathrm{U}[0, 2\pi)$. The chord subtending central angle θ has length

$$L = 2R\sin(\theta/2),$$

(the isoceles triangle with two radii and the chord, split down the middle). The condition $L > R$ reads

$$2R\sin(\theta/2) > R \iff \sin(\theta/2) > \tfrac{1}{2} \iff \theta/2 \in (\tfrac{\pi}{6}, \tfrac{5\pi}{6}) \iff \theta \in (\tfrac{\pi}{3}, \tfrac{5\pi}{3}),$$

an arc of measure $\frac{5\pi}{3} - \frac{\pi}{3} = \frac{4\pi}{3}$ out of 2π: probability $\frac{4\pi/3}{2\pi} = \frac{2}{3}$.

Sanity check: a chord equal to the radius subtends 60° (equilateral triangle with the two radii), so the chord exceeds the radius unless the second point falls within 60° on either side of the first — 120° of 360° fails, 240° succeeds: 2/3 again.

Answer: (B) 2/3.

P1.20 *Max of four dice, exactly* ♦

The max-of-dice pipeline: go through the CDF, then difference.

$$P(\max \leq k) = \left(\tfrac{k}{6}\right)^4.$$

$$P(\max = 3) = P(\max \leq 3) - P(\max \leq 2) = \left(\tfrac{3}{6}\right)^4 - \left(\tfrac{2}{6}\right)^4 = \frac{81 - 16}{1296} = \frac{65}{1296}.$$

The trap is option (D): $1/16 = 81/1296 = P(\max \leq 3)$ — the *at most* probability, which forgets to remove the outcomes where no die shows a 3 at all. The option set is built for exactly this slip: under a clock, anyone who computes $(1/2)^4$ and stops sees a clean 1/16 waiting for them.

Sanity check: $65/1296 \approx 5\%$, sensibly small — with four dice the max is usually 5 or 6 ($P(\max \geq 5) = 1 - (4/6)^4 \approx 80\%$), so "exactly 3" should be rare.

Answer: (C) 65/1296.

28.2 Solutions — §2 Bayesian Updating and Reading Flow

P2.1 *Bayes odds form, base rates* ♦

Odds form. Prior odds of infected against not: $0.001 : 0.999 = 1 : 999$. Likelihood ratio of a

positive test: $P(+ \mid \text{inf})/P(+ \mid \text{not}) = 1/0.1 = 10$. Posterior odds $= 10 : 999$, so

$$P(\text{infected} \mid +) \;=\; \frac{10}{10+999} \;=\; \frac{10}{1009} \;\approx\; 0.99\%.$$

The headline: a positive result from a test that *never misses an infection* still leaves you around 1%, because the prior was $999 : 1$ against and the false-positive channel (10% of everyone healthy) is a hundred times wider than the prevalence.

Self-check (counting version). Take 10,000 people: 10 are infected and all test positive; of the 9,990 uninfected, 10%, i.e. 999, also test positive. Among the 1,009 positives, 10 are infected: 10/1009. Sanity bound: when prevalence $\ll$ false-positive rate, $P \approx$ prevalence/FP rate $= 0.001/0.1 = 1\%$ — consistent.

Answer: $10/1009 \approx 0.99\%$.

P2.2 *Conditioning on the decisive event* ♦

Condition on the game that ends the match. A draw carries no information and simply restarts the process, so the match is decided by the first decisive game:

$$P(\text{I win}) \;=\; \frac{0.4}{0.4+0.2} \;=\; \frac{2}{3}.$$

In odds language: my win-to-loss odds are $0.4 : 0.2 = 2 : 1$ in every game, and replaying draws cannot change those odds — only renormalise them.

Self-check 1 (first-step analysis). Let W be my overall win probability. Conditioning on game one: $W = 0.4 + 0.4\,W$, so $0.6\,W = 0.4$ and $W = 2/3$.

Self-check 2 (series). $W = \sum_{n\geq 0}(0.4)^n \cdot 0.4 = 0.4/(1-0.4) = 2/3$; the opponent gets $0.2/0.6 = 1/3$, and $2/3 + 1/3 = 1$ with zero mass left on drawing forever ($0.4^n \to 0$).

Answer: 2/3.

P2.3 *Odds form, prior vs evidence* ♦

Odds form. Prior odds gag against fair: $1 : 99$. Likelihood ratio of five heads: $1/(1/2)^5 = 32$. Posterior odds $= 32 : 99$, so

$$P(\text{gag} \mid 5 \text{ heads}) \;=\; \frac{32}{32+99} \;=\; \frac{32}{131} \;\approx\; 24.4\%.$$

Five straight heads and the fair coins are *still* the favourite: a $99 : 1$ prior beats a $32 : 1$ likelihood ratio.

Self-check (direct Bayes).

$$\frac{\frac{1}{100}\cdot 1}{\frac{1}{100}\cdot 1 + \frac{99}{100}\cdot\frac{1}{32}} = \frac{1}{1+99/32} = \frac{32}{131}.$$

Sanity: a sixth head doubles the odds to $64 : 99 \approx 39\%$, and only at seven heads ($128 : 99$) does the gag coin finally become the favourite — the same shape as the notes' 1000-coin anchor, where even ten heads ($2^{10} = 1024$ vs 999) lands at barely over 50%.

Answer: $32/131 \approx 24.4\%$.

P2.4 *Sequential likelihood ratios*

The likelihood ratio of a single six is $\text{LR} = \frac{1/2}{1/6} = 3$, and rolls are conditionally independent given the die, so sequential LRs multiply.

(a) Prior odds $1 : 1$; posterior odds $3 : 1$, so $P = 3/4$.

(b) Odds $3^3 : 1 = 27 : 1$, so $P = 27/28 \approx 96.4\%$.

(c) We need posterior odds $\geq 99:1$, i.e. $3^n \geq 99$. $3^4 = 81$ gives $81/82 \approx 98.8\%$ — not enough; $3^5 = 243$ gives $243/244 \approx 99.6\%$. So $n = 5$.

(d) It drops. A non-six also updates: $P(\text{non-six} \mid \text{loaded}) = 1/2$ versus $5/6$ for the fair die, so its LR is $\frac{1/2}{5/6} = 3/5 < 1$. After three sixes and a two the odds are $27 \cdot \frac{3}{5} = 16.2$, i.e. $P = 16.2/17.2 \approx 94.2\%$. The absence of the signal is itself evidence — the flow-reading version: the aggressive trades that do *not* arrive also move your posterior.

Self-check. Recompute (b) directly: $\dfrac{(1/2)^3}{(1/2)^3 + (1/6)^3} = \dfrac{1/8}{1/8 + 1/216} = \dfrac{27}{28}$. General consistency check on the LRs: under the fair-die hypothesis the expected likelihood ratio of one roll must equal 1, and indeed $\frac{1}{6} \cdot 3 + \frac{5}{6} \cdot \frac{3}{5} = \frac{1}{2} + \frac{1}{2} = 1$.

Answer: (a) 3/4; (b) $27/28 \approx 96.4\%$; (c) 5; (d) no — odds $16.2:1$, $P \approx 94.2\%$. □

P2.5 *Correlated evidence*

Multiplying likelihood ratios is valid only when the pieces of evidence are *conditionally independent given the hypothesis*. Three child fills of one parent order are one trading decision executed in slices, not three independent draws of the account's behaviour: given that this account decided to lift, the second and third slices were near-certain *whether it is informed or noise*, so their conditional likelihood ratios are ≈ 1. The junior counted one bit of evidence three times.

A defensible update treats the burst as a single lift event: odds $= \frac{1}{9} \times 4 = \frac{4}{9}$, so $P = 4/13 \approx 31\%$. (If total order *size* itself discriminates informed from noise, that is a separate, measured likelihood term — it is not recovered by re-multiplying the same fill.)

The general failure mode is correlated evidence: three quotes copied from one pricing source, three headlines rewriting one press release, three fills from one parent order. Independence is an assumption about the *generating process*, not about the number of lines in the log.

Self-check. Bounds: the truth must lie between the no-update prior (10%) and the fully-independent computation ($64:9 \Rightarrow 64/73 \approx 87.7\%$); the one-event posterior $4/13 \approx 30.8\%$ does. Conversion check: odds $4:9 \Rightarrow 4/(4+9) = 4/13$. Note the junior's *arithmetic* was right ($64/73 \approx 88\%$) — the independence assumption was the error.

Answer: LRs multiply only for conditionally independent evidence; three slices of one iceberg are one event, so the posterior is $\approx 4/13 \approx 31\%$, not 88%.

P2.6 *Flow reading, sequential update to a threshold*

The likelihood ratio of one lift is $\text{LR} = 0.9/0.5 = 1.8$; prior odds are $5:95 = 1:19$.

(a) Posterior odds $= 1.8^2/19 = 3.24/19 \approx 0.1705$, so

$$P = \frac{0.1705}{1.1705} \approx 14.6\%.$$

Two straight lifts and it is still only $\sim$15%: with mostly-noise flow the base rate dominates early evidence — exactly the notes' base-rate trap.

(b) Need odds $> 9:1$, i.e. $1.8^n > 9 \times 19 = 171$. Powers of 1.8: $1.8^2 = 3.24$, $1.8^4 = 10.50$, $1.8^8 = 110.2$ (odds 5.80, $P \approx 85.3\%$ — not yet), $1.8^9 = 198.4$ (odds 10.44, $P \approx 91.3\%$). So $n = 9$.

(c) Raise both quotes (skew the whole market up), and widen if uncertainty about the size of the move is growing. Each fill at the stale offer is a sale to a counterparty increasingly likely to know the price is going up; the posterior is precisely the probability weight on the adverse scenario, so the expected adverse-selection cost per fill grows with it. "Keep selling at the same price" treats the 9th lift like the 1st — it ignores that the flow has already told you which side the informed money is on. Moving the market both cuts your expected loss and makes the informed trader pay through the move.

Self-check. (a) directly: $0.05 \times 0.9^2 = 0.0405$ and $0.95 \times 0.5^2 = 0.2375$; $0.0405/(0.0405 + 0.2375) =$

$0.0405/0.2780 \approx 14.6\%$. (b) by logs: $n > \ln 171/\ln 1.8 = 5.142/0.588 \approx 8.75 \Rightarrow n = 9$, agreeing with the power table.

Answer: (a) $\approx 14.6\%$; (b) 9 consecutive lifts; (c) skew both quotes up (and widen) — filling at a stale offer sells cheap insurance to informed flow.

28.3 Solutions — §3 Betting, Bankroll, Ruin and Markov Chains

P3.1 *Kelly criterion*

(a) Betting fraction f: wealth multiplies by $1+bf$ (probability p) or $1-f$ (probability $q=1-p$). Maximise the expected log growth

$$g(f) = p\ln(1+bf) + q\ln(1-f), \qquad g'(f) = \frac{pb}{1+bf} - \frac{q}{1-f} = 0 \;\Rightarrow\; f^* = \frac{bp-q}{b}.$$

Second derivative $g''(f) = -\dfrac{pb^2}{(1+bf)^2} - \dfrac{q}{(1-f)^2} < 0$ everywhere, so g is strictly concave and f^* is the unique maximum.

(b) Even money ($b=1$): $f^* = p-q = 0.10$. For $b=2$, $p=0.4$: $f^* = (0.8-0.6)/2 = 0.10$. Both bets: 10% of bankroll.

(c) $g(0.10) = 0.55\ln 1.1 + 0.45\ln 0.9 \approx 0.0524 - 0.0474 = 0.0050$, about 0.50% compounded per bet. At $2f^* = 0.20$: $g(0.20) = 0.55\ln 1.2 + 0.45\ln 0.8 \approx 0.10028 - 0.10041 = -0.00014 < 0$. Doubling Kelly annihilates the growth. Small-edge picture: $g(f) \approx (p-q)f - f^2/2$, a parabola with roots at 0 and $2(p-q) = 2f^*$ — the growth curve crosses zero at exactly twice Kelly. Moral: the arithmetic EV per bet, $f(p-q)$, is positive at *any* f, yet compounding is negative beyond $2f^*$: positive edge per trade and a shrinking bankroll are perfectly compatible.

Self-check. Differentiate back: $g'(0.10) = \frac{0.55}{1.1} - \frac{0.45}{0.9} = 0.5 - 0.5 = 0$. The parabola approximation at f^* predicts $g \approx (0.1)(0.1) - (0.1)^2/2 = 0.005$, matching the exact 0.0050.

Answer: (a) $f^* = (bp-q)/b$; (b) 10% in both cases; (c) $g(f^*) \approx +0.50\%$ per bet, $g(2f^*) \approx -0.014\%$ — past twice Kelly you compound downward despite positive EV per bet. □

P3.2 *Kelly under estimation error*

(a) Kelly requires the *true* p; $\hat{p}$ from 25 trades is extremely noisy, and the Kelly penalty is asymmetric: under-betting gives up a little growth, while betting beyond twice the true Kelly produces *negative* growth. Full Kelly on an estimated edge is structural over-betting.

(b) True $p = 0.55$ makes the true Kelly $f^* = 0.10$ with $g(0.10) = 0.55\ln 1.1 + 0.45\ln 0.9 \approx +0.0050$ per trade. His $f = 0.20$ is exactly twice the true Kelly:

$$g(0.20) = 0.55\ln 1.2 + 0.45\ln 0.8 \approx -0.00014 < 0,$$

so he slowly grinds the book to zero while holding a genuine 5-point edge. Half of his estimated Kelly is 0.10 — exactly the true optimum here. That is the fractional-Kelly defence in one example: for small edges $g(cf^*) \approx c(2-c)\,g(f^*)$, so half-Kelly keeps $\approx 75\%$ of the growth at half the volatility (a quarter of the variance), and it is robust to precisely this estimation error.

(c) $\mathrm{SE}(\hat{p}) = \sqrt{\hat{p}\hat{q}/n} = \sqrt{0.24/25} \approx 0.098$ — ten percentage points. A 95% interval is roughly 0.60 ± 0.19, i.e. $(0.41,\ 0.79)$: it contains 0.5, so the data cannot even rule out *no edge*. Discipline: half-Kelly or less, hard caps per bet, and scale up only as the sample grows.

Self-check. The rule $g(cf^*) \approx c(2-c)g(f^*)$ gives 0 at $c=2$, matching the exact $g(0.20) \approx 0$; and gives 1 at $c=1$. Binomial check of (c): the win count has SD $\sqrt{25\times 0.24} \approx 2.45$ wins, so 15 ± 4.8 wins spans $p \in (0.41, 0.79)$ — same interval.

Answer: (a) full Kelly on a 25-trade estimate; (b) $g(0.20 \mid p{=}0.55) \approx -0.014\%$ per trade vs $g(0.10) \approx +0.50\%$; half of estimated Kelly $= 0.10 =$ the true optimum; (c) SE ≈ 0.10, CI includes 0.5 — bet fractional Kelly with caps.

P3.3 *Gambler's ruin, fair case*

(a) Fair gambler's ruin from i with absorbing barriers 0 and N: the harmonic relation $p_i = \frac{1}{2}p_{i-1} + \frac{1}{2}p_{i+1}$ forces p_i linear in i, so $P = i/N = 30/100 = 0.30$.

(b) Expected duration $= i(N-i) = 30 \times 70 = 2100$ flips.

(c) Work in units of \$5: $i = 6$, $N = 20$. Win probability $6/20 = 0.30$ — unchanged: in a fair game it depends only on your *fraction* of the total capital. Duration $= 6 \times 14 = 84$ flips: 25 times fewer. Duration scales as $1/(\text{stake})^2$.

Self-check (martingale/EV argument, independent of the recursions). The game is fair, so your expected final wealth equals your initial \$30; the final wealth is \$100 with probability P and \$0 otherwise, hence $100P = 30$, $P = 0.30$ — for *any* stake size. Duration scaling: each \$5 flip moves wealth with 25 times the per-step variance of a \$1 flip while the distances to the barriers are fixed, so the expected time shrinks by 25: $2100/25 = 84$.

Answer: (a) 0.30; (b) 2100 flips; (c) win probability still 0.30; expected duration drops to 84 flips.

P3.4 *Gambler's ruin, biased*

(a) With $r = q/p = 51/49$, $P(\text{reach } N) = \dfrac{1-r^i}{1-r^N}$, here $i = 50$, $N = 100$. Since $N = 2i$,

$$\frac{1-r^i}{1-r^{2i}} = \frac{1}{1+r^i}.$$

Mental math: $\ln \frac{51}{49} = \ln \frac{1+0.02}{1-0.02} \approx 2(0.02) = 0.04$, so $r^{50} \approx e^2 \approx 7.39$ and $P \approx 1/8.39 \approx 12\%$. A 1% per-round edge against you turns a coin-flip doubling shot into roughly 1-in-8.

(b) Now the *up* probability is 0.51, so $r = 49/51 < 1$. Let $N \to \infty$ in $\frac{1-r^i}{1-r^N}$: the survival probability is $1 - r^i$, hence

$$P(\text{ruin}) = r^i = \left(\tfrac{49}{51}\right)^{20} \approx e^{-20\times 0.04} = e^{-0.8} \approx 0.45.$$

Despite a genuine positive edge, the under-capitalised desk dies before compounding about 45% of the time. For ruin $\le 1\%$: $i \times 0.04 \ge \ln 100 \approx 4.61$, so $i \ge 115.2$ — a buffer of about 116 ticks, nearly six times larger.

Self-check. Exact $\ln(51/49) = 3.9318 - 3.8918 = 0.0400$, so the $2x$ approximation is essentially exact. Recompute (a) by the full formula: $(1 - 7.39)/(1 - 54.6) = 6.39/53.6 = 0.119 = 1/8.39$. Check the 1% buffer: $e^{-116\times 0.04} = e^{-4.64} \approx 0.0097 < 1\%$. Symmetry sanity: with the edge *for* you in setup (a), success would be $1/(1 + e^{-2}) \approx 88\%$ — the mirror image of 12%, as it must be.

Answer: (a) $1/(1 + (51/49)^{50}) \approx 12\%$; (b) ruin $= (49/51)^{20} \approx e^{-0.8} \approx 45\%$; need ≈ 116 ticks for ruin below 1%.

P3.5 *First-step analysis / ruin reduction*

Derivation 1 (blocks of two points). Group points in pairs. A pair goes win-win with probability p^2 (you win the match from level), lose-lose with q^2 (you lose), and split with $2pq$ (back to level, no memory). Condition on the first decisive pair:

$$P = \frac{p^2}{p^2+q^2} = \frac{0.36}{0.36+0.16} = \frac{9}{13} \approx 69.2\%.$$

Derivation 2 (gambler's-ruin reduction). Your lead performs a random walk from 0 with absorption at ± 2 — i.e. ruin with $i = 2$, $N = 4$, $r = q/p = 2/3$:

$$P = \frac{1-r^2}{1-r^4} = \frac{1-4/9}{1-16/81} = \frac{5/9}{65/81} = \frac{9}{13}.$$

Self-check (first-step analysis, a third route). States = lead $\in \{-1, 0, +1\}$: $W_{+1} = p + qW_0$, $W_{-1} = pW_0$, $W_0 = pW_{+1} + qW_{-1}$. Substituting: $W_0 = p^2 + 2pq\,W_0$, so $W_0 = \dfrac{p^2}{1-2pq} = \dfrac{p^2}{p^2+q^2}$ (since $(p+q)^2 = 1$). Limits: $p = \frac{1}{2} \Rightarrow \frac{1}{2}$; $p = 1 \Rightarrow 1$.

Why more than 0.6: requiring two clear points forces a longer contest, and longer formats favour the better player — each extra independent point is another draw from a distribution tilted your way. Trading version: a desk with a small per-trade edge wants *many independent* expressions of it; the format (or the trade count) amplifies edge exactly as deuce amplifies 0.6 into 0.69.

Answer: $9/13 \approx 69.2\%$. □

P3.6 *First-step analysis, pattern race*

Structural argument. HH appears first if and only if the first two flips are both heads. Indeed, suppose a tail appears at any point before HH has been completed. Every later occurrence of HH begins with a head, and the *first* head to arrive after that tail completes TH immediately — so TH beats any subsequent HH. Hence HH can only win by arriving before any tail: flips 1 and 2 are HH, probability 1/4.

First-step verification. States: $\varnothing$ (start), H (last flip heads, no tail seen yet), T (a tail is pending). From T, the next head finishes TH and more tails stay in T: $P_T = 0$. From H: heads finishes HH, tails moves to T: $P_H = \frac{1}{2}(1) + \frac{1}{2}(0) = \frac{1}{2}$. From $\varnothing$: $P_\varnothing = \frac{1}{2}P_H + \frac{1}{2}P_T = \frac{1}{4}$.

Why waiting times mislead. E[wait for HH] = 6 and E[wait for TH] = 4, but the race odds are 3 : 1, not 6 : 4 — and the comparison is not even directionally reliable: HH vs HT also have waits 6 vs 4, yet that race is exactly 50:50 (once the first head arrives, the very next flip decides it). Races are governed by the overlap structure of the patterns (what happens to your progress when you "fail"), not by marginal waiting times. This is the engine of Penney's game: the second player can always pick a pattern that beats the first player's.

Self-check (enumeration of openings). The four equally likely opening pairs: HH ⇒ HH wins; HT ⇒ tail pending ⇒ TH wins eventually; TH ⇒ TH has already won; TT ⇒ tail pending ⇒ TH wins. HH wins in exactly one case out of four.

Answer: 1/4.

P3.7 *Stationary distribution, two-state chain*

(a) Two-state chain with exit probabilities $a = 0.1$ (leave trending) and $b = 0.3$ (leave choppy): $\pi = (\frac{b}{a+b}, \frac{a}{a+b}) = (\frac{0.3}{0.4}, \frac{0.1}{0.4}) = (\frac{3}{4}, \frac{1}{4})$. Time is shared inversely to how easily each state is left: choppy is exited three times more readily than trending, so it carries a third as much weight — a 1:3 split, i.e. 1/4 choppy versus 3/4 trending.

(b) Expected return time to a state is $1/\pi_{\text{state}}$ (Kac's formula — in the long run a fraction π of days are trending, so trending days are on average $1/\pi$ apart): $1/(3/4) = 4/3$ days.

(c) Next-day trending probability under π: $0.75 \times 0.9 + 0.25 \times 0.3 = 0.675 + 0.075 = 0.75$. Stationary.

Self-check (independent derivation of (b) by first-step). From a trending day: with probability 0.9 tomorrow is trending — return time 1; with probability 0.1 we fall into choppy, from which the wait to switch back is geometric with success 0.3, mean 10/3. So $E = 0.9(1) + 0.1(1 + \frac{10}{3}) = 0.9 + 0.1 \times \frac{13}{3} = \frac{4}{3}$ — Kac's formula confirmed by hand.

Answer: (a) 3/4; (b) 4/3 days; (c) $0.75(0.9) + 0.25(0.3) = 0.75$.

P3.8 *Stationary distribution on a graph, return time*

Step 1: stationary distribution of a random walk on a graph. For uniform random moves on an undirected graph, $\pi_v = \dfrac{d_v}{2E}$ (d_v = degree, E = number of edges) is stationary: each neighbour u of v contributes $\pi_u \cdot \frac{1}{d_u} = \frac{1}{2E}$ of inbound probability, and v has d_v neighbours, so $(\pi P)_v = \frac{d_v}{2E} = \pi_v$.

Step 2: Kac's return-time formula. In the long run the walk spends a fraction π_v of its steps at v,

so visits to v are on average $1/\pi_v$ steps apart:

$$E[\text{return time to } v] = \frac{1}{\pi_v} = \frac{2E}{d_v}.$$

Step 3: count. Knight moves come in 8 vectors $(\pm1, \pm2), (\pm2, \pm1)$; a vector (a, b) fits on the board in $(8 - |a|)(8 - |b|) = 7 \times 6 = 42$ positions, identically for all 8 vectors. Ordered moves $= 8 \times 42 = 336$, so $E = 168$ edges and $2E = 336$. A corner square has $d = 2$ (from a1: b3 and c2). Hence

$$E[\text{return}] = \frac{336}{2} = 168 \text{ moves}.$$

Self-check (independent degree census). The 8x8 knight-degree histogram: 4 squares of degree 2, 8 of degree 3, 20 of degree 4, 16 of degree 6, 16 of degree 8. Squares: $4+8+20+16+16 = 64$; total degree $8 + 24 + 80 + 96 + 128 = 336 = 2E$ — agreeing with the vector count. Sanity: a central square (degree 8) has return time $336/8 = 42$, much shorter than the badly-connected corner's 168, as it should be. The reversibility structure did all the work — no linear system in 64 unknowns.

Answer: 168 moves.

P3.9 *St. Petersburg, capped EV*

(a) $\mathbb{E} = \sum_{n\geq1} 2^{-n}\, 2^n = \sum_{n\geq1} 1 = \infty$: every level contributes one dollar and there are infinitely many levels.

(b) With the cap, levels $n = 1, \ldots, 20$ still contribute \$1 each ($2^{-n} \times 2^n$), totalling \$20; the event "no head in the first 20 flips" has probability 2^{-20} and pays the capped 2^{20}, contributing \$1 more:

$$\mathbb{E} = 20 + 2^{-20} \cdot 2^{20} = 21.$$

General pattern: cap at $2^K \Rightarrow$ fair price $K + 1$.

(c) Four defensible reasons to bid well under \$21 for a single play:

- **Concave utility:** $\mathbb{E}[\ln(\text{payoff})] = \sum_n 2^{-n} n \ln 2 = 2 \ln 2$, so a log agent's certainty equivalent for the raw payoff is $e^{2\ln 2} = \$4$ — Bernoulli's original resolution.
- **One-shot heavy tails:** the median payoff is \$2, and about 97% of plays pay $\leq$ \$32; the EV is carried by vanishing tail events, and with tails this heavy the sample mean converges uselessly slowly — "fair price" by EV presumes a repetition you are not getting.
- **Counterparty credit:** any valuation above the cap's contribution assumes the casino can and will pay; the cap *is* the credit limit made explicit.
- **The meta-lesson:** price $\neq$ expected value when distributions are wild — variance, bankroll and utility must enter. This is the whole point of risk management.

Self-check. (b) another way: $\mathbb{E} = \sum_{n=1}^{20} 2^{-n} 2^n + P(\text{first 20 all tails}) \cdot 2^{20} = 20 + 1 = 21$, and it matches the notes' anchor "cap $2^{30} \Rightarrow$ \$31" via $K + 1$ with $K = 30$. The log moment uses the standard $\sum_{n\geq1} n\, 2^{-n} = 2$; check small terms: $\frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} = 1.5 + 0.4375 \approx 1.94 \to 2$.

Answer: (a) infinite; (b) \$21 (cap at 2^K gives $K + 1$); (c) concave utility (log CE = \$4), one-shot heavy tails, counterparty credit — price is not EV for wild distributions. □

P3.10 *Utility shape, risk aversion* ♦

Increasing: $U'(W) > 0$ — more wealth is preferred to less (non-satiation). Without it the agent is not even trying to make money.

Concave: $U''(W) < 0$ — risk aversion. By Jensen's inequality, for any non-degenerate zero-mean gamble X,

$$\mathbb{E}[U(W + X)] < U(W + \mathbb{E}[X]) = U(W),$$

so the investor declines every fair gamble and demands a positive risk premium to carry risk. The fair \$100 flip: $\frac{1}{2}U(W+100)+\frac{1}{2}U(W-100) < U(W)$ — the chord lies below a concave curve. Marginal-utility phrasing: because U' is decreasing, the hundred dollars lost hurts more than the hundred gained helps.

Standard examples: $\ln W$, $\sqrt{W}$, $-e^{-aW}$. Connections: maximising expected log growth (Kelly) is exactly behaving as a log-utility investor — concavity is what makes over-betting penalised; and log utility gives the St. Petersburg game a finite certainty equivalent despite infinite EV (Bernoulli's resolution), since $\mathbb{E}[\ln(\text{payoff})] = 2\ln 2 < \infty$.

Self-check (numerical Jensen). $U=\sqrt{W}$, $W=10{,}000$, fair flip of $\pm\$3{,}600$: $\frac{1}{2}\sqrt{13{,}600}+\frac{1}{2}\sqrt{6{,}400} \approx \frac{1}{2}(116.6)+\frac{1}{2}(80) = 98.3 < 100 = \sqrt{10{,}000}$ — declined. Direction check: a convex (risk-loving) U flips the Jensen inequality and *accepts* fair gambles — the sign of U'' is exactly what controls the decision, confirming concavity as the defining property.

Answer: Increasing and concave: $U' > 0$ (more wealth preferred), $U'' < 0$ (risk aversion).

28.4 Solutions — §4 Statistics I: Estimators and Inference

P4.1 *Second moment of the standard normal* ♦

The fastest route is the decomposition of a second moment into variance plus squared mean:

$$E[X^2] = \mathrm{Var}(X) + (E[X])^2 = 1 + 0^2 = 1.$$

This identity is worth knowing in its general form $E[(X-c)^2] = \mathrm{Var}(X) + (E[X]-c)^2$: it is exactly the bias–variance decomposition of mean squared error, with X playing the estimator and c the target.

Self-check (compute another way). With $\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ we have $\varphi'(x) = -x\,\varphi(x)$, so integrating by parts with $u = x$ and $dv = x\varphi(x)\,dx = -d\varphi$,

$$\int_{-\infty}^{\infty} x^2\varphi(x)\,dx \;=\; \big[-x\varphi(x)\big]_{-\infty}^{\infty} + \int_{-\infty}^{\infty}\varphi(x)\,dx \;=\; 0+1 \;=\; 1,$$

confirming the identity route. A third check: $X^2 \sim \chi^2_1$, whose mean equals its degrees of freedom, 1.

Answer: $E[X^2] = 1$.

P4.2 *Bias-variance-MSE trade-off in variance estimation*

Under normality, $S/\sigma^2 \sim \chi^2_{n-1}$, so with $Q \sim \chi^2_{n-1}$ (mean $n-1$, variance $2(n-1)$):

$$E[S] = (n-1)\sigma^2, \qquad E[S^2] = \sigma^4\big(\mathrm{Var}(Q)+(EQ)^2\big) = \sigma^4\big(2(n-1)+(n-1)^2\big) = (n-1)(n+1)\,\sigma^4.$$

Expand the MSE as a function of the divisor c:

$$\mathrm{MSE}(c) = E\Big[\Big(\frac{S}{c}-\sigma^2\Big)^2\Big] = \frac{E[S^2]}{c^2} - \frac{2\sigma^2 E[S]}{c} + \sigma^4 = \sigma^4\left[\frac{(n-1)(n+1)}{c^2} - \frac{2(n-1)}{c} + 1\right].$$

Setting the derivative in c to zero: $-2(n-1)(n+1)/c^3 + 2(n-1)/c^2 = 0$, i.e. $c^* = n+1$. Plugging back in:

$$\mathrm{MSE}(n+1) = \frac{2\sigma^4}{n+1} \;<\; \mathrm{MSE}(n) = \frac{(2n-1)\sigma^4}{n^2} \;<\; \mathrm{MSE}(n-1) = \frac{2\sigma^4}{n-1}.$$

So the MSE ranking is $n+1$ (best), then n (MLE), then $n-1$ (unbiased, worst of the three). Dividing by $n+1$ shrinks the estimate toward zero; the small bias it introduces is more than paid for by the variance reduction — the same logic that makes ridge and shrinkage estimators work.

Self-check (small case $n = 2$). Then $S = (X_1 - X_2)^2/2$ and $S/\sigma^2 \sim \chi^2_1$, so $E[S^2] = 3\sigma^4 = (n-1)(n+1)\sigma^4$ with $n = 2$. The formula gives $c^* = 3$ and $\mathrm{MSE}(3) = \sigma^4(3/9 - 2/3 + 1) = \frac{2}{3}\sigma^4 = 2\sigma^4/(n+1)$. Ordering check at $n = 10$: $2/11 \approx 0.182 < 19/100 = 0.19 < 2/9 \approx 0.222$.

Answer: $c = n + 1$ minimises MSE; ranking $n + 1 \prec n \prec n - 1$. Unbiasedness is a constraint, not a virtue — a slightly biased (shrunk) estimator strictly dominates the unbiased one on MSE. □

P4.3 *Bessel's correction and the bias of s*

(a) Decompose deviations from μ around the sample mean:

$$\sum_i (X_i - \mu)^2 = \sum_i (X_i - \bar{X})^2 + n(\bar{X} - \mu)^2$$

(the cross term $2(\bar{X} - \mu)\sum_i (X_i - \bar{X})$ vanishes identically). Taking expectations, $E[\sum_i (X_i - \mu)^2] = n\sigma^2$ and $E[n(\bar{X} - \mu)^2] = n\,\mathrm{Var}(\bar{X}) = \sigma^2$, hence

$$E\Big[\sum_i (X_i - \bar{X})^2\Big] = n\sigma^2 - \sigma^2 = (n-1)\sigma^2.$$

Intuition: the sample mean chases the data, so deviations measured from $\bar{X}$ are systematically smaller than deviations from μ — by exactly one degree of freedom.

(b) With μ known, $E[\sum_i (X_i - \mu)^2] = n\sigma^2$ directly, so divide by n: no degree of freedom is spent estimating the centre. (Regression generalises this: $\hat{\sigma}^2 = \mathrm{RSS}/(n-p)$ after fitting p parameters.)

(c) No. By Jensen's inequality applied to the strictly concave square root (equivalently: $\mathrm{Var}(s) > 0$ unless s is degenerate),

$$(E[s])^2 \;<\; E[s^2] \;=\; \sigma^2 \quad\Longrightarrow\quad E[s] < \sigma.$$

So even after Bessel's correction, the sample volatility is biased *low*. Unbiasedness is not preserved under nonlinear transformations.

Self-check. For (a), take $n = 2$: $\sum_i (X_i - \bar{X})^2 = (X_1 - X_2)^2/2$, and $E[(X_1 - X_2)^2] = 2\sigma^2$, giving expectation $\sigma^2 = (2-1)\sigma^2$. For (c), compute $E[s]$ exactly at $n = 2$ for normal data: $s = |X_1 - X_2|/\sqrt{2}$ with $X_1 - X_2 \sim \mathcal{N}(0, 2\sigma^2)$, and $E|\mathcal{N}(0, \tau^2)| = \tau\sqrt{2/\pi}$, so $E[s] = \sigma\sqrt{2/\pi} \approx 0.80\,\sigma < \sigma$.

Answer: (a) $E[\sum_i (X_i - \bar{X})^2] = (n-1)\sigma^2$, so divide by $n - 1$; (b) divide by n when μ is known; (c) no — $E[s] < \sigma$ by Jensen, e.g. $E[s] \approx 0.80\,\sigma$ at $n = 2$. □

P4.4 *Sum of i.i.d. normals* ♦

Sums of independent normals are normal, and for independent variables both means and variances add:

$$X_1 + \cdots + X_k \;\sim\; \mathcal{N}(km,\; ks^2).$$

Contrast this with the classic confusion: *scaling* one variable gives $kX_1 \sim \mathcal{N}(km, k^2 s^2)$. Summing k independent copies multiplies the standard deviation by $\sqrt{k}$ only — this is exactly why risk grows like $\sqrt{T}$ while drift grows like T under i.i.d. returns.

Self-check (MGF). Each X_i has moment generating function $M(t) = \exp(mt + s^2t^2/2)$; independence makes the MGF of the sum the product,

$$M(t)^k = \exp\left(kmt + ks^2t^2/2\right),$$

which is precisely the MGF of $\mathcal{N}(km, ks^2)$. Dimension check: variance is in squared units, so it must scale linearly in the number of terms while the sd scales as $\sqrt{k}$.

Answer: $\mathcal{N}(km,\; ks^2)$ — mean km, variance ks^2 (sd $s\sqrt{k}$).

P4.5 *Sum of normals without independence* ♦

Trap: independence (or joint normality) was never stated, and without it the question cannot be answered. Two separate failures:

- *Unknown variance even if the sum were normal.* $\mathrm{Var}(X+Y) = 4+9+2\,\mathrm{Cov}(X,Y)$, and Cauchy–Schwarz only bounds $|\,\mathrm{Cov}\,| \le \sigma_X\sigma_Y = 6$, so the variance can be anywhere in $[1, 25]$.
- *The sum need not be normal at all.* Marginal normality does not imply joint normality. Standard counterexample: let $X \sim \mathcal{N}(0,1)$ and $Y = X$ on $\{|X| \le c\}$, $Y = -X$ on $\{|X| > c\}$. By the symmetry of the normal density, $Y \sim \mathcal{N}(0,1)$ marginally, yet $X+Y$ equals $2X$ when $|X| \le c$ and 0 otherwise — a distribution with a point mass at 0, certainly not normal. Shifting and scaling transplants the same construction to the given means and variances.

If the two variables *are* independent (the intended textbook reading), means and variances add:

$$X + Y \sim \mathcal{N}(5+3,\ 4+9) = \mathcal{N}(8, 13).$$

Self-check. Under independence, the MGF product gives $\exp(8t + 13t^2/2)$, the $\mathcal{N}(8,13)$ MGF. Boundary sanity: perfect positive correlation gives sd $= 2+3 = 5$, variance 25, matching the Cauchy–Schwarz upper bound; perfect negative correlation gives sd $= 1$, variance 1, the lower bound.

Answer: Not determined — it depends on the joint dependence; the variance alone ranges over $[1, 25]$ and the sum need not even be normal. If independent: $\mathcal{N}(8, 13)$.

P4.6 *Distribution recognition (binomial)* ♦

Recognition checklist: a *fixed* number of trials ($n = 50$), each a Bernoulli success/failure, *independent*, with *constant* success probability ($p = 0.7$). That is the definition of a binomial:

$$\#\text{fills} \ \sim\ \text{Binomial}(n = 50,\ p = 0.7), \qquad E = np = 35, \quad \mathrm{Var} = np(1-p) = 10.5.$$

Why the distractors fail: geometric / negative binomial count *trials until* a success quota (random n, fixed successes) — here n is fixed and successes are random; Poisson would need p small with np moderate. Worth saying out loud in an interview: in reality fill probability co-moves with queue depth, volatility and toxicity, so independence and constant p are modelling assumptions — exactly the kind you state and then question.

Self-check. Mean 35 is 70% of 50. Variance sanity: $np(1-p)$ is maximised at $p = 0.5$ (would be 12.5); at $p = 0.7$ it must be smaller, and $10.5 < 12.5$. Complement symmetry: unfilled orders $\sim \text{Binomial}(50, 0.3)$, and the two means $35 + 15 = 50$.

Answer: Binomial($n = 50, p = 0.7$); mean 35, variance 10.5.

P4.7 *Normal approximation to the binomial* ♦

Match the first two moments (de Moivre–Laplace, the CLT for Bernoulli sums):

$$\text{mean} = np = 20, \qquad \text{variance} = np(1-p) = 100 \times 0.2 \times 0.8 = 16,$$

so approximate by $\mathcal{N}(20, 16)$, i.e. sd $= 4$. The standard distractor is $\mathcal{N}(20, 20)$ — forgetting the $(1-p)$ factor (that would be the Poisson limit, where variance equals mean, valid only for small p). Rule-of-thumb validity here: $np = 20 \ge 10$ and $n(1-p) = 80 \ge 10$, so the approximation is respectable; add a continuity correction (± 0.5) for point or boundary probabilities.

Self-check. Variance must be strictly below the mean whenever p is not small: $16 < 20$, and the ratio $16/20 = 0.8 = 1 - p$. Plausibility of the sd: about 95% of outcomes in 20 ± 8, i.e. $[12, 28]$ successes out of 100 at $p = 0.2$ — sensible.

Answer: $\mathcal{N}$(mean 20, variance 16) (sd 4); the trap answer is variance 20.

P4.8 *CLT - what it does not say*

The CLT is being asked to certify exactly the things it is silent about.

- *The CLT is an asymptotic statement about the centre, not the far tails at fixed n.* It says the standardised sum converges in distribution; it says nothing at $n = 252$, and convergence is slowest precisely in the tails (Berry–Esseen error degrades with the third moment). A 4-sigma event lives in the large-deviations zone, where the tail of the *sum* is governed by the tails of the *summands*: for power-law-tailed daily P&L, the sum's extreme tail remains power-law (dominated by the single worst day), no matter how bell-shaped the middle looks.
- *Independence fails.* Daily P&L exhibits volatility clustering and serial dependence; the classical CLT needs i.i.d. (or weak-dependence variants with much smaller effective n). Dependent summands can normalise far more slowly or not at all.
- *Finite variance may effectively fail.* In stressed regimes the variance is unstable or practically unbounded; with infinite variance the limit is a heavy-tailed stable law, not a normal.
- *(Bonus) Parameter uncertainty.* Even granting normality, μ and σ are estimates; at 4 sigma the answer is exquisitely sensitive to both.

Direction: all of these fatten the true tail relative to the Gaussian, so the normal-table number *understates* the loss probability — compare the distribution-free Cantelli bound in the Sharpe/loss section, which sits orders of magnitude above $\Phi(-4)$.

Self-check (concrete case). Daily P&L with Student-t_3 tails: any finite sum of 252 such days still has a tail exponent of 3 (heavy tails are preserved under addition), so its 4-sigma exceedance probability is orders of magnitude above $\Phi(-4) \approx 3.2 \times 10^{-5}$ — consistent with the claim that the Gaussian answer is a lower bound in practice.

Answer: CLT is asymptotic and about the centre (silent at fixed n and in far tails, where the summands' tails rule); i.i.d. fails (vol clustering, dependence); finite variance may fail (stable, not normal, limits); plus parameter error at 4 sigma. The normal-table probability is biased low — the true tail is fatter.

P4.9 *Confidence interval: variance vs sd* ♦

First convert variance to standard deviation — the step the question is testing:

$$\sigma = \sqrt{100} = 10.$$

A central 95% interval for a normal is mean $\pm 1.96\sigma \approx$ mean $\pm 2\sigma$:

$$1000 \pm 2 \times 10 = [980,\ 1020].$$

The trap answer treats the variance 100 as if it were the sd, giving $1000 \pm 200 = [800, 1200]$ — off by a factor of σ itself. (Pedantry worth one sentence if pressed: for a single random variable this is a central *probability* interval; "confidence interval" properly refers to interval estimates of a parameter — the test's wording is the colloquial one.)

Self-check. $P(|Z| \le 2) \approx 95.4\%$, close enough to 95% for an "approximate" answer; the exact 1.96 multiplier gives $[980.4, 1019.6]$, which rounds to the same interval. Width sanity: total width 40 is 4 sds — the standard 95% span.

Answer: $[980, 1020]$ (sd = 10, mean ± 2 sd); the distractor $[800, 1200]$ confuses variance with sd.

P4.10 *Hypothesis testing recipe*

Follow the recipe. *Null:* daily signs are i.i.d. fair coin flips, $p = 1/2$ (up days independent of each other). *Statistic:* N_{up}, the number of up days. *Null distribution:* $N_{\text{up}} \sim \text{Binomial}(252, 0.5) \approx \mathcal{N}(126,\ 63)$, sd $= \sqrt{63} \approx 7.94$. *Compute:*

$$z = \frac{140 - 126}{7.94} \approx 1.76, \qquad p_{\text{two-sided}} = 2\big(1 - \Phi(1.76)\big) \approx 2 \times 0.039 \approx 7.8\%.$$

Conclusion: fail to reject at 5% — "obviously better than a coin flip" is not supported at the conventional threshold. A one-sided test gives $p \approx 3.9\%$ and would reject; but choosing the

one-sided alternative *after* seeing that the count came out high halves the p-value by construction — a garden-of-forking-paths move. The directional hypothesis must be fixed before looking.

Two further objections worth volunteering: daily signs are not i.i.d. (autocorrelation and volatility clustering shrink the effective n, pushing the true p-value up), and hit rate ignores magnitudes — a 55% hit rate with negatively skewed payoffs can still lose money.

Self-check. Continuity correction: $z = (139.5 - 126)/7.94 \approx 1.70$, $p_{\text{two-sided}} \approx 8.9\%$ — same conclusion. CI duality: the 95% interval for p is $\hat{p} \pm 1.96\sqrt{\hat{p}(1-\hat{p})/252} = 0.556 \pm 0.061 = [0.494, 0.617]$, which contains 0.5, consistent with non-rejection.

Answer: $z \approx 1.76$, two-sided $p \approx 8\%$: not significant at 5%. One-sided $p \approx 3.9\%$ would reject, but the direction must be pre-specified — choosing it after seeing the data is p-hacking.

P4.11 *Sharpe ratio standard error (the humility formula)*

Work at daily frequency, then annualise. Daily Sharpe $\text{SR}_d = 1.2/\sqrt{252} \approx 0.0756$; sample size $n = 3 \times 252 = 756$ days:

$$\text{SE}_d \approx \sqrt{\frac{1 + 0.0756^2/2}{756}} = \sqrt{\frac{1.0029}{756}} \approx 0.0364, \qquad \text{SE}_{\text{ann}} = \sqrt{252}\,\text{SE}_d \approx 0.58.$$

Because $\text{SR}_d^2/2$ is negligible at daily frequency, the clean rule of thumb is $\text{SE}_{\text{ann}} = \sqrt{1.0029/3} \approx 0.58$. Anchor consistency: the notes' benchmark "one year of daily data measures SR to about ± 1" scaled by $1/\sqrt{3} \approx 0.577$ matches.

Answer: $\text{SE}_{\text{ann}} \approx 0.58$. Versus 0: $z \approx 2.1$, just significant at 5%. Versus 1: $z \approx 0.35$, indistinguishable (would take ≈ 96 years). ± 0.25 requires about 16 years of daily data.

P4.12 *Multiple testing / deflated-Sharpe logic*

(a) With $n = 1000$ days, the t-statistic is effectively standard normal under the null. Per test, $P(|t| \geq 2.5) = 2(1 - \Phi(2.5)) \approx 2 \times 0.0062 = 0.0124$. Across 100 independent null tests,

$$P\big(\max_i |t_i| \geq 2.5\big) = 1 - (1 - 0.0124)^{100} \approx 1 - e^{-1.24} \approx 71\%.$$

Seeing a 2.5 somewhere among 100 noise signals is the *expected* outcome, not evidence. Equivalently, the expected maximum of N standard normal draws grows like $\sqrt{2\ln N}$: for the 200 tail events implicit in 100 two-sided tests, $\sqrt{2\ln 200} \approx 3.3$ — the best-of-100 *under pure noise* typically beats 3.

(b) Bonferroni: per-test level $0.05/100 = 5 \times 10^{-4}$ two-sided, i.e. 2.5×10^{-4} per tail, requiring $|t| \geq \Phi^{-1}(1 - 2.5 \times 10^{-4}) \approx 3.5$. The observed 2.5 is not close.

(c) The recipe: always ask "how many things were tried?"; correct with Bonferroni-style thresholds or the Deflated Sharpe Ratio (which penalises the reported Sharpe by the expected maximum under the null given the number of trials and non-normality); demand genuine out-of-sample or forward evidence; and only then discuss costs, capacity and leakage. Note the trials are rarely independent in practice — correlated variants reduce the effective number of tests, which the DSR framework handles via the variance of trial outcomes.

Self-check. Expected count of null signals clearing 2.5: $100 \times 0.0124 = 1.24 \geq 1$, so by a Poisson approximation $P(\text{at least one}) \approx 1 - e^{-1.24} \approx 71\%$ — matches the direct computation. And the $\sqrt{2\ln N}$ anchor (≈ 3.3) sits consistently just below the Bonferroni cutoff 3.5.

Answer: (a) $\approx 71\%$ — best-of-100 noise clears $|t| = 2.5$ more often than not; (b) $|t| \approx 3.5$ (Bonferroni at 5%); (c) count the trials, apply multiple-testing corrections / Deflated Sharpe, and demand out-of-sample confirmation.

28.5 Solutions — §5 Regression and Regularisation

P5.1 *OLS univariate slope and R-squared*

(a) With an intercept the slope is $\hat{\beta} = \widehat{\mathrm{Cov}}(X,Y)/\widehat{\mathrm{Var}}(X) = \rho\,\sigma_Y/\sigma_X = 0.4 \times 3/1 = 1.2$.

(b) In univariate regression $R^2 = \rho^2 = 0.16$: only 16% of Y's daily variance is explained.

(c) Predicted move $= \hat{\beta} \times 2\% = 2.4\%$.

(d) The slope carries the volatility ratio $\sigma_Y/\sigma_X = 3$: a modest correlation times a large vol ratio gives $\beta > 1$. R^2 measures tightness of the relationship, β measures magnitude per unit of X — different objects.

Self-check. Recompute the slope the long way: $\mathrm{Cov} = \rho\sigma_X\sigma_Y = 0.4 \times 0.01 \times 0.03 = 1.2 \times 10^{-4}$, $\mathrm{Var}(X) = 10^{-4}$, ratio $= 1.2$. And $R^2 = 0.16 \le 1$ no matter how large β is, confirming (d) is not a contradiction.

Answer: (a) 1.2; (b) 0.16; (c) +2.4%; (d) $\beta = \rho\,\sigma_Y/\sigma_X$ carries the vol ratio 3, so modest correlation still yields a slope above 1.

P5.2 *OLS derivation and projection*

(a) Minimise $L(\beta) = \|y - X\beta\|^2 = y^\top y - 2\beta^\top X^\top y + \beta^\top X^\top X\beta$. The gradient $\nabla_\beta L = -2X^\top y + 2X^\top X\beta = 0$ gives the normal equations $X^\top X\beta = X^\top y$; full column rank makes $X^\top X$ invertible, so $\hat{\beta} = (X^\top X)^{-1}X^\top y$. The Hessian is $2X^\top X \succ 0$ (for $v \neq 0$, $v^\top X^\top Xv = \|Xv\|^2 > 0$ by full rank), so this is the unique global minimum.

(b) $\hat{\varepsilon} = y - X\hat{\beta}$ gives $X^\top\hat{\varepsilon} = X^\top y - X^\top X(X^\top X)^{-1}X^\top y = 0$: residuals are orthogonal to every regressor. Equivalently $\hat{y} = Hy$ is the orthogonal projection of y onto $\mathrm{col}(X)$, and $y = \hat{y} + \hat{\varepsilon}$ is an orthogonal decomposition.

(c) Symmetry: $H^\top = X\big((X^\top X)^{-1}\big)^\top X^\top = H$ since $(X^\top X)^{-1}$ is symmetric. Idempotence: $H^2 = X(X^\top X)^{-1}(X^\top X)(X^\top X)^{-1}X^\top = H$ — projecting twice is projecting once. Trace: by cyclicity, $\mathrm{tr}(H) = \mathrm{tr}\left((X^\top X)^{-1}X^\top X\right) = \mathrm{tr}(I_p) = p$: the fit spends p degrees of freedom (this is what makes $\hat{\sigma}^2 = \mathrm{RSS}/(n-p)$ the right denominator).

(d) Centred univariate case: $X = x$ (one column, mean removed), so $\hat{\beta} = (x^\top x)^{-1}x^\top y = \sum_i x_i y_i / \sum_i x_i^2 = \widehat{\mathrm{Cov}}(x,y)/\widehat{\mathrm{Var}}(x)$.

Self-check. Orthogonality (b) implies Pythagoras: $\|y\|^2 = \|\hat{y}\|^2 + \|\hat{\varepsilon}\|^2$; equivalently the decomposition behind R^2 — consistent. Independently, differentiate back: plugging $\hat{\beta}$ into the gradient gives $-2X^\top y + 2X^\top X(X^\top X)^{-1}X^\top y = 0$.

Answer: $\hat{\beta}_{\mathrm{OLS}} = (X^\top X)^{-1}X^\top y$; residuals satisfy $X^\top\hat{\varepsilon} = 0$; H is the symmetric idempotent projector onto $\mathrm{col}(X)$ with $\mathrm{tr}(H) = p$ (degrees of freedom used); univariate centred case reduces to Cov/Var. □

P5.3 *Gauss–Markov assumptions and violations*

(a) Assumptions: (1) linearity / correct specification; (2) full column rank of X; (3) strict exogeneity $\mathbb{E}[\varepsilon \mid X] = 0$; (4) spherical errors $\mathrm{Var}(\varepsilon \mid X) = \sigma^2 I$ (homoskedastic *and* uncorrelated). Conclusion: $\hat{\beta}_{\mathrm{OLS}}$ is **BLUE** — among estimators *linear in* y and *unbiased*, it has minimum variance ($\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta}) \succeq 0$, hence minimal for every $c^\top\beta$). **Normality is not assumed.**

(b)(i) Leakage breaks strict exogeneity: the regressor is correlated with the error. Consequence: **bias and inconsistency** — more data does not save you; this is the fatal one. Fix: redesign the signal to be point-in-time; in the omitted-variable version, add the missing control or use an instrument.

(ii) Heteroskedasticity breaks spherical errors. OLS stays unbiased but is *inefficient* and its reported standard errors are wrong. Fix: WLS/GLS (whiten, then Gauss–Markov applies to the transformed data, restoring BLUE) or robust standard errors for inference.

(iii) Near-multicollinearity violates *no* assumption while the rank is technically full: OLS is still BLUE — but $\mathrm{Var}(\hat{\beta}) = \sigma^2(X^\top X)^{-1}$ explodes along the near-singular direction, so "best" is still terrible. Fix: ridge, or drop/combine the collinear features.

(iv) Fat tails break nothing in Gauss–Markov (normality was never assumed): OLS remains BLUE. Only exact finite-sample t/F inference is affected. Fix: rely on CLT asymptotics or bootstrap.

Self-check. Order the four by severity: only (i) attacks $\mathbb{E}[\hat{\beta}]$; (ii) and (iii) attack variance/inference; (iv) attacks only the exact sampling distribution. That matches the MSE decomposition bias^2 + variance: bias failures are unfixable by more data, variance failures are merely expensive.

Answer: Linearity, full rank, strict exogeneity, spherical errors ⇒ OLS is BLUE; normality NOT assumed. (i) exogeneity: bias + inconsistency (fatal) — redesign; (ii) heteroskedasticity: inefficiency + wrong SEs — WLS/robust; (iii) no assumption broken: variance explodes — ridge/drop; (iv) nothing broken: only exact t/F — asymptotics/bootstrap.

P5.4 *Omitted-variable bias*

(a) The short regression gives $\hat{\beta}_1 = (x_1^\top x_1)^{-1} x_1^\top y$. Substitute $y = x_1\beta_1 + x_2\beta_2 + \varepsilon$:

$$\hat{\beta}_1 = \beta_1 + \beta_2 \frac{x_1^\top x_2}{x_1^\top x_1} + \frac{x_1^\top \varepsilon}{x_1^\top x_1}.$$

Taking expectations (exogeneity kills the last term):

$$\mathbb{E}[\hat{\beta}_1] = \beta_1 + \beta_2 \frac{\mathrm{Cov}(x_1, x_2)}{\mathrm{Var}(x_1)}$$

— the true coefficient plus the omitted coefficient times the auxiliary regression slope of x_2 on x_1.

(b) $\mathbb{E}[\hat{\beta}_1] = 0 + 1 \times 0.3 = 0.3$. The backtest reports a "signal loading" of 0.3 that is nothing but market beta leaking through the correlation.

(c) The mistake: omit market exposure while your signal correlates with the market, and you rediscover beta and call it alpha. Fix: include x_2 in the regression, or *residualise* x_1 against the market (and other factors) before testing — after residualisation $\mathrm{Cov}(x_1^\perp, x_2) = 0$, so the bias term vanishes. (This is exactly what factor-residualised returns, e.g. Axioma-style, are for.)

Self-check. Two limits: if $\mathrm{Cov}(x_1, x_2) = 0$ the bias vanishes and the short regression is fine; if $x_1 = x_2$ the auxiliary slope is 1 and the short regression reports $\beta_1 + \beta_2 = 1$, i.e. the full market coefficient — exactly what regressing y on the market alone should give. Sign check: positive correlation and positive β_2 bias the estimate upward, as computed.

Answer: $\mathbb{E}[\hat{\beta}_1] = \beta_1 + \beta_2 \,\mathrm{Cov}(x_1, x_2)/\,\mathrm{Var}(x_1)$; the backtest reports 0.3 — pure market beta disguised as alpha; fix by including the market or residualising the signal against it first.

P5.5 *Gauss–Markov BLUE proof + ridge follow-up*

(a) Unbiasedness for *every* β forces $\mathbb{E}[\tilde{\beta}] = CX\beta = \beta$, i.e. $CX = I_p$. Write $C = (X^\top X)^{-1}X^\top + D$; then $CX = I_p + DX$, so $DX = 0$. With $\mathrm{Var}(y) = \sigma^2 I$,

$$\mathrm{Var}(\tilde{\beta}) = \sigma^2 CC^\top = \sigma^2\Big[(X^\top X)^{-1} + (X^\top X)^{-1}X^\top D^\top + DX(X^\top X)^{-1} + DD^\top\Big]$$

$$= \sigma^2[(X^\top X)^{-1} + DD^\top],$$

the cross terms dying because $DX = 0$ (and its transpose $X^\top D^\top = 0$). Hence $\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta}) = \sigma^2 DD^\top \succeq 0$ (a Gram matrix), with equality iff $D = 0$, i.e. iff $\tilde{\beta}$ *is* OLS. One idea: any competitor is OLS plus a piece orthogonal to the design, and that piece only adds variance. PSD ordering means $\mathrm{Var}(c^\top\tilde{\beta}) \geq \mathrm{Var}(c^\top\hat{\beta})$ for every linear functional.

(b) No contradiction. The theorem crowns OLS best *among linear unbiased* estimators. Ridge is biased, so it competes outside that class, and MSE = bias^2 + variance lets it win: there always

exists some $\lambda > 0$ for which ridge has strictly smaller MSE than OLS (Hoerl–Kennard). Under Gaussian errors OLS is the MLE and attains the Cramér–Rao bound, making it best among *all* unbiased estimators — which still says nothing about biased ones. Unbiasedness is a constraint, not a virtue.

(c) Here $X = \mathbf{1}_n$ and $\beta = \mu$. A linear estimator $\sum_i w_i y_i$ is unbiased iff $\sum_i w_i = 1$, and its variance is $\sigma^2 \sum_i w_i^2$. By Cauchy–Schwarz, $1 = (\sum_i w_i)^2 \le n \sum_i w_i^2$, so $\sum_i w_i^2 \ge 1/n$ with equality iff $w_i = 1/n$: the sample mean. And OLS on a constant regressor is exactly $(\mathbf{1}^\top \mathbf{1})^{-1} \mathbf{1}^\top y = \bar{y}$ — the theorem verified concretely.

Self-check. Part (c) is itself an independent verification of (a) in the simplest design. Additionally, set $D = 0$ in the variance formula: it collapses to $\sigma^2 (X^\top X)^{-1}$, the known OLS variance; and $DD^\top$ is $p \times p$ PSD by construction ($v^\top DD^\top v = \|D^\top v\|^2 \ge 0$).

Answer: $\mathrm{Var}(\tilde{\beta}) - \mathrm{Var}(\hat{\beta}) = \sigma^2 DD^\top \succeq 0$ with $DX = 0$; ridge does not contradict the theorem because it is biased (outside the class) and wins on MSE; for the mean, the unique best linear unbiased weights are $w_i = 1/n$ — the sample mean. □

P5.6 *Ridge derivation, invertibility, shrinkage*

(a) $L(\beta) = \|y - X\beta\|^2 + \lambda \|\beta\|_2^2$; the gradient $-2X^\top y + 2X^\top X\beta + 2\lambda\beta = 0$ gives $(X^\top X + \lambda I)\beta = X^\top y$, so

$$\hat{\beta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top y.$$

The Hessian $2(X^\top X + \lambda I) \succ 0$ strictly, so the minimiser is unique — even when OLS is not.

(b) $X^\top X \succeq 0$ has eigenvalues $d_i \ge 0$; adding λI shifts every eigenvalue to $d_i + \lambda \ge \lambda > 0$. All eigenvalues strictly positive $\Rightarrow$ invertible, always. This is the mechanical fix for multicollinearity and for $p > n$.

(c) With $X^\top X = I$: $\hat{\beta}_{\text{Ridge}} = (1+\lambda)^{-1} X^\top y = \hat{\beta}_{\text{OLS}}/(1+\lambda)$ — every coefficient shrunk by the same factor, never set exactly to zero. Numerically: $1.2/1.5 = 0.8$.

(d) Put a prior $\beta \sim \mathcal{N}(0, \tau^2 I)$ with Gaussian noise σ^2: the log-posterior is $-\|y - X\beta\|^2/(2\sigma^2) - \|\beta\|^2/(2\tau^2) + \text{const}$, so the MAP estimate is exactly ridge with $\lambda = \sigma^2/\tau^2$. Strong prior (small τ) $\Rightarrow$ large $\lambda \Rightarrow$ heavy shrinkage.

Self-check. Limits: $\lambda \to 0$ recovers OLS; $\lambda \to \infty$ shrinks everything to 0 (infinitely confident prior at zero) — both sensible. Differentiate back in the 1-D orthonormal case at $b = 0.8$: $2(b - 1.2) + 2\lambda b = 2(-0.4) + 2(0.5)(0.8) = -0.8 + 0.8 = 0$.

Answer: $\hat{\beta}_{\text{Ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$; always invertible because every eigenvalue is at least $\lambda > 0$; orthonormal case shrinks OLS by $1/(1+\lambda)$, so $1.2 \to 0.8$; MAP under a Gaussian prior with $\lambda = \sigma^2/\tau^2$. □

P5.7 *Regularisation menu* ★

The five to lead with:

- **L2 / ridge** — penalise $\|w\|_2^2$: smooth shrinkage toward zero, handles collinearity, Gaussian prior.
- **L1 / lasso** — penalise $\|w\|_1$: exact zeros, so feature selection; Laplace prior.
- **Dropout** — randomly zero units in training: prevents co-adaptation, acts as an implicit ensemble of subnetworks.
- **Early stopping** — halt training when validation loss turns: limits effective capacity along the optimisation path (for linear models, closely related to ridge).
- **Data augmentation / noise injection** — enlarge the training distribution with invariances; in finance: train across assets and regimes rather than one series.

Backups when probed: elastic net (L1 + L2); label smoothing; batch norm's implicit regularising effect; ensembling/bagging; weight decay as distinct from L2 under Adam (AdamW); and

the tree-model menu — `max_depth`/number of leaves, `min_data_in_leaf`/`min_child_weight`, L1/L2 penalties on leaf values (`lambda_l1`/`lambda_l2`), `feature_fraction`/`bagging_fraction` subsampling, small learning rate with more trees, early stopping on validation. Finance-specific: covariance *shrinkage* (Ledoit–Wolf) is regularisation of an estimator, not just of a predictive model.

Self-check. Apply the unifying test to each item: regularisation = anything that constrains effective capacity or pulls the fit toward a prior, trading bias for variance. L2/L1 add explicit priors; dropout and augmentation inject noise/invariances that shrink the hypothesis space actually reachable; early stopping bounds the optimisation path; every tree control caps how far a tree can chase noise. All pass; nothing on the list merely "improves optimisation" without constraining fit.

Answer: L2/ridge, L1/lasso, dropout, early stopping, data augmentation/noise injection — with elastic net, label smoothing, batch norm, bagging/ensembling, tree controls (depth, min leaf data, leaf-value L1/L2, feature/row subsampling, small learning rate) and Ledoit–Wolf covariance shrinkage as backups.

P5.8 *Lasso soft-thresholding*

(a) Case $b > 0$: the objective is smooth, derivative $2(b - \hat{\beta}_j) + \lambda = 0$ gives $b = \hat{\beta}_j - \lambda/2$, admissible only if $\hat{\beta}_j > \lambda/2$. Case $b < 0$: symmetrically $b = \hat{\beta}_j + \lambda/2$, admissible if $\hat{\beta}_j < -\lambda/2$. Case $b = 0$: the subdifferential of the objective at 0 is $-2\hat{\beta}_j + \lambda[-1, 1]$; it contains 0 iff $|\hat{\beta}_j| \le \lambda/2$, and the objective is convex, so $b = 0$ is then optimal. Combining:

$$\hat{\beta}_j^{\text{Lasso}} = \operatorname{sign}(\hat{\beta}_j)(|\hat{\beta}_j| - \tfrac{\lambda}{2})_+ \qquad \text{(soft-thresholding)}.$$

(b) Threshold $\lambda/2 = 0.5$: $2.0 \to 1.5$; $0.6 \to 0.1$; $-0.3 \to 0$ (inside the threshold, killed exactly).

(c) At $b = 0$ the ridge penalty's pull is $\frac{d}{db}\lambda b^2 = 2\lambda b = 0$ — the penalty exerts *no force* at zero, so any nonzero data pull moves the coefficient off zero; ridge only ever multiplies by $1/(1 + \lambda)$. The L1 penalty keeps a *constant* force λ (per unit of $|b|$, i.e. $\lambda/2$ against the data pull in this parametrisation) all the way to zero: whenever the data pull $|\hat{\beta}_j|$ is below the threshold, the penalty wins outright and pins the coefficient at exactly zero. Geometrically: the L1 ball has corners on the axes; the L2 ball is round.

Self-check. Verify (b) for $\hat{\beta}_j = 0.6$ by direct evaluation: at $b = 0.1$: $(0.5)^2 + 1(0.1) = 0.35$; at $b = 0$: 0.36; at $b = 0.2$: $0.16 + 0.2 = 0.36$. The claimed optimum 0.35 beats both neighbours. And the formula's two limits: $\lambda \to 0$ returns OLS, $\lambda \ge 2|\hat{\beta}_j|$ kills everything.

Answer: Soft-thresholding $\operatorname{sign}(\hat{\beta})(|\hat{\beta}| - \lambda/2)_+$; with $\lambda = 1$: $(2.0, 0.6, -0.3) \to (1.5, 0.1, 0)$; lasso zeros exist because the L1 penalty keeps a constant force at zero while the ridge penalty's force vanishes there.

P5.9 *Lasso vs ridge in alpha research*

(a) Penalised regression is equivalent to minimising the loss subject to a norm ball constraint. The L1 ball is a diamond (cross-polytope) with corners on the coordinate axes; the elliptical level sets of the quadratic loss typically first touch the ball at a corner — where some coordinates are exactly zero. The L2 ball is round: contact happens at a generic boundary point, so coefficients shrink but essentially never land exactly on zero.

(b) With $p = 500$ approaching $n = 750$ and heavy collinearity, plain OLS variance explodes, so regularisation is mandatory. Lasso's failure mode: within a group of highly correlated signals it tends to pick *one* representative near-arbitrarily and zero the rest, and the choice is unstable — resample the data and a different group member gets selected. For alpha research that instability is costly (turnover in which signals are "on," fragile attribution). Ridge keeps the whole group with weight shared across it (grouping behaviour) and handles the collinearity, at the cost of no selection. Additionally, when $p > n$, lasso can select at most n variables — a structural limit to know.

(c) Elastic net combines both penalties, $\lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2$: the L2 part makes the solution unique

and spreads weight across correlated groups (grouping effect), the L1 part still zeroes genuinely useless signals. That is precisely the correlated-many-signals regime, hence the default.

Self-check. The duplicate-feature thought experiment: let $x_1 = x_2$ exactly. Ridge's objective is strictly convex (thanks to the penalty), and by symmetry its unique solution splits the coefficient equally — stable. Lasso's fit and penalty are identical for any split $b_1 + b_2 = s$ with $b_1, b_2 \geq 0$: a whole continuum of minimisers, so the selected representative is arbitrary — exactly the instability claimed in (b).

Answer: L1's diamond has corners on the axes (exact zeros), L2's ball is round (shrinkage only); here ridge or elastic net — lasso's selection within correlated groups is arbitrary and resample-unstable (and caps at n selections when $p > n$); elastic net adds L2's grouping and uniqueness to L1's sparsity, the pragmatic default.

P5.10 *Inverse regression*

(a) The two regressions are different projections: Y-on-X minimises *vertical* squared distances, X-on-Y minimises *horizontal* ones. They produce two different lines (unless the data are perfectly collinear), each attenuated toward zero relative to the SD line by a factor ρ. Fitted slopes are not reciprocals.

(b) $\hat{\beta}_{X|Y} = \rho\,\sigma_X/\sigma_Y = 0.5 \times 1 = 0.5$ — the same number as the forward slope here, not 2. The shortcut overstates by $2/0.5 = 4 = 1/\rho^2$.

(c) $\hat{\beta}_{Y|X} = \rho\,\sigma_Y/\sigma_X = B$ and $\hat{\beta}_{X|Y} = \rho\,\sigma_X/\sigma_Y = D$. Multiplying: $BD = \rho^2 \leq 1$, hence $D = \rho^2/B$, which equals $1/B$ only when $|\rho| = 1$ (exactly collinear data). In general naive inversion inflates the answer by the factor $1/\rho^2$.

Self-check. Internal consistency of the given numbers: with $\sigma_X = \sigma_Y$, the forward slope should itself equal $\rho = 0.5$ — and it does, matching the stated 0.5. And the identity: $BD = 0.5 \times 0.5 = 0.25 = \rho^2$. Sanity bound: both slopes have $|\text{slope}| \leq \sigma_{\text{num}}/\sigma_{\text{den}}$, and $2 > \sigma_X/\sigma_Y = 1$ immediately flags the shortcut as impossible.

Answer: Slopes are not reciprocals: $\hat{\beta}_{Y|X}\hat{\beta}_{X|Y} = \rho^2$, so the true X-on-Y slope is $\rho^2/B = 0.5$, not 2 — the shortcut overstates by $1/\rho^2 = 4\times$; inversion is legitimate only when $|\rho| = 1$.

P5.11 *Hedge-ratio inversion, market application*

(a) Per \$1 of S, hedged return $= r_S - h\,r_A$ with h the notional ratio. Variance $= \sigma_S^2 + h^2\sigma_A^2 - 2h\rho\sigma_S\sigma_A$; minimising over h: $2h\sigma_A^2 = 2\rho\sigma_S\sigma_A$, so

$$h^* = \rho\,\frac{\sigma_S}{\sigma_A} = 0.6 \times \frac{2}{1} = 1.2 \quad \Rightarrow \quad \text{short \$1.2M of } A.$$

This is exactly the regression slope of S on A.

(b) The intern's regression gives $\hat{\beta}_{A|S} = \rho\,\sigma_A/\sigma_S = 0.6 \times 0.5 = 0.3$; inverting, $h = 1/0.3 = 3.33$, i.e. *\$3.33M* short. Over-hedge factor: $3.33/1.2 = 2.78 = 1/\rho^2$ — the inverse-regression trap in production.

(c) In $\%^2$ per day (on \$1M):

- Unhedged: $\sigma = 2\%$, i.e. \$20,000/day.
- Correct hedge $h = 1.2$: variance $= 4 + (1.2)^2(1) - 2(1.2)(0.6)(2)(1) = 4 + 1.44 - 2.88 = 2.56$, so $\sigma = 1.6\% =$ \$16,000/day.
- Intern $h = 3.33$: variance $= 4 + (3.33)^2(1) - 2(3.33)(0.6)(2)(1) = 4 + 11.11 - 8.00 = 7.11$, so $\sigma = 2.67\% \approx$ \$26,700/day.

Punchline: the over-hedged book is *riskier than not hedging at all* — the extra \$2.1M of A is unhedged exposure in its own right. Secondary lesson: even the correct hedge only cuts vol from 2.0% to 1.6%, because with $\rho = 0.6$ the residual fraction $\sqrt{1-\rho^2} = 0.8$ of the risk is unhedgeable by A.

Self-check. The correct-hedge residual should equal $\sigma_S\sqrt{1-\rho^2} = 2 \times 0.8 = 1.6\%$ — matches the direct variance computation. And the two slopes multiply to $1.2 \times 0.3 = 0.36 = \rho^2$.

Answer: (a) \$1.2M; (b) \$3.33M, over-hedged by $1/\rho^2 = 2.78\times$; (c) \$20,000 unhedged, \$16,000 correctly hedged, ≈\$26,700 for the intern — the inverted hedge is riskier than no hedge at all.

P5.12 *Beta vs correlation*

(a) False. $\beta = \rho\,\sigma_Y/\sigma_X$: beta mixes correlation with the *relative volatility*. Counterexample: $\rho = 0.2$ with $\sigma_Y/\sigma_X = 10$ gives $\beta = 2$ — a "high-beta" stock whose returns are barely aligned with the index (think an illiquid small-cap or a crypto-adjacent name against an equity index).

(b) The *beta* sizes the hedge: hedging neutralises P&L magnitude, and magnitude lives in σ_Y/σ_X. The *correlation* (through $R^2 = \rho^2$, i.e. the residual fraction $1 - \rho^2$) tells you whether hedging achieves anything: it is the share of variance the hedge can remove. Correlation says whether to hedge, beta says how much.

(c) $\beta = 0.2 \times 10/1 = 2$; explained fraction $= R^2 = \rho^2 = 4\%$. Reconciliation: the index-sized piece of the stock is large per unit of index move (beta 2), yet it is a tiny share of the stock's own huge variance — a beta-2 hedge here would be technically correctly sized and almost pointless, leaving residual vol $10\% \times \sqrt{0.96} \approx 9.8\%$.

Self-check. Bound check: $|\beta| \le \sigma_Y/\sigma_X = 10$, and $2 \le 10$. Consistency: R^2 recomputed from the regression view — explained variance $= \beta^2\sigma_X^2 = 4 \times 1 = 4$ out of $\sigma_Y^2 = 100$, i.e. 4% (matches ρ^2).

Answer: False — e.g. $\rho = 0.2$, $\sigma_Y/\sigma_X = 10$ gives $\beta = 2$; beta sizes the hedge, correlation (via R^2) decides if it is worthwhile; numbers: $\beta = 2$, explained variance 4%, residual vol ≈ 9.8% — correctly sized yet nearly pointless.

P5.13 *Choosing a hedge instrument*

(a) Hedge ratio $h = \rho\,\sigma_Y/\sigma_F$; residual vol $= \sigma_Y\sqrt{1-\rho^2}$ (the part of Y orthogonal to the hedge).

- F_1: $h_1 = 0.9 \times 4/8 = 0.45 \Rightarrow$ short \$450k. Residual $= 4\%\sqrt{1-0.81} = 4\% \times 0.436 = 1.74\%$.
- F_2: $h_2 = 0.6 \times 4/1 = 2.4 \Rightarrow$ short \$2.4M. Residual $= 4\%\sqrt{1-0.36} = 4\% \times 0.8 = 3.2\%$.

(b) Choose F_1. The residual vol of an optimally sized hedge is $\sigma_Y\sqrt{1-\rho^2}$ — it depends *only on the correlation*, not on the hedge instrument's vol or beta. Higher $|\rho| \Rightarrow$ better hedge, full stop.

(c) The rule confuses sizing with quality. F_2 carries the much larger hedge ratio (2.4 vs 0.45) — by the rule it looks like "more hedge" — yet it leaves nearly twice the residual risk (3.2% vs 1.74%). Beta only tells you *how much* to trade once the instrument is chosen; correlation tells you *which* instrument can actually absorb the risk. (Real desks then layer on liquidity and cost — but risk-wise the correlation decides.)

Self-check. Recompute F_1's residual directly: variance $= \sigma_Y^2 + h^2\sigma_F^2 - 2h\rho\sigma_Y\sigma_F = 16 + (0.45)^2(64) - 2(0.45)(0.9)(4)(8) = 16 + 12.96 - 25.92 = 3.04$, and $\sqrt{3.04} = 1.74\%$ — matches $\sigma_Y\sqrt{1-\rho^2} = 4\sqrt{0.19}$. Same check for F_2: $16 + 5.76 - 11.52 = 10.24$, $\sqrt{10.24} = 3.2\%$.

Answer: F_1: short \$450k, residual 1.74%; F_2: short \$2.4M, residual 3.2%. Choose F_1 — residual vol $\sigma_Y\sqrt{1-\rho^2}$ depends only on correlation; the higher-beta rule confuses hedge sizing with hedge quality.

P5.14 *WLS precision weighting*

(a) Take $\hat{\mu} = wx_1 + (1-w)x_2$ (unbiased for any w). Variance $= w^2(1) + (1-w)^2(4)$; setting the derivative to zero: $2w - 8(1-w) = 0$, so $w = 4/5$. Variance $= (16/25)(1) + (1/25)(4) = 20/25 = 0.8 < 1$: yes, strictly better than the good measurement alone — even a noisy second reading adds information when weighted correctly.

(b) This is OLS of the "regression" $x_i = \mu + \varepsilon_i$ on a constant with heteroskedastic noise, done properly: minimise $\sum_i (x_i - \mu)^2/\sigma_i^2$, i.e. *whiten* each observation by dividing by σ_i and run OLS on the transformed data. The first-order condition gives precision weighting: $w_i \propto 1/\sigma_i^2$ (here

$1:1/4$, i.e. 4/5 and 1/5), with combined variance $\left(\sum_i 1/\sigma_i^2\right)^{-1}$. Gauss–Markov applies to the whitened model, so WLS is BLUE.

(c) Unweighted OLS minimises raw squared residuals, so the crisis months — which have the largest residual variance — dominate the objective while carrying the *least* information per observation; the fit chases the noisiest regime and the reported standard errors are wrong on top. Fixes: WLS with weights $\propto$ inverse conditional variance (e.g. inverse realised variance / vol-scaled returns), which restores efficiency; or, if you only need valid inference with the same point estimates, heteroskedasticity-robust (White) standard errors.

Self-check. Recompute the optimal variance the closed-form way: $(1/1+1/4)^{-1} = (5/4)^{-1} = 0.8$ — matches the calculus route. And equal weights would give $(1/4)(1) + (1/4)(4) = 1.25 > 0.8$, confirming the weighting matters.

Answer: $w = 4/5$ on the sd-1 reading, combined variance 0.8 (better than either alone); this is WLS with precision weights $w_i \propto 1/\sigma_i^2$ and combined variance $(\sum 1/\sigma_i^2)^{-1}$; in crisis-heteroskedastic data unweighted OLS lets the noisiest points steer the fit — use inverse-variance WLS (or robust SEs for inference).

P5.15 *Residual weighting vs GLS*

Adjudication. Teammate 3 has the right formula for the wrong problem. Two opposite weighting philosophies are being conflated:

- **WLS/GLS (statistics):** you are estimating *one* model efficiently under heteroskedastic noise; you *down*-weight noisy observations ($w_i = 1/\sigma_i^2$) because they carry less information. Goal: efficiency.
- **Residual weighting (the ML move):** weighting by $|y_i - \hat{y}_{A,i}|$ *up*-weights the points A got wrong so that B *specialises on A's errors*. That is the **boosting** idea in disguise (safe names: boosting-style reweighting, residual fitting, cascading — not "stacking," which trains a meta-learner on model *outputs*). Goal: specialisation, i.e. bias reduction of the ensemble.

So nobody's weights are "upside-down" — they answer different questions. But two genuine fixes to Teammate 2's execution: (1) the cleaner boosting form fits B *to the residuals* $y_i - \hat{y}_{A,i}$ and combines additively, $F = A + \eta B$ with shrinkage $\eta < 1$, rather than refitting y with weights and naively averaging (averaging halves A's calibrated signal and mixes objectives); (2) B should be weak/shallow — a specialist, not a second generalist.

Backfire regime. In financial data, A's largest residuals concentrate in high-volatility regimes, where much of the error is *irreducible noise*, not structure. Up-weighting them makes B specialise on noise — overfitting crisis observations exactly where WLS logic says to down-weight. Guardrails: shrinkage, shallow B, early stopping, or weighting by residuals *after* vol-normalising returns.

Self-check. Limiting cases separate the two philosophies cleanly: if A's residuals are pure i.i.d. noise, the weights are uninformative and B chases noise — the scheme should be abandoned (and WLS logic wins); if A systematically misses one regime (structured residuals), the weights concentrate exactly there and the cascade adds real skill (boosting logic wins). The scheme is valid precisely when residuals contain structure — consistent with the adjudication.

Answer: It is not WLS: GLS/WLS down-weights noisy points ($1/\sigma_i^2$) to fit one model efficiently, while this scheme up-weights hard points so B specialises on A's errors — boosting-style residual fitting; combine as a cascade $F = A + \eta B$ (fit B on residuals, shrink), not a naive average; it backfires in high-vol regimes where large residuals are irreducible noise.

P5.16 *Dart problem, Gaussian, pitchfork*

(a) Write $a = m + u$. Then $S = g(u + d/2) + g(u - d/2)$ with $g(t) = e^{-t^2/(2\sigma^2)}$, which is even — so S is symmetric in u, and a smooth symmetric function has $S'(m) = 0$. (Directly: the two derivative terms at $u = 0$ are equal and opposite.)

(b) Classify the critical point via the second derivative. For one term, $g''(t) = g(t)(t^2/\sigma^4 - 1/\sigma^2)$; at $u = 0$ each term has $t = \pm d/2$, so

$$S''(m) = \frac{2\, e^{-d^2/(8\sigma^2)}}{\sigma^4}\Big(\frac{d^2}{4} - \sigma^2\Big).$$

For $d < 2\sigma$ this is negative: the midpoint is a maximum (indeed the global one — an equal two-component Gaussian mixture is unimodal exactly when the separation is at most 2σ, the classic bimodality threshold). For $d > 2\sigma$ it is *positive*: the midpoint becomes a local *minimum*, and two symmetric maxima split off on either side — a pitchfork bifurcation at $d = 2\sigma$. As $d/\sigma \to \infty$ the two optima migrate onto the targets themselves: you commit to one target rather than split the difference.

(c) A Gaussian log-density is a negative *squared* distance: maximising a Gaussian score is minimising squared loss. Squared loss is minimised by the **mean** — the conditional mean is what L2 regression estimates, and "aim at the mean" is the regression content of the $p = 2$ exponent. Sharp addendum: if the score were *multiplicative* across targets (equivalently, if you summed *log*-densities), the objective would be $-\sum_i (a - x_i)^2/(2\sigma^2)$, maximised at the mean for *every* d — the bifurcation exists only because we add densities, not log-densities. Precision about which objective you are optimising is the whole game.

Self-check. Units: $S''(m)$'s sign depends only on d/σ, as it must (the problem has one dimensionless parameter). Limits: $d \to 0$, the two bumps merge and the midpoint is trivially optimal; $d \gg \sigma$, aiming at a target scores ≈ 1 while the midpoint scores $2e^{-d^2/(8\sigma^2)} \to 0$, so the optimum must indeed leave the midpoint — consistent with the 2σ threshold. Boundary: at $d = 2\sigma$ exactly, $S''(m) = 0$, the degenerate pitchfork point.

Answer: The midpoint is always critical by symmetry; it is the optimum iff $d \le 2\sigma$ (since $S''(m) \propto d^2/4 - \sigma^2$), and past $d = 2\sigma$ a pitchfork sends two maxima toward the targets; the squared exponent is L2 loss — Gaussian score maximisation = squared-loss minimisation = predict the (conditional) mean.

P5.17 *Dart problem, L1, median vs endpoint*

(a) For continuous F, differentiate through the expectation: $g'(c) = \mathbb{E}[\mathrm{sign}(c - X)] = P(X < c) - P(X > c) = 2F(c) - 1$. Setting $g'(c) = 0$ gives $F(c) = 1/2$; since $|x - c|$ is convex in c, g is convex and this stationary point is the global minimum — the median. For the two-point empirical distribution, $g(c) = \frac{1}{2}|x_1 - c| + \frac{1}{2}|x_2 - c|$ is flat and minimal on the whole interval $[x_1, x_2]$: *every* point between the targets is optimal under absolute loss (any point in the interval is a median).

(b) The game flips the conclusion. For a between the targets, write $a = x_1 + t$, $t \in [0, d]$: $S = e^{-t} + e^{-(d-t)}$, a sum of two convex exponentials, hence convex in t — and a convex function on an interval attains its *maximum at an endpoint*. Outside the interval both distances grow, so S strictly decreases; the global maximum is therefore at a target. The margin:

$$S(0) - S(d/2) = \big(1 + e^{-d}\big) - 2e^{-d/2} = \big(1 - e^{-d/2}\big)^2 > 0 \quad \text{for every } d > 0.$$

With Laplace tails you aim *at* a target, never between them. Intuition: e^{-r} rewards being very close much more than squared-exponential-type spreading — the kink at zero pays you for committing.

(c) Mapping: **L2 loss** $\leftrightarrow$ conditional **mean** (outlier-sensitive), **L1 loss** $\leftrightarrow$ conditional **median** (robust) — and part (b) is a warning that "maximise a score" and "minimise a loss" can point to different aims even for the same tail shape (minimising expected absolute distance says anywhere in $[x_1, x_2]$; maximising summed Laplace density says an endpoint). The classic conflation to avoid: *losses* (what functional you predict: mean vs median) versus *penalties* (how you constrain coefficients: ridge shrinks, lasso selects). "L1" appears in both sentences and means different things.

Self-check. Numeric probe at $d = 2$: endpoint $1 + e^{-2} \approx 1.135$ vs midpoint $2e^{-1} \approx 0.736$; difference $\approx 0.399 = (1 - e^{-1})^2 \approx (0.632)^2$. Limits: $d \to 0$ gives margin $(1 - 1)^2 = 0$ — targets coincide, no preference, as it must. Convexity check: $S''(t) = e^{-t} + e^{-(d-t)} > 0$ on $(0, d)$, so no interior maximum exists. And (a) sanity: for symmetric X the median equals the mean, so L1 and L2 point estimates agree there — consistent with both derivations.

Answer: (a) $g'(c) = 2F(c) - 1$ and convexity give the median; with two points, the whole interval $[x_1, x_2]$ is optimal. (b) S is convex between the targets, so the maximum is at an endpoint, beating the midpoint by $(1 - e^{-d/2})^2 > 0$: with Laplace tails, aim at a target. (c) L2 $\leftrightarrow$ mean, L1 $\leftrightarrow$ median; keep losses (what you predict) distinct from penalties (ridge shrinks, lasso selects).

28.6 Solutions — §6 Linear Algebra, PCA and Covariance

P6.1 *Vector norms / reverse triangle inequality* ♦

By the triangle inequality, $|A| = |(A - B) + B| \le |A - B| + |B|$, so $|A - B| \ge |A| - |B|$; swapping the roles of A and B gives $|A - B| \ge |B| - |A|$. Together these are the *reverse triangle inequality*

$$|A - B| \;\ge\; \big|\,|A| - |B|\,\big| \;=\; |5 - 7| \;=\; 2.$$

Equality requires the triangle formed by A, B, $A - B$ to degenerate: A and B parallel and pointing in the *same* direction. Concretely $B = \frac{7}{5}A$ gives $A - B = -\frac{2}{5}A$, with norm $\frac{2}{5} \cdot 5 = 2$. (For contrast: the maximum of $|A - B|$ is $|A| + |B| = 12$, attained anti-parallel.)

Self-check: law of cosines: $|A - B|^2 = 5^2 + 7^2 - 2 \cdot 5 \cdot 7 \cos\theta = 74 - 70\cos\theta$, minimised at $\cos\theta = 1$, giving $|A - B|^2 = 4$, i.e. $|A - B| = 2$ — any misalignment strictly increases the distance. One-dimensional sanity case: $A = 5$, $B = 7$ on the real line, $|5 - 7| = 2$.

Answer: 2, attained when A and B are parallel in the same direction (reverse triangle inequality).

P6.2 *Matrix transformations rapid-fire* ♦

(a) SR. Operators act right-to-left on column vectors: first Rx, then $S(Rx) = (SR)x$. RS is the classic wrong-order trap (it shears first). *Self-check:* feed a concrete x through the two steps and through the product — only SR reproduces "rotate, then shear."

(b) The y-axis. The middle row and middle column are e_2, so the y-component passes through untouched, while the (x, z) block is a standard planar rotation. *Self-check:* the matrix maps $(0, 1, 0)^\top$ to itself for every t, and at $t = 0$ it is the identity.

(c) $\det(A) = 0$. Dependent columns mean the image of the unit cube is squashed into a lower-dimensional set — zero volume; equivalently A is singular. *Self-check:* $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$ (second column twice the first): $\det = 4 - 4 = 0$.

(d) $B^{-1}A^{-1}$ — inverses reverse order ("socks–shoes"). *Self-check:* $(AB)(B^{-1}A^{-1}) = I$ (the inner pair cancels first).

(e) Areas are multiplied by $|\det| = 2$ and orientation is inverted (the sign flip): every shape doubles in area and is mirrored. *Self-check:* $\mathrm{diag}(2, -1)$ has determinant -2; it stretches x by 2 (unit square $\to$ area 2) and flips the y-axis (orientation reversed).

Answer: (a) SR; (b) the y-axis; (c) $\det(A) = 0$; (d) $B^{-1}A^{-1}$; (e) areas doubled, orientation inverted.

P6.3 *Eigenvectors of a 2D rotation* ♦

The matrix is $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Its characteristic polynomial is $\lambda^2 + 1 = 0$, so $\lambda = \pm i$: no real eigenvalues, hence *no real eigenvectors*. Geometrically that is obvious: every nonzero vector is turned by 90 degrees, so none keeps its direction.

The bait option imports 3D intuition: a rotation of $\mathbb{R}^3$ does have its axis as an eigenvector (eigenvalue 1), but the "axis" of a 2D rotation is not a vector inside $\mathbb{R}^2$ — there is no axis to point along. General fact worth stating: a 2D rotation by angle θ has real eigenvectors only for $\theta = 0$ (identity, $\lambda = 1, 1$) and $\theta = \pi$ ($\lambda = -1, -1$, every vector is an eigenvector).

Self-check: two independent routes. (i) Trace/determinant: $\operatorname{tr} R = 0 = \lambda_1 + \lambda_2$ and $\det R = 1 = \lambda_1\lambda_2$ force $\lambda^2 + 1 = 0$, i.e. $\pm i$. (ii) Geometric: $Rx \perp x$ for every x, so $Rx = \lambda x$ with real $\lambda \neq 0$ would make x orthogonal to itself — impossible for $x \neq 0$ ($\lambda = 0$ is ruled out since $\det R \neq 0$).

Answer: It has no real eigenvectors — eigenvalues are $\pm i$; the "axis of rotation" option describes 3D, not $\mathbb{R}^2$.

P6.4 *Correlation-matrix feasibility*

(a) A valid correlation matrix is the covariance matrix of standardised variables, hence PSD: $x^\top \Sigma x = \operatorname{Var}(x^\top Z) \geq 0$ for all x, so all eigenvalues $\lambda_i \geq 0$ and $\det = \prod \lambda_i \geq 0$. Here, using $\det = 1 + 2abc - a^2 - b^2 - c^2$ for $\begin{pmatrix} 1 & a & b \\ a & 1 & c \\ b & c & 1 \end{pmatrix}$ with $a = 0.9$, $b = -0.9$, $c = 0.8$:

$$\det = 1 + 2(0.9)(-0.9)(0.8) - 0.81 - 0.81 - 0.64 = 1 - 1.296 - 2.26 = -2.556 < 0.$$

A negative determinant means an odd number of negative eigenvalues; since $\operatorname{tr} = 3 > 0$ rules out all three negative, exactly one eigenvalue is negative — the matrix is not PSD, so it is not a correlation matrix. Intuition: B is nearly the same bet as A, and C is nearly the opposite bet, so B and C must be strongly *negatively* correlated; $+0.8$ is geometrically impossible.

(b) Each entry was estimated on a different sample. The one-line PSD proof needs all entries to come from a *single* quadratic form — one dataset, one Σ. Pairwise-complete estimation breaks that structure: there is no single sample whose covariance produces all three numbers simultaneously, so nothing forces the assembled matrix to be PSD. (Same failure mode as asynchronous data across assets.)

(c) Treat standardised variables as unit vectors with ρ = cosine of the angle between them. Feasibility of the triple is $\det \geq 0$:

$$\det(c) = -c^2 - 2(0.81)c + 1 - 0.81 - 0.81 = -c^2 - 1.62c - 0.62 \geq 0 \iff c^2 + 1.62c + 0.62 \leq 0,$$

with roots $c = \frac{-1.62 \pm \sqrt{1.62^2 - 4(0.62)}}{2} = \frac{-1.62 \pm 0.38}{2}$, i.e. $c = -1$ and $c = -0.62$. So $\rho_{BC} \in [-1, -0.62]$: the recorded $+0.8$ is far outside. (Equivalently the general bound $\rho_{BC} \in \left[ab - \sqrt{(1-a^2)(1-b^2)},\ ab + \sqrt{(1-a^2)(1-b^2)}\right] = [-0.81 - 0.19, -0.81 + 0.19]$.) Repairs: (i) eigenvalue clipping — floor negative eigenvalues at 0, rescale the diagonal back to 1; (ii) shrinkage toward the identity, $(1-\alpha)\Sigma + \alpha I$ (the same $+\lambda I$ medicine as ridge for near-singular $X^\top X$); (iii) impose factor structure (low-rank plus diagonal).

Self-check: the two routes agree — the quadratic's roots (-1 and -0.62; Vieta: product 0.62, sum -1.62, both match) coincide with the cosine bound -0.81 ± 0.19; and evaluating det at $c = 0.8$ by cofactor expansion also gives $0.36 - 1.458 - 1.458 = -2.556$, matching the formula above.

Answer: (a) $\det = -2.556 < 0$, so one eigenvalue is negative — not PSD, not a valid correlation matrix; (b) pairwise estimation on different samples breaks the single-quadratic-form structure that guarantees PSD; (c) $\rho_{BC} \in [-1, -0.62]$; repair by eigenvalue clipping, shrinkage toward I, or factor structure.

P6.5 *PCA derivation and explained variance*

(a) The projection of the data onto w has variance $w^\top \Sigma w$. Maximise subject to $w^\top w = 1$ with a Lagrangian: $L = w^\top \Sigma w - \lambda(w^\top w - 1)$, and $\nabla_w L = 2\Sigma w - 2\lambda w = 0$ gives $\Sigma w = \lambda w$ — stationary points are *eigenvectors* of Σ. At an eigenvector the objective equals $w^\top \Sigma w = \lambda$, so the maximiser

is the eigenvector of the largest eigenvalue λ_1, and the maximised variance is λ_1. Explained fraction of PC_i: $\lambda_i / \sum_j \lambda_j$.

(b) For $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$: eigenvalues 3 (eigenvector $(1,1)^\top/\sqrt{2}$) and 1 (eigenvector $(1,-1)^\top/\sqrt{2}$). PC1 explains $\frac{3}{3+1} = \frac{3}{4} = 75\%$.

(c) Without centring, $X^\top X = n\,\bar{x}\bar{x}^\top +$ (centred scatter): the rank-one mean term is added to the true scatter, and with a large offset it dominates — the leading "component" chases the direction of the *mean vector*, not the direction of maximal variance, and its explained share is inflated. Centre first (PCA is defined on the covariance, not the raw second moment).

Self-check (b): trace $= 4 = 3 + 1$ and det $= 3 = 3 \cdot 1$, both consistent; direct verification $\Sigma(1,1)^\top = (3,3)^\top = 3\,(1,1)^\top$; and the shortcut for symmetric 2×2 with equal diagonal d and off-diagonal c — eigenvalues $d \pm c = 2 \pm 1$ with eigenvectors $(1, \pm 1)^\top$ — agrees.

Answer: (a) $w =$ top eigenvector of Σ, maximised variance $= \lambda_1$; (b) directions $(1,1)^\top/\sqrt{2}$ and $(1,-1)^\top/\sqrt{2}$, PC1 explains 75%; (c) it chases the mean direction, not the maximal-variance direction. □

P6.6 *Equicorrelated covariance / PC1 as market mode*

(a) $\mathbf{1}\mathbf{1}^\top$ is rank one with eigenvector $\mathbf{1}$ (eigenvalue n) and eigenvalue 0 on the $(n-1)$-dimensional complement $\{x : \sum_i x_i = 0\}$. Since Σ is a linear combination of I and $\mathbf{1}\mathbf{1}^\top$, it has the same eigenvectors:

$$\text{on } \mathbf{1}: \quad \lambda_1 = \sigma^2\big[(1-\rho) + \rho n\big] = \sigma^2(1 + (n-1)\rho);$$

$$\text{on the complement}: \quad \lambda = \sigma^2(1-\rho), \text{ multiplicity } n-1.$$

With $n = 100$, $\rho = 0.3$: $\lambda_1 = \sigma^2(1 + 29.7) = 30.7\,\sigma^2$ and $\lambda = 0.7\,\sigma^2$ (99 times).

(b) PC1 is the equal-weight direction $\mathbf{1}/\sqrt{n}$ — long everything, equally. Explained fraction $= \lambda_1 / \operatorname{tr}\Sigma = \sigma^2(1 + (n-1)\rho)/(n\sigma^2) = \frac{1+99(0.3)}{100} = 30.7\%$. Note the limit: as $n \to \infty$ the share tends to ρ — with $\rho = 0.3$, roughly a third of all variance in a large equicorrelated book is one single direction.

(c) PC1 has same-sign (here equal) weights on every asset: it *is* the broad-market portfolio, and any average pairwise correlation $\rho > 0$ makes it the dominant eigenvalue. A signal S with $\langle S, \mathbf{1} \rangle \neq 0$ (net long or net short) has a component along PC1, so a large slice of its P&L variance is market beta rather than selection — which is exactly why one beta-neutralises: project S off the market mode (next subsection's recipe) before sizing.

Self-check: the eigenvalues must sum to the trace: $\sigma^2(1 + (n-1)\rho) + (n-1)\sigma^2(1-\rho) = \sigma^2[1 + (n-1)\rho + (n-1) - (n-1)\rho] = n\sigma^2 = \operatorname{tr}\Sigma$. Small case $n = 2$, $\sigma^2 = 2$, $\rho = 0.5$: formulas give $\lambda = 2(1.5) = 3$ and $2(0.5) = 1$ — exactly the worked staple $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

Answer: (a) $\lambda_1 = \sigma^2(1 + (n-1)\rho)$ once (eigenvector $\mathbf{1}$), $\lambda = \sigma^2(1-\rho)$ with multiplicity $n-1$; (b) PC1 = equal weights, explaining $\frac{1+(n-1)\rho}{n} = 30.7\%$; (c) same-sign dominant eigenvector = market mode; net-directional signals carry market beta and must be projected off it.

P6.7 *PCA via the SVD / conditioning*

(a) $X^\top X = (USV^\top)^\top(USV^\top) = VSU^\top USV^\top = VS^2V^\top$ (using $U^\top U = I$). So the eigenvectors of the scatter matrix $X^\top X$ are the columns of V with eigenvalues s_i^2. The sample covariance is $C = X^\top X/(n-1) = V\frac{S^2}{n-1}V^\top$, hence the principal directions are the columns of V and the sample variance along v_i is $s_i^2/(n-1)$. (Under the $1/n$ convention it is s_i^2/n — pick one and stay consistent.)

(b) $n - 1 = 4$: variances $16/4 = 4$ and $4/4 = 1$. PC1 share $= \frac{4}{4+1} = 80\%$ (equivalently $s_1^2/(s_1^2 + s_2^2) = 16/20$ — the $(n-1)$ cancels in the ratio).

(c) Call the SVD of X. Forming $X^\top X$ explicitly *squares the condition number* ($\kappa(X^\top X) = \kappa(X)^2$), so directions belonging to small singular values drown in floating-point error, and rounding can even return tiny negative eigenvalues for what should be a PSD matrix. The SVD works on X directly and is backward stable; it also never materialises the $p \times p$ product.

Self-check: total variance two ways: sum of per-PC variances $4 + 1 = 5$ must equal $\mathrm{tr}(X^\top X)/(n-1) = (s_1^2 + s_2^2)/4 = 20/4 = 5$; shares sum to $80\% + 20\% = 100\%$. Dimension check: $X^\top X$ is $p \times p$ and V's columns live in $\mathbb{R}^p$, the space where directions should live.

Answer: (a) directions = columns of V; variance along v_i is $s_i^2/(n-1)$; (b) variances 4 and 1, PC1 explains 80%; (c) the SVD of X — forming $X^\top X$ squares the condition number. □

P6.8 *Gram-Schmidt / orthogonalising correlated signals*

(a) Projection (Gram–Schmidt step):

$$B_\perp = B - \frac{\langle A, B\rangle}{\langle A, A\rangle} A = B - \frac{\mathrm{Cov}(A,B)}{\mathrm{Var}(A)} A = B - 0.8\, A,$$

since both are demeaned with unit variance, so $\mathrm{Cov}(A,B) = \rho = 0.8$ and $\mathrm{Var}(A) = 1$. Then $\mathrm{Cov}(A, B_\perp) = \mathrm{Cov}(A,B) - 0.8\,\mathrm{Var}(A) = 0.8 - 0.8 = 0$, hence zero correlation. (This is exactly the OLS residual of B regressed on A — residuals are orthogonal to regressors.)

(b) $\mathrm{Var}(B_\perp) = \mathrm{Var}(B) - 2(0.8)\,\mathrm{Cov}(A,B) + 0.8^2\,\mathrm{Var}(A) = 1 - 1.28 + 0.64 = 0.36 = 1 - \rho^2$. So $\rho^2 = 64\%$ of B's variance was already in A — only about a third of B is new.

(c) Zero raw inner product equals zero covariance only when the means are zero: $\mathrm{Cov}(A,B) = \mathbb{E}[AB] - \mathbb{E}[A]\mathbb{E}[B]$, so with nonzero means two orthogonal vectors can be correlated and vice versa. Demean first, or equivalently include an intercept in the regression (residuals are then orthogonal to the constant column too).

(d) $B_\perp$ is the *incremental information* in B net of what A already trades. It is the right object when A is a frozen black box you must not double-count against: the combined book $A + B_\perp$ has additive variances (no covariance term), so risk aggregates cleanly and B gets credit only for what is genuinely new. If instead you were free to re-optimise the weights of A and B jointly, feeding raw B into the combiner is equivalent — orthogonalisation earns its keep precisely when the incumbent is untouchable.

Self-check: regression route: regressing unit-variance B on A has $R^2 = \rho^2 = 0.64$, so the residual variance is $1 - R^2 = 0.36$ — matches the direct expansion. Boundary sanity: $\rho = 0$ gives $B_\perp = B$ (nothing removed), $\rho = 1$ gives $\mathrm{Var}(B_\perp) = 0$ (nothing left).

Answer: (a) $B_\perp = B - 0.8A$, uncorrelated with A by construction; (b) $\mathrm{Var}(B_\perp) = 1 - \rho^2 = 0.36$; 64% of B was already in A; (c) orthogonality = zero correlation only for demeaned signals; (d) $B_\perp$ is B's incremental information — trade it when the overlapping incumbent is frozen.

P6.9 *Beta neutralisation via PC1*

(a) $\langle S, M\rangle = 2 - 1 + 1 = 2$ and $\langle M, M\rangle = 3$, so

$$S' = S - \tfrac{2}{3}(1,1,1)^\top = \left(\tfrac{4}{3}, -\tfrac{5}{3}, \tfrac{1}{3}\right)^\top.$$

(b) $\langle S', M\rangle = \frac{4}{3} - \frac{5}{3} + \frac{1}{3} = 0$. The weights sum to zero: the position is long–short balanced against the equal-weight market portfolio, so a broad move that lifts all three assets equally produces zero P&L to first order — the signal no longer loads on the dominant risk direction PC1. It is beta-neutral *by construction*, not by regression estimate.

(c) With $M = \mathbf{1}$ the projection term is $\frac{\langle S, \mathbf{1}\rangle}{n}\mathbf{1} = \bar{S}\,\mathbf{1}$: subtracting it is exactly *cross-sectional demeaning* of the signal. A real estimated PC1 only approximately has equal weights — high-beta names carry larger loadings — so projecting off the true PC1 removes a *beta-weighted* average rather than the plain average. Demeaning is the cheap first cut; PC1 projection is the refinement

that respects each name's actual market loading. (Caveat worth saying aloud: in a mixed universe or a stressed regime, PC1 need not be a clean "market" — inspect the loadings' signs before calling it that.)

Self-check: Pythagoras — projection splits norms: $\|S\|^2 = 4+1+1 = 6$ should equal $\|S'\|^2 + \frac{\langle S,M\rangle^2}{\|M\|^2} = (\frac{16}{9}+\frac{25}{9}+\frac{1}{9}) + \frac{4}{3} = \frac{42}{9}+\frac{12}{9} = \frac{54}{9} = 6$. And the demeaning view: $\bar{S} = 2/3$, so $S' = (2,-1,1)^\top - \frac{2}{3}(1,1,1)^\top$ — the same vector.

Answer: (a) $S' = (4/3, -5/3, 1/3)^\top$; (b) $\langle S', M\rangle = 0$ and the weights sum to zero — market-neutral by construction; (c) with equal-weight M the projection is exactly cross-sectional demeaning; an estimated PC1 instead removes a beta-weighted average.

P6.10 *A-orthogonal matrices* ♦

First translate the definition into one matrix equation: $\langle Qx, Qy\rangle_A = x^\top(Q^\top AQ)y$ equals $x^\top Ay$ for *all* x, y iff

$$\boxed{Q^\top AQ = A}$$

(ordinary orthogonality is the $A = I$ case). Now test each candidate against $A = \mathrm{diag}(2,3,5)$:

- Swap $e_1 \leftrightarrow e_3$: $Q^\top AQ = \mathrm{diag}(5,3,2) \neq A$. A permutation *reorders* the weights — it can only work if the swapped weights are equal.
- $Q = \mathrm{diag}(1,-1,1)$: $Q^\top AQ = \mathrm{diag}(1^2\cdot 2,\ (-1)^2\cdot 3,\ 1^2\cdot 5) = A$. Sign flips square away — axis reflections are always A-orthogonal for diagonal A.
- Swap $e_1 \leftrightarrow e_2$: $\mathrm{diag}(3,2,5) \neq A$.
- Rotation in the x–y plane: the upper 2×2 block of $Q^\top AQ$ is $R^\top \mathrm{diag}(2,3)R$, which for generic t has off-diagonal entry $\cos t \sin t \neq 0$ (and even at $t = \pi/2$ it becomes $\mathrm{diag}(3,2)$). Rotations *mix* unequal weights — they preserve $\langle\cdot,\cdot\rangle_A$ only on blocks where A is a multiple of the identity.

The concept in one line: A-orthogonality is congruence-invariance of a weighted geometry; with distinct diagonal weights, the only surviving axis symmetries are the sign flips. □

Answer: (B) $Q = \mathrm{diag}(1,-1,1)$.

28.7 Solutions — §7 Calculus for Quant Interviews

P7.1 *Derivatives toolkit: power rule* ♦

Write $\sqrt{x} = x^{1/2}$ and apply the power rule $(x^n)' = nx^{n-1}$:

$$\frac{d}{dx}\,x^{1/2} \;=\; \tfrac{1}{2}\,x^{-1/2} \;=\; \frac{1}{2\sqrt{x}}.$$

Self-check (independent route, implicit differentiation): let $y = \sqrt{x}$, so $y^2 = x$. Differentiating both sides: $2y\,y' = 1$, hence $y' = 1/(2y) = 1/(2\sqrt{x})$ — same result. *Numerical spot-check:* at $x = 4$, the difference quotient $(\sqrt{4.01} - \sqrt{4})/0.01 \approx (2.00250 - 2)/0.01 = 0.250$, matching $1/(2\cdot 2) = 0.25$.

Answer: $\dfrac{1}{2\sqrt{x}}$.

P7.2 *Derivatives toolkit: chain-rule cycle* ♦

Each differentiation applies the chain rule (a factor of 4) and advances the trig cycle $\sin \to \cos \to -\sin \to -\cos \to \sin$:

$$4\cos(4x) \;\to\; -16\sin(4x) \;\to\; -64\cos(4x) \;\to\; 256\sin(4x).$$

Sine returns to itself after exactly four derivatives, and the chain rule contributes 4 each time, so the fourth derivative is $4^4\sin(4x) = 256\sin(4x)$. *Self-check (closed form):* $\frac{d^n}{dx^n}\sin(kx) =$

$k^n \sin\!\big(kx + \frac{n\pi}{2}\big)$; with $n = 4$, $k = 4$ this gives $4^4 \sin(4x + 2\pi) = 256 \sin(4x)$ — the phase shift of a full period confirms we are back at sine. Sign sanity: the two minus signs picked up at steps 2 and 3 cancel as $(-1)^2$, so the positive coefficient is right. **Answer:** $256 \sin(4x)$.

P7.3 *Derivatives toolkit: logistic identity*

(a) Differentiate $p(x) = (1 + e^{-x})^{-1}$ by the chain rule:

$$p'(x) = \frac{e^{-x}}{(1+e^{-x})^2} = \underbrace{\frac{1}{1+e^{-x}}}_{p} \cdot \underbrace{\frac{e^{-x}}{1+e^{-x}}}_{1-p} = p(1-p),$$

using $1 - p = \dfrac{e^{-x}}{1+e^{-x}}$. **(b)** At $p = 0.80$: sensitivity $= p(1-p) = 0.8 \times 0.2 = 0.16$, i.e. about 16 cents of price move per unit of log-odds ($\Delta p \approx 0.16\,\Delta x$ for small Δx). **(c)** $p(1-p)$ is a downward parabola in p, maximised at $p = \frac{1}{2}$ with value $\frac{1}{4}$. A 50-cent contract moves the most per unit of news; contracts near 0 or 1 barely react. Practically: the same headline should reprice mid-priced contracts far more than long shots, which matters for sizing, for deciding where quoted markets are most at risk of going stale, and it is the exact analogue of an at-the-money digital having the most concentrated sensitivity. *Self-check (numerical):* $p = 0.8$ corresponds to $x = \ln 4 \approx 1.386$. At $x = 1.486$: $e^{-1.486} \approx 0.25 \times 0.905 = 0.226$, so $p \approx 1/1.226 = 0.8155$, giving slope $\approx 0.0155/0.1 = 0.155$ — consistent with 0.16 (slightly below because $p(1-p)$ decreases as p rises past 0.8). Bound sanity: $p(1-p) \le \frac{1}{4}$ always, and $0.16 < 0.25$. **Answer:** (a) $p' = p(1-p)$; (b) 0.16 per unit of log-odds; (c) at $p = \frac{1}{2}$ (sensitivity $\frac{1}{4}$), where a contract reprices most per unit of news.

P7.4 *Derivatives toolkit: optimisation recipe*

Step 1: critical point. By the product rule,

$$f'(s) = e^{-s/2} + s \cdot \big(-\tfrac{1}{2}\big) e^{-s/2} = e^{-s/2}\big(1 - \tfrac{s}{2}\big),$$

which vanishes iff $s^* = 2$ (the exponential never vanishes). **Step 2: justify the maximum** (the justification sentence is graded). Since $e^{-s/2} > 0$, the sign of f' is the sign of $1 - s/2$: positive on $(0, 2)$, negative on $(2, \infty)$, so f rises then falls — a unique interior maximum. Equivalently $f''(s) = e^{-s/2}(\frac{s}{4} - 1)$ gives $f''(2) = -\frac{1}{2e} < 0$; and the boundary behaviour $f(0) = 0$, $f(s) \to 0$ as $s \to \infty$ with a single interior critical point forces a maximum there. **Step 3: report both s^* and $f(s^*)$.** $f(2) = 2e^{-1} = 2/e \approx 0.736$ ticks per quote. *Self-check (bracketing):* $f(1) = e^{-1/2} \approx 0.607$, $f(2) \approx 0.736$, $f(3) = 3e^{-3/2} \approx 0.669$ — the middle value is the largest of the three, consistent with a peak at $s = 2$. Trading sanity: quoting too tight ($s \to 0$) earns nothing per fill, quoting too wide never fills; an interior optimum is exactly what the economics demands. **Answer:** $s^* = 2$ ticks, with maximum expected profit $f(2) = 2/e \approx 0.736$ ticks per quote.

P7.5 *Integration by parts* ♦

Integration by parts with $u = \ln x$ (by LIATE the log is always u: it dies when differentiated) and $dv = dx$, so $du = \frac{1}{x}dx$, $v = x$:

$$\int \ln x\, dx = x \ln x - \int x \cdot \frac{1}{x}\, dx = x \ln x - \int 1\, dx = x \ln x - x + C.$$

Self-check (differentiate back — ten seconds, catches most sign errors): $\frac{d}{dx}(x \ln x - x) = \ln x + x \cdot \frac{1}{x} - 1 = \ln x + 1 - 1 = \ln x$. *Definite sanity:* $\int_1^e \ln x\, dx = (e \cdot 1 - e) - (1 \cdot 0 - 1) = 1$; plausible, since the integrand rises from 0 to 1 over an interval of length $e - 1 \approx 1.72$, giving an average value $\approx 0.58 \in (0, 1)$. **Answer:** $x \ln x - x + C$.

P7.6 *Integration: substitution to Gamma + symmetry caveat*

(a) The screaming tell is a factor $x\,dx$ next to functions of x^2: substitute $u = x^2$, $du = 2x\,dx$, transforming the limits immediately ($0 \to 0$, $\infty \to \infty$). Since $x^3 dx = x^2 \cdot x\,dx = u \cdot \frac{du}{2}$,

$$\int_0^\infty x^3 e^{-x^2} dx = \frac{1}{2}\int_0^\infty u\, e^{-u} du = \frac{1}{2}\,\Gamma(2) = \frac{1!}{2} = \frac{1}{2},$$

using the named Gamma result $\int_0^\infty u^n e^{-u} du = n!$. **(b)** The integrand is odd ($(-x)^3 e^{-(-x)^2} = -x^3 e^{-x^2}$) over a symmetric interval, so the integral is 0 — and the argument is legitimate because the integrand is *absolutely integrable*: by (a), $\int_{-\infty}^{\infty} |x^3 e^{-x^2}| dx = 2 \times \frac{1}{2} = 1 < \infty$, so each half-line contributes a finite value and the two cancel exactly. For $\int_{-\infty}^{\infty} x\,dx$ both halves diverge, so the improper integral simply does not exist; only the symmetric principal value $\lim_{R\to\infty} \int_{-R}^{R} x\,dx$ is 0, which is a weaker statement — "odd so zero" needs the absolute-integrability caveat said aloud. *Self-check for (a) (differentiate back):* the antiderivative $F(x) = -\frac{1}{2}(x^2+1)e^{-x^2}$ satisfies $F'(x) = -\frac{1}{2}\big[2x\,e^{-x^2} + (x^2+1)(-2x)e^{-x^2}\big] = xe^{-x^2}(x^2+1-1) = x^3 e^{-x^2}$, and $F(\infty) - F(0) = 0 - (-\frac{1}{2}) = \frac{1}{2}$. Positivity sanity: the integrand on $(0,\infty)$ is strictly positive, and $\frac{1}{2} > 0$. **Answer:** (a) $\frac{1}{2}$; (b) 0, valid because the integrand is odd *and* absolutely integrable — for $\int x\,dx$ both halves diverge, so the integral is undefined (only its principal value is 0). □

P7.7 *Improper integrals: p-tests, comparison, log refinement*

(a) Converges. Tail at ∞ with $p = \frac{3}{2} > 1$. (Explicitly: antiderivative $-2x^{-1/2}$ gives value 2.) **(b) Converges.** The only problem point is the finite endpoint $x = 1$; substituting $u = 1 - x$ gives $\int_0^1 u^{-1/2} du$, a finite-endpoint blow-up with $p = \frac{1}{2} < 1$. (Value: 2.) **(c) Diverges.** Log refinement: $u = \ln x$ turns it into $\int_{\ln 2}^{\infty} \frac{du}{u}$, the $p = 1$ boundary case — log divergence. **(d) Diverges.** Comparison: $2 + \sin x \geq 1$, so the integrand is $\geq \frac{1}{x} \geq 0$, and $\int_1^\infty \frac{dx}{x}$ diverges. (Note the integrand $\to 0$; that proves nothing — the question is how fast.) **(e) Converges.** Exponentials beat polynomials: any $\int^\infty x^k e^{-cx} dx$ with $c > 0$ converges. (Value via $u = x/10$: $10^8 \int_0^\infty u^7 e^{-u} du = 10^8 \cdot 7!$.) *Self-check:* every verdict is backed by an explicit antiderivative or an explicit bound, not just a rule name; the two p-test convergents (a), (b) produce positive finite values; and the boundary logic is consistent — at ∞ we needed fast decay ($p > 1$: (a) passes, (c) sits exactly on the failing boundary), at a finite endpoint we can afford a mild blow-up ($p < 1$: (b) passes). $1/x$ fails at both ends, which is exactly why (c) and (d) both reduce to it and diverge. **Answer:** (a) converges (= 2); (b) converges (= 2); (c) diverges (log); (d) diverges (compare $1/x$); (e) converges ($= 10^8 \cdot 7!$).

P7.8 *Improper integrals: interior singularity*

Two independent red flags, either of which should stop you:

- **Sign:** the integrand $1/x^2$ is strictly positive, so no negative answer is possible — a positive integrand can never produce a negative or zero integral.
- **Illegal FTC:** the Fundamental Theorem of Calculus requires an antiderivative valid on the whole interval, but $1/x^2$ has an *interior* singularity at $x = 0$; $-1/x$ is not an antiderivative across it. Never apply FTC across an interior singularity — find every problem point and split there.

Correct procedure: split at 0 and require *every* piece to converge. Near the finite endpoint 0, $\int_0^1 x^{-2} dx$ has $p = 2 \geq 1$, which diverges; explicitly $\int_\varepsilon^1 x^{-2} dx = \frac{1}{\varepsilon} - 1 \to \infty$. The left half diverges identically by symmetry. Hence the integral diverges (to $+\infty$). *Self-check (bound):* on $[-1,1]$ we have $|x| \leq 1$, so $1/x^2 \geq 1$; even if the integral existed it would be ≥ 2, flatly contradicting -2 — and the explicit ε-computation confirms $+\infty$. **Answer:** The FTC was applied across the interior singularity at $x = 0$; splitting there shows each half diverges ($p = 2$ at a finite endpoint), so the integral diverges to $+\infty$ — and the negative sign was the instant giveaway.

P7.9 *Improper integrals: combined p-tests via limit comparison*

Problem points: $x = 0$ (possible blow-up from x^{-p} when $p > 0$) and $x = \infty$. Split at any interior point, say $x = 1$, and require both pieces to converge. **Near 0:** $1 + x \to 1$, so limit-compare with x^{-p}: the ratio $\frac{1/(x^p(1+x))}{1/x^p} = \frac{1}{1+x} \to 1 \in (0, \infty)$, so $\int_0^1$ behaves like $\int_0^1 x^{-p}dx$, which converges iff $p < 1$ (finite-endpoint p-test: mild blow-up allowed). **Near ∞:** $x^p(1 + x) \sim x^{p+1}$, and the ratio against $x^{-(p+1)}$ tends to 1, so $\int_1^\infty$ behaves like $\int_1^\infty x^{-(p+1)}dx$, which converges iff $p + 1 > 1$, i.e. $p > 0$ (tail p-test: need fast decay). Both conditions must hold simultaneously, so the range of convergence is $0 < p < 1$. **Boundary sanity:** $p = 0$: $\int_0^\infty \frac{dx}{1+x} = \ln(1 + x)\big|_0^\infty$ diverges (log at ∞) excluded. $p = 1$: near 0 the integrand $\sim \frac{1}{x}$, log-divergent excluded. As always, whenever an effective $p = 1$ appears, the verdict is log divergence. *Self-check (explicit interior value):* at $p = \frac{1}{2}$, substitute $u = \sqrt{x}$ ($x = u^2$, $dx = 2u\,du$):

$$\int_0^\infty \frac{2u\,du}{u\,(1 + u^2)} = 2\int_0^\infty \frac{du}{1 + u^2} = 2 \cdot \frac{\pi}{2} = \pi,$$

finite — confirming convergence strictly inside the claimed range. **Answer:** $0 < p < 1$. □

P7.10 *Moments as integrals: normalisation via Gamma*

Normalisation questions are improper-integral evaluations in probability clothing. The one workhorse needed: substituting $u = x/\theta$,

$$\int_0^\infty x^n e^{-x/\theta}dx = \theta^{n+1}\int_0^\infty u^n e^{-u}du = \theta^{n+1}\,n!.$$

Find c: with $\theta = 2$, $n = 1$: $\int_0^\infty xe^{-x/2}dx = 2^2 \cdot 1! = 4$, so $c = \frac{1}{4}$. **Mean (LOTUS):** $\mathbb{E}[X] = \frac{1}{4}\int_0^\infty x^2e^{-x/2}dx = \frac{1}{4} \cdot 2^3 \cdot 2! = \frac{16}{4} = 4$. **Second moment and variance:** $\mathbb{E}[X^2] = \frac{1}{4}\int_0^\infty x^3e^{-x/2}dx = \frac{1}{4} \cdot 2^4 \cdot 3! = \frac{96}{4} = 24$, hence $\mathrm{Var}(X) = 24 - 4^2 = 8$. *Self-check (recognise the distribution):* f is the Gamma density with shape 2 and scale 2 — the sum of two independent exponentials, each with mean 2 (rate $\lambda = \frac{1}{2}$, moments $n!/\lambda^n$). So the mean is $2 + 2 = 4$ and the variance is $4 + 4 = 8$ (each exponential has variance $1/\lambda^2 = 4$). Scale sanity: standard deviation $\sqrt{8} \approx 2.8$, the same order as the mean, as expected for a mildly right-skewed waiting-time distribution. **Answer:** $c = \frac{1}{4}$, $\mathbb{E}[X] = 4$, $\mathrm{Var}(X) = 8$. □

P7.11 *Moments as integrals: Gaussian fourth moment*

(a) Standardise: $Z = X/\sigma$, so $\mathbb{E}[X^4] = \sigma^4\,\mathbb{E}[Z^4]$ and we need $\int_{-\infty}^\infty z^4\varphi(z)\,dz$ with $\varphi(z) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}$. The key structural fact is $\varphi'(z) = -z\,\varphi(z)$, so $z\varphi(z)\,dz$ integrates to $-\varphi(z)$. Integrate by parts with $u = z^3$, $dv = z\,\varphi(z)\,dz$, $v = -\varphi(z)$:

$$\int_{-\infty}^\infty z^4\varphi(z)\,dz = \Big[-z^3\varphi(z)\Big]_{-\infty}^\infty + 3\int_{-\infty}^\infty z^2\varphi(z)\,dz = 0 + 3 \cdot 1 = 3,$$

since the boundary term dies (exponentials beat polynomials) and $\int z^2\varphi = \mathrm{Var}(Z) = 1$. Hence $\mathbb{E}[X^4] = 3\sigma^4$: Gaussian kurtosis is exactly 3. **(b)** A ratio of 9 against a Gaussian's 3 means the return distribution has *fat tails* (excess kurtosis 6): extreme moves are far more common than the model admits. One practical consequence (any one suffices): σ-based limits or VaR calibrated on the Normal will be breached much more often than advertised — under the Normal a 4σ day has probability $\approx 6 \times 10^{-5}$ (roughly once in 60 years of trading days), yet fat-tailed markets deliver them routinely. Standard fixes: heavier-tailed models (Student-t), or recognising volatility clustering — a mixture of Gaussians with time-varying σ has kurtosis above 3 even if each regime is Gaussian, so conditioning on the current vol regime removes much of the excess. *Self-check:* the same parts argument with general even power gives the recursion $\mathbb{E}[Z^{2m}] = (2m-1)\,\mathbb{E}[Z^{2m-2}]$, i.e. the double factorial $1, 3, 15, \ldots$; at $m = 1$ it returns $\mathbb{E}[Z^2] = 1 \cdot 1 = 1$, and at $m = 2$ it gives $3 \cdot 1 = 3$, agreeing with the direct computation. Symmetry sanity: all odd moments vanish, so kurtosis is the first shape parameter beyond variance — exactly why desks track it. **Answer:** (a) $\mathbb{E}[X^4] = 3\sigma^4$

(kurtosis 3); (b) sample kurtosis ≈ 9 means fat tails / vol clustering — Normal-calibrated risk limits materially understate the frequency of extreme moves. □

P7.12 *Taylor micro-kit: limit*

Taylor first (faster, shows more): $e^x = 1 + x + \frac{x^2}{2} + O(x^3)$, so the numerator is $\frac{x^2}{2} + O(x^3)$ and

$$\frac{e^x - 1 - x}{x^2} = \frac{1}{2} + O(x) \;\longrightarrow\; \frac{1}{2}.$$

Self-check 1 (L'Hopital, checking the form each time): at $x = 0$ the expression is $\frac{0}{0}$, so differentiate: $\frac{e^x - 1}{2x}$, again $\frac{0}{0}$; differentiate once more: $\frac{e^x}{2} \to \frac{1}{2}$. *Self-check 2 (numerical):* at $x = 0.1$: $(1.105171 - 1.1)/0.01 = 0.5171$; the next Taylor term predicts $\frac{1}{2} + \frac{x}{6} = 0.5167$ — the numerics match the expansion to the next order, not just the limit. **Answer:** $\frac{1}{2}$.

P7.13 *Taylor micro-kit: compounding estimate*

Route: through the exponential. $(1.02)^{30} = e^{30\ln(1.02)}$. The limit $(1 + \frac{a}{m})^m \to e^a$ with $a = 0.6$ and $m = 30$ gives the crude estimate $e^{0.6} \approx 1.822$. Refine with the second-order log expansion $\ln(1 + x) \approx x - \frac{x^2}{2}$:

$$\ln(1.02) \approx 0.02 - 0.0002 = 0.0198, \qquad 30 \times 0.0198 = 0.594,$$

$$e^{0.594} = e^{0.6} \cdot e^{-0.006} \approx 1.822 \times 0.994 \approx 1.811.$$

So $(1.02)^{30} \approx 1.81$ — the second-order term pulls the naive $e^{0.6}$ down by about 1%. *Self-check 1 (rule of 72):* at 2% per period, doubling takes about $72/2 = 36$ periods; after only 30 periods we should sit noticeably below 2.0 — and 1.81 does. *Self-check 2 (convexity bound):* compounding must beat the linear estimate $1 + 30 \times 0.02 = 1.60$, and $1.81 > 1.60$. The true value is 1.8114, so the two-term estimate is accurate to about 0.02% — the pattern to remember: convert powers to $e^{m\ln(1+x)}$ and keep two terms of the log. **Answer:** $(1.02)^{30} \approx 1.81$ (true value 1.8114).

P7.14 *Change of variables in an ODE* ♦

Pure chain rule. From $\tau = 1/t$ we get $t = 1/\tau$ and

$$\frac{dt}{d\tau} = -\frac{1}{\tau^2}, \qquad \frac{dx}{d\tau} = \frac{dx}{dt} \cdot \frac{dt}{d\tau} = x \cdot \left(-\frac{1}{\tau^2}\right) = -\frac{x}{\tau^2}.$$

Check by explicit solution: $x(t) = Ce^t = Ce^{1/\tau}$, so $\dfrac{dx}{d\tau} = Ce^{1/\tau} \cdot \left(-\frac{1}{\tau^2}\right) = -x/\tau^2$ — matches.

Answer: (C) $dx/d\tau = -x/\tau^2$.

P7.15 *Differentiate a parameter-dependent integral* ♦

The fastest route: x is a constant with respect to s, so evaluate the integral in closed form first, then differentiate:

$$F(x) = x \int_x^{2x} s\, ds = x \cdot \frac{(2x)^2 - x^2}{2} = x \cdot \frac{3x^2}{2} = \frac{3}{2}x^3 \qquad \Rightarrow \qquad F'(x) = \frac{9}{2}x^2.$$

Cross-check with the full Leibniz rule (limits and integrand both depend on x):

$$F'(x) = \underbrace{2 \cdot (2x)x}_{\text{upper limit}} - \underbrace{1 \cdot x \cdot x}_{\text{lower limit}} + \underbrace{\int_x^{2x} s\, ds}_{\partial/\partial x \text{ of integrand}} = 4x^2 - x^2 + \frac{3x^2}{2} = \frac{9}{2}x^2.$$

Both routes agree — and under a clock, evaluate-then-differentiate is the safer one whenever the integral is elementary.

Answer: (B) $\frac{9}{2}x^2$.

P7.16 *Probability by integration* ♦

Condition on X: given $X = x$, $P(Y < x^2) = x^2$ (uniform CDF). Averaging over X,

$$P(X^2 > Y) = \mathbb{E}[P(Y < X^2 \mid X)] = \mathbb{E}[X^2] = \int_0^1 x^2\,dx = \frac{1}{3}.$$

Equivalently: the probability is the area under the parabola $y = x^2$ inside the unit square.

Sanity check: $P(X^2 > Y) < P(X > Y) = \frac{1}{2}$ since $X^2 < X$ on $(0, 1)$ — and $\frac{1}{3} < \frac{1}{2}$. The trap options bait a symmetry guess (1/2) or a quarter-circle association ($\pi/4$, which belongs to $P(X^2 + Y^2 < 1)$, a different question).

Answer: (C) 1/3.

P7.17 *Newton's method: the slowest root* ♦

Find the roots. $\tanh(u) = 0$ iff $u = 0$, so the roots solve $x^4 - x^2 = x^2(x^2 - 1) = 0$: $x \in \{-1, 0, 1\}$.

Determine multiplicities. Near $x = 0$: $x^4 - x^2 \approx -x^2$ and $\tanh u \approx u$, so $f(x) \approx -x^2$ — a root of *multiplicity 2*. At $x = \pm 1$: $(x^4 - x^2)' = 4x^3 - 2x = \pm 2 \neq 0$, so both are *simple* roots.

Convergence rates. Newton is quadratic at simple roots, but only *linear* at a root of multiplicity m, with error ratio $(m - 1)/m$ per step. Derivation for the model case $f = (x - r)^m$:

$$x_{k+1} - r = x_k - r - \frac{(x_k - r)^m}{m(x_k - r)^{m-1}} = (x_k - r)\left(1 - \frac{1}{m}\right),$$

so with $m = 2$ the error only halves each iteration — linear, ratio $\frac{1}{2}$ — while at ± 1 the digits double per step. The slowest root is $x = 0$: the one with the smallest *absolute value*.

The distractor: "the smallest one" is -1 — a simple root with quadratic convergence. The option pair (smallest vs smallest absolute value) is engineered to punish reading fast. And (B) is the textbook overstatement: quadratic convergence *requires* a simple root ($f'(r) \neq 0$).

Bonus (worth saying in an interview): the fix at a known multiplicity is the modified step $x_{k+1} = x_k - m\,f/f'$, which restores quadratic convergence. □

Answer: (D) the root with the smallest absolute value ($x = 0$, multiplicity 2, linear convergence).

28.8 Solutions — §8 ML Theory: Trees, Ensembles, Bias–Variance

P8.1 *Bias-variance decomposition*

Drop the argument x_0. Insert and subtract f and $\mathbb{E}_D[\hat{f}]$:

$$y - \hat{f} = \underbrace{\varepsilon}_{\text{test noise}} + \underbrace{\left(f - \mathbb{E}[\hat{f}]\right)}_{\text{a deterministic constant}} + \underbrace{\left(\mathbb{E}[\hat{f}] - \hat{f}\right)}_{\text{mean zero over } D}.$$

Square and take $\mathbb{E}_{D,\varepsilon}$. The squared terms give σ^2, $\left(f - \mathbb{E}[\hat{f}]\right)^2 = \text{Bias}^2$, and $\mathbb{E}\left[(\hat{f} - \mathbb{E}[\hat{f}])^2\right] = \text{Var}(\hat{f})$. Cross terms: (i) both terms containing ε factor as $\mathbb{E}[\varepsilon] \cdot \mathbb{E}[\cdot] = 0$ because the test noise is independent of the training randomness and has mean zero; (ii) the remaining term is $2\left(f - \mathbb{E}[\hat{f}]\right)\mathbb{E}\left[\mathbb{E}[\hat{f}] - \hat{f}\right] = 0$ because the first factor is a constant and the second factor has mean zero by construction. Hence

$$\mathbb{E}[(y - \hat{f})^2] = \left(f - \mathbb{E}[\hat{f}]\right)^2 + \text{Var}(\hat{f}) + \sigma^2.$$

More data, same model class: the *variance* term shrinks (the fitted model concentrates around its own expectation); the *bias* term converges to the approximation error of the model class and does not vanish; σ^2 is irreducible by definition.

Self-check (constant predictor). Take $\hat{f} \equiv c$, ignoring the data. Directly: $\mathbb{E}[(y-c)^2] = \mathbb{E}[(f-c+\varepsilon)^2] = (f-c)^2 + \sigma^2$. The formula: $\text{bias}^2 = (f-c)^2$, variance $= 0$, plus σ^2 — identical.

Answer: $\mathbb{E}[(y-\hat{f})^2] = (f - \mathbb{E}[\hat{f}])^2 + \text{Var}(\hat{f}) + \sigma^2$; the ε cross terms die by independence and $\mathbb{E}[\varepsilon] = 0$, the third because $\hat{f} - \mathbb{E}[\hat{f}]$ has mean zero against a constant. More data shrinks only the variance term. □

P8.2 *Decision trees: Gini and information gain*

(a) Parent: $G = 1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = \frac{1}{2}$. Each child has proportions $\left(\frac{3}{4}, \frac{1}{4}\right)$: $G_{\text{child}} = 1 - \frac{9}{16} - \frac{1}{16} = \frac{6}{16} = \frac{3}{8}$. Both children have 8 of the 16 samples, so the weighted Gini is $\frac{1}{2} \cdot \frac{3}{8} + \frac{1}{2} \cdot \frac{3}{8} = \frac{3}{8} = 0.375$, an impurity decrease of $\frac{1}{2} - \frac{3}{8} = \frac{1}{8} = 0.125$.

(b) Parent entropy = 1 bit (a fair 50/50). Child entropy $= -\frac{3}{4}\log_2\frac{3}{4} - \frac{1}{4}\log_2\frac{1}{4} = \frac{3}{4}(2 - \log_2 3) + \frac{1}{4}\cdot 2 = 0.75 \times 0.415 + 0.5 \approx 0.811$ bits. Information gain $= 1 - 0.811 \approx 0.189$ bits.

(c) None of them. A tree splits on thresholds, i.e. on the *rank order* of the feature; a strictly monotone transform preserves exactly which partitions of the samples are achievable by a threshold, so the same split (at the transformed threshold) produces the same child sets and the same impurity numbers. This is why scaling and monotone transforms are wasted effort for tree models.

Self-check. Bounds: binary Gini lies in $[0, \frac{1}{2}]$ and binary entropy in $[0, 1]$ bit — all computed values respect them; both children are mirror images so the weighted value must equal the single-child value, which it does; and $\log_2 \frac{4}{3} = 2 - \log_2 3 = 2 - 1.585 = 0.415$ confirms the entropy arithmetic.

Answer: Gini $0.5 \to 0.375$, decrease $= 0.125$; information gain ≈ 0.189 bits; nothing changes under a monotone feature transform.

P8.3 *Trees cannot extrapolate*

A tree partitions feature space into rectangles and predicts a *constant* on each leaf. VIX = 80 falls into the open-ended rightmost cell, whose rule is roughly "VIX > (largest threshold learned, high 20s)", so the model outputs approximately the average target of the highest-VIX training days — about what it predicts for VIX ≈ 30. The prediction function is flat beyond the training range: trees (and sums of trees) *cannot extrapolate*, so realised vol is severely underpredicted exactly when it matters. A linear regression extrapolates along its fitted slope — directionally more sensible here, though extrapolation reliability is its own risk.

Fixes: (1) make the feature *stationary/relative* so extreme regimes map into the seen range — ratios (VIX over its trailing median), log-changes, rolling z-scores or ranks instead of raw levels; (2) add explicit interaction/spread features or hybridise with a linear component (fit trees on the residuals of a linear baseline, or linear-in-leaf variants such as LightGBM's `linear_tree`). A monotonic constraint ("higher VIX ⇒ higher predicted vol") is a further regularising guard: it cannot restore the missing slope but guarantees the saturation is at least monotone.

Self-check (degenerate case). A depth-0 tree predicts the global training mean for every input — the "constant beyond the training range" behaviour is just the general form of this, so the mechanism is right. And the direction matches reality: March 2020 realised vol exceeded anything in 2015–2019, so a saturated forecast is an underprediction, as claimed.

Answer: the model outputs roughly its VIX-30 prediction (top-leaf mean) because tree predictions are piecewise constant and cannot extrapolate; a linear model would extend along its slope; fixes: stationary/ratio/z-scored features and explicit interactions or linear-hybrid designs (plus monotonic constraints).

P8.4 *Ensemble variance formula and the correlation floor*

(a) $\text{Var}\left(\sum_i X_i\right) = N\sigma^2 + N(N-1)\rho\sigma^2$ (there are $N(N-1)$ ordered covariance pairs, each $\rho\sigma^2$). Dividing by N^2:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{N} + \frac{N-1}{N}\rho\sigma^2 = \rho\sigma^2 + \frac{1-\rho}{N}\sigma^2.$$

(b) As $N \to \infty$ the second term dies and $\rho\sigma^2$ survives: *correlation between learners is the floor on ensemble variance* — averaging more trees cannot cross it. Bagging alone leaves trees highly correlated (they all chase the same strong features), so random forests additionally sample a random feature subset at every split: this attacks ρ itself, accepting slightly worse individual trees, because lowering the floor beats polishing the $1/N$ term.

(c) Validity requires the equicorrelated covariance matrix to be PSD: $\mathrm{Var}\big(\sum X_i\big) = N\sigma^2[1 + (N-1)\rho] \geq 0$, i.e. $\rho \geq -\frac{1}{N-1}$. At the boundary $\rho = -\frac{1}{N-1}$:

$$\mathrm{Var}(\bar{X}) = \sigma^2\Big[-\frac{1}{N-1} + \frac{1}{N}\Big(1 + \frac{1}{N-1}\Big)\Big] = \sigma^2\Big[-\frac{1}{N-1} + \frac{1}{N}\cdot\frac{N}{N-1}\Big] = 0.$$

Maximally anti-correlated predictors cancel exactly — the diversification ideal (in book terms: negatively correlated strategies are worth more than uncorrelated ones).

(d) Need $\frac{(1-\rho)\sigma^2}{N} \leq 0.05 \iff \frac{0.8}{N} \leq 0.05 \iff N \geq 16$.

Self-check. Boundary cases of (a): $\rho = 1 \Rightarrow \mathrm{Var} = \sigma^2$ for all N (identical models, averaging useless); $\rho = 0 \Rightarrow \sigma^2/N$ (i.i.d. rate); $N = 1 \Rightarrow \sigma^2$. For (d): $N = 16$ gives $0.2 + 0.8/16 = 0.25$, exactly the floor 0.2 plus 0.05.

Answer: (a) $\rho\sigma^2 + \frac{1-\rho}{N}\sigma^2$; (b) limit $\rho\sigma^2$ — the correlation floor, which feature-subsampling attacks directly; (c) $\rho \geq -\frac{1}{N-1}$, where the ensemble variance is exactly 0; (d) $N = 16$. □

P8.5 *Bootstrap and out-of-bag*

(a) Each draw misses the point with probability $1 - \frac{1}{n}$, and draws are independent:

$$P(\text{never drawn}) = \Big(1 - \frac{1}{n}\Big)^n \xrightarrow{n\to\infty} e^{-1} \approx 0.368.$$

(b) So each tree sees on average $1 - e^{-1} \approx 63.2\%$ of the distinct points. (c) The *out-of-bag (OOB) error*: every training point is predicted using only the $\approx 36.8\%$ of trees that never saw it, which yields an honest generalisation estimate — effectively cross-validation for free, no held-out set needed — and drives OOB-based feature importances.

Self-check. Small cases: $n = 2$: $(1/2)^2 = 0.25$; $n = 5$: $(4/5)^5 = 1024/3125 \approx 0.328$; $n = 100 \approx 0.366$ — increasing towards 0.368 from below. Analytically, $\ln(1-1/n)^n = n\ln(1-1/n) \approx n\big(-\frac{1}{n} - \frac{1}{2n^2}\big) = -1 - \frac{1}{2n} \to -1$.

Answer: (a) $(1 - 1/n)^n \to 1/e \approx 36.8\%$; (b) $\approx 63.2\%$; (c) the out-of-bag error estimate (free cross-validation).

P8.6 *Boosting as functional gradient descent*

(a) Treat the model's values $F(x_i)$ at the n training points as free coordinates. Then $\frac{\partial L}{\partial F(x_i)} = -(y_i - F(x_i))$, so the negative gradient at F_{t-1} is the residual vector $y_i - F_{t-1}(x_i)$. The tree h_t is fit to approximate this vector (a projection of the gradient onto the class of trees), and $F_t = F_{t-1} + \eta h_t$ is a gradient step of length η. Residual fitting *is* steepest descent on squared loss.

(b) $-\frac{\partial}{\partial F}|y - F| = \mathrm{sign}(y - F)$: each tree is fit to the *sign* of the residual (LAD boosting; optimal leaf values become medians of the in-leaf residuals) — outlier-robust, exactly the L1-vs-L2 mean/median story.

(c) Boosting attacks *bias*: each round fits what the current ensemble still misses, so shallow high-bias trees suffice. Its overfitting knobs: learning rate η (shrinkage), number of rounds with early stopping, tree depth/leaf count, row/feature subsampling, L1/L2 penalties on leaf weights, minimum child samples. More rounds *do* eventually overfit (training error keeps falling, validation error turns). Bagging attacks *variance* of deep low-bias trees fit in parallel; adding more bagged trees never increases expected test error — the average just converges (to the $\rho\sigma^2$ floor), so "more trees" is safe in a forest but a tuning decision in boosting.

Self-check (one concrete run). Single observation, $y = 10$, $F_0 = 0$, $\eta = 0.1$; each tree can fit its target exactly. Residual recursion: $r_t = (1-\eta)^t \cdot 10$ — geometric decay to zero, exactly gradient descent with step η on a quadratic. After 2 rounds: $F_2 = 1 + 0.9 = 1.9 = 10(1 - 0.81)$. Units also check: the gradient carries units of y, matching what a regression tree predicts.

Answer: (a) $-\partial L/\partial F\big|_{F_{t-1}} = y - F_{t-1}$, the residual; (b) $\operatorname{sign}(y - F)$ with median leaf values; (c) boosting cuts bias (knobs: η, rounds/early stopping, depth, subsampling, leaf penalties) and can overfit with rounds; bagging cuts variance and more trees never hurt. □

P8.7 *Good features for tree models* ★

Organise the answer around what tree splits can and cannot do.

- *Splits are thresholds on rank order.* Monotone transforms and standardisation are irrelevant (the same partitions remain reachable), and robustness to outliers comes free. Do not spend effort scaling; spend it below.
- *Trees cannot extrapolate.* Predictions are constant beyond the training range, so prefer *stationary* constructions over raw levels: returns rather than prices, ratios to rolling baselines, cross-sectional ranks and z-scores. A feature whose live range drifts outside the training range silently saturates.
- *Splits are axis-aligned.* A tree cannot natively express relations *between* columns ($x_1 > x_2$ needs a staircase of splits), so engineer interactions explicitly: ratios, spreads, differences, imbalances (bid-ask imbalance, price over moving average, book-to-price).
- *Missing values are handled natively* (LightGBM learns a default split direction), so avoid distorting imputations.
- *Dangers:* high-cardinality ID-like categoricals are overfit fuel (the tree memorises identities); leakage — construct everything point-in-time; and where you *know* the sign of an effect, monotonic constraints encode it as regularisation.

Self-check (invariance claim on a toy). Samples $x = 1, 10, 100$ with labels $0, 0, 1$: best split "$x > 55$" isolates the third point. After $\log_{10}$: $0, 1, 2$ and split "> 1.5" produces the identical partition with identical impurity — confirming split-invariance to monotone maps. The extrapolation point is the P8.3 mechanism, verified there.

Answer: good tree features are stationary (returns, ratios, ranks/z-scores, not levels), interaction-explicit (ratios/spreads/imbalances, since splits are axis-aligned), and rank-meaningful; scaling and monotone transforms do not matter; handle missing natively; beware high-cardinality IDs and leakage; encode known signs via monotonic constraints.

28.9 Solutions — §9 Deep Learning

P9.1 *Backprop mechanics and vanishing gradients*

(a) Backprop is the chain rule organised back-to-front: the sensitivity $\delta_\ell = \partial a_L/\partial a_\ell$ is computed once per layer and *reused* for every parameter feeding that layer, so the whole gradient costs a small constant times one forward pass (backward $\approx 2\times$ forward flops), independent of parameter count. Finite differences need one perturbed forward pass per parameter: $10^7 \times 10\,\text{ms} = 10^5\,\text{s} \approx 28$ hours per gradient step, versus ~ 30 ms for forward+backward — a factor of about 10^7 (i.e. exactly the parameter count).

(b) With $z_\ell = w_\ell a_{\ell-1}$, the chain rule gives

$$\frac{\partial a_L}{\partial w_1} = \Big[\prod_{\ell=2}^{L} \sigma'(z_\ell)\, w_\ell\Big]\, \sigma'(z_1)\, a_0.$$

The sigmoid satisfies $\sigma' = \sigma(1-\sigma) \le \frac{1}{4}$ (max at $\sigma = \frac{1}{2}$). With $|w_\ell| \le 1$ and $|a_0| \le 1$, the

magnitude is at most $(1/4)^L = 4^{-20} = 2^{-40} \approx 9 \times 10^{-13}$ for $L = 20$: the early layers receive essentially zero signal. Any product of factors bounded below 1 decays exponentially — that is the vanishing-gradient mechanism (factors above 1 give the mirror-image exploding case).

(c) *ReLU:* $\sigma' = 1$ on the active half — the 1/4 shrink factor becomes 1. *Residual connections:* $\frac{\partial}{\partial x}(x + f(x)) = 1 + f'$ — an identity term keeps a product-free gradient path through depth. *LSTM gating:* the cell state updates additively through gates near 1, so the through-time product becomes a sum-like highway. *Batch norm:* keeps pre-activations z in the non-saturated region, so $\sigma'(z)$ is not driven to ≈ 0.

Self-check. $4^{20} = 2^{40} = (2^{10})^4 \approx (10^3)^4 = 10^{12}$, so $4^{-20} \approx 10^{-12}$ (more precisely 9.1×10^{-13}); the finite-difference estimate 10^5 s $= 10^5/3600 \approx 27.8$ h; and the sigmoid bound: $\sigma'(0) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, indeed the maximum of $p(1-p)$.

Answer: (a) backprop reuses per-layer sensitivities — cost $\approx$ a constant times one forward pass, vs ≈ 28 hours per step for finite differences on 10^7 parameters; (b) the gradient is a product of L factors, each at most $\frac{1}{4}$ in magnitude here, so at $L = 20$ it is bounded by $4^{-20} \approx 10^{-12}$; (c) ReLU makes the derivative factor 1, residuals add an identity path, LSTM gates make time propagation additive, batch norm keeps activations unsaturated. □

P9.2 *Self-attention and the sqrt(dk) scaling*

(a) $q \cdot k = \sum_{i=1}^{d_k} q_i k_i$. Each term has mean 0 and variance $\mathbb{E}[q_i^2 k_i^2] = \mathbb{E}[q_i^2]\,\mathbb{E}[k_i^2] = 1$ (independence), and distinct terms are uncorrelated, so $\mathrm{Var}(q \cdot k) = d_k$.

(b) Standard deviation $\sqrt{64} = 8$. Two independent logits $\sim$ scale 8 differ by about $8\sqrt{2} \approx 11$ in a typical draw. Softmax of $[11, 0]$: weight $\frac{1}{1+e^{-11}}$ with $e^{-11} \approx 1.7 \times 10^{-5}$ — effectively one-hot. Attention starts out frozen onto single random tokens.

(c) The softmax Jacobian is $\partial p_i/\partial z_j = p_i(\delta_{ij} - p_j)$. At a one-hot p, every entry is ≈ 0 (either $p_i \approx 0$ or $1 - p_j \approx 0$): the layer is saturated exactly like a sigmoid in its tails, and no gradient flows to the scores — the attention pattern cannot learn its way out.

(d) Dividing by $\sqrt{d_k}$ makes the score variance 1 *independent of head dimension*, so softmax operates in its sensitive regime at initialisation: attention distributions start diffuse and sharpen only as training gives them reason to.

Self-check. $d_k = 1$: formula gives $\mathrm{Var}(q_1 k_1) = 1$, matching the product-of-independents computation directly. The tail number: $e^{-11} = e^{-8}e^{-3} \approx 3.4 \times 10^{-4} \times 5.0 \times 10^{-2} \approx 1.7 \times 10^{-5}$. And the Jacobian check at uniform $p = (\frac{1}{2}, \frac{1}{2})$ gives entries $\pm\frac{1}{4} \neq 0$ — gradients flow when softmax is *not* saturated, isolating saturation as the culprit.

Answer: (a) $\mathrm{Var}(q \cdot k) = d_k$; (b) std $= 8$ at $d_k = 64$; typical logit gaps ≈ 11 make softmax one-hot to within $\sim 2 \times 10^{-5}$; (c) the softmax Jacobian $p_i(\delta_{ij} - p_j)$ vanishes at one-hot outputs — saturated, no learning signal; (d) unit-variance scores for any d_k, keeping softmax un-saturated at init. □

P9.3 *Transformers on tick data*

(a) $L^2 = (5 \times 10^5)^2 = 2.5 \times 10^{11}$ entries; at 2 bytes each that is 5×10^{11} bytes $= 500$ GB *per head per layer* — an order of magnitude beyond an entire A100 (40–80 GB) before counting any other activations. Naive full-day tick attention is not merely slow; it does not fit.

(b) FlashAttention never materialises the $L \times L$ matrix: it streams tiles through on-chip SRAM and recomputes what it needs, cutting attention *memory* to $O(L)$. But every query–key pair is still computed, so *compute remains* $O(L^2)$ — training over years of 500k-tick days stays quadratic and brutal. Say both halves: memory fixed, compute not.

(c) Transformer with a KV cache: appending one tick costs one query against all L cached keys, i.e. $O(L)$ per layer per new tick, and the cache itself grows with the session — cost and memory both scale with the window. SSM/recurrent models: a fixed-size state updated in $O(1)$ per tick,

constant memory, no re-attending — the right computational shape for streaming inference in live trading (S4 trains in $O(L \log L)$, Mamba in $O(L)$; both step recurrently at inference).

(d) Aggregate ticks before modelling: time/volume/dollar bars or order-flow features (OFI, imbalance, microprice deltas) cut L by 2–4 orders of magnitude while *adding* prior knowledge, after which any sequence model (or LightGBM on features) is on the table — or run an SSM on ticks if the raw stream truly carries the alpha.

Self-check. Arithmetic: $(5 \times 10^5)^2 = 25 \times 10^{10} = 2.5 \times 10^{11}$; $\times 2$ B $= 5 \times 10^{11}$ B $= 500$ GB at 10^9 B/GB. Shape check on (c): the KV cache holds L key/value vectors, so one new query costs L dot products — $O(L)$, consistent; an RNN-style state is size-fixed by construction — $O(1)$.

Answer: (a) 2.5×10^{11} entries ≈ 500 GB per head per layer — infeasible; (b) FlashAttention fixes memory ($O(L)$), not compute (still $O(L^2)$); (c) transformer: $O(L)$ per new tick with a growing KV cache; SSM: $O(1)$ per tick with fixed state; (d) aggregate ticks into bars/flow features first (or use an SSM on the raw stream).

P9.4 *The model stopped working*

The five hypotheses, each with its test:

- *Regime change / non-stationarity* — the relationship weakened as the world changed. Diagnostics: rolling IC and rolling Sharpe through time; feature-distribution drift by year (KS/PSI); refit by sub-period and compare coefficients.
- *Overfit all along* — 2005–2018 was in-sample luck. Diagnostics: strict walk-forward re-run with the current codebase; deflated Sharpe accounting for how many variants were ever tried; sensitivity of P&L to small hyperparameter perturbations (a real edge is not knife-edge).
- *Alpha decay / crowding* — the edge was real and got arbitraged away. Diagnostics: return per unit of turnover trending to zero; rising implementation shortfall and impact; correlation of the strategy's P&L with published factors and factor ETFs after the idea became public.
- *Look-ahead / leakage in the original backtest* — the historical P&L was never achievable. Diagnostics: rebuild data point-in-time (as-of joins, restatement-safe fundamentals, correct timestamps); compare early live fills against backtest assumptions (slippage, availability).
- *Structural change* — regulation, tick-size regimes, new participant mix, market microstructure shifts. Diagnostics: align the performance break date with known structural events; test the same signal in markets the change did not touch.

Check *leakage* and *overfitting* first: they answer the prior question of whether the edge ever existed. They are also the cheapest to test from your own artefacts (no new data), and every other conclusion changes if they fail — a regime analysis of a never-real edge is wasted work.

Self-check (coverage). The five split cleanly into "the edge was never real" (overfit, leakage) versus "real, then gone" (regime, crowding, structure) — both branches covered, no overlap; and the diagnostics discriminate: crowding predicts *gradual* decay with rising costs, structural change predicts a *sharp* break at a datable event, leakage predicts live-vs-backtest divergence from day one. Distinct fingerprints, so the tests can actually separate the hypotheses.

Answer: (1) regime change, (2) overfitting, (3) alpha decay/crowding, (4) look-ahead/leakage, (5) structural change — with the paired diagnostics above; test leakage and overfitting first because they determine whether the edge ever existed and are cheapest to verify.

P9.5 *Numpy vs lists, and why PyTorch exists* ★

Lists vs numpy. A Python list is an array of *pointers* to boxed PyObjects: every element carries a type tag and reference count, lives wherever the heap allocator put it, and every operation on it dispatches dynamically through the interpreter. A numpy array is *one contiguous typed buffer*; a vectorised operation runs a compiled C loop with SIMD over cache-friendly memory — no boxing, no pointer chasing, no per-element interpreter tick, no per-element refcounting. One vectorised op

is typically 10–100× a Python loop before any parallelism enters. (MCQ bait to dodge: this is *not* a GIL story — the win is single-threaded, from memory layout and compiled loops.)

Why PyTorch does not use numpy. numpy is CPU-only and has no autograd — the two things a deep-learning framework cannot live without. A `torch.Tensor` is a numpy-like buffer *plus*: (1) autograd metadata — every op records a `grad_fn` node so backprop can run; (2) a device abstraction — the same API executes CUDA kernels with asynchronous streams and a caching GPU allocator; (3) deep-learning dtypes (fp16/bf16) and fused kernels; (4) distributed primitives. Interop is zero-copy on CPU: `torch.from_numpy` shares the same memory.

Self-check (do the magnitudes cohere?). Summing 10^7 Python floats in a loop at ∼50–100 ns of dispatch per element ≈ 0.5–1 s; `np.sum` on the same data is memory-bandwidth bound: 8×10^7 bytes at $\sim 2 \times 10^{10}$ B/s ≈ 4 ms — a ∼100–250× ratio, consistent with the 10–100× claim. And zero-copy `from_numpy` confirms the storage layout is identical — the difference really is autograd + device machinery, not the buffer.

Answer: lists store pointers to boxed objects with per-element interpreter dispatch; numpy stores one contiguous typed buffer processed by compiled C loops with SIMD (not a GIL effect) — 10–100× per op. PyTorch adds what numpy lacks: autograd (grad_fn graph) and GPU/device execution (plus fp16/bf16, fused kernels, distributed); `torch.from_numpy` is zero-copy on CPU.

P9.6 *GPU vs CPU speedup* ★

Headline: **one to two orders of magnitude (10–100×)** for dense batched linear algebra. Derive it from hardware ratios rather than folklore:

- *Compute-bound* (big matmuls): A100 ≈ 19.5 TFLOPs FP32, ≈ 312 TFLOPs FP16 on tensor cores, versus a good multi-core CPU at ∼1–2 TFLOPs ⇒ ∼10–100×, more with mixed precision.
- *Memory-bound* ops (elementwise, reductions): the bandwidth ratio rules — ∼2 TB/s HBM vs ∼100–200 GB/s DDR ⇒ ∼10–20×.

Caveats to volunteer before being asked: *PCIe transfers can eat the entire win* (shipping data to the card costs more than computing on it — keep data resident); *small, sequential, or branchy workloads* can be *slower* on GPU (kernel-launch overhead, poor occupancy, no latency-hiding); and versus naive single-threaded Python the observed gap is 10^3–10^4×, but that credits the GPU for vectorisation a CPU could also have delivered — compare against a well-vectorised CPU baseline to keep the claim honest.

Self-check (two independent routes agree). FLOPs route: 19.5/1.5 ≈ 13× (FP32) up to 312/1.5 ≈ 200× (FP16 tensor cores); bandwidth route: 2000/150 ≈ 13×. Both land in or bracket the 10–100× band; and the everyday observation — a CPU training epoch of hours becoming minutes on one GPU — sits at ∼30–60×, inside the same band.

Answer: ∼10–100× (1–2 orders of magnitude) for dense batched linear algebra — FLOPs ratio when compute-bound, ∼10–20× bandwidth ratio when memory-bound; the advantage collapses for small/sequential/transfer-dominated workloads and inflates to 10^3–10^4× only against naive single-threaded Python.

P9.7 *Python class variables* ♦

A class variable is defined on the class object itself, so there is *one shared binding*: every instance that has not shadowed it sees the same value, because attribute lookup goes instance dict → class dict. Rebinding on the class (`C.x = 7`) changes what all such instances see; mutating a *mutable* class variable in place (e.g. appending to a class-level list) is likewise visible everywhere — the classic shared-mutable-default bug. The intended MCQ answer: **it is shared by the class — the same value for all objects**. Pedantic caveat worth one sentence in interview: assigning *through an instance* (`obj.x = 5`) creates a per-instance attribute that shadows the class variable

for that object only.

Self-check (mental trace). `class C: x = 0; a, b = C(), C(); C.x = 7` ⇒ `a.x == b.x == 7` (both look ups fall through to the class); then `a.x = 1` ⇒ `b.x` still `7`, confirming the shadowing caveat rather than contradicting the sharing rule.

Answer: it is shared by the class — all instances see the same value (until an instance assignment shadows it for that object alone).

P9.8 *Training bottleneck diagnosis* ♦

The job is **I/O bound** (disk-bound): throughput is limited by data delivery, not by compute — the GPU starves waiting on the input pipeline. Worth volunteering the fix list: parallel data loading with prefetch (`DataLoader` workers), local NVMe or caching instead of network storage, sharded binary formats (Parquet/TFRecord/WebDataset) instead of millions of small files, decode/augment offline or on GPU (DALI). Contrast terms to have ready: *compute-bound* (GPU pegged at 100%), *memory-bandwidth-bound*, *CPU-preprocessing-bound* (CPU pegged, disk idle).

Self-check (the utilisation signature is discriminating). If compute were the bottleneck, GPU utilisation would be high; if CPU preprocessing were, the CPU would be pegged with the disk quiet; here the *only* saturated resource is the disk, and the bottleneck is by definition the saturated resource — so I/O bound is not just consistent but uniquely identified.

Answer: I/O bound — the data pipeline (disk reads), not compute, is the bottleneck.

P9.9 *Fail fast vs fail silent* ♦

The **fail-fast principle**: an unexpected NaN in a critical column means the pipeline's output will likely be invalid, so the remaining ∼71 hours of compute would be spent manufacturing a corrupt artefact. Crashing at hour one (i) saves the compute, and (ii) — more important in a research context — prevents *silent corruption from propagating*: a model trained on quietly-skipped data does not announce the problem, it just misleads, and silently dropped rows also shift distributions and make runs unreproducible (which rows vanished this time?). An unexpected NaN is usually a symptom of an upstream join or vendor problem that affects more than the visible rows, so surface it, fix the cause, rerun. (In the MCQ, the distractor options described benefits that do not generally happen.)

Self-check (cost asymmetry). Crash-now cost: 1 hour of compute plus a fix. Continue cost: 72 hours of compute *plus* debugging a subtly wrong model later — and if the output feeds trading decisions, the cost of quietly-wrong is unbounded relative to the cost of loudly-absent. The asymmetry independently justifies the principle without appeal to authority.

Answer: to avoid wasting the remaining 3 days of compute on output that will likely be invalid, and to stop silent corruption propagating downstream — the fail-fast principle.

P9.10 *Immutability and concurrency* ♦

It **eliminates race conditions**. A data race requires a write to shared state; if data cannot be modified after creation, threads cannot overwrite what another thread is reading — there are no read-modify-write hazards, no torn reads, and *nothing to lock*. Any number of threads can share the same structure safely; an "update" builds a new version, so every thread holds a consistent snapshot. Downstream benefits: no locks means no deadlocks from lock-ordering, and code becomes referentially transparent and easy to test. The cost — allocation and copying — is what persistent data structures with structural sharing exist to amortise.

Self-check (contrapositive on the canonical race). The textbook race is two threads executing `x += 1` and losing an increment: it requires reading, modifying, and writing back *shared mutable* `x`. Make `x` immutable and that program cannot be written — each thread can only produce a *new* value, and combining them forces an explicit, visible coordination point. The hazard is not merely

mitigated; it is unrepresentable.

Answer: immutability eliminates race conditions — threads cannot overwrite shared state, so there are no read-modify-write hazards and nothing to lock.

P9.11 *Shared-node OOM forensics* ♦

The job is **consuming most of the node's available RAM**. When the node exhausts memory, the kernel's OOM-killer terminates processes to free it — and its victim selection (an `oom_score` heuristic favouring large, recent, low-priority processes, adjustable via `oom_score_adj` and cgroup limits) is *not obliged to pick the culprit.* So the hog survives while its neighbours are sacrificed: "runs without issue" is precisely what the offender looks like. Everyday SLURM reality: request memory honestly (`-mem`/`-mem-per-cpu`); with cgroup enforcement configured, the kill is contained to the offending job's cgroup and this failure mode disappears. Diagnostics: `dmesg` OOM-killer lines naming sacrificed PIDs, and `sacct` MaxRSS showing the survivor's footprint.

Self-check (eliminate the alternatives). CPU oversubscription slows neighbours but does not OOM-kill them; disk/network saturation causes stalls and timeouts, not OOM; a bug in the neighbours' code would not correlate with sharing a node with this particular job. Only memory exhaustion — with kernel victim selection decoupled from blame — explains "offender healthy, neighbours OOM-killed".

Answer: the job is eating the node's RAM, triggering the OS OOM-killer, which kills other processes (not necessarily the offender) to free memory.

28.10 Solutions — §10 Sharpe, Loss Probabilities and Model Combination

P10.1 *P(loss) = Phi(-SR) and time scaling*

(a) With $R \sim \mathcal{N}(\mu, \sigma^2)$ and $\mathrm{SR} = \mu/\sigma$,

$$P(R < 0) = P\Big(Z < -\frac{\mu}{\sigma}\Big) = \Phi(-\mathrm{SR}) = \Phi(-1.5) \approx 6.7\%.$$

(b) Under i.i.d. returns the mean scales with the horizon T and the sd with $\sqrt{T}$, so $\mathrm{SR}_T = \mathrm{SR}\sqrt{T}$. Monthly: $\mathrm{SR}_{1/12} = 1.5/\sqrt{12} \approx 0.433$, hence

$$P(\text{losing month}) = \Phi(-0.433) \approx 33\%.$$

(c) Shrinking the horizon shrinks drift linearly but noise only as the square root, so signal-to-noise degrades as $\sqrt{T}$: at short horizons the drift is invisible inside the noise and the loss frequency drifts toward 1/2. Same strategy, different clock — this is also why staring at daily P&L makes a good strategy feel bad (roughly $\Phi(-1.5/\sqrt{252}) \approx 46\%$ of days lose).

Self-check. Anchor consistency: SR 1 gives $\Phi(-1) \approx 15.9\%$ losing years; our higher SR 1.5 must lose less often, and $6.7\% < 15.9\%$. Limits: as $T \to \infty$, $\Phi(-\mathrm{SR}\sqrt{T}) \to 0$; as $T \to 0$, $\to \Phi(0) = 50\%$; the monthly 33% sits between the annual 6.7% and 50% as it must.

Answer: (a) $\Phi(-1.5) \approx 6.7\%$; (b) $\Phi(-1.5/\sqrt{12}) \approx 33\%$; (c) SR scales as $\sqrt{T}$, so shorter horizons are noise-dominated and loss frequency tends to 50%.

P10.2 *Sharpe to loss probability with assumption demolition* ★

Assumptions (state them before computing). (i) Annual P&L X is normal; (ii) the parameters are the *true* ones and constant; (iii) years are i.i.d.; (iv) horizon is one year and "losing \$100M" means *terminal* annual P&L $\leq -\$100$M (not touching $-\$100$M intra-year). Justification: normality is the maximum-entropy default given only mean and vol, and annual P&L aggregates many shorter-period P&Ls (a CLT gesture — with known weaknesses in the tails); i.i.d. and known parameters are what the given numbers implicitly assert.

Computation. Mean P&L $\mu = \text{SR} \times \sigma = 2 \times \$50\text{M} = \$100\text{M}$. A $-\$100\text{M}$ year sits at

$$z = \frac{-100 - 100}{50} = -4 \qquad \Longrightarrow \qquad P = \Phi(-4) \approx 3.2 \times 10^{-5} \approx 0.003\%.$$

Direction: higher in reality. Every assumption breaks the same way:

- *Fat tails* — real P&L distributions have excess kurtosis; Φ underprices 4-sigma events badly.
- *Vol clustering / non-constant σ* — even if each regime is conditionally normal, a scale mixture of normals is unconditionally fat-tailed.
- *Negative skew* — many strategies have short-vol-like payoff profiles: frequent small gains, rare large losses.
- *Estimated vs true Sharpe* — a reported SR 2 is typically selected (best strategy, survivor); the deflated-Sharpe argument says true μ is lower, moving the loss threshold closer.
- *Path vs terminal* — if "losing \$100M" means the drawdown ever touching $-\$100\text{M}$ during the year, the probability is strictly larger (for driftless Brownian motion, exactly $2\times$ by the reflection principle; with positive drift, between $1\times$ and $2\times$).
- *Regime change, crowding, correlations to one in crises* — the years are not exchangeable draws.

The distribution-free anchor (the interviewer's own follow-up). Cantelli's one-sided inequality needs only the mean and variance: $P(X - \mu \le -k\sigma) \le 1/(1+k^2)$. Here $k = 4$: bound $= 1/17 \approx 5.9\%$. The truth lives between 0.003% and 5.9% — a gap of over three orders of magnitude that is *entirely* the Gaussian tail assumption. Naming that gap is the point of the question.

Self-check. Recompute z: loss of \$100M means $X = -100$; $(-100 - 100)/50 = -4$. Anchor table: $\Phi(-4) \approx 3.2 \times 10^{-5}$. Monotonicity sanity: merely losing *any* money is a 2-sigma event, $\Phi(-2) \approx 2.3\%$; losing \$100M must be far rarer, and $0.003\% \ll 2.3\%$.

Answer: Under normal, i.i.d., known-parameter, terminal-P&L assumptions: $P = \Phi(-4) \approx 3.2 \times 10^{-5} \approx 0.003\%$. In reality *higher* (fat tails, vol clustering, negative skew, estimated Sharpe, path-vs-terminal, regime/crowding risk); distribution-free ceiling: Cantelli gives $\le 1/17 \approx 5.9\%$.

P10.3 *Cantelli's inequality: proof, application, tightness*

(a) Let $Y = \mu - X$, so $E[Y] = 0$, $\text{Var}(Y) = \sigma^2$, and the event is $\{Y \ge k\sigma\}$. For any $u \ge 0$: if $Y \ge k\sigma$ then $Y + u \ge k\sigma + u > 0$, so $(Y+u)^2 \ge (k\sigma + u)^2$. Markov's inequality on the nonnegative variable $(Y+u)^2$ gives

$$P(Y \ge k\sigma) \;\le\; \frac{E[(Y+u)^2]}{(k\sigma+u)^2} \;=\; \frac{\sigma^2 + u^2}{(k\sigma+u)^2}.$$

Optimise the free parameter: minimising over u (set the derivative to zero, or just try $u = \sigma/k$):

$$\frac{\sigma^2 + \sigma^2/k^2}{(k\sigma + \sigma/k)^2} = \frac{\sigma^2(k^2+1)/k^2}{\sigma^2(k^2+1)^2/k^2} = \frac{1}{1+k^2}. \qquad \blacksquare$$

(b) Losing at least \$100M means $X \le -100 = \mu - 4\sigma$ (since $(100 + 100)/50 = 4$), so $k = 4$:

$$P \;\le\; \frac{1}{1+16} = \frac{1}{17} \approx 5.9\%.$$

The Gaussian assumption gives $\Phi(-4) \approx 3.2 \times 10^{-5}$; the bound is larger by a factor of about 1800 — three-plus orders of magnitude. Neither number is "the answer": $\Phi(-4)$ is the best case of a specific thin-tailed model, 1/17 is the worst case over *all* distributions with that mean and vol, and reality sits in between depending on tail assumptions.

(c) Take the two-point distribution

$$X = \mu + \frac{\sigma}{k} \text{ w.p. } \frac{k^2}{1+k^2}, \qquad X = \mu - k\sigma \text{ w.p. } \frac{1}{1+k^2}.$$

Mean: $\frac{\sigma}{k} \cdot \frac{k^2}{1+k^2} - k\sigma \cdot \frac{1}{1+k^2} = \frac{\sigma k - \sigma k}{1+k^2} = 0$ above μ. Variance: $\frac{\sigma^2}{k^2} \cdot \frac{k^2}{1+k^2} + k^2\sigma^2 \cdot \frac{1}{1+k^2} = \sigma^2 \frac{1+k^2}{1+k^2} = \sigma^2$. And $P(X - \mu \le -k\sigma) = \frac{1}{1+k^2}$ exactly — equality holds, so no better distribution-free bound exists.

Self-check. At $k = 1$ the bound is $1/2$, strictly better than the vacuous two-sided Chebyshev ($1/k^2 = 1$), and safely above the Gaussian $\Phi(-1) \approx 15.9\%$. At $k = 4$: $1/17 \approx 5.88\%$ and $\Phi(-4) \approx 0.0032\%$ — bound respected with room to spare.

Answer: (a) Markov on $(\mu - X + u)^2$ optimised at $u = \sigma/k$ gives $1/(1+k^2)$; (b) $1/17 \approx 5.9\%$ versus Gaussian 3.2×10^{-5} — an $\approx 1800\times$ model-risk gap; (c) the two-point law at $\mu + \sigma/k$ and $\mu - k\sigma$ with masses $k^2/(1+k^2)$, $1/(1+k^2)$ attains it. □

P10.4 *Sharpe vs EV at maximum capacity* ♦

Compute the daily Sharpe (EV/sd) of each: (A) 5000/1 = 5000; (B) 0.67; (C) 0.5; (D) 0.25. Annualised ($\times\sqrt{252} \approx 15.9$): (A) astronomical; (B) ≈ 10.6; (C) ≈ 7.9; (D) ≈ 4.0. Option (A) is the bait.

The crux is the phrase *at maximum capacity.* Sharpe is the right ranking criterion only because a scalable strategy can be levered: any high-Sharpe edge can be turned into any desired expected P&L at chosen risk. The capacity clause removes exactly that move — you must compare the strategies *as they stand.* A large, diversified, well-capitalised trading firm is effectively risk-neutral at these dollar magnitudes, so it simply maximises expected value: **(D)**, \$100k/day ≈ \$25.2M/year.

Why the risk in (D) does not matter: annualised, D earns \$25.2M with a standard deviation of $400\text{k} \times \sqrt{252} \approx \$6.35\text{M}$ — a standalone annual Sharpe of ≈ 4, so a merely *flat* year is a 4σ event ($\Phi(-4) \approx 3 \times 10^{-5}$). Even a 2σ-bad year still makes $\approx \$12.5\text{M}$. The scary-looking "40% of days lose" is daily noise, irrelevant at the horizon a firm cares about. Meanwhile (A), for all its Sharpe of 5000, earns \$1.26M/year — a rounding error at firm scale. (You run (A) too, of course — it is free money — but the question asks which you *most prefer*, and at capacity preference goes by EV.)

Self-check via certainty equivalents: for a mean–variance agent to prefer (A), risk aversion would have to be so extreme that the firm would also reject a coin flip winning \$1.0M vs losing \$0.84M — no trading firm behaves that way. Cross-check of the rule itself: if capacity were *unlimited*, lever (A) $\times 10^4$ and it dominates everything — confirming that the capacity clause is precisely what flips the answer from Sharpe to EV.

Answer: (D) — at maximum capacity you cannot lever, so a well-capitalised firm maximises expected value: \$100k/day (≈ \$25M/yr, standalone annual Sharpe still ≈ 4). Picking the Sharpe-5000 strategy (A) — \$1.26M/yr — is the trap the capacity clause is designed to spring. (Note: at the live sitting the recorded answer was (A); on adjudicated review the canonical trading-firm answer is (D).)

P10.5 *Var(A+B) and the Sharpe of combined strategies*

Throughout, $\mathrm{Var}(A+B) = \mathrm{Var}(A) + \mathrm{Var}(B) + 2\,\mathrm{Cov}(A,B) = 100 + 100 + 2\rho \cdot 100 = 200(1+\rho)$ (in \$M^2), and the mean is always $10 + 10 = 20$.

(a) $\rho = 0.5$: Var = 300, vol = $\sqrt{300} \approx \$17.3\text{M}$, SR = $20/17.3 \approx 1.15$. Positive correlation is redundancy: risk is superadditive relative to the diversified case, and the Sharpe improvement over a single leg is modest.

(b) $\rho = 0$: Var = 200, vol ≈ \$14.1M, SR = $20/14.1 = \sqrt{2} \approx 1.41$ — the classic $\sqrt{2}$ diversification gain.

(c) $\rho = -0.3$: Var = 140, vol $\approx$ \$11.8M, SR = $20/11.8 \approx 1.69$. Negative correlation is hedging: combined risk falls below the sum and can fall below either leg alone.

(d) N uncorrelated strategies, each mean μ and vol σ: the sum has mean $N\mu$ and vol $\sigma\sqrt{N}$, so $\mathrm{SR}_N = \sqrt{N}\,\mathrm{SR}_1 = \sqrt{\sum_i \mathrm{SR}_i^2}$ (all equal). The quadrature formula in general, however, presumes each strategy contributes equal *risk*. A raw dollar sum weights legs by their vols: a leg with Sharpe 1 but tiny vol adds almost nothing to a raw sum's P&L, while a high-vol leg dominates the book's risk. Scaling each leg to equal vol (weights $\propto 1/\sigma_i$, "risk-balanced") restores the symmetric case and delivers $\sqrt{\sum \mathrm{SR}_i^2}$ for uncorrelated legs.

Self-check. (b) via quadrature: $\sqrt{1^2 + 1^2} = 1.414$, matching the direct computation. Limiting cases of the formula $\mathrm{SR}(\rho) = 20/\sqrt{200(1+\rho)}$: $\rho = 1$ gives vol 20, SR = 1 (no benefit — same strategy twice); $\rho \to -1$ gives vol $\to 0$, SR $\to \infty$ (perfect hedge with positive carry) — both limits sensible. Monotonicity: $1.15 < 1.41 < 1.69$ as ρ falls.

Answer: (a) vol $\approx$ \$17.3M, SR ≈ 1.15; (b) vol $\approx$ \$14.1M, SR $= \sqrt{2} \approx 1.41$; (c) vol $\approx$ \$11.8M, SR ≈ 1.69; (d) $\mathrm{SR}_N = \sqrt{N}\,\mathrm{SR}_1$, and the quadrature rule needs equal-risk (vol-scaled) weights, not a raw dollar sum. □

P10.6 *Decorrelating a new model from a black box*

(a) Two independent failures:

- *It discards genuinely predictive features.* A feature can be correlated with A's output and still carry incremental information A does not use; deleting it throws away alpha wholesale rather than removing only the overlap.
- *It does not even achieve the goal.* Pairwise feature-to-A correlations do not bound the correlation of the *trained model's output* with A. Concrete counterexample: suppose A's output effectively equals $X_1 + \cdots + X_{10}$ for i.i.d. features X_i. Each single feature has $\mathrm{Corr}(X_i, A) = 1/\sqrt{10} \approx 0.32$ — below any reasonable filter threshold — yet a model trained on all ten can reproduce A exactly (correlation 1).

(b) *Exact one-liner (post-hoc):* regress B's output on A's and trade the residual, $B_\perp = B - \hat{\beta}A$ with $\hat{\beta} = \mathrm{Cov}(A, B)/\mathrm{Var}(A)$. By construction $\mathrm{Cov}(B_\perp, A) = \mathrm{Cov}(B, A) - \hat{\beta}\,\mathrm{Var}(A) = 0$: you trade exactly B's incremental information. Trade-off: the part of B's alpha that overlapped A is stripped out — but that part was already being monetised by A, so nothing real is lost at the book level; B's standalone Sharpe falls even as the book improves.

Training-time alternative: add a correlation penalty to the loss, $\mathcal{L} = \mathrm{MSE} + \lambda\,\mathrm{Corr}(A, B)^2$, and tune λ. Trade-off: a continuous dial between B's standalone accuracy and orthogonality; costs some fit, requires A's outputs at training time, and gives approximate rather than exact orthogonality.

Why bother at all: with Corr = 0.7, $\mathrm{Var}(A + B) = \mathrm{Var}\,A + \mathrm{Var}\,B + 2 \times 0.7\,\sigma_A\sigma_B$ — heavily superadditive risk for mostly redundant signal. After orthogonalisation the cross term is zero and (with risk-balanced weights) the book's Sharpe approaches $\sqrt{\mathrm{SR}_A^2 + \mathrm{SR}_{B_\perp}^2}$.

Self-check. The residual method's exactness: $\mathrm{Cov}(B - \beta A, A) = \mathrm{Cov}(B, A) - \beta\,\mathrm{Var}(A)$, which vanishes precisely at the OLS slope $\beta = \mathrm{Cov}(B, A)/\mathrm{Var}(A)$. The counterexample in (a): $\mathrm{Corr}(X_i, \sum_j X_j) = \sigma^2/(\sigma \cdot \sigma\sqrt{10}) = 1/\sqrt{10}$ — ten individually "weak" features jointly reconstruct A.

Answer: (a) feature-dropping discards predictive information, and pairwise feature correlations do not bound the trained output's correlation with A; (b) regress B on A and trade the residual (exact, loses only the already-monetised overlap), or penalise $\mathrm{Corr}(A, B)^2$ in the training loss (tunable, approximate).

28.11 Solutions — §11 Portfolio Theory: Mean–Variance Optimisation, Diversification, and the Capital Market Line

P11.1 *Two-asset diversification: the same-mean, triple-volatility asset*

(a) The mean is linear in the weights: $0.5(4\%)+0.5(10\%) = 7\%$. The variance of the mix $(w, 1-w)$ is $w^2\sigma_1^2 + (1-w)^2\sigma_2^2 + 2w(1-w)\rho\sigma_1\sigma_2$, so at $w = 0.5$

$$0.25(0.01) + 0.25(0.09) + 2(0.25)(0.2)(0.10)(0.30) = 0.0025 + 0.0225 + 0.0030 = 0.0280,$$

a volatility of $\sqrt{0.028} = 16.73\%$, against the weighted average $0.5(10\%) + 0.5(30\%) = 20\%$. The mean is exactly the weighted average; the volatility is 3.27 points below it.

(b) Collecting powers of w, the variance is $Vw^2 - 2cw + \sigma_2^2$ with $V = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 = 0.01 + 0.09 - 0.012 = 0.088$ and $c = \sigma_2^2 - \rho\sigma_1\sigma_2 = 0.09 - 0.006 = 0.084$. Since $V > 0$ the quadratic is convex and its minimiser is $w^* = c/V = 0.084/0.088 = 0.9545$: 95.45% in asset 1, 4.55% in asset 2. The minimum variance is $\sigma_1^2\sigma_2^2(1-\rho^2)/V = (0.01)(0.09)(0.96)/0.088 = 0.009818$, volatility 9.91%, below asset 1's 10%. The mean is $0.9545(4\%) + 0.0455(10\%) = 4.27\%$.

(c) Asset 2 gets positive weight iff $w^* < 1$, i.e. $c < V$, i.e. $\sigma_2^2 - \rho\sigma_1\sigma_2 < \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$, i.e. $\rho\sigma_1\sigma_2 < \sigma_1^2$, i.e. $\rho < \sigma_1/\sigma_2 = 1/3$. At $\rho = 0.5 > 1/3$: $V = 0.1 - 0.03 = 0.07$, $c = 0.09 - 0.015 = 0.075$, $w^* = 1.071 > 1$, so the unconstrained minimiser shorts asset 2 and the long-only minimum over $[0, 1]$ is at the boundary $w = 1$: asset 1 alone, volatility 10%.

(d) The objection is wrong at $\rho = 0.2$: the test compares ρ with the volatility *ratio* $\sigma_1/\sigma_2 = 1/3$, not one volatility with the other, and $0.2 < 1/3$, so a 4.55% allocation to the triple-volatility asset lowers the volatility from 10% to 9.91%. And no: the mean of the minimum-variance mix is $4.27\% > 4\%$. Diversification changed the variance through the cross term; the mean moved (upward) only because the weights did, and a long-only mix always has mean between the two asset means.

Self-check. Evaluate the variance directly at $w = 0.9545$: $0.9112(0.01) + 0.00207(0.09) + 2(0.9545)(0.0455)(0.006) = 0.009111 + 0.000186 + 0.000521 = 0.009818$, matching the closed form. Limit check at $\rho = 0$: $w^* = \sigma_2^2/(\sigma_1^2 + \sigma_2^2) = 0.09/0.10 = 0.90$ and variance $0.0009/0.10 = 0.009$ (volatility 9.49%); raising ρ to 0.2 should push the weight toward the low-volatility asset and raise the minimum, and indeed $0.90 \to 0.9545$ and $9.49\% \to 9.91\%$.

Answer: (a) mean 7%, volatility 16.73% (versus 20% weighted average); (b) $w^* = 0.9545$ in asset 1, minimum volatility 9.91%, mean 4.27%; (c) iff $\rho < \sigma_1/\sigma_2 = 1/3$; at $\rho = 0.5$ the long-only minimum is asset 1 alone (10%); (d) the objection is wrong ($0.2 < 1/3$: hold 4.55% of asset 2), and the mix earns $4.27\% > 4\%$.

P11.2 *Eight portfolio claims: true or false, with a counterexample*

(i) False. The mean $w^\top\mu$ is linear in the weights, so a long-only mix has mean exactly the weighted average of the asset means: a 50/50 mix of assets with means 4% and 10% has mean 7% whatever the correlation. Only the variance has a correlation-dependent cross term.

(ii) False. With volatilities $\sigma_1 \le \sigma_2$, the minimum-variance mix holds a positive amount of the riskier asset whenever $\rho < \sigma_1/\sigma_2$, which permits positive correlation. Example: $\sigma_1 = 10\%$, $\sigma_2 = 30\%$, $\rho = 0.2 < 1/3$ gives minimum volatility $9.91\% < 10\%$ with 4.55% in asset 2.

(iii) False whenever the average covariance is positive. For n assets with common variance v and common correlation ρ, the equal-weight variance is $\rho v + (1-\rho)v/n \to \rho v$. With $\sigma = 20\%$ and $\rho = 0.3$ the floor is $\sqrt{0.3} \times 20\% = 10.95\%$; no n goes below it.

(iv) False. Uncorrelated strategies combine in quadrature: the best Sharpe ratio is $\sqrt{1^2 + 1^2} = \sqrt{2} = 1.41$, attained at equal risk budgets; not 2 (Sharpe ratios do not add) and not 1 (they do not average).

(v) False unless every asset has the same expected excess return. Two uncorrelated assets with volatility 10% each and excess means 2% and 6%: the minimum-variance portfolio is 50/50

with Sharpe ratio $4\%/7.07\% = 0.566$; the tangency portfolio has weights proportional to $\mu_{e,i}/\sigma_i^2$, i.e. $(0.25, 0.75)$, mean 5%, volatility $\sqrt{0.0625(0.01) + 0.5625(0.01)} = 7.91\%$ and Sharpe ratio $0.632 = \sqrt{0.2^2 + 0.6^2}$.

(vi) False. $\beta_i = \rho_{iM}\sigma_i/\sigma_M$ carries the volatility ratio. A stock with $\rho_{iM} = 0.5$ and $\sigma_i = 40\%$ against a 20%-volatility market has $\beta = 1.0$; a stock with $\rho_{iM} = 0.8$ and $\sigma_i = 15\%$ has $\beta = 0.6$: the more correlated stock has the smaller beta.

(vii) False. Risk contributions are $w_i(\Sigma w)_i/\sigma_p$, not capital weights. With $\sigma_E = 18\%$, $\sigma_B = 6\%$, $\rho = 0$ and $w = (0.6, 0.4)$ (equities first): $\Sigma w = (0.6 \times 0.0324,\ 0.4 \times 0.0036) = (0.01944, 0.00144)$, $\sigma_p^2 = 0.6(0.01944) + 0.4(0.00144) = 0.01224$, and the equity risk share is $0.011664/0.01224 = 95.3\%$: 60/40 in capital is 95/5 in risk.

(viii) False. The mean vector is the least identifiable input: the standard error of an average return shrinks only as $\sigma/\sqrt{T}$ (with $\sigma = 20\%$ and ten years, 6.3 percentage points), and the optimiser multiplies mean errors by Σ^{-1}, which amplifies them by $(1+\rho)/(1-\rho)$ along the spread direction of two correlated assets (9 at $\rho = 0.8$). A shrunk Σ with a raw $\hat{\mu}$ still produces error-maximised weights; shrink or avoid μ as well.

Self-check. Each counterexample is a number that can be recomputed by a second route: in (v), $\sqrt{0.04 + 0.36} = 0.632$ agrees with the direct ratio $5\%/7.91\% = 0.632$; in (vii), the two risk contributions $0.011664 + 0.000576 = 0.01224 = \sigma_p^2$, so the shares sum to one as the Euler decomposition requires; in (iv), risk budgets of one unit each give mean 2 and volatility $\sqrt{2}$, ratio $\sqrt{2}$.

Answer: All eight are false. (i) means are linear (7% for the 50/50 mix of 4% and 10%); (ii) the condition is $\rho < \sigma_1/\sigma_2$ ($0.2 < 1/3$ works); (iii) the floor is the average covariance (10.95% at $\sigma = 20\%$, $\rho = 0.3$); (iv) $\sqrt{2} = 1.41$; (v) 0.566 versus 0.632 in the example; (vi) $\beta = \rho\sigma_i/\sigma_M$ (1.0 versus 0.6); (vii) 95/5 in risk; (viii) $\hat{\mu}$ is the dangerous input and Σ^{-1} amplifies its error.

P11.3 *The global minimum-variance portfolio from scratch*

(a) *Candidate.* With multiplier θ, the Lagrangian is $L(w, \theta) = \frac{1}{2}w^\top\Sigma w - \theta(\mathbf{1}^\top w - 1)$. Since $\nabla_w(w^\top\Sigma w) = 2\Sigma w$ for symmetric Σ, stationarity reads $\Sigma w = \theta\mathbf{1}$, so $w = \theta\Sigma^{-1}\mathbf{1}$ (Σ is invertible because it is positive definite: no eigenvalue is zero). The budget constraint gives $1 = \theta\,\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = \theta A$, hence $\theta = 1/A$ and $w = w_g$. Note $A > 0$ because Σ^{-1} is itself positive definite (its eigenvalues are the reciprocals $1/\lambda_i > 0$) and $\mathbf{1} \neq 0$.

Certificate. Let v be any feasible portfolio and write $v = w_g + d$; feasibility of both gives $\mathbf{1}^\top d = 0$. Expanding,

$$v^\top\Sigma v = w_g^\top\Sigma w_g + 2\,d^\top\Sigma w_g + d^\top\Sigma d.$$

By the stationarity equation $\Sigma w_g = (1/A)\mathbf{1}$, the cross term is $2(1/A)\,\mathbf{1}^\top d = 0$, so $v^\top\Sigma v = w_g^\top\Sigma w_g + d^\top\Sigma d \geq w_g^\top\Sigma w_g$, with equality iff $d = 0$ by positive definiteness. Hence w_g is the unique global minimiser. Its variance is $w_g^\top\Sigma w_g = (1/A^2)\,\mathbf{1}^\top\Sigma^{-1}\Sigma\Sigma^{-1}\mathbf{1} = (1/A^2)\,\mathbf{1}^\top\Sigma^{-1}\mathbf{1} = 1/A$, and $\mathrm{Cov}(R_i, R_{w_g}) = e_i^\top\Sigma w_g = 1/A$ for every i: the equal-covariance property is the stationarity equation read row by row. (This is Theorem 11.7.)

(b) The matrix is block diagonal, so invert the blocks: $0.01^{-1} = 100$, and for the 2×2 block, determinant $0.04(0.02) - 0.02^2 = 0.0004$, inverse $\frac{1}{0.0004}\begin{pmatrix} 0.02 & -0.02 \\ -0.02 & 0.04 \end{pmatrix} = \begin{pmatrix} 50 & -50 \\ -50 & 100 \end{pmatrix}$. Row sums of Σ^{-1}: $\Sigma^{-1}\mathbf{1} = (100,\ 0,\ 50)^\top$, so $A = 150$ and

$$w_g = \left(\tfrac{2}{3},\ 0,\ \tfrac{1}{3}\right)^\top, \qquad \sigma_g^2 = \tfrac{1}{150} = 0.006667, \qquad \sigma_g = 8.16\%,$$

against 10% for the best single asset. Equal covariances: $\Sigma w_g = \left(0.01 \cdot \tfrac{2}{3},\ 0.02 \cdot \tfrac{1}{3},\ 0.02 \cdot \tfrac{1}{3}\right)^\top = (0.006667, 0.006667, 0.006667)^\top = (1/A)\mathbf{1}$.

(c) Asset 2 is asset 3 plus uncorrelated noise: $\mathrm{Cov}(R_2, R_3) = 0.02 = \mathrm{Var}(R_3)$, so $R_2 - R_3$ is uncorrelated with R_3 (and with R_1), meaning any exposure taken through asset 2 can be taken

through asset 3 with strictly less variance; equivalently, asset 2's covariance with the mix of assets 1 and 3 already equals that mix's variance, so at the margin it adds nothing.

Self-check. Since asset 2 drops out, the answer must agree with the two-asset formula on assets 1 and 3, which are uncorrelated: $w^* = \sigma_3^2/(\sigma_1^2+\sigma_3^2) = 0.02/0.03 = 2/3$ and variance $\sigma_1^2\sigma_3^2/(\sigma_1^2+\sigma_3^2) = 0.0002/0.03 = 0.006667$; both match. The block inverse also checks: $(0.04)(50) + (0.02)(-50) = 1$ and $(0.04)(-50) + (0.02)(100) = 0$.

Answer: (a) $w_g = \Sigma^{-1}\mathbf{1}/(\mathbf{1}^\top\Sigma^{-1}\mathbf{1})$, variance $1/A$, $\Sigma w_g = (1/A)\mathbf{1}$; the certificate is $v^\top\Sigma v = 1/A + d^\top\Sigma d$ for every feasible $v = w_g + d$. (b) $w_g = (2/3, 0, 1/3)$, $\sigma_g = 8.16\%$, $\Sigma w_g = 0.006667\,\mathbf{1}$. (c) Asset 2 equals asset 3 plus uncorrelated noise ($\mathrm{Cov}(R_2, R_3) = \mathrm{Var}(R_3)$), so it is dominated.

P11.4 *The diversification floor*

(a) Write $v = 0.25^2 = 0.0625$. The equal-weight portfolio has $w_i = 1/n$, so its variance is

$$\frac{1}{n^2}\Big(\sum_i \Sigma_{ii} + \sum_{i\neq j}\Sigma_{ij}\Big) = \frac{1}{n^2}\big(nv + n(n-1)\rho v\big) = \rho v + \frac{(1-\rho)v}{n}.$$

As $n \to \infty$ the second term vanishes and the variance tends to ρv, the average covariance: what every stock shares (the market) survives averaging, what is specific to each cancels at rate $1/n$.

(b) With $\rho v = 0.0125$ and $(1-\rho)v = 0.05$, the volatility is $\sqrt{0.0125 + 0.05/n}$: $n = 1$ gives $\sqrt{0.0625} = 25\%$; $n = 5$ gives $\sqrt{0.0225} = 15\%$; $n = 20$ gives $\sqrt{0.015} = 12.25\%$; the limit is $\sqrt{0.0125} = 11.18\%$. Five names already remove 10 of the 13.8 achievable points.

(c) Require $(1-\rho)v/n = \frac{1}{10}\rho v$, i.e. $0.05/n = 0.00125$, so $n = 40$ (volatility $\sqrt{0.01375} = 11.73\%$, within 0.55 points of the floor). In general $n = 10(1-\rho)/\rho$.

(d) The averaged error is $\frac{1}{n}\sum_i \varepsilon_i$ with the same covariance structure, so its variance is $\rho v + (1-\rho)v/n$: the identical formula, because an equal-weight portfolio of n correlated returns and an average of n correlated errors are the same random variable up to naming. The correlation floor is why bagging and random forests decorrelate their trees (feature subsampling) rather than just growing more of them: more trees attack only the $1/n$ term.

Self-check. At $n = 5$ compute directly: five weights of 0.2, diagonal terms $5\times0.04\times0.0625 = 0.0125$, off-diagonal terms $20 \times 0.04 \times 0.0125 = 0.01$, total 0.0225, volatility 15%, as in (b). Limit sanity: the floor volatility is $\sqrt{\rho}\,\sigma = 0.447 \times 25\% = 11.18\%$, and at $\rho = 0$ the formula gives $v/n \to 0$, the uncorrelated case.

Answer: (a) $\rho v + (1-\rho)v/n \to \rho v = 0.0125$; (b) 25%, 15%, 12.25%, limit 11.18%; (c) $n = 40$; (d) $\rho v + (1-\rho)v/n$, the same formula because an average of correlated variables is an equal-weight portfolio.

P11.5 *Frontier constants and the hyperbola of a three-asset universe*

(a) With $\Sigma^{-1} = \mathrm{diag}(25, 25, 50)$: $\Sigma^{-1}\mathbf{1} = (25, 25, 50)^\top$ and $\Sigma^{-1}\mu = (1.25, 2.25, 5.5)^\top$. Hence $A = 25 + 25 + 50 = 100$, $B = 1.25 + 2.25 + 5.5 = 9$, $C = 0.05(1.25) + 0.09(2.25) + 0.11(5.5) = 0.0625 + 0.2025 + 0.605 = 0.87$, and $D = 100(0.87) - 81 = 6$.

(b) $w_g = \Sigma^{-1}\mathbf{1}/A = (0.25, 0.25, 0.50)^\top$, the normalised precisions. Its mean is $B/A = 9\%$ (directly: $0.25(5) + 0.25(9) + 0.5(11) = 9$) and its variance $1/A = 0.01$, volatility 10%.

(c) At $m = 0.06$: $\theta_1 = (0.87 - 0.54)/6 = 0.055$, $\theta_2 = (6-9)/6 = -0.5$, so $w = 0.055(25, 25, 50) - 0.5(1.25, 2.25, 5.5) = (1.375{-}0.625,\ 1.375{-}1.125,\ 2.75{-}2.75) = (0.75, 0.25, 0)^\top$. At $m = 0.12$: $\theta_1 = (0.87 - 1.08)/6 = -0.035$, $\theta_2 = (12-9)/6 = 0.5$, so $w = (-0.875 + 0.625,\ -0.875 + 1.125,\ -1.75 + 2.75) = (-0.25, 0.25, 1.00)^\top$, a short position in asset 1. Both have variance $(100m^2 - 18m + 0.87)/6$: at $m = 0.06$, $(0.36 - 1.08 + 0.87)/6 = 0.025$; at $m = 0.12$, $(1.44 - 2.16 + 0.87)/6 = 0.025$; volatility 15.81% in both cases. The two targets sit symmetrically about the minimum-variance mean 9%, and the 12% portfolio is the efficient one: same volatility, six points more mean; $w(6\%)$ lies on the dominated lower branch.

(d) Completing the square, $\sigma^2(m) = 1/A + (A/D)(m - B/A)^2 = 0.01 + 16.67\,(m - 0.09)^2$, i.e.

$$100\,\sigma^2 - \frac{10000}{6}(m - 0.09)^2 = 1,$$

the right-hand branch of a hyperbola in the (σ, m) plane with vertex $(1/\sqrt{A}, B/A) = (10\%, 9\%)$, the minimum-variance portfolio, and asymptotes $m = 0.09 \pm \sqrt{D/A}\,\sigma$ with slope $\sqrt{0.06} = 0.245$. Against variance it would be a parabola; against volatility it is a hyperbola whose vertex sits at $\sigma = 10\% > 0$, not on the vertical axis.

Self-check. Verify $w(6\%)$ directly: mean $0.75(5) + 0.25(9) = 6\%$, budget 1, variance $0.5625(0.04) + 0.0625(0.04) = 0.025$; and $w(12\%)$: mean $-1.25 + 2.25 + 11 = 12\%$, variance $0.0625(0.04) + 0.0625(0.04) + 0.02 = 0.025$. Both agree with the formula. The constants satisfy the sign requirements $A > 0$, $C > 0$, $D > 0$, and the vertex check $\sigma^2(0.09) = 0.01 = 1/A$ passes.

Answer: (a) $A = 100$, $B = 9$, $C = 0.87$, $D = 6$; (b) $w_g = (0.25, 0.25, 0.5)$, mean 9%, volatility 10%; (c) $w(6\%) = (0.75, 0.25, 0)$, $w(12\%) = (-0.25, 0.25, 1)$, both with volatility 15.81%; $w(12\%)$ is efficient; (d) $\sigma^2 = 0.01 + 16.67(m - 0.09)^2$: hyperbola with vertex $(10\%, 9\%)$ and asymptote slope $\sqrt{0.06} = 0.245$.

P11.6 *Two-fund separation in action*

(a) Two-fund separation: the frontier map $m \mapsto w(m)$ is affine, so $w(m) = (1-\tau)\,w(m_1) + \tau\,w(m_2)$ with $\tau = (m - m_1)/(m_2 - m_1)$. With $m_1 = 6\%$, $m_2 = 12\%$: for $m = 10\%$, $\tau = 2/3$ and

$$w(10\%) = \tfrac{1}{3}(0.75, 0.25, 0) + \tfrac{2}{3}(-0.25, 0.25, 1) = (0.0833,\ 0.25,\ 0.6667)^\top;$$

for $m = 3\%$, $\tau = -1/2$ and $w(3\%) = 1.5(0.75, 0.25, 0) - 0.5(-0.25, 0.25, 1) = (1.25,\ 0.25,\ -0.50)^\top$, which shorts asset 3 and holds 150% of the 6% fund.

(b) The frontier variance is a quadratic in the target mean, $(Am^2 - 2Bm + C)/D$, whose minimum is at the minimum-variance mean B/A; a parabola takes equal values at two points symmetric about its vertex, so equal variances at 6% and 12% put the vertex at 9%. Then $w_g = w(9\%)$ is the midpoint, $\tau = 1/2$: $w_g = (0.25, 0.25, 0.50)^\top$, with variance $0.0625(0.04) + 0.0625(0.04) + 0.25(0.02) = 0.0025 + 0.0025 + 0.005 = 0.01$, volatility 10%, below the 0.025 of the two given portfolios, as a vertex must be.

(c) Since $w(m)$ is affine in m, each coordinate is affine in m; asset 2's coordinate takes the value 0.25 at two distinct targets, so it is the constant 0.25. The reason: the first-order condition for a frontier portfolio is $\Sigma w = \theta_1 \mathbf{1} + \theta_2 \mu$, which for diagonal Σ reads $\sigma_i^2 w_i = \theta_1 + \theta_2 \mu_i$ with $\theta_1 = (C - Bm)/D$ and $\theta_2 = (Am - B)/D$; the derivative of w_i in m is $(A\mu_i - B)/(\sigma_i^2 D)$, which vanishes exactly when $\mu_i = B/A$. Asset 2's mean, 9%, equals the minimum-variance mean, so raising or lowering the target is done entirely by trading asset 1 against asset 3: an asset whose mean is already the vertex mean cannot help move the mean.

(d) Any mix of the two proposed funds is $(0.5(1-\tau),\ 0.5(1-\tau),\ \tau)$, with equal weights on assets 1 and 2. On the frontier, by (c), $w_2 = 0.25$ always, while $w_1 = 0.75 - (m - 0.06)/0.06$ varies; equality $w_1 = w_2 = 0.25$ holds only at $m = 9\%$. So the pair reaches the frontier at exactly one point, $\tau = 0.5$, which is $w_g = (0.25, 0.25, 0.5)$, and nowhere else. Concretely, the 50/50 fund itself has mean 7% and variance $0.25(0.04) + 0.25(0.04) = 0.02$, whereas the frontier portfolio at 7%, $w(7\%) = (0.5833, 0.25, 0.1667)$, has variance $0.3403(0.04) + 0.0625(0.04) + 0.0278(0.02) = 0.01667$; asset 3 alone has variance 0.02 against 0.01667 for $w(11\%) = (-0.0833, 0.25, 0.8333)$. Spanning funds must themselves be on the frontier; neither of these is.

Self-check. Verify $w(10\%)$ from (a) against the constants of this universe ($A = 100$, $B = 9$, $C = 0.87$, $D = 6$): mean $0.0833(5) + 0.25(9) + 0.6667(11) = 0.417 + 2.25 + 7.333 = 10\%$; variance $0.00694(0.04) + 0.0625(0.04) + 0.4444(0.02) = 0.000278 + 0.0025 + 0.008889 = 0.011667$, and the formula $(100(0.01) - 18(0.10) + 0.87)/6 = 0.07/6 = 0.011667$ agrees. For $w(3\%)$: mean $6.25 + 2.25 - 5.5 = 3\%$; variance $1.5625(0.04) + 0.0625(0.04) + 0.25(0.02) = 0.07$, and $(0.09 - 0.54 + 0.87)/6 = 0.07$.

Answer: (a) $w(10\%) = (0.0833, 0.25, 0.6667)$, $w(3\%) = (1.25, 0.25, -0.50)$; (b) minimum-variance mean 9% (midpoint of two equal-variance targets), $w_g = (0.25, 0.25, 0.5)$, variance 0.01; (c) affine in m with two equal values, hence constant; asset 2's mean equals the minimum-variance mean B/A, so its weight never moves; (d) the pair has $w_1 = w_2$ always, which meets the frontier only at w_g; the 50/50 fund (variance 0.02 versus 0.01667 at 7%) and asset 3 (variance 0.02 versus 0.01667 at 11%) are not frontier portfolios.

P11.7 *Combining uncorrelated strategies, and the correlation hurdle*

(a) For uncorrelated strategies the maximal squared Sharpe ratio is the sum of the squares:

$$\mathrm{SR}^{*2} = 0.36 + 0.64 + 1.44 = 2.44, \qquad \mathrm{SR}^* = 1.562.$$

The tangency direction is $\Sigma^{-1}\mu_e$, whose entries are

$$\mu_i/\sigma_i^2 = \mathrm{SR}_i/\sigma_i = (0.6/0.08,\ 0.8/0.16,\ 1.2/0.04) = (7.5,\ 5,\ 30);$$

normalising by 42.5 gives capital weights (0.176, 0.118, 0.706). The risk budgets are $w_i\sigma_i = (1.41\%,\ 1.88\%,\ 2.82\%)$, proportional to $(0.6, 0.8, 1.2)$: each strategy's volatility allocation is proportional to its Sharpe ratio. Check: the means are $\mathrm{SR}_i\sigma_i = (4.8\%, 12.8\%, 4.8\%)$; combined mean $0.176(4.8) + 0.118(12.8) + 0.706(4.8) = 5.74\%$, volatility $\sqrt{0.0141^2 + 0.0188^2 + 0.0282^2} = 3.68\%$, ratio 1.562.

(b) Equal weights 1/3: mean $(4.8 + 12.8 + 4.8)/3 = 7.47\%$, variance $(0.0064 + 0.0256 + 0.0016)/9 = 0.003733$, volatility 6.11%, Sharpe ratio 1.22: a higher mean, but 22% less Sharpe ratio than the optimum, because equal capital puts three quarters of the variance in the 16%-volatility strategy, which has only the middling Sharpe ratio.

(c) For two correlated legs the best squared Sharpe ratio is $(\mathrm{SR}_1^2 - 2\rho\,\mathrm{SR}_1\mathrm{SR}_2 + \mathrm{SR}_2^2)/(1-\rho^2)$, with the position in leg 2 long iff $\mathrm{SR}_2 > \rho\,\mathrm{SR}_1$. Here $\mathrm{SR}_1 = 1.562$, $\mathrm{SR}_2 = 0.6$. At $\rho = 0.5$ the hurdle is $0.781 > 0.6$: do not add it long (the optimiser would short it, and the gain is small: $\mathrm{SR}^{*2} = (2.44 - 0.937 + 0.36)/0.75 = 2.484$, $\mathrm{SR}^* = 1.576$). At $\rho = 0.2$ the hurdle is $0.312 < 0.6$: add it long, with $\mathrm{SR}^{*2} = (2.44 - 0.375 + 0.36)/0.96 = 2.526$, $\mathrm{SR}^* = 1.589$. At $\rho = 0$ the quadrature rule would give $\sqrt{2.44 + 0.36} = 1.673$.

Self-check. The quadrature answer must beat the best single strategy (1.2), and equal risk budgets in (a) must reproduce it: with risk budgets $(0.6, 0.8, 1.2)$ units, mean $0.36 + 0.64 + 1.44 = 2.44$ and volatility $\sqrt{2.44}$, ratio $\sqrt{2.44} = 1.562$. For (c), $\rho \to 0$ in the two-leg formula gives $\sqrt{\mathrm{SR}_1^2 + \mathrm{SR}_2^2}$, the quadrature rule, and the gain at $\rho = 0.5$ has the right size: $\mathrm{SR}^{*2} - \mathrm{SR}_1^2 = (\mathrm{SR}_2 - \rho\mathrm{SR}_1)^2/(1-\rho^2) = 0.0328/0.75 = 0.044$, matching $2.484 - 2.44$.

Answer: (a) $\mathrm{SR}^* = \sqrt{2.44} = 1.562$ at capital weights $(0.176, 0.118, 0.706)$, risk budgets $\propto (0.6, 0.8, 1.2)$; (b) 1.22; (c) at $\rho = 0.5$ the hurdle 0.78 exceeds 0.6: do not add long (1.576 only by shorting); at $\rho = 0.2$ the hurdle is 0.31: add long, $\mathrm{SR}^* = 1.589$.

P11.8 *Maximum Sharpe ratio by Cauchy–Schwarz*

(a) Put $y = \Sigma^{-1}\mu_e$, so that $\Sigma y = \mu_e$ and $w^\top\mu_e = w^\top\Sigma y = \langle w, y\rangle_\Sigma$. Cauchy–Schwarz in the Σ-inner product (proof: for $y \neq 0$ the quadratic $t \mapsto (w - ty)^\top\Sigma(w - ty) = w^\top\Sigma w - 2t\, w^\top\Sigma y + t^2\, y^\top\Sigma y$ is nonnegative with positive leading coefficient, so its discriminant is nonpositive) gives

$$(w^\top\mu_e)^2 = (w^\top\Sigma y)^2 \le (w^\top\Sigma w)(y^\top\Sigma y) = (w^\top\Sigma w)\,(\mu_e^\top\Sigma^{-1}\Sigma\Sigma^{-1}\mu_e) = (w^\top\Sigma w)\,(\mu_e^\top\Sigma^{-1}\mu_e),$$

with equality iff w and y are linearly dependent (a zero discriminant gives a double root t_0 with $(w - t_0 y)^\top\Sigma(w - t_0 y) = 0$, hence $w = t_0 y$ by positive definiteness). Both factors on the right are positive (Σ and Σ^{-1} are positive definite, $w \neq 0$, $\mu_e \neq 0$), so taking square roots and dividing by $\sqrt{w^\top\Sigma w}$ gives $|\mathrm{SR}(w)| \le \sqrt{\mu_e^\top\Sigma^{-1}\mu_e}$. Equality for $\mathrm{SR}(w)$ itself needs $w = ty$ with $w^\top\mu_e = t\, y^\top\Sigma y > 0$, i.e. $t > 0$; negative multiples attain the negative of the bound,

and every positive multiple attains it because SR is invariant under $w \mapsto tw$, $t > 0$. (This is Theorem 11.23(i).)

(b) With $\Sigma^{-1} = \operatorname{diag}(\sigma_i^{-2})$, $\mu_e^\top \Sigma^{-1} \mu_e = \sum_i \mu_{e,i}^2/\sigma_i^2 = \sum_i \mathrm{SR}_i^2$, and the maximiser $\Sigma^{-1}\mu_e$ has entries $\mu_{e,i}/\sigma_i^2 = \mathrm{SR}_i/\sigma_i$, so $\sigma_i w_i \propto \mathrm{SR}_i$. For two legs with $\mathrm{SR}_1 = \mathrm{SR}_2$: $\sqrt{2\,\mathrm{SR}_1^2} = \sqrt{2}\,\mathrm{SR}_1$, attained at equal risk budgets.

(c) Write $\Sigma = \Delta K \Delta$ with $\Delta = \operatorname{diag}(\sigma_1, \sigma_2)$ and $K = \left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, and $\mu_e = \Delta s$ with $s = (\mathrm{SR}_1, \mathrm{SR}_2)^\top$. Then $\mu_e^\top \Sigma^{-1} \mu_e = s^\top K^{-1} s = (\mathrm{SR}_1^2 - 2\rho\,\mathrm{SR}_1\mathrm{SR}_2 + \mathrm{SR}_2^2)/(1-\rho^2)$ and the risk budgets $\sigma_i w_i$ are proportional to $(K^{-1}s)_i \propto \mathrm{SR}_i - \rho\,\mathrm{SR}_j$. Numbers: $(1 - 0.9 + 0.25)/0.19 = 0.35/0.19 = 1.842$, so $\mathrm{SR}^* = 1.357$. The second risk budget is proportional to $0.5 - 0.9(1.0) = -0.4 < 0$: the weaker strategy is *shorted*; capital weights are proportional to $(0.55/0.10,\ -0.4/0.20) = (5.5, -2)$, i.e. $(1.571, -0.571)$ when normalised to sum to one. The value exceeds 1.118 because at $\rho = 0.9$ the weaker leg is a cheap hedge: shorting it removes most of the shared variance of the stronger leg while giving up little mean, and Σ^{-1} finds exactly that spread. The correlation hurdle $\mathrm{SR}_2 > \rho\,\mathrm{SR}_1 = 0.9$ is failed by a wide margin, and the gain over holding strategy 1 alone is $(\mathrm{SR}_2 - \rho\mathrm{SR}_1)^2/(1-\rho^2) = 0.16/0.19 = 0.842$ in squared Sharpe.

Self-check. In (c) both strategies have excess mean 10% ($1.0 \times 10\%$ and $0.5 \times 20\%$), so $\mu_e \propto \mathbf{1}$ and the tangency portfolio must coincide with the minimum-variance portfolio. The two-asset minimum-variance weight is $(\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2) = (0.04 - 0.018)/(0.05 - 0.036) = 1.571$ in strategy 1, matching the normalised weights above; its variance is $\sigma_1^2\sigma_2^2(1-\rho^2)/0.014 = 0.0004(0.19)/0.014 = 0.005429$, volatility 7.37%, and mean 10%, ratio $0.10/0.0737 = 1.357$. In (a), the bound is attained by $w = y$ itself: $\mathrm{SR}(y) = y^\top \Sigma y/\sqrt{y^\top \Sigma y} = \sqrt{\mu_e^\top \Sigma^{-1} \mu_e}$.

Answer: (a) Cauchy–Schwarz in the Σ-inner product with $y = \Sigma^{-1}\mu_e$: $\sup \mathrm{SR} = \sqrt{\mu_e^\top \Sigma^{-1} \mu_e}$, attained exactly on the ray $t\,\Sigma^{-1}\mu_e$, $t > 0$; (b) $\mathrm{SR}^{*2} = \sum_i \mathrm{SR}_i^2$ at $\sigma_i w_i \propto \mathrm{SR}_i$, and $\sqrt{2}\,\mathrm{SR}_1$ for two equal uncorrelated legs; (c) $\mathrm{SR}^* = \sqrt{0.35/0.19} = 1.357$ with the weaker strategy shorted (weights $(1.571, -0.571)$); it beats 1.118 because the highly correlated weak leg works as a hedge.

P11.9 *Tangency, the capital market line and zero alphas*

(a) The excess-mean vector is $\mu_e = (4\%, 8\%, 10\%)^\top$ and $\Sigma^{-1} = \operatorname{diag}(25, 25, 50)$, so $\Sigma^{-1}\mu_e = (1, 2, 5)^\top$ with sum $k = 8 = B - r_f A$. The tangency portfolio is the normalisation

$$w_T = (0.125,\ 0.25,\ 0.625)^\top.$$

Its expected return is $0.125(5) + 0.25(9) + 0.625(11) = 9.75\%$, its variance $0.015625(0.04) + 0.0625(0.04) + 0.390625(0.02) = 0.000625 + 0.0025 + 0.0078125 = 0.0109375$, volatility 10.46%, and its Sharpe ratio $(9.75 - 1)/10.46 = 0.837$. Directly, $\mathrm{SR}^{*2} = \mu_e^\top \Sigma^{-1} \mu_e = 0.04(1) + 0.08(2) + 0.10(5) = 0.70$, $\mathrm{SR}^* = \sqrt{0.7} = 0.837$.

(b) Mixing cash with w_T traces mean $= r_f + \mathrm{SR}^* \sigma = 0.01 + 0.837\,\sigma$. At $\sigma = 10\%$ it delivers 9.37%, held as $a = 0.10/0.1046 = 0.956$ of wealth in w_T and 4.4% in cash. The minimum-variance portfolio has the same 10% volatility but mean 9%: the line beats the hyperbola by 0.37 points there, and it beats it everywhere except at the tangency point.

(c) $\mathrm{Cov}(R_i, R_T) = (\Sigma w_T)_i = (0.04(0.125),\ 0.04(0.25),\ 0.02(0.625)) = (0.005, 0.010, 0.0125)$, which is $\mu_e/k = \mu_e/8$. Dividing by $\mathrm{Var}(R_T) = 0.0109375$: $\beta = (0.457, 0.914, 1.143)$. Then $\beta_i(\mu_T - r_f) = \beta_i \times 8.75\% = (4.0\%, 8.0\%, 10.0\%) = \mu_e$ exactly: every alpha against the tangency portfolio is zero. The identity holds for any $w_T \propto \Sigma^{-1}\mu_e$ because $\Sigma w_T = \mu_e/k$ makes every covariance with the tangency portfolio proportional to the excess mean; no equilibrium assumption is involved.

(d) The minimum-variance portfolio has Sharpe ratio $(9 - 1)/10 = 0.80 < 0.837$. The two solve different problems: $w_g = \Sigma^{-1}\mathbf{1}/A$ never looks at μ, while $w_T \propto \Sigma^{-1}\mu_e$. They coincide iff $\Sigma^{-1}\mu_e \propto \Sigma^{-1}\mathbf{1}$, i.e. iff $\mu_e \propto \mathbf{1}$: every asset must have the same expected excess return, e.g. $\mu = (9\%, 9\%, 9\%)$ at $r_f = 1\%$, or any other constant mean vector.

(e) At $r_f = 9\% = B/A$, the minimum-variance mean, $k = B - r_f A = 0$: the maximum-Sharpe ray $\Sigma^{-1}\mu_e = (25(-0.04),\ 0,\ 50(0.02)) = (-1, 0, 1)$ sums to zero, so it cannot be normalised to a fully invested portfolio (it is a pure long–short spread with zero net capital), and no tangency point exists on the efficient frontier; for $r_f > 9\%$ the normalised point lies on the inefficient lower branch with Sharpe ratio $-\mathrm{SR}^*$. The condition for a tangency portfolio on the efficient branch is $r_f < B/A$.

Self-check. The tangency portfolio must lie on the frontier hyperbola $\sigma^2 = 1/A + (A/D)(m - B/A)^2$ with $D = AC - B^2 = 6$: at $m = 9.75\%$, $0.01 + 16.67(0.0075)^2 = 0.01 + 0.0009375 = 0.0109375$, the variance found in (a). Also $\mu_T - B/A = 0.75\% = D/(Ak) = 6/800$, and $\mathrm{SR}^{*2} = C - 2r_f B + r_f^2 A = 0.87 - 0.18 + 0.01 = 0.70$. The betas in (c) satisfy $\sum_i w_{T,i}\beta_i = 0.057 + 0.229 + 0.714 = 1.000$, as they must for a portfolio's beta to itself.

Answer: (a) $w_T = (0.125, 0.25, 0.625)$, mean 9.75%, volatility 10.46%, Sharpe $\sqrt{0.7} = 0.837$; (b) mean $= 0.01 + 0.837\sigma$; 9.37% at 10% volatility, held as 95.6% in w_T and 4.4% cash, versus 9% for the minimum-variance portfolio; (c) $\beta = (0.457, 0.914, 1.143)$, and $\beta_i \times 8.75\% = (4, 8, 10)\%$, all alphas zero; (d) $0.80 < 0.837$ because w_g ignores μ; they coincide iff all excess means are equal; (e) at $r_f = 9\% = B/A$ the tangency direction $(-1, 0, 1)$ sums to zero and no tangency portfolio exists on the efficient branch.

P11.10 *Beta, correlation and the hedge ratio*

(a) $\beta = \rho\,\sigma_i/\sigma_M$: $\beta_A = 0.5 \times 0.40/0.20 = 1.0$ and $\beta_B = 0.8 \times 0.15/0.20 = 0.6$. Stock B co-moves more tightly (correlation 0.8 against 0.5) yet has the smaller beta: beta carries the volatility ratio, and A is 2.67 times as volatile as B.

(b) Systematic variance $\beta^2\sigma_M^2$, idiosyncratic variance the remainder. A: $1.0^2(0.04) = 0.04$ (volatility 20%) and $0.16 - 0.04 = 0.12$ (volatility 34.6%), systematic share $0.04/0.16 = 25\% = \rho^2$. B: $0.36(0.04) = 0.0144$ (volatility 12%) and $0.0225 - 0.0144 = 0.0081$ (volatility 9%), systematic share $64\% = \rho^2$. The share is ρ^2 in both cases, the R^2 of the regression on the market.

(c) CAPM: $\mathbb{E}[R_i] = r_f + \beta_i(\mu_M - r_f) = 3\% + 6\%\,\beta_i$: 9% for A and 6.6% for B. Alphas are forecast minus CAPM value: $\alpha_A = 12 - 9 = 3\%$, $\alpha_B = 6.6 - 6.6 = 0$: B sits on the security market line, A is 3 points above it.

(d) The variance-minimising hedge ratio is beta: short 1.0 units of the market against A and 0.6 units against B. The hedged position has the idiosyncratic variance of (b): volatility $\sqrt{0.12} = 34.6\%$ for A (down from 40%, a 13% reduction) and $\sqrt{0.0081} = 9\%$ for B (down from 15%, a 40% reduction). The B hedge is better in the sense that matters, removing 64% of the variance against 25%, even though it uses the smaller hedge: beta sets the *size* of the hedge, correlation measures its *quality*, and the residual share is $1 - \rho^2$.

Self-check. Residual volatility computed the other way, $\sigma_i\sqrt{1-\rho^2}$: A gives $0.40\sqrt{0.75} = 34.6\%$, B gives $0.15\sqrt{0.36} = 9\%$, agreeing with (d). Under the CAPM, $\mathrm{SR}_i = \rho\,\mathrm{SR}_M$ with $\mathrm{SR}_M = 6/20 = 0.30$: A has $6\%/40\% = 0.15 = 0.5 \times 0.30$ and B has $3.6\%/15\% = 0.24 = 0.8 \times 0.30$, consistent with (c).

Answer: (a) $\beta_A = 1.0$, $\beta_B = 0.6$; B is more correlated, A has the larger beta; (b) A: $0.04 + 0.12$ (share 25%), B: $0.0144 + 0.0081$ (share 64%); (c) CAPM 9% and 6.6%; $\alpha_A = 3\%$, $\alpha_B = 0$; (d) short 1.0 and 0.6 market units; residual volatilities 34.6% and 9%; the B hedge is better because it removes $\rho^2 = 64\%$ of the variance.

P11.11 *Where a book's risk comes from: Euler contributions and risk parity*

(a) By the chain rule and $\nabla_w(w^\top\Sigma w) = 2\Sigma w$ (symmetry of Σ), $\nabla_w\sqrt{w^\top\Sigma w} = 2\Sigma w/(2\sqrt{w^\top\Sigma w}) = \Sigma w/\sigma_p$, whose i-th entry is $(\Sigma w)_i/\sigma_p$. For the decomposition, differentiate the identity $\sigma_p(tw) = t\,\sigma_p(w)$ with respect to t: the chain rule on the left gives $\sum_i w_i\,(\partial\sigma_p/\partial w_i)(tw)$, the right gives $\sigma_p(w)$; setting $t = 1$ yields $\sigma_p(w) = \sum_i w_i\,\partial\sigma_p/\partial w_i$, Euler's theorem for a function homogeneous of degree one. Cross-check with the first part: $\sum_i w_i(\Sigma w)_i/\sigma_p = w^\top\Sigma w/\sigma_p = \sigma_p$. (This is

Theorem 11.45.)

(b) $\Sigma = \begin{pmatrix} 0.0036 & 0.00324 \\ 0.00324 & 0.0324 \end{pmatrix}$ (off-diagonal $0.3 \times 0.06 \times 0.18$). With $w = (0.5, 0.5)$: $\Sigma w = (0.0018 + 0.00162,\ 0.00162 + 0.0162) = (0.00342, 0.01782)$; $\sigma_p^2 = 0.5(0.00342) + 0.5(0.01782) = 0.01062$, $\sigma_p = 10.31\%$. Marginal contributions $\Sigma w/\sigma_p = (0.0332, 0.1729)$: one extra point of equity weight adds 0.17 points of volatility, 5.2 times the bond marginal. Total contributions $w_i \times \mathrm{MRC}_i = (1.66\%, 8.65\%)$, which sum to $10.31\% = \sigma_p$; risk shares 16.1% bonds, 83.9% equities. Half the capital, over four fifths of the risk.

(c) With $(\Sigma w)_1 = w_1\sigma_1^2 + w_2\rho\sigma_1\sigma_2$ and $(\Sigma w)_2 = w_2\sigma_2^2 + w_1\rho\sigma_1\sigma_2$, the contributions are $w_1(\Sigma w)_1 = w_1^2\sigma_1^2 + w_1w_2\rho\sigma_1\sigma_2$ and $w_2(\Sigma w)_2 = w_2^2\sigma_2^2 + w_1w_2\rho\sigma_1\sigma_2$; equality forces $w_1\sigma_1 = w_2\sigma_2$ for positive weights, so $w \propto (1/\sigma_1, 1/\sigma_2)$: $w = (18, 6)/24 = (0.75, 0.25)$, three quarters in bonds. Its variance is $0.5625(0.0036) + 0.0625(0.0324) + 2(0.1875)(0.00324) = 0.002025 + 0.002025 + 0.001215 = 0.005265$, volatility 7.26% (closed form $\sigma_1\sigma_2\sqrt{2(1+\rho)}/(\sigma_1+\sigma_2) = 0.0108\sqrt{2.6}/0.24$), each side contributing 3.63%. Reaching the 10.31% of the 50/50 mix requires leverage $10.31/7.26 = 1.42$: 106% bonds and 35% equities, financed by borrowing 42%.

(d) The correlation entered both contributions through the identical term $w_1w_2\rho\sigma_1\sigma_2$, which cancelled from the equation; so the weights $(\sigma_2, \sigma_1)/(\sigma_1 + \sigma_2)$ are the same for every $\rho \in (-1, 1)$. What depends on ρ is the total: the volatility $\sigma_1\sigma_2\sqrt{2(1+\rho)}/(\sigma_1+\sigma_2)$, hence the leverage needed to hit a target (5.69% at $\rho = -0.2$, 8.05% at $\rho = 0.6$, against 7.26% at $\rho = 0.3$). With three or more assets the third-asset cross terms no longer cancel and the risk-parity weights are found numerically; inverse volatility is then only an approximation.

Self-check. At risk parity, $\Sigma w = (0.75(0.0036) + 0.25(0.00324),\ 0.75(0.00324) + 0.25(0.0324)) = (0.00351, 0.01053)$ and $w_i(\Sigma w)_i = (0.0026325, 0.0026325)$: equal, and their sum 0.005265 is the variance found in (c). In (b) the betas to the portfolio, $(\Sigma w)_i/\sigma_p^2 = (0.322, 1.678)$, satisfy $0.5(0.322) + 0.5(1.678) = 1.000$, the risk-share identity $\sum_i w_i\beta_{i,p} = 1$ in another form.

Answer: (a) $\mathrm{MRC}_i = (\Sigma w)_i/\sigma_p$ and $\sigma_p = \sum_i w_i\mathrm{MRC}_i$ by differentiating $\sigma_p(tw) = t\sigma_p(w)$ at $t = 1$; (b) $\sigma_p = 10.31\%$, MRC $= (0.033, 0.173)$, RC $= (1.66\%, 8.65\%)$, shares 16%/84%; (c) $w = (0.75, 0.25)$, $\sigma_p = 7.26\%$, leverage 1.42; (d) the ρ term cancels from the equal-contribution equation, so only the level of volatility (and the leverage) depends on ρ.

P11.12 *One percentage point of expected return: error amplification by the inverse covariance*

(a) $\Sigma = \begin{pmatrix} 0.0225 & 0.0135 \\ 0.0135 & 0.0225 \end{pmatrix}$, $\det\Sigma = 0.0225^2(1 - 0.36) = 0.000324$, so

$$\Sigma^{-1} = \frac{1}{0.000324}\begin{pmatrix} 0.0225 & -0.0135 \\ -0.0135 & 0.0225 \end{pmatrix} = \begin{pmatrix} 69.4 & -41.7 \\ -41.7 & 69.4 \end{pmatrix}.$$

For $(6\%, 6\%)$: $\Sigma^{-1}\mu_e = (1.667, 1.667)$, weights 50/50. For $(7\%, 6\%)$: $\Sigma^{-1}\mu_e = (4.861 - 2.500,\ -2.917 + 4.167) = (2.361, 1.250)$, sum 3.611, weights $(65.4\%, 34.6\%)$: one point of expected return moved 15 points of weight. For $(8\%, 4\%)$: $\Sigma^{-1}\mu_e = (5.556 - 1.667,\ -3.333 + 2.778) = (3.889, -0.556)$, sum 3.333, weights $(116.7\%, -16.7\%)$: a short position appears, a 100% directional holding plus a 16.7% long–short spread between two 60%-correlated assets.

(b) With $\Sigma = \sigma^2\left(\begin{smallmatrix} 1 & \rho \\ \rho & 1 \end{smallmatrix}\right)$, $\Sigma^{-1}\mu_e \propto (\mu_1 - \rho\mu_2,\ \mu_2 - \rho\mu_1)$. The sum of the entries is $(1 - \rho)(\mu_1 + \mu_2)$ and their difference $(1 + \rho)(\mu_1 - \mu_2)$; normalised weights are the entries over their sum, so $w_1 - w_2$ is the difference over the sum, which is the displayed formula. The eigenvalues of Σ are $\sigma^2(1 + \rho)$ on $(1, 1)$ and $\sigma^2(1 - \rho)$ on $(1, -1)$, so for $\rho \geq 0$ the factor $(1 + \rho)/(1 - \rho)$ is the condition number $\lambda_{\max}/\lambda_{\min}$; here $1.6/0.4 = 4$. Check: $(0.01/0.13) \times 4 = 0.308 = 0.654 - 0.346$, and $(0.04/0.12) \times 4 = 1.333 = 1.167 + 0.167$. At $\rho = 0$ the $(7\%, 6\%)$ case would give $(53.8\%, 46.2\%)$, a 3.8-point move; the extra factor of four is entirely the correlation.

(c) Let $\Sigma = \sum_i \lambda_i q_i q_i^\top$ with orthonormal q_i and $\lambda_i > 0$, and expand $\delta = \sum_i z_i q_i$, so $\|\delta\|^2 = \sum_i z_i^2$.

Since $\Sigma^{-1}q_i = q_i/\lambda_i$, $\Sigma^{-1}\delta = \sum_i (z_i/\lambda_i)q_i$ and by orthonormality

$$\|\Sigma^{-1}\delta\|^2 = \sum_i \frac{z_i^2}{\lambda_i^2} \le \frac{1}{\lambda_{\min}^2}\sum_i z_i^2 = \frac{\|\delta\|^2}{\lambda_{\min}^2},$$

with equality iff $z_i = 0$ for every i with $\lambda_i > \lambda_{\min}$, i.e. iff δ lies in the eigenspace of $\lambda_{\min}$. Here $\lambda_{\min} = 0.0225(0.4) = 0.009$, so $1/\lambda_{\min} = 111.1$. For $\delta = (0.01, 0)$: $\Sigma^{-1}\delta = (0.694, -0.417)$ with norm 0.810, against the bound 1.111 (the perturbation half-loads the spread direction). For $\delta = (0.01, -0.01)/\sqrt{2}$, which is $0.01\, q_{\min}$: $\Sigma^{-1}\delta = (0.786, -0.786)$ with norm 1.111, the bound attained.

(d) The average of T iid annual returns has standard error $\sigma/\sqrt{T} = 0.15/\sqrt{20} = 3.4$ percentage points. The one-point gap between $(7\%, 6\%)$ and $(6\%, 6\%)$ is less than a third of a standard error: the data cannot distinguish the two mean vectors, yet the optimiser moves 15 points of weight between them, and a one-standard-error error on each mean in opposite directions (6.7 points of gap) would carry the weights well past the $(8\%, 4\%)$ short. Remedies: shrink the covariance toward a structured target (Ledoit–Wolf), which raises the small eigenvalues that Σ^{-1} amplifies; impose long-only or position constraints (Jagannathan–Ma), which act as implicit shrinkage and remove the spread first; and shrink or avoid μ altogether, through minimum variance, risk parity, or the $1/N$ benchmark that sample-based optimisers rarely beat out of sample. Then size total leverage as fractional Kelly rather than the raw $\Sigma^{-1}\mu_e$.

Self-check. $\Sigma\Sigma^{-1}$: $0.0225(69.44) + 0.0135(-41.67) = 1.5625 - 0.5625 = 1.000$ and $0.0225(-41.67) + 0.0135(69.44) = -0.9375 + 0.9375 = 0$, so the inverse is right. In (c), $\lambda_{\max} = 0.036$ and $\lambda_{\min} = 0.009$ sum to the trace 0.045 and multiply to $\det\Sigma = 0.000324$. The eigen-decomposition also recovers (c): $\Sigma^{-1}(0.01, 0) = 0.005\,[(1,1)/0.036 + (1,-1)/0.009] = (0.139 + 0.556,\ 0.139 - 0.556) = (0.694, -0.417)$.

Answer: (a) $\Sigma^{-1} = \left(\begin{smallmatrix} 69.4 & -41.7 \\ -41.7 & 69.4 \end{smallmatrix}\right)$; weights 50/50, then $(65.4\%, 34.6\%)$, then $(116.7\%, -16.7\%)$; (b) the relative mean gap is amplified by $(1+\rho)/(1-\rho) = \lambda_{\max}/\lambda_{\min} = 4$; (c) expand in the eigenbasis: $\|\Sigma^{-1}\delta\| \le \|\delta\|/\lambda_{\min}$ with equality on the $\lambda_{\min}$ eigenspace; 0.810 versus bound 1.111 for $(0.01, 0)$, and 1.111 = bound for $(0.01, -0.01)/\sqrt{2}$; (d) standard error 3.4 points, so the one-point gap is noise; remedies: shrinkage of Σ, weight constraints, shrinking or avoiding μ ($1/N$, minimum variance, risk parity), with fractional-Kelly leverage.

28.12 Solutions — §12 Market Making I: the Economics of the Spread

P12.1 *Spread decomposition*

The spread decomposes into four components:

- **Adverse selection** — the flow that chooses to trade against your quote is biased against you (informed pickoffs); the spread is the insurance premium.
- **Inventory risk** — the price can move while you hold the position a fill gives you.
- **Order-processing costs and fees** — infrastructure, exchange fees, clearing.
- **Margin** — the maker's profit.

Volatility widens the first two: when prices move more, informed traders' private information is worth more (their pickoff profit per fill grows), and any inventory you are handed is riskier to carry. Competition compresses the margin (and squeezes operational slack out of processing costs) but cannot push the spread below the actuarial cost of the first two components — a maker quoting inside that cost loses money in expectation and exits.

Self-check (sanity bound): the Glosten–Milgrom miniature confirms the floor independently: with zero fees, zero inventory cost and zero profit margin, the breakeven spread is still $2\pi > 0$ whenever a fraction $\pi > 0$ of flow is informed. So "competition drives the spread to zero" fails exactly when adverse selection exists, which is consistent with the decomposition.

Answer: spread = adverse selection + inventory risk + order-processing/fees + margin; volatility widens the first two; competition compresses the margin.

P12.2 *Glosten-Milgrom zero-profit quotes*

(a) Condition on the side that trades. An informed trader buys only when $V = 101$; noise buys half the time regardless:

$$P(\text{buy} \mid V{=}101) = \pi + \tfrac{1-\pi}{2} = 0.65, \qquad P(\text{buy} \mid V{=}99) = \tfrac{1-\pi}{2} = 0.35, \qquad P(\text{buy}) = \tfrac{1}{2}.$$

By Bayes, $P(V{=}101 \mid \text{buy}) = \frac{0.65 \times 0.5}{0.5} = 0.65$. The zero-profit ask is the value *conditional on being lifted*:

$$a = \mathbb{E}[V \mid \text{buy}] = 99 + 2(0.65) = 100.30,$$

and by symmetry the bid is $b = \mathbb{E}[V \mid \text{sell}] = 99.70$.

(b) Breakeven spread $= a - b = 2\pi = 0.60$. The friend's error: *unconditional* flow is indeed 50/50, but the flow that trades against a *fixed* quote is conditionally biased against it — your resting ask is lifted disproportionately in the states where $V = 101$. Quotes must be set at the expected value conditional on being traded with, not at the unconditional mean. Any spread below 0.60 loses in expectation even with zero fees, zero inventory cost, and no profit margin.

Self-check (direct P&L): at ask a, conditional on being lifted, expected value delivered is $101 \times 0.65 + 99 \times 0.35 = 100.30$, so expected profit per lifted offer is $a - 100.30$: zero exactly at $a = 100.30$, negative below. Small-case check: at $\pi = 0$ the formula gives spread 0 — the friend's claim is precisely the no-informed-flow special case.

Answer: ask 100.30, bid 99.70; breakeven spread $2\pi = 0.60$. The claim ignores that the flow hitting a fixed quote is conditionally biased against it — quote the value conditional on being hit. □

P12.3 *Glosten-Milgrom with asymmetric prior*

(a) Buy side: $P(\text{buy} \mid 101) = \pi + \frac{1-\pi}{2} = 0.6$ and $P(\text{buy} \mid 99) = \frac{1-\pi}{2} = 0.4$, so

$$P(\text{buy}) = 0.6(0.6) + 0.4(0.4) = 0.52, \qquad P(101 \mid \text{buy}) = \frac{0.6 \times 0.6}{0.52} = \frac{9}{13} \approx 0.692.$$

Ask $= \mathbb{E}[V \mid \text{buy}] = 99 + 2 \cdot \frac{9}{13} = 100 + \frac{5}{13} \approx 100.385$. Sell side: $P(\text{sell} \mid 101) = 0.4$, $P(\text{sell} \mid 99) = 0.6$, so $P(\text{sell}) = 0.6(0.4) + 0.4(0.6) = 0.48$ and $P(101 \mid \text{sell}) = \frac{0.24}{0.48} = \frac{1}{2}$: bid $= 99 + 2(\frac{1}{2}) = 100.00$ exactly.

(b) Spread $= \frac{5}{13} \approx 0.385 < 2\pi = 0.4$: the asymmetric prior has *less* uncertainty (variance $4p(1-p)$ peaks at $p = \frac{1}{2}$), so there is less for the informed to know that you do not, and the insurance premium shrinks. The quote midpoint is $100 + \frac{5}{26} \approx 100.192$, *not* the prior mean 100.20. The bid updates further (0.20 down vs 0.185 up for the ask): a *sell* against the bullish prior is the more surprising event, so it moves the posterior more ($0.6 \to 0.5$ on a sell vs $0.6 \to 0.692$ on a buy).

(c) $p \to 1$: both posteriors $\to 1$, so ask = bid = 101 and the spread $\to 0$ — no private information left to protect against. $\pi \to 0$: posteriors collapse to the prior, ask = bid = 100.2, spread $\to 0$ — no informed flow. Both limits behave.

Self-check (recompute the spread another way): spread $= 2\,[P(101 \mid \text{buy}) - P(101 \mid \text{sell})] = 2\left(\frac{9}{13} - \frac{1}{2}\right) = 2 \cdot \frac{5}{26} = \frac{5}{13}$, matching ask − bid $= 100\frac{5}{13} - 100$. Also $P(\text{buy}) + P(\text{sell}) = 0.52 + 0.48 = 1$.

Answer: ask $= 100 + \frac{5}{13} \approx 100.38$, bid $= 100.00$; spread $\frac{5}{13} \approx 0.385 < 2\pi$; midpoint $100.19 \neq$ prior mean 100.20, and the bid side updates further because a sell is the more surprising trade under a bullish prior. □

P12.4 *Mark-outs*

(a) **Mark-outs**: the mid-price change t seconds after your fill, signed so that negative means the market moved against your trade, bucketed by counterparty, venue, and size. The average mark-out

on your fills *is* realized adverse selection — an estimator of the informed-flow premium (the π of the Glosten–Milgrom model) computed from your own trades. A mark-outs-by-counterparty dashboard panel is the concrete, operational answer.

(b) Counterparty X's flow is toxic: consistently negative mark-outs mean X trades when you are about to be wrong. Widen or pull *for X specifically* if the venue lets you differentiate (or against the patterns that precede X's fills); cap the size you show X. If X's flow cannot be distinguished from the rest, you must widen for everyone — indistinguishable toxic flow raises the average toxicity of all flow.

(c) Volatility doubled: **widen**. Both vol-sensitive spread components (adverse selection and inventory risk) scale up; the flow itself has not become more informed, so you keep quoting, just at a spread that re-prices the new risk.

(d) Imminent unscheduled news: **pull first**, re-quote wide once the news is out, tighten as the picture clears. Around an announcement, the informed/faster fraction of flow spikes and the price can gap through your quote; being on the wrong side of one gap costs many spreads.

Self-check (map each action to the spread component that changed): (b) adverse selection rose *for one bucket* $\rightarrow$ widen/pull for that bucket; (c) σ rose $\rightarrow$ the two vol-driven components rise $\rightarrow$ widen; (d) momentarily $\pi \rightarrow$ large, so the breakeven spread 2π can exceed any spread that would still get filled — quoting nothing *is* the zero-profit quote. Each action is exactly the decomposition's prescription.

Answer: (a) mark-outs (post-fill mid drift) by counterparty/venue/size; (b) widen/pull/size-cap against X, or widen for all if unidentifiable; (c) widen; (d) pull, re-quote wide, tighten as the picture clears.

28.13 Solutions — §13 Market Making II: Inventory and Skewing

P13.1 *Reservation price computation*

(a) The inventory shift is $q\gamma\sigma^2(T-t) = 60 \times 0.001 \times 2.5 \times 0.4 = 0.06$, so

$$r = 100 - 0.06 = 99.94.$$

(b) Quotes: bid 99.84, ask 100.04. Compared with mid-centered quotes (99.90/100.10), *both* are shifted down by 0.06: the ask undercuts, so your offer is more likely to be lifted (you sell the long more eagerly); the bid backs off, so you are less likely to be hit into an even bigger long. The book leans toward shedding inventory while still quoting two-sided. (c) With $q = -60$ the shift flips sign: $r = 100.06$, quotes 99.96 bid at 100.16 ask — now leaning to buy the short back.

Self-check: recompute the shift stepwise: $60 \times 0.001 = 0.06$; $\times 2.5 = 0.15$; $\times 0.4 = 0.06$. Sign logic: long ($q > 0$) must give $r < s$ (a long wants to sell) — it does; $q = 0$ recovers mid-centered quotes; flipping q flips the shift exactly, by linearity.

Answer: (a) $r = 99.94$; (b) 99.84 bid at 100.04 ask — skewed to sell down the long; (c) $r = 100.06$, quotes 99.96 at 100.16.

P13.2 *Reservation-price anatomy, skew vs widen*

(a) The reservation price is the price at which you are indifferent between holding and shedding a marginal unit of inventory. Carrying q units to the horizon adds P&L variance $q^2\sigma^2(T-t)$; under exponential (mean-variance) utility the *marginal* risk cost of one more unit is proportional to $q\gamma\sigma^2(T-t)$. So: q sets the direction and urgency (more inventory, bigger concession); γ converts variance into the price concession you are willing to pay; σ^2 is the rate at which risk accrues; $(T-t)$ is how long you must carry it.

(b) As $T - t \rightarrow 0$ the skew vanishes: with no time left to hold the position, inventory carries (in the model) no further risk, so you quote around mid. Internally consistent — though real desks

add an explicit terminal liquidation penalty (you cannot magically flatten at the bell for free), which keeps end-of-horizon skews alive in practice.

(c) Shifting both quotes sheds inventory through *both* fill channels at once: a lower ask raises the probability your offer is lifted (you sell), and a lower bid reduces the probability your bid is hit (you avoid buying more) — while you remain two-sided, keeping the flow, the queue position, and the information that comes with being in the market. Improving only the ask gives away edge on one side and does nothing to slow accumulation on the other.

(d) **Widening** prices *uncertainty* — adverse selection and volatility — and is symmetric: it addresses not knowing. **Skewing** manages a *known* directional position and is asymmetric: it addresses knowing exactly what you hold. A vol spike widens; an inventory build skews; toxic flow may widen or pull.

Self-check (limits and dimensions): $\gamma \to 0$ (risk-neutral) gives $r = s$ — no concession, consistent with (a); $\sigma \to 0$ likewise kills the skew. Dimensionally, $q \cdot \gamma \cdot \sigma^2 \cdot (T - t)$ carries [units] $\times$ [1/(price $\times$ units)] $\times$ [price^2/time] $\times$ [time] = [price], as a price correction must.

Answer: the correction is the marginal risk cost of carrying q to horizon; skew $\to 0$ as $T - t \to 0$ (no time left to bear risk); both quotes move so inventory sheds through both fill probabilities while staying two-sided; widening prices uncertainty, skewing manages a known position. □

P13.3 *Fermi estimation + interval calibration* ★

The point estimate is a rate $\times$ time chain said fast: 6 blocks/hour $\times 24 = 144$/day; $\times 365 \approx$ 52,600/year; $\times 14$ years (2012 to 2026) $\approx$ 735,000. The 15-second version: 50k/yr $\times 14 = 700$k. Any answer of the form "about 700–750k" is full marks on the point.

The interval is the actual test. On a number you produced in 15 seconds from a possibly-misremembered rate, an honest 95% interval must be *embarrassingly wide* — order-of-magnitude, e.g. 200k to 2M (point estimate $\times/\div$ 3 is a good default when any link in the chain is soft). What it must *not* be is a memorized-fact interval: at the actual interview the candidate said $\approx$ 100k with interval 98k–120k, and the truth ($\approx$ 735k) fell far outside — that is the graded failure, because a 95% interval that misses is a direct demonstration of miscalibration. The one wrong move is refusing to give an interval at all: the interval *is* the statement of your uncertainty. Where should the width come from? The 10-minute rate is handed to you and the 14-year window is known, so the width prices your own arithmetic under pressure and any soft facts (e.g. if you are not sure whether halvings change the block rate — they do not, difficulty retargets to hold 10 minutes — your interval must cover being wrong about that).

The drill (how to train this): pick any quantity, produce point + 95% interval in 15 seconds, repeat many times, and score *coverage*: about 19/20 intervals should contain the truth. Missing half the time means far too narrow; never missing means too wide (you are giving away edge).

Self-check (independent recomputation via minutes): $14 \times 525{,}600 \approx 7.36$M minutes; divide by $10 \Rightarrow 736$k — matches the per-day chain. Coverage check: $[200\text{k}, 2\text{M}]$ contains 736k comfortably, while $[98\text{k}, 120\text{k}]$ does not — the calibration lesson in one line.

Answer: $\approx$ 735,000 blocks (144/day $\approx$ 52.6k/yr $\times 14$); a calibrated 95% interval is order-of-magnitude wide, e.g. 200k–2M — and refusing to give one is the only losing move.

P13.4 *Make-a-market game: position and P&L accounting* ★

Bookkeeping convention first (this is where the game is won or lost): the counterparty's BUY lifts your *offer* — you SOLD one unit at your ask; their SELL hits your *bid* — you BOUGHT one unit at your bid. So the tape reads: sold at 200k, sold at 600k, sold at 1M, bought at 4M.

Position: 3 sold − 1 bought = **short 2 units**. (At the live interview the candidate answered "short 2 million" — *short* and *2* were right, but "million" conflated position size with price level. The clean answer is "short 2 units.")

Cash (millions of block-units): $+0.2 + 0.6 + 1.0 - 4.0 = -2.2$M.

Mark-to-market at the final mid 2.5M: equity = cash $+q\times$ mid $= -2.2+(-2)(2.5) = -7.2$M — the book is down badly. (Marked at the true value ≈ 0.735M instead: $-2.2+(-2)(0.735) \approx -3.7$M; down either way — three sales went off far below value before the buy printed far above it.)

The flow read: repeated buys mean your offer is cheap — jump the market in multiples, not ticks, until the flow flips (the candidate did move 100–200 → 400–600 → 900k–1M → 4–6M, which is the right behavior). The flip is the information payoff: they *bought* at your 1M offer and later *sold* at your 4M bid, so they are acting as if the value lies in $(1\text{M}, 4\text{M})$ — and the final quote 2M–3M correctly sits inside that bracket. The graded checklist: (1) width = confidence; (2) move with the flow, never repeat an offer; (3) always know your inventory — the closing question is the whole point; (4) know your rough P&L. The game grades *process*, not proximity to the true number.

Self-check (recount two independent ways): by side: offers lifted at 200k, 600k, 1M (3 sales), bid hit at 4M (1 buy), net -2. Cash: received $0.2 + 0.6 + 1.0 = 1.8$M, paid 4.0M, net -2.2M. Equity by liquidation thought-experiment: buying back 2 units at the final mid costs $2 \times 2.5 = 5.0$M, leaving $-2.2 - 5.0 = -7.2$M with a flat book — matches cash $+q\times$ mid.

Answer: short 2 *units*; cash -2.2M; mark-to-market ≈ -7.2M at the final mid 2.5M; the buy-then-sell flip brackets the counterparty's implied value inside $(1\text{M}, 4\text{M})$.

P13.5 *Pre-trade risk controls / position limits* ♦

(C). Hard, automatically enforced pre-trade position limits are the standard control: a risk layer rejects any order that would push the position beyond its limit, mechanically bounding worst-case inventory *no matter what* the upstream logic does — bug, bad data, or runaway loop. (A) and (B) are detective controls with human latency: in fast markets a runaway algorithm builds a catastrophic position in seconds, and a dashboard or alert only tells you it already happened. (D) reduces the probability of a bug but cannot bound the loss when one slips through — and one always can. Note the design detail inside (C): block only the *breaching* side; the algorithm must remain free to trade the position back down, otherwise the limit locks in the very risk it was meant to cap.

Self-check (control-theory sanity test): a control that bounds the bad state *ex ante* dominates one that reports it *ex post* whenever the state can move faster than the responder — algorithmic trading is exactly that regime, and (C) is the only ex-ante binding option. Consistent, too, with inventory economics: the loss tail scales with $|q|$, so capping $|q|$ caps the tail.

Answer: (C) — hard position limits that automatically block further trading on the breaching side.

P13.6 *Inventory hedging with perps*

Hedge externally: short the perpetual future on the same or another venue B against the long spot on A. Your delta is now approximately flat, so you can *keep quoting* — keep earning the spread and keep seeing the flow — while carrying basis instead of direction. The perp costs (or earns) the funding rate rather than directional exposure, and you can unwind the pair slowly: route future sell flow to close the spot leg while buying back the perp, or let both legs run off.

Residual risks:

- *Basis risk* — the perp-spot spread can move against the pair.
- *Funding cost* — if funding runs against your side, the hedge bleeds carry while it is on.
- *Venue/counterparty risk* — collateral now sits on venue B; where inventory and margin live overnight is a first-class risk decision.
- *Margin/liquidation risk on the short perp leg* — a violent spike can stop the hedge out exactly when it is most needed.

- *Execution cost* of putting the hedge on (spread, fees, impact on B).

The colleague's proposal fails twice: pulling quotes forfeits the franchise (flow, queue position, information) *and* does nothing about the inventory you already hold. The larger lesson: at size, inventory management becomes a *routing and financing* problem — where to hedge, what carry to pay, where collateral sits — not just a quoting problem.

Self-check (P&L decomposition): long spot + short perp = basis position + funding leg; the directional term cancels by construction (both legs carry $\approx$ unit delta), so the remaining exposures are exactly the residuals listed — nothing directional left unaccounted. Comparison bound: dumping Q into your own book costs on the order of the impact ($\propto \sqrt{Q}$, large at size) *and* signals your position; the hedge caps the cost at spread + fees + basis.

Answer: short perps against the long spot (same or another venue): stay quoting, carry funding/basis instead of direction; residual risks are basis moves, funding cost, venue/counterparty exposure, margin/liquidation on the hedge leg, and hedge execution cost.

28.14 Solutions — §14 Market Microstructure

P14.1 *Square-root impact law by dimensional analysis*

Write the units equation:

$$\text{price} \;=\; \text{shares}^{\alpha}\left(\frac{\text{shares}}{\text{day}}\right)^{\beta}\left(\frac{\text{price}}{\text{day}^{1/2}}\right)^{\gamma} \;=\; \text{shares}^{\alpha+\beta}\,\text{day}^{-\beta-\gamma/2}\,\text{price}^{\gamma}.$$

Matching exponents on each unit: price $\Rightarrow \gamma = 1$; shares $\Rightarrow \alpha + \beta = 0$; time $\Rightarrow -\beta - \frac{\gamma}{2} = 0 \Rightarrow \beta = -\frac{1}{2}$, hence $\alpha = \frac{1}{2}$. So

$$I = Y\,\sigma\sqrt{\frac{Q}{V}},$$

the square-root law, with Y dimensionless.

What the ansatz sneaked in: the entire argument rests on the premise that Q, V, σ are the *only* relevant variables (no tick size, no spread, no execution duration or participation rate as an independent quantity) and that the law is a pure power law with a universal dimensionless Y. Dimensional analysis then shows the square root is the *only* consistent power law in those three variables — but the variable-selection premise is doing the heavy lifting; it is the empirics (roughly $Y \approx 1$ across markets and epochs) that justify it.

Self-check (plug back and scale): units of $\sigma\sqrt{Q/V}$: $(\text{price}\cdot\text{day}^{-1/2}) \times (\text{shares}/(\text{shares}/\text{day}))^{1/2} = \text{price}\cdot\text{day}^{-1/2}\cdot\text{day}^{1/2} = \text{price}$. Scaling sanity: doubling both Q and V leaves impact unchanged — impact depends on your size *relative to the market*, as intuition demands.

Answer: $\gamma = 1$, $\beta = -\frac{1}{2}$, $\alpha = \frac{1}{2}$, i.e. $\Delta P = Y\sigma\sqrt{Q/V}$; the hidden assumption is that Q, V, σ (in power-law form, universal Y) are the only variables that matter. □

P14.2 *Impact vs alpha capture*

(a) $Q/V = 250\text{k}/10\text{M} = 0.025$; $\sqrt{0.025} \approx 0.158$; so

$$\Delta P \approx 2\% \times 0.158 \approx 0.32\% = 32 \text{ bp}.$$

(b) **No.** Your own impact (≈ 32 bp) exceeds the 25 bp alpha — at this size the trade destroys its own edge; even the *average* execution cost, a substantial fraction of the peak impact, leaves nothing after spread and fees. Since impact scales as $\sqrt{Q}$ while the per-share alpha does not depend on your size, shrinking the order restores positive expectancy: e.g. 25k shares gives $\sqrt{0.0025} = 0.05$, impact ≈ 10 bp against 25 bp of alpha. (c) The Almgren–Chriss trade-off. Slicing over five days does cut the impact, but (i) a *two-hour* alpha is long gone by day five — you would pay real

costs to acquire a position whose forecast has expired; and (ii) slower execution bears more risk of adverse price moves while you are still un-executed. Too big eats the alpha, too small misses it, too slow risks the move.

Self-check (recompute (a) differently): $Q/V = 1/40$, $\sqrt{1/40} = 1/\sqrt{40} \approx 1/6.32 \approx 0.158$, so $2\% \times 0.158 = 31.6$ bp. Robustness bound for (b): even using the common refinement cost $\approx \frac{2}{3}\times$ peak impact ≈ 21 bp, the margin over 25 bp alpha is ≈ 4 bp before spread and fees — effectively zero, so the conclusion stands.

Answer: (a) ≈ 32 bp; (b) no — impact is larger than the 25 bp alpha at this size (trade smaller, e.g. 25k shares for ≈ 10 bp impact); (c) the alpha decays long before the slices execute, and slower execution adds adverse-move risk.

P14.3 *Kyle's lambda and reconciliation with concave impact*

(a) λ is Kyle's lambda: the price move per unit of signed order flow — a direct measure of *illiquidity* (the inverse of depth). Uses: comparing liquidity across assets, venues, and time; pre-trade cost models and sizing (an impact budget); flagging liquidity droughts when λ spikes.

(b) (i) Linear: $4\times$ the flow $\Rightarrow 4 \times 2 = 8$ bp. (ii) Square root: $\sqrt{4} = 2\times \Rightarrow 4$ bp.

(c) Empirically the local slope is not constant in trade size: fitted per size bucket, $\lambda(Q) \propto Q^{-1/2}$, so total impact $\lambda(Q)\,Q \propto \sqrt{Q}$ — concave. A single linear regression estimates an *average* slope over whatever size distribution you sampled; extrapolating it linearly to $4\times$ the typical size overstates impact. The square-root prediction (4 bp) is the one supported by metaorder studies. Economically, large orders move the price less per share because they are deliberately worked where and when liquidity is available, and convey proportionally less information per share.

Self-check (consistency identity): if total impact is $I(Q) = c\sqrt{Q}$, then the local slope is $I(Q)/Q = c\,Q^{-1/2}$ — exactly the stated empirical scaling $\lambda(Q) \propto Q^{-1/2}$, so the two statements are one law. And both models agree at the calibration size by construction ($Q = 10$k $\Rightarrow 2$ bp), diverging only out of sample — which is why in-sample regression fit cannot pick between them; size-bucketed fits can.

Answer: (a) λ = illiquidity: price move per unit signed volume; (b) 8 bp linear vs 4 bp square-root; (c) empirically $\lambda(Q) \propto Q^{-1/2}$ so total impact $\propto \sqrt{Q}$ — trust the concave, 4 bp prediction. □

P14.4 *OBI and VAMP computation*

(a) $\text{OBI} = \frac{1000-100}{1100} = \frac{900}{1100} \approx +0.82$ (strong buying pressure);

$$P_{\text{VAMP}} = \frac{100 \times 1000 + 99 \times 100}{1100} = \frac{109{,}900}{1100} \approx 99.91.$$

(b) The 1,000-share bid wall means sellers must chew through enormous resting demand to push the price down, while a mere 100 shares of buying lifts the whole ask: the next move is far more likely up than down, so the "true" price sits near the ask. Weighting the *ask price* by the *bid size* encodes exactly this — a heavy bid pins value toward the ask. The plain mid (99.50) ignores the pressure; VAMP (≈ 99.91) reads it. (The naive same-side weighting would do the opposite and put the price near the crowded side — backwards.) (c) Precise answer: this is the *volume-weighted (imbalance-weighted) mid*, a first-order proxy. Stoikov's formal *micro-price* is defined as the limit of expected future mids conditional on the book state; the weighted mid is its first-order approximation. Making that naming distinction out loud is worth doing at an HFT-adjacent shop.

Self-check (boundary cases): $q_b = q_a$ gives OBI = 0 and VAMP = plain mid — correct neutral limit; $q_b \gg q_a$ sends VAMP $\to P_a$, matching the near-certain uptick logic of (b); and VAMP is a convex combination of P_b and P_a (weights $\frac{1000}{1100} + \frac{100}{1100} = 1$), so it always lies inside the quoted spread.

Answer: OBI $\approx +0.82$; $P_{\text{VAMP}} \approx 99.91$, pinned near the ask; it is the weighted mid — a first-order proxy for Stoikov's micro-price (expected future mid), not the formal micro-price itself.

P14.5 *OBI signal monetisation*

Three independent killers:

- **The economics do not clear the spread.** A 58% hit rate on a one-tick move is an expected edge of $(0.58 - 0.42) \times 1 = 0.16$ ticks per trade, while crossing the spread costs a full spread (≥ 1 tick) plus taker fees, *twice* per round trip. Cost exceeds edge by construction — short-horizon book signals are usually worth a fraction of the spread.
- **Latency, queues, and adverse fill timing.** By the time you observe OBI = 0.8 and your order arrives, faster participants have repriced; the thin ask you meant to lift is gone precisely when the signal was right. You get filled preferentially in the states where the move already happened or the book refilled — the live fill distribution is strictly worse than the backtest's, which assumed you trade the quote you saw.
- **Displayed depth is not commitment, and the signal double-counts.** Resting size can be flickered or spoofed, inflating backtest OBI; and OBI is highly autocorrelated, so "58% of ticks" largely re-samples the same persistent episodes rather than 58% of independent bets. Worse, the payoff is asymmetric: the occasions when the thin side is swept *through* — exactly when you buy the top — are the expensive ones.

How a maker monetises it: passively. Use OBI/VAMP as a quote-*placement* input: lean (skew) quotes toward the imbalance — quote tighter/earlier on the side the price is drifting toward, back away on the other side — so the drift is captured while *earning* the spread instead of paying it; and use extreme imbalance as a pull/fade trigger to avoid being picked off on the stale side. The general principle: a signal smaller than the spread pays the passive, not the aggressive.

Self-check (net-expectancy bound): aggressive round trip: +0.16 ticks of edge − ≥ 1 tick of spread $-2\times$ taker fees < 0, robustly negative *granting* the 58%; passive use collects spread + drift-conditional fill quality, the standard maker P&L on flow-anticipating signals — signs opposite, same signal, which is the whole point.

Answer: taking pays ≥ 1 tick + fees for ≈ 0.16 ticks of edge; live fills arrive after faster players reprice (worse than backtest by selection); displayed imbalance is spoofable/flickery and autocorrelated with asymmetric downside. Monetise it passively: skew quote placement with the imbalance and pull on extremes — earn the spread, don't pay it.

28.15 Solutions — §15 Crypto Market Structure

P15.1 *Bitcoin mechanics* ★

Answer in the four beats the question asks for.

(1) Getting a transaction included. A user signs a transfer with their private key and broadcasts it; pending transactions sit in every node's *mempool*. Each transaction attaches a fee, and block space is scarce, so miners sort candidates by fee rate — attaching a higher fee is how a user buys priority. In congestion, fees spike and low-fee transactions wait.

(2) Mining a block. Miners assemble mempool transactions into a candidate block and search for a nonce such that the SHA-256 hash of the block header falls below a difficulty target (proof of work). Finding one is a lottery with win rate proportional to hash power; the difficulty retargets every 2016 blocks so that blocks keep arriving roughly every 10 minutes — about 144 per day.

(3) The chain. Every block header contains the hash of the *previous* block header, so the blocks form a hash-linked chain: changing any historical transaction changes that block's hash and breaks every later link, so rewriting history means redoing the proof of work for the entire suffix. Nodes adopt the chain with the most cumulative work; that is why a transaction buried under several blocks (the 6-confirmation convention) is treated as final.

(4) Miner pay. Two sources: the *block subsidy* of newly minted bitcoin (halving every 210,000

blocks, roughly 4 years: $50 \to 25 \to 12.5 \to 6.25 \to 3.125\ldots$), plus the sum of the fees of the transactions included.

Self-check (numbers hang together): total supply $= 210{,}000 \times 50 \times (1 + \frac{1}{2} + \frac{1}{4} + \cdots) = 210{,}000 \times 100 =$ 21M — the famous cap drops out of the halving schedule. Cadence: 6/hr $\times$ 24 = 144 blocks/day, the same anchor as the blocks-since-2012 Fermi estimate ($\approx$ 52.6k/yr).

Trading relevance (worth one closing sentence): the 10-minute cadence plus the fee market means on-chain transfers have stochastic latency — exactly the friction cross-venue crypto arbitrage has to cross, and why inventory logistics matter as much as quoting.

Answer: Blocks are mined by a proof-of-work lottery roughly every 10 minutes; each block hashes the previous one into a chain and the most-work chain wins; users attach fees to get transactions included; miners are paid the block subsidy of newly minted coins (halving every $\sim$4 years) plus transaction fees.

P15.2 *Cross-exchange arbitrage, leg risk* ♦

Go through the options against the economics of the trade, whose edge is a few basis points per round trip.

- *Fees (a):* known constants, priced in *ex ante* — if fees exceeded the discrepancy you would simply not fire. They cannot surprise you after the fact.
- *Latency (b):* missing an opportunity is an opportunity cost with zero realized loss. Annoying, not dangerous.
- *Withdrawal times (d):* a real friction, but it is a capital-efficiency and inventory-logistics problem spread across many trades (pre-position inventory on both venues), not the dominant per-trade risk.
- *Leg risk (c):* you intended to be flat; if only the buy (or only the sell) fills, you hold a naked directional position in a volatile asset until you chase the missing leg or unwind — at whatever price the market has moved to. This is the tail that eats the strategy.

Self-check by magnitudes: suppose the edge is 3 bp and the coin runs 80%/yr volatility. Five minutes of vol is $0.80/\sqrt{365 \times 288} \approx 0.80/324 \approx 25$ bp — a one-sigma move while you scramble to complete the leg is nearly ten times the edge, so a single bad leg-out erases dozens of successful arbs. Fees, by contrast, are bounded by construction below the edge.

Answer: (c) — leg risk: a partial fill leaves a large unhedged delta position; fees and latency are secondary distractors.

P15.3 *The market-maker mandate* ♦

Neither — and recognizing that the question is a trap is the point.

A designated market maker's contract is about *liquidity*, not levels or activity: presence in the book, a maximum spread, a minimum quoted depth, an uptime percentage, across named venues. Consider what promising each forbidden thing would mean:

- *Higher prices:* committing to support a price level means buying whatever supply arrives at that level — economically, writing an uncapped put on the token for a fee. No market-neutral liquidity business can carry that, and deliberately propping a price is market manipulation (painting a floor).
- *Higher volumes:* volume is generated by other participants; the only way a single firm can *guarantee* volume is to trade with itself — wash trading, again manipulation.

What the project actually buys is *tradability*: tight spreads and real depth lower the cost for anyone to enter or exit, which may well make the token more attractive — but price and volume are *outcomes* of the market, never deliverables of the market maker. This is the compliance-correct framing and the one a trading firm wants to hear you give unprompted.

If the format forces a choice between the two (as the live MCQ's binary phrasing did): pick *higher volumes* — liquidity provision legitimately supports tradability and hence volume, while any price promise is manipulation outright — and then immediately reframe to the true deliverables (spread, depth, uptime).

Self-check via incentives: both forbidden promises reduce to the same absurdity — either unbounded inventory risk (price support) or self-dealing (volume support). A promise a rational, compliant firm cannot structure is not the promise being made.

Answer: Neither. A market maker promises liquidity — maximum spread, minimum depth, uptime — never price levels or volumes; promising either would border on market manipulation.

P15.4 *Perpetual funding arithmetic*

(a) Direction. The perp has no expiry, so funding is the cash flow that ties it to spot: the rate is (a clamp of) the perp's premium over the spot index. Perp above index ⇒ positive funding ⇒ *longs pay shorts* every interval. Paying to stay long pushes longs to close and makes shorting attractive, pressuring the perp back *down* toward the index.

(b) Annualisation. Three intervals per day:

$$0.015\% \times 3 = 0.045\%/\text{day}, \qquad 0.045\% \times 365 \approx 16.4\%/\text{yr (simple)}.$$

Compounded: $(1.00015)^{3\times 365} = e^{1095\ln(1.00015)} \approx e^{0.1642} \approx 1.179$, i.e. about 17.9%/yr.

(c) Weekly cost. 21 intervals × 0.015% = 0.315% of notional; on \$2M that is

$$2{,}000{,}000 \times 0.00315 \approx \$6{,}300 \text{ per week.}$$

Self-check: against the standard anchor 0.01%/8h ≈ 11%/yr: our rate is 1.5× that, and 1.5 × 10.95% = 16.4%. And the weekly number back-solves: \$6,300/\$2M = 0.315%, times 52 ≈ 16.4%/yr.

Answer: (a) Longs pay shorts; pressure pushes the perp down toward the index. (b) 0.045%/day ≈ 16.4%/yr simple, ≈ 17.9% compounded. (c) About \$6,300 per week on \$2M.

P15.5 *Cash-and-carry*

The trade (strongest version). This is the crypto cash-and-carry: buy spot, short an equal notional of the perp *on the same venue with the spot posted as collateral* (cross-margin), so the book is coin-for-coin delta-neutral and a rally increases collateral value in step with the short's mark-to-market loss. Collect funding every 8 hours while it is positive.

Why it is not free money:

- **Funding floats.** The rate resets every interval as a clamp of the perp premium; 30% is a two-week snapshot, not a lock. When crowded longs de-lever, the rate compresses or flips negative — and then *you* are the one paying to hold the position.
- **Basis risk.** There is no expiry forcing convergence at any horizon. You enter with the perp rich, but an early unwind at an adverse basis can surrender months of carry in a single mark.
- **Venue / counterparty risk.** Both legs and the collateral sit on an exchange; post-FTX, *where inventory sits* is a first-class risk decision. A large slice of the 30% is compensation for exactly this.
- **Margin / liquidation risk on the short leg.** If the legs sit on different venues (or the collateral is stablecoin), a sharp rally generates margin calls on the short faster than spot gains can be moved; a forced buy-in at the top converts a hedged book into a realised loss.
- **Frictions.** Fees and impact on four executions (into and out of both legs), funding leakage during exit, and the capital cost of fully-funded spot.

Positioning read. Persistent positive funding means leveraged longs are paying a fat premium to stay long: *crowded* positioning. At extremes it is a contrarian gauge — and a warning that the carry trade itself is crowded, so its unwind correlates with liquidation cascades.

Self-check by magnitudes: 30% of \$10M ≈ \$58k per week of carry; one 5% adverse basis/unwind mark is \$500k ≈ nine weeks of carry. A yield that one bad mark can erase is a risk premium, not an arbitrage.

Answer: Cash-and-carry (long spot, short perp, same-venue cross-margin) collects the funding, but the 30% is a floating risk premium, not free money: funding can compress or flip, the basis can move against an unwind, the venue/counterparty can fail, the short leg carries margin/liquidation risk in rallies, and frictions bite; persistent positive funding signals crowded leveraged longs.

P15.6 *Liquidation price and cascades*

(a) Per coin of position: initial margin $= P_0/L = 50{,}000/10 = 5{,}000$; equity at price P is $5{,}000 + (P - 50{,}000)$; the maintenance requirement is $0.005P$. Liquidation when

$$5{,}000 + P - 50{,}000 = 0.005P \quad \Longrightarrow \quad 0.995\,P = 45{,}000 \quad \Longrightarrow \quad P \approx \$45{,}226,$$

about 9.55% below entry. In general, with leverage L and maintenance fraction m,

$$P_{\text{liq}} = P_0\,\frac{1 - 1/L}{1 - m}, \qquad \text{adverse move} = \frac{1/L - m}{1 - m} \approx \frac{1}{L} - m.$$

Rule of thumb: a $1/L$ move against you wipes the margin; the maintenance buffer triggers liquidation slightly *earlier* (at a higher price for a long).

Self-check: plug back: at $P = 45{,}226$, equity per coin $= 5{,}000 - 4{,}774 = 226$ and requirement $= 0.005 \times 45{,}226 \approx 226$. And the crude $1/L$ rule gives 45,000, correctly just below the exact answer.

(b) The cascade. A liquidation is a *forced market sell* of the position. Price trading into the cluster triggers the first tranche; those sells push price down further, triggering the next tranche — mechanical positive feedback *in the direction of the move*, producing one-way flow and gaps. For a maker this flow is dangerous: it is relentless, all same-side, and marks against every fill in the short run. The standard response, in order: *widen or pull* quotes first (the flow is toxic in the mark-out sense even though it is forced rather than informed), keep inventory discipline (no knife-catching at size, ladder bids rather than one large level), then *re-provide liquidity at wide levels*: because the flow is forced and information-free about fundamentals, cascades overshoot and revert, and the liquidity premium available to whoever has risk capacity at the bottom is exactly the compensation for standing in front of it.

Answer: (a) $P_{\text{liq}} = 0.9 \times 50{,}000/0.995 \approx \$45{,}226$, a $\sim 9.5\%$ drop; in general an adverse move of about $1/L$ (less the maintenance buffer). (b) Forced sells trigger further liquidations — a cascade with the move; widen or pull first, then supply liquidity wide into the overshoot.

P15.7 *Constant-product AMM pricing and impact*

(a) $p = y/x = 200{,}000/100 = \$2{,}000$ per ETH, and $k = xy = 2 \times 10^7$.

(b) After the buy the pool holds $x' = 90$ ETH, so it must hold $y' = k/x' = 2 \times 10^7/90 = 222{,}222.22$ USDC. The trader pays

$$\Delta y = \frac{k}{x - \Delta x} - y = 222{,}222.22 - 200{,}000 = 22{,}222.22 \text{ USDC},$$

an average price of $22{,}222.22/10 = \$2{,}222.22$ per ETH. The new marginal price is $y'/x' = 222{,}222.22/90 = \$2{,}469.14$. Closed form worth knowing: buying a fraction f of reserves gives average price $p/(1-f)$ and marginal price $p/(1-f)^2$; here $f = 0.1$ gives 2,222.22 and 2,469.14 — impact is built into the curve and super-linear in size.

Self-checks: invariant: $90 \times 222{,}222.22 = 2.0 \times 10^7$. The average price must lie between the pre- and post-trade marginal prices: $2{,}000 < 2{,}222 < 2{,}469$; indeed it is exactly their geometric mean, $\sqrt{2{,}000 \times 2{,}469.14} = 2{,}222.2$.

(c) The pool now quotes 2,469 while the CEX trades at 2,000: arbitrageurs buy ETH on the CEX and sell it into the pool until the pool price returns to $\approx 2{,}000$, pocketing the difference. The pool has no view and no oracle — *external arbitrage is its price discovery.* That keeps AMM prices honest, but note who pays: the arbitrageurs' profit is extracted from the passive LPs, which is precisely the adverse-selection cost that surfaces as impermanent loss.

Answer: (a) \$2,000. (b) Pays 22,222.22 USDC, average \$2,222.22/ETH, new pool price \$2,469.14. (c) Arbitrageurs sell ETH into the pool until it re-prices to ≈\$2,000; arbitrage is the mechanism that ties the AMM to fair value, at the LPs' expense.

P15.8 *MEV sandwich attack*

Track the pool state (x, y) through the three trades, with $k = 2 \times 10^7$.

(a) The three legs.

- **Front-run** (attacker buys 10): pays $k/90 - 200{,}000 = 22{,}222.22$; pool $\to (90,\ 222{,}222.22)$.
- **Victim buys 10**: pays $k/80 - 222{,}222.22 = 250{,}000 - 222{,}222.22 = 27{,}777.78$; pool $\to (80,\ 250{,}000)$.
- **Back-run** (attacker sells 10): receives $250{,}000 - k/90 = 27{,}777.78$; pool $\to (90,\ 222{,}222.22)$.

Attacker profit $= 27{,}777.78 - 22{,}222.22 =$ \$5,555.56 (about 25% of the front-run outlay).

(b) Flat, and whose money. Attacker ETH: $+10 - 10 = 0$. The back-run exactly retraces the victim's segment of the curve (from $x = 80$ back to $x = 90$), so the attacker *receives precisely what the victim paid.* Compare the victim's counterfactual: unsandwiched, they would have paid 22,222.22; sandwiched, they paid 27,777.78 — worse by exactly 5,555.56. The attack is a pure zero-sum transfer of the victim's extra slippage to the attacker.

Self-check by conservation: final pool $(90,\ 222{,}222.22)$ is *identical* to the unsandwiched final state, so the LPs are unaffected; net USDC into the pool $= 22{,}222.22 + 27{,}777.78 - 27{,}777.78 = 22{,}222.22$, matching $200{,}000 \to 222{,}222.22$; net ETH out $= 10$, the victim's coins. Every account balances.

(c) Limits and the lesson. The victim's *slippage tolerance* caps their worst execution price, hence caps the front-run size (a competent attacker sizes to push the victim exactly to tolerance); priority-fee auctions among competing searchers hand much of the profit to validators/builders; pool fees ($3 \times 0.3\%$ here would eat a large slice) and gas set a minimum viable victim size. Mitigations: tight slippage settings, private order flow (private RPCs, order-flow auctions), batch auctions. The implication: a public mempool *broadcasts your intention to adversaries who can order transactions around yours* — on public chains, execution is an adversarial game, and MEV is its price.

Answer: (a) Profit $= 27{,}777.78 - 22{,}222.22 =$ \$5,555.56. (b) ETH-flat; the profit is exactly the victim's extra slippage (LPs end unaffected). (c) Bounded by the victim's slippage tolerance, gas/priority-fee auctions, and pool fees; a public mempool shows your order to adversaries before it executes.

P15.9 *Impermanent loss*

(a) Holdings track price. The pool constraints are $xy = k$ and marginal price $p = y/x$. Solving: $x = \sqrt{k/p}$, $y = \sqrt{kp}$. At deposit, price p_0: $x_0 = \sqrt{k/p_0}$, $y_0 = \sqrt{kp_0}$, and the equal-value condition holds automatically since $p_0 x_0 = \sqrt{kp_0} = y_0$. Initial value $V_0 = 2\sqrt{kp_0}$. After the move to $p_1 = rp_0$, arbitrageurs rebalance the pool to

$$x_1 = \sqrt{k/p_1} = \frac{x_0}{\sqrt{r}}, \qquad y_1 = \sqrt{kp_1} = y_0\sqrt{r},$$

so $V_{\text{LP}} = p_1 x_1 + y_1 = 2\sqrt{kp_1} = V_0\sqrt{r}$: *the LP's value grows like* $\sqrt{r}$*, not* r.

(b) Against the HODL benchmark. Holding the deposited coins instead: $V_{\text{HODL}} = p_1 x_0 + y_0 = y_0(r+1) = V_0 \frac{1+r}{2}$. Hence

$$\frac{V_{\text{LP}}}{V_{\text{HODL}}} = \frac{2\sqrt{r}}{1+r}, \qquad \text{IL}(r) = \frac{2\sqrt{r}}{1+r} - 1 \;\le\; 0$$

by AM–GM ($\sqrt{r} \le \frac{1+r}{2}$, equality iff $r = 1$). Values: $r = 2$: $2\sqrt{2}/3 - 1 = 0.9428 - 1 = -5.72\%$; $r = 4$: $4/5 - 1 = -20\%$.

(c) Symmetry and the small-move law. Replace $r \to 1/r$: $\frac{2/\sqrt{r}}{1+1/r} = \frac{2\sqrt{r}}{r+1}$ — identical, so $\text{IL}(r) = \text{IL}(1/r)$: a halving hurts exactly as much as a doubling (in relative terms). For $r = 1 + \varepsilon$: $\sqrt{r} \approx 1 + \frac{\varepsilon}{2} - \frac{\varepsilon^2}{8}$, so

$$\frac{2\sqrt{r}}{1+r} \approx \frac{2+\varepsilon-\varepsilon^2/4}{2+\varepsilon} \approx 1 - \frac{\varepsilon^2}{8} \qquad \Longrightarrow \qquad \text{IL} \approx -\frac{\varepsilon^2}{8}.$$

A loss *quadratic in the move*, whichever direction, is the signature of a **short gamma** position: the pool mechanically sells the appreciating asset on the way up and buys the depreciating one on the way down. The LP has sold an option; trading fees are the premium collected. Rational iff expected fees (plus incentives) exceed the expected give-away — "LP profits iff fees beat the short-gamma cost." (And "impermanent" is a misnomer: withdraw at $r \neq 1$ and the loss is permanent.)

Self-checks: $\text{IL}(1) = 2/2 - 1 = 0$; the symmetry proved in (c) is itself a consistency check; and testing the Taylor law at $r = 1.1$: exact $2\sqrt{1.1}/2.1 - 1 = -0.113\%$ vs $-\varepsilon^2/8 = -0.125\%$ (right order, slight overestimate as expected for a truncation).

Answer: $V_{\text{LP}} = V_0\sqrt{r}$, $V_{\text{LP}}/V_{\text{HODL}} = 2\sqrt{r}/(1+r)$; $\text{IL}(2) = -5.72\%$, $\text{IL}(4) = -20\%$; $\text{IL}(r) = \text{IL}(1/r)$ and $\text{IL} \approx -\varepsilon^2/8$ for small moves — the LP is short gamma and should provide liquidity only when expected fees exceed the option value given away. □

P15.10 *Impermanent-loss benchmark*

The error: wrong benchmark. "Up 41%" compares against the deposit, but the deposit was half ETH — and ETH doubled. The relevant counterfactual is *holding the deposited assets*:

$$V_{\text{HODL}} = \underbrace{5{,}000}_{\text{USDC}} + \underbrace{5{,}000 \times 2}_{\text{2.5 ETH at \$4,000}} = \$15{,}000.$$

The LP position at \$14,142 lags by \$858, i.e. $858/15{,}000 = 5.72\%$ — exactly $\text{IL}(2) = 1 - 2\sqrt{2}/3$. LPing made money in absolute terms while *costing* \$858 relative to doing nothing.

Verify the pool arithmetic from scratch: deposit $x_0 = 2.5$ ETH, $y_0 = 5{,}000$ USDC, $k = 12{,}500$. Price doubles ($r = 2$), arbitrage rebalances to $x_1 = 2.5/\sqrt{2} = 1.7678$ ETH, $y_1 = 5{,}000\sqrt{2} = 7{,}071.07$ USDC; value $= 1.7678 \times 4{,}000 + 7{,}071.07 = 7{,}071.07 + 7{,}071.07 = \$14{,}142$ — matching both the quoted figure and the closed form $V_{\text{LP}} = V_0\sqrt{r} = 10{,}000\sqrt{2}$. Note the pool now holds *fewer* ETH: it sold ETH all the way up — the short-gamma mechanics in action.

What "winning" requires. The LP genuinely beat holding iff accumulated trading fees (plus any liquidity incentives) over the period exceeded \$858. That is the entire LP business in one line: fees are the premium on the option the pool sold for you.

Two closing precisions: (i) "compared to what" matters — versus holding \$10,000 of pure USDC the LP did great; versus pure ETH (\$20,000) it did terribly; the honest benchmark is the assets actually deposited. (ii) "Impermanent" is a misnomer: withdraw now and the \$858 is permanent.

Self-check: $\text{IL}(2) \times V_{\text{HODL}} = 0.0572 \times 15{,}000 = 858$ — formula and first-principles pool computation agree.

Answer: The benchmark is wrong: holding the deposited assets would be worth \$15,000, so LPing cost \$858 = 5.72% (the $r = 2$ impermanent loss); the LP position genuinely won only if fees earned over the period exceeded \$858.

28.16 Solutions — §16 Forwards, Futures and No-Arbitrage Pricing

P16.1 *Quoted rates, compounding frequencies and the cheaper loan*

Throughout, r_m is a rate compounded m times a year and r the continuously compounded rate with the same growth. By Proposition 16.2(i) the two are tied by (16.1): $r = m\ln(1 + r_m/m)$ and $r_m = m(e^{r/m} - 1)$.

(a) With $m = 2$ and $r_m = 0.08$,

$$r = 2\ln(1.04) = 2 \times 0.0392207 = 0.0784414 = 7.844\%.$$

Over one year the deposit grows by the factor $1.04^2 = 1.0816$, so the equivalent rate with annual compounding ($m = 1$) is 8.16%.

(b) With $m = 12$ and $r = 0.045$,

$$r_m = 12(e^{0.00375} - 1) = 12 \times 0.00375704 = 0.0450845 = 4.508\%.$$

The shortcut gives $0.045 + 0.045^2/24 = 0.045 + 0.0000844 = 0.0450844$. It understates the exact figure by 1.056×10^{-7}, which lies inside the bracket of Proposition 16.2(iii), between $r^3/(6m^2) = 1.055 \times 10^{-7}$ and $e^{r/m}r^3/(6m^2) = 1.059 \times 10^{-7}$.

(c) Each loan is repaid, with all its interest, in a single payment after one year, so its cost is its growth factor over the year, and quotes with different compounding frequencies are compared after conversion to one convention. The monthly quote grows by the factor $(1 + 0.077/12)^{12} = 1.0064167^{12} = 1.079776$ in a year, so it is 7.978% with annual compounding, above the 7.90% of the first loan. The loan quoted at 7.90% with annual compounding is the cheaper one, although its headline is the higher of the two. On 100,000 for one year the interest is 7,900.00 against 7,977.64, a difference of 77.64. In continuous terms the two loans cost $\ln(1.079) = 7.603\%$ and $12\ln(1.0064167) = 7.675\%$: the same ranking.

(d) Growth by the factor 1.0003 over each 8-hour period is the growth of the quoted 32.85% compounded $m = 1095$ times a year, since $1 + 0.3285/1095 = 1.0003$, so by (16.1)

$$r = 1095\ln(1.0003) = 0.3284507 = 32.845\%.$$

The simple figure is the larger one, by 4.93×10^{-5}. By Proposition 16.2(ii) every quote with a finite compounding frequency exceeds the continuous rate of the same growth, and by the second bracket of part (iii) the gap falls short of $r_m^2/(2m) = 0.3285^2/2190 = 4.93 \times 10^{-5}$ by less than $r_m^3/(3m^2) = 9.9 \times 10^{-9}$; it is small here because m is large.

Self-check. The conversions are redone with the brackets of Proposition 16.2(iii) instead of logarithms. (a) $r \approx r_m - r_m^2/(2m) = 0.08 - 0.0016 = 7.840\%$, and the second bracket of part (iii) adds between $r_m^3/(3m^2(1 + r_m/m)^3) = 0.0000379$ and $r_m^3/(3m^2) = 0.0000427$, so r lies strictly between 7.8438% and 7.8443%, which is 7.844%. (c) By the second bracket of part (iii), the monthly 7.70% is more than $0.077 - 0.077^2/24 = 7.6753\%$ continuous. By the first bracket with $m = 1$, a continuous rate $r > 0$ is more than $r + r^2/2 + r^3/6$ with annual compounding, an expression that increases with r, so the monthly loan costs more than $0.076753 + 0.002946 + 0.000075 = 7.977\%$ with annual compounding, above 7.90%, as found. (d) $0.3285 - 0.3285^2/2190 = 0.3284507$, and the second bracket of part (iii) adds less than 10^{-8}. The growth factors agree as well: $e^{0.0784414} = 1.0816$ in (a), $(1 + 0.0450845/12)^{12} = 1.046028 = e^{0.045}$ in (b), and $e^{0.3284507/1095} = 1.0003$ in (d).

Answer: (a) 7.844% continuous, 8.16% annual; (b) 4.508% compounded monthly (exact 4.50845%, shortcut 4.50844%); (c) the loan at 7.90% annual is cheaper: the monthly quote is 7.978% annual, which is 77.64 more interest per 100,000; (d) 32.845%, below the simple 32.85%, because a rate compounded finitely often always exceeds the continuous rate of the same growth.

P16.2 *Both cash-and-carry arbitrages, and the no-arbitrage band with frictions*

(a) We argue by contradiction: each strict inequality is turned into a strategy that costs nothing and pays a positive constant, which is an arbitrage (reading (b) of Definition 16.7).

Case $F > Se^{rT}$ (cash-and-carry). Borrow S, buy one unit, and go short one forward at F.

position	cash flow at 0	value at T
borrow S	$+S$	$-Se^{rT}$
buy one unit	$-S$	$+S_T$
short one forward at F	0	$F - S_T$
net	0	$F - Se^{rT}$

The asset pays no income and costs nothing to store, so nothing is paid or received between 0 and T, and the strategy is self-financing. At T the unit is delivered under the forward for F and the loan is repaid. The payoff is the constant $F - Se^{rT} > 0$ at zero cost, an arbitrage. Hence $F \le Se^{rT}$.

Case $F < Se^{rT}$ (reverse cash-and-carry). Sell one unit short, deposit the proceeds, and go long one forward at F.

position	cash flow at 0	value at T
sell one unit short	$+S$	$-S_T$
deposit S	$-S$	$+Se^{rT}$
long one forward at F	0	$S_T - F$
net	0	$Se^{rT} - F$

At T the deposit returns Se^{rT}, of which F pays for the unit taken under the forward; the unit goes back to its lender, who is owed no income. Nothing is paid or received between 0 and T, so the strategy is self-financing. The payoff is the constant $Se^{rT} - F > 0$ at zero cost, an arbitrage again. Hence $F \ge Se^{rT}$, and the two cases together give $F = Se^{rT}$, which is Theorem 16.11. For an investment asset the second trade does not even need a short sale: a holder who sells the unit, deposits S and buys forward ends with the unit and $Se^{rT} - F$ in cash.

(b) $F = 45\,e^{0.06\times 0.75} = 45\,e^{0.045} = 47.07$. The quote 46.50 is below it, so the forward is cheap: sell the asset short at 45, deposit 45, and buy the forward at 46.50. At T the deposit is worth 47.07, the forward takes 46.50 and the unit received is returned: a sure $47.07 - 46.50 = 0.57$ per unit, whatever S_T is.

(c) Write $S^{\text{bid}} = 44.95$ and $S^{\text{off}} = 45.05$. Each arbitrage now trades on its worse side. The cash-and-carry buys at the offer and borrows at r_{bor}, so it pays only if

$$F > S^{\text{off}} e^{r_{\text{bor}} T} = 45.05\,e^{0.065\times 0.75} = 45.05\,e^{0.04875} = 47.301.$$

The reverse trade sells at the bid and lends at r_{len}, so it pays only if

$$F < S^{\text{bid}} e^{r_{\text{len}} T} = 44.95\,e^{0.055\times 0.75} = 44.95\,e^{0.04125} = 46.843.$$

Every quote in the interval $[S^{\text{bid}} e^{r_{\text{len}} T},\ S^{\text{off}} e^{r_{\text{bor}} T}] = [46.843,\ 47.301]$ admits neither arbitrage, and every quote outside it admits one; to the cent, these are the quotes from 46.85 to 47.30. The quote 46.50 lies below the interval, so the reverse cash-and-carry still works: sell at 44.95, lend the proceeds at 5.5% to hold 46.843 at T, and pay 46.50 under the forward, for a sure $46.843 - 46.50 = 0.34$ per unit, down from 0.57.

Self-check. A different route to (a) is replication. One unit of the asset together with a short forward at F costs S and pays $S_T + (F - S_T) = F$ for certain, so $S = Fe^{-rT}$ by Lemma 16.8(iv); with the numbers of (b), $47.07\,e^{-0.045} = 45.00$. The quotes can also be read as rates. Selling the asset at 45 and buying it back forward at 46.50 borrows cash at $\ln(46.50/45)/0.75 = 4.37\%$, below the 6% that the cash account pays, which is the trade of (b). Against the bid 44.95 the same rate

is $\ln(46.50/44.95)/0.75 = 4.52\%$, still below the lending rate of 5.5%, so the arbitrage survives the frictions, as found in (c). Without frictions ($r_{\text{len}} = r_{\text{bor}} = 6\%$ and bid and offer both 45) the interval shrinks to the single price 47.07, which lies inside [46.843, 47.301].

Answer: (a) $F = Se^{rT}$: the cash-and-carry excludes $F > Se^{rT}$ and the reverse cash-and-carry excludes $F < Se^{rT}$; (b) $F = 47.07$; at 46.50 sell the asset short, deposit 45 and buy the forward, for a sure 0.57 per unit at T; (c) no arbitrage exactly for quotes in [46.843, 47.301] (from 46.85 to 47.30 to the cent); the quote 46.50 is still an arbitrage, worth 0.34 per unit at T.

P16.3 *A short forward on a dividend-paying stock, priced and revalued*

(a) The present value of the dividends is

$$I = 1.20\,e^{-0.04\times 3/12} + 1.20\,e^{-0.04\times 9/12} = 1.18806 + 1.16453 = 2.35259.$$

A stock is an investment asset (Definition 16.10) that costs nothing to store, so by Proposition 16.13

$$F = (S - I)e^{rT} = (80 - 2.35259)\,e^{0.04} = 77.64741 \times 1.040811 = 80.8163,$$

so $K = 80.82$. The delivery price exceeds the forward price by $80.82 - 80.8163 = 0.0037$, so by Proposition 16.17(i) at $t = 0$, with the off-market K that its Setup allows, the short position is worth $(K - F)e^{-rT} = 0.0037 \times 0.960789 = 0.0036$ and the long -0.0036. That is 0 to the cent, the precision of the strike, so the position costs nothing to enter.

(b) Now $t = 5/12$ and $\tau = T - t = 7/12$. Only the second dividend remains, due 4 months after t, and by Proposition 16.17(ii) the forward price at t is that of Proposition 16.13 with S_t, I_t and τ in the places of S, I and T:

$$I_t = 1.20\,e^{-0.04\times 4/12} = 1.18411,$$
$$F_t = (S_t - I_t)e^{r\tau} = (76 - 1.18411)\,e^{0.04\times 7/12} = 74.81589 \times 1.023608 = 76.5821.$$

By Proposition 16.17(i) the long forward is worth $(F_t - K)e^{-r\tau}$ and the short forward the negative of that:

$$(K - F_t)e^{-r\tau} = (80.82 - 76.5821) \times 0.976937 = 4.2379 \times 0.976937 = 4.14.$$

The stock has fallen. The short side sells at 80.82 a share that can now be bought forward at 76.58, and it is ahead by the present value of the difference.

Self-check. (a) Carry each cash flow to T instead of discounting the dividends to time 0: $80\,e^{0.04} - 1.20\,e^{0.04\times 9/12} - 1.20\,e^{0.04\times 3/12} = 83.2649 - 1.2365 - 1.2121 = 80.8163$. (b) The spot form of Proposition 16.17(ii) values the long contract without the forward price: $S_t - I_t - Ke^{-r\tau} = 76 - 1.18411 - 80.82 \times 0.976937 = 74.8159 - 78.9560 = -4.14$, so the short position is worth +4.14.

Answer: (a) $K = 80.82$ (with $I = 2.3526$), and the trader's position is worth 0 to the cent when written (exactly +0.0036 to the short); (b) $F_t = 76.58$ (with $I_t = 1.1841$), and the short position is worth +4.14.

P16.4 *Three mispriced quotes: an index future, an implied foreign rate and a currency forward*

(a) An equity index is an investment asset (Definition 16.10), so Proposition 16.15(ii) applies; with $T = 1/3$,

$$F = Se^{(r-q)T} = 4{,}800\,e^{(0.045-0.018)/3} = 4{,}800\,e^{0.009} = 4{,}843.39.$$

The quote 4,830 is 13.39 too low: buy the future and sell the index basket short. By Proposition 16.15(i) a holding of $e^{-qT} = e^{-0.006} = 0.99402$ units with dividends reinvested grows to exactly one unit at T. The short seller therefore shorts 0.99402 units today and pays the dividends owed to the lender by shorting further units, so that exactly one unit is owed at T.

position	cash flow at 0	value at T
short 0.99402 units, dividends paid in units	+4,771.29	$-S_T$
deposit the proceeds	−4,771.29	+4,843.39
long one future at 4,830	0	$S_T - 4{,}830$
net	0	+13.39

The proceeds are $4{,}800\,e^{-0.006} = 4{,}771.29$. Deposited for four months, they grow by the factor $e^{0.045/3} = e^{0.015}$, to $4{,}800\,e^{0.009} = 4{,}843.39$. The futures price converges to S_T at delivery (Proposition 16.27), so the long future has earned $S_T - 4{,}830$, the unit owed costs S_T, and 13.39 per unit of index is left in every state.

(b) With the dollar as the domestic currency, Proposition 16.19 gives $F = Se^{(r-r_{\text{for}})T}$, so

$$r_{\text{for}} = r - \frac{1}{T}\ln\frac{F}{S} = 0.043 - \frac{\ln(1.2620/1.2700)}{0.5} = 0.043 + 0.01264 = 5.56\%.$$

Sterling trades at a forward discount, so it is the currency with the higher interest rate.

(c) The unit of foreign currency is 100 yen, with $S = 0.6700$ dollars, $r = 4.3\%$ and $r_{\text{for}} = 0.5\%$. By Proposition 16.19,

$$F = 0.6700\,e^{(0.043-0.005)\times 0.25} = 0.6700\,e^{0.0095} = 0.67640.$$

The quote 0.6720 is too low, so yen are cheap forward: buy yen forward and sell yen spot, with borrowed yen. A loan that repays 500,000,000 yen at T raises $500{,}000{,}000\,e^{-0.005\times 0.25} = 499{,}375{,}390$ yen today, which convert into $499{,}375{,}390 \times 0.6700/100 = 3{,}345{,}815$ dollars. Deposited at 4.3%, these grow to $3{,}345{,}815\,e^{0.01075} = 3{,}381{,}977$ dollars. The forward purchase of 500,000,000 yen costs $500{,}000{,}000 \times 0.6720/100 = 3{,}360{,}000$ dollars at T, and the yen received repay the loan.

	cash flow at 0		cash flow at $T = 0.25$	
position	USD	JPY	USD	JPY
borrow JPY at 0.5%		+499,375,390		−500,000,000
convert at 0.6700 per 100	+3,345,815	−499,375,390		
deposit USD at 4.3%	−3,345,815		+3,381,977	
buy JPY forward at 0.6720 per 100			−3,360,000	+500,000,000
net	0	0	+21,977	0

The profit is 21,977 dollars at T, with no outlay and no exposure to the exchange rate.

Self-check. Each answer is recovered by a second route. (a) Read the quote as an off-market forward: by Proposition 16.17(ii) at $t = 0$, a long contract struck at $K = 4{,}830$ is worth $Se^{-qT} - Ke^{-rT} = 4{,}771.28623 - 4{,}830\,e^{-0.015} = 4{,}771.28623 - 4{,}758.09067 = 13.19556$ today and costs nothing to enter, so locking it in leaves $13.19556\,e^{0.015} = 13.39$ at T, the profit of the table. As mental checks, the forward points are about $S(r-q)T = 4{,}800 \times 0.027/3 = 43.2$, against the exact 43.39; and the quote 4,830 would be fair only for a dividend yield of $r - \ln(4{,}830/4{,}800)/T = 0.045 - 0.01869 = 2.63\%$, above the actual 1.8%, which says again that the future is cheap. (b) To first order, $(F - S)/S \approx (r - r_{\text{for}})T$, so $r_{\text{for}} \approx 0.043 - (1.2620 - 1.2700)/(1.2700 \times 0.5) = 0.043 + 0.01260 = 5.56\%$; and sterling, at a forward discount, must carry the higher rate, as $5.56\% > 4.3\%$ confirms. Putting the rate back: $1.2700\,e^{(0.043-0.05564)\times 0.5} = 1.2700\,e^{-0.00632} = 1.2620$. (c) Per unit, the quote is $0.6763953 - 0.6720 = 0.00439533$ dollars per 100 yen too low, and 500 million yen are 5,000,000 such units: $0.00439533 \times 5{,}000{,}000 = 21{,}976.65$, that is, 21,977 dollars, the profit of the table. Read as a rate, the quote implies a yen interest rate of $0.043 - \ln(0.6720/0.6700)/0.25 = 3.11\%$, while yen are borrowed at 0.5%.

Answer: (a) $F = 4{,}843.39$; buy the future at 4,830, short 0.99402 units of the basket and deposit the proceeds: 13.39 per unit of index at delivery; (b) $r_{\text{for}} = 5.56\%$; (c) $F = 0.67640$ dollars per 100

yen; borrow 499,375,390 yen, convert into 3,345,815 dollars, deposit them, and buy 500 million yen forward at 0.6720: a profit of 21,977 dollars at delivery.

P16.5 *Storage, an upper bound and the convenience yield: silver versus crude*

(a) The present value of the storage is

$$U = 0.12 + 0.12\,e^{-0.05\times 4/12} + 0.12\,e^{-0.05\times 8/12} = 0.12 + 0.118017 + 0.116066 = 0.354083.$$

Silver is an investment commodity, so by Proposition 16.21(i) the forward price is given by an equality, and with known rates the futures price is the same number (Theorem 16.28):

$$F = (S+U)e^{rT} = 30.354083\,e^{0.05\times 10/12} = 30.354083 \times 1.042547 = 31.65.$$

(b) Here $U = 1.20\,e^{-0.0225} + 1.20\,e^{-0.045} = 1.17330 + 1.14720 = 2.32050$. Crude oil is a consumption commodity, so only the cash-and-carry operates, and Proposition 16.21(ii) gives an upper bound:

$$F \le (S+U)e^{rT} = 77.32050\,e^{0.045} = 77.32050 \times 1.046028 = 80.88.$$

No lower bound follows.

(c) There is no arbitrage. The quote 73.80 is below the bound, so the cash-and-carry would lose $80.88 - 73.80 = 7.08$ per barrel, and the reverse trade needs a holder of physical oil who sells it and buys it back forward, which a refiner that needs the oil declines to do. By Definition 16.23(ii), in its form for storage paid at known dates, $Fe^{yT} = (S+U)e^{rT}$, so

$$y = \frac{1}{T}\ln\frac{(S+U)e^{rT}}{F} = \ln\frac{80.879}{73.80} = 9.16\% \text{ a year.}$$

Since $F = 73.80 < S = 75$, the market is in backwardation (Definition 16.23(iv)).

(d) The quote 82 is above the bound, so the cash-and-carry is an arbitrage. Borrow $S+U = 77.32$, buy a barrel for 75 and store it, deposit 1.1733 and 1.1472, which grow to the two storage bills of 1.20 due at $t = 0.5$ and at $t = 1$, and sell the future at 82. At delivery the barrel is delivered at the final settlement price S_T (Proposition 16.27), the variation margin has paid $82 - S_T$ in all, its interest being ignored as the statement says, and the loan is repaid with 80.88: a sure profit of $S_T + (82 - S_T) - 80.88 = 1.12$ per barrel. The trade only buys and stores, so it is open to anyone. With the interest on the variation margin counted, one contract held unchanged does not lock the profit in: with 252 settlements a year, if the future jumped to 107 after the first day and stayed there, the 25 paid that day would cost $25(e^{0.045\times 251/252} - 1) = 1.15$ of interest by delivery, and the trade would end at -0.03. The exact trade tails the short: during day i it is short $B_i/B_n = e^{-0.045(1-i/252)}$ contracts, $B_i = e^{0.045i/252}$ being the cash account of §16.6, the position for which the settlements carried to delivery telescope (the paragraph *Where deterministic rates are used* after Theorem 16.28), so that they sum to exactly $82 - S_T$ and the profit is 1.12 whatever the path.

(e) The upper bound comes from buying and storing, which works for every storable commodity. The matching lower bound needs holders who sell spot and buy forward as soon as the forward is cheap. Holders of silver own it only for its return and do exactly that, so both bounds bind and the price is pinned. Holders of crude want the physical barrel for their plant and do not, so the price can sit anywhere below the bound, and the gap is the convenience yield. Storage is not the reason: both commodities are costly to store, and U enters both formulas in the same way.

Self-check. Instead of discounting the storage bills to today, carry every cash flow forward to delivery. Silver: $30\,e^{0.05\times 10/12} + 0.12\,e^{0.05\times 10/12} + 0.12\,e^{0.05\times 6/12} + 0.12\,e^{0.05\times 2/12} = 31.2764 + 0.1251 + 0.1230 + 0.1210 = 31.65$. Crude: $75\,e^{0.045} + 1.20\,e^{0.0225} + 1.20 = 78.4521 + 1.2273 + 1.2000 = 80.88$. Both agree with (a) and (b). For (c), $73.80\,e^{0.0916} = 73.80 \times 1.09593 = 80.88$ returns the bound; in rates, the bound is $\ln(80.879/75) = 7.55\%$ above spot and the quote is $\ln(73.80/75) = -1.61\%$, that is, 1.61% below spot, a gap of 9.16%.

Answer: (a) $U = 0.3541$ and $F = 31.65$, an equality; (b) $U = 2.3205$ and $F \leq 80.88$, an upper bound only; (c) no arbitrage; $y = 9.16\%$ a year; backwardation, since $73.80 < 75$; (d) cash-and-carry (borrow 77.32, buy and store the barrel, sell the future at 82): 1.12 per barrel at delivery, with margin interest ignored, and exactly so when the short is tailed to B_i/B_n contracts on day i; (e) holders of an investment commodity sell spot and buy forward when the forward is cheap, holders of a consumption commodity do not; storage costs enter both prices in the same way.

P16.6 *A short futures position through four settlement days*

The position is short $3 \times 1{,}000 = 3{,}000$ barrels, so a move of 0.01 in the futures price is worth 30, and the short side loses when the price rises. The initial margin is $3 \times 4{,}000 = 12{,}000$ and the maintenance margin is $3 \times 3{,}100 = 9{,}300$ (Definition 16.25).

(a) The account can lose $12{,}000 - 9{,}300 = 2{,}700$ before a call, which is $2{,}700/3{,}000 = 0.90$ per barrel: a call is triggered by a settlement price above $72.40 + 0.90 = 73.30$. Only the balance above the initial margin can be withdrawn, so withdrawing 3,000 takes a gain of 3,000, that is, a fall of 1.00 per barrel: a settlement price at or below 71.40.

(b) With F_i the settlement price of day i, the gain of day i on the short position is $(F_{i-1} - F_i) \times 3{,}000$.

day	settlement price	gain of the day	balance before call	margin call	balance
0	72.40				12,000
1	72.85	−1,350	10,650		10,650
2	73.55	−2,100	8,550	3,450	12,000
3	72.90	+1,950	13,950		13,950
4	71.60	+3,900	17,850		17,850

On day 2 the balance of 8,550 is below the maintenance margin of 9,300 (the price 73.55 is above the trigger 73.30), and the call of 3,450 restores the initial margin of 12,000, not merely the maintenance level. On days 3 and 4 the balance exceeds the initial margin, and nothing is withdrawn. The trader has deposited $12{,}000 + 3{,}450 = 15{,}450$ and holds 17,850: a total gain of 2,400.

(c) At the delivery date T, day 4, the futures price equals the spot price (Proposition 16.27), so $S_T = 71.60$, and a short forward at 72.40 pays $(72.40 - 71.60) \times 3{,}000 = 2{,}400$: the same total. What differs is the timing. The forward pays once, at T. The futures position paid 1,350 and 2,100 on days 1 and 2, which forced the trader to find 3,450 in cash on day 2, and received 1,950 and 3,900 on days 3 and 4. With interest, the two totals would differ by the interest on these interim payments.

Self-check. The daily gains telescope: $-1{,}350 - 2{,}100 + 1{,}950 + 3{,}900 = 2{,}400 = (72.40 - 71.60) \times 3{,}000$, the price change times the position, whatever the path in between. The account identity holds: deposits plus gains equal the final balance, $15{,}450 + 2{,}400 = 17{,}850$. The triggers of (a) are confirmed from the balance: at a settlement price of 73.30 the balance would be $12{,}000 - 2{,}700 = 9{,}300$, exactly the maintenance level and therefore no call, while at 73.55 it is 8,550; and at a settlement price of 71.40 the balance would be $12{,}000 + 3{,}000 = 15{,}000$, exactly 3,000 above the initial margin.

Answer: (a) a margin call above 73.30; 3,000 can be withdrawn at or below 71.40; (b) closing balances 10,650, 12,000 (after a call of 3,450 on a balance of 8,550), 13,950 and 17,850; total gain +2,400; (c) the forward pays the same 2,400, in a single payment at delivery, while the futures position settled it day by day and needed 3,450 of extra cash on day 2.

P16.7 *Futures prices equal forward prices when rates are known*

(a) We compare two self-financing strategies with the same payoff and apply the law of one price, Lemma 16.8(iii), which holds with the cash account B_i (Setup of §16.6); Steps 1 to 6 prove the

equality at time 0, and Step 7 runs the same comparison from the end of a later day. This is the proof of Theorem 16.28.

Step 1 (a rolling futures position). During day i hold B_i contracts long, for $i = 1, \dots, n$. The position for day i is set at the end of day $i-1$, and the number B_i is known then, because the rates are known at time 0. Resizing the position costs nothing.

Step 2 (one day's settlement, carried to T). The settlement of day i is $B_i(F_i - F_{i-1})$, of either sign. Deposited in the cash account until T (borrowed, if it is negative), it grows by the factor B_n/B_i and becomes

$$B_i\,(F_i - F_{i-1})\,\frac{B_n}{B_i} = (F_i - F_{i-1})\,B_n.$$

The size of the position has cancelled against the reinvestment factor.

Step 3 (telescoping). Summing over the days, $\sum_{i=1}^{n}(F_i - F_{i-1})\,B_n = (F_n - F_0)\,B_n$.

Step 4 (the futures strategy). Add a deposit of F_0 made at time 0, worth $F_0 B_n$ at T. Futures cost nothing to enter and every settlement goes through the cash account, so the strategy is self-financing. Its price is $F_0 = F_0^{\text{fut}}$ and its payoff is

$$(F_n - F_0)\,B_n + F_0 B_n = F_n B_n = S_T B_n.$$

Step 5 (the forward strategy). Go long B_n forward contracts at F, at no cost, and deposit F. The price is F, nothing is paid or received before T, and the payoff is $B_n(S_T - F) + FB_n = S_T B_n$.

Step 6 (the law of one price). The two strategies have the same payoff in every state, so they have the same price: $F_0^{\text{fut}} = F$. Spelled out: if $F_0^{\text{fut}} > F$, holding the forward strategy and selling the futures strategy short brings in $F_0^{\text{fut}} - F > 0$ today and nets to zero at T; deposited until T, the proceeds leave $(F_0^{\text{fut}} - F)B_n > 0$ from nothing, an arbitrage by reading (a) of Definition 16.7 with B_n in place of e^{rT}; if $F_0^{\text{fut}} < F$, the opposite trade is one. Both strategies consist of contracts and cash, so each can be held and sold short. A constant rate r is the special case $B_i = e^{ri\Delta t}$, in which the position of day i is $e^{ri\Delta t}$ contracts, as in (b).

Step 7 (from the end of day k). Fix k with $1 \le k < n$ (the case $k = 0$ is Steps 1 to 6), and let F' be the forward price for delivery at T written at the end of day k. Hold B_i/B_k contracts long during each day $i > k$, a number known when it is set, and deposit F_k at the end of day k. This position is worth F_k then, because a futures position is worth zero after settlement. As in Step 2 the gain of day i carried to T is $(B_i/B_k)(F_i - F_{i-1})B_n/B_i = (F_i - F_{i-1})B_n/B_k$; the gains telescope to $(F_n - F_k)B_n/B_k$, and with the deposit, which grows to $F_k B_n/B_k$, the payoff is $S_T B_n/B_k$. Going long B_n/B_k forwards at F', at no cost, and depositing F' is worth F' at the end of day k and pays $(B_n/B_k)(S_T - F') + F'B_n/B_k = S_T B_n/B_k$. Neither position adds or withdraws cash after the end of day k, and both consist of contracts and cash, so each can be held and sold short. The second comparison clause of Lemma 16.8(v), which holds with the cash account B_i (Setup of §16.6), gives $F_k = F'$ with probability one.

Where known rates are used. The number B_i of contracts for day i in Step 1, the number B_n of forwards in Step 5, and the numbers B_i/B_k and B_n/B_k in Step 7 must be known when the positions are set. With random rates B_n is not known at time 0, so no fixed number of forwards pays $S_T B_n$; equivalently, a futures position that reproduces one forward must hold B_i/B_n contracts on day i, a number that depends on rates not yet known.

(b) Here $B_1 = e^{0.0002} = 1.00020$, $B_2 = e^{0.0004} = 1.00040$ and $B_3 = e^{0.0006} = 1.00060$.

day i	$F_i - F_{i-1}$	contracts B_i	settlement $B_i(F_i - F_{i-1})$	carried to T
1	+1.0	1.00020	+1.00020	+1.00060
2	−1.5	1.00040	−1.50060	−1.50090
3	+1.3	1.00060	+1.30078	+1.30078
sum	+0.8			+0.80048

Each carried amount is $(F_i - F_{i-1})e^{0.0006}$, and the total is $0.8\,e^{0.0006} = 0.80048 = (F_3 - F_0)B_3$: the payoff of $e^{0.0006}$ forward contracts struck at 100 when $S_T = 100.8$. A constant position of one contract carries $1.0\,e^{0.0004} - 1.5\,e^{0.0002} + 1.3 = 0.80010$ to T, a number that depends on the order of the gains and losses and not only on $F_3 - F_0$. A proof that holds B_i/B_n contracts on day i instead replicates one forward rather than B_n of them; its positions here are $e^{-0.0004} = 0.99960$, $e^{-0.0002} = 0.99980$ and 1, its settlements carried to T are exactly $+1.0$, -1.5 and $+1.3$, and they sum to $0.8 = S_T - F_0$.

(c) The forward pays $S_T - 100 = 0$ on both paths. On the path $100, 104, 104, 100$ the futures contract receives 4 at the end of day 1, which earns interest for two days, and pays 4 at the end of day 3: at T it is worth $4\,e^{0.0004} - 4 = +0.0016$. On the path $100, 96, 96, 100$ it pays 4 on day 1, financed for two days, and receives 4 on day 3: $-4\,e^{0.0004} + 4 = -0.0016$. The futures contract ends ahead of the forward by 0.0016 when the gains come first and behind it by 0.0016 when the losses come first; for a short position the signs are reversed. Daily settlement therefore neither helps nor hurts a hedger systematically. It moves cash flows in time, with an effect whose sign depends on the path, and part (a) shows that this effect has no value at time 0 when rates are known. What the hedger must provide is the cash for the interim losses.

(d) The futures price is the higher one. A long futures position receives its gains when the price rises, which tends to be when rates rise, and so reinvests them at high rates, while it pays its losses when the price falls and rates tend to fall, and so finances them cheaply; the forward has no such advantage, and since both contracts cost nothing to enter, the futures price must be higher to offset it. This is the sign stated in Remark 16.30: $F^{\text{fut}} > F$ exactly when the risk-neutral covariance of the path discount factor $1/B_n$ with S_T is negative.

Self-check. The total of (b) is recomputed by running the cash account forward one day at a time instead of carrying each settlement separately. After day 1 the balance is 1.00020; after day 2 it is $1.00020\,e^{0.0002} - 1.50060 = -0.50020$; after day 3 it is $-0.50020\,e^{0.0002} + 1.30078 = 0.80048$, the same total. A limiting case checks (a): for $n = 1$ the rolling position is B_1 contracts settled once, at T, for $B_1(S_T - F_0)$, which is the payoff of B_1 forward contracts struck at F_0; the futures contract is then a forward, and the statement reduces to the uniqueness of the forward price.

Answer: (a) $F_0^{\text{fut}} = F$: holding B_i futures on day i and depositing F_0 pays $S_T B_n$, and so do B_n forwards with a deposit of F; the same comparison from the end of day k (B_i/B_k futures plus F_k against B_n/B_k forwards plus F') gives $F_k = F'$; known rates are needed to set these positions in advance; (b) positions 1.00020, 1.00040, 1.00060, total $0.8\,e^{0.0006} = 0.80048$ at T, against 0.80010 for a constant contract; (c) the future ends 0.0016 ahead on the path that rises first and 0.0016 behind on the path that falls first, so daily settlement changes the timing of cash flows without a systematic gain or loss; (d) the futures price is higher.

P16.8 *The minimum-variance hedge ratio: from statistics, from data, and the wrong regression*

(a) We complete the square. By the variance of a sum (Proposition 10.10), applied to ΔS and $-h\,\Delta F$, $\operatorname{Var}(\Delta S - h\,\Delta F) = \operatorname{Var}(\Delta S) + \operatorname{Var}(-h\,\Delta F) + 2\operatorname{Cov}(\Delta S, -h\,\Delta F)$. By bilinearity of covariance, $\operatorname{Var}(-h\,\Delta F) = h^2\sigma_F^2$ and $\operatorname{Cov}(\Delta S, -h\,\Delta F) = -h\operatorname{Cov}(\Delta S, \Delta F) = -h\rho\,\sigma_S\sigma_F$, the last step by the definition of the correlation. Hence

$$\operatorname{Var}(\Delta S - h\,\Delta F) = \sigma_S^2 - 2h\rho\,\sigma_S\sigma_F + h^2\sigma_F^2 = \sigma_F^2\Big(h - \rho\,\frac{\sigma_S}{\sigma_F}\Big)^2 + (1-\rho^2)\,\sigma_S^2.$$

The second equality is checked by expanding the square, which returns $h^2\sigma_F^2 - 2h\rho\,\sigma_S\sigma_F + \rho^2\sigma_S^2$; adding $(1-\rho^2)\sigma_S^2$ restores the term σ_S^2. The first term on the right is nonnegative and, since $\sigma_F > 0$, vanishes only at

$$h^* = \rho\,\frac{\sigma_S}{\sigma_F} = \frac{\operatorname{Cov}(\Delta S, \Delta F)}{\operatorname{Var}(\Delta F)},$$

which is therefore the unique global minimiser. The minimum variance is the second term, $(1-\rho^2)\sigma_S^2$. The unhedged variance, the value at $h = 0$, is σ_S^2, so the hedge removes the fraction ρ^2

of it, the hedge effectiveness. Finally, selling $h\,n_A/n_F$ contracts of n_F units each is a futures position of $h\,n_A$ units against n_A units of exposure, so its hedge ratio is h, and the change in the value of the hedged position is $n_A\,\Delta S - h\,n_A\,\Delta F = n_A(\Delta S - h\,\Delta F)$, with variance $n_A^2\,\mathrm{Var}(\Delta S - h\,\Delta F)$. Since $n_A^2 > 0$, this is smallest exactly when $h = h^*$, that is, with $N_F^* = h^* n_A/n_F$ contracts. This is Theorem 16.37(i)–(iii).

(b) $h^* = 0.85 \times 0.45/0.60 = 0.6375$. The hedge removes $0.85^2 = 0.7225$ of the variance, and the residual standard deviation is $\sqrt{1 - 0.7225} \times 0.45 = 0.5268 \times 0.45 = 0.237$ per unit, against 0.45 unhedged. In units the exposure equals 900,000/20,000 = 45 contracts, so by part (a) the hedge sells $N_F^* = 0.6375 \times 45 = 28.7$ contracts, in practice 29: the variance is a quadratic in the number of contracts, symmetric about 28.69, so the nearest whole number, 29, is the best one.

(c) The sample means are $\overline{\Delta F} = 6/6 = 1$ and $\overline{\Delta S} = 3/6 = 0.5$. The deviations of ΔF from its mean are $1, -2, 2, -1, -3, 3$, and those of ΔS are $0.5, -1.5, 1.5, 0.5, -2.5, 1.5$. Summing over the six periods,

$$\sum(\Delta F - 1)^2 = 1 + 4 + 4 + 1 + 9 + 9 = 28,$$
$$\sum(\Delta S - 0.5)^2 = 0.25 + 2.25 + 2.25 + 0.25 + 6.25 + 2.25 = 13.5,$$
$$\sum(\Delta F - 1)(\Delta S - 0.5) = 0.5 + 3 + 3 - 0.5 + 7.5 + 4.5 = 18.$$

The estimate of h^* is the slope of the regression of ΔS on ΔF (Corollary 5.3), the sample covariance divided by the sample variance of the regressor:

$$\hat{h} = \frac{18}{28} = \frac{9}{14} = 0.643, \qquad \hat{\rho} = \frac{18}{\sqrt{28 \times 13.5}} = \frac{18}{19.44} = 0.926,$$

and $\hat{\rho}^{\,2} = 324/378 = 6/7 = 0.857$ estimates the fraction of variance removed.

(d) The regression of ΔF on ΔS has the slope $18/13.5 = 4/3$, and its inverse is 0.75. This overstates the hedge ratio 9/14 by the factor $0.75 \times 14/9 = 7/6 = 1/\hat{\rho}^{\,2}$, about 17% (Proposition 5.21). The quantity being hedged, ΔS, must be the dependent variable.

Self-check. (a) By calculus instead of completing the square: the derivative of the quadratic in h is $2h\sigma_F^2 - 2\rho\,\sigma_S\sigma_F$, which vanishes at $h = \rho\,\sigma_S/\sigma_F$, and the second derivative $2\sigma_F^2$ is positive. (b) The variance $0.2025 - 0.459\,h + 0.36\,h^2$ takes the values 0.05670, 0.05619 and 0.05760 at $h = 0.60$, 0.6375 and 0.70: the middle value is the smallest, and it equals $(1 - 0.7225) \times 0.2025 = 0.05619$. (c) The hedged changes $\Delta S - (9/14)\,\Delta F$ must be uncorrelated with ΔF, and their summed cross-product with the deviations of ΔF is $18 - (9/14) \times 28 = 0$. The slope is also the correlation times the ratio of the sample standard deviations, $0.926 \times \sqrt{13.5/28} = 0.926 \times 0.694 = 0.643$. (d) $0.75 \times 6/7 = 0.643$.

Answer: (a) $h^* = \rho\,\sigma_S/\sigma_F = \mathrm{Cov}(\Delta S, \Delta F)/\,\mathrm{Var}(\Delta F)$, minimum variance $(1 - \rho^2)\sigma_S^2$, fraction removed ρ^2, and $N_F^* = h^* n_A/n_F$ contracts; (b) $h^* = 0.6375$, fraction removed 0.7225, residual standard deviation 0.237, and 28.7, in practice 29, contracts sold; (c) $\hat{h} = 9/14 = 0.643$ and $\hat{\rho} = 0.926$; (d) 0.75, too large by the factor $7/6 = 1/\hat{\rho}^{\,2}$.

P16.9 *Index futures: hedge, re-gear, and judge the manager*

(a) One contract is worth $V_F = 4{,}000 \times 50 = 200{,}000$, and $V_A/V_F = 50{,}000{,}000/200{,}000 = 250$. By Proposition 16.40(i) the minimum-variance hedge sells

$$N_F^* = \beta\,\frac{V_A}{V_F} = 0.9 \times 250 = 225 \text{ contracts.}$$

(b) By Proposition 16.40(ii) a target beta $\beta' = 1.4$ takes $N_F = (\beta' - \beta)V_A/V_F = (1.4 - 0.9) \times 250 = +125$ contracts: buy 125.

(c) Over three months $r\,\Delta t = 0.048 \times 0.25 = 1.2\%$. The futures return is $R_F = -6\%$, so under the model $R_M - r\,\Delta t = -6\%$, and with $\alpha = 0$ and $\varepsilon = 0$ the decomposition (16.8) gives

$$R_A = r\,\Delta t + \beta\,(R_M - r\,\Delta t) = 1.2\% + 0.9 \times (-6\%) = -4.2\%,$$

a loss of 2,100,000. The futures fall by 240 points, to 3,760, and the 225 short contracts gain $225 \times 240 \times 50 = 2{,}700{,}000$. The hedged book makes +600,000, which is 1.2% of 50 million. It earns $r\,\Delta t$, the riskless return to first order, as Proposition 16.40(iii) says: $r\,\Delta t + \alpha + \varepsilon$ with $\alpha = \varepsilon = 0$. A hedged equity book does not earn zero.

(d) For this portfolio over the year, $R_A = -6\%$, $\beta = 0.3$ against the index, $R_M = -25\%$, and the riskless simple return is $r_f = 4\%$. Taking the index as the market portfolio M of Fact 11.40, the CAPM benchmark (the security market line of Definition 11.41, read at the realised index return) is

$$r_f + \beta\,(R_M - r_f) = 4\% + 0.3 \times (-25\% - 4\%) = 4\% - 8.7\% = -4.7\%,$$

and the realised alpha is $R_A - r_f - \beta(R_M - r_f) = -6\% - (-4.7\%) = -1.3\%$. The manager lost far less than the index and still fell short of the benchmark: with a beta of 0.3, a loss of only 4.7% was to be expected. By the decomposition of Proposition 11.36, with r_f as the constant, $R_A - r_f - \beta(R_M - r_f) = \alpha' + \varepsilon$, where α' is Jensen's alpha of Definition 11.41, computed with r_f as in §16.8, and ε is the residual return: one year of returns shows Jensen's alpha and the residual return together, so -1.3% is evidence of negative skill, not a proof of it.

Self-check. (a), (b) The beta of the position with N_F contracts is $\beta + N_F V_F / V_A$, and each contract moves it by $200{,}000/50{,}000{,}000 = 0.004$: $0.9 - 225 \times 0.004 = 0$ and $0.9 + 125 \times 0.004 = 1.4$. (c) The total return of the index is $R_M = -6\% + 1.2\% = -4.8\%$. A portfolio with a beta of 0.9, no alpha and no residual return behaves like 90% index and 10% cash, which returns $0.9 \times (-4.8\%) + 0.1 \times 1.2\% = -4.32\% + 0.12\% = -4.2\%$, as found. In return form the futures leg is $N_F V_F R_F = (-225) \times 200{,}000 \times (-6\%) = +2{,}700{,}000$, and the net $-2{,}100{,}000 + 2{,}700{,}000 = 600{,}000$ is 1.2% of 50 million. (d) The benchmark is the passive mix with the same beta, 30% index and 70% riskless: $0.3 \times (-25\%) + 0.7 \times 4\% = -7.5\% + 2.8\% = -4.7\%$, and the manager trailed it by 1.3 points.

Answer: (a) sell 225 contracts; (b) buy 125 contracts; (c) portfolio $-2{,}100{,}000$, futures $+2{,}700{,}000$, hedged book $+600{,}000 = +1.2\%$, the riskless return $r\,\Delta t$ (plus alpha, here zero); (d) no: the benchmark is -4.7% and the realised alpha is -1.3%.

P16.10 *Crypto carry: a dated future against the perpetual*

(a) The future stands $65{,}600 - 64{,}000 = 1{,}600$ above spot, a premium of 2.5% for $T = 0.25$; in the sign convention of the basis, $\text{basis}_0 = S - F = -1{,}600$. Mentally, $\ln(1.025) \approx 0.025 - 0.025^2/2 = 0.02469$, so the continuously compounded premium is about $4 \times 0.02469 = 9.88\%$ a year. Exactly,

$$\frac{1}{T} \ln \frac{F}{S} = \frac{\ln(1.025)}{0.25} = \frac{0.024693}{0.25} = 9.88\%,$$

and as a simple rate the premium is $0.025/0.25 = 10.00\%$. The continuous figure exceeds the dollar rate by $9.88\% - 5\% = 4.88$ percentage points. With deterministic rates the futures price is the forward price (Theorem 16.28), so in the notation of (16.5) the quote gives $b - y = 9.88\%$, while bitcoin, with no income and no storage cost, has $b = r = 5\%$: the implied convenience yield is $y = -4.88\%$, a negative number, which the frictionless market excludes (Definition 16.23(iii)).

(b) Buy one bitcoin spot and sell one future. At expiry the futures price equals the spot price (Proposition 16.27, or the cash-settlement terms of the contract), so the spot leg has made $S_T - 64{,}000$ and the short future $65{,}600 - S_T$: a total of 1,600, whatever S_T is. At $S_T = 50{,}000$ the two legs make $-14{,}000 + 15{,}600 = 1{,}600$; at $S_T = 80{,}000$ they make $+16{,}000 - 14{,}400 = 1{,}600$. Financing the purchase costs $64{,}000\,(e^{0.05 \times 0.25} - 1) = 64{,}000 \times 0.012578 = 805$, which leaves 795 per bitcoin, interest on the variation margin aside.

(c) With three payments a day and a 365-day year there are 1095 payments, so as a simple rate the funding is $1095 \times 0.012\% = 13.14\%$ a year (Proposition 15.4(i)); compounded, with every payment reinvested, it is $1.00012^{1095} - 1 = 14.04\%$ (part (ii)). On the continuous convention of (a) it is $1095 \ln(1.00012) = 13.14\%$ against 9.88%, so the conventions are not what separates the two carries. The 9.88% of the dated future is locked in on the day of the trade, because the contract

converges to spot on a known date. The funding rate is reset every 8 hours, so 13.14% annualises one snapshot: the rate can fall or turn negative, and the perpetual has no expiry that forces its price back to spot.

(d) A rally of 25% takes the future from 65,600 to 82,000, and the short leg pays $0.25 \times 65{,}600 = 16{,}400$ per bitcoin in variation margin at once. The spot leg has gained a similar amount, but that gain is not cash until the bitcoin is sold. The short leg gets this cash back only if the futures price falls back; otherwise it is recovered when the bitcoin is sold, at expiry at the latest. By then the short leg has paid $S_T - 65{,}600$ in all, the sale brings $S_T - 64{,}000$ above cost, and the trade still nets 1,600 (at $S_T = 82{,}000$: $-16{,}400 + 18{,}000 = 1{,}600$). Until then the 16,400 has to be funded, which is the liquidity trap of §16.6. Both trades also carry venue risk: the futures leg and its margin, and often the bitcoin itself, sit on an exchange and are exposed to its credit (Remark 15.2). Liquidation risk is another answer: if a margin call on the short leg cannot be met in time, the venue closes the short and leaves the spot leg unhedged, on the dated and on the perpetual trade alike.

Self-check. (a) Going back from the rate to the price, $64{,}000\, e^{0.09877 \times 0.25} = 64{,}000 \times 1.02500 = 65{,}600$. Over one quarter the simple annualised rate is the rate compounded $m = 4$ times a year, because $(1 + r_4/4)^{4 \times 0.25} = 1.025$ gives $r_4 = 10.00\%$ (Definition 16.1(i)). The continuous and the simple figure are therefore consistent with the conversion shortcut of Proposition 16.2(iii) at $m = 4$: the simple quarterly figure exceeds the continuous one by about $r^2/(2m) = 0.0988^2/8 = 0.12\%$, which is $10.00\% - 9.88\%$. (b) The 795 is also the excess of the quote over the cash-and-carry bound: $64{,}000\, e^{0.0125} = 64{,}805$ and $65{,}600 - 64{,}805 = 795$. Through the spread of (a): $64{,}805.02\,(e^{0.048770 \times 0.25} - 1) = 64{,}805.02 \times 0.012267 = 794.98$. (c) Through the continuous rate, $1095 \ln(1.00012) = 13.139\%$ and $e^{0.13139} - 1 = 14.04\%$; to second order, $0.1314 + 0.1314^2/2 = 14.00\%$, and the cubic term $0.1314^3/6 = 0.04\%$ closes the gap.

Answer: (a) 9.88% continuously compounded, 10.00% simple; 4.88 points above the dollar rate; (b) buy spot and sell the future: 1,600 per bitcoin at expiry at any price, and 795 after 805 of financing; (c) 13.14% simple and 14.04% compounded, but the rate floats and nothing forces convergence; (d) 16,400 per bitcoin, paid at once; recovered when the bitcoin is sold, at expiry at the latest (earlier only if the futures price falls back); venue (custody) risk, or liquidation risk.

P16.11 *Ten claims about futures, forecasts and hedges: true or false*

(i) False. The asset is an investment asset with no income and no storage cost, so its forward price, which is also its futures price because rates are deterministic (Theorem 16.28), is $F = Se^{rT} = 50 \times 1.03 = 51.50$ whatever its beta (Theorem 16.11, with $e^{rT} = 1 + r_f = 1.03$). Under the CAPM its expected simple return is $r_f + \beta \times 6\%$. For $\beta = 1.5$ this is 12%, so $\mathbb{E}[S_T] = 56.00 > F$; for $\beta = -0.5$ it is 0%, so $\mathbb{E}[S_T] = 50.00 < F$. The futures price lies below the expected spot price for a positive beta and above it for a negative beta, and it is unbiased only when $\beta = 0$ (Proposition 16.31).

(ii) False. $h^* = \rho\, \sigma_S/\sigma_F = 0.6 \times 5/3 = 1$, yet the hedge removes only $\rho^2 = 0.36$ of the variance, and the hedged standard deviation is $5\sqrt{1 - 0.36} = 4$, against 5 unhedged (Theorem 16.37). The slope says how large the hedge is; the correlation says how good it is.

(iii) True. With the hedge opened at t_1 and closed at t_2, $\Delta S - \Delta F = (S_2 - F_2) - (S_1 - F_1) = \text{basis}_2 - \text{basis}_1$. If basis_2 is known when the hedge is opened, this is a constant, so $\text{Var}(\Delta S - 1 \times \Delta F) = 0$. A variance is never negative, so $h = 1$ attains the minimum, and the minimiser is unique (Theorem 16.37(i)): $h^* = 1$. The standard case is a hedge closed at the delivery date of a futures contract on the asset itself, where $\text{basis}_2 = 0$.

(iv) False. The perfect hedge realises $F_1 + \text{basis}_2 = 100 + 0 = 100$ (Proposition 16.35; the basis vanishes at delivery for futures on the asset itself); the half hedge realises $120 + 0.5 \times (100 - 120) = 110$, and no hedge 120. A hedge fixes the outcome and does not improve it.

(v) False. At delivery $\text{basis}_T = 0$ (Proposition 16.27), so the hedge locks in $F_1 = 100$, the futures

price at which it was opened, not the spot price 97 of that day.

(vi) False. Full carry is $Se^{(r+r_{\text{st}})T} = 9{,}000\,e^{0.06} = 9{,}556.53$. Gold is an investment asset, so its futures price equals full carry (Proposition 16.21(i)): at 8,820 holders would sell $e^{0.02}$ units of gold for each unit they would have held at T, deposit the proceeds and buy one unit back forward, for a sure $9{,}556.53 - 8{,}820 = 736.53$ per unit bought forward. Copper is a consumption commodity, for which only the upper bound $F \le 9{,}556.53$ holds (Proposition 16.21(ii)). The quote 8,820 admits no arbitrage and implies the convenience yield $y = b - \ln(F/S)/T = 0.06 - \ln(0.98) = 8.02\%$ (Definition 16.23).

(vii) False. $F = Se^{(r-q)T} = 5{,}000\,e^{(0.025-0.04)\times 0.5} = 5{,}000\,e^{-0.0075} = 4{,}962.64 < 5{,}000$ (Proposition 16.15(ii)). The futures price is below spot whenever $q > r$, and for a commodity whenever $y > b$, as with the copper quote of (vi).

(viii) False. By Proposition 16.17(i), $V_t = (F_t - K)e^{-r\tau} = (57 - 60)\,e^{-0.02} = -2.94$. The contract cost nothing and is now a liability.

(ix) False. The two prices coincide when interest rates are deterministic (Theorem 16.28). With random rates they can differ: the futures price is the higher one exactly when the asset tends to be high when rates are high, and the gap is small, about 0.1% of the price in the one-year example of Remark 16.30.

(x) False. Here $V_F = 5{,}000 \times 50 = 250{,}000$, the hedge sells $N_F^* = 1.2 \times 20{,}000{,}000/250{,}000 = 96$ contracts, and $r\,\Delta t = 1.25\%$. With the futures down 8%, the portfolio returns $1.25\% + 0.5\% + 1.2 \times (-8\%) = -7.85\%$, a loss of 1,570,000, and the short futures gain $96 \times 0.08 \times 250{,}000 = 1{,}920{,}000$. The hedged book makes +350,000, or 1.75%: to first order, the riskless return $r\,\Delta t$ plus the alpha, as in Proposition 16.40(iii).

Closing part. In the market's sense (Definition 16.23(iv)) the comparison is with today's spot price: $80 < 82 < 84$, the curve rises with the delivery date, and the market is in contango, not in backwardation. In Keynes's sense (Remark 16.33) the comparison is with the expected spot price: $F = 84 < \mathbb{E}[S_T] = 88$ is normal backwardation, and a long futures position expects to gain 4. The two usages give opposite answers on the same curve. The first is read off a screen; the second needs a view of $\mathbb{E}[S_T]$.

Self-check. The numbers are recomputed by second routes. (i) The expected gain of a long futures position is $\mathbb{E}[S_T] - F = S\beta \times 6\%$: $50 \times 1.5 \times 0.06 = 4.50 = 56.00 - 51.50$, and $50 \times (-0.5) \times 0.06 = -1.50 = 50.00 - 51.50$. (ii) Directly, $\text{Var}(\Delta S - \Delta F) = 25 - 2 \times 0.6 \times 5 \times 3 + 9 = 16$, a standard deviation of 4. (vi) Carried at the convenience yield, the quote returns full carry: $8{,}820\,e^{0.080203} = 9{,}556.53$. Per unit of gold sold today, the gold arbitrage is $9{,}000\,e^{0.04} - 8{,}820\,e^{-0.02} = 721.94$, which is $736.53\,e^{-0.02}$. (vii) $F - S \approx S(r-q)T = 5{,}000 \times (-0.015) \times 0.5 = -37.5$, against the exact -37.36. (x) The beta of the hedged position is $1.2 - 96 \times 250{,}000/20{,}000{,}000 = 0$, so what remains is $(r\,\Delta t + \alpha) \times 20{,}000{,}000 = 0.0175 \times 20{,}000{,}000 = 350{,}000$.

Answer: Only (iii) is true. (i) false: $F = 51.50$ against $\mathbb{E}[S_T] = 56.00$ for $\beta = 1.5$ and 50.00 for $\beta = -0.5$; (ii) false: $h^* = 1$ with a hedged standard deviation of 4 out of 5; (iii) true: a known final basis makes $\Delta S - \Delta F$ constant; (iv) false: 100 fully hedged against 110 half hedged and 120 unhedged; (v) false: it locks in the futures price 100, not the spot price 97; (vi) false: gold must trade at 9,556.53, copper may trade at 8,820 with $y = 8.02\%$; (vii) false: $4{,}962.64 < 5{,}000$; (viii) false: the contract is worth -2.94; (ix) false: equal under deterministic rates only; (x) false: $+350{,}000 = 1.75\% = r\,\Delta t + \alpha$. Closing part: contango in the market's sense ($84 > 80$), normal backwardation in Keynes's sense ($84 < 88$).

28.17 Solutions — §17 Options: Payoffs, Bounds, Put–Call Parity and Trading Strategies

P17.1 *Writing a put versus buying a call*

(a) The carry factor is $e^{0.01} = 1.01005$, so the premiums carried to T are $3.05 \times 1.01005 = 3.08$ for the call and $1.60 \times 1.01005 = 1.62$ for the put. By Definition 17.1 the writer of the put earns the carried premium less the payoff of the long side, and the buyer of the call earns the payoff less the carried premium:

$$\text{short put: } 1.62 - (45 - S_T)^+, \qquad \text{long call: } (S_T - 45)^+ - 3.08.$$

S_T	38	42	45	48	52
short put	-5.38	-1.38	$+1.62$	$+1.62$	$+1.62$
long call	-3.08	-3.08	-3.08	-0.08	$+3.92$

(b) *Short put.* The put is exercised against its writer when $S_T < 45$. The breakeven solves $45 - S_T = 1.62$, so it is $S_T = 43.38$. The gain is at most the carried premium, 1.62, earned for every $S_T \geq 45$. The loss is largest at $S_T = 0$, where the writer pays 45 for a worthless share and keeps 1.62: a loss of 43.38. *Long call.* The call is exercised when $S_T > 45$. The breakeven solves $S_T - 45 = 3.08$, so it is $S_T = 48.08$. The loss is at most the carried premium, 3.08, for every $S_T \leq 45$, and the gain has no upper limit.

(c) Subtracting the two profits,

$$\big[(S_T - 45)^+ - 3.08\big] - \big[1.62 - (45 - S_T)^+\big] = (S_T - 45)^+ + (45 - S_T)^+ - 4.70 = |S_T - 45| - 4.70,$$

where $4.70 = (3.05 + 1.60)\, e^{0.01}$. A long straddle struck at 45 (Definition 17.32) pays $|S_T - 45|$ and costs $3.05 + 1.60 = 4.65$, which is 4.70 at T, so the right-hand side is its profit. The short put is ahead exactly when this difference is negative, $|S_T - 45| < 4.70$, that is for $40.30 < S_T < 49.70$. Writing the put is the better trade if the stock stays near the strike, and buying the call is the better trade after a large move in either direction: both positions gain when the stock rises, but they are not the same trade.

(d) The identity $x^+ - (-x)^+ = x$ at $x = S_T - 45$ gives the payoff $(S_T - 45)^+ - (45 - S_T)^+ = S_T - 45$. The net premium paid is $3.05 - 1.60 = 1.45$, which is $1.45 \times 1.01005 = 1.46$ at T, so the profit is

$$S_T - 45 - 1.46 = S_T - 46.46.$$

By Corollary 17.12(i) the pair is a long forward contract with delivery price 45, bought for 1.45. Paying 1.45 now and 45 at T amounts to paying $45 + 1.45 \times 1.01005 = 46.46$ at T, which is, to the cent, the forward price $F = 46\, e^{0.01} = 46.46$ of the stock (Theorem 16.11). The third trader holds a synthetic forward bought at the forward price: the profit is linear in S_T, with no floor and no cap.

Self-check. Adding the two rows of the table must reproduce (d): the sums are -8.46, -4.46, -1.46, $+1.54$ and $+5.54$, and each equals $S_T - 46.46$. Independently of the table, the quotes satisfy put–call parity (Theorem 17.9): $C - \mathcal{P} = 1.45$, against $S - Ke^{-rT} = 46 - 45\, e^{-0.01} = 46 - 44.55 = 1.45$, to the cent, which is why the synthetic forward costs the forward price to the cent. For (c), the long-call row minus the short-put row is $+2.30$, -1.70, -4.70, -1.70, $+2.30$, which is $|S_T - 45| - 4.70$ at the five prices.

Answer: (a) Short put -5.38, -1.38, $+1.62$, $+1.62$, $+1.62$; long call -3.08, -3.08, -3.08, -0.08, $+3.92$. (b) Short put: assigned for $S_T < 45$, breakeven 43.38, maximum gain 1.62, maximum loss 43.38 (at $S_T = 0$). Long call: exercised for $S_T > 45$, breakeven 48.08, maximum loss 3.08, gain unlimited. (c) The difference is $|S_T - 45| - 4.70$, the profit of a long straddle; the short put is ahead for $40.30 < S_T < 49.70$. (d) $S_T - 46.46$: a synthetic long forward at the forward price 46.46.

P17.2 *Lower bounds and the discounted strike*

The lower bounds of Proposition 17.6(iii) for a stock without dividends use the *discounted* strike: $C \geq S - Ke^{-rT}$ and $\mathcal{P} \geq Ke^{-rT} - S$. Here $e^{-rT} = e^{-0.03} = 0.97045$ and $e^{rT} = 1.03045$.

(a) The discounted strike is $38\,e^{-0.03} = 36.877$, so the put is worth at least $36.877 - 33 = 3.877$. The quote 3.60 is below this floor: an arbitrage. Buy the dominating portfolio, the put and the share, for 36.60, and borrow the whole 36.60; at T the loan is repaid with $36.60\,e^{0.03} = 37.715$.

Trade	$t = 0$	T, if $S_T < 38$	T, if $S_T \geq 38$
Buy the put	-3.60	$38 - S_T$	0
Buy the share	-33.00	S_T	S_T
Borrow 36.60	$+36.60$	-37.715	-37.715
Net	0	0.285	$S_T - 37.715 \geq 0.285$

The share and the put are worth $\max(S_T, 38) \geq 38$ at T, so the trade costs nothing and pays at least 0.285 in every state. The present value of 0.285 is $0.285\,e^{-0.03} = 0.277 = 3.877 - 3.60$, the size of the violation.

(b) The discounted strike is $30\,e^{-0.03} = 29.113$, so the call is worth at least $33 - 29.113 = 3.887$. The quote 3.50 is below this floor: an arbitrage. Buy the call, sell the share short, and lend the net proceeds $33.00 - 3.50 = 29.50$, which return $29.50\,e^{0.03} = 30.398$ at T.

Trade	$t = 0$	T, if $S_T \leq 30$	T, if $S_T > 30$
Buy the call	-3.50	0	$S_T - 30$
Short the share	$+33.00$	$-S_T$	$-S_T$
Lend 29.50	-29.50	$+30.398$	$+30.398$
Net	0	$30.398 - S_T \geq 0.398$	0.398

If $S_T > 30$ the call is exercised, and the 30 paid buys the share that closes the short sale. If $S_T \leq 30$ the call lapses and the share is bought in the market for $S_T \leq 30$. At least 0.398 is kept in every state (0.3984 before rounding), and its present value is $0.3984\,e^{-0.03} = 0.387 = 3.887 - 3.50$, the size of the violation.

(c) An American put is worth at least its intrinsic value (Proposition 17.6(i)), here $38 - 33 = 5.00$, and the quote is 4.90: an arbitrage that needs no waiting. Buy the put for 4.90 and the share for 33.00, and exercise at once, delivering the share for 38. The cash flow today is $-4.90 - 33.00 + 38.00 = +0.10$, and nothing is left to settle.

(d) This quote is not an arbitrage. A European put cannot be exercised today, so the trade of (c) is not available: its strike arrives only at T and is worth 36.877 today. By Proposition 17.6(ii) and (iii) the European put must lie in $[Ke^{-rT} - S,\ Ke^{-rT}] = [3.877,\ 36.877]$, and 4.90 lies inside this band although it is below the intrinsic value 5.00: the time value of a European put can be negative (Proposition 17.18(i) describes the extreme case). The quote is also the Black–Scholes–Merton price (Corollary 20.23, the stock form of Theorem 20.22 in §20) at a volatility of about 20.7%, in a market that is free of arbitrage by its assumption (A6) (Remark 20.14).

Self-check. Put–call parity (Theorem 17.9) tests the same quotes by another route: a European quote is below its lower bound exactly when the option of the other type that parity implies, $C = \mathcal{P} + S - Ke^{-rT}$ or $\mathcal{P} = C - S + Ke^{-rT}$, has a negative price. In (a) the implied call is $3.60 + 33 - 36.877 = -0.277$. In (b) the implied put is $3.50 - 33 + 29.113 = -0.387$. Both are negative, by exactly the present values locked in. In (d) the implied call is $4.90 + 33 - 36.877 = 1.023$, which lies between 0 and $S = 33$, as the price of a call must: no bound is violated. Parity does not test (c): it does not hold for American options (Proposition 17.14 gives only a band, and no call is quoted), and the European floor 3.877 that $\mathcal{P}^{\text{am}} \geq \mathcal{P}$ passes on is met by 4.90. Only the intrinsic

floor of Proposition 17.6(i) catches the quote, and the shortfall below it, $5.00 - 4.90 = 0.10$, is the cash the trade of (c) collects today.

Answer: (a) Arbitrage: the floor is $38\,e^{-0.03} - 33 = 3.877 > 3.60$; buy the put and the share, borrow 36.60, and keep at least 0.285 at T (present value 0.277). (b) Arbitrage: the floor is $33 - 30\,e^{-0.03} = 3.887 > 3.50$; buy the call, short the share, lend 29.50, and keep at least 0.398 at T (present value 0.387). (c) Arbitrage: the quote is below the intrinsic value 5.00; buy the put and the share and exercise at once, for +0.10 today. (d) No arbitrage: 4.90 lies in $[3.877, 36.877]$; a European put can trade below its intrinsic value.

P17.3 *Put–call parity from scratch, in forward form, and an arbitrage on the spot*

(a) We use the law of one price: two portfolios with the same payoff in every state cost the same (Lemma 16.8(iii)). *Portfolio 1* (the fiduciary call) holds the call and a zero-coupon bond paying K at T; the bond costs Ke^{-rT} (Lemma 16.8(iv)), so the portfolio costs $C + Ke^{-rT}$. *Portfolio 2* (the protective put) holds the put and one share and costs $\mathcal{P} + S$. Both are held unchanged to T, and neither receives nor pays anything before T, because the share pays no income. Their payoffs are $(S_T - K)^+ + K$ and $(K - S_T)^+ + S_T$:

	Portfolio 1	Portfolio 2
if $S_T \geq K$	$(S_T - K) + K = S_T$	$0 + S_T = S_T$
if $S_T < K$	$0 + K = K$	$(K - S_T) + S_T = K$

Both pay $\max(S_T, K)$ in every state. If one cost less than the other, buying it and selling the other would bring in money today and leave nothing to pay at T, which is an arbitrage. Hence $C + Ke^{-rT} = \mathcal{P} + S$, that is $C - \mathcal{P} = S - Ke^{-rT}$ (Theorem 17.9).

(b) A long call together with a short put pays $(S_T - K)^+ - (K - S_T)^+ = S_T - K$: for $S_T \geq K$ this is $(S_T - K) - 0$, and for $S_T < K$ it is $0 - (K - S_T)$. The same payoff is produced without options. Enter a long forward at the forward price F, which costs nothing and pays $S_T - F$, and deposit $e^{-rT}(F - K)$ (a loan when $F < K$), which pays $F - K$. Together they pay $S_T - K$ and cost $e^{-rT}(F - K)$. By the law of one price,

$$C - \mathcal{P} = e^{-rT}(F - K),$$

which is Corollary 17.11(iii). Neither side of the comparison holds the underlying, so the argument needs only the forward. The right-hand side is zero exactly when $K = F$; it is positive for $K < F$ and negative for $K > F$. For the index, $F = 80\,e^{0.01} = 80.804$ and, at $K = 80$,

$$C - \mathcal{P} = e^{-0.03}(80.804 - 80) = 0.97045 \times 0.804 = 0.780.$$

The call is dearer than the put by 0.78, because the strike is below the forward price. The two cost the same at $K = F = 80.80$, not at $K = S = 80$.

(c) The discounted strike is $60\,e^{-0.015} = 59.107$, so parity requires $C - \mathcal{P} = 58 - 59.107 = -1.107$. The market has $2.95 - 4.80 = -1.85$. The call is cheap relative to the put by $1.85 - 1.107 = 0.743$: given the put, parity prices the call at $4.80 - 1.107 = 3.693$. Buy the cheap side and sell the dear one. Buy the call and write the put, which together are a long forward with delivery price 60 (Corollary 17.12(i)); sell the share short; and lend the net proceeds $58.00 + 4.80 - 2.95 = 59.85$, which return $59.85\,e^{0.015} = 60.7545$ at T.

Trade	$t = 0$	T, if $S_T \leq 60$	T, if $S_T > 60$
Buy the call	-2.95	0	$S_T - 60$
Write the put	$+4.80$	$-(60 - S_T)$	0
Short the share	$+58.00$	$-S_T$	$-S_T$
Lend 59.85	-59.85	$+60.7545$	$+60.7545$
Net	0	0.7545	0.7545

In both columns the share that closes the short sale is bought for exactly 60: above 60 by exercising the call, and below 60 because the put is exercised against the position. The trade costs nothing and pays 0.7545, or 0.755 to three decimals, in every state; its present value is $0.7545\,e^{-0.015} = 0.743$, the mispricing. The violation also shows up as two implied volatilities for one strike and one expiry, about 21.1% for the call and 25.6% for the put (§20), which is a parity violation and not a smile.

(d) The trade of (c) sells the share short, and on a hard-to-borrow stock the lender of the share charges a fee for the whole life of the trade. Write q_{fee} for the yearly rate of the fee (the label keeps it apart from the index yield q of (b)): over $[t, t+dt]$ the short seller pays $q_{\text{fee}} S_t\,dt$ per borrowed share, as it would pay a dividend yield to the lender. Redo the trade of (c) with the fee. Sell $e^{-q_{\text{fee}}T}$ shares short today, and pay the fee by selling $q_{\text{fee}}\,dt$ more shares short per share already short over each $[t, t+dt]$. The number of shares short then grows like a reinvested holding with yield q_{fee} and reaches exactly one share at T (the count of Proposition 16.15(i), run short), and the short sale brings in $Se^{-q_{\text{fee}}T}$ today. Buying the call, writing the put and lending $Se^{-q_{\text{fee}}T} + \mathcal{P} - C$ leaves

$$\left(Se^{-q_{\text{fee}}T} + \mathcal{P} - C\right) e^{rT} - K$$

at T in every state, because the call and the put together pay $S_T - K$ and the one share still short is bought back for S_T. This is positive exactly when $C - \mathcal{P} < Se^{-q_{\text{fee}}T} - Ke^{-rT}$, the yield form of parity (Corollary 17.11(ii)) with the fee as the yield. Here it is an inequality and not an equality: the short seller pays the fee, while a holder of the share collects it only by lending the share out. The trade therefore stops making money once

$$58\,e^{-0.5q_{\text{fee}}} \le -1.85 + 59.107 = 57.257, \qquad \text{that is} \qquad q_{\text{fee}} \ge -\frac{1}{0.5}\ln\frac{57.257}{58} = 2.58\%.$$

A fee of 2.58% a year or more absorbs the whole 0.743, and the persistent gap is then the price of the short sale, not an arbitrage. The opposite trade, a conversion (buy the share and the put, write the call), needs no short sale; it is a genuine arbitrage when the put is the cheap side, and here, since $-1.85 < -1.107$, it loses.

Self-check. For (c), read parity backwards. The quotes imply a forward price $K + e^{rT}(C - \mathcal{P}) = 60 - 1.85\,e^{0.015} = 58.122$, while buying the share and carrying it gives the forward price $58\,e^{0.015} = 58.877$. The gap, 0.755, is the amount locked in at T, found without the cash-flow table. For (b), the yield form of Corollary 17.11(ii) gives the same number: $80\,e^{-0.02} - 80\,e^{-0.03} = 78.416 - 77.636 = 0.780$. For (d), the fee acts as a yield on the short side, so at the break-even fee the implied forward equals $Se^{(r-q_{\text{fee}})T}$: $q_{\text{fee}} = r - \frac{1}{T}\ln\frac{58.122}{58} = 3\% - 0.420\% = 2.58\%$, the fee found above.

Answer: (a) The call with a bond paying K and the put with the share both pay $\max(S_T, K)$, so $C + Ke^{-rT} = \mathcal{P} + S$. (b) The call minus the put pays $S_T - K$, as does a forward entered at F with a deposit of $e^{-rT}(F - K)$, so $C - \mathcal{P} = e^{-rT}(F - K)$, zero exactly at $K = F$; for the index $F = 80.80$ and $C - \mathcal{P} = 0.780$ at $K = 80$ (the call is dearer), with equality at $K = 80.80$. (c) Parity requires -1.107 against the quoted -1.85: the call is cheap by 0.743; buy the call, write the put, short the share and lend 59.85, which leaves 0.755 at T in every state (present value 0.743). (d) The trade needs a short sale; it stops making money at a borrowing fee of about 2.58% a year (the break-even fee), and any fee at least this large makes the quotes consistent.

P17.4 *Parity with a discrete dividend*

(a) The present value of the dividend is $I = 0.90\,e^{-0.01} = 0.891$, and the discounted strike is $50\,e^{-0.02} = 49.010$. Parity with known income (Corollary 17.11(i)) requires

$$C - \mathcal{P} = S - I - Ke^{-rT} = 52 - 0.891 - 49.010 = 2.099.$$

The market has $4.60 - 2.80 = 1.80$. The call is cheap relative to the put by $2.099 - 1.80 = 0.299$.

(b) Buy the call, write the put and sell the share short, which brings in $52.00 - 4.60 + 2.80 = 50.20$. The short seller owes the lender of the share the dividend of 0.90 at three months; deposit its

present value, 0.891, until then. Lend the remaining $50.20 - 0.891 = 49.309$ until T, when it returns $49.309\,e^{0.02} = 50.305$.

Trade	$t = 0$	$t = 0.25$	T, if $S_T \le 50$	T, if $S_T > 50$
Buy the call	-4.60		0	$S_T - 50$
Write the put	$+2.80$		$-(50 - S_T)$	0
Short the share	$+52.00$	-0.90	$-S_T$	$-S_T$
Deposit 0.891 (3 months)	-0.891	$+0.90$		
Lend 49.309 (6 months)	-49.309		$+50.305$	$+50.305$
Net	0	0	0.305	0.305

At T the share that closes the short sale costs exactly 50 in both columns: through the call above 50, and through the put exercised against the position below 50. The trade costs nothing, pays the dividend out of its own deposit, and leaves $50.305 - 50 = 0.305$ at T in every state. The present value is $0.305\,e^{-0.02} = 0.299$, the mispricing.

(c) The present value of the dividend rises by $0.60\,e^{-0.01} = 0.594$, to 1.485, and the value of $C - \mathcal{P}$ required by parity falls by the same 0.594, from 2.099 to $52 - 1.485 - 49.010 = 1.505$. In forward terms, the forward price $F = (S - I)e^{rT}$ falls from 52.14 to 51.54, and $C - \mathcal{P} = e^{-rT}(F - K)$ follows it. The absence of arbitrage fixes only this difference. On each price separately it gives only bounds (Proposition 17.6), and these move with the dividend: the upper bound $S - I$ of the call falls from 51.109 to 50.515 and its lower bound from 2.099 to 1.505, while the bounds $0 \le \mathcal{P} \le 49.010$ of the put do not move (its lower bound $Ke^{-rT} + I - S$ is negative both times, -2.099 and then -1.505). No sign for either price follows. That the call falls and the put rises needs a model: under the Black–Scholes–Merton model it does (the dividend row of Table 3), and by parity the fall of the call and the rise of the put add up to 0.594.

(d) The band reads

$$52 - 0.891 - 50 = 1.109 \;\le\; C^{\text{am}} - \mathcal{P}^{\text{am}} \;\le\; 52 - 49.010 = 2.990,$$

and the quoted difference 1.80 lies inside it: the band is not violated. The trade of (b) is no longer riskless, because the written put can be exercised against the position at a time t_e that its writer does not choose, and the strike of 50 must then be paid at t_e and not at T. Before the dividend date the cash in hand, $50.20\,e^{0.04t_e} \ge 50.20$, covers it: the share received closes the short sale before it goes ex, so the dividend deposit is free as well. After the dividend has been paid, the loan is worth only $49.309\,e^{0.04t_e}$, which is 49.805 just after the dividend date and reaches 50 only at $t_e = 0.348$, about 4.2 months. An assignment in between leaves a shortfall of up to $50 - 49.805 = 0.195$, against which the position holds only the call; and a holder exercises a put early to advantage only when the stock is well below the strike (the intuition after Proposition 17.18; a model locates the boundary), which is when that call is worth little. The profit of (b) is then no longer locked in.

Self-check. The forward route reproduces (a) and (b) without the table. The forward price is $F = (52 - 0.891)\,e^{0.02} = 52.141$, so $C - \mathcal{P} = e^{-0.02}(52.141 - 50) = 2.099$. The quotes imply a forward of $50 + 1.80\,e^{0.02} = 51.836$, and the gap $52.141 - 51.836 = 0.305$ is the profit at T. The dividend deposit is also consistent: $0.891\,e^{0.01} = 0.90$. In (c), the forward falls by the dividend change carried from the dividend date to T, $0.60\,e^{0.01} = 0.606$ (from 52.141 to 51.535), and $e^{-0.02} \times 0.606 = 0.594$. In (d), the shortfall just after the dividend is the strike plus the dividend paid less the cash grown to the dividend date: $50 + 0.90 - 50.20\,e^{0.01} = 50.90 - 50.705 = 0.195$.

Answer: (a) Parity requires $C - \mathcal{P} = 52 - 0.891 - 49.010 = 2.099$ against the quoted 1.80: the call is cheap by 0.299. (b) Buy the call, write the put and short the share (+50.20); deposit 0.891 to pay the dividend and lend 49.309, which returns 50.305; the share is bought back for 50, leaving 0.305 at T (present value 0.299). (c) The parity value falls by 0.594, to 1.505; only the difference is fixed without a model (the bounds of the call drop by 0.594, those of the put do not move). (d) No violation: 1.80 lies in $[1.109, 2.990]$; early assignment of the put between the dividend date and about month 4.2 can leave a shortfall of up to 0.195.

P17.5 *Bounds on an American put from an American call, with and without dividends*

(a) Solving the two inequalities of the band (Proposition 17.14) for the put gives $C^{\text{am}} - S + Ke^{-rT} \le \mathcal{P}^{\text{am}} \le C^{\text{am}} - S + K$. With $Ke^{-rT} = 52\,e^{-0.03} = 50.463$,

$$5.80 - 55 + 50.463 = 1.263 \;\le\; \mathcal{P}^{\text{am}} \;\le\; 5.80 - 55 + 52 = 2.80.$$

The interval is $K(1 - e^{-rT}) = 1.537$ wide, the interest on the strike over the life of the options.

(b) The quote 3.00 is above 2.80. Write the put (+3.00), sell the share short (+55.00), buy the call (−5.80) and deposit 52 (−52.00): the trade collects 0.20 today, which is set aside. The call is held to T and exercised there only if it is in the money, so it pays $(S_T - 52)^+$.

- If the put is exercised against the position at a time $t_e < T$, the deposit is worth $52\,e^{0.04t_e} \ge 52$. Pay 52 out of it, receive the share and return it to its lender, which closes the short sale. What remains is the call and the interest $52(e^{0.04t_e} - 1) \ge 0$, both worth a nonnegative amount at T.
- If the put is not exercised before T, its writer owes at most $(52 - S_T)^+$ at T. The deposit is worth $52\,e^{0.03} = 53.584$, the call pays $(S_T - 52)^+$ and buying back the share costs S_T, so the position is worth at least

  $$53.584 + (S_T - 52)^+ - (52 - S_T)^+ - S_T = 53.584 + (S_T - 52) - S_T = 1.584,$$

  by the identity $x^+ - (-x)^+ = x$ at $x = S_T - 52$.

Every exercise decision leaves a nonnegative value at T on top of the 0.20 collected today, which is an arbitrage (Lemma 17.5(ii)). A put quoted above 2.80 cannot persist.

(c) We use exercise matching (Lemma 17.5(ii)), once for each edge: a position that brings in money today and is worth a nonnegative amount at T under every exercise policy of the holder of the written option is an arbitrage. Both edges use the identity $x^+ - (-x)^+ = x$.

Lower edge. Suppose that $\mathcal{P}^{\text{am}} + S > C^{\text{am}} + K + I$. Write the put, sell one share short, buy the call and deposit K; for each dividend, deposit $D_i e^{-rt_i}$ until t_i, when it has grown to D_i. These dividend deposits cost I in total, and the inflow $\mathcal{P}^{\text{am}} + S - C^{\text{am}} - K - I > 0$ is set aside. While the short sale is open, each dividend owed to the lender of the share is paid by its own deposit. The call is held to T and exercised there only if it is in the money, so it pays $(S_T - K)^+$.

- If the put is exercised against the position at a time $t_e < T$, the deposit of K is worth $Ke^{rt_e} \ge K$, because $r \ge 0$. Pay K out of it, receive the share and return it to its lender. The dividends that go ex after the share has been returned are no longer owed, and their deposits stay in the position. What remains is the call, the interest $K(e^{rt_e} - 1) \ge 0$ and those deposits: a nonnegative value at T.
- If the put is not exercised before T, every dividend has been paid by its deposit, the writer owes at most $(K - S_T)^+$, the call pays $(S_T - K)^+$, the deposit is worth Ke^{rT} and buying back the share costs S_T. The position is worth at least

  $$Ke^{rT} + (S_T - K)^+ - (K - S_T)^+ - S_T = K(e^{rT} - 1) \;\ge\; 0.$$

Every policy ends with a nonnegative value after a positive inflow, which is excluded. Hence $\mathcal{P}^{\text{am}} + S \le C^{\text{am}} + K + I$, that is $S - I - K \le C^{\text{am}} - \mathcal{P}^{\text{am}}$.

Upper edge. Suppose that $C^{\text{am}} + Ke^{-rT} > \mathcal{P}^{\text{am}} + S$. Write the call, borrow Ke^{-rT}, buy the put and buy one share; the inflow $C^{\text{am}} + Ke^{-rT} - \mathcal{P}^{\text{am}} - S > 0$ is set aside. The dividends received on the share are deposited, and the put is held to T and exercised there only if it is in the money, so it pays $(K - S_T)^+$.

- If the call is exercised against the position at a time $t_e < T$, deliver the share for K and repay the loan, which is then worth $Ke^{-r(T-t_e)} \le K$, because $r \ge 0$. What remains is the cash $K - Ke^{-r(T-t_e)} \ge 0$, the dividends already received and the put: a nonnegative value at T.

- If the call is not exercised before T, its writer owes at most $(S_T - K)^+$, the put pays $(K - S_T)^+$, the share is worth S_T, the loan costs K and the deposited dividends are worth a nonnegative amount. The position is worth at least

$$(K - S_T)^+ + S_T - K - (S_T - K)^+ = 0.$$

Hence $C^{\text{am}} + Ke^{-rT} \le \mathcal{P}^{\text{am}} + S$, that is $C^{\text{am}} - \mathcal{P}^{\text{am}} \le S - Ke^{-rT}$. The dividends lower the lower edge by I and leave the upper edge where it was: on the upper edge the share is held long, where it only collects them.

(d) The dividend has present value $I = 0.70\,e^{-0.04/3} = 0.691$. Solving the band of (c) for the put,

$$C^{\text{am}} - S + Ke^{-rT} \;\le\; \mathcal{P}^{\text{am}} \;\le\; C^{\text{am}} - S + I + K,$$

so the put lies in $[1.263,\ 5.80 - 55 + 0.691 + 52] = [1.263,\ 3.491]$. The lower end is unchanged and the upper end rises by I, so the band of (c) no longer excludes the quote 3.00 of (b) on this dividend-paying stock. The band is not the tightest bound these numbers allow. The written put is assigned either before the dividend date, when the short sale is closed before any dividend is owed and 52 in cash suffices, or after it, when the cash must also have paid the 0.70. A single deposit of $(52 + 0.70)\,e^{-0.04/3} = 52.002$ covers both cases: it is worth more than 52 until the dividend date, exactly $52.70 - 0.70 = 52$ just after the dividend is paid out of it, and more than 52 afterwards, and without an assignment it leaves $52(e^{0.04\times 5/12} - 1) = 0.874 \ge 0$ at T, as in (b). Running the lower-edge trade of (c) with this single deposit in place of the deposits $K + I = 52.691$ caps the put at $5.80 - 55 + 52.002 = 2.802$, so the quote 3.00 is still an arbitrage.

Self-check. At $I = 0$ the band of (c) is the band of (a). The two ends of (a) follow from a different route as well. On a stock without dividends $C^{\text{am}} = C$ (Theorem 17.16(i)), so put–call parity (Theorem 17.9) prices the European put at $5.80 - 55 + 50.463 = 1.263$, which is the lower end, as $\mathcal{P}^{\text{am}} \ge \mathcal{P}$ (Proposition 17.6(i)) requires; and the upper end is that value plus the interest on the strike, $1.263 + 1.537 = 2.80$. For (d), the dividend deposit returns the dividend, $0.691\,e^{0.04/3} = 0.700$, and the upper end is that of (a) raised by I: $2.80 + 0.691 = 3.491$; the call quote is still admissible there, since $5.80 \ge S - I - Ke^{-rT} = 55 - 0.691 - 50.463 = 3.846$. For (b), three paths in numbers: an assignment at $t_e = 0.25$ leaves the call and $52\,e^{0.01} - 52 = 0.52$; no assignment and $S_T = 40$ gives $53.584 - 12 - 40 = 1.584$; no assignment and $S_T = 70$ gives $53.584 + 18 - 70 = 1.584$.

Answer: (a) $1.263 \le \mathcal{P}^{\text{am}} \le 2.80$. (b) Write the put, short the share, buy the call and deposit 52, for +0.20 today; an assignment at t_e leaves the call and $52(e^{0.04t_e} - 1) \ge 0$, and without one the position is worth at least 1.584 at T. (c) Lower edge: written put, short share, call held to T, deposits of K and I; upper edge: written call, loan of Ke^{-rT}, long put, long share; each position is nonnegative at T under every exercise policy. (d) $1.263 \le \mathcal{P}^{\text{am}} \le 3.491$ from the band of (c); a single deposit of 52.002 sharpens the upper end to 2.802.

P17.6 *Calls are never exercised early; puts can be*

(a) We use exercise matching (Lemma 17.5(ii)): a position that brings in money today and is worth a nonnegative amount at T whatever the holder of the written option does is an arbitrage. Since $C^{\text{am}} \ge C$ is known, suppose that $C^{\text{am}} > C$. Write the American call and buy the European call; the inflow $C^{\text{am}} - C > 0$ is set aside.

- If the American call is exercised against the position at a time $t_e < T$, borrow one share, deliver it for K and deposit the K. Nothing is owed to the lender of the share before T, because the share pays no income. At T the deposit is worth $Ke^{r(T-t_e)} \ge K$, the European call pays $(S_T - K)^+$ and buying the share back costs S_T, so the position is worth

$$(S_T - K)^+ - S_T + Ke^{r(T-t_e)} \;\ge\; (S_T - K)^+ - (S_T - K) = (K - S_T)^+ \;\ge\; 0,$$

by the identity $x^+ - x = (-x)^+$ at $x = S_T - K$.

- If it is not exercised before T, its writer owes at most $(S_T - K)^+$ at T, which the European call pays.

Every exercise policy leaves a nonnegative value after a positive inflow, which is excluded. Hence $C^{\text{am}} \le C$, and the two prices are equal (Theorem 17.16(i)). The same argument, started at a date $t < T$, gives $C_t^{\text{am}} = C_t$ (Lemma 17.5(i)), and then, for $r > 0$,

$$C_t^{\text{am}} = C_t \;\ge\; S_t - Ke^{-r\tau} \;>\; S_t - K,$$

because $Ke^{-r\tau} < K$, as $r > 0$, $\tau > 0$ and $K > 0$: the call sells for strictly more than exercise pays.

Monotonicity in the expiry. The same kind of trade compares two European calls, by dominance at T_2 (Lemma 16.8(i)). Buy the call that expires at T_2 and write the one that expires at T_1. At T_1, if $S_{T_1} > K$, sell one share short, pay the $S_{T_1} - K$ owed to the holder out of the proceeds and deposit the remaining K; at T_2 the position is then worth

$$(S_{T_2} - K)^+ - S_{T_2} + Ke^{r(T_2 - T_1)} \;\ge\; (S_{T_2} - K)^+ - (S_{T_2} - K) = (K - S_{T_2})^+ \;\ge\; 0,$$

because $r \ge 0$ and no dividend is owed on the short share. If $S_{T_1} \le K$, nothing is done and the position is worth $(S_{T_2} - K)^+ \ge 0$. The strategy is self-financing, trades only at T_1 on what is known then, and has a nonnegative payoff, so its price is nonnegative: $C(K, T_2) - C(K, T_1) \ge 0$ (Theorem 17.16(iii)).

What exercise gives up. Compare, at T, a holder who keeps the call with one who exercises at t_e and finances the strike with a loan, ending with $S_T - Ke^{r(T-t_e)}$. The first is ahead by

$$(S_T - K)^+ - \big(S_T - Ke^{r(T-t_e)}\big) = K\big(e^{r(T-t_e)} - 1\big) + (K - S_T)^+.$$

The first term is the time value of money: the strike is paid later. The second is insurance: the call carries a put struck at K, which protects its holder if the share ends below the strike. When $r > 0$ the first term is positive, so the holder who keeps the call is strictly ahead in every state. At $r = 0$ the first term vanishes and the second remains, which is why $C^{\text{am}} = C$ holds at $r = 0$ as well.

(b) (1) Exercising and selling the share brings in $140 - 100 = 40.00$ today. (2) The call is worth at least $140 - 100\,e^{-0.04/12} = 140 - 99.667 = 40.33$, so selling it brings in at least 0.33 more, the interest on the strike for one month, and on top of that whatever the market pays for the embedded put. (3) Selling the share short for 140 and depositing the proceeds gives $140\,e^{0.04/12} = 140.47$ at T. If $S_T > 100$, the call is exercised and the 100 paid buys the share that closes the short sale, leaving 40.47. If $S_T \le 100$, the share is bought in the market, leaving $140.47 - S_T \ge 40.47$. Course (3) is worth $140\,e^{0.04/12} - 100 + (100 - S_T)^+ = 40.47 + (100 - S_T)^+$ at T, which is worth at least $140 - 100\,e^{-0.04/12} = 40.33$ today. Courses (2) and (3) both beat exercise by at least 0.33: the holder who wants out of the market sells the call, and the bearish holder who wants to stay in it sells the share short, which turns the call into a put.

(c) *The put.* Mirror the proof: write the American put and buy the European put. If the put is exercised against the position at t_e, the strike must be *paid*: borrow K, take the share, and hold the share and the European put to T. They pay $S_T + (K - S_T)^+ = \max(S_T, K)$, but when $r > 0$ the loan has grown to $Ke^{r(T-t_e)} > K$, and the position is short of money whenever $\max(S_T, K) < Ke^{r(T-t_e)}$, for instance whenever $S_T \le K$. In (a) the strike was received early and earned interest; for the put it is paid early and costs interest, so the step of (a) "at T the deposit is worth $Ke^{r(T-t_e)} \ge K$" becomes "the loan has grown to $Ke^{r(T-t_e)} > K$" when $r > 0$, and the hedge fails. The bounds show that the conclusion fails too:

$$\mathcal{P} \;\le\; Ke^{-rT} = 60\,e^{-0.12} = 53.22 \;<\; 56 = K - S \;\le\; \mathcal{P}^{\text{am}},$$

so the American put is worth at least 2.78 more than the European put (Proposition 17.18(i)). The chain needs $Ke^{-rT} < K - S$, that is $S < K(1 - e^{-rT}) = 60(1 - e^{-0.12}) = 6.78$.

A call on an asset with a yield. The step that fails is "nothing is owed to the lender of the share before T". A short position in an asset with a yield q owes the yield to the lender; if it is paid out of further short sales, one share sold short at t_e has become $e^{q(T-t_e)}$ shares to buy back at T, and $(S_T-K)^+ - S_T e^{q(T-t_e)} + Ke^{r(T-t_e)}$ is negative for large S_T. Again the bounds decide:

$$C \;\le\; Se^{-qT} = 150\,e^{-0.12} = 133.04 \;<\; 140 = S-K \;\le\; C^{\text{am}},$$

so the American call is worth at least 6.96 more than the European call (Proposition 17.20(iv)). This chain works whenever $K < S(1-e^{-qT})$, here 16.96.

(d) Let $r=0$. Since $\mathcal{P}^{\text{am}} \ge \mathcal{P}$ is known, suppose that $\mathcal{P}^{\text{am}} > \mathcal{P}$: write the American put, buy the European put and set the inflow aside. If the put is exercised against the position at a time $t_e < T$, borrow K, pay it and receive the share. At T the share and the European put pay $S_T + (K-S_T)^+ = \max(S_T, K) \ge K$, which repays the loan, still K because $r=0$. If the put is not exercised before T, its writer owes at most $(K-S_T)^+$, which the European put pays. Every policy ends with a nonnegative value after a positive inflow, which Lemma 17.5(ii) excludes, so $\mathcal{P}^{\text{am}} = \mathcal{P}$ (Proposition 17.18(iii)); the same argument, started at a date $t<T$, gives $\mathcal{P}^{\text{am}}_t = \mathcal{P}_t$ (Lemma 17.5(i)). The interest that broke the argument in (c) is gone. Exercise can still tie. At $r=0$ put–call parity at a date t reads $\mathcal{P}_t = (K-S_t) + C_t$, so selling the American put, worth $\mathcal{P}^{\text{am}}_t = \mathcal{P}_t$, beats exercising it by the price C_t of the call struck at K, and the two tie exactly when that call is worthless. That happens at $S_t = 0$: the American put is then worth at least its intrinsic value K and, being equal to the European put, at most $Ke^{-r\tau} = K$, so it is worth exactly the K that exercise pays.

Self-check. In (b), courses (2) and (3) can be compared through put–call parity (Theorem 17.9). By (a), $C^{\text{am}} = C$, so the call is worth $S - Ke^{-r\tau} + \mathcal{P} = 40.33 + \mathcal{P}$, where $\mathcal{P}$ is the one-month European put struck at 100; course (3) pays $40.47 + (100 - S_T)^+$ at T, which is worth $40.33 + \mathcal{P}$ today as well. The two agree, as the law of one price requires, and both exceed 40.00. In (c), the early-exercise premium of the put is at least $56 - 53.22 = 2.78$ and, by the American parity band (Proposition 17.14) with $C^{\text{am}} = C$, at most the interest on the strike, $K(1-e^{-rT}) = 6.78$. The two bounds are consistent, and their gap is the spot price, $6.78 - 2.78 = 4$, as the algebra $K(1-e^{-rT}) - (K - S - Ke^{-rT}) = S$ requires.

Answer: (a) Write the American call and buy the European call; after an assignment at t_e, a share sold short and a deposit of K leave $(K-S_T)^+ + K(e^{r(T-t_e)} - 1) \ge 0$ at T, so $C^{\text{am}} = C$, and $C^{\text{am}}_t \ge S_t - Ke^{-r\tau} > S_t - K$ for $r>0$; the same trade between two expiries gives $C(K,T_1) \le C(K,T_2)$. Exercise gives up the interest on the strike and the embedded put (insurance); the insurance survives at $r=0$. (b) Exercise brings 40.00; selling the call, or shorting the share against it, is worth at least 40.33. (c) Put: on assignment the hedged writer must pay the strike early and borrow it at interest ($r>0$); $\mathcal{P} \le 53.22 < 56 \le \mathcal{P}^{\text{am}}$, valid for $S < 6.78$. Yield: the short share owes the yield; $C \le 133.04 < 140 \le C^{\text{am}}$. (d) At $r=0$ the loan of K is repaid by $\max(S_T,K) \ge K$, so $\mathcal{P}^{\text{am}} = \mathcal{P}$; exercise ties with selling only when the call struck at K is worthless, as at $S_t = 0$.

P17.7 *Ex-dividend dates and the American call*

(a) Write t_i^- for the instant before the ex-date t_i, $S_{t_i^-}$ for the price with the dividend D_i still attached, and t_{i+1} for the next ex-date, or for the expiry after the last dividend. The share goes ex at $S_{t_i} = S_{t_i^-} - D_i$, and no interest accrues in between. The claim is

$$C^{\text{am}}_{t_i^-} \;\ge\; S_{t_i^-} - D_i - Ke^{-r(t_{i+1}-t_i)}.$$

We argue by an arbitrage trade started at t_i^-. If the call were cheaper, buy it, sell the share short for $S_{t_i^-}$, and split what is left of the proceeds into three parts: D_i, paid to the lender of the share when it goes ex; $Ke^{-r(t_{i+1}-t_i)}$, deposited; and a surplus, which is positive and is set aside. From t_i on the position is the one behind the known bound: the call, one share sold short, and a deposit

that has grown to K the instant before t_{i+1}. At that instant close the short sale, by exercising the call for K if the share is worth more than K, and by buying the share for at most K otherwise. The share is returned before it goes ex again, so no further dividend is owed. The surplus is kept in every state, which is an arbitrage; hence the bound (Proposition 17.20(ii)).

Exercise at t_i^- pays $S_{t_i^-} - K$, and the bound exceeds this amount by $K\bigl(1 - e^{-r(t_{i+1}-t_i)}\bigr) - D_i$. Exercise is therefore ruled out when

$$D_i \;\le\; K\bigl(1 - e^{-r(t_{i+1}-t_i)}\bigr),$$

the interest on the strike until the next date; below the threshold, selling the call pays strictly more than exercising it. *First date.* The next date is t_2, half a year later, and the threshold is $80(1 - e^{-0.025}) = 1.975$. The dividend 0.60 is below it: exercise before the first ex-date is ruled out. *Second date.* The next date is the expiry, a quarter later, and the threshold is $80(1 - e^{-0.0125}) = 0.994$. The dividend 1.50 is above it: exercise before the second ex-date is not ruled out.

(b) At most $80(1 - e^{-0.0125}) = 0.9938$, that is 0.99 in whole cents. For a second dividend below this threshold, exercise just before t_2 pays strictly less than selling the call; at the threshold itself it gains nothing over selling.

(c) At any other time t, let t' be the next ex-date or the expiry. No dividend goes ex between t and t', so the known bound gives $C_t^{\text{am}} \ge S_t - Ke^{-r(t'-t)} > S_t - K$, because $r > 0$ and $t' > t$ (Proposition 17.20(i)). Exercising at such a time pays the strike earlier than necessary and captures no dividend that exercising the instant before the next ex-date would not capture as well. Only the instants before the ex-dates, and the expiry, are candidate exercise times.

(d) Exercise pays $120 - 80 = 40.00$. The bound of (a) is $120 - 1.50 - 80\,e^{-0.0125} = 118.50 - 79.006 = 39.494$, which is below 40.00, so the bound does not decide. After the ex-date no dividend remains, so the call is then worth the European call, $C_{t_2}^{\text{am}} = C_{t_2}$ (Theorem 17.16(i) at the date t_2), on a share at $S_{t_2} = 118.50$. By put–call parity at t_2 (Theorem 17.9) this is $118.50 - 79.006 + \mathcal{P}_{t_2} = 39.494 + \mathcal{P}_{t_2}$, where $\mathcal{P}_{t_2}$ is the European put struck at 80 that expires at T, three months later. Waiting beats exercising exactly when that put is worth more than $40.00 - 39.494 = 0.506$: the dividend captured by exercising, net of the interest saved on the strike by waiting, is weighed against the insurance that exercise gives up. No arbitrage argument fixes the price of that put; a model must supply it. In a market without volatility the put is worthless and exercise is right: the first comment after Proposition 17.20 shows that market free of arbitrage for the share, the cash account and calls priced at the present values of their sure payoffs, and the same argument covers the put, whose sure payoff is 0 because $S_T = 118.50\,e^{0.0125} = 119.99 > 80$. Under the Black–Scholes–Merton model (Corollary 20.23, the stock form of Theorem 20.22 in §20, with $S_{t_2} = 118.50$ and three months left; free of arbitrage by its assumption (A6), Remark 20.14) the put is worth 0.02 at a volatility of 30%, so exercise is right, and 1.17 at 60%, so waiting is right; the two courses tie near 49.5%.

Self-check. A second model-free test confirms the first date. The American call is worth at least the European call that expires at T, which is worth at least $S_{t_1^-} - I_{t_1^-} - Ke^{-r(T-t_1)}$, with $I_{t_1^-}$ the present value at t_1^- of both dividends (Proposition 17.6(i) and (iii) at that instant); this exceeds the exercise value $S_{t_1^-} - K$ when $I_{t_1^-} < K\bigl(1 - e^{-r(T-t_1)}\bigr)$. Here $I_{t_1^-} = 0.60 + 1.50\,e^{-0.025} = 2.063$ and $80(1 - e^{-0.0375}) = 2.944$, so exercise before the first ex-date is ruled out by this route too. At the second date the same test asks for $1.50 < 0.994$, which fails, in agreement with (a). The thresholds also match the rule of thumb "interest on the strike": $80 \times 0.05 \times 0.5 = 2.0$ and $80 \times 0.05 \times 0.25 = 1.0$, slightly above 1.975 and 0.994. The gap in (d) has a second reading: the excess of the second dividend over its threshold, $1.50 - 80(1 - e^{-0.0125}) = 1.50 - 0.994 = 0.506$, which matches $40.00 - 39.494$.

Answer: (a) Just before an ex-date, $C_{t_i^-}^{\text{am}} \ge S_{t_i^-} - D - Ke^{-r(t_{i+1}-t_i)}$, so exercise is ruled out when $D \le K\bigl(1 - e^{-r(t_{i+1}-t_i)}\bigr)$. The thresholds are 1.975 at the first date (0.60 is below it: exercise ruled

out) and 0.994 at the second (1.50 is above it: not ruled out). (b) At most $80(1-e^{-0.0125}) = 0.9938$. (c) Between ex-dates $C_t^{\text{am}} \geq S_t - Ke^{-r(t'-t)} > S_t - K$. (d) Exercise pays 40.00 and the bound is 39.494, so it does not decide; waiting wins exactly when the put struck at 80, priced at t_2 with three months left, is worth more than 0.506, which needs a model.

P17.8 *Convexity in the strike: proof and arbitrage*

(a) The discount factor is $e^{-0.015} = 0.98511$. The spreads cost $6.10 - 4.40 = 1.70$, within $[0,\ 5\,e^{-0.015}] = [0, 4.93]$, and $4.40 - 0.90 = 3.50$, within $[0,\ 10\,e^{-0.015}] = [0, 9.85]$. Both pass (Proposition 17.23(i)).

(b) The weight is $w = (55-45)/(55-40) = \frac{2}{3}$, and

$$\tfrac{2}{3}(6.10) + \tfrac{1}{3}(0.90) = 4.067 + 0.300 = 4.367 \ < \ 4.40 :$$

the middle call is too dear by 0.033. Multiply the weights w, -1, $1-w$ by three: buy two 40 calls for 12.20 and one 55 call for 0.90, and write three 45 calls for 13.20. The trade brings in $13.20 - 13.10 = 0.10$ today, and it pays $2(S_T-40)^+ - 3(S_T-45)^+ + (S_T-55)^+$ at T:

S_T	35	40	45	50	55	60
Payoff	0	0	10	5	0	0

The payoff is a tent: zero up to 40, rising two for one to 10 at 45, falling one for one to zero at 55, and zero beyond, where $2(S_T-40) - 3(S_T-45) + (S_T-55) = 0$. Money today and a payoff that is never negative: an arbitrage.

(c) We use dominance: a position whose payoff is nonnegative in every state has a nonnegative price (Lemma 16.8(i), applied against the empty portfolio; otherwise buying the position would bring in money today and could not lose). For $x \geq 0$ let

$$\chi(x) = w\,(x-K_1)^+ + (1-w)\,(x-K_3)^+,$$

the payoff of w calls struck at K_1 and $1-w$ calls struck at K_3 when $S_T = x$. Since $K_1 < K_2 < K_3$, we have $0 < K_3 - K_2 < K_3 - K_1$, so $0 < w < 1$; both terms are therefore nonnegative, and $\chi(x) \geq 0$. A positive part is at least its argument, so

$$\chi(x) \ \geq \ w(x-K_1) + (1-w)(x-K_3) = x - \big(wK_1 + (1-w)K_3\big) = x - K_2,$$

because $wK_1 + (1-w)K_3 = K_3 - w(K_3-K_1) = K_3 - (K_3-K_2) = K_2$. The two bounds together give $\chi(x) \geq \max(x-K_2, 0) = (x-K_2)^+$. The butterfly that is long w calls at K_1, short one call at K_2 and long $1-w$ calls at K_3 therefore pays $\chi(S_T) - (S_T-K_2)^+ \geq 0$ in every state, and its price $w\,C(K_1) - C(K_2) + (1-w)\,C(K_3)$ is nonnegative. For puts the same two steps apply to $w\,(K_1-x)^+ + (1-w)\,(K_3-x)^+$: it is nonnegative, and it is at least $w(K_1-x) + (1-w)(K_3-x) = K_2 - x$, hence at least $(K_2-x)^+$. The put butterfly has a nonnegative payoff and therefore a nonnegative price. Every point strictly between two strikes $K_1 < K_3$ can be written $wK_1 + (1-w)K_3$ with $0 < w < 1$, and it is then a strike K_2 whose weight is $w = (K_3-K_2)/(K_3-K_1)$. So the two inequalities, holding for all $K_1 < K_2 < K_3$, say exactly that C and $\mathcal{P}$ are convex functions of the strike on $(0,\infty)$, whatever the spacing. With equal spacing, $w = \frac{1}{2}$ and the call inequality times two is $C(K_1) - 2\,C(K_2) + C(K_3) \geq 0$. This is Proposition 17.24.

(d) Parity (Theorem 17.9) gives $\mathcal{P}(K) = C(K) - S + Ke^{-rT}$:

$$\begin{aligned}
\mathcal{P}(40) &= 6.10 - 45 + 39.4045 = 0.5045,\\
\mathcal{P}(45) &= 4.40 - 45 + 44.3300 = 3.7300,\\
\mathcal{P}(55) &= 0.90 - 45 + 54.1812 = 10.0812.
\end{aligned}$$

The test reads $\frac{2}{3}(0.5045)+\frac{1}{3}(10.0812)=0.3363+3.3604=3.6967<3.7300$: a violation of 0.033, the same as for the calls. The two must agree, because $\mathcal{P}(K)-C(K)=Ke^{-rT}-S$ is an affine function of the strike, and the weights w, -1, $1-w$ annihilate affine functions: they sum to zero, and $wK_1-K_2+(1-w)K_3=0$. The put butterfly and the call butterfly on the same strikes always cost the same (Proposition 17.29(iii)).

(e) A call spread of width 5 is worth at most $5\,e^{-0.02}=4.901$, and the quote is 5.10. Sell the spread: write the 100 call, buy the 105 call, and deposit the 5.10 received, which grows to $5.10\,e^{0.02}=5.203$. At T the spread costs its seller $(S_T-100)^+-(S_T-105)^+$, which lies between 0 and 5, so at least $5.203-5=0.203$ is kept in every state (present value $5.10-4.901=0.199$). The options are European, so no leg can be assigned before T.

Digitals and the density. A call spread of width ΔK, scaled by $1/\Delta K$, pays 1 above its upper strike and 0 below its lower strike, so two such spreads bracket the digital; where the call curve is differentiable, the digital is worth $\mathrm{Dig}(K)=-\partial_K C(K)$ (Proposition 23.2). The bounds of (a) then say that $0\le\mathrm{Dig}(K)\le e^{-rT}$: the digital is a discounted probability. Multiplied by K_3-K_1, the inequality of (c) reads $(K_2-K_1)\big[C(K_3)-C(K_2)\big]\ge(K_3-K_2)\big[C(K_2)-C(K_1)\big]$, that is $\big[C(K_3)-C(K_2)\big]/(K_3-K_2)\ge\big[C(K_2)-C(K_1)\big]/(K_2-K_1)$: the slope of the call curve never decreases (the quotes of (a) fail this test, with chord slopes $-1.70/5=-0.34$ on $[40,45]$ and $-3.50/10=-0.35$ on $[45,55]$), so its curvature is nonnegative where it exists; e^{rT} times that curvature is the risk-neutral density of S_T (Theorem 22.7), and a butterfly with a negative price would be a negative density.

Self-check. The butterfly of (b) is two 40/45 call spreads less one 45/55 call spread, $2C(40)-3C(45)+C(55)=2(1.70)-3.50=-0.10$: the 0.10 brought in, found from the spreads of (a) alone. Per unit of strike the two spreads cost $1.70/5=0.34$ and $3.50/10=0.35$; since $wC(K_1)+(1-w)C(K_3)-C(K_2)=w\big[C(K_1)-C(K_2)\big]-(1-w)\big[C(K_2)-C(K_3)\big]$ and $w(K_2-K_1)=(1-w)(K_3-K_2)=(K_2-K_1)(K_3-K_2)/(K_3-K_1)=\frac{10}{3}$, the violation is $\frac{10}{3}(0.35-0.34)=0.033$, as in (b). In whole numbers the put butterfly brings in the same 0.10: $2(0.5045)+10.0812-3(3.7300)=11.090-11.190=-0.10$. The tent of (b) agrees with the general formula: a butterfly with weights w, -1, $1-w$ peaks at $w(K_2-K_1)=\frac{2}{3}\times5=3.33$, and three of them peak at 10. The inequality $\chi(x)\ge(x-K_2)^+$ of (c) can be tested with the strikes of (a) ($K_1=40$, $K_2=45$, $K_3=55$, $w=\frac{2}{3}$): at $x=50$, $\chi=\frac{2}{3}(10)=6.67\ge5$; at $x=60$, $\chi=\frac{2}{3}(20)+\frac{1}{3}(5)=15=60-45$, with equality from K_3 on, where the tent has returned to zero.

Answer: (a) The spreads cost $1.70\le4.93$ and $3.50\le9.85$: both pass. (b) $w=2/3$ and $\frac{2}{3}(6.10)+\frac{1}{3}(0.90)=4.367<4.40$, a violation of 0.033; buy two 40 calls and one 55 call and write three 45 calls, for +0.10 today and the payoff 0, 0, 10, 5, 0, 0. (c) $w(x-K_1)^+\!+(1-w)(x-K_3)^+\ge(x-K_2)^+$, so the butterfly has a nonnegative payoff and a nonnegative price; the same holds for puts. (d) The puts are 0.5045, 3.7300 and 10.0812, with the same violation 0.033, because $\mathcal{P}-C$ is affine in K. (e) The cap is $5\,e^{-0.02}=4.901<5.10$: sell the spread and deposit 5.10, keeping at least 0.203 at T; $0\le\mathrm{Dig}(K)=-\partial_K C(K)\le e^{-rT}$, and the density $e^{rT}\partial_{KK}C(K)$ is nonnegative.

P17.9 *The box spread as a loan*

(a) The bull call spread costs $345.60-244.60=101.00$ and the bear put spread costs $260.20-167.40=92.80$, so $\mathrm{Box}(4000,4200)=193.80$. Grouped by strike, the four options pay

$$\big[(S_T-4000)^+-(4000-S_T)^+\big]-\big[(S_T-4200)^+-(4200-S_T)^+\big]=(S_T-4000)-(S_T-4200)=200$$

in every state, by the identity $x^+-(-x)^+=x$ (Proposition 17.29(i)). The box is a zero-coupon bond paying 200.

(b) The implied rate is

$$r_{\mathrm{box}}=\frac{1}{0.75}\ln\frac{200}{193.80}=\frac{0.03149}{0.75}=4.20\%.$$

(c) Buying the box pays 193.80 today and receives 200 at T: it lends at 4.20%. Selling the box receives 193.80 today and pays 200 at T: it borrows at 4.20%. The desk buys the box to lend,

since $4.20\% > 3.8\%$: the same 193.80 lent at 3.8% would return $193.80\,e^{0.0285} = 199.40$, so each box gains 0.60 at T. The desk sells the box to borrow, since $4.20\% < 4.6\%$: borrowing 193.80 at 4.6% would cost $193.80\,e^{0.0345} = 200.60$ at T, so each box saves 0.60.

(d) Regroup the four options by strike. The long 4000 call and the short 4000 put pay $S_T - 4000$: a long forward with delivery price 4000 (Corollary 17.12(i)). The short 4200 call and the long 4200 put pay $4200 - S_T$: a short forward with delivery price 4200. The index cancels and 200 remains. By parity in forward form (17.3) at each strike,

$$\begin{aligned}\text{Box}(4000, 4200) &= \big[C(4000) - \mathcal{P}(4000)\big] - \big[C(4200) - \mathcal{P}(4200)\big] \\ &= e^{-rT}(F - 4000) - e^{-rT}(F - 4200) = 200\,e^{-rT}.\end{aligned}$$

Regrouped by type, the right-hand side of the first line is $[C(4000) - C(4200)] - [\mathcal{P}(4000) - \mathcal{P}(4200)]$: the bull call spread costs exactly $200\,e^{-rT}$ more than the bull put spread (Proposition 17.29(ii)). In numbers, the bull put spread costs $167.40 - 260.20 = -92.80$, a credit, and $101.00 - (-92.80) = 193.80$.

(e) By (d), $\text{Box}(4000, 4200) = 200\,e^{-rT}$, so $e^{rT} = 200/193.80 = 1.03199$, that is $r = r_{\text{box}}$; parity read backwards gives $F = K + e^{rT}(C - \mathcal{P})$. At 4000, $C - \mathcal{P} = 178.20$ and $F = 4000 + 178.20 \times 1.03199 = 4183.90$. At 4200, $C - \mathcal{P} = -15.60$ and $F = 4200 - 15.60 \times 1.03199 = 4183.90$: the same forward. Since $F = Se^{(r-q)T}$ for an index with yield q,

$$r - q = \frac{1}{0.75} \ln \frac{4183.90}{4100} = 2.70\%, \qquad q = 4.20\% - 2.70\% = 1.50\%.$$

(f) The seller of a box is short the put with the higher strike and short the call with the lower strike. With American options either can be exercised at a time that the seller does not choose, and exercising a put that is deep in the money beats waiting when $r > 0$ (Proposition 17.18(i)). The seller then pays the higher strike for the share and, delivering it through the long put with the lower strike, has paid the full width, 200, at the exercise date, before T. It is still short the 4000/4200 call spread (short the 4000 call, long the 4200 call), whose payoff at T lies between -200 and 0. The single payment of 200 at T on which the price $e^{-rT}(K_2 - K_1)$ rests no longer holds: the seller has lost at least the interest on the width, and owes the call spread on top if the share recovers above the lower strike. Proposition 17.23(ii) caps an American spread only at its undiscounted width, because a spread can be assigned at once. The dominance argument that fixes the European box at $e^{-rT}(K_2 - K_1)$ has no American counterpart, and by the assignment just described a sale of an American box above $e^{-rT}(K_2 - K_1)$ is not riskless, so it is not an arbitrage. The options of this problem are European, which is why the index box is a clean loan.

Self-check. The rate, directly: $193.80\,e^{0.042 \times 0.75} = 193.80 \times 1.0320 = 200.00$. The yield, through the other form of parity at the other strike, $C - \mathcal{P} = Se^{-qT} - Ke^{-rT}$ (Corollary 17.11(ii)), with no forward computed: at $K = 4200$, $Se^{-qT} = -15.60 + 4200 \times 0.969 = -15.60 + 4069.80 = 4054.20$, so $qT = \ln(4100/4054.20) = 0.011234$ and $q = 0.011234/0.75 = 1.50\%$, as in (e). The two forwards of (e) had to agree, because their difference is $200 - e^{rT}\,\text{Box}(4000, 4200)$, which is zero by the definition of r_{box}.

Answer: (a) The box costs $101.00 + 92.80 = 193.80$ and pays 200 in every state. (b) $r_{\text{box}} = \ln(200/193.80)/0.75 = 4.20\%$. (c) Buy the box to lend at 4.20%, a gain of 0.60 per box over lending at 3.8%; sell it to borrow at 4.20%, a saving of 0.60 per box over borrowing at 4.6%. (d) A long forward at 4000 plus a short forward at 4200; the bull call spread costs $200\,e^{-rT} = 193.80$ more than the bull put spread (101.00 against -92.80). (e) $F = 4183.90$ at both strikes, and $q = 1.50\%$. (f) A short American leg can be assigned early, so the width is paid before T.

P17.10 *Name the position*

(a) The long 95 put and the long 105 call are a long strangle; the short 100 put and the short 100 call are a short straddle (Definition 17.32). The payoff is $(95 - S_T)^+ + (S_T - 105)^+ - |S_T - 100|$, and the call butterfly pays $(S_T - 95)^+ - 2(S_T - 100)^+ + (S_T - 105)^+$:

Range of S_T	$S_T \le 95$	$95 \le S_T \le 100$	$100 \le S_T \le 105$	$S_T \ge 105$
The position	-5	$S_T - 100$	$100 - S_T$	-5
The butterfly	0	$S_T - 95$	$105 - S_T$	0

In the two outer ranges the position pays $(95 - S_T) - (100 - S_T) = -5$ and $(S_T - 105) - (S_T - 100) = -5$. In every range the first row is the second minus 5: the position is the butterfly minus a bond paying 5. Its payoff is never positive, and it is negative except at $S_T = 100$, so its price is at most zero (Lemma 16.8(i)) and strictly negative as soon as $S_T \neq 100$ has positive probability (Lemma 16.8(ii), against the empty portfolio): it is opened for a credit, here $3.73 + 4.23 - 1.74 - 2.22 = 4.00$. The profit is 4.00 plus the payoff: the maximum gain is 4.00, at $S_T = 100$; the maximum loss is $5 - 4.00 = 1.00$, for $S_T \le 95$ or $S_T \ge 105$; and the breakevens are $100 - 4 = 96$ and $100 + 4 = 104$. Desks call this position an iron butterfly.

(b) A risk reversal. It pays $(S_T - 110)^+ - (90 - S_T)^+$: the negative amount $S_T - 90$ below 90, nothing between the strikes, and $S_T - 110$ above 110. With both strikes at K the payoff is $(S_T - K)^+ - (K - S_T)^+ = S_T - K$: a synthetic long forward with delivery price K (Corollary 17.12(i)), which costs nothing exactly when K is the forward price.

(c) A slope of $+2$ above the strike needs two calls, and a slope of -1 below it needs one put: two 100 calls and one 100 put, a strap. The cost is $2(5) + 4 = 14$, as stated, and the payoff is $2(S_T - 100)^+ + (100 - S_T)^+$, which is zero at $S_T = 100$, where the whole 14 is lost. The breakevens solve $2(S_T - 100) = 14$ and $100 - S_T = 14$: they are $S_T = 107$ and $S_T = 86$.

(d) A collar. Its payoff is

$$S_T + (90 - S_T)^+ - (S_T - 115)^+ = \max(S_T, 90) - (S_T - 115)^+ = 90 + (S_T - 90)^+ - (S_T - 115)^+,$$

a bond paying 90 plus a 90/115 bull call spread. It equals 90 for $S_T \le 90$, S_T between the strikes and 115 for $S_T \ge 115$: the payoff lies in $[90, 115]$.

(e) *From calls.* Long the 80 call and short the 100 call, an 80/100 bull call spread, which pays $(S_T - 80)^+ - (S_T - 100)^+$ and is a debit. *From puts.* The same payoff is $20 - [(100 - S_T)^+ - (80 - S_T)^+]$: a bond paying 20 together with a short 100 put and a long 80 put, that is an 80/100 bull put spread, which is a credit. By ranges: for $S_T \le 80$, $20 - (100 - S_T) + (80 - S_T) = 0$; between the strikes, $20 - (100 - S_T) = S_T - 80$; above 100, 20. Equal payoffs have equal costs, so $C(80) - C(100) = 20\,e^{-rT} + [\mathcal{P}(80) - \mathcal{P}(100)]$ (Proposition 17.29(ii)). The bond costs $20\,e^{-0.02} = 19.60$, so the put spread costs $9.78 - 19.60 = -9.82$: it brings in 9.82.

Self-check. Each payoff is evaluated at one point through both of its descriptions. (a) At $S_T = 98$ the four options pay $0 + 0 - 2 = -2$, and the butterfly pays 3, minus 5. The credit also agrees with convexity (Proposition 17.24): the call butterfly is the position together with a bond paying 5, so it costs $5e^{-rT} - 4.00$, which the proposition requires to be nonnegative; the credit 4.00 is at most $5e^{-rT}$ whenever $rT \le \ln(5/4) = 0.223$. (c) At $S_T = 107$ the strap pays $2 \times 7 = 14$, and at $S_T = 86$ it pays 14: both breakevens. (d) At $S_T = 100$: $100 + 0 - 0 = 100 = 90 + 10 - 0$; at $S_T = 130$: $130 + 0 - 15 = 115 = 90 + 40 - 15$. (e) At $S_T = 90$ the calls pay $10 - 0 = 10$, and the bond and the puts pay $20 - 10 + 0 = 10$. The credit follows a second way from Proposition 17.29(i): the bull call spread together with the bear put spread is the 80/100 box, which costs $20\,e^{-0.02} = 19.60$, so the bear put spread costs $19.60 - 9.78 = 9.82$ and its reverse, the bull put spread, brings in 9.82; and 9.82 lies in $[0, 19.60]$, as Proposition 17.23(i) requires.

Answer: (a) A long strangle plus a short straddle (an iron butterfly); its payoff is the call butterfly minus 5; credit 4.00; maximum gain 4.00 at $S_T = 100$, maximum loss 1.00 for $S_T \le 95$ or $S_T \ge 105$, breakevens 96 and 104. (b) A risk reversal; at a common strike it is a synthetic long forward with delivery price K. (c) A strap (two 100 calls and one 100 put); breakevens 86 and 107. (d) A collar, equal to a bond paying 90 plus a 90/115 bull call spread; payoff in $[90, 115]$. (e) Long the 80 call and short the 100 call; or a bond paying 20, long the 80 put and short the 100 put; the put spread brings in $19.60 - 9.78 = 9.82$.

P17.11 *Build it from calls*

(a) The payoff is continuous and piecewise linear, with kinks at 90, 100, 110 and 130. Its value at zero is $f(0) = 10$, and its slopes are $a_0 = 0$ on $[0, 90]$, $a_1 = -1$ on $[90, 100]$, $a_2 = 0$ on $[100, 110]$, $a_3 = 2$ on $[110, 130]$ and $a_4 = 0$ above. By Proposition 17.35, $f(x) = f(0) + a_0 x + \sum_j (a_j - a_{j-1})(x - K_j)^+$: the weight of each call is the jump of the slope at its strike, here -1, $+1$, $+2$ and -2, so

$$f(S_T) = 10 - (S_T - 90)^+ + (S_T - 100)^+ + 2(S_T - 110)^+ - 2(S_T - 130)^+.$$

The replica is a bond paying 10, one 90 call written, one 100 call bought, two 110 calls bought and two 130 calls written; no share is needed, because $a_0 = 0$. Its price is

$$10\, e^{-0.015} - 13.79 + 7.76 + 2(3.90) - 2(0.73) = 9.85 - 13.79 + 7.76 + 7.80 - 1.46 = 10.16.$$

Checks. At $S_T = 95$ the replica pays $10 - 5 = 5$, and $f(95) = 100 - 95 = 5$. At $S_T = 120$ it pays $10 - 30 + 20 + 20 = 20$, and $f(120) = 2(120 - 110) = 20$. At $S_T = 150$ it pays $10 - 60 + 50 + 80 - 40 = 40$, and $f(150) = 40$.

(b) The first three terms satisfy $10 - (S_T - 90)^+ + (S_T - 100)^+ = (100 - S_T)^+ - (90 - S_T)^+$: both sides equal 10 for $S_T \le 90$, $100 - S_T$ between 90 and 100, and 0 above 100. The right-hand side is a 90/100 bear put spread, long the 100 put and short the 90 put. Its price is that of the pieces it replaces, by the law of one price (Lemma 16.8(iii)): $9.85 - 13.79 + 7.76 = 3.82$. The claim is therefore a 90/100 bear put spread together with two 110/130 bull call spreads: $3.82 + 2(3.90 - 0.73) = 3.82 + 6.34 = 10.16$.

(c) For $f(x) = |x - 100|$ the value at zero is $f(0) = 100$, and the slope is $a_0 = -1$ below 100 and $a_1 = +1$ above, so the weight at the kink is 2:

$$|S_T - 100| = 100 - S_T + 2(S_T - 100)^+,$$

a bond paying 100, one share sold short and two 100 calls. The price is $100\, e^{-0.015} - 100 + 2(7.76) = 98.51 - 100 + 15.52 = 14.03$. By put–call parity (Theorem 17.9) the 100 put costs $7.76 - 100 + 98.51 = 6.27$, and the straddle, a call and a put, costs $7.76 + 6.27 = 14.03$: the same.

(d) Let f be continuous on $[0, K_m]$, where K_m is the highest strike. The piecewise-linear function through the points $(0, f(0))$ and $(K_j, f(K_j))$, extended beyond K_m by the constant $f(K_m)$, is continuous and affine between consecutive strikes, so it is replicated exactly by Proposition 17.35. On each interval between two neighbouring points it lies between the values of f at the two ends, so it differs from f by at most the oscillation of f over that interval; and that oscillation tends to zero as the strikes become dense, because a continuous function on a closed bounded interval is uniformly continuous. Once the strikes are dense enough for that difference to be at most ε on $[0, K_m]$, and when f is also within ε of $f(K_m)$ above K_m (for instance constant there), the claim and the replica differ by at most ε in every state, and their prices then differ by at most εe^{-rT} (Lemma 16.8(i) and (iv)). Otherwise the match holds only for S_T in $[0, K_m]$, which is why it is a match on a range of prices. The weights are the jumps of the slope, a discrete second derivative. For a payoff with two continuous derivatives on $[0, \infty)$ (one-sided at 0), integration by parts gives, for every $x \ge 0$,

$$\int_0^x (x - K)\, f''(K)\, dK = \Big[(x - K)\, f'(K)\Big]_{K=0}^{K=x} + \int_0^x f'(K)\, dK = -x\, f'(0) + f(x) - f(0),$$

the last step by the fundamental theorem of calculus. The integrand $(x - K)^+ f''(K)$ vanishes for $K > x$, so the integral may run over $[0, \infty)$, and the sum over the strikes becomes an integral,

$$f(S_T) = f(0) + f'(0)\, S_T + \int_0^\infty f''(K)\, (S_T - K)^+\, dK,$$

which is Taylor's formula with integral remainder: a bond paying $f(0)$, $f'(0)$ shares and $f''(K)\, dK$ calls at each strike K. Lemma 22.34 in §22 states this replication expanded about the forward price, with puts below it and calls above.

Self-check. The replica of (a) reproduces f at the four kinks: 10 at 90; $10-10=0$ at 100; $10-20+10=0$ at 110; $10-40+30+40=40$ at 130. The price of (b) is confirmed through parity: $\mathcal{P}(100)=6.27$ and $\mathcal{P}(90)=13.79-100+90\,e^{-0.015}=13.79-100+88.66=2.45$, and $6.27-2.45=3.82$. The price 10.16 respects the bounds of a payoff between 0 and 40: it lies in $[0,\ 40\,e^{-0.015}]=[0,39.40]$. The integral formula of (d) can be tested on $f(x)=x^2$: $f(0)=f'(0)=0$, $f''=2$ and $\int_0^x 2(x-K)\,dK=x^2$.

Answer: (a) A bond paying 10 and calls with weights -1 at 90, $+1$ at 100, $+2$ at 110 and -2 at 130; price $9.85-13.79+7.76+7.80-1.46=10.16$; the replica pays 5, 20 and 40 at 95, 120 and 150. (b) A 90/100 bear put spread (long the 100 put, short the 90 put), price 3.82. (c) A bond paying 100, one share sold short and two 100 calls: $98.51-100+15.52=14.03$, equal to the call plus the put, $7.76+6.27$. (d) Interpolate the payoff linearly between the strikes, with call weights equal to the jumps of the slope; in the limit $f(S_T)=f(0)+f'(0)S_T+\int_0^\infty f''(K)(S_T-K)^+\,dK$.

P17.12 *Participation rate of a principal-protected note*

The budget. The bond that repays 10,000 in four years costs $10{,}000\,e^{-0.14}=8{,}693.58$ and the fee is 100, so the issuer has $10{,}000-100-8{,}693.58=1{,}206.42$ to spend on options. One at-the-money call on the notional 10,000 pays the whole gain of the index, so the participation rate is the number of such calls, or spreads, that the budget buys.

(a) The call costs 1,400, and $1{,}206.42/1{,}400=86.2\%$.

(b) The spread costs 1,060, and $1{,}206.42/1{,}060=113.8\%$. One spread pays at most 40% of the notional, 4,000, so the largest addition is $4{,}000\times 1{,}206.42/1{,}060=4{,}552.52$ and the largest payout is 14,552.52, about 14,553, reached when the index has gained 40% or more.

(c) The bond now costs $10{,}000\,e^{-0.06}=9{,}417.65$, and the budget falls to $10{,}000-100-9{,}417.65=482.35$. The call costs 1,065, so the participation rate is $482.35/1{,}065=45.3\%$. The lower rate cheapens the call, from 14.0% to 10.65%, but it shrinks the budget much more, from 1,206.42 to 482.35.

(d) *Six years.* The bond costs $10{,}000\,e^{-0.21}=8{,}105.84$, the budget is $10{,}000-100-8{,}105.84=1{,}794.16$, and the participation rate is $1{,}794.16/1{,}700=105.5\%$. *Higher volatility.* The budget is unchanged, and the participation rate is $1{,}206.42/1{,}710=70.6\%$. A longer life raises participation, here from 86.2% to 105.5%: the budget is the interest forgone on the principal, $10{,}000(1-e^{-rT})$ less the fee, and here it grows faster with T than the price of the call. A higher volatility lowers it, from 86.2% to 70.6%: the call costs more and the budget does not move. A cap raises it, from 86.2% to 113.8%: the issuer sells the gains beyond the cap, so each unit of participation costs less, and the investor pays with the upside above 40%.

(e) Measure the index in units of the notional: one unit is 10,000 of the index at today's level, so its price today is $S=10{,}000$, its value at T is S_T, the at-the-money strike is $K=S$ and the principal is S. Without a fee the budget is what the bond leaves, $S-Se^{-rT}$, which is positive because $r>0$. Without dividends, put–call parity (Theorem 17.9) at $K=S$ reads

$$C=S-Se^{-rT}+\mathcal{P}=\text{budget}+\mathcal{P},$$

so $C>0$, and the participation rate is budget$/C=1-\mathcal{P}/C$. The put pays $(S-S_T)^+\geq 0$, which is positive with positive probability as soon as $S_T<S$ has positive probability, and its price is then strictly positive (Lemma 16.8(ii), applied against the empty portfolio). Hence the call costs strictly more than the budget, and the participation rate is strictly below 100%. No volatility entered the argument. *Check.* The budget is $100(1-e^{-0.14})=13.064\%$ of the notional, and $19.25-6.19=13.06$, as parity requires to the rounding of the quotes; the participation rate is $13.064/19.25=67.9\%$, about 68%.

Self-check. The budgets follow without the bond prices from the series $1-e^{-x}=x-x^2/2+x^3/6-x^4/24+x^5/120-\ldots$, summed to the fifth power: at $x=0.14$ it gives 0.130642, so the budget is $1{,}306.42-100=1{,}206.42$; at $x=0.06$ it gives 0.0582355, so $582.35-100=482.35$; at $x=0.21$ it

gives 0.189416, so 1,894.16 − 100 = 1,794.16. For the spread of (b): its price 10.6% lies between 0 and $40\%\, e^{-0.14} = 34.77\%$ (Proposition 17.23(i)), and the call struck at 140% that it implies costs 14.0 − 10.6 = 3.4% > 0. In (e), the budget route 13.064/19.25 = 0.679 and the put route $1 - \mathcal{P}/C = 1 - 6.19/19.25 = 0.678$ agree to the rounding of the quotes. The stated prices of (a) also pass a parity test: with the dividend yield, $C - \mathcal{P} = S(e^{-qT} - e^{-rT}) = 10{,}000(e^{-0.08} - e^{-0.14}) = 537.58$ (Corollary 17.11(ii)), so the at-the-money put costs 1,400 − 537.58 = 862.42, a positive price, as it must be.

Answer: The budget is 10,000 − 100 − 8,693.58 = 1,206.42. (a) 1,206.42/1,400 = 86.2%. (b) 1,206.42/1,060 = 113.8%, with a largest payout of 14,552.52. (c) The budget is 482.35 and the participation rate 45.3%. (d) Six years: 1,794.16/1,700 = 105.5%; higher volatility: 70.6%; here a longer life and a cap raise participation, and a higher volatility lowers it. (e) $C = \text{budget} + \mathcal{P}$ with $\mathcal{P} > 0$, so the participation rate is $1 - \mathcal{P}/C < 100\%$; with the stated prices it is 13.064/19.25 = 67.9%, about 68%.

28.18 Solutions — §18 Binomial Trees and Risk-Neutral Valuation

P18.1 *One-step call and put: the hedge, the risk-neutral check and a mispriced quote*

(a) The factors are $u = 73.60/64 = 1.15$ and $d = 56/64 = 0.875$, and $e^{r\Delta t} = e^{0.01} = 1.010050$ lies strictly between them; both moves have positive probability, so the tree admits no arbitrage (Proposition 18.1). The call pays $V_u = 73.60 - 66 = 7.60$ in the up state and $V_d = 0$ in the down state. By (18.2) of Theorem 18.3, its replicating portfolio holds

$$\Delta = \frac{7.60 - 0}{73.60 - 56.00} = \frac{7.6}{17.6} = \frac{19}{44} = 0.4318 \ \text{ shares},$$
$$B = e^{-0.01}\,\frac{1.15 \times 0 - 0.875 \times 7.60}{0.275} = -24.182\, e^{-0.01} = -23.941,$$

a loan. The shares cost $\frac{19}{44} \times 64 = 27.636$, so $C = 27.636 - 23.941 = 3.695$. The payoffs confirm it: at the end of the step the loan has grown to 24.182 and the shares are worth $\frac{19}{44} \times 73.60 = 31.782$ or $\frac{19}{44} \times 56 = 24.182$, so the portfolio pays 31.782 − 24.182 = 7.60 or 24.182 − 24.182 = 0, exactly what the call pays.

(b) The risk-neutral up-probability (Definition 18.6) is

$$p = \frac{e^{r\Delta t} - d}{u - d} = \frac{1.010050 - 0.875}{0.275} = \frac{0.135050}{0.275} = 0.49109, \qquad 1 - p = 0.50891.$$

By (18.4), $C = e^{-0.01} \times 0.49109 \times 7.60 = 0.990050 \times 3.73230 = 3.6952$, the price found in (a). The put pays $V_u = 0$ and $V_d = 66 - 56 = 10$. Its replica holds $\Delta = (0 - 10)/17.6 = -25/44 = -0.5682$ shares, a short position worth 36.364, and $B = e^{-0.01} \times 1.15 \times 10/0.275 = 41.818\, e^{-0.01} = 41.402$ in the bond, a deposit; hence $\mathcal{P} = 41.402 - 36.364 = 5.038$. The weights give the same price, $\mathcal{P} = e^{-0.01} \times 0.50891 \times 10 = 5.0384$. To five decimals $C = 3.69516$ and $\mathcal{P} = 5.03845$, so

$$C - \mathcal{P} = -1.34329, \qquad S - Ke^{-r\Delta t} = 64 - 66\, e^{-0.01} = 64 - 65.34329 = -1.34329,$$

and put–call parity (Theorem 17.9) holds.

(c) The dealer's price 4.20 is above the replication cost 3.695, so the call is dear: sell it to the dealer and buy its replica.

Trade	today	up state	down state
Sell the call at 4.20	+4.200	−7.600	0
Buy 19/44 share	−27.636	+31.782	+24.182
Borrow 23.941	+23.941	−24.182	−24.182
Net	+0.505	0	0

The trades bring in $4.20 - 3.695 = 0.505$ today and leave a position worth zero in both states: an arbitrage. The real-world probability of the up-move was never needed.

Self-check. The riskless-portfolio route gives the call without solving for the bond. A position long 19/44 share and short one call is worth $31.782 - 7.60 = 24.182$ in the up state and $24.182 - 0 = 24.182$ in the down state. It is riskless, so it is worth $24.182\, e^{-0.01} = 23.941$ today, and $27.636 - C = 23.941$ gives $C = 3.695$ again. For the two options together: $C/7.60$ is the price of a claim that pays 1 in the up state and $\mathcal{P}/10$ that of a claim that pays 1 in the down state, so their sum must be the price of a bond that pays 1 in both states. Indeed $3.69516/7.60 + 5.03845/10 = 0.48621 + 0.50384 = 0.99005 = e^{-0.01}$. That identity holds for any pair of weights that sum to one, so it cannot detect a wrong p; the weight itself checks against Proposition 18.7(i): $0.49109 \times 73.60 + 0.50891 \times 56 = 36.144 + 28.499 = 64.643 = 64\, e^{0.01}$, the forward price.

Answer: (a) No arbitrage, because $0.875 < e^{0.01} = 1.01005 < 1.15$; the call is $19/44 = 0.4318$ share financed by a loan of 23.941, so $C = 3.695$. (b) $p = 0.4911$ and $C = 3.6952$ again; the put is short 0.5682 share with 41.402 in the bond, $\mathcal{P} = 5.0384$; $C - \mathcal{P} = -1.3433 = 64 - 66e^{-0.01}$. (c) Sell the call at 4.20, buy 0.4318 share, borrow 23.941: 0.505 received today, nothing owed in either state.

P18.2 *Up 120, down 80: the mental-arithmetic round*

(a) Here $u = 1.2$, $d = 0.8$ and $e^{r\Delta t} = 1$, so the risk-neutral up-probability is $p = (1 - 0.8)/(1.2 - 0.8) = \frac{1}{2}$ (Definition 18.6). The call pays 20 or 0, and by (18.4) it is worth $\frac{1}{2} \times 20 = 10$. Replication, by (18.2): $\Delta = (20 - 0)/(120 - 80) = \frac{1}{2}$ share and $B = (1.2 \times 0 - 0.8 \times 20)/0.4 = -40$. Half a share costs 50, the loan brings 40, and the cost is 10; the portfolio pays $60 - 40 = 20$ or $40 - 40 = 0$.

(b) The put pays 0 or 20 and is worth $\frac{1}{2} \times 20 = 10$. The straddle pays 20 in both states; it is riskless and, at a zero rate, worth $20 = 10 + 10$. Parity (Theorem 17.9) reads $C - \mathcal{P} = S - Ke^{-r\Delta t} = 100 - 100 = 0$: with a zero rate and the strike at the stock price, the call and the put cost the same.

(c) The call struck at 90 pays 30 or 0 and is worth $\frac{1}{2} \times 30 = 15$. The call struck at 110 pays 10 or 0 and is worth $\frac{1}{2} \times 10 = 5$.

(d) By Proposition 18.7(ii) the digital is worth $e^{-r\Delta t}p = \frac{1}{2}$: the discounted risk-neutral probability of being paid.

(e) The call is still worth 10. The portfolio of (a) pays what the call pays in both states and costs 10, and the up-probability, 0.8 or any other number strictly between 0 and 1, appears nowhere in it. The bid $16 = 0.8 \times 20$ is a real-world expected payoff discounted at the riskless rate, which is not a price. Sell the call at 16, buy half a share for 50 and borrow 40: the trade brings in $16 - 50 + 40 = 6$ today. At the end of the period the position is worth $60 - 20 - 40 = 0$ after an up-move and $40 - 0 - 40 = 0$ after a down-move. The 6 is kept in both states.

(f) Now $e^{r\Delta t} = 1.05$ and $p = (1.05 - 0.8)/0.4 = 0.625$. The call is worth $0.625 \times 20/1.05 = 12.5/1.05 = 11.905$ and the put $0.375 \times 20/1.05 = 7.5/1.05 = 7.143$. Their difference is $5/1.05 = 4.762$, and $S - Ke^{-r\Delta t} = 100 - 100/1.05 = 100 - 95.238 = 4.762$: parity holds. The rate raises the call and lowers the put, and the gap of $4.762 = 100 - 100/1.05$ is the interest that 95.238, the present value of the strike, earns over the period.

Self-check. Redo (f) by replication instead of the weights. The call's replica holds $\Delta = \frac{1}{2}$ share (the share count does not involve the rate) and $B = -(0.8 \times 20)/(0.4 \times 1.05) = -40/1.05 = -38.095$, so it costs $50 - 38.095 = 11.905$. The put's replica is short half a share and holds $(1.2 \times 20)/(0.4 \times 1.05) = 60/1.05 = 57.143$ in the bond, so it costs $57.143 - 50 = 7.143$. A second check prices the traded assets with the two state claims at the rate of (f), the digital of (d) and its mirror that pays 1 after a down-move: the claim on the up state is worth $0.625/1.05 = 0.59524$ and the claim on the down state $0.375/1.05 = 0.35714$. They sum to $0.95238 = 1/1.05$, the price of the bond, and $120 \times 0.59524 + 80 \times 0.35714 = 71.43 + 28.57 = 100$, the price of the stock. In (c), replication agrees as well: three quarters of a share and a loan of 60 cost $75 - 60 = 15$, and a

quarter of a share and a loan of 20 cost $25 - 20 = 5$.

Answer: (a) 10, with $p = \frac{1}{2}$, or half a share and a loan of 40; (b) put 10, straddle 20, and $C - \mathcal{P} = S - K = 0$; (c) 15 and 5; (d) 0.5; (e) still 10: sell at 16, buy half a share, borrow 40, and keep 6 in both states; (f) $p = 0.625$, call 11.905, put 7.143, $C - \mathcal{P} = 4.762 = 100 - 100/1.05$.

P18.3 *Re-derive one-step replication and the no-arbitrage band*

(a) The *only if* direction is proved by explicit construction of an arbitrage, the *if* direction by a sign argument on an arbitrary portfolio. *Only if.* Suppose that the condition fails. Since $d < u$, either $e^{r\Delta t} \le d$ or $e^{r\Delta t} \ge u$. In the first case borrow S and buy one share: the portfolio $(\Delta, B) = (1, -S)$ is worth 0 today, pays $S(u - e^{r\Delta t}) > 0$ in the up state, because $u > d \ge e^{r\Delta t}$, and pays $S(d - e^{r\Delta t}) \ge 0$ in the down state. In the second case short one share and lend S: the portfolio $(-1, S)$ is worth 0 today and pays $S(e^{r\Delta t} - u) \ge 0$ in the up state and $S(e^{r\Delta t} - d) > 0$ in the down state, because $d < u \le e^{r\Delta t}$. Both states have positive probability, so each portfolio is an arbitrage.

If. Assume $d < e^{r\Delta t} < u$, and let (Δ, B) have the value $\Pi_0 = \Delta S + B \le 0$ today and nonnegative payoffs. Substituting $B = \Pi_0 - \Delta S$, the payoffs are

$$\Delta S\,(u - e^{r\Delta t}) + \Pi_0 e^{r\Delta t} \quad \text{(up)}, \qquad \Delta S\,(d - e^{r\Delta t}) + \Pi_0 e^{r\Delta t} \quad \text{(down)}.$$

The first is nonnegative, so $\Delta S\,(u - e^{r\Delta t}) \ge -\Pi_0 e^{r\Delta t} \ge 0$, and $u - e^{r\Delta t} > 0$ forces $\Delta \ge 0$. The second is nonnegative, so $\Delta S\,(d - e^{r\Delta t}) \ge -\Pi_0 e^{r\Delta t} \ge 0$, and $d - e^{r\Delta t} < 0$ forces $\Delta \le 0$. Hence $\Delta = 0$, both payoffs equal $\Pi_0 e^{r\Delta t}$, which is nonnegative only if $\Pi_0 = 0$. Both payoffs then vanish, and the portfolio is not an arbitrage. This is Proposition 18.1.

(b) A portfolio replicates the derivative exactly when

$$\Delta S u + B e^{r\Delta t} = V_u, \qquad \Delta S d + B e^{r\Delta t} = V_d.$$

The determinant of this linear system is $S(u-d)e^{r\Delta t} \ne 0$, so it has exactly one solution. Subtracting the equations gives $\Delta S(u - d) = V_u - V_d$, and the first equation then gives $B e^{r\Delta t} = V_u - \Delta S u = \big((u-d)V_u - u(V_u - V_d)\big)/(u-d)$, so that

$$\Delta = \frac{V_u - V_d}{S(u-d)}, \qquad B = e^{-r\Delta t}\,\frac{uV_d - dV_u}{u-d}.$$

The cost is

$$\begin{aligned} \Delta S + B &= \frac{V_u - V_d}{u-d} + e^{-r\Delta t}\,\frac{uV_d - dV_u}{u-d} \\ &= e^{-r\Delta t}\,\frac{V_u\,(e^{r\Delta t} - d) + V_d\,(u - e^{r\Delta t})}{u-d} = e^{-r\Delta t}\big[pV_u + (1-p)V_d\big], \end{aligned}$$

because $(u - e^{r\Delta t})/(u-d) = 1 - p$. For the price, compare the derivative with the portfolio: they have the same payoff in both states, so they have the same price (the law of one price, Lemma 16.8(iii)). Explicitly, if $V > \Delta S + B$, sell the derivative and buy the portfolio: the position is worth $(\Delta S + B) - V < 0$ today and pays zero in both states, and placing the proceeds $V - (\Delta S + B)$ in the bond turns it into a position of value zero with a positive payoff in both states, an arbitrage. If $V < \Delta S + B$, buy the derivative and hold $(-\Delta, -B)$; the same argument applies. Hence $V = \Delta S + B$, which is Theorem 18.3. Conversely, at $V = \Delta S + B$ a holding of Δ' shares, B' in the bond and w units of the derivative has the price $(\Delta' + w\Delta)S + (B' + wB)$ and, in each state, the payoff of the stock-and-bond portfolio $(\Delta' + w\Delta, B' + wB)$, because one unit of the derivative pays what (Δ, B) pays. By (a) that portfolio is not an arbitrage, so neither is the holding: this price leaves no arbitrage.

(c) A position long Δ shares and short one derivative is worth $\Delta S u - V_u$ in the up state and $\Delta S d - V_d$ in the down state. The two are equal exactly when $\Delta S(u-d) = V_u - V_d$, the Δ of (b). The

position is then riskless, and a sure payment is worth its discounted value today (Lemma 16.8(iv); otherwise the position and the bond would form an arbitrage): $\Delta S - V = e^{-r\Delta t}(\Delta S u - V_u)$. Solving for V and inserting $\Delta S = (V_u - V_d)/(u-d)$,

$$\begin{aligned} V &= \Delta S\,(1 - u e^{-r\Delta t}) + e^{-r\Delta t} V_u = e^{-r\Delta t}\Big[\frac{(V_u - V_d)(e^{r\Delta t} - u)}{u-d} + V_u\Big] \\ &= e^{-r\Delta t}\Big[\frac{e^{r\Delta t} - d}{u-d}\,V_u + \frac{u - e^{r\Delta t}}{u-d}\,V_d\Big], \end{aligned}$$

the price of (b). The two routes are one equation: $e^{-r\Delta t}(\Delta S u - V_u) = -B$ by the first replication equation, so $\Delta S - V = -B$.

(d) The condition $d < e^{r\Delta t} < u$ says $0 < e^{r\Delta t} - d < u - d$, that is, $0 < p < 1$: the weights are positive exactly when there is no arbitrage. Next, $pSu + (1-p)Sd = S(d + p\,(u-d)) = S(d + e^{r\Delta t} - d) = Se^{r\Delta t}$: under the weights p and $1-p$ the stock earns the riskless rate, that is, $\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = Se^{r\Delta t}$ for the probability $\mathbb{Q}$ of Definition 18.6 (Proposition 18.7(i)). The probability p_{real} enters the price nowhere: Δ, B and p are built from S, u, d, r and the two payoffs, and p_{real} was used only in (a), to know that both states are possible.

Self-check. A pricing formula must return the prices of the traded assets. For the stock itself, $V_u = Su$ and $V_d = Sd$ give $\Delta = 1$, $B = 0$ and $V = S$; for the bond, $V_u = V_d = 1$ give $\Delta = 0$ and $V = B = e^{-r\Delta t}$. Then a numerical instance: let $S = 50$, $u = 1.2$, $d = 0.9$ and $e^{r\Delta t} = 1.05$, and take the call struck at 50, with $V_u = 10$ and $V_d = 0$. The replica holds $\Delta = 10/15 = 2/3$ share and $B = -(0.9 \times 10)/(0.3 \times 1.05) = -28.571$, and costs $33.333 - 28.571 = 4.762$. The weights give $p = (1.05 - 0.9)/0.3 = \frac{1}{2}$ and $\frac{1}{2} \times 10/1.05 = 4.762$. At time Δt the loan has grown to $28.571 \times 1.05 = 30$, so the replica pays $\frac{2}{3} \times 60 - 30 = 10$ and $\frac{2}{3} \times 45 - 30 = 0$, and the riskless position of (c) is worth $40 - 10 = 30$ in the up state and $30 - 0 = 30$ in the down state, so $33.333 - V = 30/1.05 = 28.571$ and $V = 4.762$.

Answer: (a) No arbitrage if and only if $d < e^{r\Delta t} < u$; when $e^{r\Delta t} \le d$ the portfolio $(1, -S)$ is an arbitrage, and when $e^{r\Delta t} \ge u$ the portfolio $(-1, S)$ is. (b) $\Delta = (V_u - V_d)/(S(u-d))$ and $B = e^{-r\Delta t}(uV_d - dV_u)/(u-d)$, with cost $V = e^{-r\Delta t}[pV_u + (1-p)V_d]$, $p = (e^{r\Delta t} - d)/(u-d)$; any other price is arbitraged against the replica, and this one leaves no arbitrage. (c) The same Δ makes the position long Δ shares and short the derivative riskless, and $\Delta S - V = e^{-r\Delta t}(\Delta S u - V_u)$ gives the same V. (d) $0 < p < 1$ is the no-arbitrage condition, $pu + (1-p)d = e^{r\Delta t}$, and p_{real} does not enter the price.

P18.4 *The discount rate the real world would need*

(a) The factors are $u = 1.25$ and $d = 0.85$, and $e^{r\Delta t} = e^{0.025} = 1.025315$ lies between them. The risk-neutral up-probability is $p = (1.025315 - 0.85)/0.40 = 0.175315/0.40 = 0.43829$, and $1 - p = 0.56171$. The call pays 25 or 0 and the put 0 or 15, so by (18.4)

$$\begin{aligned} C &= e^{-0.025} \times 0.43829 \times 25 = 0.975310 \times 10.9572 = 10.687, \\ \mathcal{P} &= e^{-0.025} \times 0.56171 \times 15 = 0.975310 \times 8.4257 = 8.218. \end{aligned}$$

Parity (Theorem 17.9): $C - \mathcal{P} = 2.469$ and $S - Ke^{-r\Delta t} = 100 - 97.531 = 2.469$.

(b) The stock has $\mathbb{E}[S_{\Delta t}] = 0.6 \times 125 + 0.4 \times 85 = 109$, so $e^{\mu\Delta t} = 1.09$ and $\mu = 2\ln 1.09 = 17.24\%$ a year. The call has the expected payoff $0.6 \times 25 = 15$, so $e^{\mu_V \Delta t} = 15/10.687 = 1.4036$ and $\mu_V = 2\ln 1.4036 = 67.8\%$ a year. The put has the expected payoff $0.4 \times 15 = 6$, so $e^{\mu_V \Delta t} = 6/8.218 = 0.7301$ and $\mu_V = 2\ln 0.7301 = -62.9\%$ a year: the holder of the put expects to lose money.

(c) The proof takes the real-world expectation of the replication identity. In both states $V_{\Delta t} = \Delta \cdot S_{\Delta t} + Be^{r\Delta t}$, because the portfolio replicates the derivative. Linearity of expectation (Theorem 1.2) gives $\mathbb{E}[V_{\Delta t}] = \Delta\,\mathbb{E}[S_{\Delta t}] + Be^{r\Delta t}$. The market of the stock, the bond and the derivative admits

no arbitrage, so the price is the cost of the replica, $V = \Delta S + B$ (Theorem 18.3(ii)), and $Ve^{r\Delta t} = \Delta S e^{r\Delta t} + Be^{r\Delta t}$. Subtracting, the bond terms cancel:

$$\mathbb{E}[V_{\Delta t}] - Ve^{r\Delta t} = \Delta\big(\mathbb{E}[S_{\Delta t}] - Se^{r\Delta t}\big),$$

which is (18.6) of Remark 18.8. Dividing by $V > 0$ shows that the expected excess gross return of the option is its elasticity $\Delta S/V$ times that of the stock. Here $\mathbb{E}[S_{\Delta t}] - Se^{r\Delta t} = 109 - 102.5315 = 6.4685$. The call has $\Delta = 25/40 = 0.625$: the left side is $15 - 10.687 \times 1.025315 = 15 - 10.957 = 4.043$ and the right side $0.625 \times 6.4685 = 4.043$; its elasticity is $62.5/10.687 = 5.85$. The put has $\Delta = -15/40 = -0.375$: the left side is $6 - 8.218 \times 1.025315 = 6 - 8.426 = -2.426$ and the right side $-0.375 \times 6.4685 = -2.426$; its elasticity is $-37.5/8.218 = -4.56$.

(d) The call is the stock bought on credit: its replica holds 62.5 of shares against 10.687 of capital, the rest being a loan of 51.813, so every unit invested in the call carries 5.85 units of stock exposure, and its expected excess gross return, $1.4036 - 1.0253 = 0.3783$, is 5.85 times that of the stock, $1.09 - 1.0253 = 0.0647$. The put is a short position in the stock (-37.5 of shares and 45.718 in the bond): its elasticity is negative, so on a stock that earns more than r it earns less than r, here a negative return; its holder pays for insurance. In the language of beta (Definition 11.35): the return of each option is an affine function of the return of the stock with slope $\Delta S/V$, so by bilinearity of the covariance, the constant term contributing nothing (Remark 18.8), its beta with respect to any reference return of positive variance is 5.85 times, or -4.56 times, the beta of the stock. Three assets on one tree need three discount rates, 17.24%, 67.8% and -62.9%. Discounting the expected payoff 15 of the call at r gives $15e^{-0.025} = 14.63$, and at μ it gives $15/1.09 = 13.76$: both far above the price 10.687. The rate that works is known only once the price is, which is why the price is computed under the risk-neutral weights.

Self-check. Recompute the two expected payoffs from the replicating portfolios instead of the option payoffs. The call's replica holds 0.625 share and owes $(0.85 \times 25)/0.40 = 53.125$ at the end of the step, so its expected value is $0.625 \times 109 - 53.125 = 68.125 - 53.125 = 15$. The put's replica is short 0.375 share and holds a deposit that grows to $(1.25 \times 15)/0.40 = 46.875$, so its expected value is $46.875 - 0.375 \times 109 = 46.875 - 40.875 = 6$. Discounting these end-of-step bond amounts also recovers the prices of (a) by replication: $B = -53.125e^{-0.025} = -51.813$ and $C = 62.5 - 51.813 = 10.687$ for the call, $B = 46.875e^{-0.025} = 45.718$ and $\mathcal{P} = 45.718 - 37.5 = 8.218$ for the put. The gross returns are also weighted averages of those of the stock and the bond, with the weights $\Delta S/V$ and $B/V = 1 - \Delta S/V$ of (18.8): $5.8484 \times 1.09 - 4.8484 \times 1.02532 = 6.375 - 4.971 = 1.404$ for the call and $-4.5633 \times 1.09 + 5.5633 \times 1.02532 = -4.974 + 5.704 = 0.730$ for the put, which are the ratios 1.4036 and 0.7301 of (b) to three decimals. Finally, with p_{real} replaced by $p = 0.43829$ the stock would have the expected value $102.53 = 100e^{0.025}$ and both sides of the identity would vanish: under the risk-neutral weights every asset earns r.

Answer: (a) $p = 0.4383$, $C = 10.687$, $\mathcal{P} = 8.218$, and $C - \mathcal{P} = 2.469 = 100 - 100e^{-0.025}$. (b) $\mu = 17.24\%$ for the stock, $\mu_V = 67.8\%$ for the call and -62.9% for the put. (c) Take expectations in $V_{\Delta t} = \Delta \cdot S_{\Delta t} + Be^{r\Delta t}$ and subtract $Ve^{r\Delta t} = (\Delta S + B)e^{r\Delta t}$; both sides are 4.043 for the call and -2.426 for the put, with elasticities 5.85 and -4.56. (d) The call is a levered long position and the put a short position in the stock; discounting the call's expected payoff 15 at r or at μ gives 14.63 or 13.76 instead of 10.687.

P18.5 *Two-step puts: European, American, parity and the strike for immediate exercise*

(a) Since $e^{r\Delta t} = e^{0.035} = 1.035620$ lies between $d = 0.88$ and $u = 1.12$, the risk-neutral up-probability is $p = (1.035620 - 0.88)/0.24 = 0.155620/0.24 = 0.648415$, with $1 - p = 0.351585$, and the one-step discount factor is $e^{-r\Delta t} = 0.965605$. Each amount below is computed from unrounded values and rounded separately, so a sum or difference of rounded amounts can differ by one in the last digit. The stock nodes are 60; then 67.2 and 52.8; then 75.264, 59.136 and 46.464, at which the put pays 0, 2.864 and 15.536. By the backward induction (18.9), the up node is worth $0.965605 \times 0.351585 \times 2.864 = 0.9723$, the down node $0.965605\,(0.648415 \times 2.864 +$

$0.351585 \times 15.536) = 7.0675$, and the root

$$\mathcal{P} = 0.965605\,(0.648415 \times 0.9723 + 0.351585 \times 7.0675) = 3.0081.$$

(b) The call pays $75.264 - 62 = 13.264$ at the top node and nothing elsewhere. Its up node is worth $0.965605 \times 0.648415 \times 13.264 = 8.3048$, its down node 0, and $C = 0.965605 \times 0.648415 \times 8.3048 = 5.1997$. Then $C - \mathcal{P} = 5.1997 - 3.0081 = 2.1916$ and $S - Ke^{-rT} = 60 - 62\,e^{-0.07} = 60 - 57.8084 = 2.1916$: parity holds, as it does exactly on every European tree by (18.12).

(c) Procedure 18.16 compares, at each node, the exercise value with the continuation value computed from the American values of the two children. At the up node the exercise value is $(62 - 67.2)^+ = 0$ and the continuation value 0.9723: hold. At the down node the exercise value is $62 - 52.8 = 9.2$ and the continuation value 7.0675: exercise, and the American value there is 9.2. Both children of that node are in the money, so continuing is worth $62\,e^{-0.035} - 52.8 = 59.8675 - 52.8 = 7.0675$, the value of a short forward, and waiting only gives away the interest $62\,(1 - e^{-0.035}) = 2.1325$ on the strike. At the root the continuation value is

$$0.965605\,(0.648415 \times 0.9723 + 0.351585 \times 9.2) = 3.7321,$$

above the exercise value $62 - 60 = 2$: hold. Hence $\mathcal{P}^{\text{am}} = 3.7321$ (Proposition 18.17(iii)), the put is exercised early at the down node only, and the early-exercise premium is $3.7321 - 3.0081 = 0.7240$.

(d) For $60 < K \le 67.2$ the put pays 0, $K - 59.136$ and $K - 46.464$ at the terminal nodes. At the up node the exercise value is $(K - 67.2)^+ = 0$ and the continuation value is $e^{-r\Delta t}(1-p)(K - 59.136) > 0$: hold. At the down node both children are in the money, so the continuation value is

$$e^{-r\Delta t}\big[p\,(K - 59.136) + (1-p)(K - 46.464)\big] = Ke^{-r\Delta t} - 52.8 < K - 52.8,$$

because $p \times 59.136 + (1-p) \times 46.464 = 52.8\,e^{r\Delta t}$ by Proposition 18.7(i); the put is exercised there and is worth $K - 52.8$. The continuation value at the root is therefore

$$\begin{aligned} V_{0,0}^{\text{cont}} &= e^{-r\Delta t}\Big[p\,e^{-r\Delta t}(1-p)(K - 59.136) + (1-p)(K - 52.8)\Big] \\ &= e^{-r\Delta t}(1-p)\Big[\big(1 + pe^{-r\Delta t}\big)K - \big(52.8 + 59.136\,pe^{-r\Delta t}\big)\Big]. \end{aligned}$$

With $e^{-r\Delta t}(1-p) = 0.3394919$ and $pe^{-r\Delta t} = 0.6261135$ this is $0.3394919\,(1.6261135\,K - 89.82585)$, that is,

$$V_{0,0}^{\text{cont}} = 0.5520524\,K - 30.49515.$$

Exercising at once pays $K - 60$, which is at least the continuation value exactly when $0.4479476\,K \ge 29.50485$, that is, when

$$K \ge \frac{29.50485}{0.4479476} = 65.867$$

to three decimals. Exercise is strictly better above that strike and continuing strictly better below it, so on $60 < K \le 67.2$ the American put is exercised immediately exactly for the strikes from this threshold up to 67.2; on strikes quoted in cents the first one exercised at once is 65.87. At $K = 62$ the formula returns $0.5520524 \times 62 - 30.49515 = 3.7321$, the continuation value of (c).

Self-check. The European prices follow from the closed form (18.11) without stepping backward. With $e^{-rT} = 0.9323938$, $p^2 = 0.4204426$, $2p(1-p) = 0.4559457$ and $(1-p)^2 = 0.1236117$,

$$\mathcal{P} = 0.9323938\,(0.4559457 \times 2.864 + 0.1236117 \times 15.536) = 0.9323938 \times 3.226260 = 3.0081,$$
$$C = 0.9323938 \times 0.4204426 \times 13.264 = 5.1997.$$

The American price is the risk-neutral value of the exercise rule of Proposition 18.17(ii), which attains the maximum in (18.17). That rule exercises at the down node and nowhere else before

maturity, so it pays 9.2 at time Δt after a first down-move, 2.864 at T on the path up-then-down, and nothing on the path up-up; it is worth $e^{-0.035}\times 0.351585\times 9.2 + e^{-0.07}\times 0.648415\times 0.351585\times 2.864 = 3.1233 + 0.6088 = 3.7321$. The premium is created at the down node: $e^{-r\Delta t}(1-p)(9.2-7.0675) = 0.339492\times 2.1325 = 0.7240$. For (d), run Procedure 18.16 directly at two strikes. At $K=65$ the up node has the continuation value $0.339492\times 5.864 = 1.9908$ against the exercise value 0 (hold), the down node $65\,e^{-0.035}-52.8 = 9.9644$ against 12.2 (exercise), and the root $0.965605\,(0.648415\times 1.9908 + 0.351585\times 12.2) = 5.3883$ against 5: hold. At $K=66$ the up node has $0.339492\times 6.864 = 2.3303$ (hold), the down node $66\,e^{-0.035}-52.8 = 10.9300$ against 13.2 (exercise), and the root $0.965605\,(0.648415\times 2.3303 + 0.351585\times 13.2) = 5.9403$ against 6: exercise at once. The threshold lies between the two strikes, as found.

Answer: (a) $p = 0.6484$; the European put is worth 3.0081 (0.9723 at the up node, 7.0675 at the down node). (b) $C = 5.1997$ and $C-\mathcal{P} = 2.1916 = 60-62e^{-0.07}$. (c) $\mathcal{P}^{\text{am}} = 3.7321$, exercised early at the down node only ($9.2 > 7.0675$); premium 0.7240. (d) For $60 < K \le 67.2$ the continuation value at time 0 is $0.5520524\,K - 30.49515$, and immediate exercise is at least as good exactly when $K \ge 65.867$ (to three decimals).

P18.6 *A claim on the square of the price: one step, n steps and the limit*

(a) Here $e^{r\Delta t} = e^{0.01} = 1.010050$ lies between d and u, so $p = (1.010050-0.95)/0.15 = 0.400334$ and $1-p = 0.599666$. The claim pays $V_u = 66^2 = 4356$ or $V_d = 57^2 = 3249$. By Proposition 18.7(ii),

$$\mathbb{E}^{\mathbb{Q}}[S^2_{\Delta t}] = 0.400334\times 4356 + 0.599666\times 3249 = 3692.17, \qquad V = e^{-0.01}\times 3692.17 = 3655.43.$$

Replication, by (18.2): the share holding is $\Delta = (S^2u^2 - S^2d^2)/(S(u-d)) = S(u+d) = 123$ shares, and the bond holding is $B = e^{-r\Delta t}(u\,S^2d^2 - d\,S^2u^2)/(u-d) = -S^2ud\,e^{-r\Delta t} = -3762\,e^{-0.01} = -3724.57$. The cost is $123\times 60 - 3724.57 = 3655.43$, which is the price by Theorem 18.3(ii), the same number, and in symbols

$$V = \Delta S + B = S^2\big[(u+d) - ud\,e^{-r\Delta t}\big] = 3600\,(2.05 - 1.045\times 0.990050) = 3600\times 1.015398 = 3655.43.$$

The risk-neutral expectation gives the same closed form: expanding $(u+d)(pu+(1-p)d) = pu^2 + (1-p)d^2 + ud$ and using $pu+(1-p)d = e^{r\Delta t}$ (Proposition 18.7(i)),

$$e^{-r\Delta t}\big[pu^2 + (1-p)d^2\big] = (u+d) - ud\,e^{-r\Delta t}.$$

(b) We argue by induction on the number m of remaining steps, with the claim: at node (i,j) of the tree (Setup of §18.3), with price $S_{i,j}$ and $m = n-i$ steps to go, the backward-induction value (18.9) is $V_{i,j} = S^2_{i,j}[(u+d) - ud\,e^{-r\Delta t}]^m$. For $m=0$ the value is the payoff $S^2_{n,j}$. If the claim holds with $m-1$ steps to go, the two children $(i+1,j+1)$ and $(i+1,j)$, with prices $S_{i,j}u$ and $S_{i,j}d$, carry $S^2_{i,j}u^2[\,\cdot\,]^{m-1}$ and $S^2_{i,j}d^2[\,\cdot\,]^{m-1}$, where $[\,\cdot\,]$ abbreviates the bracket, and the recursion gives

$$V_{i,j} = e^{-r\Delta t}\big[pu^2 + (1-p)d^2\big]\,S^2_{i,j}[\,\cdot\,]^{m-1} = S^2_{i,j}\big[(u+d) - ud\,e^{-r\Delta t}\big]^m,$$

by the identity that closes (a). At the root, $S_{0,0} = S$ and $m = n$, and the backward-induction value is the price (Proposition 18.13(ii)): $V_n = S^2[(u+d) - ud\,e^{-r\Delta t}]^n$. With the data of (a), $V_2 = 3600\times 1.015398^2 = 3600\times 1.031033 = 3711.72$, against $S^2e^{rT} = 3600\,e^{0.02} = 3672.72$ at $T = 0.5$: the claim is worth about 39 more.

(c) The proof expands the one-step moment generating function to first order in Δt and then compounds along a convergent sequence. Fix a real θ, and let n be large enough that $\sigma > |r|\sqrt{T/n}$, so that $p \in (0,1)$ and $\mathbb{Q}$ is a probability (Corollary 18.23). Write $\cosh y = (e^y + e^{-y})/2$ and $\sinh y = (e^y - e^{-y})/2$, the hyperbolic cosine and sine of Step 8 of the proof of Theorem 18.37. Then

$$\mathbb{E}^{\mathbb{Q}}e^{\theta R_1} = p\,e^{\theta x} + (1-p)\,e^{-\theta x} = \cosh(\theta x) + (2p-1)\sinh(\theta x),$$

because on the right the coefficients of $e^{\theta x}$ and of $e^{-\theta x}$ are $\frac12+\frac12(2p-1)=p$ and $\frac12-\frac12(2p-1)=1-p$. The expansion of the exponential (Proposition 7.34), applied to $e^{\theta x}$ and to $e^{-\theta x}$, gives, as $x\to 0$,

$$\cosh(\theta x)=1+\frac{\theta^2x^2}{2}+O(x^4),\qquad \sinh(\theta x)=\theta x\,(1+O(x^2)),$$

the odd powers cancelling in the sum and the even powers in the difference. With $x^2=\sigma^2\Delta t$ and $(2p-1)\,x=\nu\Delta t+O(\Delta t^{3/2})$ (Proposition 18.21(ii) with (D.3)),

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}&=1+\frac{\theta^2\sigma^2}{2}\,\Delta t+O(\Delta t^2)+\theta\,(\nu\Delta t+O(\Delta t^{3/2}))\,(1+O(\Delta t))\\&=1+\Big(\theta\nu+\frac{\theta^2\sigma^2}{2}\Big)\Delta t+O(\Delta t^{3/2}).\end{aligned}$$

Put $a_n=n\,(\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}-1)$. Since $n\Delta t=T$ and $n\Delta t^{3/2}=T^{3/2}n^{-1/2}$,

$$a_n\to a=\theta\nu T+\frac{\theta^2\sigma^2T}{2}.$$

It remains to show that $(1+a_n/n)^n\to e^a$ when $a_n\to a$. For $|y|\le\frac12$, Taylor's theorem at order one with the Lagrange remainder (Corollary 7.32), applied to the logarithm about 0, gives $\ln(1+y)=y-y^2/\big(2(1+y^*)^2\big)$ for a point y^* with $|y^*|\le|y|\le\frac12$; then $(1+y^*)^2\ge\frac14$ and $|\ln(1+y)-y|\le 2y^2$. A convergent sequence is bounded, so $|a_n|/n\le\frac12$ for all large n, and for such n

$$\Big|n\ln\Big(1+\frac{a_n}{n}\Big)-a_n\Big|\le n\cdot\frac{2a_n^2}{n^2}=\frac{2a_n^2}{n}\to 0.$$

Hence $n\ln(1+a_n/n)\to a$, and by the continuity of the exponential $\big(\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}\big)^n=\exp\big(n\ln(1+a_n/n)\big)\to e^a$, which is the claim. At $\theta=1$ the convergence is an equality for every n: $\mathbb{E}^{\mathbb{Q}}e^{R_1}=pu+(1-p)d=e^{r\Delta t}$ by Proposition 18.7(i), so $(\mathbb{E}^{\mathbb{Q}}e^{R_1})^n=e^{rT}=e^{\nu T+\sigma^2T/2}$. This is part (iv) of Theorem 18.37: by (18.24) the left side is the moment generating function $\mathbb{E}^{\mathbb{Q}}e^{\theta\ln(S_T/S)}$ of the log-price, and the limit is that of the law $\mathcal{N}(\nu T,\sigma^2T)$ (Lemma 18.35(i)). The compounding step extends Proposition 7.39 from a fixed a to a convergent sequence.

For the claim, $e^{-r\Delta t}\,\mathbb{E}^{\mathbb{Q}}e^{2R_1}=e^{-r\Delta t}[pu^2+(1-p)d^2]$ is the bracket of (b), so $V_n=S^2e^{-rT}\big(\mathbb{E}^{\mathbb{Q}}e^{2R_1}\big)^n$, and with $\theta=2$, where $2\nu T+2\sigma^2T=(2r+\sigma^2)T$,

$$V_n\to S^2e^{-rT}e^{(2r+\sigma^2)T}=S^2e^{(r+\sigma^2)T}.$$

Numbers: here $\sigma=0.2>0.04=|r|\sqrt{T}$, so the condition holds for every $n\ge1$ and $d<e^{r\Delta t}<u$ on every one of these trees. The limit is $3600\,e^{0.08}=3899.83$. On this tree $ud=1$, so the bracket of (b) is $u+d-e^{-r\Delta t}$. For $n=1$, $x=0.2$, $u+d=e^{0.2}+e^{-0.2}=2.0401335$ and $e^{-r\Delta t}=e^{-0.04}=0.9607894$, so the bracket is 1.0793441 and $V_1=3600\times1.0793441=3885.64$. For $n=12$, $x=0.057735$, $u+d=2.003334$ and $e^{-r\Delta t}=0.996672$, so the bracket is 1.006662 and $V_{12}=3600\times1.006662^{12}=3600\times1.082940=3898.58$.

(d) By (18.11), $V_n=e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[S_T^2]$. By the shortcut (1.1) under $\mathbb{Q}$, $\mathbb{E}^{\mathbb{Q}}[S_T^2]=\big(\mathbb{E}^{\mathbb{Q}}[S_T]\big)^2+\mathrm{Var}^{\mathbb{Q}}(S_T)$, and $\mathbb{E}^{\mathbb{Q}}[S_T]=Se^{rT}$ on every tree, because the tree prices the stock itself: on the tree of (c) this is the case $\theta=1$ at the end of (c), and on a general tree the induction of (b) with the terminal payoff $S_{n,j}$ in place of $S_{n,j}^2$ gives the root value S, because $e^{-r\Delta t}[pu+(1-p)d]=1$, which (18.11) turns into $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[S_T]=S$. Hence

$$V_n=e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[S_T^2]=S^2e^{rT}+e^{-rT}\,\mathrm{Var}^{\mathbb{Q}}(S_T)\,:$$

the difference is the discounted risk-neutral variance of S_T, which is positive because S_T is not constant. This is Jensen's inequality for the convex function $y\mapsto y^2$, with the gap made explicit: the payoff is convex, and the claim is worth more than the same function of the forward price. In (b)

it is $3711.72 - 3672.72$, about 39. In the limit of (c) it is $S^2 e^{rT}(e^{\sigma^2 T} - 1) = 3746.92 \times 0.0408108 = 152.91 = 3899.83 - 3746.92$.

Self-check. The limit of V_n can be reached from the closed form of (b) without the moment generating function. On the tree of (c), $ud = 1$ and the bracket is $u + d - e^{-r\Delta t} = (2 + x^2 + O(x^4)) - (1 - r\Delta t + O(\Delta t^2)) = 1 + (r + \sigma^2)\Delta t + O(\Delta t^2)$, so n times its excess over 1 tends to $(r + \sigma^2)T$ and the compounding step of (c) gives $V_n \to S^2 e^{(r+\sigma^2)T}$ again; at $n = 12$ the approximation $1 + 0.08/12 = 1.006667$ stands against the exact bracket 1.006662. The values increase toward the limit, $3885.64 < 3898.58 < 3899.83$. The decomposition of (d) can be checked on the one-step tree of (a): $\mathrm{Var}^{\mathbb{Q}}(S_{\Delta t}) = p(1-p)(66-57)^2 = 0.400334 \times 0.599666 \times 81 = 19.445$ (Proposition 1.10, scaled by the squared spread), its discounted value is 19.252, and $S^2 e^{r\Delta t} + 19.252 = 3636.18 + 19.25 = 3655.43$, the price of (a).

Answer: (a) $V = 3655.43$, by $p = 0.4003$ or by 123 shares and a loan of 3724.57; in closed form $V = S^2[(u + d) - ud\, e^{-r\Delta t}]$. (b) $V_n = S^2[(u + d) - ud\, e^{-r\Delta t}]^n$; $V_2 = 3711.72$ against $S^2 e^{rT} = 3672.72$. (c) $\mathbb{E}^{\mathbb{Q}} e^{\theta R_1} = 1 + (\theta\nu + \theta^2\sigma^2/2)\Delta t + O(\Delta t^{3/2})$ and compounding give the limit $e^{\theta\nu T + \theta^2\sigma^2 T/2}$; hence $V_n \to S^2 e^{(r+\sigma^2)T} = 3899.83$, with $V_1 = 3885.64$ and $V_{12} = 3898.58$. (d) $V_n - S^2 e^{rT} = e^{-rT}\,\mathrm{Var}^{\mathbb{Q}}(S_T)$, equal to $S^2 e^{rT}(e^{\sigma^2 T} - 1) = 152.91$ in the limit.

P18.7 *Delta, the cash account and the self-financing check on a two-step put*

(a) Since $e^{r\Delta t} = e^{0.0125} = 1.012578$ lies between d and u, the risk-neutral up-probability is $p = (1.012578 - 0.92)/0.18 = 0.514325$, with $1 - p = 0.485675$, and $e^{-r\Delta t} = 0.987578$. The stock nodes are 120; then 132 and 110.4; then 145.2, 121.44 and 101.568, at which the put pays 0, 3.56 and 23.432. By (18.9) the node values are

$$
\begin{aligned}
V_{1,1} &= 0.987578 \times 0.485675 \times 3.56 = 1.7075, \\
V_{1,0} &= 0.987578\,(0.514325 \times 3.56 + 0.485675 \times 23.432) = 13.0472, \\
V_{0,0} &= 0.987578\,(0.514325 \times 1.7075 + 0.485675 \times 13.0472) = 7.1253.
\end{aligned}
$$

(b) By (18.13) of Proposition 18.13, the share holding at a node is the difference of the two child values over the difference of the two child prices, and the bond holding is the node value minus the value of the shares. The three share holdings are

$$
\Delta_{0,0} = \frac{1.7075 - 13.0472}{132 - 110.4} = -0.5250,
$$

$$
\Delta_{1,1} = \frac{0 - 3.56}{145.2 - 121.44} = -0.1498, \qquad \Delta_{1,0} = \frac{3.56 - 23.432}{121.44 - 101.568} = -1,
$$

and the positions are as follows.

Node	stock price	shares	value of the shares	bond
time 0	120	-0.5250	-62.9983	$7.1253 + 62.9983 = 70.1236$
up node	132	-0.1498	-19.7778	$1.7075 + 19.7778 = 21.4853$
down node	110.4	-1	-110.4	$13.0472 + 110.4 = 123.4472$

Every holding is short shares and long the bond, as it must be for a put. At the down node both children (121.44 and 101.568) are below the strike, so from there the put pays $125 - S_T$ for certain. It is a short forward: one share short and the present value of the strike, $125\, e^{-0.0125} = 123.4472$, in the bond.

(c) The deposit of 70.1236 grows to $70.12364 \times 1.0125785 = 71.0057$. After an up-move the 0.524986 short share costs $0.524986 \times 132 = 69.2982$ to buy back, so the position is worth $71.0057 - 69.2982 = 1.7075 = V_{1,1}$. After a down-move it costs $0.524986 \times 110.4 = 57.9585$, and the position is worth $71.0057 - 57.9585 = 13.0472 = V_{1,0}$. In both cases the position carried into the node pays exactly for the holding of (b), which costs the node value: no money is added or

withdrawn. The trades are these. At the up node the trader buys back $0.524986-0.149832 = 0.3752$ share for 49.5204, paid out of the deposit, which falls to $71.0057 - 49.5204 = 21.4853$. At the down node the trader sells a further $1 - 0.524986 = 0.4750$ share for 52.4415 and deposits the proceeds, which raises the deposit to $71.0057 + 52.4415 = 123.4472$. No position set up at time 0 and left alone can replicate the put. A holding of Δ' shares and B' in the bond that is never traded is worth $\Delta' S_T + B' e^{2r\Delta t}$ at maturity, an affine function of the terminal price, while the put pays 23.432, 3.56 and 0 at 101.568, 121.44 and 145.2: the slope between consecutive terminal prices is $-19.872/19.872 = -1$ on the lower pair and $-3.56/23.76 = -0.1498$ on the upper pair, so these values lie on no line. Equivalently, by Proposition 18.13(iii) the replicating holdings are constant only for a payoff that is affine on the terminal prices, and here they move from -0.5250 to -0.1498 or -1.

(d) The highest and lowest terminal prices are 145.2 and 101.568, so

$$\Gamma \approx \frac{\Delta_{1,1} - \Delta_{1,0}}{\frac{1}{2}\left(S_{2,2} - S_{2,0}\right)} = \frac{-0.1498 + 1}{\frac{1}{2}\left(145.2 - 101.568\right)} = \frac{0.8502}{21.816} = 0.0390$$

per unit of stock price (Remark 18.15). It is positive: the short share position shrinks as the stock rises.

(e) By Procedure 18.16, at the down node the exercise value $125 - 110.4 = 14.60$ exceeds the continuation value 13.0472, so an American put is exercised there; at the up node the exercise value is 0 and the put is held. At the root the continuation value is

$$0.987578\left(0.514325 \times 1.7075 + 0.485675 \times 14.60\right) = 7.8701,$$

above the exercise value $125 - 120 = 5$. The American put is worth $\mathcal{P}^{\text{am}} = 7.8701$ by Proposition 18.17(iii), which is 0.7448 more than the European put.

Self-check. The root value follows from the closed form (18.11) without the intermediate nodes: with $T = 2\Delta t = 1$, $e^{-rT} = e^{-0.025} = 0.975310$, $2p(1-p) = 0.499590$ and $(1-p)^2 = 0.235880$, the put is worth $0.975310\left(0.499590 \times 3.56 + 0.235880 \times 23.432\right) = 0.975310 \times 7.3057 = 7.1253$. The hedge can be followed to maturity. From the up node the deposit grows to $21.4853 \times 1.012578 = 21.756$ and the 0.149832 short share costs 21.756 or 18.196 to buy back, which leaves 0 or 3.560. From the down node the deposit grows to $123.4472 \times 1.012578 = 125.000$ and the short share costs 121.44 or 101.568, which leaves 3.56 or 23.432. On all four paths the strategy ends with the payoff of the put. The American premium is created at the down node alone: $e^{-r\Delta t}(1-p)(14.60 - 13.0472) = 0.987578 \times 0.485675 \times 1.5528 = 0.7448$, and $1.5528 = 125\left(1 - e^{-0.0125}\right)$ is the interest on the strike over one step.

Answer: (a) $p = 0.5143$; the put is worth 7.1253 today, 1.7075 at the up node and 13.0472 at the down node. (b) Time 0: -0.5250 share and 70.124 in the bond; up node: -0.1498 share and 21.485; down node: -1 share and $123.447 = 125e^{-0.0125}$, because the put is a short forward there. (c) The time-0 position is worth 1.7075 or 13.0472 on arrival; the trader buys back 0.3752 share at the up node and sells 0.4750 more at the down node; no position left untouched replicates the put, whose payoff is not affine on the terminal prices (slopes -1 and -0.1498). (d) $\Gamma \approx 0.8502/21.816 = 0.0390$. (e) Exercised at the down node ($14.60 > 13.047$); $\mathcal{P}^{\text{am}} = 7.8701$.

P18.8 *Re-derive the CRR moments, then sharpen them*

(a) We compute the two moments of a two-point variable exactly and then expand p in powers of x. *Exact moments.* The mean is $\mathbb{E}^{\mathbb{Q}}[R_1] = px - (1-p)x = (2p-1)\,x$. Since $R_1^2 = x^2$ in both states, $\mathbb{E}^{\mathbb{Q}}[R_1^2] = x^2$, and the shortcut (1.1) gives

$$\text{Var}^{\mathbb{Q}}(R_1) = x^2 - (2p-1)^2 x^2 = x^2\big(1 - (2p-1)\big)\big(1 + (2p-1)\big) = 4p(1-p)\,x^2.$$

The numerator of p. With $e^{r\Delta t} = 1 + r\Delta t + O(\Delta t^2)$ and $e^{-x} = 1 - x + x^2/2 + O(x^3)$,

$$e^{r\Delta t} - e^{-x} = x - \frac{x^2}{2} + r\Delta t + O(\Delta t^{3/2}) = x\Big[1 + \Big(\frac{r}{\sigma^2} - \frac{1}{2}\Big)x + O(x^2)\Big],$$

where $O(x^3) = O(\Delta t^{3/2})$, a term $O(\Delta t^2)$ is also $O(\Delta t^{3/2})$, and $r\Delta t = (r/\sigma^2)\,x^2$. *The denominator.* The terms of even order cancel in $e^x - e^{-x} = 2x + O(x^3) = 2x\big(1 + O(x^2)\big)$. *The quotient.* The expansion of $1/(1-y)$, applied to a quantity y that is $O(x^2)$, shows that $1/\big(1+O(x^2)\big) = 1+O(x^2)$, so

$$p = \frac{1}{2}\Big[1+\Big(\frac{r}{\sigma^2}-\frac{1}{2}\Big)x+O(x^2)\Big]\big(1+O(x^2)\big) = \frac{1}{2}+\frac{1}{2}\Big(\frac{r}{\sigma^2}-\frac{1}{2}\Big)x+O(x^2) = \frac{1}{2}+\frac{(r-\sigma^2/2)\sqrt{\Delta t}}{2\sigma}+O(\Delta t),$$

because $\frac{1}{2}(r/\sigma^2 - \frac{1}{2})\,\sigma\sqrt{\Delta t} = (r-\sigma^2/2)\sqrt{\Delta t}/(2\sigma)$ and $O(x^2) = O(\Delta t)$. *Mean and variance.* Then $2p - 1 = (r-\sigma^2/2)\sqrt{\Delta t}/\sigma + O(\Delta t)$, and multiplying by $x = \sigma\sqrt{\Delta t}$ gives $\mathbb{E}^{\mathbb{Q}}[R_1] = (r-\sigma^2/2)\Delta t + O(\Delta t^{3/2})$. The mean is therefore $O(\Delta t)$ and its square $O(\Delta t^2)$, so $\mathrm{Var}^{\mathbb{Q}}(R_1) = x^2 - \big(\mathbb{E}^{\mathbb{Q}}[R_1]\big)^2 = \sigma^2\Delta t + O(\Delta t^2)$. These are parts (i) to (iii) of Proposition 18.21.

(b) Put $a = r/\sigma^2 - \frac{1}{2}$. With one more term, $e^{r\Delta t} = 1 + (r/\sigma^2)\,x^2 + O(x^4)$ and $e^{\pm x} = 1 \pm x + x^2/2 \pm x^3/6 + O(x^4)$, so

$$e^{r\Delta t} - e^{-x} = x\Big[1 + ax + \frac{x^2}{6} + O(x^3)\Big], \qquad e^x - e^{-x} = 2x\Big[1 + \frac{x^2}{6} + O(x^3)\Big].$$

The reciprocal of the second bracket is $1 - x^2/6 + O(x^3)$, by the expansion of $1/(1-y)$ with $y = -x^2/6 + O(x^3)$. Multiplying,

$$p = \frac{1}{2}\Big[1 + ax + \frac{x^2}{6} + O(x^3)\Big]\Big[1 - \frac{x^2}{6} + O(x^3)\Big] = \frac{1}{2} + \frac{1}{2}\,ax + O(x^3)\,:$$

the two terms in $x^2/6$ cancel, and every other product is $O(x^3)$. The error of the first expansion is thus $O(x^3) = O(\Delta t^{3/2})$. Consequently $\mathbb{E}^{\mathbb{Q}}[R_1] = (2p-1)\,x = ax^2 + O(x^4) = (r-\sigma^2/2)\Delta t + O(\Delta t^2)$.

(c) Parts (a) and (b) used the number r only through the expansion of $e^{r\Delta t}$, which holds for every real constant in its place. With μ in place of r the probability is p_{real}, and the same steps give

$$p_{\text{real}} = \frac{1}{2} + \frac{(\mu-\sigma^2/2)\sqrt{\Delta t}}{2\sigma} + O(\Delta t^{3/2}),$$

$$\mathbb{E}[R_1] = \Big(\mu - \frac{\sigma^2}{2}\Big)\Delta t + O(\Delta t^2), \qquad \mathrm{Var}(R_1) = \sigma^2\Delta t + O(\Delta t^2).$$

The variance of a step is $\sigma^2\Delta t$ to leading order under both probabilities, and only the drift changes: the volatility sits in u and d, which neither probability enters. This is part (iv) of the proposition, and the reason why one calibration of the tree to σ serves in the real and in the risk-neutral world.

(d) Here $x = 0.3/\sqrt{52} = 0.041603$, $x^2 = \sigma^2\Delta t = 0.00173077$, $u = 1.0424800$, $d = 0.9592510$ and $e^{r\Delta t} = 1.0003847$, so

$$p = \frac{1.0003847 - 0.9592510}{1.0424800 - 0.9592510} = \frac{0.0411337}{0.0832290} = 0.494223,$$

against $\frac{1}{2} + (0.02 - 0.045) \times 0.138675/0.6 = 0.494222$ from the expansion. The gap is 1.1×10^{-6}, about $0.015\,x^3$ (here $x^3 = 7.2 \times 10^{-5}$), far below the order $\Delta t = 0.019$ of the error bound in (a). The mean is $(2p-1)\,x = -0.011554 \times 0.041603 = -0.00048068$, against $(r - \sigma^2/2)\Delta t = -0.025/52 = -0.00048077$. The variance is $4p(1-p)\,x^2 = 0.00173054$, against $\sigma^2\Delta t = 0.00173077$; the gap is the square of the mean, 2.3×10^{-7}. Under the real-world probability, $p_{\text{real}} = (1.0023104 - 0.9592510)/0.0832290 = 0.517360$, against $\frac{1}{2} + 0.075 \times 0.138675/0.6 = 0.517334$; the mean is $(2p_{\text{real}} - 1)\,x = 0.001444$, against $(\mu - \sigma^2/2)\Delta t = 0.001442$; and the variance is $4 \times 0.517360 \times 0.482640 \times x^2 = 0.00172868$. Moving from $\mathbb{Q}$ to the real world changes the sign of the mean and triples its size, while the variance moves from 0.00173054 to 0.00172868, both within 0.13% of $\sigma^2\Delta t$.

Self-check. Part (b) also follows from a symmetry, without multiplying brackets. With $\cosh y = (e^y + e^{-y})/2$ and $\sinh y = (e^y - e^{-y})/2$,

$$2p - 1 = \frac{2e^{r\Delta t} - e^x - e^{-x}}{e^x - e^{-x}} = \frac{e^{(r/\sigma^2)x^2} - \cosh x}{\sinh x}.$$

Replacing x by $-x$ leaves the numerator unchanged and reverses the sign of the denominator, so $2p-1$ is an odd function of x. The odd powers cancel in $\cosh x = (e^x+e^{-x})/2 = 1+x^2/2+O(x^4)$, and $e^{(r/\sigma^2)x^2} = 1+(r/\sigma^2)\,x^2+O(x^4)$ by $e^y = 1+y+O(y^2)$, so the numerator is $(r/\sigma^2-\frac12)\,x^2+O(x^4) = ax^2+O(x^4)$. The even powers cancel in the denominator, $\sinh x = x(1+O(x^2))$, so $2p-1 = ax+O(x^3)$: an even function over an odd one leaves no term in x^2, which is the cancellation found in (b). In numbers, $(1.0003847-1.0008655)/0.0416145 = -0.011554$, against $ax = -0.277778\times 0.041603 = -0.011556$.

Answer: (a) $\mathbb{E}^{\mathbb{Q}}[R_1] = (2p-1)x$ and $\mathrm{Var}^{\mathbb{Q}}(R_1) = 4p(1-p)x^2$ exactly; dividing $e^{r\Delta t}-e^{-x} = x[1+(r/\sigma^2-\frac12)x+O(x^2)]$ by $e^x-e^{-x} = 2x(1+O(x^2))$ gives the three expansions. (b) With the cubic terms the $x^2/6$ terms cancel: $p = \frac12+\frac12(r/\sigma^2-\frac12)x+O(x^3)$, so the errors are $O(\Delta t^{3/2})$ for p and $O(\Delta t^2)$ for the mean. (c) The same expansions hold with μ in place of r: the drift changes, the variance stays $\sigma^2\Delta t+O(\Delta t^2)$. (d) $p = 0.494223$ against 0.494222; mean -0.00048068 against -0.00048077; variance 0.00173054 against 0.00173077; $p_{\text{real}} = 0.517360$ with variance 0.00172868.

P18.9 *Futures options: a zero-cost hedge, and a call exercised early*

(a) The factors are $u = 1.1$ and $d = 0.9$; since $d<1<u$, the tree admits no arbitrage (Proposition 18.25(iii)). The call pays $V_u = 8$ or $V_d = 0$. A position of Δ_F long contracts and B in the bond costs B today, because the contracts are free, and is worth $\Delta_F(66-60)+Be^{r\Delta t}$ in the up state and $\Delta_F(54-60)+Be^{r\Delta t}$ in the down state. Setting these equal to 8 and 0 and subtracting gives $12\,\Delta_F = 8$, so $\Delta_F = 2/3$ of a contract, and then $Be^{r\Delta t} = 0+6\times\frac23 = 4$, so $B = 4\,e^{-0.0125} = 3.9503$. The call is worth what its replica costs, 3.9503, and the replica keeps all of it in the bond. The risk-neutral weights agree: by (18.20), $p = (1-d)/(u-d) = 0.1/0.2 = \frac12$, and $e^{-0.0125}\times\frac12\times 8 = 3.9503$. One short call is hedged by 2/3 of a long contract. The stock formula $(e^{r\Delta t}-d)/(u-d) = 0.5629$ would be wrong here, and would give 4.447: a futures contract costs nothing to hold, so its price does not drift at the rate r.

(b) Here $\sigma\sqrt{\Delta t} = 0.1$, so $u = e^{0.1} = 1.105171$ and $d = 0.904837$, and by (18.20)

$$p = \frac{1-0.904837}{1.105171-0.904837} = \frac{0.095163}{0.200334} = 0.475021, \qquad 1-p = 0.524979,$$

and the one-step discount factor is $e^{-r\Delta t} = e^{-0.01} = 0.990050$. The futures nodes are 120; then 132.6205 and 108.5805; then 146.5683, 120 and 98.2477. The backward induction and the American procedure apply with this p (Proposition 18.25, carried over to several steps by Corollary 18.26).

European call. The terminal values are 31.5683, 5 and 0. The up node is worth $0.990050\,(0.475021\times 31.5683+0.524979\times 5) = 17.4452$, the down node $0.990050\times 0.475021\times 5 = 2.3515$, and the root $C = 0.990050\,(0.475021\times 17.4452+0.524979\times 2.3515) = 9.4266$.

American call. At the up node the exercise value $132.6205-115 = 17.6205$ exceeds the continuation value 17.4452: exercise. At the down node the exercise value is 0, and the call is held at 2.3515. At the root the continuation value $0.990050\,(0.475021\times 17.6205+0.524979\times 2.3515) = 9.5090$ exceeds the exercise value $120-115 = 5$: hold. Hence $C^{\text{am}} = 9.5090$, the call is exercised early at the up node, and the early-exercise premium is $9.50902-9.42656 = 0.0825$.

Puts. The terminal values are 0, 0 and $115-98.2477 = 16.7523$. The down node is worth $0.990050\times 0.524979\times 16.7523 = 8.7071$, above the exercise value $115-108.5805 = 6.4195$; the up node is worth 0; and the root is worth $0.990050\times 0.524979\times 8.7071 = 4.5256$, with exercise value 0. The put is never exercised early: $\mathcal{P}^{\text{am}} = \mathcal{P} = 4.5256$.

Parity. The European pair satisfies $C-\mathcal{P} = 9.4266-4.5256 = 4.9010$ and $e^{-rT}(F-K) = e^{-0.02}\times 5 = 4.9010$.

(c) At the up node both children are in the money, so continuing pays F_T-115 for certain, and its value is the discounted risk-neutral mean of that amount. A futures price has no risk-neutral drift, $pu+(1-p)d = 1$ (Proposition 18.25(iii)), so the mean of F_T is the futures price 132.6205 of the node and continuing is worth $e^{-r\Delta t}(132.6205-115) = 17.4452$: exactly the

exercise value discounted over one step. Waiting only postpones the receipt of 17.6205 and costs its interest, $17.6205\,(1-e^{-0.01}) = 0.1753$. For a stock without dividends the same situation gives $S - Ke^{-r\Delta t} > S - K$: the stock itself grows at r under the risk-neutral probability, and waiting saves the interest on the strike. On this tree a stock at 132.6205 would give $132.6205 - 115\,e^{-0.01} = 18.7648$, above 17.6205. A futures price behaves like an asset whose yield equals r, and Theorem 17.16 needs the absence of a yield.

Self-check. For (a), the riskless-hedge route: a short call together with 2/3 of a long contract loses $-8 + \frac{2}{3} \times 6 = -4$ in the up state and $0 + \frac{2}{3} \times (-6) = -4$ in the down state. A certain liability of 4 is covered by $4\,e^{-0.0125} = 3.9503$ today, which is therefore what the seller of the call must receive. For (b), the closed form (18.11), with $e^{-rT} = 0.980199$, $p^2 = 0.225645$, $2p(1-p) = 0.498752$ and $(1-p)^2 = 0.275603$, gives

$$C = 0.980199\,(0.225645 \times 31.5683 + 0.498752 \times 5) = 0.980199 \times 9.61699 = 9.4266,$$
$$\mathcal{P} = 0.980199 \times 0.275603 \times 16.7523 = 0.980199 \times 4.61699 = 4.5256,$$

and the two undiscounted sums, 9.61699 and 4.61699, differ by $5.0000 = F - K$, as they must when the mean of F_T is F. The American call is the risk-neutral value of the exercise rule of Proposition 18.17(ii), which attains the maximum in (18.17), both carried over to a futures price by Corollary 18.26(iii). That rule exercises at the up node and nowhere else before maturity, so it pays 17.6205 at time Δt after a first up-move and 5 at T on the path down-then-up, and it is worth $e^{-0.01} \times 0.475021 \times 17.6205 + e^{-0.02} \times 0.524979 \times 0.475021 \times 5 = 8.2868 + 1.2222 = 9.5090$.

Answer: (a) The replica is 2/3 of a long futures contract and 3.9503 in the bond; $p = \frac{1}{2}$ and the call is worth 3.9503; one short call is hedged with 2/3 of a long contract. (b) $u = 1.10517$, $d = 0.90484$, $p = 0.47502$; European call 9.4266; American call 9.5090, exercised at the up node ($17.6205 > 17.4452$), premium 0.0825; European and American put 4.5256, never exercised early; $C - \mathcal{P} = 4.9010 = e^{-0.02} \times 5$. (c) A futures price has no risk-neutral drift, so with both children in the money continuing is worth $e^{-r\Delta t}(F-K) < F - K$: the futures price acts like an asset with yield r, and the no-early-exercise theorem needs no yield.

P18.10 *A currency tree: CRR parameters, admissibility and an American call exercised early*

(a) The step is $\Delta t = 0.25$ and $\sigma\sqrt{\Delta t} = 0.05$, so $u = e^{0.05} = 1.051271$ and $d = 0.951229$. A foreign currency is an underlying with yield r_{for}, so by Proposition 18.25(ii)

$$p = \frac{e^{(r-r_{\text{for}})\Delta t} - d}{u - d} = \frac{0.9900498 - 0.9512294}{0.1000417} = 0.388042, \qquad 1 - p = 0.611958,$$

and the one-step discount factor is $e^{-r\Delta t} = e^{-0.005} = 0.995012$. Since $d = 0.951229 < e^{(r-r_{\text{for}})\Delta t} = 0.990050 < u$, the backward induction (18.9) and Procedure 18.16 apply with this p, carried over by Corollary 18.26. Amounts are computed at full precision and rounded separately. The nodes are 1.25; then 1.314089 and 1.189037; then 1.381464, 1.25 and 1.131047.

European call. The terminal values are 0.181464, 0.05 and 0. The backward induction gives

$$\text{up node:} \quad 0.995012\,(0.388042 \times 0.181464 + 0.611958 \times 0.05) = 0.10051,$$
$$\text{down node:} \quad 0.995012 \times 0.388042 \times 0.05 = 0.01931,$$
$$\text{root:} \quad C = 0.995012\,(0.388042 \times 0.100510 + 0.611958 \times 0.019305) = 0.05056.$$

American call. At the up node the exercise value $1.314089 - 1.20 = 0.11409$ exceeds the continuation value 0.10051: exercise. At the down node the exchange rate is below the strike and the call is held at 0.01931. At the root the continuation value $0.995012\,(0.388042 \times 0.11409 + 0.611958 \times 0.01931) = 0.05581$ exceeds the exercise value 0.05: hold. Hence $C^{\text{am}} = 0.05581$, with early exercise at the up node and a premium of 0.0052.

Puts. The terminal values are 0, 0 and $1.20 - 1.131047 = 0.068953$. The down node is worth $0.995012 \times 0.611958 \times 0.068953 = 0.04199$, above the exercise value $1.20 - 1.189037 = 0.01096$,

and the root $0.995012 \times 0.611958 \times 0.04199 = 0.02557$, where the exercise value is 0. The put is never exercised early: $\mathcal{P}^{\text{am}} = \mathcal{P} = 0.02557$.

Parity. To six decimals $C = 0.050563$ and $\mathcal{P} = 0.025566$, so $C - \mathcal{P} = 0.024997$, and $Se^{-r_{\text{for}}T} - Ke^{-rT} = 1.25\,e^{-0.03} - 1.20\,e^{-0.01} = 1.213057 - 1.188060 = 0.024997$.

(b) With $K = 1.18$ the terminal call values are 0.201464, 0.07 and 0. At the up node the continuation value $0.995012\,(0.388042 \times 0.201464 + 0.611958 \times 0.07) = 0.12041$ is below the exercise value 0.13409, so the American value there is 0.13409. At the down node the exercise value $1.189037 - 1.18 = 0.00904$ is below the continuation value $0.995012 \times 0.388042 \times 0.07 = 0.02703$. The continuation value at the root is

$$0.995012\,(0.388042 \times 0.13409 + 0.611958 \times 0.02703) = 0.06823 < 0.07 = 1.25 - 1.18,$$

so the holder should exercise at once, and the call is worth its exercise value 0.07. The currency earns 6% while the strike costs only 2% to pay early, so a call that is far enough in the money is worth more exercised than alive.

(c) By Corollary 18.23 with the cost of carry $b = r - r_{\text{for}}$, the probability lies in $(0, 1)$ if and only if $\sigma > |r - r_{\text{for}}|\sqrt{\Delta t}$, that is,

$$\Delta t < \Big(\frac{\sigma}{|r - r_{\text{for}}|}\Big)^2 = \Big(\frac{0.02}{0.04}\Big)^2 = 0.25 \text{ years}.$$

For $\Delta t = 1/12$: $u = 1.005790$, $d = 0.994243$ and $e^{b\Delta t} = 0.996672$, so $p = 0.002429/0.011547 = 0.2104$. For $\Delta t = 0.25$: $d = e^{-0.01} = e^{b\Delta t}$, so $p = 0$. For $\Delta t = 0.5$: $u = 1.014243$, $d = 0.985957$ and $e^{b\Delta t} = e^{-0.02} = 0.980199$, so $p = -0.005759/0.028285 = -0.2036$. A negative p means $e^{(r - r_{\text{for}})\Delta t} < d$, that is, $d\,e^{r_{\text{for}}\Delta t} > e^{r\Delta t}$: the foreign deposit beats the domestic bond even after a down-move. The arbitrage at $\Delta t = 0.5$: borrow S in domestic currency, buy one unit of foreign currency and deposit it. After the step the deposit holds $e^{0.025} = 1.025315$ units, worth $1.014243 \times 1.025315\,S = 1.03992\,S$ after an up-move and $0.985957 \times 1.025315\,S = 1.01092\,S$ after a down-move, against a debt of $e^{0.005}S = 1.00501\,S$. The profit is $0.0349\,S$ or $0.0059\,S$, from no outlay. At $\Delta t = 0.25$ the same trade pays nothing after a down-move and a profit after an up-move, which is still an arbitrage: the inequality in the condition is strict. The remedy is a shorter step, never a clipped probability.

Self-check. The European call follows from the closed form (18.11), carried over by Corollary 18.26(i): with $e^{-rT} = 0.990050$, $p^2 = 0.150577$ and $2p(1-p) = 0.474931$,

$$C = 0.990050\,(0.150577 \times 0.181464 + 0.474931 \times 0.05) = 0.990050 \times 0.051071 = 0.05056.$$

The early exercise at the up node of (a) can be read without the tree: both children are in the money, so continuing is worth a forward position, $S_{1,1}e^{-r_{\text{for}}\Delta t} - Ke^{-r\Delta t} = 1.314089 \times 0.985112 - 1.20 \times 0.995012 = 1.29452 - 1.19401 = 0.10051$, the continuation value found on the tree. It falls short of the exercise value by $0.114089 - 0.100510 = 0.013579$, which is the interest forgone on the currency, $1.314089\,(1 - e^{-0.015}) = 0.019564$, minus the interest saved on the strike, $1.20\,(1 - e^{-0.005}) = 0.005985$. Since the up node is the only exercise node, the early-exercise premium is $e^{-r\Delta t}p \times 0.013579 = 0.995012 \times 0.388042 \times 0.013579 = 0.005243 = 0.055806 - 0.050563$. For (c), the threshold is the exact condition $d < e^{b\Delta t} < u$ written out: since $b = -0.04 < 0$, $e^{b\Delta t} < 1 < u$ for every Δt, and $-\sigma\sqrt{\Delta t} < (r - r_{\text{for}})\Delta t$ is $\sqrt{\Delta t} < 0.02/0.04 = 0.5$.

Answer: (a) $u = 1.05127$, $d = 0.95123$, $p = 0.38804$; European call 0.05056; American call 0.05581, exercised at the up node ($0.1141 > 0.1005$), premium 0.0052; European and American put 0.02557; $C - \mathcal{P} = 0.0250 = 1.25e^{-0.03} - 1.20e^{-0.01}$. (b) Exercise at once: the continuation value 0.0682 is below the exercise value 0.07. (c) The probability is valid if and only if $\Delta t < (\sigma/|r - r_{\text{for}}|)^2 = 0.25$; $p = 0.2104$, 0 and -0.2036 at $\Delta t = 1/12$, 0.25 and 0.5; at $\Delta t = 0.5$, borrowing domestic currency and holding the foreign deposit earns $0.0349\,S$ or $0.0059\,S$ from no outlay.

P18.11 *Counting steps: oscillation and averaging on the CRR tree*

(a) *One step.* Here $\Delta t = 0.75$ and $\sigma\sqrt{\Delta t} = 0.216506$, so $u = 1.241731$, $d = 0.805327$ and $e^{r\Delta t} = e^{0.0225} = 1.022755$, and

$$p = \frac{1.022755 - 0.805327}{1.241731 - 0.805327} = \frac{0.217428}{0.436404} = 0.498226.$$

The call pays $60 \times 1.241731 - 60 = 14.5039$ after an up-move and nothing after a down-move, so by (18.4) $C_1 = e^{-0.0225} \times 0.498226 \times 14.5039 = 0.977751 \times 7.2262 = 7.0654$.

Two steps. Here $\Delta t = 0.375$ and $\sigma\sqrt{\Delta t} = 0.153093$, so $u = 1.165433$, $d = 0.858050$ and $e^{r\Delta t} = 1.011314$, and $p = 0.153264/0.307384 = 0.49861$. The terminal prices are $60u^2 = 81.4941$, 60 and $60d^2 = 44.1750$; the call pays 21.4941 at the top node and nothing at the other two, the middle node sitting exactly on the strike. By (18.11), $C_2 = e^{-0.0225} \times p^2 \times 21.4941 = 0.977751 \times 0.248609 \times 21.4941 = 5.2247$.

Three and four steps. The two given values of small trees come out of the same formula. With three steps, $p = 0.498825$ and the terminal prices are 87.2995, 67.9889, 52.9498 and 41.2374; the call pays 27.2995 and 7.9889 at the two upper nodes, so $C_3 = 0.977751\,(p^3 \times 27.2995 + 3p^2(1-p) \times 7.9889) = 0.977751\,(3.38844 + 2.98878) = 6.2353$. With four steps, $p = 0.498966$ and the terminal prices are 92.5137, 74.5039, 60, 48.3196 and 38.9131; the call pays 32.5137 and 14.5039 at the two upper nodes and nothing at the middle node, which again sits on the strike, so $C_4 = 0.977751\,(p^4 \times 32.5137 + 4p^3(1-p) \times 14.5039) = 0.977751\,(2.01536 + 3.61097) = 5.5011$.

(b) The errors are as follows.

n	C_n	$C_n - 5.8097$	$n \times (C_n - 5.8097)$
1	7.0654	+1.2557	+1.26
2	5.2247	−0.5850	−1.17
3	6.2353	+0.4256	+1.28
4	5.5011	−0.3086	−1.23
50	5.7841	−0.0256	−1.28
51	5.8342	+0.0245	+1.25

The error is positive for odd n and negative for even n, and n times the error stays between 1.17 and 1.28 in absolute value: the error is of order $1/n$, with a sign that alternates with the parity of n. The prices zigzag around the limit instead of approaching it from one side.

(c) On this tree $ud = 1$, so for even n the middle terminal node is $Su^{n/2}d^{n/2} = S = K$. The most likely node then sits exactly on the kink of the payoff, where the call pays nothing, and the value that outcomes just above the strike carry is not sampled: the even trees lose value at the kink. For odd n no node sits on the strike: the two central nodes $Se^{\sigma\sqrt{\Delta t}}$ and $Se^{-\sigma\sqrt{\Delta t}}$ straddle it, the payoff is nearly linear on the range of prices each of them stands for, and nothing is lost at the kink. The even trees therefore lie below the odd ones, by about $2.53/n$ (Self-check), which is the gap $1.25 + 1.28$ of the table of (b). That gap fixes the difference of the two errors, not their signs. The rest of the error, which the position of the strike does not affect to this order, sets where the pair sits: Fact 18.40 gives only the order $1/n$ of the whole error, and the table of (b) shows that this rest is about $+1.25/n$ here. So the odd trees, which carry only that part, lie above the limit, and the even trees, which lose about $2.53/n$ more, lie below it by about $1.28/n$. Since two consecutive trees err in opposite directions by nearly the same amount, their average cancels most of the error:

$$\tfrac{1}{2}\,(C_{50} + C_{51}) = \tfrac{1}{2}\,(5.7841 + 5.8342) = 5.80915,$$

which is 0.00055 below the limit (0.00056 with the unrounded tree prices), about forty-five times closer than either tree. For a single tree the pattern of (b) gives an error of about $1.25/n$ to $1.28/n$, which falls below 0.01 only for n above 125 to 128: about 130 steps. A tree of 100 steps is

still off by about $1.28/100 = 0.013$, so two trees of about fifty steps, averaged, do better than one tree of a hundred.

(d) The probability is below $\frac{1}{2}$. Exactly, $p < \frac{1}{2}$ means $e^{r\Delta t} - d < \frac{1}{2}(u-d)$, that is,

$$e^{r\Delta t} < \tfrac{1}{2}\,(u+d)\,:$$

the riskless growth factor lies below the midpoint of the two factors. For $n = 1$ the midpoint is $\frac{1}{2}(1.241731 + 0.805327) = 1.023529 > 1.022755$, and for $n = 2$ it is $1.011742 > 1.011314$, which is why $p = 0.4982$ and 0.4986 in (a). The same comparison holds on the other four trees: the midpoints are 1.007823, 1.005865, 1.000469 and 1.000460 for $n = 3, 4, 50, 51$, against $e^{r\Delta t} = 1.007528$, 1.005641, 1.000450 and 1.000441, so $p < \frac{1}{2}$ on all six trees. To leading order, by Proposition 18.21(i),

$$p = \frac{1}{2} + \frac{(r - \sigma^2/2)\sqrt{\Delta t}}{2\sigma} + O(\Delta t),$$

so for small steps the sign of $p - \frac{1}{2}$ is the sign of $r - \sigma^2/2$. Here $\sigma^2/2 = 0.03125 > r = 0.03$. A positive rate makes the mean of the price grow at the rate r, but an up-move and a down-move are symmetric in the logarithm of the price, whose mean grows at $r - \sigma^2/2$, here -0.125% a year: the tree leans slightly downward in log terms. The leading term gives $p \approx 0.4978$ for $n = 1$ and 0.4985 for $n = 2$, against the exact 0.4982 and 0.4986.

Self-check. For (a), price the puts on the same two trees and test put–call parity, which holds exactly on every tree by (18.12): $S - Ke^{-rT} = 60 - 60\,e^{-0.0225} = 60 - 58.6651 = 1.3349$. With one step the put pays $60 - 48.3196 = 11.6804$ after a down-move and is worth $0.977751 \times 0.501774 \times 11.6804 = 5.7305$, and $C_1 - 5.7305 = 1.3349$. With two steps the put pays $60 - 44.1750 = 15.8250$ at the bottom node only and is worth $0.977751 \times 0.501393^2 \times 15.8250 = 0.977751 \times 0.251395 \times 15.8250 = 3.8898$, and $C_2 - 3.8898 = 1.3349$. For (b) and (c), a rough count predicts the size of the zigzag. In the limit $\ln(S_T/S)$ is normal with mean $(r - \sigma^2/2)T = -0.0009$ and standard deviation $\sigma\sqrt{T} = 0.2165$ (Theorem 18.37(v)), so its density at $0 = \ln(K/S)$, the point of the strike, is about $0.39894/0.216506 = 1.8426$. Terminal nodes are $2x$ apart in log terms, with $x = \sigma\sqrt{\Delta t}$, so each node stands for a cell of log-width $2x$. For even n the cell around the strike has a probability of about $2x \times 1.8426$; the tree pays 0 on it, whereas the payoff, about S times the positive part of the log-return there, averages $Sx/4$ over the cell. For odd n the strike lies on the boundary between two cells, on each of which the payoff is nearly linear, so that a node at the centre of its cell loses nothing at the kink; the small curvature errors of the cells away from the strike are nearly the same for both parities and cancel in the comparison. The even trees should therefore lie below the odd trees by about

$$e^{-rT} \times \frac{Sx}{4} \times 2x \times 1.8426 = e^{-rT}\,S\sigma^2 T \times \frac{0.9213}{n} = \frac{0.977751 \times 2.8125 \times 0.9213}{n} = \frac{2.53}{n},$$

and the table shows $1.25 + 1.28 = 2.53$ for n times the gap at $n = 50$ and 51.

Answer: (a) $C_1 = 7.0654$ (with $p = 0.4982$) and $C_2 = 5.2247$ (with $p = 0.4986$); the same formula returns the given $C_3 = 6.2353$ and $C_4 = 5.5011$. (b) The errors are $+1.2557$, -0.5850, $+0.4256$, -0.3086, -0.0256 and $+0.0245$; n times the error lies between 1.17 and 1.28 in absolute value, positive for odd n and negative for even n: an error of order $1/n$ with alternating sign. (c) For even n the middle node sits on the strike and the tree underprices; for odd n the nodes straddle the strike and it overprices. The average of C_{50} and C_{51} is 5.80915, 0.00055 below the limit, while a single tree needs about 130 steps for one cent. (d) Below: $p < \frac{1}{2}$ exactly when $e^{r\Delta t} < \frac{1}{2}(u+d)$, which to leading order is $r < \sigma^2/2$, and here $0.03 < 0.03125$.

28.19 Solutions — §19 Brownian Motion, Itô's Lemma and the Lognormal Model

P19.1 *Sqrt-t vol scaling*

(a) Scaling down. With 252 trading days a year and five a week, $\sqrt{252} \approx 15.87 \approx 16$. The daily figure is Corollary 19.2 read backwards, and the weekly one is Proposition 19.1 with $n = 5$ daily periods:

$$\sigma_{\text{daily}} = \frac{20\%}{\sqrt{252}} \approx 1.26\%, \qquad \sigma_{\text{weekly}} = \sqrt{5}\,\sigma_{\text{daily}} = \frac{20\%}{\sqrt{252/5}} = \frac{20\%}{\sqrt{50.4}} \approx 2.82\%.$$

A calendar year of 52 weeks would give $20\%/\sqrt{52} = 2.77\%$ instead: the convention moves the weekly figure by 0.04 percentage points (2.817% against 2.774%). Hence the desk anchor of Corollary 19.2: "16% annual ≈ 1% a day."

(b) The trap. We re-derive Proposition 19.1 by expanding the variance of a sum. Let $R_1, \dots, R_n$ be iid daily log-returns with finite variance σ^2_{daily}, so that $\ln(S_n/S_0) = R_1 + \cdots + R_n$. Squaring the centred sum and using linearity of expectation,

$$\operatorname{Var}\Big(\sum_{i=1}^{n} R_i\Big) = \mathbb{E}\Big[\Big(\sum_{i=1}^{n} (R_i - \mathbb{E}[R_i])\Big)^2\Big] = \sum_{i=1}^{n} \operatorname{Var}(R_i) + \sum_{i \neq j} \operatorname{Cov}(R_i, R_j),$$

which is (19.2). For $i \neq j$ the returns R_i and R_j are independent with finite variances, so $\mathbb{E}[R_i R_j] = \mathbb{E}[R_i]\,\mathbb{E}[R_j]$ by the product rule (19.1), and every cross term vanishes. Hence $\operatorname{Var}(\ln(S_n/S_0)) = n\,\sigma^2_{\text{daily}}$, and the n-day volatility is $\sigma_{\text{daily}}\sqrt{n}$. Variance, not volatility, is additive (the computation uses only uncorrelated returns with a common finite variance), so variance grows like n and volatility like $\sqrt{n}$. Annualising therefore multiplies the daily volatility by $\sqrt{252} \approx 15.9$, giving back 20%; multiplying by 252 applies to the standard deviation the factor that belongs to the variance, which double-counts time. By Lemma 19.12(ii), applied to the centred returns, each cross term of (19.2) is at most σ^2_{daily}. The variance can therefore reach $252^2\,\sigma^2_{\text{daily}}$, and the volatility $252 \times 1.26\% = 317.5\% \approx 318\%$, only if every one of the 252×251 cross covariances takes that largest value, that is, only if every pair of daily returns has correlation one: the 318% figure assumes that the days never offset each other at all.

(c) Assumption and violation. The rule needs uncorrelated increments with a common finite variance (the closing paragraph of the proof of Proposition 19.1; Remark 19.3). Violations: serial correlation; momentum (positive autocorrelation) makes the cross terms of (19.2) positive, so long-horizon volatility is *larger* than the $\sqrt{n}$ extrapolation, and mean reversion makes them negative, so it is smaller (the variance ratio measures exactly this). Intraday or day-of-week seasonality, in which the variance differs from one period to the next, breaks the common-variance assumption. Volatility clustering acts differently: in a stationary GARCH model the returns stay uncorrelated with a constant unconditional variance, so the rule holds for the unconditional variance; what moves is the conditional variance, which is why a point-in-time daily estimate extrapolates poorly to long horizons.

Self-check. For (a), the variance per unit of time gives both figures by another route: the annual variance $0.20^2 = 0.04$ accrues at $0.04/252 = 0.000159$ per trading day, whose square root is 1.26%, and at $0.04 \times 5/252 = 0.000794$ per five-day week, whose square root is 2.82%; and the inversion $1.26\% \times 15.87 = 20.0\%$ closes the loop. For (b), the reported figure is the correct one times $317.52/20 = 15.88 \approx \sqrt{252}$: the error is exactly one extra factor $\sqrt{252}$, the square root of the horizon, which is what applying the variance factor 252 to a standard deviation must produce.

Answer: (a) Daily ≈ 1.26%, weekly (five trading days) ≈ 2.82% (2.77% with a 52-week year). (b) Variance is additive, not volatility: annualise volatility with $\times\sqrt{252} \approx 15.9$, not $\times 252$; the 318% would need every pair of days to be perfectly correlated. (c) Assumes uncorrelated increments with a common variance; autocorrelation (momentum or mean reversion) breaks it, and so does

seasonality in the variance; volatility clustering leaves the unconditional rule intact but makes a point-in-time daily estimate extrapolate poorly.

P19.2 *Brownian moments and quadratic variation from scratch*

(a) Both sides are symmetric in s and t, so let $s \le t$. By the first two properties of the statement, $W_s = W_s - W_0 \sim \mathcal{N}(0, s)$, so $\mathbb{E}[W_s] = 0$ and $\mathbb{E}[W_s^2] = s$ (Proposition 7.25); both also hold at $s = 0$, where $W_0 = 0$. If $s = 0$ the product is $0 = \min(0, t)$, and if $s = t$ its mean is $\mathbb{E}[W_s^2] = s$. If $0 < s < t$, write $W_t = W_s + (W_t - W_s)$: the variables $W_s - W_0$ and $W_t - W_s$ are the increments over the cells of the grid $0 < s < t$, independent by the third property and with finite variances, so by the product rule (19.1)

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}[W_s]\,\mathbb{E}[W_t - W_s] = s = \min(s, t)$$

(Proposition 19.8(i)). The means are zero, so this is the covariance. Hence $\operatorname{Corr}(W_1, W_4) = 1/\sqrt{1 \times 4} = 1/2$. The variance of a sum is the sum of the variances plus twice the sum of the covariances of the pairs (Proposition 10.10 applied to $(W_1 + W_2) + W_3$ and then to $W_1 + W_2$, with $\operatorname{Cov}(W_1 + W_2, W_3) = \operatorname{Cov}(W_1, W_3) + \operatorname{Cov}(W_2, W_3)$ by linearity of expectation), so

$$\operatorname{Var}(W_1 + W_2 + W_3) = (1 + 2 + 3) + 2\,\big(\min(1,2) + \min(1,3) + \min(2,3)\big) = 6 + 2 \times 4 = 14,$$

not 6: the three values share increments.

(b) We compute means and variances from the normal moments of the increments. Write $\Delta W_i = \sqrt{\Delta t_i}\, Z_i$, where $Z_1, \dots, Z_n$ are iid standard normal, because the increments over the cells are independent with the laws $\mathcal{N}(0, \Delta t_i)$.

The deterministic sum. Each Δt_i is at most $\Delta t_{\max}$ and the cells add up to $[0, t]$, so $\sum_i (\Delta t_i)^2 \le \Delta t_{\max} \sum_i \Delta t_i = t\, \Delta t_{\max}$.

The quadratic variation. Since $(\Delta W_i)^2 = \Delta t_i\, Z_i^2$,

$$\mathbb{E}[\mathrm{QV}_n] = \sum_i \Delta t_i\, \mathbb{E}[Z_i^2] = t, \qquad \operatorname{Var}\left((\Delta W_i)^2\right) = (\Delta t_i)^2\,\left(\mathbb{E}[Z_i^4] - 1\right) = 2(\Delta t_i)^2.$$

The squares are functions of the mutually independent increments taken one at a time, hence mutually independent by the grouping property (a) of the Setup of §19.2; they have finite variances, so any two of them are uncorrelated by the product rule (19.1). Expanding the variance of the sum as in (19.2) (Proposition 10.10 extended to n terms), every cross covariance vanishes, so $\operatorname{Var}(\mathrm{QV}_n) = \sum_i \operatorname{Var}\left((\Delta W_i)^2\right) = 2\sum_i (\Delta t_i)^2 \le 2t\, \Delta t_{\max}$ by the deterministic sum. Because the mean of QV_n is t, the mean-square error $\mathbb{E}[(\mathrm{QV}_n - t)^2]$ is this variance, which tends to 0 with the mesh.

The mixed sum. Each term $\Delta t_i\, \Delta W_i$ has mean 0 and variance $(\Delta t_i)^2\, \Delta t_i$. The terms are constant multiples of the mutually independent increments, hence pairwise uncorrelated by the same argument, so the cross terms of the expansion vanish, and the sum has mean 0 and second moment, equal to its variance because its mean is 0, $\sum_i (\Delta t_i)^3 \le \Delta t_{\max}^2 \sum_i \Delta t_i = t\, \Delta t_{\max}^2$.

The three statements are the rules $(dW)^2 = dt$, $dt\, dW = 0$ and $(dt)^2 = 0$ of Table 4: each is a statement about sums over a partition whose mesh tends to zero (Proposition 19.10).

(c) Here $\Delta t_i = 2/100 = 0.02$ for every i, so $\mathbb{E}[\mathrm{QV}_n] = 2$, $\operatorname{Var}(\mathrm{QV}_n) = 2 \times 100 \times 0.02^2 = 0.08$, and the standard deviation is $\sqrt{0.08} = 0.283$.

(d) With n equal steps $|\Delta W_i| = \sqrt{t/n}\, |Z_i|$, so

$$\mathbb{E}[\mathrm{FV}_n] = n\,\sqrt{t/n}\,\mathbb{E}|Z| = \sqrt{nt}\,\sqrt{2/\pi} = \sqrt{2nt/\pi},$$

which is $\sqrt{800/\pi} = 15.96$ at $t = 1$ and $n = 400$, and grows without bound with n. For an arbitrary partition, $\sqrt{\Delta t_i} = \Delta t_i / \sqrt{\Delta t_i} \ge \Delta t_i / \sqrt{\Delta t_{\max}}$ gives $\mathbb{E}[\mathrm{FV}_n] = \sqrt{2/\pi} \sum_i \sqrt{\Delta t_i} \ge \sqrt{2/\pi}\; t/\sqrt{\Delta t_{\max}}$,

which also tends to infinity as the mesh tends to zero. For any partition $|\Delta W_i|^3 = (\Delta t_i)^{3/2}\,|Z_i|^3$ and $(\Delta t_i)^{3/2} \le \sqrt{\Delta t_{\max}}\,\Delta t_i$, so

$$\mathbb{E}\Big[\sum_i |\Delta W_i|^3\Big] = \mathbb{E}|Z|^3 \sum_i (\Delta t_i)^{3/2} \le 1.596\,\sqrt{\Delta t_{\max}} \sum_i \Delta t_i = 1.596\,t\,\sqrt{\Delta t_{\max}},$$

which tends to 0 with the mesh; for the 400 steps it is $1.596/20 = 0.080$. Read with (b): over finer and finer partitions the sum of the absolute increments explodes, the sum of their squares settles at the elapsed time t, and the sum of their cubes vanishes. The path has infinite length (Theorem 19.14(i) proves it for almost every path, not only in the mean). And in a Taylor expansion of a function of W along a fine partition the second-order terms survive while the third-order terms do not, which is why Itô's lemma (Theorem 19.30) stops at order two.

(e) On such a path $|\Delta W_i| \le c\,\Delta t_i$ for every cell, so $\mathrm{QV}_n \le c^2 \sum_i (\Delta t_i)^2 \le c^2\,t\,\Delta t_{\max}$ by (b). The set of these paths is described by uncountably many conditions, so it is bounded by events rather than given a probability directly. Take the n equal steps of $[0,t]$, so that $\Delta t_{\max} = t/n$. For every $n \ge 2c^2 t$, $c^2\,\Delta t_{\max} \le 1/2$, so every path with the property lies in the event $A_n = \{\mathrm{QV}_n \le t/2\} \subset \{|\mathrm{QV}_n - t| \ge t/2\}$, and Chebyshev's inequality (Lemma 19.12(i)) with (b) gives

$$P(A_n) \le \frac{\mathrm{Var}(\mathrm{QV}_n)}{(t/2)^2} \le \frac{8\,\Delta t_{\max}}{t} = \frac{8}{n}.$$

The paths with the property therefore lie in the event $\bigcap_{n \ge 2c^2 t} A_n$, whose probability is at most $8/n$ for every such n, hence zero. By the mean value theorem, a path that is differentiable on $[0,t]$ with $|W'| \le c$ throughout has the property. A path with a bounded derivative on $[0,t]$ has the property for some integer $c = 1, 2, \ldots$, so it lies in the union over c of these null events, an event of probability zero by countable subadditivity (Lemma 19.12(0)). The property on $[0,t]$ implies it on $[0, 1/m]$ whenever $1/m \le t$, so one more union, over $m = 1, 2, \ldots$ of the null events for $t = 1/m$, covers every $t > 0$ at once: with probability one the path has a bounded derivative on no interval $[0,t]$ with $t > 0$; in particular it is continuously differentiable on none. Smoothness and a positive quadratic variation exclude each other. Theorem 19.14(ii) proves much more: with probability one the path is differentiable at no point.

Self-check. For (a), write the three values in independent increments: with $X_1 = W_1$, $X_2 = W_2 - W_1$ and $X_3 = W_3 - W_2$, iid standard normal, $W_1 + W_2 + W_3 = 3X_1 + 2X_2 + X_3$, of variance $9 + 4 + 1 = 14$. For (b), the one-cell partition gives $\mathrm{QV}_1 = W_t^2$, whose variance is $\mathbb{E}[W_t^4] - t^2 = 3t^2 - t^2 = 2t^2$, the formula with $n = 1$. For (c), Proposition 19.8(ii) with the constant 2 makes $\tilde{W}_s = W_{2s}/\sqrt{2}$ a standard Brownian motion, so the increments of W over the 100 cells of $[0,2]$ are $\sqrt{2}$ times those of $\tilde{W}$ over the 100 cells of $[0,1]$, and QV_n is twice the sum of Example 19.11 with $n = 100$, whose variance is $2/100$; hence $\mathrm{Var}(\mathrm{QV}_n) = 4 \times 0.02 = 0.08$ and the standard deviation is $2\sqrt{0.02} = 0.283$. For (d), the Cauchy–Schwarz inequality for sums gives $\mathrm{FV}_n^2 \le n\,\mathrm{QV}_n$ on every path, hence $(\mathbb{E}[\mathrm{FV}_n])^2 \le \mathbb{E}[\mathrm{FV}_n^2] \le n\,\mathbb{E}[\mathrm{QV}_n] = nt$ and $\mathbb{E}[\mathrm{FV}_n] \le \sqrt{nt}$, which is 20 for the 400 steps; the value $15.96 = 0.798 \times 20$ respects the bound.

Answer: (a) $\mathbb{E}[W_s W_t] = \min(s,t)$; $\mathrm{Corr}(W_1, W_4) = 1/2$; $\mathrm{Var}(W_1 + W_2 + W_3) = 14$. (b) $\mathbb{E}[\mathrm{QV}_n] = t$, $\mathrm{Var}(\mathrm{QV}_n) = 2\sum_i (\Delta t_i)^2 \le 2t\,\Delta t_{\max}$; the mixed sum has second moment $\sum_i (\Delta t_i)^3 \le t\,\Delta t_{\max}^2$; $\sum_i (\Delta t_i)^2 \le t\,\Delta t_{\max}$: the rules $(dW)^2 = dt$, $dt\,dW = 0$, $(dt)^2 = 0$. (c) Mean 2, variance 0.08, standard deviation 0.283. (d) $\mathbb{E}[\mathrm{FV}_n] = \sqrt{2nt/\pi} = 15.96$, unbounded in n; the cubes sum to at most $1.596\,t\sqrt{\Delta t_{\max}} \to 0$: infinite length, and a Taylor expansion that stops at order two. (e) $\mathrm{QV}_n \le c^2 t\,\Delta t_{\max} \to 0$ contradicts $\mathrm{QV}_n \to t$, so such paths lie in an event of probability zero: with probability one a Brownian path has a bounded derivative on no interval $[0,t]$ with $t > 0$.

P19.3 *A cash balance as a generalised Wiener process*

Throughout, $X_t - X_0 = \alpha t + \beta W_t \sim \mathcal{N}(\alpha t,\ \beta^2 t)$ (Definition 19.25(i)), so that, by standardising,

$$P(X_t < 0) = \Phi\Big(-\frac{X_0 + \alpha t}{\beta\sqrt{t}}\Big).$$

(a) After three months, $X_3 \sim \mathcal{N}(4 + 0.25 \times 3,\ 1 \times 3) = \mathcal{N}(4.75,\ 3)$, with standard deviation $\sqrt{3} = 1.73$.

(b) After twelve months, $X_{12} \sim \mathcal{N}(7,\ 12)$, with standard deviation $\sqrt{12} = 3.464$, so $P(X_{12} < 0) = \Phi(-7/3.464) = \Phi(-2.02) = 2.17\%$.

(c) The condition is $(X_0 + 3)/3.464 = \Phi^{-1}(0.99) = 2.326$, that is, $X_0 = 2.326 \times 3.464 - 3 = 8.06 - 3 = 5.06$ million.

(d) The probability is largest where the ratio $(X_0 + \alpha t)/(\beta\sqrt{t}) = X_0\, t^{-1/2}/\beta + \alpha\, t^{1/2}/\beta$ is smallest. Its derivative in t is $(-X_0\, t^{-3/2} + \alpha\, t^{-1/2})/(2\beta)$, negative for $t < X_0/\alpha$ and positive for $t > X_0/\alpha$. The minimum is at $t^* = X_0/\alpha = 16$ months, where the ratio is $2\sqrt{X_0\alpha}/\beta = 2$ and the probability is $\Phi(-2) = 2.28\%$. Before month 16 the noise, which grows like $\sqrt{t}$, gains on the cushion; afterwards the drift, which grows like t, wins. Within the first year the probability therefore rises all the way to month 12, where it is the 2.17% of (b).

(e) The balance touches zero by month 12 exactly when $\min_{0 \le t \le 12}(\alpha t + \beta W_t) \le -X_0$. The formula, which is Theorem 19.63(iii), with $\mu = 0.25$, $\sigma = 1$, $a = 4$ and $T = 12$, so that $\mu T = 3$ and $2\mu a/\sigma^2 = 2$, gives

$$\begin{aligned}\Phi\Big(\frac{-4-3}{3.464}\Big) + e^{-2}\,\Phi\Big(\frac{-4+3}{3.464}\Big) &= \Phi(-2.02) + e^{-2}\,\Phi(-0.289)\\ &= 0.0217 + 0.1353 \times 0.3863 = 0.0217 + 0.0523 = 0.0740,\end{aligned}$$

about 7.4%. This is 3.4 times the terminal probability of (b): about seven in ten of the paths that touch zero during the year are back above it at the end of the year. The factor is above the 2 of the driftless reflection principle because the drift points away from the level.

(f) Write $X_{12} = X_6 + (X_{12} - X_6)$, where, by the statement, $X_{12} - X_6 = -0.1 \times 6 + \sqrt{2}\,(W_{12} - W_6)$. The first half gives $X_6 \sim \mathcal{N}(4 + 1.5,\ 6) = \mathcal{N}(5.5,\ 6)$. The increment over the second half is $-0.6 + \sqrt{2}\,(W_{12} - W_6) \sim \mathcal{N}(-0.6,\ 2 \times 6) = \mathcal{N}(-0.6,\ 12)$, by (19.3), and it is independent of X_6 because the two are functions of the increments of W over disjoint intervals. A sum of independent normal variables is normal, with the means added and the variances added (Lemma 19.38(ii)), so

$$X_{12} \sim \mathcal{N}(4.9,\ 18), \qquad P(X_{12} < 0) = \Phi\Big(-\frac{4.9}{\sqrt{18}}\Big) = \Phi(-1.155) = 12.4\%,$$

with $\sqrt{18} = 4.243$. A lower drift and a doubled variance rate for six months raise the probability almost sixfold.

Self-check. For (d), the inequality between the arithmetic and the geometric mean gives $X_0/\sqrt{t} + \alpha\sqrt{t} \ge 2\sqrt{X_0\alpha} = 2$, with equality exactly when $X_0/\sqrt{t} = \alpha\sqrt{t}$, that is, $t = X_0/\alpha = 16$: the same horizon and the same ratio without calculus; the ratio is 2.02 at $t = 12$ and 2.05 at $t = 25$, both above 2. For (c), a start of 5.06 gives the mean 8.06 and $8.06/3.464 = 2.327$. For (e), the touching probability must exceed the terminal probability 2.17%, because a path that ends below zero has touched it, and it must lie below the driftless value $2\,\Phi(-4/3.464) = 2\,\Phi(-1.155) = 2 \times 0.1240 = 24.8\%$, because a positive drift lifts every path pointwise; 7.4% lies between the two. For (f), the average drift over the year is $(0.25 - 0.1)/2 = 0.075$ per month and the average variance rate is 1.5 per month, which give the mean $4 + 0.9 = 4.9$ and the variance 18.

Answer: (a) $X_3 \sim \mathcal{N}(4.75, 3)$. (b) $\Phi(-2.02) = 2.17\%$. (c) $X_0 = 5.06$ million. (d) $t^* = X_0/\alpha = 16$ months, where the probability is $\Phi(-2) = 2.28\%$. (e) $0.0217 + 0.0523 = 7.4\%$, about 3.4 times the terminal probability. (f) $X_{12} \sim \mathcal{N}(4.9, 18)$ and $P(X_{12} < 0) = \Phi(-1.155) = 12.4\%$.

P19.4 *Which process for a price?*

(a) Over one step the change is normal with mean equal to the drift term and standard deviation equal to the coefficient of Z. For one week, $\Delta t = 1/52$ and $\sqrt{\Delta t} = 1/7.211 = 0.1387$, so $\alpha\,\Delta t = 0.077$, $\beta\sqrt{\Delta t} = 2.22$, $\mu\,\Delta t = 0.19\%$ and $\sigma\sqrt{\Delta t} = 5.55\%$.

- Model (1): constant dollar drift and constant dollar noise. The weekly move has mean 0.077 and standard deviation 2.22 dollars at either price: 0.48% and 13.9% of a price of 16, but 0.048% and 1.39% of a price of 160.
- Model (2): constant percentage drift, constant dollar noise. The mean is 0.031 dollars at 16 and 0.31 dollars at 160 (0.19% of either price); the standard deviation is 2.22 dollars at either price (13.9% and 1.39%).
- Model (3): constant dollar drift, constant percentage noise. The mean is 0.077 dollars at either price (0.48% and 0.048%); the standard deviation is 0.89 dollars at 16 and 8.88 dollars at 160 (5.55% of either price).
- Model (4): constant percentage drift and constant percentage noise. The mean is 0.031 and 0.31 dollars and the standard deviation 0.89 and 8.88 dollars: 0.19% and 5.55% of the price, whatever the price.

A stock should follow model (4), the Euler step of the geometric Brownian motion of Definition 19.27 (the paragraph on proportional shocks in §19.4): the law of the return $\Delta S/S$ over the step, $\mathcal{N}(\mu\,\Delta t,\ \sigma^2\Delta t)$, does not depend on the price level. A two percent move means the same to a holder at a price of 16 as at a price of 160, and the unit in which a share is quoted is arbitrary: a ten-for-one split should not change the risk of the position. Model (1) says that the stock at 16 is ten times as volatile, in percentage terms, as the stock at 160, and it allows negative prices. Models (2) and (3) each keep one of the two terms in dollars: under (2) the standard deviation of the weekly return is 13.9% at a price of 16 and 1.39% at 160, under (3) its mean is 0.48% and 0.048%. After a ten-for-one split, one share at 160 becomes ten shares at 16: under (1) the weekly dollar standard deviation of the position goes from 2.22 to $10 \times 2.22 = 22.2$, under (4) it is 8.88 before and $10 \times 0.888 = 8.88$ after.

(b) By Definition 19.25(i), $S_2 \sim \mathcal{N}(40 + 4 \times 2,\ 16^2 \times 2) = \mathcal{N}(48,\ 512)$, with standard deviation $\sqrt{512} = 22.63$, so

$$P(S_2 < 0) = \Phi\Big(-\frac{48}{22.63}\Big) = \Phi(-2.12) = 1.7\%.$$

The probability is small, but it is not zero. Under the geometric Brownian motion the price is $S_0 > 0$ times an exponential, so it is positive at every time (Corollary 19.34(i)) and the probability is 0.

(c) A spread between two prices or two rates; an interest rate near or below zero; the profit and loss of a book. Each can be negative or pass through zero, and for each the current level sets no scale for its moves: a move of a given size means the same whatever the level, so shocks add and do not multiply. Bachelier's normal model is the right one for them; the geometric Brownian motion, the continuous-time form of the proportional model (4), cannot be, because started at a positive level it stays positive at every time (Corollary 19.34(i)), so it never reaches zero or changes sign.

Self-check. The four models agree at $S = 40$: $\mu S = 0.10 \times 40 = 4 = \alpha$ and $\sigma S = 0.40 \times 40 = 16 = \beta$. The weekly percentages follow from the annual ones by the square-root rule: a dollar volatility of 16 is 100% a year of a price of 16 and 10% a year of a price of 160, and $100\%/\sqrt{52} = 13.9\%$, $10\%/\sqrt{52} = 1.39\%$, $40\%/\sqrt{52} = 5.55\%$. The weekly mean percentages follow from the annual drift divided by 52: a dollar drift of 4 is 25% a year of a price of 16 and 2.5% a year of a price of 160, and $25\%/52 = 0.48\%$, $2.5\%/52 = 0.048\%$, $10\%/52 = 0.19\%$. In (b) the standardised distance to zero simplifies exactly: $48/(16\sqrt{2}) = 3/\sqrt{2} = 2.12$.

Answer: (a) (1) dollar drift and dollar noise constant; (2) percentage drift and dollar noise; (3) dollar drift and percentage noise; (4) percentage drift and percentage noise. Weekly move: under (1), mean 0.077 dollars (0.48% of 16, 0.048% of 160) and standard deviation 2.22 dollars (13.9% of 16, 1.39% of 160); under (2), mean 0.031 and 0.31 dollars (0.19% of either price) and standard deviation 2.22 dollars (13.9% and 1.39%); under (3), mean 0.077 dollars (0.48% and 0.048%) and standard deviation 0.89 and 8.88 dollars (5.55% of either price); under (4), mean 0.031 and 0.31

dollars (0.19% of either price) and standard deviation 0.89 and 8.88 dollars (5.55% of either price). A stock follows (4): returns, not dollar changes, have a law that is free of the price level. (b) $S_2 \sim \mathcal{N}(48, 512)$, $P(S_2 < 0) = \Phi(-2.12) = 1.7\%$; zero under the geometric Brownian motion. (c) A spread, a rate near or below zero, a profit and loss: quantities that can be negative or cross zero, whose moves do not scale with the level.

P19.5 *Itô's lemma on functions of a stock price*

For a function $V(S,t)$ of the price and of time, Itô's lemma (Corollary 19.31(iii)) reads

$$dV = \left(\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\,\sigma^2 S^2\,\partial_{SS} V\right) dt + \sigma S\,\partial_S V\,dW.$$

Each function below, with its derivatives, is a power of S or $\ln S$ times a bounded function of t, hence of moderate growth, so the corollary applies.

(a) For $V = S^a$: $\partial_t V = 0$, $\partial_S V = a\,S^{a-1}$ and $\partial_{SS} V = a(a-1)\,S^{a-2}$, so

$$\frac{d(S^a)}{S^a} = \left(a\mu + \tfrac{1}{2}\,a(a-1)\,\sigma^2\right) dt + a\sigma\,dW.$$

The power is again a geometric Brownian motion, by its explicit form rather than by its coefficients: by (19.10), $S_t^a = S_0^a \exp(agt + a\sigma W_t)$ with $g = \mu - \sigma^2/2$, which is Definition 19.27 with volatility $|a|\sigma$ and drift $ag + \frac{1}{2}a^2\sigma^2 = a\mu + \frac{1}{2}a(a-1)\sigma^2$, driven by W for $a > 0$ and by $-W$ for $a < 0$, a standard Brownian motion by Proposition 19.8(iii) (Example 19.33). With $\mu = 0.09$ and $\sigma^2 = 0.09$:

- for $a = 3$: drift $0.27 + 3 \times 0.09 = 54\%$, volatility 90%;
- for $a = -1$: drift $-0.09 + 0.09 = 0$ and volatility 30%, the coefficient of dW being -30%: the shock is reversed. The drift of $1/S$ is $\sigma^2 - \mu$, which vanishes here because $\sigma^2 = \mu$;
- for $a = \frac{1}{2}$: drift $0.045 + \frac{1}{2} \times \frac{1}{2} \times (-\frac{1}{2}) \times 0.09 = 0.045 - 0.01125 = 3.375\%$, volatility 15%.

(b) By (a) with $a = -1$, applied to the rate in place of the stock (its drift 0.5% and volatility 10% in place of μ and σ), the reciprocal rate drifts at $0.10^2 - 0.005 = 1\% - 0.5\% = +0.5\%$ a year, with a volatility of 10%. Both the rate and its reciprocal are expected to appreciate at 0.5% a year (Siegel's paradox). There is no contradiction: $x \mapsto 1/x$ is convex, the convexity term, for the rate at the level x, is $\frac{1}{2}\,(0.10)^2 x^2 \cdot 2x^{-3} = 0.01/x > 0$, and quoting the reciprocal does not simply flip the sign of the drift.

(c) For $V = e^{-rt}S$: $\partial_t V = -rV$, $\partial_S V = e^{-rt}$ and $\partial_{SS} V = 0$, so

$$d(e^{-rt}S) = (\mu - r)\,e^{-rt}S\,dt + \sigma\,e^{-rt}S\,dW,$$

a geometric Brownian motion, because $e^{-rt}S_t = S_0 \exp\left((\mu - r - \frac{1}{2}\sigma^2)\,t + \sigma W_t\right)$ is Definition 19.27 with drift $\mu - r = 5\%$ and volatility 30%. For $V = S\,e^{(r-q)(T-t)}$: $\partial_t V = -(r-q)\,V$, $\partial_S V = e^{(r-q)(T-t)}$ and $\partial_{SS} V = 0$, so

$$dF = (\mu - r + q)\,F\,dt + \sigma F\,dW,$$

a geometric Brownian motion with drift $9\% - 4\% + 2\% = 7\%$ and volatility 30% (Corollary 19.36). Both functions are linear in S, so there is no convexity term; only the time factor changes the drift.

(d) For $V = \ln S$: $\partial_S V = 1/S$ and $\partial_{SS} V = -1/S^2$, so

$$d\ln S = \left(\mu - \tfrac{1}{2}\,\sigma^2\right) dt + \sigma\,dW = 0.045\,dt + 0.30\,dW\,:$$

a generalised Wiener process with drift 4.5% and variance rate 9% a year (Corollary 19.34). The quoted argument is the first-order chain rule. It drops the second-order term $\frac{1}{2}\,\sigma^2 S^2 \times (-1/S^2) = -\sigma^2/2 = -4.5\%$, which is of first order in dt because $(dW)^2 = dt$. The logarithm is concave, so

the term is negative: the log price, and with it the median price, grows at 4.5% while the mean price grows at 9% (Proposition 19.42(i) and (ii)).

Self-check. The explicit formula gives every number without Itô's lemma. With $g = \mu - \sigma^2/2 = 4.5\%$, $S_t^a = S_0^a \exp(agt + a\sigma W_t)$ has the drift $ag + \frac{1}{2}a^2\sigma^2$: $0.135 + 0.405 = 0.54$ for $a = 3$, $-0.045 + 0.045 = 0$ for $a = -1$ and $0.0225 + 0.01125 = 0.03375$ for $a = \frac{1}{2}$. For the exchange rate the log drift is $0.005 - 0.01/2 = 0$; the reciprocal, whose logarithm is minus that of the rate, has the log drift 0 and so the drift $0 + \frac{1}{2} \times 0.01 = 0.5\%$. Likewise $e^{-rt}S_t = S_0 \exp\left((\mu - r - \frac{1}{2}\sigma^2)\,t + \sigma W_t\right)$ has drift $\mu - r = 5\%$, $F_t = S_0 e^{(r-q)T} \exp\left((\mu - r + q - \frac{1}{2}\sigma^2)\,t + \sigma W_t\right)$ has drift $\mu - r + q = 7\%$, and $\ln S_t = \ln S_0 + gt + \sigma W_t$ has drift $g = 4.5\%$. The general formula returns the drift μ at $a = 1$.

Answer: (a) $d(S^a)/S^a = (a\mu + \frac{1}{2}a(a-1)\sigma^2)\,dt + a\sigma\,dW$, a geometric Brownian motion with volatility $|a|\sigma$: drift 54% and volatility 90% for $a = 3$; drift 0 and volatility 30% for $a = -1$ (coefficient of dW equal to -30%: driven by $-W$); drift 3.375% and volatility 15% for $a = \frac{1}{2}$. (b) The reciprocal rate drifts at $0.10^2 - 0.005 = +0.5\%$ a year as well. (c) $e^{-rt}S$ has drift $\mu - r = 5\%$ and F has drift $\mu - r + q = 7\%$, both with volatility 30%. (d) $d\ln S = 0.045\,dt + 0.30\,dW$: the drift is $\mu - \sigma^2/2 = 4.5\%$, not 9%, because the chain rule misses the convexity term $-\sigma^2/2$.

P19.6 *Itô's lemma on Brownian motion from the sums, and which processes have zero drift*

(a) We telescope, compute three second moments from the independence of the increments, and pass to the limit.

The identities. For real numbers x and y, $(x+y)^2 - x^2 = 2xy + y^2$ and $(x+y)^3 - x^3 = 3x^2y + 3xy^2 + y^3$. Take $x = W_{t_{i-1}}$ and $y = \Delta W_i$, so that $x + y = W_{t_i}$, and sum over i. The left sides telescope to $W_t^2 - W_0^2 = W_t^2$ and to W_t^3, and the right sides are the sums of the statement. This is Step 1 of the proof of Theorem 19.30 with every term explicit, because the Taylor expansion of a polynomial stops (for the cube the Lagrange point of Step 1 is $u^* = 1/3$): the time sum vanishes, the first sum of each identity is the gradient sum, QV_n and $3\sum_i W_{t_{i-1}}(\Delta W_i)^2$ are the quadratic sums, and the sum of cubes is the remainder, which is zero for the square.

The three second moments. Write $\Delta W_i = \sqrt{\Delta t}\,Z_i$ with $Z_1, \ldots, Z_n$ iid standard normal. Every product below is a polynomial in finitely many of the variables Z_i, so it has finite moments of every order, and the mean of a product of independent factors is the product of their means (19.5). The centred square $(\Delta W_i)^2 - \Delta t = \Delta t\,(Z_i^2 - 1)$ has mean 0 and second moment $(\Delta t)^2\,(3 - 2 + 1) = 2(\Delta t)^2$.

First, $\mathrm{QV}_n - t = \sum_i \left((\Delta W_i)^2 - \Delta t\right)$ is a sum of independent terms of mean zero, so its second moment is the sum of theirs: $2n(\Delta t)^2 = 2t^2/n$.

Second, put $a_i = 6W_{t_{i-1}}$ and $\mathrm{QV}_n^a = \frac{1}{2}\sum_i a_i \left((\Delta W_i)^2 - \Delta t\right)$, the weighted sum of the statement in the notation of Step 4 of the proof of Theorem 19.30. The weight a_i is a function of the increments before the i-th cell, so it is independent of ΔW_i. For $i < j$ the product $a_i a_j \left((\Delta W_i)^2 - \Delta t\right)$ is a function of the increments before the j-th cell, so it is independent of ΔW_j, and the cross term has the mean

$$\mathbb{E}\Big[a_i a_j \left((\Delta W_i)^2 - \Delta t\right)\Big]\,\mathbb{E}\Big[(\Delta W_j)^2 - \Delta t\Big] = 0.$$

A diagonal term has the mean $\mathbb{E}[a_i^2] \times 2(\Delta t)^2 = 36\,t_{i-1} \times 2(\Delta t)^2$. Since $t_{i-1} = (i-1)\,\Delta t$ and $\sum_{i=1}^n (i-1) = n(n-1)/2$,

$$\mathbb{E}\big[(\mathrm{QV}_n^a)^2\big] = \tfrac{1}{4} \times 36 \times 2(\Delta t)^3 \times \frac{n(n-1)}{2} = \frac{9t^3(n-1)}{n^2}.$$

Third, the cubes $(\Delta W_i)^3 = (\Delta t)^{3/2} Z_i^3$ are independent, with mean zero and second moment $(\Delta t)^3\,\mathbb{E}[Z^6] = 15(\Delta t)^3$, so the second moment of their sum is $15n(\Delta t)^3 = 15t^3/n^2$.

At $t = 1$ and $n = 100$ the three numbers are $2/100 = 0.02$, $9 \times 99/10^4 = 0.0891$ and $15/10^4 = 0.0015$. All three tend to 0; the two second-order sums do so like $1/n$, the third-order sum like $1/n^2$.

The limits. Let $n \to \infty$, and solve each identity for its first sum. In the first identity,

$2\sum_i W_{t_{i-1}}\Delta W_i = W_t^2 - \mathrm{QV}_n$, and $\mathrm{QV}_n \to t$ in mean square. In the second, the middle sum is

$$3\sum_i W_{t_{i-1}}\,(\Delta W_i)^2 = \mathrm{QV}_n^a + 3\sum_i W_{t_{i-1}}\,\Delta t,$$

where $\mathrm{QV}_n^a \to 0$ in mean square and the last sum is a Riemann sum of the continuous path, which converges to $3\int_0^t W_s\,ds$ on every path; and the sum of cubes tends to 0 in mean square. Each of these convergences holds in probability, and limits in probability add, so the first sums converge in probability to $W_t^2 - t$ and to $W_t^3 - 3\int_0^t W_s\,ds$. They also converge in mean square, hence in probability, to the Itô integrals $2\int_0^t W_s\,dW_s$ and $3\int_0^t W_s^2\,dW_s$. Two limits in probability of one sequence agree with probability one (Lemma 19.12(iii)), so

$$W_t^2 = t + 2\int_0^t W_s\,dW_s, \qquad W_t^3 = 3\int_0^t W_s\,ds + 3\int_0^t W_s^2\,dW_s.$$

In shorthand, $d(W^2) = dt + 2W\,dW$ and $d(W^3) = 3W\,dt + 3W^2\,dW$: Theorem 19.30 for $f = x^2$ and $f = x^3$, whose terms $\frac{1}{2}\,\partial_{xx}f\,dt$ are dt and $3W\,dt$. The third-order sum left nothing, which is why the expansion stops at order two.

(b) For $f = e^x$: $\partial_x f = \partial_{xx} f = e^x$, so $d(e^W) = \frac{1}{2}\,e^W\,dt + e^W\,dW$. For $f = tx$: $\partial_t f = x$, $\partial_x f = t$ and $\partial_{xx} f = 0$, so $d(tW) = W\,dt + t\,dW$. For $f = \sin x$: $\partial_x f = \cos x$ and $\partial_{xx} f = -\sin x$, so $d(\sin W) = -\frac{1}{2}\sin W\,dt + \cos W\,dW$.

(c) The drift is $(\partial_t f + \frac{1}{2}\,\partial_{xx} f)(t, W_t)$.

- For $f = x^2 - t$: $-1 + 1 = 0$. Zero drift.
- For $f = x^3$: $0 + 3x$. Drift $3W_t$.
- For $f = x^3 - 3tx$: $-3x + 3x = 0$. Zero drift.
- For $f = e^{\sigma x - \sigma^2 t/2}$: $-\frac{1}{2}\sigma^2 f + \frac{1}{2}\sigma^2 f = 0$. Zero drift.
- For $f = tx$: $x + 0$. Drift W_t.
- For $f = e^{t/2}\cos x$: $\frac{1}{2}f - \frac{1}{2}f = 0$. Zero drift.

Four of the six are driftless. The processes W_t^3 and tW_t have the constant mean 0 and still a drift: zero drift means that the dt coefficient vanishes on every path, not on average. In interview language the four driftless processes are the martingales of the list.

(d) For $f = x^4$: $\partial_x f = 4x^3$ and $\partial_{xx} f = 12x^2$, so $d(W^4) = 6W^2\,dt + 4W^3\,dW$. The function x^4 is in $C_E^{1,2}([0,t]\times\mathbb{R})$, because it and its derivatives are polynomials and $|x|^k \le k!\,e^{|x|}$, so the expectation identity applies. The stochastic integral has mean zero, and the expectation identity (19.15) gives

$$\mathbb{E}[W_t^4] = 6\int_0^t \mathbb{E}[W_s^2]\,ds = 6\int_0^t s\,ds = 3t^2.$$

The check: $W_t \sim \mathcal{N}(0,t)$ has fourth moment $3\times t^2$ (Proposition 7.25(iii) with variance t).

(e) For $f = e^{2x}$: $\partial_x f = 2e^{2x}$ and $\partial_{xx} f = 4e^{2x}$, so $d(e^{2W}) = 2e^{2W}\,dt + 2e^{2W}\,dW$. The function e^{2x} and its derivatives are at most $4e^{2|x|} \le 4e^{4|x|}$ in absolute value, so $e^{2x} \in C_E^{1,2}([0,t]\times\mathbb{R})$. Taking expectations as in (d), $\chi(t) = 1 + 2\int_0^t \chi(s)\,ds$. The function χ is continuous, by the argument of Example 19.32, so by the fundamental theorem of calculus it is differentiable, with $\chi' = 2\chi$ and $\chi(0) = 1$. Then $\big(\chi(t)\,e^{-2t}\big)' = 0$, and $\chi(t) = e^{2t}$. The check: W_t has the law of $\sqrt{t}\,Z$, so $\mathbb{E}\,e^{2W_t} = \mathbb{E}\,e^{\theta Z}$ with $\theta = 2\sqrt{t}$, which is $e^{\theta^2/2} = e^{2t}$ (Lemma 19.38(i)).

Self-check. The identities of (a) can be tested on a two-step path with the values $0, 1, -1$, whose increments are 1 and -2: $2\,(0\times 1 + 1\times(-2)) + (1+4) = 1 = (-1)^2$, and $3\,(0 + 1\times(-2)) + 3\,(0 + 1\times 4) + (1 - 8) = -6 + 12 - 7 = -1 = (-1)^3$. The three formulas can be tested at $n = 1$, where $\mathrm{QV}_1 = W_t^2$ has variance $3t^2 - t^2 = 2t^2$, the weighted sum is $3W_0\,(W_t^2 - t) = 0$, and the cubic sum is W_t^3, with second moment $\mathbb{E}[W_t^6] = 15t^3$. Taking expectations in the two limits gives $\mathbb{E}[W_t^2] = t$ and $\mathbb{E}[W_t^3] = 3\int_0^t \mathbb{E}[W_s]\,ds = 0$, as the law $\mathcal{N}(0,t)$ requires. For (c), a driftless process

has a constant mean, and the four means can be computed from the law of W_t: $\mathbb{E}[W_t^2 - t] = 0$, $\mathbb{E}[W_t^3 - 3tW_t] = 0$, $\mathbb{E}\, e^{\sigma W_t - \sigma^2 t/2} = 1$, and, from the even moments $\mathbb{E}[W_t^{2k}] = t^k\,(2k)!/(2^k k!)$ that the reduction (7.2) gives, $\mathbb{E}\cos W_t = \lim_m \mathbb{E}[Y_m]$ for the partial sums $Y_m = \sum_{k=0}^m (-1)^k W_t^{2k}/(2k)!$ (a name local to this check): they tend to $\cos W_t$ on every path, and $|Y_m| \le \cosh W_t \le e^{|W_t|}$, whose fourth power has a finite mean (Lemma 19.20(iii)), so Lemma 19.12(iv) applies; and $\mathbb{E}[Y_m] = \sum_{k=0}^m (-t/2)^k/k! \to e^{-t/2}$, so that $\mathbb{E}\big[e^{t/2}\cos W_t\big] = 1$ at every t. The constant-mean test confirms only the four driftless answers, because W_t^3 and tW_t also have constant mean 0. For (b) and (c), the drifts of $W_t^2 - t$ and of W_t^3 are the dt terms that (a) obtained from the sums alone; and tW_t telescopes exactly, $t_i W_{t_i} - t_{i-1} W_{t_{i-1}} = t_{i-1}\,\Delta W_i + W_{t_i}\,\Delta t$, with no quadratic sum, so its limits $\int_0^t s\,dW_s$ (Theorem 19.23(i)) and $\int_0^t W_s\,ds$ (on every path) give $d(tW) = W\,dt + t\,dW$ and the drift W_t.

Answer: (a) The identities follow by telescoping $(x+y)^2 - x^2$ and $(x+y)^3 - x^3$; the three mean squares are $2t^2/n$, $9t^3(n-1)/n^2$ and $15t^3/n^2$, that is, 0.02, 0.0891 and 0.0015 at $t = 1$, $n = 100$; in the limit $W_t^2 = t + 2\int_0^t W_s\,dW_s$ and $W_t^3 = 3\int_0^t W_s\,ds + 3\int_0^t W_s^2\,dW_s$. (b) $d(e^W) = \frac{1}{2}e^W\,dt + e^W\,dW$; $d(tW) = W\,dt + t\,dW$; $d(\sin W) = -\frac{1}{2}\sin W\,dt + \cos W\,dW$. (c) Zero drift: $W_t^2 - t$, $W_t^3 - 3tW_t$, $e^{\sigma W_t - \sigma^2 t/2}$ and $e^{t/2}\cos W_t$; W_t^3 has drift $3W_t$ and tW_t has drift W_t. (d) $d(W^4) = 6W^2\,dt + 4W^3\,dW$ and $\mathbb{E}[W_t^4] = 3t^2$. (e) $\chi' = 2\chi$, $\chi(0) = 1$, so $\mathbb{E}\, e^{2W_t} = e^{2t}$.

P19.7 *The Gaussian tilt from scratch*

(a) We complete the square in the exponent. For all real z, $\theta z - z^2/2 = \theta^2/2 - (z-\theta)^2/2$, that is,

$$e^{\theta z}\,\varphi(z) = e^{\theta^2/2}\,\varphi(z-\theta),$$

which is (E.13). Multiply by $\psi(z)$, integrate over the line, and substitute $z \mapsto z + \theta$:

$$\begin{aligned}\mathbb{E}\big[e^{\theta Z}\psi(Z)\big] &= \int \psi(z)\,e^{\theta z}\varphi(z)\,dz = e^{\theta^2/2}\int \psi(z)\,\varphi(z-\theta)\,dz \\ &= e^{\theta^2/2}\int \psi(z+\theta)\,\varphi(z)\,dz = e^{\theta^2/2}\,\mathbb{E}\big[\psi(Z+\theta)\big].\end{aligned}$$

The integrals converge absolutely. Indeed $|\psi(z)|\,e^{\theta z} \le c'\,e^{(c'+|\theta|)|z|}$ and $|\psi(z+\theta)| \le c'\,e^{c'|\theta|}\,e^{c'|z|}$, and for every constant $c'' \ge 0$, $e^{c''|z|} \le e^{c''z} + e^{-c''z}$, whose integral against φ is $2e^{c''^2/2}$ by the square just completed with $\pm c''$ in place of θ. Taking $\psi = 1$ gives $\mathbb{E}\, e^{\theta Z} = e^{\theta^2/2}$. Taking $\psi(z) = \mathbf{1}\{z > c\}$ gives

$$\mathbb{E}\big[e^{\theta Z}\,\mathbf{1}\{Z > c\}\big] = e^{\theta^2/2}\,P(Z + \theta > c) = e^{\theta^2/2}\,P(Z > c - \theta) = e^{\theta^2/2}\,\Phi(\theta - c),$$

by the symmetry of the normal law. Weighting by $e^{\theta Z}$ shifts the variable by θ, at the price of the factor $e^{\theta^2/2}$ (Lemma 19.38(i)).

(b) We rotate the pair so that Y is carried by one variable, freeze the other, and apply (a).

Moments. Since Z_1 and Z_2 are independent with mean 0 and variance 1, $\mathbb{E}[Z_1 Z_2] = \mathbb{E}[Z_1]\,\mathbb{E}[Z_2] = 0$, the constant terms are the means, as the notation says, and expanding the products gives $\mathrm{Var}(Y) = a_{Y,1}^2 + a_{Y,2}^2$ and $\mathrm{Cov}(X,Y) = a_{X,1}a_{Y,1} + a_{X,2}a_{Y,2}$.

The rotation. Put $\sigma_Y = \sqrt{a_{Y,1}^2 + a_{Y,2}^2} > 0$ and $(c_1, c_2) = (a_{Y,1}, a_{Y,2})/\sigma_Y$, a unit vector, and let Z_1' and Z_2' be the rotated variables of the statement, which are independent standard normal. Multiplying out, $Z_1 = c_1 Z_1' - c_2 Z_2'$ and $Z_2 = c_2 Z_1' + c_1 Z_2'$. Substituting,

$$Y = \mathbb{E}[Y] + \sigma_Y\,Z_1', \qquad X = \mathbb{E}[X] + c_{XY}\,\sigma_Y\,Z_1' + \sigma_\perp\,Z_2',$$

with $c_{XY} = \mathrm{Cov}(X,Y)/\mathrm{Var}(Y)$ and $\sigma_\perp = a_{X,2}c_1 - a_{X,1}c_2$: the coefficient of Z_1' in Y is $a_{Y,1}c_1 + a_{Y,2}c_2 = \sigma_Y$ and that of Z_2' is $-a_{Y,1}c_2 + a_{Y,2}c_1 = 0$, and the coefficient of Z_1' in X is $a_{X,1}c_1 + a_{X,2}c_2 = \mathrm{Cov}(X,Y)/\sigma_Y$. This is the decomposition (E.15).

Freezing and shifting. Freeze Z_2' at a value z_2'. The function $z \mapsto \psi\big(\mathbb{E}[X] + c_{XY}\sigma_Y z + \sigma_\perp z_2'\big)$ is continuous except at finitely many points and of exponential growth: its absolute value is at most

$c' e^{c'|\mathbb{E}[X]+\sigma_\perp z_2'|}\, e^{c'|c_{XY}|\sigma_Y|z|}$, hence at most $c'' e^{c''|z|}$ with c'' the larger of the constants $c' e^{c'|\mathbb{E}[X]+\sigma_\perp z_2'|}$ and $c'|c_{XY}|\sigma_Y$ (it is a constant when $c_{XY}=0$). So (a) with $\theta=\sigma_Y$ gives

$$\begin{aligned}\mathbb{E}\Big[e^{\sigma_Y Z_1'}\,\psi\big(\mathbb{E}[X]+c_{XY}\sigma_Y Z_1'+\sigma_\perp z_2'\big)\Big]\\ = e^{\sigma_Y^2/2}\,\mathbb{E}\Big[\psi\big(\mathbb{E}[X]+c_{XY}\sigma_Y\,(Z_1'+\sigma_Y)+\sigma_\perp z_2'\big)\Big].\end{aligned}$$

The variable $e^{\sigma_Y Z_1'}\,\psi(X)$ has a finite absolute mean. Its absolute value is at most

$$c' e^{c'|\mathbb{E}[X]|}\; e^{(1+c'|c_{XY}|)\,\sigma_Y|Z_1'|}\; e^{c'|\sigma_\perp|\,|Z_2'|},$$

a product of a function of Z_1' and a function of Z_2', each with a finite mean by the bound of (a), and the freezing lemma (Lemma 19.17), applied to this nonnegative product, makes its mean the product of the two means. So the mean of $e^{\sigma_Y Z_1'}\,\psi(X)$ may be computed by averaging over Z_1' first, with Z_2' frozen (Lemma 19.17). Averaging both sides of the display over the independent variable Z_2' (on the left this is the freezing lemma, as just justified; on the right it is the same lemma used once more, for the function $(z_1',z_2')\mapsto\psi\big(\mathbb{E}[X]+c_{XY}\sigma_Y(z_1'+\sigma_Y)+\sigma_\perp z_2'\big)$, whose absolute value is bounded in the same way, so that its mean is its iterated mean) and multiplying by $e^{\mathbb{E}[Y]}$,

$$\begin{aligned}\mathbb{E}\big[e^Y\psi(X)\big] &= e^{\mathbb{E}[Y]+\sigma_Y^2/2}\,\mathbb{E}\Big[\psi\big(\mathbb{E}[X]+c_{XY}\sigma_Y Z_1'+\sigma_\perp Z_2'+c_{XY}\sigma_Y^2\big)\Big]\\ &= e^{\mathbb{E}[Y]+\sigma_Y^2/2}\,\mathbb{E}\big[\psi\big(X+\mathrm{Cov}(X,Y)\big)\big],\end{aligned}$$

because $c_{XY}\,\sigma_Y^2=\mathrm{Cov}(X,Y)$. With $\psi=1$ this reads $\mathbb{E}\,e^Y=e^{\mathbb{E}[Y]+\sigma_Y^2/2}=e^{\mathbb{E}[Y]+\mathrm{Var}(Y)/2}$, and the display is then the second identity (Lemma 19.38(iii)).

(c) Write $g=\mu-\sigma^2/2$, so that $S_T^a=S_0^a\,e^{agT}\,e^{a\sigma\sqrt{T}\,Z}$. By (a) with $\theta=a\sigma\sqrt{T}$,

$$\mathbb{E}[S_T^a]=S_0^a\exp\big(agT+\tfrac12\,a^2\sigma^2T\big)=S_0^a\exp\big(a\mu T+\tfrac12\,a(a-1)\,\sigma^2T\big),$$

because $ag+\frac12a^2\sigma^2=a\mu-\frac12a\sigma^2+\frac12a^2\sigma^2$. The reconciliation: Itô's lemma gives $d(S^a)=\big(a\mu+\frac12a(a-1)\sigma^2\big)\,S^a\,dt+a\sigma S^a\,dW$ (Example 19.33). The drift is the expected rate of change, by the expectation identity (19.15) of Corollary 19.31(ii). Its hypotheses hold by part (iii) of that corollary, because $V(x,t)=x^a$, $\partial_tV=0$, ∂_xV and $\partial_{xx}V$ are continuous and of moderate growth on $(0,\infty)\times[0,T]$ (Example 19.33), and with $\alpha_s=\mu S_s$ and $\beta_s=\sigma S_s$ its integrand is $\big(a\mu+\frac12a(a-1)\sigma^2\big)\,S_s^a$. So the function $\chi(t)=\mathbb{E}[S_t^a]$, $0\le t\le T$, satisfies

$$\chi(t)=S_0^a+\big(a\mu+\tfrac12\,a(a-1)\,\sigma^2\big)\int_0^t\chi(s)\,ds,$$

and the exponential just found, with t in the place of T, satisfies this equation: it equals S_0^a at $t=0$ and its derivative is $\big(a\mu+\frac12a(a-1)\sigma^2\big)$ times itself. The mean of S^a grows at the drift of S^a.

(d) The logarithm is increasing and $\sigma\sqrt{T}>0$, so $\{S_T>K\}=\{\ln S_0+gT+\sigma\sqrt{T}\,Z>\ln K\}=\{Z>-d\}$, of probability $\Phi(d)$. By (a) with $\theta=\sigma\sqrt{T}$ and $c=-d$,

$$\begin{aligned}\mathbb{E}\big[S_T\,\mathbf{1}\{S_T>K\}\big] &= S_0\,e^{gT}\,\mathbb{E}\big[e^{\sigma\sqrt{T}\,Z}\,\mathbf{1}\{Z>-d\}\big]\\ &= S_0\,e^{gT+\sigma^2T/2}\,\Phi\big(\sigma\sqrt{T}+d\big)=S_0\,e^{\mu T}\,\Phi\big(d+\sigma\sqrt{T}\big).\end{aligned}$$

Numbers. Here $g=0.08-0.03125=0.04875$ and $\ln(100/110)=-0.09531$, so

$$d=\frac{-0.09531+0.04875}{0.25}=-0.18624,\qquad d+\sigma\sqrt{T}=0.06376,$$

and $S_0\,e^{\mu T}=100\,e^{0.08}=108.329$. Hence $P(S_T>110)=0.42613$, $\mathbb{E}[S_T\,\mathbf{1}\{S_T>110\}]=108.329\times0.52542=56.918$, and

$$\begin{aligned}\mathbb{E}\big[(S_T-K)^+\big] &= \mathbb{E}\big[S_T\,\mathbf{1}\{S_T>K\}\big]-K\,P(S_T>K)\\ &= 56.918-110\times0.42613=56.918-46.874=10.044.\end{aligned}$$

This is the real-world expected payoff, and in general discounting it does not give the price. The price of the call is the cost of the strategy that replicates it, and that cost is a discounted expectation in which μ is replaced by the risk-free rate (§20). The two agree only when μ happens to equal that rate. The number 10.04 moves with the estimate of μ, and the price does not.

Self-check. For (a), a different route to $\mathbb{E}\,e^{\theta Z}$ is the power series: the even moments are $\mathbb{E}[Z^{2k}] = (2k)!/(2^k k!)$ by the reduction (7.2) and the odd ones vanish, so $\mathbb{E}\,e^{\theta Z} = \sum_{k\ge 0}\theta^{2k}\,\mathbb{E}[Z^{2k}]/(2k)! = \sum_{k\ge 0}(\theta^2/2)^k/k! = e^{\theta^2/2}$ (the sum and the mean may be exchanged by Fact 18.34(i), with a sum in place of one integral, because the series of absolute values has the finite mean $\mathbb{E}\,e^{|\theta|\,|Z|} \le e^{\theta^2}\,\mathbb{E}\,e^{Z^2/4} = \sqrt{2}\,e^{\theta^2}$, from $|\theta|\,|z| \le \theta^2 + z^2/4$). The indicator formula reduces to $P(Z > c) = \Phi(-c)$ at $\theta = 0$ and to the exponential moment as $c \to -\infty$. For (b), take $X = Z_1$, $Y = Z_1 + Z_2$ and $\psi(z) = \mathbf{1}\{z > 0\}$, so that $\mathbb{E}[Y] = 0$, $\mathrm{Var}(Y) = 2$ and $\mathrm{Cov}(X, Y) = 1$: (b) gives $e \times P(Z_1 + 1 > 0) = e\,\Phi(1) = 2.7183 \times 0.8413 = 2.287$, while the independence of the factors e^{Z_2} and $e^{Z_1}\mathbf{1}\{Z_1 > 0\}$ and (a) alone give $e^{1/2} \times e^{1/2}\,\Phi(1) = 2.287$, without the rotation. For (c), $a = 1$ gives $S_0 e^{\mu T}$ and $a = 2$ gives $S_0^2 e^{2\mu T + \sigma^2 T}$, the two moments behind $\mathrm{Var}(S_T) = S_0^2 e^{2\mu T}(e^{\sigma^2 T} - 1)$. For (d), a numerical quadrature of $S_T\,\mathbf{1}\{S_T > K\}$ and of $(S_T - K)^+$ against the density φ, with no closed form used, returns 56.918 and 10.044. The expected payoff of the mirror claim is $\mathbb{E}[(K - S_T)^+] = K\,\Phi(-d) - S_0 e^{\mu T}\,\Phi(-d - \sigma\sqrt{T}) = 110 \times 0.57387 - 108.329 \times 0.47458 = 63.126 - 51.411 = 11.715$, and $10.044 - 11.715 = -1.671 = 108.329 - 110 = \mathbb{E}[S_T] - K$, as $(x - K)^+ - (K - x)^+ = x - K$ requires. The bounds $K\,P(S_T > K) = 46.87 \le 56.92 \le \mathbb{E}[S_T] = 108.33$ hold as well.

Answer: (a) $e^{\theta z}\varphi(z) = e^{\theta^2/2}\varphi(z - \theta)$ gives $\mathbb{E}[e^{\theta Z}\psi(Z)] = e^{\theta^2/2}\,\mathbb{E}[\psi(Z + \theta)]$, hence $\mathbb{E}\,e^{\theta Z} = e^{\theta^2/2}$ and $\mathbb{E}[e^{\theta Z}\mathbf{1}\{Z > c\}] = e^{\theta^2/2}\Phi(\theta - c)$. (b) After the rotation, $Y = \mathbb{E}[Y] + \sigma_Y Z_1'$ and $X = \mathbb{E}[X] + c_{XY}\sigma_Y Z_1' + \sigma_\perp Z_2'$; freezing Z_2' and shifting Z_1' by σ_Y gives $\mathbb{E}[e^Y\psi(X)] = e^{\mathbb{E}[Y] + \mathrm{Var}(Y)/2}\,\mathbb{E}[\psi(X + \mathrm{Cov}(X, Y))]$. (c) $\mathbb{E}[S_T^a] = S_0^a \exp\left(a\mu T + \frac{1}{2}a(a-1)\sigma^2 T\right)$, which grows at the rate $a\mu + \frac{1}{2}a(a-1)\sigma^2$, the drift of S^a. (d) $d = -0.1862$, $P(S_T > K) = 0.4261$, $\mathbb{E}[S_T\mathbf{1}\{S_T > K\}] = 56.92$ and $\mathbb{E}[(S_T - K)^+] = 10.04$: a real-world expected payoff, which, discounted, is the price of the call only if μ equals the risk-free rate.

P19.8 *Mean, median, mode and range of a future price*

(a) Here $\sigma^2 = 0.16$, so $g = 0.10 - 0.08 = 2\%$, $gT = 0.005$, $\mu T = 0.025$, $\sigma^2 T = 0.04$ and $\sigma\sqrt{T} = 0.20$. By Theorem 19.40,

$$\ln S_T \sim \mathcal{N}\big(\ln 80 + 0.005,\ 0.04\big) = \mathcal{N}\big(4.3870,\ 0.20^2\big).$$

By Proposition 19.42:

- the mean is $S_0\,e^{\mu T} = 80\,e^{0.025} = 82.03$;
- the median is $S_0\,e^{gT} = 80\,e^{0.005} = 80.40$;
- the mode is $S_0\,e^{gT - \sigma^2 T} = 80\,e^{-0.035} = 77.25$;
- the standard deviation is $82.03 \times \sqrt{e^{0.04} - 1} = 82.03 \times 0.2020 = 16.57$;
- since $1.96 \times 0.20 = 0.392$, the central 95% interval is $\big[80\,e^{0.005 - 0.392},\ 80\,e^{0.005 + 0.392}\big] = \big[80\,e^{-0.387},\ 80\,e^{0.397}\big] = [54.33,\ 118.99]$.

The three centres are ordered, mode < median < mean: $77.25 < 80.40 < 82.03$.

(b) Now $\mu T = 0.000397$, $gT = 0.000079$, $\sigma^2 T = 0.000635$ and $\sigma\sqrt{T} = 0.40/15.87 = 0.02520$. The mean is $80\,e^{0.000397} = 80.03$; the standard deviation is $80.03 \times \sqrt{e^{0.000635} - 1} = 80.03 \times 0.02520 = 2.02$; and, with $1.96 \times 0.02520 = 0.04939$, the interval is

$$\big[80\,e^{0.000079 - 0.04939},\ 80\,e^{0.000079 + 0.04939}\big] = \big[80\,e^{-0.04931},\ 80\,e^{0.04947}\big] = [76.15,\ 84.06].$$

(c) We read the quantiles in log space. The logarithm is increasing, so for a price $x > 0$

$$P(S_T \le x) = P\big(\ln S_0 + gT + \sigma\sqrt{T}\,Z \le \ln x\big) = \Phi\Big(\frac{\ln x - \ln S_0 - gT}{\sigma\sqrt{T}}\Big).$$

The function Φ is continuous and strictly increasing, with the limits 0 and 1 at $-\infty$ and $+\infty$, so it takes each value in $(0,1)$ exactly once. It takes the value 0.975 at $\Phi^{-1}(0.975)$, which the statement rounds to 1.96, and by the symmetry of the normal law $\Phi(-\Phi^{-1}(0.975)) = 1 - 0.975 = 0.025$. So the probability equals 2.5% exactly when $\ln x = \ln S_0 + gT - \Phi^{-1}(0.975)\,\sigma\sqrt{T}$ and 97.5% exactly when $\ln x = \ln S_0 + gT + \Phi^{-1}(0.975)\,\sigma\sqrt{T}$. Exponentiating, and writing 1.96 for the quantile, gives the two ends: this is the interval of Proposition 19.42(iv). Quantiles pass through the increasing exponential, but means do not. The interval is symmetric in log space about $\ln S_0 + gT$, the logarithm of the median: its upper end is the median times $e^{1.96\sigma\sqrt{T}}$ and its lower end the median divided by the same factor (1.480 in (a)), which puts more dollars above the median than below it, 38.59 against 26.07 in (a). The mean sits above the median, by the factor $e^{\sigma^2 T/2}$, because the right tail pulls it up. About the mean the interval reaches $118.99 - 82.03 = 36.96$ above and $82.03 - 54.33 = 27.70$ below.

The symmetric range is $82.025 \pm 1.96 \times 16.570 = [49.55,\ 114.50]$ in (a), against the true $[54.33,\ 118.99]$: the lower end is 4.78 and the upper end 4.49 dollars too low. In (b) it is $80.032 \pm 1.96 \times 2.017 = [76.079,\ 83.985]$, against $[76.151,\ 84.057]$: seven cents too low at each end. Over one day the lognormal law is almost normal, because $\sigma\sqrt{T}$ is small; over a quarter at a 40% volatility it is visibly skewed.

Self-check. Three ratios test the numbers of (a) without the formulas that produced them. The two ends of the interval have the median as their geometric mean: $\sqrt{54.33 \times 118.99} = 80.40$. The mean over the median is $e^{\sigma^2 T/2} = e^{0.02} = 1.0202$, and $82.025/80.401 = 1.0202$. The median over the mode is $e^{\sigma^2 T} = e^{0.04} = 1.0408$, and $80.401/77.248 = 1.0408$. The ratio of the two ends tests the width: $118.99/54.33 = 2.190 = e^{2\times 0.392}$. The second moment gives the standard deviation by another route: $\mathbb{E}[S_T^2] = 80^2\, e^{2\times 0.025 + 0.04} = 6400\, e^{0.09} = 7002.715$, so the standard deviation is $\sqrt{7002.715 - 82.025^2} = \sqrt{274.61} = 16.57$. For (b), the discretisation $\Delta S/S = \mu\,\Delta t + \sigma\sqrt{\Delta t}\,Z$ of §19.4, with $\Delta t = 1/252$, gives the standard deviation $S_0\,\sigma\sqrt{\Delta t} = 80 \times 0.02520 = 2.02$ and the mean $80\,(1 + 0.10/252) = 80.03$.

Answer: (a) $\ln S_T \sim \mathcal{N}(4.3870,\ 0.04)$; mean 82.03, median 80.40, mode 77.25, standard deviation 16.57; interval $[54.33,\ 118.99]$. (b) Mean 80.03, standard deviation 2.02, interval $[76.15,\ 84.06]$. (c) $P(S_T \le x) = \Phi\big((\ln x - \ln S_0 - gT)/(\sigma\sqrt{T})\big)$ equals 2.5% and 97.5% at $\ln x = \ln S_0 + gT \mp 1.96\,\sigma\sqrt{T}$; the interval is symmetric in log space about the median, not about the mean; the symmetric range is $[49.55,\ 114.50]$ in (a), 4.78 and 4.49 dollars too low at its two ends, and $[76.079,\ 83.985]$ in (b), seven cents too low at each end.

P19.9 *Real-world exercise probabilities*

By Proposition 19.42(iii), with $g = \mu - \sigma^2/2 = 0.14 - 0.03125 = 10.875\%$,

$$P(S_T > K) = \Phi(d), \qquad P(S_T < K) = \Phi(-d), \qquad d = \frac{\ln(S_0/K) + gT}{\sigma\sqrt{T}}.$$

(a) Here $\ln(80/90) = -0.11778$, $gT = 0.10875 \times 0.75 = 0.08156$ and $\sigma\sqrt{T} = 0.25 \times 0.86603 = 0.21651$, so

$$d = \frac{-0.11778 + 0.08156}{0.21651} = \frac{-0.03622}{0.21651} = -0.1673.$$

The call is exercised with probability $\Phi(-0.1673) = 43.4\%$ and the put with probability $1 - 0.4336 = 56.6\%$.

(b) With $K = 2S_0$ and $T = 3$: $\ln(1/2) = -0.69315$, $gT = 0.32625$ and $\sigma\sqrt{T} = 0.43301$, so $d = (-0.69315 + 0.32625)/0.43301 = -0.8473$ and the probability is $\Phi(-0.8473) = 19.8\%$.

(c) With $K = S_0$ and $T = 1$: $d = g/\sigma = 0.10875/0.25 = 0.435$, and the probability of a loss is $P(S_1 < S_0) = \Phi(-0.435) = 33.2\%$. One year in three ends below the starting price, although the expected return is 14%.

(d) With r in the place of μ, the drift of the log price is $r - \sigma^2/2 = 0.04 - 0.03125 = 0.875\%$, and

$$d_2 = \frac{-0.11778 + 0.00875 \times 0.75}{0.21651} = \frac{-0.11778 + 0.00656}{0.21651} = -0.5137, \qquad \Phi(d_2) = 30.4\%.$$

It is lower than the 43.4% of (a) because it is computed in a world in which the stock earns 4% instead of 14%: the stock carries a risk premium of $\mu - r = 10\%$, and without it the price is less likely to finish above 90. If μ is re-estimated at 20%, then $gT = 0.16875 \times 0.75 = 0.12656$, $d = (-0.11778 + 0.12656)/0.21651 = 0.0406$, and the real-world probability rises from 43.4% to $\Phi(0.0406) = 51.6\%$. The risk-neutral probability does not contain μ and stays at 30.4%; the price of the call, a discounted risk-neutral expectation (§20), does not change either. The risk-neutral probability can be read off market prices: it is a function of S_0, K, r, T and the volatility that option prices imply. The real-world probability needs an estimate of μ, which decades of data do not pin down.

Self-check. The two normal arguments of (a) and (d) differ only through the drift, so $d - d_2 = (\mu - r)\sqrt{T}/\sigma = 0.10 \times 0.8660/0.25 = 0.3464$, and $-0.1673 - 0.3464 = -0.5137$. The tangent rule $\Phi(x) \approx 0.5 + 0.4x$ (the tangent of Φ at 0, of slope $\varphi(0) = 1/\sqrt{2\pi} = 0.399$) gives 0.433 for (a), against 0.4336. The medians give the side of one half on which each answer must fall: the median of $S_{0.75}$ is $80\,e^{0.08156} = 86.80 < 90$, so the call of (a) is exercised with probability below one half; the median doubles only after $\ln 2/g = 6.4$ years, more than three, so (b) is below one half; and $g > 0$ puts the median of S_1 above S_0, so (c) is below one half. With $\mu = 20\%$ the median is $80\,e^{0.12656} = 90.79 > 90$, and the probability is indeed above one half.

Answer: (a) $d = -0.1673$: the call is exercised with probability 43.4%, the put with probability 56.6%. (b) $\Phi(-0.8473) = 19.8\%$. (c) $\Phi(-g/\sigma) = \Phi(-0.435) = 33.2\%$. (d) $d_2 = -0.5137$ and $\Phi(d_2) = 30.4\%$, lower because the risk premium $\mu - r = 10\%$ is removed; at $\mu = 20\%$ the real-world probability rises to 51.6% while $\Phi(d_2)$ and the price of the call are unchanged; only the risk-neutral probability can be read off market prices.

P19.10 *Average versus compounded returns, and the leveraged product*

(a) The arithmetic mean is $(18 - 12 + 27 - 6 + 8)/5 = 35/5 = 7.0\%$. One dollar grows to $1.18 \times 0.88 \times 1.27 \times 0.94 \times 1.08 = 1.3388$, so the geometric mean return is $1.3388^{1/5} - 1 = 6.01\%$. The deviations from the mean are 11, −19, 20, −13 and 1 points, with squares adding up to 1052, so the variance is $1052/5 = 210.4$ squared points, that is, 0.02104, a standard deviation of 14.5%. The estimate is $7.0\% - \frac{1}{2} \times 0.02104 = 7.0\% - 1.05\% = 5.95\%$, against the exact 6.01%. The geometric mean lies below the arithmetic mean, as Proposition 19.46(i) says it must.

(b) By the same second-order rule the investors compounded at about $18\% - \frac{1}{2} \times 0.30^2 = 18\% - 4.5\% = 13.5\%$ a year. The rule is rough at a 30% volatility: if the annual gross return is lognormal with mean 1.18 and standard deviation 0.30, then by Proposition 19.42(i), with $T = 1$, the squared ratio of the standard deviation to the mean, $(0.30/1.18)^2 = 0.0646$, is e raised to the variance of the log return, minus one; that variance is therefore $\ln 1.0646 = 0.0626$, and the compound rate, the median gross return minus one, is $1.18\,e^{-0.0626/2} - 1 = 1.18\,e^{-0.0313} - 1 = 14.4\%$ as a simple annual rate. As a continuously compounded rate it is $\ln 1.18 - 0.0313 = 0.1655 - 0.0313 = 13.4\%$, so the rule's 13.5% lies between the two, as in Example 19.48. Either way the investors earned about four points a year less than the reported figure. The arithmetic average estimates the expected return of a single year, not the rate at which a dollar grew: a fund that makes +100% and then −50% averages +25% a year and has earned nothing.

(c) The growth rate is $g = \mu - \sigma^2/2 = 0.07 - 0.0162 = 5.38\%$, and by Proposition 19.44

$$\hat{g}_5 \sim \mathcal{N}(g,\ \sigma^2/5) = \mathcal{N}(5.38\%,\ (8.05\%)^2), \qquad \sigma/\sqrt{5} = 0.18/2.236 = 8.05\%.$$

The 95% interval is $5.38\% \pm 1.96 \times 8.05\% = 5.38\% \pm 15.78\% = [-10.4\%,\ 21.2\%]$, and $P(\hat{g}_5 < 0) = \Phi(-5.38/8.05) = \Phi(-0.668) = 25.2\%$: one five-year period in four shows a loss. For a half-width of 1% the horizon must satisfy $1.96\,\sigma/\sqrt{T} = 0.01$, that is, $T = (1.96 \times 0.18/0.01)^2 = 35.28^2 = 1{,}245$ years.

(d) The process $V_t = V_0 \exp\left((L\mu - \frac{1}{2}L^2\sigma^2)\,t + L\sigma W_t\right)$ is a geometric Brownian motion with drift $L\mu$ and volatility $|L|\sigma$, so it satisfies $dV = L\mu V\,dt + L\sigma V\,dW = LV\,dS/S$ (Corollary 19.49). Its logarithm has the mean $(L\mu - \frac{1}{2}L^2\sigma^2)\,T$, so

$$g_L = L\mu - \tfrac{1}{2}\,L^2\sigma^2 = 0.07\,L - 0.0162\,L^2.$$

Hence $g_1 = 5.38\%$, $g_2 = 14\% - 6.48\% = 7.52\%$, $g_3 = 21\% - 14.58\% = 6.42\%$, $g_{-1} = -7\% - 1.62\% = -8.62\%$ and $g_{-2} = -14\% - 6.48\% = -20.48\%$. The derivative $\mu - L\sigma^2$ changes sign from positive to negative at $L^* = \mu/\sigma^2 = 0.07/0.0324 = 2.16$, where $g_{L^*} = \mu^2/(2\sigma^2) = 0.0049/0.0648 = 7.56\%$. The growth is zero at $L = 0$ and at $L = 2\mu/\sigma^2 = 4.32$: beyond 4.32 times leverage the product has a positive expected return and a negative growth rate. The $3\times$ product grows more slowly than the $2\times$ product, and the inverse products shrink faster than the index grows.

(e) A product with leverage L returns $+10L\%$ and then $-10L\%$, so one dollar becomes $(1 + 0.1L)(1 - 0.1L) = 1 - 0.01\,L^2$: 0.99 in the index, 0.96 in the $2\times$ product, 0.99 in the $-1\times$ product and 0.91 in the $3\times$ product. The index is down 1%; the $2\times$ product is down 4%, not 2%; the $3\times$ product is down 9%, not 3%; and the inverse product is down 1%, not up 1%. The loss grows like L^2, the discrete form of the term $-\frac{1}{2}L^2\sigma^2$.

(f) From (d), $\ln(V_T/V_0) = (L\mu - \frac{1}{2}L^2\sigma^2)\,T + L\sigma W_T$ and $\ln(S_T/S_0) = (\mu - \frac{1}{2}\sigma^2)\,T + \sigma W_T$, so

$$\ln\frac{V_T}{V_0} - L\ln\frac{S_T}{S_0} = -\tfrac{1}{2}\,L^2\sigma^2T + \tfrac{1}{2}\,L\sigma^2T = -\tfrac{1}{2}\,L(L-1)\,\sigma^2T, \qquad \frac{V_T}{V_0} = \Big(\frac{S_T}{S_0}\Big)^L\,e^{-L(L-1)\sigma^2T/2}.$$

On a flat year, $S_T = S_0$ and $T = 1$, the product is worth $e^{-L(L-1)\sigma^2/2}$ per dollar. For $L = 2$ and for $L = -1$, $L(L-1) = 2$ and the loss is $1 - e^{-0.0324} = 3.2\%$. For $L = 3$ and for $L = -2$, $L(L-1) = 6$ and the loss is $1 - e^{-0.0972} = 9.3\%$. The decay depends on L only through $L(L-1)$, which takes the same value at L and at $1 - L$.

Self-check. For (a), the logarithms of the five gross returns are 0.1655, −0.1278, 0.2390, −0.0619 and 0.0770; their mean is $0.2918/5 = 0.0584$, and $e^{0.0584} - 1 = 6.01\%$. For (c), the horizon scales like the square of the ratio of the volatility to the half-width: Example 19.45 needs $19.6^2 = 384.16$ years for a 20% volatility and a half-width of 2%, so here $384.16 \times (0.18/0.20)^2 \times (0.02/0.01)^2 = 384.16 \times 0.81 \times 4 = 1{,}244.7$ years. Also $P(\hat{g}_5 < 0) = P(S_5 < S_0) = \Phi(-d)$ with $d = gT/(\sigma\sqrt{T}) = 0.269/0.4025 = 0.668$, by Proposition 19.42(iii). For (d), the factored form $g_L = \frac{1}{2}\sigma^2\,L\,(2\mu/\sigma^2 - L) = 0.0162\,L\,(4.32 - L)$ is a parabola with zeros 0 and 4.32, so its maximum is at their midpoint 2.16, where it equals $0.0162 \times 2.16^2 = 7.56\%$; and it returns $0.0162 \times 2 \times 2.32 = 7.52\%$, $0.0162 \times 3 \times 1.32 = 6.42\%$ and $0.0162 \times (-1) \times 5.32 = -8.62\%$. For (e) and (f) together, the two days realise the variance $0.1^2 + 0.1^2 = 0.02$, and the formula of (f) with $\sigma^2T = 0.02$ and $S_T/S_0 = 0.99$ gives $0.99^2\,e^{-0.02} = 0.961$, $0.99^{-1}\,e^{-0.02} = 0.990$ and $0.99^3\,e^{-0.06} = 0.914$, against the exact 0.96, 0.99 and 0.91 of daily rebalancing.

Answer: (a) Arithmetic mean 7.0%, geometric mean 6.01%, second-order estimate $7.0\% - 1.05\% = 5.95\%$. (b) About 13.5% by the second-order rule (14.4% if the annual gross return is lognormal): the average of annual returns overstates the compound rate by about half the variance. (c) $\hat{g}_5 \sim \mathcal{N}(5.38\%, (8.05\%)^2)$, interval $[-10.4\%,\ 21.2\%]$, $P(\hat{g}_5 < 0) = 25.2\%$; about 1,245 years. (d) $g_L = L\mu - \frac{1}{2}L^2\sigma^2$: 5.38%, 7.52%, 6.42%, −8.62% and −20.48%; $L^* = \mu/\sigma^2 = 2.16$ with $g_{L^*} = 7.56\%$; zero growth at $L = 4.32$ (besides $L = 0$). (e) 0.99, 0.96, 0.99 and 0.91. (f) 3.2% for the $2\times$ and the $-1\times$ products, 9.3% for the $3\times$ and the $-2\times$ products: the decay factor is $e^{-L(L-1)\sigma^2T/2}$.

P19.11 *Correlated Brownian motions: construction, ratios and the FX triangle*

(a) The weight is $w = \sqrt{1 - \rho^2} = \sqrt{0.84} = 0.9165$. We check the defining properties of a standard Brownian motion. The process starts at $W_{2,0} = 0$, and its paths are continuous as combinations of continuous paths. For $0 \le s < t$ put $Z_1' = (W_{1,t} - W_{1,s})/\sqrt{t-s}$ and $Z_\perp' = (W_{\perp,t} - W_{\perp,s})/\sqrt{t-s}$ (primed, so that Z_1 and $Z_\perp$ keep their values 1.2 and −0.7), independent standard normal variables

as functions of the two independent motions. Then

$$W_{2,t} - W_{2,s} = \sqrt{t-s}\,(\rho\, Z_1' + \sqrt{1-\rho^2}\, Z_\perp'),$$

and a combination of two independent standard normal variables with coefficients whose squares add up to one is standard normal (Lemma 19.38(ii)), so the increment has the law $\mathcal{N}(0, t-s)$. For a weight w, Lemma 19.38(ii) gives the increment the variance $(\rho^2 + w^2)(t-s)$, which equals $t-s$ only for $w^2 = 1-\rho^2$, so $w = \sqrt{1-\rho^2}$ is the only positive weight. Over the n cells of a grid, the increments of W_2 are functions of the pairs of increments of $(W_1, W_\perp)$ over the same cells. The increments of W_1 over the cells are mutually independent, so are those of $W_\perp$, and the two vectors are independent because the motions are; hence the $2n$ increments are mutually independent (as in the proof of Proposition 19.52), and the n pairs, being disjoint groups of them, are mutually independent by the grouping property (a) of the Setup of §19.2. So the increments of W_2 are mutually independent. For the covariance, both values have mean zero and

$$\mathbb{E}[W_{1,t}\, W_{2,t}] = \rho\, \mathbb{E}[W_{1,t}^2] + \sqrt{1-\rho^2}\, \mathbb{E}[W_{1,t}]\, \mathbb{E}[W_{\perp,t}] = \rho\, t = -0.4\, t.$$

This is Proposition 19.52(i). The draw is $Z_2 = \rho\, Z_1 + \sqrt{1-\rho^2}\, Z_\perp = -0.4 \times 1.2 + 0.9165 \times (-0.7) = -0.48 - 0.6416 = -1.1216$: the pair is $(1.2,\ -1.12)$.

(b) The sum is $X_{1,t} + X_{2,t} = (X_{1,0} + X_{2,0}) + (\alpha_1 + \alpha_2)\, t + \beta_1 W_{1,t} + \beta_2 W_{2,t}$. The process $\beta_1 W_1 + \beta_2 W_2$ starts at zero and has continuous paths. Its increment over $[s,t]$ is a combination of a jointly normal pair, hence normal (Lemma 19.38(ii)), with mean zero and, by Proposition 10.10, variance $(\beta_1^2 + \beta_2^2 + 2\rho\,\beta_1\beta_2)(t-s)$; and its increments over the cells of a grid are mutually independent, as functions of the mutually independent pairs $(\Delta W_{1,i}, \Delta W_{2,i})$ (the definition of correlation). If the constant $\beta_1^2 + \beta_2^2 + 2\rho\,\beta_1\beta_2$ is positive, dividing by its square root gives a standard Brownian motion $\tilde{W}$, and

$$X_{1,t} + X_{2,t} = (X_{1,0} + X_{2,0}) + (\alpha_1 + \alpha_2)\, t + \sqrt{\beta_1^2 + \beta_2^2 + 2\rho\,\beta_1\beta_2}\, \tilde{W}_t :$$

a generalised Wiener process with drift $\alpha_1 + \alpha_2$ and variance rate $\beta_1^2 + \beta_2^2 + 2\rho\,\beta_1\beta_2$ (Definition 19.25(i)). Drifts add, and variance rates add with the cross term $2\rho\,\beta_1\beta_2$. If the constant is zero, the noise has mean zero and variance zero, so for each t it vanishes with probability one (Lemma 19.12(i) and (0)), and the sum is the line $(X_{1,0} + X_{2,0}) + (\alpha_1 + \alpha_2)\, t$, the case of a zero diffusion coefficient in Definition 19.25(i).

(c) No. *The identity.* If $\ln X$ is normal, write $\ln X = \mathbb{E}[\ln X] + \sqrt{\mathrm{Var}(\ln X)}\; Z$ with Z standard normal; then $X^k = e^{k\,\mathbb{E}[\ln X]}\, e^{\theta Z}$ with $\theta = k\sqrt{\mathrm{Var}(\ln X)}$, and the Gaussian shift (Lemma 19.38(i)) gives $\mathbb{E}[X^k] = \exp\big(k\,\mathbb{E}[\ln X] + \frac{1}{2}\,k^2\,\mathrm{Var}(\ln X)\big)$. The logarithm of $(\mathbb{E}[X^2])^3$ is therefore $3 \times \big(2\,\mathbb{E}[\ln X] + 2\,\mathrm{Var}(\ln X)\big)$, and that of $(\mathbb{E}[X])^3\,\mathbb{E}[X^3]$ is $3 \times \big(\mathbb{E}[\ln X] + \frac{1}{2}\,\mathrm{Var}(\ln X)\big) + 3\,\mathbb{E}[\ln X] + \frac{9}{2}\,\mathrm{Var}(\ln X)$. Both are $6\,\mathbb{E}[\ln X] + 6\,\mathrm{Var}(\ln X)$.

The moments of the portfolio. For integers $a, b \geq 0$, the variable $a\,\sigma_1 W_{1,1} + b\,\sigma_2 W_{2,1}$ is normal with mean zero and variance $a^2\sigma_1^2 + 2ab\,\rho\,\sigma_1\sigma_2 + b^2\sigma_2^2$ (the definition of correlation, Lemma 19.38(ii) and Proposition 10.10), so by the same lemma, with $g_k = \mu_k - \sigma_k^2/2$,

$$\mathbb{E}\big[S_{1,1}^a\, S_{2,1}^b\big] = 50^{a+b} \exp\Big(a\mu_1 + b\mu_2 + \tfrac{1}{2}\big(a(a-1)\,\sigma_1^2 + b(b-1)\,\sigma_2^2 + 2ab\,\rho\,\sigma_1\sigma_2\big)\Big),$$

because $a\,g_1 + \frac{1}{2}a^2\sigma_1^2 = a\mu_1 + \frac{1}{2}a(a-1)\sigma_1^2$, and likewise for b. With $\sigma_1^2 = 0.0625$, $\sigma_2^2 = 0.1225$ and $\rho\,\sigma_1\sigma_2 = 0.04375$, the exponents are 0.06 and 0.10 for $(a,b) = (1,0)$ and $(0,1)$; 0.1825, 0.20375 and 0.3225 for $(2,0)$, $(1,1)$ and $(0,2)$; and 0.3675, 0.37, 0.47 and 0.6675 for $(3,0)$, $(2,1)$, $(1,2)$ and $(0,3)$. The binomial expansion of $(S_{1,1} + S_{2,1})^k$, with the coefficients $1,2,1$ and $1,3,3,1$ and with each term the exponential of the exponent just listed, gives, for $X/50$,

$$\begin{aligned}
\mathbb{E}[X]/50 &= 1.061837 + 1.105171 = 2.1670,\\
\mathbb{E}[X^2]/50^2 &= 1.200214 + 2 \times 1.225992 + 1.380575 = 5.0328,\\
\mathbb{E}[X^3]/50^3 &= 1.444120 + 3 \times 1.447735 + 3 \times 1.599994 + 1.949358 = 12.5367.
\end{aligned}$$

In dollars, computed before rounding, $\mathbb{E}[X] = 108.35$, $\mathbb{E}[X^2] = 12{,}581.9$ and $\mathbb{E}[X^3] = 1.56708 \times 10^6$. The ratio does not depend on the unit of X:

$$\frac{\left(\mathbb{E}[X^2]\right)^3}{\left(\mathbb{E}[X]\right)^3 \mathbb{E}[X^3]} = \frac{5.0328^3}{2.1670^3 \times 12.5367} = 0.9992,$$

not 1 (0.99921 before rounding). The gap, 7.9×10^{-4}, is more than seven times what the rounding of the three four-decimal inputs could produce, which is at most about 1.1×10^{-4}. So X is not lognormal. A geometric Brownian motion is lognormal at every date (Theorem 19.40), hence the portfolio is not one (Example 19.60).

What goes wrong. Theorem 19.55(iii), which applies to any pair of correlated motions by its part (iv), with $V = x_1 + x_2$, whose second derivatives vanish, gives $d(S_1 + S_2) = (\mu_1 S_1 + \mu_2 S_2)\,dt + \sigma_1 S_1\,dW_1 + \sigma_2 S_2\,dW_2$. The variance rate of the noise, divided by the square of the portfolio value, is $w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2\rho\, w_1 w_2\, \sigma_1\sigma_2$ with the weights $w_k = S_k/(S_1 + S_2)$: a volatility of 26.1% at equal weights, 24.5% when the first stock is 70% of the portfolio and 29.0% when it is 30%. The percentage volatility moves with the weights, and the weights move with the prices, whereas a geometric Brownian motion has a constant one. This explains the failure; the proof is the moment test.

(d) The function $V = x_1/x_2$ has $\partial_t V = 0$, $\partial_1 V = 1/x_2$, $\partial_2 V = -x_1/x_2^2$, $\partial_{11} V = 0$, $\partial_{12} V = -1/x_2^2$ and $\partial_{22} V = 2x_1/x_2^3$, all products of powers and so of moderate growth. With $\alpha_{k,t} = \mu_k S_{k,t}$ and $\beta_{k,t} = \sigma_k S_{k,t}$, Theorem 19.55(iii), applicable to the pair by its part (iv), gives the drift

$$\mu_1 S_1 \cdot \frac{1}{S_2} - \mu_2 S_2 \cdot \frac{S_1}{S_2^2} - \rho\,\sigma_1\sigma_2\, S_1 S_2 \cdot \frac{1}{S_2^2} + \tfrac{1}{2}\,\sigma_2^2 S_2^2 \cdot \frac{2S_1}{S_2^3} = \left(\mu_1 - \mu_2 + \sigma_2^2 - \rho\,\sigma_1\sigma_2\right)\frac{S_1}{S_2}$$

and the noise $\sigma_1(S_1/S_2)\,dW_1 - \sigma_2(S_1/S_2)\,dW_2$, so

$$\frac{d(S_1/S_2)}{S_1/S_2} = \left(\mu_1 - \mu_2 + \sigma_2^2 - \rho\,\sigma_1\sigma_2\right)dt + \sigma_1\,dW_1 - \sigma_2\,dW_2.$$

The noise term has the variance rate $\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2$, by $dW_1\,dW_2 = \rho\,dt$, so the volatility of the ratio is $\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2}$, and that of the product is $\sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\,\sigma_1\sigma_2}$: Proposition 19.57. With the numbers, $\sigma_1^2 + \sigma_2^2 = 0.185$ and $2\rho\,\sigma_1\sigma_2 = 0.07$, so the ratio has volatility $\sqrt{0.115} = 33.9\%$ and the product $\sqrt{0.255} = 50.5\%$.

(e) The less volatile stock is the first. The ratio is less volatile than it when $\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2 < \sigma_1^2$, that is, $\sigma_2^2 < 2\rho\,\sigma_1\sigma_2$, that is,

$$\rho > \frac{\sigma_2}{2\sigma_1} = \frac{0.35}{0.50} = 0.7.$$

(f) Solving $\sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2$ for the correlation, with $\sigma_1 = 8\%$, $\sigma_2 = 9\%$ and $\sigma = 6\%$,

$$\rho = \frac{\sigma_1^2 + \sigma_2^2 - \sigma^2}{2\sigma_1\sigma_2} = \frac{64 + 81 - 36}{2 \times 72} = \frac{109}{144} = 0.757.$$

Self-check. For (a), the squares of the weights add up to $0.16 + 0.84 = 1$. For (b), $\mathrm{Var}(X_{1,t} + X_{2,t}) = \beta_1^2 t + \beta_2^2 t + 2\beta_1\beta_2\rho\,t$ by Proposition 10.10, the variance rate found. For (c), the variance route gives the second moment independently. By Proposition 19.42(i), $\mathrm{Var}(S_{1,1}) = 2500\,e^{0.12}(e^{0.0625} - 1) = 181.793$ and $\mathrm{Var}(S_{2,1}) = 2500\,e^{0.20}(e^{0.1225} - 1) = 397.930$; the product is a geometric Brownian motion with drift $\mu_1 + \mu_2 + \rho\,\sigma_1\sigma_2 = 20.375\%$ (Proposition 19.57(ii)), so $\mathbb{E}[S_{1,1}S_{2,1}] = 2500\,e^{0.20375}$ by Proposition 19.42(i), and $\mathrm{Cov}(S_{1,1}, S_{2,1}) = 2500\,e^{0.20375} - 2500\,e^{0.16} = 131.202$, the second term being $\mathbb{E}[S_{1,1}]\,\mathbb{E}[S_{2,1}]$. Hence $\mathrm{Var}(X) = 181.793 + 397.930 + 2 \times 131.202 = 842.127$ (Proposition 10.10) and $\mathbb{E}[X^2] = 842.127 + 108.3504^2 = 12{,}581.9$, as above. The test returns exactly 1 when the two stocks are the same lognormal variable, $X = 2S_{1,1}$, as it must. For (d), the log

route needs no Itô lemma: $\ln(S_1/S_2)$ has the drift $g_1 - g_2$ and the noise $\sigma_1 W_1 - \sigma_2 W_2$, of variance rate $\sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2$, so the ratio is a geometric Brownian motion with drift $g_1 - g_2 + \frac{1}{2}\sigma^2 = \mu_1 - \mu_2 + \sigma_2^2 - \rho\,\sigma_1\sigma_2$, the same expression. For (e), at $\rho = 0.7$ the ratio has variance $0.185 - 0.1225 = 0.0625$, a volatility of exactly 25%. For (f), putting $\rho = 109/144$ back gives $\sqrt{64 + 81 - 109} = \sqrt{36} = 6$ percent, and the three volatilities are the sides of a triangle, $|8 - 9| \le 6 \le 8 + 9$, which is what keeps ρ between -1 and 1.

Answer: (a) $w = \sqrt{1 - \rho^2} = 0.9165$; the increments of W_2 are independent with the laws $\mathcal{N}(0, t - s)$, and the covariance of $W_{1,t}$ and $W_{2,t}$ is $-0.4\,t$; the draw is (1.2, -1.12). (b) A generalised Wiener process with drift $\alpha_1 + \alpha_2$ and variance rate $\beta_1^2 + \beta_2^2 + 2\rho\,\beta_1\beta_2$. (c) No: $\mathbb{E}[X] = 108.35$, $\mathbb{E}[X^2] = 12{,}581.9$, $\mathbb{E}[X^3] = 1.56708 \times 10^6$ and $(\mathbb{E}[X^2])^3/((\mathbb{E}[X])^3\,\mathbb{E}[X^3]) = 0.9992 \neq 1$, so X is not lognormal; the percentage volatility of the portfolio moves with the weights. (d) Drift $\mu_1 - \mu_2 + \sigma_2^2 - \rho\,\sigma_1\sigma_2$ and volatility $\sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2} = 33.9\%$ for the ratio; $\sqrt{\sigma_1^2 + \sigma_2^2 + 2\rho\,\sigma_1\sigma_2} = 50.5\%$ for the product. (e) $\rho > \sigma_2/(2\sigma_1) = 0.7$. (f) $\rho = 109/144 = 0.757$.

28.20 Solutions — §20 The Black–Scholes–Merton Model

P20.1 *Deriving the Black–Scholes–Merton equation*

(a) By Itô's lemma for a function of the price (Corollary 19.31), with $dS = \mu S\,dt + \sigma S\,dW$,

$$dV = (\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V)\,dt + \sigma S\,\partial_S V\,dW.$$

The portfolio $\Pi = -V + (\partial_S V)\,S$, with the holding $\partial_S V$ kept fixed over dt, changes by

$$d\Pi = -dV + \partial_S V\,dS = -(\partial_t V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V)\,dt,$$

because $\partial_S V\,dS = \mu S\,\partial_S V\,dt + \sigma S\,\partial_S V\,dW$ cancels both the dW term and the μ term of dV. The portfolio is riskless over dt, so it must earn the rate r: $d\Pi = r\Pi\,dt = r(-V + S\,\partial_S V)\,dt$. Equating the two expressions for $d\Pi$,

$$-\partial_t V - \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V = -rV + rS\,\partial_S V, \qquad \text{that is,} \qquad \mathcal{L}V = 0.$$

Where the absence of arbitrage enters: in the sentence "a riskless portfolio must earn r". If it earned more, one would borrow at r and buy it; if less, sell it and lend the proceeds. *Why the portfolio is rebalanced:* the dW terms cancel for the holding $\partial_S V(S_t, t)$ of that instant only. The ratio moves with S and with t, and a hedge that is left alone keeps an exposure to the curvature $\partial_{SS} V$. *Why μ drops out:* μ enters dV and $\partial_S V\,dS$ through the same term $\mu S\,\partial_S V\,dt$, which the hedge removes together with the risk. Two investors who disagree about μ and agree on σ agree on the price.

(b) The words "kept fixed over dt" are made exact by a strategy. Fix $T' < T$ and work on $[0, T']$, where V and its three derivatives have moderate growth, so that Corollary 19.31 applies to the function $(S, t) \mapsto e^{-rt}V(S, t)$. For the discounted stock, $d(e^{-rt}S) = e^{-rt}\big[(\mu - r)S\,dt + \sigma S\,dW\big]$. For the discounted derivative, Itô's lemma applied to the function $(S, t) \mapsto e^{-rt}V(S, t)$ gives

$$d(e^{-rt}V) = e^{-rt}\Big[(\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V - rV)\,dt + \sigma S\,\partial_S V\,dW\Big].$$

The functions $\mathcal{L}V$ and $\mathcal{L}V\,\partial_S V$ are sums and products of continuous functions of moderate growth, so the two holdings are continuous functions of (t, W_t) of exponential growth on $[0, T']$ (Lemma 19.20(i) and (iii)) and are admissible; the strategy holds them on $[0, T')$ and keeps its whole value in the bank from T' on. Insert the holdings $h_{V,t} = \mathcal{L}V$ and $h_{S,t} = -\mathcal{L}V\,\partial_S V$ into the self-financing identity. The dW terms add up to

$$e^{-rt}\sigma S\,(\mathcal{L}V\,\partial_S V - \mathcal{L}V\,\partial_S V) = 0$$

identically, and the dt terms add up to

$$e^{-rt}\,\mathcal{L}V\left[\partial_t V+\mu S\,\partial_S V+\tfrac12\sigma^2S^2\,\partial_{SS}V-rV-(\mu-r)S\,\partial_S V\right]=e^{-rt}\,(\mathcal{L}V)^2,$$

because the bracket is $\mathcal{L}V$. Hence $d(e^{-rt}\,\Pi_t)=e^{-rt}(\mathcal{L}V)^2\,dt$, and with $\Pi_0=0$ this is the stated integral: an ordinary integral of a nonnegative continuous function along the path (the Itô integral of the identically zero integrand vanishes with probability one), so, for every $t\le T'$, $\Pi_t\ge0$ with probability one.

Suppose that $\mathcal{L}V(S^*,t^*)\neq0$ at some point. The function $\mathcal{L}V$ is continuous, so t^* may be taken positive, and there are $\varepsilon>0$ and $\varepsilon'>0$, with $t^*+\varepsilon'<T$ and $\varepsilon'<S^*$, such that $|\mathcal{L}V|\ge\varepsilon$ on the rectangle $(S^*-\varepsilon',S^*+\varepsilon')\times[t^*,t^*+\varepsilon']$. Run the strategy with $T'=t^*+\varepsilon'$. The price S_{t^*} is lognormal (Theorem 19.40), so the event $|S_{t^*}-S^*|<\varepsilon'/2$ has positive probability. On that event the continuous path stays inside the rectangle for a positive length of time, during which the integrand is at least $e^{-rs}\varepsilon^2$, so $\Pi_{T'}>0$. The strategy starts from nothing, loses nothing with probability one and wins with positive probability: an arbitrage, which is excluded. Hence $\mathcal{L}V=0$ everywhere, which is Theorem 20.3.

(c) Here $r+\sigma^2=0.09$ and $r-\frac12\sigma^2=0.03$, and $\partial_t\,e^{-r\tau}=re^{-r\tau}$ because $\tau=T-t$.

(1) $V=e^{-r\tau}$: $\partial_tV=rV$ and the derivatives in S vanish, so $\mathcal{L}V=rV-rV=0$. It passes: the bond that pays 1 at T.

(2) $V=S^2$: $\partial_tV=0$, $rS\,\partial_SV=2rS^2$ and $\frac12\sigma^2S^2\,\partial_{SS}V=\sigma^2S^2$, so $\mathcal{L}V=(2r+\sigma^2-r)S^2=(r+\sigma^2)S^2=0.09\,S^2\neq0$. It fails.

(3) For a power, $\mathcal{L}S^a=\big(ra+\frac12\sigma^2a(a-1)-r\big)S^a=(a-1)\big(r+\frac12\sigma^2a\big)S^a$. At $a=-2.5$ the second factor is $0.05-0.02\times2.5=0$; term by term, $-0.125\,V+0.175\,V-0.05\,V=0$. It passes: the claim that pays $S_T^{-2.5}$ at T, whose price does not depend on the time left.

(4) $V=\ln S$: $rS\,\partial_SV=r$ and $\frac12\sigma^2S^2\,\partial_{SS}V=-\frac12\sigma^2$, so $\mathcal{L}V=r-\frac12\sigma^2-r\ln S=0.03-0.05\ln S$, which vanishes only at $S=e^{0.6}=1.82$. It fails.

(5) $V=e^{-r\tau}\big[\ln S+0.03\,\tau\big]$: $\partial_tV=rV-0.03\,e^{-r\tau}$, $rS\,\partial_SV=0.05\,e^{-r\tau}$ and $\frac12\sigma^2S^2\,\partial_{SS}V=-0.02\,e^{-r\tau}$, so $\mathcal{L}V=rV+(-0.03+0.05-0.02)\,e^{-r\tau}-rV=0$. It passes, and at $\tau=0$ it equals $\ln S$: the claim that pays $\ln S_T$.

Each of the five functions is smooth on $(0,\infty)\times[0,T)$, and it and its three derivatives have moderate growth ($|\ln S|\le S+S^{-1}$), so the test of (b) applies: (2) and (4) are not prices. The equation is a necessary test. The payoffs 1, $S^{-2.5}$ and $\ln S$ are continuous and of moderate growth, so they are in the class $\mathcal{G}$ and Theorem 20.7 applies to them: (1), (3) and (5) are prices, and not only solutions, because each is the discounted risk-neutral expectation of its payoff, as the self-check verifies.

Self-check. For (b), the undiscounted bookkeeping gives the same result. Gains plus interest on the bank holding read $d\Pi=h_V\,dV+h_S\,dS+r(\Pi-h_VV-h_SS)\,dt$. With the two holdings, $h_V\,dV+h_S\,dS=\mathcal{L}V\,(dV-\partial_SV\,dS)=\mathcal{L}V\,(\partial_tV+\frac12\sigma^2S^2\,\partial_{SS}V)\,dt$ and $-r(h_VV+h_SS)=-r\,\mathcal{L}V\,(V-S\,\partial_SV)$, and the two add up to $(\mathcal{L}V)^2$. So $d\Pi=(\mathcal{L}V)^2\,dt+r\Pi\,dt$, which is $d(e^{-rt}\,\Pi)=e^{-rt}(\mathcal{L}V)^2\,dt$ again. For (c), price the claims by expectation instead of by the equation. Under the risk-neutral law $S_T=S\exp(0.03\,\tau+0.2\sqrt{\tau}\,Z)$ with $Z\sim\mathcal{N}(0,1)$, and $\mathbb{E}\,e^{\theta Z}=e^{\theta^2/2}$, so the claim that pays S_T^a is worth $S^a\exp\big((-0.05+0.03\,a+0.02\,a^2)\tau\big)$. The exponent is $-0.05-0.075+0.125=0$ at $a=-2.5$, which confirms (3); it is -0.05 at $a=0$, the claim that pays 1, which confirms the bond $e^{-r\tau}$ of (1); and it is $-0.05+0.06+0.08=0.09$ at $a=2$: the claim that pays S_T^2 is worth $S^2e^{0.09\tau}$, not S^2, and 0.09 is the residual coefficient of (2). The claim that pays $\ln S_T$ is worth $e^{-r\tau}\,\mathbb{E}\big[\ln S+0.03\,\tau+0.2\sqrt{\tau}\,Z\big]=e^{-r\tau}\big[\ln S+0.03\,\tau\big]$, which is function (5) and not function (4).

Answer: (a) $d\Pi=-(\partial_tV+\frac12\sigma^2S^2\,\partial_{SS}V)\,dt=r(-V+S\,\partial_SV)\,dt$ gives $\partial_tV+rS\,\partial_SV+\frac12\sigma^2S^2\,\partial_{SS}V=rV$; the absence of arbitrage enters through "riskless earns r", the hedge ratio ∂_SV moves with S and t, and μ cancels together with the risk. (b) $d(e^{-rt}\,\Pi_t)=e^{-rt}(\mathcal{L}V)^2\,dt$, so a violation at any point is an arbitrage and $\mathcal{L}V=0$. (c) Prices: (1) the bond paying 1, (3) the

claim paying $S_T^{-2.5}$, (5) the claim paying $\ln S_T$. Not prices: (2), with residual $0.09\,S^2$, and (4), with residual $0.03 - 0.05 \ln S$.

P20.2 *The yield equation by a change of variables; power and log claims on a stock with a yield*

(a) Write $\tilde{S} = Se^{-q(T-t)}$; the derivatives of $\tilde{V}$ below are evaluated at $(\tilde{S}, t)$. Because the partial derivatives $\partial_t \tilde{V}$ and $\partial_{\tilde{S}} \tilde{V}$ are continuous, $\tilde{V}$ is differentiable, and the chain rule applies to its composition with $(S,t) \mapsto (Se^{-q(T-t)}, t)$, which sends $(0,\infty) \times [0,T)$ into itself; the second derivative in S uses the existence and continuity of $\partial_{\tilde{S}\tilde{S}} \tilde{V}$ in the same way. Differentiating $V(S,t) = \tilde{V}(Se^{-q(T-t)}, t)$ in S by the chain rule,

$$\partial_S V = e^{-q(T-t)}\, \partial_{\tilde{S}} \tilde{V}, \qquad \partial_{SS} V = e^{-2q(T-t)}\, \partial_{\tilde{S}\tilde{S}} \tilde{V},$$

so that $S\, \partial_S V = \tilde{S}\, \partial_{\tilde{S}} \tilde{V}$ and $S^2\, \partial_{SS} V = \tilde{S}^2\, \partial_{\tilde{S}\tilde{S}} \tilde{V}$. In t, at a fixed S, the variable $\tilde{S}$ grows at the rate $q\tilde{S}$, because the derivative of $e^{-q(T-t)}$ in t is $q\, e^{-q(T-t)}$. Hence

$$\partial_t V = \partial_t \tilde{V} + q\tilde{S}\, \partial_{\tilde{S}} \tilde{V} = \partial_t \tilde{V} + qS\, \partial_S V, \qquad \text{that is,} \qquad \partial_t \tilde{V} = \partial_t V - qS\, \partial_S V.$$

Substituting the three relations into the equation for $\tilde{V}$,

$$(\partial_t V - qS\, \partial_S V) + rS\, \partial_S V + \tfrac{1}{2}\sigma^2 S^2\, \partial_{SS} V = rV,$$

which is the equation with the yield. This is the general clause of Corollary 20.45(i), from which its first clause, for a payoff in $\mathcal{G}$, follows by Corollary 20.41(i)(a) and Theorem 20.7(i); no new hedging argument is needed, because the reinvested holding is an asset without income.

(b) The computation is that of (a) with $\tilde{F}$ in the place of $\tilde{S}$, F in the place of S, r in the place of q and T_F in the place of T, the map $(F,t) \mapsto (Fe^{-r(T_F - t)}, t)$ again sending $(0,\infty) \times [0,T)$ into itself: $F\, \partial_F V^{\text{fut}} = \tilde{F}\, \partial_{\tilde{F}} \tilde{V}$, $F^2\, \partial_{FF} V^{\text{fut}} = \tilde{F}^2\, \partial_{\tilde{F}\tilde{F}} \tilde{V}$, and at a fixed F the variable $\tilde{F}$ grows at the rate $r\tilde{F}$, so $\partial_t \tilde{V} = \partial_t V^{\text{fut}} - rF\, \partial_F V^{\text{fut}}$. Substituting,

$$(\partial_t V^{\text{fut}} - rF\, \partial_F V^{\text{fut}}) + rF\, \partial_F V^{\text{fut}} + \tfrac{1}{2}\sigma^2 F^2\, \partial_{FF} V^{\text{fut}} = rV^{\text{fut}},$$

and the two terms $rF\, \partial_F V^{\text{fut}}$ cancel. This is the equation of (a) with $q = r$: a futures price behaves like an asset whose yield equals the interest rate, that is, with zero cost of carry, because a futures position costs nothing to enter (equation (20.14) of Corollary 20.45(ii)).

(c) The risk-neutral log-drift is $\nu = r - q - \frac{1}{2}\sigma^2 = 0.03 - 0.01 - 0.045 = -0.025$.

The cube. With $\theta = 3\sigma\sqrt{\tau}$, $\mathbb{E}^{\mathbb{Q}} S_T^3 = S^3 e^{3\nu\tau}\, \mathbb{E}\, e^{3\sigma\sqrt{\tau}\, Z} = S^3 e^{3\nu\tau + 9\sigma^2\tau/2}$, so the price function is

$$V(S,t) = e^{-r\tau} S^3 e^{3\nu\tau + 9\sigma^2\tau/2} = S^3 e^{c\tau}, \qquad c = -r + 3\big(r - q - \tfrac{1}{2}\sigma^2\big) + \tfrac{9}{2}\sigma^2 = 2r - 3q + 3\sigma^2 = 0.30.$$

At $S = 50$ and $\tau = 2$ the claim is worth $125{,}000\, e^{0.60} = 125{,}000 \times 1.822119 = 227{,}765$. In the equation, $\partial_t V = -cV$, $(r-q)S\, \partial_S V = 3(r-q)V$ and $\frac{1}{2}\sigma^2 S^2\, \partial_{SS} V = 3\sigma^2 V$, so the left side is $(-0.30 + 0.06 + 0.27)V = 0.03\, V = rV$; in numbers, $-68{,}329.5 + 13{,}665.9 + 61{,}496.5 = 6{,}832.9 = 0.03 \times 227{,}764.85$.

The logarithm. Since $\mathbb{E}\, Z = 0$, $\mathbb{E}^{\mathbb{Q}} \ln S_T = \ln S + \nu\tau$, and

$$V(S,t) = e^{-r\tau}\big[\ln S + \nu\tau\big] = e^{-0.06}\,(3.912023 - 0.05) = 0.941765 \times 3.862023 = 3.637.$$

In the equation, $\partial_t V = rV - \nu e^{-r\tau}$, $(r-q)S\, \partial_S V = (r-q)e^{-r\tau}$ and $\frac{1}{2}\sigma^2 S^2\, \partial_{SS} V = -\frac{1}{2}\sigma^2 e^{-r\tau}$; the three sum to $rV + \big(-\nu + r - q - \frac{1}{2}\sigma^2\big)e^{-r\tau} = rV$. In numbers, $0.1327 + 0.0188 - 0.0424 = 0.1091 = 0.03 \times 3.6371$.

(d) For $V = S^a$ the time derivative vanishes, and the equation reads $\big[(r-q)a + \frac{1}{2}\sigma^2 a(a-1) - r\big]S^a = 0$, that is,

$$\tfrac{1}{2}\sigma^2 a^2 + \big(r - q - \tfrac{1}{2}\sigma^2\big)a - r = 0, \qquad 0.045\, a^2 - 0.025\, a - 0.03 = 0.$$

The roots are $a = \big(0.025 \pm \sqrt{0.000625 + 0.0054}\big)/0.09 = (0.025 \pm 0.077621)/0.09$, that is, $a = 1.140$ and $a = -0.585$. With $q = 0$ the quadratic factors as $(a-1)\big(\frac{1}{2}\sigma^2 a + r\big)$, with the roots 1 and $-2r/\sigma^2 = -0.667$: the yield moves both roots to the right.

At $a = 1$ and $q > 0$ the left side of the equation minus the right side is $(r-q)S - rS = -qS$, which is -0.50 a year at $S = 50$. The function S is the price of the share, and the share earns part of its return as dividends; a claim that paid no income and were worth S_t at every date would be dominated by the share itself. The position without income that ends as one share is the reinvested holding, worth $Se^{-q\tau}$ (49.01 at $S = 50$ and $\tau = 2$), and that function does satisfy the equation: $\partial_t V = qV$ and $(r-q)S\,\partial_S V = (r-q)V$ sum to rV. A time-independent power needs $a = 1.140 > 1$, so that the convexity term $\frac{1}{2}\sigma^2 a(a-1)$ makes up for the missing yield.

Self-check. Part (a) gives a second route to (c). In the market of the reinvested holding, which pays no income, the claims pay $\tilde{S}_T^3$ and $\ln \tilde{S}_T$, because $\tilde{S}_T = S_T$, and their price functions are the $q = 0$ formulas $\tilde{S}^3 e^{(2r+3\sigma^2)\tau}$ and $e^{-r\tau}\big[\ln \tilde{S} + (r - \frac{1}{2}\sigma^2)\tau\big]$. With $\tilde{S} = 50\,e^{-0.02} = 49.0099$: $49.0099^3 \times e^{0.66} = 117{,}720.6 \times 1.934792 = 227{,}765$, and $0.941765 \times (3.892023 - 0.03) = 0.941765 \times 3.862023 = 3.637$. Both agree with (c). For (d), the roots must have the sum $-(r-q-\frac{1}{2}\sigma^2)/(\frac{1}{2}\sigma^2) = 0.025/0.045 = 0.5556$ and the product $-r/(\frac{1}{2}\sigma^2) = -0.6667$: indeed $1.1402 - 0.5847 = 0.5555$ and $1.1402 \times (-0.5847) = -0.6667$.

Answer: (a) and (b): by the chain rule, $S\,\partial_S V = \tilde{S}\,\partial_{\tilde{S}}\tilde{V}$, $S^2\,\partial_{SS}V = \tilde{S}^2\,\partial_{\tilde{S}\tilde{S}}\tilde{V}$ and $\partial_t \tilde{V} = \partial_t V - qS\,\partial_S V$, which turns the equation without income into $\partial_t V + (r-q)S\,\partial_S V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS}V = rV$; for a futures price $q = r$ and the first-order term cancels. (c) The claim paying S_T^3 is worth $S^3 e^{(2r-3q+3\sigma^2)\tau} = 125{,}000\,e^{0.60} = 227{,}765$; the claim paying $\ln S_T$ is worth $e^{-r\tau}[\ln S + (r - q - \frac{1}{2}\sigma^2)\tau] = 3.637$; both satisfy the equation. (d) $a = 1.140$ and $a = -0.585$ (against 1 and -0.667 at $q = 0$); S leaves the residual $-qS$, because the share's price leaves out its dividends, and the income-free position is $Se^{-q\tau}$.

P20.3 *A call and a put from the formula*

(a) By Corollary 20.23, with $\sigma\sqrt{T} = 0.25 \times 0.70711 = 0.17678$,

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}} = \frac{-0.04879 + 0.07125 \times 0.5}{0.17678} = \frac{-0.01317}{0.17678} = -0.0745,$$
$$d_2 = -0.0745 - 0.1768 = -0.2513.$$

Since $\Phi(-y) = 1 - \Phi(y)$, $\Phi(d_1) = 1 - 0.5297 = 0.4703$ and $\Phi(d_2) = 1 - 0.5992 = 0.4008$. The discounted strike is $Ke^{-rT} = 84\,e^{-0.02} = 82.3367$. Then

$$C = S\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2) = 80 \times 0.4703 - 82.3367 \times 0.4008 = 37.624 - 33.001 = 4.62,$$
$$\mathcal{P} = Ke^{-rT}\,\Phi(-d_2) - S\,\Phi(-d_1) = 82.3367 \times 0.5992 - 80 \times 0.5297 = 49.336 - 42.376 = 6.96.$$

Before rounding, the two differences are 4.6235 and 6.9601.

(b) Parity (Theorem 17.9) requires $C - \mathcal{P} = S - Ke^{-rT} = 80 - 82.3367 = -2.34$, and indeed $4.62 - 6.96 = -2.34$. The put is the dearer option because the strike lies above the forward price $F = 80\,e^{0.02} = 81.62$.

(c) The stock pays no dividends and $r > 0$, so the American call is never exercised early and is worth the same as the European call: $C^{\text{am}} = C = 4.62$ (Theorem 17.16(i), and part (ii) for the strict suboptimality of early exercise when $r > 0$).

(d) At expiry the loan has grown to $4.6235\,e^{0.02} = 4.717$, and the call pays $(S_T - 84)^+$. The purchase makes money when $S_T - 84 > 4.717$, that is, when

$$S_T > K + Ce^{rT} = 84 + 4.72 = 88.72,$$

a rise of 10.9% in six months. (Without the interest on the premium the figure would be $84 + 4.62 = 88.62$.)

Self-check. A different route: start from the at-the-money-forward call and move the strike. At $K = F = 81.62$ the call is worth $e^{-rT}F\,[2\,\Phi(\sigma\sqrt{T}/2) - 1] = 80 \times (2\,\Phi(0.0884) - 1) = 80 \times 0.0704 = 5.63$, with $\Phi(0.0884) = 0.5352$ (Proposition 20.29). The slope of the call price in the strike is minus the digital price, $-e^{-rT}\,\Phi(d_2)$ (the identity $\mathrm{Dig}(K) = -\partial_K C(K)$ of Proposition 23.2 in §23, evaluated with Theorem 20.17): $-0.9802 \times 0.4648 = -0.456$ at $K = F$, where $d_2 = -0.0884$, and $-0.9802 \times 0.4008 = -0.393$ at $K = 84$. Over the $84 - 81.62 = 2.38$ between the two strikes the mean of the two end slopes, -0.424 (a trapezoid estimate of the average slope, whose exact value is -0.4239), lowers the price by about 1.01, to $5.63 - 1.01 = 4.62$, in agreement with (a). The put also respects its model-free bound, $6.96 \geq Ke^{-rT} - S = 2.34$.

Answer: (a) $d_1 = -0.0745$, $d_2 = -0.2513$, $C = 4.62$ and $\mathcal{P} = 6.96$. (b) $C - \mathcal{P} = -2.34 = 80 - 84\,e^{-0.02}$. (c) $C^{\mathrm{am}} = C = 4.62$: no early exercise without dividends. (d) Break-even at $S_T = K + Ce^{rT} = 88.72$.

P20.4 *The formula assembled from digitals*

(a) Take logarithms, the logarithm being strictly increasing, and divide by $\sigma\sqrt{T} > 0$:

$$F_T > K \iff -\tfrac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z > \ln\frac{K}{F} \iff Z > \frac{\ln(K/F) + \sigma^2 T/2}{\sigma\sqrt{T}} = -d_2.$$

Since $\varphi(-z) = \varphi(z)$,

$$\mathbb{Q}(F_T > K) = \int_{-d_2}^{\infty} \varphi(z)\,dz = \int_{-\infty}^{d_2} \varphi(z)\,dz = \Phi(d_2),$$

and the same computation gives $P(Z > -y) = \Phi(y)$ for every real y; since φ integrates to 1 over the line (its constant $1/\sqrt{2\pi}$ is the one fixed by Theorem 7.5), also $\Phi(-y) = 1 - \Phi(y)$. The expectation of an indicator is the probability of its event, so $\mathrm{Dig}(K) = e^{-rT}\,\Phi(d_2)$: the risk-neutral probability of exercise, discounted (Theorem 20.17).

(b) Put $\theta = \sigma\sqrt{T}$, so that $F_T = Fe^{\theta Z - \theta^2/2}$ and $\{F_T > K\} = \{Z > -d_2\}$. Then

$$\mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T > K\}\big] = F\int_{-d_2}^{\infty} e^{\theta z - \theta^2/2}\,\varphi(z)\,dz.$$

The exponent of the integrand is $\theta z - \frac{1}{2}\theta^2 - \frac{1}{2}z^2 = -\frac{1}{2}(z-\theta)^2$: the square is complete, and $e^{\theta z - \theta^2/2}\,\varphi(z) = \varphi(z-\theta)$. Substituting $y = z - \theta$,

$$\int_{-d_2}^{\infty} \varphi(z-\theta)\,dz = \int_{-d_2-\theta}^{\infty} \varphi(y)\,dy = \int_{-d_1}^{\infty} \varphi(y)\,dy = \Phi(d_1),$$

because $d_2 + \theta = d_1$. Hence $\mathrm{AoN}(K) = e^{-rT}F\,\Phi(d_1)$ (Proposition 20.19). Over the whole line the same completed square gives $\mathbb{E}^{\mathbb{Q}}F_T = F\int_{-\infty}^{\infty}\varphi(z-\theta)\,dz = F$, because φ integrates to 1 over the line, as in (a).

(c) For every $x > 0$, $(x-K)^+ = x\,\mathbf{1}\{x > K\} - K\,\mathbf{1}\{x > K\}$: both sides equal $x - K$ when $x > K$ and 0 otherwise. With $x = F_T$ and the linearity of expectation,

$$C = \mathrm{AoN}(K) - K\,\mathrm{Dig}(K) = e^{-rT}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big].$$

For the put, $(K-x)^+ = K\,\mathbf{1}\{x \leq K\} - x\,\mathbf{1}\{x \leq K\}$. By (a), $\mathbb{Q}(F_T \leq K) = 1 - \Phi(d_2) = \Phi(-d_2)$, and, since $F_T\,\mathbf{1}\{F_T \leq K\} = F_T - F_T\,\mathbf{1}\{F_T > K\}$, linearity, $\mathbb{E}^{\mathbb{Q}}F_T = F$ and (b) give $\mathbb{E}^{\mathbb{Q}}\big[F_T\,\mathbf{1}\{F_T \leq K\}\big] = F - F\,\Phi(d_1) = F\,\Phi(-d_1)$. Hence

$$\mathcal{P} = e^{-rT}\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big].$$

Subtracting, and using $\Phi(y) + \Phi(-y) = 1$ twice,

$$C - \mathcal{P} = e^{-rT}\Big[F\big(\Phi(d_1) + \Phi(-d_1)\big) - K\big(\Phi(d_2) + \Phi(-d_2)\big)\Big] = e^{-rT}(F - K),$$

with no σ left: the formulas satisfy put–call parity identically (Theorem 20.22).

(d) The difference $d_1 - d_2 = \sigma\sqrt{T}$ is positive and Φ is strictly increasing, its derivative φ being positive; so $\Phi(d_1) > \Phi(d_2)$. By (a), $\Phi(d_2) = \mathbb{Q}(F_T > K)$ is the risk-neutral probability that the call is exercised. By (b), $\Phi(d_1) = \mathbb{E}^{\mathbb{Q}}\big[(F_T/F)\,\mathbf{1}\{F_T > K\}\big]$ is the probability of exercise when each outcome is reweighted by F_T/F, a positive weight with mean one (Proposition 20.26). The weights are largest where F_T is high, which is where the call is exercised, so the reweighted probability is the larger one. In the formula the strike is paid with the weight $\Phi(d_2)$ and the asset is received with the weight $\Phi(d_1)$.

(e) Here $\sigma\sqrt{T} = 0.30 \times 0.70711 = 0.21213$, $\ln(50/45) = 0.10536$ and $\sigma^2 T/2 = 0.0225$, so $d_1 = 0.12786/0.21213 = 0.6027$ and $d_2 = 0.3906$; the discount factor is $e^{-0.015} = 0.98511$. Then

$$\begin{aligned}
\mathrm{Dig}(45) &= 0.98511 \times 0.65196 = 0.64225,\\
\mathrm{AoN}(45) &= 0.98511 \times 50 \times 0.72666 = 35.79,\\
F \times \mathrm{Dig}(45) &= 50 \times 0.64225 = 32.11,\\
C &= 35.79 - 45 \times 0.64225 = 35.79 - 28.90 = 6.89,\\
\mathcal{P} &= 0.98511 \times \big(45 \times 0.34804 - 50 \times 0.27334\big) = 0.98511 \times 1.9948 = 1.97.
\end{aligned}$$

The asset-or-nothing claim is worth 3.68 more than F digitals, because it pays most in the highest states. Parity holds: before rounding, $6.891 - 1.965 = 4.926 = 0.98511 \times 5$.

Self-check. Three checks that do not repeat the derivation. (1) The Gaussian shift $\mathbb{E}\big[e^{\theta Z}\psi(Z)\big] = e^{\theta^2/2}\,\mathbb{E}\,\psi(Z+\theta)$ of Lemma 19.38(i), with $\psi(z) = \mathbf{1}\{z > -d_2\}$, gives (b) with no integral: $P(Z + \theta > -d_2) = \Phi(d_2 + \theta) = \Phi(d_1)$. (2) The digital is minus the slope of the call price in the strike. Repricing the call of (e) at the strikes 44.9 and 45.1 gives 6.95519 and 6.82674 (the four values of Φ this needs are not among those supplied and were computed to full precision), and $(6.95519 - 6.82674)/0.2 = 0.64225$, the digital. (3) The ratio $\mathrm{AoN}/\mathrm{Dig} = F\,\Phi(d_1)/\Phi(d_2) = 50 \times 0.72666/0.65196 = 55.73$ is the risk-neutral mean of F_T over the exercise states; it must exceed both the strike 45 and the forward price 50, and it does.

Answer: (a) $\{F_T > K\} = \{Z > -d_2\}$, so $\mathbb{Q}(F_T > K) = \Phi(d_2)$ and $\mathrm{Dig}(K) = e^{-rT}\,\Phi(d_2)$. (b) $e^{\theta z - \theta^2/2}\varphi(z) = \varphi(z-\theta)$ with $\theta = \sigma\sqrt{T}$, so $\mathrm{AoN}(K) = e^{-rT}F\,\Phi(d_1)$. (c) $C = \mathrm{AoN}(K) - K\,\mathrm{Dig}(K)$, $\mathcal{P} = e^{-rT}[K\,\Phi(-d_2) - F\,\Phi(-d_1)]$, and $C - \mathcal{P} = e^{-rT}(F - K)$ for every σ. (d) $\Phi(d_2)$ is the risk-neutral probability of exercise and $\Phi(d_1)$ the same probability reweighted by F_T/F; $d_1 > d_2$. (e) Dig $= 0.6423$, AoN $= 35.79$ against $F \times \mathrm{Dig} = 32.11$ (gap 3.68), $C = 6.89$ and $\mathcal{P} = 1.97$.

P20.5 *Three exercise probabilities and a cash digital*

(a) Here $\ln(S/K) = \ln(50/55) = -0.09531$ and $\sigma\sqrt{T} = 0.30$.

Real-world. The terminal price is lognormal with the log-drift $\mu - \frac{1}{2}\sigma^2$, so (Proposition 19.42(iii))

$$\begin{aligned}
P(S_T > K) &= \Phi\Big(\frac{\ln(S/K) + (\mu - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}\Big) = \Phi\Big(\frac{-0.09531 + 0.075}{0.30}\Big)\\
&= \Phi(-0.0677) = 1 - 0.52699 = 0.473.
\end{aligned}$$

Risk-neutral. The same expression with μ replaced by r is d_2:

$$d_2 = \frac{-0.09531 + 0.04 - 0.045}{0.30} = -0.3344, \qquad \mathbb{Q}(S_T > K) = \Phi(d_2) = 1 - 0.63096 = 0.369.$$

Reweighted. $d_1 = d_2 + \sigma\sqrt{T} = -0.0344$ and $\Phi(d_1) = 1 - 0.51372 = 0.486$.

The put is exercised when $S_T < K$, with the complementary probabilities: 0.527 under the real-world law and 0.631 under the risk-neutral law.

(b) The price is $C = S\,\Phi(d_1) - Ke^{-rT}\,\Phi(d_2)$ (Corollary 20.23), so $\Phi(d_2) = 0.369$ and $\Phi(d_1) = 0.486$ enter it, and the real-world 0.473 does not: μ appears nowhere in the price. The replicating

strategy of Theorem 20.7(iii) holds $\partial_S C$ shares, and $\partial_S C = \Phi(d_1) = 0.486$, the delta of the call (Proposition 21.2 with $q = 0$); the shares are worth $0.48628 \times 50 = 24.31$, against a loan of $Ke^{-rT}\,\Phi(d_2) = 55 \times 0.96079 \times 0.36904 = 19.50$; its value, $24.31 - 19.50 = 4.81$, is the call price. The number 0.473 plays no part in pricing or hedging.

(c) By Theorem 20.17, with the forward price $F = Se^{rT}$, the digital that pays 1 is worth $e^{-rT}\,\Phi(d_2)$, so the digital that pays 100 is worth

$$100 \times e^{-0.04} \times 0.36904 = 100 \times 0.96079 \times 0.36904 = 35.46.$$

(d) The number $\Phi(d_1) = 0.486$, the literature's $N(d_1)$, is not the probability of finishing in the money under either law. Under the risk-neutral law that probability is $\Phi(d_2) = 0.369$; under the real-world law it is 0.473, and it depends on μ. The number $\Phi(d_1)$ is the exercise probability reweighted by S_T/F (Proposition 20.26), and it is the hedge ratio. That 0.486 lies close to the real-world 0.473 is an accident of $\mu = 12\%$: at $\mu = 4\%$ the real-world probability would be 0.369.

Self-check. Medians give the order of the probabilities with no table. The real-world median of S_T is $Se^{(\mu-\sigma^2/2)T} = 50\,e^{0.075} = 53.89$, a little below the strike 55, so the real-world probability of exercise is a little below one half: 0.473. The risk-neutral median is $50\,e^{-0.005} = 49.75$, further below the strike, so the risk-neutral probability is smaller: 0.369. The two arguments of Φ differ by $(\mu - r)\sqrt{T}/\sigma = 0.08/0.30 = 0.2667$, and indeed $-0.0677 - (-0.3344) = 0.2667$. The forward form of Theorem 20.22 recomputes d_2 from another input: $F = 50\,e^{0.04} = 52.04$, $\ln(F/K) = -0.05531$ and $d_2 = (-0.05531 - 0.045)/0.30 = -0.3344$, as in (a); the call is $e^{-0.04}(52.04 \times 0.48628 - 55 \times 0.36904) = 0.96079 \times (25.31 - 20.30) = 4.81$, as in (b). For (c), the risk-neutral median 49.75 lies below the strike, so $\mathbb{Q}(S_T > 55) < \frac{1}{2}$ and the digital is worth less than $100\,e^{-0.04}/2 = 48.04$; it is 35.46.

Answer: (a) Call: 0.473 real-world, 0.369 risk-neutral, and $\Phi(d_1) = 0.486$; put: 0.527 real-world and 0.631 risk-neutral. (b) $\Phi(d_2) = 0.369$ and $\Phi(d_1) = 0.486$ enter the price; $\Phi(d_1) = 0.486$ is the number of shares in the hedge; the real-world 0.473 enters nothing. (c) $100\,e^{-0.04}\,\Phi(d_2) = 35.46$. (d) The risk-neutral probability of finishing in the money is $\Phi(d_2) = 0.369$ (the real-world one is 0.473); $\Phi(d_1) = 0.486$ is the reweighted probability and the hedge ratio.

P20.6 *The at-the-money rule, with its error bar*

(a) *The exact value.* At $K = F$, $\ln(F/K) = 0$, so $d_1 = \sigma\sqrt{T}/2$ and $d_2 = -\sigma\sqrt{T}/2$. With $\Phi(-y) = 1 - \Phi(y)$,

$$C = e^{-rT}F\left[\Phi\Big(\frac{\sigma\sqrt{T}}{2}\Big) - \Phi\Big(-\frac{\sigma\sqrt{T}}{2}\Big)\right] = e^{-rT}F\left[2\,\Phi\Big(\frac{\sigma\sqrt{T}}{2}\Big) - 1\right],$$

and $\mathcal{P} = C - e^{-rT}(F - K) = C$.

The function. Put $\chi(y) = \Phi(y) - \frac{1}{2}$, so that $C = 2e^{-rT}F\,\chi(\sigma\sqrt{T}/2)$. Since $\Phi' = \varphi$ and $\varphi'(y) = -y\,\varphi(y)$,

$$\chi' = \varphi, \qquad \chi''(y) = -y\,\varphi(y), \qquad \chi'''(y) = (y^2 - 1)\,\varphi(y),$$

and $\chi(0) = 0$, $\chi'(0) = \varphi(0) = 1/\sqrt{2\pi}$, $\chi''(0) = 0$. All three derivatives are continuous, so χ has three continuous derivatives on the real line, the hypothesis of Theorem 7.31 with $n = 2$.

The upper bound. For $y \geq 0$, $\chi(y) = \int_0^y \varphi(z)\,dz \leq \varphi(0)\,y$, because $\varphi(z) \leq \varphi(0)$.

The lower bound. By Taylor's theorem with the integral remainder (Theorem 7.31), at the expansion point 0 and with the polynomial of degree two,

$$\chi(y) = \varphi(0)\,y + \frac{1}{2}\int_0^y (y - z)^2\,\chi'''(z)\,dz.$$

For every z, $\chi'''(z) = z^2\varphi(z) - \varphi(z) \geq -\varphi(z) \geq -\varphi(0)$, and for $y > 0$ the weight $(y-z)^2$ is nonnegative, so the remainder is at least $-\frac{1}{2}\varphi(0)\int_0^y (y-z)^2\,dz = -\varphi(0)\,y^3/6$. Hence

$$\varphi(0)\,y\left(1 - \frac{y^2}{6}\right) \;\leq\; \chi(y) \;\leq\; \varphi(0)\,y \qquad \text{for } y \geq 0.$$

Conclusion. Put $y = \sigma\sqrt{T}/2$ and multiply by $2e^{-rT}F$: $2\,\varphi(0)\,y = \sigma\sqrt{T}/\sqrt{2\pi}$ and $y^2/6 = \sigma^2 T/24$, which gives the two bounds. Since $1/\sqrt{2\pi} = 0.3989$, the call and the put are each worth about $0.4\,e^{-rT}F\,\sigma\sqrt{T}$, and the linear rule overstates the price by at most the fraction $\sigma^2 T/24$ of itself (Proposition 20.29). Subtracting, $0 \leq C^{\text{lin}} - C \leq C^{\text{lin}}\sigma^2 T/24 = e^{-rT}F(\sigma\sqrt{T})^3/(24\sqrt{2\pi})$, a constant multiple of $(\sigma\sqrt{T})^3$. When $\sigma^2 T < 24$ the lower bound gives $C \geq C^{\text{lin}}(1 - \sigma^2 T/24) > 0$, and dividing the upper estimate of $C^{\text{lin}} - C$ by this lower estimate of C gives $(C^{\text{lin}} - C)/C \leq (\sigma^2 T/24)/(1 - \sigma^2 T/24)$.

(b) Here $\sigma\sqrt{T} = 0.15 \times 0.5 = 0.075$ and $F\sigma\sqrt{T} = 300$. The 0.4 rule gives $0.4 \times 300 = 120.00$, and the rule with 0.39894228 gives $C^{\text{lin}} = 300 \times 0.39894228 = 119.682684$, or 119.68. The exact value is $4{,}000 \times (2 \times 0.5149568299 - 1) = 4{,}000 \times 0.0299136598 = 119.654639$, or 119.65. The error of the second rule is $119.682684 - 119.654639 = 0.028045$, and the bound is $C^{\text{lin}}\,\sigma^2 T/24 = 119.682684 \times 0.005625/24 = 0.028051$: the error is inside the bound, and almost fills it, because for a small y the remainder is close to $-\varphi(0)\,y^3/6$. The data resolve the gap of 6×10^{-6}: a ten-decimal Φ fixes the exact value to within $4{,}000 \times 2 \times 5 \times 10^{-11} = 4 \times 10^{-7}$. The 0.4 rule is 0.35 too high, of which 0.32 comes from rounding 0.3989 up to 0.4.

(c) Here $\sigma\sqrt{T} = 0.32/\sqrt{52} = 0.32/7.2111 = 0.044376$. The straddle is two options, so the rule gives $0.8 \times 200 \times 0.044376 = 7.10$. The exact value is $2 \times 200 \times (2 \times 0.50885101 - 1) = 400 \times 0.01770202 = 7.08$. The gap of 0.02 is again the rounding of $2/\sqrt{2\pi} = 0.7979$ up to 0.8; the cubic term is below 0.001. Read through Corollary 20.30(i), with $r = 0$ the straddle is exactly the risk-neutral expected absolute move of the forward price over the week, $\mathbb{E}^{\mathbb{Q}}|F_T - F| = 7.08$, or 3.5% of 200.

(d) The second half of the identity needs no model: by parity, $\frac{1}{2}(C - \mathcal{P}) = \frac{1}{2}(S - Ke^{-rT}) = \frac{1}{2}(50 - 47.088) = 1.456$. The first half is replaced by $0.4\,S\sigma\sqrt{T} = 0.4 \times 50 \times 0.30 = 6.00$, where $e^{-rT}F = S$ on a stock without dividends. Hence

$$C \approx 6.00 + 1.46 = 7.46, \qquad \mathcal{P} \approx 6.00 - 1.46 = 4.54.$$

Exactly, $d_1 = (0.06 + 0.045)/0.30 = 0.35$ and $d_2 = 0.05$, so

$$C = 50 \times 0.63683 - 47.088 \times 0.51994 = 31.842 - 24.483 = 7.36, \qquad \mathcal{P} = 7.36 - 2.91 = 4.45.$$

Each estimate is 0.097 too high; the error is the same for both options because the half-carry term is exact. Its three sources are the following. First, 0.4 in the place of 0.39894228: $0.00105772 \times 15 = 0.016$. Second, the cubic term: the rule $0.39894228 \times 15 = 5.984$ against the exact at-the-money-forward half-straddle $50 \times (2 \times 0.55962 - 1) = 5.962$, a gap of 0.022. Third, the strike sits at the spot price and not at the forward price 53.09: the exact half-straddle at $K = S$ is $(7.3585 + 4.4468)/2 = 5.903$, which is 0.059 below 5.962. The three add up to $0.016 + 0.022 + 0.059 = 0.097$.

Self-check. A series gives the bounds and the numbers by another route. Integrating $e^{-z^2/2} = 1 - z^2/2 + z^4/8 - \dots$ term by term, $\chi(y) = \varphi(0)\,(y - y^3/6 + y^5/40 - \dots)$, an alternating series whose terms decrease for $0 < y < 1$; its value therefore lies between $\varphi(0)\,y$ and $\varphi(0)\,(y - y^3/6)$, which are the bounds of (a). For (b), with $y = 0.0375$: $4{,}000 \times 2 \times 0.39894228 \times (0.0375 - 0.0000088) = 3{,}191.54 \times 0.0374912 = 119.6546$, the exact value without a table; the next term, $y^5/40 = 1.9 \times 10^{-9}$, adds $3{,}191.54 \times 1.9 \times 10^{-9} = 6 \times 10^{-6}$, which is the gap between the error and the bound in (b). For (c), with $y = 0.022188$: $400 \times 2 \times 0.39894228 \times (0.022188 - 0.0000018) = 319.154 \times 0.0221862 = 7.08$. For (d), the put computed directly from Corollary 20.23, $\mathcal{P} = 47.088 \times \Phi(-0.05) - 50 \times \Phi(-0.35) = 47.088 \times 0.48006 - 50 \times 0.36317 = 22.605 - 18.159 = 4.45$, agrees with the value found by parity; and the half-carry from the series of e^{-rT}, $\frac{1}{2}S\,(rT - (rT)^2/2 + (rT)^3/6) = 25 \times 0.058236 = 1.456$, agrees with $\frac{1}{2}(50 - 47.088)$.

Answer: (a) $C = \mathcal{P} = e^{-rT}F\,[2\,\Phi(\sigma\sqrt{T}/2)-1]$, and $\varphi(0)\,y\,(1-y^2/6) \le \Phi(y)-\frac{1}{2} \le \varphi(0)\,y$ for $y \ge 0$, at $y = \sigma\sqrt{T}/2$, gives $C^{\text{lin}}(1-\sigma^2 T/24) \le C \le C^{\text{lin}}$, so $0 \le C^{\text{lin}} - C \le e^{-rT}F(\sigma\sqrt{T})^3/(24\sqrt{2\pi})$. (b) 120.00 (the 0.4 rule), 119.68 (the rule with $1/\sqrt{2\pi}$) and 119.65 (exact); the error 0.028045 is inside the bound 0.028051. (c) About 7.10, against 7.08 exactly. (d) $C \approx 7.46$ and $\mathcal{P} \approx 4.54$, against 7.36 and 4.45; the error 0.097 is 0.016 (the rounding of 0.3989 to 0.4) plus 0.022 (the cubic term) plus 0.059 (the strike at the spot price and not at the forward price).

P20.7 *Implied volatility by Newton's method*

(a) The start is $\sigma^{(0)} = 2.506628 \times 30/100 = 0.7520$. At the money forward $d_1 = \sigma\sqrt{T}/2$ and $d_2 = -d_1$, so here $C^{\text{BSM}}(\sigma) = 100\,(2\,\Phi(\sigma/2)-1)$ and $\partial_\sigma C^{\text{BSM}}(\sigma) = 100\,\varphi(\sigma/2)$ (Propositions 20.29 and 20.35).

First step. $C^{\text{BSM}}(0.7520) = 100\,(2 \times 0.6465416 - 1) = 29.3083$ and the vega is 37.1715, so

$$\sigma^{(1)} = 0.7520 + \frac{30 - 29.3083}{37.1715} = 0.7520 + 0.0186 = 0.7706.$$

Second step. $C^{\text{BSM}}(0.7706) = 100\,(2 \times 0.6499924 - 1) = 29.99848$ and the vega is 37.0402, so

$$\sigma^{(2)} = 0.7706 + \frac{30 - 29.99848}{37.0402} = 0.7706 + 0.000041 = 0.770641.$$

The exact value. At the money forward the formula inverts in closed form: $30 = 100\,(2\,\Phi(\sigma_{\text{imp}}/2)-1)$ gives $\Phi(\sigma_{\text{imp}}/2) = 0.65$ and $\sigma_{\text{imp}} = 2\,\Phi^{-1}(0.65) = 0.770641$, or 77.06%. Two steps reproduce it to six decimals.

The iterates rise: $0.7520 < 0.7706 < 0.770641$, and none passes the root. At the money forward the price is concave in σ (the vega falls, from 37.17 to 37.04 here), and the start is the point where the linear rule, the tangent at $\sigma = 0$, meets the level of the quote; that tangent lies above the concave curve, so it meets the level to the left of the root. Every later Newton tangent also lies above the concave curve, so every step stops short of the root: the iterates increase monotonically to σ_{imp} (Proposition 20.38(iii)).

(b) The start is $\sigma^{(0)} = 2.506628 \times 3.50/(100 \times 0.70711) = 0.1241$. With the tabulated values,

$$\sigma^{(1)} = 0.1241 + \frac{3.50 - 0.8916}{19.30} = 0.1241 + 0.1352 = 0.2593,$$
$$\sigma^{(2)} = 0.2593 + \frac{3.50 - 4.1448}{26.57} = 0.2593 - 0.0243 = 0.2350,$$
$$\sigma^{(3)} = 0.2350 + \frac{3.50 - 3.5053}{26.04} = 0.2350 - 0.0002 = 0.2348.$$

The last correction is 0.0002. The vega rises by $(26.57 - 26.04)/0.0243 = 21.8$ per unit of volatility between the last two tabulated points, so by the error identity (7.5) in the proof of Theorem 7.42 the next correction is about $0.0002^2 \times 21.8/(2 \times 26.04) = 2 \times 10^{-8}$, far below the rounding unit. The implied volatility is therefore 0.2348, or 23.48%. The first step *overshoots*: it jumps from 0.1241, far below the root, to 0.2593, above it. The start is built for an at-the-money option and is much too low for a strike 8% above the forward price; and on this stretch the price bends upward in σ: the tabulated vegas increase with σ (2.225, 19.30, 26.04, 26.57 at 0.05, 0.1241, 0.2350, 0.2593), so the tangent at 0.1241 lies below the curve and reaches the level of the quote beyond the root. The monotone guarantee of (a) is lost. From 0.2593 the iterates come down to the root from above, each step stopping just short of it.

The fixed guess. From $\sigma = 0.05$ the step is

$$0.05 + \frac{3.50 - 0.0146}{2.225} = 0.05 + 1.5665 = 1.6165,$$

a volatility of 162%, seven times the root: at 0.05 the option is so far out of the money in volatility units that its vega is a twelfth of the vega near the root, and dividing the price error by it sends

the iterate far away. A safeguard keeps a bracket $[\sigma', \sigma'']$ with $C^{\mathrm{BSM}}(\sigma') < C^{\mathrm{mkt}} < C^{\mathrm{BSM}}(\sigma'')$. The table supplies one, $[0.0500, 0.2593]$, because $0.0146 < 3.50 < 4.1448$. The step to 1.6165 falls outside it, so it is rejected and replaced by the midpoint 0.1547 (a bisection step); the root stays bracketed, and Newton's steps resume once they land inside the bracket (step 4 of Procedure 20.39).

(c) None. Here $e^{-rT}F = 100$, and the formula is strictly increasing in σ with the limit $e^{-rT}F$ as $\sigma \to \infty$, so $C^{\mathrm{BSM}}(\sigma) < 100$ for every σ: no volatility reproduces 101 (Proposition 20.38(ii)). The quote is an arbitrage. Sell the call for 101, enter one long forward at 100, which costs nothing, and deposit 100 (at $r = 0$ it stays 100); keep the remaining 1. At T the forward and the deposit are worth F_T, and the call costs $(F_T - 100)^+$, so the position is worth $F_T - (F_T - 100)^+ = \min(F_T, 100) \geq 0$, because $F_T > 0$. The trade locks in at least 1 with no risk.

Self-check. For (a), the size of the first correction can be predicted without running it. Expanding the price to second order about $\sigma^{(0)}$, as in the proof of Theorem 7.42, the error after one Newton step is about $(\sigma_{\mathrm{imp}} - \sigma^{(0)})^2\,|\partial_{\sigma\sigma}C^{\mathrm{BSM}}|/(2\,\partial_\sigma C^{\mathrm{BSM}})$. Here $\partial_{\sigma\sigma}C^{\mathrm{BSM}}(\sigma) = -100 \times \frac{1}{2} \times (\sigma/2)\,\varphi(\sigma/2)$, which is $-50 \times 0.38 \times 0.37115 = -7.05$ at $\sigma = 0.76$, so the predicted error is $0.01864^2 \times 7.05/(2 \times 37.17) = 0.000033$; the actual one, against the unrounded first iterate $0.7520 + 0.69168/37.1715 = 0.7520 + 0.018608 = 0.770608$, is $0.770641 - 0.770608 = 0.000033$. For (b), a secant through the two tabulated points nearest the root, which uses no vega, gives $0.2350 + (3.50 - 3.5053) \times (0.2593 - 0.2350)/(4.1448 - 3.5053) = 0.2350 - 0.0002 = 0.2348$; and the average slope between those points, $0.6395/0.0243 = 26.3$, lies between the two tabulated vegas 26.04 and 26.57, as the mean value theorem requires when the vega increases on that stretch. For (c), even an absurd volatility stays below the bound: at $\sigma = 10$ the formula gives $100\,(2\,\Phi(5) - 1) = 99.99994 < 100 < 101$. The model-free payoff comparison $(F_T - K)^+ \leq F_T$ gives $C^{\mathrm{mkt}} \leq e^{-rT}F = 100$, which 101 violates.

Answer: (a) $\sigma^{(0)} = 0.7520$, $\sigma^{(1)} = 0.7706$, $\sigma^{(2)} = 0.770641 = 2\,\Phi^{-1}(0.65)$; the iterates rise monotonically to the root, because the at-the-money price is concave in σ and the start lies below the root. (b) $0.1241 \to 0.2593$ (an overshoot) $\to 0.2350 \to 0.2348$, so $\sigma_{\mathrm{imp}} = 23.48\%$; from the fixed guess 0.05 the step lands at 1.6165, which a safeguard rejects, taking the midpoint of the bracket $[0.0500, 0.2593]$ instead. (c) The quote 101 exceeds $e^{-rT}F = 100$, so it has no implied volatility: sell the call, hold the forward and a deposit of 100, and keep at least 1.

P20.8 *Index and currency options, and the forward implied by parity*

(a) The forward price is $F = Se^{(r-q)T} = 5{,}200\,e^{0.0075} = 5{,}239.15$, and $\sigma\sqrt{T} = 0.08$, so

$$d_1 = \frac{\ln(F/K)}{\sigma\sqrt{T}} + \frac{\sigma\sqrt{T}}{2} = \frac{0.0269181}{0.08} + 0.04 = 0.376476, \qquad d_2 = d_1 - 0.08 = 0.296476.$$

The discounted index level is $Se^{-qT} = 5{,}180.54$ and the discounted strike is $Ke^{-rT} = 5{,}042.95$ (the factors are $e^{-0.00375}$ and $e^{-0.01125}$; the display below carries two more decimals). By Corollary 20.41(i),

$$C = 5{,}180.5365 \times 0.6467185 - 5{,}042.9465 \times 0.6165667 = 3{,}350.3488 - 3{,}109.3129 = 241.04,$$

$$\mathcal{P} = 5{,}042.9465 \times 0.3834333 - 5{,}180.5365 \times 0.3532815 = 1{,}933.6336 - 1{,}830.1877 = 103.45.$$

The lower bound of the call is $Se^{-qT} - Ke^{-rT} = 5{,}180.54 - 5{,}042.95 = 137.59$ (Proposition 17.6(iii)), which is $e^{-rT}(F - K)$. The call exceeds it by $241.04 - 137.59 = 103.45$, exactly the put, as parity requires.

(b) Parity in forward form, $C - \mathcal{P} = e^{-rT}(F - K)$, contains no volatility and no dividend forecast. Read backwards (Remark 20.46),

$$F = K + e^{rT}(C - \mathcal{P}) = 5{,}100 + 1.011314 \times 137.59 = 5{,}100 + 139.15 = 5{,}239.15,$$

and since $F = Se^{(r-q)T}$,

$$q = r - \frac{\ln(F/S)}{T} = 4.5\% - \frac{\ln(5{,}239.15/5{,}200)}{0.25} = 4.5\% - \frac{0.007501}{0.25} = 4.5\% - 3.0\% = 1.5\%.$$

(c) A foreign deposit is an asset with the yield r_{for}, so the formulas are those of (a) with $q = r_{\text{for}}$ (Corollary 20.43). The forward rate is $F = 0.68\, e^{(0.045-0.02)\times 0.75} = 0.68\, e^{0.01875} = 0.692870$, below the strike, so that $\ln(F/K) = -0.0102375$, and $\sigma\sqrt{T} = 0.11 \times 0.866025 = 0.095263$. Hence

$$d_1 = \frac{-0.0102375}{0.095263} + 0.047631 = -0.107466 + 0.047631 = -0.059835,$$
$$d_2 = d_1 - 0.095263 = -0.155098.$$

Hence $\Phi(d_1) = 1 - 0.523856 = 0.476144$ and $\Phi(d_2) = 1 - 0.561628 = 0.438372$. With $Se^{-r_{\text{for}}T} = 0.68\, e^{-0.015} = 0.669876$ and $Ke^{-rT} = 0.70\, e^{-0.03375} = 0.676769$,

$$C = 0.669876 \times 0.476144 - 0.676769 \times 0.438372 = 0.318957 - 0.296677 = 0.02228,$$
$$\mathcal{P} = 0.676769 \times 0.561628 - 0.669876 \times 0.523856 = 0.380092 - 0.350919 = 0.02917,$$

in units of domestic currency per unit of foreign currency. Parity holds: $C - \mathcal{P} = -0.00689 = 0.669876 - 0.676769$.

(d) For every $S_T > 0$,

$$(K - S_T)^+ = K\, S_T \Big(\frac{1}{S_T} - \frac{1}{K}\Big)^+,$$

because $KS_T > 0$, $c\, x^+ = (cx)^+$ for every $c > 0$, and $KS_T(1/S_T - 1/K) = K - S_T$. A call on one unit of domestic currency struck at $1/K$ pays $(1/S_T - 1/K)^+$ units of foreign currency, each worth S_T units of domestic currency at T; so K such calls pay $(K - S_T)^+$ in domestic currency, the payoff of the put. Two positions with the same payoff have the same price (Lemma 16.8(iii)).

For the foreign investor the exchange rate is $1/S = 1.470588$, the strike is $1/K = 1.428571$, the discounting rate is r_{for} and the yield of the underlying is r; the volatility is the same σ: if $S_t = S_0 \exp\big((\mu - \frac{1}{2}\sigma^2)t + \sigma W_t\big)$, with μ the real-world drift of (A1), then $1/S_t = (1/S_0)\exp\big((\sigma^2 - \mu - \frac{1}{2}\sigma^2)t + \sigma(-W_t)\big)$, and $-W$ is a standard Brownian motion, so (A1) holds for the exchange rate seen from abroad, with the drift $\sigma^2 - \mu$, which no price uses, and the traded instrument is a domestic deposit. The numbers d_1 and d_2 of this call are

$$d_1' = \frac{\ln(K/S) + (r_{\text{for}} - r + \sigma^2/2)T}{\sigma\sqrt{T}} = -d_2 = 0.155098, \qquad d_2' = -d_1 = 0.059835,$$

and Corollary 20.43 gives, in foreign currency,

$$\begin{aligned}\frac{1}{S}\, e^{-rT}\, \Phi(d_1') - \frac{1}{K}\, e^{-r_{\text{for}}T}\, \Phi(d_2') \\ &= 1.470588 \times 0.966813 \times 0.561628 - 1.428571 \times 0.985112 \times 0.523856 \\ &= 0.798513 - 0.737224 = 0.061289.\end{aligned}$$

So K calls cost $0.70 \times 0.061289 = 0.042902$ units of foreign currency, that is, $0.042902 \times 0.68 = 0.02917$ units of domestic currency: the price of the put. In symbols, $KS\big[S^{-1}e^{-rT}\,\Phi(-d_2) - K^{-1}e^{-r_{\text{for}}T}\,\Phi(-d_1)\big] = Ke^{-rT}\,\Phi(-d_2) - Se^{-r_{\text{for}}T}\,\Phi(-d_1) = \mathcal{P}$.

Self-check. For (a), start from the at-the-money-forward call and move the strike. At $K = F$ the call is worth $e^{-rT}F\,[2\,\Phi(0.04) - 1] = 5{,}180.54 \times 0.031907 = 165.29$, with $\Phi(0.04) = 0.515953$. The slope of the call price in the strike is $-e^{-rT}\,\Phi(d_2)$ (minus the digital: Proposition 23.2 in §23 with Theorem 20.17): $-0.988813 \times 0.484047 = -0.4786$ at $K = F$ and $-0.988813 \times 0.616567 = -0.6097$ at $K = 5{,}100$. Lowering the strike by 139.15 at the mean of the two end slopes, -0.54415, adds 75.72, which gives 241.01, within 0.03 of the formula (the slope is not exactly linear in the strike). For (c), the forward route gives the same parity value as the spot route: $e^{-rT}(F - K) = 0.966813 \times (0.692870 - 0.70) = -0.00689$. And the half-straddle $(0.02228 + 0.02917)/2 = 0.0257$ is close to the at-the-money estimate $0.4\, Se^{-r_{\text{for}}T}\,\sigma\sqrt{T} = 0.4 \times 0.669876 \times 0.095263 = 0.0255$, as it should be for a strike within about 1% of the forward rate.

Answer: (a) $C = 241.04$ and $\mathcal{P} = 103.45$; the lower bound of the call is $Se^{-qT} - Ke^{-rT} = 137.59 = C - \mathcal{P}$. (b) $F = K + e^{rT}(C - \mathcal{P}) = 5{,}239.15$ and $q = r - \ln(F/S)/T = 1.5\%$. (c) $C = 0.02228$ and $\mathcal{P} = 0.02917$ per unit of foreign currency. (d) $(K - S_T)^+ = KS_T(1/S_T - 1/K)^+$; the call on the domestic currency costs 0.061289 in foreign currency, and $K \times 0.061289 \times S = 0.02917$.

P20.9 *Discrete dividends and the American call*

(a) The present value of the dividends is

$$I = 0.80\, e^{-0.05\times 2/12} + 0.80\, e^{-0.05\times 8/12} = 0.7934 + 0.7738 = 1.567,$$

so $S - I = 43.433$ and the forward price is $F = (S - I)e^{rT} = 43.433 \times 1.042547 = 45.28$ (Proposition 16.13). With $\sigma\sqrt{T} = 0.30 \times 0.912871 = 0.27386$,

$$d_1 = \frac{\ln(43.433/45) + (0.05 + 0.045) \times 10/12}{0.27386} = \frac{-0.03545 + 0.07917}{0.27386} = 0.1596,$$
$$d_2 = d_1 - 0.27386 = -0.1142$$

(the supplied values are at $d_1 = 0.15965$ and at $-d_2 = 0.11422$, computed from unrounded inputs; $\Phi(d_2) = 1 - \Phi(-d_2)$), and $Ke^{-rT} = 45 \times 0.959189 = 43.164$. By Corollary 20.41(ii), with $\Phi(d_2) = 1 - 0.545468 = 0.454532$,

$$C = 43.433 \times 0.563422 - 43.164 \times 0.454532 = 24.471 - 19.619 = 4.85,$$

and by parity $\mathcal{P} = C - (S - I - Ke^{-rT}) = 4.852 - 0.269 = 4.58$. The dividends lower the call by $5.77 - 4.85 = 0.92$ and raise the put by $4.58 - 3.94 = 0.64$. Nothing is paid to, or withheld from, the option holder: the dividends lower the forward price, from $45\, e^{rT} = 46.91$ to 45.28, and a call pays less, and a put more, on a lower forward price. By parity the two changes add up to $I = 1.567$ (up to the rounding of the four prices).

(b) Before the first ex-dividend date: $K\big(1 - e^{-r(t_2 - t_1)}\big) = 45\,(1 - e^{-0.025}) = 1.111 > 0.80$, so exercising just before t_1 gains nothing; the interest on the strike until the next ex-date is worth more than the dividend captured. Before the second: $K\big(1 - e^{-r(T - t_2)}\big) = 45\,(1 - e^{-0.05\times 2/12}) = 0.373 < 0.80$, so exercise just before t_2 cannot be ruled out (Proposition 17.20). Between ex-dates exercise is never optimal when $r > 0$, so the instant just before t_2 is the only date at which early exercise can pay.

(c) For the call expiring just before t_2, only D_1 falls in its life: it is a call on $45 - 0.7934 = 44.207$ struck at 45 with maturity 8/12 (Corollary 20.41(ii) with the horizon t_2 in the place of T). With $\sigma\sqrt{t_2} = 0.30 \times 0.816497 = 0.24495$ and $Ke^{-rt_2} = 45 \times 0.967216 = 43.525$,

$$d_1 = \frac{\ln(44.207/45) + 0.095 \times 8/12}{0.24495} = \frac{-0.01779 + 0.06333}{0.24495} = 0.1859,$$
$$d_2 = d_1 - 0.24495 = -0.0590$$

(0.18594 and -0.05901 from unrounded inputs), and its price is $44.207 \times 0.573754 - 43.525 \times 0.476472 = 25.364 - 20.738 = 4.626$, or 4.63 (4.6254 from unrounded inputs). Black's approximation of the American call is $\max(4.85, 4.63) = 4.85$.

(d) Write I_t for the present value at t of the dividends still to be paid before T, so that the escrowed part $S_t - I_t$ is the geometric Brownian motion of the model with horizon T; it starts at $S - I = 43.433$. Just before t_2 the stock is the escrowed part plus the dividend about to be paid, $S_{t_2^-} = (S_{t_2} - I_{t_2}) + D_2$. Exercise just before t_2 therefore pays $\big((S_{t_2} - I_{t_2}) + D_2 - K\big)^+$: a call on the escrowed part struck at $K - D_2 = 44.20$, with maturity 8/12. The escrowed part is the price of a traded position that pays no income (the share and the loans of the proof of Corollary 20.41(ii)), so Corollary 20.23 prices this call with that position in the role of the stock and t_2 in the role of

T. With $(K - D_2)e^{-rt_2} = 44.20 \times 0.967216 = 42.751$,

$$d_1 = \frac{\ln(43.433/44.20) + 0.06333}{0.24495} = \frac{-0.01751 + 0.06333}{0.24495} = 0.1871,$$
$$d_2 = d_1 - 0.24495 = -0.0579$$

(0.18708 and -0.05787 from unrounded inputs), and its price is $43.433 \times 0.574201 - 42.751 \times 0.476926 = 24.939 - 20.389 = 4.55$. The answer does not change: $\max(4.85, 4.55) = 4.85$. The two readings differ by 0.08 because the model of the call's own life lets the volatility act on 44.21, which includes the present value of D_2, and the single model lets it act on 43.43. In the single model both European prices are lower bounds on the American price (each is the value of one exercise policy; Proposition 17.6(i) with $T' = t_2^-$ and with $T' = T$), so 4.85 is a lower bound there; the two-model number 4.63 carries no such guarantee (Remark 20.48).

Self-check. The option is almost at the money forward ($F = 45.28$ against $K = 45$), so the mental rule applies: the half-straddle is about $0.4\,(S - I)\,\sigma\sqrt{T} = 0.4 \times 43.433 \times 0.27386 = 4.758$, and half the forward value is $\frac{1}{2}(S - I - Ke^{-rT}) = 0.135$, which gives $C \approx 4.758 + 0.135 = 4.89$ and $\mathcal{P} \approx 4.758 - 0.135 = 4.62$, each 0.04 above the formula. Of that, 0.027 is the upward bias of the rule at $K = F$: the 0.4 rule exceeds $C^{\text{lin}} = 4.745$, which exceeds the exact at-the-money-forward value 4.730 by at most $C^{\text{lin}}\sigma^2 T/24 = 0.015$ (Proposition 20.29); the other 0.013 is the fall of the half-straddle from $K = F = 45.28$ to $K = 45$. The put formula of Corollary 20.41(ii), used directly, gives $\mathcal{P} = Ke^{-rT}\,\Phi(-d_2) - (S - I)\,\Phi(-d_1) = 43.164 \times 0.545468 - 43.433 \times 0.436578 = 23.545 - 18.962 = 4.58$, the value found by parity. For (d), the single-model call is the call of (c) with the underlying lowered by the present value of D_2, 0.7738, and the strike lowered by $D_2 = 0.80$. To first order the call of (c) moves by $\Phi(0.18594) = 0.5738$ per unit of underlying (its delta, Proposition 21.2) and by $-e^{-rt_2}\,\Phi(-0.05901) = -0.967216 \times 0.476472 = -0.4609$ per unit of strike (minus the digital: Proposition 23.2 with Theorem 20.17). The change is $-0.5738 \times 0.7738 + 0.4609 \times 0.80 = -0.4440 + 0.3687 = -0.0753$, and $4.6254 - 0.0753 = 4.5501$, the 4.55 of (d): the call gains less from the lower strike than it loses from the lower underlying.

Answer: (a) $I = 1.567$, $F = 45.28$, $C = 4.85$ and $\mathcal{P} = 4.58$, against 5.77 and 3.94 without dividends: the call loses 0.92 and the put gains 0.64, because the dividends lower the forward price. (b) $1.111 > 0.80$: no exercise before t_1; $0.373 < 0.80$: exercise just before t_2 cannot be ruled out. (c) $\max(4.85, 4.63) = 4.85$. (d) A call on $S - I$ struck at $K - D_2 = 44.20$ with maturity 8/12, worth 4.55; the approximation is still 4.85.

P20.10 *Black's formula and a futures-option parity arbitrage*

(a) For European futures options, parity reads $C - \mathcal{P} = e^{-rT}(F - K)$ (Theorem 20.52(iii); it is model-free, Proposition 20.54(i)). Here

$$e^{-rT}(F - K) = e^{-0.05} \times 2 = 0.951229 \times 2 = 1.9025,$$

while the quotes give $5.10 - 3.00 = 2.10$: the call is dear relative to the put by $2.10 - 1.9025 = 0.1975$.

Sell the dear side and buy the cheap one. Sell the call, buy the put, enter one long forward for delivery at T at the price 60, and deposit 1.9025. With deterministic rates the forward price is the futures price (Theorem 16.28), and the forward costs nothing to enter.

Trade	today	T, if $F_T \le 58$	T, if $F_T > 58$
Sell the call	$+5.10$	0	$-(F_T - 58)$
Buy the put	-3.00	$58 - F_T$	0
Long forward at 60	0	$F_T - 60$	$F_T - 60$
Deposit 1.9025	-1.9025	$+2.00$	$+2.00$
Net	$+0.1975$	0	0

The short call and the long put together pay $58 - F_T$ in every state, the forward pays $F_T - 60$, and the deposit pays the missing 2. The position leaves 0.198 today and nothing to pay at T: an arbitrage. If the hedge is held in the futures contract itself, hold $e^{-r(T-s)}$ contracts during each day, where s is the settlement time at the end of that day, and carry each daily settlement at the rate r to T: each day's settlement grows to that day's change in the futures price, and the changes telescope to $F_T - 60$. This is the rolling position of the proof of Theorem 16.28 divided by the growth factor e^{rT} of the cash account over the year.

(b) By Proposition 20.38(ii), which holds for options on a futures price, the implied volatility of the put quote 3.00 is the implied volatility of the call price $3.00 + e^{-rT}(F - K) = 3.00 + 1.9025 = 4.9025$. The formula is strictly increasing in σ (part (i) of that proposition), so the call quote and the put quote have the same implied volatility exactly when $5.10 = 4.9025$, that is, exactly when the quotes satisfy parity; for options on the underlying itself this is Proposition 22.3 in §22. Here the implied volatilities are 18.24% and 17.34%, so the two quotes cannot both be arbitrage-free: the gap of 0.90 volatility points is the parity violation of (a) in other units. At the call's implied volatility the formula prices the put at $5.10 - 1.9025 = 3.1975$, so against the call the put is 0.1975 too cheap; at the put's implied volatility it prices the call at $3.00 + 1.9025 = 4.9025$, so against the put the call is 0.1975 too dear.

(c) At $K = F = 60$, $d_1 = \sigma\sqrt{T}/2 = 0.125$ and $d_2 = -0.125$, so (Theorem 20.52(iii) with Proposition 20.29)

$$C = \mathcal{P} = e^{-rT} F \left[2\,\Phi(0.125) - 1\right] = 0.951229 \times 60 \times 0.099476 = 5.68.$$

The colleague is wrong. By parity $C - \mathcal{P} = e^{-rT}(F - K) = 0$ at $K = F$, whatever the carry of the commodity: the contango is already inside the futures price 60, and the options are written on that price.

Self-check. For (a), read the forward price off the quotes instead: $F = K + e^{rT}(C - \mathcal{P}) = 58 + 1.051271 \times 2.10 = 60.208$, against 60 in the futures market. Buying the forward at 60 and selling it synthetically at 60.208 locks in 0.208 at T, which is $0.208 \times 0.951229 = 0.198$ today, the same profit. For (c), the linear rule $C^{\text{lin}} = e^{-rT} F\,\sigma\sqrt{T}/\sqrt{2\pi} = 57.074 \times 0.25 \times 0.398942 = 5.6923$ exceeds the exact price 5.67750 (computed with the unrounded $\Phi(0.125) = 0.5497382$) by 0.01479, just inside the bound $C^{\text{lin}}\,\sigma^2 T/24 = 5.6923 \times 0.0625/24 = 0.01482$ (Proposition 20.29), and the 0.4 rule gives 5.71.

Answer: (a) Parity requires $C - \mathcal{P} = e^{-0.05} \times 2 = 1.9025$, against the quoted 2.10. Sell the call, buy the put, go long one forward at 60 and deposit 1.9025: the position pays nothing at T in every state and leaves 0.198 today. (b) One strike and one expiry must carry one implied volatility; 18.24% against 17.34% is the same parity violation. (c) $C = \mathcal{P} = 5.68$; the call and the put are equal at $K = F$ whatever the carry.

P20.11 *When to exercise an American futures call*

(a) The European call pays $(F_T - K)^+$, which is at least $F_T - K$ and at least 0. The claim that pays F_T is worth $e^{-rT}F$ and a deposit that pays K is worth Ke^{-rT}, so by dominance $C \geq e^{-rT}(F - K)^+$. The American call has a nonnegative payoff under every exercise policy, so $C^{\text{am}} \geq 0$, and it can be exercised at once for $F - K$, so $C^{\text{am}} \geq F - K$ (otherwise buy it and exercise): together $C^{\text{am}} \geq (F - K)^+$. For $F > K$ and $r > 0$ the European bound lies *below* the exercise value, here $e^{-0.03} \times 20 = 19.41 < 20$, whereas for a call on a stock without dividends the bound $S - Ke^{-rT}$ lies above $S - K$. By parity $C = e^{-rT}(F - K) + \mathcal{P}$, so

$$C < F - K \iff \mathcal{P} < (1 - e^{-rT})(F - K),$$

which is $\mathcal{P} < 0.029554 \times 20 = 0.591$ here (Proposition 20.54(i)).

(b) The discount factor is $e^{-0.03} = 0.970446$ and $\ln(F/K) = \ln 1.2 = 0.182322$.

At $\sigma = 25\%$: $\sigma\sqrt{T} = 0.176777$, $d_1 = (0.182322 + 0.015625)/0.176777 = 1.11975$ and $d_2 = 0.94298$, so

$$C = 0.970446 \times \big(120 \times 0.868590 - 100 \times 0.827154\big) = 0.970446 \times 21.5154 = 20.88 > 20.$$

Since $C^{\text{am}} \geq C > F - K$, the American call is worth more alive than exercised: immediate exercise is ruled out.

At $\sigma = 10\%$: $\sigma\sqrt{T} = 0.070711$, $d_1 = (0.182322 + 0.0025)/0.070711 = 2.61377$ and $d_2 = 2.54306$, so

$$C = 0.970446 \times \big(120 \times 0.995523 - 100 \times 0.994506\big) = 0.970446 \times 20.0122 = 19.42 < 20.$$

Holding to expiry is worth less than exercising now, so immediate exercise is not ruled out, and $C^{\text{am}} \geq 20 > C$: the early-exercise premium is at least $20 - 19.421 = 0.579$. The comparison does not by itself say that today is the best date to exercise; it says that the policy "never exercise before T" is beaten.

(c) By (a), the European call equals $F - K$ exactly when $\mathcal{P} = (1 - e^{-rT})(F - K)$. At $F = 126.87$: $0.029554 \times 26.87 = 0.794$, the put at $\sigma = 25\%$. At $F = 110.00$: $0.029554 \times 10.00 = 0.296$, the put at $\sigma = 10\%$. As F rises the put falls (its payoff $(K - F_T)^+$ falls, because F_T is F times a factor that does not depend on F) and $(1 - e^{-rT})(F - K)$ rises, so the two cross once: the European call is worth less than $F - K$ for every futures price above 126.9 at $\sigma = 25\%$, and above 110.0 at $\sigma = 10\%$.

For every σ: by Black's formula $\mathcal{P} = e^{-rT}\big[K\,\Phi(-d_2) - F\,\Phi(-d_1)\big] \leq e^{-rT}K\,\Phi(-d_2)$, and as $F \to \infty$, $d_2 \to +\infty$ and $\Phi(-d_2) \to 0$, so $\mathcal{P} \to 0$, while $(1 - e^{-rT})(F - K) \to \infty$ because $r > 0$. So for every F large enough, $C < F - K$. In such a state a policy that never exercises before T collects at most $(F_T - K)^+$ at T, the payoff of the European call, which costs C; the holder who exercises at once and buys the European call holds the same payoff and $F - K - C > 0$ in cash. Holding to expiry is therefore not optimal (Proposition 20.54(ii)). The higher the volatility, the deeper in the money the call must be: the put rises with σ (its vega is positive, Proposition 20.35) and falls with F, so the zero of $\mathcal{P} - (1 - e^{-rT})(F - K)$ moves up as σ rises, from 110.0 at 10% to 126.9 at 25%.

(d) Both bounds are proved by exercise matching, with $r \geq 0$ and a positive futures price: a positive inflow today followed by a nonnegative value under every exercise policy of the counterparty is an arbitrage (Lemma 17.5(ii)). Call the claim that pays F_T at T the *forward claim*; it is worth $e^{-r(T-t)}F_t$ at t.

Lower bound. Suppose $\mathcal{P}^{\text{am}} - C^{\text{am}} > K - Fe^{-rT}$. Write the put, buy the call, sell one forward claim short for Fe^{-rT} and deposit K: the inflow $\mathcal{P}^{\text{am}} - C^{\text{am}} + Fe^{-rT} - K$ is positive. If the put is exercised against the position at a time $t_e \leq T$, pay $K - F_{t_e}$ and take the long futures position, worth zero and closed at no cost; buy the forward claim back for $e^{-r(T-t_e)}F_{t_e} \leq F_{t_e}$; withdraw the deposit, worth $Ke^{rt_e} \geq K$. The cash left, at least $-(K - F_{t_e}) - F_{t_e} + K = 0$, is deposited until T, and the call is held to T and exercised there only if it is in the money, paying $(F_T - K)^+ \geq 0$. If the put is never exercised, at T the call pays $(F_T - K)^+ \geq F_T - K$, the forward claim costs F_T and the deposit holds Ke^{rT}: what is left is at least $K(e^{rT} - 1) \geq 0$. So $C^{\text{am}} - \mathcal{P}^{\text{am}} \geq Fe^{-rT} - K$.

Upper bound. Suppose $C^{\text{am}} - \mathcal{P}^{\text{am}} > F - Ke^{-rT}$. Write the call, buy the put, buy e^{rT} forward claims for $e^{rT} \times e^{-rT}F = F$ and borrow Ke^{-rT}: the inflow $C^{\text{am}} - \mathcal{P}^{\text{am}} - F + Ke^{-rT}$ is positive. If the call is exercised against the position at $t_e \leq T$, pay $F_{t_e} - K$ and take the short futures position, worth zero and closed at no cost; sell the forward claims for $e^{rT}e^{-r(T-t_e)}F_{t_e} = e^{rt_e}F_{t_e} \geq F_{t_e}$; repay the loan, $Ke^{-r(T-t_e)} \leq K$. The cash left, at least $-(F_{t_e} - K) + F_{t_e} - K = 0$, is deposited until T, and the put is held to T and exercised there only if it is in the money, paying $(K - F_T)^+ \geq 0$. If the call is never exercised, at T the put pays $(K - F_T)^+ \geq K - F_T$, the forward claims pay $e^{rT}F_T$ and the loan costs K: what is left is at least $(e^{rT} - 1)F_T \geq 0$. So $C^{\text{am}} - \mathcal{P}^{\text{am}} \leq F - Ke^{-rT}$.

Here the band is $120 \times 0.970446 - 100 = 16.45 \leq C^{\text{am}} - \mathcal{P}^{\text{am}} \leq 120 - 97.04 = 22.96$. The European difference $e^{-rT}(F - K) = 19.41$ lies inside it.

(e) The exercise values are $F_t - K$ for the futures call and $S_t - K$ for the spot call, $K - F_t$ for the futures put and $K - S_t$ for the spot put. With $F_t \geq S_t$ at every date, the futures call's exercise value is at least the spot call's at every date, and the futures put's is at most the spot put's. By exercise matching (if the option whose exercise value is the larger at every date were the cheaper one, buy it, write the other and deposit the difference; when the other is exercised at t_e, exercise the first at once, the net cash flow being $(F_{t_e} - K) - (S_{t_e} - K) = F_{t_e} - S_{t_e} \geq 0$ for the calls and $(K - S_{t_e}) - (K - F_{t_e}) = F_{t_e} - S_{t_e} \geq 0$ for the puts; if the other is never exercised, let the first lapse; Lemma 17.5(ii) then gives the inequality), the American futures call is worth at least the American spot call, and the American futures put at most the American spot put (Remark 20.56(ii)). In an inverted market, $F_t \leq S_t$, both inequalities reverse; for European options with the delivery date at T the two are equal, because $F_T = S_T$.

Self-check. As an arithmetic check, the prices of (b) follow from the puts by parity, using the complements $1 - \Phi$ of the same values: at $\sigma = 25\%$, $\mathcal{P} = 0.970446 \times (100 \times 0.172846 - 120 \times 0.131410) = 0.970446 \times 1.5154 = 1.471$ and $C = 19.409 + 1.471 = 20.88$; at $\sigma = 10\%$, $\mathcal{P} = 0.970446 \times (100 \times 0.005494 - 120 \times 0.004477) = 0.970446 \times 0.0122 = 0.012$ and $C = 19.409 + 0.012 = 19.42$. The test of (a) agrees: $1.471 > 0.591$ in the first case, $0.012 < 0.591$ in the second. Both prices lie in the model-free bracket $e^{-rT}(F - K)^+ \leq C < e^{-rT}F$ of Proposition 20.54(i), here $19.409 \leq C < 116.45$, and C rises with σ, $20.880 - 19.421 = 1.459 > 0$, as the positive vega of Proposition 20.35 requires. For (d), three limits: at $r = 0$ both ends of the band equal $F - K$, the parity of options that are never exercised early; deep in the money $C^{\text{am}} \approx F - K$ and $\mathcal{P}^{\text{am}} \approx 0$, and $F - K$ lies between $Fe^{-rT} - K$ and $F - Ke^{-rT}$; deep out of the money $C^{\text{am}} \approx 0$ and $\mathcal{P}^{\text{am}} \approx K - F$, which gives $F - K$ again. The width of the band is $(F + K)(1 - e^{-rT}) = 220 \times 0.029554 = 6.50$, and indeed $22.955 - 16.453 = 6.50$.

Answer: (a) $C \geq e^{-rT}(F - K)^+$ and $C^{\text{am}} \geq (F - K)^+$; the European bound 19.41 is below the exercise value 20, and $C < F - K$ exactly when $\mathcal{P} < (1 - e^{-rT})(F - K) = 0.591$. (b) $C = 20.88 > 20$ at 25%: immediate exercise is ruled out. $C = 19.42 < 20$ at 10%: it is not, and holding to expiry is beaten. (c) $(1 - e^{-rT})(F - K)$ equals the put at $F = 126.87$ ($\sigma = 25\%$) and at $F = 110.00$ ($\sigma = 10\%$), the crossover prices; for every σ the put tends to 0 as $F \to \infty$, so $C < F - K \leq C^{\text{am}}$ for F large enough, and holding to expiry is then not optimal. (d) Exercise matching with the claim paying F_T gives $Fe^{-rT} - K \leq C^{\text{am}} - \mathcal{P}^{\text{am}} \leq F - Ke^{-rT}$, here $[16.45, 22.96]$. (e) With $F_t \geq S_t$ the American futures call is worth at least the spot call, and the American futures put at most the spot put.

P20.12 *Nine claims about the Black–Scholes–Merton model: true or false*

(i) False. The risk-neutral probability of exercise is $\Phi(d_2) = 0.495$, below one half, not $\Phi(d_1) = 0.579$ (Proposition 20.26). The number $\Phi(d_1)$ is the exercise probability reweighted by S_T/F, and the number of shares in the hedge. The real-world probability is Φ at d_2 with μ in the place of r, which is 0.505 at $\mu = 5\%$ and 0.598 at $\mu = 15\%$: a third number again.

(ii) False. The expected return μ appears neither in the pricing equation (Theorem 20.3) nor in the formula: at fixed S and σ the call is worth 5.63 whether μ is 5% or 15%. The hedge removes the exposure to μ, and today's S already reflects it.

(iii) False. As $\sigma \to \infty$, $d_1 \to +\infty$ and $d_2 \to -\infty$, so $C \to e^{-rT}F = S$, the upper bound (Remark 20.28(3)). Here the call is worth 42.84 at $\sigma = 300\%$ and 59.98 at $\sigma = 1000\%$, against $S = 60$, while the risk-neutral probability of exercise falls to 0.147 and then to 0.0002: the holder almost never pays the strike and ends up with the value of the asset.

(iv) False. At $K = F$ the price is $e^{-rT}F\,[2\,\Phi(\sigma\sqrt{T}/2) - 1]$ (Proposition 20.29), which is strictly concave in σ (Proposition 20.38(iii)) and so less than doubles: with $y = \sigma\sqrt{T}/2$, $2\,[\Phi(y) - \frac{1}{2}] - [\Phi(2y) - \frac{1}{2}] = \int_0^y [\varphi(z) - \varphi(z+y)]\,dz > 0$, because φ is decreasing on $[0, \infty)$. Only the linear rule C^{lin} doubles exactly. On the stock above the at-the-money-forward strike is $K = F = 60\,e^{0.02} = 61.21$, where $e^{-rT}F = S = 60$: the call is worth 5.068 at 30% against 10.080 at 60%, a ratio of 1.989; and 16.580 at 100% against 31.230 at 200%, a ratio of 1.884.

(v) True. A dividend lowers the forward price, $F = (S - I)e^{rT}$, and the put pays more on a lower forward price; nothing is paid to the option holder (Corollary 20.41(ii)). With the dividend of 1.50 at $t = 0.25$, $I = 1.50\,e^{-0.01} = 1.485$, the forward price falls from 61.21 to 59.70, the put rises from 4.45 to 5.11 and the call falls from 5.63 to 4.81.

(vi) True. Parity for futures options reads $C - \mathcal{P} = e^{-rT}(F - K)$, which is zero at $K = F$ (Theorem 20.52(iii)). The carry of the underlying is already inside the futures price.

(vii) False whenever $r > 0$. Deep in the money the European futures call is worth about $e^{-rT}(F - K)$, less than the $F - K$ that exercise pays at once (Proposition 20.54). With $F = 90$, $K = 60$, $r = 4\%$, $T = 0.5$ and $\sigma = 30\%$, the put is worth 0.16 and, by parity, the European call $e^{-0.02} \times 30 + 0.16 = 29.41 + 0.16 = 29.57 < 30$.

(viii) False. For every σ the formula stays strictly below $e^{-rT}F$, which is S for a stock without dividends, so a call quoted above the stock price has no implied volatility. It is an arbitrage: sell the call and buy the share (Proposition 20.38(ii)).

(ix) False. The digital and the digital put together pay 1 at T, which is worth e^{-rT} today, so the digital put is worth $e^{-rT} - \mathrm{Dig}(K)$ (Corollary 20.20). Here $\mathrm{Dig}(60) = e^{-0.02} \times 0.4953 = 0.4855$ and the digital put is worth $0.9802 - 0.4855 = 0.4947$, not 0.5145.

Self-check. The numbers of (i), (iii), (iv), (v), (vii) and (ix) have a second route. In (i), by the mean value theorem $\Phi(d_1) - \Phi(d_2)$ is $\sigma\sqrt{T} = 0.2121$ times a value of φ between d_2 and d_1, about $\varphi(0.094) = 0.397$: $0.2121 \times 0.397 = 0.084 = 0.5794 - 0.4953$. In (iii), $e^{-rT}F - C = S\,\Phi(-d_1) + Ke^{-rT}\,\Phi(d_2)$ (Remark 20.28(5)); at 300% this is $60 \times 0.1423 + 58.81 \times 0.1466 = 8.54 + 8.62 = 17.16 = 60 - 42.84$. In (iv), the cubic correction of the at-the-money expansion reproduces the 60% price: $60 \times 2\,\varphi(0)\,(y - y^3/6)$ at $y = 0.2121$ is $47.87 \times 0.21054 = 10.08$. In (v), parity must move by the present value of the dividend: $C - \mathcal{P}$ goes from $S - Ke^{-rT} = 1.188$ to $1.188 - 1.485 = -0.297$, and indeed $4.81 - 5.11 = -0.30$. In (vii), the test $\mathcal{P} < (1 - e^{-rT})(F - K)$ of Proposition 20.54(i) reads $0.16 < 0.0198 \times 30 = 0.59$. In (ix), $0.4855 + 0.4947 = 0.9802 = e^{-0.02}$.

Answer: (i) False: the risk-neutral probability is $\Phi(d_2) = 0.495$; $\Phi(d_1) = 0.579$ is the reweighted probability and the hedge ratio. (ii) False: μ is not in the price. (iii) False: $C \to e^{-rT}F = S = 60$. (iv) False: the price is concave in σ (5.068 at 30% against 10.080 at 60%). (v) True: the dividend lowers the forward price (put $4.45 \to 5.11$). (vi) True: $C - \mathcal{P} = e^{-rT}(F - K) = 0$. (vii) False for $r > 0$ ($29.57 < 30$). (viii) False: no implied volatility exists above $e^{-rT}F$; the quote is an arbitrage. (ix) False: the digital put is $e^{-rT} - \mathrm{Dig}(K)$ (0.4947, not 0.5145).

28.21 Solutions — §21 The Greeks and Hedging

P21.1 *Hedging a written option book through a move*

(a) By Definition 21.1(i) the delta is $\Delta = \partial_S V$, the rate at which the value of the option changes with the price of the stock. The call gains about 0.37 when the stock rises by one point and loses about 0.37 when it falls by one point: over a small move it behaves like 0.37 of a share. (It is not the probability that the call is exercised.) The 18,000 written calls have the delta $-18{,}000 \times 0.37 = -6{,}660$. By Definition 21.1(iii) a written option is hedged by holding $+\Delta$ shares, so the block is delta-hedged by a purchase of 6,660 shares.

(b) Greeks add across positions (Definition 21.1(ii)):

$$\begin{aligned}\Delta_\Pi &= -18{,}000\,(0.37) - 9{,}000\,(-0.47) + 7{,}000\,(0.33) + 6{,}000\,(-0.23)\\ &= -6{,}660 + 4{,}230 + 2{,}310 - 1{,}380 = -1{,}500,\\ \Gamma_\Pi &= -18{,}000\,(0.031) - 9{,}000\,(0.033) + 7{,}000\,(0.021) + 6{,}000\,(0.018)\\ &= -558 - 297 + 147 + 108 = -600 \text{ per point.}\end{aligned}$$

A share has delta 1 and gamma 0 (Proposition 21.4(i) and (iv)), so the dealer buys 1,500 shares. The hedged book has delta 0, and its gamma is still -600.

(c) Gamma is the rate of change of the delta, so after the jump the delta of the hedged book is about $\Gamma_\Pi \, \Delta S = -600 \times 2 = -1{,}200$. The dealer buys 1,200 more shares, at 82, and then holds 2,700.

(d) By Proposition 21.14, with a zero delta, no passage of time and no change of volatility, the value of the hedged book changes by about

$$\tfrac{1}{2}\Gamma_\Pi \, (\Delta S)^2 = \tfrac{1}{2} \, (-600) \, (2)^2 = -1{,}200 :$$

a loss of about 1,200. Leg by leg, the options change in value by $-1{,}500 \times 2 + \frac{1}{2} \, (-600) \, (4) = -4{,}200$ and the 1,500 shares gain 3,000. After a fall to 78 the squared move is the same, so the loss is again about 1,200; only the re-hedge changes sign. The delta of the hedged book is then about $-600 \times (-2) = +1{,}200$, and the dealer sells 1,200 shares at 78. A book with negative gamma buys after a rise and sells after a fall, and it loses on a move in either direction.

Self-check. A round trip gives the loss by pure bookkeeping. Suppose that, after the re-hedge at 82, the stock returns to 80. The options and the first 1,500 shares are then worth what they were at the start, the delta of the book is back at $-1{,}500$, and the extra 1,200 shares are sold at 80. The whole profit and loss of the round trip is therefore the loss on these shares, $1{,}200 \times (80 - 82) = -2{,}400$: two gamma losses of 1,200, one for each two-point move, as (d) predicts. A full revaluation agrees. The Greeks of the table are Black–Scholes–Merton values at a volatility of 30% and a zero rate, rounded; repricing the four positions with that formula changes the value of the options by $-4{,}317$ at 82 and by $+1{,}888$ at 78, so the book hedged with 1,500 shares loses 1,317 and 1,112, which is 1,214 on average. (The two losses differ mainly because the unrounded delta of the book is $-1{,}547$: the 1,500 shares leave -47 of delta unhedged, worth -94 on the way up and $+94$ on the way down. The remaining 9 each way is the third-order term of the expansion: the gamma of the book grows in size as the stock rises, from -584 at 78 to -609 at 80 and -612 at 82.)

Answer: (a) The call gains or loses about 0.37 per one-point move of the stock; the written block has delta $-6{,}660$ and is hedged by buying 6,660 shares. (b) $\Delta_\Pi = -1{,}500$ and $\Gamma_\Pi = -600$ per point; buy 1,500 shares. (c) The delta is about $-1{,}200$; buy 1,200 more shares at 82 (2,700 held). (d) A loss of about $\frac{1}{2} \, (600) \, (2)^2 = 1{,}200$; the same loss after a fall to 78, where 1,200 shares are sold instead.

P21.2 *Deltas from the key identity, the at-the-money delta, and a digital near expiry*

(a) *The identity.* Since $d_1 - d_2 = \sigma\sqrt{\tau}$ and $d_1 + d_2 = 2 \, [\ln(S/K) + (r - q)\tau]/(\sigma\sqrt{\tau})$,

$$\tfrac{1}{2}(d_1^2 - d_2^2) = \tfrac{1}{2} \, (d_1 - d_2)(d_1 + d_2) = \ln(S/K) + (r - q)\tau.$$

Hence $\varphi(d_2)/\varphi(d_1) = e^{(d_1^2 - d_2^2)/2} = (S/K) \, e^{(r-q)\tau}$, so $Se^{-q\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$ (Lemma 20.34).

The call. We differentiate the price formula and cancel the density terms with the identity. Since $d_1 - d_2$ does not depend on S, $\partial_S d_1 = \partial_S d_2 = 1/(S\sigma\sqrt{\tau})$. The product and chain rules, with $\Phi' = \varphi$, give

$$\begin{aligned} \partial_S C &= e^{-q\tau}\Phi(d_1) + Se^{-q\tau}\varphi(d_1) \, \partial_S d_1 - Ke^{-r\tau}\varphi(d_2) \, \partial_S d_2 \\ &= e^{-q\tau}\Phi(d_1) + \frac{Se^{-q\tau}\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)}{S\sigma\sqrt{\tau}} = e^{-q\tau}\Phi(d_1), \end{aligned}$$

because the numerator of the fraction is zero by the identity. Since $0 < \Phi(d_1) < 1$, the call delta lies strictly between 0 and $e^{-q\tau}$.

The put, directly. Since $\varphi(-x) = \varphi(x)$, the chain rule gives $\partial_S\Phi(-d_2) = -\varphi(d_2) \, \partial_S d_2$ and $\partial_S\Phi(-d_1) = -\varphi(d_1) \, \partial_S d_1$, so

$$\partial_S \mathcal{P} = -Ke^{-r\tau}\varphi(d_2) \, \partial_S d_2 - e^{-q\tau}\Phi(-d_1) + Se^{-q\tau}\varphi(d_1) \, \partial_S d_1 = -e^{-q\tau}\Phi(-d_1),$$

the two density terms cancelling as before.

The put, from parity. Since $\Phi(x)+\Phi(-x)=1$, subtracting the two price formulas gives $C-\mathcal{P}=Se^{-q\tau}-Ke^{-r\tau}$. Differentiating in S, $\partial_S C-\partial_S\mathcal{P}=e^{-q\tau}$, so $\partial_S\mathcal{P}=e^{-q\tau}(\Phi(d_1)-1)=-e^{-q\tau}\Phi(-d_1)$, which lies strictly between $-e^{-q\tau}$ and 0 because $0<\Phi(-d_1)<1$. This is Proposition 21.2.

(b) Here $\sigma\sqrt{\tau}=0.30\sqrt{0.75}=0.2598$ and $\ln(S/K)=0$, so

$$d_1=\frac{(0.05+0.045)(0.75)}{0.2598}=\frac{0.07125}{0.2598}=0.2742,\qquad d_2=0.2742-0.2598=0.0144.$$

The call has $\Delta_C=\Phi(0.2742)=0.608$, the put has $\Delta_{\mathcal{P}}=0.608-1=-0.392$, and the risk-neutral probability of exercise is $\Phi(0.0144)=0.506$. For the strike $K=F=80e^{0.0375}=83.06$, the forward price, $\ln(F/K)=0$ and $d_1=\sigma\sqrt{\tau}/2=0.1299$, so the delta is $\Phi(0.1299)=0.552$. The delta is not the probability of exercise (0.608 against 0.506), and it is one half neither at the money spot nor at the money forward.

(c) The rule gives $0.5+0.4\,(0.2742)=0.610$ against 0.608, $0.5+0.4\,(0.0144)=0.506$ against 0.506, and $0.5+0.4\,(0.1299)=0.552$ against 0.552. It is the tangent of Φ at 0: $\Phi(0)=0.5$ and $\Phi'(0)=\varphi(0)=1/\sqrt{2\pi}=0.399$. Since $\varphi'(0)=0$, the first term the tangent neglects is of third order, $-\varphi(0)\,x^3/6$, which is -0.0014 at $x=0.274$. Rounding the slope 0.399 up to 0.4 adds $0.0011\,x$, here 0.0003, and the two together give the observed gap of 0.0016. The rule stays within 0.005 of Φ for $|x|$ up to about 0.4.

(d) With $\partial_S d_2=1/(S\sigma\sqrt{\tau})$, the delta of the digital (Theorem 20.17, with τ in the place of T) is

$$\partial_S\,\mathrm{Dig}(K)=\frac{e^{-r\tau}\varphi(d_2)}{S\sigma\sqrt{\tau}}.$$

Differentiating once more, with $\varphi'(x)=-x\varphi(x)$ and $d_2+\sigma\sqrt{\tau}=d_1$,

$$\partial_{SS}\,\mathrm{Dig}(K)=\frac{e^{-r\tau}}{\sigma\sqrt{\tau}}\Big(-\frac{d_2\,\varphi(d_2)}{S^2\sigma\sqrt{\tau}}-\frac{\varphi(d_2)}{S^2}\Big)=-\frac{e^{-r\tau}\varphi(d_2)\,d_1}{S^2\sigma^2\tau}.$$

The gamma is positive where $d_1<0$ and negative where $d_1>0$. Since $\partial_S d_1=1/(S\sigma\sqrt{\tau})>0$, the function d_1 increases strictly in S and vanishes at exactly one price, $S=Ke^{-(r-q+\sigma^2/2)\tau}$: the gamma is positive below that price and negative above it.

Numbers. Here $\sigma\sqrt{\tau}=0.3175/\sqrt{252}=0.0200$, $\sigma^2\tau=0.0004$, and the gamma changes sign at $50\,e^{-0.0002}=49.99$. At $S=50$: $d_1=0.01$, $d_2=-0.01$ and $\varphi(d_2)=0.3989$, so the delta is $0.3989/(50\times0.02)=0.399$ per unit of payout. At $S=49.5$: $\ln(49.5/50)=-0.01005$, so $d_1=-0.5025+0.01=-0.4925$, $d_2=-0.5125$, $\varphi(d_2)=0.3498$, and the gamma is

$$\frac{0.3498\times0.4925}{49.5^2\times0.0004}=\frac{0.1723}{0.9801}=+0.176.$$

At $S=50.5$: $\ln(50.5/50)=0.00995$, so $d_1=0.5075$, $d_2=0.4875$, $\varphi(d_2)=0.3542$, and the gamma is $-0.3542\times0.5075/(50.5^2\times0.0004)=-0.1798/1.0201=-0.176$. The ordinary call has the gamma $\varphi(d_1)/(S\sigma\sqrt{\tau})=0.3989/(50\times0.02)=0.399$ at $S=50$ (Proposition 21.6(i)): the delta of the digital at the strike equals the gamma of the call, because $\varphi(d_2)=\varphi(d_1)$ when $d_2=-d_1$. The delta of a digital has the shape of a gamma, and its own gamma swings from $+0.176$ to -0.176 within one point. For a payout of 1,000,000 the hedge at $S=50$ is 399,000 shares, worth about 20 times the payout.

Self-check. Bumping the inputs and repricing gives the same Greeks without any differentiation. In (b), the call is worth 9.9789 at $S=80.5$ and 9.3709 at $S=79.5$: the central difference is 0.6080, against $\Phi(d_1)=0.6081$. In (d), the digital is worth 0.4562 at $S=49.9$ and 0.5358 at $S=50.1$: the central difference is 0.398, against the delta 0.399. For the gamma, the digital is worth 0.269730, 0.304145 and 0.340313 at $S=49.4$, 49.5 and 49.6: the second difference $(0.269730-2\times0.304145+0.340313)/0.1^2=0.175$, against the gamma 0.176 at 49.5.

Answer: (a) $Se^{-q\tau}\varphi(d_1) = Ke^{-r\tau}\varphi(d_2)$ cancels the density terms: $\Delta_C = e^{-q\tau}\Phi(d_1) \in (0, e^{-q\tau})$, and $\Delta_{\mathcal{P}} = -e^{-q\tau}\Phi(-d_1) \in (-e^{-q\tau}, 0)$, both directly and from $\Delta_C - \Delta_{\mathcal{P}} = e^{-q\tau}$. (b) $d_1 = 0.274$, $\Delta_C = 0.608$, $\Delta_{\mathcal{P}} = -0.392$, $\Phi(d_2) = 0.506$; forward-struck delta 0.552. (c) The rule gives 0.610, 0.506 and 0.552; it is the tangent of Φ at 0, of slope $\varphi(0) = 0.399$. (d) Delta $e^{-r\tau}\varphi(d_2)/(S\sigma\sqrt{\tau})$, gamma $-e^{-r\tau}\varphi(d_2)\, d_1/(S^2\sigma^2\tau)$, which changes sign at $S = Ke^{-(r-q+\sigma^2/2)\tau} = 49.99$; delta 0.399 at 50, gamma +0.176 at 49.5 and −0.176 at 50.5; the call's gamma at 50 is 0.399.

P21.3 *Hedging with spot, forwards and futures; the Greeks of a futures option*

(a) Here $\sigma\sqrt{\tau} = 0.16\sqrt{1/3} = 0.09238$, $\ln(F/K) = \ln(2{,}400/2{,}450) = -0.02062$ and $\sigma^2\tau/2 = 0.00427$, so

$$d_1 = \frac{-0.02062 + 0.00427}{0.09238} = -0.1770, \qquad d_2 = -0.1770 - 0.0924 = -0.2694,$$

with $\Phi(d_1) = 0.429745$, $\Phi(d_2) = 0.393812$ and $e^{-r\tau} = e^{-0.01333} = 0.98676$. The call is worth

$$C = 0.98676\,(2{,}400 \times 0.429745 - 2{,}450 \times 0.393812) = 0.98676\,(1{,}031.39 - 964.84) = 65.67.$$

The futures call is the yield formula at $x = F$ and $q = r$ (Theorem 20.52(iii)), so Proposition 21.2 with these substitutions gives its delta with respect to the futures price,

$$\partial_F C = e^{-r\tau}\Phi(d_1) = 0.98676 \times 0.429745 = 0.4241.$$

The written position has the delta $-2{,}000 \times 0.4241 = -848.1$: it loses about 848 for each rise of one dollar in the six-month futures price.

(b) (i) A futures contract on one ounce is credited the change of its futures price at the daily settlement, so its delta with respect to its own futures price is 1: the dealer buys six-month futures on 848.1 ounces. For the other instruments, write the position as a function of the spot price. The six-month futures price is $F = Se^{r\tau_F}$ with $\tau_F = 0.5$ (Proposition 16.15(ii) with $q = 0$), so by the chain rule the delta of the position with respect to S is

$$\Delta_\Pi = -848.1 \times e^{0.02} = -865.2.$$

By Proposition 21.4 with $q = 0$, with τ_F now the delivery time of the hedging contract (0.5 for the six-month contracts, 1 for the one-year futures), the position is made delta-neutral by $-\Delta_\Pi$ units of the underlying, by $-\Delta_\Pi e^{q\tau_F}$ forward contracts or by $-\Delta_\Pi e^{-(r-q)\tau_F}$ futures contracts: (ii) 865.2 ounces of spot gold; (iii) $865.2\, e^{-0.04} = 831.3$ ounces of the one-year futures; (iv) 865.2 ounces of the six-month forward. The same formula returns (i): $865.2\, e^{-0.02} = 848.1$.

The two six-month contracts have the same price, $Se^{0.02}$, and both prices move by $e^{0.02}$ when the spot moves by 1. The gain on the futures is paid at once, through the daily settlement, so the futures has the delta $e^{0.02} = 1.0202$ per ounce. The gain on the forward is paid at delivery, in six months, so it is worth its present value today, $e^{-0.02} \times e^{0.02} = 1$ per ounce. Fewer futures than forwards are needed, by the factor $e^{-r\tau_F} = 0.9802$.

(c) We differentiate the price in σ at fixed F and cancel with the key identity in forward form, $F\varphi(d_1) = K\varphi(d_2)$ (Lemma 20.34):

$$\partial_\sigma C = e^{-r\tau}\left[F\varphi(d_1)\,\partial_\sigma d_1 - K\varphi(d_2)\,\partial_\sigma d_2\right] = e^{-r\tau}F\varphi(d_1)\,\partial_\sigma(d_1 - d_2) = e^{-r\tau}F\varphi(d_1)\sqrt{\tau}.$$

This is Proposition 21.9(i) evaluated at the price F with $q = r$. With $\varphi(-0.1770) = 0.3927$,

$$\mathcal{V} = 0.98676 \times 2{,}400 \times 0.3927 \times 0.5774 = 537.0 \text{ per unit of volatility},$$

which is 5.37 per volatility point for one call and $-2{,}000 \times 5.37 = -10{,}740$ per volatility point for the position.

(d) With F held fixed, d_1 and d_2 do not contain r, which enters the price only through the discount factor. Hence

$$\mathcal{R} = \partial_r C = -\tau e^{-r\tau}\big[F\,\Phi(d_1) - K\,\Phi(d_2)\big] = -\tau\,C = -\tfrac{1}{3}\,(65.67) = -21.89 \text{ per unit of rate,}$$

which is -0.219 per percentage point for one call and $+437.8$ per percentage point for the written position: a higher rate lowers the present value of what the writer owes. The yield-formula rho with $q = r$ substituted would give $K\tau e^{-r\tau}\Phi(d_2) = +317.4$, with the wrong sign. The reason is what is held fixed. Write the yield formula as a function of its price variable x, its rate and its yield,

$$C^{\text{yld}}(x, r, q) = xe^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2), \qquad d_1 = \frac{\ln(x/K) + (r - q + \sigma^2/2)\tau}{\sigma\sqrt{\tau}}, \quad d_2 = d_1 - \sigma\sqrt{\tau},$$

so that the futures call is $C = C^{\text{yld}}(F, r, r)$. The rho of Proposition 21.9(ii) is $\partial_r C^{\text{yld}}$, the derivative in r at fixed x and fixed q, along which the formula's forward price $xe^{(r-q)\tau}$ rises with r. For a futures option the futures price is held fixed and the yield is the rate itself, so the substitution $q = r$ has to be made before differentiating, not after. By the chain rule,

$$\mathcal{R} = \frac{d}{dr}\,C^{\text{yld}}(F, r, r) = \partial_r C^{\text{yld}}(F, r, r) + \partial_q C^{\text{yld}}(F, r, r).$$

With x, r and σ fixed, $\partial_q d_1 = \partial_q d_2 = -\sqrt{\tau}/\sigma$, so

$$\partial_q C^{\text{yld}} = -\tau x e^{-q\tau}\Phi(d_1) - \frac{\sqrt{\tau}}{\sigma}\big[xe^{-q\tau}\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)\big] = -\tau x e^{-q\tau}\Phi(d_1),$$

the bracket vanishing by Lemma 20.34 in its yield form, as in the proof of Proposition 21.9(ii). At $x = F$ and $q = r$,

$$\mathcal{R} = K\tau e^{-r\tau}\Phi(d_2) - \tau F e^{-r\tau}\Phi(d_1) = -\tau\,C,$$

in numbers $317.35 - 339.24 = -21.89$.

Self-check. The spot delta of (b) is confirmed by repricing in the spot. Spot gold is $2{,}400\,e^{-0.02} = 2{,}352.48$; a rise of one dollar moves the six-month futures price to 2,401.020 and a fall moves it to 2,398.980, where the call is worth 66.10275 and 65.23751, so the position changes by $-2{,}000 \times (66.10275 - 65.23751)/2 = -865.2$ per dollar of spot. All four hedges of (b) must then gain the same amount, 865.2, when the spot price rises by one dollar. The spot position gains 865.2; the six-month futures gain $848.1 \times e^{0.02} = 865.2$ at the settlement; the one-year futures gain $831.3 \times e^{0.04} = 865.2$; and the forward position gains $865.2 \times e^{0.02}$ at delivery, worth 865.2 today. The Greeks are confirmed by repricing. At $F = 2{,}410$ and $F = 2{,}390$ the call is worth 69.997 and 61.516: the central difference is 0.4241. At $r = 5\%$ with F unchanged it is worth 65.451, a change of -0.219. At $\sigma = 17\%$ it is worth 71.047, a change of $+5.38$, against the vega 5.37 per point.

Answer: (a) $C = 65.67$; $\partial_F C = e^{-r\tau}\Phi(d_1) = 0.4241$, so the position has delta -848.1 with respect to the six-month futures price. (b) Buy 848.1 ounces of the six-month futures, or 865.2 ounces of spot gold, or 831.3 ounces of the one-year futures, or 865.2 ounces of the six-month forward; a futures gain is paid today and a forward gain at delivery, so their deltas differ by $e^{r\tau_F}$. (c) $\mathcal{V} = e^{-r\tau}F\varphi(d_1)\sqrt{\tau} = 537.0$; the position has $-10{,}740$ per volatility point. (d) $\mathcal{R} = -\tau C = -21.89$; the position has $+437.8$ per percentage point. The yield-formula rho holds its price variable and q fixed and lets the formula's forward move with r; here F is fixed and $q = r$ moves with r.

P21.4 *The Greeks of a written currency-option block, interpreted*

Write $q = r_{\text{for}} = 0.025$ and $\tau = 5/12$. The formulas are those of Corollary 20.43, the yield case of Corollary 20.41, and of Propositions 21.2, 21.6 and 21.9.

(a) Here $\sqrt{\tau} = 0.6455$, $\sigma\sqrt{\tau} = 0.05164$, $\ln(S/K) = \ln(1.15/1.17) = -0.017242$ and $(r - q + \sigma^2/2)\tau = (0.015 + 0.0032)(0.41667) = 0.007583$, so

$$d_1 = \frac{-0.017242 + 0.007583}{0.05164} = -0.1870, \qquad d_2 = -0.1870 - 0.0516 = -0.2387,$$

with $\Phi(d_1) = 0.4258164$, $\Phi(d_2) = 0.4056787$ and $\varphi(d_1) = 0.39202$. The discounted spot and strike are $Se^{-q\tau} = 1.15\,e^{-0.010417} = 1.1380830$ and $Ke^{-r\tau} = 1.17\,e^{-0.016667} = 1.1506616$. Then:

$$C = 1.1380830 \times 0.4258164 - 1.1506616 \times 0.4056787 = 0.4846144 - 0.4667989 = 0.0178155,$$

$$\Delta = e^{-q\tau}\Phi(d_1) = 0.98964 \times 0.42582 = 0.4214,$$

$$\Gamma = \frac{e^{-q\tau}\varphi(d_1)}{S\sigma\sqrt{\tau}} = \frac{0.98964 \times 0.39202}{1.15 \times 0.05164} = 6.533 \text{ per unit of the exchange rate},$$

$$\mathcal{V} = Se^{-q\tau}\varphi(d_1)\sqrt{\tau} = 1.13808 \times 0.39202 \times 0.6455 = 0.2880 \text{ per unit of volatility},$$

$$\Theta = -\frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}} + qSe^{-q\tau}\Phi(d_1) - rKe^{-r\tau}\Phi(d_2)$$

$$= -0.027647 + 0.012115 - 0.018672 = -0.034204 \text{ a year},$$

$$\mathcal{R} = K\tau e^{-r\tau}\Phi(d_2) = 0.41667 \times 0.466799 = 0.1945 \text{ per unit of rate}.$$

(The price is a difference of two nearly equal terms, so its factors are carried to seven digits; to six decimals $C = 0.017816$, the value used below.)

(b) With S, r, σ and τ fixed, $\partial_q d_1 = \partial_q d_2 = -\sqrt{\tau}/\sigma$. Differentiating the call formula in q,

$$\partial_q C = -\tau Se^{-q\tau}\Phi(d_1) + \left[Se^{-q\tau}\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)\right]\partial_q d_1 = -\tau Se^{-q\tau}\Phi(d_1),$$

the bracket being zero by the key identity (Lemma 20.34). In numbers, $-0.41667 \times 0.484614 = -0.20192$ per unit of rate: a higher foreign rate lowers the forward exchange rate and with it the call.

(c) The position is short calls on 40 million units, so each number of (a) and (b) is multiplied by $-40{,}000{,}000$. The premium received is 712,600 of domestic currency.

- *Delta:* -16.86 million units of foreign currency. The position behaves like a short position of 16.86 million units, and it is made delta-neutral by buying 16.86 million units of the foreign currency spot, for 19.38 million of domestic currency (one unit of the underlying has delta 1, Proposition 21.4(i)).
- *Gamma:* -261.3 million per unit of the exchange rate, that is, -2.61 million units per move of 0.01. If the rate rises by 0.01 the position becomes 2.61 million units shorter, and that amount must be bought as well.
- *Vega:* $-40{,}000{,}000 \times 0.2880/100 = -115{,}200$ of domestic currency per volatility point.
- *Theta:* $+1.368$ million a year, which is $+5{,}429$ per trading day: the writer collects the time decay.
- *Rho:* $-40{,}000{,}000 \times 0.1945/100 = -77{,}800$ per percentage point of the domestic rate.
- *Foreign rate:* $+40{,}000{,}000 \times 0.20192/100 = +80{,}770$ per percentage point of the foreign rate.

(d) At $S = 1.16$, $\ln(S/K) = \ln(1.16/1.17) = -0.0085837$, so

$$d_1 = \frac{-0.0085837 + 0.0075833}{0.0516398} = -0.019373, \qquad d_2 = -0.071013,$$

with $\Phi(d_1) = 0.4922718$ and $\Phi(d_2) = 0.4716938$. The discounted spot is $1.16\,e^{-0.010417} = 1.1479794$, so the call is worth

$$1.1479794 \times 0.4922718 - 1.1506616 \times 0.4716938 = 0.5651179 - 0.5427600 = 0.0223579,$$

that is 0.022358. The central difference of the two revalued prices is

$$\frac{0.022358 - 0.013925}{0.02} = 0.4217,$$

against the delta 0.4214, and their second difference is

$$\frac{0.022358 - 2\,(0.017816) + 0.013925}{0.01^2} = \frac{0.000651}{0.0001} = 6.51,$$

against the gamma 6.53. The small gaps are the higher-order terms of the Taylor expansion and the rounding of the three prices to six decimals.

Self-check. The Greeks of (a) must satisfy the theta–gamma identity of Corollary 21.8 with $q = r_{\text{for}}$, which no step of the computation used:

$$\begin{aligned}\Theta + (r-q)S\Delta + \tfrac{1}{2}\sigma^2 S^2\Gamma &= -0.034204 + 0.015 \times 1.15 \times 0.4214 + 0.004232 \times 6.5329 \\ &= -0.034204 + 0.007269 + 0.027647 = 0.000712,\end{aligned}$$

and $rC = 0.04 \times 0.017816 = 0.000713$, equal up to the rounding of the last digit. The vega also agrees with Proposition 21.9(iii): $\sigma\tau S^2\Gamma = 0.08 \times 0.41667 \times 1.3225 \times 6.533 = 0.2880$. Both checks are blind to the density value, which cancels from the identity and from the vega ratio, so two further checks cover what they cannot. The key identity of Lemma 20.34 checks the density: $Ke^{-r\tau}\varphi(d_2) = 1.15066 \times 0.38774 = 0.44616$ against $Se^{-q\tau}\varphi(d_1) = 1.13808 \times 0.39202 = 0.44615$. A common shift of r and q leaves the forward $F = Se^{(r-q)\tau}$ unchanged, so in the forward form of Corollary 20.43 it changes only the discount factor $e^{-r\tau}$; hence $\partial_r C + \partial_q C = -\tau C$, and indeed $0.19450 - 0.20192 = -0.00742 = -0.41667 \times 0.017816$. This checks the rho and part (b) without differentiating d_1.

Answer: (a) $C = 0.017816$, $\Delta = 0.4214$, $\Gamma = 6.533$, $\mathcal{V} = 0.2880$ per unit of volatility, $\Theta = -0.0342$ a year, $\mathcal{R} = 0.1945$. (b) $\partial_{r_{\text{for}}} C = -\tau S e^{-r_{\text{for}}\tau}\Phi(d_1) = -0.2019$. (c) Delta -16.86 million units (buy 16.86 million units spot), gamma -2.61 million units per 0.01, vega $-115{,}200$ per volatility point, theta $+5{,}429$ per trading day, rho $-77{,}800$ and foreign-rate sensitivity $+80{,}770$ per percentage point. (d) The call is worth 0.022358 at 1.16; the bumps give a delta of 0.4217 and a gamma of 6.51, against 0.4214 and 6.53.

P21.5 *Theta is rent for gamma: the identity, its variants, parity Greeks and the run into expiry*

(a) *The call.* We substitute the three Greeks:

$$\begin{aligned}\Theta_C + (r-q)S\Delta_C + \tfrac{1}{2}\sigma^2 S^2\Gamma &= -\frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}} + qSe^{-q\tau}\Phi(d_1) - rKe^{-r\tau}\Phi(d_2) \\ &\quad + (r-q)Se^{-q\tau}\Phi(d_1) + \frac{\sigma^2 S^2}{2}\cdot\frac{e^{-q\tau}\varphi(d_1)}{S\sigma\sqrt{\tau}}.\end{aligned}$$

The last term equals $Se^{-q\tau}\varphi(d_1)\,\sigma/(2\sqrt{\tau})$ and cancels the first, and the two terms in $\Phi(d_1)$ add up to $rSe^{-q\tau}\Phi(d_1)$. What remains is $r\big[Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)\big] = rC$.

The forward. For $V = Se^{-q\tau} - Ke^{-r\tau}$, with $\partial_t e^{-q(T-t)} = qe^{-q\tau}$, the Greeks are $\Theta = qSe^{-q\tau} - rKe^{-r\tau}$, $\Delta = e^{-q\tau}$ and $\Gamma = 0$, so

$$\Theta + (r-q)S\Delta + \tfrac{1}{2}\sigma^2 S^2\Gamma = qSe^{-q\tau} - rKe^{-r\tau} + (r-q)Se^{-q\tau} = r\big(Se^{-q\tau} - Ke^{-r\tau}\big) = rV.$$

The put and the book. Both sides of the identity are linear in the position. If the instruments of values V_j satisfy it, then the book $\sum_j w_j V_j$, whose Greeks are the w-weighted sums of theirs (Definition 21.1(ii)), satisfies it as well. By put–call parity, $\mathcal{P} = C - (Se^{-q\tau} - Ke^{-r\tau})$: the put is the call minus the forward, so it satisfies the identity. The strike K and the date T were arbitrary, so every call, every put and every forward of a book, each with its own strike or delivery price and its own date, satisfies the identity at every $S > 0$ and every t before its own expiry or delivery date. One unit of cash is worth e^{rt}, with $\Theta = re^{rt} = rV$ and $\Delta = \Gamma = 0$, so it satisfies it too. When $q = 0$, one share ($\Theta = 0$, $\Delta = 1$, $\Gamma = 0$, value S) gives $rS = rV$ on the left, so shares may be added as well, and a book delta-hedged with shares satisfies $\Theta + \frac{1}{2}\sigma^2 S^2\Gamma = r\Pi$, where Π is its value. Adding up gives the identity for every book of calls, puts, forwards and cash at every t before the earliest expiry and delivery date in the book, each option being valued at the volatility σ that appears in the identity. This is Corollary 21.8, whose proof obtains the option case from the pricing equation instead of by substitution.

(b) *Currency options.* A currency option is the yield case with $q = r_{\text{for}}$ (Corollary 20.43), and a currency forward is worth $Se^{-r_{\text{for}}\tau} - Ke^{-r\tau}$ (Proposition 21.4(ii) with $q = r_{\text{for}}$). Part (a) holds for every q, so

$$\Theta + (r - r_{\text{for}})S\Delta + \tfrac{1}{2}\sigma^2 S^2\Gamma = rV.$$

Futures options. The call on a futures price is worth $e^{-r\tau}[F\,\Phi(d_1) - K\,\Phi(d_2)]$ with $d_1 = [\ln(F/K) + \sigma^2\tau/2]/(\sigma\sqrt{\tau})$ (Theorem 20.52(iii)). This is the function C of the statement evaluated at the price F with $q = r$, so its derivatives in F, and in t at fixed F, are the three formulas of the statement with $S = F$ and $q = r$, and part (a) with $q = r$ reads

$$\Theta + \tfrac{1}{2}\sigma^2 F^2\Gamma = rV.$$

The delta term has dropped out, because a futures position costs nothing to carry. The put on the futures is the call minus $e^{-r\tau}(F - K)$, a quantity with $\Theta = re^{-r\tau}(F - K)$ and $\Gamma = 0$, which satisfies the same identity; books follow by linearity. This is the futures equation of Corollary 20.45(ii), read as a relation between Greeks.

(c) Parity reads $C - \mathcal{P} = Se^{-q\tau} - Ke^{-r\tau}$, and its right side is linear in S and contains no σ. Differentiating it once and twice in S, in σ and in t (Proposition 21.11):

$$\Delta_{\mathcal{P}} = \Delta_C - e^{-q\tau}, \qquad \Gamma_{\mathcal{P}} = \Gamma_C, \qquad \mathcal{V}_{\mathcal{P}} = \mathcal{V}_C, \qquad \Theta_{\mathcal{P}} = \Theta_C - qSe^{-q\tau} + rKe^{-r\tau}.$$

(d) At $r = q = 0$ the identity of (a) reads $\Theta = -\frac{1}{2}\sigma^2 S^2\Gamma$ for the book, whatever its delta (shares add neither theta nor gamma). Hence

$$\Gamma = -\frac{2\Theta}{\sigma^2 S^2} = -\frac{18{,}000}{0.09 \times 2{,}500} = -80 \text{ per point.}$$

A call or a put has a positive gamma, so the book is short options on balance: it is short gamma. On a day without a move it earns $\Theta\,\Delta t = 9{,}000/252 = 35.7$. By Proposition 21.14 with a zero delta and an unchanged volatility, a day with a move ΔS earns about $\frac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t = -40\,(\Delta S)^2 + 35.7$, which is zero at $|\Delta S| = \sqrt{35.7/40} = 0.945$ points; this is $\sigma S\sqrt{\Delta t} = 0.30 \times 50/\sqrt{252}$, or 1.89% of the price. On a two-point day the book loses $160 - 35.7 = 124.3$. The position that profits when neither the price nor the implied volatility moves is thus the short-gamma one: it collects the rent that the long-gamma side pays.

(e) At the money with $r = q = 0$, $d_1 = \sigma\sqrt{\tau}/2$ and $\Theta = -S\sigma\varphi(d_1)/(2\sqrt{\tau})$. One month before expiry, $\sqrt{\tau} = 0.2887$, $d_1 = 0.0433$ and $\varphi(d_1) = 0.3986$, so

$$\Theta = -\frac{50 \times 0.30 \times 0.3986}{2 \times 0.2887} = -10.36 \text{ a year} = -0.0411 \text{ per trading day.}$$

One trading day before expiry, $\sqrt{\tau} = 0.0630$, $d_1 = 0.0094$ and $\varphi(d_1) = 0.3989$, so $\Theta = -5.9839/0.12599 = -47.50$ a year, which is -0.188 per trading day: 4.6 times as much. Up to the factor $e^{-\sigma^2\tau/8}$ in $\varphi(d_1)$, which contributes 1.0009 to the ratio, this is the ratio $\sqrt{21} = 4.58$ of the two values of $\tau^{-1/2}$. Theta accelerates because the at-the-money value is about $0.4\,S\sigma\sqrt{\tau}$ (Proposition 20.29), a concave function of τ whose slope grows like $\tau^{-1/2}$; equivalently, it is the rent for a gamma of about $0.4/(S\sigma\sqrt{\tau})$ that grows at the same rate (Proposition 21.6(iii)). The call is worth 1.727 one month out and 0.377 one day out, so the linearised decay of the last day is half of what is left. Away from the strike, at $r = q = 0$, gamma and theta both die out instead.

Self-check. Numbers test the identity where the algebra could hide a slip. Take $S = 50$, $K = 52$, $r = 4\%$, $q = 1\%$, $\sigma = 30\%$ and $\tau = 0.5$. The call has $C = 3.6717$, $\Delta = 0.4943$, $\Gamma = 0.037424$ and $\Theta = -4.8047$, and

$$\begin{aligned} -4.8047 + 0.03 \times 50 \times 0.4943 + \tfrac{1}{2}\,(0.09)(2{,}500)(0.037424) &= -4.8047 + 0.7414 + 4.2102 \\ &= 0.1469 = 0.04 \times 3.6717. \end{aligned}$$

The put, evaluated from its own closed forms and not through parity, has $\mathcal{P} = 4.8914$, $\Delta = -0.5007$ and $\Theta = -3.2634$, and $-3.2634 - 0.7511 + 4.2102 = 0.1957 = 0.04 \times 4.8914$. The futures call with $F = 50$ and the same K, r, σ and τ is worth 3.3142, with $\Gamma = 0.036754$ and $\Theta = -4.0022$, and $-4.0022 + \frac{1}{2}(0.09)(2{,}500)(0.036754) = -4.0022 + 4.1348 = 0.1326 = 0.04 \times 3.3142$. For (e), direct repricing: the one-month call is worth 1.7269, and 1.6853 one trading day later at the same price, a decay of 0.0416 against the theta 0.0411.

Answer: (a) Substitution leaves $r[Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)] = rC$; the forward satisfies the identity, the put is the call minus the forward, and books follow by linearity. (b) $\Theta + (r - r_{\text{for}})S\Delta + \frac{1}{2}\sigma^2 S^2\Gamma = rV$ for currency options and $\Theta + \frac{1}{2}\sigma^2 F^2\Gamma = rV$ for futures options. (c) $\Delta_{\mathcal{P}} = \Delta_C - e^{-q\tau}$, $\Gamma_{\mathcal{P}} = \Gamma_C$, $\mathcal{V}_{\mathcal{P}} = \mathcal{V}_C$, $\Theta_{\mathcal{P}} = \Theta_C - qSe^{-q\tau} + rKe^{-r\tau}$. (d) $\Gamma = -80$ per point: short options; it earns 35.7 on a quiet day, breaks even at a move of 0.945 points (1.89%) and loses 124.3 on a two-point day. (e) -0.0411 per trading day one month out and -0.188 one day out; theta grows like $\tau^{-1/2}$, as the rent for a gamma that grows at the same rate.

P21.6 *Explaining a day's profit and loss*

(a) At $r = q = 0$ the theta–gamma identity for a book hedged with shares reads $\Theta = -\frac{1}{2}\sigma^2 S^2\Gamma$ (Corollary 21.8(a) and (b)). Here $\frac{1}{2}(0.09)(2{,}500)(786.40) = 112.5 \times 786.40 = 88{,}470$, which is the stated theta. Per trading day the theta is $88{,}470/252 = 351.1$.

(b) By Proposition 21.14, $\Delta V \approx \Delta\,\Delta S + \frac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t + \mathcal{V}\,\Delta\sigma$, with $\Delta S = -1.5$, $\Delta t = 1/252$, and $\Delta\sigma = 0.02$ in scenario A and 0 in scenario B. The delta term is zero, and

$$\begin{aligned} \tfrac{1}{2}\Gamma\,(\Delta S)^2 &= \tfrac{1}{2}(-786.40)(2.25) = -884.7, \\ \Theta\,\Delta t &= 88{,}470/252 = +351.1, \\ \mathcal{V}\,\Delta\sigma &= -147{,}450 \times 0.02 = -2{,}949.0 \quad \text{(scenario A only)}. \end{aligned}$$

Scenario A: $-884.7 + 351.1 - 2{,}949.0 = -3{,}482.6$. Scenario B: $-884.7 + 351.1 = -533.6$.

(c) The residuals are $-3{,}821.4 - (-3{,}482.6) = -338.8$ in scenario A, which is 10% of the explained loss, and $-547.3 - (-533.6) = -13.7$ in scenario B. The two named terms of scenario A are

$$\begin{aligned} \partial_{S\sigma}V\,\Delta S\,\Delta\sigma &= 12{,}335 \times (-1.5) \times 0.02 = -370.05, \\ \tfrac{1}{2}\,\partial_{SS\sigma}V\,(\Delta S)^2\,\Delta\sigma &= \tfrac{1}{2}(1{,}343)(2.25)(0.02) = +30.2. \end{aligned}$$

Together they give -339.8, against the residual -338.8. The cross term carries the residual, and the gamma–volatility term takes back 30 of it. The other terms of the omitted part $\mathcal{E}$ are each far smaller than the cross term, but several are larger than the gamma–volatility term: the third-order price term $\frac{1}{6}\,\partial_{SSS}V\,(\Delta S)^3$ is -54.7, the term $\frac{1}{2}\,\partial_{\sigma\sigma}V\,(\Delta\sigma)^2$ is -47.9, the term $\partial_{St}V\,\Delta S\,\Delta t$ is $+44.1$ and the term $\partial_{t\sigma}V\,\Delta t\,\Delta\sigma$ is $+34.8$. They largely cancel, and all the remaining terms net to $+1.1$ here, so the close agreement of the two named terms with the residual is partly a coincidence of these numbers. In scenario B the volatility does not move, so the residual consists of terms of the orders $(\Delta S)^3$, $\Delta S\,\Delta t$ and $(\Delta t)^2$ (Proposition 21.14(ii)); the first of them is negative here, because the gamma of the written puts grows as the price falls toward the strike. The sign of the cross term has a plain reading. A higher volatility makes the delta of an out-of-the-money put more negative, so the book of written puts gets longer the stock exactly when the stock is falling.

(d) The vega term, $-2{,}949$, is the largest: 85% of the explained loss. It is removed by making the book vega-neutral with the traded call (Procedure 21.31, one target): buy

$$w_1 = -\frac{\mathcal{V}_\Pi}{\mathcal{V}_1} = \frac{1{,}474.5}{0.1403} = 10{,}510 \text{ calls}.$$

They add $10{,}510 \times 0.0374 = 393.1$ of gamma, so the gamma of the book becomes $-786.4 + 393.1 = -393.3$: halved, not removed, because one option fixes one target. (The call has twice the vega-to-gamma ratio of the book, $0.1403/0.0374 = 3.75$ against $1{,}474.5/786.4 = 1.875$ per volatility

point, as $\mathcal{V} = \sigma\tau S^2\Gamma$ requires of six months against three; Proposition 21.9(iii).) The calls also bring $10{,}510 \times 0.542 = 5{,}696$ of delta, so 5,696 more shares are sold short, about 10,066 in all. The cross term stays. It is a different derivative, $\partial_{S\sigma}V$, and a vega of zero does not make it zero: an at-the-money call has the small vanna $-\varphi(d_1)\,d_2/\sigma = 0.140$ (Example 21.13), so the purchase adds about +1,474 to the 12,335 of the book and makes the cross term larger. Removing the cross term instead takes options of negative vanna, such as out-of-the-money puts: the written puts have the vanna -0.617 each, and buying back $12{,}335/0.617 = 20{,}000$ of them would close the book.

(e) With no time to re-hedge, the delta–gamma estimate is $\frac{1}{2}\Gamma\,(\Delta S)^2 = \frac{1}{2}\,(-786.40)(25) = -9{,}830$ for a gap of 5 points up or down: $9{,}830/351.1 = 28$ trading days of theta. A book with a zero delta is neutral only for small moves. A large negative gamma means that the book has collected a small daily rent in exchange for a loss that grows with the square of the move, in either direction, and that cannot be hedged away once the price has gapped; after the gap the hedger must sell after a fall and buy after a rise. The estimate is itself local. Full revaluation at the same instant, with no time elapsed, gives a loss of 11,743 after a fall to 45, where the gamma of the written puts has grown to −1,179, and of 7,883 after a rise to 55.

Self-check. Three independent routes. First, by Proposition 21.17 the hedged book earns $\frac{1}{2}\Gamma\,[(\Delta S)^2 - \sigma_{\text{imp}}^2 S^2\,\Delta t]$ when the volatility does not move. The implied variance of one day is $0.09 \times 2{,}500/252 = 0.893$, so scenario B gives $\frac{1}{2}\,(-786.40)(2.25 - 0.893) = -533.6$, the explain of (b); the breakeven move is $\sqrt{0.893} = 0.945$ points, and the stock moved 1.5. Second, the stated revaluations can be rebuilt leg by leg. One put is worth 1.01086 at the start, 1.52962 in scenario A and 1.36592 in scenario B, so the written puts lose $20{,}000 \times 0.51876 = 10{,}375$ and $20{,}000 \times 0.35506 = 7{,}101$, while the short shares gain $4{,}369.2 \times 1.5 = 6{,}554$: the book changes by −3,821 and by −547. Third, the hedge of (d) follows from the ratios alone: the call's vega-to-gamma ratio is twice the book's, so $w_1 = (786.4/0.0374) \times (1.875/3.75) = 10{,}513$, which is the 10,510 of (d) up to the rounding of the call's Greeks, and it leaves $786.4 \times (1 - \frac{1}{2}) = 393.2$ of short gamma.

Answer: (a) $\frac{1}{2}\sigma^2 S^2\,|\Gamma| = 112.5 \times 786.40 = 88{,}470 = \Theta$. (b) Scenario A: $-884.7 + 351.1 - 2{,}949.0 = -3{,}482.6$; scenario B: $-884.7 + 351.1 = -533.6$. (c) Residuals −338.8 and −13.7; in scenario A the cross term is −370.05 and the gamma-volatility term +30.2, together −339.8. (d) The vega term; buying 10,510 of the calls makes the book vega-neutral, halves its gamma to −393.3 and requires selling 5,696 more shares; the cross term is not removed. (e) About 9,830 either way, 28 days of theta: a zero delta protects only against small moves, and a short-gamma book loses with the square of the move.

P21.7 *Breakeven moves and gamma scalping*

(a) The bracket of the leading term vanishes when $(\Delta S)^2 = \sigma_{\text{imp}}^2 S^2\,\Delta t$, that is, at the move $|\Delta S|/S = \sigma_{\text{imp}}\sqrt{\Delta t}$ (Proposition 21.17); the straddle has a positive gamma, so it gains beyond this move and loses inside it. For one day $\sqrt{\Delta t} = 1/\sqrt{252} = 1/15.87$, and for five days $\sqrt{5/252} = 1/7.10$:

Implied volatility	16%	25%	40%
Breakeven move, one day	1.01%	1.57%	2.52%
Breakeven move, one week	2.25%	3.52%	5.63%

The weekly figures are the daily ones times $\sqrt{5} = 2.24$, not times 5.

(b) Sell. A rally of 2% is $\Delta S = 1.6$ points, and it raises the delta of the 1,000 straddles by about $1{,}000 \times 0.0609 \times 1.6 = 97$ shares, so the trader sells about 97 shares at 81.6. A long-gamma position sells rallies and buys dips.

(c) After each close the position is short $1{,}000\,\Delta_{t_i}$ shares, so it starts short 65 shares, and its profit and loss over the next day is $1{,}000\,[(V_{t_{i+1}} - V_{t_i}) - \Delta_{t_i}\,\Delta S_i]$. The implied variance of one

day, $\sigma_{\text{imp}}^2 S_{t_i}^2\,\Delta t = 0.16\,S_{t_i}^2/252$, is 4.063, 4.374, 4.166 and 4.217 squared points at the closes 80, 83, 81 and 81.5.

- *Day 1* ($\Delta S = +3$): the straddles gain 339 and the 65 short shares lose 195, a profit of 144. The formula gives $\frac{1}{2}\,(60.9)(9 - 4.063) = 150.3$. At the close the delta is 243: sell 178 shares at 83.
- *Day 2* ($\Delta S = -2$): the straddles lose 495 and the 243 short shares gain 486, a loss of 9. The formula gives $\frac{1}{2}\,(56.8)(4 - 4.374) = -10.6$. Buy 118 shares at 81, which leaves 125 short.
- *Day 3* ($\Delta S = +0.5$): the straddles lose 58 and the 125 short shares lose 62.5, a loss of 120.5. The formula gives $\frac{1}{2}\,(61.0)(0.25 - 4.166) = -119.4$. Sell 31 shares at 81.5, which leaves 156 short.
- *Day 4* ($\Delta S = -3.5$): the straddles lose 288 and the 156 short shares gain 546, a profit of 258. The formula gives $\frac{1}{2}\,(61.0)(12.25 - 4.217) = 245.0$. Buy 224 shares at 78: the delta of the straddles is now negative, and the position ends long 68 shares.

The four days add up to $144-9-120.5+258 = +272.5$, against $150.3-10.6-119.4+245.0 = +265.3$ from the formula: the two agree within 3%. The difference, 7.2, is the sum over the four days of the remainders $\mathcal{E}$ of Proposition 21.17, about 6.3 (the largest, +12.7, on day 4, the largest move), plus about 0.9 from the rounding of the table to three decimals. The straddles themselves lost 502 of value over the four days; the hedge made 774.5, by selling at 83 and 81.5 and buying at 81 and 78.

(d) For one straddle, $\Gamma S^2\sigma^2 T = 0.0609 \times 6{,}400 \times 0.16 \times \frac{1}{6} = 10.394$. With $n = 42$ the estimate is $10.394/\sqrt{84} = 1.134$, and with $n = 294$ it is $10.394/\sqrt{588} = 0.429$. For the 1,000 straddles this is about 1,130 against 430, or 10.9% and 4.1% of the premium of 10,412. Rebalancing seven times as often divides the error by $\sqrt{7} = 2.65$, the $n^{-1/2}$ law of Proposition 21.27(iii). (At $r = q = 0$ the delta of the put is that of the call minus one, so the two legs of the straddle have the same hedging error, and both the error and the gamma of the straddle are twice those of the call. At the money the frozen-gamma constant runs low by a factor of about $\sqrt{\pi/2} = 1.25$, Example 21.28, so about 1,420 and 540 are the better estimates.)

Self-check. For (a), the first entry is the rule that 16 vol is 1% a day (Corollary 19.2), and the others scale with the volatility: $1.0079\% \times 25/16 = 1.5748\%$ and $1.0079\% \times 40/16 = 2.5198\%$. For (b), the table checks the rule gamma times move: on the 3-point rise of day 1 the delta of the 1,000 straddles went from 65 to 243, up 178 shares, against $1{,}000 \times 0.0609 \times 3 = 183$ from the day-0 gamma; and the delta of a straddle, $2\Phi(d_1) - 1$ at $r = q = 0$, is 0.161 at 81.6 with two months to expiry, a rise of 96 shares for the position against the 97 of (b). For (c), an aggregate check that uses no daily mark: the squared moves add up to $9 + 4 + 0.25 + 12.25 = 25.5$, against an implied $4.063+4.374+4.166+4.217 = 16.82$, so the four days realised $40\% \times \sqrt{25.5/16.82} = 49\%$, above the implied volatility, and a long-gamma position must gain. With the average gamma 59.9 of the four starting dates, $\frac{1}{2}\,(59.9)(25.5 - 16.82) = +260$, close to both totals. For (d), Proposition 21.9(iii) gives $\Gamma S^2\sigma^2 T = \mathcal{V}\sigma$; the vega of the straddle is $2 \times 80 \times \varphi(0.0816) \times \sqrt{1/6} = 25.97$, and $25.97 \times 0.4 = 10.39$, the same number up to the rounding of the gamma.

Answer: (a) One day: 1.01%, 1.57% and 2.52% at 16, 25 and 40 vol; one week: 2.25%, 3.52% and 5.63%. (b) Sell about 97 shares. (c) Daily profit and loss +144, −9, −120.5 and +258, total +272.5 against +265.3 from the formula; trades at the closes: sell 178, buy 118, sell 31, buy 224. (d) About 1.13 per straddle with daily rebalancing and 0.43 with seven rebalancings a day (1,130 and 430 for the position), a ratio of $\sqrt{7}$.

P21.8 *Variance risk premium, delta-hedged P&L*

(a) Gamma rent against theta rent. Take $r = q = 0$: over one month the rate and the dividend yield of the index are neglected, and Proposition 21.17 needs both to vanish. The straddle is one call and one put with strike K and expiry T, both marked at σ_{imp}, so its value V is a book of the kind of Corollary 21.8. Fix one rebalancing interval from t to $t + \Delta t < T$, over which the index

moves from $S > 0$ to $S + \Delta S > 0$; the position holds $-\Delta$ units of the index, Δ being the delta of the straddle at the start. We combine the profit-and-loss explain with the theta–gamma identity, in three steps.

Step 1 (the explain). Each formula (21.2) is infinitely differentiable for every $\tau > 0$ and $\sigma > 0$, so V is infinitely differentiable in (S, t, σ) on the open convex set $(0, \infty) \times (-\infty, T) \times (0, \infty)$, which contains the start $(S, t, \sigma_{\text{imp}})$ and the end $(S + \Delta S, t + \Delta t, \sigma_{\text{imp}})$ of the move. Proposition 21.14 with $\Delta\sigma = 0$ gives $\Delta V = \Delta\,\Delta S + \frac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t + \mathcal{E}$, where $\mathcal{E} = O(|\Delta S|^3 + |\Delta S|\,\Delta t + (\Delta t)^2)$ by its part (ii).

Step 2 (the hedge). The $-\Delta$ units of the index gain $-\Delta\,\Delta S$. At $r = q = 0$ the cash that finances the position earns nothing and the index pays no income, so the P and L of the interval is $\frac{1}{2}\Gamma\,(\Delta S)^2 + \Theta\,\Delta t + \mathcal{E}$: a gamma gain and a theta decay.

Step 3 (the rent). The book is priced at the one volatility σ_{imp} and $t < T$, so Corollary 21.8(a) gives $\Theta = -\frac{1}{2}\sigma_{\text{imp}}^2 S^2 \Gamma$: theta is exactly the rent on the gamma, priced at the implied volatility. Hence

$$\text{P and L of the interval} = \tfrac{1}{2}\Gamma\,\big[(\Delta S)^2 - \sigma_{\text{imp}}^2 S^2\,\Delta t\big] + \mathcal{E},$$

which is Proposition 21.17. Summed over the days it gives the lifetime sum of Remark 21.21,

$$\text{total} \;\approx\; \sum_i \tfrac{1}{2}\Gamma_{t_i}\big[(\Delta S_i)^2 - \sigma_{\text{imp}}^2 S_{t_i}^2\,\Delta t\big],$$

a *gamma-weighted* sum of realised-minus-implied variance: the one-line meaning of "trading volatility". With realised 20% against implied 25%, the daily $(\Delta S)^2$ falls short of the theta bill on average: **the position loses**. The weights $\Gamma_{t_i} S_{t_i}^2$ make the sum path-dependent: variance realised while the price is near the strike counts most, so the loss is larger than average if the shortfall in realised volatility came while the price sat near the strike, and smaller if the quiet days came while it was far from it.

Exactly, in the driftless constant-volatility model with $\sigma_{\text{real}} = 20\%$, the expected P and L is $V^{\text{BSM}}(S_0, 0; \sigma_{\text{real}}) - V^{\text{BSM}}(S_0, 0; \sigma_{\text{imp}})$ for daily or any other rebalancing (Proposition 21.19), where S_0 is the index level at the start. It is negative because the vega $2S\varphi(d_1)\sqrt{T}$ of the straddle is positive at every volatility (Proposition 21.9(i)), so its price increases strictly with σ. The position therefore loses on average. Under continuous rebalancing it loses on every path (Proposition 21.22(ii), applied to the call and to the put); with daily rebalancing the hedging noise can turn single paths positive.

(b) Size of the loss. The straddle premium $\approx 0.8\,S\,\sigma_{\text{imp}}\sqrt{T}$ is *linear in* σ. Had the market priced 20 instead of 25, the straddle would cost $0.8\,S\,\sigma_{\text{real}}\sqrt{T}$. By (a), the expected P and L of the hedged straddle is exactly its price at σ_{real} minus its price at σ_{imp}, for daily or any other rebalancing (Proposition 21.19); with the price linear in σ, the expected loss is minus that difference:

$$\text{loss} \approx 0.8\,S\sqrt{T}\,(\sigma_{\text{imp}} - \sigma_{\text{real}}) = \text{premium} \times \left(1 - \frac{20}{25}\right) = 20\% \text{ of premium}.$$

Concretely with $S = 100$: premium $\approx 0.8 \times 100 \times 0.25 \times 0.289 \approx 5.8$; loss ≈ 1.15, about 1.2% of spot.

(c) The variance risk premium. Implied vol systematically exceeds subsequently realised vol (equity indices: a few vol points on average), so delta-hedged long-option positions lose on average: that gap is the *variance risk premium* (Fact 21.29). It persists because options are insurance: hedgers pay up for protection; the short side's losses arrive precisely in high-vol crisis states (when marginal utility is highest), jump risk cannot be hedged away by trading the underlier, however often the hedge is rebalanced, and limits to arbitrage (margin calls, gamma losses when vol spikes) cap how hard sellers can lean in. The trader on the other side, short volatility, earns a steady stream of small carry profits punctuated by rare, violent drawdowns: "picking up nickels in front of a steamroller," or, sized honestly, being paid an insurance premium to bear catastrophe states.

Self-check. Via vega: straddle vega $= 2S\varphi(d_1)\sqrt{T} \approx 2 \times 100 \times 0.399 \times 0.289 \approx 23$ per unit of vol $= 0.23$ per vol point; times 5 points $= 1.15$, which matches the premium-ratio route. By exact repricing: in the driftless constant-volatility model the expected P and L of the hedged position is the Black–Scholes–Merton price at the realised vol minus the price at the implied vol, for every rebalancing grid (Proposition 21.19); with $S = 100$ the straddle is worth 5.76 at 25 vol and 4.61 at 20 vol, a difference of 1.15 again. Consistency of (a): if $\sigma_{\text{real}} = \sigma_{\text{imp}}$, the expected P and L is exactly 0 for every grid (Proposition 21.19), and under continuous rebalancing the P and L is 0 on every path (Proposition 21.22(i) and (ii)): the Black–Scholes replication result.

Answer: (a) The position loses on average: the daily P and L is $\frac{1}{2}\Gamma\,[(\Delta S)^2 - \sigma_{\text{imp}}^2 S^2\,\Delta t]$ up to a third-order remainder, a gamma-weighted realised-minus-implied variance, negative in expectation when realised 20% is below implied 25%. (b) Roughly $(1 - 20/25) = 20\%$ of the premium ($\approx 1.2\%$ of spot). (c) The variance risk premium: implied exceeds realised on average because sellers are paid to insure crash states; the short-vol counterparty earns steady carry with rare large losses.

P21.9 *Delta-, gamma- and vega-neutral with two traded options*

(a) Greeks add across positions (Definition 21.1(ii)):

$$\begin{aligned}
\Delta_\Pi &= -4{,}000\,(0.53) + 2{,}000\,(-0.20) - 1{,}000\,(0.30) = -2{,}120 - 400 - 300 = -2{,}820,\\
\Gamma_\Pi &= -4{,}000\,(0.00115) + 2{,}000\,(0.00080) - 1{,}000\,(0.00100) = -4.6 + 1.6 - 1.0 = -4.0,\\
\mathcal{V}_\Pi &= -4{,}000\,(6.9) + 2{,}000\,(4.8) - 1{,}000\,(6.0) = -27{,}600 + 9{,}600 - 6{,}000 = -24{,}000.
\end{aligned}$$

The book is short 2,820 index units of delta, short gamma and short vega.

(b) By Procedure 21.31 (one target), buy $w_1 = -\Gamma_\Pi/\Gamma_1 = 4.0/0.0020 = 2{,}000$ of option 1. The vega becomes $-24{,}000 + 2{,}000 \times 4.0 = -16{,}000$ per volatility point, and the delta $-2{,}820 + 2{,}000 \times 0.52 = -1{,}780$: buy 1,780 index units. A vega of $-16{,}000$ is left.

(c) Buy $w_2 = -\mathcal{V}_\Pi/\mathcal{V}_2 = 24{,}000/9.6 = 2{,}500$ of option 2. The gamma becomes $-4.0 + 2{,}500 \times 0.0008 = -2.0$, and the delta $-2{,}820 + 2{,}500 \times 0.42 = -1{,}770$: buy 1,770 index units. A gamma of -2.0 is left.

(d) The two targets give the system (21.12),

$$0.0020\,w_1 + 0.0008\,w_2 = 4.0, \qquad 4.0\,w_1 + 9.6\,w_2 = 24{,}000,$$

whose determinant is $0.0020 \times 9.6 - 0.0008 \times 4.0 = 0.0192 - 0.0032 = 0.0160$. By Cramer's rule (21.13),

$$\begin{aligned}
w_1 &= \frac{4.0 \times 9.6 - 0.0008 \times 24{,}000}{0.016} = \frac{38.4 - 19.2}{0.016} = 1{,}200,\\
w_2 &= \frac{0.0020 \times 24{,}000 - 4.0 \times 4.0}{0.016} = \frac{48 - 16}{0.016} = 2{,}000.
\end{aligned}$$

Buy 1,200 of option 1 and 2,000 of option 2. The delta becomes $-2{,}820 + 1{,}200 \times 0.52 + 2{,}000 \times 0.42 = -2{,}820 + 624 + 840 = -1{,}356$, so the last trade is a purchase of 1,356 index units. The delta is fixed last, because every option bought for its gamma or its vega brings its own delta, while the index has neither gamma nor vega (Proposition 21.4(iv)).

(e) With one further option of Greeks Γ_3 and $\mathcal{V}_3$, the gamma is removed by $w_3 = -\Gamma_\Pi/\Gamma_3$ and the vega by $w_3 = -\mathcal{V}_\Pi/\mathcal{V}_3$, and the two agree exactly when the option has the vega-to-gamma ratio of the book: $\mathcal{V}_3/\Gamma_3 = \mathcal{V}_\Pi/\Gamma_\Pi = 24{,}000/4.0 = 6{,}000$ per volatility point. Under one flat volatility, $\mathcal{V} = \sigma\tau S^2\Gamma$ per unit of volatility for every European call and put (Proposition 21.9(iii)), so with the vega per volatility point the ratio is $\mathcal{V}/\Gamma = \sigma\tau S^2/100 = 0.2 \times \tau \times 4{,}000{,}000/100 = 8{,}000\,\tau$ here. The ratio depends on the expiry alone, and it equals 6,000 exactly when $\tau = 0.75$: any nine-month European option on the index, call or put and of any strike, valued at the same flat volatility, does it, because the book consists of nine-month options. With the nine-month call struck at 2,000, buy $4.0/0.00115 = 3{,}478$ (to the nearest option): they bring $3{,}478 \times 6.9 = 23{,}998$

of vega, the 24,000 of the book up to the rounding to whole options, and then 977 index units restore the delta, since $-2{,}820 + 3{,}478 \times 0.53 = -977$. The two listed options have the ratios 2,000 and 12,000, so neither works alone, and two options of one expiry could not separate gamma from vega at all: their system has the determinant zero.

Self-check. The answer of (d) can be read off the ratios without solving the system. By (21.14), w_1 is the gamma-only hedge of (b) times $(12{,}000 - 6{,}000)/(12{,}000 - 2{,}000) = 0.6$, that is $2{,}000 \times 0.6 = 1{,}200$, and w_2 is the gamma-only hedge with option 2, $4.0/0.0008 = 5{,}000$, times $(6{,}000 - 2{,}000)/(12{,}000 - 2{,}000) = 0.4$, that is 2,000. The ratio of the book lies between those of the two options, which is why both are purchases. Substituting back: the gamma is $-4.0 + 1{,}200 \times 0.0020 + 2{,}000 \times 0.0008 = -4.0 + 2.4 + 1.6 = 0$, the vega is $-24{,}000 + 4{,}800 + 19{,}200 = 0$, and the delta is $-2{,}820 + 624 + 840 + 1{,}356 = 0$. The determinant also has the form $\sigma S^2 \, \Gamma_1 \Gamma_2 \, (\tau_2 - \tau_1)/100 = 0.2 \times 4{,}000{,}000 \times 0.0020 \times 0.0008 \times 1.25/100 = 0.016$.

Answer: (a) $\Delta_\Pi = -2{,}820$, $\Gamma_\Pi = -4.0$, $\mathcal{V}_\Pi = -24{,}000$ per volatility point. (b) Buy 2,000 of option 1 and 1,780 index units; a vega of $-16{,}000$ is left. (c) Buy 2,500 of option 2 and 1,770 index units; a gamma of -2.0 is left. (d) Buy 1,200 of option 1, 2,000 of option 2 and 1,356 index units. (e) When the option has the vega-to-gamma ratio of the book, 6,000 per volatility point, which under a flat volatility means the same expiry: any nine-month European option, for instance 3,478 of the calls struck at 2,000, then 977 index units.

P21.10 *The stop-loss rule is not free; synthetic options along two paths*

(a) The call starts out of the money, so the rule starts with no capital and no share. At the first monitoring date at which the price is above K it buys one share; at the next date at which the price is at or below K it sells the share; and so on until T, when it holds the share exactly if the last observed price before T was above the strike. The rule looks free because, if every purchase and every sale were made at the price K, the writer would end with the share when the call is exercised and with nothing otherwise, having paid nothing on the way. It is a poor hedge because a crossing of the strike is seen only after it has happened: the rule buys above K and sells below K, each round trip loses the two overshoots, and the number of crossings grows as the monitoring is refined. Part (b) shows that on average the rule falls short by the whole time value of the option, for every grid. In simulation the spread of the shortfall hardly shrinks as the monitoring is refined (Example 21.35), whereas the error of a delta hedge shrinks like $n^{-1/2}$ (Proposition 21.27).

(b) We integrate out each future move, in three steps.

Step 1 (the payoff). Since $W_T \sim \mathcal{N}(0, T)$, the price at expiry is $S_T = S_0 \exp(-\sigma^2 T/2 + \sigma\sqrt{T}\, Z)$ with Z standard normal: a lognormal variable with mean $\mathbb{E}\, S_T = S_0 e^{-\sigma^2 T/2}\, \mathbb{E}\, e^{\sigma\sqrt{T}\, Z} = S_0$ (Lemma 18.35(i) with $\theta = \sigma\sqrt{T}$), the forward price at $r = q = 0$. Its call payoff has the mean $\mathbb{E}\,(S_T - K)^+ = S_0\, \Phi(d_1) - K\, \Phi(d_2) = C$ (Theorem 20.22 at $r = 0$), as the statement recalls. The payoff is integrable, because $0 \le (S_T - K)^+ \le S_T$ and $\mathbb{E}\, S_T = S_0$.

Step 2 (each hedge term has mean zero). Fix i and write $S_{t_{i+1}} - S_{t_i} = S_{t_i}\,(Y_i - 1)$, where

$$Y_i = \exp\big(-\sigma^2 (t_{i+1} - t_i)/2 + \sigma\,(W_{t_{i+1}} - W_{t_i})\big).$$

The holding h_i and the price S_{t_i} are functions of W_{t_i}, while Y_i is a function of the increment $W_{t_{i+1}} - W_{t_i}$, which is independent of the path up to t_i (Proposition 19.8(iv)) and has the law of $\sqrt{t_{i+1} - t_i}\, Z$. With $\theta = \sigma\sqrt{t_{i+1} - t_i}$,

$$\mathbb{E}\, Y_i = e^{-\theta^2/2}\, \mathbb{E}\, e^{\theta Z} = e^{-\theta^2/2}\, e^{\theta^2/2} = 1$$

(Lemma 18.35(i)). The hedge term is integrable, because $|h_i\,(S_{t_{i+1}} - S_{t_i})| \le S_{t_{i+1}} + S_{t_i}$, whose mean is $2S_0$, because $\mathbb{E}\, S_t = S_0$ for every t by the same computation with $\theta = \sigma\sqrt{t}$. Integrate Y_i out first, with the earlier price frozen (Lemma 19.17): for a frozen value x of S_{t_i}, the mean of $\mathbf{1}\{x > K\}\, x\,(Y_i - 1)$ is $\mathbf{1}\{x > K\}\, x\,(\mathbb{E}\, Y_i - 1) = 0$. Hence $\mathbb{E}\big[h_i\,(S_{t_{i+1}} - S_{t_i})\big] = 0$.

Step 3 (conclusion). The shortfall is a sum of integrable terms, so its mean is the sum of their means (Theorem 1.2, in the general form declared in §19.2): $C - (S_0 - K)^+ - 0$. The monitoring dates entered through Step 2 alone, which holds for every grid. Step 2 goes through unchanged for any holding h_i that is a bounded function of $(S_{t_0}, \dots, S_{t_i})$: freeze the vector $(W_{t_1}, \dots, W_{t_i})$, which is independent of Y_i by Proposition 19.8(iv), and bound $|h_i\,(S_{t_{i+1}} - S_{t_i})|$ by $c\,(S_{t_{i+1}} + S_{t_i})$, where the constant c bounds $|h_i|$. So every rule that starts from the intrinsic value and trades on past prices alone has the same expected shortfall. This is Proposition 21.34.

(c) Here $\sigma\sqrt{T} = 0.1$ and

$$d_1 = \frac{\ln(0.95) + 0.005}{0.1} = \frac{-0.05129 + 0.005}{0.1} = -0.4629, \qquad d_2 = -0.5629,$$

so $C = 95\,\Phi(-0.4629) - 100\,\Phi(-0.5629) = 95\,(0.32171) - 100\,(0.28674) = 30.562 - 28.674 = 1.888$. The call has no intrinsic value, so the time value is 1.888, and by (b) this is the expected shortfall. Each round trip buys at 100.05 and sells at 99.95 and so loses 0.10; a last purchase that is not followed by a sale, when the call ends in the money, is delivered against the strike and loses 0.05. Every trade therefore costs 0.05, the expected number of trades is about $1.888/0.05 = 38$, and the expected number of round trips is about $(1.888 - 0.05\,\Phi(d_2))/0.10 = (1.888 - 0.014)/0.10 = 18.7$, about 19. The count is an estimate, as in Remark 21.36; what is exact is the mean shortfall.

(d) The path that swings costs more. The replicating position holds the delta of the call, which rises when the price rises and falls when it falls, because the gamma is positive: the replicator buys after every rise and sells after every fall, and each swing realises a loss of about $\frac{1}{2}\Gamma\,(\Delta S)^2$. The cost of the replication beyond what the pricing volatility has paid for is the gamma-weighted excess of realised over implied variance along the path, about $\sum_i \frac{1}{2}\Gamma_{t_i}\big[(\Delta S_i)^2 - \sigma_{\text{imp}}^2 S_{t_i}^2\,\Delta t\big]$ with σ_{imp} the pricing volatility (Remark 21.21; in continuous time, Proposition 21.22(ii) and Corollary 21.24). Both paths end at 60 and owe the same payoff, 15. The steady path realises little variance, so the replication started from the price of the call ends with money to spare. The wild path realises more variance than was paid for, much of it near the strike where the gamma is large, and the replication falls short.

(e) The writer of a call hedges by holding $+\Delta$ shares, so that the hedge and the written option together are flat. The hedge alone therefore behaves like a bought call: run without the written option and financed by the premium, the same strategy manufactures the call. Creating an option synthetically is running the hedge of a short position in it; the dealer who has sold an option and hedges it is creating a long option to set against the one sold. Hedging a bought option, in turn, creates a written one. Proposition 21.39 is this statement for a put.

Self-check. With a single interval ($n = 1$) the result of (b) follows from put–call parity, with no use of Step 2. If $S_0 \le K$, the rule holds nothing and the shortfall is the payoff $(S_T - K)^+$, of mean C, the time value. If $S_0 > K$, the rule holds the share throughout and the shortfall is $(S_T - K)^+ - (S_0 - K) - (S_T - S_0) = (S_T - K)^+ - (S_T - K) = (K - S_T)^+$, the payoff of the put, whose mean is $\mathcal{P} = C - (S_0 - K)$ by put–call parity at $r = 0$: the time value again. In numbers, for $S_0 = 105$ with the other data of (c), $C = 7.064$ and $\mathcal{P} = 2.064 = 7.064 - 5$. For (c) itself, parity gives the same time value from the put: $\mathcal{P} = 100\,\Phi(0.5629) - 95\,\Phi(0.4629) = 6.888$, and $6.888 - (100 - 95) = 1.888$.

Answer: (a) The rule buys the share whenever an observed price is above K and sells it whenever one is back below; it looks free if every trade is at K, but it buys above and sells below the strike at every crossing. (b) $\mathbb{E}\,(S_T - K)^+ = C$ and each term $h_i\,(S_{t_{i+1}} - S_{t_i})$ has mean zero, because h_i depends on the past and the ratio $S_{t_{i+1}}/S_{t_i}$ is independent of it with mean 1; so the expected shortfall is $C - (S_0 - K)^+$ for every grid. (c) Time value 1.888; about 19 round trips, or 38 trades. (d) The wild path: the replicator buys high and sells low on every swing, and the cost beyond the premium is the gamma-weighted realised-minus-implied variance. (e) The hedge of a written option, run on its own, is a synthetic long option.

P21.11 *Insuring a low-beta portfolio: index puts, then synthetically*

(a) *The mapping.* The total return of the portfolio is $(V_{A,T} - V_A + q_A V_A \tau)/V_A$. Setting it equal to $r\tau + \beta\,(S_T/S - 1 + q\tau - r\tau)$ and solving for $V_{A,T}$ gives $V_{A,T} = V_A\,(a + \beta\,S_T/S)$ with $a = 1 + r\tau - q_A\tau - \beta\,(1 + r\tau - q\tau)$ (Proposition 21.37(i)). Here $r\tau = 0.03$, $q_A\tau = 0.015$ and $q\tau = 0.01125$, so

$$a = 1.015 - 0.8 \times 1.01875 = 0.200, \qquad V_{A,T} = 25\left(0.2 + 0.8\,\frac{S_T}{5{,}000}\right) \text{ million.}$$

The puts. The portfolio reaches the floor when $0.2 + 0.8\,S_T/5{,}000 = 23/25 = 0.92$, that is, at $S_T = 5{,}000 \times 0.72/0.8 = 4{,}500$, which is the strike. Below it the portfolio loses $\beta V_A/S$ per index point, so the puts are on

$$n_{\text{put}} = \frac{\beta V_A}{S} = \frac{0.8 \times 25{,}000{,}000}{5{,}000} = 4{,}000 \text{ index units, struck at } K = 4{,}500$$

(Proposition 21.37(ii)).

The cost. With $\sigma\sqrt{\tau} = 0.18\sqrt{0.75} = 0.155885$ and $(r - q + \sigma^2/2)\tau = (0.025 + 0.0162)(0.75) = 0.0309$,

$$d_1 = \frac{\ln(5{,}000/4{,}500) + 0.0309}{0.155885} = \frac{0.105361 + 0.0309}{0.155885} = 0.87411, \qquad d_2 = 0.71823.$$

The put of Corollary 20.41(i) is worth

$$\begin{aligned}\mathcal{P} &= 4{,}500\,e^{-0.03}\,\Phi(-0.71823) - 5{,}000\,e^{-0.01125}\,\Phi(-0.87411)\\ &= 4{,}367.00 \times 0.236309 - 4{,}944.07 \times 0.191029 = 1{,}031.96 - 944.46 = 87.50\end{aligned}$$

per index unit. The insurance costs $4{,}000 \times 87.50 = 350{,}000$, which is 1.4% of the portfolio.

(b) With $\beta = 1$, $a = 1.015 - 1.01875 = -0.00375$, so $V_{A,T} = 25\,(-0.00375 + S_T/5{,}000)$ million; the strike is $K = 5{,}000\,(0.92 + 0.00375) = 4{,}618.75$, and the puts are on $n_{\text{put}} = 25{,}000{,}000/5{,}000 = 5{,}000$ index units. Then

$$d_1 = \frac{\ln(5{,}000/4{,}618.75) + 0.0309}{0.155885} = \frac{0.079314 + 0.0309}{0.155885} = 0.70702, \qquad d_2 = 0.55114,$$

and $\mathcal{P} = 4{,}618.75\,e^{-0.03}\,(0.290770) - 4{,}944.07\,(0.239776) = 1{,}303.30 - 1{,}185.47 = 117.83$. The insurance costs $5{,}000 \times 117.83 = 589{,}150$, about 0.589 million, which is 2.4% of the portfolio. The ratio of the costs is $589/350 = 1.68$, above the ratio of the betas, 1.25: a higher beta needs more puts, and it also lifts the strike toward the index, from 4,500 to 4,619, so each put is dearer as well. (The strike rises with beta because the floor, $0.92\,V_A$, lies below $V_A\,(1 + (r - q_A)\tau) = 1.015\,V_A$.) The low-beta portfolio needs fewer puts, struck further out of the money.

(c) By Proposition 21.2 the put of (a) has

$$\Delta_{\mathcal{P}} = e^{-q\tau}\,(\Phi(d_1) - 1) = 0.98881 \times (-0.191029) = -0.1889.$$

The replication of the 4,000 puts (Proposition 21.39) is short $4{,}000 \times 0.1889 = 755.6$ index units, which is 3.78 million of index exposure. It starts from the capital of 0.35 million, the price of the puts, and holds $4{,}000 \times Ke^{-r\tau}\Phi(-d_2) = 4{,}000 \times 1{,}031.96 = 4.13$ million in bills: the proceeds of the short sale plus the capital. In delta, and on the CAPM line, the short sale is the sale of $3.778/0.8 = 4.72$ million of the portfolio itself into bills, which is 18.9% of it, the fraction $|\Delta_{\mathcal{P}}|$; 20.28 million stays invested.

With futures the portfolio is kept. A one-year futures has $\tau_F = 1$ and the delta $e^{(r-q)\tau_F}$ per index unit (Proposition 21.4(iii)), so the short position is $755.6\,e^{-0.025} = 736.9$ index units, which is 73.7 contracts of 10: sell 74. In the terms of Proposition 16.40, the futures price is $5{,}000\,e^{0.025} = 5{,}126.6$, one contract has the value 51,266, the full hedge is $0.8 \times 25{,}000{,}000/51{,}266 = 390.1$ contracts, and

$0.1889 \times 390.1 = 73.7$. The beta of the position falls from 0.8 to $0.8 \times (1 - 0.1889) = 0.65$. As the index falls, $|\Delta_{\mathcal{P}}|$ grows and more is sold; as it recovers, the position is bought back.

(d) The floor rests on the hypothesis that the portfolio lies on the CAPM line with no residual. A real portfolio carries the idiosyncratic residual of Proposition 11.36, which index puts do not insure, its beta is an estimate, and the mapping is a one-period model. For the index itself the mapping is an identity: $\beta = 1$, $q_A = q$ and there is no residual, so $a = 0$ and the hedge is exact. For any other portfolio the floor rests on the CAPM hypothesis. Dynamic insurance, for its part, requires selling as prices fall, and its replication argument assumes continuous prices and trades that can be made near the model's prices without moving the market. In a crash prices gap, the selling cannot be executed near those prices, and many insurers following the same rule push prices lower and trigger further sales (Remarks 21.41 and 15.7), so the portfolios end below their floors, whereas a bought put pays regardless.

Self-check. Scenarios test the floor without the formulas for the strike and the number of puts. If the index ends at 4,000, the portfolio with $\beta = 0.8$ is worth $25\,(0.2 + 0.64) = 21.0$ million and the puts pay $4{,}000 \times 500 = 2.0$ million: 23.0 million. At 4,500 the portfolio is worth $25\,(0.2 + 0.72) = 23.0$ million and the puts pay nothing; at 5,500 it is worth 27.0 million. For $\beta = 1$ at 4,000: $25\,(-0.00375 + 0.8) = 19.906$ million plus $5{,}000 \times 618.75 = 3.094$ million, again 23.0 million. The put prices are checked in the forward form, which recomputes d_1 from the first expression of (21.1) and uses $Se^{-q\tau} = e^{-r\tau}F$ in (21.2), so that $\mathcal{P} = e^{-r\tau}\big(K\Phi(-d_2) - F\Phi(-d_1)\big)$. The forward is $F = 5{,}000\,e^{0.01875} = 5{,}094.63$ and $e^{-r\tau} = 0.970446$. At $K = 4{,}500$, $d_1 = (0.124111 + 0.012150)/0.155885 = 0.87411$ and $\mathcal{P} = 0.970446 \times (1{,}063.389 - 973.222) = 0.970446 \times 90.167 = 87.50$; at $K = 4{,}618.75$, $d_1 = (0.098064 + 0.012150)/0.155885 = 0.70702$ and $\mathcal{P} = 0.970446 \times (1{,}342.993 - 1{,}221.573) = 0.970446 \times 121.420 = 117.83$. The bills of (c) are checked by addition: $755.6 \times 5{,}000 = 3.778$ million of proceeds plus 0.350 million of capital is 4.128 million.

Answer: (a) $V_{A,T} = 25\,(0.2 + 0.8\,S_T/5{,}000)$ million; puts on 4,000 index units struck at 4,500, each worth 87.50: 0.35 million, 1.4% of the portfolio. (b) For $\beta = 1$: $V_{A,T} = 25\,(-0.00375 + S_T/5{,}000)$ million; puts on 5,000 units struck at 4,618.75, each worth 117.83: 0.589 million, 2.4%; the cost ratio 1.68 exceeds the beta ratio 1.25. (c) $\Delta_{\mathcal{P}} = -0.1889$: short 755.6 index units (3.78 million) with 4.13 million in bills, equivalent to selling 4.72 million (18.9%) of the portfolio; or sell 73.7, in practice 74, one-year futures contracts. (d) The floor holds only on the CAPM line, the residual is not insured; dynamic insurance needs sales into a falling market, which gaps, illiquidity and the feedback of many insurers selling at once defeat.

28.22 Solutions — §22 Volatility: Smiles, Surfaces, Estimation and Variance Swaps

P22.1 *Call and put at different implied volatilities*

(a) European put–call parity (Theorem 17.9) requires

$$C - \mathcal{P} = S - Ke^{-rT} = 80 - 78.809 = 1.191.$$

The quotes give $5.986 - 5.462 = 0.524$. Given the call, parity prices the put at $5.986 - 1.191 = 4.795$, and it trades at 5.462: the put is dear relative to the call by $5.462 - 4.795 = 0.667$. (Equivalently the call is cheap: given the put, parity prices it at $5.462 + 1.191 = 6.653$.) Sell the dear side, buy the cheap one, and remove the exposure to the share. Write the put and buy the call, which together are a long forward contract with delivery price 80 (Corollary 17.12(i)); sell the share short; and lend 78.809, which returns 80 at T. The trade is called a reversal.

Trade	today	T, if $S_T \le 80$	T, if $S_T > 80$
Write the put	$+5.462$	$-(80 - S_T)$	0
Buy the call	-5.986	0	$S_T - 80$
Short the share	$+80.000$	$-S_T$	$-S_T$
Lend 78.809	-78.809	$+80$	$+80$
Net	$+0.667$	0	0

In both columns the share that closes the short sale is bought for exactly 80, by exercising the call if $S_T > 80$ and because the put is exercised against the position if $S_T \le 80$, and the loan repays the 80. The trade brings in 0.667 today and leaves nothing to pay at T in any state: an arbitrage.

(b) With $\ln(F/K) = rT = 0.015$, $\sigma\sqrt{T} = 0.24 \times 0.7071 = 0.1697$ and $\sigma^2 T/2 = 0.0144$,

$$d_1 = \frac{0.015 + 0.0144}{0.1697} = 0.173,$$
$$\mathcal{V} = e^{-rT} F\, \varphi(d_1)\sqrt{T} = S\, \varphi(d_1)\sqrt{T} = 80 \times 0.393 \times 0.7071 = 22.23$$

per unit of volatility, the same for the call and the put (Proposition 20.35). The formula itself satisfies parity at every trial volatility (Step 2 of the proof of Proposition 22.3), so at 24%, the volatility that reprices the call, it prices the put at $5.986 - 1.191 = 4.795$, the parity-consistent value. The quoted put sits 3 volatility points above 24%, worth about $22.23 \times 0.03 = 0.667$: the amount of (a). The vega barely moves between the two volatilities (at 27%, $d_1 = 0.174$ and $\mathcal{V} = 22.23$ again), which is why the first-order estimate is exact to the digits shown.

(c) No. The trade of (a) rests on payoffs alone: a long call and a short put pay $S_T - K$ in every state, whatever the law of S_T. The Black–Scholes–Merton formula enters only as the map that turns each price into a volatility. By Proposition 22.3, for arbitrage-free prices the pricing errors of the formula for the call and for the put are equal at every trial volatility, so the two options always carry one implied volatility; two different implied volatilities, both computed at the market forward, are a restatement of a parity violation and not a view on volatility.

(d) No. For American options on a stock without dividends parity is a band (Proposition 17.14),

$$S - K \;\le\; C^{\text{am}} - \mathcal{P}^{\text{am}} \;\le\; S - Ke^{-rT}, \qquad \text{here} \qquad 0 \;\le\; C^{\text{am}} - \mathcal{P}^{\text{am}} \;\le\; 1.191,$$

and the quoted difference 0.524 lies inside it. Nothing else rules the quotes out either: they are the prices of a model with two states. Let the share jump at once to 92.415 or to 70.247, with the probabilities 0.43995 and 0.56005, so that its mean after the jump is 80, and then grow at the rate r. In the high state the share stays above 80, the put is worthless and the call is worth $92.415 - 78.809 = 13.606$. In the low state $70.247\, e^{0.015} = 71.31 < 80$, so the call is worthless, and the put is exercised at once, because $80 - 70.247$ exceeds $80\, e^{-rt} - 70.247$, the value today of exercise at any later date t. Weighted by the two probabilities these values are $0.43995 \times 13.606 = 5.986$ and $0.56005 \times (80 - 70.247) = 5.462$, the two quotes. In this model every price is the probability-weighted value of the claim just after the jump, and after the jump nothing is random, so a position that costs nothing today has a weighted value of zero just after the jump and cannot be worth a nonnegative amount in both states and a positive amount in one: the quotes admit no arbitrage. The reversal of (a) is no longer riskless, because the written put can be exercised against the position at a date $t_e < T$: the position must then pay 80 for the share while its loan is worth only $80\, e^{-r(T - t_e)} < 80$. Read the other way, the band allows the American put any price from $5.986 - 1.191 = 4.795$ to 5.986. A volatility of 27% obtained by putting the American put price into the European formula measures the value of early exercise as well, and no riskless trade follows from the gap.

Self-check. Read parity backwards, without the cash-flow table. The two quotes imply the forward price $K + e^{rT}(C - \mathcal{P}) = 80 + 1.01511 \times 0.524 = 80.532$ (Remark 20.46), while buying the share

and carrying it gives $F = 80\,e^{0.015} = 81.209$. Selling the share forward at 81.209 through the cash market and buying it forward at 80.532 through the options leaves 0.677 at T, whose present value is $0.677\,e^{-0.015} = 0.667$, the amount of (a). The vega route of (b) gives 0.667 a third time. For (d), $4.795 \le 5.462 \le 5.986$ confirms that the put quote is inside the American band.

Answer: (a) Parity requires $C - \mathcal{P} = 1.191$ against the quoted 0.524: the put is dear (the call cheap) by 0.667; write the put, buy the call, sell the share short and lend 78.809, which brings in 0.667 today and leaves 0 at T in every state. (b) $\mathcal{V} = 22.23$ and $22.23 \times 0.03 = 0.667$. (c) No: parity is model-free, and the formula only converts prices into volatilities. (d) No: the American band $0 \le C^{\text{am}} - \mathcal{P}^{\text{am}} \le 1.191$ contains 0.524, and early exercise of the written put breaks the trade; a two-state model reproduces both quotes, so they admit no arbitrage.

P22.2 *Earnings tomorrow: the binary frown*

(a) The market is a one-step tree (Definition 18.6) with up-factor $46/40 = 1.15$ and down-factor $36/40 = 0.9$, and $0.9 < e^{r\Delta t} = 1 < 1.15$. The risk-neutral up-probability p makes the stock earn the zero rate, $40 = 46p + 36(1-p)$:

$$p = \frac{40 - 36}{46 - 36} = 0.4, \qquad 1 - p = 0.6.$$

(b) By Proposition 18.7(ii) at $r = 0$, each option is worth its $\mathbb{Q}$-expected payoff. For $36 \le K \le 46$ a call pays only in the up state, so $C(K) = 0.4\,(46 - K)$:

$$C(38) = 3.20, \qquad C(40) = 2.40, \qquad C(42) = 1.60, \qquad C(44) = 0.80.$$

Parity at $r = 0$ and $F = S = 40$ reads $\mathcal{P} = C - (40 - K)$ (Theorem 17.9), so

$$\mathcal{P}(38) = 1.20, \qquad \mathcal{P}(40) = 2.40, \qquad \mathcal{P}(42) = 3.60, \qquad \mathcal{P}(44) = 4.80.$$

Directly, a put pays only in the down state: $\mathcal{P}(38) = 0.6 \times 2 = 1.20$ and $\mathcal{P}(44) = 0.6 \times 8 = 4.80$.

(c) *At the money.* The rule of Proposition 20.29, $2.40 \approx 0.4 \times 40 \times \sigma_{\text{imp}}\sqrt{T}$, gives $\sigma_{\text{imp}}\sqrt{T} \approx 0.150$, and with the constant $1/\sqrt{2\pi} = 0.3989$ in place of 0.4 it gives 0.1504. By (20.11) this linear rule overstates the price by at most the fraction $\sigma^2 T/24 = 0.0009$, so the exact total volatility lies between 0.1504 and $0.1504/(1 - 0.0009) = 0.1505$, which confirms the quoted 15.05%.

The shape. The profile 13.17, 15.05, 15.13, 13.50 is highest near the money and lower in both wings: a frown. A two-point law has no tails: it puts no mass below 36 or above 46, whereas a lognormal law with the at-the-money total volatility does. One volatility for every strike would therefore charge for moves that cannot happen: at the total volatility 0.1505 the formula prices the 38 call at 3.47 and the 44 call at 1.01, against 3.20 and 0.80. Matching the cheaper wings takes lower volatilities.

The edges. The time value of the call is $0.4\,(46 - K) - (40 - K) = 0.6\,(K - 36)$ for $36 \le K \le 40$, and its whole value $0.4\,(46 - K)$ for $40 \le K \le 46$. Strictly between 36 and 46 the price lies strictly inside the bounds $(F - K)^+ < C < F$, so the implied volatility exists and is unique (Proposition 20.38(ii)). The time value tends to 0 at both edges, and the implied volatility tends to 0 with it: for any fixed volatility $\sigma' > 0$ the formula's time value $K \mapsto C^{\text{BSM}}(K, T; \sigma') - (F - K)^+$ is continuous in the strike (d_1, d_2 and Φ are continuous) and positive at every strike, edges included, because by Proposition 20.38(i) the formula's price strictly increases in the volatility from its limit $(F - K)^+$ as $\sigma \to 0$. It is therefore at least some $\varepsilon > 0$ on a neighbourhood of each edge. There the market's time value, $0.6\,(K - 36)$ or $0.4\,(46 - K)$, is eventually below ε, so the market price is below $C^{\text{BSM}}(K, T; \sigma')$, and the strict monotonicity of the formula in the volatility (Proposition 20.38(i)) forces $\sigma_{\text{imp}} < \sigma'$. The volatility σ' was arbitrary, so $\sigma_{\text{imp}} \to 0$ at both edges. At $K \le 36$ the call is worth $40 - K$, and at $K \ge 46$ it is worth 0: in both cases the price sits on the lower bound $(F - K)^+$, and no implied volatility exists (Proposition 20.38(ii)); desks quote it as 0.

(d) At $r = 0$ a put is worth the area under the distribution function to the left of its strike and a call the area under the tail probability to the right of it, $\mathcal{P}(K) = \int_0^K \mathbb{Q}(S_T \le x)\,dx$ and $C(K) = \int_K^\infty \mathbb{Q}(S_T > x)\,dx$: take expectations in $(K - S_T)^+ = \int_0^K \mathbf{1}\{S_T \le x\}\,dx$ and $(S_T - K)^+ = \int_K^\infty \mathbf{1}\{S_T > x\}\,dx$, exchanging the order of integration by Fact 18.34(i). Take as comparison the lognormal law with mean F and the at-the-money implied volatility, which prices the at-the-money option exactly, as in (c). A tail heavier than that law's makes the out-of-the-money options on its side dearer than the formula at the at-the-money volatility. The formula's price strictly increases in the volatility (Proposition 20.38(i)), so their implied volatility lies above the at-the-money level; a thinner tail puts it below.

(i) Both tails thinner: both wings below the at-the-money volatility, a *frown*, as in (c).
(ii) Heavier right tail, thinner left tail: high strikes dear and low strikes cheap, an *upward skew*.
(iii) As long as the band is believed to hold, the law has no mass beyond its edges: both tails are cut off, a *frown*, with the implied volatility falling towards zero at the edges as in (c).
(iv) A volatility that rises as the price falls fattens the left tail and thins the right one (the leverage effect): low strikes dear, a *downward skew*, the equity case of Fact 22.5.

Self-check. Price two calls by replication instead of by expectation. The 40 call pays 6 or 0: hold $(6-0)/(46-36) = 0.6$ shares and borrow $0.6 \times 36 = 21.6$; the portfolio pays $0.6 \times 46 - 21.6 = 6$ or $0.6 \times 36 - 21.6 = 0$ and costs $24 - 21.6 = 2.40$. The 44 call pays 2 or 0: hold 0.2 shares and borrow 7.2, at a cost of $8 - 7.2 = 0.80$. Both agree with (b). The risk-neutral mean is $0.4 \times 46 + 0.6 \times 36 = 40 = F$, as it must be. For (c), the exact at-the-money price is $F\,[2\,\Phi(\sigma\sqrt{T}/2) - 1]$, and $2.40/40 = 0.06$ requires $\Phi(\sigma\sqrt{T}/2) = 0.53$, that is $\sigma\sqrt{T}/2 = 0.0753$ and $\sigma\sqrt{T} = 0.1505$, the quoted value.

Answer: (a) $p = 0.4$. (b) Calls 3.20, 2.40, 1.60, 0.80; puts 1.20, 2.40, 3.60, 4.80. (c) $2.40/(0.4 \times 40) = 0.150$ confirms 15.05%; the profile is a frown, because a two-point law has no tails; the implied volatility tends to 0 as K decreases to 36 or increases to 46, and at or beyond those strikes the price is on the lower bound and no implied volatility exists (quoted as 0). (d) (i) frown; (ii) upward skew; (iii) frown; (iv) downward skew.

P22.3 *Breeden-Litzenberger*

(a) Difference quotients, with no limit taken under an integral sign. Take the hypotheses of the Setup of §22.2: under $\mathbb{Q}$ the law of $S_T > 0$ has a density $f_{\mathbb{Q}}$ that is continuous on $(0, \infty)$ (the words "risk-neutral density" presuppose a density; its continuity is what makes the identity hold at every strike), with a finite mean, and every European claim that pays a function of S_T at T, with finite $\mathbb{Q}$-expected absolute payoff, is worth its discounted $\mathbb{Q}$-expected payoff. Since $0 \le (x-K)^+ \le x$ for $x > 0$ and the mean is finite, Fact 7.22 applies to the payoff $x \mapsto (x-K)^+$ and gives

$$C(K) = e^{-rT} \int_K^\infty (x - K)\, f_{\mathbb{Q}}(x)\,dx,$$

and the digital, whose payoff is the indicator $\mathbf{1}\{S_T > K\}$, is worth $\mathrm{Dig}(K) = e^{-rT}\,\mathbb{Q}(S_T > K)$.

Step 1: an exact difference. Fix $K > 0$ and $\Delta K > 0$. Every integral below is finite: on $[K, \infty)$ the integrands $(x-K)f_{\mathbb{Q}}(x)$ and $f_{\mathbb{Q}}(x)$ are nonnegative and at most $x\,f_{\mathbb{Q}}(x)$ and $f_{\mathbb{Q}}(x)$, whose integrals over $(0, \infty)$ are the finite mean and 1. Write $x - K - \Delta K = (x-K) - \Delta K$ in the integral for $C(K + \Delta K)$, and split the integral for $C(K)$ at $K + \Delta K$:

$$e^{rT}\,C(K + \Delta K) = \int_{K+\Delta K}^\infty (x - K)\, f_{\mathbb{Q}}(x)\,dx - \Delta K \int_{K+\Delta K}^\infty f_{\mathbb{Q}}(x)\,dx,$$

$$e^{rT}\,C(K) = \int_K^{K+\Delta K} (x - K)\, f_{\mathbb{Q}}(x)\,dx + \int_{K+\Delta K}^\infty (x - K)\, f_{\mathbb{Q}}(x)\,dx.$$

Subtracting, the common integral cancels, $\int_{K+\Delta K}^\infty f_{\mathbb{Q}}(x)\,dx = \mathbb{Q}(S_T > K + \Delta K)$, and

$$C(K + \Delta K) - C(K) = -\,e^{-rT}\Big[\Delta K\;\mathbb{Q}(S_T > K + \Delta K) + \int_K^{K+\Delta K} (x - K)\, f_{\mathbb{Q}}(x)\,dx\Big].$$

Step 2: two sandwiches. On $[K, K+\Delta K]$ the weight $x-K$ lies between 0 and ΔK, so the last integral lies between 0 and $\Delta K\ \mathbb{Q}(K < S_T \le K+\Delta K)$. Dividing by ΔK and using $\mathbb{Q}(S_T > K+\Delta K) + \mathbb{Q}(K < S_T \le K+\Delta K) = \mathbb{Q}(S_T > K)$,

$$-\,e^{-rT}\,\mathbb{Q}(S_T > K) \;\le\; \frac{C(K+\Delta K) - C(K)}{\Delta K} \;\le\; -\,e^{-rT}\,\mathbb{Q}(S_T > K+\Delta K).$$

The same inequality with $K-\Delta K$ in place of K, for $0 < \Delta K < K$, is

$$-\,e^{-rT}\,\mathbb{Q}(S_T > K-\Delta K) \;\le\; \frac{C(K) - C(K-\Delta K)}{\Delta K} \;\le\; -\,e^{-rT}\,\mathbb{Q}(S_T > K).$$

Step 3: the first derivative. The density is continuous on the closed bounded interval $[K/2, 2K]$, hence bounded there by a constant c. For $0 < \Delta K \le K/2$ the probabilities $\mathbb{Q}(K < S_T \le K+\Delta K)$ and $\mathbb{Q}(K-\Delta K < S_T \le K)$ are integrals of the density over intervals of length ΔK inside $[K/2, 2K]$, so each is at most $c\,\Delta K$. Since $\mathbb{Q}(S_T > K+\Delta K) = \mathbb{Q}(S_T > K) - \mathbb{Q}(K < S_T \le K+\Delta K)$ and $\mathbb{Q}(S_T > K-\Delta K) = \mathbb{Q}(S_T > K) + \mathbb{Q}(K-\Delta K < S_T \le K)$, the outer bounds of the two sandwiches lie within $e^{-rT} c\,\Delta K$ of $-e^{-rT}\,\mathbb{Q}(S_T > K)$, and so do both difference quotients. Letting $\Delta K \to 0$, the right-hand and the left-hand derivatives exist and agree:

$$\partial_K C(K) = -\,e^{-rT}\,\mathbb{Q}(S_T > K), \qquad \text{so} \qquad \mathrm{Dig}(K) = e^{-rT}\,\mathbb{Q}(S_T > K) = -\,\partial_K C(K).$$

The digital is the negative slope of the call curve.

Step 4: the second derivative. For a fixed reference strike $K' > 0$ and every $K > 0$, $\mathbb{Q}(S_T > K) = \mathbb{Q}(S_T > K') - \int_{K'}^{K} f_{\mathbb{Q}}(x)\,dx$, the integral being read as $-\int_{K}^{K'}$ when $K < K'$ (split $\{S_T > \min(K, K')\}$ into the disjoint events $\{\min(K, K') < S_T \le \max(K, K')\}$ and $\{S_T > \max(K, K')\}$). The density is continuous, so by the fundamental theorem of calculus the right side is differentiable in K with derivative $-f_{\mathbb{Q}}(K)$. Differentiating the slope once more,

$$\partial_{KK} C(K) = e^{-rT} f_{\mathbb{Q}}(K) \qquad \Longrightarrow \qquad f_{\mathbb{Q}}(K) = e^{rT}\,\partial_{KK} C(K).$$

This is Theorem 22.7, with (22.2) and (22.3).

(b) Finite differences with $\Delta K = 5$, $r = 0$:

$$\mathrm{Dig}(100) \approx -\frac{C(105) - C(95)}{2\,\Delta K} = -\frac{2-7}{10} = 0.50,$$

$$f_{\mathbb{Q}}(100) \approx \frac{C(95) - 2\,C(100) + C(105)}{(\Delta K)^2} = \frac{7-8+2}{25} = 0.04$$

per unit of price. The two one-sided slope estimates are $(7-4)/5 = 0.60$ and $(4-2)/5 = 0.40$. By the sandwiches of Step 2 at $K = 100$, that is by (H.4), they bracket the digital itself, $0.40 \le \mathrm{Dig}(100) \le 0.60$, and, when the density is also continuously differentiable on an open interval containing $[95, 105]$, the centred estimate is second-order accurate: Theorem 7.31 at order two, applied to C about 100 at $100 \pm \Delta K$, bounds its error by $(\Delta K)^2/6$ times the largest value of $|\partial_{KKK} C(K)|$ on $[95, 105]$. Quote 0.50 and know the bracket.

(c) No-arbitrage and uses. The numerator $C(95) - 2\,C(100) + C(105) = 1 > 0$ is the price of the 95/100/105 *butterfly* held as one 95 call and one 105 call bought and two 100 calls sold (twice the $w = \frac{1}{2}$ butterfly of Proposition 17.24, which costs 0.50); butterflies have non-negative payoffs, so no-arbitrage forces their price ≥ 0 (Proposition 17.24) — which is precisely convexity of C in K, hence a non-negative density. (Similarly the call-spread bounds of Proposition 17.23(i), $0 \le (C(K_1) - C(K_2))/(K_2 - K_1) \le e^{-rT}$ for $K_1 < K_2$, keep both one-sided slopes, and with them the bracket of the digital, inside $[0, e^{-rT}]$, which is $[0, 1]$ at $r = 0$: here 0.60 and 0.40 both qualify.) In practice the strike curve of a liquid options market (a listed crypto options venue, for instance) is differentiated to extract the full risk-neutral density and price *any* European payoff at

any strike (digitals matching a prediction-market contract, the options-venue-to-prediction-market comparison; tail probabilities; moments), and to run no-arbitrage checks on a volatility surface.

Self-check. Two routes that avoid the centred formulas. For (b), the second difference is the change of the one-sided slopes per unit of strike and the centred slope is their average: $(0.60-0.40)/5 = 0.04$ and $(0.60 + 0.40)/2 = 0.50$; equivalently the quadratic interpolant $C^{\text{fit}}(K) = 4 - 0.5\,(K - 100) + 0.02\,(K - 100)^2$ (a local fit, not the call curve) gives $C^{\text{fit}}(95) = 4 + 2.5 + 0.5 = 7$, $C^{\text{fit}}(105) = 4 - 2.5 + 0.5 = 2$, a slope of -0.50 and a curvature of 0.04 at 100. For (a), test the identities on a law whose call curve is explicit: the exponential law with mean F, density $e^{-x/F}/F$, gives $C(K) = e^{-rT} F\, e^{-K/F}$ (substitute $x = K + y$ in (22.1)), so $-\partial_K C(K) = e^{-rT} e^{-K/F} = e^{-rT}\,\mathbb{Q}(S_T > K)$ and $\partial_{KK} C(K) = e^{-rT} e^{-K/F}/F = e^{-rT} f_{\mathbb{Q}}(K)$; with $F = K = 100$, $r = 0$ and $\Delta K = 5$ the centred differences give 0.3680 and 0.003680 against the exact 0.3679 and 0.003679. Sanity: digital $\in [0, 1]$ and C decreasing in K $(7 > 4 > 2)$.

Answer: Digital $= -\partial_K C(K) = e^{-rT}\,\mathbb{Q}(S_T > K)$; $f_{\mathbb{Q}}(K) = e^{rT}\,\partial_{KK} C(K)$. Numerically $\text{Dig}(100) \approx 0.50$ (bracketed by 0.40 and 0.60) and $f_{\mathbb{Q}}(100) \approx 0.04$; non-negativity is the butterfly no-arbitrage (convexity) condition; used to extract risk-neutral densities from strike curves and price digitals at arbitrary strikes.

P22.4 *A digital off the skew*

(a) By Theorem 20.17 at $r = 0$ the flat-volatility digital is $\Phi(d_2)$. With $\sigma\sqrt{T} = 0.18 \times 0.7071 = 0.1273$ and $\sigma^2 T/2 = 0.0081$,

$$d_2 = \frac{-0.04879 - 0.0081}{0.1273} = -0.447, \qquad \Phi(d_2) = 1 - 0.6726 = 0.3274 :$$

about 0.327.

(b) Here $d_1 = d_2 + \sigma\sqrt{T} = -0.320$, and the vega of the 210 call at its own volatility is (Proposition 20.35)

$$\mathcal{V} = e^{-rT} F\, \varphi(d_1)\sqrt{T} = 200 \times 0.3790 \times 0.7071 = 53.6$$

per unit of volatility. By Corollary 22.10,

$$\text{Dig}(210) = e^{-rT}\,\Phi(d_2) - \mathcal{V}\,\sigma'_{\text{imp}}(210) = 0.3274 + 53.6 \times 0.0015 = 0.3274 + 0.0804 = 0.408.$$

The skew adds 8 cents to a 33-cent digital, a quarter of its flat-volatility price.

(c) The digital is minus the slope of the call curve (Theorem 22.7), and the call spread over a width of 10 approximates that slope by a centred difference. With each call at its own volatility,

$$\frac{C(205) - C(215)}{10} = \frac{8.388 - 4.311}{10} = 0.408,$$

which agrees with (b). At a flat 18% the same spread is $(7.968 - 4.681)/10 = 0.329$, which agrees with (a) up to the error of a centred difference over a width of 10.

(d) By digital parity at $r = 0$ (Corollary 20.20(i)), the digital that pays at or below 210 is worth 1 minus the digital that pays above it: $1 - 0.327 = 0.673$ at a flat volatility and $1 - 0.408 = 0.592$ on the smile. The downward skew makes the upside digital dearer and the downside digital cheaper, by the same 0.080.

For the reading, write the call as the area under the tail probability to the right of its strike: $(S_T - K)^+ = \int_K^\infty \mathbf{1}\{S_T > x\}\,dx$, so at $r = 0$, exchanging the expectation and the integral of a nonnegative integrand (Fact 18.34(i)), $C(K) = \int_K^\infty \mathbb{Q}(S_T > x)\,dx$. The 210 call has the same price under the market's law and under the lognormal law at 18%, because 18% is its own implied volatility, so the two tail-probability curves enclose the same area to the right of 210. The market's curve starts higher there, 0.408 against 0.327. It must therefore fall below the lognormal curve further out, and it does: the 215 call, the area to the right of 215, costs 4.311 on the smile against 4.681. The market's law puts more mass just above 210 and less far above it: a finish above

210 is more likely, and a finish far above it less likely. This is the shape of Figure 47, where the equity-skew mixture lies above its flat law just above the forward and below it in the far right tail. That is a thin right tail, the mirror of the fat left tail that a downward skew prices.

Self-check. The one-sided call spreads must bracket the digital, by (H.4). To five decimals $d_1 = -0.31969$ and $d_2 = -0.44697$, so $\Phi(d_1) = 0.37460$, $\Phi(d_2) = 0.32745$, and the 210 call costs $200 \times 0.37460 - 210 \times 0.32745 = 6.156$. Then

$$\frac{C(210) - C(215)}{5} = \frac{6.156 - 4.311}{5} = 0.369, \qquad \frac{C(205) - C(210)}{5} = \frac{8.388 - 6.156}{5} = 0.446,$$

so that $0.369 \le \mathrm{Dig}(210) \le 0.446$. The corrected 0.408 lies inside the bracket and the flat-volatility 0.327 lies outside it: on this smile the single-volatility formula gives a price that the listed spreads exclude. For (d), price the downside digital from puts instead. At $r = 0$ parity gives, on the smile, $\mathcal{P}(205) = 8.388 + 5 = 13.388$ and $\mathcal{P}(215) = 4.311 + 15 = 19.311$. Taking expectations in $(K - S_T)^+ = \int_0^K \mathbf{1}\{S_T \le x\}\, dx$, as for the call in (d), gives $\mathcal{P}(K) = \int_0^K \mathbb{Q}(S_T \le x)\, dx$, whose slope is $\mathbb{Q}(S_T \le K)$; the put spread $(19.311 - 13.388)/10 = 0.592$ approximates it at 210, in agreement with the smile price.

Answer: (a) $d_2 = -0.447$, flat-volatility digital 0.327. (b) $\mathcal{V} = 53.6$, correction +0.080, digital 0.408. (c) The spread on the smile gives $(8.388 - 4.311)/10 = 0.408$; at a flat 18% it gives 0.329. (d) The digital that pays at or below 210 is worth 0.673 at a flat volatility and 0.592 on the smile: the downward skew cheapens it by 0.080, because the market's law puts more mass just above 210 and less far above it (a thin right tail).

P22.5 *Black-76 digital, prediction-market link*

(a) The formula. Under Black-76 with $r \approx 0$ the digital is worth $e^{-rT}\Phi(d_2) \approx \Phi(d_2)$ (Theorem 20.17), "the risk-neutral probability, discounted," with

$$d_2 = \frac{\ln(F/K) - \sigma_{\mathrm{imp}}^2 T/2}{\sigma_{\mathrm{imp}}\sqrt{T}}.$$

Pieces: $T = 30/365 = 0.0822$, $\sqrt{T} = 0.2867$, so $\sigma_{\mathrm{imp}}\sqrt{T} = 0.6 \times 0.2867 = 0.172$ (a 17.2% one-month vol); $\ln(F/K) = -\ln(7/6) \approx -0.154$; $\sigma_{\mathrm{imp}}^2 T/2 = 0.36 \times 0.0822/2 = 0.0148$. Then

$$d_2 = \frac{-0.154 - 0.015}{0.172} \approx -0.98, \qquad \Phi(d_2) = 1 - \Phi(-d_2) \approx 1 - 0.837 = 0.163.$$

The digital is worth about *16 cents*: the options market prices roughly a 16% risk-neutral probability of settling above 70k. (To four decimals $d_2 = -0.9822$ and $\Phi(0.9822) = 0.8370$, which is the value that the hint of the statement carries; at exactly 0.98 the distribution function is 0.8365.)

(b) The gap. The prediction market implies 25%; the options market implies 16%. If the options surface is trusted, the binary is rich: sell it on the prediction market and buy a tight call spread just below 70k on the options venue, long $1/\Delta K$ calls at $70{,}000 - \Delta K$ and short $1/\Delta K$ at 70,000. The spread pays 1 wherever the digital does, and by (H.4) it costs at least the digital, and little more when it is tight: at a flat 60%, 17.4 cents for $\Delta K = 1{,}000$ and 16.4 cents for $\Delta K = 100$. Breeden–Litzenberger (Theorem 22.7) makes the comparison general: differentiating the strike curve prices a digital at *any* strike, the options-venue-to-prediction-market comparison of the Trading application in §22.2. Caveats before trading:

- *Risk-neutral vs real-world:* $\Phi(d_2)$ is a $\mathbb{Q}$-probability, under which the futures price has no drift; under the real-world law of the model it drifts at the risk premium $\mu - r$, which adds $(\mu - r)T/(\sigma_{\mathrm{imp}}\sqrt{T})$ to d_2. Over one month this is small in units of total volatility: a premium of 20% a year adds 0.096 and lifts the probability from 16.3% to 18.8%, while closing the whole gap to 25% would take a premium of about 64% a year. Crypto risk premia are not automatically negligible, so some of the gap can be legitimate (the trap on "the risk-neutral probability, discounted" in §20.3).

- *Skew:* using a single implied vol ignores the slope $\sigma'_{\text{imp}}(K)$ of the smile; the exact digital is minus the slope of the *smile-consistent* call curve, which adds the vega × skew correction $-\mathcal{V}\,\sigma'_{\text{imp}}(K)$ to $\Phi(d_2)$ (Corollary 22.10). Differentiate the actual strike curve rather than plugging one vol.
- *Vol clock:* crypto volatility has strong intraday and day-of-week seasonality, so calendar T misprices short-dated contracts (Remark 22.12); over 30 days the pattern largely averages out, so here it is a second-order check, but it matters for the hours-to-days binaries that prediction markets also list.
- *Contract wording and plumbing:* whether the contract resolves on the price at the date or on any touch of 70k before it (a touch contract pays on every path on which the settlement is above 70k, and on others, so it is worth at least the digital); the settlement definitions (which index, what time); fees, capital lockup and counterparty/venue risk on both legs, which can eat a 9-cent edge.

Self-check. Price the digital as a tight call spread instead of by the closed form (Theorem 22.7: the digital is minus the slope of the call curve). At a flat 60% the Black-76 calls struck at 69,500 and 70,500 cost 1,209.07 and 1,045.97, so the spread per unit of width is $(1{,}209.07 - 1{,}045.97)/1{,}000 = 0.163$, the same 16 cents. Size check: the required log-move is 0.154 against a one-month standard deviation of 0.172, about 0.90 standard deviations; adding the $-\sigma_{\text{imp}}^2 T/2$ drift of the risk-neutral log-price pushes it to 0.98 standard deviations, and a one-standard-deviation upper tail is $\Phi(-1) = 0.159$, in the mid-teens. Bounds: the price lies in $(0, 1)$ and well below the $\Phi(-\sigma_{\text{imp}}\sqrt{T}/2) = \Phi(-0.086) = 0.466$ of a digital struck at the forward at the same flat 60%.

Answer: (a) $d_2 \approx -0.982$, digital $= \Phi(d_2) \approx 0.163$, i.e. about 16 cents. (b) The prediction market's 25 cents implies a higher probability than options' 16%; check risk-neutral-vs-real-world drift, the skew correction to $\Phi(d_2)$, crypto vol seasonality, and settlement/fee/venue frictions before trading the gap.

P22.6 *The calendar condition: prove it, then trade it*

(a) *Prices.* We argue by dominance (Lemma 16.8(i)). At time 0 buy the call expiring at T_2 and write the call expiring at T_1, both struck at K; the position costs $C(K, T_2) - C(K, T_1)$. At T_1 one of two cases holds, and which one is known then.

- If $S_{T_1} > K$, the written call is exercised and $S_{T_1} - K$ is owed. Sell one unit of the underlying short, which brings in S_{T_1}, pay the holder $S_{T_1} - K$ and keep the remaining K in cash; at the zero rate it is still K at T_2, and the borrowed unit pays no income because $q = 0$. At T_2 the bought call pays $(S_{T_2} - K)^+$ and closing the short sale costs S_{T_2}, so the position is worth

$$(S_{T_2} - K)^+ - S_{T_2} + K = (S_{T_2} - K)^+ - (S_{T_2} - K) = (K - S_{T_2})^+ \;\geq\; 0.$$

- If $S_{T_1} \leq K$, the written call expires worthless, nothing is traded, and at T_2 the position is worth $(S_{T_2} - K)^+ \geq 0$.

No cash is added or withdrawn after time 0, and the one later trade depends only on what is known at T_1, so the strategy is self-financing. Its payoff at T_2 is nonnegative in every state. The empty strategy has price 0 and payoff 0, so Lemma 16.8(i) gives $C(K, T_2) - C(K, T_1) \geq 0$. This is Theorem 17.16(iii) at $r = 0$, which Proposition 22.15(i) restates.

Total variance. At $r = q = 0$ the forward price is S at both expiries, and the Black-76 price

$$C^{\text{BSM}}(K, T; \sigma) = S\,\Phi(d_1) - K\,\Phi(d_2), \qquad d_1, d_2 = \frac{\ln(S/K) \pm (\sigma\sqrt{T})^2/2}{\sigma\sqrt{T}},$$

contains σ and T only through the total volatility $\sigma\sqrt{T}$: $C^{\text{BSM}}(K, T; \sigma) = C^{\text{BSM}}(K, 1; \sigma\sqrt{T})$. The function $\sigma \mapsto C^{\text{BSM}}(K, 1; \sigma)$ has the positive derivative $S\,\varphi(d_1)$, the vega of Proposition 20.35, so it is strictly increasing, by the mean value theorem. By the definition of the implied volatility, the

inequality of prices reads

$$C^{\mathrm{BSM}}(K,1;\sigma_{\mathrm{imp}}(K,T_1)\sqrt{T_1}) \;\le\; C^{\mathrm{BSM}}(K,1;\sigma_{\mathrm{imp}}(K,T_2)\sqrt{T_2}).$$

If the first total volatility were larger than the second, strict monotonicity would give the opposite strict inequality. Hence $\sigma_{\mathrm{imp}}(K,T_1)\sqrt{T_1} \le \sigma_{\mathrm{imp}}(K,T_2)\sqrt{T_2}$, and squaring the two nonnegative sides gives the claim, which is Proposition 22.15(ii). Dividing the difference of the two total variances by $T_2 - T_1 > 0$, the forward variance at the strike K is nonnegative, so the forward volatility is real: Proposition 22.15(iii) with zero carry.

(b) The total implied variances are $0.30^2 \times 0.25 = 0.0225$ and $0.22^2 \times 0.5 = 0.0242$. By Definition 22.13 the forward variance is $(0.0242 - 0.0225)/0.25 = 0.0068$ and the forward volatility is

$$\sigma_{\mathrm{fwd}}(0.25, 0.5) = \sqrt{0.0068} = 8.2\%.$$

The two quotes price the second quarter at 8.2 vol, against 30 for the first.

(c) At 20% the six-month total variance is $0.20^2 \times 0.5 = 0.0200 < 0.0225$, and the forward variance is $(0.0200 - 0.0225)/0.25 = -0.0100$: no forward volatility exists. The three-month call has $\sigma\sqrt{T}/2 = 0.30 \times 0.5/2 = 0.0750$ and the six-month call $0.20 \times 0.7071/2 = 0.0707$, so

$$C(100, 0.25) = 100\,(2 \times 0.5299 - 1) = 5.98, \qquad C(100, 0.5) = 100\,(2 \times 0.5282 - 1) = 5.64.$$

The nearer call costs more than the farther one, which (a) forbids. Sell the three-month call, buy the six-month call and collect $5.98 - 5.64 = 0.34$; then run the strategy of (a). At three months, if the price is above 100, sell one share short, pay the holder of the near call and keep 100 in cash; otherwise do nothing. At six months the position is worth $(100 - S_{0.5})^+$ in the first case and $(S_{0.5} - 100)^+$ in the second, never a negative amount. The 0.34 is a riskless profit. The six-month total variance may fall to 0.0225 and no further, so the lowest six-month volatility is $\sqrt{0.0225/0.5} = 30\% \times \sqrt{0.25/0.5} = 21.2\%$.

(d) The expiry $T = 0.375$ lies halfway between the two nodes. Linear in total variance: $(0.0225 + 0.0242)/2 = 0.02335$, and

$$\sigma_{\mathrm{imp}} = \sqrt{0.02335/0.375} = \sqrt{0.06227} = 24.95\%.$$

Linear in volatility: $(30\% + 22\%)/2 = 26\%$, a total variance of $0.26^2 \times 0.375 = 0.02535$. This exceeds the six-month total variance 0.0242: between 4.5 and 6 months the total variance would fall, the forward variance would be $(0.0242 - 0.02535)/0.125 = -0.0092$, and, because the price is strictly increasing in $\sigma\sqrt{T}$ (the second half of (a)), the 4.5-month call would cost more than the six-month call, which the first half of (a) forbids. With $\sigma\sqrt{T}/2 = 0.07961$ and 0.07778, and $\Phi(0.07961) = 0.531726$ and $\Phi(0.07778) = 0.530999$, the formula of (c) gives 6.345 against 6.200. Selling the 4.5-month call at 26%, buying the six-month call and running the strategy of (a) collects 0.145 with no risk. A desk interpolates the total variance: a linear interpolation between nondecreasing node values is nondecreasing, so the calendar condition holds between the nodes, whereas interpolation that is linear in volatility has just broken it between two nodes that are themselves consistent ($0.0225 \le 0.0242$).

(e) With carry the forward price moves with the expiry, $F(T) = Se^{rT}$, and the calendar condition compares strikes at the same forward moneyness $K_1/F(T_1) = K_2/F(T_2)$ (Proposition 22.15(iv)). Here

$$F(0.25) = 100\,e^{0.0125} = 101.26, \qquad F(0.5) = 100\,e^{0.025} = 102.53,$$

so the three-month call struck at $K_1 = 101.26$ is compared with the six-month call struck at $K_2 = K_1 e^{r(T_2 - T_1)} = 102.53$, each at the money forward: $C(102.53, 0.5) \ge C(101.26, 0.25)$ (one far call against one near call, because $q = 0$), and the total implied variance at 102.53 and six months is at least that at 101.26 and three months. Comparing the two expiries at one fixed strike would compare two different moneynesses.

Self-check. Variance adds across periods, so the forward volatility of (b) must rebuild the six-month quote: $0.25 \times 0.30^2 + 0.25 \times 0.0068 = 0.0225 + 0.0017 = 0.0242 = 0.5 \times 0.22^2$. The bound of (c) by a second route: the two at-the-money calls cost the same when $2\,\Phi(\sigma\sqrt{0.5}/2) - 1 = 2\,\Phi(0.075) - 1$, that is when $\sigma\sqrt{0.5} = 0.15$ and $\sigma = 21.2\%$; the quote of 22% is above this bound and the quote of 20% below it, in line with (b) and (c). The strategy of (c) on three paths: a price of 108 at three months and 97 at six leaves $0 - 97 + 100 = 3 = (100 - 97)^+$; 108 then 125 leaves $25 - 125 + 100 = 0$; 94 then 103 leaves 3. No path gives a negative amount. Interpolation in total variance spreads the forward variance of (b) evenly over the two halves: $(0.02335 - 0.0225)/0.125 = (0.0242 - 0.02335)/0.125 = 0.0068$, so the forward volatility is 8.2% on both and the total variance never falls, whereas the volatility interpolation of (d) gives -0.0092 on the second half. In (e), $101.26/F(0.25) = 102.53/F(0.5) = 1$: both strikes are at the money forward.

Answer: (a) Long the T_2 call and short the T_1 call, with one share sold short at T_1 if $S_{T_1} > K$, is worth $(K - S_{T_2})^+$ or $(S_{T_2} - K)^+$ at T_2, so $C(K, T_2) \geq C(K, T_1)$; the price is strictly increasing in $\sigma\sqrt{T}$ alone, so the total implied variance cannot fall. (b) $\sigma_{\text{fwd}} = \sqrt{0.0068} = 8.2\%$. (c) Forward variance -0.0100; the calls cost 5.98 and 5.64; sell the three-month call, buy the six-month call and collect 0.34; the lowest six-month volatility is 21.2%. (d) 24.95% in total variance; 26% linearly in volatility, whose total variance 0.02535 exceeds the six-month 0.0242, a calendar arbitrage; a desk interpolates the total variance. (e) The strikes 101.26 and 102.53, each at the money forward.

P22.7 *Sticky strike or sticky delta: hedging a skewed put*

(a) With $\sigma\sqrt{T} = 0.23 \times 0.5 = 0.115$ and $\sigma^2 T/2 = 0.0529 \times 0.25/2 = 0.00661$,

$$d_1 = \frac{\ln(S/K) + \sigma^2 T/2}{\sigma\sqrt{T}} = \frac{0.05129 + 0.00661}{0.115} = 0.5035.$$

The delta of the put is $\Phi(d_1) - 1 = 0.6927 - 1 = -0.307$ (Proposition 21.2 at $q = 0$), and its vega is

$$\mathcal{V} = S\,\varphi(d_1)\sqrt{T} = 100 \times 0.3514 \times 0.5 = 17.57$$

per unit of volatility (Proposition 21.9(i)), about 17.6.

(b) The market put is the formula's put at the implied volatility of its strike (Proposition 22.3), so by the chain rule (22.5), written for the put,

$$\frac{d\mathcal{P}}{dS} = (\Phi(d_1) - 1) + \mathcal{V}\,\frac{d\sigma_{\text{imp}}}{dS},$$

where $d\sigma_{\text{imp}}/dS$ is the derivative, at the fixed strike, of the implied volatility with respect to the spot. Under sticky strike (Definition 22.17(i)) that derivative is 0 and the smile-adjusted delta is the Black–Scholes–Merton delta, -0.307. Under sticky delta (Definition 22.17(ii)) the implied volatility of the strike K is $0.20 - 0.6\,(K/S - 1)$, so

$$\frac{d\sigma_{\text{imp}}}{dS} = \frac{0.6\,K}{S^2} = \frac{0.6 \times 95}{100^2} = 0.0057 \text{ per unit of spot},$$
$$\frac{d\mathcal{P}}{dS} = -0.307 + 17.57 \times 0.0057 = -0.307 + 0.100 = -0.207.$$

The same number comes from the general relation $d\sigma_{\text{imp}}/dS = -(K/S)\,\partial_K \sigma_{\text{imp}} = -0.95 \times (-0.006)$. Under sticky delta the downward skew raises the delta of the put by 0.100, a third of its size.

(c) Under sticky strike the 95 strike keeps 23.000%, and the put is worth 3.085. Under sticky delta the moneyness is $95/98 = 0.96939$ and the implied volatility is $0.20 - 0.6 \times (0.96939 - 1) = 0.20 + 0.01837 = 21.837\%$, so the put is worth 2.870. The value at 24.163% belongs to neither rule: it moves the volatility by the right amount, 1.163 points, in the wrong direction. The put gains $3.085 - 2.408 = 0.677$ under sticky strike and $2.870 - 2.408 = 0.462$ under sticky delta.

(d) The put has the delta -0.307, so the hedge is a holding of 0.307 shares, which loses $0.307 \times 2 = 0.614$ on the fall. The book makes

$$0.677 - 0.614 = +0.063 \quad \text{under sticky strike,} \qquad 0.462 - 0.614 = -0.152 \quad \text{under sticky delta.}$$

Under sticky strike the hedge is the right one and the book earns its gamma. Under sticky delta the implied volatility of the 95 strike falls with the spot, the put gains less than its Black–Scholes–Merton delta predicts, and the book is over-hedged: the right holding was 0.207 shares, with which it would have made $0.462 - 0.414 = +0.048$. One put, one price and one smile today give two hedge ratios, -0.307 and -0.207. A delta is a derivative of the price along an assumed motion of the smile, so it is not defined until the rule is.

Self-check. For the sticky-delta delta, use the homogeneity relation $d\mathcal{P}/dS = (\mathcal{P} - K\,\partial_K\mathcal{P})/S$, which needs no derivative in the spot. With $d_2 = d_1 - 0.115 = 0.3885$ and $\Phi(-0.3885) = 0.3488$, the slope of the put curve on the smile is $\partial_K\mathcal{P} = \Phi(-d_2) + \mathcal{V}\,\partial_K\sigma_{\text{imp}} = 0.3488 - 17.57 \times 0.006 = 0.2434$, and $(2.408 - 95 \times 0.2434)/100 = -0.207$. For the repricing, expand to second order. The gamma is $\varphi(d_1)/(S\sigma\sqrt{T}) = 0.3514/11.5 = 0.0306$ (Proposition 21.6(i)), so under sticky strike the put should gain about $0.307 \times 2 + \frac{1}{2} \times 0.0306 \times 2^2 = 0.614 + 0.061 = 0.675$, against 0.677 from the quoted prices, and the hedged book should make the gamma term, 0.061, against 0.063. Under sticky delta the volatility falls by 0.0116, which costs $17.57 \times 0.0116 = 0.204$: a gain of about $0.675 - 0.204 = 0.471$, against 0.462 from the quoted prices. Most of the difference is the cross effect of the two moves: at $S = 98$ the vega has risen to about 18.5, so the same fall in volatility costs $3.085 - 2.870 = 0.215$ there, not 0.204; with the 0.002 that the second-order estimate misses under sticky strike, this accounts for the gap of 0.009.

Answer: (a) Delta -0.307, vega 17.6. (b) -0.307 under sticky strike; -0.207 under sticky delta, where $d\sigma_{\text{imp}}/dS = +0.0057$. (c) 23.000% and 3.085 under sticky strike; 21.837% and 2.870 under sticky delta. (d) $+0.063$ under sticky strike and -0.152 under sticky delta, per put; the hedge ratio depends on the assumed motion of the smile, so the rule comes first.

P22.8 *Volatility from ten closes*

(a) The squares of the ten returns, in percent squared, are

$$2.519, \quad 3.197, \quad 0.994, \quad 2.481, \quad 1.390, \quad 0.349, \quad 3.073, \quad 1.852, \quad 2.490, \quad 1.910,$$

and they sum to 20.255. The zero-mean estimate of Definition 22.19(i) is

$$\hat{\sigma}_{\text{per}} = \sqrt{\frac{20.255}{10}} = \sqrt{2.0255} = 1.423\% \text{ a day,} \qquad \hat{\sigma} = 1.423\% \times 15.87 = 22.6\% \text{ a year.}$$

For the demeaned estimate, the returns sum to 1.979%, so $\bar{R} = 0.198\%$, and $\sum_i (R_i - \bar{R})^2 = \sum_i R_i^2 - n\bar{R}^2 = 20.255 - 10 \times 0.198^2 = 20.255 - 0.392 = 19.863$. Hence

$$s = \sqrt{\frac{19.863}{9}} = \sqrt{2.207} = 1.486\% \text{ a day,} \qquad 1.486\% \times 15.87 = 23.6\% \text{ a year.}$$

The two differ by one volatility point: the divisor 9 in place of 10 adds 1.2 points, and removing the squared mean, 0.392 out of 20.255, takes 0.2 back.

(b) By Proposition 22.20(ii) and (iv) the standard error of the volatility estimate is, to first order, the volatility divided by $\sqrt{2n}$, and annualisation does not change the ratio; with the estimate in place of the unknown volatility,

$$\frac{\hat{\sigma}}{\sqrt{2n}} = \frac{22.6}{\sqrt{20}} = 22.6 \times 0.2236 = 5.1 \text{ volatility points.}$$

Ten returns give 22.6 ± 5.1. The large-n interval $22.6\,(1 \pm 1.96 \times 0.2236)$, from 12.7% to 32.5%, is only indicative at $n = 10$, since the chapter derives it for large n. At $n = 10$ the first-order value

is itself rough (the bound of part (iv) on its error is of the same size); the exact standard error under the normal model is about 5.0 points.

(c) The requirement $25/\sqrt{2n} \le 1$ gives $n \ge 25^2/2 = 312.5$, so 313 daily returns, about a year and a quarter of trading days, as in Example 22.21.

(d) As one period each. With no holiday, ten consecutive daily returns are two full weeks of five returns, and each week contains one return from a Friday close to a Monday close. Return variance accrues while the market trades: measured from close to close, a span that contains a weekend carries little more variance than an ordinary trading day (Fact 22.22), so time is counted in trading days and the annualisation uses 252 of them (Corollary 19.2). Counting each weekend return as three periods would put 14 periods in the sample and give $\sqrt{20.255/14} = 1.20\%$ per calendar day. That spreads each weekend's one day of variance over three days: it understates the volatility of a trading day by the factor $\sqrt{10/14} = 0.85$ and overstates that of a weekend, $1.20\% \times \sqrt{3} = 2.08\%$ against 1.42%. Annualised consistently, with $\sqrt{365} = 19.10$, the same figure gives 23.0%, close to (a): the damage done by the calendar clock is to the variance of individual days, such as that of an option held over a weekend, not to a long-run annual average.

Self-check. The log-returns telescope, so their sum must be $\ln(51.0/50.0) = 1.980\%$: the listed returns sum to 1.979%, the difference being rounding. Recompute the estimate from simple returns instead of log-returns: the daily changes $0.8/50.0$, $-0.9/50.8$, ... are, in percent, 1.600, −1.772, 1.002, 1.587, −1.172, 0.593, 1.768, −1.351, −1.566, 1.392, their squares sum to 20.289, and $\sqrt{2.0289} = 1.424\%$ a day, 22.6% a year: the same answer, as it should be for returns of this size. Computed directly, the squared deviations $(R_i - 0.198)^2$ sum to 19.863, as the shortcut of (a) gave. For (c), $25/\sqrt{2 \times 313} = 0.9992 \le 1$ while $25/\sqrt{2 \times 312} = 1.0008 > 1$.

Answer: (a) Zero-mean: 1.42% a day, 22.6% a year; demeaned: 1.49% a day, 23.6% a year. (b) $22.6/\sqrt{20} = 5.1$ volatility points. (c) 313 daily returns. (d) One period each: variance accrues in trading time, and the annualisation counts 252 trading days.

P22.9 *EWMA and GARCH: update, prove the forecast, use it*

(a) Yesterday's variance was $0.014^2 = 0.000196$. The return is $R = \ln(78.40/80.00) = \ln 0.98 = -0.02020$, with $R^2 = 0.000408$: a fall of $2.02/1.4 = 1.4$ standard deviations. By (22.7),

$$\sigma_i^2 = 0.94 \times 0.000196 + 0.06 \times 0.000408 = 0.0001842 + 0.0000245 = 0.0002087, \qquad \sigma_i = 1.445\%,$$

and by (22.9),

$$\begin{aligned} \sigma_i^2 &= 0.0000025 + 0.06 \times 0.000408 + 0.92 \times 0.000196 \\ &= 0.0000025 + 0.0000245 + 0.0001803 = 0.0002073, \qquad \sigma_i = 1.440\%. \end{aligned}$$

The persistence is $\alpha + \beta = 0.98$; the long-run variance of Definition 22.24 is $\bar{\sigma}^2 = 0.0000025/0.02 = 0.000125$: a long-run volatility of 1.118% a day, or $1.118\% \times 15.87 = 17.7\%$ a year.

(b) We proceed by induction on k, computing the one new expectation of each step with the product rule for independent variables.

Step 1: the variances are functions of the earlier shocks. Substituting $R_{i+j} = \sigma_{i+j} Z_{i+j}$ into the rule gives, for every $j \ge 0$,

$$\sigma_{i+j+1}^2 = \omega + (\alpha\, Z_{i+j}^2 + \beta)\, \sigma_{i+j}^2.$$

Starting from the known number σ_i^2, induction on j shows that σ_{i+j}^2 is a function of the shocks $Z_i, \ldots, Z_{i+j-1}$ alone, and that it is nonnegative, because ω, α and β are.

Step 2: one step of the rule in expectation. We claim that for every $k \ge 1$ the mean $\mathbb{E}[\sigma_{i+k}^2]$ is finite and

$$\mathbb{E}[\sigma_{i+k}^2] = \omega + (\alpha + \beta)\, \mathbb{E}[\sigma_{i+k-1}^2].$$

Suppose that $\mathbb{E}[\sigma_{i+k-1}^2]$ is finite, which holds at $k = 1$, where it is the number σ_i^2. The only new expectation is that of $Z_{i+k-1}^2\, \sigma_{i+k-1}^2$. A shock with mean 0 and variance 1 has $\mathbb{E}[Z^2] = 1$, by (1.1).

For $k = 1$ the variance σ_i^2 is a constant and $\mathbb{E}[Z_i^2\,\sigma_i^2] = \sigma_i^2$. For $k \geq 2$ put $X = (Z_i, \ldots, Z_{i+k-2})$ and $Y = Z_{i+k-1}$. The shocks are mutually independent, so X and Y are independent (the joint law of $(Z_i, \ldots, Z_{i+k-1})$ is the product of the k marginal laws, and grouping the first $k-1$ factors writes it as the law of X times the law of Y), and by Step 1 $\sigma_{i+k-1}^2 = \sigma_{i+k-1}^2(X)$. Apply Lemma 19.17 to the nonnegative function $\psi(x, y) = y^2\,\sigma_{i+k-1}^2(x)$. With x frozen its mean over Y is $\bar{\psi}(x) = \sigma_{i+k-1}^2(x)\,\mathbb{E}[Y^2] = \sigma_{i+k-1}^2(x)$, so

$$\mathbb{E}\big[Z_{i+k-1}^2\,\sigma_{i+k-1}^2\big] = \mathbb{E}\,\psi(X, Y) = \mathbb{E}\,\bar{\psi}(X) = \mathbb{E}\big[\sigma_{i+k-1}^2\big].$$

The constant ω and the terms $\alpha\,Z_{i+k-1}^2\sigma_{i+k-1}^2$ and $\beta\,\sigma_{i+k-1}^2$ have finite means, by the display above and the induction hypothesis, so taking expectations in the display of Step 1 at $j = k-1$ and using linearity (Theorem 1.2),

$$\mathbb{E}\big[\sigma_{i+k}^2\big] = \omega + \alpha\,\mathbb{E}\big[\sigma_{i+k-1}^2\big] + \beta\,\mathbb{E}\big[\sigma_{i+k-1}^2\big] = \omega + (\alpha + \beta)\,\mathbb{E}\big[\sigma_{i+k-1}^2\big],$$

which is finite. By induction the claim holds for every $k \geq 1$. Only the signs $\omega \geq 0$, $\alpha \geq 0$ and $\beta \geq 0$ were used, to make the variances and ψ nonnegative; neither $\omega > 0$ nor $\alpha + \beta < 1$ entered.

Step 3: the deviation shrinks geometrically. The definition of $\bar{\sigma}^2$, rearranged, reads $\bar{\sigma}^2 = \omega + (\alpha + \beta)\,\bar{\sigma}^2$. Subtracting it from the identity of Step 2,

$$\mathbb{E}\big[\sigma_{i+k}^2\big] - \bar{\sigma}^2 = (\alpha + \beta)\,\Big(\mathbb{E}\big[\sigma_{i+k-1}^2\big] - \bar{\sigma}^2\Big).$$

At $k = 0$ the deviation is $\sigma_i^2 - \bar{\sigma}^2 = (\alpha + \beta)^0(\sigma_i^2 - \bar{\sigma}^2)$, by the convention, and each further day multiplies it by $\alpha + \beta$; by induction on k it equals $(\alpha + \beta)^k(\sigma_i^2 - \bar{\sigma}^2)$, which is the formula. Since $0 \leq \alpha + \beta < 1$, the forecast tends to $\bar{\sigma}^2$ as k grows. A number x satisfies $x = \omega + (\alpha + \beta)\,x$ exactly when $(1 - \alpha - \beta)\,x = \omega$, that is, when $x = \bar{\sigma}^2$, because $1 - \alpha - \beta \neq 0$: the long-run variance is the unique fixed point of this map, and by the display each application multiplies the distance to it by $\alpha + \beta < 1$, so it is attracting. Together these are (22.11) and part (ii) of Proposition 22.27.

(c) Today's deviation from the long-run variance is $0.0002073 - 0.000125 = 0.0000823$. By (b),

$$\mathbb{E}\big[\sigma_{i+10}^2\big] = 0.000125 + 0.8171 \times 0.0000823 = 0.00019225, \qquad \sqrt{0.00019225} = 1.387\%,$$
$$\mathbb{E}\big[\sigma_{i+50}^2\big] = 0.000125 + 0.3642 \times 0.0000823 = 0.00015497, \qquad \sqrt{0.00015497} = 1.245\%.$$

The deviation is halved when $0.98^k = 1/2$, that is after $\ln 2/(-\ln 0.98) = 0.6931/0.02020 = 34.3$ trading days (Corollary 22.28(iii)). For the options, average the formula of (b) over $j = 0, \ldots, k-1$ with the geometric series, as in (22.12):

$$\frac{1}{k}\sum_{j=0}^{k-1}\mathbb{E}\big[\sigma_{i+j}^2\big] = \bar{\sigma}^2 + (\sigma_i^2 - \bar{\sigma}^2)\,\frac{1 - (\alpha + \beta)^k}{k\,(1 - \alpha - \beta)}.$$

For $k = 20$ the averaging factor is $(1 - 0.6676)/(20 \times 0.02) = 0.831$, the average variance is $0.000125 + 0.831 \times 0.0000823 = 0.0001934$, and the volatility is $\sqrt{252 \times 0.0001934} = 22.1\%$. For $k = 100$ the factor is $(1 - 0.1326)/2 = 0.4337$, the average variance is $0.000125 + 0.4337 \times 0.0000823 = 0.0001607$, and the volatility is $\sqrt{252 \times 0.0001607} = 20.1\%$. The moving average is the same rule with $\omega = 0$, $\alpha = 0.06$ and $\beta = 0.94$, so that $\alpha + \beta = 1$: Step 2 of (b) used only the signs of ω, α and β, so it applies and gives $\mathbb{E}[\sigma_{i+k}^2] = \mathbb{E}[\sigma_{i+k-1}^2]$; by induction on k, $\mathbb{E}[\sigma_{i+k}^2] = \sigma_i^2$ for every k. Its forecast is flat, and both options get $\sqrt{252 \times 0.0002087} = 22.9\%$.

(d) With $R = 0$,

$$\text{moving average:} \quad 0.94 \times 0.0002087 = 0.0001962, \qquad \sqrt{0.0001962} = 1.401\%;$$
$$\text{GARCH:} \quad 0.0000025 + 0.92 \times 0.0002073 = 0.0001932, \qquad \sqrt{0.0001932} = 1.390\%.$$

After a day without a move the moving average keeps the fraction $\lambda = 0.94$ of its variance and nothing else. GARCH keeps only $\beta = 0.92$ of its variance and adds ω, the weight $1-\alpha-\beta = 0.02$ on the long-run variance 0.000125; the weight $\alpha = 0.06$ falls on a squared return of zero. The smaller weight on yesterday's variance makes GARCH fall faster here, but it falls towards a floor: after many quiet days it would settle at $\omega/(1-\beta) = 0.00003125$, a volatility of 0.56% a day, whereas the variance of the moving average would halve every $\ln 2/(-\ln 0.94) = 11.2$ days and decay to zero. On an average day, with a squared return equal to its variance, the moving average does not move at all, and GARCH closes 2% of its distance to the long-run variance.

Self-check. Iterate the rule of Step 2 instead of using the closed form: $\mathbb{E}[\sigma_{i+1}^2] = 0.0000025 + 0.98 \times 0.0002073 = 0.00020565$ and $\mathbb{E}[\sigma_{i+2}^2] = 0.0000025 + 0.98 \times 0.00020565 = 0.00020404$, while the formula of (b) gives $0.000125 + 0.9604 \times 0.0000823 = 0.00020404$. The average over two days is $(0.0002073 + 0.00020565)/2 = 0.00020648$, and the averaging factor $(1 - 0.9604)/(2 \times 0.02) = 0.99$ gives $0.000125 + 0.99 \times 0.0000823 = 0.00020648$. Order check: today's variance is above the long-run one, so every forecast must lie between them and the term structure must fall with the horizon, and indeed $1.440\% > 1.387\% > 1.245\% > 1.118\%$ a day and $22.9\% > 22.1\% > 20.1\% > 17.7\%$ a year (the one-day GARCH volatility is $1.440\% \times 15.87 = 22.9\%$).

Answer: (a) Moving average 1.445% a day; GARCH 1.440% a day, with a long-run volatility of 1.118% a day, 17.7% a year. (b) Each day the forecast keeps the fraction $\alpha + \beta$ of its deviation from $\bar{\sigma}^2$, because $\mathbb{E}[Z^2\sigma^2] = \mathbb{E}[\sigma^2]$ for a shock independent of the past; $\bar{\sigma}^2$ is the unique fixed point of $x \mapsto \omega + (\alpha+\beta)\,x$, and it is attracting. (c) 1.387% and 1.245% a day; half-life 34.3 days; 22.1% for 20 days and 20.1% for 100 days, against a flat 22.9% for the moving average. (d) 1.401% (moving average) and 1.390% (GARCH): GARCH falls further, because the moving average only decays by λ, while GARCH decays by β and is pulled towards its long-run variance.

P22.10 *The log-variance strike from five quotes*

(a) We split the floating leg, by an exact identity, into a part that a trading strategy produces at no cost and a European claim on S_T, and we price the claim with the expansion.

Step 1: the identity. Since $e^{R_i} = S_{t_i}/S_{t_{i-1}}$, each term satisfies $e^{R_i} - 1 = (S_{t_i} - S_{t_{i-1}})/S_{t_{i-1}}$, and the log-returns telescope, $\sum_{i=1}^n R_i = \ln S_T - \ln S_0$. Multiplying both by 2 and subtracting gives the identity. Only the positivity of the prices was used.

Step 2: the dynamic leg costs nothing. At each date t_{i-1}, $i = 1, \dots, n$, borrow $2/T$ and buy $2/(T\,S_{t_{i-1}})$ units of the underlier, which costs exactly $2/T$. At t_i sell them for $(2/T)\,S_{t_i}/S_{t_{i-1}}$ and repay the loan, at the zero rate; this leaves $(2/T)(S_{t_i} - S_{t_{i-1}})/S_{t_{i-1}}$, kept in cash until T (a negative amount is a loan). The number of units bought at t_{i-1} is known then, and no cash is added or withdrawn: the strategy is self-financing, it costs nothing, and at T it is worth

$$\frac{2}{T}\sum_{i=1}^{n}\frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}}.$$

Step 3: the log claim is a strip. Apply the expansion to $f(x) = -\ln(x/F)$, for which $f(F) = 0$, $f'(F) = -1/F$ and $f''(K) = 1/K^2$:

$$-\ln\frac{x}{F} = -\frac{x-F}{F} + \int_0^F \frac{(K-x)^+}{K^2}\,dK + \int_F^\infty \frac{(x-K)^+}{K^2}\,dK \qquad (x > 0).$$

The two integrals are nonnegative, and their sum $-\ln(x/F) + (x-F)/F$ is at most $|\ln x| + |\ln F| + x/F + 1$, which has a finite $\mathbb{Q}$-mean at $x = S_T$ by the hypotheses. Put $x = S_T$ and take $\mathbb{Q}$-expectations (Theorem 1.2). The first term has mean zero, because $\mathbb{E}^{\mathbb{Q}}[S_T] = F$. Each expectation is an integral against the density (Fact 7.22) and the integrands are nonnegative, so the expectation and the integration over K may be exchanged (Fact 18.34(i)); the inner expectations are $\mathbb{E}^{\mathbb{Q}}[(K - S_T)^+] = \mathcal{P}(K)$ and $\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] = C(K)$. Hence

$$\mathbb{E}^{\mathbb{Q}}\Big[-\ln\frac{S_T}{F}\Big] = \int_0^F \frac{\mathcal{P}(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK,$$

and this is the price today of the *log claim* that pays $-\ln(S_T/F)$ at T.

Step 4: the strike. By Step 1 and $F = S_0$, the floating leg of the swap is the payoff of the strategy of Step 2 plus the payoff of $2/T$ log claims. Running that strategy and buying $2/T$ log claims at time 0 is self-financing, pays the floating leg on every path, and costs $0 + (2/T)\,\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F)]$. Adding a loan of the strike, repaid at T at the zero rate, gives a strategy that pays the floating leg minus the strike, the payoff of the swap, at the price $(2/T)\,\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F)]$ minus the strike. By the law of one price (Lemma 16.8(iii)) a swap with any given strike is worth that amount, and it is zero exactly when the strike is $K_{\log} = (2/T)\,\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F)]$; Step 3 turns $K_{\log}$ into the strip. This is Theorem 22.36(i) and (ii), with (22.14) and (22.15); the expansion is Lemma 22.34. No model of the path was used.

(b) The five values of price/K^2 are

$$\frac{0.843}{4{,}900} = 0.00017204, \qquad \frac{2.985}{7{,}225} = 0.00041315, \qquad \frac{7.966}{10{,}000} = 0.00079660,$$
$$\frac{2.256}{13{,}225} = 0.00017059, \qquad \frac{0.385}{16{,}900} = 0.00002278,$$

which sum to 0.0015752. Multiplying by $\Delta K = 15$ and by $2/T = 2$,

$$K_{\log} \approx 2 \times 15 \times 0.0015752 = 0.0473, \qquad \sqrt{0.0473} = 21.7 \text{ volatility points.}$$

(c) With a flat smile the exact strip is $\sigma^2 = 0.0400$ (Corollary 22.38), so the five-strike grid overstates it by $0.0431 - 0.0400 = 0.0031$, the net of two errors of opposite sign: the five buckets cover the strikes from 62.5 to 137.5; inside that range the coarse sum overstates the exact integral by 0.0038, because the bucket of width 15 at the money covers more than the area under the peak of the out-of-the-money prices, and the wings beyond it, worth 0.0007, are missing. The fine integration isolates the skew: $0.0443 - 0.0400 = 0.0043$. Therefore

$$0.0473 \;=\; \underbrace{0.0400}_{\text{at-the-money variance}} + \underbrace{0.0043}_{\text{skew}} + \underbrace{0.0030}_{\text{discretisation}},$$

and the two estimates of each effect agree to within the interaction of the two effects, about 0.0002: the discretisation error is $0.0473 - 0.0443 = 0.0030$ under the skew and 0.0031 under the flat smile, and on the grid the skew adds $0.04726 - 0.04313 = 0.0041$. The best estimate of the strike is 0.0443, which is 21.0 volatility points, one point above the at-the-money 20. It depends on the wings: the flat extension is a convention for the quadrature (it leaves a kink in the smile at 70, where the implied call curve is not convex), and continuing the lower line below 70 instead gives 0.0455.

(d) *The weights.* By the expansion, a payoff f is replicated with $f''(K)\,dK$ options at each strike K, and for the log claim $-(2/T)\ln(x/F)$ this is $2/(T\,K^2)$. The reason the log claim is the right one is its gamma. In the Black–Scholes–Merton model at a volatility σ the claim is worth $V(S,t) = -(2/T)\,[\ln(S/F) - \sigma^2(T-t)/2]$ (Theorem 20.7: the mean of $\ln S_T$ given the price S at t is $\ln S - \sigma^2(T-t)/2$), so $\partial_S V = -2/(T\,S)$ and $\frac{1}{2}S^2\,\partial_{SS}V = 1/T$: its dollar gamma is the same at every spot and every date, so it earns realised variance at the same rate wherever the path goes, which is what a variance swap pays. A single option has its dollar gamma concentrated near its strike, and among strips with power weights K^{-c} only $c = 2$ makes the total independent of the spot.

The skew. Compare the quotes with the flat-20% prices of the same options, 0.248, 2.161, 3.062 and 1.009 at the strikes 70, 85, 115 and 130. The puts add $(0.843 - 0.248)/4{,}900 = 0.0001214$ and $(2.985 - 2.161)/7{,}225 = 0.0001140$ to the sum of (b), and the calls take away $(3.062 - 2.256)/13{,}225 = 0.0000609$ and $(1.009 - 0.385)/16{,}900 = 0.0000369$: a net 0.0001376, which is 0.0041 after the factor 30. The dear puts outweigh the cheap calls. On the grid the raw price changes nearly balance, +1.419 for the puts against −1.430 for the calls, and the weights

$1/K^2$ tip the sum towards the puts. In the continuum the first-order effect of a tilt linear in the strike cancels (the second smile in the paragraph after Corollary 22.38), but the second-order effect is not small for a tilt of six points over thirty: a smile linear in the strike at the put-side slope, 26, 23, 20, 17 and 14% at the five strikes and flat beyond, has a fine strip of 0.0434. The shallower call side of this smile (-2.5 and -4 points at 115 and 130, against $+3$ and $+6$ at 85 and 70) adds the remaining 0.0009. This is a property of the shape, not of every downward slope: the strike $K_{\log}$ is an average of the whole smile, and a smile that is almost flat below the forward and falls steeply above it has a strip below the at-the-money variance (the third smile in the paragraph after Corollary 22.38, 19.4 against 20 volatility points).

The squared-return strike. In the model the claim that pays $\sigma^2_{\text{real}} = \frac{1}{T}\sum_i R_i^2$ at T is worth its $\mathbb{Q}$-expectation (Corollary 20.11, as in the proof of Theorem 22.36(iii)), so $K_{\text{var}} = \mathbb{E}^{\mathbb{Q}}[\sigma^2_{\text{real}}]$. Under the law of Definition 20.6 with $r = q = 0$ each log-return is $R_i = -\frac{1}{2}\sigma^2\Delta t + \sigma\sqrt{\Delta t}\,Z_i$ with $Z_i \sim \mathcal{N}(0,1)$, so $\mathbb{E}^{\mathbb{Q}}[R_i^2] = \sigma^2\Delta t + \frac{1}{4}\sigma^4(\Delta t)^2$ and, with $n\,\Delta t = T$,

$$K_{\text{var}} = \frac{n}{T}\Big(\sigma^2\Delta t + \frac{\sigma^4(\Delta t)^2}{4}\Big) = \sigma^2 + \frac{\sigma^4\,\Delta t}{4}, \qquad K_{\log} = \frac{2}{T}\,\mathbb{E}\Big[\frac{\sigma^2 T}{2} - \sigma\sqrt{T}\,Z\Big] = \sigma^2.$$

The gap is $\sigma^4\Delta t/4 = 0.21^4/(4\times 252) = 0.00194/1{,}008 = 1.9\times 10^{-6}$, about 0.000002 (Theorem 22.36(iii)): more than a thousand times smaller than the discretisation error of (c). It is with jumps that the gap becomes material (the reading after Theorem 22.36).

Self-check. The identity on a path with two dates, $100 \to 104 \to 99$: $R_1 = 0.0392207$ and $R_2 = -0.0492710$, so the left side is $2\,(0.04 - 0.0392207) + 2\,(-0.0480769 + 0.0492710) = 0.001559 + 0.002388 = 0.003947$; on the right, $2\,(0.04 - 0.0480769) = -0.016154$ and $-2\ln 0.99 = 0.020101$, which sum to 0.003947. The expansion of Step 3 at $x = 110$, $F = 100$: the put integral vanishes and $\int_{100}^{110}(110 - K)/K^2\,dK = \big[-110/K - \ln K\big]_{100}^{110} = 0.1 - \ln 1.1$, so the right side is $-0.1 + 0.1 - \ln 1.1 = -\ln 1.1$, as it must be. The flat-smile figure of (c) from the flat prices: $0.248/4{,}900 + 2.161/7{,}225 + 7.966/10{,}000 + 3.062/13{,}225 + 1.009/16{,}900 = 0.0014375$, and $30\times 0.0014375 = 0.0431$. The sum of (b) is also the flat sum plus the four skew differences of (d): $0.0014375 + 0.0001376 = 0.0015751$, which is 0.0015752 up to the rounding of the quotes. The gap of (d) period by period: $\mathbb{E}^{\mathbb{Q}}[e^{R_i}] = 1$ gives $\mathbb{E}^{\mathbb{Q}}[2\,(e^{R_i} - 1 - R_i)] = -2\,\mathbb{E}^{\mathbb{Q}}[R_i] = \sigma^2\Delta t$, against $\mathbb{E}^{\mathbb{Q}}[R_i^2] = \sigma^2\Delta t + \sigma^4(\Delta t)^2/4$; summed over the 252 periods and divided by $T = 1$, the excess is $0.21^4/(4\times 252) = 1.9\times 10^{-6}$.

Answer: (a) The floating leg is a zero-cost dynamic position plus $2/T$ log claims, and the log claim is the strip with $1/K^2$ options at each strike, so $K_{\log} = (2/T)\,\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F)]$ is the strip. (b) $K_{\log} \approx 30\times 0.0015752 = 0.0473$, which is 21.7 volatility points. (c) $0.0473 = 0.0400$ (at the money) $+ 0.0043$ (skew) $+ 0.0030$ (discretisation), the skew measured by the fine integration (measured on the grid instead, 0.0041 and 0.0031); the best estimate is 0.0443, 21.0 volatility points. (d) The weights $1/K^2$ give the strip a dollar gamma that does not depend on the spot; this steep skew lifts the strip, because weighted by $1/K^2$ the dear puts outweigh the cheap calls (a smile linear in the strike at the put-side slope already gives 0.0434, and the shallower call side adds 0.0009), which is not true of every downward-sloping smile; in the model $K_{\text{var}} - K_{\log} = \sigma^4\Delta t/4 = 0.000002$.

P22.11 *Selling variance: notionals and P and L*

(a) The vega notional is the sensitivity of the payoff to the realised volatility at the strike, so $N_{\text{vega}} = 2\sqrt{K_{\text{var}}}\,N_{\text{var}}$ (Definition 22.32(e)). In points,

$$N_{\text{var}} = \frac{N_{\text{vega}}}{2\sqrt{K_{\text{var}}}} = \frac{50{,}000}{2\times 25} = 1{,}000 \text{ per variance point.}$$

(b) The dealer is short and receives $N_{\text{var}} = 1{,}000$ times the strike minus the realised variance, in variance points. The linear estimate is 50,000 times the strike minus the realised volatility, in volatility points: $50{,}000\times 5$, 0 and $50{,}000\times(-15)$.

realised volatility (points)	$625 - 10^4\,\sigma_{\text{real}}^2$ (points)	variance swap	linear estimate	difference
20	$625 - 400 = +225$	+225,000	+250,000	−25,000
25	$625 - 625 = 0$	0	0	0
40	$625 - 1{,}600 = -975$	−975,000	−750,000	−225,000

The seller gains less than the linear estimate when the market is calm and loses more when it is turbulent. The shortfall is N_{var} times the squared distance of the realised volatility from the strike, $1{,}000 \times 5^2 = 25{,}000$ and $1{,}000 \times 15^2 = 225{,}000$, the payoff identity of Proposition 22.40.

(c) Realised variance adds in time: over the whole period it is the average, weighted by time, of the variance realised in each half. With 31 of the 62 daily returns in each half, the two weights are exactly $\frac{1}{2}$. The first half is known, $30^2 = 900$ points. The second half can be locked at the new strike, $28^2 = 784$ points, by buying a new swap on half the variance notional, which costs nothing; the floating legs of the second half then cancel, and the dealer's payoff at expiry is the known amount

$$1{,}000 \times \Big(625 - \big(\tfrac{1}{2} \times 900 + \tfrac{1}{2} \times 784\big)\Big) = 1{,}000 \times (625 - 842) = -217{,}000.$$

A known amount at expiry is worth itself at a zero rate (Lemma 16.8(v)), so the position is marked at −217,000; the floating leg is locked at 842 variance points, which is 29.0 vol.

(d) The short volatility swap would lose $50{,}000 \times (40 - 25) = 750{,}000$, against 975,000 for the variance swap. The variance swap loses 225,000 more, because its payoff is quadratic in the realised volatility while that of the volatility swap is linear: a short variance swap is a short volatility swap with the same vega notional plus a short position in $N_{\text{var}}\,(100\,\sigma_{\text{real}} - 25)^2$, with $N_{\text{var}} = 1{,}000$ per variance point, which loses whenever the realised volatility ends away from the strike, on either side. At 20% the same term makes the variance seller's gain 25,000 smaller than the volatility seller's 250,000. The comparison is of payoffs at a common strike: a fair volatility swap would be struck below 25 (Proposition 22.40), and the gap is the price of the convexity term.

Self-check. Redo the conversion in decimal units. The vega notional is 5,000,000 per unit of volatility, so $N_{\text{var}} = 5{,}000{,}000/(2 \times 0.25) = 10{,}000{,}000$ per unit of variance; at 20% the dealer receives $10{,}000{,}000 \times (0.0625 - 0.04) = 225{,}000$ and at 40% pays $10{,}000{,}000 \times (0.16 - 0.0625) = 975{,}000$, as in (b). For (c), read the mark through the completed square behind Proposition 22.40, the linear estimate plus the convexity term at the locked level. The floating leg is locked at $\sqrt{842} = 29.01724$ vol, and in points $625 - 842 = 2 \times 25 \times (25 - 29.01724) - (29.01724 - 25)^2$, so the vega estimate is $50{,}000 \times (25 - 29.01724) = -200{,}862$ and the convexity term is $-1{,}000 \times 4.01724^2 = -16{,}138$; together −217,000, as in (c).

Answer: (a) 1,000 per variance point. (b) +225,000, 0 and −975,000, against the linear estimates +250,000, 0 and −750,000. (c) The floating leg is locked at 842 variance points and the position is marked at −217,000. (d) The volatility swap would lose 750,000; the variance swap loses 225,000 more, because its payoff is convex in the realised volatility.

P22.12 *The volatility swap is worth less than the square root*

(a) *First proof: the variance shortcut.* By (1.1) applied to σ_{real} under $\mathbb{Q}$,

$$0 \;\le\; \text{Var}^{\mathbb{Q}}(\sigma_{\text{real}}) = \mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}^2] - \big(\mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}]\big)^2 = K_{\text{var}} - K_{\text{vol}}^2.$$

The realised volatility is nonnegative, so its mean K_{vol} is nonnegative, and taking square roots,

$$K_{\text{vol}} = \sqrt{K_{\text{var}} - \text{Var}^{\mathbb{Q}}(\sigma_{\text{real}})} \;\le\; \sqrt{K_{\text{var}}},$$

with equality exactly when $\text{Var}^{\mathbb{Q}}(\sigma_{\text{real}}) = 0$.

Second proof: a tangent line. For every value of σ_{real}, completing the square,

$$\sigma_{\text{real}}^2 - K_{\text{var}} - 2\sqrt{K_{\text{var}}}\,(\sigma_{\text{real}} - \sqrt{K_{\text{var}}}) = \sigma_{\text{real}}^2 - 2\,\sigma_{\text{real}}\sqrt{K_{\text{var}}} + K_{\text{var}} = (\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2 \;\geq\; 0,$$

which is the stated inequality: the parabola lies above its tangent at the strike. Take $\mathbb{Q}$-expectations of this identity; they are linear, and σ_{real} and σ_{real}^2 are integrable because the second moment is finite and $\sigma_{\text{real}} \leq (1 + \sigma_{\text{real}}^2)/2$. The first two terms have mean $K_{\text{var}} - K_{\text{var}} = 0$, so

$$2\sqrt{K_{\text{var}}}\,(\sqrt{K_{\text{var}}} - K_{\text{vol}}) = \mathbb{E}^{\mathbb{Q}}\Big[(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2\Big] \;\geq\; 0.$$

If $K_{\text{var}} > 0$, dividing by $2\sqrt{K_{\text{var}}}$ gives $K_{\text{vol}} \leq \sqrt{K_{\text{var}}}$, with equality exactly when $\mathbb{E}^{\mathbb{Q}}[(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2] = 0$, that is, when the realised volatility equals $\sqrt{K_{\text{var}}}$ with probability one. If $K_{\text{var}} = 0$, complete the same square at any point $c > 0$: $(\sigma_{\text{real}} - c)^2 \geq 0$ gives $\sigma_{\text{real}} \leq (\sigma_{\text{real}}^2 + c^2)/(2c)$, hence $K_{\text{vol}} \leq c/2$ for every $c > 0$. Since $K_{\text{vol}} \geq 0$, both strikes are zero and equality holds. This is Proposition 22.40.

(b) The square root has the derivatives $1/(2\sqrt{x})$ and $-1/(4x^{3/2})$, so to second order about K_{var} (Theorem 7.31; Proposition 7.34 for the power $\frac{1}{2}$)

$$\sigma_{\text{real}} \approx \sqrt{K_{\text{var}}} + \frac{\sigma_{\text{real}}^2 - K_{\text{var}}}{2\sqrt{K_{\text{var}}}} - \frac{(\sigma_{\text{real}}^2 - K_{\text{var}})^2}{8\,K_{\text{var}}^{3/2}}.$$

Under $\mathbb{Q}$ the linear term has mean zero and the square has mean $\text{Var}^{\mathbb{Q}}(\sigma_{\text{real}}^2) = 0.03^2 = 0.0009$, so

$$K_{\text{vol}} \approx \sqrt{K_{\text{var}}} - \frac{\text{Var}^{\mathbb{Q}}\,(\sigma_{\text{real}}^2)}{8\,K_{\text{var}}^{3/2}} = 0.30 - \frac{0.0009}{8 \times 0.027} = 0.30 - 0.0042 = 29.58\%.$$

The fair volatility strike is 0.42 volatility points below the quoted 30.

(c) With $N_{\text{vega}} = 2\sqrt{K_{\text{var}}}\,N_{\text{var}}$, the long variance swap pays $N_{\text{var}}(\sigma_{\text{real}}^2 - K_{\text{var}})$ and the short volatility swap pays $-2\sqrt{K_{\text{var}}}\,N_{\text{var}}(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})$. By the identity of (a) their sum is

$$N_{\text{var}}\,(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2 \;\geq\; 0$$

on every path. The position pays whenever the realised volatility ends away from the strike, in either direction, and nothing when it ends at the strike: the trader is long the volatility of volatility. It is not an arbitrage, because the volatility swap is not struck at its fair strike. Its strike $\sqrt{K_{\text{var}}}$ is at least K_{vol}, by (a), so the short side of that swap is worth

$$N_{\text{vega}}\,(\sqrt{K_{\text{var}}} - K_{\text{vol}}) = N_{\text{var}}\,\mathbb{E}^{\mathbb{Q}}\Big[(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2\Big] \;\geq\; 0,$$

with equality only when $\sigma_{\text{real}} = \sqrt{K_{\text{var}}}$ with probability one, and then the combined payoff is zero with probability one. When it is positive, no counterparty takes the long side without being paid that amount at inception. In every case the position costs at inception exactly the $\mathbb{Q}$-expectation of what it pays, so it is not an arbitrage. (The equality is the expectation of the payoff identity, the variance swap being fair.) The trader pays for the convexity exactly what it is worth. With the numbers of (b) the price is about $N_{\text{vega}} \times 0.0042$, that is, about $0.0025\,N_{\text{var}}$.

(d) The adjustment of (b) is $\text{Var}^{\mathbb{Q}}(\sigma_{\text{real}}^2)/(8\,K_{\text{var}}^{3/2})$, and the exact gap $\sqrt{K_{\text{var}}} - K_{\text{vol}}$ of (a) vanishes exactly when $\text{Var}^{\mathbb{Q}}(\sigma_{\text{real}}) = 0$. The two conditions coincide: because $\sigma_{\text{real}} \geq 0$, the variable σ_{real}^2 is constant with probability one exactly when σ_{real} is. The realised volatility is then known in advance under the pricing law, and the two swaps are the same bet. The Black–Scholes–Merton model approaches this case as the monitoring becomes continuous, since its realised variance converges to σ^2 in mean square (Theorem 22.36(iii)). Indeed $|\sigma_{\text{real}} - \sigma| = |\sigma_{\text{real}}^2 - \sigma^2|/(\sigma_{\text{real}} + \sigma) \leq |\sigma_{\text{real}}^2 - \sigma^2|/\sigma$, and a variance is at most the mean squared distance to any constant (by (1.1) applied to $\sigma_{\text{real}} - \sigma$),

so $\mathrm{Var}^{\mathbb{Q}}(\sigma_{\text{real}}) \le \mathbb{E}^{\mathbb{Q}}[(\sigma^2_{\text{real}} - \sigma^2)^2]/\sigma^2 \to 0$. As soon as the realised volatility is uncertain, the volatility swap is struck below $\sqrt{K_{\text{var}}}$, by the price of the volatility of volatility.

Self-check. Use the exact formula of (a) in place of the expansion of (b). To first order a standard deviation of 0.03 for σ^2_{real} is a standard deviation of $0.03/(2 \times 0.30) = 0.05$ for σ_{real}, a variance of 0.0025, and $\sqrt{0.09 - 0.0025} = \sqrt{0.0875} = 0.2958$, the same 29.58%. A concrete law gives the same size: if σ^2_{real} is 0.06 or 0.12 with probability one half each, its mean is 0.09 and its standard deviation 0.03, and $K_{\text{vol}} = (\sqrt{0.06} + \sqrt{0.12})/2 = (0.2449 + 0.3464)/2 = 0.2957$. For (c), a path with $\sigma_{\text{real}} = 0.45$ pays $N_{\text{var}}(0.2025 - 0.09) - 0.6\, N_{\text{var}}(0.45 - 0.30) = (0.1125 - 0.0900)\, N_{\text{var}} = 0.0225\, N_{\text{var}} = 0.15^2\, N_{\text{var}}$.

Answer: (a) $K_{\text{vol}} = \sqrt{K_{\text{var}} - \mathrm{Var}^{\mathbb{Q}}(\sigma_{\text{real}})} \le \sqrt{K_{\text{var}}}$, and also by taking expectations in $\sigma^2_{\text{real}} - K_{\text{var}} - 2\sqrt{K_{\text{var}}}(\sigma_{\text{real}} - \sqrt{K_{\text{var}}}) = (\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2 \ge 0$; equality exactly when σ_{real} is constant under $\mathbb{Q}$. (b) $K_{\text{vol}} \approx 0.30 - 0.0009/(8 \times 0.027) = 29.58\%$. (c) The payoff is $N_{\text{var}}(\sigma_{\text{real}} - \sqrt{K_{\text{var}}})^2 \ge 0$: the trader is long the volatility of volatility, and pays $N_{\text{vega}}(\sqrt{K_{\text{var}}} - K_{\text{vol}})$ at inception for it, so there is no arbitrage. (d) When $\mathrm{Var}^{\mathbb{Q}}(\sigma^2_{\text{real}}) = 0$, which for $\sigma_{\text{real}} \ge 0$ is the same as $\mathrm{Var}^{\mathbb{Q}}(\sigma_{\text{real}}) = 0$.

28.23 Solutions — §23 Exotic Options and Numerical Methods

P23.1 *Hedging a short digital with call spreads*

(a) Write $x = S_T$. Half the call spread from 98 to 100 (the *upper spread*) and half the call spread from 100 to 102 (the *lower spread*) pay

$$\text{upper spread:} \quad \frac{(x-98)^+ - (x-100)^+}{2} = \begin{cases} 0, & x \le 98, \\ (x-98)/2, & 98 < x < 100, \\ 1, & x \ge 100, \end{cases}$$

$$\text{lower spread:} \quad \frac{(x-100)^+ - (x-102)^+}{2} = \begin{cases} 0, & x \le 100, \\ (x-100)/2, & 100 < x < 102, \\ 1, & x \ge 102, \end{cases}$$

and the digital pays $\mathbf{1}\{x > 100\}$. For $x \le 100$ the digital and the lower spread pay 0 and the upper spread pays a nonnegative amount. For $100 < x < 102$ the digital and the upper spread pay 1 and the lower spread pays less than 1. For $x \ge 102$ all three pay 1. So, in every state,

$$\text{upper spread} \;\ge\; \mathbf{1}\{S_T > 100\} \;\ge\; \text{lower spread},$$

and the dominance lemma (Lemma 16.8(i)) orders the three prices in the same way. Prices add, so the upper spread costs $(4.10 - 3.00)/2 = 0.55$ and the lower spread $(3.00 - 2.10)/2 = 0.45$:

$$0.55 \;\ge\; \mathrm{Dig}(100) \;\ge\; 0.45.$$

This is Proposition 23.2 with $\varepsilon = 2$.

(b) The seller needs a position that pays at least the digital in every state. Of the two spreads of (a) only the upper spread does: the lower spread pays less than 1 when $100 < S_T < 102$, where the digital pays 1. For each unit of payout it buys half a call struck at 98 and sells half a call struck at 100; for a payout of 1,000,000 it buys 500,000 calls struck at 98 and sells 500,000 calls struck at 100, at a cost of $500{,}000 \times (4.10 - 3.00) = 550{,}000$. The midpoint of the bracket is 0.50, a value of 500,000 for the digitals, so the hedge costs 50,000 more: 5 cents per unit of payout, half the width of the bracket. In exchange the desk keeps the residual payoff $500{,}000\,(S_T - 98)$ when $98 < S_T \le 100$ and zero otherwise, which is never negative, and it has nothing to trade before expiry.

(c) The bid 0.60 lies above the upper bound 0.55. Sell one digital at 0.60 and buy the upper spread (long half a call struck at 98, short half a call struck at 100) for 0.55.

- Today: $+0.60 - 0.55 = +0.05$.
- At T, if $S_T \le 98$: the spread pays 0 and the digital costs 0; net 0.
- At T, if $98 < S_T \le 100$: the spread pays $(S_T - 98)/2 > 0$ and the digital pays nothing (at $S_T = 100$ exactly it is not in the money); net $(S_T - 98)/2 > 0$.
- At T, if $S_T > 100$: the spread pays 1 and the digital costs 1; net 0.

The position brings in 0.05 today and is never negative at T: an arbitrage, of 0.05 per digital (and 50,000 if the bid were good for 1,000,000 of payout). (Against an offer below 0.45 the mirror trade buys the digital and sells the lower spread.)

(d) With the spacing ε the bounds are

$$\frac{C(100-\varepsilon) - C(100)}{\varepsilon} = -\,\frac{C(100-\varepsilon) - C(100)}{-\varepsilon},$$
$$\frac{C(100) - C(100+\varepsilon)}{\varepsilon} = -\,\frac{C(100+\varepsilon) - C(100)}{\varepsilon},$$

two difference quotients of the call curve at 100 with the sign changed. Both tend to $-\partial_K C(100)$. The number Dig(100) does not depend on ε and lies between them for every ε with $0 < \varepsilon < 100$ (Proposition 23.2), so $\mathrm{Dig}(100) = -\partial_K C(100)$: the digital is minus the slope of the call curve, whatever the model. With the three quotes, the central difference estimates the slope at $(2.10 - 4.10)/4 = -0.50$, so Dig(100) is estimated at 0.50, the midpoint of the bracket.

(e) Differentiating $\Phi(d_2)$ in S, with $\partial_S d_2 = 1/(S\sigma\sqrt{\tau})$, the delta of one digital is $\varphi(d_2)/(S\sigma\sqrt{\tau})$. Here $\sigma\sqrt{\tau} = 0.3/15.8745 = 0.018898$, so $S\sigma\sqrt{\tau} = 0.94491$ and, at $S = K$, $d_2 = -\sigma\sqrt{\tau}/2 = -0.0094$, where $\varphi(d_2) = 0.39894\, e^{-0.0094^2/2} = 0.39892$. The delta is

$$\frac{0.39892}{0.94491} = 0.4222 \text{ per unit of payout},$$

so the desk, short the digitals and holding no spread, would buy about 844,400 shares, worth 42.2 million: more than twenty times the largest amount the digitals can ever pay. This position must also be rebalanced with every move of the stock (the delta is the height of a narrow bell centred at the strike), whereas a static hedge such as the upper spread of (b) is bought once and needs no trading at all.

Self-check. The quotes are consistent: the butterfly $C(98) - 2\,C(100) + C(102) = 0.20$ has a nonnegative price, as Proposition 17.24 requires, and half of it, 0.10, is the width $0.55 - 0.45$ of the bracket, so the bracket is not empty. The arbitrage of (c) can be checked at sample prices: the net payoff is 0 at $S_T = 96$, 0.5 at 99, 1 at 100, 0 at 101 and 0 at 104. For (e), a finite difference replaces the derivative: the digital is worth $\Phi(d_2) = 0.51732$ at $S = 50.05$ and 0.47513 at $S = 49.95$, and $(0.51732 - 0.47513)/0.1 = 0.4219$, against 0.4222.

Answer: (a) $0.45 \le \mathrm{Dig}(100) \le 0.55$. (b) Hold the upper spread: buy 500,000 calls struck at 98 and sell 500,000 struck at 100 for 550,000, which is 50,000 (5 cents per unit of payout) above the midpoint value 500,000. (c) Sell the digital at 0.60 and buy half a 98/100 call spread at 0.55: $+0.05$ per digital today, $(S_T - 98)/2 > 0$ at T when $98 < S_T \le 100$ and 0 otherwise. (d) Both bounds tend to $-\partial_K C(100)$, so $\mathrm{Dig}(100) = -\partial_K C(100)$ (estimated at 0.50 here). (e) Delta 0.4222 per unit of payout: about 844,400 shares, worth 42.2 million against a payout of 2 million.

P23.2 *Gap and chooser options by parity*

(a) For every $x \ge 0$ and strikes K_1 (trigger) and K_2 (payment),

$$(x - K_2)\,\mathbf{1}\{x > K_1\} = (x - K_1)^+ - (K_2 - K_1)\,\mathbf{1}\{x > K_1\} = x\,\mathbf{1}\{x > K_1\} - K_2\,\mathbf{1}\{x > K_1\}:$$

all three vanish for $x \le K_1$ and equal $x - K_2$ for $x > K_1$. Equal payoffs have equal prices (Lemma 16.8(iii)), so, writing $\mathrm{Gap}(K_1, K_2)$ for the price of the gap call,

$$\mathrm{Gap}(K_1, K_2) = C(K_1) - (K_2 - K_1)\,\mathrm{Dig}(K_1) = \mathrm{AoN}(K_1) - K_2\,\mathrm{Dig}(K_1),$$

which is Proposition 23.5. Under BSM, at the trigger 40, $\mathrm{Dig}(40) = e^{-rT}\Phi(d_2) = 0.96319 \times 0.50576 = 0.487143$ and $\mathrm{AoN}(40) = S\,\Phi(d_1) = 40 \times 0.60805 = 24.3220$. Hence

$$\mathrm{Gap}(40, 45) = 24.3220 - 45 \times 0.487143 = 24.3220 - 21.9214 = 2.4006,$$

and the first form gives the same, $4.8363 - 5 \times 0.487143 = 4.8363 - 2.4357 = 2.4006$. The price $\mathrm{AoN}(40) - K_2\,\mathrm{Dig}(40)$ is negative exactly when

$$K_2 > \frac{\mathrm{AoN}(40)}{\mathrm{Dig}(40)} = \frac{24.3220}{0.487143} = 49.93 :$$

beyond that payment strike the obligation to pay $K_2 - S_T$ when $40 < S_T < K_2$ outweighs the right.

(b) Let $\tau = T - t_1 = 0.5$ and let C_{t_1}, $\mathcal{P}_{t_1}$ be the values at t_1 of the call and the put with strike $K = 40$ and expiry T. Put–call parity at t_1, for a stock without dividends, reads $\mathcal{P}_{t_1} - C_{t_1} = Ke^{-r\tau} - S_{t_1} = \tilde{K} - S_{t_1}$. The holder takes the better of the two options, so at t_1 the chooser is worth

$$\max\left(C_{t_1}, \mathcal{P}_{t_1}\right) = C_{t_1} + \left(\mathcal{P}_{t_1} - C_{t_1}\right)^+ = C_{t_1} + \left(\tilde{K} - S_{t_1}\right)^+.$$

The first term is the value at t_1 of the call with strike K and expiry T, and the second is the payoff at t_1 of a put with strike $\tilde{K}$ that expires at t_1. A portfolio of that call and that put, bought today, is therefore worth exactly as much as the chooser at t_1, in every state, and by the law of one price (Lemma 16.8(iii), both positions being sold at t_1) it costs the same today:

$$\text{chooser} = C(40, 0.75) + \mathcal{P}(39.0124, 0.25) = 4.8363 + 1.6841 = 6.5204.$$

This is Proposition 23.10 with $q = 0$. The straddle costs $4.8363 + 3.3641 = 8.200$: the chooser gives the better of the two options, not both, and is 1.68 cheaper.

(c) By (b) the holder declares the call exactly when $S_{t_1} > \tilde{K}$ (at $S_{t_1} = \tilde{K}$ the two options are worth the same). Since $(S_T - K)^+ - (K - S_T)^+ = S_T - K$, the chooser pays at T, in every state,

$$(K - S_T)^+ + \mathbf{1}\{S_{t_1} > \tilde{K}\}\,(S_T - K).$$

Buy today one put with strike K and expiry T and one call with strike $\tilde{K}$ and expiry t_1. At t_1 the call pays $(S_{t_1} - \tilde{K})^+$. On the event $\{S_{t_1} > \tilde{K}\}$, buy one share and borrow $Ke^{-r\tau} = \tilde{K}$ until T: this costs $S_{t_1} - \tilde{K}$, exactly what the call pays, so no cash is added or withdrawn. At T the share is worth S_T and the loan is repaid with K. The strategy therefore pays $(K - S_T)^+$ from the put, plus $S_T - K$ on the event $\{S_{t_1} > \tilde{K}\}$: the payoff of the chooser. By Lemma 16.8(iii),

$$\text{chooser} = \mathcal{P}(40, 0.75) + C(39.0124, 0.25) = 3.3641 + 3.1563 = 6.5204,$$

the value found in (b). The two decompositions are tied by parity today: $C(\tilde{K}, t_1) - \mathcal{P}(\tilde{K}, t_1) = S - \tilde{K}e^{-rt_1} = S - Ke^{-rT} = C(K, T) - \mathcal{P}(K, T)$, and indeed $3.1563 - 1.6841 = 1.4722 = 4.8363 - 3.3641$. With a continuous yield q the same construction uses $e^{-q\tau}$ calls struck at $\tilde{K} = Ke^{-(r-q)\tau}$: on the event $\{S_{t_1} > \tilde{K}\}$ buy $e^{-q\tau}$ units of the asset with the dividends reinvested (one unit at T, Proposition 16.15(i)) and borrow $Ke^{-r\tau} = e^{-q\tau}\tilde{K}$. This costs $e^{-q\tau}(S_{t_1} - \tilde{K})$, exactly what the calls pay, and gives $\mathcal{P}(K, T) + e^{-q\tau}C(\tilde{K}, t_1)$, the last display of the proof of Proposition 23.10.

(d) No. A holder who declares at a date before t_1 owns, at t_1, either the call or the put, worth C_{t_1} or $\mathcal{P}_{t_1}$; a holder who waits owns $\max(C_{t_1}, \mathcal{P}_{t_1})$, which is at least as much in every state. A dealer who has sold the variant and holds the simple chooser declares at t_1 the better of the two options, sells it for $\max(C_{t_1}, \mathcal{P}_{t_1})$ and buys the option the buyer declared, for C_{t_1} or $\mathcal{P}_{t_1}$. The position is worth at least 0 at t_1 in every state, so by Lemma 16.8(i) (both positions sold at t_1) the variant is worth at most the simple chooser; and it is worth at least as much, because its holder may wait until t_1. The two are worth the same, 6.5204: waiting only adds information.

Self-check. A different route for the chooser: its value is $e^{-rt_1}\,\mathbb{E}^{\mathbb{Q}}\max(C_{t_1},\mathcal{P}_{t_1})$, and a numerical integration of the two BSM values at t_1 over the lognormal law of S_{t_1} returns 6.5204. The price also sits between the bounds of Proposition 23.10: $\max(C,\mathcal{P}) = 4.836 \le 6.520 \le 8.200 = C+\mathcal{P}$. For the gap call, setting the payment strike equal to the trigger must return the vanilla call, and $\text{AoN}(40) - 40\,\text{Dig}(40) = 24.3220 - 40\times 0.487143 = 24.3220 - 19.4857 = 4.8363 = C(40,0.75)$.

Answer: (a) $\text{Gap}(40,45) = C(40) - 5\,\text{Dig}(40) = \text{AoN}(40) - 45\,\text{Dig}(40) = 2.4006$; the price is negative for payment strikes above $\text{AoN}(40)/\text{Dig}(40) = 49.93$. (b) Chooser $= C(40,0.75) + \mathcal{P}(39.0124,0.25) = 4.8363 + 1.6841 = 6.5204$, against 8.200 for the straddle. (c) Chooser $= \mathcal{P}(40,0.75) + C(39.0124,0.25) = 3.3641 + 3.1563 = 6.5204$. (d) No: waiting until t_1 is at least as good as any earlier declaration, so the variant is also worth 6.5204.

P23.3 *Forward-start options with a percentage strike*

(a) In the formula $C(S,K;\tau) = Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)$, the numbers $d_{1,2} = [\ln(S/K) + (r - q \pm \sigma^2/2)\tau]/(\sigma\sqrt{\tau})$ depend on S and K only through the ratio S/K, which does not change when both are multiplied by c. So $\Phi(d_1)$ and $\Phi(d_2)$ are unchanged and each of the two terms is multiplied by c: this is Lemma 23.7, and the put is treated in the same way. At t_1 the holder owns a call with spot S_{t_1}, strike S_{t_1} and life 1, worth

$$C(S_{t_1},S_{t_1};1) = S_{t_1}\,C(1,1;1) = c_\eta\,S_{t_1}, \qquad c_\eta = \frac{C(40,40;1)}{40} = \frac{5.7316}{40} = 0.14329,$$

where homogeneity was used twice, with $c = S_{t_1}$ and with $c = 40$, and where $c_\eta = C(1,\eta;T-t_1)$ is the call per unit of spot whose strike is the fraction η of the spot, here with $\eta = 1$.

(b) A claim worth $c_\eta S_{t_1}$ at t_1 is c_η shares delivered at t_1. Holding shares and reinvesting every dividend in the stock makes the number of shares grow at the rate q, so $c_\eta e^{-qt_1}$ shares bought today become c_η shares at t_1. The forward-start call therefore costs

$$e^{-qt_1}\,c_\eta\,S = e^{-qt_1}\,C(40,40;1) = 0.98511\times 5.7316 = 5.646$$

today (Proposition 23.8). The replicating position is $0.14329\times 0.98511 = 0.1412$ shares bought today, growing with the reinvested dividends to 0.1433 shares at t_1; at t_1 the shares are worth exactly the at-the-money call, which is then hedged by its own delta strategy until T.

(c) At a time $t < t_1$ the replicating position is worth $c_\eta e^{-q(t_1-t)}S_t$, which is linear in the stock price. The delta today is $c_\eta e^{-qt_1} = 0.1412$ and the gamma is zero. The risk before t_1 is therefore not to the stock price, which the linear holding hedges exactly, but to c_η: to the volatility that will prevail between t_1 and T (the forward volatility and, when there is a smile, the forward smile).

(d) The same argument with the put and the ratio $\eta = 0.9$: at t_1 the put is worth $\mathcal{P}(S_{t_1},0.9\,S_{t_1};1) = S_{t_1}\,\mathcal{P}(40,36;1)/40 = 0.08071\,S_{t_1}$, so today

$$e^{-qt_1}\,\mathcal{P}(40,36;1) = 0.98511\times 3.2283 = 3.180.$$

(e) Only the factor e^{-qt_1} depends on the start date. With $t_1 = 2$ the call is worth $e^{-0.03}\times 5.7316 = 0.98511^2\times 5.7316 = 5.562$: a later start is cheaper only because the dividends paid before t_1 go to a holder of the stock and not to the holder of the option. With $q = 0$ the factor is 1 for every t_1, and the forward-start call is worth today's one-year at-the-money call whatever its start date (with $q = 0$ and the other inputs unchanged that call is worth 6.086).

Self-check. By the pricing rule for payoffs fixed on finitely many dates (Corollary 20.11, with the two dates t_1 and T, applied to the reinvested holding), the price is also a risk-neutral expectation, which gives a route without replication. Under $\mathbb{Q}$, $S_T = S_{t_1}R$ with $R = \exp\big((r-q-\tfrac12\sigma^2)(T-t_1) + \sigma\sqrt{T-t_1}\,Z\big)$, where $Z\sim\mathcal{N}(0,1)$ is built from the increment of $W^{\mathbb{Q}}$ over $[t_1,T]$ and is therefore independent of S_{t_1}. Since $S_{t_1} > 0$, $(S_T - S_{t_1})^+ = S_{t_1}(R-1)^+$, so

$$e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_T - S_{t_1})^+\big] = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[S_{t_1}]\,\mathbb{E}\big[(R-1)^+\big] = e^{-rT}\cdot Se^{(r-q)t_1}\cdot e^{r(T-t_1)}c_\eta = e^{-qt_1}c_\eta S,$$

because $e^{-r(T-t_1)}\,\mathbb{E}(R-1)^+$ is the call with spot 1, strike 1 and life $T-t_1$. A simulation of 4,000,000 paths gives 5.644 for the call, with standard error 0.005, and 3.182 for the put, with standard error 0.003.

Answer: (a) $c_\eta = C(40,40;1)/40 = 0.14329$ per unit of S_{t_1}. (b) $e^{-0.015}\times 5.7316 = 5.646$, replicated until t_1 by 0.1412 shares with the dividends reinvested (0.1433 shares at t_1). (c) Delta 0.1412, gamma 0; the exposure is to the forward volatility between t_1 and T. (d) $e^{-0.015}\times 3.2283 = 3.180$. (e) $e^{-0.03}\times 5.7316 = 5.562$ for a start in two years; with $q=0$ the price would not depend on the start date.

P23.4 *Barrier rapid-fire: parity, degenerate barriers and monitoring*

(a) Write V^{in} and V^{out} for the prices of the down-and-in and the down-and-out call, and $C(S,K;\tau)$ for the vanilla call with spot S, strike K and life τ. The payoff $(x-K)^+$ vanishes for $x \le H$ because $H \le K$, so Theorem 23.16(iii) applies: the down-and-in call is the image term. The exponent is $1-2(r-q)/\sigma^2 = 1-0.10/0.0625 = -0.6$ and the reflected spot is $H^2/S = 8100/100 = 81$, so

$$V^{\text{in}} = \Big(\frac{100}{90}\Big)^{-0.6} C(81,100;1) = 0.93874\times 3.4352 = 3.225,$$

and by in–out parity (Proposition 23.15(i)) $V^{\text{out}} = 12.336 - 3.225 = 9.111$. With daily observation, $\Delta t = 1/252$ and $\sigma\sqrt{\Delta t} = 0.25/15.8745 = 0.015749$, so Fact 23.21 moves the barrier away from the spot, to

$$90\,e^{-0.5826\times 0.015749} = 90\,e^{-0.009175} = 90\times 0.99087 = 89.18.$$

A lower barrier is reached on fewer paths: the down-and-out call rises from 9.111 to 9.527, and by parity the down-and-in call falls from 3.225 to $12.336 - 9.527 = 2.809$.

(b) (i) True. On a path that survives, $S_T \ge S_T^{\min} > H \ge K$, so the put pays nothing; the claim pays zero in every state (Proposition 23.15(ii)), and the down-and-in put is the vanilla put.

(ii) False. A path that rises from S without ever coming down to H pays the vanilla call and not the knock-in. With $K=80$, $H=90$ and the other inputs of (a), the vanilla call is worth 25.41 and the down-and-in call 9.77. The degenerate pairs are the down puts with $K \le H$ and the up calls with $H \le K$.

(iii) True. On a path that survives, $S_T \le S_T^{\max} < H \le K$, so the call pays nothing (Proposition 23.15(iii)).

(iv) False. The pair carries two exercise rights: the knock-out can be exercised only before the touch and the knock-in only after it, and exercising the knock-out early leaves the knock-in alive. So the pair can follow any exercise policy of the American vanilla (exercising the knock-out when the vanilla would be exercised before the touch, the knock-in otherwise) and still hold a claim worth at least zero: it is worth at least the American vanilla, and in general more. The identity is one between European payoffs.

(v) False. The monthly dates are among the daily ones, so the daily minimum is at most the monthly minimum on every path: more observation can only knock out more paths, and the down-and-out call is *cheaper* (Remark 23.26). In (a) it is worth 9.111 with continuous monitoring, 9.527 with daily monitoring and about 10.68 with monthly monitoring by the corrected formula (barrier moved to 86.29); a direct numerical integration over the twelve monthly dates gives 10.665.

(vi) False. A lower observed minimum raises $S_T - S_T^{\min}$ on every path: the floating lookback call is *dearer*. Knock-outs and lookbacks move in opposite directions.

(vii) False. Neither grid contains the other: day 7 is observed only by the coarser grid and day 3 only by the finer one (day 252, the expiry, is on both). A path that closes below H on day 7 only and ends above K is knocked out by the seven-day grid and survives on the three-day grid, where it pays $(S_T-K)^+ > 0$. No comparison holds path by path.

(viii) False. Before T the futures price is $S_t e^{(r-q)(T-t)} > S_t$, so its path lies above the spot path and reaches H on fewer paths: the option with the barrier on the futures price is dearer. With

the inputs of (a), Theorem 23.16 with zero carry, at the futures price $100\,e^{0.05} = 105.13$, gives 10.47 against 9.111.

(ix) False. Near the barrier more volatility means more knock-outs. With $S = K = 100$, $H = 120$, $r = 5\%$, $q = 0$ and $T = 1$, the up-and-out call is worth 2.12 at $\sigma = 15\%$, 1.18 at 20% and 0.69 at 25% (Theorem 23.16(iv)).

Self-check. The Φ form of Corollary 23.17 gives the down-and-in call of (a) by a second computation: $y = [\ln(0.81) + 0.08125]/0.25 = -0.5179$ and $y - \sigma\sqrt{T} = -0.7679$, so

$$V^{\text{in}} = 100 \times 0.9^{2.6}\,\Phi(-0.5179) - 100\,e^{-0.05} \times 0.9^{0.6}\,\Phi(-0.7679) = 22.984 - 19.759 = 3.225.$$

Simulation agrees with both monitoring rules: 2,000,000 paths with the Brownian-bridge correction give 9.097 for the continuously monitored down-and-out call, with standard error 0.012, and 6,000,000 paths observed on the 252 daily dates give 9.528, with standard error 0.007. The prices of (ix) were confirmed in the same way (2.120 and 0.689 at 15% and 25%, with standard errors 0.003 and 0.002).

Answer: (a) Down-and-in 3.225, down-and-out 9.111; with daily monitoring the barrier moves to 89.18, the down-and-out call rises to 9.527 and the down-and-in call falls to 2.809. (b) (i) true; (ii) false; (iii) true; (iv) false; (v) false, it is cheaper; (vi) false, it is dearer; (vii) false, the grids are not nested; (viii) false, the barrier on the futures price is hit on fewer paths; (ix) false, the vega is negative near the barrier.

P23.5 *A one-touch is worth about two digitals*

(a) On the risk-neutral path $(S_t - S_0)/\sigma_{\text{N}} = W_t$, so the price reaches H by T exactly when W reaches the level $\ell = (H - S_0)/\sigma_{\text{N}} > 0$. By (R), which is Theorem 19.63(i),

$$\mathbb{Q}(S_T^{\max} \ge H) = 2\,P(W_T \ge \ell) = 2\,\Phi\Big(-\frac{\ell}{\sqrt{T}}\Big) = 2\,\Phi\Big(\frac{S_0 - H}{\sigma_{\text{N}}\sqrt{T}}\Big),$$

and $P(W_T \ge \ell) = \mathbb{Q}(S_T \ge H)$.

(b) *The function.* Put $y = (S - H)/(\sigma_{\text{N}}\sqrt{T-t})$, so that $V^{\text{out}} = 1 - 2\,\Phi(y)$, with $\partial_S y = 1/(\sigma_{\text{N}}\sqrt{T-t})$ and $\partial_t y = y/(2(T-t))$. Using $\varphi'(y) = -y\,\varphi(y)$,

$$\partial_t V^{\text{out}} = -\frac{y\,\varphi(y)}{T-t}, \qquad \partial_S V^{\text{out}} = -\frac{2\,\varphi(y)}{\sigma_{\text{N}}\sqrt{T-t}}, \qquad \partial_{SS} V^{\text{out}} = \frac{2\,y\,\varphi(y)}{\sigma_{\text{N}}^2\,(T-t)},$$

so $\partial_t V^{\text{out}} + \frac{1}{2}\sigma_{\text{N}}^2\,\partial_{SS} V^{\text{out}} = 0$. On the barrier $y = 0$ and $V^{\text{out}}(H,t) = 1 - 2\,\Phi(0) = 0$. For $S < H$, $y \to -\infty$ as $t \to T$, so $V^{\text{out}}(S,t) \to 1$. (The function is the digital put $\Phi\big((H-S)/(\sigma_{\text{N}}\sqrt{T-t})\big)$ minus its mirror image about H.)

The strategy. Now let $S_t = S_0 + \mu_{\text{N}} t + \sigma_{\text{N}} W_t$ be the real-world path, and let t_H be the first time at which it reaches H: the first time at which W reaches the straight line $\ell - (\mu_{\text{N}}/\sigma_{\text{N}})t$, with $\ell = (H - S_0)/\sigma_{\text{N}}$, which starts at $\ell > 0 = W_0$, so t_H is a hitting time of Definition 19.22(c). Fix $T' < T$. The function $(t,x) \mapsto V^{\text{out}}(S_0 + \mu_{\text{N}} t + \sigma_{\text{N}} x,\, t)$ is in $C_E^{1,2}([0,T'] \times \mathbb{R})$ (Definition 19.19(iii)): by the formulas above, V^{out} and its partial derivatives are continuous and bounded on $\mathbb{R} \times [0,T']$, because $T - t \ge T - T'$ and $\varphi(y)$, $y\,\varphi(y)$ are bounded. Its derivative in t is $\partial_t V^{\text{out}} + \mu_{\text{N}}\,\partial_S V^{\text{out}}$, and its first and second derivatives in x are $\sigma_{\text{N}}\,\partial_S V^{\text{out}}$ and $\sigma_{\text{N}}^2\,\partial_{SS} V^{\text{out}}$. The stopped clause of Itô's lemma (Theorem 19.30(iii)) therefore gives, for every $t \le T'$ and with probability one,

$$\begin{aligned} V^{\text{out}}(S_{t\wedge t_H}, t \wedge t_H) = V^{\text{out}}(S_0,0) &+ \int_0^t \mathbf{1}\{s < t_H\}\big(\partial_t V^{\text{out}} + \tfrac{1}{2}\sigma_{\text{N}}^2\,\partial_{SS} V^{\text{out}}\big)\,ds \\ &+ \int_0^t \mathbf{1}\{s < t_H\}\,\partial_S V^{\text{out}}\,(\mu_{\text{N}}\,ds + \sigma_{\text{N}}\,dW_s), \end{aligned}$$

every derivative being evaluated at (S_s, s), and the first integral vanishes by the equation just verified. With $r = 0$ the last integral is the gain of holding $\partial_S V^{\text{out}}(S_s, s)$ units of the underlying

up to t_H, the bank paying no interest. So the strategy is worth $V^{\mathrm{out}}(S_{t\wedge t_H}, t\wedge t_H)$ at every $t<T$, and the drift μ_{N} has dropped out. If the barrier is reached before T, the value at that moment is $V^{\mathrm{out}}(H,t_H)=0$, nothing is held afterwards, and the strategy ends with 0. If it is not reached, then $S_T<H$ (the event $S_T=H$ has probability zero), and the value $V^{\mathrm{out}}(S_t,t)$ tends to 1 as $t\to T$, because $(S_t-H)/(\sigma_{\mathrm{N}}\sqrt{T-t})\to-\infty$ when $S_t\to S_T<H$. The values are bounded by 1, so the limit also holds in mean square (Lemma 19.12(iv)). The terminal value is $\mathbf{1}\{S_T^{\max}<H\}$: the strategy replicates the no-touch, and absence of arbitrage makes its cost the price,

$$V^{\mathrm{out}}(S_0,0)=1-2\,\Phi\Big(\frac{S_0-H}{\sigma_{\mathrm{N}}\sqrt{T}}\Big).$$

The one-touch pays one unit of cash at T, worth 1 today at a zero rate, minus the no-touch, so $\mathrm{OT}(H)=2\,\Phi((S_0-H)/(\sigma_{\mathrm{N}}\sqrt{T}))$. The same argument without the image and without stopping, applied to $\Phi((S-H)/(\sigma_{\mathrm{N}}\sqrt{T-t}))$, which solves the same equation and tends to $\mathbf{1}\{S>H\}$, gives $\mathrm{Dig}(H)=\Phi((S_0-H)/(\sigma_{\mathrm{N}}\sqrt{T}))$. Hence $\mathrm{OT}(H)=2\,\mathrm{Dig}(H)$, exactly.

(c) *Up barrier.* The no-touch pays $f(F_T)=\mathbf{1}\{F_T<H\}$ when the barrier is not reached, and f vanishes for $x\ge H$. The European claim is the digital put, $V(x,0)=e^{-rT}\Phi(-d_2(x))$ with $d_2(x)=[\ln(x/H)-\sigma^2T/2]/(\sigma\sqrt{T})$. At the reflected price,

$$d_2\Big(\frac{H^2}{F}\Big)=\frac{\ln(H/F)-\sigma^2T/2}{\sigma\sqrt{T}}=-d_1,\qquad\text{so}\qquad V\Big(\frac{H^2}{F},0\Big)=e^{-rT}\Phi(d_1).$$

By (M), which is Theorem 23.16(iv) with zero carry, the no-touch is worth $e^{-rT}[\Phi(-d_2)-(F/H)\,\Phi(d_1)]$. A unit paid at T is worth e^{-rT}, so

$$\mathrm{OT}(H)=e^{-rT}-e^{-rT}\Big[\Phi(-d_2)-\frac{F}{H}\,\Phi(d_1)\Big]=e^{-rT}\Big[\Phi(d_2)+\frac{F}{H}\,\Phi(d_1)\Big].$$

Since $C(H)/H=e^{-rT}[(F/H)\,\Phi(d_1)-\Phi(d_2)]$, the bracket is $2\,\mathrm{Dig}(H)+C(H)/H$.

Down barrier. Now $f(x)=\mathbf{1}\{x>H\}$, which vanishes for $x\le H$; the European claim is the digital, $V(x,0)=e^{-rT}\Phi(d_2(x))$, and again $d_2(H^2/F)=-d_1$. The no-touch is worth $e^{-rT}[\Phi(d_2)-(F/H)\,\Phi(-d_1)]$ and

$$\mathrm{OT}(H)=e^{-rT}\Big[\Phi(-d_2)+\frac{F}{H}\,\Phi(-d_1)\Big]=2\,e^{-rT}\Phi(-d_2)-\frac{\mathcal{P}(H)}{H},$$

twice the digital put less $\mathcal{P}(H)/H$, because $\mathcal{P}(H)/H=e^{-rT}[\Phi(-d_2)-(F/H)\,\Phi(-d_1)]$.

(d) Here $\sigma\sqrt{T}=0.09$ and $\sigma^2T/2=0.00405$.

Up barrier 2,200. Here $\ln(F/H)=-0.09531$, so $d_2=(-0.09531-0.00405)/0.09=-1.1040$ and $d_1=-1.0140$. The digital is worth $\Phi(-1.1040)=0.1348$ and

$$\mathrm{OT}=0.13480+\frac{2000}{2200}\times0.15529=0.1348+0.1412=0.2760,$$

against 0.2696 for two digitals: the excess 0.0064 is $C(H)/H$ (the call is worth about 14.0).

Down barrier 1,800. Here $\ln(F/H)=0.10536$, so $d_2=(0.10536-0.00405)/0.09=1.1257$ and $d_1=1.2157$. The digital put is worth $\Phi(-1.1257)=0.13015$ and

$$\mathrm{OT}=0.13015+\frac{2000}{1800}\times0.11205=0.13015+0.12450=0.25465,$$

against 0.26030 for two digital puts: the shortfall 0.00565 is $\mathcal{P}(H)/H$ (the put is worth about 10.2). At full precision these are 0.254658, 0.260304 and 0.005647, that is, $0.2547=0.2603-0.0056$.

(e) Under $\mathbb{Q}$ the futures price has no drift, so its logarithm drifts at $-\sigma^2/2$: away from an upper barrier and toward a lower one. A path that has just touched an upper barrier therefore ends

above it with probability $\Phi(-\sigma\sqrt{\tau}/2)$, a little less than one half, so touching is a little more than twice as likely as ending beyond, and the up one-touch exceeds two digitals. At a lower barrier the drift helps the path stay beyond, and the one-touch falls short of two digital puts. In the arithmetic model of (a) the price itself has no drift and the factor is exactly two.

At the touch, $F = H$ and the sold one-touch has become a sure payment of 1 at T, a liability of $e^{-r\tau}$. Each digital has $d_2 = -\sigma\sqrt{\tau}/2$ and is worth $e^{-r\tau}\Phi(-\sigma\sqrt{\tau}/2)$. The position is worth

$$2\,e^{-r\tau}\Phi\big(-\tfrac{1}{2}\sigma\sqrt{\tau}\big) - e^{-r\tau} = -\,e^{-r\tau}\big[2\,\Phi\big(\tfrac{1}{2}\sigma\sqrt{\tau}\big) - 1\big],$$

a loss equal to the at-the-money value of $C(H)/H$, about $0.4\,\sigma\sqrt{\tau}$. With the volatility $\sigma = 18\%$ of (d) it is $2\,\Phi(0.045) - 1 = 0.036$ with three months left, and $2\,\Phi(0.0318) - 1 = 0.025$ with $\tau = 0.125$. Adding $1/H$ calls struck at H to the two digitals closes the gap at every date up to the touch: by (c), with τ in the place of T, that package is worth exactly the one-touch for every $F < H$, and $e^{-r\tau}$ at the touch, where it is sold and the proceeds are held in the bank until T.

Self-check. The up barrier can be priced from the reflection formula with drift instead of the image formula: for $X_t = \nu t + W_t$ and $\ell > 0$, $P(\max_{t\le T} X_t \ge \ell) = \Phi\big((\nu T - \ell)/\sqrt{T}\big) + e^{2\nu\ell}\,\Phi\big((-\ell - \nu T)/\sqrt{T}\big)$ (Theorem 19.63(ii)). Under $\mathbb{Q}$, $\ln(F_t/F)/\sigma = -(\sigma/2)t + W_t$, so $\nu = -0.09$ and $\ell = \ln(1.1)/0.18 = 0.5295$; then $(\nu T - \ell)/\sqrt{T} = (-0.0225 - 0.5295)/0.5 = -1.1040 = d_2$, $(-\ell - \nu T)/\sqrt{T} = -1.0140 = d_1$ and $e^{2\nu\ell} = e^{-0.09531} = 1/1.1 = F/H$: the same 0.2760. A simulation that draws the terminal price and then the running extreme from its exact law given the endpoints (20,000,000 draws) gives 0.27592 and 0.25466 for the two one-touches, each with standard error 0.0001. Finally, the arithmetic model with $\sigma_{\mathrm{N}} = 0.18 \times 2{,}000 = 360$ puts both barriers $200/180 = 1.111$ standard deviations away and gives $\mathrm{OT} = 2\,\Phi(-10/9) = 2\,\Phi(-1.1111) = 0.2665$ for each, between the two lognormal values 0.2760 and 0.2547, in line with the opposite signs of the two corrections.

Answer: (a), (b) $\mathbb{Q}(\text{touch}) = 2\,\mathbb{Q}(S_T \ge H) = 2\,\Phi\big((S_0 - H)/(\sigma_{\mathrm{N}}\sqrt{T})\big)$, and it is the price because the delta hedge of V^{out}, stopped at the barrier, replicates the no-touch: $\mathrm{OT}(H) = 2\,\mathrm{Dig}(H)$. (c) Up: $\mathrm{OT}(H) = e^{-rT}[\Phi(d_2) + (F/H)\,\Phi(d_1)] = 2\,\mathrm{Dig}(H) + C(H)/H$; down: $\mathrm{OT}(H) = e^{-rT}[\Phi(-d_2) + (F/H)\,\Phi(-d_1)]$, twice the digital put less $\mathcal{P}(H)/H$. (d) Up barrier 2,200: digital 0.1348, one-touch $0.2760 = 0.2696 + 0.0064$; down barrier 1,800: digital put 0.13015, one-touch $0.2547 = 0.2603 - 0.0056$ ($0.25465 = 0.26030 - 0.00565$ with the values of Φ given). (e) The log price drifts at $-\sigma^2/2$ under $\mathbb{Q}$; at the touch the two digitals are worth $2e^{-r\tau}\Phi(-\sigma\sqrt{\tau}/2) < e^{-r\tau}$, a shortfall of $e^{-r\tau}[2\,\Phi(\sigma\sqrt{\tau}/2) - 1] \approx 0.4\,\sigma\sqrt{\tau}$, which $1/H$ calls struck at H would cover up to the touch.

P23.6 *Asian options: effective volatility, bounds and average-price parity*

(a) Under $\mathbb{Q}$, $\ln S_t = \ln S + (b - \frac{1}{2}\sigma^2)t + \sigma W_t$, so $\ln G_T = \frac{1}{2}(\ln S_{T/2} + \ln S_T)$ is a constant plus $\frac{1}{2}\sigma\,(W_{T/2} + W_T)$. With the independent increments $\Delta W_1 = W_{T/2}$ and $\Delta W_2 = W_T - W_{T/2}$, each of variance $T/2$,

$$W_{T/2} + W_T = 2\,\Delta W_1 + \Delta W_2, \qquad \mathrm{Var}\,(W_{T/2} + W_T) = (4+1)\,\frac{T}{2} = 2.5\,T,$$

and $\mathrm{Var}\ln G_T = \frac{1}{4}\sigma^2 \times 2.5\,T = 0.625\,\sigma^2 T$. The general factor at $J = 2$ is $3 \times 5/24 = 0.625$, as it should be; the effective volatility is $0.35\sqrt{0.625} = 27.67\%$. The first increment counts twice because it moves both fixings.

(b) By Proposition 23.28, with $J = 4$ the factor is $5 \times 9/96 = 0.46875$, so

$$\sigma_G^2 = 0.1225 \times 0.46875 = 0.057422, \qquad \sigma_G = 23.96\%.$$

The mean of $\ln G_T$ is $\ln S + (b - \frac{1}{2}\sigma^2)T\,(J+1)/(2J) = \ln 50 + (0.02 - 0.06125) \times 0.625 = \ln 50 - 0.025781$, and adding half the variance, 0.028711, gives $\ln F_G = \ln 50 + 0.00293$:

$$F_G = 50 \times 1.00293 = 50.147.$$

Then $d_1 = (0.00293 + 0.028711)/0.23963 = 0.13204$ and $d_2 = d_1 - 0.23963 = -0.10759$, and the geometric average-price call is worth

$$e^{-rT}\big[F_G\,\Phi(d_1) - K\,\Phi(d_2)\big] = 0.97045 \times \big(50.147 \times 0.55252 - 50 \times 0.45716\big) = 0.97045 \times 4.8492 = 4.706.$$

(c) On every path $(\bar S_T - K)^+ \le \frac{1}{4}\sum_j (S_{t_j} - K)^+$, and the right side is the payoff of holding, for each fixing date, $e^{-r(T-t_j)}/4$ vanilla calls that expire on that date and depositing what they pay until T. By Proposition 23.27(i),

$$\begin{aligned} C^{\text{avg}} &\le \tfrac{1}{4}\big(0.97775 \times 3.5947 + 0.98511 \times 5.1266 + 0.99253 \times 6.3102 + 7.3099\big) \\ &= \tfrac{1}{4}\,(3.5147 + 5.0503 + 6.2631 + 7.3099) = 5.5345. \end{aligned}$$

The geometric mean never exceeds the arithmetic mean, so the geometric call of (b) is a lower bound (Proposition 23.27(iii)):

$$4.706 \;\le\; C^{\text{avg}} \;\le\; 5.5345,$$

and both bounds are below the vanilla call, 7.310.

(d) The factor $(1 + 1/J)(2 + 1/J)/6$ tends to $1/3$, so $\sigma_G \to \sigma/\sqrt{3} = 20.21\%$. With four fixings the variance of $\ln G_T$ is 0.469 of that of $\ln S_T$, against one third in the limit, because its fixings lie later on average (at $T(J+1)/(2J) = 0.625\,T$ against $0.5\,T$); its forward is slightly higher as well, 50.147 against 49.990. On this grid, with each fixing at the end of its period, fewer fixings mean fixings later on average and a dearer option: the geometric price is 7.310 for $J = 1$ (the vanilla call), 5.540 for $J = 2$, 4.706 for $J = 4$, 4.166 for $J = 12$ and 3.900 in the limit.

(e) For any numbers a and a', $(a - a')^+ - (a' - a)^+ = a - a'$: if $a \ge a'$ the second term is 0 and the first is $a - a'$, and otherwise the first is 0 and the second is $a' - a$. Applied three times, on every path,

$$\begin{aligned} (\bar S_T - K)^+ - (K - \bar S_T)^+ &= \bar S_T - K, \\ (S_T - \bar S_T)^+ - (\bar S_T - S_T)^+ &= S_T - \bar S_T, \\ (S_T - K)^+ - (K - S_T)^+ &= S_T - K. \end{aligned}$$

The first two right sides add up to the third. So the portfolio that is long the average-price call and the average-strike call and short the two corresponding puts pays, in every state, what the portfolio long the vanilla call and short the vanilla put pays. The six options are assumed to trade, as in every model-free statement of the section, and each static position in them is a self-financing strategy. By the law of one price (Lemma 16.8(iii)), $C^{\text{avg}} - \mathcal{P}^{\text{avg}} + C^{\text{as}} - \mathcal{P}^{\text{as}} = C - \mathcal{P}$, which rearranges to the identity. No model was used.

(f) Once j of the J fixings are known, only the fraction $(J - j)/J$ of the average still moves with the price: the claim is $(J - j)/J$ average-price calls on the remaining fixings, struck at an effective strike (Remark 23.31), so its delta is at most $(J - j)/J$ and its gamma is scaled down by the same fraction. In the last quarter it is a quarter of a vanilla call struck at the effective strike, with a delta of at most 0.25, and its hedge moves a quarter as much as that of the vanilla; only with continuous averaging do the delta and the gamma tend to zero. A vanilla option near the money keeps a delta that swings between 0 and 1 until the last moment, with a gamma that grows without bound. The same averaging makes the Asian payoff hard to move by pushing the price at settlement.

Self-check. The variance factor can be computed from covariances instead of increments:

$$\operatorname{Var}\Big(\sum_j W_{t_j}\Big) = \sum_{j,k} \min(t_j, t_k) = \frac{T}{J}\sum_{j,k} \min(j, k).$$

For $J = 2$ the four terms are $1 + 1 + 1 + 2 = 5$, a factor $5/(2^2 \times 2) = 0.625$; for $J = 4$ the sixteen terms add up to 30, a factor $30/(4^2 \times 4) = 0.46875$. A simulation of 4,000,000

paths gives 4.708 for the geometric call, with standard error 0.004, and, with the geometric call as a control variate, 4.970 for the arithmetic call, with standard error 0.0002: inside the bracket of (c). The parity of (e) checks on forward values: $\mathbb{E}^{\mathbb{Q}}\,\bar{S}_T = \frac{1}{4}\sum_j 50\,e^{0.02t_j} = 50.630$, so $C^{\text{avg}} - \mathcal{P}^{\text{avg}} = e^{-0.03}(50.630 - 50) = 0.611$ and $C^{\text{as}} - \mathcal{P}^{\text{as}} = 50\,e^{-0.01} - e^{-0.03} \times 50.630 = 0.369$, and their sum 0.980 is $C - \mathcal{P} = 50\,e^{-0.01} - 50\,e^{-0.03} = 0.980$.

Answer: (a) $\operatorname{Var}\ln G_T = 0.625\,\sigma^2 T$ (effective volatility 27.67%). (b) $\sigma_G = 23.96\%$, $F_G = 50.147$, geometric call 4.706. (c) $4.706 \le C^{\text{avg}} \le 5.5345$, below the vanilla 7.310. (d) $\sigma/\sqrt{3} = 20.21\%$; with four fixings the variance factor is 0.469 against 1/3, because the fixings lie later on average ($0.625\,T$ against $0.5\,T$), so $4.706 > 3.900$. (e) The payoff differences are $\bar{S}_T - K$, $S_T - \bar{S}_T$ and $S_T - K$, and the first two add up to the third. (f) With j of J fixings known, only $(J-j)/J$ of the average still moves, so the delta is capped at $(J-j)/J$ (0.25 in the last quarter) and the gamma is scaled by the same fraction, while a vanilla's delta swings between 0 and 1 to the end.

P23.7 *Exchanging one asset for another*

(a) *Factor out the asset given up.* Put $X = \ln(S_{1,T}/S_{2,T})$ and $Y = \ln S_{2,T}$, so that the payoff is $S_{2,T}\,(S_{1,T}/S_{2,T} - 1)^+ = e^Y \psi(X)$ with $\psi(x) = (e^x - 1)^+ \le e^x$. Both are constants plus linear combinations of Z_1 and $Z_\perp$:

$$\begin{aligned} Y &= \ln S_{2,0} + (r - q_2 - \tfrac{1}{2}\sigma_2^2)T + \sigma_2\sqrt{T}\,Z_2, \\ X &= \ln\frac{S_{1,0}}{S_{2,0}} + (q_2 - q_1 - \tfrac{1}{2}\sigma_1^2 + \tfrac{1}{2}\sigma_2^2)T + \sigma_1\sqrt{T}\,Z_1 - \sigma_2\sqrt{T}\,Z_2. \end{aligned}$$

The rate has already cancelled in X, because both prices grow at r.

Moments. Since $\operatorname{Var} Z_1 = \operatorname{Var} Z_2 = 1$ and $\operatorname{Cov}(Z_1, Z_2) = \rho$,

$$\operatorname{Var} X = (\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)T = \sigma^2 T, \qquad \operatorname{Cov}(X, Y) = (\rho\sigma_1\sigma_2 - \sigma_2^2)T,$$

and, by (G) with $\mathbb{E}\,Y = \ln S_{2,0} + (r - q_2 - \frac{1}{2}\sigma_2^2)T$ and $\operatorname{Var} Y = \sigma_2^2 T > 0$, $\mathbb{E}\,e^Y = S_{2,0}e^{(r-q_2)T}$. By (G) the variable X is normal, so the shifted variable $X + \operatorname{Cov}(X, Y)$ is normal with variance $\sigma^2 T$ and mean

$$\ln\frac{S_{1,0}}{S_{2,0}} + (q_2 - q_1)T - \tfrac{1}{2}\sigma_1^2 T + \tfrac{1}{2}\sigma_2^2 T + \rho\sigma_1\sigma_2 T - \sigma_2^2 T = \ln x_0 - \tfrac{1}{2}\sigma^2 T, \qquad x_0 = \frac{S_{1,0}e^{-q_1 T}}{S_{2,0}e^{-q_2 T}}.$$

So $X + \operatorname{Cov}(X, Y) = \ln x_0 - \frac{1}{2}\theta^2 + \theta Z$ with $\theta = \sigma\sqrt{T} > 0$ and Z standard normal.

Shift and integrate. By (G), which is Lemma 19.38(iii), and then (L) with $x = x_0$ and $\theta = \sigma\sqrt{T}$,

$$\mathbb{E}^{\mathbb{Q}}\big[(S_{1,T} - S_{2,T})^+\big] = \mathbb{E}\,e^Y\;\mathbb{E}\big[(x_0\,e^{-\theta^2/2 + \theta Z} - 1)^+\big] = S_{2,0}e^{(r-q_2)T}\,\big[x_0\,\Phi(d_1) - \Phi(d_2)\big].$$

Multiplying by e^{-rT} removes the last trace of the rate:

$$\begin{aligned} e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_{1,T} - S_{2,T})^+\big] &= S_{2,0}e^{-q_2 T}\,x_0\,\Phi(d_1) - S_{2,0}e^{-q_2 T}\,\Phi(d_2) \\ &= S_{1,0}e^{-q_1 T}\,\Phi(d_1) - S_{2,0}e^{-q_2 T}\,\Phi(d_2), \end{aligned}$$

with $d_{1,2} = (\ln x_0 \pm \sigma^2 T/2)/(\sigma\sqrt{T})$. This is the formula of Theorem 23.32; the rate cancels once inside the ratio $S_{1,T}/S_{2,T}$ and once between the growth e^{rT} of asset 2 and the discount factor.

(b) *The key identity.* Since $d_1 - d_2 = \sigma\sqrt{T-t}$ and $d_1 + d_2 = 2\ln(x_1/x_2)/(\sigma\sqrt{T-t})$,

$$\frac{\varphi(d_2)}{\varphi(d_1)} = \exp\Big(\frac{d_1^2 - d_2^2}{2}\Big) = \exp\Big(\frac{(d_1 - d_2)(d_1 + d_2)}{2}\Big) = \exp\Big(\ln\frac{x_1}{x_2}\Big) = \frac{x_1}{x_2},$$

that is, $x_1\,\varphi(d_1) = x_2\,\varphi(d_2)$.

The deltas. Both d_1 and d_2 have the derivative $1/(x_1\sigma\sqrt{T-t})$ in x_1 and $-1/(x_2\sigma\sqrt{T-t})$ in x_2. Hence

$$\partial_1 V = \Phi(d_1) + \frac{x_1\,\varphi(d_1) - x_2\,\varphi(d_2)}{x_1\sigma\sqrt{T-t}} = \Phi(d_1), \qquad \partial_2 V = -\Phi(d_2) - \frac{x_1\,\varphi(d_1) - x_2\,\varphi(d_2)}{x_2\sigma\sqrt{T-t}} = -\Phi(d_2),$$

the fractions vanishing by the key identity. Therefore

$$x_1\,\partial_1 V + x_2\,\partial_2 V = x_1\,\Phi(d_1) - x_2\,\Phi(d_2) = V,$$

which is Euler's identity for a function that is homogeneous of degree one in (x_1, x_2).

The pricing equation. Differentiating the deltas once more, with the derivatives of d_1 and d_2 found above,

$$\partial_{11} V = \varphi(d_1)/(x_1\sigma\sqrt{T-t}), \qquad \partial_{12} V = -\varphi(d_1)/(x_2\sigma\sqrt{T-t}),$$
$$\partial_{22} V = \varphi(d_2)/(x_2\sigma\sqrt{T-t}) = x_1\varphi(d_1)/(x_2^2\sigma\sqrt{T-t}),$$

the last by the key identity. In t only d_1 and d_2 move, so by the key identity

$$\partial_t V = x_1\varphi(d_1)\,\partial_t d_1 - x_2\varphi(d_2)\,\partial_t d_2 = x_1\varphi(d_1)\,\partial_t(d_1 - d_2) = x_1\varphi(d_1)\,\partial_t(\sigma\sqrt{T-t}) = -\frac{x_1\varphi(d_1)\,\sigma}{2\sqrt{T-t}}.$$

The bracket is

$$\begin{aligned}\tfrac{1}{2}(\sigma_1^2 x_1^2\,\partial_{11} V + 2\rho\sigma_1\sigma_2 x_1 x_2\,\partial_{12} V + \sigma_2^2 x_2^2\,\partial_{22} V) &= \frac{x_1\varphi(d_1)\,(\sigma_1^2 - 2\rho\sigma_1\sigma_2 + \sigma_2^2)}{2\sigma\sqrt{T-t}} \\ &= \frac{x_1\varphi(d_1)\,\sigma}{2\sqrt{T-t}} = -\partial_t V\,:\end{aligned}$$

the correlation and the two volatilities combine into exactly the ratio variance, and the drift term vanishes.

Reading. The delta hedge holds $\partial_1 V = \Phi(d_1)$ units of the reinvested holding of asset 1 and $\partial_2 V = -\Phi(d_2)$ units of that of asset 2. Their combined value is $x_1\,\partial_1 V + x_2\,\partial_2 V$, and the hedge is worth V; the difference, which is what sits in the bank, is zero at every date. A strategy that never borrows or lends neither earns nor pays the rate, so r cannot enter the price. The yields do enter, because the instruments of the hedge are the reinvested holdings $S_{k,t}e^{-q_k(T-t)}$: each unit of them is fewer than one share today when $q_k > 0$. (With the pricing equation, the two-dimensional Itô lemma (Theorem 19.55), applied to V at the discounted holdings $e^{-rt}x_1$ and $e^{-rt}x_2$, writes $e^{-rt}V$ as its initial value plus the gains of the holdings $\partial_1 V$ and $\partial_2 V$ alone, which is the self-financing property; the homogeneity of V lets the discount factor pass through it. The growth conditions, the limit at T and the arbitrage argument are Step 4 of the proof of Theorem 23.32.)

(c) Take asset 1 to be one share of A and asset 2 to be two shares of B. The reinvested holdings are worth 60 (no dividends) and $2 \times 30 \times e^{-0.02\times 0.5} = 60 \times 0.99005 = 59.403$. The ratio volatility is

$$\sigma^2 = 0.0625 + 0.1225 - 2 \times 0.6 \times 0.25 \times 0.35 = 0.08, \qquad \sigma = 28.28\%, \qquad \sigma\sqrt{T} = 0.2.$$

With $\ln x_0 = \ln(60/59.403) = 0.01$, $d_1 = (0.01 + 0.02)/0.2 = 0.15$ and $d_2 = -0.05$, so $\Phi(d_2) = 1 - 0.51994 = 0.48006$ and the option is worth

$$60 \times 0.55962 - 59.403 \times 0.48006 = 33.577 - 28.517 = 5.060,$$

whatever the interest rate. The replicating portfolio today is long 0.5596 shares of A, worth 33.58, and short 0.4801 units of the reinvested holding of two B shares, each unit worth 59.40, that is $0.4801 \times 2 \times 0.99005 = 0.9506$ shares of B, worth 28.52; it holds no cash, and $33.58 - 28.52 = 5.06$.

(d) For any two numbers $\max(a, a') = a' + (a - a')^+$. The claim therefore pays two shares of B at T plus the payoff of the exchange option, and (Remark 23.34)

$$\text{best-of} = 59.40 + 5.06 = 64.46.$$

Self-check. A simulation of 8,000,000 pairs of terminal prices gives 5.062, with standard error 0.002, at $r = 0$ and exactly the same number at $r = 8\%$: with the normals held fixed, both terminal prices carry the factor e^{rT} and the discount removes it path by path. A back-of-envelope value comes from the at-the-money rule in units of asset 2: a call on the ratio, struck at 1, with $x_0 = 1.01005$ and $\theta = 0.2$ is worth about $\theta/\sqrt{2\pi} + (x_0 - 1)/2 = 0.0798 + 0.0050 = 0.0848$ units (the at-the-money value plus a delta of one half on the excess of the ratio over 1), that is $0.0848 \times 59.40 = 5.04$. And if B paid no dividend, $x_0 = 1$, $d_1 = -d_2 = 0.1$ and the option would be worth $60\,(2\,\Phi(0.1) - 1) = 4.78$: the yield on the asset given up makes the option dearer, as it should.

Answer: (a) $S_{1,0}e^{-q_1 T}\Phi(d_1) - S_{2,0}e^{-q_2 T}\Phi(d_2)$ with the ratio volatility $\sigma^2 = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$; the rate cancels in the ratio and against the discount factor. (b) $\partial_1 V = \Phi(d_1)$, $\partial_2 V = -\Phi(d_2)$ and $V = x_1\,\partial_1 V + x_2\,\partial_2 V$, and the drift term vanishes because $\sigma_1^2 - 2\rho\sigma_1\sigma_2 + \sigma_2^2 = \sigma^2$: the hedge holds the two reinvested assets and no cash, so r cannot appear, while the yields enter through the reinvested holdings. (c) $\sigma = 28.28\%$, $d_1 = 0.15$, $d_2 = -0.05$, price 5.060 for every r; hedge: long 0.5596 shares of A, short 0.4801 units of the reinvested two-share holding of B (0.9506 shares of B), no cash. (d) $59.40 + 5.06 = 64.46$.

P23.8 *The Monte Carlo error budget*

(a) By Proposition 23.37, the standard error is $s/\sqrt{n} = 22.4/200 = 0.112$, and the approximate 95% interval is the estimate $\pm\,1.96 \times 0.112 = \pm\,0.22$.

(b) Solving $s/\sqrt{n} = 0.02$ gives

$$n = \Big(\frac{22.4}{0.02}\Big)^2 = 1120^2 = 1{,}254{,}400,$$

about 1.25 million paths, 31.36 times as many as were run. The error falls like $n^{-1/2}$, so halving it takes four times the paths, not twice.

(c) By (23.13) of Proposition 23.39,

$$\operatorname{Var}\Big(\frac{X - Y}{\Delta S}\Big) = \frac{\operatorname{Var} X + \operatorname{Var} Y - 2\operatorname{Cov}(X, Y)}{(\Delta S)^2}.$$

(i) With independent random numbers the covariance is zero: the variance is $(180 + 170)/0.25 = 1{,}400$ for one pair, and, by Proposition 23.37(ii) applied to the 10,000 independent pairs, the standard error of the delta is $\sqrt{1400/10{,}000} = 0.374$. (ii) With common random numbers, $\operatorname{Cov}(X, Y) = 0.999 \times \sqrt{180 \times 170} = 0.999 \times 174.93 = 174.75$, so the numerator is $350 - 349.507 = 0.4927$, the variance is $0.4927/0.25 = 1.97$, and the standard error is $\sqrt{1.97/10{,}000} = 0.014$. Reusing the random numbers divides the variance by about 710 and the standard error by about 27: with independent numbers a delta of order one is drowned in an error of 0.37.

(d) Build the array row by row, so that each row has unit length and the rows have the required inner products.

- Row 1: $(1, 0, 0)$.
- Row 2: $\gamma_{21} = \rho_{12} = 0.6$ and $\gamma_{22} = \sqrt{1 - 0.6^2} = 0.8$, so $(0.6, 0.8, 0)$.
- Row 3: $\gamma_{31} = \rho_{13} = 0.3$; then $\gamma_{21}\gamma_{31} + \gamma_{22}\gamma_{32} = \rho_{23}$ gives $\gamma_{32} = (0.5 - 0.18)/0.8 = 0.4$; and unit length gives $\gamma_{33} = \sqrt{1 - 0.09 - 0.16} = \sqrt{0.75} = 0.866$. So $(0.3, 0.4, 0.866)$.

Verification: the squared lengths are 1, $0.36 + 0.64 = 1$ and $0.09 + 0.16 + 0.75 = 1$; the inner products are 0.6 (rows 1 and 2), 0.3 (rows 1 and 3) and $0.18 + 0.32 = 0.5$ (rows 2 and 3). The

three combinations are Z_1, $0.6\,Z_1 + 0.8\,Z_2$ and $0.3\,Z_1 + 0.4\,Z_2 + 0.866\,Z_3$. Each is normal with mean zero, as a linear combination of independent normal variables (Lemma 19.38(ii), one term at a time), and by the bilinearity of the covariance their variances and covariances are these squared lengths and inner products (the second claim behind Procedure 23.36, proved in Appendix I.7). So they are standard normal with the prescribed correlations, and the diagonal entries 1, 0.8 and 0.866 are positive.

Self-check. For (b), scale instead of solving: the error must fall from 0.112 to 0.02, a factor 5.6, which costs $5.6^2 = 31.36$ times the paths, $40{,}000 \times 31.36 = 1{,}254{,}400$. For (c), the numerator can be split differently: with standard deviations 13.416 and 13.038,

$$\operatorname{Var}(X - Y) = (13.416 - 13.038)^2 + 2\,(1 - 0.999) \times 174.93 = 0.143 + 0.350 = 0.493,$$

the first term coming from the difference of scale of the two payoffs and the second from their imperfect correlation. For (d), the product of the squared diagonal entries, $1 \times 0.64 \times 0.75 = 0.48$, must equal the determinant of the correlation matrix, $1 + 2 \times 0.6 \times 0.3 \times 0.5 - 0.36 - 0.09 - 0.25 = 0.48$.

Answer: (a) Standard error 0.112; 95% interval $\pm\,0.22$. (b) 1,254,400 paths (31.36 times as many); halving an error takes four times the paths. (c) Independent numbers: variance 1,400 per pair, standard error 0.374; common numbers: variance 1.97, standard error 0.014. (d) Rows $(1, 0, 0)$, $(0.6, 0.8, 0)$ and $(0.3, 0.4, 0.866)$.

P23.9 *Antithetic pairs and control variates*

(a) By linearity of expectation, $\mathbb{E}[X - \beta\,(Y - \mathbb{E}\,Y)] = \mathbb{E}\,X - \beta\,(\mathbb{E}\,Y - \mathbb{E}\,Y) = \mathbb{E}\,X$ for every β. Adding the constant $\beta\,\mathbb{E}\,Y$ does not change a variance, so by the formula for the variance of a sum (Proposition 10.10), applied to X and $-\beta Y$, with $\operatorname{Var}(-\beta Y) = \beta^2 \operatorname{Var} Y$ and $\operatorname{Cov}(X, -\beta Y) = -\beta \operatorname{Cov}(X, Y)$ (the covariance is bilinear),

$$\operatorname{Var}\left(X - \beta\,(Y - \mathbb{E}\,Y)\right) = \operatorname{Var} X - 2\beta \operatorname{Cov}(X, Y) + \beta^2 \operatorname{Var} Y.$$

We complete the square. With $\beta^* = \operatorname{Cov}(X, Y)/\operatorname{Var} Y$,

$$\operatorname{Var} X - 2\beta \operatorname{Cov}(X, Y) + \beta^2 \operatorname{Var} Y = \operatorname{Var} Y\,(\beta - \beta^*)^2 + \operatorname{Var} X - \frac{\operatorname{Cov}(X, Y)^2}{\operatorname{Var} Y},$$

because $\operatorname{Var} Y\,(\beta - \beta^*)^2 = \beta^2 \operatorname{Var} Y - 2\beta \operatorname{Cov}(X, Y) + \operatorname{Cov}(X, Y)^2/\operatorname{Var} Y$, using $\beta^* \operatorname{Var} Y = \operatorname{Cov}(X, Y)$. The first term is positive for every $\beta \neq \beta^*$, because $\operatorname{Var} Y > 0$, and zero at β^*: the variance is smallest at β^* and only there. Its smallest value is

$$\operatorname{Var} X - \frac{\operatorname{Cov}(X, Y)^2}{\operatorname{Var} Y} = \operatorname{Var} X \left(1 - \frac{\operatorname{Cov}(X, Y)^2}{\operatorname{Var} X\ \operatorname{Var} Y}\right) = (1 - \rho^2) \operatorname{Var} X.$$

This is Proposition 23.45; β^* is the slope of the regression of X on Y, and what remains is the variance of the regression residual.

(b) The correlation is $\rho = 54.6/(8 \times 7) = 0.975$ and the coefficient is $\beta^* = 54.6/49 = 1.114$. Then $1 - \rho^2 = 1 - 0.950625 = 0.049375$, and the variance per path is

$$64 \times 0.049375 = 3.16 \ \text{ at } \beta^*, \qquad 64 - 2 \times 54.6 + 49 = 3.80 \ \text{ at } \beta = 1.$$

The variance of a sample mean is the variance per path divided by the number of paths, so at β^* the same accuracy needs $64/3.16 = 20.3$ times fewer paths, about twenty-fold; the choice $\beta = 1$ gives $64/3.80 = 16.8$.

(c) The variable $-Z$ has the law of Z, so $f(-Z)$ has the mean and the variance of $f(Z)$, and the pair average has the mean $\mathbb{E}\,f(Z)$. By Proposition 10.10,

$$\operatorname{Var}\left(f(Z) + f(-Z)\right) = 2 \operatorname{Var} f(Z) + 2 \operatorname{Cov}\left(f(Z), f(-Z)\right) = 2 \operatorname{Var} f(Z)\,(1 + \rho_A),$$

the last step because $f(Z)$ and $f(-Z)$ have the same standard deviation, so $\operatorname{Cov}(f(Z), f(-Z)) = \rho_A \operatorname{Var} f(Z)$; dividing by 4 gives $\frac{1}{2} \operatorname{Var} f(Z)\,(1+\rho_A)$. Two independent draws have zero covariance, so their average has the variance $\frac{1}{2} \operatorname{Var} f(Z)$. For the same two payoff evaluations the pair is better exactly when $\rho_A < 0$. A payoff that is monotone in the normal (a call, a put, a digital) has $\rho_A \le 0$ (Proposition 23.43(iii)), so for it the pair never does worse than two independent draws; the same holds for a path payoff that is monotone in each of its normals, such as an Asian call.

(d) (i) The digital pays when $Z > 0.5$ and its mirror image when $Z < -0.5$, so the two never pay together and $\mathbb{E}[f(Z)f(-Z)] = 0$. With $p = \Phi(-0.5) = 1 - 0.69146 = 0.30854$, the covariance is $-p^2$ and the variance is $p\,(1-p)$, so

$$\rho_A = -\frac{p}{1-p} = -\frac{0.30854}{0.69146} = -0.446,$$

and the pair has $1+\rho_A = 0.554$ of the variance of two independent draws. (ii) Here $f(-Z) = f(Z)$, so $\rho_A = +1$: the pair average is $f(Z)$ itself, with the variance $\operatorname{Var} f(Z)$, twice that of two independent draws. The second evaluation is wasted, and antithetic sampling doubles the variance for the same work. A straddle on the price itself is not exactly symmetric in the normal, but it is close: for an at-the-money straddle at a volatility of 20% over one year, with zero rates, Example 23.44 gives $\rho_A = +0.899$, so the variance of a pair is 1.90 times that of two independent draws.

(e) Four: the target and the control at Z, and both again at $-Z$. The value of a trial is

$$\tfrac{1}{2}\Big[\big(X(Z) - \beta\,(Y(Z) - \mathbb{E}\,Y)\big) + \big(X(-Z) - \beta\,(Y(-Z) - \mathbb{E}\,Y)\big)\Big],$$

writing $X(Z)$ and $Y(Z)$ for the two payoffs as functions of the normal. Different trials are independent; the two members of a pair are not. The standard error is therefore the sample standard deviation of the n trial values divided by $\sqrt{n}$, never a figure computed from the $2n$ corrected payoffs as if they were independent.

Self-check. The minimum of (a) can be found by calculus instead of completing the square: the derivative of the variance in β is $-2\operatorname{Cov}(X,Y) + 2\beta \operatorname{Var} Y$, which vanishes at β^*, and the second derivative $2 \operatorname{Var} Y$ is positive. On the numbers of (b), the minimum computed without the correlation is $64 - 54.6^2/49 = 64 - 60.84 = 3.16$, and the completed square returns the variance at $\beta = 1$ from the minimum: $3.16 + 49 \times (1 - 1.1143)^2 = 3.16 + 0.64 = 3.80$, the value found directly. For (d)(i), the probabilities can be listed: the pair $(f(Z), f(-Z))$ is $(1,0)$ with probability 0.30854, $(0,1)$ with probability 0.30854 and $(0,0)$ with probability 0.38292, so the pair average has the variance $\frac{1}{4} \times 0.61708 - 0.30854^2 = 0.05907$, and $\frac{1}{2} \times 0.30854 \times 0.69146 \times 0.554 = 0.0591$.

Answer: (a) The variance is $\operatorname{Var} X - 2\beta \operatorname{Cov}(X,Y) + \beta^2 \operatorname{Var} Y$; it is smallest only at $\beta^* = \operatorname{Cov}(X,Y)/\operatorname{Var} Y$, where it equals $(1-\rho^2)\operatorname{Var} X$. (b) $\rho = 0.975$, $\beta^* = 1.114$; variance 3.16 at β^* (a factor 0.049, about 20 times fewer paths) against 3.80 at $\beta = 1$. (c) The pair average has the variance $\frac{1}{2} \operatorname{Var} f(Z)\,(1+\rho_A)$ against $\frac{1}{2} \operatorname{Var} f(Z)$ for two independent draws: it helps exactly when $\rho_A < 0$; a monotone payoff has $\rho_A \le 0$, so for it the pair never hurts. (d) Digital: $\rho_A = -0.446$, a variance factor 0.554; $|z|$: $\rho_A = +1$, the variance doubles. (e) Four payoff evaluations per trial; the standard error comes from the n independent trial values.

P23.10 *One explicit finite-difference step and the stability limit*

(a) Multiply the scheme by Δt and collect the three values of the level $i+1$:

$$(1 + r\,\Delta t)\,V_j^i = \underbrace{\frac{1}{2}\Big(\frac{\sigma^2 \Delta t}{\Delta x^2} + \frac{\nu\,\Delta t}{\Delta x}\Big)}_{p_u} V_{j+1}^{i+1} + \underbrace{\Big(1 - \frac{\sigma^2 \Delta t}{\Delta x^2}\Big)}_{p_m} V_j^{i+1} + \underbrace{\frac{1}{2}\Big(\frac{\sigma^2 \Delta t}{\Delta x^2} - \frac{\nu\,\Delta t}{\Delta x}\Big)}_{p_d} V_{j-1}^{i+1}.$$

Here $\nu = 0.05 - 0.045 = 0.005$. For $\Delta t = 0.02$, $\sigma^2 \Delta t/\Delta x^2 = 0.09 \times 0.02/0.0036 = 0.5$ and $\nu\,\Delta t/\Delta x = 0.005 \times 0.02/0.06 = 0.001667$, so

$$p_u = \tfrac{1}{2}(0.5 + 0.001667) = 0.2508, \qquad p_m = 0.5, \qquad p_d = \tfrac{1}{2}(0.5 - 0.001667) = 0.2492.$$

They sum to 1. Read as the probabilities of a step of the log price by $+\Delta x$, 0 or $-\Delta x$, they give the mean $\Delta x\,(p_u - p_d) = 0.06 \times 0.001667 = 0.0001 = \nu\,\Delta t$ and the second moment $\Delta x^2\,(p_u + p_d) = 0.0036 \times 0.5 = 0.0018 = \sigma^2 \Delta t$: one step of a trinomial tree, discounted by $1/(1 + r\,\Delta t) = 1/1.001$ (Proposition 23.48(i)).

(b) The middle weight is nonnegative when $\sigma^2 \Delta t \le \Delta x^2$, that is,

$$\Delta t \le \frac{\Delta x^2}{\sigma^2} = \frac{0.0036}{0.09} = 0.04,$$

and the two outer weights are nonnegative when $|\nu|\,\Delta x \le \sigma^2$, which holds with room to spare ($0.0003 \le 0.09$). For $\Delta t = 0.05$ the ratio $\sigma^2 \Delta t / \Delta x^2$ is 1.25 and $\nu\,\Delta t / \Delta x = 0.004167$:

$$p_u = 0.6271, \qquad p_m = -0.25, \qquad p_d = 0.6229,$$

a negative "probability". On the sawtooth,

$$p_u\,(-1)^{j+1} + p_m\,(-1)^j + p_d\,(-1)^{j-1} = (p_m - p_u - p_d)\,(-1)^j = (1 - 2 \times 1.25)\,(-1)^j = -1.5\,(-1)^j,$$

so one step multiplies the error by -1.5 before the discount and by $-1.5/1.0025 = -1.496$ after it (Proposition 23.48(iv)). The error changes sign and grows by half at every step: after 40 steps an error of 10^{-6} has been multiplied by 1.4963^{40}, about 10^7, and has become about 10, more than the two-year European put struck at 50 on a stock at 50, which is worth 5.84 at these parameters (forty steps of 0.05 make two years).

(c) On $w_j = (-1)^j$ the second difference is $w_{j+1} - 2w_j + w_{j-1} = (-1 - 2 - 1)\,(-1)^j = -4\,w_j$ and the centred first difference is $w_{j+1} - w_{j-1} = 0$. So

$$(Aw)_j = \Big(\tfrac{1}{2}\sigma^2\,\frac{-4}{\Delta x^2} - r\Big) w_j = \Lambda(\pi)\,w_j, \qquad \Lambda(\pi) = -\frac{2 \times 0.09}{0.0036} - 0.05 = -50.05.$$

Looking for V^i as a multiple of the sawtooth, the implicit scheme gives $(1 - \Delta t\,\Lambda(\pi))V^i = w$ and the Crank–Nicolson scheme $(1 - \tfrac{1}{2}\Delta t\,\Lambda(\pi))V^i = (1 + \tfrac{1}{2}\Delta t\,\Lambda(\pi))\,w$. With $\Delta t\,\Lambda(\pi) = -2.5025$ the factors are

$$\frac{1}{1 + 2.5025} = 0.286 \quad \text{(implicit)}, \qquad \frac{1 - 1.25125}{1 + 1.25125} = -0.112 \quad \text{(Crank–Nicolson)}.$$

By Proposition 23.51(iii) these multiples of the sawtooth are the only bounded solutions of the two steps. Both factors are below one in absolute value, so the sawtooth dies out in both schemes (with alternating sign under Crank–Nicolson), at a time step for which the explicit scheme blows up.

(d) The payoffs at the five nodes are 5.6540, 2.9118, 0, 0 and 0. With the weights of (a):

- At 47.0882: $(0.5 \times 2.9118 + 0.24917 \times 5.6540)/1.001 = (1.4559 + 1.4088)/1.001 = 2.862$. This is below the intrinsic value 2.912, so the grid exercises and the node takes the value 2.912.
- At 50: $0.24917 \times 2.9118/1.001 = 0.725$, above the intrinsic value 0, so the node keeps 0.725.

The exact European value at 47.0882 is 2.939, which is *above* the intrinsic value 2.912; the American put is worth at least as much, so exercising there is premature. The grid exercises only because one coarse step beside the kink of the payoff undervalues the option (2.862 against 2.939): a discretisation artefact, which a finer grid removes. A full grid carries the value 0 on its top boundary row and, on its bottom row, at the lowest node $S_{\min}$, the intrinsic value $50 - S_{\min}$ of the American put (Setup (c) of §23.7); the two rows are placed about four to five standard deviations $\sigma\sqrt{T}$ of the log price from the spot.

(e) The limit becomes $\Delta t \le 0.03^2/0.09 = 0.01$, a quarter of 0.04. Over the same range of log prices, halving Δx doubles the number of nodes, and cutting Δt to the new limit quadruples the number of time steps: eight times the work.

Self-check. The weights can be recovered without the scheme, from the two moments they must match: $p_u + p_d = \sigma^2\Delta t/\Delta x^2 = 0.5$ and $p_u - p_d = \nu\,\Delta t/\Delta x = 0.001667$ give $p_u = 0.2508$ and $p_d = 0.2492$, and $p_m = 1 - 0.5$. The explicit factor of (b) also follows from $\Lambda(\pi)$ of (c): in vector form the explicit step is $(1 + r\,\Delta t)V^i = (I + \Delta t\,A_0)V^{i+1}$ with $A_0 = A + rI$, whose multiplier on the sawtooth is $\Lambda_0(\pi) = \Lambda(\pi) + r = -50$, and $(1 - 0.05 \times 50)/1.0025 = -1.496$. At $\Delta t = 0.02$, where $\sigma^2\Delta t/\Delta x^2 = \frac{1}{2}$, the same formula gives $(1 - 0.02 \times 50)/1.001 = 0$: at this step the sawtooth is removed in one step, and anywhere within the limit its factor has modulus at most $1/(1 + r\,\Delta t)$. For (d), the exact European values at the nodes 47.0882 and 50 are 2.939 and 0.821, against 2.862 and 0.725 on the grid: one coarse step is low by 0.08 and 0.10 at these two nodes beside the kink, and a binomial tree with 4,000 steps puts the American value at 47.0882 at 2.958, above the intrinsic value. On the refined grid of (e), four explicit steps of 0.005 give 2.9445 at 47.0882, above the intrinsic value 2.9118: the refined grid no longer exercises there.

Answer: (a) $p_u = 0.2508$, $p_m = 0.5$, $p_d = 0.2492$; sum 1, mean $\nu\,\Delta t = 0.0001$, second moment $\sigma^2\Delta t = 0.0018$. (b) $\Delta t \le \Delta x^2/\sigma^2 = 0.04$; at $\Delta t = 0.05$ the weights are 0.6271, -0.25, 0.6229 and the sawtooth is multiplied by -1.5 (-1.496 with the discount), so 10^{-6} grows to about 10 in 40 steps. (c) $\Lambda(\pi) = -50.05$; implicit factor 0.286, Crank–Nicolson factor -0.112. (d) At 47.0882 the step gives $2.862 < 2.912$, so the grid exercises (value 2.912), although the European value 2.939 shows that exercise is premature there: an artefact of the coarse grid; at 50 the value is 0.725; the boundary rows are 0 at the top and the intrinsic value at the bottom. (e) $\Delta t \le 0.01$; eight times the work.

P23.11 *Which model makes which smile, and Merton's formula*

(a) The log price at $T = 1$ is $\ln S$ plus two independent normal variables, with variances $0.18^2 \times 0.25 = 0.0081$ and $0.26^2 \times 0.75 = 0.0507$. A sum of independent normal variables is normal, and the means and the variances add, so $\ln S_T$ is normal with variance 0.0588 and mean $\ln S - \frac{1}{2} \times 0.0588$: the price S_T is lognormal with the mean S, the forward price. That is the law of the BSM model with the volatility

$$\sigma_{\text{avg}} = \sqrt{0.0588} = 24.25\%,$$

the root of the time-averaged *variance* (Proposition 23.58 with a constant σ_{avg}). Variances add over time and volatilities do not; the time-weighted average volatility, $0.25 \times 18\% + 0.75 \times 26\% = 24\%$, is smaller, because the root of a mean of squares is at least the mean. At the money, the call is worth $100\,(2 \times 0.54825 - 1) = 9.650$ at 24.25%, against $100\,(2 \times 0.54776 - 1) = 9.552$ at 24%.

(b) (1)–(B): a deterministic σ_t keeps the price lognormal at each maturity, with the root-mean-square volatility of (a), so the smile is flat at a level that depends on the maturity. (2)–(D): mixing BSM prices over an independent volatility fattens both tails alike. (3)–(C): a jump matters most when little diffusion has accumulated, and its effect averages out over long horizons. (4)–(E): a volatility that rises when the price falls skews the law at every horizon. (5)–(A): local volatility is calibrated to reproduce every vanilla price today, and what that fit costs is a rigid evolution of the smile, with forward smiles that flatten. The pairings (2), (3) and (4) are the observations of Remark 23.64; (1) is the deterministic-volatility case discussed after Proposition 23.58, and (5) is the trap on fitting every vanilla price that follows Proposition 23.56.

(c) By Proposition 23.58 the call is the average of the two BSM prices:

$$\tfrac{1}{2} \times 100\,(2 \times 0.51994 - 1) + \tfrac{1}{2} \times 100\,(2 \times 0.55962 - 1) = \tfrac{1}{2}\,(3.988 + 11.924) = 7.956.$$

Its implied volatility solves $2\,\Phi(\frac{1}{2}\sigma_{\text{imp}}) - 1 = 0.07956$, that is $\Phi(\frac{1}{2}\sigma_{\text{imp}}) = 0.53978$, so $\frac{1}{2}\sigma_{\text{imp}} = 0.09988$ and $\sigma_{\text{imp}} = 19.98\%$. The root-mean-square volatility is $\sqrt{(0.01 + 0.09)/2} = 22.36\%$: at the money the price is a concave function of the variance, so averaging over the variance lowers it. At the strike 125 the implied volatility, 22.91%, is higher than at the money. For the strike $80 = 100^2/125$, each of the two scenario prices satisfies the symmetry, hence so does their average: the mixture put struck at 80 is worth the mixture call struck at 125 divided by 1.25. The BSM

prices at the single volatility 22.91% satisfy the same relation, so the BSM put struck at 80 at that volatility is worth the mixture put: the implied volatility of the put at 80 is 22.91%. Put–call parity holds for the mixture prices, being an average of two parities, so the call struck at 80 has the same implied volatility (Proposition 22.3). The smile reads 22.91%, 19.98%, 22.91% at the strikes 80, 100, 125: symmetric in $\ln(K/S)$, and lowest at the money of the three strikes, the minimum near the money of Remark 23.64.

(d) *Derivation.* We condition on the number of jumps. On the event $\{N_T = j\}$ the exponent is

$$X_j = (r - q - \lambda\kappa - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z_0 + \sum_{i=1}^{j} \big(\ln(1+\kappa) - \tfrac{1}{2}\delta^2 + \delta Z_i\big),$$

a constant plus a sum of independent normal variables. It is normal with the variance $\sigma^2 T + j\,\delta^2 = \sigma_{(j)}^2 T$ and the mean

$$(r - q - \lambda\kappa - \tfrac{1}{2}\sigma^2)T + j\ln(1+\kappa) - \tfrac{1}{2}j\,\delta^2 = (r_{(j)} - q - \tfrac{1}{2}\sigma_{(j)}^2)T,$$

so Se^{X_j} has the lognormal law of the BSM model with the rate $r_{(j)}$ and the volatility $\sigma_{(j)}$. The count N_T is independent of $(Z_0, \dots, Z_j)$, so

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}\big[(S_T - K)^+\,\mathbf{1}\{N_T = j\}\big] &= P(N_T = j)\,\mathbb{E}\big[(Se^{X_j} - K)^+\big] \\ &= e^{-\lambda T}\,\frac{(\lambda T)^j}{j!}\,e^{r_{(j)}T}\,C^{\mathrm{BSM}}\big(S, K; \sigma_{(j)}, r_{(j)}\big),\end{aligned}$$

because the BSM call at the rate $r_{(j)}$ is $e^{-r_{(j)}T}$ times that mean. Discounting at r and using $e^{r_{(j)}T} = e^{rT}e^{-\lambda\kappa T}(1+\kappa)^j$,

$$e^{-rT}\cdot e^{-\lambda T}\,\frac{(\lambda T)^j}{j!}\cdot e^{r_{(j)}T} = e^{-\lambda(1+\kappa)T}\,\frac{\big(\lambda T(1+\kappa)\big)^j}{j!} = e^{-\lambda' T}\,\frac{(\lambda' T)^j}{j!},$$

so $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+\,\mathbf{1}\{N_T = j\}]$ is the term of index j of the series. The weights are Poisson probabilities at the intensity λ', not λ, because each scenario formula discounts at its own rate $r_{(j)}$.

Summing over j. It remains to exchange the infinite sum with the expectation, which is done through the density of S_T. Let $f^{(j)}$ be the lognormal density of Se^{X_j} (the superscript is a label). For an open interval $\mathcal{O}$ of $(0,\infty)$ the events $\{N_T = j,\ Se^{X_j} \in \mathcal{O}\}$, $j = 0, 1, 2, \dots$, are disjoint with union $\{S_T \in \mathcal{O}\}$, so countable additivity and the product rule for the independent N_T and $(Z_0, \dots, Z_j)$ give

$$P(S_T \in \mathcal{O}) = \sum_{j=0}^{\infty} P(N_T = j)\int_{\mathcal{O}} f^{(j)}(x)\,dx = \int_{\mathcal{O}} \sum_{j=0}^{\infty} P(N_T = j)\,f^{(j)}(x)\,dx,$$

the exchange being Fact 18.34(i) for a nonnegative integrand, with a sum in the discrete coordinate j. So S_T has the density $\sum_j P(N_T = j)\,f^{(j)}(x)$. The mean of Se^{X_j} is $Se^{(r_{(j)} - q)T}$ (Lemma 18.35(i)), and $P(N_T = j)\,e^{(r_{(j)} - r)T} = e^{-\lambda' T}(\lambda' T)^j/j!$ by the display above, so

$$\sum_{j=0}^{\infty} P(N_T = j)\,\mathbb{E}\big[Se^{X_j}\big] = Se^{(r-q)T}\sum_{j=0}^{\infty} e^{-\lambda' T}\,\frac{(\lambda' T)^j}{j!} = Se^{(r-q)T}.$$

The payoff $(x - K)^+$ is nonnegative and at most x, so each integral $\int_0^\infty (x-K)^+ f^{(j)}(x)\,dx = \mathbb{E}(Se^{X_j} - K)^+$ is finite (Fact 7.22), and Fact 18.34(i) gives

$$\int_0^\infty (x-K)^+ \sum_{j=0}^{\infty} P(N_T = j)\,f^{(j)}(x)\,dx = \sum_{j=0}^{\infty} P(N_T = j)\,\mathbb{E}\big[(Se^{X_j} - K)^+\big] \le Se^{(r-q)T}.$$

The integral converges, so Fact 7.22 makes its left side $\mathbb{E}^{\mathbb{Q}}(S_T - K)^+$. Discounting term by term with the weights found above gives the series (Proposition 23.60).

Numbers. Here $\lambda' = 0.95$ and $\lambda' T = 0.475$. The weights are $e^{-0.475} = 0.6219$, then $0.6219 \times 0.475 = 0.2954$, $0.2954 \times 0.475/2 = 0.0702$ and $0.0702 \times 0.475/3 = 0.0111$ (and 0.00132, 0.00013 for $j = 4, 5$). With one jump, $\sigma_{(1)}^2 = 0.04 + 0.01/0.5 = 0.06$, so $\sigma_{(1)} = 24.49\%$, and $r_{(1)} = 0.05 + (-0.05129)/0.5 = -5.26\%$; with no jump the rate is $r_{(0)} = +5\%$, the drift that pays for the jumps that did not come. The series is

$$\begin{aligned} &0.62189 \times 6.8887 + 0.29540 \times 5.7413 + 0.07016 \times 4.8665 \\ &\quad + 0.01111 \times 4.1656 + 0.00132 \times 3.5889 + 0.00013 \times 3.1066, \end{aligned}$$

that is $4.2840 + 1.6960 + 0.3414 + 0.0463 + 0.0047 + 0.0004 = 6.373$; the terms beyond $j = 5$ add less than 0.0001. By the at-the-money formula this is an implied volatility of 22.6%, between the diffusion volatility 20% and the volatility 23.1% of the whole log return.

The put. Applied to the payoff S_T itself, the same density and Fact 18.34(i) give $\mathbb{E}^{\mathbb{Q}} S_T = \sum_j P(N_T = j)\, \mathbb{E}[Se^{X_j}] = Se^{(r-q)T} = 100$, by the sum computed above: the term $-\lambda\kappa T$ in the exponent compensates the mean $e^{\lambda\kappa T}$ of the product of the jump factors. Since $(x - K)^+ - (K - x)^+ = x - K$, put–call parity holds, $C - \mathcal{P} = S - K = 10$, and the put struck at 90 is worth $12.428 - 10 = 2.428$.

(e) Vanilla prices fix the risk-neutral law of the price at each expiry, one date at a time; a forward-start option depends on the law of the ratio S_T/S_{t_1} given S_{t_1}, that is, on the smile that will prevail at t_1. A local-volatility model supplies one particular answer: its volatility is a function of the price level and the date, so the law of the ratio depends on the level (the homogeneity behind Proposition 23.8 fails) and its forward smiles flatten. A stochastic-volatility or jump model fitted to the same vanilla prices keeps a smile in the future and gives a different price. The fit to vanillas therefore does not validate the forward-start price.

Self-check. For (d), a simulation of 8,000,000 draws of the law of S_T gives 6.374 for the at-the-money call, with standard error 0.004, 12.429 for the call struck at 90, with standard error 0.005, and a mean terminal price of 99.998, with standard error 0.006; and the series written for the put struck at 90 sums to 2.4283, as parity requires. For (c), Jensen's inequality can be seen on numbers: the BSM call at the root-mean-square volatility 22.36% is $100\,(2\,\Phi(0.1118) - 1) = 8.90$, above the mixture price 7.956; and the mixture put struck at 80, computed directly, is worth 1.787, which is the mixture call struck at 125, 2.234, divided by 1.25. For (a), the at-the-money approximation $C \approx S\sigma\sqrt{T}/\sqrt{2\pi}$ gives 9.674 at 24.25% and 9.575 at 24%, a gap of 0.099 against the exact $9.650 - 9.552 = 0.098$; and the second moment of the price, $\mathbb{E}^{\mathbb{Q}} S_T^2 = S^2 e^{0.0081} e^{0.0507} = 1.06056\, S^2$, is that of a lognormal price at 24.25% ($e^{0.0588}$), not at 24% ($e^{0.0576} = 1.05929$).

Answer: (a) The root-mean-square volatility $\sqrt{0.0588} = 24.25\%$, not the average 24% (at-the-money call 9.650 against 9.552). (b) (1)–(B), (2)–(D), (3)–(C), (4)–(E), (5)–(A). (c) Mixture call 7.956, implied volatility 19.98%, below the root-mean-square 22.36%; implied volatility 22.91% at the strike 80 by symmetry. (d) $\lambda' T = 0.475$; weights 0.622, 0.295, 0.070, 0.011; $\sigma_{(1)} = 24.49\%$, $r_{(1)} = -5.26\%$; at-the-money call 6.373 (implied volatility 22.6%); put at 90: $12.428 - 10 = 2.428$, because $\mathbb{E}^{\mathbb{Q}} S_T = Se^{(r-q)T}$. (e) Vanillas fix only the law of the price at each expiry; the forward-start depends on the law of S_T/S_{t_1}, which local volatility ties to the price level, flattening the forward smile.

28.24 Solutions — §24 Algorithmic Cost Models

P24.1 *Matmul operation count* ★

Each of the mp entries of the product is a dot product of length n: n multiplications and $n - 1$ additions. Totals: mnp multiplications and $mp(n - 1)$ additions — "mnp multiply-adds" is

the interview-grade summary. Square case: n^3 multiplications and $n^3 - n^2$ additions, so $O(n^3)$. Convention: quote the *multiplication* count — multiplications historically cost more CPU cycles than additions, and the two counts agree to leading order anyway.

Below cubic: Strassen's algorithm recursively multiplies 2×2 blocks with 7 multiplications instead of 8, giving $O(n^{\log_2 7}) \approx O(n^{2.81})$ — practical only for large n (bigger constants, more memory traffic, weaker numerical stability). The theoretical frontier is $\approx O(n^{2.37})$, but those algorithms are "galactic" — constants so large they never win at feasible sizes. Practical BLAS remains cubic: the real-world wins are cache blocking, SIMD, and tensor cores, not asymptotics.

Self-check: smallest case $m = n = p = 2$: formula says $2 \cdot 2 \cdot 2 = 8$ multiplications and $mp(n-1) = 4$ additions; writing out a 2×2 product confirms four length-2 dot products = 8 mults, 4 adds — and Strassen's whole point is doing that block product in 7, seeding the recursion.

Answer: $(m \times n)(n \times p)$ costs mnp multiplications (plus $mp(n-1)$ additions); square is $O(n^3)$; multiplications are the conventional count (costlier cycles); Strassen $O(n^{2.81})$ is real but large-n only, $O(n^{2.37})$ is galactic, and practical speed comes from cache/SIMD/tensor cores.

P24.2 *Why count multiplications* ♦

Because multiplications generally cost more CPU cycles than additions — historically by a wide margin — so the multiply count dominated the running-time estimate and the convention stuck. Two supporting observations: (i) in an $(m \times n)(n \times p)$ product the two counts are nearly equal anyway (mnp vs $mp(n-1)$, identical to leading order), so tracking one number loses nothing asymptotically; (ii) on modern hardware a fused multiply-add executes both in one instruction, which is why "FLOPs" today is often quoted as $2mnp$ — but the reason the *multiplication* became the unit is its historically higher cycle cost.

Self-check: the difference between the two counts is $mnp - mp(n-1) = mp$, a lower-order term — confirming that choosing which to count cannot change the complexity class, only the convention.

Answer: Multiplications cost more CPU cycles than additions, so their count dominates the time estimate (and the two counts agree to leading order anyway).

P24.3 *Low-rank decomposition crossover* ★

(a) NPR multiplications — each of the NR output entries is a length-P dot product.

(b) Association order is the hidden half of the question.

- $(UV)B$: forming UV costs NKP (an $N \times K$ by $K \times P$ product), producing the dense $N \times P$ matrix, then NPR for the product with B. Total $NKP + NPR$ — *strictly worse than the direct NPR for any $K \geq 1$*: re-multiplying the factors together first recreates the dense cost and adds the assembly bill. Never do this.
- $U(VB)$: VB costs KPR (result is $K \times R$ — small), then $U(VB)$ costs NKR. Total $KPR + NKR = KR(N+P)$.

Factored beats dense iff

$$KR(N+P) < NPR \iff K(N+P) < NP \iff \boxed{K < \frac{NP}{N+P}}.$$

Note R cancels: the threshold does not depend on how much data you push through. $NP/(N+P)$ is half the harmonic mean of N and P; for square $N = P$ it reads $K < N/2$. This is the LoRA economics — rank-K adapters with $K \ll \min(N, P)$ make the factored path far cheaper, provided you always multiply the data into V first.

Self-check: boundary cases. $N = P = 2$, $K = 1$, $R = 1$: dense = 4; factored $= 1 \cdot 1 \cdot (2+2) = 4$ — exact break-even, and indeed the threshold $NP/(N+P) = 1$ sits right at $K = 1$. Full rank $K = P$ (square): $KR(N+P) = PR(N+P) > NPR$ always — a factorisation with no rank reduction can never win. Units check: all three costs scale linearly in R, so the comparison is per-column of data, as it should be.

Answer: (a) NPR multiplications; (b) $(UV)B$ costs $NKP + NPR$ (always worse — never reassemble the dense matrix), $U(VB)$ costs $KR(N+P)$; the factored form wins iff $K < NP/(N+P)$ (half the harmonic mean; $K < N/2$ when square). □

P24.4 *Low-rank speedup, numeric*

(a) Dense matrix–vector product: $NP = 4096 \times 4096 = 16{,}777{,}216$ multiplications. Factored, with the correct association $U(Vx)$: $K(N + P) = 64 \times (4096 + 4096) = 64 \times 8192 = 524{,}288$. Ratio: $16{,}777{,}216/524{,}288 = 32$ — the claim is exactly right.

(b) Break-even at $K = \frac{NP}{N+P} = \frac{4096^2}{8192} = 2048$ (half of 4096, as always for square layers); any $K < 2048$ is a win in raw multiplications.

(c) Approximation error: the rank-64 layer equals the dense layer only if the dense weights truly had rank ≤ 64. In general it is a lossy compression, and the accuracy given up is the real price — the flop count is silent about it. (Second-order practicalities: two kernel launches and different memory-access patterns can eat part of the theoretical 32×.)

Self-check: compute the speedup a second way: ratio $= \frac{NP}{K(N+P)} = \frac{NP/(N+P)}{K} = \frac{2048}{64} = 32$ — consistent with (b): the speedup is (break-even rank)/(actual rank).

Answer: (a) verified — exactly 32× fewer multiplications (16.78M vs 524K); (b) break-even rank 2048; (c) the low-rank approximation error (fidelity), which flop counting ignores.

P24.5 *Membership: scan vs sort vs hash* ★

Costs:

- **Scan:** $O(N)$ per query, $O(qN)$ total. No preprocessing, no extra memory.
- **Sort + binary search:** $O(N \log N)$ once, then $O(\log N)$ per query: $O((N + q) \log N)$ total.
- **Hash set:** $O(N)$ expected build, $O(1)$ expected per query: $O(N + q)$ *expected* total.

Crossover (scan vs sort): sorting pays off when

$$qN > N \log N + q \log N \iff q(N - \log N) > N \log N \iff q > \frac{N \log N}{N - \log N} \approx \log N \quad (N \text{ large}).$$

So after roughly $\log N$ queries, pre-sorting wins — a strikingly small number.

When each wins. *Scan:* one or two queries, tiny N, or data you stream past exactly once — zero setup and perfect cache behaviour make it hard to beat at small scale. *Sort:* many queries when you also want *order* — range queries, predecessor/nearest lookups, sorted output — or deterministic worst-case guarantees. *Hash:* many pure membership queries; asymptotically the best at $O(N + q)$ expected, and the phrase to say is "expected $O(1)$ lookups, amortised build" — paying for it with extra memory, loss of ordering information, and worst-case degradation (collisions/adversarial keys).

Self-check: plug $N = 1024$: crossover $q > \frac{1024 \times 10}{1024 - 10} \approx 10.1$, and $\log_2 N = 10$ — the approximation is tight. Endpoints: $q = 1$: scan N beats sort $N \log N$; $q \to \infty$ with N fixed: hash $O(q)$ beats binary search $O(q \log N)$ beats scan $O(qN)$. House pattern: write both cost expressions, set them equal, solve for the variable asked about.

Answer: scan $O(qN)$; sort-then-search $O((N+q) \log N)$, paying off once $q \gtrsim \log N$; hash set $O(N+q)$ expected — best for many queries, at the price of memory, no ordering, and worst-case degradation.

P24.6 *CS fundamentals rapid-fire* ♦

(a) *Memoization* — caching a pure function's outputs keyed by input. "Dynamic programming" is the bait: memoization is an ingredient of DP (DP = recursion over overlapping subproblems +

memoization/tabulation), but caching by itself, with no subproblem structure, is just memoization. *Self-check:* nothing in the setup mentions subproblems or recurrences — only repeat calls.

(b) It roughly *doubles*: one comparison per record is $O(N)$ per query, so $2N$ records cost $2\times$ the time. *Self-check:* linear cost model — constant work per record, no early exit assumed.

(c) Trace it: after Push(10), Push(20) the stack is $[10, 20]$; val = Pop() = 20, stack [10]; Push(40), Push(30) gives $[10, 40, 30]$; Push(val) pushes 20 back: $[10, 40, 30, 20]$; Pop() returns 20; Pop() returns 30. Last popped: **30**. *Self-check:* conservation count — 5 pushes (10, 20, 40, 30, val) and 3 pops in total, so $5 - 3 = 2$ elements must remain, and indeed the final stack is $[10, 40]$.

(d) $O(N)$ space: each active call keeps a frame (locals, return address) on the call stack, and without tail-call optimisation frames cannot be reused even for a tail-position call. (CPython never performs TCO — deep recursion hits the recursion limit around depth 10^3 by default.) *Self-check:* at maximum depth exactly N frames are simultaneously live — linear, not constant.

(e) Left bracket, by distributivity of $\vee$ over $\wedge$: $(A \vee B) \wedge (A \vee \neg B) = A \vee (B \wedge \neg B) = A \vee \text{false} = A$. Right bracket: $(C \wedge D) \wedge (C \wedge \neg D) = C \wedge (D \wedge \neg D) = C \wedge \text{false} = \text{false}$. Whole expression: $A \vee \text{false} = A$. *Self-check:* truth-table spots — A false, B true: left = (T) $\wedge$ (F) = F = A; A false, B false: (F) $\wedge$ (T) = F; A true: T $\wedge$ T = T; the right bracket needs D and $\neg D$ simultaneously — always false.

Answer: (a) memoization; (b) it roughly doubles ($O(N)$); (c) 30; (d) $O(N)$ stack space; (e) A.

28.25 Solutions — §25 Forecasting Returns: the Open-Ended Question

P25.1 *Open-ended process question*

Answer with a *process*, not a model name. A model answer hits five stages, in order:

- **Define the target.** Log vs simple returns; the horizon (next-day here) and why it matters — signal decay, tradability, and noise level all change with horizon. Also pin down what the forecast is scored on: point forecast, direction, or distribution.
- **Explore, then build features in layers.** EDA first: stationarity, outliers, regime breaks, the correlation matrix, PCA for factor structure. Feature layers: endogenous (lagged returns, realised vol, volume, spreads); exogenous lead-lag (the US close leads the European open; overnight futures, rates, FX, commodities); composition-specific (sector weights make particular macro series relevant — e.g. a heavy energy weight makes Brent a feature); sentiment/news if the horizon supports it.
- **Model: baseline first, then a fork.** Start with a trivial baseline (zero / historical mean) and a Ridge regression on standardised features; escalate to LightGBM or sequence models only if the linear information coefficient justifies the added variance. Name the direct-vs-factorial fork explicitly: model the index series directly, or model constituents and aggregate (often cleaner — see P25.3).
- **Validate honestly.** Walk-forward / time-series splits, never shuffled; metrics that respect the use case: IC, hit rate, and the Sharpe of the implied strategy *after costs* — raw MSE is direction-blind.
- **Name the risks unprompted.** Look-ahead bias, overfitting ("a too-beautiful Sharpe is a bug"), transaction costs, non-stationarity/regime change. Volunteering these is what separates a process answer from a model-name answer.

Self-check (rubric): re-read your narrated answer and test three gates — (i) was the target defined before any model was named? (ii) is there a baseline and an out-of-sample protocol? (iii) did costs and look-ahead appear without prompting? Independently, sanity-check any performance you would claim: daily direction on a broad index sits near 50–55% for good pipelines, so a backtest printing far above that should be treated as a leakage bug, not a triumph.

Answer: A five-stage process — define the target; layered EDA/features (endogenous, lead-lag, composition, sentiment); baseline-then-model with the direct-vs-factorial fork stated; walk-forward validation on IC/hit-rate/net Sharpe; and unprompted naming of look-ahead, overfitting, costs, and non-stationarity.

P25.2 *Backtest critique* ♦

Two answers deserve the label "main issue", and a complete response gives both. Read literally, "the main issue in *executing*" points at extbftransaction costs; the deeper *validity* critique is that the backtest's profit is *beta, not timing alpha.*

extbfValidity: the profit is beta. The strategy is long ETH ten hours out of every twenty-four, over a seven-year sample in which ETH rose by an order of magnitude. Any always-long-by-day rule mechanically inherits a large share of that drift; the specific 9am–7pm window need carry no information at all. Diagnosis tests: (i) benchmark against buy-and-hold scaled to the same average exposure; (ii) randomise the entry/exit clock times — if random ten-hour windows backtest about as well, the timing content is zero; (iii) sub-period analysis: does it survive the drawdown years, or is the P&L concentrated in bull regimes?

extbfExecution: the cost drag is enormous. About 365 round trips per year is roughly 730 fee-paying sides; at 5–10 bp per side in taker fee plus spread plus slippage, that is on the order of 50% of notional per year in friction — enough to destroy a modest edge on its own. Also worth a sentence: a fixed clock-time execution is crowd-visible and easy to front-run.

Self-check (quantitative): ETH went from a few hundred dollars in 2019 to thousands in 2026, i.e. a 10×-order move; a rule long roughly 10/24 of each day rides a substantial fraction of that log drift with no skill, so "backtests very well" is exactly what pure beta predicts. Fee drag arithmetic: $730 \times 7.5\,\text{bp} \approx 5{,}500\,\text{bp} \approx 55\%$ per year — consistent with the cost concern being material, not cosmetic. Both checks point the same way: the burden of proof is on the timing, and the backtest as described has not met it.

Answer: The performance is long-ETH beta harvested over a seven-year bull sample, not timing alpha; the secondary issue is fee/slippage drag from ∼365 round trips a year. Test against buy-and-hold and randomised windows before believing the timing.

P25.3 *Open-ended design choice: direct vs factorial*

Case for the factorial route.

- *Cross-section multiplies data:* 40 related series instead of 1 gives far more observations of the shared structure per unit of calendar time.
- *Decomposition isolates the forecastable part:* write $R_i = \beta_i R_{\text{mkt}} + \alpha_i$. The market component is the hardest thing in finance to predict; the beta-neutralised residuals carry cleaner, slower structure (lead-lag between related names, residual mean-reversion) and are where most systematic edge lives.
- *Information enters at the right level:* Brent belongs in the TotalEnergies model, China sentiment in the luxury names — at stock level these are sharp features; averaged into an index feature they dilute.

Where PCA enters. Eigendecompose the constituent correlation matrix: PC1 ≈ the market mode, the next few components ≈ sector modes. Uses: (a) build the market factor for residualisation; (b) compress 40 correlated series into a few orthogonal drivers to forecast; (c) check the factor structure is *stable* across subperiods before trusting anything built on it.

Failure modes of the aggregation.

- *Cap-weighted aggregation cancels relative alpha.* The index is dominated by PC1; forecast skill on the α_i largely nets out in the weighted sum. You can build an excellent long-short stock model and a nearly useless index model from the same forecasts.

- *Correlated errors do not diversify.* The variance of the weighted error sum carries the full covariance matrix; if the common-factor forecast is wrong, all 40 stock errors point the same way and aggregation buys you nothing.
- *Plumbing risk:* weight and composition drift, corporate actions, and asynchronous closes create silent look-ahead in the aggregation step.

Self-check (toy case): two stocks, equal weights, $R_i = R_m + \alpha_i$ with $\alpha_2 = -\alpha_1$. The index return is exactly R_m: a *perfect* alpha model contributes zero index forecasting power — confirming the cancellation failure concretely. Conversely it shows where the factorial route genuinely helps an index mandate: a better-estimated market mode and risk model, not the relative alphas.

Answer: The constituent route wins on data volume and on forecasting beta-neutralised structure, with PCA supplying the market/sector factors; but cap-weighted aggregation cancels relative alpha and correlated forecast errors do not diversify — for a pure index target the market-factor forecast still does the heavy lifting.

28.26 Solutions — §26 Computer Science for Research Engineers: Python Data Structures, Parallelism, and Time

P26.1 *Pricing Python's containers, qualifiers included*

(a) `L.append(x)`: $O(1)$ *amortised*; the pointer array grows geometrically, so the occasional $O(n)$ copy is paid for by the cheap appends around it. `D[k]`: $O(1)$ *expected*, $O(n)$ worst case; the average is over the randomised hash, and colliding keys degrade a probe chain to a scan. `dq.popleft()`: $O(1)$ *worst case*; a deque is a linked list of fixed blocks, so nothing shifts. `L.insert(0, x)`: $\Theta(n)$ *worst case*; contiguity forces every pointer behind index 0 to move. `heapify`: $O(n)$ *worst case*, at most $2n$ swaps, because the sift-downs from height h number at most $n/2^h$ and $\sum_h h/2^h = 2$. `x in S`: $O(1)$ *expected*, $O(n)$ worst case, for the same reason as the dict.

(b) Removing the front of a list of current length j moves $j-1$ pointers, so draining from n costs $(n-1)+(n-2)+\cdots+0 = n(n-1)/2$ moves: $200{,}000 \times 199{,}999/2 = 19{,}999{,}900{,}000 \approx 2.0 \times 10^{10}$. A deque moves nothing: n block-local pops, $O(1)$ each. On the star, the list holds $V-1 = 99{,}999$ leaves after the centre is expanded, and removing them costs $(V-1)(V-2)/2 = 99{,}999 \times 99{,}998/2 = 4{,}999{,}850{,}001 \approx 5.0 \times 10^9$ pointer moves, against $V + E = 199{,}999 \approx 2 \times 10^5$ operations for the deque: a factor of 2.5×10^4, and $\Theta(V^2)$ against $\Theta(V+E)$ in general, because the queue is $\Theta(V)$ long when the pops happen.

(c) Strings are immutable, so the i-th concatenation copies the $(i-1)L$ characters already accumulated plus the new piece, iL characters in all; summing, $L(1+2+\cdots+n) = L\,n(n+1)/2 = 20 \times 50{,}000 \times 50{,}001/2 = 25{,}000{,}500{,}000 \approx 2.5 \times 10^{10}$ characters. `join` sums the lengths, allocates once and copies each of the $nL = 10^6$ characters once: a ratio of 25,000. For a local accumulator CPython's in-place specialisation may extend the string without copying when it has a single owner, which hides the quadratic cost in a micro-benchmark; for a global, an attribute or a container item the cost is fully visible, and the optimisation is an implementation choice rather than a guarantee, so the answer to quote is the quadratic one.

Self-check. The drain count is also $\binom{n}{2}$ pairs of (item, later pop): $\binom{200{,}000}{2} = 1.99999 \times 10^{10}$, matching. For the concatenation, the average accumulated length is about $nL/2 = 5 \times 10^5$ characters, copied $n = 5 \times 10^4$ times: 2.5×10^{10}, matching the exact sum to four figures.

Answer: (a) append amortised $O(1)$; dict get expected $O(1)$ (worst $O(n)$); deque popleft worst-case $O(1)$; insert(0) worst-case $\Theta(n)$; heapify worst-case $O(n)$; set membership expected $O(1)$ (worst $O(n)$). (b) $n(n-1)/2 \approx 2.0 \times 10^{10}$ moves against 0; on the star $\approx 5.0 \times 10^9$ moves against $\approx 2 \times 10^5$ operations. (c) 2.5×10^{10} characters against 10^6, a ratio of 25,000; a local accumulator may dodge it by an implementation accident, not by guarantee.

P26.2 *Dynamic arrays from scratch: geometric against additive growth*

(a) We use the aggregate method: bound the total cost of the whole sequence, then divide by n. Let $s_0 < s_1 < \cdots < s_M$ be the sizes at which resizes occur; a resize at size s_j copies s_j pointers. After the resize at s_j the capacity is at least gs_j, so the next resize happens at a size $s_{j+1} \geq gs_j$; going backwards from the last, $s_j \leq s_M/g^{M-j}$. The last resize is triggered by an append into a full array of size s_M, so $s_M \leq n-1 < n$. Hence

$$\sum_{j=0}^{M} s_j \;\leq\; s_M \sum_{i=0}^{M} g^{-i} \;<\; s_M \cdot \frac{1}{1-1/g} \;=\; s_M \,\frac{g}{g-1} \;<\; \frac{g\,n}{g-1},$$

summing the geometric series. Adding the n direct writes and dividing by n, an append costs less than $1 + g/(g-1)$ amortised, a constant independent of n.

(b) With $c_0 = c$ and additive growth the array is full exactly at the sizes $c, 2c, \ldots, Mc$ with $M = \lfloor (n-1)/c \rfloor$, and the j-th resize copies jc pointers. By the arithmetic series the copies total $c\,M(M+1)/2$. Since $(n-1)/c - 1 < M \leq (n-1)/c$, this lies between $(n-1)(n-1-c)/(2c)$ and $(n-1)(n-1+c)/(2c)$, and both bounds equal $n^2/(2c) + O(n)$: quadratic in n.

(c) With $c_0 = 4$ and doubling the array is full at the sizes $4 \cdot 2^j$; a resize occurs at every such size at most $n-1 = 2{,}999{,}999$. Since $4 \cdot 2^{19} = 2{,}097{,}152 \leq 2{,}999{,}999 < 4 \cdot 2^{20} = 4{,}194{,}304$, resizes occur at $4, 8, \ldots, 2^{21}$: 20 of them, copying $4(1+2+\cdots+2^{19}) = 4(2^{20}-1) = 4{,}194{,}300$ pointers. Per append that is $4{,}194{,}300/3 \times 10^6 = 1.398$ copies plus one write, 2.40 writes, inside the bound $1 + 2/1 = 3$ of (a); the copy total 4.19×10^6 is inside $gn/(g-1) = 2n = 6 \times 10^6$. With $g = 9/8$ the bound is $1 + (9/8)/(1/8) = 10$ writes per append: the same order, a larger constant, bought with at most one eighth of slack memory after a resize instead of up to one half. Additive growth by $c = 1000$: $M = \lfloor 2{,}999{,}999/1000 \rfloor = 2999$ resizes copying $1000 \times 2999 \times 3000/2 = 4{,}498{,}500{,}000 \approx 4.5 \times 10^9$ pointers, which is $n^2/(2c) = 9 \times 10^{12}/2000 = 4.5 \times 10^9$ to three figures and about 1070 times the geometric total.

Self-check. The potential method gives the $g = 2$ case independently: with $\Psi = 2\,\mathrm{size} - \mathrm{capacity}$, a cheap append costs 1 and raises Ψ by 2, and a resizing append at size s costs $s+1$ and changes Ψ from s to 2, so both have amortised cost 3; the total for n appends is at most $3n = 9 \times 10^6$ writes, and the actual $3 \times 10^6 + 4{,}194{,}300 = 7{,}194{,}300$ lies below it. The geometric sum also checks directly: $4 + 8 + \cdots + 2^{21} = 2^{22} - 4 = 4{,}194{,}300$.

Answer: (a) copies $< gn/(g-1)$, hence at most $1 + g/(g-1)$ writes per append amortised; (b) $c\,M(M+1)/2$ with $M = \lfloor (n-1)/c \rfloor$, which is $n^2/(2c) + O(n)$; (c) resizes at $4 \cdot 2^j$ for $j = 0, \ldots, 19$ (20 resizes), 4,194,300 copies, 2.40 writes per append against the bounds 3 ($g = 2$) and 10 ($g = 9/8$); additive growth by 1000 copies $4{,}498{,}500{,}000 \approx 4.5 \times 10^9$ pointers.

P26.3 *Resizes, heaps and binary search by the numbers*

(a) With index size s the table holds up to $\lfloor 2s/3 \rfloor$ keys, so the resize thresholds are $\lfloor 2 \cdot 8/3 \rfloor = 5$, then $10, 21, 42, 85, 170, 341, 682, 1365, 2730, 5461, 10922, 21845, 43690, 87381, 174762, 349525, 699050$, and the next one, 1,398,101, is never reached by 10^6 keys. That is 18 resizes; each re-inserts the keys stored at that moment, which is the threshold itself, so the re-insertions total $5 + 10 + 21 + \cdots + 699{,}050 = 1{,}398{,}087$, about 1.40 per insertion: amortised $O(1)$ by the same geometric argument as for lists. The index doubled 18 times from $8 = 2^3$, so it ends at $2^{21} = 2{,}097{,}152$ slots, of which 1,398,101 are usable and 10^6 used.

(b) Define the height of a node as the length of the longest downward path from it. In a full tree of 31 nodes the heights $0, 1, 2, 3, 4$ hold $16, 8, 4, 2, 1$ nodes, and a sift-down from height h makes at most h swaps, so `heapify` makes at most $8 \times 1 + 4 \times 2 + 2 \times 3 + 1 \times 4 = 26$ swaps, inside the general bound $2n = 62$. Pushing a strictly decreasing sequence makes every new item climb to the root: the i-th push sits at index $i-1$, at depth $\lfloor \log_2 i \rfloor$, and swaps that many times, so the total is $\sum_{i=1}^{31} \lfloor \log_2 i \rfloor = 0 + 2 \times 1 + 4 \times 2 + 8 \times 3 + 16 \times 4 = 98$ swaps: almost four times `heapify`'s worst case, and $\Theta(n \log n)$ in general.

(c) The heap method compares each item with the root and, when it wins, replaces the root and sifts down through a heap of k: $n \log_2 k = 2 \times 10^7 \times 5.64 = 1.13 \times 10^8$ heap steps and 50 slots. Sorting costs $n \log_2 n = 2 \times 10^7 \times 24.25 = 4.85 \times 10^8$ comparisons and 2×10^7 slots: 4.3 times the time and 4×10^5 times the memory.

(d) The interval at least halves each iteration, so at most $\lfloor \log_2 n \rfloor + 1 = 19 + 1 = 20$ iterations for $n = 10^6$. Trace for $t = 50$, $n = 12$: $(0, 12, 6)$, $a[6] = 49 < 50$, so $lo = 7$; $(7, 12, 9)$, $a[9] = 100 \geq 50$, so $hi = 9$; $(7, 9, 8)$, $a[8] = 81 \geq 50$, so $hi = 8$; $(7, 8, 7)$, $a[7] = 64 \geq 50$, so $hi = 7$. The loop exits after 4 iterations, exactly the bound $\lfloor \log_2 12 \rfloor + 1 = 4$, with $lo = hi = 7$: seven elements are below 50 (those up to 49), 7 is the insertion point (the `bisect_left` value), and since $a[7] = 64 \neq 50$ the target is absent.

(e) The recursion reaches depth 1000 on the 1000-th vertex of the path and raises `RecursionError`; nothing beyond it is visited. The iterative version pushes and pops each of the 2500 vertices once and scans each of the 2499 edges once: $V + E = 4999$ units of work, and it finishes.

Self-check. (a) The thresholds are two thirds of successive powers of two, so their sum is about $\frac{2}{3}(2^{21} - 8) = 1{,}398{,}096$ before the floors, within 10 of the exact 1,398,087; and 1.40 per insertion is the $g/(g-1) = 2$ of doubling scaled by the 2/3 load factor, $2 \times 2/3 = 1.33$, plus the effect of the floors. (b) A direct run of the sift-downs on the reversed sequence makes exactly 26 swaps, meeting the bound; the push total also follows from $n = 15$ giving 34 $(0 + 2 + 8 + 24)$ plus $16 \times 4 = 64$ for the last level. (d) The number of elements strictly below 50 is 7 by inspection, matching the exit value.

Answer: (a) 18 resizes, 1,398,087 re-insertions, 1.40 per insertion amortised, final index $2^{21} =$ 2,097,152 slots; (b) at most 26 swaps for `heapify` (bound $2n = 62$) against 98 for the pushes; (c) 1.13×10^8 heap steps and 50 slots against 4.85×10^8 comparisons and 2×10^7 slots; (d) at most 20 iterations; trace $(0, 12, 6) \to (7, 12, 9) \to (7, 9, 8) \to (7, 8, 7)$, exit $lo = 7$ after 4 iterations, the count of elements below 50 and the insertion point, target absent; (e) `RecursionError` at depth 1000; the iterative search does $V + E = 4999$ units and finishes.

P26.4 *Waves, work and span on a ten-task DAG*

(a) In-degrees: A 0; B, C, D 1 each; E 2; F 2; G 1; H 2; I 1; J 2 (they sum to 13, the edge count). Wave 1 is {A}. Scanning A decrements B, C, D to zero: wave 2 is $\{B, C, D\}$. Scanning B, C, D in that order: E reaches zero on C (after B), F reaches zero on D (after C), G reaches zero on D: wave 3 is $\{E, F, G\}$. Scanning E, F, G: H reaches zero on F, I on G: wave 4 is $\{H, I\}$. Scanning H, I: J reaches zero on I: wave 5 is {J}. Each task is enqueued once and each edge decrements one counter: $V + E = 10 + 13 = 23$ container operations (plus the $O(E)$ in-degree initialisation).

(b) The work is the task count, $T_1 = 10$. The waves are the levels (wave t holds the tasks whose longest ending path has t tasks), so the span is the number of waves, $T_\infty = 5$, and the parallelism is $10/5 = 2$. Critical paths are the paths of 5 tasks from A to J: through H they come via E (from B or C) or via F (from C or D), four paths; through I they come via G from D, one path; five critical paths in all: A–B–E–H–J, A–C–E–H–J, A–C–F–H–J, A–D–F–H–J and A–D–G–I–J.

(c) $P = 2$: step 1 {A}; step 2 $\{B, C\}$; step 3 $\{D, E\}$ (E is ready since B and C are done); step 4 $\{F, G\}$ (H is not ready, F being unfinished); step 5 $\{H, I\}$; step 6 {J}. Six steps: four complete (steps 2 to 5, eight tasks) and two incomplete (steps 1 and 6), against the bound $\lfloor 10/2 \rfloor + 5 = 10$ and the lower bounds $\max(10/2, 5) = 5$. Five steps would need every step complete, but only A is ready at step 1, so any 2-processor schedule has at least one incomplete step and at least $\lceil (10 - 1)/2 \rceil + 1 = 6$ steps: the greedy schedule is optimal. (A greedy choice of $\{B, D\}$ at step 2, then $\{C, G\}$, then $\{E, I\}$, leaves only F ready at step 5 and finishes in 7 steps, still within the bound: greediness guarantees the bound, not optimality.) $P = 3$: {A}, $\{B, C, D\}$, $\{E, F, G\}$, $\{H, I\}$, {J}: five steps, two complete and three incomplete, against the bound $\lfloor 10/3 \rfloor + 5 = 8$; the length equals T_∞, so the span law is tight and a fourth processor buys nothing.

Self-check. Levels computed directly as longest ending paths: A 1; B, C, D 2; E, F, G 3 (each has

a level-2 predecessor); H 4 (via E or F), I 4 (via G); J 5: the same five sets as the waves, so the wave count and the span agree. In the $P = 2$ schedule every incomplete step lowers the remaining span by exactly one ($5 \to 4$ at step 1, $1 \to 0$ at step 6), as the greedy argument requires, and the four complete steps account for $8 \le \lfloor 10/2 \rfloor \times 2$ tasks.

Answer: (a) in-degrees A 0, B C D 1, E 2, F 2, G 1, H 2, I 1, J 2; waves $\{A\}$, $\{B, C, D\}$, $\{E, F, G\}$, $\{H, I\}$, $\{J\}$; 23 container operations. (b) $T_1 = 10$, $T_\infty = 5$, parallelism 2, five critical paths. (c) $P = 2$: six steps (four complete, two incomplete), optimal because only A is ready at step 1, bound 10; $P = 3$: five steps $= T_\infty$ (two complete, three incomplete), bound 8; a fourth processor buys nothing.

P26.5 *Reading the two clocks, then Amdahl*

(a) CPU time $U + S = 72$ s, so $r = 72/240 = 0.30$ and the utilisation is $72/(4 \times 240) = 0.075$. Since the process is on at least one core whenever it accrues CPU time, the set of instants at which it is on some core has measure at most $rW = 72$ s, so it was on no core for at least $(1 - r)W = 0.7 \times 240 = 168$ s: it waited (I/O, a lock, a GPU, a network reply) or was runnable but descheduled; the accounting cannot tell which. Spreading the 72 s of compute over four cores gives $72/4 = 18$ s, plus the unchanged 168 s: 186 s, a speedup of $240/186 = 1.29$. This is Amdahl's law with parallel fraction $p = 0.30$, whose ceiling is $1/(1 - p) = 1/0.7 = 1.43$: more cores cannot touch the 168 s, and the lever is to overlap or remove the waiting.

(b) CPU time 18.5 s over 2.5 s: $r = 7.4$ and utilisation $18.5/20 = 0.925$. The inequality $U + S \le \min(C, m)\, W$ requires $18.5 \le \min(8, m) \times 2.5$, that is $\min(8, m) \ge 7.4$, which holds with $C = 8$: consistent. It also forces $m \ge 8$ threads, since $m \le 7$ would cap CPU time at 17.5 s; the Python program created none of them, the BLAS library underneath numpy did. A second copy would put sixteen BLAS threads on eight cores that are already 92.5% busy: oversubscription, with the two runs sharing the cores and finishing in roughly 5 s, not 2.5; the machine is saturated, so the next lever is algorithmic.

(c) Thread 1 is busy $5+3 = 8$ s and thread 2 $6+2 = 8$ s, so the CPU time is 16 s, $r = 16/12 = 1.33$ and the utilisation is $16/24 = 0.667$. Both busy on $[2, 5]$ and $[10, 12]$: 5 s; one busy on $[0, 2]$, $[5, 8]$ and $[9, 10]$: 6 s; none on $[8, 9]$: 1 s. The pieces reconcile: $2 \times 5 + 1 \times 6 = 16$ s of CPU and $5 + 6 + 1 = 12$ s of wall.

(d) $1 - p = (12/6 - 1)/(12 - 1) = 1/11 = 0.0909$. The predicted speedup on 24 is $1/(0.0909 + 0.9091/24) = 1/(0.0909 + 0.0379) = 1/0.1288 = 7.76$, not 12. A measured 7.0 on 24 reads as $1 - p = (24/7 - 1)/23 = 2.4286/23 = 0.106$: the "serial fraction" grew from 0.091 to 0.106 with the process count. A fixed serial part gives a constant reading, so a reading that grows with P signals overhead that scales with P (communication, contention, load imbalance) rather than a fixed serial part, and the fix is in the coordination, not in the serial code.

Self-check. In (a) the Amdahl form gives the same wall time: $T_4 = (1-p)T_1 + pT_1/4 = 168 + 18 = 186$ s. In (b) $r = 7.4 \le C = 8$ and the utilisation is below one, both as the inequality demands. In (c) the CPU integral computed from the core count, $\int k(t)\, dt = 2 \times 5 + 1 \times 6 + 0 \times 1 = 16$, equals the sum of the threads' busy lengths, $8 + 8$. In (d) the identity inverts correctly: with $1 - p = 1/11$ and $P = 12$, $1/S = 1/11 + (10/11)/12 = (12 + 10)/132 = 1/6$, so $S = 6$.

Answer: (a) $r = 0.30$, utilisation 0.075, off-core at least 168 s (waiting or descheduled); four-way compute gives 186 s, speedup 1.29, ceiling 1.43. (b) consistent ($18.5 \le 20$); at least 8 threads, created by BLAS; a second copy oversubscribes eight saturated cores (roughly 5 s for both). (c) CPU 16 s, $r = 1.33$, utilisation 0.667; two cores busy 5 s, one 6 s, none 1 s. (d) $1 - p = 1/11 = 0.091$, predicted $S(24) = 7.76$; 7.0 on 24 reads as 0.106, and the growth signals P-dependent overhead, not a fixed serial part.

P26.6 *The greedy-scheduler bound from scratch*

(a) We count. Each step runs at most P tasks, so $T_1 \le P\, T_P$, which is the work law. The tasks of a longest path are each a predecessor of the next, so they run in distinct steps in path order,

which takes at least T_∞ steps: the span law.

(b) We classify the steps of a greedy schedule as complete, running exactly P tasks, or incomplete. There are at most $\lfloor T_1/P \rfloor$ complete steps, since each consumes P of the T_1 tasks. Consider an incomplete step and the DAG R of tasks not yet run before it. Fewer than P tasks are ready, so the greedy rule runs every ready task. Take any longest path in R: its first task has no unrun predecessor (a predecessor would extend the path within R), so it is ready and runs in this step. Any step runs only tasks with no unrun predecessor, and a path in R contains at most one such task, its first; so every path in R loses at most one task during the step, every longest path loses exactly one, and the span of the remaining DAG falls by exactly one at every incomplete step. The span starts at T_∞ and cannot fall below 0, so there are at most T_∞ incomplete steps. Adding the two counts, $T_P \le \lfloor T_1/P \rfloor + T_\infty$.

(c) Each term of the bound in (b) is at most $\max(T_1/P, T_\infty)$, so $T_P \le 2\max(T_1/P, T_\infty)$; by (a) an optimal schedule has length at least $\max(T_1/P, T_\infty)$, so the greedy length is at most twice optimal. If $P \le \varepsilon T_1/T_\infty$ then $T_\infty \le \varepsilon T_1/P$, so $T_P \le T_1/P + \varepsilon T_1/P = (1+\varepsilon)T_1/P$ and $T_1/T_P \ge P/(1+\varepsilon)$.

(d) A task of duration d is a chain of d unit tasks, and the chains are independent, so $T_1 = 40$ and $T_\infty = t_{\max} = 9$; the bound of (b) reads $\lfloor 40/3 \rfloor + 9 = 22$, or $T_1/P + t_{\max} = 13.33 + 9 = 22.33$ in the non-preemptive form, and the lower bound of (a) is $\lceil 40/3 \rceil = 14$ since finish times are integers. Longest first: loads after $9, 7, 6$ are $(9, 7, 6)$; the 5 goes to the worker at 6 (load 11); the first 4 to the worker at 7 (11); the second 4 to the worker at 9 (13); the 3 to one of the workers at 11 (14); the 2 to the other (13). Finish time $\max(13, 14, 13) = 14$, which meets the lower bound: optimal. Shortest first: $2, 3, 4$ give $(2, 3, 4)$; $4 \to (6, 3, 4)$; $5 \to (6, 8, 4)$; $6 \to (6, 8, 10)$; $7 \to (13, 8, 10)$; $9 \to (13, 17, 10)$. Finish time 17. Both are within the bound 22.33; the shortest-first order loses because the largest task is handed out last, starting at time 8 when the others are nearly done, which is the $t_{\max}$ term of the bound made visible, while longest first hands out the large tasks while every worker is still busy and uses the small ones to level the loads.

Self-check. In (d) the loads sum to 40 in both schedules ($13 + 14 + 13$ and $13 + 17 + 10$), so no work was dropped. Graham's non-preemptive argument gives the same bound directly: if the last-finishing task j starts at s_j, no worker was idle before s_j, so $Ps_j \le T_1 - d_j$ and $T_P \le (T_1 - d_j)/P + d_j \le T_1/P + t_{\max}$; for the shortest-first run, j is the 9 with $s_j = 8$, and $3 \times 8 = 24 \le 40 - 9 = 31$ holds. The lower bound 14 is attained, so no 3-worker schedule beats longest first.

Answer: (a) $T_P \ge T_1/P$ (at most P tasks per step) and $T_P \ge T_\infty$ (a longest path runs in distinct steps); (b) at most $\lfloor T_1/P \rfloor$ complete steps and at most T_∞ incomplete ones, since each incomplete step lowers the remaining span by exactly one; (c) $T_P \le 2\max(T_1/P, T_\infty) \le 2\,T_P^{\text{opt}}$, and speedup $\ge P/(1+\varepsilon)$ when $P \le \varepsilon T_1/T_\infty$; (d) longest first finishes at 14 (optimal, equal to $\lceil 40/3 \rceil$), shortest first at 17, bound $13.33 + 9 = 22.33$; the late large task is the $t_{\max}$ term.

P26.7 *Which reductions parallelise, and why the answer is the same*

(a) *Associative and commutative:* integer sum; max; "argmax, ties to the smaller index", because it is the maximum under the total order "larger value first, smaller index on ties", and a maximum under any total order is associative and commutative. *Associative only:* string concatenation ("ab" $\ne$ "ba"); last non-null, since $(a \oplus b) \oplus c$ and $a \oplus (b \oplus c)$ both return c when c is non-null and otherwise both reduce to $a \oplus b$, but $5 \oplus 7 = 7 \ne 5 = 7 \oplus 5$; the matrix product, associative by the definition of composition, but with $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$, $AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \ne BA = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$. *Neither:* subtraction, $(8 - 3) - 5 = 0 \ne 8 - (3 - 5) = 10$ and $8 - 3 \ne 3 - 8$, repaired by the state (first element, sum of the rest) merged as $(f_1, s_1) \oplus (f_2, s_2) = (f_1, s_1 + f_2 + s_2)$, since the fold equals $x_1 - \sum_{i \ge 2} x_i$; the pairwise mean, $((1+3)/2+5)/2 = 3.5 \ne (1+(3+5)/2)/2 = 2.5$, repaired by (sum, count); the median, for which the median of a union is not a function of the medians and counts of the parts: $\{1, 2, 9\}$ and $\{1, 2, 100\}$ both have count 3 and median 2, yet their unions with $\{50, 60, 70\}$ have medians $(9 + 50)/2 = 29.5$ and $(50 + 60)/2 = 55$; so no bounded lifted state

exists, and it is not repairable without shipping the values or accepting an approximation such as a mergeable quantile sketch; the number of distinct elements, since $|A \cup B| \neq |A| + |B|$ whenever A and B overlap, repaired by the state "the set of distinct elements" under union (exact, $O(d)$ memory) or by a sketch. *Commutative but not associative:* floating-point sum, whose roundings depend on the bracketing (part (c)) while $a + b = b + a$ exactly.

(b) Left fold: $8 - 3 = 5$, $5 - 5 = 0$, $0 - 2 = -2$, $-2 - 7 = -9$, $-9 - 1 = -10$. The tree: $(5 - 3) - 6 = -4$. Different, so subtraction admits no parallel evaluation as written. Lifted: chunk $[8, 3, 5]$ has state $(8, 8)$ and chunk $[2, 7, 1]$ has $(2, 8)$; merging gives $(8, 8 + 2 + 8) = (8, 18)$, and $8 - 18 = -10$, the left fold.

(c) $10^{16} + 1$ lies halfway between 10^{16} and $10^{16} + 2$ and rounds to the even neighbour 10^{16}; adding 1 again gives 10^{16}. But $1 + 1 = 2$ exactly and $10^{16} + 2$ is representable, so $10^{16} + (1 + 1) = 10^{16} + 2$. Sequential: each $+1$ rounds back, result 10^{16}. Two workers: chunk $[10^{16}, 1, 1]$ folds to 10^{16}, chunk $[1, 1]$ to 2, merge $10^{16} + 2$. Three workers: chunks fold to 10^{16}, 2 and 1; in chunk order $(10^{16} + 2) + 1 = 10^{16} + 3$, a tie between $10^{16} + 2$ (odd significand) and $10^{16} + 4$ (even), which rounds to $10^{16} + 4$, the exact sum; in the order (chunk 1 + chunk 3) + chunk 2, $(10^{16} + 1) + 2 = 10^{16} + 2$. So the same five numbers give 10^{16}, $10^{16} + 2$ or $10^{16} + 4$ depending on the worker count and the arrival order, and only one of the three is exact. A test asserting equality of the 2-worker and 3-worker sums fails; the correct test uses a tolerance, or the sum is fixed by a fixed reduction tree, compensated summation, or integer accumulation.

Self-check. The lifted subtraction state applied to the whole list at once gives $(8, 3+5+2+7+1) = (8, 18)$ and -10, matching both the left fold and the chunked merge. The floating-point results are consistent with the spacing: every computed value is a multiple of 2, as any double near 10^{16} must be, and the sequential result 10^{16} is the one the bound $(n-1)u \sum |x_i| \approx 4 \times 1.1 \times 10^{-16} \times 10^{16} \approx 4.4$ allows, since it errs by 4.

Answer: (a) associative and commutative: integer sum, max, argmax with index tie-break; associative only: concatenation, last non-null, matrix product; neither: subtraction (lift to (first, sum of rest)), pairwise mean (lift to (sum, count)), median (no bounded lifted state; ship values or approximate), distinct count (lift to the set of distinct elements, or a sketch); floating-point sum is commutative but not associative. (b) left fold -10, tree -4, lifted chunks $(8, 8) \oplus (2, 8) = (8, 18)$, giving -10. (c) $(10^{16} + 1) + 1 = 10^{16}$, $10^{16} + (1 + 1) = 10^{16} + 2$; sequential 10^{16}, two workers $10^{16} + 2$, three workers $10^{16} + 4$ in chunk order (exact) and $10^{16} + 2$ in the other order; an equality test fails.

P26.8 *Merging shard statistics and shard gradients*

(a) Merge shards 1 and 2 first. With $n = 4 + 6 = 10$ and $\delta = 15 - 10 = 5$, the merged mean is $10 + 5 \times 6/10 = 13$ and $M_2 = 20 + 54 + 5^2 \times 4 \times 6/10 = 74 + 60 = 134$. Merge with shard 3: $n = 12$, $\delta = 30 - 13 = 17$, mean $13 + 17 \times 2/12 = 13 + 2.8333 = 15.8333$, and $M_2 = 134 + 8 + 17^2 \times 10 \times 2/12 = 142 + 289 \times 20/12 = 142 + 481.667 = 623.667$. Population variance $623.667/12 = 51.97$, sample variance $623.667/11 = 56.70$. The unweighted mean of means is $(10 + 15 + 30)/3 = 18.33$, against the true 15.83: the size-2 shard is over-weighted. The shard population variances are $20/4 = 5$, $54/6 = 9$ and $8/2 = 4$, averaging 6, against 51.97: the average of within-shard variances omits the between-shard term $\sum_j n_j (m_j - m)^2 = 541.667$ entirely.

(b) The true value is $\sum x^2 - nm^2 = M_2 = (-1)^2 + 0 + 1^2 = 2$ (mean $10^8 + 2$), giving variance $2/3$. But $\sum x^2 \approx 3 \times 10^{16}$ and $nm^2 \approx 3 \times 10^{16}$ are each rounded to a multiple of 4, after intermediate roundings of the squares (each $(10^8 + k)^2 = 10^{16} + 2k \times 10^8 + k^2$ is odd for odd k where doubles are spaced 2), so their difference is a multiple of 4 carrying an error of several units: the true 2 is below the resolution and the returned value is a rounding artefact, 0.0 on a standard run. The deviation state of (a) never forms $\sum x^2$ and returns $2/3$ exactly.

(c) The batch gradient is the size-weighted mean of the shard means: $(5 \times 2.0 + 3 \times (-2.0))/8 = 4/8 = 0.5$; the unweighted mean 0 is wrong, the same error as the unweighted shard means of (a). Per GPU the ring moves $2 \times 3/4 \times 4 = 6$ GB. At 300 GB/s that is 20 ms, a step of 220

ms and an efficiency of $200/220 = 90.9\%$; at 25 GB/s it is 240 ms, a step of 440 ms and an efficiency of $200/440 = 45.5\%$: the synchronisation term is the span of the step, and on a slow link it dominates.

Self-check. The summaries in (a) are those of the data $\{7, 9, 11, 13\}$, $\{12, 12, 12, 18, 18, 18\}$ and $\{28, 32\}$; pooled, the twelve values sum to 190, mean $190/12 = 15.8333$, and $\sum x^2 = 420 + 1404 + 1808 = 3632$, so $M_2 = 3632 - 12 \times 15.8333^2 = 3632 - 3008.33 = 623.67$, matching the merge. Merging in the other order (shard 3 with shard 2, then shard 1) gives the same $(12, 15.8333, 623.667)$, as commutativity requires. In (c) the weighted gradient is also $(1/8)\sum_i \nabla \ell_i$ with 5 terms averaging 2 and 3 averaging -2: $(10 - 6)/8 = 0.5$.

Answer: (a) merged $(12, 15.833, 623.667)$; population variance 51.97, sample variance 56.70; unweighted mean 18.33 and unweighted variance 6 are wrong. (b) true $\sum x^2 - nm^2 = 2$ (variance 2/3), below the spacing 4 of the operands, so the naive form returns a rounding artefact (0.0). (c) full-batch gradient 0.5 (unweighted 0 is wrong); 6 GB per GPU; 20 ms, 220 ms, 90.9% at 300 GB/s; 240 ms, 440 ms, 45.5% at 25 GB/s.

P26.9 *Threads, processes, pools and a data loader, priced*

(a) Six threads executing only bytecode are serialised by the lock, so the wall time is at least the 60 s of bytecode work whatever the core count, in practice slightly more from lock hand-offs, and $r \leq 1$, near 1. Six processes each have their own interpreter: $8 + 52/6 = 8 + 8.67 = 16.67$ s, a speedup of $60/16.67 = 3.6$, and the ceiling is the serial part alone, $60/8 = 7.5$, however many processes are used (Amdahl with $1 - p = 8/60$).

(b) Serially the requests take $200 \times (0.4 + 0.005) = 81$ s. With 25 threads the waits overlap, since the lock is released during each wait: $200/25 = 8$ rounds of 0.4 s, about 3.2 s of wall time (between 3.2 and 4.2 s, the 1 s of CPU being serialised by the lock but largely hidden under the waits). Total CPU time 200×5 ms $= 1$ s, so $r \approx 1/3.2 = 0.3$: the threads helped despite the lock, and the low r says the run is still wait-dominated, as an I/O job should be.

(c) Pickling copies the 2 GB to each of the 16 workers: 32 GB, about 16 s at 2 GB/s before any work starts; fork inheritance (copy-on-write) or a shared-memory block copies nothing per worker. One task per item costs $10^6 \times 75\ \mu$s $= 75$ s of pure overhead; 100 chunks cost $100 \times 75\ \mu$s $= 7.5$ ms, a factor of 10^4.

(d) By the arithmetic–geometric mean inequality $cP + W_0/P \geq 2\sqrt{cW_0}$ with equality when $cP = W_0/P$, so $P^* = \sqrt{W_0/c} = \sqrt{400} = 20$ and $T^* = 2\sqrt{400} = 40$ s. $T(10) = 10 + 40 = 50$ s, $T(40) = 40 + 10 = 50$ s, $T(80) = 80 + 5 = 85$ s: eighty workers are more than twice as slow as twenty, against $T(1) = 401$ s.

(e) The loaders deliver $3 \times 6 = 18$ batches per second and the GPU can consume $1/0.040 = 25$ per second; the pipeline runs at the slowest stage, 18 batches per second, and the GPU is busy $18/25 = 72\%$ of the time while the training process shows a low r. The GPU becomes the bottleneck at $\lceil 25/6 \rceil = 5$ workers ($30 > 25$ per second). Batching: latency $c + B\tau$ and throughput $B/(c + B\tau)$ give, for $B = 10^4$, $2 + 0.5 = 2.5$ ms and $10^4/0.0025 = 4.0 \times 10^6$ items per second; for $B = 10^5$, $2 + 5 = 7$ ms and 1.43×10^7 per second: 3.6 times the throughput for 2.8 times the latency. Overhead equals compute at $B^* = c/\tau = 2 \times 10^{-3}/5 \times 10^{-8} = 40{,}000$ items.

Self-check. In (a) the process figure is Amdahl's law with $p = 52/60 = 0.867$ and $P = 6$: $1/(0.133 + 0.867/6) = 1/(0.133 + 0.144) = 3.6$, matching 60/16.67. In (d) the two terms at $P^* = 20$ are 20 s and 20 s, equal as the crossover requires, and $T(10) = T(40)$ because the two are reflections of each other about P^* on the log scale ($10 \times 40 = 20^2$). In (e) the throughput at $B = 10^5$ is close to the ceiling $1/\tau = 2 \times 10^7$, as it should be for $B \gg B^*$.

Answer: (a) six threads: about 60 s, $r \approx 1$; six processes: 16.67 s, speedup 3.6; ceiling 7.5. (b) serial 81 s; threaded about 3.2 s (at most 4.2); CPU 1 s; $r \approx 0.3$. (c) 32 GB, about 16 s; nothing per worker when inherited or shared; 75 s of overhead against 7.5 ms. (d) $P^* = 20$, $T^* = 40$ s; $T(10) = T(40) = 50$ s, $T(80) = 85$ s. (e) 18 batches per second, GPU 72% busy, 5 workers;

$B = 10^4$: 2.5 ms and 4.0×10^6 per second; $B = 10^5$: 7 ms and 1.43×10^7 per second; $B^* = 40{,}000$.

P26.10 *How low can a racy counter go?*

(a) We prove the invariant "at every moment c is at most the number of stores completed so far". It holds initially, $0 \le 0$. A store writes temp $+ 1$ where the temporary was loaded earlier, at a moment when c was at most the number of stores completed by then, itself at most the number completed just before this store; so the value written is at most that number plus one, which is the number of stores completed once this store is done. The run has KN stores, so $v \le KN$.

(b) Every store writes at least 1, because every loaded value is at least 0; so once any store has completed, $c \ge 1$ for ever. The last store of the run is some thread's N-th store. In that thread's own order its N-th load comes after its $(N-1)$-th store, which exists because $N \ge 2$; at the moment of that load at least one store had completed, so the thread loaded a value at least 1 and its final store wrote at least 2. Hence $v \ge 2$.

(c) Ending at 2 (threads A and B): A loads 0. B runs its first and second increments to completion, $c = 2$. A adds and stores 1, so $c = 1$. B loads 1, the start of its third increment, and stops. A runs its second and third increments in isolation: load 1, store 2; load 2, store 3; $c = 3$. B adds its stale temporary to 2 and stores 2, the last store of the run. Six increments were executed and $c = 2$. Ending at 6: run A's three increments, then B's, with no interleaving; each load reads the previous store and $c = 6$.

(d) With $N = 1$ every thread's only store writes 1 plus what it loaded, and no load can follow a store of the same thread, so the lower-bound argument of (b) does not apply. Reachable values are $\{1, 2, 3\}$: 3 by running the threads one after another; 1 by letting all three load 0 before any store, then storing 1 three times; 2 by letting A and B load 0, then A store 1 and B store 1, then C load 1 and store 2. In general j threads loading 0 before any store, the rest running sequentially, gives $K - j + 1$, and the invariant of (a) forbids anything above K.

(e) The lock serialises all 8×10^7 increments, so the locked version takes at least $8 \times 10^7 \times 25$ ns $= 2.0$ s however many workers there are, before contention adds its cache-line transfers and waits. One thread unlocked takes $8 \times 10^7 \times 20$ ns $= 1.6$ s: the locked parallel version is slower than the serial one, a speedup of 0.8. Private counters: each worker performs 10^7 local increments in 0.2 s, in parallel, and the merge is one eight-term sum, so about 0.2 s and a speedup near 8. A lock guarding 4% of each worker's time is a serial fraction of 0.04, so Amdahl caps the speedup at $1/0.04 = 25$ regardless of the worker count.

Self-check. A brute-force enumeration of all interleavings for $K = 2$, $N = 3$ gives the reachable set $\{2, 3, 4, 5, 6\}$: the minimum 2 and maximum 6 of (c) are attained and nothing below 2 appears, as (b) requires; for $K = 3$, $N = 1$ it gives $\{1, 2, 3\}$, as in (d). Tracing the schedule of (c) against the invariant of (a): when B stores 2 at the end, six stores have completed and $2 \le 6$. In (e) the eight-worker locked figure can only be a lower bound because contention is ignored; the direction of the comparison (locked parallel slower than serial unlocked) is therefore robust.

Answer: (a) $v \le KN$ by the invariant "$c \le$ stores completed"; (b) $v \ge 2$ because the thread making the last store loaded after its own earlier store, when $c \ge 1$; (c) A loads 0; B completes two increments ($c = 2$); A stores 1; B loads 1; A completes two more increments ($c = 3$); B stores 2: final 2; sequential execution gives 6; (d) $\{1, 2, 3\}$, since with $N = 1$ no thread loads after its own store; (e) locked at least 2.0 s (speedup 0.8 against 1.6 s serial unlocked), private counters about 0.2 s (speedup near 8), a 4% locked section caps the speedup at 25.

P26.11 *Deadlock with three locks, and what is safe to retry*

(a) Interleaving: A takes L_1; B takes L_2; C takes L_3; A requests L_2, held by B, and waits; B requests L_3, held by C, and waits; C requests L_1, held by A, and waits. The wait-for graph is the cycle $A \to B \to C \to A$. Coffman: each lock has a single holder (mutual exclusion); each thread holds one lock while waiting for another (hold-and-wait); no lock is taken from its holder (no preemption); the wait-for graph has a cycle (circular wait). No pair deadlocks alone: with C

absent, B holds L_2 and wants L_3, which is free, so B completes and releases L_2, and A proceeds; with B absent, A wants L_2, which is free; with A absent, C wants L_1, which is free. The cycle needs all three edges, so the bug is invisible in any two-thread test.

(b) Only C violates the order, acquiring L_3 before L_1; rewrite it to acquire L_1 then L_3. Run: A takes L_1; B takes L_2; C requests L_1 and blocks while holding nothing; A requests L_2 and blocks; B requests L_3, which is free, takes it, finishes, and releases L_2 and L_3; A takes L_2, finishes, releases L_1 and L_2; C takes L_1, then L_3, and finishes. The threads finish in the order B, A, C. In general, a thread waiting while holding locks waits only for a lock of higher rank than every lock it holds, so a wait-for cycle would give a strictly increasing sequence of ranks that returns to its start, which is impossible.

(c) (i) Idempotent: the second execution overwrites the entry with the identical value. (ii) Not idempotent: the list gains a duplicate record set. (iii) Not idempotent: the partial is added twice. (iv) Idempotent: the rename replaces the file with identical content, and atomically, so no reader sees a half-written file. (v) Idempotent: $\max(\max(b, x), x) = \max(b, x)$. (vi) Not idempotent: two lines for one shard. (vii) Not idempotent: the account is credited 20. Numbers: $40 \times 250 = 10{,}000$ records plus 250 duplicates, 10,250; the word's total is over by 7, because the shard's partial count of 7 for that word is added a second time.

Self-check. In (a) the three "absent" cases can be read off the graph: removing any vertex from a 3-cycle leaves a path, which has no cycle, so by the Coffman conditions no deadlock. In (b) the ordered run never has a thread waiting while holding a lock of rank above the one it wants: C waits for L_1 holding nothing, A waits for L_2 holding L_1, and B never waits. In (c) the idempotent actions are exactly those whose effect is a function of the input alone, *set* rather than *add*: (i), (iv) and (v) set a value, the others accumulate.

Answer: (a) A takes L_1, B takes L_2, C takes L_3, then each requests the next lock: cycle $A \to B \to C \to A$; all four Coffman conditions hold; no two threads can deadlock alone. (b) C acquires L_1 then L_3; the run completes in the order B, A, C. (c) idempotent: (i), (iv), (v); not: (ii), (iii), (vi), (vii); 10,250 records, total over by 7.

P26.12 *Top-k across shards and the shuffle*

(a) Let words a and b occur 3 times each on every shard, and let each shard carry two private words occurring 4 times each: shard 1 has c, d; shard 2 has e, f; shard 3 has g, h. Globally a and b have 9 each and every other word 4, so the global top-2 is $\{a, b\}$. Locally every shard ranks its two private words (4) above a and b (3), so the local top-2 lists are $\{c, d\}$, $\{e, f\}$, $\{g, h\}$: no tie is involved, and neither a nor b appears in the union. Select-then-aggregate returns a wrong answer whatever it does next.

(b) After partitioning by word, a's complete count 9 lives on one machine, b's 9 on one machine, and each private word's 4 on one machine; every machine ranks its own words by complete counts, so wherever a and b land they head their machines' local top-2 lists, and the coordinator, merging lists of complete counts, returns $\{a, b\}$. The invariant: *after key-partitioning, the local top-k lists are globally exhaustive candidates.* Proof: let x be any global top-k word, so at most $k - 1$ words precede x in the global order; x's owner ranks by the same order on the same complete counts, so at most $k - 1$ of the owner's words precede x, and x is in the owner's local top-k; hence every global top-k word is in the union, with its exact count, and the merge selects correctly. The repartition sends one record per (shard, distinct word on that shard): each shard holds 4 distinct words (a, b and two private ones), so $\sum_j d_j = 12$ records move, against $d = 8$ distinct words and $n = 3 \times 14 = 42$ raw occurrences.

(c) Each machine maps $n/M = 2 \times 10^7$ words at 100 ns per update: 2 s. Without a combiner every occurrence crosses the network: 4×10^9 records, 64 GB, of which each machine sends 2×10^7 records, 320 MB, 1.6 s at 200 MB/s. With a combiner a machine sends at most one record per distinct word it holds, at most $\min(n/M, d) = 5 \times 10^5$: 8 MB and 0.04 s per machine, at most 10^8 records and 1.6 GB in total. The combiner turns a shuffle costing 80% of the map into one

costing 2% of it.

(d) Without a combiner, 8×10^8 records (12.8 GB) converge on the hot word's owner, which receives for $12.8 \times 10^3/200 = 64$ s while every other machine finishes its shuffle in about 1.6 s: a straggler by construction, and hash partitioning cannot help, since it balances words, not occurrences. With the combiner the owner receives at most $M = 200$ partials, one per mapper. With salt $J = 8$ the word becomes eight sub-keys on up to eight machines, each receiving at most 200 partials, and the second round merges the eight sub-key totals into one count.

(e) Merging M sorted lists of k with a heap of M heads pops and pushes once per output: $Mk \log_2 M = 2000 \times 7.64 \approx 15{,}300$ heap steps, negligible against the shuffle; the coordinator is never the bottleneck. With the seed unset, each Python process hashes strings with its own random key, so `hash(word) % M` sends the same word to different owners from different mappers, no machine holds a complete count, and the invariant of (b) fails silently. Fixes: fix the key with `PYTHONHASHSEED`, or partition by a stable hash from `hashlib` (or any deterministic function of the word).

Self-check. In (a) the global totals sum to $9+9+6\times4 = 42$, matching the $3\times(3+3+4+4) = 42$ raw occurrences, so no count was miscounted; and lowering the private words to 3 would create ties that a tie-break could resolve in a's favour, which is why they are set to 4. In (c) the combiner bound $Md = 10^8$ is also 2.5% of n, consistent with the per-machine ratio 8/320. In (d) the hot word's 8×10^8 occurrences are 20% of the corpus, so its owner would receive 20% of the uncombined 64 GB, 12.8 GB, as computed.

Answer: (a) a, b at 3 per shard on all three shards; two private words at 4 per shard; local top-2 lists $\{c,d\},\{e,f\},\{g,h\}$ miss the global top-2 $\{a,b\}$ (9 each). (b) after key-partitioning, the local top-k lists are globally exhaustive candidates, because the owner ranks by the same order on complete counts; the repartition moves 12 records against 8 distinct words and 42 occurrences. (c) map 2 s; 4×10^9 records (64 GB) against at most 10^8 (1.6 GB); per machine 320 MB and 1.6 s against 8 MB and 0.04 s. (d) 8×10^8 records, 12.8 GB, 64 s against 1.6 s; at most 200 partials with the combiner; the second round merges 8 sub-key totals. (e) about 15,300 heap steps; per-process hash randomisation scatters a word across owners; fix the seed or use a stable hash.

28.27 Solutions — Puzzle Classics

PZ.1 *Monty Hall*

Switch. Your initial pick holds the car with probability $\frac{1}{3}$, a goat with probability $\frac{2}{3}$ — and nothing the host does changes the probability that your *initial* pick was right, because he can *always* open a goat door regardless of where the car is. Switching wins exactly when the initial pick was wrong:

$$P(\text{win} \mid \text{switch}) = P(\text{initial pick wrong}) = \tfrac{2}{3}.$$

The trick: the host's reveal is *informative about the other door, not about yours.* He never opens the car door, so his choice funnels the remaining $\frac{2}{3}$ of probability onto the one door he pointedly did not open. An equivalent bookkeeping: over the three equally likely car positions, list what switching does — it loses in the one world where you started on the car and wins in the other two.

The protocol matters (the second question). If the host opens one of the two other doors *uniformly at random* and it happens to show a goat, the answer changes: now the event "a goat was revealed" is evidence about your own door too. Bayes: the random host shows a goat with probability 1 when your pick is the car and probability $\frac{1}{2}$ otherwise, so the posterior odds on your pick being right are $\frac{1}{3}\cdot 1 : \frac{2}{3}\cdot\frac{1}{2} = 1{:}1$ — switching is then worth exactly $\frac{1}{2}$, no better than staying. Saying this aloud — *the answer depends on the host's protocol, not just on what you saw* — is what separates a memorized answer from an understood one.

Answer: Switch: $P(\text{win}) = \frac{2}{3}$ under the standard always-reveals-a-goat host; with a random host who happened to show a goat it drops to $\frac{1}{2}$. □

PZ.2 *Two ropes, 45 minutes*

Light rope A at *both* ends and rope B at *one* end, simultaneously. Rope A's two flames meet after exactly 30 minutes (wherever the meeting point is), at which moment rope B has 30 minutes of burn time left. Now light rope B's *other* end: its remaining burn time halves, so it finishes 15 minutes later. Total: $30 + 15 = 45$ minutes.

The trick: lighting both ends halves the remaining *burn time*, never the length — non-uniformity is irrelevant because the two flames jointly consume rope at twice the one-flame rate, wherever the fast and slow stretches lie. One rope therefore gives you 30 or 60; a second rope, re-lit at the right moment, lets you halve any *remaining* time. (The same idea generates 15, 22.5, ... — each extra both-ends ignition is another halving.)

Answer: Light A both ends and B one end; when A dies (30 min), light B's other end; B dies at 45 min.

PZ.3 *Clock-hands angle* ♦

The hour hand moves 30° per hour plus 0.5° per minute: at 09:30 it sits at $9 \times 30 + 30 \times 0.5 = 270 + 15 = 285°$. The minute hand at 30 minutes sits at 180°. Difference: $285 - 180 = 105°$ (already $\leq 180°$, so no reflex correction needed). The classic error is forgetting that the hour hand has moved half way from 9 towards 10, which gives the wrong answer 90°.

Self-check: one-line formula $|30H - 5.5M| = |30 \times 9 - 5.5 \times 30| = |270 - 165| = 105$. Sanity: at 9:00 the (minor) angle is 90° and over the next 30 minutes the minute hand gains 5.5°/min on the hour hand, sweeping past alignment territory — 105° on the other side is plausible in size.

Answer: 105°.

PZ.4 *100 lockers*

Locker n is toggled once by every divisor of n, so it ends *open* iff n has an *odd* number of divisors.

The trick: divisors come in pairs $d \leftrightarrow n/d$, which makes the divisor count even — *except* when some divisor pairs with itself, $d = n/d$, i.e. $n = d^2$. So the divisor count is odd exactly for perfect squares.

The perfect squares up to 100 are $1, 4, 9, 16, 25, 36, 49, 64, 81, 100$ — ten of them.

Small-case check: locker 12 has divisors $1, 2, 3, 4, 6, 12$ (six toggles, ends closed); locker 9 has $1, 3, 9$ (three toggles, ends open) — the pairing $1{\cdot}12, 2{\cdot}6, 3{\cdot}4$ versus $1{\cdot}9, 3{\cdot}3$ shows the mechanism directly.

Answer: 10 lockers open — exactly the perfect squares $1, 4, 9, \ldots, 100$. □

PZ.5 *25 horses, 5 lanes*

Seven races suffice. Run five heats of five (races 1–5), then race the five heat winners (race 6). Label the heats A–E by their winner's finish in race 6, so $A_1 > B_1 > C_1 > D_1 > E_1$ (subscripts = position within the heat).

Race 6's winner A_1 is the fastest horse overall — it beat its own heat and the other heats' best. Now prune who can still be second or third: a horse is eliminated if three horses are already provably faster. D_1, E_1 and their heats are out (three winners beat them); C_2 and below are out (behind C_1, who is behind B_1 and A_1); B_3 and below are out (behind B_2, B_1, A_1); A_4, A_5 are out. Exactly five candidates remain: A_2, A_3, B_1, B_2, C_1. Race them (race 7); the first two across the line are the overall second and third.

Why six races cannot work: after the five heats, ranking information exists only *within* heats; an adversary argument closes it. Every horse must race at least once (an unraced horse could be placed anywhere), so with 25 horses and 5 lanes the first five races are forced to be a perfect partition into disjoint heats. The sixth race must then contain every heat winner: a missing

winner has never been compared outside its heat, so it could consistently be the overall fastest or merely fourth — indeterminate either way. That fixes race 6 as exactly the five winners, and after it the second place overall is still unresolved: B_1 and A_2 have never met and no chain of results orders them. Six races cannot pin down the podium, so 7 is minimal.

The trick: heats + a winners' race eliminates whole *tail groups* at once (transitivity does the work), and the candidate count for places 2–3 collapses to exactly one final race.

Answer: 7 races. □

PZ.6 *Sixteen gamers, four per game* ♦

Lower bound. A game eliminates at most 3 players from contention (the losers); to crown 1 champion you must eliminate 15, so at least $\lceil 15/3 \rceil = 5$ games are needed.

Construction achieving it. Four first-round games of 4 produce 4 winners; one final among the winners decides it. Because the best player always wins, the champion is guaranteed to win their heat and then the final — 5 games suffice.

Self-check: small case: 4 players need $(4-1)/3 = 1$ game; the general formula, games $= (n-1)/(g-1)$ for n entrants and g per game, gives $15/3 = 5$. Contrast with the 25-horses classic: there you need the *top 3* and only relative order per race, which is why horses cost 7 races; here the deterministic-winner assumption and the "best only" demand make the naive knockout optimal.

Answer: 5 games.

PZ.7 *Prisoners' hats in a line*

$n-1$ **guaranteed; the first speaker is a coin flip.**

Strategy: agree beforehand that the first caller (backmost) announces the *parity* of the black hats they see — say "black" for odd, "white" for even. That caller is right about their own hat only by luck. Every subsequent prisoner maintains a running parity: they know the parity announced, they can see all black hats in front of them, and they have heard every hat color already (correctly) revealed behind them — so the parity of black hats *excluding their own head* is computable, and comparing it with the announced parity reveals their own color exactly. By induction each of the remaining $n-1$ calls is correct.

The trick: the whole line needs only *one shared bit* of information — a parity check — agreed in advance; every later prisoner then solves one equation in one unknown. It is a one-bit error-correcting scheme dressed as a puzzle.

Generalisation worth saying aloud: with k hat colors, announce the sum of the visible colors mod k (colors encoded $0, \ldots, k-1$); again everyone after the first is saved — the structure is a checksum, not anything specific to two colors.

Answer: $n-1$ guaranteed (the backmost announces the parity of black hats seen; everyone else deduces their own); the first speaker survives with probability $\frac{1}{2}$. □

PZ.8 *Hats: guess or pass*

Strategy. If you see two hats of the *same* colour, name the *opposite* colour; if you see one of each, pass.

Winning probability. The 8 hat configurations are equally likely.

- All-same (RRR, BBB — 2 configurations): every player sees a matching pair and guesses the opposite of it; all three guess, all three are wrong. Team loses.
- Mixed (6 configurations): exactly one player wears the minority colour. That player sees two same-coloured hats and names the opposite — which is their own colour: correct. The other two players each see one hat of each colour and pass. Team wins.

So $P(\text{win}) = 6/8 = 3/4$.

Optimality. Whatever the strategy, any particular guess is right with probability exactly 1/2: your own hat is independent of what you see, and your action depends only on what you see. So, summed over the 8 configurations, the number of correct guess-instances equals the number of wrong guess-instances, say $R = W$. Each winning configuration uses at least one correct instance, so $R \geq w$ where w is the number of winning configurations; every wrong instance must sit in a losing configuration (a winning configuration has zero wrong guesses by definition), and a configuration can host at most 3 wrong guesses, so $W \leq 3(8-w)$. Combining: $w \leq R = W \leq 3(8-w)$, hence $4w \leq 24$, i.e. $w \leq 6$ and $P \leq 3/4$. You cannot make guesses *righter* — you can only pile the inevitable wrongness into few configurations (here, the two all-same configurations; for 7 players the same idea via the Hamming code gives 7/8).

Self-check: count the instances in our strategy: the 2 losing configurations absorb $3 \times 2 = 6$ wrong guesses; the 6 winning ones use $1 \times 6 = 6$ correct guesses — $R = W = 6$ exactly, so the counting bound is tight, confirming both the 3/4 computation and that no slack remains.

Answer: See-a-pair-guess-the-opposite wins with probability 3/4, and 3/4 is optimal for three players.

PZ.9 *Broken stick triangle*

Let the stick have length 1 and the break points be $X, Y \sim \mathrm{U}(0,1)$ independent. The three pieces form a triangle iff every piece is shorter than $\frac{1}{2}$ (a piece of length $\geq \frac{1}{2}$ is at least as long as the other two combined, violating the triangle inequality; conversely all pieces $< \frac{1}{2}$ satisfies all three inequalities).

The trick: draw the unit square of (X, Y). Take the case $X < Y$ (probability $\frac{1}{2}$, the triangle above the diagonal); the pieces are X, $Y - X$, $1 - Y$. The conditions

$$X < \tfrac{1}{2}, \qquad Y - X < \tfrac{1}{2}, \qquad Y > \tfrac{1}{2}$$

carve out, inside that half-square triangle, a smaller triangle with vertices $(0, \frac{1}{2}), (\frac{1}{2}, \frac{1}{2}), (\frac{1}{2}, 1)$ of area $\frac{1}{8}$. By symmetry the $X > Y$ case contributes another $\frac{1}{8}$, so

$$P(\text{triangle}) = \tfrac{1}{8} + \tfrac{1}{8} = \tfrac{1}{4}.$$

Sanity check: the answer must be well below $\frac{1}{2}$ — a uniform second break point frequently leaves one long piece; and the complementary events "piece i is too long" each have probability $\frac{1}{4}$ and are disjoint, giving $1 - 3 \cdot \frac{1}{4} = \frac{1}{4}$ directly — a second derivation that agrees.

Answer: $\frac{1}{4}$. □

PZ.10 *Four points on a quarter arc*

Witness decomposition, exactly as in the semicircle classic. For each point i let A_i be the event that all other $n-1$ points lie in the arc of length f (as a fraction of the circumference) running *clockwise from point i*. If the points fit in some arc of length f, the "first" point of the cluster is such a witness, so $\{\text{all in some } f\text{-arc}\} = \bigcup_i A_i$.

Disjointness (this is where $f \leq 1/2$ is used). If A_i and A_j both held for $i \neq j$, then j lies within clockwise-distance f of i and i within clockwise-distance f of j; those two distances sum to the whole circle, giving $1 \leq 2f$ — impossible for $f < 1/2$, and a probability-zero tie at $f = 1/2$. So the A_i are (almost surely) disjoint.

Probability. Each other point independently lands in the clockwise f-arc with probability f, so $P(A_i) = f^{\,n-1}$ and

$$P(\text{all in some } f\text{-arc}) = \sum_{i=1}^{n} P(A_i) = n f^{\,n-1}.$$

For $n = 4$, $f = 1/4$: $4 \times (1/4)^3 = 4/64 = 1/16$.

Self-check: two independent limits. (i) $f = 1/2$ recovers the semicircle classic $n/2^{n-1}$ — for $n = 4$ that is $1/2$, the known answer. (ii) $n = 2$, $f = 1/4$: the formula gives $2 \times 1/4 = 1/2$; directly, the clockwise gap from one point to the other is uniform on $[0, 1)$ and the two points share a quarter-arc iff that gap is $\leq 1/4$ or $\geq 3/4$ — probability $1/2$.

Answer: $1/16$ for four points in a quarter-circle arc; in general $nf^{\,n-1}$ for $f \leq 1/2$. □

PZ.11 *Random walk on a solid*

Collapse states by distance to the target, exactly as for the cube. Let a be the expected steps from the start (the target's antipode) and b from any of the 4 "middle" vertices (each adjacent to both start and target).

From the start, all 4 neighbours are middle vertices:

$$a = 1 + b.$$

From a middle vertex, the 4 neighbours are: the target (absorb), the start, and 2 other middle vertices:

$$b = 1 + \tfrac{1}{4} \cdot 0 + \tfrac{1}{4}a + \tfrac{2}{4}b \quad \Longrightarrow \quad \tfrac{1}{2}b = 1 + \tfrac{1}{4}a \quad \Longrightarrow \quad b = 2 + \tfrac{a}{2}.$$

Substituting: $a = 3 + a/2$, so $a = 6$ and $b = 5$.

Self-check (independent route, via return times): the walk on a regular graph has uniform stationary distribution, so the long-run fraction of time at the target is $1/6$ and the expected return time to it is 6. A return trip is one step to a uniformly random neighbour of the target — always a middle vertex — followed by the hitting time back: $6 = 1 + b$, giving $b = 5$, and then $a = 1 + b = 6$. Sanity against the cube: the cube (8 vertices, degree 3, diameter 3) needs 10 steps; the octahedron is smaller, better connected, and shorter in diameter, so $6 < 10$ is the right direction.

Answer: 6 steps in expectation (5 from a middle vertex). □

PZ.12 *Bold vs timid play*

(a) Bold play. One bet: $P = 18/38 \approx 0.474$.

(b) Timid play. Gambler's ruin with $i = 100$, $N = 200$, $r = q/p = (20/38)/(18/38) = 10/9$:

$$P = \frac{1 - r^{100}}{1 - r^{200}} = \frac{1 - r^{100}}{(1 - r^{100})(1 + r^{100})} = \frac{1}{1 + r^{100}},$$

the clean form because the target is exactly double the stake. Now $r^{100} = (10/9)^{100} = e^{100\ln(10/9)} \approx e^{10.54} \approx 3.8 \times 10^4$, so

$$P \approx \frac{1}{38{,}000} \approx 2.7 \times 10^{-5}.$$

Bold play is roughly 18,000 times likelier to succeed.

Lesson. With the edge against you, every additional bet gives the edge another bite; bold play minimises total money staked, timid play grinds you into the house edge with near certainty. Flip the sign and the prescription flips: with a *positive* edge you bet small and often, and your success probability tends to 1.

Trading analogue. Crossing the spread as a taker is a negative-edge bet — trade as rarely as your objective allows and in size when you do. A market maker earning a small positive edge per fill wants the opposite: maximum trade count, small size, and the ruin formula is exactly why a small *negative* edge on a high-frequency book means near-certain ruin — the sign of the per-trade edge, not its size, decides your fate at scale.

Self-check: (i) fair-coin limit: as $p \to 1/2$, $r \to 1$ and $1/(1 + r^{100}) \to 1/2$, matching bold play's $1/2$ and the fair-ruin answer $i/N = 100/200$. (ii) Monotonicity in bet size: with \$50 bets ($i = 2$, $N = 4$), $P = 1/(1 + r^2) = 81/181 \approx 0.448$ — between the \$1 and \$100 answers and below the

single bold bet, as it must be. (iii) magnitude: $(9/10)^{100} = e^{-100\times 0.10536} = e^{-10.54} \approx 2.7 \times 10^{-5}$, consistent.

Answer: Bold: $18/38 \approx 0.474$. Timid: $1/(1 + (10/9)^{100}) \approx 2.7 \times 10^{-5}$ — about 18,000 times worse. Negative edge: bet boldly and rarely; positive edge: grind small and often.

PZ.13 *Fair coin to probability 1/3* ★

Construction (rejection sampling). Flip the coin twice. HH → the event occurs; HT or TH → it does not; TT → discard the pair and flip twice again. Conditional on the pair not being TT, the three surviving outcomes are equally likely, so the event has probability exactly 1/3.

Explicitly: $P(\text{HH is the first non-TT pair}) = \sum_{k\geq 0}(1/4)^k \cdot (1/4) = \frac{1/4}{1-1/4} = \frac{1}{3}$.

Expected flips. The number of two-flip rounds is geometric with success probability 3/4, so $\mathbb{E}[\text{rounds}] = 4/3$ and, by Wald, $\mathbb{E}[\text{flips}] = 2 \times 4/3 = 8/3 \approx 2.67$.

Fixed number of flips: impossible — say this aloud. With n predetermined flips, every event is a union of atoms of probability 2^{-n}, so its probability is $k/2^n$ for an integer k. Then $k/2^n = 1/3$ would require $2^n = 3k$, but 2^n is never divisible by 3. Exactness *requires* an unbounded (rejection) scheme; volunteering the impossibility argument is what turns a correct answer into a strong one.

Same family, worth having loaded: (i) three flips with target set {HHT, HTH, THH} and the other five outcomes rejected also yields exactly 1/3 per surviving outcome, but at $3 \times 8/3 = 8$ expected flips — the two-flip scheme dominates (in the live interview the candidate reached the construction via this 3-flip, 3/8 hint path); (ii) von Neumann's biased-to-fair trick: flip twice, HT → H, TH → T, HH/TT → repeat — works for any unknown bias since $P(\text{HT}) = P(\text{TH}) = pq$; (iii) simulating a die with a coin: 3 flips give 8 outcomes, reject 2, giving 6 equally likely.

Self-check (symmetry, no series needed): each round produces HH, HT, TH with equal probability and the procedure treats rounds identically, so conditioned on eventual acceptance the three outcomes are exchangeable — each must get probability exactly 1/3; the geometric series above confirms it numerically. And 8/3 flips sits sensibly above the trivial lower bound of 2 (a single fixed round can never carve out exactly 1/3, by the divisibility argument, so some rejection overhead is unavoidable).

Answer: Flip twice: HH = the event, HT/TH = not, TT = reject and repeat — probability exactly 1/3; expected flips 8/3; a fixed number of flips can never do it since $3 \nmid 2^n$.

Part X

Appendix: Long Proofs

This appendix holds the complete proofs that the chapters only sketch. A proof that runs for more than about three quarters of a page would interrupt the chapter it belongs to, so the chapter keeps the statement, a sketch of the argument and a pointer, and the proof stands here. Nothing is left unproved by this arrangement: every result sketched in a chapter is proved in full below.

The proofs are grouped by section and follow the order of their statements. Each one opens with the page of its statement and uses the notation and the standing assumptions of the subsection in which the statement appears. Equation numbers that begin with a letter refer to this appendix.

A Proofs for §11: Portfolio Theory: Mean–Variance Optimisation, Diversification, and the Capital Market Line

A.1 Proof of Proposition 11.4: Two-asset variance, the minimum-variance mix, and the degenerate lines

The statement is on page 105. The notation and the standing assumptions are those of §11.1.

Proof. We specialise Proposition 11.2 to two assets, locate the minimum by differentiating a quadratic with a second-derivative certificate, and treat the degenerate correlations by completing the square.

Part (i). Formula (11.1) with $w_1 = w$, $w_2 = 1 - w$ has one diagonal term per asset and two equal off-diagonal terms, giving (11.2). Collecting powers of w,

$$\sigma_p^2(w) = Vw^2 - 2cw + \sigma_2^2, \qquad V = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2, \quad c = \sigma_2^2 - \rho\sigma_1\sigma_2. \tag{A.1}$$

Proposition 10.10 applied to R_1 and $-R_2$ gives $\operatorname{Var}(R_1 - R_2) = \sigma_1^2 + \sigma_2^2 - 2\operatorname{Cov}(R_1, R_2) = V$, and $V = (\sigma_1 - \sigma_2)^2 + 2(1 - \rho)\sigma_1\sigma_2 > 0$ when $|\rho| < 1$ (the first summand is nonnegative and the second strictly positive). The coefficient of ρ in (11.2) is $2w(1 - w)\sigma_1\sigma_2$, positive for $w \in (0, 1)$, so the variance increases strictly with ρ there.

Part (ii). We differentiate (A.1): $\frac{d}{dw}\sigma_p^2(w) = 2Vw - 2c$, which vanishes at the single point $w^* = c/V$; the second derivative is $2V = 2\operatorname{Var}(R_1 - R_2) > 0$, so the stationary point is the strict global minimum of the quadratic. Writing c/V out gives the stated w^*. Substituting back, $\sigma_p^2(w^*) = \sigma_2^2 - c^2/V = (V\sigma_2^2 - c^2)/V$, and the numerator collapses in two lines:

$$\begin{aligned} V\sigma_2^2 - c^2 &= \sigma_1^2\sigma_2^2 + \sigma_2^4 - 2\rho\sigma_1\sigma_2^3 - (\sigma_2^4 - 2\rho\sigma_1\sigma_2^3 + \rho^2\sigma_1^2\sigma_2^2) \\ &= \sigma_1^2\sigma_2^2(1 - \rho^2). \end{aligned}$$

Part (iii). Since $V > 0$, $w^* < 1$ if and only if $c < V$, i.e. $\sigma_2^2 - \rho\sigma_1\sigma_2 < \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$, i.e. $\rho\sigma_1\sigma_2 < \sigma_1^2$, i.e. $\rho < \sigma_1/\sigma_2$. In that case $1 - w^* = (V - c)/V = \sigma_1(\sigma_1 - \rho\sigma_2)/V > 0$ and $w^* = \sigma_2(\sigma_2 - \rho\sigma_1)/V > 0$ because $\rho < \sigma_1/\sigma_2 \le 1 \le \sigma_2/\sigma_1$; so $w^* \in (0, 1)$. When $\rho \ge \sigma_1/\sigma_2$, $w^* \ge 1$ and $1 - w^* \le 0$. On the long-only range $[0, 1]$ the convex quadratic is then decreasing on $[0, w^*] \supseteq [0, 1]$, so its minimum over $[0, 1]$ is at $w = 1$, asset 1 alone.

Part (iv). We complete the square. At $\rho = +1$, (11.2) reads $w^2\sigma_1^2 + (1 - w)^2\sigma_2^2 + 2w(1 - w)\sigma_1\sigma_2 = \big(w\sigma_1 + (1 - w)\sigma_2\big)^2$; at $\rho = -1$ the cross term changes sign and the square is $\big(w\sigma_1 - (1 - w)\sigma_2\big)^2$. Taking square roots gives the two absolute values. The first vanishes when $w(\sigma_1 - \sigma_2) = -\sigma_2$, i.e. $w = \sigma_2/(\sigma_2 - \sigma_1)$, which exceeds 1 when $\sigma_1 < \sigma_2$ and is negative when $\sigma_1 > \sigma_2$ — a short position either way; the second vanishes when $w\sigma_1 = (1 - w)\sigma_2$, i.e. $w = \sigma_2/(\sigma_1 + \sigma_2) \in (0, 1)$. For the geometry, $\mu_p(w)$ is affine in w with slope $\mu_1 - \mu_2$, and $\sigma_p(w)$ is the absolute value of an affine function of w with slope $\sigma_1 - \sigma_2$ at $\rho = +1$ and $\sigma_1 + \sigma_2$ at $\rho = -1$. When both slopes are nonzero the image of $w \mapsto (\sigma_p(w), \mu_p(w))$ is two rays meeting where the affine function vanishes, on the axis $\sigma = 0$: a V, folded outside $[0, 1]$ at $\rho = +1$ and inside $(0, 1)$ at $\rho = -1$. A zero slope in σ_p freezes the volatility (a vertical line) and a zero slope in μ_p freezes the mean (a horizontal ray, covered once by each arm). At either correlation $\det\Sigma = \sigma_1^2\sigma_2^2(1 - \rho^2) = 0$, so a riskless combination exists, and if a risk-free rate r_f is available, any such combination whose mean differs from r_f is an arbitrage. □

B Proofs for §16: Forwards, Futures and No-Arbitrage Pricing

B.1 Proof of Proposition 16.2: Converting between compounding frequencies

The statement is on page 154. The notation and the standing assumptions are those of §16.1.

Proof. We take logarithms for (i), compare integrands for (ii), and apply Taylor's theorem with the integral remainder for (iii).

Part (i). Fix $T > 0$. Both sides of $(1 + r_m/m)^{mT} = e^{rT}$ are positive, because $1 + r_m/m > 0$, and ln is strictly increasing on the positive reals, so the equation holds if and only if the logarithms of its two sides agree, $mT \ln(1 + r_m/m) = rT$. Dividing by $T > 0$, this is $r = m \ln(1 + r_m/m)$, a condition that does not involve T. Equality at one horizon therefore gives the condition, and the condition gives equality at every horizon. For the second form, divide the condition by m to get $r/m = \ln(1 + r_m/m)$. The exponential function is the inverse of ln, so this holds if and only if $e^{r/m} = 1 + r_m/m$, that is, $r_m = m(e^{r/m} - 1)$. In this second form the requirement $r_m > -m$ holds automatically, because $e^{r/m} > 0$.

Part (ii), Step 1 (an auxiliary function). Define $\chi(x) = (e^x - 1)/x$ for $x \neq 0$ and $\chi(0) = 1$. (The letter is χ and not g, because g is the expected log-growth rate throughout the Part.) For every real x,

$$\chi(x) = \int_0^1 e^{xs}\, ds. \tag{B.1}$$

For $x \neq 0$ the function $s \mapsto e^{xs}/x$ is an antiderivative of $s \mapsto e^{xs}$, so by the fundamental theorem of calculus the integral equals $(e^x - e^0)/x = \chi(x)$; for $x = 0$ the integrand is 1 and the integral is $1 = \chi(0)$. The variable of integration is s, here and in part (iii), because t is a time in §16.

Step 2 (a positivity principle, and χ is strictly increasing). The principle is this: a continuous function ζ on a closed interval $[s_1, s_2]$ with $s_1 < s_2$ that is nonnegative on the interval and positive at one of its points s_0 has a positive integral over the interval. Indeed, by continuity at s_0 there is an $\varepsilon > 0$ such that $\zeta(s) \geq \zeta(s_0)/2$ for every s of the interval with $|s - s_0| \leq \varepsilon$. Those s form the closed interval $[s_3, s_4]$ with $s_3 = \max(s_1, s_0 - \varepsilon)$ and $s_4 = \min(s_2, s_0 + \varepsilon)$, whose length $s_4 - s_3$ is positive because $s_1 < s_2$ and $s_1 \leq s_0 \leq s_2$. On it the integral of ζ is at least $(s_4 - s_3)\,\zeta(s_0)/2$; on the rest of $[s_1, s_2]$ the integrand is nonnegative. Hence $\int_{s_1}^{s_2} \zeta(s)\, ds \geq (s_4 - s_3)\,\zeta(s_0)/2 > 0$.

Now let $x < x'$. For every $s \in (0, 1]$ we have $xs < x's$, hence $e^{xs} < e^{x's}$, because the exponential function is strictly increasing. The difference $e^{x's} - e^{xs}$ is therefore continuous in s, nonnegative on $[0, 1]$ and positive on $(0, 1]$, and by (B.1) and the positivity principle

$$\chi(x') - \chi(x) = \int_0^1 (e^{x's} - e^{xs})\, ds > 0.$$

No derivative of an integral has been taken.

Step 3 (the limit at 0). For $x \neq 0$, $\chi(x) = (e^x - e^0)/(x - 0)$ is the difference quotient of the exponential function at 0, so by the definition of the derivative $\chi(x) \to \exp'(0) = e^0 = 1$ as $x \to 0$.

Step 4 (back to rates). Since $r \neq 0$, $r_m = m(e^{r/m} - 1) = r\,\chi(r/m)$. Suppose first that $r > 0$. As m increases, $x = r/m$ is positive and decreases strictly, with limit 0. By Step 2, $\chi(r/m) > \chi(0) = 1$ and $m \mapsto \chi(r/m)$ is strictly decreasing; by Step 3 its limit is 1. Multiplying by $r > 0$ gives $r_m > r$, with r_m strictly decreasing in m and tending to r. Suppose now that $r < 0$. As m increases, $x = r/m$ is negative and increases strictly, with limit 0. By Step 2, $\chi(r/m) < \chi(0) = 1$ and $m \mapsto \chi(r/m)$ is strictly increasing; by Step 3 its limit is 1. Multiplying by $r < 0$ reverses both statements: $r_m = r\,\chi(r/m) > r$, and r_m is strictly decreasing in m and tends to r.

Part (iii), Step 1 (the two remainders). Theorem 7.31, applied with order 2 at the point 0 and written with the integration variable s, says that a function with three continuous derivatives on an open interval containing 0 and x equals its Taylor polynomial of order 2 at 0 plus the remainder (7.3), which is one half of the integral from 0 to x of $(x - s)^2$ times the third derivative at s. The hypothesis holds on the whole line for the exponential function and on $(-1, \infty)$ for $x \mapsto \ln(1 + x)$. Every derivative of e^x is e^x, with value 1 at 0, so

$$e^x = 1 + x + \frac{x^2}{2} + \frac{1}{2}\int_0^x (x - s)^2 e^s\, ds \qquad (x \text{ real}). \tag{B.2}$$

The first three derivatives of $\ln(1+x)$ are $(1+x)^{-1}$, $-(1+x)^{-2}$ and $2(1+x)^{-3}$, as computed in the proof of Proposition 7.34; the function and its first two derivatives take the values 0, 1 and -1 at 0, so

$$\ln(1+x) = x - \frac{x^2}{2} + \int_0^x (x-s)^2 (1+s)^{-3}\, ds \qquad (x > -1). \tag{B.3}$$

Step 2 (strict brackets for $x > 0$). Let $x > 0$. The weight $(x-s)^2$ is continuous in s on $[0,x]$ and positive on $[0,x)$, and $\int_0^x (x-s)^2\, ds = x^3/3$. For $0 < s < x$ we have $1 < e^s < e^x$, because the exponential function is strictly increasing, and $(1+x)^{-3} < (1+s)^{-3} < 1$, because $s \mapsto (1+s)^{-3}$ is strictly decreasing on $(-1,\infty)$. Each of the four differences

$$(x-s)^2(e^s - 1), \quad (x-s)^2(e^x - e^s), \quad (x-s)^2\big((1+s)^{-3} - (1+x)^{-3}\big), \quad (x-s)^2\big(1 - (1+s)^{-3}\big)$$

is therefore continuous in s, nonnegative on $[0,x]$ and positive on $(0,x)$, so by the positivity principle of part (ii), Step 2, each has a positive integral over $[0,x]$. Writing these four integrals out,

$$\frac{x^3}{3} < \int_0^x (x-s)^2 e^s\, ds < e^x \frac{x^3}{3}, \qquad \frac{x^3}{3(1+x)^3} < \int_0^x (x-s)^2 (1+s)^{-3}\, ds < \frac{x^3}{3}.$$

The remainder in (B.2) therefore lies strictly between $x^3/6$ and $e^x x^3/6$, and the remainder in (B.3) strictly between $x^3/(3(1+x)^3)$ and $x^3/3$.

Step 3 (back to rates). Put $x = r/m$ in (B.2), subtract 1 and multiply by m:

$$r_m = m\big(e^{r/m} - 1\big) = r + \frac{r^2}{2m} + \frac{m}{2} \int_0^{r/m} \Big(\frac{r}{m} - s\Big)^2 e^s\, ds.$$

For $r > 0$, Step 2 with $x = r/m > 0$ places the last term strictly between $m\,(r/m)^3/6 = r^3/(6m^2)$ and $e^{r/m}\, r^3/(6m^2)$, which is the first bracket. Next put $x = r_m/m$ in (B.3), which is allowed because $r_m > -m$, multiply by m and use $r = m\ln(1 + r_m/m)$ from (16.1):

$$r = r_m - \frac{r_m^2}{2m} + m \int_0^{r_m/m} \Big(\frac{r_m}{m} - s\Big)^2 (1+s)^{-3}\, ds.$$

For $r_m > 0$, Step 2 with $x = r_m/m > 0$ places the last term strictly between $m\,(r_m/m)^3/(3(1 + r_m/m)^3) = r_m^3/(3m^2(1 + r_m/m)^3)$ and $r_m^3/(3m^2)$, which is the second bracket.

Step 4 (either sign). By the bound (7.4) at order 2, the remainder at x is at most $c\,|x|^3/6$ in absolute value, where c is any bound for the absolute value of the third derivative between 0 and x. For the exponential function, every s between 0 and x has $e^s \le e^{|x|}$, so $c = e^{|x|}$ serves and the remainder in (B.2) is at most $e^{|x|}|x|^3/6$ in absolute value. For the logarithm with $|x| \le \frac{1}{2}$, every s between 0 and x has $1 + s \ge \frac{1}{2}$, hence $0 < 2(1+s)^{-3} \le 2 \times 2^3 = 16$, so $c = 16$ serves and the remainder in (B.3) is at most $16|x|^3/6 = 8|x|^3/3$ in absolute value. Multiplying by m at $x = r/m$ and at $x = r_m/m$, as in Step 3, gives

$$\Big| r_m - r - \frac{r^2}{2m} \Big| \le m\, e^{|r|/m}\, \frac{|r|^3}{6m^3} = e^{|r|/m}\, \frac{|r|^3}{6m^2},$$
$$\Big| r - r_m + \frac{r_m^2}{2m} \Big| \le m\, \frac{8\,|r_m|^3}{3m^3} = \frac{8\,|r_m|^3}{3m^2} \qquad \Big(|r_m| \le \frac{m}{2}\Big).$$

For $|r| \le m$ the first bound is at most $\big(e/(6m^2)\big)|r|^3$, and the second is the constant $8/(3m^2)$ times $|r_m|^3$; these are the two O-statements. □

B.2 Proof of Lemma 16.8: Dominance and the law of one price

The statement is on page 159. The notation and the standing assumptions are those of §16.2.

Proof. We argue by contradiction: each violation of a conclusion is turned into an arbitrage in the sense of Definition 16.7, which the standing assumption excludes.

Part (i). Suppose that $\Pi_0^A < \Pi_0^B$. Hold strategy A, hold the negative of strategy B (sell it short), and deposit $\Pi_0^B - \Pi_0^A > 0$ in the cash account. By the bookkeeping rules of the Setup of §16.2 this is a self-financing strategy. Its price is

$$\Pi_0^A - \Pi_0^B + (\Pi_0^B - \Pi_0^A) = 0,$$

and, because $\Pi_T^A - \Pi_T^B \geq 0$ with probability one, its payoff satisfies

$$\Pi_T^A - \Pi_T^B + (\Pi_0^B - \Pi_0^A)e^{rT} \;\geq\; (\Pi_0^B - \Pi_0^A)e^{rT} \;>\; 0 \qquad \text{with probability one.}$$

A strategy with price zero and a payoff that is positive with probability one is an arbitrage. Hence $\Pi_0^A \geq \Pi_0^B$.

Part (ii). Suppose that $\Pi_0^A \leq \Pi_0^B$. The same combination, now with the deposit $\Pi_0^B - \Pi_0^A \geq 0$, has price zero, and its payoff is at least $\Pi_T^A - \Pi_T^B$, which is nonnegative with probability one and positive with positive probability: an arbitrage. Hence $\Pi_0^A > \Pi_0^B$.

Part (iii). Equal payoffs give both $\Pi_T^A \geq \Pi_T^B$ and $\Pi_T^B \geq \Pi_T^A$ with probability one, and part (i), applied in both directions, gives $\Pi_0^A \geq \Pi_0^B \geq \Pi_0^A$.

Part (iv). Depositing ce^{-rT} in the cash account at time 0 (borrowing $-ce^{-rT}$ if $c < 0$) is a self-financing strategy with price ce^{-rT} and payoff $ce^{-rT}e^{rT} = c$. The given strategy has the same payoff with probability one, so by part (iii) it has the same price.

Part (v). We trade only on the event of a violation. From t on, holding the positions of A and following its trading is worth Π_t^A at t and pays Π_T^A, and likewise for B. Consider the strategy that holds nothing before t and that, at t and only on the event $\{\Pi_t^A < \Pi_t^B\}$, which is known at t, holds A, sells B short and deposits $\Pi_t^B - \Pi_t^A$ in the cash account. On the event its value at t is $\Pi_t^A - \Pi_t^B + (\Pi_t^B - \Pi_t^A) = 0$, and off the event it holds nothing. It is therefore a strategy with a late start in the sense of the Setup of §16.2, and it is self-financing by the bookkeeping rules for positions held from t. Off the event its payoff is zero. On the event its payoff is

$$\Pi_T^A - \Pi_T^B + (\Pi_t^B - \Pi_t^A)e^{r(T-t)} \;\geq\; (\Pi_t^B - \Pi_t^A)e^{r(T-t)} \;>\; 0$$

outside the event $\{\Pi_T^A < \Pi_T^B\}$, which has probability zero. If $P(\Pi_t^A < \Pi_t^B)$ were positive, the strategy would be an arbitrage, by reading (c) of Definition 16.7. Hence $P(\Pi_t^A < \Pi_t^B) = 0$, which is the inequality $\Pi_t^A \geq \Pi_t^B$ with probability one. Under equal payoffs the inequality holds in both directions, which gives the equality, as in part (iii).

For the last clause, let B be the deposit of $ce^{-r(T-t)}$ made at t (a loan if $c < 0$). It is a position opened at t whose size is known at t; it is worth $ce^{-r(T-t)}$ at t and pays c at T. The payoffs of A and B are equal with probability one, so $\Pi_t^A = ce^{-r(T-t)}$ with probability one. Spelled out: on the event $\{\Pi_t^A < ce^{-r(T-t)}\}$ the violation trade holds A, borrows $ce^{-r(T-t)}$ and deposits the surplus $ce^{-r(T-t)} - \Pi_t^A$. Its value at t is zero; at T it receives c from A, repays c on the loan and keeps $(ce^{-r(T-t)} - \Pi_t^A)e^{r(T-t)} > 0$ from the deposit. On the reverse event $\{\Pi_t^A > ce^{-r(T-t)}\}$ it sells A short and deposits the proceeds Π_t^A in two parts, $ce^{-r(T-t)}$ and the surplus $\Pi_t^A - ce^{-r(T-t)}$. At T the first part has grown to the amount c owed on the short position, and the second part leaves $(\Pi_t^A - ce^{-r(T-t)})e^{r(T-t)} > 0$. □

B.3 Proof of Proposition 16.21: Commodity forwards: equality for investment commodities, a bound for consumption commodities

The statement is on page 169. The notation and the standing assumptions are those of §16.5.

Proof. We prove (ii) first, by contradiction, with a cash-and-carry arbitrage that only buys and stores; (i) then follows by replicating a sure payment, as in Proposition 16.13 and Proposition 16.15.

Part (ii), discrete storage. Suppose that $F > (S+U)e^{rT}$. At time 0 borrow $S+U$ from the cash account, buy one unit and store it, deposit $U_j e^{-rt_j}$ for each storage date t_j, and go short one forward at F.

position	cash flow at 0	value at T
borrow $S+U$	$+(S+U)$	$-(S+U)e^{rT}$
buy one unit and store it	$-S$	$+S_T$
deposit $U_j e^{-rt_j}$ for each j	$-U$	0
short one forward at F	0	$F-S_T$
net	0	$F-(S+U)e^{rT}$

The j-th deposit has grown to exactly $U_j e^{-rt_j}e^{rt_j} = U_j$ at the date t_j, where it pays the j-th storage cost; the deposits sum to $\sum_j U_j e^{-rt_j} = U$ at time 0 and are exhausted by the last storage date. No cash is added or withdrawn, so the strategy is self-financing. At T the unit is delivered under the forward for F and the loan, which has grown to $(S+U)e^{rT}$, is repaid. The price is 0 and the payoff is the constant $F-(S+U)e^{rT} > 0$: an arbitrage, by reading (b) of Definition 16.7. Hence $F \le (S+U)e^{rT}$.

Part (ii), proportional storage. Suppose that $F > Se^{(r+r_{\mathrm{st}})T}$. At time 0 borrow $Se^{r_{\mathrm{st}}T}$, buy $e^{r_{\mathrm{st}}T}$ units and store them, paying the storage as it falls due by selling units, and go short one forward at F. The holding is the reinvested holding of Proposition 16.15(i) with $q = -r_{\mathrm{st}}$: it starts with $e^{-qT} = e^{r_{\mathrm{st}}T}$ units, it holds $e^{-q(T-t)} = e^{r_{\mathrm{st}}(T-t)}$ units at time t whatever path the price takes, it is self-financing, and it holds exactly one unit at T. At T that unit is delivered under the forward for F, and the loan, which has grown to $Se^{r_{\mathrm{st}}T}e^{rT}$, is repaid. The strategy has price 0 and the constant payoff

$$S_T + (F-S_T) - Se^{r_{\mathrm{st}}T}e^{rT} = F - Se^{(r+r_{\mathrm{st}})T} > 0,$$

an arbitrage again. Hence $F \le Se^{(r+r_{\mathrm{st}})T}$.

Both arbitrages buy the commodity, store it and borrow cash; neither sells a unit short. They are therefore available for every storable commodity, and (ii) holds for consumption commodities as well.

Part (i), discrete storage. Consider the package of the first table without its loan: one unit bought and stored, the deposits $U_j e^{-rt_j}$ that pay the storage, and one short forward at F. It is self-financing, it costs $S+U$, and it pays $S_T + (F-S_T) = F$ with probability one. For an investment commodity the package can also be reversed (rule (2) of the Setup of §16.5): one unit is sold short or sold by a holder, $U_j e^{-rt_j}$ is borrowed for each j and repaid at t_j out of the storage cost that the lender of the unit passes on, or, when a holder sells, out of the storage cost that the holder no longer pays, and one forward is bought at F. Both the package and its reversal being available, Lemma 16.8(iv) applies and gives $S+U = Fe^{-rT}$, that is, $F = (S+U)e^{rT}$.

The direction that part (ii) does not cover can be spelled out. If $F < (S+U)e^{rT}$, a holder sells one unit for S and deposits the proceeds, deposits each storage cost U_j at the date t_j at which it would have been paid, and buys one forward at F. At T the deposits stand at

$$Se^{rT} + \sum_j U_j e^{r(T-t_j)} = \Big(S + \sum_j U_j e^{-rt_j}\Big)e^{rT} = (S+U)e^{rT}.$$

The holder pays F under the forward, takes delivery, and owns the unit again together with $(S+U)e^{rT} - F > 0$ in cash, where storing the unit would have left the unit alone. Holding the commodity outright is then dominated, and Definition 16.10 says that holders of an investment commodity make the switch.

Part (i), proportional storage. An investment commodity that pays no income and is stored at the proportional rate r_{st} is an investment asset with the known yield $q = -r_{\mathrm{st}}$ (Setup of §16.5), so Proposition 16.15(ii) gives $F = Se^{(r-q)T} = Se^{(r+r_{\mathrm{st}})T}$. Its replicating package is the holding of

$e^{r_{\mathrm{st}}T}$ units used in part (ii), with the storage paid in units, together with one short forward: it costs $Se^{r_{\mathrm{st}}T}$ and pays F, and it can be reversed because the commodity is an investment commodity. □

B.4 Proof of Theorem 16.28: Futures and forward prices coincide under deterministic rates

The statement is on page 174. The notation and the standing assumptions are those of §16.6.

Proof. We compare two self-financing strategies with the same payoff, a rolling futures position (the argument of Cox, Ingersoll and Ross, 1981) and a forward position, and apply Lemma 16.8(iii); Steps 1 to 6 prove the equality at time 0, and Step 7 runs the same comparison from the end of a later day with Lemma 16.8(v).

Step 1 (the rolling position). During day i, for $i = 1, \ldots, n$, hold B_i futures contracts long. The position for day i must be set at the end of day $i-1$ (at time 0 for $i = 1$), and the number B_i is known then, because the rates, and hence B_i, are deterministic. Changing the position from B_{i-1} to B_i contracts costs nothing, because a futures position is worth zero after each settlement (Definition 16.25). The position is closed at the last settlement, at the end of day n, again at no cost.

Step 2 (one day's gain, carried to T). The settlement at the end of day i pays $B_i(F_i - F_{i-1})$, an amount of either sign. Deposit it in the cash account until T, or borrow it if it is negative. By the Setup of §16.6, an amount deposited at the end of day i grows by the factor B_n/B_i until T, so the gain of day i has become

$$B_i\,(F_i - F_{i-1})\,\frac{B_n}{B_i} = (F_i - F_{i-1})\,B_n$$

at T. The size B_i of the position has cancelled against the reinvestment factor.

Step 3 (summing). The terms telescope:

$$\sum_{i=1}^{n} (F_i - F_{i-1})\,B_n = (F_n - F_0)\,B_n.$$

Step 4 (the futures strategy). Add to the rolling position a deposit of $F_0 = F_0^{\mathrm{fut}}$ made at time 0, which grows to $F_0 B_n$ at T. Every settlement goes into or out of the cash account, so the strategy is self-financing. Futures cost nothing to enter, so its price is F_0^{fut}. Its payoff is

$$(F_n - F_0)\,B_n + F_0 B_n = F_n B_n = S_T B_n \qquad \text{with probability one,}$$

because $F_n = F_T^{\mathrm{fut}} = S_T$ with probability one (Proposition 16.27).

Step 5 (the forward strategy). At time 0 go long B_n forward contracts struck at the forward price F, a number of contracts that is known at time 0, and deposit F in the cash account. The forwards cost nothing (Definition 16.5), so the price of the strategy is F. It produces no cash flow before T, and its payoff is

$$B_n\,(S_T - F) + F B_n = S_T B_n.$$

Step 6 (the law of one price). The two self-financing strategies have equal payoffs with probability one. Each consists of futures or forwards and of positions in the cash account, so each can be held and sold short, whatever the underlying is. Lemma 16.8(iii), which holds with the cash account B_i (Setup of §16.6), gives equal prices: $F_0^{\mathrm{fut}} = F$.

Step 7 (from a later settlement date). Fix a day k with $1 \le k < n$, and let F' be the forward price for delivery at T written at the end of day k, a quantity known then. Consider first the futures position held from the end of day k: during each day $i > k$ hold B_i/B_k contracts, a

number known when it is set, and deposit F_k in the cash account at the end of day k. As in Step 2 the gain of day i carried to T is $(B_i/B_k)(F_i - F_{i-1})B_n/B_i = (F_i - F_{i-1})B_n/B_k$, the gains telescope to $(F_n - F_k)B_n/B_k$, and the deposit grows to F_kB_n/B_k. The payoff is therefore $F_nB_n/B_k = S_TB_n/B_k$ with probability one, by Proposition 16.27, and the value at the end of day k is F_k, the futures positions being worth zero. Consider next the forward position held from the end of day k: go long B_n/B_k forwards struck at F', at no cost, and deposit F'. Its value at the end of day k is F', and its payoff is $(B_n/B_k)(S_T - F') + F'B_n/B_k = S_TB_n/B_k$. Neither position adds or withdraws cash after the end of day k; both consist of contracts and cash, so each can be held and sold short, as in Step 6; and their payoffs are equal with probability one. The second comparison clause of Lemma 16.8(v), which holds with the cash account B_i (Setup of §16.6), gives $F_k = F'$ with probability one. □

C Proofs for §17: Options: Payoffs, Bounds, Put–Call Parity and Trading Strategies

C.1 Proof of Proposition 17.6: Model-free bounds and comparisons

The statement is on page 193. The notation and the standing assumptions are those of §17.2.

Proof. We use dominance at T (Lemma 16.8(i)) for the European claims and exercise matching (Lemma 17.5(ii)) for the American ones, in four steps.

Step 1 (case (N), explicit portfolios). Each row of the table compares two portfolios held from time 0 to T. They are self-financing, because in case (N) nothing they hold pays or costs anything before T. "Bond" is the zero-coupon bond paying K at T, and the empty portfolio costs nothing and pays nothing.

Portfolio	Cost at 0	Payoff at T		Portfolio	Cost at 0	Payoff at T
share	S	S_T	$\geq$	call	C	$(S_T - K)^+$
bond	Ke^{-rT}	K	$\geq$	put	$\mathcal{P}$	$(K - S_T)^+$
call and bond	$C + Ke^{-rT}$	$\max(S_T, K)$	$\geq$	share	S	S_T
put and share	$\mathcal{P} + S$	$\max(S_T, K)$	$\geq$	bond	Ke^{-rT}	K
call	C	$(S_T - K)^+$	$\geq$	empty	0	0
put	$\mathcal{P}$	$(K - S_T)^+$	$\geq$	empty	0	0

The inequality between the two payoffs of a row holds in every state. In the first row, if $S_T \geq K$ then $(S_T - K)^+ = S_T - K \leq S_T$ because $K > 0$, and if $S_T < K$ then $(S_T - K)^+ = 0 \leq S_T$ by limited liability. In the second row, if $S_T \leq K$ then $(K - S_T)^+ = K - S_T \leq K$ by limited liability, and if $S_T > K$ then $(K - S_T)^+ = 0 < K$. In the third and fourth rows, the two cases $S_T \geq K$ and $S_T < K$ give $(S_T - K)^+ + K = \max(S_T, K)$ and $(K - S_T)^+ + S_T = \max(S_T, K)$, and a maximum is at least each of its two arguments. In the last two rows a positive part is nonnegative. By Lemma 16.8(i) the cost on the left of each row is at least the cost on the right:

$$C \leq S, \qquad \mathcal{P} \leq Ke^{-rT}, \qquad C \geq S - Ke^{-rT}, \qquad \mathcal{P} \geq Ke^{-rT} - S, \qquad C \geq 0, \qquad \mathcal{P} \geq 0.$$

Since $F = Se^{rT}$ in case (N), these are the European statements of (i), (ii) and (iii) in that case. The call-and-bond portfolio is called a *fiduciary call* and the put-and-share portfolio a *protective put*; both return in §17.3.

Step 2 (any income, forward form). Let the underlying be in any of the three income cases, with forward price F. The last two rows of the table use nothing about income, so $C \geq 0$ and $\mathcal{P} \geq 0$ in every case. The remaining bounds rest on the identity $x^+ - x = (-x)^+$, valid for every real x: for $x \geq 0$ both sides are 0, and for $x < 0$ both sides are $-x$.

Call, lower bound. Hold one call and one short forward contract with delivery price K, which pays $K - S_T$ at T. The payoff of the pair is $(S_T - K)^+ - (S_T - K) = (K - S_T)^+ \geq 0$, by the identity at $x = S_T - K$. A long forward with delivery price K is worth $e^{-rT}(F - K)$ at time 0 by Proposition 16.17(i), and the short forward is worth the negative of that. The pair therefore costs $C - e^{-rT}(F - K)$, and Lemma 16.8(i), applied against the empty portfolio, gives $C \geq e^{-rT}(F - K)$. Together with $C \geq 0$ this is $C \geq e^{-rT}(F - K)^+$.

Call, upper bound. The prepaid forward of the Setup of §17.2 pays S_T, which is at least $(S_T - K)^+$ by the first row of the table, and costs $e^{-rT}F$. Lemma 16.8(i) gives $C \leq e^{-rT}F$.

Put, lower bound. Hold one put and one long forward contract with delivery price K. The payoff is $(K - S_T)^+ + (S_T - K) = (S_T - K)^+ \geq 0$, by the identity at $x = K - S_T$, and the cost is $\mathcal{P} + e^{-rT}(F - K)$. So $\mathcal{P} \geq e^{-rT}(K - F)$, and with $\mathcal{P} \geq 0$ this is $\mathcal{P} \geq e^{-rT}(K - F)^+$.

Put, upper bound. The second row of the table uses only limited liability, so the bond dominates the put in every income case and $\mathcal{P} \leq Ke^{-rT}$.

Substituting $e^{-rT}F = S$, $S - I$ or Se^{-qT} gives the three written-out forms of (ii) and (iii).

Step 3 (American options, by exercise matching). For each claim, suppose that it fails. We exhibit one position that brings in a positive amount at time 0 and is worth a nonnegative amount at its horizon (T, or T_2 in (e)) under every exercise policy, which Lemma 17.5(ii) excludes. The positive amount is set aside in each case, and the rest of the position is followed to its horizon.

(a) American against European. Suppose that a European option with expiry $T' \leq T$ trades above the American option of the same type and strike. Buy the American option and write the European one; the difference of the two prices is the positive inflow. At T', if the European option finishes in the money, exercise the American option. For calls: pay K, receive the share and sell it for $S_{T'}$, which leaves $S_{T'} - K$, exactly the amount owed to the holder of the European call. For puts: buy the share for $S_{T'}$ and deliver it for K, which leaves $K - S_{T'}$, exactly the amount owed to the holder of the European put. If the European option does not finish in the money, nothing is owed and the American option is never exercised. Either way no cash flow occurs after T', and the position, the inflow apart, is worth 0 at T. When $T' = t_i^-$ both options settle on the cum-dividend price $S_{t_i^-}$, and the argument is unchanged. The only written option is European, so no exercise policy other than the arbitrageur's own is involved. The case $T' = T$ gives $C^{\mathrm{am}} \geq C$ and $\mathcal{P}^{\mathrm{am}} \geq \mathcal{P}$.

(b) Intrinsic value. If an American option had a negative price, buying it would bring in a positive amount, and never exercising it would leave nothing to settle; so $C^{\mathrm{am}} \geq 0$ and $\mathcal{P}^{\mathrm{am}} \geq 0$. If $C^{\mathrm{am}} < S - K$, buy the call, exercise it at once by paying K, and sell the share received for S: the inflow is $S - K - C^{\mathrm{am}} > 0$ and nothing is left to settle. If $\mathcal{P}^{\mathrm{am}} < K - S$, buy the put and the share and deliver the share at once for K: the inflow is $K - S - \mathcal{P}^{\mathrm{am}} > 0$ and nothing is left to settle. Hence $C^{\mathrm{am}} \geq (S - K)^+$ and $\mathcal{P}^{\mathrm{am}} \geq (K - S)^+$, and by Definition 17.2 the time value of an American option is nonnegative.

(c) The bound $C^{\mathrm{am}} \leq S$. Suppose $C^{\mathrm{am}} > S$. Write the American call and buy one share; the inflow is $C^{\mathrm{am}} - S > 0$. Income only adds to the position: dividends $D_i > 0$ are deposited in case (I), and in case (Y) one share with the yield $q \geq 0$ reinvested is e^{qT} copies of the strategy of Proposition 16.15(i), so that it holds $e^{qT}e^{-q(T-t)} = e^{qt} \geq 1$ shares at the date t. If the call is exercised at some time $t_e \leq T$, deliver one share and deposit the K received: the position then holds a positive amount of cash and, in case (Y), the remaining $e^{qt_e} - 1 \geq 0$ shares, which are worth a nonnegative amount at T by limited liability. If the call is never exercised, nothing is owed and the position holds at least one share, worth at least $S_T \geq 0$. Every policy ends nonnegative.

(d) The bound $\mathcal{P}^{\mathrm{am}} \leq K$. Suppose $\mathcal{P}^{\mathrm{am}} > K$. Write the American put and deposit K; the inflow that remains is $\mathcal{P}^{\mathrm{am}} - K > 0$. If the put is exercised at some time $t_e \leq T$, the deposit is worth $Ke^{rt_e} \geq K$, because $r \geq 0$: pay K out of it, receive the share and sell it at once for $S_{t_e} \geq 0$. The cash $K(e^{rt_e} - 1) + S_{t_e} \geq 0$ that remains is deposited until T. If the put is never exercised, nothing

is owed and the deposit is worth $Ke^{rT} > 0$ at T. Every policy ends nonnegative.

(e) Maturity. Let $T_1 < T_2$ and suppose that the American option with expiry T_2 is cheaper than the American option of the same type and strike with expiry T_1. Buy the T_2 option and write the T_1 option; the difference is the positive inflow, and the horizon of the comparison is T_2. If the T_1 option is exercised against the position at some time $t_e \le T_1$, exercise the T_2 option at the same instant, which is allowed because $t_e < T_2$. For calls, the K received from the one exercise is the K paid in the other, and the share received is the share delivered; for puts the same holds with the two roles exchanged. The two exercises cancel and nothing is held afterwards. If the T_1 option is never exercised, nothing is owed, and the T_2 option is exercised at T_2 if it is then in the money and left to lapse otherwise, a nonnegative payoff. Every policy ends nonnegative.

Step 4 (later dates). Each argument of Steps 1 to 3 has the property required by Lemma 17.5(i). When a European bound fails, the position is the dominating portfolio held long and the dominated one held short; when an American bound fails, it is the position of Step 3. In both cases it is specified by the spot price, the forward price, the income still to come, the remaining lives, K, r and the quoted prices; its holdings change only at the exercise times of its options, each exercised at most once, at the date T' of Step 3(a) and, in case (I), at the ex-dividend dates t_i, where a dividend is paid into the cash account or repays a loan, so at most a number of times fixed in advance; it brings in a positive amount; and its value at the horizon (T, or T_2 in Step 3(e)) was bounded below by 0 on every path with nonnegative prices, under every exercise policy, using only the standing assumptions. The arguments were written for arbitrary values of these inputs, and the standing assumptions hold at every date, so the same arguments apply at any date at which the prices are quoted. Lemma 17.5(i) therefore gives every bound at every date t earlier than the expiries it involves. □

C.2 Proof of Theorem 17.16: No early exercise of an American call without dividends

The statement is on page 201. The notation and the standing assumptions are those of §17.4.

Proof. We use exercise matching (Lemma 17.5(ii)) for part (i), the bounds of Proposition 17.6 at a later date for part (ii), and dominance at the far expiry (Lemma 16.8(i)) for part (iii). All three parts rest on the identity $x^+ - x = (-x)^+$, valid for every real x (Step 2 of the proof of Proposition 17.6).

Part (i). Proposition 17.6(i) gives $C^{\mathrm{am}} \ge C$. Suppose that $C^{\mathrm{am}} > C$. Write the American call and buy the European call; the difference $C^{\mathrm{am}} - C > 0$ is the inflow, which is set aside. The rest of the position is followed to T under every exercise policy of the holder of the American call.

- If the American call is exercised against the position at some time $t_e < T$, borrow one share, deliver it to the holder for K and deposit the K received. From t_e on the position holds the European call, a short position of one share and the deposit. Nothing is owed to the lender of the share before T, because the share pays no income in case (N). At T the deposit is worth $Ke^{r(T-t_e)} \ge K$, because $r \ge 0$; the European call pays $(S_T - K)^+$; and buying the share back to return it costs S_T. The position is therefore worth

$$(S_T - K)^+ - S_T + Ke^{r(T-t_e)} \;\ge\; (S_T - K)^+ - (S_T - K) \;=\; (K - S_T)^+ \;\ge\; 0,$$

  by the identity at $x = S_T - K$.
- If the American call is not exercised before T, its writer owes at most $(S_T - K)^+$ at T (Setup of §17.2), and the European call pays exactly that amount. The position is worth a nonnegative amount at T.

Every exercise policy leaves a nonnegative value at T after a positive inflow, which Lemma 17.5(ii) excludes. Hence $C^{\mathrm{am}} \le C$, and with $C^{\mathrm{am}} \ge C$ the two prices are equal.

Part (ii). Fix a date $t < T$. Part (i) holds at the date t by Lemma 17.5(i). Its proof has the property required there: for any values of the inputs, the position it opens when $C^{\text{am}} > C$ is specified by K, r and the two quoted prices, changes its holdings only at the exercise time of the American call, which comes at most once, brings in a positive amount, and is worth a nonnegative amount at T on every path with nonnegative prices, under every exercise policy. The reverse inequality $C_t^{\text{am}} \geq C_t$ is Proposition 17.6(i) at the date t. The lower bound of Proposition 17.6(iii) in case (N) holds at t by the last sentence of that proposition. With probability one, therefore,

$$C_t^{\text{am}} = C_t \;\geq\; S_t - Ke^{-r\tau} \;>\; S_t - K,$$

where the last step uses $Ke^{-r\tau} < K$, true because $r > 0$, $\tau > 0$ and $K > 0$. The price of the call exceeds its exercise value, which is the definition of strict suboptimality in the Setup of §17.4.

For the second claim, compare the two courses of action at T. A holder who exercises at t and borrows the strike holds one share and owes $Ke^{r\tau}$ at T, a position worth $S_T - Ke^{r\tau}$. A holder who keeps the call to T receives $(S_T - K)^+$. Since a positive part is at least its argument and $Ke^{r\tau} > K$,

$$(S_T - K)^+ \;\geq\; S_T - K \;>\; S_T - Ke^{r\tau}$$

in every state.

Part (iii). Let $T_1 < T_2$, and take T_2 as the horizon of Lemma 16.8. No American option is involved. Consider the strategy that, at time 0, buys the call that expires at T_2 and writes the call that expires at T_1; its price is $C(K, T_2) - C(K, T_1)$. At T_1 the strategy acts as follows.

- If $S_{T_1} > K$, the written call is exercised against it. Sell one share short for S_{T_1}, pay the holder the $S_{T_1} - K$ owed out of the proceeds, and deposit the remaining K. At T_2 the bought call pays $(S_{T_2} - K)^+$, the deposit is worth $Ke^{r(T_2 - T_1)} \geq K$ because $r \geq 0$, and closing the short sale costs S_{T_2}; no dividend is owed on the borrowed share in case (N). The strategy is worth

  $$(S_{T_2} - K)^+ - S_{T_2} + Ke^{r(T_2 - T_1)} \;\geq\; (S_{T_2} - K)^+ - (S_{T_2} - K) \;=\; (K - S_{T_2})^+ \;\geq\; 0,$$

  by the identity at $x = S_{T_2} - K$.
- If $S_{T_1} \leq K$, the written call lapses and nothing is done. At T_2 the strategy is worth $(S_{T_2} - K)^+ \geq 0$.

The strategy trades only at T_1, on what is known then, and no cash is added or withdrawn, so it is self-financing. Its payoff at T_2 is nonnegative in every state, which is the payoff of the empty strategy or more. Lemma 16.8(i) gives $C(K, T_2) - C(K, T_1) \geq 0$. □

C.3 Proof of Proposition 17.18: The American put carries an early-exercise premium

The statement is on page 202. The notation and the standing assumptions are those of §17.4.

Proof. We read part (i) off the bounds of Proposition 17.6, prove part (ii) by comparing values at T, and prove part (iii) by exercise matching (Lemma 17.5(ii)).

Part (i). Proposition 17.6(ii) gives $\mathcal{P} \leq Ke^{-rT}$. The inequality $Ke^{-rT} < K - S$ is the hypothesis $S < K(1 - e^{-rT})$, rearranged. That hypothesis also gives $S < K$, so Proposition 17.6(i) gives $\mathcal{P}^{\text{am}} \geq (K - S)^+ = K - S$. This is the chain displayed in part (i) of Proposition 17.18, and it gives $\mathcal{P}^{\text{am}} > \mathcal{P}$.

For the comparison of policies, consider a policy that never exercises before T. At T it exercises or lets the put lapse, and it receives $K - S_T$ or 0, in either case at most $(K - S_T)^+$. Now exercise at once instead: buy the share for S and deliver it for K, which leaves $K - S$, and buy the European

put for $\mathcal{P}$. The cash left is $K - S - \mathcal{P} > 0$ by the chain, and the European put pays $(K - S_T)^+$ at T, at least what the waiting policy receives. The comparison does not need a European put to trade: the $K - S$ received, deposited until T, grows to $(K - S)e^{rT} > K$, because $K - S > Ke^{-rT}$, while the waiting policy receives at most $(K - S_T)^+ \leq K$ by limited liability.

At a date $t < T$ the bounds of Proposition 17.6 hold with S_t and τ (Lemma 17.5(i)), and the comparison of policies is the same argument started at t on the event $\{S_t < K(1 - e^{-r\tau})\}$.

Part (ii). Work on the event $\{S_t = 0\}$. Exercising at t pays $K - S_t = K$, which is worth $Ke^{r\tau}$ at T once deposited. Take any other policy. In a state in which it also exercises at t, it receives the same K. In a state in which it does not, it exercises at some time $t_e \in (t, T]$ or never. Exercise at t_e pays $K - S_{t_e} \leq K$ by limited liability, and that amount is worth at most $Ke^{r(T-t_e)}$ at T, which is strictly less than $Ke^{r\tau}$ because $r > 0$ and $T - t_e < \tau$. Never exercising pays $0 < Ke^{r\tau}$. Immediate exercise is therefore worth at least as much at T as every other policy, and strictly more in every state in which that policy waits. For the price, parts (i) and (ii) of Proposition 17.6 at the date t (Lemma 17.5(i)) give, with probability one on the event,

$$K = (K - S_t)^+ \;\leq\; \mathcal{P}_t^{\mathrm{am}} \;\leq\; K.$$

Part (iii). Let $r = 0$. Proposition 17.6(i) gives $\mathcal{P}^{\mathrm{am}} \geq \mathcal{P}$. Suppose that $\mathcal{P}^{\mathrm{am}} > \mathcal{P}$. Write the American put and buy the European put; the difference $\mathcal{P}^{\mathrm{am}} - \mathcal{P} > 0$ is the inflow, which is set aside.

- If the American put is exercised against the position at some time $t_e < T$, borrow K, pay it to the holder and receive one share. Hold the share and the European put to T. The loan is repaid as K at T, because $r = 0$, and the share and the put pay

  $$S_T + (K - S_T)^+ = \max(S_T, K) \;\geq\; K,$$

  which is enough to repay it. The position is worth $\max(S_T, K) - K \geq 0$ at T.
- If the American put is not exercised before T, its writer owes at most $(K - S_T)^+$ at T (Setup of §17.2), and the European put pays exactly that amount.

Every exercise policy leaves a nonnegative value at T after a positive inflow, which Lemma 17.5(ii) excludes. Hence $\mathcal{P}^{\mathrm{am}} = \mathcal{P}$. With $r > 0$ the loan grows to $Ke^{r(T-t_e)} > K$ when $t_e < T$, the share and the put may not cover it, and the argument fails, as part (i) says it must. □

C.4 Proof of Proposition 17.20: American calls on assets with income

The statement is on page 204. The notation and the standing assumptions are those of §17.4.

Proof. We use exercise matching with the share, started at a later date, for parts (i) and (ii): the violation trade of the proof of Lemma 17.5(i), with reading (c) of Definition 16.7. Parts (iii) and (iv) are read off the bounds of Proposition 17.6 at a later date, followed by a comparison of policies.

Part (i). Let A be the event that $C_t^{\mathrm{am}} < S_t - Ke^{-r(t'-t)}$; it is known at t. No dividend goes ex in the open interval (t, t'): for $i \geq 1$ the dividend D_i went ex at $t_i \leq t$, and the next one, D_{i+1}, goes ex at t' itself (when $i = n$ there is no next one and $t' = T$). Consider the strategy that holds nothing before t and, at t and on A only, buys the American call, sells one share short and deposits the proceeds $S_t - C_t^{\mathrm{am}}$. Its value at t is zero, so it is a strategy with a late start. On A the deposit splits as

$$S_t - C_t^{\mathrm{am}} = Ke^{-r(t'-t)} + (S_t - Ke^{-r(t'-t)} - C_t^{\mathrm{am}}),$$

the present value of the strike at t' and a surplus that is positive on A. At the instant t'^- immediately before t' (at T itself when $i = n$) the first part has grown to K and the surplus to its value with interest. The strategy then closes the short sale.

- If $S_{t'-} > K$, exercise the call: pay the K, receive the share and return it to its lender.
- If $S_{t'-} \le K$, buy the share in the market for $S_{t'-} \le K$ out of the K and return it to its lender; the call is not exercised then. When $t' < T$ the call is still held, and it can only add to the final value, because never exercising it costs nothing.

In both cases the borrowed share is returned before it goes ex at t', so no dividend was owed to its lender. What remains is at least the grown surplus, which is deposited until T. The payoff is zero off A and positive on A. By reading (c) of Definition 16.7 the strategy is an arbitrage unless $P(A) = 0$. Hence $C_t^{\mathrm{am}} \ge S_t - Ke^{-r(t'-t)}$ with probability one. The right-hand side exceeds $S_t - K$ because $r > 0$, $t' > t$ and $K > 0$, so the price of the call exceeds its exercise value.

Part (ii). The construction is the same, started at t_i^- with the share cum dividend. Let A be the event that $C^{\mathrm{am}}_{t_i^-} < S_{t_i^-} - D_i - Ke^{-r(t_{i+1}-t_i)}$, known at t_i^-. At t_i^- and on A only, buy the American call, sell one share short for $S_{t_i^-}$ and deposit the proceeds, which split as

$$S_{t_i^-} - C^{\mathrm{am}}_{t_i^-} = D_i + Ke^{-r(t_{i+1}-t_i)} + (S_{t_i^-} - D_i - Ke^{-r(t_{i+1}-t_i)} - C^{\mathrm{am}}_{t_i^-}),$$

with a surplus that is positive on A. When the share goes ex at t_i, the short seller owes its lender the dividend, and the D_i is paid out of the deposit; no interest has accrued. From t_i on the position is that of part (i) started at the date t_i with $t' = t_{i+1}$: the call, one share sold short, the amount $Ke^{-r(t'-t_i)}$ on deposit, and the surplus. The argument of part (i) runs from there and ends with at least the grown surplus, which is positive on A. Hence $P(A) = 0$, which is the bound. The bound minus the exercise value $S_{t_i^-} - K$ is

$$K\bigl(1 - e^{-r(t_{i+1}-t_i)}\bigr) - D_i,$$

which is positive exactly when D_i lies below the threshold, so that the price then exceeds the exercise value, and zero when D_i equals the threshold, so that the price is then at least the exercise value.

Part (iii). Limited liability at t_i and the exact drop give $S_{t_i^-} = S_{t_i} + D_i \ge D_i > K$, so the call is in the money just before the ex-date. Proposition 17.6(ii) at the date t_i (Lemma 17.5(i)) gives $C^{\mathrm{am}}_{t_i} \le S_{t_i}$, and $S_{t_i} = S_{t_i^-} - D_i < S_{t_i^-} - K$ because $D_i > K$.

For the premium, apply Proposition 17.6(ii) in case (I) to the European call at the instant t_i^- (Lemma 17.5(i)): $C_{t_i^-} \le S_{t_i^-} - I_{t_i^-}$. The present value $I_{t_i^-}$ of the dividends still to come is at least D_i, because D_i goes ex at t_i with no interest accruing before it and the later dividends only add to it. Hence, with Proposition 17.6(i) at t_i^- for the last step,

$$C_{t_i^-} \;\le\; S_{t_i^-} - D_i \;<\; S_{t_i^-} - K \;\le\; C^{\mathrm{am}}_{t_i^-}.$$

For the pathwise comparison, let the exerciser pay the strike with a loan of K taken at t_i^- and repaid out of the dividend at t_i, when no interest has accrued. From t_i on the exerciser holds the share and the cash $D_i - K > 0$ on deposit, and collects every later dividend. A policy that continues past t_i^- and exercises at some $t_e \in [t_i, T]$ receives the exercise value $S_{t_e} - K$ at t_e. At that time the exerciser holds

$$S_{t_e} + (D_i - K)e^{r(t_e - t_i)} \;>\; S_{t_e} \;>\; S_{t_e} - K,$$

or more. A policy that never exercises receives 0, and the exerciser holds at least $S_T + (D_i - K)e^{r(T-t_i)} > 0$ at T. In every state the exerciser is strictly ahead.

Part (iv). At the date t (Lemma 17.5(i)), Proposition 17.6(ii) in case (Y) and Proposition 17.6(i) give $C_t \le S_t e^{-q\tau}$ and $C_t^{\mathrm{am}} \ge (S_t - K)^+ \ge S_t - K$. The middle inequality $S_t e^{-q\tau} < S_t - K$ is the hypothesis $K < S_t(1 - e^{-q\tau})$, rearranged.

For the comparison of policies, a policy that never exercises before T receives at most $(S_T - K)^+$ at T, which the European call delivers for C_t. Exercising at t, selling the share and buying the European call out of the proceeds leaves $S_t - K - C_t > 0$ in cash on top of that payoff, as for the put in Proposition 17.18(i). The comparison does not need a European call to trade. Exercise at t, keep $e^{-q\tau}$ of the share received with its yield reinvested, which is one share at T (Proposition 16.15(i), started at t), and sell the remaining $1 - e^{-q\tau}$ of it for $S_t(1 - e^{-q\tau})$, out of which the strike is paid. The cash left is $S_t(1 - e^{-q\tau}) - K > 0$, and the holding is worth $S_T \geq (S_T - K)^+$ at T. □

C.5 Proof of Proposition 17.23: Call-spread and put-spread bounds: the strike curve is monotone and Lipschitz

The statement is on page 207. The notation and the standing assumptions are those of §17.5.

Proof. We use dominance at T (Lemma 16.8(i)) for the European options and exercise matching (Lemma 17.5(ii)) for the American ones.

Part (i). The *call spread*, long the K_1 call and short the K_2 call, is held from 0 to T and pays $(S_T - K_1)^+ - (S_T - K_2)^+$. On the three ranges of S_T this is

$$0 \ \ (S_T \leq K_1), \qquad S_T - K_1 \ \ (K_1 < S_T < K_2), \qquad (S_T - K_1) - (S_T - K_2) = K_2 - K_1 \ \ (S_T \geq K_2),$$

so the payoff lies in $[0, K_2 - K_1]$ in every state. It is at least the payoff of the empty portfolio, and Lemma 16.8(i) gives $C(K_1) - C(K_2) \geq 0$. It is at most the payoff of a zero-coupon bond paying $K_2 - K_1$ at T, which costs $e^{-rT}(K_2 - K_1)$ by Lemma 16.8(iv), and Lemma 16.8(i) gives $C(K_1) - C(K_2) \leq e^{-rT}(K_2 - K_1)$. The *put spread*, long the K_2 put and short the K_1 put, pays $(K_2 - S_T)^+ - (K_1 - S_T)^+$, which is

$$(K_2 - S_T) - (K_1 - S_T) = K_2 - K_1 \ \ (S_T \leq K_1), \qquad K_2 - S_T \ \ (K_1 < S_T < K_2), \qquad 0 \ \ (S_T \geq K_2),$$

again a payoff in $[0, K_2 - K_1]$, and the same two comparisons give the bounds for the puts. The options pay nothing before T, so the argument holds in every income case.

The first inequality says that C is nonincreasing; the second says that $|C(K_1) - C(K_2)| \leq e^{-rT}|K_1 - K_2|$, which is the Lipschitz property. For the derivative, let C be differentiable at a strike K. For every strike $K_2 > K$ the two inequalities, divided by $K_2 - K > 0$, read

$$-e^{-rT} \ \leq \ \frac{C(K_2) - C(K)}{K_2 - K} \ \leq \ 0,$$

and letting K_2 tend to K gives $-e^{-rT} \leq \partial_K C(K) \leq 0$. The same division applied to the put gives $0 \leq \partial_K \mathcal{P}(K) \leq e^{-rT}$.

Part (ii). Each of the four inequalities is proved by supposing that it fails and exhibiting a position that brings in a positive amount and is worth a nonnegative amount at T under every exercise policy of the holder of the written option. The positive amount is set aside in each case. Every matching exercises the bought option at the instant at which the written one is exercised, so the share received in one exercise is the share delivered in the other, and the income of the underlying plays no part.

Calls, upper bound. Suppose $C^{\mathrm{am}}(K_1) > C^{\mathrm{am}}(K_2) + (K_2 - K_1)$. Write the K_1 call, buy the K_2 call and deposit $K_2 - K_1$; the inflow that remains is $C^{\mathrm{am}}(K_1) - C^{\mathrm{am}}(K_2) - (K_2 - K_1) > 0$. If the K_1 call is exercised against the position at some time $t_e \leq T$, exercise the K_2 call at t_e: pay K_2, receive the share and deliver it for K_1. The shortfall $K_2 - K_1$ is covered by the deposit, which is worth $(K_2 - K_1)e^{rt_e} \geq K_2 - K_1$ because $r \geq 0$, and what is left of it is nonnegative. If the K_1 call is never exercised, nothing is owed; the K_2 call is kept and the deposit is worth $(K_2 - K_1)e^{rT} > 0$ at T.

Calls, monotonicity. Suppose $C^{\mathrm{am}}(K_1) < C^{\mathrm{am}}(K_2)$. Buy the K_1 call and write the K_2 call; the inflow is $C^{\mathrm{am}}(K_2) - C^{\mathrm{am}}(K_1) > 0$. If the K_2 call is exercised against the position at t_e, exercise the K_1 call at t_e: pay K_1, receive the share and deliver it for K_2, which leaves $K_2 - K_1 > 0$. If it is never exercised, nothing is owed, and the K_1 call is kept; exercised at T only if it is then in the money, it pays a nonnegative amount.

Puts, upper bound. Suppose $\mathcal{P}^{\mathrm{am}}(K_2) > \mathcal{P}^{\mathrm{am}}(K_1) + (K_2 - K_1)$. Write the K_2 put, buy the K_1 put and deposit $K_2 - K_1$; the inflow that remains is positive. If the K_2 put is exercised against the position at t_e, pay K_2 and receive the share, then exercise the K_1 put at t_e and deliver the share for K_1. The shortfall $K_2 - K_1$ is covered by the deposit, worth $(K_2 - K_1)e^{rt_e} \geq K_2 - K_1$. If the K_2 put is never exercised, nothing is owed, and the K_1 put and the deposit are worth a nonnegative amount at T.

Puts, monotonicity. Suppose $\mathcal{P}^{\mathrm{am}}(K_2) < \mathcal{P}^{\mathrm{am}}(K_1)$. Buy the K_2 put and write the K_1 put; the inflow is positive. If the K_1 put is exercised against the position at t_e, pay K_1 and receive the share, then exercise the K_2 put at t_e and deliver the share for K_2, which leaves $K_2 - K_1 > 0$. If it is never exercised, nothing is owed and the K_2 put is kept.

In each case every exercise policy leaves a nonnegative value at T after a positive inflow, which Lemma 17.5(ii) excludes. □

D Proofs for §18: Binomial Trees and Risk-Neutral Valuation

D.1 Proof of Proposition 18.11: European price on an n-step tree

The statement is on page 229. The notation and the standing assumptions are those of §18.3.

Proof. Part (i) is a backward induction on i that applies Theorem 18.3 at each node. Part (ii) is an induction on the number m of remaining steps that merges two binomial sums with Pascal's rule.

Part (i). The price of the derivative at time $i\Delta t$ is a quantity known at $i\Delta t$; a priori it may depend on the whole path so far, and not only on the node that the path has reached. The induction runs over i from n down to 0, and its claim at level i is that on every path the price at time $i\Delta t$ is $V_{i,j}$, where (i, j) is the node that the path has reached.

At level n the derivative pays $f(S_{n,j}) = V_{n,j}$ at once on a path that ends at node (n, j). Suppose that on some path its price at time T differed from $f(S_T)$. On the event that the price is below $f(S_T)$, buy the derivative with money borrowed from the cash account and collect the payment; on the event that it is above, sell the derivative and pay $f(S_T)$ out of the proceeds. Once the loan is repaid, this strategy, which holds nothing before T, leaves the absolute difference between the price and $f(S_T)$, a positive amount, on the union of the two events, and nothing elsewhere. That union contains a path, so it has positive real-world probability, and the strategy is an arbitrage by reading (c) of Definition 16.7, which allows the date $t = T$. Hence the price at T is $V_{n,j}$ on every path, which is the claim at level n.

Suppose that the claim holds at level $i + 1 \leq n$, and let a path be at node (i, j) at time $i\Delta t$. Seen from this node, the next step is a one-step tree: the stock, now at $S_{i,j}$, moves to $S_{i,j}u = S_{i+1,j+1}$ or to $S_{i,j}d = S_{i+1,j}$, and the derivative, bought at the node and sold one step later, fetches $V_{i+1,j+1}$ or $V_{i+1,j}$, by the induction hypothesis, whichever moves led to the node. It is therefore a one-step claim with $V_u = V_{i+1,j+1}$ and $V_d = V_{i+1,j}$. By Theorem 18.3(i), with $S_{i,j}$ in place of S, the portfolio (18.2) formed at the node pays exactly these two amounts one step later, and by (18.3) and (18.9) it costs

$$e^{-r\Delta t}\big[pV_{i+1,j+1} + (1-p)V_{i+1,j}\big] = V_{i,j}.$$

Now compare two positions held from time $i\Delta t$, in the sense of §16.2. The first buys the derivative at time $i\Delta t$ and sells it one step later. The second buys, at the node (i, j) that the path has reached, the portfolio just described, and closes it one step later. Each deposits its proceeds in the cash account until T, and each is managed with no cash added or withdrawn and with holdings that depend only on what is known when they are chosen. One step later the two fetch the same amount, $V_{i+1,j+1}$ after an up-move and $V_{i+1,j}$ after a down-move, so their payoffs at T are equal on every path. Lemma 16.8(v) therefore gives them equal values at time $i\Delta t$ with probability one: the price of the derivative equals the cost $V_{i,j}$ of the portfolio. (The proof of that lemma is the arbitrage of reading (c) of Definition 16.7, started at time $i\Delta t$ on the event that the two values differ.) Every path has positive real-world probability, so "with probability one" means "on every path", and this is the claim at level i.

Part (ii). Pascal's rule states that, for integers $1 \le k \le m-1$,

$$\binom{m-1}{k-1} + \binom{m-1}{k} = \frac{(m-1)!}{(k-1)!\,(m-k)!} + \frac{(m-1)!}{k!\,(m-1-k)!} = \frac{(m-1)!\,(k+(m-k))}{k!\,(m-k)!} = \binom{m}{k},$$

where the two fractions were brought to the common denominator $k!\,(m-k)!$. We prove (18.10) by induction on m. For $m = 0$ the node is terminal, the sum has the single term $k = 0$, and the formula reads $V_{n,j} = f(S_{n,j})$, which is the definition. Let $m \ge 1$, suppose that the formula holds at every node with $m-1$ steps remaining, and let (i, j) be a node with $n - i = m$. Its children $(i+1, j+1)$ and $(i+1, j)$ have $m-1$ steps remaining and carry the prices $S_{i,j}u$ and $S_{i,j}d$. By the hypothesis at the first child, and after renaming $k+1$ as k,

$$\begin{aligned} pV_{i+1,j+1} &= e^{-r(m-1)\Delta t} \sum_{k=0}^{m-1} \binom{m-1}{k} p^{k+1}(1-p)^{m-1-k} f(S_{i,j}u^{k+1}d^{m-1-k}) \\ &= e^{-r(m-1)\Delta t} \sum_{k=1}^{m} \binom{m-1}{k-1} p^{k}(1-p)^{m-k} f(S_{i,j}u^{k}d^{m-k}), \end{aligned}$$

and by the hypothesis at the second child,

$$(1-p)V_{i+1,j} = e^{-r(m-1)\Delta t} \sum_{k=0}^{m-1} \binom{m-1}{k} p^{k}(1-p)^{m-k} f(S_{i,j}u^{k}d^{m-k}).$$

Both sums are now combinations of the same quantities $p^k(1-p)^{m-k} f(S_{i,j}u^k d^{m-k})$. The term $k = m$ appears only in the first sum, with coefficient $\binom{m-1}{m-1} = 1 = \binom{m}{m}$; the term $k = 0$ appears only in the second, with coefficient $\binom{m-1}{0} = 1 = \binom{m}{0}$; and for $1 \le k \le m-1$ the two coefficients add up to $\binom{m}{k}$ by Pascal's rule. Adding the two displays and multiplying by $e^{-r\Delta t}$, the recursion (18.9) gives (18.10) at node (i, j). Only the defining recursion was used, so part (ii) rests on no trading assumption.

At the root, $i = j = 0$, $m = n$, $S_{0,0} = S$ and $n\Delta t = T$, which is the first equality of (18.11). For the second, group the 2^n paths by their number j of up-moves. There are $\binom{n}{j}$ paths with j up-moves; each has $\mathbb{Q}$-probability $p^j(1-p)^{n-j}$ and ends at $S_T = Su^j d^{n-j}$. Hence $\mathbb{Q}(Y_n = j) = \binom{n}{j} p^j (1-p)^{n-j}$, which is the $\mathrm{Bin}(n, p)$ mass of Definition 1.9, as it must be for the number of up-moves in n independent moves that are each up with probability p; and

$$\mathbb{E}^{\mathbb{Q}}[f(S_T)] = \sum_{j=0}^{n} \binom{n}{j} p^j (1-p)^{n-j} f(Su^j d^{n-j}). \qquad \square$$

D.2 Proof of Proposition 18.13: Dynamic replication on an n-step tree

The statement is on page 231. The notation and the standing assumptions are those of §18.4.

Proof. We proceed by induction on i, solving at each node the linear system of Theorem 18.3; the law of one price then gives the price, and a second induction, on the number of remaining steps, gives (iii).

Step 1 (one node). Fix a non-terminal node (i,j) and consider the one-step tree that starts there: the price $S_{i,j}$ moves to $S_{i,j}u$ or to $S_{i,j}d$, and the two payoffs are taken to be the numbers $V_u = V_{i+1,j+1}$ and $V_d = V_{i+1,j}$. By Theorem 18.3(i), which uses no price of any derivative, the system (18.5) with these data has exactly one solution. Its share holding is $(V_{i+1,j+1} - V_{i+1,j})/(S_{i,j}(u-d)) = \Delta_{i,j}$, and its cost is, by (18.3) and then by the defining recursion (18.9),

$$e^{-r\Delta t}\big[pV_{i+1,j+1} + (1-p)V_{i+1,j}\big] = V_{i,j}.$$

Its bond holding is the cost minus the cost of the shares, $V_{i,j} - \Delta_{i,j}S_{i,j} = B_{i,j}$. Hence the holding $(\Delta_{i,j}, B_{i,j})$ of (18.13) costs $V_{i,j}$ at node (i,j) and satisfies

$$\Delta_{i,j}S_{i,j}u + B_{i,j}e^{r\Delta t} = V_{i+1,j+1}, \qquad \Delta_{i,j}S_{i,j}d + B_{i,j}e^{r\Delta t} = V_{i+1,j}. \tag{D.1}$$

Step 2 (induction along a path). The claim at level i is that, on every path, the value of the strategy entering the node (i,j) reached at time $i\Delta t$ is $V_{i,j}$. At level 0 the entering value is the initial capital $V_{0,0}$. Suppose that the claim holds at level $i < n$, and let the path be at node (i,j). By Step 1 the entering value $V_{i,j}$ is exactly the cost of the holding $(\Delta_{i,j}, B_{i,j})$, so that holding is bought with no money added or withdrawn: the strategy is self-financing at this node, and its value there is $V_{i,j}$. One step later the holding is worth the left-hand side of (D.1), that is, $V_{i+1,j+1}$ after an up-move and $V_{i+1,j}$ after a down-move. This is the value entering the next node, which is the claim at level $i+1$. By induction the strategy is self-financing and worth $V_{i,j}$ at every node, and at level n its value is $V_{n,j} = f(S_{n,j}) = f(S_T)$ on every path. This proves (i); no step used a price of the derivative.

Step 3 (the price). Buying the derivative at time 0 and holding it to maturity is a self-financing strategy with payoff $f(S_T)$. The strategy of (i) is a self-financing strategy with price $V_{0,0}$ and the same payoff on every path. By Lemma 16.8(iii) the price of the derivative is $V_{0,0}$, and the arbitrage behind the lemma is the following. If the derivative sells for $x > V_{0,0}$ at time 0, sell it, run the strategy from the capital $V_{0,0}$ and deposit $x - V_{0,0}$: at maturity the strategy pays the amount $f(S_T)$ owed on the derivative, and the deposit is worth $(x - V_{0,0})e^{rT} > 0$, an arbitrage by reading (b) of Definition 16.7. If $x < V_{0,0}$, buy the derivative, run the negative of the strategy (every holding reversed), which brings in $V_{0,0}$ at time 0 and owes $f(S_T)$ at maturity, and deposit $V_{0,0} - x$: the derivative pays what is owed, and the deposit is worth $(V_{0,0} - x)e^{rT} > 0$. Both trades buy or sell the derivative at time 0 and never trade it again.

Step 4 (constant holdings). Suppose first that $f(S_{n,j}) = c + c'S_{n,j}$ for every j. We show by induction on the number $m = n - i$ of remaining steps that $V_{i,j} = ce^{-rm\Delta t} + c'S_{i,j}$ at every node with m steps remaining. For $m = 0$ this is the hypothesis. If it holds with $m-1$ steps remaining, the children of a node (i,j) with $n - i = m$ carry $ce^{-r(m-1)\Delta t} + c'S_{i,j}u$ and $ce^{-r(m-1)\Delta t} + c'S_{i,j}d$, and (18.9), with $p + (1-p) = 1$ and Proposition 18.7(i), gives

$$V_{i,j} = e^{-r\Delta t}\Big[ce^{-r(m-1)\Delta t} + c'S_{i,j}\big(pu + (1-p)d\big)\Big] = ce^{-rm\Delta t} + c'S_{i,j}.$$

Then (18.13), in which the terms in c of the two children cancel, gives $\Delta_{i,j} = c'(S_{i,j}u - S_{i,j}d)/(S_{i,j}(u-d)) = c'$ at every non-terminal node, and $B_{i,j} = V_{i,j} - c'S_{i,j} = ce^{-r(n-i)\Delta t}$. The share holding never changes, and the bond holding chosen at each node is the holding $ce^{-r(n-i+1)\Delta t}$ of the previous node grown by $e^{r\Delta t}$, so no bond is bought or sold either: the position held at time 0 is never traded again. Conversely, suppose that $\Delta_{i,j} = c'$ at every non-terminal node, and look at the nodes $(n-1,j)$, $0 \le j \le n-1$, whose children are terminal. Since $S_{n,j+1} - S_{n,j} = S_{n-1,j}u - S_{n-1,j}d = S_{n-1,j}(u-d)$, the definition (18.13) of $\Delta_{n-1,j}$ reads $f(S_{n,j+1}) - f(S_{n,j}) = c'(S_{n,j+1} - S_{n,j})$. Adding these identities for the indices $0, 1, \dots, j-1$, the left-hand sides telescope and give $f(S_{n,j}) = f(S_{n,0}) + c'(S_{n,j} - S_{n,0})$, which is the affine form with $c = f(S_{n,0}) - c'S_{n,0}$. □

D.3 Proof of Proposition 18.17: Backward induction gives the American price

The statement is on page 235. The notation and the standing assumptions are those of §18.5.

Proof. We proceed by induction on i, as in Proposition 18.13, with the transfers as the one new feature; the price then follows from two explicit arbitrages, and the maximum from a backward induction over path prefixes.

Step 1 (one node). Fix a node (i,j) with $i<n$ and consider the one-step tree that starts there, with the two payoffs $V_u = V^{\mathrm{am}}_{i+1,j+1}$ and $V_d = V^{\mathrm{am}}_{i+1,j}$. By Theorem 18.3(i), which uses no price of any derivative, the system (18.5) with these data and with $S_{i,j}$ in place of S has exactly one solution. By (18.2) its share holding is the $\Delta^{\mathrm{am}}_{i,j}$ of (18.16), and by (18.3) and the definition (18.14) its cost is

$$e^{-r\Delta t}\bigl[pV^{\mathrm{am}}_{i+1,j+1}+(1-p)V^{\mathrm{am}}_{i+1,j}\bigr] = V^{\mathrm{cont}}_{i,j}.$$

Its bond holding is the cost minus the cost of the shares, $V^{\mathrm{cont}}_{i,j} - \Delta^{\mathrm{am}}_{i,j}S_{i,j}$. Hence the holding named in (i) costs $V^{\mathrm{cont}}_{i,j}$ at node (i,j), and one step later it is worth

$$V^{\mathrm{am}}_{i+1,j+1} \ \text{ after an up-move,} \qquad V^{\mathrm{am}}_{i+1,j} \ \text{ after a down-move.} \tag{D.2}$$

Step 2 (the seller's hedge). The claim at level i is: on every path on which the holder has not exercised before step i, the hedging portfolio is worth $V^{\mathrm{am}}_{i,j}$ on entering the node (i,j) that the path reaches at time $i\Delta t$. At level 0 the portfolio is the initial capital $V^{\mathrm{am}}_{0,0}$. Suppose that the claim holds at level $i<n$, and that the holder does not exercise at node (i,j). By (18.15), $V^{\mathrm{am}}_{i,j} \ge V^{\mathrm{cont}}_{i,j}$, so the seller can move $V^{\mathrm{am}}_{i,j} - V^{\mathrm{cont}}_{i,j} \ge 0$ to the separate deposit, and the remaining $V^{\mathrm{cont}}_{i,j}$ is, by Step 1, exactly the cost of the holding of (i). By (D.2) that holding is worth the American value of the next node one step later, which is the claim at level $i+1$. The transfer moves money from one part of the seller's position to the other, and the holding is bought with what remains, so no money is added or withdrawn.

It remains to see that exercise is covered. Let the holder exercise at a node (i,j), at whatever step $i\le n$ and for whatever reason; at $i=n$ this is the payment of $f(S_{n,j})$ at maturity. The portfolio entering that node is worth $V^{\mathrm{am}}_{i,j}$, and $V^{\mathrm{am}}_{i,j} \ge f(S_{i,j})$: by (18.15) when $i<n$, and with equality when $i=n$. The seller pays $f(S_{i,j})$ out of the portfolio and adds the remainder $V^{\mathrm{am}}_{i,j} - f(S_{i,j}) \ge 0$ to the deposit. From then on the seller's position is the separate deposit alone, and every amount that entered it was nonnegative. This proves (i).

Step 3 (the holder's rule). Let the holder follow the rule of (ii), and let i^* be its exercise step on a given path. At a node (i,j) of the path with $i<i^*$ the rule continued, so $f(S_{i,j}) < V^{\mathrm{cont}}_{i,j}$, hence $V^{\mathrm{am}}_{i,j} = V^{\mathrm{cont}}_{i,j}$ by (18.15), and the transfer of (i) is zero. At the node (i^*,j) where the rule exercises, $V^{\mathrm{am}}_{i^*,j} = f(S_{i^*,j})$: if $i^*<n$, because $f(S_{i^*,j}) \ge V^{\mathrm{cont}}_{i^*,j}$ makes the maximum in (18.15) equal to $f(S_{i^*,j})$; if $i^*=n$, by step 2 of Procedure 18.16. By Step 2 the portfolio entering that node is worth $V^{\mathrm{am}}_{i^*,j} = f(S_{i^*\Delta t})$, and since nothing was transferred before, the hedge has been a self-financing strategy in the stock and the bond alone, started from $V^{\mathrm{am}}_{0,0}$. This proves (ii).

Step 4 (the price). Let the claim sell for x at time 0. The two arbitrages below are built as in the proof of Lemma 16.8; the lemma itself is not quoted, because an American claim has no fixed payoff: what it pays depends on its holder.

If $x > V^{\mathrm{am}}_{0,0}$, sell the claim, run the hedge of (i) from the capital $V^{\mathrm{am}}_{0,0}$ and deposit the surplus $x - V^{\mathrm{am}}_{0,0}$. The position costs nothing at time 0. Whatever the holder does, part (i) shows that the exercise value is paid out of the hedging portfolio, after which the seller holds the separate deposit, which is nonnegative, and the surplus, which is worth $(x - V^{\mathrm{am}}_{0,0})e^{rT} > 0$ at time T. The payoff is positive on every path: an arbitrage (Definition 16.7).

If $x < V^{\mathrm{am}}_{0,0}$, buy the claim, exercise it by the rule of (ii), run the negative of the hedge of (i) along that rule (every holding reversed) and deposit $V^{\mathrm{am}}_{0,0} - x$. The reversed hedge brings in $V^{\mathrm{am}}_{0,0}$ at time 0, so the position costs nothing. By (ii) the reversed hedge owes exactly $f(S_{i^*\Delta t})$ at the step i^* of

the rule, which is what exercising the claim pays at that step; the two cancel, and the deposit is worth $(V^{\mathrm{am}}_{0,0} - x)e^{rT} > 0$ at time T: an arbitrage again. Both trades deal in the claim at time 0 only. Hence, if there is no arbitrage, $x = V^{\mathrm{am}}_{0,0}$.

Step 5 (the maximum). Fix an exercise rule. Call a path prefix of length i *live* if the rule has not exercised before step i on the paths that begin with it, that is, if each of its shorter prefixes carries the decision "continue". A *continuation* of the prefix is a sequence of $n - i$ further moves, which completes it to a path. Its *weight* is the product of p for each of its up-moves and $1 - p$ for each of its down-moves, and the weights of the 2^{n-i} continuations sum to $\big(p + (1-p)\big)^{n-i} = 1$. The *value of the rule from a live prefix* of length i is the sum, over the continuations of the prefix, of

$$(\text{weight of the continuation}) \times e^{-r(i^*-i)\Delta t} f(S_{i^*\Delta t}),$$

where $i^* \geq i$ and $S_{i^*\Delta t}$ are evaluated on the completed path. We show by backward induction on i that the value of the rule from a live prefix that ends at node (i, j) is at most $V^{\mathrm{am}}_{i,j}$.

For $i = n$ the only continuation is the empty one, of weight 1, and $i^* = n$, so the value is $f(S_{n,j}) = V^{\mathrm{am}}_{n,j}$. Let $i < n$, and suppose that the bound holds for live prefixes of length $i + 1$. If the rule exercises at the prefix, then $i^* = i$ on every continuation, the factor $f(S_{i,j})$ is common to all the terms, and the weights sum to one: the value is $f(S_{i,j}) \leq V^{\mathrm{am}}_{i,j}$. If the rule continues at the prefix, its two one-step extensions are live prefixes of length $i + 1$, which end at the nodes $(i + 1, j + 1)$ and $(i + 1, j)$. Write y for the value of the rule from the prefix and y_u, y_d for its values from the up-extension and the down-extension. Every continuation of the prefix is a first move followed by a continuation of the corresponding extension; its weight is p or $1 - p$ times the weight of the latter, and $e^{-r(i^*-i)\Delta t} = e^{-r\Delta t} e^{-r(i^*-(i+1))\Delta t}$. Grouping the continuations by their first move,

$$y = e^{-r\Delta t}\big[p\, y_u + (1-p)\, y_d\big] \;\leq\; e^{-r\Delta t}\big[pV^{\mathrm{am}}_{i+1,j+1} + (1-p)V^{\mathrm{am}}_{i+1,j}\big] = V^{\mathrm{cont}}_{i,j} \;\leq\; V^{\mathrm{am}}_{i,j},$$

by the induction hypothesis, (18.14) and (18.15). The empty prefix is live, a continuation of it is a path, and the weight of that continuation is the $\mathbb{Q}$-probability of the path, so the value of the rule from the empty prefix is its risk-neutral value $\mathbb{E}^{\mathbb{Q}}[e^{-ri^*\Delta t} f(S_{i^*\Delta t})]$. The case $i = 0$ of the bound therefore reads: the risk-neutral value of every exercise rule is at most $V^{\mathrm{am}}_{0,0}$.

For the rule of (ii) every inequality above is an equality, by the same induction with "is at most" replaced by "equals". At $i = n$ the value is $V^{\mathrm{am}}_{n,j}$. At a live prefix of length $i < n$ where the rule exercises, $f(S_{i,j}) \geq V^{\mathrm{cont}}_{i,j}$, so the value $f(S_{i,j})$ is the maximum $V^{\mathrm{am}}_{i,j}$. Where it continues, $f(S_{i,j}) < V^{\mathrm{cont}}_{i,j}$, and the value is $y = e^{-r\Delta t}[pV^{\mathrm{am}}_{i+1,j+1} + (1-p)V^{\mathrm{am}}_{i+1,j}] = V^{\mathrm{cont}}_{i,j} = V^{\mathrm{am}}_{i,j}$. At the root the risk-neutral value of the rule of (ii) is therefore $V^{\mathrm{am}}_{0,0}$, which proves (18.17). □

D.4 Proof of Proposition 18.21: Moments of the CRR step; Cox, Ross and Rubinstein, 1979

The statement is on page 239. The notation and the standing assumptions are those of §18.6.

Proof. We compute the mean and the variance of a two-point variable exactly, and then expand p in powers of $x = \sigma\sqrt{\Delta t}$.

Step 1 (exact moments). The variable R_1 equals x with probability p and $-x$ with probability $1 - p$, so

$$\mathbb{E}^{\mathbb{Q}}[R_1] = px - (1-p)x = (2p-1)\, x. \tag{D.3}$$

Since $R_1^2 = x^2$ in both states, $\mathbb{E}^{\mathbb{Q}}[R_1^2] = x^2$, and the shortcut (1.1) gives

$$\mathrm{Var}^{\mathbb{Q}}(R_1) = x^2 - \big(\mathbb{E}^{\mathbb{Q}}[R_1]\big)^2 = x^2\big(1 - (2p-1)^2\big) = 4p(1-p)\, x^2, \tag{D.4}$$

because $1 - (2p-1)^2 = \big(1 - (2p-1)\big)\big(1 + (2p-1)\big) = (2-2p)(2p)$. Both formulas are exact.

Step 2 (the numerator of p). Proposition 7.34, whose remainders are controlled by Corollary 7.32, gives $e^y = 1 + y + O(y^2)$ as $y \to 0$ and $e^{-x} = 1 - x + x^2/2 + O(x^3)$ as $x \to 0$, the higher terms of that proposition being absorbed in the O-terms. With $y = r\Delta t$,

$$e^{r\Delta t} - e^{-x} = \big(1 + r\Delta t + O(\Delta t^2)\big) - \Big(1 - x + \frac{x^2}{2} + O(x^3)\Big) = x - \frac{x^2}{2} + r\Delta t + O(\Delta t^{3/2}),$$

because $O(x^3) = O(\Delta t^{3/2})$ and a term $O(\Delta t^2)$ is $O(\Delta t^{3/2})$. Substituting $r\Delta t = (r/\sigma^2)\,x^2$,

$$e^{r\Delta t} - e^{-x} = x\Big[1 + \Big(\frac{r}{\sigma^2} - \frac{1}{2}\Big)x + O(x^2)\Big].$$

Step 3 (the denominator). By the same proposition, $e^x = 1 + x + x^2/2 + O(x^3)$ and $e^{-x} = 1 - x + x^2/2 + O(x^3)$, so

$$e^x - e^{-x} = 2x + O(x^3) = 2x\big(1 + O(x^2)\big).$$

Step 4 (the quotient). Dividing the result of Step 2 by that of Step 3,

$$p = \frac{1 + (r/\sigma^2 - 1/2)\,x + O(x^2)}{2\,(1 + O(x^2))}.$$

By the expansion $1/(1-y) = 1 + y + O(y^2)$ of Proposition 7.34, applied to a quantity y that is $O(x^2)$, the factor $1/(1 + O(x^2))$ is $1 + O(x^2)$. Multiplying out,

$$p = \frac{1}{2} + \frac{1}{2}\Big(\frac{r}{\sigma^2} - \frac{1}{2}\Big)x + O(x^2) = \frac{1}{2} + \frac{(r - \sigma^2/2)\sqrt{\Delta t}}{2\sigma} + O(\Delta t),$$

because $\frac{1}{2}\,(r/\sigma^2 - 1/2)\,\sigma\sqrt{\Delta t} = (r - \sigma^2/2)\sqrt{\Delta t}/(2\sigma)$ and $O(x^2) = O(\Delta t)$. This is (i).

Step 5 (mean and variance). By (i), $2p - 1 = (r - \sigma^2/2)\sqrt{\Delta t}/\sigma + O(\Delta t)$, and by (D.3),

$$\mathbb{E}^{\mathbb{Q}}[R_1] = (2p - 1)\,\sigma\sqrt{\Delta t} = (r - \sigma^2/2)\Delta t + O(\Delta t^{3/2}),$$

which is (ii). In particular $\mathbb{E}^{\mathbb{Q}}[R_1] = O(\Delta t)$, so its square is $O(\Delta t^2)$, and (D.4) gives $\mathrm{Var}^{\mathbb{Q}}(R_1) = x^2 + O(\Delta t^2) = \sigma^2\Delta t + O(\Delta t^2)$, which is (iii).

Step 6 (any rate, any step). Steps 1 to 5 used the number r only through the expansion $e^{r\Delta t} = 1 + r\Delta t + O(\Delta t^2)$ of Step 2, which holds for every real constant in place of r. With μ in place of r the probability is p_{real}, and the same steps give

$$p_{\text{real}} = \frac{1}{2} + \frac{(\mu - \sigma^2/2)\sqrt{\Delta t}}{2\sigma} + O(\Delta t),$$
$$\mathbb{E}[R_1] = (\mu - \sigma^2/2)\Delta t + O(\Delta t^{3/2}), \qquad \mathrm{Var}(R_1) = \sigma^2\Delta t + O(\Delta t^2).$$

This is (iv). Finally, under $\mathbb{Q}$ each move is up with probability p whatever the earlier moves, and under P with probability p_{real}, so each R_i takes the values x and $-x$ with the same probabilities as R_1. □

D.5 Proof of Proposition 18.25: Risk-neutral probability with a yield, for a currency and for a futures price

The statement is on page 242. The notation and the standing assumptions are those of §18.7.

Proof. For (i) and (ii) we apply Theorem 18.3 to a traded asset that pays no income, the underlying held with its income reinvested; for (iii) we solve the replication system with futures contracts in place of shares.

Part (i). Consider the holding that starts with one unit of the underlying and reinvests all income. It costs S today, pays nothing before time Δt, and then consists of $e^{q\Delta t}$ units, worth $Sue^{q\Delta t}$ in the up state and $Sde^{q\Delta t}$ in the down state. It is therefore an asset without income on a one-step tree with the factors $ue^{q\Delta t}$ and $de^{q\Delta t}$, and by the Setup of §18.7, in which the yield is reinvested in the underlying, every position in the underlying over the step is a position in this holding. For these factors the condition (18.1) reads $de^{q\Delta t} < e^{r\Delta t} < ue^{q\Delta t}$, which is the hypothesis $d < e^{(r-q)\Delta t} < u$, so Theorem 18.3 applies with $ue^{q\Delta t}$ and $de^{q\Delta t}$ in place of u and d. By (18.2) the replica holds

$$\frac{V_u - V_d}{S\,(ue^{q\Delta t} - de^{q\Delta t})} = e^{-q\Delta t}\,\frac{V_u - V_d}{S(u-d)}$$

units of the reinvested holding, that is, this many units of the underlying at the start of the step; at its end they have grown by the factor $e^{q\Delta t}$ to $(V_u - V_d)/(S(u-d))$ units. Its bond holding is

$$e^{-r\Delta t}\,\frac{ue^{q\Delta t}V_d - de^{q\Delta t}V_u}{ue^{q\Delta t} - de^{q\Delta t}} = e^{-r\Delta t}\,\frac{uV_d - dV_u}{u-d}.$$

By (18.4) and (18.3) the price is $e^{-r\Delta t}[pV_u + (1-p)V_d]$ with

$$p = \frac{e^{r\Delta t} - de^{q\Delta t}}{ue^{q\Delta t} - de^{q\Delta t}} = \frac{e^{r\Delta t} - de^{q\Delta t}}{(u-d)\,e^{q\Delta t}} = \frac{e^{(r-q)\Delta t} - d}{u-d},$$

which lies in $(0,1)$ by the hypothesis. Finally $\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = pSu + (1-p)Sd = S\big(d + p\,(u-d)\big) = Se^{(r-q)\Delta t}$. A forward contract with delivery price K and delivery at Δt (Definition 16.5) is the derivative with $V_u = Su - K$ and $V_d = Sd - K$. By (18.19) and Theorem 1.2, applied to $\mathbb{E}^{\mathbb{Q}}$, it is worth

$$e^{-r\Delta t}\big[p\,(Su-K) + (1-p)(Sd-K)\big] = e^{-r\Delta t}\big(\mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] - K\big),$$

which vanishes exactly at $K = \mathbb{E}^{\mathbb{Q}}[S_{\Delta t}] = Se^{(r-q)\Delta t}$. This is therefore the forward price on the tree, in agreement with the forward price $Se^{(r-q)T}$ of Proposition 16.15(ii) for delivery at $T = \Delta t$.

Part (ii). One unit of foreign currency placed in a foreign deposit grows to $e^{r_{\text{for}}\Delta t}$ units, whatever the exchange rate does. A foreign currency is therefore an underlying whose yield $q = r_{\text{for}}$ is paid in units of itself, which is how Proposition 16.19 is proved, and (i) applies, its forward-price computation included.

Part (iii), the position. A position of Δ_F long futures contracts and B in the bond costs B today, because a futures position is free. At time Δt it is worth the settlement of the contracts plus the bond:

$$\Delta_F F(u-1) + Be^{r\Delta t} \ \text{ in the up state,} \qquad \Delta_F F(d-1) + Be^{r\Delta t} \ \text{ in the down state.} \tag{D.5}$$

Part (iii), no arbitrage. If $d \geq 1$, one long contract costs nothing and pays $F(u-1) > 0$ or $F(d-1) \geq 0$. If $u \leq 1$, one short contract costs nothing and pays $F(1-u) \geq 0$ or $F(1-d) > 0$. Both states have positive probability, so each position is an arbitrage. Conversely, let $d < 1 < u$, and let a position (Δ_F, B) cost $B \leq 0$ and have nonnegative payoffs. The up-state payoff gives $\Delta_F F(u-1) \geq -Be^{r\Delta t} \geq 0$, hence $\Delta_F \geq 0$, because $F(u-1) > 0$; the down-state payoff gives $\Delta_F F(d-1) \geq -Be^{r\Delta t} \geq 0$, hence $\Delta_F \leq 0$, because $F(d-1) < 0$. So $\Delta_F = 0$, both payoffs equal $Be^{r\Delta t} \leq 0$, hence $B = 0$ and both payoffs vanish: the position is not an arbitrage. This is the argument of Proposition 18.1, with the gain $F(u-1)$ or $F(d-1)$ of a free contract in place of the gain $S(u - e^{r\Delta t})$ or $S(d - e^{r\Delta t})$ of a share bought with borrowed money.

Part (iii), replication. Setting the two payoffs (D.5) equal to V_u and V_d gives a linear system in Δ_F and B with determinant $F(u-1)e^{r\Delta t} - F(d-1)e^{r\Delta t} = F(u-d)e^{r\Delta t}$, which is nonzero because $F > 0$ and $u > d$, hence with exactly one solution. Subtracting the two equations gives $\Delta_F F(u-d) = V_u - V_d$. Substituting into the first,

$$Be^{r\Delta t} = V_u - \frac{(V_u - V_d)(u-1)}{u-d} = \frac{V_u(u-d) - V_u(u-1) + V_d(u-1)}{u-d} = \frac{V_u(1-d) + V_d(u-1)}{u-d}.$$

The coefficient of V_u is $p = (1-d)/(u-d)$ and that of V_d is $(u-1)/(u-d) = 1-p$, so the replica costs $B = e^{-r\Delta t}[pV_u + (1-p)V_d]$, and $0 < p < 1$ because $0 < 1-d < u-d$. The derivative and the replica have the same payoff in both states, so they have the same price by Lemma 16.8(iii): $V = B$. The arbitrage behind the lemma is that of Step 3 of the proof of Theorem 18.3. Finally $\mathbb{E}^{\mathbb{Q}}[F_{\Delta t}] = pFu + (1-p)Fd = F\big(d + p\,(u-d)\big) = F(d+1-d) = F$, and the p of (18.19) at $q = r$ is $(e^0 - d)/(u-d) = (1-d)/(u-d)$. □

D.6 Proof of Corollary 18.26: Pricing and hedging on an n-step tree with a yield or on a futures price

The statement is on page 243. The notation and the standing assumptions are those of §18.7.

Proof. We rerun, one Step each, the five results that the statement carries over: the backward induction (18.9), Proposition 18.11, Proposition 18.13, Procedure 18.16 and Proposition 18.17. The recursion and the Procedure are recipes on the tree that need only the new p, and each of the three propositions uses the one-step market of §18.1 at a single line, which Step 1 below replaces by the same two facts, proved for the new underlyings. As in the statement, F stands in place of S for a futures price.

Step 1 (one node). Since $u - d > 0$, the hypothesis $d < e^{b\Delta t} < u$ reads $0 < e^{b\Delta t} - d < u - d$, so that $p \in (0,1)$ and $1 - p = (u - e^{b\Delta t})/(u-d) \in (0,1)$. Fix a non-terminal node (i,j) and two numbers V_u and V_d, the amounts to be held on entering its up child and its down child, and put $\Delta = (V_u - V_d)/(S_{i,j}(u-d))$, with $F_{i,j}$ in place of $S_{i,j}$ for a futures price; the node price is positive in both cases, so Δ is defined. Consider the following holding at the node: with a yield, $e^{-q\Delta t}\Delta$ units and $e^{-r\Delta t}(uV_d - dV_u)/(u-d)$ in the bond; with a futures price, Δ contracts and $e^{-r\Delta t}[pV_u + (1-p)V_d]$ in the bond. Steps 2 and 6 show that it is the holding of (ii), or of (iii), when V_u and V_d are the values of the children. We show two facts: one step later the holding is worth V_u after an up-move and V_d after a down-move, with no money added or withdrawn during the step (the *value fact*); and it costs $e^{-r\Delta t}[pV_u + (1-p)V_d]$ at the node (the *cost fact*). For a stock without dividends these are the two facts that Theorem 18.3(i) supplies, through (18.5) and (18.3). The two holdings are the replicas of Proposition 18.25(i) and (iii) at the node price, and the two facts are checked here directly, because that proposition states its replicas under a no-arbitrage hypothesis that the facts do not need.

With a yield, the $e^{-q\Delta t}\Delta$ units have grown to Δ units one step later, by case (a) of the statement, and are worth $\Delta S_{i,j}u = u(V_u - V_d)/(u-d)$ after an up-move and $\Delta S_{i,j}d = d(V_u - V_d)/(u-d)$ after a down-move, while the bond is then worth $(uV_d - dV_u)/(u-d)$. The two sums are

$$\frac{uV_u - uV_d + uV_d - dV_u}{u-d} = V_u, \qquad \frac{dV_u - dV_d + uV_d - dV_u}{u-d} = V_d.$$

The reinvested units are a single position (Setup of §16.2) and the bond only grows, so nothing is added or withdrawn: this is the value fact. Writing $e^{-q\Delta t} = e^{-r\Delta t}e^{b\Delta t}$, the cost at the node is

$$e^{-q\Delta t}\Delta S_{i,j} + e^{-r\Delta t}\,\frac{uV_d - dV_u}{u-d} = e^{-r\Delta t}\,\frac{(e^{b\Delta t} - d)V_u + (u - e^{b\Delta t})V_d}{u-d} = e^{-r\Delta t}\big[pV_u + (1-p)V_d\big],$$

which is the cost fact.

With a futures price, the contracts are free, so the holding costs its bond holding $e^{-r\Delta t}[pV_u + (1-p)V_d]$: this is the cost fact. At the end of the step the contracts are settled, the settlement $\Delta F_{i,j}(u-1)$ or $\Delta F_{i,j}(d-1)$ is paid into the bond, and the contracts are then worth zero and can be closed at no cost (Definition 16.25), so nothing is added or withdrawn. With $b = 0$, $p = (1-d)/(u-d)$ and $1-p = (u-1)/(u-d)$, and the bond has grown to $pV_u + (1-p)V_d$. The value entering the up child is therefore

$$\Delta F_{i,j}(u-1) + pV_u + (1-p)V_d = \frac{(u-1)(V_u - V_d) + (1-d)V_u + (u-1)V_d}{u-d} = \frac{(u-d)V_u}{u-d} = V_u,$$

and the value entering the down child is

$$\Delta F_{i,j}(d-1) + pV_u + (1-p)V_d = \frac{(d-1)(V_u - V_d) + (1-d)V_u + (u-1)V_d}{u-d} = \frac{(u-d)V_d}{u-d} = V_d,$$

which is the value fact.

Step 2 (the backward induction (18.9)). The recursion builds numbers from the terminal prices, the factor $e^{-r\Delta t}$ and the weights p and $1-p$, with no trading assumption, so with the p of the statement and the node prices of the new underlying it defines $V_{i,j}$ at every node, from $V_{n,j} = f(S_{n,j})$. At a non-terminal node (i,j), take $V_u = V_{i+1,j+1}$ and $V_d = V_{i+1,j}$ in Step 1. Then $\Delta = \Delta_{i,j}$ by (18.13), and by (18.9)

$$e^{-r\Delta t}\big[pV_{i+1,j+1} + (1-p)V_{i+1,j}\big] = V_{i,j}.$$

The holding of Step 1 is therefore the holding of (ii). With a yield, its bond holding $e^{-r\Delta t}(uV_{i+1,j} - dV_{i+1,j+1})/(u-d)$ is the right-hand side of the display in (ii), and by the cost fact it equals the cost $V_{i,j}$ minus the cost $e^{-q\Delta t}\Delta_{i,j}S_{i,j}$ of the units, which is the left-hand side; with a futures price, its bond holding is $V_{i,j}$, by the display above. By Step 1 this holding costs $V_{i,j}$ and is worth $V_{i+1,j+1}$ after an up-move and $V_{i+1,j}$ after a down-move, with no money added or withdrawn. Steps 3 and 4 use this node fact where the original proofs use Theorem 18.3(i), and Step 6 uses its American form.

Step 3 (Proposition 18.11). In the proof of part (i), the argument at level n uses only that the derivative pays $f(S_T)$ at T and reading (c) of Definition 16.7, so it is unchanged. In the induction step one sentence changes: the portfolio (18.2) formed at the node (i,j), which Theorem 18.3(i) shows to pay $V_{i+1,j+1}$ or $V_{i+1,j}$ one step later and to cost $V_{i,j}$, is replaced by the holding of Step 2, which pays the same two amounts and has the same cost. The rest of the induction step compares two positions held from time $i\Delta t$, the derivative and this holding, each closed one step later with its proceeds deposited until T; both are managed with no cash added or withdrawn, by Step 2 for the holding, and their payoffs at T are equal on every path, so Lemma 16.8(v) and the positive real-world probability of every path give the claim at level i as before. The proof of part (ii) uses only the recursion (18.9), Pascal's rule and the definition of $\mathbb{Q}$ by the products $p^j(1-p)^{n-j}$, none of which depends on the value of p; run with the node prices $S_{i,j}u^k d^{m-k}$ of the new underlying, it gives (18.10) and (18.11) with the new p, and $Y_n \sim \mathrm{Bin}(n,p)$ under $\mathbb{Q}$. This proves (i).

Step 4 (Proposition 18.13, parts (i) and (ii)). The Setup of §18.4 is read with the holdings of (ii): the cost at a node of the holding chosen there is the cost of its units plus its bond holding with a yield, and its bond holding alone with a futures price, and its value one step later, the value entering the next node, is the one given by Step 1. Step 1 of the proof of that proposition, which obtains from Theorem 18.3(i) that the holding chosen at (i,j) costs $V_{i,j}$ and satisfies (D.1), is replaced by Step 2 above: the holding of (ii) costs $V_{i,j}$ and is worth $V_{i+1,j+1}$ after an up-move and $V_{i+1,j}$ after a down-move. Step 2 of that proof (the induction along a path) uses only these two facts and the definition of a self-financing strategy, and Step 3 (the price) uses only part (i), deposits in the cash account, the negative of the strategy and Lemma 16.8(iii); both are unchanged. This proves (ii).

Step 5 (Procedure 18.16). The Procedure uses the tree only through p, the factor $e^{-r\Delta t}$ and the node prices at which f is evaluated, so with the new p and node prices each of its steps is defined, and the European value $V_{0,0}$ of its step 4 is that of Step 3. The two inequalities of (iii) are the two properties recorded after the Procedure, and their argument uses only (18.15), (18.14), (18.9) and the positivity of p and $1-p$, which holds by Step 1. Indeed, $V^{\mathrm{am}}_{i,j} \geq f(S_{i,j})$ is read off (18.15) for $i < n$ and holds with equality at $i = n$. The values $V^{\mathrm{am}}_{n,j}$ and $V_{n,j}$ are both $f(S_{n,j})$, and if $V^{\mathrm{am}}_{i+1,j+1} \geq V_{i+1,j+1}$ and $V^{\mathrm{am}}_{i+1,j} \geq V_{i+1,j}$, then, because $p > 0$ and $1-p > 0$,

$$V^{\mathrm{am}}_{i,j} \;\geq\; V^{\mathrm{cont}}_{i,j} \;\geq\; e^{-r\Delta t}\big[pV_{i+1,j+1} + (1-p)V_{i+1,j}\big] = V_{i,j},$$

so backward induction on i gives $V^{\mathrm{am}}_{i,j} \geq V_{i,j}$ at every node.

Step 6 (Proposition 18.17). At a node (i,j) with $i<n$, take $V_u = V^{\text{am}}_{i+1,j+1}$ and $V_d = V^{\text{am}}_{i+1,j}$ in Step 1. Then $\Delta = \Delta^{\text{am}}_{i,j}$ by (18.16), and by (18.14)

$$e^{-r\Delta t}\bigl[pV^{\text{am}}_{i+1,j+1} + (1-p)V^{\text{am}}_{i+1,j}\bigr] = V^{\text{cont}}_{i,j}.$$

As in Step 2, the holding of Step 1 is therefore the holding named in (iii): with a yield, its bond holding is $e^{-r\Delta t}(uV^{\text{am}}_{i+1,j} - dV^{\text{am}}_{i+1,j+1})/(u-d)$, which by the cost fact equals $V^{\text{cont}}_{i,j} - e^{-q\Delta t}\Delta^{\text{am}}_{i,j}S_{i,j}$; with a futures price, it is $V^{\text{cont}}_{i,j}$. By Step 1 this holding costs $V^{\text{cont}}_{i,j}$ and one step later is worth $V^{\text{am}}_{i+1,j+1}$ after an up-move and $V^{\text{am}}_{i+1,j}$ after a down-move, which is (D.2), with no money added or withdrawn. Step 1 of the proof of that proposition, which obtains this cost and (D.2) from Theorem 18.3(i), is replaced by this. Steps 2 and 3 of that proof (the seller's hedge, the holder's rule) use only these two facts, (18.15) and transfers between the hedging portfolio and the separate deposit. Step 4 (the price) uses only parts (i) and (ii), the negative of the hedge and deposits in the cash account. Step 5 (the maximum) uses only (18.14), (18.15) and weights of continuations that are products of p and $1-p$, which are positive and sum to one for every $p \in (0,1)$. All four are unchanged, and together with Step 5 above this proves (iii). □

D.7 Proof of Proposition 18.29: Futures and forward prices on a tree with random rates; Cox, Ingersoll and Ross, 1981

The statement is on page 246. The notation and the standing assumptions are those of §18.7.

Proof. We argue by induction on the number of steps that remain, with sums over paths and no conditional expectation. (Proposition 1.7 conditions only on a discrete variable and only through $\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X]$; it cannot take the path-dependent factor $1/B_n$ out of a conditional expectation, and the path sums need nothing of the kind.)

Step 1 (node averages). Fix a node at time $k\Delta t$, $0 \le k \le n$. A *continuation* of the node is a sequence of $n-k$ further moves; the node followed by a continuation is a path. The *weight* of a continuation is the product, over the steps $i = k+1, \dots, n$, of p_i if the move of step i is up and of $1-p_i$ if it is down, each p_i being evaluated on that path. For a function Y of the path, the *node average* of Y at the node is

$$\bar{Y} = \sum_{\text{continuations}} (\text{weight of the continuation}) \times Y(\text{the node followed by the continuation}),$$

a bar marking an average, the convention of the Part *Derivatives*. Three properties are read off the definition. At a terminal node ($k=n$) the only continuation is the empty one, of weight 1, so $\bar{Y} = Y$. At the root ($k=0$) the continuations are the paths and their weights are the $\mathbb{Q}$-probabilities, so $\bar{Y} = \mathbb{E}^{\mathbb{Q}}[Y]$. If a function of the path takes the same value on all the paths through the node, as do $S_{k\Delta t}$, the rates $r_1, \dots, r_{k+1}$ and the amounts B_k and B_{k+1}, then the node average of its product with Y is that value times $\bar{Y}$, because the value is a common factor of the sum.

Let $k<n$, and write $\bar{Y}_u$ and $\bar{Y}_d$ for the node averages of Y at the two children of the node, as V_u and V_d denote values at the children. Then

$$\bar{Y} = p_{k+1}\,\bar{Y}_u + (1-p_{k+1})\,\bar{Y}_d, \tag{D.6}$$

where p_{k+1} is the up-probability set at the node. Indeed, a continuation of the node is a first move followed by a continuation of the child to which that move leads, and both complete to the same path. Its weight is the factor of the first move, p_{k+1} for an up-move and $1-p_{k+1}$ for a down-move, a number fixed at the node, times the weight of the continuation of the child. Splitting the sum according to the first move gives (D.6). With $Y=1$, backward induction on k from $k=n$ shows that the node average of the constant 1 is $p_{k+1} + (1-p_{k+1}) = 1$ at every node: the weights of the

continuations of a node sum to one, and in particular the $\mathbb{Q}$-probabilities of the 2^n paths sum to one.

Step 2 (node values). Let X be a function of the path and put $Y = X/B_n$. The *node value* of the claim X at a node at time $k\Delta t$ is

$$V = B_k\,\bar{Y},$$

where B_k is evaluated at the node. Since $B_k/B_n = e^{-\Delta t(r_{k+1}+\cdots+r_n)}$, the node value is the sum, over the continuations of the node, of the weight of the continuation times its discount factors $e^{-r_i\Delta t}$, $i = k+1,\dots,n$, times X on the whole path. At a terminal node $V = B_nX/B_n = X$, and at the root $V = \mathbb{E}^{\mathbb{Q}}[X/B_n]$. Let $k < n$, and let V_u and V_d be the node values at the two children. Both children sit at time $(k+1)\Delta t$ and share the same $B_{k+1} = B_k e^{r_{k+1}\Delta t}$, because r_{k+1} is set at the node; hence $V_u = B_{k+1}\bar{Y}_u$ and $V_d = B_{k+1}\bar{Y}_d$, and (D.6) gives

$$V = B_k\big[p_{k+1}\bar{Y}_u + (1-p_{k+1})\bar{Y}_d\big] = \frac{B_k}{B_{k+1}}\big[p_{k+1}V_u + (1-p_{k+1})V_d\big] = e^{-r_{k+1}\Delta t}\big[p_{k+1}V_u + (1-p_{k+1})V_d\big]. \tag{D.7}$$

Step 3 (replication, part (i)). We show by induction on $m = n-k$ that at every node at time $k\Delta t$ there is a self-financing portfolio of the asset and one-period deposits, set up at the node at the cost V, the node value, that pays X at time T on every continuation of the node. No trading of the claim after time 0 is assumed. For $m = 0$ the node is a path, $V = X$, and the portfolio is the cash amount X. Suppose that the statement holds for m, and take a node at time $k\Delta t$ with $n-k = m+1$, with price $S_{k\Delta t}$, rate r_{k+1} and probability p_{k+1}; its children carry the node values V_u and V_d and, by the hypothesis, portfolios that cost these amounts. Apply Theorem 18.3(i) at this node with r_{k+1} in place of r and $S_{k\Delta t}$ in place of S: Steps 1 and 2 of its proof use only the growth factor of the bond over the step, here $e^{r_{k+1}\Delta t}$, the condition $d < e^{r_{k+1}\Delta t} < u$, and a positive price, here $S_{k\Delta t} = Su^jd^{k-j} > 0$ for the number j of up-moves so far, so that the system (18.5) at the node has the nonzero determinant $S_{k\Delta t}(u-d)e^{r_{k+1}\Delta t}$. The holding of

$$\frac{V_u - V_d}{S_{k\Delta t}(u-d)} \text{ shares} \qquad \text{and} \qquad e^{-r_{k+1}\Delta t}\,\frac{uV_d - dV_u}{u-d} \text{ in the one-period deposit}$$

is worth V_u after an up-move and V_d after a down-move, and it costs $e^{-r_{k+1}\Delta t}[p_{k+1}V_u + (1-p_{k+1})V_d]$, which is V by (D.7). On arriving at a child the holding is worth exactly the node value of that child, which buys the portfolio of that child. The portfolio "this holding, then the portfolio of the child reached" is therefore self-financing, costs V at the node and pays X at time T on every continuation. This is the induction of Proposition 18.13, with the rate of each node.

At the root the portfolio is a self-financing strategy that costs $\mathbb{E}^{\mathbb{Q}}[X/B_n]$ and pays X on every path. Buying the claim at time 0 and holding it is a strategy with the same payoff. By Lemma 16.8(iii), used with the bank account of this tree, the two have the same price. The arbitrage behind the lemma trades the claim at time 0 only: sell the claim if it is dear, or buy it if it is cheap, take the opposite position in the portfolio, and deposit the difference, which is worth the difference times $B_n > 0$ at time T.

Step 4 (the asset and the bond). For $X = S_T$ the node value at a node at time $k\Delta t$ is the price $S_{k\Delta t}$ there, by backward induction on k. At $k = n$ the node value is $X = S_T$. If the node values at the two children are their prices $S_{k\Delta t}u$ and $S_{k\Delta t}d$, then by (D.7)

$$V = e^{-r_{k+1}\Delta t}\,S_{k\Delta t}\big[p_{k+1}u + (1-p_{k+1})d\big] = S_{k\Delta t},$$

because $p_{k+1}u + (1-p_{k+1})d = d + p_{k+1}(u-d) = e^{r_{k+1}\Delta t}$, the computation of Proposition 18.7(i) with the rate of the node. At the root this reads $\mathbb{E}^{\mathbb{Q}}[S_T/B_n] = S$, as it must: buying the asset and holding it pays S_T. For $X = 1$ the claim is the zero-coupon bond that pays 1 at time T, and its price is $\mathbb{E}^{\mathbb{Q}}[1/B_n]$, a positive number because every path has positive $\mathbb{Q}$-probability and $B_n > 0$.

Step 5 (futures, part (ii)). Fix a node at time $k\Delta t$ with $k < n$. Let the futures price be $F^{\mathrm{fut}}_{k\Delta t}$ at the node and F^{fut}_u, F^{fut}_d at its two children. A long futures contract entered at the node costs

nothing, receives the change in the futures price at the next settlement, $F_u^{\text{fut}} - F_{k\Delta t}^{\text{fut}}$ or $F_d^{\text{fut}} - F_{k\Delta t}^{\text{fut}}$, and is then worth zero again, so it can be closed at no cost (Definition 16.25). Held over this one step, it is a one-step claim with these two payoffs. By Theorem 18.3(i) with the rate of the node, as in Step 3, the replica of that claim in the asset and the one-period deposit costs

$$x = e^{-r_{k+1}\Delta t}\Big[p_{k+1}\big(F_u^{\text{fut}} - F_{k\Delta t}^{\text{fut}}\big) + (1-p_{k+1})\big(F_d^{\text{fut}} - F_{k\Delta t}^{\text{fut}}\big)\Big].$$

We claim that $x = 0$. Consider, at every node at time $k\Delta t$, the long contract and its replica, each held over the step, then closed, with its proceeds deposited until T. These are two positions held from time $k\Delta t$, and their payoffs at T are equal on every path, because the replica pays the two settlement amounts of the contract. Lemma 16.8(v), used with this tree's bank account (Setup for Proposition 18.29 in §18.7), therefore gives them equal values at time $k\Delta t$ with probability one. The path is at the given node with positive real-world probability (the same Setup), so there the replica is worth what the contract is worth, which is zero: $x = 0$. (The proof of that lemma is the arbitrage of reading (c) of Definition 16.7, started at time $k\Delta t$ on the event that the two values differ.) Since $e^{-r_{k+1}\Delta t} > 0$ and the two weights sum to one, $x = 0$ reads

$$F_{k\Delta t}^{\text{fut}} = p_{k+1}F_u^{\text{fut}} + (1-p_{k+1})F_d^{\text{fut}}, \tag{D.8}$$

which is the weighted-average statement.

Now let $Y = S_T$. By backward induction on k, the futures price at every node is the node average $\bar{Y}$ of S_T there. At $k = n$ the futures price is S_T (Proposition 16.27, whose proof trades at the delivery date only and so does not involve the rates), and $\bar{Y} = Y$ at a terminal node. If the futures prices at the two children are $\bar{Y}_u$ and $\bar{Y}_d$, then by (D.8) and (D.6) the futures price at the node is $p_{k+1}\bar{Y}_u + (1-p_{k+1})\bar{Y}_d = \bar{Y}$. At the root, $F^{\text{fut}} = \mathbb{E}^{\mathbb{Q}}[S_T]$. This is the recursion of Step 2 without the discount factors.

Step 6 (parts (iii) and (iv)). A forward contract with delivery price K pays $S_T - K$ at time T (Definition 16.5). By part (i), the linearity of the path sum (Theorem 1.2) and Step 4, it is worth

$$\mathbb{E}^{\mathbb{Q}}\Big[\frac{S_T - K}{B_n}\Big] = \mathbb{E}^{\mathbb{Q}}\Big[\frac{S_T}{B_n}\Big] - K\,\mathbb{E}^{\mathbb{Q}}\Big[\frac{1}{B_n}\Big] = S - K\,\mathbb{E}^{\mathbb{Q}}\Big[\frac{1}{B_n}\Big]$$

at time 0. The forward price is the delivery price at which the contract costs nothing, and the expression vanishes exactly at $K = S/\mathbb{E}^{\mathbb{Q}}[1/B_n]$, which is (iii). Subtracting (iii) from (ii), and using $S = \mathbb{E}^{\mathbb{Q}}[S_T/B_n]$ once more,

$$F^{\text{fut}} - F = \mathbb{E}^{\mathbb{Q}}[S_T] - \frac{\mathbb{E}^{\mathbb{Q}}[S_T/B_n]}{\mathbb{E}^{\mathbb{Q}}[1/B_n]} = \frac{\mathbb{E}^{\mathbb{Q}}[1/B_n]\,\mathbb{E}^{\mathbb{Q}}[S_T] - \mathbb{E}^{\mathbb{Q}}[S_T/B_n]}{\mathbb{E}^{\mathbb{Q}}[1/B_n]} = -\frac{\operatorname{Cov}^{\mathbb{Q}}(1/B_n,\ S_T)}{\mathbb{E}^{\mathbb{Q}}[1/B_n]},$$

by the definition of the covariance. The denominator is positive, so the gap has the sign of minus the covariance. If the rates are deterministic, $1/B_n$ is a constant: then $\mathbb{E}^{\mathbb{Q}}[S_T/B_n] = \mathbb{E}^{\mathbb{Q}}[S_T]/B_n$ and $\mathbb{E}^{\mathbb{Q}}[1/B_n] = 1/B_n$, because the $\mathbb{Q}$-probabilities sum to one (Step 1), the covariance is zero, and $F^{\text{fut}} = F = SB_n$. □

D.8 Proof of Proposition 18.32: Independent steps, and the mean and variance of the CRR log-price

The statement is on page 249. The notation and the standing assumptions are those of §18.8.

Proof. For (i) we group the paths of a finite sum; (ii) follows from (i) by expanding a square; (iii) is Proposition 18.21 multiplied by n.

Part (i), two factors. Call the steps that X reads the first set, the steps that X' reads the second set, and all the other steps the remaining steps. A path is a choice of moves for the first set, a

choice for the second set and a choice for the remaining steps, and every combination of three such choices is a path. The $\mathbb{Q}$-probability of a path is a product of one factor per step, so, grouping the factors, it is the product of three *weights*, one for each set, the weight of a choice of moves for a set of steps being the product of the factors p or $1-p$ of those steps. The value of X depends only on the choice for the first set, and that of X' only on the choice for the second set. Hence

$$\mathbb{E}^{\mathbb{Q}}[XX'] = \sum_{\substack{\text{choices for}\\ \text{the first set}}} \ \sum_{\substack{\text{choices for}\\ \text{the second set}}} \ \sum_{\substack{\text{choices for the}\\ \text{remaining steps}}} (\text{first weight})\, X \cdot (\text{second weight})\, X' \cdot (\text{third weight}).$$

The innermost sum involves the third weight only. It equals the product, over the remaining steps, of $p+(1-p)=1$, because expanding that product gives exactly one term for each choice of moves; so it equals 1. What is left is a product of two sums,

$$\mathbb{E}^{\mathbb{Q}}[XX'] = \Big(\sum_{\substack{\text{choices for}\\ \text{the first set}}} (\text{first weight})\, X \Big) \Big(\sum_{\substack{\text{choices for}\\ \text{the second set}}} (\text{second weight})\, X' \Big).$$

The same grouping applied to X alone, where the sum over the moves of all the steps outside the first set equals 1, shows that the first bracket is $\mathbb{E}^{\mathbb{Q}}[X]$; likewise the second bracket is $\mathbb{E}^{\mathbb{Q}}[X']$. This is (18.22).

Part (i), several factors. For three or more factors we argue by induction on their number. The product of all the factors but the last is computed from the moves of the union of their sets of steps, and this union is disjoint from the set of the last factor. By (18.22) the expectation of the whole product is the expectation of that partial product times the expectation of the last factor, and the induction hypothesis splits the former.

Part (i), the three particular cases. For any step i and any function ψ_1 of one variable, the grouping above with the single step i as the first set gives $\mathbb{E}^{\mathbb{Q}}\psi_1(R_i) = p\,\psi_1(x) + (1-p)\,\psi_1(-x)$, which does not depend on i: every R_i has the law of R_1. Since $\psi_1(R_i)$ reads step i only and $\psi_2(R_k)$ reads step $k \neq i$ only, (18.22) gives (18.23). Next, $e^{\theta(R_1+\cdots+R_n)}$ is the product of the n factors $e^{\theta R_i}$, which read n different steps and have the same expectation $\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}$; the rule for several factors gives (18.24). Finally, on every path

$$\psi(X, R_k) = \psi(X, x)\,\mathbf{1}\{R_k = x\} + \psi(X, -x)\,\mathbf{1}\{R_k = -x\},$$

because R_k takes only the values x and $-x$. Each term is a variable computed from the moves of steps $1, \dots, k-1$ times a variable computed from the move of step k, and $\mathbb{E}^{\mathbb{Q}}\mathbf{1}\{R_k = x\} = \mathbb{Q}(R_k = x) = p$ by Lemma 1.4. By Theorem 1.2 and (18.22),

$$\mathbb{E}^{\mathbb{Q}}\psi(X, R_k) = p\,\mathbb{E}^{\mathbb{Q}}\psi(X, x) + (1-p)\,\mathbb{E}^{\mathbb{Q}}\psi(X, -x) = \mathbb{E}^{\mathbb{Q}}\big[\, p\,\psi(X, x) + (1-p)\,\psi(X, -x)\,\big],$$

which is (18.25). For $k = 1$ the variable X is a constant, and the identity reads $\mathbb{E}^{\mathbb{Q}}\psi(X, R_1) = p\,\psi(X, x) + (1-p)\,\psi(X, -x)$.

Part (ii). On every path $R_i = x\,(2\cdot\mathbf{1}\{\text{step } i \text{ is up}\} - 1)$, so $R_1 + \cdots + R_n = x\,(2Y_n - n)$. On the other hand $S_T = S u^{Y_n} d^{n-Y_n}$ with $u = e^x$ and $d = e^{-x}$, so $\ln(S_T/S) = Y_n x - (n - Y_n)x = (2Y_n - n)\,x$ as well. By (D.3), $\mathbb{E}^{\mathbb{Q}}R_i = \mathbb{E}^{\mathbb{Q}}R_1 = (2p-1)\,x$ for every i, and Theorem 1.2 gives the mean $n(2p-1)\,x$. For the variance, subtract the mean of each step, so that $\ln(S_T/S) - n(2p-1)\,x$ is the sum of the n centred steps $R_i - (2p-1)\,x$, and expand the square of that sum into its n squares and its $n(n-1)$ cross terms (the expansion that proves Proposition 10.10 for two terms, carried out here for n):

$$\operatorname{Var}^{\mathbb{Q}}\Big(\ln\frac{S_T}{S}\Big) = \sum_{i=1}^{n} \mathbb{E}^{\mathbb{Q}}\Big[\big(R_i - (2p-1)\,x\big)^2\Big] + \sum_{i \neq k} \mathbb{E}^{\mathbb{Q}}\Big[\big(R_i - (2p-1)\,x\big)\big(R_k - (2p-1)\,x\big)\Big].$$

Each cross term vanishes: by (18.23) with $\psi_1(y) = \psi_2(y) = y - (2p-1)\,x$, it equals the square of $\mathbb{E}^{\mathbb{Q}}[R_1 - (2p-1)\,x] = 0$. Each square has the expectation $\operatorname{Var}^{\mathbb{Q}}(R_1) = 4p(1-p)\,x^2$, by (D.4);

equivalently, R_1 is $2x$ times a Bernoulli(p) indicator minus the constant x, and its variance is $4x^2$ times the Bernoulli variance $p(1-p)$ of Proposition 1.10. This gives the variance $4np(1-p)\,x^2$. For the count, $Y_n = \big(\ln(S_T/S)/x + n\big)/2$. Theorem 1.2 gives $\mathbb{E}^{\mathbb{Q}} Y_n = \big(n(2p-1)+n\big)/2 = np$, and then $Y_n - np = \big(\ln(S_T/S) - n(2p-1)\,x\big)/(2x)$, whose square has the expectation $4np(1-p)\,x^2/(4x^2) = np(1-p)$.

Part (iii). By Proposition 18.21(ii) and (iii), $(2p-1)\,x = \nu\Delta t + O(\Delta t^{3/2})$ and $4p(1-p)\,x^2 = \sigma^2\Delta t + O(\Delta t^2)$ as $\Delta t \to 0$. Put $\Delta t = T/n$ and multiply by n: since $n\Delta t = T$, $n\Delta t^{3/2} = T^{3/2}n^{-1/2}$ and $n\Delta t^2 = T^2 n^{-1}$, the exact moments (18.26) are $\nu T + O(n^{-1/2})$ and $\sigma^2 T + O(n^{-1})$. □

D.9 Proof of Lemma 18.35: The normal moment generating function, and composing Gaussian smoothings

The statement is on page 251. The notation and the standing assumptions are those of §18.8.

Proof. For (i) we complete the square; for (ii) we rotate the plane, writing the rotation as three shears.

Part (i), the identity. For real θ and z,

$$\theta z - \frac{z^2}{2} = \frac{\theta^2}{2} - \frac{(z-\theta)^2}{2},$$

so $e^{\theta z}\varphi(z) = e^{\theta^2/2}\,\varphi(z-\theta)$, and

$$\int_{-\infty}^{\infty} e^{\theta z}\varphi(z)\,dz = e^{\theta^2/2}\int_{-\infty}^{\infty}\varphi(z-\theta)\,dz = e^{\theta^2/2}\int_{-\infty}^{\infty}\varphi(z)\,dz = e^{\theta^2/2},$$

by the substitution $z \mapsto z+\theta$ and the normalisation $\int\varphi = 1$ (Theorem 7.5, in the form of Example 7.6). In particular the integral is finite. The integrand is positive, so the integral converges absolutely, and Fact 7.22 gives (18.27).

Part (i), the three consequences. For $X = \mu_X + \sigma_X Z$ we have $e^{\theta X} = e^{\theta\mu_X}\,e^{(\theta\sigma_X)Z}$, and (18.27) with $\theta\sigma_X$ in place of θ gives $\mathbb{E}\,e^{\theta X} = e^{\theta\mu_X}\,e^{\theta^2\sigma_X^2/2}$. Next, for $c \geq 0$ and every real z,

$$e^{c|z|} \leq e^{cz} + e^{-cz},$$

because one of the two terms on the right equals $e^{c|z|}$ and the other is positive. By (18.27) with $\theta = c$ and with $\theta = -c$, each term has the Gaussian integral $e^{c^2/2}$, so the continuous positive function $e^{c|z|}\varphi(z)$ has an integral of at most $2e^{c^2/2}$ (Proposition 7.16), and this integral is $\mathbb{E}\,e^{c|Z|}$. Finally, let ψ be as in the statement and $X = \mu_X + \sigma_X Z$. For every real z,

$$|\psi(\mu_X + \sigma_X z)|^k \leq c^k\,e^{kc|\mu_X + \sigma_X z|} \leq c^k\,e^{kc|\mu_X|}\,e^{kc\sigma_X|z|}.$$

By the previous bound, applied with the constant $kc\sigma_X$, the right side times $\varphi(z)$ has a finite integral. The function $z \mapsto |\psi(\mu_X + \sigma_X z)|^k\varphi(z)$ is continuous except at finitely many points and is dominated by that integrable function, so its integral over the real line is an ordinary improper integral, split at the finitely many exceptional points, and it converges absolutely by Proposition 7.16. Fact 7.22 therefore applies, and $\mathbb{E}|\psi(X)|^k$ is this finite integral.

Part (ii), absolute convergence. For all real z and z',

$$|\psi(a_1 z + a_2 z')| \leq c\,e^{c|a_1 z + a_2 z'|} \leq c\,e^{c|a_1||z|}\,e^{c|a_2||z'|}.$$

The iterated integral of $|\psi(a_1 z + a_2 z')|\,\varphi(z)\varphi(z')$ is therefore at most

$$c\int e^{c|a_1||z|}\varphi(z)\,dz \cdot \int e^{c|a_2||z'|}\varphi(z')\,dz' \leq c\cdot 2e^{c^2a_1^2/2}\cdot 2e^{c^2a_2^2/2} = 4c\,e^{c^2(a_1^2+a_2^2)/2},$$

by (i), and this is finite. So the left side of (18.28) converges absolutely, and by Fact 18.34(i) it equals the integral of $\psi(a_1z + a_2z')\,\varphi(z)\varphi(z')$ over the plane. The right side of (18.28) converges absolutely by the last consequence of (i), applied with $k = 1$ to the normal variable $(a_1^2 + a_2^2)^{1/2}Z$.

Part (ii), the rotation. If $a_1 = a_2 = 0$, both sides of (18.28) equal $\psi(0)$, because $\int \varphi = 1$. Otherwise put $a = (a_1^2 + a_2^2)^{1/2} > 0$. We rotate the plane so that $a_1z + a_2z'$ becomes a times the first coordinate, and we write the rotation as three shears, so that only translations of a line and Fact 18.34(i) are used. In this proof a function $\zeta \geq 0$ on the plane is *admissible* if it is continuous outside finitely many lines, and $\iint \zeta$ denotes the common value, possibly infinite, of its two iterated integrals, which Fact 18.34(i) provides.

(a) Shears. For every admissible ζ and every real number a',

$$\iint \zeta(z + a'z', z')\,dz\,dz' = \iint \zeta(z, z')\,dz\,dz' = \iint \zeta(z, z' + a'z)\,dz\,dz'.$$

For the first equality, integrate over z first: for each fixed z' the substitution $z \mapsto z - a'z'$ is a translation of the line, and a translation leaves the integral over the line of a nonnegative function unchanged. For the second, integrate over z' first and translate z' for each fixed z. By Fact 18.34(i) each of the three double integrals may be computed in either order. The two functions on the outside are admissible again: each is ζ composed with the linear map $(z, z') \mapsto (z + a'z', z')$ or $(z, z') \mapsto (z, z' + a'z)$, which has determinant 1, so the points that it sends into a line form a line.

(b) Rotations. For every admissible ζ,

$$\iint \zeta\Big(\frac{a_1z + a_2z'}{a},\ \frac{-a_2z + a_1z'}{a}\Big)\,dz\,dz' = \iint \zeta(y, y')\,dy\,dy'. \tag{D.9}$$

Suppose first that $a_1 \neq -a$, and put $a' = a_2/(a + a_1)$. Then

$$\frac{a'a_2}{a} = \frac{a_2^2}{a(a + a_1)} = \frac{a^2 - a_1^2}{a(a + a_1)} = \frac{a - a_1}{a},$$

so that $1 - a'a_2/a = a_1/a$ and $a'(1 + a_1/a) = a'(a + a_1)/a = a_2/a$. Apply to a point (z, z') the shear $(z, z') \mapsto (z + a'z', z')$, then the shear $(z, z') \mapsto (z, z' - (a_2/a)z)$, then the first shear again. The second shear sends $(z + a'z', z')$ to $(z + a'z',\ (a_1/a)z' - (a_2/a)z)$, and the third sends that point to

$$\Big(\frac{a_1}{a}\,z + a'\Big(1 + \frac{a_1}{a}\Big)z',\ \frac{-a_2z + a_1z'}{a}\Big) = \Big(\frac{a_1z + a_2z'}{a},\ \frac{-a_2z + a_1z'}{a}\Big).$$

So, with the admissible functions $\zeta_1(z, z') = \zeta(z + a'z', z')$ and $\zeta_2(z, z') = \zeta_1(z, z' - (a_2/a)z)$, the integrand on the left of (D.9) is $\zeta_2(z + a'z', z')$, and (a), used with a', with $-a_2/a$ in place of a', and with a' again, gives

$$\iint \zeta_2(z + a'z', z')\,dz\,dz' = \iint \zeta_2 = \iint \zeta_1 = \iint \zeta.$$

If $a_1 = -a$, then $a_2 = 0$ and the integrand on the left is $\zeta(-z, -z')$. The substitution $z \mapsto -z$ for each fixed z', then, after the order of integration is exchanged by Fact 18.34(i), the substitution $z' \mapsto -z'$ for each fixed z, give (D.9) in this case as well.

(c) Application to ψ. The two functions $\zeta_\pm(y, y') = \max(\pm\psi(ay), 0)\,\varphi(y)\varphi(y')$, one for each sign, are admissible: they are nonnegative, and continuous outside the finitely many lines $\{y = y_j/a\}$, where the y_j are the points at which ψ may be discontinuous. At the point $y = (a_1z + a_2z')/a$, $y' = (-a_2z + a_1z')/a$ we have $ay = a_1z + a_2z'$, and the sum of the squares is preserved:

$$y^2 + y'^2 = \frac{(a_1z + a_2z')^2 + (-a_2z + a_1z')^2}{a^2} = \frac{(a_1^2 + a_2^2)(z^2 + z'^2)}{a^2} = z^2 + z'^2,$$

the two cross terms $\pm 2a_1a_2zz'$ cancelling. Hence $\varphi(y)\varphi(y') = (2\pi)^{-1}e^{-(y^2+y'^2)/2} = \varphi(z)\varphi(z')$, and (D.9), applied to ζ_+ and to ζ_-, reads

$$\iint \max\big(\pm\psi(a_1z + a_2z'), 0\big)\,\varphi(z)\varphi(z')\,dz\,dz' = \iint \max\big(\pm\psi(ay), 0\big)\,\varphi(y)\varphi(y')\,dy\,dy',$$

one identity for each sign. The left sides are finite by the bound of the absolute-convergence step, because $\max(\pm\psi, 0) \le |\psi|$, so the right sides are finite as well. Subtracting the identity with the minus sign from the one with the plus sign, and using $\max(\psi, 0) - \max(-\psi, 0) = \psi$,

$$\iint \psi(a_1z + a_2z')\,\varphi(z)\varphi(z')\,dz\,dz' = \iint \psi(ay)\,\varphi(y)\varphi(y')\,dy\,dy'.$$

The iterated integral of $|\psi(ay)|\,\varphi(y)\varphi(y')$ is $\int |\psi(ay)|\varphi(y)\,dy \cdot \int \varphi(y')\,dy'$, which is finite. By Fact 18.34(i) once more, the right side is the iterated integral in which y' is integrated first:

$$\int \psi(ay)\,\varphi(y)\Big[\int \varphi(y')\,dy'\Big]dy = \int \psi(ay)\,\varphi(y)\,dy.$$

This proves (18.28).

Part (ii), independent variables. Let Z and Z' be independent standard normal variables. Fact 18.34(ii), applied with $X = Z$ and $Y = Z'$ to the nonnegative function $|\psi(a_1z + a_2z')|$, shows that $\mathbb{E}|\psi(a_1Z + a_2Z')|$ is the iterated integral bounded above, so it is finite. The same Fact then applies to $\psi(a_1z + a_2z')$ itself: $\mathbb{E}\,\psi(a_1Z + a_2Z') = \mathbb{E}\,\bar\psi(Z)$, where $\bar\psi(z) = \mathbb{E}\,\psi(a_1z + a_2Z') = \int \psi(a_1z + a_2z')\varphi(z')\,dz'$. So $\mathbb{E}\,\psi(a_1Z + a_2Z')$ is the left side of (18.28), and the right side is $\mathbb{E}\,\psi(aZ)$: this is (18.29). Now fix a real y and take for ψ the indicator of $(-\infty, y]$, which is continuous except at y and bounded by 1. Then (18.29) reads $P(a_1Z + a_2Z' \le y) = P(aZ \le y)$. The two variables have the same distribution function, hence the same law, and $aZ \sim \mathcal{N}(0, a^2)$ by the convention of (i) with $\mu_X = 0$ and $\sigma_X = a$.

Part (ii), the shift. For a real y_0 the function $y \mapsto \psi(y_0 + y)$ is continuous except at finitely many points, and $|\psi(y_0 + y)| \le c\,e^{c|y_0|}\,e^{c|y|}$, so it satisfies the growth bound with the constant $\max(c, c\,e^{c|y_0|})$. Applying (18.28) and (18.29) to it puts y_0 inside ψ on both sides. □

D.10 Proof of Theorem 18.37: Convergence of the CRR tree: the law of the log-price and European prices; Cox, Ross and Rubinstein, 1979

The statement is on page 252. The notation and the standing assumptions are those of §18.8.

Proof. We use Lindeberg's replacement argument (Lindeberg, 1922): the n tree steps are replaced, one at a time, by normal variables with the same mean and variance. Steps 1 to 5 prove (i), Step 6 proves (ii) for the put, Step 7 proves (iii) and (ii) for the call, Step 8 proves (iv), and Step 9 proves (v).

Step 1 (the step moments). Each R_i equals x with $\mathbb{Q}$-probability p and $-x$ with $\mathbb{Q}$-probability $1 - p$. Exactly,

$$\mathbb{E}^{\mathbb{Q}}R_1 = (2p-1)\,x, \qquad \mathbb{E}^{\mathbb{Q}}R_1^2 = x^2 = \sigma^2\Delta t,$$
$$\mathrm{Var}^{\mathbb{Q}}\,R_1 = 4p(1-p)\,x^2, \qquad \mathbb{E}^{\mathbb{Q}}|R_1|^3 = x^3 = \sigma^3\Delta t^{3/2}\,:$$

the first and the third are (D.3) and (D.4), and the other two hold because $R_1^2 = x^2$ and $|R_1|^3 = x^3$ after either move. By Proposition 18.21(ii) and (iii), read at $\Delta t = T/n$, there are a constant c and an integer n_0, depending only on r, σ and T, with $\sigma > |r|\sqrt{T/n_0}$, such that for every $n \ge n_0$

$$\big|(2p-1)\,x - \nu\Delta t\big| \le c\,\Delta t^{3/2} \qquad \text{and} \qquad \big|4p(1-p)\,x^2 - \sigma^2\Delta t\big| \le c\,\Delta t^2. \tag{D.10}$$

In Steps 2 to 5, ψ is a function with bounded continuous derivatives ψ', ψ'', ψ'''. Steps 2 to 4 hold for every n with $\sigma > |r|\sqrt{T/n}$, because they use only $0 < p < 1$; Step 5, which uses (D.10), takes $n \ge n_0$.

Step 2 (Gaussian comparison functions). For a continuous function χ on the real line of at most exponential growth, $|\chi(y)| \le c' e^{c'|y|}$ for all real y and some constant c', and for $k = 0, 1, \dots, n$, put

$$\Psi_k[\chi](y) = \mathbb{E}\,\chi\Big(y + (n-k)(2p-1)\,x + 2x\big((n-k)\,p(1-p)\big)^{1/2} Z\Big). \tag{D.11}$$

This is the function χ smoothed by the normal law that has the mean $(n-k)(2p-1)\,x$ and the variance $4(n-k)\,p(1-p)\,x^2$ of the last $n-k$ tree steps (Proposition 18.32(ii), applied to $n-k$ steps); the expectation is finite by Lemma 18.35(i). Write $\Psi_k = \Psi_k[\psi]$. For $k = n$ no step is left, and $\Psi_n = \psi$. The definition applies to ψ, ψ' and ψ'': the last two are continuous and bounded, and ψ is continuous with

$$|\psi(y)| \le |\psi(0)| + \sup|\psi'|\,|y| \le c'\,(1 + |y|) \le c' e^{c'|y|}, \qquad c' = \max\big(|\psi(0)|, \sup|\psi'|, 1\big),$$

the first inequality by the mean value theorem (Corollary 7.32 at order zero) and the last because $1 + |y| \le e^{|y|}$ and $c' \ge 1$. The same mean value theorem gives $|\psi(y_1+z) - \psi(y_2+z)| \le \sup|\psi'|\,|y_1 - y_2|$ for all real y_1, y_2 and z; taking expectations, $|\Psi_k(y_1) - \Psi_k(y_2)| \le \sup|\psi'|\,|y_1 - y_2|$. So every Ψ_k is continuous and grows at most linearly.

Let

$$R' = (2p-1)\,x + 2x\big(p(1-p)\big)^{1/2} Z',$$

with Z' standard normal: a normal variable with the mean and the variance of one tree step, because $\mathbb{E}Z' = 0$ and $\mathbb{E}Z'^2 = 1$ (Proposition 7.25) give $\mathbb{E}R' = (2p-1)\,x$ and $\operatorname{Var} R' = 4p(1-p)\,x^2$. We claim that for $1 \le k \le n$ and every real y,

$$\mathbb{E}\,\Psi_k(y + R') = \Psi_{k-1}(y)\,: \tag{D.12}$$

replacing one tree step by its normal surrogate moves from Ψ_k to Ψ_{k-1}. Indeed, with both expectations written as integrals, the left side is

$$\int \Big[\int \psi\Big(y + (n-k+1)(2p-1)\,x + 2x\big(p(1-p)\big)^{1/2} z' + 2x\big((n-k)\,p(1-p)\big)^{1/2} z\Big)\varphi(z)\,dz\Big]\varphi(z')\,dz'.$$

This is the left side of (18.28) for the shifted function $\psi(y + (n-k+1)(2p-1)\,x + \cdot\,)$, with z' as the outer variable of the lemma, at the scale $2x(p(1-p))^{1/2}$, and z as its inner variable, at the scale $2x((n-k)\,p(1-p))^{1/2}$. The squares of the two scales add up to $4x^2(n-k+1)\,p(1-p)$, so Lemma 18.35(ii) turns the iterated integral into

$$\int \psi\Big(y + (n-k+1)(2p-1)\,x + 2x\big((n-k+1)\,p(1-p)\big)^{1/2} z\Big)\varphi(z)\,dz = \Psi_{k-1}(y).$$

Step 3 (one replacement). Put

$$\mathcal{M}_k = \mathbb{E}^{\mathbb{Q}}\,\Psi_k(R_1 + \cdots + R_k) \quad (1 \le k \le n), \qquad \mathcal{M}_0 = \Psi_0(0).$$

This is the hybrid expectation in which the first k steps are those of the tree and the last $n-k$ are Gaussian. In particular

$$\mathcal{M}_n = \mathbb{E}^{\mathbb{Q}}\psi\Big(\ln\frac{S_T}{S}\Big), \qquad \mathcal{M}_0 = \mathbb{E}\,\psi\Big(n(2p-1)\,x + 2x\big(np(1-p)\big)^{1/2} Z\Big).$$

Fix k with $1 \le k \le n$. The partial sum $R_1 + \cdots + R_{k-1} = \ln(S_{(k-1)\Delta t}/S)$ is computed from the moves of steps $1, \dots, k-1$, and it is zero when $k = 1$. In this proof the letter X names only the

kernel of Step 6, so the partial sum is written out. By (18.25), with the partial sum in the role of the variable called X there and with the function $(y,z) \mapsto \Psi_k(y+z)$ of two variables,

$$\mathcal{M}_k = \mathbb{E}^{\mathbb{Q}}\Big[\, p\,\Psi_k\Big(\ln\frac{S_{(k-1)\Delta t}}{S} + x\Big) + (1-p)\,\Psi_k\Big(\ln\frac{S_{(k-1)\Delta t}}{S} - x\Big)\Big],$$

and by the definition of $\mathcal{M}_{k-1}$ and (D.12),

$$\mathcal{M}_{k-1} = \mathbb{E}^{\mathbb{Q}}\,\Psi_{k-1}\Big(\ln\frac{S_{(k-1)\Delta t}}{S}\Big), \qquad \Psi_{k-1}(y) = \mathbb{E}\,\Psi_k(y+R').$$

Both are sums, over the paths of the first $k-1$ steps, of a probability times a function of the partial sum, and the probabilities sum to one. Since $p\,\Psi_k(y+x) + (1-p)\,\Psi_k(y-x) = \mathbb{E}^{\mathbb{Q}}\Psi_k(y+R_1)$, it follows that

$$|\mathcal{M}_k - \mathcal{M}_{k-1}| \le \sup_y\, |\mathbb{E}^{\mathbb{Q}}\Psi_k(y+R_1) - \mathbb{E}\,\Psi_k(y+R')|, \tag{D.13}$$

the supremum being over all real y.

Fix a real y, and let

$$y' = y + (n-k)(2p-1)\,x + 2x\big((n-k)\,p(1-p)\big)^{1/2}Z$$

be the point at which ψ is evaluated in the definition (D.11) of $\Psi_k(y) = \mathbb{E}\,\psi(y')$; it is a function of Z. Corollary 7.32 at order two, applied to ψ about the point y' with an increment z, gives

$$\psi(y'+z) = \psi(y') + \psi'(y')\,z + \tfrac{1}{2}\psi''(y')\,z^2 + \tfrac{1}{6}\psi'''(y'^*)\,z^3$$

for a point y'^* between y' and $y'+z$. (The point that the corollary supplies, called ξ in §7.5, is starred in the Part *Derivatives*, because ξ is the Itô integrand of §19.) Hence the difference

$$\psi(y'+z) - \psi(y') - \psi'(y')\,z - \tfrac{1}{2}\psi''(y')\,z^2,$$

which is a continuous function of Z, is at most $\frac{1}{6}\sup|\psi'''|\,|z|^3$ in absolute value. Each of its four terms has a finite expectation over Z, because ψ grows at most linearly and ψ', ψ'' are bounded. Taking the expectation over Z term by term, and noting that $y'+z$ is the point at which ψ is evaluated in $\Psi_k(y+z)$,

$$\Psi_k(y+z) = \Psi_k(y) + \Psi_k[\psi'](y)\,z + \tfrac{1}{2}\Psi_k[\psi''](y)\,z^2 + \mathcal{E}_k(y,z), \qquad |\mathcal{E}_k(y,z)| \le \tfrac{1}{6}\sup|\psi'''|\,|z|^3, \tag{D.14}$$

for every real z. No derivative of Ψ_k is needed: the coefficients $\Psi_k[\psi'](y)$ and $\Psi_k[\psi''](y)$ do not depend on z. The error $\mathcal{E}_k(y,z)$ is a continuous function of z, because Ψ_k is continuous (Step 2).

Now average (D.14) over $z = R_1$ under $\mathbb{Q}$, and over $z = R'$:

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}\Psi_k(y+R_1) &= \Psi_k(y) + \Psi_k[\psi'](y)\,\mathbb{E}^{\mathbb{Q}}R_1 + \tfrac{1}{2}\Psi_k[\psi''](y)\,\mathbb{E}^{\mathbb{Q}}R_1^2 + \mathbb{E}^{\mathbb{Q}}\mathcal{E}_k(y,R_1),\\
\mathbb{E}\,\Psi_k(y+R') &= \Psi_k(y) + \Psi_k[\psi'](y)\,\mathbb{E}R' + \tfrac{1}{2}\Psi_k[\psi''](y)\,\mathbb{E}R'^2 + \mathbb{E}\,\mathcal{E}_k(y,R').
\end{aligned}$$

The constant, linear and quadratic terms of the two lines agree, because $\mathbb{E}^{\mathbb{Q}}R_1 = \mathbb{E}R'$ and, by the shortcut (1.1),

$$\mathbb{E}^{\mathbb{Q}}R_1^2 = \mathrm{Var}^{\mathbb{Q}}\,R_1 + (\mathbb{E}^{\mathbb{Q}}R_1)^2 = \mathrm{Var}\,R' + (\mathbb{E}R')^2 = \mathbb{E}R'^2.$$

Subtracting the two lines and using the bound on $\mathcal{E}_k$,

$$|\mathbb{E}^{\mathbb{Q}}\Psi_k(y+R_1) - \mathbb{E}\,\Psi_k(y+R')| \le \tfrac{1}{6}\sup|\psi'''|\,(\mathbb{E}^{\mathbb{Q}}|R_1|^3 + \mathbb{E}|R'|^3).$$

Here $\mathbb{E}^{\mathbb{Q}}|R_1|^3 = x^3$. For the normal step, note that $(y_1+y_2)^3 \le 4\,(y_1^3+y_2^3)$ for nonnegative numbers y_1 and y_2, because $4(y_1^3+y_2^3) - (y_1+y_2)^3 = 3(y_1-y_2)^2(y_1+y_2) \ge 0$. Therefore

$$|R'|^3 \le \Big(|2p-1|\,x + 2x(p(1-p))^{1/2}|Z'|\Big)^3 \le 4\,\Big(|2p-1|^3x^3 + 8(p(1-p))^{3/2}x^3\,|Z'|^3\Big).$$

Now $|2p-1| \le 1$ and $p(1-p) = \frac14 - (p-\frac12)^2 \le \frac14$, so $8(p(1-p))^{3/2} \le 1$; and, integrating by parts with the antiderivative $-\varphi(z)$ of $z\varphi(z)$,

$$\mathbb{E}|Z'|^3 = 2\int_0^\infty z^3\varphi(z)\,dz = 2\Big(\big[-z^2\varphi(z)\big]_{z=0}^{z=\infty} + 2\int_0^\infty z\,\varphi(z)\,dz\Big) = 4\varphi(0) = 1.5958,$$

because the bracket vanishes at both ends and $\int_0^\infty z\,\varphi(z)\,dz = \big[-\varphi(z)\big]_{z=0}^{z=\infty} = \varphi(0)$. So $\mathbb{E}|R'|^3 \le 4x^3\,(1+4\varphi(0))$, and with (D.13),

$$|\mathcal{M}_k - \mathcal{M}_{k-1}| \le \tfrac16 \sup|\psi'''|\,\big(x^3 + 4x^3(1+4\varphi(0))\big) = \tfrac16\big(5+16\varphi(0)\big)\,\sup|\psi'''|\,x^3. \tag{D.15}$$

Step 4 (summing the replacements). Telescoping and (D.15) give

$$\begin{aligned}|\mathcal{M}_n - \mathcal{M}_0| \le \sum_{k=1}^{n}|\mathcal{M}_k - \mathcal{M}_{k-1}| &\le \tfrac16\big(5+16\varphi(0)\big)\,\sup|\psi'''|\,nx^3\\ &= \tfrac16\big(5+16\varphi(0)\big)\,\sigma^3T^{3/2}\,\sup|\psi'''|\,n^{-1/2},\end{aligned}$$

since $nx^3 = n\sigma^3(T/n)^{3/2} = \sigma^3T^{3/2}n^{-1/2}$. The constant $\frac16(5+16\varphi(0))$ is 1.897.

Step 5 (the normal comparison). It remains to compare $\mathcal{M}_0$, a normal expectation with the exact mean and variance of the tree, with the normal expectation of the statement. By the mean value theorem (Corollary 7.32 at order zero), for every real z,

$$\begin{aligned}&\Big|\psi\Big(n(2p-1)\,x + 2x\big(np(1-p)\big)^{1/2}z\Big) - \psi\big(\nu T + \sigma\sqrt{T}\,z\big)\Big|\\ &\qquad\le \sup|\psi'|\,\Big(\big|n(2p-1)\,x - \nu T\big| + \Big|2x\big(np(1-p)\big)^{1/2} - \sigma\sqrt{T}\Big|\,|z|\Big).\end{aligned}$$

Taking the expectation over $z = Z$, with $\mathbb{E}|Z| = 2\int_0^\infty z\varphi(z)\,dz = 2\varphi(0)$ (an antiderivative of $z\varphi(z)$ is $-\varphi(z)$),

$$\big|\mathcal{M}_0 - \mathbb{E}\,\psi\big(\nu T + \sigma\sqrt{T}\,Z\big)\big| \le \sup|\psi'|\,\Big(\big|n(2p-1)\,x - \nu T\big| + 2\varphi(0)\,\Big|2x\big(np(1-p)\big)^{1/2} - \sigma\sqrt{T}\Big|\Big).$$

By (D.10) multiplied by n, the first difference is at most $c\,n\Delta t^{3/2} = cT^{3/2}n^{-1/2}$, and the squares $4np(1-p)\,x^2$ and σ^2T differ by at most $c\,n\Delta t^2 = cT^2/n$. A difference of square roots is the difference of the squares divided by the sum of the roots, and that sum is at least $\sigma\sqrt{T}$; so the two square roots differ by at most $cT^{3/2}/(\sigma n) \le cT^{3/2}n^{-1/2}/\sigma$. Hence

$$\big|\mathcal{M}_0 - \mathbb{E}\,\psi\big(\nu T + \sigma\sqrt{T}\,Z\big)\big| \le cT^{3/2}\Big(1 + \frac{2\varphi(0)}{\sigma}\Big)\sup|\psi'|\,n^{-1/2}.$$

Adding the bound of Step 4, and recalling that $\mathcal{M}_n = \mathbb{E}^{\mathbb{Q}}\psi(\ln(S_T/S))$, proves (18.30) with

$$c_0 = \max\Big(\tfrac16\big(5+16\varphi(0)\big)\,\sigma^3T^{3/2},\ cT^{3/2}\Big(1+\frac{2\varphi(0)}{\sigma}\Big)\Big),$$

which depends only on r, σ and T. This is (i).

Step 6 (the put: smoothing the kink). The put payoff has a kink at the strike, so (i) does not apply to it directly. We smooth the kink with a kernel. Let X be a random variable with the density

$$f_X(y) = \tfrac{35}{32}\,(1-y^2)^3 \ \text{ for } |y| \le 1, \qquad f_X(y) = 0 \ \text{ otherwise.}$$

It is a density, because $f_X \ge 0$ and $\int_{-1}^1(1-y^2)^3\,dy = \int_{-1}^1(1-3y^2+3y^4-y^6)\,dy = 2\,(1-1+\frac35-\frac17) = \frac{32}{35}$. The variable X takes its values in $[-1,1]$; $\mathbb{E}X = 0$, because $yf_X(y)$ is an odd function (Proposition 7.10); and $\mathbb{E}|X| = \frac{35}{16}\int_0^1 y(1-y^2)^3\,dy = \frac{35}{16}\cdot\frac18 = \frac{35}{128}$, since $-(1-y^2)^4/8$ is an antiderivative of $y(1-y^2)^3$. Its distribution function is $F_X(y) = P(X \le y) = \int_{-\infty}^y f_X$, which

equals 0 on $(-\infty,-1]$ and 1 on $[1,\infty)$; the subscript names the variable, as in f_X, and F_X is not a forward price.

The kernel is smooth. The function f_X has two continuous derivatives on the real line: for $|y|\le 1$,

$$f_X'(y)=-\tfrac{105}{16}\,y\,(1-y^2)^2,\qquad f_X''(y)=-\tfrac{105}{16}\,(1-y^2)(1-5y^2),$$

and both vanish for $|y|\ge 1$. Inside $(-1,1)$ these are the derivatives of the polynomial, and outside $[-1,1]$ the function is zero. At $y=1$ the difference quotients of f_X and of f_X' from the left are $-\frac{35}{32}(1-y)^2(1+y)^3$ and $\frac{105}{16}\,y\,(1-y)(1+y)^2$, which tend to 0 as $y\to 1$, and those from the right are 0; so $f_X'(1)=f_X''(1)=0$, the values that the two formulas take at $y=1$, and both derivatives are continuous there. The point $y=-1$ is the same by symmetry. Moreover $0\le f_X\le\frac{35}{32}$; $|f_X'|\le 1.88$, because $|y|(1-y^2)^2$ is largest at $y^2=\frac15$, where it equals $16/(25\sqrt5)=0.2862$; and $|f_X''|\le\frac{105}{16}$, because the quadratic $(1-t)(1-5t)=1-6t+5t^2$ lies between $-\frac45$ and 1 for $0\le t=y^2\le 1$. Since f_X is continuous, the fundamental theorem of calculus gives $F_X'=f_X$: the function F_X has the three bounded continuous derivatives f_X, f_X' and f_X'', and $0\le F_X\le 1$.

The smoothed positive part. For real z put $(z)_1^+=\mathbb{E}\,(z-X)^+$. Three properties hold.

(a) For $|z|\ge 1$ the smoothing changes nothing: $(z)_1^+=z^+$. For $z\ge 1$, $z-X\ge 0$ and $(z)_1^+=\mathbb{E}(z-X)=z$; for $z\le -1$, $(z-X)^+=0$.

(b) For every real z, $0\le (z)_1^+-z^+\le\mathbb{E}|X|=\frac{35}{128}$. Indeed $(z-X)^+$ is at least 0 and at least $z-X$, so $(z)_1^+$ is at least 0 and at least $z-\mathbb{E}X=z$, that is, at least z^+; and $(z-X)^+\le z^++|X|$, because both 0 and $z-X$ are at most $z^++|X|$, so that $(z)_1^+\le z^++\mathbb{E}|X|$ after taking expectations.

(c) For every real z, $(z)_1^+=\int_{-\infty}^z F_X(y)\,dy$. For $z\le -1$ both sides are 0. For $-1\le z\le 1$, Fact 7.22 and one integration by parts with the antiderivative F_X of f_X give

$$(z)_1^+=\int_{-1}^z(z-y)\,f_X(y)\,dy=\Big[(z-y)\,F_X(y)\Big]_{y=-1}^{y=z}+\int_{-1}^z F_X(y)\,dy=\int_{-1}^z F_X(y)\,dy,$$

because the bracket vanishes at $y=z$ and, since $F_X(-1)=0$, at $y=-1$. For $z\ge 1$, $\int_{-1}^z F_X(y)\,dy=\int_{-1}^1 F_X(y)\,dy+(z-1)=(1)_1^++z-1=z$, by the case just proved and (a).

By (c), the fundamental theorem of calculus and the continuity of F_X, the function $z\mapsto(z)_1^+$ has the derivative $F_X(z)$ at every real z. Its first three derivatives are therefore F_X, f_X and f_X', which are continuous and bounded; the fourth, f_X'', is continuous and bounded as well, and is not needed. For $\varepsilon>0$ put

$$(z)_\varepsilon^+=\varepsilon\,(z/\varepsilon)_1^+=\varepsilon\,\mathbb{E}\,(z/\varepsilon-X)^+.$$

Since $z^+=\varepsilon\,(z/\varepsilon)^+$, properties (a) to (c) give: $(z)_\varepsilon^+=z^+$ for $|z|\ge\varepsilon$; $0\le(z)_\varepsilon^+-z^+\le\frac{35}{128}\,\varepsilon\le\varepsilon$ for every real z; and $(\cdot)_\varepsilon^+$ has bounded continuous derivatives of orders one to three, the derivative of order j at z being ε^{1-j} times the derivative of order j of $(\cdot)_1^+$ at z/ε. These three derivatives vanish for $z\le-\varepsilon$.

The smoothed put. Fix $\varepsilon>0$, write $\chi(z)=(z)_\varepsilon^+$ for the length of this step, and let

$$\psi_\varepsilon(y)=\big(K-Se^y\big)_\varepsilon^+=\chi(K-Se^y).$$

By the chain rule and the product rule,

$$\begin{aligned}
\psi_\varepsilon'(y)&=-\chi'(K-Se^y)\,Se^y,\\
\psi_\varepsilon''(y)&=\chi''(K-Se^y)\,S^2e^{2y}-\chi'(K-Se^y)\,Se^y,\\
\psi_\varepsilon'''(y)&=-\chi'''(K-Se^y)\,S^3e^{3y}+3\chi''(K-Se^y)\,S^2e^{2y}-\chi'(K-Se^y)\,Se^y.
\end{aligned}$$

All three are continuous, and all three are bounded: where $Se^y\ge K+\varepsilon$ the argument $K-Se^y$ is at most $-\varepsilon$ and the derivatives of χ vanish there, and where $Se^y<K+\varepsilon$ every factor

Se^y is below $K+\varepsilon$ while the derivatives of χ are bounded. So (i) applies to ψ_ε. Moreover $|\psi_\varepsilon(y)-(K-Se^y)^+|\le\varepsilon$ for every real y.

By (18.11) with the payoff of the put, $\mathcal{P}_n=e^{-rT}\,\mathbb{E}^{\mathbb{Q}}(K-S_T)^+$, and $S_T=Se^{\ln(S_T/S)}$. Replacing the put payoff by ψ_ε changes the tree expectation by at most ε and the normal expectation by at most ε, so

$$\Big|\mathcal{P}_n-e^{-rT}\,\mathbb{E}\big(K-Se^{\nu T+\sigma\sqrt{T}\,Z}\big)^+\Big|\le e^{-rT}\Big(2\varepsilon+\Big|\,\mathbb{E}^{\mathbb{Q}}\psi_\varepsilon\Big(\ln\frac{S_T}{S}\Big)-\mathbb{E}\,\psi_\varepsilon\big(\nu T+\sigma\sqrt{T}\,Z\big)\Big|\Big).$$

For each fixed ε the last term tends to 0 as $n\to\infty$, by (i). The left side therefore stays below $3\varepsilon e^{-rT}$ for all large n. Since $\varepsilon>0$ was arbitrary, the left side tends to 0: this is the first limit of (18.31), part (ii) for the put.

Step 7 (the call, through parity). The payoff of the call is unbounded, so the call is reached through the put. For every real z, $(z-K)^+-(K-z)^+=z-K$. By (18.11) for the two payoffs and Theorem 1.2,

$$C_n-\mathcal{P}_n=e^{-rT}\big(\mathbb{E}^{\mathbb{Q}}S_T-K\big),\qquad \mathbb{E}^{\mathbb{Q}}S_T=S\left(\mathbb{E}^{\mathbb{Q}}e^{R_1}\right)^n=S\left(pu+(1-p)d\right)^n=Se^{rn\Delta t}=Se^{rT},$$

by (18.24) with $\theta=1$ and Proposition 18.7(i). This is (18.32), part (iii): the identity (18.12) of §18.3, read on the CRR tree and proved here a second way, through (18.24). The same identity holds in the limit. The variable $(Se^{\nu T+\sigma\sqrt{T}\,Z}-K)^+$ lies between 0 and $Se^{\nu T+\sigma\sqrt{T}\,Z}$, whose expectation is finite by Lemma 18.35(i), and the put payoff is bounded by K; so both expectations below are finite, and by the linearity of the Gaussian integral and Lemma 18.35(i),

$$\mathbb{E}\big(Se^{\nu T+\sigma\sqrt{T}\,Z}-K\big)^+-\mathbb{E}\big(K-Se^{\nu T+\sigma\sqrt{T}\,Z}\big)^+=S\,\mathbb{E}\,e^{\nu T+\sigma\sqrt{T}\,Z}-K=Se^{\nu T+\sigma^2T/2}-K=Se^{rT}-K.$$

By (iii) and Step 6, $C_n=\mathcal{P}_n+S-Ke^{-rT}$ converges to

$$e^{-rT}\,\mathbb{E}\big(K-Se^{\nu T+\sigma\sqrt{T}\,Z}\big)^+ + S-Ke^{-rT}=e^{-rT}\,\mathbb{E}\big(Se^{\nu T+\sigma\sqrt{T}\,Z}-K\big)^+,$$

which is the second limit of (18.31), part (ii) for the call.

Step 8 (moment generating functions: compounding along a convergent sequence). Fix a real θ. By (18.24), $\mathbb{E}^{\mathbb{Q}}e^{\theta\ln(S_T/S)}=(\mathbb{E}^{\mathbb{Q}}e^{\theta R_1})^n$. The *hyperbolic cosine* and *sine* are

$$\cosh y=\frac{e^y+e^{-y}}{2},\qquad \sinh y=\frac{e^y-e^{-y}}{2},$$

as in the solutions of Problems P18.6 and P18.8. With them,

$$\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}=p\,e^{\theta x}+(1-p)\,e^{-\theta x}=\cosh(\theta x)+(2p-1)\sinh(\theta x),$$

as one checks by collecting the coefficients of $e^{\theta x}$ and of $e^{-\theta x}$ on the right: they are $\frac12+\frac12(2p-1)=p$ and $\frac12-\frac12(2p-1)=1-p$. By Proposition 7.34 applied to $e^{\theta x}$ and to $e^{-\theta x}$, as $x\to0$,

$$\cosh(\theta x)=1+\frac{\theta^2x^2}{2}+O(x^4),\qquad \sinh(\theta x)=\theta x\,(1+O(x^2)),$$

the odd powers cancelling in the first sum and the even powers in the second. By (D.10), $(2p-1)\,x=\nu\Delta t+O(\Delta t^{3/2})$, which is $O(\Delta t)$. With $x^2=\sigma^2\Delta t$,

$$\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}=1+\frac{\theta^2\sigma^2}{2}\,\Delta t+O(\Delta t^2)+\theta\,(2p-1)\,x\,(1+O(\Delta t))=1+\Big(\theta\nu+\frac{\theta^2\sigma^2}{2}\Big)\Delta t+O(\Delta t^{3/2}),$$

the constants in the O-terms depending on θ as well as on r, σ and T. Put $a_n=n\,(\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}-1)$, so that $\mathbb{E}^{\mathbb{Q}}e^{\theta R_1}=1+a_n/n$. Since $n\Delta t=T$ and $n\Delta t^{3/2}=T^{3/2}n^{-1/2}$,

$$a_n\to a=\theta\nu T+\frac{\theta^2\sigma^2T}{2};$$

the letters a_n and a follow Proposition 7.39, which treats a fixed a. For $|y| \le \frac{1}{2}$, Corollary 7.32 at order one, applied to the logarithm about 0, gives

$$\ln(1+y) = y - \frac{y^2}{2\,(1+y^*)^2}$$

for a point y^* with $|y^*| \le |y| \le \frac{1}{2}$; then $(1+y^*)^2 \ge \frac{1}{4}$ and $|\ln(1+y) - y| \le 2y^2$. The sequence a_n converges, so it is bounded, and $|a_n|/n \le \frac{1}{2}$ for all large n. For such n,

$$\Big| n \ln\Big(1 + \frac{a_n}{n}\Big) - a_n \Big| \le n \cdot 2\,\frac{a_n^2}{n^2} = \frac{2a_n^2}{n} \to 0,$$

so $n \ln(1 + a_n/n) \to a$, and by the continuity of the exponential

$$(\mathbb{E}^{\mathbb{Q}} e^{\theta R_1})^n = \exp\Big(n \ln\Big(1 + \frac{a_n}{n}\Big)\Big) \to e^a = e^{\theta\nu T + \theta^2\sigma^2 T/2} = \mathbb{E}\, e^{\theta(\nu T + \sigma\sqrt{T}\,Z)},$$

the last equality by Lemma 18.35(i), applied to the normal variable $\nu T + \sigma\sqrt{T}\,Z$, whose mean is νT and whose standard deviation is $\sigma\sqrt{T}$. This extends Proposition 7.39 from a fixed a to a convergent sequence $a_n \to a$. At $\theta = 1$, $\mathbb{E}^{\mathbb{Q}} e^{R_1} = pu + (1-p)d = e^{r\Delta t}$ exactly (Proposition 18.7(i)), so the left side is e^{rT} for every n, and the limit is $e^{\nu T + \sigma^2 T/2} = e^{rT}$ as well. This is (iv). It is not used for (ii); it covers payoffs such as S_T^2, whose derivatives in the log variable are unbounded and so fall outside (i).

Step 9 (distribution functions: smoothing the indicator). Fix a real y_0 and $\varepsilon > 0$. With the distribution function F_X of Step 6, put

$$\psi_+(y) = 1 - F_X\Big(\frac{y - y_0 - \varepsilon}{\varepsilon}\Big), \qquad \psi_-(y) = 1 - F_X\Big(\frac{y - y_0 + \varepsilon}{\varepsilon}\Big).$$

Both have bounded continuous derivatives of orders one to three, by Step 6, so (i) applies to them. Since F_X takes its values in $[0,1]$, equals 0 on $(-\infty, -1]$ and equals 1 on $[1, \infty)$, the function ψ_+ takes its values in $[0,1]$, equals 1 for $y \le y_0$ and 0 for $y \ge y_0 + 2\varepsilon$, while ψ_- takes its values in $[0,1]$, equals 1 for $y \le y_0 - 2\varepsilon$ and 0 for $y \ge y_0$. Hence, for every real y,

$$\mathbf{1}\{y \le y_0 - 2\varepsilon\} \le \psi_-(y) \le \mathbf{1}\{y \le y_0\} \le \psi_+(y) \le \mathbf{1}\{y \le y_0 + 2\varepsilon\}.$$

Evaluating the middle inequalities at $y = \ln(S_T/S)$ and taking $\mathbb{Q}$-expectations (Lemma 1.4),

$$\mathbb{E}^{\mathbb{Q}}\psi_-\Big(\ln\frac{S_T}{S}\Big) \le \mathbb{Q}\Big(\ln\frac{S_T}{S} \le y_0\Big) \le \mathbb{E}^{\mathbb{Q}}\psi_+\Big(\ln\frac{S_T}{S}\Big).$$

By (i) the outer terms converge, as $n \to \infty$, to $\mathbb{E}\,\psi_-(\nu T + \sigma\sqrt{T}\,Z)$ and $\mathbb{E}\,\psi_+(\nu T + \sigma\sqrt{T}\,Z)$. The whole chain of inequalities gives $\mathbf{1}\{y \le y_0 - 2\varepsilon\} \le \psi_-(y) \le \psi_+(y) \le \mathbf{1}\{y \le y_0 + 2\varepsilon\}$ for every real y; evaluated at $y = \nu T + \sigma\sqrt{T}\,Z$ and followed by expectations (Lemma 1.4), it shows that these two limits lie between

$$P\big(\nu T + \sigma\sqrt{T}\,Z \le y_0 - 2\varepsilon\big) = \Phi\Big(\frac{y_0 - 2\varepsilon - \nu T}{\sigma\sqrt{T}}\Big) \qquad \text{and}$$
$$P\big(\nu T + \sigma\sqrt{T}\,Z \le y_0 + 2\varepsilon\big) = \Phi\Big(\frac{y_0 + 2\varepsilon - \nu T}{\sigma\sqrt{T}}\Big).$$

The derivative of Φ is $\varphi \le \varphi(0)$, so by the mean value theorem both numbers are within $2\varepsilon\varphi(0)/(\sigma\sqrt{T})$ of $\Phi((y_0 - \nu T)/(\sigma\sqrt{T}))$. Hence, for all large n, $\mathbb{Q}(\ln(S_T/S) \le y_0)$ is within $3\varepsilon\varphi(0)/(\sigma\sqrt{T})$ of $\Phi((y_0 - \nu T)/(\sigma\sqrt{T}))$. Since $\varepsilon > 0$ was arbitrary, this proves (v). $\square$

E Proofs for §19: Brownian Motion, Itô's Lemma and the Log-normal Model

E.1 Proof of Proposition 19.8: Covariance, scaling, symmetry and restart of Brownian motion

The statement is on page 263. The notation and the standing assumptions are those of §19.2.

Proof. We expand a product for (i) and check the four properties of Definition 19.5 for (ii) to (iv).

Part (i). For $t > 0$, properties (1) and (2) give $W_t = W_t - W_0 \sim \mathcal{N}(0, t)$, so $\mathbb{E}[W_t] = 0$ and $\mathbb{E}[W_t^2] = t$ by Proposition 7.25(i) and (ii); both equalities also hold at $t = 0$, where $W_0 = 0$. The means being zero, the covariance equals $\mathbb{E}[W_s W_t]$. Both $\mathbb{E}[W_s W_t]$ and $\min(s, t)$ are symmetric in s and t, so it suffices to take $s \le t$. If $s = 0$ the product is $0 = \min(0, t)$, and if $s = t$ its mean is $\mathbb{E}[W_s^2] = s$. If $0 < s < t$, write $W_t = W_s + (W_t - W_s)$ and expand:

$$\mathbb{E}[W_s W_t] = \mathbb{E}[W_s^2] + \mathbb{E}\big[W_s\,(W_t - W_s)\big] = s + \mathbb{E}[W_s]\,\mathbb{E}[W_t - W_s] = s.$$

The middle step holds because $W_s = W_s - W_0$ and $W_t - W_s$ are the increments of W over the grid $0 < s < t$, which are independent by property (3) and have finite variances, so the mean of their product is the product of their means by the product rule (19.1) of §19.1. The last step uses $\mathbb{E}[W_s] = 0$. Dividing by the standard deviations $\sqrt{s}$ and $\sqrt{t}$ gives the correlation $s/\sqrt{st} = \sqrt{s/t}$.

Parts (ii) to (iv). Write $\tilde{W}$ for the new process in each case: $\tilde{W}_t = c^{-1/2}\,W_{ct}$, or $\tilde{W}_t = -W_t$, or $\tilde{W}_t = W_{u+t} - W_u$.

Property (1). In each case $\tilde{W}_0 = 0$, because $W_0 = 0$ and $W_u - W_u = 0$.

Property (2). For $0 \le s < t$ the increment $\tilde{W}_t - \tilde{W}_s$ equals

$$c^{-1/2}\,(W_{ct} - W_{cs}), \qquad -(W_t - W_s), \qquad W_{u+t} - W_{u+s}$$

in the three cases: a constant multiple of an increment of W over an interval of length $c(t - s)$, $t - s$ and $t - s$. By (19.3) of the Setup of §19.2, applied with the multiplier $c^{-1/2}$, -1 or 1 in the role of its constant c and with $c' = 0$, its law is normal with mean zero and variance $(c^{-1/2})^2 \cdot c(t - s) = t - s$, $(-1)^2(t - s) = t - s$ and $t - s$.

Property (3). Let $0 \le t_0 < t_1 < \cdots < t_k$. The increments of $\tilde{W}$ over this grid are constant multiples of the increments of W over the grid $ct_0 < \cdots < ct_k$, or $t_0 < \cdots < t_k$, or $u + t_0 < \cdots < u + t_k$. Those are mutually independent by property (3) of W, and functions of them taken one at a time are mutually independent by the grouping property (a) of the Setup of §19.2, with one increment in each group.

Property (4). The path of $\tilde{W}$ is the continuous path of W composed with the continuous map $t \mapsto ct$ or $t \mapsto u + t$, then multiplied by a constant or shifted by the constant W_u; it is continuous.

The independence clause of (iv). Take values $W_{s'_1}, \ldots, W_{s'_m}$ with $0 \le s'_j \le u$ and finitely many increments of $\tilde{W}$; the latter are increments of W over intervals contained in $[u, \infty)$. Merge the times 0, $s'_1, \ldots, s'_m$, u and the end points of those intervals into one increasing grid. By property (3) the increments of W over this grid are mutually independent. Each value $W_{s'_j} = W_{s'_j} - W_0$ is the sum of the grid increments over the cells inside $[0, s'_j]$, which lie in $[0, u]$, and each chosen increment of $\tilde{W}$ is the sum of the grid increments over cells inside $[u, \infty)$. The two groups of grid increments have no member in common, so the vector of the chosen values and the vector of the chosen increments of $\tilde{W}$, functions of the two groups, are independent by the grouping property (a) of the Setup of §19.2. □

E.2 Proof of Lemma 19.12: Continuity of probability, Markov, Cauchy–Schwarz and the modes of convergence

The statement is on page 265. The notation and the standing assumptions are those of §19.2.

Proof. We disjointify for (0), bound an indicator for (i), expand a square for (ii), and split on a small event for (iii) and (iv).

Part (0). Only countable additivity (the Setup of §19.2) and finite additivity are used. Let $A_1 \subset A_2 \subset \cdots$, and put $A'_1 = A_1$ and $A'_k = A_k \setminus A_{k-1}$ for $k \geq 2$. These events are disjoint: for $j < k$, $A'_j \subset A_j \subset A_{k-1}$, while A'_k avoids A_{k-1}. Since $A_n = A_{n-1} \cup A'_n$, induction gives $A'_1 \cup \cdots \cup A'_n = A_n$ for every n, so the events A'_k and the events A_n have the same union. Countable additivity, then finite additivity, give

$$P\Big(\bigcup_{n=1}^{\infty} A_n\Big) = \sum_{k=1}^{\infty} P(A'_k) = \lim_{n\to\infty} \sum_{k=1}^{n} P(A'_k) = \lim_{n\to\infty} P(A_n).$$

Let now $A_1 \supset A_2 \supset \cdots$. The complements A_n^{c} increase, their union is the complement of $\bigcap_n A_n$, and $P(A^{\mathrm{c}}) = 1 - P(A)$, so by the case just proved

$$P\Big(\bigcap_n A_n\Big) = 1 - \lim_{n\to\infty} P(A_n^{\mathrm{c}}) = \lim_{n\to\infty} P(A_n).$$

For arbitrary events put $A'_1 = A_1$ and $A'_n = A_n \setminus (A_1 \cup \cdots \cup A_{n-1})$ for $n \geq 2$. These are disjoint, and they have the same union as the A_n, because an outcome of the union belongs to A_n for a smallest index n and then to A'_n. Moreover $A'_n \subset A_n$, so $P(A'_n) \leq P(A_n)$ by finite additivity. Hence $P\big(\bigcup_n A_n\big) = \sum_n P(A'_n) \leq \sum_n P(A_n)$.

Part (i). Pointwise, $a\,\mathbf{1}\{X \geq a\} \leq X$: on the event $\{X \geq a\}$ the left side is $a \leq X$, and off it the left side is $0 \leq X$. Expectation is linear and monotone (the Setup of §19.2), and the mean of an indicator is the probability of its event (Lemma 1.4), so $a\,P(X \geq a) \leq \mathbb{E}[X]$; the bound says nothing when $\mathbb{E}[X]$ is infinite. For Chebyshev's inequality note that $\{|X - Y| \geq \varepsilon\} = \{(X-Y)^2 \geq \varepsilon^2\}$ and apply Markov's inequality to the nonnegative variable $(X-Y)^2$ with $a = \varepsilon^2$. The proof of Theorem 10.8 used Markov's inequality without stating it.

Part (ii). Since $(|X| - |Y|)^2 \geq 0$, we have $|XY| \leq (X^2 + Y^2)/2$ pointwise, so $\mathbb{E}|XY| \leq (\mathbb{E}[X^2] + \mathbb{E}[Y^2])/2$ is finite. For every real c, by linearity,

$$0 \leq \mathbb{E}\big[(|X| - c\,|Y|)^2\big] = \mathbb{E}[X^2] - 2c\,\mathbb{E}|XY| + c^2\,\mathbb{E}[Y^2].$$

If $\mathbb{E}[Y^2] > 0$, choose $c = \mathbb{E}|XY|/\mathbb{E}[Y^2]$, the value that minimises the right side: the inequality becomes $0 \leq \mathbb{E}[X^2] - (\mathbb{E}|XY|)^2/\mathbb{E}[Y^2]$, which is the claim. If $\mathbb{E}[Y^2] = 0$, the inequality reads $2c\,\mathbb{E}|XY| \leq \mathbb{E}[X^2]$ for every $c > 0$, which forces $\mathbb{E}|XY| = 0$, and the claim holds again. For the triangle inequality expand the square, use $XY \leq |XY|$, and apply the inequality just proved:

$$\begin{aligned}\mathbb{E}[(X+Y)^2] &= \mathbb{E}[X^2] + 2\,\mathbb{E}[XY] + \mathbb{E}[Y^2] \\ &\leq \mathbb{E}[X^2] + 2\big(\mathbb{E}[X^2]\,\mathbb{E}[Y^2]\big)^{1/2} + \mathbb{E}[Y^2] = \Big(\big(\mathbb{E}[X^2]\big)^{1/2} + \big(\mathbb{E}[Y^2]\big)^{1/2}\Big)^2.\end{aligned}$$

Take square roots. Applied to $X - X'$ and $X' - Y$, whose sum is $X - Y$, this reads $\big(\mathbb{E}[(X-Y)^2]\big)^{1/2} \leq \big(\mathbb{E}[(X-X')^2]\big)^{1/2} + \big(\mathbb{E}[(X'-Y)^2]\big)^{1/2}$ for any third variable X': the triangle inequality for the mean-square distance.

Part (iii). Mean square implies in probability. By Chebyshev's inequality (i), $P(|X_n - X| > \varepsilon) \leq \mathbb{E}[(X_n - X)^2]/\varepsilon^2 \to 0$.

Almost surely implies in probability. Fix $\varepsilon > 0$ and let A_n be the event that $|X_m - X| > \varepsilon$ for some $m \geq n$. The events A_n decrease. An outcome that belongs to every A_n has $|X_m - X| > \varepsilon$

for infinitely many m, so X_m does not converge to X there; thus $\bigcap_n A_n$ lies in the event of non-convergence, which has probability zero. By (0), $P(A_n) \to P(\bigcap_n A_n) = 0$, and $P(|X_n - X| > \varepsilon) \le P(A_n)$.

Uniqueness. Let $X_n \to X$ and $X_n \to X'$ in probability, and fix $\varepsilon > 0$. Since $|X - X'| \le |X_n - X| + |X_n - X'|$,

$$\{|X - X'| > 2\varepsilon\} \subset \{|X_n - X| > \varepsilon\} \cup \{|X_n - X'| > \varepsilon\} \qquad \text{for every } n.$$

The probability of the right side tends to 0 and the left side does not depend on n, so $P(|X - X'| > 2\varepsilon) = 0$. The event $\{X \ne X'\}$ is the union over $k = 1, 2, \dots$ of the events $\{|X - X'| > 2/k\}$, each of probability zero; by (0) it has probability zero.

Sums. The same inclusion with $(X_n + Y_n) - (X + Y) = (X_n - X) + (Y_n - Y)$ gives $P(|X_n + Y_n - X - Y| > 2\varepsilon) \le P(|X_n - X| > \varepsilon) + P(|Y_n - Y| > \varepsilon) \to 0$.

Products. Write

$$X_n Y_n - XY = (X_n - X)(Y_n - Y) + X\,(Y_n - Y) + Y\,(X_n - X).$$

By the rule for sums it is enough that each of the three terms tends to 0 in probability. The first exceeds ε in absolute value only if $|X_n - X| > \sqrt{\varepsilon}$ or $|Y_n - Y| > \sqrt{\varepsilon}$, and both probabilities tend to 0. For the second, for every integer $M \ge 1$,

$$P\big(|X|\,|Y_n - Y| > \varepsilon\big) \le P(|X| > M) + P\big(|Y_n - Y| > \varepsilon/M\big),$$

because the product is at most ε when $|X| \le M$ and $|Y_n - Y| \le \varepsilon/M$. The events $\{|X| > M\}$, $M = 1, 2, \dots$, decrease and their intersection is empty, because X takes real values; so $P(|X| > M) \to 0$ by (0). Given $\varepsilon' > 0$, choose first M with $P(|X| > M) < \varepsilon'/2$ and then n_0 with $P(|Y_n - Y| > \varepsilon/M) < \varepsilon'/2$ for all $n \ge n_0$: the left side is below ε' for $n \ge n_0$. The third term is treated in the same way, with the roles of the two sequences exchanged.

Part (iv). Two applications of the elementary inequality $(a + b)^2 \le 2a^2 + 2b^2$ give $(a - b)^4 \le (2a^2 + 2b^2)^2 \le 8a^4 + 8b^4$ for real a and b, so

$$\mathbb{E}\big[(X_n - X)^4\big] \le c, \qquad c = 8\Big(\sup_m \mathbb{E}[X_m^4] + \mathbb{E}[X^4]\Big),$$

for every n. A finite fourth moment gives a finite second moment: by (ii) applied to X^2 and the constant 1, $\big(\mathbb{E}[X^2]\big)^2 \le \mathbb{E}[X^4]$, and likewise for X_n and for $X_n - X$. Fix $\varepsilon > 0$ and split on the event $\{|X_n - X| > \varepsilon\}$. Pointwise, $(X_n - X)^2 \le \varepsilon^2 + (X_n - X)^2\,\mathbf{1}\{|X_n - X| > \varepsilon\}$. Taking expectations and applying (ii) to $(X_n - X)^2$ and the indicator, whose square is itself,

$$\mathbb{E}\big[(X_n - X)^2\big] \le \varepsilon^2 + \Big(\mathbb{E}\big[(X_n - X)^4\big]\Big)^{1/2} P\big(|X_n - X| > \varepsilon\big)^{1/2} \le \varepsilon^2 + c^{1/2}\,P\big(|X_n - X| > \varepsilon\big)^{1/2}.$$

The last term tends to 0 as $n \to \infty$, so $\mathbb{E}[(X_n - X)^2] \le 2\varepsilon^2$ for all large n. Since ε was arbitrary, $\mathbb{E}[(X_n - X)^2] \to 0$.

For the means, monotonicity and (ii) applied to $X_n - X$ and the constant 1 give

$$\big|\mathbb{E}[X_n] - \mathbb{E}[X]\big| \le \mathbb{E}|X_n - X| \le \Big(\mathbb{E}\big[(X_n - X)^2\big]\Big)^{1/2} \to 0.$$

For the second moments, $X_n^2 - X^2 = (X_n - X)(X_n + X)$, so by (ii)

$$\big|\mathbb{E}[X_n^2] - \mathbb{E}[X^2]\big| \le \Big(\mathbb{E}\big[(X_n - X)^2\big]\Big)^{1/2}\Big(\mathbb{E}\big[(X_n + X)^2\big]\Big)^{1/2}.$$

The first factor tends to 0. By the triangle inequality of (ii) and the bound on second moments just noted, the second factor is at most

$$\big(\mathbb{E}[X_n^2]\big)^{1/2} + \big(\mathbb{E}[X^2]\big)^{1/2} \le \Big(\sup_m \mathbb{E}[X_m^4]\Big)^{1/4} + \big(\mathbb{E}[X^4]\big)^{1/4},$$

which does not depend on n. Hence $\mathbb{E}[X_n^2] \to \mathbb{E}[X^2]$. □

E.3 Proof of Theorem 19.14: Brownian paths have unbounded variation and are nowhere differentiable (Paley, Wiener and Zygmund, 1933)

The statement is on page 266. The notation and the standing assumptions are those of §19.2.

Proof. Part (i) is forced by the quadratic variation, and part (ii) by a counting bound on three consecutive small increments, the argument of Dvoretzky, Erdős and Kakutani (1961). The proof has four steps.

Step 1: part (i) on one interval, for the dyadic sums. Fix $t > 0$ and consider the dyadic partitions of $[0, t]$ into $n = 2, 4, 8, \dots$ equal cells (n counts the cells, as in the Setup of §19.2), so that $\Delta t_{\max} = t/n$; let QV_n and FV_n be their sums. Each squared increment is the product of its absolute value with itself, and one factor is at most the largest increment, so

$$\mathrm{QV}_n = \sum_{i=1}^{n} |\Delta W_i|\,|\Delta W_i| \le \Big(\max_{1\le i\le n} |\Delta W_i|\Big)\,\mathrm{FV}_n.$$

Fix $M > 0$. If $\mathrm{FV}_n \le M$ and $\mathrm{QV}_n > t/2$, then $\max_i |\Delta W_i| \ge \mathrm{QV}_n/M > t/(2M)$. Hence the event $\{\mathrm{FV}_n \le M\}$ lies in $\{\mathrm{QV}_n \le t/2\} \cup \{\max_i |\Delta W_i| > t/(2M)\}$, and

$$P(\mathrm{FV}_n \le M) \le P\big(\mathrm{QV}_n \le t/2\big) + P\Big(\max_{1\le i\le n} |\Delta W_i| > \frac{t}{2M}\Big). \tag{E.1}$$

For the first term, $\mathrm{QV}_n \le t/2$ implies $|\mathrm{QV}_n - t| \ge t/2$, so by Proposition 19.10(i) and Chebyshev's inequality (Lemma 19.12(i)),

$$P\big(\mathrm{QV}_n \le t/2\big) \le P\big(|\mathrm{QV}_n - t| \ge t/2\big) \le \frac{\mathrm{Var}(\mathrm{QV}_n)}{(t/2)^2} \le \frac{4 \cdot 2t\,\Delta t_{\max}}{t^2} = \frac{8}{n}.$$

For the second term, each fourth power $(\Delta W_i)^4$ has mean $3(t/n)^2$ by Proposition 7.25(iii), and the largest increment exceeds a number $a > 0$ only if some $(\Delta W_i)^4$ is at least a^4. Finite subadditivity over the n cells and Markov's inequality, Lemma 19.12(i), therefore give

$$P\Big(\max_{1\le i\le n} |\Delta W_i| > a\Big) \le \sum_{i=1}^{n} P\big((\Delta W_i)^4 \ge a^4\big) \le n \cdot \frac{3(t/n)^2}{a^4} = \frac{3t^2}{n\,a^4} \qquad \text{for every } a > 0.$$

With $a = t/(2M)$ the second term of (E.1) is at most $3t^2(2M)^4/(n\,t^4) = 48M^4/(n\,t^2)$. (The uniform continuity of a continuous path on the closed interval $[0, t]$ gives the same limit path by path, without a rate.) Hence

$$P(\mathrm{FV}_n \le M) \le \frac{8}{n} + \frac{48M^4}{n\,t^2} \to 0$$

as $n \to \infty$ along the dyadic sequence, for every $M > 0$.

Step 2: part (i) on every interval. By the definition of a supremum, $\mathrm{TV}_t \ge \mathrm{FV}_n$ for every dyadic n. (Refining a partition can only increase its first-variation sum, by the triangle inequality, so the dyadic sums in fact increase with n.) Hence, for each integer $M \ge 1$, every path with $\mathrm{TV}_t \le M$ belongs to the event $\bigcap_n \{\mathrm{FV}_n \le M\}$, the intersection running over $n = 2, 4, 8, \dots$. The probability of this intersection is at most $P(\mathrm{FV}_n \le M)$ for every dyadic n, hence zero by Step 1. The union of these intersections over $M = 1, 2, \dots$ is an event of probability zero (Lemma 19.12(0)), and it contains every path with $\mathrm{TV}_t < \infty$.

The total variation TV_t is nondecreasing in t, because a partition of $[0, t]$ becomes a partition of a longer interval when the new right end point is added, which adds a nonnegative term to the sum. So a path of finite variation on some $[0, t]$, $t > 0$, has finite variation on $[0, 1/m]$ for some integer $m \ge 1$. The union over m of the null events just constructed for $t = 1/m$ is again an event of probability zero, and outside it $\mathrm{TV}_t = \infty$ for every $t > 0$.

Now let $u \ge 0$. The restarted motion $s \mapsto W_{u+s} - W_u$ is a standard Brownian motion (Proposition 19.8(iv)). For every $s' > u$ its increments over a partition of $[0, s'-u]$ are the increments of W over the shifted partition of $[u, s']$, so its total variation on $[0, s'-u]$ is that of W on $[u, s']$. By the previous paragraph, applied to the restarted motion, there is an event of probability zero outside which W has infinite total variation on $[u, s']$ for every $s' > u$. Take the union of these events over the rational numbers $u \ge 0$, a countable family; it has probability zero. Outside it, let $[s, s']$ with $0 \le s < s'$ be any interval and choose a rational u with $s \le u < s'$. Every partition of $[u, s']$, with the point s added when $s < u$, is a partition of $[s, s']$ whose sum is at least as large, so the total variation on $[s, s']$ is at least that on $[u, s']$, which is infinite. Finally, a function that is monotone on $[s, s']$ has increments of one sign there, so each of its sums equals $|W_{s'} - W_s|$ and its total variation is finite. Outside the null event the path is therefore monotone on no interval.

Step 3: part (ii) on $[0,1)$, a necessary event. Suppose that the path is differentiable at some point $s \in (0,1)$, or has a right derivative at $s = 0$, and write $W'(s)$ for that derivative (a prime on a function of one variable is a derivative). Then there are integers $M \ge 1$ and $n_0 \ge 1$ such that

$$|W_{s'} - W_s| \le M\,|s' - s| \qquad \text{whenever } s' \ge 0 \text{ and } |s' - s| \le 4/n_0 : \tag{E.2}$$

take $M \ge |W'(s)| + 1$, and n_0 so large that the difference quotient $(W_{s'} - W_s)/(s' - s)$ stays within 1 of $W'(s)$ for all $s' \ge 0$ with $0 < |s' - s| \le 4/n_0$. Let $n \ge n_0$ and consider the grid $t_i = i/n$, $i = 0, 1, 2, \ldots$, of step $1/n$ on $[0, \infty)$, with increments $\Delta W_i = W_{i/n} - W_{(i-1)/n}$. Let k be the integer part of ns, so that $k/n \le s < (k+1)/n$ and $0 \le k \le n-1$. Since $|k/n - s| < 1/n$, the grid point $(k+i)/n$ lies within $(i+1)/n$ of s, for $i = 0, 1, 2, 3$. The increment ΔW_{k+j}, $j = 1, 2, 3$, joins the grid points $(k+j-1)/n$ and $(k+j)/n$, which lie within j/n and $(j+1)/n$ of s, both at most $4/n \le 4/n_0$. By (E.2) and the triangle inequality,

$$|\Delta W_{k+j}| \le |W_{(k+j)/n} - W_s| + |W_s - W_{(k+j-1)/n}| \le \frac{M(j+1)}{n} + \frac{Mj}{n} \le \frac{7M}{n}, \qquad j = 1, 2, 3.$$

Hence, for every $n \ge n_0$, the path belongs to the event

$$A_{n,M} = \bigcup_{k=0}^{n-1} \Big\{ |\Delta W_{k+1}| \le \frac{7M}{n},\ |\Delta W_{k+2}| \le \frac{7M}{n},\ |\Delta W_{k+3}| \le \frac{7M}{n} \Big\}.$$

Step 4: the bound. Fix integers $M \ge 1$ and $n \ge 1$. For each k the increments ΔW_{k+1}, ΔW_{k+2} and ΔW_{k+3} are independent with the law $\mathcal{N}(0, 1/n)$ (Definition 19.5), the law of $Z/\sqrt{n}$. The density φ never exceeds $\varphi(0) = 1/\sqrt{2\pi}$, so

$$P\Big(|\Delta W_{k+j}| \le \frac{7M}{n}\Big) = P\Big(|Z| \le \frac{7M}{\sqrt{n}}\Big) = \int_{-7M/\sqrt{n}}^{7M/\sqrt{n}} \varphi(z)\,dz \le \frac{14M}{\sqrt{n}}\,\varphi(0) = \frac{14M}{\sqrt{2\pi n}}.$$

The three increments are mutually independent by property (3) of Definition 19.5, so by the product rule of the Setup of §19.2 the probability that all three are this small is the product of the three probabilities, at most the cube of the bound. Finite subadditivity over the n values of k then gives

$$P(A_{n,M}) \le n\,\Big(\frac{14M}{\sqrt{2\pi n}}\Big)^3 = \frac{(14M)^3}{(2\pi)^{3/2}\sqrt{n}} = \frac{c\,M^3}{\sqrt{n}}, \qquad c = \frac{14^3}{(2\pi)^{3/2}} = 174.2,$$

which tends to 0 as $n \to \infty$. By Step 3, the paths that are differentiable at some point of $(0,1)$, or have a right derivative at 0, all belong to

$$\bigcup_{M=1}^{\infty} \bigcup_{n_0=1}^{\infty} \bigcap_{n \ge n_0} A_{n,M}.$$

For fixed M and n_0 the intersection is contained in $A_{n,M}$ for every $n \ge n_0$, so its probability is at most $c\,M^3/\sqrt{n}$ for every such n, hence zero. A countable union of events of probability zero has probability zero (Lemma 19.12(0)).

For the points of $[m, m+1)$, $m = 1, 2, \ldots$, apply Steps 3 and 4 to the restarted motion $\tilde{W}_s = W_{m+s} - W_m$, a standard Brownian motion by Proposition 19.8(iv). If W is differentiable at $m+s$ with $0 < s < 1$, then $\tilde{W}$ is differentiable at s; if W is differentiable at m, then $\tilde{W}$ has a right derivative at 0. (Step 3 at an interior point uses grid points on both sides of it and so needs the two-sided derivative there, which is why the two cases are stated apart.) So the paths of W that are differentiable at some point of $[m, m+1)$ belong to an event of probability zero, and one more countable union, over m, proves (ii). The event of the theorem is the union of the null events of Steps 2 and 4. □

E.4 Proof of Lemma 19.20: Moderate growth in prices is exponential growth in Brownian coordinates

The statement is on page 270. The notation and the standing assumptions are those of §19.3.

Proof. We bound explicitly, one coordinate at a time.

One constant suffices. Let $0 \le a \le b$. For real y, $e^{a|y|} \le e^{b|y|}$. For $x > 0$,

$$(x^b + x^{-b}) - (x^a + x^{-a}) = (x^b - x^a)(1 - x^{-a-b}) \ge 0,$$

because both factors are nonnegative when $x \ge 1$ and both are nonpositive when $x < 1$. Hence a bound $c_1\, e^{c_2(|x_1|+\cdots+|x_d|)}$, or a bound $c_1 \prod_j (x_j^{c_2} + x_j^{-c_2})$, with two constants $c_1, c_2 \ge 0$ holds with the single constant $c = \max(c_1, c_2)$ in both places. The bounds produced below have two constants, and this last step is not repeated.

Part (iii), sums and products. Write $a = |x_1| + \cdots + |x_d| \ge 0$. If two functions are bounded by $c_1 e^{c_1 a}$ and $c_2 e^{c_2 a}$, their sum is bounded by $(c_1 + c_2)\, e^{\max(c_1, c_2)\, a}$ and their product by $c_1 c_2\, e^{(c_1+c_2)\, a}$.

Part (iii), moments. Let f satisfy $|f| \le c\, e^{c(|x_1|+\cdots+|x_d|)}$ and let $k \ge 1$ be a moment order. A product of d numbers that are at least 1 is at most the d-th power of the largest, which is at most the sum of the d-th powers, so

$$|f|^k \le c^k \prod_{j=1}^{d} e^{kc|x_j|} \le c^k \max_{1 \le j \le d} e^{dkc|x_j|} \le c^k \sum_{j=1}^{d} e^{dkc|x_j|}.$$

Evaluate at $x_j = W_{s_j}$ and take expectations; expectation is linear and monotone. If $s_j > 0$, then $W_{s_j} = \sqrt{s_j}\, Z_j$, where $Z_j = W_{s_j}/\sqrt{s_j}$ is standard normal, and Lemma 18.35(i), in the form $\mathbb{E}\, e^{c'|Z|} \le 2e^{c'^2/2}$ with $c' = dkc\sqrt{s_j}$, gives

$$\mathbb{E}\, e^{dkc|W_{s_j}|} \le 2\, e^{(dkc)^2 s_j/2} \le 2\, e^{(dkc)^2 T/2}.$$

If $s_j = 0$, the left side is 1. Hence the k-th absolute moment is at most $2d\, c^k\, e^{(dkc)^2 T/2}$, a bound free of the times. Only the law of each single value W_{s_j} entered, so the bound holds whether the values come from one motion or from two.

Part (i), one price. Suppose $|V(x,t)| \le c\,(x^c + x^{-c})$. At $x = S(t,y)$ with $0 \le t \le T$,

$$x^c + x^{-c} = S_0^c\, e^{cgt}\, e^{c\sigma y} + S_0^{-c}\, e^{-cgt}\, e^{-c\sigma y} \le (S_0^c + S_0^{-c})\, e^{c|g|T}\, e^{c\sigma|y|},$$

so $|V(S(t,y),t)| \le c_1 e^{c_2|y|}$ with $c_1 = c\,(S_0^c + S_0^{-c})\, e^{c|g|T}$ and $c_2 = c\sigma$. This is exponential growth. Conversely, suppose $|V(S(t,y),t)| \le c\, e^{c|y|}$ for all $t \in [0,T]$ and all real y. Given $x > 0$ and $t \in [0,T]$, the number $y = (\ln(x/S_0) - gt)/\sigma$ satisfies $S(t,y) = x$ and $|y| \le (|\ln x| + |\ln S_0| + |g|T)/\sigma$. Since $e^{a|\ln x|} = \max(x^a, x^{-a}) \le x^a + x^{-a}$ for $a \ge 0$,

$$|V(x,t)| \le c\, e^{c|y|} \le c\, e^{c(|\ln S_0| + |g|T)/\sigma}\, (x^{c/\sigma} + x^{-c/\sigma}),$$

which is moderate growth.

Part (i), several prices; sums and products. For a function of d prices and a time, apply the two computations to each coordinate $x_j = S(t_j, y_j)$ and multiply; write σ_j for the volatility constant of the j-th coordinate (all equal to σ when the coordinates share their constants). A bound $c\prod_j (x_j^c + x_j^{-c})$ becomes the product of c, of d constants and of the factors $e^{c\sigma_j|y_j|}$, which is at most the same product of constants times $e^{c\sigma'(|y_1|+\cdots+|y_d|)}$ with $\sigma' = \max_j \sigma_j$. A bound $c\,e^{c(|y_1|+\cdots+|y_d|)} = c\prod_j e^{c|y_j|}$ becomes a constant times $\prod_j (x_j^{c/\sigma_j} + x_j^{-c/\sigma_j})$, which is at most the same constant times $\prod_j (x_j^{c''} + x_j^{-c''})$ with $c'' = c/\min_j \sigma_j$, by the first paragraph of this proof (*One constant suffices*). No constant depends on the times. For sums and products of functions of moderate growth, pass to exponential growth with $S_0 = 1$, $g = 0$ and $\sigma = 1$, so that $x_j = e^{y_j}$, use (iii), and come back.

Part (ii). This is (i) with $S_0 = 1$, $g = 0$ and $\sigma = 1$ in every coordinate, so that $x_j = e^{y_j}$ and $y_j = \ln x_j$: moderate growth of f is exponential growth of $\psi(y_1, \ldots, y_J) = f(e^{y_1}, \ldots, e^{y_J})$. The hyperplanes $\{x_j = \ell\}$ and $\{y_j = \ln \ell\}$ correspond under these maps. The maps $y \mapsto e^y$ and $x \mapsto \ln x$ are continuous, and a composition of continuous maps is continuous, so ψ is continuous at a point $(y_1, \ldots, y_J)$ exactly when f is continuous at $(e^{y_1}, \ldots, e^{y_J})$. □

E.5 Proof of Theorem 19.23: The Itô integral and the Itô isometry (Itô, 1944)

The statement is on page 272. The notation and the standing assumptions are those of §19.3.

Proof. We compute exactly for left-endpoint sums, show that they form a Cauchy sequence in mean square, and take the limit that Fact 19.16 provides. The proof has five steps. Steps 2 to 5 are written for a process ξ in the span of $\mathcal{H}$: a finite sum of elements of $\mathcal{H}$, called its *summands*, each with its own vector Y and its own stopping time. An element of $\mathcal{H}$ is the case of one summand. The left-endpoint sum of such a ξ over a partition is $\mathcal{I}_n(\xi) = \sum_i \xi_{t_{i-1}}\,\Delta W_i$, the sum of the left-endpoint sums of the summands.

Step 1: the next increment is independent of the past. Fix $0 \le s < u \le T$. Let X be a finite vector of values of W, and of $W_\perp$ when it is present, at times at most s. Let $\ell_1, \ldots, \ell_J$ be finitely many continuous boundaries as in Definition 19.22(c), not necessarily distinct, each approached from below or from above, and let $s'_1, \ldots, s'_J$ be times at most s. The claim of this step is that the increment $W_u - W_s$ is independent of the random vector

$$(X,\ \mathbf{1}\{t_{\ell_1} > s'_1\},\ \ldots,\ \mathbf{1}\{t_{\ell_J} > s'_J\}), \tag{E.3}$$

and that, when $W_\perp$ is present, so is the pair of increments $(W_u - W_s,\ W_{\perp,u} - W_{\perp,s})$, whose two coordinates are moreover independent of each other. Several hitting times are allowed because an element of the span carries the indicators of several of them, and so do the trading strategies of §20.

Values. Write $\Delta W = W_u - W_s$ and $X^{(1)}$ for the vector of the values of W in X, and, when $W_\perp$ is present, $\Delta W_\perp = W_{\perp,u} - W_{\perp,s}$ and $X^{(\perp)}$ for the vector of the values of $W_\perp$ in X (proof-local names). By Proposition 19.8(iv), which rests on the independent increments of Definition 19.5(3), ΔW is independent of $X^{(1)}$, and, applied to $W_\perp$, $\Delta W_\perp$ is independent of $X^{(\perp)}$. Without $W_\perp$ the vector X is $X^{(1)}$, and the claim for values is proved. With $W_\perp$, the pairs $(\Delta W, X^{(1)})$ and $(\Delta W_\perp, X^{(\perp)})$ are functions of finite vectors of values of W and of $W_\perp$ respectively, so they are independent of each other (the Setup of §19.3). In particular ΔW is independent of $\Delta W_\perp$, and $X^{(1)}$ of $X^{(\perp)}$, because an event determined by ΔW or by $X^{(1)}$ is determined by the pair $(\Delta W, X^{(1)})$, and likewise for the second pair. The first of these is the independence of the two coordinates of the pair of increments. It remains to show that the pair $(\Delta W, \Delta W_\perp)$ is independent of $(X^{(1)}, X^{(\perp)})$, which is X up to the order of its coordinates. This is the grouping property (a) of the Setup of §19.2 for the four vectors ΔW, $X^{(1)}$, $\Delta W_\perp$, $X^{(\perp)}$; here it follows from Lemma 19.17, as follows. Let A be an event determined by $(\Delta W, \Delta W_\perp)$ and A' an event determined by $(X^{(1)}, X^{(\perp)})$, and

write $\mathbf{1}\{A\} = \psi_1(\Delta W, \Delta W_\perp)$ and $\mathbf{1}\{A'\} = \psi_2(X^{(1)}, X^{(\perp)})$, with ψ_1 and ψ_2 the indicator functions of the two sets. Apply (19.4) to the independent pairs $(\Delta W, X^{(1)})$ and $(\Delta W_\perp, X^{(\perp)})$ and to the nonnegative function $\psi_1(y_1, y_2)\,\psi_2(x_1, x_2)$. With (y_1, x_1) frozen, the mean over $(\Delta W_\perp, X^{(\perp)})$ is

$$\mathbb{E}\big[\psi_1(y_1, \Delta W_\perp)\,\psi_2(x_1, X^{(\perp)})\big] = \bar\psi_1(y_1)\,\bar\psi_2(x_1),$$
$$\bar\psi_1(y_1) = \mathbb{E}\,\psi_1(y_1, \Delta W_\perp), \qquad \bar\psi_2(x_1) = \mathbb{E}\,\psi_2(x_1, X^{(\perp)}),$$

by the product rule (19.5) for the bounded functions $\psi_1(y_1, \cdot)$ of $\Delta W_\perp$ and $\psi_2(x_1, \cdot)$ of $X^{(\perp)}$, which are independent. Hence, by (19.4), then by (19.5) for the bounded functions $\bar\psi_1$ of ΔW and $\bar\psi_2$ of $X^{(1)}$, which are independent,

$$\begin{aligned} P(A \cap A') = \mathbb{E}\big[\psi_1(\Delta W, \Delta W_\perp)\,\psi_2(X^{(1)}, X^{(\perp)})\big] &= \mathbb{E}\big[\bar\psi_1(\Delta W)\,\bar\psi_2(X^{(1)})\big] \\ &= \mathbb{E}\,\bar\psi_1(\Delta W)\,\mathbb{E}\,\bar\psi_2(X^{(1)}). \end{aligned}$$

Finally $\mathbb{E}\,\bar\psi_1(\Delta W) = \mathbb{E}\,\psi_1(\Delta W, \Delta W_\perp) = P(A)$ and $\mathbb{E}\,\bar\psi_2(X^{(1)}) = \mathbb{E}\,\psi_2(X^{(1)}, X^{(\perp)}) = P(A')$, by (19.4) for the independent pairs ΔW, $\Delta W_\perp$ and $X^{(1)}$, $X^{(\perp)}$. So $P(A \cap A') = P(A)\,P(A')$: the pair of increments is independent of X.

Hitting times. Let A be an event determined by the increment, or by the pair of increments, and let A' be an event determined by X. Put

$$A'' = \{t_{\ell_1} > s'_1\} \cap \cdots \cap \{t_{\ell_J} > s'_J\}.$$

We first prove that

$$P(A \cap A' \cap A'') = P(A)\,P(A' \cap A''). \tag{E.4}$$

By (19.7), the event $\{t_{\ell_j} > s'_j\}$ says that the continuous function $\ell_j - W$ is positive on $[0, s'_j]$ (the function $W - \ell_j$, for a boundary approached from above). For integers $k \geq 1$ and $n = 2, 4, 8, \ldots$, let $A_{k,n}$ be the event that, for every j, this function is at least $1/k$ at each point of the dyadic grid of $[0, s'_j]$ with n cells. Then

$$A'' = \bigcup_{k=1}^{\infty} \bigcap_{n} A_{k,n},$$

the intersection running over $n = 2, 4, 8, \ldots$. Indeed, on A'' each of the J functions is positive on a closed interval, where it attains its minimum; an integer k with $1/k$ below the J minima puts the path in $A_{k,n}$ for every n. Conversely, on a path of $\bigcap_n A_{k,n}$ each function is at least $1/k$ at every dyadic point of its interval. These points are dense in the interval and the function is continuous, so it is at least $1/k$ on the whole interval, and the path belongs to A''.

Each $A_{k,n}$ is determined by finitely many values of W at times at most s, so $A' \cap A_{k,n}$ is determined by a finite vector of such values, and the first part of this step gives $P(A \cap A' \cap A_{k,n}) = P(A)\,P(A' \cap A_{k,n})$. For fixed k the events $A_{k,n}$ decrease as n doubles, because the finer dyadic grid contains the coarser one. Continuity of probability along decreasing events (Lemma 19.12(0)), applied to both sides, gives

$$P\Big(A \cap A' \cap \bigcap_n A_{k,n}\Big) = P(A)\,P\Big(A' \cap \bigcap_n A_{k,n}\Big).$$

The events $\bigcap_n A_{k,n}$ increase with k, because the requirement "at least $1/(k+1)$" is weaker than "at least $1/k$". Continuity of probability along increasing events gives (E.4).

Now prescribe a value, 0 or 1, for each of the J indicators. The indicator of the event that the j-th one takes its prescribed value is $\mathbf{1}\{t_{\ell_j} > s'_j\}$ when that value is 1, and $1 - \mathbf{1}\{t_{\ell_j} > s'_j\}$ when it is 0. The product of these J expressions, expanded, is a sum with signs ± 1 of indicators of events of the form A'', each built from some of the pairs (ℓ_j, s'_j); the product over no pair is 1, the indicator of the sure event. Multiply by $\mathbf{1}\{A\}\,\mathbf{1}\{A'\}$ and take expectations. By linearity and Lemma 1.4, the probability of $A \cap A'$ intersected with the event that the J indicators take their prescribed

values is the same signed sum of probabilities $P(A \cap A' \cap A'')$. Each of these factorises by (E.4), and summing back gives $P(A)$ times the probability of A' intersected with that event. Finally, an event determined by the vector (E.3) is a disjoint union, over the 2^J ways of prescribing the values, of events of the kind just treated, and finite additivity gives the product rule for it. This proves the claim.

Step 2: step sums have mean zero and obey the isometry. Fix a partition $t_0 < t_1 < \cdots < t_n = t$. A *step sum* over it is a sum $\sum_{i=1}^{n} a_i \, \Delta W_i$ whose weights have two properties: a_i is a function of finitely many values of W (and of $W_\perp$) at times at most t_{i-1} and of finitely many indicators $\mathbf{1}\{t_{\ell_j} > s'\}$ with $s' \le t_{i-1}$, for short a function of the past of t_{i-1}; and $\mathbb{E}[a_i^2] < \infty$. We show that

$$\mathbb{E}\Big[\sum_{i=1}^{n} a_i \, \Delta W_i\Big] = 0 \qquad \text{and} \qquad \mathbb{E}\Big[\Big(\sum_{i=1}^{n} a_i \, \Delta W_i\Big)^2\Big] = \sum_{i=1}^{n} \mathbb{E}[a_i^2] \, \Delta t_i. \tag{E.5}$$

By Step 1, a_i is a function of a random vector that is independent of ΔW_i. Its mean is finite, since $\mathbb{E}|a_i| \le (\mathbb{E}[a_i^2])^{1/2}$ by the Cauchy–Schwarz inequality with the constant 1 (Lemma 19.12(ii)), and the increment $\Delta W_i \sim \mathcal{N}(0, \Delta t_i)$ has finite moments. The product rule (19.5) gives

$$\mathbb{E}[a_i \, \Delta W_i] = \mathbb{E}[a_i] \, \mathbb{E}[\Delta W_i] = 0, \qquad \mathbb{E}\big[a_i^2 \, (\Delta W_i)^2\big] = \mathbb{E}[a_i^2] \, \mathbb{E}\big[(\Delta W_i)^2\big] = \mathbb{E}[a_i^2] \, \Delta t_i.$$

Summing the first identity over i proves the first half of (E.5). For $i < j$, the product $a_i \, \Delta W_i \, a_j$ is a function of values at times at most t_{j-1} (those behind a_i and a_j, together with $W_{t_{i-1}}$ and W_{t_i}) and of indicators at times at most t_{j-1}. By Step 1 it is a function of a vector independent of ΔW_j. It is integrable, by the Cauchy–Schwarz inequality:

$$\mathbb{E}|a_i \, \Delta W_i \, a_j| \le \big(\mathbb{E}[a_i^2 (\Delta W_i)^2]\big)^{1/2} \big(\mathbb{E}[a_j^2]\big)^{1/2} < \infty.$$

The product rule gives $\mathbb{E}[a_i \, \Delta W_i \, a_j \, \Delta W_j] = \mathbb{E}[a_i \, \Delta W_i \, a_j] \, \mathbb{E}[\Delta W_j] = 0$. Expand the square of the sum into its n diagonal terms and its cross terms, all integrable; linearity then proves the second half of (E.5).

The same proof gives (E.5) for a step sum $\sum_i b_i \, \Delta W_{\perp,i}$ against $W_\perp$, with weights b_i of the same kind, because it used only that the increment has the law $\mathcal{N}(0, \Delta t_i)$ and is independent of the past in the sense of Step 1. Moreover, two step sums over the same partition against the two motions are uncorrelated:

$$\mathbb{E}\Big[\Big(\sum_{i=1}^{n} a_i \, \Delta W_i\Big)\Big(\sum_{j=1}^{n} b_j \, \Delta W_{\perp,j}\Big)\Big] = 0. \tag{E.6}$$

Take the terms of the product one at a time. If $i = j$, the weight product $a_i b_i$, integrable by the Cauchy–Schwarz inequality, is a function of a vector independent of the pair $(\Delta W_i, \Delta W_{\perp,i})$, whose coordinates are independent of each other with mean zero. Two applications of the product rule give $\mathbb{E}[a_i b_i \, \Delta W_i \, \Delta W_{\perp,i}] = \mathbb{E}[a_i b_i] \, \mathbb{E}[\Delta W_i] \, \mathbb{E}[\Delta W_{\perp,i}] = 0$. If $i < j$, the factor $a_i \, \Delta W_i \, b_j$ is a function of the past of t_{j-1}, and $\Delta W_{\perp,j}$ is independent of it with mean zero. If $i > j$, the factor $b_j \, \Delta W_{\perp,j} \, a_i$ is a function of the past of t_{i-1}, and ΔW_i is independent of it with mean zero. In both cases the term vanishes as in the first part.

The left-endpoint sums of ξ. The weight $a_i = \xi_{t_{i-1}}$ is of the required kind. Each summand of $\xi_{t_{i-1}}$ is a function of $W_{t_{i-1}}$ (and of $W_{\perp,t_{i-1}}$), of its vector Y, whose times are at most $t_0 \le t_{i-1}$, and of its indicator $\mathbf{1}\{t_\ell > t_{i-1}\}$. For the moments, $|\xi_s|$ is at most the sum, over the summands, of $|\xi(s, W_s, Y)|$. Viewed as a function of W_s and of all the vectors Y together, this sum has exponential growth (Lemma 19.20(iii), sums), and it is evaluated at Brownian values at times in $[0, T]$. By Lemma 19.20(iii) its moments are bounded uniformly in s; in particular

$$\sup_{t_0 \le s \le t} \mathbb{E}[\xi_s^4] < \infty, \qquad \text{and} \qquad \big(\mathbb{E}[\xi_s^2]\big)^2 \le \mathbb{E}[\xi_s^4] \tag{E.7}$$

by the Cauchy–Schwarz inequality applied to ξ_s^2 and the constant 1. So (E.5) applies and gives

$$\mathbb{E}[\mathcal{I}_n(\xi)] = 0, \qquad \mathbb{E}[\mathcal{I}_n(\xi)^2] = \sum_{i=1}^{n} \mathbb{E}[\xi_{t_{i-1}}^2]\,\Delta t_i. \tag{E.8}$$

Step 3: mean-square continuity, which is (v). Fix $s \in [t_0, t]$ and a sequence $u_k \to s$ in $[t_0, t]$. Consider one summand, a process $\xi(u, W_u, Y)\,\mathbf{1}\{u < t_\ell\}$ of $\mathcal{H}$; until the summands are added again, the function $\xi(\cdot,\cdot,\cdot)$, the vector Y and the time t_ℓ are those of that summand. On every path, $\xi(u_k, W_{u_k}, Y) \to \xi(s, W_s, Y)$, because the function is continuous in its first two arguments on $[t_0, t] \times \mathbb{R} \times \mathcal{D}$, for the box $\mathcal{D}$ that contains Y, and the path is continuous. (With two motions the function is continuous in the value of $W_\perp$ as well, and both paths are continuous.) The indicator $\mathbf{1}\{u_k < t_\ell\}$ converges to $\mathbf{1}\{s < t_\ell\}$ unless $t_\ell = s$: if $s < t_\ell$, then $u_k < t_\ell$ for all large k, and if $s > t_\ell$, then $u_k > t_\ell$ for all large k. The event $\{t_\ell = s\}$ has probability zero by (19.6). Adding the summands, $\xi_{u_k} \to \xi_s$ outside a finite union of events of probability zero, that is, almost surely, and hence in probability (Lemma 19.12(iii)). The fourth moments of ξ_{u_k} and of ξ_s are bounded by (E.7). The fourth-moment lemma (Lemma 19.12(iv)) therefore gives

$$\mathbb{E}\big[(\xi_{u_k} - \xi_s)^2\big] \to 0 \qquad \text{and} \qquad \mathbb{E}[\xi_{u_k}^2] \to \mathbb{E}[\xi_s^2].$$

The sequence was arbitrary, so $s \mapsto \xi_s$ is continuous in mean square and $s \mapsto \mathbb{E}[\xi_s^2]$ is continuous; the latter is bounded by (E.7). If ζ is a second process of the span, then $\xi + \zeta$ and $\xi - \zeta$ are in the span, and

$$\mathbb{E}[\xi_s \zeta_s] = \tfrac{1}{4}\Big(\mathbb{E}\big[(\xi_s + \zeta_s)^2\big] - \mathbb{E}\big[(\xi_s - \zeta_s)^2\big]\Big)$$

is bounded and continuous in s as well. A continuous function on $[t_0, t]$ is Riemann integrable, which is the last assertion of (v).

The continuity is uniform: for every $\varepsilon > 0$ there is an $\varepsilon' > 0$ such that

$$\mathbb{E}\big[(\xi_u - \xi_s)^2\big] \le \varepsilon \qquad \text{whenever } u, s \in [t_0, t] \text{ and } |u - s| \le \varepsilon'. \tag{E.9}$$

This is the Heine–Cantor argument of calculus. If (E.9) failed for some $\varepsilon > 0$, there would be points $u_k, s_k \in [t_0, t]$ with $|u_k - s_k| \le 1/k$ and $\mathbb{E}[(\xi_{u_k} - \xi_{s_k})^2] > \varepsilon$. A bounded sequence of real numbers has a convergent subsequence; along one, $s_k \to s^* \in [t_0, t]$, and then $u_k \to s^*$ too. The triangle inequality for the mean-square distance (Lemma 19.12(ii)) gives

$$\Big(\mathbb{E}\big[(\xi_{u_k} - \xi_{s_k})^2\big]\Big)^{1/2} \le \Big(\mathbb{E}\big[(\xi_{u_k} - \xi_{s^*})^2\big]\Big)^{1/2} + \Big(\mathbb{E}\big[(\xi_{s^*} - \xi_{s_k})^2\big]\Big)^{1/2},$$

and the right side tends to 0 along the subsequence, by the continuity at s^* just proved. This contradicts the lower bound ε.

Step 4: the left-endpoint sums are Cauchy. Take two partitions of $[t_0, t]$, with meshes $\Delta t_{\max}$ and $\Delta t'_{\max}$ and with left-endpoint sums $\mathcal{I}$ and $\mathcal{I}'$ of ξ. For $s \in [t_0, t)$ let $\underline{s}$ be the last point of the first partition at or before s, and $\underline{s}'$ the last point of the second (proof-local names), so that $0 \le s - \underline{s} \le \Delta t_{\max}$ and $0 \le s - \underline{s}' \le \Delta t'_{\max}$. Merge the points of the two partitions into one partition $t_0 = t''_0 < t''_1 < \cdots < t''_{n''} = t$, their common refinement. A cell $[t_{i-1}, t_i]$ of the first partition is a union of cells of the refinement, on each of which $\underline{s} = t_{i-1}$, and the increments of W over these cells add up to ΔW_i. So $\mathcal{I}$ is the sum over j of $\xi_{\underline{s}}\,(W_{t''_j} - W_{t''_{j-1}})$ with $s = t''_{j-1}$, and likewise for $\mathcal{I}'$ with $\underline{s}'$. The difference is therefore a step sum over the refinement,

$$\mathcal{I} - \mathcal{I}' = \sum_{j=1}^{n''} a_j\,(W_{t''_j} - W_{t''_{j-1}}), \qquad a_j = \xi_{\underline{s}} - \xi_{\underline{s}'} \ \text{ at } s = t''_{j-1}.$$

Its weights are functions of values and of indicators at times at most t''_{j-1}, with finite second moments by (E.7). By (E.5),

$$\mathbb{E}\big[(\mathcal{I} - \mathcal{I}')^2\big] = \sum_{j=1}^{n''} \mathbb{E}[a_j^2]\,(t''_j - t''_{j-1}) = \int_{t_0}^{t} \mathbb{E}\big[(\xi_{\underline{s}} - \xi_{\underline{s}'})^2\big]\,ds,$$

because the function $s \mapsto \mathbb{E}[(\xi_{\underline{s}} - \xi_{\underline{s}'})^2]$ is constant on each cell $[t''_{j-1}, t''_j)$ of the refinement, where it equals $\mathbb{E}[a_j^2]$. Since $(x-y)^2 \le 2(x-z)^2 + 2(z-y)^2$ for real numbers, applied with $z = \xi_s$,

$$\mathbb{E}\big[(\mathcal{I} - \mathcal{I}')^2\big] \le 2\,\mathcal{E} + 2\,\mathcal{E}', \qquad \mathcal{E} = \int_{t_0}^{t} \mathbb{E}\big[(\xi_{\underline{s}} - \xi_s)^2\big]\,ds,$$

where $\mathcal{E}$ is the discretisation error of the first partition and $\mathcal{E}'$, defined with $\underline{s}'$ in place of $\underline{s}$, that of the second. (On each cell of its partition, where $\underline{s}$ is fixed, the integrand of $\mathcal{E}$ is continuous in s, by Step 3 and the triangle inequality; so $\mathcal{E}$ is an ordinary integral.) Given $\varepsilon > 0$, let ε' be as in (E.9). If $\Delta t_{\max} \le \varepsilon'$, then $s - \underline{s} \le \varepsilon'$ for every s, and $\mathcal{E} \le (t - t_0)\,\varepsilon$; the same holds for $\mathcal{E}'$ if $\Delta t'_{\max} \le \varepsilon'$. Hence

$$\mathbb{E}\big[(\mathcal{I} - \mathcal{I}')^2\big] \le 4(t - t_0)\,\varepsilon \qquad \text{whenever both meshes are at most } \varepsilon'. \tag{E.10}$$

Step 5: the limit and its properties. The six parts below prove (i) to (iv) and (vi) for ξ in the span.

(a) Existence and uniqueness. Take a sequence of partitions of $[t_0, t]$ whose meshes tend to zero. In this step $\mathcal{I}_n(\xi)$ denotes the left-endpoint sum of the n-th partition of the sequence. By (E.10), $\mathbb{E}[(\mathcal{I}_n(\xi) - \mathcal{I}_{n'}(\xi))^2] \to 0$ as $n, n' \to \infty$, and the second moments are finite by (E.8). Fact 19.16 provides a random variable, written $\mathcal{I}(\xi)$ in this proof, with a finite second moment and $\mathbb{E}[(\mathcal{I}_n(\xi) - \mathcal{I}(\xi))^2] \to 0$. Now fix $\varepsilon > 0$, with ε' as in (E.9), and let $\mathcal{I}'$ be the left-endpoint sum of *any* partition of mesh at most ε'. By the triangle inequality of Lemma 19.12(ii) and by (E.10), for all n so large that the n-th mesh is at most ε',

$$\Big(\mathbb{E}\big[(\mathcal{I}' - \mathcal{I}(\xi))^2\big]\Big)^{1/2} \le \big(4(t - t_0)\,\varepsilon\big)^{1/2} + \Big(\mathbb{E}\big[(\mathcal{I}_n(\xi) - \mathcal{I}(\xi))^2\big]\Big)^{1/2},$$

and letting $n \to \infty$ gives $\mathbb{E}[(\mathcal{I}' - \mathcal{I}(\xi))^2] \le 4(t - t_0)\,\varepsilon$. So the left-endpoint sums converge to $\mathcal{I}(\xi)$ in mean square as the mesh tends to zero, along every sequence of partitions. A limit obtained from another sequence is then a second mean-square limit of sums that converge to $\mathcal{I}(\xi)$, and two mean-square limits of one sequence agree with probability one (Fact 19.16). This is (i), with $\mathcal{I}(\xi) = \int_{t_0}^{t} \xi_s\,dW_s$.

(b) Mean zero and the isometry. By monotonicity and the Cauchy–Schwarz inequality with the constant 1 for the first line, and by the triangle inequality for the second (Lemma 19.12(ii)),

$$\big|\mathbb{E}[\mathcal{I}_n(\xi)] - \mathbb{E}[\mathcal{I}(\xi)]\big| \le \mathbb{E}\big|\mathcal{I}_n(\xi) - \mathcal{I}(\xi)\big| \le \Big(\mathbb{E}\big[(\mathcal{I}_n(\xi) - \mathcal{I}(\xi))^2\big]\Big)^{1/2},$$

$$\Big|\big(\mathbb{E}[\mathcal{I}_n(\xi)^2]\big)^{1/2} - \big(\mathbb{E}[\mathcal{I}(\xi)^2]\big)^{1/2}\Big| \le \Big(\mathbb{E}\big[(\mathcal{I}_n(\xi) - \mathcal{I}(\xi))^2\big]\Big)^{1/2},$$

and the right sides tend to 0. With (E.8), $\mathbb{E}[\mathcal{I}(\xi)] = 0$, which is (ii), and

$$\mathbb{E}[\mathcal{I}(\xi)^2] = \lim_{n\to\infty} \sum_i \mathbb{E}[\xi_{t_{i-1}}^2]\,\Delta t_i = \int_{t_0}^{t} \mathbb{E}[\xi_s^2]\,ds,$$

which is (19.8): the sums are Riemann sums, with left-endpoint tags, of the function $s \mapsto \mathbb{E}[\xi_s^2]$, which is continuous on $[t_0, t]$ by Step 3, and uniformly continuous there by (E.9) and the Cauchy–Schwarz inequality (Lemma 19.12(ii)):

$$\big|\mathbb{E}[\xi_u^2] - \mathbb{E}[\xi_s^2]\big| \le \Big(\mathbb{E}\big[(\xi_u - \xi_s)^2\big]\Big)^{1/2}\Big(\mathbb{E}\big[(\xi_u + \xi_s)^2\big]\Big)^{1/2},$$

where the second factor is bounded on $[t_0, t]$ by Step 3, because $(\xi_u + \xi_s)^2 \le 2\xi_u^2 + 2\xi_s^2$. So the Riemann sums converge to its integral as the mesh tends to zero.

(c) Linearity and the product form. Let ξ and ζ be in the span and let c, c' be real constants. Then $c\xi + c'\zeta$ is in the span, and $\mathcal{I}_n(c\xi + c'\zeta) = c\,\mathcal{I}_n(\xi) + c'\,\mathcal{I}_n(\zeta)$ on every partition. The left

side converges in mean square to $\mathcal{I}(c\,\xi + c'\zeta)$ and the right side, by the triangle inequality, to $c\,\mathcal{I}(\xi) + c'\,\mathcal{I}(\zeta)$; the two limits agree with probability one. In particular the integral of an element of the span is the sum of the integrals of its summands, which is the extension by linearity; parts (a) and (b), with the product form proved next, are (i) to (iii) on the span. The identity $4xy = (x+y)^2 - (x-y)^2$, applied first to $x = \mathcal{I}(\xi)$ and $y = \mathcal{I}(\zeta)$ and then to $x = \xi_s$ and $y = \zeta_s$, gives the product form:

$$\begin{aligned}\mathbb{E}\big[\mathcal{I}(\xi)\,\mathcal{I}(\zeta)\big] &= \tfrac{1}{4}\Big(\mathbb{E}\big[\mathcal{I}(\xi+\zeta)^2\big] - \mathbb{E}\big[\mathcal{I}(\xi-\zeta)^2\big]\Big)\\ &= \tfrac{1}{4}\int_{t_0}^{t}\Big(\mathbb{E}\big[(\xi_s+\zeta_s)^2\big] - \mathbb{E}\big[(\xi_s-\zeta_s)^2\big]\Big)\,ds = \int_{t_0}^{t}\mathbb{E}[\xi_s\zeta_s]\,ds.\end{aligned}$$

Here $\xi \pm \zeta$ may carry the indicators of two different hitting times, which Step 1 allows.

(d) Two independent motions, which is (vi). Steps 2 to 4 and parts (a) to (c) apply word for word to the left-endpoint sums $\mathcal{I}_n^{\perp}(\xi)$ against $W_\perp$, because Step 1 covers the increments of $W_\perp$ and Step 2 proved (E.5) for them. They define $\mathcal{I}^{\perp}(\xi) = \int_{t_0}^{t}\xi_s\,dW_{\perp,s}$, with the same properties. For the cross expectation, take one sequence of partitions with meshes tending to zero, and abbreviate $x_n = \mathcal{I}_n(\xi)$, $x = \mathcal{I}(\xi)$, $y_n = \mathcal{I}_n^{\perp}(\zeta)$ and $y = \mathcal{I}^{\perp}(\zeta)$. By (E.6), $\mathbb{E}[x_n y_n] = 0$ for every n. Since $x_n y_n - xy = (x_n - x)\,y_n + x\,(y_n - y)$, the Cauchy–Schwarz inequality gives

$$\big|\mathbb{E}[x_n y_n] - \mathbb{E}[xy]\big| \le \big(\mathbb{E}[(x_n-x)^2]\big)^{1/2}\big(\mathbb{E}[y_n^2]\big)^{1/2} + \big(\mathbb{E}[x^2]\big)^{1/2}\big(\mathbb{E}[(y_n-y)^2]\big)^{1/2},$$

which tends to 0, the second moments $\mathbb{E}[y_n^2]$ being bounded because they converge. Hence $\mathbb{E}[xy] = 0$.

(e) Additivity. Let $t_0 < t' < t$. Restricted to $[t_0, t']$ or to $[t', t]$, the process ξ is in the span of the class of that interval: the times of each vector Y are at most $t_0 \le t'$, and the stopping times are computed from time 0. A partition of $[t_0, t]$ that contains the point t' is a partition of $[t_0, t']$ followed by a partition of $[t', t]$, and its left-endpoint sum is the sum of theirs. Along a sequence of such partitions with meshes tending to zero the three sums converge in mean square to the three integrals, and the limit of a sum is the sum of the limits, with probability one.

(f) Stopping. Let $t_0 \le t' < t$, let ξ be in the span with $\xi_s = 0$ whenever $s \ge t_\ell$, and put $A = \{t_\ell \le t'\}$. When each summand of ξ carries the factor $\mathbf{1}\{s < t_\ell\}$, each summand vanishes for $s \ge t_\ell$, and so does their sum. In a left-endpoint sum $\mathcal{I}_n$ over a partition of $[t', t]$, every term is $\xi_{t_{i-1}}\,\Delta W_i$ with $t_{i-1} \ge t'$, and on A the weight $\xi_{t_{i-1}}$ is zero, because $t_{i-1} \ge t' \ge t_\ell$ there. So $\mathcal{I}_n\,\mathbf{1}\{A\} = 0$, and with $\mathcal{I}$ the integral over $[t', t]$,

$$\mathbb{E}\big[\mathcal{I}^2\,\mathbf{1}\{A\}\big] = \mathbb{E}\big[(\mathcal{I}-\mathcal{I}_n)^2\,\mathbf{1}\{A\}\big] \le \mathbb{E}\big[(\mathcal{I}-\mathcal{I}_n)^2\big] \to 0.$$

Hence $\mathbb{E}[\mathcal{I}^2\,\mathbf{1}\{A\}] = 0$, and Markov's inequality (Lemma 19.12(i)) gives, for $k = 1, 2, \ldots$,

$$P\big(\mathcal{I}^2\,\mathbf{1}\{A\} \ge 1/k\big) \le k\,\mathbb{E}\big[\mathcal{I}^2\,\mathbf{1}\{A\}\big] = 0.$$

The union of these events over k is the event that $\mathcal{I} \ne 0$ and $t_\ell \le t'$. It has probability zero by Lemma 19.12(0). Parts (e) and (f) hold for the integral against $W_\perp$ with the same proofs, which completes (vi). □

E.6 Proof of Theorem 19.30: Itô's lemma (Itô, 1951)

The statement is on page 280. The notation and the standing assumptions are those of §19.5.

Proof. We telescope over a partition, expand each step by Taylor's theorem (§7.5), to first order in time and to second order in space, and identify the limit of each of the four sums that result; the quadratic sum is Proposition 19.10 with weights. The proof has eight steps. Steps 1 to 6 prove (i) for a fixed $t \in (0, T]$. For $t = 0$ both sides of (19.11) equal $f(0,0)$, because $W_0 = 0$.

Step 1: telescoping and Taylor's theorem. Fix a partition $0 = t_0 < t_1 < \cdots < t_n = t$ of $[0,t]$. Since $W_{t_0} = W_0 = 0$, on every path

$$f(t,W_t) - f(0,0) = \sum_{i=1}^{n} \Big(f(t_i, W_{t_i}) - f(t_{i-1}, W_{t_i}) \Big) + \sum_{i=1}^{n} \Big(f(t_{i-1}, W_{t_i}) - f(t_{i-1}, W_{t_{i-1}}) \Big) :$$

the i-th terms of the two sums add up to $f(t_i, W_{t_i}) - f(t_{i-1}, W_{t_{i-1}})$, and these differences telescope. A term of the first sum is a move in time at the new Brownian value; a term of the second is a move in space at the old time.

The time bracket. Fix i and a real number x. The function $s \mapsto f(s,x)$ is continuous on the closed cell $[t_{i-1}, t_i]$ and has the continuous derivative $\partial_t f(s,x)$ there, one-sided at the end points; the last cell may end at T, beyond which f is not defined. The fundamental theorem of calculus, which is the zeroth-order case of Theorem 7.31 and uses only the closed interval, gives

$$f(t_i, x) - f(t_{i-1}, x) = \int_{t_{i-1}}^{t_i} \partial_t f(s,x)\, ds.$$

The integral lies between Δt_i times the minimum and Δt_i times the maximum of the continuous function $s \mapsto \partial_t f(s,x)$ on the cell. By the intermediate value theorem, as in the proof of Corollary 7.32, it equals $\partial_t f(t^*, x)\, \Delta t_i$ for a point t^* of the closed cell. (Theorem 7.31 and its corollary are stated for a function on an open interval, which is why the closed cell is treated directly.) With $x = W_{t_i}$ this gives a point $t_i^* \in [t_{i-1}, t_i]$, which depends on the path, such that

$$f(t_i, W_{t_i}) - f(t_{i-1}, W_{t_i}) = \partial_t f(t_i^*, W_{t_i})\, \Delta t_i.$$

The space bracket. Fix i and a path, and put

$$\chi(u) = f\big(t_{i-1},\ W_{t_{i-1}} + u\, \Delta W_i\big), \qquad u \in \mathbb{R}.$$

This function of one real variable is twice continuously differentiable on the whole line, which is the open interval that Theorem 7.31 asks for, with

$$\chi'(u) = \partial_x f\big(t_{i-1},\ W_{t_{i-1}} + u\, \Delta W_i\big)\, \Delta W_i, \qquad \chi''(u) = \partial_{xx} f\big(t_{i-1},\ W_{t_{i-1}} + u\, \Delta W_i\big)\, (\Delta W_i)^2.$$

Theorem 7.31 at order one, with the expansion point 0 and the Lagrange form of the remainder (Corollary 7.32), evaluated at $u = 1$, gives a point $u^* \in [0,1]$ with $\chi(1) = \chi(0) + \chi'(0) + \frac{1}{2}\chi''(u^*)$. That is,

$$f(t_{i-1}, W_{t_i}) - f(t_{i-1}, W_{t_{i-1}}) = \partial_x f(t_{i-1}, W_{t_{i-1}})\, \Delta W_i + \tfrac{1}{2}\, \partial_{xx} f(t_{i-1}, x_i^*)\, (\Delta W_i)^2,$$

where $x_i^* = W_{t_{i-1}} + u^*\, \Delta W_i$ lies between $W_{t_{i-1}}$ and W_{t_i}. Only the space variable is expanded to second order, because the class $C_E^{1,2}$ provides no second derivative in time. Proposition 21.14 in §21 reuses this device, a function χ of one variable along the segment that joins two points.

The four sums. Adding the two brackets over i gives, on every path,

$$f(t,W_t) - f(0,0) = \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4, \tag{E.11}$$

where

$$\mathcal{T}_1 = \sum_{i=1}^{n} \partial_t f(t_i^*, W_{t_i})\, \Delta t_i \qquad \text{(time sum)},$$

$$\mathcal{T}_2 = \sum_{i=1}^{n} \partial_x f(t_{i-1}, W_{t_{i-1}})\, \Delta W_i \qquad \text{(gradient sum)},$$

$$\mathcal{T}_3 = \tfrac{1}{2} \sum_{i=1}^{n} \partial_{xx} f(t_{i-1}, W_{t_{i-1}})\, (\Delta W_i)^2 \qquad \text{(quadratic sum)},$$

$$\mathcal{T}_4 = \tfrac{1}{2} \sum_{i=1}^{n} \Big(\partial_{xx} f(t_{i-1}, x_i^*) - \partial_{xx} f(t_{i-1}, W_{t_{i-1}}) \Big)\, (\Delta W_i)^2 \qquad \text{(remainder)}.$$

The starred points depend on the path, but each of the four sums is a function of the Brownian values at the partition points alone: $\mathcal{T}_1$ is the first sum of the telescoping identity, $\mathcal{T}_2$ and $\mathcal{T}_3$ contain no starred point, and $\mathcal{T}_4$ is the second sum of that identity minus $\mathcal{T}_2$ and $\mathcal{T}_3$.

Step 2: the time sum, path by path. Fix a path. It is continuous on $[0,t]$, hence bounded there; let $M_W = \max_{0\le s\le t}|W_s|$, a bound that depends on the path. A continuous function on a closed bounded box is uniformly continuous (the Heine–Cantor theorem). The argument after (E.9) in Step 3 of the proof of Theorem 19.23 applies, with the absolute difference of the function in place of the mean-square distance and with points of the box in place of times. On the rectangle $[0,t]\times[-M_W,M_W]$ it reads as follows. If uniform continuity of $\partial_t f$ failed there for some $\varepsilon>0$, there would be points (s_k,x_k) and (s'_k,x'_k) of the rectangle with $|s'_k-s_k|\le 1/k$, $|x'_k-x_k|\le 1/k$ and $|\partial_t f(s'_k,x'_k)-\partial_t f(s_k,x_k)|>\varepsilon$. A bounded sequence of real numbers has a convergent subsequence; taking one for the first coordinate and then, along it, one for the second, the points (s_k,x_k) converge along a subsequence to a point (s^*,x^*), which lies in the closed rectangle, and the points (s'_k,x'_k) converge to the same point. The continuity of $\partial_t f$ at (s^*,x^*) then makes $|\partial_t f(s'_k,x'_k)-\partial_t f(s_k,x_k)|$ tend to 0 along the subsequence, which contradicts the lower bound ε. The same argument, one coordinate at a time, applies on a closed bounded box of any dimension, and in one variable it gives the uniform continuity of the path. So $\partial_t f$ is uniformly continuous on $[0,t]\times[-M_W,M_W]$, and the path is uniformly continuous on $[0,t]$. Given $\varepsilon>0$, there is therefore an $\varepsilon'>0$ such that

$$|\partial_t f(s',x')-\partial_t f(s,x)|\le\varepsilon \qquad \text{whenever } |s'-s|\le\varepsilon' \text{ and } |x'-x|\le\varepsilon',$$

for points of the rectangle, and there is an ε'' with $0<\varepsilon''\le\varepsilon'$ such that $|W_{s'}-W_s|\le\varepsilon'$ whenever $s,s'\in[0,t]$ and $|s'-s|\le\varepsilon''$. Let the partition have $\Delta t_{\max}\le\varepsilon''$. For s in the cell $[t_{i-1},t_i]$, both $|t_i^*-s|$ and $|t_i-s|$ are at most $\Delta t_i\le\varepsilon''$, so $|\partial_t f(t_i^*,W_{t_i})-\partial_t f(s,W_s)|\le\varepsilon$. The term $\partial_t f(t_i^*,W_{t_i})\,\Delta t_i$ is the integral of a constant over the cell, and therefore

$$\Big|\mathcal{T}_1-\int_0^t \partial_t f(s,W_s)\,ds\Big| = \Big|\sum_{i=1}^n\int_{t_{i-1}}^{t_i}\Big(\partial_t f(t_i^*,W_{t_i})-\partial_t f(s,W_s)\Big)\,ds\Big|\le\varepsilon\,t.$$

The integrand $s\mapsto\partial_t f(s,W_s)$ is continuous, so the integral on the left is a Riemann integral. Hence $\mathcal{T}_1\to\int_0^t\partial_t f(s,W_s)\,ds$ as $\Delta t_{\max}\to 0$, on every path.

Step 3: the gradient sum. The process $\xi_s=\partial_x f(s,W_s)$ is a Brownian integrand on $[0,t]$ in the sense of Definition 19.22: its generating function $\partial_x f(s,x)$ is continuous on $[0,t]\times\mathbb{R}$ and of exponential growth by hypothesis, there is no vector Y, and there is no stopping. The gradient sum is its left-endpoint sum, $\mathcal{T}_2=\mathcal{I}_n(\xi)$. By Theorem 19.23(i),

$$\mathcal{T}_2\to\int_0^t\partial_x f(s,W_s)\,dW_s \qquad \text{in mean square, as } \Delta t_{\max}\to 0.$$

Step 4: the quadratic sum, or $(dW)^2=dt$ with weights. Put $a_i=\partial_{xx}f(t_{i-1},W_{t_{i-1}})$, a weight fixed at the start of the i-th cell, and split

$$\mathcal{T}_3=\mathrm{QV}_n^a+\tfrac12\sum_{i=1}^n a_i\,\Delta t_i, \qquad \mathrm{QV}_n^a=\tfrac12\sum_{i=1}^n a_i\,((\Delta W_i)^2-\Delta t_i).$$

We first show that $\mathbb{E}[(\mathrm{QV}_n^a)^2]\to 0$. Every product that appears below is a function of exponential growth of finitely many Brownian values at times in $[0,T]$: a polynomial has exponential growth, because $|x|^k\le k!\,e^{|x|}$ for every integer $k\ge 0$, and sums and products of functions of exponential growth have exponential growth (Lemma 19.20(iii)). By the same lemma each product has finite moments of every order, so the expectations below exist and linearity applies. Expanding the square,

$$\mathbb{E}\big[(\mathrm{QV}_n^a)^2\big]=\tfrac14\sum_{i=1}^n\mathbb{E}\Big[a_i^2\,((\Delta W_i)^2-\Delta t_i)^2\Big]+\tfrac12\sum_{i<j}\mathbb{E}\Big[a_ia_j\,((\Delta W_i)^2-\Delta t_i)((\Delta W_j)^2-\Delta t_j)\Big].$$

Cross terms. Let $i < j$. The product $a_i a_j\,((\Delta W_i)^2 - \Delta t_i)$ is a function of the values $W_{t_{i-1}}$, W_{t_i} and $W_{t_{j-1}}$, all at times at most t_{j-1}. The increment ΔW_j is independent of the vector of these three values (Proposition 19.8(iv), with the motion restarted at the time t_{j-1}, which rests on Definition 19.5(3)), and $\mathbb{E}[(\Delta W_j)^2 - \Delta t_j] = 0$ by Proposition 7.25(ii). The product rule (19.5) of Lemma 19.17 gives

$$\mathbb{E}\Big[a_i a_j\,((\Delta W_i)^2 - \Delta t_i)\,((\Delta W_j)^2 - \Delta t_j)\Big] = \mathbb{E}\Big[a_i a_j\,((\Delta W_i)^2 - \Delta t_i)\Big]\,\mathbb{E}\Big[(\Delta W_j)^2 - \Delta t_j\Big] = 0.$$

Diagonal terms. The weight a_i is a function of $W_{t_{i-1}}$, of which ΔW_i is independent. By the same product rule and by Proposition 7.25(ii) and (iii),

$$\mathbb{E}\Big[a_i^2\,\big((\Delta W_i)^2 - \Delta t_i\big)^2\Big] = \mathbb{E}[a_i^2]\,\Big(3(\Delta t_i)^2 - 2(\Delta t_i)^2 + (\Delta t_i)^2\Big) = 2\,\mathbb{E}[a_i^2]\,(\Delta t_i)^2.$$

Therefore, with $\sum_i (\Delta t_i)^2 \le t\,\Delta t_{\max}$ (Proposition 19.10(iii)),

$$\mathbb{E}[(\mathrm{QV}_n^a)^2] = \tfrac{1}{2}\sum_{i=1}^{n} \mathbb{E}[a_i^2]\,(\Delta t_i)^2 \le \tfrac{1}{2}\,t\,\Delta t_{\max} \sup_{0\le s\le t} \mathbb{E}[\partial_{xx} f(s, W_s)^2].$$

The supremum is finite by Lemma 19.20(iii), whose moment bounds are uniform in the time. So $\mathrm{QV}_n^a \to 0$ in mean square. With every weight equal to 2 this is the computation of Proposition 19.10(i), because then $\mathrm{QV}_n^a = \mathrm{QV}_n - t$.

The other part, $\frac{1}{2}\sum_i a_i\,\Delta t_i$, is a Riemann sum, with left end points as tags, of the continuous function $s \mapsto \frac{1}{2}\,\partial_{xx} f(s, W_s)$. On a fixed path, the argument of Step 2, applied to $\partial_{xx} f$ in place of $\partial_t f$ and to the tags t_{i-1} in place of the pairs (t_i^*, t_i), bounds its distance from $\frac{1}{2}\int_0^t \partial_{xx} f(s, W_s)\,ds$ by $\varepsilon\,t/2$ once the mesh is small enough. It converges to that integral on every path.

Convergence in mean square implies convergence in probability, and so does convergence on every path, which is in particular almost sure (Lemma 19.12(iii)). By the rule for sums of that lemma,

$$\mathcal{T}_3 \to \tfrac{1}{2}\int_0^t \partial_{xx} f(s, W_s)\,ds \qquad \text{in probability, as } \Delta t_{\max} \to 0.$$

Step 5: the remainder. Put

$$\mathrm{osc}_n = \max_{1\le i\le n}\ \max_{0\le u\le 1}\ \Big|\partial_{xx} f\big(t_{i-1},\ W_{t_{i-1}} + u\,\Delta W_i\big) - \partial_{xx} f(t_{i-1}, W_{t_{i-1}})\Big|,$$

the largest oscillation of $\partial_{xx} f$ along the segments of Step 1; it does not involve the starred points. Since $x_i^* = W_{t_{i-1}} + u^*\,\Delta W_i$ with $u^* \in [0,1]$,

$$|\mathcal{T}_4| \le \tfrac{1}{2}\,\mathrm{osc}_n \sum_{i=1}^{n} (\Delta W_i)^2 = \tfrac{1}{2}\,\mathrm{osc}_n\,\mathrm{QV}_n.$$

Fix a path and $\varepsilon > 0$, and let ε' and ε'' be as in Step 2, now for the function $\partial_{xx} f$. A point $W_{t_{i-1}} + u\,\Delta W_i$ with $0 \le u \le 1$ lies between two values of the path, hence in $[-M_W, M_W]$, and its distance from $W_{t_{i-1}}$ is at most $|\Delta W_i|$, which is at most ε' when $\Delta t_{\max} \le \varepsilon''$. So $\mathrm{osc}_n \le \varepsilon$ for such partitions: $\mathrm{osc}_n \to 0$ on every path, hence in probability. By Proposition 19.10(i), $\mathrm{QV}_n \to t$ in mean square, hence in probability. The rule for products of Lemma 19.12(iii) gives $\mathrm{osc}_n\,\mathrm{QV}_n \to 0 \cdot t = 0$ in probability, and since $P(|\mathcal{T}_4| > \varepsilon) \le P(\frac{1}{2}\,\mathrm{osc}_n\,\mathrm{QV}_n > \varepsilon)$ for every $\varepsilon > 0$, also $\mathcal{T}_4 \to 0$ in probability.

Step 6: conclusion of (i). Take any sequence of partitions of $[0, t]$ whose meshes tend to zero. By (E.11),

$$\mathcal{T}_2 = f(t, W_t) - f(0,0) - \mathcal{T}_1 - \mathcal{T}_3 - \mathcal{T}_4.$$

By Steps 2, 4 and 5 and the rule for sums of limits in probability, the right side converges in probability to

$$f(t, W_t) - f(0,0) - \int_0^t \Big(\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\Big)(s, W_s)\, ds.$$

By Step 3 the left side converges in mean square, hence in probability, to $\int_0^t \partial_x f(s, W_s)\, dW_s$. Two limits in probability of one sequence agree with probability one (Lemma 19.12(iii)). This is (19.11).

Step 7: the parameter clause (ii). If $t_0 = t$, both sides of (19.12) are $f(t_0, W_{t_0}, Y)$. Let $t_0 < t$, take a partition $t_0 < t_1 < \cdots < t_n = t$ of $[t_0, t]$, and repeat Steps 1 to 6 with $f(s, x, Y)$ in place of $f(s, x)$, the vector Y being frozen at the value it takes on the path. The telescoping sum now starts from $f(t_0, W_{t_0}, Y)$.

- *Steps 1, 2 and 5.* These steps work on one path, on which Y is a fixed vector y of one of the boxes. For that y, the function $(s, x) \mapsto f(s, x, y)$ and its derivatives $\partial_t f$, $\partial_x f$ and $\partial_{xx} f$ are continuous on $[t_0, t] \times \mathbb{R}$, which is all that the two expansions of Step 1 and the uniform continuity on the rectangle $[t_0, t] \times [-M_W, M_W]$ in Steps 2 and 5 use.
- *Step 3.* The integrand $\xi_s = \partial_x f(s, W_s, Y)$ is in $\mathcal{H}$ on $[t_0, t]$ by Definition 19.22(a) and (b), so its left-endpoint sums converge in mean square to $\int_{t_0}^t \partial_x f(s, W_s, Y)\, dW_s$.
- *Step 4.* The weight $a_i = \partial_{xx} f(t_{i-1}, W_{t_{i-1}}, Y)$ is a function of values of W at times at most t_{i-1}, because every $s_j \le t_0 \le t_{i-1}$. For $i < j$ the product $a_i a_j\,((\Delta W_i)^2 - \Delta t_i)$ is a function of values of W at times at most t_{j-1}, of which ΔW_j is independent (Proposition 19.8(iv)); likewise ΔW_i is independent of the values behind a_i. Exponential growth in (x, y) keeps every moment finite and bounded uniformly in the time (Lemma 19.20(iii)). So the cross terms vanish, the diagonal terms are $2\,\mathbb{E}[a_i^2]\,(\Delta t_i)^2$, and the bound of Step 4 holds with $\sup_s \mathbb{E}[\partial_{xx} f(s, W_s, Y)^2] < \infty$ and $t - t_0$ in place of t. The Riemann-sum part converges path by path, as before.
- *Step 6.* Unchanged.

Step 8: the stopped clause (iii). For $t = 0$ both sides of (19.13) equal $f(0,0)$, because $t_\ell > 0$ by (19.6). Let $t \in (0, T]$, take a partition $0 = t_0 < t_1 < \cdots < t_n = t$, and let t^+ be the first partition point at or after $t \wedge t_\ell$. We claim that, on every path,

$$\sum_{i=1}^{n} \mathbf{1}\{t_{i-1} < t_\ell\} \Big(f(t_i, W_{t_i}) - f(t_{i-1}, W_{t_{i-1}})\Big) = f(t^+, W_{t^+}) - f(0,0), \tag{E.12}$$

and that $0 \le t^+ - t \wedge t_\ell \le \Delta t_{\max}$. Since $t_0 = 0 < t_\ell$, the indices i with $t_{i-1} < t_\ell$ are $1, \ldots, k$, where $k \ge 1$ is the largest of them, and the sum telescopes to $f(t_k, W_{t_k}) - f(0,0)$. If $t_\ell > t$, every index qualifies, so $k = n$ and $t_k = t = t \wedge t_\ell$. If $t_\ell \le t$, then $t_{k-1} < t_\ell \le t_k$; the second inequality holds because either $k = n$ and $t_k = t \ge t_\ell$, or $k < n$ and the index $k + 1$ does not qualify. In both cases t_k is the first partition point at or after $t \wedge t_\ell$, that is, $t_k = t^+$, and $t^+ - t \wedge t_\ell$ is at most the length of the k-th cell.

Expand each difference in (E.12) as in Step 1. The left side becomes $\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4$, where in this step each of the four sums of Step 1 carries the factor $\mathbf{1}\{t_{i-1} < t_\ell\}$ in its i-th term. Steps 2 to 6 are repeated with these sums.

- *The right side of (E.12).* As $\Delta t_{\max} \to 0$, $t^+ \to t \wedge t_\ell$, so $f(t^+, W_{t^+}) \to f(t \wedge t_\ell, W_{t \wedge t_\ell})$ on every path, by the continuity of f and of the path.
- *The time sum.* On a fixed path the indicator selects the cells $1, \ldots, k$, which make up $[0, t^+]$. The estimate of Step 2, summed over these cells only, shows that $\mathcal{T}_1$ is within $\varepsilon\, t$ of $\int_0^{t^+} \partial_t f(s, W_s)\, ds$ once $\Delta t_{\max} \le \varepsilon''$. The continuous function $s \mapsto \partial_t f(s, W_s)$ is bounded on $[0, t]$, by a constant c that depends on the path, so that integral differs from the integral over $[0, t \wedge t_\ell]$ by at most $c\, \Delta t_{\max}$. Hence, on every path,

$$\mathcal{T}_1 \to \int_0^{t \wedge t_\ell} \partial_t f(s, W_s)\, ds = \int_0^t \mathbf{1}\{s < t_\ell\}\, \partial_t f(s, W_s)\, ds.$$

- *The gradient sum.* Now $\mathcal{T}_2 = \mathcal{I}_n(\xi)$ with $\xi_s = \mathbf{1}\{s < t_\ell\}\,\partial_x f(s, W_s)$, which is in $\mathcal{H}$ by Definition 19.22(c). By Theorem 19.23(i), $\mathcal{T}_2$ converges in mean square to the stopped integral $\int_0^t \mathbf{1}\{s < t_\ell\}\,\partial_x f(s, W_s)\,dW_s$.
- *The quadratic sum.* The weights are now $\mathbf{1}\{t_\ell > t_{i-1}\}\,a_i$. For $i < j$, the product of the i-th weight, the j-th weight and $(\Delta W_i)^2 - \Delta t_i$ is a function of the vector

  $$(W_{t_{i-1}},\ W_{t_i},\ W_{t_{j-1}},\ \mathbf{1}\{t_\ell > t_{i-1}\},\ \mathbf{1}\{t_\ell > t_{j-1}\}),$$

  of which ΔW_j is independent by Step 1 of the proof of Theorem 19.23. Its absolute value is at most that of the product without the indicators, so it is integrable, and the product rule (19.5) makes the cross term vanish as in Step 4. By the same independence a diagonal term equals $2\,\mathbb{E}[\mathbf{1}\{t_\ell > t_{i-1}\}\,a_i^2]\,(\Delta t_i)^2 \le 2\,\mathbb{E}[a_i^2]\,(\Delta t_i)^2$. So the bound of Step 4 on $\mathbb{E}[(\mathrm{QV}_n^a)^2]$ holds for the stopped weights. The other part of $\mathcal{T}_3$, $\frac{1}{2}\sum_i \mathbf{1}\{t_{i-1} < t_\ell\}\,a_i\,\Delta t_i$, converges on every path to $\frac{1}{2}\int_0^t \mathbf{1}\{s < t_\ell\}\,\partial_{xx} f(s, W_s)\,ds$, by the argument just given for the time sum.
- *The remainder.* An indicator is at most 1, so $|\mathcal{T}_4| \le \frac{1}{2}\,\mathrm{osc}_n\,\mathrm{QV}_n$ as in Step 5, and $\mathcal{T}_4 \to 0$ in probability.
- *Conclusion.* As in Step 6, $\mathcal{T}_2 = f(t^+, W_{t^+}) - f(0,0) - \mathcal{T}_1 - \mathcal{T}_3 - \mathcal{T}_4$ converges in probability to

  $$f\big(t \wedge t_\ell,\ W_{t \wedge t_\ell}\big) - f(0,0) - \int_0^t \mathbf{1}\{s < t_\ell\}\,\Big(\partial_t f + \tfrac{1}{2}\,\partial_{xx} f\Big)(s, W_s)\,ds,$$

  and in mean square to the stopped integral. The two limits agree with probability one, which is (19.13). □

E.7 Proof of Corollary 19.31: Itô's lemma for a function of an Itô process

The statement is on page 281. The notation and the standing assumptions are those of §19.5.

Proof. The chain rule of calculus reduces the corollary to Theorem 19.30, applied to the composite.

Derivatives of the composite. Because f takes its values in $\mathcal{O}$, the composite $\tilde{V}(y,t) = V(f(t,y),t)$ is defined on all of $\mathbb{R} \times [0,T]$. Write $x = f(t,y)$. In the variable y only the first argument of V moves, and the chain and product rules for functions of one variable give

$$\partial_y \tilde{V} = \partial_x V(x,t)\,\partial_y f(t,y), \qquad \partial_{yy}\tilde{V} = \partial_{xx} V(x,t)\,\big(\partial_y f(t,y)\big)^2 + \partial_x V(x,t)\,\partial_{yy} f(t,y).$$

In the variable t both arguments of V move, along the curve $t \mapsto (f(t,y),t)$, and the chain rule along a curve is proved as follows. For $\Delta t \neq 0$ with $t + \Delta t \in [0,T]$, split the increment into a move of the state at the new time and a move of the time at the old state:

$$\begin{aligned}\tilde{V}(y, t+\Delta t) - \tilde{V}(y,t) = &\Big(V\big(f(t+\Delta t, y),\ t+\Delta t\big) - V\big(f(t,y),\ t+\Delta t\big)\Big)\\ &+ \Big(V\big(f(t,y),\ t+\Delta t\big) - V\big(f(t,y),\ t\big)\Big).\end{aligned}$$

The function $x' \mapsto V(x', t+\Delta t)$ is continuously differentiable on the open interval $\mathcal{O}$, so by the mean value theorem (Corollary 7.32 at order zero) the first bracket equals $\partial_x V(x^*, t+\Delta t)\,(f(t+\Delta t, y) - f(t,y))$ for a point x^* between $f(t,y)$ and $f(t+\Delta t, y)$. As $\Delta t \to 0$, $f(t+\Delta t, y) \to f(t,y) = x$, so $x^* \to x$ and $\partial_x V(x^*, t+\Delta t) \to \partial_x V(x,t)$ by the continuity of $\partial_x V$, while $(f(t+\Delta t, y) - f(t,y))/\Delta t \to \partial_t f(t,y)$. The second bracket, divided by Δt, tends to $\partial_t V(x,t)$. Hence

$$\partial_t \tilde{V} = \partial_t V(x,t) + \partial_x V(x,t)\,\partial_t f(t,y),$$

one-sided at $t = 0$ and at $t = T$. The three derivatives are sums and products of continuous functions composed with the continuous map $(t,y) \mapsto (f(t,y),t)$, so they are continuous on $\mathbb{R} \times [0,T]$. With the growth hypothesis, $(t,y) \mapsto \tilde{V}(y,t)$ is in $C_E^{1,2}([0,T] \times \mathbb{R})$.

Part (i), the chain rule (19.14). By the three formulas, with every derivative of V evaluated at $(x,t) = (f(t,y),t)$,

$$\partial_t \tilde{V} + \tfrac{1}{2}\,\partial_{yy}\tilde{V} = \partial_t V + \partial_x V\left(\partial_t f + \tfrac{1}{2}\,\partial_{yy} f\right) + \tfrac{1}{2}\,\partial_{xx} V\,(\partial_y f)^2.$$

At $y = W_s$ and $t = s$ the right side is $(\partial_t V + \alpha_s\,\partial_x V + \frac{1}{2}\,\beta_s^2\,\partial_{xx} V)(X_s, s)$, and $\partial_y \tilde{V}$ is $\beta_s\,\partial_x V(X_s, s)$. Theorem 19.30(i), applied to the function $(t,y) \mapsto \tilde{V}(y,t)$, with $\tilde{V}(W_t, t) = V(X_t, t)$ and $\tilde{V}(0,0) = V(f(0,0),0) = V(X_0, 0)$, is (19.14). The drift is continuous in s on every path and the integrand of the stochastic integral is in $\mathcal{H}$, as Definition 19.25(ii) requires.

Part (ii), the expectation identity (19.15). Write $\alpha_s^V = (\partial_t \tilde{V} + \frac{1}{2}\,\partial_{yy}\tilde{V})(W_s, s)$ for the drift of $V(X_t, t)$ at time s (a name local to this proof). We show that each term on the right of (19.14) has a finite mean, and compute the means. The path $s \mapsto \alpha_s^V$ is continuous, and α_s^V is a function of exponential growth evaluated at W_s, so $\mathbb{E}|\alpha_s^V|$ is bounded uniformly in $s \in [0,t]$ (Lemma 19.20(iii)) and $\int_0^t \mathbb{E}|\alpha_s^V|\,ds < \infty$. By Fact 18.34(iii), applied to the nonnegative process $|\alpha_s^V|$, the variable $\int_0^t |\alpha_s^V|\,ds$ has this finite mean. Since $|\int_0^t \alpha_s^V\,ds| \le \int_0^t |\alpha_s^V|\,ds$, the ds-integral has a finite mean, by monotonicity, and Fact 18.34(iii), now in its integrable case, gives $\mathbb{E}\int_0^t \alpha_s^V\,ds = \int_0^t \mathbb{E}[\alpha_s^V]\,ds$. The stochastic integral has a finite second moment (Theorem 19.23(i)), hence a finite mean (the Cauchy–Schwarz inequality of Lemma 19.12(ii) with the constant 1), and its mean is zero (Theorem 19.23(ii)). So the left side $V(X_t, t)$ of (19.14), the sum of $V(X_0, 0)$ and these two terms, has a finite mean, and taking expectations, by linearity, gives (19.15).

Part (iii), geometric Brownian motion. Let $f(t,y) = S_0\,e^{gt+\sigma y}$, so that $f(t, W_t) = S_t$ by (19.10). Then $\partial_t f = g\,f$, $\partial_y f = \sigma f$ and $\partial_{yy} f = \sigma^2 f$. On $[0,T] \times \mathbb{R}$ each of f and these three derivatives is at most $S_0 \max(1, |g|, \sigma, \sigma^2)\,e^{|g|T}\,e^{\sigma|y|}$ in absolute value, which is exponential growth; so $f \in C_E^{1,2}([0,T] \times \mathbb{R})$, with values in $(0, \infty)$. The coefficients are $\alpha_t = (g + \sigma^2/2)\,S_t = \mu S_t$ and $\beta_t = \sigma S_t$. With $x = f(t,y)$, the formulas for the derivatives of the composite read

$$\partial_t \tilde{V} = \partial_t V + g\,x\,\partial_x V, \qquad \partial_y \tilde{V} = \sigma\,x\,\partial_x V, \qquad \partial_{yy}\tilde{V} = \sigma^2(x^2\,\partial_{xx} V + x\,\partial_x V),$$

the right sides being evaluated at (x,t). The functions x and x^2 have moderate growth, and sums and products of functions of moderate growth have moderate growth (Lemma 19.20(i)). So if V, $\partial_t V$, $\partial_x V$ and $\partial_{xx} V$ have moderate growth, then V and the three right sides are functions of (x,t) of moderate growth, and Lemma 19.20(i) turns moderate growth in $x = S_0\,e^{gt+\sigma y}$ into exponential growth in y. This is the hypothesis on the composite. With $\alpha_t = \mu S_t$, $\beta_t = \sigma S_t$ and $\partial_S V = \partial_x V$, $\partial_{SS} V = \partial_{xx} V$, the shorthand of (i) is the display of (iii).

Part (iv), parameter and stopped clauses. Apply part (ii) or part (iii) of Theorem 19.30 to the composite in place of part (i). In the parameter clause, for each frozen value y the functions $f(\cdot,\cdot,y)$ and $V(\cdot,\cdot,y)$ satisfy the hypotheses of the corollary on $[t_0, t]$, so the derivatives of the composite are computed as above, with y held fixed, and the integrands of Theorem 19.30(ii) are those of (19.14), with α_s and β_s computed from $f(\cdot,\cdot,Y)$. For the expectation identity under stopping, the argument of Part (ii) is repeated term by term on the right side, and the left side is then integrable as a sum of integrable terms. This order matters: the left side $V(X_{t\wedge t_\ell}, t \wedge t_\ell)$ is a value at a random time, to which the moment bounds of Lemma 19.20(iii), stated at fixed times, do not apply directly. The stopped drift $\mathbf{1}\{s < t_\ell\}\,\alpha_s^V$ is piecewise continuous in s on every path, with at most one jump, at t_ℓ, and

$$\Big|\int_0^t \mathbf{1}\{s < t_\ell\}\,\alpha_s^V\,ds\Big| \le \int_0^t |\alpha_s^V|\,ds,$$

whose mean $\int_0^t \mathbb{E}|\alpha_s^V|\,ds$ is finite by Part (ii). So the stopped ds-integral has a finite mean, and Fact 18.34(iii), in its integrable case, gives it the mean $\int_0^t \mathbb{E}[\mathbf{1}\{s < t_\ell\}\,\alpha_s^V]\,ds$. The stopped stochastic integral, whose integrand is in $\mathcal{H}$ by Definition 19.22(c), has a finite second moment, hence a finite mean, and mean zero (Theorem 19.23(i) and (ii)). Therefore $V(X_{t\wedge t_\ell}, t \wedge t_\ell)$, which by the stopped form of (19.14) is $V(X_0, 0)$ plus these two terms, has a finite mean, and taking expectations gives the stopped form of (19.15). □

E.8 Proof of Lemma 19.38: Gaussian shift, rotation and the jointly normal pair

The statement is on page 287. The notation and the standing assumptions are those of §19.6.

Proof. We complete the square for (i), rotate the plane for (ii), and combine the two for (iii). The proof has three steps.

Step 1: the shift (i). For all real z, $\theta z - z^2/2 = \theta^2/2 - (z-\theta)^2/2$, that is,

$$e^{\theta z}\,\varphi(z) = e^{\theta^2/2}\,\varphi(z-\theta). \tag{E.13}$$

Multiply by $\psi(z)$ and integrate. By Fact 7.22, by (E.13) and by the substitution $z \mapsto z+\theta$,

$$\begin{aligned}\mathbb{E}\big[e^{\theta Z}\psi(Z)\big] &= \int \psi(z)\,e^{\theta z}\varphi(z)\,dz = e^{\theta^2/2}\int \psi(z)\,\varphi(z-\theta)\,dz \\ &= e^{\theta^2/2}\int \psi(z+\theta)\,\varphi(z)\,dz = e^{\theta^2/2}\,\mathbb{E}\big[\psi(Z+\theta)\big].\end{aligned}$$

Fact 7.22 applies to the two functions $z \mapsto \psi(z)\,e^{\theta z}$ and $z \mapsto \psi(z+\theta)$, because both integrals converge absolutely. Indeed, with c the constant of ψ, $|\psi(z)|\,e^{\theta z} \le c\,e^{(c+|\theta|)|z|}$ and $|\psi(z+\theta)| \le c\,e^{c|\theta|}\,e^{c|z|}$. For every constant $c' \ge 0$, the integral $\int e^{c'|z|}\varphi(z)\,dz$, which is $\mathbb{E}\,e^{c'|Z|}$, is at most $2e^{c'^2/2}$ by Lemma 18.35(i). The same bounds make the two means of (i) finite. The case $\psi = 1$ is $\mathbb{E}\,e^{\theta Z} = e^{\theta^2/2}$. It is Lemma 18.35(i), which was proved by the same square and by $\int \varphi = 1$ (Theorem 7.5 through Example 7.6).

Step 2: the rotation (ii). Let (c_1, c_2) be a unit vector and put

$$Z_1' = c_1 Z_1 + c_2 Z_2, \qquad Z_2' = -c_2 Z_1 + c_1 Z_2.$$

We show that, for every test function ψ on the plane,

$$\mathbb{E}\,\psi(Z_1', Z_2') = \iint \psi(z_1', z_2')\,\varphi(z_1')\,\varphi(z_2')\,dz_1'\,dz_2' : \tag{E.14}$$

the pair (Z_1', Z_2') has the joint density $\varphi(z_1')\,\varphi(z_2')$ of two independent standard normal variables. First, every integral below converges absolutely. Since $|c_1|, |c_2| \le 1$, the composite $(z_1, z_2) \mapsto \psi(c_1 z_1 + c_2 z_2,\ -c_2 z_1 + c_1 z_2)$ of ψ with the rotation is at most $c\,e^{2c|z_1|}\,e^{2c|z_2|}$ in absolute value, with c the constant of ψ. The two factors of this bound have finite means at Z_1 and at Z_2 by the bound of Step 1, so the composite has a finite absolute mean at (Z_1, Z_2) by the product rule (19.5), and the same bound makes the iterated integral of its absolute value against $\varphi(z_1)\,\varphi(z_2)$ finite. Since Z_1 and Z_2 are independent, each with density φ, Lemma 19.17 (integrate out Z_2 first) and Fact 7.22, used once for each variable, write the left side as an iterated integral, which Fact 18.34(i), in its integrable case, turns into a double integral over the plane:

$$\begin{aligned}\mathbb{E}\,\psi(Z_1', Z_2') &= \int \Big[\int \psi(c_1 z_1 + c_2 z_2,\ -c_2 z_1 + c_1 z_2)\,\varphi(z_2)\,dz_2\Big]\varphi(z_1)\,dz_1 \\ &= \iint \psi(c_1 z_1 + c_2 z_2,\ -c_2 z_1 + c_1 z_2)\,\varphi(z_1)\,\varphi(z_2)\,dz_1\,dz_2.\end{aligned}$$

The rotation $(z_1, z_2) \mapsto (z_1', z_2') = (c_1 z_1 + c_2 z_2,\ -c_2 z_1 + c_1 z_2)$ is the rotation of identity (D.9) in the proof of Lemma 18.35(ii), with $a_1 = c_1$, $a_2 = c_2$ and $a = 1$. There it is written as three shears when $c_1 \ne -1$, and as the reflection of each coordinate when $c_1 = -1$, and shown to leave unchanged the double integral of every nonnegative function on the plane that is continuous outside finitely many lines, using only translations and reflections of a line and Fact 18.34(i). It also preserves the sum of squares:

$$(z_1')^2 + (z_2')^2 = (c_1^2 + c_2^2)\,z_1^2 + (c_2^2 + c_1^2)\,z_2^2 + 2c_1c_2\,z_1z_2 - 2c_1c_2\,z_1z_2 = z_1^2 + z_2^2.$$

Hence $\varphi(z_1)\,\varphi(z_2) = (2\pi)^{-1} e^{-(z_1^2+z_2^2)/2} = \varphi(z_1')\,\varphi(z_2')$. Apply (D.9) to the two functions $\zeta_\pm(y, y') = \max\big(\pm\psi(y, y'), 0\big)\,\varphi(y)\,\varphi(y')$, one for each sign (names local to this step, as in that proof), which are nonnegative and continuous outside the finitely many lines on which ψ may be discontinuous. By the identity of densities just shown, $\zeta_\pm(z_1', z_2') = \max\big(\pm\psi(z_1', z_2'), 0\big)\,\varphi(z_1)\,\varphi(z_2)$, so (D.9) reads

$$\iint \max\big(\pm\psi(z_1', z_2'), 0\big)\,\varphi(z_1)\,\varphi(z_2)\,dz_1\,dz_2 = \iint \max\big(\pm\psi(y, y'), 0\big)\,\varphi(y)\,\varphi(y')\,dy\,dy',$$

with z_1' and z_2' the functions of (z_1, z_2) above. Each left side is at most the double integral of the absolute value of the composite against $\varphi(z_1)\,\varphi(z_2)$, which is finite by the bound above, so both sides of each identity are finite. Subtracting the identity with the minus sign from the one with the plus sign, and using $\max(\psi, 0) - \max(-\psi, 0) = \psi$, turns the double integral over (z_1, z_2) obtained above into the right side of (E.14), with no change-of-variables theorem for double integrals. Lemma 18.35(ii) states only the law of $c_1 Z_1 + c_2 Z_2$; identity (D.9) of its proof holds for a function of both rotated coordinates, and (E.14) records this for test functions.

Now take $\psi(z_1', z_2') = \mathbf{1}\{z_1' \in \mathcal{D}_1\}\,\mathbf{1}\{z_2' \in \mathcal{D}_2\}$ for two intervals $\mathcal{D}_1$ and $\mathcal{D}_2$, a bounded test function. By (E.14) and Fact 18.34(i),

$$P(Z_1' \in \mathcal{D}_1,\ Z_2' \in \mathcal{D}_2) = \int_{\mathcal{D}_1} \varphi(z)\,dz \int_{\mathcal{D}_2} \varphi(z)\,dz.$$

With $\mathcal{D}_2$ the whole line this reads $P(Z_1' \in \mathcal{D}_1) = \int_{\mathcal{D}_1} \varphi(z)\,dz$: the variable Z_1' is standard normal. With $\mathcal{D}_1$ the whole line, so is Z_2'. The display is then the product rule $P(Z_1' \in \mathcal{D}_1, Z_2' \in \mathcal{D}_2) = P(Z_1' \in \mathcal{D}_1)\,P(Z_2' \in \mathcal{D}_2)$ for all intervals $\mathcal{D}_1$ and $\mathcal{D}_2$. By property (b) of the Setup of §19.2, the product rule for intervals suffices: Z_1' and Z_2' are independent. Identity (E.14) is the form in which Step 3 uses this independence.

Consequences. Let c_1 and c_2 be real numbers, not both zero, and put $c = (c_1^2 + c_2^2)^{1/2} > 0$. Applied to the unit vector $(c_1/c,\ c_2/c)$, the first part gives $c_1 Z_1 + c_2 Z_2 = c\,Z_1'$ with Z_1' standard normal, and a constant multiple of a standard normal variable is normal with the variance multiplied by the square of the constant, by (19.3) in the Setup of §19.2: $c_1 Z_1 + c_2 Z_2 \sim \mathcal{N}(0, c^2)$. If $c_1 = c_2 = 0$, the combination is the constant 0. Next, let (X, Y) be jointly normal, with the coefficient vectors a_X and a_Y of the Setup of §19.6. Then

$$cX + c'Y = \big(c\,\mathbb{E}[X] + c'\,\mathbb{E}[Y]\big) + \big(c\,a_{X,1} + c'a_{Y,1}\big)\,Z_1 + \big(c\,a_{X,2} + c'a_{Y,2}\big)\,Z_2$$

is a constant plus a combination of Z_1 and Z_2, hence normal. Finally, let X_1 and X_2 be independent normal variables with positive variances. The standardised variables $(X_k - \mathbb{E}[X_k])/\sqrt{\mathrm{Var}(X_k)}$, $k = 1, 2$, are standard normal by (19.3), and independent as functions of independent variables (the grouping property (a) of the Setup of §19.2), so the first part applies to them in place of Z_1 and Z_2. The sum $X_1 + X_2$ is the constant $\mathbb{E}[X_1] + \mathbb{E}[X_2]$ plus $\sqrt{\mathrm{Var}(X_1)}$ and $\sqrt{\mathrm{Var}(X_2)}$ times these two variables, and therefore has the law $\mathcal{N}\big(\mathbb{E}[X_1] + \mathbb{E}[X_2],\ \mathrm{Var}(X_1) + \mathrm{Var}(X_2)\big)$. If one of the variances is zero, that variable is a constant, and adding it shifts the mean.

Step 3: the pair (iii). Write $X = \mathbb{E}[X] + a_{X,1}Z_1 + a_{X,2}Z_2$ and $Y = \mathbb{E}[Y] + a_{Y,1}Z_1 + a_{Y,2}Z_2$. Since Z_1 and Z_2 are independent with mean 0 and variance 1, $\mathbb{E}[Z_1 Z_2] = 0$, and expanding the products gives

$$\mathrm{Var}(Y) = a_{Y,1}^2 + a_{Y,2}^2, \qquad \mathrm{Cov}(X, Y) = a_{X,1}a_{Y,1} + a_{X,2}a_{Y,2}.$$

Put $\sigma_Y = \sqrt{\mathrm{Var}(Y)} > 0$ and $(c_1, c_2) = (a_{Y,1}, a_{Y,2})/\sigma_Y$, a unit vector, and let Z_1' and Z_2' be the rotated variables of Step 2. Multiplying out and using $c_1^2 + c_2^2 = 1$,

$$c_1 Z_1' - c_2 Z_2' = Z_1, \qquad c_2 Z_1' + c_1 Z_2' = Z_2.$$

Substituting these into Y and X,

$$Y = \mathbb{E}[Y] + \sigma_Y\,Z_1', \qquad X = \mathbb{E}[X] + c_{XY}\,\sigma_Y\,Z_1' + \sigma_\perp\,Z_2', \tag{E.15}$$

with $c_{XY} = \operatorname{Cov}(X,Y)/\operatorname{Var}(Y)$ and $\sigma_\perp = a_{X,2}c_1 - a_{X,1}c_2$: the coefficient of Z_1' in Y is $a_{Y,1}c_1 + a_{Y,2}c_2 = \sigma_Y$, that of Z_2' in Y is $-a_{Y,1}c_2 + a_{Y,2}c_1 = 0$, and that of Z_1' in X is $a_{X,1}c_1 + a_{X,2}c_2 = \operatorname{Cov}(X,Y)/\sigma_Y = c_{XY}\,\sigma_Y$. The constant $\sigma_\perp$ may have either sign. So Y is carried by Z_1' alone, and X is c_{XY} times the centred Y plus a term carried by Z_2' alone.

Let ψ be a test function, with constant c. The function

$$\psi_2(z_1', z_2') = e^{\mathbb{E}[Y]+\sigma_Y z_1'}\,\psi\big(\mathbb{E}[X] + c_{XY}\sigma_Y z_1' + \sigma_\perp z_2'\big)$$

(a name local to this step) is a test function on the plane. It is continuous except on the finitely many lines on which the argument of ψ is a point of discontinuity of ψ, and

$$|\psi_2(z_1', z_2')| \le c\, e^{\mathbb{E}[Y]+c\,|\mathbb{E}[X]|}\; e^{(1+c\,|c_{XY}|)\,\sigma_Y\,|z_1'|}\; e^{c\,|\sigma_\perp|\,|z_2'|},$$

which is at most $c'' e^{c''(|z_1'|+|z_2'|)}$ for a large enough constant c''. By (E.15), $\mathbb{E}\big[e^Y\psi(X)\big] = \mathbb{E}\,\psi_2(Z_1', Z_2')$. By (E.14) and by Fact 18.34(i), in its integrable case, which lets z_1' be integrated first,

$$\mathbb{E}\big[e^Y\psi(X)\big] = e^{\mathbb{E}[Y]}\int \psi_1(z_2')\,\varphi(z_2')\,dz_2',$$

where

$$\psi_1(z_2') = \int e^{\sigma_Y z_1'}\,\psi\big(\mathbb{E}[X] + c_{XY}\sigma_Y z_1' + \sigma_\perp z_2'\big)\,\varphi(z_1')\,dz_1'.$$

In the inner average $\psi_1(z_2')$ the value z_2' is frozen, and the integral converges absolutely for every z_2', by the bound of Step 1. By (E.13) with $\theta = \sigma_Y$ and the substitution $z_1' \mapsto z_1' + \sigma_Y$, as in Step 1,

$$\psi_1(z_2') = e^{\sigma_Y^2/2}\int \psi\big(\mathbb{E}[X] + c_{XY}\sigma_Y\,(z_1' + \sigma_Y) + \sigma_\perp z_2'\big)\,\varphi(z_1')\,dz_1'.$$

Insert this into the outer integral and use Fact 18.34(i) and (E.14) once more, now for the function $(z_1', z_2') \mapsto \psi\big(\mathbb{E}[X] + c_{XY}\sigma_Y^2 + c_{XY}\sigma_Y z_1' + \sigma_\perp z_2'\big)$, which is again a test function on the plane:

$$\begin{aligned}\mathbb{E}\big[e^Y\psi(X)\big] &= e^{\mathbb{E}[Y]+\sigma_Y^2/2}\,\mathbb{E}\Big[\psi\big(\mathbb{E}[X] + c_{XY}\sigma_Y^2 + c_{XY}\sigma_Y Z_1' + \sigma_\perp Z_2'\big)\Big]\\ &= e^{\mathbb{E}[Y]+\sigma_Y^2/2}\,\mathbb{E}\big[\psi\big(X + c_{XY}\,\sigma_Y^2\big)\big].\end{aligned}$$

Here $c_{XY}\,\sigma_Y^2 = \operatorname{Cov}(X,Y)$. Taking $\psi = 1$ gives $\mathbb{E}\,e^Y = e^{\mathbb{E}[Y]+\sigma_Y^2/2} = e^{\mathbb{E}[Y]+\operatorname{Var}(Y)/2}$, and the display is then the second identity of (iii), every mean in it being finite. □

E.9 Proof of Corollary 19.49: Growth of a continuously rebalanced leveraged position

The statement is on page 295. The notation and the standing assumptions are those of §19.7.

Proof. We verify each claim with Corollary 19.31; no differential equation is solved.

Part (i), the explicit path. By (19.10), $(S_t/S_0)^L = \exp(Lgt + L\sigma W_t)$ with $g = \mu - \sigma^2/2$, and

$$Lg - \tfrac12\,L(L-1)\,\sigma^2 = L\mu - \tfrac12\,L\sigma^2 - \tfrac12\,L^2\sigma^2 + \tfrac12\,L\sigma^2 = L\mu - \tfrac12\,L^2\sigma^2.$$

So $V_t = V_0\exp\big((L\mu - \frac12 L^2\sigma^2)\,t + L\sigma W_t\big)$, and the formula for V_T/V_0 is the definition of V at $t = T$. For $L > 0$ the path is (19.10) with the parameters $(L\mu, L\sigma)$ and the starting value V_0. For $L < 0$, $L\sigma W_t = |L|\sigma\,(-W_t)$, and $-W$ is a standard Brownian motion (Proposition 19.8(iii)): the path is (19.10) with the parameters $(L\mu, |L|\sigma)$, driven by $-W$.

Part (ii), the trading gains. The function $V(x,t)$ is a constant times x^L times a function of t that is bounded on $[0,T]$. Its derivatives satisfy

$$\partial_t V = -\tfrac12\,L(L-1)\,\sigma^2\,V, \qquad x\,\partial_x V = L\,V, \qquad x^2\,\partial_{xx} V = L(L-1)\,V.$$

Since $x^a \le x^{|a|} + x^{-|a|}$ for every real a, the functions V, $\partial_t V$, $\partial_x V = LV/x$ and $\partial_{xx} V = L(L-1)\,V/x^2$ are continuous and of moderate growth on $(0,\infty)\times[0,T]$. Corollary 19.31(iii) applies, with $\alpha_t = \mu S_t$ and $\beta_t = \sigma S_t$. The drift of V_t is

$$\partial_t V + \mu x\,\partial_x V + \tfrac{1}{2}\,\sigma^2 x^2\,\partial_{xx} V = \Big(-\tfrac{1}{2}\,L(L-1)\,\sigma^2 + L\mu + \tfrac{1}{2}\,\sigma^2 L(L-1)\Big)\,V = L\mu\,V,$$

and its diffusion coefficient is $\sigma x\,\partial_x V = L\sigma\,V$, both at $x = S_t$. So $V_t = V_0 + \int_0^t L\mu\,V_s\,ds + \int_0^t L\sigma\,V_s\,dW_s$, and $L\mu\,V_s = h_s\,\mu S_s$ and $L\sigma\,V_s = h_s\,\sigma S_s$ with $h_s = L\,V_s/S_s$.

Part (iii), the growth rate. By the explicit path, which is Corollary 19.34(i) for the geometric Brownian motion V, $\ln(V_T/V_0) = (L\mu - \frac{1}{2}L^2\sigma^2)\,T + L\sigma W_T$, whose mean is $g_L\,T$. As a function of L, $g_L = -\frac{1}{2}\sigma^2 L^2 + \mu L$ is a quadratic with the negative leading coefficient $-\sigma^2/2$ and the zeros 0 and $2\mu/\sigma^2$. Such a quadratic is positive strictly between its zeros and negative outside the closed interval that they bound; when $\mu = 0$ the zeros coincide and $g_L = -L^2\sigma^2/2 < 0$ for $L \neq 0$. Its derivative $\mu - L\sigma^2$ is positive for $L < \mu/\sigma^2$ and negative for $L > \mu/\sigma^2$, so g_L is largest at $L^* = \mu/\sigma^2$, where it equals $\mu^2/\sigma^2 - \mu^2/(2\sigma^2) = \mu^2/(2\sigma^2)$.

Part (iv), the financing clause. The verification is the same. For the new V, $\partial_t V = \big((1-L)\,r - \frac{1}{2}L(L-1)\sigma^2\big)\,V$, while $x\,\partial_x V = LV$ and $x^2\,\partial_{xx} V = L(L-1)V$ as before; the time factor is bounded on $[0,T]$, so the four functions have moderate growth as before. The drift is

$$\Big((1-L)\,r - \tfrac{1}{2}\,L(L-1)\,\sigma^2 + L\mu + \tfrac{1}{2}\,\sigma^2 L(L-1)\Big)\,V = \big(r + L(\mu - r)\big)\,V = h\,\mu S + r\,(V - hS),$$

because $hS = LV$, and the diffusion coefficient is $L\sigma\,V = h\,\sigma S$. Along the path,

$$\ln\frac{V_T}{V_0} = L\ln\frac{S_T}{S_0} + \Big((1-L)\,r - \tfrac{1}{2}\,L(L-1)\,\sigma^2\Big)\,T = \Big(r + L(\mu - r) - \tfrac{1}{2}\,L^2\sigma^2\Big)\,T + L\sigma W_T,$$

by the computation of the first paragraph with the term $(1-L)\,r$ added. This is (19.10) with drift $r + L(\mu - r)$ and volatility $|L|\sigma$, driven by W when $L > 0$ and by $-W$ when $L < 0$, and its mean is $g_L\,T$ with $g_L = r + L(\mu - r) - \frac{1}{2}L^2\sigma^2$. The derivative of g_L in L is $\mu - r - L\sigma^2$, positive below $L^* = (\mu - r)/\sigma^2$ and negative above it, and $g_{L^*} = r + (\mu - r)^2/\sigma^2 - (\mu - r)^2/(2\sigma^2) = r + (\mu - r)^2/(2\sigma^2)$. □

E.10 Proof of Proposition 19.52: Cholesky construction and cross-variation of correlated Brownian motions

The statement is on page 298. The notation and the standing assumptions are those of §19.8.

Proof. We check the four properties of Definition 19.5 for (i), rotate a jointly normal pair for (iii), and expand a square for (ii). For a general pair, part (ii) rests on part (iii), so (iii) is proved first.

The increments of two independent motions. Let W_1 and $W_\perp$ be independent and let $0 \le t_0 < t_1 < \cdots < t_n$ be a grid. The vector $(\Delta W_{1,1}, \dots, \Delta W_{1,n})$ is a function of the values of W_1 at the grid times, and the vector $(\Delta W_{\perp,1}, \dots, \Delta W_{\perp,n})$ is a function of the values of $W_\perp$ there. The two vectors are independent of each other, because the motions are, and the coordinates of each are mutually independent (Definition 19.5(3)). So the probability that each of the $2n$ increments lies in a set prescribed for it factorises, first across the two motions and then within each: the $2n$ increments $\Delta W_{1,1}, \dots, \Delta W_{1,n}, \Delta W_{\perp,1}, \dots, \Delta W_{\perp,n}$ are mutually independent. By Definition 19.5(2), $\Delta W_{1,i}$ and $\Delta W_{\perp,i}$ have the law $\mathcal{N}(0, \Delta t_i)$.

Part (i): W_2 is a standard Brownian motion. Property (1) holds because $W_{2,0} = \rho\cdot 0 + \sqrt{1-\rho^2}\cdot 0 = 0$, and property (4) because a combination of two continuous paths is continuous. For property (2), let $0 \le s < t$ and put

$$Z_1 = \frac{W_{1,t} - W_{1,s}}{\sqrt{t-s}}, \qquad Z_2 = \frac{W_{\perp,t} - W_{\perp,s}}{\sqrt{t-s}}.$$

These are standard normal variables, by (19.3), and they are independent, the first being a function of values of W_1 and the second of values of $W_\perp$. The increment of W_2 is

$$W_{2,t} - W_{2,s} = \sqrt{t-s}\,(\rho\, Z_1 + \sqrt{1-\rho^2}\, Z_2).$$

The vector $(\rho, \sqrt{1-\rho^2})$ has length one, so $\rho Z_1 + \sqrt{1-\rho^2}\, Z_2$ is standard normal by Lemma 19.38(ii), and by (19.3) $W_{2,t} - W_{2,s} \sim \mathcal{N}(0, t-s)$. For property (3), take a grid $0 \le t_0 < \cdots < t_n$. The increment $\Delta W_{2,i} = \rho\, \Delta W_{1,i} + \sqrt{1-\rho^2}\, \Delta W_{\perp,i}$ is a function of the pair $(\Delta W_{1,i}, \Delta W_{\perp,i})$. These n pairs are disjoint groups of the $2n$ mutually independent increments, and functions of disjoint groups of mutually independent variables are mutually independent (the grouping property (a) of the Setup of §19.2).

Part (i): the correlation. With the same Z_1 and Z_2,

$$(W_{1,t} - W_{1,s},\ W_{2,t} - W_{2,s}) = \sqrt{t-s}\,(Z_1,\ \rho Z_1 + \sqrt{1-\rho^2}\, Z_2)\,:$$

both coordinates are affine functions of the pair (Z_1, Z_2) of independent standard normal variables, so the pair of increments is jointly normal. Both coordinates have mean zero, so their covariance is the mean of their product,

$$(t-s)\left(\rho\, \mathbb{E}[Z_1^2] + \sqrt{1-\rho^2}\, \mathbb{E}[Z_1 Z_2]\right) = \rho\,(t-s),$$

because $\mathbb{E}[Z_1^2] = 1$ and $\mathbb{E}[Z_1 Z_2] = \mathbb{E}[Z_1]\,\mathbb{E}[Z_2] = 0$ by the product rule (19.5). This is condition (1) of Definition 19.51. For condition (2), the pair $(\Delta W_{1,i}, \Delta W_{2,i})$ is a function of the pair $(\Delta W_{1,i}, \Delta W_{\perp,i})$, and these pairs are mutually independent, as in the proof of property (3). Finally, $s = 0$ in the covariance gives $\mathrm{Cov}(W_{1,t}, W_{2,t}) = \rho\, t$ for $t > 0$, and both sides vanish at $t = 0$.

Part (iii). Let (W_1, W_2) have correlation ρ with $|\rho| < 1$, and define $W_\perp$ as in the statement; solving for W_2 gives $W_2 = \rho\, W_1 + \sqrt{1-\rho^2}\, W_\perp$. Properties (1) and (4) of Definition 19.5 hold for $W_\perp$ as they did for W_2 in (i). Fix a grid $0 \le t_0 < \cdots < t_n$ and a cell i. By Definition 19.51(1) the pair $(\Delta W_{1,i}, \Delta W_{2,i})$ is jointly normal, with means zero, variances Δt_i and covariance $\rho\, \Delta t_i$. In the notation of the Setup of §19.6, there are independent standard normal variables Z_1 and Z_2 and coefficient vectors a_X and a_Y with

$$\Delta W_{1,i} = a_{X,1} Z_1 + a_{X,2} Z_2, \qquad \Delta W_{2,i} = a_{Y,1} Z_1 + a_{Y,2} Z_2,$$

the constant terms being the means, zero. Expanding the products and using $\mathbb{E}[Z_1^2] = \mathbb{E}[Z_2^2] = 1$ and $\mathbb{E}[Z_1 Z_2] = 0$, the variances are the squared lengths $|a_X|^2 = |a_Y|^2 = \Delta t_i$ and the covariance is the scalar product $a_X \cdot a_Y = \rho\, \Delta t_i$. Hence

$$\Delta W_{\perp,i} = \frac{\Delta W_{2,i} - \rho\, \Delta W_{1,i}}{\sqrt{1-\rho^2}} = a_{\perp,1} Z_1 + a_{\perp,2} Z_2, \qquad a_\perp = \frac{a_Y - \rho\, a_X}{\sqrt{1-\rho^2}},$$

with

$$a_X \cdot a_\perp = \frac{\rho\, \Delta t_i - \rho\, \Delta t_i}{\sqrt{1-\rho^2}} = 0, \qquad |a_\perp|^2 = \frac{\Delta t_i - 2\rho^2\, \Delta t_i + \rho^2\, \Delta t_i}{1-\rho^2} = \Delta t_i.$$

So $(c_1, c_2) = a_X / \sqrt{\Delta t_i}$ is a unit vector, and $a_\perp / \sqrt{\Delta t_i}$ is a unit vector orthogonal to it. The vectors orthogonal to the nonzero vector (c_1, c_2) are the multiples of $(-c_2, c_1)$: if $c_1 y_1 + c_2 y_2 = 0$ and $c_1 \ne 0$, then $(y_1, y_2) = (y_2/c_1)(-c_2, c_1)$, and if $c_2 \ne 0$, then $(y_1, y_2) = (-y_1/c_2)(-c_2, c_1)$. The unit ones are $\pm(-c_2, c_1)$, so, for one of the two signs,

$$\Delta W_{1,i} = \sqrt{\Delta t_i}\; Z_1', \qquad \Delta W_{\perp,i} = \pm\sqrt{\Delta t_i}\; Z_2', \qquad Z_1' = c_1 Z_1 + c_2 Z_2, \quad Z_2' = -c_2 Z_1 + c_1 Z_2.$$

By Lemma 19.38(ii), Z_1' and Z_2' are independent standard normal variables. Two conclusions follow. First, $\Delta W_{\perp,i} \sim \mathcal{N}(0, \Delta t_i)$, by (19.3); applied to the one-cell grid $s < t$ this is property

(2) for $W_\perp$. Second, $\Delta W_{1,i}$ and $\Delta W_{\perp,i}$ are independent of each other, being functions of the independent variables Z_1' and Z_2'.

The n pairs $(\Delta W_{1,i}, \Delta W_{\perp,i})$ are functions of the pairs $(\Delta W_{1,i}, \Delta W_{2,i})$, which are mutually independent by Definition 19.51(2), so they are mutually independent, by the grouping property (a) of the Setup of §19.2; and the two coordinates of each pair are independent. Therefore the probability that each of the $2n$ variables $\Delta W_{1,1}, \dots, \Delta W_{1,n}, \Delta W_{\perp,1}, \dots, \Delta W_{\perp,n}$ lies in a set prescribed for it factorises, first across the cells and then within each cell: the $2n$ variables are mutually independent. In particular the increments of $W_\perp$ over the grid are mutually independent, which is property (3), and $W_\perp$ is a standard Brownian motion. For the independence of the two motions, take finitely many times for W_1 and finitely many for $W_\perp$, and merge them, together with the time 0, into one grid starting at $t_0 = 0$. Each chosen value of W_1 is a sum of increments $\Delta W_{1,i}$ and each chosen value of $W_\perp$ is a sum of increments $\Delta W_{\perp,i}$. By the grouping property (a), the vector of the chosen values of W_1 is independent of the vector of the chosen values of $W_\perp$.

Part (ii) for the pair of the construction. Let $W_2 = \rho\, W_1 + \sqrt{1-\rho^2}\, W_\perp$ with W_1 and $W_\perp$ independent, $-1 \le \rho \le 1$. Since $\Delta W_{2,i} = \rho\, \Delta W_{1,i} + \sqrt{1-\rho^2}\, \Delta W_{\perp,i}$ and $\sum_i \Delta t_i = t$,

$$\sum_{i=1}^{n} \Delta W_{1,i}\, \Delta W_{2,i} - \rho\, t = \rho \sum_{i=1}^{n} \big((\Delta W_{1,i})^2 - \Delta t_i\big) + \sqrt{1-\rho^2} \sum_{i=1}^{n} \Delta W_{1,i}\, \Delta W_{\perp,i}.$$

Square this and take expectations. Every product that appears is a product of powers of the $2n$ mutually independent normal increments, which have moments of every order, so by the product rule (19.5), applied repeatedly, its mean is the product of the means of its independent factors. Three kinds of terms arise.

- The square of the first sum has mean $\mathrm{Var}(\mathrm{QV}_n) = 2\sum_i (\Delta t_i)^2$, by Proposition 19.10(i) applied to W_1, whose quadratic-variation sum has mean t.
- In the square of the second sum, a diagonal term has mean $\mathbb{E}[(\Delta W_{1,i})^2]\, \mathbb{E}[(\Delta W_{\perp,i})^2] = (\Delta t_i)^2$, and a term with $i \ne j$ has mean $\mathbb{E}[\Delta W_{1,i}]\, \mathbb{E}[\Delta W_{\perp,i}]\, \mathbb{E}[\Delta W_{1,j}]\, \mathbb{E}[\Delta W_{\perp,j}] = 0$. So the square has mean $\sum_i (\Delta t_i)^2$.
- A term of the product of the two sums is $\big((\Delta W_{1,i})^2 - \Delta t_i\big)\, \Delta W_{1,j}\, \Delta W_{\perp,j}$. The factor $\Delta W_{\perp,j}$ is independent of the pair $(\Delta W_{1,i}, \Delta W_{1,j})$ and has mean zero, so the term has mean zero.

Adding the three contributions with their coefficients ρ^2, $1-\rho^2$ and $2\rho\sqrt{1-\rho^2}$,

$$\mathbb{E}\Big[\Big(\sum_{i=1}^{n} \Delta W_{1,i}\, \Delta W_{2,i} - \rho\, t\Big)^2\Big] = (2\rho^2 + 1 - \rho^2) \sum_{i=1}^{n} (\Delta t_i)^2 = (1+\rho^2) \sum_{i=1}^{n} (\Delta t_i)^2,$$

and $\sum_i (\Delta t_i)^2 \le t\, \Delta t_{\max}$ by Proposition 19.10(iii).

Part (ii) for any pair. Let (W_1, W_2) have correlation ρ. If $|\rho| < 1$, part (iii) writes $W_2 = \rho\, W_1 + \sqrt{1-\rho^2}\, W_\perp$ with $W_\perp$ a standard Brownian motion independent of W_1, path by path, so the pair is the pair of the construction and the previous paragraph applies. If $\rho = \pm 1$, then by Proposition 10.10 and Definition 19.51(1)

$$\mathrm{Var}\,(\Delta W_{2,i} - \rho\, \Delta W_{1,i}) = \Delta t_i + \rho^2\, \Delta t_i - 2\rho \cdot \rho\, \Delta t_i = 0,$$

so $\Delta W_{2,i} = \rho\, \Delta W_{1,i}$ with probability one: by Chebyshev's inequality (Lemma 19.12(i)) the difference exceeds $1/k$ in absolute value with probability zero for every $k = 1, 2, \dots$, and a countable union of events of probability zero has probability zero (Lemma 19.12(0)). The sum of products is then $\rho\, \mathrm{QV}_n$ with probability one, where QV_n is the quadratic-variation sum of W_1, and since $\rho^2 = 1$,

$$\mathbb{E}\Big[\big(\rho\, \mathrm{QV}_n - \rho\, t\big)^2\Big] = \mathbb{E}\big[(\mathrm{QV}_n - t)^2\big] = 2\sum_{i=1}^{n} (\Delta t_i)^2 = (1+\rho^2) \sum_{i=1}^{n} (\Delta t_i)^2,$$

by Proposition 19.10(i). □

E.11 Proof of Lemma 19.54: Equality of mixed partial derivatives (Schwarz, 1873)

The statement is on page 299. The notation and the standing assumptions are those of §19.8.

Proof. We apply the mean value theorem twice to a double difference, once in each order. Fix a point (x_1, x_2) of the set and an open square centred at it that lies inside the set, and let Δx_1 and Δx_2 be nonzero numbers, each smaller in absolute value than half the side of the square. Every point used below then lies in the square. The *double difference* is

$$\psi(x_1 + \Delta x_1,\ x_2 + \Delta x_2) - \psi(x_1 + \Delta x_1,\ x_2) - \psi(x_1,\ x_2 + \Delta x_2) + \psi(x_1, x_2).$$

Step 1: the first variable. Put

$$\chi(u) = \psi(x_1 + u\,\Delta x_1,\ x_2 + \Delta x_2) - \psi(x_1 + u\,\Delta x_1,\ x_2).$$

This function is defined for all u with $|u\,\Delta x_1|$ smaller than half the side of the square, an open interval that contains $[0, 1]$, and the double difference is $\chi(1) - \chi(0)$. In each of its two terms only the first argument of ψ moves, so by the chain rule for functions of one variable

$$\chi'(u) = \Delta x_1 \Big(\partial_1\psi(x_1 + u\,\Delta x_1,\ x_2 + \Delta x_2) - \partial_1\psi(x_1 + u\,\Delta x_1,\ x_2)\Big),$$

which is continuous in u because $\partial_1\psi$ is continuous. Corollary 7.32 at order zero, the mean value theorem, gives a point $u^* \in [0, 1]$ with $\chi(1) - \chi(0) = \chi'(u^*)$. Put $x_1^* = x_1 + u^*\,\Delta x_1$.

Step 2: the second variable. The bracket in $\chi'(u^*)$ is the increment, between x_2 and $x_2 + \Delta x_2$, of the function $\partial_1\psi(x_1^*, \cdot)$ of one variable. That function has the continuous derivative $\partial_2\partial_1\psi(x_1^*, \cdot)$ on an open interval that contains both end points. The mean value theorem gives a point x_2^* between x_2 and $x_2 + \Delta x_2$ at which the increment equals $\Delta x_2\,\partial_2\partial_1\psi(x_1^*, x_2^*)$. Hence the double difference equals

$$\Delta x_1\,\Delta x_2\ \partial_2\partial_1\psi(x_1^*, x_2^*).$$

Step 3: the other order. The double difference is also the increment, between x_2 and $x_2 + \Delta x_2$, of the function $\psi(x_1 + \Delta x_1, \cdot) - \psi(x_1, \cdot)$ of one variable. Steps 1 and 2, with the roles of the two variables exchanged, give a point (x_1^{**}, x_2^{**}) of the closed rectangle with corners (x_1, x_2) and $(x_1 + \Delta x_1, x_2 + \Delta x_2)$ at which the double difference equals

$$\Delta x_1\,\Delta x_2\ \partial_1\partial_2\psi(x_1^{**}, x_2^{**}).$$

Step 4: the limit. Dividing the two expressions for the double difference by $\Delta x_1\,\Delta x_2 \neq 0$,

$$\partial_2\partial_1\psi(x_1^*, x_2^*) = \partial_1\partial_2\psi(x_1^{**}, x_2^{**}).$$

Both points lie in the closed rectangle of Step 3, so they tend to (x_1, x_2) as $\Delta x_1 \to 0$ and $\Delta x_2 \to 0$. The two mixed derivatives are continuous at (x_1, x_2), and the limit of the last display is $\partial_2\partial_1\psi(x_1, x_2) = \partial_1\partial_2\psi(x_1, x_2)$.

Three or more variables. Hold all variables but two fixed. The set of values of the two free variables for which the point lies in the open domain is an open set of the plane, and on it the function of the two free variables satisfies the hypotheses of the first part.

The boundary $t = 0$. Let the function be defined at the points of an open set $\mathcal{O}$ with $t \geq 0$. The points of $\mathcal{O}$ with $t > 0$ form an open set, on which the two mixed derivatives agree by the previous paragraph. A point of $\mathcal{O}$ with $t = 0$ is the limit of points of $\mathcal{O}$ with $t > 0$, obtained by raising t by a small positive amount, because $\mathcal{O}$ is open. Both mixed derivatives are continuous up to $t = 0$, so at that point each is the limit of its values at those points, and the two limits agree. □

E.12 Proof of Theorem 19.55: Two-dimensional Itô lemma (Itô, 1951)

The statement is on page 300. The notation and the standing assumptions are those of §19.8.

Proof. We repeat the proof of Theorem 19.30 with a Taylor expansion in two space variables; the only new computation is one cross-variation sum. The proof has six steps. Steps 1 to 3 prove (i) for a fixed $t \in (0,T]$; for $t = 0$ both sides of (19.16) equal $f(0,0,0)$. Fix a partition $0 = t_0 < t_1 < \cdots < t_n = t$ of $[0,t]$. In the sums of Steps 1 to 3 a derivative of f written without an argument is evaluated at the left point $(t_{i-1}, W_{1,t_{i-1}}, W_{\perp,t_{i-1}})$ of its cell.

Step 1: the expansion. The step begins with a chain rule along a segment, which is then applied on each cell of the partition.

A chain rule along a segment. Let $\psi(x_1,x_2)$ have continuous first partial derivatives on an open rectangle of the plane, that is, on a product of two open intervals, bounded or not; the whole plane is one. Let (x_1,x_2) be a point and $(\Delta x_1, \Delta x_2)$ a pair of increments, and put $\chi(u) = \psi(x_1 + u\,\Delta x_1,\ x_2 + u\,\Delta x_2)$ for the real numbers u at which this point lies in the rectangle; they form an open interval. Then χ is continuously differentiable, with

$$\chi'(u) = \Delta x_1\,\partial_1\psi + \Delta x_2\,\partial_2\psi, \qquad \text{both derivatives at } (x_1 + u\,\Delta x_1,\ x_2 + u\,\Delta x_2). \tag{E.16}$$

Indeed, let $\Delta u \neq 0$ be an increment with $u' = u + \Delta u$ in the interval. A rectangle that contains two points contains every point whose coordinates lie between theirs, so all the points used below lie in it. Split $\chi(u') - \chi(u)$ into a move of the first argument and a move of the second:

$$\begin{aligned}\chi(u') - \chi(u) = {}& \Big(\psi(x_1 + u'\,\Delta x_1,\ x_2 + u'\,\Delta x_2) - \psi(x_1 + u\,\Delta x_1,\ x_2 + u'\,\Delta x_2)\Big)\\ &+ \Big(\psi(x_1 + u\,\Delta x_1,\ x_2 + u'\,\Delta x_2) - \psi(x_1 + u\,\Delta x_1,\ x_2 + u\,\Delta x_2)\Big).\end{aligned}$$

By the mean value theorem in the first variable (Corollary 7.32 at order zero), the first bracket equals $\Delta u\,\Delta x_1\,\partial_1\psi$ at a point whose first coordinate lies between $x_1 + u\,\Delta x_1$ and $x_1 + u'\,\Delta x_1$ and whose second coordinate is $x_2 + u'\,\Delta x_2$. By the mean value theorem in the second variable, the second bracket equals $\Delta u\,\Delta x_2\,\partial_2\psi$ at a point whose first coordinate is $x_1 + u\,\Delta x_1$ and whose second coordinate lies between $x_2 + u\,\Delta x_2$ and $x_2 + u'\,\Delta x_2$. As $\Delta u \to 0$ both points tend to $(x_1 + u\,\Delta x_1,\ x_2 + u\,\Delta x_2)$. Dividing by Δu and using the continuity of $\partial_1\psi$ and $\partial_2\psi$ gives (E.16), and χ' is continuous as a combination of continuous functions of u.

Telescoping and the two brackets. Telescope as in Step 1 of the proof of Theorem 19.30, with the pair $(W_{1,t_i}, W_{\perp,t_i})$ in place of W_{t_i}: on every path, $f(t, W_{1,t}, W_{\perp,t}) - f(0,0,0)$ is the sum over the cells of a move in time at the new pair of Brownian values,

$$f(t_i, W_{1,t_i}, W_{\perp,t_i}) - f(t_{i-1}, W_{1,t_i}, W_{\perp,t_i}),$$

and a move in space at the old time, $f(t_{i-1}, W_{1,t_i}, W_{\perp,t_i}) - f(t_{i-1}, W_{1,t_{i-1}}, W_{\perp,t_{i-1}})$. With the two space arguments held fixed, the time bracket is treated as in that proof, word for word: by the fundamental theorem of calculus on the closed cell and the intermediate value theorem it equals $\partial_t f(t_i^*, W_{1,t_i}, W_{\perp,t_i})\,\Delta t_i$ for a point $t_i^* \in [t_{i-1}, t_i]$ that depends on the path. For the space bracket, fix i and a path, and put

$$\chi_i(u) = f\big(t_{i-1},\ W_{1,t_{i-1}} + u\,\Delta W_{1,i},\ W_{\perp,t_{i-1}} + u\,\Delta W_{\perp,i}\big), \qquad u \in \mathbb{R},$$

the function f along the line through the old and the new pair of Brownian values. Apply (E.16), on the whole plane, to $\psi = f(t_{i-1},\cdot,\cdot)$, and then to $\psi = \partial_1 f(t_{i-1},\cdot,\cdot)$ and to $\psi = \partial_2 f(t_{i-1},\cdot,\cdot)$, which have continuous first partial derivatives by hypothesis. The function χ_i is twice continuously differentiable on the whole line, with

$$\begin{aligned}\chi_i'(u) &= \partial_1 f\,\Delta W_{1,i} + \partial_2 f\,\Delta W_{\perp,i},\\ \chi_i''(u) &= \partial_{11} f\,(\Delta W_{1,i})^2 + (\partial_2\partial_1 f + \partial_1\partial_2 f)\,\Delta W_{1,i}\,\Delta W_{\perp,i} + \partial_{22} f\,(\Delta W_{\perp,i})^2\\ &= \partial_{11} f\,(\Delta W_{1,i})^2 + 2\,\partial_{12} f\,\Delta W_{1,i}\,\Delta W_{\perp,i} + \partial_{22} f\,(\Delta W_{\perp,i})^2,\end{aligned}$$

the derivatives of f in these two displays being evaluated at $(t_{i-1},\ W_{1,t_{i-1}} + u\,\Delta W_{1,i},\ W_{\perp,t_{i-1}} + u\,\Delta W_{\perp,i})$. The two cross terms of χ_i'' are equal by Lemma 19.54. Theorem 7.31 at order one, with the expansion point 0 and the Lagrange form of the remainder (Corollary 7.32), evaluated at $u = 1$, gives a point $u_i^* \in [0,1]$ with

$$f(t_{i-1}, W_{1,t_i}, W_{\perp,t_i}) - f(t_{i-1}, W_{1,t_{i-1}}, W_{\perp,t_{i-1}}) = \chi_i(1) - \chi_i(0) = \chi_i'(0) + \tfrac{1}{2}\,\chi_i''(u_i^*).$$

The four sums. Adding the brackets over i, on every path

$$f(t, W_{1,t}, W_{\perp,t}) - f(0,0,0) = \mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3 + \mathcal{T}_4, \tag{E.17}$$

where, as in the proof of Theorem 19.30, $\mathcal{T}_1$ is the time sum, $\mathcal{T}_2$ the gradient sum, $\mathcal{T}_3$ the quadratic sum and $\mathcal{T}_4$ the remainder:

$$\begin{aligned}
\mathcal{T}_1 &= \sum_{i=1}^{n} \partial_t f(t_i^*, W_{1,t_i}, W_{\perp,t_i})\,\Delta t_i,\\
\mathcal{T}_2 &= \sum_{i=1}^{n} \chi_i'(0) = \sum_{i=1}^{n} \partial_1 f\,\Delta W_{1,i} + \sum_{i=1}^{n} \partial_2 f\,\Delta W_{\perp,i},\\
\mathcal{T}_3 &= \tfrac{1}{2}\sum_{i=1}^{n} \chi_i''(0) = \tfrac{1}{2}\sum_{i=1}^{n} \partial_{11} f\,(\Delta W_{1,i})^2 + \sum_{i=1}^{n} \partial_{12} f\,\Delta W_{1,i}\,\Delta W_{\perp,i} + \tfrac{1}{2}\sum_{i=1}^{n} \partial_{22} f\,(\Delta W_{\perp,i})^2,\\
\mathcal{T}_4 &= \tfrac{1}{2}\sum_{i=1}^{n} (\chi_i''(u_i^*) - \chi_i''(0)).
\end{aligned}$$

As in the one-dimensional proof, each of the four sums is a function of the Brownian values at the partition points alone, although the starred points depend on the path.

Step 2: the cross sum. Put $a_i = \partial_{12} f(t_{i-1}, W_{1,t_{i-1}}, W_{\perp,t_{i-1}})$ in this step. We show that

$$\mathbb{E}\Big[\Big(\sum_{i=1}^{n} a_i\,\Delta W_{1,i}\,\Delta W_{\perp,i}\Big)^2\Big] = \sum_{i=1}^{n} \mathbb{E}[a_i^2]\,(\Delta t_i)^2 \le t\,\Delta t_{\max} \sup_{0\le s\le t} \mathbb{E}\big[\partial_{12} f(s, W_{1,s}, W_{\perp,s})^2\big],$$

which tends to 0 with the mesh; the supremum is finite by Lemma 19.20(iii). Expand the square. By Step 1 of the proof of Theorem 19.23, the pair of increments $(\Delta W_{1,j}, \Delta W_{\perp,j})$ is independent of every vector of values of the two motions at times at most t_{j-1}, and its two coordinates are independent of each other, each with the law $\mathcal{N}(0, \Delta t_j)$. A diagonal term is the product of a_i^2, a function of such values with $j = i$, and of $(\Delta W_{1,i})^2\,(\Delta W_{\perp,i})^2$. Two applications of the product rule (19.5) of Lemma 19.17 give

$$\mathbb{E}\big[a_i^2\,(\Delta W_{1,i})^2\,(\Delta W_{\perp,i})^2\big] = \mathbb{E}[a_i^2]\,\mathbb{E}\big[(\Delta W_{1,i})^2\big]\,\mathbb{E}\big[(\Delta W_{\perp,i})^2\big] = \mathbb{E}[a_i^2]\,(\Delta t_i)^2.$$

For $i < j$, the factor $a_i a_j\,\Delta W_{1,i}\,\Delta W_{\perp,i}$ is a function of values of the two motions at times at most t_{j-1}, with a finite mean, and in the same way

$$\mathbb{E}\big[a_i a_j\,\Delta W_{1,i}\,\Delta W_{\perp,i}\,\Delta W_{1,j}\,\Delta W_{\perp,j}\big] = \mathbb{E}\big[a_i a_j\,\Delta W_{1,i}\,\Delta W_{\perp,i}\big]\,\mathbb{E}[\Delta W_{1,j}]\,\mathbb{E}[\Delta W_{\perp,j}] = 0.$$

This is Proposition 19.52(ii) at $\rho = 0$, with weights. So the cross sum of $\mathcal{T}_3$ tends to 0 in mean square.

Step 3: the other sums and the conclusion of (i). The time sum, the two diagonal sums of $\mathcal{T}_3$, the gradient sums and the remainder are handled by Steps 2 to 5 of the proof of Theorem 19.30, with four changes. First, the pair $(W_{1,s}, W_{\perp,s})$ takes the place of W_s, and the box $[0,t] \times [-M_W, M_W]^2$, where M_W bounds both paths on $[0,t]$, takes the place of the rectangle. A continuous function on this box is uniformly continuous, by the Heine–Cantor argument written out in Step 2 of the proof of Theorem 19.30, with points of the box in place of points of the rectangle: if pairs of points of the box at distance at most $1/k$ in each coordinate had values differing by more than ε,

then a subsequence along which both points converge to one point of the box (taking convergent subsequences one coordinate at a time) would contradict continuity at that point. Second, in the diagonal sum in the increments of W_1, and in the one in the increments of $W_\perp$, the increment of the j-th cell is independent of the vector of values of both motions at times at most t_{j-1} (Step 1 of the proof of Theorem 19.23), and the moments are bounded uniformly in time by Lemma 19.20(iii), which covers values of two independent motions. Third, the gradient sums are the left-endpoint sums, against W_1 and against $W_\perp$, of $\partial_1 f(s, W_{1,s}, W_{\perp,s})$ and $\partial_2 f(s, W_{1,s}, W_{\perp,s})$, which are in $\mathcal{H}$ with two motions (Definition 19.22), so Theorem 19.23(i) and (vi) and the triangle inequality of Lemma 19.12(ii) apply. Fourth, let osc_n be the largest value, over the cells, over $u \in [0,1]$ and over the three derivatives $\partial_{11} f$, $\partial_{12} f$ and $\partial_{22} f$, of the absolute difference between the derivative at the point of the segment of Step 1 with parameter u and the same derivative at the left point of the cell. By the formula for χ_i'' and the inequality $2|y_1 y_2| \le y_1^2 + y_2^2$,

$$|\mathcal{T}_4| \le \tfrac{1}{2}\,\mathrm{osc}_n \sum_{i=1}^{n} \Big((\Delta W_{1,i})^2 + 2\,|\Delta W_{1,i}\,\Delta W_{\perp,i}| + (\Delta W_{\perp,i})^2 \Big) \le \mathrm{osc}_n \left(\mathrm{QV}_n^{(1)} + \mathrm{QV}_n^{(\perp)} \right),$$

where $\mathrm{QV}_n^{(1)}$ and $\mathrm{QV}_n^{(\perp)}$ are the quadratic-variation sums of W_1 and $W_\perp$. Each tends to t in mean square (Proposition 19.10(i)), and $\mathrm{osc}_n \to 0$ on every path, by the uniform continuity of the second derivatives on the box, proved in the first change, and that of the two paths on $[0,t]$ (Step 2 of the proof of Theorem 19.30). With these changes the arguments of Steps 2 to 5 of that proof give, as $\Delta t_{\max} \to 0$, the integrands being evaluated at $(s, W_{1,s}, W_{\perp,s})$,

$$\mathcal{T}_1 \to \int_0^t \partial_t f\, ds \ \text{ on every path}, \qquad \mathcal{T}_2 \to \int_0^t \partial_1 f\, dW_{1,s} + \int_0^t \partial_2 f\, dW_{\perp,s} \ \text{ in mean square},$$
$$\mathcal{T}_3 \to \tfrac{1}{2} \int_0^t (\partial_{11} f + \partial_{22} f)\, ds \ \text{ and } \ \mathcal{T}_4 \to 0 \ \text{ in probability},$$

where the limit of $\mathcal{T}_3$ uses Step 2 for its cross sum and the rule for sums of Lemma 19.12(iii). The conclusion is that of Step 6 of the same proof. Take a sequence of partitions of $[0,t]$ whose meshes tend to zero. By (E.17), $\mathcal{T}_2 = f(t, W_{1,t}, W_{\perp,t}) - f(0,0,0) - \mathcal{T}_1 - \mathcal{T}_3 - \mathcal{T}_4$; the right side converges in probability to

$$f(t, W_{1,t}, W_{\perp,t}) - f(0,0,0) - \int_0^t \Big(\partial_t f + \tfrac{1}{2}\,\partial_{11} f + \tfrac{1}{2}\,\partial_{22} f \Big)(s, W_{1,s}, W_{\perp,s})\, ds,$$

and the left side to the two stochastic integrals. Two limits in probability of one sequence agree with probability one (Lemma 19.12(iii)). This is (19.16).

Step 4: correlated motions, part (ii). Both paragraphs of this step use the inequality

$$|\rho\, y_1 + \sqrt{1-\rho^2}\, y_2| \le |y_1| + |y_2| \qquad \text{for all real } y_1, y_2, \tag{E.18}$$

which holds because $|\rho| \le 1$ and $\sqrt{1-\rho^2} \le 1$. By it, a function bounded by $c\, e^{c(|x_1|+|x_2|)}$, evaluated at $x_1 = y_1$ and $x_2 = \rho\, y_1 + \sqrt{1-\rho^2}\, y_2$, is bounded by $c\, e^{2c(|y_1|+|y_2|)}$, and a function bounded by $c\, e^{c|x|}$, evaluated at $x = \rho\, y_1 + \sqrt{1-\rho^2}\, y_2$, is bounded by $c\, e^{c(|y_1|+|y_2|)}$: exponential growth is kept.

The integral against W_2. Let ξ be in the span of $\mathcal{H}$ with two motions. On every partition, since $\Delta W_{2,i} = \rho\, \Delta W_{1,i} + \sqrt{1-\rho^2}\, \Delta W_{\perp,i}$,

$$\sum_{i=1}^{n} \xi_{t_{i-1}}\, \Delta W_{2,i} = \rho \sum_{i=1}^{n} \xi_{t_{i-1}}\, \Delta W_{1,i} + \sqrt{1-\rho^2} \sum_{i=1}^{n} \xi_{t_{i-1}}\, \Delta W_{\perp,i}.$$

The two sums on the right converge in mean square to $\int_0^t \xi_s\, dW_{1,s}$ and to $\int_0^t \xi_s\, dW_{\perp,s}$ (Theorem 19.23(i), (iv) and (vi)), so by the triangle inequality the left side converges in mean square to the integral against W_2 defined in the statement. That integral has mean zero because both of its parts have (Theorem 19.23(ii) and (vi)). If $\xi_s = \xi(s, W_{2,s})$ with $\xi(s,x)$ continuous and of

exponential growth, then ξ is a Brownian integrand for the motion W_2, and Theorem 19.23(i), applied to W_2, gives its integral as the mean-square limit of the same left-endpoint sums. As a function of the values of W_1 and $W_\perp$, $\xi_s = \xi(s, \rho\, W_{1,s} + \sqrt{1-\rho^2}\, W_{\perp,s})$ is continuous and of exponential growth as well, by (E.18), so it is in $\mathcal{H}$ with two motions. Two mean-square limits of one sequence agree with probability one, so the two integrals agree.

The formula. Put

$$\tilde{f}(t, y_1, y_2) = f(t,\ y_1,\ \rho\, y_1 + \sqrt{1-\rho^2}\, y_2),$$

so that $\tilde{f}(t, W_{1,t}, W_{\perp,t}) = f(t, W_{1,t}, W_{2,t})$ and $\tilde{f}(0,0,0) = f(0,0,0)$; the tilde marks the function f in transformed variables. In the variable y_2 only the last argument of f moves, and in the variable y_1 the last two arguments move along a line, so the chain rule of one variable and (E.16) give, with every derivative of f evaluated at $\bigl(t, y_1, \rho\, y_1 + \sqrt{1-\rho^2}\, y_2\bigr)$,

$$\begin{aligned}
&\partial_t \tilde{f} = \partial_t f, && \partial_1 \tilde{f} = \partial_1 f + \rho\, \partial_2 f, && \partial_2 \tilde{f} = \sqrt{1-\rho^2}\, \partial_2 f,\\
&\partial_{11} \tilde{f} = \partial_{11} f + 2\rho\, \partial_{12} f + \rho^2\, \partial_{22} f, && \partial_{22} \tilde{f} = (1-\rho^2)\, \partial_{22} f, && \partial_{12} \tilde{f} = \sqrt{1-\rho^2}\, (\partial_{12} f + \rho\, \partial_{22} f),
\end{aligned}$$

the mixed derivative of $\tilde{f}$ taking the same value in both orders. These functions are continuous, as combinations of continuous functions composed with a continuous map. They have exponential growth, by (E.18) and because sums and constant multiples keep exponential growth (Lemma 19.20(iii)). So $\tilde{f}$ satisfies the hypotheses of the theorem, and (i) applies to it. Its drift is

$$\begin{aligned}
\partial_t \tilde{f} + \tfrac{1}{2}\, \partial_{11} \tilde{f} + \tfrac{1}{2}\, \partial_{22} \tilde{f} &= \partial_t f + \tfrac{1}{2}\, \partial_{11} f + \rho\, \partial_{12} f + \tfrac{1}{2}\, \rho^2\, \partial_{22} f + \tfrac{1}{2}\, (1-\rho^2)\, \partial_{22} f\\
&= \partial_t f + \tfrac{1}{2}\, \partial_{11} f + \rho\, \partial_{12} f + \tfrac{1}{2}\, \partial_{22} f,
\end{aligned}$$

and by the linearity of the two integrals (Theorem 19.23(iv)) its stochastic integrals recombine as

$$\begin{aligned}
\int_0^t \partial_1 \tilde{f}\, dW_{1,s} + \int_0^t \partial_2 \tilde{f}\, dW_{\perp,s} &= \int_0^t \partial_1 f\, dW_{1,s} + \rho \int_0^t \partial_2 f\, dW_{1,s} + \sqrt{1-\rho^2} \int_0^t \partial_2 f\, dW_{\perp,s}\\
&= \int_0^t \partial_1 f\, dW_{1,s} + \int_0^t \partial_2 f\, dW_{2,s},
\end{aligned}$$

all derivatives of f being evaluated at $(s, W_{1,s}, W_{2,s})$. This is (19.17). The cases $\rho = \pm 1$ are included: then $W_2 = \pm W_1$ and the motion $W_\perp$ drops out.

Step 5: the process form, part (iii). The chain rule of calculus reduces (iii) to (ii), applied to the composite. Because X_k takes its values in $\mathcal{O}_k$, the composite $\tilde{V}(y_1, y_2, t) = V(X_1(t, y_1), X_2(t, y_2), t)$ is defined on $\mathbb{R}^2 \times [0, T]$. Write $x_k = X_k(t, y_k)$. The derivatives of the composite are computed as in the proof of Corollary 19.31, one coordinate at a time. In the variable y_1 only the first argument of V moves, in the variable y_2 only the second, and in the variable t the increment $\tilde{V}(y_1, y_2, t + \Delta t) - \tilde{V}(y_1, y_2, t)$ splits into three brackets, in each of which one argument of V moves; the mean value theorem on $\mathcal{O}_1$, on $\mathcal{O}_2$ and the definition of $\partial_t V$ treat them in turn. This gives

$$\begin{aligned}
&\partial_{y_1} \tilde{V} = \partial_1 V\ \partial_y X_1, && \partial_{y_1 y_1} \tilde{V} = \partial_{11} V\ (\partial_y X_1)^2 + \partial_1 V\ \partial_{yy} X_1, && \partial_{y_1 y_2} \tilde{V} = \partial_{12} V\ \partial_y X_1\ \partial_y X_2,\\
&\partial_t \tilde{V} = \partial_t V + \partial_1 V\ \partial_t X_1 + \partial_2 V\ \partial_t X_2,
\end{aligned}$$

and the same formulas with the indices exchanged for $\partial_{y_2} \tilde{V}$ and $\partial_{y_2 y_2} \tilde{V}$, with the derivatives of V at (x_1, x_2, t), those of X_k at (t, y_k), and one-sided derivatives in t at 0 and at T. Differentiating $\partial_{y_2} \tilde{V}$ in y_1 gives $\partial_1 \partial_2 V\ \partial_y X_2\ \partial_y X_1$, the same function by Lemma 19.54. All these derivatives are sums and products of continuous functions composed with the continuous map $(y_1, y_2, t) \mapsto (X_1(t, y_1), X_2(t, y_2), t)$, so they are continuous, and by hypothesis they and $\tilde{V}$ have exponential growth. The function $(t, y_1, y_2) \mapsto \tilde{V}(y_1, y_2, t)$ therefore satisfies the hypotheses of the theorem.

The formula. Apply (ii) to this function. At $y_k = W_{k,s}$ and $t = s$, where $x_k = X_{k,s}$, the definitions of $\alpha_{k,s}$ and $\beta_{k,s}$ and the formulas above give

$$\begin{aligned}\partial_t\tilde V + \tfrac12\,\partial_{y_1y_1}\tilde V + \rho\,\partial_{y_1y_2}\tilde V + \tfrac12\,\partial_{y_2y_2}\tilde V &= \partial_t V + \partial_1 V\left(\partial_t X_1 + \tfrac12\,\partial_{yy}X_1\right) + \partial_2 V\left(\partial_t X_2 + \tfrac12\,\partial_{yy}X_2\right)\\ &\quad + \tfrac12\,\partial_{11}V\,(\partial_y X_1)^2 + \rho\,\partial_{12}V\,\partial_y X_1\,\partial_y X_2 + \tfrac12\,\partial_{22}V\,(\partial_y X_2)^2\\ &= \partial_t V + \alpha_{1,s}\,\partial_1 V + \alpha_{2,s}\,\partial_2 V\\ &\quad + \tfrac12\,\beta_{1,s}^2\,\partial_{11}V + \rho\,\beta_{1,s}\beta_{2,s}\,\partial_{12}V + \tfrac12\,\beta_{2,s}^2\,\partial_{22}V,\end{aligned}$$

and $\partial_{y_k}\tilde V = \beta_{k,s}\,\partial_k V$. Since $\tilde V(W_{1,t}, W_{2,t}, t) = V(X_{1,t}, X_{2,t}, t)$ and $\tilde V(0,0,0) = V(X_{1,0}, X_{2,0}, 0)$, identity (19.17) for this function is (19.18). The equation $dX_2 = \alpha_{2,t}\,dt + \beta_{2,t}\,dW_2$ of the statement comes from Theorem 19.30(i) for the Brownian motion W_2 (Proposition 19.52(i)), and its stochastic integral is the integral against W_2 of (ii), by the agreement proved in Step 4.

Part (iii)(a). Let $X_k(t,y) = S_{k,0}\,e^{g_k t + \sigma_k y}$. By Part (iii) of the proof of Corollary 19.31, X_k is in $C_E^{1,2}([0,T]\times\mathbb{R})$ with values in $(0,\infty)$, with $\partial_t X_k = g_k X_k$, $\partial_y X_k = \sigma_k X_k$ and $\partial_{yy}X_k = \sigma_k^2 X_k$; so $\alpha_{k,t} = (g_k + \sigma_k^2/2)\,S_{k,t} = \mu_k S_{k,t}$ and $\beta_{k,t} = \sigma_k S_{k,t}$. With $x_k = X_k(t,y_k)$ the derivatives of the composite read

$$\begin{aligned}\partial_t\tilde V &= \partial_t V + g_1x_1\,\partial_1 V + g_2x_2\,\partial_2 V, & \partial_{y_k}\tilde V &= \sigma_k x_k\,\partial_k V,\\ \partial_{y_ky_k}\tilde V &= \sigma_k^2(x_k^2\,\partial_{kk}V + x_k\,\partial_k V), & \partial_{y_1y_2}\tilde V &= \sigma_1\sigma_2\,x_1x_2\,\partial_{12}V.\end{aligned}$$

The functions x_k and x_k^2 have moderate growth, and sums and products of functions of moderate growth have moderate growth (Lemma 19.20(i)). So if V and its six derivatives have moderate growth, then V and the right sides are functions of (x_1,x_2,t) of moderate growth, and Lemma 19.20(i), applied coordinate by coordinate, turns moderate growth in the two prices into exponential growth in (y_1,y_2). This is the hypothesis on the composite.

Part (iii)(b). The function $V(x_1,x_2,t) = x_1x_2$ is continuous with $\partial_1 V = x_2$, $\partial_2 V = x_1$, $\partial_{12}V = 1$ and $\partial_t V = \partial_{11}V = \partial_{22}V = 0$. Its composite is $X_1(t,y_1)\,X_2(t,y_2)$, and each derivative of the composite listed in the statement is a sum of products of one of the functions X_1, $\partial_t X_1$, $\partial_y X_1$, $\partial_{yy}X_1$ of (t,y_1) with one of the corresponding functions of (t,y_2). These factors have exponential growth because $X_k \in C_E^{1,2}$, and sums and products of functions of exponential growth have exponential growth (Lemma 19.20(iii)). So the hypothesis on the composite holds, and (19.18) is the product rule.

Step 6: every correlated pair, part (iv). For $|\rho| < 1$, Proposition 19.52(iii) gives the stated $W_\perp$ and the identity $W_2 = \rho\,W_1 + \sqrt{1-\rho^2}\,W_\perp$, path by path, so (ii) and (iii) apply as proved. Let $\rho = \pm1$. For each $s \ge 0$, Proposition 10.10 and Definition 19.51(1) on the interval $[0,s]$ give $\operatorname{Var}(W_{2,s} - \rho\,W_{1,s}) = s + \rho^2 s - 2\rho\cdot\rho\,s = 0$, so $W_{2,s} = \rho\,W_{1,s}$ with probability one, by Chebyshev's inequality and a countable union (Lemma 19.12(i) and (0)). The rational times $s \ge 0$ are countably many, so by countable subadditivity (Lemma 19.12(0)) there is one event of probability one on which $W_{2,s} = \rho\,W_{1,s}$ for every rational $s \ge 0$. On it the continuous paths W_2 and $\rho\,W_1$ agree at every $s \ge 0$, each real s being a limit of rational times. On this event $\xi_s = \xi(s, W_{1,s}, \rho\,W_{1,s})$, whose generating function is continuous and of exponential growth in $W_{1,s}$ because $|\rho\,x| = |x|$, so ξ is a Brownian integrand for W_1; and $\sum_i \xi_{t_{i-1}}\,\Delta W_{2,i} = \rho\sum_i \xi_{t_{i-1}}\,\Delta W_{1,i}$, which converges in mean square to $\rho\int_0^t \xi_s\,dW_{1,s}$ (Theorem 19.23(i)). For the formula, put $f_\rho(t,x) = f(t,x,\rho\,x)$, the subscript being a label. By (E.16) with the increments $(1,\rho)$, applied to $f(t,\cdot,\cdot)$ and then to $\partial_1 f(t,\cdot,\cdot)$ and $\partial_2 f(t,\cdot,\cdot)$, and by Lemma 19.54,

$$\partial_x f_\rho = \partial_1 f + \rho\,\partial_2 f, \qquad \partial_{xx} f_\rho = \partial_{11} f + 2\rho\,\partial_{12} f + \rho^2\,\partial_{22} f,$$

at $(t,x,\rho\,x)$, and $\partial_t f_\rho = \partial_t f$ there. These functions are continuous and of exponential growth, so $f_\rho \in C_E^{1,2}([0,T]\times\mathbb{R})$. Theorem 19.30(i) for f_ρ, with $f_\rho(t,W_{1,t}) = f(t,W_{1,t},W_{2,t})$ on the event above, $\rho^2 = 1$ and the linearity of the integral, is (19.17) with $\int_0^t \partial_2 f\,dW_{2,s} = \rho\int_0^t \partial_2 f\,dW_{1,s}$. Part (iii) follows from this as in Step 5; the integral of Theorem 19.23 built on W_2 alone is the mean-square limit of the same left-endpoint sums, so it agrees with this one. □

E.13 Proof of Proposition 19.57: Volatility of a ratio and of a product of geometric Brownian motions

The statement is on page 303. The notation and the standing assumptions are those of §19.8.

Proof. We write the logarithm of the ratio explicitly and identify the process that drives it. By Definition 19.27, in the form of Corollary 19.34(i) for each of the two prices,

$$\ln \frac{S_{1,t}}{S_{2,t}} = \ln \frac{S_{1,0}}{S_{2,0}} + (g_1 - g_2)\, t + \sigma_1 W_{1,t} - \sigma_2 W_{2,t}.$$

The driving process. The process $t \mapsto \sigma_1 W_{1,t} - \sigma_2 W_{2,t}$ starts at zero and has continuous paths. Its increment over $[s,t]$ is the combination $\sigma_1(W_{1,t} - W_{1,s}) - \sigma_2(W_{2,t} - W_{2,s})$ of a jointly normal pair, so it is normal (Lemma 19.38(ii)), with mean zero. By Proposition 10.10, applied to $\sigma_1(W_{1,t} - W_{1,s})$ and $-\sigma_2(W_{2,t} - W_{2,s})$, whose covariance is $-\sigma_1\sigma_2\,\rho\,(t-s)$, its variance is

$$(\sigma_1^2 + \sigma_2^2 - 2\rho\,\sigma_1\sigma_2)\,(t-s) = \sigma^2\,(t-s).$$

This is the quadratic that appears as the leading coefficient in Proposition 11.4(i), the variance of a difference of two returns. Over the cells of a grid the increments are functions of the pairs $(\Delta W_{1,i}, \Delta W_{2,i})$, which are mutually independent (Definition 19.51(2)), so they are mutually independent. Moreover

$$\sigma^2 = (\sigma_1 - \sigma_2)^2 + 2\,(1-\rho)\,\sigma_1\sigma_2$$

is a sum of two nonnegative terms, because $\rho \le 1$ and $\sigma_1\sigma_2 > 0$. So $\sigma = 0$ exactly when both terms vanish, that is, when $\sigma_1 = \sigma_2$ and $\rho = 1$.

The case $\sigma > 0$. Put $\tilde{W}_t = (\sigma_1 W_{1,t} - \sigma_2 W_{2,t})/\sigma$. By the previous paragraph $\tilde{W}$ starts at zero, has continuous paths, and has independent increments with the law $\mathcal{N}(0, t-s)$, by (19.3): it is a standard Brownian motion (Definition 19.5). Hence

$$\frac{S_{1,t}}{S_{2,t}} = \frac{S_{1,0}}{S_{2,0}} \exp\big((g_1 - g_2)\,t + \sigma\,\tilde{W}_t\big),$$

which is the process (19.10) of Definition 19.27, driven by $\tilde{W}$, with volatility σ and with the drift μ' determined by $\mu' - \sigma^2/2 = g_1 - g_2$. Expanding,

$$\mu' = g_1 - g_2 + \tfrac{1}{2}\,\sigma^2 = \mu_1 - \tfrac{1}{2}\,\sigma_1^2 - \mu_2 + \tfrac{1}{2}\,\sigma_2^2 + \tfrac{1}{2}\,\sigma_1^2 + \tfrac{1}{2}\,\sigma_2^2 - \rho\,\sigma_1\sigma_2 = \mu_1 - \mu_2 + \sigma_2^2 - \rho\,\sigma_1\sigma_2.$$

The case $\sigma = 0$. For each t the variable $\sigma_1 W_{1,t} - \sigma_2 W_{2,t}$ has mean zero and variance zero. By Chebyshev's inequality (Lemma 19.12(i)) it exceeds $1/k$ in absolute value with probability zero, for every $k = 1, 2, \ldots$, so by a countable union it vanishes with probability one (Lemma 19.12(0)). Since $\sigma_1 = \sigma_2$ in this case, $g_1 - g_2 = \mu_1 - \mu_2$, and the ratio equals $(S_{1,0}/S_{2,0})\,e^{(\mu_1 - \mu_2)t}$ with probability one.

The product. The proof is the same with a plus sign. Now $\ln(S_{1,t}S_{2,t}) = \ln(S_{1,0}S_{2,0}) + (g_1 + g_2)\,t + \sigma_1 W_{1,t} + \sigma_2 W_{2,t}$, the increments of $\sigma_1 W_1 + \sigma_2 W_2$ are independent and normal with variance $(\sigma_1^2 + \sigma_2^2 + 2\rho\,\sigma_1\sigma_2)(t-s)$, and

$$\sigma_1^2 + \sigma_2^2 + 2\rho\,\sigma_1\sigma_2 = (\sigma_1 - \sigma_2)^2 + 2\,(1+\rho)\,\sigma_1\sigma_2$$

vanishes exactly when $\sigma_1 = \sigma_2$ and $\rho = -1$. If it is positive, dividing $\sigma_1 W_1 + \sigma_2 W_2$ by its square root gives a standard Brownian motion, and the product is a geometric Brownian motion with that volatility and with drift

$$g_1 + g_2 + \tfrac{1}{2}(\sigma_1^2 + \sigma_2^2 + 2\rho\,\sigma_1\sigma_2) = \mu_1 + \mu_2 + \rho\,\sigma_1\sigma_2.$$

If it vanishes, $\sigma_1 W_{1,t} + \sigma_2 W_{2,t} = 0$ with probability one for each t, and $g_1 + g_2 = \mu_1 + \mu_2 - \sigma_1^2$ because $\sigma_1 = \sigma_2$. □

E.14 Proof of Lemma 19.61: Gaussian smoothing solves the backward equation

The statement is on page 306. The notation and the standing assumptions are those of §19.9.

Proof. We write Ψ as an integral against a Gaussian kernel, check the equation on the kernel, and justify differentiation under the integral sign with Taylor's theorem and with bounds on the kernel that are uniform near a fixed point; no convergence theorem for integrals is used. The proof has six steps. Throughout, $\tau = T - t$, and c' and c'' are constants whose values may change from one line to the next. The expectation that defines Ψ makes sense for every $\tau > 0$, that is, for every $t < T$, negative t included. Steps 1 to 4 treat all $\tau > 0$, so the derivatives of Ψ at $t = 0$ are ordinary two-sided derivatives.

Step 1: the kernel form. For $\tau > 0$ and real x and z,

$$\bigl|\psi(x + \nu\tau + \sigma\sqrt{\tau}\, z)\bigr| \le c\, e^{c|x| + c|\nu|\tau}\, e^{c\sigma\sqrt{\tau}\,|z|},$$

and $\mathbb{E}\, e^{c\sigma\sqrt{\tau}\,|Z|}$ is finite by Lemma 18.35(i). So the integral below converges absolutely, and Fact 7.22 gives $\Psi(t,x) = \int \psi(x + \nu\tau + \sigma\sqrt{\tau}\, z)\, \varphi(z)\, dz$. The substitution $y = x + \nu\tau + \sigma\sqrt{\tau}\, z$ turns this into

$$\Psi(t,x) = \int \psi(y)\, \varphi_{\sigma\sqrt{\tau}}(y - x - \nu\tau)\, dy, \tag{E.19}$$

where $\varphi_s(w) = (2\pi s^2)^{-1/2}\, e^{-w^2/(2s^2)}$ is the density of $\mathcal{N}(0, s^2)$. The *kernel* is the function

$$(\tau, w) \mapsto \varphi_{\sigma\sqrt{\tau}}(w - \nu\tau) = \frac{1}{\sigma\sqrt{2\pi\tau}}\, \exp\Bigl(-\frac{(w - \nu\tau)^2}{2\sigma^2\tau}\Bigr), \qquad \tau > 0,\ w \in \mathbb{R};$$

for fixed τ it is the density of $\mathcal{N}(\nu\tau, \sigma^2\tau)$ at w. It is built from polynomials, powers of τ and the exponential function, so it has continuous partial derivatives of every order, and by Lemma 19.54 the order of the differentiations does not matter. For integers $i, j \ge 0$ write $\varphi^{(i,j)}(\tau, w)$ for the kernel differentiated i times in τ and j times in w, so that $\varphi^{(0,0)}(\tau, w) = \varphi_{\sigma\sqrt{\tau}}(w - \nu\tau)$. The superscript counts derivatives, as in the $f^{(k)}$ of §7.5, and the kernel keeps the letter of the normal density.

Step 2: the kernel solves an equation. The logarithm of the kernel is $-\frac{1}{2}\ln(2\pi\sigma^2\tau) - (w - \nu\tau)^2/(2\sigma^2\tau)$, and the derivative of a positive function is the function times the derivative of its logarithm. Differentiating the logarithm in w gives the first line below, differentiating the first line once more in w gives the second, and differentiating the logarithm in τ gives the third:

$$\begin{aligned}
\varphi^{(0,1)} &= -\frac{w - \nu\tau}{\sigma^2\tau}\, \varphi^{(0,0)}, \\
\varphi^{(0,2)} &= \Bigl(\frac{(w - \nu\tau)^2}{\sigma^4\tau^2} - \frac{1}{\sigma^2\tau}\Bigr)\, \varphi^{(0,0)}, \\
\varphi^{(1,0)} &= \Bigl(-\frac{1}{2\tau} + \frac{\nu\,(w - \nu\tau)}{\sigma^2\tau} + \frac{(w - \nu\tau)^2}{2\sigma^2\tau^2}\Bigr)\, \varphi^{(0,0)}.
\end{aligned}$$

The bracket of the third line is $-\nu$ times the factor of the first line plus $\sigma^2/2$ times the bracket of the second. Hence

$$\varphi^{(1,0)} = -\nu\, \varphi^{(0,1)} + \tfrac{1}{2}\, \sigma^2\, \varphi^{(0,2)} \qquad \text{for all } \tau > 0 \text{ and real } w. \tag{E.20}$$

Step 3: kernel derivatives and their envelope. Every $\varphi^{(i,j)}$ is the kernel $\varphi^{(0,0)}$ times a polynomial in w, τ and $1/\tau$, with coefficients that depend on ν and σ only. This holds for $\varphi^{(0,0)}$, with the polynomial 1. If it holds for $\varphi^{(i,j)}$, differentiate the product in w or in τ: the derivative of the polynomial is again such a polynomial, because the derivative of $(1/\tau)^m$ in τ is $-m\,(1/\tau)^{m+1}$, and the derivative of the kernel is the kernel times such a polynomial, by the first and the third line of Step 2. Induction on $i + j$ proves the claim.

Now fix $\tau_0 > 0$ and a real number x_0, and consider the times $\tau \in [\tau_0/2, 2\tau_0]$ and the points x with $|x - x_0| \le 1$. For real z_1 and z_2, $(z_1 - z_2)^2 \ge z_1^2/2 - z_2^2$, because the difference of the two sides is $(z_1 - 2z_2)^2/2$. With $z_1 = y - x_0$ and $z_2 = x - x_0 + \nu\tau$, for which $|z_2| \le 1 + 2|\nu|\tau_0$,

$$(y - x - \nu\tau)^2 \ge \tfrac{1}{2}\,(y - x_0)^2 - \big(1 + 2|\nu|\tau_0\big)^2 \qquad \text{for all real } y.$$

On the range considered $1/(2\sigma^2\tau) \ge 1/(4\sigma^2\tau_0)$ and $(2\pi\sigma^2\tau)^{-1/2} \le (\pi\sigma^2\tau_0)^{-1/2}$, and these two bounds give

$$\varphi^{(0,0)}(\tau, y - x) \le \frac{1}{\sigma\sqrt{\pi\tau_0}}\, \exp\Big(\frac{(1 + 2|\nu|\tau_0)^2}{4\sigma^2\tau_0}\Big)\, \exp\Big(-\frac{(y - x_0)^2}{8\sigma^2\tau_0}\Big).$$

The prefactors are compared by the second bound. For the exponents, the exponent on the right is the bracket $\frac{1}{2}(y - x_0)^2 - (1 + 2|\nu|\tau_0)^2$ multiplied by $-1/(4\sigma^2\tau_0)$, and two cases arise. When the bracket is negative, the exponent of the kernel, $-(y - x - \nu\tau)^2/(2\sigma^2\tau)$, is at most 0, while the exponent on the right is positive. When the bracket is nonnegative, the inequality above and the first bound, applied to this nonnegative quantity, give

$$\frac{(y - x - \nu\tau)^2}{2\sigma^2\tau} \ge \frac{\frac{1}{2}(y - x_0)^2 - (1 + 2|\nu|\tau_0)^2}{2\sigma^2\tau} \ge \frac{\frac{1}{2}(y - x_0)^2 - (1 + 2|\nu|\tau_0)^2}{4\sigma^2\tau_0},$$

which is the comparison of the exponents after a change of sign. The polynomial factor of $\varphi^{(i,j)}$, evaluated at $w = y - x$, is at most a constant times $1 + |y - x_0|^m$, where m is its degree in w, because $|y - x| \le |y - x_0| + 1$ and because τ and $1/\tau$ are bounded on the range considered. The function $z \mapsto (1 + |z|^m)\, e^{-z^2/(16\sigma^2\tau_0)}$ is bounded on the real line: it is continuous and tends to zero at infinity (the pointwise bound in the proof of Corollary 7.18). Writing $e^{-(y-x_0)^2/(8\sigma^2\tau_0)}$ as the square of $e^{-(y-x_0)^2/(16\sigma^2\tau_0)}$ and using one of the two factors to absorb the polynomial,

$$|\varphi^{(i,j)}(\tau, y - x)| \le c'\, \exp\Big(-\frac{(y - x_0)^2}{16\sigma^2\tau_0}\Big) \tag{E.21}$$

for all real y, all $\tau \in [\tau_0/2, 2\tau_0]$ and all x with $|x - x_0| \le 1$. The constant c' depends on i, j, τ_0 and x_0, but not on y, x or τ. The right side is an *envelope* centred at the fixed point x_0, not at the variable x, which is what Step 4 needs.

The envelope is integrable against ψ. Since $|\psi(y)| \le c\, e^{c|x_0|}\, e^{c|y - x_0|}$, and $c\,|z| \le z^2/(32\sigma^2\tau_0) + 8\sigma^2\tau_0\, c^2$ for real z (the difference of the two sides is a square),

$$|\psi(y)|\, \exp\Big(-\frac{(y - x_0)^2}{16\sigma^2\tau_0}\Big) \le c\, e^{c|x_0| + 8\sigma^2\tau_0 c^2}\, \exp\Big(-\frac{(y - x_0)^2}{32\sigma^2\tau_0}\Big).$$

The right side is a constant multiple of the density of $\mathcal{N}(x_0, 16\sigma^2\tau_0)$, whose integral over the line is 1 (Example 7.6, after a shift and a change of scale). By the direct comparison test (Proposition 7.16), applied on each of the finitely many intervals on which ψ is continuous,

$$\int |\psi(y)|\, \exp\Big(-\frac{(y - x_0)^2}{16\sigma^2\tau_0}\Big)\, dy < \infty. \tag{E.22}$$

Step 4: differentiation under the integral sign. For integers $i, j \ge 0$ consider the function

$$(\tau, x) \mapsto \int \psi(y)\, \varphi^{(i,j)}(\tau, y - x)\, dy, \qquad \tau > 0,\ x \in \mathbb{R},$$

whose integral converges absolutely by (E.21) and (E.22). We claim that it is continuous, that it has partial derivatives in τ and in x, and that these are obtained by differentiating the kernel under the integral sign: its derivative in τ is $\int \psi(y)\, \varphi^{(i+1,j)}(\tau, y - x)\, dy$, and its derivative in x is $-\int \psi(y)\, \varphi^{(i,j+1)}(\tau, y - x)\, dy$.

The derivative in x. Fix (τ_0, x_0) and let $0 < |\Delta x| \le 1$. For each real y the function $x \mapsto \varphi^{(i,j)}(\tau_0, y-x)$ of one variable has the first derivative $-\varphi^{(i,j+1)}(\tau_0, y-x)$ and the second derivative $\varphi^{(i,j+2)}(\tau_0, y-x)$, both continuous. Corollary 7.32 at order one gives a point x^* between x_0 and $x_0 + \Delta x$, which depends on y, with

$$\varphi^{(i,j)}(\tau_0, y - x_0 - \Delta x) = \varphi^{(i,j)}(\tau_0, y - x_0) - \Delta x\, \varphi^{(i,j+1)}(\tau_0, y - x_0) + \tfrac{1}{2}\,(\Delta x)^2\, \varphi^{(i,j+2)}(\tau_0, y - x^*).$$

Since $|x^* - x_0| \le 1$, the envelope (E.21) of $\varphi^{(i,j+2)}$ bounds the last term, in absolute value, by $\frac{1}{2}\,(\Delta x)^2$ times the right side of (E.21). Multiply by $\psi(y)$, integrate over y and divide by Δx. The difference quotient

$$\frac{1}{\Delta x}\Big(\int \psi(y)\, \varphi^{(i,j)}(\tau_0, y - x_0 - \Delta x)\, dy - \int \psi(y)\, \varphi^{(i,j)}(\tau_0, y - x_0)\, dy \Big)$$

of the function of the claim differs from $-\int \psi(y)\, \varphi^{(i,j+1)}(\tau_0, y - x_0)\, dy$ by at most

$$\tfrac{1}{2}\,|\Delta x|\; c' \int |\psi(y)|\, \exp\Big(-\frac{(y-x_0)^2}{16\sigma^2\tau_0} \Big)\, dy$$

in absolute value. This bound tends to zero with Δx, by (E.22). So the difference quotient in x converges, with an error of at most a constant times $|\Delta x|$, and its limit is the claimed derivative at (τ_0, x_0).

The derivative in τ. Let $0 < |\Delta\tau| \le \tau_0/2$. For each real y the function $\tau \mapsto \varphi^{(i,j)}(\tau, y - x_0)$ is defined on the open interval $(0, \infty)$, and Corollary 7.32 at order one gives a point τ^* between τ_0 and $\tau_0 + \Delta\tau$, hence in $[\tau_0/2, 2\tau_0]$, with

$$\varphi^{(i,j)}(\tau_0 + \Delta\tau, y - x_0) = \varphi^{(i,j)}(\tau_0, y - x_0) + \Delta\tau\, \varphi^{(i+1,j)}(\tau_0, y - x_0) + \tfrac{1}{2}\,(\Delta\tau)^2\, \varphi^{(i+2,j)}(\tau^*, y - x_0).$$

The envelope of $\varphi^{(i+2,j)}$ bounds the last term, and the argument just given, with $\Delta\tau$ in place of Δx, gives the derivative in τ.

Continuity. Let $|\tau - \tau_0| \le \tau_0/2$ and $|x - x_0| \le 1$. The mean value theorem (Corollary 7.32 at order zero), applied in τ and then in x, gives points τ^* between τ_0 and τ and x^* between x_0 and x with

$$\varphi^{(i,j)}(\tau, y - x) - \varphi^{(i,j)}(\tau_0, y - x_0) = (\tau - \tau_0)\, \varphi^{(i+1,j)}(\tau^*, y - x) - (x - x_0)\, \varphi^{(i,j+1)}(\tau_0, y - x^*).$$

By the envelopes of $\varphi^{(i+1,j)}$ and $\varphi^{(i,j+1)}$, the right side is at most $|\tau - \tau_0| + |x - x_0|$ times the right side of (E.21) in absolute value. Integrating against $|\psi|$ shows that the function of the claim changes by at most a constant times $|\tau - \tau_0| + |x - x_0|$ near (τ_0, x_0). It is continuous there.

Conclusion: (i) without its bound, and (ii). By (E.19), $(\tau, x) \mapsto \Psi(T - \tau, x)$ is the function of the claim with $i = j = 0$. By induction on the number of differentiations it has partial derivatives of every order, all continuous, and its derivative of order i in τ and j in x is

$$(-1)^j \int \psi(y)\, \varphi^{(i,j)}(\tau, y - x)\, dy,$$

in whatever order the differentiations are performed. Since $t = T - \tau$, a derivative in t is minus a derivative in τ. So Ψ is infinitely differentiable for $t < T$, in particular on $[0, T) \times \mathbb{R}$, and

$$\partial_t \Psi = -\int \psi(y)\, \varphi^{(1,0)}\, dy, \qquad \partial_x \Psi = -\int \psi(y)\, \varphi^{(0,1)}\, dy, \qquad \partial_{xx} \Psi = \int \psi(y)\, \varphi^{(0,2)}\, dy,$$

the kernel derivatives being evaluated at $(\tau, y - x)$. By (E.20),

$$\partial_t \Psi + \nu\, \partial_x \Psi + \tfrac{1}{2}\,\sigma^2\, \partial_{xx} \Psi = \int \psi(y) \Big(-\varphi^{(1,0)} - \nu\, \varphi^{(0,1)} + \tfrac{1}{2}\,\sigma^2\, \varphi^{(0,2)} \Big)(\tau, y - x)\, dy = 0,$$

which is (19.19).

Step 5: the growth bounds. Undo the substitution of Step 1 in the integral $(-1)^j \int \psi(y)\,\varphi^{(i,j)}(\tau, y-x)\,dy$. With $y = x + \nu\tau + \sigma\sqrt{\tau}\,z$ the argument of the kernel is $w = y - x = \nu\tau + \sigma\sqrt{\tau}\,z$, and $\varphi^{(0,0)}(\tau, w)\,dy = \varphi(z)\,dz$. By Step 3 the factor that multiplies the kernel in $\varphi^{(i,j)}$ is a polynomial in w, τ and $1/\tau$. At $w = \nu\tau + \sigma\sqrt{\tau}\,z$ it becomes a polynomial $\chi(z)$ in z whose coefficients are polynomials in $\sqrt{\tau}$ and $1/\sqrt{\tau}$: continuous functions of $\tau > 0$ that do not depend on ψ or on x. Hence every partial derivative of Ψ in (t, x) has the form

$$\pm\,\mathbb{E}\Big[\psi(x + \nu\tau + \sigma\sqrt{\tau}\,Z)\,\chi(Z)\Big]. \tag{E.23}$$

For example, the first two lines of Step 2, with $w - \nu\tau = \sigma\sqrt{\tau}\,z$, give

$$\partial_x \Psi = \frac{\mathbb{E}\big[\psi(x + \nu\tau + \sigma\sqrt{\tau}\,Z)\,Z\big]}{\sigma\sqrt{\tau}}, \qquad \partial_{xx}\Psi = \frac{\mathbb{E}\big[\psi(x + \nu\tau + \sigma\sqrt{\tau}\,Z)\,(Z^2 - 1)\big]}{\sigma^2\tau}.$$

Let $T' < T$ and $0 \le t \le T'$, so that $T - T' \le \tau \le T$. Then $|\psi(x+\nu\tau+\sigma\sqrt{\tau}\,z)| \le c\,e^{c|x|}\,e^{c|\nu|T}\,e^{c\sigma\sqrt{T}\,|z|}$. The coefficients of χ are bounded on the closed interval $[T - T', T]$, and each power obeys $|z|^k \le k!\,e^{|z|}$ (one term of the exponential series), so $|\chi(z)| \le c'\,e^{|z|}$ for a constant c'. By (E.23), each partial derivative of Ψ is therefore at most

$$c\,c'\,e^{c|\nu|T}\;\mathbb{E}\,e^{(c\sigma\sqrt{T}+1)\,|Z|}\;e^{c|x|}$$

in absolute value on $[0, T'] \times \mathbb{R}$, and the expectation is finite by Lemma 18.35(i). This is the bound of (i), with the exponent c of ψ. The same computation without χ gives, for $0 \le t < T$,

$$|\Psi(t,x)| \le c\,e^{c|\nu|T}\;\mathbb{E}\,e^{c\sigma\sqrt{T}\,|Z|}\;e^{c|x|} \le 2c\,e^{c|\nu|T + c^2\sigma^2 T/2}\;e^{c|x|},$$

and $|\Psi(T,x)| = |\psi(x)| \le c\,e^{c|x|}$. This is the bound of (iii), with $c'' = 2c\,e^{c|\nu|T + c^2\sigma^2 T/2}$.

Step 6: continuity at the horizon. Let ψ be continuous at x^* and let $\varepsilon > 0$. Choose $\varepsilon' > 0$ with $|\psi(y) - \psi(x^*)| < \varepsilon$ whenever $|y - x^*| < \varepsilon'$. At the horizon itself, $\Psi(T, x) = \psi(x)$ is within ε of $\psi(x^*)$ as soon as $|x - x^*| < \varepsilon'$. Let now $t < T$ and $|x - x^*| \le 1$, write $Y = x + \nu\tau + \sigma\sqrt{\tau}\,Z$ (a name local to this step), and let $A_{t,x}$ be the event $\{|Y - x^*| \ge \varepsilon'\}$. Off this event $|\psi(Y) - \psi(x^*)| < \varepsilon$, and on it $|\psi(Y) - \psi(x^*)| \le |\psi(Y)| + |\psi(x^*)|$. By the monotonicity of expectation and the Cauchy–Schwarz inequality (Lemma 19.12(ii)), applied to $|\psi(Y)| + |\psi(x^*)|$ and to the indicator of $A_{t,x}$,

$$|\Psi(t,x) - \psi(x^*)| \le \mathbb{E}\,|\psi(Y) - \psi(x^*)| \le \varepsilon + \Big(\mathbb{E}\big[\big(|\psi(Y)| + |\psi(x^*)|\big)^2\big]\Big)^{1/2} P(A_{t,x})^{1/2}.$$

The first factor is bounded by a constant c' that does not depend on (t, x): since $|x| \le |x^*| + 1$ and $\tau \le T$, $|\psi(Y)| \le c\,e^{c(|x^*|+1) + c|\nu|T}\,e^{c\sigma\sqrt{T}\,|Z|}$, whose square has a finite mean by Lemma 18.35(i). For the second factor, suppose that $|x - x^*| + |\nu|\tau < \varepsilon'/2$. Then $|Y - x^*| < \varepsilon'/2 + \sigma\sqrt{\tau}\,|Z|$, so $A_{t,x}$ is contained in the event $\{\sigma\sqrt{\tau}\,|Z| \ge \varepsilon'/2\}$, and Chebyshev's inequality (Lemma 19.12(i)) gives

$$P(A_{t,x}) \le \frac{\sigma^2\tau\;\mathbb{E}[Z^2]}{(\varepsilon'/2)^2} = \frac{4\sigma^2\tau}{\varepsilon'^2}.$$

Hence $|\Psi(t,x) - \psi(x^*)| \le \varepsilon + 2c'\sigma\sqrt{\tau}/\varepsilon'$, which is at most 2ε once $\tau = T - t$ and $|x - x^*|$ are small enough. Since ε was arbitrary, $\Psi(t,x) \to \psi(x^*)$ as $(t,x) \to (T, x^*)$. □

E.15 Proof of Theorem 19.63: The reflection principle with drift (Bachelier, 1900, for zero drift)

The statement is on page 307. The notation and the standing assumptions are those of §19.9.

Proof. We use the method of images, with the stopped clause of Theorem 19.30. The proof has six steps. Steps 1 to 4 prove (a) for a fixed ψ; write $\sup|\psi|$ for a bound on $|\psi|$.

Step 1: the image function. For $t < T$ let

$$\Psi(t,x) = \mathbb{E}\,\psi\big(x + \nu(T-t) + \sqrt{T-t}\,Z\big), \qquad Z \sim \mathcal{N}(0,1),$$

and $\Psi(T,x) = \psi(x)$: the smoothing of Lemma 19.61 with the drift ν and $\sigma = 1$. Its growth hypothesis holds, because a bounded function satisfies $|\psi(y)| \le c \le c\,e^{c|y|}$ with $c = \sup|\psi|$. So Ψ is infinitely differentiable for $t < T$ (Lemma 19.61(i)), it solves $\partial_t\Psi + \nu\,\partial_x\Psi + \frac{1}{2}\,\partial_{xx}\Psi = 0$ on $[0,T)\times\mathbb{R}$ (part (ii)), and $|\Psi| \le \sup|\psi|$ by the monotonicity of expectation. For $t < T$ and real x put

$$\Psi^{\mathrm{im}}(t,x) = e^{-2\nu(x-\ell)}\,\Psi(t, 2\ell - x), \qquad \Psi^{\mathrm{out}} = \Psi - \Psi^{\mathrm{im}}.$$

The function Ψ^{im} is the *image* of Ψ: the same function seen in the mirror $x \mapsto 2\ell - x$ placed at the level, with an exponential weight. By the chain and product rules of one variable, with Ψ and its derivatives evaluated at $(t, 2\ell - x)$,

$$\begin{aligned}
\partial_t\Psi^{\mathrm{im}} &= e^{-2\nu(x-\ell)}\,\partial_t\Psi,\\
\partial_x\Psi^{\mathrm{im}} &= -e^{-2\nu(x-\ell)}\,(2\nu\,\Psi + \partial_x\Psi),\\
\partial_{xx}\Psi^{\mathrm{im}} &= e^{-2\nu(x-\ell)}\,(4\nu^2\,\Psi + 4\nu\,\partial_x\Psi + \partial_{xx}\Psi).
\end{aligned}$$

Substituting,

$$\begin{aligned}
\partial_t\Psi^{\mathrm{im}} + \nu\,\partial_x\Psi^{\mathrm{im}} + \tfrac{1}{2}\,\partial_{xx}\Psi^{\mathrm{im}} &= e^{-2\nu(x-\ell)}\Big(\partial_t\Psi - 2\nu^2\,\Psi - \nu\,\partial_x\Psi + 2\nu^2\,\Psi + 2\nu\,\partial_x\Psi + \tfrac{1}{2}\,\partial_{xx}\Psi\Big)\\
&= e^{-2\nu(x-\ell)}\Big(\partial_t\Psi + \nu\,\partial_x\Psi + \tfrac{1}{2}\,\partial_{xx}\Psi\Big)(t, 2\ell - x) = 0.
\end{aligned}$$

The mirror reverses the sign of the first derivative in x, and the factor $e^{-2\nu(x-\ell)}$ restores the drift term. So Ψ^{out} is infinitely differentiable for $t < T$ and solves the same equation on $[0,T)\times\mathbb{R}$,

$$\partial_t\Psi^{\mathrm{out}} + \nu\,\partial_x\Psi^{\mathrm{out}} + \tfrac{1}{2}\,\partial_{xx}\Psi^{\mathrm{out}} = 0, \qquad \text{and} \qquad \Psi^{\mathrm{out}}(t,\ell) = 0 \quad \text{for every } t < T, \tag{E.24}$$

because the two terms of Ψ^{out} coincide at $x = \ell$. At the horizon, let $x^* < \ell$ be a point at which ψ is continuous. As $(t,x) \to (T,x^*)$, $\Psi(t,x) \to \psi(x^*)$ by Lemma 19.61(iii). The mirrored point $2\ell - x$ tends to $2\ell - x^* > \ell$, and ψ vanishes on $[\ell,\infty)$, so it is continuous at $2\ell - x^*$ with the value 0; by the same lemma $\Psi(t, 2\ell - x) \to 0$, while the factor $e^{-2\nu(x-\ell)}$ stays bounded. Hence

$$\Psi^{\mathrm{out}}(t,x) \to \psi(x^*) \qquad \text{as } (t,x) \to (T,x^*) \text{ with } t < T. \tag{E.25}$$

Step 2: a bound below the level. Let $t < T$ and $x \le \ell$, and write $\tau = T - t$ and $z = \ell - x \ge 0$. Since ψ vanishes on $[\ell,\infty)$, the variable $\psi(\ell + z + \nu\tau + \sqrt{\tau}\,Z)$ vanishes unless $\sqrt{\tau}\,Z < -z - \nu\tau$, so

$$\big|\Psi^{\mathrm{im}}(t,x)\big| = e^{2\nu z}\,\big|\Psi(t,\ell+z)\big| \le \sup|\psi|\;e^{2\nu z}\;P\big(\sqrt{\tau}\,Z < -z - \nu\tau\big) = \sup|\psi|\;e^{2\nu z}\;\Phi\Big(-\frac{z+\nu\tau}{\sqrt{\tau}}\Big).$$

If $z + \nu\tau < 0$, then $\nu < 0$, so $e^{2\nu z} \le 1$ and the right side is at most $\sup|\psi|$. If $z + \nu\tau \ge 0$, use the Gaussian tail bound

$$\Phi(-y) \le \tfrac{1}{2}\,e^{-y^2/2} \qquad \text{for } y \ge 0.$$

To prove it, write $\Phi(-y) = \int_y^\infty \varphi(y')\,dy'$. For $y' \ge y \ge 0$, $y'^2 = y^2 + 2y\,(y'-y) + (y'-y)^2$, so $\varphi(y') = \varphi(y)\,e^{-y(y'-y) - (y'-y)^2/2} \le \varphi(y)\,e^{-(y'-y)^2/2}$, and

$$\Phi(-y) \le \varphi(y)\int_y^\infty e^{-(y'-y)^2/2}\,dy' = \varphi(y)\,\sqrt{\pi/2} = \tfrac{1}{2}\,e^{-y^2/2},$$

the integral being half of $\int e^{-y'^2/2}\,dy' = \sqrt{2\pi}$ (Example 7.6). With $y = (z+\nu\tau)/\sqrt{\tau}$,

$$|\Psi^{\mathrm{im}}(t,x)| \le \tfrac{1}{2}\,\sup|\psi|\;\exp\Big(2\nu z - \frac{(z+\nu\tau)^2}{2\tau}\Big) = \tfrac{1}{2}\,\sup|\psi|\;\exp\Big(-\frac{(z-\nu\tau)^2}{2\tau}\Big) \le \sup|\psi|,$$

because $4\nu z\tau - (z+\nu\tau)^2 = -(z-\nu\tau)^2$. In both cases $|\Psi^{\mathrm{im}}(t,x)| \le \sup|\psi|$, and with $|\Psi| \le \sup|\psi|$,

$$\big|\Psi^{\mathrm{out}}(t,x)\big| \le 2\,\sup|\psi| \qquad \text{for all } t<T \text{ and all } x \le \ell. \tag{E.26}$$

Above the level no such bound holds when $\nu < 0$, because of the factor $e^{-2\nu(x-\ell)}$; the stopping of Step 3 keeps the path at or below the level.

Step 3: Itô's lemma, stopped at the level. The hitting time $t_\ell = \inf\{s \in [0,T] : X_s \ge \ell\}$ of the Setup of §19.9 is the first time at which W reaches the continuous boundary $s \mapsto \ell - \nu s$, approached from below, whose value at $s=0$ is $\ell > 0$: the situation of Definition 19.22(c). Fix $T' < T$ and put

$$f(t,y) = \Psi^{\mathrm{out}}(t, \nu t + y), \qquad 0 \le t \le T',\ y \in \mathbb{R},$$

so that $f(t, W_t) = \Psi^{\mathrm{out}}(t, X_t)$. By the chain rule (E.16), applied to Ψ^{out} on the open rectangle of the points (t,x) with $t<T$, along the direction $(1,\nu)$ for the derivative in t,

$$\partial_t f = (\partial_t \Psi^{\mathrm{out}} + \nu\,\partial_x \Psi^{\mathrm{out}})(t,\nu t+y), \qquad \partial_y f = \partial_x \Psi^{\mathrm{out}}(t,\nu t+y), \qquad \partial_{yy} f = \partial_{xx}\Psi^{\mathrm{out}}(t,\nu t+y),$$

all continuous. These functions and f have exponential growth on $[0,T'] \times \mathbb{R}$. Indeed, by Lemma 19.61(i) and (iii), Ψ and each of its partial derivatives are bounded by $c'\,e^{c|x|}$ on $[0,T'] \times \mathbb{R}$; the formulas of Step 1 express Ψ^{im} and its derivatives as $e^{-2\nu(x-\ell)}$, which is at most $e^{2|\nu|\ell}\,e^{2|\nu||x|}$, times combinations of Ψ and its derivatives at $(t, 2\ell - x)$, where $|2\ell - x| \le 2\ell + |x|$; sums and products keep exponential growth (Lemma 19.20(iii)); and $|\nu t + y| \le |\nu| T + |y|$. So $f \in C^{1,2}_E([0,T'] \times \mathbb{R})$, and by (E.24)

$$\partial_t f + \tfrac{1}{2}\,\partial_{yy} f = \Big(\partial_t \Psi^{\mathrm{out}} + \nu\,\partial_x \Psi^{\mathrm{out}} + \tfrac{1}{2}\,\partial_{xx}\Psi^{\mathrm{out}}\Big)(t,\nu t+y) = 0.$$

Apply the stopped clause, Theorem 19.30(iii), on $[0,T']$ at the time $t = T'$, with the boundary $s \mapsto \ell - \nu s$. Up to the time T' it makes no difference whether the infimum that defines the hitting time runs over $[0,T']$ or over $[0,T]$: the two give the same $T' \wedge t_\ell$ and the same indicators $\mathbf{1}\{s < t_\ell\}$ for $s \le T'$. The ds-integral of (19.13) vanishes, and $f(0,0) = \Psi^{\mathrm{out}}(0,0)$, so with probability one

$$\Psi^{\mathrm{out}}\big(T' \wedge t_\ell,\ X_{T' \wedge t_\ell}\big) = \Psi^{\mathrm{out}}(0,0) + \int_0^{T'} \mathbf{1}\{s < t_\ell\}\,\partial_x \Psi^{\mathrm{out}}(s, X_s)\,dW_s.$$

The stopped integral has mean zero (Theorem 19.23(ii)), and the left side is bounded, by (E.26): the stopped path stays at or below the level, by (19.7) before t_ℓ and by (19.6) at t_ℓ. Taking expectations,

$$\mathbb{E}\,\Psi^{\mathrm{out}}\big(T' \wedge t_\ell,\ X_{T' \wedge t_\ell}\big) = \Psi^{\mathrm{out}}(0,0) \qquad \text{for every } T' < T. \tag{E.27}$$

Step 4: letting T' increase to T. Let $T'_k = T\,(1 - 1/(k+1))$, $k = 1, 2, \ldots$, a sequence that increases to T. Three events cover all paths.

- On $\{t_\ell < T\}$, as soon as $T'_k \ge t_\ell$ the stopped value is $\Psi^{\mathrm{out}}(t_\ell, X_{t_\ell}) = \Psi^{\mathrm{out}}(t_\ell, \ell) = 0$, by (19.6), which gives $X_{t_\ell} = \ell$, and by (E.24).
- On $\{t_\ell > T\}$, where the level is not reached by time T, the stopped value is $\Psi^{\mathrm{out}}(T'_k, X_{T'_k})$, and $X_T < \ell$ by (19.7). The path is continuous, so $(T'_k, X_{T'_k}) \to (T, X_T)$, and by (E.25) the stopped values tend to $\psi(X_T)$ whenever X_T is a point of continuity of ψ. This fails with probability zero, because X_T has a continuous distribution function and ψ has finitely many discontinuities.
- The event $\{t_\ell = T\}$ has probability zero, by (19.6).

So the stopped values converge almost surely, hence in probability, to $\psi(X_T)\,\mathbf{1}\{t_\ell > T\}$. They are bounded by $2\sup|\psi|$, and the limit by $\sup|\psi|$, so their fourth moments are bounded, and the fourth-moment lemma (Lemma 19.12(iv)) lets the limit pass through the expectation in (E.27):

$$\mathbb{E}\big[\psi(X_T)\,\mathbf{1}\{t_\ell > T\}\big] = \Psi^{\text{out}}(0,0) = \Psi(0,0) - e^{2\nu\ell}\,\Psi(0,2\ell).$$

By (19.7), $\{t_\ell > T\}$ is the event that $X_s < \ell$ for every $s \in [0,T]$, which is $\{X_T^{\max} < \ell\}$ because the maximum is attained. Finally $\sqrt{T}\,Z$ has the law of W_T, so $\Psi(0,x) = \mathbb{E}\,\psi(x + \nu T + W_T)$: $\Psi(0,0) = \mathbb{E}\,\psi(X_T)$ and $\Psi(0,2\ell) = \mathbb{E}\,\psi(2\ell + \nu T + W_T)$. This is (19.20).

Step 5: parts (b), (ii) and (i). Part (b). Let $x < \ell$ and take $\psi(y) = \mathbf{1}\{y < x\}$. This function is bounded, it has one discontinuity, and it vanishes on $[\ell,\infty)$ because $x < \ell$. Because $W_T \sim \mathcal{N}(0,T)$ (Definition 19.5(2)), the two variables in (19.20) have the laws $X_T \sim \mathcal{N}(\nu T, T)$ and $2\ell + \nu T + W_T \sim \mathcal{N}(2\ell + \nu T, T)$, so that identity reads

$$P\big(X_T^{\max} < \ell,\ X_T < x\big) = \Phi\Big(\frac{x - \nu T}{\sqrt{T}}\Big) - e^{2\nu\ell}\,\Phi\Big(\frac{x - 2\ell - \nu T}{\sqrt{T}}\Big).$$

Subtract this from $P(X_T < x) = \Phi\big((x - \nu T)/\sqrt{T}\big)$: the probability of $\{X_T^{\max} \ge \ell,\ X_T < x\}$ is the right side of (19.21). The distribution function of X_T is continuous, so $P(X_T = x) = 0$, and $X_T < x$ may be replaced by $X_T \le x$. For $x = \ell$, the events $\{X_T^{\max} \ge \ell,\ X_T \le \ell - 1/k\}$ increase with k to $\{X_T^{\max} \ge \ell,\ X_T < \ell\}$. By the continuity of probability (Lemma 19.12(0)) and of Φ, the probability of the limit event is $e^{2\nu\ell}\,\Phi\big((-\ell - \nu T)/\sqrt{T}\big)$, and $P(X_T = \ell) = 0$ again.

Part (ii). On the event $\{X_T > \ell\}$ the maximum is at least X_T, hence above ℓ. So

$$P\big(X_T^{\max} \ge \ell\big) = P\big(X_T^{\max} \ge \ell,\ X_T \le \ell\big) + P(X_T > \ell) = e^{2\nu\ell}\,\Phi\Big(\frac{-\ell - \nu T}{\sqrt{T}}\Big) + \Phi\Big(\frac{\nu T - \ell}{\sqrt{T}}\Big),$$

by (b) at $x = \ell$ and because $P(X_T > \ell) = 1 - \Phi\big((\ell - \nu T)/\sqrt{T}\big) = \Phi\big((\nu T - \ell)/\sqrt{T}\big)$.

Part (i). Let $\nu = 0$. Then (b) reads $P(W_T^{\max} \ge \ell,\ W_T \le x) = \Phi\big((x - 2\ell)/\sqrt{T}\big) = P(W_T \le x - 2\ell) = P(W_T \ge 2\ell - x)$, the last step by the symmetry of the law $\mathcal{N}(0,T)$, and (ii) reads $P(W_T^{\max} \ge \ell) = 2\,\Phi(-\ell/\sqrt{T}) = 2\,P(W_T \ge \ell)$. The right side is $P(|W_T| \ge \ell)$. So, for every $y > 0$,

$$P\big(W_T^{\max} < y\big) = 1 - 2\,\Phi\Big(-\frac{y}{\sqrt{T}}\Big) = 2\,\Phi\Big(\frac{y}{\sqrt{T}}\Big) - 1 = P(|W_T| < y).$$

Both variables are nonnegative, since $W_T^{\max} \ge W_0 = 0$. Their distribution functions therefore agree: each equals $2\,\Phi(y/\sqrt{T}) - 1$ for $y \ge 0$ and 0 for $y < 0$, the value at a point $y \ge 0$ being the limit of the display from the right, by the continuity of probability and of Φ. This function is continuous, and on $(0,\infty)$ it has the continuous derivative $2\,\varphi(y/\sqrt{T})/\sqrt{T}$, the density. By Fact 7.22 and the substitution $y = \sqrt{T}\,z$,

$$\mathbb{E}\,W_T^{\max} = \int_0^\infty y\,\frac{2}{\sqrt{T}}\,\varphi\Big(\frac{y}{\sqrt{T}}\Big)\,dy = 2\sqrt{T}\int_0^\infty z\,\varphi(z)\,dz = 2\sqrt{T}\,\varphi(0) = \sqrt{\frac{2T}{\pi}},$$

the last integral being evaluated with the antiderivative $-\varphi$ of $z\,\varphi(z)$. At $T = 1$ this is the value $\mathbb{E}|Z| = \sqrt{2/\pi}$ of Example 19.11; Corollary 20.30 in §20 computes the same integral itself, for the at-the-money straddle, and refers to this theorem for the running maximum.

Step 6: part (iii), the Fact of §10.2. Divide the process by $-\sigma$:

$$-\frac{\mu t + \sigma W_t}{\sigma} = \Big(-\frac{\mu}{\sigma}\Big)\,t + (-W_t),$$

a Brownian motion with drift $-\mu/\sigma$, because $-W$ is a standard Brownian motion (Proposition 19.8(iii)). The minimum of $\mu t + \sigma W_t$ over $[0,T]$ is at most $-a$ exactly when the maximum of this process reaches a/σ. So (ii), with $\nu = -\mu/\sigma$ and $\ell = a/\sigma$, gives

$$\Phi\Big(\frac{-\mu T/\sigma - a/\sigma}{\sqrt{T}}\Big) + e^{-2\mu a/\sigma^2}\,\Phi\Big(\frac{-a/\sigma + \mu T/\sigma}{\sqrt{T}}\Big),$$

which is the display of (iii); at $\mu = 0$ both terms are $\Phi(-a/(\sigma\sqrt{T}))$.

The ratio. In the standardised variables, $(-a-\mu T)/(\sigma\sqrt{T}) = -\tilde{\mu} - \tilde{\ell}$, $(-a+\mu T)/(\sigma\sqrt{T}) = \tilde{\mu} - \tilde{\ell}$ and $\mu a/\sigma^2 = \tilde{\mu}\tilde{\ell}$. The path probability is $\Phi(-\tilde{\mu}-\tilde{\ell}) + e^{-2\tilde{\mu}\tilde{\ell}}\,\Phi(\tilde{\mu}-\tilde{\ell})$, and the terminal probability is $P(\mu T + \sigma W_T \le -a) = \Phi(-\tilde{\mu}-\tilde{\ell})$; dividing gives the ratio of the statement. Its monotonicity rests on the inequality

$$\frac{\varphi(y)}{\Phi(y)} > -y \qquad \text{for every real } y,$$

a form of Mills' inequality. For $y \ge 0$ the left side is positive and the right side is not. For $y < 0$,

$$\Phi(y) = \int_{|y|}^{\infty} \varphi(y')\,dy' < \int_{|y|}^{\infty} \frac{y'}{|y|}\,\varphi(y')\,dy' = \frac{\varphi(|y|)}{|y|} = \frac{\varphi(y)}{|y|},$$

by the antiderivative $-\varphi$ of $y'\,\varphi(y')$ and because φ is even, and this is the claim. Now fix $\tilde{\ell} > 0$. The ratio exceeds one by the positive quantity $e^{-2\tilde{\mu}\tilde{\ell}}\,\Phi(\tilde{\mu}-\tilde{\ell})/\Phi(-\tilde{\mu}-\tilde{\ell})$, whose logarithm is $-2\tilde{\mu}\tilde{\ell} + \ln\Phi(\tilde{\mu}-\tilde{\ell}) - \ln\Phi(-\tilde{\mu}-\tilde{\ell})$. Since $\Phi' = \varphi$, the derivative of this logarithm in $\tilde{\mu}$ is

$$-2\tilde{\ell} + \frac{\varphi(\tilde{\mu}-\tilde{\ell})}{\Phi(\tilde{\mu}-\tilde{\ell})} + \frac{\varphi(-\tilde{\mu}-\tilde{\ell})}{\Phi(-\tilde{\mu}-\tilde{\ell})} > -2\tilde{\ell} + (\tilde{\ell}-\tilde{\mu}) + (\tilde{\mu}+\tilde{\ell}) = 0,$$

by the inequality at $y = \tilde{\mu}-\tilde{\ell}$ and at $y = -\tilde{\mu}-\tilde{\ell}$. So the logarithm, and with it the ratio, is strictly increasing in $\tilde{\mu}$. At $\tilde{\mu} = 0$ the ratio equals $1 + \Phi(-\tilde{\ell})/\Phi(-\tilde{\ell}) = 2$. Hence it exceeds 2 exactly when $\tilde{\mu} > 0$, that is, when $\mu > 0$, and it is below 2 when $\mu < 0$. Finally let $\tilde{\mu} + \tilde{\ell} > 0$. The inequality at $y = -(\tilde{\mu}+\tilde{\ell})$, with φ even, gives $\Phi(-\tilde{\mu}-\tilde{\ell}) < \varphi(\tilde{\mu}+\tilde{\ell})/(\tilde{\mu}+\tilde{\ell})$, so the ratio minus one exceeds

$$\frac{\Phi(\tilde{\mu}-\tilde{\ell})\,(\tilde{\mu}+\tilde{\ell})\,e^{-2\tilde{\mu}\tilde{\ell}}}{\varphi(\tilde{\mu}+\tilde{\ell})} = \Phi(\tilde{\mu}-\tilde{\ell})\,(\tilde{\mu}+\tilde{\ell})\,\sqrt{2\pi}\,e^{(\tilde{\mu}-\tilde{\ell})^2/2},$$

because $(\tilde{\mu}+\tilde{\ell})^2/2 - 2\tilde{\mu}\tilde{\ell} = (\tilde{\mu}-\tilde{\ell})^2/2$. The right side tends to infinity as $\tilde{\mu} \to \infty$, since $\Phi(\tilde{\mu}-\tilde{\ell}) \to 1$.

The conditional reading. The minimum of the path is at most its final value, so the event $\{\mu T + \sigma W_T \le -a\}$ is contained in the event $\{\min_{0\le t\le T}(\mu t + \sigma W_t) \le -a\}$, whose probability is positive by the display of (iii). The conditional probability of the first event given the second is therefore the probability of the first divided by that of the second, which is one over the ratio. It is below $1/2$ exactly when the ratio exceeds 2, that is, exactly when $\mu > 0$. □

F Proofs for §20: The Black–Scholes–Merton Model

F.1 Proof of the three checks of Definition 20.1: Trading strategies, self-financing and arbitrage in continuous time

The statement is on page 313. The notation and the standing assumptions are those of §20.1.

Proof. We check the three claims on which Definition 20.1 relies, citing its parts as (i), (ii) and (iv): the integrals in (ii) exist, the value at a switch date is as (ii) states, and the closure properties of (iv) hold.

Check 1: the integrals in (ii) exist. Fix a cell $[t_{j-1}, t_j)$ and $T' < t_j$. A holding is a finite sum of terms $\xi(s, W_s, Y)\,\mathbf{1}\{s < t_\ell\}$ in which ξ has exponential growth and is continuous on each box of values of Y. The factor $e^{-rs}\beta_{k,s} = e^{-rs}\,\partial_x a_k(s, W_s)$ is a continuous function of (s, W_s) of exponential growth on $[0, T']$, and products of functions of exponential growth have exponential growth (Lemma 19.20(iii)). So $h_s\,e^{-rs}\beta_{k,s}$ is again a finite sum of terms of the same form: it lies in the span of $\mathcal{H}$ on $[t_{j-1}, T']$, and its Itô integral is defined (Theorem 19.23(i) and (iv)). On every path the integrand $h_s\,e^{-rs}(\alpha_{k,s} - rA_{k,s})$ of the other integral is continuous in s except at the finitely many hitting times carried by the terms of h, so its Riemann integral exists.

Check 2: the value at a switch date. Suppose that the holdings of the cell $[t_{j-1}, t_j)$ are admissible on the closed interval $[t_{j-1}, t_j]$ and, when $t_j = T$, that every instrument they hold has a price function a_k in $C_E^{1,2}([0,T]\times\mathbb{R})$, as the stock of (A1) does by the Setup of §20.1 (with two motions, the class of that Setup with $[0,T]$ in the place of $[0,T']$); for $t_j < T$ that Setup already gives $a_k \in C_E^{1,2}([0,t_j]\times\mathbb{R})$. Then (20.1) holds at $t = t_j$ for these instruments, by Theorem 19.30(i) on $[0,t_j]$ (Theorem 19.55(i) with two motions), Check 1 applies on $[t_{j-1}, t_j]$, the limit required in (ii) exists, and Π_{t_j} is given by (20.3) at $t = t_j$. The capital $\Pi_{t_{j-1}}$ at the start of the cell has a finite second moment: an initial capital has moments of every order by Lemma 19.20(iii), and the value at an earlier switch date is a mean-square limit. Let $t < t_j$. By the additivity of both integrals over adjacent intervals (Theorem 19.23(iv)), the right side of (20.3) at t_j minus its value at t is $\sum_k \int_t^{t_j} h_{k,s}\, d(e^{-rs}A_{k,s})$. Each Itô integral in this sum has second moment $\int_t^{t_j} \mathbb{E}\big[(h_{k,s}\, e^{-rs}\beta_{k,s})^2\big]\, ds \le c\,(t_j - t)$, by the isometry and the boundedness of the second moment of an integrand (Theorem 19.23(iii) and (v)). Each Riemann integral $\int_t^{t_j} X_s\, ds$, with $X_s = h_{k,s}\, e^{-rs}(\alpha_{k,s} - rA_{k,s})$, satisfies

$$\Big(\int_t^{t_j} X_s\, ds\Big)^2 \le (t_j - t)\int_t^{t_j} X_s^2\, ds$$

on every path: with $\bar{X}$ the average of X_s over $[t, t_j]$, the inequality is $\int_t^{t_j}(X_s - \bar{X})^2\, ds \ge 0$ expanded. Its second moment is therefore at most $(t_j - t)\int_t^{t_j} \mathbb{E}X_s^2\, ds \le c\,(t_j - t)^2$, where the expectation enters the integral by Fact 18.34(iii) (the integrand X_s^2 is nonnegative) and $\mathbb{E}X_s^2$ is bounded uniformly in s by Lemma 19.20(iii). By the triangle inequality of Lemma 19.12(ii), $e^{-rt}\,\Pi_t$ converges in mean square, as t increases to t_j, to the right side of (20.3) at t_j. That limit has a finite second moment and the factor e^{rt} is continuous, so Π_t converges in mean square to e^{rt_j} times the limit, which is the claim.

Check 3: the closure properties of (iv). Sums. Inserting a further switch date t'' inside a cell $[t_{j-1}, t_j)$ changes no value: the holdings restricted to $[t_{j-1}, t'']$ are admissible on that closed interval, so by Check 2 the value at t'' is (20.3) at $t = t''$, and by the additivity of both integrals over adjacent intervals (Theorem 19.23(iv)) the identity restarted at t'' gives the same values on $[t'', t_j)$. So two strategies may be given common switch dates, the union of theirs. On each common cell, holdings, initial capitals and values then add, by the linearity of both integrals in (20.2) (Theorem 19.23(iv)), and at each switch date the mean-square limit of a sum is the sum of the limits, by the triangle inequality of Lemma 19.12(ii). Scalar multiples are treated in the same way. *Concatenation.* The cells of the joined strategy are those of its two pieces. On the cells before t' its values are those of the first piece, and the first piece's value $\Pi_{t'}$ exists as a mean-square limit. On the cells after t' identity (20.3) restarts from $e^{-rt'}\,\Pi_{t'}$, which equals the discounted initial capital of the second piece with probability one, so the values are those of the second piece and the limits required in (ii) exist. *Stopping.* A term $\xi(s, W_s, Y)$ that carries no indicator becomes $\xi(s, W_s, Y)\,\mathbf{1}\{s < t_\ell\}$, which is of the kind allowed in (i). Let a term carry an indicator $\mathbf{1}\{s < t_{\ell'}\}$ whose boundary is approached from the same side as ℓ, say from below. Then $\min(t_\ell, t_{\ell'})$ is the infimum of the union of the sets $\{s \in [0,T] : W_s \ge \ell(s)\}$ and $\{s \in [0,T] : W_s \ge \ell'(s)\}$, which is the set $\{s \in [0,T] : W_s \ge \min(\ell(s), \ell'(s))\}$; the function $\min(\ell, \ell')$ is continuous with a positive value at 0, so $\min(t_\ell, t_{\ell'})$ is again a hitting time of Definition 19.22(c), and $\mathbf{1}\{s < t_\ell\}\,\mathbf{1}\{s < t_{\ell'}\} = \mathbf{1}\{s < \min(t_\ell, t_{\ell'})\}$. For two boundaries approached from above, the same argument applies to $\max(\ell, \ell')$. Finally, for $t_{j-1} \le t' \le t < t_j$, on the event $\{t_\ell \le t'\}$ both integrals of a stopped holding over $[t', t]$ vanish with probability one, the Riemann integral because its integrand is zero and the Itô integral by Theorem 19.23(iv). So the stopped strategy holds no instrument from t_ℓ on, and its discounted value is constant from t_ℓ on, the values at switch dates being limits of constant values. □

F.2 Proof of Theorem 20.3: The Black–Scholes–Merton equation

The statement is on page 314. The notation and the standing assumptions are those of §20.1.

Proof. We first run the riskless-hedge (delta-hedging) argument at the whiteboard, then make it exact by turning any violation of the equation into an arbitrage.

Step 1: the whiteboard argument (motivation; no later step relies on it). By Corollary 19.31, whose hypothesis for a geometric Brownian motion is the moderate growth assumed of V and its derivatives,

$$dV = (\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V)\,dt + \sigma S\,\partial_S V\,dW.$$

The portfolio $\Pi = -V + (\partial_S V)\,S$, short one derivative and long $\partial_S V$ shares, with the holding $\partial_S V$ kept fixed over dt, changes by

$$d\Pi = -dV + \partial_S V\,dS = -(\partial_t V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V)\,dt,$$

because $\partial_S V\,dS = \mu S\,\partial_S V\,dt + \sigma S\,\partial_S V\,dW$: the dW terms cancel, and so do the μ terms. A riskless portfolio must earn the rate r (otherwise borrow to buy it, or sell it and lend the proceeds), so $d\Pi = r\Pi\,dt = r(-V + S\,\partial_S V)\,dt$. Equating the two expressions for $d\Pi$ gives $\mathcal{L}V = 0$. The words "kept fixed over dt" are not yet a statement about a strategy of Definition 20.1; Steps 2 and 3 replace them by one.

Step 2: the residual as a position. The instruments are the stock and the derivative, with price functions

$$a_1(t,x) = S(t,x) = S_0 e^{(\mu-\sigma^2/2)t+\sigma x} \qquad \text{and} \qquad a_2(t,x) = V\big(S(t,x),t\big).$$

Each is in $C_E^{1,2}([0,T'] \times \mathbb{R})$ for every $T' < T$. For a_1 this was checked in the Setup of §20.1. For a_2, the chain rule gives

$$\partial_t a_2 = \partial_t V + (\mu - \tfrac{1}{2}\sigma^2) S\,\partial_S V, \qquad \partial_x a_2 = \sigma S\,\partial_S V, \qquad \partial_{xx} a_2 = \sigma^2 S^2\,\partial_{SS} V + \sigma^2 S\,\partial_S V,$$

where $S = S(t,x)$ and the derivatives of V are evaluated at $(S(t,x),t)$. Each of the three is continuous. Each has moderate growth in S on $[0,T']$, because S and S^2 have moderate growth and sums and products of functions of moderate growth have moderate growth; and moderate growth in S is exponential growth in x (both by Lemma 19.20(i)).

Fix $T' < T$, and consider the strategy on $[0,T)$ with the switch dates $0 < T' < T$ and zero initial capital that, on the cell $[0,T')$, holds

$$h_{V,t} = \mathcal{L}V(S_t,t) \;\text{ units of the derivative and }\; h_{S,t} = -\mathcal{L}V(S_t,t)\,\partial_S V(S_t,t) \;\text{ shares,}$$

the bank taking the rest, and that holds no instrument on $[T',T)$. The functions $\mathcal{L}V$ and $\mathcal{L}V\,\partial_S V$ are sums of products of functions of moderate growth, so the two holdings are continuous functions of (t,W_t) of exponential growth on $[0,T']$ (Lemma 19.20(i) and (iii)). They carry no parameter Y and no stopping indicator, so they are in $\mathcal{H}$ on $[0,T']$, and the holdings are admissible on the closed interval $[0,T']$.

The coefficients of the two discounted prices in (20.1) follow from the chain-rule formulas. For the derivative, $\alpha_{2,t} = \partial_t a_2 + \frac{1}{2}\,\partial_{xx} a_2 = \partial_t V + \mu S\,\partial_S V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS} V$, so that

$$e^{-rt}(\alpha_{2,t} - rA_{2,t}) = e^{-rt}(\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V - rV), \qquad e^{-rt}\beta_{2,t} = e^{-rt}\sigma S\,\partial_S V;$$

this is also Corollary 19.31 with the outer function $(S,t) \mapsto e^{-rt}V(S,t)$. For the stock, $\alpha_{1,t} = (\mu - \frac{1}{2}\sigma^2)S + \frac{1}{2}\sigma^2 S = \mu S$ and $\beta_{1,t} = \sigma S$, so that

$$e^{-rt}(\alpha_{1,t} - rA_{1,t}) = e^{-rt}(\mu - r)S, \qquad e^{-rt}\beta_{1,t} = e^{-rt}\sigma S.$$

Everything is evaluated at $S = S_t$. Insert the two holdings into (20.3) on the cell $[0,T')$. The sum of the two dW integrands is

$$e^{-rt}(h_{V,t}\,\sigma S\,\partial_S V + h_{S,t}\,\sigma S) = e^{-rt}\sigma S\,(\mathcal{L}V\,\partial_S V - \mathcal{L}V\,\partial_S V) = 0$$

identically. By linearity (Theorem 19.23(iv)) the two Itô integrals add up to the integral of this sum, every left-endpoint sum of which is zero; so the stochastic part of the value vanishes with probability one. The sum of the two ds integrands is

$$e^{-rt}\,\mathcal{L}V\left[\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS}V - rV - (\mu - r)S\,\partial_S V\right] = e^{-rt}\,(\mathcal{L}V)^2,$$

because the bracket is $\partial_t V + rS\,\partial_S V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS}V - rV = \mathcal{L}V$. Hence, for every $t \le T'$, with probability one,

$$e^{-rt}\,\Pi_t = \int_0^t e^{-rs}\,\big(\mathcal{L}V(S_s, s)\big)^2\,ds \;\ge\; 0: \tag{F.1}$$

a pathwise Riemann integral of a continuous function, with no stochastic integral left in it. At $t = T'$ this is the value at a switch date, by Check 2 of Definition 20.1, the holdings being admissible on $[0, T']$. On $[T', T)$ the whole value sits in the bank, $\Pi_t = e^{r(t-T')}\,\Pi_{T'}$, so every limit that Definition 20.1(ii) requires exists, and the holdings define a strategy.

Step 3: positivity. Suppose, for contradiction, that $\mathcal{L}V(S^*, t^*) \neq 0$ at some point of $(0, \infty) \times [0, T)$. The function $\mathcal{L}V$ is continuous there, being built from the continuous functions V, $\partial_t V$, $\partial_S V$ and $\partial_{SS}V$. So it is nonzero at (S^*, t) for every t near t^*, and t^* may be taken in $(0, T)$. By continuity again there are $\varepsilon > 0$ and $\varepsilon' > 0$, with $t^* + \varepsilon' < T$ and $\varepsilon' < S^*$, such that

$$|\mathcal{L}V| \ge \varepsilon \quad \text{on the rectangle } (S^* - \varepsilon', S^* + \varepsilon') \times [t^*, t^* + \varepsilon'].$$

Run the strategy of Step 2 with $T' = t^* + \varepsilon'$. The price S_{t^*} is lognormal (Theorem 19.40, with $t^* > 0$): $\ln S_{t^*}$ is normal with the positive variance $\sigma^2 t^*$, so S_{t^*} falls in every interval of $(0, \infty)$ with positive probability, Φ being strictly increasing. In particular the event

$$A = \{\, |S_{t^*} - S^*| < \varepsilon'/2 \,\}$$

has positive probability. On A, the path $s \mapsto S_s$ is continuous, so there is a number $\varepsilon'' \in (0, \varepsilon']$, depending on the path, with $S_s \in (S^* - \varepsilon', S^* + \varepsilon')$ for all $s \in [t^*, t^* + \varepsilon'']$. On that stretch the integrand of (F.1) is at least $e^{-rs}\varepsilon^2$, and elsewhere it is nonnegative, so by (F.1) at $t = T'$

$$e^{-rT'}\,\Pi_{T'} \;\ge\; \int_{t^*}^{t^*+\varepsilon''} e^{-rs}\,\varepsilon^2\,ds \;\ge\; \varepsilon^2\,\min(1, e^{-rT})\,\varepsilon'' \;>\; 0,$$

where the minimum appears because r may be negative: $e^{-rs} \ge \min(1, e^{-rT})$ for $0 \le s \le T$. So the strategy has zero initial capital, and its value at the deterministic time $T' < T$ is nonnegative with probability one (Step 2) and positive on A, apart from an event of probability zero. It is an arbitrage in the sense of Definition 20.1(v), which (A6) excludes. Hence $\mathcal{L}V = 0$ on all of $(0, \infty) \times [0, T)$. □

F.3 Proof of Theorem 20.7: Risk-neutral valuation of European payoffs, by replication (Black and Scholes, 1973; Merton, 1973)

The statement is on page 318. The notation and the standing assumptions are those of §20.2.

Proof. We proceed by verification: Gaussian smoothing produces a smooth solution of the equation, Itô's lemma shows that its delta strategy delivers it, and the law of one price fixes the time-0 price.

Step 1: log variables; smoothness and the equation. Put $\psi(x) = f(e^x)$ for real x. It is continuous except at finitely many points, and $|\psi(x)| \le c\,(e^{cx} + e^{-cx}) \le 2c\,e^{c|x|}$ (Lemma 19.20(ii)), so the growth hypothesis of Lemma 19.61 holds with the constant $2c$. Let

$$\Psi(t, x) = \mathbb{E}\,\psi\big(x + \nu(T - t) + \sigma\sqrt{T - t}\,Z\big) \quad \text{for } t < T, \qquad \Psi(T, x) = \psi(x).$$

Writing $S\exp(\cdots) = \exp(\ln S + \cdots)$ in (20.5) shows that

$$V(S,t) = e^{-r(T-t)}\,\Psi(t, \ln S) \qquad \text{on } (0,\infty)\times[0,T]. \tag{F.2}$$

Lemma 19.61, with the drift ν, the volatility σ and calendar time t, gives four properties. The function Ψ is infinitely differentiable on $[0,T)\times\mathbb{R}$, and for every $T' < T$ each of its partial derivatives is bounded by $c'e^{c'|x|}$ on $[0,T']\times\mathbb{R}$, with a constant c' that depends on T' and on the derivative. It solves

$$\partial_t\Psi + \nu\,\partial_x\Psi + \tfrac{1}{2}\sigma^2\,\partial_{xx}\Psi = 0 \qquad \text{on } [0,T)\times\mathbb{R}.$$

It satisfies $|\Psi(t,x)| \le c'e^{c'|x|}$ on $[0,T]\times\mathbb{R}$. And $\Psi(t,x)\to\psi(x^*)$ as $(t,x)\to(T,x^*)$ at every continuity point x^* of ψ.

Step 2: back to the price variable. By (F.2) and the chain rule, with $x = \ln S$,

$$\partial_t V = e^{-r(T-t)}\big(r\Psi + \partial_t\Psi\big), \qquad S\,\partial_S V = e^{-r(T-t)}\,\partial_x\Psi, \qquad S^2\,\partial_{SS}V = e^{-r(T-t)}\big(\partial_{xx}\Psi - \partial_x\Psi\big),$$

each function of Ψ evaluated at $(t,\ln S)$; the third follows from $\partial_S V = e^{-r(T-t)}\,\partial_x\Psi/S$ by one more differentiation. Hence

$$\begin{aligned}\mathcal{L}V &= e^{-r(T-t)}\Big[r\Psi + \partial_t\Psi + r\,\partial_x\Psi + \tfrac{1}{2}\sigma^2\big(\partial_{xx}\Psi - \partial_x\Psi\big) - r\Psi\Big]\\ &= e^{-r(T-t)}\Big[\partial_t\Psi + \nu\,\partial_x\Psi + \tfrac{1}{2}\sigma^2\,\partial_{xx}\Psi\Big] = 0,\end{aligned}$$

because $r - \sigma^2/2 = \nu$. The function V is infinitely differentiable on $(0,\infty)\times[0,T)$, as a product and composition of infinitely differentiable functions.

For the growth of its derivatives, call a function of (S,t) a *term* if it is a constant, times $e^{-r(T-t)}$, times a power S^{-i} with an integer $i \ge 0$, times a partial derivative of Ψ of some order (Ψ itself included) evaluated at $(t,\ln S)$. Every partial derivative of V is a finite sum of terms. The three formulas above show this for the first derivatives. Differentiating a term in t gives r times the term, plus the same term with its derivative of Ψ differentiated once more in t. Differentiating it in S gives $-i/S$ times the term, plus the same term with S^{-i} replaced by S^{-i-1} and its derivative of Ψ differentiated once more in x. Both results are sums of terms, so the claim follows by induction on the order of the derivative. On $(0,\infty)\times[0,T']$ each term has moderate growth: $e^{-r(T-t)} \le e^{|r|T}$; by Step 1 the derivative of Ψ in it is at most $c'e^{c'|\ln S|} = c'\max(S^{c'}, S^{-c'}) \le c'\,(S^{c'} + S^{-c'})$ in absolute value; and dividing by S^i keeps moderate growth with a larger constant, since $S^{-i}(S^{c'} + S^{-c'}) \le 2\,(S^{c'+i} + S^{-c'-i})$. Sums of functions of moderate growth have moderate growth (Lemma 19.20(i)). This proves (i).

For (ii), the bound of Step 1 on $[0,T]\times\mathbb{R}$ gives $|V(S,t)| \le e^{|r|T}c'\,(S^{c'} + S^{-c'})$ for $t < T$, and $|V(S,T)| = |f(S)| \le c\,(S^c + S^{-c})$; one larger constant serves both. For the limit, let $(S,t)\to(S^*,T)$ with S^* a continuity point of f. Then $\ln S^*$ is a continuity point of ψ and $\ln S\to\ln S^*$, so $\Psi(t,\ln S)\to\psi(\ln S^*) = f(S^*)$ by Step 1, while $e^{-r(T-t)}\to 1$. This proves (ii).

Step 3: replication before maturity. By (A1) the stock is the geometric Brownian motion of Definition 19.27, with drift coefficient μS_t and diffusion coefficient σS_t. Fix $T' < T$. By (i), the outer function $(S,t)\mapsto e^{-rt}V(S,t)$ and its partial derivatives $e^{-rt}(\partial_t V - rV)$, $e^{-rt}\,\partial_S V$ and $e^{-rt}\,\partial_{SS}V$ are continuous and of moderate growth on $(0,\infty)\times[0,T']$. Corollary 19.31 applies to this outer function and gives, for every $t \le T'$, with probability one,

$$e^{-rt}V(S_t,t) = V(S_0,0) + \int_0^t e^{-rs}\big(\partial_t V + \mu S\,\partial_S V + \tfrac{1}{2}\sigma^2S^2\,\partial_{SS}V - rV\big)\,ds + \int_0^t e^{-rs}\sigma S\,\partial_S V\,dW_s,$$

with $S = S_s$ and the derivatives of V evaluated at (S_s,s). By (i), $\partial_t V + \frac{1}{2}\sigma^2S^2\,\partial_{SS}V - rV = -rS\,\partial_S V$, so the ds integrand is $e^{-rs}(\mu - r)S\,\partial_S V$. The discounted stock has the coefficients $e^{-rs}(\mu - r)S_s$ and $e^{-rs}\sigma S_s$ in (20.1), as computed in Step 2 of the proof of Theorem 20.3. By

the definition (20.2) of the integral of a holding, the two integrals together are the integral of the holding $\partial_S V(S_s, s)$ against the discounted stock:

$$e^{-rt}V(S_t,t) = V(S_0,0) + \int_0^t \partial_S V(S_s,s)\, d(e^{-rs}S_s). \tag{F.3}$$

This is the defining identity (20.3) for the strategy with one instrument, the stock, with no switch date before T, with the holding $h_s = \partial_S V(S_s, s)$ and with the initial capital $V(S_0, 0)$. The holding is a continuous function of (s, W_s), of exponential growth on $[0, T']$ for every $T' < T$ by (i) and Lemma 19.20(i), so it is admissible. Since $T' < T$ was arbitrary, the value of the strategy is $V(S_t, t)$ at every $t < T$, and its bank holding is $B_t = V(S_t, t) - S_t\, \partial_S V(S_t, t)$.

Step 4: the terminal value. On every path $S_t \to S_T$ as $t \to T$, the path being continuous. The price S_T is lognormal (Theorem 19.40), so it takes each of the finitely many discontinuity points of f with probability zero; with probability one it is a continuity point of f, and then (ii) gives $V(S_t, t) \to f(S_T)$ as $t \to T$. Next, by (ii) and the inequality $(a+b)^4 \le 8(a^4+b^4)$,

$$V(S_t,t)^4 \le c'^4 \left(S_t^{c'} + S_t^{-c'}\right)^4 \le 8c'^4 \left(S_t^{4c'} + S_t^{-4c'}\right) \qquad \text{for } 0 \le t \le T,$$

and for every real exponent θ, $\mathbb{E}S_t^\theta = S_0^\theta \exp\left(\theta(\mu - \frac{1}{2}\sigma^2)t + \frac{1}{2}\theta^2\sigma^2 t\right)$ (Lemma 18.35(i)), which is bounded for $t \in [0, T]$. So $\sup_{t<T} \mathbb{E}V(S_t,t)^4 < \infty$ and $\mathbb{E}f(S_T)^4 < \infty$. Along any sequence of times increasing to T the convergence holds with probability one, hence in probability, and the fourth-moment lemma (Lemma 19.12(iv)) upgrades it to mean square: $\mathbb{E}\big(V(S_t,t) - f(S_T)\big)^2 \to 0$ as $t \to T$. The terminal value of Definition 20.1(ii) therefore exists and equals $f(S_T)$. This proves (iii).

Step 5: the price, part (iv). Suppose that the claim traded at time 0 at a quoted price $V^{\mathrm{mkt}} > V(S_0, 0)$. Sell one claim (a static position, Definition 20.1(iii)), run the strategy of (iii) from the capital $V(S_0, 0)$, and deposit $V^{\mathrm{mkt}} - V(S_0, 0)$ in the bank (a strategy with no holdings); the sum is a strategy by the closure clause of Definition 20.1(iv). The package has the initial capital $-V^{\mathrm{mkt}} + V(S_0,0) + (V^{\mathrm{mkt}} - V(S_0,0)) = 0$, and its terminal value

$$-f(S_T) + f(S_T) + \big(V^{\mathrm{mkt}} - V(S_0,0)\big)e^{rT}$$

is positive with probability one: an arbitrage (Definition 20.1(v)), excluded by (A6). If $V^{\mathrm{mkt}} < V(S_0, 0)$, buy the claim, run the negative of the strategy (a strategy by the same closure clause) and deposit $V(S_0,0) - V^{\mathrm{mkt}}$: the initial capital is again zero and the terminal value is $(V(S_0,0) - V^{\mathrm{mkt}})e^{rT} > 0$. So $V^{\mathrm{mkt}} = V(S_0, 0)$. Finally, with no dividends $S_T^{\mathbb{Q}} = S_0 \exp(\nu T + \sigma W_T)$, and W_T has the law of $\sqrt{T}\, Z$ (Definition 19.5(2)), so (20.5) at $t = 0$ reads $V(S_0,0) = e^{-rT}\,\mathbb{E}f(S_T^{\mathbb{Q}}) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}} f(S_T)$, by Definition 20.6.

Step 6: the parameter clause, part (v). This step is needed only where a payoff depends on earlier fixings (Corollary 20.11 and the results that rest on it); a first reading may pass to Proposition 20.8. Fix y off the hyperplanes $\{y_j = \ell\}$ of f. Then $f(\cdot\,; y)$ is in $\mathcal{G}$: it is continuous in S except at the finitely many levels of the hyperplanes $\{S = \ell\}$ of f, and

$$|f(S;y)| \le M(y)\,(S^c + S^{-c}), \qquad M(y) = c\prod_{j=1}^m (y_j^c + y_j^{-c}).$$

So Steps 1 and 2 apply to $f(\cdot\,; y)$ and give (i) and (ii) for $V(\cdot, \cdot\,; y)$. Three further properties are needed before Steps 3 and 4 can be run with the random Y in the place of y.

(a) Bounds that are linear in the multiplier. Write $\psi(x; y) = f(e^x; y)$ and $\Psi(t, x; y)$ for its smoothing, as in Step 1. Step 5 of the proof of Lemma 19.61 rewrites the derivatives of the Gaussian kernel in the variable z of the normal density and shows that every partial derivative of Ψ in (t, x) has the form

$$\pm\,\mathbb{E}\big[\psi(x + \nu(T-t) + \sigma\sqrt{T-t}\, Z;\, y)\, \chi(Z)\big],$$

where χ is a polynomial whose coefficients are continuous functions of $T-t>0$ and do not depend on ψ or on x; for example $\partial_x\Psi=\mathbb{E}[\psi(\cdots)\,Z]/(\sigma\sqrt{T-t})$ and $\partial_{xx}\Psi=\mathbb{E}[\psi(\cdots)\,(Z^2-1)]/(\sigma^2(T-t))$. Since $|\psi(x;y)|\le 2M(y)\,e^{c|x|}$, such an expectation is at most

$$2M(y)\,e^{c|x|}\,e^{c|\nu|T}\,\mathbb{E}\big[e^{c\sigma\sqrt{T}\,|Z|}\,|\chi(Z)|\big]$$

in absolute value. The last expectation is finite, and bounded for $T-t\in[T-T',T]$: the coefficients of χ are bounded on that closed interval, each power obeys $|z|^k\le k!\,e^{|z|}$ (one term of the exponential series), so $e^{c\sigma\sqrt{T}\,|z|}\,|\chi(z)|$ is at most a constant times $e^{(c\sigma\sqrt{T}+1)|z|}$, and $\mathbb{E}e^{(c\sigma\sqrt{T}+1)|Z|}$ is finite by Lemma 18.35(i). So each partial derivative of $\Psi(\cdot,\cdot\,;y)$ is bounded on $[0,T']\times\mathbb{R}$ by $c'M(y)\,e^{c|x|}$ with a constant c' that does not depend on y, and by the formulas of Step 2 each partial derivative of $V(\cdot,\cdot\,;y)$ in (S,t) is bounded on $(0,\infty)\times[0,T']$ by $c'M(y)\,(S^{c'}+S^{-c'})$: moderate growth jointly in (S,y). The same argument without derivatives gives $|V(S,t;y)|\le c'M(y)\,(S^{c'}+S^{-c'})$ on $(0,\infty)\times[0,T]$.

(b) Continuity in the parameter. The hyperplanes $\{y_j=\ell\}$ cut $(0,\infty)^m$ into finitely many open boxes, products of open intervals. On each open box, V and its partial derivatives in (S,t) are continuous in (S,t,y) for $t<T$. Indeed, by (a) and Step 2 each of them is $e^{-r(T-t)}$ times a finite sum of terms, each a constant times a nonpositive integer power of S times an expectation $\pm\,\mathbb{E}\big[f\big(Se^{\nu(T-t)+\sigma\sqrt{T-t}\,Z};y\big)\,\chi(Z)\big]$. Let $(S^{(i)},t^{(i)},y^{(i)})\to(S,t,y)$ inside the box, with $t<T$. For every value of Z outside the finitely many at which $Se^{\nu(T-t)+\sigma\sqrt{T-t}\,Z}$ equals the level of a hyperplane $\{S=\ell\}$, the function f is continuous at the limit point, and the coefficients of χ are continuous in $T-t$; so the quantity inside the expectation converges with probability one. Its fourth moments are bounded along the sequence, by the pointwise bound of (a) raised to the fourth power, which is again a constant times an exponential of $|Z|$: the multiplier M is continuous on the box, hence bounded along $(y^{(i)})$, the numbers $S^{(i)}$ stay in a closed interval inside $(0,\infty)$, and the numbers $T-t^{(i)}$ stay away from zero. Lemma 19.12(iv) passes the limit through the expectation.

(c) The holding is admissible. Write Y through Brownian values, $S_{s_j}=S_0\exp\big((\mu-\frac12\sigma^2)s_j+\sigma W_{s_j}\big)$. In the Brownian coordinates the hyperplane $\{y_j=\ell\}$ becomes the hyperplane

$$\Big\{W_{s_j}=\frac{\ln(\ell/S_0)-(\mu-\frac12\sigma^2)s_j}{\sigma}\Big\},$$

and these finitely many hyperplanes cut $\mathbb{R}^m$ into finitely many disjoint boxes: the open ones, on which (b) gives continuity, and those lying inside a hyperplane, on which $V=0$ by definition. By (a) and Lemma 19.20(i), applied coordinate by coordinate, $\partial_S V(S_t,t;Y)$ is therefore a function $\xi(t,W_t,(W_{s_1},\dots,W_{s_m}))$ of exponential growth, continuous on $[t_0,T']\times\mathbb{R}\times\mathcal{D}$ for each box $\mathcal{D}$ and every $T'<T$: the form required by Definition 19.22(a)–(b). It is an admissible holding on $[t_0,T']$ for every $T'<T$. The parameter clause of Corollary 19.31, applied with the outer function $(S,t)\mapsto e^{-rt}V(S,t;y)$, asks of the composite

$$(s,w)\mapsto e^{-rs}\,V\big(S_0e^{(\mu-\sigma^2/2)s+\sigma w},s;y\big),$$

regarded also as a function of the Brownian values that determine the prices y, what Theorem 19.30(ii) asks of its function: on each box $\mathcal{D}$ of Brownian values, the composite and its partial derivatives in s, in w and twice in w are continuous in s, w and the Brownian values on $[t_0,T']\times\mathbb{R}\times\mathcal{D}$, and these four functions have exponential growth in w and in the Brownian values. Here w is the Brownian coordinate of the current price, not a log price. By the chain rule, with $S=S_0e^{(\mu-\sigma^2/2)s+\sigma w}$, the three derivatives are $e^{-rs}\big(\partial_tV-rV+(\mu-\frac12\sigma^2)S\,\partial_SV\big)$, $e^{-rs}\sigma S\,\partial_SV$ and $e^{-rs}\sigma^2\big(S^2\,\partial_{SS}V+S\,\partial_SV\big)$. Continuity: on an open box it follows from (b), the factors S and S^2 being continuous; on a box inside a hyperplane all four functions vanish. Growth: by (a), V, ∂_tV, ∂_SV and $\partial_{SS}V$ have moderate growth jointly in S and in the prices y on $(0,\infty)\times[t_0,T']$; multiplying by S or S^2 keeps moderate growth, and Lemma 19.20(i), applied

coordinate by coordinate, turns moderate growth in the prices into exponential growth in their Brownian coordinates. Finally, the initial capital $V(S_{t_0}, t_0; Y)$ is a function of exponential growth of Brownian values at times $\le t_0$, as Definition 20.1(i) requires.

Replication. Run Step 3 on $[t_0, T)$ with the parameter clause of Corollary 19.31, in which Y is frozen, in place of the corollary itself. For every y the function $V(\cdot, \cdot\,; y)$ satisfies $\mathcal{L}V = 0$: by (i) when y lies off the hyperplanes, and because $V = 0$ when it lies on one. So the computation that led to (F.3) gives, for $t_0 \le t < T$, with probability one,

$$e^{-rt}V(S_t, t; Y) = e^{-rt_0}V(S_{t_0}, t_0; Y) + \int_{t_0}^{t} \partial_S V(S_s, s; Y)\, d\big(e^{-rs}S_s\big),$$

which is (20.3) for the strategy of (v): its value is $V(S_t, t; Y)$ on $[t_0, T)$. For the terminal value, run Step 4. With probability one Y lies off the hyperplanes, because each S_{s_j} with $s_j > 0$ is lognormal. The discontinuity points of $f(\cdot\,; Y)$ lie among the finitely many fixed levels of the hyperplanes $\{S = \ell\}$ whenever Y lies off the hyperplanes $\{y_j = \ell\}$, and S_T is lognormal, so with probability one S_T is a continuity point of $f(\cdot\,; Y)$. Then (ii) for $V(\cdot, \cdot\,; Y)$ gives $V(S_t, t; Y) \to f(S_T; Y)$ as $t \to T$, with probability one. By (a), $|V(S_t, t; Y)| \le c' M(Y)\,(S_t^{c'} + S_t^{-c'})$ for $t_0 \le t \le T$: a function of moderate growth of the prices $(S_t, S_{s_1}, \dots, S_{s_m})$, hence of exponential growth of the Brownian values $(W_t, W_{s_1}, \dots, W_{s_m})$ (Lemma 19.20(i)), whose moments of every order are finite and bounded uniformly in t (Lemma 19.20(iii)). The fourth-moment lemma (Lemma 19.12(iv)) gives the mean-square limit $f(S_T; Y)$, which is the terminal value. □

F.4 Proof of Corollary 20.11: Payoffs observed on finitely many dates

The statement is on page 322. The notation and the standing assumptions are those of §20.2.

Proof. By backward induction on j: we show that each stage value is a payoff of the class to which Theorem 20.7(v) applies, chain the stage strategies, and compute the initial capital.

(a) Each $V^{(j)}$ is in $\mathcal{G}_j$. This holds for $j = J$ by hypothesis. Suppose that $V^{(j)} \in \mathcal{G}_j$ for some $j \ge 2$, with $|V^{(j)}| \le c \prod_{i \le j}(x_i^c + x_i^{-c})$. *Continuity.* Let $(x_1, \dots, x_{j-1})$ lie off the hyperplanes $\{x_i = \ell\}$, $i \le j-1$, of $V^{(j)}$, and let a sequence of points converge to it. For every value of Z outside the finitely many at which $x_{j-1}\exp(\nu\Delta t_j + \sigma\sqrt{\Delta t_j}\,Z)$ equals the level of a hyperplane $\{x_j = \ell\}$, the function $V^{(j)}$ is continuous at the limit point $(x_1, \dots, x_{j-1}, x_{j-1}\exp(\cdots))$, so the quantity inside the expectation that defines $V^{(j-1)}$ converges with probability one. Its fourth moments are bounded along the sequence, by the growth bound and Lemma 18.35(i). Lemma 19.12(iv) gives the convergence of the expectations: $V^{(j-1)}$ is continuous outside finitely many hyperplanes. *Growth.* For $x > 0$, by Lemma 18.35(i),

$$\mathbb{E}\Big[\big(x\, e^{\nu\Delta t_j + \sigma\sqrt{\Delta t_j}\,Z}\big)^{\pm c}\Big] = x^{\pm c}\, e^{\pm c\nu\Delta t_j + c^2\sigma^2\Delta t_j/2} \le c'' x^{\pm c}, \qquad c'' = e^{c|\nu|\Delta t_j + c^2\sigma^2\Delta t_j/2},$$

so the expectation of the last factor $x_j^c + x_j^{-c}$ of the bound, at $x_j = x_{j-1}\exp(\cdots)$, is at most $c''(x_{j-1}^c + x_{j-1}^{-c})$. Hence

$$|V^{(j-1)}| \le e^{|r|\Delta t_j}\, c\, c'' \prod_{i \le j-2} (x_i^c + x_i^{-c}) \cdot \big(x_{j-1}^c + x_{j-1}^{-c}\big)^2,$$

and $(x^c + x^{-c})^2 \le 2\,(x^{2c} + x^{-2c})$, while $x^c + x^{-c} \le 2\,(x^{2c} + x^{-2c})$. So $V^{(j-1)}$ has moderate growth, with c replaced by $2c$ and a larger multiplier, and $V^{(j-1)} \in \mathcal{G}_{j-1}$.

(b) The stage strategies. Let $2 \le j \le J$. By (a), $(S, y) \mapsto V^{(j)}(y_1, \dots, y_{j-1}, S)$ is in $\mathcal{G}_j$, which is the hypothesis of Theorem 20.7(v) on $[t_{j-1}, t_j)$ with the maturity t_j, the start date t_{j-1} and the $m = j-1$ earlier prices $Y = (S_{t_1}, \dots, S_{t_{j-1}})$, observed at the positive times $t_1 < \cdots < t_{j-1}$. The replication cost given there is $V(S_{t_{j-1}}, t_{j-1}; Y)$, where for y off the hyperplanes

$$V(y_{j-1}, t_{j-1}; y) = e^{-r\Delta t_j}\, \mathbb{E}\, V^{(j)}\big(y_1, \dots, y_{j-1},\ y_{j-1}e^{\nu\Delta t_j + \sigma\sqrt{\Delta t_j}\,Z}\big) = V^{(j-1)}(y_1, \dots, y_{j-1})$$

by the definition of $V^{(j-1)}$. So the capital that the stage-j strategy needs at t_{j-1} equals $V^{(j-1)}(S_{t_1},\dots,S_{t_{j-1}})$ with probability one: the two can differ only when Y lies on a hyperplane, where (v) sets V to zero, and that event has probability zero because $0<t_1$. Its terminal value at t_j is $V^{(j)}(S_{t_1},\dots,S_{t_j})$. For $j=1$, $V^{(1)}\in\mathcal{G}$ and Theorem 20.7(iii) with the maturity t_1 gives a strategy on $[0,t_1)$ with the initial capital $V^{(0)}$ and the terminal value $V^{(1)}(S_{t_1})$.

(c) Concatenation. By (b), the terminal value at t_j of the stage-j strategy equals, with probability one, the capital that the stage-$(j+1)$ strategy needs, for $j=1,\dots,J-1$; equality with probability one is enough, because values are defined up to events of probability zero. So the stage strategies, run one after the other, form a strategy (Definition 20.1(iv), with the switch dates $t_1,\dots,t_{J-1}$) with the initial capital $V^{(0)}$ and the terminal value $V^{(J)}(S_{t_1},\dots,S_{t_J})=f(S_{t_1},\dots,S_{t_J})$.

(d) The initial capital is the risk-neutral expectation. Under Definition 20.6 with no dividends, $S^{\mathbb{Q}}_{t_j}=S^{\mathbb{Q}}_{t_{j-1}}\exp\left(\nu\Delta t_j+\sigma(W_{t_j}-W_{t_{j-1}})\right)$, where the increment has the law of $\sqrt{\Delta t_j}\,Z$ and is independent of $(S^{\mathbb{Q}}_{t_1},\dots,S^{\mathbb{Q}}_{t_{j-1}})$ (Definition 19.5(2) and (3)). Lemma 19.17, which integrates out the independent increment first, therefore gives, for $j=J,\dots,2$,

$$\mathbb{E}\big[e^{-rt_j}\,V^{(j)}(S^{\mathbb{Q}}_{t_1},\dots,S^{\mathbb{Q}}_{t_j})\big]=\mathbb{E}\big[e^{-rt_{j-1}}\,V^{(j-1)}(S^{\mathbb{Q}}_{t_1},\dots,S^{\mathbb{Q}}_{t_{j-1}})\big],$$

the integrability that the lemma asks for coming from moderate growth and the lognormal moments (Lemma 19.20(iii)). Indeed, with the earlier prices frozen at $(x_1,\dots,x_{j-1})$, the expectation over the increment is

$$\begin{aligned}e^{-rt_j}\,\mathbb{E}\,V^{(j)}\big(x_1,\dots,x_{j-1},\ x_{j-1}e^{\nu\Delta t_j+\sigma\sqrt{\Delta t_j}\,Z}\big)&=e^{-rt_j}\,e^{r\Delta t_j}\,V^{(j-1)}(x_1,\dots,x_{j-1})\\&=e^{-rt_{j-1}}\,V^{(j-1)}(x_1,\dots,x_{j-1}),\end{aligned}$$

by the definition of $V^{(j-1)}$. Also $\mathbb{E}\big[e^{-rt_1}\,V^{(1)}(S^{\mathbb{Q}}_{t_1})\big]=V^{(0)}$ by the definition of $V^{(0)}$. Chaining these identities from $j=J$ down to $j=1$ gives $e^{-rT}\,\mathbb{E}f(S^{\mathbb{Q}}_{t_1},\dots,S^{\mathbb{Q}}_{t_J})=V^{(0)}$, which is the stated formula.

(e) The price. The arbitrage argument of Step 5 of the proof of Theorem 20.7, run with the concatenated strategy of (c) in place of the strategy of its part (iii), shows that any time-0 price other than $V^{(0)}$ is an arbitrage. □

F.5 Proof of Proposition 20.29: At-the-money-forward approximation

The statement is on page 335. The notation and the standing assumptions are those of §20.5.

Proof. We apply Taylor's theorem (Theorem 7.31) to the normal distribution function at 0, in six steps.

Step 1: the exact value. At $K=F$, $\ln(F/K)=0$, so $d_1=\sigma\sqrt{T}/2$ and $d_2=-\sigma\sqrt{T}/2$. By (20.8) and $\Phi(-y)=1-\Phi(y)$,

$$C=e^{-rT}F\left[\Phi\Big(\frac{\sigma\sqrt{T}}{2}\Big)-\Phi\Big(-\frac{\sigma\sqrt{T}}{2}\Big)\right]=e^{-rT}F\left[2\,\Phi\Big(\frac{\sigma\sqrt{T}}{2}\Big)-1\right]=2e^{-rT}F\,\chi\Big(\frac{\sigma\sqrt{T}}{2}\Big),$$

with $\chi(y)=\Phi(y)-\frac{1}{2}$. By the parity clause of Theorem 20.22, $C-\mathcal{P}=e^{-rT}(F-K)=0$, so the put equals the call; for $T_F=T$ and $r\ge 0$ this is also the last sentence of Corollary 17.11(iii).

Step 2: the derivatives of χ. Since $\Phi(y)=\int_{-\infty}^{y}\varphi(z)\,dz$ and φ is continuous, $\chi'=\varphi$. Differentiating $\varphi(y)=e^{-y^2/2}/\sqrt{2\pi}$ gives $\varphi'(y)=-y\,\varphi(y)$, so

$$\chi''(y)=-y\,\varphi(y),\qquad \chi'''(y)=-\varphi(y)+y^2\varphi(y)=(y^2-1)\,\varphi(y),$$

and χ has three continuous derivatives on the real line. At the expansion point, $\chi(0)=\Phi(0)-\frac{1}{2}=0$, $\chi'(0)=\varphi(0)=1/\sqrt{2\pi}=0.3989$ and $\chi''(0)=0$.

Step 3: the upper bound. For $y \geq 0$,

$$\chi(y) = \Phi(y) - \Phi(0) = \int_0^y \varphi(z)\,dz \leq \varphi(0)\,y,$$

because $\varphi(z) \leq \varphi(0)$ for every z.

Step 4: the extremes of χ'''. One more differentiation gives the derivative of χ''':

$$2y\,\varphi(y) - (y^2-1)\,y\,\varphi(y) = y\,(3-y^2)\,\varphi(y),$$

which is positive on $(0,\sqrt{3})$ and negative on $(\sqrt{3},\infty)$. So on $[0,\sqrt{3}]$ the function χ''' increases from $\chi'''(0) = -\varphi(0)$ to $\chi'''(\sqrt{3}) = 2\,\varphi(\sqrt{3}) = 0.178$, and on $[\sqrt{3},\infty)$ it decreases and stays positive, since $y^2 > 1$ there. Hence $-\varphi(0) \leq \chi'''(y) \leq 2\,\varphi(\sqrt{3})$ for $y \geq 0$, and for every real y because χ''' is even. On the real line χ''' has its minimum $-\varphi(0) = -0.399$ at 0 and its maximum 0.178 at $\pm\sqrt{3}$; in particular $|\chi'''| \leq \varphi(0)$ everywhere.

Step 5: the lower bound. By (7.3) with the Taylor polynomial of degree 2 at the expansion point 0, and by Step 2,

$$\chi(y) = \chi(0) + \chi'(0)\,y + \tfrac{1}{2}\chi''(0)\,y^2 + \frac{1}{2}\int_0^y (y-z)^2\,\chi'''(z)\,dz = \varphi(0)\,y + \frac{1}{2}\int_0^y (y-z)^2\,\chi'''(z)\,dz.$$

For $y > 0$ the weight $(y-z)^2$ is nonnegative and $\chi''' \geq -\varphi(0)$ by Step 4, so the remainder is at least

$$-\frac{\varphi(0)}{2}\int_0^y (y-z)^2\,dz = -\frac{\varphi(0)\,y^3}{6}.$$

Hence $\chi(y) \geq \varphi(0)\,y\,(1 - y^2/6)$ for $y \geq 0$. (Since $|\chi'''| \leq \varphi(0)$, the bound (7.4) with $M = \varphi(0)$ gives the same information without the sign: $|\chi(y) - \varphi(0)\,y| \leq \varphi(0)\,|y|^3/6$ for every real y.)

Step 6: conclusion. Put $y = \sigma\sqrt{T}/2$ in Steps 3 and 5 and multiply by $2e^{-rT}F$. Since $2\,\varphi(0)\,y = \sigma\sqrt{T}/\sqrt{2\pi}$ and $y^2/6 = \sigma^2 T/24$, Step 1 gives

$$e^{-rT}F\,\frac{\sigma\sqrt{T}}{\sqrt{2\pi}}\left(1 - \frac{\sigma^2 T}{24}\right) \leq C \leq e^{-rT}F\,\frac{\sigma\sqrt{T}}{\sqrt{2\pi}},$$

which is (20.11). The right side is C^{lin}, and the two sides differ by $C^{\mathrm{lin}}\,\sigma^2 T/24$, so

$$0 \leq C^{\mathrm{lin}} - C \leq C^{\mathrm{lin}}\,\frac{\sigma^2 T}{24} = e^{-rT}F\,\frac{(\sigma\sqrt{T})^3}{24\sqrt{2\pi}}.$$

This is the absolute error bound. It is a constant multiple of $(\sigma\sqrt{T})^3$, the constant being $e^{-rT}F/(24\sqrt{2\pi})$, which gives the O-statement. Dividing by $C^{\mathrm{lin}} > 0$ gives the relative error measured against the linear rule, $(C^{\mathrm{lin}} - C)/C^{\mathrm{lin}} \leq \sigma^2 T/24$. For the relative error measured against the price, let $\sigma^2 T < 24$. The left side of (20.11) then gives $C \geq C^{\mathrm{lin}}(1 - \sigma^2 T/24) > 0$, and dividing the upper estimate of $C^{\mathrm{lin}} - C$ by this lower estimate of C gives

$$\frac{C^{\mathrm{lin}} - C}{C} \leq \frac{C^{\mathrm{lin}}\,\sigma^2 T/24}{C^{\mathrm{lin}}\,(1 - \sigma^2 T/24)} = \frac{\sigma^2 T/24}{1 - \sigma^2 T/24}.$$

□

F.6 Proof of Proposition 20.38: Existence and uniqueness of implied volatility, and the at-the-money Newton guarantee

The statement is on page 341. The notation and the standing assumptions are those of §20.6.

Proof. We combine the positive vega of Proposition 20.35 with the intermediate value theorem for (i) and (ii), and for (iii) we use the concavity of the at-the-money price to trap the Newton iterates below the root.

Part (i): continuity and monotonicity. In the form of the Setup of §20.5, d_1 and d_2 are continuous functions of σ on $(0,\infty)$, and Φ is continuous, so (20.8) is continuous in σ. By Proposition 20.35 it is differentiable with a positive derivative, which is continuous in σ. Let $0<\sigma'<\sigma''$. By Corollary 7.32 at order zero (the mean value theorem) there is a $\sigma^*\in[\sigma',\sigma'']$ with

$$C^{\mathrm{BSM}}(\sigma'')-C^{\mathrm{BSM}}(\sigma')=\partial_\sigma C^{\mathrm{BSM}}(\sigma^*)\,(\sigma''-\sigma')>0,$$

so C^{BSM} is strictly increasing.

Part (i): the two limits. They are limits (1) and (3) of Remark 20.28, proved again here so that the proposition rests on no remark. Recall that $\Phi(y)\to 1$ as $y\to\infty$, that $\Phi(y)\to 0$ as $y\to-\infty$ and that $\Phi(0)=\frac12$. As $\sigma\to 0$, the term $\sigma\sqrt{T}/2$ tends to 0. If $F>K$, then $\ln(F/K)/(\sigma\sqrt{T})\to+\infty$, so d_1 and d_2 tend to $+\infty$ and $C^{\mathrm{BSM}}(\sigma)\to e^{-rT}(F-K)$. If $F<K$, then $\ln(F/K)/(\sigma\sqrt{T})\to-\infty$, so d_1 and d_2 tend to $-\infty$ and $C^{\mathrm{BSM}}(\sigma)\to 0$. If $F=K$, then $d_1=\sigma\sqrt{T}/2$ and $d_2=-\sigma\sqrt{T}/2$ both tend to 0, and $C^{\mathrm{BSM}}(\sigma)\to e^{-rT}(\frac12F-\frac12K)=0$. In all three cases the limit is $e^{-rT}(F-K)^+$. As $\sigma\to\infty$, $\ln(F/K)/(\sigma\sqrt{T})\to 0$ and $\sigma\sqrt{T}/2\to\infty$, so $d_1\to+\infty$ and $d_2\to-\infty$, and $C^{\mathrm{BSM}}(\sigma)\to e^{-rT}(F\cdot 1-K\cdot 0)=e^{-rT}F$.

Part (ii): the formula stays strictly inside the bounds. Let $\sigma>0$. For every $\sigma'<\sigma/2$, part (i) gives $C^{\mathrm{BSM}}(\sigma')<C^{\mathrm{BSM}}(\sigma/2)$; letting $\sigma'\to 0$ gives $e^{-rT}(F-K)^+\le C^{\mathrm{BSM}}(\sigma/2)$, and $C^{\mathrm{BSM}}(\sigma/2)<C^{\mathrm{BSM}}(\sigma)$. In the same way $C^{\mathrm{BSM}}(\sigma)<C^{\mathrm{BSM}}(2\sigma)\le e^{-rT}F$. Hence

$$e^{-rT}(F-K)^+\;<\;C^{\mathrm{BSM}}(\sigma)\;<\;e^{-rT}F\qquad\text{for every }\sigma>0,$$

and a price on or outside the bounds is not a value of C^{BSM}: it has no implied volatility. A price equal to a bound is a limit of the formula that no volatility attains.

Part (ii): a price outside the bounds is an arbitrage. We compare payoffs with Lemma 16.8, which places no condition on the sign of r. The package of forwards and deposit in the proof of Corollary 20.20(i) pays F_T at T and costs $e^{-rT}F$; its construction uses no sign of r. Since $F_T\ge 0$, its payoff dominates the call payoff, $F_T\ge(F_T-K)^+$. The call payoff in turn dominates the payoff F_T-K of that package less a deposit that pays K, which costs $e^{-rT}F-Ke^{-rT}$ by Lemma 16.8(iv), and it dominates the payoff 0 of the empty portfolio. Lemma 16.8(i) therefore gives

$$C^{\mathrm{mkt}}\le e^{-rT}F,\qquad C^{\mathrm{mkt}}\ge e^{-rT}(F-K),\qquad C^{\mathrm{mkt}}\ge 0,$$

that is, $e^{-rT}(F-K)^+\le C^{\mathrm{mkt}}\le e^{-rT}F$, and the proof of that lemma turns a quote that violates one of these inequalities into an arbitrage: if $C^{\mathrm{mkt}}>e^{-rT}F$, sell the call, hold the package and deposit the difference; if $C^{\mathrm{mkt}}<e^{-rT}(F-K)$, buy the call, sell the package short, deposit Ke^{-rT} and keep the difference, the position paying $(F_T-K)^+-F_T+K=(K-F_T)^+\ge 0$ at T; if $C^{\mathrm{mkt}}<0$, buy the call. For $r\ge 0$ and $T_F=T$ the package does the work of the prepaid forward of the Setup of §17.2, and the three inequalities are those of Proposition 17.6(ii)–(iii).

Part (ii): existence and uniqueness. Let C^{mkt} lie strictly inside the bounds. Since $C^{\mathrm{BSM}}(\sigma)\to e^{-rT}(F-K)^+<C^{\mathrm{mkt}}$ as $\sigma\to 0$, there is a σ' with $C^{\mathrm{BSM}}(\sigma')<C^{\mathrm{mkt}}$. Since $C^{\mathrm{BSM}}(\sigma)\to e^{-rT}F>C^{\mathrm{mkt}}$ as $\sigma\to\infty$, there is a σ'' with $C^{\mathrm{BSM}}(\sigma'')>C^{\mathrm{mkt}}$, and $\sigma'<\sigma''$ because C^{BSM} is increasing. The function is continuous on $[\sigma',\sigma'']$, so by the intermediate value theorem it takes the value C^{mkt} at some point of (σ',σ''). A strictly increasing function takes each value at most once, so that point is the only implied volatility.

Part (ii): the put. By the parity clause of Theorem 20.22, an identity between the two formulas, $\mathcal{P}^{\mathrm{BSM}}(\sigma)=C^{\mathrm{BSM}}(\sigma)-e^{-rT}(F-K)$ for every σ. So σ is an implied volatility of the put price $\mathcal{P}^{\mathrm{mkt}}$ if and only if it is an implied volatility of the call price $\mathcal{P}^{\mathrm{mkt}}+e^{-rT}(F-K)$. By the call case there is exactly one such σ when

$$e^{-rT}(F-K)^+\;<\;\mathcal{P}^{\mathrm{mkt}}+e^{-rT}(F-K)\;<\;e^{-rT}F,$$

and none otherwise. Subtract $e^{-rT}(F-K)$ from the three members. Since $x^+-x=(-x)^+$ for every real x, the left member becomes $e^{-rT}(K-F)^+$, and the right member becomes $e^{-rT}K$: the

condition is $e^{-rT}(K-F)^+ < \mathcal{P}^{\mathrm{mkt}} < e^{-rT}K$, the put bounds (those of Proposition 17.6(ii)–(iii) when $r \geq 0$ and $T_F = T$) with strict inequalities.

Part (iii), Step 1: concavity. At $K = F$, Proposition 20.29 gives $C^{\mathrm{BSM}}(\sigma) = e^{-rT}F\,[2\,\Phi(\sigma\sqrt{T}/2) - 1]$, and Proposition 20.35 gives $\partial_\sigma C^{\mathrm{BSM}}(\sigma) = e^{-rT}F\,\sqrt{T}\,\varphi(\sigma\sqrt{T}/2)$. Since $\varphi'(y) = -y\,\varphi(y)$ (Step 2 of the proof of Proposition 20.29), one more differentiation gives

$$\partial_{\sigma\sigma} C^{\mathrm{BSM}}(\sigma) = e^{-rT}F\,\sqrt{T}\cdot\frac{\sqrt{T}}{2}\cdot\Big(-\frac{\sigma\sqrt{T}}{2}\Big)\varphi\Big(\frac{\sigma\sqrt{T}}{2}\Big) = -\,e^{-rT}F\,\frac{T}{2}\,\frac{\sigma\sqrt{T}}{2}\,\varphi\Big(\frac{\sigma\sqrt{T}}{2}\Big) \;<\; 0$$

for every $\sigma > 0$, and this second derivative is continuous in σ. By Corollary 7.32 at order zero, applied to $\partial_\sigma C^{\mathrm{BSM}}$ as in part (i), the vega is strictly decreasing on $(0,\infty)$.

Part (iii), Step 2: the start lies at or below the root. At $K = F$ the bounds of (ii) read $0 < C^{\mathrm{mkt}} < e^{-rT}F$, so σ_{imp} exists, and $\sigma^{(0)} > 0$. The upper bound of (20.11), applied at the volatility $\sigma^{(0)}$, and the definition of $\sigma^{(0)}$ give

$$C^{\mathrm{BSM}}(\sigma^{(0)}) \;\leq\; e^{-rT}F\,\frac{\sigma^{(0)}\sqrt{T}}{\sqrt{2\pi}} = C^{\mathrm{mkt}} = C^{\mathrm{BSM}}(\sigma_{\mathrm{imp}}).$$

If $\sigma^{(0)}$ exceeded σ_{imp}, part (i) would give $C^{\mathrm{BSM}}(\sigma^{(0)}) > C^{\mathrm{BSM}}(\sigma_{\mathrm{imp}})$. Hence $\sigma^{(0)} \leq \sigma_{\mathrm{imp}}$.

Part (iii), Step 3: one Newton step. We show that $0 < \sigma^{(k)} \leq \sigma_{\mathrm{imp}}$ implies $\sigma^{(k)} \leq \sigma^{(k+1)} \leq \sigma_{\mathrm{imp}}$. Suppose $0 < \sigma^{(k)} \leq \sigma_{\mathrm{imp}}$. By part (i), $C^{\mathrm{BSM}}(\sigma^{(k)}) \leq C^{\mathrm{BSM}}(\sigma_{\mathrm{imp}}) = C^{\mathrm{mkt}}$, and the vega is positive, so the fraction in the iteration is nonnegative: $\sigma^{(k+1)} \geq \sigma^{(k)}$. If $\sigma^{(k)} = \sigma_{\mathrm{imp}}$, the fraction is zero and $\sigma^{(k+1)} = \sigma_{\mathrm{imp}}$. Otherwise Corollary 7.32 at order zero gives a $\sigma^* \in [\sigma^{(k)}, \sigma_{\mathrm{imp}}]$ with

$$C^{\mathrm{mkt}} - C^{\mathrm{BSM}}(\sigma^{(k)}) = C^{\mathrm{BSM}}(\sigma_{\mathrm{imp}}) - C^{\mathrm{BSM}}(\sigma^{(k)}) = \partial_\sigma C^{\mathrm{BSM}}(\sigma^*)\,(\sigma_{\mathrm{imp}} - \sigma^{(k)}).$$

By Step 1 the vega is decreasing and $\sigma^* \geq \sigma^{(k)}$, so $\partial_\sigma C^{\mathrm{BSM}}(\sigma^*) \leq \partial_\sigma C^{\mathrm{BSM}}(\sigma^{(k)})$, and therefore

$$C^{\mathrm{mkt}} - C^{\mathrm{BSM}}(\sigma^{(k)}) \;\leq\; \partial_\sigma C^{\mathrm{BSM}}(\sigma^{(k)})\,(\sigma_{\mathrm{imp}} - \sigma^{(k)}) :$$

the tangent at $\sigma^{(k)}$ lies above the concave graph at σ_{imp}. Dividing by the positive vega at $\sigma^{(k)}$ gives $\sigma^{(k+1)} - \sigma^{(k)} \leq \sigma_{\mathrm{imp}} - \sigma^{(k)}$, that is, $\sigma^{(k+1)} \leq \sigma_{\mathrm{imp}}$.

Part (iii), Step 4: convergence. By Step 2 and by induction on k with Step 3, every iterate is defined and $0 < \sigma^{(0)} \leq \sigma^{(k)} \leq \sigma^{(k+1)} \leq \sigma_{\mathrm{imp}}$ for every k. A nondecreasing sequence of real numbers that is bounded above converges, so the iterates have a limit $\sigma^{(\infty)}$ in $[\sigma^{(0)}, \sigma_{\mathrm{imp}}]$, a positive number. The functions C^{BSM} and $\partial_\sigma C^{\mathrm{BSM}}$ are continuous at $\sigma^{(\infty)}$, and the vega is positive there, so letting $k \to \infty$ in the iteration gives

$$\sigma^{(\infty)} = \sigma^{(\infty)} + \frac{C^{\mathrm{mkt}} - C^{\mathrm{BSM}}(\sigma^{(\infty)})}{\partial_\sigma C^{\mathrm{BSM}}(\sigma^{(\infty)})}.$$

Hence $C^{\mathrm{BSM}}(\sigma^{(\infty)}) = C^{\mathrm{mkt}}$, and $\sigma^{(\infty)} = \sigma_{\mathrm{imp}}$ by the uniqueness in (ii). □

F.7 Proof of Corollary 20.41: Options on an asset with a yield or known dividends; Merton, 1973

The statement is on page 344. The notation and the standing assumptions are those of §20.7.

Proof. We prove (i) by reduction to a non-dividend asset, in six steps, and (ii) by the same reduction.

Step 1: the reinvested holding is a non-dividend asset. By Proposition 16.15(i), the strategy that holds $e^{-q(T-t)}$ units of the asset at each time t, all income being reinvested, is self-financing and pays no income; it is worth $\tilde{S}_t = S_t e^{-q(T-t)}$ at t and $\tilde{S}_T = S_T$ at T. Under (A1), $\tilde{S}_t =$

$\tilde S_0 \exp\big((\mu+q-\frac12\sigma^2)t+\sigma W_t\big)$ with $\tilde S_0 = S_0 e^{-qT}$ (Setup of §20.7): the geometric Brownian motion of Definition 19.27 with the drift $\mu+q$ and the volatility σ. Its price function $(t,x)\mapsto \tilde S_0 e^{(\mu+q-\sigma^2/2)t+\sigma x}$ is in $C_E^{1,2}$ for the reason given for the stock in the Setup of §20.1, so $\tilde S$ is an instrument of Definition 20.1. The market made of the bank and of $\tilde S$ therefore satisfies (A1)–(A6), with $\tilde S$ in the role of the stock and $\mu+q$ in the role of μ.

Step 2: the theorem for the reinvested holding. Theorem 20.7 holds for every constant drift in (A1), and the drift appears in none of its conclusions. Apply it to the market of Step 1 and to the payoff $f(\tilde S_T) = f(S_T)$. Its price function (20.5), with the price variable written $\tilde S$, is

$$\tilde V(\tilde S,t) = e^{-r(T-t)}\,\mathbb{E}\,f\big(\tilde S\exp\big((r-\tfrac12\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z\big)\big) \quad \text{for } t<T, \qquad \tilde V(\tilde S,T) = f(\tilde S).$$

By parts (i) and (ii) of the theorem, $\tilde V$ is infinitely differentiable on $(0,\infty)\times[0,T)$, each of its partial derivatives has moderate growth on $(0,\infty)\times[0,T']$ for every $T'<T$, it solves equation (20.4) in the variable $\tilde S$, it satisfies $|\tilde V(\tilde S,t)|\le c'\,(\tilde S^{c'}+\tilde S^{-c'})$ on $(0,\infty)\times[0,T]$, and $\tilde V(\tilde S,t)\to f(S^*)$ as $(\tilde S,t)\to(S^*,T)$ at every continuity point S^* of f. By part (iii), the strategy with the initial capital $\tilde V(\tilde S_0,0)$ that holds $\partial_{\tilde S}\tilde V(\tilde S_t,t)$ units of $\tilde S$ has the value $\tilde V(\tilde S_t,t)$ at every $t<T$ and the terminal value $f(\tilde S_T)=f(S_T)$. By part (iv), $\tilde V(\tilde S_0,0)$ is the arbitrage-free time-0 price of the claim.

Step 3: back to the price of the asset; (a), (c) and (d). Put $\tilde S = Se^{-q(T-t)}$ in the formula for $\tilde V$. Since

$$Se^{-q(T-t)}\exp\big((r-\tfrac12\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z\big) = S\exp\big((r-q-\tfrac12\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z\big),$$

and $r-q-\sigma^2/2=\nu$, the result is (20.12): $\tilde V(Se^{-q(T-t)},t)=V(S,t)$ for $t<T$, and at $t=T$ both sides are $f(S)$. This is the identity of (a); its call and put formulas are proved in Step 5. The value of the strategy of Step 2 is therefore $\tilde V(\tilde S_t,t)=V(S_t,t)$. By the chain rule applied to (a),

$$\partial_S V(S,t) = e^{-q(T-t)}\,\partial_{\tilde S}\tilde V(Se^{-q(T-t)},t),$$

so the holding is $\partial_{\tilde S}\tilde V(\tilde S_t,t) = e^{q(T-t)}\,\partial_S V(S_t,t)$ units of $\tilde S$. Each unit of $\tilde S$ is $e^{-q(T-t)}$ units of the asset, so the strategy holds $\partial_S V(S_t,t)$ units of the asset. This is (c). At $t=0$, $V(S_0,0)=\tilde V(\tilde S_0,0)$ is the price of the claim, and by (20.12)

$$V(S_0,0) = e^{-rT}\,\mathbb{E}\,f\big(S_0\exp(\nu T+\sigma\sqrt{T}\,Z)\big) = e^{-rT}\,\mathbb{E}\,f(S_T^{\mathbb{Q}}) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}f(S_T),$$

because $S_T^{\mathbb{Q}} = S_0\exp(\nu T+\sigma W_T)$ by Definition 20.6(i) with the yield q, and W_T has the law of $\sqrt{T}\,Z$ (Definition 19.5(2)). This is (d).

Step 4: the properties of V; (b). This step is bookkeeping of growth bounds, which other proofs cite through (b); a first reading may pass to Step 5. Call a function of (S,t) a *term* if it is a constant, times a power $e^{-iq(T-t)}$, times a power S^j, with integers $i\ge0$ and $j\ge0$, times a partial derivative of $\tilde V$ of some order ($\tilde V$ itself included) evaluated at $(Se^{-q(T-t)},t)$. By (a), V is a term. The map $(S,t)\mapsto(Se^{-q(T-t)},t)$ sends $(0,\infty)\times[0,T)$ into itself, and the derivative of $e^{-q(T-t)}$ in t is $q\,e^{-q(T-t)}$. Differentiating a term in S gives the same term with S^j replaced by jS^{j-1} (zero when $j=0$), plus the same term with i raised by one and its derivative of $\tilde V$ differentiated once more in $\tilde S$. Differentiating a term in t gives iq times the term, plus the same term with its derivative of $\tilde V$ differentiated once more in t, plus q times the term with i and j each raised by one and its derivative of $\tilde V$ differentiated once more in $\tilde S$. Both results are finite sums of terms. By induction on the order, V is infinitely differentiable on $(0,\infty)\times[0,T)$ and every partial derivative of V is a finite sum of terms; for instance

$$\partial_t V = \partial_t\tilde V + q\,Se^{-q(T-t)}\,\partial_{\tilde S}\tilde V \qquad\text{and}\qquad \partial_{SS}V = e^{-2q(T-t)}\,\partial_{\tilde S\tilde S}\tilde V,$$

the derivatives of $\tilde V$ being evaluated at $(Se^{-q(T-t)},t)$.

Each term has moderate growth on $(0,\infty)\times[0,T']$. For $0\le t\le T$ the factor $e^{-q(T-t)}$ lies between $e^{-|q|T}$ and $e^{|q|T}$. So $e^{-iq(T-t)}\le e^{i|q|T}$; the power S^j is at most S^j+S^{-j}; and by Step 2 the derivative of $\tilde V$ in the term is at most

$$c\Big(\big(Se^{-q(T-t)}\big)^c+\big(Se^{-q(T-t)}\big)^{-c}\Big)\;\le\;c\,e^{c|q|T}\,\big(S^c+S^{-c}\big)$$

in absolute value, with a constant c that depends on T' and on the derivative. Products and sums of functions of moderate growth have moderate growth (Lemma 19.20(i)). In the same way the bound of Step 2 on $[0,T]$ gives $|V(S,t)|=|\tilde V(Se^{-q(T-t)},t)|\le c'e^{c'|q|T}\,(S^{c'}+S^{-c'})$ on $(0,\infty)\times[0,T]$, a bound of the form stated in (b) with the constant $c'e^{c'|q|T}\ge c'$ in the place of c', because $x^a+x^{-a}\le x^b+x^{-b}$ for $x>0$ and $0\le a\le b$: the difference $(x^b+x^{-b})-(x^a+x^{-a})=(x^b-x^a)(1-x^{-a-b})$ is a product of two factors of the same sign, each being zero or of the sign of $\ln x$. Finally, when $(S,t)\to(S^*,T)$, the point $(Se^{-q(T-t)},t)$ tends to (S^*,T), so $V(S,t)=\tilde V(Se^{-q(T-t)},t)\to f(S^*)$ at every continuity point S^* of f, by Step 2. This is (b).

Step 5: the call and the put. The forward price of the asset for delivery at T is $F=Se^{(r-q)T}$ (Proposition 16.15(ii)). Take $T_F=T$ in Definition 20.16, so that $F_T=S_T$. By Definition 20.6(i), under $\mathbb{Q}$

$$S_T=S\exp\big((r-q-\tfrac12\sigma^2)T+\sigma\sqrt{T}\,Z\big)=F\exp\big(-\tfrac12\sigma^2T+\sigma\sqrt{T}\,Z\big),$$

which is the law (20.6), and by (d) the pricing rule of Definition 20.16(ii) holds. So the Black-76 model holds for the asset, and Theorem 20.22 gives (20.8) and (20.9) with this F. Substituting $e^{-rT}F=Se^{-qT}$ and $\ln(F/K)=\ln(S/K)+(r-q)T$ gives the stated formulas.

At a time $t<T$, since $\nu(T-t)=(r-q)(T-t)-\frac12\sigma^2(T-t)$, the price function (20.12) of the call is

$$V(S,t)=e^{-r(T-t)}\,\mathbb{E}\Big(Se^{(r-q)(T-t)}\exp\big(-\tfrac12\sigma^2(T-t)+\sigma\sqrt{T-t}\,Z\big)-K\Big)^+.$$

The variable inside the expectation is the right side of (20.6) with $Se^{(r-q)(T-t)}$ in the place of F and $T-t$ in the place of T, and the computation that proved Theorem 20.22 used only that expression, as in the paragraph after Corollary 20.23. So $V(S,t)=e^{-r(T-t)}\big[Se^{(r-q)(T-t)}\,\Phi(d_1)-K\,\Phi(d_2)\big]$, with $\ln(S/K)+(r-q)(T-t)$ in the place of $\ln(F/K)$ and $T-t$ in the place of T in d_1 and d_2; this is the call formula of (a). The put formula of (a) is the same computation with the put payoff. Under $\mathbb{Q}$ the asset itself drifts at the rate $r-q$, and $\mathbb{E}^{\mathbb{Q}}S_T=F$ by Lemma 19.38(i), as after Definition 20.16.

Step 6: earlier fixings and several dates; (e). This step is used only where a yield meets earlier fixings or several dates; a first reading may pass to part (ii). Both reductions rest on one remark: at every date $s\le T$ the price $S_s=\tilde S_se^{q(T-s)}$ is a fixed multiple of $\tilde S_s$.

Earlier fixings. Let t_0, the times $0<s_1<\cdots<s_m\le t_0$, the vector $Y=(S_{s_1},\dots,S_{s_m})$ and the payoff $f(S_T;Y)$ be as in Theorem 20.7(v). Put $\tilde Y=(\tilde S_{s_1},\dots,\tilde S_{s_m})$ and

$$\tilde f(\tilde S;\tilde y)=f\big(\tilde S;\;\tilde y_1e^{q(T-s_1)},\dots,\tilde y_me^{q(T-s_m)}\big),$$

so that $f(S_T;Y)=\tilde f(\tilde S_T;\tilde Y)$, because $\tilde S_T=S_T$. The function $\tilde f$ is in $\mathcal{G}_{m+1}$. A positive rescaling of a coordinate moves each hyperplane $\{y_j=\ell\}$ of f to the hyperplane $\{\tilde y_j=\ell e^{-q(T-s_j)}\}$ and keeps continuity off the hyperplanes; and each factor of the growth bound obeys $(\tilde y_je^{q(T-s_j)})^c+(\tilde y_je^{q(T-s_j)})^{-c}\le e^{c|q|T}(\tilde y_j^c+\tilde y_j^{-c})$, so $\tilde f$ has moderate growth. Theorem 20.7(v), applied in the market of Step 1 to $\tilde f$, gives a function $\tilde V(\tilde S,t;\tilde y)$, defined by (20.5) with $\tilde f(\cdot\,;\tilde y)$, and a strategy that starts at t_0 with the capital $\tilde V(\tilde S_{t_0},t_0;\tilde Y)$, holds $\partial_{\tilde S}\tilde V(\tilde S_t,t;\tilde Y)$ units of $\tilde S$, has the value $\tilde V(\tilde S_t,t;\tilde Y)$ on $[t_0,T)$ and the terminal value $\tilde f(\tilde S_T;\tilde Y)=f(S_T;Y)$. Let $y_j=\tilde y_je^{q(T-s_j)}$. The substitution of Step 3 gives

$$\tilde V\big(Se^{-q(T-t)},t;\tilde y\big)=e^{-r(T-t)}\,\mathbb{E}\,f\big(S\exp\big(\nu(T-t)+\sigma\sqrt{T-t}\,Z\big);\;y\big),$$

the price function $V(S,t;y)$ of Theorem 20.7(v) with $\nu=r-q-\sigma^2/2$; and $\tilde y$ lies on a hyperplane of $\tilde f$ exactly when y lies on a hyperplane of f, where both functions are zero. For each y off the

hyperplanes the function $\tilde V(\cdot,\cdot\,;\tilde y)$ has properties (i) and (ii) of the theorem, so the argument of Step 4, run with $\tilde V(\cdot,\cdot\,;\tilde y)$ in the place of $\tilde V$, gives the properties of (b) for $V(\cdot,\cdot\,;y)$, and the general clause of Corollary 20.45(i) gives (20.13). So the replication cost at t_0 is $V(S_{t_0},t_0;Y)$, the value on $[t_0,T)$ is $V(S_t,t;Y)$, and by the chain rule of Step 3 the holding is $e^{q(T-t)}\,\partial_S V(S_t,t;Y)$ units of $\tilde S$.

Several dates. In this step $t_1,\dots,t_J$ are the fixing dates of Corollary 20.11; the yield model has no ex-dividend dates. Let $0<t_1<\cdots<t_J=T$ and $f\in\mathcal{G}_J$ be as in that corollary. Then $f(S_{t_1},\dots,S_{t_J})=\tilde f(\tilde S_{t_1},\dots,\tilde S_{t_J})$ with

$$\tilde f(y_1,\dots,y_J)=f\big(y_1e^{q(T-t_1)},\dots,y_Je^{q(T-t_J)}\big),$$

and $\tilde f\in\mathcal{G}_J$ by the argument just given: each hyperplane $\{x_j=\ell\}$ of f becomes $\{y_j=\ell e^{-q(T-t_j)}\}$, and moderate growth is kept with a larger constant. Corollary 20.11, applied in the market of Step 1 to $\tilde f$, prices the claim and replicates it in stages, each stage holding $\tilde S$. Let $\tilde V^{(j)}$ be its value functions, with $\tilde V^{(J)}=\tilde f$ and the recursion

$$\tilde V^{(j-1)}(y_1,\dots,y_{j-1})=e^{-r\Delta t_j}\,\mathbb{E}\,\tilde V^{(j)}\Big(y_1,\dots,y_{j-1},\ y_{j-1}\exp\big((r-\tfrac12\sigma^2)\Delta t_j+\sigma\sqrt{\Delta t_j}\,Z\big)\Big),$$

and define the value functions in the prices of the asset by

$$V^{(j)}(x_1,\dots,x_j)=\tilde V^{(j)}\big(x_1e^{-q(T-t_1)},\dots,x_je^{-q(T-t_j)}\big).$$

Then $V^{(J)}=f$. Put $y_i=x_ie^{-q(T-t_i)}$ in the recursion. Since $T-t_{j-1}=(T-t_j)+\Delta t_j$,

$$x_{j-1}e^{-q(T-t_{j-1})}\exp\big((r-\tfrac12\sigma^2)\Delta t_j+\sigma\sqrt{\Delta t_j}\,Z\big)=\Big[x_{j-1}\exp\big(\nu\Delta t_j+\sigma\sqrt{\Delta t_j}\,Z\big)\Big]\,e^{-q(T-t_j)},$$

so the last argument of $\tilde V^{(j)}$ is $x_je^{-q(T-t_j)}$ with $x_j=x_{j-1}\exp(\nu\Delta t_j+\sigma\sqrt{\Delta t_j}\,Z)$, and the recursion reads

$$V^{(j-1)}(x_1,\dots,x_{j-1})=e^{-r\Delta t_j}\,\mathbb{E}\,V^{(j)}\Big(x_1,\dots,x_{j-1},\ x_{j-1}\exp\big(\nu\Delta t_j+\sigma\sqrt{\Delta t_j}\,Z\big)\Big)\,:$$

the recursion of Corollary 20.11 with $\nu=r-q-\sigma^2/2$. In the same way $\tilde S_0\exp\big((r-\frac12\sigma^2)t_1+\sigma\sqrt{t_1}\,Z\big)=S_0\exp(\nu t_1+\sigma\sqrt{t_1}\,Z)\,e^{-q(T-t_1)}$, so the price is $\tilde V^{(0)}=e^{-rt_1}\,\mathbb{E}\,V^{(1)}\big(S_0\exp(\nu t_1+\sigma\sqrt{t_1}\,Z)\big)$, the number $V^{(0)}$ of the corollary with this ν. The risk-neutral path of $\tilde S$ is $\tilde S^{\mathbb{Q}}_t=\tilde S_0\exp\big((r-\frac12\sigma^2)t+\sigma W_t\big)$ (Definition 20.6(i) with no yield), and

$$\tilde S^{\mathbb{Q}}_t\,e^{q(T-t)}=S_0\exp\big((r-q-\tfrac12\sigma^2)t+\sigma W_t\big)=S^{\mathbb{Q}}_t,$$

the path of Definition 20.6(i) with the yield q. Hence $\tilde f(\tilde S^{\mathbb{Q}}_{t_1},\dots,\tilde S^{\mathbb{Q}}_{t_J})=f(S^{\mathbb{Q}}_{t_1},\dots,S^{\mathbb{Q}}_{t_J})$, and the price given by the corollary is $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}f(S_{t_1},\dots,S_{t_J})$. On the stage $[t_{j-1},t_j)$ the price function in the market of Step 1 is (20.5) with the maturity t_j and the payoff $\tilde V^{(j)}$, the earlier values frozen. The substitution $\tilde S=Se^{-q(T-t)}$ turns it into

$$e^{-r(t_j-t)}\,\mathbb{E}\,V^{(j)}\Big(x_1,\dots,x_{j-1},\ S\exp\big(\nu(t_j-t)+\sigma\sqrt{t_j-t}\,Z\big)\Big),$$

because $-q(T-t)+(r-\frac12\sigma^2)(t_j-t)=\nu(t_j-t)-q(T-t_j)$: the stage function of Corollary 20.11 with this ν. By the chain rule the stage holds $e^{q(T-t)}$ times its derivative in S, in units of $\tilde S$. This proves (e) and completes (i).

Part (ii). We reduce to a non-dividend asset again. Consider the position of the proof of Proposition 16.13: one share, and for each i a loan of $D_ie^{-rt_i}$ taken at time 0, which has grown to D_i at t_i and is repaid then by the dividend. It is self-financing and pays no net income. At time t it is worth the share minus the loans still outstanding, each grown to $D_ie^{-r(t_i-t)}$: that is S_t-I_t, and at T no loan is left, so it is worth $S_T-I_T=S_T$. In the escrowed-dividend model its

price is the geometric Brownian motion $(S-I)\exp\big((\mu-\frac12\sigma^2)t+\sigma W_t\big)$, with volatility σ, so it is an instrument of Definition 20.1, and the market made of the bank and of this position satisfies (A1)–(A6) with the position in the role of the stock. Theorem 20.7(iv), applied to it and to the payoff $f(S_T-I_T)=f(S_T)$, gives the price $e^{-rT}\,\mathbb{E}\,f\big((S-I)\exp\big((r-\frac12\sigma^2)T+\sigma\sqrt{T}\,Z\big)\big)$. With $F=(S-I)e^{rT}$, the forward price of Proposition 16.13, this says that the Black-76 model holds with $T_F=T$ and $F_T=S_T$: the argument of f is $F\exp(-\frac12\sigma^2T+\sigma\sqrt{T}\,Z)$. Theorem 20.22 then gives the call and the put, and the substitutions $e^{-rT}F=S-I$ and $\ln(F/K)=\ln\big((S-I)/K\big)+rT$ give the stated formulas, which are those of Corollary 20.23 with $S-I$ in the place of S. □

F.8 Proof of Theorem 20.52: Black's model for futures options; Black, 1976

The statement is on page 354. The notation and the standing assumptions are those of §20.8.

Proof. We reduce to Theorem 20.7 with the prepaid forward as the traded asset, in six steps.

Step 1: the market and the payoff in the prepaid variable. By the check in the Setup of §20.8 that the prepaid forward is a traded instrument, $\tilde F$ is an instrument of Definition 20.1 that pays no income, and it is a geometric Brownian motion with the drift μ_F+r and the volatility σ. The market made of the bank and of $\tilde F$ therefore satisfies (A1)–(A6), with $\tilde F$ in the role of the stock and μ_F+r in the role of μ. Since $F_T=e^{r(T_F-T)}\,\tilde F_T$, the claim pays

$$f(F_T)=\tilde f(\tilde F_T)\qquad\text{with}\qquad \tilde f(\tilde F)=f\big(e^{r(T_F-T)}\,\tilde F\big).$$

The function $\tilde f$ is in $\mathcal{G}$. It is continuous except at the finitely many points $e^{-r(T_F-T)}\ell$, where ℓ runs over the discontinuities of f. And if $|f(x)|\le c\,(x^c+x^{-c})$, then

$$|\tilde f(\tilde F)|\le c\,\Big(e^{cr(T_F-T)}\,\tilde F^c+e^{-cr(T_F-T)}\,\tilde F^{-c}\Big)\le c\,e^{c|r|(T_F-T)}\,\big(\tilde F^c+\tilde F^{-c}\big).$$

Step 2: the theorem for the prepaid forward. Theorem 20.7 holds for every constant drift in (A1), and the drift appears in none of its conclusions. Apply it to the market of Step 1 and to the payoff $\tilde f$. Its price function (20.5), with the price variable written $\tilde F$, is

$$\tilde V(\tilde F,t)=e^{-r(T-t)}\,\mathbb{E}\,\tilde f\big(\tilde F\exp\big((r-\tfrac12\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z\big)\big)\quad\text{for } t<T,\qquad \tilde V(\tilde F,T)=\tilde f(\tilde F).$$

By its parts (i) and (ii), $\tilde V$ is infinitely differentiable on $(0,\infty)\times[0,T)$, with partial derivatives of moderate growth on $(0,\infty)\times[0,T']$ for every $T'<T$; it solves equation (20.4) in the variable $\tilde F$; it satisfies $|\tilde V(\tilde F,t)|\le c'\,(\tilde F^{c'}+\tilde F^{-c'})$ on $(0,\infty)\times[0,T]$; and $\tilde V(\tilde F,t)\to\tilde f(\tilde F^*)$ as $(\tilde F,t)\to(\tilde F^*,T)$ at every continuity point $\tilde F^*$ of $\tilde f$. By its part (iii), the strategy with the initial capital $\tilde V(\tilde F_0,0)$ that holds $\partial_{\tilde F}\tilde V(\tilde F_t,t)$ units of $\tilde F$ has the value $\tilde V(\tilde F_t,t)$ at every $t<T$ and the terminal value $\tilde f(\tilde F_T)=f(F_T)$. By its part (iv), $\tilde V(\tilde F_0,0)$ is the arbitrage-free time-0 price of the claim.

Step 3: back to the futures price. Put $\tilde F=Fe^{-r(T_F-t)}$ in the formula for $\tilde V$. The argument of f becomes

$$e^{r(T_F-T)}\,Fe^{-r(T_F-t)}\exp\big((r-\tfrac12\sigma^2)(T-t)+\sigma\sqrt{T-t}\,Z\big)=F\exp\big(-\tfrac12\sigma^2(T-t)+\sigma\sqrt{T-t}\,Z\big),$$

because $r(T_F-T)-r(T_F-t)+r(T-t)=0$. Hence

$$\tilde V\big(Fe^{-r(T_F-t)},t\big)=V^{\text{fut}}(F,t)\qquad\text{on } (0,\infty)\times[0,T],$$

the case $t=T$ being $\tilde f(Fe^{-r(T_F-T)})=f(F)$. Since $\tilde F_t=F_te^{-r(T_F-t)}$, the value of the strategy of Step 2 is $\tilde V(\tilde F_t,t)=V^{\text{fut}}(F_t,t)$. By the chain rule,

$$\partial_F V^{\text{fut}}(F,t)=e^{-r(T_F-t)}\,\partial_{\tilde F}\tilde V\big(Fe^{-r(T_F-t)},t\big),$$

so the holding of the strategy is $\partial_{\tilde F}\tilde V(\tilde F_t,t)=e^{r(T_F-t)}\,\partial_F V^{\text{fut}}(F_t,t)$ units of $\tilde F$. This is the first sentence of (ii).

Step 4: the properties of V^{fut}; (i). The properties pass from $\tilde{V}$ to V^{fut} by the chain rule, exactly as in Step 4 of the proof of Corollary 20.41(i), with $\tilde{F}$, F and $e^{-r(T_F-t)}$ in the places of $\tilde{S}$, S and $e^{-q(T-t)}$. The derivative of $e^{-r(T_F-t)}$ in t is $r\,e^{-r(T_F-t)}$, so every partial derivative of V^{fut} is a finite sum of terms, each a constant, times a power of $e^{-r(T_F-t)}$, times a power of F, times a partial derivative of $\tilde{V}$ evaluated at $(Fe^{-r(T_F-t)}, t)$. For $0 \le t \le T$ the factor $e^{-r(T_F-t)}$ lies between $e^{-|r|T_F}$ and $e^{|r|T_F}$, so moderate growth at the point $Fe^{-r(T_F-t)}$ is moderate growth in F, and products and sums of functions of moderate growth have moderate growth (Lemma 19.20(i)). For the bound on $[0,T]$, Step 2 gives

$$|V^{\text{fut}}(F,t)| = |\tilde{V}\big(Fe^{-r(T_F-t)},t\big)| \le c'\Big(\big(Fe^{-r(T_F-t)}\big)^{c'} + \big(Fe^{-r(T_F-t)}\big)^{-c'}\Big) \le c'e^{c'|r|T_F}\,\big(F^{c'}+F^{-c'}\big),$$

a bound of the form stated in (i) with the constant $c'e^{c'|r|T_F} \ge c'$, because $x^a + x^{-a} \le x^b + x^{-b}$ for $x > 0$ and $0 \le a \le b$. For the terminal limit, let $(F,t) \to (F^*,T)$ with F^* a continuity point of f. Then $Fe^{-r(T_F-t)} \to F^*e^{-r(T_F-T)}$, which is a continuity point of $\tilde{f}$, so by Step 2

$$V^{\text{fut}}(F,t) = \tilde{V}\big(Fe^{-r(T_F-t)},t\big) \to \tilde{f}\big(F^*e^{-r(T_F-T)}\big) = f(F^*).$$

Finally $\tilde{V}$ has continuous partial derivatives and solves (20.4) in the variable $\tilde{F}$, which is the hypothesis of Corollary 20.45(ii); so V^{fut} solves (20.14).

Step 5: the price; (ii). At $t=0$, $V^{\text{fut}}(F,0) = \tilde{V}(\tilde{F}_0,0)$ is the arbitrage-free price of the claim, by Steps 2 and 3. By (20.15) and Proposition 20.51(ii),

$$V^{\text{fut}}(F,0) = e^{-rT}\,\mathbb{E}\,f\big(F\exp(-\tfrac{1}{2}\sigma^2T + \sigma\sqrt{T}\,Z)\big) = e^{-rT}\,\mathbb{E}\,f(F_T^{\mathbb{Q}}) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}f(F_T).$$

So both parts of Definition 20.16 hold for the futures price: the law (i) is Proposition 20.51(ii), and the pricing rule (ii) has just been proved.

Step 6: calls and puts; (iii). Definition 20.16 is the only hypothesis of the formulas of Theorem 20.22, which therefore gives (20.8) and (20.9) with F the current futures price, and the first sentence of its parity clause gives $C - \mathcal{P} = e^{-rT}(F-K)$. The volatility in the formulas is that of the futures price. When $F_t = S_te^{(r-q)(T_F-t)}$, the logarithm of the futures price is the logarithm of S_t plus a deterministic function of t, so the two volatilities are equal. At a time $t < T$, the variable inside the expectation of (20.15) is the right side of (20.6) with $T-t$ in the place of T, and the computation that proved Theorem 20.22 used only that expression; so for the call and the put payoffs, (20.15) is (20.8) or (20.9) with $T-t$ in the place of T, which are the price functions of (iii). □

F.9 Proof of Proposition 20.54: American futures calls can be exercised early

The statement is on page 356. The notation and the standing assumptions are those of §20.8.

Proof. We price the claim that pays F_T by replication and compare payoffs with Lemma 16.8. Steps 1–4 use no sign of r.

Step 1: the claim paying F_T at T is worth $e^{-rT}F$. Hold $e^{r(T_F-T)}$ units of the prepaid forward $\tilde{F}$ of the Setup of §20.8 (the check there that it is a traded instrument): that is, $e^{r(T_F-T)}$ long forwards for delivery at T_F struck at F, which cost nothing, and $e^{r(T_F-T)}\,e^{-rT_F}F = e^{-rT}F$ in the bank. At T the forwards are closed, where each is worth $(F_T-F)e^{-r(T_F-T)}$ by Proposition 16.17(i), so together they are worth $F_T - F$, and the deposit has grown to F. The package costs $e^{-rT}F$ and pays F_T; by Lemma 16.8(iii), every claim paying F_T at T has the price $e^{-rT}F$. No model of the futures price was used.

Step 2: the upper bound. The payoff of Step 1 dominates the call payoff strictly: $F_T - (F_T-K)^+ = \min(F_T,K) > 0$, because $K > 0$ and the futures price is positive. By Lemma 16.8(ii), $C < e^{-rT}F$.

Step 3: parity and the lower bound. One call minus one put pays $(F_T-K)^+-(K-F_T)^+=F_T-K$ at T, the payoff of the package of Step 1 less a deposit that pays K. By Lemma 16.8(iii) and (iv), $C-\mathcal{P}=e^{-rT}F-Ke^{-rT}$. For $r\geq 0$, the standing assumption of §17.2, this is the parity of Corollary 17.11(iii), with the claim paying F_T in the role of the prepaid forward. The payoffs of the call and of the put are nonnegative, so $C\geq 0$ and $\mathcal{P}\geq 0$ (Lemma 16.8(i), against the empty portfolio). Hence $C\geq e^{-rT}(F-K)$ and $C\geq 0$, that is, $C\geq e^{-rT}(F-K)^+$.

Step 4: the American call. Its payoff is nonnegative under every exercise policy, so $C^{\text{am}}\geq 0$. If $C^{\text{am}}<F-K$, buy the call and exercise it at once: by Definition 20.50 this brings the cash amount $F-K$ and a long futures position at the latest settlement price, which is worth zero and is closed at no cost. The profit $F-K-C^{\text{am}}>0$ is riskless, an arbitrage. So $C^{\text{am}}\geq(F-K)^+$; this is the analogue of the exercise-value bound of Proposition 17.6(i), with the futures position in the place of the share.

Step 5: the comparison. By Step 3, $C<F-K$ if and only if $e^{-rT}(F-K)+\mathcal{P}<F-K$, that is, $\mathcal{P}<(1-e^{-rT})(F-K)$. This proves (i). Now let $r>0$. When $K\leq F(1-e^{-rT})$, Step 2 alone gives $C<e^{-rT}F\leq F-K$. Under Black's model the put is given by (20.9), so $\mathcal{P}\leq e^{-rT}K\,\Phi(-d_2)$, the subtracted term $e^{-rT}F\,\Phi(-d_1)$ being nonnegative. As $F\to\infty$ with K, σ and T fixed, $d_2\to+\infty$ and $\Phi(-d_2)\to 0$, so $\mathcal{P}\to 0$, while $(1-e^{-rT})(F-K)\to+\infty$ because $r>0$. So the condition holds for every F large enough, whatever σ is. This proves the first sentence of (ii).

Step 6: the rest of (ii). In a state in which $C<F-K$, Step 4 gives $C^{\text{am}}\geq F-K>C$. A policy that never exercises before T receives at most $(F_T-K)^+$ at T, the payoff of the European call, which can be bought for C. The holder who exercises at once and buys the European call out of the proceeds holds that same payoff and, in addition, $F-K-C>0$ in cash. So waiting until T is strictly worse than exercising at once, and a policy that never exercises before T is not optimal. □

G Proofs for §21: The Greeks and Hedging

G.1 Proof of Proposition 21.6: Gamma and theta of European options

The statement is on page 363. The notation and the standing assumptions are those of §21.2.

Proof. We differentiate the closed forms, cancelling the density terms with the key identity (21.4) as in the proof of Proposition 21.2, and then bound the normal density for the limits.

Part (i). By Proposition 21.2, $\Delta_C=e^{-q\tau}\Phi(d_1)$, and $\partial_S d_1=1/(S\sigma\sqrt{\tau})$, so the chain rule gives $\partial_S\Delta_C=e^{-q\tau}\varphi(d_1)/(S\sigma\sqrt{\tau})$. The put delta is $\Delta_C-e^{-q\tau}$, which differs from the call delta by a quantity that does not depend on S, so its derivative in S is the same.

Part (ii). With S, σ and r fixed, the prices depend on t through $\tau=T-t$ only, so $\Theta=\partial_t V=-\partial_\tau V$. Differentiating the call formula in τ by the product and chain rules,

$$\partial_\tau C=-qSe^{-q\tau}\Phi(d_1)+Se^{-q\tau}\varphi(d_1)\,\partial_\tau d_1+rKe^{-r\tau}\Phi(d_2)-Ke^{-r\tau}\varphi(d_2)\,\partial_\tau d_2.$$

By (21.4), $Ke^{-r\tau}\varphi(d_2)=Se^{-q\tau}\varphi(d_1)$, so the two density terms combine to

$$Se^{-q\tau}\varphi(d_1)\,\partial_\tau(d_1-d_2)=Se^{-q\tau}\varphi(d_1)\,\partial_\tau\bigl(\sigma\sqrt{\tau}\bigr)=\frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}}.$$

Hence $\Theta_C=-\partial_\tau C$ is the stated expression. For the put, differentiate the parity identity (21.3) in t. Since $\partial_t e^{-q(T-t)}=qe^{-q\tau}$ and $\partial_t e^{-r(T-t)}=re^{-r\tau}$,

$$\Theta_C-\Theta_{\mathcal{P}}=qSe^{-q\tau}-rKe^{-r\tau}.$$

Subtracting this from Θ_C replaces $\Phi(d_1)$ by $\Phi(d_1)-1=-\Phi(-d_1)$ and $\Phi(d_2)$ by $\Phi(d_2)-1=-\Phi(-d_2)$, which is the stated $\Theta_{\mathcal{P}}$.

An elementary inequality. Both estimates of part (iii) rest on one inequality. For every real y, Corollary 7.32 at order one, applied to the exponential function at the point 0, gives a point y^* between 0 and y with $e^y = 1 + y + \frac{1}{2}e^{y^*}y^2$. The last term is nonnegative, so

$$e^y \geq 1 + y \qquad \text{for every real } y. \tag{G.1}$$

Two consequences follow. For $y \geq 0$, (G.1) at $-y$ reads $e^{-y} \geq 1 - y$, and $e^{-y} \leq 1$ because the exponential is increasing. For $y > 0$, $e^{-y} = 1/e^y \leq 1/(1+y) < 1/y$.

Part (iii), at the money forward. Here $d_1 = \sigma\sqrt{\tau}/2$, so $\varphi(d_1) = e^{-d_1^2/2}/\sqrt{2\pi} = e^{-\sigma^2\tau/8}/\sqrt{2\pi}$. Substituting in (i) gives the stated Γ, and the two-sided bound on $e^{-\sigma^2\tau/8}$ is the first consequence of (G.1) with $y = \sigma^2\tau/8$. The factor $e^{-\sigma^2\tau/8}$ therefore differs from 1 by at most $\sigma^2\tau/8$, which is the term $O(\sigma^2\tau)$, and $1/\sqrt{2\pi} = 0.3989$ gives the rule $0.4/(S\sigma\sqrt{\tau})$. When $r = q = 0$, the forward price is $F = S$, and the thetas of (ii) reduce to their common density term, $-S\varphi(d_1)\sigma/(2\sqrt{\tau}) = -S\sigma e^{-\sigma^2\tau/8}/(2\sqrt{2\pi\tau})$; here $1/(2\sqrt{2\pi}) = 0.1995$ gives the rule $-0.2\,S\sigma/\sqrt{\tau}$. As $\tau \to 0$ the exponential factor tends to 1, so both quantities grow like $\tau^{-1/2}$.

Part (iii), away from the strike. Fix $S \neq K$ and put

$$c = \frac{|\ln(S/K)|}{2\sigma} > 0, \qquad \tau_0 = \frac{|\ln(S/K)|}{2|r-q| + \sigma^2} > 0.$$

By (21.1),

$$d_1 = \frac{\ln(S/K)}{\sigma\sqrt{\tau}} + \frac{(r - q + \sigma^2/2)\sqrt{\tau}}{\sigma}, \qquad d_2 = \frac{\ln(S/K)}{\sigma\sqrt{\tau}} + \frac{(r - q - \sigma^2/2)\sqrt{\tau}}{\sigma}.$$

For $0 < \tau \leq \tau_0$ the second term of each is at most $(|r-q| + \sigma^2/2)\tau/(\sigma\sqrt{\tau}) \leq |\ln(S/K)|/(2\sigma\sqrt{\tau})$ in absolute value, which is half the absolute value of the first term. Hence, for $0 < \tau \leq \tau_0$, both d_1 and d_2 have the sign of $\ln(S/K)$ and

$$|d_1| \geq \frac{c}{\sqrt{\tau}}, \qquad |d_2| \geq \frac{c}{\sqrt{\tau}}.$$

The second consequence of (G.1), with $y = c^2/(2\tau)$, now bounds the density:

$$\frac{\varphi(d_1)}{\sqrt{\tau}} \leq \frac{e^{-c^2/(2\tau)}}{\sqrt{2\pi\tau}} < \frac{2\tau}{c^2} \cdot \frac{1}{\sqrt{2\pi\tau}} = \frac{2\sqrt{\tau}}{c^2\sqrt{2\pi}} \longrightarrow 0 \qquad (\tau \to 0).$$

Since $\Gamma = \frac{e^{-q\tau}}{S\sigma} \cdot \frac{\varphi(d_1)}{\sqrt{\tau}}$ and the density term of theta is $-\frac{1}{2}Se^{-q\tau}\sigma \cdot \frac{\varphi(d_1)}{\sqrt{\tau}}$, both tend to 0. Next, if $S > K$ then $d_1, d_2 \geq c/\sqrt{\tau} \to +\infty$, and if $S < K$ then $d_1, d_2 \leq -c/\sqrt{\tau} \to -\infty$. A distribution function tends to 1 at $+\infty$ and to 0 at $-\infty$, so

$$\Phi(d_1),\ \Phi(d_2) \to \mathbf{1}\{S > K\}, \qquad \Phi(-d_1),\ \Phi(-d_2) \to \mathbf{1}\{S < K\}.$$

Together with $e^{-q\tau} \to 1$ and $e^{-r\tau} \to 1$, the last two terms of Θ_C in (ii) tend to $(qS - rK)\mathbf{1}\{S > K\}$, and those of $\Theta_{\mathcal{P}}$ to $(rK - qS)\mathbf{1}\{S < K\}$. At $r = q = 0$ both limits are 0. Finally, by the computation that gave $\Theta_C - \Theta_{\mathcal{P}}$, the theta of a long forward worth $Se^{-q\tau} - Ke^{-r\tau}$ is $qSe^{-q\tau} - rKe^{-r\tau}$, which tends to $qS - rK$; the theta of a short forward is its negative. □

G.2 Proof of Corollary 21.8: The theta–gamma identity for a book

The statement is on page 365. The notation and the standing assumptions are those of §21.2.

Proof. We verify the identity for each instrument and add up, by linearity in the positions.

Step 1 (each option). Take an option of the book, with strike K and expiry T. Its payoff, $f(x) = (x - K)^+$ or $f(x) = (K - x)^+$, is continuous with $|f(x)| \leq x + K$. With $c = 1 + K$, this

bound gives $|f(x)| \le c\,(x^c + x^{-c})$ for every $x > 0$: for $x \ge 1$, $x + K \le (1+K)x \le c\,x^c$, and for $0 < x < 1$, $x + K < 1 + K = c < c\,x^{-c}$. So f belongs to the class $\mathcal{G}$ of Definition 19.19(iv). Its price function is the function $V(S,t)$ of Corollary 20.41(i) for this payoff, which by the Setup of §21.1 is the closed form (21.2), and by Corollary 20.45(i) that function satisfies the pricing equation with a yield,

$$\partial_t V + (r-q)S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V = rV \qquad \text{on } (0,\infty)\times[0,T).$$

The reasons behind this citation are two. By Theorem 20.7(i), the price written in the reinvested variable $\tilde{S} = Se^{-q(T-t)}$ solves the equation (20.4) of Theorem 20.3; and Corollary 20.45 turns that equation into the one above by the chain rule. Both are statements about the price function alone, valid at every $\sigma > 0$.

Step 2 (forwards and cash). A long forward contract with delivery price K and delivery date T_F is worth $V = Se^{-q\tau_F} - Ke^{-r\tau_F}$ with $\tau_F = T_F - t$ (Proposition 16.17(ii), which applies because the underlying is an investment asset). Its derivatives are $\partial_t V = qSe^{-q\tau_F} - rKe^{-r\tau_F}$, $\partial_S V = e^{-q\tau_F}$ and $\partial_{SS} V = 0$, so

$$\partial_t V + (r-q)S\,\partial_S V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V = qSe^{-q\tau_F} - rKe^{-r\tau_F} + (r-q)Se^{-q\tau_F} = rV.$$

One unit of cash deposited at time 0 is worth $V = e^{rt}$ at time t, with $\partial_t V = re^{rt} = rV$ and $\partial_S V = \partial_{SS} V = 0$, so it satisfies the same equation.

Step 3 (the book). By Definition 21.1(i), $\Theta = \partial_t V$, $\Delta = \partial_S V$ and $\Gamma = \partial_{SS} V$. Steps 1 and 2 therefore say that every instrument of the book satisfies (21.5) at every $S > 0$ and every t before its own expiry or delivery date. Multiply the identity of instrument j by its position w_j and add. By Definition 21.1(ii) the value and the Greeks of the book are the w-weighted sums of those of the instruments, so the sum is (21.5) for the book. Part (a) is the case $r = q = 0$. For part (b), let $q = 0$. One share has value S, with $\Theta = 0$, $\Delta = 1$ and $\Gamma = 0$, so the left side of (21.5) is rS, which is r times its value: shares may be added to the book. A book that is delta-hedged with shares has $\Delta = 0$, and (21.5) reads $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma = r\Pi$.

Check by substitution. Step 1 can be verified for the call without the pricing equation. Substituting Propositions 21.2 and 21.6,

$$\begin{aligned}\Theta_C + (r-q)S\Delta_C + \tfrac{1}{2}\sigma^2 S^2\Gamma &= -\frac{Se^{-q\tau}\varphi(d_1)\,\sigma}{2\sqrt{\tau}} + qSe^{-q\tau}\Phi(d_1) - rKe^{-r\tau}\Phi(d_2)\\ &\quad + (r-q)Se^{-q\tau}\Phi(d_1) + \frac{\sigma^2 S^2}{2}\cdot\frac{e^{-q\tau}\varphi(d_1)}{S\sigma\sqrt{\tau}}.\end{aligned}$$

The last term equals $Se^{-q\tau}\varphi(d_1)\sigma/(2\sqrt{\tau})$ and cancels the first, and the two terms in $\Phi(d_1)$ add up to $rSe^{-q\tau}\Phi(d_1)$. What remains is $r\big[Se^{-q\tau}\Phi(d_1) - Ke^{-r\tau}\Phi(d_2)\big] = rC$. The put follows from the call and Step 2, because by (21.3) a put is a call minus a forward with delivery price K and delivery date T. □

G.3 Proof of Proposition 21.9: Vega and rho of European options; vega as gamma times maturity

The statement is on page 367. The notation and the standing assumptions are those of §21.3.

Proof. We cite part (i) and differentiate the closed forms for (ii), cancelling the density terms with the key identity (21.4) as in the proof of Proposition 21.2; part (iii) divides the vega by the gamma and bounds one exponential factor.

Part (i). By the Setup of §21.1, the prices (21.2) are the formulas (20.8) and (20.9) of Theorem 20.22 with T read as τ and $F = Se^{(r-q)\tau}$. At fixed S, r and q this forward price does not contain σ, so the

derivative in σ at fixed S is the derivative at fixed F of Proposition 20.35. That proposition gives the yield form $Se^{-q\tau}\varphi(d_1)\sqrt{\tau}$, the same for the call and the put and positive because $\varphi > 0$. At the money forward, $F = K$, the at-the-money clause of the same proposition, with $e^{-r\tau}F = Se^{-q\tau}$, gives the value $Se^{-q\tau}\varphi(\sigma\sqrt{\tau}/2)\sqrt{\tau}$ and its two-sided bound, between $\varphi(0)\,Se^{-q\tau}\sqrt{\tau}\,(1-\sigma^2\tau/8)$ and $\varphi(0)\,Se^{-q\tau}\sqrt{\tau}$. Since $\varphi(\sigma\sqrt{\tau}/2) = \varphi(0)\,e^{-\sigma^2\tau/8}$ and $\varphi(0) = 1/\sqrt{2\pi} = 0.3989$, the value is $Se^{-q\tau}\sqrt{\tau/(2\pi)}\,e^{-\sigma^2\tau/8}$, dividing the bound by $\varphi(0)\,Se^{-q\tau}\sqrt{\tau} > 0$ places the factor $e^{-\sigma^2\tau/8}$ between $1-\sigma^2\tau/8$ and 1, and the constant 0.3989 gives the rule $0.4\,Se^{-q\tau}\sqrt{\tau}$. The approximation of Proposition 20.29 is $e^{-r\tau}F\sigma\sqrt{\tau}/\sqrt{2\pi} = Se^{-q\tau}\sigma\sqrt{\tau}/\sqrt{2\pi}$, whose derivative in σ is $Se^{-q\tau}\sqrt{\tau/(2\pi)}$: the vega at the money forward without the factor $e^{-\sigma^2\tau/8}$.

Part (ii). With S, q, σ and τ fixed, (21.1) gives $\partial_r d_1 = \partial_r d_2 = \tau/(\sigma\sqrt{\tau}) = \sqrt{\tau}/\sigma$. Differentiating the call formula in r,

$$\begin{aligned}\partial_r C &= Se^{-q\tau}\varphi(d_1)\,\partial_r d_1 + \tau K e^{-r\tau}\Phi(d_2) - Ke^{-r\tau}\varphi(d_2)\,\partial_r d_2 \\ &= K\tau e^{-r\tau}\Phi(d_2) + \frac{\sqrt{\tau}}{\sigma}\Big(Se^{-q\tau}\varphi(d_1) - Ke^{-r\tau}\varphi(d_2)\Big) = K\tau e^{-r\tau}\Phi(d_2),\end{aligned}$$

the bracket being zero by (21.4): only the explicit derivative of the factor $e^{-r\tau}$ survives. For the put, differentiate (21.3) in r: the right side has derivative $K\tau e^{-r\tau}$, so $\mathcal{R}_C - \mathcal{R}_{\mathcal{P}} = K\tau e^{-r\tau}$ and $\mathcal{R}_{\mathcal{P}} = K\tau e^{-r\tau}(\Phi(d_2) - 1) = -K\tau e^{-r\tau}\Phi(-d_2)$. The signs follow from $0 < \Phi < 1$.

Part (iii). Divide (i) by the gamma of Proposition 21.6(i):

$$\frac{\mathcal{V}}{\Gamma} = Se^{-q\tau}\varphi(d_1)\sqrt{\tau}\cdot\frac{S\sigma\sqrt{\tau}}{e^{-q\tau}\varphi(d_1)} = \sigma\tau S^2.$$

At the money forward, Proposition 21.6(iii) and part (i) give the stated Γ and $\mathcal{V}$, with the common factor e^y, where $y = -(q+\sigma^2/8)\tau$. By (G.1), $e^y \geq 1 + y \geq 1 - (|q| + \sigma^2/8)\tau$; and $y \leq |q|\tau$, so $e^y \leq e^{|q|\tau}$ because the exponential is increasing. Both bounds tend to 1 as $\tau \to 0$, and so does e^y. Apart from this factor, Γ is the constant $1/(S\sigma\sqrt{2\pi})$ times $\tau^{-1/2}$, and $\mathcal{V}$ is the constant $S/\sqrt{2\pi}$ times $\tau^{1/2}$. □

G.4 Proof of Proposition 21.14: Profit-and-loss explain

The statement is on page 371. The notation and the standing assumptions are those of §21.4.

Proof. We apply Taylor's theorem to the value along the segment from the start to the end, and use Lemma 19.54 for the cross terms.

Step 1 (the segment). Consider the set of real numbers u for which the point $(S + u\Delta S,\ t + u\Delta t,\ \sigma + u\Delta\sigma)$ lies in $\mathcal{O}$. This point depends continuously on u and $\mathcal{O}$ is open, so the set is open. It contains 0 and 1, because the start and the end lie in $\mathcal{O}$, and therefore all of $[0,1]$, because $\mathcal{O}$ is convex. Being open, it contains an interval $(-\varepsilon', \varepsilon')$ around 0 and an interval $(1-\varepsilon', 1+\varepsilon')$ around 1 for some $\varepsilon' > 0$, hence the open interval $(-\varepsilon', 1+\varepsilon')$. On that interval define

$$\chi(u) = V(S + u\Delta S,\ t + u\Delta t,\ \sigma + u\Delta\sigma),$$

so that $\chi(0)$ is the value at the start, $\chi(1)$ the value at the end, and $\Delta V = \chi(1) - \chi(0)$. The chain rule, for a function with continuous partial derivatives composed with a straight path, differentiates such a composition as $\frac{d}{du}\,\psi(x + u\Delta x) = \sum_j \partial_j\psi(x + u\Delta x)\,\Delta x_j$. Applied to $\psi = V$, then to each $\psi = \partial_j V$, then to each $\psi = \partial_k\partial_j V$, all of which have continuous partial derivatives because V has continuous partial derivatives up to order three, it shows that χ is three times continuously differentiable, with

$$\chi'(u) = \sum_j \partial_j V\,\Delta x_j, \qquad \chi''(u) = \sum_{j,k}\partial_k\partial_j V\,\Delta x_j\Delta x_k, \qquad \chi'''(u) = \sum_{j,k,l}\partial_l\partial_k\partial_j V\,\Delta x_j\Delta x_k\Delta x_l, \tag{G.2}$$

the partial derivatives being evaluated at the point of the segment with parameter u.

Step 2 (Taylor). Theorem 7.31 at order two, applied to χ on the interval $(-\varepsilon', 1+\varepsilon')$ with expansion point 0 and evaluated at 1, gives

$$\chi(1) = \chi(0) + \chi'(0) + \tfrac{1}{2}\chi''(0) + \mathcal{E}_3, \qquad \mathcal{E}_3 = \frac{1}{2}\int_0^1 (1-u)^2\,\chi'''(u)\,du,$$

where $\mathcal{E}_3$ is the remainder (7.3).

Step 3 (the terms). By (G.2) at $u = 0$ and Definition 21.1(i),

$$\chi'(0) = \partial_S V\,\Delta S + \partial_t V\,\Delta t + \partial_\sigma V\,\Delta\sigma = \Delta\,\Delta S + \Theta\,\Delta t + \mathcal{V}\,\Delta\sigma,$$

and $\chi''(0) = \sum_{j,k} \partial_k\partial_j V\,\Delta x_j \Delta x_k$ is the quadratic form of the nine second partial derivatives at the start in the increment vector Δx. The function V has continuous partial derivatives up to order two on the open set $\mathcal{O}$, so Lemma 19.54 gives $\partial_k\partial_j V = \partial_j\partial_k V$: each of the three mixed derivatives appears twice in the form, with one value. Therefore

$$\begin{aligned}\tfrac{1}{2}\chi''(0) = {} & \tfrac{1}{2}\partial_{SS}V\,(\Delta S)^2 + \tfrac{1}{2}\partial_{tt}V\,(\Delta t)^2 + \tfrac{1}{2}\partial_{\sigma\sigma}V\,(\Delta\sigma)^2 \\ & + \partial_{St}V\,\Delta S\,\Delta t + \partial_{S\sigma}V\,\Delta S\,\Delta\sigma + \partial_{t\sigma}V\,\Delta t\,\Delta\sigma.\end{aligned}$$

The first term is the gamma term $\frac{1}{2}\Gamma\,(\Delta S)^2$, and the other five are exactly the second-order terms listed in $\mathcal{E}$. Without the lemma, half of the form would carry $\frac{1}{2}(\partial_\sigma\partial_S V + \partial_S\partial_\sigma V)\,\Delta S\,\Delta\sigma$ in place of $\partial_{S\sigma}V\,\Delta S\,\Delta\sigma$, and likewise for the other two pairs. Substituting $\chi'(0)$ and $\frac{1}{2}\chi''(0)$ in Step 2 gives (21.6) with the stated $\mathcal{E}$.

Step 4 (the remainder). For $0 \le u \le 1$ the point with parameter u lies on the segment, where every third partial derivative of V, in every order of differentiation, is at most M_3 in absolute value; no symmetry is needed here. By (G.2), whose third sum has 27 terms,

$$|\chi'''(u)| \le M_3 \sum_{j,k,l} |\Delta x_j|\,|\Delta x_k|\,|\Delta x_l| = M_3\big(|\Delta S| + |\Delta t| + |\Delta\sigma|\big)^3 \qquad (0 \le u \le 1).$$

The bound (7.4) at order two, with the right side of the last display as its constant and with $|x - x_0| = 1$, gives the stated bound on $\mathcal{E}_3$, since $3! = 6$.

Step 5 (the two consequences). For (i), each second derivative is at most M_2 in absolute value at the start, and $\varepsilon^4 \le \varepsilon^3$ because $\varepsilon \le 1$. The five second-order terms of $\mathcal{E}$ are therefore bounded, in the order in which they are listed, by

$$M_2\varepsilon^3, \qquad \tfrac{1}{2}M_2\varepsilon^4 \le \tfrac{1}{2}M_2\varepsilon^3, \qquad M_2\varepsilon^3, \qquad M_2\varepsilon^4 \le M_2\varepsilon^3, \qquad \tfrac{1}{2}M_2\varepsilon^4 \le \tfrac{1}{2}M_2\varepsilon^3,$$

which add up to $4M_2\varepsilon^3$. Moreover $|\Delta S| + |\Delta t| + |\Delta\sigma| \le \varepsilon + 2\varepsilon^2 \le 3\varepsilon$, so $|\mathcal{E}_3| \le \frac{27}{6}M_3\varepsilon^3 = \frac{9}{2}M_3\varepsilon^3$ by Step 4. Hence $|\mathcal{E}| \le 4M_2\varepsilon^3 + \frac{9}{2}M_3\varepsilon^3 \le \frac{9}{2}(M_2 + M_3)\varepsilon^3$.

For (ii), with $\Delta\sigma = 0$ only $\partial_{St}V\,\Delta S\,\Delta t$ and $\frac{1}{2}\partial_{tt}V\,(\Delta t)^2$ remain beside $\mathcal{E}_3$, and they are bounded by $M_2|\Delta S|\,\Delta t$ and $\frac{1}{2}M_2(\Delta t)^2$. For the remainder, every pair of numbers $y, z \ge 0$ satisfies $(y+z)^3 \le 4(y^3 + z^3)$, because

$$4(y^3+z^3) - (y+z)^3 = (y+z)(4y^2 - 4yz + 4z^2 - y^2 - 2yz - z^2) = 3(y+z)(y-z)^2 \ge 0.$$

With $y = |\Delta S|$ and $z = \Delta t$, Step 4 gives $|\mathcal{E}_3| \le \frac{M_3}{6}\cdot 4\big(|\Delta S|^3 + (\Delta t)^3\big) = \frac{2}{3}M_3\big(|\Delta S|^3 + (\Delta t)^3\big)$, and adding the three bounds gives the inequality of (ii). Since $(\Delta t)^3 \le (\Delta t)^2$ for $0 \le \Delta t \le 1$, its right side is at most $(M_2 + \frac{2}{3}M_3)\big(|\Delta S|^3 + |\Delta S|\,\Delta t + (\Delta t)^2\big)$.

The constants in the $O(\cdot)$ form. The constants M_2 and M_3 above depend on the segment, hence on the move. For the statement with $O(\cdot)$, fix the start and take a closed box around it inside $\mathcal{O}$, which exists because $\mathcal{O}$ is open. The second and third partial derivatives of V are continuous on the box, which is closed and bounded, so by the extreme value theorem their absolute values have maxima there; let M_2 be the largest of these maxima over the second derivatives and M_3 the largest over the third derivatives. A box is convex, so every segment that starts at the fixed point and ends inside the box lies in the box, and one pair of constants serves for every sufficiently small move. $\square$

G.5 Proof of Proposition 21.19: The expected P and L of a delta-hedged option, for every rebalancing grid

The statement is on page 376. The notation and the standing assumptions are those of §21.5.

Proof. We integrate out each future move with Lemma 19.17.

Step 1 (the P and L). At $r = 0$ no cash balance earns or pays interest, so the P and L at T is the payoff, minus the premium, plus the gains of the hedge, and the short position of h_i units gains $-h_i\,(S_{t_{i+1}} - S_{t_i})$ over $(t_i, t_{i+1}]$:

$$f(S_T) - V^{\mathrm{BSM}}(S_0, 0; \sigma_{\mathrm{imp}}) - \sum_{i=0}^{n-1} h_i\,(S_{t_{i+1}} - S_{t_i}).$$

Every term is integrable. For the payoff, $|f(x)| \le c\,(x^c + x^{-c})$ and S_T is lognormal, so $\mathbb{E}\,S_T^{\pm c}$ is finite (Lemma 18.35(i)). The hedge terms are integrable by hypothesis. The expectation of the P and L is therefore the sum of the expectations of its terms (Theorem 1.2, proved for discrete variables, in the general form declared in §19.2).

Step 2 (the payoff). By Definition 19.5, $W_T \sim \mathcal{N}(0, T)$, so $Z = W_T/\sqrt{T}$ is a standard normal variable and $S_T = S_0 \exp\bigl(-\sigma_{\mathrm{real}}^2 T/2 + \sigma_{\mathrm{real}}\sqrt{T}\,Z\bigr)$. At the rate $r = 0$ and the volatility σ_{real}, the number ν of Theorem 20.7 is $-\sigma_{\mathrm{real}}^2/2$, and its definition (20.5) at $t = 0$ reads $V^{\mathrm{BSM}}(S_0, 0; \sigma_{\mathrm{real}}) = \mathbb{E}\,f\bigl(S_0 \exp(-\sigma_{\mathrm{real}}^2 T/2 + \sigma_{\mathrm{real}}\sqrt{T}\,Z)\bigr)$. Hence

$$\mathbb{E}\,f(S_T) = V^{\mathrm{BSM}}(S_0, 0; \sigma_{\mathrm{real}}).$$

Only the definition of the function is used here, at the volatility σ_{real}; nothing is assumed about prices in a market.

Step 3 (each hedge term has mean zero). Fix i and write $S_{t_{i+1}} - S_{t_i} = S_{t_i}(Y_i - 1)$ with

$$Y_i = \exp\bigl(-\sigma_{\mathrm{real}}^2 (t_{i+1} - t_i)/2 + \sigma_{\mathrm{real}}(W_{t_{i+1}} - W_{t_i})\bigr).$$

The vector $X = (S_{t_0}, \dots, S_{t_i})$ is a function of the values $W_{t_1}, \dots, W_{t_i}$, and Y_i is a function of the increment $W_{t_{i+1}} - W_{t_i}$, which is independent of these values (Proposition 19.8(iv), restarted at t_i); so X and Y_i are independent. The increment $W_{t_{i+1}} - W_{t_i}$ has the law of $\sqrt{t_{i+1} - t_i}\,Z$, so Lemma 18.35(i) with $\theta = \sigma_{\mathrm{real}}\sqrt{t_{i+1} - t_i}$ gives

$$\mathbb{E}\,Y_i = e^{-\theta^2/2}\,\mathbb{E}\,e^{\theta Z} = e^{-\theta^2/2} e^{\theta^2/2} = 1.$$

Apply Lemma 19.17 to $\psi(x, y) = h_i(x)\,x_i\,(y - 1)$, where $x = (x_0, \dots, x_i)$ stands for the values of X: the function ψ is Borel measurable because h_i is, the variable $\psi(X, Y_i) = h_i\,(S_{t_{i+1}} - S_{t_i})$ is integrable by hypothesis, and

$$\bar{\psi}(x) = \mathbb{E}\,\psi(x, Y_i) = h_i(x)\,x_i\,\mathbb{E}\,(Y_i - 1) = 0.$$

Hence $\mathbb{E}\bigl[h_i\,(S_{t_{i+1}} - S_{t_i})\bigr] = \mathbb{E}\,\bar{\psi}(X) = 0$.

Step 4 (conclusion, and the delta hedge). By Steps 1 to 3 the expected P and L is $V^{\mathrm{BSM}}(S_0, 0; \sigma_{\mathrm{real}}) - V^{\mathrm{BSM}}(S_0, 0; \sigma_{\mathrm{imp}}) - 0$, which is the claim. It remains to check the two hypotheses for the delta hedge. The dates $t_0, \dots, t_{n-1}$ lie in $[0, t_{n-1}]$ and $t_{n-1} < T$, and $\partial_S V^{\mathrm{BSM}}(\cdot, \cdot\,; \sigma_{\mathrm{imp}})$ is infinitely differentiable, hence continuous, and has moderate growth on $(0, \infty) \times [0, t_{n-1}]$ (Theorem 20.7(i), a statement about the function (20.5) at any volatility, proved without (A1)–(A6)). So h_i is a continuous, hence Borel measurable, function of S_{t_i}, and $|h_i| \le c\,(S_{t_i}^c + S_{t_i}^{-c})$. Lognormal prices have moments of every order (Lemma 19.20(iii)), so $\mathbb{E}\,h_i^2$ and $\mathbb{E}\,(S_{t_{i+1}} - S_{t_i})^2$ are finite, and the Cauchy–Schwarz inequality (Lemma 19.12(ii)) gives $\mathbb{E}\,|h_i\,(S_{t_{i+1}} - S_{t_i})| < \infty$. For $h_i = 0$ there is nothing to check. □

G.6 Proof of Proposition 21.22: Continuous-time hedged P and L at constant realised volatility

The statement is on page 377. The notation and the standing assumptions are those of §21.5.

Proof. We apply Itô's lemma to the marked option and use the pricing equation at the marking volatility.

Step 1 (the bookkeeping, and Itô's lemma for the marked option). Fix $T' < T$. The option is an instrument of Definition 20.1 quoted at its mark: its price $V(S_t, t)$ is the function $(t, x) \mapsto V(S_0 e^{(\mu - \sigma_{\text{real}}^2/2)t + \sigma_{\text{real}} x}, t)$ evaluated at (t, W_t), and this function lies in $C_E^{1,2}([0, T'] \times \mathbb{R})$ because V, $\partial_t V$, $\partial_S V$ and $\partial_{SS} V$ are continuous and of moderate growth on $(0, \infty) \times [0, T']$ (Theorem 20.7(i)) and moderate growth in the price is exponential growth in x (Lemma 19.20(i)). The holdings 1 and $-\partial_S V(S_t, t)$ are continuous functions of (t, W_t) of exponential growth on $[0, T']$, so they are admissible. With the initial capital 0, the defining identity of Definition 20.1(ii) reads $e^{-rt}\,\Pi_t = \int_0^t 1\, d(e^{-rs} V(S_s, s)) - \int_0^t \partial_S V(S_s, s)\, d(e^{-rs} S_s)$, and by the decomposition of the discounted price of an instrument in the Setup of that definition, applied to the option, the first integral is $e^{-rt}\, V(S_t, t) - V(S_0, 0)$. So, for every $t \le T'$, with probability one,

$$e^{-rt}\,\Pi_t = \bigl[e^{-rt}\, V(S_t, t) - V(S_0, 0)\bigr] - \int_0^t \partial_S V(S_s, s)\, d(e^{-rs} S_s)\,: \tag{G.3}$$

the mark of the option, minus the borrowed premium, minus the gains of the hedge, all discounted. The outer function $(x, t) \mapsto e^{-rt} V(x, t)$ has the continuous partial derivatives $e^{-rt}(\partial_t V - rV)$, $e^{-rt}\,\partial_S V$ and $e^{-rt}\,\partial_{SS} V$, all of moderate growth on $(0, \infty) \times [0, T']$, so Corollary 19.31, along the geometric Brownian motion S with drift μS and diffusion coefficient $\sigma_{\text{real}} S$, gives for every $t \le T'$, with probability one,

$$\begin{aligned} e^{-rt}\, V(S_t, t) - V(S_0, 0) = &\int_0^t e^{-rs} \bigl(\partial_t V + \mu S\, \partial_S V + \tfrac{1}{2}\sigma_{\text{real}}^2 S^2\, \partial_{SS} V - rV\bigr)(S_s, s)\, ds \\ &+ \int_0^t e^{-rs} \sigma_{\text{real}} S_s\, \partial_S V(S_s, s)\, dW_s\,: \end{aligned}$$

the realised volatility enters through the diffusion coefficient of the underlier. The hedge integral is, by the definition of the integral of a holding against a discounted price in the Setup of Definition 20.1, with the drift μS_s and the diffusion coefficient $\sigma_{\text{real}} S_s$ of the underlier (Corollary 19.34(ii)),

$$\begin{aligned} \int_0^t \partial_S V(S_s, s)\, d(e^{-rs} S_s) = &\int_0^t \partial_S V(S_s, s)\, e^{-rs} (\mu - r) S_s\, ds \\ &+ \int_0^t \partial_S V(S_s, s)\, e^{-rs} \sigma_{\text{real}} S_s\, dW_s. \end{aligned}$$

Step 2 (the pricing equation at the marking volatility). By Theorem 20.7(i), V solves (20.4) at the volatility σ_{imp}: $\partial_t V + rS\, \partial_S V + \frac{1}{2}\sigma_{\text{imp}}^2 S^2\, \partial_{SS} V = rV$ on $(0, \infty) \times [0, T)$. Hence $\partial_t V - rV = -rS\, \partial_S V - \frac{1}{2}\sigma_{\text{imp}}^2 S^2\, \partial_{SS} V$, and the ds-integrand of the first display of Step 1 is

$$e^{-rs}\Bigl[(\mu - r) S\, \partial_S V + \tfrac{1}{2}(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2) S^2\, \partial_{SS} V\Bigr](S_s, s).$$

Step 3 (the hedge cancels the rest). Substitute the two displays of Step 1 in (G.3). The two Itô integrals have the same integrand, $e^{-rs} \sigma_{\text{real}} S_s\, \partial_S V(S_s, s)$, and opposite signs, so they cancel (linearity, Theorem 19.23(iv)). The ds-integral of the hedge cancels the term in $(\mu - r)$ of Step 2. What is left is

$$e^{-rt}\,\Pi_t = \int_0^t e^{-rs}\, \tfrac{1}{2}(\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2) S_s^2\, \partial_{SS} V(S_s, s)\, ds,$$

for every $t \le T'$, and $T' < T$ is arbitrary: this is (i). The integrand is continuous in s on $[0, t]$ on every path, so this is a Riemann integral.

Step 4 (part (ii)). The right side. For a call or a put with $q = 0$, V is the closed form (21.2) at the volatility σ_{imp} (Corollary 20.41(i)), so Proposition 21.6(i) and $0 < \varphi \le 1/\sqrt{2\pi}$ give

$$0 < \Gamma_s S_s^2 = \frac{S_s\,\varphi(d_1)}{\sigma_{\text{imp}}\sqrt{T-s}} \le \frac{S_s}{\sigma_{\text{imp}}\sqrt{2\pi(T-s)}} \qquad (0 \le s < T), \tag{G.4}$$

with d_1 evaluated at the price S_s and the time to expiry $T - s$. A continuous path is bounded on $[0, T]$, and $\int_0^T (T-s)^{-1/2}\,ds = 2\sqrt{T}$ is finite. So the integrand of (21.8) is bounded in absolute value by a constant, which depends on the path, times $(T-s)^{-1/2}$, and the integral over $[0, t]$ converges as $t \to T$ on every path. The convergence is monotone, because by (G.4) the integrand keeps the sign of $\sigma_{\text{real}}^2 - \sigma_{\text{imp}}^2$; for the same reason the limit is positive if $\sigma_{\text{real}} > \sigma_{\text{imp}}$ and negative if $\sigma_{\text{real}} < \sigma_{\text{imp}}$.

The left side. We show that each of the three terms of (G.3) converges in mean square as $t \to T$.

First, the mark. The payoff f of a call or a put is continuous, so by Theorem 20.7(ii) and the continuity of the path, $e^{-rt}V(S_t, t) \to e^{-rT}f(S_T)$ on every path. The fourth moments are bounded. Indeed, by (20.5), $V \ge 0$ because $f \ge 0$; for the call, $f(x) \le x$ gives $V(S,t) \le e^{-r(T-t)}S\,\mathbb{E}\,e^{\nu(T-t)+\sigma_{\text{imp}}\sqrt{T-t}\,Z} = S$, where $\nu = r - \sigma_{\text{imp}}^2/2$ is the log-drift of (20.5) at the volatility σ_{imp}, since $\mathbb{E}\,e^{\theta Z} = e^{\theta^2/2}$ (Lemma 18.35(i)) with $\theta = \sigma_{\text{imp}}\sqrt{T-t}$; for the put, $f \le K$ gives $V(S,t) \le Ke^{-r(T-t)} \le Ke^{|r|T}$. So $0 \le V(S,t) \le S + Ke^{|r|T}$ on $(0,\infty) \times [0,T]$, while $\mathbb{E}\,S_t^4 = S_0^4\,e^{(4\mu+6\sigma_{\text{real}}^2)t}$ is bounded on $[0,T]$ by the same lemma. By Lemma 19.12(iv), applied along any sequence of times increasing to T, the convergence holds in mean square.

Second, the ds-part of the hedge integral. Put $X_s = \partial_S V(S_s, s)\,e^{-rs}(\mu - r)S_s$. Since $|\partial_S V| \le 1$ for a call or a put with $q = 0$ (Proposition 21.2), $|X_s| \le c\,S_s$ with $c = |\mu - r|\,e^{|r|T}$. On every path X is continuous and bounded on $[0,T)$, so $\int_0^t X_s\,ds$ converges to $\int_0^T X_s\,ds$. For the mean square, $\big(\int_t^T X_s\,ds\big)^2 \le c^2\big(\int_t^T S_s\,ds\big)^2 \le c^2(T-t)\int_t^T S_s^2\,ds$, the last step because the average m of S_s over $[t,T]$ satisfies $0 \le \int_t^T (S_s - m)^2\,ds = \int_t^T S_s^2\,ds - \big(\int_t^T S_s\,ds\big)^2/(T-t)$. The path $s \mapsto S_s^2$ is continuous and nonnegative on the closed interval $[t,T]$, so Fact 18.34(iii) gives

$$\mathbb{E}\Big(\int_t^T X_s\,ds\Big)^2 \le c^2(T-t)\int_t^T \mathbb{E}\,S_s^2\,ds \le c^2(T-t)^2 \sup_{s\le T}\mathbb{E}\,S_s^2 \longrightarrow 0 \qquad (t \to T).$$

Third, the dW-part. Put $\mathcal{M}_t = \int_0^t e^{-rs}\sigma_{\text{real}}S_s\,\partial_S V(S_s, s)\,dW_s$ for $t < T$. For $t' < t < T$, the additivity and the isometry of the Itô integral (Theorem 19.23(iii)–(iv)) and $|\partial_S V| \le 1$ give

$$\mathbb{E}\,(\mathcal{M}_t - \mathcal{M}_{t'})^2 = \int_{t'}^t e^{-2rs}\sigma_{\text{real}}^2\,\mathbb{E}\big[S_s^2\,(\partial_S V(S_s,s))^2\big]\,ds \le \sigma_{\text{real}}^2\,e^{2|r|T}\int_{t'}^t \mathbb{E}\,S_s^2\,ds,$$

where the factor $e^{2|r|T}$ bounds e^{-2rs} for either sign of r. The right side tends to 0 as t' and t tend to T. Along any sequence of times increasing to T the variables $\mathcal{M}_t$ therefore form a Cauchy sequence in mean square, and Fact 19.16 provides a limit $\mathcal{M}_T$ in mean square. Interleaving two such sequences gives a third, whose limit agrees with both, so $\mathcal{M}_T$ does not depend on the sequence, up to an event of probability zero.

By the triangle inequality of Lemma 19.12(ii), $e^{-rt}\,\Pi_t$ converges in mean square to $e^{-rT}f(S_T) - V(S_0,0) - \int_0^T X_s\,ds - \mathcal{M}_T$. This limit has a finite second moment and $e^{rt} \to e^{rT}$, so Π_t converges in mean square to

$$\Pi_T = f(S_T) - e^{rT}\Big[V(S_0,0) + \int_0^T X_s\,ds + \mathcal{M}_T\Big],$$

which is the terminal value of Definition 20.1(ii): the payoff of the option, minus the borrowed premium with its interest, minus the accumulated gains of the hedge.

The two limits agree. Let t_k, $k = 1, 2, \ldots$, increase to T (proof-local names). For each k, identity (i) at t_k holds outside an event of probability zero, and the union of these countably many events has probability zero (Lemma 19.12(0)). Outside it, $e^{-rt_k}\,\Pi_{t_k}$ equals the right side of

(21.8) at t_k for every k, so it converges to $\int_0^T e^{-rs}\,\frac{1}{2}\Gamma_s S_s^2\,(\sigma_{\text{real}}^2-\sigma_{\text{imp}}^2)\,ds$. The same sequence converges in mean square to $e^{-rT}\,\Pi_T$. Almost sure and mean-square convergence both imply convergence in probability, and two limits in probability of one sequence agree with probability one (Lemma 19.12(iii)). This is the formula of (ii).

Step 5 (part (iii)). Let $\mu = r = 0$. Then $X_s = 0$ and $\Pi_T = f(S_T) - V(S_0,0) - \mathcal{M}_T$. Each $\mathcal{M}_t$ has mean zero (Theorem 19.23(ii)), and mean-square convergence carries the mean to the limit: by the Cauchy–Schwarz inequality with the second variable equal to 1 (Lemma 19.12(ii)), $|\mathbb{E}\,\mathcal{M}_{t_k} - \mathbb{E}\,\mathcal{M}_T| \le \left(\mathbb{E}\,(\mathcal{M}_{t_k}-\mathcal{M}_T)^2\right)^{1/2} \to 0$. So $\mathbb{E}\,\mathcal{M}_T = 0$; no fourth moment of the integrals is needed. Finally $\mathbb{E}\,f(S_T) = V^{\text{BSM}}(S_0,0;\sigma_{\text{real}})$ by Step 2 of the proof of Proposition 21.19, and $V(S_0,0) = V^{\text{BSM}}(S_0,0;\sigma_{\text{imp}})$. □

G.7 Proof of Corollary 21.24: Continuous-time hedged P and L when the volatility is re-set along the path

The statement is on page 380. The notation and the standing assumptions are those of §21.5.

Proof. We run the proof of Proposition 21.22 on each cell with the earlier Brownian values frozen, and add the cells.

Step 1 (each cell is a parameter case). By induction on j, starting from $S_{t_0} = S_0$ and $W_{t_0} = 0$, the definition of the underlier gives, for $t \in [t_{j-1}, t_j]$,

$$\ln S_t = \ln S_0 + \sum_{i=1}^{j-1}\Big[(\mu-\tfrac{1}{2}\sigma_i^2)(t_i - t_{i-1}) + \sigma_i\,(W_{t_i} - W_{t_{i-1}})\Big] + (\mu-\tfrac{1}{2}\sigma_j^2)(t-t_{j-1}) + \sigma_j\,(W_t - W_{t_{j-1}}).$$

Each σ_i with $i \le j$ is a function of $(W_{t_1},\dots,W_{t_{i-1}})$, which is part of Y_j. So $\ln S_t$ is an explicit function of (t, W_t, Y_j). Extend the boxes of each σ_i to boxes in the coordinates of Y_j by letting the remaining coordinates range over the whole line. The intersections of one box of each of $\sigma_1,\dots,\sigma_j$ are again boxes, finitely many and disjoint, and they cover the range of Y_j. On each of them every σ_i is continuous, so $\ln S_t$ is continuous in (t, W_t, Y_j). Every volatility is at most c, and each of $W_{t_1},\dots,W_{t_{j-1}}$ appears in at most two terms of the sum, so

$$|\ln S_t| \le c'(1 + |W_t| + |W_{t_1}| + \cdots + |W_{t_{j-1}}|), \qquad c' = \max\left(|\ln S_0| + (|\mu| + c^2/2)\,T,\ 2c\right).$$

Hence, for every real exponent c'',

$$S_t^{c''} = e^{c''\ln S_t} \le e^{|c''|c'}\,e^{|c''|c'(|W_t|+|W_{t_1}|+\cdots+|W_{t_{j-1}}|)}:$$

every power of the price has exponential growth in the Brownian coordinates (Definition 19.19(i)). Two consequences are used below. A function of moderate growth in the price, evaluated at S_t, has exponential growth in (W_t, Y_j), which is the conclusion of Lemma 19.20(i) for this price. And the integrand $\partial_S V(S_s, s)\,e^{-rs}\sigma_i S_s$ of (21.9), on the cell i, is a function of (s, W_s, Y_i) that is continuous on each box and of exponential growth, because $\partial_S V$ is continuous and of moderate growth on $(0,\infty)\times[0,T']$ for every $T' < T$ (Theorem 20.7(i), a statement about the price function at any volatility, proved without (A1)–(A6)). It is therefore an admissible integrand on $[t_{i-1}, t]$ for every $t < T$ in the closed cell (Definition 19.22(a)–(b)).

Step 2 (Itô's lemma on a cell). Fix a cell j and a time t with $t_{j-1} < t \le t_j$ and $t < T$; so $t = t_j$ is allowed except on the last cell. By Step 1, on $[t_{j-1}, t]$ the price is $S_s = a(s, W_s, Y_j)$ with $a(s,x,y) = S_{t_{j-1}}\exp\left((\mu-\sigma_j^2/2)(s-t_{j-1}) + \sigma_j\,(x - y_{j-1})\right)$, where $S_{t_{j-1}}$ and σ_j are the functions of y found in Step 1 and y_{j-1}, the last coordinate of y, is the value of $W_{t_{j-1}}$ (for $j = 1$, y is absent and $y_0 = 0$). By the chain rule, the derivatives of $(s,x,y) \mapsto e^{-rs}V(a(s,x,y),s)$ in s, in x and twice in x are e^{-rs} times $\partial_t V - rV + (\mu - \frac{1}{2}\sigma_j^2)S\,\partial_S V$, $\sigma_j S\,\partial_S V$ and $\sigma_j^2(S^2\,\partial_{SS}V + S\,\partial_S V)$, evaluated at $S = a(s,x,y)$: those of the geometric-Brownian-motion clause of Corollary 19.31, with σ_j in the place of σ. By Step 1,

$\sigma_j \le c$ and the moderate growth of V and its partial derivatives on $(0,\infty)\times[0,t]$ (Theorem 20.7(i)), they are continuous on each box and of exponential growth in (x,y), which is the hypothesis of the parameter clause of Itô's lemma (Theorem 19.30(ii)), here with Y_j, a vector of Brownian values at times at most t_{j-1}, frozen. That clause gives the Itô identity of Step 1 of the proof of Proposition 21.22 on $[t_{j-1},t]$, starting from $e^{-rt_{j-1}}V(S_{t_{j-1}},t_{j-1})$, with σ_j, a constant once Y_j is frozen, in place of σ_{real}, and Steps 2 and 3 of that proof then hold on the cell line for line: the pricing equation at σ_{imp} turns the drift of $e^{-rs}V(S_s,s)$ into $e^{-rs}\big[(\mu-r)S\,\partial_S V+\frac12(\sigma_j^2-\sigma_{\text{imp}}^2)S^2\,\partial_{SS}V\big]$, the j-th bracket of (21.9) has the same dW-integrand $e^{-rs}\sigma_j S\,\partial_S V$ and the same term in $(\mu-r)$, and on subtracting it the Itô integrals cancel (Theorem 19.23(iv)) and so do the terms in $(\mu-r)$. So

$$e^{-rt}\,V(S_t,t)-e^{-rt_{j-1}}\,V(S_{t_{j-1}},t_{j-1})-\big[j\text{-th bracket of (21.9)}\big]=\int_{t_{j-1}}^{t} e^{-rs}\,\tfrac12\Gamma_s S_s^2\,(\sigma_j^2-\sigma_{\text{imp}}^2)\,ds \tag{G.5}$$

with probability one.

Step 3 (adding the cells). Let $t<T$ lie in the cell $[t_{j-1},t_j)$. Apply (G.5) to each cell $i<j$ with its right end t_i in the place of t, which is allowed because $t_i<T$, and to the cell j with t itself; if $t=t_{j-1}$ the last cell contributes nothing. Add the j identities. On the left the values of $e^{-rs}\,V(S_s,s)$ at the inner dates $t_1,\dots,t_{j-1}$ telescope, leaving $e^{-rt}\,V(S_t,t)-V(S_0,0)$ minus the whole sum of (21.9), which is $e^{-rt}\,\Pi_t$. On the right the integrals add up to $\int_0^t e^{-rs}\,\frac12\Gamma_s S_s^2\,(\sigma(s)^2-\sigma_{\text{imp}}^2)\,ds$. The finitely many exceptional events have probability zero together (Lemma 19.12(0)). This is (a).

Step 4 (part (b)). We repeat Step 4 of the proof of Proposition 21.22 on the last cell, with the bound c in the place of σ_{real}. The bound (G.4) is a property of the function V, and $|\sigma(s)^2-\sigma_{\text{imp}}^2|\le c^2+\sigma_{\text{imp}}^2$, so the right side converges on every path as there. On the left side two inputs change.

The moments of the price stay bounded. Fix j and $t\in[t_{j-1},t_j]$. The increment $W_t-W_{t_{j-1}}$ is independent of $Y_j=(W_{t_1},\dots,W_{t_{j-1}})$ (Proposition 19.8(iv), restarted at t_{j-1}), and $S_{t_{j-1}}$ and σ_j are functions of Y_j. Integrating the increment out first (Lemma 19.17, for a nonnegative function), with $\mathbb{E}\,e^{4\sigma_j(W_t-W_{t_{j-1}})}=e^{8\sigma_j^2(t-t_{j-1})}$ for a frozen σ_j (Lemma 18.35(i)),

$$\mathbb{E}\,S_t^4=\mathbb{E}\Big[S_{t_{j-1}}^4\,e^{(4\mu+6\sigma_j^2)(t-t_{j-1})}\Big]\le e^{(4|\mu|+6c^2)(t-t_{j-1})}\,\mathbb{E}\,S_{t_{j-1}}^4.$$

By induction on j, $\mathbb{E}\,S_t^4\le S_0^4\,e^{(4|\mu|+6c^2)t}$ for every $t\le T$, and $\mathbb{E}\,S_t^2\le(\mathbb{E}\,S_t^4)^{1/2}$ by the Cauchy–Schwarz inequality.

And for the dW-part of the last cell and $t_{J-1}\le t'<t<T$, the isometry gives

$$\mathbb{E}\Big(\int_{t'}^{t} e^{-rs}\sigma_J S_s\,\partial_S V(S_s,s)\,dW_s\Big)^2=\int_{t'}^{t}\mathbb{E}\big[e^{-2rs}\sigma_J^2 S_s^2\,(\partial_S V)^2\big]\,ds\le c^2e^{2|r|T}\int_{t'}^{t}\mathbb{E}\,S_s^2\,ds\longrightarrow 0.$$

With these two bounds the mark, the ds-parts and the dW-part converge in mean square as there; the brackets of the earlier cells do not depend on t and have finite second moments. So Π_t converges in mean square to a limit Π_T, the value with the option settled at its payoff, and this limit agrees with the pathwise one with probability one, by the argument that closes Step 4 there. For the sign, $e^{-rs}\,\frac12\Gamma_s S_s^2>0$ by (G.4), so the integrand of (b) is nonnegative on a cell where $\sigma_j\ge\sigma_{\text{imp}}$ and positive on a cell where $\sigma_j>\sigma_{\text{imp}}$, and symmetrically.

Step 5 (part (c)). Let $\mu=r=0$. The ds-parts of (21.9) vanish, and each Itô integral in it has mean zero (Theorem 19.23(ii)). The mean passes to the mean-square limit at T by Lemma 19.12(ii), as in Step 5 there. Hence $\mathbb{E}\,\Pi_T=\mathbb{E}\,f(S_T)-V(S_0,0)$. □

G.8 Proof of Proposition 21.27: The discrete-hedging error: exact mean, exact variance and the $n^{-1/2}$ law, to expiry included (Boyle and Emanuel, 1980; Bertsimas, Kogan and Lo, 2000)

The statement is on page 382. The notation and the standing assumptions are those of §21.5.

Proof. We integrate out each move with Lemma 19.17 for (i) and (ii), use the Itô isometry for the variance, and make a Taylor computation on each interval for (iii), with a localisation near the strike when $T' = T$; the proof has six steps.

Step 1 (a one-step identity). For $x > 0$ and $0 \le t' < t'' \le T'$ put $Y = \exp\big(-\sigma^2(t''-t')/2 + \sigma\sqrt{t''-t'}\,Z\big)$. Then

$$\mathbb{E}\,V(xY, t'') = V(x, t') \qquad \text{and} \qquad \mathbb{E}\big[x\,(Y-1)\big] = 0. \tag{G.6}$$

By the definition (20.5) of V in Theorem 20.7 at $r = 0$, where $\nu = -\sigma^2/2$,

$$V(x, t') = \mathbb{E}\, f\big(x \exp(-\sigma^2(T-t')/2 + \sigma\sqrt{T-t'}\,Z)\big).$$

(This is the function (20.5); clauses (i) and (ii) of the theorem, cited below, are statements about it that are proved without (A1)–(A6).) Suppose first that $t'' < T$, and put $\psi(y) = f\big(x\exp(-\sigma^2(T-t')/2 + \sigma y)\big)$. This function is continuous except at finitely many points, and the growth bound of the class $\mathcal{G}$ gives $|\psi(y)| \le c' e^{c'|y|}$ for a constant c'. Lemma 18.35(ii), with its two scale constants equal to $\sqrt{t''-t'}$ and $\sqrt{T-t''}$, whose squares add up to $T-t'$, gives

$$\mathbb{E}\,\psi\big(\sqrt{t''-t'}\,Z + \sqrt{T-t''}\,Z'\big) = \mathbb{E}\,\psi\big(\sqrt{T-t'}\,Z\big) = V(x, t').$$

Since $(t''-t') + (T-t'') = T-t'$, the argument of f on the left is xYY', where $Y' = \exp\big(-\sigma^2(T-t'')/2 + \sigma\sqrt{T-t''}\,Z'\big)$ is independent of Y. So $\mathbb{E}\,f(xYY') = V(x, t')$, the variable $f(xYY')$ being integrable by the growth bound. Integrate Y' out first (Lemma 19.17): for a frozen value y of Y, $\mathbb{E}\,f(xyY') = V(xy, t'')$ by the definition (20.5) at the time t'', so $\mathbb{E}\,f(xYY') = \mathbb{E}\,V(xY, t'')$. This is the first identity. If $t'' = T$, then $V(\cdot, T) = f$ and the display for $V(x, t')$ is the first identity as it stands. The second identity is $\mathbb{E}\,Y = 1$, which is Lemma 18.35(i) with $\theta = \sigma\sqrt{t''-t'}$.

Step 2 (part (i) and the orthogonality). The variable ε_i is a function of S_{t_i} and of the ratio $Y_i = S_{t_{i+1}}/S_{t_i} = \exp\big(-\sigma^2\Delta t/2 + \sigma(W_{t_{i+1}} - W_{t_i})\big)$:

$$\varepsilon_i = V(S_{t_i}Y_i, t_{i+1}) - V(S_{t_i}, t_i) - \partial_S V(S_{t_i}, t_i)\, S_{t_i}\,(Y_i - 1).$$

The ratio Y_i is a function of the increment $W_{t_{i+1}} - W_{t_i}$, which is independent of $W_{t_1}, \dots, W_{t_i}$ (Proposition 19.8(iv), restarted at t_i), so Y_i is independent of $X = (S_{t_0}, \dots, S_{t_i})$, a function of these values; and Y_i has the law of the variable Y of Step 1 with $t' = t_i$ and $t'' = t_{i+1}$. Each ε_i has a finite second moment: V has moderate growth on $(0,\infty)\times[0,T]$ and $\partial_S V$ has moderate growth at the dates $t_i \le t_{n-1} < T$ (Theorem 20.7(i)–(ii)), and lognormal prices have moments of every order (Lemma 19.20(iii)). Let ψ_1 be any Borel measurable function of X with $\mathbb{E}\,|\psi_1(X)\,\varepsilon_i| < \infty$. Lemma 19.17, applied to $\psi_1(X)\,\varepsilon_i$ as a function of X and Y_i (Borel measurable, because ψ_1 is, V and $\partial_S V$ are continuous before T, and $V(\cdot, T) = f$ is continuous outside finitely many points), and the two identities (G.6) give

$$\mathbb{E}\big[\psi_1(X)\,\varepsilon_i\big] = \mathbb{E}\big[\psi_1(X)\,\bar\varepsilon(S_{t_i})\big], \qquad \bar\varepsilon(x) = \mathbb{E}\,V(xY_i, t_{i+1}) - V(x, t_i) - \partial_S V(x, t_i)\,\mathbb{E}\big[x\,(Y_i - 1)\big] = 0,$$

so $\mathbb{E}[\psi_1(X)\,\varepsilon_i] = 0$. The choice $\psi_1 = 1$ gives $\mathbb{E}\,\varepsilon_i = 0$. For $j < i$, the choice $\psi_1 = \varepsilon_j$, a function of $S_{t_0}, \dots, S_{t_{j+1}}$, which are coordinates of X, and a continuous one, hence Borel measurable, since $t_{j+1} \le t_i < T' \le T$ and V is infinitely differentiable before T, gives $\mathbb{E}[\varepsilon_j\varepsilon_i] = 0$; the product is integrable by the Cauchy–Schwarz inequality (Lemma 19.12(ii)). The sum $\sum_i \varepsilon_i$ telescopes to

$V(S_{T'},T') - V(S_0,0) - \sum_i \Delta_{t_i}\,(S_{t_{i+1}} - S_{t_i}) = \mathcal{E}_n$, which gives (i) by linearity. Since $\mathbb{E}\,\mathcal{E}_n = 0$, expanding the square of the sum gives

$$\operatorname{Var}\mathcal{E}_n = \mathbb{E}\,\mathcal{E}_n^2 = \sum_i \sum_j \mathbb{E}[\varepsilon_i \varepsilon_j] = \sum_{i=0}^{n-1} \mathbb{E}\,\varepsilon_i^2,$$

the form of Proposition 10.10 for n uncorrelated terms.

Step 3 (the exact variance). Consider first a cell with $t_{i+1} < T$. At $r = 0$ the pricing equation is $\partial_t V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS} V = 0$ (Theorem 20.7(i)), and along the driftless geometric Brownian motion S the drift of $V(S_t,t)$ in Corollary 19.31 is exactly the left side of this equation. So that corollary, on $[0,t_{i+1}]$, where V and its derivatives have moderate growth, gives $V(S_t,t) = V(S_0,0) + \int_0^t \Delta_s\,\sigma S_s\,dW_s$ for every $t \le t_{i+1}$, with probability one. Subtracting this identity at t_i from the one at t_{i+1}, with the additivity of the integral (Theorem 19.23(iv)),

$$V(S_{t_{i+1}},t_{i+1}) - V(S_{t_i},t_i) = \int_{t_i}^{t_{i+1}} \Delta_s\,\sigma S_s\,dW_s, \qquad S_{t_{i+1}} - S_{t_i} = \int_{t_i}^{t_{i+1}} \sigma S_s\,dW_s,$$

the second by Corollary 19.34(ii) with $\mu = 0$ in the same way. The factor Δ_{t_i}, fixed at t_i, passes inside the second integral. Indeed, $\Delta_{t_i}\,\sigma S_s$ is an admissible integrand on $[t_i,t_{i+1}]$ with the earlier value W_{t_i} frozen (Definition 19.22(a)–(b)), and each of its left-endpoint sums is Δ_{t_i} times the left-endpoint sum of σS_s. The former sums converge in mean square to $\int_{t_i}^{t_{i+1}} \Delta_{t_i}\,\sigma S_s\,dW_s$, the latter to $\int_{t_i}^{t_{i+1}} \sigma S_s\,dW_s$, and multiplying a sequence that converges in probability by the fixed variable Δ_{t_i} preserves the convergence; two limits in probability of one sequence agree with probability one (Lemma 19.12(iii)). Hence, by linearity and the isometry (Theorem 19.23(iii)–(iv)),

$$\varepsilon_i = \int_{t_i}^{t_{i+1}} (\Delta_s - \Delta_{t_i})\,\sigma S_s\,dW_s, \qquad \mathbb{E}\,\varepsilon_i^2 = \sigma^2 \int_{t_i}^{t_{i+1}} \mathbb{E}\big[(\Delta_s - \Delta_{t_i})^2\,S_s^2\big]\,ds.$$

On the last cell when $T' = T$ (a call or a put), the same argument holds on $[t_{n-1},t]$ for every $t < T$: the variable $V(S_t,t) - V(S_{t_{n-1}},t_{n-1}) - \Delta_{t_{n-1}}\,(S_t - S_{t_{n-1}})$ has the second moment $\sigma^2 \int_{t_{n-1}}^t \mathbb{E}[(\Delta_s - \Delta_{t_{n-1}})^2 S_s^2]\,ds$. As $t \to T$, this variable converges to ε_{n-1} on every path, because $V(S_t,t) \to f(S_T)$ (Theorem 20.7(ii); f is continuous) and $S_t \to S_T$. Its fourth moments are bounded, since $0 \le V(S,t) \le S + K$ for a call or a put at $r = 0$ (Step 4 of the proof of Proposition 21.22) and $|\Delta_{t_{n-1}}| \le 1$ (Proposition 21.2); so it converges in mean square and its second moment converges to $\mathbb{E}\,\varepsilon_{n-1}^2$ (Lemma 19.12(iv)). The right side increases to the integral over $[t_{n-1},T)$, which is finite because $|\Delta_s - \Delta_{t_{n-1}}| \le 1$ and $\mathbb{E}\,S_s^2$ is bounded. So the formula for $\mathbb{E}\,\varepsilon_i^2$ holds on the last cell too. Summing over the cells, with $t(s) = t_i$ on the cell i, gives (21.10).

Step 4 (the leading term on a cell). Fix a cell $[t_i,t_{i+1}]$ and a time s in it, and define the remainder $\mathcal{E}_s^\Delta$ by

$$\Delta_s - \Delta_{t_i} = \Gamma_{t_i}\,(S_s - S_{t_i}) + \mathcal{E}_s^\Delta.$$

The ratio $Y_s = S_s/S_{t_i} = \exp\big(-\sigma^2(s-t_i)/2 + \sigma(W_s - W_{t_i})\big)$ is independent of S_{t_i} (Proposition 19.8(iv), restarted at t_i), and $S_s - S_{t_i} = S_{t_i}(Y_s - 1)$, $S_s = S_{t_i}Y_s$. The product rule of Lemma 19.17 gives

$$\mathbb{E}\big[\Gamma_{t_i}^2\,(S_s - S_{t_i})^2\,S_s^2\big] = \mathbb{E}\big[\Gamma_{t_i}^2 S_{t_i}^4\big]\,\mathbb{E}\big[(Y_s - 1)^2\,Y_s^2\big].$$

Since $\mathbb{E}\,Y_s^k = e^{k(k-1)\sigma^2(s-t_i)/2}$ for $k = 2,3,4$ (Lemma 18.35(i)), expanding $(Y_s-1)^2Y_s^2 = Y_s^4 - 2Y_s^3 + Y_s^2$ gives

$$\mathbb{E}\big[(Y_s-1)^2\,Y_s^2\big] = e^{6\sigma^2(s-t_i)} - 2e^{3\sigma^2(s-t_i)} + e^{\sigma^2(s-t_i)} = \sigma^2(s-t_i) + \tfrac{19}{2}\sigma^4(s-t_i)^2 + O\big((s-t_i)^3\big), \quad \text{(G.7)}$$

by the expansion of the exponential in Proposition 7.34: the constant terms are $1 - 2 + 1 = 0$, the terms of first order $(6 - 6 + 1)\,\sigma^2(s-t_i)$, and those of second order $(18 - 9 + \frac{1}{2})\,\sigma^4(s-t_i)^2$. The bound is uniform: the function $y \mapsto e^{6y} - 2e^{3y} + e^y$ has a bounded third derivative on $[0,\sigma^2 T]$,

so by (7.4) at order two the term $O((s-t_i)^3)$ is at most a constant multiple of $(s-t_i)^3$ for $0 \le s - t_i \le T$. For the same reason, by the same bound at order zero, the whole expression is at most a constant multiple of $s - t_i$.

Step 5 (the limit when $T' < T$). In this step c is a generic constant: a number that depends only on σ, T, T', S_0 and f, not on n, i or s, and whose value may change from one bound to the next.

Three bounds. First, the derivatives of V. Formula (20.5) involves t and T only through $T-t$, so it defines $V(S,t)$ for $-1 \le t < 0$ as well, and $V(S,t)$ is the function of Theorem 20.7 for the maturity $T+1$, evaluated at the time $t+1$. By clause (i) of that theorem, V is infinitely differentiable on $(0,\infty)\times[-1,T)$, and each of the finitely many partial derivatives of V used in this step is bounded in absolute value by $c\,(x^c + x^{-c})$ at every point (x,t) of $(0,\infty)\times[-1,T']$. The extension to negative times serves one purpose: a segment that starts at time 0 then lies in an open set on which V is smooth. Second, the moments of the price: $\mathbb{E}\,S_t^{c''} = S_0^{c''} e^{c''(c''-1)\sigma^2 t/2}$ for every real exponent c'' (Lemma 18.35(i)), which is bounded on $[0,T]$. Third, the ratio of Step 4: with $s - t_i \le \Delta t \le T$,

$$\mathbb{E}\,(Y_s - 1)^8 \le c\,(s-t_i)^4. \tag{G.8}$$

To see this, note that Y_s has the law of e^y with $y = -\sigma^2(s-t_i)/2 + \sigma\sqrt{s-t_i}\,Z$. By Corollary 7.32 at order zero, $e^y - 1 = y\,e^{y^*}$ for a point y^* between 0 and y, so $|e^y - 1| \le |y|\,e^{|y|}$. And $|y| \le \sqrt{s-t_i}\,(\sigma^2\sqrt{T}/2 + \sigma|Z|)$. Hence $(Y_s-1)^8 \le (s-t_i)^4$ times a function of Z of exponential growth, whose mean is finite by Lemma 18.35(i).

Taylor's theorem along the segment. For u in an open interval that contains $[0,1]$, define

$$\chi(u) = \partial_S V\big(S_{t_i} + u\,(S_s - S_{t_i}),\ t_i + u\,(s-t_i)\big),$$

so that $\chi(0) = \Delta_{t_i}$ and $\chi(1) = \Delta_s$; such an interval exists because the points of the segment lie in the open set $(0,\infty)\times(-1,T)$. As in Step 1 of the proof of Proposition 21.14, the chain rule gives

$$\begin{aligned}
\chi'(u) &= \partial_{SS}V\,(S_s - S_{t_i}) + \partial_{St}V\,(s-t_i),\\
\chi''(u) &= \partial_{SSS}V\,(S_s - S_{t_i})^2 + 2\,\partial_{SSt}V\,(S_s - S_{t_i})(s-t_i) + \partial_{Stt}V\,(s-t_i)^2,
\end{aligned}$$

the derivatives of V being evaluated at the point of the segment with parameter u. Let M_3 be the largest absolute value taken on the segment by the three third derivatives $\partial_{SSS}V$, $\partial_{SSt}V$ and $\partial_{Stt}V$. Then $|\chi''(u)| \le M_3\,(|S_s - S_{t_i}| + s - t_i)^2$ for $0 \le u \le 1$, and Theorem 7.31 at order one and its bound (7.4) give $\chi(1) = \chi(0) + \chi'(0) + \mathcal{E}_s''$ with $|\mathcal{E}_s''| \le \frac{1}{2} M_3\,(|S_s - S_{t_i}| + s - t_i)^2$. Since $\chi'(0) = \Gamma_{t_i}\,(S_s - S_{t_i}) + \partial_{St}V(S_{t_i},t_i)\,(s-t_i)$, the remainder of Step 4 is

$$\mathcal{E}_s^{\Delta} = \partial_{St}V(S_{t_i},t_i)\,(s-t_i) + \mathcal{E}_s'', \qquad |\mathcal{E}_s''| \le M_3\,\big(|S_s - S_{t_i}| + s - t_i\big)^2.$$

On the segment the price lies between S_{t_i} and S_s, so its c-th power is at most $S_{t_i}^c + S_s^c$ and its $(-c)$-th power at most $S_{t_i}^{-c} + S_s^{-c}$. By the first bound, $M_3 \le c\,(S_{t_i}^c + S_{t_i}^{-c} + S_s^c + S_s^{-c})$.

The size of the remainder. By (G.8), the product rule of Lemma 19.17 and the moment bound, $\mathbb{E}\,(S_s - S_{t_i})^8 = \mathbb{E}\,S_{t_i}^8\ \mathbb{E}\,(Y_s-1)^8 \le c\,(s-t_i)^4$, so $\mathbb{E}\,(|S_s - S_{t_i}| + s - t_i)^8 \le 2^7\big(\mathbb{E}\,(S_s - S_{t_i})^8 + (s-t_i)^8\big) \le c\,(s-t_i)^4$. The Cauchy–Schwarz inequality (Lemma 19.12(ii)) and the moment bound then give

$$\mathbb{E}\big[(\mathcal{E}_s'')^2\,S_s^2\big] \le \Big(\mathbb{E}\big[M_3^4\,S_s^4\big]\Big)^{1/2}\Big(\mathbb{E}\,\big(|S_s - S_{t_i}| + s - t_i\big)^8\Big)^{1/2} \le c\,(s-t_i)^2,$$

and $\mathbb{E}\big[(\partial_{St}V(S_{t_i},t_i))^2\,S_s^2\big] \le c$. Since $(y+z)^2 \le 2y^2 + 2z^2$,

$$\mathbb{E}\big[(\mathcal{E}_s^{\Delta})^2\,S_s^2\big] \le c\,(s-t_i)^2.$$

By Step 4, $\mathbb{E}[\Gamma_{t_i}^2\,(S_s - S_{t_i})^2\,S_s^2] \le c\,(s-t_i)$, because $\mathbb{E}[\Gamma_{t_i}^2 S_{t_i}^4] \le c$ by the first two bounds. One more use of the Cauchy–Schwarz inequality bounds the cross term:

$$\Big|\,2\,\mathbb{E}\big[\Gamma_{t_i}\,(S_s - S_{t_i})\,\mathcal{E}_s^{\Delta}\,S_s^2\big]\Big| \le 2\Big(\mathbb{E}\big[\Gamma_{t_i}^2\,(S_s - S_{t_i})^2\,S_s^2\big]\Big)^{1/2}\Big(\mathbb{E}\big[(\mathcal{E}_s^{\Delta})^2 S_s^2\big]\Big)^{1/2} \le c\,(s-t_i)^{3/2}.$$

One cell. Squaring the decomposition of Step 4 and using (G.7), with all three bounds uniform in i and n,

$$\mathbb{E}\big[(\Delta_s - \Delta_{t_i})^2\, S_s^2\big] = \sigma^2 (s - t_i)\, \mathbb{E}\big[\Gamma_{t_i}^2 S_{t_i}^4\big] + O\big((s-t_i)^{3/2}\big).$$

By Step 3 and $\int_{t_i}^{t_{i+1}} (s - t_i)\, ds = (\Delta t)^2/2$,

$$\mathbb{E}\,\varepsilon_i^2 = \sigma^2 \int_{t_i}^{t_{i+1}} \Big[\sigma^2 (s-t_i)\, \mathbb{E}\big[\Gamma_{t_i}^2 S_{t_i}^4\big] + O\big((s-t_i)^{3/2}\big)\Big]\, ds = \tfrac{1}{2}\sigma^4 (\Delta t)^2\, \mathbb{E}\big[\Gamma_{t_i}^2 S_{t_i}^4\big] + O\big((\Delta t)^{5/2}\big).$$

All cells. By Step 2, $n \operatorname{Var} \mathcal{E}_n = n \sum_i \mathbb{E}\,\varepsilon_i^2$; with $n\,\Delta t = T'$ and $n \cdot n \cdot (\Delta t)^{5/2} = T'^{5/2}\, n^{-1/2}$, the estimate of one cell gives

$$n \operatorname{Var} \mathcal{E}_n = n \sum_{i=0}^{n-1} \mathbb{E}\,\varepsilon_i^2 = \tfrac{1}{2}\sigma^4\, T' \sum_{i=0}^{n-1} \Delta t\, \mathbb{E}\big[\Gamma_{t_i}^2 S_{t_i}^4\big] + O(n^{-1/2}).$$

The sum is a left Riemann sum of the function $t \mapsto \mathbb{E}[\Gamma_t^2 S_t^4]$ on $[0, T']$. This function is Lipschitz. Indeed, the outer function $(x,t) \mapsto (\partial_{SS} V(x,t))^2\, x^4$ is smooth, with derivatives of moderate growth on $(0,\infty) \times [0,T']$, so the expectation identity of Corollary 19.31 writes $\mathbb{E}[\Gamma_t^2 S_t^4]$ as $\Gamma_0^2 S_0^4$ plus the integral over $[0,t]$ of the expected drift, and the expected drift is bounded on $[0,T']$ by the moment bound. For a Lipschitz function with constant c, the left Riemann sum differs from the integral by at most $\sum_i \int_{t_i}^{t_{i+1}} c\,(t - t_i)\, dt = c\,T'\,\Delta t/2$, which is $O(1/n)$. Hence

$$n \operatorname{Var} \mathcal{E}_n = \tfrac{1}{2}\sigma^4\, T' \int_0^{T'} \mathbb{E}\big[\Gamma_t^2 S_t^4\big]\, dt + O(n^{-1/2}),$$

which is (iii) for $T' < T$.

Step 6 (the limit when $T' = T$, for a call or a put). Take the call. The put has the same Γ, and by parity at $r = q = 0$ its $\partial_S V$ is that of the call minus the constant 1, so $\Delta_s - \Delta_{t_i}$ and, by (21.10), the variance are the same. Let z_t be the value of d_1 at the price S_t and the time to expiry $T - t$,

$$z_t = \frac{\ln(S_t/K) + \sigma^2 (T-t)/2}{\sigma\sqrt{T-t}}, \qquad \text{so that} \qquad \Delta_t = \Phi(z_t), \quad \Gamma_t S_t = \frac{\varphi(z_t)}{\sigma\sqrt{T-t}}$$

(Propositions 21.2 and 21.6(i) with $r = q = 0$). In this step c, c' and c'' are three fixed constants, defined in (a), (b) and (c) below; they depend only on σ, T and S_0.

(a) Localisation. For $T/2 \le t < T$,

$$\mathbb{E}\big[\varphi(z_t)^2\, S_t^2\big] \le \frac{S_0^2\, e^{\sigma^2 T}}{2\pi\sqrt{T}}\, \sqrt{T-t}. \tag{G.9}$$

The variable $\ln S_t$ is normal with mean $\ln S_0 - \sigma^2 t/2$ and variance $\sigma^2 t$, so its density at x is $\varphi\big((x - \ln S_0 + \sigma^2 t/2)/(\sigma\sqrt{t})\big)/(\sigma\sqrt{t})$. Completing the square in the exponent,

$$2x - \frac{(x - \ln S_0 + \sigma^2 t/2)^2}{2\sigma^2 t} = -\frac{(x - \ln S_0 - 3\sigma^2 t/2)^2}{2\sigma^2 t} + 2\ln S_0 + \sigma^2 t,$$

so e^{2x} times this density is $S_0^2 e^{\sigma^2 t}$ times the normal density with mean $\ln S_0 + 3\sigma^2 t/2$ and the same variance $\sigma^2 t$. A normal density with variance $\sigma^2 t$ is at most $1/(\sigma\sqrt{2\pi t})$, and $t \ge T/2$, so

$$\frac{e^{2x}}{\sigma\sqrt{t}}\, \varphi\Big(\frac{x - \ln S_0 + \sigma^2 t/2}{\sigma\sqrt{t}}\Big) \le \frac{S_0^2\, e^{\sigma^2 T}}{\sigma\sqrt{\pi T}} \qquad \text{for every real } x.$$

As a function of $x = \ln S_t$, z_t is $z(x) = (x - \ln K + \sigma^2(T-t)/2)/(\sigma\sqrt{T-t})$, and the substitution $x \mapsto z(x)$ has $dx = \sigma\sqrt{T-t}\, dz$. By Fact 7.22,

$$\mathbb{E}\big[\varphi(z_t)^2\, S_t^2\big] = \int \varphi(z(x))^2\, \frac{e^{2x}}{\sigma\sqrt{t}}\, \varphi\Big(\frac{x - \ln S_0 + \sigma^2 t/2}{\sigma\sqrt{t}}\Big)\, dx \le \frac{S_0^2\, e^{\sigma^2 T}}{\sigma\sqrt{\pi T}}\, \sigma\sqrt{T-t} \int \varphi(z)^2\, dz,$$

and $\int \varphi(z)^2\,dz = \frac{1}{2\pi}\int e^{-z^2}\,dz = \frac{1}{2\sqrt{\pi}}$ (Theorem 7.5). This is (G.9). Consequently

$$\mathbb{E}\big[\Gamma_t^2 S_t^4\big] = \frac{\mathbb{E}[\varphi(z_t)^2 S_t^2]}{\sigma^2(T-t)} \le c\,(T-t)^{-1/2} \qquad \text{for } T/2 \le t < T, \qquad c = \frac{S_0^2\, e^{\sigma^2 T}}{2\pi\sigma^2\sqrt{T}}. \tag{G.10}$$

On $[0, T/2]$ the same quantity is at most $\mathbb{E}\,S_t^2/(\pi\sigma^2 T)$, because $\varphi^2 \le 1/(2\pi)$ and $T - t \ge T/2$; so it is bounded there. By the argument of Step 5, applied on every interval $[0, t]$ with $t < T$, the function $t \mapsto \mathbb{E}[\Gamma_t^2 S_t^4]$ is continuous on $[0, T)$, and $\int_{T/2}^{T}(T-t)^{-1/2}\,dt$ is finite. So the limit integral of (iii) is finite for $T' = T$.

(b) A bound near expiry. For $T/2 \le t < s < T$,

$$\mathbb{E}\big[(\Delta_s - \Delta_t)^2\, S_s^2\big] \le \frac{c'\,(s-t)}{\sqrt{T-t}}, \qquad c' = 4\sigma^2\big(S_0^2\, e^{\sigma^2 T}\sqrt{T} + c\big). \tag{G.11}$$

Differentiate the pricing equation $\partial_t V + \frac{1}{2}\sigma^2 S^2\,\partial_{SS}V = 0$ in S. Since V is infinitely differentiable before expiry (Theorem 20.7(i)), Lemma 19.54 gives $\partial_S\partial_t V = \partial_t\partial_S V$, written $\partial_{St}V$ as in Step 5, so

$$\partial_{St}V + \sigma^2 S\,\partial_{SS}V + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SSS}V = 0.$$

Now freeze Δ_t, a function of the earlier value W_t, and consider the outer function $(x, s') \mapsto (\partial_S V(x, s') - \Delta_t)^2\,x^2$ on $(0,\infty)\times[t, s]$, where s' is a second time variable. No letter is given to this function. Its drift along the driftless geometric Brownian motion is $\partial_{s'} + \frac{1}{2}\sigma^2 x^2\,\partial_{xx}$ applied to it. With the derivatives of V evaluated at (x, s'), the two parts are

$$\begin{aligned}
\partial_{s'}\big[(\partial_S V - \Delta_t)^2 x^2\big] &= 2x^2\,(\partial_S V - \Delta_t)\,\partial_{St}V \\
&= -2\sigma^2 x^3\,(\partial_S V - \Delta_t)\,\partial_{SS}V - \sigma^2 x^4\,(\partial_S V - \Delta_t)\,\partial_{SSS}V, \\
\tfrac{1}{2}\sigma^2 x^2\,\partial_{xx}\big[(\partial_S V - \Delta_t)^2 x^2\big] &= \sigma^2 x^2\Big[x^2\,(\partial_{SS}V)^2 + x^2\,(\partial_S V - \Delta_t)\,\partial_{SSS}V \\
&\qquad + 4x\,(\partial_S V - \Delta_t)\,\partial_{SS}V + (\partial_S V - \Delta_t)^2\Big],
\end{aligned}$$

the second line by the differentiated equation. In the sum the terms in $\partial_{SSS}V$ cancel, and what is left is $\sigma^2\big[x^4\,(\partial_{SS}V)^2 + 2x^3\,(\partial_S V - \Delta_t)\,\partial_{SS}V + x^2\,(\partial_S V - \Delta_t)^2\big]$. So the drift is the square

$$\sigma^2\,\big(x\,(\partial_S V - \Delta_t) + x^2\,\partial_{SS}V\big)^2.$$

The outer function and its derivatives have moderate growth on $(0,\infty)\times[t, s]$, uniformly in the frozen value, because $s < T$ and $|\Delta_t| \le 1$; and $\Delta_t = \Phi(z_t)$ is a continuous function of the frozen value W_t. Corollary 19.31, in its parameter clause with the frozen value W_t, therefore gives its integral identity on $[t, s]$: with probability one, $(\Delta_s - \Delta_t)^2 S_s^2$ is the value 0 of the outer function at (S_t, t), plus the integral over $[t, s]$ of the drift above at $(S_{s'}, s')$, plus an Itô integral. That corollary states its expectation identity without the frozen value, so we take expectations here directly. The left side is at most $4S_s^2$, so it has a finite mean. The integrand of the Itô integral is a function of $(s', W_{s'}, W_t)$ of exponential growth, continuous in its three arguments, which is a Brownian integrand with W_t frozen (Definition 19.22(a)–(b)); so the Itô integral has mean zero (Theorem 19.23(ii)). The drift is nonnegative and continuous in s' on every path, so Fact 18.34(iii) moves the expectation inside the ds'-integral. Hence

$$\mathbb{E}\big[(\Delta_s - \Delta_t)^2\, S_s^2\big] = \sigma^2\int_t^s \mathbb{E}\Big[\big(S_{s'}\,(\Delta_{s'} - \Delta_t) + \Gamma_{s'} S_{s'}^2\big)^2\Big]\,ds' \le 2\sigma^2\int_t^s \Big(\mathbb{E}\,S_{s'}^2 + \mathbb{E}\big[\Gamma_{s'}^2 S_{s'}^4\big]\Big)\,ds',$$

using $|\Delta_{s'} - \Delta_t| \le 1$ and $(y+z)^2 \le 2y^2 + 2z^2$. Here $\mathbb{E}\,S_{s'}^2 = S_0^2\,e^{\sigma^2 s'} \le S_0^2\,e^{\sigma^2 T}\sqrt{T}\,(T - s')^{-1/2}$, because $(T-s')^{-1/2} \ge T^{-1/2}$, and the second term is at most $c\,(T-s')^{-1/2}$ by (G.10). So the last integrand is at most $(S_0^2\,e^{\sigma^2 T}\sqrt{T} + c)(T-s')^{-1/2}$, and

$$\int_t^s (T-s')^{-1/2}\,ds' = 2\big(\sqrt{T-t} - \sqrt{T-s}\big) = \frac{2\,(s-t)}{\sqrt{T-t} + \sqrt{T-s}} \le \frac{2\,(s-t)}{\sqrt{T-t}},$$

which gives (G.11).

(c) Splitting at $T-\varepsilon'$. Fix $\varepsilon' \in (0, T/4)$, a small number of the proof, and let n be so large that $\Delta t < \varepsilon'$. For the cells contained in $[0, T-\varepsilon']$ the cell estimate of Step 5 holds, with constants that depend on ε', because every derivative of V has moderate growth on $(0,\infty)\times[0,T-\varepsilon']$. Their contribution to $n \operatorname{Var} \mathcal{E}_n$ is $\frac{1}{2}\sigma^4 T$ times a left Riemann sum of $t \mapsto \mathbb{E}[\Gamma_t^2 S_t^4]$ over an interval that ends within Δt of $T-\varepsilon'$, plus $O(n^{-1/2})$, so it converges to $\frac{1}{2}\sigma^4 T \int_0^{T-\varepsilon'} \mathbb{E}[\Gamma_t^2 S_t^4]\, dt$. Each remaining cell $[t_i, t_{i+1}]$ has $t_{i+1} > T-\varepsilon'$. Write $T - t_i = j\,\Delta t$, where the integer $j \geq 1$ counts the cells back from expiry; then $(j-1)\,\Delta t < \varepsilon'$, and $t_i > T - \varepsilon' - \Delta t > T/2$. By Step 3 and (G.11) with t_i in the place of t, the last cell included,

$$\mathbb{E}\,\varepsilon_i^2 \leq \sigma^2 \int_{t_i}^{t_{i+1}} \frac{c'\,(s-t_i)}{\sqrt{j\,\Delta t}}\, ds = c''\,(\Delta t)^{3/2}\, j^{-1/2}, \qquad c'' = \sigma^2 c'/2.$$

Let J' be the number of the remaining cells, so that $(J'-1)\,\Delta t < \varepsilon'$. Since $j^{-1/2} \leq \int_{j-1}^{j} y^{-1/2}\, dy = 2(\sqrt{j} - \sqrt{j-1})$, the sum $\sum_{j=1}^{J'} j^{-1/2}$ is at most $2\sqrt{J'}$, and the remaining cells contribute to $\operatorname{Var} \mathcal{E}_n$ at most

$$2c''\,(\Delta t)^{3/2}\sqrt{J'} = 2c''\,\Delta t\,\sqrt{J'\,\Delta t} \leq 2c''\,\Delta t\,\sqrt{\varepsilon' + \Delta t},$$

which is at most $2c''\,T\sqrt{\varepsilon' + \Delta t}$ after multiplication by $n = T/\Delta t$.

(d) Conclusion. The difference between $n \operatorname{Var} \mathcal{E}_n$ and the limit of (iii) is at most the sum of three terms: the difference between the contribution of the cells in $[0, T-\varepsilon']$ and $\frac{1}{2}\sigma^4 T \int_0^{T-\varepsilon'} \mathbb{E}[\Gamma_t^2 S_t^4]\, dt$, which tends to 0 by (c); the contribution of the remaining cells, bounded in (c); and the tail $\frac{1}{2}\sigma^4 T \int_{T-\varepsilon'}^{T} \mathbb{E}[\Gamma_t^2 S_t^4]\, dt$, bounded by (G.10). Hence

$$\begin{aligned}\limsup_{n\to\infty} \Big| n \operatorname{Var} \mathcal{E}_n - \tfrac{1}{2}\sigma^4 T \int_0^T \mathbb{E}\big[\Gamma_t^2 S_t^4\big]\, dt \Big| &\leq 2c''\,T\sqrt{\varepsilon'} + \tfrac{1}{2}\sigma^4 T \int_{T-\varepsilon'}^{T} c\,(T-t)^{-1/2}\, dt \\ &= \big(2c''\,T + \sigma^4\,T\,c\big)\sqrt{\varepsilon'},\end{aligned}$$

and ε' is arbitrary. This is (iii) for $T' = T$. No rate is claimed in this case. □

G.9 Proof of Proposition 21.34: The stop-loss rule leaves the time value as expected shortfall

The statement is on page 389. The notation and the standing assumptions are those of §21.7.

Proof. We integrate out each future move with Lemma 19.17, in three steps.

Step 1 (the payoff). At $r = q = 0$ the risk-neutral price path of Definition 20.6(i) is $S_0 \exp(-\sigma^2 t/2 + \sigma W_t)$, which is the path of the underlier itself. The risk-neutral expectation of a quantity computed from the path is therefore its ordinary expectation, and $\mathbb{E}^{\mathbb{Q}}$ is written $\mathbb{E}$ in the rest of the proof. By Definition 19.5, $W_T \sim \mathcal{N}(0, T)$, so $Z = W_T/\sqrt{T}$ is a standard normal variable and

$$S_T = S_0 \exp\big(-\sigma^2 T/2 + \sigma\sqrt{T}\,Z\big).$$

This is the lognormal law (20.6) with $F = S_0$, the forward price at $r = q = 0$. Theorem 20.22 computes the expectation of the call payoff under that law: at $r = 0$ its formula (20.8) reads

$$\mathbb{E}\,(S_T - K)^+ = S_0\,\Phi(d_1) - K\,\Phi(d_2) = C,$$

where d_1 and d_2 are those of (21.1) with $r = q = 0$, $S = S_0$ and $\tau = T$, and the last equality is (21.2). By Theorem 20.7(iv) this expectation is the arbitrage-free price of the call at $r = 0$, which is why it is called C; the proof uses its value only. The payoff is integrable, because $0 \leq (S_T - K)^+ \leq S_T$ and $\mathbb{E}\,S_T = S_0$ (Lemma 18.35(i) with $\theta = \sigma\sqrt{T}$).

Step 2 (each hedge term has mean zero). Fix i. The holding $h_i = \mathbf{1}\{S_{t_i} > K\}$ is a bounded function of S_{t_i}, continuous outside the hyperplane $\{S_{t_i} = K\}$ and hence Borel measurable, and $|h_i\,(S_{t_{i+1}} - S_{t_i})| \le S_{t_{i+1}} + S_{t_i}$, whose mean is $2S_0$ because $\mathbb{E}\,S_t = S_0$ for every t (Lemma 18.35(i) with $\theta = \sigma\sqrt{t}$, as in Step 1); so the hedge term is integrable. The price path is the driftless geometric Brownian motion of Proposition 21.19 with $\sigma_{\text{real}} = \sigma$, and Step 3 of the proof of that proposition holds for every holding that is a Borel measurable function of $(S_{t_0}, \dots, S_{t_i})$ with an integrable hedge term: it writes $S_{t_{i+1}} - S_{t_i} = S_{t_i}(Y_i - 1)$, where the ratio $Y_i = S_{t_{i+1}}/S_{t_i}$ is independent of the prices up to t_i and has mean 1, and integrates Y_i out with Lemma 19.17. Hence $\mathbb{E}[h_i\,(S_{t_{i+1}} - S_{t_i})] = 0$.

Step 3 (conclusion). The shortfall is a sum of $n+2$ integrable terms: the payoff, the constant $-(S_0 - K)^+$ and the n hedge terms. By linearity of expectation (Theorem 1.2, in the general form declared in §19.2) and Steps 1 and 2, its mean is $C - (S_0 - K)^+ - 0$, which is (21.15). The grid entered through Step 2 alone, and Step 2 holds for every grid.

The same result from the buyer's side. Proposition 21.19, with $\sigma_{\text{real}} = \sigma_{\text{imp}} = \sigma$ and the holdings h_i of the stop-loss rule, says that the trader who buys the call for $V^{\text{BSM}}(S_0, 0; \sigma)$, which is C by Step 1, and holds $-h_i$ units over $(t_i, t_{i+1}]$ has the expected P and L $C - C = 0$:

$$\mathbb{E}\Big[(S_T - K)^+ - C - \sum_{i=0}^{n-1} h_i\,(S_{t_{i+1}} - S_{t_i})\Big] = 0.$$

Adding $C - (S_0 - K)^+$ to both sides gives (21.15) again. The writer who runs the stop-loss rule holds the other side of that trader's hedge. □

G.10 Proof of Proposition 21.39: Synthetic put

The statement is on page 394. The notation and the standing assumptions are those of §21.7.

Proof. We cite the replication theorem in its yield form, and then add strategies and deltas.

The replication. The put payoff $f(x) = (K - x)^+$ is continuous, and $0 \le f(x) \le K \le K\,(x^K + x^{-K})$ because $x^K + x^{-K} \ge 2$ for $x > 0$ (the arithmetic–geometric mean inequality, since $x^K \cdot x^{-K} = 1$); so f belongs to the class $\mathcal{G}$ of Definition 19.19(iv), with the constant $c = K$. Let V be its price function, the function $V(S,t)$ defined in Corollary 20.41(i). Its part (c), the yield case of Theorem 20.7(iii) obtained by reduction to the reinvested holding, states that the strategy of Definition 20.1 with the initial capital $V(S_0, 0)$ that holds $e^{q(T-t)}\,\partial_S V(S_t, t)$ units of $\tilde{S}$ at every time $t < T$ has the value $V(S_t, t)$ at every $t < T$ and the terminal value $f(S_T)$. By part (a) of the same corollary, for the put payoff and every $t < T$, $V(S,t) = Ke^{-r(T-t)}\Phi(-d_2) - Se^{-q(T-t)}\Phi(-d_1)$, with $T - t$ in the place of T in d_1 and d_2: this is the function $\mathcal{P}(S,t)$ of (21.2). Proposition 21.2 then gives $\partial_S V = \partial_S \mathcal{P} = \Delta_{\mathcal{P}}$. The strategy of the statement is therefore this replicating strategy, which proves the two claims about its value.

Its position, in units of the asset, is $\Delta_{\mathcal{P}}\,e^{q\tau} \times e^{-q\tau} = \Delta_{\mathcal{P}}$, which is negative. By Definition 20.1(ii) its bank holding is its value minus the value of its holding, $\mathcal{P} - \Delta_{\mathcal{P}}\,e^{q\tau} \times Se^{-q\tau} = \mathcal{P} - \Delta_{\mathcal{P}} S$. Since $\Delta_{\mathcal{P}} S = -Se^{-q\tau}\Phi(-d_1)$ (Proposition 21.2), the put formula in (21.2) gives $\mathcal{P} - \Delta_{\mathcal{P}} S = Ke^{-r\tau}\Phi(-d_2)$, which is positive.

Part (a). The owner's position holds one unit of the asset at every time, that is, $e^{q(T-t)}$ units of $\tilde{S}$ at the time t, and it starts from some capital, for instance the price S_0 of the unit. This holding is a continuous and bounded function of the time alone, so it is admissible on the closed interval $[0, T]$, and the owner's position is a strategy with a terminal value (Definition 20.1(i)–(ii) and Check 2 of that definition, on the value at a switch date). By Definition 20.1(iv) the sum of two strategies is a strategy, and the holdings, the initial capitals, the values and the terminal values add. The sum therefore holds $(1 + \Delta_{\mathcal{P}})\,e^{q\tau}$ units of $\tilde{S}$, which is $1 + \Delta_{\mathcal{P}}$ units of the asset. Its bank holding, which is its value minus the value of its holding, is the owner's bank balance

plus $\mathcal{P} - \Delta_{\mathcal{P}} S$. Its value is the owner's value plus $\mathcal{P}(S_t, t)$ at every $t < T$, and plus $(K - S_T)^+$ at T. Finally $1 + \Delta_{\mathcal{P}} = 1 - e^{-q\tau}\Phi(-d_1)$ lies strictly between 0 and 1, because $0 < e^{-q\tau} \le 1$ for $q \ge 0$ and $0 < \Phi(-d_1) < 1$. This is the additivity of delta (Definition 21.1(ii)) read as a holding: the unit has delta 1 and the put has delta $\Delta_{\mathcal{P}}$.

Part (b). This part is a statement about deltas. By Proposition 21.4(i) and (iii) and by additivity, one unit of the asset together with a short position of w futures contracts has the delta $1 - w\, e^{(r-q)\tau_F}$. It equals $1 + \Delta_{\mathcal{P}}$ exactly when

$$w = -\Delta_{\mathcal{P}}\, e^{-(r-q)\tau_F} = e^{-q\tau}\big(1 - \Phi(d_1)\big)\, e^{-(r-q)\tau_F}.$$

For the insured portfolio, the n_{put} puts have the delta $n_{\text{put}}\Delta_{\mathcal{P}}$ in index units (Definition 21.1(ii)), the delta of a short position of $n_{\text{put}}\,|\Delta_{\mathcal{P}}|$ index units. By the computation just made, the same delta is carried by a short position in futures on $n_{\text{put}}\,|\Delta_{\mathcal{P}}|\, e^{-(r-q)\tau_F}$ index units. One contract is on a number of index units equal to its multiplier, and its contract value in the sense of §16.8 is $V_F = S e^{(r-q)\tau_F} \times \text{multiplier}$, the futures price times the multiplier (the position itself is worth zero after each settlement). With $n_{\text{put}} = \beta V_A / S$ (Proposition 21.37(ii)), the number of contracts is

$$\frac{n_{\text{put}}\,|\Delta_{\mathcal{P}}|\, e^{-(r-q)\tau_F}}{\text{multiplier}} = |\Delta_{\mathcal{P}}|\, \frac{\beta V_A}{S e^{(r-q)\tau_F} \times \text{multiplier}} = |\Delta_{\mathcal{P}}|\, \frac{\beta V_A}{V_F} = |\Delta_{\mathcal{P}}|\, N_F^*. \qquad \square$$

H Proofs for §22: Volatility: Smiles, Surfaces, Estimation and Variance Swaps

H.1 Proof of Theorem 22.7: Breeden–Litzenberger

The statement is on page 403. The notation and the standing assumptions are those of §22.2.

Proof. We compute the difference quotient of the call curve exactly and squeeze it between two probabilities; no limit is taken under an integral sign.

Step 1: an exact difference. Fix $K > 0$ and $\Delta K > 0$. On $[K, \infty)$ the integrands $(x - K) f_{\mathbb{Q}}(x)$ and $f_{\mathbb{Q}}(x)$ are nonnegative, and their integrals are finite: the first integrand is at most $x\, f_{\mathbb{Q}}(x)$ there, and $\int_0^\infty x\, f_{\mathbb{Q}}(x)\, dx = \mathbb{E}^{\mathbb{Q}}[S_T]$ and $\int_0^\infty f_{\mathbb{Q}}(x)\, dx = 1$ are finite. The integrals below may therefore be split at $K + \Delta K$ and recombined term by term. By (22.1) at the strikes $K + \Delta K$ and K,

$$\begin{aligned} e^{rT}\, C(K + \Delta K) &= \int_{K+\Delta K}^{\infty} (x - K - \Delta K)\, f_{\mathbb{Q}}(x)\, dx \\ &= \int_{K+\Delta K}^{\infty} (x - K)\, f_{\mathbb{Q}}(x)\, dx \;-\; \Delta K \int_{K+\Delta K}^{\infty} f_{\mathbb{Q}}(x)\, dx, \\ e^{rT}\, C(K) &= \int_{K}^{K+\Delta K} (x - K)\, f_{\mathbb{Q}}(x)\, dx \;+\; \int_{K+\Delta K}^{\infty} (x - K)\, f_{\mathbb{Q}}(x)\, dx. \end{aligned}$$

Subtract the second line from the first. The integral of $(x - K) f_{\mathbb{Q}}(x)$ over $[K + \Delta K, \infty)$ cancels, and $\int_{K+\Delta K}^{\infty} f_{\mathbb{Q}}(x)\, dx = \mathbb{Q}(S_T > K + \Delta K)$, so

$$C(K + \Delta K) - C(K) \;=\; -\,e^{-rT}\Big[\Delta K\; \mathbb{Q}(S_T > K + \Delta K) + \int_{K}^{K+\Delta K} (x - K)\, f_{\mathbb{Q}}(x)\, dx\Big]. \tag{H.1}$$

Step 2: two sandwiches. On $[K, K + \Delta K]$ the weight $x - K$ lies between 0 and ΔK, and $f_{\mathbb{Q}} \ge 0$, so

$$0 \;\le\; \int_{K}^{K+\Delta K} (x - K)\, f_{\mathbb{Q}}(x)\, dx \;\le\; \Delta K \int_{K}^{K+\Delta K} f_{\mathbb{Q}}(x)\, dx \;=\; \Delta K\; \mathbb{Q}(K < S_T \le K + \Delta K).$$

Insert these two bounds into (H.1), divide by $\Delta K > 0$, and use $\mathbb{Q}(S_T > K + \Delta K) + \mathbb{Q}(K < S_T \le K + \Delta K) = \mathbb{Q}(S_T > K)$. The result is the *right-hand sandwich*

$$-\,e^{-rT}\,\mathbb{Q}(S_T > K) \;\le\; \frac{C(K+\Delta K) - C(K)}{\Delta K} \;\le\; -\,e^{-rT}\,\mathbb{Q}(S_T > K + \Delta K). \tag{H.2}$$

The same inequality with $K - \Delta K$ in place of K, which requires $0 < \Delta K < K$, is the *left-hand sandwich*

$$-\,e^{-rT}\,\mathbb{Q}(S_T > K - \Delta K) \;\le\; \frac{C(K) - C(K-\Delta K)}{\Delta K} \;\le\; -\,e^{-rT}\,\mathbb{Q}(S_T > K). \tag{H.3}$$

Since $\mathrm{Dig}(K) = e^{-rT}\,\mathbb{Q}(S_T > K)$, the first inequality of (H.2) and the second inequality of (H.3), each multiplied by -1, bracket the digital between two call spreads:

$$\frac{C(K) - C(K+\Delta K)}{\Delta K} \;\le\; \mathrm{Dig}(K) \;\le\; \frac{C(K-\Delta K) - C(K)}{\Delta K} \qquad (0 < \Delta K < K). \tag{H.4}$$

Step 3: the first derivative. The density $f_{\mathbb{Q}}$ is continuous on the closed bounded interval $[K/2, 2K]$, hence bounded there by some constant c. Let $0 < \Delta K \le K/2$. The intervals $[K, K + \Delta K]$ and $[K - \Delta K, K]$ lie inside $[K/2, 2K]$, so

$$\mathbb{Q}(K < S_T \le K + \Delta K) = \int_K^{K+\Delta K} f_{\mathbb{Q}}(x)\,dx \le c\,\Delta K,$$
$$\mathbb{Q}(K - \Delta K < S_T \le K) = \int_{K-\Delta K}^{K} f_{\mathbb{Q}}(x)\,dx \le c\,\Delta K.$$

Now $\mathbb{Q}(S_T > K + \Delta K) = \mathbb{Q}(S_T > K) - \mathbb{Q}(K < S_T \le K + \Delta K)$ and $\mathbb{Q}(S_T > K - \Delta K) = \mathbb{Q}(S_T > K) + \mathbb{Q}(K - \Delta K < S_T \le K)$, so (H.2) and (H.3) give

$$0 \;\le\; \frac{C(K+\Delta K) - C(K)}{\Delta K} + e^{-rT}\,\mathbb{Q}(S_T > K) \;\le\; e^{-rT}\,c\,\Delta K,$$
$$-\,e^{-rT}\,c\,\Delta K \;\le\; \frac{C(K) - C(K-\Delta K)}{\Delta K} + e^{-rT}\,\mathbb{Q}(S_T > K) \;\le\; 0.$$

As $\Delta K \to 0$ both difference quotients tend to $-e^{-rT}\,\mathbb{Q}(S_T > K)$. The right-hand and the left-hand derivatives of C at K therefore exist and agree: C is differentiable at K, with $\partial_K C(K) = -e^{-rT}\,\mathbb{Q}(S_T > K)$, which is (22.2). Since $\mathrm{Dig}(K) = e^{-rT}\,\mathbb{Q}(S_T > K)$ by the Setup of §22.2, $\mathrm{Dig}(K) = -\,\partial_K C(K)$.

Step 4: the second derivative. Fix a reference strike $K' > 0$. For every $K > 0$,

$$\mathbb{Q}(S_T > K) \;=\; \mathbb{Q}(S_T > K') - \int_{K'}^{K} f_{\mathbb{Q}}(x)\,dx,$$

where for $K < K'$ the integral is read as $-\int_K^{K'}$. Indeed, for $K \ge K'$ the event $\{S_T > K'\}$ is the disjoint union of $\{K' < S_T \le K\}$ and $\{S_T > K\}$, and for $K < K'$ the event $\{S_T > K\}$ is the disjoint union of $\{K < S_T \le K'\}$ and $\{S_T > K'\}$. Because $f_{\mathbb{Q}}$ is continuous on $(0, \infty)$, the fundamental theorem of calculus says that $K \mapsto \int_{K'}^{K} f_{\mathbb{Q}}(x)\,dx$ is differentiable on $(0, \infty)$ with derivative $f_{\mathbb{Q}}(K)$. Hence $K \mapsto \mathbb{Q}(S_T > K)$ is differentiable with derivative $-f_{\mathbb{Q}}(K)$, and differentiating (22.2) once more,

$$\partial_{KK} C(K) \;=\; -\,e^{-rT} \times \big(-\,f_{\mathbb{Q}}(K)\big) \;=\; e^{-rT} f_{\mathbb{Q}}(K),$$

which is (22.3). □

H.2 Proof of Corollary 22.10: The digital off the smile: flat-volatility price plus a vega-times-skew correction

The statement is on page 404. The notation and the standing assumptions are those of §22.2.

Proof. We differentiate the identity $C(K) = C^{\mathrm{BSM}}(K; \sigma_{\mathrm{imp}}(K))$ by the chain rule and compute the two partial derivatives.

Step 1: the digital is a slope. By (22.2), $\mathrm{Dig}(K) = -\,\partial_K C(K)$.

Step 2: the chain rule along the smile. The function $C^{\mathrm{BSM}}(K;\sigma) = e^{-rT}[F\,\Phi(d_1) - K\,\Phi(d_2)]$ is built from Φ, the logarithm, products, and quotients whose denominator $\sigma\sqrt{T}$ does not vanish, so its partial derivatives $\partial_K C^{\mathrm{BSM}}$ and $\partial_\sigma C^{\mathrm{BSM}}$ exist on $(0,\infty)^2$; Steps 3 and 4 compute them and show that they are continuous there. We claim that

$$\partial_K C(K) \;=\; \partial_K C^{\mathrm{BSM}} \;+\; \partial_\sigma C^{\mathrm{BSM}}\,\sigma'_{\mathrm{imp}}(K), \tag{H.5}$$

where the left side is the slope of the market curve and the two partial derivatives on the right are those of the formula, taken at the point $(K, \sigma_{\mathrm{imp}}(K))$. To verify this, let $\Delta K \neq 0$ (in this proof only, a strike increment of either sign, since the limit is two-sided) be small enough for $K + \Delta K$ to lie in the interval, and split the increment of C into a move of the strike at the new volatility and a move of the volatility at the old strike:

$$\begin{aligned} C(K+\Delta K) - C(K) = {} & \Big[C^{\mathrm{BSM}}\big(K+\Delta K; \sigma_{\mathrm{imp}}(K+\Delta K)\big) - C^{\mathrm{BSM}}\big(K; \sigma_{\mathrm{imp}}(K+\Delta K)\big)\Big] \\ & + \Big[C^{\mathrm{BSM}}\big(K; \sigma_{\mathrm{imp}}(K+\Delta K)\big) - C^{\mathrm{BSM}}\big(K; \sigma_{\mathrm{imp}}(K)\big)\Big]. \end{aligned}$$

By the mean value theorem in the strike, the first bracket equals $\partial_K C^{\mathrm{BSM}}(K^*; \sigma_{\mathrm{imp}}(K+\Delta K))\,\Delta K$ for some K^* between K and $K + \Delta K$. By the mean value theorem in the volatility, the second equals $\partial_\sigma C^{\mathrm{BSM}}(K;\sigma^*)\,(\sigma_{\mathrm{imp}}(K+\Delta K) - \sigma_{\mathrm{imp}}(K))$ for some σ^* between $\sigma_{\mathrm{imp}}(K)$ and $\sigma_{\mathrm{imp}}(K+\Delta K)$. Divide by ΔK and let $\Delta K \to 0$. A differentiable function is continuous, so $\sigma_{\mathrm{imp}}(K+\Delta K) \to \sigma_{\mathrm{imp}}(K)$, and with it $K^* \to K$ and $\sigma^* \to \sigma_{\mathrm{imp}}(K)$. The partial derivatives are continuous, so the two derivative factors tend to their values at $(K, \sigma_{\mathrm{imp}}(K))$, while $(\sigma_{\mathrm{imp}}(K+\Delta K) - \sigma_{\mathrm{imp}}(K))/\Delta K \to \sigma'_{\mathrm{imp}}(K)$. This is (H.5).

Step 3: the strike derivative. Since $d_1 = [\ln F - \ln K + \sigma^2 T/2]/(\sigma\sqrt{T})$ and $d_2 = d_1 - \sigma\sqrt{T}$, both satisfy $\partial_K d_1 = \partial_K d_2 = -1/(K\sigma\sqrt{T})$. Differentiating $C^{\mathrm{BSM}} = e^{-rT}[F\,\Phi(d_1) - K\,\Phi(d_2)]$ in K by the product and chain rules of Fact 7.1, with $\Phi' = \varphi$,

$$\begin{aligned} \partial_K C^{\mathrm{BSM}} &= e^{-rT}\Big[F\,\varphi(d_1)\,\partial_K d_1 - \Phi(d_2) - K\,\varphi(d_2)\,\partial_K d_2\Big] \\ &= -\,e^{-rT}\,\Phi(d_2) \;-\; e^{-rT}\,\frac{F\,\varphi(d_1) - K\,\varphi(d_2)}{K\sigma\sqrt{T}} \;=\; -\,e^{-rT}\,\Phi(d_2), \end{aligned}$$

because $F\,\varphi(d_1) = K\,\varphi(d_2)$ by Lemma 20.34. The strike derivative is minus the flat-volatility digital of Theorem 20.17.

Step 4: the volatility derivative. By Proposition 20.35, $\partial_\sigma C^{\mathrm{BSM}} = e^{-rT} F\,\varphi(d_1)\sqrt{T} = \mathcal{V}$. Both partial derivatives are continuous in (K,σ) on $(0,\infty)^2$, as Step 2 required, because d_1 and d_2 are, and Φ and φ are continuous.

Substituting Steps 3 and 4 into (H.5) at $\sigma = \sigma_{\mathrm{imp}}(K)$, and the result into Step 1,

$$\mathrm{Dig}(K) \;=\; -\,\partial_K C(K) \;=\; e^{-rT}\,\Phi(d_2) - \mathcal{V}\,\sigma'_{\mathrm{imp}}(K). \qquad \square$$

H.3 Proof of Proposition 22.15: The calendar condition: call prices and total implied variance are nondecreasing in the expiry; forward volatility is real

The statement is on page 409. The notation and the standing assumptions are those of §22.3.

Proof. We compare payoffs by dominance to order the prices, and then carry the order over to the implied volatilities, using the fact that the Black-76 price is strictly increasing in the total volatility.

Step 1: part (i), by citation. With $q = 0$ the underlying pays no income, which is case (N) of §17.2, and $r = 0$ is allowed there. Part (i) is therefore Theorem 17.16(iii) at $r = 0$. The strategy of that proof writes the T_1 call and buys the T_2 call. If $S_{T_1} > K$ at T_1, it sells one share short, pays the holder of the near call $S_{T_1} - K$ out of the proceeds and keeps the remaining K in cash, so that at T_2 it is worth $(S_{T_2} - K)^+ - S_{T_2} + K = (K - S_{T_2})^+ \geq 0$; otherwise it is worth $(S_{T_2} - K)^+ \geq 0$. Lemma 16.8(i) then gives $C(K, T_2) - C(K, T_1) \geq 0$. The result is restated as Proposition 22.15(i), not proved a second time, because "calendar condition" is the volatility desk's name for it and parts (ii) to (iv) build on it.

Step 2: part (i) implies part (ii). Let $r = q = 0$, so that $F = S$ at both expiries. By Proposition 20.35 with $T = 1$ and $r = 0$, the function $\sigma \mapsto C^{\mathrm{BSM}}(K, 1; \sigma)$ has the derivative $F\,\varphi(d_1) > 0$ on $(0, \infty)$; a function with a positive derivative on an interval is strictly increasing there, by the mean value theorem. By the definition of the implied volatility and by (22.4),

$$C(K, T_j) = C^{\mathrm{BSM}}\big(K, T_j; \sigma_{\mathrm{imp}}(K, T_j)\big) = C^{\mathrm{BSM}}\big(K, 1; \sigma_{\mathrm{imp}}(K, T_j)\sqrt{T_j}\big) \qquad (j = 1, 2).$$

Part (i) therefore reads

$$C^{\mathrm{BSM}}\big(K, 1; \sigma_{\mathrm{imp}}(K, T_1)\sqrt{T_1}\big) \;\leq\; C^{\mathrm{BSM}}\big(K, 1; \sigma_{\mathrm{imp}}(K, T_2)\sqrt{T_2}\big).$$

If $\sigma_{\mathrm{imp}}(K, T_1)\sqrt{T_1}$ were larger than $\sigma_{\mathrm{imp}}(K, T_2)\sqrt{T_2}$, strict monotonicity would give the opposite strict inequality. Hence $\sigma_{\mathrm{imp}}(K, T_1)\sqrt{T_1} \leq \sigma_{\mathrm{imp}}(K, T_2)\sqrt{T_2}$, and squaring the two nonnegative sides gives (ii).

Step 3: part (iv), carry. We repeat the dominance argument of Step 1 with scaled positions. The condition $K_1/F(T_1) = K_2/F(T_2)$ and $F(T_2) = F(T_1)e^{(r-q)(T_2-T_1)}$ give

$$K_2 = K_1\, e^{(r-q)(T_2-T_1)}.$$

At time 0 buy one T_2 call struck at K_2 and write $e^{-q(T_2-T_1)}$ calls expiring at T_1 and struck at K_1; the position costs $C(K_2, T_2) - e^{-q(T_2-T_1)}\,C(K_1, T_1)$. At T_1 two cases arise, and which one holds is known at T_1.

- If $S_{T_1} > K_1$, the written calls are owed $e^{-q(T_2-T_1)}(S_{T_1} - K_1)$. Sell $e^{-q(T_2-T_1)}$ units of the underlying short, which brings in $e^{-q(T_2-T_1)}S_{T_1}$; pay the holders of the near calls; and deposit the remaining $e^{-q(T_2-T_1)}K_1$ at the rate r until T_2, where it is worth $e^{-q(T_2-T_1)}K_1\,e^{r(T_2-T_1)} = K_2$. The trades at T_1 net to zero cash. The short seller owes the yield of the borrowed units to their lender, and pays it by selling more units short as it falls due: a short holding of n_t units owes $qS_tn_t\,dt$ over $[t, t + dt]$ and covers it by selling $qn_t\,dt$ more units, so the short holding is a reinvested holding run short. Its unit count therefore solves $dn_t/dt = q\,n_t$, the equation (16.2), on $[T_1, T_2]$ with $n_{T_1} = e^{-q(T_2-T_1)}$. The computation in the proof of Proposition 16.15(i), carried out on that interval, gives $n_t = e^{-q(T_2-t)}$, so the position is short exactly one unit at T_2. At T_2 the long call pays $(S_{T_2} - K_2)^+$, closing the short position costs S_{T_2}, and the deposit returns K_2, so the position is worth

$$(S_{T_2} - K_2)^+ - (S_{T_2} - K_2) = (K_2 - S_{T_2})^+ \;\geq\; 0.$$

- If $S_{T_1} \leq K_1$, the written calls expire worthless, nothing is traded, and at T_2 the position is worth $(S_{T_2} - K_2)^+ \geq 0$.

The holdings of the strategy change only at T_1, on what is known then: the reinvested short holding, whose unit count is a function of time fixed once it is opened, counts as one position and uses none of the changes allowed to a strategy (Setup of §16.2). No cash is added or withdrawn

after time 0, so the strategy is self-financing, and its payoff at T_2 is nonnegative with probability one. Lemma 16.8(i), with T_2 as the horizon and the empty strategy (price 0, payoff 0) as the dominated one, gives

$$C(K_2,T_2) - e^{-q(T_2-T_1)}\,C(K_1,T_1) \;\geq\; 0.$$

Multiply this inequality by $e^{rT_2}/F(T_2) > 0$. On the right-hand side,

$$\frac{e^{rT_2}\,e^{-q(T_2-T_1)}}{F(T_2)} = \frac{e^{rT_2-q(T_2-T_1)-(r-q)(T_2-T_1)}}{F(T_1)} = \frac{e^{rT_1}}{F(T_1)},$$

which gives the normalised form $e^{rT_1}C(K_1,T_1)/F(T_1) \leq e^{rT_2}C(K_2,T_2)/F(T_2)$.

For the implied volatilities, divide the Black-76 price of §22.1 by $e^{-rT}F$:

$$\frac{e^{rT}\,C^{\mathrm{BSM}}(K,T;\sigma)}{F} = \Phi(d_1) - \frac{K}{F}\,\Phi(d_2), \qquad d_1,d_2 = \frac{-\ln(K/F) \pm (\sigma\sqrt{T})^2/2}{\sigma\sqrt{T}}.$$

The right-hand side depends on K, F, T and σ only through the moneyness K/F and the total volatility $\sigma\sqrt{T}$: it is the Black-76 price computed with forward 1, strike K/F, expiry 1, rate 0 and volatility $\sigma\sqrt{T}$. By Proposition 20.35 with those inputs, its derivative in the total volatility is $\varphi(d_1) > 0$, so at a fixed moneyness it is strictly increasing in the total volatility. The two strikes have the same moneyness, so the normalised form of the inequality compares two values of this one increasing function, at the total volatilities $\sigma_{\mathrm{imp}}(K_1,T_1)\sqrt{T_1}$ and $\sigma_{\mathrm{imp}}(K_2,T_2)\sqrt{T_2}$. The argument of Step 2 gives $\sigma_{\mathrm{imp}}(K_1,T_1)\sqrt{T_1} \leq \sigma_{\mathrm{imp}}(K_2,T_2)\sqrt{T_2}$, and squaring gives the last claim of (iv).

Step 4: part (iii). The numerator of the forward variance of Definition 22.13(ii) is nonnegative, by (ii) with zero carry (where $K_1 = K_2 = K$) and by (iv) with carry, both implied volatilities existing. Dividing by $T_2 - T_1 > 0$ keeps the sign, so the forward variance is nonnegative and its square root is real. □

H.4 Proof of Proposition 22.20: Unbiasedness, standard error and bias of the historical estimator

The statement is on page 414. The notation and the standing assumptions are those of §22.4.

Proof. We compute the moments of the squared returns up to order four, and then transfer them to the square root by an exact identity. Part (iii) is proved before part (ii), which uses it.

Part (i). By Proposition 7.25(ii), $\mathbb{E}[R_i^2] = \sigma_{\mathrm{per}}^2$, and linearity of expectation (Theorem 1.2) gives $\mathbb{E}[\hat{\sigma}_{\mathrm{per}}^2] = \frac{1}{n}\sum_i \mathbb{E}[R_i^2] = \sigma_{\mathrm{per}}^2$. By (1.1) and Proposition 7.25(iii),

$$\mathrm{Var}(R_i^2) = \mathbb{E}[R_i^4] - \left(\mathbb{E}[R_i^2]\right)^2 = 3\,\sigma_{\mathrm{per}}^4 - \sigma_{\mathrm{per}}^4 = 2\,\sigma_{\mathrm{per}}^4.$$

For $i \neq j$ the squares are uncorrelated. Indeed R_i and R_j are independent, and the function $\psi_1(x) = x^2 - \sigma_{\mathrm{per}}^2$ satisfies $\mathbb{E}|\psi_1(R_i)| \leq 2\sigma_{\mathrm{per}}^2 < \infty$, so the product rule of Lemma 19.17, with $\psi_2 = \psi_1$, gives

$$\mathrm{Cov}(R_i^2, R_j^2) = \mathbb{E}[\psi_1(R_i)\,\psi_1(R_j)] = \mathbb{E}[\psi_1(R_i)]\,\mathbb{E}[\psi_1(R_j)] = 0.$$

The variance of the sum now follows by induction on the number of terms. If $\mathrm{Var}\left(\sum_{i=1}^m R_i^2\right) = 2m\,\sigma_{\mathrm{per}}^4$ for some $m \geq 1$, which holds at $m = 1$, then Proposition 10.10 gives

$$\mathrm{Var}\Big(\sum_{i=1}^{m+1} R_i^2\Big) = \mathrm{Var}\Big(\sum_{i=1}^{m} R_i^2\Big) + \mathrm{Var}\left(R_{m+1}^2\right) + 2\,\mathrm{Cov}\Big(\sum_{i=1}^{m} R_i^2,\; R_{m+1}^2\Big) = 2(m+1)\,\sigma_{\mathrm{per}}^4,$$

because the covariance is the sum of the covariances $\mathrm{Cov}(R_i^2, R_{m+1}^2) = 0$, by linearity of expectation. Hence $\mathrm{Var}\left(\sum_{i=1}^n R_i^2\right) = 2n\,\sigma_{\mathrm{per}}^4$, and dividing by n^2 gives $\mathrm{Var}(\hat{\sigma}_{\mathrm{per}}^2) = 2\sigma_{\mathrm{per}}^4/n$.

Part (iii). By the difference of squares, $\hat{\sigma}_{\text{per}}^2 - \sigma_{\text{per}}^2 = (\hat{\sigma}_{\text{per}} - \sigma_{\text{per}})(\hat{\sigma}_{\text{per}} + \sigma_{\text{per}})$, and $\hat{\sigma}_{\text{per}} + \sigma_{\text{per}} \geq \sigma_{\text{per}} > 0$ because $\hat{\sigma}_{\text{per}} \geq 0$. Hence

$$|\hat{\sigma}_{\text{per}} - \sigma_{\text{per}}| = \frac{|\hat{\sigma}_{\text{per}}^2 - \sigma_{\text{per}}^2|}{\hat{\sigma}_{\text{per}} + \sigma_{\text{per}}} \;\leq\; \frac{|\hat{\sigma}_{\text{per}}^2 - \sigma_{\text{per}}^2|}{\sigma_{\text{per}}}. \tag{H.6}$$

Square both sides, take expectations and use (i): since $\mathbb{E}[\hat{\sigma}_{\text{per}}^2] = \sigma_{\text{per}}^2$, the mean of $(\hat{\sigma}_{\text{per}}^2 - \sigma_{\text{per}}^2)^2$ is the variance of $\hat{\sigma}_{\text{per}}^2$, so

$$\mathbb{E}\big[(\hat{\sigma}_{\text{per}} - \sigma_{\text{per}})^2\big] \;\leq\; \frac{\text{Var}\,(\hat{\sigma}_{\text{per}}^2)}{\sigma_{\text{per}}^2} = \frac{2\,\sigma_{\text{per}}^2}{n}.$$

Part (ii). The identity (22.6) is algebra. Writing $\hat{\sigma}_{\text{per}} + \sigma_{\text{per}} = (\hat{\sigma}_{\text{per}} - \sigma_{\text{per}}) + 2\sigma_{\text{per}}$ in the difference of squares,

$$\frac{\hat{\sigma}_{\text{per}}^2 - \sigma_{\text{per}}^2}{2\,\sigma_{\text{per}}} = \frac{(\hat{\sigma}_{\text{per}} - \sigma_{\text{per}})\big((\hat{\sigma}_{\text{per}} - \sigma_{\text{per}}) + 2\sigma_{\text{per}}\big)}{2\,\sigma_{\text{per}}} = (\hat{\sigma}_{\text{per}} - \sigma_{\text{per}}) + \frac{(\hat{\sigma}_{\text{per}} - \sigma_{\text{per}})^2}{2\,\sigma_{\text{per}}},$$

and moving the last term to the left gives (22.6). The identity is the Taylor expansion of order one of the square root at the point σ_{per}^2 (Theorem 7.31: the derivative of $\sqrt{x}$ at σ_{per}^2 is $1/(2\sigma_{\text{per}})$), with its remainder written in closed form. By (i), the linear term has mean 0 and variance

$$\frac{\text{Var}\,(\hat{\sigma}_{\text{per}}^2)}{4\,\sigma_{\text{per}}^2} = \frac{\sigma_{\text{per}}^2}{2n},$$

so its standard deviation is $\sigma_{\text{per}}/\sqrt{2n}$. The remainder $-(\hat{\sigma}_{\text{per}} - \sigma_{\text{per}})^2/(2\sigma_{\text{per}})$ is nonpositive, and by (iii) its mean is at least $-(2\sigma_{\text{per}}^2/n)/(2\sigma_{\text{per}}) = -\sigma_{\text{per}}/n$. Taking expectations in (22.6), the linear term contributes 0 and the remainder a number in $[-\sigma_{\text{per}}/n, 0]$, which is the bias statement.

Part (iv). Write ε_{lin} and ε_{rem} for the two terms on the right of (22.6), names local to this proof, so that $\hat{\sigma}_{\text{per}} = \sigma_{\text{per}} + \varepsilon_{\text{lin}} + \varepsilon_{\text{rem}}$.

Step 1: a fourth moment. Put $Y_i = R_i^2 - \sigma_{\text{per}}^2$, so that

$$\hat{\sigma}_{\text{per}}^2 - \sigma_{\text{per}}^2 = \frac{1}{n}\sum_{i=1}^{n} Y_i, \qquad \mathbb{E}[Y_i] = 0, \qquad \mathbb{E}[Y_i^2] = \text{Var}(R_i^2) = 2\,\sigma_{\text{per}}^4$$

by (i). The reduction (7.2) at the orders six and eight gives $\mathbb{E}[R_i^6] = 5\sigma_{\text{per}}^2 \times 3\sigma_{\text{per}}^4 = 15\,\sigma_{\text{per}}^6$ and $\mathbb{E}[R_i^8] = 7\sigma_{\text{per}}^2 \times 15\sigma_{\text{per}}^6 = 105\,\sigma_{\text{per}}^8$; every even moment of a normal variable is finite. The binomial expansion of $(R_i^2 - \sigma_{\text{per}}^2)^4$ and linearity of expectation then give

$$\begin{aligned}\mathbb{E}[Y_i^4] &= \mathbb{E}[R_i^8] - 4\,\sigma_{\text{per}}^2\,\mathbb{E}[R_i^6] + 6\,\sigma_{\text{per}}^4\,\mathbb{E}[R_i^4] - 4\,\sigma_{\text{per}}^6\,\mathbb{E}[R_i^2] + \sigma_{\text{per}}^8 \\ &= (105 - 60 + 18 - 4 + 1)\,\sigma_{\text{per}}^8 = 60\,\sigma_{\text{per}}^8.\end{aligned}$$

Now expand $\big(\sum_i Y_i\big)^4$ as the sum of the products $Y_a Y_b Y_c Y_d$ over the n^4 ordered quadruples of indices; the letters a, b, c, d are indices local to this proof. The quadruples fall into three classes.

- *Some index occurs exactly once in the quadruple*, say a. The returns are mutually independent: their joint density is the product of the n normal densities, which is the density of R_a times the joint density of the others. So R_a is independent of the vector formed by the other returns that occur in the term. The term is $\psi_1(R_a)$ times a function ψ_2 of that vector, the product of the three other factors. Both have finite means: $\mathbb{E}|\psi_1(R_a)| \leq 2\sigma_{\text{per}}^2$, and for real numbers y_1, y_2, y_3,

  $$|y_1 y_2 y_3| \leq \max\big(|y_1|, |y_2|, |y_3|\big)^3 \leq |y_1|^3 + |y_2|^3 + |y_3|^3,$$

  and $|y|^3 \leq 1 + y^4$ for every real y, so $\mathbb{E}|\psi_2| \leq 3\,(1 + 60\,\sigma_{\text{per}}^8)$. The product rule of Lemma 19.17 factors the mean of the term as $\mathbb{E}[Y_a] = 0$ times a finite number: the term has mean 0.

- *All four indices are equal.* There are n such terms, Y_a^4, each of mean $60\,\sigma_{\mathrm{per}}^8$.
- *Two distinct indices occur twice each.* The term is $Y_a^2 Y_b^2$ with $a \neq b$. There are $n(n-1)/2$ unordered pairs $\{a, b\}$ and $\binom{4}{2} = 6$ ways to place the two copies of a among the four positions, hence $3n(n-1)$ terms. Each has mean $\mathbb{E}[Y_a^2]\,\mathbb{E}[Y_b^2] = 4\,\sigma_{\mathrm{per}}^8$, by the same product rule with $\psi_1(x) = \psi_2(x) = (x^2 - \sigma_{\mathrm{per}}^2)^2$.

No other class exists: a quadruple in which no index occurs exactly once has every index at least twice, so it has one index four times or two indices twice each. Therefore

$$\mathbb{E}\big[(\hat{\sigma}_{\mathrm{per}}^2 - \sigma_{\mathrm{per}}^2)^4\big] = \frac{60\,n + 12\,n(n-1)}{n^4}\,\sigma_{\mathrm{per}}^8 = \sigma_{\mathrm{per}}^8\Big(\frac{12}{n^2} + \frac{48}{n^3}\Big). \tag{H.7}$$

Step 2: the remainder is small in mean square. By (H.6) raised to the fourth power,

$$\varepsilon_{\mathrm{rem}}^2 = \frac{(\hat{\sigma}_{\mathrm{per}} - \sigma_{\mathrm{per}})^4}{4\,\sigma_{\mathrm{per}}^2} \;\leq\; \frac{(\hat{\sigma}_{\mathrm{per}}^2 - \sigma_{\mathrm{per}}^2)^4}{4\,\sigma_{\mathrm{per}}^6},$$

so by (H.7)

$$\mathbb{E}\big[\varepsilon_{\mathrm{rem}}^2\big] \;\leq\; \sigma_{\mathrm{per}}^2\Big(\frac{3}{n^2} + \frac{12}{n^3}\Big).$$

Step 3: the triangle inequality. A standard deviation is the root-mean-square of the variable minus its mean. All three of $\hat{\sigma}_{\mathrm{per}}$, $\varepsilon_{\mathrm{lin}}$ and $\varepsilon_{\mathrm{rem}}$ have finite second moments, by (i), (ii) and Step 2, and by linearity

$$\hat{\sigma}_{\mathrm{per}} - \mathbb{E}[\hat{\sigma}_{\mathrm{per}}] = (\varepsilon_{\mathrm{lin}} - \mathbb{E}[\varepsilon_{\mathrm{lin}}]) + (\varepsilon_{\mathrm{rem}} - \mathbb{E}[\varepsilon_{\mathrm{rem}}]).$$

The triangle inequality for the mean-square distance (Lemma 19.12(ii)), applied to this sum, gives $\sqrt{\mathrm{Var}(\hat{\sigma}_{\mathrm{per}})} \leq \sqrt{\mathrm{Var}(\varepsilon_{\mathrm{lin}})} + \sqrt{\mathrm{Var}(\varepsilon_{\mathrm{rem}})}$. Applied to the rearrangement $\varepsilon_{\mathrm{lin}} - \mathbb{E}[\varepsilon_{\mathrm{lin}}] = (\hat{\sigma}_{\mathrm{per}} - \mathbb{E}[\hat{\sigma}_{\mathrm{per}}]) - (\varepsilon_{\mathrm{rem}} - \mathbb{E}[\varepsilon_{\mathrm{rem}}])$, it gives $\sqrt{\mathrm{Var}(\varepsilon_{\mathrm{lin}})} \leq \sqrt{\mathrm{Var}(\hat{\sigma}_{\mathrm{per}})} + \sqrt{\mathrm{Var}(\varepsilon_{\mathrm{rem}})}$. Together,

$$\Big|\sqrt{\mathrm{Var}(\hat{\sigma}_{\mathrm{per}})} - \sqrt{\mathrm{Var}(\varepsilon_{\mathrm{lin}})}\Big| \;\leq\; \sqrt{\mathrm{Var}(\varepsilon_{\mathrm{rem}})} \;\leq\; \sqrt{\mathbb{E}[\varepsilon_{\mathrm{rem}}^2]},$$

the last step because a variance is at most the second moment, by (1.1). Finally $\sqrt{\mathrm{Var}(\varepsilon_{\mathrm{lin}})} = \sigma_{\mathrm{per}}/\sqrt{2n}$ by (ii), and Step 2 bounds the right-hand side by $\sigma_{\mathrm{per}}\sqrt{3/n^2 + 12/n^3}$. Since $12/n^3 \leq 12/n^2$, this bound is at most $\sqrt{15}\,\sigma_{\mathrm{per}}/n$, a term of order $1/n$; dividing by σ_{per} gives the statement on the relative standard error. □

H.5 Proof of Proposition 22.27: GARCH(1,1) variance forecasts revert geometrically to the long-run level

The statement is on page 421. The notation and the standing assumptions are those of §22.6.

Proof. We proceed by induction on k, computing the one new expectation of each step with the product rule for independent variables; part (ii) then follows from part (i) by a second induction.

Preliminary: the variances are functions of the shocks. Substituting $R_{i+j} = \sigma_{i+j} Z_{i+j}$ into the recursion gives, for every $j \geq 0$,

$$\sigma_{i+j+1}^2 = \omega + (\alpha\, Z_{i+j}^2 + \beta)\,\sigma_{i+j}^2. \tag{H.8}$$

By induction on j, the variance j days ahead is therefore a function of the first j shocks, $\sigma_{i+j}^2 = \sigma_{i+j}^2(Z_i, \ldots, Z_{i+j-1})$. The functions are defined on real variables $z_i, z_{i+1}, \ldots$ by

$$\sigma_{i+j+1}^2(z_i, \ldots, z_{i+j}) = \omega + (\alpha\, z_{i+j}^2 + \beta)\,\sigma_{i+j}^2(z_i, \ldots, z_{i+j-1}),$$

starting from the known number σ_i^2, a function of no variable. The variance and the function that generates it share one letter, as the Itô integrand of §19 does. Every one of these functions is nonnegative, again by induction on j: $\sigma_i^2 > 0$, and the right-hand side is a sum of products of nonnegative numbers, because $\omega \geq 0$, $\alpha \geq 0$ and $\beta \geq 0$.

Part (i), base case $k = 0$. The variance σ_i^2 is a known number, so $\mathbb{E}[\sigma_i^2] = \sigma_i^2$ is finite.

Part (i), induction step. Let $k \geq 1$, and suppose that $\mathbb{E}[\sigma_{i+k-1}^2]$ is finite. Nothing else is assumed about it.

Step (a): the new expectation. We show that

$$\mathbb{E}\big[Z_{i+k-1}^2\,\sigma_{i+k-1}^2\big] = \mathbb{E}\big[\sigma_{i+k-1}^2\big].$$

A shock with mean 0 and variance 1 has $\mathbb{E}[Z^2] = \operatorname{Var}(Z) + (\mathbb{E}Z)^2 = 1$, by (1.1). For $k = 1$ the variance σ_i^2 is a constant, and the left side is $\sigma_i^2\,\mathbb{E}[Z_i^2] = \sigma_i^2$. For $k \geq 2$, put

$$X = (Z_i, \ldots, Z_{i+k-2}), \qquad Y = Z_{i+k-1}.$$

The shocks are mutually independent, which means that the joint law of $(Z_i, \ldots, Z_{i+k-1})$ is the product of the k marginal laws. Grouping the first $k-1$ factors, it is the product of the joint law of X and the law of Y, so X and Y are independent random vectors. Apply Lemma 19.17 to the nonnegative function

$$\psi(x, y) = y^2\,\sigma_{i+k-1}^2(x) \;\geq\; 0.$$

With x frozen, $\sigma_{i+k-1}^2(x)$ is a constant, so the frozen mean is $\bar{\psi}(x) = \sigma_{i+k-1}^2(x)\,\mathbb{E}[Y^2] = \sigma_{i+k-1}^2(x)$, and the lemma gives

$$\mathbb{E}\big[Z_{i+k-1}^2\,\sigma_{i+k-1}^2\big] = \mathbb{E}\,\psi(X, Y) = \mathbb{E}\,\bar{\psi}(X) = \mathbb{E}\big[\sigma_{i+k-1}^2\big],$$

which is finite by the induction hypothesis. The lemma is applied to the shocks themselves, so no step of the form "a function of independent variables is independent" is needed.

Step (b): one step of the recursion in expectation. Take expectations in (H.8) at $j = k-1$. The constant ω and the two random terms on the right have finite means, by Step (a) and the induction hypothesis, so linearity of expectation (Theorem 1.2) applies:

$$\mathbb{E}[\sigma_{i+k}^2] = \omega + \alpha\,\mathbb{E}\big[Z_{i+k-1}^2\,\sigma_{i+k-1}^2\big] + \beta\,\mathbb{E}\big[\sigma_{i+k-1}^2\big] = \omega + (\alpha + \beta)\,\mathbb{E}\big[\sigma_{i+k-1}^2\big],$$

which is (22.10); in particular $\mathbb{E}[\sigma_{i+k}^2]$ is finite. This completes the induction for part (i). Only the signs $\omega \geq 0$, $\alpha \geq 0$ and $\beta \geq 0$ were used, to make ψ and the variances nonnegative; no condition on $\alpha + \beta$ entered.

Step (c): part (ii), the deviation from the long-run level. Let now $\omega > 0$ and $\alpha + \beta < 1$; this is the only place where the inequality, through $\bar{\sigma}^2$, enters. The sign $\omega > 0$ is not used in part (ii). It is kept, as in Definition 22.24, so that $\bar{\sigma}^2 > 0$, which the proof of Corollary 22.28 uses (the bound $\min(\bar{\sigma}^2, x) > 0$ on the radicand). We prove (22.11) by induction on k. At $k = 0$ both sides equal σ_i^2, by the convention $(\alpha + \beta)^0 = 1$. Let $k \geq 1$ and suppose (22.11) holds with $k-1$ in place of k. The definition $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$, rearranged, reads $\bar{\sigma}^2 = \omega + (\alpha + \beta)\,\bar{\sigma}^2$. Subtracting this from (22.10) of part (i) and using the induction hypothesis,

$$\mathbb{E}[\sigma_{i+k}^2] - \bar{\sigma}^2 = (\alpha + \beta)\Big(\mathbb{E}[\sigma_{i+k-1}^2] - \bar{\sigma}^2\Big) = (\alpha + \beta)\,(\alpha + \beta)^{k-1}(\sigma_i^2 - \bar{\sigma}^2),$$

which is (22.11) for k. The induction is complete.

The fixed point and the limit. A number x satisfies $x = \omega + (\alpha + \beta)\,x$ exactly when $(1 - \alpha - \beta)\,x = \omega$, that is, when $x = \bar{\sigma}^2$; the solution is unique because $1 - \alpha - \beta \neq 0$. By Step (c), each application of the map multiplies the distance to $\bar{\sigma}^2$ by $\alpha + \beta$. Since $0 \leq \alpha + \beta < 1$, the power $(\alpha + \beta)^k$ tends to 0 as $k \to \infty$, so the fixed point is attracting and $\mathbb{E}[\sigma_{i+k}^2] \to \bar{\sigma}^2$. □

H.6 Proof of Corollary 22.28: Average variance over k days, the GARCH term structure, and the half-life

The statement is on page 423. The notation and the standing assumptions are those of §22.6.

Proof. We average the forecast formula with a geometric series, and compare horizons through the monotonicity of an average and of one derivative.

Part (i), the average. By linearity of expectation (Theorem 1.2), the expectation of the average daily variance is the average of the expectations. By (22.11) and the geometric series $\sum_{j=0}^{k-1}(\alpha+\beta)^j = \big(1-(\alpha+\beta)^k\big)/(1-\alpha-\beta)$, which is valid because $\alpha+\beta \neq 1$,

$$\frac{1}{k}\sum_{j=0}^{k-1}\mathbb{E}[\sigma_{i+j}^2] = \bar{\sigma}^2 + (\sigma_i^2-\bar{\sigma}^2)\,\frac{1}{k}\sum_{j=0}^{k-1}(\alpha+\beta)^j = \bar{\sigma}^2 + (\sigma_i^2-\bar{\sigma}^2)\,\frac{1-(\alpha+\beta)^k}{k\,(1-\alpha-\beta)}.$$

The slope claims of (i) are proved after the averaging factor has been studied.

Part (ii), the averaging factor. Write w_k for the averaging factor at the horizon k. It is the weight that the expected average variance puts on today's variance, since by (22.12) that average equals $(1-w_k)\,\bar{\sigma}^2 + w_k\,\sigma_i^2$. By the geometric series,

$$w_k = \frac{1}{k}\sum_{j=0}^{k-1}(\alpha+\beta)^j,$$

the average of the first k terms of the strictly decreasing positive sequence $1,\ \alpha+\beta,\ (\alpha+\beta)^2, \ldots$. Every term is positive, so $w_k > 0$; no term exceeds the first, which is 1, so $w_k \leq 1$. Appending a term to an average changes it by

$$w_{k+1} - w_k = \frac{k\,w_k + (\alpha+\beta)^k}{k+1} - w_k = \frac{(\alpha+\beta)^k - w_k}{k+1}.$$

The average w_k is at least its smallest term, $(\alpha+\beta)^{k-1}$, which is strictly larger than $(\alpha+\beta)^k$ because $0 < \alpha+\beta < 1$. Hence $w_{k+1} < w_k$: the averaging factor is strictly decreasing in k. By (22.12) the expected average variance is $\bar{\sigma}^2 + w_k\,(\sigma_i^2-\bar{\sigma}^2)$, an affine function of σ_i^2 with slope w_k, which is the statement about the variance.

Part (i), the slope of the term structure. The expected average variance $\bar{\sigma}^2 + w_k\,(\sigma_i^2-\bar{\sigma}^2)$ is the constant $\bar{\sigma}^2$ plus a strictly decreasing positive factor times $\sigma_i^2-\bar{\sigma}^2$. It is strictly decreasing in k if $\sigma_i^2 > \bar{\sigma}^2$, strictly increasing if $\sigma_i^2 < \bar{\sigma}^2$ and constant if the two are equal. It is positive, because it equals $(1-w_k)\,\bar{\sigma}^2 + w_k\,\sigma_i^2$ with $0 < w_k \leq 1$. The square root is strictly increasing on $(0,\infty)$, so $\sigma_{\mathrm{GARCH}}(k)$ moves with k in the same direction.

Part (ii), the volatility. The volatility $\sigma_{\mathrm{GARCH}}(k)$ depends on k through w_k under a square root, so the claim does not follow from the variance statement alone. Write $\sigma_{\mathrm{GARCH}}(k;x)$ for the model volatility when today's variance is $\sigma_i^2 = x > 0$:

$$\sigma_{\mathrm{GARCH}}(k;x) = \sqrt{252\,\big(\bar{\sigma}^2 + w_k\,(x-\bar{\sigma}^2)\big)}.$$

The radicand is 252 times $(1-w_k)\,\bar{\sigma}^2 + w_k\,x \geq \min(\bar{\sigma}^2, x) > 0$, so the function is differentiable in x, with

$$\partial_x\,\sigma_{\mathrm{GARCH}}(k;x) = \frac{\sqrt{252}}{2}\,\frac{w_k}{\sqrt{\bar{\sigma}^2 + w_k\,(x-\bar{\sigma}^2)}} = \frac{\sqrt{252}}{2}\,\chi(w_k),$$

where, for the fixed x, χ is the local auxiliary function of a weight w,

$$\chi(w) = \frac{w}{\sqrt{\bar{\sigma}^2 + w\,(x-\bar{\sigma}^2)}}, \qquad 0 < w \leq 1.$$

Differentiating by the quotient rule,

$$\chi'(w) = \frac{\sqrt{\bar\sigma^2 + w\,(x-\bar\sigma^2)} - \dfrac{w\,(x-\bar\sigma^2)}{2\sqrt{\bar\sigma^2 + w\,(x-\bar\sigma^2)}}}{\bar\sigma^2 + w\,(x-\bar\sigma^2)} = \frac{\bar\sigma^2 + \frac{1}{2}\,w\,(x-\bar\sigma^2)}{\left(\bar\sigma^2 + w\,(x-\bar\sigma^2)\right)^{3/2}}.$$

The numerator is positive: it is at least $\bar\sigma^2$ if $x \geq \bar\sigma^2$, and if $x < \bar\sigma^2$ it is at least $\bar\sigma^2 - \frac{1}{2}(\bar\sigma^2 - x) = \frac{1}{2}(\bar\sigma^2 + x)$, because $w \leq 1$. So $\chi' > 0$ on $(0,1]$, and χ is strictly increasing there, by the mean value theorem. Since w_k is strictly decreasing in k, the derivative $\partial_x\,\sigma_{\mathrm{GARCH}}(k;x)$ is positive and strictly decreasing in k, at every $x > 0$.

A finite move of today's variance from x_0 to $x_1 > x_0$ changes the volatility by the integral of this derivative, which is continuous in x, by the fundamental theorem of calculus:

$$\sigma_{\mathrm{GARCH}}(k;x_1) - \sigma_{\mathrm{GARCH}}(k;x_0) = \int_{x_0}^{x_1} \partial_x\,\sigma_{\mathrm{GARCH}}(k;x)\,dx.$$

The integrand is positive, and at every x of $[x_0, x_1]$ it is strictly smaller at the horizon $k+1$ than at the horizon k. The integral is therefore positive and strictly smaller at $k+1$ than at k.

Part (iii). By (22.11), the deviation $\mathbb{E}[\sigma^2_{i+k}] - \bar\sigma^2$ is $(\alpha+\beta)^k$ times its value at $k=0$. It is halved when $(\alpha+\beta)^k = 1/2$, that is, taking logarithms, when $k = \ln(1/2)/\ln(\alpha+\beta)$; both logarithms are negative, so the ratio is positive. The solution is in general not an integer, and the half-life is read as a time scale. □

H.7 Proof of Theorem 22.36: The fair variance strike (Neuberger, 1994; Carr and Madan, 1998; Demeterfi, Derman, Kamal and Zou, 1999)

The statement is on page 431. The notation and the standing assumptions are those of §22.7.

Proof. We split the floating leg, by an exact pathwise identity, into a dynamically replicated part that is worth zero and a European log claim, and we price the log claim by static replication with Lemma 22.34.

Step 1: part (i), by algebra. Since $e^{R_i} = S_{t_i}/S_{t_{i-1}}$,

$$e^{R_i} - 1 = \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}}.$$

The log-returns telescope: $\sum_{i=1}^n R_i = \sum_{i=1}^n (\ln S_{t_i} - \ln S_{t_{i-1}}) = \ln S_T - \ln S_0 = \ln(S_T/S_0)$. Multiplying both facts by 2 and subtracting the second sum from the first gives (22.14). Only the positivity of the prices was used.

Step 2: the dynamic leg is worth zero. Consider the following strategy. At each date t_{i-1}, $i = 1, \ldots, n$, borrow $2/T$ and buy $2/(T\,S_{t_{i-1}})$ units of the underlier, which costs exactly $2/T$. At t_i sell them for $(2/T)\,S_{t_i}/S_{t_{i-1}}$ and repay the loan of $2/T$, at the zero rate. This leaves

$$\frac{2}{T}\,\frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}},$$

which is kept in the cash account until T; a negative amount is a loan carried at the zero rate. The number of units bought at t_{i-1} is known at t_{i-1}, the underlier pays no income, and no cash is added or withdrawn, so the strategy is self-financing in the sense of §16.2. Its price is zero, because the first purchase is financed in full by the first loan. Its payoff at T is

$$\frac{2}{T}\sum_{i=1}^{n}\frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}}.$$

By the law of one price (Lemma 16.8(iii)), any strategy with this payoff, in particular the holding of a claim that pays this amount at T, has the price 0.

Step 3: the log claim is worth the strip. Apply Lemma 22.34 to $f(x) = -\ln(x/F)$, which is twice continuously differentiable on $(0,\infty)$, with $f(F) = 0$, $f'(F) = -1/F$ and $f''(K) = 1/K^2 > 0$. With the two nonnegative local functions

$$\chi_1(x) = \int_0^F \frac{(K-x)^+}{K^2}\,dK, \qquad \chi_2(x) = \int_F^\infty \frac{(x-K)^+}{K^2}\,dK,$$

identity (22.13) reads, for every $x > 0$,

$$-\ln\frac{x}{F} = -\frac{x-F}{F} + \chi_1(x) + \chi_2(x). \tag{H.9}$$

(a) Integrability. By (H.9), $0 \le \chi_1(x)+\chi_2(x) = -\ln(x/F)+(x-F)/F \le |\ln x|+|\ln F|+x/F+1$. The right side has a finite integral against $f_{\mathbb{Q}}$, by the Setup of §22.7: $\int_0^\infty |\ln x|\, f_{\mathbb{Q}}(x)\,dx < \infty$, $\int_0^\infty x\, f_{\mathbb{Q}}(x)\,dx = F$ and $\int_0^\infty f_{\mathbb{Q}}(x)\,dx = 1$. Hence χ_1 and χ_2, each nonnegative and at most their sum, have finite integrals against $f_{\mathbb{Q}}$, and Fact 7.22 applies to $\chi_1(S_T)$, to $\chi_2(S_T)$ and to $\ln(S_T/F)$.

(b) Exchanging the two integrations. By Fact 7.22,

$$\mathbb{E}^{\mathbb{Q}}\big[\chi_1(S_T)\big] = \int_0^\infty \chi_1(x)\, f_{\mathbb{Q}}(x)\,dx = \int_0^\infty \Big[\int_0^F \frac{(K-x)^+}{K^2}\,dK\Big] f_{\mathbb{Q}}(x)\,dx.$$

The integrand $(K-x)^+ f_{\mathbb{Q}}(x)/K^2$ is continuous and nonnegative on $(0,\infty)\times(0,F)$, so Fact 18.34(i) (Tonelli) allows the order of the two integrations to be exchanged:

$$\mathbb{E}^{\mathbb{Q}}\big[\chi_1(S_T)\big] = \int_0^F \frac{1}{K^2}\Big[\int_0^\infty (K-x)^+ f_{\mathbb{Q}}(x)\,dx\Big]\,dK = \int_0^F \frac{\mathcal{P}(K)}{K^2}\,dK,$$

because the inner integral is $\mathbb{E}^{\mathbb{Q}}[(K-S_T)^+] = \mathcal{P}(K)$, by Fact 7.22 for the bounded payoff $x \mapsto (K-x)^+$ and by the pricing rule. In the same way, with the integrand $(x-K)^+ f_{\mathbb{Q}}(x)/K^2 \ge 0$ on $(0,\infty)\times(F,\infty)$ and with (22.1) at $r = 0$ for the inner integral,

$$\mathbb{E}^{\mathbb{Q}}\big[\chi_2(S_T)\big] = \int_F^\infty \frac{C(K)}{K^2}\,dK.$$

Both strip integrals are finite, by (a).

(c) The price. Evaluate (H.9) at $x = S_T$ and take $\mathbb{Q}$-expectations. The three terms on the right have finite expectations, and $\mathbb{E}^{\mathbb{Q}}[(S_T-F)/F] = 0$ because $\mathbb{E}^{\mathbb{Q}}[S_T] = F$. By linearity of expectation (Theorem 1.2) and (b),

$$\mathbb{E}^{\mathbb{Q}}\Big[-\ln\frac{S_T}{F}\Big] = \int_0^F \frac{\mathcal{P}(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK.$$

Multiplying by $2/T$ gives the second equality of (22.15); the first is the definition of $K_{\log}$. By the pricing rule, the *log claim* that pays $-\ln(S_T/F)$ at T trades today at $\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F)]$.

(d) Any law. Parts (a) to (c) used only that the law of S_T has a density continuous on $(0,\infty)$, the mean F and a finite mean of $|\ln S_T|$, with $C(K)$ and $\mathcal{P}(K)$ the expectations of the call and the put payoffs, which is what the pricing rule makes them; the pricing rule itself entered only in the last sentence of (c). This proves the last sentence of part (ii).

Step 4: part (ii). With the unit notional, the floating leg of the log-variance swap is $1/T$ times the left side of (22.14). By Step 1 and $F = S_0$ it equals

$$\frac{1}{T}\sum_{i=1}^n 2\,(e^{R_i} - 1 - R_i) = \frac{2}{T}\sum_{i=1}^n \frac{S_{t_i} - S_{t_{i-1}}}{S_{t_{i-1}}} + \frac{2}{T}\Big(-\ln\frac{S_T}{F}\Big):$$

the payoff of the strategy of Step 2 plus the payoff of $2/T$ log claims. The strategy that runs Step 2 and buys $2/T$ log claims at time 0 is self-financing, pays the floating leg exactly, and costs $0 + (2/T)\,\mathbb{E}^{\mathbb{Q}}[-\ln(S_T/F)] = K_{\log}$, by Steps 2 and 3. Add a loan of the strike, repaid at T at the zero rate: the resulting strategy pays the floating leg minus the strike, which is the payoff of the long swap with the unit notional, and its price is $K_{\log}$ minus the strike. By the law of one price (Lemma 16.8(iii)) the swap, whose price is zero by contract, must have the same price as this strategy. So the swap costs nothing exactly when its strike is $K_{\log}$; a notional N_{var} scales both sides. This is (ii). The argument used the positivity of the path, trading at the monitoring dates, and prices of European claims on S_T; it did not use a model of the path.

Step 5: part (iii), the Black–Scholes–Merton model. Under Definition 20.6 with $r = q = 0$, the risk-neutral path is $S_t^{\mathbb{Q}} = S_0 \exp(-\frac{1}{2}\sigma^2 t + \sigma W_t)$. The parts of the Setup of §22.7 that Steps 2 to 4 use hold in this model. Under $\mathbb{Q}$ the terminal price is $S_T = F \exp\big(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z\big)$ with $Z \sim \mathcal{N}(0,1)$ and $F = S_0$ (Proposition 20.8(i)). Its density is the lognormal density computed in the proof of Proposition 19.42(ii), with $\mu = 0$ and the initial price F,

$$f_{\mathbb{Q}}(x) = \frac{1}{x\,\sigma\sqrt{T}}\,\varphi\Big(\frac{\ln(x/F) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}\Big),$$

which is continuous on $(0,\infty)$. Its mean is F (Proposition 20.8(ii)), and $\mathbb{E}^{\mathbb{Q}}|\ln S_T| \le |\ln F| + \frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,\mathbb{E}|Z| < \infty$. The claims used in Steps 2 to 4, the calls, the puts, the log claim and the asset, have payoffs of the class $\mathcal{G}$ and trade at their $\mathbb{Q}$-expectations (Theorem 20.7). Since $\ln(S_T/F) = -\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z$ and $\mathbb{E}[Z] = 0$,

$$K_{\log} = \frac{2}{T}\Big(\frac{\sigma^2 T}{2} - \sigma\sqrt{T}\,\mathbb{E}[Z]\Big) = \sigma^2,$$

the computation of Example 20.10(3).

The two swap legs are priced by their expectations. Corollary 20.11 prices a claim on finitely many dates whose payoff is of the class $\mathcal{G}_n$ of Definition 19.19(iv): continuous outside finitely many hyperplanes and of moderate growth. Write $x_1, \dots, x_n > 0$ for the prices at the monitoring dates and $x_0 = S_0$. As a function of these prices, $\sigma_{\text{real}}^2 = \frac{1}{T}\sum_i \big(\ln(x_i/x_{i-1})\big)^2$ is continuous on $(0,\infty)^n$. It has moderate growth. Indeed $|\ln y| \le y + 1/y$ for every $y > 0$: for $y \ge 1$, $0 \le \ln y \le y - 1$, because $y - 1 - \ln y$ vanishes at $y = 1$ and has the derivative $1 - 1/y \ge 0$ on $[1,\infty)$, and for $0 < y < 1$ the same bound applied to $1/y$ gives $|\ln y| \le 1/y$. Hence

$$\big|\ln(x_i/x_{i-1})\big| \;\le\; \frac{x_i}{x_{i-1}} + \frac{x_{i-1}}{x_i},$$

a sum of two products of powers of the prices. Each product has moderate growth in the sense of Definition 19.19(ii): $x_i/x_{i-1} \le (x_i + 1/x_i)(x_{i-1} + 1/x_{i-1})$, the factors $x_j + 1/x_j$ of the other prices are at least 2, and the fixed price $x_0 = S_0$ contributes a constant. Sums and products of functions of moderate growth have moderate growth (Lemma 19.20(i)), so σ_{real}^2 has moderate growth. The square root σ_{real} is continuous as well, and $0 \le \sigma_{\text{real}} \le 1 + \sigma_{\text{real}}^2$. Both payoffs are therefore in $\mathcal{G}_n$, and under (A1)–(A6) Corollary 20.11 gives the claims that pay σ_{real}^2 and σ_{real} at T the arbitrage-free prices $\mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}^2]$ and $\mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}]$ at $r = 0$. The fixed leg of each swap, a sure amount at T, is worth its strike at $r = 0$ (Lemma 16.8(iv)), so each swap is worth nothing exactly at the strike of Definition 22.32(d): K_{var} and K_{vol} are the arbitrage-free strikes, the first claim of (iii). Under $\mathbb{Q}$ each log-return is

$$R_i = \nu\,\Delta t + \sigma\,\Delta W_i, \qquad \nu = -\tfrac{1}{2}\sigma^2, \qquad \Delta W_i = W_{t_i} - W_{t_{i-1}} = \sqrt{\Delta t}\,Z_i,$$

where ν is the log-drift under $\mathbb{Q}$ at $r = q = 0$, as in Corollary 20.11, and the $Z_i = \Delta W_i/\sqrt{\Delta t}$ are iid $\mathcal{N}(0,1)$ (Definition 19.5(2) and (3)). Expanding the square and the fourth power,

$$R_i^2 = \nu^2(\Delta t)^2 + 2\,\nu\,\sigma\,(\Delta t)^{3/2} Z_i + \sigma^2\,\Delta t\,Z_i^2,$$
$$R_i^4 = \nu^4(\Delta t)^4 + 4\,\nu^3\sigma\,(\Delta t)^{7/2} Z_i + 6\,\nu^2\sigma^2(\Delta t)^3 Z_i^2 + 4\,\nu\,\sigma^3(\Delta t)^{5/2} Z_i^3 + \sigma^4(\Delta t)^2 Z_i^4.$$

By Proposition 7.25, the odd moments of Z_i vanish, $\mathbb{E}[Z_i^2]=1$ and $\mathbb{E}[Z_i^4]=3$, so

$$\mathbb{E}^{\mathbb{Q}}[R_i^2]=\sigma^2\,\Delta t+\nu^2(\Delta t)^2=\sigma^2\,\Delta t+\tfrac{1}{4}\sigma^4(\Delta t)^2,$$
$$\mathbb{E}^{\mathbb{Q}}[R_i^4]=3\,\sigma^4(\Delta t)^2+6\,\nu^2\sigma^2(\Delta t)^3+\nu^4(\Delta t)^4,$$

and, by (1.1) and $\nu^2=\sigma^4/4$,

$$\operatorname{Var}^{\mathbb{Q}}\left(R_i^2\right)=\mathbb{E}^{\mathbb{Q}}[R_i^4]-\left(\mathbb{E}^{\mathbb{Q}}[R_i^2]\right)^2=2\,\sigma^4(\Delta t)^2+4\,\nu^2\sigma^2(\Delta t)^3=2\,\sigma^4(\Delta t)^2+\sigma^6(\Delta t)^3.$$

The floating leg $\sigma_{\text{real}}^2=\frac{1}{T}\sum_i R_i^2$ is priced by its $\mathbb{Q}$-expectation, as just shown. By linearity and $n\,\Delta t=T$,

$$K_{\text{var}}=\frac{n}{T}\Big(\sigma^2\,\Delta t+\tfrac{1}{4}\sigma^4(\Delta t)^2\Big)=\sigma^2+\frac{\sigma^4\,\Delta t}{4}.$$

For the mean-square distance, the squares of two different returns are uncorrelated. Indeed, for $i\neq j$ the increments ΔW_i and ΔW_j are independent (Definition 19.5(3)), and the function $\psi_1(z)=(\nu\,\Delta t+\sigma z)^2-\mathbb{E}^{\mathbb{Q}}[R_i^2]$ has $\mathbb{E}|\psi_1(\Delta W_i)|\le 2\,\mathbb{E}^{\mathbb{Q}}[R_i^2]<\infty$, so the product rule of Lemma 19.17, with $\psi_2=\psi_1$, gives

$$\operatorname{Cov}^{\mathbb{Q}}\left(R_i^2,R_j^2\right)=\mathbb{E}\big[\psi_1(\Delta W_i)\,\psi_1(\Delta W_j)\big]=\mathbb{E}\big[\psi_1(\Delta W_i)\big]\,\mathbb{E}\big[\psi_1(\Delta W_j)\big]=0.$$

The variance of the sum is then the sum of the variances, by Proposition 10.10 and the induction on the number of terms written out in Part (i) of the proof of Proposition 22.20; this is the computation of Proposition 19.10(i) with the drift kept. Hence

$$\operatorname{Var}^{\mathbb{Q}}\left(\sigma_{\text{real}}^2\right)=\frac{n}{T^2}\Big(2\,\sigma^4(\Delta t)^2+\sigma^6(\Delta t)^3\Big)=\frac{2\,\sigma^4\,\Delta t+\sigma^6(\Delta t)^2}{T}.$$

Finally, the mean square of a variable about a constant is its variance plus its squared bias: expanding $\left(\sigma_{\text{real}}^2-K_{\text{var}}+K_{\text{var}}-\sigma^2\right)^2$, the cross term has mean zero because $\mathbb{E}^{\mathbb{Q}}[\sigma_{\text{real}}^2]=K_{\text{var}}$, so

$$\mathbb{E}^{\mathbb{Q}}\Big[\left(\sigma_{\text{real}}^2-\sigma^2\right)^2\Big]=\operatorname{Var}^{\mathbb{Q}}\left(\sigma_{\text{real}}^2\right)+\left(K_{\text{var}}-\sigma^2\right)^2=\frac{2\,\sigma^4\,\Delta t+\sigma^6(\Delta t)^2}{T}+\frac{\sigma^8(\Delta t)^2}{16}.$$

Both terms tend to 0 as $\Delta t\to 0$.

Step 6: part (iii), the general gap. Corollary 7.32 at order two, for the exponential function at the expansion point 0, gives for every real R a point R^* between 0 and R with

$$e^R=1+R+\frac{R^2}{2}+\frac{e^{R^*}R^3}{6},\qquad\text{that is,}\qquad 2\left(e^R-1-R\right)=R^2+\tfrac{1}{3}\,e^{R^*}R^3.$$

The point supplied by the corollary is starred, as everywhere in the Part *Derivatives*. Apply this to each return and sum:

$$\frac{1}{T}\sum_{i=1}^{n}2\left(e^{R_i}-1-R_i\right)=\sigma_{\text{real}}^2+\frac{1}{3\,T}\sum_{i=1}^{n}e^{R_i^*}R_i^3.$$

Under the stated hypothesis the fair strike of each swap is the $\mathbb{Q}$-expectation of its floating leg. For the log-variance swap that strike is $K_{\log}$, because by (ii) the strike at which the swap costs nothing is unique; for the squared-return swap it is K_{var}. Both expectations are finite, so linearity of expectation gives $K_{\log}=K_{\text{var}}+\frac{1}{3T}\,\mathbb{E}^{\mathbb{Q}}\big[\sum_i e^{R_i^*}R_i^3\big]$, which is (22.16). □

I Proofs for §23: Exotic Options and Numerical Methods

I.1 Proof of Proposition 23.8: Forward-start option (Rubinstein, 1991)

The statement is on page 443. The notation and the standing assumptions are those of §23.1.

Proof. We replicate in two stages, a static holding up to t_1 and a delta hedge after it, and join them at t_1. Write $\tau = T - t_1$, and let $\tilde{S}_t = S_t e^{-q(T-t)}$ be the reinvested holding of the symbol notices at the start of §23: by Proposition 16.15(i) it is the price of a traded position that pays no income, with $\tilde{S}_T = S_T$; under (A1) it is a geometric Brownian motion with volatility σ (§20.7). It is an instrument of Definition 20.1, and the only one the proof uses.

Step 1 (the value needed at t_1 is linear in S_{t_1}). At t_1 the holder owns a call with spot S_{t_1}, strike ηS_{t_1} and life τ. By Lemma 23.7 with $c = S_{t_1}$,

$$C(S_{t_1}, \eta S_{t_1}; \tau) = S_{t_1}\, C(1, \eta; \tau) = c_\eta S_{t_1},$$

a constant multiple of the price of the asset at t_1.

Step 2 (first stage, static). On $[0, t_1)$ hold the constant number $h = c_\eta e^{q\tau}$ of units of $\tilde{S}$, starting from the capital $h\tilde{S}_0$. A constant is an admissible holding, on the closed interval $[0, t_1]$ as well, and by the decomposition of the discounted price in the Setup of Definition 20.1 the integral of the constant 1 against $d(e^{-rs}\tilde{S}_s)$ over $[0, t]$ is $e^{-rt}\tilde{S}_t - \tilde{S}_0$ (the computation of Example 20.2). The value Π_t of the strategy therefore satisfies

$$e^{-rt}\Pi_t = h\tilde{S}_0 + h(e^{-rt}\tilde{S}_t - \tilde{S}_0) = e^{-rt}h\tilde{S}_t, \qquad 0 \le t \le t_1,$$

that is, $\Pi_t = h\tilde{S}_t = c_\eta e^{q\tau} S_t e^{-q(T-t)} = c_\eta e^{-q(t_1-t)} S_t$. In words: the strategy holds $c_\eta e^{-q(t_1-t)}$ units of the asset at time t, the number growing at the rate q as dividends are reinvested. Its initial capital is $c_\eta e^{-qt_1} S$ and its value at t_1 is $c_\eta S_{t_1}$.

Step 3 (second stage, dynamic). Since $S_{t_1} = e^{q\tau}\tilde{S}_{t_1}$, the payoff is a function of the terminal value of the instrument and of one earlier value of it:

$$(S_T - \eta S_{t_1})^+ = f(\tilde{S}_T;\, \tilde{S}_{t_1}), \qquad f(x; y) = \bigl(x - \eta e^{q\tau} y\bigr)^+.$$

The function f is continuous on $(0, \infty)^2$ and $0 \le f(x; y) \le x \le (x + x^{-1})(y + y^{-1})$, so f belongs to the class $\mathcal{G}_2$ of Definition 19.19(iv). Theorem 20.7(v) therefore applies to the non-dividend asset $\tilde{S}$, with $t_0 = t_1$ and the single earlier price $Y = \tilde{S}_{t_1}$ (this is the yield form of that clause, recorded in Corollary 20.41(i)). With

$$\tilde{V}(x, t; y) = e^{-r(T-t)}\, \mathbb{E}\, f\bigl(x \exp\bigl((r - \tfrac{1}{2}\sigma^2)(T-t) + \sigma\sqrt{T-t}\, Z\bigr);\, y\bigr), \qquad t_1 \le t < T,$$

it provides a strategy on $[t_1, T)$ that starts with the capital $\tilde{V}(\tilde{S}_{t_1}, t_1; \tilde{S}_{t_1})$, holds $\partial_x \tilde{V}(\tilde{S}_t, t; \tilde{S}_{t_1})$ units of $\tilde{S}$, and has the terminal value $f(\tilde{S}_T; \tilde{S}_{t_1}) = (S_T - \eta S_{t_1})^+$. Its starting capital is, by $\tilde{S}_{t_1} = S_{t_1} e^{-q\tau}$,

$$\tilde{V}(\tilde{S}_{t_1}, t_1; \tilde{S}_{t_1}) = e^{-r\tau}\, \mathbb{E}\bigl(S_{t_1} e^{(r-q-\sigma^2/2)\tau + \sigma\sqrt{\tau} Z} - \eta S_{t_1}\bigr)^+ = C(S_{t_1}, \eta S_{t_1}; \tau),$$

the price of Corollary 20.41(i) for the call with spot S_{t_1}, strike ηS_{t_1} and life τ; and the strategy is the delta strategy of that call, held in the reinvested asset.

Step 4 (concatenation and price). By Step 1 the capital that the second stage needs at t_1, $C(S_{t_1}, \eta S_{t_1}; \tau) = c_\eta S_{t_1}$, is exactly the value of the first stage at t_1 (Step 2). By Definition 20.1(iv) the two stages concatenate, at the deterministic date t_1, into one strategy with initial capital $e^{-qt_1} c_\eta S$ and terminal value $(S_T - \eta S_{t_1})^+$. The arbitrage argument of Step 5 of the proof of Theorem 20.7, run with this strategy, shows that at any other time-0 price of the forward-start call there is an arbitrage: if the claim is dearer, sell it, run the strategy and deposit the difference; if it is cheaper, do the opposite. So the price is $e^{-qt_1} c_\eta S$, and $c_\eta S = C(S, \eta S; \tau)$ by Lemma 23.7 with $c = S$. For the forward-start put, replace f by $(\eta e^{q\tau} y - x)^+ \le \eta e^{q\tau} y$, which is again in $\mathcal{G}_2$, and C by $\mathcal{P}$ in every step. $\square$

I.2 Proof of Proposition 23.10: Simple chooser by put–call parity (Rubinstein, 1991)

The statement is on page 444. The notation and the standing assumptions are those of §23.1.

Proof. We use put–call parity at the choice date and make the resulting decomposition static.

Step 1 (when the put is chosen). Put–call parity holds at the date t_1, for every sign of r: one put held long and one call held short from t_1 pay $(K-S_T)^+-(S_T-K)^+=K-S_T$ at T, and so does the position opened at t_1 that deposits $Ke^{-r\tau}$ and sells short $e^{-q\tau}$ units of the asset with the income reinvested (Proposition 16.15(i): a reinvested holding counts $e^{-q(T-t)}$ units at each date t, so the short position owes exactly one unit at T); Lemma 16.8(v) gives

$$\mathcal{P}_{t_1}-C_{t_1}=Ke^{-r\tau}-S_{t_1}e^{-q\tau}=e^{-q\tau}\,(\tilde{K}-S_{t_1})$$

(for $r\geq 0$ this is Corollary 17.11(ii), read at t_1 by Lemma 17.5(i)). So the holder declares the put exactly on the event $\{S_{t_1}<\tilde{K}\}$, which is known at t_1. On that event the chooser pays $(K-S_T)^+$ at T and otherwise $(S_T-K)^+$; by the identity $(K-S_T)^+-(S_T-K)^+=K-S_T$ used above, its payoff at T is, in every state,

$$(S_T-K)^+ + \mathbf{1}\{S_{t_1}<\tilde{K}\}\,(K-S_T). \tag{I.1}$$

Step 2 (a strategy with the same payoff). Buy today one call with strike K and expiry T and $e^{-q\tau}$ puts with strike $\tilde{K}$ and expiry t_1. At t_1 the puts pay

$$e^{-q\tau}(\tilde{K}-S_{t_1})^+=\left(Ke^{-r\tau}-S_{t_1}e^{-q\tau}\right)\mathbf{1}\{S_{t_1}<\tilde{K}\}.$$

On the event $\{S_{t_1}<\tilde{K}\}$, deposit $Ke^{-r\tau}$ in the cash account and sell short $e^{-q\tau}$ units of the asset, reinvesting (that is, owing) the dividends, so that exactly one unit is owed at T (Proposition 16.15(i)). The deposit costs $Ke^{-r\tau}$ and the short sale brings in $S_{t_1}e^{-q\tau}$, so the two trades cost exactly what the puts pay, and no cash is added or withdrawn: the strategy is self-financing. At T the deposit is worth K and the short position owes S_T. The strategy therefore pays $(S_T-K)^+$ from the call, plus $K-S_T$ on the event $\{S_{t_1}<\tilde{K}\}$: the amount (I.1), in every state.

Step 3 (the price and the upper bound). Equal payoffs give equal prices by Lemma 16.8(iii), which is (23.3). For the upper bound, the chooser pays either $(S_T-K)^+$ or $(K-S_T)^+$, hence at most their sum, the payoff of one call and one put; Lemma 16.8(i) gives chooser $\leq C(K,T)+\mathcal{P}(K,T)$.

Step 4 (the lower bound). A put has a nonnegative payoff and hence a nonnegative price (Lemma 16.8(i) against the empty portfolio), so (23.3) is at least $C(K,T)$. For the other half, apply parity today twice, to the options expiring at T and to those expiring at t_1 (the same dominance argument at time 0, Lemma 16.8(iii) with the expiry in the place of T; for $r\geq 0$, Corollary 17.11(ii)):

$$C(K,T)-\mathcal{P}(K,T)=Se^{-qT}-Ke^{-rT},\qquad \mathcal{P}(\tilde{K},t_1)=C(\tilde{K},t_1)+\tilde{K}e^{-rt_1}-Se^{-qt_1}.$$

Since $e^{-q\tau}\tilde{K}e^{-rt_1}=Ke^{-r\tau}e^{-rt_1}=Ke^{-rT}$ and $e^{-q\tau}e^{-qt_1}=e^{-qT}$, multiplying the second identity by $e^{-q\tau}$ and adding the first gives

$$C(K,T)+e^{-q\tau}\,\mathcal{P}(\tilde{K},t_1)=\mathcal{P}(K,T)+e^{-q\tau}\,C(\tilde{K},t_1)\ \geq\ \mathcal{P}(K,T),$$

because the call expiring at t_1 also has a nonnegative price. □

I.3 Proof of Theorem 23.16: Barrier options by the method of images (Merton, 1973; Reiner and Rubinstein, 1991)

The statement is on page 448. The notation and the standing assumptions are those of §23.2.

Proof. We use the method of images to build a solution of the pricing equation that vanishes on the barrier, and then replicate the claim by the delta hedge of that solution, stopped at the barrier; for (v) the same stopped Itô identity is run on the risk-neutral path and averaged. Three properties of V are used. Each holds for both kinds of price: by Corollary 20.41(i) and Corollary 20.45(i) for an asset, and by Theorem 20.52(i) and Corollary 20.45(ii) for a futures price, whose price function is written V^{fut} there and is obtained from that of an asset without income by the change of variables $F \mapsto Fe^{-r(T_F-t)}$, whose factor stays between two positive constants on $[0,T]$.

(V1) The function V is infinitely differentiable on $(0,\infty)\times[0,T)$, and each of its partial derivatives has moderate growth (Definition 19.19(ii)) on $(0,\infty)\times[0,T']$, for every $T'<T$.

(V2) There is a constant c with $|V(S,t)|\le c(S^c+S^{-c})$ on $(0,\infty)\times[0,T]$, and $V(S,t)\to f(S^*)$ as $(S,t)\to(S^*,T)$ at every continuity point S^* of f.

(V3) On $(0,\infty)\times[0,T)$, the function V solves $\partial_t V + bS\,\partial_S V + \frac12\sigma^2 S^2\,\partial_{SS}V = rV$.

Step 1 (image symmetry). For $t<T$ and $S>0$ write $x=H^2/S$ for the reflected spot and

$$V^{\mathrm{im}}(S,t)=\Big(\frac{S}{H}\Big)^{1-2b/\sigma^2}V(x,t)$$

for the image term of (23.4). We show that V^{im} solves (23.5). Since $\partial_S x=-H^2/S^2$, we have $S\,\partial_S x=-x$, and so, by the chain rule, applying the operator $S\,\partial_S$ to a function of (x,t) is the same as applying $-x\,\partial_x$ to it. The power factor satisfies $S\,\partial_S(S/H)^{1-2b/\sigma^2}=(1-2b/\sigma^2)(S/H)^{1-2b/\sigma^2}$. By the product rule, with V and its derivatives evaluated at (x,t),

$$S\,\partial_S V^{\mathrm{im}}=\Big(\frac{S}{H}\Big)^{1-2b/\sigma^2}\Big[\Big(1-\frac{2b}{\sigma^2}\Big)V-x\,\partial_x V\Big]. \tag{I.2}$$

Applying $S\,\partial_S$ once more, and using $-x\,\partial_x(x\,\partial_x V)=-x\,\partial_x V-x^2\,\partial_{xx}V$,

$$S\,\partial_S(S\,\partial_S V^{\mathrm{im}})=\Big(\frac{S}{H}\Big)^{1-2b/\sigma^2}\Big[\Big(1-\frac{2b}{\sigma^2}\Big)^2V-2\Big(1-\frac{2b}{\sigma^2}\Big)x\,\partial_x V+x\,\partial_x V+x^2\,\partial_{xx}V\Big].$$

Since $S^2\,\partial_{SS}=S\,\partial_S(S\,\partial_S)-S\,\partial_S$, subtracting (I.2) gives

$$S^2\,\partial_{SS}V^{\mathrm{im}}=\Big(\frac{S}{H}\Big)^{1-2b/\sigma^2}\Big[-\frac{2b}{\sigma^2}\Big(1-\frac{2b}{\sigma^2}\Big)V+\frac{4b}{\sigma^2}\,x\,\partial_x V+x^2\,\partial_{xx}V\Big]. \tag{I.3}$$

The time derivative does not see the reflection, and (V3) at the point (x,t) gives

$$\partial_t V^{\mathrm{im}}=\Big(\frac{S}{H}\Big)^{1-2b/\sigma^2}\partial_t V=\Big(\frac{S}{H}\Big)^{1-2b/\sigma^2}\Big[rV-bx\,\partial_x V-\tfrac12\sigma^2x^2\,\partial_{xx}V\Big].$$

Now add $\partial_t V^{\mathrm{im}}$, b times (I.2) and $\frac12\sigma^2$ times (I.3), and subtract rV^{im}. Inside the bracket the terms in $x^2\,\partial_{xx}V$ cancel $(-\frac12\sigma^2+\frac12\sigma^2)$, the terms rV cancel, the coefficient of $x\,\partial_x V$ is

$$-b-b+\tfrac12\sigma^2\cdot\frac{4b}{\sigma^2}=0,$$

and the coefficient of V is

$$b\Big(1-\frac{2b}{\sigma^2}\Big)-\tfrac12\sigma^2\cdot\frac{2b}{\sigma^2}\Big(1-\frac{2b}{\sigma^2}\Big)=\Big(1-\frac{2b}{\sigma^2}\Big)(b-b)=0.$$

So $\partial_t V^{\mathrm{im}}+bS\,\partial_S V^{\mathrm{im}}+\frac12\sigma^2S^2\,\partial_{SS}V^{\mathrm{im}}=rV^{\mathrm{im}}$: the image term solves the equation, and so does $V^{\mathrm{out}}=V-V^{\mathrm{im}}$, because the equation is linear and V solves it. The exponent $1-2b/\sigma^2$ is exactly the one that makes the two coefficients vanish: the same computation with a real exponent a in its place gives $(a^2-a)V+(2-2a)\,x\,\partial_x V+x^2\,\partial_{xx}V$ inside the bracket of (I.3), hence the coefficient $-2b+\sigma^2(1-a)$ for $x\,\partial_x V$, which is zero only for $a=1-2b/\sigma^2$. In the variable $\ln(S/H)$ the equation has constant coefficients and the image is the reflection of that variable through zero

together with a multiplication by an exponential of it, which is the computation of Step 1 of the proof of Theorem 19.63.

Step 2 (boundary values, terminal values, growth). At $S = H$ the reflected spot is $x = H$ and the factor $(S/H)^{1-2b/\sigma^2}$ equals 1, so $V^{\text{out}}(H,t) = V(H,t) - V(H,t) = 0$ for every $t < T$. Let $S^* > H$ be a continuity point of f, and let $(S,t) \to (S^*, T)$ with $t < T$. Then $V(S,t) \to f(S^*)$ by (V2). The reflected spot H^2/S tends to H^2/S^*, which is smaller than H; since f vanishes identically on $(0,H]$, every point of the open interval $(0,H)$ is a continuity point of f, with value 0, and (V2) gives $V(H^2/S,t) \to f(H^2/S^*) = 0$. The factor $(S/H)^{1-2b/\sigma^2}$ stays bounded, so the image term tends to 0 and $V^{\text{out}}(S,t) \to f(S^*)$. With Step 1 this proves (i).

Growth bounds are needed in Steps 3 and 4. By (I.2), (I.3) and the formula for $\partial_t V^{\text{im}}$, each of V^{im}, $\partial_t V^{\text{im}}$, $\partial_S V^{\text{im}}$ and $\partial_{SS} V^{\text{im}}$ is a finite sum of terms, each a constant times a power of S times a partial derivative of V evaluated at $(H^2/S,t)$. A power S^a, a real, has moderate growth, because $S^a \le S^{|a|} + S^{-|a|}$; and a function of moderate growth evaluated at H^2/S has moderate growth in S, because

$$\Big(\frac{H^2}{S}\Big)^c + \Big(\frac{H^2}{S}\Big)^{-c} = H^{2c}S^{-c} + H^{-2c}S^c \le \big(H^{2c} + H^{-2c}\big)\big(S^c + S^{-c}\big).$$

Sums and products of functions of moderate growth have moderate growth (Lemma 19.20(i)). With (V1), the functions V^{out}, $\partial_t V^{\text{out}}$, $\partial_S V^{\text{out}}$ and $\partial_{SS} V^{\text{out}}$ are therefore continuous and of moderate growth on $(0,\infty) \times [0,T']$ for every $T' < T$; and by (V2), the same display and $f \in \mathcal{G}$, there is a constant c' with

$$|V^{\text{out}}(S,t)| \le c'\big(S^{c'} + S^{-c'}\big) \qquad \text{on } (0,\infty) \times [0,T]. \tag{I.4}$$

Step 3 (stopped replication on $[0,T']$, $T' < T$). The real-world path $S_t = S_0 \exp(gt + \sigma W_t)$ is an explicit function of (t, W_t). (For an asset this is (A1) with $g = \mu - \sigma^2/2$. For a futures price it is (A1) for the futures price itself, as in the Setup of §20.8; when an asset with yield q stands behind the futures, Corollary 19.36 shows that the futures price is such a process.) Since $S_t \le H$ exactly when $W_t \le \ell(t)$, with

$$\ell(t) = \frac{\ln(H/S_0) - gt}{\sigma},$$

the time t_H is the first time at which W reaches the straight line ℓ. The line is continuous and starts at $\ell(0) = \ln(H/S_0)/\sigma < 0 = W_0$, so it is approached from above: t_H is a hitting time in the sense of Definition 19.22(c).

The hedge instrument $\tilde{S}_t = S_t e^{-(r-b)(T-t)} = \tilde{S}_0 \exp((g + r - b)t + \sigma W_t)$ is a geometric Brownian motion with volatility σ that pays no income (Setup (e) of §23.2), hence an instrument of Definition 20.1. Its drift and diffusion coefficients are $\alpha_t = (g + r - b + \sigma^2/2)\tilde{S}_t$ and $\beta_t = \sigma \tilde{S}_t$ (Corollary 19.34(ii)). Write the knock-out function in the variable of the hedge instrument,

$$\tilde{V}^{\text{out}}(x,t) = V^{\text{out}}\big(xe^{(r-b)(T-t)},\, t\big), \qquad x > 0,\ 0 \le t < T,$$

so that $\tilde{V}^{\text{out}}(\tilde{S}_t, t) = V^{\text{out}}(S_t, t)$; from here on x stands for a price of the hedge instrument. At $S = xe^{(r-b)(T-t)}$ the chain rule gives

$$\partial_t \tilde{V}^{\text{out}} = \partial_t V^{\text{out}} - (r-b)S\,\partial_S V^{\text{out}}, \qquad x\,\partial_x \tilde{V}^{\text{out}} = S\,\partial_S V^{\text{out}}, \qquad x^2\,\partial_{xx}\tilde{V}^{\text{out}} = S^2\,\partial_{SS} V^{\text{out}},$$

and substituting these in (23.5) gives

$$\partial_t \tilde{V}^{\text{out}} + rx\,\partial_x \tilde{V}^{\text{out}} + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx}\tilde{V}^{\text{out}} = r\tilde{V}^{\text{out}}, \tag{I.5}$$

the equation (20.4) of a claim on an asset without income (this is Corollary 20.45 read backwards). By Step 2, and because the factor $e^{(r-b)(T-t)}$ stays between two positive constants on $[0,T]$, the functions $\tilde{V}^{\text{out}}$, $\partial_t \tilde{V}^{\text{out}}$, $\partial_x \tilde{V}^{\text{out}}$ and $\partial_{xx}\tilde{V}^{\text{out}}$ are continuous and of moderate growth on $(0,\infty) \times [0,T']$.

Apply Corollary 19.31, with the stopped clause of Theorem 19.30, to the geometric Brownian motion $\tilde{S}$ and the outer function $(x,t) \mapsto e^{-rt}\tilde{V}^{\mathrm{out}}(x,t)$ on $[0,T']$. The drift coefficient of $e^{-rt}\tilde{V}^{\mathrm{out}}(\tilde{S}_t,t)$ is

$$e^{-rt}\Big[\partial_t\tilde{V}^{\mathrm{out}} + \alpha_t\,\partial_x\tilde{V}^{\mathrm{out}} + \tfrac{1}{2}\beta_t^2\,\partial_{xx}\tilde{V}^{\mathrm{out}} - r\tilde{V}^{\mathrm{out}}\Big] = e^{-rt}(\alpha_t - r\tilde{S}_t)\,\partial_x\tilde{V}^{\mathrm{out}},$$

by (I.5) at $x=\tilde{S}_t$, and its diffusion coefficient is $e^{-rt}\beta_t\,\partial_x\tilde{V}^{\mathrm{out}}$, every derivative being evaluated at $(\tilde{S}_t,t)$. Now $e^{-rt}(\alpha_t - r\tilde{S}_t)$ and $e^{-rt}\beta_t$ are the drift and diffusion coefficients of the discounted instrument $e^{-rt}\tilde{S}_t$ in Definition 20.1. Hence, in the notation of that definition, for every $t \le T'$ and with probability one,

$$e^{-r(t\wedge t_H)}\,V^{\mathrm{out}}(S_{t\wedge t_H},\,t\wedge t_H) = V^{\mathrm{out}}(S_0,0) + \int_0^t \mathbf{1}\{s<t_H\}\,\partial_x\tilde{V}^{\mathrm{out}}(\tilde{S}_s,s)\,d(e^{-rs}\tilde{S}_s), \tag{I.6}$$

where $t\wedge t_H$ is the smaller of the two times. The right side is the discounted value at t of the strategy with initial capital $V^{\mathrm{out}}(S_0,0)$ and the holding

$$\mathbf{1}\{s<t_H\}\,\partial_x\tilde{V}^{\mathrm{out}}(\tilde{S}_s,s) = \mathbf{1}\{s<t_H\}\,\partial_S V^{\mathrm{out}}(S_s,s)\,e^{(r-b)(T-s)}$$

of $\tilde{S}$. This holding is, by Lemma 19.20(i), a continuous function of (s,W_s) of exponential growth on $[0,T']$, stopped at a hitting time, so it is admissible (Definition 20.1(i)); by the stopping clause of Definition 20.1(iv), from t_H on the strategy holds no instrument and its whole value sits in the bank. Since $T'<T$ was arbitrary, (I.6) says that the value Π_t of this strategy satisfies

$$e^{-rt}\,\Pi_t = e^{-r(t\wedge t_H)}\,V^{\mathrm{out}}(S_{t\wedge t_H},\,t\wedge t_H) \qquad \text{for every } t<T.$$

Step 4 (the terminal value). Let T' increase to T along a sequence, and consider three events. On $\{t_H<T\}$: as soon as $T'\ge t_H$ the stopped point is $(S_{t_H},t_H)=(H,t_H)$, because the continuous path sits on the barrier at the hitting time, and $V^{\mathrm{out}}(H,t_H)=0$ by (i); so $\Pi_{T'}=0$ from then on. On $\{t_H>T\}$, the event that the barrier is not reached by T: $S_t>H$ for every $t\le T$, in particular $S_T>H$, and $\Pi_{T'}=V^{\mathrm{out}}(S_{T'},T')$ tends to $f(S_T)$ by (i) and the continuity of the path, provided that S_T is a continuity point of f. That fails with probability zero, because S_T has a density (Theorem 19.40) and f has finitely many discontinuities. The remaining event $\{t_H=T\}$ lies in $\{S_T=H\}$, which has probability zero. So, with probability one,

$$\Pi_{T'} \to f(S_T)\,\mathbf{1}\{t_H>T\} \qquad \text{as } T'\to T.$$

To turn this into convergence in mean square we bound fourth moments. The stopped price is never below the barrier: it equals H on $\{t_H\le T'\}$ and $S_{T'}>H$ on $\{t_H>T'\}$. So $S_{T'\wedge t_H}^{-c'}\le H^{-c'}$ and $S_{T'\wedge t_H}^{c'}\le H^{c'}+S_{T'}^{c'}$, and (I.4) gives

$$|\Pi_{T'}| \le e^{|r|T}c'(H^{-c'}+H^{c'}+S_{T'}^{c'}).$$

By Lemma 18.35(i), $\mathbb{E}\,S_{T'}^{4c'} = S_0^{4c'}\exp(4c'gT' + 8c'^2\sigma^2T')$, which is bounded for $T'\le T$; so $\sup_{T'<T}\mathbb{E}\,\Pi_{T'}^4<\infty$, and $\mathbb{E}\,f(S_T)^4<\infty$ in the same way because $f\in\mathcal{G}$. Lemma 19.12(iv) upgrades the convergence with probability one to convergence in mean square. By Definition 20.1(ii), the terminal value of the strategy is therefore $f(S_T)\,\mathbf{1}\{t_H>T\}$. Finally $\{t_H>T\}=\{S_T^{\min}>H\}$, because a continuous path stays above H on $[0,T]$ exactly when its minimum over $[0,T]$ exceeds H. The strategy replicates the down-and-out claim.

Step 5 (the price; the other cases). Suppose that the down-and-out claim traded today at a price above $V^{\mathrm{out}}(S_0,0)$. Selling it (a static position, Definition 20.1(iii)), running the strategy of Step 3 and depositing the difference in the bank is a package with zero initial capital whose terminal value is the deposit with its interest, a positive amount in every state: an arbitrage, which (A6) excludes. If the claim traded below $V^{\mathrm{out}}(S_0,0)$, buy it, run the negative of the strategy and

deposit the difference. This is the argument of Step 5 of the proof of Theorem 20.7, and it shows that $V^{\text{out}}(S_0,0)$ is the arbitrage-free price. If $f \ge 0$ and $V^{\text{out}}(S_0,0)$ were negative, the strategy of Step 3 together with a deposit of $-V^{\text{out}}(S_0,0) > 0$ would have zero initial capital and a terminal value at least equal to the deposit with its interest, again an arbitrage; so the price is nonnegative. This proves (ii).

For (iii), the down-and-in claim pays $f(S_T) - f(S_T)\,\mathbf{1}\{S_T^{\min} > H\}$. The vanilla claim is replicated from the capital $V(S_0,0)$ by the strategy that holds $\partial_S V(S_t,t)\,e^{(r-b)(T-t)}$ units of $\tilde{S}$ (Corollary 20.41(i) for an asset; Theorem 20.52(ii) for a futures price, whose $e^{r(T_F-t)}\,\partial_S V$ units of the prepaid forward are $e^{r(T-t)}\,\partial_S V$ units of $\tilde{S}$). That strategy minus the strategy of Step 3 is a strategy (Definition 20.1(iv)) with initial capital $V(S_0,0) - V^{\text{out}}(S_0,0)$, which is the image term at $(S_0,0)$, and with terminal value $f(S_T)\,\mathbf{1}\{S_T^{\min} \le H\}$; the arbitrage argument makes its initial capital the price. This agrees with Proposition 23.15(i), which gives the same price without a model.

For (iv), the proof is the mirror image. Step 1 does not depend on the side of the barrier. In Step 2, a continuity point $S^* < H$ is reflected to $H^2/S^* > H$, a point of the open interval (H,∞) on which f now vanishes identically, so the image term again tends to zero. In Step 3, $\ell(0) = \ln(H/S_0)/\sigma > 0$ and the line is approached from below, the other case of Definition 19.22(c). In Step 4 the stopped price is never above the barrier, so its positive powers are at most $H^{c'}$ and its negative powers at most $H^{-c'} + S_{T'}^{-c'}$, whose fourth moment is bounded in the same way; and $\{t_H > T\} = \{S_T^{\max} < H\}$. Step 5 is unchanged.

Step 6 (the risk-neutral representation, (v)). We rerun Steps 3 and 4 on the risk-neutral path and take expectations. Let $S_t^{\mathbb{Q}} = S_0 \exp((b - \frac{1}{2}\sigma^2)t + \sigma W_t)$ be the risk-neutral path (Definition 20.6(i) for an asset, with $b = r - q$; Proposition 20.51(i) for a futures price, with $b = 0$), and let $t_H^{\mathbb{Q}}$ be the first time at which it reaches H. As in Step 3, $t_H^{\mathbb{Q}}$ is the first time at which W reaches the straight line $(\ln(H/S_0) - (b - \sigma^2/2)t)/\sigma$, which starts below 0: a hitting time of Definition 19.22(c). The process $S^{\mathbb{Q}}$ is a geometric Brownian motion with drift coefficient $bS_t^{\mathbb{Q}}$ and diffusion coefficient $\sigma S_t^{\mathbb{Q}}$ (Corollary 19.34(ii)), and the outer function $(x,t) \mapsto e^{-rt}V^{\text{out}}(x,t)$ has continuous partial derivatives of moderate growth on $(0,\infty)\times[0,T']$ for every $T' < T$ (Step 2). Corollary 19.31, with the stopped clause of Theorem 19.30, gives the drift coefficient

$$e^{-rt}\Big[\partial_t V^{\text{out}} + bS\,\partial_S V^{\text{out}} + \tfrac{1}{2}\sigma^2 S^2\,\partial_{SS} V^{\text{out}} - rV^{\text{out}}\Big]\Big|_{S=S_t^{\mathbb{Q}}} = 0$$

by (23.5), and therefore, for every $T' < T$ and with probability one,

$$e^{-r(T'\wedge t_H^{\mathbb{Q}})}\,V^{\text{out}}\big(S^{\mathbb{Q}}_{T'\wedge t_H^{\mathbb{Q}}},\, T'\wedge t_H^{\mathbb{Q}}\big) = V^{\text{out}}(S_0,0) + \int_0^{T'} \mathbf{1}\{s < t_H^{\mathbb{Q}}\}\,e^{-rs}\sigma S_s^{\mathbb{Q}}\,\partial_S V^{\text{out}}(S_s^{\mathbb{Q}},s)\,dW_s.$$

The integrand is a continuous function of (s,W_s) of exponential growth on $[0,T']$ (Lemma 19.20(i)), stopped at a hitting time, so it belongs to the class $\mathcal{H}$ and the integral has mean zero (Theorem 19.23(ii)). Hence the left side has mean $V^{\text{out}}(S_0,0)$ for every $T' < T$. Step 4 applies word for word with $S^{\mathbb{Q}}$ in place of S, the drift g being replaced by $b - \sigma^2/2$ in the moment bound: as $T' \to T$ the left side tends, with probability one, to $e^{-rT}f(S_T^{\mathbb{Q}})\,\mathbf{1}\{t_H^{\mathbb{Q}} > T\}$, its fourth moments are bounded uniformly in T', and $\mathbb{E}\,f(S_T^{\mathbb{Q}})^4 < \infty$. Lemma 19.12(iv) lets the means converge, so

$$V^{\text{out}}(S_0,0) = e^{-rT}\,\mathbb{E}\big[f(S_T^{\mathbb{Q}})\,\mathbf{1}\{t_H^{\mathbb{Q}} > T\}\big] = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[f(S_T)\,\mathbf{1}\{S_T^{\min} > H\}\big],$$

the last step by Definition 20.6, which computes the running minimum from $S^{\mathbb{Q}}$, and by path continuity, as at the end of Step 4. The formula for V in the statement gives $V(S_0,0) = e^{-rT}\,\mathbb{E}\,f(S_T^{\mathbb{Q}}) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}f(S_T)$, because W_T has the law of $\sqrt{T}\,Z$; subtracting proves the formula for $V^{\text{in}}(S_0,0)$. For an up barrier the argument is the same, with the stopped price bounded as in (iv). In particular a nonnegative f gives a nonnegative price directly, without the arbitrage argument of Step 5. □

I.4 Proof of Proposition 23.19: A one-touch is worth about two digitals

The statement is on page 451. The notation and the standing assumptions are those of §23.2.

Proof. For (i) we read the probabilities off the reflection principle and then show, by a replication stopped at the barrier, that they are prices; (ii) is Theorem 23.16 applied to the no-touch.

(i), the probabilities. The risk-neutral path satisfies $(S_t - S_0)/\sigma_{\mathrm{N}} = W_t$, so $S_T^{\max} \geq H$ exactly when $W_T^{\max} \geq \ell$, with the level $\ell = (H - S_0)/\sigma_{\mathrm{N}} > 0$. By Theorem 19.63(i),

$$\mathbb{Q}(S_T^{\max} \geq H) = P(W_T^{\max} \geq \ell) = 2\,P(W_T \geq \ell) = 2\,\Phi\Big(\frac{-\ell}{\sqrt{T}}\Big) = 2\,\Phi\Big(\frac{S_0 - H}{\sigma_{\mathrm{N}}\sqrt{T}}\Big),$$

and $P(W_T \geq \ell) = \mathbb{Q}(S_T \geq H)$. Figure 50 shows the pairing of paths behind the factor 2.

(i), the prices. The real-world price $S_t = S_0 + \mu_{\mathrm{N}} t + \sigma_{\mathrm{N}} W_t$ is an explicit function of (t, W_t) that grows linearly, with constant partial derivatives, so it is in the class $C_E^{1,2}$ of Definition 19.19(iii) and the underlying is an instrument of Definition 20.1, with drift coefficient μ_{N} and diffusion coefficient σ_{N}. Since $r = 0$, prices are their own discounted values and the bank pays no interest. Put

$$X_t = \frac{S_t - S_0}{\sigma_{\mathrm{N}}} = \frac{\mu_{\mathrm{N}}}{\sigma_{\mathrm{N}}}\, t + W_t.$$

The barrier is reached when X reaches the level ℓ, that is, when W reaches the straight line $\ell - (\mu_{\mathrm{N}}/\sigma_{\mathrm{N}})t$, which starts at $\ell > 0 = W_0$ and is approached from below: t_H is a hitting time of Definition 19.22(c).

Step 1 (the image function). Apply Lemma 19.61, with zero drift and unit volatility, to the bounded function $\psi(y) = \mathbf{1}\{y < \ell\}$: the smoothing

$$\Psi(t,x) = \mathbb{E}\,\psi\big(x + \sqrt{T-t}\,Z\big) = \Phi\Big(\frac{\ell - x}{\sqrt{T-t}}\Big), \qquad t < T,$$

is infinitely differentiable on $[0,T) \times \mathbb{R}$, each of its partial derivatives has exponential growth on $[0,T'] \times \mathbb{R}$ for every $T' < T$ (it is bounded by $c' e^{c|x|}$, by part (i) of the lemma), and $\partial_t \Psi + \frac{1}{2}\,\partial_{xx}\Psi = 0$. Here Ψ is explicit, and its derivatives are even bounded on $[0,T'] \times \mathbb{R}$: for example $\partial_x \Psi(t,x) = -\varphi\big((\ell - x)/\sqrt{T-t}\big)/\sqrt{T-t}$, at most $1/\sqrt{2\pi(T-T')}$ in absolute value. As in Step 1 of the proof of Theorem 19.63 with zero drift, subtract the mirror image about the level:

$$\Psi^{\mathrm{out}}(t,x) = \Psi(t,x) - \Psi(t, 2\ell - x) = \Phi\Big(\frac{\ell - x}{\sqrt{T-t}}\Big) - \Phi\Big(\frac{x - \ell}{\sqrt{T-t}}\Big) = 1 - 2\,\Phi\Big(\frac{x - \ell}{\sqrt{T-t}}\Big).$$

The mirror image solves the same equation, because the two derivatives in x absorb the two changes of sign; so $\partial_t \Psi^{\mathrm{out}} + \frac{1}{2}\,\partial_{xx}\Psi^{\mathrm{out}} = 0$. Moreover $\Psi^{\mathrm{out}}(t,\ell) = 1 - 2\,\Phi(0) = 0$ for every $t < T$; $|\Psi^{\mathrm{out}}| \leq 1$; and $\Psi^{\mathrm{out}}(t,x) \to 1$ as $(t,x) \to (T, x^*)$ for every $x^* < \ell$, because then $(x - \ell)/\sqrt{T-t} \to -\infty$.

Step 2 (stopped replication of the no-touch). Fix $T' < T$ and apply the stopped clause of Itô's lemma (Theorem 19.30(iii)) on $[0,T']$ to the function $(t,y) \mapsto \Psi^{\mathrm{out}}\big(t, (\mu_{\mathrm{N}}/\sigma_{\mathrm{N}})t + y\big)$, which is in $C_E^{1,2}([0,T'] \times \mathbb{R})$ because it and its partial derivatives, combinations of those of Ψ^{out} with constant coefficients evaluated at a point shifted by at most $|\mu_{\mathrm{N}}/\sigma_{\mathrm{N}}|\,T$, have exponential growth (Step 1). Its derivative in t plus half its second derivative in y is

$$\partial_t \Psi^{\mathrm{out}} + \frac{\mu_{\mathrm{N}}}{\sigma_{\mathrm{N}}}\,\partial_x \Psi^{\mathrm{out}} + \tfrac{1}{2}\,\partial_{xx}\Psi^{\mathrm{out}} = \frac{\mu_{\mathrm{N}}}{\sigma_{\mathrm{N}}}\,\partial_x \Psi^{\mathrm{out}},$$

by Step 1, and its derivative in y is $\partial_x \Psi^{\mathrm{out}}$, all evaluated at (t, X_t). So, for every $t \leq T'$ and with probability one,

$$\Psi^{\mathrm{out}}\big(t \wedge t_H,\, X_{t \wedge t_H}\big) = \Psi^{\mathrm{out}}(0,0) + \int_0^t h_s\,\mu_{\mathrm{N}}\,ds + \int_0^t h_s\,\sigma_{\mathrm{N}}\,dW_s, \qquad h_s = \mathbf{1}\{s < t_H\}\,\frac{\partial_x \Psi^{\mathrm{out}}(s, X_s)}{\sigma_{\mathrm{N}}}.$$

The two integrals together are the integral of the holding h_s against the price of the underlying, in the sense of Definition 20.1 with $r = 0$. Hence the strategy with initial capital $\Psi^{\mathrm{out}}(0,0)$ that holds h_s units of the underlying has the value $\Psi^{\mathrm{out}}(t \wedge t_H, X_{t\wedge t_H})$ at every $t < T$, whatever the real-world drift μ_{N}.

Step 3 (the terminal value). Let T' increase to T. On $\{t_H < T\}$ the value is $\Psi^{\mathrm{out}}(t_H, \ell) = 0$ as soon as $T' \geq t_H$. On $\{t_H > T\}$ the path stays below the level, $X_T < \ell$, and the value $\Psi^{\mathrm{out}}(T', X_{T'})$ tends to 1 by Step 1. The event $\{t_H = T\}$ lies in $\{X_T = \ell\}$, which has probability zero. The values are bounded by 1 in absolute value, so Lemma 19.12(iv) gives convergence in mean square, as in Step 4 of the proof of Theorem 19.63. The terminal value of the strategy is therefore $\mathbf{1}\{t_H > T\} = \mathbf{1}\{S_T^{\max} < H\}$, the payoff of the no-touch, and its cost is

$$\Psi^{\mathrm{out}}(0,0) = 1 - 2\,\Phi\Big(\frac{-\ell}{\sqrt{T}}\Big) = \mathbb{Q}(S_T^{\max} < H).$$

Step 4 (the digital, the one-touch and the prices). The same argument without the image term and without stopping, applied to $\psi(y) = \mathbf{1}\{y > \ell\}$, uses $\Psi(t,x) = \Phi((x-\ell)/\sqrt{T-t})$ and the holding $\partial_x \Psi(s, X_s)/\sigma_{\mathrm{N}}$. Its value $\Psi(t, X_t)$ tends to $\mathbf{1}\{X_T > \ell\} = \mathbf{1}\{S_T > H\}$, with probability one (because $X_T \neq \ell$ with probability one) and in mean square, so it replicates the digital at the cost $\Psi(0,0) = \Phi(-\ell/\sqrt{T}) = \mathbb{Q}(S_T > H)$. The one-touch pays one unit of cash at T minus the payoff of the no-touch, so it is replicated by one unit held in the bank minus the no-touch strategy (Definition 20.1(iv)), at the cost $1 - \Psi^{\mathrm{out}}(0,0) = 2\,\Phi(-\ell/\sqrt{T})$. The arbitrage argument of Step 5 of the proof of Theorem 20.7, run under (A6) with these strategies, makes each replication cost the price: $\mathrm{Dig}(H) = \Phi(-\ell/\sqrt{T})$ and $\mathrm{OT}(H) = 2\,\Phi(-\ell/\sqrt{T}) = 2\,\mathrm{Dig}(H)$.

(ii), up barrier. The no-touch pays $f(F_T)\,\mathbf{1}\{F_T^{\max} < H\}$ with $f(x) = \mathbf{1}\{x < H\}$, where $F_T^{\max}$ and $F_T^{\min}$ are the running maximum and minimum of the futures price on $[0,T]$. This f is bounded with one discontinuity, so it is in $\mathcal{G}$, and it vanishes on $[H,\infty)$: Theorem 23.16(iv) applies, with $S_t = F_t$, $b = 0$ and exponent 1. The European claim paying $f(F_T)$ is the digital put, whose price function at time 0 is $V(x,0) = e^{-rT}\,\Phi(-d_2(x))$, with $d_2(x) = [\ln(x/H) - \sigma^2 T/2]/(\sigma\sqrt{T})$ (Theorem 20.17 with Corollary 20.20). At the reflected point,

$$d_2\Big(\frac{H^2}{F}\Big) = \frac{\ln(H/F) - \sigma^2 T/2}{\sigma\sqrt{T}} = -\,\frac{\ln(F/H) + \sigma^2 T/2}{\sigma\sqrt{T}} = -d_1,$$

so $V(H^2/F, 0) = e^{-rT}\,\Phi(d_1)$ and the no-touch is worth

$$V^{\mathrm{out}}(F,0) = e^{-rT}\Big[\Phi(-d_2) - \frac{F}{H}\,\Phi(d_1)\Big].$$

The one-touch pays 1 at T minus the no-touch, and a unit paid at T is worth e^{-rT}; by the law of one price,

$$\mathrm{OT}(H) = e^{-rT} - V^{\mathrm{out}}(F,0) = e^{-rT}\Big[\Phi(d_2) + \frac{F}{H}\,\Phi(d_1)\Big],$$

using $1 - \Phi(-d_2) = \Phi(d_2)$. Finally $C(H) = e^{-rT}[F\,\Phi(d_1) - H\,\Phi(d_2)]$ (Theorem 20.22) gives $C(H)/H = e^{-rT}[(F/H)\,\Phi(d_1) - \Phi(d_2)]$, and $\mathrm{Dig}(H) = e^{-rT}\,\Phi(d_2)$, so the bracket is $2\,\mathrm{Dig}(H) + C(H)/H$.

(ii), down barrier. Now the no-touch pays $f(F_T)\,\mathbf{1}\{F_T^{\min} > H\}$ with $f(x) = \mathbf{1}\{x > H\}$, which vanishes on $(0,H]$, and Theorem 23.16(ii) applies. The European claim is the digital, $V(x,0) = e^{-rT}\,\Phi(d_2(x))$, and again $d_2(H^2/F) = -d_1$, so

$$\mathrm{OT}(H) = e^{-rT} - e^{-rT}\Big[\Phi(d_2) - \frac{F}{H}\,\Phi(-d_1)\Big] = e^{-rT}\Big[\Phi(-d_2) + \frac{F}{H}\,\Phi(-d_1)\Big].$$

With $\mathcal{P}(H) = e^{-rT}[H\,\Phi(-d_2) - F\,\Phi(-d_1)]$, so that $\mathcal{P}(H)/H = e^{-rT}[\Phi(-d_2) - (F/H)\,\Phi(-d_1)]$, and with the digital put $e^{-rT} - \mathrm{Dig}(H) = e^{-rT}\,\Phi(-d_2)$, this is $2(e^{-rT} - \mathrm{Dig}(H)) - \mathcal{P}(H)/H$.

(ii), a second route for the up barrier. By Theorem 23.16(v) the no-touch is worth $e^{-rT}\,\mathbb{Q}(F_T^{\max} < H)$, so $\mathrm{OT}(H) = e^{-rT}\,\mathbb{Q}(F_T^{\max} \ge H)$, and the probability can be read off the drifted reflection formula. Under $\mathbb{Q}$ the futures price is $F\exp(-\frac{1}{2}\sigma^2 t + \sigma W_t)$ (Proposition 20.51(i)), so $\ln(F_t/F)/\sigma = -(\sigma/2)t + W_t$, which reaches the level $\ell' = \ln(H/F)/\sigma > 0$ exactly when the futures price reaches H. Theorem 19.63(ii) with drift $-\sigma/2$ and level ℓ' gives

$$\mathbb{Q}(F_T^{\max} \ge H) = \Phi\Big(\frac{-\sigma T/2 - \ell'}{\sqrt{T}}\Big) + e^{-\sigma\ell'}\,\Phi\Big(\frac{-\ell' + \sigma T/2}{\sqrt{T}}\Big) = \Phi(d_2) + \frac{F}{H}\,\Phi(d_1),$$

because $(-\sigma T/2 - \ell')/\sqrt{T} = [\ln(F/H) - \sigma^2 T/2]/(\sigma\sqrt{T}) = d_2$, $(-\ell' + \sigma T/2)/\sqrt{T} = d_1$ and $e^{-\sigma\ell'} = F/H$. This is the formula of (ii) again. □

I.5 Proof of Proposition 23.28: Geometric Asian in closed form (Kemna and Vorst, 1990)

The statement is on page 458. The notation and the standing assumptions are those of §23.3.

Proof. We reduce the price to an expectation, compute the law of $\ln G_T$ with finite sums only, and evaluate a lognormal expectation.

Step 1 (the price is an expectation). The payoff is $(G_T - K)^+ = f(S_{t_1}, \dots, S_{t_J})$ with

$$f(x_1, \dots, x_J) = \big((x_1 x_2 \cdots x_J)^{1/J} - K\big)^+,$$

a continuous function on $(0,\infty)^J$. It has moderate growth: $0 \le f(x_1,\dots,x_J) \le \prod_j x_j^{1/J}$, and $x^{1/J} \le \max(1,x) \le x + x^{-1}$ for every $x > 0$, so $0 \le f \le \prod_j (x_j + x_j^{-1})$. Hence $f \in \mathcal{G}_J$ (Definition 19.19(iv)).

Corollary 20.11 is stated for an asset without income, and we apply it to the reinvested holding $\tilde S_t = S_t e^{-q(T-t)}$, which pays no income (Proposition 16.15(i)) and, under (A1), is a geometric Brownian motion with volatility σ. Each fixing is a fixed multiple of a value of that instrument, $S_{t_j} = \tilde S_{t_j} e^{q(T-t_j)}$, so the payoff is

$$\tilde f(\tilde S_{t_1}, \dots, \tilde S_{t_J}), \qquad \tilde f(y_1, \dots, y_J) = f\big(y_1 e^{q(T-t_1)}, \dots, y_J e^{q(T-t_J)}\big).$$

The function $\tilde f$ is again continuous, and $0 \le \tilde f \le e^{|q|T}\prod_j (y_j + y_j^{-1})$, because the factors $e^{q(T-t_j)/J}$ multiply to at most $e^{|q|T}$; so $\tilde f \in \mathcal{G}_J$. By Corollary 20.11, the claim is replicated by trading $\tilde S$ and the bank, and its arbitrage-free price is e^{-rT} times the expectation of $\tilde f$ along the risk-neutral path of $\tilde S$. That path is $\tilde S_0 \exp((r - \frac{1}{2}\sigma^2)t + \sigma W_t)$ with $\tilde S_0 = S e^{-qT}$ (Definition 20.6(i) with zero yield), and multiplying it by $e^{q(T-t)}$ gives

$$S e^{-qT}\, e^{q(T-t)} \exp\big((r - \tfrac{1}{2}\sigma^2)t + \sigma W_t\big) = S \exp\big((b - \tfrac{1}{2}\sigma^2)t + \sigma W_t\big),$$

the risk-neutral path of the asset with yield q. So the price is $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(G_T - K)^+]$, with the $\mathbb{E}^{\mathbb{Q}}$ of Definition 20.6(i) for the yield q. This reduction is the one recorded in Corollary 20.41(i), and it applies in the same way to every payoff of $\mathcal{G}_J$.

Step 2 (the law of $\ln G_T$). Under $\mathbb{Q}$, by Setup (d) of §23.3,

$$\ln G_T = \frac{1}{J}\sum_{j=1}^{J} \ln S_{t_j} = \ln S + (b - \tfrac{1}{2}\sigma^2)\,\frac{1}{J}\sum_{j=1}^{J} t_j + \frac{\sigma}{J}\sum_{j=1}^{J} W_{t_j}.$$

The dates add up to $\sum_j t_j = (T/J)\sum_{j=1}^{J} j = (T/J)\cdot J(J+1)/2 = T(J+1)/2$, which gives the constant term $\ln S + (b - \frac{1}{2}\sigma^2)T(J+1)/(2J)$. For the random term, write each Brownian value as

a sum of increments, $W_{t_j} = \sum_{k=1}^{j} \Delta W_k$, and exchange the two sums. The increment ΔW_k occurs in W_{t_j} for $j = k, k+1, \dots, J$, that is, $J - k + 1$ times, so

$$\sum_{j=1}^{J} W_{t_j} = \sum_{j=1}^{J} \sum_{k=1}^{j} \Delta W_k = \sum_{k=1}^{J} (J - k + 1)\, \Delta W_k.$$

The increments $\Delta W_1, \dots, \Delta W_J$ are mutually independent, each with the law $\mathcal{N}(0, T/J)$ (Definition 19.5, properties (2) and (3)). A sum of two independent normal variables is normal, with the means and the variances added (Lemma 19.38(ii)). Adding the terms $(J - k + 1)\, \Delta W_k$ one at a time, each new term being independent of the sum of the earlier ones because every event about that sum is an event about the earlier increments, shows that the weighted sum is normal with mean zero and variance

$$\frac{T}{J} \sum_{k=1}^{J} (J - k + 1)^2 = \frac{T}{J} \sum_{i=1}^{J} i^2 = \frac{T}{J} \cdot \frac{J(J+1)(2J+1)}{6} = \frac{T(J+1)(2J+1)}{6}.$$

(The sum of the first J squares is $J(J+1)(2J+1)/6$: this holds for $J = 1$, and adding $(J+1)^2$ gives $(J+1)(2J^2+7J+6)/6 = (J+1)(J+2)(2J+3)/6$, the formula at $J+1$.) Multiplying by $(\sigma/J)^2$, the variance of $\ln G_T$ is $\sigma^2 T(J+1)(2J+1)/(6J^2) = \sigma_G^2 T$. Therefore, with a standard normal variable Z and the definition (23.8) of F_G,

$$G_T = S \exp\Big((b - \tfrac{1}{2}\sigma^2) T\, \frac{J+1}{2J} + \sigma_G \sqrt{T}\, Z \Big) = F_G \exp\big(-\tfrac{1}{2}\sigma_G^2 T + \sigma_G \sqrt{T}\, Z \big),$$

and $\mathbb{E}^{\mathbb{Q}}\, G_T = F_G$, because $\mathbb{E}\, e^{\theta Z} = e^{\theta^2/2}$ with $\theta = \sigma_G \sqrt{T}$ (Lemma 18.35(i)). This proves (i).

Step 3 (the lognormal expectation). Put $\theta = \sigma_G \sqrt{T}$. By Step 2, $G_T = F_G \exp(-\frac{1}{2}\theta^2 + \theta Z)$: this is the law (20.6) of the terminal price in the Black-76 model of Definition 20.16, with the forward F_G and the volatility σ_G, and there $\mathbb{E}^{\mathbb{Q}}$ is an ordinary expectation over Z. Theorem 20.22 evaluates e^{-rT} times the expectation of the call payoff under that law; read at a zero rate, so that no discount factor enters, it gives

$$\mathbb{E}\big[(F_G e^{-\theta^2/2 + \theta Z} - K)^+ \big] = F_G\, \Phi(d_1) - K\, \Phi(d_2), \tag{I.7}$$

with the d_1 and d_2 of (23.9). The identity uses only that F_G, K and θ are positive numbers, a remark that §23.4 uses twice. With Step 1, (I.7) proves (ii).

Step 4 (the limit). As $J \to \infty$, $(J+1)(2J+1)/(6J^2) = (1 + 1/J)(2 + 1/J)/6 \to 1/3$ and $(J+1)/(2J) \to 1/2$. Hence $\sigma_G^2 \to \sigma^2/3$ and

$$\ln F_G \to \ln S + \tfrac{1}{2}(b - \tfrac{1}{2}\sigma^2) T + \tfrac{1}{6}\sigma^2 T = \ln S + (\tfrac{1}{2} b - \tfrac{1}{12}\sigma^2) T.$$

The formula (23.9) is a continuous function of the pair (F_G, σ_G) on the set $F_G > 0$, $\sigma_G > 0$, because d_1 and d_2 are continuous there and Φ is continuous; and the limit $\sigma/\sqrt{3}$ of σ_G is positive. So the prices converge to the formula evaluated at the limits, which is (iii). □

I.6 Proof of Theorem 23.32: Margrabe's formula (Margrabe, 1978)

The statement is on page 461. The notation and the standing assumptions are those of §23.4.

Proof. We compute the risk-neutral expectation by factoring out the asset given up and applying a Gaussian shift (Steps 1 to 3), and then verify, by a replication that holds no bond and rests on the two-dimensional Itô lemma, that this expectation is the price (Step 4). Steps 1 to 3 produce the candidate (23.12) as $e^{-rT}\, \mathbb{E}^{\mathbb{Q}}[(S_{1,T} - S_{2,T})^+]$; Step 4 shows that the stated strategy replicates the option. Throughout,

$$x_0 = \frac{\tilde{S}_{1,0}}{\tilde{S}_{2,0}} = \frac{S_{1,0} e^{-q_1 T}}{S_{2,0} e^{-q_2 T}},$$

so that the d_1 and d_2 of the statement are $d_1(x_0, 0)$ and $d_2(x_0, 0)$.

Step 1 (factor out the asset given up). Under $\mathbb{Q}$ put

$$X = \ln \frac{S_{1,T}}{S_{2,T}}, \qquad Y = \ln S_{2,T}.$$

Then $(S_{1,T} - S_{2,T})^+ = S_{2,T}\,(S_{1,T}/S_{2,T} - 1)^+ = e^Y \psi(X)$ with $\psi(x) = (e^x - 1)^+$, a continuous function with $0 \le \psi(x) \le e^x$. By Setup (a) and (b) of §23.4, with the independent standard normal variables $Z_1 = W_{1,T}/\sqrt{T}$ and $Z_\perp = W_{\perp,T}/\sqrt{T}$,

$$\begin{aligned} Y &= \ln S_{2,0} + (r - q_2 - \tfrac{1}{2}\sigma_2^2)T + \sigma_2\sqrt{T}\,(\rho Z_1 + \sqrt{1-\rho^2}\,Z_\perp), \\ X &= \ln \frac{S_{1,0}}{S_{2,0}} + (q_2 - q_1 - \tfrac{1}{2}\sigma_1^2 + \tfrac{1}{2}\sigma_2^2)T + \sigma_1\sqrt{T}\,Z_1 - \sigma_2\sqrt{T}\,(\rho Z_1 + \sqrt{1-\rho^2}\,Z_\perp). \end{aligned}$$

Each is a constant plus a linear combination of the same two independent standard normal variables. So (X, Y) is a jointly normal pair in the sense of Lemma 19.38, and X and Y are normal (part (ii) of that lemma). The rate r has already cancelled in X.

Step 2 (reweight by a Gaussian shift). The moments are read off the two displays. First, $\mathbb{E}\,Y = \ln S_{2,0} + (r - q_2 - \frac{1}{2}\sigma_2^2)T$ and $\operatorname{Var} Y = \sigma_2^2 T\,(\rho^2 + 1 - \rho^2) = \sigma_2^2 T > 0$. Next, $\operatorname{Cov}(W_{1,T}, W_{2,T}) = \rho T$ (Proposition 19.52(i)), $\operatorname{Var} W_{1,T} = \operatorname{Var} W_{2,T} = T$, and the covariance is bilinear, so

$$\begin{aligned} \operatorname{Cov}(X, Y) &= \operatorname{Cov}\left(\sigma_1 W_{1,T} - \sigma_2 W_{2,T},\ \sigma_2 W_{2,T}\right) = (\rho\sigma_1\sigma_2 - \sigma_2^2)T, \\ \operatorname{Var} X &= \sigma_1^2 T + \sigma_2^2 T - 2\rho\sigma_1\sigma_2 T = \sigma^2 T, \end{aligned}$$

the second line by Proposition 10.10 applied to the sum of $\sigma_1 W_{1,T}$ and $-\sigma_2 W_{2,T}$. Lemma 19.38(iii) now gives

$$\mathbb{E}^{\mathbb{Q}}\big[(S_{1,T} - S_{2,T})^+\big] = \mathbb{E}\big[e^Y \psi(X)\big] = \mathbb{E}\big[e^Y\big]\ \mathbb{E}\big[\psi(X + \operatorname{Cov}(X, Y))\big],$$

with $\mathbb{E}\,e^Y = e^{\mathbb{E}Y + \operatorname{Var}(Y)/2} = S_{2,0} e^{(r-q_2)T}$. The shifted variable $X' = X + \operatorname{Cov}(X, Y)$ is normal, with the variance $\sigma^2 T$ of X and with the mean

$$\begin{aligned} \mathbb{E}\,X + \operatorname{Cov}(X, Y) &= \ln \frac{S_{1,0}}{S_{2,0}} + (q_2 - q_1)T - \tfrac{1}{2}\sigma_1^2 T + \tfrac{1}{2}\sigma_2^2 T + \rho\sigma_1\sigma_2 T - \sigma_2^2 T \\ &= \ln \frac{S_{1,0} e^{-q_1 T}}{S_{2,0} e^{-q_2 T}} - \tfrac{1}{2}(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)T = \ln x_0 - \tfrac{1}{2}\sigma^2 T. \end{aligned}$$

Step 3 (a lognormal expectation; the rate cancels). By Step 2, $e^{X'} = x_0 \exp(-\frac{1}{2}\sigma^2 T + \sigma\sqrt{T}\,Z)$ with a standard normal Z, so $\mathbb{E}\,\psi(X') = \mathbb{E}[(x_0 e^{-\sigma^2 T/2 + \sigma\sqrt{T}Z} - 1)^+]$. This is the lognormal expectation (I.7), with x_0, 1 and σ in the places of F_G, K and σ_G; in the language of Theorem 20.22, a call with the forward x_0, the strike 1, the total variance $\sigma^2 T$ and a zero rate. Hence

$$\mathbb{E}\,\psi(X') = x_0\,\Phi(d_1) - \Phi(d_2) = \tilde{V}(x_0, 0).$$

Multiplying by the discount factor and by $\mathbb{E}\,e^Y$,

$$\begin{aligned} e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\big[(S_{1,T} - S_{2,T})^+\big] &= e^{-rT}\,S_{2,0} e^{(r-q_2)T}\,\left[x_0\,\Phi(d_1) - \Phi(d_2)\right] \\ &= S_{1,0} e^{-q_1 T}\,\Phi(d_1) - S_{2,0} e^{-q_2 T}\,\Phi(d_2), \end{aligned}$$

which is the left side of (23.12); it equals $\tilde{S}_{2,0}\,\tilde{V}(x_0, 0) = V(\tilde{S}_{1,0}, \tilde{S}_{2,0}, 0)$. The rate has cancelled, and Step 4 explains why: the replicating strategy holds the two prepaid forwards and never the bank account.

Step 4 (verification by replication with no bond). The step has six parts.

(a) Properties of $\tilde{V}$. For $t < T$, the lognormal expectation (I.7), now with x, 1 and $\sigma\sqrt{T-t}$ in the places of F_G, K and $\sigma_G\sqrt{T}$, shows that

$$\tilde{V}(x,t) = \mathbb{E}\Big[\Big(x\exp\big(-\tfrac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}\,Z\big) - 1\Big)^+\Big].$$

This is the function (20.5) of Theorem 20.7 for the payoff $f(x) = (x-1)^+$, the rate zero and the volatility σ. The payoff is continuous with $0 \le f(x) \le x$, so $f \in \mathcal{G}$. Parts (i) and (ii) of that theorem concern the function alone and hold for every rate and every positive volatility; they give three properties.

- The function $\tilde{V}$ is infinitely differentiable on $(0,\infty)\times[0,T)$, and each of its partial derivatives has moderate growth on $(0,\infty)\times[0,T']$, for every $T' < T$.
- It solves the pricing equation at a zero rate,

$$\partial_t\tilde{V} + \tfrac{1}{2}\sigma^2x^2\,\partial_{xx}\tilde{V} = 0 \qquad \text{on } (0,\infty)\times[0,T), \tag{I.8}$$

  which has the form of the equation for a futures price in Corollary 20.45(ii), taken at $r = 0$.
- As $(x,t) \to (x^*,T)$, $\tilde{V}(x,t) \to (x^*-1)^+$, for every $x^* > 0$, because the payoff is continuous at every point.

Moreover $0 \le \tilde{V}(x,t) \le x$ for $x > 0$ and $0 \le t \le T$: the payoff lies between 0 and its argument, and $\mathbb{E}[x\exp(-\frac{1}{2}\sigma^2(T-t) + \sigma\sqrt{T-t}\,Z)] = x$ (Lemma 18.35(i)).

(b) Homogeneity and the derivatives of V. By (23.11), $V(cx_1,cx_2,t) = c\,V(x_1,x_2,t)$ for every $c > 0$: the function V is homogeneous of degree one in (x_1,x_2). Write $x = x_1/x_2$, so that $\partial_1 x = 1/x_2$ and $\partial_2 x = -x/x_2$. For $t < T$ the chain rule gives, with $\tilde{V}$ and its derivatives evaluated at (x,t),

$$\partial_1 V = \partial_x\tilde{V}, \qquad \partial_2 V = \tilde{V} - x\,\partial_x\tilde{V}, \qquad \partial_t V = x_2\,\partial_t\tilde{V}, \tag{I.9}$$

and therefore Euler's identity for a homogeneous function,

$$x_1\,\partial_1 V + x_2\,\partial_2 V = x_1\,\partial_x\tilde{V} + x_2\tilde{V} - x_2x\,\partial_x\tilde{V} = x_2\tilde{V} = V, \tag{I.10}$$

because $x_2x = x_1$. Differentiating (I.9) once more,

$$\partial_{11}V = \frac{\partial_{xx}\tilde{V}}{x_2}, \qquad \partial_{12}V = -\frac{x\,\partial_{xx}\tilde{V}}{x_2}, \qquad \partial_{22}V = \frac{x^2\,\partial_{xx}\tilde{V}}{x_2}; \tag{I.11}$$

for the last one, $\partial_2(\tilde{V} - x\,\partial_x\tilde{V}) = \partial_x\tilde{V}\,\partial_2 x - \partial_2 x\,\partial_x\tilde{V} - x\,\partial_{xx}\tilde{V}\,\partial_2 x = x^2\,\partial_{xx}\tilde{V}/x_2$. The first derivatives are explicit. Differentiating (23.10), with $\partial_x d_1 = \partial_x d_2 = 1/(x\sigma\sqrt{T-t})$,

$$\partial_x\tilde{V} = \Phi(d_1) + \frac{x\,\varphi(d_1) - \varphi(d_2)}{x\sigma\sqrt{T-t}} = \Phi(d_1),$$

because $x\,\varphi(d_1) = \varphi(d_2)$ (Lemma 20.34 at the forward x, the strike 1 and the life $T-t$). Hence, with $d_{1,2} = d_{1,2}(x_1/x_2,t)$,

$$\partial_1 V = \Phi(d_1), \qquad \partial_2 V = \tilde{V} - x\,\Phi(d_1) = -\Phi(d_2). \tag{I.12}$$

Both are bounded, and both depend on (x_1,x_2) only through the ratio x_1/x_2.

(c) The discounted instruments. By Setup (b) and (c) of §23.4, the prepaid forwards and their discounted values are

$$\begin{aligned} \tilde{S}_{k,t} &= S_{k,0}e^{-q_kT}\exp\big((g_k+q_k)t + \sigma_kW_{k,t}\big), \\ e^{-rt}\,\tilde{S}_{k,t} &= S_{k,0}e^{-q_kT}\exp\big((g_k+q_k-r)t + \sigma_kW_{k,t}\big), \qquad k = 1,2, \end{aligned}$$

geometric Brownian motions in W_k with the volatilities σ_k. As geometric Brownian motions driven by W_k, the two instruments have the drift and diffusion coefficients

$$\alpha_{k,t} = (g_k + q_k + \tfrac{1}{2}\sigma_k^2)\,\tilde{S}_{k,t}, \qquad \beta_{k,t} = \sigma_k\,\tilde{S}_{k,t}$$

(Corollary 19.34(ii), with $g_k + q_k + \frac{1}{2}\sigma_k^2$ in the place of μ). For $k = 1$ these are the coefficients of Definition 20.1. For $k = 2$ that definition, which writes the price as a function of $(t, W_{1,t}, W_{\perp,t})$ through $W_{2,t} = \rho W_{1,t} + \sqrt{1-\rho^2}\,W_{\perp,t}$, has the same drift (the time derivative contributes $g_2 + q_2$ and the two second derivatives $\frac{1}{2}\sigma_2^2(\rho^2 + 1 - \rho^2) = \frac{1}{2}\sigma_2^2$, times the price, which is $\alpha_{2,t}$) and the two diffusion coefficients $\rho\beta_{2,t}$ against W_1 and $\sqrt{1-\rho^2}\,\beta_{2,t}$ against $W_\perp$. So the discounted prepaid forward has the drift $e^{-rt}(\alpha_{k,t} - r\tilde{S}_{k,t}) = (g_k + q_k - r + \frac{1}{2}\sigma_k^2)\,e^{-rt}\tilde{S}_{k,t}$ and the diffusion coefficient $e^{-rt}\beta_{k,t} = \sigma_k e^{-rt}\tilde{S}_{k,t}$, and these are the two integrands of the holding integral of Definition 20.1: for a holding h of the instrument k,

$$\int_0^t h_s\,d(e^{-rs}\tilde{S}_{k,s}) = \int_0^t h_s\,e^{-rs}(\alpha_{k,s} - r\tilde{S}_{k,s})\,ds + \int_0^t h_s\,e^{-rs}\beta_{k,s}\,dW_{k,s}.$$

For $k = 1$ this is the definition as written, with the motion W_1. For $k = 2$ the definition writes the diffusion part as two integrals, of $h_s\,\rho\sigma_2 e^{-rs}\tilde{S}_{2,s}$ against W_1 and of $h_s\sqrt{1-\rho^2}\,\sigma_2 e^{-rs}\tilde{S}_{2,s}$ against $W_\perp$. Their sum is ρ times the integral of $h_s e^{-rs}\beta_{2,s}$ against W_1 plus $\sqrt{1-\rho^2}$ times its integral against $W_\perp$, which is, by definition, the integral of $h_s e^{-rs}\beta_{2,s}$ against W_2 (Theorem 19.55(ii)). By the homogeneity of V, with $c = e^{-rt}$,

$$e^{-rt}\,V(\tilde{S}_{1,t}, \tilde{S}_{2,t}, t) = V(e^{-rt}\tilde{S}_{1,t},\, e^{-rt}\tilde{S}_{2,t},\, t). \tag{I.13}$$

(d) Itô's lemma: the drift term vanishes. Fix $T' < T$. On $(0,\infty)^2 \times [0, T']$ the function V and its derivatives $\partial_t V$, $\partial_1 V$, $\partial_2 V$, $\partial_{11} V$, $\partial_{12} V$ and $\partial_{22} V$ are continuous, and by (23.11), (I.9) and (I.11) each is a product of powers of x_1 and x_2 with $\tilde{V}$, or one of its derivatives, evaluated at $(x_1/x_2, t)$. A function of moderate growth evaluated at x_1/x_2 has moderate growth in (x_1, x_2), because

$$\Big(\frac{x_1}{x_2}\Big)^c + \Big(\frac{x_1}{x_2}\Big)^{-c} \le (x_1^c + x_1^{-c})(x_2^c + x_2^{-c}),$$

and products of functions of moderate growth have moderate growth (Lemma 19.20(i)). So, by (a), V and these six derivatives have moderate growth on $(0,\infty)^2 \times [0, T']$. With $X_k(t, y) = S_{k,0}e^{-q_k T}\exp\big((g_k + q_k - r)t + \sigma_k y\big)$, so that $\partial_y X_k = \sigma_k X_k$, $\partial_{yy} X_k = \sigma_k^2 X_k$ and $\partial_t X_k = (g_k + q_k - r)X_k$, the composite $V(X_1(t, y_1), X_2(t, y_2), t)$, its partial derivative in t and its first and second partial derivatives in y_1 and y_2 are, by the chain rule, sums of products of these seven functions, evaluated at $(X_1(t, y_1), X_2(t, y_2), t)$, with constants and with powers of X_1 and X_2. They have moderate growth in the prices, hence exponential growth in (y_1, y_2) by Lemma 19.20(i): this is the hypothesis of Theorem 19.55(iii), applied with T' in the place of T. Applied on $[0, T']$ to V and the two discounted prepaid forwards, and combined with (I.13), the theorem gives, in shorthand,

$$\begin{aligned} d[e^{-rt}\,V(\tilde{S}_{1,t}, \tilde{S}_{2,t}, t)] = &\sum_{k=1}^{2} \partial_k V\,\Big(e^{-rt}(\alpha_{k,t} - r\tilde{S}_{k,t})\,dt + e^{-rt}\beta_{k,t}\,dW_k\Big) \\ &+ \Big[\partial_t V + \tfrac{1}{2}(\sigma_1^2 x_1^2\,\partial_{11} V + 2\rho\sigma_1\sigma_2 x_1 x_2\,\partial_{12} V + \sigma_2^2 x_2^2\,\partial_{22} V)\Big]dt, \end{aligned}$$

where V and its derivatives are evaluated at $(x_1, x_2, t) = (e^{-rt}\tilde{S}_{1,t}, e^{-rt}\tilde{S}_{2,t}, t)$, and where $e^{-rt}\beta_{k,t} = \sigma_k x_k$ at that point, the diffusion coefficient of the k-th discounted prepaid forward, has been used in the bracket. By (I.9) and (I.11), with $x = x_1/x_2$ and therefore $x_1^2/x_2 = x^2 x_2$ and $x_1 x = x^2 x_2$, the bracket equals

$$\begin{aligned} x_2\,\partial_t\tilde{V} + \tfrac{1}{2}\Big(\sigma_1^2\,\frac{x_1^2}{x_2} - 2\rho\sigma_1\sigma_2\,x_1 x + \sigma_2^2\,x_2 x^2\Big)\partial_{xx}\tilde{V} &= x_2\Big[\partial_t\tilde{V} + \tfrac{1}{2}(\sigma_1^2 - 2\rho\sigma_1\sigma_2 + \sigma_2^2)x^2\,\partial_{xx}\tilde{V}\Big] \\ &= x_2\Big[\partial_t\tilde{V} + \tfrac{1}{2}\sigma^2 x^2\,\partial_{xx}\tilde{V}\Big] = 0, \end{aligned}$$

by (I.8): the combination of the two volatilities and the correlation that Itô's lemma produces is exactly the ratio variance σ^2. Only the two terms of the first line remain. Written as an integral identity: for every $t \le T'$, with probability one,

$$e^{-rt}\,V(\tilde S_{1,t},\tilde S_{2,t},t) = V(\tilde S_{1,0},\tilde S_{2,0},0) + \int_0^t h_{1,s}\,d(e^{-rs}\tilde S_{1,s}) + \int_0^t h_{2,s}\,d(e^{-rs}\tilde S_{2,s}), \tag{I.14}$$

with the holdings $h_{k,s} = \partial_k V(e^{-rs}\tilde S_{1,s}, e^{-rs}\tilde S_{2,s}, s)$, $k = 1, 2$. Since $\partial_1 V$ and $\partial_2 V$ depend only on the ratio of their first two arguments, in which the discount factor cancels, (I.12) gives

$$h_{1,s} = \Phi(d_1), \qquad h_{2,s} = -\Phi(d_2), \qquad d_{1,2} = d_{1,2}(\tilde S_{1,s}/\tilde S_{2,s},\, s).$$

These are bounded continuous functions of $(s, W_{1,s}, W_{\perp,s})$ on $[0, T']$, hence admissible holdings (Definition 20.1(i)). Identity (I.14) is the self-financing identity of Definition 20.1(ii) for the strategy with the initial capital $V(\tilde S_{1,0}, \tilde S_{2,0}, 0)$ and these holdings. Since $T' < T$ was arbitrary, the value of that strategy is $\Pi_t = V(\tilde S_{1,t}, \tilde S_{2,t}, t)$ for every $t < T$, and its bank holding is

$$\Pi_t - h_{1,t}\,\tilde S_{1,t} - h_{2,t}\,\tilde S_{2,t} = V - \tilde S_{1,t}\,\partial_1 V - \tilde S_{2,t}\,\partial_2 V = 0,$$

by Euler's identity (I.10) at the point $(\tilde S_{1,t}, \tilde S_{2,t}, t)$. The strategy never holds cash.

(e) The terminal value. Let t increase to T along any sequence. The paths are continuous and positive, so $x_t = \tilde S_{1,t}/\tilde S_{2,t}$ tends to $x_T = S_{1,T}/S_{2,T}$ and $\tilde S_{2,t}$ tends to $S_{2,T}$ on every path, and by the third property in (a)

$$\Pi_t = \tilde S_{2,t}\,\tilde V(x_t, t) \;\to\; S_{2,T}\Big(\frac{S_{1,T}}{S_{2,T}} - 1\Big)^+ = (S_{1,T} - S_{2,T})^+$$

on every path. By (a), $0 \le V(x_1, x_2, t) = x_2\,\tilde V(x_1/x_2, t) \le x_1$, so $0 \le \Pi_t \le \tilde S_{1,t}$, and

$$\mathbb{E}\,\tilde S_{1,t}^4 = \big(S_{1,0}e^{-q_1 T}\big)^4 \exp\big(4(g_1 + q_1)t + 8\sigma_1^2 t\big)$$

(Lemma 18.35(i)) is bounded for $t \le T$. Likewise the fourth moment of $(S_{1,T} - S_{2,T})^+$ is at most $\mathbb{E}\,S_{1,T}^4 < \infty$. Lemma 19.12(iv) upgrades the convergence on every path to convergence in mean square, so the terminal value of the strategy, in the sense of Definition 20.1(ii), is $(S_{1,T} - S_{2,T})^+$.

(f) The price. The strategy of (d) starts from the capital $V(\tilde S_{1,0}, \tilde S_{2,0}, 0)$ and ends with the payoff of the exchange option. If the option traded today above that capital, then selling it, running the strategy and depositing the difference would be an arbitrage; if it traded below, buying it, running the negative of the strategy and depositing the difference would be one. This is the argument of Step 5 of the proof of Theorem 20.7, and (A6) excludes both. So the arbitrage-free price is $V(\tilde S_{1,0}, \tilde S_{2,0}, 0) = \tilde S_{2,0}\,\tilde V(x_0, 0)$, which is (23.12) and, by Step 3, equals $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[(S_{1,T} - S_{2,T})^+]$. The interest rate never appears, because the strategy holds no bond. □

I.7 Proof of the three claims behind Procedure 23.36: Risk-neutral Monte Carlo

The statement is on page 464. The notation and the standing assumptions are those of §23.5.

Proof. We show in turn what Procedure 23.36 estimates, why its triangular factor gives the right correlations, and when that factor exists.

What Procedure 23.36 estimates. The target is the risk-neutral expectation V of Setup (a) of §23.5. For a payoff $f(S_{t_1}, \dots, S_{t_J})$ of one asset observed on the grid, with $f \in \mathcal{G}_J$, that number is the arbitrage-free price of the claim, by Corollary 20.11 (with a yield, through Corollary 20.41(i)). The simulated path has exactly the law of the risk-neutral path of Definition 20.6 at the grid dates: in both, the logarithm of the price moves from one grid date to the next by $(b - \frac{1}{2}\sigma^2)\Delta t$

plus an independent $\mathcal{N}(0,\sigma^2\Delta t)$ variable, which for the risk-neutral path is $\sigma(W_{k\Delta t}-W_{(k-1)\Delta t})$ (Definition 19.5, properties (2) and (3)). A continuously monitored payoff is approximated by its version on the grid, which for the barriers and lookbacks of §23 is a payoff of $\mathcal{G}_J$ priced by Corollary 20.11. For a continuously monitored single barrier the target is itself a price, by Theorem 23.16(v). For a single-barrier down call or up put, knock-in or knock-out, the gap between the two prices is the one described by Fact 23.21; for other barrier payoffs the shifted barrier is only a practical approximation, with no error bound given in this document. For a payoff of two assets, V is the risk-neutral expectation of Definition 20.6(ii). The Part *Derivatives* identifies it with a price only for the exchange option (Theorem 23.32) and the best-of and worst-of claims built from it (Remark 23.34). The general two-asset case would need the backward induction of Corollary 20.11 run with Theorem 19.55, which is not carried out in this document. For more than two assets the model is not set up in the Part *Derivatives*, and the simulated number is a model value.

Why the factor gives the right correlations. Let $Z_1,\dots,Z_m$ be independent standard normal variables, and let the factor have the entries γ_{ik}, with $\gamma_{ik}=0$ for $k>i$. The i-th product is $\sum_k \gamma_{ik}Z_k$. Since $\mathrm{Cov}(Z_k,Z_l)$ is 1 for $k=l$ and 0 otherwise, the bilinearity of the covariance (the expansion behind Proposition 10.10) gives

$$\mathrm{Cov}\Big(\sum_k \gamma_{ik}Z_k,\ \sum_l \gamma_{jl}Z_l\Big)=\sum_k \gamma_{ik}\gamma_{jk}=\rho_{ij},$$

the (i,j) entry of the product of the factor with its transpose. Each product is normal with mean zero (Lemma 19.38(ii), one term at a time) and has the variance $\rho_{ii}=1$. So the products are standard normal variables with the prescribed correlations. For two variables the factor has the rows $(1,0)$ and $(\rho,\sqrt{1-\rho^2})$: the construction of Proposition 19.52.

When the triangular factor of Procedure 23.36 exists. The entries of the factor are the γ_{ik} of Setup (f) of §23.5.

A positive-definite matrix has a factor. We argue by induction on the dimension m, for a symmetric positive-definite matrix with entries a_{ij}; a correlation matrix is the case $a_{ii}=1$. Positive definiteness means that $\sum_{i,j} a_{ij}x_ix_j>0$ for every nonzero vector $(x_1,\dots,x_m)$. The pivot a_{11} is positive, because it is the value of this quadratic form at $(1,0,\dots,0)$; for $m=1$ the factor is the number $\sqrt{a_{11}}$. Let $m\ge 2$. The first column of the factor is $\gamma_{i1}=a_{i1}/\sqrt{a_{11}}$, $1\le i\le m$, and the *reduced block* that remains has the entries $a_{ij}-\gamma_{i1}\gamma_{j1}=a_{ij}-a_{i1}a_{j1}/a_{11}$, $2\le i,j\le m$. It is symmetric. For a vector $(y_2,\dots,y_m)$ put $x_1=-\sum_{j\ge2}a_{1j}y_j/a_{11}$. Then $a_{11}x_1^2=\big(\sum_{j\ge2}a_{1j}y_j\big)^2/a_{11}$ and $2x_1\sum_{j\ge2}a_{1j}y_j=-2\big(\sum_{j\ge2}a_{1j}y_j\big)^2/a_{11}$, so

$$\sum_{i,j\ge2}\Big(a_{ij}-\frac{a_{i1}a_{j1}}{a_{11}}\Big)y_iy_j=a_{11}x_1^2+2x_1\sum_{j\ge2}a_{1j}y_j+\sum_{i,j\ge2}a_{ij}y_iy_j. \tag{I.15}$$

The right side is the quadratic form of the whole matrix at the vector $(x_1,y_2,\dots,y_m)$, which is positive unless every y_j is zero. So the reduced block is positive definite, of dimension $m-1$, and by the induction hypothesis it has a lower-triangular factor: numbers γ_{ik}, $2\le k\le i\le m$, with $\sum_{k\ge2}\gamma_{ik}\gamma_{jk}=a_{ij}-\gamma_{i1}\gamma_{j1}$ for $i,j\ge2$. Together with the first column, and with $\gamma_{1k}=0$ for $k\ge2$, these form a lower-triangular array with

$$\sum_{k\ge1}\gamma_{ik}\gamma_{jk}=\gamma_{i1}\gamma_{j1}+(a_{ij}-\gamma_{i1}\gamma_{j1})=a_{ij}\quad(i,j\ge2),\qquad \sum_{k\ge1}\gamma_{1k}\gamma_{jk}=\sqrt{a_{11}}\,\frac{a_{j1}}{\sqrt{a_{11}}}=a_{1j}\quad(j\ge1):$$

the factor of the whole matrix.

When the construction fails. The identities of the last display do not use positive definiteness. By the same induction they hold whenever every pivot met on the way is positive, and the next paragraph extends them to zero pivots whose row and column vanish. When the construction

runs to its end in this way, the matrix is the product of the factor with its transpose, and then its quadratic form at a vector x is $\sum_k \left(\sum_i \gamma_{ik} x_i\right)^2 \geq 0$: the matrix is positive semidefinite (this is the computation of Remark 6.8; Theorem 6.4 gives the equivalent test on the eigenvalues). Consequently, for a matrix that is not positive semidefinite the construction must stop at some step: at a pivot that is negative (no real square root), or at a zero pivot whose row does not vanish (a division by zero that no zero column can repair). The unit-diagonal matrix

$$\begin{pmatrix} 1 & 1 & 0.5 \\ 1 & 1 & 0 \\ 0.5 & 0 & 1 \end{pmatrix}$$

is an example. Its determinant is -0.25, so it has a negative eigenvalue (Proposition 6.3), is not positive semidefinite (Theorem 6.4) and is not a correlation matrix: it makes the first two variables perfectly correlated, and yet gives them different correlations with the third. Its first column is $(1, 1, 0.5)$, the reduced block is

$$\begin{pmatrix} 1-1 & 0-0.5 \\ 0-0.5 & 1-0.25 \end{pmatrix} = \begin{pmatrix} 0 & -0.5 \\ -0.5 & 0.75 \end{pmatrix},$$

and the construction stops at a zero pivot whose row, $(0, -0.5)$, does not vanish.

Singular positive semidefinite matrices. A positive semidefinite matrix that is singular, such as a correlation matrix with an off-diagonal entry ± 1, can also reach a zero pivot. Every block that the construction reaches is again positive semidefinite. After a positive pivot, this is (I.15), whose right side is a value of the quadratic form of the previous block and therefore nonnegative. After a zero pivot, the block that remains is the principal submatrix obtained by deleting the first row and column, and its quadratic form is the form of the previous block at vectors whose first coordinate is 0, again nonnegative. In a positive semidefinite block every pivot is nonnegative, being the value of the form at the first unit vector, and a zero pivot forces the rest of its row and column to vanish: at the vector with first coordinate x, j-th coordinate 1 and zeros elsewhere, the quadratic form is $2a_{1j}x + a_{jj}$, which is negative for a suitable real x unless $a_{1j} = 0$. The corresponding column of the factor is then set to zero, and the construction continues on the block that remains. So, for a positive semidefinite matrix, every pivot met is nonnegative, a zero pivot comes with a vanishing row and column, and the construction runs to its end. The identities of the last display still hold: at a zero pivot the reduced row is zero, so the earlier columns of the factor already reproduce that row of the matrix, and the zero column adds nothing. The variable of a row with a zero pivot is $\sum_k \gamma_{ik} Z_k$ with k ranging over earlier columns only, so no normal is drawn for it. Among the earlier rows, those with positive pivots express the normals used so far through a lower-triangular system with a positive diagonal (the columns of zero pivots being zero, no other normal occurs); solving that system one row at a time writes these normals, and hence the variable of the current row, as a fixed linear combination of the earlier variables. □

I.8 Proof of Proposition 23.37: Monte Carlo error

The statement is on page 464. The notation and the standing assumptions are those of §23.5.

Proof. We compute the mean and the variance of a sample mean, apply the central limit theorem, and, for the interval with s, combine Chebyshev's inequality with a sandwich between two events.

(i). By linearity of expectation (Theorem 1.2), $\mathbb{E}\,\hat{V}_n = \frac{1}{n}\sum_{i=1}^n \mathbb{E}\,X_i = V$.

(ii). Expanding the variance of $X_1 + \cdots + X_n$ by Proposition 10.10, one term at a time, gives the sum of the n variances plus twice the sum of the covariances of the pairs. Every covariance vanishes: for $i \neq j$ the variables X_i and X_j are independent with finite means (a finite variance gives a finite mean, by the Cauchy–Schwarz inequality of Lemma 19.12(ii)), so $\mathbb{E}[X_i X_j] = \mathbb{E}\,X_i\,\mathbb{E}\,X_j$ by

the product rule for independent variables (Lemma 19.17), and $\operatorname{Cov}(X_i, X_j) = 0$; the rest is the expansion of Corollary 10.12. So $\operatorname{Var}\big(\sum_i X_i\big) = n\sigma_X^2$ and $\operatorname{Var}\hat{V}_n = n\sigma_X^2/n^2 = \sigma_X^2/n$. The identity $\mathbb{E}\, s^2 = \sigma_X^2$ is Theorem 4.5.

(iii), with σ_X. Fix $a > 0$. The event $\{|\hat{V}_n - V| \le a\,\sigma_X/\sqrt{n}\}$ says that the standardised error $\sqrt{n}\,(\hat{V}_n - V)/\sigma_X$ lies in $[-a, a]$. By Fact 4.8, the probability that the standardised error is at most y tends to $\Phi(y)$ for every real y. The probability that it is strictly below $-a$ lies between the probabilities that it is at most $-a - \varepsilon$ and at most $-a$, for every $\varepsilon > 0$; these tend to $\Phi(-a - \varepsilon)$ and $\Phi(-a)$, and Φ is continuous, so it tends to $\Phi(-a)$. Hence

$$P(|\hat{V}_n - V| \le a\,\sigma_X/\sqrt{n}) \to \Phi(a) - \Phi(-a) = 2\,\Phi(a) - 1 \qquad \text{for every } a > 0, \tag{I.16}$$

and $a = 1.96$ gives the first claim, with $2\,\Phi(1.96) - 1 = 0.950004$.

(iii), with s. We show that s/σ_X tends to 1 in probability and then sandwich the event between two events of the form (I.16). First, by Chebyshev's inequality (Lemma 19.12(i)) and part (ii), $P(|\hat{V}_n - V| \ge \varepsilon) \le \sigma_X^2/(n\varepsilon^2) \to 0$ for every $\varepsilon > 0$, so $\hat{V}_n \to V$ in probability. The squares $X_1^2, \dots, X_n^2$ are again independent and identically distributed, with mean $\mathbb{E}\, X_1^2$ and the finite variance $\mathbb{E}\, X_1^4 - (\mathbb{E}\, X_1^2)^2$, so the same argument, applied to them, shows that $\frac{1}{n}\sum_i X_i^2 \to \mathbb{E}\, X_1^2$ in probability. Next, expanding the square in the definition of s^2 and using $\sum_i X_i = n\hat{V}_n$,

$$s^2 = \frac{1}{n-1}\Big(\sum_{i=1}^{n} X_i^2 - n\hat{V}_n^2\Big) = \frac{n}{n-1}\Big(\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \hat{V}_n^2\Big).$$

Products and sums of sequences that converge in probability converge in probability to the products and sums of the limits (Lemma 19.12(iii)), and the numbers $n/(n-1)$ tend to 1; so $s^2 \to \mathbb{E}\, X_1^2 - V^2 = \sigma_X^2$ in probability. For $z \ge 0$, $|\sqrt{z} - 1| = |z - 1|/(\sqrt{z} + 1) \le |z - 1|$; with $z = s^2/\sigma_X^2$ this gives $|s/\sigma_X - 1| \le |s^2 - \sigma_X^2|/\sigma_X^2$, so $s/\sigma_X \to 1$ in probability. Finally, fix $0 < \varepsilon < 1$ and let A_n be the event $\{|\hat{V}_n - V| \le 1.96\,s/\sqrt{n}\}$ and A_n' the event $\{|s/\sigma_X - 1| > \varepsilon\}$, whose probability tends to 0. Outside A_n' the bound $1.96\,s/\sqrt{n}$ lies between $1.96(1-\varepsilon)\,\sigma_X/\sqrt{n}$ and $1.96(1+\varepsilon)\,\sigma_X/\sqrt{n}$, so

$$\{|\hat{V}_n - V| \le 1.96(1-\varepsilon)\,\sigma_X/\sqrt{n}\} \setminus A_n' \;\subseteq\; A_n \;\subseteq\; \{|\hat{V}_n - V| \le 1.96(1+\varepsilon)\,\sigma_X/\sqrt{n}\} \cup A_n'.$$

Taking probabilities and applying (I.16) with $a = 1.96(1 \mp \varepsilon)$, the lower limit of $P(A_n)$ is at least $2\,\Phi(1.96(1-\varepsilon)) - 1$ and the upper limit at most $2\,\Phi(1.96(1+\varepsilon)) - 1$. Letting $\varepsilon \to 0$ and using the continuity of Φ, $P(A_n) \to 2\,\Phi(1.96) - 1$. □

I.9 Proof of Proposition 23.43: Antithetic variates

The statement is on page 468. The notation and the standing assumptions are those of §23.6.

Proof. We compute the variance of a sum, and obtain the sign of the covariance by comparing two independent draws, first for one normal and then, by induction on the dimension, for a vector.

Step 1 (mean and variance: (i)). The vector $-Z$ has the law of Z. Indeed each $-Z_i$ is standard normal, because $P(-Z_i \le y) = P(Z_i \ge -y) = 1 - \Phi(-y) = \Phi(y)$ for every real y, and the variables $-Z_1, \dots, -Z_m$ are independent because $Z_1, \dots, Z_m$ are. Hence $f(-Z)$ has the law of $f(Z)$: the same mean and the same variance. By linearity of expectation (Theorem 1.2) the pair average has the mean $\frac{1}{2}(\mathbb{E}\, f(Z) + \mathbb{E}\, f(-Z)) = \mathbb{E}\, f(Z)$. By Proposition 10.10,

$$\begin{aligned}\operatorname{Var}\big(f(Z) + f(-Z)\big) &= \operatorname{Var} f(Z) + \operatorname{Var} f(-Z) + 2\operatorname{Cov}\big(f(Z), f(-Z)\big)\\ &= 2\operatorname{Var} f(Z) + 2\operatorname{Cov}\big(f(Z), f(-Z)\big),\end{aligned}$$

and dividing by 4 gives the first form of the variance. The two standard deviations are equal, so $\operatorname{Cov}(f(Z), f(-Z)) = \rho_A \operatorname{Var} f(Z)$, which gives the second form.

Step 2 (the comparison: (ii)). If Z' is independent of Z, then $f(Z)$ and $f(Z')$ are independent, their covariance is zero, and Proposition 10.10 gives $\operatorname{Var}\big(\frac{1}{2}(f(Z)+f(Z'))\big) = \frac{1}{4}\cdot 2\operatorname{Var} f(Z) = \operatorname{Var} f(Z)/2$. Since $\operatorname{Var} f(Z) > 0$, the pair variance $\frac{1}{2}\operatorname{Var} f(Z)\,(1+\rho_A)$ is smaller than this exactly when $\rho_A < 0$.

Step 3 (the sign for one normal). Let Z be a scalar and f nondecreasing. (If f is nonincreasing, apply what follows to $-f$, which is nondecreasing; this changes neither $\operatorname{Cov}(f(Z), f(-Z))$ nor ρ_A, because both factors change sign.) The mirror image $\tilde f(z) = f(-z)$ is then nonincreasing, so for all real z and z'

$$\big(f(z) - f(z')\big)\big(\tilde f(z) - \tilde f(z')\big) \le 0 :$$

if $z \ge z'$ the first factor is nonnegative and the second nonpositive, and if $z < z'$ the signs are exchanged. Evaluate this at $z = Z$ and $z' = Z'$, an independent copy, and take the mean. Each of the four products obtained by expanding has a finite mean, by the Cauchy–Schwarz inequality (Lemma 19.12(ii)) and $\mathbb{E}\, f(Z)^2 = \mathbb{E}\, \tilde f(Z)^2 < \infty$. The two products in which the same variable occurs twice have the same mean, $\mathbb{E}[f(Z)\tilde f(Z)]$, because Z' has the law of Z. The two mixed products factor by the product rule for independent variables (Lemma 19.17): $\mathbb{E}[f(Z)\tilde f(Z')] = \mathbb{E}[f(Z')\tilde f(Z)] = \mathbb{E}\, f(Z)\ \mathbb{E}\, \tilde f(Z)$. A variable that is never positive has a mean that is not positive, so

$$0 \ge \mathbb{E}\Big[\big(f(Z)-f(Z')\big)\big(\tilde f(Z)-\tilde f(Z')\big)\Big] = 2\,\mathbb{E}\big[f(Z)\tilde f(Z)\big] - 2\,\mathbb{E}\, f(Z)\ \mathbb{E}\, \tilde f(Z) = 2\operatorname{Cov}\big(f(Z), f(-Z)\big).$$

Dividing by $2\operatorname{Var} f(Z) > 0$ gives $\rho_A \le 0$.

Step 4 (an association inequality). We claim that, for every $m \ge 1$, if $Z = (Z_1, \dots, Z_m)$ has independent standard normal coordinates and ψ_1, ψ_2 are functions on $\mathbb{R}^m$ of exponential growth that are nondecreasing in each coordinate, then

$$\operatorname{Cov}\big(\psi_1(Z), \psi_2(Z)\big) \ge 0. \tag{I.17}$$

All the means involved are finite. If $|\psi_1(z)| \le c\, e^{c(|z_1|+\cdots+|z_m|)}$, then for every $k \ge 1$

$$\mathbb{E}\,|\psi_1(Z)|^k \le c^k\,\mathbb{E}\Big[\prod_{i=1}^m e^{kc|Z_i|}\Big] = c^k \prod_{i=1}^m \mathbb{E}\, e^{kc|Z_i|} \le c^k\big(2e^{k^2c^2/2}\big)^m < \infty,$$

by the product rule for independent variables (Lemma 19.17, applied $m-1$ times) and the bound $\mathbb{E}\, e^{c|Z|} \le 2e^{c^2/2}$ of Lemma 18.35(i). The same holds for ψ_2, and the product $\psi_1\psi_2$ again has exponential growth (Lemma 19.20(iii)), so $\psi_1(Z)\psi_2(Z)$ has a finite mean as well.

We prove (I.17) by induction on m. For $m = 1$ it is the argument of Step 3 with two nondecreasing functions: now $(\psi_1(z) - \psi_1(z'))(\psi_2(z) - \psi_2(z')) \ge 0$ for all real z, z', and the mean of this product at (Z, Z') is $2\operatorname{Cov}(\psi_1(Z), \psi_2(Z))$.

Let $m \ge 2$, and suppose that (I.17) holds in dimension $m-1$. Write $z = (z_1, \dots, z_{m-1})$ for the first $m-1$ coordinates of a point of $\mathbb{R}^m$ and y for the last one, and put

$$\bar\psi_1(z) = \mathbb{E}\,\psi_1(z, Z_m), \qquad \bar\psi_2(z) = \mathbb{E}\,\psi_2(z, Z_m).$$

(a) These means are finite at every z, and $\bar\psi_1$, $\bar\psi_2$ have exponential growth on $\mathbb{R}^{m-1}$:

$$|\bar\psi_1(z)| \le \mathbb{E}\,|\psi_1(z, Z_m)| \le c\, e^{c(|z_1|+\cdots+|z_{m-1}|)}\ \mathbb{E}\, e^{c|Z_m|} \le 2c\, e^{c^2/2}\, e^{c(|z_1|+\cdots+|z_{m-1}|)}.$$

(b) They are nondecreasing in each of their $m-1$ coordinates. If one coordinate of z is raised, $\psi_1(z, y)$ does not fall, for every y; so $\psi_1(z, Z_m)$ does not fall in any state, and its mean does not fall.

(c) At each fixed z, the functions $y \mapsto \psi_1(z, y)$ and $y \mapsto \psi_2(z, y)$ are nondecreasing and of exponential growth in the single variable y, with the constant $\max\big(c, c\, e^{c(|z_1|+\cdots+|z_{m-1}|)}\big)$. The case $m = 1$ therefore gives

$$\mathbb{E}\big[\psi_1(z, Z_m)\,\psi_2(z, Z_m)\big] \ge \bar\psi_1(z)\,\bar\psi_2(z).$$

Now apply Lemma 19.17, with the independent vectors $(Z_1, \dots, Z_{m-1})$ and Z_m in the roles of its two variables, to the function $\psi_1\psi_2$, which is integrable: the mean of $\psi_1(Z)\psi_2(Z)$ is the mean, at $z = (Z_1, \dots, Z_{m-1})$, of the left side of the inequality in (c). By (a) and Lemma 19.20(iii) the product $\bar\psi_1\bar\psi_2$ has exponential growth on $\mathbb{R}^{m-1}$, so its value at $(Z_1, \dots, Z_{m-1})$ has a finite mean (the moment bound above, in dimension $m-1$), and monotonicity of expectation applies to the inequality in (c). Therefore

$$\begin{aligned}\mathbb{E}[\psi_1(Z)\,\psi_2(Z)] &\ge \mathbb{E}[\bar\psi_1(Z_1, \dots, Z_{m-1})\;\bar\psi_2(Z_1, \dots, Z_{m-1})] \\ &\ge \mathbb{E}\,\bar\psi_1(Z_1, \dots, Z_{m-1})\;\mathbb{E}\,\bar\psi_2(Z_1, \dots, Z_{m-1}) = \mathbb{E}\,\psi_1(Z)\;\mathbb{E}\,\psi_2(Z).\end{aligned}$$

The second inequality is the induction hypothesis, which applies to $\bar\psi_1$ and $\bar\psi_2$ by (a) and (b). The final equality is Lemma 19.17 again, applied to ψ_1 and to ψ_2 separately. This is (I.17) in dimension m.

Step 5 (the sign for a vector). Let f have exponential growth and be nondecreasing in each coordinate. Put $\psi_1 = f$ and $\psi_2 = -\tilde f$, that is, $\psi_2(z) = -f(-z)$. Raising a coordinate of z lowers that coordinate of $-z$, which does not raise $f(-z)$ and so does not lower $-f(-z)$: the function ψ_2 is nondecreasing in each coordinate, and it has exponential growth with the constant of f. By (I.17), $\operatorname{Cov}(f(Z), -f(-Z)) \ge 0$, that is, $\operatorname{Cov}(f(Z), f(-Z)) \le 0$, and $\rho_A \le 0$. If f is nonincreasing in each coordinate, apply this to $-f$, as in Step 3.

Step 6 (the two path payoffs). In Procedure 23.36 the price at the grid date $k\,\Delta t$ is

$$S_{k\Delta t} = S\exp\Big(\big(b - \tfrac12\sigma^2\big)k\,\Delta t + \sigma\sqrt{\Delta t}\,(Z_1 + \cdots + Z_k)\Big), \qquad k = 1, \dots, J,$$

with $m = J$ normals. Each price is a nondecreasing function of each Z_i: it increases with Z_i for $i \le k$ and does not depend on it for $i > k$. The average of the fixings and the largest of the observed prices are nondecreasing in each price, and $x \mapsto (x-K)^+$ is nondecreasing; so the discounted payoffs $e^{-rT}(\bar S_T - K)^+$ and $e^{-rT}(G_T - K)^+$ of the arithmetic and geometric average-price calls (Definition 23.25, with the grid prices as the fixings) and $e^{-rT}(\max_k S_{k\Delta t} - K)^+$ are nondecreasing in each coordinate. Each of them lies between 0 and $e^{-rT}\max_k S_{k\Delta t}$, and

$$\max_k S_{k\Delta t} \le S\,e^{|b-\sigma^2/2|T}\,e^{\sigma\sqrt{\Delta t}\,(|Z_1|+\cdots+|Z_J|)},$$

which is exponential growth; this is the bound that Lemma 19.20(i) gives for every payoff of moderate growth in the prices. None of these payoffs is constant with probability one, so the second sentence of (iii) applies to them. □

I.10 Proof of Proposition 23.51: Stability and accuracy of the implicit and Crank–Nicolson schemes (Crank and Nicolson, 1947)

The statement is on page 477. The notation and the standing assumptions are those of §23.7.

Proof. We prove an energy estimate by summation by parts, deduce stability from it with the Cauchy–Schwarz inequality, compute the amplification factors by substituting a mode, bound the truncation errors by Taylor's theorem, and combine stability with the truncation errors (the Lax argument).

Step 1 (summation by parts: (i)). Let w vanish at the boundary nodes, and write $\Delta w_j = w_{j+1} - w_j$ for $j = -j_{\max}, \dots, j_{\max} - 1$, the differences over all the pairs of adjacent nodes. At an interior node, $w_{j+1} - 2w_j + w_{j-1} = \Delta w_j - \Delta w_{j-1}$, so

$$\sum_{|j|<j_{\max}} (w_{j+1} - 2w_j + w_{j-1})\,w_j = \sum_{j=-j_{\max}+1}^{j_{\max}-1} \Delta w_j\,w_j - \sum_{j=-j_{\max}+1}^{j_{\max}-1} \Delta w_{j-1}\,w_j.$$

In the first sum the missing term $j=-j_{\max}$ is $\Delta w_{-j_{\max}}\,w_{-j_{\max}}=0$, so the sum may start at $-j_{\max}$. In the second sum put $k=j-1$, which runs from $-j_{\max}$ to $j_{\max}-2$; the missing term $k=j_{\max}-1$ is $\Delta w_{j_{\max}-1}\,w_{j_{\max}}=0$, so the sum may end at $j_{\max}-1$. Hence

$$\sum_{|j|<j_{\max}} (w_{j+1}-2w_j+w_{j-1})\,w_j = \sum_{k=-j_{\max}}^{j_{\max}-1} \Delta w_k\,(w_k-w_{k+1}) = -\sum_{k=-j_{\max}}^{j_{\max}-1} (\Delta w_k)^2.$$

For the first differences,

$$\sum_{|j|<j_{\max}} (w_{j+1}-w_{j-1})\,w_j = \sum_{j=-j_{\max}+1}^{j_{\max}-1} w_{j+1}w_j - \sum_{j=-j_{\max}+1}^{j_{\max}-1} w_j w_{j-1}.$$

The last term of the first sum, $w_{j_{\max}}w_{j_{\max}-1}$, and the first term of the second, $w_{-j_{\max}+1}w_{-j_{\max}}$, vanish. What is left of the second sum is $\sum_{j=-j_{\max}+2}^{j_{\max}-1} w_j w_{j-1}$, which, with $k=j-1$, is $\sum_{k=-j_{\max}+1}^{j_{\max}-2} w_{k+1}w_k$: exactly what is left of the first. The difference is zero: the centred first difference is antisymmetric and contributes nothing. Multiplying the two results by $\Delta x\,\sigma^2/(2\,\Delta x^2)$ and $\Delta x\,\nu/(2\,\Delta x)$, and subtracting $r\langle w,w\rangle$, gives (23.19). It is nonpositive because $r\ge 0$.

Step 2 (stability: (ii)). Let $M=I-\Delta t\,A$, a square matrix acting on grid vectors that vanish at the boundary. By (i),

$$\langle Mw,w\rangle = \|w\|^2 - \Delta t\,\langle Aw,w\rangle \ge \|w\|^2.$$

The Cauchy–Schwarz inequality for the weighted inner product (Lemma 11.6(ii) with the positive-definite matrix $\Delta x\,I$), $\langle w,z\rangle^2 \le \|w\|^2\|z\|^2$, gives $\langle Mw,w\rangle \le \|Mw\|\,\|w\|$. Hence $\|Mw\|\,\|w\| \ge \|w\|^2$, and

$$\|Mw\| \ge \|w\| \qquad \text{for every such } w. \tag{I.18}$$

The same holds for the transpose: $\langle M^{\top}w,w\rangle = \langle w,Mw\rangle = \langle Mw,w\rangle \ge \|w\|^2$, so $\|M^{\top}w\|\ge\|w\|$. The matrix $MM^{\top}$ is symmetric, and its quadratic form at w is the squared Euclidean length of $M^{\top}w$, which is positive for $w\ne 0$ by what was just shown; so $MM^{\top}$ is positive definite, and it is invertible by Lemma 11.6(i). Consequently, for every z the vector $w=M^{\top}(MM^{\top})^{-1}z$ solves $Mw=z$. The solution is unique, because $Mw=Mw'$ implies $\|w-w'\|\le\|M(w-w')\|=0$ by (I.18); and it satisfies $\|w\|\le\|Mw\|=\|z\|$. So M is invertible and its inverse does not increase the norm.

Implicit scheme. In (23.17), split V^i into its interior part, which vanishes at the boundary, and its boundary part, which is known. By linearity the step reads $M(\text{interior part of } V^i) = V^{i+1} + \Delta t\,A(\text{boundary part of } V^i)$ on the interior nodes, a system with the invertible matrix M: it has exactly one solution. For two solutions with the same boundary values the boundary parts cancel, the difference w^i vanishes at the boundary and $Mw^i=w^{i+1}$ on the interior nodes; by (I.18), $\|w^i\|\le\|w^{i+1}\|$.

Crank–Nicolson scheme. The matrix $I-\frac12\Delta t\,A$ is M with $\Delta t/2$ in the place of Δt, so it is invertible and (23.18) has exactly one solution, by the same splitting. For the difference of two solutions with the same boundary values, write $z=w^i$ and $w=w^{i+1}$. Then $(I-\frac12\Delta t\,A)z=(I+\frac12\Delta t\,A)w$, that is, $z-w=\frac12\Delta t\,A(z+w)$. Take the inner product with $z+w$, which vanishes at the boundary. The left side gives $\langle z-w,z+w\rangle=\|z\|^2-\|w\|^2$, the mixed terms cancelling by symmetry, and the right side gives $\frac12\Delta t\,\langle A(z+w),z+w\rangle\le 0$, by (i). So $\|z\|\le\|w\|$.

Maximum norm for the implicit step. Let $|\nu|\,\Delta x\le\sigma^2$. At an interior node j,

$$(Mw)_j = \Big(1+\Delta t\Big(\frac{\sigma^2}{\Delta x^2}+r\Big)\Big)w_j - \Delta t\Big(\frac{\sigma^2}{2\,\Delta x^2}+\frac{\nu}{2\,\Delta x}\Big)w_{j+1} - \Delta t\Big(\frac{\sigma^2}{2\,\Delta x^2}-\frac{\nu}{2\,\Delta x}\Big)w_{j-1}.$$

The two coefficients $\frac12\sigma^2/\Delta x^2 \pm \frac12\nu/\Delta x$ are nonnegative, by the hypothesis, and their sum is $\sigma^2/\Delta x^2$. Choose an interior node j at which $|w_j|$ is largest. By the triangle inequality,

$$|(Mw)_j| \ge \Big(1+\Delta t\,\frac{\sigma^2}{\Delta x^2}+r\,\Delta t\Big)|w_j| - \Delta t\,\frac{\sigma^2}{\Delta x^2}\,|w_j| = (1+r\,\Delta t)\,|w_j|,$$

because $|w_{j\pm1}| \le |w_j|$ (a neighbour that is a boundary node has the value zero). With $w = w^i$ and $Mw^i = w^{i+1}$ this gives $\max_j |w_j^{i+1}| \ge (1 + r\,\Delta t)\max_j |w_j^i|$.

Step 3 (amplification factors: (iii)). For the mode $w_j = e^{\mathrm{i}j\omega}$, the product rule $e^{\mathrm{i}(j\pm1)\omega} = e^{\pm\mathrm{i}\omega}e^{\mathrm{i}j\omega}$ and Euler's formula give

$$w_{j+1} - 2w_j + w_{j-1} = (e^{\mathrm{i}\omega} - 2 + e^{-\mathrm{i}\omega})w_j = (2\cos\omega - 2)\,w_j,$$
$$w_{j+1} - w_{j-1} = (e^{\mathrm{i}\omega} - e^{-\mathrm{i}\omega})w_j = 2\,\mathrm{i}\sin\omega\; w_j.$$

Substituting in the definition of A_0,

$$(A_0 w)_j = \Big[\frac{\sigma^2}{2\,\Delta x^2}\,(2\cos\omega - 2) + \frac{\nu}{2\,\Delta x}\,2\,\mathrm{i}\sin\omega\Big]w_j = \Lambda_0(\omega)\,w_j,$$

and $(Aw)_j = (\Lambda_0(\omega) - r)\,w_j = \Lambda(\omega)\,w_j$. Since $1 - \cos\omega \ge 0$, $\mathrm{Re}\,\Lambda_0 \le 0$ and $\mathrm{Re}\,\Lambda \le -r \le 0$. If $V^{i+1} = w$, the explicit scheme gives $(1 + r\,\Delta t)V^i = (1 + \Delta t\,\Lambda_0)\,w$. For the other two schemes, try V^i equal to a multiple of w: the implicit scheme requires $(1 - \Delta t\,\Lambda)V^i = w$ and the Crank–Nicolson scheme $(1 - \frac{1}{2}\Delta t\,\Lambda)V^i = (1 + \frac{1}{2}\Delta t\,\Lambda)\,w$. These give the three factors, the divisions being allowed by the bounds that follow, and the multiples of w so obtained are bounded.

For the implicit factor, $|1 - \Delta t\,\Lambda| \ge \mathrm{Re}(1 - \Delta t\,\Lambda) = 1 - \Delta t\,\mathrm{Re}\,\Lambda \ge 1$, so its modulus is at most one. For the Crank–Nicolson factor, note that for real numbers a and a'

$$|1 + a + \mathrm{i}a'|^2 - |1 - a - \mathrm{i}a'|^2 = (1 + a)^2 + a'^2 - (1 - a)^2 - a'^2 = 4a;$$

with $a + \mathrm{i}a' = \frac{1}{2}\Delta t\,\Lambda$, whose real part a is at most zero, the numerator has a modulus that is at most that of the denominator, and the denominator has a modulus that is at least $1 - a \ge 1$. At $\omega = \pi$, $\cos\omega = -1$ and $\sin\omega = 0$, so $\Lambda_0(\pi) = -2\sigma^2/\Delta x^2$ and the explicit factor is $(1 - 2\sigma^2\,\Delta t/\Delta x^2)/(1 + r\,\Delta t)$. It is below -1 if and only if $1 - 2\sigma^2\,\Delta t/\Delta x^2 < -(1 + r\,\Delta t)$, that is, $\sigma^2\,\Delta t/\Delta x^2 > 1 + \frac{1}{2}r\,\Delta t$. The mode at $\omega = \pi$ is $e^{\mathrm{i}j\pi} = (-1)^j$, the sawtooth.

The only bounded solution. The difference of two bounded solutions of one implicit or Crank–Nicolson step from the same V^{i+1} is a bounded vector z on the infinite grid with $(I - \Delta t\,A)z = 0$, or with $(I - \frac{1}{2}\Delta t\,A)z = 0$; we show that such a z is zero. In terms of the numbers p_u and p_d of (23.16), which here need not be nonnegative and whose sum is $\sigma^2\,\Delta t/\Delta x^2$, the two equations read (the second after multiplication by 2)

$$D\,z_j = p_u\,z_{j+1} + p_d\,z_{j-1} \qquad \text{for every integer } j,$$

where $D = 1 + \sigma^2\,\Delta t/\Delta x^2 + r\,\Delta t$ for the implicit step and $D = 2 + \sigma^2\,\Delta t/\Delta x^2 + r\,\Delta t$ for the Crank–Nicolson step (the letter D is local to this paragraph); in both cases $D > p_u + p_d$. If $p_u \ge 0$ and $p_d \ge 0$, the triangle inequality gives $D\,|z_j| \le (p_u + p_d)\sup_k |z_k|$ for every j; taking the supremum over j, $D\sup_k |z_k| \le (p_u + p_d)\sup_k |z_k|$, which forces $\sup_k |z_k| = 0$. Otherwise one of the two numbers is negative and the other positive, their sum being positive. The quadratic $p_u y^2 - Dy + p_d$ then has two real roots y_1 and y_2, its discriminant $D^2 - 4p_u p_d$ being positive. Their product p_d/p_u is negative, so they have opposite signs; their sum D/p_u is not zero, so their moduli differ; and neither has modulus one, because the quadratic equals $p_u + p_d - D < 0$ at $y = 1$ and $p_u + p_d + D > 0$ at $y = -1$. Since $p_u \ne 0$ and $p_d \ne 0$, the recursion determines z on all the integers from z_0 and z_1, forward and backward, so its solutions form a space of dimension two. The solutions y_1^j and y_2^j span that space, because their values at $j = 0, 1$ are $(1, y_1)$ and $(1, y_2)$, with $y_1 \ne y_2$. Write $z_j = a_1 y_1^j + a_2 y_2^j$, with complex numbers a_1 and a_2, and number the roots so that $|y_1| < |y_2|$. If $|y_2| > 1$, then $z_j/y_2^j = a_1(y_1/y_2)^j + a_2$ tends to a_2 as $j \to \infty$, and also to 0, because z is bounded; so $a_2 = 0$, and $a_1 y_1^j$, with $0 < |y_1| \ne 1$, is unbounded as $j \to \infty$ or as $j \to -\infty$ unless $a_1 = 0$. If $|y_2| < 1$, then $z_j/y_1^j = a_1 + a_2(y_2/y_1)^j$ tends to a_1 and to 0 as $j \to -\infty$; so $a_1 = 0$, and $a_2 y_2^j$ is unbounded as $j \to -\infty$ unless $a_2 = 0$. In every case $z = 0$.

Step 4 (truncation errors: the first half of (iv)). The local truncation errors are defined on the interior nodes by

$$\begin{aligned}
\Delta t\,\varepsilon^i &= (1 + r\,\Delta t)\,\tilde V^i - (I + \Delta t\,A_0)\,\tilde V^{i+1} && \text{(explicit)},\\
\Delta t\,\varepsilon^i &= (I - \Delta t\,A)\,\tilde V^i - \tilde V^{i+1} && \text{(implicit)},\\
\Delta t\,\varepsilon^i &= (I - \tfrac12\Delta t\,A)\,\tilde V^i - (I + \tfrac12\Delta t\,A)\,\tilde V^{i+1} && \text{(Crank–Nicolson)}.
\end{aligned}$$

Let c' be a common bound, on the rectangle, for the absolute values of the derivatives of the hypothesis. Every expansion below is Taylor's theorem in one variable with the Lagrange form of the remainder (Theorem 7.31 and Corollary 7.32), applied to a function χ that has the required number of continuous derivatives because $2k + m \le 6$ for each derivative that occurs; the intermediate points are starred.

In space. Fix a time $t \in [0, T]$ and an interior node x_j, and let $\chi(x) = \tilde V(x, t)$. Expanding to the third order with a fourth-order remainder,

$$\chi(x_j \pm \Delta x) = \chi(x_j) \pm \Delta x\,\chi'(x_j) + \tfrac12\Delta x^2\,\chi''(x_j) \pm \tfrac16\Delta x^3\,\chi'''(x_j) + \tfrac{1}{24}\Delta x^4\,\chi''''(x_\pm^*),$$

with x_+^* between x_j and $x_j + \Delta x$ and x_-^* between $x_j - \Delta x$ and x_j. Adding the two expansions, the odd terms cancel:

$$\frac{\chi(x_j + \Delta x) - 2\chi(x_j) + \chi(x_j - \Delta x)}{\Delta x^2} = \chi''(x_j) + \frac{\Delta x^2}{24}\big(\chi''''(x_+^*) + \chi''''(x_-^*)\big).$$

Expanding to the second order with a third-order remainder and subtracting, the even terms cancel:

$$\frac{\chi(x_j + \Delta x) - \chi(x_j - \Delta x)}{2\,\Delta x} = \chi'(x_j) + \frac{\Delta x^2}{12}\big(\chi'''(x_+^{**}) + \chi'''(x_-^{**})\big),$$

with intermediate points $x_\pm^{**}$ of the same kind. The derivatives of χ are $\partial_x^m \tilde V(\cdot, t)$, $m \le 4$, so the two remainders are at most $\frac{1}{12}c'\,\Delta x^2$ and $\frac16 c'\,\Delta x^2$ in absolute value. Hence, for the grid vector $\tilde V(t)$ of the values $\tilde V(x_j, t)$,

$$\Big|\big(A_0\tilde V(t)\big)_j - \big(\tfrac12\sigma^2\,\partial_{xx}\tilde V + \nu\,\partial_x\tilde V\big)(x_j, t)\Big| \le c''\,\Delta x^2, \qquad c'' = \Big(\frac{\sigma^2}{24} + \frac{|\nu|}{6}\Big)c', \tag{I.19}$$

at every interior node and every time; the same bound holds with A on the left and the term $-r\tilde V$ added on the right, because A and A_0 differ by exactly that term.

In time, implicit scheme. Let $\chi(t) = \tilde V(x_j, t)$. Then $\chi(t_{i+1}) = \chi(t_i) + \Delta t\,\chi'(t_i) + \frac12\Delta t^2\,\chi''(t^*)$, so $(\tilde V_j^{i+1} - \tilde V_j^i)/\Delta t = \partial_t\tilde V(x_j, t_i) + \frac12\Delta t\,\partial_{tt}\tilde V(x_j, t^*)$. By the definition of ε^i and (I.19) at $t = t_i$,

$$-\varepsilon_j^i = \frac{\tilde V_j^{i+1} - \tilde V_j^i}{\Delta t} + \big(A\tilde V^i\big)_j = \big(\partial_t\tilde V + \tfrac12\sigma^2\,\partial_{xx}\tilde V + \nu\,\partial_x\tilde V - r\tilde V\big)(x_j, t_i) + \text{remainders},$$

where the bracket vanishes by (23.15) and the remainders are at most $\frac12 c'\,\Delta t + c''\,\Delta x^2$ in absolute value.

In time, explicit scheme. Expand about t_{i+1}: $\chi(t_i) = \chi(t_{i+1}) - \Delta t\,\chi'(t_{i+1}) + \frac12\Delta t^2\,\chi''(t^*)$, so $(\tilde V_j^{i+1} - \tilde V_j^i)/\Delta t = \partial_t\tilde V(x_j, t_{i+1}) - \frac12\Delta t\,\partial_{tt}\tilde V(x_j, t^*)$; and, by the mean value theorem (Corollary 7.32 at order zero), $\tilde V_j^i = \tilde V_j^{i+1} - \Delta t\,\partial_t\tilde V(x_j, t^{**})$. Therefore

$$-\varepsilon_j^i = \frac{\tilde V_j^{i+1} - \tilde V_j^i}{\Delta t} + \big(A_0\tilde V^{i+1}\big)_j - r\,\tilde V_j^i = \big(\partial_t\tilde V + \tfrac12\sigma^2\,\partial_{xx}\tilde V + \nu\,\partial_x\tilde V - r\tilde V\big)(x_j, t_{i+1}) + \text{remainders},$$

where the bracket vanishes and the remainders are at most $(\frac12 + r)\,c'\,\Delta t + c''\,\Delta x^2$.

In time, Crank–Nicolson scheme. Expand about the midpoint $t_{i+1/2} = t_i + \frac12\Delta t$. With $\chi(t) = \tilde V(x_j, t)$, to the second order with a third-order remainder,

$$\chi(t_{i+1/2} \pm \tfrac12\Delta t) = \chi \pm \tfrac12\Delta t\,\chi' + \tfrac18\Delta t^2\,\chi'' \pm \tfrac{1}{48}\Delta t^3\,\chi'''(t_\pm^*),$$

the functions without argument being evaluated at the midpoint. Subtracting, $(\tilde V_j^{i+1}-\tilde V_j^i)/\Delta t = \partial_t\tilde V(x_j,t_{i+1/2}) + \frac{1}{48}\Delta t^2\,(\chi'''(t_+^*)+\chi'''(t_-^*))$, with a remainder of at most $\frac{1}{24}c'\,\Delta t^2$. Next let $\chi(t) = (\frac12\sigma^2\,\partial_{xx}\tilde V + \nu\,\partial_x\tilde V - r\tilde V)(x_j,t)$, which is twice continuously differentiable in t, with $|\chi''| \le (\frac12\sigma^2+|\nu|+r)\,c'$, because $\partial_{tt}\partial_{xx}\tilde V$, $\partial_{tt}\partial_x\tilde V$ and $\partial_{tt}\tilde V$ are among the derivatives of the hypothesis. Expanding to the first order with a second-order remainder,

$$\tfrac12\big(\chi(t_i)+\chi(t_{i+1})\big) = \chi(t_{i+1/2}) + \tfrac{1}{16}\Delta t^2\,(\chi''(t_+^{**})+\chi''(t_-^{**})),$$

with a remainder of at most $\frac18(\frac12\sigma^2+|\nu|+r)\,c'\,\Delta t^2$. By (I.19), $\frac12(A\tilde V^i)_j + \frac12(A\tilde V^{i+1})_j$ differs from $\frac12(\chi(t_i)+\chi(t_{i+1}))$ by at most $c''\,\Delta x^2$. Altogether

$$\begin{aligned} -\varepsilon_j^i &= \frac{\tilde V_j^{i+1}-\tilde V_j^i}{\Delta t} + \tfrac12\big(A\tilde V^i\big)_j + \tfrac12\big(A\tilde V^{i+1}\big)_j \\ &= \big(\partial_t\tilde V + \tfrac12\sigma^2\,\partial_{xx}\tilde V + \nu\,\partial_x\tilde V - r\tilde V\big)(x_j,t_{i+1/2}) + \text{remainders}, \end{aligned}$$

where the bracket vanishes and the remainders are at most $\big(\frac{1}{24}+\frac18(\frac12\sigma^2+|\nu|+r)\big)c'\,\Delta t^2 + c''\,\Delta x^2$. In the three cases the bound has the stated form, with c the largest of the coefficients that occur.

Step 5 (convergence, the Lax argument: the second half of (iv)). The global error $\mathcal{E}^i = \tilde V^i - V^i$ vanishes at the boundary nodes, where $\tilde V$ takes the values of the scheme, and on the terminal row: $\mathcal{E}^J = 0$. Subtract the scheme from the definition of the truncation error. The boundary parts cancel, and on the interior nodes

$$\begin{aligned} (I-\Delta t\,A)\,\mathcal{E}^i &= \mathcal{E}^{i+1} + \Delta t\,\varepsilon^i && \text{(implicit)},\\ (I-\tfrac12\Delta t\,A)\,\mathcal{E}^i &= (I+\tfrac12\Delta t\,A)\,\mathcal{E}^{i+1} + \Delta t\,\varepsilon^i && \text{(Crank–Nicolson)}. \end{aligned}$$

The norm obeys the triangle inequality, $\|w+z\| \le \|w\|+\|z\|$, because $\|w+z\|^2 = \|w\|^2 + 2\langle w,z\rangle + \|z\|^2 \le (\|w\|+\|z\|)^2$ by the Cauchy–Schwarz inequality. For the implicit scheme, Step 2 (the inverse of M does not increase the norm) gives

$$\|\mathcal{E}^i\| \le \|\mathcal{E}^{i+1} + \Delta t\,\varepsilon^i\| \le \|\mathcal{E}^{i+1}\| + \Delta t\,\|\varepsilon^i\|.$$

For the Crank–Nicolson scheme, $\mathcal{E}^i$ is the sum of the solution z of $(I-\frac12\Delta t\,A)z = (I+\frac12\Delta t\,A)\mathcal{E}^{i+1}$, for which $\|z\| \le \|\mathcal{E}^{i+1}\|$ by Step 2, and of $(I-\frac12\Delta t\,A)^{-1}\Delta t\,\varepsilon^i$, whose norm is at most $\Delta t\,\|\varepsilon^i\|$; so the same inequality holds. Adding it over $i = 0,\dots,J-1$, with $\mathcal{E}^J = 0$ and $J\,\Delta t = T$,

$$\|\mathcal{E}^0\| \le \Delta t\sum_{i=0}^{J-1}\|\varepsilon^i\| \le T\,\max_i\|\varepsilon^i\|, \qquad \|\varepsilon^i\| = \Big(\Delta x\sum_{|j|<j_{\max}}(\varepsilon_j^i)^2\Big)^{1/2} \le (2j_{\max}\,\Delta x)^{1/2}\,\max_j|\varepsilon_j^i|,$$

there being fewer than $2j_{\max}$ interior nodes. For the bound at a node, $\|\mathcal{E}^0\|^2 = \Delta x\sum_k(\mathcal{E}_k^0)^2 \ge \Delta x\,(\mathcal{E}_j^0)^2$.

For the explicit scheme under the positivity condition, the error satisfies

$$(1+r\,\Delta t)\,\mathcal{E}_j^i = p_u\,\mathcal{E}_{j+1}^{i+1} + p_m\,\mathcal{E}_j^{i+1} + p_d\,\mathcal{E}_{j-1}^{i+1} + \Delta t\,\varepsilon_j^i$$

with nonnegative weights that sum to one (Proposition 23.48(i)–(ii)). As in the proof of part (iii) of that proposition, and since $1+r\,\Delta t \ge 1$,

$$\max_j|\mathcal{E}_j^i| \le \max_j|\mathcal{E}_j^{i+1}| + \Delta t\,\max_j|\varepsilon_j^i|,$$

and adding over i gives $\max_j|\mathcal{E}_j^0| \le T\max_{i,j}|\varepsilon_j^i|$. For the implicit scheme with $|\nu|\,\Delta x \le \sigma^2$, the last clause of Step 2, applied to $w = \mathcal{E}^i$ with $M\mathcal{E}^i = \mathcal{E}^{i+1} + \Delta t\,\varepsilon^i$, gives $(1+r\,\Delta t)\max_j|\mathcal{E}_j^i| \le \max_j|\mathcal{E}_j^{i+1}| + \Delta t\max_j|\varepsilon_j^i|$, and the same recursion follows. □

I.11 Proof of Proposition 23.56: Dupire's formula from the forward equation (Dupire, 1994)

The statement is on page 481. The notation and the standing assumptions are those of §23.8.

Proof. We differentiate the call price under the integral, with an explicit bound, and then integrate by parts twice. Fix $K > 0$ and an expiry T of the interval, and write

$$\chi(x) = \sigma_{\mathrm{loc}}(x,T)^2\, x^2\, f_{\mathbb{Q}}(x,T), \qquad x > 0,$$

which is twice continuously differentiable by (a); in this proof (a)–(d) are the hypotheses of Proposition 23.56, not the items of the Setup of §23.8.

Step 1 (the derivative in T). Let ε and ψ be those of (c) at T, and let $0 < |\Delta T| \le \varepsilon$. By (d) the integrals that define $C(K,T)$ and $C(K,T+\Delta T)$ converge, so

$$\frac{C(K,T+\Delta T) - C(K,T)}{\Delta T} = \int_K^\infty (x-K)\, \frac{f_{\mathbb{Q}}(x,T+\Delta T) - f_{\mathbb{Q}}(x,T)}{\Delta T}\, dx.$$

For each fixed x, Corollary 7.32 applied to the function $s \mapsto f_{\mathbb{Q}}(x,s)$, to the first order with a second-order remainder, gives a point T^* between T and $T+\Delta T$ with

$$f_{\mathbb{Q}}(x,T+\Delta T) - f_{\mathbb{Q}}(x,T) - \Delta T\, \partial_T f_{\mathbb{Q}}(x,T) = \tfrac{1}{2}\, \Delta T^2\, \partial_{TT} f_{\mathbb{Q}}(x,T^*),$$

and therefore, by (c),

$$\Big| \frac{f_{\mathbb{Q}}(x,T+\Delta T) - f_{\mathbb{Q}}(x,T)}{\Delta T} - \partial_T f_{\mathbb{Q}}(x,T) \Big| \le \tfrac{1}{2}\, |\Delta T|\, \psi(x) \qquad \text{for every } x > 0. \tag{I.20}$$

The integral $\int_K^\infty (x-K)\, \partial_T f_{\mathbb{Q}}(x,T)\, dx$ converges absolutely. Indeed its integrand is continuous in x, by (a), and by (I.20) with $\Delta T = \varepsilon$ its absolute value is at most

$$(x-K)\, \frac{f_{\mathbb{Q}}(x,T+\varepsilon) + f_{\mathbb{Q}}(x,T)}{\varepsilon} + \tfrac{1}{2}\, \varepsilon\, (x-K)\, \psi(x),$$

whose integral over (K,∞) is finite by (d) and by (c), since $0 \le x - K \le 1 + x$ there. Multiplying (I.20) by $x - K \ge 0$ and integrating over (K,∞),

$$\Big| \frac{C(K,T+\Delta T) - C(K,T)}{\Delta T} - \int_K^\infty (x-K)\, \partial_T f_{\mathbb{Q}}(x,T)\, dx \Big| \le \tfrac{1}{2}\, |\Delta T| \int_K^\infty (1+x)\, \psi(x)\, dx,$$

which tends to 0 with ΔT. So C is differentiable in T, and by the forward equation (a)

$$\partial_T C(K,T) = \int_K^\infty (x-K)\, \partial_T f_{\mathbb{Q}}(x,T)\, dx = \frac{1}{2} \int_K^\infty (x-K)\, \chi''(x)\, dx. \tag{I.21}$$

Step 2 (first integration by parts). Let $y > K$. Integrating by parts on $[K,y]$,

$$\int_K^y (x-K)\, \chi''(x)\, dx = \Big[(x-K)\, \chi'(x) \Big]_{x=K}^{x=y} - \int_K^y \chi'(x)\, dx = (y-K)\, \chi'(y) - \int_K^y \chi'(x)\, dx.$$

The bracket vanishes at $x = K$. As $y \to \infty$ it tends to zero, because $0 \le y - K \le y$ and $y\, \chi'(y) \to 0$ by (b).

Step 3 (second integration: the fundamental theorem of calculus). The remaining integral is $-\int_K^y \chi'(x)\, dx = \chi(K) - \chi(y)$, and $\chi(y) \to 0$ as $y \to \infty$, by (b). With Step 2,

$$\int_K^y (x-K)\, \chi''(x)\, dx \;\to\; \chi(K) = \sigma_{\mathrm{loc}}(K,T)^2\, K^2\, f_{\mathbb{Q}}(K,T) \qquad \text{as } y \to \infty.$$

The integral over (K,∞) in (I.21) converges, so it is this limit, and

$$\partial_T C(K,T) = \tfrac{1}{2}\,\sigma_{\mathrm{loc}}(K,T)^2\,K^2\,f_{\mathbb{Q}}(K,T).$$

Step 4 (the density is the second strike derivative). For the fixed expiry T, the proof of (22.3) in Theorem 22.7 uses from its Setup only that the density $f_{\mathbb{Q}}(\cdot,T)$ is continuous, that $\mathbb{E}^{\mathbb{Q}}\,S_T$ is finite and that $C(K,T)$ is the integral of $(x-K)^+$ against the density; the pricing of other claims enters only its digital identity. All three hold here with $r=0$: continuity by hypothesis, the finite mean by (d), and the integral by the definition of C. By (22.3) the call price is twice differentiable in K and $f_{\mathbb{Q}}(K,T) = \partial_{KK}C(K,T)$. Substituting gives the equation of the statement. Where $\partial_{KK}C(K,T) > 0$ we may divide by $\frac{1}{2}K^2\,\partial_{KK}C$, which is (23.21); its right side is computed from the call surface alone, so two local variances that are compatible with the same surface agree at every such point. □

I.12 Proof of Proposition 23.60: Merton's jump-diffusion price (Merton, 1976)

The statement is on page 484. The notation and the standing assumptions are those of §23.8.

Proof. We condition on the number of jumps. For $j = 0,1,2,\dots$ let

$$X_j = (r - q - \lambda\kappa - \tfrac{1}{2}\sigma^2)T + \sigma\sqrt{T}\,Z_0 + \sum_{i=1}^{j}\big(\ln(1+\kappa) - \tfrac{1}{2}\delta^2 + \delta Z_i\big)$$

be the exponent of (23.22) with exactly j jumps, so that $S_T = Se^{X_j}$ on the event $\{N_T = j\}$.

Step 1 (the law given j jumps). The variable X_j is a constant plus a sum of the $j+1$ independent normal variables $\sigma\sqrt{T}\,Z_0$ and $\delta Z_1,\dots,\delta Z_j$. Adding them one at a time, each sum of two independent normal variables being normal with the means and the variances added (Lemma 19.38(ii)), shows that X_j is normal, with the variance $\sigma^2T + j\,\delta^2 = \sigma_{(j)}^2 T$ and the mean

$$(r - q - \lambda\kappa - \tfrac{1}{2}\sigma^2)T + j\,\big(\ln(1+\kappa) - \tfrac{1}{2}\delta^2\big) = (r_{(j)} - q - \tfrac{1}{2}\sigma_{(j)}^2)T,$$

because $r_{(j)}T = (r-\lambda\kappa)T + j\ln(1+\kappa)$ and $\frac{1}{2}\sigma_{(j)}^2 T = \frac{1}{2}\sigma^2 T + \frac{1}{2}j\,\delta^2$. So Se^{X_j} has the law of $S\exp\big((r_{(j)} - q - \frac{1}{2}\sigma_{(j)}^2)T + \sigma_{(j)}\sqrt{T}\,Z\big)$, with Z standard normal: the lognormal law of the BSM model with the rate $r_{(j)}$, the volatility $\sigma_{(j)}$ and the yield q. Note that $\sigma_{(j)} \geq \sigma > 0$.

Step 2 (freezing the count). Fix j. The count N_T is independent of the vector $(Z_0,\dots,Z_j)$, of which X_j is a function, and $f(Se^{X_j})$ has a finite mean, because f has moderate growth and X_j is normal (Lemma 18.35(i)). The product rule for independent variables (Lemma 19.17) and the Poisson probabilities of Definition 1.9 give

$$\begin{aligned}
\mathbb{E}^{\mathbb{Q}}\big[f(S_T)\,\mathbf{1}\{N_T = j\}\big] &= \mathbb{E}\big[\mathbf{1}\{N_T = j\}\,f(Se^{X_j})\big] = P(N_T = j)\,\mathbb{E}\,f(Se^{X_j}) \\
&= e^{-\lambda T}\,\frac{(\lambda T)^j}{j!}\,\mathbb{E}\,f\Big(S\exp\big((r_{(j)} - q - \tfrac{1}{2}\sigma_{(j)}^2)T + \sigma_{(j)}\sqrt{T}\,Z\big)\Big) \\
&= e^{-\lambda T}\,\frac{(\lambda T)^j}{j!}\,e^{r_{(j)}T}\,V^{\mathrm{BSM}}\big(S,0;\sigma_{(j)},r_{(j)}\big).
\end{aligned}$$

The second line uses Step 1: the mean of $f(Se^{X_j})$ depends only on the law of X_j. The last line is the definition of the price function in Theorem 20.7, in the yield form of Corollary 20.41(i), with the rate $r_{(j)}$ and the volatility $\sigma_{(j)} > 0$: that function is $e^{-r_{(j)}T}$ times the mean in the second line.

Step 3 (the discount factors). Since $e^{r_{(j)}T} = e^{rT}\,e^{-\lambda\kappa T}\,(1+\kappa)^j$,

$$e^{-rT}\cdot e^{-\lambda T}\,\frac{(\lambda T)^j}{j!}\cdot e^{r_{(j)}T} = e^{-\lambda(1+\kappa)T}\,\frac{\big(\lambda T\,(1+\kappa)\big)^j}{j!} = e^{-\lambda' T}\,\frac{(\lambda' T)^j}{j!}.$$

So $e^{-rT}\,\mathbb{E}^{\mathbb{Q}}[f(S_T)\,\mathbf{1}\{N_T=j\}]$ is the term of index j of the series in (ii).

Step 4 (summing over the count). It remains to show that $\mathbb{E}^{\mathbb{Q}}\,f(S_T)$ is the sum over j of the means of Step 2. This exchanges an infinite sum with an expectation, and we obtain it through the density of S_T. Let $f_{\mathbb{Q}}^{(j)}(x)$ be the lognormal density of Se^{X_j} at the price x, given by Step 1 (the superscript is a label, not a derivative). Let $\mathcal{O}$ be an open interval of $(0,\infty)$. The events $\{N_T=j,\ Se^{X_j}\in\mathcal{O}\}$, $j=0,1,2,\dots$, are disjoint, and their union is the event $\{S_T\in\mathcal{O}\}$. By the countable additivity of P, declared in the Setup of §19.2, and the product rule for the independent N_T and $(Z_0,\dots,Z_j)$ (Lemma 19.17, applied to two indicator functions),

$$\begin{aligned}P(S_T\in\mathcal{O}) &= \sum_{j=0}^{\infty} P(N_T=j)\,P(Se^{X_j}\in\mathcal{O}) = \sum_{j=0}^{\infty} P(N_T=j)\int_{\mathcal{O}} f_{\mathbb{Q}}^{(j)}(x)\,dx\\ &= \int_{\mathcal{O}} \sum_{j=0}^{\infty} P(N_T=j)\,f_{\mathbb{Q}}^{(j)}(x)\,dx,\end{aligned}$$

the last exchange being Fact 18.34(i) in the coordinates x and j, the second of them discrete, for a nonnegative integrand. So S_T has the density $f_{\mathbb{Q}}(x,T)=\sum_j P(N_T=j)\,f_{\mathbb{Q}}^{(j)}(x)$.

Next, a bound. By Lemma 18.35(i) and Step 1, for a real exponent θ,

$$\mathbb{E}\,e^{\theta X_j} = \exp\Big(\theta\,(r-q-\lambda\kappa-\tfrac12\sigma^2)T+\tfrac12\theta^2\sigma^2T\Big)\cdot\Big((1+\kappa)^{\theta}\,e^{\theta(\theta-1)\delta^2/2}\Big)^{j},$$

a constant times the j-th power of a positive number. With the growth bound $|f(x)|\le c\,(x^c+x^{-c})$ and $\theta=\pm c$, there are therefore constants $c'\ge0$ and $a>0$ with $\mathbb{E}\,|f(Se^{X_j})|\le c'\,a^j$ for every j. Each integral $\int_0^\infty |f(x)|\,f_{\mathbb{Q}}^{(j)}(x)\,dx$ is finite: the substitution $x=S\exp\big((r_{(j)}-q-\tfrac12\sigma_{(j)}^2)T+\sigma_{(j)}\sqrt{T}\,z\big)$, which by Step 1 carries the standard normal density to $f_{\mathbb{Q}}^{(j)}$, turns it into

$$\int_{-\infty}^{\infty}\Big|f\Big(S\exp\big((r_{(j)}-q-\tfrac12\sigma_{(j)}^2)T+\sigma_{(j)}\sqrt{T}\,z\big)\Big)\Big|\,\varphi(z)\,dz.$$

This converges absolutely by Lemma 18.35(i), because the function of z inside the absolute value is continuous except at finitely many points and has exponential growth in z (Lemma 19.20(ii), the exponent being an affine function of z). So Fact 7.22 applies to $|f|$ and to f for each density $f_{\mathbb{Q}}^{(j)}$. Fact 18.34(i) for the nonnegative integrand $|f(x)|\,P(N_T=j)\,f_{\mathbb{Q}}^{(j)}(x)$, and the law of the unconscious statistician (Fact 7.22) for each density $f_{\mathbb{Q}}^{(j)}$, give

$$\begin{aligned}\int_0^\infty |f(x)|\,f_{\mathbb{Q}}(x,T)\,dx &= \sum_{j=0}^{\infty} P(N_T=j)\int_0^\infty |f(x)|\,f_{\mathbb{Q}}^{(j)}(x)\,dx = \sum_{j=0}^{\infty} P(N_T=j)\,\mathbb{E}\,|f(Se^{X_j})|\\ &\le c'\,e^{-\lambda T}\sum_{j=0}^{\infty}\frac{(\lambda Ta)^j}{j!} = c'\,e^{(a-1)\lambda T}<\infty.\end{aligned}$$

The integral converges absolutely, so Fact 7.22 applies to $f(S_T)$, and Fact 18.34(i) for an integrand of either sign whose iterated integral of the absolute value has just been shown finite, together with Fact 7.22 for each $f_{\mathbb{Q}}^{(j)}$, gives

$$\begin{aligned}\mathbb{E}^{\mathbb{Q}}\,f(S_T) &= \int_0^\infty f(x)\sum_{j=0}^{\infty} P(N_T=j)\,f_{\mathbb{Q}}^{(j)}(x)\,dx = \sum_{j=0}^{\infty} P(N_T=j)\int_0^\infty f(x)\,f_{\mathbb{Q}}^{(j)}(x)\,dx\\ &= \sum_{j=0}^{\infty} P(N_T=j)\,\mathbb{E}\,f(Se^{X_j}).\end{aligned}$$

By Step 2 the term of index j is $\mathbb{E}^{\mathbb{Q}}[f(S_T)\,\mathbf{1}\{N_T=j\}]$, and by Step 3 its discounted value is the term of index j of the series in (ii). This gives (ii), and the call is the case $f(x)=(x-K)^+$.

Step 5 (the forward: (i)). Apply (ii) to $f(x) = x$, which is in $\mathcal{G}$. The BSM price of the claim paying S_T, with any rate and any volatility, is

$$V^{\mathrm{BSM}}\big(S, 0; \sigma_{(j)}, r_{(j)}\big) = e^{-r_{(j)}T}\, S e^{(r_{(j)} - q - \sigma_{(j)}^2/2)T}\, \mathbb{E}\, e^{\sigma_{(j)}\sqrt{T}\, Z} = S e^{-qT},$$

by Lemma 18.35(i), and the weights $e^{-\lambda' T}(\lambda' T)^j / j!$ sum to one, by the exponential series. So $e^{-rT}\, \mathbb{E}^{\mathbb{Q}}\, S_T = S e^{-qT}$. In words: the factor $e^{-\lambda \kappa T}$ in (23.22) compensates the mean of the product of the jump factors, which is $\sum_j e^{-\lambda T}(\lambda T)^j (1+\kappa)^j / j! = e^{\lambda \kappa T}$. □

Operation	Cost	Qualifier and reason
`L[i]`, `L[i] = x`	$O(1)$	worst case; contiguous pointer array (Fact 26.2(i))
`L.append(x)`	$O(1)$	amortised; geometric resize (Theorem 26.3(i))
`L.pop()`	$O(1)$	amortised; no shift, but the allocation shrinks below half capacity (Fact 26.2(i))
`L.insert(0, x)`, `L.pop(0)`	$\Theta(n)$	worst case; every pointer moves (Theorem 26.3(iii))
`x in L`	$O(n)$	worst case; linear scan
`L[a:b]` of length ℓ	$O(\ell)$	worst case; pointers copied
`T[i]` (tuple)	$O(1)$	worst case; inline pointer array, no spare capacity
`D[k]`, `D[k] = v`, `del D[k]`	$O(1)$ / $O(n)$	expected / worst under collisions (Fact 26.2(ii)); insertion also amortised over resizes
iterate `D`	$O(n)$	worst case; insertion order (3.7 guarantee)
`S.add(x)`, `x in S`	$O(1)$ / $O(n)$	expected / worst; no order (Fact 26.2(iii))
`A | B` (union)	$O(\|A\| + \|B\|)$	expected
`A & B` (intersection)	$O(\min(\|A\|, \|B\|))$	expected; the smaller set is iterated
`dq.append`, `dq.popleft` (both ends)	$O(1)$	worst case; linked blocks (Fact 26.2(iv))
`dq[i]` (middle)	$O(n)$	worst case; block walk
`heappush`, `heappop`	$O(\log n)$	worst case; one root-to-leaf path
`heapify`	$O(n)$	worst case (Theorem 26.7)
`h[0]` (peek)	$O(1)$	worst case; the invariant puts the minimum at the root
`sorted`, `L.sort()`	$O(n \log n)$	worst case; $O(n)$ on presorted input; stable (Fact 26.10)
`bisect_left`	$O(\log n)$	worst case (Proposition 26.14, proved in §26.2)
`insort`	$O(n)$	worst case; the shift dominates the search
`s += t` in a loop	$\Theta(n^2 L)$	total, not guaranteed away (Proposition 26.11, Fact 26.12)
`''.join(pieces)`	$\Theta(nL)$	total length copied once

Table 5: Costs of Python's containers as CPython implements them. Every row names its qualifier from Definition 26.1: "expected" rows average over the randomised hash and can degrade to $O(n)$ on colliding keys; "amortised" rows contain an occasional $O(n)$ resize paid for by the cheap operations around it; "worst case" rows hold for every call.

r	Reading
near 0	the process slept or waited nearly the whole interval: I/O, a lock, a network reply, a *blocking* GPU wait, `sleep`
$0 < r < 1$	one thread that waits part of the time; the waiting fraction is at least $1 - r$
$r < 1$, machine loaded	the machine, not the program: runnable but descheduled; a load average above the core count or many involuntary context switches point at oversubscription or other tenants
near 1	one core saturated: CPU-bound pure Python under the interpreter lock of Fact 26.61, stated in §26.6 — or one host thread spinning on a device wait (Fact 26.35 below)
$\kappa > 1$	κ cores busy on average: BLAS threads, multiprocessing, or lock-releasing extensions
large system share	system-call or page-fault churn: many small reads, allocation, pickling
near C	the machine is saturated; adding workers cannot help (Theorem 26.37, proved in §26.4)

Table 6: Reading the CPU-to-wall ratio r of Theorem 26.26. GPU busy time is not CPU time, but a spinning wait for the GPU is: under the CUDA runtime's default spin-wait a GPU-bound loop reads r near 1 on one core, under a blocking wait it reads r well below 1, and neither reading is conclusive — the device must be profiled separately (Fact 26.35 below).